Simcenter Nastran Quick Reference Guide

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Table of Contents

nastran Command and NASTRAN Statement 1-1
The nastran Command 1-1
The NASTRAN Statement 1-14
NASTRAN 1-16

File Management Statements 2-1
Key to Descriptions 2-1
The File Management Section (FMS) 2-2
File Management Statement Summary 2-3
File Management Statement Descriptions 2-4
WHERE and CONVERT Clauses 2-6
$ 2-8
ACQUIRE 2-9
ASSIGN 2-10
CONNECT 2-16
DBCLEAN 2-18
DBDICT 2-19
DBDIR 2-31
DBFIX 2-33
DBLOAD 2-34
DBLOCATE 2-38
DBSETDEL 2-42
DBUNLOAD 2-43
DBUPDATE 2-46
DEFINE 2-48
ENDJOB 2-51
EXPAND 2-52
INCLUDE 2-54
INIT 2-56
PROJ 2-60
RESTART 2-62

Executive Control Statements 3-1
Key to Descriptions 3-1
Executive Control Section 3-2
Executive Control Statement Summary 3-3
Executive Control Statement Descriptions 3-3
$ 3-5
ALTER 3-6
APP 3-10
CEND 3-11
COMPILE 3-12
COMPILER 3-16

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DIAG 3-18
DOMAINSOLVER 3-22
ECHO 3-23
ENDALTER 3-24
GEOMCHECK 3-25
ID 3-32
INCLUDE 3-33
LINK 3-35
MALTER 3-37
SOL 3-41
SOL 402 3-45
TIME 3-51

Case Control Commands 4-1
Key to Descriptions 4-1
The Case Control Section 4-2
Format of Case Control Command Descriptions 4-3
Case Control Command Summary 4-5
Case Control Command Descriptions 4-17
$ 4-17
A2GG 4-18
ACCELERATION 4-20
ACCONV 4-24
ACINTENSITY 4-25
ACORDCHK 4-27
ACPOTENTIAL 4-28
ACPOWER 4-30
ACTEMP 4-33
ACVELOCITY 4-34
ACVPO4STR 4-36
ADAMSMNF 4-37
ADAPTERR 4-48
ADMRECVR 4-51
AECONFIG 4-54
AEROF 4-55
AESYMXY 4-56
AESYMXZ 4-57
ALOAD 4-58
ANALYSIS 4-59
APRESSURE 4-64
ATVOUT 4-65
AUXCASE 4-67
AUXMODEL 4-68
B2GG 4-69
B2PP 4-71
BC 4-73
BCRESULTS 4-74
BCSET 4-77

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BEGIN BULK 4-80
BGRESULTS 4-82
BGSET 4-84
BOLTLD 4-87
BOLTRESULTS 4-90
BOUTPUT 4-92
BVTSET 4-94
CKGAP 4-95
CLOAD 4-97
CMETHOD 4-98
CRSTRN 4-99
CSMSET 4-101
CSSCHD 4-102
CYCFORCES 4-103
CYCSET 4-105
CZRESULTS 4-106
DAMPING 4-108
DEFORM 4-110
DESGLB 4-111
DESOBJ 4-112
DESSUB 4-116
DISPLACEMENT 4-117
DIVERG 4-121
DLOAD 4-122
DMTRCOEF 4-124
DMTRLOSS 4-126
DRSPAN 4-128
DSAPRT 4-130
DSYM 4-132
DTEMP 4-133
ECHO 4-134
EDE 4-136
EFLOAD 4-139
EKE 4-141
ELAR 4-145
ELAROUT 4-147
ELSDCON 4-148
ELSTRN 4-150
ELSUM 4-154
ENTHALPY 4-156
ERP 4-157
ESE 4-164
ESHLPNL 4-169
EXTSEOUT 4-171
FHAC 4-181
FHAR 4-184
FLSFSEL 4-186
FLSPOUT 4-188
FLSTCNT 4-190

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FLUX 4-193
FLXSLI 4-195
FLXRESULTS 4-196
FMETHOD 4-197
FMH3D 4-198
FORCE 4-199
FREQUENCY 4-203
FRFIN 4-204
FRFOUT 4-205
GCRSTRN 4-208
GELSTRN 4-210
GKRESULTS 4-212
GMESTRN 4-215
GPFORCE 4-217
GPKE 4-221
GPLSTRN 4-223
GPRSORT 4-225
GPSDCON 4-226
GPSTRAIN 4-227
GPSTRESS 4-229
GRDCON 4-231
GROUNDCHECK 4-234
GSTRAIN 4-236
GSTRESS 4-238
GTHSTRN 4-240
GUST 4-242
HARCOEFF 4-243
HARMONICS 4-245
HARTAB 4-247
HDOT 4-249
HOUTPUT 4-250
IC 4-252
IMPERF 4-255
INCLUDE 4-256
INITS 4-258
INPOWER 4-260
JCONSET 4-263
JRESULTS 4-264
JINTEG 4-265
K2GG 4-266
K2PP 4-268
K42GG 4-270
LABEL 4-272
LINE 4-273
LOAD 4-274
LOADSET 4-276
M2GG 4-278
M2PP 4-280
MASTER 4-282

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MAXLINES 4-284
MAXMIN..... 4-285
MBDEXPORT..... 4-288
MBDRECVR..... 4-312
MEFFMASS..... 4-316
MESTRN..... 4-321
METHOD 4-323
MFLUID 4-325
MLOAD..... 4-326
MODALE 4-327
MODCON..... 4-331
MODES 4-334
MODSEL..... 4-336
MODTRAK 4-338
MONITOR..... 4-339
MONVAR..... 4-340
MPC 4-342
MPCFORCES..... 4-343
MPRES 4-346
NLARCL..... 4-348
NLCNTL 4-349
NLLOAD 4-350
NLPARM 4-351
NLSTRESS..... 4-352
NONLINEAR 4-354
NOUTPUT 4-355
NSM..... 4-356
OFREQUENCY 4-357
OLOAD..... 4-359
OMEGA..... 4-363
OMODES 4-364
OPRESS..... 4-366
OSTNINI..... 4-368
OTEMP..... 4-370
OTIME 4-372
OTMFORC..... 4-374
OUTMGT..... 4-376
OUTPUT 4-377
P2G..... 4-379
PANCON..... 4-381
PARAM 4-384
PARTN 4-386
PEAKOUT..... 4-387
PFRESULTS..... 4-388
PLOTID 4-390
PLSTRN..... 4-391
PRESSURE..... 4-393
RANDOM..... 4-396
RCROSS..... 4-400

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REPCASE 4-402
RESVEC 4-403
RIGID 4-406
RMAXMIN 4-410
RMETHOD 4-413
RSAPPLY 4-414
RSMETHOD 4-415
SACCELERATION 4-417
SDAMPING 4-419
SDISPLACEMENT 4-420
SEALL 4-422
SEDR 4-424
SEDV 4-426
SEEXCLUDE 4-428
SEFINAL 4-430
SEKETAB 4-432
SEKREDUCE 4-434
SELGENERATE 4-436
SELREDUCE 4-438
SEMGENERATE 4-440
SEMREDUCE 4-442
SEQDEP 4-444
SERESP 4-446
SET 4-448
SET 4-451
SETMC 4-454
SETMCNAME 4-456
SETS DEFINITION 4-458
SFORCE 4-459
SHELLTHK 4-461
SKIP 4-462
SMETHOD 4-463
SPC 4-464
SPCFORCES 4-465
STATSUB 4-469
STATVAR 4-471
STRAIN 4-473
STRESS 4-479
STRFIELD 4-486
SUBCASE 4-488
SUBCOM 4-489
SUBSEQ 4-491
SUBTITLE 4-493
SUPER 4-494
SUPORT1 4-496
SURFACE 4-497
SVECTOR 4-500
SVELOCITY 4-502
SYM 4-504

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SYMCOM 4-505
SYMSEQ 4-507
TEMPERATURE 4-508
TFL 4-513
THERMAL 4-515
THSTRN 4-517
TITLE 4-520
TRIM 4-521
TRLOSS 4-522
TRPOWER 4-524
TSTEP 4-527
TSTEPNL 4-528
TSTRU 4-529
ULOAD 4-531
VATVOUT 4-532
VELOCITY 4-534
VOLUME 4-538
WEIGHTCHECK 4-540
Case Control Applicability Tables 4-542

Case Control X-Y PLOT Commands 5-1

Case Control X-Y PLOT Commands Overview 5-1
X-Y Output Command Summary 5-3
ALLEDGE TICS 5-6
BALL EDGE TICS 5-7
BLEFT TICS 5-8
BRIGHT TICS 5-9
CAMERA 5-10
CLEAR 5-11
CSCALE 5-12
CURVELINESYMBOL 5-13
DENSITY 5-15
LEFT TICS 5-16
LONG 5-17
LOWER TICS 5-18
PENSIZE 5-19
PLOTTER 5-20
RIGHT TICS 5-21
TALL EDGE TICS 5-22
TCURVE 5-23
TLEFT TICS 5-24
TRIGHT TICS 5-25
UPPER TICS 5-26
XAXIS 5-27
XBAXIS 5-28
XBGRID LINES 5-29
XDIVISIONS 5-30
XGRID LINES 5-31

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XINTERCEPT 5-32
XLOG 5-33
XMAX 5-34
XMIN 5-35
XPAPER 5-36
XTGRID LINES 5-37
XTITLE 5-38
XYPAPLOT 5-39
XYPEAK 5-40
XYPLOT 5-41
XYPRINT 5-45
XYPUNCH 5-46
XVALUE PRINT SKIP 5-47
YAXIS 5-48
YBDIVISIONS 5-49
YBINTERCEPT 5-50
YBGRID LINES 5-51
YBLOG 5-52
YBMAX 5-53
YBMIN 5-54
YBTITLE 5-55
YBVALUE PRINT SKIP 5-56
YDIVISIONS 5-57
YINTERCEPT 5-58
YGRID LINES 5-59
YLOG 5-60
YMAX 5-61
YMIN 5-62
YPAPER 5-63
YTAXIS 5-64
YTDIVISIONS 5-65
YTGRID LINES 5-66
YTINTERCEPT 5-67
YTITLE 5-68
YTLOG 5-69
YTMAX 5-70
YTMIN 5-71
YTTITLE 5-72
YTVALUE PRINT SKIP 5-73
YVALUE PRINT SKIP 5-74

Parameters and Parameter Applicability Tables 6-1

Parameter Descriptions 6-1

A – Parameters 6-2
B – Parameters 6-11
C – Parameters 6-15
D – Parameters 6-26
E – Parameters 6-32
F – Parameters 6-38

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G – Parameters....6-45
H – Parameters....6-48
I – Parameters....6-49
K – Parameters....6-52
L – Parameters....6-55
M – Parameters....6-58
N – Parameters....6-66
O – Parameters....6-72
P – Parameters....6-81
Q & R – Parameters....6-95
S – Parameters....6-104
T – Parameters....6-118
U – Parameters....6-122
V – Parameters....6-128
W & X – Parameters....6-132
Parameter Applicability Tables....6-139

Item Codes....7-1
Item Codes Overview....7-1
Element Stress (or Strain) Item Codes....7-3
Element Force Item Codes....7-70
Fluid Virtual Mass Pressure Item Codes....7-91
Slideline Contact Item Codes....7-92
Element Strain Energy Item Codes....7-93

Degree-of-Freedom Sets....8-1
Degree-of-Freedom Set Definitions....8-1
Degree-of-Freedom Set Bulk Data Entries....8-4
User Defined Degree-of-Freedom Sets....8-5

Bulk Entries....9-1
Key to Descriptions....9-1
The Bulk Data Section....9-1
Bulk Entry Descriptions....9-2
Bulk Data Syntax Rules....9-3
Continuations....9-7
Bulk Entry Summary....9-12
Geometry....9-13
Elements....9-16
Materials....9-27
Constraints and Partitioning....9-31
Loads....9-34
Solution Control....9-39
Miscellaneous....9-45
Element Utilization Summary....9-47
Key to Element Applicability Table....9-49

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Element Applicability Table..... 9-50
Element-Property-Material Cross-Reference Tables.....9-52
$.....9-60
/.....9-61

Bulk Entries A—B..... 10-1
ACADAPT..... 10-1
ACCEL.....10-2
ACCEL1.....10-4
ACCONV..... 10-6
ACCONVD.....10-7
ACDUCT.....10-8
ACFAN.....10-10
ACMODL.....10-12
ACMODPE..... 10-17
ACNDUCT..... 10-20
ACORDER..... 10-21
ACPLNW..... 10-22
ACPNVEL..... 10-24
ACPOLE1..... 10-26
ACPOLE2..... 10-29
ACPRESS.....10-31
ACRADS.....10-33
ACSPO2..... 10-35
ACSRCE..... 10-37
ACTEMP..... 10-39
ACTRAD.....10-41
ACTRTM..... 10-43
ACUPOLE.....10-44
ACVPO4.....10-45
ACVSTT..... 10-47
ADUMi 10-50
AECOMP 10-51
AECOMPL 10-53
AEDW 10-54
AEFACT 10-55
AEFORCE 10-56
AELINK 10-58
AELIST 10-60
AEPARM 10-61
AEPRESS 10-62
AERO 10-63
AEROS 10-65
AESTAT 10-67
AESURF 10-69

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AESURFS 10-71
ALOAD 10-72
AMLREG 10-73
ASET 10-75
ASET1 10-76
ATVBK 10-78
ATVBULK 10-80
ATVFS 10-81
BAROR 10-82
BCONP 10-84
BCPROP 10-87
BCPROPS 10-88
BCRPARA 10-89
BCTADD 10-91
BCTPAR2 10-93
BCTPARM 10-109
BCTSET 10-126
BDYOR 10-130
BEAMOR 10-132
BEDGE 10-134
BFRIC 10-136
BGADD 10-138
BGPARM 10-139
BGSET 10-146
BLSEG 10-150
BNDFIX 10-152
BNDFIX1 10-154
BNDFREE 10-156
BNDFREE1 10-158
BNDGRID 10-160
BOLT 10-162
BOLTFRC 10-169
BOLTLD (bulk entry) 10-172
BOLTSEQ 10-175
BOUTPUT 10-177
BSET 10-179
BSET1 10-181
BSURF 10-183
BSURFS 10-185
BVTADD 10-187
BVTPAR 10-189
BVTSET 10-195
BWIDTH 10-199

Bulk Entries CA—CM 11-1

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CAABSF 11-1
CAERO1 11-3
CAERO2 11-6
CAERO3 11-8
CAERO4 11-10
CAERO5 11-12
CBAR 11-16
CBARAO 11-22
CBEAM 11-24
CBEAR 11-29
CBEAR2 11-30
CBEND 11-31
CBUSH 11-34
CBUSH1D 11-39
CBUSH2 11-41
CCHOCK3 11-42
CCHOCK4 11-45
CCHOCK6 11-48
CCHOCK8 11-51
CDAMP1 11-54
CDAMP2 11-56
CDAMP3 11-58
CDAMP4 11-59
CDAMP5 11-60
CDUMi 11-61
CELAS1 11-62
CELAS2 11-64
CELAS3 11-66
CELAS4 11-68
CFAST 11-70
CGAP 11-72
CGEAR 11-75
CHBDYE 11-76
CHBDYG 11-81
CHBDYP 11-85
CHEXA 11-89
CHEXCZ 11-93
CJOINT 11-95
CLINK 11-97
CLOAD 11-100
CMASS1 11-102
CMASS2 11-104
CMASS3 11-106
CMASS4 11-107

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Bulk Entries CO—CY 12-1
CONM1 12-1
CONM2 12-3
CONROD 12-6
CONV 12-8
CONVM 12-10
CORD1C 12-12
CORD1R 12-14
CORD1S 12-16
CORD2C 12-18
CORD2R 12-20
CORD2S 12-22
CORD3G 12-24
CPENTA 12-26
CPENTCZ 12-29
CPLSTN3 12-31
CPLSTN4 12-34
CPLSTN6 12-37
CPLSTN8 12-40
CPLSTS3 12-43
CPLSTS4 12-47
CPLSTS6 12-51
CPLSTS8 12-55
CPYRAM 12-59
CQUAD 12-61
CQUAD4 12-63
CQUAD8 12-67
CQUADR 12-75
CQUADX4 12-79
CQUADX8 12-82
CRAC2D 12-85
CRAC3D 12-87
CRAKTP 12-91
CREEP 12-92
CROD 12-97
CSET 12-99
CSET1 12-101
CSHEAR 12-103
CSMADD 12-105
CSMSET 12-106
CSSCHD 12-107
CSUPER 12-109
CSUPEXT 12-111
CTETRA 12-113
CTRAX3 12-116

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CTRAX6 12-119
CTRIA3 12-122
CTRIA6 12-126
CTRIAR 12-133
CTUBE 12-136
CVISC 12-137
CWELD 12-138
CYAX 12-148
CYCADD 12-150
CYCAXIS 12-152
CYCISC 12-153
CYCSET 12-155
CYJOIN 12-159
CYSUP 12-161
CYSYM 12-163

Bulk Entries D—E 13-1
DAMPING 13-1
DAMPTAB 13-4
DAREA 13-7
DCONADD 13-9
DCONSTR 13-10
DDAMCTR 13-13
DDVAL 13-16
DEFORM 13-18
DEFUSET 13-19
DELAY 13-20
DEQATN 13-21
DESC 13-26
DESVAR 13-27
DIVERG 13-29
DLINK 13-30
DLOAD 13-32
DMI 13-35
DMIG 13-40
DMIG,UACCEL 13-45
DMIJ 13-47
DMIJI 13-51
DMIK 13-55
DMNCON 13-59
DMRLAW 13-64
DOPTPRM 13-67
DPHASE 13-73
DRESP1 13-74
DRESP2 13-85

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DRESP3 13-90
DRIVER 13-94
DSCREEN 13-96
DTABLE 13-98
DTEMP 13-99
DTEMPEX 13-101
DTI 13-103
DTI,DFRFNC 13-105
DTI,DISTL 13-107
DTI,ESTDATA 13-108
DTI,INDTA 13-112
DTI,SELOAD 13-115
DTI,SETREE 13-117
DTI,SPECSEL 13-119
DTI,SPSEL 13-120
DTI,UNITS 13-122
DVBSHAP 13-126
DVCREL1 13-127
DVCREL2 13-129
DVEREL1 13-131
DVGRID 13-134
DVMREL1 13-136
DVMREL2 13-138
DVPREL1 13-140
DVPREL2 13-143
DVSHAP 13-146
DVTREL1 13-148
ECHOOFF 13-151
ECHOON 13-152
EIGB 13-153
EIGC 13-156
EIGP 13-166
EIGR 13-167
EIGRL 13-172
ELAR 13-178
ELAR2 13-182
ELARADD 13-187
ELIST 13-188
ENDDATA 13-189
EPOINT 13-190
EXTRN 13-191

Bulk Entries F—L 14-1
FLFACT 14-1
FLUTTER 14-3

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FLXADD....14-5
FLXSLI....14-6
FMH3D....14-8
FORCDST....14-11
FORCE....14-14
FORCE1....14-16
FORCE2....14-18
FOU3....14-20
FREQ....14-21
FREQ1....14-23
FREQ2....14-25
FREQ3....14-27
FREQ4....14-30
FREQ5....14-33
FREQH....14-35
FREQV....14-37
FRFFLEX....14-38
FRFOMAP....14-42
FRFOTM....14-44
FRFSTIF....14-48
GENEL....14-52
GRAV....14-56
GRDSET....14-58
GRID....14-60
GROUP....14-62
GUST....14-64
INCMOPT....14-66
INCMPFL....14-70
IMPERF....14-72
IMPRADD....14-74
INCLUDE....14-76
INITADD....14-78
INITS....14-79
INITSO....14-83
IPLANE....14-86
ITER....14-88
JCON....14-93
JCONADD....14-95
LOAD....14-96
LOADCYH....14-98
LOADCYN....14-100
LOADCYT....14-102
LSEQ....14-104

Bulk Entries M—N....15-1

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MAT1....15-1
MAT1F....15-5
MAT2....15-7
MAT2F....15-10
MAT3....15-12
MAT4....15-15
MAT5....15-17
MAT8....15-19
MAT9....15-22
MAT9F....15-24
MAT10....15-26
MAT10C....15-28
MAT11....15-30
MATCID....15-32
MATCRP....15-34
MATCZ....15-37
MATDMG....15-40
MATF10C....15-45
MATFT....15-46
MATG....15-50
MATHE....15-53
MATHEM....15-58
MATHEV....15-59
MATHP....15-62
MATOVR....15-66
MATPOR....15-67
MATS1....15-71
MATSR....15-79
MATT1....15-81
MATT2....15-84
MATT3....15-86
MATT4....15-88
MATT5....15-90
MATT8....15-92
MATT9....15-95
MATT11....15-97
MATTC....15-100
MATVST....15-101
MDSBULK....15-102
MFLUID....15-104
MICPNT....15-106
MKAERO1....15-107
MKAERO2....15-109
MLINK....15-110
MODTRAK....15-113

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MOMENT....15-115
MOMENT1....15-117
MOMENT2....15-119
MONPNT1....15-121
MONPNT2....15-123
MONPNT3....15-161
MPC....15-164
MPCADD....15-166
MPLAS....15-167
MUCRP....15-186
MUMAT....15-189
MVPLAS....15-191
NLARCL....15-199
NLCNTL....15-201
NLCNTL2....15-221
NLCNTLG....15-237
NLPARM....15-247
NLPCI....15-253
NLRGAP....15-256
NOLIN1....15-258
NOLIN2....15-260
NOLIN3....15-262
NOLIN4....15-264
NSM....15-266
NSM1....15-268
NSMADD....15-270
NSML....15-271
NSML1....15-273

Bulk Entries O—P....16-1
OMEGA....16-1
OMEGA1....16-3
OMEGA2....16-5
OMIT....16-7
OMIT1....16-8
OUTMGT....16-10
PAABSF....16-17
PAABSF1....16-19
PACDUCT....16-21
PACFAN....16-23
PACSPO2....16-24
PACTRAD....16-25
PACTRTM....16-27
PACVPO4....16-29
PAERO1....16-31

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PAERO2 16-32
PAERO3 16-34
PAERO4 16-36
PAERO5 16-39
PANEL..... 16-43
PARAM 16-45
PBAR..... 16-47
PBARL.....16-51
PBCOMP 16-57
PBEAM..... 16-62
PBEAML.....16-68
PBEAR..... 16-75
PBEAR2..... 16-85
PBEND 16-99
PBUSH.....16-105
PBUSH1D.....16-108
PBUSH2.....16-113
PBUSHT.....16-119
PCHOCK..... 16-122
PCOMP.....16-124
PCOMPG.....16-131
PCOMPG1.....16-138
PCOMPS..... 16-142
PCONV..... 16-148
PCONVM..... 16-150
PDAMP 16-152
PDAMP5 16-153
PDAMPT 16-154
PDUMi 16-155
PEAKOUT.....16-156
PELAS.....16-160
PELAST 16-162
PFAST.....16-164
PGAP 16-166
PGEAR.....16-170
PGPLSN..... 16-173
PHBDY 16-175
PJOINT.....16-177
PJOINT2.....16-179
PLOAD.....16-181
PLOAD1.....16-184
PLOAD2 16-187
PLOAD4.....16-189
PLOADE1.....16-192
PLOADFP 16-197

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PLOADX1 16-201
PLOTEL 16-205
PLOTEL3 16-206
PLOTEL4 16-207
PLOTEL6 16-208
PLOTEL8 16-209
PLOTHEX 16-210
PLOTPEN 16-211
PLOTPYR 16-212
PLOTTET 16-213
PLPLANE 16-214
PLSOLID 16-216
PMASS 16-217
PMIC 16-218
POINT 16-219
PPLANE 16-220
PRAC2D 16-223
PRAC3D 16-225
PROD 16-226
PSHEAR 16-228
PSHELL 16-231
PSHL3D 16-235
PSHLPNL 16-236
PSOLCZ 16-241
PSOLID 16-244
PTUBE 16-255
PVISC 16-257
PWELD 16-258

Bulk Entries Q—S 17-1
QBDY1 17-1
QBDY2 17-3
QBDY3 17-5
QHBDY 17-7
QSET 17-9
QSET1 17-10
QVECT 17-12
QVOL 17-15
RADBC 17-17
RADBND 17-19
RADCAV 17-21
RADLST 17-23
RADM 17-25
RADMT 17-27
RADMTX 17-28

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RADSET 17-29
RANDPEX 17-30
RANDPS 17-32
RANDT1 17-34
RBAR 17-35
RBE1 17-37
RBE2 17-39
RBE3 17-41
RCROSS 17-44
RCROSSC 17-46
RDMODES 17-48
RELEASE 17-50
RLCYC1 17-52
RLCYC2 17-55
RFORCE 17-58
RFORCE1 17-64
RFORCE2 17-71
RLHARM 17-75
RLOAD1 17-77
RLOAD2 17-81
RLOADEX 17-85
ROTORB 17-88
ROTORD 17-90
ROTORG 17-97
ROTPARM 17-99
ROTSE 17-101
RROD 17-103
RSAPPLY 17-105
RSPECTR 17-107
RSPLINE 17-109
RSPLOC 17-111
RSPOPT 17-113
RSSCON 17-116
RTRPLT 17-119
RVDOF 17-121
RVDOF1 17-122
RVEL 17-124
SEBNDRY 17-126
SEBSET 17-127
SEBSET1 17-129
SEBULK 17-131
SECONCT 17-133
SECSET 17-135
SECSET1 17-137
SEDAMP 17-139

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SEELT 17-141
SEEXCLD 17-143
SELABEL 17-144
SELMODE 17-145
SELOAD 17-147
SELOC 17-149
SEMAT 17-151
SEMCF 17-152
SEMPLN 17-154
SENAME 17-155
SENQSET 17-156
SENSCLC 17-157
SEOUT 17-160
SEQEP 17-162
SEQGP 17-163
SEQSEP 17-164
SEQSET 17-166
SEQSET1 17-168
SESET 17-170
SESUP 17-172
SET1 17-173
SET2 17-175
SET3 17-177
SETREE 17-179
SEUNIT 17-181
SEUSET 17-182
SEUSET1 17-184
SLOAD 17-186
SLVCNTL 17-187
SNORM 17-189
SPC 17-191
SPC1 17-193
SPCADD 17-195
SPCD 17-196
SPCF 17-199
SPCOFF 17-202
SPCOFF1 17-203
SPDIR 17-205
SPEBE2 17-206
SPLINE1 17-209
SPLINE2 17-211
SPLINE3 17-213
SPLINE4 17-216
SPLINE5 17-218
SPOINT 17-221

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SUPORT....17-223
SUPORT1....17-225
SWLDPRM....17-227

Bulk Entries T—V....18-1
TABDMP1....18-1
TABLED1....18-4
TABLED2....18-8
TABLED3....18-11
TABLED4....18-14
TABLED6....18-16
TABLEH2....18-20
TABLEM1....18-22
TABLEM2....18-25
TABLEM3....18-28
TABLEM4....18-31
TABLEM5....18-33
TABLES1....18-36
TABLEST....18-38
TABRND1....18-41
TABRNDG....18-44
TEMP....18-46
TEMPBC....18-48
TEMPD....18-50
TEMPEX....18-52
TEMPP1....18-54
TEMPP3....18-57
TEMPRB....18-58
TF....18-61
TIC....18-63
TLOAD1....18-65
TLOAD2....18-69
TLOAD3....18-72
TRIM....18-74
TSTEP....18-76
TSTEP1....18-78
TSTEPNL....18-82
UNBA....18-87
USET....18-89
USET1....18-91
UXVEC....18-93
VATVBK....18-94
VATVFS....18-95
VCEV....18-96
VIEW....18-98

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VIEW3D 18-100
VISGRP 18-102

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1. nastran Command and NASTRAN Statement

The nastran Command

Simcenter Nastran is executed with a command called nastran. (Your system administrator may assign a different name to the command.) The nastran command permits the specification of keywords used to request options affecting a Simcenter Nastran job execution. The format of the nastran command is:

```
nastran input_data_file [keyword1 = value1 keyword2 = value2 ...]
```

where input_data_file is the name of the file containing the input data and $keyword_i$=$value_i$ is one or more optional keyword assignment arguments. For example, to run a job using the data file example1.dat, enter the following command:

```
nastran example1
```

Keyword assignments can be specified as command line arguments and/or included in RC files.

In addition to the RC file located in *install_dir*\conf\nastr.rcf, there are two local RC file options.

- You can create an RC file in your home directory.
- You can create an RC file in the same directory as the input file. This RC file is ignored when the “rcf” keyword is used.

The following syntax rules exist for RC files.

- The tilde (~) character is not recognized within RC files.
- Environment variables are only recognized when used in the context of a logical symbol (see “Using Filenames and Logical Symbols” in the *Simcenter Nastran Installation and Operations Guide*).
- When a keyword is specified on the command line, embedded spaces or special characters that are significant to the shell must be properly quoted; quotes should not be used within RC files.

The keywords listed below are the most common ones for various computers, but they are not available on all computers. Also, the defaults may be site dependent. Please consult the **Keywords and Environment Variables** in the *Simcenter Nastran Installation and Operations Guide* for keyword applicability, specialized keywords, and further discussion of keywords and their defaults. Also see the *Simcenter Nastran Parallel Processing User’s Guide* for details on all parallel processing keywords. Keywords that use yes/no values accept partial specification and case-independent values. For example, “yes” may be specified as “y”, “ye”, or “yes” using uppercase or lowercase letters.

after	after=time	Default: None

Holds the job until the time specified by time. See the description of the “at” command in your system documentation for the format of time.

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Example: `nastran example after=10:00`

The job is held until 10:00 A.M.

altdcmp altdcmp={"bcslib","pardiso"} Default: None

Selects an alternate sparse solver to be used instead of the standard sparse solver. Choices include BCSLIB or PARDISO. The system cells 429 or 555 can also be used to request the BCSLIB or PARDISO solvers, respectively. The keyword altdcmp takes precedence over the system cell settings.

Example: `nastran example altdcmp=pardiso`

append append={yes|no} Default: no

Controls the combining of the F04, F06, and LOG files into a single file after the run completes. If “no” is specified, the files are not combined. If “yes” is specified, the files are combined into one file with the suffix “.out”.

Example: `nastran example append=yes`

The F04, F06, and LOG files are combined into a file named example.out.

authqueue authqueue=*number of minutes* Default: 0

When Simcenter Nastran attempts to run, but cannot because a license is unavailable, the job will be retried every minute up to the value of the AUTHQUEUE keyword.

batch batch={yes|no} Default: yes

Indicates how the job is to be run. If “yes” is specified, the job is run as a background process. If “no” is specified, the job is run in the foreground. If the “aft” or “queue” keywords are specified, the batch keyword is ignored.

> **Note**
> If the job is already running in an NQS or NQE batch job, the default is “no”.

Example: `nastran example batch=no`

The job is run in the foreground.

buffpool buffpool=*value* The default is 51 buffers. The value defined in the installation RC file is buffpool=20X.

Specifies the memory size of the buffer pool. The buffer pool is a portion of the total memory used to cache database memory. *value* can be defined as a percentage of the total memory requested with the keyword **memory**, or as a specific value.

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- To define as a percentage of the total memory requested with the keyword **memory**, enter

  `buffpool=<n>X`

  where *n* is a percentage 0<*n*<100, and X represents the total memory requested with the keyword **memory**. For example, if **memory**=5Gb is defined, and **buffpool**=20X is defined, the memory requested for the buffer pool is 20% of 5Gb, or 1Gb.
- When *value* is defined as a specific value, you either enter the number of buffers (default), or a number followed by one of the modifiers below. In the conversion column, the *bytes_per_word* is 8. The modifier may be specified using any case combination.
- The buffpool keyword is ignored by SOLs 402 and 414.

Modifier	**Conversion to number of words**
G or Gw	Multiply *entered_value* by 1024**3.
Gb	Multiply *entered_value* by (1024**3)/*bytes_per_word*.
M or Mw	Multiply *entered_value* by 1024**2.
Mb	Multiply *entered_value* by (1024**2)/*bytes_per_word*.
K or Kw	Multiply *entered_value* by 1024.
Kb	Multiply *entered_value* by 1024/*bytes_per_word*.
w	Use *entered_value* as is.
b	Divide *entered_value* by *bytes_per_word*.

where *bytes_per_word* is 8. The modifier may be specified using any case combination.

Example: `nastran example buffpool=200`

This example requests 200 buffers for the buffer pool since no modifier exists.

buffsize =*value* The default is 8193 words. The value defined in the installation RC file is buffsize=32769.

Specifies the memory size of a buffer in words.

cl_dcmp =*value*

Enables GPU for FRRD1 module computations.

Example: `nastran.exe cl_dcmp=1 input_file.dat`

=1 Enables GPU for DCMP module computations with the first available AMD or NVIDIA GPU.

=2 Enables GPU for DCMP module computations with the first available AMD GPU.

=1 and sys531=1 Enables GPU for DCMP module computations with the first available NVIDIA GPU (both cl_dcmp=1 and sys531=1 are required for this option).

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cl_frrd =*value*

Enables GPU for FRRD1 module computations.

Example: `nastran.exe cl_frrd=1 input_file.dat`

=1 Enables GPU for FRRD1 module computations with the first available AMD or NVIDIA GPU.

=2 Enables GPU for FRRD1 module computations with the first available AMD GPU.

=3 Enables GPU for FRRD1 module computations with the first available NVIDIA GPU.

coremem coremem=*value* The USER OPENCORE is computed by the software by default.

For SOL 401 with the MUMPS solver only, memory to reserve for the USER OPENCORE memory, where *value* can be defined as a percentage of the total memory requested with the keyword **memory**, or as a specific value.

When you request the MUMPS solver, by default, Nastran allocates a percentage of the USER OPENCORE memory to MUMPS. You can use the COREMEM keyword to control how much memory is allocated to MUMPS, and to allocate memory for retaining stiffness data.

See MUMPS memory option (SOL 401) in the Multi-Step Nonlinear User's Guide.

- To define as a percentage of the total memory requested with the keyword **memory**, enter

  `coremem=<`*n*`>X`

  where *n* is a percentage 0<*n*<100, and X represents the total memory requested with the keyword **memory**. For example, if **memory**=5Gb is defined, and **coremem**=20X is defined, the memory requested for USER OPENCORE memory is 20% of 5Gb, or 1Gb.
- When *value* is defined as a specific value, you enter a number followed by one of the modifiers below. In the conversion column, the *bytes_per_word* is 8. The modifier may be specified using any case combination.

Modifier	**Conversion to number of words**
G or Gw	Multiply *entered_value* by 1024**3.
Gb	Multiply *entered_value* by (1024**3)/*bytes_per_word*.
M or Mw	Multiply *entered_value* by 1024**2.
Mb	Multiply *entered_value* by (1024**2)/*bytes_per_word*.
K or Kw	Multiply *entered_value* by 1024.
Kb	Multiply *entered_value* by 1024/*bytes_per_word*.
w	Use *entered_value* as is.
b	Divide *entered_value* by *bytes_per_word*.

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where *bytes_per_word* is 8. The modifier may be specified using any case combination.

Examples:

The following example requests 30 percent of the total memory for USER OPENCORE memory.

```
nastran example coremem=30X
```

The following example requests a USER OPENCORE memory size of 25 megawords, or 25 600 kilowords, or 26 214 400 words.

```
nastran example coremem=25mw
```

Note: When you define **coremem** as a percentage of the total memory or in units of words or bytes, the software will convert the value to a *number of buffers*. As a result, the value you enter will be adjusted such that **coremem** / **buffsize** is an integer. The value used by the software is reported in the .f04 file. See "Managing Memory" in the *Configuration and Operations Guide*.

dbs dbs=*pathname* Default=input_data_file directory

Creates database files using an alternate file prefix. (See "Default DBsets and Their Default Attribute" in the *Simcenter Nastran Installation and Operations Guide*.) If "dbs" is not specified, database files are created in the current directory using the base name of the input data file as the prefix. If the "dbs" value is a directory, database files are created in the specified directory using the base name of the input data file as the filename.

Note

If "dbs" is specified and "scratch=yes" is specified, a warning will be issued and "scratch=no" assumed. This is a change from the behavior of prior releases, which would ignore the "dbs" keyword if "scratch=yes" was set.

In the following examples, assume the following directory contents:

```
example.dat mydir/ other/ other/example.dat
```

where "mydir" and "other" are directories.

Example: `nastran example`

or: `nastran other/example`

Database files are created in the current directory with name example, e.g., ./example.DBALL.

Example: `nastran example dbs=myfile`

Database files are created in the current directory with name myfile, e.g., ./myfile.DBALL.

Example: `nastran example dbs=mydir`

Database files are created in the mydir directory with name example, e.g., mydir/example.DBALL.

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Example: `nastran example dbs=mydir/myfile`

Database files are created in the mydir directory with name myfile, e.g., mydir/myfile.DBALL.

dmparallel dmparallel=value Default=0

Distributed memory parallel (DMP) processing selection.

Specifies the number of tasks for a Distributed Memory Parallel (DMP) analysis.

The value must be null or zero to cancel DMP processing, or an integer greater than zero to enable DMP processing. This value may only be set on the command line. See the *Parallel Processing User's Guide*.

For distributed-memory parallel (DMP) processing in SOL 402 and SOL 414, see the keyword **mpi402**.

For distributed-memory parallel (DMP) processing in SOL 414, see the keyword mpi414 and Running a rotor dynamics simulation in parallel.

gpgpu =*value*

Enables GPU for both FRRD1 and DCMP module computations.

Example: `nastran.exe gpgpu=any input_file.dat`

=none Disables GPU for both FRRD1 and DCMP module computations.

=any Enables GPU for both FRRD1 and DCMP module computations with the first available AMD or NVIDIA GPU.

=amd Enables GPU for both FRRD1 and DCMP module computations with the first available AMD GPU.

krylov krylov={yes|no} Default=no

Enables Padé via Lanczos (PVL) approach in SOL 108 for fast direct frequency response analysis.

Example: `$ nastran example krylov=yes`

memory memory=*memory_size* Default=estimate.
The value defined in the installation RC file is 0.45*physical, where physical is the total machine physical memory.

Specifies the amount of open core memory to allocate, where *memory_size* can be defined as a fraction of the machine physical memory, as "estimate", or as a specific value.

- To define *memory_size* as a fraction of the machine physical memory, enter:

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`memory=<n>*physical`

where *n* is a fraction 0<*n*<1.0.

- If memory=estimate is defined, ESTIMATE will be used to determine the memory size.
- When *memory_size* is defined as a specific value, you either enter the number of words, or a number followed by one of the modifiers below. In the conversion column, the *bytes_per_word* is 8. The modifier may be specified using any case combination.
- For SOL 402, the total memory requested is allocated for the input data processing and for the computational part. The details of this allocation is reported in the .f06 file. In this output, the memory conversion from Mb to the number of words is *memory_size* * 1000**2/*bytes_per_word*.

Modifier	**Conversion to number of words**
G or Gw	Multiply *memory_size* by 1024**3.
Gb	Multiply *memory_size* by (1024**3)/*bytes_per_word*.
M or Mw	Multiply *memory_size* by 1024**2.
Mb	Multiply *memory_size* by (1024**2)/*bytes_per_word*.
K or Kw	Multiply *memory_size* by 1024.
Kb	Multiply *memory_size* by 1024/*bytes_per_word*.
w	*memory_size* is the number of words.
b	Divide *memory_size* by *bytes_per_word*.
Example:	`nastran example memory=25mw` The job is run using an open core memory size of 25 megawords, or 25 600 kilowords, or 26 214 400 words.

The maximum *memory_size* limits are included in the *Simcenter Nastran Instillation and Operations Guide* in the section titled “Maximum Memory Size”.

mpi402 mpi402=value Default=0

For SOL 402 and SOL 414 solutions only.

For SOL 414 solutions, you can use the mpi414 keyword or the mpi402 keyword, but you cannot use them both in the solution.

Specifies the number of CPUs selected for distributed-memory parallel (DMP) processing. The default is 0, which specifies no DMP processing.

For SOLs 414,110 and 414,111, use mpi402 to request DMP processing for simultaneous calculations of different rotation speeds or excitation frequencies.

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mpi414 mpi414=value Default=0

For SOL 414 solutions only.

For SOL 414 solutions, you can use the mpi414 keyword or the mpi402 keyword, but you cannot use them both in the solution.

Specifies the number of CPUs selected for distributed-memory parallel (DMP) processing. The default is 0, which specifies no DMP processing.

The mpi402 and mpi414 keywords function the same in all SOL 414 solutions except for the following:

- SOL 414,110 Complex Modal Analysis
  - mpi402 distributes all rotating speed steps between the defined number of processors.
  - mpi414 uses the defined number of processors for each rotation speed step.
- SOL 414,111 Harmonic Response with a rotor defined (RMETHOD)
  - mpi402 distributes all frequency steps between the defined number of processors.
  - mpi414 uses the defined number of processors for each frequency step.

For more information, see Running a rotor dynamics simulation in parallel in the *Simcenter Nastran SOL 414 Rotor Dynamics User's Guide*.

mumpsmem mumpsmem=value Default=50

For SOLs 402 and 414 only.

When MUMPS is the solver choice, by default, the software allocates half of the total memory you specify with the memory keyword to the main executable and the other half to the MUMPS. You can adjust the percentage of memory allocated to MUMPS with the mumpsmem=*value* keyword, where *value* is an integer between 1 and 99. Particularly if MUMPS does not have enough memory to work in-core.

news news={yes|no} Default=yes

Displays the news file in the F06 file. If “yes” is specified, the news file is displayed in the F06 file regardless of when it was last changed. If “no” is specified, the news file is not displayed in the F06 file.

Example: `nastran example news=yes`

The news file is displayed in the F06 file after the title page block.

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notify notify={yes|no} Default=yes

Sends notification when the job is completed. See also the “ncmd” keyword in the *Simcenter Nastran Installation and Operations Guide*.

Note

If the job is queued using the queue keyword, or the job is already running in an NQS batch job, the default is “no”.

Example: `nastran example notify=yes`

A message is sent when the job is completed.

old old={yes|no} Default=yes

Saves previous copies of the F04, F06, LOG, OP2, OUT, PCH, and PLT output files using sequence numbers. Sequence numbers are appended to the keyword filename and are separated by a period.

If “yes” is specified, the highest sequence number of each of the output files is determined. The highest sequence number found is incremented by one to become the new sequence number. Then, all current output files that do not include sequence numbers are renamed using the new sequence number as a suffix.

Example: `nastran v2401 old=yes`

For example, the user’s directory contains the following files:

v2401.dat	v2401.log
v2401.f04	v2401.log.1
v2401.f04.1	v2401.log.2
v2401.f04.2	v2401.log.3
v2401.f06	v2401.f06.1

Apparently, the user ran the job four times, but deleted some of the files. When the next job is run, the following files are renamed: v2401.f04 is renamed to v2401.f04.4, v2401.f06 is renamed to v2401.f06.4, and v2401.log is renamed to v2401.log.4. The sequence number 4 is used because it is one greater than the highest sequence number of all of the selected files (the highest being v2401.log.3). Using this method, all files related to a single run will have the same sequence number.

out out=pathname Default=current directory

Saves the output files using a different file prefix or in a different directory. If “out” is not specified, the output files are saved in the current directory using the base name of the input data file as a prefix. For example, if your current directory is `/home`,

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and your input file `test.dat` is in the directory `/input`, and you run a job with `install_path/bin/nastran.exe /input/test.dat`, the output for this job will be in your current directory, which is `/home`.

If the "out" value is a directory, output files are created in the specified directory using the base name of the input data file as the filename. In the following examples, assume the following directory contents:

```
example.data mydir/ other/other/example.dat
```

where "mydir" and "other" are directories.

Example: `nastran example`

or: `nastran other/example`

Output files are created in the current directory with name example, e.g., ./example.f06.

Example: `nastran example out=myfile`

Output files are created in the current directory with name myfile, e.g., ./myfile.f06.

Example: `nastran example out=mydir`

Output files are created in the mydir directory with name example, e.g., mydir/example.f06.

Example: `nastran example out=mydir/myfile`

Output files are created in the mydir directory with name myfile, e.g., mydir/myfile.f06.

parallel — Same as the **smp** keyword.

pmimode =*value* — Default: local

Requests authentication using the Hydra Process Manager when running DMP parallel on Windows. The keyword pmimode only applies to DMP parallel runs on Intel/AMD 64-bit Windows systems. Also see the keyword wmpiauth.

Example: `nastran.exe pmimode=cluster input_file.dat`

= local (default) No Hydra authentication occurs.

= cluster Hydra authentication is used.

rcf — rcf=pathname — Default=no

Specifies the name of a local RC file.

The RC file referenced by the rcf keyword is one of many possible RC files. You can create RC files in the following locations, and Simcenter Nastran uses all definitions from all files. The software processes the multiple RC files in the following order. As a result, if

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you define the same keyword, parameter, or system cell definition in multiple RC files, the software will use the definition in the last processed RC file.

1. RC file referenced by the environment variable NXN_RCF.

2. RC file included with the installation in the *conf* directory.

3. RC file in your user home directory relative to your user account on the Windows or Linux file system.

4. RC file referenced by the rcf keyword. When you define the rcf command line keyword, the software does not process the RC file described in 5 below.

5. RC file in the same directory as your input file (*.dat).

See Customizing the Runtime Configuration Files in the *Installations and Operations Guide*.

Example:
```
$ nastran example rcf=nast.rc
```

scratch

scratch={yes|no|mini} Default=no

Deletes the database files at the end of the run. If the database files are not required, “scratch=yes” can be used to remove them. If “mini” is specified, a reduced size database (that can only be used for data recovery restarts) will be created. See “Mini-Database Restarts” in the *Simcenter Nastran User’s Guide* for further details on the “mini” database.

Example:
```
nastran example scratch=yes
```

All database files created by the run are deleted at the end of the job in the same way as the FMS statement INIT MASTER(S).

sdirectory

sdirectory=*directory* Default: See the description below.

Specifies the directory to use for temporary scratch files created during the run. As Simcenter Nastran can create very large scratch files, the scratch directory should contain sufficient space. You must have read, write, and execute privileges to the directory.

See “Determining Resource Requirements” in the *Simcenter Nastran Installation and Operations Guide* for information on estimating a job’s total disk space requirements.

The default value is taken from the TMPDIR environment variable on Linux or the TEMP environment variable on Windows. Otherwise, the computer’s default temporary file directory is chosen; this is usually /tmp.

Example:
```
nastran example sdir=/scratch
```

Scratch files are created in the directory /scratch.

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smemory smemory=*value* The default is 100 buffers. The value defined in the installation RC file is smemory=20X.

Specifies the memory to reserve for scratch memory, where *value* can be defined as a percentage of the total memory requested with the keyword **memory**, or as a specific value. The keyword **smemory** can be overridden by the FMS statement INIT SCRATCH(MEM=*value*).

- To define as a percentage of the total memory requested with the keyword **memory**, enter

  `smemory=<n>X`

  where *n* is a percentage 0<*n*<100, and X represents the total memory requested with the keyword **memory**. For example, if **memory**=5Gb is defined, and **smemory**=20X is defined, the memory requested for scratch memory is 20% of 5Gb, or 1Gb.
- When *value* is defined as a specific value, you either enter the number of buffers (default), or a number followed by one of the modifiers below. In the conversion column, the *bytes_per_word* is 8. The modifier may be specified using any case combination.
- The smemory keyword is ignored by SOLs 402 and 414.

Modifier	**Conversion to number of words**
G or Gw	Multiply *entered_value* by 1024**3.
Gb	Multiply *entered_value* by (1024**3)/*bytes_per_word*.
M or Mw	Multiply *entered_value* by 1024**2.
Mb	Multiply *entered_value* by (1024**2)/*bytes_per_word*.
K or Kw	Multiply *entered_value* by 1024.
Kb	Multiply *entered_value* by 1024/*bytes_per_word*.
w	Use *entered_value* as is.
b	Divide *entered_value* by *bytes_per_word*.

where *bytes_per_word* is 8. The modifier may be specified using any case combination.

Example: `nastran example smem=200`

This example reserves 200 buffers for scratch memory since no modifier exists.

Example: `nastran example smemory=25mw`

This example will use a scratch memory size of 25 megawords, or 25 600 kilowords, or 26 214 400 words.

Note: When you define **smemory** as a percentage of the total memory or in units of words or bytes, the software will convert the value to *number of buffers*. As a result, the value you enter will be adjusted such that **smemory** / **buffsize** is an integer. The value used by the software is reported in the .f04 file. See “Managing Memory” in the *Installations and Operations Guide*.

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smp smp=value Default=0

Same as the **parallel** keyword.

Example: `nastran example smp=2`

Specifies the maximum number of CPUs selected for shared-memory parallel (SMP) processing in several numeric modules. SMP processing reduces elapsed time at the expense of increased CPU time. The default is 0, which specifies no SMP processing. If “parallel=1", the parallel algorithms are used on one processor.

Note: If you need to vary the number of SMP CPUs during a job, you must set either the “smp” keyword or SYSTEM(107) on a NASTRAN statement to the maximum number of SMP CPUs that will be requested. Some systems cannot process a DMAP request for CPUs in excess of this initial value.

See the *Parallel Processing User’s Guide*.

symbol symbol=*symbolic_name=string* Default: *None*

Defines a symbolic (or logical) name used on ASSIGN and INCLUDE statements and in command line arguments. This statement can only be specified in initialization or RC files. It *cannot* be specified on the command line (although logical symbols defined using this keyword may be used on the command line). Symbolic names must be 16 characters or less, the value assigned to the symbolic name must be 256 characters or less. If the symbolic name used in ASSIGN or INCLUDE statements or in command line arguments is not defined, it is left in the filename specification as is.

For example, many of the TPL and DEMO input data files have ASSIGN statements, such as the following:

```
ASSIGN MASTER=DBSDIR:abc.master
```

The string “DBSDIR:” specifies a symbolic name that is to be replaced by another string. The replaced string is defined by the “symbol” keyword in the initialization or RC file or as an environment variable. For example,

```
SYMBOL=DBSDIR=/dbs
```

When the previous ASSIGN statement is processed, the filename assigned to the logical name MASTER is “/dbs/abc.master”. An alternate way of defining symbolic names is through the use of environment variables. For example, issuing the following command at a Bourne or Korn shell prompt:

```
DBSDIR=/dbs; export DBSDIR
```

or the following at a C shell prompt:

```
setenv DBSDIR /dbs
```

is equivalent to the above “symbol” keyword.

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Note

If a symbolic name is defined by both an RC file and an environment variable, the symbol statement value will be used.

Keywords and Environment Variables in the *Simcenter Nastran Installation and Operations Guide* contains a list of environment variables that are automatically created by the driver program. Of particular importance to the logical symbol feature are the OUTDIR and DBSDIR variables. These variables refer to the directory that will contain the output files (set using the "out" keyword) and the directory that will contain the permanent database files (set using the "dbs" keyword), respectively.

wmpiauth =*value* Default: password

Selects the type of Windows authentication when running DMP parallel with the Hydra Process Manager. Behavior of this keyword depends on the pmimode keyword setting. The pmimode and wmpiauth keywords only apply to DMP parallel runs on Intel/AMD 64-bit Windows systems.

Example:

```
nastran.exe pmimode=cluster wmpiauth=delegate input_file.dat
```

When pmimode=local, no authentication is required and the wmpiauth keyword is ignored.

When pmimode=cluster, the Hydra authentication is used. The type of Hydra authentication is then defined with the wmpiauth keyword as follows.

wmpiauth=password (default) The Hydra authentication method uses a password.

wmpiauth=delegate The Hydra authentication method uses domain (Active Directory) login credentials.

wmpiauth=impersonate The Hydra authentication method uses limited domain authentication, which does not allow access to network drives.

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The NASTRAN Statement

The NASTRAN statement is used to specify values for certain Executive System operational parameters. These parameters are also called system cells. The NASTRAN statement is used for exceptional circumstances and is therefore not needed in most runs. The NASTRAN statement may also be specified in the Runtime Configuration (RC) files at the system, user, and job level as described in the *Simcenter Nastran Installation and Operations Guide*.

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NASTRAN

Executive System Parameter Modification

Specifies values for certain Executive System operational parameters called system cells.

Format:

NASTRAN cellname1=expression1, ..., cellnamen=expression*n*

or

NASTRAN SYSTEM(i)=expression1, ..., SYSTEM(n)=expression*n*

Describers

Describer	Meaning
cellnamei	System cell names from Table 1-1.
SYSTEM	Specifies the system cell number.
expression	See DEFINE statement for description.
i	System cell number from Table 1-1.

Remarks:

1. The NASTRAN statements may appear anywhere in the File Management Section. The NASTRAN statement may also be specified in Runtime Configuration (RC) files. See “Customizing the Runtime Configuration Files” in the *Simcenter Nastran Installation and Operations Guide*.
2. System cell values and their associated cell names may also be set with the DEFINE statement. They may also be set or values returned with the DMAP PUTSYS and GETSYS functions and the PARAM module. See “PUTSYS, GETSYS” of the *Simcenter Nastran DMAP Programmer’s Guide*.
3. More than one NASTRAN statement and/or DEFINE statement may be present and, if a system cell is specified more than once among these statements, then the last specification takes precedence.
4. The expression will use type conversion rules based on the type (i.e., integer, real, or logical) of the cellname, as defined on a previous DEFINE statement (see the DEFINE statement for conversion rules).
5. If the expression is omitted, the system cell associated with the cellname will be assigned the value as set on a previous DEFINE statement.
6. Only integers are allowed when using the system cell numbers. Thus, when a real number is requested, use the system cell name.

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Examples:

1. Either of the following statements could be used to change the default value for block size.

   ```
   NASTRAN SYSTEM (1)=4097 or NASTRAN BUFFSIZE=4097
   ```

   Or, if a prior DEFINE statement had defined a keyword MY_SYSBUF to the value 4097 then the following code could be used:

   ```
   NASTRAN SYSTEM(1)=MY_SYSBUF
   ```

   or

   ```
   NASTRAN BUFFSIZE=MY_SYSBUF
   ```

   The following statement is used to request execution of MSGMESH:

   ```
   NASTRAN MESH
   ```

2. Table 1-1 gives a summary of the recommended system cells.

Table 1-1: System Cell Summary

System Cell Number	System Cell Name	Function and Reference
1	BUFFSIZE	Specifies the number of words in a physical record. Also called block length.
2	F06	Specifies FORTRAN unit number for standard output file (.f06). (Integer > 0; Default = 6)
9	NLINES	Specifies the number of lines printed per page of output. See LINE.
14	MAXLINES	See MAXLINES.
19	ECHO	See ECHO.
20	METIME	Minimum time for execution summary table message.
21	APP	Approach Flag. See APP of the Executive Control section. If APP HEAT is specified, then this system cell is set to 1.
22	MACHTYPE	Machine type.
25	DIAGA	Alternate method to set DIAGs 1 through 32. See DIAG.
28	CONFIG	Machine subtype.
31	MESH	Requests execution of MSGMESH.
46 - 54	ADUMi	Dummy element flag., i=1 through 9.
56	HEAT	See APP. 0: Structural analysis. (Default) 1: Heat transfer.

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System Cell Number	System Cell Name	Function and Reference
57	HICORE	Working Memory. See Managing Memory in the *Simcenter Nastran Installation and Operations Guide*.
61	DIAGB	Alternate method to set diagnostics 33 through 64. See **DIAG** .
64	PUNCH	Specifies FORTRAN unit number for PUNCH file (.f07). (Default = 7)
66	MPYAD	Selects/deselects multiplication methods. See MPYAD in the *Simcenter Nastran DMAP Programmer's Guide*.
69	SOLVE	Controls matrix decomposition. Same as DECOMP in the *Simcenter Nastran DMAP Programmer's Guide* and Option Selection in the *Simcenter Nastran Numerical Methods User's Guide*. 0 or -1: Prints up to 50 messages for null columns and zero diagonals. (Default = -1) 1: Terminates execution when first null column is encountered. 2: Suppresses printing of message when a null column is encountered. 4: Terminates execution when first zero diagonal term is encountered. 16: Places 1.0 in diagonal position for all null columns and proceeds with the decomposition. 32: Terminates execution on zero diagonal term. 64: Exits after execution of preface for symmetric decomposition.
70	FBSOPT	Selects forward-backward substitution methods. FBS in the *Simcenter Nastran DMAP Programmer's Guide* and the FBS Method Selection in the *Simcenter Nastran Numerical Methods User's Guide*.
77	DELFF	Deletes form feeds.
81	REAL	Specifies the amount of open core memory that certain numerical modules will be restricted to.
- - -	DBSET	Database neutral file set. SubDMAP DBFETCH in the *Simcenter Nastran DMAP Programmer's Guide* .
82	DMAP	Allows NOGO to operate. See Processing of User Errors in the *Simcenter Nastran DMAP Programmer's Guide*.
86	F04	Specifies FORTRAN unit number for Execution Summary Table (.f04). See Understanding the .f04 File in the *Simcenter Nastran User's Guide*. (Integer > 0; Default = 4)

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System Cell Number	System Cell Name	Function and Reference
87	RADMTX	Type of radiation exchange coefficient, See **RADMTX**.
88	RADLST	Print radiation area summary. See **RADLST**.
107	PARALLEL	Parallel processing deselection (deactivation) for matrix operations. See Keywords in the *Simcenter Nastran Installation and Operations Guide*. 0: Deactivate parallel processing. 1 through 1023: Number of processors. The desired number of processors is summed with the values below in order to deactivate parallel processing methods in the following matrix operations: 1024: Forward-backward substitution. 2048: Decomposition. 4096: Multiplication. 8192: Householder in eigenvalue extraction. 65536: Element matrix assembly. 262144: Sparse decomposition. 524288: Sparse forward-backward substitution. If PARALLEL is set simply to the number of processors, then parallel processing is selected in all operations above.
108	NEWHESS	Request complex eigenvalue method. See **EIGC**.
109	- - -	Controls DMAP execution: 0: Do not execute DMAP instruction if all outputs are previously computed. 1: Always execute DMAP instruction. (Default)
114	BUFFPOOL	Sets bufferpool size. See “Keywords and Environment Variables” in the *Simcenter Nastran Installation and Operations Guide*.
124	ATTDEL	Controls the automatic assignment of the delivery database. See “Delivery Database” in the *Simcenter Nastran User’s Guide* and “Creating and Attaching Alternate Delivery Databases” in the *Simcenter Nastran Installation and Operations Guide*. 0: Enables automatic assigning. (Default) -1: Disables automatic assigning.
125	NOKEEP	Controls NOKEEP option of the RESTART File Management statement. 0: Disable NOKEEP

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System Cell Number	System Cell Name	Function and Reference
		1: Enable NOKEEP
126	SPARSE	Selects the sparse matrix method. For unsymmetrical sparse matrix decomposition method selection, see cell number 209. The following values may be summed in order to select sparse matrix methods in the operations listed below: 0: Deactivate sparse methods. 1: Multiplication. 8: Symmetric decomposition. 16: Forward-backward substitution. The default is 25, which is the sum of all values.
128	UPDTTIM	Specifies database directory update time. See **DBUPDATE** . 0: Do not update. >0: Time, in minutes, between database directory updates.
129	SMPYAD67	Triple Multiply Method Selection in the SMPYAD module. See SMPYAD in the *Simcenter Nastran DMAP Programmer's Guide*. 0: Use current method (Default). 1: Use two MPYADs method. 5: Use pre-Simcenter Nastran 1.0 method.
- - -	MAXDBSET	The maximum number of online DBsets attached to the run.
133	AUTOASGN	Controls auto-assigning of dbsets. Sum the desired values. (Default = 7). See "Database Autoassignment" in the *Simcenter Nastran User's Guide*. 0: No databases are automatically assigned. 1: Only the primary database is automatically assigned. 2: Only the delivery database is automatically assigned. 4: Only located databases are automatically assigned.
135	TSTAMP	Controls timestamp checking of DBsets. 0: Do not check. 1: Check. (Default) 2: Same as 1 and print diagnostics.

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System Cell Number	System Cell Name	Function and Reference
141	QUADINT	Specifies quadratic or linear interpolation for the line search method in nonlinear analysis. 0: Quadratic interpolation. (Default) 1: Linear interpolation.
142	SCR300	Requests creation of SCR300 partition on SCRATCH DBset. See **"INIT"** . 1: Do not create SCR300 partition. 2: Create SCR300 partition. (Default)
143	LOCBULK	LOCBULK=1 or 2 specifies that Bulk Data is being obtained via the DBLOCATE FMS statement. NASTRAN LOCBULK=2 is specified when no Bulk Data entries, except for PARAM entries, are to be deleted or added. All PARAM entries must be respecified. All other entries will be ignored and, if present, may increase CPU times in XSORT and IFP. With LOCBULK=2, the XSORT and IFP modules will not reprocess the Bulk Data Section stored in the SEMAP run. Also, GP1, TASNP2, SEP1 and SEP1X modules will be skipped. 0 is the default, which assumes the RESTART FMS statement. This system cell is recognized only in SOLs 101 through 200, and SOL 402.
144	- - -	RESTART FMS statement existence flag. Set to 1 if RESTART statement is present.
145	BFGS	Selects strategies of BFGS updates for the arc-length methods in nonlinear analysis. Please see the *Simcenter Nastran Nonlinear Handbook*. 0: Update Δu_R and Δu_P at every iteration with γ^*. (Default) 1: Update Δu_R only with γ. 2: Update Δu_R only with γ^*.
146	- - -	Reserves (n x BUFFSIZE x 3) memory for faster solution in the Lanczos method of eigenvalue extraction. Default=1. A value of 2 increases the memory reserved by 200%, a value of 3 increases the memory reserved by 300%, etc. See **EIGRL** . See system cell 229 for a disk space saving feature.
147	UWM	SYSTEM(147)=1 issues User Warning Message for a DMAP parameter appearing on a CALL statement that has an inconsistent authorization in the called subDMAP. 0 is the default, which means no message is issued.
148	DBVERCHK	In general, databases are not compatible between major releases, thus a check is performed in Simcenter Nastran to ensure that the major version which created the database is the same as

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System Cell Number	System Cell Name	Function and Reference
		that being executed. Since specific data on the database may be compatible, SYSTEM(148) allows this check to be circumvented. However, circumventing the check may lead to problems later in the run. 0 (Default): Check is performed. 1: Check is not performed.
150	SCR300DEL	Sets minimum number of blocks of SCR300 partition of SCRATCH DBset at which it is deleted. See **"INIT"**. (Default = 100)
151	- - -	Requests spill or no spill of the SCR300 partition of SCRATCH DBset. See**"INIT"** . FMS statement. (Default = 1)
155	CBLAMKD	Sets differential stiffness formulation for CBEAM element. 0: Current formulation. (Default) 1: Pre-Simcenter Nastran 1.0 formulation.
166	- - -	Controls sparse symmetric decomposition. Sum the desired values. (See the section "Matrix Decomposition" in the *Numerical Methods User's Guide* for more options) 0: No action (Default) 1: If insufficient core is encountered, then switch to conventional decomposition and continue. 2: Print diagnostics 4: Do not issue fatal message if maximum ratios are exceeded. Although high maximum ratios may be printed, they will not cause job termination. This applies to the DCMP, DECOMP, REIGL, and LANCZOS modules. 128: Diagonal scaling.
- - -	OLDQ4K	Requests the pre-Simcenter Nastran 1.0 CQUAD4 element stiffness formulation. No value is required after the keyword. Equivalent to SYSTEM(173)=1.
170	LDQRKD	Selects the differential stiffness method for CQUAD4 and CTRIA3 elements. 0: Pre-Simcenter Nastran 1.0, improved method. (Default) 1: Pre-Simcenter Nastran 1.0 method.
173	- - -	0: Default. 1: Requests pre-Simcenter Nastran 1.0 QUAD4 Formulation.

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System Cell Number	System Cell Name	Function and Reference
		2: Requests pre-Simcenter Nastran 1.0 - V70.5 QUAD4 Formulation.
178		Defines a grid identification number offset for CWELD elements. See the parameter **OSWPPT**. (Integer > 0; Default = 101,000,000)
182		Defines a constraint element identification number offset for CWELD elements. See the parameter **OSWELM**. (Integer > 0; Default = 100,001,001)
189	Q4TAPER	Specifies the maximum allowable value of taper for the CQUAD4 element. Taper is computed by connecting opposite grid points and computing the area of the enclosed triangles. Another way to think of taper is the ratio of the areas on the two sides of a diagonal. (Real ≥ 0.0; Default = 0.5)
190	Q4SKEW	Specifies the minimum allowable value of skew for the CQUAD4 element. Skew is the angle measured in degrees between the lines that join opposite midsides. (Real ≥ 0.0; Default = 30.0)
191	TETRAAR	Specifies the maximum allowable aspect ratio of the longest edge to the shortest altitude for the CTETRA element. (Real ≥ 0.0; Default = 100.0)
193	DISKSAVE	Lanczos High Performance Option: Controls whether the matrix/vector multiplication is saved in a scratch file or recomputed at every iteration. 0: Save (Default for all machines) 1: No save 2: Save 3: No Save
194	FASTIO	Lanczos High Performance Option: Controls input/output in orthogonalization/renormalization routines. 0: UNPACK/PACK 1: GINO READ/WRITE (Default)
195	FRQSEQ	Lanczos High Performance Option: 100 < Exponent for the rational function used to determine segment boundaries. See also the ALPH field on the See **"EIGRL"**. (Integer; Default = 0, which means equal segments.)
196	SCRSAVE	Lanczos High Performance Option: Controls reuse of scratch files in segment logic. 0: Do not reuse. (Default) 1: Reuse.

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System Cell Number	System Cell Name	Function and Reference
197	NUMSEG	Lanczos High Performance Option: Selects number of segments. See also the NUMS field on the **"EIGRL"**. (Default = 1)
198	MINFRONT	Lanczos High Performance Option: Sets minimum front size. (The default value is machine dependent.) This can also be set with the "rank" keyword. See "Keywords and Environment Variables" for more information on the "rank" keyword, and Computer-Dependent Defaults in the *Simcenter Nastran Installation and Operations Guide* for the machine dependent default values.
199	MASSBUF	Lanczos High Performance Option: Sets half the number of buffers to set aside for storing the mass matrix in core. (Default = 1, which means 2 buffers will be used.)
204	CORDM	Specifies the default value for CORDM field on the PSOLID entry. (Integer ≥ -1; Default = 0)
205	- - -	Defines the number of rows that will be simultaneously updated during a sparse symmetric decomposition. This can also be set with the "rank" keyword. See "Keywords and Environment Variables" and for more information on the "rank" keyword, and Computer-Dependent Defaults in the *Simcenter Nastran Installation and Operations Guide* for the machine dependent default values.
206	DCMPSEQ	Selects ordering method for sparse matrix decomposition. Note that the EXTREME method has two submethods; BEND and AMF. If EXTREME is used when SYSTEM(206) = 0 (default), then either BEND or AMF is automatically selected by the software depending on the size of the model. If SYSTEM(206) = 4, then BEND is used. 0: EXTREME for 3D, Metis or MMD for 2D (Default) 1: MMD – definite matrices 2: MMD – indefinite matrices 3: No sequencing 4: EXTREME 8: Metis 9: Better of Metis and MMD 32: MLV + 64 (adding 64 to any of the above values): Turns on Supernodal Compression Scheme. For example, sys(206)=72 would indicate Metis (8) + Supernodal Compression Scheme (64).

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System Cell Number	System Cell Name	Function and Reference
209	USPARSE	Unsymmetric sparse matrix method selection for the decomposition and forward-backward substitution operations. 0: Deactivate. 1: Activate. (Default)
210	- - -	Option to revert back to the “old” punch algorithm. 0: Writes the “old” punch file. Default until version 5.0. This punch file does not use NDDL definitions and hence some data in the punch file can come out incorrectly. 1: Writes the "new" punch file. Default in NX Nastran version 5.0 and beyond. This uses NDDL definition and hence produces a correct punch file.
213	DISTORT	Element distortion fatal termination override. Applies to the TETRA h-elements. 0: Terminate run. (Default) 1: Do not terminate run.
216	ITER	Controls execution of global iterative solver in SOLs 101,106,108,111,153 and the element iterative solver in SOL 101. YES: Execute global iterative solver. NO: Do not execute global iterative solver (Default). See bulk card ITER. To run the CASI element iterative solution, see system cell 399.
218	T3SKEW	Allows the user to control the minimum vertex angle for TRIA3 elements at which USER WARNING MESSAGE 5491 is issued. See the description of CTRIA3.
223		Stores information used for internal processing on boundary grids for superelements. Default = 2,000,000,000 when large grid IDs are disabled (that is, when SYSTEM(525) = 0 or it is not specified) Default = 2,000,000,000,000 when large grid IDs are enabled (that is, when SYSTEM(525) = 1)
224		Stores information used for internal processing on QSET points for superelements. Default = 2,147,483,001 when large grid IDs are disabled (that is, when SYSTEM(525) = 0 or it is not specified)

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System Cell Number	System Cell Name	Function and Reference
		Default = 2,000,999,999,999 when large grid IDs are enabled (that is, when SYSTEM(525) = 1)
229	MEMSAVE	Specifies space-saving method for the old Lanczos method of eigenvalue extraction (system(273)=1). **"EIGRL"**. 0: No space savings. (Default) 1: Does not write factors to disk which reduces scratch space usage by 67%. However, CPU costs will increase.
242	- - -	Controls module BEGN and END messages in .f04 file. 0: Print everything. (Default) 1: Print major modules only. 2: Print sub-modules only. 3: Do not print.
252	- - -	Used to select or deselect triple loop kernels in triple MPYAD operations. 0: Do not use 1: Use triple loop numerical kernels
253 - 262	- - -	SYSTEM(253) to (262) have been set aside for user DMAPS. The software does not use these values in its code in present or future versions. The SSSAlter library may use this range.
270	QUARTICDLM	A value of 1 selects the new quartic formulation of the doublet lattice kernel (N5KQ) while 0 selects the original quadratic form. (Default=0)
273	- - -	A value of 1 selects the old Lanczos shift logic from V70 and prior systems. (Default=0)
303	MINDEF	Indefinite Mass Matrix Check, the Default=1 does not perform the check. If MINDEF>0, then check is not performed. If MINDEF<0, then epsilon = -1.E(MINDEF). If MINDEF=0, then MINDEF defaults to -6.
304	MPERTURB	Perturbation factor for indefinite mass matrix. The default=1 does not perturb the mass. If MPERTURB>0, then the mass is not perturbed. If MPERTURB<0, then the mass 1.E**(MPERTURB) is added to the diagonal terms of the mass matrix.

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System Cell Number	System Cell Name	Function and Reference
		If MPERTURB=0, then MPERTURB defaults to -6. The perturbed mass matrix is used in the subsequent eigenvalue analysis.
309	- - -	If set to 1, requests the pre-Simcenter Nastran 1.0 HEXA8 element stiffness formulation. (Default=0)
310	- - -	If set to 1, requests a legacy RBE3 formulation from versions of Nastran prior to Simcenter Nastran 1.0. (Default =0)
319	XMSG	If set to 1, gives extended error messages. If set to 0, only message summaries are reported. (Default=1)
320	- - -	Controls the conversion of DAREA Bulk Data entries for grid and scalar points to equivalent FORCE/MOMENT/SLOAD Bulk Data entries as appropriate. 0: Perform the conversion, but do not give details of the conversion. (Default) N: Perform the conversion and give details of the first N such conversions. -1: Do not perform the conversion.
321	Q4_WARP	GEOMCHECK check value for CQUAD4 and CQUADR – surface warping factor. Default =0.05
322	Q4_IAMIN	GEOMCHECK check value for CQUAD4 and CQUADR – minimum interior angle. Default =30.0
323	Q4_IAMAX	GEOMCHECK check value for CQUAD4 and CQUADR – maximum interior angle. Default =150.0
325	T3_IAMAX	GEOMCHECK check value for CTRIA3 and CTRIAR – maximum interior angle. Default =160.0
327	BEAM_OFF	GEOMCHECK check value for CBEAM – element offset length ratio. Default =0.15
328	BAR_OFF	GEOMCHECK check value for CBAR – element offset length ratio. Default =0.15
329	MSGLIMIT	Maximum number of messages produced for each element type.

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System Cell Number	System Cell Name	Function and Reference
		Default =100
330	MSGTYPE	Controls the messages that are produced when geometry tests exceed tolerance values. 1: INFORM option produces informative messages 2: WARN option produces warning messages 3: FATAL option produces fatal messages Default =1
332	TET_EPLR	GEOMCHECK check value for CTETRA – edge point length ratio. Default =0.5
334	HEX_AR	GEOMCHECK check value for CHEXA – longest edge to shortest edge aspect ratio. Default =100.0
335	HEX_EPLR	GEOMCHECK check value for CHEXA – edge point length ratio. Default =0.5
337	HEX_WARP	GEOMCHECK check value for CHEXA – face warp coefficient. Default =0.707
338	PEN_AR	GEOMCHECK check value for CPENTA – longest edge to shortest edge aspect ratio. Default =100.0
339	PEN_EPLR	GEOMCHECK check value for CPENTA – edge point length ratio. Default =0.5
341	PEN_WARP	GEOMCHECK check value for CPENTA – quadrilateral face warp coefficient. Default =0.707
357	RSEQCONT	Resequence all continuations (useful to eliminate conflicts with auto-continuation capability). 0: Do not resequence continuations. (Default) 1: Resequence all continuations.
359	- - -	= 0: Use the user-requested eigensolution method. <> 0: When LAN is requested, switch to AHOU if the number of DOFs sent to the eigensolver is ≤ “nswitch”, an input parameter to the READ module. This parameter has an MPL default of 20. It may

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System Cell Number	System Cell Name	Function and Reference
		be set to other values in the solution sequences, depending on the context. When HOU, MHOU, or AHOU is selected, switch to the new Householder-QL solution. (Default=1)
368	- - -	Reserved.
370	QRMETH	Selects the formulation used by the QUADR/TRIAR elements. 0: Selects the QUADR/TRIAR membrane formulation with active out of plane rotational dof. (default). 1: Selects the older, limited QUADR/TRIAR formulation. Complex stresses and strains are not computed. 3: Converts QUADR/TRIAR into QUAD4/TRIA3 formulation. 5: Converts QUAD4/TRIA3 into the newer QUADR/TRIAR formulation.
395-398	- - -	Reserved.
399	ELEMITER	Requests the CASI element iterative solver in SOL 101. NO or 0: Do not execute the CASI element iterative solver (Default). YES, 1, or 2: Execute the CASI element iterative solver. For SOL 101, both SYSTEM(216)=YES (or ITER=YES) and SYSTEM(399)=YES (or ELEMITER=YES) are required to run the CASI element iterative solver. See the ITER bulk entry and system cell 216. For SOL 401, the NLCNTL parameter setting SOLVER=ELEMITER invokes the CASI element iterative solver. See the NLCNTL bulk entry.
400-401	- - -	Reserved.
402	DPBLKTOL	Tolerance to determine the removal of duplicate GRID IDs and/or CORD1C, CORD1R, CORD1S, CORD2C, CORD2R, CORD2S. (Real) Set DPBLKTOL < 0.0 (default) to have the software abort a solve if a model contains duplicate GRID point IDs and/or coordinate system definitions. Set DPBLKTOL = 0.0 to have the software ignore duplicate GRID point IDs and/or coordinate system definitions and proceed with the solve. Set DPBLKTOL > 0.0 to specify a tolerance value that determines whether the software ignores duplicate GRID IDs and/or coordinate systems and proceeds with the solve. Grids: Duplicates will be removed if their coordinates are within the specified tolerance, one (or none) has a local coordinate system defined, and all belong to the same superelement. If all conditions are met, the GRID using the local coordinate system will be used and

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System Cell Number	System Cell Name	Function and Reference
		the others ignored. If all conditions are not met, the solution stops with an error. Coordinate Systems: Duplicate coordinate system definitions will be removed if the coordinate system types are the same and their transformation matrices are within the specified tolerance and they are on the same superelement. If all conditions are met, the first coordinate system will be kept and subsequent duplicates ignored. If all conditions are not met, the solution will stop with an error.
404	- - -	Debug text file (.txt) generation for RFI/MNF via MBDEXPORT/ ADAMSMNF case control commands. 0: Do not generate text file (Default). 1: Generate text file.
405	- - -	Complex eigenvalue solver options using ISRR method. 0: Use ISRR special solver (higher memory requirement). (Default) 1: Use Simcenter Nastran general solver (small memory requirement).
406	- - -	Complex eigenvalue solver options using ISRR method. 0: Do not calculate left eigenvectors (better performance). (Default) 1: Calculate left eigenvectors (extra computational cost).
407	- - -	Determines if the version stamp header is written on the first line of the OP2 file. 0: Version stamp written. (Default) 1: Turns off the version stamp.
409	- - -	Changes how the RBE3 check for unconnected grid points is done. Normally the check will produce a fatal error when grids referenced by an RBE3 are unconnected, but if this system cell is set to 1, only a warning will be issued and the solution continues. The check for an unconnected node on a RBE3 element is not performed for a superelement or dmp solution.
412	OLDGAPS	Controls how CGAPs are treated in SOL 101 when linear contact is defined. (See the section on linear contact in the *Simcenter Nastran User's Guide* for more information.) 0: CGAPs are treated as contact elements. (Default) 1: CGAPs are treated as linear springs.
413	- - -	OP2 file format (32-bit or 64-bit).

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System Cell Number	System Cell Name	Function and Reference
		0: let the software chooses the format (Default). 1: 32-bit format.
414-416		Reserved.
417	REDORTH	Controls if the reduced-orthogonalization option during Lanczos solutions is turned on or off. See Chapter 6 in the *Numerical Methods User's Guide*. 0: Reduced-orthogonalization option is turned off. (Default) 1: Reduced-orthogonalization option is turned on.
419	SVDSPC	Determines which AUTOSPC method is used. 0: Use the eigenvalue method (Default) 1: Use the Singular Value Decomposition (SVD) method
421	SPARSEDR	Selects if the sparse data recovery option is used for modal response solutions 0: Sparse data recovery option not used. 1: Sparse data recovery option used. (default)
422	ENFMOTN	Controls which formulation is used for enforced motion response analysis (and mode acceleration, if requested). 0: Constraint mode method of enforced motion formulation (and new mode acceleration method). (default) 2: Absolute displacement enforced motion formulation is used, but includes a modal viscous damping coupling term to improve accuracy.
424	SPCHOL	Select if sparse Cholesky decomposition technique is used in linear solutions. 0: Sparse Cholesky technique is not used. 1: Use Sparse Cholesky technique. If fail, fall back to option 0. (default) 4: TAUCS is used. If fail, fall back to option 0. 5: TAUCS is used. If fail, fall back to option 1.
425	SELOPT	Specifies the optimizer to use for design optimization. (Note: previous system cell names are also valid.) 0: Use DOT

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System Cell Number	System Cell Name	Function and Reference
		1: Use SDO - Siemens Design Optimizer (Default, except for topology optimization*) 3: Use SPOT * By default, SOL 200 topology optimization uses SPOT when the number of design constraints is 100 or less, and uses SDO when the number of design constraints is more than 100. A setting of system(425)=1 overrides the default behavior and forces the use of SDO from the start. Topology optimization ignores a system(425)=0 or a system(425)=3 setting and instead uses the default behavior. A topology optimization run is requested with the DVTREL1 bulk entry.
426	REDMULT	Selects the REDMULT performance option when solving vibration problems with the Lanczos method. See Chapter 6 in the *Numerical Methods User's Guide*. 0: REDMULT is not used.(default) 1: REDMULT is used.
429		Selects the BCSLIB sparse solver to be used instead of the standard sparse solver. 0: Do not use the BCSLIB sparse solver in DECOMP/DCMP (Default). 3: Select both the symmetric and un-symmetric BCSLIB sparse solver in DECOMP/DCMP. 2: Select only the symmetric BCSLIB sparse solver in DECOMP/DCMP. 1: Select only the unsymmetric BCSLIB sparse solver in DECOMP/DCMP. A system(555) setting takes precedence over a system(429) setting. The keyword altdcmp takes precedence over both system(429) and system(555).
442	- - -	Controls if the Solution Monitor files are generated for Simcenter 3D Pre/Post and Simcenter Femap. It is only valid on the platforms supported by Simcenter 3D Pre/Post and Simcenter Femap. 0: Do not generate (Default). 1: Generate the original solution monitor files and delete after solution completes. -1: Generate the original solution monitor files and keep after solution completes. 2: Launches the monitor in Simcenter 3D Pre/Post. In case of a DMP run, the monitor will only be displayed on the primary node.

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System Cell Number	System Cell Name	Function and Reference
444	TEMPWARN	Controls if temperature definitions on undefined grid points produce a fatal error or a warning. 0: A warning occurs. (Default) 1: A fatal error occurs.
445	NREC	Defines the number of substructures in RDMODES. We do not recommend that you define NREC either with this system cell or on the RDMODES bulk entry because the software automatically computes a reasonable number. In addition, you should activate RDMODES by defining the RDMODES bulk entry, and not by defining NREC.
446	RDSCALE	A factor to modify the selected frequency range in the EIGRL specification for eigensolutions of each substructure in RDMODES. The default value is 5.0. RDSCALE can also be defined on the RDMODES bulk entry. The bulk entry definition takes precedence over the system cell definition.
447	QDX_SKEW	GEOMCHECK check value for CQUADX4, CQUADX8, CPLSTS4, CPLSTS8, CPLSTN4, and CPLSTN8 – skew angle. Default =30.0
448	QDX_TAPR	GEOMCHECK check value for CQUADX4, CQUADX8, CPLSTS4, CPLSTS8, CPLSTN4, and CPLSTN8 – taper ratio. Default =0.5
449	QDX_IAMN	GEOMCHECK check value for CQUADX4, CQUADX8, CPLSTS4, CPLSTS8, CPLSTN4, and CPLSTN8 – minimum interior angle. Default =30.0
450	QDX_IAMX	GEOMCHECK check value for CQUADX4, CQUADX8, CPLSTS4, CPLSTS8, CPLSTN4, and CPLSTN8 – maximum interior angle. Default =150.0
451	QDX_AR	GEOMCHECK check value for CQUADX4, CQUADX8, CPLSTS4, CPLSTS8, CPLSTN4, and CPLSTN8 – longest edge to shortest edge aspect ratio. Default =100.0
452	QDX_EPLR	GEOMCHECK check value for CQUADX8, CPLSTS8, and CPLSTN8 – edge point length ratio. Default =0.5

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System Cell Number	System Cell Name	Function and Reference
454	TRX_IAMN	GEOMCHECK check value for CTRAX3, CTRAX6, CPLSTS3, CPLSTS6, CPLSTN3, and CPLSTN6 – minimum interior angle. Default =30.0
455	TRX_IAMX	GEOMCHECK check value for CTRAX3, CTRAX6, CPLSTS3, CPLSTS6, CPLSTN3, and CPLSTN6 – maximum interior angle. Default =160.0
456	TRX_AR	GEOMCHECK check value for CTRAX3, CTRAX6, CPLSTS3, CPLSTS6, CPLSTN3, and CPLSTN6 – longest edge to shortest edge aspect ratio. Default =100.0
457	TRX_EPLR	GEOMCHECK check value for CTRAX6, CPLSTS6, and CPLSTN6 – edge point length ratio. Default =0.5
459		Reserved.
461	- - -	For each constraint equation (user-defined or generated from a rigid element), the software checks the magnitude of the coefficient of the dependent dof. If the value is relatively small, a warning message is output. If the value is very small, fatal message 5289 is issued. The reason for the check is that this value will be used in a denominator of an equation and, if very small, can lead to numerical round-off errors and poor answers. In rare occasions it might be permissible to allow this to happen and thus if system cell 461 is set to a "1", the message will only be a warning. 0: A fatal error occurs (default). 1: A warning occurs.
462	- - -	Methods for SOL 111 0 (Default): The software automatically selects either the original FRRD1 method or the in-core FRRDRU method. • The software selects the original FRRD1 method if the number of modes < PARAM,FRUMIN (default=500), or if the model includes any of the following: frequency dependent properties or materials, unsymmetric damping, or consist of only fluid elements. • The software selects the in-core FRRDRU method if the model does not include any of the items listed above and the number of modes ≥ PARAM,FRUMIN.

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System Cell Number	System Cell Name	Function and Reference
		Note: You can optionally set PARAM,FRRU,NO and SYSTEM(462)=0 to turn off the automatic selection and the original FRRD1 method is always used.
		1: The software automatically selects either the original FRRD1 method, the in-core FRRD1 method, or the in-core FRRDRU method. • The software selects the original FRRD1 method if the model includes any of the following: frequency dependent properties or materials, unsymmetric damping, or consist of only fluid elements. • If the model does not include any of the items listed above and the number of modes < PARAM,FRUMIN (default=500), the software selects the in-core FRRD1 method if there is sufficient memory to solve in-core. Otherwise, the original FRRD1 method is selected. • The software selects the in-core FRRDRU method if the model does not include any of the items listed above and the number of modes ≥ PARAM,FRUMIN. Note: If you define SYSTEM(262)=YES and the in-core FRRD1 method is selected by the software, an in-core iterative method is used.
		2: Original FRRDRU method is used. This method cannot handle multiple eigenvalues (either repeated roots or rigid body modes). If multiple eigenvalues occur, the method defaults to the original FRRD1 method.
		4: Enhanced FRRDRU method is used. This method allows multiple eigenvalues and sets KDAMP=-1.
463	- - -	Determines if CBEAM elements with offset vectors are permitted in solutions 106, 129, 153, and 159. See remarks on the CBEAM bulk entry. Also determines if shell elements with ZOFFS should be permitted in nonlinear solutions 106 and 129 in combination with thermal loads. See remarks on the CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, and CQUADR bulk entries. 0: A fatal error occurs (default). 1: Solution is permitted to continue, but CBEAM offset vectors remain parallel to their original orientation.
464	- - -	Determines the CGAP normal and transverse stiffness when a BCSET case control command exists (contact conditions exist). 0: (default) Stiffness is determined from the properties on the PGAP bulk data entry. (See remarks on the PGAP bulk data entry.)

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System Cell Number	System Cell Name	Function and Reference
		1: Stiffness is determined from PENN and PENT on the BCTPARM bulk data entry.
466	PYR_AR	GEOMCHECK check value for CPYRAM – longest edge to shortest edge aspect ratio. Default =100.0
467	PYR_EPLR	GEOMCHECK check value for CPYRAM – edge point length ratio. Default =0.5
469	PYR_WARP	GEOMCHECK check value for CPYRAM – face warp coefficient. Default =0.707
474	- - -	Determines the CROD and CBAR element lumped mass calculation. The “old” calculation before NX Nastran 6.1 was inconsistent with the CBEAM. 0: CROD and CBAR lumped mass calculation is consistent with CBEAM element (default). 1: CBAR will use the old method. 2: CROD will use the old method. 3: CBAR and CROD will use the old method.
476	- - -	Determines if a non-converged contact condition in solution 101 continues with a warning, or ends with an error. 0: (default) A warning is reported in the *.f06 file, the solution continues and results creation occurs. 1: A fatal error is reported in the *.f06 file, and the solution ends immediately with no results creation.
480	- - -	Determines the linear contact processing defaults. See the BCTPARM bulk entry. (default): – The elastic modulus of the softer material in each contact pair is used. – PENN and PENT are automatically calculated. – The "REFINE=2" source-refinement algorithm is used. 1: – The modulus averaged over the entire model is used. – The automatic penalty factor calculation is turned off and PENN and PENT are used. – The NX Nastran 6.1 source-refinement algorithm is used.

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System Cell Number	System Cell Name	Function and Reference
489	PERPTOL	Determines the perpendicularity tolerance for shell element material orientations. (Real, Default = 1.0E-8 degrees) . A fatal error will occur when the angle between a shell element face normal and the x-direction of the material coordinate system (MCID) is less than the value of SYSTEM(489).
491	- - -	When the GLUETYPE=2 option is used on the BGPARM bulk entry, reverts the surface-to-surface glue stiffness back to the NX Nastran 6.1 defaults. 3 (default): Use current stiffness defaults. 1: use the NX Nastran 6.1 stiffness defaults.
492	- - -	Determines whether the factor caching for the Lanczos real eigensolver is used. 0: No factor caching (default). 1: The factor caching is used.
495	- - -	Determines if a shell element thickness value of 0.0 causes a fatal error. 0 (default): A fatal error will occur if any Ti value of 0.0 exists on the elements CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, CQUADR. This thickness check default began in NX Nastran 7.1. 1: The solution proceeds as it did before version 7.1.
500	- - -	Determines which strains are reported as total strains in the results under the label “NONLINEAR STRESSES”. 0 (default): Report the sum of mechanical strains and thermal strains as total strains. 1: Report only the mechanical strains as total strains. This is the default behavior of versions prior to NX Nastran 8.
503	- - -	On the ACSRCE bulk entry, determines if the scale factor A is included inside or outside of the radical. See the ACSRCE bulk entry for the source strength equation. 1 (default): The scale factor A is included outside of the radical. 0: The scale factor A is included inside of the radical. This is the behavior before version 8.
509	- - -	When ND is specified on the EIGRL card, determines whether to increase ND automatically to include all roots in a cluster. If ND is not permitted to increase (default), only part of a cluster of multiple eigenvalues is computed, the modal space will not be uniquely defined. In this scenario, results computed from that modal

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System Cell Number	System Cell Name	Function and Reference
		space may be unstable (frequency response, transient response, CMS solution, etc.). 0 (default): Compute ND roots as requested. 1: If necessary, increase ND automatically.
516	- - -	When gluing solid element faces, determines if the solid element face normal check occurs. See remarks on BGSET case control command for more information. 0 (default): Solid element face normal check occurs. 1: Solid element face normal check is off.
519	- - -	0 (default): Automatically supply blank fields 5 through 8 of a wide field card if only fields 1 through 4 are provided. 1: Do not automatically supply fields 5 through 8 (retained to preserve old behavior of the software).
524	RANFRF	Flag to output frequency response in addition to any PSD output 0 (default): Do not output frequency response in addition to PSD output. 1: Output frequency response in addition to PSD output.
525	- - -	Enables integer inputs to be 11 characters. 0 (default): Integer inputs are limited to 8 digits. The maximum ID of grid points or elements is then 99999999. 1: Integer inputs can be defined up to 11 digits when large field formatted bulk entries are used. The maximum ID of grid points or elements is then 99999999999. Note: When SYSTEM(525)=1 is defined, the OP2FMT parameter is ignored, and the software writes the OP2 file as 64-bit. See **OP2FMT**.
526	- - -	In NX Nastran 8.1, the torsional constant calculation for the cross-section types “CHAN1” and “H” on the PBARL/PBEAML bulk entries was updated. 0 (default): The new torsional constant calculation occurs. 1: Reverts to the previous torsional constant calculation.
528	- - -	Allows you to revert to the glue stiffness calculations before NX Nastran 8.5. 0 (default): Use the average modulus (E) in the source region when computing the glue stiffness. Also use the improved linear shell glue stiffness.

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System Cell Number	System Cell Name	Function and Reference
		1: Use the average modulus (E) of the entire model when computing the glue stiffness. Also use the old linear shell glue stiffness. This is the behavior before version 8.5.
539	- - -	Selects optimization enhancement level. 0: Selects NX Nastran 8.1 enhancement level. 1: Selects NX Nastran 8.5 enhancement level. 2: Selects NX Nastran 8.1 enhancement level plus selective modification of design variable bounds. 3 (default): Selects NX Nastran 8.5 enhancement level plus selective modification of design variable bounds. For details on these enhancement levels, consult the appropriate *NX Nastran Release Guides*.
542	- - -	Selects SSG1 method. 0: Original method. (Expect slow performance when this method is used with models that contain very large numbers of PLOAD2 bulk entries.) 3 (default): Enhanced method. (Better suited for models that contain very large numbers of PLOAD2 bulk entries.)
544	HEX_TP	GEOMCHECK check value for CHEXA – Ratio of longest edge to shortest edge in stacking direction for CHEXA elements that reference PCOMPS entries only. Default =1.2
545	PEN_TP	GEOMCHECK check value for CPENTA – Ratio of longest edge to shortest edge in stacking direction for CPENTA elements that reference PCOMPS entries only. Default =1.2
548	HEX_TK	GEOMCHECK check value for CHEXA – Ratio of the difference between thickness in stacking direction as defined by grids and as defined by ply thickness specification to the thickness in the stacking direction as defined by grids for CHEXA elements that reference PCOMPS entries only. Default =0.05
549	PEN_TK	GEOMCHECK check value for CPENTA – Ratio of the difference between thickness in stacking direction as defined by grids and as defined by ply thickness specification to the thickness in the stacking direction as defined by grids for CPENTA elements that reference PCOMPS entries only.

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System Cell Number	System Cell Name	Function and Reference
		Default =0.05
550	RMSVM	Determines if the RMS von Mises stress is computed when the stress RMS output is requested with the RMS describer on the STRESS case control command. The data blocks OESXRMS1 and OESXRMS2 are written when the RMS von Mises stress is computed. 0 (default): The RMS von Mises stress is computed, and the data blocks OESXRMS1 and OESXRMS2 are written. 1: The RMS von Mises stress is not computed, and the data blocks OESRMS1 and OESRMS2 are written.
552	F56	Defines the Fortran unit number for the .f56 output file. The .f56 file is requested with the PARAM,F56,YES setting in the bulk data. Default=56
553	QAD_AR	GEOMCHECK check value for CQUAD4 element - longest edge to shortest edge aspect ratio. Default =100.0
555		Selects the PARDISO sparse solver to be used instead of the standard sparse solver. 0: Do not use the PARDISO sparse solver in DECOMP/DCMP (Default). 3: Select both the symmetric and unsymmetric PARDISO sparse solver in DECOMP/DCMP. 2: Select only the symmetric PARDISO sparse solver in DECOMP/DCMP. 1: Select only the unsymmetric PARDISO sparse solver in DECOMP/DCMP. A system(555) setting takes precedence over a system(429) setting. The keyword altdcmp takes precedence over both system(429) and system(555).
560	PCHLN	Determines if the line numbers are included in the PUNCH file. 0 (default): The line numbers are on. 1: The line numbers are turned off.
563	Q8_TAPER	GEOMCHECK check value for CQUAD8 element - taper ratio. Default =0.5
564	Q8_IAMIN	GEOMCHECK check value for CQUAD8 element - minimum interior angle.

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System Cell Number	System Cell Name	Function and Reference
		Default =30.0
565	Q8_IAMAX	GEOMCHECK check value for CQUAD8 element - maximum interior angle. Default =150.0
566	Q8_AR	GEOMCHECK check value for CQUAD8 element - longest edge to shortest edge aspect ratio. Default =100.0
567	Q8_EPLR	GEOMCHECK check value for CQUAD8 element - edge point length ratio. Default =0.5
568	Q8_SKEW	GEOMCHECK check value for CQUAD8 element - skew angle. Default =30.0
569	TA6_AR	GEOMCHECK check value for CTRIA6 – longest edge to shortest edge aspect ratio. Default =100.0
570	TA6_EPLR	GEOMCHECK check value for CTRIA6 – edge point length ratio. Default =0.5
571	TA6_IAMN	GEOMCHECK check value for CTRIA6 – minimum interior angle. Default =10.0
572	TA6_IAMX	GEOMCHECK check value for CTRIA6 – maximum interior angle. Default =160.0
579	FREQVM	Determines whether the von Mises stress and strain are computed for a deterministic frequency response analysis in SOL 108 or SOL 111. 0 (default): von Mises stress and strain are computed and, along with the other stress and strain results, written to the OESVM and OSTRVM output data blocks. 1: von Mises stress and strain are not computed. Other stress and strain results are written to the OES and OSTR output data blocks.
580	- - -	Suppresses warning messages from the DMAP compiler that arise because of undefined input or output to DMAP commands. 0 (default): Allow warning messages 1: Suppress warning messages

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System Cell Number	System Cell Name	Function and Reference
581	- - -	Turns off the error trap which occurs when an SPCD entry is defined with a bolt preload. 0 (default): A fatal error occurs when an SPCD bulk entry and a bolt preload are both defined. 1: No fatal error occurs. Solution proceeds when an SPCD is defined with a bolt preload.
583	- - -	Controls the maximum number of flutter iterations in SOL 145. Default = 50.
587	- - -	Determines the sectional properties for the axisymmetric elements CTRAX3, CQUADX4, CTRAX6, and CQUADX8. 0 (default): Stiffness, mass and loads are based on a 2*PI section (beginning with NX Nastran 10). 1: Stiffness, mass and loads are computed on a per radian basis (behavior prior to NX Nastran 10), except for heat transfer solutions with CTRAX3, CQUADX4, CTRAX6, and CQUADX8 which remain on a 2*PI basis. See “Axisymmetric Element Sectional Properties” in the *Element Library Reference*.
589	- - -	Controls the drilling degree-of-freedom on QUADR/TRIAR elements. 0: The drilling degree-of-freedom is active for all QUADR/TRIAR elements in the model (Default). 1: The drilling degree-of-freedom is deactivated for all QUADR/TRIAR elements in the model. 2: The drilling degree-of-freedom is deactivated for the QUADR/TRIAR elements which have membrane stiffness only (MID2 and MID3 are blank on the PSHELL entry).
592	SMPORT0	Reserved. Used when sys442=2.
593	SMPORT1	Reserved. Used when sys442=2.
599		Controls the internal force calculation method of SOL 401 when SMP is requested. 0 (default): Use OpenMP to parallelize element internal force calculation. 1: Use serial element internal force calculation.
610	- - -	**For SOLs 101, 103, 105, 107-112:**

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System Cell Number	System Cell Name	Function and Reference
		0 (default): Glue based method is used for ETYPE=2 solid bolts. Bolt forces can be recovered from the glue results requested with the BGRESULTS command. 10: Pre-NX Nastran 12 MPC based method is used for ETYPE=2 solid bolts. Bolt forces can be recovered from the MPC force results requested with the MPCFORCES command. May not be valid for bolts with significant bending. Note: The glue approach is not supported by the element iterative solver in SOL 101. If you run with the element iterative solver in SOL 101 and you have defined the ETYPE=2 bolt, the software will use the MPC approach.
		For SOL 401: When a bolt is defined with ETYPE=2 for SOL 401 and the glue approach is selected with SYSTEM(750)=0 or 1, the software holds the grid pairs on each side of the cut bolt in their relative deformed state using a stiff glue condition. You can use this system cell to replace the tangential glue stiffness with tangential MPC conditions. The axial stiffness remains as a glue condition. 0 (default): The software uses a tangential stiff glue condition. 10: The software uses a tangential MPC condition. The MPCs are generated based on the initial configuration and are not updated during the solution. As a result, these MPC constraints are only valid when the rotations in the bolt are small. By using the tangential MPC condition, you are coupling the grid point pairs such that they always move together in the tangential direction. For example, if the default tangential stiffness is not large enough and the bolt halves slide relative to each other, and you are confident that the bolt will not experience large rotations, you can consider defining SYSTEM(610)=10. Note that SYSTEM(610) has no effect if SYSTEM(750)=2, which is the default.
612	- - -	Requests the form of the mass matrix for a static subcase in SOL401. 0 (default): Use a consistent mass matrix. 1: Use a lumped mass (form used before version 10).
617	ACFORM	Selects acoustic formulations. 0: Simcenter Nastran base acoustic capabilities including element formulation are used. The features such as AML boundary condition, infinite plane, porous materials and microphone elements are NOT

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System Cell Number	System Cell Name	Function and Reference
		available. Set the system cell to this value to obtain results identical to prior versions or releases of Simcenter Nastran. 1 (default): Allows extended acoustic capabilities such as AML, porous materials and microphone elements. All extended capabilities are available only in SOLs 108 and 111.
625		Requests the FRRUD3 solution method for SOL 111. The system cell 625 settings described below only apply when the system cell 462 default setting is used (SYS462=1). 0 (default): The FRRUD1 method is used. 1: The FRRUD3 method is used. The FRRUD3 solution method is available to reduce the solve time when your model includes a large fluid-structure coupling, you are solving for a large number of fluid modes (> 3000), and your number of solution frequencies is >100.
634	- - -	Controls the number of times the software prints the message number 2083.3 into the .f06 file. Message number 2083.3 is printed for a shell element that incorrectly has an associated PSHELL with the MID3 field that references a MATS1 bulk entry. Only the MID1 and MID2 should reference a MATS1. > 0 Limits the number of messages. Default = 100. = 0 The message is not written. = -1 The message is written for all.
635	Q8T6_ANG	Controls the maximum allowable angle between normals to corner grids for CQUAD8 and CTRIA6 elements. USER WARNING MESSAGE 5276 is issued if the maximum allowable angle is exceeded. Default =30.0
636	TET_EPIA	GEOMCHECK check value for CTETRA – edge point included angle. Default =30.0
637	HEX_EPIA	GEOMCHECK check value for CHEXA and CHEXCZ – edge point included angle. Default =30.0
638	PEN_EPIA	GEOMCHECK check value for CPENTA and CPENTCZ – edge point included angle. Default =30.0
639	PYR_EPIA	GEOMCHECK check value for CPYRAM – edge point included angle.

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System Cell Number	System Cell Name	Function and Reference
		Default =30.0
640	PRESOUT	Determines the case control command to be used for acoustic pressure output at fluid grids. 0: The PRESSURE and DISPLACEMENT case control commands are interchangeable. Set the system cell to this value to revert to pre-NX Nastran 12 behavior. 1 (default): The PRESSURE case control command requests pressure output at fluid grid points only and the DISPLACEMENT case control command requests displacement at structural grid points only.
649		Controls the GDSTAT parallel solution in SOL 401 for fast static analyses of large models. The performance of GDSTAT depends on the size of the boundary produced by the graph-based (GPART = 1) domain partition. When the boundary size is small, GDSTAT is most efficient. For most cases, DMP = 2 or DMP = 4 is sufficient. 0 (default): Does not select GDSTAT. That is, Simcenter Nastran performs serial processing. 1: Selects GDSTAT with null columns in the stiffness matrix. Use this for most parallel processing cases. 2: Selects GDSTAT after null columns are removed from the stiffness matrix. For example, for models with large open contact patches, which generally include a large number of null columns, you may see a performance increase with sys649 = 2. For more information, see the *Parallel Processing User's Guide*.
655		In NX Nastran 12, the automatic normal penalty calculation (PENN) for SOL 101 contact was reduced by one order. Since PENT = PENN / 10, this change also reduces the transverse direction penalty factor. In addition, the SOL 101 contact friction algorithm was improved in NX Nastran 12. You can use SYSTEM(655) to revert to the previous behavior (NX Nastran 11.2) for the sparse solver. The element iterative solver always uses the SYSTEM(655)=0 behavior. 2 (default): The new penalty calculation and improved contact friction algorithm introduced in NX Nastran 12 is used. 0: Reverts to the previous penalty calculation and friction algorithm.
665	BSHDCP	Turns off the computation of the CBUSH coupling moments which are computed when the connecting grid points are not coincident. System cell 665 applies to all solutions except for SOL 402.

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System Cell Number	System Cell Name	Function and Reference
		0 (default): The coupling moments are computed. 1: The coupling moments are not computed.
674		When an enforced displacement is defined with the SPCD entry, SOL 402 does not require a companion SPC entry. This system cell is used to control the behavior of SOL 402 with and without companion SPC entries. Note that if you are running SOL 402 from Simcenter 3D Pre/Post and have defined an enforced displacement, do not define this system cell. The value of this system cell is chosen automatically by Simcenter 3D Pre/Post . 0 (Default): You must use this option when companion SPC entries are included for SOL 402. Otherwise, the enforced displacement defined with the SPCD entry will be ignored. This option is useful if you are converting an already created input file which includes companion SPC entries to SOL 402. 1: The SOL 402 computation is most efficient when there are no companion SPC entries and SYSTEM(674)=1 is defined. If there are no companion SPC entries and SYSTEM(674)=0 is defined, results will be correct but the computation time will be less efficient because all fixed constraints will be treated as imposed displacements with a value of zero instead of true fixed constraints. Note: All other Simcenter Nastran solution types ignore this system cell. Except for SOL 402, all other Simcenter Nastran solution types do require a companion SPC entry.
675		Verbosity level for SOL 402. 0 (default): Minimum verbosity. 1: Low verbosity. List the output of the pre-processing (Bacon) and post-processing (Fac) steps of SOL402 solver. This is done automatically in case of warning or errors during those steps. 2: High verbosity. In addition to sys675=1, list the content of the model read during the pre-processing step. 3: High verbosity. In addition to sys675=2, call the DMAP module TABPRT to print the content of all the output datablocks (except rotation matrices TRMBU and TRMBD). 4: High verbosity. In addition to sys675=3, call the DMAP module TABPRT to print the content of all the output datablocks (very large output, not recommended).

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<table>
<tr><th>System Cell Number</th><th>System Cell Name</th><th>Function and Reference</th></tr>
<tr><td>679</td><td>KRYLOV</td><td>Controls whether the Padé via Lanczos (PvL) method in SOL 108 is activated.

YES: PvL method is used.

NO (default): PvL method is not used.</td></tr>
<tr><td>683</td><td></td><td>Temporarily turn off printing USER INFORMATION MESSAGE 4114 (OUTPX2) during writing the datablock in the scratch OP2 file. Automatically turned on when needed.</td></tr>
<tr><td>687</td><td></td><td>Specifies the modulus of elasticity that is used for rigid Lagrange elements in the model when no material properties are present in the input file. Assigns 1.0E12. (Real, Default)</td></tr>
<tr><td>693</td><td></td><td>Requests the Pardiso solver for SOLs 108 and 111. Note that the settings on the SLVCNTL bulk entry override this system cell.

0 (Default): SOL 108 automatically selects the MUMPS or the SPARSE solver.

SOL 111 automatically selects the Pardiso solver based on the density of the resulting input matrices (sparsity < 10%). Otherwise, the Sparse solver is used.

-1: Neither SOLs 108 and 111 use the Pardiso solver.

1: Both SOLs 108 and 111 select the Pardiso solver if sys738 is not set as 1, i.e. MUMPS is not selected.

The following table summarizes the SOL 108 and SOL 111 solver choice based on the SYSTEM(693) and SYSTEM(738) settings.
<table>
<tr><th></th><th>738=0</th><th>738=-1</th><th>738=1</th></tr>
<tr><td>693=0</td><td>Automatic selection</td><td>SPARSE/PARDISO</td><td>MUMPS</td></tr>
<tr><td>693=-1</td><td>SPARSE/MUMPS</td><td>SPARSE</td><td>MUMPS</td></tr>
<tr><td>693=1</td><td>PARDISO</td><td>PARDISO</td><td>MUMPS</td></tr>
</table>
</td></tr>
<tr><td>698</td><td></td><td>Controls the number of warning or error messages printed to the .f06 file. (Default = 100)

N < 0 Prints all warning and error messages. This is the pre-NX Nastran 12 behavior.

N > 0 Prints N messages or errors of a particular type.

Simcenter Nastran prints a warning message when the number of messages of a particular type exceeds N. Message types which have not exceeded N will continue to be printed.</td></tr>
</table>

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System Cell Number	System Cell Name	Function and Reference
699		Specifies the number of displacement curves that appear in a report. (Default = 3) The list of nodes that will be considered for displaying the curves is defined with the SETMC Case Control entry. If more than 2 curves are requested, only the curves for the minimum, the maximum, and the average values of all the selected nodes will be displayed.
700		Controls the CQUAD8 and CTRIA6 element formulations. -1 (default): Use the heterosis formulation for the CQUAD8 and the reformulated CTRIA6. 1: Use the pre-Simcenter Nastran 2019.2 formulations for the CQUAD8 and CTRIA6. For SOLs 401 and 402, the software ignores the SYSTEM(700) setting because only the reformulated versions of the elements are available. For hyperelastic analysis in SOLs 106 and 129, the software ignores the SYSTEM(700) setting because only the pre-Simcenter Nastran 2019.2 formulations can account for hyperelasticity.
701		Reserved for SOL 401 and SOL 402 restarts.
707		Controls the element matrix assembly method when SMP is requested. 0 (default): Use OpenMP to parallelize element matrix assembly. 1: Use serial element matrix assembly.
709		Reserved for SOL 401 and SOL 402 restarts.
710		Reserved for SOL 401 and SOL 402 restarts.
715		Controls transverse shear formulation for composite shells in SOL 401. 0 (default): Use the pre-Simcenter Nastran 2019.1 formulation. 1: Use the Simcenter Nastran 2019.1 formulation that uses a frame-invariant approach to calculate the interlaminar stresses.
717		Controls the convergence measures used in SOL 401. 0 (default): Use existing convergence measures and defaults. 1: SOL 401 uses convergence measures that are consistent with SOL 402: ● Default Convergence criterion (CONV): UP ● Default convergence norms for U and P (NORMU & NORMP): L1

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System Cell Number	System Cell Name	Function and Reference
		● Default error tolerance in U (EPSU): 1E-2 ● Default error tolerance in P (EPSP): 1E-3
721		Overrides the individual extrapolation option settings on TABLEDi (i = 1, 2, 3, and 6) bulk entries. 1: Sets the extrapolation option for all of the TABLEDi (i = 1, 2, 3, and 6) bulk entries to EXTRAP = 1. 2: Sets the extrapolation option for all of the TABLEDi (i = 1, 2, 3, and 6) bulk entries to EXTRAP = 2.
722	FEMAO	Controls whether the FEM Adaptive Order (FEMAO) method is activated. NO (default): FEMAO method is not used. Accelerates the processing of input files that are not intended for FEMAO computation. YES: FEMAO method is used.
734		Controls the Pardiso solver memory allocation method for SOL 401. (Integer; Default=0) 0 (default): Simcenter Nastran allocates a percentage of its open core memory to the Pardiso solver. The system cell 735 defines the percentage. 1: The Pardiso solver will use the available in-core memory that is not allocated to any application, including Simcenter Nastran. This is the software behavior before Simcenter Nastran 2020.1.
735		When the Pardiso solver is requested for SOL 401 by defining SOLVER=PARDISO on the NLCNTL bulk entry and SYSTEM(734)=0 (default), Simcenter Nastran allocates a percentage of its open core memory to the Pardiso solver. The system cell 735 can be used to change this percentage. (Integer; Default=75)
736		Activates the NUSOL external subroutine for SOL 401. (Integer; Default=0) 0 (default): NUSOL is not activated. 1: NUSOL is activated. 2: NUSOL is activated and the solution/time information is printed in the f06 file.
738		Requests the MUltifrontal Massively Parallel Sparse direct Solver (MUMPS) for SOLs 101, 108 and 111. Note that the settings on the SLVCNTL bulk entry override this system cell for SOLs 108 and 111. 0 (Default): SOL 101 uses either the Sparse solver or an iterative solver. See SYSTEM(216) and SYSTEM(399) for iterative solver

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<table>
<tr><th>System Cell Number</th><th>System Cell Name</th><th>Function and Reference</th></tr>
<tr><td></td><td></td><td>options.
SOL 108 automatically selects MUMPS or the SPARSE solver.
SOL 111 automatically selects the Pardiso solver based on the density of the resulting input matrices (sparsity < 10%). Otherwise, the Sparse solver is used.

-1: Neither SOLs 108 and 111 use MUMPS.
Ignored by SOL 101 (default behavior).

1: SOLs 101, 108, and 111 use MUMPS.

The following table summarizes the SOL 108 and SOL 111 solver choice based on the SYSTEM(738) and SYSTEM(693) settings.
<table>
<tr><th></th><th>693=0</th><th>693=-1</th><th>693=1</th></tr>
<tr><td>738=0</td><td>Automatic selection</td><td>SPARSE/MUMPS</td><td>PARDISO</td></tr>
<tr><td>738=-1</td><td>SPARSE/PARDISO</td><td>SPARSE</td><td>PARDISO</td></tr>
<tr><td>738=1</td><td>MUMPS</td><td>MUMPS</td><td>MUMPS</td></tr>
</table>
</td></tr>
<tr><td>740</td><td></td><td>For SOL 401, controls when the contact modulus is updated for material nonlinear problems. Only applicable when PENTYP=1 (default) is used on the BCTPARM bulk entry.

0 (default): The initial contact modulus computed at the start of the solution is used throughout the solution.

1: The contact modulus is updated if the underlying element undergoes plastic deformations.

See Contact Penalty Factors in the Simcenter Nastran Multi-Step Nonlinear User's Guide.</td></tr>
<tr><td>742</td><td></td><td>In a modal frequency response solution (SOL 111), when frequency dependent fluid exists, and the PANCON case control command is defined to request acoustic panel contribution results, SYSTEM(742) controls if the METHOD(FLUID) request is honored, and consequently, if the fluid response is computed in physical coordinates or in modal coordinates. (Integer; Default=0)

0 (default): The METHOD(FLUID) case control command is ignored, and the fluid response is computed in physical coordinates.

1: The METHOD(FLUID) case control command is honored, and the fluid response is computed in modal coordinates.</td></tr>
<tr><td>746</td><td></td><td>For SOL 401, selects the element architecture in a small strain analysis. When you run with large strains off (PARAM,LGSTRN,0), the</td></tr>
</table>

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System Cell Number	System Cell Name	Function and Reference
		improved architecture is used by default. The following elements are applied to the improved architecture: ● The 3D solid elements CHEXA, CTETRA, CPENTA, and CPYRAM that reference a PSOLID property. ● The axisymmetric elements CTRAX3, CTRAX4, CQUADX6, and CQUADX8. ● The plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, and CPLSTS8. ● The plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, and CPLSTN8. ● The shell elements CTRIA3, CTRIAR, CQUAD4, CQUADR, CTRIA6, and CQUAD8 that reference a PSHELL property. ● The spring elements CELAS1, CELAS2, CBUSH1D, and CBUSH. ● The beam elements CBEAM and CBAR. ● The rigid elements RBE2 and RBAR. ● The mass elements CMASSi, CONM1, and CONM2. 0: The pre-version 2020.2 architecture is used. 1 (default): The improved architecture is used. The following restrictions exist. The software automatically uses the SYSTEM(746)=0 option if any of the following are present. ● Dynamic subcase. ● Collapsed CHEXA element used in the crack simulation. ● J-integral output for a crack simulation. ● Adaptive meshing output requested with the ADAPTERR case control command. ● Element addition and removal requested with the ELAR case control command. You can use the improved architecture with a restart, although you must use the same architecture in both the initial run and in the restart runs. For example, if your initial run includes only static subcases, but the restart run includes a dynamic subcase, you will need to define SYSTEM(746)=0 for both runs because the improved architecture is not supported with a dynamic subcase. See Improved architecture for small strain problems (SOL 401) for more information.
750		For SOLs 401 and 402, controls if a fatal error occurs for the specific bolt preload conditions summarized in the table below. In addition for SOL 401 with ETYPE=2 bolts, it selects between the nonlinear MPC preload method (default) and the glue preload method.
Bolt preload conditions table		

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<table>
<tr><th>System Cell Number</th><th>System Cell Name</th><th>Function and Reference</th></tr>
<tr><td colspan="3">A. For SOLs 401 and 402 with ETYPE=2, if the grid points (Gi) defining the cross section are not coplanar, or if the normal vector to the plane defined by Gi is not parallel with the resulting bolt axis coordinate (IDIR).

B. For SOL 401 only with ETYPE=2, if a TYPE=DISP bolt preload is defined on the BOLTFRC bulk entry. SOL 402 always allows the TYPE=DISP bolt preload.

C. For SOL 401 only with ETYPE=2, if a bolt sequence is defined with the BOLTSEQ bulk entry. SOL 402 does not support the BOLTSEQ entry.

"-NA-" Not applicable to an ETYPE=1 bolt.

<table>
<tr><th></th><th></th><th>ETYPE=1</th><th>ETYPE=2</th><th>ETYPE=3</th></tr>
<tr><td rowspan="2">SYS(750)=0</td><td>SOL 401</td><td>-NA-</td><td>Glue method is used.
Fatal if A exists.
Fatal if B exists.
Allow C.</td><td>Fatal for ETYPE=3.</td></tr>
<tr><td>SOL 402</td><td>-NA-</td><td>Fatal if A exists.</td><td>Fatal for ETYPE=3.</td></tr>
<tr><td rowspan="2">SYS(750)=1</td><td>SOL 401</td><td>-NA-</td><td>Glue method is used.
Allow A.
Allow B.
Allow C.</td><td>Allow ETYPE=3.</td></tr>
<tr><td>SOL 402</td><td>-NA-</td><td>Allow A.</td><td>Allow ETYPE=3.</td></tr>
<tr><td rowspan="2">SYS(750)=2 (default)</td><td>SOL 401</td><td>-NA-</td><td>Nonlinear MPC method is used.
Fatal if A exists.
Allow B.
Glue method is used if C exists.</td><td>Fatal for ETYPE=3.</td></tr>
<tr><td>SOL 402</td><td>-NA-</td><td>Fatal if A exists.</td><td>Fatal for ETYPE=3.</td></tr>
</table>
</td></tr>
<tr><td>754</td><td></td><td>For SOL 200, requests the P-Norm aggregation for constrained stress response. The default is 100, which means that when a stress response defined with the DRESP1 bulk entry references 100 elements or more, the software uses the P-Norm aggregation.

0: P-Norm aggregation is disabled.

≥1: Minimum number of referenced elements for a DRESP1 to be converted to P-Norm aggregation.

The P-Norm aggregation computes a global unique response G from the many local stress responses (s_1, s_2, ..., s_n). The intent is to reduce the n constraints $s_i < s_{max}$ to a single constraint G_s:</td></tr>
</table>

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System Cell Number	System Cell Name	Function and Reference
		$G_s = (1/C)PNorm(s_i) = (1/C)({s_1}^k + {s_2}^k + \ldots + {s_n}^k)^{1/k}$ where *C* is a normalization constant, and *k* is the P-Norm exponent.
757		Selects the CBEAM/CBAR mass formulation. 1: Use the pre-Simcenter Nastran 2020.2 v1938 formulation that ignores the mass moments of inertia I_{yy} and I_{zz} in the mass matrix. 0 (default): Use the new formulation that includes the mass moments of inertia I_{yy} and I_{zz} in the mass matrix.
760		Controls Jacobian checks for SOL 401. 0 (default): In SOL 401, always perform Gauss and corner point Jacobian checks and terminate the solution if any element fails either check. 1: In SOL 401, perform Jacobian check according to GEOMCHECK options (pre-Simcenter Nastran 2021.1 behavior and the same behavior as all other solution sequences).
763		For SOL 200 topology optimization, requests the QP-relaxation method. 0: QP-relaxation is disabled. 1 (default): QP-relaxation is enabled. The QP-relaxation method resolves singularity problems from the stress response of low density elements. These elements are out of interest because they are no longer part of the structure, but stress can still remain high and perturb convergence. To avoid the issue, the software scales the original stress value s_0 by a factor dependent on the element density. This is an alternative penalty method instead of the classical penalty method (SIMP or RAMP). The scaled stress is computed with: $s_i = {\rho_i}^{(p-q)} * s_{0i}$ where ρ_i is the element density, *p* is the SIMP exponent, and *q* is the relaxation exponent typically chosen such that (*p*-*q*)=0.5 .
767 (has two purposes; A and B)		**Purpose A** For SOL 402 (beginning with Simcenter Nastran 2021.2) and for SOL 414,129 (beginning with Simcenter Nastran 2206), the requirements for mechanical and thermal loads are as follows:

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System Cell Number	System Cell Name	Function and Reference
		• You must define time-assigned loads globally (above the subcases). A fatal error occurs by default if you define either the DLOAD or the DTEMP case control command in a subcase. • The time-unassigned loads are always ramped in a subcase. A fatal error occurs by default if either LVAR=STEP or TVAR=STEP is defined on the NLCNTL2 entry. You can use system cell 767 to turn off the fatal errors: 0: Selects the old behavior (pre-Simcenter Nastran 2021.2 for SOL 402, and pre-Simcenter Nastran 2206 for SOL 414,129). The software will not produce a fatal error if a DLOAD or DTEMP command is defined in a subcase, or if LVAR=STEP or TVAR=STEP are defined. 1 (default): Selects the new behavior (Simcenter Nastran 2021.2 for SOL 402, and Simcenter Nastran 2206 for SOL 414,129). The load requirements are enforced. **Note** For SOL 414,103 with a transient response subcase and SOL 414,111 with a frequency response subcase, you can assign the loads to the subcase.
		Purpose B For SOLs 402 and 414, before Simcenter Nastran 2212, when you combined rotational inertia loads defined with the RFORCE or RFORCE1 on the LOAD or DLOAD bulk entries, the software scaled the angular velocity ($\vec{\omega}$) and acceleration ($\vec{\alpha}$) by the time varying scale factor *f(t)*, and by the scale factors S and Si on the LOAD or DLOAD entry, which are used to compute an inertia force at each grid point using: $$\vec{P}_{gp} = m_{gp}\left[\vec{\omega}_s \times \left(\vec{\omega}_s \times \vec{r}_{gp}\right) + \vec{\alpha}_s \times \vec{r}_{gp}\right]$$ where $\vec{\omega}_s$ is the angular velocity scaled, $$\vec{\omega}_s = S\ S_i\ f(t)\ \vec{\omega}$$ $\vec{\alpha}_s$ is the angular acceleration scaled, $$\vec{\alpha}_s = S\ S_i\ f(t)\ \vec{\alpha}$$

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System Cell Number	System Cell Name	Function and Reference
		m_{gp} is the mass at the grid point, $\vec{r}_{gp}$ is a vector from the center of rotation to the grid point, *f(t)* is defined on the associated TABLEDi for a time-assigned load, and *f(t)* is computed by the software when it ramps a time-unassigned load in a subcase. Because $\vec{\omega}_s$ was squared in the force computation, the resulting scaling from S and S_i was also squared and not linearly related to the computed force. Now by default, the software uses S and S_i on the LOAD or DLOAD entry to scale the rotational force, and not the angular velocity and acceleration. The angular velocity and acceleration are still scaled by the time varying scale factor *f(t)*. The software applies S and S_i the same way for all loads. The total load from *i* load sets is computed using: $$\{P\} = S\sum_i S_i \{P_{Li}\}$$ where $\{P_{Li}\}$ represents all load sets referenced on the LOAD or DLOAD entry, including rotational inertia loads. You can revert to the previous scaling by defining SYSTEM(767)=0. 0: Selects the old behavior (pre-Simcenter Nastran 2212). 1 (default): Selects the new behavior (Simcenter Nastran 2212)
769		For SOL 401 static and dynamic subcase types, controls if displacement and residual force results are sent to the Simcenter 3D Solution Control Monitor, which is a 3D visualization tool within the Simcenter 3D Solution Monitor. The Simcenter 3D Solution Monitor must be activated by setting SYSTEM(442)=2 and defining UGII_MONITOR_DIR appropriately. 0: No results will be sent to the Solution Control Monitor. 1 (default): The results will be sent to the Solution Control Monitor after every iteration. 2: The results will be sent to the Solution Control Monitor at the output intervals defined with the N_{OUT} field on the TSTEP1 bulk entry.
771		For SOL 401, beginning with Simcenter Nastran 2021.2, a new GPFORCE output format was introduced. Previously, the software

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System Cell Number	System Cell Name	Function and Reference
		summed the applied loads at a grid point and listed this in the "APP-LOAD" row in the grid point force output. With the new output format, the software separates the applied load into NOD_LOAD (nodal loads) and ELE_LOAD (element loads). The element loads only include the rotational loads defined with the RFORCE and RFORCE1 bulk entries. All other applied loads, except for thermal loads, are included as nodal loads. SOL 401 includes the thermal loads with the internal element forces. In addition, a new body-load computation method was introduced for RFORCE and RFORCE1 loads. With this method, the load vector is computed on an element basis rather than a grid point basis. The system cell 771 is available to change both the GPFORCE output format and the body-load computation method. 0 (default): The new GPFORCE output format and the new element-based body-load computation is used. 1: The old GPFORCE output format and the new element-based body-load computation is used. 2: The old GPFORCE output format and the old grid point body-load computation is used. Note that options 1 and 2 will not produce the refined free body results needed for Simcenter 3D post processing.
773		For SOL 401, requests that the contact force convergence check occur on the contact pair basis. The software computes a contact force ratio which indicates how the contact traction is changing relative to the previous iterations. It is computed by: $$Frat = \sqrt{\sum_{c=1}^{nce} \frac{\left(\mathrm{T}_c^i - \mathrm{T}_c^{i-1}\right)\left(\mathrm{T}_c^i - \mathrm{T}_c^{i-1}\right)}{\mathrm{T}_c^i \mathrm{T}_c^i}}$$ where *c* is a contact element, *nce* is the number of contact elements, T_c^i is the total contact traction at a contact element, and *i* is the iteration number.

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System Cell Number	System Cell Name	Function and Reference
		0 (default): The software computes a single contact force ratio (FRAT) over all contact elements from all of the pairs (contact sets) in the model. When the single contact force ratio (FRAT) is less than the contact force tolerance that you define with the RCTOL parameter, the software considers the contact force ratio converged. 1: The software computes a contact force ratio (FRAT) for each pair using the contact elements specific to a pair. The convergence check against the RCTOL parameter is then enforced for each contact pair. The global contact convergence is only satisfied when every pair has converged. The largest pair based force ratio computed when SYSTEM(773)=1 will be greater than the single global force ratio computed over all of the contact elements when SYSTEM(773)=0. As a result, the pair based force ratio is a stricter convergence check and will require more iterations to converge. See SOL 401 - Contact convergence in the *Simcenter Nastran Multi-Step Nonlinear User's Guide*.
776		For SOL 402, selects the computation used for the force convergence error. 0 (default): The original force convergence error computation is used. 1: The new force convergence error computation is used. The new computation with SYSTEM(776)=1 changes how the contact forces are included when the software computes the total force value at each increment. The new computation can aid in convergence for models that include contact conditions. Sensitive contact problems that did not converge may converge. You will likely see the number of iterations and number of rejected or accepted time steps change. In some cases, the converged solution can be more accurate with the new computation using the same force tolerance value. See SOL 402 - Error functions and weighted normalization in the *Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*.
777		For SOLs 402 and 414,129, selects the default solver behavior. Before Simcenter Nastran 2022.1, the sparse solver was used by default for sequential and SMP runs (that is, runs that are not using DMP), and MUMPS was used by default for DMP runs (that is, parallel runs requested with the keyword MPI402). Now all runs use MUMPS by default. You can revert to the previous default behavior by defining the system cell setting SYSTEM(777)=0. You can also

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System Cell Number	System Cell Name	Function and Reference
		explicitly select the solver by defining RESO=2 for sparse or RESO=6 for MUMPS, which overrides the SYSTEM(777) setting. 0: The default solver behavior before Simcenter Nastran 2022.1 is used. 1 (default): All runs use MUMPS.
778		For SOL 401, controls how contact separation output is computed. Contact separation output is requested with the BCRESULTS case control command. 0: Report the face-averaged values at the grid points. This is the pre-Simcenter Nastran 2206 behavior. 1 (default): Output the true gap value from the contact Gauss point closest to the grid point.
779		For SOL 401, selects the Newton-Raphson iteration strategy. 0: The legacy Newton-Raphson strategy is used. 1 (default): The consistent Newton-Raphson iteration strategy is used. See SOL 401 - Newton-Raphson iteration strategy in the *Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*.
780		For all solutions except for SOLs 144–146, 402, and 414, selects the shell characteristic length calculation used for glue definitions. For SOL 401 only, selects the shell characteristic length calculation used for contact definitions. The software computes a characteristic length for contact and glue pairs and uses the value for the penalty stiffness calculation as well as penetration tolerance calculations. 0: The characteristic length is computed using the element thickness over all the shell elements in the regions. This is the pre-Simcenter Nastran 2206 behavior. 1 (default): The characteristic length is computed using the face geometry of all shell and solid element faces in the regions. It is computed by averaging the square roots of the element face areas in the source and target regions.
784		For SOL 401, edge-to-surface and edge-to-edge glue conditions on shell elements are not updated when large displacements are requested (LGDISP=1). By default the solution will end with a fatal error when shell edge glue conditions and large displacements are requested.

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System Cell Number	System Cell Name	Function and Reference
		If the glue condition is in a location of the model that will not experience large rotations, the glue condition should be reasonably accurate. For this case, you can define SYSTEM(784)=1 to allow the solution to continue. 0 (default): A fatal error occurs when shell edge glue conditions and large displacements are requested. 1: The software allows the shell edge glue conditions with large displacements, and a message is reported in the .f06 file warning that the edge glue conditions are not updated when large displacements occur. For SOLs 106 and 129, all types of glue conditions are not updated when large displacements are requested (LGDISP=1). By default the solution will end with a fatal error when glue conditions and large displacements are requested. If the glue condition is in a location of the model that will not experience large rotations, the glue condition should be reasonably accurate. For this case, you can define SYSTEM(784)=1 to allow the solution to continue. 0 (default): A fatal error occurs when glue conditions and large displacements are requested. 1: The software allows the glue conditions with large displacements, and a message is reported in the .f06 file warning that glue conditions are not updated when large displacements occur.
792		For SOL 401, for CBAR and CBEAM elements with a distributed load defined with the PLOAD1 bulk entry, the computation of force, stress, and strain results on the connecting nodes was improved in Simcenter Nastran 2212. You can use system cell 792 to revert to the pre-Simcenter Nastran 2212 results computation. 0 (default): The software uses the improved force, stress, and strain results computation introduced in Simcenter Nastran 2212. The software uses the SDRX module to compute the improved output. -1: The software uses the pre-Simcenter Nastran 2212 force, stress, and strain results computation.
806		For SOL 401, selects whether von Mises strain or effective strain are output for elastic, thermal, and total strains. 0 (default): Output von Mises strain. 2: Output effective strain.

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System Cell Number	System Cell Name	Function and Reference
		For plastic, creep, and mechanical strains, the software always outputs effective strain. For initial strain, the software always outputs von Mises strain. For more information about how the software calculates von Mises and effective strain, see Types of strain output in the Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402).
807		For SOL 401, the 3D solid, axisymmetric, plane stress, and plane strain elements have rotational degrees of freedom, but when the elements are either not connected, or connected only to each other, the rotational degrees of freedom have no stiffness. Rotational degrees of freedom without stiffness do not affect the solution. To improve performance, for the elements listed below, the software applies a fixed constraint to the rotational degrees of freedom on the grid points that are not connected to elements with rotational stiffness. The performance improvement is not available when the element iterative solver is requested with SOLVER=ELEMITER on the NLCNTL bulk entry. • 3D solid elements CHEXA, CTETRA, CPENTA, and CPYRAM • Axisymmetric elements CTRAX3, CQUADX4, CTRAX6, and CQUADX8 • Plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, and CPLSTS8 • Plane strain and generalized plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, and CPLSTN8 • Chocking elements CCHOCK3, CCHOCK4, CCHOCK6, and CCHOCK8 • Cohesive elements CHEXCZ and CPENTCZ 0: The software does not apply the fixed constraint to the rotational degrees of freedom that have no stiffness. 1 (default): The software applies the fixed constraint to the rotational degrees of freedom that have no stiffness. Examples of elements that do have rotational stiffness include the CBAR, CBEAM, CBUSH, CBUSH1D, CELAS1, CELAS2, CONM1, CONM2, CTRIA3, CQUAD4, CTRIA6, CQUAD8, CROD, CSHEAR, MPC, RBAR, RBE2, and RBE3.
809		For SOL 401, specifies if user-defined MPC conditions or RBE2, RBE3, RBAR elements are allowed in a Fourier modes subcase when harmonic numbers k>0 are requested. 0 (default): The solution ends with a fatal message.

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System Cell Number	System Cell Name	Function and Reference
		1: The solution is allowed to continue with a warning message. Note that this system cell does not change the solution behavior as a result of the MPC conditions automatically created by the software for harmonic number k=1. See Fourier harmonic solution.
810		For SOLs 103, 105, 111, 112, and 401, selects the solver used for the factorization when performing a real eigenvalue analysis with the Lanczos method. 1 (default): Use the MUltifrontal Massively Parallel sparse direct Solver (MUMPS). 0: Use the sparse solver. For SOLs 111 and 112, you use system cell 810 to select the solver for the factorization with the modes computation and the SLVCNTL bulk entry to select the solver for the response computation. For SOL 401, you use system cell 810 to select the solver for the factorization in the normal modes, cyclic modes, and Fourier modes subcases and the SOLVER and USOLVER parameters on the NLCNTL bulk entry to select the symmetric and unsymmetric solver for the statics, preload, and transient dynamics subcases.
811		For SOL 401, before Simcenter Nastran 2406, some parameters were supported in either the bulk data and/or case control section of the input file. Those parameters were previously designated with "E" in the **parameter applicability table**. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. They are now designated with "$B^{(1)}$" in the parameter applicability table. You can revert to the previous behavior by defining SYSTEM(811)=1 which allows these parameters to be defined in the bulk data and/or the case control without the fatal error. Note that the SOL 401 parameters with "B" in the applicability table (that is, no $^{(1)}$ footnote) must always be defined in the bulk data regardless of the system cell 811 setting. 0 (default): The solution ends with a fatal message if a parameter is defined in the case control. 1: Allows parameters with $B^{(1)}$ in the parameter applicability table to be defined in the bulk data and/or the case control without a fatal error.
814		For SOL 401 with the default Newton-Raphson iteration strategy (SYSTEM(779)=1), if convergence fails for a time step, the software bisects the time step, and if you *do not* have KUPDATE= -1 defined on

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System Cell Number	System Cell Name	Function and Reference
		the NLCNTL bulk entry, it computes an initial stiffness for the bisected step. The system cell setting determines how the software computes the initial stiffness. 0 (default): The software computes the stiffness based on the last converged time step. 1: The software computes the initial stiffness based on the first iteration of the bisected time step. The SYSTEM(814)=1 setting can help avoid ill-conditioned stiffness and may help the convergence for the rejected time step, especially when creep materials are defined. We recommend that you use it when a lot of bisections are occuring and the following warning messages are reported: `ERROR IN THE MATERIAL LIBRARY OCCURS DURING STIFFNESS UPDATE, IT WILL LIKELY RESULT IN AN ILL-CONDITIONED STIFFNESS MATRIXNEGATIVE JACOBIAN IN LARGE STRAIN ANALYSIS OCCURS DURING STIFFNESS UPDATE, IT WILL LIKELY RESULT IN AN ILL-CONDITIONED STIFFNESS MATRIX` The SYSTEM(814)=1 setting is not yet supported when elements are defined using bubble modes for a small strain analysis (PARAM,LGSTRN,0). For example, the default integration option on the PSOLID entry for a CHEXA element uses bubble modes. For this case, the software automatically switches to the default SYSTEM(814)=0 setting and reports a warning message. See SOL 401 - Newton-Raphson iteration strategy for more information.
821		For SOL 401, you can use this system cell to revert to the Simcenter Nastran 2406 version of the CASI iterative solver. In Simcenter Nastran 2412, the CASI iterative solver was improved and is used by default when you request the iterative solver by defining SOLVER=ELEMITER on the NLCNTL bulk entry. You can define SYSTEM(821)=0 to revert to the Simcenter Nastran 2406 version of the CASI iterative solver. 1 (default): The software uses the Simcenter Nastran 2412 version of the CASI iterative solver. 0: Revert to the Simcenter Nastran 2406 version of the CASI iterative solver.
826		Controls the calculation of von Mises strain results in SOLs 108 and 111.

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System Cell Number	System Cell Name	Function and Reference
		0: Output the von Mises strain results as calculated by the Charron method. 1 (default): Output the product of von Mises strain results as calculated by the Charron method and 2/3, which accounts for a Poisson's ratio of 0.5. 2: Output the product of von Mises strain results as calculated by the Charron method and $\left(\frac{1}{1+\nu}\right)$, where ν is the Poisson's ratio specified on the MATi bulk entry for isotropic materials or 0.5 for anisotropic materials. For information about the Charron method, which does not account for Poisson's ratio in the von Mises strain calculation, see Charron, Donato, and Fontaine, *Exact Calculation of Minimum Margin of Safety for Frequency Response Analysis Stress Results Using Yielding or Failure Theories*, MSC 1993 World Users' Conference Proceedings.
827		For SOL 401, you can use this system cell to request the output of stiffness and load matrices to a *.pch file. 0 (default): The matrices are not output. 1: The stiffness and load matrices are written to a *.pch file.

Remarks related to SOL 402:

1. **System Cell** 400 is forced to 1.
2. **System Cell** 413 is forced to 1.
3. If **System Cell** 442 has a non-zero value, this value is forced to 2.

Related Topics

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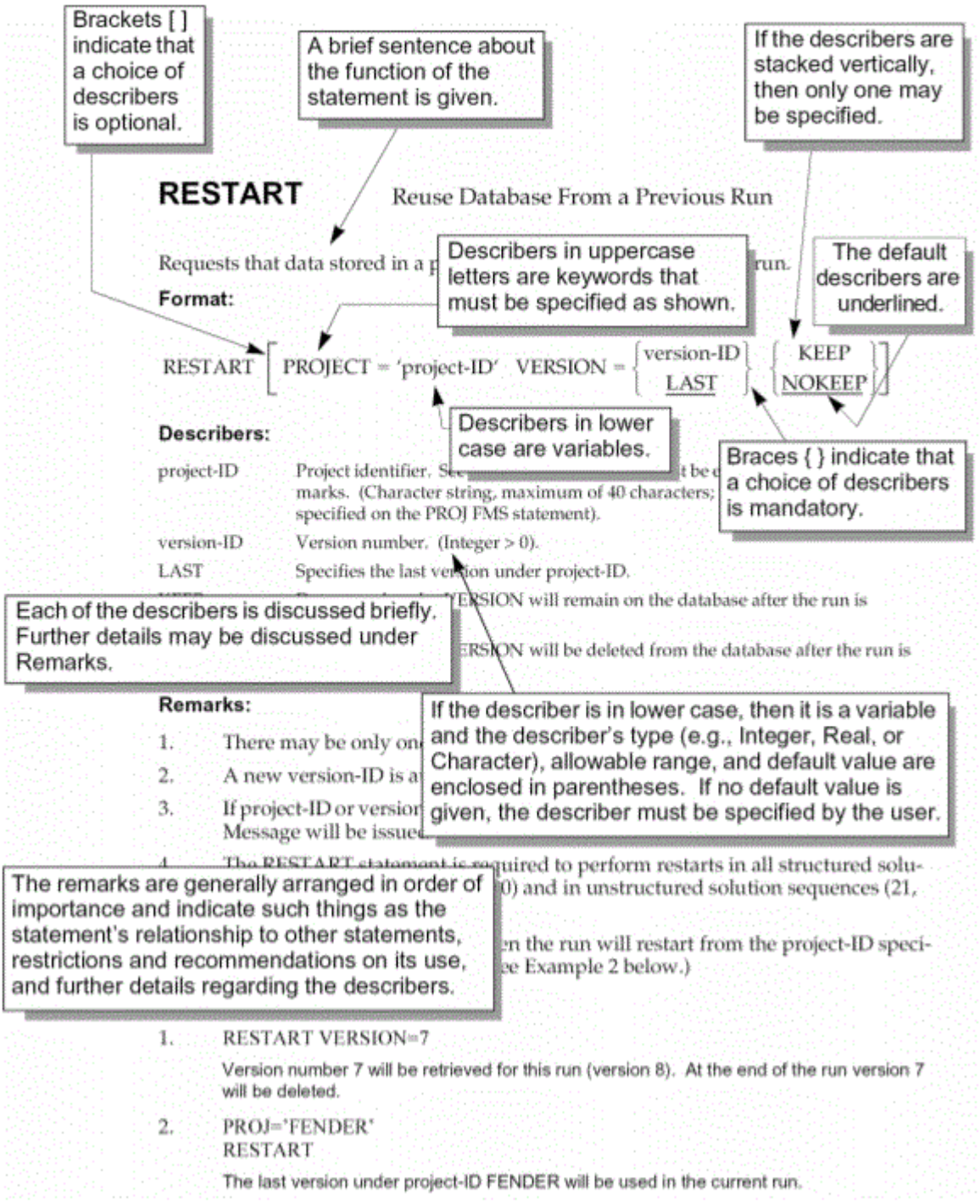
2. File Management Statements

Key to Descriptions



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The File Management Section (FMS)

The File Management Section (FMS) is primarily intended for the attachment and initialization of Database sets (DBsets) and FORTRAN files. The initialization of DBsets includes specification of their maximum size, member names, and physical filenames. The initialization of FORTRAN files includes the specification of their filenames, FORTRAN unit numbers, and FORTRAN attributes.

In most classes of problems that use Simcenter Nastran solution sequences (SOL), no File Management statements are required because a default File Management Section is executed at the beginning of every run. If a restart is desired, then the RESTART statement is required in SOL 101 through 200. All other solutions may not be restarted. If the problem is large in terms of requiring significant amounts of memory or disk space, then the INIT, ASSIGN, and EXPAND statements may be required. If any FORTRAN files are required, then the ASSIGN statement is required; for example, the OUTPUT2 DMAP module. The ASSIGN statement is also required to assign databases for DBLOCATE, DBLOAD, and DBUNLOAD. Special database operations are performed by the DBLOCATE, DBLOAD, DBUNLOAD, DBLCLEAN, ACQUIRE, DBDIR, DBDICT, DBFIX, DBSETDEL, DBUPDATE, and PROJECT statements.

Related Topics

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File Management Statement Summary

The following is a summary of all File Management statements:

$	Comment statement.
ACQUIRE	Selects NDDL schema and Simcenter Nastran Delivery Database.
ASSIGN	Assigns physical files to DBset members or special FORTRAN files.
CONNECT	Groups geometry data by evaluator and database.
DBCLEAN	Deletes selected database version(s) and/or projects.
DBDICT	Prints the database directory in user-defined format.
DBDIR	Prints the database directory.
DBFIX	Identifies and optionally corrects errors found in the database.
DBLOAD	Loads a database previously unloaded by DBUNLOAD.
DBLOCATE	Obtains data blocks and parameters from databases.
DBSETDEL	Deletes DBsets.
DBUNLOAD	Unloads a database for compression, transfer, or archival storage.
DBUPDATE	Specifies the time between updates of the database directory.
ENDJOB	Terminates a job upon completion of FMS statements.
EXPAND	Concatenates additional DBset members to an existing DBset.
INCLUDE	Inserts an external file in the input file.
INIT	Creates a temporary or permanent DBset.
NASTRAN	Specifies values for system cells.
PROJ	Defines the current or default project identifier.

The FMS statements are executed in the following order regardless of their order of appearance in the input file:

NASTRAN, DEFINE

RFINCLUDE, INCLUDE

ASSIGN, INIT, EXPAND, DBUPDATE

PROJECT

DBCLEAN

DBFIX

DBDIR (1), DBDICT(1)

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DBSETDEL

ACQUIRE

RESTART

DBLOCATE

DBUNLOAD

DBLOAD

DBDIR (2), DBDICT(2)

ENDJOB

If DBDIR or DBDICT is specified before any of the FMS statements DBSETDEL through DBLOAD, then the directory printout will reflect the processing of DBCLEAN and DBFIX only. If DBDIR or DBDICT is specified after DBSETDEL through DBLOAD, then the directory printout will reflect the processing of all statements in the FMS Section. It is recommended that the DBDIR and DBDICT statements be specified last in the FMS Section. Multiple DBLOCATE, DBLOAD, or DBUNLOAD statements are processed in the order in which they appear. If the ENDJOB statement is specified, then only the File Management Section is processed and the Executive Control, Case Control, and Bulk Data Sections are ignored.

Related Topics

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File Management Statement Descriptions

File Management statements may be abbreviated down to the first four characters as long as the abbreviation is unique relative to all other statements. Each statement is described as follows:

Description

A brief sentence about the function of the statement is given.

Format

Describers in uppercase are keywords that must be specified as shown. In addition, describers in lowercase indicate that the user must provide a value.

Braces { } indicate that a choice of describers is mandatory. If the describers are stacked vertically, then only one may be specified.

Brackets [ ] indicate that a choice of describers is optional. If the describers are stacked vertically, then only one may be specified.

Describers that are shaded indicate the defaults.

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If the statement line is longer than 72 columns, then it may be continued to the next line with a comma as long as the comma is preceded by one or more spaces. For example:

```
DBLOCATE DATABLK=(KAA)  ,
   WHERE(PROJECT='FRONT BUMPER' AND  ,
   SEID>0 AND VERSION=4)  ,
   LOGI=MASTER3
```

However, if a filename is to be continued on the next line, no space must precede the comma, and the continuation line must have no leading spaces.

For example:

```
ASSIGN     SDB='/jw/johannes/Projects/secret/Aero/Tests/wing/Modes/w,
ing_modal.MASTER'
```

Note that all quotation marks shown under formats and examples are right-handed single quotation marks and must be entered as such. For example:

```
PROJ='MYJOB'
```

Examples

One or more examples are given.

Describers

Each of the describers is briefly discussed. The describer's type (e.g., Integer, Real, or Character), allowable range, and default value are enclosed in parentheses. The describer must be specified by the user if no default value is given.

Remarks

The remarks are generally arranged in order of importance and indicate such things as the FMS statement's relationship to other commands, restrictions and recommendations on its use, and further descriptions of the describers.

Related Topics

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WHERE and CONVERT Clauses

The WHERE clause is used in the selection of items (data blocks and parameters) on the DBDICT, DBLOCATE, DBLOAD, and DBUNLOAD statements. The CONVERT clause modifies qualifier values of items selected by the WHERE clause on the DBLOCATE and DBLOAD statements.

The WHERE and CONVERT clauses specify values for PROJECT, VERSION, qualifiers, and DBSET. PROJECT specifies the project-ID that is originally defined on the PROJECT FMS statement at the time the project is created. VERSION specifies the desired version-ID under the project-ID. Qualifiers are used to uniquely identify items on the database with the same name. For example, data block KAA has SEID as one of its qualifiers, which is the superelement ID. An item may have more than one qualifier and the collection of all qualifiers assigned to an item is called a path.

All data blocks and parameters with qualifiers are defined in the NDDL (NASTRAN Data Definition Language) sequence. See the *Simcenter Nastran DMAP Programmer's Guide*. Data blocks and parameters are defined on the DATABLK and PARAM NDDL statements. The DATABLK and PARAM statements specify the name of the data block, parameter, and also its pathname. The pathnames are defined on the PATH NDDL statement, which lists the qualifiers assigned to the path. Qualifiers are defined on the QUAL NDDL statement. DBSET specifies the desired DBset. The DBset of an item is specified after the LOCATION keyword on the DATABLK and PARAM NDDL statement.

The format of the WHERE clause is:

WHERE (where-expr)

where-expr is a logical expression that specifies the desired values of qualifiers, PROJECT, VERSION, and DBSET. If the result of the logical expression is TRUE for an item on the database then the item is selected. For example, WHERE(VERSION=4 AND SEID < > 2 AND SEID > 0) selects all items under version 4 for all values of SEID greater than 0 except 2.

A simple where-expr is a comparison using the following relational operators = , > , < , < = , > =, >< or <>. For example, SEID > 0 means if SEID is greater than zero, then the logical expression is true. Several simple where expressions may be joined into a single where-expression using the following logical operators: AND, OR, XOR, and EQV. The NOT operator may be used to negate a where expression. For example, NOT(SEID > 0) is the same as SEID < = 0. Arithmetic operations and DMAP functions may also be specified in the where-expression. (See the *Simcenter Nastran DMAP Programmer's Guide* .)

If a qualifier in a where-expr is not a qualifier in the path of a specified item, then the where-expr is set to FALSE. If the where-expr does not contain a specification for all qualifiers in the path of an item, then the unspecified qualifiers will be wildcarded, (i.e., quali= * , all values will be selected). The default values of qualifiers, PROJECT, VERSION, and DBSET are described under the statement in which the WHERE clause is specified.

Examples of the WHERE clause are:

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1. Select all items in the database for all superelements except 10 and 30 from Version 1.

```
WHERE (VERSION=1 AND SEID>=0 AND NOT(SEID=10 OR SEID=30))
```

2. Select all entries in the database on DBSET=DBALL from all projects and versions.

```
WHERE(PROJECT=PROJECT AND VERSlON>0 AND DBSET='DBALL')
```

The CONVERT clause modifies project- and version-ID, DBset-name (see INIT statement), and qualifier values of items selected by the WHERE clause on the DBLOCATE and DBLOAD statements. It contains one or more assignment statements separated by semicolons. The format of the CONVERT clause is:

CONVERT(PROJECT=project-expr; VERSION=version-expr; ,

The PROJECT and VERSION statements modify the project-ID (see PROJECT FMS statement) and version-ID. The DBSET statement modifies the DBset-name. The value of quali will be replaced by qual-expri for selected items that have quali in their path. qual-expri is any valid expression (see "Expressions and Operators" in the *Simcenter Nastran DMAP Programmer's Guide*) containing constants or any qualifier name defined in the path of the item. If qual-expri contains names of qualifiers not in the path of the selected item, then a fatal message is issued. If project-expr and/or version-expr produces a project- or version-ID which does not exist, then one will be created. Also, all version-IDs less than version-expr that do not exist will be created; but they will be "empty."

Examples of the CONVERT clause are:

1. Set qualifiers SEID, PEID, and SPC to constants 10, 20, 102 respectively.

```
CONVERT(SEID=10;PEID=20;SPC=102)
```

   If more than one value of a qualifier is found for an item by the WHERE clause, then each value is processed in qual-expri to define the new qualifier value for each of the selected items. In the example below, if the original values of PEID were 1, 2, and 3; then the new values for the SEID qualifier will be 2, 4, and 6.

2. Set all values of qualifier SEID to be twice that of the values of qualifier PEID.

```
CONVERT(SEID=2*PEID)
```

Related Topics

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$

Comment

Used to insert comments into the input file. Comment statements may appear anywhere within the input file.

Format:

$ followed by any characters out to column 80.

Examples:

```
$ TEST FIXTURE-THIRD MODE
```

Remarks:

1. Comments are ignored by the program.
2. Comments will appear only in the unsorted echo of the Bulk Data.

Related Topics

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ACQUIRE

Selects NDDL Schema

Selects the NDDL schema and Simcenter Nastran delivery database to be used for primary database creation.

Format:

$$\text{ACQUIRE } \underline{\text{NDDL}}= \begin{Bmatrix} \text{NDDL} \\ \text{nddl} - \text{name} \end{Bmatrix}$$

Describers:

Describer	Meaning
NDDL	Simcenter Nastran NDDL schema.
nddl-name	Name of a user NDDL schema specified on a COMPILE NDDL statement when the user NDDL was stored.

Remarks:

This statement is used to specify the delivery database when the user wishes to create his or her own solution sequence, yet use the subDMAP objects or NDDL schema or both from the delivery database supplied with the software.

Examples:

The following requests the Simcenter Nastran NDDL schema to be used when creating a new database.

```
ACQUIRE NDDL
SOL MYDMAP
COMPILE DMAP=MYDMAP,SOUOUT=USROBJ
.
.
.
LINK MYDMAP,SOLOUT=USROBJ
```

Related Topics

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ASSIGN

Assigns Physical File

Assigns physical file names to DBset members, special FORTRAN files that are used by other FMS statements or DMAP modules, or HDF5 files.

Format:

Format 1

Assign a DBset member name:

$$\text{ASSIGN log} - \text{name} = \text{'filename1'} \left[\text{TEMP DELETE SYS} = \text{'sys} - \text{spec'}\right]$$

Format 2

Assign a FORTRAN file:

$$\text{ASSIGN logical} - \text{key} = \text{'filename2'},$$
$$\left[\text{STATUS} = \left\{\begin{matrix}\text{NEW}\\ \text{OLD}\\ \text{UNKNOWN}\end{matrix}\right\}\right],$$
$$\left[\text{FORM} = \left\{\begin{matrix}\text{FORMATTED}\\ \text{UNFORMATTED}\\ \text{LITTLEENDIAN}\end{matrix}\right\}\right],$$
$$[\text{UNIT} = \text{u}], [\text{TEMP}], [\text{DELETE}], [\text{SYS} = \text{' sys} - \text{spec'}]$$

Format 3

Assign an HDF5 file:

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$$\text{ASSIGN logical} - \text{key} = \text{'filename3'},$$
$$\left[\text{STATUS} = \left\{\begin{array}{c}\text{NEW}\\ \text{OLD}\\ \text{UNKNOWN}\end{array}\right\}\right],$$
$$\left[\text{FORM} = \left\{\begin{array}{c}\text{FORMATTED}\\ \text{UNFORMATTED}\\ \text{LITTLEENDIAN}\end{array}\right\}\right],$$
$$\text{UNIT} = \text{u}, \text{ DEFER}$$

Examples:

1. Assign the DBALL DBset:

```
ASSIGN DB1='filename of member DB1'
```

```
INIT DBALL LOGI=(DB1)
```

2. Assign FORTRAN file 12 to the OUTPUT4 module using the ASCII option:

```
ASSIGN OUTPUT4='filename of FORTRAN file',
```

```
UNIT=12, FORM=FORMATTED
```

3. Assign HDF5 file to unit number 101:

```
ASSIGN SC_H5='filename of HDF5 file .sc_h5 extension typical',
```

```
UNIT=101, DEFER
```


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Describers:

Describer	Meaning
log-name	The name of a DBset member name. log-name is also referenced on an INIT statement after the LOGICAL keyword.
filenamei	The physical filename assigned to the DBset member.
TEMP	Requests that filenamei is deleted at the end of the run.
DELETE	Requests that filenamei, if it exists before the start of the run, is to be deleted.
logical-key	Specifies defaults for STATUS, UNIT, and FORM of FORTRAN files for other FMS statements, DMAP modules, punching and plotting operations.
filename2	The physical file name assigned to the FORTRAN file.
filename3	The physical file name assigned to the HDF5 file with commonly a .sc_h5 extension.
STATUS	Specifies whether the FORTRAN file is being created (STATUS=NEW) or has been created prior to the run (STATUS=OLD). If its status is not known, then STATUS=UNKNOWN is specified.
UNIT=u	u is the unit number of the file. See Remark 9.
FORM	Indicates whether the FORTRAN file is written in ASCII (FORM=FORMATTED), binary (FORM=UNFORMATTED), or little endian binary (LITTLEENDIAN).
DEFER	The specified file will not be opened/created during initialization. The file must be opened by the module or DMAP accessing the file.
sys-spec	System specific or machine-dependent controls. See Remark 8 .

Remarks:

1. The ASSIGN statement and its applications are discussed further in “Introduction to Database Concepts” in the *Simcenter Nastran User’s Guide*.
2. The logical-key names and their default attributes that may be assigned by the user are as follows:

logical-key Name	Default Status	Default Unit	Default Form	Application
DBC	NEW	40	UNFORMATTED	DBC module — PARAM,POST,0
INPUTT2	OLD	none	UNFORMATTED	INPUTT2 module
INPUTT4	OLD	none	UNFORMATTED	INPUTT4 module
OUTPUT2	NEW	none	UNFORMATTED	OUTPUT2 module
OUTPUT4	NEW	none	UNFORMATTED	OUTPUT4 module
DBUNLOAD	NEW	50	UNFORMATTED	DBUNLOAD FMS statement
DBLOAD	OLD	51	UNFORMATTED	DBLOAD FMS statement
USERFILE	none	none	none	User-defined

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logical-key Name	Default Status	Default Unit	Default Form	Application
HDF	NEW	52	UNFORMATTED	HDF5 output file with commonly a .hdf extension
SC_H5	OLD	none	UNFORMATTED	HDF5 output file with commonly a .sc_h5 extension. DEFER must always be included. This logical-key relates to the RANDPEX, RLOADEX, SPCF, and MDSBULK bulk entries.
EXTSAM	OLD	none	UNFORMATTED	Samcef superelement creation database with a .sdb extension.

The defaults may be overridden on the ASSIGN statement.

3. Certain reserved names may not be used for log-names or logical-key names: SEMTRN, LNKSWH, MESHFL, LOGFL, INPUT, PRINT, INCLD1, and CNTFL. If they are used, then a fatal message is issued. Also, unit numbers 1 through 10, 14 and 16 should not be assigned. PUNCH and PLOT may be used but are not recommended.
4. If one of the logical-key names indicated in Remarks 2. and 3. is not specified on this statement, then it is assumed to be a DBset member log-name as shown in Format 1.
5. The logical-key names DBUNLOAD and DBLOAD may be assigned only once in the FMS section. The others may be assigned as many times as needed for the application. However, in all logical-key assignments, the unit number u must be unique. If it is necessary to execute the INPUTT4 and OUTPUT4 modules on the same unit, then specify ASSIGN OUTPUT4 only. The same is recommended for the INPUTT2 and OUTPUT2 modules.
6. STATUS, UNIT, and FORM are ignored if assigning a log-name (DBset member name).
7. FORM=FORMATTED must be specified for a unit when:
   - Ascii output is desired on the INPUTT4 and OUTPUT4 DMAP modules that process the unit. See the *Simcenter Nastran DMAP Programmer's Guide*.
   - FORMAT=NEUTRAL is selected on the DBUNLOAD and DBLOAD FMS statements that process the unit. See "Database Archival and Retrieval" in the *Simcenter Nastran User's Guide*.
   - The neutral file format is desired for the INPUTT2 and OUTPUT2 modules.
8. See the *Simcenter Nastran Installation and Operations Guide* for further information on machine-dependent aspects of the ASSIGN statement.
9. The following unit numbers are reserved for specific tasks. Only use these numbers when performing the associated task. For example, UNIT=7 is used for punch files.

```
ASSIGN PUNCH='results_output.pch',NEW,UNIT=7
```

UNIT NO	DESCRIPTION
1	Default UNIT for SCR.F01
2	Default UNIT for SCR.F02

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3	Mesh file SCR.F03
4	Log file (typically standard out)
5	Input (deck) file
6	Print file (f06 file)
7	Punch file
8	Node-locked license file
9	Include files
10	Control file or the ASG file
11	INPUTT2 unit
12	OP2 file
14	Plot file
15	Mesh file
16	Assembly file
18	Acoustics coupling file - 1
19	Acoustics coupling file - 2
20	Data base migration files
21	Adina input file
22	Adina output file - 1
23	Adina output file - 2
24–29	BCS linear solver
24–33	BCS eigensolver
31–35	Reserved for AMLS files
40	DBC (xdb) File1
41	Reserved for SML (SAMCEF Material Library)
42	Reserved for SML (SAMCEF Material Library)
43	Reserved for SML (SAMCEF Material Library)
44	Reserved for SML (SAMCEF Material Library)
45	Reserved for SML (SAMCEF Material Library)
46	Reserved for AMLS files
50	DBC (xdb) File2

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51	DBC (xdb) File3
52	HDF File (hdf)
56	Print file (f56 file)
66	Linear contact refinement file
70	Reserved for Akusmod files
71-80	Reserved for BUN files (TEMPEX and DTEMPEX)
81-90	Reserved for SML (SAMCEF Material Library)
101-110	Reserved for HDF5 files (commonly .sc_h5), but they are not mandatory.
112-113	Reserved for SOL 200 use
115	Reserved for stiffness matrix [K] and load matrix [P] output from SOL401 when SYSTEM(827)=1.
151–154	Solution monitor files

10. Windows systems have a default maximum path and filename length of 260 characters. If you are defining a path relative to your current working directory, all characters in the complete path should be considered. For example, if you run a job in *C:\workdir* and the ASSIGN statement *ASSIGN INPUTT2='.\user\filename.op2' UNIT=201* is defined in your input file, the path you should consider is *C:\workdir\.\user\filename.op2* which is 30 characters.

Related Topics

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CONNECT

Group Evaluator Data

Defines a group of beam cross-section entities, or a DRESP3 second order response function external evaluator. These entities should belong to the same evaluator-class (set of routines that process them). The PBARL, PBEAML, and DRESP3 Bulk Data entries refer to the groups defined here.

Format:

$$\text{CONNECT} \begin{bmatrix} \text{BEAMEVAL} \\ \text{DRESP3} \end{bmatrix} \text{group evaluator 'path' 'data'}$$

Describers:

Describer	Meaning
group	Group name referenced by the GROUP field on PBARL, PBEAML, and DRESP3 Bulk Data entries.
evaluator	Identifies the particular class of evaluator to which the beam cross-section entities belong, or is an alias for the path name to the DRESP3 evaluator. Entities belonging to one evaluator-class are handled by the same set of routines (either provided with the software or user-provided). For beam cross-sections, the class MSCBML (Simcenter Nastran Beam Library) is provided internally. For the DRESP3 response evaluation, template subroutines are provided with Simcenter Nastran, so that the users can customize them accordingly, modifying them and/or adding more subroutines called from these. Users may develop custom evaluator libraries for beam cross-sections, or DRESP3 functions and configure them for use with Simcenter Nastran. See Remarks 2, 3 and, 4.
path	Optional pathname or filename used by evaluator. Path must be enclosed by single quotation marks if it contains lowercase characters.
data	Optional character string passed to the evaluator. Data must be enclosed within single quotation marks if it contains lowercase characters or embedded blanks.

Remarks:

1. CONNECT requests:
   - an external database or evaluator,
   - a user defined grouping for beam cross-section data defined by PBARL and PBEAML entries.

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2. A single reserved group name, MSCBML0 has been predefined for beam cross-section entities. It may be used in the PBARL and PBEAML entries, without being defined explicitly by means of a CONNECT FMS statement. It corresponds to the MSCBML (Beam-Library) evaluator.
3. Custom beam cross-section or DRESP3 evaluator libraries developed by users should comply with the guidelines outlined in the *Simcenter Nastran User's Guide*.
4. Once developed, an evaluator may be configured as:
   - internal, where the evaluator routines are linked with the rest of the Simcenter Nastran object modules to comprise the Simcenter Nastran executable program,
   - or external, where the evaluator routines are linked with a provided server program to constitute an independent geometry server, or DRESP3 server.
5. The Simcenter Nastran provided DRESP3 external program development capability will result in an external program with the default name dr3serv.exe.
6. The file which has the resolution of the alias for the DRESP3 evaluator path name should be referred to from the nastran command line by the gmconn keyword. For example, if the file name is xyz, then gmconn = xyz should be added to the nastran command. Thus, the nastran command refers to the file which contains the resolution of the alias for the path to the evaluator, and the alias is contained in the CONNECT statement, completing the information necessary to refer to the evaluator program from inside the data file.
7. An example for the content of the file that resolves the alias for the DRESP3 evaluator path name is:

   EXTRESP,-,/network_drive/user_dir/dr3srv/dr3serv

   where, for this example, EXTRESP is the evaluator alias used in the CONNECT statement (see also under Examples).

Examples:

1. CONNECT BEAMEVAL HOIST, NEWBEAMS

   In this case, the user is requesting that all calculations on PBARL and PBEAML Bulk Data entries that are grouped as HOIST use the NEWBEAMS evaluator. The user must supply the NEWBEAMS beam cross-section evaluator library, and configure it to function with the Simcenter Nastran executable program.

2. CONNECT DRESP3 TESTGRP EXTRESP

   In this case, the user is requesting that all DRESP3 calculations that are grouped as TESTGRP use the evaluator pointed at by the evaluator alias EXTRESP. EXTRESP is defined as in Remark **7** above, and the file containing this definition is pointed at as defined in Remark **6** above. The external program, with the default name dr3serv, contains Simcenter Nastran supplied routines which the user may modify, and any user added routines called by the first set of routines.

Related Topics

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DBCLEAN

Delete Database Versions and/or Projects

Deletes one or more versions and/or projects from the database.

Format:

DBCLEAN VERSION = {version-ID,*} [PROJECT={'project-ID',*}]

Describers:

Describer	Meaning
version-ID	Version identifier of the database to be deleted.
*	Wildcard. All versions or projects to be deleted.
project-ID	Project identifier of the project to be deleted. (See the FMS statement, **"PROJ"** .)

Remarks:

1. There may be up to ten DBCLEAN statements in the FMS Section.
2. If no project-ID is given, the current project-ID is assumed.

Examples:

```
DBCLEAN VERS=7 PROJ = 'OUTER WING - LEFT'
```

The above example would delete from the database all data blocks and parameters stored under Version 7 of the project identified as OUTER WING – LEFT.

Related Topics

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DBDICT

Prints Database Directory Tables:

Prints database directory tables.

Database Directory Tables

- Data blocks described by an NDDL DATABLK statement.
- Parameters described by an NDDL PARAM statement.
- All unique paths (KEYs) and their qualifiers values.
- Qualifiers and their current values.
- Data blocks not described by an NDDL DATABLK statement.
- Parameters not described by an NDDL PARAM statement.
- Project and version information.

Basic Format:

The basic format of DBDICT specifies which tables to print and prints all items (data blocks and parameters) found in the directory. Also, the attributes (colnames) to be printed and the print format are predefined. Note that more than one table may be specified on the same DBDICT statement.

DBDICT [DATABLK PARAM PROJVERS QUALCURR QUALIFIERS]

Examples:

```
DBDICT
DBDICT PARAM PROJVERS
```

Full Format:

The full format permits the selection of items by name and/or by the WHERE describer. The full format also permits the attributes to be printed using the SELECT describer. In addition, the print format can be specified with the SORT, FORMAT, and LABEL describers. Note that the full format only allows the specification of a single table on a DBDICT statement.

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$$\text{DBDICT}\left(\left[\begin{array}{c}\left(\left[\begin{array}{c}\underline{\text{DATABLK}}\\ \text{DATABLK (LOCAL)}\end{array}\right]=\left[\begin{array}{c}\underline{*}\\ \text{datablk}-\text{list}\end{array}\right]\right)\\ \left(\left[\begin{array}{c}\text{PARAM}\\ \text{PARAM (LOCAL)}\end{array}\right]=\left[\begin{array}{c}*\\ \text{param}-\text{list}\end{array}\right]\right)\\ \text{PROJVERS}\\ \text{QUALCURR}\\ \text{QUALIFIERS}\end{array}\right]\text{WHERE (where}-\text{expr)},\right)$$

$$\text{SELECT}\left(\text{colname}\left[-'\text{col}-\text{label}'\right]\ldots\right),$$

FORMAT (FWIDTH = w [.d] DWIDTH = w [.d] AWIDTH = a IWIDTH = i,
LWIDTH = k COLSPACE = c VALUE = w,
colname = col-width, . . .),

$$\text{SORT}\left(\text{colname}=\left[\begin{array}{c}\underline{\text{A}}\\ \text{D}\end{array}\right],\ldots\right),$$

$$\text{LABEL}\left('\text{page}-\text{title}'\left[\begin{array}{c}\text{RIGHT}\\ \text{CENTER}\\ \text{LEFT}\end{array}\right]\right)$$

Describers:

Describer	Meaning
DATABLK	Print the data blocks. datablk-list specifies a list of NDDL-defined data blocks separated by commas. If LOCAL is specified, the non-NDDL-defined data blocks are printed.
PARAM	Print the parameter table. param-list specifies a list of parameters separated by commas. If LOCAL is specified, the non-NDDL-defined parameters are printed.
PROJVERS	Print the project-version table.
QUALIFIERS	Print the qualifier table.
QUALCURR	Print the current values of the qualifiers. SORT is ignored.
where-expr	Logical expression that specifies the desired values of colnames described below. For example, WHERE(VERSION=4 AND SEID < > 2 AND SEID > 0) selects all items under version 4 for all values of SEID greater than 0 except 2. See the beginning of this section for a further description. The default for VERSION is the

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Describer	Meaning
	last version, and PROJECT is the current project. The default for qual is *, which is all qualifier values found on the database. See also Remark 12.
SELECT	Specifies a list of column names to be printed. The order of the specified colnames will be printed from left to right. If a colname is not specified, then all columns will be printed.
colname	Column name. colname specifies a particular attribute of the database item; such as, data block name (NAME), creation date (CDATE), number of blocks (SIZE), or qualifier name (SEID, SPC, etc.). The allowable colnames are given in the Remarks.
col-label	The label to be printed above the column identified by colname. The default for col-label is the colname. col-label may not be specified for colnames QUALSET, QUALALL, and TRAILER.
FWIDTH=w.d	Specifies the default width for single-precision real numbers in real and complex qualifiers. (Integers: w > 0 and d > 0, Default = 12.5)
DWIDTH=w.d	Specifies the default width for double-precision real numbers in real and complex qualifiers. (Integers: w > 0 and d > 0, Default = 17.10)
AWIDTH=a	Specifies the default width for character string qualifiers. Character strings are printed with enclosing single quotation marks, even if the string is blank. (Integer > 0, Default = 8)
IWIDTH=i	Specifies the default width for integer qualifiers. (Integer > 0.) See Remarks for defaults.
LWIDTH=k	Specifies the default width for logical qualifiers. Logical values are printed as either “T” for TRUE or “F” for FALSE. (Integer > 0, Default = 1)
COLSPACE=c	Specifies the default number of spaces between columns. (Integer > 0) See Remarks for defaults.
VALUE=w	Specifies the default width for parameter values. The values are printed as character strings with left justification. (Integer > 0, Default = 40)
col-width	The print width of the data under colname or qual-name. For real numbers, specify w.d where w is the width of the field and d is the number of digits in the mantissa. For integers and character strings, specify w where w is the width of the field. col-width may not be specified for colnames QUALSET, QUALALL, and TRAILER.
SORT	Specifies how the rows are sorted. The sort is performed in order according to each colname specified in the list. A “D” following the colname causes the sort to be in descending order. An “A” following the colname causes the sort to be in ascending order. Colnames QUALSET, QUALALL, and TRAILER may not be specified under SORT. Each colname specified in SORT must be separated by commas.

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Describer	Meaning
page-title	A title to be printed on each page of the directory output.
RIGHT, CENTER, LEFT	Print justification of the page title.

Remarks:

1. DBDICT prints seven different tables according to a default or a user-defined format. The tables are:

Table 2-2: DBDICT Tables

Describer	Description	Default Page-Title	See Remark
DATABLK	Data blocks described by a NDDL DATABLK statement.	NDDL DATABLOCKS	2
PARAM	Parameters described by a NDDL PARAM statement.	NDDL PARAMETERS	3
QUALCURR	Current Qualifiers and their values.	CURRENT QUALIFIERS	4
QUALIFIERS	Qualifiers and their values for each key number.	QUALIFIERS	5
DATABLK(LOCAL)	Data blocks not described by a NDDL DATABLK statement.	LOCAL DATABLOCKS	6
PARAM(LOCAL)	Parameters not described by a NDDL PARAM statement.	LOCAL PARAMETERS	7
PROJVERS	Project-Version.	PROJECT-VERSION	8

   If DBDICT is specified without any describers then the NDDL Data blocks Table will be printed. See Remark 2.

   DATABLK(LOCAL) and PARAM(LOCAL) produce no output, and QUALCURR produces the default values specified on the NDDL QUAL statement.

   The defaults and allowable colnames for SELECT, FORMAT, SORT, and LABEL depend on the table. The defaults are described in the following remarks and tables.

2. The default print of the NDDL Data Blocks Table is obtained by:

```
DBDICT
```

   or

```
DBDICT DATABLK
```


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and is equivalent to:

```
DBDICT DATABLK ,
     SELECT(NAME,DATABASE,DBSET,PROJ,VERS,CDATE,CTIME,
          SIZE,KEY,PURGED='PU',EQUIVD='EQ',
          POINTER='FILE',QUALSET) ,
     FORMAT(NAME=8,DBSET=8,CDATE=6,CTIME=6,SIZE=5,
          KEY=4 ,PURGED=4,EQUIVD=4,POINTER=8,
          IWIDTH=5,COLSPACE=1) ,
     SORT(PROJ=A,VERS=A,DBSET=A,NAME=A) ,
     LABEL('NDDL DATABLOCKS' CENTER)
```

and looks like:

Figure 2-1: DBDICT DATABLK Example

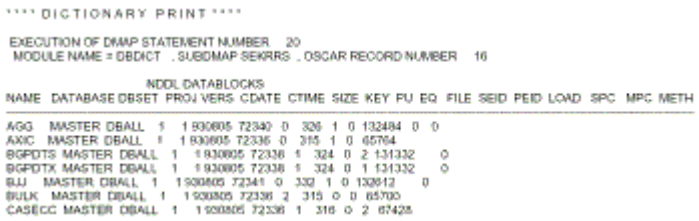

Table 2-3 gives the allowable colnames along with a description that may be specified in the FORMAT, SELECT, and SORT describers.

Table 2-3: DBDICT DATABLK Colnames

colname	Default col-width	Default col-label	Description
PROJECT	40	PROJECT NAME	Project name defined by PROJECT statement
PROJ	4	PROJ NO	Project number associated with PROJECT
VERS	4	VERSION	Version number
CDATE	6	CDATE	Creation date
CTIME	6	CTIME	Creation time
NAME	8	NAME	Parameter name
DATABASE	8	DATABASE	MASTER DBset name
DBSET	8	DBSET	DBset name
RDATE	6	RDATE	Revision date
RTIME	6	RTIME	Revision time
SIZE	5	SIZE	Number of blocks
qual-name	See Note.	qualifier name	Qualifier name
KEY	4	KEY	Key number
TRLi	8	TRLi	i-th word in the trailer
TRAILER	8	TRLi	All 10 trailer words

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colname	Default col-width	Default col-label	Description
EXTNAME	8	EXTNAME	Extended name
EQUIVD	4	EQ	Equivalenced flag
PURGED	4	PU	Purged flag
EQFLAG	4	EF	Scratch equivalenced flag
SCRFLAG	4	SF	Scratch DBSET flag
POINTER	8	POINTER	Directory pointer
DBENTRY	8	DBENTRY	Database entry pointer
FEQCHAIN	8	FEQCHAIN	Forward equivalence chain
BEQCHAIN	8	BEQCHAIN	Backward equivalence chain
DBDIR20	9	DBDIR(20)	Directory word 20
QUALALL	See Note.	qualifier name	All qualifiers
QUALSET	See Note.	qualifier name	Predefined subset of all qualifiers

Note

Default widths for qualifiers are DWIDTH=17.10, IWIDTH=5, LWIDTH=1, AWIDTH=8, and FWIDTH=12.5.

3. The default print of the NDDL Parameter Table is obtained by:

```
DBDICT PARAM
```

and is equivalent to:

```
DBDICT PARAM,
     SELECT(NAME,DATABASE,DBSET,PROJ,VERS,CDATE,CTIME,
          KEY,VALUE,QUALSET),
     FORMAT(NAME=8,DATABASE=8,DBSET=8,CDATE=6,CTIME=6,
          KEY=4,VALUE=40,IWIDTH=5,COLSPACE=1),
     SORT(PROJ=A,VERS=A,DBSET=A,NAME=A),
     LABEL('NDDL PARAMETERS' CENTER)
```

and looks like:

Figure 2-2: DBDICT PARAM Example

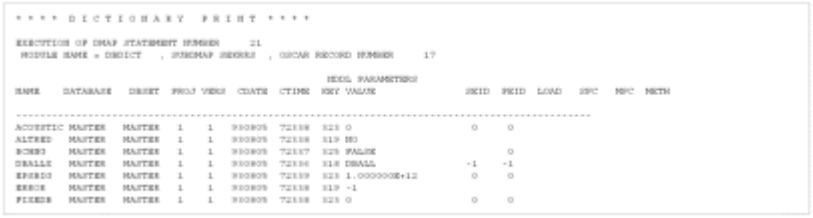

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Table 2-4 gives the allowable colnames along with a description that may be specified in the FORMAT, SELECT, and SORT describers.

Table 2-4: DBDICT PARAM Colnames

colname	Default col-width	Default col-label	Description
PROJECT	40	PROJECT NAME	Project name defined by PROJECT statement
PROJ	5	PROJ	Project number associated with PROJECT
VERS	4	VERS	Version number
CDATE	6	CDATE	Creation date
CTIME	6	CTIME	Creation time
NAME	8	NAME	Parameter name
DATABASE	8	DATABASE	MASTER DBset name
DBSET	8	DBSET	DBset name
RDATE	6	RDATE	Revision date
RTIME	6	RTIME	Revision time
POINTER	8	POINTER	Directory pointer
VALUE	40	VALUE	Parameter value
KEY	4	KEY	Key number
qual-name	See Note.	qualifier name	Qualifier name
QUALALL	See Note.	qualifier name	All qualifiers
QUALSET	See Note.	qualifier name	Predefined subset of all qualifiers

Note

Default widths for qualifiers are DWIDTH=17.10, AWIDTH=8, IWIDTH=5, LWIDTH=1, and FWIDTH=12.5.

4. The default print of the Qualifier Table is obtained by:

```
DBDICT QUALIFIERS
```

and is equivalent to:

```
DBDICT QUALIFIERS ,
     SELECT(KEY QUALALL) ,
     FORMAT(DWIDTH=17.10 AWIDTH=8 IWIDTH=5 LWIDTH=1 ,
          FWIDTH=12.5  COLSPACE=2) SORT(KEY=A) ,
     LABEL('QUALIFIERS' CENTER)
```

and looks like:

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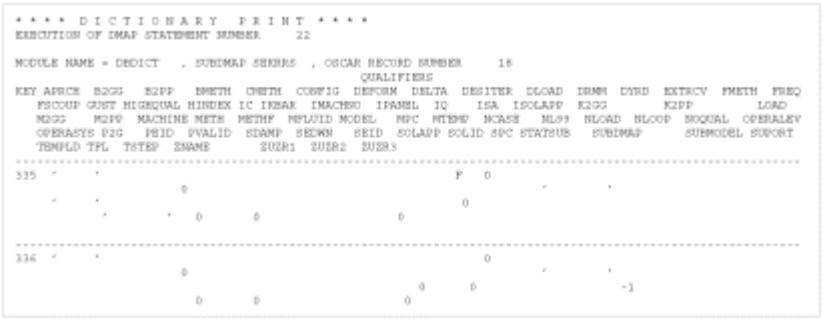
Figure 2-3: DBDICT QUALIFIERS Example



QUALALL selects all qualifiers to be printed. The qualifiers will be printed in alphabetic order. QUALSET selects the qualifiers SEID, PEID, SPC, MPC, LOAD, and METH to be printed.

Table 2-5 gives the allowable colnames along with a description that may be specified in the FORMAT, SELECT, and SORT describers. QUALALL and QUALSET may not be specified in the FORMAT or SORT describers. The qualifier names and values are not printed one per row, but rather from left to right as one logical line that is allowed to wrap after 132 columns.

Table 2-5: DBDICT QUALIFIERS Colnames

colname	Default col-width	Default col-label	Description
KEY	5	KEY	Key number
qual-name	See Note.	qualifier name	Qualifier name
QUALALL	See Note.	qualifier name	All qualifiers
QUALSET	See Note.	qualifier name	Predefined subset of all qualifiers

Note

Default widths for qualifiers are DWIDTH=17.10, IWIDTH=5, LWIDTH=1, and FWIDTH=12.5. AWIDTH defaults to the length specified on the QUAL statement in the NDDL sequence.

5. The default print of the Current Qualifier Table is obtained by:

```
DBDICT QUALCURR
```

and is equivalent to:

```
DBDICT QUALCURR SELECT(QUALALL),
     FORMAT(AWIDTH=8,IWIDTH=5,LWIDTH=1,COLSPACE=2),
     LABEL=('CURRENT QUALIFIERS' CENTER)
```

and looks like:

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Figure 2-4: DBDICT QUALCURR Example

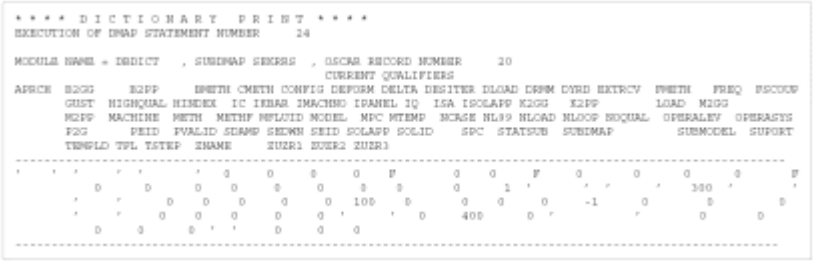

Table 2-6 gives the allowable colnames along with a description that may be specified in the SELECT describers.

Table 2-6: DBDICT QUALCURR Colnames

colname	Default col-width	Default col-label	Description
qual-name	See Note.	qualifier name	Qualifier name
QUALALL	See Note.	qualifier name	All qualifiers
QUALSET	See Note.	qualifier name	Predefined subset of all qualifiers

> **Note**
>
> Default widths for qualifiers are DWIDTH=17.10, IWIDTH=5, LWIDTH=1, and FWIDTH=12.5. AWIDTH defaults to the length specified on the QUAL statement in the NDDL sequence.

6. The default print of the Local Data Block Table is obtained by:

```
DBDICT DATABLK(LOCAL)
```

and is equivalent to:

```
DBDICT DATABLK(LOCAL),
     SELECT(NAME,SUBDMAP,SIZE='BLOCKS',PURGED='PU',
          EQUIVD='EQ',POINTER,TRL1,TRL2,TRL3,TRL4,
          TRL5,TRL6,TRL7),
     FORMAT(NAME=8,SUBDMAP=8,IWIDTH=8,COLSPACE=2),
     SORT(NAME=A) LABEL('LOCAL DATABLOCKS' CENTER)
```

and looks like:

Figure 2-5: DBDICT DATABLK(LOCAL) Example

TRLi specifies the data block trailer word i where 1 ≤ i ≤ 10. TRAILER selects all 10 data block trailer words.

Table 2-7 gives the allowable colnames along with a description that may be specified in the FORMAT, SELECT, and SORT describers.

Table 2-7: DBDICT DATABLK(LOCAL) Colnames

colname	Default col-width	Default col-label	Description
NAME	8	NAME	Parameter name
SUBDMAP	8	SUBDMAP	SubDMAP name
SIZE	8	BLOCKS	Number of blocks
EQUIVD	8	EQ	Equivalenced flag
PURGED	8	PU	Scratch flag
POINTER	8	POINTER	Directory pointer
TRLi	8	TRLi	i-th word in the trailer
TRAILER	8	TRLi	All 10 trailer words
EXTNAME	8	EXTNAME	Extended name

7. The default print of the Local Parameter Table is obtained by:

```
DBDICT PARAM(LOCAL)
```

and is equivalent to:

```
DBDICT PARAM(LOCAL) SELECT(NAME,SUBDMAP,VALUE),
     FORMAT(COLSPACE=4,VALUE=40,AWIDTH=8),
     SORT(NAME=A) LABEL(' LOCAL PARAMETERS' CENTER)
```

and looks like:

Figure 2-6: DBDICT PARAM(LOCAL) Example

Table 2-8 gives the allowable colnames along with a description that may be specified in the FORMAT, SELECT, and SORT describers.

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Table 2-8: DBDICT PARAM(LOCAL) Colnames

colname	Default col-width	Default col-label	Description
NAME	8	NAME	Parameter name
SUBDMAP	8	SUBDMAP	SubDMAP name
VALUE	40	VALUE	Parameter name

8. The default print of Project-Version Table is obtained by:

```
DBDICT PROJVERS
```

and is equivalent to:

```
DBDICT PROJVERS ,
     SELECT(PROJECT='PROJECT NAME',PROJ='PROJ NO.' ,
          VERS='VERSION',DELFLG='DELETED' ,
          CDATE='CREATION DATE' CTIME='CREATION
          TIME') ,
     FORMAT(PROJECT=40,PROJ=10,VERS=10,DELFLG=7,
          COLSPACE=1 ,CDATE=13,CTIME=13) ,
     LABEL('PROJECT-VERSION',CENTER) ,
          SORT(PROJ=A,VERS=A)
```

Table 2-9 gives the allowable colnames along with a description that may be specified in the FORMAT, SELECT, and SORT describers.

Table 2-9: DBDICT PROJVERS Colnames

colname	Default col-width	Default col-label	Description
PROJECT	40	PROJECT NAME	Project name defined by PROJECT statement
PROJ	10	PROJ NO	Project number associated with PROJECT
VERS	10	VERSION	Version number
DELFLG	7	DELETED	Flag indicating whether this project/version has been deleted by the RESTART NOKEEP or DBCLEAN statements.
CDATE	13	CREATION DATE	Creation date
CTIME	13	CREATION TIME	Creation time

CDATE is printed as YYMMDD where YY, MM, and DD are the year, month, and date, respectively. CTIME is HHMMSS where HH, MM, and SS are the hour, minute, and second, respectively.

9. If a parameter or qualifier value is defined to be character string, then the value will be printed with enclosing single quotation marks. Blank strings will also be printed with single quotation marks.

10. If a given qualifier is not in the path of a given data block or parameter, then blank spaces will be printed.

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11. A line will wrap if additional columns need to be printed and not enough space is available on the output. The first column of each additional line is to be indented by the width of the first column printed for the entry.
12. The where-expr has the following rules:
    - If the where-expr specifies a colname that is not assigned to the data block or parameter, then no directory information will be printed for that data block or parameter. For example, given that SPC is not a qualifier for KGG, then the following DBDICT statement will produce no output:

      ```
      DBDICT DATABLK=KGG WHERE(SPC=10)
      ```

    - If the where-expr does not specify a colname that is assigned to the data block (or parameter), then the qualifier is wildcarded. For example, given that SEID is a qualifier for KAA, then the following DBDICT statements are equivalent:

      ```
      DBDICT DATABLK=KAA
      DBDICT DATABLK=KAA WHERE(SEID = *)
      ```

13. A colname specified in the where-expr must be specified in the SELECT clause if the SELECT clause is also specified.

Examples:

1. Print the Project Version Table with a title.

   ```
   DBDICT  PROJVERS SORT(PROJ,VERSION) LABEL('PROJECT
        VERSION TABLE' LEFT)
   ```

2. Print a directory of all data blocks qualified with PEID=10 or SEID=10. Print columns for the NAME and DBSET, and the qualifiers SPC, MPC, and LOAD.

   ```
   DBDICT  DATABLK SELECT(NAME,SPC,MPC,LOAD,DBSET,SIZE,
             SEID,PEID) ,
        SORT(NAME,SIZE=D)  WHERE(SEID=10 OR PEID=10)
   ```

Related Topics

Browse Simcenter Nastran documentation

Simcenter Nastran Quick Reference Guide
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DBDIR

Prints Database Directory Tables

Prints the database directory.

Format:

DBDIR [VERSION={version-ID,*}] [PROJECT={'project-ID',*}] [FORMAT={format-no}]

Describers:

Describer	Meaning
version-ID	Version identifier.
project-ID	Project identifier. (See **PROJ**.)
*	Wildcard. All versions or projects will be printed.
format-no	Specifies the type(s) of directory information to be printed. If several items are desired, then the sum of the following values is required. 63 is the default, which is the sum of the first six values below: 1 Project Version Table 2 NDDL Data Blocks 4 NDDL Parameters 8 NDDL Empty Data Blocks 16 NDDL Data Block Trailers 32 NDDL Path Value Table for NDDL entries 64 Scratch Data Blocks

Remarks:

1. Trailers are not printed unless NDDL data blocks, empty data blocks, or local scratch data blocks are also printed.
2. The path value table is not printed unless NDDL data blocks, empty data blocks, or parameters are also printed.
3. If no version-ID is given, all versions will be printed. If no project-ID is given, the current project will be printed.
4. The directory print includes all items found on the primary database and all databases attached by the DBLOCATE and DBLOAD FMS statements.

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5. Only one DBDIR statement is permitted in the FMS Section, and this statement should appear last in the FMS Section.

Examples:

```
DBDIR
```

The above example would print all versions of the current project-ID in the database.

Related Topics

Browse Simcenter Nastran documentation

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DBFIX

Database Directory Error Detection

Detects and optionally corrects errors in the database directory.

Format:

$$\text{DBFIX}\left[\left\{\begin{matrix}\text{LIST}\\ \underline{\text{NOLIST}}\end{matrix}\right\}\left\{\begin{matrix}\underline{\text{CORRECT}}\\ \text{NOCORRECT}\end{matrix}\right\}\right]$$

Describers:

Describer	Meaning
LIST	Requests a debug listing of the database directory pointers.
NOLIST	Suppresses a debug listing of the database directory.
CORRECT	Corrects the database if any errors are found.
NOCORRECT	Suppresses the correction of the database.

Remarks:

1. It is recommended that a backup copy of the database be made before this statement is used since corrections of the database are achieved through the deletion of data. Data blocks and parameters are deleted from the database if they have (1) incorrect paths (different from those listed in the NDDL), (2) incorrect names (two or more names that are not equivalenced and reference the same data), or (3) incorrect directory pointers.
2. NOLIST does not suppress the listing of any corrections made to the database.

Examples:

```
DBFIX LIST,NOCORRECT
```

The example above requests a printout of the directory pointers and any errors, but not the corrections.

Related Topics

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DBLOAD

Load a Database from a FORTRAN File

Recovers data blocks or parameters from a database created by the DBUNLOAD statement.

Format:

$$\text{DBLOAD}\left[\text{DATABLK}=\left[\underset{(\text{datablk}-\text{list})}{\underline{*}}\right]\text{PARAM}=\left[\underset{(\text{param}-\text{list})}{\underline{*}}\right]\text{WHERE}\,(\text{where}-\text{expr}),\right.$$

$$\left.\text{CONVERT}\,(\text{convert}-\text{expr})\text{UNIT}=\text{unit}\,\text{FORMAT}\left\{\begin{matrix}\underline{\text{BINARY}}\\ \text{NEUTRAL}\end{matrix}\right\}\left\{\begin{matrix}\text{OVRWRT}\\ \underline{\text{NOOVRWRT}}\end{matrix}\right\}\right]$$

Examples:

1. Load the database stored in ASCII format on FORTRAN unit 12.

```
DBLOAD      UNIT=12 FORMAT=NEUTRAL
ASSIGN      DBLOAD='physical file name of unloaded database'UNIT=12
            FORMATTED
```

2. Load version 1 of KAA under project FRONT BUMPER and store it on the primary database under version 5 and project BUMPER. Overwrite duplicates found on the primary database.

```
DBLOAD      DATABLK=(KAA) WHERE(PROJECT='FRONT
            BUMPER' AND ,SEID=10 AND VERSION=1) CONVERT(VERSION=5;
            , PROJECT='BUMPER') OVRWRT
ASSIGN      DBLOAD='physical file name of unloaded
            database'
```

Describers:

Describer	Meaning
datablk-list	Specifies a list of data blocks separated by commas. The default is * which selects all data blocks. The loaded data block may be renamed in the primary database by specifying a slash after the

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Describer	Meaning
	old name followed by the new name. For example, if KLL is to be renamed to KLL1, then DATABLK=(KLL/KLL1) is specified.
param-list	Specifies a list of parameters separated by commas. The default is *, which selects all parameters. The loaded parameter may be renamed in the primary database by specifying a slash after the old name followed by the new name. For example, if LUSETS is to be renamed to LUSET, then PARAM=(LUSETS/LUSET) is specified.
where-expr	A logical expression that specifies the desired values of qualifiers, PROJECT, VERSION, and DBSET. For example, WHERE(VERSION=4 AND SEID<>2 AND SEID>0) selects all items under version 4 for all values of SEID greater than 0 except 2. See the beginning of this section on WHERE and CONVERT clauses. The default for VERSION is * for all versions; PROJECT is * for all projects; and DBSET is * for all DBsets. The default for qual is *, which is all qualifier values found on the loaded database. See also Remark 8.
convert-expr	Modifies the values for PROJECT, VERSION, DBSET, and qualifiers selected by the where-expr. The format of convert-expr is: PROJECT=project-expr; VERSION=version-expr; DBSET=DBset-name; quali=qual-expri[;..] For example, CONVERT (SEID=100 + SEID; SPC=102). See the beginning of this section on WHERE and CONVERT clauses. The default action for VERSION and PROJECT is to use the same version IDs and project IDs; i.e., CONVERT(PROJECT=PROJECT; VERSION=VERSION). But if either PROJECT or VERSION is specified in the convert-expr, then both must be specified. The default action for qualifiers and DBSET is to use the same values as long as they are defined in both databases. If not, see Remark 8.
unit	Specifies the FORTRAN unit number of the database to be loaded. The unit must be specified on an ASSIGN statement that references the physical filename of the loaded database. The default is 51.
OVRWRT NOOVRWRT	By default, if duplicate data blocks or parameters exist on the loaded and primary databases, then a fatal message is issued. A duplicate means that a data block or parameter has not only the same name but also the same qualifier values, PROJECT, VERSION, and DBSET as a data block or parameter on the primary database.
NEUTRAL, BINARY	The database to be loaded may be in BINARY or NEUTRAL format. BINARY indicates the database to be loaded is in binary or FORTRAN

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Describer	Meaning
	unformatted format. NEUTRAL indicates the database to be loaded is in ASCII format. The default is BINARY.

Remarks:

1. The DBLOAD statement and its applications are discussed further in “Database Archival and Retrieval” in the *Simcenter Nastran User’s Guide*.
2. If the DATABLK keyword is specified and PARAM is not specified, then only data blocks may be loaded. If the PARAM keyword is specified and DATABLK is not specified, then only parameters may be loaded. If neither DATABLK nor PARAM is specified, then all data blocks and parameters may be loaded.
3. The DB keyword is equivalent to DATABLK, and the PARM keyword is equivalent to PARAM.
4. The database to be loaded is attached as read-only. In other words, items can only be fetched and not stored on this database.
5. If more than one DBLOAD statement is specified, then the statements will be processed in the order in which they appear. If a duplicate data block or parameter is found on two or more different DBLOAD statements, then the last duplicate will be used.
6. If NEUTRAL is specified, then the FORMATTED keyword must be specified on the corresponding ASSIGN statement.
7. If a data block or parameter is being renamed, then the new name must be defined in the NDDL of the primary database.
8. If the database to be loaded and the primary database have different NDDL schemes and CONVERT is not used, then the following is performed:
   - If a qualifier in the NDDL of the database to be loaded is not in the NDDL of the primary database, then all of its values are converted to the null value corresponding to its type. For example, if the qualifier is integer, real, double-precision, complex or character, then the value is converted to 0, 0., 0.D0, (0.,0.), or blank, respectively. If this conversion results in a duplicate data block(s) or parameter(s), then a User Warning Message is printed and the duplicates are not loaded.
   - If a DBset-name in the NDDL of the database to be loaded is not in the NDDL of the primary database, then its values will be converted to the PARAM default value in the NDDL of the database to be loaded.
9. Data blocks that are equivalenced on the database to be loaded remain equivalenced as long as they are loaded in the same DBLOAD statement or in consecutive DBLOAD statements with the same unit number. Otherwise, a separate copy is produced for the secondary data block.
10. It is not possible to restart from a database created by DBLOAD in the same run.
11. SOL 190 (or DBTRANS) is also required with DBLOAD if:
    - The database to be loaded has a different BUFFSIZE.
    - The database to be loaded is in neutral format or is being transferred between different machine types.

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See also Migrating Databases in the *Simcenter Nastran User's Guide*.

Related Topics

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DBLOCATE

Attaches Secondary Databases

Obtains data blocks or parameters from prior versions of the primary database, or other databases. DBLOCATE may also be used to compress the primary database and to migrate databases created in prior versions of the software.

Format:

$$\text{DBLOCATE}\left[\text{DATABLK}=\left[\begin{matrix}*\\(\text{datablk}-\text{list})\end{matrix}\right]\text{PARAM}=\left[\begin{matrix}*\\(\text{param}-\text{list})\end{matrix}\right]\text{WHERE}\,(\text{where}-\text{expr}),\right.$$

$$\left.\text{CONVERT}\,(\text{convert}-\text{expr})\,\text{LOGICAL}=\text{dbname}\left[\begin{matrix}\text{OVRWRT}\\\text{NOOVRWRT}\end{matrix}\right]\text{COPY}\right]$$

Example:

1. Locate in version 4 of MASTER3 all data blocks named KAA for all superelements with IDs greater than 0.

```
DBLOCATE    DATABLK=(KAA) WHERE(PROJECT='FRONT
            BUMPER'  ,AND SEID>0 AND VERSION=4) LOGI=MASTER3
ASSIGN      MASTER3='physical file name of master
            DBset'
```

2. Copy all data blocks and parameters from the last version of MASTER3 to the primary database. For all items with the qualifier SEID, change the SEID to twice the old ID number.

```
DBLOCATE    CONVERT(SEID=2*SEID) COPY LOGI=MASTER3
ASSIGN      MASTER3='physical file name of master
            DBset'
```

3. Compress a database with multiple versions. All versions under the current project-ID (see PROJ statement) will be copied from the database OLDDB to NEWDB.

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```
ASSIGN      MASTER3='physical filename of new
            master DBset'

ASSIGN      OLDDB='physical filename of old master
            DBset'

DBLOCATE    LOGI=OLDDB COPY WHERE(VERSION=*) ,
            CONVERT(VERSION=VERSION;PROJECT=PROJECT)
```

Describers:

Describer	Meaning
datablk-list	Specifies a list of data blocks separated by commas. The default is *, which selects all data blocks. The located data block may be renamed in the primary database by specifying a slash after the old name followed by the new name. For example, if KLL is to be renamed to KLL1, then DATABLK = (KLL/KLL1) is specified.
param-list	Specifies a list of parameters separated by commas. The default is *, which selects all parameters. The located parameter may be renamed in the primary database by specifying a slash after the old name followed by the new name. For example, if LUSETS is to be renamed to LUSET, then PARAM=(LUSETS/LUSET) is specified.
where-expr	A logical expression that specifies the desired values of qualifiers, PROJECT, VERSION, and DBSET. For example, WHERE(VERSION=4 AND SEID < > 2 AND SEID > 0) selects all items under version 4 for all values of SEID greater than 0 except 2. See the beginning of this section on WHERE and CONVERT clauses. The default for VERSION is the last version-ID and PROJECT is the current project-ID. The default for qual is *, which is all qualifier values found on the located database. See also Remark 9 .
convert-expr	Modifies the values for PROJECT, VERSION, DBSET, and qualifiers selected by the where-expr. The format of convert-expr is: PROJECT=project-expr; VERSION=version-expr; DBSET=DBset-name; quali=qual-expri[;...] For example, CONVERT (SEID=100+SEID; SPC=102). See the beginning of this section on WHERE and CONVERT clauses. The default action for VERSION and PROJECT is to convert to the current version-ID and current project-ID. But if either PROJECT or VERSION is specified in the convert-expr, then both must be specified. See Example 3. The default action for qualifiers and DBSET is to use the same values as long as both databases have the same NDDL scheme. If not, see Remark 9 .

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Describer	Meaning
dbname	Specifies the logical name of the master directory DBset of the located database. dbname must be specified on an ASSIGN statement, which references the physical file name. By default, the located database is also the primary database. (If dbname is specified for the primary database, then dbname must be MASTER.)
OVRWRT NOOVRWRT	By default, duplicate data blocks or parameters on the located database will take precedence over those on the primary database. A duplicate means that a data block or parameter has not only the same name but also the same qualifier values, PROJECT, VERSION, and DBSET as the data block or parameter on the primary database. If NOOVRWRT is specified, then a fatal message is issued.
COPY	Requests that the located data blocks or parameters be copied to the primary database.

Remarks:

1. The DBLOCATE statement and its applications are discussed further in “Database Concepts” in the *Simcenter Nastran User’s Guide*.
2. If the DATABLK keyword is specified and PARAM is not specified, then only data blocks may be located. If the PARAM keyword is specified and DATABLK is not specified, then only parameters may be located. If neither DATABLK nor PARAM is specified, then all data blocks and parameters may be located.
3. The DB keyword is equivalent to DATABLK, and the PARM keyword is equivalent to PARAM.
4. If more than one DBLOCATE statement is specified, then the statements will be processed in the order in which they appear. If a duplicate data block or parameter is found on two or more different DBLOCATE statements, then the last duplicate will be used.
5. If the located database is not the primary database, then it is attached for read-only purposes. In other words, items can only be fetched and not stored on the located database.
6. If the RESTART FMS statement is also specified, then located data blocks and parameters are treated as if they exist in the restart version. In other words, restart equivalences will be made on located items at the beginning of the run and can be subsequently broken as a result of regeneration and/or NDDL dependencies.
7. If a data block or parameter is being renamed, then the new name must be defined in the NDDL of the primary database.
8. If LOGICAL refers to the primary database and one version is to be copied to another, then the items are equivalenced.
9. If the located database and the primary database have different NDDL schemes and CONVERT is not used, then the following is performed:
   - If a qualifier in the NDDL of the located database is not in the NDDL of the primary database, then all of its values are converted to the null value corresponding to its type. For example, if the qualifier is integer, real, double precision, complex or character, then the value is converted to 0, 0., 0.D0, (0.,0.), or blank, respectively. If this conversion results in a duplicate data block(s) or parameter(s), then a User Warning Message is printed and the duplicates are not located.

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- If a dbset-name in the NDDL of the located database is not in the NDDL of the primary database, then its values will be converted to the PARAM default value in the NDDL of the located database.

Related Topics

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DBSETDEL

Deletes DBsets

Deletes DBsets, all of its members, and associated physical files.

Format:

DBSETDEL dbsetnamei

Describers:

Describer	Meaning
dbsetnamei	Specifies the name(s) of DBset(s) to be deleted. The DBset names MASTER, OBJSCR, or SCRATCH may not be specified.

Remarks:

1. The DBSETDEL statement and its applications are discussed further in “Deleting DBsets” in the *Simcenter Nastran User’s Guide*.
2. If dbsetnamei does not exist, then no action is taken.
3. After a DBset has been deleted with this statement, it may be recreated with the INIT statement in a subsequent run.

Examples:

Delete DBset DBUP20 from the database.

```
DBSETDEL DBUP20
```

Related Topics

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DBUNLOAD

Unload a Database to a FORTRAN File

Stores data blocks or parameters from the primary database onto a FORTRAN file in a binary or neutral format, for purposes of database compression or database transfer between different computers.

Format:

$$\text{DBUNLOAD}\left[\text{DATABLK}=\left[\underset{(\text{datablk}-\text{list})}{\underline{*}}\right]\text{PARAM}=\left[\underset{(\text{param}-\text{list})}{\underline{*}}\right]\text{WHERE}\,(\text{where}-\text{expr}),\right.$$

$$\left.\text{UNIT}=\text{unit}\,\text{FORMAT}=\left\{\begin{matrix}\underline{\text{BINARY}}\\ \text{NEUTRAL}\end{matrix}\right\}\left\{\begin{matrix}\text{REWIND}\\ \underline{\text{NOREWIND}}\end{matrix}\right\}\right]$$

Examples:

1. Unload the database in ASCII format onto FORTRAN unit 12.

```
DBUNLOAD    UNIT=12 FORMAT=NEUTRAL
ASSIGN      DBUNLOAD='physical file name of
            FORTRAN unit 12'

            UNIT=12 FORMATTED
```

2. Unload version 1 of KAA under project FRONT BUMPER.

```
DBUNLOAD    DATABLK=(KAA) WHERE(PROJECT='FRONT
            BUMPER' ,AND SEID=10 AND VERSION=1)
ASSIGN      DBUNLOAD='physical file name of
            FORTRAN unit 50'
```


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Describers:

Describer	Meaning
datablk-list	Specifies a list of data blocks separated by commas. The default is * which selects all data blocks.
param-list	Specifies a list of parameters separated by commas. The default is *, which selects all parameters.
where-expr	Logical expression which specifies the desired values of qualifiers, PROJECT, VERSION, and DBSET. For example, WHERE(VERSION=4 AND SEID<>2 AND SEID>0) selects all items under version 4 for all values of SEID greater than 0 except 2. See the beginning of this section on WHERE and CONVERT Clauses. The default for VERSION is * for all versions; PROJECT is * for all projects; and DBSET is * for all DBsets. The default for qual is *, which is all qualifier values found on the primary database.
unit	Specifies the FORTRAN unit number to unload the database. The unit must be specified on an ASSIGN statement, which references its physical filename. The default is 50.
NEUTRAL BINARY	The database may be unloaded in BINARY or NEUTRAL format. BINARY indicates the database is to be unloaded in binary or FORTRAN unformatted. NEUTRAL indicates the database is to be unloaded in ASCII format. The default is BINARY.
NOREWIND REWIND	By default, if DBUNLOAD is executed more than once for the same unit, then the unit is not rewound. REWIND requests that the unit be rewound prior to unloading.

Remarks:

1. The DBUNLOAD statement and its applications are discussed further in “Database Archival and Retrieval” in the *Simcenter Nastran User’s Guide*.
2. If the DATABLK keyword is specified and PARAM is not specified, then only data blocks may be unloaded. If the PARAM keyword is specified and DATABLK is not specified, then only parameters may be unloaded. If neither DATABLK nor PARAM is specified, then all data blocks and parameters may be unloaded.
3. The DB keyword is equivalent to DATABLK, and the PARM keyword is equivalent to PARAM.
4. If more than one DBUNLOAD statement is specified, then the statements will be processed in the order in which they appear.
5. If NEUTRAL is specified, then the FORMATTED keyword must be specified on the corresponding ASSIGN statement.
6. If NEUTRAL is specified, then only data blocks with an NDDL description are unloaded. (See the *Simcenter Nastran DMAP Programmer’s Guide* under the DATABLK statement.) An NDDL description is required for TYPE=TABLE and none is required for TYPE=MATRIX. The data block must be unloaded in BINARY if TYPE=UNSTRUCTURED, KDICT, or KELM.

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7. Data blocks that are equivalenced on the primary database remain equivalenced as long as they are unloaded in the same DBUNLOAD statement or in consecutive DBUNLOAD statements with the same unit number. Otherwise, a separate copy for the secondary data block is produced.

Related Topics

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DBUPDATE

Specify Database Directory Update Interval

Specifies the maximum length of CPU time between database directory updates to the MASTER DBset. This statement is intended to be used if the INIT MASTER(RAM=r) option is specified.

Format:

DBUPDATE [=] update-time

Describers:

Describer	Meaning
update-time	CPU time interval in minutes (real or integer) between database directory updates.

Remarks:

1. The difference in CPU time from the last update is checked after the execution of each DMAP instruction. The database directory is updated if this difference is greater than update-time. Update-time and CPU time are accurate to the nearest whole second only.
2. If update-time < 0, then database directory updates are only performed at the end of the run.
3. Defaults for update-time are machine dependent and may be found in the *Simcenter Nastran Installation and Operations Guide*.
4. Periodic updates of the directory tables to MASTER DBset increase the integrity of the database during system crashes (for example, crashes due to insufficient time or space).
5. In addition to those specified by DBUPDATE, directory updates are performed automatically at various points in the execution of the DMAP. An asterisk appears after the word “BEGN” in the Executive Summary Table whenever an update occurs. These updates occur whenever a permanent data block or parameter DMAP equivalence or restart equivalence is broken. Updates also occur upon deletions. Additions to the database do not automatically cause a directory update to take place.
6. This statement is in effect only when INIT MASTER(RAM=r) is being used. INIT MASTER(S) and INIT MASTER(NORAM) disable periodic and automatic updates.
7. Update-time may also be changed with the DMAP instruction PUTSYS(update-time, 128) or the NASTRAN SYSTEM(128)=update-time statement. (The update-time must be a real, single-precision value specified in minutes.)

Examples:

```
DBUPDATE=5.5
```


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The above example would call for a database directory update at the end of a DMAP module execution, after five and one-half minutes of CPU time have elapsed from the last update.

Related Topics

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DEFINE

Parameter Definition

Assigns user defined keywords (or cellnames) to NASTRAN system cells. (See the NASTRAN statement for a description of “cellname”). In addition, the DEFINE statement provides a mechanism to set default values for system cells.

Format:

DEFINE keyword [=expression] [LOCATION=SYSTEM(i)] [TYPE=type]

Describers:

Describer	Meaning
keyword	User defined name, 1 through 24 characters in length. The first character must be alphabetic. Only the following characters can be used for keywords: A-Z, 0-9, ’ , _ , and “. All other characters are invalid.
expression	Expression produces a single value from a set of constant and/or variable parameters separated by operators. The value is assigned to the “keyword” and is also used to set the value for the NASTRAN system cell specified by “LOCATION”. The TYPE determines both the type of the result and the type conversions that will be applied to the constants and variables within the expression—mixed mode expressions are allowed (see Remark **6**). The parentheses can be used to change the order of precedence. Operations within parentheses are performed first with the usual order of precedence being maintained within the parentheses. The variable parameters within the expression must be keywords previously defined on a DEFINE statement. The following operations are allowed

Parameter Type	Operator	Operation
Integer or Real	+	Addition
	–	Subtraction
	×	Multiplication
	/	Division
Logical	+	Bit-wise OR
Logical	-	Bit clear. For example, the result of *a*-*b* is equal to the value of *a* with the bits associated with *b* set to 0.

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Describer	Meaning
SYSTEM(I)	Specifies the NASTRAN system cell number to be associated with the keyword.
type	The type of the expression result and the type conversions that will be applied to the constants and variables within the expression. Allowable data types are as follows:

Description	Type
Integer (default)	I
Real	R
Logical	LOGICAL

Remarks:

1. If TYPE, LOCATION and EXPRESSION are omitted, the default data type is integer and the default value is zero.
2. If expression is omitted, an internal default will be assigned to the keyword/cellname based on the LOCATION. (See **"NASTRAN"** for a list of internal default values).
3. A DEFINE statement that specifies a LOCATION sets the default for a NASTRAN system cell and therefore it is not necessary to set the system cell value on a subsequent NASTRAN statement, unless the user wishes to override the previous DEFINE statement setting. Also, since more than one DEFINE statement may be present for the same "keyword" the last specification takes precedence. "Keywords" referenced on a NASTRAN statement or in an expression on the DEFINE statement are automatically substituted by the last specification of the "keyword" prior to the current statement being processed.
4. DEFINE statements may also be specified in Runtime Configuration (RC) files. See the *Simcenter Nastran Installation and Operations Guide*.
5. System cells may also be set with the NASTRAN statement. In addition, they may be set or values returned with the DMAP PUTSYS and GETSYS functions and the PARAM module. See the *Simcenter Nastran DMAP Programmer's Guide*
6. Each operand within the expression will be converted to the result type prior to the arithmetic operation. For example: the statement "DEFINE JJ=2.5 + 3.6 TYPE=I" would result in 2.5 and 3.6 being converted to 2 and 3 respectively and the result of 5 would be assigned to JJ.

Examples:

1. Change the default value for block size.

```
DEFINE BUFFSIZE=4097  LOCATION=SYSTEM(1)
```


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2. Set the sparse matrix selection to forward-backward substitution only.

```
DEFINE SPARSE=16 LOCATION=SYSTEM(126)
```

3. Define the system cell keyword and default value for the maximum output line count and then reset it to another value on a NASTRAN statement. Note: The DEFINE statement would typically be placed in an RC file and the NASTRAN statement would be placed in the File Management Section whenever the user wants to override the DEFINE statement default setting.

```
DEFINE    MAXLINES=999999999 LOCATION=SYSTEM(9)
NASTRAN   MAXLINES=100000
```

4. Define system cells that behave like “toggles”, turning some feature on or off.

```
DEFINE    MESH=2 LOCATION=(31)
DEFINE    NOMESH=0 LOCATION=(31)
NASTRAN   MESH
```

Note: Since each subsequent DEFINE statement redefines the default value, the second DEFINE of system cell location 31 sets the default value to 0. A NASTRAN statement can then be inserted in the input file to reset the MESH system cell back to a value of 2. This same technique can be used with any system cell where the user wishes to simply refer to the system cell keyword and have the system cell set to a previous DEFINE statement default.

5. Invalid usage of the DEFINE and NASTRAN statement:

```
DEFINE    BUFFSIZE=4097
NASTRAN   BUFFSIZE=2048
```

Valid usage:

```
DEFINE    BUFFSIZE=4097 LOCATION=SYSTEM(1)
NASTRAN   BUFFSIZE=2048
```

Related Topics

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ENDJOB

Terminate Job

Terminates the job at a user-specified location in the FMS Section.

Format:

ENDJOB

Remarks:

ENDJOB is normally used after a DBDICT or DBDIR statement or after database initialization.

Examples:

```
DBDICT
ENDJOB
```

Related Topics

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EXPAND

Concatenate New DBset Members

Concatenates additional DBset members on an existing permanent DBset previously defined with an INIT statement.

Format:

EXPAND dbset-name LOGICAL=(log-namei [(max-sizei)]...)

Describers:

Describer	Meaning
dbset-name	The name of a DBset previously defined with an INIT statement.
log-namei	Specifies the logical name of a DBset member. log-namei may also be referenced on an ASSIGN statement which refers to the physical file name of the DBset member.
max-sizei	Specifies the maximum size, in blocks, of a DBset member. For storage units specified in words or bytes, the size must be followed by one of the unit keywords below

Unit Keyword	Storage Unit
KW	Kilowords (thousand)
KB	Kilobytes (thousand)
MW	Megawords (million)
MB	Megabytes (million)
GW	Gigawords (billion)
GB	Gigabytes (billion)

For example, 100MB is 100 megabytes. The size of a block in words is defined by BUFFSIZE.

Remarks:

1. On all computers with dynamic file allocation, the physical filename of a DBset member may be specified on an ASSIGN statement:

```
ASSIGN log-name='physical filename'
```


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If an ASSIGN statement is not specified for the member, then a name is automatically assigned. The naming convention is described in “Introduction to Database Concepts” in the *Simcenter Nastran User’s Guide*.

Examples:

```
ASSIGN    DBMEM02=’physical file name’
EXPAND    DBALL LOGICAL=(DBMEM02)
```

This would create and add the DBset member DBMEM02 to the already existing DBset DBALL.

Related Topics

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INCLUDE

Insert External File

Inserts an external file into the input file. The INCLUDE statement may appear anywhere within the input data file.

Format:

INCLUDE 'filename'

Describers:

Describer	Meaning
filename	External file to be inserted. The '*directory_path/filename*' must begin and end with the single quote character.

Examples:

The following INCLUDE statement is used to obtain the Bulk Data from another file called MYBULK.DATA:

```
SOL 101
CEND
TITLE = STATIC ANALYSIS
LOAD=100
BEGIN BULK
INCLUDE 'MYBULK.DATA'
ENDDATA
```

Remarks:

1. INCLUDE statements may be nested; that is, INCLUDE statements may appear inside the external file. The nested depth level must not be greater than 10.
2. Each line has a 72 character limit. Multiple lines can be used when file names are long. The entire string must begin and end with the single quote character.

   For example,

   D:\folder1\folder2\folder3\folder4\folder5\folder6\folder7\folder8\folder9\mydata.dat

   can be defined on multiple lines:

   ```
   INCLUDE 'D:\folder1\folder2\folder3\folder4\folder5\folder6\
   folder7\folder8\folder9\mydata.dat'
   ```


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The following input format is also supported:

```
INCLUDE ‘D:\folder1\folder2\folder3\
         folder4\folder5\
     folder6\folder7\
        folder8\folder9\mydata.dat'
```

Related Topics

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INIT

Create a DBset

Creates a temporary or permanent DBset. For the SCRATCH and MASTER DBsets, all or some of their space may be allocated to real memory.

Format 1:

Initialize any DBset except MASTER and SCRATCH:

$$\text{INIT DBset} - \text{name } [\text{LOGICAL} = (\log - \text{namei}\,(\text{maxsizei}),\ldots)$$

$$\text{BUFFSIZE} = \text{b } \text{CLUSTER} = \text{c}]$$

Format 2:

Initialize the MASTER DBset:

$$\text{INIT MASTER}\left[\begin{pmatrix}\text{RAM} = \text{r}\\ \text{NORAM}\end{pmatrix}, \quad \text{S}\right)\text{LOGICAL} = (\log - \text{name}\,(\text{max} - \text{sizei}),\ldots)$$

$$\text{BUFFSIZE} = \text{b } \text{CLUSTER} = \text{c}]$$

Format 3:

Initialize the SCRATCH DBset:

$$\text{INIT MASTER}\left[\begin{pmatrix}\text{MEM} = \text{m}\\ \text{NOMEM}\end{pmatrix}, \quad \text{S}\right)\text{LOGICAL} = (\log - \text{name}\,(\text{max} - \text{sizei}),\ldots)$$

$$\text{SCR300} = (\log - \text{namei}\,(\text{max} - \text{sizei}),\ldots)\,\text{BUFFSIZE} = \text{b}\,\text{CLUSTER} = \text{c}]$$

Examples:

1. Modify the default allocation of the DBALL DBset to 50000 blocks.

```
INIT DBALL LOGI=(DBALL(50000))
```

2. Do not allocate any real memory for the MASTER and SCRATCH DBsets.

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```
INIT        MASTER(NORAM)
INIT        SCRATCH(NOMEM)
```

3. Create a new DBset called DBUP with two members DBUP1 and DBUP2.

```
INIT        DBUP LOGI=(DBUP1, DBUP2)
ASSIGN      DBUP1 =’physical filename 1’
ASSIGN      DBUP2=’physical filename 2’
```

Describers:

Describer	Meaning
dbset-name MASTER SCRATCH	The name of a temporary or permanent DBset.
log-namei	Specifies the logical name of a DBset member. log-namei may also be referenced on an ASSIGN statement, which refers to the physical file name of the DBset member. If no log-namei is specified, then the DBset will have one member and the log-name will be the same as the DBset-name. A maximum of twenty log-names may be specified. For the SCRATCH DBset see also Remark 8 . SCR300 is a special keyword that indicates that the log-names are members reserved for DMAP module internal scratch files.
max-sizei	Specifies the maximum size, in blocks, words, or bytes, of a DBset member. For storage units specified in words or bytes, the size must be followed by one of the unit keywords below.

Unit Keyword	Storage Unit
KW	Kilowords (1024)
KB	Kilobytes (1024)
MW	Megawords (1048576)
MB	Megabytes (1048576)
GB	Gigabytes (1073741824)
GW	Gigawords (1073741824)

For example, 100MB is 100 megabytes. The size of a block in words is defined by BUFFSIZE. The default for DBALL and SCRATCH is 250,000 blocks.

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Describer	Meaning
RAM NORAM	RAM=r requests that r words of real memory are to be allocated for the MASTER DBset. The default is RAM or RAM=30000. NORAM or RAM=0 specifies no real memory is to be allocated.
S	If the primary database is being created in the run, this option requests that all DBsets in the primary database will be automatically deleted at the end of the run. INIT MASTER(S) is equivalent to specifying scr = yes on the "nastran" command. See the **"nastran Command and NASTRAN Statement"** . If the run is a restart, then this option is ignored.
MEM NOMEM	MEM=m specifies that m blocks of real memory are to be allocated for the SCRATCH DBset. The default m is machine dependent and may be found in the *Simcenter Nastran Installation and Operations Guide*. NOMEM or MEM = 0 requests that no real memory is to be allocated.
BUFFSIZE	BUFFSIZE = b specifies the number of words per block in the DBset and will override the value specified by the BUFFSIZE keyword on the NASTRAN statement. The default for b is obtained from the NASTRAN BUFFSIZE statement. See the *Simcenter Nastran Installation and Operations Guide*.
CLUSTER	CLUSTER = c specifies the number of blocks per cluster in the DBset. The default is 1, and any other value is not recommended.

Remarks:

1. The INIT statement and its applications are discussed further in "Introduction to Database Concepts" in the *Simcenter Nastran User's Guide*.
2. There are four DBsets that are predefined and automatically allocated by the program. Their DBset-names are MASTER, DBALL, SCRATCH, and OBJSCR, and they are described in "Introduction to Database Concepts" in the *Simcenter Nastran User's Guide*.
3. On all computers with dynamic file allocation, the physical filename of a DBset member may specified on an ASSIGN statement:

   ```
   ASSIGN log-name='physical filename'
   ```

   If an ASSIGN statement is not specified for the member, then a name is automatically assigned.
4. It is recommended that there be sufficient physical space to hold a DBset member should it reach its maximum size (max-sizei). The max-sizei may be converted to words by multiplying by b. A summary of space usage and allocation is printed at the end of the Execution Summary Table.
5. In restart runs, the INIT statement is ignored for preexisting permanent DBsets. The INIT statement is intended to be specified only in the run in which the DBset is being created. If more DBset members need to be added to the DBset, then the EXPAND statement is used.

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6. If RAM or RAM = r is specified and the run terminates because the computer operating system detects insufficient space or time or the computer halts due to a power outage or operator interruption, then it may not be possible to restart from the database. See the DBUPDATE FMS statement.

7. BUFFSIZE = b and CLUSTER = c must satisfy the following inequality:

$$b \leq \frac{64000}{c} + 5$$

8. By default, the SCRATCH DBset is divided into two partitions: LOGICAL and SCR300. The LOGICAL partition, log-names after the LOGICAL keyword, are reserved for DMAP scratch data blocks and the SCR300 partition for DMAP module internal scratch files.

   - The maximum total number of log-names for LOGICAL and SCR300 is 20. For example, if LOGICAL has 8 log-names, then SCR300 can have no more than 12 log-names.
   - If NASTRAN SYSTEM(142) = 1 is specified, then the SCR300 partition is not created and internal scratch files, along with DMAP scratch data blocks, will reside on the LOGICAL partition. The default is SYSTEM(142) = 2.
   - If NASTRAN SYSTEM(151)=1 is specified and the LOGICAL partition has reached its maximum size, then the SCR300 partition will be used. The default is SYSTEM(151) = 1.
   - By default, the space specified for the SCR300 partition is released to the operating system after each DMAP module is executed as long as the module used more than 100 blocks for internal scratch files. If 100 blocks is not a desirable threshold, then it may be changed by specifying NASTRAN SYSTEM(150) = t, where t is the number of blocks for the threshold.

9. BUFFSIZE = b is predefined for DBset-names MSCOBJ, OBJSCR, and USROBJ and may not be changed by BUFFSIZE on this statement or the NASTRAN BUFFSIZE bmax statement. The default for b is recommended for all except very large problems. bmax must reflect the maximum of b specified for all DBsets attached to the run, including the delivery database. See *Simcenter Nastran Installation and Operations Guide* for the defaults of b and bmax.

10. If INIT MASTER(RAM = r) and INIT SCRATCH(MEM = m) are specified, then BUFFSIZE for these DBsets must be the same. If not, a warning message is issued, and the BUFFSIZE for the SCRATCH DBset is reset to that of the MASTER DBset.

11. Only one INIT statement per dbset-name may be specified in the File Management Section.

Related Topics

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PROJ

Define Database Project-Identifier

Defines the current or default project identifier, project-ID.

Format:

PROJ [=] 'project-ID'

Describers:

Describer	Meaning
project-ID	Project identifier. Must be enclosed in single quotes. (Character string, maximum of 40 characters; Default = blank)

Remarks:

1. There may be only one PROJECT statement in the File Management Section. The PROJECT statement must be specified before all DBCLEAN, DBDIR, DBDICT, RESTART, DBLOCATE, and DBLOAD statements where project-ID is not specified by the user.
2. This statement is optional and specifies that all data blocks and parameters stored on or accessed from the database in the current run shall also be identified by project-ID. Therefore, in subsequent runs that may access this data through other FMS statements such as RESTART, the project-ID must be specified.
3. Project-ID is the default on DBCLEAN, DBDIR, DBDICT, and RESTART FMS statements and in the WHERE and CONVERT clause of the DBLOCATE statement.
4. Leading blanks and trailing blanks enclosed within single quotes are ignored. All other blanks are considered part of the project-ID.
5. Project-ID is saved with only the first 40 characters specified.

Examples:

1. PROJ = 'MY JOB'
2. The following project-ID will be truncated to 40 characters:

```
PROJ 'CAR MODEL 1999 BODY FRAME SYM - PROTYP B RUN'
```

and all subsequent restarts must have the statement.

```
PROJ 'CAR MODEL 1999 BODY FRAME SYM - PROTYP B'
```


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Related Topics

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RESTART

Reuse Database From a Previous Run

Requests that data stored in a previous run be used in the current run.

Format:

$$\text{RESTART}\left[\text{PROJECT}=\text{'project}-\text{ID'}\ \text{VERSION}=\begin{bmatrix}\text{version}-\text{ID}\\ \underline{\text{LAST}}\end{bmatrix}\right.$$

$$\left.\begin{bmatrix}\text{KEEP}\\ \underline{\text{NOKEEP}}\end{bmatrix}\ \text{LOGICAL}=\text{dbname}\right]$$

Describers:

Describer	Meaning
project-ID	Project identifier. See PROJ FMS statement. Must be enclosed in single right-hand quotation marks (’). (Character string, maximum of 40 characters; default is the project-ID specified on the PROJ FMS statement).
version-ID	Version number. (Integer > 0)
LAST	Specifies the last version under project-ID.
KEEP	Data stored under VERSION will remain on the database after the run is completed.
NOKEEP	Data stored under VERSION will be deleted from the database after the run is completed.
dbname	Specifies the logical name of an existing MASTER (master directory) DBset to be used for restart purposes. This MASTER and its associated database will be opened in a read-only mode to perform the restart, any new data will be written to the database for the current run.

Remarks:

1. There may be only one RESTART statement in the File Management Section.
2. A new version-ID is automatically assigned whenever a restart is performed.
3. If project-ID or version-ID or both are specified and cannot be found, then a fatal message will be issued.
4. The RESTART statement is required to perform restarts in solution sequences 101 through 200.

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5. If PROJECT is not specified, then the run will restart from the project-ID specified on the PROJ statement. (See Example 2 below.)
6. Databases created in one version cannot be directly restarted into a different version. Restrictions are typically documented in the current release guide; however, a DBLOCATE type restart might work.

Examples:

1.
```
RESTART VERSION=7
```
Version number 7 will be retrieved for this run (version 8). At the end of the run, version 7 will be deleted.

2.
```
PROJ='FENDER'
RESTART
```
The last version under project-ID FENDER will be used in the current run.

3.
```
ASSIGN RUN1='run1.MASTER'
RESTART LOGICAL=RUN1
```
The run1.MASTER and its associated database will be used (read only) for restart purposes.

Related Topics

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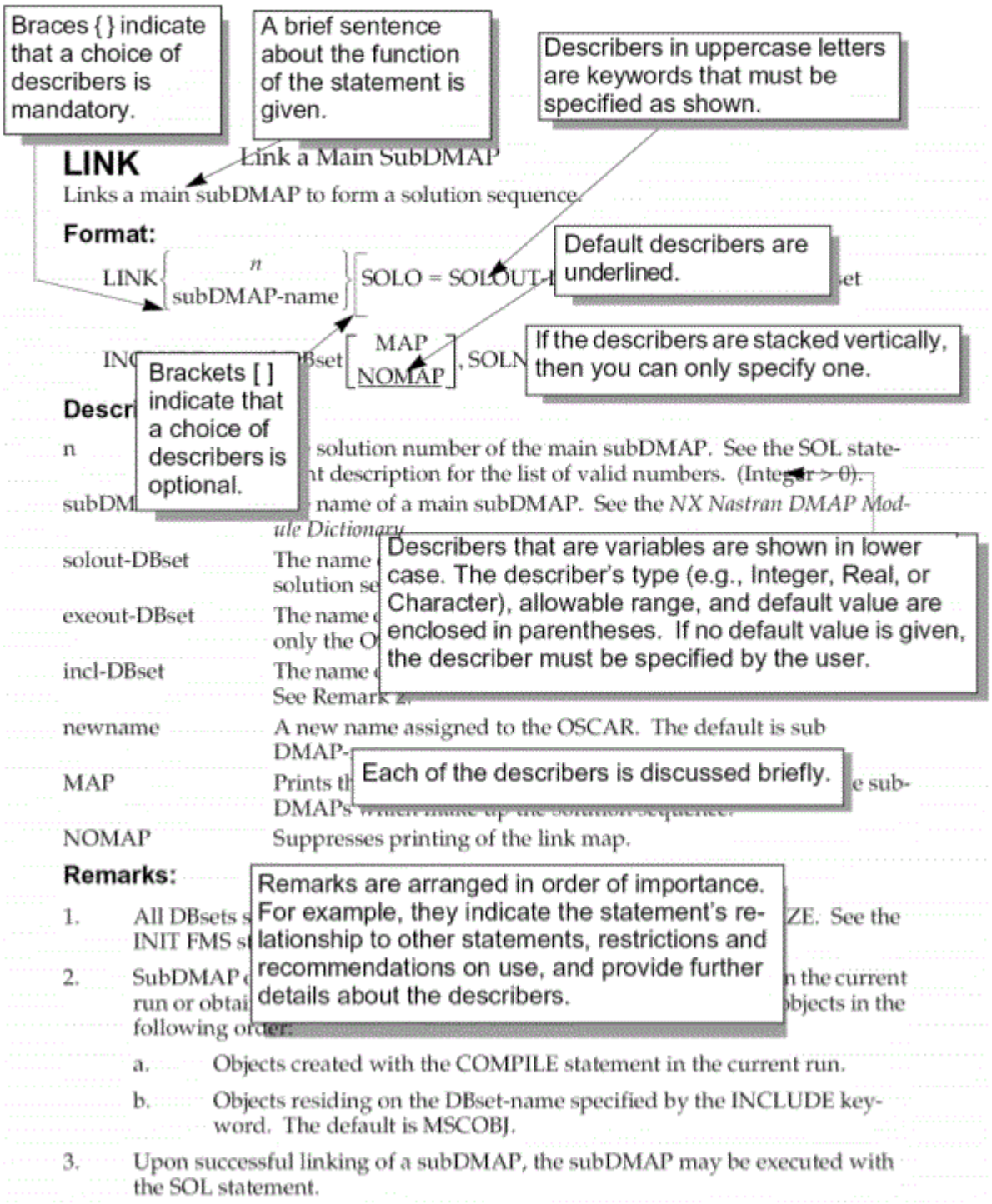
3. Executive Control Statements

Key to Descriptions



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Executive Control Section

This section describes the Executive Control statements. These statements select a solution sequence and various diagnostics.

Most Executive Control statements are order independent. The exceptions are the COMPILE, COMPILER, ALTER, ENDALTER, and LINK statements. If used, the LINK statement must appear after all COMPILE statements. The COMPILER statement (or equivalent DIAGs) must appear before all COMPILE statements. The COMPILER statement also sets the defaults for subsequent COMPILE statements.

Related Topics

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Executive Control Statement Summary

The Executive Control statements are summarized as follows:

ALTER	Specifies deletion and/or insertion of following DMAP statements.
APP	Specifies an approach in a solution sequence.
CEND	Designates the end of the Executive Control statements.
COMPILE	Requests compilation of specified subDMAPs or the NDDL file.
COMPILER	Specifies DMAP compilation diagnostics.
DIAG	Requests diagnostic output or modifies operational parameters.
ECHO	Controls the echo of Executive Control statements.
ENDALTER	Designates the end of a DMAP sequence headed by an ALTER.
GEOMCHECK	Specifies tolerance values and options for optional finite element geometry tests.
ID	Specifies a comment.
LINK	Requests the link of a main subDMAP.
SOL	Requests execution of a solution sequence or DMAP program.
TIME	Sets the maximum allowable execution time.

Related Topics

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Executive Control Statement Descriptions

You can abbreviate Executive Control statements down to their first four characters as long as the abbreviation is unique relative to all other statements. Each statement is described as follows:

Description

A brief sentence about the function of the statement is given.

Format

Describers in uppercase are keywords that must be specified as shown. In addition, describers in lowercase indicate that the user must provide a value.

Braces { } indicate that a choice of describers is mandatory. If the describers are stacked vertically, then only one may be specified.

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Brackets [ ] indicate that a choice of describers is optional. If the describers are stacked vertically, then only one may be specified.

Describers that are shaded indicate the defaults.

If the statement line is longer than 72 columns, then it may be continued to the next line with a comma. For example:

```
COMPILE  SEDRCVR  SOUIN=MSCSOU,
  NOREF    NOLIST
```

Examples

One or more examples are given.

Describers

Each of the describers is briefly discussed. The describer's type (e.g., Integer, Real, or Character), its allowable range, and its default value are enclosed in parentheses. The describer must be specified by the user if no default value is given.

Remarks

The remarks in the Remarks section are generally arranged in order of importance and indicate such things as the Executive Control statement's relationship to other statements, restrictions and recommendations on its use, and further descriptions of the describers.

Related Topics

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$

Comment

Used to insert comments into the input file. Comment statements may appear anywhere within the input file.

Format:

$ followed by any characters out to column 80.

Examples:

```
$ TEST FIXTURE-THIRD MODE
```

Remarks:

1. Comments are ignored by the program.
2. Comments will appear only in the unsorted echo of the Bulk Data.

Related Topics

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ALTER

Insert and/or Delete DMAP Statements

Inserts and/or deletes DMAP statements in a subDMAP.

Format:

```
ALTER k1 [,k2]
```

or

```
ALTER ‘string1’[(occurrence,offset)] ,[‘string2’[(occurrence,offset)] ]
```

or

```
ALTER k1 , [‘string2’[(occurrence,offset)] ]
```

or

```
ALTER ‘string1’[(occurrence,offset)] , [k2]
```

Describers:

Describer	Meaning
k1	If k2 or ‘string2’ is not specified, the subsequent DMAP statements will be inserted after either the statement number k1 or the ‘string1’, [(occurrence,offset)] reference point.
k1, k2	DMAP statements numbered k1 through k2 will be deleted and may be replaced with subsequent DMAP statements.
‘string1’	If ‘string2’ or k2 is not specified, the subsequent DMAP statements will be inserted after the first occurrence of ‘string1’.
‘string1’, ‘string2’	DMAP statements beginning with the first occurrence of ‘string1’ through DMAP statements containing the first occurrence of ‘string2’ will be deleted and may be replaced with subsequent DMAP statements.
occurrence	This flag indicates which occurrence of the proceeding string is to be used, starting at the beginning of the subDMAP. (Integer > 0; Default = 1)
offset	This flag indicates the offset from the reference DMAP statement. Depending on the sign, the specific DMAP statement may be above (-offset) or below (+offset) the referenced DMAP statement. (Integer; Default = 0)

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Remarks:

1. The ALTER statement must be used in conjunction with the COMPILE Executive Control statement. Note: ALTER statements cannot be used in conjunction with a MALTER statement, and therefore, should never immediately follow this statement.
2. If a MALTER statement is used in conjunction with the ALTER statement, then the MALTER should be placed above the COMPILE statements. Failure to place the MALTER in this position may cause ALTER statements to be ignored.
3. The ALTERs can reference the DMAP statements in any order within a subDMAP. Two restrictions on ordering are:
   - K2 or 'string2'(occurence, offset) references must refer to a DMAP statement number that is greater than or equal to the k1 or 'string1'(occurrence,offset) reference within a single ALTER statement.
   - K1 or 'string1' and k2 or 'string2' cannot overlap DMAP line positions with another ALTER that references the same subDMAP.
4. The 'string1' or 'string2' used as a search pattern will apply to one complete DMAP statement-i.e., a multiline DMAP statement will be searched for a pattern match as if each 72 character line of the DMAP statement were concatenated together into one string. All blanks and comments (either embedded or immediately preceding the DMAP statement) will be retained. However, comments are ignored for the following type of alter:

```
alter '^ *gp0'
```

5. Within a SUBDMAP, both 'string1' and 'string2' will be used to search for a pattern match starting at the beginning of the subDMAP-not at the current position of the last string match.
6. The special characters (meta characters) used for string searching are described in Remark 9. The characters <,>, and $, which are common DMAP characters, are also special meta characters. If they are to be used in the search string as regular characters, then they must be preceded by a backward slash (\). For example, to find the string

```
IF(DDRMM >=-1)
```

   the command is

```
ALTER 'IF (DDRMM \>=-1)' $
```

7. The ALTER statement must not exceed 72 characters (no continuations are allowed).
8. 'string2' (r2,02) can be defaulted to 'string1' (r1,01) by using a null string (''). For example, the alter statement

```
ALTER 'string1'(r1,01),''
```


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is equivalent to

```
ALTER ‘string1’(r1,01),‘string1’(r1,01)
```

The defaults for (r2,02) using the null string can be overridden by specifying (r2,02).

As another example, the alter statement

```
ALTER ‘string1’(r1,01),‘’(r2,02)
```

is equivalent to

```
ALTER ‘string1’(r1,01),‘string1’(r2,02)
```

9. Meta characters:

.	Matches any *single* character except *newline*.
*	Matches any number (including zero) of the single character (including a character specified by a regular expression) that immediately precedes it. For example, since “.” (dot) means any character, “.*” means “match any number of characters”.
[...] or < >	Matches any *one* of the characters enclosed between the brackets. For example, “[AB]” matches either “A” or “B”. A range of consecutive characters can be specified by separating the first and last characters in the range with a hyphen. For example “[A-Z]” will match any uppercase letter from A to Z and “[0-9]” will match any digit from 0 to 9. Some meta characters lose special meaning inside brackets. A circumflex (^) as the first character in the bracket tries to match any one character *not* in the list.
^ or ! or -|	Requires that the following regular expression be found at the beginning of the line. Note, that these meta characters may lead to UFM 802 if the preceding line is a comment.
$	Requires that the preceding regular expression be found at the end of the line.
\	Treats the following special character as an ordinary character. For example “ \ .” stands for a period and “ *” for an asterisk. Also, to search for a tic (’), the search string must be single quotes.
’	Marks the beginning and end of a pattern to be matched.
	Note: Nonportable characters such as ^ and [ ] should be replaced (e.g., ^ → ! and [] → <>) if portability is required. However, all the above characters are recognized by Simcenter Nastran.

10. If a string-based alter uses the “!” in the expression (find occurrence at the beginning of line), it is possible Simcenter Nastran will fail with USER FATAL MESSAGE 802.

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Examples:

1. The following alter will insert a MATPRN DMAP statement after the first occurrence of the string ‘SDR2’ in subDMAP DSASTAT.

```
SOL 101
COMPILE DSASTAT $
ALTER ’SDR2’ $
MATPRN OESDS1//$
CEND
```

2. The following alter will delete the second occurrence of the OUTPUT4 DMAP statement in subDMAP DSASTAT and replace it with a MATPRN DMAP statement.

```
SOL 101
COMPILE DSASTAT $
ALTER ‘OUTPUT4’(2),‘OUTPUT4’(2) $
$ OR
$ ALTER ‘OUTPUT4’(2),‘’ $
MATPRN OESDS1//$
CEND
```

Related Topics

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APP

Specify Solution Sequence Approach

Selects heat transfer analysis in the linear static SOL 101 or a coupled analysis combining heat transfer and structural analysis in SOL 153.

Format:

APP approach.

Describers:

Describer	Meaning	
approach	Specifies one of the following:	
	HEAT	Indicates that heat transfer is to be performed in SOL 101.
	COUPLED	Indicates a coupled analysis combining heat transfer and structural analysis in SOL 153.

Remarks:

1. The APP statement is optional.
2. The APP HEAT statement applies only to linear static SOL 101. The APP HEAT statement is not required in SOLs 153 and 159, or in SOL 101 if PARAM,HEATSTAT,YES is specified.
3. The NASTRAN HEAT=101 statement is an alternate specification of APP HEAT. See **"nastran Command and NASTRAN Statement"**.

Examples:

The following requests a heat transfer rather than a structural analysis in Solution Sequence 101.

```
SOL 101
APP HEAT
```

Related Topics

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CEND

End of Executive Control Delimiter

Designates the end of the Executive Control Section.

Format:

CEND

Remarks:

CEND is required unless an ENDJOB statement appears in the File Management Section.

Related Topics

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COMPILE

Compile DMAP Statements

Requests the compilation of a subDMAP, subDMAP alter, or NDDL sequence.

Format 1:

Compiles a subDMAP or subDMAP alter sequence

$$\text{COMPILE}\left[\begin{matrix}\text{SUBDMAP}\\ \underline{\text{DMAP}}\end{matrix}\right]\text{subDMAP}-\text{name}\left[\text{SOUIN}=\text{souin}-\text{DBset}\,\text{SOUOUT}=\text{souout}-\text{DBset}\right.$$

$$\text{OBJOUT}=\text{objout}-\text{DBset}\left[\begin{matrix}\text{LIST}\\ \underline{\text{NOLIST}}\end{matrix}\right]\left[\begin{matrix}\text{REF}\\ \underline{\text{NOREF}}\end{matrix}\right]\left[\begin{matrix}\text{DECK}\\ \underline{\text{NODECK}}\end{matrix}\right]\Bigg]$$

Format 2:

Compiles an NDDL sequence

$$\text{COMPILE}\,\text{NDDL}=\text{nddl}-\text{name}\left(\left\{\begin{matrix}\text{SOUIN}=\text{souin}-\text{dbset}\\ \text{SOUOUT}=\text{souout}-\text{dbset}\end{matrix}\right\}\left\{\begin{matrix}\text{LIST}\\ \underline{\text{NOLIST}}\end{matrix}\right\}\left\{\begin{matrix}\text{REF}\\ \underline{\text{NOREF}}\end{matrix}\right\}\left\{\begin{matrix}\text{DECK}\\ \underline{\text{NODECK}}\end{matrix}\right\}\right)$$

Examples:

1. The following compiles an alter in subDMAP PHASEIDR.

```
COMPILE PHASE1DR
ALTER 'CALL PHASE1A'
CEND
```

2. The following compiles a subDMAP called MYDMAP. (SUBDMAP and END are DMAP statements; see *Simcenter Nastran DMAP Programmer's Guide*.)

```
COMPILE MYDMAP LIST REF
SUBDMAP MYDMAP $
.
.
.
END $
CEND
```


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3. The following obtains a listing of the NDDL.

```
ACQUIRE NDDL
COMPILE NDDL=NDDL LIST
CEND
```

Describers:

Describer	Meaning
subDMAP-name	The name of a subDMAP sequence. SubDMAP-name must be 1 to 8 alphanumeric characters in length and the first character must be alphabetic. The keywords DMAP and SUBDMAP are optional and do not have to be specified.
nddl-name	The name of an NDDL sequence. (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.) The keyword NDDL must be specified.
souin-DBset	The name of a DBset from which the subDMAP or NDDL source statements will be retrieved. (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.) The default is MSCSOU if the next statement is not a subDMAP statement.
souout-DBset	The name of a DBset on which the subDMAP or NDDL source statements will be stored. (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.) The default is the SCRATCH DBset.
objout-DBset	The name of a DBset on which the subDMAP object code will be stored. (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.) The default is the OBJSCR DBset.
LIST, NOLIST	LIST requests a compiled listing of the subDMAP or NDDL sequence. NOLIST suppresses the listing. NOLIST is the default.
REF, NOREF	REF requests a compiled cross reference of the subDMAP or NDDL sequence. NOREF suppresses the cross reference. NOREF is the default.
DECK, NODECK	DECK requests the subDMAP or NDDL source statements to be written to the PUNCH file. NODECK suppresses the writing to the PUNCH file. NODECK is the default.

Remarks:

1. SubDMAP names for Simcenter Nastran solution sequences are given in the SOL statement description. The “COMPILER LIST REF” statement may be used to determine the appropriate subDMAP-name.
2. If a subDMAP is being compiled and SOUIN=souin-DBset is specified, then an ALTER Executive Control statement or an INCLUDE statement which contains an ALTER statement as the first non comment

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line, must appear immediately after these. If not, then the SUBDMAP DMAP statement must appear immediately after these statements. See the *Simcenter Nastran DMAP Programmer's Guide*.

3. DBsets USRSOU and USROBJ are no longer automatically created. They must be initialized by the INIT FMS statement and then may be specified for souin-dbset (or souout-dbset) and objout-dbset, respectively. They may be used to store the subDMAP source statements and object code on the primary database for re-execution in a subsequent run. For example:

   In the first run, the following COMPILE statement compiles and stores a subDMAP called MYDMAP.

```
COMPILE MYDMAP SOUOUT=USRSOU OBJOUT=USROBJ
SUBDMAP MYDMAP $
.
.
END $
CEND
```

   In the second run, the SOL statement is used to execute the MYDMAP stored in the previous run. The LINK statement is required to retrieve the object code from the USROBJ DBset.

```
SOL MYDMAP
LINK MYDMAP INCL=USROBJ
CEND
```

   In the third run, the COMPILE statement is used to alter MYDMAP and execute.

```
SOL MYDMAP
COMPILE MYDMAP SOUIN=USRSOU
ALTER...
.
.
.
CEND
```

4. If SOUOUT or OBJOUT is specified and a subDMAP with the same name as the subDMAP-name already exists on the database, then its source statements or object code will be replaced.

5. A COMPILE statement is required for each subDMAP to be compiled. If two or more COMPILE statements reference the same subDMAP name, then only the last is used in the linking of the object code. If the COMPILE statement is being used only to alter a subDMAP and two or more COMPILE statements reference the same subDMAP name, then the multiple alters are assembled and the subDMAP is compiled only once.

6. Only one COMPILE statement for an NDDL sequence may be specified in the input file.

   - SOUIN=souin-DBset requests only a compilation of the NDDL sequence stored on souin-DBset for purposes of obtaining a listing or a cross reference, and it cannot be modified with the ALTER statement. See Remark 3 . COMPILE NDDL=NDDL SOUIN=MSCSOU LIST requests a listing of the Simcenter Nastran NDDL sequence. The ACQUIRE FMS statement or the SOL statement must be specified in order to attach the corresponding Delivery Database.

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- To alter the Simcenter Nastran NDDL sequences, the entire modified NDDL sequence is included after the COMPILE statement and SOUIN=souin-DBset is not specified.
- SOUOUT=souout-DBset requests the storage of the NDDL source statements on the souout-DBset and may not be specified with SOUIN=souin-DBset.

7. The COMPILER statement may be used to override the defaults of NOLIST, NOREF, and NODECK. In other words, if LIST or NOLIST, REF or NOREF, or DECK or NODECK is not specified, then the corresponding option on the COMPILER statement will be used. In the following example, REF on the COMPILER statement will override the default of NOREF on the COMPILE statement.

```
COMPILER REF
COMPILE MYDMAP
```

8. MSCSOU and MSCOBJ, specified with SOUOUT and OBJOUT, are special DBsets similar to USRSOU and USROBJ except that they are used in the creation or modification of a delivery database. For an example application, see the *Simcenter Nastran Installation and Operations Guide*.

Related Topics

Browse Simcenter Nastran documentation

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COMPILER

DMAP Compiler Output Options

Requests compilation of a DMAP sequence and/or overrides defaults on the COMPILE statement.

Format:

$$\text{COMPILER } [=] \left[\begin{matrix}\text{LIST}\\ \underline{\text{NOLIST}}\end{matrix}\right] \left[\begin{matrix}\text{DECK}\\ \underline{\text{NODECK}}\end{matrix}\right] \left[\begin{matrix}\text{REF}\\ \underline{\text{NOREF}}\end{matrix}\right] \left[\begin{matrix}\underline{\text{GO}}\\ \text{NOGO}\end{matrix}\right] \left[\begin{matrix}\underline{\text{SORT}}\\ \text{NOSORT}\end{matrix}\right]$$

Describers:

Describer	Meaning
LIST, NOLIST	LIST requests the compilation listing of the solution sequence. NOLIST suppresses the listing.
DECK, NODECK	DECK requests that the DMAP source statements of the solution sequence be written to the PUNCH file. NODECK suppresses the DECK option.
REF, NOREF	REF requests a compilation cross reference. NOREF suppresses a compilation cross reference.
GO, NOGO	GO requests the execution of the solution sequence following compilation. NOGO requests termination following compilation.
SORT, NOSORT	SORT compiles subDMAPs in alphabetical order. NOSORT compiles subDMAPs in calling sequence order.

Remarks:

1. REF is equivalent to DIAG 4. LIST is equivalent to DIAG 14. DECK is equivalent to DIAG 17.
2. NOGO is an alternative to NOEXE on the SOL statement.
3. This statement provides the user a means of obtaining a compilation or source listing, or both, of a complete solution sequence, including all the component subDMAPs.
4. See the COMPILE statement to compile a single subDMAP.
5. This statement also requests the automatic link of the solution sequence. Therefore, all objects must be created in the current run or obtained from the DBset such as USROBJ. See the COMPILE statement for how to create and store objects.
6. The COMPILER statement may be used to override the defaults of NOLIST, NOREF, NODECK on the COMPILE entry when they are not explicitly specified. However, COMPILER LIST produces a list of

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the entire solution sequence. If a listing of only specific subdmaps are desired, then COMPILER LIST should not be specified and the LIST request should be made on the COMPILE entry.

```
COMPILER REF
COMPILE MYDMAP
```

Examples:

```
COMPILER=LIST
```

Related Topics

Browse Simcenter Nastran documentation

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DIAG

Request Diagnostic Output

Requests diagnostic output or special options.

Format:

DIAG [=] k1[k2, ..., kn]

Describers:

Describer	Meaning
ki	A list separated by commas and/or spaces of desired diagnostics.

Remarks:

1. The DIAG statement is optional.
2. Multiple DIAG statements are allowed.
3. The following table lists the possible values for ki and their corresponding actions:

k=1	Dumps memory when a nonpreface fatal message is generated.
k=2	Prints database directory information before and after each DMAP statement. Prints bufferpooling information.
k=3	Prints “DATABASE USAGE STATISTICS” after execution of each functional module. This message is the same as the output that appears after the run terminates.
k=4	Prints cross-reference tables for compiled sequences. Equivalent to the COMPILER REF statement.
k=5	Prints the BEGIN time on the operator’s console for each functional module.
k=6	Prints the END time for each functional module in the log file or day file and on the operator’s console. Modules that consume less time than the threshold set by SYSTEM(20) do not create a message.
k=7	Prints eigenvalue extraction diagnostics for the Complex Determinate method.
k=8	Prints matrix trailers as the matrices are generated in the Execution Summary Table.
k=9	Not used.
k=10	Uses alternate nonlinear loading in linear transient analysis. Replaces N_{n+1} with $(N_{n+1} + N_n + N_{n-1})/3$.
k=11	DBLOAD, DBUNLOAD, and DBLOCATE diagnostics.

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k=12 Prints eigenvalue extraction diagnostics for complex Inverse Power, complex Lanczos and ISRR methods.

k=13 Prints the open core length (the value of REAL on VAX computers).

k=14 Prints the solution sequence. Equivalent to the COMPILER LIST statement.

k=15 Prints table trailers.

k=16 Traces real inverse power eigenvalue extraction operations.

k=17 Punches solution sequences. Equivalent to the COMPILER DECK statement.

k=18 In aeroelastic analysis, prints internal grid points specified on SET2 Bulk Data entries.

k=19 Prints data for MPYAD and FBS method selection in the Execution Summary Table.

k=20 Similar to DIAG 2 except the output appears in the Execution Summary Table and has a briefer and more user-friendly format. However, the .f04 file will be quite large if DIAG 20 is specified with a Simcenter Nastran solution sequence. A DMAP Alter with DIAGON(20) and DIAGOFF(20) is recommended. DIAG 20 also prints DBMGR, DBFETCH, and DBSTORE subDMAP diagnostics. See the *Simcenter Nastran DMAP Programmer's Guide*.

k=21 Prints diagnostics of DBDIR and DBENTRY table.

k=22 Outputs EQUIV and EQUIVX module diagnostics.

k=23 Not used.

k=24 Prints files that are left open at the end of a module execution. Also prints DBVIEW diagnostics.

k=25 Outputs internal plot diagnostics.

k=26 Outputs dynamic file allocation diagnostics.

k=27 Prints Input File Processor (IFP) table. See section 4.3 of the *Simcenter Nastran Programmer's Guide* for more information on IFP.

k=28 Punches the link specification table (XBSBD). The Bulk Data and Case Control Sections are ignored, and no analysis is performed.

k=29 Process link specification table update. The Bulk Data and Case Control Sections are ignored, and no analysis is performed.

k=30 In link 1, punches the XSEMii data (i.e., set ii via DIAG 1 through 15). The Bulk Data and Case Control Sections are ignored, and no analysis is performed. After link 1, this turns on BUG output. Used also by MATPRN module. See also Remark 5 on the **"TSTEP"** Bulk Data entry.

k=31 Prints the link specification table and module properties list (MPL) data. The Bulk Data and Case Control Sections are ignored, and no analysis is performed.

k=32 Prints diagnostics for XSTORE and PVA expansion.

k=33 Not used.

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k=34 Turns off plot line optimization.

k=35 Prints diagnostics for 3-D slideline contact analysis in SOLs 106 and 129.

k=36 Legacy option which is no longer supported.

k=37 Disables the superelement congruence test option and ignores User Fatal Messages 4277 and 4278. A better alternative is available with PARAMeter CONFAC. See **"Parameters and Parameter Applicability Tables"**.

k=38 Prints material angles for CQUAD4, CQUAD8, CTRIA3, and CTRIA6 elements. The angle is printed only for elements that specify MCID in field 8 of the connection entry.

k=39 Traces module FA1 operations and aerodynamic splining in SOLs 145 and 146.

k=40 Prints constraint override/average information for edges and faces in p-adaptive analysis.

k=41 Traces GINO OPEN/CLOSE operations.

k=42 Not used.

k=43 Not used.

k=44 Prints a mini-dump for fatal errors and suppresses user message exit.

k=45 Prints the same database directory information as DIAG 2 except that it prints only after each DMAP statement.

k=46 Used by Siemens development for GINO printout.

k=47 Prints DBMGR, DBFETCH, and DBSTORE subDMAP diagnostics.

k=48 Used by Siemens development for GINO printout.

k=49 Prints the DMAP execution time summary table.

k=50 Traces the nonlinear solution in SOLs 106, 129, 153, and 159. Prints subcase status; echoes NLPARM, NLPCI, and TSTEPNL entry fields; and prints initial arc-length. Prints iteration summary only in SOLs 129, and 159.

In static aeroelastic analysis (SOL 144), prints transformation information associated with the generation of the DJX matrix in the ADG module and intermediate solutions information in the ASG module.

k=51 Prints intermediate displacement, load error vectors, and additional iteration information helpful to debugging in SOLs 106, 129, 153, and 159.

k=52 Disables the printing of errors at each time step in SOLs 129 and 159.

k=53 Prints the MESSAGE module output in the execution summary table.

k=54 Print linker debug diagnostics.

k=55 Output performance timing.

k=56 Prints all DMAP statements and RESTART deletions. (Extended print of Execution Summary table).

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k=57	Outputs executive table (XDIRLD) performance timing and last-time-used (LTU) diagnostics.
k=58	Outputs data block deletion debug and timing constants echo.
k=59	Prints buffpool debug.
k=60	Prints diagnostics for data block cleanup at the end of each module execution in subroutines DBCLN, DBEADD, and DBERPL.
k=61	GINO block allocator diagnostics.
k=62	GINO block manager diagnostics.
k=63	Prints each item checked by the RESTART module and its NDDL description.
k=64	Not used.

Examples:

```
DIAG 8,53
```

or

```
DIAG 8
DIAG 53
```

See the “Simcenter Nastran Output Files” in the *Simcenter Nastran User’s Guide*.

Related Topics

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DOMAINSOLVER

Domain Decomposition Solution Method

This is an obsolete interface to various domain decomposition methods. If you are looking for FDMODES or GDMODES, use the numseg and dmp keywords. For GDSTAT/LDSTAT, use the dmp and dstat keywords. For ACMS, use the RDMODES method instead using the RDMODES bulk entry. See the *Parallel Processing Guide* for details.

Related Topics

Browse Simcenter Nastran documentation

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ECHO

Control Printed Echo

Controls the echo (printout) of the Executive Control Section.

Formats:

```
ECHOOFF
ECHOON
```

Remarks:

1. The ECHO statement is optional.
2. ECHOOFF suppresses the echo of subsequent Executive Control statements. ECHOON reactivates the echo after an ECHOOFF statement.

Related Topics

Browse Simcenter Nastran documentation

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ENDALTER

End of DMAP Alter

Designates the end of an alter.

Format:

ENDALTER

Remarks:

The ENDALTER statement is required when using an alter unless the alter package ends with the CEND, COMPILE, or LINK statement.

Related Topics

Browse Simcenter Nastran documentation

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GEOMCHECK

Specifies Geometry Check Options

Specifies tolerance values for (optional) finite element geometry tests.

Format:

$$\text{GEOMCHECK}\ \text{test_keyword}\ [=\text{tol_value}],[\text{MSGLIMIT}=\text{n}],\left[\text{MSGTYPE}=\begin{matrix}\text{FATAL}\\ \underline{\text{INFORM}}\\ \text{WARN}\end{matrix}\right],[\text{SUMMARY}],[\text{NONE}]$$

Describers:

Describer	Meaning
test_keyword	A keyword associated with the particular element geometry test. See Remark 2 for a list of acceptable selections.
tol_value	Tolerance value to be used for the specified test. See Remark 2 for default values of the test tolerances.
n	The maximum number of messages that will be produced. The default is 100 messages for each element type. See Remark 3.
FATAL	Geometry tests that exceed tolerance values produce fatal messages. See Remark 4.
INFORM	Geometry tests that exceed tolerance values produce informative messages. See Remark 4.
WARN	Geometry tests that exceed tolerance values produce warning messages. See Remark 4.
SUMMARY	A summary table of the geometry tests performed is produced. No individual element information messages are output.
NONE	None of the optional element geometry tests will be performed. System controlled checks are performed depending on the solution type. See Remark 1.

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Examples:

1. Adjust the tolerance for the CQUAD4 element skew angle test to 15.0 degrees and limit messages to 50.

```
GEOMCHECK   Q4_SKEW=15.0,MSGLIMIT=50
```

2. Limit messages to 500 for each element type.

```
GEOMCHECK  MSGLIMIT=500
```

3. Adjust the tolerance for the CQUAD4 element taper and set the message type to fatal for all tests.

```
GEOMCHECK  Q4_TAPER=0.4,MSGTYPE=FATAL
```

4. Request summary table output only using default tolerance values.

```
GEOMCHECK  SUMMARY
```

Remarks:

1. There are two categories of element checks in a Simcenter Nastran solution:

   - The system controlled checks are documented in "System Element Checks" in the Simcenter Nastran User's Guide. Most are related to the Element Jacobian checks. These checks will be performed as follows:
     ● For all solutions except SOLs 401, 402, and 414, the system geometry checks are performed unless GEOMCHECK=NONE. If any tests exceed the tolerance, the solution behavior is the same as for the optional tests and is governed by the MSGTYPE setting.
     ● For SOL 401, the system geometry checks are always performed. If any tests exceed the tolerance, the solution will issue a fatal error. You can select the pre-Simcenter Nastran 2021.1 system check by setting NASTRAN SYSTEM(760)=1, which does not include the Jacobian check at the CTETRA element corners.
     ● See **Remark related to SOLs 402 and 414**.

   - GEOMCHECK are the optional, user controlled checks, and can optionally generate a fatal error if it finds an element which fails the user defined criteria. By default, GEOMCHECK does not produce fatal errors. One exception is that the aspect ratio checks (test_keyword = TET_AR, HEX_AR, PYR_AR, PEN_AR) for 3D solid elements will produce a fatal message by default (MSGTYPE=FATAL).

     Element checks which are controlled with the GEOMCHECK statement are summarized in Remark 2. See "User Controlled Element Checks" in the *Simcenter Nastran User's Guide* for detailed descriptions of all checks. The GEOMCHECK element checks will occur using default values when the GEOMCHECK statement is not included. GEOMCHECK controls are available for the shell elements CQUAD4, CQUADR, CTRIA3, CTRIAR, the solid elements CHEXA, CPENTA, CPYRAM, CTETRA, CHEXCZ, CPENTCZ the axisymmetric elements CTRAX3, CTRAX6, CQUADX4, CQUADX8, the

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plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8, the plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8, the chocking elements CCHOCK3, CCHOCK4, CCHOCK6, CCHOCK8, the CBAR element, and the CBEAM element. Multiple GEOMCHECK directives may be present. Continuations are acceptable.

Note

Be aware that when MSGTYPE=inform, GEOMCHECK may flag a poor quality element with an informational message, the element may pass the system check, and the solve may complete. MSGTYPE should be set appropriately if you expect fatal errors in these cases. Fatal errors can also be forced by assigning a negative value to MSGLIMIT, for example, MSGLIMIT=-100. This is also useful since a specific "FAIL" message will be reported for each element failing the GEOMCHECK criteria. See Remark 3 for more information on MSGLIMIT.

2. The **test_keyword** describer can optionally be used to modify the default thresholds. The following table lists the **test_keywords** and their defaults. The **test_keyword** inputs are Real ≥ 0.0. See "User Controlled Element Checks" in the *Simcenter Nastran User's Guide* for detailed descriptions of all element checks.

test_keyword	Default Threshold	Summary
CQUAD4 and CQUADR Shell Element Checks		
Q4_SKEW	Failure when < 30.0	Skew angle in degrees
Q4_TAPER	Failure when > 0.5	Taper ratio
Q4_WARP	Failure when > 0.05	Surface warping factor
Q4_IAMIN	Failure when < 30.0	Minimum Interior Angle in degrees
Q4_IAMAX	Failure when > 150.0	Maximum Interior Angle in degrees
QAD_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio (CQUAD4)
CTRIA3 and CTRIAR Shell Element Checks		
T3_SKEW	Failure when < 10.0	Skew angle in degrees
T3_IAMAX	Failure when > 160.0	Maximum Interior Angle in degrees
CQUAD8 Shell Element Checks		
Q8_SKEW	Failure when < 30.0	Skew angle in degrees
Q8_TAPER	Failure when > 0.5	Taper ratio
Q8_IAMIN	Failure when < 30.0	Minimum Interior Angle in degrees
Q8_IAMAX	Failure when > 150.0	Maximum Interior Angle in degrees
Q8_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio
Q8_EPLR	Failure when < 0.5	Edge point length ratio

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test_keyword	Default Threshold	Summary
CTRIA6 Shell Element Checks		
TA6_IAMN	Failure when < 10.0	Minimum Interior Angle in degrees
TA6_IAMX	Failure when > 160.0	Maximum Interior Angle in degrees
TA6_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio
TA6_EPLR	Failure when < 0.5	Edge point length ratio
CTETRA Solid Element Checks		
TET_AR	Failure when >100.0	Longest edge to shortest height aspect ratio
TET_EPIA	Failure when > 30.0	Edge point included angle
TET_EPLR	Failure when < 0.5	Edge point length ratio
CHEXA Solid Element Checks		
HEX_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio
HEX_EPIA	Failure when > 30.0	Edge point included angle
HEX_EPLR	Failure when < 0.5	Edge point length ratio
HEX_TK	Failure when > 0.05	Ratio of the difference between thickness in stacking direction as defined by grids and as defined by ply thickness specification to the thickness in the stacking direction as defined by grids for CHEXA elements that reference PCOMPS entries only
HEX_TP	Failure when > 1.2	Ratio of longest edge to shortest edge in stacking direction for CHEXA elements that reference PCOMPS entries only
HEX_WARP	Failure when < 0.707	Face warp coefficient
CPYRAM Solid Element Checks		
PYR_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio
PYR_EPIA	Failure when > 30.0	Edge point included angle
PYR_EPLR	Failure when < 0.5	Edge point length ratio
PYR_WARP	Failure when < 0.707	Face warp coefficient
CPENTA Solid Element Checks		
PEN_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio
PEN_EPIA	Failure when > 30.0	Edge point included angle

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test_keyword	Default Threshold	Summary
PEN_EPLR	Failure when < 0.5	Edge point length ratio
PEN_TK	Failure when > 0.05	Ratio of the difference between thickness in stacking direction as defined by grids and as defined by ply thickness specification to the thickness in the stacking direction as defined by grids for CPENTA elements that reference PCOMPS entries only
PEN_TP	Failure when > 1.2	Ratio of longest edge to shortest edge in stacking direction for CPENTA elements that reference PCOMPS entries only
PEN_WARP	Failure when < 0.707	Quadrilateral face warp coefficient
CHEXCZ Solid Element Checks		
HEX_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio
HEX_EPIA	Failure when > 30.0	Edge point included angle
HEX_EPLR	Failure when < 0.5	Edge point length ratio
HEX_WARP	Failure when < 0.707	Face warp coefficient
CPENTCZ Solid Element Checks		
PEN_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio
PEN_EPIA	Failure when > 30.0	Edge point included angle
PEN_EPLR	Failure when < 0.5	Edge point length ratio
PEN_WARP	Failure when < 0.707	Face warp coefficient
Checks for the Axisymmetric Elements CTRAX3 and CTRAX6, the Plane Stress Elements CPLSTS3 and CPLSTS6, the Plane Strain Elements CPLSTN3 and CPLSTN6, the Chocking Elements CCHOCK3 and CCHOCK6		
TRX_IAMN	Failure when < 30.0	Minimum Interior Angle in degrees
TRX_IAMX	Failure when > 160.0	Maximum Interior Angle in degrees
TRX_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio
TRX_EPLR	Failure when < 0.5	Edge point length ratio (CTRAX6, CPLSTS6, CPLSTN6, CCHOCK6)
Checks for the Axisymmetric Elements CQUADX4 and CQUADX8, the Plane Stress Elements CPLSTS4 and CPLSTS8, the Plane Strain Elements CPLSTN4 and CPLSTN8, the Chocking Elements CCHOCK4 and CCHOCK8		
QDX_IAMN	Failure when < 30.0	Minimum Interior Angle in degrees

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test_keyword	Default Threshold	Summary
QDX_IAMX	Failure when > 150.0	Maximum Interior Angle in degrees
QDX_AR	Failure when > 100.0	Longest edge to shortest edge aspect ratio
QDX_SKEW	Failure when < 30.0	Skew angle in degrees
QDX_TAPR	Failure when > 0.5	Taper ratio
QDX_EPLR	Failure when < 0.5	Edge point length ratio (CQUADX8, CPLSTS8, CPLSTN8, CCHOCK8)
CBEAM and CBAR Element Checks		
BEAM_OFF	Failure when > 0.15	CBEAM element offset length ratio
BAR_OFF	Failure when > 0.15	CBAR element offset length ratio

3. A single line summarizing the results of all tests for an element will be output if any of the geometry tests exceeds the test tolerance. Only the first n of these messages will be produced. A summary of the test results indicating the number of tolerances exceeded as well as the element producing the worst violation is also output. If the SUMMARY keyword has been specified, only the summary table is produced and none of the single line element messages will be output.

Solutions which fail with numerical problems, i.e. negative Jacobian, report differently based on the sign of MSGLIMIT. When troubleshooting a failed analysis, you may find it useful to switch between the following two strategies:

Value of MSGLIMIT	Meaning when numerical fatal errors occur
> 0 (positive)	If numerical problems are found resulting from poor quality elements, a FATAL ERROR message is generated without the offending element ID, the analysis terminates, but the GEOMCHECK criteria *is* processed and output for all elements. This strategy works when only a few elements fail the GEOMCHECK. However, if there are hundreds or thousands of failures, diagnosing the critical elements could become difficult. In addition, the offending element ID may or may not have failed the GEOMCHECK criteria depending on the criteria you entered.
< 0 (negative)	If numerical problems are found resulting from poor quality elements, a FATAL ERROR message is generated which includes the offending element ID, the analysis terminates, yet GEOMCHECK *is not* processed for all elements. The downside of using a negative MSGLIMIT is if the model contains multiple offending elements, the procedure of fixing/rerunning needs to be repeated until all offending elements have been corrected.

4. When SUMMARY is not specified, each geometry test that exceeds the tolerance will be identified in the single line output summary by an indicator based on the specification for MSGTYPE. For the FATAL option, the indicator is “FAIL”; for the INFORM option, it is “xxxx”; for the WARN option, it is “WARN”. If the FATAL option is specified and any test fails, the run is terminated.

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Remark related to SOLs 402 and 414:

1. SOLs 402 and 414 do not support either the optional or the system checks.

Related Topics

Browse Simcenter Nastran documentation

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ID

Comment

Specifies a comment.

Format:

ID [=] i1, i2

Describers:

Describer	Meaning
i1, i2	Character strings (1 to 8 characters in length and the first character must be alphabetic).

Remarks:

The ID statement is optional and not used by the program.

Related Topics

Browse Simcenter Nastran documentation

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INCLUDE

Insert External File

Inserts an external file into the input file. The INCLUDE statement may appear anywhere within the input data file.

Format:

INCLUDE ‘filename’

Describers:

Describer	Meaning
filename	External file to be inserted. The ‘*directory_path/filename*’ must begin and end with the single quote character.

Examples:

The following INCLUDE statement is used to obtain the Bulk Data from another file called MYBULK.DATA.

```
SOL 101
INCLUDE ‘MYEXEC.DATA’
CEND
TITLE = STATIC ANALYSIS
LOAD=100
BEGIN BULK
ENDDATA
```

Remarks:

1. INCLUDE statements may be nested; that is, INCLUDE statements may appear inside the external file. The nested depth level must not be greater than 10.
2. Each line has a 72 character limit. Multiple lines can be used when file names are long. The entire string must begin and end with the single quote character.

   For example,

   D:\folder1\folder2\folder3\folder4\folder5\folder6\folder7\folder8\folder9\mydata.dat

   can be defined on multiple lines:

   ```
   INCLUDE ‘D:\folder1\folder2\folder3\folder4\folder5\folder6\folder7\fold
   er8\folder9\mydata.dat'
   ```


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The following input format is also supported:

```
INCLUDE ‘D:\folder1\folder2\folder3\
         folder4\folder5\
     folder6\folder7\
        folder8\folder9\mydata.dat'
```

Related Topics

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LINK

Link a Main SubDMAP

Links a main subDMAP to form a solution sequence.

Format:

$$\text{LINK}\ \left\{\begin{matrix}\text{n}\\ \text{subDMAP}-\text{name}\end{matrix}\right\}\ [\text{SOLOUT}=\text{solout}-\text{DBset}\ \text{EXECOUT}=\text{execout}-\text{DBset}$$

$$\text{INCLUDE}=\text{incl}-\text{DBset}\ \left[\begin{matrix}\text{MAP}\\ \text{NOMAP}\end{matrix}\right]\ \text{SOLNAME}=\ \text{newname}]$$

Describers:

Describer	Meaning
n	The solution number of the main subDMAP. See **"SOL"** for the list of valid numbers. (Integer>0)
subDMAP-name	The name of a main subDMAP. See the *Simcenter Nastran DMAP Programmer's Guide*. (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.)
solout-DBset	The name of a DBset where the solution sequence executable and the link table of the solution sequence may be stored. See Remark **6** . (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.)
exeout-DBset	The name of an alternate DBset different than solout-DBset where only the solution sequence executable may be stored. See Remark **6** . (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.)
incl-DBset	The name of a DBset where other subDMAP objects are obtained. See Remark **2** . (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.)
newname	A new name which is referenced by the SOL statement. (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic; default is subDMAP-name.)
MAP	Prints the link map. A link map will give the name of all the subDMAPs that make up the solution sequence.
NOMAP	Suppresses printing of the link map.

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Remarks:

1. All DBsets specified on this statement must have the same BUFFSIZE. See **"INIT"** .
2. SubDMAP objects are created with the COMPILE statement either in the current run or obtained from previous runs. The LINK statement collects objects in the following order:
   - Objects created with the COMPILE statement in the current run.
   - Objects residing on the DBset-name specified by the INCLUDE keyword. The default is MSCOBJ.
3. Upon successful linking of a subDMAP, the subDMAP may be executed with the SOL statement.
4. The LINK statement must appear after all the COMPILE packages, if any. A compile package begins with the COMPILE statement and is delimited by the ENDALTER, CEND, LINK, or another COMPILE statement.
5. The link table is necessary for COMPILER (or DIAG 4, 14, 17) Executive statement requests and the automatic link process.
6. EXEOUT is useful in building delivery databases where executables are not to be saved. EXEOUT defaults to the same DBset as specified by SOLOUT.

Examples:

1. `LINK STATICS`

   Links the STATICS main subDMAP. The program links any subDMAPs compiled in this run, with any other subDMAP objects called in STATICS and stored on the MSCOBJ DBset.

2. `LINK MYDMAP,SOLNAM=STATICS,SOLOUT=USROBJ,`
   `NOMAP,INCLUDE=USROBJ`

   Links MYDMAP and renames the solution sequence executable to STATICS. The executable will be saved on the USROBJ DBset. The order of search for subDMAP objects is:
   - Compiled subDMAP in this run.
   - USROBJ DBset.

Related Topics

Browse Simcenter Nastran documentation

Simcenter Nastran Quick Reference Guide
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MALTER

Insert and/or Delete DMAP Statements in Solution Sequences

Inserts or deletes DMAP statements by allowing a global “string” search across all subDMAPs within the current solution sequence.

Format:

```
MALTER ‘string1’[(occurrence,offset)] , [‘string2’[(occurrence,offset)] ]
```

or

```
MALTER ‘string1’[(occurrence,offset)] , [k2]
```

Describers:

Describer	Meaning
‘string1’	If ‘string2’ or k2 is not specified, the subsequent DMAP statements will be inserted after the first occurrence of ‘string1’.
‘string1’, ‘string2’	DMAP statements beginning with the first occurrence of ‘string1’ through DMAP statements containing the first occurrence of ‘string2’ will be deleted and may be replaced with subsequent DMAP statements.
k2	If k2 is specified, it is applied to the subDMAP in which ‘string1’ was found. (Integer > 0)
occurrence	This flag indicates which occurrence of the preceding string is to be used, starting at the beginning of the subDMAP. (Integer > 0, Default = 1)
offset	This flag indicates the offset from the referenced DMAP statement. Depending on the sign, the specific DMAP statement may be above (-offset) or below (+offset) the referenced DMAP statement. (Integer, Default = 0)

Remarks:

1. If a MALTER statement is used in conjunction with the ALTER statement, then the MALTER should be placed above the COMPILE statements. Failure to place the MALTER in this position may cause ALTER statements to be ignored.
2. The MALTER statement can reference the DMAP statements in any order within a subDMAP. Two restrictions on ordering are:
   - k2 or ‘string2’(occurence,offset) references must refer to a DMAP line number that is greater than or equal to the k1 or ‘string1’(occurrence,offset) reference within a single MALTER statement.

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- ‘string1’ and k2 or ‘string2’ cannot overlap DMAP line positions with another MALTER that references the same subDMAP.

3. The ‘string1’ or ‘string2’ used as a search pattern will apply to one complete DMAP statement-i.e., a multiline DMAP statement will be searched for a pattern match as if each 72 character line of the DMAP statement were concatenated together into one string. All blanks and comments (either embedded or immediately preceding the DMAP statement) will be retained.

4. The special characters used for string searching are described in Remark 9 . The characters <, >, and $, which are common DMAP characters, are also special meta characters. If they are to be used in the search string as regular characters, then they must be preceded by a backward slash (\). For example, to find the string

```
IF (DDRMM >=-1)
```

the command is

```
ALTER ‘IF (DDRMM \>=-1)’ $
```

5. ‘string2’ (r2,02) can be defaulted to ‘string1’ (r1,01) by using a null string (‘’). For example, the alter statement

```
MALTER ‘string1’(r1,01),‘’
```

is equivalent to

```
MALTER ‘string1’(r1,01),‘string1’(r1,01)
```

The defaults for (r2,02) using the null string can be overridden by specifying (r2,02).

As another example, the alter statement

```
MALTER ‘string1’(r1,01),‘’(r2,02)
```

is equivalent to

```
MALTER ‘string1’(r1,01),‘string1’(r2,02)
```

6. The existing COMPILE statement options, such as LIST, XREF, SOUIN, etc., cannot be directly specified on the new MALTER statement. They are obtained as follows:

- If a COMPILE statement exists for the subDMAP referenced by the MALTER, then options from this COMPILE statement will be used.
- Else, they will be taken from the COMPILER statement, provided the LIST, and SORT option is always on.

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7. The MALTER string search order is as follows:
   - All COMPILE statement references that are part of the existing solution sequence (i.e., SOL=) are searched first.
   - Then, all remaining subDMAPs in the solution sequence are searched in ascending alphabetical order.
   - Within a subDMAP, both 'string1' and 'string2' will be used to search for a pattern match starting at the beginning of the subDMAP, not at the current position of the last string match.
8. The MALTER statement must not exceed 72 characters (no continuations are allowed).
9. Meta characters:

.	Matches any *single* character except *newline*.
*	Matches any number (including zero) of the single character (including a character specified by a regular expression) that immediately precedes it. For example, since "." (dot) means any character, ".*" means "match any number of characters."
[...] or < >	Matches any *one* of the characters enclosed between the brackets. For example, "[AB]" matches either "A" or "B". A range of consecutive characters can be specified by separating the first and last characters in the range with a hyphen. For example "[A-Z]" will match any uppercase letter from A to Z and "[0-9]" will match any digit from 0 to 9. Some meta characters lose special meaning inside brackets. A circumflex (^) as the first character in the bracket tries to match any one character *not* in the list.
^ or ! or ¬|	Requires that the following regular expression be found at the beginning of the line.
$	Requires that the preceding regular expression be found at the end of the line.
\	Treats the following special character as an ordinary character. For example " \." stands for a period and " *" for an asterisk. Also, to search for a tic ('), the search string must be " \' ".
'	Marks the beginning and end of a pattern to be matched.

Note

Nonportable characters such as ^ and [ ] should be replaced (e.g., ^ → ! and [] → <>) if portability is required. However, all the above characters are recognized by Simcenter Nastran.

10. Labels for use with the MALTER have been included in the solution sequences. See **Table 3-10**. These labels will be maintained in future versions and it is strongly suggested that alters which use the MALTER command take advantage of the unique MALTER labels. Use of the MALTER labels will significantly reduce the time required to convert alters between versions.

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Table 3-10: DMAP Labels and Corresponding SubDMAP Positions

DMAP MALTER Labels
$MALTER:AFTER PREFACE MODULES
$MALTER:TOP OF PHASE 1 SUPERELEMENT LOOP, AFTER PARAMETERS AND QUALIFIERS SET
$MALTER:AFTER SUPERELEMENT STIFFNESS, VISCOUS DAMPING, MASS, AND ELEMENT STRUCTURAL DAMPING GENERATION (KJJZ, BJJZ, MJJZ, K4JJ)
$MALTER:AFTER X2GG MATRICES READ (K2JJ, M2JJ, B2JJ)
$MALTER:AFTER TOTAL SUPERELEMENT STIFFNESS, VISCOUS DAMPING, AND MASS FORMULATED, STRUCTURAL + DIRECT INPUT
$MALTER:AFTER SUPERELEMENT LOAD GENERATION (PJ)
$MALTER:AFTER UPSTREAM SUPERELEMENT MATRIX AND LOAD ASSEMBLY (KGG, BGG, MGG, K4GG, PG)
$MALTER:AFTER SUPERELEMENT MATRIX AND LOAD REDUCTION TO A-SET, STATIC AND DYNAMIC (KAA, KLAA, MAA, MLAA, BAA, K4AA, PA)
$MALTER:BOTTOM OF PHASE 1 SUPERELEMENT LOOP
$MALTER:AFTER X2PP MATRICES READ (K2PP, M2PP, B2PP)
$MALTER:AFTER SUPERELEMENT DISPLACEMENT RECOVERY (UG)
$MALTER:AFTER ELEMENT STRESS, STRAIN, ETC. DATA RECOVERY, SORT1 (OUGV1, OES1, OEF1, ETC.)
$MALTER:AFTER ELEMENT STRESS, STRAIN, ETC. DATA RECOVERY, SORT2 (OUGV2, OES2, OEF2, ETC.)
$MALTER:BOTTOM OF SUPERELEMENT DATA RECOVERY LOOP
$MALTER:USERDMAP - AFTER CALL PREFACE

Examples:

1. The following MALTER will insert a MATPRN DMAP statement to print the KJJ matrix for each superelement.

```
SOL 101
MALTER ‘MALTER:AFTER SUPERELEMENT STIFFNESS .* GENERATION’
MESSAGE //‘SEID=’/SEID $
MATPRN KJJZ/ $
```

2. The following MALTER will add a user DMAP after the PREFACE modules in SOL 100 (USERDMAP).

```
SOL 101
MALTER ‘AFTER CALL PREFACE’
.
.
.
```

Related Topics

Browse Simcenter Nastran documentation

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SOL

Execute a Solution Sequence

Specifies the solution sequence or main subDMAP to be executed.

Format:

$$\text{SOL} \left\{ \begin{matrix} \text{n} \\ \text{subDMAP} - \text{name} \end{matrix} \right\} [\text{SOLIN} = \text{obj} - \text{DBset} \quad \text{NOEXE}]$$

Describers:

Describer	Meaning
n	Solution number. See Remark 6 . for the list of valid numbers. (Integer > 0)
subDMAP-name	The name of a main subDMAP. See the *Simcenter Nastran DMAP Programmer's Guide*. (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.)
obj-DBset	The character name of a DBset where the OSCAR is stored. See Remarks 1 and 2. (Character; 1 to 8 alphanumeric characters in length and the first character must be alphabetic.)
NOEXE	Suppresses execution after compilation and/or linkage of the solution is complete. Also, the Bulk Data and Case Control Sections are not read or processed.

Remarks:

1. If a SOLIN keyword is not given and if there are no LINK statements within the input data, the program will perform an automatic link. The program will first collect the objects created in the current run by the COMPILE statement and the remaining objects stored in the MSCOBJ DBset. The program will then perform an automatic link of the collected objects.
2. If the SOLIN keyword is not given but a LINK statement is provided, the SOLIN default will be obtained from the SOLOUT keyword on the LINK statement.
3. The OSCAR (Operation Sequence Control ARray) defines the problem solution sequence. The OSCAR consists of a sequence of entries with each entry containing all of the information needed to execute one step of the problem solution. The OSCAR is generated from information supplied by the user's entries in the Executive Control Section.
4. The SOLIN keyword will skip the automatic link and execute the OSCAR on the specified DBset.
5. The DOMAINSOLVER may be used in conjunction with Solution Sequences 101, 103, 108, and 111 to select domain decomposition solution methods.

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6. The following Solution Sequences are currently available in Simcenter Nastran:

Table 3-11: Structured Solution Sequences

SOL Number	SOL Name	Description
101	SESTATIC	Statics with Options: Linear Steady State Heat Transfer Alternate Reduction (See "Superelement Alternate Statics Solution" in the *Simcenter Nastran User's Guide*.) Inertia Relief
103	SEMODES	Normal Modes
105	SEBUCKL	Buckling with options: Static Analysis Alternate Reduction Inertia Relief
106	NLSTATIC	Nonlinear or Linear Statics
107	SEDCEIG	Direct Complex Eigenvalues
108	SEDFREQ	Direct Frequency Response
109	SEDTRAN	Direct Transient Response
110	SEMCEIG	Modal Complex Eigenvalues
111	SEMFREQ	Modal Frequency Response
112	SEMTRAN	Modal Transient Response
114	CYCSTATX	Cyclic Statics with Option: Alternate Reduction
115	CYCMODE	Cyclic Normal Modes
116	CYCBUCKL	Cyclic Buckling
118	CYCFREQ	Cyclic Direct Frequency Response
129	NLTRAN	Nonlinear or Linear Transient Response
144	AESTAT	Static Aeroelastic Response
145	SEFLUTTR	Aerodynamic Flutter
146	SEAERO	Aeroelastic Response
153	NLSCSH	Static Structural and/or Steady State Heat Transfer Analysis with Options: Linear or Nonlinear Analysis
159	NLTCSH	Transient Structural and/or Transient Heat Transfer Analysis with Options: Linear or Nonlinear Analysis
187	RESDDAM	DDAM analysis.
190	DBTRANS	Database Transfer.
200	DESOPT	Design Optimization with option for only Sensitivity Analysis. See Remark 7.
401	NLSTEP	Multistep, structural solution which supports a combination of static (linear or nonlinear) subcases and modal (real eigenvalue) subcases.

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<table>
<tr><th>SOL Number</th><th>SOL Name</th><th>Description</th></tr>
<tr><td>402</td><td>NLSTPKIN</td><td>Multistep, structural solution which supports a combination of subcase types (static linear, static nonlinear, nonlinear dynamic, preload, modal, Fourier, buckling) and supports large rotation kinematics.</td></tr>
<tr><td>414,N</td><td>ROTDYN</td><td>Rotor dynamic analysis. N represents a valid solution type. The available types are:
<table>
<tr><th>N</th><th>SOL 414 solution type</th></tr>
<tr><td>101</td><td>Maneuvers (linear static)</td></tr>
<tr><td>103</td><td>Eigenvalues and Superelement Reduction</td></tr>
<tr><td>110</td><td>Complex Modal Analysis</td></tr>
<tr><td>111</td><td>Harmonic Response</td></tr>
<tr><td>129</td><td>Transient Response</td></tr>
</table>
</td></tr>
</table>

7. Design sensitivity is supported with SOL 200 using the DSAPRT case control command. As a result, the following legacy inputs for design sensitivity analysis with SOL 101, 103, and 105 have been undocumented:
   – Case control commands SENSITY, SET2
   – Parameters EIGD, NORM
   – Bulk entries DSCONS, DVAR, DVSET
   Since design sensitivity analysis is automatically done during design optimization, it is not necessary to use the DSAPRT case control unless specific sensitivity information is requested as part of the output, or only design sensitivity analysis is desired.

Examples:

1. In the following example, SOL 103 is executed from MSCOBJ.

```
SOL 103
```

2. In the following example, the PHASE0 subDMAP is altered, SOL 103 is relinked onto the OBJSCR DBset (which is the default for SOLOUT), and SOL 103 is executed.

```
SOL 103
COMPILE PHASE1
ALTER 'DTIIN'
TABPT SETREE,,,,// $
.
.
.
ENDALTER $
```

3. In the following example, the solution sequence called DYNAMICS is executed from the USROBJ DBset.

```
SOL DYNAMICS SOLIN = USROBJ
```


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Related Topics

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SOL 402

Executes a Simcenter Nastran Multi-Step Nonlinear Kinematics solution

Format:

SOL 402

Remarks:

1. The ANALYSIS case control command defines the subcase analysis type for SOL 402. The following subcase types are available.

STATICS	(Non-Linear) static analysis
DYNAMICS	(Non-Linear) dynamic analysis, with velocity and inertia effects
MODES	Normal Modes
PRELOAD	Bolt Preload subcase computation
CYCMODES	Cyclic Normal Modes
FOURIER	Fourier Normal Modes.
BUCKLING	Buckling Modes (Incremental Stability)
CEIG	Complex Eigenvalues Extraction

2. A SOL 402 job may be terminated gracefully at any equilibrium iteration (or time step) by modifying the RUN file *<NAME>*_*<PID>*_me.run, where *<NAME>* is the problem name and *<PID>* is replaced by a unique number during execution.

   To dynamically update any numerical parameter for the next time step, the software reads the content of this file at the end of every computation time step, only up to the EXIT line. In particular, moving the line containing ISTO before the EXIT line and setting this parameter to 999 will stop the computation. When you do this instead of killing the job, you ensure that all results are saved and all files are properly closed.

   For more information on all the parameters that be can defined in the RUN file, see the **NLCNTL2** bulk entry.

3. Distributed-memory parallel processing can be requested by the MPI402 nastran keyword. Shared-memory parallel processing can be request by the parallel (or smp) nastran keyword.

   - The MPI402 keyword specifies the number of CPUs selected for distributed-memory parallel processing.

     The .f06 file will include a message similar to the following indicating the number of processors used by the solution:

     ```
     NUMBER OF PROCESSORS USED.....(NPROC) = n
     ```

   - The parallel keyword specifies the number of MKL threads.

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4. The total memory requested with the memory keyword is shared by input data processing and by the actual SOL 402 computation. By default, the input data processing will use as much memory as it needs out of the total, and the SOL 402 computation will receive the rest. For larger problems, the SOL 402 computation will need much more memory than what is required for input data processing. In that case, you can use the environment variable NXNS_MEMORY to specify in MB the memory used for the computation:

Environment Variable	Description
NXNS_MEMORY	Specifies memory in MB for the solution only.

Note

When defining NXNS_MEMORY, include the numerical value in MB, but do not include the "MB" text describer.
For example:

- Valid definition:
  NXNS_MEMORY=4000
- Invalid definition:
  NXNS_MEMORY=4000MB

5. See the Case Control Applicability Tables in the Simcenter Nastran Quick Reference Guide for the supported commands in SOL 402.

Note

The TSTEP case control command defines the set identification number of a TSTEP1 or a TSTEP bulk entry. You cannot use both TSTEP1 and TSTEP in the same model.

The NLCNTL case control command defines the set identification number of an NLCNTL2 bulk entry, not an NLCNTL bulk entry.

6. See the User Interface topic in the *Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)* for the list of bulk entries supported by SOL 402.

Note

Not all fields in each bulk entry are used by the software. Restrictions are given in the remarks for each bulk entry.

Note

- Only the diagonal terms of the mass matrix defined by CONM1 or CONM2 are used.
- To define an orthotropic material, the use of MAT11 is preferred to the use of MAT2 or MAT9.
- CROD and CONROD define a truss element with no torsional stiffness.
- Transverse properties of the PGAP element are not supported.
- Logarithmic XAXIS is not supported for TABLEM1 and TABLED1.

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7. See the Parameter Applicability Tables in the Simcenter Nastran Quick Reference Guide for the parameters supported by SOL 402. Parameters with a "B" in the table can only be specified in the Bulk Data. Parameters with an "E" in the table can be specified in either the Bulk Data or in the Case Control.

   Note

   PARAM,POST,-2 must be specified to generate both geometry and results data blocks in the output .op2 file. This is done automatically when a job is run from Simcenter. PARAM,POST,-1 must be specified to generate only results data blocks, as required by PATRAN. Otherwise, no output .op2 file is generated.

8. You can use the following NASTRAN System Cells.

System Cell (Name)	Function and Reference
57 (HICORE)	Working memory
81	Specifies the amount of open core memory to which certain numerical modules will be restricted.
107 (PARALLEL)	Parallel processing deselection (deactivation) for matrix operations. See Keywords in the *Simcenter Nastran Installation and Operations Guide*. 0: Deactivate parallel processing N: Number of processors
143 (LOCBULK)	LOCBULK=1 or 2 specifies that Bulk Data is being obtained with the DBLOCATE FMS statement. NASTRAN LOCBULK=2 is specified when no bulk entries, except for PARAM entries, are to be deleted or added. All PARAM entries must be respecified. All other entries will be ignored and, if present, may increase CPU times in XSORT and IFP. With LOCBULK=2, the XSORT and IFP modules will not reprocess the Bulk Data Section stored in the SEMAP run. Also, GP1, TASNP2, SEP1 and SEP1X modules will be skipped. 0 is the default, which assumes the RESTART FMS statement.
144	RESTART FMS statement existence flag. Set to 1 if RESTART statement is present.
370 (QRMETH)	Selects the formulation used by the QUADR/TRIAR elements. 0: Selects the QUADR/TRIAR membrane formulation with active out of plane rotational DOF (default) 1: Selects the older, limited QUADR/TRIAR formulation. Complex stresses and strains are not computed 3: Converts QUADR/TRIAR into QUAD4/TRIA3 formulation 5: Converts QUAD4/TRIA3 into the newer QUADR/TRIAR formulation
400	Reserved

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401	Reserved
413	OP2 file format (32-bit or 64-bit). 0: let the software chooses the format (Default). 1: 32-bit format.
442	Controls if Solution Monitor files are generated for Simcenter 3D and FEMAP. It is only valid on the platforms supported by Simcenter 3D and FEMAP. 0: Do not generate (default) 1: Generate files and delete after solution completes -1: Generate files and keep after solution completes 2: reserved
525	Enables integer inputs to be 11 characters.
587	Determines the sectional properties for the axisymmetric elements CTRAX3, CQUADX4, CTRAX6, and CQUADX8. 0 (default): Stiffness, mass, and loads are based on a 2*PI section (beginning with NX Nastran 10). 1: Stiffness, mass, and loads are computed on a per radian basis (behavior prior to NX Nastran 10), except for heat transfer solutions with CTRAX3, CQUADX4, CTRAX6, and CQUADX8, which remain on a 2*PI basis.
592	Reserved. Used when sys442=2.
593	Reserved. Used when sys442=2.
674	Controls the export of decks. 0: The deck contains one SPC (or SPC1) companion card for every imposed displacement SPCD (default for all Nastran solutions except SOL 402). 1: The deck does not contain any companion card. The computation will then be more efficient. However, if you manually set the system cell to 1 on a deck that contains companion cards (for example, one that was generated in Simcenter 3D using SOL 401, but then later converted to SOL 402), any imposed displacements on a DOF that is fixed with SPC/SPC1 will be ignored (default for SOL 402).
675	Verbosity level for SOL 402. 0 (default) = minimum verbosity. 1 = Low verbosity. List the output of the pre-processing (Bacon) and post-processing (Fac) steps of SOL402 solver. This is done automatically in case of warning or errors during those steps.

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	2 = High verbosity. In addition to sys675=1, list the content of the model read during the pre-processing step. 3 = High verbosity. In addition to sys675=2, call the DMAP module TABPRT to print the content of all the output datablocks (except rotation matrices TRMBU and TRMBD). 4 = High verbosity. In addition to sys675=3, call the DMAP module TABPRT to print the content of all the output datablocks (very large output, not recommended).
683	Temporarily turns off printing USER INFORMATION MESSAGE 4114 (OUTPX2) during writing the datablock in the scratch OP2 file. Automatically turned on when needed.
699	Specifies the number of displacement curves that appear in a report. (Default = 3) The list of nodes that will be considered for displaying the curves is defined with the SETMC Case Control entry. If more than 2 curves are requested, only the curves for the minimum, maximum, and average values of all the selected nodes will be displayed.
701	Checks for the existence of NLCNTLG,RSTGEN to request restart information generation.
709	Checks for the existence of NLCNTLG,RSTFROM to request restart information generation.
710	Checks for the existence of NLCNTLG,EXEFORM to request restart information generation.

9. Structural damping is not supported for individual elements: the GE field in the supported bulk entries is not used. Instead, Rayleigh damping is supported and specified with PARAM ALPHA1 and ALPHA2.

10. Several limitations apply to beam elements, for example, beam offsets, tapered beams, shear center. Please see remarks in the CBAR, CBEAM, PBAR, PBARL, PBEAM, and PBEAML entries.

11. Depending on the PARAM,LGDISP value, the use of follower force/moment in FORCE1, FORCE2, MOMENT1, and MOMENT2 entries varies.

12. The following are some important remarks on how SOL402 handle nonlinearities (PARAM,LGDISP):

- When PARAM,LGDISP,1 is defined, you enable geometric nonlinear effects in your model.

  For example, if you define at one end of a beam a follower force perpendicular to the beam, when PARAM,LGDISP,1 is defined, the force will remain perpendicular to the beam as the beam deforms.

  When PARAM,LGDISP,-1 is defined, geometric nonlinear effects are disabled. For the beam example above, the force will not remain perpendicular to the beam as the beam deforms. Instead, it remains along the same initial direction relative to the undeformed beam.

- PARAM,LGDISP,1 is a prerequisite:

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- If you want to take into account contact with large sliding (DISP=2 formulation in the BCTPAR2 entry).
- When you perform a modal analysis (MODES, CYCMODES, FOURIER, or BUCKLING) and you want to take into account the static loading that you compute in a previous subcase of the type ANALYSIS=STATICS or ANALYSIS=DYNAMICS.

- The rigid elements RBAR, RBE2, and RBE3 support large rotations.
- Contact or glue conditions are fully nonlinear, so that large displacements/strains are taken into account in contact equations, and not only the undeformed shape.
- When PARAM,LGDISP,1 is defined, subcases with ANALYSIS=STATICS and subcases with ANALYSIS=DYNAMICS will use the same nonlinear capabilities.

13. Use PARAM, LGSTRN, 1 if your model can undergo large strains.

> **Note**
>
> Enabling large strains will modify the strain-stress {ϵ,σ} measures in your material law from {Green-Lagrange, Piola-Kirchoff 2} to {Natural, Kirchhoff}.

14. SOL 402 will only run on long-word machines.

Related Topics

Browse Simcenter Nastran documentation

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TIME

Sets the maximum elapsed and I/O time.

Format:

TIME[=]t1[,t2]]

Describers:

Describer	Meaning
t1	Maximum allowable execution time in elapsed minutes. (Real or Integer; 0 < t1 < 3.15E7 minutes; Default = 3.15E7 minutes). Values entered with an exponent are interpreted as Real and must include a decimal (for example, 1.E2).
t2	Maximum allowable I/O limit in minutes. (Real or Integer > 0; Default is infinity, which is machine dependent.)

Remarks:

1. The TIME statement is optional.
2. If t2 is specified then t1 must be specified.

Examples:

1. The following example designates a runtime of 8 hours:

```
TIME 480
```

2. The following example designates 90 seconds:

```
TIME 1.5
```

Related Topics

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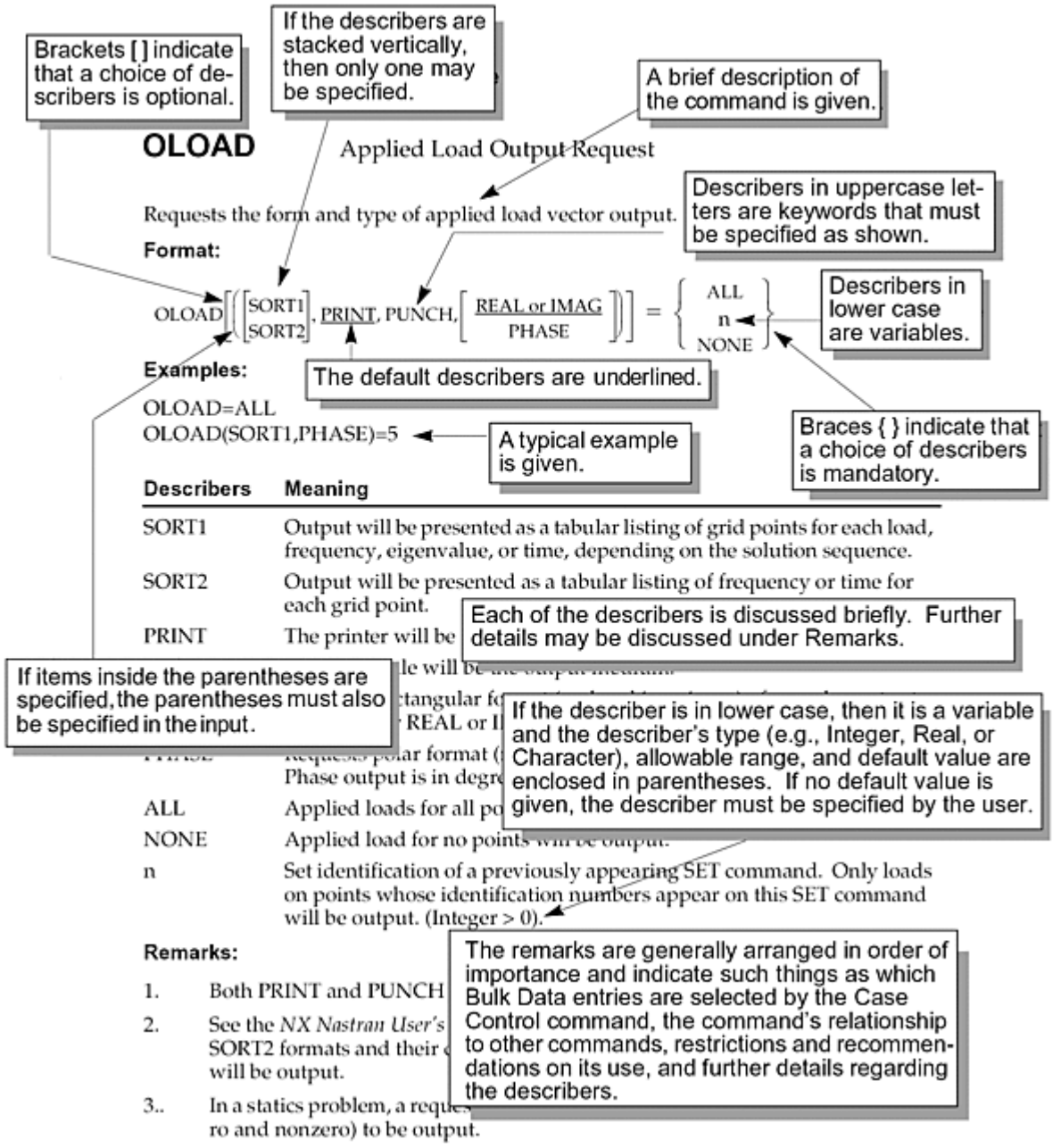
4. Case Control Commands

Key to Descriptions



Related Topics

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The Case Control Section

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The Case Control Section has several basic functions; specifically, it:

- Selects loads and constraints.
- Requests printing, plotting, and/or punching of input and output data (see “Plotting” in the *Simcenter Nastran User’s Guide*.)
- Defines the subcase structure for the analysis.

The applicability tables at the end of this section indicate the applicability of each command in all solution sequences.

Related Topics

Browse Simcenter Nastran documentation

Format of Case Control Command Descriptions

Case Control commands may be abbreviated down to the first four characters as long as the abbreviation is unique relative to all other commands. Each command is described as follows:

Description

A brief sentence about the function of the command is stated.

Format

Describers in uppercase are keywords that must be specified as shown. In addition, describers in lowercase indicate that the user must provide a value.

Braces { } indicate that a choice of describers is mandatory. If the describers are stacked vertically, then only one may be specified.

Brackets [ ] indicate that a choice of describers is optional. If the describers are stacked vertically, then only one may be specified.

Describers that are shaded indicate the defaults.

If the command line is longer than 72 columns, then it may be continued to the next line with a comma. For example:

```
SET 1=5, 6, 7, 8, 9,
        10 THRU 55
```


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Examples

One or more examples are given.

Describers

Each of the describers is briefly discussed. The describer's type (e.g., Integer, Real, or Character), allowable range, and default value are enclosed in parentheses. The describer must be specified by the user if no default value is given.

Remarks

The Remarks are generally arranged in order of importance and indicate such things as which Bulk Data entries are selected by the Case Control command; the command's relationship to other commands, restrictions and recommendations on its use; and further descriptions of the describers.

Related Topics

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Case Control Command Summary

This section contains a summary of all Case Control commands categorized as follows:

- Subcase Definition
- Data Selection
- Output Selection
- Superelement Control
- Multi-Body Dynamics Interface Control
- Miscellaneous

Subcase Definition

1. Output Request Delimiters

OUTPUT	Delimits the various types of commands for the structure plotter, curve plotter, grid point stress, and MSGSTRESS.
OUTPUT(PLOT)	Beginning of structure plotter output request.
OUTPUT(POST) or SETS DEFINITION	Beginning of grid point stress output requests.
OUTPUT(XYOUT) or OUTPUT(XYPLOT)	Beginning of curve plotter output request.
OUTPUT(CARDS)	Suppresses processing of Bulk Data entries (MSGMESH).

2. Subcase Delimiters

REPCASE	Delimits and identifies a repeated output request subcase.
SUBCASE	Delimits and identifies a subcase.
SUBCOM	Delimits and identifies a combination subcase.
SYM	Delimits and identifies a symmetry subcase.
SYMCOM	Delimits and identifies a symmetry combination subcase.

3. Subcase Control

MASTER	Allows the redefinition of a MASTER subcase.
MODES	Repeats a subcase.

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SUBSEQ	Gives the coefficients for forming a linear combination of the previous subcases.
SYMSEQ	Gives the coefficients for combining the symmetry subcases into the total structure.

Back to top

Data Selection

1. Static Load Selection

BOLTLD	Selects a BOLTLD or BOLTFRC bulk entry for bolt preload processing.
CLOAD	Requests a CLOAD Bulk Data entry that defines a list of superelement loads and their scale factors in nonlinear static analysis only.
DEFORM	Selects the element deformation set.
EFLOAD	Converts surface loads from an external field solution to structural loads.
LOAD	Selects an external static loading set.

2. Dynamic Load Selection

ALOAD	Selects acoustic load set in a frequency response problem.
DLOAD	Selects a dynamic load or an acoustic source to be applied in a transient or frequency response problem.
LOADSET	Selects a sequence of static load sets to be applied to the structural model. The load sets may be referenced by dynamic load commands.
MLOAD	Selects the misalignment load in SOL 414,111 and SOL 414,129 rotor dynamic analysis.
NONLINEAR	Selects nonlinear dynamic load set for transient problems.
ULOAD	Selects the unbalance load for SOL 414 rotor dynamic analysis.

3. Constraint Selection

BC	Identifies multiple boundary conditions for normal modes, buckling, and flutter analysis in SOLs 103, 105, 145, and 200.
BCSET	Selects the contact set.
BGSET	Selects the glue contact set.
CYCSET	Selects a cyclic symmetric boundary coupling in SOLs 401, 402, and 414.
DSYM	Provides for either one or two planes of overall symmetry in DIH-type cyclic symmetry problems.

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FMH3D	Selects the coupling of a Fourier multi-harmonic region with a 3D structure in SOL 414.
MPC	Selects a multipoint constraint set.
RIGID	Selects the rigid element processing method for RBAR, RBE1, RBE2, RBE3, RROD, and RTRPLT elements.
SPC	Selects a single-point constraint set to be applied.
STATSUB	Selects the static solution to use in forming the differential stiffness for buckling analysis, normal modes, complex eigenvalue, frequency response and transient response analysis.
SUPORT1	Selects the fictitious support set (SUPORT1 entries only) to be applied to the model.

4. Fluid-Structure Interaction

FLSFSEL	Selects frequency content for controlling fluid-structure interaction.
FLSTCNT	Specifies miscellaneous control parameters for fluid-structure interaction.
MFLUID	Selects the MFLUID Bulk Data entries to be used to specify the fluid-structure interface.

5. Thermal Field Selection

TEMPERATURE	Selects the temperature set to be used in either material property calculations or thermal loading in heat transfer and structural analysis.
TEMPERATURE(INITIAL)	Selects initial temperature distribution for temperature-dependent material properties and heat transfer problems.
TEMPERATURE(LOAD)	Selects temperature set for static thermal load.
TEMPERATURE(MATERIAL)	Selects temperature set for temperature-dependent material properties.
TSTRU	Defines a temperature set ID for a structural run based on a heat transfer subcase.

6. Static Solution Conditions

SMETHOD	Selects iterative solver parameters.

7. Dynamic Solution Conditions

ACTEMP	Selects spatially-varying intensive properties for the fluid in standard FEM and FEMAO acoustic analysis.
CMETHOD	Selects complex eigenvalue extraction parameters.
DAMPING	Selects the damping definition and, for transient response analysis or a modal-based analysis that writes an interface file for external

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	multibody dynamics or control system software, the structural damping to viscous damping conversion method.
FREQUENCY	Selects the set of forcing frequencies to be solved in frequency response problems.
IC	Selects the initial conditions for direct transient analysis (SOLs 109, 129, and 159).
METHOD	Selects the real eigenvalue extraction parameters.
MODSEL	Selects mode numbers to include in a modal dynamic response solution.
NSM	Selects non-structural mass (NSM) set for mass generation.
OMEGA	Selects the set of rotational speeds to be used in a SOL 414,110 complex modal analysis
RANDOM	Selects the RANDPS, RANDPEX, and RANDT1 Bulk Data entries to be used in random analysis.
RESVEC	Controls the computation of residual vectors.
RMETHOD	Selects the ROTORD bulk data entry used for rotor dynamics analysis.
RSAPPLY	Selects the data for a response spectrum application run.
RSMETHOD	Selects the real eigenvalue extraction parameters in the component mode reduction of the residual structure for direct and modal solutions.
SDAMPING	Requests modal damping as a function of natural frequency in modal solutions or viscoelastic materials as a function of frequency in direct frequency response analysis.
SMETHOD	Selects iterative solver override options in frequency response analysis.
TSTEP	Selects integration and output time steps for linear or nonlinear transient analysis.

8. Direct Input Matrix Selection

A2GG	Selects direct input acoustic/fluid-structure coupling matrices.
B2GG	Selects direct input damping matrices.
B2PP	Selects direct input damping matrices.
FRFIN	Selects direct input, frequency-dependent dynamic stiffness or dynamic flexibility matrix for frequency response analysis.
K2GG	Selects direct input stiffness matrices.
K2PP	Selects direct input stiffness matrices, which are not included in normal modes.
M2GG	Selects direct input mass matrices.

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M2PP	Selects direct input mass matrices, which are not included in normal modes.
TFL	Selects the transfer function set(s) to be added to the direct input matrices.
P2G	Selects direct input load matrices.

9. Nonlinear Analysis

BVTSET	Selects the virtual thread set for SOL 402.
ELAR	Selects the element add/removal set.
FLXSLI	Selects an external flexible sliders set for SOL 402.
IMPERF	Selects grid point imperfections for SOL 401.
INITS	Selects initial stress or strain set or offset strain set for SOL 401
JCONSET	Selects an external kinematic joint constraint set for SOL 402.
NLARCL	Selects nonlinear buckling solution in SOL 401
NLPARM	Selects the parameters used for nonlinear static analysis.
SMETHOD	Selects iterative solver override parameters in nonlinear static analysis.
TSTEP	Selects integration and output time steps for linear or nonlinear transient analysis.
TSTEPNL	Selects transient time step set for nonlinear analysis.

10. Aeroelastic Analysis

AECONFIG	Assigns the aerodynamic configuration parameter used to locate the associated data blocks on the aerodynamic and aeroelastic databases.
AESYMXY	Aerodynamic XY plane of symmetry flag.
AESYMXZ	Aerodynamic XZ plane of symmetry flag.
AEUXREF	Defines the Reference Aerodynamic Extra Point (Controller) Vector.
CSSCHD	Aerodynamic Control Surface Schedule.
DIVERG	Selects the divergence parameters in a static aeroelastic divergence problem.
FMETHOD	Selects the parameters to be used in an aerodynamic flutter analysis.
GUST	Selects the gust field in an aerodynamic response problem.
TRIM	Selects trim variable constraints in static aeroelastic response.

11. Design Optimization and Sensitivity (SOL 200)

ANALYSIS	Specifies the type of analysis being performed for the current subcase.
AUXCASE	Delimits Case Control commands for an auxiliary model in SOL 200.

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AUXMODEL	References an auxiliary model for generation of boundary shapes in shape optimization.
BEGIN BULK	Designates the end of the Case Control Section and/or the beginning of a Bulk Data Section.
DESGLB	Selects the design constraints to be applied at the global level in a design optimization task.
DESOBJ	Selects the DRESP1, DRESP2, or DRESP3 entry to be used as the design objective.
DESSUB	Selects the design constraints to be used in a design optimization task for the current subcase.
DSAPRT	Specifies design sensitivity output parameters.
DRSPAN	Assigns a set of DRESP1 responses to a specific subcase.
MODTRAK	Selects mode tracking options in design optimization (SOL 200).

Back to top

Output Selection

1. Output Control

ACORDCHK	Outputs the maximum frequency and element order information for solutions with the FEMAO method and terminates the run.
ECHO	Controls echo (printout) of the Bulk Data.
ESHLPNL	Requests output for tension-only shell elements in SOL 401.
FLSPOUT	Controls modal participation output for the fluid-structure interaction.
HARCOEFF	Selects a **SET** case control command that contains a list of coefficients (real numbers) that are factors of the reference harmonics.
HARTAB	Selects the number of harmonics in a SOL 414, 103 eigenvalues analysis.
LABEL	Defines a character string that will appear on the third heading line of each page of the printer output.
LINE	Defines the maximum number of output lines per printed page.
MAXLINES	Sets the maximum number of output lines.
OUTMGT	Selects XY plot requests in a SOL 402, SOL 414,103 with a transient response subcase, SOL 414,111, or SOL 414,129 analysis.
PAGE	Causes a page eject in the echo of the Case Control Section.
PLOTID	Defines a character string that will appear on the first frame of any plotter output.

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SEKETAB	Specifies elements for which energy balances are output to the .f06 file in SOL 414,110 and SOL 414,111.
SKIP	Activates or deactivates the execution of subsequent commands in the Case Control (including plot commands).
SUBTITLE	Defines a subtitle that will appear on the second heading line of each page of the printer output.
TITLE	Defines a character string that will appear on the first heading line of each page of the Simcenter Nastran printer output.

2. Set Definition

MAXMIN	Specifies options for max/min surveys of certain output data associated with grid points.
OFREQUENCY	Selects a set of frequencies for output requests.
OMODES	Selects a set of modes for output requests.
OTIME	Selects a set of times for output requests.
PARTN	Specifies a list of grid point identification numbers that will be partitioned with the DMAP module MATMOD (Option 17). In SOLs 111 and 200, the PARTN command specifies the points at which modal participation factors are to be computed.
SET	Defines a set of element or grid point numbers to be plotted.
SETMC	Defines sets for modal contribution results.
SURFACE	Defines a surface for the calculation of grid point stresses, strains, or mesh stress discontinuities.
VOLUME	Defines a volume for the calculation of grid point stresses, strains, or mesh stress discontinuities.

3. Physical Set Output Requests

ACCELERATION	Requests the form and type of acceleration vector output.
ACPOWER	Requests acoustic power output.
ACPOTENTIAL	Acoustic potential results output request for AML regions or acoustic free faces.
ACVPO4STR	Acoustic volume quadrupole strength results output request for fluid solid elements.
BCRESULTS	Requests contact result output.
BGRESULTS	Requests glue result output.
BOUTPUT	Selects contact regions for output.
CKGAP	Requests gap output for chocking elements. Valid for SOL 401 only.

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CZRESULTS	Requests results output for cohesive elements. Valid for SOL 401 only.
DISPLACEMENT	Requests the form and type of displacement output.
DMTRCOEF	Requests the duct modes transmission coefficients.
DMTRLOSS	Requests the duct modes transmission loss output.
EDE	Requests the output of the energy loss per cycle in selected elements.
EKE	Requests the output of the kinetic energy in selected elements.
ELSDCON	Requests mesh stress discontinuities based on element stresses (see STRESS).
ENTHALPY	Requests form of enthalpy vector output in transient heat transfer analysis for SOL 159.
ERP	Requests the output of equivalent radiated power for selected panels and the elements comprising the selected panels.
ESE	Requests the output of the strain energy in selected elements.
FORCE	Requests the form and type of element force output or particle velocity output in coupled fluid-structural analysis.
FLUX	Requests the form and type of gradient and flux output in heat transfer analysis.
FLXRESULTS	Requests flexible slider joint output for SOL 402.
GKRESULTS	Requests the form and type of gasket result output in SOL 402.
GPFORCE	Requests grid point force balance at selected grid points.
GPKE	Requests the output of the kinetic energy at selected grid points in normal modes analysis only.
GPRSORT	Requests that the output of composites ply results on elements referencing a PCOMPG be sorted by global ply ID, then by element ID.
GPSDCON	Requests mesh stress discontinuities based on grid point stresses (see GPSTRESS).
GPSTRAIN	Requests grid points strains for printing only.
GPSTRESS	Requests grid point stresses for printing only.
GRDCON	Requests acoustic contribution results for the grids of structural panels and for the residual.
HDOT	Requests form of rate of change of enthalpy vector output in transient heat transfer analysis for SOL 159.
INPOWER	Requests incident acoustic power output.
JINTEG	Request computation of the j-integral.

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JRESULTS	Requests output for kinematic joints in SOL 402.
MEFFMASS	Requests the output of the modal effective mass, participation factors, and modal effective mass fractions in normal modes analysis.
MODALE	Requests the type of energy to output for SOL 111.
MODCON	Requests modal contribution results for the residual.
MONITOR	Requests integrated load monitor point results be printed in the .f06 file.
MPCFORCES	Requests the form and type of multipoint force of constraint vector output.
NLSTRESS	Requests the form and type of nonlinear element stress output in SOL 106.
NOUTPUT	Requests physical output in cyclic symmetry problems.
OLOAD	Requests the form and type of applied load vector output.
OPRESS	Requests the output of pressure from SOL 401 and 402, and the Simcenter 3D Multiphysics environment.
OTMFORC	Requests output of force and moment results from FRFOTM bulk entries
PANCON	Requests acoustic panel contribution results.
PEAKOUT	Requests response peak output for SOL 108 and 111 frequency response analysis.
PFRESULTS	Requests progressive failure results output for composite solid elements. Valid for SOL 401 only.
PRESSURE	Requests the form and type of pressure output.
RCROSS	Requests computation and output of cross-power spectral density and cross-correlation functions in random analysis.
RMAXMIN	Defines parameters to output the minimum, maximum, absolute value maximum, average, and RMS value of stress, force, and displacement results for SOLs 101, 109, and 112.
SHELLTHK	Requests the form of shell thickness output for SOL 402.
SPCFORCES	Requests the form and type of single-point force of constraint vector output.
STRAIN	Requests the form and type of strain output.
STRESS	Requests the form and type of element stress output.
STRFIELD	Requests the computation of grid point stresses for graphical postprocessing and mesh stress discontinuities.
SVECTOR	Requests the form and type of solution set eigenvector output.

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THERMAL	Requests the form and type of temperature output.
TRLOSS	Requests acoustic power transmission loss output.
TRPOWER	Requests transmitted acoustic power output.
VELOCITY	Requests the form and type of velocity vector output.

4. Solution Set Output Requests

AEROF	Requests the aerodynamic loads on aerodynamic control points.
APRESSURE	Requests the aerodynamic pressures in static aeroelastic response.
FHAC	Requests the solution constrained Fourier multi-harmonics in SOL 414.
FHAR	Requests the solution Fourier multi-harmonics in SOL 414.
HARMONICS	Controls the number of harmonics output in axisymmetric shell or axisymmetric fluid problems; controls the number of harmonics to be used for analysis in cyclic symmetry problems.
HOUTPUT	Requests harmonic output in cyclic symmetry problems.
MPRES	Requests the pressure for selected surface elements in fluid-structure interaction problems.
NLLOAD	Requests the form and type of nonlinear load output for transient problems.
SACCELERATION	Requests the form and type of solution set acceleration output.
SDISPLACEMENT	Requests the form and type of solution set displacement output.
SVELOCITY	Requests the form and type of solution set velocity output.

5. Model Checkout

ADAPTERR	Requests stress norm and strain energy norm error estimates.
ELSUM	Requests a summary of element properties for output.
GROUNDCHECK	Performs grounding check analysis on stiffness matrix to expose unintentional constraints by moving the model rigidly.
WEIGHTCHECK	At each stage of the mass matrix reduction, computes rigid body mass and compares it with the rigid body mass of the g-set.

Back to top

Superelement Control

EXTSEOUT	Specifies creation of an external superelement.
FRFOUT	Specifies creation of a frequency-dependent external superelement for frequency response analysis.

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SEALL	Specifies the superelement identification numbers of Phase 1 processing in which all matrices and loads are generated and assembled. Controls execution of the solution sequence.
SEDR	Specifies the superelement identification numbers for which data recovery will be performed.
SEDV	Specifies the superelement identification numbers for which the design variables will be processed.
SEEXCLUDE	Specifies the superelement identification numbers for which all matrices and loads will not be assembled into the downstream superelement.
SEFINAL	Specifies the superelement identification number of the final superelement to be assembled.
SEKREDUCE	Specifies the superelement identification numbers for which stiffness matrices are assembled and reduced.
SELGENERATE	Specifies the superelement identification numbers for which static loads will be generated.
SELREDUCE	Specifies the superelement identification numbers for which the static load matrices will be assembled and reduced.
SEMGENERATE	Specifies the superelement identification numbers for which stiffness, mass, and damping matrices will be generated.
SEMREDUCE	Specifies the superelement identification numbers for which the mass and damping matrices will be assembled and reduced. In buckling analysis, the differential stiffness matrices are assembled and reduced.
SERESP	Specifies the superelement identification numbers for which the design sensitivity matrices will be generated.
SUPER	Assigns a subcase(s) to a superelement or set of superelements.

Back to top

Multi-Body Dynamics Interface Control

ADAMSMNF	Generates Adams interface Modal Neutral File (MNF) during a SOL 103, 111, or 112 run.
ADMRECVR	Recovers stress results from an Adams flexbody analysis.
MBDEXPORT	Generates interface file for multi-body dynamics and control system software during a SOL 103, 111, or 112 run.
MBDRECVR	Recovers stress results from multi-body dynamics analysis.

Back to top

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Miscellaneous

$	Used to insert comments into the input file. Comment statements may appear anywhere within the input file.
ATVOUT	Specifies requirements for the creation of an acoustic transfer vector (ATV).
INCLUDE	Inserts an external file into the input file. The INCLUDE statement may appear anywhere within the input data file.
OUTPUT	Delimits the various types of commands for the structure plotter, curve plotter, grid point stress, and MSGSTRESS.
PARAM	Specifies values for parameters.
VATVOUT	Defines requirements for the creation of a vibro-acoustic transfer vector (VATV).

Back to top

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Related Topics

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Case Control Command Descriptions

$

Comment

Used to insert comments into the input file. Comment statements may appear anywhere within the input file.

Format:

$ followed by any characters out to column 80.

Examples:

```
$ TEST FIXTURE-THIRD MODE
```

Remarks:

1. Comments are ignored by the program.
2. Comments will appear only in the unsorted echo of the Bulk Data.

Related Topics

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A2GG

Direct Input Acoustic/Fluid-Structure Matrix Selection

Selects direct input acoustic/fluid-structure coupling matrix or matrices.

Format:

$$\text{A2GG} = \{\text{name}\}$$

Examples:

```
A2GG=ADMIG
A2GG=ADMIG1, ADMIG2, ADMIG3
A2GG=1.25*ADMIG1, 1.0*ADMIG2, 0.75*ADMIG3
SET 100=A1, A2
A2GG=100
```


```
A2GG=ADMIG
A2GG=ADMIG1, ADMIG2, ADMIG3
A2GG=1.25*ADMIG1, 1.0*ADMIG2, 0.75*ADMIG3
SET 100=A1, A2
A2GG=100
```


Describers:

Describer	Meaning
name	Name of direct input acoustic/fluid-structure coupling matrix $[A^2_{gg}]$ defined using DMIG bulk entries. (Character) Scale factors may be included (see Remarks 4 and 5).

Remarks:

1. DMIG matrices are not used unless selected using A2GG.
2. If PARAM,ASCOUP,YES, the direct input acoustic/fluid-structure coupling matrix selected using A2GG is added to the computed acoustic/fluid-structure coupling matrix. If PARAM,ASCOUP,NO, the direct input acoustic/fluid-structure coupling matrix selected using A2GG replaces the computed acoustic/fluid-structure coupling matrix.
3. The matrix must be square and field 4 of the DMIG bulk entry must contain the integer 1. When filling out the DMIG bulk entries, the GJ column index corresponds to fluid points, CJ is zero, the Gi row index corresponds to structural points, Ci corresponds to DOF, and Ai are the area values.

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4. The associated DMIG matrices can be scaled using either in-line scale factors entered on A2GG (for example, A2GG=1.25*ADMIG1), or the parameter CA2 (for example, PARAM,CA2,1.25), or both. For information regarding the CA2 parameter, see **"Parameter Descriptions"**.

5. Multiple matrices separated by a comma or a blank are additive. When multiple matrices and in-line scale factors are used together, each matrix name in the list must include a scale factor. 1.0 should be entered for matrices in the list that are not scaled. For example, if A2GG=1.25*ADMIG1,1.0*ADMIG2,0.75*ADMIG3 is specified, the result is A2GG=1.25*ADMIG1 + ADMIG2 + 0.75*ADMIG3.

   Specifying the CA2 parameter scales all the A2GG. For example, if both PARAM,CA2,1.30 and A2GG=1.25*ADMIG1,1.0*ADMIG2,0.75*ADMIG3 are specified, the result is A2GG=1.30(1.25*ADMIG1 + ADMIG2 + 0.75*ADMIG3).

6. A2GG is supported in dynamic solutions with acoustic/fluid-structure coupling.

7. Only one A2GG case control command should be used in an input file and it should appear above any subcases.

Related Topics

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ACCELERATION

Acceleration Output Request

Requests form and type of acceleration vector output.

Format:

$$\text{ACCELERATION}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\left[\begin{matrix}\underline{\text{ABS}}\\ \text{REL}\end{matrix}\right],\right.\right.$$

$$\left.\left.\left[\begin{matrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{matrix}\right],\left[\begin{matrix}\underline{\text{RPRINT}}\\ \text{NORPRINT}\end{matrix}\right],[\text{RPUNCH}],\text{SOLUTION}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{setf}\\ \text{PEAKOUT}\end{matrix}\right\}\right)\right]$$

$$=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
ACCELERATION=5
ACCELERATION(SORT2, PHASE)=ALL
ACCELERATION(SORT1, PRINT, PUNCH, PHASE)=17
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence. (Default)
SORT2	Output a tabular listing of frequency or time for each grid point.
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.

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Describer	Meaning
PLOT	Generate, but do not print, acceleration results.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
ABS	For enforced motion dynamic analysis, acceleration results are output as absolute acceleration. (Default)
REL	For enforced motion dynamic analysis, acceleration results are output relative to the enforced motion input.
PSDF	Requests power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7.
ATOC	Requests autocorrelation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7.
CRMS	Requests cumulative root mean square function be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7.
RMS	Requests root mean square and zero crossing functions be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7. (Default)
RALL	Requests all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7.
RPRINT	Writes random analysis results to the print file. See Remark 7. (Default)
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 7.
RPUNCH	Writes random analysis results to the punch file. See Remark 7.
SOLUTION = ALL	Output acceleration for all frequencies. See Remark 12. (Default)
SOLUTION = setf	Output acceleration only for the frequencies listed on a previously appearing SET command with setf as the identification number. See Remark 12. (Integer > 0)
SOLUTION = PEAKOUT	Output acceleration only for frequencies identified with the PEAKOUT command. See Remark 12.
ALL	Output acceleration for all points.
n	Output acceleration only for the points listed on a previously appearing SET command with n as the identification number. (Integer > 0)
NONE	Do not output acceleration for any points. (Default)

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Remarks:

1. Both PRINT and PUNCH can be requested.
2. Acceleration output is only available for transient and frequency response problems.
3. The defaults for SORT1 and SORT2 depend on the analysis type:
   - SORT1 is the default in static analysis, frequency response, steady state heat transfer analysis, real and complex eigenvalue analysis, flutter analysis, and buckling analysis. If SORT2 is selected in a frequency response solution for one or more of the commands ACCE, DISP, FORC, GPFO, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands are also output in SORT2 format.
   - SORT2 is the default in transient response analysis (structural and heat transfer). SORT2 is not available for real eigenvalue (including buckling), complex eigenvalue, or flutter analysis. If SORT1 is selected in a transient solution for one or more of the commands ACCE, DISP, ENTH, FORC, GPFO, HDOT, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands are also output in SORT1 format.
   - XY plot requests force SORT2 format, thus overriding SORT1 format requests.
4. ACCELERATION = NONE allows overriding an overall output request.
5. The PLOT option is used when curve plots are desired in the magnitude/phase representation and no printer request is present for magnitude/phase representation.
6. Acceleration results are printed and/or punched in the global coordinate system (see field CD on the GRID bulk data entry). The coordinate system for plotted acceleration output depends on the PARAM,POST setting. See the parameter **POST**.
7. The following applies to random solutions:
   - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, PUNCH, PLOT describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
   - The SORT1 and SORT2 describers only control the output format for the frequency response output. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
   - Any combination of the PSDF, ATOC, RMS, and CRMS describers can be selected. The RALL describer selects all four.
   - Autocorrelation (ATOC) calculations require the RANDT1 bulk entry.
8. When doing enforced motion dynamic analysis and relative output is requested (using the REL describer), the output is relative to the input as described by the equation:

$$u_f = y_f - K_{ff}^{-1} K_{fs} u_s$$

where u_f is absolute displacement, y_f is relative displacement, and u_s is enforced motion.

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9. During a SOL 108 or 111 frequency-dependent external superelement creation run, on the SET case control command that is referenced by the ACCELERATION case control command, include any points at which you want to recover results during the system run.
10. The SOLUTION describer is only valid in a SOL 108 or 111 frequency response analysis.
11. When an OFREQ case control command is present, the SOLUTION describer is used as follows:
    - If SOLUTION=ALL, output results for all frequencies specified by the OFREQ command.
    - If SOLUTION=setf, output results for all frequencies specified by the OFREQ command that are listed on a previously appearing SET command with setf as the identification number.
    - If SOLUTION=PEAKOUT and a PEAKOUT case control command is also present in the input file, output results for all frequencies specified by the OFREQ command that satisfy the PEAKOUT filtering criteria.
12. The SOLUTION describer is not valid for superelements or frequency-dependent components.

Remarks related to SOL 402:

1. Output is restricted to REAL and SORT1 format. SORT2, IMAG, PHASE, ABS, REL, PSDF, ATOC, CRMS, RMS, RALL, NOPRINT and RPUNCH are ignored.

Related Topics

Browse Simcenter Nastran documentation

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ACCONV

Acoustic Convected Flow Property Selection

Selects the acoustic convected flow properties to be used in a SOL 108 FEMAO Acoustic analysis.

Format:

$$\text{ACCONV} = \{\text{n}\}$$

Examples:

```
ACCONV=5
```

Describers:

Describer	Meaning
n	Acoustic convected flow property identification number defined on the ACCONVD and ACCONV bulk entries. (Integer>0)

Remarks:

1. Only a single ACCONV case control command can be defined, and it must be specified above the subcase level (global).
2. ACCONV should not be specified in an acoustic analysis with incident sources.

Related Topics

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ACINTENSITY

Acoustic Intensity Output

Acoustic intensity output request at fluid grids referenced by microphone elements.

Format:

$$\text{ACINTENSITY} \left[\left(\left[\frac{\text{SORT1}}{\text{SORT2}}\right], \left[\frac{\text{PRINT}}{\text{NOPRINT}}\right], \text{PUNCH}, \left[\frac{\text{REAL or IMAG}}{\text{PHASE}}\right]\right)\right] = \left\{\begin{matrix} \text{n} \\ \text{ALL} \\ \text{NONE} \end{matrix}\right\}$$

Examples:

```
SET 123=5,10,15,25
```

```
ACINTENSITY=123
```

```
SET 5=6,8,11
```

```
ACINTENSITY(PRINT,PUNCH,PHASE)=5
```

Describers:

Describer	Meaning
SORT1	Present output as a tabular listing of grids for each frequency. (Default)
SORT2	Present output as a tabular listing of frequency for each grid.
PRINT	The print file (.f06) will be the output medium. (Default)
NOPRINT	Generates, but does not print intensity results.
PUNCH	The standard punch file (.pch) will be the output medium.

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Describer	Meaning
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Default)
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
n	ID of SET case control command containing the list of identification numbers of fluid grids referenced by microphone elements at which to calculate acoustic intensity results. (Integer > 0)
ALL	Requests acoustic intensity results for all microphone elements in the model.
NONE	Turns this output request off; useful for controlling output requests across subcases.

Remarks:

1. The PRESSURE case control command settings determine whether the acoustic intensity results are based on total or scattered pressure.
2. Acoustic intensity results will be written to the op2 file, if the value of PARAM, POST is -1 or -2.
3. When the fluid domain is represented by an acoustic transfer vector (ATV), acoustic particle velocity and acoustic intensity results cannot be computed, and the ACINTENSITY and ACVELOCITY case control commands are ignored.

Related Topics

Browse Simcenter Nastran documentation

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ACORDCHK

FEMAO Acoustics and Vibro-Acoustic Check Option

Outputs the maximum frequency and element order information for solutions with the FEMAO method and terminates the run.

Format:

$$\text{ACORDCHK} = \text{STOP}$$

Examples:

```
ACORDCHK=STOP
```

Remarks:

1. The ACORDCHK = STOP case control command is only supported by Sol 108 FEMAO and Sol 111 FEMAO solutions. These solutions may contain vibro-acoustic transfer vectors (VATV).
2. If ACORDCHK = STOP exists, the software writes the maximum frequency and element order information (OEFMXORD data block) to the OP2 file and terminates the run before the frequency response computation.
3. If ACORDCHK = STOP does not exist, the software still writes the maximum frequency and element order information to the OP2 file and proceeds with the frequency response computation.
4. If ACORDCHK = STOP exists in a standard FEM solution, the software ignores the case control command and does not write the maximum frequency or element order information to the OP2 file.
5. When both ACORDCHK = STOP and the SKINOUT = STOP describer setting is specified on the FLSTCNT case control command, the software writes the following to the OP2 file before terminating the run:
   - Maximum frequency
   - Element order information
   - Coupling quality information

Related Topics

Browse Simcenter Nastran documentation

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ACPOTENTIAL

Acoustic Potential Results Output

Acoustic potential results output request for AML regions or specific AML faces.

Format:

$$\text{ACPOTENTIAL}\left[\left(\text{AMLREG}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{NONE}\\ \text{na}\end{matrix}\right\},\text{FACES}=\text{nf}\right)\right]=\left\{\begin{matrix}\underline{\text{YES}}\\ \text{NO}\end{matrix}\right\}$$

Examples:

```
SET 101 = 2
```

```
ACPOTENTIAL (AMLREG = NONE, FACES = 101)
```

```
ACPOTENTIAL = NO
```

Describers:

Describer	Meaning
AMLREG = ALL	Calculate acoustic potential results for all AMLREG bulk entries. (Default)
AMLREG = NONE	Do not calculate acoustic potential results for any AMLREG bulk entries.
AMLREG = na	na (integer>0) is the identification number of the SET case control command that lists the identification numbers of AMLREG bulk entries at which to calculate acoustic potential results. See Remark.
FACES =nf	nf (integer>0) is the identification number of the SET case control command that lists the identification numbers of BSURFS bulk entries that contain the acoustic AML faces at which to calculate acoustic potential results. See Remark.
YES	Honor describer settings, calculate, and output acoustic potential results. (Default)
NO	Ignore describer settings, and do not calculate or output acoustic potential results.

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Remark:

The software processes either AML regions or acoustic AML faces referenced in BSURFS bulk entries, but not both. Thus, FACES = nf is ignored when AMLREG = ALL or AMLREG = na is specified. For the software, to take FACES = nf into account, AMLREG must be NONE.

Related Topics

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ACPOWER

Acoustic Power Results Output

Acoustic power results output request for AML regions or 2D microphone elements (SORT2 results only).

Format:

$$\text{ACPOWER}\left[\left(\text{AMLREG}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{NONE}\\ \text{na}\end{matrix}\right\},\ \text{GROUP}=\text{ng},\ \left[\begin{matrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{matrix}\right],\ \text{PUNCH},\right.\right.$$
$$\left.\left.\left[\begin{matrix}\underline{\text{REAL or IMAG}}\\ \text{PHASE}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{RPRINT}}\\ \text{NORPRINT}\end{matrix}\right],\ \text{RPUNCH}\right)\right]=\left\{\begin{matrix}\underline{\text{YES}}\\ \text{NO}\end{matrix}\right\}$$

Examples:

```
SET 123=3,6,9
```

```
ACPOWER(AMLREG=123)
```

```
SET 5=5,10
```

```
ACPOWER(GROUP=5,PRINT,PUNCH,PHASE)
```

```
ACPOWER=NO
```

```
ACPOWER
```

Describers:

Describer	Meaning
AMLREG = ALL	Calculate acoustic power results for all AMLREG bulk entries. (Default)
AMLREG = NONE	Do not calculate acoustic power results for any AMLREG bulk entries.

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Describer	Meaning
AMLREG =na	na (integer>0) is the identification number of the SET case control command that lists the identification numbers of AMLREG bulk entries at which to calculate acoustic power results. See Remark 3.
GROUP =ng	ng (integer>0) is the identification number of the SET case control command that lists the identification numbers of GROUP bulk entries that contain the 2D microphone elements at which to calculate acoustic power results. See Remark 3.
PRINT	Write the acoustic power results to the print (.f06) file. (Default)
NOPRINT	Do not print the acoustic power results.
PUNCH	Write the acoustic power results to the standard punch (.pch) file.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Default)
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
RPRINT	Write the random analysis results to the print (.f06) file. See Remark 5. (Default)
NORPRINT	Do not write the random analysis results to the print file. See Remark 5.
RPUNCH	Write the random analysis results to the punch (.pch) file. See Remark 5.
YES	Honor describer settings and calculate and output acoustic power results. (Default)
NO	Ignore describer settings and do not calculate or output acoustic power results. See Remark ACPOWER.

Remarks:

1. The PRESSURE case control command settings determine whether the acoustic power results are total or scattered.
2. Both PRINT and PUNCH may be requested.
3. Only AML regions or 2D microphone elements referenced in GROUP bulk entries are processed for acoustic power results. Either or both of the AMLREG and GROUP describers must be specified. Microphone elements must reference a PMIC property.
4. Acoustic power results are written to the op2 file, if PARAM, POST,-1 or PARAM,POST,-2.
5. The following applies to random solutions:
   - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, NOPRINT, PUNCH describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
   - Only SORT2 output format is available for the frequency response and random results output.
   - Only the power spectral density function (PSDF) is calculated. For RANDPS (or RANDPEX) bulk entries that refer to *n* subcases, when $i = j$ the results in W/Hz are computed as follows:

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$$ACPOWER_{psdf}\left(f\right) = \sum_{i=1}^{n} RANDPS_{ii}\left(f\right) \cdot ACPOWER_{i}\left(f\right)$$

If $i \neq j$, then a warning message is issued and the RANDPS bulk entries are ignored.

For more information, see the **RANDPS** bulk entry.

- Use ACPOWER=NO within a subcase to override a ACPOWER specification above the subcase level.

Related Topics

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ACTEMP

Acoustic Fluid Property Selection

Selects spatially varying fluid property data for acoustic analysis with standard FEM and FEMAO.

Format:

$$\mathrm{ACTEMP} = \mathrm{n}$$

Examples:

```
ACTEMP=5
```

Describers:

Describer	Meaning
n	Set identification number of ACTEMP bulk entry. (Integer > 0)

Remarks:

1. A single ACTEMP case control command can be placed above the subcases. If an ACTEMP case control command is placed in a subcase, a fatal error occurs.
2. An ACTEMP case control command can reference a single ACTEMP bulk entry only.

Related Topics

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ACVELOCITY

Acoustic Velocity Output

Acoustic velocity output request at microphone points.

Format:

$$\text{ACVELOCITY}\left[\left(\left[\begin{matrix}\text{SORT1}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\text{PRINT}\\ \text{NOPRINT}\end{matrix}\right],\text{PUNCH},\left[\begin{matrix}\text{REAL or IMAG}\\ \text{PHASE}\end{matrix}\right]\right)\right]$$
$$=\left\{\begin{matrix}n\\ \text{ALL}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
SET 123=5,10,15,25
```

```
ACVELOCITY=123
```

```
SET 5=6,8,11
```

```
ACVELOCITY(PRINT,PUNCH,PHASE)=5
```

Describers:

Describer	Meaning
SORT1	Present output as a tabular listing of grids for each frequency. (Default)
SORT2	Present output as a tabular listing of frequency for each grid.
PRINT	The print file (.f06) will be the output medium. (Default)
NOPRINT	Generates, but does not print, acoustic velocity.
PUNCH	The standard punch file (.pch) will be the output medium.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Default)

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Describer	Meaning
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
n	ID of SET case control command containing the list of identification numbers of fluid grids referenced by microphone elements at which to calculate acoustic velocity results. (Integer > 0)
ALL	Request acoustic velocity results for all microphone elements in the model.
NONE	Turns this output request off; useful for controlling output requests across subcases.

Remarks:

1. The PRESSURE case control command settings determine whether the acoustic velocity results are based on total or scattered pressure.
2. Acoustic velocity results will be written to the op2 file, if the value of PARAM, POST is -1 or -2.
3. When the fluid domain is represented by an acoustic transfer vector (ATV), acoustic particle velocity and acoustic intensity results cannot be computed, and the ACINTENSITY and ACVELOCITY case control commands are ignored.
4. Acoustic velocity results are presented in the basic coordinate system.

Related Topics

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ACVPO4STR

Acoustic Volume Quadrupole Strength Results Output

Acoustic volume quadrupole strength results output request for fluid solid elements.

Format:

$$\text{ACVPO4STR}\left[\left(\left[\begin{matrix}\text{PRINT}\\ \text{NOPRINT}\end{matrix}\right]\right)\right]=\left\{\begin{matrix}\text{n}\\ \text{ALL}\\ \text{NONE}\end{matrix}\right\}$$

Example:

```
SET 101 = 68201 THRU 68204, 68242, 68245 THRU 68246
```

```
ACVPO4STR (PRINT = 101)
```

Describers:

Describer	Meaning
PRINT	Prints results to the output medium (.f06 file). (Default)
NOPRINT	Calculates but does not print the results.
n	Identification number of the SET case control command that lists the identification numbers of fluid solid elements at which to calculate acoustic volume quadrupole strength results. (Integer > 0)
ALL	Requests acoustic volume quadrupole strength results for all fluid solid elements in the model.
NONE	Turns this output request off, which is useful for controlling output requests across subcases.

Related Topics

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ADAMSMNF

Generates Adams Interface Modal Neutral File

Generates Adams Interface Modal Neutral File (MNF) during SOL 103, 111, or 112.

Format:

$$\text{ADAMSMNF}\left[\text{FLEXBODY}=\left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],\left[\text{FLEXONLY}=\left\{\begin{matrix}\underline{\text{YES}}\\ \text{NO}\end{matrix}\right\}\right],$$
$$\left[\text{OUTGSTRS}=\left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],\left[\text{OUTGSTRN}=\left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],$$
$$\left[\text{MINVAR}=\left\{\begin{matrix}\underline{\text{PARTIAL}}\\ \text{CONSTANT}\\ \text{FULL}\\ \text{NONE}\end{matrix}\right\}\right],\left[\text{PSETID}=\left\{\begin{matrix}\underline{\text{NONE}}\\ \text{setid}\\ \text{ALL}\\ \text{sktunit}\end{matrix}\right\}\right],$$
$$\left[\text{ADMOUT}=\left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],\left[\text{CHECK}=\left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],\left[\text{NONCUP}=\left\{\begin{matrix}\underline{\text{-1}}\\ \text{-2}\end{matrix}\right\}\right]$$

Examples:

```
ADAMSMNF FLEXBODY=YES
```

Describers:

Describer		Meaning
FLEXBODY		Requests the generation of MNF.
	NO	Standard Simcenter Nastran solution without MNF creation (default).
	YES	MNF generation requested.
FLEXONLY		Determines if standard DMAP solution runs or not after MNF creation is complete.
	YES	Only MNF creation occurs (default).
	NO	MNF file creation occurs along with standard DMAP solution.
OUTGSTRS		Determines if grid point stress is written to MNF.

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Describer		Meaning
	NO	Do not write grid point stress to MNF (default).
	YES	Write grid point stress to MNF.
OUTGSTRN		Determines if grid point strain is written to MNF.
	NO	Do not write grid point strain to MNF (default).
	YES	Write grid point strain to MNF.
MINVAR		Determines how mass invariants are computed.
	PARTIAL	Mass invariants 5 and 9 are not computed.
	CONSTANT	Mass invariants 1,2,6 and 7 are computed.
	FULL	All nine mass invariants are computed.
	NONE	No mass invariants are computed.
PSETID		Selects a set of elements defined in the OUTPUT(PLOT) section (including PLOTEL) or on a sketch file whose connectivity is exported to face geometry to be used in Adams. See Remark 16.
	NONE	All grids, geometry and associated modal data is written to MNF (default).
	setid	The connectivity of a specific element set is used to export face geometry.
	ALL	The connectivity of all element sets are used to export face geometry.
sktunit		The connectivity of element faces defined on a sketch file is used to export face geometry. Note that the value must be a negative number to distinguish it from a setid value.
ADMOUT		Requests that the FLEXBODY run output a Simcenter Nastran OP2 file for use in post processing of Adams/Flex results.
	NO	OP2 file will not be generated (default).
	YES	OP2 file will be generated.
CHECK		Requests debug output be written to the f06 file when ADMOUT=YES. See Remark 20.
	NO	No debug output will be written (default).
	YES	Debug output will be written.
NONCUP		Modal damping output control. See Remark 22.
	–1	Output the full equivalent modal viscous damping matrix (default).
	–2	Output only diagonal values of the equivalent modal viscous damping matrix.

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Remarks:

1. The creation of the Adams MNF, which is applicable in a non-restart SOL 103, 111, or 112 analysis only, is initiated by ADAMSMNF FLEXBODY=YES (other describers are optional) along with the inclusion of the bulk data entry DTI,UNITS. The MNF file naming convention is as follows: 'jid_seid.mnf', where seid is the integer number of the superelement (0 for residual-only run). The location of these files is the same directory as the jid.f06 file.
2. ADAMSMNF must appear above the subcase level.
3. Since Adams is not a unitless code, the Data Table Input bulk entry DTI,UNITS is required for an MBDEXPORT ADAMS FLEXBODY=YES run. The DTI,UNITS entry specifies the system of units of the original Simcenter Nastran input file, and is then included with the data written to the MNF file. Simcenter Nastran does not do a units conversion of the nastran data when writing the MNF file. Once identified, the units will apply to all superelements in the model. The complete format is:

```
DTI     UNITS   1       MASS    FORCE   LENGTH  TIME
```

All entries are required. Acceptable character strings are listed below.

Mass:

KG - kilogram

LBM – pound-mass (0.45359237 kg)

SLUG – slug (14.5939029372 kg)

GRAM – gram (1E-3 kg)

OZM – ounce-mass (0.02834952 kg)

KLBM – kilo pound-mass (1000 lbm) (453.59237 kg)

MGG – megagram (1E3 kg)

MG – milligram (1E-6 kg)

MCG – microgram (1E-9 kg)

NG – nanogram (1E-12 kg)

UTON – U.S. ton (907.18474 kg)

SLI – slinch (175.1268352 kg)

Force:

N – Newton

LBF – pound-force (4.44822161526 N)

KGF – kilograms-force (9.80665 N)

OZF – ounce-force (0.2780139 N)

DYNE – dyne (1E-5 N)

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KN – kilonewton (1E3 N)

KLBF – kilo pound-force (1000 lbf) (4448.22161526 N)

MN – millinewton (1E-3 N)

MCN – micronewton (1E-6 N)

NN – nanonewton (1E-9 N)

Length:

M – meter

KM – kilometer (1E3 m)

CM – centimeter (1E-2 m)

MM – millimeter (1E-3 m)

MI – mile (1609.344 m)

FT – foot (0.3048 m)

IN – inch (25.4E-3 m)

MCM – micrometer (1E-6 m)

NM – nanometer (1E-9 m)

A – Angstrom (1E-10 m)

YD – yard (0.9144 m)

ML – mil (25.4E-6 m)

MCI – microinch (25.4E-9 m)

Time:

S – second

H – hour (3600.0 sec)

MIN-minute (60.0 sec)

MS – millisecond (1E-3 sec)

MCS – microsecond (1E-6 sec)

NS – nanosecond (1E-9 sec)

D – day (86.4E3 sec)

4. Since DTI,UNITS determines all units for the MNF, the units defined in WTMASS, which are important for units consistency in Simcenter Nastran, are ignored in the output to the MNF. For example, if the model mass is kilograms, force in Newtons, length in meters, and time in seconds, then WTMASS would equal 1 ensuring that Simcenter Nastran works with the consistent set of kg, N, and m. The units written to the MNF would be: “DTI,UNITS,1,KG,N,M,S”.

5. You can create flexible body attachment points by defining the component as a superelement or part superelement, in which case the physical external (a-set) grids become the attachment points; or for

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a residual-only type model, you can use standard Simcenter Nastran ASET Bulk Data entries to define the attachment points.

6. The nine mass variants are:

$$ {}^{1}I_{1\,x\,1} = \sum_{p=1}^{N} m_p $$

$$ {}^{2}I_{3\,x\,1} = \sum_{p=1}^{N} m_p s_p $$

$$ {}^{3}I_{j\,3\,x\,M} = \sum_{p=1}^{N} m_p \Phi_p \quad j = 1, \ldots, M $$

$$ {}^{4}I_{3\,x\,M} = \sum_{p=1}^{N} m_p \tilde{s}_p \bar{\Phi}_p + I_p \Phi_p^{*} $$

$$ {}^{5}I_{j\,3\,x\,M} = \sum_{p=1}^{N} m_p \tilde{\phi}_{pj} \Phi_p \quad j = 1, \ldots, M $$

$$ {}^{6}I_{M\,x\,M} = \sum_{p=1}^{N} m_p \Phi_p^{T} \Phi_p + \Phi_p^{*T} I_p \Phi_p^{*} $$

$$ {}^{7}I_{3\,x\,3} = \sum_{p=1}^{N} m_p \tilde{s}_p^{T} \tilde{s}_p + I_p $$

$$ {}^{8}I_{j\,3\,x\,3} = \sum_{p=1}^{N} m_p \tilde{s}_p \tilde{\phi}_{pj} \quad j = 1, \ldots, M $$

$$ {}^{9}I_{jk\,3\,x\,3} = \sum_{p=1}^{N} m_p \tilde{\phi}_{pj} \phi_{pk} \quad j, k = 1, \ldots, M $$

$s_p = [xyz]^T$ are the coordinates of grid point p in the basic coordinate system.

$$ \tilde{s}_p = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix} $$ = skew symmetric vector cross product operator.

φ_p = partitioned orthogonal modal matrix that corresponds to the translational degrees of freedom of grid *p*.

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I_p = inertia tensor p.

φ_p^* = partitioned orthogonal modal matrix that corresponds to the rotational degrees of freedom of grid p.

$\tilde{\phi}_{pf}$ = skew-symmetric matrix formed for each grid translational degree of freedom for each mode.

M=number of modes.

N=number of grids.

7. To accurately capture the mode shapes when supplying SPOINT/QSET combinations, the number of SPOINTS (ns) should be at least ns = n+(6+p), assuming that residual flexibility is on. In the above equation for ns, the number of modes (n) is specified on the EIGR (METHOD=LAN) or EIGRL Bulk Data entries; the number of load cases is p. In general, you can't have too many SPOINTs, as excess ones will simply be truncated with no performance penalty.
8. For FLEXBODY=YES runs, residual vectors for the component should always be calculated as they result in a more accurate representation of the component shapes at little additional cost.
9. OMIT or OMIT1 Bulk Data entries are not supported.
10. Lumped mass formulation (default) is required. Either leave PARAM,COUPMASS out of the input file or supply PARAM,COUPMASS,-1 (default) to ensure lumped mass.
11. CBEND elements are not allowed because they always use a coupled mass formulation. Likewise, the MFLUID fluid structure interface is not allowed because the virtual mass matrix it generates is not diagonal.
12. PARAM,WTMASS,value with a value other than 1.0 may be used with a Simcenter Nastran run generating an MNF. It must have consistent units with regard to the DTI,UNITS Bulk Data entry. Before generating the MNF, Simcenter Nastran will appropriately scale the WTMASS from the physical mass matrix and mode shapes.
13. There is a distinction between how an ADAMSMNF FLEXBODY=YES run handles element-specific loads (such as a PLOAD4 entry) versus those that are grid-specific (such as a FORCE entry), especially when superelements are used. The superelement sees the total element-specific applied load. For grid-specific loads, the loads attached to an external grid will move downstream with the grid. That is to say, it is part of the boundary and not part of the superelement. This distinction applies to a superelement run and not to a residual-only or parts superelement run.
14. The loads specified in Simcenter Nastran generally fall into two categories: non-follower or fixed direction loads (non-circulatory) and follower loads (circulatory). The follower loads are nonconservative in nature. Examples of fixed direction loads are the FORCE entry or a PLOAD4 entry when its direction is specified via direction cosines. Examples of follower loads are the FORCE1 entry or the PLOAD4 entry when used to apply a normal pressure. By default in Simcenter Nastran, the follower loads are always active in SOL 103 and will result in follower stiffness being added to the differential stiffness and elastic stiffness of the structure. In a run with ADAMSMNF FLEXBODY=YES and superelements, if the follower force is associated with a grid description (such as a FORCE1) and the grid is external to the superelement, the follower load will move downstream with the grid. Thus, the downstream follower contribution to the component's stiffness will be lost, which could yield poor results. This caution only applies to a superelement run and not to a residual-only or a part superelement run.

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15. OUTGSTRS and OUTGSTRN entries require the use of standard Simcenter Nastran STRESS= or STRAIN= used in conjunction with GPSTRESS= or GPSTRAIN= commands to produce grid point stress or strain. GPSTRESS(PLOT)= or GPSTRAIN(PLOT)= will suppress grid stress or strain print to the Simcenter Nastran .f06 file.

16. To reduce the FE mesh detail for dynamic simulations, PSETID (on the ADAMSMNF Case Control command) defined with a SET entry (i.e. setid) is used to define a set of PLOTELs or other elements used to select grids to display the components in Adams. This option can significantly reduce the size of the MNF without compromising accuracy in the Adams simulation providing that the mass invariant computation is requested. With superelement analysis, for any of these elements that lie entirely on the superelement boundary (all of the elements' grids attached only to a-set or exterior grids), a SEELT Bulk Data entry must be specified to keep that display element with the superelement component. This can also be accomplished using PARAM, AUTOSEEL,YES. The SEELT entry is not required with parts superelements, as boundary elements stay with their component.

    If the SET entry points to an existing set from the OUTPUT(PLOT) section, this single set is used explicitly to define elements used to select grids to display the component in Adams. If PSETID does not find the set ID in OUTPUT(PLOT), it will search sets in the case control for a matching set ID. This matching set ID list then represents a list of OUTPUT(PLOT) defined elements' sets, the union of which will be used to define a set of PLOTELs or other elements used to select grids to display the component in Adams. If the user wishes to select all of the sets in the OUTPUT(PLOT) section, then use PSETID=ALL.

    The following element types are not supported for writing to an MNF, nor are they supported as a 'type' entry in a set definition in OUTPUT(PLOT): CAABSF, CAEROi, CDUMi, CHBDYx, CDAMP3, CDAMP4, CELAS3, CELAS4, CMASS3, CMASS4, CRAC2D, CRAC3D, CTWIST, CWEDGE, CWELD, and GENEL.

    PSETID can also point to a sketch file using PSETID=-sktunit, where sktunit references an ASSIGN statement of the form

```
ASSIGN SKT='sketch_file.dat',UNIT=sktunit.
```

    The grids defined for the elements' faces in the sketch file, along with *all* external (i.e. boundary) grids for the superelements, will be the only grids (and their associated data) written to the MNF.

    The format of the sketch file, which describes the mesh as a collection of faces, must be as follows:

```
face_count
face_1_node_count face_1_nodeid_1 face_1_nodeid_2 ...
face_2_node_count face_2_nodeid_1 face_2_nodeid_2 ...

<etc>
```

    Faces must have a node count of at least two. For example, a mesh comprised of a single brick element might be described as follows:

```


6
```


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```
4 1000 1001 1002 1003
4 1007 1006 1005 1004
4 1000 1004 1005 1001
4 1001 1005 1006 1002
4 1002 1006 1007 1003
4 1003 1007 1004 1000
```

Alternatively, the mesh might be described as a stick figure using a collection of lines (two node faces), as shown below:

```
8
2 101 102
2 102 103
2 103 104
2 104 105
2 105 106
2 106 107
2 107 108
2 108 109
```

17. Typical Simcenter Nastran data entry requirements are described below.

Typical Parameters:

- PARAM,RESVEC,character_value – controls calculation of residual flexibility (including inertia relief) modes. In SOL 103, residual flexibility is on by default for only component modes (o-set).
- PARAM,GRDPNT, value - mass invariants ^{1}I, ^{2}I, and ^{7}I will be computed using results of Simcenter Nastran grid point weight generator execution in the basic coordinate system.

Typical Case Control:

- ADAMSMNF FLEXBODY=YES is required for MNF generation.
- METHOD=n is required before or in the first subcase for modal solutions.
- SUPER=n,SEALL=n is useful with multiple superelement models to select an individual superelement as a flexible body. Cannot be used with a linear STATSUB(PRELOAD) run.
- OUTPUT(PLOT) is necessary to define elements used to select grids to display the component in Adams when PSETID=ALL or setid.

  SET n=list of elements (including PLOTELs) is used to select grids to display the component.
- OUTPUT(POST) is necessary to define volume and surface for grid stress or strain shapes.

  SET n=list is a list of elements for surface definition for grid stress or strain shapes.

  Stress and strain data in the MNF is limited to the six components (i.e. 3 normal and 3 shear) for a grid point for a given mode.

  SURFACE n SET n NORMAL z3 is used to define a surface for writing stress and strain data. Only one FIBER selection is allowed for each SURFACE, thus the use of the FIBRE ALL keyword on the SURFACE case control command will write stresses to the MNF at the Z1 fiber location only.

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Since the FIBRE keyword only applies to stresses, strain data will always be written to the MNF at the MID location.

Stress and strain data at grid points can only be written to the MNF for surface and volume type elements (e.g. CQUAD and CHEXA).

VOLUME n SET n is a volume definition.

The default SYSTEM BASIC is required with SURFACE or VOLUME.

- STRESS(PLOT) is necessary for stress shapes.
- STRAIN(PLOT) is necessary for strain shapes.
- GPSTRESS(PLOT) is necessary for grid point stress shapes to be included in the MNF.
- GPSTRAIN(PLOT) is necessary for grid point strain shapes to be included in the MNF.

Typical Bulk Data:

- DTI,UNITS,1,MASS,FORCE,LENGTH,TIME is required for MNF generation. For input files containing superelements, this command must reside in the main bulk data section.
- SPOINT,id_list defines and displays modal amplitude.SESET,SEID,grid_list defines a superelement (see GRID and BEGIN BULK SUPER=). The exterior grids will represent the attachment points along with the q-set.
- SEELT,SEID,element_list reassigns superelement boundary elements to an upstream superelement.
- RELEASE,SEID,C,Gi is an optional entry that removes DOFs from an attachment grid for which no constraint mode is desired. For example, this allows the removal of rotational degrees of freedom from an analysis where only translational degrees of freedom are required.
- SEQSET,SEID,spoint_list defines modal amplitudes of a superelement (see SEQSET1).
- SENQSET,SEID,N defines modal amplitudes of a part superelement. It must reside in the main Bulk Data Section.
- ASET,IDi,Ci defines attachment points for a residual-only run (see ASET1).
- QSET1,C,IDi defines modal amplitudes for the residual structure or modal amplitudes for a part superelement (see QSET).
- PLOTEL,EID,Gi can be used, along with existing model elements, to define elements used to select grids to display the components in Adams.
- EIGR,SID,METHOD,... obtains real eigenvalue extraction (see EIGRL).

18. ADAMSMNF and MBDEXPORT case control entries cannot be used in the same analysis run. In other words, an Adams MNF file or a RecurDyn RFI file can be generated during a particular Simcenter Nastran execution, but not both files at the same time. Attempting to generate both files in the same analysis will cause an error to be issued and the execution to be terminated.

19. The ADMOUT=YES option is used when you would like results recovery (using the ADMRECVR case control entry) from an Adams/Flex analysis. This option requires the following assignment command:

ASSIGN OUTPUT2='name.out' STATUS=UNKNOWN UNIT=20 FORM=UNFORM

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inserted into the file management section of the Simcenter Nastran input file. It will cause an OP2 file with a .out extension to be generated, which then can be used as input into a Simcenter Nastran SOL 103 run using the ADMRECVR case control capability to perform results recovery from an Adams/Flex analysis. FLEXBODY=YES is required with its use.

The data blocks output are:

MGGEW - physical mass external sort with weight mass removed
MAAEW - modal mass
KAAE - modal stiffness
CMODEXT - component modes.

This capability is limited to no more than one superelement per Simcenter Nastran model. Residual-only analyses are supported.

If differential stiffness is included, the static portion of the results will not be included in the recovered results when using ADMRECVR or MBDRECVR.

20. Setting CHECK=YES (which is only available when ADMOUT=YES) is *not* recommended for models of realistic size due to the amount of data that will be written to the f06.

21. The ADAMSMNF data routines use the environment variable TMPDIR for temporary storage during the processing of mode shape data. As a result, TMPDIR must be defined when using ADAMSMNF. TMPDIR should equate to a directory string for temporary disk storage, preferably one with a large amount of free space.

22. If any damping is defined in the model, an equivalent modal viscous damping will be determined for each mode and written to the MNF. This equivalent modal viscous damping is defined as:

$D = \psi^T B_e \psi$

where D is the equivalent modal viscous damping matrix, ψ is the eigenvector matrix, and B_e is the equivalent viscous damping matrix.

The equivalent viscous damping matrix is given by:

$$B_e = B^1_{AA} + B^2_{AA} + \frac{G}{W3} K^1_{AA} + \frac{1}{W4} K^4_{AA}$$

where *G*, *W3*, and *W4* are structural damping-related parameters described in the “Parameter Descriptions” section of this guide.

By default, the full equivalent modal viscous damping matrix is written to the MNF. To write only the diagonal values of the equivalent modal viscous damping matrix to the MNF, specify NONCUP=−2 or specify PARAM,NONCUP,-2.

If both the NONCUP describer and the NONCUP parameter are specified, the NONCUP describer specification takes precedence.

23. Preload conditions are not supported.

24. Cylindrical and spherical nodal displacement coordinate systems (i.e. CD on the GRID bulk entry) are not supported.

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Related Topics

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ADAPTERR

Error estimator output request for a SOL 401 statics subcase.

Controls the computation and output of error estimates from a statics subcase.

Format:

$$\text{ADAPTERR}\left[\left(\text{STRESS},\underline{\text{STNERGY}},\text{STEP},\left[\begin{matrix}\text{PRINT}\\\text{PLOT}\end{matrix}\right]\right)\right]=\left\{\begin{matrix}\text{ALL}\\\text{NONE}\end{matrix}\right\}$$

Examples:

```
ADAPTERR(STRESS)=ALL
ADAPTERR(STRESS,STNERGY,PUNCH)=ALL
ADAPTERR(STEP)=ALL
```

Describers:

Describer	Meaning
STRESS	Request the stress norm and the stress error norm. See Remarks 3 and 4.
STNERGY	Request the strain energy norm and strain energy error norm. See Remarks 3 and 4. (Default)
STEP	Request the output at the output increment steps defined with the TSTEP1 entries. See Remark 5.
PRINT	Compute and write output to the print (.f06) file. (Default)
PLOT	Compute output only.
ALL	Compute error estimates.
NONE	Do not compute error estimates.

Remarks:

1. The ADAPTERR case control command is only supported in SOL 401. The error estimates are only computed in the statics subcase type (ANALYSIS=STATICS). The ADAPTERR command must be defined above the subcases (globally).
2. The calculation of error estimates is supported for the following element types:

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Solid elements	CHEXA, CPENTA, CPYRAM, CTETRA (excludes CHEXA and CPENTA elements referencing PCOMPS bulk entries)
Axisymmetric elements	CQUADX4, CQUADX8, CTRAX3, CTRAX6
Plane strain elements	CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8 (includes elements referencing PGPLSN bulk entries)
Plane stress elements	CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8

3. The stress norm is calculated from:

$$||u||_{L_2} = \sqrt{\frac{1}{\Omega}\int_{\Omega} \sigma^{T}_{\text{unaveraged}}\,\sigma_{\text{unaveraged}}\,d\Omega}$$

The stress error norm is calculated from:

$$||e_{\sigma}|| = \sqrt{\frac{1}{\Omega}\int_{\Omega} (\sigma_{\text{unaveraged}} - \sigma_{\text{averaged}})^{T}(\sigma_{\text{unaveraged}} - \sigma_{\text{averaged}})d\Omega}$$

The strain energy norm is calculated from:

$$||u|| = \sqrt{\int_{\Omega} \sigma^{T}_{\text{unaveraged}}\,D^{-1}\sigma_{\text{unaveraged}}\,d\Omega}$$

The strain energy error norm is calculated from:

$$||e_{U}|| = \sqrt{\int_{\Omega} (\sigma_{\text{unaveraged}} - \sigma_{\text{averaged}})^{T} D^{-1}(\sigma_{\text{unaveraged}} - \sigma_{\text{averaged}})d\Omega}$$

where

Ω is the element volume,

$\sigma_{unaveraged}$ is the unaveraged stress vector,

$\sigma_{averaged}$ is an averaged stress vector computed at a grid point using the stress vectors from elements connected to the grid point,

D matrix is the constitutive relation.

When computing $\sigma_{averaged}$, stress values are not averaged across different element families, material properties, material coordinate systems, orientation angles in 2D solid elements, and thicknesses in plane stress elements.

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4. You can specify both the STRESS and STNERGY describers to request stress norm, stress error normal, strain energy norm and strain energy error norm output.
5. The software always outputs the maximum value on each element, for each output type requested, by comparing the values from all solution steps. In addition, if you specify the STEP describer, the software will output what you have requested at the output increment steps defined with the TSTEP1 entries.

Related Topics

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ADMRECVR

Adams stress recovery.

Recovers stress results from an Adams/Flex analysis.

Format:

$$\text{ADMRECVR}\left[\text{ADMFORM}=\left\{\begin{matrix}\underline{\text{BINARY}}\\ \text{ASCII}\end{matrix}\right\}\right],$$
$$\left[\text{MSRMODE}=\left\{\begin{matrix}\underline{0}\\ 1\\ 2\end{matrix}\right\}\right],\ \left[\text{RGBODY}=\left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\\ \text{BOTH}\end{matrix}\right\}\right],$$
$$\left[\text{MSGLVL}=\left\{\begin{matrix}\underline{0}\\ 1\text{ - }4\end{matrix}\right\}\right],\ \left[\text{ADMCHK}=\left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right]$$

Examples:

```
ASSIGN INPUTT2=`adams_results.mdf' UNIT=13
...
CEND

STRESS(PLOT)=100
ADMRECVR
```

Describers:

Describer		Meaning
ADMFORM		Specifies the format of the Adams/Flex modal deformations file (see remarks 1 and 2)
	BINARY	OUTPUT2 file (default)
	ASCII	PUNCH file.
MSRMODE		Specifies stress recovery type (see remarks 6 and 7).
	0	Component modal definitions are stored in an OUTPUT2 file (specifically, a *.out file was created by using ADMOUT=YES on the ADAMSMNF case control in a pre-Adams/Flex Simcenter Nastran run). The OUTPUT2 files used in this case do not contain data blocks used for MNF creation (default).

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Describer		Meaning
	1	Same as option 0, except that the OUTPUT2 file will contain 10 additional data blocks used for MNF creation by an Adams pre-processor (specifically, a *.out file created through use of the mnfx.alt DMAP alter capability).
	2	No file reference (specifically, component modal definitions will be recomputed)
RGBODY		Requests the addition of rigid body motion with modal deformations (see remark 5).
	NO	Do not include rigid body motion (default).
	YES	Include rigid body motion.
	BOTH	Generate two output data blocks; one containing rigid body motion and the other without.
MSGLVL		Level of diagnostic output from Lanczos eigensolver when component modal definitions are determined (applies only when MSRMODE=2).
	0	No output (default).
	1	Warning and fatal messages.
	2	Summary output.
	3	Detailed output on cost and convergence.
	4	Detailed output on orthogonalization.
ADMCHK		Requests debug output be written to the f06 file (See Remark 9).
	NO	No debug output will be written (default).
	YES	Debug output will be written.

Remarks:

1. When modal deformations to be read are in binary (OUTPUT2) format (specifically, ADMFORM=BINARY), the following statement needs to be specified near the top of the Simcenter Nastran input file in the file management section:

   ASSIGN INPUTT2=`<MDFilename>' UNIT=13

   where <MDFilename> is the name of the modal deformations file from Adams.

2. To input the modal deformations file from Adams in ASCII (Punch) format (specifically, ADMFORM=ASCII), the following statement needs to be included in the bulk data section:

   INCLUDE `<MDFilename>' where `<MDFilename>' is the name of the modal deformations file.

3. Dynamic stress/strain output can either be in .f06, PUNCH, and/or OUTPUT2 according to standard Simcenter Nastran functionality. However, stress recovery in Simcenter Nastran from Adams/Flex results does not support XYPLOT output.

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4. If displacements, stresses, and/or strains are to be available for post processing, one or more of the following statements must appear in the case control section of the Simcenter Nastran input file:

   DISP(PLOT) = <set id>

   STRAIN(FIBER,PLOT) = <set id>

   STRESS(PLOT) = <set id>

   Even though the recovery process is performed in SOL 103, the derived results are transient response results. The default output format for transient results is SORT2. If no SORT2 results are desired, then SORT1 must be requested for all output requests.

5. Rigid body motions from an Adams simulation are included in the modal deformation file, but they are not applied unless the RGBODY keyword is set to YES or BOTH. Including rigid body motion affects the display and animation of the flexible component, but it has no effect on dynamic stresses. For RGBODY=YES, the displacements output data block OUGV1 or OUGV2 will contain a '0' in word 3 of each header record. For RGBODY=NO, OUGV1 or OUGV2 will contain a '1' in word 3 of each header record. For RGBODY=BOTH, two OUGV1 or OUGV2 data blocks will be generated; one for RGBODY=YES and one for RGBODY=NO.

6. For MSRMODE=0 or 1, stress recovery references the OUTPUT2 file obtained from the initial CMS analysis (specifically, ADMOUT=YES on the ADAMSMNF case control entry or use of the mnfx.alt DMAP alter capability). No other files are required. The geometric data needs to be included in the bulk data of the Simcenter Nastran input file because geometry is missing from the OUTPUT2 file. This mode of stress recovery is faster than the MSRMODE=2 mode. To reference this OUTPUT2 file the following line needs to be included in the file specification section of the Simcenter Nastran input file:

   ASSIGN INPUTT2=`<OUTPUT2_filename>' UNIT=20

7. For MSRMODE=2, no files are referenced for stress recovery. Instead, a full CMS reanalysis is performed to build the reference data for the stress recovery analysis. Obviously, the analysis time is significantly far greater compared to the MSRMODE=0 or 1 method, but this method frees up disk space. There is also risk in using this method. If the reanalysis generates slightly different component eigenvalues or eigenvectors than were generated during the creation of the Adams MNF in the initial Simcenter Nastran run, then the Adams results in the Adams MDF (modal deformation file) will be inconsistent and incorrect results will be recovered. Something as simple as a sign change for one eigenvector will cause incorrect results to be recovered. It is, therefore, *highly recommended* that MSRMODE=0 or 1 *always* be used.

8. This capability must be performed in SOL 103 and is limited to no more than one superelement per Simcenter Nastran model. Residual-only analyses are supported.

9. Setting ADMCHK=YES is *not* recommended for models of realistic size due to the amount of data that will be written to the f06.

Related Topics

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AECONFIG

Aeroelastic Configuration Name

Assigns the aerodynamic configuration parameter used to locate the associated datablocks on the aerodynamic and aeroelastic databases. The configuration name is the Aerodynamic Supergroup identified as part of the aeroelastic model in the Aeroelasticity module prior to analysis submission. The standard location of the AECONFIG entry will be above the SUBCASE entries. An AECONFIG located within a SUBCASE entry will override the global AECONFIG; this requires attachment to an existing aerodynamic database. If the AECONFIG is not found, new qualified datablocks will be created from the data in the Bulk Data Section.

Format:

Assigns a DBset member name

AECONFIG	=config-name.

Examples:

Assigns a MASTER file for the aerodynamic and aeroelastic DBsets.

AECONFIG	=CONFIG_A

Describers:

Describer	Meaning
config-name	The configuration name. This is the Aerodynamic SuperGroup name identified as part of the aeroelastic model.

Related Topics

Browse Simcenter Nastran documentation

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AEROF

Aerodynamic Force Output Request

Requests the aerodynamic loads on aerodynamic control points.

Format:

$$\text{AEROF} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{NONE} \end{matrix} \right\}$$

Examples:

```
AEROF=ALL
AEROF=5
```

Describers:

Describer	Meaning
n	Set identification of a previously appearing SET command. (Integer>0)
ALL	Output forces at all points.
NONE	Do not output aerodynamic loads.

Remarks:

1. The SET command references box or body element identification numbers.
2. Output is in the units of force or moment.
3. Only aerodynamic forces on points specified on the SET command will be output.

Related Topics

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AESYMXY

Aerodynamic Flow Symmetry About XY Plane

Aerodynamic XY plane of symmetry flag. This is used to indicate whether the aerodynamic model has symmetry with respect to the ground. This selection is typically done within each subcase, but a case control default can be defined by placing an entry above the subcase.

Format:

$$\text{AESYMXY} = \left\{ \begin{matrix} \text{SYMMETRIC} \\ \text{ANTISYMMETRIC} \\ \text{ASYMMETRIC} \end{matrix} \right\}$$

Example 1:

AESYMXY=ASYMMETRIC

Describers:

Describer	Meaning
SYMMETRIC	Indicates that the aerodynamic model is moving in a symmetric manner with respect to the XY plane.
ANTISYMMETRIC	Indicates that the aerodynamic model is moving in an antisymmetric manner with respect to the XY plane.
ASYMMTRIC	Indicates that the aerodynamic model has no reflection about the XY plane.

Remarks:

1. If the AESYMXY and AESYMXZ commands are not present for a particular subcase, aerodynamic XY symmetry will be determined from the SYMXY field of the AEROS Bulk Data entry for static aeroelastic analysis and from the SYMXY field of the AERO Bulk Data entry for flutter and dynamic aeroelastic analysis.
2. Symmetric implies ground effect while asymmetric implies free air analysis.

Related Topics

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AESYMXZ

Aerodynamic Flow XZ Symmetry About XZ Plane

Aerodynamic XZ plane of symmetry flag. This is used to support symmetric models about the centerline. Default is ASYMMETRIC (no symmetry). (Subcase level inheritance from AESYMXY applies.)

Format:

$$\text{AESYMXZ} = \left\{ \begin{matrix} \text{SYMMETRIC} \\ \text{ANTISYMMETRIC} \\ \text{ASYMMETRIC} \end{matrix} \right\}$$

Example 1:

AESYMXZ=SYMMETRIC

Describers:

Describer	Meaning
SYMMETRIC	Indicates that a half span aerodynamic model is moving in a symmetric manner with respect to the XZ plane.
ANTISYMMETRIC	Indicates that a half span aerodynamic model is moving in an antisymmetric manner with respect to the XZ plane.
ASYMMETRIC	Indicates that a full aerodynamic model is provided. (Default.)

Remarks:

If the AESYMXZ and AESYMXY commands are not present, aerodynamic XZ symmetry is determined from the SYMXZ field of the AEROS Bulk Data entry for static aeroelastic analysis and from the SYMXZ field of the AERO Bulk Data entry for flutter and dynamic aeroelastic analysis.

Related Topics

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ALOAD

Acoustic Load Set Selection

Selects an acoustic load or source to be applied in a frequency response problem.

Format:

ALOAD=n

Examples:

```
ALOAD=57
```

Describers:

Describer	Meaning
n	Set identification of an ACDUCT, ACFAN, ALOAD, ACPOLE1, ACPOLE2, ACPLNW, ACPNVEL, ACPRESS, ACSPO2, or ACVPO4 bulk entry. (Integer > 0)

Remarks:

None.

Related Topics

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ANALYSIS

Analysis Discipline Subcase Assignment

Specifies the type of analysis being performed for the current subcase.

Format:

ANALYSIS=type

Examples:

```
ANALYSIS=STATICS
ANALYSIS=MODES
```

Describers:

Table 4-12: SOL 200 Analysis Types

Describer	Meaning		
type	Analysis type. Allowable values and applicable solution sequences (Character):		
	STATICS	Statics	SOL 200
	MODES	Normal Modes	
	BUCK	Buckling	
	DFREQ	Direct Frequency	
	MFREQ	Modal Frequency	
	MTRAN	Modal Transient	
	DCEIG	Direct Complex Eigenvalue Analysis	
	MCEIG	Modal Complex Eigenvalue Analysis	
	SAERO	Static Aeroelasticity	
	DIVERGE	Static Aeroelastic Divergence	
	FLUTTER	Flutter	

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Table 4-13: SOL 401 Analysis Types

Describer	Meaning		
type	Analysis type. Allowable values and applicable solution sequences (Character):		
	STATICS	Statics	SOL 401
	DYNAMICS	Dynamics	
	MFREQ	Modal Frequency	
	MODES	Normal Modes	
	PRELOAD	Bolt Preload	
	CYCMODES	Cyclic Modes	
	FOURIER	Fourier Modes	

Table 4-14: SOL 402 Analysis Types

Describer	Meaning		
type	Analysis type. Allowable values and applicable solution sequences (Character):		
	STATICS	(Non-Linear) static analysis	SOL 402
	DYNAMICS	(Non-Linear) dynamic analysis, with velocity and inertia effects	
	MODES	Normal Modes	
	PRELOAD	Bolt Preload subcase	
	CYCMODES	Cyclic Normal Modes	
	FOURIER	Fourier Normal Modes	
	BUCKLING	Buckling	
	CEIG	Complex Eigenvalue Extraction	

Table 4-15: SOLs 108, 110, 111, and 112 Analysis Types

Describer	Meaning		
type	Analysis type. Allowable values and applicable solution sequences (Character):		
	MODES	Normal Modes	SOLs 110, 111, and 112 only.
	RANDOM	Random Analysis	SOLs 108 and 111 only.

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Table 4-16: SOLs 153 and 159 Analysis Types

Describer	Meaning		
type	Analysis type. Allowable values and applicable solution sequences (Character):		
	HEAT	Heat Transfer Analysis	SOLs 153 and 159 only.
	STRUCTURE	Structural Analysis	

Remarks:

1. ANALYSIS = STRUC is the default in SOLs 153 and 159.
2. In SOL 200, all subcases, including superelement subcases, must be assigned by an ANALYSIS command either in the subcase or above all subcases. Also, all subcases assigned by ANALYSIS=MODES must contain a DESSUB request. If a SOL 200 job contains both ANALYSIS=STATICS and ANALYSIS=BUCK subcases, the STATICS subcases should come before ANALYSIS=BUCK. A SOL 200 job may not contain both DFREQ and MFREQ subcases at the same time.
3. ANALYSIS=DIVERG is only available for analysis in SOL 200. Sensitivity and optimization are not supported for this analysis type.
4. In order to obtain normal modes data recovery in SOLs 110, 111, and 112, ANALYSIS = MODES must be specified under one or more separate subcase(s) which contains requests for data recovery intended for normal modes only. For example, in SOL 111:

```
METH=40
SPC=1
SUBCASE 1 $ Normal Modes
  ANALYSIS=MODES
  DISP=ALL
SUBCASE 2 $ Frequency response
  STRESS=ALL
  DLOAD=12
  FREQ=4
```

   All commands that control the boundary conditions (SPC, MPC, and SUPORT) and METHOD selection should be copied inside the ANALYSIS=MODES subcase or specified above the subcase level.
5. Use ANALYSIS = RANDOM and either or both of the RANDOM and RCROSS case control commands to perform multiple random calculations from the same frequency responses. When you use multiple frequency responses in a random calculation:
   - For SOL 108 or 111, each frequency response must reference the same set of frequencies.
   - For SOL 111, if SDAMPING is specified, each frequency response must reference the same SDAMPING specification.

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For example, in SOL 111:

```
SDAMPING=101
SPC=2
ACCELERATION(RPRINT)=ALL
STRESS(SORT2,PSDF)=ALL
RCROSS(PRINT)=1
METHOD=1
$
SUBCASE 1 $ Normal Modes
  ANALYSIS=MODES
  DISP=ALL
SUBCASE 2 $ Frequency Response using FREQ set 13
  FREQUENCY=13
  DLOAD=111
SUBCASE 3 $ Frequency Response using FREQ set 13
  FREQUENCY=13
  DLOAD=211
SUBCASE 4 $ Random Response ref FRF from subcases 2 and 3
  ANALYSIS=RANDOM
  RANDOM=100
SUBCASE 5 $ Random Response ref FRF from subcases 2 and 3
  ANALYSIS=RANDOM
  RANDOM=200
$ Change frequencies
SUBCASE 11 $ Frequency Response using FREQ set 23
  FREQUENCY=23
  DLOAD=111
SUBCASE 12 $ Frequency Response using FREQ set 23
  FREQUENCY=23
  DLOAD=211
SUBCASE 14 $ Random Response ref FRF from subcases 11 and 12
  ANALYSIS=RANDOM
  RANDOM=1100
SUBCASE 15 $ Random Response ref FRF from subcases 11 and 12
  ANALYSIS=RANDOM
  RANDOM=1200
```

Remarks related to SOL 402:

1. Subcases with ANALYSIS= MODES, CYCMODES, FOURIER, CEIG, and BUCKLING that follow a subcase with ANALYSIS=STATICS, ANALYSIS=PRELOAD, or ANALYSIS=DYNAMICS must include the PARAM,LGDISP,1. Otherwise, the stress-stiffening or damping effects computed in the previous STATICS, PRELOAD, or DYNAMICS subcase will not be taken into account for the modal or linear stability computation.
2. ANALYSIS=BUCKLING must follow a ANALYSIS=STATICS.
3. For input files which do not include a subcase (global subcase only), or include a single subcase ('Subcase ID' is defined), ANALYSIS=STATICS is the default for this input file. The purpose of this default is to simplify the conversion of a legacy SOL 601 statics input file to SOL 402. For all other situations, the ANALYSIS command is required.

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Related Topics

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APRESSURE

Aerodynamic Pressure Output Request

Requests the aerodynamic pressures in static aeroelastic response.

Format:

$$\text{APRES} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{NONE} \end{matrix} \right\}$$

Examples:

APRES=ALL

APRES=6

Describers:

Describer	Meaning
n	Set identification number of a previously appearing SET command. Only aerodynamic pressures on the referenced aerodynamic boxes will be output. (Integer>0)
ALL	Pressures at all points will be output.
NONE	Do not output aerodynamic pressures.

Related Topics

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ATVOUT

ATV Creation Specification

Specifies requirements for the creation of an acoustic transfer vector (ATV).

Format:

$$\text{ATVOUT}\,(\text{ATVOP2} = unit,\ \text{ATVFS} = sid) = \left\{\begin{matrix}\text{ALL}\\ \text{n}\end{matrix}\right\}$$

Examples:

```
ATVOUT(ATVOP2=33,ATVFS=110)=200
```

Describers:

Describer	Meaning
ATVOP2 = *unit*	*unit* (integer>0) is the Fortran unit number for the .op2 file to which the ATV results are written. See Remark 2.
ATVFS = *sid*	*sid* (integer>0) is the identification number of the ATVFS bulk entry that defines the set of fluid element faces that define the coupling interface.
ALL	Output ATV results for all microphone elements. See Remark 5. (Default)
n	*n* (integer>0) is the identification number of a previously appearing SET case control command that contains microphone element identification numbers at which to output the ATV results. See Remark 5.

Remarks:

1. ATVOUT is valid for SOLs 108 and 111 only. If ATVOUT is included in the input file for any other solution, the software issues a user fatal message.
2. For specifying ATVOP2 = *unit*, an appropriate ASSIGN OUTPUT2 statement must be present in the File Management Section of the input file.
3. If ATVOUT is specified at the global level (above the subcases) and subcase levels, the software honors the output request at the global level only.
4. Only one ATVOUT can be included in an input file.
5. ATV output is controlled by the specification of ALL or *n* only. Any other output-related case control commands like DISPLACEMENT or PRESSURE are ignored.

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Related Topics

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AUXCASE

Auxiliary Model Case Control Delimiter

Delimits Case Control commands for an auxiliary model in SOL 200.

Format:

AUXCASE

Examples:

AUXCAS

AUXC

Remarks:

1. AUXCASE indicates the beginning of Case Control commands for an auxiliary model. AUXCASE commands must follow the primary model Case Control commands.
2. All Case Control commands following this entry are applicable until the next AUXCASE or BEGIN BULK command. Commands from preceding Case Control Sections are ignored.
3. Each auxiliary model Case Control must be delimited with the AUXCASE command.
4. The AUXMODEL command is used to associate the auxiliary model Case Control with a particular auxiliary model.

Related Topics

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AUXMODEL

Auxiliary Model Identification Number

References an auxiliary model for generation of boundary shapes in shape optimization.

Format:

AUXMODEL=n

Examples:

```
AUXMODEL=4
AUXM=4
```

Describers:

Describer	Meaning
n	Auxiliary model identification number. (Integer>0)

Remarks:

1. AUXMODEL references a particular auxiliary model for analysis and may only be specified in the auxiliary model Case Control Section.
2. See the BEGIN BULK command for the Bulk Data definition of an auxiliary model.

Related Topics

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B2GG

Direct Input Damping Matrix Selection

Selects direct input damping matrices.

Format:

B2GG=name

Examples:

```
B2GG=BDMIG
B2GG=BDMIG1, BDMIG2, BDMIG3
B2GG=1.25*BDMIG1, 1.0*BDMIG2, 0.75*BDMIG3
SET 100=B1, B2
B2GG=100
```

Describers:

Describer	Meaning
name	Name of [B^2_{gg}] matrix that is input on the DMIG Bulk Data entry. Scale factors may be included (see Remarks 4 and 5). See "Matrix Assembly Operations in SubDMAP SEMG" in the *Simcenter Nastran User's Guide*. (Character)

Remarks:

1. DMIG matrices will not be used unless selected.
2. Terms are added to the damping matrix before any constraints are applied.
3. The matrix must be symmetric and field 4 on the DMIG,name entry must contain the integer 6.
4. The associated DMIG matrices can be scaled using either in-line scale factors on B2GG (for example, B2GG=1.25*BDMIG1), using the parameter CB2 (for example, PARAM,CB2,1.25), or both. See **"Parameter Descriptions"**.
5. Multiple matrices separated by a comma or a blank are additive. When multiple matrices and in-line scale factors are used together, each matrix name in the list must include a scale factor. A scale factor of 1.0 should be used for matrices in the list which are not scaled. For example,

   ```
   B2GG=1.25*BDMIG1,1.0*BDMIG2,0.75*BDMIG3
   ```

   The parameter CB2 when defined will scale all B2GG. For example, if PARAM,CB2,1.30 is defined with the B2GG example above, the result would be B2GG=1.30(1.25*BDMIG1 + 1.0*BDMIG2 + 0.75*BDMIG3).

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6. The B2GG command is limited to use in the first subcase only.

Remarks related to SOL 401:

1. You can define the parameters CM2, CK2, and CB2 to scale the linear DMIG matrices defining mass, stiffness, and damping. The parameters CM1, CK1, and CB1 are not supported.
2. When you include DMIG matrices in restart solutions, for both the initial and restart solution, you must reference the DMIG entries from the global case control using the appropriate K2GG, M2GG, and B2GG command.

Related Topics

Browse Simcenter Nastran documentation

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B2PP

Direct Input Damping Matrix Selection

Selects direct input damping matrices.

Format:

B2PP=name

Examples:

```
B2PP=BDMIG
B2PP=BDMIG1, BDMIG2, BDMIG3
B2PP=1.25*BDMIG1, 1.0*BDMIG2, 0.75*BDMIG3
B2PP=(1.25,0.5)*BDMIG1, (1.0,0.0)*BDMIG2, (0.75,-2.2)*BDMIG3
SET 100=B1, B2
B2PP=100
```

Describers:

Describer	Meaning
name	Name of $[B^2_{pp}]$ matrix that is input on the DMIG Bulk Data entry. Scale factors may be included (see remarks 7 and 8). See “Formulation of Dynamic Equations in SubDMAP GMA” in the *Simcenter Nastran User’s Guide*. (Character)

Remarks:

1. DMIG entries will not be used unless selected.
2. B2PP is used only in dynamics problems.
3. The matrix must be square or symmetric and field 4 on the DMIG,name entry must contain a 1 or 6.
4. It is recommended that PARAM,AUTOSPC,NO be specified. See “Constraint and Mechanism Problem Identification in SubDMAP SEKR” in the *Simcenter Nastran User’s Guide*.
5. The B2PP command is supported across subcases. A B2PP command selecting a different DMIG matrix can be defined for each subcase.
6. The associated DMIG matrices can be scaled using in-line scale factors on B2PP (for example, B2PP=1.25*BDMIG1).

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7. Multiple matrices separated by a comma or a blank are additive. When multiple matrices and in-line scale factors are used together, each matrix name in the list must include a scale factor. A scale factor of 1.0 should be used for matrices in the list which are not scaled. For example,

```
B2PP=1.25*BDMIG1,1.0*BDMIG2,0.75*BDMIG3
```

Related Topics

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BC

Boundary Condition Identification

Identifies multiple boundary conditions for normal modes, buckling, and flutter analysis in SOLs 103, 105, 145, and 200.

Format:

BC=n

Examples:

BC=23

Describers:

Describer	Meaning
n	Identification number. (Integer>0)

Remarks:

1. In SOLs 103, 105, 145, and 200 BC is required in each subcase if multiple boundary conditions are specified for normal modes, buckling, and flutter analysis.
2. If only one boundary condition is specified, then BC does not have to be specified and n defaults to zero.

Related Topics

Browse Simcenter Nastran documentation

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BCRESULTS

Contact Result Output Request (SOLs 101, 103, 111, 112, 401, and 402)

Format:

$$\text{BCRESULTS}\left[\left(\underline{\text{TRACTION}},\text{FORCE},\text{SEPDIS},\text{STATUS},\left[\begin{matrix}\underline{\text{PRINT}}\\ \text{PLOT}\\ \text{PUNCH}\end{matrix}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \underline{\text{NONE}}\end{matrix}\right\}$$

Examples:

```
BCRESULTS=ALL
BCRESULTS(FORCE,PLOT)=ALL
BCRESULTS(TRACTION,FORCE,PLOT)=ALL
```

Describers:

Describer	Meaning
TRACTION	Contact pressure (scalar) and in-plane contact tractions (vector in basic coordinate system with units of force per area) are output for each contact grid point.
FORCE	Contact force vector in basic coordinates is output for each contact grid point.
SEPDIS	For SOLs 101, 103, 111, and 112, requests the initial and final separation distance for grids on the source region. For SOL 401, requests the final separation distance, and the total and incremental slide distance for grids on the source and target regions. Also requests the initial separation distance for grids on the source region. See Remarks 1 and 2.
STATUS	Requests the status of contact elements in SOL 401. See Remark 4.
PRINT	The printer will be the output medium.
PLOT	Computes and puts contact results in OP2 file only.
PUNCH	The punch file is the output media.

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Describer	Meaning
ALL	Contact results at all contact grid points will be output.
NONE	Contact results will not be output.
n	Set identification of a previously appearing SET command. Only contact grid points with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks :

1. For SOLs 101, 103, 111, and 112, the SEPDIS describer requests both the initial and final separation distance for grids on the source region. The separation distance is a scalar quantity representing the source side normal distance to the target. During the solution, the separation distance is known at the element integration points, but is written to the grids when output. The result at each source grid is the value of separation distance at the closest contact element. If there are two or more contact elements equidistant from the grid, then the minimum value of separation distance is used at the grids rather than the average, since the average gives unexpected results for coarse meshes.
2. For SOL 401, the SEPDIS describer requests the initial separation distance for grids on the source region and the final separation distance for grids on both the source and target regions. For grids on the source region, the separation distance is a scalar quantity representing the source side distance to the target in the source normal direction. For grids on the target region, the separation distance is a scalar quantity representing the target side distance to the source in the source normal direction. The normal for the projection is evaluated at the contact element location on the source face.

   The initial separation is computed at the start of the solution in the undeformed state. The contact definition must be either applied globally, or in the first subcase in order for the software to compute the initial separation. The initial separation is not computed again, even if additional contact pairs are introduced in later subcases, or if a non-sequentially dependent subcase is defined with the SEQDEP=NO command.

   The final separation distance is computed at the end of each time step in which output occurs. It is computed based on the current deformed configuration.

   When the software recovers the contact separation distance output to the grid points, it computes the gap on an element face as the average of the gap at all of the contact Gauss points within that face. The grid point result that is output is then computed as the average of the element face (average) values for all contact region element faces attached to that grid. The face averaged approach is appropriate when contact is distributed across the element faces, but when contact is very localized such that some locations on a specific element face are in contact and some are not, it is better to recover the unaveraged values. You can define SYSTEM(778)=1, and the software will use the unaveraged value from the Gauss point closest to the grid point.

   In addition for SOL 401, the SEPDIS describer requests the total and incremental slide distance for grids on both the source and target regions. The slide distance is reported in the basic coordinate system as the algebraic sum of the incremental slip (elastic and plastic) from each increment. When a contact grid goes inactive, the total slide distance output resets to zero. The incremental slide distance is the difference in the total slide distance between two output intervals.

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3. For edge-to-edge contact, force, traction, and separation distance output can be requested. By default, the contact force for axisymmetric elements is based on a 2π section. The **system cell 587** can optionally be set to 1 to select the per radian section basis. The contact force unit is consistent with the applied and reaction force unit for each element.
4. When the STATUS describer is included on the BCRESULTS command in SOL 401, an integer value indicating the contact status is output on each grid point included in a contact source or target region. The status values are:

   0: No contact exists.

   1: A sticking contact condition exist.

   2: A sliding contact condition exist.

Remarks related to SOL 402:

1. The TRACTION describer outputs the Samcef pressure results code (1305) and the Samcef friction stress results code (1306). These results are associated to the face in contact and not to grid points.
2. The SEPDIS describer outputs the Samcef normal distance results code (1307). In addition, it outputs the total and incremental sliding results. These results are computed from the initial position, not from the first contact position. The sliding results do not take into account the shell thickness and offset if they are defined. Instead, they are computed relative to the midplane of the shell. As a result, the sliding results output may be different from the actual sliding when bending of the shell occurs.

Related Topics

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BCSET

Contact Set Selection (SOLs 101, 103, 105, 108, 111, 112, 401, and 402)

Selects the contact set for SOLs 101, 103, 105, 108, 111, 112, 401, and 402.

Format:

BCSET=n

Examples:

```
BCSET=5
```

Describers:

Describer	Meaning
n	Contact set identification of a BCTSET, BCTADD or BCTPARM Bulk Data entry. (Integer>0)

Remarks:

1. BCSET references a BCTSET Bulk Data entry if there is only one contact set. When multiple contact sets are used, BCSET must reference a BCTADD bulk data entry.
2. The following table summarizes all supported contact conditions.

Table 4-17: Contact Summary

Type	Description	Solution Support
Edge-to-Edge	Contact between the edges of axisymmetric elements (CTRAX3, CQUADX4, CTRAX6, CQUADX8), plane stress elements (CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8), and plane strain elements (CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8).	Solution 101 and consecutive solutions 103, 105, 108, 111, and 112 (elements in XZ or XY plane). Solution 401 (elements in XZ or XY plane).
Surface-to-Surface	Contact between shell element faces (CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, CQUADR) and solid element faces (CHEXA, CPYRAM, CPENTA, CTETRA).	Solution 101 and consecutive solutions 103, 105, 108, 111, and 112. Solution 401.

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3. A converged SOL 101 contact condition can be included in a consecutive normal mode solution (SOL 103). The normal mode solution results, which included the SOL 101 contact conditions, can then be used in an optional dynamic response calculation (SOLs 111 and 112).

   Similarly, a converged SOL 101 contact condition can be included in a consecutive linear buckling solution (SOL 105), or in a direct frequency response solution (SOL 108).

   See the chapter on contact in the *Simcenter Nastran User's Guide* for more information.

4. When a BCSET case control command exists, CGAP elements are treated as linear contact elements if the system cell (412) OLDGAPS is set to 0 (default). You can use CGAP elements this way with surface-to-surface contact defined (BCTSET bulk entries exist), or without (no BCTSET bulk entries exist). In the case where no surface-to-surface contact is defined, "n" can point to a BCTPARM bulk entry which optionally defines PENN, PENT or PENTYP for the CGAP/linear contact elements, or to nothing if a BCTPARM bulk entry does not exist (an integer value for "n" is still required in this case). See the chapter on surface contact in the Simcenter Nastran User's Guide for more information.

5. If the contact solution is having problems converging, the fields on the BCTPARM bulk data entry can be modified from their defaults. See the BCTPARM bulk data entry, and the chapter on surface contact in the Simcenter Nastran User's Guide for more information.

6. Contact conditions can be included in SMP and RDMODES solutions. See "Recursive Domain Normal Modes Analysis (RDMODES)" in the *Parallel Processing Guide*. Contact conditions cannot be included in a GDMODES, FDMODES, or HDMODES DMP solution.

7. When defining contact regions and pairs on geometry which is not tangent continuous, creating single contact regions which cross corner transitions can result in non-uniform stress results around the corners. It is recommended to break these areas into multiple regions and pairs. See the chapter on contact in the *Simcenter Nastran User's Guide* for more information.

8. Contact definitions defined on composite solid faces which are perpendicular to the stack direction (edge faces) may produce poor stress continuity. If the contact definition is between edge faces belonging to different PCOMPS definitions, and if the number of plies on each PCOMPS definition is small and the same, and the ply thicknesses are similar, the stress continuity should be fairly smooth. This also applies to the results requested with the BCRESULTS case control command.

9. If multiple eigenvalue subcases (solutions 103 or 105) include STATSUB commands to select linear static subcases, and the linear static subcases include glue and/or linear contact definitions, the BCSET and/or BGSET case control commands must be specified in the global case. Failure to do this can result in erroneous results.

10. Contact conditions can be included in a model with main bulk data type superelements, but the contact regions must be in the residual. Contact conditions are not supported in a model with partitioned type superelements. Main bulk data and partitioned type superelements are both described in Chapter 2 of the *Simcenter Nastran Superelement User's Guide*.

11. You cannot restart from a solution which included contact conditions. You cannot add contact conditions when restarting.

12. The combination of a bolt preload in which the bolt is meshed with solid elements, the element iterative solver, and contact conditions is not supported.

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Related Topics

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BEGIN BULK

Case Control and Bulk Data Delimiter

Designates the end of the Case Control Section and/or the beginning of a Bulk Data Section.

Format:

$$\text{BEGIN}\ [\text{BULK}]\ \begin{bmatrix}\text{AUXMODEL} = \text{auxmid} \\ \text{SUPER} = \text{seid}\end{bmatrix}$$

Examples:

```
BEGIN BULK
BEGIN AUXMODEL=22
```

Describers:

Describer	Meaning
AUXMODEL	Indicates the beginning of an auxiliary model Bulk Data Section.
auxmid	Auxiliary model identification number. (Integer>0)
SUPER	Indicates the beginning of partitioned superelement Bulk Data Section.
seid	Superelement identification number. (Integer ≥ 0)

Remarks:

1. BEGIN BULK is not required if there are no Bulk Data entries and there is no ENDDATA entry. The BEGIN SUPER or BEGIN AUXMODEL Bulk Data must lie between BEGIN BULK and ENDDATA entries.
2. For an auxiliary model, AUXMID is referenced by the AUXMODEL Case Control command.
3. Partitioned Bulk Data Sections defined by BEGIN SUPER are used to define only one superelement each. Bulk Data commands which define superelements are ignored in partitioned Bulk Data Sections.

   Superelements specified by a BEGIN SUPER entry can be automatically attached to other superelements based on relative location of grid points. For connection to the downstream superelement, the global coordinate directions of the attachment grid points of the upstream superelement will be internally transformed to the global coordinate directions of the grid points of the downstream superelement. For displacement data recovery, the output will be in the original global coordinate directions.

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Related Topics

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BGRESULTS

Glue Result Output Request (SOLs 101, 103, 105, 401, and 402)

Format:

$$\text{BGRESULTS}\left[\left(\underline{\text{TRACTION}},\text{FORCE},\text{SEPDIS},\left[\begin{matrix}\underline{\text{PRINT}}\\\text{PLOT}\\\text{PUNCH}\end{matrix}\right],\left[\begin{matrix}\underline{\text{SORT1}}\\\text{SORT2}\end{matrix}\right]\right)\right]$$
$$=\left\{\begin{matrix}\text{ALL}\\\text{n}\\\underline{\text{NONE}}\end{matrix}\right\}$$

Examples:

```
BGRESULTS=ALL
BGRESULTS(FORCE,PLOT)=ALL
BGRESULTS(TRACTION,FORCE,PLOT)=ALL
```

Describers:

Describer	Meaning
TRACTION	Glue normal traction (scalar) and in-plane glue tractions (vector in basic coordinate system) are output for each glue grid point. Traction units are force per area.
FORCE	Glue force vector is output for each glue grid point.
SEPDIS	For SOLs 401 and 402 only. Requests the final separation distance, and the total and incremental slide distance for grids on the source and target regions.
PRINT	The printer will be the output medium.
PLOT	Computes and puts glue results in OP2 file only.
PUNCH	The punch file is the output media.
SORT1	Output will be presented as a tabular listing of grid points for each load, frequency, eigenvalue or time, depending of the solution sequence.
SORT2	Output will be presented as a tabular listing of load,frequency or time for each grid point (not supported for SOL 401).
ALL	Glue results at all contact grid points will be output.

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Describer	Meaning
NONE	Glue results will not be output.
n	Set identification of a previously appearing SET command. Only glue grid points with identification numbers that appear on this SET command will be output. (Integer>0)

REMARKS:

1. The glue traction request is supported in solutions 101, 103 (but only for a static preload subcase if present), 105, and 401 (static subcase only). The glue results from SOL 105 are a result of the applied static loads which are not necessarily the loads at which buckling occurs.
2. Sliding glue can be requested in SOL 401 with the parameter setting SLIDE=1 on the BGPARM entry. Sliding glue includes a normal stiffness, but no tangential stiffness. When sliding glue is requested, the SEPDIS describer can be used to request slide distance output. The slide distance represents the tangential distance traveled by the source and target grids relative to each other. The total and incremental slide distance output is written in the basic coordinate system on grid points for both the source and target regions. The total slide distance is the vector sum of the incremental slide distances from all time steps. If a subcase is sequentially dependent, the total slide distance will also include the total slide distance from a previous static subcase. Any gaps between the glue edges and surfaces are preserved as slidiing occurs.

Remarks related to SOL 402:

1. The BGRESULTS command is not supported In SOL 402. You can use the MPCFORCES command to output the glue results.

Related Topics

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BGSET

Glue Contact Set Selection

Selects the glue contact set.

Format:

BGSET=n

Examples:

```
BGSET=5
```

Describers:

Describer	Meaning
n	Glue contact set identification of a BGSET Bulk Data entry. (Integer>0)

Remarks:

1. The default search distance defined on the BGSET bulk entry will glue the closest sections of the source and target regions. This value can be adjusted to control the parts of a region to be glued. The software searches in both the positive and negative normal directions from the source region.
2. The following table summarizes all supported glue conditions.

Table 4-18: Glue Summary

Type	Description	Solution Support
Edge-to-Edge	Glue between the edges of shell, axisymmetric, plane stress, and plane strain elements. Elements being glued must be in the same plane.	All solutions except 144 -146. SOL 401 does not support gluing of shell edges. In heat transfer solutions, they are treated as constant conductivity connections. See Remark 3. They cannot be used to represent fluid-to-fluid glue connections in a coupled acoustic analysis.
Edge-to-Surface	Glue between shell element edges and shell or solid element faces.	All solutions except 144 -146, and 401.

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		In heat transfer solutions, they are treated as constant conductivity connections. See Remark 3. They cannot be used to represent fluid-to-fluid glue connections in a coupled acoustic analysis.
Surface-to-Surface	Glue between shell or solid element faces and shell or solid element faces.	All solutions except for SOLs 144–146. In heat transfer solutions, they are treated as constant conductivity connections. See Remark 3. They can be used to represent fluid-to-fluid glue connections in a coupled acoustic analysis.

3. Glue is supported in the heat transfer solutions 153 and 159, and in a SOL 101 heat transfer/structural combined solution.

   The SOL 101 combined solution requires both a heat transfer and a structure subcase. See the HEATSTAT parameter for information. Since these subcases represent different analysis types and glue has a different meaning in each, to define a unique set of glue parameters (penalty values) for each subcase, you must create unique glue sets for each subcase. Specifically, you must create a unique BGSET case control, BGSET bulk entry, and BGPARM bulk entry for each subcase. You select the same glue regions in each of the glue sets.

4. When defining glue regions and pairs on geometry which is not tangent continuous, creating single glue regions which cross corner transitions can result in non-uniform stress results around the corners. It is recommended to break these areas into multiple regions and pairs. See the chapter on constraints in the *Simcenter Nastran User's Guide* for more information on glue definitions.

5. Glue definitions defined on composite solid faces which are perpendicular to the stack direction (edge faces) may produce poor stress continuity. If the glue definition is between edge faces belonging to different PCOMPS definitions, and if the number of plies on each PCOMPS definition is small and the same, and the ply thicknesses are similar, the stress continuity should be fairly smooth. This also applies to the results requested with the BGRESULTS case control command.

6. Inertia relief (PARAM,INREL) and glue conditions can be included together when using the sparse solver (default), but not the element iterative solver.

7. Shell offsets are not accounted for in edge-to-surface gluing. The gluing occurs at the grid locations and not at the offset location.

8. If multiple eigenvalue subcases (solutions 103 or 105) include STATSUB commands to select linear static subcases, and the linear static subcases include glue and/or linear contact definitions, the BCSET and/or BGSET case control commands must be specified in the global case. Failure to do this can result in erroneous results.

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9. Glue conditions can be included in a model with main bulk data type superelements. Each glue pair must be contained in a single superelement or in the residual. That is, the glue regions cannot cross superelement boundaries. Glue conditions are not supported in a model with partitioned type superelements. Main bulk data and partitioned type superelements are both described in Chapter 2 of the *Simcenter Nastran Superelement User's Guide*.
10. When gluing solid element faces, a glue condition is not created if the dot product of the element face normals is greater or equal to 0. Including SYSTEM(516)=1 will override this and allow the glue condition to be created if the search distance is satisfied.
11. You cannot add glue conditions when restarting. You can restart from a solution which included glue conditions, but only to request addition output.
12. Glue definitions model linear glue behavior in solution 106. The glue condition does not update in regard to geometry updates, nonlinear material changes, or orientation changes. A glue definition should only be defined in portions of the structure that exhibit linear behavior.

Related Topics

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BOLTLD

Bolt Preload Set Selection

Selects a bolt preload set ID for bolt preload processing.

Format:

BOLTLD=n

Examples:

```
BOLTLD=5
```

Describers:

Describer	Meaning
n	In SOLs 101, 103, 105, 107 through 112, set identification number (SID) of a BOLTFRC or BOLTLD bulk entry. (Integer>0) In SOLs 401, set identification number (SID) of a BOLTFRC, BOLTSEQ, or BOLTLD bulk entry. (Integer>0) In SOLs 402, set identification number (SID) of a BOLTFRC or BOLTLD bulk entry. (Integer>0)

Remarks:

1. Bolt preload is supported in SOLs 101, 103, 105, 107 through 112, 401, and 402.

   SOLs 101, 103, 105, 107 through 112 support bolts modeled with the 1D elements CBAR and CBEAM, and the 3D solid elements CHEXA, CPENTA, and CTETRA.

   SOLs 401 and 402 support bolts modeled with the 1D elements CBAR and CBEAM, the 3D solid elements CHEXA, CPENTA, and CTETRA, the plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, and CPLSTS8, and the axisymmetric elements CTRAX3, CTRAX6, CQUADX4, and CQUADX8.

   See the **BOLT** bulk entry.

2. Bolt preload is not supported in a solution RESTART.

3. The following remarks appy to SOLs 101, 103, 105, 107 through 112.

   - In SOLs 101, 103, 105, 107 through 112, for the special case where the sparse solver is used, the model does not contain contact, and the keyword scratch = yes, BOLTLD case control commands in the subcases must be ordered. For example, if a model contains three subcases with BOLTLD = 1 used twice and BOLTLD = 2 used once, the BOLTLD case control commands should be ordered with:

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- The first and second subcases containing BOLTLD = 1 and the third subcase containing BOLTLD = 2.

- The first subcase containing BOLTLD = 2 and the second and third subcases containing BOLTLD = 1.

- Superelements with preloaded bolts are allowed. However, the elements used to define the bolts along with the bolt preload forces must be in the residual.
- Inertia relief (PARAM,INREL) is not supported with bolt preload.
- For solid bolts, the case control section must include an SPC set reference. The SPC set may be empty (no DOF assigned).
- The combination of a bolt preload in which the bolt is meshed with solid elements, the element iterative solver, and contact conditions is not supported.
- Enforced displacements defined with non-zero SPC or SPCD bulk entries can be included with a bolt preload. When an SPCD is defined with a bolt preload, system cell 581 must be set to 1 to override an error trap that exists in the software. The software behaviour changes during the bolt preload solution steps for various input scenarios (with and without contact defined; enforced displacements defined with SPCD entries or with SPC entries). See “Enforced Displacements and Bolt Preloads” in the *User’s Guide* for a description of the software behaviour.
- For the dynamic solutions 103, and 107 through 112, a static subcase containing the BOLTLD case control command must be referenced by a STATSUB case control command. This is the only method for a bolt preload to be part of a dynamics solution.
- When a bolt preload is defined on CBEAM or CBAR elements and contact is in the model, the software runs an initial preload solution to compute the strain in the bolt before applying any service loads. When RESET=1 is defined on the BCTPARM bulk entry for the contact condition, for all consecutive solutions for the same subcase as the preload or for consecutive subcases, the software applies the computed bolt strain from the initial preload solution, and it uses the final contact status from the initial bolt preload solution as the starting contact status.
- When contact conditions are defined and the set ID of any one of the BOLTLD, SPC, or MPC case control commands changes from one subcase to the next, the contact conditions will be recomputed in the new subcase. That is, the contact status in the new subcase will not be the final contact status from the previous subcase. This will occur regardless of how the RESET parameter is defined on the BCTPARM bulk entry.

4. For solution 401, the software iterates until the resulting bolt force matches, within a tolerance, the bolt preload you requested. The final bolt strain required to achieve the desired bolt force includes the effects of any other defined loads (thermal loads, contact forces, and conditions from a previous subcase). For example, if you compare the results from a solution with only bolt preloads defined with the results from the same model with the addition of thermal loads and contact, the bolt force will be the same, but the final bolt strain needed to achieve the bolt force will be different.

5. For solution 401, a bolt preload subcase requires the ANALYSIS=PRELOAD and BOLTLD=n commands. A subcase with ANALYSIS=STATICS and the BOLTLD=n will cause a fatal error. Multiple bolt preload subcases can be defined, they can be defined as sequentially dependent or nonsequentially dependent, and can optionally include the new bolt preload sequence described below. As a result, you can define bolt preload subcases to apply or remove bolt forces in any sequence.

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6. The following rules apply when BOLTFRC, BOLTLD, or BOLTSEQ bulk entries are defined using the same SID.
   - If a BOLTSEQ entry is referenced by the BOLTLD case control command, BOLTFRC and BOLTLD entries with the same SID as the BOLTSEQ are only used if they are referenced in a sequence defined on the BOLTSEQ entry.
   - If a BOLTLD bulk entry is referenced by the BOLTLD case control command or by the BOLTSEQ bulk entry, BOLTFRC entries with the same SID as the BOLTLD entry are used if they are referenced on the BOLTLD entry.
   - The software will fatal if multiple BOLTFRC entries with the same SID are referenced.
   - Only one BOLTSEQ entry can be referenced. The software will fatal if multiple BOLTSEQ entries with the same SID are referenced.
   - Only one BOLTLD entry can be referenced. The software will fatal if multiple BOLTLD entries with the same SID are referenced.

Remarks related to SOL 402:

1. Bolt loads can be defined in a preload subcase (ANALYSIS=PRELOAD), and in statics and dynamics subcases (ANALYSIS=STATICS and ANALYSIS=DYNAMICS).
2. Multiple bolt preload subcases can be defined.

Related Topics

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BOLTRESULTS

Requests bolt results output in SOLs 401 and 402.

Requests bolt results output in SOLs 401 and 402.

Format:

$$\text{BOLTRESULTS}\left[\left(\underline{\text{PRINT}},\text{PUNCH},\text{PLOT}\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \underline{\text{NONE}}\end{matrix}\right\}$$

Examples:

```
BOLTRESULTS=ALL
BOLTRESULTS(PRINT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all bolt IDs.
n	Set identification number of a previously appearing SET command. The referenced SET lists the bolt IDs in which output is requested. (Integer>0)
NONE	Output is not computed. (Default)

Remarks:

1. The BOLTRESULTS case control is only supported in SOLs 401 and 402.
2. The output requested with the BOLTRESULTS case control command includes the axial force, bending moment, the shear forces, and the axial strain for each bolt ID listed on the SET command. The output is computed relative to the coordinate system used to define the bolt axis. The computed results for a bolt are written to each element defining that bolt. For example, if you display the bending moment in a post processor, the bolt will display as a constant.

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3. For SOL 401, when a ETYPE=1 is used on the BOLT bulk entry, if multiple CBAR or CBEAM elements are used to model a bolt, only one of the elements needs to be listed on the BOLT entry. Although, if you are requesting results with the BOLTRESULTS case control command and you would like these results stored for all of these CBAR or CBEAM elements, then you should list all of the elements on the BOLT entry.

Remarks related to SOL 402:

1. The BOLTRESULTS command output is the sum of forces and moments across the bolt cross section. The computed total force and moment is output on each element touching the bolt cut.
2. When ETYPE=1 is used on the BOLT bulk entry, if multiple CBAR and CBEAM elements are used to model a bolt, only one of the elements needs to be listed on the BOLT entry. Enter the element ID in the EID1 field. If multiple CBAR and CBEAM elements are listed, the software will only use the element in the EID1 field for the bolt preload calculation.

   If you are requesting results with the BOLTRESULTS case control command, regardless of whether you list one or multiple elements on the BOLT entry, the software only stores these results on the element in the EID1 field.

Related Topics

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BOUTPUT

Contact Output Requests

Selects contact regions for output.

Format:

$$\text{BOUTPUT}\left[\left(\begin{bmatrix}\text{SORT1}\\ \text{SORT2}\end{bmatrix},\ \begin{bmatrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{bmatrix}\right)\right] = \begin{Bmatrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{Bmatrix}$$

Examples:

```
BOUTPUT=ALL
BOUTPUT=5
```

Describers:

Describer	Meaning
SORT1	Output is presented as a tabular listing of secondary nodes for each load or time depending on the solution sequence.
SORT2	Output is presented as a tabular listing of load or time for each secondary node.
PRINT	The print file is the output media.
PUNCH	The punch file is the output media.
PLOT	Generate output histories for secondary nodes, but do not print.
ALL	Histories of all the secondary nodes listed in all the BOUTPUT Bulk Data entries are output. If no BOUTPUT Bulk Data entries are specified, histories of all the secondary nodes in all the contact regions are output.
n	Set identification of previously appearing SET command. Only contact regions with identification numbers that appear on the SET command are selected for output. If there is a BOUTPUT Bulk Data entry for a contact region selected via the set command, histories for secondary nodes listed in the Bulk Data entry are output. If there is no BOUTPUT Bulk Data entry for a contact region selected via the set command, histories for all the secondary nodes in that contact region are output.
NONE	Result histories for secondary nodes are not calculated or output.

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Remarks:

1. BOUTPUT is processed in SOLs 106, 129, 153, and 159 only.
2. SORT1 is the default in SOLs 106 and 153. SORT2 is the default in SOLs 129 and 159.

Related Topics

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BVTSET

Virtual Thread Set Selection (SOL 402)

Selects the virtual thread set for SOL 402.

Format:

BVTSET=n

Examples:

```
BVTSET=5
```

Describers:

Describer	Meaning
n	Virtual thread set identification of the BVTSET, BVTADD or BVTPAR bulk entries. (Integer>0)

Remarks:

1. BVTSET command must be defined above the subcase level (globally).

Related Topics

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CKGAP

Gap Result Output Request

Requests gap result output for chocking elements in SOL 401.

Format:

$$\text{CKGAP}\left[\left(\left[\begin{matrix}\underline{\text{GRID}}\\ \text{GAUSS}\\ \text{BOTH}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{PRINT}}\\ \text{PUNCH}\\ \text{PLOT}\end{matrix}\right]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
CKGAP(GAUSS)=ALL
```

Describers:

Describer	Meaning
GRID	Requests gap results for corner grids. (Default)
GAUSS	Requests gap results for Gauss points.
BOTH	Requests gap results for corner grids and Gauss points.
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
PLOT	Generates results for requested set but no printer output.
ALL	Results are output for all chocking elements.
n	Identification number of a SET case control command that lists the chocking elements for which results are output. (Integer>0)
NONE	No results are output for any chocking elements.

Remarks:

1. The software uses the gap result at the gauss points to determine if the element is chocked or unchocked.
2. Gap results at the grid points is computed using a linear extrapolation of the gap results at the gauss points. Since gap results can vary across the elements, the linear extrapolation can introduce

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inaccurate gap results at the grid points. This becomes more significant when there is a large variation of gap across an element, which is possible with a coarse mesh.

Related Topics

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CLOAD

Static Load Request for Upstream Superelement Loads

Requests a CLOAD Bulk Data entry that defines a list of superelement loads and their scale factors in nonlinear static analysis only.

Format:

CLOAD=n

Examples:

```
CLOAD=15
```

Describers:

Describer	Meaning
n	Identification number of a unique CLOAD Bulk Data entry. (Integer>0)

Remarks:

1. This command may only appear in the residual structure subcases (see **"SUPER"**) and if used it must be specified in all of them.
2. The CLOAD Bulk Data entry must reference previously processed LSEQ (load sequence) Bulk Data that was requested by LOADSET Case Control commands on the upstream (SUPER ≠ 0) subcases.
3. The resulting load is added to those produced by LOAD and TEMP(LOAD) Case Control commands in the residual structure subcases.

Related Topics

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CMETHOD

Complex Eigenvalue Extraction Method Selection

Selects complex eigenvalue extraction parameters.

Format:

CMETHOD=n

Examples:

```
CMETHOD=77
```

Describers:

Describer	Meaning
n	Set identification number of EIGC (or EIGP) bulk entry. (Integer>0)

Remarks:

1. The CMETHOD command must be specified in order to compute complex eigenvalues.
2. See the description for the parameter, UNSYMF, to perform complex eigenvalue analysis in Solution 106.

Related Topics

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CRSTRN

Requests creep strain output at grid points for SOLs 401 and 402.

Requests creep strain at grid points.

Format:

$$\text{CRSTRN}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
CRSTRN=ALL
CRSTRN(PRINT,PLOT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

Remarks:

1. Both PRINT and PUNCH may be requested.
2. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.
3. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

Remarks related to SOL 402:

1. The output results are the Samcef creep strain results (codes *300).

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2. CRSTRN is not supported for CBEAM, CROD, and CBAR elements.

Related Topics

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CSMSET

Wetted Face and Edge Set Selection

Selects a set of wetted faces and edges for a SOL 401 FSI Co-Simulation with STAR CCM+.

Format:

CSMSET=n

Examples:

```
CSMSET=200
```

Describers:

Describer	Meaning
n	Set identification number selecting a CSMSET or a CSMADD bulk entry that defines the wetted faces and edges for a SOL 401 FSI Co-Simulation with STAR CCM+. (Integer > 0)

Remarks:

1. CSMSET references a CSMSET bulk entry if there is only one CSMSET entry defined. When multiple CSMSET entries are defined, CSMSET ID references a CSMADD bulk entry, which combines the multiple CSMSET entries into a single set.
2. The CSMSET case control, CSMSET bulk entry, and CSMADD bulk entry are supported only in the context of SOL 401 Multistep Nonlinear STAR CCM+ Co-simulation.
3. The CSMSET case control command can only be defined above the subcase level (global).

Related Topics

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CSSCHD

Aerodynamic Control Surface Schedule

Selects control system schedule information.

Format:

CSSCHD=n

Examples:

```
CSSCHD=10
```

Describers:

Describer	Meaning
n	Set identification of a control system schedule that appears on a CSSCHD Bulk Data entry.

Remarks:

One or more CSSCHD entries can be invoked by this entry.

Related Topics

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CYCFORCES

Requests MPC force output at the grid points selected for the automatic coupling in SOLs 401 and 402 cyclic symmetry analysis.

Requests MPC force output at the grid points selected for the automatic coupling in a SOL 401 cyclic symmetry analysis.

Format:

$$\text{CYCFORCES}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
CYCFORCES=ALL
CYCFORCES(PRINT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. The SET command is a list of grid point IDs. (Integer>0)
NONE	Output is not computed.

Remarks:

1. Only supported in a static subcase for SOL 401.
2. The CYCFORCES output only occurs on the grid points included in a coupling source and target region. The grid points listed on the referenced SET command should be included in a coupling region. Grid points listed in a referenced SET which are not in a source or target region will be ignored.

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Remarks related to SOL 402:

1. The CYCFORCES command is not supported In SOL 402. You can use the MPCFORCES command instead.

Related Topics

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CYCSET

Selects Cyclic Symmetric Boundary Coupling

Selects a cyclic symmetric boundary coupling in SOL 401, SOL 402, and all SOL 414 solution types.

Format:

CYCSET=n

Examples:

```
CYCSET=5
```

Describers:

Describer	Meaning
n	Set identification number of a CYCSET or CYCADD bulk entry. See Remark 1. (Integer>0)

Remarks:

1. The set identification number, n, can reference either a single CYCSET bulk entry or a CYCADD bulk entry that combines multiple CYCSET bulk entries.
2. The CYCSET case control must be defined above the subcase level. Thus, the MPCs generated by the automatic coupling are used in every subcase (cyclic modes, static, and normal modes subcases).
3. To solve cyclic symmetry in SOL 414,103 and SOL 414,111 solutions, no rotor (RMETHOD case control command and ROTORD bulk entry) can be defined.

   You can define a rotor in SOL 414,101, SOL 414,110, and SOL 414,129 solutions.

Related Topics

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CZRESULTS

Cohesive Element Output Request for SOLs 401 and 402

Requests results output for cohesive elements.

Format:

$$\text{CZRESULTS}\left[\left(\underline{\text{TRACTION}},\text{RLTDIS},\text{DAMAGE},\left[\begin{matrix}\underline{\text{PRINT}}\\ \text{PUNCH}\\ \text{PLOT}\end{matrix}\right]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
CZRESULTS(DAMAGE,PUNCH)=100
```

Describers:

Describer	Meaning
TRACTION	Requests normal and in-plane traction results. See Remarks 1 and 2. (Default)
RLTDIS	Requests normal and in-plane relative displacement results. See Remarks 1 and 2.
DAMAGE	Requests damage results. See Remark 1.
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
PLOT	Generates results for requested set but no printer output.
ALL	Results are output for all cohesive elements.
n	Identification number of a SET case control command that lists the cohesive elements for which results are output. (Integer>0)
NONE	No results are output for any cohesive elements.

Remarks:

1. All results are reported at the corner grid points.
2. Traction and relative displacement are both output into three components. The components are output in the material coordinate system relative to the deformed state. Tractions have units of force per unit area. Relative deformation has units of length.

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Related Topics

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DAMPING

Damping Selection

Selects the damping definition and, for transient response analysis or a modal-based analysis that writes an interface file for external multibody dynamics or control system software, the structural damping to viscous damping conversion method.

Format:

$$\text{DAMPING}\left[\frac{\text{STRUCTURE}}{\text{FLUID}}\right]=\text{n}$$

Examples:

```
DAMPING=33
DAMPING(FLUID)=34
```

Describers:

Describer	Meaning
STRUCTURE or FLUID	The referenced DAMPING bulk entry is applied to the structural or fluid portion of the model. (Default = STRUCTURE)
n	Set identification number of a DAMPING bulk entry. (Integer>0)

Remarks:

1. DAMPING case control commands must be placed above the subcases.
2. A single DAMPING case control command can reference a single DAMPING bulk entry only.
3. In a coupled fluid-structure analysis, include a DAMPING case control command to specify the damping for the structural portion of the model and a second DAMPING case control command to specify the damping for the fluid portion of the model.

Remarks related to SOL 401:

1. For SOL 401, you can include the DAMPING=n case control command in the modal frequency response subcase to select the DAMPTAB bulk entry and the DAMPING bulk entry. You can define both the DAMPTAB entry and the DAMPING entry, and they can share the same SID. See Damping definition in the Modal frequency response subcase (SOL 401) topic in the Multi-Step Nonlinear User's Guide for details on all supported damping in this subcase.

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Remarks related to SOLs 402 and 414:

1. Because the DAMPING case control command and DAMPING bulk entry are not supported in SOLs 402 and 414, for these solution sequences you define damping using parameter specifications such as PARAM,G, PARAM,W3.

Related Topics

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DEFORM

Element Deformation Static Load

Selects the element deformation set.

Format:

DEFORM=n

Examples:

```
DEFORM=27
```

Describers:

Describer	Meaning
n	Set identification number of DEFORM Bulk Data entries. (Integer>0)

Remarks:

1. DEFORM Bulk Data entries will not be used unless selected by the DEFORM command in the Case Control Section.
2. DEFORM is only applicable in linear statics, inertia relief, differential stiffness, and buckling problems (SOLs 101, 105, 114, and 200), and will produce a fatal message in other solution sequences.
3. The total load applied will be the sum of external (LOAD), thermal (TEMP(LOAD)), element deformation (DEFORM) and constrained displacement loads (SPC, SPCD).
4. Static, thermal, and element deformation loads should have unique identification numbers.
5. In the superelement solution sequences, if this command is used in a cold start, it must also be specified in the restart.

Related Topics

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DESGLB

Request Design Constraints at the Global Level

Selects the design constraints to be applied at the global level in a design optimization task.

Format:

DESGLB=n

Examples:

```
DESGLB=10
DESG=25
```

Describers:

Describer	Meaning
n	Set identification of a set of DCONSTR or a DCONADD Bulk Data entry identification number. (Integer>0)

Remarks:

1. If used, this command must occur before the first SUBCASE.
2. A DESGLB command is optional. It should be used to invoke either constraints on responses that are independent of a particular subcase (such as WEIGHT or VOLUME), or constraints that involve a combination of responses from various subcases (see also DRSPAN Case Control command).
3. DESGLB constraints on DRESP2 or DRESP3 responses that are independent of DRESP1 responses are also acceptable.
4. When DESGLB constraints refer to second level responses (DRESP2 and/or DRESP3) that are based on combinations of DRESP1 responses from more than one subcase, each such DRESP1 must have been assigned to a particular subcase by way of the DRSPAN Case Control command. Such DESGLB constraints are output for the last analyzed subcase associated with the second level response in question.

Related Topics

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DESOBJ

Design Objective

Selects the DRESP1, DRESP2, or DRESP3 entry to be used as the design objective.

Formats:

Format for global definition:

$$\text{DESOBJ}\left[\left(\begin{matrix}\underline{\text{MIN}}\\ \text{MAX}\end{matrix}\right), \text{SCSET} = \left(\begin{matrix}\underline{\text{FIRST}}\\ \text{ALL}\\ \text{m}\end{matrix}\right), \text{SCFUNC} = \left(\begin{matrix}\underline{\text{MAX}}\\ \text{MIN}\\ \text{SUM}\\ \text{AVG}\\ \text{SSQ}\\ \text{RSS}\end{matrix}\right), \text{ENFUNC} = \left(\begin{matrix}\underline{\text{NONE}}\\ \text{MAX}\\ \text{MIN}\\ \text{SUM}\\ \text{AVG}\\ \text{SSQ}\\ \text{RSS}\end{matrix}\right)\right] = \text{n}$$

Format for subcase definition:

$$\text{DESOBJ}\left[\left(\begin{matrix}\underline{\text{MIN}}\\ \text{MAX}\end{matrix}\right), \text{ENFUNC} = \left(\begin{matrix}\underline{\text{NONE}}\\ \text{MAX}\\ \text{MIN}\\ \text{SUM}\\ \text{AVG}\\ \text{SSQ}\\ \text{RSS}\end{matrix}\right)\right] = \text{n}$$

Examples:

```
DESOBJ=10
DESO=25
DESOBJ(MIN,SCSET=ALL,SCFUNC=MAX)=10
```


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Describers:

Describer	Meaning
General Describers	
MIN	Specifies that the objective is to be minimized.
MAX	Specifies that the objective is to be maximized.
n	ID number of a DRESP1, DRESP2, or DRESP3 bulk entry. The DRESP2 and DRESP3 options are valid only for the simpler DESOBJ command (see Remark 11). (Integer>0)
SCSET Describers	
FIRST	The objective response is computed only for the first subcase (legacy behavior).
ALL	Spans the objective response across all compatible subcases (Default).
m	Spans the objective response across the subcases listed in SET m.
SCFUNC and ENFUNC Describers	
MAX	Combine the responses from the different subcases/entities by taking the maximum value (Default, see remarks 13 and 14).
MIN	Combine the responses from the different subcases/entities by taking the minimum value (See remarks 13 and 14).
SUM	Combine the responses from the different subcases/entities by computing the sum.
AVG	Combine the responses from the different subcases/entities by computing the average.
SSQ	Combine the responses from the different subcases/entities by computing the sum of the squares.
RSS	Combine the responses from the different subcases/entities by computing square root of the sum of the squares.
ENFUNC only Describers	
NONE	No combination is made across entities listed on the response (legacy behavior).

Remarks:

1. A DESOBJ command is required for a design optimization task and is optional for a sensitivity task. No more than one DESOBJ may appear in Case Control.
2. If the DESOBJ command is specified within a SUBCASE, the identified DRESPi Bulk Data entry uses a response only from that subcase. If DESOBJ appears above all SUBCASE commands and there are multiple subcases, it uses a global response (See also Remarks 4, 5, 9, 10, 11 and 12).
3. The minimization or maximization is on the algebraic value of the objective function. Thus, be careful if you expect negative values or if there is a possibility for the objective function to change sign

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during optimization. For example, to minimize the absolute value of a displacement component, the absolute value or the square of the relevant response may need to be specified with a DRESP2 response that references the DRESP1 for that particular displacement component. If it is possible for the objective function to change sign, as in the case of a displacement component, the change of sign can cause the objective function to move in an undesirable direction. When such a possibility exists, it is best to place appropriate constraints on the value of the response that is used for the objective function.

4. A global response, as mentioned in Remark 2, may be either a subcase-independent response or a subcase-dependent response (see remarks 9, 10, 11 and 12) or one that includes a combination of DRESP1 responses from more than one subcase (see also DRSPAN Case Control command, and the Remarks for DESGLB).

5. When the global response includes a combination of DRESP1 responses from more than one subcase, each such DRESP1 response must have been assigned to a particular subcase using the DRSPAN Case Control command. A DESOBJ referencing such a global response is output for the last analyzed subcase associated with the second level response in question.

6. When the design is infeasible due to violated constraints, the objective function may move in the direction opposite to that desired in an attempt to reach the feasible region.

7. The best design found is not necessarily the one in the last design cycle or the one with the lowest value of the objective function. If there are one or more feasible designs (very small positive within specified or default tolerance, or negative maximum normalized constraint values), the best design is the feasible design with the lowest value of the objective function. If there are no feasible designs (all positive non-trivial values of the maximum normalized constraint), the best infeasible design found is that with the smallest value of the maximum normalized constraint, without regard for the value of the objective function. However, for topology optimization, see also Remark 8.

8. For topology optimization jobs, if there are no maximum size, casting or additive manufacturing constraint, the determination of the best design follows the logic described in Remark 7. Otherwise, the comparisons for the best design selection is based only on those design cycles where manufacturing constraints have been activated (see also Parameter BDMNCON).

9. SCSET, SCFUNC and ENFUNC options can only be used if n refers to subcase-dependent DRESP1 responses. They do not apply to responses related to frequency (DFREQ and MFREQ), transient (MTRAN) and aeroelasticity (SAERO and FLUTTER) subcases types. They are not applicable to DRESP1 responses such as WEIGHT, VOLUME, FRDISP or TSTRESS, or to DRESP2 or DRESP3 responses.

10. The SCSET, SCFUNC and ENFUNC options are not supported with superelements.

11. SCFUNC can be used such that the objective function is evaluated for each of the selected (SCSET) and relevant subcases. The values resulting from each subcase are then combined using the operator specified after SCFUNC to compute the objective value. This option can only be used when DESOBJ card is placed at global level and not inside a subcase.

12. ENFUNC applies to DRESP1, which can reference several entities (for example, a displacement component over a number of grid points, stress over several elements). These operators allow combining all the listed entities onto the design objective.

13. The default for SCFUNC and ENFUNC option is MAX. However, to preserve the consistency of the optimization problem, this default is switched to MIN in case the objective is requested to be maximized (for example, DESOBJ(MAX)=n).

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14. MAX and MIN operators (SCFUNC and ENFUNC) consider response algebraic values (with their sign) alike corresponding DRESP2 integrated functions. Hence, if the objective response corresponds to both positive and negative values across selected subcases and/or entities, MAX always selects the positive value with the largest magnitude and MIN selects the negative value with the largest absolute value.

Related Topics

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DESSUB

Design Constraints Request at the Subcase Level

Selects the design constraints to be used in a design optimization task for the current subcase.

Format:

DESSUB=n

Examples:

```
DESSUB=10
DESS=25
```

Describers:

Describer	Meaning
n	Set identification of a set of DCONSTR and/or a DCONADD Bulk Data entry identification number. (Integer ≥ 0)

Remarks:

1. A DESSUB command is required for every subcase for which constraints are to be applied. An exception to this is 'global constraints', which are selected by the DESGLB command.
2. All DCONSTR and DCONADD Bulk Data entries with the selected set ID will be used.
3. A DESSUB command appearing before all subcases is applicable for all subcases except for those subcases where a different DESSUB command appears.

Related Topics

Browse Simcenter Nastran documentation

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DISPLACEMENT

Displacement Output Request

Requests the form and type of displacement vector output.

Format:

$$\text{DISPLACEMENT}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{REAL or IMAG}}\\ \text{PHASE}\end{matrix}\right],\left[\begin{matrix}\underline{\text{ABS}}\\ \text{REL}\end{matrix}\right],\left[\begin{matrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{matrix}\right],\right.\right.$$
$$\left.\left.\left[\begin{matrix}\underline{\text{RPRINT}}\\ \text{NORPRINT}\end{matrix}\right],\ [\text{RPUNCH}],\ \text{SOLUTION}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{setf}\\ \text{PEAKOUT}\end{matrix}\right\}\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
DISPLACEMENT=5
DISPLACEMENT(REAL)=ALL
DISPLACEMENT(SORT2, PUNCH, REAL)=ALL
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence. (Default)
SORT2	Output a tabular listing of load, frequency or time for each grid point.
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
PLOT	Generate, but do not print, displacement results.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
ABS	For enforced motion dynamic analysis, displacement results are output as absolute displacement. (Default)

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Describer	Meaning
REL	For enforced motion dynamic analysis, displacement results are output relative to the enforced motion input.
PSDF	Requests power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 8.
ATOC	Requests autocorrelation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 8.
CRMS	Requests cumulative root mean square function be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 8.
RMS	Requests root mean square and zero crossing functions be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 8. (Default)
RALL	Requests all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 8.
RPRINT	Writes random analysis results to the print file. See Remark 8. (Default)
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 8.
RPUNCH	Writes random analysis results to the punch file. See Remark 8.
SOLUTION = ALL	Output displacement for all frequencies. See Remark 13. (Default)
SOLUTION = setf	Output displacement only for the frequencies listed on a previously appearing SET command with setf as the identification number. See Remark 13. (Integer > 0)
SOLUTION = PEAKOUT	Output displacement only for frequencies identified with the PEAKOUT command. See Remark 13.
ALL	Output displacement for all points.
n	Output displacement only for the points listed on a previously appearing SET command with n as the identification number. (Integer > 0)
NONE	Do not output displacement for any points. (Default)

Remarks:

1. Both PRINT and PUNCH can be requested.
2. The defaults for SORT1 and SORT2 depend on the analysis type:
   - SORT1 is the default in static analysis, frequency response, steady state heat transfer analysis, real and complex eigenvalue analysis, flutter analysis, and buckling analysis. If SORT2 is selected

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in a frequency response solution for one or more of the commands ACCE, DISP, FORC, GPFO, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands are also output in SORT2 format.

- SORT2 is the default in transient response analysis (structural and heat transfer). SORT2 is not available for real eigenvalue (including buckling), complex eigenvalue, or flutter analysis. If SORT1 is selected in a transient solution for one or more of the commands ACCE, DISP, ENTH, FORC, GPFO, HDOT, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands are also output in SORT1 format.
- XY plot requests force SORT2 format, thus overriding SORT1 format requests.

3. By default, the PRESSURE case control command requests pressure output at fluid grid points only and the DISPLACEMENT case control command requests displacement at structural grid points only. In versions prior to NX Nastran12, the PRESSURE and DISPLACEMENT case control commands are interchangeable. Use SYSTEM(640) to revert to the pre-NX Nastran12 behavior.
4. DISPLACEMENT = NONE overrides an overall output request.
5. The PLOT option is used when curve plots are desired in the magnitude/phase representation and no printer output request is present for the magnitude/phase representation.
6. The units of translation are the same as the units of length of the model. Rotations are in units of radians.
7. Displacement results are printed and/or punched in the global coordinate system (see field CD on the GRID bulk data entry). The coordinate system for plotted displacement output depends on the PARAM,POST setting. See the parameter **POST**.
8. The following applies to random solutions:
   - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, PUNCH, PLOT describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
   - The SORT1 and SORT2 describers only control the output format for the frequency response output. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
   - Any combination of the PSDF, ATOC, RMS, and CRMS describers can be selected. The RALL describer selects all four.
   - Autocorrelation (ATOC) calculations require the RANDT1 bulk entry.
9. When doing enforced motion dynamic analysis and relative output is requested (using the REL describer), the output is relative to the input as described by the equation:

$$u_f = y_f - K_{ff}^{-1} K_{fs} u_s$$

where u_f is absolute displacement, y_f is relative displacement, and u_s is enforced motion.

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10. During a SOL 108 or 111 frequency-dependent external superelement creation run, on the SET case control command that is referenced by the DISPLACEMENT case control command, include any points at which you want to recover results during the system run.
11. The SOLUTION describer is only valid in a SOL 108 or 111 frequency response analysis.
12. When an OFREQ case control command is present, the SOLUTION describer is used as follows:
    - If SOLUTION=ALL, output results for all frequencies specified by the OFREQ command.
    - If SOLUTION=setf, output results for all frequencies specified by the OFREQ command that are listed on a previously appearing SET command with setf as the identification number.
    - If SOLUTION=PEAKOUT and a PEAKOUT case control command is also present in the input file, output results for all frequencies specified by the OFREQ command that satisfy the PEAKOUT filtering criteria.
13. The SOLUTION describer is not valid for superelements or frequency-dependent components.

Remarks related to SOL 402:

1. Output is restricted to REAL and SORT1 format. SORT2, IMAG, PHASE, ABS, REL, PSDF, ATOC, CRMS, RMS, RALL, NOPRINT and RPUNCH are ignored.

Related Topics

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DIVERG

Static Aeroelastic Divergence Request

Selects the divergence parameters in a static aeroelastic divergence problem.

Format:

DIVERG=n

Examples:

```
DIVERG=70
```

Describers:

Describer	Meaning
n	Set identification of a DIVERG Bulk Data entry. (Integer>0)

Remarks:

Static aeroelastic divergence analysis can be performed in SOLs 144 and 200.

Related Topics

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DLOAD

Dynamic Load Set Selection

Selects a dynamic load or an acoustic source to be applied in a transient or frequency response problem.

For SOLs 401 and 402, DLOAD selects a time-dependent load for nonlinear analysis.

For SOL 414, DLOAD selects time-dependent or frequency-dependent loads.

Format:

DLOAD=n

Examples:

```
DLOAD=73
```

Describers:

Describer	Meaning
n	Set identification of a DLOAD, RLOAD1, RLOAD2, TLOAD1, TLOAD2, or ACSRCE Bulk Data entry. (Integer>0)
n (SOLs 401 or 402)	Set identification of a DLOAD or TLOAD1 Bulk Data entry. (Integer>0)

Remarks related to transient and frequency response:

1. RLOAD1 and RLOAD2 may only be selected in a frequency response problem.
2. TLOAD1 and TLOAD2 may be selected in a transient or frequency response problem.
3. Either a RLOADi or TLOADi entry (but not both) must be selected in an aeroelastic response problem. If RLOADi is selected, a frequency response is calculated. If TLOADi is selected, then transient response is computed by Fourier transform. When there are only gust loads (GUST entry), the DLOAD selects a TLOADi or RLOADi entry with zero load along with field 3 of the GUST command.
4. The DLOAD command will be ignored if specified for upstream superelements in dynamic analysis. To apply loads to upstream superelements, see the **LOADSET** case control command.

Remarks related to SOL 402:

1. Beginning with Simcenter Nastran 2021.2, you must define the DLOAD command globally (above all subcases). A fatal error occurs if you define the DLOAD command in a subcase. You can use the system cell setting NASTRAN SYSTEM(767)= 0 to turn off the fatal error and revert to the previous behavior.

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Remarks related to SOL 414:

1. DLOAD is used in SOL 414,129 Transient Response, SOL 414,111 Harmonic Response with a frequency response subcase, and SOL 414,103 Eigenvalues and Superelement Reduction with a transient response subcase.
   - In SOL 414,129, you must define DLOAD globally for the solution.
   - In SOL 414,111, you must define DLOAD in the frequency response subcase.
   - In SOL 414,103 Eigenvalues and Superelement Reduction with a transient response subcase, you must assign DLOAD to the subcase.

Related Topics

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DMTRCOEF

Duct modes transmission coefficient output request

Requests output of duct modes transmission coefficients.

Format:

$$\text{DMTRCOEF}\left(\left[\begin{matrix}\text{PRINT}\\ \text{NOPRINT}\end{matrix}\right], \text{PUNCH}, \left[\frac{\text{REAL, IMAG}}{\text{PHASE}}\right]\right) = \left\{\begin{matrix}\text{YES}\\ \text{NO}\end{matrix}\right\}$$

Examples:

```
DMTRCOEF(PRINT,PHASE)
```

Describers:

Describer	Meaning
PRINT	Transmission coefficients are written to the print (.f06) file. (Default)
NOPRINT	Transmission coefficients are not written to the print (.f06) file.
PUNCH	Transmission coefficients are written to the standard punch (.pch) file.
REAL or IMAG	Complex output is written in rectangular format (real and imaginary). REAL or IMAG results in the same output. (Default)
PHASE	Complex output is written in polar format (magnitude and phase). Phase output is in degrees.
YES	Describer settings are honored. Transmission coefficients are computed and output.
NO	Describer settings are ignored. Transmission coefficients are not computed or output. See Remark 3.

Remarks:

1. Duct modes are only available in SOL 108 uncoupled acoustic and coupled vibro-acoustic solutions and requires the FEM Adaptive Order (FEMAO) method.
2. Output is always written in SORT1 format.
3. DMTRCOEF=NO can be defined in a subcase to override a DMTRCOEF that is defined globally (above the subcase level).
4. Computed transmission coefficient results are written to the .op2 file.

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5. PRINT output occurs by default when the DMTRCOEF command is defined, except when PUNCH output is requested. To request both PRINT and PUNCH output, you must explicitly define both PRINT and PUNCH. For example,

```
DMTRCOEF(PRINT,PUNCH)
```

Related Topics

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DMTRLOSS

Duct modes transmission loss output request

Requests output of duct modes transmission loss.

Format:

$$\text{DMTRLOSS}\left(\left[\begin{matrix}\text{PRINT}\\ \overline{\text{NOPRINT}}\end{matrix}\right], \text{PUNCH}\right) = \left\{\begin{matrix}\text{YES}\\ \text{NO}\end{matrix}\right\}$$

Examples:

```
DMTRLOSS(PRINT)
```

Describers:

Describer	Meaning
PRINT	Transmission loss is written to the print (.f06) file. (Default)
NOPRINT	Transmission loss is not written to the print (.f06) file.
PUNCH	Transmission loss is written to the standard punch (.pch) file.
YES	Describer settings are honored. Transmission loss is computed and output.
NO	Describer settings are ignored. Transmission loss is not computed or output. See Remark 4.

Remarks:

1. Duct modes are only available in SOL 108 uncoupled acoustic and coupled vibro-acoustic solutions and requires the FEM Adaptive Order (FEMAO) method.
2. The DMTRLOSS command can only be used to request transmission loss for models with duct modes defined. To request transmission loss for acoustic models without duct modes, you should use the TRLOSS command. The TRLOSS command is not supported when duct modes are defined.
3. DMTRLOSS results are always real values.
4. Only SORT2 output format is available for the frequency response
5. DMTRLOSS=NO can be defined in a subcase to override a DMTRLOSS that is defined globally (above the subcase level).
6. Computed transmission loss results are written to the .op2 file.

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7. PRINT output occurs by default when the DMTRCOEF command is defined, except when PUNCH output is requested. To request both PRINT and PUNCH output, you must explicitly define both PRINT and PUNCH. For example,

```
DMTRLOSS(PRINT,PUNCH)
```

Related Topics

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DRSPAN

Exclusively assigns DRESP1 Responses to a Specific Subcase

Exclusively assigns a set of DRESP1 responses to a specific subcase, by referencing an existing SET. This set should then contain the list of the DRESP1 to be assigned to the subcase for which the DRSPAN data references the set.

Format:

DRSPAN = SETID

Example:

```
DRSPAN=375
```

Describers:

Describer	Meaning
SETID	The ID number of a previously defined SET containing the list of DRESP1 responses that should be assigned to the subcase within which this DRSPAN data resides.

Remarks:

1. A DRESP1 response assigned to a particular subcase may not be assigned to another subcase.
2. A DRSPAN referenced SET may not be referenced by another DRSPAN command.
3. A DRSPAN referenced SET may not refer to ALL.
4. A DRESP2 or DRESP3 response that references DRSPAN based DRESP1 responses, may not contain non-DRSPAN based DRESP1.
5. A DRESP2 or DRESP3 response that references DRSPAN based DRESP1 responses may not be nested within another DRESP2 or DRESP3, or have nested DRESP2. If nesting is done with DRSPAN, the job may produce unreliable results if it does not fail. See Remark 6 below for an exception.
6. One exception to Remark 5 above is the case of referenced (nested) DRESP2, which are actually DRESP1 that are mathematical functions (integrated responses) defined in Remark 20 for DRESP1. In accordance with Remark 20 of DRESP1, these responses should be referred to as DRESP2 from a referencing DRESP2 (should not be referenced from a DRESP3). Such responses actually being DRESP1, however, may therefore be specified in DRSPAN sets.
7. An objective function referencing a DRSPAN related second level response (DRESP2 or DRESP3) should be above the subcase level.
8. A constraint referencing a DRSPAN related second level response (DRESP2 or DRESP3) should be referenced by a DESGLB command, and therefore should be above the subcase level.

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9. The DRSPAN command can be used with superelements, as long as all DRESP1 responses in a DRSPAN related DRESP2 or DRESP3 are from the same superelement.

Related Topics

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DSAPRT

Design Sensitivity Output Parameters

Specifies design sensitivity output parameters.

Format:

$$\text{DSPART}\left[\left(\left[\begin{matrix}\underline{\text{FORMATTED}}\\ \text{UNFORMATTED}\\ \text{NOPRINT}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{NOEXPORT}}\\ \text{EXPORT}\end{matrix}\right],\right.\right.$$
$$\left.\left.[\text{START}=\text{i}],\ [\text{BY}=\text{y}],\ [\text{END}=\text{k}]\right)\right] = \left\{\begin{matrix}\underline{\text{ALL}}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
DSAPRT(FORMATTED,EXPORT)
DSAPRT(FORMATTED,START=FIRST,BY=3,END=LAST)=101
DSAPRT(UNFORMATTED,START=FIRST)
DSAPRT(UNFORMATTED,EXPORT)
DSAPRT(FORMATTED,END=4)=ALL
DSAPRT(UNFORMATTED,END=SENS)=ALL
DSAPRT(NOPRINT, EXPORT)
```

Describers:

Describer	Meaning
FORMATTED	Output will be presented with headings and labels.
UNFORMATTED	Output will be printed as a matrix print (see the description for the MATPRN module in the *Simcenter Nastran DMAP Programmer's Guide*.).
NOPRINT	No output will be printed.
EXPORT	Output will be exported to an external FORTRAN readable file specified by PARAM,IUNIT.
NOEXPORT	Output will not be exported to an external FORTRAN readable file.
START=i	Specifies the first design cycle for output. (Integer>0 or Character: "FIRST" or "LAST"; Default = 1 or "FIRST")

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Describer	Meaning
BY=j	Specifies the design cycle interval for output. (Integer ≥ 1, or >0; Default = 0) See Remark 2 .
END=k	Specifies the last design cycle for output. (Integer>0 or Character: “FIRST”, “LAST”, or “SENS”; Default = “LAST”)
ALL	All design responses (defined in DRESP1, DRESP2, and DRESP3 entries) will be output.
n	Set identification of a previously appearing SET command. Only sensitivities of responses with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

1. Only one DSAPRT may appear in the Case Control Section and should appear above all SUBCASE commands.
2. Sensitivity data will be output at design cycles i, i+j, i+2j, ..., k. Note that the BY=0 default implies no sensitivity analysis at the intermediate design cycles.
3. END = SENS requests design sensitivity analysis, and no optimization will be performed.
4. If both DSAPRT and PARAM,OPTEXIT, 4, -4, or 7 are specified, the DSAPRT definition overrides PARAM,OPTEXIT, 4, -4, or 7. PARAM,OPTEXIT values and the equivalent DSAPRT commands are as follows:

OPTEXIT	Equivalent DSAPRT Command
4	DSAPRT(UNFORMATTED, END=SENS)
-4	DSAPRT(NOPRINT, EXPORT, END=SENS)
7	DSAPRT(UNFORMATTED, START=LAST)

   The EXPORT describer exports data to an external file in the same way as PARAM,OPTEXIT, -4. See the OPTEXIT parameter description and Design Sensitivity Output in the Design Sensitivity and Optimization User’s Guide for more details.
5. Formatted sensitivity output is not supported for aeroelastic response types (DRESP1 RTYPE=TRIM, STABDER or FLUTTER). Unformatted output must be requested.

Related Topics

Browse Simcenter Nastran documentation

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DSYM

Dihedral Symmetry Option in Cyclic Symmetry

Provides for either one or two planes of overall symmetry in DIH-type cyclic symmetry problems.

Format:

$$\text{DSYM} = \left\{ \begin{matrix} \text{S} \\ \text{A} \\ \text{SS} \\ \text{SA} \\ \text{AS} \\ \text{AA} \end{matrix} \right\}$$

Examples:

```
DSYM=AS
```

Describers:

Describer	Meaning
S, A	The problem is assumed to be symmetrical (S) or antisymmetrical (A) with respect to the plane containing Side 1 of segment 1.
SS, SA, AS, AA	The problem is assumed to be symmetrical (or antisymmetrical) with respect to the plane containing Side 1 of segment 1 (denoted by the first symbol), and also with respect to a plane perpendicular to Side 1 (denoted by the second symbol).

Related Topics

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DTEMP

Time-assigned temperature set selection for SOLs 401, 402, and 414,129.

Selects a time-assigned temperature set to be used for temperature dependent material properties and thermal loading.

Format:

DTEMP =n

Examples:

```
DTEMP=5
```

Describers:

Describer	Meaning
n	Set identification number of only the DTEMP bulk entry, only the DTEMPEX bulk entry, or both DTEMPEX and TEMPD bulk entries. (Integer > 0)

Remarks related to SOL 401:

1. Only supported in static subcases.
2. If both DTEMP and TEMP(LOAD) are specified, the last one defined takes precedence. For example, if you define the TEMP(LOAD) command globally, and you define the DTEMP command in a subcase, the DTEMP command is used for that subcase.
3. When the DTEMP command selects the DTEMPEX bulk entry, you can select the TEMPD bulk entry to define a temperature for the grid points not included in the BUN file.

   When the DTEMP command selects the DTEMP bulk entry, you cannot select the TEMPD bulk entry directly using the DTEMP set ID. Although, the individual set IDs listed on the DTEMP entry can select unique TEMPD entries.

Related Topics

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ECHO

Bulk Data Echo Request

Controls echo (i.e., printout) of the Bulk Data.

Format:

$$\text{ECHO} = \left\{ \left[\text{UNSORT} \left[\begin{matrix} \underline{\text{SORT}} \left[(\left[\text{EXCEPT}\right] \text{cdni}, \ldots) \right] \\ \text{NOSORT} \end{matrix} \right] \right], \text{PUNCH} \left[\begin{pmatrix} \text{BSTBULK} \\ \text{NEWBULK} \end{pmatrix} \right] \right\}$$
$$\text{BOTH}$$
$$\text{NONE}$$

Examples:

```
ECHO=NOSORT
ECHO=BOTH
ECHO=PUNCH,SORT(MAT1,PARAM)
ECHO=SORT(EXCEPT DMI,DMIG)
ECHO=PUNCH(BSTBULK)
ECHO=PUNCH(NEWBULK)
ECHO=SORT,PUNCH(BSTBULK)
```

Describers:

Describer	Meaning
UNSORT	The unsorted Bulk Data will be printed. If SORT is also not specified then the sorted Bulk Data will not be printed.
SORT	The sorted (arranged in alphanumeric order) Bulk Data will be printed.
NOSORT	The sorted Bulk Data will not be printed. If UNSORT is also not specified then the unsorted Bulk Data will not be printed.
cdni,...	Bulk Data entry names to be included or excluded by EXCEPT, in the sorted echo printout. The PUNCH file is not affected by cdni.
EXCEPT	Exclude cdni Bulk Data entries from sorted echo printout. See Remark 6.
BOTH	Both sorted and unsorted Bulk Data will be printed. This is equivalent to ECHO=SORT, UNSORT.
NONE	Neither sorted nor unsorted Bulk Data will be printed.
PUNCH	The entire Bulk Data will be written to the punch file in sorted form.
BSTBULK	For SOL 200 design optimization runs, an unsorted updated bulk data file will be written to the punch file, using the updated design variables and properties as well

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Describer	Meaning
	as any mode-tracked eigenvalue DRESP1 (i.e. the updated design data) for the BEST design cycle obtained and indicated in the SOL 200 run. See Remark 7.
NEWBULK	For SOL 200 design optimization runs, an unsorted updated bulk data file will be written to the punch file, using the updated design variables and properties as well as any mode-tracked eigenvalue DRESP1 (i.e. the updated design data) for the LAST design cycle obtained and indicated in the SOL 200 run. See Remark 7.

Remarks:

1. If no ECHO command appears, a sorted Bulk Data will be printed.
2. Comments will appear at the front of the sorted file if ECHO=PUNCH.
3. Portions of the unsorted Bulk Data can be selectively echoed by including the commands ECHOON and ECHOOFF at various places within the Bulk Data. ECHOOFF stops the unsorted echo until an ECHOON command is encountered. Many such pairs of commands may be used. The ECHOON and ECHOOFF commands may be used in the Executive and Case Control Sections; however, ECHOOF should not be the first entry as continuation entries will not be handled correctly.
4. If the SORT (cdni,...) is specified in a restart in SOLs 101 through 200, then the continuation entries will not be printed.
5. If the SORT (cdni,...) describer is used, then it must appear as the last describer, as in the example above.
6. If EXCEPT is specified then it must be specified before all cdni. All Bulk Data entry types will be listed except those given for cdn1, cdn2, etc. If EXCEPT is not specified, then only those Bulk Data entry types listed under cdn1, cdn2, etc. will be listed.
7. For SOL 200, often the last design cycle and the best design cycle coincide. However, BSTBULK is useful for cases where the best design cycle is actually an earlier cycle. In contrast, NEWBULK may be useful when the user knows that the job will be continued later starting from the last design cycle.

Related Topics

Browse Simcenter Nastran documentation

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EDE

Element Energy Loss Per Cycle Output Request

Requests the output of the energy loss per cycle in selected elements.

Format:

$$\text{EDE}\left[\left(\left[\begin{matrix}\underline{\text{PRINT}}, \text{PUNCH} \\ \text{PLOT}\end{matrix}\right], \left[\begin{matrix}\underline{\text{AVERAGE}} \\ \text{AMPLITUDE} \\ \text{PEAK}\end{matrix}\right], \left[\begin{matrix}\underline{\text{RMAG}} \\ \text{REAL or IMAG} \\ \text{PHASE}\end{matrix}\right], \left[\begin{matrix}\underline{\text{SORT1}} \\ \text{SORT2}\end{matrix}\right], [\text{THRESH} = \text{p}]\right)\right] = \left\{\begin{matrix}\text{ALL} \\ \text{n}\end{matrix}\right\}$$

Examples:

```
EDE=ALL
EDE(PUNCH, THRESH=.0001)=19
```

Describers:

Describer	Meaning
PRINT	Write energies to the print file (default).
PUNCH	Write energies to the punch file.
PLOT	Do not write energies to either the punch file or the print file.
AVERAGE	Requests average energy in frequency response analysis only.
AMPLITUDE	Requests amplitude of energy in frequency response analysis only.
PEAK	Requests peak energy for frequency response analysis only. PEAK is the sum of AVERAGE and AMPLITUDE.
RMAG	Outputs the energy magnitude in real data format.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Only valid with the AVERAGE option.)
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees. (Only valid with the AVERAGE option.)
SORT1	Output will be presented as a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence.

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Describer	Meaning
SORT2	Output will be presented as a tabular listing of frequency or time for each grid point.
THRESH	Suppresses energies for elements having an energy value of less than p% in all output files. THRESH overrides the value of TINY described in Remark 1. (Default=0.001)
ALL	Computes energy for all elements.
n	Set identification number. Energy for all elements specified on the SET n command will be computed. The SET n command must be specified in the same subcase as the EDE command or above all subcases. (Integer>0)
NONE	Element energy will not be output.

Remarks:

1. If THRESH = p is not specified, then p defaults to the values specified by user parameter TINY.
2. The energy calculations include the contribution of initial thermal strain.
3. Energy density (element energy divided by element volume) is also computed in some solution sequences. It can be suppressed by use of PARAM,EST,-1.
4. The equations used to calculate elemental damping energy components are given below.

   Average Damping Energy:

$$E_{avg} = \Pi\omega\left[\{u_r\}^T[B_e]\{u_r\} + \{u_i\}^T[B_e]\{u_i\}\right]$$

   where:

E	=	elemental energy
$\{u_r\}$	=	displacement - real part
$\{u_i\}$	=	displacement - imaginary part
$[B_e]$	=	elemental damping

   Real Part of Damping Energy Amplitude:

$$E_{amp_real} = \Pi\omega\left[\{u\}^T[B_e]\{u_r\} - \{u_i\}^T[B_e]\{u_i\}\right]$$

   Imaginary Part of Damping Energy Amplitude:

$$E_{amp_imag} = -2\Pi\omega\{u_r\}^T[B_e]\{u_i\}$$

   Magnitude of Damping Energy Amplitude:

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$$E_{amp_mag} = \sqrt{E_{amp_real}^2 + E_{amp_imag}^2}$$

Phase of Damping Energy Amplitude:

$$\theta_{amp} = \arctan\left(\frac{E_{amp_imag}}{E_{amp_real}}\right)$$

Peak Damping Energy:

$$E_{peak} = E_{avg} + E_{amp_mag}$$

5. In SOL 111, EDE can only be requested if PARAM,DDRMM,-1 is used.
6. Only damping from the viscous dampers (e.g., CVISC, CDAMPi, etc.) are included. Structural damping is not included in the calculation.
7. When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the energy loss per cycle results are based on the nominal values for stiffness and damping.

Related Topics

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EFLOAD

External Field Surface Loading

Converts surface loads from an external field solution to structural loads.

Format:

$$\text{EFLOAD}\,[\text{NDIR} = \text{n},\, \text{SCID} = \text{m}] = \left\{ \begin{matrix} \text{YES} \\ \text{NO} \end{matrix} \right\}$$

Examples:

```
EFLOAD=YES
EFLOAD(NDIR=2)=YES
EFLOAD(NDIR=3,SCID=14)=Yes
EFLOAD(SCID=7)=YES
```

Describers:

Describer	Meaning
NDIR=*n*	*n* (integer>0) specifies the normal direction of the surface patch defined in the same subcase. Allowable values for *n* are 1, 2, and 3. See Remarks 2 and 3.
SCID=*m*	*m* (integer>0) specifies the identification number of a local coordinate system. See Remarks 4 and 5.
YES	Convert external field surface loads to structural loads.
NO	Do not convert external field surface loads to structural loads.

Remarks:

1. A subcase that includes an EFLOAD case control command is required for each surface patch that external field surface loads are converted to structural loads. The last subcase cannot contain an EFLOAD case control command, but only output requests and other structural loads (if any).
2. The table lists the allowable values for *n* and the corresponding normal directions.

n	Rectangular CSYS	Cylindrical CSYS
1 (default)	z	z
2	y	θ
3	x	r

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3. If NDIR=*n* is not specified, NDIR=1 is used by default.
4. If SCID=*m* is not specified, the normal direction to the surface patch and the coordinates in the corresponding external force points matrix are defined relative to the basic coordinate system.
5. If SCID=*m* is specified, the normal direction to the surface patch and the coordinates in the corresponding external force points matrix are defined relative to a cylindrical coordinate system whose:
   - Origin coincides with the basic coordinate system.
   - Cylindrical axis coincides with the z-axis of the basic coordinate system.
   - Tangential coordinate, θ, is measured from the x-axis of the basic coordinate system. ($0 \le \theta \le 360$ degrees)

   The local coordinate system with *m* as the identification number does not need to be cylindrical. By specifying any local coordinate system, the cylindrical coordinate system described above is used as the coordinate basis.
6. For a direct frequency response (SOL 108) solution, the structural loads converted from the external field surface loads are used as scaling factor components of the dynamic load.
7. For a statically prestressed normal modes analysis, two runs are required. The first run is a linear static (SOL 101) solution that converts the external field surface loads to structural loads. During this run, a punch (.pch) file is automatically written with the converted external field surface loads. The second run is a normal modes (SOL 103) solution. For this run, the first subcase, which is the static subcase, must include EFLOAD=YES. The second subcase, which is the normal modes subcase, must include STATSUB=1. The bulk data must include an INCLUDE bulk entry that references the .pch file written during the first run.
8. For additional information on how to use the EFLOAD case control command to convert surface loads from an external field solution to structural loads, see External Force Fields in the *Simcenter Nastran User's Guide*.
9. An external force field can be included in a rotor dynamic analysis with the direct complex eigenvalue solution sequence (SOL 107). Three solution steps are required, and are described in External Force Fields in the *Simcenter Nastran User's Guide*.

Related Topics

Browse Simcenter Nastran documentation

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EKE

Element Kinetic Energy Output Request

Requests the output of the kinetic energy in selected elements.

Format:

$$\text{EKE}\left[\left(\left[\begin{matrix}(\underline{\text{PRINT}},\ \text{PUNCH})\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{AVERAGE}}\\ \text{AMPLITUDE}\\ \text{PEAK}\end{matrix}\right],\left[\begin{matrix}\underline{\text{RMAG}}\\ \text{REAL or IMAG}\\ \text{PHASE}\end{matrix}\right],\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],[\text{THRESH}=\text{p}]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\end{matrix}\right\}$$

Examples:

```
EKE=ALL
EKE(PUNCH, THRESH=.0001)=19
```

Describers:

Describer	Meaning
PRINT	Write energies to the print file (default).
PUNCH	Write energies to the punch file.
PLOT	Do not write energies to either the punch file or the print file.
AVERAGE	Requests average energy in frequency response analysis only.
AMPLITUDE	Requests amplitude of energy in frequency response analysis only.
PEAK	Requests peak energy for frequency response analysis only. PEAK is the sum of AVERAGE and AMPLITUDE.
RMAG	Outputs the energy magnitude in real data format.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Only valid with the AVERAGE option.)
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees. (Only valid with the AVERAGE option.)

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Describer	Meaning
SORT1	Output will be presented as a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output will be presented as a tabular listing of frequency or time for each grid point.
THRESH	Suppresses energies for elements having an energy value of less than p% in all output files. THRESH overrides the value of TINY as described in the remarks. (Default=0.001)
ALL	Computes energy for all elements.
n	Set identification number. Energy for all elements specified on the SET n command will be computed. The SET n command must be specified in the same subcase as the EKE command or above all subcases. (Integer>0)
NONE	Element energy will not be output.

Remarks:

1. If THRESH = p is not specified, then p defaults to the values specified by user parameter TINY.
2. The energy calculations include the contribution of initial thermal strain.
3. Energy density (element energy divided by element volume) is also computed in some solution sequences. It can be suppressed by use of PARAM,EST,-1.
4. The equations used to calculate elemental kinetic energy components are given below.

   Average Kinetic Energy:

$$E_{avg} = \frac{1}{4}\left[\{v_r\}^T[M_e]\{v_r\} + \{v_i\}^T[M_e]\{v_i\}\right]$$

   where:

E	=	elemental energy
$\{v_r\}$	=	velocity - real part
$\{v_i\}$	=	velocity - imaginary part
$[M_e]$	=	elemental mass

   Real Part of Kinetic Energy Amplitude:

$$E_{amp_real} = \frac{1}{4}\left[\{v_r\}^T[M_e]\{v_r\} - \{v_i\}^T[M_e]\{v_i\}\right]$$

   Imaginary Part of Kinetic Energy Amplitude:

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$$E_{amp_imag} = -\frac{1}{2}\{v_r\}^T [M_e]\{v_i\}$$

Magnitude of Kinetic Energy Amplitude:

$$E_{amp_mag} = \sqrt{E^2_{amp_real} + E^2_{amp_imag}}$$

Phase of Kinetic Energy Amplitude:

$$\theta_{amp} = \arctan\left(\frac{E_{amp_imag}}{E_{amp_real}}\right)$$

Peak Kinetic Energy:

$$E_{peak} = E_{avg} + E_{amp_mag}$$

5. In SOL 111, EKE can only be requested if PARAM,DDRMM,-1 is used.
6. When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the kinetic energy results are based on the nominal values for stiffness and damping.

Remark related to SOLs 401 and 402:

1. Kinetic energy requested with the EKE command is only output from a normal modes subcase.

Remarks related to SOLs 402 and 414:

1. In SOL 402, only PRINT, PUNCH, PLOT, REAL, THRESH and SORT1 are used. All the other describers are ignored.

   As a consequence, in complex modes analyses, only the REAL part can be post-processed. The IMAG part is always zeroed and the PHASE part is useless.
2. In SOL 402, the THRESH parameter has no default. In addition, SOL 402 does not support the PARAM,TINY definition.
3. SOLs 402 and 414 compute kinetic energy as follows.

   Kinetic energy ratio of element *e*:

$$KER = \frac{x_e^{\mathrm{T}} M_e x_e}{x^{\mathrm{T}} M x}$$

   Kinetic energy density of element *e*:

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$$KED = (KER)\frac{V}{V_e}$$

where,

M and *K* are the mass and stiffness matrices of a volume *V*,
M_e and K_e are the mass and stiffness matrices of element *e* of volume V_e,
and x_e is the displacement vector of element *e* for mode *x*.

Related Topics

Browse Simcenter Nastran documentation

Simcenter Nastran Quick Reference Guide
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ELAR

Element Add/Remove Set Selection

Selects an element add/remove set (SOL 401)

Format:

ELAR=n

Examples:

```
ELAR=5
```

Describers:

Describer	Meaning
n	Element add/remove set identification of an ELAR, ELAR2, or ELARADD bulk entry. (Integer>0)

Remarks:

1. For the static and dynamic subcases:
   - Element addition and removal by time, and element removal by strain are supported.
   - The ELAR case control command can be defined above the subcase level (global) or in a subcase.

     An ELAR command in a static or dynamic subcase and a definition at the global level are equivalent. That is, if you define the ELAR command in a subcase, the software internally converts it to a global definition. As a result, the element addition or removal will occur in the static or dynamic subcase where the requested time or strain occurs, and not necessarily the subcase in which the ELAR command was defined.

     If you define multiple ELAR definitions globally and/or in subcases, the software combines and applies all ELAR sets (global + subcase definitions) as a global set.
   - Element addition or removal is not supported in a statics subcase in which nonlinear buckling is requested with the NLARCL=ID case control command.
2. For preload subcases:
   - Only element addition by time is supported.
   - The ELAR command must be defined in the preload subcase.
   - A preload subcase in general (with and without the ELAR command) can include a new thermal load set (TEMPERATURE command) and/or a new contact set (BCSET command). No other new

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load sets are allowed in a preload subcase. Load sets defined globally or in previous subcases are maintained in the preload subcase.

A preload subcase can include an element addition set (ELAR command) or a bolt preload set (BOLDLD command), but it cannot include both.

- Elements included in a bolt definition cannot be added or removed in any subcase (static, dynamic, preload).

3. Element addition and removal cannot occur in a modal, cyclic, or Fourier subcase. Although, the addition and removal status is maintained for these subcases if they are sequentially dependent. For example, if an element is added in a static subcase and the next subcase is a sequentially dependent modal subcase, the added element will be active in the modal subcase.
4. Element addition and removal is not supported when large strain is requested.

Related Topics

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ELAROUT

Element Add/Remove Status Output

Element Add/Remove status output (SOL 401)

Format:

$$\text{ELAROUT}\left[\left(\underline{\text{PRINT}},\text{PUNCH},\text{PLOT}\right)\right]=\left\{\begin{matrix}\text{YES}\\\underline{\text{NO}}\end{matrix}\right\}$$

Examples:

```
ELAROUT=YES
```

Describers:

Describer	Meaning
PRINT	Output is written to the .f06 file and the .op2 file. (Default)
PUNCH	Output is written to the .pch file and the .op2 file.
PLOT	Output is written to the .op2 file.
YES	Request output.
NO	No output requested. (Default)

Remarks:

1. The ELAROUT command outputs a status of 0 for elements that have not been added yet, and a status of 1 for elements that are removed. No status is written for elements that are active.
2. To request output to only the .op2 file, include only PLOT without PRINT or PUNCH.

Related Topics

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ELSDCON

Element Stress Discontinuity Output Request

Requests mesh stress discontinuities based on element stresses.

Format:

$$\text{ELSDCON}\left[\begin{matrix}\text{PRINT}\\ \text{PLOT}\end{matrix}\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
ELSDCON=ALL
ELSDCON=19
```

Describers:

Describer	Meaning
ALL	Stress discontinuity requests for all SURFACE and VOLUME commands defined in the OUTPUT(POST) section will be output.
n	Set identification number of a previously appearing SET command. Only surfaces and volumes with identification numbers that appear on this SET command will be included in the stress discontinuity output request. (Integer>0)
NONE	No element stress discontinuity output.

Remarks:

1. This output is available in linear static analysis in SOLs 101 and 144 only. Output will be presented for each surface or volume as a tabular listing of stress discontinuities for each subcase.
2. Only elements used to define the surface or volume are output. See the descriptions of **"SURFACE"** and **"VOLUME"** .
3. Element stress output (STRESS) must be requested for elements referenced on SURFACE and VOLUME commands. Also, the GPSTRESS command must be present for printed output and the STRFIELD command for postprocessing output using the .xdb file (PARAM,POST,0) for the same surfaces and volumes.

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Related Topics

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ELSTRN

Requests elastic strain output at grid points on elements for SOL 401 and SOL 402. Also requests elastic strain for SOLs 101, 103, 105, 106, 107-112.

Requests elastic strain on elements.

Format:

$$\text{ELSTRN}\left[\left(\left[\begin{matrix}\underline{\text{STRCUR}}\\ \text{FIBER}\end{matrix}\right],\left[\begin{matrix}\underline{\text{CENTER}}\\ \text{CORNER or BILIN}\\ \text{SGAGE}\\ \text{CUBIC}\end{matrix}\right],\underline{\text{PRINT}},\text{PUNCH},\text{PLOT}\right)\right]$$
$$=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
ELSTRN=ALL
ELSTRN(PRINT,PUNCH)=17
ELSTRN(STRCUR,CORNER,PRINT) = ALL
```

Describers for SOLs 401 & 402:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all elements.
n	Set identification number of a previously appearing SET command. Only elements with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

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Describers for SOLs 101, 103, 105, 106, 107-112:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all elements.
n	Set identification number of a previously appearing SET command. Only elements with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.
STRCUR	Elastic strain at the reference plane and curvatures is output for shell elements.
FIBER	Elastic strain at locations Z1, Z2 is computed for shell elements.
CENTER	Outputs elastic strains at the center of the element only.
CORNER or BILIN	Outputs elastic strains at the center and corners of the element.
SGAGE	Outputs elastic strains at the center and corners of the element using strain gage approach.
CUBIC	Outputs elastic strains at the center and corners of the element using cubic bending correction.

General Remarks:

1. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

Remarks for SOL 401:

1. ELSTRN outputs elastic strain at the grid point location on the elements.
2. Only supported in a static or preload subcase.
3. Both PRINT and PUNCH may be requested.
4. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.

Remarks for SOLs 101, 103, 105, 106, 107-112:

1. For SOL 106, ELSTRN is only supported for the shell elements CQUAD4, CTRIA3, CQUAD8, CTRIA6, CTRIAR, and CQUADR that reference a PCOMP or PCOMPG entry.

   For SOL 101, ELSTRN is only supported for the shell elements CQUAD4, CTRIA3 , CQUAD8, CTRIA6, CTRIAR, and CQUADR that reference a PSHELL, PCOMP or PCOMPG entry, and the 3D solid elements CTETRA, CHEXA, CPENTA and CPYRAM that reference a PSOLID entry.

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For SOLs 103, 105, 107-112, ELSTRN is only supported for the shell elements CQUAD4, CTRIA3 , CQUAD8, CTRIA6, CTRIAR, and CQUADR that reference a PSHELL entry, and the 3D solid elements CTETRA, CHEXA, CPENTA and CPYRAM that reference a PSOLID entry.

2. For shell elements referencing a composite property defined with the PCOMP or PCOMPG entries, the software automatically creates an equivalent PSHELL and MATi entry. You can use the parameter NOCOMPS to control if stress and/or strain is recovered at the composite ply layers (default) and/or on the equivalent PSHELL.

   For strain output on the composite ply layers, the STRCUR, FIBER, CENTER, CORNER, SGAGE, CUBIC describers are ignored, and strain is always reported at the center of the element (in the element plane) and at the center of each ply (in the thickness direction).

   For the strain output on the equivalent PSHELL, STRCUR is the default, and the behaviour for the CENTER, CORNER, SGAGE, CUBIC describers is the same as described in the following remark. Note that if FIBER is defined on a STRAIN, ELSTRN, or THSTRN command, the FIBER option applies to all STRAIN, ELSTRN and THSTRN commands.

3. For shell elements referencing a PSHELL entry:

   For the CENTER option, elastic strains are output at the center of the element for CQUAD4, CQUADR, and CTRIAR elements.

   For the CORNER (or BILIN) option, elastic strains are output at the center and at the corner for CQUAD4, CQUADR, and CTRIAR elements.

   For the SGAGE option, elastic strains are output at the center and at the corner for CQUAD4 elements.

   For the CUBIC option, elastic strains are output at the center and at the corner for CQUAD4 and CQUADR elements.

4. If the STRAIN command is defined when the ELSTRN and/or the THSTRN commands are also defined, the CENTER/CORNER(BILIN)/SGAGE/CUBIC option on the STRAIN command overrides the corresponding option on the ELSTRN and/or THSTRN command.

Remarks for SOL 402:

1. ELSTRN is not supported for CBEAM, CROD, and CBAR elements.
2. As SOL 402 ignores any of the STRCUR, FIBER, CENTER, CORNER, BILIN, SGAGE, or CUBIC describers, the location of the output request will be automatically selected based on the element type:
   - For solids, the output location is the corner nodes.
   - For shells, the output location is the top and bottom of the shell.
   - For multilayered shells, the output location is the top and bottom of each ply of the stacking sequence.
   - For multilayered solids, the output location is the mid-plane of each ply of the stacking sequence.

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Related Topics

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ELSUM

Element Summary Output Request

Requests output of an element property summary.

Format:

$$\text{ELSUM}\left[(\underline{\text{EID}},\text{PID},\text{BOTH},\text{EIDSUM},\right.$$
$$\left.\text{PIDSUM},\text{SUMMARY},\text{NSMCONT})\right] = \left\{\begin{array}{c}\text{ALL}\\ \text{n}\\ \text{NONE}\end{array}\right\}$$

Examples:

```
ELSUM = ALL
ELSUM(PID) = 9
ELSUM(EIDSUM) = ALL
```

Describers:

Describer	Meaning
EID	Group element summary output by element type. (Default)
PID	Group element summary output by element property type.
BOTH	Output element summaries for both the EID and PID groupings.
EIDSUM	Output only a summary of mass totals for the EID grouping.
PIDSUM	Output only a summary of mass totals for the PID grouping.
SUMMARY	Output only a summary of mass totals for both the EID and PID groupings.
NSMCONT	Add non-structural mass from NSM, NSM1, NSML, and NMSL1 bulk entries to the EID and/or PID element summaries.
ALL	Element summary output for all elements.
n	Set identification of a previously appearing SET command. Only produces output for elements whose identification numbers appear on this SET command. (Integer>0)
NONE	No element summary output.

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Remarks:

1. The ELSUM command is ignored in heat transfer solutions.
2. ELSUM output is only available in the F06 file. PUNCH and PLOT options are not supported.
3. The ELSUM command produces a summary of properties for elements. The properties include element-id, material-id, length or thickness, area, volume, structural mass, non-structural mass, total mass, and the product of the total mass and the WTMASS parameter. Total mass is the sum of structural and non-structural mass.
4. For elements referencing temperature dependent materials defined with MATTi entries, the ELSUM output is approximated using temperature=0.0.
5. Certain element types produce only partial data. For these element types, output will contain element-id, length or thickness, volume, and area without mass data. Totals will thus not include those elements for which mass data was not generated.
6. Mass data will be computed for the following elements: CBAR, CBEAM, CBEND, CBUSH1D, CHEXA, CMASSi, CONMi, CONROD, CPENTA, CPLSTNi, CPLSTSi, CPYRAM, CQUAD4, CQUAD8, CQUADR, CQUADX4, CQUADX8, CROD, CSHEAR, CTETRA, CTRAX3, CTRAX6, CTRIA3, CTRIA6, CTRIAR, CTUBE.
7. EIDSUM takes precedence over EID if both describers are used.
8. PIDSUM takes precedence over PID if both describers are used.
9. If the NSMCONT describer is specified in combination with the EID describer, a table is produced that identifies the contribution of each non-structural mass bulk entry to the total element non-structural mass. If the NSMCONT describer is specified in combination with the PID describer, a table is produced that identifies non-structural mass from all NSM type bulk entries for all element property types by property identification number.

Related Topics

Browse Simcenter Nastran documentation

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ENTHALPY

Heat Transfer Enthalpy Output Request

Requests form of enthalpy vector output in transient heat transfer analysis (SOL 159).

Format:

$$\text{ENTHALPY}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{PRINT}},\ \text{PUNCH}\\ \text{PLOT}\end{matrix}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
ENTHALPY=5
```

Describers:

Describer	Meaning
SORT1	Output will be presented as a tabular listing of grid points for each time.
SORT2	Output will be presented as a tabular listing of time for each grid point.
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
PLOT	Generates but does not print enthalpies.
ALL	Enthalpy for all points will be output.
NONE	Enthalpy for no points will be output.
n	Set identification of previously appearing SET command. Only enthalpies of points with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

ENTHALPY=NONE is used to override a previous ENTHALPY=n or ENTHALPY=ALL command.

Related Topics

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ERP

Requests Equivalent Radiated Power Output

Requests equivalent radiated power output for selected panels and the shell or visualization elements that comprise the selected panels in frequency response analysis, and in the equivalent ANALYSIS=DFREQ or ANALYSIS=MFREQ options with SOL 200.

Format:

$$\text{ERP}\left[\left(\left[\begin{matrix}\text{SORT1}\\ \text{SORT2}\end{matrix}\right], \underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}, \text{SOLUTION}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{setf}\end{matrix}\right\},\right.\right.$$
$$\left.\left.\text{ERPCOEFF}=\text{c}, \text{ELEMOUT}=\left\{\begin{matrix}\text{YES}\\ \underline{\text{NO}}\end{matrix}\right\}\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{setp}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
ERP(PRINT,SOLUTION=53,ERPCOEFF=0.75)=55
```

Describers:

Describer	Meaning
SORT1	Output equivalent radiated power results as a tabular listing at each forcing frequency. See Remarks 3 and 4. (Default)
SORT2	Output equivalent radiated power results as a tabular listing for each panel. See Remarks 3 and 5.
PRINT	Write output to the print file. (Default)
PUNCH	Write output to the punch file.
PLOT	Do not write ERP data to either the print or punch file.
SOLUTION = ALL	Calculate equivalent radiated power at all frequencies specified by FREQUENCY case control commands. (Default)
SOLUTION = setf	Calculate equivalent radiated power at all frequencies specified by a SET case control command having the identification number *setf*. (Integer > 0)
ERPCOEFF = c	Coefficient used to scale the equivalent radiated power. See Remark 3. (Real > 0.0; Default = 1.0)

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Describer	Meaning
ELEMOUT = YES	Output equivalent radiated power by element for elements that comprise the selected panels. See Remarks **6**, **7**, and **8**.
ELEMOUT = NO	Do not output equivalent radiated power by element for elements that comprise the selected panels. (Default)
ALL	Output equivalent radiated power for all panels.
setp	Output equivalent radiated power for selected panels. *setp* selects the set identification of a previously appearing SET case control command. The SET command lists the names (character entry) of panels defined on the PANEL bulk entry. The SET command must be specified in the same subcase as the ERP command or above all subcases. (Integer > 0) Note that for ERP output requests, the panels defined on the PANEL bulk entry must reference SET3 bulk entries only.
NONE	Do not output equivalent radiated power.

Remarks:

1. Equivalent radiated power (ERP) calculation is supported on the shell elements CQUAD4, CQUAD8, CQUADR, CTRIA3, CTRIA6, and CTRIAR, or on solid element faces which are surface coated with the PLOTEL3, PLOTEL4, PLOTEL6, and PLOTEL8 visualization elements. The input requirements are as follows:

   - For either shell or PLOTEL element selection, you can:
     - List the shell or PLOTEL element IDs on a SET3 bulk entry with "ELEM" defined in the TYPE field.
     - List grid point IDs used by the shell or PLOTEL elements on a SET3 bulk entry with "GRID" defined in the TYPE field. The software then automatically computes the output on the associated elements.

     For only shell element selection, you can list the shell element property IDs on a SET3 bulk entry with "PROP" defined in the TYPE field.
   - You select the SET3 entry IDs on the PANEL bulk entry in the SID*i* fields.
   - You list the names (character entry) of the panels defined on the PANEL bulk entry on a SET case control command. The SET command must be specified in the same subcase as the ERP command or above all subcases.
   - You define the ID of the SET command on the ERP case control command with the *setp* describer, or you can use the *ALL* describer to request the output on all panels.
2. ERP output is supported in SOL 108 and 111, and in the equivalent ANALYSIS=DFREQ or ANALYSIS=MFREQ options with solution 200. You can request ERP output for models that contain structural elements only, or models that include both structural and fluid elements.
3. The equivalent radiated power attributable to the i^{th} element at frequency ω_k is given by:

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$$ERPE_i\left(\omega_k\right) = \frac{1}{2}c\int_{S_i} v_n\left(\mathbf{x},\omega_k\right)v_n^*\left(\mathbf{x},\omega_k\right)ds$$

where:

c is the scaling coefficient specified with the ERPCOEFF describer, which is usually taken to be the product of density and speed of sound in the fluid medium.

$v_n\left(\mathbf{x},\omega_k\right)$ is the normal velocity as a function of position, $\mathbf{x}$, at frequency ω_k.

Frequencies ω_k are specified by the SOLUTION describer.

$v_n^*\left(\mathbf{x},\omega_k\right)$ is the complex conjugate of $v_n\left(\mathbf{x},\omega_k\right)$.

S_i is the surface area of the i^{th} element.

The equivalent radiated power density for the i^{th} element at frequency ω_k is given by:

$$ERPED_i\left(\omega_k\right) = \frac{ERPE_i\left(\omega_k\right)}{S_i}$$

The equivalent radiated power for the j^{th} panel at frequency ω_k is the sum of the equivalent radiated power for each element comprising the j^{th} panel at frequency ω_k and is given by:

$$ERPP_j\left(\omega_k\right) = \sum_i ERPE_i\left(\omega_k\right)$$

The equivalent radiated power density for the j^{th} panel at frequency ω_k and is given by:

$$ERPD_j\left(\omega_k\right) = \frac{ERPP_j\left(\omega_k\right)}{A_j}$$

where A_j is the surface area of the j^{th} panel.

When the SORT1 (default) describer is specified, the values in the "FRACTION" column represent the fraction of the total equivalent radiated power for all selected panels that is attributable to the j^{th} panel at frequency ω_k. These values are computed as follows:

$$\overline{ERPP_j}\left(\omega_k\right) = \frac{ERPP_j\left(\omega_k\right)}{\sum_{m=1}^{M} ERPP_m\left(\omega_k\right)}$$

where *M* is the total number of selected panels.

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The output is then grouped according to frequency and sorted by panel in descending order of equivalent radiated power.

For a given frequency, the value for equivalent radiated power density in the "TOTAL" row is the ratio of the total equivalent radiated power for all the selected panels at that frequency to the total area of those panels.

When the SORT2 describer is specified, for an individual panel, the values in the "FRACTION" column represent the fraction of the total equivalent radiated power for all frequencies that is attributable to the j^{th} panel at frequency ω_k. These values are computed as follows:

$$\overline{ERPP_j}\left(\omega_k\right) = \frac{ERPP_j\left(\omega_k\right)}{\sum_{n=1}^{N} ERPP_j\left(\omega_n\right)}$$

where *N* is the total number of frequencies specified by the SOLUTION describer.

In the "PANEL=TOTAL" section of the SORT2 results, the values in the "FRACTION" column represent the fraction of the total equivalent radiated power for all panels at all frequencies that is attributable to all the panels at frequency ω_k. These values are computed as follows:

$$\overline{ERPPT}\left(\omega_k\right) = \frac{\sum_{m=1}^{M} ERPP_m\left(\omega_k\right)}{\sum_{n=1}^{N}\left(\sum_{m=1}^{M} ERPP_m\left(\omega_n\right)\right)}$$

4. The following sample of SORT1 output for panels is from a model with three panels selected for output and two forcing frequencies.

```
FREQUENCY =  3.000000E+03
                     EQUIVALENT RADIATED POWER

PANEL NAME     ERP         FRACTION   NORMAL AREA  ERP DENSITY
   PANEL3 1.491040E+11 4.901360E-01 3.000000E+04 4.970134E+06
   PANEL1 1.392054E+11 4.575974E-01 1.250000E+04 1.113644E+07
   PANEL2 1.590001E+10 5.226666E-02 1.750000E+04 9.085721E+05
   TOTAL  3.042095E+11 1.000000E+00 6.000000E+04 5.070158E+06

FREQUENCY =  4.000000E+03
                     EQUIVALENT RADIATED POWER

PANEL NAME     ERP         FRACTION   NORMAL AREA  ERP DENSITY
   PANEL3 1.874405E+10 4.868036E-01 3.000000E+04 6.248018E+05
   PANEL1 1.761857E+10 4.575737E-01 1.250000E+04 1.409486E+06
   PANEL2 2.141717E+09 5.562273E-02 1.750000E+04 1.223838E+05
   TOTAL  3.850434E+10 1.000000E+00 6.000000E+04 6.417390E+05
```


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5. The following sample of SORT2 output for panels is from a model with three panels selected for output and two forcing frequencies.

```
PANEL=PANEL3(AREA=3.000000E+04 ERP. MAXIMUM=1.491040E+11)
                    EQUIVALENT RADIATED POWER
        FREQUENCY       ERP        FRACTION   ERP DENSITY
       3.000000E+03 1.491040E+11 8.883273E-01 4.970134E+06
       4.000000E+03 1.874405E+10 1.116727E-01 6.248018E+05



PANEL=PANEL1(AREA=1.250000E+04 ERP. MAXIMUM=1.392054E+11)
                    EQUIVALENT RADIATED POWER
        FREQUENCY       ERP        FRACTION   ERP DENSITY
       3.000000E+03 1.392054E+11 8.876539E-01 1.113644E+07
       4.000000E+03 1.761857E+10 1.123461E-01 1.409486E+06



PANEL=PANEL2(AREA=1.750000E+04 ERP. MAXIMUM=1.590001E+10)
                    EQUIVALENT RADIATED POWER
        FREQUENCY       ERP        FRACTION   ERP DENSITY
       3.000000E+03 1.590001E+10 8.812909E-01 9.085721E+05
       4.000000E+03 2.141717E+09 1.187091E-01 1.223838E+05



PANEL=TOTAL (AREA=6.000000E+04 ERP. MAXIMUM=1.590001E+10)
                    EQUIVALENT RADIATED POWER
        FREQUENCY       ERP        FRACTION   ERP DENSITY
       3.000000E+03 3.042095E+11 8.876487E-01 5.070158E+06
       4.000000E+03 3.850434E+10 1.123513E-01 6.417390E+05
```

6. Only absolute equivalent radiated power values are available for element output.

7. The following sample of SORT1 output for elements is from a model where the panels selected for output are constructed from PLOTEL3 and PLOTEL4 elements, and there are two forcing frequencies.

```
FREQUENCY =  3.000000E+03
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL3)
  ELEMENT ID            ERP                ERP DENSITY
          5          2.196778E+10          1.757423E+07
          6          3.317672E+10          2.654138E+07
         29          3.317672E+10          2.654138E+07

FREQUENCY =  3.000000E+03
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL4)
  ELEMENT ID            ERP                ERP DENSITY
          2          8.997186E+09          3.598874E+06
         14          7.782757E+06          3.113103E+03
         25          3.932071E+06          1.572828E+03
         28          3.932071E+06          1.572828E+03

FREQUENCY =  4.000000E+03
```


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```
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL3)
  ELEMENT ID             ERP                 ERP DENSITY
         5          2.744031E+09           2.195225E+06
         6          4.020963E+09           3.216771E+06
        29          4.020963E+09           3.216771E+06

FREQUENCY =  4.000000E+03
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL4)
  ELEMENT ID             ERP                 ERP DENSITY
         2          1.214752E+09           4.859006E+05
        14          6.009078E+05           2.403631E+02
        25          2.783943E+06           1.113577E+03
        28          2.783943E+06           1.113577E+03
```

8. The following sample of SORT2 output for elements is from a model where the panels selected for output are constructed from PLOTEL3 and PLOTEL4 elements, and there are two forcing frequencies.

```
ELEMENT ID = 5
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL3)
  FREQUENCY                ERP                 ERP DENSITY
3.000000E+03          2.196778E+10           1.757423E+07
4.000000E+03          2.744031E+09           2.195225E+06

ELEMENT ID = 6
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL3)
  FREQUENCY                ERP                 ERP DENSITY
3.000000E+03          3.317672E+10           2.654138E+07
4.000000E+03          4.020963E+09           3.216771E+06

ELEMENT ID = 29
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL3)
  FREQUENCY                ERP                 ERP DENSITY
3.000000E+03          3.317672E+10           2.654138E+07
4.000000E+03          4.020963E+09           3.216771E+06

ELEMENT ID = 2
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL4)
  FREQUENCY                ERP                 ERP DENSITY
3.000000E+03          8.997186E+09           3.598874E+06
4.000000E+03          1.214752E+09           4.859006E+05

ELEMENT ID = 14
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL4)
  FREQUENCY                ERP                 ERP DENSITY
3.000000E+03          7.782757E+06           3.113103E+03
4.000000E+03          6.009078E+05           2.403631E+02

ELEMENT ID = 25
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL4)
  FREQUENCY                ERP                 ERP DENSITY
3.000000E+03          3.932071E+06           1.572828E+03
4.000000E+03          2.783943E+06           1.113577E+03

ELEMENT ID = 28
```


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```
        ELEMENT EQUIVALENT RADIATED POWER (PLOTEL4)
  FREQUENCY                ERP                ERP DENSITY
3.000000E+03          3.932071E+06          1.572828E+03
4.000000E+03          2.783943E+06          1.113577E+03
```

Related Topics

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ESE

Element Strain Energy Output Request

Requests the output of the strain energy in selected elements.

Format:

$$\text{ESE}\left[\left(\left[\begin{matrix}(\underline{\text{PRINT}},\ \text{PUNCH})\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{AVERAGE}}\\ \text{AMPLITUDE}\\ \text{PEAK}\end{matrix}\right],\left[\begin{matrix}\underline{\text{RMAG}}\\ \text{REAL or IMAGE}\\ \text{PHASE}\end{matrix}\right],\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],[\text{THRESH}=\text{p}]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\end{matrix}\right\}$$

Examples:

```
ESE=ALL
ESE (PUNCH, THRESH=.0001)=19
```

Describers:

Describer	Meaning
PRINT	Write energies to the print file (default).
PUNCH	Write energies to the punch file.
PLOT	Do not write energies to either the punch file or the print file.
AVERAGE	Requests average energy in frequency response analysis only.
AMPLITUDE	Requests amplitude of energy in frequency response analysis only.
PEAK	Requests peak energy for frequency response analysis only. PEAK is the sum of AVERAGE and AMPLITUDE.
RMAG	Outputs the energy magnitude in real data format.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Only valid with the AVERAGE option.)
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees. (Only valid with the AVERAGE option.)

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Describer	Meaning
SORT1	Output will be presented as a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output will be presented as a tabular listing of frequency or time for each grid point.
THRESH	Suppresses energies for elements having an energy value of less than p% in all output files. THRESH overrides the value of TINY described in Remark 1. (Default=0.001)
ALL	Computes energy for all elements.
n	Set identification number. Energy for all elements specified on the SET n command will be computed. The SET n command must be specified in the same subcase as the ESE command or above all subcases. (Integer>0)
NONE	Element strain energy will not be output.

Remarks:

1. If THRESH = p is not specified then p defaults to the values specified by user parameter TINY.
2. The energy calculations include the contribution of initial thermal strain.
3. Energy density (element strain energy divided by element volume) is also computed in some solution sequences. It can be suppressed by use of PARAM,EST,-1.
4. The equations used to calculate elemental strain energy components are given below.

   Average Strain Energy:

$$E_{avg} = \frac{1}{4}\left[\{u_r\}^T [K_e]\{u_r\} + \{u_i\}^T [K_e]\{u_i\}\right]$$

   where:

E	=	elemental energy
$\{u_r\}$	=	displacement - real part
$\{u_i\}$	=	displacement - imaginary part
$[K_e]$	=	elemental stiffness

   Real Part of Strain Energy Amplitude:

$$E_{amp_real} = \frac{1}{4}\left[\{u_r\}^T [K_e]\{u_r\} - \{u_i\}^T [K_e]\{u_i\}\right]$$

   Imaginary Part of Strain Energy Amplitude:

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$$E_{amp_imag} = -\frac{1}{2}\{u_r\}^T [K_e]\{u_i\}$$

Magnitude of Strain Energy Amplitude:

$$E_{amp_mag} = \sqrt{E^2_{amp_real} + E^2_{amp_imag}}$$

Phase of Strain Energy Amplitude:

$$\theta_{amp} = \arctan\left(\frac{E_{amp_imag}}{E_{amp_real}}\right)$$

Peak Strain Energy:

$$E_{peak} = E_{avg} + E_{amp_mag}$$

5. In SOL 111, ESE can only be requested if PARAM,DDRMM,-1 is used.
6. Element data recovery for thermal loads is not currently implemented in dynamics.
7. Element strain energy is available for nonlinear static analysis (SOL 106), but the INTOUT entry on the NLPARM bulk entry must be set to "YES" since the strain energy at the intermediate steps is required to correctly integrate the strain energy over all load steps. All other nonlinear solution sequences do not support element strain energy output.
8. The strain energy for nonlinear elements is calculated by integrating the specific energy rate, which is the inner product of strain rate and stress, over element volume and time.

Equation 4-1:

$$E = \int_0^t \int_V \dot{\varepsilon}^T \sigma dV d\tau$$

where:

σ	=	stress tensor
$\dot{\varepsilon}$	=	tensor of the strain rate
V	=	element volume
t	=	actual time in the load history

Loads from temperature changes are included in **Equation 4-1**. If we assume a linear variation of temperatures from subcase to subcase, then the strain energy in **Equation 4-1**, for the special case of linear material and geometry becomes

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Equation 4-2:

$$E = \frac{1}{2}u^T K_e u - \frac{1}{2}u^T P_{et}$$

where P_{et} is the element load vector for temperature differences.

For linear elements, the default calculation of strain energy is also **Equation 4-2**. You can optionally set the parameter XFLAG to 0 to change the strain energy calculation to

Equation 4-3:

$$E = \frac{1}{2}u^T K_e u - u^T P_{et}$$

In **Equation 4-3**, the temperatures are assumed to be constant within a subcase. See the XFLAG parameter in the *Simcenter Nastran Quick Reference Guide*.

Remark related to SOLs 401 and 402:

1. Strain energy requested with the ESE command is only output from a normal modes subcase.

Remarks related to SOLs 402 and 414:

1. In SOL 402, only PRINT, PUNCH, PLOT, REAL, THRESH and SORT1 are used. All the other describers are ignored.

   As a consequence, in complex modes analyses, only the REAL part can be post-processed. The IMAG part is always zeroed and the PHASE part is useless.
2. The output corresponds to the $u^T * K_{tan} * u$ value for the given mode.
3. In SOL 402, the THRESH parameter has no default. In addition, SOL 402 does not support the PARAM,TINY definition.
4. SOLs 402 and 414 compute strain energy as follows.

   Strain energy ratio of element *e*:

$$SER = \frac{x_e^{\mathrm{T}} K_e x_e}{x^{\mathrm{T}} K x}$$

   Strain energy density of element *e*:

$$SED = (SER)\frac{V}{V_e}$$

   where,

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M and K are the mass and stiffness matrices of a volume V,
M_e and K_e are the mass and stiffness matrices of element e of volume V_e,
and x_e is the displacement vector of element e for mode x.

Related Topics

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ESHLPNL

Tension-only Shell Element Output

Tension-only shell element output (SOL 401)

Format:

$$\text{ESHLPNL}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{matrix}\text{YES} \\ \underline{\text{NO}}\end{matrix}\right\}$$

Examples:

```
ESHLPNL=YES
```

Describers:

Describer	Meaning
PRINT	Output is written to the .f06 file and the .op2 file. (Default)
PUNCH	Output is written to the .pch file and the .op2 file.
PLOT	Output is written to the .op2 file.
YES	Request output.
NO	No output requested. (Default)

Remarks:

1. Tension-only shell elements are defined with either of the following element and property combinations:

   - CQUAD4 or CQUADR elements that reference PSHLPNL bulk entries.

     A CQUAD4 or CQUADR element that references a PSHLPNL bulk entry initially behave as a shell element, but during the solution can convert to behave as a shear panel.

   - CSHEAR elements that reference PSHEAR bulk entries.

     A CSHEAR element that references a PSHEAR bulk entry behaves as a shear panel throughout the solution.

   The ESHLPNL command outputs the status (shell or shear panel behavior) and, if applicable, the conversion time for each tension-only shell element at each time point. The conversion time is the time point during the solution when the tension-only shell element status changes from shell to shear panel.

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- For CQUAD4 or CQUADR elements that reference PSHLPNL bulk entries, STATUS = 0 denotes that the element behaves as a shell element up to that time point. For such a case, NONE is listed as the corresponding conversion time. STATUS = 1 denotes that the element behaves as a shell element up to the corresponding conversion time and then behaves as a shear panel for the remainder of the solution unless you start a new sequentially independent subcase.
- For CSHEAR elements that reference PSHEAR bulk entries, because they behave as a shear panel throughout the solution, STATUS =1 and the corresponding conversion time is 0.0.

An example of ESHLPNL command output in an .f06 file is as follows:

TIME = 5.000000E-01

S H E A R P A N E L E L E M E N T S T A T U S
(STATUS: 0 - SHELL, 1 - SHEAR PANEL)

ELEMENT ID	STATUS	CONVERSION TIME	ELEMENT ID	STATUS	CONVERSION TIME	ELEMENT ID	STATUS	CONVERSION TIME
3	0	NONE	4	0	NONE	5	0	NONE
6	1	2.500000E-01	7	1	2.500000E-01	8	1	2.500000E-01
9	1	5.000000E-01	41	1	0.000000E+00	42	1	0.000000E+00

In this example:

- Elements 3, 4, and 5 behave as shell elements through the time point for the output of 0.5.
- Elements 6, 7, and 8 converted to behave as shear panels at the previous time point of 0.25.
- Element 9 converted to behave as a shear panel at the time point for the output of 0.5.
- Because elements 41 and 42 have STATUS =1 and a corresponding conversion time of 0.0, they are CSHEAR elements that reference PSHEAR bulk entries. The other elements are CQUAD4 or CQUADR elements that reference PSHLPNL bulk entries.

Related Topics

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EXTSEOUT

External Superelement Creation Specification

Specify the various requirements for the creation of an external superelement.

Format:

$$\text{EXTSEOUT}\left[\left(\left[\underline{\text{STIFFNESS}},\ \underline{\text{MASS}},\ \underline{\text{DAMPING}},\ \underline{\text{K4DAMP}},\ \underline{\text{LOADS}},\right.\right.$$

$$\left.\text{GEOM},\ \text{MODACC},\ \text{FSCOUP},\ \text{ASMBULK},\ \text{EXTBULK},\ \text{EXTID}=\text{seid},\right.$$

$$\left.\text{DMIGSFIX}=\left\{\begin{matrix}\text{ccccccc}\\ \text{EXTID}\end{matrix}\right\}\right],\left[\begin{matrix}\underline{\text{MATDB (or MATRIXDB)}}\\ \text{DMIGDB}\\ \text{DMIGOP2}=\text{unit}\\ \text{DMIGPCH}\\ \text{MATOP4 (or MATRIXOP4)}=\text{unit}\end{matrix}\right]\left.\right)\left.\right]$$

Examples:

```
EXTSEOUT
EXTSEOUT(ASMBULK,EXTID=100)
EXTSEOUT(ASMBULK,EXTBULK,EXTID=200)
EXTSEOUT(EXTBULK,EXTID=300)
EXTSEOUT(DMIGDB)
EXTSEOUT(ASMBULK,EXTID=400,DMIGOP2=21)
EXTSEOUT(EXTID=500,DMIGPCH)
EXTSEOUT(ASMBULK,EXTBULK,EXTID=500,DMIGSFIX=XSE500,DMIGPCH)
EXTSEOUT(ASMBULK,EXTBULK,EXTID=500,DMIGSFIX=EXTID,DMIGPCH)
EXTSEOUT(STIF,MASS,DAMP,EXTID=600,ASMBULK,EXTBULK,MATDB)
EXTSEOUT(STIF,MASS,DAMP,GEOM,EXTID=600)
```

Describers:

Describer	Meaning
STIFFNESS	Store the boundary stiffness matrix. See Remarks 2, 3, and 1.
MASS	Store the boundary mass matrix. See Remarks 2 and 1.

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Describer	Meaning
DAMPING	Store the boundary viscous damping matrix. See Remarks 2, 3, and 1.
K4DAMP	Store the boundary structural damping matrix. See Remarks 2, 3, and 1.
LOADS	Store the loads matrix and associated DTI,SELOAD entries. See Remarks 2, 3, and 22.
GEOM	Store geometry. See Remark 21.
MODACC	Create the external superelement output transformation matrices using the mode acceleration method. See Remarks 26 and 27.
FSCOUP	Create the a-set acoustic area matrix. See Remark 28.
ASMBULK	Generate bulk entries related to the subsequent superelement assembly process and store them on the assembly punch file (.asm). This data is to be included in the main bulk portion of the subsequent assembly solution. See Remarks 5 and 14.
EXTBULK	Generate and store bulk entries for the external superelement on the standard punch file (.pch) when used in combination with one of either MATDB, DMIGDB, or DMIGOP2. This data is used in the BEGIN SUPER portion of the bulk section of the subsequent assembly solution. EXTBULK is ignored if either DMIGPCH or MATOP4 is specified. If EXTBULK is not specified, the subsequent assembly solution retrieves the required data for the external superelement from the medium on which the boundary matrices are stored. See Remarks 6 and 7.
EXTID = *seid*	*seid* (integer>0) is the superelement ID to be used in the SEBULK and SECONCT bulk entries stored on the assembly punch file (.asm) if ASMBULK is specified and in the BEGIN SUPER bulk entry stored on the standard punch file (.pch) if DMIGPCH or MATOP4 is specified. See Remarks 4, 5, 6, and 8.
DMIGSFIX = *cccccc*	*cccccc* is the suffix (up to six characters and must not = any EXTSEOUT keyword) that is to be employed in the names of the DMIG matrices stored on the standard punch file (.pch) if the DMIGPCH keyword is specified. See Remarks 9 through 12.
DMIGSFIX = EXTID	The *seid* defined by the EXTID keyword is the suffix that is to be employed in the names of the DMIG matrices stored on the standard punch file (.pch) if the DMIGPCH keyword is specified. See Remarks 9 through 12.
MATDB (or MATRIXDB)	Store the boundary matrices and other information on the database (default).
DMIGDB	Similar to MATDB (or MATRIXDB) except that the boundary matrices are stored as DMIG bulk entries on the database.
DMIGOP2 = *unit*	Store the boundary matrices as DMIG bulk entries on an OUTPUT2 file whose Fortran unit number if given by *unit* (integer>0). See Remark 15.

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Describer	Meaning
DMIGPCH	Store the boundary matrices as DMIG bulk entries on the standard punch file (.pch). See Remarks 7 through 14.
MATOP4 = *unit* (or MATRIXOP4 = *unit*)	Store the boundary matrices on an OP4 file whose Fortran unit number is given by *unit*. (Integer ≠ 0) If *unit* > 0, matrices are written to the OP4 file in sparse format. If *unit* < 0, matrices are written to the OP4 file in full matrix format. See Remarks 4, 6, 7, 14, 15, and 24.

Remarks:

1. You can create an external superelement using either a SOL 101 static reduction or a SOL 103 modal reduction.

   After you create one or more external superelements, you can include them with your residual structure in an assembly solution.

   You can include external superelements in SOLs 101, 103, 107-112, 129, 144-146, 159, 187, 402, and 414 for an assembly solution.

   See External Superelements in the *Superelement User's Guide*.

   See SOL 402 - External superelements in the *Multi-Step Nonlinear User's Guide*.

   See SOL 414 - Modeling with external superelements in the *Rotor Dynamics User's Guide (SOL 414)*.

2. If none of the describers STIFFNESS, MASS, DAMPING, K4DAMP, and LOADS are specified, then all of the boundary matrices are stored by default. If any subset of the describers STIFFNESS, MASS, DAMPING, K4DAMP, and LOADS are specified, then only the boundary matrices specified are stored.

3. STIFFNESS, DAMPING, K4DAMP, and LOADS may be abbreviated to STIF, DAMP, K4DA, and LOAD, respectively.

4. EXTID and an *seid* value must be specified if one or more of ASMBULK, EXTBULK, DMIGPCH, or MATOP4 are specified. If the DMIGSFIX=EXTID form is employed along with the DMIGPCH keyword, the value *seid* may not exceed 999999, since this value becomes part of the names given to the DMIG matrices generated on the standard punch file (.pch). See Remark 12.

5. If ASMBULK is specified, the following bulk entries are generated and stored on the assembly punch file (.asm):

   SEBULK *seid* ...

   SECONCT *seid* ...

   GRID entries for the boundary points

   CORD2x entries associated with the above GRID entries

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6. If DMIGPCH is not specified, but MATOP4 or EXTBULK (in combination with MATDB, DMIGDB, or DMIGOP2) is specified, the following bulk entries are generated and stored on the standard punch file (.pch):

   BEGIN SUPER *seid*

   GRID entries for the boundary points

   GRID entries for the interior points referenced by PLOTEL entries

   CORD2x entries associated with the above GRID entries

   EXTRN

   ASET/ASET1

   QSET/QSET1

   SPOINT

   PLOTEL

7. If DMIGPCH or MATOP4 is specified, then EXTBULK is ignored even if it is specified.

8. If DMIGPCH is specified, the following bulk entries are generated and stored on the standard punch file (.pch):

   BEGIN SUPER seid

   GRID entries for the boundary points

   CORD2x entries associated with the above GRID entries

   ASET/ASET1

   SPOINT

   DMIG entries for the requested boundary matrices

9. The DMIGSFIX keyword is ignored if DMIGPCH is not specified.

10. If DMIGPCH is specified without the DMIGSFIX keyword, then the boundary DMIG matrices generated and stored on the standard punch file (.pch) will have names of the following form:

    KAAX (boundary stiffness matrix)

    MAAX (boundary mass matrix)

    BAAX (boundary viscous damping matrix)

    K4AAX (boundary structural damping matrix)

    PAX (boundary load matrix)

11. If the DMIGSFIX = *cccccc* form is employed along with the DMIGPCH keyword, then the boundary DMIG matrices generated and stored on the standard punch file (.pch) will have names of the following form:

    K*cccccc* (boundary stiffness matrix)

    M*cccccc* (boundary mass matrix)

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B*cccccc* (boundary viscous damping matrix)

K4*cccccc* (boundary structural damping matrix)

P*cccccc* (boundary load matrix)

12. If the DMIGSFIX = EXTID form is employed along with the DMIGPCH keyword, then the boundary DMIG matrices generated and stored on the standard punch file (.pch) will have names of the following form:

    K*seid* (boundary stiffness matrix)

    M*seid* (boundary mass matrix)

    B*seid* (boundary viscous damping matrix)

    K4*seid* (boundary structural damping matrix)

    P*seid* (boundary load matrix)

13. If the DMIGPCH option is specified, the boundary DMIG matrices generated and stored on the standard punch file (.pch) may not be as accurate as the boundary matrices resulting from other options (MATDB/MATRIXDB or DMIGOP2 or MATOP4/MATRIXOP4). Accordingly, this may result in decreased accuracy from the subsequent assembly job utilizing these DMIG matrices.

14. The punch output resulting from EXTSEOUT usage is determined by ASMBULK, EXTBULK, DMIGPCH, and MATOP4 as follows:

    - No ASMBULK, EXTBULK, DMIGPCH, or MATOP4 results in no punch output.
    - ASMBULK, but no DMIGPCH, MATOP4, or EXTBULK (in combination with MATDB, DMIGDB, or DMIGOP2) results in punch output being generated and stored on the assembly punch file (.asm). See Remark 5.
    - No ASMBULK, but DMIGPCH, MATOP4, or EXTBULK (in combination with MATDB, DMIGDB, or DMIGOP2) results in punch output being generated and stored on the standard punch file (.pch). See Remarks 6 or 8, as appropriate.
    - ASMBULK and DMIGPCH, MATOP4, or EXTBULK (in combination with MATDB, DMIGDB, or DMIGOP2) results in punch output consisting of two distinct and separate parts. One part is generated and stored on the assembly punch file (.asm) as indicated in Remark 5. The other part is generated and stored on the standard punch file (.pch) as indicated in Remark 6 or 8, as appropriate.

15. If DMIGOP2=*unit* or MATOP4=*unit* is specified, an appropriate ASSIGN OUTPUT2 or ASSIGN OUTPUT4 statement must be present in the File Management Section (FMS) for the absolute value of *unit*. An OUTPUT2 file created during a LP-64 Nastran executable superelement creation run cannot be used in an ILP-64 Nastran executable system run and vice versa. Use the SEOP2CV parameter to control the binary format of the OUTPUT2 file. Note that beginning in NX Nastran12, only the ILP-64 executable is available.

16. The creation of an external superelement using EXTSEOUT involves running a Simcenter Nastran job with the following additional data:

    - The data for the creation of the external superelement is specified by the EXTSEOUT case control command, which must appear above the subcase level.

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- The boundary points of the external superelement are specified by ASET/ASET1 bulk entries.
- If the creation involves component mode reduction, the required generalized coordinates are specified using QSET/QSET1 bulk entries. The boundary data for the component mode reduction may be specified using the BNDFIX/BNDFIX1 and BNDFREE/BNDFREE1 bulk entries (or their equivalent BSET/BSET1 and CSET/CSET1 bulk entries). (The default scenario assumes that all boundary points are fixed for the component mode reduction.)
- The output for the external superelement is generated in the assembly job. This output consists of displacements, velocities, accelerations, SPC forces, MPC forces, grid point force balances, stresses, strains, and element forces. However, in order for this output to be generated in the assembly job, the output requests must be specified in the external superelement creation run. Normally, the only output requests for the external superelement that are honored in the assembly job are those that are specified in the creation run. There is, however, one important exception to this: the displacement, velocity, acceleration, SPC forces, and MPC forces output for the boundary grid points as well as for all grid points associated with the PLOTEL, PLOTEL3, PLOTEL4, PLOTEL6, PLOTEL8, PLOTHEX, PLOTTET, PLOTPEN, and PLOTPYR bulk entries can be obtained in the assembly job *even if there is no output request specified for these points in the creation run*.
- If the assembly job involves the use of PARAM bulk entries, then the following points should be noted:
  - PARAM entries specified in the main bulk portion of the input data apply *only to the residual and not to the external superelement*.
  - PARAM entries specified in the BEGIN SUPER portion of the bulk section for an external superelement apply *only to the superelement*.
  - The most convenient way of ensuring that PARAM entries apply not only to the residual but also to all external superelements is to specify such PARAM entries in Case Control, not in the main bulk section. This is particularly relevant for such PARAMs as POST.

17. Output transformation matrices are generated for the following outputs requested in the in external superelement run with EXTSEOUT:

- DISPLACEMENT
- VELOCITY
- ACCELERATION
- SPCFORCE
- MPCFORCES
- GPFORCE
- STRESS
- STRAIN
- FORCE

Only these external superelement results can be output in the system analysis run. PARAM,OMID,YES is not applicable to the output transformation matrices.

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18. If a PARAM,EXTOUT or PARAM,EXTUNIT also exist, they will be ignored. The existence of the EXTSEOUT case control command takes precedence over PARAM,EXTOUT and PARAM,EXTUNIT.

19. When an external superelement was created in SOL 101 and then used in a SOL 101 system run, the static offset output transformation matrices that were created for the external superelement are appropriately scaled during the external superelement results recovery. The scaling is based on P2G case control scaling or LOAD bulk data scaling. If an external superelement created in SOL 101 is used in a system run other than a SOL 101 run, the static offset output transformation matrices are ignored in the superelement results recovery.

If you create an external superelement in SOL 101 and include thermal loads, the thermal loads are available in the system run via the o-set output transformation matrices. When you request thermal loads in the system run, the grid point results such as displacements are correct, but element-based results such as stress, strain, and forces are not correct because the software cannot remove thermal strain from the modal-based output transformation matrices for these quantities.

Fluid and acoustic elements such as absorbers are not supported in an external superelement, although they can be included as part of the residual in a system run. For SOL 103, this capability creates the external superelement and also creates the component results and data blocks that are written to an .op2 file if requested. For the other solution sequences, this capability only creates the external superelement and no other analysis is performed. Superelement results can be recovered in the second step (i.e. superelement assembly, analysis, and data recovery) for SOLs 101, 103, 105, 107-112, and 144-146.

20. The run creating an external superelement is not a superelement assembly run. The bulk entries SUPER, SEALL, SESET, and BEGIN SUPER should not be defined for the external superelement creation run.

21. The GEOM describer will output geometry data blocks GEOM1EXA, GEOM2EXA, and GEOM4EXA containing all of the external superelement geometry to support post-processing. This describer only works for the MATDB (or MATRIXDB), DMIGDB, and DMIGOP2 storage options. By default, the full geometry will not be exported; the GEOM describer must be explicitly defined to have these geometry data blocks written.

22. The LOADS describer will output load information in the $[P_a]$ matrix along with associated DTI,SELOAD bulk entries for each load represented in the $[P_a]$ matrix. DTI,SELOAD bulk entries will not be output for the DMIGPCH option; the use of the DMIGPCH option requires the use of the P2G case control command in the system analysis in order to access the load information defined in the $[P_a]$ matrix that is stored in DMIG format. Thermal loads and enforced motion loads using the SPCD bulk data definition method are not supported. The load information and DTI,SELOAD bulk entries that are output depend on the method in which loads are defined and referenced.

Static solutions. For example, SOL 101:

- In a creation run, the LOAD = *n* case control command will create a single load. The $[P_a]$ matrix will have a single column that contains the load values referenced by *n*. The corresponding LIDSE and EXCSE values on the DTI,SELOAD bulk entry will be the value *n*.

  To select this load in a system run, the LIDSE field on the SELOAD bulk entry should equal the value of *n* from the creation run.

- In a creation run, the LOADSET = *n* case control command is generally used to create multiple loads. Each definition of an LSEQ bulk entry will create a column in the $[P_a]$ matrix regardless of whether or not the LSEQ is referenced by the LOADSET = *n* case control command. However, an

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unreferenced LSEQ will generate a null column in the [P_a] matrix. The corresponding LIDSE and EXCSE on an DTI,SELOAD bulk entry will be the value of LID and EXCITEID, respectively, on an LSEQ bulk entry.

To select these loads in a system run, the LIDSE field on the SELOAD bulk entry should equal the value of an EXCITEID from the creation run.

- If both LOADSET and LOAD appear in the case control, LOADSET takes precedence.

Dynamic solutions. For example, SOL 103:

- In a creation run, a column in the [P_a] matrix is created for each load (not enforced motion) defined on RLOADi and TLOADi bulk entries whether or not they are referenced in the case control. The corresponding LIDSE and EXCSE on the DTI,SELOAD bulk entries will both be the value of EXCITEID on the RLOADi or TLOADi bulk entry.

  To select these loads in a system run, the LIDSE field on the SELOAD entry should equal the value of an EXCITEID from the creation run.

- In a creation run, the LOADSET = n case control command can be used to create multiple loads. Each definition of an LSEQ bulk entry will create a column in the [Pa] matrix. A LSEQ bulk entry must be referenced by a LOADSET case control command in order to generate a column in the [P_a] matrix. The corresponding LIDSE and EXCSE on an DTI,SELOAD bulk entry will be the value of LID and EXCITEID, respectively, on an LSEQ bulk entry.

  To select these loads in a system run, the LIDSE field on the SELOAD entry should equal the value of an EXCITEID from the creation run.

23. To include differential stiffness in the definition of an external superelement, two subcases are required in the creation run. The first subcase is a static subcase. The second subcase performs the analysis and generation of the external superelement. To obtain the displacement field used to generate the differential stiffness effects, include a STATSUB case control command in the second subcase that references the first subcase. Always include the EXTSEOUT case control command above the subcase level. An example of the required setup is as follows:

```
…
SOL 103
CEND
TITLE = …
EXTSEOUT(…)
$
SUBCASE 1
     $ STATIC SUBCASE
     LOAD = 11
$
SUBCASE 2
     $ MODAL SUBCASE
     TSTEP = 100
     STATSUB = 1
     METHOD = 10
BEGIN BULK
…
```


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For accuracy and consistency, the loads used to generate differential stiffness for the external superelement during the creation run should be the same loads used in the system run without any scaling. If the loads are scaled by a non-unity scaling factor from a case control command like P2G or a bulk entry like LOAD, the differential stiffness portion of the external superelement stiffness matrix will no longer be consistent with the applied loads.

The ability to generate an external superelement including differential stiffness effects is available for SOLs 103, 107-112, and 187. For SOL 112, the IC(STATSUB,DIFFK) or IC(TZERO,DIFFK) case control commands can be used to generate differential stiffness effects instead of the STATSUB case control command.

24. For the MATOP4 (or MATRIXOP4) option, the number of digits of precision for matrix data is controlled by the DIGITS parameter.

25. See **Parameters and Superelements** for considerations regarding parameter definitions in a system solution which includes external superelements.

26. The MODACC describer generates all the output transformation matrices discussed in Remarks **16** and **17** using the mode acceleration method. The MODACC describer is only applicable when the external superelement is dynamically reduced using component mode synthesis. When any other reduction method such as Guyan reduction is used, the software issues a fatal error.

27. When the MODACC describer is specified, residual vectors are automatically turned off.

28. The FSCOUP describer reduces the g-set acoustic area matrix, AGG, to the user-defined a-set DOF. The FSCOUP describer is valid only when the DMIGPCH describer is specified. When any other output media is specified, the FSCOUP describer is ignored.

29. Specifying PARAM,WMODAL,YES causes hysteretic damping to be converted to viscous damping, which is useful when the external superelement is used in a system transient analysis.

Remarks related to SOLs 402 and 414:

1. Note the following when you include an external superelement in SOLs 402 and 414:

- The STIFFNESS matrix is required.
- The MASS matrix is optional.
- The DAMPING (viscous damping) matrix is optional.

  The software will take viscous damping into account in SOL 402 Multi-Step Nonlinear Kinematics and SOL 414 rotor dynamics solutions.
- The K4DAMP (structural/hysteretic damping) matrix is ignored.

  Even if you define structural/hysteretic damping, the software will not take it into account in SOL 402 Multi-Step Nonlinear Kinematics and SOL 414 rotor dynamics solutions.

Remarks related to SOL 414:

1. For SOL 414 external superelements, the software automatically stores all the boundary matrices in the .sdb and .u18 files. To write the boundary matrices to the .f06 file, use the SEMATPRT parameter.

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2. In a SOL 414 analysis, a superelement created by a SOL 101 or SOL 103 solution in .op2 format always represents a non-rotating part. The results of those superelements cannot be recovered on the original structure. Only superelements created by a SOL 414,103 can be recovered.

Related Topics

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FHAC

Removes Fourier Harmonic Numbers

Removes harmonic numbers for Fourier multi-harmonics in SOLs 402 and 414.

Format:

$$\text{FHAC} = \text{n}$$

Examples:

```
FHAC=101
```

Describers:

Describer	Meaning
n	Set identification number. The referenced SET lists the harmonic numbers to remove from the solution. (Integer > 0)

Remarks:

1. The FHAC command must be defined inside subcases.
2. Use the FHAC command to remove a subset of the harmonic numbers requested with the FHAR command. If a FHAC command is not present, the software accounts for all the harmonics requested with the FHAR command.
3. Every subcase of the same type such as static subcases or complex modal subcases, must request the same harmonic numbers. Thus, when you include a FHAC command in a subcase, you must also include identical FHAC commands in all other subcases of the same type.

   The following examples demonstrate the use of the FHAC command in the various types of SOL 414 rotor dynamic analysis.

   - For SOL 414,101, you can include one or more static subcases only. Thus, if you want to specify a FHAC command in one of the subcases, you must also include identical FHAC command specifications in all of the other subcases.

```
......
SET 3 = 0,1,2
FHAR = 3
SET 4 = 1,2
......
SUBCASE 1
  LABEL = Subcase - Static Loads 1
  LOAD = 23
```


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```
  FHAC = 4
SUBCASE 2
  LABEL = Subcase - Static Loads 2
  LOAD = 24
  FHAC = 4
SUBCASE 3
  LABEL = Subcase - Static Loads 3
  LOAD = 25
  FHAC = 4
......
```

- For SOL 414,103 you can include one static or transient preload subcase and one modal or frequency response subcase. The static or transient preload subcase must precede the modal or frequency response subcase. Thus, you can include a FHAC command specification in one or both of the subcases. If you include FHAC command specifications in both, they do not have to be identical to one another.

```
......
SET 3 = 0,1,2
FHAR = 3
SET 4 = 1,2
SET 5 = 2
SUBCASE 1
  LABEL = Subcase - Static Loads 1
  LOAD = 23
  FHAC = 4
SUBCASE 2
  LABEL = Subcase – Eigenvalues and Superelement
  LOAD = 24
  FHAC = 5
......
```

- For SOL 414,110 and SOL 414,111, you can include one static subcase and one modal or frequency response subcase. The static subcase must precede the modal or frequency response subcase. Thus, you can include a FHAC command specification in one or both of the subcases. If you include FHAC command specifications in both, they do not have to be identical to one another.

Note

The example is for SOL 414,110, but the same subcase organization is used for SOL 414,111.

```
......
SET 3 = 0,1,2
FHAR = 3
SET 4 = 1,2
SET 5 = 2
SUBCASE 1
  LABEL = Subcase - Static Loads 1
  LOAD = 23
  FHAC = 4
SUBCASE 2
  LABEL = Subcase – Modal Complex
```


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```
  LOAD = 24
  FHAC = 5
......
```

- For SOL 414,129, you can include a single nonlinear transient subcase only. For this case, the zeroth and first harmonic are requested because the FHAC command excludes the second harmonic.

```
......
SET 3 = 0,1,2
FHAR = 3
SET 6 = 2
SUBCASE 1
  LABEL = Subcase - NonLinear Transient
  LOAD = 23
  FHAC = 6
......
```

Related Topics

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FHAR

Requests Harmonic Behavior for Fourier elements

Requests the harmonic numbers of harmonic behavior for Fourier elements in SOLs 402 and 414.

Format:

$$\mathrm{FHAR} = \mathrm{n}$$

Examples:

```
FHAR=101
```

Describers:

Describer	Meaning
n	Set identification number. The referenced SET lists the harmonic numbers of harmonic behavior. (Integer > 0)

Remarks:

1. The FHAR command must be defined above the subcases (globally). Thus, it applies to all subcases.
2. You use the FHAR case control command to request the harmonic numbers of harmonic behavior for Fourier axisymmetric elements. The FHAR references the ID of a SET case control command in which the SET command lists the harmonic numbers.
3. When the FHAR command is defined, the software automatically considers all axisymmetric elements as Fourier elements.
4. The following example shows a FHAR command specification and the corresponding SET specification. The SET command lists the numbers of the harmonics to use.

   ```
   ......
   FHAR=101
   SET 101=3,5,8,12
   ......
   ```

5. When your model includes a bolt preload and/or a contact condition on axisymmetric elements, only the 0th harmonic is supported. If you request harmonics higher than 0, the solution will end with a fatal error.

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Related Topics

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FLSFSEL

Fluid-Structure Frequency Selection

Frequency selection for controlling the fluid-structure interaction.

Format:

$$\mathrm{FLSFSEL}\left[\mathrm{LFREQFL}=\left\{\frac{0.0}{\mathrm{f}_1}\right\}\right],\ \left[\mathrm{HFREQFL}=\left\{\frac{1.0\mathrm{E}+30}{\mathrm{f}_1}\right\}\right],$$
$$\left[\mathrm{LFREQF}=\left\{\frac{0.0}{\mathrm{f}_1}\right\}\right],\ \left[\mathrm{HFREQF}=\left\{\frac{1.0\mathrm{E}+30}{\mathrm{f}_1}\right\}\right],$$
$$\left[\mathrm{LMODESFL}=\left\{\begin{matrix}0\\ \mathrm{m}_1\end{matrix}\right\}\right],\ \left[\mathrm{LMODES}=\left\{\begin{matrix}0\\ m_1\end{matrix}\right\}\right],$$
$$\left[\mathrm{FLUIDSE}=\left\{\begin{matrix}0\\ \mathrm{seid}\end{matrix}\right\}\right]$$

Examples:

```
FLSFSEL LFREQL=1.0 HFREQFL=2.0
```

Describers:

Describer	Meaning
LFREQFL	The lower bound modal frequency for fluid modes in hertz. Default is 0.0 unless f_1 is specified.
HFREQFL	The upper bound modal frequency for fluid modes in hertz. Default is 1.0E+30 unless f_1 is specified.
LFREQ	The lower bound modal frequency for structural modes in hertz. Default is 0.0 unless f_1 is specified.
HFREQF	The upper bound modal frequency for structural modes in hertz. Default is 1.0E+30 unless f_1 specified.
LMODESFL	The number of lowest fluid modes to be used. If 0, then LFREQFL and HFREQFL will be used to determined number of modes. Default is 0 unless m_1 is specified.

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Describer	Meaning
LMODES	The number of lowest structural modes to be used. If 0, then LFREQFL and HFREQFL will be used to determined number of modes. Default is 0 unless m_1 is specified.
FLUIDSE	Defines a superelement for fluids only. Default is 0, unless *seid* is specified.

Remarks:

All the entries specified in this case control statements are available as PARAMETER statements.

Related Topics

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FLSPOUT

Fluid-Structure Modal Participation Output

Modal participation output control for the fluid-structure interaction.

Format:

$$\text{FLSPOUT}\ \left[\text{FLUIDMP} = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \underline{\text{NONE}}\end{matrix}\right\}\right],\ \left[\text{STRUCTMP} = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \underline{\text{NONE}}\end{matrix}\right\}\right],$$
$$\left[\text{GRIDFMP} = \left\{\frac{\text{ALL}}{\text{setf}}\right\}\right],\ \left[\text{GRIDMP} = \left\{\begin{matrix}\text{ALL}\\ \text{setg}\\ \underline{\text{NONE}}\end{matrix}\right\}\right],$$
$$\left[\text{OUTFMP} = \left\{\begin{matrix}\text{ALL}\\ \text{p}\\ \underline{\text{NONE}}\end{matrix}\right\}\right],\ \left[\text{OUTSMP} = \left\{\begin{matrix}\text{ALL}\\ \text{p}\\ \underline{\text{NONE}}\end{matrix}\right\}\right],$$
$$\left[\text{PANELMP} = \left\{\begin{matrix}\text{ALL}\\ \text{setp}\\ \underline{\text{NONE}}\end{matrix}\right\}\right],\ \left[\text{FEPS} = \left\{\frac{\text{1.0E - 11}}{\text{eps}}\right\}\right],$$
$$\left[\text{SEPS} = \left\{\frac{\text{1.0E - 11}}{\text{eps}}\right\}\right],\ \left[\text{ARF} = \left\{\frac{0.95}{\text{a}}\right\}\right],\ \left[\text{ARS} = \left\{\frac{0.95}{\text{a}}\right\}\right],$$
$$\left[\text{PSQRT} = \left\{\begin{matrix}\underline{\text{ABSOLUTE}}\\ \text{REAL}\\ \text{IMAGINARY}\end{matrix}\right\},\ \left\{\frac{\underline{\text{DESCENDING}}}{\text{ASCENDING}}\right\}\right]$$

Examples:

```
FLSPOUT FLUIDMP=ALL PANELMP=ALL STRUCTMP=ALL
```


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Describers:

Describer	Meaning
FLUIDMP	First “n” fluid modes to be used for modal participation requests. See FLUIDMP parameter. Default=NONE.
STRUCTMP	First “n” structural modes to be used for modal participation requests. See FLUIDMP parameter. Default=NONE.
GRIDFMP	Fluid grid set for modal participation output. Default is ALL unless setf is specified in which case the modal participation factors will be output for all fluid grids in setf.
GRIDMP	Structure grid set for modal participation output. Default is NONE unless setg is specified in which case the modal participation factors will be output for all structural grids.
OUTFMP	Request “p” highest fluid mode participation factors be output. Default=NONE.
OUTSMP	Request “p” highest structural mode participation factors be output. Default=NONE.
PANELMP	Include panel id's which are included in a “setp” case control set for modal participation calculations. Default=NONE.
FEPS	Threshold for filtering fluid modal participation factors from being printed. Default=1.0E-11.
SEPS	Threshold for filtering structure modal participation factors from being printed. Default=1.0E-11.
ARF	Determines which fluid participation factors are printed. The default is ARF=0.95, which means only values above 95% of the highest value for each mode are printed.
ARS	Determines which structural participation factors are printed. The default is ARS=0.95, which means only values above 95% of the highest value for each mode are printed.
PSORT	Requests how the modal participation output is sorted. The request must occur in pairs. For example, PSORT= (REAL,ASCENDING).

Remarks:

All the entries specified in this case control statements are available as PARAMETER statements.

Related Topics

Browse Simcenter Nastran documentation

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FLSTCNT

Control Parameters for Fluid-Structure Interaction

Miscellaneous control parameters for fluid-structure interaction.

Format:

$$\mathrm{FLSTCNT}\left[\mathrm{ACSYM}=\left\{\begin{matrix}\underline{\mathrm{YES}}\\ \mathrm{NO}\end{matrix}\right\}\right],\left[\mathrm{ACOUT}=\left\{\begin{matrix}\underline{\mathrm{PEAK}}\\ \mathrm{RMS}\end{matrix}\right\}\right],$$
$$\left[\mathrm{ASCOUP}=\left\{\begin{matrix}\underline{\mathrm{YES}}\\ \mathrm{NO}\end{matrix}\right\}\right],\left[\mathrm{PREFDB}=\left\{\begin{matrix}\underline{1.0}\\ \mathrm{P}\end{matrix}\right\}\right],$$
$$\left[\mathrm{SKINOUT}=\left\{\begin{matrix}\underline{\mathrm{NONE}}\\ \mathrm{PUNCH}\\ \mathrm{FREEFACE}\\ \mathrm{STOP}\end{matrix}\right\}\right],\left[\mathrm{AGGPCH}=\left\{\begin{matrix}\underline{\mathrm{NO}}\\ \mathrm{YES}\end{matrix}\right\}\right],$$
$$\left[\mathrm{SFEF70}=\left\{\begin{matrix}\underline{\mathrm{NO}}\\ \mathrm{YES}\end{matrix}\right\}\right]$$

Examples:

```
FLSTCNT ACOUT=RMS PREFDB=1.0E-06
```

Describers:

Describer	Meaning
ACSYM	Requests the symmetric solution for fluid structure analysis, when YES. When NO, non-symmetric solution is requested. (Character; Default = YES)
ACOUT	PEAK or RMS output with the FORCE Case Control request. (Character; Default = PEAK)
ASCOUP	Requests a fluid-structure coupled solution when YES. When NO, the fluid and the structure are decoupled (Character; Default = YES). See Remark 2.
PREFDB	Specifies the reference pressure. (Real; Default = 1.0)

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Describer	Meaning
SKINOUT	Specifies if coupling data with the pairing information is output. SKINOUT=FREEFACE writes coupling data to a .dat and punch files, and writes the coupling datablock to the .op2 file. The solution proceeds after the data is written. Both coupled and uncoupled faces are written. SKINOUT=STOP works like SKINOUT=FREEFACE, but the solution stops immediately after the debug files are created. SKINOUT=PUNCH writes coupling data to .dat and punch files. The solution proceeds after the data is written. It will not write the coupling datablock to the .op2 file. Only coupled free faces are written. SKINOUT=NONE does not output coupling data. (Character; Default = NONE)
AGGPCH	Requests the output of the fluid-structure coupling matrix AGG to the punch file when YES. When NO, the coupling matrix is not written. (Character; Default = NO)
SFEF70	Requests the import of a fluid-structure coupling matrix created by SFE AKUSMOD when YES. When NO, the coupling matrix is not imported. (Character; Default = NO)
NONE	No debug deck or pairing information is generated.
PUNCH	Debug pairing information file is created. The name of the files will be the base name of the deck appended with "_acdbg.dat" and "_acdbg.pch".
FREEFACE	All free faces are written into a debug file.
STOP	The run is terminated as soon as the debug file with pairing information is created.

Remarks:

1. All the entries specified in this case control statement are available as PARAMETER statements.
2. ASCOUP is only supported by standard FEM. It is not supported by the FEM Adaptive Order (FEMAO) method.
3. The coupling data files output by the SKINOUT describer include dummy shell elements. The PSHELL IDs referenced be these elements have the following meaning.
   - Shell elements representing the *coupled* structural free faces reference PSHELL ID = 1.
   - Shell elements representing the *coupled* fluid free faces reference PSHELL ID = 2.
   - Shell elements representing the *uncoupled* structural free faces reference PSHELL ID = 3.
   - Shell elements representing the *uncoupled* fluid free faces reference PSHELL ID = 4.
4. If SFEF70 = YES, Simcenter Nastran does not compute the coupling, and instead uses the SFE AKUSMOD coupling definition from the external file. Simcenter Nastran expects the AKUSMOD file in the same directory where the job is being run, and expects the file name to be fort.70. An ASSIGN

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statement that uses UNIT=70 must be defined in the file management section of your input file if the coupling file is not named fort.70 or if it is in a location other than where the job is run. For example,

ASSIGN OUTPUT2='/directory_path/user_file_name.70' UNIT=70

Related Topics

Browse Simcenter Nastran documentation

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FLUX

Flux, Gradient, Heat flow, or Particle Velocity Output Request

Requests element temperature gradients and heat flow for CHBDYi elements in heat transfer solutions 153 and 159. Also requests particle velocity output in a coupled fluid-structural analysis with solutions 103, 107, 108, 109, 110, 111, 112 and 200.

Format:

$$\text{FLUX}\left[\left(\underline{\text{PRINT}}, \text{PLOT}, \text{PUNCH}\right)\right] = \left\{\begin{array}{c}\text{ALL}\\ \text{n}\\ \text{NONE}\end{array}\right\}$$

Examples:

```
FLUX=ALL
FLUX(PUNCH,PRINT)=17
FLUX=25
```

Describers:

Describer	Meaning
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
PLOT	The output will be sent to the plot file.
ALL	Flux for all elements will be output.
NONE	Flux for no elements will be output.
n	Set identification of a previously appearing SET command. Only fluxes of elements with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

1. FLUX=ALL in SOL 159 may produce excessive output.
2. FLUX=NONE overrides an overall request.

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Related Topics

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FLXSLI

Flexible Sliders Set Selection (SOL 402)

Selects an external flexible sliders set.

Format:

FLXSLI=n

Examples:

```
FLXSLI=15
```

Describers:

Describer	Meaning
n	Set identification of a FLXSLI Bulk Data entry. (Integer>0)

Remark:

1. FLXSLI case Control command is global and cannot vary from one subcase to another.

Related Topics

Browse Simcenter Nastran documentation

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FLXRESULTS

Flexible Silders Result Output Request (SOL 402)

Requests flexible slider joint output.

Format:

$$\text{FLXRESULTS}\left[\left(\text{CURVDISP},\ \left[\frac{\text{PRINT}}{\text{PLOT}}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \underline{\text{NONE}}\end{matrix}\right\}$$

Example:

```
FLXRESULTS=ALL
```

Describers:

Describer	Meaning
CURVDISP	Relative displacement along the curvilinear abscissa of flexible sliders.
PRINT	The printer will be the output medium.
PLOT	Puts contact results in OP2 file only.
ALL	Results at all grid points will be output.
NONE	Results will not be output.
n	Set identification of a previously appearing SET command. Only grid points that support a sensor or a driver and with identification numbers that appear on this SET command will be output. (Integer>0)

Related Topics

Browse Simcenter Nastran documentation

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FMETHOD

Flutter Analysis Method Parameter Selection

Selects the parameters to be used by the aerodynamic flutter analysis.

Format:

FMETHOD=n

Examples:

```
FMETHOD=72
```

Describers:

Describer	Meaning
n	Set identification number of a FLUTTER Bulk Data entry. (Integer>0)

Remarks:

1. An FMETHOD command is required for flutter analysis.
2. A CMETHOD command in also required for the K-method of flutter analysis.
3. If this entry is being used in SOL 200 in conjunction with flutter design conditions, the METHOD selected on the FLUTTER Bulk Data entry must be “PK” or “PKNL”.

Related Topics

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FMH3D

Defines 2D-3D Coupling in SOLs 402 and 414

Defines the coupling of a Fourier multi-harmonic region with a 3D structure in SOLs 402 and 414. Not supported in a superelement creation run.

Format:

FMH3D=n

Examples:

```
FMH3D=17
```

Describers:

Describer	Meaning
n	Set identification number of FMH3D bulk entries. (Integer > 0)

Remarks:

1. All FMH3D bulk entries with the same set identification number are used.
2. The FMH3D case control command must be defined above the subcase level. As a result, the constraints generated by the automatic coupling are used in every subcase.

Related Topics

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FORCE

Element Force Output or Particle Velocity Request

Requests the form and type of element force output or particle velocity output in coupled fluid-structural analysis.

Format:

$$\text{FORCE}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\right.\right.$$

$$\left.\left.\left[\begin{matrix}\underline{\text{CENTER}}\\ \text{CORNER or BILIN}\\ \text{SGAGE}\\ \text{CUBIC}\end{matrix}\right],\left[\begin{matrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{matrix}\right],\left[\begin{matrix}\underline{\text{RPRINT}}\\ \text{NOPRINT}\end{matrix}\right],[\text{PUNCH}]\right)\right]$$

$$=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
FORCE=ALL
FORCE(REAL, PUNCH, PRINT)=17
FORCE=25
```


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Describers:

Describer	Meaning
SORT1	Output will be presented as a tabular listing of elements for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output will be presented as a tabular listing of frequency or time for each element type.
PLOT	Generates force output for requested set but no printed output.
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output.
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
PSDF	Requests the power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 6.
ATOC	Requests the autocorrelation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 6.
CRMS	Requests the cumulative root mean square function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 6.
RMS	Requests the root mean square and zero crossing functions be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 6.
RALL	Requests all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 6.
RPRINT	Writes random analysis results to the print file. (Default) See Remark 6.
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 6.
RPUNCH	Writes random analysis results to the punch file. See Remark 6.
CENTER	Output CQUAD4, CQUADR, CTRIAR element forces at the center only.
CORNER or BILIN	Output CQUAD4, CQUADR, CTRIAR element forces at the center and grid points using strain gage approach with bilinear extrapolation.
SGAGE	Output CQUAD4 element forces at center and grid points using strain gage approach.

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Describer	Meaning
CUBIC	Output CQUAD4 element forces at center and grid points using cubic bending correction.
ALL	Forces for all elements will be output.
n	Set identification of a previously appearing SET command. Only forces of elements with identification numbers that appear on this SET command will be output. See Remark 2 for SOL 414. (Integer>0)
NONE	Forces for no elements will be output.

Remarks:

1. ALL should be used with caution in transient solutions as it can produce a large amount of output.
2. The defaults for SORT1 and SORT2 depend on the type of analysis:
   - SORT1 is the default in static analysis, frequency response, steady state heat transfer analysis, real and complex eigenvalue analysis, flutter analysis, and buckling analysis. If SORT2 is selected in a frequency response solution for one or more of the commands ACCE, DISP, FORC, GPFO, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT2 format.
   - SORT2 is the default in transient response analysis (structural and heat transfer). SORT2 is not available for real eigenvalue (including buckling), complex eigenvalue, or flutter analysis. If SORT1 is selected in a transient solution for one or more of the commands ACCE, DISP, ENTH, FORC, GPFO, HDOT, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT1 format.
   - XY plot requests will force SORT2 format thus overriding SORT1 format requests.
3. FORCE=NONE overrides an overall request.
4. In nonlinear transient analysis, this request is ignored for nonlinear elements.
5. The options CENTER, CORNER, CUBIC, SGAGE, and BILIN are recognized only in the first subcase and determine the option to be used in all subsequent subcases with the STRESS, STRAIN, and FORCE commands. Consequently, options specified in subcases other than the first subcase will be ignored.
   - If the STRESS command is specified in the first subcase then the option on the STRESS command is used in all subcases with STRESS, STRAIN, and FORCE commands.
   - If the STRAIN command and no STRESS command is specified in the first subcase, then the option on the STRAIN command is used in all subcases containing STRESS, STRAIN, and FORCE commands.
   - If the FORCE command and no STRESS or STRAIN command is specified in the first subcase, then the option on the FORCE command is used in all subcases containing STRESS, STRAIN, and FORCE commands.
   - If STRESS, STRAIN, and FORCE commands are not specified in the first subcase, then the CENTER option is used in all subcases containing STRESS, STRAIN, and FORCE commands.
6. The following applies to random solutions:

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- By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, PUNCH, PLOT describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
- The SORT1 and SORT2 describers only control the output format for the frequency response output. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
- Any combination of the PSDF, ATOC, RMS, and CRMS describers can be selected. The RALL describer selects all four.

Remarks related to SOL 401:

1. Output requested with the STRESS, STRAIN, and FORCE commands is only at the element corners. The CENTER, CUBIC, or SGAGE options are ignored.
2. See Shell element output summary.

Remarks related to SOL 402:

1. Output is restricted to REAL and SORT1 format. SORT2, IMAG, PHASE, CUBIC, SGAGE, PSDF, ATOC, RMS, and RALL describers are ignored.
2. Output of CQUAD8 and CTRIA6 elements forces are output at grid points, regardless the CENTER/CORNER describer presence.

Remarks related to SOL 414:

1. The FORCE case control command (FORCE=ALL) applies to all SOL 414 solutions.
2. For SOL 414,101 Maneuvers solutions, you can request a set using FORCE=*<n>* to refer to a group of the following type of elements:

   CBEAR, CBEAR2, CBUSH, CBUSH1D, CBUSH2, CELAS1, CELAS2, CDAMP1, CDAMP2, CONM1, RBE3

   All other elements are ignored.

   The force output values are written to the .f06 file.

Related Topics

Browse Simcenter Nastran documentation

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FREQUENCY

Frequency Set Selection

Selects the set of forcing frequencies to be solved in frequency response problems.

Format:

FREQUENCY=n

Examples:

```
FREQUENCY=17
```

Describers:

Describer	Meaning
n	Set identification number of FREQ, FREQ1, FREQ2, FREQ3, FREQ4, and FREQ5 Bulk Data entries. (Integer>0)

Remarks:

1. A frequency set selection is required for a frequency response problem.
2. A frequency set selection is required for transient response by Fourier methods (SOL 146).
3. All FREQ, FREQ1, FREQ2, FREQ3, FREQ4, and FREQ5 entries with the same frequency set identification numbers will be used. Duplicate frequencies will be ignored. f_N and f_{N-1} are considered duplicated if
   - $|f_N - f_{N-1}|<10^{-5}|f_{MAX} - f_{MIN}|$

Related Topics

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FRFIN

Frequency-Dependent Frequency Response Function Matrix Input

Selects FRFSTIF, FRFFLEX, or FRFOTM bulk entries to define frequency-dependent frequency response function matrices.

Format:

$$\text{FRFIN} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \underline{\text{NONE}} \end{matrix} \right\}$$

Examples:

```
SET 99=230,231,233
FRFIN=99
```

Describers:

Describer	Meaning
ALL	Select all FRFSTIF, FRFFLEX, and FRFOTM bulk entries.
n	Set identification number of a previously appearing SET command. Only FRFSTIF, FRFFLEX, and FRFOTM bulk entries with identification numbers that appear on this SET command are selected. (Integer>0)
NONE	No FRFSTIF, FRFFLEX, and FRFOTM bulk entries are selected. (Default)

Remarks:

1. A single FRFIN case control command can be placed above the subcases. If a FRFIN case control command is placed in a subcase, a fatal error occurs.

Related Topics

Browse Simcenter Nastran documentation

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FRFOUT

Frequency-Dependent External Superelement Creation Specification

Creates a frequency-dependent external superelement using Schur complement reduction.

Format:

$$\text{FRFOUT}\left[(\text{ASMBULK},\ \text{EXTBULK},\ \text{GEOM},\ \text{SOLVE},\ \text{EXTID} = \text{seid},\ \underline{\text{DMIGOP2} = \text{unit}})\right]$$

Examples:

```
FRFOUT
FRFOUT(ASMBULK,EXTID=100)
FRFOUT(ASMBULK,EXTID=100,GEOM)
FRFOUT(ASMBULK,EXTID=100,DMIGOP2=99)
```

Describers:

Describer	Meaning
ASMBULK	Generate bulk entries related to the subsequent superelement assembly process and store them on the assembly punch file (.asm). This data is to be included in the main bulk portion of the subsequent assembly solution. See Remarks 3 and 5.
EXTBULK	Generate and store bulk entries for the external superelement on the standard punch file (.pch). This data is used in the BEGIN SUPER portion of the bulk section of the subsequent assembly solution. If EXTBULK is not specified, the subsequent assembly solution retrieves the required data for the external superelement from the OP2 file. See Remarks 4 and 5.
GEOM	Store full model geometry. See Remark 10.
SOLVE	Compute the response to the input load with the reduced FRF matrices and loads. Data recovery to the analysis DOFs uses the same back transformations that are used for the output transformation matrix (OTM) generation. Data recovery is performed the same way in a modal solution as in a direct solution.
EXTID = *seid*	*seid* (integer > 0) is the frequency-dependent superelement ID to be used in the SEBULK and SECONCT bulk entries stored on the assembly punch file (.asm) if ASMBULK is specified. See Remarks 2, 3, and 4.
DMIGOP2 = *unit*	Store the boundary matrices as DMIG bulk entries on an OUTPUT2 file whose Fortran unit number is given by *unit* (integer>0). See Remarks 6 and 7.

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Remarks:

1. The default is to output the frequency-dependent impedance and loads. Use the NOLOAD option to disable the load output.
2. EXTID and an *seid* value must be specified if one or more of ASMBULK or EXTBULK are specified.
3. If ASMBULK is specified, the following bulk entries are generated and stored on the assembly punch file (.asm):

   SEBULK *seid* ...

   SECONCT *seid* ...

   GRID entries for the boundary points

   CORD2x entries associated with the above GRID entries
4. If EXTBULK is specified, the following bulk entries are generated and stored on the standard punch file (.pch):

   BEGIN SUPER *seid* ...

   GRID entries for the boundary points

   CORD2x entries associated with the above GRID entries

   EXTRN

   ASET/ASET1
5. The punch output resulting from FRFOUT usage is determined by ASMBULK and EXTBULK as follows:
   - No ASMBULK or EXTBULK results in no punch output.
   - ASMBULK, but no EXTBULK, results in punch output being generated and stored on the assembly punch file (.asm). See Remark 3.
   - ASMBULK and EXTBULK results in punch output consisting of two distinct and separate parts. One part is generated and stored on the assembly punch file (.asm) as indicated in Remark 3. The other part is generated and stored on the standard punch file (.pch) as indicated in Remark 4.
6. If DMIGOP2 = *unit* is specified, an appropriate ASSIGN OUTPUT2 statement must be present in the File Management Section (FMS) for the absolute value of *unit*. An OUTPUT2 file created during a LP-64 Nastran executable superelement creation run cannot be used in a ILP-64 Nastran executable system run and vice versa. Note that beginning in version 12, only the ILP-64 executable is available.
7. The default unit number for the DMIGOP2 describer is 30. To change the default unit number, use the EXTUNIT parameter.
8. The creation of a frequency-dependent external superelement using FRFOUT involves running a Simcenter Nastran job with the following additional data:
   - The data for the creation of the frequency-dependent external superelement is specified by the FRFOUT case control command, which must appear above the subcase level.
   - The boundary points of the frequency-dependent external superelement are specified by ASET/ASET1, CSET/CSET1, BNDFIX/BNDFIX1, and BNDFREE/BNDFREE1 bulk entries.

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- If the creation involves component mode reduction, the optional generalized coordinates are specified using QSET/QSET1 bulk entries. The boundary data for the component mode reduction may be specified using the BNDFIX/BNDFIX1 and BNDFREE/BNDFREE1 bulk entries (or their equivalent BSET/BSET1 and CSET/CSET1 bulk entries). (The default scenario assumes that all boundary points are fixed for the component mode reduction.)

9. To include differential stiffness in the definition of a frequency-dependent external superelement, two subcases are required in the creation run. The first subcase is a static subcase. The second subcase performs the analysis and generation of the frequency-dependent external superelement. To obtain the displacement field used to generate the differential stiffness effects, include a STATSUB case control command in the second subcase that references the first subcase. Always include the EXTSEOUT case control command above the subcase level. An example of the required setup is as follows:

```
…
SOL 111
CEND
TITLE = …
EXTSEOUT(…)
$
SUBCASE 1
     $ STATIC SUBCASE
$
SUBCASE 2
     $ DYNAMIC SUBCASE
     TSTEP = 100
     STATSUB = 1
     METHOD = 10
     DLOAD = 20
BEGIN BULK
…
```

For accuracy and consistency, the loads used to generate differential stiffness for the frequency-dependent external superelement during the creation run should be the same loads used in the system run without any scaling. If the loads are scaled by a non-unity scaling factor from a case control command like P2G or a bulk entry like LOAD, the differential stiffness portion of the frequency-dependent external superelement impedance will no longer be consistent with the applied loads.

10. The GEOM describer outputs geometry data blocks GEOM1EXA, GEOM2EXA, and GEOM4EXA that contain all of the frequency-dependent external superelement geometry to support post-processing. By default, the full geometry is not exported; the GEOM describer must be explicitly defined to have these geometry data blocks written.

11. The OFREQ command should not be specified and is ignored for the frequency-dependent external superelement creation.

Related Topics

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GCRSTRN

Requests creep strain output at Gauss points for SOLs 401 and 402.

Requests creep strain at Gauss points.

Format:

$$\text{GCRSTRN}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{array}{c}\text{ALL}\\ \text{n}\\ \text{NONE}\end{array}\right\}$$

Examples:

```
GCRSTRN=ALL
GCRSTRN(PRINT,PLOT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

Remarks:

1. The following elements support the Gauss point output:
   - The plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.
   - The plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8.
   - The axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6.
   - The 3D solids elements CTETRA, CHEXA, CPENTA, CPYRAM.

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2. You can define the number of integration points for an element with the IN field on the PSOLID bulk entry, or the FOROPT field on the PPLANE bulk entry. Regardless of the number of integration points you request, the software outputs the same number of gauss point results as the number of corner grid points. In general, the gauss point closest to a corner grid point is used for the output. See Gauss point output in the *Multi-Step Nonlinear User's Guide* for the tables showing the corner grid point to Gauss point mapping for each element type and integration point option.
3. Both PRINT and PUNCH may be requested.
4. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.
5. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

Related Topics

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GELSTRN

Requests elastic strain output at Gauss points for SOLs 401 and 402.

Requests elastic strain at Gauss points.

Format:

$$\text{GELSTRN}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
GELSTRN=ALL
GELSTRN(PRINT,PLOT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

Remarks:

1. The following elements support the Gauss point output:
   - The plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.
   - The plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8.
   - The axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6.
   - The 3D solids elements CTETRA, CHEXA, CPENTA, CPYRAM.

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2. You can define the number of integration points for an element with the IN field on the PSOLID bulk entry, or the FOROPT field on the PPLANE bulk entry. Regardless of the number of integration points you request, the software outputs the same number of gauss point results as the number of corner grid points. In general, the gauss point closest to a corner grid point is used for the output. See Gauss point output in the *Multi-Step Nonlinear User's Guide* for the tables showing the corner grid point to Gauss point mapping for each element type and integration point option.
3. Both PRINT and PUNCH may be requested.
4. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.
5. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

Related Topics

Browse Simcenter Nastran documentation

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GKRESULTS

Gasket Result Output Request for SOL 402

Requests the form and type of gasket result output for SOL 402.

Format:

$$\text{GKRESULTS}\left[\left(\left[\frac{\text{PRINT}}{\text{PLOT}}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \underline{\text{NONE}}\end{matrix}\right\}$$

Examples:

```
GKRESULTS=ALL
GKRESULTS(PLOT)=ALL
```

Describers:

Describer	Meaning
PRINT	The printer will be the output medium.
PLOT	Computes and puts gasket results in OP2 file only.
ALL	Gasket results will be output for all gasket elements.
NONE	Gasket results will not be output.

Remarks:

1. Gasket results consist of gasket pressure, gasket closure strain, gasket yield pressure, plastic gasket closure strain, and gasket status. Figure 4-7 depicts a gasket load/unload characteristic defined using a MATG bulk entry where TABLD, TABLUi, and YPRESS refer to fields on the MATG bulk entry.

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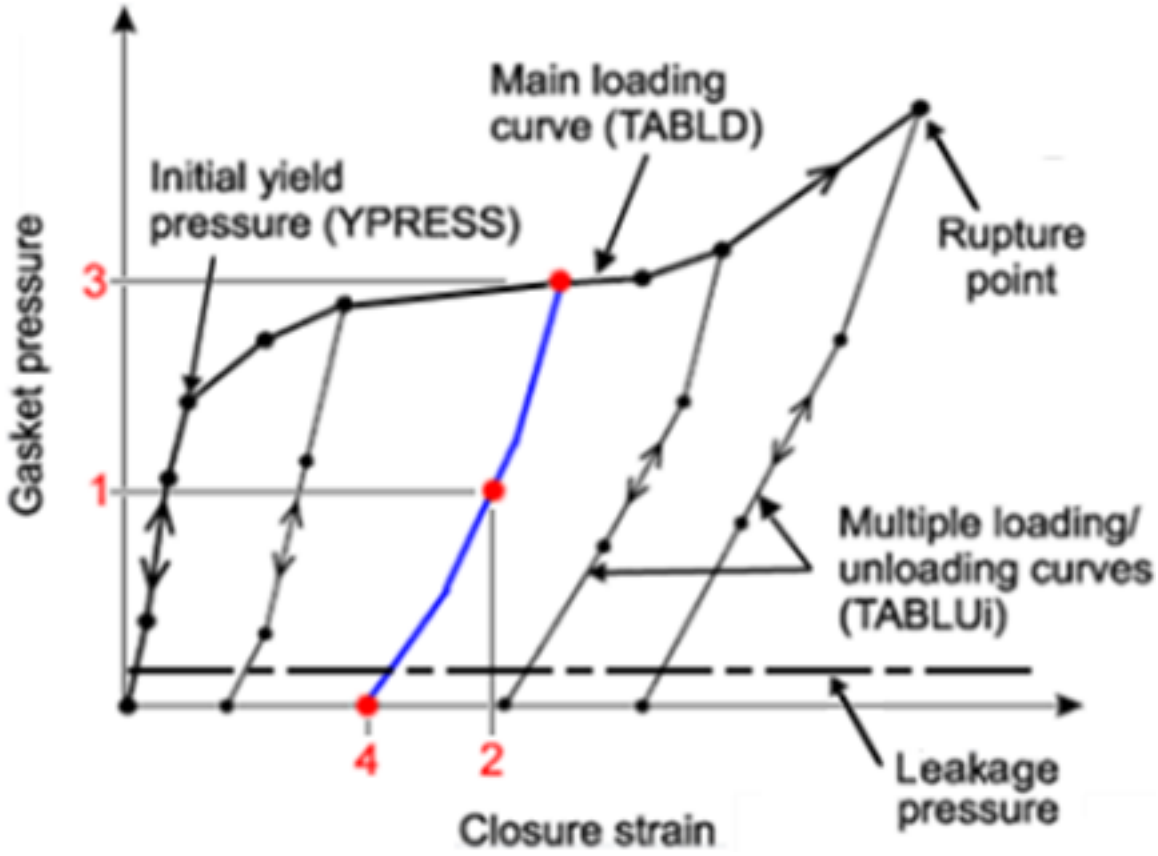
Figure 4-7: Gasket load/unload characteristic. Gasket pressure (1), gasket closure strain (2), gasket yield pressure (3), plastic gasket closure strain (4).



2. The leakage pressure is automatically set to 1% of the initial yield pressure.
3. Gasket status is defined as follows:

Status	Definition	Integer in .op2 (OGK datablock)
Open	The gasket pressure is less than the leakage pressure.	1
Closed	The gasket pressure is higher than the leakage pressure but has not yet caused plasticity.	2
Sealed	There has been plastic gasket deformation and the current pressure is above the gasket leakage pressure.	3

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Leaked	After plastic deformation, the gasket pressure has dropped below gasket leakage pressure.	4
Crushed	Gasket closure strain has exceeded the rupture value.	5

Related Topics

Browse Simcenter Nastran documentation

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GMESTRN

Requests mechanical strain output at Gauss points for SOL 401.

Requests mechanical strain at Gauss points.

Format:

$$\text{GMESTRN}\left[(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT})\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
GMESTRN=ALL
GMESTRN(PRINT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all elements.
n	Set identification number of a previously appearing SET command. Only elements with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Do not compute output.

Remarks:

1. The output includes the mechanical strain components and the effective mechanical strain. The mechanical strain components are the sum of the elastic, plastic, and creep strain contributions for each component. The effective mechanical strain is the sum of the effective elastic, effective plastic, and effective creep strains.
2. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.
3. The following elements support the output:

   - Plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.

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- Plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8.

- Axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6.

- 3D solids elements CTETRA, CHEXA, CPENTA, CPYRAM.

- Chocking elements CCHOCK3, CCHOCK4, CCHOCK6, CCHOCK8.

- Generalized plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8 that reference a PGPLSN entry.

- Shell elements CQUAD8, CTRIA6, CTRIAR, CQUADR that reference a PSHELL entry.

- Tension-only shell element CQUADR that references a PSHLPNL entry.

- Composite shell elements CQUAD8, CTRIA6, CTRIAR, CQUADR that reference a composite property defined with the PCOMPG1, PCOMP or PCOMPG entries.

- Composite solid elements CHEXA, CPENTA that reference a composite property defined with the PCOMPS entries.

4. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.
5. You can requeset both PRINT and PUNCH.
6. You can define the number of integration points for an element using the IN field on the PSOLID bulk entry, or the FOROPT field on the PPLANE bulk entry. Regardless of the number of integration points you request, the software outputs the same number of gauss point results as the number of corner grid points. In general, the gauss point closest to a corner grid point is used for the output. See Gauss point output in the *Multi-Step Nonlinear User's Guide* for the tables showing the corner grid point to Gauss point mapping for each element type and integration point option.

Related Topics

Browse Simcenter Nastran documentation

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GPFORCE

Grid Point Force Output Request

Requests grid point force balance at selected grid points.

Format:

$$\text{GPFORCE}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{matrix}\right],\text{PUNCH}\right],\left[\begin{matrix}\underline{\text{REAL}}\ \text{or}\ \underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\right.$$
$$\left.\text{SOLUTION}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{setf}\\ \text{PEAKOUT}\end{matrix}\right\}\right)\Bigg]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
GPFORCE=ALL
GPFORCE=17
GPFORCE(SORT2,PRINT,PUNCH,PHASE) = 123
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of grid points for each load, frequency, eigenvalue, or time depending on solution sequence. (Default)
SORT2	Output a tabular listing of load, frequency, eigenvalue, or time for each grid point.
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
NOPRINT	Generate, but do not print, grid point force balance results.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
SOLUTION = ALL	Output grid point force balance for all frequencies. See Remark 13. (Default)
SOLUTION = setf	Output grid point force balance only for the frequencies listed on a previously appearing SET command with setf as the identification number. See Remark 13. (Integer > 0)

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Describer	Meaning
SOLUTION = PEAKOUT	Output grid point force balance only for frequencies identified with the PEAKOUT command. See Remark 13.
ALL	Output grid point force balance for all grid points.
n	Output grid point force balance only for the grid points listed on a previously appearing SET command with n as the identification number. (Integer > 0)
NONE	Do not output grid point force balance for any grid points.

Remarks:

1. Both PRINT and PUNCH can be requested.
2. PARAM,DDRMM,-1 is required in the modal solution sequences 111, 112, 146, and 200.
3. The printing of grid point forces is suppressed when PARAM,NOGPF,-1 appears in the Bulk Data.
4. The Bulk Data entry PARAM,NOELOF,+1 causes the output of grid point forces to be aligned with the edges of the two-dimensional elements. The default value of -1 suppresses this output.
5. The Bulk Data entry PARAM,NOELOP,+1 causes the output of the sum of the forces parallel to the edges of adjacent elements. The default value of -1 suppresses this output.
6. The output of grid point forces that are aligned with the edges of elements is available for the following elements: CBAR, CROD, CBEAM, CSHEAR, CONROD, CTRIA3

   The positive direction for grid point forces aligned with the edges of elements is from the reference point to the load point as indicated on the printed output.
7. Grid point force balance is computed from linear and nonlinear elements as the sum of applied and thermal loads, MPC and SPC forces, and element forces. Effects not accounted for include those from general elements and boundary loads from upstream superelements. These effects can lead to an apparent lack of equilibrium at the grid point level. The following table summarizes those effects that are considered and those effects that are ignored in the calculation of grid point forces in the global coordinate system:

Contributions Included	Contributions Ignored
Applied Loads (including DMI Forces)	GENEL Forces
Thermal Loads	Boundary Loads from Upstream Superelements
SPC Forces	
MPC Forces	
Element Elastic Forces	
Element Viscous Damping Forces(1)	
Element Inertial Forces(1)	
PARAM,G Structural Damping Forces(1)	

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Contributions Included	Contributions Ignored
GE Field Structural Damping Forces(1)	
Rayleigh Damping Forces(1)	
DMIG (K2GG, B2GG, M2GG, and K42GG) Forces(1)	
Differential Stiffness Forces(1)	
FRF Components Forces(2)	
FRF External Superelement Forces(2)	
(1)Included in grid point force balance for frequency response analysis and transient response analysis only. To exclude these contributions from the grid point force balance, specify PARAM,FULLGPF,NO. (2)Included in grid point force balance for frequency response analysis only. To exclude these contributions from the grid point force balance, specify PARAM,FULLGPF,NO.	

8. Only the element elastic forces are included when the grid point forces are aligned with the edges of elements.

   For SOLs 108, 109, 111, and 112, the FULLGPF parameter controls the which forces are used in the grid point force output and grid point force balances.

   - For SOLs 108 and 111 frequency response analysis, specify PARAM,FULLGPF,YES to include the element inertia forces, viscous damping forces from elements, PARAM,G structural damping forces, GE field structural damping forces, Rayleigh damping forces, DMIG forces, differential stiffness forces, FRF component forces, and FRF external superelement forces with the element elastic forces in the grid point force output and grid point force balances.
   - For SOLs 109 and 112 transient response analysis, specify PARAM,FULLGPF,YES to include element inertia forces, viscous damping forces from elements, viscous damping forces from converted PARAM,G and GE field structural damping, Rayleigh damping forces, DMIG forces, and differential stiffness forces with the element elastic forces in the grid point force output and grid point force balances.

   Specify PARAM,FULLGPF,NO to retain the pre-Simcenter Nastran 2021.1 behavior and output the element elastic forces only.

9. In inertia relief analysis, the GPFORCE output related to SPC forces and applied loads is interpreted differently for SOLs 101 and 200. In SOLs 101 and 200, the SPC force and applied load output includes both the effect of inertial loads and applied loads.

10. GPFORCE is supported in a nonlinear static analysis. It is not supported in SOL 129. PARAM,NOELOF and PARAM,NOELOP are not supported in nonlinear static analysis. Thus, Remarks 4, 5, 6, and 8 do not apply to SOL 106.

11. The SOLUTION describer is only valid in a SOL 108 or 111 frequency response analysis.

12. When an OFREQ case control command is present, the SOLUTION describer is used as follows:

    - If SOLUTION=ALL, output results for all frequencies specified by the OFREQ command.

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- If SOLUTION=setf, output results for all frequencies specified by the OFREQ command that are listed on a previously appearing SET command with setf as the identification number.
- If SOLUTION=PEAKOUT and a PEAKOUT case control command is also present in the input file, output results for all frequencies specified by the OFREQ command that satisfy the PEAKOUT filtering criteria.

13. The SOLUTION describer is not valid for external superelements and direct-input, frequency-dependent components.
14. When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the grid point force results are based on the nominal values for stiffness.
15. For SOL 401, beginning with Simcenter Nastran 2021.2, a new GPFORCE output format is available. Previously, the software summed the applied loads at a grid point and listed this in the "APP-LOAD" row in the grid point force output. With the new output format, the software separates the applied load into NOD_LOAD (grid point loads) and ELE_LOAD (element loads). The element loads include only the rotational loads defined with the RFORCE and RFORCE1 bulk entries. All other applied loads, except for thermal loads, are included as grid point loads. SOL 401 includes the thermal loads with the internal element forces. In addition, a new body-load computation method is available for RFORCE and RFORCE1 loads. With this method, the load vector is computed on an element basis rather than a grid point basis. The system cell 771 is available to change both the GPFORCE output format and the body-load computation method to the pre-Simcenter Nastran 2021.2 behavior. For more information, see **system cell 771**.

Related Topics

Browse Simcenter Nastran documentation

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GPKE

Grid Point Kinetic Energy Output Request

Requests the output of the kinetic energy at selected grid points in normal modes analysis only.

Format:

$$\text{GPKE}\left[\left(\left[\begin{matrix}\text{PRINT}\\ \text{NOPRINT}\end{matrix}\right],\text{PUNCH},\text{THRESH}=\text{e}\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
GPKE=ALL
GPKE (PRINT, PUNCH)=19
```

Describers:

Describer	Meaning
PRINT	The printer will be the output medium.
NOPRINT	Generates, but does not print, grid point kinetic energy output.
PUNCH	The punch file will be the output medium.
e	Minimum energy threshold. Only energies above this value will be printed and/or punched.
ALL	Grid point kinetic energy for all grid points will be output.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Grid point kinetic energy for no points will be output.

Remarks:

1. Grid point kinetic energy is only available for normal modes analysis.
2. Both PRINT and PUNCH may be requested.
3. GPKE=NONE overrides an overall output request.

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4. For models using the lumped mass formulation, the grid point kinetic energy can be used to examine the distribution of kinetic energy among the grid points. It is computed as:

$$E_{k_g} = \Phi_g^{mass} \otimes [M_{gg}\Phi_g^{mass}]$$

where φ_g^{mass} is the mass-normalized eigenvector so that the total grid point kinetic energy is scaled to be unity. Note that the operator shown in the previous equation indicates term-wise matrix multiplication.

5. The grid point kinetic energy output has limited meaning for a coupled mass formulation. Since this mass formulation produces a coupling of mass across grid points, the sharing of kinetic energy among grid points can occur. In general, this obscures the meaning of the computation as a means of identifying important model parameters to control modal behavior.

6. When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the grid point kinetic energy results are based on the nominal values for stiffness and damping.

Related Topics

Browse Simcenter Nastran documentation

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GPLSTRN

Requests plastic strain output at Gauss points for SOLs 401 and 402.

Requests plastic strain at Gauss points.

Format:

$$\text{GPLSTRN}\left[(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT})\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
GPLSTRN=ALL
GPLSTRN(PRINT,PLOT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

Remarks:

1. The following elements support the Gauss point output:
   - The plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.
   - The plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8.
   - The axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6.
   - The 3D solids elements CTETRA, CHEXA, CPENTA, CPYRAM.

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2. You can define the number of integration points for an element with the IN field on the PSOLID bulk entry, or the FOROPT field on the PPLANE bulk entry. Regardless of the number of integration points you request, the software outputs the same number of gauss point results as the number of corner grid points. In general, the gauss point closest to a corner grid point is used for the output. See Gauss point output in the *Multi-Step Nonlinear User's Guide* for the tables showing the corner grid point to Gauss point mapping for each element type and integration point option.
3. Both PRINT and PUNCH may be requested.
4. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.
5. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

Related Topics

Browse Simcenter Nastran documentation

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GPRSORT

Composites Ply Results Sorted Output

Requests that the output of composites ply results (stress, strain and failure indices) on elements referencing a PCOMPG be sorted by global ply ID, then by element ID, versus by element ID, then by global ply ID.

Formats:

$$\text{GPRSORT} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \end{matrix} \right\}$$

Examples:

```
GPRSORT = ALL
GPRSORT = 16
```

Describers:

Describer	Meaning
n	Set identification number of a previously appearing SET command.
ALL	All composite elements referencing a PCOMPG property entry type. See Remarks.

Remarks:

1. Composite element output will be sorted by global ply ID, and by element ID. This sorted output is only available for composite elements referencing a PCOMPG property entry, which includes global ply IDs.
2. Composite elements referencing the PCOMP property entry will be excluded from the sorted output.

Related Topics

Browse Simcenter Nastran documentation

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GPSDCON

Grid Point Stress Discontinuity Output Request

Requests mesh stress discontinuities based on grid point stresses.

Format:

$$\text{GPSDCON}\left[\left(\left[\frac{\text{PRINT}}{\text{PLOT}}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\end{matrix}\right\}$$

Examples:

```
GPSDCON=ALL
GPSDCON=19
```

Describers:

Describer	Meaning
ALL	Stress discontinuity requests for all SURFACE and VOLUME commands defined in the OUTPUT(POST) section will be output.
n	Set identification number of a previously appearing SET command. Only surfaces and volumes with identification numbers that appear on this SET command will be included in the stress discontinuity output request. (Integer>0)
NONE	No grid point stress discontinuity output.

Remarks:

1. This output is available in linear static analysis SOLs 101 and 144 only. Output will be presented for each surface or volume as a tabular listing of stress discontinuities for each subcase.
2. Only elements used to define the surface or volume are output. See the description for the SURFACE or VOLUME commands.
3. Element stress output (STRESS) must be requested for elements referenced on SURFACE and VOLUME commands. Also, the GPSTRESS command must be present for printed output and the STRFIELD command for postprocessing output using the .xdb file (PARAM,POST,0) for the same surfaces and volumes.

Related Topics

Browse Simcenter Nastran documentation

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GPSTRAIN

Grid Point Strain Output Request for Printing Only

Requests grid point strains for printing only.

Format:

$$\text{GPSTRAIN}\left[\left(\left[\frac{\text{PRINT}}{\text{PLOT}}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
GPSTRAIN=ALL
GPSTRAIN=19
```

Describers:

Describer	Meaning
ALL	Grid point strain requests for all SURFACE and VOLUME commands defined in the OUTPUT(POST) section will be output.
n	Set identification number of a previously appearing SET command. Only surfaces and volumes with identification numbers that appear on this SET command will be included in the grid point strain output request. (Integer>0)
NONE	No grid point strain output.

Remarks:

1. For statics, normal modes and transient analysis output will be presented for each surface or volume as a tabular listing of grid point strains for each load, eigenvalue, and time step. (See **"DISPLACEMENT"** for a discussion of SORT1 and SORT2.)
2. Only grid points connected to elements used to define the surface or volume are output. See the description for **"SURFACE"** and **"VOLUME"** .
3. Element strain output (STRAIN) must be requested for elements referenced on SURFACE and VOLUME commands.
4. In nonlinear transient analysis, grid point strains are computed only if parameter LGDISP is -1, which is the default, and for elements with linear material properties only.

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5. For the graphical display of grid point strains in the computation of mesh strain discontinuities, the STRFIELD command must be specified.
6. When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the grid point strain results are based on the nominal values for stiffness and damping.
7. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

Related Topics

Browse Simcenter Nastran documentation

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GPSTRESS

Grid Point Stress Output Request

Requests grid point stresses.

Format:

$$\text{GPSTRESS}\left[\left(\left[\frac{\text{PRINT}}{\text{PLOT}}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
GPSTRESS=ALL
GPSTRESS=19
```

Describers:

Describer	Meaning
ALL	Grid point stress requests for all SURFACE and VOLUME commands defined in the OUTPUT(POST) section will be output.
n	Set identification number of a previously appearing SET command. Only surfaces and volumes with identification numbers that appear on this SET command will be included in the grid point stress output request. (Integer>0)
NONE	No grid point stress output.

Remarks:

1. For statics, normal modes and transient analysis output will be presented for each surface or volume as a tabular listing of grid point stresses for each load, eigenvalue, and time step. (See **"DISPLACEMENT"** for a discussion of SORT1 and SORT2.)
2. Only grid points connected to elements used to define the surface or volume are output. See the description for **"SURFACE"** and **"VOLUME"** .
3. Element stress output (STRESS) must be requested for elements referenced on SURFACE and VOLUME commands.
4. In nonlinear transient analysis, grid point stresses are computed only if parameter LGDISP is -1, which is the default, and for elements with linear material properties only. Grid point stresses are not computed for the hyperelastic elements.

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5. For the graphical display of grid point stresses in the computation of mesh stress discontinuities, the STRFIELD command must be specified.
6. Grid point stress is not output for midside nodes.
7. When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the grid point stress results are based on the nominal values for stiffness and damping.

Related Topics

Browse Simcenter Nastran documentation

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GRDCON

Acoustic Contribution Output Request for Grids

Requests the form and type of acoustic contribution output for the grids of structural panels and for the residual.

Format:

$$\text{GRDCON}\left[\left(\begin{bmatrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{bmatrix},\ \begin{bmatrix}\underline{\text{REAL}}\ \text{or}\ \underline{\text{IMAG}}\\ \text{PHASE}\end{bmatrix},\ \begin{bmatrix}\text{PRINT}\\ \underline{\text{NOPRINT}}\end{bmatrix},\ \text{PUNCH},\right.\right.$$
$$\left.\begin{bmatrix}\underline{\text{ABS}}\\ \text{NORM}\\ \text{BOTH}\end{bmatrix},\ \text{TOPG}=\begin{Bmatrix}\text{pg}\\ \underline{\text{ALL}}\end{Bmatrix},\ \text{SOLUTION}=\begin{Bmatrix}\underline{\text{ALL}}\\ \text{setf}\\ \text{PEAKOUT}\end{Bmatrix},\right.$$
$$\left.\left.\text{GRID}=\begin{Bmatrix}\underline{\text{ALL}}\\ \text{setg}\end{Bmatrix}\right)\right]=\begin{Bmatrix}\text{n}\\ \underline{\text{ALL}}\\ \text{NONE}\end{Bmatrix}$$

Examples:

```
GRDCON=123
GRDCON(SORT1,PHASE,PRINT,PUNCH,BOTH,TOPG=5)=ALL
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of grids for each frequency. (Default)
SORT2	Output a tabular listing of frequency for each grid.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
PRINT	Write acoustic grid contribution output to the print (.f06) file. (Default)
NOPRINT	Do not print the acoustic grid contribution results.
PUNCH	Write acoustic grid contribution output to the standard punch (.pch) file.

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Describer	Meaning
ABS	Output acoustic grid contributions in absolute terms. (Default)
NORM	Output acoustic grid contributions in normalized terms.
BOTH	Output acoustic grid contributions in both absolute and normalized terms.
TOPG = pg	The number of structural grids to list in the output that have the greatest contribution to the response at each frequency. If pg > 1, the output is sorted in descending order from the structural grid that has the greatest contribution. If pg = 0, no structural grid contributions are output, only totals. (Integer ≥ 0)
TOPG = ALL	List all of the structural grids in the output. Sort the output in descending order from the structural grid that has the greatest contribution. (Default)
SOLUTION = ALL	Perform the contribution calculations at all frequencies defined by FREQUENCY case control commands. See Remark 10. (Default)
SOLUTION = setf	Perform the contribution calculations only at the frequencies listed on a previously appearing SET case control command with setf as the identification number. See Remark 10. (Integer > 0)
SOLUTION = PEAKOUT	Perform the contribution calculations only at the frequencies identified with the PEAKOUT command. See Remark 10.
GRID = ALL	Output contributions for all structural grids that are part of the acoustic coupling matrix. (Default)
GRID = setg	Output contributions for the grids listed on a previously appearing SET case control command with setg as the identification number. Any grid in the set that is not part of the coupling matrix is ignored. (Integer > 0)
n	Calculate grid contributions for the responses listed on a previously appearing SETMC case control command with n as the identification number. Any response listed in this SETMC case control command that is not an acoustic response is ignored. (Integer > 0)
ALL	Calculate grid contributions for all the responses listed on the SETMC case control commands specified in and above the current subcase. Any response listed in these SETMC case control commands that is not an acoustic response is ignored. (Default)
NONE	Do not calculate grid contributions.

Remarks:

1. Both PRINT and PUNCH can be requested.
2. GRDCON = NONE overrides an overall output request. Thus, to turn off contribution output for a specific subcase, place GRDCON = NONE in the subcase.
3. SOL 108 and 111 are supported.
4. The parameters LFREQ, LFREQFL, HFREQ, HFREQFL, LMODES, and LMODESFL are supported.

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5. The SOLUTION describer can be abbreviated to SOLU.
6. The SET case control command referenced by SOLUTION = setf must contain real values for frequencies. Using integer values may lead to unintended results.
7. GRDCON supports results for microphone points.
8. The SOLUTION describer is only valid in a SOL 108 or 111 frequency response analysis.
9. When an OFREQ case control command is present, the SOLUTION describer is used as follows:
   - If SOLUTION=ALL, output results for all frequencies specified by the OFREQ command.
   - If SOLUTION=setf, output results for all frequencies specified by the OFREQ command that are listed on a previously appearing SET command with setf as the identification number.
   - If SOLUTION=PEAKOUT and a PEAKOUT case control command is also present in the input file, output results for all frequencies specified by the OFREQ command that satisfy the PEAKOUT filtering criteria.
10. The SOLUTION describer is not valid for superelements or frequency-dependent components.

Related Topics

Browse Simcenter Nastran documentation

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GROUNDCHECK

Rigid Body Motion Grounding Check

Perform grounding check analysis on stiffness matrix to expose unintentional constraints by moving the model rigidly.

Format:

$$\text{GROUNDCHECK}\left[\left(\left[\begin{matrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{matrix}\right],\text{PUNCH},\text{SET}=\left(\left\{\begin{matrix}\underline{\text{G}},\text{N},\text{N}+\text{AUTOSPC},\text{F},\text{A}\\ \text{ALL}\end{matrix}\right\}\right),\right.$$
$$\left.\text{GRID}=\text{gid},\text{THRESH}=\text{e},\text{DATAREC}=\left[\begin{matrix}\text{YES}\\ \underline{\text{NO}}\end{matrix}\right],\ (\text{RTHRESH}=\text{r})\right)\Bigg]=\left\{\begin{matrix}\text{YES}\\ \text{NO}\end{matrix}\right\}$$

Examples:

```
GROUNDCHECK=YES
GROUNDCHECK(GRID=12,SET=(G,N,A),THRESH=1.E-5,DATAREC=YES)=YES
```

Describers:

Describer	Meaning
PRINT	Writes output to the print file. (Default)
NOPRINT	Does not write output to the print file.
PUNCH	Writes output to the punch file.
SET	Selects degree-of-freedom set(s). (Default: SET=G).
gid	Reference grid point for the calculation of the rigid body motion.
e	Maximum strain energy which passes the check. The default value is computed by dividing the largest term in the stiffness matrix by 1.E10.
DATAREC	Requests data recovery of grounding forces. (Default: DATAREC=NO)
r	Grounding forces which are larger than the product of (r x largest grounding force) will be printed if DATAREC=YES. (Default=0.1) (0. < r < 1.)

Remarks:

1. GROUNDCHECK must be specified above the subcase level.

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2. SET=N+AUTOSPC uses the stiffness matrix for the n-set with the rows corresponding to degrees-of-freedom constrained by the PARAM,AUTOSPC operation zeroed out. If AUTOSPC was not performed, then this check is redundant with respect to SET=N.

Related Topics

Browse Simcenter Nastran documentation

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GSTRAIN

Requests strain output at Gauss points for SOLs 401 and 402.

Requests strain at Gauss points.

Format:

$$\text{GSTRAIN}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{array}{c}\text{ALL}\\ \text{n}\\ \text{NONE}\end{array}\right\}$$

Examples:

```
GSTRAIN=ALL
GSTRAIN(PRINT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

Remarks:

1. The following elements support the Gauss point output:
   - The plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.
   - The plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8.
   - The axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6.
   - The 3D solids elements CTETRA, CHEXA, CPENTA, CPYRAM.

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2. You can define the number of integration points for an element with the IN field on the PSOLID bulk entry, or the FOROPT field on the PPLANE bulk entry. Regardless of the number of integration points you request, the software outputs the same number of gauss point results as the number of corner grid points. In general, the gauss point closest to a corner grid point is used for the output. See Gauss point output in the *Multi-Step Nonlinear User's Guide* for the tables showing the corner grid point to Gauss point mapping for each element type and integration point option.
3. Both PRINT and PUNCH may be requested.
4. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.

Related Topics

Browse Simcenter Nastran documentation

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GSTRESS

Requests stress output at Gauss points for SOLs 401 and 402.

Requests stress at Gauss points.

Format:

$$\text{GSTRESS}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{array}{c}\text{ALL}\\ \text{n}\\ \text{NONE}\end{array}\right\}$$

Examples:

```
GSTRESS=ALL
GSTRESS(PRINT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

Remarks:

1. The following elements support the Gauss point output:
   - The plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.
   - The plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8.
   - The axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6.
   - The 3D solids elements CTETRA, CHEXA, CPENTA, CPYRAM.

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2. You can define the number of integration points for an element with the IN field on the PSOLID bulk entry, or the FOROPT field on the PPLANE bulk entry. Regardless of the number of integration points you request, the software outputs the same number of gauss point results as the number of corner grid points. In general, the gauss point closest to a corner grid point is used for the output. See Gauss point output in the *Multi-Step Nonlinear User's Guide* for the tables showing the corner grid point to Gauss point mapping for each element type and integration point option.
3. Both PRINT and PUNCH may be requested.
4. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.

Related Topics

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GTHSTRN

Requests thermal strain output at Gauss points for SOLs 401 and 402.

Requests thermal strain at Gauss points.

Format:

$$\text{GTHSTRN}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{array}{c}\text{ALL}\\ \text{n}\\ \text{NONE}\end{array}\right\}$$

Examples:

```
GTHSTRN=ALL
GTHSTRN(PRINT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

Remarks:

1. The following elements support the Gauss point output:
   - The plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.
   - The plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8.
   - The axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6.
   - The 3D solids elements CTETRA, CHEXA, CPENTA, CPYRAM.

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2. You can define the number of integration points for an element with the IN field on the PSOLID bulk entry, or the FOROPT field on the PPLANE bulk entry. Regardless of the number of integration points you request, the software outputs the same number of gauss point results as the number of corner grid points. In general, the gauss point closest to a corner grid point is used for the output. See Gauss point output in the *Multi-Step Nonlinear User's Guide* for the tables showing the corner grid point to Gauss point mapping for each element type and integration point option.
3. Both PRINT and PUNCH may be requested.
4. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.
5. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

Related Topics

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GUST

Aerodynamic Gust Load Requests

Selects the gust field in an aeroelastic response problem.

Format:

GUST=n

Examples:

```
GUST=73
```

Describers:

Describer	Meaning
n	Set identification of a GUST Bulk Data entry. (Integer>0)

Remarks:

The choice of transient or frequency response GUST depends upon the type of TLOAD or RLOAD referenced on the selected GUST entry.

Related Topics

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HARCOEFF

Selects coefficients (multipliers) of the reference frequency to define multiple harmonics in SOL 414,111 Harmonic Response solutions

Selects a SET case control command that contains a list of coefficients (real numbers) that are factors of the reference harmonics.

Format:

HARCOEFF=n

Examples:

```
SET 1=0.0, 1.0, 2.0
HARCOEFF=1
```

Describers:

Describer	Meaning
n	SET identification number for a set definition of general form. See Remark 3. (Integer>0)

Remarks:

1. HARCOEFF is used in SOL 414,111 Harmonic Response solutions for solving non-symmetric assemblies of a rotor and stator. You can also use it to define loads corresponding to different frequencies (using RLOAD1/RLOAD2 and RLHARM) in the fixed and rotating reference frame. You can provide one or more coefficients.

   HARCOEFF is valid in these types of analyses:

   - In a fixed or rotating reference frame when you use it to apply loads at different frequencies.
     - In a fixed or rotating reference frame, when you apply loads (using RLOAD1/RLOAD2 and RLHARM) at different frequencies, you can define harmonic coefficients in HARCOEFF. The coefficients are real numbers that you can define as whole numbers (1.0) or fractions (1.5).
     - In a fixed reference frame, if you apply an unbalance load (ULOAD/UNBA), misalignment load (MLOAD/MLINK), or bow case (NLCNTLG BOWCASE), you must add a coefficient of 1.0 in HARCOEFF.
   - In a combination of fixed and rotating reference frame to solve a non-symmetric model.

     For a non-symmetric rotor and stator that are linked by a CLINK bulk entry with OPT = 3, the software combines the fixed and rotating reference frames. In this case, you must define HARCOEFF with 0.0, 1.0, and 2.0, even if you define no loads using RLOAD1/RLOAD2.

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Note

If the model or bearings are very non-symmetric and therefore have strong anisotropic characteristics, we recommend that you add coefficients 3.0, 4.0, and so on. The coefficients must be real numbers in whole form (such as 3.0, 4.0, 5.0, and so on), not fractional form (such as 3.5, 4.3, 5.1, and so on).

2. The software requires that all coefficients up to the highest number in the SET are requested. For example, if you want a coefficient of 5.0, you must create a SET that contains 0.0, 1.0, 2.0, 3.0, 4.0, and 5.0.
3. The SET ID is an integer, but the values in the SET must be real numbers in ascending order, as in this example:

   ```
   SET 1=0.0, 1.0, 2.0
   ```

   The ascending order is especially important when you also use the RLHARM bulk entry with RLOAD1/RLOAD2.
4. HARCOEFF is not valid for models with multiple rotors.

Related Topics

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HARMONICS

Requests Solution Harmonics

Requests the solution harmonics for cyclic symmetry and axisymmetric models in solutions 114, 115, 116, 118, 401, 402, and 414.

Format:

$$\text{HARMONICS} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \end{matrix} \right\}$$

Examples:

```
HARMONICS=ALL
HARMONICS=32
```

Describers:

Describer	Meaning
ALL	All harmonics will be used.
n	Set identification number. The referenced SET lists the harmonic numbers to use. (Integer>0)

Remarks:

1. For a cyclic symmetry analysis in SOLs 114, 115, 116, 118, 401, and 402, either HARMONICS=ALL or HARMONICS=n are required.
2. For a Fourier normal modes analysis in SOL 401 or 402, because there is an infinite number of harmonics in a Fourier normal modes analysis, only HARMONICS=n is supported.
3. *n* is the ID of a SET command which lists the harmonic numbers. For example,

   ```
   ...
   HARMONICS = 101
   SET 101 = 3,5,8,12
   ...
   ```

4. When an ANALYSIS=FOURIER subcase is defined for SOL 402, the HARMONICS command must be defined above the subcases (globally).
5. The HOUTPUT case control command can be used to request the harmonic numbers for results output.

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Remarks for SOL 414

1. For a cyclic symmetry analysis with the SOL 414,110 complex modal solution or a SOL 414,111 harmonic response solution, the zeroth harmonic is the default. Other harmonics are requested by HARMONICS=n. HARMONICS=ALL is not supported. The software ends with a fatal error if you request a harmonic index larger than N/2 for an even number of segments N or larger than (N-1)/2 for an odd number of segments.
2. For a cyclic symmetry analysis with the SOL 414,101 and SOL 414,129 solutions, only the zeroth harmonic is used.

Related Topics

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HARTAB

Decompose Modes in Harmonics

Decompose modes in harmonics in a SOL 414,103 eigenvalues analysis.

Format:

HARTAB=n

Examples:

```
HARTAB=18
```

Describers:

Describer	Meaning
n	Sets the number of harmonics you want to be decomposed for each mode. (Integer>0)

Remarks:

1. The rotor can be an axisymmetric rotor with respect to the rotation axis, such as a shaft with circular disks, or it can be a cyclic symmetric structure (CYCSET).
   - The primary use case is a full 3D rotor, but you can use a cyclic symmetric structure if the cyclic sector consists of several blades. For example, if you have a disk that consists of 24 blades, you could create a sector of 4 blades. The total number of sectors NSEG in CYCSET bulk entry would be 6.
   - Although you can solve a cyclic symmetry sector of a disk or of one blade, you can request only one harmonic for cyclic symmetry in SOL 414,103 Eigenvalues solutions. Therefore, the results for a cyclic sector or a disk or a blade include only the diameter modes that correspond to the requested harmonic. For example, if you request harmonic 2, the decomposition results contain only the 2-diameter modes. Cyclic sectors that consist of multiple blades, however, produce additional diameter modes for the same requested harmonic.
2. If you are requesting harmonic decomposition for cyclic symmetry, the number of harmonics you request must be at least half the number of the cyclic symmetry sectors. For example, if the cyclic symmetry consists of six cyclic sectors (also called *segments*), the minimum number of harmonics you can request is three.
3. The model can be made of 1D, 2D shell, or 3D elements. The model cannot use any 2D axisymmetric elements.
4. If the model contains cyclic symmetry, it cannot contain a CYCISC bulk entry (cyclic symmetry interstage connection).

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5. The model cannot contain 2D Fourier elements (for the shaft, for example), and it cannot use the FMH3D (Fourier-3D) connection.
6. The axis of rotation must be the global X-axis, Y-axis, or Z-axis that is deduced from the inertial properties.
7. The output is in the MODAL DECOMPOSITION and the MODAL PAIRINGS tables in the .f06 file.

Related Topics

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HDOT

Heat Transfer Rate of Change of Enthalpy Output Request

Requests form of rate of change of enthalpy vector output in transient heat transfer analysis (SOL 159).

Format:

$$\text{HDOT}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \underline{\text{SORT2}}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{PRINT}},\ \text{PUNCH}\\ \text{PLOT}\end{matrix}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
HDOT=5
```

Describers:

Describer	Meaning
SORT1	Output will be presented as a tabular listing of grid points for each time.
SORT2	Output will be presented as a tabular listing of time for each grid point.
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
PLOT	Generates but does not print rate of change of enthalpy.
ALL	Rate of change of enthalpy for all points will be output.
NONE	Rate of change of enthalpy for no points will be output.
n	Set identification of previously appearing SET command. Only rates of change of enthalpy for points with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

HDOT=NONE is used to override a previous HDOT=n or HDOT=ALL command.

Related Topics

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HOUTPUT

Harmonic Output Request

Requests harmonic output for cyclic symmetry and axisymmetric models in solutions 114, 115, 116, 118, 401 and 402.

Format for SOLs 114, 115, 116, 118:

$$\text{HOUTPUT}\left[\left(\underline{\text{C}},\underline{\text{S}},\underline{\text{C}^*},\underline{\text{S}^*}\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{K}\end{matrix}\right\}$$

Format for SOLs 401, 402:

$$\text{HOUTPUT} = \left\{\begin{matrix}\text{ALL}\\ \text{K}\end{matrix}\right\}$$

Examples:

```
HOUTPUT=ALL
HOUTPUT=5
```

Describers:

Describer	Meaning
ALL	All harmonics will be output.
k	Set identification number. The referenced SET lists the harmonics for output. (Integer>0)
C, S, C*, S*	Harmonic coefficients for SOLs 114, 115, 116, 118. Not supported by SOLs 401 and 402. See Remark 4.

Remarks:

1. *k* is the ID of a SET command which lists the harmonic numbers for output. For example,

```
...
HOUTPUT = 101
SET 101 = 3,5,8,12
...
```


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2. HOUTPUT=ALL requests output for all harmonics specified on the HARMONICS command.
3. In SOLs 114, 115, 116, 118, either the HOUTPUT or NOUTPUT command is required to obtain data recovery in cyclic symmetry analysis.

   In SOL s401 and 402, the HOUTPUT command is required to obtain harmonic data recovery in a ANALYSIS=CYCMODES or ANALYSIS=FOURIER subcase.
4. In SOLs 114, 115, 116, 118, C and S correspond to the cosine and sine coefficients when the STYPE field is ROT or AXI on the CYSYM Bulk Data entry. C, S, C*, and S* correspond to the cosine symmetric, sine symmetric, cosine antisymmetric, and sine antisymmetric coefficients respectively, when the STYPE field is DIH on the CYSYM Bulk Data entry.

Related Topics

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IC

Transient Initial Condition Set Selection

Selects the initial conditions for transient solutions (SOLs 109, 112, 129, 159, 401, and 402).

Format:

$$\text{IC}\left[\begin{array}{c}\underline{\text{PHYSICAL}}\\ \text{MODAL}\\ \text{STATSUB}\,[,\,\text{DIFFK}]\\ \text{TZERO}\,[,\,\text{DIFFK}]\end{array}\right]=\text{n}$$

Examples:

```
IC = 17
IC(PHYSICAL) = 10
IC(MODAL) = 20
IC(STATSUB) = 30
IC(STATSUB,DIFFK) = 1030
IC(TZERO)= 1
IC(TZERO,DIFFK)= 1
```

Describers:

Describer	Meaning
PHYSICAL	Use the TIC bulk entries selected by set *n* as the initial conditions for coordinates involving grid and scalar points (default). See Remarks 4 and 5.
MODAL	Use the TIC bulk entries selected by set *n* as the initial conditions for modal coordinates. See Remarks 4 and 5.
STATSUB	Use the solution of the static analysis subcase *n* as the initial conditions. See Remark 4.
TZERO	Use the static deflection resulting from the loading at time = 0 as the initial condition for a direct transient analysis (SOL 109) or a modal transient analysis (SOL 112). For this option, *n* is not used by the software, although a dummy value must be defined. See Examples. See Remark 4.
DIFFK	Include the effects of differential stiffness in the solution. See Remarks 3 and 4.
n	For the PHYSICAL option, *n* is the set identification number of TIC bulk entries for structural analysis (SOLs 109 and 129) or TEMP and TEMPD bulk entries for heat transfer analysis (SOL 159). For the MODAL option, *n* is the set identification

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Describer	Meaning
	number of TIC bulk entries for modal transient analysis (SOL 112). For the STATSUB option, *n* is the ID of a static analysis subcase (SOL 109 and 112). For the TZERO option, *n* is not used by the software, although a dummy value must be defined. (Integer>0)

Remarks:

1. TIC bulk entries will not be used (therefore, no initial conditions) unless selected in the case control section.
2. The IC case control command must be defined in the global subcase. However, when the TZERO describer is specified, the initial condition for each subcase is defined by the loads included in that subcase.
3. The DIFFK describer is meaningful only when used in conjunction with the STATSUB or TZERO describers.
4. The following table summarizes the solution sequences in which the describers may be specified.

	109	112	129	159	401	402
PHYSICAL	X	X	X	X	X	X
MODAL		X				
STATSUB*	X	X				
TZERO*	X	X				
DIFFK*	X	X				

* Not applicable in a DMP solution.

5. Initial condition definitions on extra points are not supported and will be ignored.
6. For SOL 401, you can request initial displacement and velocity conditions for a dynamic subcase with the IC=n case control command. The integer n on the IC case control references a TIC bulk entry. Only the PHYSICAL option (default) on the IC command is available, which is the option for defining initial conditions on grid points.

   The IC command can be defined either in a dynamic subcase, or if all subcases include ANALYSIS=DYNAMICS, you can define the IC command above the subcase level (globally). For either definition, the software will only apply an initial condition to a dynamic subcase that is either the first subcase, or if it is not the first, it must be non-sequentially dependent (SEQDEP=NO). The software ignores an initial condition when a dynamic subcase does not satisfy these rules, and instead starts from the state of displacement, velocity, and acceleration from the previous preload, static, or dynamic subcase.

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Remarks related to SOL 402:

1. In SOL 402, the IC case control command can be defined in any subcase.
   - If a subcase is sequentially dependent on the previous subcase (SEQDEP=YES), the initial conditions are obtained from the end of the previous subcase.
   - Initial conditions are especially useful for ANALYSIS=DYNAMICS subcases which are not sequentially dependent on the previous subcase (SEQDEP=NO).

Related Topics

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IMPERF

SOL 401 Grid Point Position Deltas

Selects grid point position deltas in SOL 401.

Format:

IMPERF=n

Examples:

```
IMPERF=33
```

Describers:

Describer	Meaning
n	Set identification number of an IMPRADD or IMPERF bulk entry. (Integer>0)

Remarks:

1. This command is only supported in SOL 401.
2. The IMPERF case control command must be defined globally (above the subcases). As a result, the updated location of the referenced grid points applies to all subcases.
3. You can use the IMPERF bulk entry to adjust the X,Y,Z location of a grid point with a positive or negative delta.
4. The set identification number n can refer to a IMPRADD entry which then combines multiple IMPERF entries, or it can refer to an individual IMPERF entry.
5. The deltas are also referred to as imperfections since they can be used in a nonlinear buckling solution to induce a buckling condition.

Related Topics

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INCLUDE

Insert External File

Inserts an external file into the input file. The INCLUDE statement may appear anywhere within the input data file.

Format:

INCLUDE ‘filename’

Describers:

Describer	Meaning
filename	External file to be inserted. The ‘*directory_path/filename*’ must begin and end with the single quote character.

Examples:

The following INCLUDE statement is used to obtain the Bulk Data from another file called MYBULK.DATA:

```
SOL 101
CEND
TITLE=STATIC ANALYSIS
LOAD=100
INCLUDE ’MYCASE.DATA’
BEGIN BULK
ENDDATA
```

Remarks:

1. INCLUDE statements may be nested; that is, INCLUDE statements may appear inside the external file. The nested depth level must not be greater than 10.
2. Each line has a 72 character limit. Multiple lines can be used when file names are long. The entire string must begin and end with the single quote character.

   For example,

   D:\folder1\folder2\folder3\folder4\folder5\folder6\folder7\folder8\folder9\mydata.dat

   can be defined on multiple lines:

   ```
   INCLUDE ‘D:\folder1\folder2\folder3\folder4\folder5\folder6\folder7\fold
   er8\folder9\mydata.dat'
   ```


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The following input format is also supported:

```
INCLUDE ‘D:\folder1\folder2\folder3\
         folder4\folder5\
     folder6\folder7\
        folder8\folder9\mydata.dat'
```

Related Topics

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INITS

Selects an initial stress or strain set or an offset strain set.

Selects an initial stress or strain set or an offset strain set.

General Format:

INITS=n

Offset Strain Format:

INITS(OFFSET)=n

Examples:

```
INITS=12
```

```
INITS(OFFSET)=12
```

Describers:

Describer	Meaning
INITS=n	Set identification number of an INITS or INITADD bulk entry. (Integer>0)
INITS(OFFSET)=n	Set identification number of an INITSO or INITADD bulk entry. (Integer>0)

Remarks:

1. INITS and INITS(OFFSET) are only supported in SOL 401.
2. The INITS selects an initial stress or strain set in general and for the special balanced initial stress or strain method.

   The INITS(OFFSET) selects a strain set for the special balanced initial stress or strain method which requires that both INITS and INITS(OFFSET) commands.
3. The INITS and INITS(OFFSET) case control must be defined above the subcases (globally), and are reapplied in each static subcase. If you want to include the effects of initial stress or strain including any other load in a normal modes subcase, you should include a static subcase before the sequentially dependent normal modes subcase.
4. The INITS command can reference either an INITS or an INITADD bulk entry. If you reference an INITADD bulk entry with the INITS command, the INITADD bulk entry must only reference INITS entries. That is, it cannot reference any INITSO entries.

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5. The INITS(OFFSET) command can reference either an INITSO or an INITADD bulk entry. If you reference an INITADD bulk entry with the INITS(OFFSET) command, the INITADD bulk entry must only reference INITSO entries. That is, it cannot reference any INITS entries.
6. The INITS(OFFSET) case control must be accompanied by an INITS case control command.
7. **Balanced stress or strain method:**

   An unbalanced initial stress-strain results in deformation when applied to an unconstrained body with the possibility of residual stress. You can use the balancing method to balance an unbalanced initial stress-strain. The method removes the unbalanced part by removing the strains which produce a deformation. After you use the balancing method, your initial stress-strain will produce a self-equilibrating stress state and no deformations. The method requires that a part of your unbalanced initial stress-strain is actually balanced. That is, when you apply your unbalanced initial stress-strain to the unconstrained model, stress does exist.

   a. The balancing method requires that you first run a static offset-solution to obtain the total strain output as a result of your unbalanced initial stress-strain with the model unconstrained. You only apply the unbalanced initial stress-strain in the offset solution using the INITS command. No temperature loads, mechanical loads, or enforced displacements are applied for this step.

      To allow the static offset-solution to complete with the unconstrained condition, a matrix stabilization option is available. Setting the parameter MSTAB to 1 on the NLCNTL bulk entry will turn on the option. In addition, the MSFAC parameter is available to define a scale factor for matrix stabilization. Specifically, when you define MSTAB=1, the software scales the diagonal terms by the factor (1+MSFAC).

   b. Once your offset strains are computed, you can run the consecutive balanced solution that includes your initial stress-strain requested with the INITS command, your offset strain requested with the INITS(OFFSET), and any other loads (temperature loads, mechanical loads, enforced displacements).

      The software does not need to associate a balanced initial stress-strain definition with an offset strain definition in the balanced solution. As a result, the SID defined on the INITS(OFFSET) command selecting your offset strains, and the SID defined on the INIT command selecting your unbalanced initial stress-strain can match, or not match. There is no software requirement either way.

Related Topics

Browse Simcenter Nastran documentation

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INPOWER

Incident Acoustic Power Results Output

Incident acoustic power results output request for 2D microphone elements or acoustic free faces (SORT2 results only).

Format:

$$\text{INPOWER}\left[\left(\text{GROUP} = \text{ng}, \text{FACES} = \text{nf}, \left[\begin{matrix}\underline{\text{PRINT}} \\ \text{NOPRINT}\end{matrix}\right], \text{PUNCH}, \left[\begin{matrix}\underline{\text{REAL}} \text{ or } \underline{\text{IMAG}} \\ \text{PHASE}\end{matrix}\right], \left[\begin{matrix}\underline{\text{RPRINT}} \\ \text{NOPRINT}\end{matrix}\right], \text{RPUNCH}, \text{SYMM}\right)\right] = \left\{\begin{matrix}\underline{\text{YES}} \\ \text{NO}\end{matrix}\right\}$$

Examples:

```
SET 5=5,10
```

```
INPOWER(GROUP=5,PRINT,PUNCH,PHASE)
```

```
SET 6=100
```

```
INPOWER(FACES=6)
```

```
INPOWER=NO
```

```
INPOWER(FACES=201,PRINT,REAL,SYMM)
```

Describers:

Describer	Meaning
GROUP =ng	*ng* (integer>0) is the identification number of the SET case control command that lists the identification numbers of GROUP bulk entries that contain the 2D

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Describer	Meaning
	microphone elements at which to calculate incident acoustic power results. See Remark 2.
FACES =nf	*nf* (integer>0) is the identification number of the SET case control command that lists the identification numbers of BSURFS bulk entries that contain the acoustic free faces at which to calculate incident acoustic power results. See Remark 2.
PRINT	Write the incident acoustic power results to the print (.f06) file. (Default)
NOPRINT	Do not print the incident acoustic power results.
PUNCH	Write the incident acoustic power results to the standard punch (.pch) file.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Default)
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
RPRINT	Write the random analysis results to the print (.f06) file. See Remark 4. (Default)
NORPRINT	Do not write the random analysis results to the print file. See Remark 4.
RPUNCH	Write the random analysis results to the punch (.pch) file. See Remark 4.
SYMM	Requests to ignore the symmetry plane that coincides with the coupling surface. See Remarks 5 and 6.
YES	Honor describer settings and calculate and output incident acoustic power results. (Default)
NO	Ignore describer settings and do not calculate or output incident acoustic power results. See Remark7.

Remarks:

1. Both PRINT and PUNCH may be requested.
2. Only 2D microphone elements referenced in GROUP bulk entries or acoustic free faces at the structural boundary referenced on BSURFS bulk entries are processed for incident acoustic power results. Either the GROUP describer or the FACES describer must be specified, but not both. Microphone elements must reference a PMIC property.
3. Incident acoustic power results are written to the op2 file, if PARAM, POST,-1 or PARAM,POST,-2.
4. The following applies to random solutions:
   - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, NOPRINT, PUNCH describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
   - Only SORT2 output format is available for the frequency response and random results output.

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- Only the power spectral density function (PSDF) is calculated. For RANDPS (or RANDPEX) bulk entries that refer to *n* subcases, when *i* = *j* the results in W/Hz are computed as follows:

$$INPOWER_{psdf}\left(f\right) = \sum_{i=1}^{n} RANDPS_{ii}\left(f\right) \cdot INPOWER_{i}\left(f\right)$$

  If $i \neq j$, then a warning message is issued and the RANDPS bulk entries are ignored.

  For more information, see the **RANDPS** bulk entry.

5. SYMM is supported in FEMAO solutions only.
6. You should request SYMM in typical panel transmission loss scenarios, where you model acoustic fluid domains on either side of a structure with an AML, including a symmetry plane. To accurately calculate the incident acoustic power and, by extension, the transmission loss, request the SYMM describer to ignore the symmetry plane.

   For more information, see *Coincidental symmetry planes and coupling surfaces* in Incident and transmitted acoustic power and acoustic power transmission loss output.
7. Use INPOWER=NO within a subcase to override a INPOWER specification above the subcase level.
8. The computed incident power takes into account the incident acoustic energy from the ACPOLE1, ACPOLE2, and ACPLNW acoustic sources only, and not from acoustic sources like ACPNVEL that can contain a scattered component.

Related Topics

Browse Simcenter Nastran documentation

Simcenter Nastran Quick Reference Guide
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JCONSET

Joint Constraint Set Selection (SOL 402 and SOL 414,129)

Selects an external kinematic joint constraint set.

Format:

JCONSET=n

Examples:

```
JCONSET=15
```

Describers:

Describer	Meaning
n	Set identification of a JCON bulk data entry. (Integer>0)

Related Topics

Browse Simcenter Nastran documentation

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JRESULTS

Kinematic Joints Result Output Request (SOL 402)

Format:

$$\text{JRESULTS}\left[(\text{FORC},\text{MOME},\underline{\text{DISP}},\underline{\text{ANGL}},\text{VELO},\text{ANGV},\text{PLOT})\right]=\left\{\begin{array}{c}\text{ALL}\\ \text{n}\\ \underline{\text{NONE}}\end{array}\right\}$$

Examples:

```
JRESULTS=ALL
JRESULTS(FORC,PLOT)=ALL
JRESULTS(FORC,MOME,PLOT)=ALL
```

Describers:

Describer	Meaning
FORC	Force vector in basic coordinates is output for each kinematic joint.
MOME	Moment vector in basic coordinates is output for each kinematic joint.
DISP	Relative position between the kinematic joint nodes definition in basic coordinates is output for each kinematic joint.
ANGL	Relative rotation between the kinematic joint nodes definition in basic coordinates is output for each kinematic joint.
VELO	Relative velocity between the kinematic joint nodes definition in basic coordinates is output for each kinematic joint.
ANGV	Relative angular velocity between the kinematic joint nodes definition in basic coordinates is output for each kinematic joint.
PLOT	Requests results to the *.op2 and *.unv files.
ALL	Results at all contact grid points will be output.
NONE	Results will not be output.
n	Set identification of a previously appearing SET command. Only kinematic joints with identification numbers that appear on this SET command will be output. (Integer>0).

Related Topics

Browse Simcenter Nastran documentation

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JINTEG

J-Integral computation and output for SOL 401.

Controls the computation and output of the j-integral.

Format:

$$\text{JINTEG}\left[\left(\left[\frac{\text{PRINT}}{\text{PLOT}}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
JINTEG=ALL
JINTEG(PLOT)=ALL
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print (.f06) file. (Default)
PLOT	Compute output only.
ALL	Compute j-integral.
NONE	Do not compute j-integral.

Remarks:

1. Only supported in a static subcases for SOL 401.

Related Topics

Browse Simcenter Nastran documentation

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K2GG

Direct Input Stiffness Matrix Selection

Selects direct input stiffness matrices.

Format:

K2GG=name

Examples:

```
K2GG=KDMIG
K2GG=KDMIG1, KDMIG2, KDMIG3
K2GG=1.25*KDMIG1, 1.0*KDMIG2, 0.75*KDMIG3
SET 100=K1, K2
K2GG=100
```

Describers:

Describer	Meaning
name	Name of a $[k^2_{gg}]$ matrix that is input on the DMIG Bulk Data entry. (Character) Scale factors may be included (see Remarks 4 and 5). See "Matrix Assembly Operations in SubDMAP SEMG" in the *Simcenter Nastran User's Guide*.

Remarks:

1. DMIG matrices will not be used unless selected.
2. Terms are added to the stiffness matrix before any constraints are applied.
3. The matrix must be symmetric and field 4 on the DMIG,name entry must contain the integer 6.
4. The associated DMIG matrices can be scaled using either in-line scale factors on K2GG (for example, K2GG=1.25*KDMIG1), using the parameter CK2 (for example, PARAM,CK2,1.25), or both. See **"Parameter Descriptions"**.
5. Multiple matrices separated by a comma or a blank are additive. When multiple matrices and in-line scale factors are used together, each matrix name in the list must include a scale factor. A scale factor of 1.0 should be used for matrices in the list which are not scaled. For example,

   ```
   K2GG=1.25*KDMIG1,1.0*KDMIG2,0.75*KDMIG3
   ```

   The parameter CK2 when defined will scale all K2GG. For example, if PARAM,CK2,1.30 is defined with the K2GG example above, the result would be K2GG=1.30(1.25*KDMIG1 + 1.0*KDMIG2 + 0.75*KDMIG3).

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6. The K2GG command is limited to use in the first subcase only.

Remarks related to SOL 401:

1. You can define the parameters CM2, CK2, and CB2 to scale the linear DMIG matrices defining mass, stiffness, and damping. The parameters CM1, CK1, and CB1 are not supported.
2. When you include DMIG matrices in restart solutions, for both the initial and restart solution, you must reference the DMIG entries from the global case control using the appropriate K2GG, M2GG, and B2GG command.

Related Topics

Browse Simcenter Nastran documentation

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K2PP

Direct Input Stiffness Matrix Selection

Selects direct input stiffness matrices, which are not included in normal modes.

Format:

K2PP=name

Examples:

```
K2PP=KDMIG
K2PP=KDMIG1, KDMIG2, KDMIG3
K2PP=1.25*KDMIG1, 1.0*KDMIG2, 0.75*KDMIG3
K2PP=(1.25,0.5)*KDMIG1, (1.0,0.0)*KDMIG2, (0.75,-2.2)*KDMIG3
SET 100=K1, K2
K2PP=100
```

Describers:

Describer	Meaning
name	Name of a $[k^2_{pp}]$ matrix that is input on the DMIG Bulk Data entry. (Character) Scale factors may be included (see remarks 6 and 7). See "Formulations of Dynamic Equations in SubDMAP GMA" in the *Simcenter Nastran User's Guide*.

Remarks:

1. DMIG entries will not be used unless selected by the K2PP command.
2. The matrix must be square or symmetric and field 4 on the DMIG,name entry must contain a 1 or 6.
3. It is recommended that PARAM,AUTOSPC,NO be specified. See "Constraint and Mechanism Problem Identification in SubDMAP SEKR" in the *Simcenter Nastran User's Guide*.
4. K2PP matrices are used only in dynamic response problems. They are not used in normal modes.
5. The K2PP command is supported across subcases. A K2PP command selecting a different DMIG matrix can be defined for each subcase.
6. The associated DMIG matrices can be scaled using in-line scale factors on K2PP (for example, K2PP=1.25*KDMIG1).

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7. Multiple matrices separated by a comma or a blank are additive. When multiple matrices and in-line scale factors are used together, each matrix name in the list must include a scale factor. A scale factor of 1.0 should be used for matrices in the list which are not scaled. For example,

```
K2PP=1.25*KDMIG1,1.0*KDMIG2,0.75*KDMIG3
```

Related Topics

Browse Simcenter Nastran documentation

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K42GG

Direct Input Structural Damping Matrix Selection

Selects direct input structural damping matrices.

Format:

K42GG=name

Examples:

```
K42GG=KDMIG
K42GG=KDMIG1, KDMIG2, KDMIG3
K42GG=1.25*KDMIG1, 1.0*KDMIG2, 0.75*KDMIG3
SET 100=K1, K2
K42GG=100
```

Describers:

Describer	Meaning
name	Name of a $[k^{4,2}{}_{gg}]$ matrix that is input on the DMIG Bulk Data entry. (Character) Scale factors may be included (see Remarks 4 and 5). See "Matrix Assembly Operations in SubDMAP SEMG" in the *Simcenter Nastran User's Guide*.

Remarks:

1. DMIG matrices will not be used unless selected.
2. Terms are added to the structural damping matrix before any constraints are applied.
3. The matrix must be symmetric and field 4 on the DMIG,name entry must contain the integer 6.
4. The associated DMIG matrices can be scaled using either in-line scale factors on K42GG (for example, K42GG=1.25*KDMIG1), using the parameter CK42 (for example, PARAM,CK42,1.25), or both. See **"Parameter Descriptions"**.
5. Multiple matrices separated by a comma or a blank are additive. When multiple matrices and in-line scale factors are used together, each matrix name in the list must include a scale factor. A scale factor of 1.0 should be used for matrices in the list which are not scaled. For example,

   ```
   K42GG=1.25*KDMIG1,1.0*KDMIG2,0.75*KDMIG3
   ```

   The parameter CK42 when defined will scale all K42GG. For example, if PARAM,CK42,1.30 is defined with the K42GG example above, the result would be K42GG=1.30(1.25*KDMIG1 + 1.0*KDMIG2 + 0.75*KDMIG3).

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6. The K42GG command is limited to use in the first subcase only.

Related Topics

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LABEL

Output Label

Defines a character string that will appear on the third heading line of each page of printer output.

Format:

LABEL=label

Examples:

```
LABEL=DEMONSTRATION PROBLEM
```

Describers:

Describer	Meaning
label	Any character string.

Remarks:

1. LABEL appearing at the subcase level will label output for that subcase only.
2. LABEL appearing before all subcases will label any outputs that are not subcase dependent.
3. If no LABEL command is supplied, the label line will be blank.
4. LABEL information is also placed on plotter output as applicable. Only the first 65 characters will appear.

Related Topics

Browse Simcenter Nastran documentation

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LINE

Maximum Lines Per Printed Page

Defines the maximum number of output lines per printed page.

Format:

LINE=n

Examples:

```
LINE=35
```

Describers:

Describer	Meaning
n	Maximum number of output lines per page. (Integer>0; Default = 50)

Remarks:

1. For 11 inch paper, 50 lines per page is recommended; for 8-1/2 inch paper, 35 lines per page is recommended.
2. The NASTRAN statement keyword NLINES may also be used to set this value. See the **"nastran Command and NASTRAN Statement"** .

Related Topics

Browse Simcenter Nastran documentation

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LOAD

External Static Load Set Selection

Selects an external static load set.

For SOL 402, selects a constant load or a driver.

Format:

LOAD=n

Examples:

```
LOAD=15
```

Describers:

Describer	Meaning
n	Set identification of at least one external load Bulk Data entry. The set identification must appear on at least one ACCEL, ACCEL1, DAREA, FORCE, FORCE1, FORCE2, GRAV, MOMENT, MOMENT1, MOMENT2, PLOAD, PLOAD1, PLOAD2, PLOAD4, PLOADE1, PLOADX1, QBDY1, QBDY2, QBDY3, QHBDY, QVECT, QVOL, RFORCE, RFORCE1, SELOAD, or SLOAD entry. (Integer>0)
n (For SOL 402)	Set identification of at least one external load Bulk Data entry. The set identification must appear on at least one FORCE, FORCE1, FORCE2, GRAV, MOMENT, MOMENT1, MOMENT2, LOAD, PLOAD, PLOAD1, PLOAD2, PLOAD4, RFORCE, SPCD or DRIVER entry. (Integer>0)

Remarks:

1. ACCEL, ACCEL1, GRAV entries cannot have the same set identification number as any other loading entries. If you want to combine these with other static loads, you must use the LOAD bulk entry.
2. LOAD is only applicable in linear and nonlinear statics, inertia relief, differential stiffness, buckling, and heat transfer problems.
3. The total load applied will be the sum of external (LOAD), thermal (TEMP(LOAD)), element deformation (DEFORM), and constrained displacement (SPC) loads.
4. Static, thermal, and element deformation loads should have unique set identification numbers.

Remarks related to SOL 402:

1. LOAD may be used in a static or transient analysis.

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2. Both DLOAD and LOAD case control commands can be specified. DLOAD is used to select time-dependent loads and LOAD is used to select constant loads.

Related Topics

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LOADSET

Static Load Set Selection

Selects a sequence of static load sets to be applied to the structural model. The load sets may be referenced by dynamic load commands.

Format:

LOADSET=n

Examples:

```
LOADSET=100
```

Describers:

Describer	Meaning
n	Set identification number of at least one LSEQ Bulk Data entry. (Integer>0)

Remarks:

1. When used in superelement analysis, this command must be used for all superelements. The number of static load vectors created for each superelement depends upon the type of analysis. In static analysis, the number of vectors created is equal to the number of unique EXCITEID IDs on all LSEQ entries in the Bulk Data; in dynamic analysis, the number of vectors created is equal to the number of unique DAREA IDs on all RLOAD1, RLOAD2, TLOAD1, TLOAD2 and ACSRCE entries in the Bulk Data.
2. When the LOADSET command is used in superelement statics, the residual structure should have as many loading conditions as the number of unique EXCITEID sets defined on all LSEQ entries. The subcases after the first should contain only SUBTITLE and LABEL information and residual structure output requests. SUBTITLE and LABEL information for all superelements will be obtained from the residual structure.
3. In SOL 101, the design sensitivity output will identify all expanded subcases by a sequence of unique integers beginning with n.
4. In the nonlinear static solution sequences (SOLs 106 and 153) the LOADSET must appear above all subcases and only one LOADSET may be specified.
5. Only one LOADSET command is allowed per superelement.
6. It is no longer necessary to employ LOADSET/LSEQ data to specify static loading data for use in dynamic analysis. In the absence of LOADSET request in Case Control, all static loads whose load set IDs match the EXCITEID IDs on all RLOAD1, RLOAD2, TLOAD1, TLOAD2 and ACSRCE entries in the Bulk Data are automatically processed.

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Related Topics

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M2GG

Direct Input Mass Matrix Selection

Selects direct input mass matrices.

Format:

M2GG=name

Examples:

```
M2GG=MDMIG
M2GG=MDMIG1, MDMIG2, MDMIG3
M2GG=1.25*MDMIG1, 1.0*MDMIG2, 0.75*MDMIG3
SET 100=M1, M2
M2GG=100
```

Describers:

Describer	Meaning
name	Name of a [M^2_{gg}] matrix that is input on the DMIG Bulk Data entry. Scale factors may be included (see Remarks 4 and 5). See "Matrix Assembly Operations in SubDMAP SEMG" in the *Simcenter Nastran User's Guide*. (Character)

Remarks:

1. DMIG matrices will not be used unless selected.
2. Terms are added to the mass matrix before any constraints are applied.
3. The matrix must be symmetric and field 4 on the DMIG, name entry must contain a 6.
4. The associated DMIG matrices can be scaled using either in-line scale factors on M2GG (for example, M2GG=1.25*MDMIG1), using the parameter CM2 (for example, PARAM,CM2,1.25), or both. See **"Parameter Descriptions"**.
5. Multiple matrices separated by a comma or a blank are additive. When multiple matrices and in-line scale factors are used together, each matrix name in the list must include a scale factor. A scale factor of 1.0 should be used for matrices in the list which are not scaled. For example,

   ```
   M2GG=1.25*MDMIG1,1.0*MDMIG2,0.75*MDMIG3
   ```

   The parameter CM2 when defined will scale all M2GG. For example, if PARAM,CM2,1.30 is defined with the M2GG example above, the result would be M2GG=1.30(1.25*MDMIG1 + 1.0*MDMIG2 + 0.75*MDMIG3).

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6. M2GG input is not affected by PARAM,WTMASS. M2GG input must either be in consistent mass units or PARAM,CM2 may be used.
7. The M2GG command is limited to use in the first subcase only.

Remarks related to SOL 401:

1. You can define the parameters CM2, CK2, and CB2 to scale the linear DMIG matrices defining mass, stiffness, and damping. The parameters CM1, CK1, and CB1 are not supported.
2. When you include DMIG matrices in restart solutions, for both the initial and restart solution, you must reference the DMIG entries from the global case control using the appropriate K2GG, M2GG, and B2GG command.

Remark related to SOLs 402 and 414:

1. For both SOL 402 and SOL 414, the mass scaling parameters CM1 and CM2 are not supported.
2. For SOL 402, when you include DMIG matrices in restart solutions, if you referenced DMIG entries in the initial solution, the same DMIG entries will automatically be included in the restart solution. You can leave them defined in the restart input file, but this is not a requirement.

   SOL 414 does not support restarts.

Related Topics

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M2PP

Direct Input Mass Matrix Selection

Selects direct input mass matrices, which are not included in normal modes.

Format:

M2PP=name

Examples:

```
M2PP=MDMIG
M2PP=MDMIG1, MDMIG2, MDMIG3
M2PP=1.25*MDMIG1, 1.0*MDMIG2, 0.75*MDMIG3
M2PP=(1.25,0.5)*MDMIG1, (1.0,0.0)*MDMIG2, (0.75,-2.2)*MDMIG3
SET 100=M1, M2
M2PP=100
```

Describers:

Describer	Meaning
name	Name of a [M^2_{pp}] matrix that is input on the DMIG Bulk Data entry. Scale factors may be included (see remarks 7 and 8). See "Dynamic Reduction and Component Mode Synthesis in SubDMAP SEMR3" in the *Simcenter Nastran User's Guide*. (Character)

Remarks:

1. DMIG entries will not be used unless selected by the M2PP input.
2. M2PP input is not affected by PARAM,WTMASS. M2PP input must be in consistent mass units.
3. The matrix must be square or symmetric and field 4 on the DMIG, name entry must contain a 1 or 6.
4. It is recommended that PARAM,AUTOSPC,NO be specified. See "Constraint and Mechanism Problem Identification in SubDMAP SEKR" in the *Simcenter Nastran User's Guide*.
5. M2PP matrices are used only in dynamic response problems. They are not used in normal modes.
6. The M2PP command is supported across subcases. A M2PP command selecting a different DMIG matrix can be defined for each subcase.
7. The associated DMIG matrices can be scaled using in-line scale factors on M2PP (for example, M2PP=1.25*MDMIG1).

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8. Multiple matrices separated by a comma or a blank are additive. When multiple matrices and in-line scale factors are used together, each matrix name in the list must include a scale factor. A scale factor of 1.0 should be used for matrices in the list which are not scaled. For example,

```
M2PP=1.25*MDMIG1,1.0*MDMIG2,0.75*MDMIG3
```

Related Topics

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MASTER

Redefine the MASTER Subcase

Allows the redefinition of a MASTER subcase.

Format:

SUBCASE n

MASTER

Examples:

```
SUBCASE 10
MASTER
```

Remarks:

1. All commands in a MASTER subcase apply to the following subcases until a new MASTER subcase is defined.
2. Suppose that superelement 10 has SPC set 10, MPC set 10, and LOAD sets 101 and 102. Suppose also that superelement 20 has SPC set 20, MPC set 20, and LOAD sets 201 and 202.

   The following setup specifies the required subcase structure:

```
   TITLE=MY MODEL
   DISP=ALL
   SEALL=ALL
SUBCASE 101
   MASTER
   SPC=10
   MPC=10
   SUPER=10, 1
   LOAD=101
   LABEL=SUPER 10
   ESE=ALL
SUBCASE 102
   LOAD=102
   SUPER=10, 2
SUBCASE 201
   MASTER
   SPC=20
   MPC=20
   SUPER=20, 1
   LOAD=201
   LABEL=SUPER 20
```


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```
SUBCASE 202
   LOAD=202
   SUPER=20, 2
```

3. MASTER may also be used to advantage with multiple boundary condition Case Control setups. Suppose that constraint sets 10 and 20 are to be solved with three loading conditions each: 1, 2, and 3 and 4, 5, and 6, respectively. The following Case Control Section may be used

```
TITLE=MULTIPLE BOUNDARY CONDITIONS
DISP=ALL
SYM 1
MASTER
   SPC=10
   LOAD=1
SYM 2
   LOAD=2
SYM 3
   LOAD=3
SYM 4
   MASTER
   SPC=20
   LOAD=4
SYM 5
   LOAD=5
SYM 6
   LOAD=6
SYMCOM 10
   SYMSEQ=1., 1., 1., -1., -1., -1.
SYMCOM 20
   SYMSEQ=-1., -1., -1., 1., 1., 1.
```

4. The MASTER command must appear immediately after a SUBCASE or SYM command.

Related Topics

Browse Simcenter Nastran documentation

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MAXLINES

Maximum Number of Output Lines

Sets the maximum number of output lines.

Format:

MAXLINES=n

Examples:

```
MAXLINES=150000
```

Describers:

Describer	Meaning
n	Maximum number of output lines allowed. (Integer>0; Default=999999999)

Remarks:

1. Any time MAXLINES is exceeded, the program will terminate.
2. MAXLINES does not override any system parameters such as those on Job Control Language commands.
3. MAXLINES may also be specified on the NASTRAN statement with SYSTEM(14). See the **"nastran Command and NASTRAN Statement"**..
4. The code actually counts the number of pages and assumes that the number of lines output is the number of lines allowed per page, specified by the "LINES" command, times the number of pages.

Related Topics

Browse Simcenter Nastran documentation

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MAXMIN

MAXMIN Survey Output Request

Specifies options for max/min surveys of certain output data associated with grid points. (SOLs 101, 103, 105, 109, 114, 115, 129, 144, 145, 146, 187, and 200)

Format:

$$\text{MAXMIN}\left(\left[\left\{\begin{matrix}\text{MAX}\\ \underline{\text{BOTH}}\\ \text{MIN}\\ \text{VMAG}\end{matrix}\right\}[=\text{num}]\right],\ \left[\text{CID}=\left\{\begin{matrix}\text{GLOBAL}\\ \underline{\text{BASIC}}\\ \text{cid}\end{matrix}\right\}\right],\right.$$
$$\left.\text{oplist},\ [\text{COMP}=\text{list}]\right)=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
MAXMIN (BOTH=10,CID=1000,DISP,COMP=T1/T3)=501
```

Describers:

Describer	Meaning
MAX	Specifies only maximum values are to be output. See Remark 1 .
MIN	Specifies only minimum values are to be output. See Remark 1 .
BOTH	Specifies both maximum and minimum values are to be output. See Remark 1 .
VMAG	Specifies vector magnitude resultants are to be output. See Remark 2 .
num	The maximum number of values that will be output. See Remark 3 . (Integer>0 , default=5)
CID	Specifies the coordinate system frame of reference in which the max/min values will be output. See Remarks 1 and 3.
GLOBAL	Requests output in the global coordinate system frame of reference.

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Describer	Meaning
BASIC	Requests output in the basic coordinate system frame of reference.
cid	Requests output in the local coordinate system defined by cid. (Integer>0)
oplist	Specifies a list of one or more standard Case Control output requests for which max/min results are to be produced. The list may include any of DISP, SPCDF, OLOAD, MPCF, VELO, ACCE or ALL. See Remark 6 . (Character, no default)
list	Specifies a list of grid point degree of freedom (DOF) component directions that will be included in the max/min survey output. The components are separated by slashes and are selected from T1, T2, T3, R1, R2, R3. See Remarks 4 and 5. (Character, default=/T1 /T2 /T3 /R1 /R2 /R3)
ALL	MAXMIN survey results for all points will be output.
NONE	MAXMIN survey results for no points will be output.
n	Set identification of a previously appearing SET command. The max/min results survey will be output only for the points specified in SET n. (Integer>0)

Remarks:

1. The MAXMIN command produces an algebraically ascending sorted list of the output quantities specified for all of the points in the selected set. MAX refers to the largest magnitude positive values, while MIN refers to the largest magnitude negative values. The output format is similar to that of displacement output. All components will be output for a grid point and the order of the grid points will be in sort on the particular component that was surveyed. The output title contains the identification number of the SET of points participating in the max/min output, the coordinate system frame of reference, the number of MAX and MIN values output and the component that was surveyed. When the output being surveyed is in the global output coordinate system reference frame and BASIC or a local output coordinate system is specified as cid, both the sorted system output and the original reference system output are displayed for the grid point if these systems are different.
2. Vector magnitude results are computed for both translations and rotations and displayed under the T1 and R1 column headings. The presence of the COMP keyword is ignored.
3. The default value of 5 generates a minimum of 10 output lines for the BOTH option. There will be 5 maximum values and 5 minimum values produced. In addition, if coordinate system are involved, both surveyed and original data will be output. This could result in as many as 10 more lines of output for each surveyed component.
4. Multiple MAXMIN commands may be specified for a subcase. This permits different output quantities to have different MAXMIN specification within a subcase. For example,

```
SET 501=1,3,5,7 THRU 99, 1001,2001
MAXMIN (DISP, COMP=T3)=501
MAXMIN (SPCF, COMP=T1/R3)=ALL
```

specifies different components and output sets for displacements and forces of single point constraint. When multiple component searches are specified using the COMP keyword, separate

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output tables are produced that contain the results of the max/min survey for each of the components in the list. For example, COMP=T1/R3 produces two surveys , one with the T1 component surveyed and the other with the R3 component surveyed.

5. Scalar point output is included only if component T1 is included in the list.

6. MAXMIN output will only be generated for items in the oplist when there is an associated case control command present. For example, a DISP Case Control command must be present in order for the MAXMIN (DISP)=ALL command to produce output. Use of the ALL keywords for the oplist requests MAXMIN output for all of the output commands acceptable to MAXMIN that are present in case control.

Related Topics

Browse Simcenter Nastran documentation

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MBDEXPORT

Multibody Dynamics Export

Creates interface file for multibody dynamics and control system software.

Format:

$$\text{MBDEXPORT}\ \begin{bmatrix}\underline{\text{SCMOTION}}\\ \text{RECURDYN}\\ \text{ADAMS}\\ \text{SIMPACK}\\ \text{OP4} = \text{unit}\\ \text{MATLAB}\end{bmatrix},\ \left[\begin{matrix}\underline{\text{STANDARD}}\\ \text{STATESPACE}\end{matrix}\right],\ \left[\text{FLEXBODY} = \left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],$$

$$\left[\text{FLEXONLY} = \left\{\begin{matrix}\underline{\text{YES}}\\ \text{NO}\end{matrix}\right\}\right],\ \left[\text{MINVAR} = \left\{\begin{matrix}\underline{\text{PARTIAL}}\\ \text{CONSTANT}\\ \text{FULL}\\ \text{NONE}\end{matrix}\right\}\right],$$

$$\left[\text{PSETID} = \left\{\begin{matrix}\underline{\text{NONE}}\\ \text{setid}\\ \text{ALL}\\ \text{sktunit}\end{matrix}\right\}\right],\ \left[\text{OUTGSTRS} = \left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],\ \left[\text{OUTGSTRN} = \left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],$$

$$\left[\text{RECVROP2} = \left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],\ \left[\text{CHECK} = \left\{\begin{matrix}\underline{\text{NO}}\\ \text{YES}\end{matrix}\right\}\right],\ \left[\text{NONCUP} = \left\{\begin{matrix}\underline{\text{-1}}\\ 2\end{matrix}\right\}\right]$$

Examples:

```
MBDEXPORT SCMOTION FLEXBODY=YES FLEXONLY=NO
MBDEXPORT ADAMS STANDARD FLEXBODY=YES FLEXONLY=NO MINVAR=FULL
MBDEXPORT OP4=22 STANDARD FLEXBODY=YES
MBDEXPORT OP4=22 STATESPACE FLEXBODY=YES
MBDEXPORT MATLAB STANDARD FLEXBODY=YES
MBDEXPORT MATLAB STATESPACE FLEXBODY=YES
```


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Interface File Request Describers:

Describer		Interface file
SCMOTION		Output Simcenter 3D Motion solver OP2 file. (Default)
RECURDYN		Output RecurDyn Flex Input (RFI) file.
ADAMS		Output ADAMS Interface Modal Neutral File (MNF).
SIMPACK		Output SIMPACK Flexible Body Input (FBI) file.
OP4		Output OP4 file.
	unit	The OP4 file is written to the specified logical unit number. (Integer ≠ 0) If unit > 0, matrices are written to the OP4 file in sparse format. If unit < 0, matrices are written to the OP4 file in full matrix format. The absolute value of the logical unit number must match the unit number on an ASSIGN statement.
MATLAB		Output MATLAB script file.

Other Describers:

Describer		Meaning
STANDARD		Matrices are based on the second-order representation of the equation of motion. (Default)
STATESPACE		Matrices are based on first-order representation of the equation of motion. Valid only when the OP4 or MATLAB interface file request describer is specified.
FLEXBODY		Controls whether the interface file is output. Specification of the FLEXBODY describer is required.
	NO	Simcenter Nastran solution without output of the interface file. (Default)
	YES	Simcenter Nastran solution with output of the interface file.
FLEXONLY		Controls whether the DMAP solution and data recovery occurs after the interface file is output.
	YES	Interface file output only. (Default)
	NO	Interface file output and DMAP solution and data recovery.
MINVAR		Controls how mass invariants are computed.

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Describer		Meaning
		Valid only when the RECURDYN or ADAMS interface file request describers are specified.
	PARTIAL	Mass invariants 6 and 8 are not computed when the RECURDYN interface file request describer is specified. (Default) Mass invariants 5 and 9 are not computed when the ADAMS interface file request describer is specified. (Default)
	CONSTANT	Mass invariants 1, 2, 3, and 8 are computed when the RECURDYN interface file request describer is specified. Mass invariants 1, 2, 6, and 7 are computed when the ADAMS interface file request describer is specified.
	FULL	All mass invariants are computed.
	NONE	No mass invariants are computed.
PSETID		When the RECURDYN or SIMPACK interface file request describer is specified, selects a set of elements that is exported to the interface file. The set of elements is defined in the OUTPUT(PLOT) section (including PLOTEL) of the Simcenter Nastran input file. When the ADAMS interface file request describer is specified, selects a set of elements whose connectivity is exported to face geometry. The set of elements is defined in the OUTPUT(PLOT) section (including PLOTEL) of the Simcenter Nastran input file or on an ADAMS sketch file. Not valid when the SCMOTION, OP4, or MATLAB interface file request describers are specified.
	NONE	Export all grids, geometry and associated modal data. (Default)
	setid	Export face geometry for elements included in the SET case control command with the same identification number as setid.
	ALL	Export face geometry for all elements included in SET case control commands.
	sktunit	Logical unit number for the ADAMS sketch file. Specify sktunit < 0 to distinguish it from the setid. Valid only when the ADAMS interface file request describers is specified.
OUTGSTRS		Controls whether grid point stress is written to interface file. Valid only when the RECURDYN or ADAMS interface file request describers are specified.
	NO	Do not write grid point stress to interface file. (Default)

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Describer		Meaning
	YES	Write grid point stress to interface file.
OUTGSTRN		Controls whether grid point strain is written to interface file. Valid only when the RECURDYN or ADAMS interface file request describers are specified.
	NO	Do not write grid point strain to interface file. (Default)
	YES	Write grid point strain to interface file.
RECVROP2		Controls whether the FLEXBODY run outputs a Simcenter Nastran OP2 file for use in post processing of results.
	NO	OP2 file is not output. (Default)
	YES	OP2 file is output.
CHECK		Controls whether debug output is written to the f06 file when RECVROP2=YES.
	NO	Debug output is not written. (Default)
	YES	Debug output is written.
NONCUP		Controls modal damping output. Valid only when the SCMOTION, ADAMS, OP4 or MATLAB interface file request describers are specified.
	-1	Output the full equivalent modal viscous damping matrix. (Default)
	-2	Output the diagonal values of the equivalent modal viscous damping matrix only.

General Remarks:

1. Only one choice of SCMOTION, RECURDYN, ADAMS, SIMPACK, OP4, or MATLAB is allowed and must immediately follow the MBDEXPORT command.
2. The describers can be truncated to the first 4 characters.
3. MBDEXPORT must appear above the subcase level.
4. Simcenter Nastran is a unitless code. RecurDyn, ADAMS, and SIMPACK are not. Thus, for a RecurDyn, ADAMS, or SIMPACK FLEXBODY=YES run, you must include a DTI,UNITS bulk entry in the Simcenter Nastran input file. The DTI,UNITS bulk entry specifies the units that are associated with the numerical data in the Simcenter Nastran input file.

   For RecurDyn and ADAMS FLEXBODY=YES runs, Simcenter Nastran writes the units contained on the DTI,UNITS bulk entry directly to the interface file. When RecurDyn or ADAMS read the interface file, they associate the units with the numerical data contained in the interface file.

   For SIMPACK FLEXBODY=YES runs, Simcenter Nastran uses the units contained on the DTI,UNITS bulk entry to convert the numerical data that it writes to the interface file to SI (kg-m-sec) units. When

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SIMPACK reads the interface file, it recognizes that the numerical data contained in the interface file is already in SI (kg-m-sec) units.

The format of the DTI,UNITS bulk entry is as follows:

```
DTI     UNITS   1       MASS    FORCE   LENGTH  TIME
```

All entries are required. Acceptable character strings are listed below.

Mass:

KG - kilogram

LBM – pound-mass (0.45359237 kg)

SLUG – slug (14.5939029372 kg)

GRAM – gram (1E-3 kg)

OZM – ounce-mass (0.02834952 kg)

KLBM – kilo pound-mass (1000 lbm) (453.59237 kg)

MGG – megagram (1E3 kg)

MG – milligram (1E-6 kg)

MCG – microgram (1E-9 kg)

NG – nanogram (1E-12 kg)

UTON – U.S. ton (907.18474 kg)

SLI – slinch (175.1268352 kg)

Force:

N – Newton

LBF – pound-force (4.44822161526 N)

KGF – kilograms-force (9.80665 N)

OZF – ounce-force (0.2780139 N)

DYNE – dyne (1E-5 N)

KN – kilonewton (1E3 N)

KLBF – kilo pound-force (1000 lbf) (4448.22161526 N)

MN – millinewton (1E-3 N)

MCN – micronewton (1E-6 N)

NN – nanonewton (1E-9 N)

CN – centinewton (1E–2 N) (Not valid for ADAMS)

P – poundal (0.138254954 N) (Not valid for ADAMS)

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Length:

M – meter

KM – kilometer (1E3 m)

CM – centimeter (1E-2 m)

MM – millimeter (1E-3 m)

MI – mile (1609.344 m)

FT – foot (0.3048 m)

IN – inch (25.4E-3 m)

MCM – micrometer (1E-6 m)

NM – nanometer (1E-9 m)

A – Angstrom (1E-10 m)

YD – yard (0.9144 m)

ML – mil (25.4E-6 m)

MCI – microinch (25.4E-9 m)

Time:

S – second

H – hour (3600.0 sec)

MIN-minute (60.0 sec)

MS – millisecond (1E-3 sec)

MCS – microsecond (1E-6 sec)

NS – nanosecond (1E-9 sec)

D – day (86.4E3 sec)

Do not include a DTI,UNITS bulk entry in the Simcenter Nastran input file when the SCMOTION, OP4, or MATLAB interface file request describer is specified. If you do include a DTI,UNITS bulk entry, Simcenter Nastran ignores it.

5. Because in a RecurDyn, ADAMS, or SIMPACK FLEXBODY=YES run the DTI,UNITS bulk entry defines the units, the scaling defined in WTMASS, which is important for units consistency in Simcenter Nastran, is ignored in the output to the interface file. For example, if the model mass is in kilograms, force in Newtons, length in meters, and time in seconds, then WTMASS would equal 1. To ensure that Simcenter Nastran works with the consistent set of kg, N, and m, the DTI,UNITS bulk entry is:

```
DTI     UNITS   1       KG      N       M       S
```

The remarks from this point on are specific to each interface type.

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Simcenter 3D Motion Remarks:

1. The generation of standard matrices and the writing of them to a Simcenter 3D Motion OP2 file via OUTPUT2, which is only applicable in a non-restart, residual-only SOL 103 analysis, is initiated by MBDEXPORT SCMOTION FLEXBODY=YES. A specific ASSIGN statement is not required; the data will be written to the standard OP2 file via PARAM,POST,-2. This implies that the user must include PARAM,POST,-2 in the bulk data.
2. The parameter LMODES can be used to control which modes are used to derive the standard matrices.
3. The mode shape output will be reduced to the DOF defined in the DISPLACEMENT case control output request.

RecurDyn Remarks:

1. The creation of the RecurDyn Flex Input file is applicable in a non-restart SOL 103, 111, or 112 analysis only. RFI files are named 'jid_seid.rfi', where seid is the integer number of the superelement (0 for residual). These files are located in the same directory as the jid.f06 file.
2. You can create flexible body attachment points by defining the component as a superelement or part superelement, in which case the physical external (a-set) grids become the attachment points; or for a residual-only type model, you can use Simcenter Nastran ASET bulk entries to define the attachment points.
3. The eight mass variants are:

$$ {}^{1}I_{1x1} = \sum_{p=1}^{N} m_p $$

$$ {}^{2}I_{3x1} = \sum_{p=1}^{N} m_p s_p $$

$$ {}^{3}I_{3x3} = \sum_{p=1}^{N} (m_p \tilde{s}_p \tilde{s}_p - I_p) $$

$$ {}^{4}I_{3xM_j} = \sum_{p=1}^{N} m_p \Phi_p \quad j = 1,\ldots,M $$

$$ {}^{5}I_{3x3_j} = \sum_{p=1}^{N} m_p \tilde{s}_p \tilde{\phi}_{pj} \quad j = 1,\ldots,M $$

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$$[6]\ \underset{3x3}{I_{jk}} = \sum_{p=1}^{N} m_p \tilde{\phi}_{pj} \phi_{pk} \quad j,k = 1,\ldots,M$$

$$[7]\ \underset{3xM}{I} = \sum_{p=1}^{N} m_p \tilde{s}_p \Phi_p + \sum_{p=1}^{N} I_p \Phi_p^*$$

$$[8]\ \underset{3xM}{I_j} = \sum_{p=1}^{N} m_p \tilde{\phi}_{pj} \Phi_p \quad j = 1,\ldots,M$$

$s_p = [xyz]^T$ are the coordinates of grid point p in the basic coordinate system.

$\tilde{s}_p = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}$ = skew symmetric vector cross product operator.

φ_p = partitioned orthogonal modal matrix that corresponds to the translational degrees of freedom of grid *p*.

I_p = inertia tensor *p*.

φ_p^* = partitioned orthogonal modal matrix that corresponds to the rotational degrees of freedom of grid *p*.

$\tilde{\phi}_{pf}$ = skew-symmetric matrix formed for each grid translational degree of freedom for each mode.

M = number of modes.

N = number of grids.

4. To accurately capture the mode shapes when supplying SPOINT/QSET combinations, the number of SPOINTS (ns) should be at least ns=n+(6+p), assuming that residual flexibility is on. In the above equation for ns, the number of modes (n) is specified on the EIGR (METHOD=LAN) or EIGRL bulk entries; the number of load cases is p. In general, you cannot have too many SPOINTs, as excess ones will be truncated with no performance penalty.
5. For FLEXBODY=YES runs, residual vectors for the component should always be calculated as they result in a more accurate representation of the component shapes with little additional computational effort.
6. OMIT or OMIT1 bulk entries are not supported.
7. Lumped mass formulation (default) is required. Either leave PARAM,COUPMASS out of the input file or supply PARAM,COUPMASS,-1 (default) to ensure lumped mass.
8. CBEND elements are not allowed because they always use a coupled mass formulation. Likewise, the MFLUID fluid structure interface is not allowed because the virtual mass matrix it generates is not diagonal.

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9. PARAM,WTMASS,value with a value other than 1.0 may be used with a Simcenter Nastran run generating an RFI. It must have consistent units with regard to the DTI,UNITS bulk entry. Before generating the RFI, Simcenter Nastran will appropriately scale the WTMASS from the physical mass matrix and mode shapes.

10. There is a distinction between how an MBDEXPORT RECURDYN FLEXBODY=YES run handles element-specific loads (such as a PLOAD4 entry) versus those that are grid-specific (such as a FORCE entry), especially when superelements are used. The superelement sees the total element-specific applied load. For grid-specific loads, the loads attached to an external grid will move downstream with the grid. That is to say, it is part of the boundary and not part of the superelement. This distinction applies to a superelement run and not to a residual-only or parts superelement run.

11. The loads specified in Simcenter Nastran generally fall into two categories: non-follower or fixed direction loads (non-circulatory) and follower loads (circulatory). The follower loads are nonconservative in nature. Examples of fixed direction loads are the FORCE entry or a PLOAD4 entry when its direction is specified via direction cosines. Examples of follower loads are the FORCE1 entry or the PLOAD4 entry when used to apply a normal pressure. By default in Simcenter Nastran, the follower loads are always active in SOL 103 and will result in follower stiffness being added to the differential stiffness and elastic stiffness of the structure. In a run with MBDEXPORT RECURDYN FLEXBODY=YES and superelements, if the follower force is associated with a grid description (such as a FORCE1) and the grid is external to the superelement, the follower load will move downstream with the grid. Thus, the downstream follower contribution to the component's stiffness will be lost, which could yield poor results. This caution only applies to a superelement run and not to a residual-only or a part superelement run.

12. OUTGSTRS and OUTGSTRN entries require the use of standard Simcenter Nastran STRESS= or STRAIN= used in conjunction with GPSTRESS= or GPSTRAIN= commands to produce grid point stress or strain. GPSTRESS(PLOT)= or GPSTRAIN(PLOT)= will suppress grid stress or strain print to the Simcenter Nastran .f06 file.

13. To reduce the FE mesh detail for dynamic simulations, PSETID can include the ID of a SET entry. The SET entry lists PLOTEL or element IDs, whose connectivity is exported into the RFI to display the components in RecurDyn. This option can significantly reduce the size of the RFI without compromising accuracy in the FunctionBay simulation providing that the mass invariant computation is requested. With superelement analysis, for any of these elements that lie entirely on the superelement boundary (all of the elements' grids are attached only to a-set or exterior grids), a SEELT bulk entry must be specified to keep that display element with the superelement component. This can also be accomplished using PARAM, AUTOSEEL,YES. The SEELT entry is not required with parts superelements, as boundary elements stay with their component.

    If the SET entry points to an existing set from the OUTPUT(PLOT) section, this single set is used explicitly to define elements that are used to select grids to display the component in RecurDyn. If PSETID does not find the set ID in OUTPUT(PLOT), it will search sets in the case control for a matching set ID. This matching set ID then represents a list of OUTPUT(PLOT) defined elements' sets. The union of which will be used to define a set of PLOTELs or other elements used to select grids to display the component in RecurDyn. If you wish to select all of the sets in the OUTPUT(PLOT) section, then use PSETID=ALL.

    The following element types are not supported for writing to an RFI, nor are they supported as a 'type' entry in a set definition in OUTPUT(PLOT): CAABSF, CAEROi, CDUMi, CHBDYx, CDAMP3, CDAMP4, CELAS3, CELAS4, CMASS3, CMASS4, CRAC2D, CRAC3D, CTWIST, CWEDGE, CWELD, and GENEL.

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14. Typical Simcenter Nastran data entry requirements are described below.

Typical Parameters:

- PARAM,RESVEC,character_value – controls calculation of residual vector modes.
- PARAM,GRDPNT,value - mass invariants ^{1}I, ^{2}I, and ^{3}I will be computed using results of Simcenter Nastran grid point weight generator execution in the basic coordinate system.

Typical Case Control:

- MBDEXPORT RECURDYN FLEXBODY=YES is required for RFI generation.
- METHOD=n is required before or in the first subcase for modal solutions.
- SUPER=n,SEALL=n is useful with multiple superelement models to select an individual superelement as a flexible body. Cannot be used with a linear STATSUB(PRELOAD) run.
- OUTPUT(PLOT) is necessary to define elements used to select grids to display the component in RecurDyn when PSETID=ALL or setid.

  SET n=list of elements (including PLOTELs) is used to select grids to display the component.
- OUTPUT(POST) is necessary to define volume and surface for grid stress or strain shapes.

  SET n=list is a list of elements for surface definition for grid stress or strain shapes.

  Stress and strain data in the RFI is limited to the six components (that is, 3 normal and 3 shear) for a grid point for a given mode.

  SURFACE n SET n NORMAL z3 is used to define a surface for writing stress and strain data. Only one FIBER selection is allowed for each SURFACE, thus the use of the FIBER ALL keyword on the SURFACE case control command will write stresses to the RFI at the Z1 fiber location only.

  Because the FIBER keyword only applies to stresses, strain data will always be written to the RFI at the MID location.

  Stress and strain data at grid points can only be written to the RFI for surface and volume type elements (for example, CQUAD and CHEXA).

  VOLUME n SET n is a volume definition.

  The default SYSTEM BASIC is required with SURFACE or VOLUME.
- STRESS(PLOT) is necessary for stress shapes.
- STRAIN(PLOT) is necessary for strain shapes.
- GPSTRESS(PLOT) is necessary for grid point stress shapes to be included in the RFI.
- GPSTRAIN(PLOT) is necessary for grid point strain shapes to be included in the RFI.

  Typical Bulk Data:
- DTI,UNITS,1,MASS,FORCE,LENGTH,TIME is required for RFI generation. For input files containing superelements, this command must reside in the main bulk data section.
- SPOINT,id_list defines and displays modal amplitude.
- SESET,SEID,grid_list defines a superelement (see GRID and BEGIN BULK SUPER=). The exterior grids will represent the attachment points along with the q-set.

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- SEELT,SEID,element_list reassigns superelement boundary elements to an upstream superelement.
- RELEASE,SEID,C,Gi is an optional entry that removes DOFs from an attachment grid for which no constraint mode is desired. For example, this allows the removal of rotational degrees of freedom from an analysis where only translational degrees of freedom are required.
- SEQSET,SEID,spoint_list defines modal amplitudes of a superelement (see SEQSET1).
- SENQSET,SEID,N defines modal amplitudes of a part superelement. It must reside in the main bulk data section.
- ASET,IDi,Ci defines attachment points for a residual-only run (see ASET1).
- QSET1,C,IDi defines modal amplitudes for the residual structure or modal amplitudes for a part superelement (see QSET).
- PLOTEL,EID,Gi can be used, along with existing model elements, to define elements used to select grids to display the components in RecurDyn.
- EIGR,SID,METHOD,... obtains real eigenvalue extraction (see EIGRL).

15. MBDEXPORT and ADAMSMNF case control entries cannot be used in the same analysis run. In other words, a RecurDyn RFI file or an Adams MNF file can be generated during a particular Simcenter Nastran execution, but not both files at the same time. Attempting to generate both files in the same analysis will cause an error to be issued and the execution to be terminated.

16. The RECVROP2=YES option is used when you would like results recovery (using the MBDRECVR case control entry) from an RecurDyn/Flex analysis. This option requires the following assignment command:

ASSIGN OUTPUT2='name.out' STATUS=UNKNOWN UNIT=20 FORM=UNFORM

be inserted into the file management section of the Simcenter Nastran input file. It will cause an OP2 file with a .out extension to be generated, which then can be used as input into a Simcenter Nastran SOL 103 run using the MBDRECVR case control capability to perform results recovery from an RecurDyn/Flex analysis. FLEXBODY=YES is required with its use.

The data blocks output are:

MGGEW - physical mass external sort with weight mass removed
MAAEW - modal mass
KAAE - modal stiffness
CMODEXT - component modes.

This capability is limited to no more than one superelement per Simcenter Nastran model. Residual-only analyses are supported.

If differential stiffness is included, the static portion of the results will not be included in the recovered results when using MBDRECVR.

17. Setting CHECK=YES (which is only available when RECVROP2=YES) is *not* recommended for large models due to the amount of data that will be written to the f06.

18. The MBDEXPORT data routines use the environment variable TMPDIR for temporary storage during the processing of mode shape data. As a result, TMPDIR must be defined when using MBDEXPORT. TMPDIR should equate to a directory string for temporary disk storage, preferably one with a large amount of free space.

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19. Preload conditions are not supported.

20. To request differential stiffness, include a static subcase that contains the stress-stiffening loads. In another subcase include STATSUB = n where n is the number of the static subcase.

ADAMS Remarks:

1. The creation of the Adams MNF, which is applicable in a non-restart SOL 103, 111, or 112 analysis only, is initiated by MBDEXPORT ADAMS FLEXBODY=YES (other describers are optional) and the inclusion of the bulk entry DTI,UNITS. MNF files are named 'jid_seid.mnf', where seid is the integer number of the superelement (0 for residual). The location of these files is the same directory as the jid.f06 file.

2. You can create flexible body attachment points by defining the component as a superelement or part superelement, in which case the physical external (a-set) grids become the attachment points. For a residual-only type model, you can use standard Simcenter Nastran ASET bulk entries to define the attachment points.

3. The nine mass variants are:

$$ {}^{1}I_{1\,x\,1} = \sum_{p=1}^{N} m_p $$

$$ {}^{2}I_{3\,x\,1} = \sum_{p=1}^{N} m_p s_p $$

$$ {}^{3}I_{j\,3\,x\,M} = \sum_{p=1}^{N} m_p \Phi_p \quad j = 1, \ldots, M $$

$$ {}^{4}I_{3\,x\,M} = \sum_{p=1}^{N} m_p \tilde{s}_p \bar{\Phi}_p + I_p \Phi_p^* $$

$$ {}^{5}I_{j\,3\,x\,M} = \sum_{p=1}^{N} m_p \tilde{\phi}_{pj} \Phi_p \quad j = 1, \ldots, M $$

$$ {}^{6}I_{M\,x\,M} = \sum_{p=1}^{N} m_p \Phi_p^T \Phi_p + \Phi_p^{*T} I_p \Phi_p^* $$

$$ {}^{7}I_{3\,x\,3} = \sum_{p=1}^{N} m_p \tilde{s}_p^T \tilde{s}_p + I_p $$

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$$ {}^{8}I_{j}_{3\,x\,3} = \sum_{p=1}^{N} m_p s_p \tilde{\phi}_{pj} \quad j = 1, \ldots, M $$

$$ {}^{9}I_{jk}_{3\,x\,3} = \sum_{p=1}^{N} m_p \tilde{\phi}_{pj} \phi_{pk} \quad j, k = 1, \ldots, M $$

$s_p = [xyz]^T$ are the coordinates of grid point p in the basic coordinate system.

$$ s_p = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix} $$ = skew symmetric vector cross product operator.

φ_p = partitioned orthogonal modal matrix that corresponds to the translational degrees of freedom of grid *p*.

I_p = inertia tensor *p*.

φ_p^* = partitioned orthogonal modal matrix that corresponds to the rotational degrees of freedom of grid *p*.

$\tilde{\phi}_{pf}$ = skew-symmetric matrix formed for each grid translational degree of freedom for each mode.

M = number of modes.

N = number of grids.

4. To accurately capture the mode shapes when supplying SPOINT/QSET combinations, the number of SPOINTS (ns) should be at least ns=n+(6+p), assuming that residual flexibility is on. In the above equation for ns, the number of modes (n) is specified on the EIGR (METHOD=LAN) or EIGRL bulk entries; the number of load cases is p. In general, you cannot have too many SPOINTs, as excess ones are truncated with no performance penalty.
5. For FLEXBODY=YES runs, residual vectors for the component should always be calculated as they result in a more accurate representation of the component shapes at little additional cost.
6. OMIT or OMIT1 bulk entries are not supported.
7. Lumped mass formulation (default) is required. Either leave PARAM,COUPMASS out of the input file or supply PARAM,COUPMASS,-1 (default) to ensure lumped mass.
8. CBEND elements are not allowed because they always use a coupled mass formulation. Likewise, the MFLUID fluid structure interface is not allowed because the virtual mass matrix it generates is not diagonal.
9. PARAM,WTMASS,value with a value other than 1.0 may be used with a Simcenter Nastran run generating an MNF. It must have consistent units with regard to the DTI,UNITS bulk entry. Before generating the MNF, Simcenter Nastran will appropriately scale the WTMASS from the physical mass matrix and mode shapes.
10. There is a distinction between how an MBDEXPORT ADAMS FLEXBODY=YES run handles element-specific loads (such as a PLOAD4 entry) versus those that are grid-specific (such as a FORCE entry), especially when superelements are used. The superelement sees the total element-specific applied

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load. For grid-specific loads, the loads attached to an external grid will move downstream with the grid. That is to say, it is part of the boundary and not part of the superelement. This distinction applies to a superelement run and not to a residual-only or parts superelement run.

11. The loads specified in Simcenter Nastran generally fall into two categories: non-follower or fixed direction loads (non-circulatory) and follower loads (circulatory). The follower loads are nonconservative in nature. Examples of fixed direction loads are the FORCE entry or a PLOAD4 entry when its direction is specified via direction cosines. Examples of follower loads are the FORCE1 entry or the PLOAD4 entry when used to apply a normal pressure. By default in Simcenter Nastran, the follower loads are always active in SOL 103 and will result in follower stiffness being added to the differential stiffness and elastic stiffness of the structure. In a run with MBDEXPORT ADAMS FLEXBODY=YES and superelements, if the follower force is associated with a grid description (such as a FORCE1) and the grid is external to the superelement, the follower load will move downstream with the grid. Thus, the downstream follower contribution to the component's stiffness will be lost, which could yield poor results. This caution only applies to a superelement run and not to a residual-only or a part superelement run.

12. OUTGSTRS and OUTGSTRN entries require the use of standard Simcenter Nastran STRESS= or STRAIN= used in conjunction with GPSTRESS= or GPSTRAIN= commands to produce grid point stress or strain. GPSTRESS(PLOT)= or GPSTRAIN(PLOT)= will suppress grid stress or strain print to the Simcenter Nastran .f06 file.

13. To reduce the FE mesh detail for dynamic simulations, PSETID (on the MBDEXPORT Case Control command) defined with a SET entry (i.e. setid) is used to define a set of PLOTELs or other elements used to select grids to display the components in Adams. This option can significantly reduce the size of the MNF without compromising accuracy in the Adams simulation providing that the mass invariant computation is requested. With superelement analysis, for any of these elements that lie entirely on the superelement boundary (all of the elements' grids attached only to a-set or exterior grids), a SEELT bulk entry must be specified to keep that display element with the superelement component. This can also be accomplished using PARAM, AUTOSEEL,YES. The SEELT entry is not required with parts superelements, as boundary elements stay with their component.

    If the SET entry points to an existing set from the OUTPUT(PLOT) section, this single set is used explicitly to define elements used to select grids to display the component in Adams. If PSETID does not find the set ID in OUTPUT(PLOT), it will search sets in the case control for a matching set ID. This matching set ID list then represents a list of OUTPUT(PLOT) defined elements' sets, the union of which will be used to define a set of PLOTELs or other elements used to select grids to display the component in Adams. If the user wishes to select all of the sets in the OUTPUT(PLOT) section, then use PSETID=ALL.

    The following element types are not supported for writing to an MNF, nor are they supported as a 'type' entry in a set definition in OUTPUT(PLOT): CAABSF, CAEROi, CDUMi, CHBDYx, CDAMP3, CDAMP4, CELAS3, CELAS4, CMASS3, CMASS4, CRAC2D, CRAC3D, CTWIST, CWEDGE, CWELD, and GENEL.

    PSETID can also point to a sketch file using PSETID= – sktunit, where sktunit references an ASSIGN statement of the form:

```
ASSIGN SKT='sketch_file.dat',UNIT=sktunit.
```


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The grids defined for the elements' faces in the sketch file, along with *all* external (i.e. boundary) grids for the superelements, will be the only grids (and their associated data) written to the MNF.

The format of the sketch file, which describes the mesh as a collection of faces, must be as follows:

```
face_count
face_1_node_count face_1_nodeid_1 face_1_nodeid_2 ...
face_2_node_count face_2_nodeid_1 face_2_nodeid_2 ...

<etc>
```

Faces must have a node count of at least two. For example, a mesh comprised of a single brick element might be described as follows:

```
6
4 1000 1001 1002 1003
4 1007 1006 1005 1004
4 1000 1004 1005 1001
4 1001 1005 1006 1002
4 1002 1006 1007 1003
4 1003 1007 1004 1000
```

Alternatively, the mesh might be described as a stick figure using a collection of lines (two node faces), as shown below:

```
8
2 101 102
2 102 103
2 103 104
2 104 105
2 105 106
2 106 107
2 107 108
2 108 109
```

14. Typical Simcenter Nastran data entry requirements are described below.

Typical Parameters:

- PARAM,RESVEC,character_value – controls calculation of residual vector modes.
- PARAM,GRDPNT, value - mass invariants ^{1}I, ^{2}I, and ^{7}I will be computed using results of Simcenter Nastran grid point weight generator execution in the basic coordinate system.

Typical Case Control:

- MBDEXPORT ADAMS FLEXBODY=YES is required for MNF generation.
- METHOD=n is required before or in the first subcase for modal solutions.

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- SUPER=n,SEALL=n is useful with multiple superelement models to select an individual superelement as a flexible body. Cannot be used with a linear STATSUB(PRELOAD) run.
- OUTPUT(PLOT) is necessary to define elements used to select grids to display the component in Adams when PSETID=ALL or setid.

  SET n=list of elements (including PLOTELs) is used to select grids to display the component.
- OUTPUT(POST) is necessary to define volume and surface for grid stress or strain shapes.

  SET n=list is a list of elements for surface definition for grid stress or strain shapes.

  Stress and strain data in the MNF is limited to the six components (i.e. 3 normal and 3 shear) for a grid point for a given mode.

  SURFACE n SET n NORMAL z3 is used to define a surface for writing stress and strain data. Only one FIBER selection is allowed for each SURFACE, thus the use of the FIBRE ALL keyword on the SURFACE case control command will write stresses to the MNF at the Z1 fiber location only.

  Because the FIBRE keyword only applies to stresses, strain data will always be written to the MNF at the MID location.

  Stress and strain data at grid points can only be written to the MNF for surface and volume type elements (e.g. CQUAD and CHEXA).

  VOLUME n SET n is a volume definition.

  The default SYSTEM BASIC is required with SURFACE or VOLUME.
- STRESS(PLOT) is necessary for stress shapes.
- STRAIN(PLOT) is necessary for strain shapes.
- GPSTRESS(PLOT) is necessary for grid point stress shapes to be included in the MNF.
- GPSTRAIN(PLOT) is necessary for grid point strain shapes to be included in the MNF.

  Typical Bulk Data:
- DTI,UNITS,1,MASS,FORCE,LENGTH,TIME is required for MNF generation. For input files containing superelements, this command must reside in the main bulk data section.
- SPOINT,id_list defines and displays modal amplitude.SESET,SEID,grid_list defines a superelement (see GRID and BEGIN BULK SUPER=). The exterior grids will represent the attachment points along with the q-set.
- SEELT,SEID,element_list reassigns superelement boundary elements to an upstream superelement.
- RELEASE,SEID,C,Gi is an optional entry that removes DOFs from an attachment grid for which no constraint mode is desired. For example, this allows the removal of rotational degrees of freedom from an analysis where only translational degrees of freedom are required.
- SEQSET,SEID,spoint_list defines modal amplitudes of a superelement (see SEQSET1).
- SENQSET,SEID,N defines modal amplitudes of a part superelement. It must reside in the main bulk data section.
- ASET,IDi,Ci defines attachment points for a residual-only run (see ASET1).

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- QSET1,C,IDi defines modal amplitudes for the residual structure or modal amplitudes for a part superelement (see QSET).
- PLOTEL,EID,Gi can be used, along with existing model elements, to define elements used to select grids to display the components in Adams.
- EIGR,SID,METHOD,... obtains real eigenvalue extraction (see EIGRL).

15. The RECVROP2=YES option is used when you would like results recovery (using the MBDRECVR case control entry) from an Adams/Flex analysis. This option requires the following assignment command:

ASSIGN OUTPUT2='name.out' STATUS=UNKNOWN UNIT=20 FORM=UNFORM

be inserted into the file management section of the Simcenter Nastran input file. It will cause an OP2 file with a .out extension to be generated, which then can be used as input into a Simcenter Nastran SOL 103 run using the MBDRECVR case control capability to perform results recovery from an Adams/Flex analysis. FLEXBODY=YES is required with its use.

The data blocks output are:

MGGEW - physical mass external sort with weight mass removed
MAAEW - modal mass
KAAE - modal stiffness
CMODEXT - component modes.

This capability is limited to no more than one superelement per Simcenter Nastran model. Residual-only analyses are supported.

If differential stiffness is included, the static portion of the results will not be included in the recovered results when using MBDRECVR.

16. Setting CHECK=YES (which is only available when RECVROP2=YES) is *not* recommended for models of realistic size due to the amount of data that will be written to the f06 file.

17. The MBDEXPORT data routines use the environment variable TMPDIR for temporary storage during the processing of mode shape data. As a result, TMPDIR must be defined when using MBDEXPORT. TMPDIR should equate to a directory string for temporary disk storage, preferably one with a large amount of free space.

18. If any damping is defined in the model, an equivalent modal viscous damping will be determined for each mode and written to the MNF. This equivalent modal viscous damping is defined as:

$D = \psi^T B_e \psi$

where D is the equivalent modal viscous damping matrix, ψ is the eigenvector matrix, and B_e is the equivalent viscous damping matrix.

The equivalent viscous damping matrix is given by:

$$B_e = B^1_{AA} + B^2_{AA} + \frac{G}{W3} K^1_{AA} + \frac{1}{W4} K^4_{AA}$$

where *G*, *W3*, and *W4* are structural damping-related parameters described in the “Parameter Descriptions” section of this guide.

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By default, the full equivalent modal viscous damping matrix is written to the MNF. To write only the diagonal values of the equivalent modal viscous damping matrix to the MNF, specify NONCUP=−2 or specify PARAM,NONCUP,-2.

If both the NONCUP describer and the NONCUP parameter are specified, the NONCUP describer specification takes precedence.

19. Preload conditions are not supported.

20. To request differential stiffness, include a static subcase that contains the stress-stiffening loads. In another subcase include STATSUB = n where n is the number of the static subcase.

21. Cylindrical and spherical nodal displacement coordinate systems (i.e. CD on the GRID bulk entry) are not supported.

SIMPACK Remarks:

1. The creation of a SIMPACK Flexible Body Input (FBI) file is applicable in a non-restart SOL 103, 111, and 112 analysis only. FBI files are named 'jid_seid.fbi', where seid is the integer number of the superelement (0 for residual). The location of these files is the same directory as the jid.f06 file.

2. The creation of the FBI file is initiated by MBDEXPORT SIMPACK FLEXBODY=YES (other describers are optional) and the inclusion of the bulk entry DTI,UNITS. This is only valid for a Component Mode Synthesis (CMS) analysis. Thus, it is necessary to define the modal coordinates using the SPOINT bulk entry, and to define them to be in the q-set using the QSET/QSET1 or SEQSET/SEQSET1 bulk entries as appropriate.

3. You can create flexible body attachment points by defining the component as a superelement or part superelement, in which case the physical external (a-set) grids become the attachment points; or for a residual-only type model, you can use Simcenter Nastran ASET bulk entries to define the attachment points. Note that the values corresponding to these attachment points in the CMS-reduced mass and stiffness matrices written to the FBI file will be defined in the nodal displacement coordinate systems of these attachment points. The user must account for these coordinate systems when loading or restraining these attachment points within the SIMPACK run.

4. To accurately capture the mode shapes when supplying SPOINT/QSET combinations, the number of SPOINTs (ns) should be at least ns=n+(6+p). In the above equation for ns, the number of modes (n) is specified on the EIGR (METHOD=LAN) or EIGRL bulk entries; the number of load cases is p. In general, you cannot have too many SPOINTs. Excess SPOINTs will be truncated with no performance penalty.

5. OMIT and OMIT1 bulk entries are not supported.

6. Lumped mass formulation (default) is required. Either leave PARAM,COUPMASS out of the input file or supply PARAM,COUPMASS,-1 (default) to ensure lumped mass formulation.

7. CBEND elements are not allowed because they always use a coupled mass formulation. Likewise, the MFLUID fluid structure interface is not allowed because the virtual mass matrix it generates is not diagonal.

8. PARAM,WTMASS,value with a value other than 1.0 may be used with a Simcenter Nastran run generating an FBI file. It must have consistent units with regard to the DTI,UNITS bulk entry. Before generating the FBI file, Simcenter Nastran will appropriately scale the WTMASS from the physical mass matrix and mode shapes.

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9. There is a distinction between how an MBDEXPORT SIMPACK FLEXBODY=YES run handles element-specific loads (such as a PLOAD4 entry) versus those that are grid-specific (such as a FORCE entry), especially when superelements are used. The superelement sees the total element-specific applied load. For grid-specific loads, the loads attached to an external grid will move downstream with the grid. That is to say, it is part of the boundary and not part of the superelement. This distinction applies to a superelement run and not to a residual-only or parts superelement run.

10. The loads specified in Simcenter Nastran generally fall into two categories: non-follower or fixed direction loads (non-circulatory) and follower loads (circulatory). The follower loads are nonconservative in nature. Examples of fixed direction loads are the FORCE entry or a PLOAD4 entry when its direction is specified via direction cosines. Examples of follower loads are the FORCE1 entry or the PLOAD4 entry when used to apply a normal pressure. By default in Simcenter Nastran, the follower loads are always active in SOL 103 and will result in follower stiffness being added to the differential stiffness and elastic stiffness of the structure. In a run with MBDEXPORT SIMPACK FLEXBODY=YES and superelements, if the follower force is associated with a grid description (such as a FORCE1) and the grid is external to the superelement, the follower load will move downstream with the grid. Thus, the downstream follower contribution to the component's stiffness will be lost, which could yield poor results. This caution only applies to a superelement run and not to a residual-only or a part superelement run.

11. To reduce the FE mesh detail for dynamic simulations, PSETID can include the ID of a SET entry. PSETID is also used to define the grids to be included in the recovery matrix that is written to the FBI file. The SET entry lists PLOTEL or element IDs, whose connectivity is exported into the FBI file to display the components in SIMPACK. This option can significantly reduce the size of the FBI file without compromising accuracy in the SIMPACK simulation. With superelement analysis, for any of these elements that lie entirely on the superelement boundary (all of the elements' grids are attached only to a-set or exterior grids), a SEELT bulk entry must be specified to keep that display element with the superelement component. This can also be accomplished using PARAM, AUTOSEEL,YES. The SEELT entry is not required with parts superelements, as boundary elements stay with their component.

    If the SET entry points to an existing set from the OUTPUT(PLOT) section, this single set is used explicitly to define elements that are used to select grids to display the component in SIMPACK. If PSETID does not find the set ID in OUTPUT(PLOT), it will search sets in the case control for a matching set ID. This matching set ID then represents a list of OUTPUT(PLOT) defined elements' sets, the union of which will be used to define a set of PLOTELs or other elements used to select grids to display the component in SIMPACK. If you wish to select all of the sets in the OUTPUT(PLOT) section, then use PSETID=ALL.

    The following element types are not supported for writing to an FBI file, nor are they supported as a 'type' entry in a set definition in OUTPUT(PLOT): CAABSF, CAEROi, CDUMi, CHBDYx, CDAMP3, CDAMP4, CELAS3, CELAS4, CMASS3, CMASS4, CPYRAM, CRAC2D, CRAC3D, CTWIST, CWEDGE, CWELD, and GENEL.

12. Typical Simcenter Nastran data entry requirements are described below.

    Typical Case Control:

    - MBDEXPORT SIMPACK FLEXBODY=YES is required for FBI file generation.
    - METHOD=n is required before or in the first subcase for modal solutions.

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- SUPER=n,SEALL=n is useful with multiple superelement models to select an individual superelement as a flexible body. Cannot be used with a linear STATSUB(PRELOAD) run.

  Typical Bulk Data:

- DTI,UNITS,1,MASS,FORCE,LENGTH,TIME is required for FBI file generation. For input files containing superelements, this command must reside in the main bulk data section.
- SPOINT,id_list defines and displays modal amplitude.
- SESET,SEID,grid_list defines a superelement (see GRID and BEGIN BULK SUPER=). The exterior grids will represent the attachment points along with the q-set.
- SEELT,SEID,element_list reassigns superelement boundary elements to an upstream superelement.
- RELEASE,SEID,C,Gi is an optional entry that removes DOFs from an attachment grid for which no constraint mode is desired. For example, this allows the removal of rotational degrees of freedom from an analysis where only translational degrees of freedom are required.
- SEQSET,SEID,spoint_list defines modal amplitudes of a superelement (see SEQSET1).
- SENQSET,SEID,N defines modal amplitudes of a part superelement. It must reside in the main bulk data section.
- ASET,IDi,Ci defines attachment points for a residual-only run (see ASET1).
- QSET1,C,IDi defines modal amplitudes for the residual structure or modal amplitudes for a part superelement (see QSET).
- PLOTEL,EID,Gi can be used, along with existing model elements, to define elements used to select grids to display the components in SIMPACK.
- EIGR,SID,METHOD,... obtains real eigenvalue extraction (see EIGRL).

13. MBDEXPORT and ADAMSMNF case control entries cannot be used in the same analysis run. In other words, a SIMPACK FBI file or an Adams MNF file can be generated during a particular Simcenter Nastran execution, but not both files at the same time. Attempting to generate both files in the same analysis will cause an error to be issued and the solve to be terminated.

14. The RECVROP2=YES option is used when you would like results recovery (using the MBDRECVR case control entry) from a SIMPACK analysis. This option requires the following assignment command:

ASSIGN OUTPUT2='name.out' STATUS=UNKNOWN UNIT=20 FORM=UNFORM

be inserted into the file management section of the Simcenter Nastran input file. It will cause an OP2 file with a .out extension to be generated, which then can be used as input into a Simcenter Nastran SOL 103 run using the MBDRECVR case control capability to perform results recovery from a SIMPACK analysis. FLEXBODY=YES is required with its use.

The data blocks output are:

MGGEW - physical mass external sort with weight mass removed
MAAEW - modal mass
KAAE - modal stiffness
CMODEXT - component modes.

This capability is limited to no more than one superelement per Simcenter Nastran model. Residual-only analyses are supported.

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If differential stiffness is included, the static portion of the results will not be included in the recovered results when using MBDRECVR.

15. Setting CHECK=YES (which is only available when RECVROP2=YES) is *not* recommended for models of realistic size due to the amount of data that will be written to the f06 file.

16. The MBDEXPORT data routines use the environment variable TMPDIR for temporary storage during the processing of mode shape data. As a result, TMPDIR must be defined when using MBDEXPORT. TMPDIR should equate to a directory string for temporary disk storage, preferably one with a large amount of free space.

17. Preload conditions are not supported.

18. To request differential stiffness, include a static subcase that contains the stress-stiffening loads. In another subcase include STATSUB = n where n is the number of the static subcase.

OP4 Remarks:

1. The generation of standard or state-space matrices and the writing of them to an OP4 file via OUTPUT4, which is applicable in a non-restart SOL 103, 111, or 112 analysis only, is initiated by MBDEXPORT OP4=unit STANDARD FLEXBODY=YES, or MBDEXPORT OP4=unit STATESPACE FLEXBODY=YES (other describers are optional) and the inclusion of the ASSIGN file management statement. This ASSIGN statement must be of the form:

   ASSIGN OUTPUT4='filename',UNIT=n,etc.

   where 'n' matches the absolute value for unit on the MBDEXPORT OP4=unit case control command.

   The number of digits of precision for matrix data is controlled by the DIGITS parameter.

   For a model with superelements, only one OP4 file will be generated. This OP4 file will be generated for the first superelement (or the residual) that satisfies the conditions defined in Remarks 3 and 4. For standard matrices, if user-defined set U8 is not defined, the residual will be written to the OP4.

2. The parameters LFREQ/HFREQ or LMODES can be used to control which modes are used to derive the standard or state-space matrices.

3. For state-space matrices, user-defined set U7 is used for input DOF. User-defined set U8 is used for output DOF. Refer to the USET/USET1 bulk entries for partitioned superelements and refer to the SEUSET/SEUSET1 bulk entries for non-partitioned superelements.

4. For standard matrices, user-defined set U8 is used for output DOF. The mode shape output will be reduced to the DOF defined in DOF set U8. If DOF set U8 is not defined, the mode shape data for all DOF will be written. Refer to the USET/USET1 bulk entries for partitioned superelements and refer to the SEUSET/SEUSET1 bulk entries for non-partitioned superelements.

5. For the state-space option, the OP4 file contains the [A], [B], [C], and [E] state-space matrices. They are defined as AMAT, BMAT, CMAT, and EMAT, respectively. The input and output DOF are defined as U7DOF and U8DOF, respectively with the first column being the grid ID and the second column being the direction code (1 through 6).

6. For the standard option, the OP4 file contains the modal mass, equivalent modal viscous damping, modal stiffness, mode shapes, and modal forces defined as MMASS, MDAMP, MSTIF, U8PHIX, and MFORC, respectively. The physical DOF corresponding one-to-one with the rows of U8PHIX are

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defined as U8DOF. The first column contains the grid ID and the second column contains the direction code (1 through 6).

7. The RECVROP2=YES option is used when you would like results recovery (using the MBDRECVR case control entry) from a system analysis. This option requires the following assignment command:

   ASSIGN OUTPUT2='name.out' STATUS=UNKNOWN UNIT=20 FORM=UNFORM

   be inserted into the file management section of the Simcenter Nastran input file. It will cause an OP2 file with a .out extension to be generated, which can then be used as an input into a Simcenter Nastran SOL 103 run using the MBDRECVR case control command. FLEXBODY=YES is required when specifying RECVROP2=YES.

   The data blocks output are:

   MGGEW – physical mass external sort with weight mass removed

   MAAEW – modal mass

   KAAE – modal stiffness

   CMODEXT – component modes

   This capability is limited to one superelement per Simcenter Nastran model. Residual-only analyses are supported.

   If differential stiffness is included, the static portion of the results will not be included in the recovered results when using MBDRECVR.

8. Setting CHECK=YES (which is only available when RECVROP2=YES) is *not* recommended for models of realistic size due to the amount of data that will be written to the f06 file.

9. Differential stiffness is only supported for standard second-order system representation. To request differential stiffness, include a static subcase that contains the stress-stiffening loads. In another subcase include STATSUB = n where n is the number of the static subcase.

10. By default, the full equivalent modal viscous damping matrix is written to standard or state-space OP4 files. To write only the diagonal values of the equivalent modal viscous damping matrix to OP4 files, specify NONCUP=–2, or specify PARAM,NONCUP,-2.

    If both the NONCUP describer and the NONCUP parameter are specified, the NONCUP describer specification takes precedence.

MATLAB Remarks:

1. The generation of standard or state-space matrices and the writing of them to a MATLAB script file, which is applicable in a non-restart SOL 103, 111, or 112 analysis only, is initiated by MBDEXPORT MATLAB STANDARD FLEXBODY=YES, or MBDEXPORT MATLAB STATESPACE FLEXBODY=YES (other describers are optional). The MATLAB script files are named jid_seid.m where seid is the integer number of the superelement (0 for residual). The location of the MATLAB script files is the same directory as the jid.f06 file.

2. The parameters LFREQ/HFREQ or LMODES can be used to control which modes are used to derive the standard or state-space matrices.

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3. For state-space matrices, user-defined set U7 is used for input DOF. User-defined set U8 is used for output DOF. Refer to the USET/USET1 bulk entries for partitioned superelements and refer to the SEUSET/SEUSET1 bulk entries for non-partitioned superelements.

4. For standard matrices, user-defined set U8 is used for output DOF. The mode shape output will be reduced to the DOF defined in DOF set U8. If DOF set U8 is not defined, the mode shape data for all DOF will be written. Refer to the USET/USET1 bulk entries for partitioned superelements and refer to the SEUSET/SEUSET1 bulk entries for non-partitioned superelements.

5. For the state-space option, the MATLAB script file contains the [A], [B], [C], and [E] state-space matrices. They are defined as AMAT, BMAT, CMAT, and EMAT, respectively. The input and output DOF are defined as U7DOF and U8DOF, respectively with the first column being the grid ID and the second column being the direction code (1 through 6).

6. For the standard option, the MATLAB script file contains the modal mass, equivalent modal viscous damping, modal stiffness, mode shapes, and modal forces defined as MMASS, MDAMP, MSTIF, MSHAP, and MFORC, respectively. The physical DOF corresponding one-to-one with the rows of MSHAP are defined as U8DOF. The first column contains the grid ID and the second column contains the direction code (1 through 6).

7. The RECVROP2=YES option is used when you would like results recovery (using the MBDRECVR case control entry) from a system analysis. This option requires the following assignment command:

   ASSIGN OUTPUT2='name.out' STATUS=UNKNOWN UNIT=20 FORM=UNFORM

   be inserted into the file management section of the Simcenter Nastran input file. It will cause an OP2 file with a .out extension to be generated, which can then be used as an input into a Simcenter Nastran SOL 103 run using the MBDRECVR case control command. FLEXBODY=YES is required when specifying RECVROP2=YES.

   The data blocks output are:

   MGGEW – physical mass external sort with weight mass removed

   MAAEW – modal mass

   KAAE – modal stiffness

   CMODEXT – component modes

   This capability is limited to one superelement per Simcenter Nastran model. Residual-only analyses are supported.

   If differential stiffness is included, the static portion of the results will not be included in the recovered results when using MBDRECVR.

8. Setting CHECK=YES (which is only available when RECVROP2=YES) is *not* recommended for models of realistic size due to the amount of data that will be written to the f06 file.

9. To request differential stiffness, include a static subcase that contains the stress-stiffening loads. In another subcase include STATSUB = n where n is the number of the static subcase.

10. By default, the full equivalent modal viscous damping matrix is written to standard or state-space MATLAB script files. To write only the diagonal values of the equivalent modal viscous damping matrix to MATLAB script files, specify NONCUP=−2 or specify PARAM,NONCUP,-2.

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If both the NONCUP describer and the NONCUP parameter are specified, the NONCUP describer specification takes precedence.

Related Topics

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MBDRECVR

Multi-Body Dynamics Results Recovery

Imports required files from third-party multi-body dynamics codes to perform results recovery.

Format:

$$\text{MBDRECVR}\left[\left\{\frac{\text{BINARY}}{\text{ASCII}}\right\}\right],\ \left[\text{MSRMODE}=\left\{\begin{matrix}\underline{0}\\1\\2\end{matrix}\right\}\right],\ \left[\text{RGBODY}=\left\{\begin{matrix}\underline{\text{NO}}\\\text{YES}\\\text{BOTH}\end{matrix}\right\}\right],$$
$$\left[\text{MSGLVL}=\left\{\begin{matrix}\underline{0}\\1\text{ - }4\end{matrix}\right\}\right],\ \left[\text{CHECK}=\left\{\begin{matrix}\underline{\text{NO}}\\\text{YES}\end{matrix}\right\}\right]$$

Examples:

RecurDyn Example:

```
ASSIGN INPUTT2=‘rfi_results.mdf' UNIT=13
...
CEND
STRESS(PLOT)=100
MBDRECVR
```

Adams Example:

```
ASSIGN INPUTT2=`adams_results.mdf' UNIT=13
...
CEND
STRESS(PLOT)=100
MBDRECVR
```

Describers:

Describer	Meaning
BINARY	OUTPUT2 file format for Adams modal deformations file (default). (See Remarks 1 and 2)
ASCII	PUNCH file format for RecurDyn or Adams modal deformations file. (See Remarks 1 and 2).
MSRMODE	Specifies stress recovery type (see Remarks 6 and 7).

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Describer		Meaning
	0	Component definitions are stored in an OUTPUT2 file (specifically, an *.out file created by using RECVROP2=YES on the MBDEXPORT ADAMS or RECURDYN case control in a pre-Adams/Flex or RecurDyn, respectively, Simcenter Nastran run. The OUTPUT2 files used in this case do not contain data blocks used for MNF or RFI creation (default).
	1	Same as option 0, except that the OUTPUT2 file will contain 10 additional data blocks used for RFI or MNF creation by a RecurDyn or Adams pre-processor (specifically, a *.out file created through use of the mnfx.alt DMAP alter capability).
	2	No file reference (specifically, component definitions will be recomputed)
RGBODY		Requests the addition of rigid body motion with modal deformations (see Remark 5).
	NO	Do not include rigid body motion (default).
	YES	Include rigid body motion.
	BOTH	Generate two output data blocks; one containing rigid body motion and the other without.
MSGLVL		Level of diagnostic output from Lanczos eigensolver when component definitions are determined (applies only when MSRMODE=2).
	0	No output (default).
	1	Warning and fatal messages.
	2	Summary output.
	3	Detailed output on cost and convergence.
	4	Detailed output on orthogonalization.
CHECK		Requests debug output be written to the f06 file (See Remark 9).
	NO	No debug output will be written (default).
	YES	Debug output will be written.

Remarks:

1. The Simcenter Nastran results recovery requires displacement, velocity, and acceleration data from a multi-body dynamics software product stored in a modal deformation file (MDF). Simcenter Nastran interprets any missing data to be 0.0. The MDF format is described in “Multi-body Dynamics and Control System Software Interfaces” in the Advanced Dynamic Analysis User’s Guide.
2. When modal deformations to be read are in binary (OUTPUT2) format (specifically, BINARY), the following statement needs to be specified near the top of the Simcenter Nastran input file in the file management section:

   ASSIGN INPUTT2=`<MDFilename>' UNIT=13

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where <MDFilename> is the name of the modal deformations file from Adams or RecurDyn.

3. To input the modal deformations file from Adams or RecurDyn in ASCII (Punch) format (specifically, ASCII), the following statement needs to be included in the bulk data section:

   INCLUDE `<MDFilename>' where `<MDFilename>' is the name of the modal deformations file.

4. Dynamic stress/strain output can either be in .f06, PUNCH, and/or OUTPUT2 according to standard Simcenter Nastran functionality. However, stress recovery in Simcenter Nastran from Adams or RecurDyn results do not support XYPLOT output.

5. If displacements, stresses, and/or strains are to be available for post processing, one or more of the following statements must appear in the case control section of the Simcenter Nastran input file:

   DISP(PLOT) = <set id>

   STRAIN(FIBER,PLOT) = <set id>

   STRESS(PLOT) = <set id>

   Even though the recovery process is performed in SOL 103, the derived results are transient response results. The default output format for transient results is SORT2. If no SORT2 results are desired, then SORT1 must be requested for all output requests.

6. Rigid body motions from an Adams or RecurDyn simulation are included in the modal deformation file, but they are not applied unless the RGBODY keyword is set to YES or BOTH. Including rigid body motion affects the display and animation of the flexible component, but it has no effect on dynamic stresses. For RGBODY=YES, the displacements output data blocks OUGV1 or OUGV2 will contain a '0' in word 3 of each header record. For RGBODY=NO, OUGV1 or OUGV2 will contain a '1' in word 3 of each header record. For RGBODY=BOTH, two OUGV1 or OUGV2 data blocks will be generated; one for RGBODY=YES and one for RGBODY=NO.

7. For MSRMODE=0 or 1, stress recovery references the OUTPUT2 file obtained from the initial CMS analysis (specifically, RECVROP2=YES on the MBDEXPORT ADAMS or RECURDYN case control entry or use of the mnfx.alt (Adams only) DMAP alter capability). No other files are required. The geometric data needs to be included in the bulk data of the Simcenter Nastran input file because geometry is missing from the OUTPUT2 file. This mode of stress recovery is faster than the MSRMODE=2 mode. To reference this OUTPUT2 file the following line needs to be included in the file specification section of the Simcenter Nastran input file:

   ASSIGN INPUTT2=`<OUTPUT2_filename>' UNIT=20

8. For MSRMODE=2, no files are referenced for stress recovery. Instead, a full CMS reanalysis is performed to build the reference data for the stress recovery analysis. Obviously, the analysis time is significantly far greater compared to the MSRMODE=0 or 1 method, but this method frees up disk space. There is also risk in using this method. If the reanalysis generates slightly different component eigenvalues or eigenvectors than were generated during the creation of the Adams MNF or RecurDyn RFI in the initial Simcenter Nastran run, then the Adams or RecurDyn results in the MDF (modal deformation file) will be inconsistent and incorrect results will be recovered. Something as simple as a sign change for one eigenvector will cause incorrect results to be recovered. It is, therefore, *highly recommended* that MSRMODE=0 or 1 *always* be used.

9. This capability must be performed in SOL 103 and is limited to no more than one superelement per Simcenter Nastran model. Residual-only analyses are supported.

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10. Setting CHECK=YES is *not* recommended for models of realistic size due to the amount of data that will be written to the f06.

Related Topics

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MEFFMASS

Modal Effective Mass Output Request

Requests the output of the modal effective mass, participation factors, and modal effective mass fractions in normal modes analysis. Optionally can limit mode output by effective mass fraction.

Format:

$$\text{MEFFMASS}\left(\left[\left(\left[\begin{matrix}\underline{\text{PRINT}}\\ \text{PUNCH}\\ \text{PLOT}\end{matrix}\right]\right),\ [\text{GRID}=\text{g}],\ [\text{THRESH}=\text{f}],\right.\right.$$
$$[\text{MINT1}=\text{f1}],\ [\text{MINT2}=\text{f2}],\ [\text{MINT3}=\text{f3}],\ [\text{MAXIT}=\text{n}],$$
$$\left.\left.\left[\begin{matrix}\text{STRUCTURE}\\ \text{FLUID}\\ \text{SLOSH}\\ \underline{\text{COMBINED}}\end{matrix}\right],\left[\begin{matrix}\underline{\text{SUMMARY}},\text{PARTFAC},\\ \text{MEFFM},\text{MEFFW},\\ \text{FRACSUM},\text{ALL}\end{matrix}\right]\right]\right)=\left\{\begin{matrix}\text{YES}\\ \text{NO}\end{matrix}\right\}$$

Examples:

```
MEFFMASS
MEFFMASS(GRID=12,SUMMARY,PARTFAC)
MEFFMASS(PLOT,ALL,THRESH=0.001)=YES
```

Describers:

Describer	Meaning
PRINT	Writes output to the print file. (Default)
PUNCH	Writes output to the punch file.
PLOT	Writes output to the output2 file.
GRID=*g*	Reference grid point for the calculation of the rigid body mass matrix. (Integer; Default is the origin of the basic coordinate system)
THRESH=*f*	Excludes modes that have effective mass fraction less than *f* from mode output. (Real $0.0 \le f \le 1.0$; Default is all modes are output) See Remark 8 and Remark 10.

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Describer	Meaning
MINT1=*f*1	Calculation of modes continue until either the total effective mass fraction in the X-direction exceeds *f*1 or the maximum number of iterations is reached. (Real 0.0 ≤ *f*1 ≤ 1.0; No default) See Remark 9 and Remark 10.
MINT2=*f*2	Calculation of modes continue until either the total effective mass fraction in the Y-direction exceeds *f*2 or the maximum number of iterations is reached. (Real 0.0 ≤ *f*2 ≤ 1.0; No default) See Remark 9 and Remark 10.
MINT3=*f*3	Calculation of modes continue until either the total effective mass fraction in the Z-direction exceeds *f*3 or the maximum number of iterations is reached. (Real 0.0 ≤ *f*3 ≤ 1.0; No default) See Remark 9 and Remark 10.
MAXIT=*n*	Maximum number of iterations to achieve MINT1, MINT2, and MINT3 specification. (Integer > 0; Default = 5)
STRUCTURE	Requests modal effective mass results for structural degrees-of-freedom attributable to the mass of the structure only. See Remark 13.
FLUID	Requests modal effective mass results for structural degrees-of-freedom attributable to the mass of the incompressible fluid only. See Remark 13.
SLOSH	Requests modal effective mass results for the sloshing free surface degrees-of-freedom attributable to the mass associated with the sloshing free surface only. See Remark 13.
COMBINED	Requests modal effective mass results for structural degrees-of-freedom attributable to the mass of the structure and fluid, and modal effective mass results for the sloshing free surface degrees-of-freedom attributable to the mass associated with the sloshing free surface only. (Default) See Remark 13.
SUMMARY	Requests calculation of total effective mass fraction, modal effective mass matrix, and the rigid body mass matrix. (Default)
PARTFAC	Requests calculation of modal participation factors.
MEFFM	Requests calculation of modal effective mass in units of mass.
MEFFW	Requests calculation of modal effective mass in units of weight.
FRACSUM	Requests calculation of modal effective mass fraction.
ALL	Requests calculation of total effective mass fraction, modal effective mass matrix, rigid body mass matrix, modal participation factors, modal effective mass in units of mass, modal effective mass in units of weight, and modal effective mass fraction.

Remarks:

1. The modal participation matrix [ϵ] is computed as:

$$[\varepsilon] = [m]^{-1}[\varphi]^{T}[M][D]$$

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where,

m is the generalized mass matrix,

φ are the eigenvectors,

M is the mass matrix reduced to the a-set (g-set for superelements),

and *D* is the rigid body transformation matrix with respect to the a-set.

For a single mode *i*, the modal participation (ϵ_{ir}) is computed as:

$$\varepsilon_{ir} = m_{ii}^{-1}\{\varphi_i\}^T [M] \{D_r\}$$

where,

i represents the mode number,

and *r* represents the rigid body direction (1 - 6).

The modal effective mass ($M_{eff,\ ir}$) for a single mode and rigid body direction is computed as:

$$M_{eff,ir} = m_{ii}\varepsilon_{ir}^2$$

The modal effective mass fraction is computed by dividing the modal effective mass by the rigid body mass, where the rigid body mass is computed as:

$$M_r = \{D_r\}^T M \{D_r\}$$

The total modal effective mass fraction is the sum of all the individual fractions and would total to 1.0 if all modes were computed. But in general, mode truncation results in a total modal effective mass less than 1.0.

2. The PARTFAC describer outputs the modal participation factors table: ϵ.
3. The MEFFM describer outputs the modal effective mass table: ϵ^2, the term-wise square of the modal participation factors table.
4. The MEFFW describer outputs the modal effective weight table; i.e., the modal effective mass divided by the user parameter WTMASS.
5. The FRACSUM describer outputs the modal effective mass fraction table; i.e., the generalized mass matrix (diagonal term) multiplied by the modal effective mass and then divided by the rigid body mass matrix (diagonal term).
6. For superelements the MEFFMASS command uses the residual structure eigenvalues and eigenvectors, by default. If however, PARAM, FIXEDB, -1 is specified then MEFFMASS command uses the component mode eigenvalues and eigenvectors.
7. Effective mass is computed in the basic coordinate system.
8. Modes which have an effective mass fraction greater than the value of THRESH in at least one translational or rotational direction are output in the .f06 and .op2 files. THRESH does not limit modes for consecutive dynamic response solutions (SOL 111 or 112). Use the MODSEL case control command to select/deselect modes in these solutions.

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9. If any combination of MINT1, MINT2, and MINT3 are specified, the calculation of modes continue until either the total effective mass fraction of all those specified exceeds their respective *f* - value or the maximum number of iterations is reached.
10. The THRESH describer cannot be specified in combination with the MINT1, MINT2, or MINT3 describers.
11. On the EIRG and EIGRL bulk entries, you can use combinations of the F1 (V1), F2 (V2), and ND fields to specify:
    - The frequency range for mode calculation.
    - The number of modes to calculate.
    - The number of modes to calculate above a frequency limit.

    If any of the MINT1, MINT2, and MINT3 describers is specified, the software calculates additional modes when either of the following conditions is true:
    - The upper limit to the frequency range is reached prior to satisfying the describer specification. The upper limit is increased to accommodate additional mode calculation.
    - The number of modes calculated does not satisfy the describer specification.
12. For SOL 110, MEFFMASS is not applicable to models that contain fluid elements.
13. Applicable when performing a SOL 103 real eigenvalues or SOL 111 modal frequency response analysis on a structural model that contains an incompressible fluid and is modeled using the defined fluid mass method. If the fluid volume does not have a free surface or the free surface is non-sloshing, the software ignores a SLOSH describer specification. For a model that is structural only, the software ignores a STRUCTURE, FLUID, SLOSH, or COMBINED describer specification.

Remarks for SOLs 402 and 414:

1. For SOL 402, MEFFMASS is supported in the modal and cyclic modes subcases. Only the PRINT describer is supported. The MEFFMASS output is written to both the .f06 file and a .csv file.
2. For SOLs 402 and 414, for the cyclic modes subcase for harmonics 0 and 1, the output contains modal participation factor and effective mass results for the entire, recombined structure.

   For harmonics > 1, the output contains:

   • Modal participation factors and effective mass results for only one sector and using only the real part of the mode.

   • Modal participation factor results for both single-stage and multi-stage cyclic symmetry.

   • Effective mass results for only single-stage cyclic symmetry, and only for the translational components.
3. For SOL 414:
   - MEFFMASS is applicable to the following solution types:
     - SOL 414,103 (eigenvalue analysis) with the **METHOD** case control (real eigenvalue extraction) only. The **CMETHOD** case control (complex eigenvalue extraction) is not valid for SOL 414 MEFFMASS.

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  - SOL 414,103 (superelement reduction).
  - SOL 414,111 (harmonic response) with no **RMETHOD** (rotor) case control command defined and with output set to use the **EIGRL** (real modes) bulk entry.
- The only describer that SOL 414 uses is PRINT. The solver uses the default values for all other describers. For example, the following is valid for SOL 414:

```
MEFFMASS = YES
MEFFMASS(PRINT) = YES
```

- Results are available as follows:
  - In the .f06 file under `MODAL PARTICIPATION FACTOR`.
  - In comma-delimited file name *<solution_name>_mpf.csv*.
  - For XY plots, in the Simcenter Samcef (.des) or Simcenter data (.scd5) results files. You can view these files in Simcenter 3D Pre/Post or another post-processing application.

Related Topics

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MESTRN

Requests mechanical strain output at grid points on elements for SOL 401.

Requests mechanical strain at grid points on elements.

Format:

$$\text{MESTRN}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
MESTRN=ALL
MESTRN(PRINT,PUNCH)=17
```

Describers for SOL 401:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all elements.
n	Set identification number of a previously appearing SET command. Only elements with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Do not compute output.

Remarks:

1. The output includes the mechanical strain components and the effective mechanical strain. The mechanical strain components are the sum of the elastic, plastic, and creep strain contributions for each component. The effective mechanical strain is the sum of the effective elastic, effective plastic, and effective creep strains.
2. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.
3. The following elements support the output:

   - Plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.

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- Plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8.

- Axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6.

- 3D solids elements CTETRA, CHEXA, CPENTA, CPYRAM.

- Chocking elements CCHOCK3, CCHOCK4, CCHOCK6, CCHOCK8.

- Generalized plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8 that reference a PGPLSN entry.

- Shell elements CQUAD8, CTRIA6, CTRIAR, CQUADR that reference a PSHELL entry.

- Tension-only shell element CQUADR that references a PSHLPNL entry.

- Composite shell elements CQUAD8, CTRIA6, CTRIAR, CQUADR that reference a composite property defined with the PCOMPG1, PCOMP or PCOMPG entries.

- Composite solid elements CHEXA, CPENTA that reference a composite property defined with the PCOMPS entries.

4. You can requeset both PRINT and PUNCH.

5. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.

Related Topics

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METHOD

Real Eigenvalue Extraction Method Selection

Selects the real eigenvalue extraction parameters.

Format:

$$\text{METHOD}\left[\frac{\text{STRUCTURE}}{\text{FLUID}}\right]=\text{n}$$

Examples:

```
METHOD=33
METHOD(FLUID)=34
```

Describers:

Describer	Meaning
STRUCTURE or FLUID	The referenced EIGB, EIGR or EIGRL Bulk Data entry is applied to the structural or fluid portion of the model. (Default = STRUCTURE)
n	Set identification number of an EIGR or EIGRL Bulk Data entry for normal modes, buckling modes or modal formulation, or an EIGB or EIGRL entry for buckling. (Integer>0)

Remarks:

1. An eigenvalue extraction method must be selected when extracting real eigenvalues using DMAP modules READ or REIGL.
2. If the set identification number selected is present on both EIGRL and EIGR and/or EIGB entries, the EIGRL entry will be used. This entry requests the Lanczos eigensolution method.
3. When a fluid material is included in the model:
   - METHOD(STRUCTURE)=n or METHOD=n selects the eigenvalue method for the structure, and METHOD(FLUID)=n selects the eigenvalue method for the fluid.
   - When METHOD(STRUCTURE)=n and METHOD(FLUID)=n are both defined for a modal response solution, the software will perform a modal reduction for both the structure and the fluid.
   - When METHOD(STRUCTURE)=n is defined but METHOD(FLUID)=n is not defined for a modal response solution, the software performs a modal reduction for the structure only. When computing the response, the software uses the modal degrees-of-freedom computed for the structure, and the physical degrees-of-freedom belonging to the fluid.

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- When a fluid only model is used in a modal response solution, any of METHOD=n, METHOD(FLUID)=n, or METHOD(STRUCTURE)=n can be specified to select the eigenvalue method for the fluid.
- The METHOD(FLUID) and METHOD(STRUCTURE) may be specified simultaneously in the same subcase for the residual structure only. Do not specify METHOD(FLUID) in a superelement subcase even if the superelement contains fluid elements.

Remarks related to SOL 402:

1. Only METHOD(STRUCTURE) is supported.
2. n can only reference the EIGRL, EIGR, or EIGB bulk entries.
3. METHOD must be used for all ANALYSIS=MODES and ANALYSIS=BUCKLING subcases.
4. If more than one of the EIGB, EIGR and EIGRL bulk entries share the same SID, and that SID is selected by a METHOD(STRUCTURE) command, the following order of precedence is used.
   - For subcases of type ANALYSIS=MODES, the EIGRL entry has priority over the EIGR entry, and the EIGR entry has priority over the EIGB entry.
   - For subcases of type ANALYSIS=BUCKLING, the EIGRL entry has priority over the EIGB entry, and the EIGB entry has priority over the EIGR entry.

Related Topics

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MFLUID

Fluid Boundary Element Selection

Selects the MFLUID Bulk Data entries to be used to specify the fluid-structure interface.

Format:

MFLUID=n

Examples:

```
MFLUID=919
```

Describers:

Describer	Meaning
n	Set identification number of one or more MFLUID Bulk Data entries. (Integer>0)

Remarks:

1. For a further discussion, see the *Simcenter Nastran User's Guide*.
2. MFLUID must exist above all subcases.

Related Topics

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MLOAD

Misalignment Load Selection

Selects the misalignment load in SOL 414 rotor dynamic analysis.

Format:

$$\text{MLOAD} = \text{n}$$

Examples:

```
MLOAD=20
```

Describers:

Describer	Meaning
n	Identification number of an MLINK bulk entry. (Integer>0)

Remarks:

1. To define misalignment loads on a subcase level, place MLOAD case control commands within the subcases of the following SOL 414 solution types:
   - SOL 414,101 Maneuvers
   - SOL 414,103 Eigenvalues and Superelement Reduction for static or transient response preload subcases
   - SOL 414,110 Complex Modal Analysis for static preload subcases
   - SOL 414,111 Harmonic Response for static or frequency response subcases

   For SOL 414,129 Transient Response, misalignment loads are supported on the global level only. However, you can enter the MLOAD case control commands on a subcase level as long as they all reference the same MLINK bulk entry.

Related Topics

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MODALE

Energy output selection for SOL 111

Energy output selection for SOL 111. This is not supported for the constraint mode method of enforced motion.

Format:

$$\text{MODALE}\left[\left(\left[\text{TYPE}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{KE}\\ \text{SE}\\ \text{BOTH}\\ \text{TOTAL}\end{matrix}\right\}\right],\left[\begin{matrix}\underline{\text{BOTH}}\\ \text{CONST}\\ \text{OSCILL}\end{matrix}\right],\left[\begin{matrix}\underline{\text{SORT2}}\\ \text{SORT1}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{NOPRINT}\end{matrix}\right],\right.$$
$$\left.\left[\begin{matrix}\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\left[\text{FREQ}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{r}\end{matrix}\right\}\right]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
MODALE=ALL
SET 200=1,3,4,5,7
SET 300=0.1 0.3 0.5
MODALE(TYPE=BOTH,FREQ=300)=200
```

Describers:

Describer	Meaning
TYPE	Type of energy output: All=Strain energy, kinetic energy, and total energy will be output. (Default) SE=Only strain energy will be output. KE=Only kinetic energy will be output.

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Describer	Meaning
	BOTH=Both strain and kinetic energy will be output. TOTAL=Only total energy will be output.
BOTH	Both constant and oscillating energy results will be output. (Default)
CONST	Only constant energy results will be output.
OSCILL	Only oscillating energy results will be output.
SORT1	Output will be presented as a tabular listing of energy values for the set of frequencies defined by the FREQ describer at each mode in the specified set of modes.
SORT2	Output will be presented as a tabular listing of energy values for the specified set of modes at each frequency defined by the FREQ describer. (Default)
PRINT	Prints the modal energy results to the .f06 file. (Default)
NOPRINT	Writes the modal energy results to the OP2 file without writing to the f06 or punch file.
PUNCH	Writes the modal energy results to the punch (.pch) file.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Default)
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
FREQ	Computes energies at ALL frequencies or the set of frequencies defined by SET r. (Default=ALL)
ALL, n, NONE	Computes modal energies for (1) all modes, (2) the modes defined on SET n, or (3) no modes.

Remarks:

1. The FREQ describer selects from the set of forcing frequencies. If a frequency in the selected set is not identical to a frequency in the set of forcing frequencies, energy values at the closest frequency will be provided. If a frequency in the selected set is greater or less than the range of forcing frequencies, energy values will be provided at the highest or lowest forcing frequency, respectively.
2. Modal energy will be calculated and reported for (1) constant energy and/or (2) oscillating energy (depending on whether BOTH, CONST, or OSCILL is requested) at each frequency for each mode and at each frequency for all modes summed together. The frequencies used are those requested by the FREQ describer; the modes used are those requested by ALL/n (i.e. SET n). Combined with the value of the TYPE describer, one or more of the following equations will be output (the modal energy computation is valid for the arbitrary loading case denoted by superscript ‘s’, and the arbitrary excitation radian frequency denoted by subscript ‘r’):

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$$MSE_{j,const} = \frac{1}{4}k_j(\xi_{j,re}^2 + \xi_{j,im}^2)$$

$$MSE_{j,osc} = \sqrt{(MSE_{j,osc,cos}^2 + MSE_{j,osc,sin}^2)}$$

$$\phi_{j,MSE} = \text{atan}\left(\frac{MSE_{j,osc,cos}}{MSE_{j,osc,sin}}\right)$$

$$MKE_{j,const} = \frac{1}{4}m_j(\xi_{j,re}^2 + \xi_{j,im}^2)$$

$$MKE_{j,osc} = \sqrt{(MKE_{j,osc,cos}^2 + MKE_{j,osc,sin}^2)}$$

$$\phi_{j,MKE} = \text{atan}\left(\frac{MKE_{j,osc,cos}}{MKE_{j,osc,sin}}\right)$$

$$ME_{j,const} = MSE_{j,const} + MKE_{j,const}$$

$$ME_{j,osc}\cos(2\omega t + \phi_{j,ME}) = MSE_{j,osc}\cos(2\omega t + \phi_{j,MSE}) + MKE_{j,osc}\cos(2\omega t + \phi_{j,MKE})$$

$${}_r^sMSE_{const} = \sum_{j=1}^{n} {}_r^sMSE_{j,const}$$

$${}_r^sMKE_{const} = \sum_{j=1}^{n} {}_r^sMKE_{j,const}$$

$${}_r^sMSE_{osc} = \sqrt{({}_r^sMSE_{osc,cos}^2 + {}_r^sMSE_{osc,sin}^2)}$$

$${}_r^sMKE_{osc} = \sqrt{({}_r^sMKE_{osc,cos}^2 + {}_r^sMKE_{osc,sin}^2)}$$

$${}_r^s\phi^{MSE} = \text{atan}\frac{{}_r^sMSE_{osc,sin}}{{}_r^sMSE_{osc,cos}}$$

$${}_r^s\phi^{MKE} = \text{atan}\frac{{}_r^sMKE_{osc,sin}}{{}_r^sMKE_{osc,cos}}$$

$${}_r^sME_{const} = {}_r^sMSE_{const} + {}_r^sMKE_{const}$$

$${}_r^sME_{osc}\cos(2\omega t + {}_r^s\phi^{ME}) = {}_r^sMSE_{osc}\cos(2\omega t + {}_r^s\phi^{MSE}) + {}_r^sMKE_{osc}\cos(2\omega t + {}_r^s\phi^{MKE})$$

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3. Both PRINT and PUNCH may be requested.

Related Topics

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MODCON

Modal Contribution Output Request

Requests the form and type of modal contribution output for the residual.

Format:

$$\text{MODCON}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{REAL}}\ \text{or}\ \underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{matrix}\right],\ \text{PUNCH}\right],\ \left[\begin{matrix}\underline{\text{ABS}}\\ \text{NORM}\\ \text{BOTH}\end{matrix}\right],\ \text{TOPS}=\text{ps},\ \text{TOPF}=\text{pf},\ \text{SOLUTION}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{setout}\\ \text{PEAKOUT}\end{matrix}\right\},$$
$$\text{PANELMC}=\left\{\begin{matrix}\underline{\text{NONE}}\\ \text{setp}\\ \text{ALL}\end{matrix}\right\}\right)\Bigg]=\left\{\begin{matrix}\text{n}\\ \text{ALL}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
MODCON=123
MODCON(SORT1,PHASE,PRINT,PUNCH,BOTH,TOPS=5)=ALL
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of modal DOF for each frequency or time. (Default)
SORT2	Output a tabular listing of frequency or time for each modal DOF. This option is not available for SOL 110.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
PRINT	Write modal contribution output to the print (.f06) file. (Default)
PUNCH	Write modal contribution output to the standard punch (.pch) file.
NOPRINT	Generate, but do not print, modal contribution results.
ABS	Output modal contributions in absolute terms. (Default)

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Describer	Meaning
NORM	Output modal contributions in normalized terms.
BOTH	Output modal contributions in both absolute and normalized terms.
TOPS (or TOP) = ps	The number of structural modes to list in the output that have the greatest contribution to the response at each frequency or time. If ps > 1, the output is sorted in descending order from the structural mode that has the greatest contribution. If ps = 0, no structural mode contributions are output, only totals. See Remark 1. (Integer ≥ 0; Default ps = 5)
TOPF = pf	The number of fluid modes to list in the output that have the greatest contribution to the response at each frequency or time. If pf > 1, the output is sorted in descending order from the fluid mode having the greatest contribution. If pf = 0, no fluid mode contributions are output, only totals. See Remark 1. (Integer ≥ 0; Default pf = 5)
SOLUTION = ALL	Perform the modal contribution calculations at all frequencies or times that are defined by the FREQUENCY or TSTEP case control commands, respectively. See Remark 12. (Default)
SOLUTION = setout	Perform the modal contribution calculations at the frequencies or times listed on a previously appearing SET case control command with setout as the identification number. See Remark 12. (Integer > 0)
SOLUTION = PEAKOUT	Perform the modal contribution calculations only at the frequencies identified with the PEAKOUT command. See Remark 12.
PANELMC = NONE	Requests that no modal contributions are calculated for panels. See Remarks 1 and 9. (Default)
PANELMC = setp	Requests that modal contributions are calculated for the panels listed on a previously appearing SET case control command with setp as the identification number. See Remarks 1 and 9. (Integer > 0)
PANELMC = ALL	Requests that modal contributions be calculated for all panels defined by PANEL bulk entries. See Remarks 1 and 9.
n	Calculate modal contributions for the responses listed on a previously appearing SETMC case control command with n as the identification number. (Integer > 0)
ALL	Calculate modal contributions for all the responses listed on the SETMC case control commands specified in and above the current subcase.
NONE	Do not calculate modal contributions.

Remarks:

1. SOL 110, 111, 112, and SOL 146 are supported.

- For SOL 111, MODCON is supported by the FEM Adaptive Order (FEMAO) method with the following exceptions:
  - The number of structural modes (TOPS describer) is not supported.

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    - The modal pressure contributions are computed on all panels. You cannot specify a subset of panels.
  - For SOL 110, modal contributions for superelements are not supported.
  - The TOPF and PANELMC describers are only supported for SOL 111.
  - The SOLUTION describer is only supported for SOL 111, 112, and 146.
2. Both PRINT and PUNCH can be requested.
3. MODCON = NONE overrides an overall output request. Thus, to turn off contribution output for a specific subcase, place MODCON = NONE in the subcase.
4. MODCON supports results for microphone points.
5. Results for SPC forces do not include the effect of any enforced motion applied at the DOF.
6. The parameters LFREQ, LFREQFL, HFREQ, HFREQFL, LMODES, and LMODESFL are supported.
7. The SOLUTION and PANELMC describers can be abbreviated to SOLU and PANE, respectively.
8. The SET case control command referenced by SOLUTION = setout must contain real values for frequencies or times. Using integer values may lead to erroneous results.
9. The PANELMC describer applies to acoustic responses and the contributions from structural modes.
10. The SOLUTION describer is only valid in a SOL 108 or 111 frequency response analysis.
11. When an OFREQ case control command is present, the SOLUTION describer is used as follows:
  - If SOLUTION=ALL, output results for all frequencies specified by the OFREQ command.
  - If SOLUTION=setout, output results for all frequencies specified by the OFREQ command that are listed on a previously appearing SET command with setf as the identification number.
  - If SOLUTION=PEAKOUT and a PEAKOUT case control command is also present in the input file, output results for all frequencies specified by the OFREQ command that satisfy the PEAKOUT filtering criteria.
12. The SOLUTION describer is not valid for superelements or frequency-dependent components.

Related Topics

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MODES

Subcase Repeater

Repeats a subcase.

Format:

MODES=n

Examples:

```
MODES=3
```

Describers:

Describer	Meaning
n	Number of times the subcase is to be repeated. (Integer>0)

Remarks:

1. This command can be illustrated by an example. Suppose stress output is desired for the first five modes only and displacements for the next two modes and forces for the remaining modes. The following example would accomplish this.

```
SUBCASE 1 $ FOR MODES 1 THRU 5
   MODES=5
   STRESS=ALL
SUBCASE 6 $ FOR MODES 6 AND 7
   DISPLACEMENTS=
   MODES=2
SUBCASE 8 $ FOR MODE 8 AND REMAINING MODES
   FORCE=ALL
```

2. This command causes the results for each mode to be considered as a separate, successively numbered subcase, beginning with the subcase number containing the MODES command. In the example above, this means that subcases 1, 2, 3, etc. are assigned to modes 1, 2, 3, etc., respectively.
3. If this command is not used, eigenvalue results are considered to be a part of a single subcase. Therefore, any output requests for the single subcase will apply for all eigenvalues.

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4. All eigenvectors with mode numbers greater than the number of subcases defined in Case Control are printed with the descriptors of the last subcase. For example, to suppress all printout for modes beyond the first three, the following Case Control could be used:

```
SUBCASE 1
   MODES=3
   DISPLACEMENTS=ALL
SUBCASE 4
   DISPLACEMENTS=NONE
BEGIN BULK
```

5. This command may be of no use in non eigenvalue analysis and may cause data recovery processing to be repeated.

Related Topics

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MODSEL

Mode Selection

Used to select mode numbers to include in a modal dynamic response solution.

Format:

$$\text{MODSEL}\left(\begin{matrix}\text{STRUCTURAL}\\ \text{FLUID}\end{matrix}\right)=\left\{\begin{matrix}\text{ALL}\\ \text{n}\end{matrix}\right\}$$

Examples:

```
MODSEL = 3
MODSEL(FLUID) = 4
MODSEL(STRUCTURAL) = ALL
MODSEL = -10
```

Describers:

Describer	Meaning
STRUCTURAL	Specifies the structural modes to include in the response solution. (Default)
FLUID	Specifies the fluid modes to include in the response solution.
ALL	Designates that all the structural or fluid modes be used in the response solution. (Default)
n	Identification number of the SET case control command containing a list of either structural or fluid modes to be used in the response solution. The mode numbers not included in the SET case control command are removed from the modal space. By preceding the identification number of the SET case control command with a negative sign, the mode numbers listed in the SET case control command are omitted from the response solution. Mode numbers listed in the SET case control command that are larger than the number of computed modes are ignored. (Integer≠0)

Remarks:

1. All structural and fluid modes are used in the response solution if a MODSEL case control command is not included in the input file.
2. Depending on the solution the type, MODSEL is supported at the subcase level and for restarts:
   - For SOLs 103, 110, and 187, MODSEL must be above the subcase level.

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- For SOLs 103, 110, and 187, MODSEL is supported for restarts, but not at the subcase level.
- For SOLs 414,103 (superelement and eigenvalue analysis), 414,110 (complex modal analysis), and 414,111 (harmonic response), MODSEL is supported at the subcase level, but not for restarts.
- For SOLs 111, 112, 145, and 146, MODSEL is supported for restarts and at the subcase level.

3. Multiple MODSEL case control commands can be included in any subcase.

Related Topics

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MODTRAK

Mode Tracking Request

Selects mode tracking options in design optimization (SOL 200).

Format:

MODTRAK=n

Examples:

```
MODTRAK=100
```

Describers:

Describer	Meaning
n	Set identification of a MODTRAK Bulk Data entry. (Integer>0)

Remarks:

Selection of a MODTRAK Bulk Data entry with the MODTRAK Case Control command activates mode tracking for the current subcase. This request is limited to normal modes subcases (ANALYSIS = MODES) in design optimization (SOL 200).

Related Topics

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MONITOR

Print Selection for Monitor Data

Requests the type of monitor data to be output to the .f06 file in SOL 101 and SOL 103.

Format:

$$\text{MONITOR}\,(\text{NOPNT2, NOPNT3}) = \left\{ \begin{matrix} \text{ALL or YES} \\ \underline{\text{NONE}} \text{ or } \underline{\text{NO}} \end{matrix} \right\}$$

Examples:

```
MONITOR(NOPNT2) = ALL
```

Describers:

Describer	Meaning
NOPNT2	Do not include MONPNT2 results in the MONITOR point prints. (Default = Provide these prints)
NOPNT3	Do not include MONPNT3 results in the MONITOR point prints. (Default = Provide these prints)
ALL or YES	Print all monitor point results except for the selected items.
NONE or NO	Do not print the monitor point results. (Default)

Remarks:

1. The MONITOR command is required to obtain MONITOR results in the printed output.

Related Topics

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MONVAR

Displacement Monitor Plot Request

Requests grid and degree-of-freedom to be monitored in a preprocessor, such as Pre/Post.

Format:

$$\text{MONVAR} = \text{n}$$

Examples:

```
SETMC 21 = DISP/ 1001(T1), 1002(T1), 1204(T1)
SETMCNAME 21 = Blade 3
SETMC 25 = DISP/ 1001(T2), 1002(T2), 1204(T2)
SETMCNAME 25 = Compressor Disk
SET 1 = 21, 25
MONVAR = 1
```

Describers:

Describer	Meaning
n	Set identification number of a previously appearing SET command that references SETMC IDs. The SETMC command lists a grid point ID and component for each degree-of-freedom to be included in the displacement monitoring. See the examples. (Integer>0)

Remarks:

1. MONVAR is supported for static and dynamic nonlinear subcases in SOLs 401, 402, 414,111, and 414,129 only.
2. MONVAR must be placed above all subcases.
3. A displacement monitor plot is created for each SETMC ID selected. Each displacement monitor plot includes a plot of the maximum, the minimum, and the average displacement. The software uses the displacement values for all degree-of-freedom listed on a particular SETMC to determine the maximum and minimum displacement, and to compute the average displacement for each converged time step. If you want to monitor the displacement of a single degree-of-freedom, you should select a SETMC which includes only this degree-of-freedom.
4. You can optionally define a title for each displacement monitor plot using the SETMCNAME command. The ID on the SETMCNAME command should match the ID on the SETMC command. See the examples. When the SETMCNAME is undefined, the ID of the SETMC is used for the plot title.

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Related Topics

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MPC

Multipoint Constraint Set Selection

Selects a multipoint constraint set.

Format:

MPC=n

Examples:

```
MPC=17
```

Describers:

Describer	Meaning
n	Set identification number of a multipoint constraint set. This set identification number must appear on at least one MPC or MPCADD Bulk Data entry. (Integer>0)

Remarks:

1. In cyclic symmetry analysis, this command must appear above the first SUBCASE command.
2. Multiple boundary (MPC sets) conditions are not allowed in superelement analysis. If more than one MPC set is specified per superelement (including the residual), then the second and subsequent sets will be ignored.

Remarks related to SOL 402:

1. This command must appear once above the first SUBCASE command. As a result, a single Multipoint Constraint Set is used for all subcases.

Related Topics

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MPCFORCES

Multipoint Forces of Constraint Output Request

Requests the form and type of multipoint force of constraint vector output.

Format:

$$\text{MPCFORCES}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{PRINT}},\ \text{PUNCH}\\ \text{PLOT}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{REAL}}\ \text{or}\ \underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\ \left[\begin{matrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{RPRINT}}\\ \text{NOPRINT}\end{matrix}\right],\ [\text{RPUNCH}]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
MPCFORCES=5
MPCFORCES(SORT2,PUNCH,PRINT,IMAG)=ALL
MPCFORCES(PHASE)=NONE
```

Describers:

Describer	Meaning
SORT1	Output will be presented as a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output will be presented as a tabular listing of frequency or time for each grid point.
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
PLOT	Generates, but does not print, multipoint constraint forces.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output.
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.

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Describer	Meaning
PSDF	Requests the power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 9.
ATOC	Requests the autocorrelation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 9.
CRMS	Requests the cumulative root mean square function be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 9.
RMS	Requests the root mean square and zero crossing functions be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 9.
RALL	Requests all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 9.
RPRINT	Writes random analysis results to the print file. (Default) See Remark 9.
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 9.
RPUNCH	Writes random analysis results to the punch file. See Remark 9.
ALL	Multipoint forces of constraint for all points will be output. See Remark 3 and Remark 6.
NONE	Multipoint forces of constraint for no points will be output.
n	Set identification of a previously appearing SET command. Only multipoint forces constraint for points with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

1. Both PRINT and PUNCH may be requested.
2. The defaults for SORT1 and SORT2 depend on the type of analysis:
   - SORT1 is the default in static analysis, frequency response, steady state heat transfer analysis, real and complex eigenvalue analysis, flutter analysis, and buckling analysis. If SORT2 is selected in a frequency response solution for one or more of the commands ACCE, DISP, FORC, GPFO, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT2 format.
   - SORT2 is the default in transient response analysis (structural and heat transfer). SORT2 is not available for real eigenvalue (including buckling), complex eigenvalue, or flutter analysis. If SORT1 is selected in a transient solution for one or more of the commands ACCE, DISP, ENTH, FORC, GPFO, HDOT, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT1 format.

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- XY plot requests will force SORT2 format thus overriding SORT1 format requests.

3. In a statics problem, a request for SORT2 causes loads at all points (zero and nonzero) to be output.
4. MPCFORCES=NONE overrides an overall output request.
5. In SORT1 format, MPCFORCES recovered at consecutively numbered scalar points are printed in groups of six (sextets) per line of output. But if a scalar point is not consecutively numbered, then it will begin a new sextet on a new line of output. If a sextet can be formed and all values are zero, then the line will not be printed. If a sextet cannot be formed, then zero values may be output.
6. MPCFORCES results are output in the global coordinate system (see field CD on the GRID Bulk Data entry).
7. MPCFORCES results are not available in SOL 129.
8. In inertia relief analysis, the MPCFORCES output includes both the effects of applied and inertial loads.
9. The following applies to random solutions:
   - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, PUNCH, PLOT describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
   - The SORT1 and SORT2 describers only control the output format for the frequency response output. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
   - Any combination of the PSDF, ATOC, RMS, and CRMS describers can be selected. The RALL describer selects all four.
   - When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the multi-point constraint force results are based on the nominal values for stiffness and damping.

Remarks related to SOL 402:

1. Output is restricted to REAL and SORT1 format. SORT2, IMAG, PHASE, ABS, REL, PSDF, ATOC, CRMS, RMS, RALL, NORPRINT, and RPUNCH are ignored.
2. The output also includes reaction forces for contact and gluing elements, in RBE2 and RBE3 rigid bodies, and MPC.

Related Topics

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MPRES

Fluid Pressure Output Request

Requests the pressure for selected surface elements in fluid-structure interaction problems.

Format:

$$\text{MPRES}\left[\left[\begin{matrix}\text{PRINT, PUNCH}\\ \underline{\text{PLOT}}\end{matrix}\right],\left[\begin{matrix}\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right]\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
MPRES=5
MPRES(IMAG)=ALL
```

Describers:

Describer	Meaning
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
PLOT	Generates, but does not print or punch, data.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output.
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
ALL	Fluid pressures for all elements will be output.
NONE	Fluid pressures for no elements will be output.
n	Set identification number of a previously appearing SET command. Only fluid pressures for elements in this set will be output. (Integer>0)

Remarks:

PARAM,DDRMM,-1 is also required in the modal solution sequences 111, 112, 146, and 200.

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Related Topics

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NLARCL

SOL 401 nonlinear buckling request

Requests a nonlinear buckling solution in a SOL 401 statics subcase

Format:

NLARCL=n

Examples:

```
NLARCL=33
```

Describers:

Describer	Meaning
n	Set identification number of a NLARCL bulk entry. (Integer>0)

Remarks:

1. To request the nonlinear buckling analysis in SOL 401, your statics subcase should include the standard ANALYSIS=STATICS command and the NLARCL command. The ID on the NLARCL command selects the NLARCL bulk entry which defines the nonlinear buckling parameters.
2. The NLARCL command in the subcase is the trigger which the software uses to start the nonlinear buckling analysis. The referenced NLARCL bulk entry is also required, even when the default values are used.
3. The nonlinear buckling statics subcase must be either the first subcase, or the last in a sequence of static subcases. A nonlinear buckling statics subcase can only be followed by a modal subcase.
   a. If the nonlinear buckling statics subcase is the first subcase, all of the loads defined in the current subcase are incrementally applied by the software during the arc-length solution.
   b. If the nonlinear buckling statics subcase is the last subcase and it is sequentially dependent, the loads applied in the previous subcase are held constant in the current subcase. The difference between the load defined in the nonlinear buckling statics subcase and the load from the previous subcase is computed. This load difference is incrementally applied by the software during the arc-length solution.

Related Topics

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NLCNTL

Analysis parameter selection for SOLs 401, 402, and 414.

Selects the parameters used in SOLs 401, 402, and 414.

Format:

NLCNTL=n

Examples:

```
NLCNTL=10
```

Describers:

Describer	Meaning
n (For SOL 401)	Set identification of NLCNTL bulk entries. (Integer>0)
n (For SOL 402)	Set identification of NLCNTL2 bulk entries. (Integer>0)
n (For SOL 414)	Set identification of NLCNTL2 bulk entries. (Integer>0)

Remarks:

1. Only supported in SOLs 401, 402, and 414.
2. NLCNTL may be defined above the subcases (globally) or within a subcase.
3. NLCNTL or NLCNTL2 entries in the Bulk Data will not be used unless selected with a NLCNTL case control command.
4. For SOL 414, NLCNTL and NLCNTL2 are supported in SOL 414,103 Eigenvalues and Superelement Reduction with a transient preload subcase, SOL 414,111 Harmonic Response with a frequency response subcase, and SOL 414,129 Transient Response with a nonlinear dynamics subcase.

Related Topics

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NLLOAD

Nonlinear Load Output Request

Requests the form and type of nonlinear load output for transient problems.

Format:

$$\text{NLLOAD}\left[(\underline{\text{PRINT}}, \text{PUNCH})\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
NLLOAD=ALL
```

Describers:

Describer	Meaning
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
ALL	Nonlinear loads for all solution points will be output.
NONE	Nonlinear loads will not be output.
n	Set identification of a previously appearing SET command. Only nonlinear loads for points with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

1. Nonlinear loads are output only in the solution (d or h) set.
2. The output is available in SORT2 format only.
3. Both PRINT and PUNCH may be used.
4. NLLOAD=NONE allows overriding an overall output request.

Related Topics

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NLPARM

Nonlinear Static Analysis Parameter Selection

Selects the parameters used for nonlinear static analysis.

Format:

NLPARM=n

Examples:

```
NLPARM=10
```

Describers:

Describer	Meaning
n	Set identification of NLPARM and NLPCI Bulk Data entries. (Integer>0)

Remarks:

1. NLPARM and NLPCI entries in the Bulk Data will not be used unless selected.
2. NLPARM may appear above or within a subcase.

Related Topics

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NLSTRESS

Nonlinear Element Stress Output Request

Requests the form and type of nonlinear element stress output in SOL 106.

Format:

$$\text{NLSTRESS}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{matrix}\right]\right)\right]=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
NLSTRESS=5
NLSTRESS (SORT1,PRINT,PUNCH,PHASE)=15
NLSTRESS(PLOT)=ALL
```

Describers:

Describer	Meaning
SORT1	Output will be presented as a tabular listing of elements for each load.
SORT2	Output will be presented as a tabular listing of load for each element type.
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
PLOT	Generates nonlinear element stresses for requested set but no printer output.
ALL	Stresses for all nonlinear elements will be output. (Default)
n	Set identification of a previously appearing SET command. Only stresses for elements with identification numbers that appear on this SET command will be output. (Integer>0)
NONE	No nonlinear element stress will be output.

Remarks:

1. Both PRINT and PUNCH may be requested.
2. ALL should be used with caution in transient solutions as it can produce a large amount of output.
3. See **"DISPLACEMENT"** for a discussion of SORT1 and SORT2.

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4. SOLs 106 and 129 support two output formats for stress: the linear output format and the nonlinear output format. The formats have a different data organization. For example, the nonlinear format includes stresses together with strains, and it provides more information with regard to nonlinear material laws (effective strain, equivalent stress). See the *Handbook of Nonlinear Analysis* for details on the computation of effective strain and equivalent stress.

   The following summarizes the output requests for these formats.

   - To output both linear and nonlinear stress formats to the same media, you only need to define the STRESS command. For example,

     ```
     STRESS(PLOT)=ALL
     ```

   - To output only the linear stress format, you need the STRESS command and NLSTRESS=NONE. For example,

     ```
     STRESS(PLOT)=ALL
     NLSTRESS=NONE
     ```

   - To output only the nonlinear stress format, you only define the NLSTRESS command without defining the STRESS command. For example,

     ```
     NLSTRESS(PLOT)=ALL
     ```

   - To output both linear and nonlinear stress formats but to different media, you would need the STRESS and the NLSTRESS commands, and each defined with a different media. For example,

     ```
     STRESS(PLOT)=ALL
     NLSTRESS(PUNCH)=ALL
     ```

5. For the output coordinate system of CQUAD4 and CTRIA3 elements, see **"CQUAD4"** and **"CTRIA3"**. For the output coordinate system of CHEXA, CPENTA, CPYRAM, and CTETRA elements, see **"PSOLID"** and **"PLSOLID"**.

Related Topics

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NONLINEAR

Nonlinear Dynamic Load Set Selection

Selects a nonlinear dynamic load set for transient problems.

Format:

NONLINEAR=n

Examples:

```
NONLINEAR=75
```

Describers:

Describer	Meaning
n	Set identification of NOLINi or NLRGAP Bulk Data entry. (Integer>0)

Remarks:

NOLINi Bulk Data entry will be ignored unless selected in the Case Control Section.

Related Topics

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NOUTPUT

Normal Output Request in Cyclic Symmetry Problems

Requests physical output in cyclic symmetry problems.

Format:

$$\text{NOUTPUT} \left\{ \text{k}, \frac{\text{R}}{\text{L}} \right\} = \left\{ \begin{matrix} \text{ALL} \\ \text{m} \end{matrix} \right\}$$

Examples:

```
NOUTPUT (R)=ALL
NOUTPUT (2)=5
NOUTPUT (4,L)=10
```

Describers:

Describer	Meaning
ALL	Output for all segments is desired.
m	Output for segments specified in SET m is desired. (Integer>0)
k	Used in eigenvalue analysis to request eigenvector and internal force output for harmonics specified in SET k. (Integer>0)
R, L	Output for only the right- or left-half of segments specified as ALL or in SET m. R and L are used in dihedral symmetry only.

Remarks:

1. Sets k and m are defined on SET commands.
2. In cyclic symmetry analysis, this command or the HOUTPUT command is required to obtain data recovery.

Related Topics

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NSM

Nonstructural Mass

Selects Nonstructural Mass (NSM) set for mass generation.

Format:

NSM=n

Examples:

```
NSM=5
```

Describers:

Describer	Meaning
n	Set identification number of a nonstructural mass that appears on a NSM, NSM1, NSML, NSML1, or NSMADD bulk data entry. (Integer > 0)

Remarks:

1. In a non-superelement analysis, the NSM request must be consistent in all subcases. This is achieved with a single NSM request above the subcase level, or by including the same NSM request in each subcase. A fatal error will occur if the NSM request changes in any subcase. In a superelement analysis, each superelement subcase can have a different NSM request, but the NSM request must be consistent in all residual subcases.

Related Topics

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OFREQUENCY

Output Frequency Set

Selects a set of frequencies for output requests.

Format:

$$\text{OFREQUENCY} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \end{matrix} \right\}$$

Examples:

```
OFREQUENCY=ALL
OFREQUENCY=15
```

Describers:

Describer	Meaning
ALL	Output for all frequencies will be computed.
n	Set identification of a previously appearing SET command. Output for frequencies closest to those given on this SET command will be output. (Integer>0)

Remarks:

1. In real eigenvalue, buckling, and complex eigenvalue analyses, the OMODES Case Control command allows for an alternate way of selecting the modes to be output based on their mode numbers. In these cases, if both the OMODES and OFREQUENCY requests appear, the OMODES request takes precedence.
2. If this command is not specified in the Case Control Section (or, in the case of real eigenvalue, buckling, and complex eigenvalue analyses, if neither the OMODES nor the OFREQUENCY request is specified), then output will be generated for all frequencies.
3. The number of solutions selected will always be equal to the number of quantities in the selected set. The closest values are used. This is accomplished by working in increasing order of magnitude in the OFREQ set, assigning the closest frequency value from the frequency set for each value in the OFREQ set, and if already previously assigned, then assigning the next closest value. Thus, for example, if the frequency set is 2.5; 3.5; ...; 14.5; 15.5, and the OFREQ set is 3.8; 17.4; 21.7, the program will select 3.5; 15.5; 14.5 from the frequency set in that order, and will sort to end up with 3.5; 14.5; 15.5.
4. In flutter analysis (SOL 145), the selected set refers to the imaginary part of the complex eigenvalues. The physical interpretation of this quantity depends on the method of flutter analysis as follows:
   - K- or KE-method: Velocity (input units).

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- PK-method: Frequency.

5. In aeroelastic response analysis (SOL 146) with RLOAD selection, the selected set refers to the frequency (cycles per unit time).
6. In complex eigenvalue analysis (SOLs 107 and 110), the selected set refers to the imaginary part of the complex eigenvalues.
7. In the modal solution sequences, if this command is specified in more than one subcase, then it is recommended that the first subcase contain OFREQ=ALL and subsequent subcases contain OFREQ=n. Instead of OFREQ=ALL, a super-set of all OFREQ sets to follow may be used in the first subcase (see exceptions described in Remark 8). Also, data recovery requests should be specified only in the subsequent subcases. For example:

```
SUBCASE 1
   OFREQ=ALL $ 0.0  through 0.5
SUBCASE 2
   SET 10=0.0 0.1 0.3
   OFREQ=10
   DISP=ALL
SUBCASE3
   SET 20=0.4 0.5
   OFREQ=20
   STRESS=ALL
```

8. An OFREQ selection in a given subcase is valid for all subsequent subcases of a given batch of frequency response subcases, until a new OFREQ selection is encountered. However, with SOL 200, when a new boundary condition and/or a new frequency set is encountered, a new batch of frequency response subcases starts, and a new OFREQ super-set (see Remark 7) needs to be entered with the first subcase of this new batch, if there was a previous definition of OFREQ.

Related Topics

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OLOAD

Applied Load Output Request

Requests the form and type of applied load vector output.

Format:

$$\text{OLOAD}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\right.\right.$$

$$\left.\left.\left[\begin{matrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{matrix}\right],\left[\begin{matrix}\underline{\text{RPRINT}}\\ \text{NORPRINT}\end{matrix}\right],[\text{RPUNCH}]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
OLOAD=ALL
OLOAD(SORT1,PHASE)=5
```

Describers:

Describer	Meaning
SORT1	Output will be presented as a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output will be presented as a tabular listing of frequency or time for each grid point.
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.

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Describer	Meaning
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output.
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
PSDF	Requests the power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 10.
ATOC	Requests the autocorrelation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 10.
CRMS	Requests the cumulative root mean square function be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 10.
RMS	Requests the root mean square and zero crossing functions be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 10.
RALL	Requests all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 10.
RPRINT	Writes random analysis results to the print file. (Default) See Remark 10.
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 10.
RPUNCH	Writes random analysis results to the punch file. See Remark 10.
ALL	Applied loads for all points will be output. See Remark 2 and Remark 7.
NONE	Applied load for no points will be output.
n	Set identification of a previously appearing SET command. Only loads on points with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

1. Both PRINT and PUNCH may be requested.
2. The defaults for SORT1 and SORT2 depend on the type of analysis:
   - SORT1 is the default in static analysis, frequency response, steady state heat transfer analysis, real and complex eigenvalue analysis, flutter analysis, and buckling analysis. If SORT2 is selected in a frequency response solution for one or more of the commands ACCE, DISP, FORC, GPFO, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT2 format.

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- SORT2 is the default in transient response analysis (structural and heat transfer). SORT2 is not available for real eigenvalue (including buckling), complex eigenvalue, or flutter analysis. If SORT1 is selected in a transient solution for one or more of the commands ACCE, DISP, ENTH, FORC, GPFO, HDOT, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT1 format.
- XY plot requests will force SORT2 format thus overriding SORT1 format requests.

3. In a statics problem, a request for SORT2 causes loads at all requested points (zero and nonzero) to be output.
4. OLOAD=NONE overrides an overall output request.
5. In the statics superelement solution sequences, and in the dynamics SOLs 107 through 112, 118, 145, 146, and 200, OLOADs are available for superelements and the residual structure. Only externally applied loads are printed, and not loads transmitted from upstream superelements. Transmitted loads can be obtained with GPFORCE requests.
   - In the nonlinear transient analysis solution sequences SOLs 129 and 159, OLOADs are available only for residual structure points and include loads transmitted by upstream superelements.
6. Loads generated via the SPCD Bulk Data entry do not appear in OLOAD output.
7. In SORT1 format, OLOADs recovered at consecutively numbered scalar points are printed in groups of six (sextets) per line of output. But if a scalar point is not consecutively numbered, then it will begin a new sextet on a new line of output. If a sextet can be formed and it is zero, then the line will not be printed. If a sextet cannot be formed, then zero values may be output.
8. OLOAD results are output in the global coordinate system (see field CD on the GRID Bulk Data entry).
9. In inertia relief analysis, the OLOAD output includes both the inertia loads and applied loads.
10. The following applies to random solutions:
    - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, PUNCH, PLOT describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
    - The SORT1 and SORT2 describers only control the output format for the frequency response output. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
    - Any combination of the PSDF, ATOC, RMS, and CRMS describers can be selected. The RALL describer selects all four.
11. OLOAD output from SOL 401 does not include thermal loads.

Remarks for SOL 414

1. When you request OLOAD output for SOL 414,111 Harmonic Response solutions with a Nonlinear Preload subcase for cyclic symmetry sector, the entire applied load (OLOAD) of the whole structure is applied on the single cyclic symmetry sector.

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In SOL 414,111 Harmonic Response results, the applied load (OLOAD) is available in the recombined results only. In the recombined results, the applied load is correctly applied to all cyclic symmetry sectors.

Related Topics

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OMEGA

Rotational Speed Set Selection

Selects the set of rotational speeds to be solved in a SOL 414,110 complex modal analysis.

Format:

OMEGA=n

Examples:

```
OMEGA=17
```

Describers:

Describer	Meaning
n	Set identification number of OMEGA, OMEGA1, and OMEGA2 bulk entries. (Integer > 0)

Remarks:

1. All OMEGA, OMEGA1, and OMEGA2 bulk entries with the same set identification number are used.

   From the set of rotational speeds, the software automatically searches for duplicate values. The software considers rotational speeds to be duplicates if the rotational speeds satisfy the following inequality:

   $$|W_n - W_{n-1}| < DFREQ \cdot |W_{\max} - W_{\min}|$$

   where:

   W_n, W_{n-1} Adjacent rotational speeds when the rotational speeds are ordered according to magnitude

   $DFREQ$ Value for the DFREQ parameter (default = 10^{-5})

   $W_{\min}, W_{\max}$ The minimum and maximum rotational speeds in the set of rotational speeds

   When a duplicate is found, the software keeps the lower rotational speed in the pair and removes the higher rotational speed.

Related Topics

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OMODES

Output Modes Set

Selects a set of modes for output requests.

Format:

$$\text{OMODES} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \end{matrix} \right\}$$

Examples:

```
OMODES=ALL
OMODES=20
```

Describers:

Describer	Meaning
ALL	Output for all extracted modes will be computed. (Default)
n	Set identification of a previously appearing SET command. Output for those extracted modes appearing on this SET command will be computed.

Remarks:

1. OMODES is supported in SOLs 103, 105, 107, 110, 111, 112, 200, and 401. It is ignored in all other solution types.
2. In contrast to the OFREQENCY Case Control request, which affords an alternate way of selecting the modes to be output based on their frequencies, the OMODES command allows mode selection based on integer mode ID. For example:

```
SUBCASE 10
      …
      SET 11=1,3,5,7
      OMODES=11
      DISP=ALL
      …
SUBCASE 20
      …
      SET 21=25., 28., 31.
      OFREQ=21
```


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```
       DISP=ALL
…
```

The “THRU” option on the SET definition can be used with the OMODES command. For example:

```
SET 20 = 5 THRU 30
OMODES=20
```

3. If both the OMODES and the OFREQUENCY requests appear, the OMODES request takes precedence.
4. If neither the OMODES nor the OFREQUENCY request is specified, output will be generated for all modes.
5. It is important to note that the OMODES request has no effect whatsoever on the number of modes computed. Instead, it only provides a means for selecting a subset of the computed modes for which output is to be generated.
6. In superelement analysis, the set definition of an OMODES request for an upstream superelement will not be honored unless a similar request appears for its downstream neighbor. The downstream request should either be comprised of the union of all upstream requests, or left blank, as the OMODES default is ALL. Note that the code does not check to see if this condition is satisfied.
7. If the parameter SECOMB is set to “YES”, then OMODES must be the same for all superelements.
8. In SOL 401, for the cyclic and Fourier subcase types, a unique modal solution occurs for the requested harmonic numbers. As a result of the inherent symmetry with these solutions, mode pairs exist for specific harmonic numbers. For a cyclic subcase, mode pairs exist for harmonic numbers 1 through N/2 -1. For the Fourier subcase, mode pairs exist for all harmonic numbers > 0. The software automatically outputs the mode pairs for these subcase types for each mode number referenced by the OMODES command.

   For example, if you define OMODES=1, where SET 1 = 1,2,5,7 and you request 10 modes on the EIGRL bulk entry:

   For a cyclic subcase:

   For harmonic index 0 and N/2, 10 modes are computed. Modes 1, 2, 5, and 7 are output.

   For harmonic numbers 1 through N/2 -1, 20 modes are computed (10 distinct modes). Modes 1, 2, 3, 4, 9, 10, 13, and 14 are output (corresponding to the 1st, 2nd, 5th and 7th unique frequency).

   For a Fourier subcase:

   For harmonic index 0, 10 modes are computed. Modes 1, 2, 5, and 7 are output.

   For harmonic index > 0, 20 modes are computed (10 distinct modes). Modes 1, 2, 3, 4, 9, 10, 13, and 14 are output (corresponding to the 1st, 2nd, 5th and 7th unique frequency).

Related Topics

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OPRESS

Requests the output of pressure from SOL 401 and 402, and the Simcenter 3D Multiphysics environment.

For SOLs 401 and 402 including the Simcenter 3D Multiphysics environment, requests output of the fluid penetrating pressure load defined with the PLOADFP bulk entry.

For the Simcenter 3D Multiphysics environment running SOL 401, requests the output of pressure computed by the Simcenter 3D Thermal solver.

Format:

$$\text{OPRESS}\left[\left(\underline{\text{COUPLED}},\ \text{FPP},\ \left[\frac{\text{PRINT}}{\text{PLOT}}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \underline{\text{NONE}}\end{matrix}\right\}$$

Examples:

```
OPRESS=ALL
OPRESS(COUPLED)=ALL
OPRESS(FPP)=17
```

Describers:

Describer	Meaning
COUPLED	For the Simcenter 3D Multiphysics environment only, requests output of the pressures computed by the Simcenter 3D Thermal solver. (Default)
FPP	For SOLs 401 and 402 including the Simcenter 3D Multiphysics environment, requests output of the fluid penetrating pressure load defined with the PLOADFP bulk entry.
PRINT	Compute and write output to the print file (f06). (Default)
PLOT	Compute output.
ALL	Requests output for all elements.
n	Set identification number of a previously appearing SET command. Only elements with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed. (Default)

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Remarks:

1. You can use the FPP describer to request output of the fluid penetrating pressure defined with the PLOADFP entry. The FPP describer is supported with SOLs 401 and 402, and with SOL 401 within the Simcenter 3D Multiphysics environment.
2. In the Simcenter 3D Multiphysics environment, you can optionally request that the pressures computed by the Simcenter 3D Thermal solver be passed to the SOL 401 structural solution as a mechanical loading. These pressures are passed to Simcenter Nastran directly and not written to the Simcenter Nastran input file.

   You can use the COUPLED describer to request output of these applied pressure loads as a result of this data exchange with the Simcenter 3D Thermal solver. The COUPLED describer is only supported with SOL 401 in the context of a coupled structural-thermal analysis within the Simcenter 3D Multiphysics environment.
3. The OPRESS command output does not include pressures defined with the PLOAD, PLOAD2, PLOAD4, PLOADE1, and PLOADX1 entries. You can use the OLOAD case control command to request the output of all applied loads.
4. Only SORT1 data is output.

Related Topics

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OSTNINI

Requests initial strain output in SOL 401.

Requests initial strain output in SOL 401.

Format:

$$\text{OSTNINI}\left[\left(\left[\begin{matrix}\underline{\text{GRID}}\\ \text{GAUSS}\\ \text{BOTH}\end{matrix}\right], \underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
OSTNINI=ALL
OSTNINI(PRINT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
GRID	Output at grid points on elements.
GAUSS	Output at gauss points on elements.
BOTH	Output at both grid and gauss points on elements.
ALL	Requests output for all elements.
n	Set identification number of a previously appearing SET command. The SET command lists element IDs. (Integer>0)
NONE	No output requested.

Remarks:

1. Only supported in SOL 401.
2. The OSTNINI command should be defined above the subcases (globally). The output only occurs once, at the beginning of the solution before any subcase solutions begin.
3. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

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4. Initial strains are written to the .op2 file in the OES data block in SORT1 format.
5. The strain components are output in the basic coordinate system.

Related Topics

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OTEMP

Requests temperature output on grid points for SOLs 401, 402, and 414.

Requests temperature output on grid points.

Format:

$$\text{OTEMP}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{array}{c}\text{ALL}\\ \text{n}\\ \text{NONE}\end{array}\right\}$$

Examples:

```
OTEMP=ALL
OTEMP(PRINT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (.f06). (Default)
PUNCH	Compute and write output to the punch file (.pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

Remarks:

1. For SOL 401, OTEMP is supported in the following subcase types: ANALYSIS=STATICS, DYNAMICS and PRELOAD. It is not supported in the modal subcase types (ANALYSIS=MODES, CYCMODES and FOURIER).

   For SOL 402, OTEMP is only supported in a static subcase (ANALYSIS=STATICS).
2. Both PRINT and PUNCH may be requested.
3. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.

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4. For SOL 414,103 Eigenvalues and Superelement Reduction with a Nonlinear Preload subcase and SOL 414,129 Transient Response solutions, you can request output using OTEMP if TEMP(LOAD) and TEMP(INIT) are defined in the solution.

Related Topics

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OTIME

Output Time Set

Selects a set of times for output requests.

Format:

$$\text{OTIME} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \end{matrix} \right\}$$

Examples:

```
OTIME=ALL
OTIME=15
```

Describers:

Describer	Meaning
ALL	Output for all times will be computed.
n	Set identification number of a previously appearing SET command. Output for times closest to those given on this SET command will be computed. (Integer>0)

Remarks:

1. If the OTIME command is not supplied in the Case Control Section, then output for all times will be computed.
2. This command is particularly useful for requesting a subset of the output (e.g., stresses at only peak times, etc.).
3. This command can be used in conjunction with the MODACC module to limit the times for which modal acceleration computations are performed.
4. If this command is specified in more than one subcase in the modal solution sequences, then it is recommended that the first subcase contain OTIME=ALL and subsequent subcases contain OTIME=n. Also, data recovery requests should be specified only in the subsequent subcases. For example:

```
SUBCASE 1
    OTIME ALL
SUBCASE 2
    OTIME=10
    SET10=. . .
```


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```
    DISP=ALL
SUBCASE 3
    OTIME=20
    SET 20=. . .
    STRESS=ALL
```

5. The OTIME command is not effective in nonlinear transient analysis.
6. In superelement analysis, the set definition of an OTIME request for an upstream superelement will not be honored unless a similar request appears for its downstream neighbor. The downstream request should either be comprised of the union of all upstream requests, or left blank, as the OTIME default is ALL. Note that the program does not check to see if this condition is satisfied.

Related Topics

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OTMFORC

FRFOTM Load Results Output Request

Requests output of force and moment results from FRFOTM bulk entries.

Format:

$$\text{OTMFORC}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{REAL}}\ \text{or}\ \underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{matrix}\right],\ \text{PUNCH}\right)\right] = \left\{\begin{matrix}n\\ \underline{\text{ALL}}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
OTMFORC=ALL
OTMFORC(PUNCH,PHASE)=100
```

Describers:

Describer	Meaning
SORT1	Present force and moment output as a tabular listing of grid or scalar points for each frequency. (Default)
SORT2	Present force and moment output as a tabular listing of frequency for each grid or scalar point.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Default)
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
PRINT	Write results to the print (.f06) file. (Default)
NOPRINT	Do not print the results.
PUNCH	Write results to the standard punch (.pch) file.
n	Output force and moment results for the grid and scalar points listed on the SET case control command that has set identification number *n*. (Integer>0)

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Describer	Meaning
ALL	Output force and moment results for the grid and scalar points listed on all the SET case control commands specified in and above the current subcase. (Default)
NONE	Do not output force and moment results. This is useful to turn off OTM force and moment output for a specific subcase.

Remarks:

1. If a grid or scalar point is listed on a SET case control command for which output is requested, the output for the grid or scalar point depends on whether the grid or scalar point is designated as a load on an FRFOMAP bulk entry:
   - If the grid or scalar point is designated as a load on an FRFOMAP bulk entry, the value from the FRFOTM calculation is output.
   - If the grid or scalar point is not designated as a load on an FRFOMAP bulk entry, zero is output.
2. OTMFORC commands can be entered on the subcase level.

Related Topics

Browse Simcenter Nastran documentation

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OUTMGT

XY Plot Selection

Selects XY plot requests in a SOL 402, SOL 414,103 Eigenvalues and Superelement Reduction with a Nonlinear Preload subcase, SOL 414,111 Harmonic Response, or a SOL 414,129 Transient Response solution.

Format:

OUTMGT=n

Examples:

```
OUTMGT=17
```

Describers:

Describer	Meaning
n	Set identification number of OUTMGT bulk entries. (Integer > 0)

Remarks:

1. The OUTMGT case control command is not related to or used with the legacy Simcenter Nastran X-Y Plot commands. Likewise, the legacy Simcenter Nastran X-Y Plot commands are not used with the OUTMGT plotting capability.
2. All OUTMGT bulk entries with the same set identification number are used.

Related Topics

Browse Simcenter Nastran documentation

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OUTPUT

Case Control Delimiter

Delimits the various types of commands for the structure plotter, curve plotter, grid point stress, and MSGSTRESS.

Format:

$$\text{OUTPUT}\left[\left(\left\{\begin{array}{c}\text{PLOT}\\ \text{POST}\\ \text{XYOUT}\\ \text{XYPLOT}\\ \text{CARDS}\end{array}\right\}\right)\right]$$

Examples:

```
OUTPUT
OUTPUT(PLOT)
OUTPUT(XYOUT)
```

Describers:

Describer	Meaning
PLOT	Beginning of the structure plotter request. This command must precede all structure plotter control commands.
POST	Beginning of grid point stress SURFACE and VOLUME commands. This command must precede all SURFACE and VOLUME commands.
XYOUT or XYPLOT	Beginning of curve plotter request. This command must precede all curve plotter control commands. XYPLOT and XYOUT are entirely equivalent. Curve plotter commands are described in **"X-Y PLOT Commands"** .
CARDS	The OUTPUT(CARDS) packet is used by the MSGSTRESS program. These commands have no format rules. This package must terminate with the command ENDCARDS (starting in column 1).

Remarks:

1. The structure plotter request OUTPUT(PLOT), the curve plotter request OUTPUT(XYOUT or XYPLOT), and the grid point stress requests (OUTPUT(POST)) must follow the standard Case Control commands.

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2. If OUTPUT is specified without a describer, then the subsequent commands are standard Case Control commands.
3. Case Control commands specified after OUTPUT(POST) are SURFACE and VOLUME.

Related Topics

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P2G

Direct Input Load Matrix Selection

Selects direct input load matrices.

Format:

P2G=name

Examples:

```
P2G=LDMIG
P2G=L1, L2, L3
SET 100=LDMIG, L1, L8
P2G=100
P2G=1.25*L1, 1.0*L2, 0.75*L3
```

Describers:

Describer	Meaning
name	Name of a [P^2_g] matrix that is input on the DMIG Bulk Data entry. (Character) Scale factors may be included (see Remarks 4 and 5). See "Applied Static Loads in SubDMAPs SELG, SELR, and SELRRS" in the *Simcenter Nastran User's Guide*.

Remarks:

1. DMIG matrices will not be used unless selected.
2. Terms are added to the load matrix before any constraints are applied.
3. The matrix must be columnar in form (field 4 on the DMIG entry must contain the integer 9).
4. The associated DMIG matrices can be scaled using either in-line scale factors on P2G (for example, P2G =1.25*L1), using the parameter CP2 (for example, PARAM,CP2,1.25), or both. See **"Parameter Descriptions"**.
5. Multiple matrices separated by a comma or a blank are additive. When multiple matrices and in-line scale factors are used together, each matrix name in the list must include a scale factor. A scale factor of 1.0 should be used for matrices in the list which are not scaled. For example,

   ```
   P2G=1.25*L1, 1.0*L2, 0.75*L3
   ```

   The parameter CP2 when defined will scale all P2G. For example, if PARAM,CP2,1.30 is defined with the P2G example above, the result would be P2G=1.30(1.25*L1 + 1.0*L2 + 0.75*L3).

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6. The P2G command is supported across subcases. A P2G command selecting a different DMIG matrix can be defined for each subcase.
7. If you are defining the P2G command for a statics solution (SOL 101 or ANALYSIS=STATICS), each subcase must explicitly include a DMIG load definition. This is required whether the same loads are needed in each subcase or the loads are unique in each subcase. There are two options.
   - You can define a single P2G definition above the subcases (global definition), the value of the NCOL field on the single referenced DMIG bulk entry must be equal to the number of subcases, and a load column must be defined for each subcase.
   - You can define a P2G definition in every subcase. The DMIG entry referenced by the P2G command in each subcase must have a single column and NCOL=1. The single column on each DMIG entry should define the loads for that subcase.

Related Topics

Browse Simcenter Nastran documentation

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PANCON

Acoustic Contribution Output Request for Panels

Requests the form and type of acoustic contribution output for structural panels and the residual.

Format:

$$\text{PANCON}\left[\left(\begin{bmatrix}\underline{\text{SORT1}}\\ \underline{\text{SORT2}}\end{bmatrix},\ \begin{bmatrix}\underline{\text{REAL}}\ \text{or}\ \underline{\text{IMAG}}\\ \text{PHASE}\end{bmatrix},\ \begin{bmatrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{bmatrix},\ \text{PUNCH}\right],\ \begin{bmatrix}\underline{\text{ABS}}\\ \text{NORM}\\ \text{BOTH}\end{bmatrix},\right.$$

$$\left.\text{TOPP}=\begin{Bmatrix}\text{pp}\\ \underline{\text{ALL}}\end{Bmatrix},\ \text{SOLUTION}=\begin{Bmatrix}\underline{\text{ALL}}\\ \text{setf}\\ \text{PEAKOUT}\end{Bmatrix},\ \text{PANEL}=\begin{Bmatrix}\underline{\text{ALL}}\\ \text{setp}\end{Bmatrix}\right)\Bigg]$$

$$=\begin{Bmatrix}\text{n}\\ \underline{\text{ALL}}\\ \text{NONE}\end{Bmatrix}$$

Examples:

```
PANCON=123
PANCON(SORT1,PHASE,PRINT,PUNCH,BOTH,TOPP=5)=ALL
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of panels for each frequency.
SORT2	Output a tabular listing of frequency for each panel. (Default)
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
PRINT	Write acoustic panel contribution output to the print (.f06) file. (Default)
NOPRINT	Do not print the acoustic panel contribution results.
PUNCH	Write acoustic panel contribution output to the standard punch (.pch) file.
ABS	Output acoustic panel contributions in absolute terms. (Default)

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Describer	Meaning
NORM	Output acoustic panel contributions in normalized terms.
BOTH	Output acoustic panel contributions in both absolute and normalized terms.
TOPP = ALL	List all of the structural panels in the output. Sort the output in descending order from the structural panel that has the greatest contribution. See Remark 2.
TOPP = pp	The number of structural panels to list in the output that have the greatest contribution to the response at each frequency. If pp > 1, the output is sorted in descending order from the structural panel that has the greatest contribution. If pp = 0, no structural mode contributions are output, only totals. See Remark 2. (Integer ≥ 0; Default pp = 5)
SOLUTION = ALL	Perform the contribution calculations at all frequencies defined by FREQUENCY case control commands. See Remark 12. (Default)
SOLUTION = setf	Perform the contribution calculations only at the frequencies listed on a previously appearing SET case control command with setf as the identification number. See Remark 12. (Integer > 0)
SOLUTION = PEAKOUT	Perform the contribution calculations only at the frequencies identified with the PEAKOUT command. See Remark 12.
PANEL = ALL	Output contributions for all panels defined by PANEL bulk entries. (Default)
PANEL = setp	Output contributions for the panels listed on a previously appearing SET case control command with setp as the identification number. (Integer > 0)
n	Calculate panel contributions for the responses listed on a previously appearing SETMC case control command with n as the identification number. Any response listed in this SETMC case control command that is not an acoustic response is ignored. (Integer > 0)
ALL	Calculate panel contributions for all the responses listed on the SETMC case control commands specified in and above the current subcase. Any response listed in these SETMC case control commands that is not an acoustic response is ignored. (Default)
NONE	Do not calculate panel contributions.

Remarks:

1. SOL 108 and 111 are supported.
2. For SOL 108 and SOL 111, PANCON is supported by the FEM Adaptive Order (FEMAO) method with the exception of the number of structural panels (TOPP describer).
3. Both PRINT and PUNCH can be requested.
4. PANCON = NONE overrides an overall output request. Thus, to turn off contribution output for a specific subcase, place PANCON = NONE in the subcase.
5. PANCON supports results for microphone points.
6. The parameters LFREQ, LFREQFL, HFREQ, HFREQFL, LMODES, and LMODESFL are supported.

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7. The SOLUTION and PANEL describers can be abbreviated to SOLU and PANE, respectively.
8. The SET case control command referenced by SOLUTION = setf must contain real values for frequencies. Using integer values may lead to unintended results.
9. The SET case control command referenced by PANEL = setp must contain the alphanumeric name of panels defined by PANEL bulk entries.
10. The SOLUTION describer is only valid in a SOL 108 or 111 frequency response analysis.
11. When an OFREQ case control command is present, the SOLUTION describer is used as follows:
    - If SOLUTION=ALL, output results for all frequencies specified by the OFREQ command.
    - If SOLUTION=setf, output results for all frequencies specified by the OFREQ command that are listed on a previously appearing SET command with setf as the identification number.
    - If SOLUTION=PEAKOUT and a PEAKOUT case control command is also present in the input file, output results for all frequencies specified by the OFREQ command that satisfy the PEAKOUT filtering criteria.
12. The SOLUTION describer is not valid for superelements or frequency-dependent components.

Related Topics

Browse Simcenter Nastran documentation

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PARAM

Parameter Specification

Specifies values for parameters. Parameters are described in **Parameters and Parameter Applicability Tables**.

Format:

PARAM,n,V1,V2

Examples:

```
PARAM,GRDPNT,0
PARAM,K6ROT,1.0
```

Describers:

Describer	Meaning
n	Parameter name (1 to 8 alphanumeric characters, the first of which is alphabetic).
V1, V2	Parameter value based on parameter type, as follows:

Type	V1	V2
Integer	Integer	Blank
Real, single precision	Real	Blank
Character	Character	Blank
Real, double precision	Real, Double Precision	Blank
Complex, single precision	Real or Blank	Real or Blank
Complex, double precision	Real, Double Precision	Real, Double Precision

Remarks:

1. The PARAM command is normally used in the Bulk Data Section and is described in **Bulk Data Entries**.
2. The parameter values that may be defined in the Case Control Section are described in **Parameters and Parameter Applicability Tables**. Case Control PARAM commands in user-written DMAPs require the use of the PVT module, described in the *Simcenter Nastran DMAP Programmer's Guide*.
3. The following should be considered when external superelements or part superelements are included in the Bulk Data section of your input file.

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- Parameters specified in the Bulk Data, but not in a BEGIN SUPER portion of the Bulk Data, apply only to the residual and not to external or part superelements.
- Parameters specified in a BEGIN SUPER portion of the Bulk Data apply only to that external or part superelement.
- To ensure that parameters apply to the residual and all superelements, you should specify them in the Case Control section above the subcases. For example, specifying the PARAM,POST setting above the subcases will request a consistent output format for the residual and the superelements.

Related Topics

Browse Simcenter Nastran documentation

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PARTN

Partitioning Vector Specifications

Specifies a list of grid point identification numbers that will be partitioned with the DMAP module MATMOD (Option 17). In SOLs 111 and 200, the PARTN command specifies the points at which modal participation factors are to be computed.

Format:

PARTN=n

Examples:

```
PARTN=10
```

Describers:

Describer	Meaning
n	Set identification number of a previously appearing SET command. Only modal participation factors for points with identification numbers that appear on the SET=n command will be output. (Integer>0)

Remarks:

1. The PARTN command and the DMAP module MATMOD provide a convenient method for building a partitioning vector for use in the DMAP modules such as PARTN and MERGE.
2. Modal participation factors are computed automatically in SOLs 111 and 200 when coupled fluid-structure analysis is performed. See the *Simcenter Nastran User's Guide*.

Related Topics

Browse Simcenter Nastran documentation

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PEAKOUT

Peak Response Frequency Set

Generates a set of frequencies where response peaks occur from SOL 108 and 111 frequency response analysis results.

Format:

$$\text{PEAKOUT} = \text{n}$$

Examples:

```
PEAKOUT=100
```

Describers:

Describer	Meaning
n	Set identification number of PEAKOUT bulk entries. (Integer > 0)

Remarks:

1. A single PEAKOUT case control command can be placed above the subcases and a single PEAKOUT case control command can be placed in each frequency response subcase.
2. When a PEAKOUT case control command is placed above the subcases, it is applied to all the subcases. When a PEAKOUT case control command is placed above the subcases and PEAKOUT case control commands are placed inside subcases, the PEAKOUT case control commands inside the subcases take precedence.
3. A PEAKOUT case control command can reference multiple PEAKOUT bulk entries that have the same identification number. When this occurs, the set of frequencies at which peak responses occur for a DOF is the union of the peak responses that are identified by each PEAKOUT bulk entry for the DOF.

   For example, suppose that a PEAKOUT case control command references two PEAKOUT bulk entries. If 100 Hz, 145 Hz, 176 Hz, and 232 Hz satisfy the criteria of one PEAKOUT bulk entry for a given DOF, and 85 Hz, 133 Hz, 176 Hz, and 210 Hz satisfy the criteria of the other PEAKOUT bulk entry for the same DOF, the set of frequencies at which peak responses occur for the DOF is 85 Hz, 100 Hz, 133 Hz, 145 Hz, 176 Hz, 210 Hz, and 232 Hz.

Related Topics

Browse Simcenter Nastran documentation

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PFRESULTS

Progressive Failure Output Request for SOLs 401 and 402

Requests progressive failure results output for composite shell and solid elements.

Format:

$$\text{PFRESULTS}\left[\left(\underline{\text{DAMAGE}},\text{STATUS},\text{ENERGY},\text{CRKDSTY},\left[\begin{matrix}\underline{\text{PRINT}}\\ \text{PUNCH}\\ \text{PLOT}\end{matrix}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
PFRESULTS(DAMAGE,ENERGY,PLOT)=ALL
```

Describers:

Describer	Meaning
DAMAGE	Requests damage value results. See Remark 1. (Default)
STATUS	Requests damage status. See Remark 3.
ENERGY	Requests damage energy results. See Remark 4.
CRKDSTY	Requests crack density.
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
PLOT	Generates results for requested set but no printer output.
ALL	Results are output for all composite solid elements.
n	Identification number of a SET case control command that lists the composite solid elements for which results are output. (Integer>0)
NONE	No results are output for any composite solid elements.

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Remarks:

1. Damage values are calculated at the middle of each ply, on the Gauss point locations, then extrapolated to the corner grid points. At the Gauss points, the calculated damage can range from 0.0 and 0.999. A damage value of 0.999 indicates that the elastic modulus is reduced to 0.1% of its original value. The software limits the damage to 0.999 to avoid numerical problems that can arise if the damage is 1.0. However, because the values at the Gauss points are extrapolated to the corner grids, the reported damage values at the corner grids can be less than 0.0 and greater than 0.999.
2. Crack density is calculated at the middle of each ply, on the Gauss point locations, then extrapolated to the corner grid points. The software outputs a negative crack density as zero.
3. Damage status is reported on an individual element basis as follows:

   "0" indicates that the laminate is undamaged.

   "1" indicates that some plies in the laminate are damaged, but have not failed.

   "2" indicates that some plies in the laminate have failed and other plies in the laminate may be damaged.

   "3: indicates that all plies in the laminate have failed.

   A ply has failed when the software calculates that the ply is damaged at all Gauss points. A ply is damaged when the software calculates that the ply is damaged at one or more, but not all of the Gauss points.
4. Damage energy is the energy that is expended to cause the damage in the laminate. Damage energy is output on an element-by-element basis.

Remarks related to SOL 402:

1. STATUS and ENERGY describers are ignored.

Related Topics

Browse Simcenter Nastran documentation

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PLOTID

Plotter Identification

Defines a character string that will appear on the first frame of any plotter output.

Format:

PLOTID=title

Examples:

```
PLOTID=BLDG. 125 BOX 91
```

Describers:

Describer	Meaning
title	Any character string.

Remarks:

1. PLOTID must appear before the OUTPUT(PLOT) or OUTPUT(XYOUT) commands.
2. The presence of PLOTID causes a special header frame to be plotted with the supplied identification plotted several times. The header frame allows plotter output to be identified easily.
3. If no PLOTID command appears, no ID frame will be plotted.
4. The PLOTID header frame will not be generated for the table plotters.

Related Topics

Browse Simcenter Nastran documentation

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PLSTRN

Requests plastic strain output at grid points for SOLs 401 and 402.

Requests plastic strain at grid points.

Format:

$$\text{PLSTRN}\left[\left(\underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \left\{\begin{array}{c}\text{ALL}\\ \text{n}\\ \text{NONE}\end{array}\right\}$$

Examples:

```
PLSTRN=ALL
PLSTRN(PRINT,PLOT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all grid points.
n	Set identification number of a previously appearing SET command. Only grid points with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

Remarks:

1. Both PRINT and PUNCH may be requested.
2. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.
3. For SOL 402, PLSTRN is not supported for CBEAM, CROD, and CBAR elements.
4. PLSTRN output includes the plastic strain tensor and the effective plastic strain $d\bar{\varepsilon}^{p}$. Note that the effective plastic strain is computed from the incremental plastic strain, and not the current plastic strain.

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5. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

Related Topics

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PRESSURE

Pressure Output Request for Fluid Grid Points

Requests the form and type of pressure output for fluid grid points.

Format:

$$\text{PRESSURE}\left[\left(\left[\frac{\underline{\text{SORT1}}}{\text{SORT2}}\right],\left[\frac{\underline{\text{PRINT}},\text{PUNCH}}{\text{PLOT}}\right],\left[\frac{\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}}{\text{PHASE}}\right],\left[\frac{\underline{\text{TOTAL}}}{\text{SCATR}}\right],\left[\begin{matrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{matrix}\right],\left[\frac{\underline{\text{RPRINT}}}{\text{NORPRINT}}\right],[\text{PUNCH}],\text{SOLUTION}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{setf}\\ \text{PEAKOUT}\end{matrix}\right\}\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
PRESSURE=5
PRESSURE(REAL)=ALL
PRESSURE(SORT2,PUNCH,REAL)=ALL
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence. (Default)
SORT2	Output a tabular listing of load, frequency or time for each grid point.
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
PLOT	Generate, but do not print, pressure results.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)

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Describer	Meaning
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
TOTAL	Output total pressure for the incident-scattered formulation. (Default)
SCATR	Output scattered pressure for the incident-scattered formulation.
PSDF	Requests power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 4.
ATOC	Requests autocorrelation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 4.
CRMS	Requests cumulative root mean square function be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 4.
RMS	Requests root mean square and zero crossing functions be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 4. (Default)
RALL	Requests all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 4.
RPRINT	Writes random analysis results to the print file. See Remark 4. (Default)
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 4.
RPUNCH	Writes random analysis results to the punch file. See Remark 4.
SOLUTION = ALL	Output acoustic pressure for all frequencies. See Remark 9. (Default)
SOLUTION = setf	Output acoustic pressure only for the frequencies listed on a previously appearing SET command with setf as the identification number. See Remark 9. (Integer > 0)
SOLUTION = PEAKOUT or PEAK	Output acoustic pressure only for frequencies identified with the PEAKOUT command. See Remark 9.
ALL	Output acoustic pressure for all points.
n	Output acoustic pressure only for the points listed on a previously appearing SET command with n as the identification number. (Integer > 0)
NONE	Do not output acoustic pressure for any points. (Default)

Remarks:

1. Both PRINT and PUNCH can be requested.

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2. By default, the PRESSURE case control command requests pressure output at fluid grid points only and the DISPLACEMENT case control command requests displacement at structural grid points only. In versions prior to NX Nastran 12, the PRESSURE and DISPLACEMENT case control commands are interchangeable. Use SYSTEM(640) to revert to the pre-NX Nastran 12 behavior.
3. PRESSURE = NONE overrides an overall output request.
4. The following applies to random solutions:
   - By default, frequency response results are not output. If you want both random output and frequency response output, specify SYSTEM(524) = 1 or RANFRF = 1 in the input file. The PRINT, PUNCH, and PLOT describers control the frequency response output. The RPRINT, NORPRINT, and RPUNCH describers control the random output.
   - The SORT1 and SORT2 describers control the output format for the frequency response output only. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
   - You can select any combination of the PSDF, ATOC, RMS, and CRMS describers. The RALL describer selects all four.
   - Autocorrelation (ATOC) calculations require the RANDT1 bulk entry.
5. Any structural points that you include in SET = n are ignored.
6. During a SOL 108 or SOL 111 frequency-dependent external superelement creation run, on the SET case control command that is referenced by the PRESSURE case control command, include any points at which you want to recover results during the system run.
7. The SOLUTION describer is only valid in a SOL 108 or 111 frequency response analysis.
8. When an OFREQ case control command is present, the SOLUTION describer is used as follows:
   - If SOLUTION=ALL, output results for all frequencies specified by the OFREQ command.
   - If SOLUTION=setf, output results for all frequencies specified by the OFREQ command that are listed on a previously appearing SET command with setf as the identification number.
   - If SOLUTION=PEAKOUT and a PEAKOUT case control command is also present in the input file, output results for all frequencies specified by the OFREQ command that satisfy the PEAKOUT filtering criteria.
9. The SOLUTION describer is not valid for superelements or frequency-dependent components.
10. To obtain pressure results at microphone points, reference the fluid grid ID attached to the microphone points.
11. Pressure output is supported when you model structures that contain incompressible fluids with the defined mass method. It is not supported when you use the virtual mass method.

Related Topics

Browse Simcenter Nastran documentation

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RANDOM

Random Analysis Set Selection

Selects the RANDPS, RANDPEX, and RANDT1 bulk entries to be used in random analysis.

Format:

RANDOM[(RPOSTS1 = i,RMSINT = j,RMSSF = r)] = n

Examples:

```
RANDOM=177
RANDOM(RPOSTS1=1,RMSINT=1,RMSSF=2.0)=123
```

Describers:

Describer	Meaning
RPOSTS1 = i	Specifies the output format for random results. (Integer; Default = 0) RPOSTS1 = 0 for SORT2 output format. RPOSTS1 = 1 for SORT1 output format.
RMSINT = j	Specifies the interpolation method for numerical integration when computing RMS response and the zero-mean crossing for the RMS von Mises stress from PSDF. (Integer; Default = 0) RMSINT = 0 for interpolation in Cartesian space. RMSINT = 1 for interpolation in log-log space.
RMSSF = r	Specifies a scaling factor for RMS and CRMS random results. (Real > 0.0; Default = 1.0)
n	Set identification number of RANDPS, RANDPEX, and RANDT1 bulk entries to be used in random analysis. (Integer>0; No default)

Remarks:

1. RANDOM must select RANDPS or RANDPEX bulk entries to perform random analysis.
2. If a describer is specified, it takes precedence over the corresponding parameter.
3. If RANDPS or RANDPEX bulk entries are used in a superelement analysis, the RANDOM command may be specified above the subcase level if a condensed subcase structure (SUPER=ALL) is used. If a condensed subcase structure is not used, then a unique RANDOM selection of a unique RANDPS entry must be specified within each of the desired superelement subcases.
4. How you organize subcases for a random analysis depends on whether you include ANALSIS = RANDOM in the subcases that contain the RANDOM command. For example, suppose that a structure

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is excited by two loads, and you want to evaluate the random response of the structure for two PSD functions using SOL 111.

Subcase organization using ANALYSIS = RANDOM

When using ANALYSIS = RANDOM in a subcase, place the subcase that contains the ANALSIS = RANDOM and RANDOM commands after the subcases that calculate the frequency responses.

```
SUBCASE 1
$
$ Subcase 1 calculates the normal modes
$
ANALYSIS=MODES
DISP=ALL
$
SUBCASE 2
$
$ Subcase 2 calculates the frequency response of the
$ structure to the loading specified by DLOAD 111 at
$ the frequencies specified by FREQUENCY set 13
$
FREQUENCY=13
DLOAD=111
$
SUBCASE 3
$
$ Subcase 3 calculates the frequency response of the
$ structure to the loading specified by DLOAD 211 at
$ the frequencies specified by FREQUENCY set 13
$
FREQUENCY=13
DLOAD=211
$
SUBCASE 4
$
$ Subcase 4 uses the frequency responses from Subcases
$ 2 and 3 to calculate the random response of the
$ structure for the PSD function specified by RANDOM 100.
$ RANDOM 100 references RANDPS and RANDPEX bulk entries
$ with SID 100.
$
ANALYSIS=RANDOM
RANDOM=100
$
SUBCASE 5
$
$ Subcase 5 uses the frequency responses from Subcases
$ 2 and 3 to calculate the random response of the
$ structure for the PSD function specified by RANDOM 200.
$ RANDOM 200 references RANDPS and RANDPEX bulk entries
$ with SID 200.
$
ANALYSIS=RANDOM
RANDOM=200
```


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Subcase organization without using ANALYSIS = RANDOM

When not using ANALYSIS = RANDOM in a subcase, place the subcase that contains the RANDOM command in the first subcase that calculates a frequency response. Each random spectrum requires a unique frequency identification number, even if you want to evaluate the random response over the same set of frequencies. Thus, in this example, FREQUENCY=13 and FREQUENCY=23 reference FREQi bulk entries that contain the same set of frequencies.

```
SUBCASE 1
$
$ Subcase 1 calculates the normal modes
$
ANALYSIS=MODES
DISP=ALL
$
SUBCASE 2
$
$ Subcase 2 calculates the frequency response of the
$ structure to the loading specified by DLOAD 111 at
$ the frequencies specified by FREQUENCY set 13 and
$ requests the software calculate the random response
$ of the structure for the PSD function specified by
$ RANDOM 100. RANDOM 100 references RANDPS and RANDPEX
$ bulk entries with SID 100.
$
RANDOM=100
FREQUENCY=13
DLOAD=111
$
SUBCASE 3
$
$ Subcase 3 calculates the frequency response of the
$ structure to the loading specified by DLOAD 211 at
$ the frequencies specified by FREQUENCY set 13
$
FREQUENCY=13
DLOAD=211
$
SUBCASE 4
$
$ Subcase 4 recalculates the frequency response of the
$ structure to the loading specified by DLOAD 111 at
$ the frequencies specified by FREQUENCY set 23 and
$ requests the software calculate the random response
$ of the structure for the PSD function specified by
$ RANDOM 200. RANDOM 200 references RANDPS and RANDPEX
$ bulk entries with SID 200.
$
RANDOM=200
FREQUENCY=23
DLOAD=111
$
SUBCASE 5
$
```


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```
$ Subcase 5 recalculates the frequency response of the
$ structure to the loading specified by DLOAD 211 at
$ the frequencies specified by FREQUENCY set 23
$
FREQUENCY=23
DLOAD=211
```

Notice that in this subcase organization, the software must recalculate the frequency responses for each PSD function. Thus, to minimize the number of calculations that the software must perform, use the subcase organization that includes ANALYSIS = RANDOM.

Related Topics

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RCROSS

Cross-Power Spectral Density and Cross-correlation Function Output Requests

Requests computation and output of cross-power spectral density and cross-correlation functions in random analysis.

Format:

$$\text{RCROSS}\left[\left(\left[\begin{matrix}\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{matrix}\right],\ [\text{PUNCH}],\ \left[\begin{matrix}\underline{\text{PSDF}}\\ \text{CORF}\\ \text{RALL}\end{matrix}\right]\right)\right] = \text{n}$$

Examples:

```
RCROSS(PHASE)=10
RCROSS(PSDF,NOPRINT,PUNCH)=20
```

Describers:

Describer	Meaning
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output (for cross-power spectral density function). Use of either REAL or IMAG yields the same output. (Default)
PHASE	Requests polar format (magnitude and phase) of complex output (for cross-power spectral density function). Phase output is in degrees.
PRINT	Writes output to print file. (Default)
NOPRINT	Does not write output to the print file.
PUNCH	Writes output to punch file.
PSDF	Requests the cross-power spectral density function be calculated and output for random analysis post-processing. (Default)
CORF	Requests the cross-correlation function be calculated and output for random analysis post-processing.
RALL	Requests both the cross-power spectral density and cross-correlation function be calculated and output for random analysis post-processing.
n	Identification number of RCROSS and RCROSSC bulk data entries to be used in random analysis. (Integer > 0)

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Remarks:

1. The case control RCROSS entry must be used in conjuction with the case control RANDOM entry. See remarks under the RANDOM case control entry.
2. Response quantities, such as DISPLACEMENT, STRESS, and FORCE, must be requested by corresponding case control entries in order to compute cross-power spectral density and cross-correlation functions between the two response quantities specified by the RCROSS bulk data entry. It is recommended that the DISPLACEMENT, STRESS, and FORCE requests be put above the subcase level to ensure that these response quantities exist when the random analysis post-processing occurs.
3. The response quantities must belong to the same superelement. The cross-power spectral density and cross-correlation functions between two responses that belong to different superelements are not supported.

Related Topics

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REPCASE

Repeat Output Subcase Delimiter

Delimits and identifies a repeated output subcase.

Format:

REPCASE=n

Examples:

```
REPCASE=137
```

Describers:

Describer	Meaning
n	Subcase identification number. (Integer>1)

Remarks:

1. n must be strictly increasing (i.e., must be greater than all previous subcase identification numbers).
2. REPCASE defines a subcase that is used to make additional output requests for the previous real subcase. This command is required because multiple output requests for the same item are not permitted in the same subcase.
3. REPCASE may only be used in statics and normal modes problems. However, in normal modes, only output for the first mode is obtained. This feature is not supported in SOLs 106 or 153.
4. One or more repeated subcases (REPCASEs) must immediately follow the subcase (SUBCASE) to which they refer.
5. If the referenced subcases contain thermal loads or element deformations, the user must define the temperature field in the REPCASE with a TEMP(LOAD) Case Control command or the element deformation state with a DEFORM command.

Related Topics

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RESVEC

Residual Vector Request

Used to control the computation of residual vectors.

Format:

$$\text{RESVEC}\left[\left(\left[\frac{\text{APPLOD}}{\text{NOAPPL}}\right],\ \left[\frac{\text{RVDOF}}{\text{NORVDO}}\right],\ \left[\frac{\text{RVEL}}{\text{NORVEL}}\right],\ \left[\frac{\text{INRLOD}}{\text{NOINRL}}\right],\ \left[\frac{\text{DAMPLOD}}{\text{NODAMP}}\right],\ \left[\frac{\text{DYNRSP}}{\text{NODYNRSP}}\right]\right)\right] = \left\{\begin{array}{c}\text{SYSTEM}\\ \text{NOSYSTEM}\\ \text{COMPONENT}\\ \text{NOCOMPONENT}\\ \text{BOTH or YES}\\ \text{NO}\end{array}\right\}$$

Examples:

```
RESVEC = SYSTEM
RESVEC(NOINRL) = COMPONENT
RESVEC = YES
```

Describers:

Describer	Meaning
APPLOD	Compute residual vectors for applied loads. See Remark 5. (Default)
NOAPPL	Do not compute residual vectors for applied loads.
RVDOF	Compute residual vectors for degree-of-freedom included in RVDOF and RVDOF1 bulk entries. See Remark 6. (Default)
NORVDO	Do not compute residual vectors for degree-of-freedom included in RVDOF and RVDOF1 bulk entries.
RVEL	Compute residual vectors for degree-of-freedom defining CBUSHi, CDAMPi, CELASi, and CVISC elements referenced by a RVEL bulk entry. See Remarks 7 and 9. (Default)

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Describer	Meaning
NORVEL	Do not compute residual vectors for degree-of-freedom defining CBUSHi, CDAMPi, CELASi, and CVISC elements referenced by a RVEL bulk entry.
INRLOD	Compute residual vectors for rigid body inertia loads. See Remark 4. (Default)
NOINRL	Do not compute residual vectors for inertia loads.
DAMPLOD	Compute residual vectors for the degree-of-freedom in which viscous damping is defined on the CBUSHi, CDAMPi, and CVISC elements. See Remark 9. (Default)
NODAMP	Do not compute residual vectors for the degree-of-freedom in which viscous damping is defined on the CBUSHi, CDAMPi, and CVISC elements.
DYNRSP	Include the dynamic effect of residual vector modes in the forced response solution. See Remark 8. (Default)
NODYNRSP	Do not include the dynamic effect of residual vector modes in the forced response solution.
The defaults for the following describers are solution dependent. The defaults apply even when the RESVEC case control entry is undefined. See Remark 10.	
SYSTEM	Request residual vectors for residual structure degree-of-freedom only (a-set).
NOSYSTEM	Do not compute residual vectors for residual structure degree-of-freedom (a-set).
COMPONENT	Request residual vectors for superelement degree-of-freedom only (o-set). (Default for SOLs 103, 106 with PARAM, NMLOOP=0, 110, 115, 153 with PARAM, NMLOOP=0, 187, and 200 with ANALYSIS=MODES)
NOCOMPONENT	Do not compute residual vectors for superelement degree-of-freedom (o-set).
BOTH (or YES)	Request residual vectors for both residual structure DOF (a-set) and for superelement DOF (o-set). (Default for SOLs 111, 112, 118, 146, and 200 with ANALYSIS=MCEIG, MTRAN, or MFREQ)
NO	Turns off calculation of residual vectors.

Remarks:

1. If the RESVEC case control command is present, the parameters RESVEC and RESVINER are ignored.
2. A unique RESVEC case control definition can be defined in different superelement subcases. You can use the NOSYSTEM and NOCOMPONENT describers in a superelement subcase to turn off a setting which may have been defined globally.
3. If a RESVEC case control command is present in a cold start analysis, then only the RESVEC case control command can be used in a restart. Similarly, if parameter RESVEC/RESVINER is used in a cold start, then only the same can be used in a restart.
4. INRLOD designates that inertia load residual vectors are to be calculated. The software automatically generates and applies inertia loads for each of the six (three translational and three rotational)

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rigid body motions. This results in a residual vector computed for each of the six loads. INRLOD is equivalent to PARAM,RESVINER,YES.

5. APPLOD designates that the applied load residual vectors are to be calculated. There are two input scenarios to determine the applied loads for computing residual vectors.

   a. The load set IDs selected with the EXCITEID field on all RLOAD1, RLOAD2, TLOAD1, TLOAD2, ACSRCE, and SELOAD entries in the bulk data are processed. No DLOAD case control is required in this case. If a LOADSET case control command exists, these are all ignored.

   b. If a LOADSET case control command exists, and it selects an LSEQ bulk entry, the load set ID selected in the LID field in the LSEQ bulk entry is processed.

6. The operation of the residual vector calculation with RVDOF is functionally equivalent to PARAM,RESVEC,YES when USET, U6 DOF are present. The RVDOF/RVDOF1 bulk entries can select both a-set and o-set DOF. This is different than the USET, U6 capability which requires USET, U6 to select a-set DOF, but requires SEUSET, U6 to select o-set DOF.

   The unit loads applied to the interior points of a superelement due to RVDOFi bulk entries are passed downstream to the residual for the purpose of residual vector processing by all superelements in its downstream path. This produces more accurate results as compared to the results produced when USETi,U6 or SEUSETi,U6 bulk entries are used for residual vector processing. When USETi,U6 or SEUSETi,U6 bulk entries are used, unit loads on a superelement are not passed downstream for residual vector processing by the downstream superelements.

7. If a CBUSHi, CDAMPi, CELASi, and CVISC element is grounded, the unit load is applied to only the ungrounded end.

8. DYNRSP/NODYNRSP is only applicable to modal forced response solutions (SOLs 111, 112, 146, and 200 when ANALYSIS=MFREQ or MTRAN). For models containing superelements, DYNRSP/NODYNRSP is only applicable to the residual structure.

9. DAMPLOD requests residual vectors for the degree-of-freedom in which viscous damping is defined on the CBUSHi, CDAMPi, and CVISC elements. RVEL requests residual vectors for the degree-of-freedom associated with the CBUSHi, CDAMPi, CELASi, and CVISC elements selected on a RVEL bulk entry, regardless if damping is defined or not. If DAMPLOD and RVEL are both defined (default), and RVEL bulk data entries are defined, residual vectors will be requested for the combined degree-of-freedom list.

10. For SOLs 111, 112, 118, 146, and 200 (ANALYSIS=MCEIG, MTRAN, or MFREQ), the default is RESVEC=YES. For SOLs 103, 106 (PARAM,NMLOOP,0), 110, 115, 153 (PARAM,NMLOOP,0), 187, and 200 (ANALYSIS=MODES), the default is RESVEC=COMPONENT. These defaults apply even when the RESVEC case control entry is undefined.

11. Residual vectors are not supported in the rotor dynamics capability.

Related Topics

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RIGID

Rigid Element Method

For SOLs 101, 103, 106 through 112, selects the rigid element processing method for RBAR, RBE1, RBE2, RBE3, RROD and RTRPLT elements.

For SOL 401, selects the rigid element processing method for RBAR and RBE2 elements.

Format:

Format for SOLs 101, 103, 106 through 112:

$$\text{RIGID} = \left\{ \begin{matrix} \underline{\text{LINEAR}} \\ \text{LAGRAN} \end{matrix} \right\}$$

Format for SOL 401::

$$\text{RIGID} = \left\{ \begin{matrix} \underline{\text{AUTO}} \\ \text{STIFF} \\ \text{LINEAR} \\ \text{NONLIN} \end{matrix} \right\}$$

Example:

```
RIGID=LAGRAN
```

Describers:

Describers for SOLs 101, 103, 106 through 112:

Describer	Meaning
LINEAR	Selects elimination method for enforcing linearized constraint. (Default)
LAGRAN	Selects Lagrange multiplier method for enforcing linearized constraint.

Describers for SOL 401:

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Describer	Meaning
AUTO	The software automatically determines the RBE2 and RBAR element behavior based on the PARAM,LGDISP setting. See Remark 4. (Default)
STIFF	The constraint is replaced with a stiff beam accounting for large rotations. RBE2 and RBAR elements include thermal expansion.
LINEAR	Selects elimination method for enforcing linearized constraint. Large displacement effects and thermal expansion are not included.
NONLIN	Selects elimination method for enforcing the nonlinear constraint which includes large rotation effects.

Remarks:

1. The RIGID command must be above the SUBCASE level.
2. The RIGID command is supported in SOLs 101, 103, 106 through 112, and 401. For all other solution sequences, the RIGID command is ignored and RIGID = LINEAR is used.
3. For SOLs 101, 103, 106 through 112:
   - If RIGID = LAGRAN is selected, the RROD, RBAR, RTRPLT, RBE1, RBE2, and RBE3 element IDs are limited to 10^8.
   - The LAGRAN method allows for the thermal expansion of the rigid elements.
   - If the RIGID command is not specified, RIGID = LINEAR is used.
   - LINEAR processing will not compute the thermal loads. Also, in SOLs 103, 106 through 112, LAGRAN method must be used to compute the differential stiffness due to the thermal expansion of the rigid elements.
   - The TEMP(LOAD) and TEMP(INIT) value used on RBAR, RROD, and RTRPLT elements is an average calculated from the grid point values. On RBE1, RBE2 and RBE3 elements, an average TEMP(LOAD) and TEMP(INIT) is calculated for each leg of the element using the values on the independent/dependent grid pairs such that each leg can have a different thermal strain if the temperatures vary at the grids.

     The rigid element thermal strains are calculated from

     $\epsilon_{thermal} = \alpha(AVGTEMP(LOAD) - AVGTEMP(INIT))$

     If TEMP(LOAD) or TEMP(INIT) are not defined, they are assumed to be zero.
   - When using RIGID = LAGRAN, K6ROT must be defined as non-zero.
   - For additional information, see “Rigid Element Processing Options” in the *Element Library Reference*.
   - When doing a restart analysis, you must use the same RIGID command setting in the restart run that was used in the iniitial analysis before the restart.
4. In SOL 401, the RIGID command and the LGDISP parameter settings determine the RBE2 and RBAR element behavior.

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- When RIGID=AUTO (default) and PARAM,LGDISP,-1 (default), the software automatically applies the RIGID=LINEAR option. Specifically, the RBE2 and RBAR elements do not include large displacement effects or thermal expansion.
- When RIGID=AUTO (default) and PARAM,LGDISP,1, the software automatically applies the behavior consistent with RIGID=STIFF and PARAM,LGDISP,1. Specifically, the RBE2 and RBAR elements use a stiff element method to compute large displacement effects, and they include thermal expansion.
- When RIGID=STIFF and PARAM,LGDISP,-1 (default), the RBE2 and RBAR elements do not include large displacement effects, but they do include thermal expansion.
- When RIGID=STIFF and PARAM,LGDISP,1, the RBE2 and RBAR elements use a stiff element method to compute large displacement effects, and they include thermal expansion.
- When RIGID=STIFF is used in a dynamic subcase, if the independent grid point on the RBAR or RBE2 has no mass defined, momentum is not transferred to the dependent grid point. For example, an initial velocity defined on the independent grid point will not be transfered to the dependent grid point. The workaround is to define a small mass with the CONM1 entry on the independent grid, or use the RIGID=NONLIN option described below.
- When RIGID=NONLIN and PARAM,LGDISP,-1 (default), the RBE2 and RBAR elements do not include large displacement effects or thermal expansion. The behavior is the same as RIGID=LINEAR.
- When RIGID=NONLIN and PARAM,LGDISP,1, the RBE2 and RBAR elements use a nonlinear MPC method to compute large displacement effects, but they do not include thermal expansion.
- When RIGID=NONLIN, the coefficients for the constraint equation can vary at each iteration based on the change in the rotations for the independent grid. As a result, the multipoint constraint equations and the multipoint constraint transformations are recomputed. For models which include many nonlinear rigid elements with many iterations, this will increase the solution time.
- When RIGID=LINEAR, the RBE2 and RBAR elements do not include large displacement effects or thermal expansion. This behavior is independent of the PARAM,LGDISP setting.
- The following occurs when thermal expansion is requested with the RIGID=STIFF option.

  For the RBAR element, the software computes an average TEMP(LOAD) and TEMP(INIT) for the element from the grid point values.

  For the RBE2 element, the software computes an average TEMP(LOAD) and TEMP(INIT) for each leg of the element using the values on the independent/dependent grid pairs such that each leg can have a different thermal strain if the temperatures vary at the grids.

  The rigid element thermal strains are calculated from

  $\epsilon_{thermal} = \alpha(AVGTEMP(LOAD) - AVGTEMP(INIT))$

  If TEMP(LOAD) or TEMP(INIT) are not defined, they are assumed to be zero.
- MPCFORCE and GPFORCE output are supported with all of the rigid elements. Since the software internally replaces an RBAR or RBE2 with a stiff beam or spring element when RIGID=STIFF, these elements are no longer represented as MPC equations. As a result, MPCFORCE output is not applicable to these elements. GPFORCE and FORCE output is applicable.
- For additional information, see Rigid Element Processing in the *Multistep Nonlinear User's Guide*.

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Related Topics

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RMAXMIN

Requests output of MIN/MAX and RMS values from solutions 101, 109, and 112.

Defines parameters to output the minimum, maximum, absolute value maximum, average, and RMS value of stress, force and displacement results generated during solutions 101, 109, and 112.

Format:

$$\text{RMAXMIN}\left(\{\text{FORCE, STRESS, DISP}\},\ \begin{bmatrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{bmatrix},\ [\text{PUNCH, PLOT}],\ \begin{bmatrix}\text{ABSOLUTE}\\ \text{MINIMUM}\\ \underline{\text{MAXIMUM}}\end{bmatrix},\right.$$
$$\left.\text{RMXTRAN} = \begin{Bmatrix}\text{YES}\\ \underline{\text{NO}}\end{Bmatrix}, \text{NPAVC} = \text{n}, \text{START} = \text{a}, \text{END} = \text{b}\right) = \begin{Bmatrix}\text{YES}\\ \text{NO}\end{Bmatrix}$$

Examples:

```
RMAXMIN(NOPRINT,STRESS,ABSOLUTE)=YES
RMAXMIN(STRESS,PRINT,MAX,NPAVG=3,START=.3,END=1.2) = yes
```

Describers:

Describer		Meaning
FORCE, STRESS, DISP		Result tables to be searched.
PRINT		Writes output to the print file. (Default)
NOPRINT		Does not write output to the print file.
PUNCH		Writes output to the punch file.
PLOT		Writes output to .plt file.
ABSOLUTE		Specifies output of absolute maximum values. See Remarks 1, 2 & 4.
MINIMUM		Specifies output of minimum values. See Remarks 1 & 2.
MAXIMUM		Specifies output of maximum values. (Default) See Remarks 1 & 2.
RMXTRAN		Turns on or off the normal printing of output data when RMAXMIN output is requested. See Remark 7.
	= YES	Results will be output according to the DISPLACEMENT, STRESS, and FORCE case control entries.
	= NO	Results are not output if RMAXMIN is activated.

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Describer	Meaning
NPAVG	Defines the number of extremes in a subcase history (SOL 101) or peaks in a time history (SOL 109 and 112) to be averaged. See Remarks 2 & 3. (Default=1)
START=a,END=b	Limits the RMAXMIN evaluation over a specific range in the subcase history (SOL 101) or the time history (SOL 109 and 112). See Remarks 1 & 4.
YES	Activates RMAXMIN.
NO	Disables RMAXMIN.

Remarks:

1. For RMAXMIN to process force output, the FORCE describer should be defined on the RMAXMIN command, and force output must be requested with either the FORCE case control command or using the XYPUNCH/XYPLOT commands.

   For RMAXMIN to process stress output, the STRESS describer should be defined on the RMAXMIN command, and stress output must be requested with either the STRESS case control command or using the XYPUNCH/XYPLOT commands.

   For RMAXMIN to process one of displacement, velocity, or acceleration output, the DISP describer should be defined on the RMAXMIN command, and only one of displacement, velocity, or acceleration output must be requested with either the DISPLACEMENT, VELOCITY, or ACCELERATION case control commands or using the XYPUNCH/XYPLOT commands.

   RMAXMIN will process displacement, velocity, or acceleration output, if requested, together as a single output type, and not separately. As a result, you should only request one of displacement, velocity, or acceleration output at a time for RMAXMIN processing.

2. RMAXMIN outputs maximum/minimum/absolute tables and RMS tables. Both tables are evaluated using the START and/or END values when they are defined. Only one of MAXIMUM (default), MINIMUM and ABSOLUTE can be selected, and the one selected determines the values reported in the maximum/minimum/absolute tables. The RMS tables are independent of this selection.

3. NPAVG is the number of extremes in a subcase range (SOL 101) or the number of peak responses (see Remark 3) in a time range (SOLs 109 and 112) to be averaged when calculating the maximum/minimum/absolute tables. The extremes/peak responses used depend on which of MAXIMUM, MINIMUM, or ABSOLUTE is defined. When NPAVG=1, an extreme value is reported with no averaging. NPAVG is ignored by the RMS calculation.

4. In solutions 109 and 112, a peak search algorithm is used to determine the peak responses in the time range. This algorithm is comparing peaks relative to one another to find “NPAVG” peak responses for averaging.

5. When ABSOLUTE is defined, the software does the following.
   a. NPAVG maximum peaks and NPAVG minimum peaks are found.
   b. NPAVG maximum peaks are determined from the absolute values of all peaks found in “a”.

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6. The START=a and END=b describers can be used to limit the RMAXMIN evaluation to a specific subcase or time range. By default, START is the beginning, and END is the final, such that one can be defined without the other.
7. RMAXMIN output is always SORT1.
8. The parameter RMXTRAN can also be used to control the output of transient results for STRESS, FORCE, DISP case control when RMAXMIN is activated, but only if the RMXTRAN input on RMAXMIN is not defined. The parameter default, RMXTRAN=NO will suppress output of transient results. The RMXTRAN describer on the RMAXMIN case control command, when defined, takes precedence over the RMXTRAN parameter.
9. The output datablocks generated by RMAXMIN are OES1MX, OEF1MX, and OUGV1MX for stress, force and displacement respectively.
10. The maximum, minimum, absolute, and RMS results are captured and stored on a component basis in the OES1MX, OEF1MX, and OUGV1MX datablocks. For example, the maximum displacement is computed and stored for all six DOF on a grid and each value may correspond to different times in the transient history.
11. When the RMAXMIN case control command is defined in a SOL 112 run, the parameter entry PARAM,RMXPANEL,PSIZE can be used to process the data recovery in a series of smaller “panels” rather than the entire output set at once. This will reduce the overall amount of scratch disk space required for the run.

    “PSIZE” is the number of timesteps to process in a single panel. The highwater disk space used will be a function of psize, and not of the total output request. A good initial value for psize is the greater of 100 or the square root of the number of timesteps. Setting psize =0 (default) will disable the panel method and process the data recovery for all the time steps at once. The START, END, and RMXTRAN=YES keyword options on the RMAXMIN command are not supported when using RMXPANEL. The RMXPANEL method is not compatible with PARAM,DDRMM,-1.

Related Topics

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RMETHOD

ROTORD Selection

Selects the ROTORD bulk entry used for rotor dynamics analysis (SOL 414 and legacy). Optionally selects complex modal reduction when using SOL 107 for rotor dynamics analysis.

Formats:

RMETHOD[(CMR=m)]=n

Examples:

```
RMETHOD=77
```

```
RMETHOD(CMR=3)=77
```

Describers:

Describer	Meaning
CMR = m	Request complex modal reduction using the complex eigenvalue method and parameters specified on the EIGC bulk entry having identification number m. See Remark 4. (Integer ≥ 0; If m = 0, complex modal reduction is not used; Default = 0)
n	Identification number of a ROTORD bulk entry. (Integer>0)

Remarks:

1. ROTORD bulk entries will not be used unless selected in the case control section.
2. RMETHOD must be located above any subcases.
3. RMETHOD is supported in SOLs 101, 107 through 112, and 414.
4. The CMR = m describer is ignored for any solution sequence other than SOL 107.

Related Topics

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RSAPPLY

Response Spectrum Data Selection

Selects the data for a response spectrum application run.

Format:

$$\text{RSAPPLY} = \text{n}$$

Examples:

```
RSAPPLY=5
```

Describers:

Describer	Meaning
n	Set identification number of an RSAPPLY bulk entry. (Integer>0)

Remarks:

1. The RSAPPLY case control command can be placed above the subcases or within subcases. If both are present, the RSAPPLY case control command in the subcase takes precedence over the RSAPPLY case control command placed above the subcases.

Related Topics

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RSMETHOD

Real Eigenvalue Extraction Method Selection

Selects the real eigenvalue extraction parameters in the component mode reduction of the residual structure for direct and modal solutions.

Format:

RSMETHOD = n

Examples:

```
RSMETHOD=33
```

Describers:

Describer	Meaning
n	Set identification number of an EIGR or EIGRL Bulk Data entry. (Integer>0)

Remarks:

1. RSMETHOD specifies the set identification number of an EIGR or EIGRL entry to be used in the computation of the normal modes on the o-set (fixed boundary modes) or v-set (mixed boundary modes) of the residual structure depending on the b-set and/or c-set entries. These modes are calculated for a component mode reduction of the residual structure when QSET type entries exist. If QSET type entries do not exist in the residual structure a component mode reduction for SE 0 is not performed.
2. Component mode reduction of a residual structure is used to account for local residual flexibility in the residual. Similar to component mode reduction of a superelement in which residual flexibility effects are obtained for the exterior DOF, a component mode reduction on the residual accounts for residual flexibility at DOF identified with ASET, BSET, and CSET type entries.
3. If the set identification number selected is present on both EIGRL and EIGR entries, the EIGRL entry will be used. This entry requests the Lanczos eigensolution method.
4. For CMR of RS the user must have QSET type entries along with ASET, BSET, and CSET type entries. In contrast, when using CMR of a superelement other than 0, the user must use SEBSET, SECSET, SEQSET, or SESET type entries.
5. If any of the following conditions exist, a Guyan reduction is executed and the results may be poor because of the absence of a dynamic portion of the reduction:
   - DOF referenced by QSET do not exist.
   - RSMETHOD specified but QSET does not exist.

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6. For SOLs 103, 111, and 112, when modeling a fluid volume with MFLUID bulk entries, you can use the RSMETHOD command to reference an eigenmethod (EIGR/EIGRL selection) that specifies the eigenvalue extraction parameters that the software uses to calculate the dry modes, where the dry modes are the modes of the structural portion of the model only. The software uses the dry modes to transform the structural model into a reduced modal representation. The software adds the fluid mass into the modal representation and then computes the modes for the entire model. The resulting modes are referred to as the wet modes for the model because they account for the presence of the fluid.

Related Topics

Browse Simcenter Nastran documentation

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SACCELERATION

Solution Set Acceleration Output Request

Requests the form and type of modal acceleration output.

Format:

$$\text{SACCELERATION}\left[\left(\begin{bmatrix}\text{SORT1}\\ \text{SORT2}\end{bmatrix}, \begin{bmatrix}\underline{\text{PRINT}}, \text{PUNCH}\\ \text{PLOT}\end{bmatrix}, \begin{bmatrix}\underline{\text{REAL}} \text{ or } \underline{\text{IMAG}}\\ \text{PHASE}\end{bmatrix}\right)\right] = \begin{Bmatrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{Bmatrix}$$

Examples:

```
SACCELERATION=ALL
SACCELERATION(PUNCH,IMAG)=142
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output a tabular listing of frequency or time for each grid point (or mode number).
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
PLOT	Generate, but do not print, acceleration results.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
ALL	Output acceleration for solution set points (modes).
n	Output acceleration only for the points (modes) listed on a previously appearing SET command with n as the identification number. (Integer > 0)
NONE	Do not output acceleration for any points (modes).

Remarks:

1. Both PRINT and PUNCH may be requested.

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2. The defaults for SORT1 and SORT2 depend on the type of analysis and are discussed. For additional information, see the remarks for the ACCELERATION command.

   If SORT1 is specified on either an SACCELERATION, SDISPLACEMENT, or SVELOCITY command, then the other commands are also SORT1.

3. SACCELERATION=NONE allows an overall output request to be overridden.

4. Due to the differences in formulations, the modal responses for the constraint mode method of enforced motion will be different than the modal responses for the absolute displacement method of enforced motion. In order to highlight this difference, the f06 output for the constraint mode method will be labeled *alternative* solution set.

5. Acceleration output is only available for transient and frequency response problems.

Related Topics

Browse Simcenter Nastran documentation

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SDAMPING

Structural Damping Selection

Requests modal damping as a function of natural frequency in modal solutions or viscoelastic materials as a function of frequency in direct frequency response analysis.

Format:

$$\text{SDAMPING}\left[\left(\frac{\text{STRUCTURE}}{\text{FLUID}}\right)\right]=\text{n}$$

Examples:

```
SDAMPING=77
```

Describers:

Describer	Meaning
STRUCTURE or FLUID	Modal damping is requested for the structural or fluid portion of the model.
n	Set identification number of a TABDMP1 (modal solutions such as 111 or MFREQ in SOL 200) or TABLEDi (direct solutions such as SOL 108 or DFREQ in SOL 200) bulk data entry. (Integer>0) See Remarks 1 and 2.

Remarks:

1. In the modal solutions (e.g., SOLs 110, 111, 112, 145, 146, 200, 414,103, 414,110, and 414,111), SDAMPING must reference a TABDMP1 entry.
2. In direct frequency response (e.g., SOL 108), SDAMPING must reference a TABLEDi entry with ID=n and another TABLEDi entry with ID=n+1, which define viscoelastic (frequency-dependent) material properties. See the *Simcenter Nastran Advanced Dynamic Analysis User's Guide*.
3. SDAMPING may be requested for superelements as long as PARAM,SESDAMP,YES is specified.
4. In SOL 414,103, SDAMPING used with PARAM,SESDAMP,YES updates the superelement damping matrix by adding the defined modal damping.
5. The SDAMPING command is supported across subcases. A SDAMPING command selecting a different TABDMP1 or TABLEDi bulk data entry can be defined for each subcase.

Related Topics

Browse Simcenter Nastran documentation

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SDISPLACEMENT

Solution Set Displacement Output Request

Requests the form and type of modal displacement output.

Format:

$$\text{SDISPLACEMENT}\left[\left(\begin{bmatrix}\text{SORT1}\\ \text{SORT2}\end{bmatrix}, \begin{bmatrix}\underline{\text{PRINT}}, \text{PUNCH}\\ \text{PLOT}\end{bmatrix}, \begin{bmatrix}\underline{\text{REAL}} \text{ or } \underline{\text{IMAG}}\\ \text{PHASE}\end{bmatrix}\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
SDISPLACEMENT=ALL
SDISPLACEMENT(SORT2,PUNCH,PHASE)=NONE
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output a tabular listing of frequency or time for each grid point (or mode number).
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
PLOT	Generate, but do not print, displacement results.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
ALL	Output displacement for solution set points (modes). See Remark 6.

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Describer	Meaning
n	Output displacement only for the points (modes) listed on a previously appearing SET command with n as the identification number. (Integer > 0)
NONE	Do not output displacement for any points (modes).

Remarks:

1. Both PRINT and PUNCH may be requested.
2. The defaults for SORT1 and SORT2 depend on the type of analysis and are discussed. For additional information, see the remarks for the **DISPLACEMENT** command.

   If SORT1 is specified on either an SACCELERATION, SDISPLACEMENT, or SVELOCITY command, then the other commands are also SORT1.
3. SDISPLACEMENT=NONE allows an overall output request to be overridden.
4. Due to the differences in formulations, the modal responses for the constraint mode method of enforced motion will be different than the modal responses for the absolute displacement method of enforced motion. In order to highlight this difference, the f06 output for the constraint mode method will be labeled *alternative* solution set.
5. The SDISPLACEMENT command is required to output normalized complex eigenvectors.
6. For SOL 414 solutions, SDISPLACEMENT=ALL computes the contribution of each mode to the displacement (real and imaginary parts) and force (real and imaginary parts) for each frequency.

   The results are available in the .f06 file and in the *_modprt.csv* file that are created in the same folder as the simulation files.
7. SDISPLACEMENT=<*n*>, where <*n*> is a set of modes, is not valid for SOL 414 solutions.

Related Topics

Browse Simcenter Nastran documentation

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SEALL

Superelement Generation and Assembly

Specifies the superelement identification numbers of Phase 1 processing in which all matrices and loads are generated and assembled. Controls execution of the solution sequence.

Format:

$$\text{SEALL} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SEALL=ALL
SEALL=7
```

Describers:

Describer	Meaning
ALL	Generate and assemble all superelements.
n	Set identification number of a previously appearing SET command. Only superelements with identification numbers that appear on this SET command will be generated and assembled. (Integer>0)
i	Identification number of a single superelement that will be generated and assembled. (Integer>0)

Remarks:

1. If this command is present, then it must be located before the first SUBCASE command.
2. Zero (0) is the identification number of the residual structure and can only appear as a member of a SET.
3. For a further discussion of this command, see the *Simcenter Nastran Superelement User's Guide*.
4. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.
5. This command combines, in one command, the functions of the SEMG, SELG, SEKR, SELR, and SEMR commands.

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6. This command does not control superelement data recovery (Phase 3). See **SEDR**.
7. SEALL=ALL is the default in the Structured Solution Sequences (SOLs 101 through 200). This default can be overridden by specifying any of the following Phase 1 commands: SEALL, SEMG, SELG, SEKR, SELR, or SEMR.

Related Topics

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SEDR

Superelement Data Recovery

Specifies the superelement identification numbers for which data recovery will be performed.

Format:

$$\text{SEDR} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SEDR=ALL
SEDR=7
```

Describers:

Describer	Meaning
ALL	Performs data recovery for all superelements.
n	Set identification number of a previously appearing SET command. Data recovery will be performed for superelements with identification numbers that appear on this SET command. (Integer>0)
i	Identification number of a single superelement for which data recovery will be performed. (Integer>0)

Remarks:

1. If this command is present, then it must be located before the first SUBCASE command.
2. Zero (0) is the identification number for the residual structure and can only appear as a member of a SET.
3. For a further discussion of this command, see the *Simcenter Nastran Superelement User's Guide*.
4. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.
5. If this command is not present, data recovery is performed for all superelements for which there are output requests (i.e., the default for this command is SEDR=ALL).

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Related Topics

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SEDV

Superelement Design Variable Processing

Specifies the superelement identification numbers for which the design variables will be processed.

Format:

$$\text{SEDV} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SEDV=ALL
SEDV=18
```

Describers:

Describer	Meaning
ALL	Requests design variable processing for all superelements.
n	Set identification number of a previously appearing SET command. Design variable processing will be performed for superelements with identification numbers that appear on this SET command. (Integer>0)
i	Identification number of a single superelement for which design variable processing will be performed. (Integer>0)

Remarks:

1. If this command is present, then it must be located before the first SUBCASE command.
2. Zero (0) is the identification number of the residual structure and can only appear as a member of a SET.
3. For a further discussion of superelement sensitivity analysis, see the *Simcenter Nastran Design Sensitivity and Optimization User's Guide*.
4. Zero (0) can only appear as a member of a SET.
5. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.

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6. If both the SEDV and SERESP commands are not present, then the design variable processing and design sensitivity matrix generation will be performed for all superelements.

Related Topics

Browse Simcenter Nastran documentation

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SEEXCLUDE

Superelement Matrix and Load Assembly Exclusion

Specifies the superelement identification numbers for which all matrices and loads will not be assembled into the downstream superelement.

Format:

$$\text{SEEXCLUDE} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SEEXCLUDE=ALL
SEEXCLUDE=18
```

Describers:

Describer	Meaning
ALL	All upstream superelements will be excluded.
n	Set identification number of a previously appearing SET command. Only those superelements with identification numbers that appear on this SET command will be excluded. (Integer>0)
i	Identification number of a single superelement for which matrices will be excluded. (Integer>0)

Remarks:

1. If this command is present, then it must be located before the first SUBCASE command.
2. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.
3. This command is not meaningful when applied to the residual structure.
4. For a further discussion of this command, see the *Simcenter Nastran Superelement User's Guide*.
5. If the SEEXCLUDE command is specified in a restart of SOLs 101 through 200, then PARAM,SERST,MANUAL must be specified. Also the SEKR command must be specified for the superelement immediately downstream from the excluded superelement. For example, if superelement 10 is excluded in the superelement tree:

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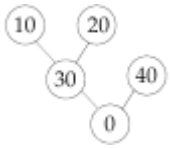

then you must specify the following commands in the Case Control Section:

```
SEKR=30
PARAM,SERST,MANUAL
```

Related Topics

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SEFINAL

Final Superelement for Assembly

Specifies the superelement identification number for the final superelement to be assembled.

Format:

$$\text{SEFINAL} = \begin{Bmatrix} n \\ i \end{Bmatrix}$$

Examples:

```
SEFINAL=14
```

Describers:

Describer	Meaning
n	Set identification of a previously appearing SET command. Each superelement identification number appearing on the SET command must belong to a disjoint model. (Integer>0)
i	Identification number of the final superelement to be assembled. (Integer>0)

Remarks:

1. If this command is not present, the program selects the order of the superelements for assembly operations.
2. If this command is present, then it must be located before the first SUBCASE command.
3. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.
4. This command can be used on restarts to ensure that minor modeling changes do not also change the processing order. For this usage, inspect the SEMAP table to determine which superelements were final superelements on the prior run.
5. See the *Simcenter Nastran Superelement User's Guide* for a further discussion of this command.
6. If all of the superelements in the model are on a single level, (i.e. No superelement connections except to residual structure), then SEFINAL is ignored and superelement processing order is set according to the superelement ID.

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Related Topics

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SEKETAB

Strain and Kinetic Energy Output

Specifies elements for which energy balances are output to the .f06 file and to a .csv file in SOL 402, SOL 414,103, SOL 414,110, and SOL 414,111.

Format:

$$\text{SEKETAB} = \text{n}$$

Examples:

```
SEKETAB=5
```

Describers:

Describer	Meaning
n	Set identification number of a previously appearing SET command. The referenced SET lists GROUP bulk entries that contain the elements for which energy balances are written to the .f06 file and a .csv file. (Integer > 0)

Remarks:

1. If you do not include a SEKETAB command specification in the input file, energies are not computed.
2. Energy balances are computed for groups of elements. In case the defined GROUP bulk entries also contain grid points, the latter are ignored.
3. The energy computation per mode and element is as follows.

   For SOL 402, the kinetic energy ratio of element *e*:

   $$KER = \frac{x_e^{\mathrm{T}} M_e x_e}{x^{\mathrm{T}} M x}$$

   For SOL 414, the kinetic energy ratio of element *e*:

   $$KER = \frac{x_e^{\mathrm{T}} M_e x_e + \Omega x_e^{\mathrm{T}} G_e x_e}{x^{\mathrm{T}} M x + \Omega x^{\mathrm{T}} G x}$$

   For both SOLs 402 and 414, strain energy ratio of element *e*:

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$$SER = \frac{x_e^{\mathrm{T}} K_e x_e}{x^{\mathrm{T}} K x}$$

where,

M and K are the mass and stiffness matrices of a volume V,
M_e and K_e are the mass and stiffness matrices of element e of volume V_e,
x_e is the displacement vector of element e for mode x,
and for SOL 414, Ω and G are the rotor speed and gyroscopic matrix, respectively.

For a rigid body mode, only the kinetic energy is relevant.

Related Topics

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SEKREDUCE

Superelement Stiffness Matrix Assembly and Reduction

Specifies the superelement identification numbers for which stiffness matrices are assembled and reduced.

Format:

$$\text{SEKREDUCE} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SEKREDUCE=ALL
SEKREDUCE=9
```

Describers:

Describer	Meaning
ALL	Assembles and reduces matrices for all superelements.
n	Set identification number of a previously appearing SET command. Matrices will only be assembled for superelements with identification numbers that appear on this SET command. (Integer>0)
i	Identification number of a single superelement for which the stiffness matrix will be assembled and reduced. (Integer>0)

Remarks:

1. If this command is present, then it must be located before the first SUBCASE command.
2. Zero (0) is the identification number for the residual structure and can only appear as a member of a SET.
3. For a further discussion of this command, see the *Simcenter Nastran Superelement User's Guide*.
4. SEKREDUCE is an alternate form and is entirely equivalent to the obsolete command SEMASSEMBLE.
5. SEALL=ALL is the default in the Structured SOLs 101 through 200. This default can be overridden by specifying any of the following Phase 1 commands: SEALL, SEMG, SELG, SEKR, SELR, or SEMR.

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Related Topics

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SELGENERATE

Superelement Load Generation

Specifies the superelement identification numbers for which static loads will be generated.

Format:

$$\text{SELGENERATE} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SELGENERATE=ALL
SELGENERATE=18
```

Describers:

Describer	Meaning
ALL	Generates static loads for all superelements.
n	Set identification number of a previously appearing SET command. Static load matrices will only be generated for superelements with identification numbers that appear on this SET command. (Integer>0)
i	Identification number of a single superelement for which load matrices will be generated. (Integer>0)

Remarks:

1. If this command is present, then it must be located before the first SUBCASE command.
2. Zero (0) is the identification number of the residual structure and can only appear as a member of a SET.
3. For a further discussion of this command, see the *Simcenter Nastran Superelement User's Guide*.
4. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.
5. SEALL=ALL is the default in the Structured SOLs 101 through 200. This default can be overridden by specifying any of the following Phase 1 commands: SEALL, SEMG, SELG, SEKR, SELR, or SEMR.

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Related Topics

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SELREDUCE

Superelement Load Assembly and Reduction

Specifies the superelement identification numbers for which the static load matrices will be assembled and reduced.

Format:

$$\text{SELREDUCE} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SELREDUCE=ALL
SELREDUCE=9
```

Describers:

Describer	Meaning
ALL	Assembles and reduces matrices for all superelements.
n	Set identification number of a previously appearing SET command. Matrices will be assembled only for superelements with identification numbers that appear on this SET command. (Integer>0)
i	Identification number of a single superelement for which the load matrices will be assembled and reduced. (Integer>0)

Remarks:

1. If this command is present, then it must be located before the first SUBCASE command.
2. Zero (0) is the identification number for the residual structure and can only be appear as a member of a SET.
3. For a further discussion of this command, see the *Simcenter Nastran Superelement User's Guide*.
4. This command is used on restarts to selectively assemble and reduce load matrices.
5. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.
6. In superelement static analysis, SELREDUCE is equivalent to SELASSEMBLE.

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7. In dynamic analysis, SELASSEMBLE combines the functions of SELREDUCE and SEMREDUCE.
8. SEALL=ALL is the default in the Structured SOLs 101 through 200. This default can be overridden by specifying any of the following Phase 1 commands: SEALL, SEMG, SELG, SEKR, SELR, or SEMR.

Related Topics

Browse Simcenter Nastran documentation

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SEMGENERATE

Superelement Matrix Generation

Specifies the superelement identification numbers for which stiffness, mass, and damping matrices will be generated.

Format:

$$\text{SEMGENERATE} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SEMGENERATE=ALL
SEMGENERATE=7
```

Describers:

Describer	Meaning
ALL	Generates structural matrices for all superelements.
n	Set identification number of a previously appearing SET command. Structural matrices will only be generated for superelements with identification numbers that appear on this SET command. (Integer>0)
i	Identification number of a single superelement for which structural matrices will be generated. (Integer>0)

Remarks:

1. If this command is present, then is must be located before the first SUBCASE command.
2. Zero (0) is the identification number for the residual structure and can only appear as a member of a SET.
3. For a further discussion of this command, see the *Simcenter Nastran Superelement User's Guide*.
4. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.
5. SEALL=ALL is the default in the Structured SOLs 101 through 200. This default can be overridden by specifying any of the following Phase 1 commands: SEALL, SEMG, SELG, SEKR, SELR, or SEMR.

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Related Topics

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SEMREDUCE

Superelement Mass and Damping Assembly and Reduction

Specifies the superelement identification numbers for which the mass and damping matrices will be assembled and reduced. In buckling analysis, the differential stiffness matrices will be assembled and reduced.

Format:

$$\text{SEMREDUCE} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SEMREDUCE=ALL
SEMREDUCE=9
```

Describers:

Describer	Meaning
ALL	Assembles and reduces matrices for all superelements.
n	Set identification number of a previously appearing SET command. Matrices will only be assembled for superelements with identification numbers that appear on this SET command. (Integer>0)
i	Identification number of a single superelement for which the load matrices or the mass and damping matrices will be assembled and reduced. (Integer>0)

Remarks:

1. If this command is present, then it must be located before the first SUBCASE command.
2. Zero (0) is the identification number for the residual structure and can only appear as a member of a set.
3. This command is used on restart to selectively assemble and reduce mass and damping matrices. For a further discussion of this command, see the *Simcenter Nastran Superelement User's Guide*.
4. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.
5. In dynamic analysis, SELASSEMBLE combines the functions of SELREDUCE and SEMREDUCE.

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6. This command has no function in static analysis.
7. SEALL=ALL is the default in the Structured SOLs 101 through 200. This default can be overridden by specifying any of the following Phase 1 commands: SEALL, SEMG, SELG, SEKR, SELR, or SEMR.

Related Topics

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SEQDEP

Solution sequence dependency selection for SOLs 401 and 402

Defines whether or not a subcase is sequentially dependent on the previous subcase.

Format:

$$\text{SEQDEP} = \left\{ \begin{matrix} \text{YES} \\ \text{NO} \end{matrix} \right\}$$

Examples:

```
SEQDEP = NO
```

Describers:

Describer	Meaning
SEQDEP=YES (Default)	Subcases are sequentially dependent.
SEQDEP=NO	Subcases are non-sequentially dependent.

Remarks:

1. Only supported in SOLs 401 and 402.
2. SEQDEP may be defined above the subcases (globally) or within a subcase. In the first situation, the same SEQDEP value is applied to all subcases. In the second situation, the SEQDEP value can vary from a subcase to another.
3. SEQDEP is ignored in the first subcase of a cold start. That is, in a solution which is not a restart.
4. SEQDEP=YES is the default. The default applies to a subcase when the SEQDEP case control entry is undefined.
5. A subcase for which SEQDEP = YES is sequentially dependent (SD). A subcase for which SEQDEP = NO is non-sequentially dependent (NSD). See Subcase sequencing in the Multi-Step Nonlinear User's Guide.

Remarks related to SOL 401:

1. A sequentially dependent static subcase (SEQDEP=YES) uses the end time from the previous static subcase for its start time. In addition, a sequentially dependent static or modal subcase uses the displacement/stress/strain state from the previous static subcase for its starting state.

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2. A non-sequentially dependent static subcase (SEQDEP=NO) has a start time of zero. In addition, a non-sequentially dependent static or modal subcase does not use the displacement/stress/strain state from the previous static subcase.
3. Except for the first subcase of a cold start (where SEQDEP is ignored), SEQDEP=YES is required when SOL 401 is used in a Simcenter 3D Multiphysics solution.

Remarks related to SOL 402:

1. The start time of a static or transient subcase is always the end time from the previous time-based subcase, regardless of the sequential dependency.
2. The starting state of a SD static, transient, modal or buckling subcase uses the displacement/stress/strain fields from the previous time-based subcase.
3. The starting state of an NSD static, transient, modal or buckling subcase can be either the initial state (default) or the state at the end of any previous subcase whose number n is to be specified in the RSUB field of the NLCNTL2 card. n can have the values -1, 0, >0:

   -1 Restart from the end of the previous time-based subcase.

   0 Restart from the initial state (default).

   >0 Restart from the end of subcase n, n being strictly smaller than the current subcase number.

Related Topics

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SERESP

Superelement Response Sensitivity

Specifies the superelement identification numbers for which the design sensitivity matrices will be generated.

Format:

$$\text{SERESP} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \\ \text{i} \end{matrix} \right\}$$

Examples:

```
SERESP=ALL
SERESP=18
```

Describers:

Describer	Meaning
ALL	Requests design sensitivity matrix generation for all superelements.
n	Set identification number of a previously appearing SET command. Design sensitivity matrices will be generated for superelements with identification numbers that appear on this SET command. (Integer>0)
i	Identification number of a single superelement for which the design sensitivity matrix will be generated.

Remarks:

1. If this command is present, then it must be located before the first SUBCASE command.
2. Zero (0) is the identification number of the residual structure and can only appear as a member of a SET.
3. For a further discussion of this command, see the *Simcenter Nastran Design Sensitivity and Optimization User's Guide*.
4. If i is used, the superelement identification number must be unique with respect to any SET identification numbers used.
5. If both SEDV and SERESP commands are not present, then the design variable processing and design sensitivity matrix generation will be performed for all superelements.

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Related Topics

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SET

Set Definition, General Form

Sets are used to define the following lists:

Lists

1. Identification numbers (point, element, or superelement) for processing and output requests.
2. Times for which output will be printed in transient response problems using the OTIME case control command or the MODCON case control command with the SOLUTION = *setout* describer specified.
3. Frequencies for which output will be printed in frequency response problems using the OFREQ case control command or the MODCON case control command with the SOLUTION = *setout* describer specified or the PANCON case control command with the SOLUTION = *setf* describer specified.
4. Panels for which output will be printed in frequency response problems using either the PANCON or ERP case control commands.
   - For the PANCON command, the PANEL = *setp* describer references the identification number of the SET that lists the alphanumeric names of panels defined by PANEL bulk entries.
   - For the ERP command, the ERP = *setp* describer references the identification number of the SET that lists the alphanumeric names of panels defined by PANEL bulk entries.
5. Surface or volume identification numbers to be used in GPSTRESS or STRFIELD case control commands.
6. With SOL 200, DRESP1 design responses that are exclusively assigned to a specific subcase via a DRSPAN case control command which refers to a particular SET.
7. For SOL 414,111 Harmonic Response solutions, harmonic coefficients that are referenced using the HARCOEFF case control command.

Formats:

$$\mathrm{SET\ n} = \{\mathrm{i1},\ [\mathrm{i2, i3, THRU\ i4, EXCEPT\ i5, i6, i7, i8, THRU\ i9}]\}$$
$$\mathrm{SET\ n} = \{\mathrm{r1},\ [\mathrm{r2, r3, r4}]\}$$
$$\mathrm{SET\ n} = \{\mathrm{name1},\ [\mathrm{name2, name3, name4}]\}\ \mathrm{SET\ n = ALL}$$

Examples:

```
SET 77=5
SET 88=5, 6, 7, 8, 9, 10 THRU 55 EXCEPT 15, 16, 77, 78, 79, 100 THRU 300
SET 99=1 THRU 100000
SET 101=1.0, 2.0, 3.0
SET 105=1.009, 10.2, 13.4, 14.0, 15.0
SET 5=PANEL1,PANEL3,PANEL4
```


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Describers:

Describer	Meaning
n	Set identification number. SETs specified under a SUBCASE command are recognized for that SUBCASE only. Any set defined in one subcase may be redefined in another subcase by reassigning its identification number. (Integer>0)
i_1, i_2, etc.	Identification numbers. If no such identification number exists, the request is ignored. (Integer ≥ 0)
i_3 THRU i_4	Identification numbers (i_4>i_3). (Integer>0)
EXCEPT	Set identification numbers following EXCEPT will be deleted from the output list as long as they are in the range of the set defined by the immediately preceding THRU. An EXCEPT list may not include a THRU list or ALL.
r_1, r_2, etc.	Frequencies or times for output. The nearest solution frequency or time will be output. EXCEPT and THRU cannot be used. (Real≥0.0 for HARCOEFF; Real>0.0 for MODCON, OFREQ, OTIME, and PANCON) The following case control commands require the values to be listed in ascending sequences (r_1>r_2>r_3>r_4...etc.). Otherwise some output may be missing. • HARCOEFF • MODCON (with SOLUTION = *setout* describer specified) • OFREQ • OTIME • PANCON (with SOLUTION = *setf* describer specified)
$name_1$, $name_2$, etc.	Alphanumeric names of panels defined by PANEL bulk entries. (Character)
ALL	All members of the set will be processed. This option may not be used in a DRSPAN referenced set.

Remarks:

1. A SET command may be more than one physical command. A comma at the end of a physical command signifies a continuation command. Commas may not end a set. THRU may not be used for continuation. Place a number after the THRU.
2. Set identification numbers following EXCEPT within the range of the THRU must be in ascending order.
3. In SET 88 in the example section above, the numbers 77, 78, etc., are included in the set because they are outside the prior THRU range.

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Remarks related to SOL 402:

1. Only grid point or element identification numbers can be listed for processing and output requests.

Remarks related to SOL 414:

1. HARCOEFF applies to SOL 414,111 Harmonic Response solutions.
2. When you use HARCOEFF in a fixed or rotating reference frame to define multiple harmonics associated to a load definition, you can define the SET with real numbers in fractional form. They must be sorted by ascending values.
3. Only in models using the CLINK OPT 3 in the harmonic domain are you required to define the SET with a minimum of 0.0, 1.0, and 2.0 coefficients. The coefficients must be real numbers in whole form (such as 3.0, 4.0, 5.0, and so on), not fractional form (such as 3.5, 4.3, 5.1, and so on).

   The software requires that all coefficients up to the highest number in the SET are defined. For example, if you want a coefficient of 5.0, you must define a SET of 0.0, 1.0, 2.0, 3.0, 4.0, and 5.0.

Related Topics

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SET

Set Definition OUTPUT(PLOT)

Defines a set of element or grid point numbers to be plotted.

Format:

$$\text{SET n} = \left[\text{ALL}\left[\begin{matrix}\underline{\text{ELEMENTS}}\\ \text{GRID POINTS}\end{matrix}\right]\left[\text{EXCEPT}\left\{\begin{matrix}\text{type1} & \text{type2} & \dots & \text{typej}\\ \text{k1} & \text{k2} & \dots & \text{kj THRU kk BY incj}\end{matrix}\right\}\right]\right],$$
$$\left[\left[\begin{matrix}\underline{\text{INCLUDE}}\\ \text{EXCLUDE}\end{matrix}\right]\left[\begin{matrix}\underline{\text{ELEMENTS}}\\ \text{GRID POINTS}\end{matrix}\right]\left\{\begin{matrix}\text{type1} & \text{type2} & \dots & \text{typej}\\ \text{k1} & \text{k2} & \dots & \text{kj THRU kk BY incj}\end{matrix}\right\},\right.$$
$$\left.\left[\text{EXCEPT}\left\{\begin{matrix}\text{type1} & \text{typem} & \dots & \text{typen}\\ \text{k1} & \text{km} & \dots & \text{kn THRU ko BY incn}\end{matrix}\right\}\right]\right]$$

Examples:

1. SET 1 consists of elements 1, 5, 10, 11, 13, 14, 15, 20, 22, 24, and 26.

```
SET 1=INCLUDE 1, 5, 10 THRU 15 EXCEPT 12, INCLUDE 20
THRU 26 BY 2
```

2. SET 2 consists of all CTRIA3 and CQUAD4 elements except element 21.

```
SET 2=QUAD4 TRIA3 EXCEPT 21
```

3. SET 10 includes all CTRIAR elements plus elements 70 through 80.

```
SET 10  TRIAR  INCLUDE ELEMENTS 70 THRU 80
```

4. SET 15 includes all elements from 15 to 20 and 26 to 100.

```
SET 15=15 THRU 100 EXCEPT 21 THRU 25
```

5. SET 2 includes all elements except CTETRA elements.

```
SET 2=ALL EXCEPT TETRA
```


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Describers:

Describer	Meaning
n	Set identification number. (0<Integer<999999)
ALL	Selects all elements or grid points. See typei.
ELEMENTS	Specifies that all identification numbers refer to elements.
GRID POINTS	Specifies that all identification numbers refer to grid points.
INCLUDE	Includes specified element or grid point identification numbers or elements in the set.
EXCLUDE	Excludes specified element or grid point identification numbers or element types in the set.
EXCEPT	Modifies a prior ALL, INCLUDE, or EXCLUDE specification.
typei	Element types. The allowed element types are (Character):

typei	Element Entry Name
TRIA3	CTRIA3
TRIA6	CTRIA6
TRIAR	CTRIAR
QUAD4	CQUAD4
QUAD8	CQUAD8
QUADR	CQUADR
HEXA	CHEXA
PENTA	CPENTA
TETRA	CTETRA

THRU	Specifies a range of identification numbers.
BY	Specifies an increment for a THRU specification.
inci	Increment for THRU range. (Integer>0)

Remarks:

1. This form of the SET command must and can only be specified after an OUTPUT(PLOT) delimiter.
2. The INCLUDE, EXCLUDE, and EXCEPT specifications may be specified more than once in the same set. See Examples above.
3. Commas or spaces may be used as separators.

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4. Not all of the identification numbers in a THRU range have to correspond to elements or grid points. For example, elements 2, 4, 7, and 9 may be selected with 2 THRU 9, even if elements 3, 5, 6, and 8 do not exist. This is called an open set. It should be noted that large open sets can cause higher computational costs.

Related Topics

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SETMC

Modal, Panel, and Grid Contribution Set Definition

Set definitions for modal, panel, and grid contribution results.

Format:

SETMC *n* = rtype / id1 (item1), id2 (item2), ...

Examples:

```
SETMC 121 = ACCE/99(T3),1200(T1),1399(R2)
SETMC 222 = STRESS/134(22)
SETMC 343 = ACCE/99(T3),1200(T1),1399(R2),STRESS/134(22)
```

Describers:

Describer	Meaning	
n	Set identification number. Any set may be redefined by reassigning its identification number. SETMC commands specified under a SUBCASE command are recognized for that SUBCASE only. (Integer > 0; No default)	
rtype	Response type (Character; No default)	
	ACCE	Acceleration
	ACPO	Acoustic power
	DISP	Displacement
	ELFO	Element force
	PRESSURE	Pressure
	SPCF	Single-point force of constraint
	STRA	Strain
	STRE	Stress
	VELO	Velocity
idj	Element or grid point identification number for j-th value. (Integer > 0; No default)	
itemj	Item code for j-th value. (Character, integer > 0, or blank; No default)	

For elements, the code represents a component of the element stress or force and is described in the *Item Codes* topic in this guide.

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Describer	Meaning
	For grid points, the code is one of the mnemonics T1, T2, T3, R1, R2, or R3, where Ti stands for the i-th translational component and Ri stands for the i-th rotational component.

Note

For rtype:

- = PRES, use itemj =T1.
- = ACPO, do not use itemj. See Remark 3.

Remarks:

1. The SETMC command may be continued on the next line as long as you specify SETMC *n* = rtype / on the first line.

   For example, the following SETMC specification can be continued:

   ```
   SETMC 33 = DISP/
   ```

2. A SETMC command may be more than one physical command. A comma at the end of a physical command signifies a continuation command. Commas may not end a set. A slash (that is, "/") or a parenthesis (that is, "(" or ")") may not begin a continuation line.
3. For acoustic power (rtype = ACPO), the idj is the ID of GROUP representing a 2D microphone mesh. This only applies to the acoustic transfer vector (ATV) response of a vibro-acoustic model for PANCON results.

Related Topics

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SETMCNAME

Title of Displacement Monitor Plot

Specifies the title of a displacement monitor plot in a preprocessor, such as Simcenter Pre/Post.

Format:

$$\text{SETMCNAME n} = \text{title of plot}$$

Examples:

```
SETMC 21 = DISP/ 1001(T1), 1002(T1), 1204(T1)
SETMCNAME 21 = Blade 3
SETMC 25 = DISP/ 1001(T2), 1002(T2), 1204(T2)
SETMCNAME 25 = Compressor Disk
SET 1 = 21, 25
MONVAR = 1
```

Describers:

Describer	Meaning
n	ID of a corresponding SETMC case control command. The SETMC and SETMCNAME commands with matching IDs can be identified by a SET command. The MONVAR case control command only references the SET command. See the examples.
plot title	Displacement monitor plot title. (Character string)

Remarks:

1. SETMCNAME is only supported for a SOL 401 and 402 statics subcase.
2. A displacement monitor plot is created for each SETMC ID selected by the MONVAR case control command. Each displacement monitor plot includes a plot of the maximum, the minimum, and the average displacement. The software uses the displacement values for all degree-of-freedom listed on a particular SETMC to determine the maximum and minimum displacement, and to compute the average displacement for each converged time step. If you want to monitor the displacement of a single degree-of-freedom, you should select a SETMC which includes only this degree-of-freedom.
3. You can optionally define a title for each displacement monitor plot using the SETMCNAME command. The ID on the SETMCNAME command should match the ID on the SETMC command. See the examples. When the SETMCNAME is undefined, the ID of the SETMC is used for the plot title.

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Related Topics

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SETS DEFINITION

Case Control Processing Delimiter

Delimits the various type of commands under grid point stress. This command is synonymous with OUTPUT(POST)

Format:

SETS DEFINITION

Examples:

```
SETS DEFINITION
```

Remarks:

SETS DEFINITION is synonymous with OUTPUT(POST). Either SETS DEFINITION or OUTPUT(POST) may be specified but not both.

Related Topics

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SFORCE

For SOL 401, requests the output of applied loads in modal coordinates.

Requests the output of applied loads in modal coordinates. Only supported by the SOL 401 modal frequency response subcase (MFREQ).

Format:

$$\text{SFORCE}\left[\left(\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{REAL}}\ \text{or}\ \underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
SFORCE=ALL
SFORCE(PUNCH,PHASE)=25
```

Describers:

Describer	Meaning
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
PLOT	Generate, but do not print results.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
ALL	Output results for all mode numbers.
n	Output results for the mode numbers listed on a previously appearing SET command with n as the identification number. (Integer > 0)
NONE	Do not output SFORCE results.

Remarks:

1. Both PRINT and PUNCH may be requested.
2. SOL 401 only supports SORT1 output.

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Related Topics

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SHELLTHK

Shell Thickness Output Request

Requests the form of shell thickness output for SOL 402.

Format:

$$\text{SHELLTHK}\left[\left(\left[\frac{\text{PRINT}}{\text{PLOT}}\right]\right)\right] = \left\{\begin{matrix}\text{ALL}\\ \underline{\text{NONE}}\end{matrix}\right\}$$

Examples:

```
SHELLTHK(PLOT)=ALL
```

Describers:

Describer	Meaning
PRINT	The printer will be the output medium.
PLOT	Computes and puts shell thickness results in op2 file only.
ALL	Shell thickness results will be output for all applicable shell elements.
NONE	Shell thickness results will not be output.

Remarks:

1. The shell thickness output is not available for linear elastic materials without temperature dependency.
2. Shell thickness results are output at nodes on elements.

Related Topics

Browse Simcenter Nastran documentation

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SKIP

Case Control Processing Delimiter

Activates or deactivates the execution of subsequent commands in the Case Control (including plot commands).

Format:

$$\text{SKIP} \left\{ \begin{matrix} \text{ON} \\ \text{OFF} \end{matrix} \right\}$$

Examples:

```
SKIPOFF
```

Remarks:

1. SKIPON and SKIPOFF commands may appear as many times as needed in the Case Control.
2. Commands that are skipped will be printed.
3. SKIPON ignores subsequent commands until either a SKIPOFF or BEGIN BULK command is encountered. This allows the user to omit requests without deleting them from his data. In the following example, plot commands will be skipped.

```
TITLE = EXAMPLE
SPC = 5
LOAD = 6
SKIPON $ SKIP PLOT REQUEST
OUTPUT (PLOT)
SET 1 INCLUDE ALL
FIND
PLOT
BEGIN BULK
```

Related Topics

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SMETHOD

Iterative Solver Method Selection

Selects iterative solver parameters.

Format:

SMETHOD=n

Examples:

```
SMETHOD=77
```

Describers:

Describer	Meaning
n	Set identification of an ITER Bulk Data entry. (Integer>0)

Remarks:

1. The iterative solver is requested by specifying ITER=YES on the NASTRAN statement. The SMETHOD command is optional and only required to override defaults shown on the ITER Bulk Data entry.
2. SMETHOD is only available in SOLs 101, 106, 108, 111, and 401.

Related Topics

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SPC

Single-Point Constraint Set Selection

Selects a single-point constraint set to be applied.

Format:

SPC=n

Examples:

```
SPC=10
```

Describers:

Describer	Meaning
n	Set identification number of a single-point constraint that appears on a SPC, SPC1, or SPCADD Bulk Data entry. (Integer>0)

Remarks:

1. In cyclic symmetry analysis, this command must appear above the first SUBCASE command.
2. Multiple boundary conditions are only supported in SOLs 101, 103, 105, 106, 145 and 200. Multiple boundary conditions are not allowed for upstream superelements. The BC command must be specified to define multiple boundary conditions for the residual structure in SOLs 103, 105, 145 and 200.

Related Topics

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SPCFORCES

Single-Point Forces of Constraint Output Request

Requests the form and type of single-point force of constraint vector output.

Format:

$$\text{SPCFORCES}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\left[\begin{matrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{matrix}\right],\left[\begin{matrix}\underline{\text{RPRINT}}\\ \text{NORPRINT}\end{matrix}\right],[\text{RPUNCH}]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
SPCFORCES=5
SPCFORCES(SORT2,PUNCH,PRINT,IMAG)=ALL
SPCFORCES(PHASE)=NONE
```

Describers:

Describer	Meaning
SORT1	Output will be presented as a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output will be presented as a tabular listing of frequency or time for each grid point.
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
PLOT	Generates, but does not print, single-point forces of constraint.

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Describer	Meaning
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output.
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
PSDF	Requests the power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 11.
ATOC	Requests the autocorrelation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 11.
CRMS	Requests the cumulative root mean square function be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 11.
RMS	Requests the root mean square and zero crossing functions be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 11.
RALL	Requests all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 11.
RPRINT	Writes random analysis results to the print file. (Default) See Remark 11.
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 11.
RPUNCH	Writes random analysis results to the punch file. See Remark 11.
ALL	Single-point forces of constraint for all points will be output. See Remark 2 and Remark 5.
NONE	Single-point forces of constraint for no points will be output.
n	Set identification of a previously appearing SET command. Only single-point forces constraint for points with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

1. Both PRINT and PUNCH may be requested.
2. The defaults for SORT1 and SORT2 depend on the type of analysis:
   - SORT1 is the default in static analysis, frequency response, steady state heat transfer analysis, real and complex eigenvalue analysis, flutter analysis, and buckling analysis. If SORT2 is selected in a frequency response solution for one or more of the commands ACCE, DISP, FORC, GPFO, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT2 format.

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- SORT2 is the default in transient response analysis (structural and heat transfer). SORT2 is not available for real eigenvalue (including buckling), complex eigenvalue, or flutter analysis. If SORT1 is selected in a transient solution for one or more of the commands ACCE, DISP, ENTH, FORC, GPFO, HDOT, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT1 format.
- XY plot requests will force SORT2 format thus overriding SORT1 format requests.

3. In a statics solution, a SORT2 request will output at the grids which have both zero and nonzero SPCFORCES. A SORT1 (default) request will only output at the grids which have nonzero SPCFORCES.
4. SPCFORCES=NONE overrides an overall output request.
5. In SORT1 format, SPCFORCESs recovered at consecutively numbered scalar points are printed in groups of six (sextets) per line of output. But if a scalar point is not consecutively numbered, then it will begin a new sextet on a new line of output. If a sextet can be formed and all values are zero, then the line will not be printed. If a sextet cannot be formed, then zero values may be output.
6. SPCFORCES results are output in the global coordinate system (see field CD on the GRID Bulk Data entry).
7. In SOLs 129 and 159, SPCFORCES results do not include the effects of mass and damping elements.
8. In all solution sequences except SOLs 129 and 159, SPCFORCES results do include the effects of mass and damping, except damping selected by SDAMPING command. PARAM,DYNSPCF,OLD may be specified to obtain SPCFORCES results, which do not include mass and damping effects.
9. In inertia relief analysis, the SPCFORCE output includes both the effects due to inertial loads and applied loads.
10. In SOL 106, the moments in the SPCFORCE output are computed in the undeformed configuration. They do not include the effects of deformations.
11. The following applies to random solutions:
    - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, PUNCH, PLOT describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
    - The SORT1 and SORT2 describers only control the output format for the frequency response output. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
    - Any combination of the PSDF, ATOC, RMS, and CRMS describers can be selected. The RALL describer selects all four.
    - When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the single-point constraint force results are based on the nominal values for stiffness and damping.

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Remarks related to SOL 402:

1. SPCFORCES is not supported for SOL 402 for an ANALYSIS=CEIG analysis.

Related Topics

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STATSUB

Static Solution Selection for Differential Stiffness

Selects the static solution to use in forming the differential stiffness for buckling analysis, normal modes, complex eigenvalue, frequency response and transient response analysis.

Format:

$$\text{STATSUB}\left[\begin{pmatrix}\text{BUCKLING}\\ \text{PRELOAD}\end{pmatrix}\right] = \text{n}$$

Examples:

```
STATSUB=23
STAT=4
STATSUB(PREL)=7
```

Describers:

Describer	Meaning
BUCKLING	Subcase id number corresponding to static subcase of buckling or varying load. (Default in buckling analysis.)
PRELOAD	Subcase id number corresponding to static subcase of preload or constant load. (Default in dynamic analysis.)
n	Subcase identification number of a prior SUBCASE specified for static analysis (Integer>0)

Remarks:

1. STATSUB may be used in:
   - SOLs 103, 105, 107, 108, 109, 110, 111, 112, 115, 116, and 187.
   - SOL 200 with an ANALYSIS=BUCKLING subcase.
2. STATSUB must be specified in the same subcase that contains the METHOD selection for buckling or normal modes, CMETHOD for complex eigenvalue analysis, TSTEP for transient response and FREQ for frequency response.
3. In SOL 105, if it is intended that results from the first static subcase are used to compute the differential stiffness, then the STATSUB command is not required. In other words, the default for STATSUB is the first static subcase identification. In the other supported solutions, STATSUB must reference a separate static subcase.

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4. In dynamic analysis, only one STATSUB command may be specified in each dynamic subcase. In buckling analysis with a pre-load, both STATSUB (BUCKLING) and STATSUB(PRELOAD) must be specified in each buckling subcase.
5. If superelements are used , the static subcase which is identified in the STATSUB command must include the residual superelement or the solution will fail with an error.
6. If multiple superelements are used with the STATSUB command, each must be used in a static subcase or their differential stiffness will not be included. All superelements can be used in a single static subcase using SUPER=ALL, or a separate subcase can exist for each one (order of subcases is not important). In either case, the STATSUB command should point to the subcase which references the residual (see above remark).

Related Topics

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STATVAR

Requests output of state variables for SOLs 401 and 402.

Requests output of state variables used by a user material subroutine with SOLs 401 and 402.

Format:

$$\text{STATVAR}\left[\left(\begin{bmatrix}\underline{\text{GRID}}\\ \text{GAUSS}\\ \text{BOTH}\end{bmatrix}, \underline{\text{PRINT}}, \text{PUNCH}, \text{PLOT}\right)\right] = \begin{Bmatrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{Bmatrix}$$

Examples:

```
STATVAR=ALL
STATVAR(PRINT,PUNCH)=17
```

Describers:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
GRID	Output at grid points on elements.
GAUSS	Output at gauss points on elements.
BOTH	Output at both grid and gauss points on elements.
ALL	Requests output for all elements.
n	Set identification number of a previously appearing SET command. The SET command lists element IDs. (Integer>0)
NONE	No output requested.

Remarks:

1. Only supported in SOLs 401 or 402.
2. SOL 402 can output a maximum of 64 state variables. If the number of state variables for output is larger than 64:

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• When SETID=0 (default) is used on the MUMAT or MUCRP entries, the software will output the first 64 variables defined on the MUMAT or MUCRP entries, and report a warning.

• When SETID references a SET1 entry listing more than 64 variables for output, the software will output the first 64 listed on the SET1 entry, and report a warning.

Related Topics

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STRAIN

Element Strain Output Request

Requests the form and type of strain output.

Format:

$$\text{STRAIN}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\left[\begin{matrix}\underline{\text{VON MISES}}\\ \text{MAXS or SHEAR}\end{matrix}\right],\right.\right.$$

$$\left[\begin{matrix}\underline{\text{SRTCUR}}\\ \text{FIBER}\end{matrix}\right],\left[\begin{matrix}\underline{\text{CENTER}}\\ \text{CORNER or BILIN}\\ \text{SGAGE}\\ \text{CUBIC}\end{matrix}\right],\left[\begin{matrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{matrix}\right],\left[\begin{matrix}\text{RPRINT}\\ \underline{\text{NORPRINT}}\end{matrix}\right],[\text{RPUNCH}],$$

$$\left.\left.\left[\begin{matrix}\underline{\text{CPLYMID}}\\ \text{CPLYBT}\\ \text{CPLYBMT}\end{matrix}\right]\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
STRAIN=5
STRAIN(CORNER)=ALL
STRAIN(PRINT,PHASE)=15
STRAIN(PLOT)=ALL
```


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Describers:

Describer	Meaning
SORT1	Output will be presented as a tabular listing of elements for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output will be presented as a tabular listing of frequency or time for each element.
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
PLOT	Generates strain for the requested set but no printer output.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output.
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
PSDF	Requests the power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 10.
ATOC	Requests the autocorrelation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 10.
CRMS	Requests the cumulative root mean square function be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 10.
RMS	Requests the root mean square and zero crossing functions be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 10.
RALL	Requests all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 10.
RPRINT	Writes random analysis results to the print file. (Default) See Remark 10.
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 10.
RPUNCH	Writes random analysis results to the punch file. See Remark 10.
VONMISES	von Mises strain is output.
MAXS or SHEAR	Maximum shear strains are output.
STRCUR	Strain at the reference plane and curvatures is output for plate elements.
FIBER	Strain at locations Z1, Z2 is computed for plate elements.
CENTER	Outputs strains at the center of the element only. See Remark 16.

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Describer	Meaning
CORNER or BILIN	Outputs strains at the center and corner grid points using extrapolation. See Remark 17.
SGAGE	Outputs strains at the center and corner grid points using strain gage approach. See Remark 18.
CUBIC	Outputs strains at the center and corner grid points using cubic bending correction. See Remark 19.
CPLYMID	Requests element strains at the middle of each ply for elements referencing PCOMPS property entries. See Remark 11, Remark 12, Remark 13, and Remark 15.
CPLYBT	Requests element strains at the bottom and top of the ply for elements referencing PCOMPS property entries. See Remark 11, Remark 12, Remark 13, and Remark 15.
CPLYBMT	Requests element strains at the bottom, middle, and top of each ply for elements referencing PCOMPS property entries. See Remark 11, Remark 12, Remark 13, and Remark 15
ALL	Output strain for all elements.
n	Set identification of a previously appearing SET command. Only strain for elements with identification numbers that appear on this SET command will be output. (Integer>0)
NONE	No element strain will be output.

Remarks:

1. In SOLs 106 and 129, nonlinear strains for nonlinear elements are requested by the STRESS/NLSTRESS commands and appear in the nonlinear stress output. The STRAIN command will generate additional output for total strain except for hyperelastic elements. The additional STRAIN output request will also be ignored for nonlinear material elements when the parameter LGDISP is -1, which is the default (strains will appear in the nonlinear stress output).
2. Both PRINT and PUNCH may be requested.
3. STRAIN=NONE overrides an overall output request.
4. The PLOT option is used when strains are requested for postprocessing but no printed output is desired.
5. Definitions of stress, strain, curvature, and output locations are given in the *Simcenter Nastran Element Library Reference Manual*.
6. If the STRCUR option is selected, the values of Z1 will be set to 0.0 and Z2 will be set to -1.0 on the output.
7. The VONMISES, MAXS, and SHEAR describers are ignored in the complex eigenvalue and frequency response solution sequences. Although, von Mises stress and strain are computed by default in the frequency response solutions 108 and 111 when stress and strain results are requested. The system

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cell setting NASTRAN SYSTEM(579)=1 can be defined to disable the von Mises stress and strain request in SOLs 108 and 111.

8. The options CENTER, CORNER, CUBIC, SGAGE, and BILIN are recognized only in the first subcase and determine the option to be used in all subsequent subcases with the STRESS, STRAIN, and FORCE commands. (In superelement analysis, the first subcase refers to the first subcase of each superelement. Therefore, it is recommended that these options be specified above all subcases.) Consequently, options specified in subcases other than the first subcase will be ignored. See also the **FORCE** command for further discussion. These options are discussed in "Understanding Plate and Shell Element Output" in the *Simcenter Nastran Element Library Reference*.

9. The defaults for SORT1 and SORT2 depend on the type of analysis:

   - SORT1 is the default in static analysis, frequency response, steady state heat transfer analysis, real and complex eigenvalue analysis, flutter analysis, and buckling analysis. If SORT2 is selected in a frequency response solution for one or more of the commands ACCE, DISP, FORC, GPFO, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT2 format.
   - SORT2 is the default in transient response analysis (structural and heat transfer). SORT2 is not available for real eigenvalue (including buckling), complex eigenvalue, or flutter analysis. If SORT1 is selected in a transient solution for one or more of the commands ACCE, DISP, ENTH, FORC, GPFO, HDOT, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands will also be output in SORT1 format.
   - XY plot requests will force SORT2 format thus overriding SORT1 format requests.

10. The following applies to random solutions:

    - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, PUNCH, PLOT describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
    - The SORT1 and SORT2 describers only control the output format for the frequency response output. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
    - Any combination of the PSDF, ATOC, RMS, and CRMS describers can be selected. The RALL describer selects all four.

11. If some combination of the CPLYMID, CPLYBT, or CPLYBMT describers are specified, the describer producing the most output data is used.

12. If different CPLYMID, CPLYBT, or CPLYBMT describers are specified on STRESS and STRAIN case control commands in the same input file, the describer specified on the STRESS case control command takes precedence.

13. Failure indices and strength ratios for in-plane ply failure are output for the locations corresponding to the CPLYMID/CPLYBT/CPLYBMT specification and the CENTER/CORNER (or BILIN)/SGAGE/CUBIC specification. Failure indices and strength ratios for inter-laminar failure are always output at the top and bottom of each ply at the locations corresponding to the CENTER/CORNER (or BILIN)/SGAGE/CUBIC specification.

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14. A PSHELL entry with a MID1 or MID2 greater than or equal to 10^8 requires the parameter NOCOMPS to be 0 or -1 for any stress or strain recovery. See the parameter **NOCOMPS** for more information.
15. For elements referencing PCOMPS property entries, the REAL, IMAG, PHASE, VONMISES, MAXS, SHEAR, STRCUR, FIBER, SGAGE, CUBIC, PSDF, ATOC, CRMS, RALL, RPRINT, NOPRINT, and RPUNCH describers are not supported.
16. For the CENTER option, strains are output at the center of the element for CQUAD4, CQUADR, and CTRIAR elements that reference a PSHELL entry. For CQUAD4, CQUADR, CTRIA3, CTRIAR, CTRIA6, and CQUAD8 elements that reference a PCOMP or PCOMPG entry, ply strains are always reported at the center of the element. For CHEXA and CPENTA elements that reference a PCOMPS entry, the ply strains are output at the center of the element for each ply.
17. For the CORNER (or BILIN) option, strains are output at the center and grid points for CQUAD4, CQUADR, and CTRIAR elements that reference a PSHELL entry. For CHEXA and CPENTA elements that reference a PCOMPS entry, the ply strains are output at the center and corner grid locations for each ply.
18. For the SGAGE option, strains are output at the center and grid points for CQUAD4 elements that reference a PSHELL entry. For CHEXA and CPENTA elements that reference a PCOMPS entry, the output is the same as that obtained by specifying CORNER or BILIN.
19. For the CUBIC option, strains are output at the center and grid points for CQUAD4 and CQUADR elements that reference a PSHELL entry. For CHEXA and CPENTA elements that reference a PCOMPS entry, the output is the same as that obtained by specifying CORNER or BILIN.
20. Shear strains are output as engineering shear strains, which are twice the tensor shear strains.
21. SOL 106 outputs engineering stress and strain.
22. When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the strain results are based on the nominal values for stiffness and damping.

Remarks related to SOL 401:

1. Output requested with the STRESS, STRAIN, and FORCE commands is only at the element corners. The CENTER, CUBIC, or SGAGE options are ignored.
2. See Shell element output summary.
3. See SOLs 401 and 402 - Stress-strain measures.

Remarks related to SOL 402:

1. Output is restricted to REAL and SORT1 format. SORT2, IMAG, PHASE, CUBIC, SGAGE, PSDF, ATOC, RMS, and RALL are ignored.
2. Strains are output at corners of the elements, regardless the CENTER/CORNER describer presence.
3. For composite solid elements, strains are output at corners of top and bottom plies (CPLYBT), regardless the CPLYMID/CPLYBT/CPLYBMT describer presence .

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4. Independently of the linearity or non-linearity of the model, elements results are output in the op2 file using the same data block.
5. Strain output recovery is not supported for beam and rod elements.

Related Topics

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STRESS

Element Stress Output Request

Requests the form and type of element stress output.

Format:

$$\text{STRESS}\left[\left(\left[\begin{matrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{matrix}\right],\left[\begin{matrix}\underline{\text{PRINT}},\text{PUNCH}\\ \text{PLOT}\end{matrix}\right],\left[\begin{matrix}\underline{\text{REAL}}\text{ or }\underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\left[\begin{matrix}\underline{\text{VON MISES}}\\ \text{MAXS or SHEAR}\end{matrix}\right],\right.\right.$$

$$\left.\left.\left[\begin{matrix}\underline{\text{CENTER}}\\ \text{CORNER or BILIN}\\ \text{SGAGE}\\ \text{CUBIC}\end{matrix}\right],\left[\begin{matrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{matrix}\right],\left[\begin{matrix}\text{RPRINT}\\ \text{NORPRINT}\end{matrix}\right],[\text{RPUNCH}],\left[\begin{matrix}\underline{\text{CPLYMID}}\\ \text{CPLYBT}\\ \text{CPLYBMT}\end{matrix}\right]\right)\right]$$

$$=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
STRESS=5
STRESS(CORNER)=ALL
STRESS (SORT1,PRINT,PUNCH,PHASE)=15
STRESS(PLOT)=ALL
```

Describers:

Describer	Meaning
SORT1	Output will be presented as a tabular listing of elements for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output will be presented as a tabular listing of frequency or time for each element type.
PRINT	The printer will be the output medium.

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Describer	Meaning
PUNCH	The punch file will be the output medium.
PLOT	Generates stresses for requested set but no printer output.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output.
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
PSDF	Requests that the power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 12.
ATOC	Requests that the auto-correlation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 12.
CRMS	Requests that the cumulative root mean square function be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 12.
RMS	Requests that the root mean square and zero crossing functions be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 12.
RALL	Requests that all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the case control. See Remark 12.
RPRINT	Writes random analysis results to the print file. (Default) See Remark 12.
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 12.
RPUNCH	Writes random analysis results to the punch file. See Remark 12.
VONMISES	von Mises stress is output.
MAXS or SHEAR	Requests maximum shear in the plane for shell elements and octahedral stress for solid elements.
CENTER	Outputs stresses at the center of the element only. See Remark 18.
CORNER or BILIN	Outputs stresses at the center and corner grid points using extrapolation. See Remark 19.
SGAGE	Outputs stresses at the center and corner grid points using strain gauge approach. See Remark 20.
CUBIC	Outputs stresses at the center and corner grid points using cubic bending correction. See Remark 21.
CPLYMID	Requests element stresses at the middle of each ply for elements referencing PCOMPS property entries. See Remark 13, Remark 14, Remark 15, and Remark 17.

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Describer	Meaning
CPLYBT	Requests element stresses at the bottom and top of the ply for elements referencing PCOMPS property entries. See Remark 13, Remark 14, Remark 15, and Remark 17.
CPLYBMT	Requests element stresses at the bottom, middle, and top of each ply for elements referencing PCOMPS property entries. See Remark 13, Remark 14, Remark 15, and Remark 17.
ALL	Stresses for all elements will be output.
n	Sets identification of a previously appearing SET command. Only stresses for elements with identification numbers that appear on this SET command will be output. (Integer >0)
NONE	No element stress will be output.

Remarks:

1. Both PRINT and PUNCH may be requested.
2. ALL should be used with caution in transient solutions as it can produce a large amount of output.
3. The defaults for SORT1 and SORT2 depend on the type of analysis:
   - SORT1 is the default in static analysis, frequency response, steady state heat transfer analysis, real and complex eigenvalue analysis, flutter analysis, and buckling analysis. If SORT2 is selected in a frequency response solution for one or more of the commands ACCE, DISP, FORC, GPFO, MPCF, OLOA, SPCF, STRA, STRE, and VELO, then the remaining commands will also be output in SORT2 format.
   - SORT2 is the default in transient response analysis (structural and heat transfer). SORT2 is not available for real eigenvalue (including buckling), complex eigenvalue, or flutter analysis. If SORT1 is selected in a transient solution for one or more of the commands ACCE, DISP, ENTH, FORC, GPFO, HDOT, MPCF, OLOA, SPCF, STRA, STRE, and VELO, then the remaining commands will also be output in SORT1 format.
   - XY plot requests will force SORT2 format thus overriding SORT1 format requests.
4. STRESS=NONE overrides an overall output request.
5. SOLs 106 and 129 support two output formats for stress: the linear output format and the nonlinear output format. The formats have a different data organization. For example, the nonlinear format includes stresses together with strains, and it provides more information with regard to nonlinear material laws (effective strain, equivalent stress). See the *Handbook of Nonlinear Analysis* for details on the computation of effective strain and equivalent stress.

   The following summarizes the output requests for these formats.

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- To output both linear and nonlinear stress formats to the same media, you only need to define the STRESS command. For example,

```
STRESS(PLOT)=ALL
```

- To output only the linear stress format, you need the STRESS command and NLSTRESS=NONE. For example,

```
STRESS(PLOT)=ALL
NLSTRESS=NONE
```

- To output only the nonlinear stress format, you only define the NLSTRESS command without defining the STRESS command. For example,

```
NLSTRESS(PLOT)=ALL
```

- To output both linear and nonlinear stress formats but to different media, you would need the STRESS and the NLSTRESS commands, each defined with a different media. For example,

```
STRESS(PLOT)=ALL
NLSTRESS(PUNCH)=ALL
```

6. The VONMISES option is ignored for ply stresses.
7. The VONMISES, MAXS, and SHEAR describers are ignored in the complex eigenvalue and frequency response solution sequences. Although, von Mises stress and strain are computed by default in the frequency response solutions 108 and 111 when stress and strain results are requested. The system cell setting NASTRAN SYSTEM(579)=1 can be defined to disable the von Mises stress and strain request in SOLs 108 and 111.
8. The options CENTER, CORNER, CUBIC, SGAGE, and BILIN are recognized only in the first subcase and determine the option to be used in all subsequent subcases with the STRESS, STRAIN, and FORCE commands. (In superelement analysis, the first subcase refers to the first subcase of each superelement. Therefore, it is recommended that these options be specified above all subcases.) Consequently, options specified in subcases other than the first subcase will be ignored. See also the **FORCE** command for further discussion. These options are discussed in the section “Understanding Plate and Shell Element Output” in Chapter 4 of the *Simcenter Nastran Element Library Reference*.
9. For composite ply output, the grid point option for CQUAD4 elements will be reset to the default option (CENTER).
10. For nonlinear analysis, the grid point option for CQUAD4 elements will be reset to the default (CENTER) option for nonlinear elements.
11. MAXS for shell elements is not an equivalent stress.
12. The following applies to random solutions:
    - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT,

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PUNCH, PLOT describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.

- The SORT1 and SORT2 describers only control the output format for the frequency response output. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
- Any combination of the PSDF, ATOC, RMS, and CRMS describers can be selected. The RALL describer selects all four.

13. If some combination of the CPLYMID, CPLYBT, or CPLYBMT describers are specified, the describer producing the most output data is used.

14. If different CPLYMID, CPLYBT, or CPLYBMT describers are specified on STRESS and STRAIN case control commands in the same input file, the describer specified on the STRESS case control command takes precedence.

15. Failure indices and strength ratios for in-plane ply failure are output for the locations corresponding to the CPLYMID/CPLYBT/CPLYBMT specification and the CENTER/CORNER (or BILIN)/SGAGE/CUBIC specification. Failure indices and strength ratios for inter-laminar failure are always output at the top and bottom of each ply at the locations corresponding to the CENTER/CORNER (or BILIN)/SGAGE/CUBIC specification.

16. A PSHELL entry with a MID1 or MID2 greater than or equal to 10^8 requires the parameter NOCOMPS to be 0 or -1 for any stress or strain recovery. See the parameter **NOCOMPS** for more information.

17. For elements referencing PCOMPS property entries, the REAL, IMAG, PHASE, VONMISES, MAXS, SHEAR, SGAGE, CUBIC, PSDF, ATOC, CRMS, RALL, RPRINT, NOPRINT, and RPUNCH describers are not supported.

18. For the CENTER option, stresses are output at the center of the element for CQUAD4, CQUADR, and CTRIAR elements that are referenced by a PSHELL entry. For CQUAD4, CQUADR, CTRIA3, CTRIAR, CTRIA6, and CQUAD8 elements that reference a PCOMP or PCOMPG entry, ply stresses are always reported at the center of the element. For CQUAD4, CQUADR, and CTRIAR elements that reference a PCOMP or PCOMPG entry and PARAM,NOCOMPS is greater than or equal to 0 (see PARAM,NOCOMPS), the homogeneous stresses (based on the smeared representation of the laminate properties) will be output at the center of the element. For CHEXA and CPENTA elements that reference a PCOMPS entry, the ply stresses are output at the center of the element for each ply.

19. For the CORNER (or BILIN) option, stresses are output at the center and grid points for CQUAD4, CQUADR, and CTRIAR elements that reference a PSHELL entry. For CHEXA and CPENTA elements that reference a PCOMPS entry, the ply stresses are output at the center and corner grid locations for each ply. For CQUAD4, CQUADR, and CTRIAR elements that reference a PCOMP or PCOMPG entry and PARAM,NOCOMPS is greater than or equal to 0 (see PARAM,NOCOMPS), the homogeneous stresses (based on the smeared representation of the laminate properties) will be output at the center and grid points.

20. For the SGAGE option, stresses are output at the center and grid points for CQUAD4 elements that reference a PSHELL entry. For CHEXA and CPENTA elements that reference a PCOMPS entry, the output is the same as that obtained by specifying CORNER or BILIN. For CQUAD4, CQUADR, and CTRIAR elements that reference a PCOMP or PCOMPG entry and PARAM,NOCOMPS is greater than or equal to 0 (see PARAM,NOCOMPS), the homogeneous stresses (based on the smeared representation of the laminate properties) will be output at the center and grid points of the element.

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21. For the CUBIC option, stresses are output at the center and grid points for CQUAD4 and CQUADR elements that reference a PSHELL entry. For CHEXA and CPENTA elements that reference a PCOMPS entry, the output is the same as that obtained by specifying CORNER or BILIN. For CQUAD4, CQUADR, and CTRIAR elements that reference a PCOMP or PCOMPG entry and PARAM,NOCOMPS is greater than or equal to 0 (see PARAM,NOCOMPS), the homogeneous stresses (based on the smeared representation of the laminate properties) will be output at the center and grid points of the element.
22. For the output coordinate system of CQUAD4 and CTRIA3 elements, see **CQUAD4** and **CTRIA3**. For the output coordinate system of CHEXA, CPENTA, CPYRAM, and CTETRA elements, see **PSOLID** and **PLSOLID**.
23. In SOL 401, for elements in which plasticity is defined, the von Mises stress defines the yield surface. When kinematic or combined hardening is used and there is loading and unloading which causes the center of the yield surface to translate, the von Mises stress output is computed relative to the new yield surface origin. The stress components are computed from the original yield surface origin. In this case, the von Mises stress computed in a post processor using the component stresses may not match the von MIses stress reported by Simcenter Nastran in the .f06 file.
24. SOL 106 outputs engineering stress and strain.
25. When frequency-dependent elements are present in a frequency response analysis or nonlinear bearing elements are present in a rotor dynamic analysis, the stress results are based on the nominal values for stiffness and damping.

Remarks related to SOL 401:

1. Output requested with the STRESS, STRAIN, and FORCE commands is only at the element corners. The CENTER, CUBIC, or SGAGE options are ignored.
2. See Shell element output summary.
3. See SOLs 401 and 402 - Stress-strain measures.

Remarks related to SOL 402:

1. Output is restricted to REAL and SORT1 format. SORT2, IMAG, PHASE, CUBIC, SGAGE, PSDF, ATOC, RMS, and RALL are ignored.
2. Stresses are output at the corners of the elements, regardless of the CENTER/CORNER describer presence.
3. For composite solid elements, stresses are output at the corners of top and bottom plies (CPLYBT), regardless of the CPLYMID/CPLYBT/CPLYBMT describer presence.
4. Independently of the linearity or non-linearity of the model, elements results are output in the op2 file using the same data block.
5. When the material exhibits a plastic behavior with kinematic or mixed hardening (i.e., MATS1 with HR=2 or 3), the equivalent Von Mises stress computed by the solver takes the back stress (updated center of plastic surface) into account. Recomputing the equivalent stress from the stress tensor would not necessarily produce the same result.

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Related Topics

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STRFIELD

Grid Point Stress Output Request

Requests the computation of grid point stresses for graphical postprocessing and mesh stress discontinuities.

Format:

$$\text{STRFIELD} = \left\{ \begin{matrix} \text{ALL} \\ \text{n} \end{matrix} \right\}$$

Examples:

```
STRFIELD=ALL
STRFIELD=21
```

Describers:

Describer	Meaning
ALL	Grid point stress requests for all surfaces and volumes defined in the OUTPUT(POST) section will be saved for postprocessing.
n	Set identification number of a previously appearing SET command. Only surfaces and volumes with identification numbers that appear on this SET command and in the OUTPUT(POST) section will be included in the grid point stress output request for postprocessing. (Integer>0)

Remarks:

1. The STRFIELD command is required for the graphical display of grid point stresses in postprocessors that use the .xdb file (PARAM,POST,0), or when the GPSDCON or ELSDCON commands are specified. The GPSTRESS command can be used to obtain printed output.
2. Only grid points connected to elements used to define the surface or volume are output. See the SURFACE and VOLUME commands.
3. Element stress output (STRESS) must be requested for elements referenced on requested SURFACE and VOLUME commands.
4. In nonlinear static and transient analysis, grid point stresses are computed only if parameter LGDISP is -1, which is the default. Also, in nonlinear transient analysis, grid point stresses are computed for elements with linear material properties only.

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Related Topics

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SUBCASE

Subcase Delimiter

Delimits and identifies a subcase.

Format:

SUBCASE=n

Examples:

```
SUBCASE=101
```

Describers:

Describer	Meaning
n	Subcase identification number. (Integer>0)

Remarks:

1. The subcase identification number, n, must be greater than all previous subcase identification numbers.
2. Plot requests, RANDPS, and RANDPEX requests refer to n.
3. See the MODES command for use of this command in normal modes analysis.
4. If a comment follows n, then the first few characters of the comment will appear in the subcase label in the upper right-hand corner of the output.
5. For more information about subcases, see Defining Subcases in the *Simcenter Nastran User's Guide*.

Remarks related to SOL 402:

1. For each subcase, the ANALYSIS case control command must be specified to define the analysis type for the subcase.

Related Topics

Browse Simcenter Nastran documentation

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SUBCOM

Combination Subcase Delimiter

Delimits and identifies a combination subcase.

Format:

SUBCOM = n

Examples:

```
SUBCOM = 125
```

Describers:

Describer	Meaning
n	SUBCOM identification number. (Integer>2)

Remarks:

1. The SUBCOM identification number, n, must be greater than all previous subcase and SUBCOM identification numbers.
2. A SUBSEQ command must follow this command.
3. SUBCOM may only be used in statics problems.
4. Output requests above the subcase level will be used.
5. If the referenced subcases contain thermal loads or element deformations, the user must define the temperature field in the SUBCOM with a TEMP(LOAD) command or the element deformations with a DEFORM command.
6. SUBCOMs may be specified in superelement analysis with the following recommendations:
   - For each superelement, specify its SUBCASEs consecutively, directly followed by its SUBCOM(s).
   - Specify a SUPER command with a new load sequence number under each SUBCOM command.

   The following example is given for a model with one superelement and one load combination:

```
SUBCASE 101
SUPER=1,1
LOAD=100
SUBCASE 102
SUPER=1,2
LOAD=200
SUBCOM 110
```


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```
LABEL=COMBINE SUBCASES 101 AND 102
SUPER=1,3
SUBSEQ=1.,1.
SUBCASE 1001
SUBCASE 1002
SUBCOM 1010
LABEL=COMBINE SUBCASES 1001 AND 1002
SUBSEQ=1.,1.
```

Related Topics

Browse Simcenter Nastran documentation

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SUBSEQ

Subcase Sequence Coefficients

Gives the coefficients for forming a linear combination of the previous subcases.

Format:

SUBSEQ=R1 [, R2, R3, ..., Rn]

Examples:

```
SUBSEQ=1.0, -1 .0, 0.0, 2.0
```

Describers:

Describer	Meaning
Ri	Coefficients of the previously occurring subcases. See Remark 4 . (Real)

Remarks:

1. The SUBSEQ command can only appear after a SUBCOM command.
2. This command may only be used in statics problems.
3. This command list is limited to a maximum of 200 numbers.
4. R1 to Rn refer to the immediately preceding subcases. In other words Rn is applied to the most recently appearing subcase and R(n - 1) is applied to the second most recently appearing subcase, and so on. The comments ($) describe the following example:

```
DISPL=ALL
SUBCASE 1
SUBCASE 2
SUBCOM 3
SUBSEQ=1.0, -1.0 $ SUBCASE 1 - SUBCASE 2
SUBCASE 11
SUBCASE 12
SUBCOM 13
SUBSEQ=0.0, 0.0, 1.0, -1 .0 $ SUBCASE 11 - SUBCASE 12
or
SUBSEQ=1.0, - 1.0 $ EQUIVALENT TO PRECEDING COMMAND.
USE ONLY ONE.
```


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Related Topics

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SUBTITLE

Output Subtitle

Defines a subtitle that will appear on the second heading line of each page of printer output.

Format:

SUBTITLE=subtitle

Examples:

```
SUBTITLE=PROBLEM NO. 5-1A
```

Describers:

Describer	Meaning
subtitle	Any character string.

Remarks:

1. SUBTITLE appearing under a SUBCASE command will appear in the output for that subcase only.
2. SUBTITLE appearing before all SUBCASE commands will appear in the output for all subcases except those in Remark 1 .
3. If no SUBTITLE command is present, the subtitle line will be blank.
4. The subtitle also appears on plotter output.

Remark for SOLs 402 and 414:

1. The SUBTITLE command can only be used at the global level (above the subcases), for example, to name the solution. To assign a name to each subcase, you can use the LABEL command instead.

Related Topics

Browse Simcenter Nastran documentation

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SUPER

Superelement Subcase Assignment

Assigns a subcase(s) to a superelement or set of superelements.

Format:

$$\text{SUPER} = \left\{ \begin{array}{c} \underline{\text{ALL}} \\ \left\{ \begin{array}{c} \text{n} \\ \text{i} \end{array} \right\} \left[\text{, l} \right] \end{array} \right\}$$

Examples:

```
SUPER=17, 3
SUPER=15
SUPER=ALL
```

Describers:

Describer	Meaning
i	Superelement identification number. (Integer≥0)
ALL	The subcase is assigned to all superelements and all loading conditions. (Default)
n	Set identification number of a previously appearing SET command. The subcase is assigned to all superelements with identification numbers that appear on this SET command. (Integer>0)
l	Load sequence number. (Integer>0; Default = 1)

Remarks:

1. All subcases with requests for specific superelement(s) must contain the SUPER command. If no SUPER command is specified in the Case Control Section then all subcases will be assigned to all superelements. In other words, SUPER=ALL is the default.
2. All subcases associated with superelements must precede those for the residual structure except when SUPER=ALL or SUPER=n and the selected set includes the residual structure.
3. The load sequence number is only used in static analysis and frequency response analysis when there are multiple loading conditions. Also, the residual structure must have a subcase specified for each unique load condition. This is required because the number of residual structure subcases is used to determine the number of load conditions for all superelements.

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4. The load sequence number is associated with the order of the subcases for the residual structure; i.e., the third loading condition is associated with the third subcase for the residual structure.
5. Subcases are required for superelements when there is a load, constraint, or output request.
6. If a set is referenced by n, then the SET identification number must be unique with respect to any superelement identification numbers. In addition, the same sets must be used for all loading conditions.
7. If the ALL option is used, it must be used for all loading conditions.
8. When i=0, the residual structure is used instead of a superelement. See the *Simcenter Nastran Superelement User's Guide* for a complete description of this option.

Related Topics

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SUPORT1

Fictitious Support Set Selection

Selects the fictitious support set (SUPORT1 entries only) to be applied to the model.

Format:

SUPORT1=n

Examples:

```
SUPORT1=15
SUPO=4
```

Describers:

Describer	Meaning
n	Set identification of fictitious support set defined on the SUPORT1 Bulk Data entry. (Integer>0)

Remarks:

1. SUPORT1 entries will not be used unless selected in Case Control by the SUPORT1 command.
2. SUPORT entries will be applied in all subcases.

Related Topics

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SURFACE

Surface Definition

Defines a surface for the calculation of grid point stresses, strains, or mesh stress discontinuities.

Format:

$$\text{SURFACE id SET sid,} \left[\underline{\text{FIBRE}} \left\{\begin{matrix}\underline{\text{ALL}}\\ \text{Z1}\\ \text{Z2}\\ \text{MID}\end{matrix}\right\}\right],$$

$$\left[\text{SYSTEM} \left\{\begin{matrix}\underline{\text{ELEMENT}}\\ \text{BASIC}\\ \text{CORD cid}\end{matrix}\right\}\right], \left[\underline{\text{AXIS}} \left\{\begin{matrix}\underline{\text{X1}}\\ \text{X2}\\ \text{X3}\end{matrix}\right\}\right], \left[\underline{\text{NORMAL}}\ \text{[M]} \left\{\begin{matrix}\underline{\text{R}}\\ \text{X1}\\ \text{X2}\\ \text{X3}\end{matrix}\right\}\right],$$

$$\left[\begin{matrix}\underline{\text{TOPOLOGICAL}}\\ \text{GEOMETRIC}\end{matrix}\right] \text{[TOLERANCE \{THETA\}],}$$

$$\left[\underline{\text{BRANCH}} \left[\left[\left\{\begin{matrix}\text{MESSAGE}\\ \underline{\text{NOMESSAGE}}\end{matrix}\right\}\right], \left[\begin{matrix}\text{BREAK}\\ \underline{\text{NOBREAK}}\end{matrix}\right]\right]\right]$$

Examples:

```
SURFACE 10 SET 9 NORMAL X3
```

Describers:

Describer	Meaning
id	Surface identification number (required).
SET	References a SET command that defines the elements in the surface (required). Either form of the SET command may be used.
sid	Set identification number.
FIBRE	Specifies the fiber location at which stresses will be calculated.

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Describer	Meaning
ALL	Requests output at all fiber locations; i.e., z=Z1, Z2, and MID.
Z1	Requests output z=Z1 only.
Z2	Requests output z=Z2 only.
MID	Requests output z=(Z1+Z2)/2 only.
SYSTEM	Specifies the coordinate system to be used to define the output coordinate system.
ELEMENT	Specifies the element coordinate system for output.
CORD cid	Specifies the coordinate system defined on a CORDij Bulk Data entry for output.
BASIC	Specifies the basic coordinate system for output.
AXIS	Specifies the axis of the coordinate system to be used as the x output axis and the local x-axis when geometric interpolation is used.
X1, X2, X3	Specifies the direction of the axis or the normal. X, Y, and Z may be substituted for X1, X2, and X3, respectively.
NORMAL	Specifies the reference direction for positive fiber and shear stress output, but has no effect when ELEMENT is specified.
M	Specifies the reverse of the direction given by R, X1, X2, or X3.
R	Specifies the radius vector from the origin of reference coordinate system to the grid point.
TOPOLOGICAL GEOMETRIC	Specifies the method to calculate the average grid point stress or strain. The default is TOPOLOGICAL.
theta	Specifies the interelement slope difference tolerance (in degrees) to detect stress discontinuity between elements, (not used with TOPOLOGICAL). (Real; Default = 0.0)
BRANCH	Selects whether multiple element intersections (BREAK/NOBREAK) are to be treated as discontinuities and if warning messages (MESSAGE/NOMESSAGE) are to be issued.
BREAK NOBREAK	Multiple element intersections are (or are not) to be treated as stress discontinuities.
MESSAGE NOMESSAGE	A warning message will (or will not) be issued when multiple element intersections are encountered.

Remarks:

1. SURFACE commands must be specified after OUTPUT(POST).

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2. The surface identification number must be referenced on a SET command, specifically the SET command which comes after OUTPUT(POST). This SET identification number may then be referenced on the GPSTRESS, GPSTRAIN, STRFIELD, ELSDCON, and GPSDCON commands.
3. The surface normal is also used in the definition of the local reference surface for geometric interpolation. Two options are available. In the first option, the radius vector (R) from the origin of the reference coordinate system to the grid point is used. In the second option, one axis (X1, X2, or X3) of the coordinate system is used. The direction can be reversed using the modification parameter, M. The positive side of an element is defined as a side from which the NORMAL direction emerges rather than the side determined by the connection specified on the element connection entries.
4. When the parameter ELEMENT is present, the element stresses or strains are used unmodified (defaults to output stresses in the output system). The CORD keyword references a CORDij Bulk Data entry with coordinate system identification number cid.
5. When theta=0, no testing is made. When theta is negative, grid point stresses will be calculated for each element connected to an exception point; otherwise, the best estimation of the grid point stress will be output.
6. BREAK is the default if theta is nonzero.
7. For all elements defined in SET 9 in the Examples section:
   - All fiber locations are output.
   - The basic output system is used.
   - The x-axis is x-axis of the basic system.
   - The surface normal direction point is z-axis of the basic system.
   - The topological interpolation method is used.
   - No tolerance test is made.
   - No branch test is made.

   The example illustrates a good choice for regular two-dimensional problems in the x-y plane.

Related Topics

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SVECTOR

Solution Set Eigenvector Output Request

Requests the form and type of solution set eigenvector output.

Format:

$$\text{SVECTOR}\left[\left(\underline{\text{PRINT}},\text{PUNCH}\right)\right]=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
SVECTOR=ALL
SVECTOR(PUNCH)=NONE
```

Describers:

Describer	Meaning
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
ALL	Displacements for all points (modes) will be output.
NONE	Displacements for no points (modes) will be output.
n	Set identification of a previously appearing SET command. Only displacements of points with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

1. Both PRINT and PUNCH may be requested.
2. SVECTOR=NONE overrides an overall output request.
3. Output will be presented as a tabular listing of grid points for each eigenvector.
4. SVECTOR can be useful, for example, if you would like to output eigenvectors for all grids in the OP2 file, yet only print the eigenvectors for a specific grid set to the f06 file. In this example, both DISPLACEMENT(PLOT) = ALL and SVECTOR(PRINT) = 101 would be included in the case control.

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Related Topics

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SVELOCITY

Solution Set Velocity Output Request

Requests the form and type of modal velocity output.

Format:

$$\text{SVELOCITY}\left[\left(\begin{bmatrix}\text{SORT1}\\ \text{SORT2}\end{bmatrix}, \begin{bmatrix}\underline{\text{PRINT}}, \text{PUNCH}\\ \text{PLOT}\end{bmatrix}, \begin{bmatrix}\underline{\text{REAL}} \text{ or } \underline{\text{IMAG}}\\ \text{PHASE}\end{bmatrix}\right)\right] = \begin{Bmatrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{Bmatrix}$$

Examples:

```
SVELOCITY=5
SVELOCITY(SORT2,PUNCH,PRINT,PHASE)=ALL
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence.
SORT2	Output a tabular listing of frequency or time for each grid point (or mode number).
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.
PLOT	Generate, but do not print, velocity results.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
ALL	Output velocity for solution set points (modes).

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Describer	Meaning
n	Output velocity only for the points (modes) listed on a previously appearing SET command with n as the identification number. (Integer > 0)
NONE	Do not output velocity for any points (modes).

Remarks:

1. Both PRINT and PUNCH may be requested.
2. The defaults for SORT1 and SORT2 depend on the type of analysis and are discussed. For additional information, see the remarks for the **VELOCITY** command.

   If SORT1 is specified on either an SACCELERATION, SDISPLACEMENT, or SVELOCITY command, then the other commands are also SORT1.
3. SVELOCITY=NONE overrides an overall output request.
4. Due to the differences in formulations, the modal responses for the constraint mode method of enforced motion will be different than the modal responses for the absolute displacement method of enforced motion. In order to highlight this difference, the f06 output for the constraint mode method will be labeled *alternative* solution set.
5. Velocity output is only available for transient and frequency response problems.

Related Topics

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SYM

Symmetry Subcase Delimiter

Delimits and identifies a symmetry subcase.

Format:

SYM=n

Examples:

```
SYM=123
```

Describers:

Describer	Meaning
n	Subcase identification number. (Integer>0)

Remarks:

1. The subcase identification number, n, must be greater than all previous subcase identification numbers.
2. Plot commands should refer to n.
3. Overall output commands will not propagate into a SYM subcase (i.e., any output desired must be requested within the subcase).
4. SYM may only be used in statics or inertia relief problems.

Related Topics

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Simcenter Nastran Quick Reference Guide
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SYMCOM

Symmetry Combination Subcase Delimiter

Delimits and identifies a symmetry combination subcase.

Format:

SYMCOM=n

Examples:

```
SYMCOM=123
```

Describers:

Describer	Meaning
n	Subcase identification number. (Integer>2)

Remarks:

1. The subcase identification number, n, must be greater than all previous subcase identification numbers.
2. SYMCOM may only be used in statics problems.
3. If the referenced subcases contain thermal loads or element deformations, the user must define the temperature field in the SYMCOM by use of a TEMP(LOAD) command, or the element deformations by a DEFORM command.
4. An alternate command is the SUBCOM command.
5. SYMCOMs may be specified in superelement analysis with the following recommendations:
   - For each superelement, specify its SUBCASEs consecutively, directly followed by its SYMCOM(s).
   - Specify a SUPER command with a new load sequence number under each SYMCOM command.

   The following example is given for a model with one superelement and one load combination:

```
SUBCASE 101
SUPER=1,1
LOAD=100
SUBCASE 102
SUPER=1,2
LOAD=200
SYMCOM 110
LABEL=COMBINE SUBCASES 101 AND 102
SUPER=1,3
```


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```
SYMSEQ=1.,1.
SUBCASE 1001
SUBCASE 1002
SYMCOM 1010
LABEL=COMBINE SUBCASES 1001 AND 1002
SYMSEQ=1.,1.
```

Related Topics

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SYMSEQ

Symmetry Sequence Coefficients

Gives the coefficients for combining the symmetry subcases into the total structure.

Format:

SYMSEQ=R1 [,R2,R3,..., Rn]

Examples:

```
SYMSEQ=1.0, -2.0, 3.0, 4.0
```

Describers:

Describer	Meaning
Ri	Coefficients of the previously occurring n SYM subcases. (Real)

Remarks:

1. SYMSEQ may only appear after a SYMCOM command.
2. The default value for the coefficients is 1.0 if no SYMSEQ command appears.
3. SYMSEQ may only be used in statics or inertia relief.
4. Ri is limited to a maximum of 200 numbers.

Related Topics

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TEMPERATURE

Temperature Set Selection

Selects the temperature set to be used in either material property calculations or thermal loading in heat transfer and structural analysis.

Format:

$$\text{TEMPERATURE}\left[\left(\begin{array}{c}\text{INITIAL}\\ \text{MATERIAL}\\ \text{LOAD}\\ \underline{\text{BOTH}}\end{array}\right)\right]=\text{n}$$

Examples:

```
TEMPERATURE(LOAD)=15
TEMPERATURE(MATERIAL)=7
TEMPERATURE=7
```

Describers:

Describer	Meaning
MATERIAL	The selected temperature set will be used to determine temperature-dependent material properties indicated on the MATTi Bulk Data entries. See Remarks 6 and 7.
LOAD	The selected temperature set will be used to determine an equivalent static load and to update material properties in a nonlinear analysis. See Remarks 6 and 7.
BOTH	Both MATERIAL and LOAD will use the same temperature set.
n	Set identification number of TEMP, TEMPD, TEMPP1, TEMPRB, or TEMPEX bulk entries. (Integer>0)
INITIAL	The selected temperature set will be used to determine initial temperature distribution. See Remarks 6 and 7.

Remarks:

1. In a linear, non-superelement analysis, only one temperature-dependent material request can be made, and should be specified above the subcase level. A fatal error will occur if multiple temperature-dependent material requests are made. See also Remarks 6 and 7.

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In a superelement analysis, each superelement subcase can have a different TEMPERATURE(MATERIAL), but consecutive load cases must have consistent TEMPERATURE(MATERIAL).

2. The total load applied will be the sum of the external (LOAD command), thermal (TEMP(LOAD) command), element deformation (DEFORM command) and constrained displacement (SPC command) loads.
3. Static, thermal, and element deformation loads should have unique set identification numbers.
4. INITIAL is used in steady state heat transfer analysis for conduction material properties and provides starting values for iteration.
5. In superelement data recovery restarts, TEMPERATURE(LOAD) requests must be respecified in the Case Control Section.
6. TEMPERATURE(MATERIAL) and TEMPERATURE(INITIAL) cannot be specified simultaneously in the same run.
7. TEMPERATURE(BOTH) cannot be specified simultaneously with TEMPERATURE(LOAD) or TEMPERATURE(MATERIAL).
8. The TEMPERATURE case control inputs for thermal strain and temperature dependent material properties in linear solutions are described as follows.

   Thermal strain in linear solutions is calculated from:

   $\epsilon = \alpha(T_2 - T_1)$

   where T_2=TEMPERATURE(LOAD) and T_1=TEMPERATURE(INITIAL) if it is defined, otherwise T_1=TREF. TREF is defined on material bulk entries (MAT1, MAT2, ..., MATi) or the PCOMP and PCOMPG bulk entries and defaults to "0.0".

   Linear Solution Input Combinations

   - If TEMPERATURE(LOAD) and TEMPERATURE(INITIAL) are defined, T_1=TEMPERATURE(INITIAL), and MATTi temperature dependent material properties are evaluated at TEMPERATURE(INITIAL).

     $\epsilon = \alpha_{init}(T_{load} - T_{init})$
   - If TEMPERATURE(LOAD) and TEMPERATURE(MATERIAL) are defined, T_1=TREF, and MATTi temperature dependent material properties are evaluated at TEMPERATURE(MATERIAL).

     $\epsilon = \alpha_{mat}(T_{load} - T_{ref})$
   - If TEMPERATURE(BOTH) is defined, T_1=TREF, and MATTi temperature dependent material properties are evaluated at TEMPERATURE(BOTH).

     $\epsilon = \alpha_{mat}(T_{load} - T_{ref})$
   - If TEMPERATURE(LOAD) is defined and neither of TEMPERATURE(INITIAL) or TEMPERATURE(MATERIAL) are defined, T_1=TREF, and MATTi temperature dependent material properties are ignored. Material properties are obtained from the MATi entries.

     $\epsilon = \alpha(T_{load} - T_{ref})$
     where α on the MATi entry is used.

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- If TEMPERATURE(LOAD) is not defined, material properties are evaluated at TEMPERATURE(MATERIAL) or TEMPERATURE(INITIAL) if either is defined, and no thermal strains result for that subcase.
- If none of TEMPERATURE(LOAD), TEMPERATURE(INITIAL) or TEMPERATURE(MATERIAL) are defined, no thermal strains result for that subcase, and MATTi temperature dependent material properties are ignored. Material properties are obtained from the MATi entries.

9. The TEMPERATURE case control inputs for thermal strain and temperature dependent material properties in solutions 106, 401, 402, and 414 are described as follows.

Thermal strain in the nonlinear solutions is calculated from:

$\epsilon = \alpha_{load}(T_{load} - T_{ref}) - \alpha_{init}(T_{init} - T_{ref})$

where α_{load} is evaluated at TEMPERATURE(LOAD) and α_{init} is evaluated at TEMPERATURE(INIT).

Input Combinations for Nonlinear Solutions 106, 401, 402, and 414

- For SOLs 106, 401, 402, and 414, TEMPERATURE(INITIAL) is required when TEMPERATURE(LOAD) is defined.

  For SOL 106, TEMPERATURE(INITIAL) must be defined above the subcase level (globally).

  For SOLs 401, 402, and 414, TEMPERATURE(INITIAL) must be the same for all subcases. Typically, it is defined globally.
- For SOLs 106, 401, 402, and 414, the specification of TEMPERATURE(MATERIAL) or TEMPERATURE(BOTH) will cause a fatal error.
- For SOLs 106, 401, 402, and 414, TEMPERATURE(LOAD) is used to update temperature-dependent material properties. Temperature-dependent material properties are specified with MATi, MATTi, MATS1, and/or TABLEST bulk entries.
- For SOLs 106, 401, 402, and 414, if TEMPERATURE(INITIAL) is defined but TEMPERATURE(LOAD) is not defined, temperature dependent material properties defined with the MATTi entries are evaluated at TEMPERATURE(INITIAL), and no thermal strains result for that subcase.
- If neither TEMPERATURE(LOAD) or TEMPERATURE(INITIAL) are defined, temperature dependent material properties defined with the MATTi entries are evaluated as follows:

  SOL 106 will ignore the temperature dependent material properties.

  SOL 401 will end with a fatal error if MATTi entries are defined.

  SOLs 402 and 414 will evaluate temperature dependent material properties at 0.0.
- For SOLs 401, 402, and 414, the TVAR parameter controls if the temperature loads are ramped, or applied as a constant for each subcase.

  For SOL 401, the TVAR parameter is on the NLCNTL bulk entry.

  For SOL 402 and 414, the TVAR parameter is on the NLCNTL2 bulk entry.
- For SOLs 401, 402, and 414, you can use the MATCTE parameter to define a single coefficient of thermal expansion (α_{matcte}) that overrides all values defined in the material definitions (MATi & MATTi bulk entries) and for the rigid elements (RBEi and RBAR bulk entries). When MATCTE is defined, the thermal strain is calculated from:

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$\epsilon = \alpha_{matcte}(T_{load} - T_{init})$

- For SOL 401, the thermal strain computation can be turned off by defining the parameter setting THRMST=NO (default=YES) on the NLCNTL bulk entry. This is useful for temperature dependent material evaluation without thermal loading.

10. For SOLs 401, 402, and 414, the TEMPEX bulk entry can be selected by TEMPERATURE(LOAD) and TEMPERATURE(INITIAL).

11. For all solutions except solution 401, if multiple TEMPERATURE(INITIAL) case control commands are present, Simcenter Nastran applies the TEMPERATURE(INITIAL) *used* in the last subcase to all of the subcases. The following examples demonstrate.

- In the following example, since TEMP(INIT)=400 is used in the last subcase (subcase 3), the software applies TEMP(INIT)=400 to all three subcases.

```
.....
   TEMP(INIT)=100
Subcase 1
   TEMP(INIT)=200
Subcase 2
   TEMP(INIT)=300
Subcase 3
   TEMP(INIT)=400
.....
```

- In the following example, since TEMP(INIT) is undefined in subcase 3, subcase 3 uses the globally defined TEMP(INIT)=100. As a result, the software applies TEMP(INIT)=100 to all three subcases.

```
.....
   TEMP(INIT)=100
Subcase 1
   TEMP(INIT)=200
Subcase 2
   TEMP(INIT)=300
Subcase 3
.....
```

- In the following example, since TEMP(INIT) is undefined in subcase 3 and TEMP(INIT) is not defined globally, subcase 3 has no initial temperature to use. As a result, the software applies no initial temperature to any of the subcases.

```
.....
Subcase 1
   TEMP(INIT)=200
Subcase 2
   TEMP(INIT)=300
Subcase 3
.....
```


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Remarks related to SOLs 402 and 414:

1. See Remark 9 above.
2. For TEMPERATURE(INITIAL)=*n*, *n* can reference the TEMP, TEMPD, or TEMPEX bulk entry.
3. Temperature loads can be used in both static and transient analysis.
4. TEMPERATURE(LOAD) is used to select a constant temperature load. For a time-dependent temperature load, the DTEMP case control command is used instead.
5. If TREF is blank on the material entry, the reference temperature for the calculation of thermal loads is obtained from the TEMPD entry referenced by the TEMPERATURE(INITIAL) set selection. If TEMPD is also not specified, TREF=0.0 is used.
6. Temperature dependent properties defined on the MATTi entries are evaluated at the temperatures referenced by the TEMPERATURE(LOAD) set selection, although, the temperature dependent density T(RHO) is an exception. If T(RHO) is defined on the MATTi entry, a constant mass density is computed as follows:
   - If TREF is defined on the MATi entry, the mass density T(RHO) is evaluated at temperature TREF.
   - If TREF is undefined on the MATi entry, a default temperature defined with the TEMPD bulk entry must be referenced by the TEMPERATURE(INITIAL) set selection. Then, the mass density T(RHO) is evaluated at temperature TREF=TEMPD. If TEMPD is not specified either, a FATAL error will be generated.

Related Topics

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TFL

Transfer Function Set Selection

Selects the transfer function set(s) to be added to the direct input matrices.

Formats:

TFL={n_1, [n_2, n_3 THRU n_4, EXCEPT n_5, n_6, n_7]}

Examples:

```
TFL=77
TFL=1, 25, 77
TFL=5, 6, 7, 8, 9, 10 THRU 55 EXCEPT 15, 16, 77, 78, 79
TFL=1 THRU 10
```

Describers:

Describer	Meaning
n_1, n_2, etc.	Identification numbers of TF Bulk Data entries. (Integer>0)
n_3 THRU n_4	Identification numbers of TF bulk data entries (n_3>n_4). (Integer>0)
EXCEPT	Identification numbers following EXCEPT will be deleted from the list as long as they are in the range of the set defined by the immediately preceding THRU. An EXCEPT list may not include a THRU.

Remarks:

1. Transfer functions will not be used unless selected in the Case Control Section.
2. Transfer functions are supported in dynamics problems only.
3. Transfer functions are described in the *Simcenter Nastran Advanced Dynamic Analysis User's Guide*.
4. It is recommended that PARAM,AUTOSPC,NO be specified when using transfer functions. See the *Simcenter Nastran User's Guide*.
5. The transfer functions are additive if multiple TF values are referenced on the TFL command.
6. Identification numbers following EXCEPT within the range of the THRU must be in ascending order.
7. In the third example in the Examples section, the numbers 77, 78, 79 are included because they are outside the prior THRU range.
8. The TFL command is supported across subcases. A TFL command selecting different TF bulk data entries can be defined for each subcase.

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Related Topics

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THERMAL

Temperature Output Request

Requests the form and type of temperature output.

Format:

$$\text{THERMAL}\left[\left(\begin{bmatrix}\text{SORT1}\\ \text{SORT2}\end{bmatrix},\ \begin{bmatrix}\underline{\text{PRINT}},\ \text{PUNCH}\\ \text{PLOT}\end{bmatrix}\right)\right] = \begin{Bmatrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{Bmatrix}$$

Examples:

```
THERMAL=5
THER(PRINT,PUNCH)=ALL
```

Describers:

Describer	Meaning
SORT1	Output is presented as a tabular listing of point temperatures for each load or time step.
SORT2	Output is presented as a tabular listing of loads or time steps for each point temperature.
PRINT	The printer will be the output medium.
PUNCH	The punch file will be the output medium.
PLOT	Compute temperatures but do not print.
ALL	Temperatures for all points will be output.
NONE	Temperatures for no points will be output.
n	Set identification of a previously appearing SET command. Only temperatures of points with identification numbers that appear on this SET command will be output. (Integer>0)

Remarks:

1. The THERMAL output request is designed for use with the heat transfer option. The printed output will have temperature headings. The PUNCH option produces TEMP Bulk Data entries, and the SID on the entries will be the subcase number (=1 if no SUBCASES are specified).

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2. SORT1 is the default in steady state heat transfer analysis. SORT2 is the default in transient heat transfer analysis.
3. In a transient heat transfer analysis, the SID on the punched TEMP Bulk Data entries, equals the time step number.

Related Topics

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THSTRN

Requests thermal strain output at grid points on elements for SOLs 401 and 402. Also requests thermal strain for SOLs 101, 103, 105, 106, 107-112.

Requests thermal strain on elements.

Format:

$$\text{THSTRN}\left[\left(\left[\begin{matrix}\underline{\text{STRCUR}}\\ \text{FIBER}\end{matrix}\right],\left[\begin{matrix}\underline{\text{CENTER}}\\ \text{CORNER or BILIN}\\ \text{SGAGE}\\ \text{CUBIC}\end{matrix}\right],\underline{\text{PRINT}},\text{PUNCH},\text{PLOT}\right)\right]$$
$$=\left\{\begin{matrix}\text{ALL}\\ \text{n}\\ \text{NONE}\end{matrix}\right\}$$

Examples:

```
THSTRN=ALL
THSTRN(PRINT,PUNCH)=17
THSTRN(STRCUR,CORNER,PRINT) = ALL
```

Describers for SOL 401:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all elements.
n	Set identification number of a previously appearing SET command. Only elements with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.

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Describers for SOLs 101 and 106:

Describer	Meaning
PRINT	Compute and write output to the print file (f06). (Default)
PUNCH	Compute and write output to the punch file (pch).
PLOT	Compute output.
ALL	Requests output for all elements.
n	Set identification number of a previously appearing SET command. Only elements with identification numbers that appear on this SET command will be included in the output. (Integer>0)
NONE	Output is not computed.
STRCUR	Thermal strain at the reference plane and curvatures is output for shell elements.
FIBER	Thermal strain at locations Z1, Z2 is computed for shell elements.
CENTER	Outputs thermal strains at the center of the element only.
CORNER or BILIN	Outputs thermal strains at the center and corners of the element.
SGAGE	Outputs thermal strains at the center and corners of the element using strain gage approach.
CUBIC	Outputs thermal strains at the center and corners of the element using cubic bending correction.

General Remarks:

1. Shear strain is output as engineering shear strain, which is twice the tensor shear strain.

Remarks related to SOL 401:

1. Only supported in static and preload subcase.
2. Both PRINT and PUNCH may be requested.
3. OP2 file output requires PARAM,POST,-1 or PARAM,POST,-2.

Remarks for SOLs 101, 103, 105, 106, 107-112:

1. For SOL 106, THSTRN is only supported for the shell elements CQUAD4, CTRIA3, CQUAD8, CTRIA6, CTRIAR, and CQUADR that reference a PCOMP or PCOMPG entry.

   For SOL 101, THSTRN is only supported for the shell elements CQUAD4, CTRIA3 , CQUAD8, CTRIA6, CTRIAR, and CQUADR that reference a PSHELL, PCOMP or PCOMPG entry, and the 3D solid elements CTETRA, CHEXA, CPENTA and CPYRAM that reference a PSOLID entry.

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For SOLs 103, 105, 107-112, THSTRN is only supported for the shell elements CQUAD4, CTRIA3 , CQUAD8, CTRIA6, CTRIAR, and CQUADR that reference a PSHELL entry, and the 3D solid elements CTETRA, CHEXA, CPENTA and CPYRAM that reference a PSOLID entry.

2. For shell elements referencing a composite property defined with the PCOMP or PCOMPG entries, the software automatically creates an equivalent PSHELL and MATi entry. You can use the parameter NOCOMPS to control if stress and/or strain is recovered at the composite ply layers (default) and/or on the equivalent PSHELL.

   For strain output on the composite ply layers, the STRCUR, FIBER, CENTER, CORNER, SGAGE, CUBIC describers are ignored, and strain is always reported at the center of the element (in the element plane) and at the center of each ply (in the thickness direction).

   For the strain output on the equivalent PSHELL, STRCUR is the default, and the behaviour for the CENTER, CORNER, SGAGE, CUBIC describers is the same as described in the following remark. Note that if FIBER is defined on a STRAIN, ELSTRN, or THSTRN command, the FIBER option applies to all STRAIN, ELSTRN and THSTRN commands.

3. For shell elements referencing a PSHELL entry:

   For the CENTER option, thermal strains are output at the center of the element for CQUAD4, CQUADR, and CTRIAR elements.

   For the CORNER (or BILIN) option, thermal strains are output at the center and at the corner for CQUAD4, CQUADR, and CTRIAR elements.

   For the SGAGE option, thermal strains are output at the center and at the corner for CQUAD4 elements.

   For the CUBIC option, thermal strains are output at the center and at the corner for CQUAD4 and CQUADR elements.

4. If the STRAIN command is defined when the ELSTRN and/or the THSTRN commands are also defined, the CENTER/CORNER(BILIN)/SGAGE/CUBIC option on the STRAIN command overrides the corresponding option on the ELSTRN and/or THSTRN command.

Remarks related to SOL 402:

1. STRCUR, FIBER, BILIN, SGAGE and CUBIC are ignored.
2. THSTRN is not supported for CBEAM, CROD, and CBAR elements.
3. The von Mises value is not computed.

Related Topics

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TITLE

Output Title

Defines a character string that will appear on the first heading line of each page of Simcenter Nastran printer output.

Format:

TITLE=title

Examples:

```
TITLE=RIGHT WING, LOAD CASE 3.
```

Describers:

Describer	Meaning
title	Any character string.

Remarks:

1. If this command appears under a SUBCASE command, then the title appears in the output for that subcase only.
2. If this command appears before all SUBCASE commands, then the title is used in all subcases without a TITLE command.
3. If no TITLE command is present, then the title line will contain data and page numbers only.
4. The title also appears on plotter output.

Remark for SOLs 402 and 414:

1. The TITLE command can only be used at the global level (above the subcases), for example, to name the solution. To assign a name to each subcase, you can use the LABEL command instead.

Related Topics

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TRIM

Aerodynamic Trim Variable Constraint Selection

Selects trim variable constraints in static aeroelastic response.

Format:

TRIM=n

Examples:

```
TRIM=1
```

Describers:

Describer	Meaning
n	Set identification number of a TRIM Bulk Data entry. (Integer>0)

Remarks:

Aerodynamic extra points (trim variables) that are not constrained by a TRIM Bulk Data entry will be free during the static aeroelastic response solution.

Related Topics

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TRLOSS

Acoustic Transmission Loss Results Output

Acoustic transmission loss results output request (SORT2 results only).

Format:

$$\text{TRLOSS}\left[\left(\left[\begin{matrix}\text{PRINT}\\ \text{NOPRINT}\end{matrix}\right],\text{PUNCH},\left[\begin{matrix}\text{RPRINT}\\ \text{NORPRINT}\end{matrix}\right],\text{RPUNCH}\right)\right]=\left\{\begin{matrix}\text{YES}\\ \text{NO}\end{matrix}\right\}$$

Examples:

```
TRLOSS(PRINT,PUNCH)
```

```
TRLOSS=NO
```

```
TRLOSS
```

Describers:

Describer	Meaning
PRINT	Write the transmitted acoustic power results to the print (.f06) file. (Default)
NOPRINT	Do not print the transmitted acoustic power results.
PUNCH	Write the transmitted acoustic power results to the standard punch (.pch) file.
RPRINT	Write the random analysis results to the print (.f06) file. See Remark 7. (Default)
NORPRINT	Do not write the random analysis results to the print file. See Remark 7.
RPUNCH	Write the random analysis results to the punch (.pch) file. See Remark 7.
YES	Honor describer settings and calculate and output transmitted acoustic power results. (Default)
NO	Ignore describer settings and do not calculate or output transmitted acoustic power results. See Remark 8.

Remarks:

1. Both PRINT and PUNCH may be requested.
2. TRLOSS results are always real values.

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3. To produce transmission loss results for a specific subcase, the subcase must request both incident acoustic power (INPOWER) and transmitted acoustic power (TRPOWER) results. TRLOSS can be requested for frequency response or random subcases.
4. For frequency response analysis, transmission loss is computed from:

$$TL = 10\log_{10}\left(\frac{\Pi_{incident}}{\Pi_{outlet}}\right) = 10\log_{10}\left(\frac{\mathrm{Re}(INPOWER)}{\mathrm{Re}(TRPOWER)}\right)$$

5. For random PSD analysis, the transmission loss is computed from:

$$TL = 10\log_{10}\left(\frac{PSD(INPOWER)}{PSD(TRPOWER)}\right)$$

6. Transmission loss results are written to the op2 file if PARAM, POST,-1 or PARAM,POST,-2.
7. The following applies to random solutions:
   - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, NOPRINT, PUNCH describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
   - Only SORT2 output format is available for the frequency response and random results output.
8. Use TRLOSS=NO within a subcase to override a TRLOSS specification above the subcase level.
9. The computed incident power takes into account the incident acoustic energy from the ACPOLE1, ACPOLE2, and ACPLNW acoustic sources only, and not from acoustic sources like ACPNVEL that can contain a scattered component.

Related Topics

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TRPOWER

Transmitted Acoustic Power Results Output

Transmitted acoustic power results output request for AML regions or 2D microphone elements (SORT2 results only).

Format:

$$\text{TRPOWER}\left[\left(\text{AMLREG}=\left\{\begin{matrix}\underline{\text{ALL}}\\ \text{NONE}\\ \text{na}\end{matrix}\right\},\ \text{GROUP}=\text{ng},\ \left[\begin{matrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{matrix}\right],\ \text{PUNCH},\right.\right.$$
$$\left.\left.\left[\begin{matrix}\underline{\text{REAL}}\ \text{or}\ \underline{\text{IMAG}}\\ \text{PHASE}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{RPRINT}}\\ \text{NORPRINT}\end{matrix}\right],\ \text{RPUNCH}\right)\right]=\left\{\begin{matrix}\underline{\text{YES}}\\ \text{NO}\end{matrix}\right\}$$

Examples:

```
SET 123=3,6,9
```

```
TRPOWER(AMLREG=123)
```

```
SET 5=5,10
```

```
TRPOWER(GROUP=5,PRINT,PUNCH,PHASE)
```

```
TRPOWER=NO
```

```
TRPOWER
```

Describers:

Describer	Meaning
AMLREG = ALL	Calculate transmitted acoustic power results for all AMLREG bulk entries. (Default)
AMLREG = NONE	Do not calculate transmitted acoustic power results for any AMLREG bulk entries.

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Describer	Meaning
AMLREG = na	na (integer>0) is the identification number of the SET case control command that lists the identification numbers of AMLREG bulk entries at which to calculate transmitted acoustic power results. See Remark 2.
GROUP = ng	ng (integer>0) is the identification number of the SET case control command that lists the identification numbers of GROUP bulk entries that contain the 2D microphone elements at which to calculate transmitted acoustic power results. See Remark 2.
PRINT	Write the transmitted acoustic power results to the print (.f06) file. (Default)
NOPRINT	Do not print the transmitted acoustic power results.
PUNCH	Write the transmitted acoustic power results to the standard punch (.pch) file.
REAL or IMAG	Requests rectangular format (real and imaginary) of complex output. Use of either REAL or IMAG yields the same output. (Default)
PHASE	Requests polar format (magnitude and phase) of complex output. Phase output is in degrees.
RPRINT	Write the random analysis results to the print (.f06) file. See Remark 4. (Default)
NORPRINT	Do not write the random analysis results to the print file. See Remark 4.
RPUNCH	Write the random analysis results to the punch (.pch) file. See Remark 4.
YES	Honor describer settings and calculate and output transmitted acoustic power results. (Default)
NO	Ignore describer settings and do not calculate or output transmitted acoustic power results. See Remark 5.

Remarks:

1. Both PRINT and PUNCH may be requested.
2. Only AML regions and 2D microphone elements referenced in GROUP bulk entries are processed for transmitted acoustic power results. The AMLREG describer and the GROUP describer can be specified simultaneously. Microphone elements must reference a PMIC property. Any other entities that are contained in a GROUP that is referenced by TRPOWER are ignored.
3. Transmitted acoustic power results are written to the op2 file if PARAM, POST,-1 or PARAM,POST,-2.
4. The following applies to random solutions:
   - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, NOPRINT, PUNCH describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
   - Only SORT2 output format is available for the frequency response and random results output.

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- Only the power spectral density function (PSDF) is calculated. For RANDPS (or RANDPEX) bulk entries that refer to *n* subcases, when *i* = *j* the results in W/Hz are computed as follows:

$$TRPOWER_{psdf}\left(f\right) = \sum_{i=1}^{n} RANDPS_{ii}\left(f\right) \cdot TRPOWER_{i}\left(f\right)$$

If $i \neq j$, then a warning message is issued and the RANDPS bulk entries are ignored.

For more information, see the **RANDPS** bulk entry.

5. Use TRPOWER=NO within a subcase to override a TRPOWER specification above the subcase level.

Related Topics

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TSTEP

Transient Time Step Set Selection

Selects integration and output time steps for linear or nonlinear transient analyses.

For SOLs 401 and 402: Selects time stepping for static and transient analyses.

Format:

TSTEP=n

Examples:

```
TSTEP=731
```

Describers:

Describer	Meaning
n	Sets the identification number of a TSTEP or TSTEPNL bulk entry. (Integer>0)
n (For SOLs 401 and 402)	Sets the identification number of a TSTEP1 bulk entry. (Integer>0)

Remarks:

1. A TSTEP entry must be selected to execute a linear transient analysis (SOLs 109 or 112) and TSTEPNL for a nonlinear transient analysis (SOLs 129 and 159).
2. A TSTEPNL entry must be selected in each subcase to execute a nonlinear transient problem.
3. For the application of time-dependent loads in modal frequency response analyses (SOLs 111 and 146), a TSTEP entry must be selected by the TSTEP command. The time-dependent loads will be recomputed in frequency domain by a Fourier Transform.

Remarks related to SOL 402:

1. When more than one subcase is specified in the case control, the TSTEP command is required in all subcases assigned by ANALYSIS = STATICS, ANALYSIS = PRELOAD, and ANALYSIS = TRANSIENT

Related Topics

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TSTEPNL

Transient Time Step Set Selection for Nonlinear Analysis

See the description for **TSTEP**.

Related Topics

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TSTRU

Temperature Set ID for a Structures Run

Defines a temperature set ID for a structures run based on a heat transfer subcase.

Format:

TSTRU=n

Examples:

```
TSTRU=999
```

Describers:

Describer	Meaning
n	Set identification for use on TEMP(LOAD)=n or TEMP(INIT)=n

Remarks:

1. TSTRU should be placed in the heat transfer subcase.
2. If TSTRU does not explicitly appear in the heat transfer subcase, it is defaulted to TSTRU=Heat Transfer Subcase ID.
3. In a structures run, a temperature set generated from a heat transfer run will override an existing temperature set with an identical set ID defined with TEMP, TEMPD, TEMPP1, TEMPRB or any combination.
4. TSTRU may be placed in the first subcase of a PARAM,HEATSTAT,YES run.
5. TSTRUs may be placed in each subcase of an APPHEAT run. The associated structures run then requires the following:

   ```
   ASSIGN heat_run='heat transfer job name.MASTER'
   DBLOC DATABLK=(UG,EST,BGPDTS,CASECCR/CASEHEAT) LOGICAL=heat_run
   ```

6. Heat transfer runs and structure runs must have the same mesh.
7. For nonlinear heat transfer SOL 106 or SOL 153 the INOUT field on the NLPARM Bulk Data entry must be blank or NO if the results of the run are to be transferred to a linear structures run.

   PARAM,NLHTLS,-1 should be placed in the nonlinear heat run. This will place UG heat transfer on the database.

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Related Topics

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ULOAD

Unbalance Load Set Selection

Selects the unbalance load for SOL 414 rotor dynamic analysis.

Format:

$$\text{ULOAD} = \text{n}$$

Examples:

```
ULOAD=15
```

Describers:

Describer	Meaning
n	Load set identification of UNBA bulk entries. (Integer>0)

Related Topics

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VATVOUT

VATV Creation Specification

Defines requirements for the creation of a Vibro-Acoustic Transfer Vector (VATV).

Format:

$$\text{VATVOUT}\left(\text{VATVOP2} = \text{unit},\ \text{VATVFS} = \text{id}\right) = \left\{\begin{matrix}\text{ALL} \\ \text{n}\end{matrix}\right\}$$

Examples:

```
VATVOUT(VATVOP2=33,VATVFS=110)=200
```

Describers:

Describer	Meaning
VATVOP2 = *unit*	*unit* (integer > 0) is the Fortran unit number for the .op2 file to which the VATV results are written. See Remark 2.
VATVFS = *id*	*id* (integer > 0) is the identification number of the VATVFS bulk entry that defines the set of structural element faces that will be used to apply pressure.
ALL	Output VATV results for all microphone elements. See Remark 5. (Default)
n	*n* (integer > 0) is the identification number of a previously appearing SET case control command that contains microphone element identification numbers at which to output the VATV results. See Remark 5.

Remarks:

1. VATVOUT is valid for SOL 108 and SOL 111 only. If VATVOUT is included in the input file for any other solution, the software issues a user fatal message.
2. For specifying VATVOP2 = *unit*, an appropriate ASSIGN OUTPUT2 statement must be present in the File Management Section (FMS) for the absolute value of *unit*.
3. If VATVOUT is specified at the global level (above the subcases) and subcase levels, the software honors the output request at the global level only.
4. Only one VATVOUT can be included in an input file.
5. VATV matrix output is controlled by the specification of ALL or *n* only. Any other output-related case control commands like DISPLACEMENT or PRESSURE are ignored.

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Related Topics

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VELOCITY

Velocity Output Request

Requests the form and type of velocity vector output.

Format:

$$\text{VELOCITY}\left[\left(\begin{bmatrix}\underline{\text{SORT1}}\\ \text{SORT2}\end{bmatrix},\ \begin{bmatrix}\underline{\text{PRINT}},\ \text{PUNCH}\\ \text{PLOT}\end{bmatrix},\ \begin{bmatrix}\underline{\text{REAL}}\ \text{or}\ \underline{\text{IMAG}}\\ \text{PHASE}\end{bmatrix},\ \begin{bmatrix}\underline{\text{ABS}}\\ \text{REL}\end{bmatrix},\right.\right.$$

$$\left.\left.\begin{bmatrix}\text{PSDF}\\ \text{ATOC}\\ \text{CRMS}\\ \underline{\text{RMS}}\\ \text{RALL}\end{bmatrix},\ \begin{bmatrix}\underline{\text{RPRINT}}\\ \text{NORPRINT}\end{bmatrix},\ [\text{RPUNCH}],\ \text{SOLUTION} = \begin{Bmatrix}\underline{\text{ALL}}\\ \text{setf}\\ \text{PEAKOUT}\end{Bmatrix}\right)\right]$$

$$= \begin{Bmatrix}\text{ALL}\\ n\\ \text{NONE}\end{Bmatrix}$$

Examples:

```
VELOCITY=5
VELOCITY(SORT2,PHASE,PUNCH)=ALL
```

Describers:

Describer	Meaning
SORT1	Output a tabular listing of grid points for each load, frequency, eigenvalue, or time, depending on the solution sequence. (Default)
SORT2	Output a tabular listing of frequency or time for each grid point.
PRINT	The printer is the output medium. (Default)
PUNCH	The punch file is the output medium.

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Describer	Meaning
PLOT	Generate, but do not print, velocity results.
REAL or IMAG	Requests real/imaginary format for complex output. Specify either REAL or IMAG. (Default)
PHASE	Requests magnitude/phase format for complex output. Phase output is in degrees.
ABS	For enforced motion dynamic analysis, velocity results are output as absolute velocity. (Default)
REL	For enforced motion dynamic analysis, velocity results are output relative to the enforced motion input.
PSDF	Requests power spectral density function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7.
ATOC	Requests autocorrelation function be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7.
CRMS	Requests cumulative root mean square function be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7.
RMS	Requests root mean square and zero crossing functions be calculated for random analysis post-processing. Request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7. (Default)
RALL	Requests all of PSDF, ATOC, RMS, and CRMS be calculated for random analysis post-processing. The request must be made above the subcase level and RANDOM must be selected in the Case Control. See Remark 7.
RPRINT	Writes random analysis results to the print file. See Remark 7. (Default)
NORPRINT	Disables the writing of random analysis results to the print file. See Remark 7.
RPUNCH	Writes random analysis results to the punch file. See Remark 7.
SOLUTION = ALL	Output velocity for all frequencies. See Remark 12. (Default)
SOLUTION = setf	Output velocity only for the frequencies listed on a previously appearing SET command with setf as the identification number. See Remark 12. (Integer > 0)
SOLUTION = PEAKOUT	Output velocity only for frequencies identified with the PEAKOUT command. See Remark 12.
ALL	Output velocity for all points.
n	Output velocity only for the points listed on a previously appearing SET command with n as the identification number. (Integer > 0)
NONE	Do not output velocity for any points. (Default)

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Remarks:

1. Both PRINT and PUNCH can be requested.
2. Velocity output is only available for transient and frequency response problems.
3. The defaults for SORT1 and SORT2 depend on the analysis type:
   - SORT1 is the default in static analysis, frequency response, steady state heat transfer analysis, real and complex eigenvalue analysis, flutter analysis, and buckling analysis. If SORT2 is selected in a frequency response solution for one or more of the commands ACCE, DISP, FORC, GPFO, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands are also output in SORT2 format.
   - SORT2 is the default in transient response analysis (structural and heat transfer). SORT2 is not available for real eigenvalue (including buckling), complex eigenvalue, or flutter analysis. If SORT1 is selected in a transient solution for one or more of the commands ACCE, DISP, ENTH, FORC, GPFO, HDOT, MPCF, OLOA, SPCF, STRA, STRE, and VELO then the remaining commands are also output in SORT1 format.
   - XY plot requests force SORT2 format, thus overriding SORT1 format requests.
4. VELOCITY=NONE overrides an overall output request.
5. The PLOT option is used when curve plots are desired in the magnitude/phase representation and no printer request is present for magnitude/phase representation.
6. Velocity results are printed and/or punched in the global coordinate system (see field CD on the GRID bulk data entry). The coordinate system for plotted velocity output depends on the PARAM,POST setting. See the parameter **POST**.
7. The following applies to random solutions:
   - By default, frequency response results are not output. If in addition to random output, frequency response output is desired, specify SYSTEM(524)=1 or RANFRF=1 in the input file. The PRINT, PUNCH, PLOT describers control the frequency response output. The RPRINT, NORPRINT, RPUNCH describers control the random output.
   - The SORT1 and SORT2 describers only control the output format for the frequency response output. The output format for random results is controlled using the RPOSTS1 describer on the RANDOM case control command or the parameter RPOSTS1, except for RMS results, which are only available in SORT1 format.
   - Any combination of the PSDF, ATOC, RMS, and CRMS describers can be selected. The RALL describer selects all four.
   - Autocorrelation (ATOC) calculations require the RANDT1 bulk entry.
8. When doing enforced motion dynamic analysis and relative output is requested (using the REL describer), the output is relative to the input as described by the equation:

$$u_f = y_f - K_{ff}^{-1} K_{fs} u_s$$

where u_f is absolute displacement, y_f is relative displacement, and u_s is enforced motion.

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9. During a SOL 108 or 111 frequency-dependent external superelement creation run, on the SET case control command that is referenced by the VELOCITY case control command, include any points at which you want to recover results during the system run.
10. The SOLUTION describer is only valid in a SOL 108 or 111 frequency response analysis.
11. When an OFREQ case control command is present, the SOLUTION describer is used as follows:
    - If SOLUTION=ALL, output results for all frequencies specified by the OFREQ command.
    - If SOLUTION=setf, output results for all frequencies specified by the OFREQ command that are listed on a previously appearing SET command with setf as the identification number.
    - If SOLUTION=PEAKOUT and a PEAKOUT case control command is also present in the input file, output results for all frequencies specified by the OFREQ command that satisfy the PEAKOUT filtering criteria.
12. The SOLUTION describer is not valid for superelements or frequency-dependent components.

Remarks realted to SOL 402:

1. Output is restricted to REAL and SORT1 format. SORT2, IMAG, PHASE, ABS, REL, PSDF, ATOC, CRMS, RMS, RALL, NORPRINT, and RPUNCH are ignored.

Related Topics

Browse Simcenter Nastran documentation

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VOLUME

Volume Definition

Defines a volume for the calculation of grid point stresses, strains, or mesh stress discontinuities.

Format:

$$\text{VOLUME id SET sid, } \left[\underline{\text{PRINCIPAL}}, \underline{\text{DIRECT STRESS}}\right], \left[\underline{\text{SYSTEM}} \left\{\begin{array}{c}\text{ELEMENT}\\ \text{CORD cid}\\ \underline{\text{BASIC}}\end{array}\right\}\right]$$

Examples:

```
VOLUME  21 SET 2
```

Describers:

Describer	Meaning
id	Volume identification number.
sid	Set identification number of a SET command that defines the elements in the volume. Either form of the SET command may be used. The default is all elements.
PRINCIPAL	Requests principal stresses or strains, direction cosines, mean pressure, and von Mises equivalent stresses or strains to be calculated. If neither PRINCIPAL nor DIRECT is specified, then the default is to output both.
DIRECT	Requests direct stress or strains, mean pressure stress and von Mises equivalent stress to be calculated. If neither PRINCIPAL nor DIRECT is specified, then the default is to output both.
SYSTEM	Used to specify the reference coordinate system to be used to define the output stress orientation coordinate system.
ELEMENT	Specifies the element coordinate system.
CORD cid	Specifies the coordinate system specified on a CORDij entry.
BASIC	Specifies the basic coordinate system.

Remarks:

1. VOLUME commands must be specified after OUTPUT(POST).

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2. The volume identification number must be referenced on a SET command, specifically the SET command which comes after OUTPUT(POST). This SET identification number may then be referenced on the GPSTRESS, GPSTRAIN, STRFIELD, ELSDCON, and GPSDCON commands.
3. If ELEMENT is specified, element stresses or strains are not transformed.
4. In the example given above, for all elements in SET 2:
   - Both PRINCIPAL and DIRECT stress are output.
   - The BASIC output system is used.

Related Topics

Browse Simcenter Nastran documentation

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WEIGHTCHECK

Rigid Body Mass Reduction Check

At each stage of the mass matrix reduction, compute the rigid body mass and compare with the rigid body mass of the g-set.

Format:

$$\text{WEIGHTCHECK}\left[\left(\left[\begin{matrix}\underline{\text{PRINT}}\\ \text{NOPRINT}\end{matrix}\right],\ \underline{\text{SET}}=\left(\left\{\begin{matrix}\underline{\text{G}},\ \text{N},\ \text{N}+\text{AUTOSPC},\ \text{F},\ \text{A},\ \text{V}\\ \text{ALL}\end{matrix}\right\}\right),\right.$$
$$\left.\text{GRID}=\text{gid},\ \text{CGI}=\left[\begin{matrix}\underline{\text{YES}}\\ \underline{\text{NO}}\end{matrix}\right],\ \left[\begin{matrix}\underline{\text{WEIGHT}}\\ \text{MASS}\end{matrix}\right]\right)\right]=\left\{\begin{matrix}\text{YES}\\ \text{NO}\end{matrix}\right\}$$

Examples:

```
WEIGHTCHECK=YES
WEIGHTCHECK(GRID=12,SET=(G,N,A),MASS)=YES
```

Describers:

Describer	Meaning
PRINT	Writes output to the print file (default).
NOPRINT	Does not write output to the print file.
SET	Selects degree-of-freedom set(s). (Default SET=G)
gid	Reference grid point for the calculation of rigid body motion. The default is the origin of the basic coordinate system.
CGI	For SET = N, N+AUTOSPC, F, A, or V, CGI = YES requests output of center of gravity and mass moments of inertia. This output is always printed for SET = G. (Default: CGI = NO).
WEIGHT/MASS	Selects output in units of weight or mass. (Default=WEIGHT)

Remarks:

1. WEIGHTCHECK must be specified above the subcase level.
2. For SET=N, N+AUTOSPC, F, or A, the WEIGHTCHECK command also outputs a percentage loss or gain in the reduced rigid body mass matrix (e.g., MAA) as compared to the g-set rigid body mass matrix (e.g., MGG). G must also be requested to obtain this comparison; e.g., WEIGHTCHECK(SET=(G,A))=YES.

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3. SET=N+AUTOSPC uses the mass matrix for the n-set with the rows corresponding to degrees-of-freedom constrained by the PARAM, AUTOSPC operation zeroed out. If AUTOSPC was not performed then this check is redundant with respect to SET=N.
4. WEIGHTCHECK is available in all SOLs. However, in SOLs 101, 105, 114, and 116, because no mass reduction is performed, only WEIGHTCHECK(SET=G) is available.

Related Topics

Browse Simcenter Nastran documentation

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Case Control Applicability Tables

The following tables describe the applicability of Case Control commands to Solution Sequences:

Table 4-19	SOLs 101 through 115
Table 4-20	SOLs 116 through 414

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Table 4-19: Case Control Command Applicability – Solutions 101 through 115

Command Name	Solution Number (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
A2GG	X	X	X	X	X	X	X	X	X	X	X	X
ACCELERATION		X				X	X		X	X		
ACCONV						X						
ACINTENSITY						X			X			
ACORDCHK						X			X			
ACPOWER						X			X			
ACPOTENTIAL						X			X			
ACTEMP						X			X			
ACVELOCITY						X			X			
ACVPO4STR						X			X			
ADAMSMNF		X										
ADAPTERR												
ADMRECVR		X										
AECONFIG												
AEROF												
AESYMXY												
AESYMXZ												
ALOAD						X			X			
ANALYSIS												
APRESSURE												
ATVOUT						X						
AUXCASE												
AUXMODEL												
B2GG	X	X	X	X	X	X	X	X	X	X	X	X
B2PP					X	X	X	X	X	X		
BC		X										
BCRESULTS	X	X	X									
BCSET	X	X							X	X		
BEGIN BULK	X	X	X	X	X	X	X	X	X	X	X	X
BGRESULTS	X	X	X			X						
BGSET	X	X	X	X	X	X	X	X	X	X	X	X
BOLTLD	X	X	X		X	X	X	X	X	X		
BOUTPUT				X								

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Command Name	Solution Number (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
BVTSET												
CKGAP												
CLOAD				X								
CMETHOD					X			X				
CSSCHD												
CZRESULTS												
DAMPING					X	X	X	X	X	X		
DEFORM	X		X								X	
DESGLB												
DESOBJ												
DESSUB												
DISPLACEMENT	X	X	X	X	X	X	X	X	X	X	X	X
DIVERG												
DLOAD					X	X	X	X	X	X		
DMTRCOEF						X						
DMTRLOSS						X						
DRSPAN												
DSAPRT												
DSYM											X	X
ECHO	X	X	X	X	X	X	X	X	X	X	X	X
ECHOOFF	X	X	X	X	X	X	X	X	X	X	X	X
ECHOON	X	X	X	X	X	X	X	X	X	X	X	X
EDE						X			X	X		X
EFLOAD	X	X				X						
ELAR												
ELSTRN	X	X	X	X	X	X	X	X	X	X		
EKE		X				X			X	X		X
ELSDCON	X		X									
ELSUM	X	X	X	X	X	X	X	X	X	X	X	X
ENTHALPY												
ERP						X			X			
ESE	X	X	X			X			X	X	X	X
ESHLPNL												
EXTSEOUT	X	X			X	X	X	X	X	X		

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Command Name	Solution Number (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
FHAC												
FHAR												
FLSFSEL						X			X			
FLSPOUT						X			X			
FLSTCNT						X			X			
FLUX	X										X	
FLXRESULTS												
FLXSLI												
FMETHOD												
FMH3D												
FORCE	X	X	X	X	X	X	X	X	X	X	X	X
FREQUENC						X			X			
FRFIN						X			X			
FRFOUT						X			X			
GCRSTRN												
GELSTRN												
GPLSTRN												
GKRESULTS												
GMESTRN												
GPFORCE	X	X	X	X		X	X		X	X	X	X
GPKE		X							X*			X
GPRSORT	X	X	X	X	X	X	X	X	X	X	X	X
GPSDCON	X		X								X	
GPSTRAIN	X	X		X			X			X	X	X
GPSTRESS	X	X		X			X			X	X	X
GRDCON						X			X			
GROUNDCHECK	X	X	X	X	X	X	X	X	X	X	X	X
GSTRAIN												
GSTRESS												
GTHSTRN												
GUST												
HARCOEFF												
HARMONICS	X	X	X		X	X	X	X	X	X	X	X
HARTAB												

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Command Name	Solution Number (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
HDOT												
HOUTPUT											X	X
IC							X					
IMPERF												
INCLUDE	X	X	X	X	X	X	X	X	X	X	X	X
INITS												
INPOWER						X			X			
JCONSET												
JINTEG												
JRESULTS												
K2GG	X	X	X	X	X	X	X	X	X	X	X	X
K2PP					X	X	X	X	X	X		
K42GG	X	X	X	X	X	X	X	X	X	X	X	X
LABEL	X	X	X	X	X	X	X	X	X	X	X	X
LINE	X	X	X	X	X	X	X	X	X	X	X	X
LOAD	X		X	X		X	X		X	X	X	
LOADSET	X		X	X		X	X		X	X	X	
M2GG	X	X	X	X	X	X	X	X	X	X	X	X
M2PP					X	X	X	X	X	X		
MASTER	X	X	X	X	X	X	X	X	X	X	X	X
MAXLINES	X	X	X	X	X	X	X	X	X	X	X	X
MAXMIN	X	X	X				X				X	X
MBDEXPORT		X										X
MBDRECVR		X										X
MEFFMASS		X						X	X	X		
MESTRN												
METHOD		X		X	X	X	X	X	X	X		X
MFLUID		X			X	X	X	X	X	X		X
MLOAD												
MODALE									X			
MODCON								X	X	X		
MODES		X										X
MODSEL		X	X	X				X	X	X		X
MODTRAK												

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Command Name	Solution Number (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
MONITOR	X	X										
MONVAR												
MPC	X	X	X	X	X	X	X	X	X	X	X	X
MPCFORCES	X	X	X		X	X	X	X	X	X	X	X
MPRES		X			X	X	X	X	X	X		X
NLARCL												
NLCNTL												
NLLOAD							X			X		
NLPARM				X								
NLSTRESS				X								
NONLINEAR							X			X		
NOUTPUT											X	X
NSM	X	X	X	X	X	X	X	X	X	X	X	X
OFREQUENCY						X			X			
OLOAD	X		X	X		X	X		X	X	X	
OMEGA												
OMODES		X	X					X*	X*	X*		X
OPRESS												
OTEMP												
OTIME							X			X	X	
OTMFORC						X			X			
OUTMGT												
OUTPUT(blank)	X	X	X	X	X	X	X	X	X	X	X	X
OUTPUT(PLOT)	X	X	X	X	X	X	X	X	X	X	X	X
OUTPUT (POST) or SETS DEFINITION	X	X	X	X	X	X	X	X	X	X	X	X
OUTPUT(XYPLOT)	X	X	X	X	X	X	X	X	X	X	X	X
P2G	X		X	X		X	X		X	X	X	
PAGE	X	X	X	X	X	X	X	X	X	X	X	X
PANCON						X			X			
PARAM	X	X	X	X	X	X	X	X	X	X	X	X
PARTN	X	X	X	X	X	X	X	X	X	X	X	X
PEAKOUT						X			X			
PFRESULTS												
PLOTID	X	X	X	X	X	X	X	X	X	X	X	X

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Command Name	Solution Number (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
PRESSURE		X			X	X	X		X	X		
RANDOM						X			X			
RCROSS						X			X			
REPCASE	X	X										
RESVEC		X		X				X	X	X		
RIGID	X	X	X	X	X	X	X	X	X	X		
RMAXMIN	X						X			X		
RMETHOD	X				X	X	X	X	X	X		
RSAPPLY		X										
RSMETHOD		X			X	X	X	X	X	X		
SACCELERATION						X	X		X	X		
SDAMPING								X	X	X		
SDISPLACEMENT		X	X		X	X	X		X	X		X
SEALL	X	X	X	X	X	X	X	X	X	X	X	X
SEDR	X	X	X	X	X	X	X	X	X	X	X	X
SEDV												
SEEXCLUDE	X	X	X	X	X	X	X	X	X	X	X	X
SEFINAL	X	X	X	X	X	X	X	X	X	X	X	X
SEKETAB												
SEKREDUCE	X	X	X	X	X	X	X	X	X	X	X	X
SELGENERATE	X		X	X		X	X		X	X	X	
SELREDUCE	X		X	X		X	X		X	X	X	
SEMGENERATE	X	X	X	X	X	X	X	X	X	X	X	X
SEMREDUCE		X	X		X	X	X	X	X	X		X
SEQDEP												
SERESP												
SET	X	X	X	X	X	X	X	X	X	X	X	X
SETMC								X	X	X		
SFORCE												
SHELLTHK												
SKIP	X	X	X	X	X	X	X	X	X	X	X	X
SMETHOD	X					X						
SPC	X	X	X	X	X	X	X	X	X	X	X	X
SPCFORCES	X	X	X	X	X	X	X	X	X	X	X	X

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Command Name	Solution Number (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
STATSUB		X	X		X	X	X	X	X	X		X
STRAIN	X	X	X	X	X	X	X	X	X	X	X	X
STRESS	X	X	X	X	X	X	X	X	X	X	X	X
STRFIELD	X	X	X				X			X	X	X
SUBCASE	X	X	X	X	X	X	X	X	X	X	X	X
SUBCOM	X											
SUBSEQ	X											
SUBTITLE	X	X	X	X	X	X	X	X	X	X	X	X
SUPER	X	X	X	X	X	X	X	X	X	X	X	X
SUPORT1	X	X	X	X	X	X	X	X	X	X	X	X
SURFACE		X	X					X	X	X	X	X
SVECTOR	X	X		X			X			X		X
SVELOCITY						X	X		X	X		
SYM	X											
SYMCOM	X											
SYMSEQ	X											
TEMP(INIT)	X	X	X	X	X	X	X	X	X	X	X	X
TEMP(LOAD)	X		X	X		X	X		X	X	X	
TEMP(MATE)	X	X	X		X	X	X	X	X	X	X	X
TFL					X	X	X	X	X	X		
THERMAL	X										X	
THSTRN	X	X	X	X	X	X	X	X	X	X		
TITLE	X	X	X	X	X	X	X	X	X	X	X	X
TRIM												
TRLOSS						X			X			
TRPOWER						X			X			
TSTEP							X			X		
TSTRU	X											
ULOAD												
VATVOUT						X			X			
VELOCITY						X	X		X	X		
VOLUME	X	X		X			X			X	X	X
WEIGHTCHECK	X	X	X	X	X	X	X	X	X	X	X	X
* For modal portion of solution only.												

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Table 4-20: Case Control Command Applicability – Solutions 116 through 414

Command Name	Solution Number (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
A2GG	X	X	X	X	X	X	X	X	X	X			
ACCELERATION		X	X			X			X	X	X	X	X
ACCONV													
ACINTENSITY													
ACORDCHK													
ACPOWER													
ACPOTENTIAL													
ACTEMP													
ACVELOCITY													
ACVPO4STR										X			
ADAMSMNF													
ADAPTERR											X		
ADMRECVR													
AECONFIG				X	X					X			
AEROF				X		X				X			
AESYMXY				X	X	X				X			
AESYMXZ				X	X	X				X			
ALOAD													
ANALYSIS							X			X	X	X	
APRESS				X						X			
ATVOUT													
AUXCASE										X			
AUXMODEL										X			
B2GG	X	X	X	X	X	X	X	X	X	X	X		X
B2PP		X	X		X	X		X		X			
BC									X				
BCRESULTS											X	X	
BCSET											X	X	X
BEGIN BULK	X	X	X	X	X	X	X	X	X	X	X	X	X
BGRESULTS			X				X	X			X		X
BGSET	X	X	X				X	X	X	X	X	X	X
BOLTLD											X	X	
BOLTRESULTS											X	X	

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Command Name	Solution Number (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
BOUTPUT			X				X	X					
BVTSET												X	
CKGAP											X		
CLOAD							X						
CMETHOD					X								X
CRSTRN											X	X	
CSSCHD				X									
CYCFORCES											X	X	
CYCSET											X	X	X
CZRESULTS											X	X	
DAMPING		X	X		X	X		X					
DEFORM	X			X						X			
DESGLB										X			
DESOBJ										X			
DESSUB										X			
DISPLACEMENT	X	X	X	X	X	X	X	X	X	X	X	X	X
DIVERG				X						X			
DLOAD		X	X		X	X		X		X	X	X	X
DMTRCOEF													
DMTRLOSS													
DRSPAN										X			
DSAPRT										X			
DSYM	X	X											
DTEMP											X	X	X
ECHO	X	X	X	X	X	X	X	X	X	X	X		X
ECHOOFF	X	X	X	X	X	X	X	X	X	X			
ECHOON	X	X	X	X	X	X	X	X	X	X			
EDE		X								X			
EFLOAD													
EKE		X								X	X	X	X
ELAR											X		
ELAROUT											X		
ELSDCON				X						X			
ELSTRN											X		X

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Command Name	Solution Number (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
ELSUM	X	X	X	X	X	X			X	X	X	X	
ENTHALPY								X					
ERP										X			
ESE	X	X		X						X	X	X	X
ESHLPNL											X		
EXTSEOUT			X	X	X	X		X	X				X
FHAC												X	X
FHAR												X	X
FLSFSEL										X			
FLSPOUT													
FLSTCNT													
FLUX							X	X		X			
FLXRESULTS												X	
FLXSLI												X	
FMETHOD					X					X			
FMH3D												X	X
FORCE	X	X	X	X	X	X	X	X	X	X	X	X	X
FREQUENC		X				X				X			
FRFIN													
FRFOUT													
GCRSTRN											X	X	
GELSTRN											X	X	
GPLSTRN											X	X	
GKRESULTS												X	
GMESTRN											X		
GPFORCE	X	X		X		X				X	X	X	X
GPKE										X	X		
GPLSTRN											X	X	
GPRSORT	X	X	X	X	X	X	X	X	X	X			
GPSDCON				X			X			X			
GPSTRAIN			X	X					X				
GPSTRESS			X	X									
GRDCON													
GROUNDCHECK	X	X	X	X	X	X			X	X	X		

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Command Name	Solution Number (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
GSTRAIN											X	X	
GSTRESS											X	X	
GTHSTRN											X	X	
GUST						X	X						
HARCOEFF													X
HARMONICS	X	X								X	X	X	X
HARTAB													X
HDOT								X					
HOUTPUT	X	X									X	X	
IC								X			X	X	
IMPERF											X		
INCLUDE	X	X	X	X	X	X	X	X	X	X	X		
INITS											X		
INPOWER													
JCONSET												X	
JINTEG											X		
JRESULTS												X	
K2GG	X	X	X	X	X	X	X	X	X	X	X		X
K2PP		X	X		X	X		X		X			
K42GG	X	X	X	X	X	X	X	X	X	X			
LABEL	X	X	X	X	X	X	X	X	X	X	X	X	X
LINE	X	X	X	X	X	X	X	X	X	X	X		
LOAD	X	X	X	X		X	X			X	X	X	X
LOADSET	X	X	X	X		X	X	X		X			
M2GG	X	X	X	X	X	X	X	X	X	X	X	X	X
M2PP		X	X		X	X		X		X			
MASTER	X	X	X	X	X	X	X	X	X	X			
MAXLINES	X	X	X	X	X	X	X	X	X	X	X		
MAXMIN			X	X	X	X			X	X			
MBDEXPORT													
MBDRECVR													
MEFFMASS					X	X			X	X	X	X	X
MESTRN											X		
METHOD	X	X	X	X	X	X	X	X	X	X	X	X	X

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Command Name	Solution Number (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
MFLUID	X	X	X	X	X	X			X	X			
MLOAD													X
MODALE													
MODCON						X							
MODES	X									X			
MODSEL	X				X	X			X				X
MODTRAK													
MONITOR													
MONVAR											X	X	X
MPC	X	X	X	X	X	X	X	X	X	X	X	X	X
MPCFORCES	X	X		X	X	X			X	X	X	X	X
MPRES		X		X	X	X			X	X			
NLARCL											X		
NLCNTL											X	X	X
NLLOAD								X					
NLPARM							X						
NLSTRESS													
NONLINEAR			X					X					
NOUTPUT	X	X											
NSM	X	X	X	X	X	X	X	X	X	X	X		
OFREQUENCY		X				X				X			
OLOAD	X	X	X	X		X	X	X		X	X	X	X
OMEGA													X
OMODES									X	X	X		
OPRESS											X	X	
OSTNINI											X		
OTEMP											X	X	X
OTIME	X	X	X	X		X	X	X	X	X			
OTMFORC													
OUTMGT												X	X
OUTPUT (blank)	X	X	X	X	X	X	X	X	X	X			
OUTPUT(PLOT)	X	X	X	X	X	X	X	X	X	X			
OUTPUT (POST) or SETS DEFINITION	X	X	X	X	X	X	X	X	X	X			
OUTPUT(XYPLOT)	X	X	X	X	X	X	X	X	X	X			

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Command Name	Solution Number (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
P2G	X	X	X	X		X	X			X			
PAGE	X	X	X	X	X	X	X	X	X	X			
PANCON													
PARAM	X	X	X	X	X	X	X	X	X	X	X		
PARTN	X	X	X	X	X	X	X	X	X	X			
PEAKOUT													
PFRESULTS											X	X	
PLSTRN											X	X	
PLOTID	X	X	X	X	X	X	X	X	X	X			
PRESSURE										X			
RANDOM		X				X				X			
RCROSS		X				X							
REPCASE									X	X			
RESVEC						X			X	X			
RIGID											X		
RMAXMIN											X		
RMETHOD													X
RSAPPLY													
RSMETHOD		X	X		X	X		X	X	X			
SACCELERATION		X	X			X				X			
SDAMPING					X	X				X			X
SDISPLACEMENT		X	X		X	X		X	X	X	X		
SEALL	X	X	X	X	X	X	X	X	X	X			
SEDR	X	X	X	X	X	X	X	X	X	X			
SEDV										X			
SEEXCLUDE	X	X	X	X	X	X	X	X	X	X			
SEFINAL	X	X	X	X	X	X	X	X	X	X			
SEKETAB													X
SEKREDUCE	X	X	X	X	X	X	X	X	X	X			
SELGENERATE	X	X	X	X		X	X	X		X			
SELREDUCE	X	X	X	X		X	X	X		X			
SEMGENERATE	X	X	X	X	X	X	X	X	X	X			
SEMREDUCE	X	X	X	X	X	X		X	X	X			
SEQDEP											X	X	

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Command Name	Solution Number (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
SERESP										X			
SET	X	X	X	X	X	X	X	X	X	X	X	X	X
SETMC						X					X	X	X
SETMCNAME											X	X	
SFORCE											X		
SHELLTHK													X
SKIP	X	X	X	X	X	X	X	X	X	X			
SMETHOD											X		
SPC	X	X	X	X	X	X	X	X	X	X	X	X	X
SPCFORCES	X	X	X	X	X	X	X	X	X	X	X	X	X
STATSUB	X								X	X			
STATVAR											X	X	
STRAIN	X	X	X	X	X	X	X	X	X	X	X	X	X
STRESS	X	X	X	X	X	X	X	X	X	X	X	X	X
STRFIELD	X			X					X	X			
SUBCASE	X	X	X	X	X	X	X	X	X	X	X	X	X
SUBCOM										X			
SUBSEQ										X			
SUBTITLE	X	X	X	X	X	X	X	X	X	X	X	X	X
SUPER	X	X	X	X	X	X	X	X	X	X			
SUPORT1	X	X	X	X	X	X	X	X	X	X			
SURFACE	X								X	X			
SVECTOR	X	X			X		X		X	X			
SVELOCITY		X	X			X		X		X			
SYM										X			
SYMCOM										X			
SYMSEQ										X			
TEMP(INIT)	X	X	X	X	X	X	X	X	X	X	X	X	X
TEMP(LOAD)		X		X		X				X	X	X	X
TEMP(MATE)	X	X	X	X	X	X	X	X	X	X			
TFL		X	X		X	X		X		X			
THERMAL							X	X	X	X			
THSTRN											X	X	X
TITLE	X	X	X	X	X	X	X	X	X	X	X	X	X

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Command Name	Solution Number (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
TRIM				X						X			
TRLOSS													
TRPOWER													
TSTEP			X			X		X		X	X	X	X
TSTRU													
ULOAD													X
VATVOUT													
VELOCITY		X	X			X			X	X	X	X	X
VOLUME	X								X	X			
WEIGHTCHECK	X	X	X	X	X	X			X	X	X		
* For modal portion of solution only.													

Related Topics

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5. Case Control X-Y PLOT Commands

Case Control X-Y PLOT Commands Overview

The X-Y output request packet of the Case Control Section includes all commands between either OUTPUT(XYPLOT) or OUTPUT(XYOUT) and either BEGIN BULK or OUTPUT(PLOT). The remainder of this section describes the X-Y output commands.

A single set of plotted X-Y pairs is known as a curve. Curves are the entities that the user requests to be plotted. The surface (paper, microfilm frame, etc.) on which one or more curves is plotted is known as a frame. Curves may be plotted on a whole frame, an upper-half frame, or a lower-half frame. Grid lines, tic marks, axes, axis labeling and other graphic control options may be chosen by the user. The program will select defaults for parameters not selected by the user.

Only two commands are required for an X-Y output request.

1. Only one of OUTPUT(XYPLOT) or OUTPUT(XYOUT) at the beginning of the X-Y output command packet.
2. At least one of the commands XYPLOT, XYPEAK, XYPRINT, XYPUNCH, or XYPAPLOT.

The commands OUTPUT(XYPLOT) and OUTPUT(XYOUT) are equivalent. If the X-Y output is to be printed and/or punched, a PLOTTER command is not required.

If only the required commands are used, the graphic control options will all assume default values. Curves using all default parameters have the following general characteristics:

1. Tic marks are drawn on all edges of the frame. Five spaces are provided on each edge of the frame.
2. All tic marks are labeled with their values.
3. Linear scales are used.
4. Scales are selected such that all points fall within the frame.
5. The plotter points are connected with straight lines.
6. The plotted points are not identified with symbols.

The above characteristics may be modified by inserting any of the parameter definition commands in the next section, ahead of the XY____ command(s). The use of a parameter definition command sets the value of that parameter for all following command operation commands unless the CLEAR command is inserted. If grid lines are requested, they will be drawn at the locations of all tic marks that result from defaults or user requests. The locations of tic marks (or grid lines) for logarithmic scales cannot be selected by the user. Values for logarithmic spacing are selected by the program. The values for the number of tic marks (or grid lines) per cycle depend on the number of logarithmic cycles required for the range of the plotted values.

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The definition and rules for the X-Y output commands follow. The form of X-Y output commands differs in many instances from that of similar commands used in the OUTPUT(PLOT) section.

Related Topics

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X-Y Output Command Summary

Commands Applied To All Curves

PLOTTER	Selects format of plot file for interpretation by plotter post-processor.
CAMERA	Selects plotter media.
PENSIZE	Selects pen number.
DENSITY	Selects the line density for microfilm plotters only.
XPAPER	Defines the size of the paper in x-direction.
YPAPER	Defines the size of the paper in y-direction.
XMIN	Specifies the minimum value on the x-axis.
XMAX	Specifies the maximum value on the x-axis.
XLOG	Selects logarithmic or linear x-axis.
YAXIS	Controls the plotting of the y-axis on all curves.
XINTERCEPT	Specifies the location of the x-axis on the y-axis.
UPPER TICS	Specifies how to draw tic marks on the upper edge.
LOWER TICS	Specifies how to draw tic marks on the lower edge.
CURVELINE	Selects lines and/or symbols to be drawn through the x-y points.
XDIVISIONS	Specifies spacing of tic marks on the x-axis for all curves.
XVALUE SKIP	Specifies how often to print the x-values alongside the x-axis tic marks.
CLEAR	Resets X-Y PLOT commands to their default value.
XTITLE	Defines a character string that will appear along the x-axis.
TCURVE	Defines a character string that will appear at the top of the plot frame.
LONG	Controls amount of curve's summary printout.
CSCALE	Defines the scale factor for characters in the plot frame.

Commands Applied to Whole Frame Curves Only

YMIN	Specifies the minimum value on the y-axis.
YMAX	Specifies the maximum value on the y-axis.
XAXIS	Controls the plotting of the x-axis.
YINTERCEPT	Specifies the location of the y-axis on the x-axis.
YLOG	Selects the logarithmic or linear y-axis.

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LEFT TICS	Specifies how to draw tic marks on the left edge of the frame.
RIGHT TICS	Specifies how to draw tic marks on the right edge of the frame.
ALLEDGE TICS	Specifies how to draw tic marks on all edges of the frame.
YDIVISIONS	Specifies spacing of tic marks on the y-axis.
YVALUE PRINT	Specifies how often to print the y-values alongside the y-axis tic marks.
XGRID LINES	Controls the drawing of the grid lines parallel to the y-axis at the x-axis tic marks.
YGRID LINES	Controls the drawing of the grid lines parallel to the x-axis at the y-axis tic marks.
YTITLE	Defines a character string that will appear along the y-axis.

Commands Applied to Upper Half Frame Curves Only

YTMIN	Specifies the minimum value on the y-axis.
YTMAX	Specifies the maximum value on the y-axis.
YTAXIS	Controls the plotting of the y-axis.
YTINTERCEPT	Specifies the location of the y-axis on the x-axis.
YTLOG	Selects the logarithmic or linear y-axis.
TLEFT TICS	Specifies how to draw tic marks on the left edge.
TRIGHT TICS	Specifies how to draw tic marks on the right edge.
TALL EDGE TIC	Specifies how to draw tic marks on all edges.
YTDIVISIONS	Specifies spacing of tic marks on the y-axis.
YTVALUE PRINT	Specifies how often to print the y-values alongside the y-axis tic marks.
XTGRID LINES	Controls the drawing of the grid lines parallel to the y-axis at the x-axis tic marks.
YTGRID LINES	Controls the drawing of the grid lines parallel to the x-axis at the y-axis tic marks.
YTTITLE	Defines a character string that will appear along the y-axis.

Commands Applied to Lower Half Frame Curves Only

YBMIN	Specifies the minimum value on the y-axis.
YBMAX	Specifies the maximum value on the y-axis.
XBAXIS	Controls the plotting of the x-axis.

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YBINTERCEPT	Specifies the location of the y-axis on the x-axis.
YBLOG	Selects the logarithmic or linear y-axis.
BLEFT TICS	Specifies how to draw tic marks on left edge.
BRIGHT TICS	Specifies how to draw tic marks on right edge.
BALL EDGE TIC	Specifies how to draw tic marks on all edges.
YBDIVISIONS	Specifies spacing of tic marks on the y-axis.
YBVALUE PRINT	Specifies how often to print the y-values alongside the y-axis tic marks.
XBGRID LINES	Controls the drawing of the grid lines parallel to the y-axis at the x-axis tic marks.
YBGRID LINES	Controls the drawing of the grid lines parallel to the x-axis at the y-axis tic marks.
YBTITLE	Defines a character string that will appear along the y-axis.

X-Y Plot Generation Commands

XYPAPLOT	Generates X-Y plots for a printer.
XYPEAK	Prints only the summary for all curves.
XYPLOT	Generates X-Y plots for a plotter.
XYPRINT	Generates a table of X-Y pairs for a printer.
XYPUNCH	Generates a table of X-Y pairs for the PUNCH file.

Related Topics

Browse Simcenter Nastran documentation

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ALLEDGE TICS

Controls Drawing of Tic Marks on All Edges

Specifies how to draw tic marks on all edges of the frame.

Format:

ALLEDGE TICS tic

Examples:

```
ALLEDGE -1
```

Describers:

Describer	Meaning	
tic	Specifies how to draw tic marks. (Integer, Default = 0)	
	-1	Draw tic marks only.
	0	Do not draw tic marks or associated values. (Default)
	1	Draw tic marks and associated values.

Remarks:

1. ALLEDGE TICS applies to whole frame curves only.
2. To determine if on any given edge (a) tic marks will be drawn without values, (b) no tic marks or values will be drawn, or (c) tic marks with values will be drawn, the following sum must be computed by the user. Add the tic values of the edge in question to its associated ALLEDGE TICS, TALL EDGE TICS, or BALL EDGE TICS tic values. If the resulting value is less than zero, tic marks will be drawn without values. If the resulting value is zero, no tic marks or values will be drawn. If the resulting value is greater than zero, tic marks with values will be drawn. The user should be careful in his use of the ALLEDGE TICS, TALL EDGE TICS, or BALL EDGE TICS commands. For example, the use of only the ALLEDGE TICS = -1 command will result in no tic marks or values being drawn since the default values for individual edges is +1. Tic values input may only be -1, 0, or 1.

Related Topics

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BALL EDGE TICS

Controls Drawing of Tic Marks on Lower Half

Specifies how to draw tic marks on lower half of frame.

Format:

BALL EDGE TICS tic

Examples:

```
BALL EDGE TICS -1
```

Describers:

Describer	Meaning	
tic	Specifies how to draw tic marks. (Integer, Default = 0)	
	-1	Draw tic marks only.
	0	Do not draw tic marks or associated values. (Default)
	1	Draw tic marks and associated values.

Remarks:

1. BALL EDGE TICS applies to lower frame curves only.
2. See **ALLEDGE TICS**.

Related Topics

Browse Simcenter Nastran documentation

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BLEFT TICS

Controls Drawing of Tic Marks on Left Edge

Specifies how to draw tic marks on the left edge of the lower half of the frame.

Format:

BLEFT TICS tic

Examples:

```
BLEFT TICS -1
```

Describers:

Describer	Meaning
tic	Specifies how to draw tic marks. (Integer, Default = 1)
	-1 Draw tic marks only.
	0 Do not draw tic marks or associated values.
	1 Draw tic marks and associated values. (Default)

Remarks:

1. BLEFT TICS applies to lower frame curves only.
2. See **ALLEDGE TICS**.
3. See related command **BRIGHT TICS**.

Related Topics

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BRIGHT TICS

Controls Drawing of Tic Marks on Right Edge

Specifies how to draw tic marks on the right edge of the lower half of the frame.

Format:

BRIGHT TICS tic

Examples:

```
BRIGHT TICS -1
```

Describers:

Describer	Meaning	
tic	Specifies how to draw tic marks. (Integer, Default = 1)	
	-1	Draw tic marks only.
	0	Do not draw tic marks or associated values.
	1	Draw tic marks and associated values. (Default)

Remarks:

1. BRIGHT TICS applies to lower frame curves only.
2. See **ALLEDGE TICS**.

Related Topics

Browse Simcenter Nastran documentation

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CAMERA

Plotter Media Selection

Selects plotter media.

Format:

CAMERA ctype

Examples:

```
CAMERA  1
```

Describers:

Describer	Meaning	
ctype	Camera type. (Integer 1, 2, or 3; Default = 2)	
	1	Film
	2	Paper (Default)
	3	Both

Remarks:

If the CAMERA command is not specified then CAMERA 2 is assumed.

Related Topics

Browse Simcenter Nastran documentation

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CLEAR

Resets X-Y PLOT Commands

Resets X-Y PLOT commands to their default value.

Format

CLEAR

Remarks:

All commands except XTITLE, YTITLE, YTTITLE, YBTITLE, and TCURVE will revert to their default values.

Related Topics

Browse Simcenter Nastran documentation

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CSCALE

Character Scale Factor

Defines scale factor for characters in the plot frame.

Format:

CSCALE cs

Examples:

```
CSCA 2.0
```

Describer	Meaning
cs	Scale factor applied to characters in the plot frame.

Remarks:

1. CSCALE is used to control the spacing of characters when plots are made with the NASTRAN plotter and they are postprocessed with the NX/NASPLOT routine. For example, if the SCALE FACTOR on the NASPLOT data command is 2.0, a value for cs of 0.5 will result in characters of default size (.07 inches) at the regular spacing. A value of 1.8 produces good spacing when using the post-processing plotter programs NASTPLT, TEKPLT, and NEUPS. On the other hand, if the user wishes to double the size of both the plot and the characters, the SCALE FACTOR and the CSCALE FACTOR on the NASPLOT data command should both be set equal to 2.0.
2. The CSCALE command must immediately precede the PLOTTER command. If a second CSCALE command is specified then a second PLOTTER command must also be specified.

Related Topics

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CURVELINESYMBOL

Curve, Line and Symbol Selection

Selects lines and/or symbols to be drawn through the x-y points.

Format:

CURVELINESYMBOL symtype

Examples:

```
CURV  4
```

Describers:

Describer	Meaning
symtype	Specifies the symbol drawn at the x-y points. If symtype is 0 then only lines will be drawn through the points with no symbol. If symtype is less than zero then only the symbol and not the lines will be drawn. If symtype is greater than zero then both the symbol and the lines will be drawn. (-9 ≤ Integer ≤ 9, Default = 0)

symtype	Symbol
0	none
1	X
2	*
3	+
4	—
5	.
6	×
7	[ ]
8	< >
9	/ \

Remarks:

If more than one curve is plotted per frame then the symbol number is incremented by 1 for each curve.

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Related Topics

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DENSITY

Microfilm Plotter Line Density

Selects the line density for microfilm plotters only.

Format

DENSITY d

Example

```
DENS  3
```

Describers:

Describer	Meaning
d	Specifies line density scale factor for microfilm plotters. A line density of d is d times heavier than a line density of 1. (Integer ≥ 0, Default = 1)

Related Topics

Browse Simcenter Nastran documentation

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LEFT TICS

Controls Drawing of Tic Marks on Left Edge

Specifies how to draw tic marks on the left edge of whole frame curves.

Format:

LEFT TICS tic

Examples:

```
LEFT -1
```

Describers:

Describer	Meaning	
tic	Specifies how to draw tic marks. (Integer, Default = 1)	
	-1	Draw tic marks only.
	0	Do not draw tic marks or associated values.
	1	Draw tic marks and associated values. (Default)

Remarks:

1. LEFT TICS applies to whole frame curves only.
2. See **ALLEDGE TICS**.
3. See related command **RIGHT TICS**.

Related Topics

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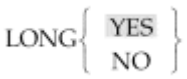
LONG

Summary Print Control

Controls amount of curve's summary printout.

Format:



Describers:

Describer	Meaning
YES	One page for each curve's summary. (Default)
NO	Condensed curve summary.

Remarks:

If LONG is not specified then LONG=NO is assumed.

Related Topics

Browse Simcenter Nastran documentation

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LOWER TICS

Controls Drawing of Tic Marks on Lower Edge

Specifies how to draw tic marks on lower edge.

Format:

LOWER TICS tic

Examples:

```
LOWER -1
```

Describers:

Describer	Meaning
tic	Specifies how to draw tic marks. (Integer, Default = 1)
-1	Draw tic marks only.
0	Do not draw tic marks or associated values.
1	Draw tic marks and associated values. (Default)

Remarks:

1. LOWER TICS applies to all curves.
2. See **ALLEDGE TICS**.

Related Topics

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PENSIZE

Pen Selection

Selects pen number.

Format:

PENSIZE p

Examples:

```
PENS    3
```

Describers:

Describer	Meaning
p	Specifies pen number that is used to generate the plot. (Integer > 0, Default = 1)

Related Topics

Browse Simcenter Nastran documentation

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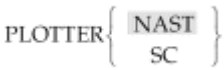
PLOTTER

X-Y Plot File Format

Selects format of plot file for interpretation by plotter post-processor.

Format:



Examples:

PLOTTER NAST

Describers:

Describer	Meaning
NAST	Specifies format suitable for postscript plotters. (Default)
SC	Specifies Stromberg-Carlson microfilm plotter format.

Related Topics

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RIGHT TICS

Controls Drawing of Tic Marks on Right Edge

Specifies how to draw tic marks on the right edge of the frame.

Format:

RIGHT TICS tic

Examples:

```
RIGHT -1
```

Describers:

Describer	Meaning
tic	Specifies how to draw tic marks. (Integer, Default = 1)
-1	Draw tic marks only.
0	Do not draw tic marks or associated values.
1	Draw tic marks and associated values. (Default)

Remarks:

1. RIGHT TICS applies to whole frame curves only.
2. See **ALLEDGE TICS**.
3. See related command **LEFT TICS**.

Related Topics

Browse Simcenter Nastran documentation

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TALL EDGE TICS

Controls Drawing of Tic Marks on All Edges

Specifies how to draw tic marks on all edges of the upper half of the frame.

Format:

TALL EDGE TICS tic

Examples:

```
TALL -1
```

Describers:

Describer	Meaning
tic	Specifies how to draw tic marks. (Integer, Default = 0)
-1	Draw tic marks only.
0	Do not draw tic marks or associated values. (Default)
1	Draw tic marks and associated values.

Remarks:

1. TALL EDGE TICS applies to upper half frame curves only.
2. See **ALLEDGE TICS**.

Related Topics

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TCURVE

Curve Title

Defines a character string that will appear at the top of the plot frame.

Format:

TCURVE ctitle

Examples:

```
TCUR  RIGHT WING -- LOAD CASE 3
```

Describers:

Describer	Meaning
ctitle	Any character string. (Character, Default = blank)

Remarks:

TCURVE may not be continued to the next command line.

Related Topics

Browse Simcenter Nastran documentation

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TLEFT TICS

Controls Drawing of Tic Marks on the Left-Edge

Specifies how to draw tic marks on the left edge of the upper half of the frame.

Format:

TLEFT TICS tic

Examples:

```
TLEFT -1
```

Describers:

Describer	Meaning
tic	Specifies how to draw tic marks. (Integer, Default = 1)
-1	Draw tic marks only.
0	Do not draw tic marks or associated values.
1	Draw tic marks and associated values. (Default)

Remarks:

1. TLEFT TICS applies to upper half frame curves only.
2. See **ALLEDGE TICS**.
3. See related command **TRIGHT TICS**.

Related Topics

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TRIGHT TICS

Controls Drawing of Tic Marks on the Right Edge

Specifies how to draw tic marks on the right-edge of the upper half of the frame.

Format:

TRIGHT TICS tic

Examples:

```
TRIGHT -1
```

Describers:

Describer	Meaning
tic	Specifies how to draw tic marks. (Integer, Default = 1)
-1	Draw tic marks only.
0	Do not draw tic marks or associated values.
1	Draw tic marks and associated values. (Default)

Remarks:

1. TRIGHT TICS applies to upper half frame curves only.
2. See **ALLEDGE TICS**.
3. See related command **TLEFT TICS**.

Related Topics

Browse Simcenter Nastran documentation

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UPPER TICS

Controls Drawing Of Tic Marks On Upper Edge

Specifies how to draw tic marks on the upper edge.

Format:

UPPER TICS tic

Examples:

```
UPPER -1
```

Describers:

Describer	Meaning
tic	Specifies how to draw tic marks. (Integer, Default = 1)
-1	Draw tic marks only.
0	Do not draw tic marks or associated values.
1	Draw tic marks and associated values. (Default)

Remarks:

1. UPPER TICS applies to all curves.
2. See **ALLEDGE TICS**.
3. See related command **LOWER TICS**.

Related Topics

Browse Simcenter Nastran documentation

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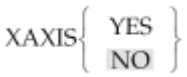
XAXIS

X-Axis Plot Control

Controls the plotting of the x-axis on whole frame curves only.

Format:



Describers:

Describer	Meaning
YES	Plot the x-axis.
NO	Do not plot the x-axis. (Default)

Remarks:

1. XAXIS applies to whole frame curves only.
2. See related command YAXIS.

Related Topics

Browse Simcenter Nastran documentation

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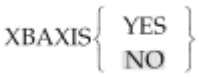
XBAXIS

X-Axis Plot Control

Controls the plotting of the x-axis on lower half frame curves only.

Format:



Describers:

Describer	Meaning
YES	Plot the x-axis.
NO	Do not plot the x-axis. (Default)

Remarks:

XBAXIS applies to lower half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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XBGRID LINES

Plot X-Axis Grid Lines

Controls the drawing of the grid lines parallel to the y-axis at the x-axis tic marks on lower half frame curves only.

Format:

Describers:

Describer	Meaning
YES	Plot the x-axis grid lines.
NO	Do not plot the x-axis grid lines. (Default)

Remarks:

1. XBGRID applies to lower half frame curves only.
2. See the related command **YBGRID LINES**.

Related Topics

Browse Simcenter Nastran documentation

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XDIVISIONS

Tic Spacing on X-Axis

Specifies spacing of tic marks on the x-axis for all curves.

Format:

XDIVISIONS xd

Examples:

```
XDIV 10
```

Describers:

Describer	Meaning
xd	Number of spaces between tic marks on x-axis. (Integer > 0, Default = 5)

Remarks:

1. XDIVISIONS applies to all curves and to the commands: UPPER TICS, LOWER TICS, and YINTERCEPT.
2. XDIVISIONS is ignored for logarithmic x-axes.

Related Topics

Browse Simcenter Nastran documentation

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XGRID LINES

Plot X-Axis Grid Lines

Controls the drawing of the grid lines parallel to the y-axis at the x-axis tic marks on whole frame curves only.

Format:

$$\text{XGRID LINES} \left\{ \begin{matrix} \text{YES} \\ \text{NO} \end{matrix} \right\}$$

Describers:

Describer	Meaning
YES	Plot the x-axis grid lines.
NO	Do not plot the x-axis grid lines. (Default)

Remarks:

1. XGRID applies to whole frame curves only.
2. See the related command YGRID LINES .

Related Topics

Browse Simcenter Nastran documentation

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XINTERCEPT

Location of X-Axis on Y-Axis

Specifies the location of the x-axis on the y-axis.

Format:

XINTERCEPT xi

Examples:

```
XINT 50.
```

Describers:

Describer	Meaning
xi	Location of x-axis on the y-axis. (Real, Default = 0.0)

Related Topics

Browse Simcenter Nastran documentation

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XLOG

Logarithmic or Linear X-Axis

Selects logarithmic or linear x-axis.

Format:

$$\text{XLOG}\left\{\begin{matrix}\text{YES}\\ \text{NO}\end{matrix}\right\}$$

Describers:

Describer	Meaning
YES	Plot a logarithmic x-axis.
NO	Plot a linear x-axis. (Default)

Remarks:

1. XLOG applies to all curves.
2. The default value for tic division interval depends on the number of log cycles. The default values for tic divisions are given below but will range over whole cycles

Number of Cycles	Intermediate Values
1, 2	2., 3., 4., 5., 6., 7., 8., 9.
3	2., 3., 5., 7., 9.,
4	2., 4., 6., 8.,
5	2., 5., 8.
6, 7	3., 6.
8, 9, 10	3.

Related Topics

Browse Simcenter Nastran documentation

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XMAX

Maximum X-Axis Value

Specifies the maximum value on the x-axis.

Format:

XMAX xmax

Examples:

```
XMAX 100.
```

Describers:

Describer	Meaning
xmax	Maximum value on the x-axis. (Real)

Remarks:

1. If XMAX is not specified then the maximum value is set to the highest value of x.
2. See related commands **XMIN**, **YMIN**, and **YMAX**.

Related Topics

Browse Simcenter Nastran documentation

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XMIN

Minimum X-Axis Value

Specifies the minimum value on the x-axis.

Format:

XMIN xmin

Examples:

```
XMIN 100.
```

Describers:

Describer	Meaning
xmin	Minimum value on the x-axis. (Real)

Remarks:

1. XMIN applies to all curves.
2. If XMIN is not specified then the minimum value is set to the lowest value of x.
3. See related commands **XMAX**, **YMIN**, and **YMAX**.

Related Topics

Browse Simcenter Nastran documentation

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XPAPER

Paper Size in the X-Direction

Defines the size of the paper in the x-direction.

Format:

XPAPER xsize

Examples:

```
XPAP 10.
```

Describers:

Describer	Meaning
xsize	Size of paper in the x-direction and in inches. (Real, Default = 20.0)

Remarks:

1. The default paper size is 20 by 20 inches.
2. See related command YPAPER.

Related Topics

Browse Simcenter Nastran documentation

Simcenter Nastran Quick Reference Guide
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XTGRID LINES

Plot X-Axis Grid Lines

Controls the drawing of the grid lines parallel to the y-axis at the x-axis tic marks on upper half frame curves only.

Format:

Describers:

Describer	Meaning
YES	Plot the x-axis grid lines.
NO	Do not plot the x-axis grid lines. (Default)

Remarks:

XTGRID applies to upper half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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XTITLE

X-Axis Title

Defines a character string that will appear along the x-axis.

Format:

XTITLE xtit

Examples:

```
XTIT   RIGHT WING CASE 3 - TIME
```

Describers:

Describer	Meaning
xtit	Any character string. (Character, Default = blank)

Remarks:

1. XTITLE may not be continued to the next command line.
2. XTITLE applies to all curves.

Related Topics

Browse Simcenter Nastran documentation

Simcenter Nastran Quick Reference Guide
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XYPAPLOT

Generate X-Y Plots for a Printer

Generates X-Y plots for a printer.

See XYPLOT for Format, Describers, and additional Remarks.

Remarks:

1. The x-axis moves vertically along the page and the y-axis moves horizontally along the page.
2. An asterisk (*) identifies the points associated with the first curve of a frame, then for successive curves on a frame, the points are designated by symbols O, A, B, C, D, E, F, G and H.

Related Topics

Browse Simcenter Nastran documentation

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XYPEAK

Print Curve Summary

Prints only the summary for all curves. The summary output is titled:

"X Y - O U T P U T S U M M A R Y"

and is also printed under XYPLOT, XYPUNCH, XYPRINT, and XYPAPLOT. This output contains the maximum and minimum values of y for the range of x.

See **XYPLOT** for Format, Describers, and additional Remarks.

Related Topics

Browse Simcenter Nastran documentation

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XYPLOT

Generates X-Y Plots for a Plotter

Generates X-Y Plots for a Plotter.

Format

XYPLOT	yvtype ptype [i1, i2, i3,...] /
	id11 (itemu11 [, iteml11]) , id12 (itemu12 [, iteml12]) , ... /
	id21 (itemu21 [, iteml21]) , id22 (itemu22 [, iteml22]) , ... / ....

Describers:

Describer	Meaning	
yvtype	Type of y-value to be plotted: (Character)	
	ACCE	Acceleration in the physical set
	BOUT	Slideline contact output
	DISP	Displacement in the physical set
	ELFORCE	Element force
	ENTHALPY	Enthalpy in the physical set
	FLUX	Element Heat flux
	HDOT	Rate of change of enthalpy in the physical set
	NONLINEAR	Nonlinear applied load
	OLOAD	Applied Load
	PRESSURE	Pressure of fluid-structure body
	SACCE	Acceleration in the solution set
	SDISP	Displacement in the solution set
	SPCF	Single-point force of constraint
	STEMP	Temperature in the solution set
	STRESS	Element stress
	SVELO	Velocity in the solution set
	TEMP	Temperature in the physical set

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Describer	Meaning	
	VECTOR	Displacement in the physical set
	VELO	Velocity in the physical set
	VG	Flutter analysis
ptype	Plot type defining the meaning of i1, i2, ... etc., idi, itemui and itemli. (Character, Default = “RESPONSE”)	
	RESPONSE	Time or frequency in SORT2 format or grid identification numbers in SORT1 format. (Default)
	AUTO	Autocorrelation function on whole frame curves only.
	PSDF	Power spectral density function on whole frame curves only.
	SPECTRAL	Response spectrum on whole frame curves only.
i1, i2,...	Subcase identification numbers for ptype = RESPONSE. The list must be specified in ascending order. For ptype = SPECTRAL, the subcase refers to the RECNO in the DTI,SPSEL Bulk Data entry. The list is ignored for ptype = AUTO and PSDF. (Integer ≥ 0, Default is all subcases)	
idij	Element, grid, scalar, or extra point identification number for y-value for frame i. For yvtype = VG, idij refers to the loop count of a flutter analysis. (Integer > 0)	
itemuij,	Item code for y-value. itemuij is for the upper half or whole itemlij curves on frame i and itemlij is for lower half curves only on frame i. If itemlij is not specified, then whole frame curves will be plotted with itemuij. itemlij is ignored for ptype = “AUTO”, “PSDF”, and “SPECTRAL”. (Character or Integer > 0)	

For elements, the code represents a component of the element stress, strain, or force and is described in the *Item Codes* topic in this guide. For ptype = “AUTO” and “PSDF”, the complex stress or strain item codes need to be used. As the output quantities are real, you can use either the real or the imaginary item code. Both will give the same result.

For grid points and ptype = “RESPONSE”, the code is one of the mnemonics T1, T2, T3, R1, R2, R3, T1RM, T2RM, T3RM, R1RM, R2RM, R3RM, T1IP, T2IP, T3IP, R1IP, R2IP, or R3IP, where Ti stands for the i-th translational component, Ri stands for the i-th rotational component, RM means real or magnitude, and IP means imaginary or phase. For scalar or extra points or heat transfer analysis, use T1, T1RM, or T1IP.

For grid points and ptype = “AUTO” or “PSDF”, the code is one of the mnemonics T1, T2, T3, R1, R2, R3. For scalar or extra points, use T1.

For yvtype = VG, itemui and/or itemli can be “F” for frequency or “G” for damping.

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Remarks:

1. Multiple XYPLOT, XYPUNCH, XYPRINT, XYPEAK and/or XYPAPLOT commands may be specified in the OUTPUT(XYPLOT) section.
2. Solution set requests are more efficient, because the time consuming recovery of the dependent displacements can be avoided.
3. The item codes appear in printed summaries under the “CURVE ID” column for grid points as well as element data. For grid points, the component number in this listing will have the values 3 through 14, corresponding to T1RM, T2RM, T3RM, R1RM, R2RM, R3RM, T1IP, T2IP, T3IP, R1IP, R2IP, R3IP, respectively. For scalar points, the component value is always 3. The grid point output component numbers are listed in the following table.

Input	Output Grid Component Number
T1RM	3
T2RM	4
T3RM	5
R1RM	6
R2RM	7
R3RM	8
T1IP	9
T2IP	10
T3IP	11
R1IP	12
R2IP	13
R3IP	14

4. The information after each slash (/) specifies the curve(s) that are to be plotted on the same frame. The describer idij identifies the grid point j or element j associated with the frame number i. All plot requests on one command are sorted on idij to improve the efficiency of the plotting process. Symbols are assigned in order by idij.
5. If any of the item codes, itemlij or itemuij, are not specified; e.g. (8,) or (,5), the corresponding half frame curve is not plotted. If both the comma (,) and itemlij not specified; e.g., (8), then whole frame curves will be plotted. Also, for any single frame, the specifications of “(itemuij,itemlij)” must be consistently half frame (upper and/or lower) or whole frame. For example on half frame curves, if iteml11 and the comma is not specified then either iteml12 or itemu12 must not be specified and on whole frame curves, the commas, iteml11, and iteml12 must not be specified. In other words, the curves on each plot frame must be all whole or half (upper and/or lower).

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6. The XYPLOT command may be continued on the next line as long as “XYPLOT yvtype ptype [i1, i2, i3,...] /” is specified on the first line.
7. Specifying a nonexistent grid point may cause the program to exit in the XYTRAN module and missing plots to occur.

Related Topics

Browse Simcenter Nastran documentation

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XYPRINT

Generate Table of X-Y Pairs for a Printer

Generates tabular printer output of the X-Y pairs.

See XYPLOT for Format, Describers, and additional Remarks.

Related Topics

Browse Simcenter Nastran documentation

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XYPUNCH

Generate Table of X-Y Pairs for the PUNCH File

Generates tabular punch output of the X-Y pairs. This is the same as XYPRINT, except that the output is written to the PUNCH file.

See **XYPLOT** for Format, Describers, and additional Remarks.

Related Topics

Browse Simcenter Nastran documentation

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XVALUE PRINT SKIP

Print Values on X-Axis Tic Marks

Specifies how often to print the x-values alongside the x-axis tic marks.

Format:

XVALUE PRINT SKIP xvps

Examples:

```
XVAL  5
```

Describers:

Describer	Meaning
xvps	Number of tic marks to be skipped between labeled tic marks with their corresponding values. (Integer ≥ 0)

Remarks:

XVALUE applies to all curves.

Related Topics

Browse Simcenter Nastran documentation

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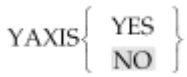
YAXIS

Y-Axis Plot Control

Controls the plotting of the y-axis on all curves.

Format:



Describers:

Describer	Meaning
YES	Plot the y-axis.
NO	Do not plot the y-axis. (Default)

Related Topics

Browse Simcenter Nastran documentation

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YBDIVISIONS

Tic Spacing on Y-Axis

Specifies spacing of tic marks on the y-axis for lower half frame curves only.

Format:

YBDIVISIONS ybd

Examples:

```
YBDI 10
```

Describers:

Describer	Meaning
ybd	Number of spaces between tic marks on y-axis. (Integer > 0, Default = 5)

Remarks:

1. YBDIVISIONS applies to lower half frame curves only.
2. YBDIVISIONS is ignored for logarithmic y-axes.

Related Topics

Browse Simcenter Nastran documentation

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YBINTERCEPT

Location of Y-Axis on X-Axis

Specifies the location of the y-axis on the x-axis for lower half frame curves only.

Format:

YBINTERCEPT ybi

Examples:

```
YBINT 50
```

Describers:

Describer	Meaning
ybi	Location of y-axis on the x-axis. (Real, Default = 0.0)

Remarks:

YBINTERCEPT applies to lower half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YBGRID LINES

Plot Y-Axis Grid Lines

Controls the drawing of the grid lines parallel to the x-axis at the y-axis tic marks on lower half frame curves only.

Format:

Describers:

Describer	Meaning
YES	Plot the y-axis grid lines.
NO	Do not plot the y-axis grid lines. (Default)

Remarks:

YBGRID applies to lower half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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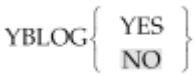
YBLOG

Logarithmic or Linear Y-Axis

Selects logarithmic or linear y-axis for lower half frame curves only.

Format:



Describers:

Describer	Meaning
YES	Plot a logarithmic y-axis.
NO	Plot a linear y-axis. (Default)

Remarks:

1. YBLOG applies to lower half frame curves only.
2. See XLOG.

Related Topics

Browse Simcenter Nastran documentation

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YBMAX

Maximum Y-Axis Value

Specifies the maximum value on the y-axis for lower half frame curves only.

Format:

YBMAX ymax

Examples:

```
YBMAX 100
```

Describers:

Describer	Meaning
ymax	Maximum value on the y-axis. (Real)

Remarks:

1. YBMAX applies to lower half frame curves only.
2. If YBMAX is not specified, then the maximum value is set to the highest value of y.
3. See the related command **YBMIN**.

Related Topics

Browse Simcenter Nastran documentation

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YBMIN

Minimum Y-Axis Value

Specifies the minimum value on the y-axis for lower half frame curves only.

Format:

YBMIN ymin

Examples:

```
YBMIN 100
```

Describers:

Describer	Meaning
ymin	Minimum value on the y-axis. (Real)

Remarks:

1. YBMIN applies to lower half frame curves only.
2. If YBMIN is not specified, then the minimum value is set to the lowest value of y.
3. See the related command YBMAX.

Related Topics

Browse Simcenter Nastran documentation

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YBTITLE

Y-Axis Title

Defines a character string that will appear along the y-axis for lower half frame curves only.

Format:

YBTITLE ytit

Examples:

```
YBTIT   RIGHT WING LOADS - CASE 3
```

Describers:

Describer	Meaning
ytit	Any character string. (Character, Default = blank)

Remarks:

1. YBTITLE may not be continued to the next command line.
2. YBTITLE applies to lower half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YBVALUE PRINT SKIP

Print Values on Y-Axis Tic Marks

Specifies how often to print the y-values alongside the y-axis tic marks on lower half frame curves only.

Format:

YBVALUE PRINT SKIP yvps

Examples:

```
YBVAL  5
```

Describers:

Describer	Meaning
yvps	Number of tic marks to be skipped between labeled tic marks with their corresponding values. (Integer ≥ 0)

Remarks:

YBVALUE applies to lower half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YDIVISIONS

Tic Spacing on Y-Axis

Specifies spacing of tic marks on the y-axis for whole frame curves only.

Format:

YDIVISIONS yd

Examples:

```
YDIV 10
```

Describers:

Describer	Meaning
yd	Number of spaces between tic marks on the y-axis. (Integer > 0, Default = 5)

Remarks:

1. YDIVISIONS applies to whole frame curves only and to the commands: LEFT TICS, RIGHT TICS, and XINTERCEPT.
2. YDIVISIONS is ignored for logarithmic y-axes.

Related Topics

Browse Simcenter Nastran documentation

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YINTERCEPT

Location of Y-Axis on X-Axis

Specifies the location of the y-axis on the x-axis for whole frame curves only.

Format:

YINTERCEPT yi

Examples:

```
YINT  50
```

Describers:

Describer	Meaning
yi	Location of y-axis on the x-axis. (Real, Default = 0.0)

Remarks:

YINTERCEPT applies to lower half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YGRID LINES

Plot Y-Axis Grid Lines

Controls the drawing of the grid lines parallel to the x-axis at the y-axis tic marks on whole frame curves only.

Format:

$$\text{YGRID LINES} \left\{ \begin{matrix} \text{YES} \\ \text{NO} \end{matrix} \right\}$$

Describers:

Describer	Meaning
YES	Plot the y-axis grid lines.
NO	Do not plot the y-axis grid lines. (Default)

Remarks:

YGRID applies to whole frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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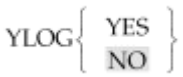
YLOG

Logarithmic or Linear Y-Axis

Selects logarithmic or linear y-axis for whole frame curves only.

Format:



Describers:

Describer	Meaning
YES	Plot a logarithmic y-axis.
NO	Plot a linear y-axis. (Default)

Remarks:

1. YLOG applies to whole frame curves only.
2. See XLOG.

Related Topics

Browse Simcenter Nastran documentation

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YMAX

Maximum Y-Axis Value

Specifies the maximum value on the y-axis.

Format:

YMAX ymax

Examples:

```
YMAX 100
```

Describers:

Describer	Meaning
ymax	Maximum value on the y-axis. (Real)

Remarks:

1. If YMAX is not specified, then the maximum value is set to the highest value of y.
2. See the related command **YMIN**.

Related Topics

Browse Simcenter Nastran documentation

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YMIN

Minimum Y-Axis Value

Specifies the minimum value on the y-axis.

Format:

YMIN ymin

Examples:

```
YMIN 100
```

Describers:

Describer	Meaning
ymin	Minimum value on the y-axis. (Real)

Remarks:

1. YMIN applies to all curves.
2. If YMIN is not specified, then the minimum value is set to the lowest value of y.
3. See the related command YMAX.

Related Topics

Browse Simcenter Nastran documentation

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YPAPER

Paper Size in Y-Direction

Defines the size of the paper in the y-direction.

Format:

YPAPER ysize

Examples:

```
YPAP 10
```

Describers:

Describer	Meaning
ysize	Size of the paper in the y-direction and in inches. (Real, Default = 20.0)

Remarks:

1. The default paper size is 20 by 20 inches.
2. See the related command XPAPER.

Related Topics

Browse Simcenter Nastran documentation

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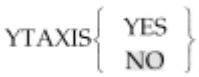
YTAXIS

Y-Axis Plot Control

Controls the plotting of the y-axis on upper half frame curves only.

Format:



Describers:

Describer	Meaning
YES	Plot the x-axis.
NO	Do not plot the x-axis. (Default)

Remarks:

YTAXIS applies to upper half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YTDIVISIONS

The Spacing on Y-Axis

Specifies spacing of tic marks on the y-axis for upper half frame curves only.

Format:

YTDIVISIONS ytd

Examples:

```
YTDI 10
```

Describers:

Describer	Meaning
ytd	Number of spaces between tic marks on y-axis. (Integer > 0, Default = 5)

Remarks:

1. YTDIVISIONS applies to upper half frame curves only.
2. YTDIVISIONS is ignored for logarithmic y-axes.

Related Topics

Browse Simcenter Nastran documentation

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YTGRID LINES

Plot Y-Axis Grid Lines

Controls the drawing of the grid lines parallel to the x-axis at the y-axis tic marks on upper half frame curves only.

Format:

$$\text{TYGRID LINES} \left\{ \begin{matrix} \text{YES} \\ \text{NO} \end{matrix} \right\}$$

Describers:

Describer	Meaning
YES	Plot the y-axis grid lines.
NO	Do not plot the y-axis grid lines. (Default)

Remarks:

YTGRID applies to upper half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YTINTERCEPT

Location of Y-Axis on X-Axis

Specifies the location of the y-axis on the x-axis for upper half frame curves only.

Format:

YTINTERCEPT yti

Examples:

```
YTINT  50
```

Describers:

Describer	Meaning
yti	Location of y-axis on the x-axis. (Real, Default = 0.0)

Remarks:

YTINTERCEPT applies to upper half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YTITLE

Y-Axis Title

Defines a character string that will appear along the y-axis for whole frame curves only.

Format:

YTITLE ytit

Examples:

```
YTIT   RIGHT WING LOADS - CASE 3
```

Describers:

Describer	Meaning
ytit	Any character string. (Character, Default = blank)

Remarks:

1. YTITLE may not be continued to the next command line.
2. YTITLE applies to whole frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YTLOG

Logarithmic or Linear Y-Axis

Selects logarithmic or linear y-axis for upper half frame curves only.

Format:

$$\text{YT(LOG)} \left\{ \begin{matrix} \text{YES} \\ \text{NO} \end{matrix} \right\}$$

Describers:

Describer	Meaning
YES	Plot a logarithmic y-axis.
NO	Plot a linear y-axis. (Default)

Remarks:

1. YTLOG applies to upper half frame curves only.
2. See XLOG.

Related Topics

Browse Simcenter Nastran documentation

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YTMAX

Maximum Y-Axis Value

Specifies the maximum value on the y-axis for upper half frame curves only.

Format:

YTMAX ymax

Examples:

```
YTMAX 100
```

Describers:

Describer	Meaning
ymax	Maximum value on the y-axis. (Real)

Remarks:

1. YTMAX applies to upper half frame curves only.
2. If YTMAX is not specified, then the maximum value is set to the highest value of y.
3. See the related command **YTMIN**.

Related Topics

Browse Simcenter Nastran documentation

Simcenter Nastran Quick Reference Guide
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YTMIN

Minimum Y-Axis Value

Specifies the minimum value on the y-axis for upper half frame curves only.

Format:

YTMIN ymin

Examples:

```
YTMIN 100
```

Describers:

Describer	Meaning
ymin	Minimum value on the y-axis. (Real)

Remarks:

1. YTMIN applies to upper half frame curves only.
2. If YTMIN is not specified, then the minimum value is set to the lowest value of y.
3. See the related command YTMAX.

Related Topics

Browse Simcenter Nastran documentation

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YTTITLE

Y-Axis Title

Defines a character string that will appear along the y-axis for upper half frame curves only.

Format:

YTTITLE ytit

Examples:

```
YTTIT   RIGHT WING LOADS - CASE 3
```

Describers:

Describer	Meaning
ytit	Any character string. (Character, Default = blank)

Remarks:

1. YTTITLE may not be continued to the next command line.
2. YTTITLE applies to upper half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YTVALUE PRINT SKIP

Print Values on Y-Axis Tic Marks

Specifies how often to print the y-values alongside the y-axis tic marks on upper half frame curves only.

Format:

YTVALUE PRINT SKIP yvps

Examples:

```
YTVAL  5
```

Describers:

Describer	Meaning
yvps	Number of tic marks to be skipped between labeled tic marks with their corresponding values. (Integer ≥ 0)

Remarks:

YTVALUE applies to upper half frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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YVALUE PRINT SKIP

Print Values on Y-Axis Tic Marks

Specifies how often to print the y-values alongside the y-axis tic marks on whole frame curves only.

Format:

YVALUE PRINT SKIP yvps

Examples:

```
YVAL  5
```

Describers:

Describer	Meaning
yvps	Number of tic marks to be skipped between labeled tic marks with their corresponding values. (Integer ≥ 0)

Remarks:

YVALUE applies to whole frame curves only.

Related Topics

Browse Simcenter Nastran documentation

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6. Parameters and Parameter Applicability Tables

Parameter Descriptions

Parameters are used extensively in the solution sequences for input of scalar values and for requesting special features. Parameters values are specified on PARAM Bulk Data entries or PARAM Case Control commands. For more information on the PARAM Bulk Data entry, see **PARAM**. For more information on the PARAM Case Control command, see **PARAM**. A complete list of PARAMeter names and their solution support is given in the **Parameter Applicability Tables**.

Simcenter Nastran allows the specification of PARAM statements as follows:

1. Inside of the input file.
2. In the nastrrc (Linux) and nastr.rcf (WINDOWS).
3. Parameters can be assigned a user defined keyword. The new keyword can then be used to specify a value for the parameter on the command line or in the nastran resource file. The keywords can be defined in the "nastran.params" file in the architecture directory.

See the section "Parameter Specification" in chapter 3 of the *Simcenter Nastran Installation and Operations Guide* for details.

Parameters and Superelements

The following should be considered when external superelements or part superelements are included in the Bulk Data section of your input file.

- Parameters specified in the Bulk Data, but not in a BEGIN SUPER portion of the Bulk Data, apply only to the residual and not to external or part superelements.
- Parameters specified in a BEGIN SUPER portion of the Bulk Data apply only to that external or part superelement.
- To ensure that parameters apply to the residual and all superelements, you should specify them in the Case Control section above the subcases. For example, specifying the PARAM,POST setting above the subcases will request a consistent output format for the residual and the superelements.

Related Topics

Browse Simcenter Nastran documentation

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A – Parameters

ACOUT

Default = PEAK

ACOUT specifies the type of output to be used with the FORCE Case Control command in coupled fluid-structural analysis (see Performing a Coupled Acoustic-Structural Analysis in the *Simcenter Nastran Acoustics Simcenter Nastran User's Guide*). ACOUT=RMS requests root-mean-square output.

To obtain sound pressure level in units of dB and dBA given by the FORCE command, a peak reference pressure must be specified with PARAM, PREFDB. The dB level is defined as:

$$\mathrm{dB} = 20 \cdot \log\left(\frac{P}{\mathrm{PREFDB}}\right)$$

ACSYM

Default = YES

By default, the dynamic equations for coupled fluid-structure analysis in frequency response are symmetrized for efficiency. PARAM,ACSYM,NO requests the pre-Simcenter Nastran 1.0 formulation which involves no symmetrization and will require more CPU time. See the Formulation of Dynamic Equations in SubDMAP GMA in the *Simcenter Nastran User's Guide*.

If the iterative solver is selected (see the ITER=YES keyword on the **NASTRAN** statement) then the external work diagnostic will be different between ACSYM=YES and ACSYM=NO.

ADPCON

Default = 1.0

Initial penalty values used in contact analysis are calculated automatically by the program and are given by k · SFAC · |ADPCON| where k is a number selected for each secondary node based on the diagonal stiffness matrix coefficients that are in the contact region, and SFAC is the value specified by the user in the SFAC field of the BCONP Bulk Data entry. The ADPCON value applies to all the contact regions in the model. During the analysis, if convergence problems are encountered, penalty values are automatically reduced. Still there may be some problems where convergence cannot be achieved with default values. In such cases, analysis may be restarted with a lower value of ADPCON.

In some cases the default penalty values may be low. In such situations, analysis may be restarted with a higher value of ADPCON.

Generally, penalty values are recalculated every time there is a change in stiffness. However, if ADPCON is negative, penalty values are calculated only at the beginning

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of each subcase, and penalty values are not adjusted during analysis. This is useful if the contact between two elastic bodies is being analyzed.

ADSTAT

Default = YES (SOL 109 and 112 only)

Used in transient analysis data recovery with differential stiffness (see the Case Control Command, **STATSUB**) to request whether the static solution (displacements, SPC forces, and MPC forces) is to be included (YES) or excluded (NO) in the transient results.

ADTOQDLD

Default = NO

For SOL 401 only, the shear panel element and the tension-only shell element in shear panel status both compute extensional forces when the effectiveness factors (F1, F2, F3, F4) are defined on the property entries (PSHEAR and PSHLPNL bulk entries). The software does not include the resulting extensional force with the actual element output.

You can define CBAR, CBEAM, or CROD elements on the perimeter of the shear panel and the tension-only shell elements and define the ADTOQDLD parameter to request for the software to add the extensional force computed for the shear panel and the tension-only shell elements, to the axial force computed for the line elements along the shear panel edge.

YES: The extensional force computed for the shear panel and the tension-only shell elements is added to the axial force output on the CBAR, CBEAM, or CROD elements defined on the perimeter. The total axial force on the CBAR, CBEAM, or CROD elements will include their own axial force contribution, plus the extensional force computed for all shear panel and the tension-only shell elements sharing the same edge.

NO: The extensional force is not output. The output for the CBAR, CBEAM, or CROD elements will only include their own axial force contribution.

Note that the stiffness you define for the extensional edges, and the stiffness you define for the CBAR, CBEAM, or CROD elements are independent of each other. You can increase or decrease the stiffness for any of these elements independently, which then contributes to the global stiffness.

AESDISC

Default = 1.E-8

Tolerance for discarding generalized coordinates in the RITZ method (see **AESMETH**) which are not linearly independent.

AESMAXIT

Default = 15

Maximum number of iterations for the ITER method (see **AESMETH**).

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AESMETH Default = SELECT

Solution method for static aeroelastic analysis.

SELECT	selects the DIRECT method on models with fewer than 5000 DOF in the solution set; otherwise selects AUTO.
AUTO	selects the reduced basis method for an approximate solution, which is used as starting vectors for an ITER solution.
DIRECT	selects the direct solution.
RITZ	selects the reduced basis approximate solution.
ITER	selects the iterative solution.

AESRNDM Default = 2

Number of random vectors to use as generalized functions in the RITZ method (see **AESMETH**).

AESTOL Default = 1.E-10

Convergence criteria for the iterative solver.

AFNORM Default = 1.0

Scale factor used in AF (amplitude-frequency) normalization. For additional information, see Mode Shape Normalization in the *Basic Dynamic Analysis User's Guide*.

AFZERO Default = 1.0 hertz

Frequency threshold in units of hertz used by the SOL 401 cyclic and Fourier subcase types to determine if a mode is a rigid body mode when AF normalization is requested. The software considers a mode to be a rigid body mode if its frequency is below the value of AFZERO. For the modes considered as rigid body modes, the software sets ω = 1 when computing the AF normalization factor. See "Subcase Types" in the *Multi-Step Nonlinear User's Guide*.

ALPHA1, ALPHA2 Default = (0.0,0.0)

Specifies scale factors for Rayleigh (proportional) damping in a frequency or transient response analysis when either PARAM,ALPHA1 or PARAM,ALPHA2 is not equal to complex zero. The software adds the Rayleigh damping to the other viscous damping contributions.

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Note

For solutions where the use of the DAMPING case control command and DAMPING bulk entry are supported and the STRUCTURE describer (the default) is specified on the DAMPING case control command, if the input file contains both an explicit parameter specification such as PARAM,ALPHA1 and a DAMPING bulk entry that is referenced by the DAMPING(STRUCTURE) case control command, the value in the ALPHA1 field (or the default value for the ALPHA1 field if left blank) of the DAMPING bulk entry takes precedence.

Note

Although the specification for the ALPHA1 and ALPHA2 parameters is complex, you must specify the imaginary part as zero.

The scale factors are applied to the d-set or h-set matrices. The contribution to the viscous damping for a structure is as follows:

$$\mathrm{ALPHA1} \cdot \left[M^1\right] + \mathrm{ALPHA2} \cdot \left(\left[K^1\right] + \left[K^d\right]\right)$$

where:

$\mathrm{ALPHA1}$ is the mass matrix scaling factor for Rayleigh damping (PARAM,ALPHA1 or the ALPHA1 field on a DAMPING bulk entry).

$\mathrm{ALPHA2}$ is the stiffness matrix scaling factor for Rayleigh damping (PARAM,ALPHA2 or the ALPHA2 field on a DAMPING bulk entry).

$\left[M^1\right]$ is the mass matrix plus M2GG direct input mass.

$\left[K^1\right]$ is the stiffness matrix plus K2GG direct input stiffness matrix

$\left[K^d\right]$ is the structural differential stiffness matrix.

In SOL 402, you can use ALPHA2 only if the nonlinear large displacement capability is turned off (PARAM,LGDISP,-1).

ALPHA1FL, ALPHA2FL

Default = (0.0,0.0)

Specifies scale factors for Rayleigh (proportional) damping in a frequency or transient response analysis when either PARAM,ALPHA1FL or PARAM,ALPHA2FL is not equal to complex zero. The software adds the Rayleigh damping to the other viscous damping contributions.

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Note

For solutions where the use of the DAMPING case control command and DAMPING bulk entry are supported and the FLUID describer is specified on the DAMPING case control command, if the input file contains both an explicit parameter specification such as PARAM,ALPHA1FL and a DAMPING bulk entry that is referenced by the DAMPING(FLUID) case control command, the value in the ALPHA1 field (or the default value for the ALPHA1 field if left blank) of the DAMPING bulk entry takes precedence.

Note

Although the specification for the ALPHA1FL and ALPHA2FL parameters is complex, you must specify the imaginary part as zero.

The scale factors are applied to the d-set or h-set matrices. The contribution to the viscous damping for a fluid is as follows:

$$\mathrm{ALPHA1FL} \cdot \left[M^1\right] + \mathrm{ALPHA2FL} \cdot \left[K^1\right]$$

where:

$\mathrm{ALPHA1FL}$ is the mass matrix scaling factor for Rayleigh damping (PARAM,ALPHA1FL or the ALPHA1 field on a DAMPING bulk entry).

$\mathrm{ALPHA2FL}$ is the stiffness matrix scaling factor for Rayleigh damping (PARAM,ALPHA2FL or the ALPHA2 field on a DAMPING bulk entry).

$\left[M^1\right]$ is the mass matrix plus M2GG direct input mass.

$\left[K^1\right]$ is the stiffness matrix plus K2GG direct input stiffness matrix

ALTRED — Default = NO

ALTRED=YES requests the alternate stiffness and load reduction technique for superelement analysis in SOLs 101 and 114. This technique is described in Static Condensation in SubDMAPs SEKR and SEMR2 and Data Recovery Operations in SubDMAP SEDISP in the *Simcenter Nastran User's Guide*.

AMPCZ — Default = 1.0e-6

Zero tolerance for pivot elements in the AUTOMPC process. You can optionally increase the tolerance if AUTOMPC has problems resolving redundant constraint equations.

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ASCOUP Default = YES

In coupled fluid-structure analysis, if PARAM,ASCOUP,YES is specified in or omitted from the input file, and the A2GG case control command is also omitted from the input file, coupling for the stiffness and mass is computed. If PARAM,ASCOUP,NO is specified and A2GG is omitted, the coupling is not computed.

When ASCOUP and A2GG are both specified, direct input acoustic/fluid-structure coupling matrices can be added to the computed coupling matrix or the computed coupling matrix can be ignored altogether in favor of direct input matrices. In either case, the coupling matrices can be scaled by:

- Specifying one or both of the CA1 and CA2 parameters.
- Specifying in-line scale factors using the A2GG case control command.
- Specifying both in-line scale factors using the A2GG case control command and one or both of the CA1 and CA2 parameters.

If PARAM,ASCOUP,YES and A2GG are both specified, the total acoustic/fluid-structure coupling matrix is:

$$\left[A_{jj}\right]=CA1\cdot\left[A^{x}_{jj}\right]+CA2\cdot\left[A^{2}_{jj}\right]$$

where [A^{x}_{jj}] is the computed acoustic/fluid-structure coupling matrix and [A^{2}_{jj}] is the direct input acoustic/fluid-structure coupling matrix specified using A2GG.

If PARAM,ASCOUP,NO and A2GG are both specified, the computed coupling matrix is ignored and the total acoustic/fluid coupling matrix is:

$$\left[A_{jj}\right]=CA2\cdot\left[A^{2}_{jj}\right]$$

where [A^{2}_{jj}] is the direct input acoustic/fluid-structure coupling matrix specified using A2GG.

For additional information, see Performing a Coupled Acoustic-Structural Analysis in the *Simcenter Nastran Acoustics Simcenter Nastran User's Guide*

ASING Default = 0

ASING specifies the action to take when singularities (null rows and columns) exist in the [K_{ll}] in statics). If ASING=-1, then a User Fatal Message will result.

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If ASING=0 (the default), singularities are removed by appropriate techniques depending on the type of solution being performed. The procedures used are described in Formulation of Dynamic Equations in SubDMAP GMA and Real Eigenvalue Analysis in SubDMAPs SEMR3 and MODERS in the *Simcenter Nastran User's Guide*.

AUNITS — Default = 1.0

AUNITS is used in SOL 144 to convert accelerations specified in units of gravity on the TRIM Bulk Data entry to units of distance per time squared.

AUTOADJ — Default = YES

In SOL 200, a value of YES will automatically select the direct or the adjoint method for sensitivity analysis based on the performance criteria. AUTOADJ=NO requests that the adjoint method not be selected for sensitivity analysis and the direct method is enforced. The default should be preferred in all cases. However, the presence of this parameter allows investigation of the alternative of using direct methods in place of adjoint methods in the sensitivity calculations.

AUTOMPC — Default = NO

AUTOMPC=YES specifies that the software automatically selects the m-set DOFs rather than use the m-set values as specified on MPC or RIGID element cards (RBE*, RBAR, RROD, etc.) in the bulk data definition except as noted below.

AUTOMPC is not supported in SOL 414 solutions.

This option relieves the need to carefully define rigid elements (or MPCs) so as to not have a conflict of the m-sets between elements. In addition, any redundant constraints will be eliminated.

This option is not available in some circumstances and will be automatically set to NO for a design optimization solution (SOL=200) with DVGRID data.

The AUTOMPC option is not recommended for use in models with RSSCON elements connected to CPENTA elements.

If an RBE3 element contains UM information on the m-set data, those DOFs will be used in the m-set. SPOINTS in MPC equations will always be made part of the m-set.

When using AUTOMPC, you can prevent the software from placing DOF in the m-set by including the DOF in the U5 set. Conversely, you can force the software to include DOF in the m-set by including the DOF in the U4 set. To place DOF in the U4 or U5 sets, use the USET/USET1 bulk entries.

AUTOSEEL — Default = NO

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Requests that elements connected totally to external (boundary) grids (b-set and c-set) of super elements be included in the super element rather than the residual structure. AUTOSEEL = NO (default) means such elements are included in the residual structure. AUTOSEEL = YES means they are included in the super element.

AUTOSPC

Default = YES (In all SOLs, except 106, 129, 153, and 159)

Default = NO (In SOLs 106, 129, 153, and 159)

Note

AUTOSPC is not supported in SOL 414 solutions. To detect zero pivots below a specified threshold, use the PRECPIVO parameter in the NLCNTLG bulk data entry.

AUTOSPC specifies the action to take when singularities exist in the stiffness matrix $[K_{gg}]$. AUTOSPC=YES means that singularities will be constrained automatically. AUTOSPC=NO means that singularities will not be constrained. If AUTOSPC=NO is set, then the user should take extra caution analyzing the results of the grid point singularity table and epsilon.

See Constraint and Mechanism Problem Identification in SubDMAP SEKR in the *Simcenter Nastran User's Guide* for details of singularity identification and constraint. Singularity ratios smaller than PARAM,EPPRT (default = 1.E-8) are listed as potentially singular. If PARAM,AUTOSPC has the value YES, identified singularities with a ratio smaller than PARAM,EPZERO (default = 1.E-8) will be automatically constrained with single-point constraints. If PARAM,EPPRT has the same value as PARAM,EPZERO (the default case), all singularities are listed. If PARAM,EPPRT is larger than PARAM,EPZERO, the printout of singularity ratios equal to exactly zero is suppressed. If PARAM,PRGPST is set to NO (default), the printout of singularities is suppressed, except when singularities are not going to be removed. If PARAM,SPCGEN is set to 1 (default = 0), the automatically generated SPCs are placed in SPCi Bulk Data entry format on the PUNCH file.

AUTOSPC provides the correct action for Superelements in all contexts. It is ignored for the residual structure in SOLs 106, 129, 153 and 159 when the solution is either material nonlinear or geometric nonlinear. PARAM, AUTOSPCR, not AUTOSPC, is used for the o-set (omitted set) in the residual structure in SOLs 106 and 153.

AUTOSPCR

Default = NO (SOLs 106 and 153 only)

AUTOSPCR specifies the action to take when singularities exist in the linear stiffness matrix of the residual structure after multipoint constraints are processed. AUTOSPCR=YES means that singularities will be constrained and AUTOSPCR=NO means they will not be constrained. It is recommended that all degrees-of-freedom attached to nonlinear elements be specified on ASETi entries. Parameters EPPRT, EPZERO, PRGPST, and SPCGEN may be used with AUTOSPCR.

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AUTOSPRT	Default = YES for SOLs 103, 110, 111, 112 Default = NO for SOLs 107, 108, 109 By default, models that are not fully constrained and involve residual vectors, constraint modes, or attachment modes will be automatically supported as long as F1 ≤ 0.0 on the EIGR bulk data entry or V1 ≤ 0.0 on the EIGRL bulk data entry (for SOLs 103 and 110, this only applies if residual vectors are requested). The auto-SUPORT capability may be deactivated by specifying PARAM,AUTOSPRT,NO or F1 / V1 > 0.0. The parameter FZERO is the maximum frequency (Hz) assumed for a rigid body mode. FZERO is used by the auto-SUPORT capability to extract rigid body frequencies. For direct solutions with SOLs 108 and 109, massless mechanisms will be sought and supported if there are no SUPORT entries defined and PARAM,AUTOSPRT,YES is specified. For a SOL 103, 107, 108, 109, 110, 111, or 112 run with a model that is not fully constrained, the model will be supported in a static subcase that computes differential stiffness if PARAM,AUTOSPRT,YES is on by default or is specified.

Related Topics

Browse Simcenter Nastran documentation

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B – Parameters

BAILOUT

Default = 0

In a normal modes analysis, Sturm sequence number techniques are used to check the agreement with the number of modes actually computed in the range. If a disagreement occurs, the computed modes are likely incorrect and the run is terminated with a fatal message by default. PARAM, BAILOUT, -1 may be used to allow the solution to proceed after the Sturm sequence check error.

See also **MAXRATIO**.

BDMNCON

Integer ≥ 1; Default = lesser of 10 or (DESMAX-3)

Defines the number of design cycles in which the software will delay the application of any manufacturing constraints in a SOL 200 topology optimization solution. This delay has been found to improve the results of the manufacturing constraints. The DESMAX parameter, which is defined on the DOPTPRM bulk entry, specifies the maximum number of design cycles, in general, for a SOL 200 solution. As a result, the BDMNCON parameter default is the smaller of 10 cycles, or the value of the DESMAX parameter minus 3, but the software default is never smaller than 2. BDMNCON=1 requests no delay. Manufacturing constraints are defined with the DMNCON bulk entry.

BETA

Default = (0.333333,0.0)

BETA specifies the transient integration control factor for the Newmark method in SOLs 129 and 159. The stability limits are $0.25 \leq \beta < 0.5$.

BIGER, BIGER1, BIGER2

Default = 0.0

See **S1**.

BLTGLSCL, BLTTNSCL

(Real>0.0; Default=1.0)

When you define a bolt preload with ETYPE=2 on the BOLT bulk entry for SOL 401, the software holds the grid pairs at the bolt cut in their relative deformed (preloaded) state using a stiff glue condition in the consecutive subcases. The default computed glue stiffness is three orders of magnitude larger than your material stiffness.

You can scale the axial bolt glue stiffness with the BLTGLSCL parameter and scale the tangential (two coordinates normal to the axis) bolt glue stiffness with the BLTTNSCL parameter. Both parameters default to 1.0, resulting in no scaling of the default stiffness. You decrease the stiffness by defining values less than 1.0, or increase the stiffness by defining values greater than 1.0.

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The default stiffness should work for most conditions. Although, if you experience conditioning and convergence issues, you can consider reducing the bolt glue stiffness.

BLTGLSCL - Scales the default axial stiffness at the bolt cut for the service load subcases.

BLTTNSCL - Scales the default tangential stiffness at the bolt cut for both preload and service load subcases.

BOLTFACT Real ≥ 0.0, default = 1.0E7

Factor used to reduce the bolt stiffness during the first phase of a bolt preload analysis. If BOLTFACT = 0.0, the axial stiffness of the bolts is zero during the first phase of a bolt preload analysis. Generally, BOLTFACT = 0.0 produces better results. However, a singular stiffness matrix will occur if eliminating the axial stiffness of the bolts cause the structure to be unstable.

BRSYMFAC Default = 0.0

BRSYMFAC controls the stiffness, damping, and inertia that the software uses for CBEAR elements.

- If BRSYMFAC = 0.0 (default), use the stiffness, damping, and inertia as defined on the PBEAR bulk entries.
- If BRSYMFAC = 1.0, use the symmetrized stiffness, damping, and inertia. The software calculates the symmetrized stiffness, damping, and inertia as follows:

$$K_{sym} = \frac{1}{2}\left(K + K^T\right) B_{sym} = \frac{1}{2}\left(B + B^T\right)$$

$$M_{sym} = \frac{1}{2}\left(M + M^T\right)$$

  where *K*, *B*, and *M* are the stiffness, damping, and mass, respectively, as defined on the PBEAR bulk entries.
- If 0.0 < BRSYMFAC < 1.0, the stiffness and damping that the software uses depends on the unsymmetry ratios, r_K, r_B, and r_M.
  - If $r_K \geq$ BRSYMFAC, use the stiffness as defined on the PBEAR bulk entries.

    If $r_B \geq$ BRSYMFAC, use the damping as defined on the PBEAR bulk entries.

    If $r_M \geq$ BRSYMFAC, use the inertia as defined on the PBEAR bulk entries.
  - If $r_K <$ BRSYMFAC, use the symmetrized stiffness.

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If r_B < BRSYMFAC, use the symmetrized damping.

If r_M < BRSYMFAC, use the symmetrized inertia.

The unsymmetry ratios are calculated as follows:

$$r_K = \frac{\max\left(|\Delta k_{ij}|\right)}{\max\left(|k_{mn}|\right)} \; r_B = \frac{\max\left(|\Delta b_{ij}|\right)}{\max\left(|b_{mn}|\right)}$$

$$r_M = \frac{\max\left(|\Delta m_{ij}|\right)}{\max\left(|m_{mn}|\right)}$$

where k_{ij}, b_{ij}, and m_{ij} are entries in K, B, and M, respectively, and Δk_{mn}, Δb_{mn}, and Δm_{mn} are entries in the following matrices:

$$\Delta K = K - K_{sym} \; \Delta B = B - B_{sym}$$

$$\Delta M = M - M_{sym}$$

For example, if:

$$K = \begin{bmatrix} 2 & -2 \\ -4 & 6 \end{bmatrix}$$

then:

$$K_{sym} = \begin{bmatrix} 2 & -1 \\ -1 & 6 \end{bmatrix}$$

and:

$$\Delta K = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$$

Thus:

$$r_K = \frac{3}{6} = 0.5$$

The parameter BRSYMFAC is not applicable to SOL 101 rotordynamic analysis, where the software always uses the symmetrized version of the stiffness, damping, and mass for CBEAR elements.

BSHDAMP Default = DIFF

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Determines if the PBUSH/PBUSHT bulk entry fields GE2-GE6/TGEID2-TGEID6 are used.

SAME: The fields GE2-GE6/TGEID2-TGEID6 are ignored, and only fields GE1/TGEID1 are considered.

DIFF: The fields GE2-GE6/TGEID2-TGEID6 are used. If any of the fields GE2-GE6 and TGEID2-TGEID6 are defined, then all PBUSH and PBUSHT bulk entries are assumed to have all GE1-GE6 and TGEID1-TGEID6 fields defined, and any blank fields are considered a zero value.

BSHFORCE — Default = NO

For the CELAS1, CELAS2, CDAMP1, and CDAMP2 elements, requests spring-damper force output in the basic coordinate system in SOLs 108 and 111.

NO: Spring-damper force output is in the global coordinate system.

YES: Spring-damper force output is in the basic coordinate system.

BUCKLE — Default = -1

BUCKLE=1 requests a nonlinear buckling analysis in a restart run of SOLs 106 or 153. See the *Simcenter Nastran Handbook for Nonlinear Analysis*.

BUCKLE=2 requests buckling in a SOL 106 cold start run.

Related Topics

Browse Simcenter Nastran documentation

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C – Parameters

CA1, CA2

Default =(1.0, 0.0)

CA1 and CA2 specify factors for scaling the total acoustic/fluid-structure coupling matrix. The total acoustic/fluid-structure coupling matrix is:

$$[A_{jj}] = CA1 \cdot [A^{x}_{jj}] + CA2 \cdot [A^{2}_{jj}]$$

where [A^{x}_{jj}] is the computed acoustic/fluid-structure coupling matrix and [A^{2}_{jj}] is the direct input acoustic/fluid-structure coupling matrix specified using the A2GG case control command. CA1 and CA2 are only effective if A2GG is specified in the case control section.

CB1, CB2

Default =(1.0, 0.0)

CB1 and CB2 specify factors for the total damping matrix. The total damping matrix is:

$$[B_{jj}] = CB1 \cdot [B^{x}_{jj}] + CB2 \cdot [B^{2}_{jj}]$$

where B^{2}_{jj} is selected using the B2GG case control command and B^{x}_{jj} comes from CDAMPi or CVISC element bulk entries. These parameters are effective only if B2GG is selected in the case control section.

CDIF

Default = YES for shape optimization with or without property optimization.

Default = NO for property optimization only.

CDIF may be used to override the default finite difference scheme used in the calculation of pseudoloads in SOL 200. PARAM,CDIF,YES forces the selection of the central difference scheme used in the semianalytic approach regardless of the type of optimization requested. PARAM,CDIF,NO forces the selection of the forward difference scheme.

CDITER

Default = 0

See **CDPRT**.

CDPCH

Default = NO

See **CDPRT**.

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CDPRT Default = YES

If CDITER>0, perform constrained displacement iterations in SOL 101. The value is the maximum number of iterations. If CDPRT=YES, print those negative displacements and tension forces which do not satisfy constraints. If CDPCH=YES, punch DMIG CDSHUT entries for final state; by default all gaps are closed. These can be used for initial conditions for a restart. Potential contact points must be specified on the SUPORTi entries. The SUPORTi points must be in the residual structure. Optional DMIG entries to define the initial shut vector may be specified. Degrees-of-freedom that are specified on the SUPORT entry and have a value of 1.0 defined on the DMIG,CDSHUT entry will be considered closed initially. If the DMIG,CDSHUT entry is not supplied, then all degrees-of-freedom specified on the SUPORT entries will be considered shut initially. A fatal message will be issued if this parameter is used and PARAM,INREL is specified.

CHECKOUT Default = NO (structured solutions only)

CHECKOUT=YES requests a model checkout in SOLs 101 through 200. See "Geometry Processing in SubDMAP PHASEO" in the *Simcenter Nastran User's Guide*. The run will terminate prior to phase 1 of superelement analysis. The PARAM,POST options are also available with PARAM,CHECKOUT,YES. The following options and their user parameters are also available with PARAM,CHECKOUT,YES,

1. PARAM,PRTGPL,YES

   Prints a list of external grid and scalar point numbers in internal sort. It also lists external grid and scalar point numbers along with the corresponding sequence numbers in internal sort. The sequence numbers are defined as (1000*external number) and will reflect any user-requested resequencing.

2. PARAM,PRTEQXIN,YES

   Prints a list of external and internal grid and scalar numbers in external sort. It also lists external grid and scalar numbers with the corresponding coded SIL number in external sort. The coded SIL numbers are defined as:

   $$10 \cdot \text{SIL} + \begin{cases} 1 \text{ for grid point} \\ 2 \text{ for scalar point} \end{cases}$$

   The SIL numbers correspond to degrees-of-freedom, i.e., one SIL number for scalar point and six SIL numbers for a grid point.

3. PARAM,PRTGPDT,YES

   Prints, for each grid and scalar point, the following information in the internal sort:

   - Coordinate system ID in which grid point geometry is defined (ID=-1 for scalar points).
   - Spatial location of grid points in the "CP" coordinate system. For scalar points, all entries are zero.

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- Coordinate system ID for grid point displacements, forces, and constraints (ID=0 for scalar points).
- Permanent single-point constraints defined on GRID Bulk Data entries. A zero is entered for scalar points.

4. PARAM,PRTCSTM,YES

Prints for each coordinate system type the transformation matrix from the global to the basic coordinate system, and the origin of the indicated coordinate system in the basic coordinate system. Coordinate system types are: 1 = rectangular; 2 = cylindrical; 3 = spherical.

5. PARAM,PRTBGPDT,YES

Prints all grid and scalar points listed in internal sort with their x, y, and z coordinates in the basic coordinate system. In addition, the coordinate system ID for grid point displacements, forces, and constraints is indicated for each grid point (ID=-1 for scalar points). The x, y, and z coordinates of scalar points are zero.

6. PARAM,PRTGPTT,YES

Prints, for each temperature load set, information on element and grid point temperatures.

7. PARAM,PRTMGG,YES

Prints the g-size mass matrix labeled by grid point/degree-of-freedom.

8. PARAM,PRTPG,YES

Prints the g-size load vectors labeled by grid point/degree-of-freedom.

9. The summation of forces and moments of applied loads in the basic coordinate system is automatically output for each loading condition requested in the Case Control Section. Related parameters are GPECT, PROUT, and EST.

CK1, CK2 Default =(1.0, 0.0)

CK1 and CK2 specify factors for the total stiffness matrix. The total stiffness matrix (exclusive of GENEL entries) is

$$[K^{x}_{jj}] = \text{CK1} \cdot [K^{z}_{jj}] + \text{CK2} \cdot [K^{2}_{jj}]$$

where $[K^{2}_{jj}]$ is selected via the Case Control command K2GG and $[K^{z}_{jj}]$ is generated from structural element (e.g., CBAR) entries in the Bulk Data. These are effective only if K2GG is selected in Case Control. A related parameter is CK3. Note that stresses and element forces are not factored by CK1, and must be adjusted manually.

CK3 Default = (1.0, 0.0)

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CK3 specifies a factor for the stiffness derived from GENEL Bulk Data entries. The total stiffness matrix is

$$[K_{jj}] = [K^{x}_{jj}] + \text{CK3} \cdot [K^{y}_{jj}]$$

where $[K^{y}_{jj}]$ comes from the GENEL Bulk Data entries and $[K^{x}_{jj}]$ is derived using PARAMs CK1 and CK2. CK3 is effective only if GENEL entries are defined. Related parameters include CK1 and CK2.

CK41, CK42 Default =(1.0, 0.0)

CK41 and CK42 specify factors for the total structural damping matrix. The total structural damping matrix is

$$\left[K^{4,x}_{jj}\right] = \text{CK41} \cdot \left[K^{4,z}_{jj}\right] + \text{CK42} \cdot \left[K^{4,2}_{jj}\right]$$

where $[K^{4,2}_{jj}]$ is selected via the Case Control command K42GG and $[K^{4,z}_{jj}]$ is generated from the stiffness of structural element (e.g., CBAR) entries in the Bulk Data times the structural damping coefficient GE on material entries (e.g. MAT1). These are effective only if K42GG is selected in Case Control. Note that stresses and element forces are not factored by CK41, and must be adjusted manually.

CLOSE Default = 1.0

See **SCRSPEC**.

CM1, CM2 Default = (1.0, 0.0)

CM1 and CM2 specify factors for the total mass matrix. The total mass matrix is

$$[M_{jj}] = \text{CM1} \cdot \left[M^{x}_{jj}\right] + \text{CM2} \cdot \left[M^{2}_{jj}\right]$$

where $[M^{2}_{jj}]$ is selected via the Case Control command M2GG and $[M^{x}_{jj}]$ is derived from the mass element entries in the Bulk Data Section. These are effective only if M2GG is selected in the Case Control Section.

CNSTRT No longer required.

CNTASET Default = NO

When contact conditions are defined in SOL 101, CNTASET determines if a static condensation (A-set reduction) occurs with the contact degrees-of-freedom.

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When CNTASET=YES, a static condensation is performed to reduce the full K_{gg} matrix to the K_{aa} matrix containing only the contact degrees-of-freedom. The contact iterations are then performed using this reduced matrix.

When CNTASET=NO, the static condensation of contact degrees-of-freedom does not occur.

CNTSCL

Default=100.0

Scales the contact stiffness for a normal modes (SOL 103), linear buckling (SOL 105), direct frequency response (SOL 108), modal frequency response (SOL 111), or modal transient response (SOL 112) solution. The CNTSCL parameter must be defined in the bulk data section of the input file.

The inputs for these solutions are consistent with differential stiffness solutions that require a linear statics subcase. The linear statics subcase should include the BCSET case control command to define a contact set. When defining the normal modes, linear buckling, or dynamics subcase, a STATSUB case control command must be included to reference the subcase ID containing the contact definition. The contact solution in the linear statics subcase must fully converge before the software begins the next subcase solution.

See Contact conditions in modal, buckling, and dynamic solutions in the Simcenter Nastran User's Guide.

COLEMAN

Default = 0

Activates the Coleman transformation for SOL 414,110 Complex Modal Analysis.

If COLEMAN = 0, the results of a cyclic symmetric structure are computed in a rotating reference frame for the specified harmonics. To solve cyclic symmetry (CYCSET) in SOL 414,110 Complex Modal Analysis solutions when the Coleman transformation is not defined, you must define the METHOD case control command and EIGRL bulk entry in addition to the CMETHOD case control command and EIGC bulk entry. Otherwise, the software writes an error message.

If COLEMAN = 1, the results of a cyclic symmetric structure are computed in a rotating reference frame and converted to results for a fixed reference frame using the Coleman transformation for all harmonics up to the limit set by the **HARMLIM** parameter.

For more information, see Cyclic symmetry with the Coleman transformation in the *Rotor Dynamics User's Guide (SOL 414)*.

COLPHEXA

Default=NO

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Specify YES to allow collapsed CHEXA elements for crack propagation analysis in SOL 401. If PARAM,COLPHEXA,YES is not specified, collapsed CHEXA elements are disallowed by the input file checks. Note that the collapsed CHEXA is not supported in a glue or contact region.

COMPMATT Default=NO

During SOL 106 solutions, composite materials compute temperature-dependent properties for the plies only at the reference temperature given on the PCOMP or PCOMPG bulk entries. The resulting ply properties are smeared and used for all load steps, regardless of whether the temperature is changing through application of thermal loads. If the parameter equals YES, the temperature-dependent properties for the plies are updated and smeared at the current temperature for each load step. This parameter only applies to SOL 106, and CQUAD4/CTRIA3 elements only.

CONFAC Default = 1.E-5

In superelement analysis, CONFAC specifies the tolerance factor used in checking the congruence of the location and displacement coordinate systems of the boundary points between image superelements and their primaries (see the Bulk Data entry, **CSUPER**). Specification of this parameter is recommended instead of DIAG 37 (DIAG 37 ignores User Fatal Messages 4277 and 4278).

CORROPT Default = NO

This parameter is valid only for SOL 103 when used with the Simcenter 3D FE Model Update product. The documentation for this product is included in the Help.

NO: Output for the Simcenter 3D FE Model Update product is deactivated.

YES: Output for the Simcenter 3D FE Model Update product is activated.

COUPMASS Default = -1

COUPMASS>0 requests the generation of coupled rather than lumped mass matrices for elements with coupled mass capability, as listed in “Element Summary – Small Strain Elements” in the *Simcenter Nastran Element Library*. This option applies to both structural and nonstructural mass for the following elements: CBAR, CBEAM, CONROD, CHEXA, CPENTA, CPYRAM, CQUAD4, CQUAD8, CQUADR, CQUADX4, CQUADX8, CRAC2D, CRAC3D, CROD, CTETRA, CTRAX3, CTRAX6, CTRIA3, CTRIA6, CTRIAR, CTUBE. A negative value (the default) causes the generation of lumped mass matrices (translational components only) for all of the above elements.

CBEND elements are always generated with coupled mass and are not affected by COUPMASS.

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CP1, CP2 Default = (1.0, 0.0)

The load vectors are generated from the equation

$$\{P_j\} = \mathrm{CP1} \cdot \left\{P_j^x\right\} + \mathrm{CP2} \cdot \left\{P_j^2\right\}$$

where

$\{P^2_j\}$ is selected via the Case Control command P2G, and $\{P^x_j\}$ comes from Bulk Data static load entries.

CURV Default = -1

PARAM,CURV,1 requests that the CTRIA3 and CQUAD4 element stress and/or strain output be computed in a material coordinate system (normal output is in the element or basic coordinate system) and/or to interpolate it to grid points. (CQUAD4 element corner stress output is not supported.)

The integer parameter OG controls the calculation of stress and/or strain data at grid points. If OG is set to -1, the calculation for stresses and/or strain data at grid points is not performed. The default value of zero provides the calculation of these quantities at those grid points to which the selected elements connect.

User parameters S1G, S1M, S1AG, and S1AM, set to 1, request the printout of stresses at grid points, stresses in the material coordinate system, strains at grid points and strains in the material coordinate system, respectively.

The integer parameter OUTOPT may be set in accordance with the below options to select print, punch, and/or plotter output for stress and/or strain data that are computed in user-defined material coordinate systems.

OUTOPT Value	Description
0	Default-standard Simcenter Nastran device codes are used.
1	Print only
2	Plot only
4	Punch only

The above values may be combined additively to select two or more forms of output. For example, OUTOPT=6 requests both plot and punch output. Related parameters include BIGER, CURVPLOT, DOPT, NUMOUT, NINTPTS, S1G, S1M.

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For stress and/or strain/curvature output in a user-defined material coordinate system MCSID must be defined on MAT1 and MAT2 Bulk Data entries. The values of MCSID reference CORDiR, CORDiC, and CORDiS Bulk Data entries. A value of zero for MCSID does not imply the basic coordinate and will eliminate all elements which reference the MATi from the subject calculations.

1. If these data are requested at the element centers, the program will compute the unit vector $\vec{i}_w$ along the T1 or x-axis of the material coordinate system, and compare $|\vec{n} \cdot \vec{i}_m|$ for each element that references the material coordinate system, where n is the normal to the surface of the element. If $|\vec{n} \cdot \vec{i}_m|^2 \geq 4$ the projection of the y-axis on the surface of the element is taken as the reference axis. Otherwise, the projection of the x-axis on the surface of the element is taken as the reference axis. The angle between the x-axis of the element coordinate system and the projection of the selected reference axis of the material coordinate system is used to transform the stress and/or strain data into the material coordinate system at the element centers.
2. If, on the other hand, the user requests these data at the grid points to which the elements connect, the program will interpolate the results from (a) to the grid points to which the elements connect. The parameter NlNTPTS = N, the stress and/or strain data at the N closest element centers to the grid point in question will be used in the interpolation. The program may include more that N points in the interpolation if the distance of other element centers is not more than 10% greater than the closest N element centers.

The following specifies the output headings for stresses and/or strains in the material coordinate system.

Element stresses (PARAM,S1M,1) are:

1. Available in CQUAD4 and CTRlA3 elements.
2. In page headings:

   STRESSES IN QUADRILATERAL ELEMENTS (CQUAD4)

   STRESSES IN TRIANGULAR ELEMENTS (CTRlA3)
3. Under the column FlBER DISTANCE:

   Z1 is replaced by MCSID.

   Z2 is replaced by 1.0 if the x-axis of the material coordinate system is selected as the reference axis, and by 2.0 if the y-axis of the material coordinate system is selected as the reference axis.

Grid point stresses (PARAM,S1G,1 and PARAM,OG,1)

1. Available for CQUAD4 and CTRIA3 elements.
2. In page heading:

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STRESSES AT GRID POINTS

3. Under the $\left\{\begin{matrix}\text{MAT1 – COORD1 – ID}\\ \text{PROJ-CODE}\end{matrix}\right\}$ column:

   Z1 is replaced by MCSID:

   Z2 = A+10*N where A is 1.0, 2.0, or 3.0, depending on whether the x-, y-, or z-axis of the material coordinate system is most nearly normal to the projected plane of the field of elements involved in the calculation.

Element strains (PARAM,S1AM,1) are:

1. Available for CQUAD4 and CTRIA3 elements.
2. In page headings:

   STRAINS IN QUADRILATERAL ELEMENTS (CQUAD4)

   STRAINS IN TRIANGULAR ELEMENTS (CTRIA3)
3. Under the column FIBER DISTANCE:

   Z1 is replaced by MCSID.

   Z2 is replaced by 1.0 if the x-axis of the material coordinate system is selected as the reference axis, and by 2.0 if the y-axis of the material coordinate system is selected as the reference axis.

Grid point strains (PARAM,S1AG,1 and PARAM,OG,1):

1. Available for CQUAD4 and CTRIA3 elements.
2. In page heading:

   STRAINS AND CURVATURES AT GRID POINTS
3. Under the $\left\{\begin{matrix}\text{MAT1 – COORD1 – ID}\\ \text{PROJ-CODE}\end{matrix}\right\}$ column:

   Z1 is replaced by MCSID.

   Z2 = A+10*N where A is 1.0, 2.0, or 3.0, depending on whether the x-, y-, or z-axis of the material coordinate system is most nearly normal to the projected plane of the field of elements involved in the calculation.

CURVPLOT Default=-1

PARAM,CURVPLOT,1 requests x-y (or curve) plots whose abscissas are a sequence of grid points and whose ordinates may be displacements, loads, SPC forces, or grid point stresses. To obtain stress plots, set the CURV parameter to +1. The default for DOPT is the length between grid points

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Specify the XYOUTPUT Case Control command in the usual manner, replacing the point ID with the SID of SET1 Bulk Data entries.

The SET1 Bulk Data entries must contain unique SIDs for each set of grid points to be plotted.

User requests for xy-plots of output quantities appear in the Case Control Section in the standard form. For example,

```
.
.
.
OUTPUT(XYOUT)
.
.
XYPLOT DISP 1/4(T3)
.
.
XYPLOT SPCF 2/5(T1)
.
.
BEGIN BULK
```

The first XYPLOT command will produce an xy-plot from the displacement output of subcase 1. The abscissa of the curve will reflect the grid point IDs listed on the SET1 entry with a SID of 4, and the ordinate will reflect the T3 component of displacement at these grid points. The second XYPLOT command will produce an xy-plot whose ordinates are the T1 components of the forces of constraint in subcase 2 at the grid points listed on the SET1 entry with a SID of 5.

The user has some degree of control over the scaling of the abscissas on these xy-plots. This control is exercised through the parameter DOPT on a PARAM Bulk Data entry. The legal values of this parameter provide the following scaling options for the abscissas.

Value of DOPT	Scaling for Abscissa
0 (default)	$\|\|g_j - g_i\|\|$
1	$\|x_j - x_i\|$
2	$\|y_j - y_i\|$
3	$\|z_j - z_i\|$
4	1.

Thus, the default value of DOPT will place the first grid point listed on the referenced SET1 card at the origin, and subsequent grid points will be located along

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the abscissa at intervals proportional to the distance between that grid point and its predecessor. Values of DOPT equal to 1, 2, or 3 will scale the abscissa so that the interval between adjacent grid points is proportional to the difference in the X, the Y, and the Z components of the subject grid points respectively. DOPT = 4 will space the grid points equally along the abscissa.

Related Topics

Browse Simcenter Nastran documentation

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D – Parameters

DBALL — Default = DBAL

By default, all data to be stored on the database for restart purposes will be located on the DBALL database set DBset. These parameters permit the storage of some data blocks on DBsets other than DBALL, which are defined by the user and specified on the INIT File Management statement. Any or all of these parameters may be set to SCRATCH in order to reduce overall disk space usage; e.g., PARAM,DBUP,SCRATCH or PARAM,DBALL,SCRATCH. However, automatic restarts will be less efficient because data normally assigned to a permanent DBset will have to be recomputed.

A unique value for each superelement may be specified in the Case Control Section for the parameters DBALL, DBDN, DBRCV, and DBUP. Certain DBsets may be taken offline depending on which phase of superelement analysis is being performed (see “Summary of Solution Sequence Operations” in the *Simcenter Nastran User’s Guide*). PARAM,DBALL specifies the default value for parameters DBDN, DBUP, and DBRCV.

The DBDN DBset contains data blocks necessary for “downstream” processing. For example, the stiffness, mass, damping, and static loads matrices that have been reduced to the boundary of the superelement are stored in this DBset.

The DBRCV DBset contains data blocks that must be online during the first pass through data recovery (Phase 3). These data blocks are used to recover the total displacement vector u_g of the superelement. This operation is performed by the SSG3 and SDR1 modules. On subsequent data recovery restarts, this DBset may be taken offline. Its default is determined from the value of DBUP.

The DBUP DBset contains data blocks necessary for “upstream” processing. For example, the geometry and property tables along with the stiffness, mass, damping, and static loads matrices related to the interior grid points of the superelement are stored in this DBset. These matrices and tables must be online during the reduction (Phase 1) and data recovery (Phase 3) of the superelement.

The IFP DBset contains data blocks that are required for all phases of the analysis. These data blocks are related to the entire model; examples are Bulk Data, superelement map, IFP module outputs, and resequenced grid points. This DBset must be online for all runs.

DBCCONV — Default = XL

See **PARAM, POST, 0**.

DBCDIAG — Default = 0

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See **PARAM, POST, 0**.

DBCOVWRT — Default = YES

See **PARAM, POST**.

DBDICT — Default = -1

Controls the printout of the database directory at the beginning and end of the run. See the **DBDICT** file management statement description. If DBDICT = 0, then the database directory will be printed at the start of the run. If DBDICT=1, then the directory will be printed at the end of the run. If DBDICT ≥ 2, then it will be printed at the beginning and end of the run. If DBDICT = -1 (the default), the directory is not printed at the beginning or end of the run.

If multiple versions and/or projects exist on the database, then the parameters DBDRPRJ and DBDRVER allow the user to select the desired project and version, respectively. The appropriate values may be found in the Project/Version Table that is printed upon restart. If DBDRVER = 0 (or DBDRPRJ=0), then the current version (or project) is selected. If DBDRPRJ=-1 (or DBDRVER = -1), then all projects (or versions) are selected.

DBDN — Default = value of PARAM,DBALL.

See **DBALL**.

DBDRPRJ — Default = 0

Specifies the desired project-identification number. See **DBDICT**.

DBDRVER — Default = 0

Specifies the desired version-identification number. See **DBDICT**.

DBRCV — Default=value of PARAM,DBUP.

See **DBALL**.

DBUP — Default=value of PARAM,DBALL.

See **DBALL**.

DDRMM — Default = 0

By default, the matrix method of data recovery is used in the modal transient and frequency response solutions. DDRMM=-1 will force calculation of complete g-set

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solution vectors by the mode displacement method. See “Dynamic Data Recovery in Modal Frequency Response Analysis” in the *Basic Dynamic Analysis User's Guide*.

DESPCH Default = 0

DESPCH specifies in SOL 200 when the optimized (updated) Bulk Data entries are written to the PUNCH file. Currently, all the property entries, material entries, and connectivity entries that can be designed and DESVAR, DRESP1, and GRID entries can be written.

DESPCH < 0 never.

DESPCH = 0 at the last design cycle only. (Default)

DESPCH > 0 at every design cycle that is a multiple of DESPCH and the last design cycle. For example, if n=2 and the maximum number of design cycles is 5 (DESMAX=5 on the DOPTPRM entry), then, DESVAR, DRESP1 (when modified due to mode tracking: see DESPCH1 below), and GRID entries at design cycle 2, 4, and 5 are written in the punch file.

DESPCH1 Default=6

DESPCH1 specifies in SOL 200 the amount of data to be written to the.pch file. A positive DESPCH1 value requests large field formats while a negative value requests small field formats. Any updated EIGN or FREQ type DRESP1 written to the punch file due to mode tracking will be in small field format regardless of the sign of DESPCH1. No DRESP1 data will be written with the DESPCH1 request unless mode tracking has resulted in a change in such data. Note that the ECHO command provides more options for bulk data output into the punch file. For a shape optimization job, if DESPCH1<>0, the updated GRID entries of the whole model will be written in the.pch file.

Descriptions of various DESPCH1 values are given below:

DESPCH1 = 0, writes no data.

DESPCH1 = ± 1, writes the property entries that are designed.

DESPCH1 = ± 2, writes all the property entries of a given type when one property of that type is designed.

DESPCH1 = ± 4, writes DESVAR and DRESP1 entries.

DESPCH1 = ± n, writes combine quantities by summing the DESPCH1 values. For example, n=1+4=5 requests writing all the designed property entries, DESVAR andDRESP1 entries to the.pch file.

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DFREQ Default = 10^{-5}

Specifies the threshold for the elimination of duplicate frequencies (or rotational speeds) on all FREQi (or OMEGAi) bulk entries. When a duplicate is found, the software keeps the lower frequency (or rotational speed) in the pair and removes the higher frequency (or rotational speed). Two frequencies (or rotational speeds), f_1 and f_2, are considered duplicates if:

$$|f_1 - f_2| < \text{DFREQ} \cdot |f_{MAX} - f_{MIN}|$$

where f_{MAX} and f_{MIN} are the maximum and minimum excitation frequencies (or rotational speeds) of the combined FREQi (or OMEGAi) entries.

DIGITS Default = 15

Controls the number of digits of precision for matrix data written to an OP4 file for the EXTSEOUT and MBDEXPORT case control commands. DIGITS refers to the number of digits after the decimal point. Thus, for the default value of 15, there are actually 16 digits of precision because of the one digit that precedes the decimal point.

DOF123 Default = 0

Valid only when PARAM,CORROPT,YES exists.

0: Nodes in Display Set will have all DOFs (123456) available in back expansion matrix (GOA).

1: Nodes in Display Set will have translational DOFs (123) available in back expansion matrix (GOA). If SPOINTs are present in the Display Set, DOF123 is forced to 0 so that all DOFs are made available.

DOPT Default = 0

See **CURVPLOT**.

DPEPS Default = 1.0E-4

In SOL 200, if any difference between the property value on the property entries and the value calculated from the design variable values on the DESVAR entry (through DVPRELi relations) is greater than DPEPS, the design model values override the analysis values. If all the differences are less than DPEPS, analysis results from a previous run are accepted in a subsequent sensitivity/optimization task, thereby avoiding a reanalysis. The PTOL parameter on the DOPTPRM entry is a related parameter that checks the maximum difference.

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DPREONLY Default = NO

DPREONLY is used to signal the SOL 187 analysis to stop execution before the Fortran NAVSHOK program is started. This option is used when you want to run the Fortran program outside the Nastran job.

If PARAM,DPREONLY,YES is specified, the DDAM solution will stop after the OUTPUT4 files are written, but just before the Fortran program is executed.

If PARAM,DPREONLY,NO is specified, the DDAM solution will start the Fortran NAVSHOK program once the OUTPUT4 files have been created.

DSNOKD Default = 0.0

DSNOKD specifies a scale factor to the differential stiffness matrix in buckling design sensitivity analysis. If DSNOKD>0.0, the effect of the differential stiffness matrix is included in buckling the design sensitivity analysis.

If PARAM,DSNOKD > 0 is specified in SOL 105, under the original Design Sensitivity Analysis (DSA), the differential stiffness sensitivity calculation is performed under the assumption that all the displacements are enforced; i.e., the change in the stiffness matrix due to the changes in the displacements are not computed. Therefore, PARAM,DSNOKD,0.0 is recommended in SOL 105. If PARAM,DSNOKD,1.0 is specified in SOL 200, the differential stiffness sensitivity calculation is performed more accurately; i.e., the change in the stiffness matrix due to the changes in the displacements are computed. However, the calculation is more expensive than with PARAM,DSNOKD,0.0.

Non-zero values of PARAM,DSNOKD cannot be used in SOL 200 with multiple buckling design subcases unless each subcase contains the same STATSUB command.

DSZERO Default = 0.0

DSZERO specifies the minimum absolute value for the printout of design sensitivities.

DUCTFMAX Real ≥ 1.0; Default=1.2

In a FEMAO SOL108 analysis with duct modes, the product (DUCTFMAX * maximum excitation frequency) is the maximum duct mode frequency.

DYNSPCF Default = NEW (structured solutions only)

PARAM,DYNSPCF,NEW requests that mass and damping coupled to ground be included in the SPCForce calculations for the linear dynamic solutions: SOLs 103,

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107 through 112, 115, 118, 145, 146, and 200. OLD neglects these effects and gives the same SPCForce results obtained in versions prior to Simcenter Nastran 1.0.

Related Topics

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E – Parameters

ELITASPC — Default = NO

Set to YES to perform the autospc in the element iterative solver. Normally the element iterative solver does not perform an autospc function as it is usually not necessary. For solid elements, the rotational dofs are eliminated directly. If K6ROT is specified for linear shell elements, there is no issue either. But for CQUAD8 and CTRIA6 elements and possibly other special cases, the autospc function is required. The drawback of this option is that it requires the assembly of the KGG matrix which is used in the autospc and this can have a significant impact on performance. This parameter will also generate the rigid body mass properties.

If MPCFORCES output is requested, this parameter will automatically be set to YES since a partition of the assembled stiffness matrix is needed for the calculation of the mpc forces.

EPPRT — Default = 1.E-8

Specifies the maximum value of singularities that will be printed. See **AUTOSPC**.

EPZERO — Default = 1.E-8

Specifies the minimum value that indicates a singularity. See **AUTOSPC**.

ERROR — Default = -1

If ERROR is set to 0, errors in superelement generation or assembly (e.g., a singular matrix, a mechanism, a nonpositive definite matrix, no stiffness, or all upstream superelements not available) will not terminate the run. The remaining computations for that superelement will be skipped, and additional superelements will be attempted. The default value causes a User Fatal Message to terminate the run. A related parameter is MAXRATIO.

EST — Default = 2

EST = +1 requests that the measure (length, area or volume) and volume be computed and printed for each element in the model, or in the case of superelement analysis, for the SEMG-selected superelements. The default value of 2 causes this data to be computed but not printed, making it available for element strain energy density calculations. If EST is set to -1, the calculation is skipped, and the density data will not be calculated.

EXTBEMI — Default = 0

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Specify PARAM,EXTBEMI,1 to accept and convert information from boundary element method software.

Specify PARAM,EXTBEMI,0 (default) to not accept and convert information from boundary element method software.

EXTBEMO Default = 0

Specify PARAM,EXTBEMO,1 to output information to boundary element method software.

Specify PARAM,EXTBEMO,0 (default) to not output information to boundary element method software.

EXTDR Default = NO. See **EXTOUT**

EXTDROUT Default = NO. See **EXTOUT**

EXTDRUNT Default = 31. See **EXTOUT**

EXTOUT Default = NO

When inputting the matrices for a reduced external superelement (SEBULK, CSUPER), there are four options that can be selected using the parameter EXTOUT. EXTOUT must be placed in the Case Control Section above any subcase or in the main Bulk Data Section. The options for Step 1 (see the table at the end of this discussion) are as follows:

If EXTOUT is set to MATRIXDB, the reduced structural matrices and loading are stored on the database.

If EXTOUT is set to DMIGDB, the reduced structural matrices and loading are stored on the database in a format which allows automatic connection to the analysis model if the identification numbers of the reduction grid points and scalar points are the same as the grid points and scalar points used in the analysis model.

If EXTOUT is set to DMIGOP2, the reduced structural matrices and loading are written in OUTPUT2 format to a tape unit specified by the parameter EXTUNIT (default=30). The storage format is the same as the DMIGDB option and allows automatic connection to the analysis model if the identification numbers of the reduction grid points and scalar points are the same as the grid points and scalar points used in the analysis model. The output unit can be assigned to a specific file by using an ASSIGN command in the File Management Section.

If EXTOUT is set to DMIGPCH, the reduced structural matrices and loading are output on the punch file (.pch) in DMIG format.

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The procedure for accessing the external superelement information depends on the option used to output the external superelement in Step 1. The methods are as follows:

1. If EXTOUT was MATRIXDB or DMIGDB in Step 1, use the following commands in the File Management Section:

```
ASSIGN SEXXX='step1.MASTER'
DBLOCATE DATABLK=(EXTDB)CONVERT(SEID=xxx),
LOGICAL=SEXXX
```

   where

   step1.MASTER is the database from Step 1.

   and

   xxx is the superelement identification number given to the partitioned Bulk Data Section for the external superelement.

2. If EXTOUT was DMIGOP2 in Step 1, then use the following commands in the File Management Section:

```
ASSIGN INPUTT2='step1_output2_file',UNIT=extunit
```

   where

   step1_output2_file is the OUTPUT2 file from Step 1.

   and

   extunit is the unit number specified by the parameter EXTUNIT (default = 30).

3. If EXTOUT was DMIGPCH in Step 1, then include the punch file from Step 1 in the partitioned Bulk Data Section. In addition, add the following Case Control commands in the subcase for the external superelement:

```
K2GG=KAAX
P2G=PAX
```

   The SEBULK entry defining the superelement as an external superelement and the EXTRN entry in the partitioned Bulk Data Section should not be specified.

If data recovery is desired for the external component in SOLs 101, 103, and 107 through 112, there are three methods of transmitting the displacements of the reduced model to the external full model. The method is selected by the parameter EXTDROUT in the partitioned Bulk Data Section. The options are as follows:

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1. EXTDROUT set to MATRIXDB. The displacements of the reduced component model are stored directly on the database. The sequencing of the displacement degrees-of-freedom corresponds to the sequencing in the reduced model.
2. EXTDROUT set to DMIGDB. The displacements of the reduced model are stored on the database in a format which allows automatic connection to the reduced component model if the reduction grid points and scalar points are the same grid points and scalar points used in the analysis model. This option can only be used if EXTOUT was set to DMIGDB or DMIGOP2.
3. EXTDROUT set to DMIGOP2. The same as EXTDROUT set to DMIGDB except that the displacements of the reduced model are written in OUTPUT2 format to a tape unit specified by parameter EXTDRUNT (default = 31). The output unit can be assigned to a specific file by using an ASSIGN command in the File Management Section. This option can only be used if EXTOUT was set to DMIGDB or DMIGOP2.

Data recovery for the external component is limited to SOLs 101 and 103 and 107 through 112. Data recovery is accomplished using a restart procedure from the data base created in Step 1 and setting parameter EXTDR to YES. The method on inputting the reduced displacements into the component model depends on the method used to output the external component in Step 2. The input methods are as follows:

1. If EXTDROUT was MATRIXDB or DMIGDB in Step 2, then add the following commands in the File Management Section:

```
ASSIGN SEXX='step1.MASTER'
RESTART LOGICAL=SEXX
ASSIGN SEYYY='step2.MASTER'
DBLOCATE DATABLK=(EXTDB) WHERE(SEID=yyy),
LOGICAL=SEYYY
```

   where:

   step1.MASTER is the database from the Step 1.

   step2.MASTER is the database from the Step 2.

   and

   yyy is the superelement identification number given to the partitioned Bulk Data Section for the external superelement in Step 2.

2. If EXTDROUT was DMIGOP2 in Step 2, then add the following commands in the File Management Section:

```
ASSIGN INPUTT2='step2_output2_file',UNIT=extdrunt
```


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where

step2_output2_file is the OUTPUT2 file from Step 2.

and

extdrunt is the unit number specified by the parameter EXTDRUNT (default = 31).

For SOL 101, the Case Control structure must match the system model subcase structure in the numbers of loading conditions. The loading used in Step 1 to generate the loads transmitted to the analysis model must also be specified in this step. If the analysis model had more loading conditions than the component model, then the loadings defined in Step 1 must be specified first.

For SOL 103 and 107 through 112, the Case Control structure must match the analysis model subcase structure in the number of eigenvalue extractions, FREQ/DLOAD or TSTEP/DLOAD subcases.

Step 1 **Create External SE**	**Step 2** **Perform Analysis**	**Step 3** **Data Recovery for External SE**
PARAM, EXTOUT, MATRIXDB	**1.** ASSIGN SEXX='step1.MASTER' DBLOCATE DATABLK=(EXTDB), CONVERT (SEID=xxx), LOGICAL=SEXX **2.** PARAM,EXDROUT,MATRIXDB	**1.** ASSIGN SEXX='step1.MASTER' RESTART LOGICAL=SEXX ASSIGN SEYYY='step2.MASTER' DBLOCATE DATABLK=(EXTDB), WHERE(SEID=YYY), LOGICAL=SEYYY **2.** PARAM,EXTDR,YES
PARAM, EXTOUT, DMIGDB	**1.** ASSIGN SEXX='step1.MASTER' DBLOCATE DATABLK=(EXTDB), CONVERT (SEID=xxx), LOGICAL=SEXX **2.** PARAM,EXDROUT,MATRIXDB or PARAM,EXDROUT,DMIGDB or PARAM,EXDROUT,DMIGOP2	**1.** ASSIGN SEXX='step1.MASTER' RESTART LOGICAL=SEXX ASSIGN SEYYY='step2.MASTER' DBLOCATE DATABLK=(EXTDB), WHERE(SEID=YYY), LOGICAL=SEYYY **2.** ASSIGN INPUTT2='step2_output_file', UNIT=Extdrunt **3.** PARAM,EXTDR,YES
PARAM, EXTOUT, DMIGOP2	**1.** ASSIGN INPUTT2='step1_output2_file', Unit = extunit **2.** PARAM,EXDROUT,MATRIXDB or PARAM,EXDROUT,DMIGDB or PARAM,EXDROUT,DMIGOP2	**1.** ASSIGN SEXX='step1.MASTER' RESTART LOGICAL=SEXX ASSIGN SEYYY='step2.MASTER' DBLOCATE DATABLK=(EXTDB), WHERE(SEID=YYY), LOGICAL=SEYYY **2.** ASSIGN INPUTT2='step2_output_file',

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Step 1 Create External SE	Step 2 Perform Analysis	Step 3 Data Recovery for External SE
		UNIT=Extdrunt 3. PARAM,EXTDR,YES
	Include the.PCH file in partitioned Bulk Data Section and for external SE subcase 1. K2GG=KAAX P2G=PAX	

EXTRCV Default = 0

EXTRCV > 0 indicates that data recovery is to be performed on an external superelement. In this type of run, the database for the external superelement must be attached as the primary database (See “Database Concepts” in the *Simcenter Nastran User’s Guide*), and the database that contains the solution vectors, superelement map, and external/internal grid point equivalence table for its downstream superelement must be attached via the DBLOCATE statements. The value of EXTRCV must also be specified in the CONVERT clause of the DBLOCATE statement for the EMAP data block.

The following example shows the DBLOCATE statements for external superelement data recovery in SOL 101.

```
ASSIGN DOWNSE=...  $ DOWNSTREAM SUPERELEMENT DATABASE
DBLOC DB=EMAP CONVERT (EXTRCV=500) LOGICAL=DOWNSE
DBLOC DB=UL WHERE (SEID=0) LOGICAL=DOWNSE
DBLOC DB=UG WHERE (SEID=0) LOGICAL=DOWNSE
DBLOC DB=BGPDTS WHERE (PEID=0) LOGICAL=DOWNSE
SOL 101
CEND
SEDR=...  $ EXTERNAL SUPERELEMENT ID
.
.
.
BEGIN BULK
PARAM,EXTRCV,500
```

EXTUNIT Default = 30.

Related Topics

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F – Parameters

F56 Default = NO

Directs the software to write an output file that contains warning messages, error messages, and convergence information only.

When F56 = YES, an .f56 file is created that contains all of the warning messages, error messages, and convergence information associated with a job.

When F56 = NO, an .f56 file is not created.

Optionally use system cell 552 to change the FORTRAN unit number from the default value of 56 to some other value.

FACTOR Default = 10000

See **OLDSEQ**.

FCTCi, FCTMi, FCTZi Default = 1.0

By default, coupling terms are included in the equations of motion for a rotor dynamic analysis. Use FCTCi, FCTMi, and FCTZi to selectively exclude coupling terms from the equations of motion.

FCTC1 = 0.0 excludes sin(ωt), cos(ωt), sin(2ωt), and cos(2ωt) terms from the Coriolis matrix components.
FCTC2 = 0.0 excludes sin(ωt) terms from the Coriolis matrix components.
FCTC3 = 0.0 excludes cos(ωt) terms from the Coriolis matrix components.
FCTC4 = 0.0 excludes sin(2ωt) terms from the Coriolis matrix components.
FCTC5 = 0.0 excludes cos(2ωt) terms from the Coriolis matrix components.

FCTM1 = 0.0 excludes sin(ωt), cos(ωt), sin(2ωt), and cos(2ωt) terms from the mass matrix components.
FCTM2 = 0.0 excludes sin(ωt) terms from the mass matrix components.
FCTM3 = 0.0 excludes cos(ωt) terms from the mass matrix components.
FCTM4 = 0.0 excludes sin(2ωt) terms from the mass matrix components.
FCTM5 = 0.0 excludes cos(2ωt) terms from the mass matrix components.

FCTZ1 = 0.0 excludes sin(ωt), cos(ωt), sin(2ωt), and cos(2ωt) terms from the centrifugal matrix components.
FCTZ2 = 0.0 excludes sin(ωt) terms from the centrifugal matrix components.
FCTZ3 = 0.0 excludes cos(ωt) terms from the centrifugal matrix components.
FCTZ4 = 0.0 excludes sin(2ωt) terms from the centrifugal matrix components.
FCTZ5 = 0.0 excludes cos(2ωt) terms from the centrifugal matrix components.

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FDINTOPT Default = 2

Specifies the approximating method for frequency-dependent stiffness, viscous damping, and mass matrices in direct and modal frequency response analysis.

FDINTOPT = 1 to approximate the mass by a constant, the viscous damping as linearly varying, and the stiffness as quadratically varying over each subinterval.

FDINTOPT = 2 to approximate the mass, viscous damping, and stiffness by a single frequency-dependent matrix over each subinterval.

For more information, see "Frequency-Dependent Structural Properties" in the *Simcenter Nastran Basic Dynamic Analysis User's Guide*.

FHPNST Integer ≥ 12, Default = 12

When FHPOST=1, you can use FHPNST to specify the number of angle increments in a 90 degree quadrant in which the software computes the 3D harmonic results for the axisymmetric Fourier elements. This output will allow you to view the non-axisymmetric results on a full revolute model.

FHPOST Default = 5

For SOL 414, specifies the results output option when the axisymmetric Fourier elements are requested with the FHAR case control command.

Currently this output is only written to *.des and *.fac Samcef output files, and not to the .op2 file. Results for solid elements and the 360 degree axisymmetric Fourier element results include "3D" in their file name. In addition, results from statics, dynamics, and preload subcases include "_me." in their file name, and results from modes, complex modes, and cyclic modes subcases include "_dy." in their file name. You can import these results files into Simcenter 3D post processing to view the output.

FHPOST=1 Requests 3D harmonic results for the axisymmetric Fourier elements at incremental angle locations around the complete 360 degree circumference. You use the FHPNST parameter to define the angle increments per 90 degrees. In addition, 2D results on the modeled axisymmetric geometry and 3D results on the modeled solid element geometry are also output.

FHPOST=4 Requests 2D results on the modeled axisymmetric geometry.

FHPOST=5 Requests 2D results on the modeled axisymmetric geometry and 3D results on the modeled solid element geometry.

FIXDIFFK Default = Do not apply the zero-strain energy or modal zero-strain energy method

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Applicable to SOL 103 when a structural model is unconstrained and statically preloaded.

FIXDIFFK specifies whether to use the zero-strain energy or modal zero-strain energy method to condition the differential stiffness matrix so that six near-zero eigenvalues are calculated for rigid body modes.

Specify PARAM,FIXDIFFK,ZEROSTR to specify the zero-strain energy method, which conditions the differential stiffness matrix in physical space before the eigensolution in the SOL 103 solve.

Specify PARAM,FIXDIFFK,MZEROSTR to specify the modal zero-energy method, which conditions the differential stiffness matrix in modal space after an initial modal reduction, but before the eigensolution in the SOL 103 solve.

Note

The FIXDIFFK parameter is supported for a SOL 103 external superelement creation run but it is not supported for a SOL 103 system run.

The FIXDIFFK parameter is not supported for a SOL 103 model that includes incompressible fluids.

FIXEDB

Default = 0

FIXEDB is used to reduce the cost of superelement checkout.

FIXEDB = -2 (SOL 101 only) is used on the initial runs when the user suspects that the superelement may contain errors and that only operations necessary for fixed-boundary solutions need be performed. In particular, the generation of the $[G_{oa}]$ matrix is branched over in the SEKR operation and $[P_a]$ is not generated in the SELR operation. These operations typically result in 50% of the reduction cost and are not needed in the fixed-boundary data recovery operations described in the next paragraph. After this operation has been completed, the keyword SELANG will appear in the database dictionary, indicating that the $[P_a]$ stored there is incomplete, and should not be summed into the downstream superelement, because System Fatal Message 4252 will be issued.

FIXEDB ≤ -1 (SOLs 101 and 103 only) allows uncoupled solutions for any superelement with conventional output requests. This output may be obtained as soon as the superelement is successfully generated and reduced and does not require that the entire model be assembled. In superelement statics, the solution is the component due to the $\{u^o_o\}$ vector, i.e., a fixed-boundary solution. In superelement modes, the solution is the uncoupled eigenvectors of the component. If PARAM,FIXEDB,-1 is specified in the Bulk Data or in the residual structure subcase, the modes of the residual structure will not be computed. For a printout or plotting of the component mode eigenvectors it is recommended

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that PARAM,FIXEDB,-1 be specified in the Bulk Data Section or above the subcase level in Case Control. If the modes of the residual structure are desired, then PARAM,FIXEDB,0 should be specified in the residual structure subcase. Exterior degrees-of-freedom listed on SECSETi and SESUP entries are free, and those on SEBSETi degrees-of-freedom are fixed. Data recovery for the residual structure should not be requested for this option.

FIXEDB = +1 (SOL 101 only) is used after the superelement has been proven valid. In the SEKR and SELR operations, it provides a branch over all operations already completed in the SEKR and SELR phases and completes the generation of the $[G_{oa}]$ matrix and the boundary stiffness and load matrices. It is also a method to recover the factor of the $[K_{oo}]$ matrix if the run aborted while computing $[G_{oa}]$.

FKSYMFAC

Default = 1.0

FKSYMFAC controls the symmetrization of the follower force stiffness in SOL 106. By default, the follower force stiffness K_f is symmetrized as:

$$K_{fs} = \frac{1}{2}(K_f + K_f^T)$$

and the symmetric part K_{fs} is used for efficiency. If FKSYMFAC= 0. is specified, the original follower force stiffness K_f is used. If a value of 0. < FKSYMFAC < 1. is specified, the non-symmetric part of the follower force stiffness is calculated as:

$$K_{fn} = K_f - K_{fs}$$

and the ratio of unsymmetry:

$$r = \frac{\|K_{fn}\|}{\|K_f\|}$$

is compared with the user specified value of FKSYMFAC. The norm ||·|| is the absolute maximum number of the matrix.

If r < FKSYMFAC, the symmetric stiffness K_{fs} is used.

If r > FKSYMFAC, the original unsymmetrical stiffness K_f is used.

For most cases, the symmetrized follower force stiffness (default) will give sufficiently accurate answers. If the influence of the non-symmetric part of the follower force stiffness is important, a value of FKSYMFAC=1.e-9 is recommended.

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The parameter FKSYMFAC is applicable to SOL 106 only. All other solution sequences symmetrize the follower force stiffness. See parameter **FOLLOWK** for a list of solution sequences which calculate the follower force stiffness.

FLEXINCR Default = NO

In SOL 144, a value of YES will cause the TRIM subcases to be ignored. Instead, the TRIM Bulk Data will be used to obtain the set of Mach, Dynamic pressure and symmetry values for Unit Solutions (Flexible Increments). These data can be archived in the aeroelastic database for subsequent reuse. (Flexible Increments are always computed. This param merely avoids the TRIM subcase if these increments are all that is required.)

FLEXINV Default = DECOMP

Selects the inversion method that the software uses when it inverts FRFFLEX bulk entry data.

Specify PARAM,FLEXINV,DECOMP to use the standard decomposition method.

Specify PARAM,FLEXINV,SVD to use the singular value decomposition method.

FLUIDMP Default = 0

In SOLs 111 and 200, under ANALYSIS=MFREQ with fluid-structure models (see Performing a Coupled Acoustic-Structural Analysis in the *Simcenter Nastran Acoustics Simcenter Nastran User's Guide*), fluid modal participation factors are computed and output for the first n modes of the fluid defined by user parameters LFREQFL, HFREQFL, and LMODESFL, where n is specified by PARAM,FLUIDMP,n. Structural, load, and panel factors are computed and output for the first n modes of the structure defined by user parameters LFREQ, HFREQ, and LMODES, where n is specified by PARAM,STRUCTMP,n. By default, panel factors are computed for all modes specified by PARAM,STRUCTMP. PARAM,PANELMP,-1 suppresses the calculation and printout of panel participation factors. If n = -1 for STRUCTMP or FLUIDMP, then factors will be output for all modes.

By default, factors are computed for all forcing frequencies at all fluid grid points in the residual structure. If this is not desired, then the OFREQ Case Control command may be used to specify the desired forcing frequencies, and the PARTN Case Control command may be used to specify the desired fluid grid points.

FLUIDSE Default = 0

PARAM,FLUIDSE,seidf specifies a special superelement reserved for fluid elements. Frequency dependent fluid elements must still be in the residual. The newer partitioned superelements are not supported.

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FOLLOWK Default = YES

In SOLs 103, 105, 106, 107, 108, 109, 110, 111, 112, 115, and 116, FOLLOWK=YES (default) requests the inclusion of follower force stiffness in the differential stiffness. FOLLOWK=NO requests that the follower force stiffness not be included. For FOLLOWK=YES in SOLs 103, 107, 108, 109, 110, 111, 112, 115, and 116, a separate static subcase is required and the STATSUB command is also required in the eigenvalue subcase. In nonlinear analysis (SOL 106), the follower force is included if PARAM,LGDISP,1 is specified. FOLLOWK is ignored in SOL 106 if LGDISP is not specified.

FRQDEPO Default = NO

By default, frequency-dependent elements cannot be connected to o-set degrees-of-freedom. PARAM,FRQDEPO,YES allows frequency-dependent elements to be connected to o-set degrees-of-freedom. However, results may not be reliable.

FRRU Default = YES

Turns off the SOL 111 model frequency response automatic method selection.

YES: For models in which stiffness and damping are not frequency dependent and SYSTEM(462) = 0, the software automatically chooses either the FRRD1 method or the FRRDRU method.

NO: Turns off the automatic selection and the software always uses the FRRD1 method.

FRUMIN Default = 500

For a modal frequency response solution (SOL 111), when you run with SYSTEM 462=1 (default), the software automatically selects the in-core FRRDRU method or a FRRD1 method. The software will select the in-core FRRDRU method if the model does not include frequency dependent properties or materials, unsymmetric damping, or consists of only fluid elements, and the number of modes requested ≥ PARAM,FRUMIN.

FULLGPF Default = YES

For SOLs 108 and 111 frequency response analysis, specify PARAM,FULLGPF,YES to include the element inertia forces, viscous damping forces from elements, PARAM,G structural damping forces, GE field structural damping forces, Rayleigh damping forces, DMIG forces, differential stiffness forces, FRF component forces, and FRF external superelement forces with the element elastic forces in the grid point force output and grid point force balances.

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For SOLs 109 and 112 transient response analysis, specify PARAM,FULLGPF,YES to include element inertia forces, viscous damping forces from elements, viscous damping forces from converted PARAM,G and GE field structural damping, Rayleigh damping forces, DMIG forces, and differential stiffness forces with the element elastic forces in the grid point force output and grid point force balances.

Specify PARAM,FULLGPF,NO to retain the pre-Simcenter Nastran 2021.2 behavior and output the element elastic forces only.

The grid point force balance is the sum of the element forces, SPC and MPC forces, applied loads, and thermal loads, where the contributions to the element forces depend on the FULLGPF parameter setting.

FZERO — Default = 1.0E-1

Tolerance for zero frequency rigid body modes. Frequencies (in Hz) below this value are considered zero. See the **AUTOSPRT** and **WMODAL** parameters.

Related Topics

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G – Parameters

G,GFL

Default = 0.0

G and GFL specify the uniform structural and fluid-damping coefficient in the formulation of dynamics problems. In coupled fluid-structure analysis, G is applied to the structural portion of the model and GFL to the fluid portion of the model.

To obtain the damping coefficient G (or GFL), multiply the critical damping ratio ζ by 2.0. That is, $G = 2\zeta = 2(b/b_{cr})$. See Viscous and Structural Damping in the *Simcenter Nastran Basic Dynamic Analysis User's Guide*.

PARAM,G and GFL are not recommended for use in hydroelastic or heat-transfer problems. If PARAM,G (or GFL) is used in transient analysis, PARAM,W3 (or W3FL) must be greater than zero or PARAM,G (or GFL) will be ignored. See "Formulation of Dynamic Equations in SubDMAP GMA" in the *Simcenter Nastran User's Guide*.

GFL is supported by the FEM Adaptive Order (FEMAO) method.

G*x*
(*x* is an integer)

Real, Default = (0.0,0.0)

G*x* is used by NX to retain node and element groups during the exporting and importing of bulk data files and .op2 results files. It is not used by Simcenter Nastran, and it has no impact on the Simcenter Nastran solution sequence.

For example, in the parameter setting

PARAM,G5,6.0,8.0

5 is the original NX group ID, 6 is a SET ID containing nodes, and 8 is a SET ID containing elements.

GDAMPF

Character, Default = YES

Determines if the global structural damping defined with PARAM,G is included in the stiffness when computing force and stress in frequency and transient response solutions for CELAS1, CELAS2, CELAS3, CELAS4, and CBUSH elements.

= NO PARAM,G is not included when computing force and stress.

= YES PARAM,G is included when computing force and stress.

GDAMPF is not supported for SOL 401, 402, or 414. For SOL 401 when ANALYSIS = DYNAMICS, the effect of PARAM,G is included in the force and stress computation.

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For SOL 414, the effect of PARAM,G is not included in the force and stress computation. For SOL 402, PARAM,G is not supported.

Note

CELAS3 and CELAS4 are not supported in SOLs 402 and 414.

GEOMU Default = 40

See **PARAM, POST, 0**.

GPACAO Default = YES

In a FEM Adaptive Order (FEMAO) acoustic analysis, the parameter GPACAO controls if the model quality check is performed. When the quality check is requested, the results of the check are output to the GPACAO data block into the .op2 file and available for post-processing. The quality check processes geometry and element information, and it includes solution diagnostics such as the maximum frequency per element and element order information. Switching off the quality check can improve the overall performance.

= YES The model quality check is performed and the check results are written to the .op2 file.

= NO The model quality check is not performed.

GPECT Default = -1

GPECT controls the printout of all elements connected to each grid point. GPECT=+1 requests the printout. In superelement analysis, the list is printed if PARAM,CHECKOUT,YES is specified or the SEMG or SEALL Case Control command selects the superelement. GPECT=-1 suppresses the printout.

GRDPNT Default = -1

GRDPNT>-1 will cause the grid point weight generator to be executed. The default value (GRDPNT=-1) suppresses the computation and output of this data. GRDPNT specifies the identification number of the grid point to be used as a reference point. If GRDPNT=0 or is not a defined grid point, the reference point is taken as the origin of the basic coordinate system. All fluid-related masses and masses on scalar points are ignored.

The Simcenter Nastran principal mass axes have a specific meaning and should not be confused with the principal axes discussed in text books. The principal mass axes in Simcenter Nastran have no coupling terms between the translational masses, but most importantly, they are with respect to the basic coordinate system.

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See Overview of the Grid Point Weight Generator in the *Simcenter Nastran Basic Dynamic Analysis User's Guide* for more information.

The following weight and balance information is automatically printed following the execution of the grid point weight generator.

- Reference point.
- Rigid body mass matrix [MO] relative to the reference point in the basic coordinate system.
- Transformation matrix [S] from the basic coordinate system to principal mass axes.
- Principal masses (mass) and associated centers of gravity (X-C.G., Y-C.G., Z-C.G.).
- Inertia matrix I(S) about the center of gravity relative to the principal mass axes. Note: Change the signs of the off-diagonal terms to produce the "inertia tensor."
- Principal inertia I(Q) about the center of gravity.
- Transformation matrix [Q] between S-axes and Q-axes. The columns of [Q] are the unit direction vectors for the corresponding principal inertia.

In superelement static or geometric nonlinear analysis, GRDPNT> -1 also specifies the grid point to be used in computing resultants, in the basic coordinate system, of external loads and single point constraint forces applied to each superelement. If GRDPNT is not a grid point (including the default value of -1), then the resultants are computed about the origin of the basic coordinate system. In superelement analysis, weights and resultants are computed for each superelement without the effects of its upstream superelements.

For the CQUDX4, CQUADX8, CTRAX3, and CTRAX6 elements, the mass and inertia are reported for the entire model of revolution but the center of gravity is reported for the axisymmetric cross section.

GUSTAERO — Default = 1

If gust loads are to be computed, for example on restart, set GUSTAERO to -1. The default is recommended if no gust loads are to be computed after the flutter analysis.

Related Topics

Browse Simcenter Nastran documentation

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H – Parameters

HARMLIM

Default = 1 (Integer)

When the Coleman transformation is activated (**PARAM,COLEMAN,1**), sets the maximum harmonic (wave number) that the Coleman transformation will consider.

For more information, see Cyclic symmetry with the Coleman transformation in the *Rotor Dynamics User's Guide (SOL 414)*.

HEATSTAT

Default = NO

In SOL 101, if PARAM,HEATSTAT,YES is entered, then temperatures are computed in a linear steady state heat transfer and then applied as thermal loads in a subsequent thermal stress analysis. Two subcases are required. The first defines the temperature loads, boundary conditions, and output requests for the heat transfer analysis and the second subcase defines the thermal loads, boundary conditions, and output requests for the thermal stress analysis. Thermal loads in the second subcase are requested through the command

```
TEMP(LOAD) = Heat Transfer Subcase ID
```

If this default is not acceptable, then in the heat transfer subcase add the Case Control word TSTRU=SID and in the structures subcase here

```
TEMP(LOAD) = SID
```

See the Case Control command, **TSTRU**. PARAM,NESET is no longer used.

HFREQ, HFREQFL

Default = 1.+30

The parameters LFREQ, HFREQ, LFREQFL, and HFREQFL specify the frequency range in cycles per unit time of the modes to be used in the modal formulations. (LFREQ and LFREQFL are the lower limits and HFREQ and HFREQFL are the upper limits.) In coupled fluid-structure analysis, HFREQ and LFREQ are applied to the structural portion of the model and HFREQFL and LFREQFL are applied to the fluid portion of the model. Note that the default for HFREQ and HFREQFL will usually include all vectors computed. Related parameters are LMODES and LMODESFL.

Related Topics

Browse Simcenter Nastran documentation

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I – Parameters

IFP

Default = value of PARAM,DBALL

See **DBALL**

IFTM

Default = 0

IFTM specifies the method for Inverse Fourier Transformation in SOLs 111 and 146. See the *Simcenter Nastran Aeroelastic Analysis User's Guide* for further discussion. Permissible values are 0, 1, and 2. The default value is recommended.

0: constant (default)

1: piecewise linear

2: cubic spline

INEMETH

Default = 1

Controls the coupling of mass and mass moments of inertia terms in the rotor dynamics solutions (107, 108 and 109) when using the coupled method between the rotating and fixed part (selected by PARAM, ROTCOUP). In the 6x6 mass matrix, the upper left 3x3 are the mass terms (M_{ij}), and the lower right 3x3 are the mass moments of inertia terms (I_{ij}).

1: Both the mass terms and the mass moments of inertia terms are coupled.

0: Only the mass terms are coupled.

INP4FMT

Default = 32

Selects the type of OP4 file you are inputting.

If INP4FMT=32 (default), a 32-bit OP4 file is the input, and the software automatically converts the 32-bit data to 64-bit.

If INP4FMT=64, a 64-bit OP4 file is the input. No conversion is required, and the 64-bit data is read directly.

INREL

Default = 0

Controls the calculation of inertia relief or enforced acceleration.

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- Specify INREL = -2 to request automatic inertia relief. Do not use SUPORT or SUPORT1 bulk entries when INREL = -2 is specified. INREL = -2 does not support enforced accelerations. INREL = -2 is valid for SOL 101 only.

  Note

  INREL = -2 is only applicable to completely unconstrained (free-free) models that possess six rigid body modes.

- Specify INREL = -1 to request inertia relief or enforced acceleration. When INREL = -1, use SUPORT or SUPORT1 bulk entries to specify one or more grid points at which to constrain the model. The total number of DOF specified on SUPORT and SUPORT1 bulk entries must be six or less. INREL = -1 is valid for SOLs 101, 105, and 200.

  For SOL 105, use SUPORT1 bulk entries only and specify a SUPORT1 case control command in a separate static subcase.

  The loads that arise due to unit rigid body accelerations at the point referenced by the GRDPNT parameter are computed and appended to the external loads. If PARAM,GRDPNT is specified in superelement analysis, then the point must be interior to the residual structure and exterior to all superelements. To specify unit rigid body accelerations, use DMIG,UACCEL bulk entries.

- INREL = 0 requests a static analysis without inertia relief or enforced acceleration.

When the sparse solver is used, the INREL parameter is supported for use with models that include linear contact and glue.

Inertia relief is not supported when bolt preload (BOLTLD case control command) is present.

For more information, see:

- “Data Recovery Operations in SubDMAP SEDISP” in the *Simcenter Nastran User’s Guide*
- “Buckling Analysis in SubDMAP MODERS” in the *Simcenter Nastran User’s Guide*
- “Automatic Inertia Relief” in the *Simcenter Nastran User’s Guide*

INRLM — Default = 0

INRLM=-1 requests that component (superelement) inertia relief shapes be appended to the generalized dynamic modes as described in the Simcenter Nastran User's Guide, Solution Sequences chapter, Superelement Analysis section. If this feature is used, the component must have at least six independent rigid body mass degrees-of-freedom; otherwise, duplicate modes will be generated, causing a singular matrix.

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IRES Default = -1

IRES=1 requests that the residual load vectors RULV and RUOV be output in all solution sequences. In superelement analysis, the parameters PRPA and PRPJ may also be used to request output of the partial load vectors $\{P_a\}$ and $\{P_j\}$, respectively. In geometric nonlinear analysis, PARAM,IRES,1 will cause the printing of the residual vector as follows:

$$\{\Delta P_f\} = [K_{ff}]\left\{u_f^{n+1} - u_f\right\} + \{F_f\} - \{P_f\}$$

ITAPE Default = -1

ITAPE specifies the output status of the DSCMR matrix in SOLs 101, 103, and 105; and the DSCMCOL table and the DSCM2 matrix in SOL 200. (See the OUTPUT2 and OUTPUT4 module descriptions in the *Simcenter Nastran DMAP Programmer's Guide*.)

IUNIT Default = 11

IUNIT specifies the FORTRAN unit number on which the DSCMR matrix in Design Sensitivity SOLs 101, 103, and 105 and the DSCMCOL table and the DSCM2 matrix in SOL 200 will be written. (See the OUTPUT2 and OUTPUT4 module descriptions in the *Simcenter Nastran DMAP Programmer's Guide*.)

Related Topics

Browse Simcenter Nastran documentation

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K – Parameters

K6ROT Default = 100.0

K6ROT is used in the calculation of the stiffness for the out-of-plane rotational DOF for CQUAD4, CQUAD8, CTRIA3, and CTRIA6 elements. K6ROT is always ignored for CQUADR and CTRIAR elements.

Specifying a value between 1.0 and 100.0 for K6ROT is recommended. A large value for K6ROT may be required in nonlinear analysis and eigenvalue analysis. K6ROT is forced to 0.0 for elements that have membrane stiffness only.

K6ROT is applicable in all solution sequences. Depending on the solution type, element type, and SYSTEM(700) setting, it may be ignored as follows:

- Linear solutions

  CQUAD4 and CTRIA3: K6ROT is used.

  CQUAD8 and CTRIA6: K6ROT is used unless any of the following apply:

  SYSTEM(700)=1 is specified.

  The elements possess membrane stiffness only.

  The elements possess bending and transverse shear stiffness only.

  The elements are plane strain.

  For these cases, K6ROT is ignored.

- SOLs 106 and 129

  CQUAD4 and CTRIA3: K6ROT is used.

  CQUAD8 and CTRIA6: K6ROT is used unless any of the following apply:

  SYSTEM(700)=1 is specified.

  The material is hyperelastic.

  The elements possess membrane stiffness only.

  The elements possess bending and transverse shear stiffness only.

  The elements are plane strain.

  For these cases, K6ROT is ignored.

- SOLs 401 and 402

  CQUAD4 and CTRIA3: K6ROT is ignored.

  CQUAD8 and CTRIA6: K6ROT is used.

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KDAMP, KDAMPFL Default = 1

For the SOLs 110, 111, 112, 145, 200:

If KDAMP (or KDAMPFL) = 1, values entered on TABDMP1 entry are used to compute modal viscous damping.

If KDAMP (or KDAMPFL) = -1, values entered on TABDMP1 entry are used to compute modal structural damping.

In coupled fluid-structure analysis, KDAMP is applied to the structural portion of the model and KDAMPFL to the fluid portion of the model.

For more information, see “Formulation of Dynamic Equations in SubDMAP GMA” in the *Simcenter Nastran User’s Guide*.

For a SOL 103, when PARAM, POST is set as -1 or -2, and PARAM, MDLDAMP is set to "YES":

KDAMP = 1 (default) Structural damping (PARAM,G and MAT GE) will be converted to viscous damping for output.

KDAMP = -1 Structural damping (PARAM,G and MAT GE) and viscous damping will remain as separate outputs.

KDIAG Default = -1.0 (SOLs 106 and 153 only)

In nonlinear static analysis, KDIAG may be used to eliminate spurious mechanisms and singularities in the nonlinear stiffness matrix. The absolute value of KDIAG will be added to some or all of the diagonal terms in the nonlinear stiffness matrix as follows:

If KDIAG < 0.0, then add the absolute value of KDIAG to the null diagonal terms only. (Default)

If KDIAG = 0.0, then no action is taken.

If KDIAG > 0.0, then add the value of KDIAG to all diagonal terms.

KGGCPCH Default=0

Requests the final contact stiffness matrix in DMIG format from a SOL 101 linear contact solution. The solver writes the contact stiffness matrix from the final contact iteration (not necessarily converged) into a PUNCH file in the format required for DMIG (Direct Matrix Input at Grids). This option is only available with the default sparse solver. The iterative solvers do not support this option.

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If KGGCPCH=0, contact stiffness matrix is not output.

If KGGCPCH=1, contact stiffness matrix is output.

KGGLPCH Default=0

Requests the glue stiffness matrix in DMIG format in any solution except 144-146, and 401. The solver writes the glue stiffness matrix into a PUNCH file in the format required for DMIG (Direct Matrix Input at Grids). This option is only available with the default sparse solver. The iterative solvers do not support this option.

If KGGLPCH=0, glue stiffness matrix is not output.

If KGGLPCH=1, glue stiffness matrix is output.

Related Topics

Browse Simcenter Nastran documentation

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L – Parameters

LAMLIM — Default = 0.001

LAMLIM is used by the NX Response Dynamics Capability to identify rigid body modes for computing the viscous damping ratio matrix which is written to the OP2 file when RSOPT =1. The frequencies below the LAMLIM value will be considered a rigid body mode. The column and rows of the viscous damping ratio matrix associated with the rigid body will be set to zero.

LANGLE — Default = 1

LANGLE specifies the method for processing large rotations in nonlinear analysis. By default, large rotations are computed with the gimbal angle method in nonlinear analyses SOLs 106, 129, 153, and 159 with geometric non linearity (PARAM,LGDISP,1). If PARAM,LANGLE,2 is specified, then they are computed with the Rotation Vector method. The value of LANGLE cannot be changed in a subsequent restart.

LFREQ, LFREQFL — Default = 0.0

See **HFREQ, HFREQFL**.

LGDISP — Default = -1

If LGDISP = 1, then large displacement effects (updated element coordinates and follower forces) are assumed for all the nonlinear element types that have a large displacement capability in SOLs 106, 129, 153, 159, 401, 402, 414,103, 414,111, and 414,129.

If LGDISP = -1, then no large displacement effects are considered.

If LGDISP = 2, then follower force effects are ignored but large displacement effects are considered.

If LGDISP ≥ 0, then the differential stiffness is computed for the linear elements and added to the differential stiffness of the nonlinear elements.

SOLs 401, 402, and 414 support only LGDISP = 1 or -1.

See Nonlinear Effects in SOL 401 and Nonlinear Effects in SOL 402.

For SOL 414, see Nonlinear capabilities and effects in the *Rotor Dynamics User's Guide (SOL 414)*.

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LGSTRN

LGSTRN is supported by SOLs 401, 402, 414,103, 414,111, and 414,129 only.

If LGSTRN = 0, small strains are assumed. (Default)

If LGSTRN = 1, large strains, displacements and rotations are assumed (i.e., LGDISP is automatically set to 1).

See Nonlinear Effects in SOL 401 and Nonlinear Effects in SOL 402.

For SOL 414, see Nonlinear capabilities and effects in the *Rotor Dynamics User's Guide (SOL 414)*.

LMDYN

Integer input; LMDYN default = -1; LMSTAT default = 6

With Lagrange RBE3, RBE2, RBAR and RTRPLT elements, these parameters scale the artificial stiffness term c_k according to the following equations:

Dynamic solutions: $$c_K = \frac{1.0}{10^{LMDYN}}$$

Static solutions: $$c_K = \frac{1.0}{10^{LMSTAT}}$$

If either LMSTAT or LMDYN is defined as -1, c_K is set to 0.0, and the equations for c_K are not applied. See “Rigid element processing options” in the *Element Library Reference*.

LMFACT

Default = 1.0E+6

Scale factor used to adjust the stiffness matrix of Lagrange rigid elements (see the case control command “**RIGID**” for more information). If their stiffness is too small, the results will be inaccurate. If their stiffness is too large, numerical problems can occur (fatal errors during the solution may occur). Best results are obtained when their stiffness is similar to the rest of the model.

LMODES, LMODESFL

Default = 0

LMODES and LMODESFL are the number of lowest modes to use in a modal formulation. In coupled fluid-structure analysis, LMODES specifies the lowest modes of the structural portion of the model and LMODESFL the modes of the fluid portion of the model. If LMODES (or LMODESFL) = 0, the retained modes are determined by the parameters LFREQ and HFREQ (or LFREQFL and HFREQFL).

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	In SOL 103, LMODES may be used to reduce the number of eigenvectors to be processed in data recovery which may significantly reduce the CPU and storage costs.
LMSTAT	See the parameter **LMDYN**.
LOADU	Default = -1 See **PARAM, POST, -1**.
LOOPID	Default = 0 LOOPID defines the desired loop number for initial conditions in a restart of SOLs 106, 129, 153, and 159. By default in SOLs 106 and 153, the restart proceeds from the last loop ID of the subcase defined by PARAM,SUBID.
LSTRN	Replaced by the **STRAIN** Case Control command.

Related Topics

Browse Simcenter Nastran documentation

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M – Parameters

MACH — Default = 0.0

Mach number. If more than one Mach number was used to compute aerodynamic matrices, the one closest to MACH will be used in dynamic aeroelastic response analysis. The default causes the matrices computed at the lowest MACH number to be used.

MATCTE — Real; No Default.

For SOLs 401, 402, and 414, specifies a uniform coefficient of thermal expansion (α_{matcte}) that overrides all of the values defined in the material definitions (MATi and MATTi bulk entries) and in the rigid element definitions (RBEi and RBAR bulk entries). When MATCTE is undefined, the coefficients defined in the material and rigid element definitions are used.

When MATCTE is defined, the thermal strain is calculated from:

$$\epsilon = \alpha_{matcte}(T_{load} - T_{init})$$

See the **TEMPERATURE** case control command.

MATNL — Default = –1

If MATNL = -1, turns off capabilities for material nonlinearity; the materials behave linearly.

If MATNL = 1, turns on capabilities for material nonlinearity.

For SOLs 401 and 402, MATNL = 1 turns on the material nonlinear capabilities listed below.

For SOL 414, see Nonlinear capabilities and effects in the *Rotor Dynamics User's Guide (SOL 414)*.

For SOL 401, PARAM, MATNL, 1 is required to turn on any nonlinear material behavior, including:

- Nonlinear elasticity defined on the MATS1 bulk entry.
- Plasticity defined on the MATS1 or MPLAS bulk entries.
- Strain rate dependent plasticity defined on the MATS1 and MATSR bulk entries.
- Creep defined on the MATCRP or MVPLAS bulk entries.

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- Damage for cohesive elements defined on the MATCZ bulk entry.
- Damage for progressive ply failure defined on the MATDMG bulk entry.
- User-defined creep defined on the MUCRP bulk entry.
- User-defined nonlinear materials defined on the MUMAT bulk entry.

Note that the case where MUMAT is used to define Hooke's matrix for the initial elastic stiffness (IOPER=3 in the NXUMAT arguments) is a linear material definition that is independent of PARAM, MATNL.

For SOL 402, PARAM,MATNL,1 is required to turn on the following:

- Plasticity defined on the MATS1 or MPLAS bulk entries.
- Strain rate dependent plasticity defined on the MATS1 and MATSR bulk entries.
- Creep defined on the CREEP, MATCRP, or MVPLAS bulk entries.
- Gasket material defined on the MATG bulk entry.
- Damage for progressive ply failure defined on the MATDMG bulk entry.
- User-defined creep defined on the MUCRP bulk entry.
- User-defined nonlinear materials defined on the MUMAT bulk entry.

Note that SOL 402 does not require PARAM, MATNL, 1 to turn on material nonlinearities due to the following:

- Hyperelasticity defined on the MATHE or MATHP bulk entries.
- Damage for cohesive elements defined on the MATCZ bulk entry.

MAXLP — Default = 5

MAXLP specifies the maximum number of iterations for element relaxation and material point subincrement processes in SOLs 129 and 159. MAXLP is 10 in SOLs 106 and 153 and cannot be changed by the user.

MAXRATIO — Default = 1.E7. Default = 1E10 if glue, linear contact or Lagrange rigid elements are present in the model.

Default = 1.E8 for SOL 402.

The ratios of terms on the diagonal of the stiffness matrix to the corresponding terms on the diagonal of the triangular factor are computed. If, for any row, this ratio is greater than MAXRATIO, the matrix will be considered to be nearly singular (having mechanisms). If any diagonal terms of the factor are negative, the stiffness matrix is considered implausible (nonpositive definite). The ratios greater than MAXRATIO and less than zero and their associated external grid identities will

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be printed out. The program will then take appropriate action as directed by the parameter BAILOUT.

By default, in the superelement solution sequences the program will terminate processing for that superelement. A negative value for BAILOUT directs the program to continue processing the superelement. Although forcing the program to continue with near-singularities is a useful modeling checkout technique, it may lead to solutions of poor quality or fatal messages later in the run. It is recommended that the default values be used for production runs. A related parameter is ERROR.

The value -1 of BAILOUT causes the program to continue processing with near singularities and a zero value will cause the program to exit if near singularities are detected.

In SOLs 101 through 200 when PARAM,CHECKOUT,YES is specified, PARAM,MAXRATIO sets the tolerance for detecting multipoint constraint equations with poor linear independence. (See “Superelement Analysis” in the *Simcenter Nastran User’s Guide*.)

BAILOUT and MAXRATIO may be specified in the Case Control Section in order to provide different values between superelements but not between different boundary conditions.

MDOF — See the parameter **MGRID**.

MDLDAMP — Default=NO

MDLDAMP is used by the Simcenter 3D Mode Sets capability, which uses SOL 103 to compute the Mode Sets.

In a SOL 103 analysis, when PARAM, POST is set as -1 or -2, and PARAM, MDLDAMP is set to "YES", modal viscous and structural damping diagonal matrices will be output to the .op2 file.

You use the parameter KDAMP to control the type of damping output:

KDAMP = 1 (default) Structural damping (PARAM,G and MAT GE) will be converted to viscous damping for output.

KDAMP = -1 Structural damping (PARAM,G and MAT GE) and viscous damping will remain as separate outputs.

MDSTYPE — Default = 0

Defines the processing method for binary mode sets in SOL 103, 108, and 111 acoustic analysis.

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MDSTYPE = 0 for automatic selection of the processing method.

MDSTYPE = 1 for creation of MPC equations. This is the only method available in pre-Simcenter Nastran 2021.1 versions.

MDSTYPE = 2 for direct generation of constraint matrix.

Direct generation of constraint matrices is typically more computationally efficient than creating and processing of MPC equations.

MECHFIL Default: 1.0E-6

Criterion for discarding massless mechanism modes with small generalized mass. A smaller value will result in more marginal constrained modes being retained. This parameter is associated with the MECHFIX parameter.

MECHFIX Default = AUTO

Control for fixing the massless mechanism problem. This capability is provided automatically by default. It is executed only when the solution does not provide answers because of symptoms consistent with the presence of massless mechanisms. If MECHFIX is set to YES, the constrained modes are removed before attempting a solution. When set to NO, this capability is blocked, and the eigensolution portion uses the rule of three failed shifts and a fatal exit. See the parameter **MMFIL**.

MECHPRT Default = NO

For SOL 103 only. If massless mechanisms are found, the constrained modes are printed with a format similar to eigenvectors when this parameter is set to YES. They are labeled CONSTRAINED MODE, and are numbered sequentially. Grid points with only zero values in a mode are not printed. This parameter should be used when performing initial checkout of a model with the goal to remove all massless mechanisms before starting production analysis. The number of each “mode” matches the corresponding GID,C pair in the high ratio message. If there are many (thousands) of such modes, the output file will be large. There is no method to plot these shapes.

MEMSTRES Default = NO

When stress output is requested in a DDAM analysis, you can request "membrane only" stresses to be output in addition to the regular stress output. The additional membrane only stresses can be output for the following elements:

- Shell elements (CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, CQUADR)
- Shell element composites

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- Bar and Beam elements (axial and shear stress)

When MEMSTRES=YES, the software writes the additional membrane only stress for the elements listed above.

When MEMSTRES=NO, the additional stress output does not occur.

MESH — Default = NO

If MESH=YES is specified, then a summary of the shading status and the subelement mesh for each CHBDYi element referencing a VIEW Bulk Data entry is printed.

METHCMRS — Default = 0

In dynamic analysis (SOLs 103, 107, 108, 109, 110, 111, 112, 145, 146, and 200), METHCMRS specifies the set identification number of an EIGR or EIGRL entry to be used in the computation of the normal modes on the o-set (fixed boundary modes) or v-set (mixed boundary modes) of the residual structure depending on the b-set and/or c-set entries.

These modes are calculated for a component mode reduction of the residual structure when q-set exists. If no q-set exists in the residual structure, a component modal reduction is not performed.

The case control entry **RSMETHOD** has the same effect as the METHCMRS parameter. If both of them exist, the RSMETHOD takes precedence. If neither exists, a warning message is issued since only Guyan reduction will be executed and the results are poor. In the latter case, the residual structure fixed boundary normal modes or mixed boundary normal modes will be computed based on the **METHOD** Case Control command and the appropriate b-set and c-set entries.

MGRID — The parameters MGRID and MDOF only apply to the solution monitor when a direct frequency response (SOL 108) or a direct transient response (SOL 109) is used. MGRID selects the grid (node) ID, and MDOF selects the DOF (123456) for plotting the solution progress within the user interface. By default, the first dof in the L-set is used (See Displacement Vector Sets in the *Simcenter Nastran Basic Dynamic Analysis User's Guide* for a definition of the L-set).

MMFIL — Default = 1.0E-10

Filter value used to distinguish between massless mechanism modes and rigid body modes. The default value has been found to be effective; a smaller value may discard rigid body modes.

MODACC — Default = -1

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MODACC = -1 do not use the mode acceleration method for data recovery in dynamic analysis.

MODACC = 0 use the mode acceleration method for data recovery in dynamic analysis. See “Formulation of Dynamic Equations in SubDMAP GMA” in the *Simcenter Nastran User’s Guide* for further discussion.

MODACC = 1 is the same as MODACC = 0 except if SUPORTi entries are present then the displacements are corrected for rigid body motion and the local flexibility at the SUPORTi degrees-of-freedom.

MODACC ≥ 0 is not recommended for use in hydroelastic problems.

MODACC is not supported for external superelements.

MODEL Default = 0

This parameter also allows several models to be stored in the same graphics database created by PARAM,POST,0.

MODTRK Default = 2

This parameter is used in a legacy rotor dynamics solution to select the mode tracking method.

If MODTRK = 1, the pre-NX Nastran 7 method is used. Outer loop over rotor speed, inner loop over degrees of freedom. This method does not work well for the direct method (SOL 107) because there, new solutions can come in and old solutions can leave the solution space.

If MODTRK = 2, the method introduced in version 7 is used. Outer loop over degrees of freedom, inner loop over rotor speed. Process is repeated until all solutions have been tracked.

If MODTRK = 3, the method introduced in NX Nastran 8.5 is used. Eigenvectors and eigenvalues are used to track the modes. This method is applicable to models having any combination of unsymmetric stiffness, unsymmetric viscous damping, and structural damping.

If MODTRK = 4, the method introduced in NX Nastran 11 is used. The initial reference rotor modes are used as the base vectors for complex eigenvalue calculations at subsequent reference rotor speeds. Mode tracking begins at the first nonzero reference rotor speed. Modes for the reference rotor speed of zero are excluded from the Campbell diagram results. Strain and kinetic energy filtering of modes can be used with this mode tracking option. For information on how to filter modes, see the **ROTPARM** bulk entry.

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The following parameters apply if MODTRK = 2: **MTREPSI**, **MTREPSR**, **MTRFCTD**, **MTRFCTV**, **MTRFMAX**, **MTRRMAX**, and **MTRSKIP**.

The following parameters apply if MODTRK = 3: **MTRFMAX** and **MTRRMAX**.

If the tolerances MTREPSR and MTREPSI are chosen too small, the solution may be lost for some speed values. If the tolerance is too large, there may be lines crossing from one solution to another. Problems may occur for turbines with many elastic blades with equal frequencies. Then clusters of lines and crossings may occur.

MODTRK is not supported in SOL 414 rotor dynamics analyses.

MPCX

Default = 0

See **OLDSEQ**.

MPCZERO

Default = 1.0E–11

Filter value to remove small “noise” values from the multipoint constraint matrix. To turn this filtering off, set to a value less than or equal to 0.0.

MPIBUFF

Default = 250,000

For DMP runs in SOLs 108 and 111, defines the maximum buffer size in words given to each processor for the merging process. The value can not be larger than 268,435,456.

MTREPSI

Default = 0.01

Applies when MODTRK = 2. Relative tolerance for the imaginary part (eigenfrequency). When a root has been found in the extrapolation, a check is made if the correct solution has been found. If the present value is outside of this tolerance, the solution is skipped.

MTREPSR

Default = 0.05

Applies when MODTRK = 2. Relative tolerance for the real part (eigenfrequency). When a root has been found in the extrapolation, a check is made if the correct solution has been found. If the present value is outside of this tolerance, the solution is skipped.

MTRFCTD

Default = 0.5

Applies when MODTRK = 2. Threshold value for damping. This is in order to disregard the real part for solutions with low damping.

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MTRFCTV Default = 0.0

Applies when MODTRK = 2. Reference value for converting aerodynamic speed to rotor speed for wind turbines. Used only for wind turbines in SOL 145.

MTRFMAX Default = 0.0

Applies when MODTRK = 2 or 3. Maximum frequency to consider. If zero, all frequencies are used. With this parameter high frequency solutions can be filtered out.

MTRRMAX Default = 0.0

Applies when MODTRK = 2 or 3. Maximum absolute value of real part of solution to consider. If zero, all real part values are used. With this parameter solutions with high real parts can be filtered out. It can be useful to disregard numerical solutions which are not physically relevant.

MTRSKIP Default = 5

Applies when MODTRK = 2. If roots are not found for a specific speed, this speed is skipped and the next speed is analyzed. If there are more than MTRSKIP values missing, the curve is not considered. The solution will be marked as unused and will be considered in the next loop.

Related Topics

Browse Simcenter Nastran documentation

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N – Parameters

NASPDV Default = 0

NASPDV controls if the SOL 200 topology optimization normalized material density design variables are written to the .f06 file. The software automatically creates the normalized material density design variable for the elements that you select for the optimization. This output can be substantial for large models.

NASPDV = 0 (Default) The topology optimization design variables are not written to the .f06 file.

NASPDV = 1 The topology optimization design variables are written to the .f06 file.

NASPRT Default = 0

NASPRT specifies how often data recovery is performed and output in SOL 200. The NASPRT behavior is different for a non-topology optimization run and for a topology optimization run. A topology optimization run occurs when the DVTREL1 bulk entry is defined. Each scenario is described below.

NASPRT descriptions for a non-topology optimization run:

- NASPRT = 0 (Default): Output occurs for the 0th and the last design cycle.
- NASPRT = N where N>0: Output occurs for the 0th, for every Nth increment, and for the last design cycle.

  For example, if PARAM,NASPRT,2 and the job has run for 5 cycles, then data recovery is done at the design cycles numbered 0, 1, 3, and 5. Design cycle 1 data recovery is done since cycle 1 is the second cycle starting with cycle 0.

  Similarly, if PARAM,NASPRT,3, and the job has run for 9 cycles, then data recovery is done at cycles 0, 2, 5, 8, and 9.

  (*) Due to legacy reasons, cycle numbers are printed out earlier within the f06 file during the flow of the program and are ahead of the numbers in the “Summary of Design Cycle History” table by a count of one before analysis takes place.
- NASPRT < 0: No output occurs.

NASPRT descriptions for a topology optimization run:

- NASPRT = 0 (Default): Output occurs for the 0th and the best design cycle.
- NASPRT = N where N>0: Output occurs for the 0th cycle and for every Nth increment.
- NASPRT = -1: No output occurs.
- NASPRT = -2: Output occurs for every improved design cycle.

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Note that for topology optimization, none of the NASPRT settings specifically requests output for the last design cycle. As a result, output for the last design cycle only occurs when the last design cycle happens to coincide with one of the NASPRT requests described above.

NDAMP — Default = 0.01

In SOLs 129 and 159, defines numerical damping in order to achieve numerical stability. A value of zero requests no numerical damping. The recommended range of values is from 0.0 to 0.1.

NEWSEQ — Renamed OLDSEQ.

NFDINTPL — Default = 0

Controls the use of approximations for frequency-dependent mass, viscous damping, and stiffness matrices in direct and modal frequency response analysis.

NFDINTPL = n, where n is a positive integer, divides the solution frequencies into subintervals, where the first subinterval contains n solution frequencies, and interpolates the frequency-dependent mass, viscous damping, and stiffness over each subinterval from the mass, viscous damping, and stiffness values at the endpoints and midpoint of the subinterval. The interpolation method is specified with the FDINTOPT parameter.

NFDINTPL = 0 (default) calculates the frequency-dependent mass, viscous damping, and stiffness at each solution frequency.

For more information, see "Frequency-Dependent Structural Properties" in the *Simcenter Nastran Basic Dynamic Analysis User's Guide*.

NINTPTS — Default = 10

NINTPTS requests interpolation over the NINTPTS elements closest to each grid point. NINTPTS=0 requests interpolation over all elements, which can be computationally intensive. Related parameters include BIGER, CURV, NUMOUT, OG, OUTOPT, S1G, S1M, S1AG and S1AM.

NLAYERS — Default=5 (minimum=1, maximum=12) (SOLs 106, 129, 153, 159, 401, and 402)

NLAYERS is used in material nonlinear analysis only and is the number of layers used to integrate through the thickness of CQUAD4 and CTRIA3 elements with isotropic material properties. Set NLAYERS=1 for efficiency if no bending is selected (MID2=0 or -1 on all PSHELL data entries). Do not specify NLAYERS=1 if MID2 is greater than zero. A larger value of NLAYERS will give greater accuracy at the cost of computing time and storage requirements.

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For SOL 401, NLAYERS applies to the CQUAD4, CTRIA3, CQUAD8, and CTRIA6 elements when a plastic material is defined with the MATS1 or MPLAS entries. The allowed values are 3,5,7, or 9.

NLHTLS Default=0

See Remarks under Case Control command, **TSTRU.**.

NLMAX Default = 60

The number of suspected massless mechanisms is determined from the number of high ratio messages. If this number exceeds NLMAX, the number of trial massless mechanisms is reduced to NLMAX. This is a means to avoid an expensive run when there may be thousands of massless mechanisms due to modeling errors. The value of this parameter may be increased on initial runs where it is possible that there are many high ratio DOFs in order to see them all at once.

NLMIN Default = 10

If there are only a few high ratio DOFs there may be more mechanism modes present. More trial mechanism mode vectors are used, and those that do not indicate true problems are discarded. A smaller value could be considered on a stable mode undergoing small modeling changes.

NLTOL Default = 2 (SOL 106 only)

Default = 0 (SOL 153, nonlinear heat transfer)

NLTOL establishes defaults for the CONV, EPSU, EPSP and EPSW fields of NLPARM Bulk Data entry for nonlinear static and heat transfer analysis according to the following table.

NLTOL	Level of Accuracy
0	Very high
1	High
2	Engineering Design
3	Preliminary Design

See the **NLPARM** entry for further details and corresponding default NLPARM field values.

NMLOOP Default = 0

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In SOLs 106 and 153, nonlinear static analysis, normal modes will be computed with the updated nonlinear stiffness if PARAM,NMLOOP is not equal to zero. The nonlinear normal modes will be computed at the last iteration loop of each subcase in which a METHOD command appears.

NOCOMPS Default = +1

NOCOMPS controls if stress and strain is computed for the composite elements, the equivalent homogeneous elements, or both. Composite elements reference a PCOMP or PCOMPG entry. Equivalent homogeneous elements, which are typically created by the software, reference a PSHELL entry and are designated with a MID1 or MID2 ID greater than or equal to 10^8. The STRESS and/or STRAIN case control commands are also required for any of these options.

NOCOMPS	Composite Stress/Strain	Equivalent Element Stress/Strain
1	YES	NO
0	YES	YES
-1	NO	YES

Stresses on the equivalent homogeneous elements are based upon a smeared representation of the laminate's properties and in general will be significantly different than the more accurate lamina stresses available on the PCOMP-based elements. Any PSHELL entries which have a MID1 or MID2 ID greater than or equal to 10^8 are considered by the software to be for equivalent homogeneous elements and require NOCOMPS 0 or -1 for any stress or strain recovery.

NOELOF Default = -1

If NOELOF > 0, then the grid point forces (GPFORCE Case Control command) are computed along the edges of the two-dimensional elements. The default value will suppress this output.

NOELOP Default = -1

If NOELOP > 0, then the sum of the grid point forces (GPFORCE Case Control command) are computed parallel to the edges of adjacent elements. The effect of CBAR element offsets is not considered in the calculation of the forces. The default value will suppress this output.

NOFISR Default = 0

NOFISR controls the output of the composite failure indices and strength ratios.

For solutions other than SOL 402 (see **Parameter applicabilty tables**):

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0: The failure indices and strength ratios are output to the *.f06 files.

> 0: The failure indices and strength ratios are not output to the *.f06 file.

For SOL 402:

0: The composite failure indices are output to both the *.op2 and *.f06 files.

1: The composite failure indices are output only to the *.op2 file.

2: The composite failure indices are not output.

NOGPF Default = 1

NOGPF controls the printout of grid point forces (GPFORCE Case Control command). NOGPF > 0 specifies that the grid point forces are to be printed. NOGPF < 0 suppresses the printing of the grid point forces.

NOMSGSTR Default = 0

If NOMSGSTR = -1, the MSGSTRES module will be skipped even though Bulk Data entries provided for it.

NONCUP Default = -1

In SOLs 111 and 112, NONCUP selects either a coupled or uncoupled solution algorithm in modal frequency and modal transient response analyses. See "Frequency Response and Random Analysis in SubDMAP FREQRS" and "Transient Response in SubDMAPs SEDTRAN and SEMTRAN" in the *Simcenter Nastran User's Guide*.

NONCUP = -1 requests the coupled algorithm; however, if the dynamic matrices KHH, MHH, and BHH are diagonal, the uncoupled algorithm is used.

NONCUP= -2 requests the uncoupled algorithm regardless of the existence of off-diagonal terms in the dynamic matrices.

User Information Message 5222 indicates which algorithm is used in the analysis.

When using the ADAMSMNF or MBDEXPORT case control commands to request an Adams MNF file, a standard or state-space MATLAB file, or a standard or state-space OP4 file, the full equivalent modal viscous damping matrix is written to the interface file by default. To write only the diagonal values of the equivalent modal viscous damping matrix to the interface file, specify NONCUP=–2 on the ADAMSMNF or MBDEXPORT case control command, or specify PARAM,NONCUP,–2 in the bulk data section of the input file.

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	If both the NONCUP describer and the NONCUP parameter are specified, the NONCUP describer specification takes precedence.
NQSET	Default = 0 If NQSET > 0, and the PARAM entry is in Case Control, all part superelements that do not contain QSET entries, or are not referenced by SENQSET entries in the main Bulk Data Section, have NQSET generalized coordinates assigned to them. These QSET variables are functionally equivalent to those generated by SENQSET entries.
NUMOUT, NUMOUT1, NUMOUT2	See **S1, S1G, S1M**.

Related Topics

Browse Simcenter Nastran documentation

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O – Parameters

OCMP Default = YES

See **POST**.

ODS Default = NO

Requests an efficient RDMODES restart method which reduces the amount of eigenvector restart data.

= NO The efficient RDMODES restart method is not requested.

= YES The efficient RDMODES restart method is used when you run SOL 103 or 111 with RDMODES. As a result, the size of the restart data is smaller and is organized for an RDMODES restart solution to begin efficiently.

OEE Default = YES

See **POST**.

OEF Default = YES

See **POST**.

OEFX Default = YES

See **POST**.

OEPT Default = YES

See **POST**.

OES Default = YES

See **POST**.

OESE Default = YES

See **POST**.

OESX Default = YES

See **POST**.

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OG Default = 0

See **CURV**.

OGEOM Default = YES

See **POST**.

OGPF Default = YES

See **POST**.

OGPS Default = YES

See **POST**.

OIBULK Default = NO

See **PARAM,POST,<0**.

OLDSEQ Default = -1 for non-iterative distributed parallel solutions

Default = 5 for iterative solutions using parallel methods (NASTRAN ITER = 1 and system (231) > 0)

Default = 6 if Super = 2

OLDSEQ selects from the following options for resequencing:

–1: No resequencing is performed.

1: Use the active/passive option.

2: Use the band option.

3: For the active/passive and the band option select the option giving the lowest RMS value of the active columns for each group of grid points.

4: Use the wavefront (Levy) option.

5: Use the Gibbs-King option even if the CPU estimate is higher than for nonsequencing.

6: Use the automatic nested dissection option even if the CPU estimate is higher than for no resequencing. See the following SUPER=2 description.

8: Semiautomatic selection. The program will compute estimates for two options that are suitable for the decomposition method selected by the PARALLEL and SPARSE keywords on the NASTRAN statement and select the option with

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the lowest estimate. The following table shows the suitable options for each decomposition method.

Decomposition Method	Suitable Options
regular	1 and 4
parallel	2 and 5
sparse	6 and 7

9: The Extreme partitioning is selected to generate domains for DMP solutions.

10: METIS with supernodal or grid compression is selected to generate domains for DMP solutions (see restrictions below).

11: MLV with supernodal or grid compression is selected to generate domains for DMP solutions.

110: Metis with supernodal compression is selected to generate domains for DMP solutions.

111: MLV with supernodal compression is selected to generate domains for DMP solutions.

210: Metis with grid compression is selected to generate domains for DMP solutions.

211: MLV with grid compression is selected to generate domains for DMP solutions.

In the GPARTN and GPARTNS modules, if at least two processors are available and both supernodal and grid compression are requested (OLDSEQ=10 or 11), then GPARTN/GPARTNS selects the compression resulting in the smallest global boundary size. If only one processor is available, the default is supernodal compression.

The default OLDSEQ value is 11. Only the MLV is supported for multilevel RDMODES. System(294)=1 requests that diagnostic information be printed to the f06 file.

Note that the wavefront option does not support superelement resequencing or starting nodes. Also note that the automatic nested dissection option uses starting nodes only to establish the root of the initial connectivity tree.

If the value of OLDSEQ is changed in superelement analysis, a SEALL=ALL restart is required.

PARAM,FACTOR is used to generate the sequenced identification number (SEQlD) on the SEQGP entry as follows:

```
SEQlD=FACTOR*GRP+SEQ
```


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where

SEQ = generated sequence number

and

GRP = group sequence number

If GRP=0, use GRP(MAX)+1 where GRP(MAX) is the largest group sequence number in the database.

PARAM,MPCX controls whether the grid point connectivity created by the MPC and MPCADD entries and/or the rigid element entries (e.g., RBAR) is considered during resequencing:

-1: Do not consider the connectivity of the MPC, MPCADD, or rigid element entries.

0: Consider the connectivity of the rigid element entries only. (Default).

>0: Consider the connectivity of the rigid element entries and the MPC and MPCADD entries with the set identification number set to the value of this parameter.

PARAM,SEQOUT controls the output options as follows:

0: Do not generate any printed or punched output for the new sequence. (Default)

1: Print a table of the internal/external sequence in internal order.

2: Write the SEQGP entries to the PUNCH file.

3: Perform SEQOUT=1 and 2.

PARAM,START specifies the number of the grid points at the beginning of the input sequence. The input sequence will be the sorted order of the grid point numbers including the effect of any SEQGP entries input by the user. A single SEQGP entry can be input to select the starting point for the new sequence. Otherwise, the first point of lowest connectivity will be used as the starting point.

If PARAM,SUPER<0, all grid points from the connection table that are not part of the group currently being processed are deleted. This option provides for sequencing only the interior points of a superelement. If any superelements are present, the residual structure is not resequenced. If all the grid points are in the residual structure, they are resequenced.

If PARAM,SUPER=0 or 1, all grid points in the connection table are considered. This option provides for the recognition of passive columns.

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If PARAM,SUPER=2, then all points that are connected to multipoint constraints (via MPC entries) or rigid elements (e.g., the RBAR entry) are placed in a special group at the end of the sequence. This option also forces OLDSEQ=6 and may not be selected with other values of OLDSEQ. This option is intended primarily for models that have many active columns due to MPCs or rigid elements; e.g., a model with disjoint structures connected only by MPCs or rigid elements.

OMAXR

Default = 2*BUFFSIZE

OMAXR specifies the maximum record length of data written by the OUTPUT2 module. OUTPUT2 data is requested when PARAM,POST,<0 is defined. BUFFSIZE is a machine-dependent value defined in the *Simcenter Nastran Installation and Operations Guide*. For further information, see the MAXR parameter description under the OUTPUT2 module description in *Simcenter Nastran DMAP Programmer's Guide*.

OMID

Default = NO

To print or punch the results in the material coordinate system, set the parameter OMID to yes. Applicable to forces, strains, and stresses for CTRIA3, CQUAD4, CTRIA6, CQUAD8, CQUADR and CTRIAR elements. Other elements and outputs are not supported. This capability is not supported by pre-processors (xdb and OP2 output do not change) and grid point stress output which assume output is in the element coordinate system.

See "Using the OMID Parameter to Output Shell Element Results" in the *Simcenter Nastran Element Library*

OMPT

Default = YES

See **POST**.

OP2FMT

Default* = 32

Determines if OUTPUT2 results files are written as 32-bit or 64-bit.

=32, the software writes the OP2 file as 32-bit.

=64, the software writes the OP2 file as 64-bit.

=0, the software automatically checks if a PARAM, POST,*n* entry exists. If *n* is less than 0, the OP2 file is written as 32-bit. If not, the OP2 file is written as 64-bit. See the SOL 402 behavior below.

* Default exceptions:

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- When SYSTEM(525)=1 is defined, the OP2FMT parameter is ignored, and the software writes the OP2 file as 64-bit.
- For SOL 401, when an initial restart run is requested with RSTGEN=YES on the NLCNTLG bulk entry, the OP2FMT parameter is ignored, and the software writes the OP2 file as 64-bit.
- The SOL 402 default is OP2FMT=0. In that case, the solver always writes a 64-bit OP2 and PARAM,POST,*n* is ignored.

  You can define OP2FMT=32 and the software will write the OP2 file as 32-bit. An exception is when SYSTEM(525)=1 is defined, the OP2FMT parameter is ignored, and the software writes the OP2 file as 64-bit.

  Note that restarts and the generation of intermediate results files while SOL 402 is running requires the 64-bit OP2 file.

When you request the 64-bit OP2 file, the integers and floating point data are written out with 64-bit precision. With the increase in precision, the OP2 file size also increases.

OP4FMT — Default = 32

Determines if OUTPUT4 results files are written as 32-bit or 64-bit.

If OP4FMT=32 (default), the OP4 file is always written as 32-bit.

If OP4FMT=64, the OP4 file is always written as 64-bit.

OPG — Default = YES

See **POST**.

OPGEOM — Default = -1

OPGEOM > -1 prints the set definition for all degrees of freedom, including the aerodynamic degrees of freedom. OPGEOM is similar to the USETPRT parameter.

OPGTKG — Default = -1

OPGTKG > -1 prints the matrix for the interpolation between the structural and aerodynamic degrees-of-freedom.

OPPHIB — Default = -1

In the flutter (SOLs 145 and 200) and aeroelastic (SOLs 146 and 200) solution sequences, OPPHIB > -1 and a DISPLACEMENT request in the Case Control Section

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will output the real vibration modes with the structural displacement components transformed to the basic coordinate system.

OPPHIPA Default = -1

In the flutter (SOLs 145 and 200) and the dynamic aeroelastic (SOL 146) solution sequences, OPPHIPA > -1 and a DISPLACEMENT command in the Case Control Section will output the real vibration modes at all degrees-of-freedom, including the aerodynamic degrees-of-freedom.

OPTEXIT Default = 0

In SOL 200, especially during the checkout of the analysis model and the design optimization input data (design model), it may be desirable to exit the solution sequence at certain points before proceeding with full optimization. OPTEXIT may be set to values of 0 through 8 and -4. The DSAPRT Case Control command overrides the specification of PARAM,OPTEXIT,4, -4, or 7. The description of OPTEXIT values follow.

OPTEXIT Value	Description
0	Do not exit. Proceed with optimization.
1	Exit after the initialization of the analysis and design model but before finite element analysis begins.
2	Exit after finite element analysis and initial design response and shape basis vector processing.
3	Exit after design constraint evaluation and screening.
4	Exit after design sensitivity analysis and print the matrix of design sensitivity coefficients (DSCM2). This is equivalent to the DSAPRT (UNFORM,END=SENS) Case Control command.
-4	Exit after design sensitivity analysis and write the data blocks related to sensitivity coefficients (DSCM2 and DSCMCOL) to an external file using the OUTPUT2 and OUTPUT4 modules. This is equivalent to the DSAPRT (NOPRINT,EXPORT, END=SENS) Case Control command. See related parameters ITAPE, IUNIT, and OMAXR.
5	Exit after the first approximate optimization of the design model.
6	Exit after the first update of the analysis model based on the first approximate optimization of the design model.

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OPTEXIT Value	Description
7	Compute and output design sensitivity coefficients at the end of normal program termination: hard convergence, soft convergence, or maximum design cycles. This is equivalent to the DSAPRT (UNFORM,START=LAST) Case Control command.
8	Exit after writing out the data blocks and matrices requested for the Simcenter 3D FE Model Update product. The documentation for this product is included in the Simcenter 3D help.

OPTION

Default = "ABS"

Specifies the modal combination summation method.

"ABS" for the absolute value method.

"SRSS" for the square-root-sum-of-squares method.

"NRL" for the Naval Research Laboratory method.

"NRLO" for the legacy Naval Research Laboratory method.

For more information, see SCRSPEC.

OQG

Default = YES

See POST.

OSWELM

Default=-1

Defines the offset for identification numbers of internally generated m-set constraints for CWELD elements which use the ELEMID or GRIDID formats.
= -1 (default) The numbering starts with SYSTEM(182) + 1. SYSTEM(182) defaults to 100,001,001.

> 0 The numbering starts with OSWELM + 1.

For each CWELD element, two RWELD constraint elements are generated if MSET=ON is specified. See the MSET field on the PWELD bulk entry.

OSWPPT

Default= -1

Defines the offset for identification numbers of the internally generated grids GA and GB for CWELD elements which use the ELEMID or GRIDID formats.

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= -1 The numbering starts with SYSTEM(178) + 1. SYSTEM(178) defaults to 101,000,000.

> 0 The numbering starts with OSWPPT + 1.

For each CWELD element, the grid points GA and GB are generated internally if GA and GB are not specified. See the Bulk Data entry **CWELD**.

OUG — Default = YES

See **POST**.

OUGCORD — See **POST**.

OUMU — Default = YES

See **POST**.

OUNIT1 — Default = Value of OUNIT2

See **POST**.

OUNIT2 — Default = 12

See **POST**.

OUTOPT — Default = 0

See **CURV**.

Related Topics

Browse Simcenter Nastran documentation

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P – Parameters

PANELMP Default = 0

See **FLUIDMP**.

PATVER Default = 3.0

See **PARAM, POST,-1**.

PBRPROP Default = NO

Specifies the method used to calculate cross sectional properties for PBARL and PBEAML entries.
If PBRPROP = YES, the mesh-based Pilkey method is used.
If PBRPROP = NO, hard-coded formulas are used.

For additional information, see the "1D Elements" section of the *Element Library Reference*.

PDRMSG Default = 1

PDRMSG controls the printout of messages associated with deformed plots, including error messages. PDRMSG = 0 suppresses the printout. Contour values will not be displayed unless the default value is used.

PEWOPT Default = 1

Controls the computation of FEMAO SOL 108 and SOL 111 analyses that contain poro-elastic materials with an elastic frame. PEWOPT lets you choose specific wave types or a combination of wave types to be included in the error indicator computation of the FEMAO analysis. The default value 1 provides optimum performance.

This option is available only for FEMAO SOL108 and SOL 111 analyses that contain poro-elastic materials with an elastic frame.

If PEWOPT is:

1, includes only slow compressional waves.

2, includes only slow and fast compressional waves.

3, includes all waves.

4, includes only fast compressional waves.

5, includes only shear waves.

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6, includes only slow compressional and shear waves.

7, includes only fast compressional and shear waves.

Note

In releases prior to Simcenter Nastran 2406, the software includes all waves in the computation. To retain the pre-Simcenter Nastran 2406 behavior, set PEWOPT to 3.

For information about:

- FEMAO, see Finite Element Method Adaptive Order (FEMAO).
- Poro-elastic material, see Defining porous or equivalent porous materials.

PHIBGN Default = 0.0

Beginning azimuth angle (in degrees)

PHIBGN, PHIDEL, and PHINUM define the range of azimuth angle over which the equations of motion with time-dependent coupling terms are solved during a rotor dynamic analysis.

PHIDEL Default = 0.0

Azimuth angle increment (in degrees)

See **PHIBGN**.

PHINUM Default = 1

Number of azimuth angle increments

See **PHIBGN**.

PKRSP Default = -1

If PKRSP=0, the magnitude of the output quantities at the time of peak acceleration of the modal variables is output. The SRSS technique that is used for response spectra is described in “Shock and Response Spectrum Analysis” in the *Simcenter Nastran Advanced Dynamic Analysis User’s Guide*. This option is available only for modal transient analysis.

PLTMSG Default = 1

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PARAM,PLTMSG,0 suppresses messages associated with undeformed plot requests, including error messages.

POST	Default = 1

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PARAM,POST,0, then the following parameters and discussion apply:

The data blocks often used for pre- and post-processing will be stored in the database and also converted, by the DBC module (see the *Simcenter Nastran DMAP Programmer's Guide*) to a format suitable for processing. These data blocks include input data related to geometry, connectivity, element and material properties, and static loads; they also include output data requested through the Case Control commands OLOAD, SPCF, DISP, VELO, ACCE, THERMAL, ELSTRESS, ELFORCE, FLUX, GPSTRESS, GPFORCE, ESE, GPSDCON, and ELSDCON.

The converted data is written to logical FORTRAN units, which may be assigned to physical files in the File Management Section. The FORTRAN unit numbers are specified by the parameters GEOMU, POSTU, and LOADU. By default, all data is written to the logical FORTRAN unit indicated by GEOMU. If LOADU > 0, static load data may be diverted to another unit indicated by LOADU. If POSTU > 0, then output data requested with the Case Control commands listed above will be diverted to the logical unit indicated by POSTU. See "Database Concepts" in the *Simcenter Nastran User's Guide* for the procedure for assigning physical files.

By default, if converted data already exists in the files indicated by GEOMU, POSTU, and LOADU, then the DBC module will overwrite the old data. If this is not desirable, then PARAM,DBCOVWRT,NO must be entered. The parameters MODEL and SOLID may be used to store more than one model and solution in the graphics database. These parameters are not supported by MSC.Patran.

PARAM,DBCDIAG > 0 requests the printing of various diagnostic messages from the DBC module (see *Simcenter Nastran DMAP Programmer's Guide*) during data conversion. By default, no messages are printed.

Simcenter 3D Pre/Post does not support the Simcenter Nastran .xdb file format. We recommend that you use the PARAM,POST,-2 setting with Simcenter 3D Pre/Post. Some Simcenter Nastran .xdb files might import into Simcenter 3D Pre/Post using the MSC Nastran .xdb import option. This depends on the version of Simcenter Nastran used to create the .xdb file and what was defined in the original input file. For example, an .xdb file that includes the new parabolic shell formulation for CQUAD8 and CTRIA6 elements will not import. For this case, you can revert to the pre-Simcenter Nastran 2019.2 parabolic shell formulation by defining SYSTEM(700)=1. See the system cell 700 description for restrictions related to SOLs 106, 129, 401, and 402.

PARAM,POST,<0, then the following parameters and discussion apply:

PARAM,POST,-1 outputs the appropriate files for FEMAP and MSC.Patran. PARAM,POST,-2 outputs the appropriate files for the Simcenter 3D Pre/Post and I-DEAS Dataloader program. PARAM,POST,-4 outputs the files indicated below along with OPHIG for the MSC_NF interface by LMS International. PARAM,POST,-5 outputs the files indicated in the table below along with LAMA and OPHG1 for the FemTools interface by Dynamic Design Solutions. PARAM, POST=-4 and -5 are intended for SOL 103 only.

An OUTPUT2 file for FORTRAN unit 12 in binary format is automatically created in the same directory and with the same name as the input file and with the extension ".op2". For example, if the input file is fender.dat, the OUTPUT2 file will be named fender.op2.

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An ASSIGN statement is required in the FMS Section only if neutral file format is desired as follows:

ASSIGN OP2='filename of FORTRAN file' FORM

Geometry data blocks are output with PARAM,OGEOM,YES (except with PARAM,PATVER<3.0) and are written to a FORTRAN unit specified by PARAM,OUNIT1 (Default = OUNIT2) for PARAM, POST = -1, -2, and -4. PARAM,OUNIT2K (default = 91) specifies the unit number for KELM and KDICT with PARAM,POST,-5. PARAM,OUNIT2M (default = 92) specifies the unit number for MELM and MDICT with PARAM,POST,-5. See the following table for the specific geometry data blocks written for different values for POST.

By default, the EPT and MPT data blocks are output to the .op2 file when either PARAM,POST,-1 or PARAM,POST,-2 or PARAM,POST,-4 is used. PARAM,OEPT,NO can be used to disable the output of the EPT data block to the .op2 file. PARAM,OMPT,NO can be used to disable the output of the MPT data block to the .op2 file.

See also the PARAM,POSTEXT description for additional data blocks written to the .op2 file.

POST				Geometry Data Block	Description
-1	-2	-4	-5		
YES	NO	NO	NO	GEOM1S GEOM1VU	Grid Point Definitions (Superelement)
NO	YES	YES	NO	CSTM	Coordinate SystemTransformations
NO	YES	YES	NO	GPL	Grid Point List
NO	YES	YES	NO	GPDT	Grid Point Definitions
YES	YES	YES	NO	EPT	Element Properties
YES	YES	YES	NO	MPT	Material Properties
YES	YES	YES	NO	GEOM2	Element Definitions
YES	YES	NO	NO	GEOM3	Load Definitions
YES	YES	NO	NO	GEOM4	Constraint Definitions
YES	NO	NO	NO	DIT	Dynamic Table Input
YES	NO	NO	NO	DYNAMICS	Dynamic Loads Definition
NO	NO	YES	YES	KDICT	Element Stiffness Dictionary
NO	NO	YES	YES	KELM	Element Stiffness Matrices
NO	NO	YES	YES	MDICT	Element Mass Dictionary
NO	NO	YES	YES	MELM	Element Mass Matrices

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POST				Geometry Data Block	Description
-1	-2	-4	-5		
NO	NO	NO	NO	ECTS	Element Connections
YES	NO	NO	NO	VIEWTB	View Element Table
YES	YES	NO	NO	EDOM	Design Model Input
YES	NO	NO	NO	GEOM2S GEOM2VU	Same as GEOM2 for superelements
YES	NO	NO	NO	CSTMS	Same as CSTM for superelements
YES	NO	NO	NO	EPTS	Same as EPT for superelements
YES	NO	NO	NO	MPTS	Same as MPT for superelements
YES	YES	NO	NO	CONTACT	Surface contact definition
YES	YES	NO	NO	EDT	Element deformation table
Note: *With PARAM,PATVER,v3.0 (default)					

PARAM,OMACHPR,NO (default) selects the pre-MSC.Nastran Version 69 format for AXIC, BGPDT, CSTM, GEOM1, and GPDT, or the pre-MSC.Nastran 2001 format for EPT and GEOM4. PARAM,OMACHPR,YES selects the current format for AXIC, BGPDT, CSTM, EPT, GEOM1, GEOM4, and GPDT.

For PARAM,POST = -1 and -2, results data blocks are output to a FORTRAN unit specified by PARAM,OUNIT2 (Default = 12). This parameter is allowed to vary between superelements. In buckling solution sequences (SOL 105), a unique value of OUNIT2 should be specified for the buckling subcase. See also the related parameter **OMAXR**.

For PARAM,POST = -1 and -2, PARAM,OIBULK,YES can be defined to include the IBULK datablock. (Default = NO) The IBULK datablock is an unsorted copy of the original bulk data including comments.

The CASECC datablock is always output for both PARAM,POST = -1 and -2.

Note that before Simcenter Nastran 2406, for PARAM,POST = -2, the CASECC datablock was not output when both PARAM,OIBULK,NO and PARAM,OGEOM,NO were defined. Now it is always output.

By default, under PARAM,PATVER, ≥ 3.0, the displacements are output in the global coordinate system. To output in the basic coordinate system, specify PARAM,OUGCORD,BASIC. Under PARAM,PATVER,<3.0, the opposite is true.

PARAM,POST,-1: Results Data Blocks for FEMAP and MSC.Patran

By default, the following data blocks are output under PARAM,POST,-1. The following parameters may be used to disable the output of data blocks to the OUTPUT2 file. For example, if PARAM,OQG,NO is

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specified, then the SPCFORCE output is not written to the OUTPUT2 file. PARAM,PATVER selects the appropriate version of MSC.Patran (Default = 3.0).

PARAM, PATVER		Parameter Name	Case Control	Data Block Name	Description
<3.0	≥3.0				
YES	YES	OQG	SPCFORCE	OQG1	Forces of single-point constraints
YES	NO	OUG	DISP	OUGV1PAT	Displacements in the basic coordinate system
YES	YES	OUG	DISP	OUGV1	Displacements in the global coordinate system
YES	NO	OES	STRESS	OES OESVM	Element stresses (linear elements only)
YES	NO	OEF	FORCE	OEF1	Element forces or heat flux (linear elements only)
YES	YES	OEE	STRAIN	OSTR1	Element strains
YES	YES	OGPS	GPSTRESS	OGS1	Grid point stresses
YES	YES	OESE	ESE	ONRGY1	Element strain energy
YES	YES	OGPF	GPFORCE	OGPFB1	Grid point force balance table
NO	YES	OEFX	FORCE	OEF1X	Element forces with intermediate (CBAR and CBEAM) station forces and forces on nonlinear elements
NO	YES	OESX	STRESS	OES1X	Element stresses with intermediate (CBAR and CBEAM) station stresses and stresses on nonlinear elements
NO	YES	OPG	OLOAD	OPG1	Applied static loads
NO	YES	OCMP	STRESS	OES1C	Ply stresses
NO	YES	OCMP	STRAIN	OSTR1C	Ply strains
NO	YES		DISP SPCFORCE FORCE STRESS STRAIN	OUPV1 OQP1 D OEF1 D OES1 D OSTR1	Scaled Response Spectra
				LAMA	Nonlinear Buckling
NO	YES		DISP OLOAD	OCRUG OCRPG	
NO	YES		NLSTRESS	OESNLXR	Nonlinear static stresses

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PARAM, PATVER		Parameter Name	Case Control	Data Block Name	Description
<3.0	≥3.0				
NO	YES		BOUTPUT	OESNLBR	Slideline stresses
NO	YES		NLLOAD	OPNL1	Nonlinear loads
NO	YES		STRESS	OESNLXD	Nonlinear transient stresses
YES	YES		MODCON	OUGMC OEFMC OESMC OSTRMC OQGMC	Modal contributions
YES	YES		PANCON	OUGPC OUGGC OUGRC	Panel contributions
YES	YES			OVG	XY plotting data for SOL 145 flutter analysis
YES	YES			OAEROF	Aeroelastic forces for SOL 144 and 145
YES	YES			OAEROP	Aeroelastic pressure for SOL 144 and 145
YES	YES			OAERCSHM	Aeroelastic control surface position and hinge moment for SOL 144
YES	YES			OAEROHMD	Aeroelastic hinge moment derivative coefficients for SOL 144
YES	YES			OAEROSCD	Aeroelastic stability and control derivative coefficients for SOL 144
YES	YES			OAEROTV	Aeroelastic trim variable for SOL 144

PARAM,OMACHPR,NO (default) selects the pre-NX Nastran 8.5 format for the MAT10 entry. PARAM,OMACHPR,YES selects the current format for the MAT10 entry.

PARAM,POST,-2: Results Data Blocks for Simcenter 3D Pre/Post and I-DEAS

The displacements are output in the basic coordinate system by default for most solutions. To output in the global coordinate system, specify PARAM,OUGCORD,GLOBAL. An exception is SOL 401 which always outputs displacements in the global coordinate system.

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The following data blocks are output when PARAM,POST,-2 is defined. The parameters listed in the first column can be used to disable the output of data blocks to the OUTPUT2 file. For example, if PARAM,OQG,NO is specified, then the SPCFORCE output is not written to the OUTPUT2 file.

PARAMeter Name	Case Control	Results Data Block Name	Description
	BCRESULTS	OBC OSPDS	Contact Result Output Request
	GKRESULTS	OGK	Gasket element results
	SHELLTHK	OSHT	Shell thickness results
	RMAXMIN	OUGV1MX OEF1MX OES1MX	Displacement, force, and stress extreme values
	OLOAD	OPG1	Applied loads.
	THERMAL	TOUGV1	Temperature output
	NLSTRESS	OESNLXR	Nonlinear stress
OGPS	GPSTRESS	OGS1	Grid point stresses
OGPF	GPFORCE	OGPFB1	Grid point forces
OQG	SPCFORCE	OQG1	Forces of single-point constraint
OUG	DISPLACEMENT	BOUGV1	Displacements in the basic coordinate system
		BOPHIG	Eigenvectors in the basic coordinate system
		OUGV1	Displacements in the global coordinate system
		TOUGV1	Grid point temperatures
OES	STRESS	OES OESVM	Element stresses (linear elements only)
OEF	FORCE	OEF1	Element forces (linear elements only)
	FLUX	HOEF1	Element heat flux
OEE	STRAIN	OSTR1	Element strains
OESE	ESE	ONRGY1	Element strain energy
OCMP	STRESS	OEFIT	Failure indices
	STRESS	OES1C	Ply stresses
	STRAIN	OSTR1C	Ply strains
OUMU	ESE	ONRGY2	Element kinetic energy
OEFX	FORCE	OEF1X	Element forces (nonlinear elements only)
OESX	STRESS	OES1X	Element stresses (nonlinear elements only)
		ODELBGPD	Shape optimization geometry changes

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PARAMeter Name	Case Control	Results Data Block Name	Description
	MODCON	OUGMC OEFMC OESMC OSTRMC OQGMC	Modal contributions
	PANCON	OUGPC OUGGC OUGRC	Panel contributions
	ELSTRN	ELSTRN	Elastic strain at grid points on elements
	GELSTRN	GELSTRN	Elastic strain at Gauss points.
	GSTRAIN	GSTRAIN	Strain at Gauss points.
	GSTRESS	GSTRESS	Stress at Gauss points.
	GTHSTRN	GTHSTRN	Thermal strain at Gauss points.
	OPRESS	OPRESS	Temperature output on grid points.
	OTEMP	OTEMP	Temperature output on grid points.
	THSTRN	THSTRN	Thermal strain at grid points on elements.
		OVG	XY plotting data for SOL 145 flutter analysis
		OAEROF	Aeroelastic forces for SOL 144 and 145
		OAEROP	Aeroelastic pressure for SOL 144 and 145
		OAERCSHM	Aeroelastic control surface position and hinge moment for SOL 144
		OAEROHMD	Aeroelastic hinge moment derivative coefficients for SOL 144
		OAEROSCD	Aeroelastic stability and control derivative coefficients for SOL 144
		OAEROTV	Aeroelastic trim variable for SOL 144

PARAM,OMACHPR,NO (default) selects the pre-NX Nastran 8.5 format for the MAT10 entry. PARAM,OMACHPR,YES selects the current format for the MAT10 entry.

PARAM, POST, -4: Results Data blocks for LMS International/MSC_NF

By default, the following data blocks are output under PARAM,POST,-4. The following parameters may be used to disable the output of data blocks to the OUTPUT2 file. For example, PARAM,OUG,NO requests that eigenvectors not be written to the OUTPUT2 file.

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PARAMeter Name	Case Control	Data Block Name	Description
OUG	DISPLAC	OPHIG	Eigenvectors in the global coordinate system

PARAM, POST, -5: Results Data blocks for Dynamic Design Solutions/FemTools

By default, the following data blocks are output under PARAM,POST,-5. The following parameters may be used to disable the output of data blocks to the OUTPUT2 file. For example, PARAM,OUG,NO requests that eigenvectors not be written to the OUTPUT2 file. PARAM,OUNIT2O (default=51) specifies the unit number of the OUTPUT2 file.

PARAMeter name	Case Control	Data Block Name	Descriptions
OUG	DISPLAC	OUGV1	Eigenvectors in the global coordinate system
		LAMA	Eigenvalue summary

POSTHDF5 Default = 1

For SOL 414 only, requests output in Simcenter data file format (.scd5), which is based on the HDF5 format. You can use the .scd5 files for post-processing and for viewing the contents of the file. The contents of the .scd5 format are easier to understand than the contents of the .des, .fac, and .sdb files created by SOL 414. The .scd5 files are faster to post-process than other file formats, and you can use them in Simcenter 3D Pre/Post as well as in a variety of other post-processors.

If POSTHDF5 = 1, outputs SOL 414 results in .scd5 format in addition to the Samcef .des, .fac, and .sdb file formats.

If POSTHDF5 = 0, outputs SOL 414 results in the .des, .fac, and .sdb file formats.

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POSTEXT Default=YES

Under PARAM,POST,-1 and -2, the following data blocks are also written to the .OP2 file specified by PARAM,OUNIT2. If these data blocks are not required, then specify PARAM,POSTEXT,NO.

Data Block Name	Description
FRL	Frequency response list
BHH	Modal damping matrix
KHH	Modal stiffness matrix
BGPDT*	Basic grid point definition table
PVT0	User parameter value table
EQEXIN(S)	Equivalence external to internal grid ID table
CLAMA	Complex eigenvalue table
OEDE1	Element energy loss output table
OEKE1	Element kinetic energy output table
OUGV2	Displacement output table in SORT2
PSDF	Power spectral density table
OGPWG	Grid point weight generator output table
TOL	Time output list
PSDFH	Power spectral density table for modal coordinates
DSCM2	Design sensitivity coefficient matrix
DSCMCOL	Design sensitivity parameters table
DBCOPT	Design optimization history table
OPHSA	Displacement output table in SORT1
* Always written for PARAM,POST,–2 regardless of POSTEXT.	

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POSTOPT Default = 2

The POSTOPT parameter is only supported by SOL 401 in the context of the Simcenter 3D Multiphysics environment. In this environment, you can request results output at specific time steps. The POSTOPT parameter then controls if Simcenter Nastran writes the requested results after each time step in which output was requested, or at the end of the subcase for the time steps in which output was requested. This control pertains to F06, OP2, and PUNCH file output requests. If a solution is successful at all time steps, the resulting output files will be the same whether you had selected POSTOPT=1 or POSTOPT=2.

1: Simcenter Nastran writes the requested results to the appropriate output files after each time step in which output was requested.

2: Simcenter Nastran writes the requested results to the appropriate output files at the end of the subcase for the time steps in which output was requested.

POSTU Default = -1

See **PARAM, POST,0**.

PREFDB Default = 1.0

See **ACOUT**.

PRGPST Default = NO

Controls the printout of singularities. See **AUTOSPC**.

PRINT Default = YES PARAM,PRINT,NO suppresses the automatic printing of the flutter summary in flutter analysis.

PROUT Default = -1

PARAM,PROUT,-1 suppresses execution and printout from the ELTPRT module. PARAM,PROUT,1 prints a list of all elements sorted on EID and summary tables giving the range of element identification numbers for each element type.

PRPA, PRPJ Default = 1.0E37

PRPA and PRPJ control the printout of intermediate load matrices for diagnostic purposes during superelement assembly. If the value of PRPA (or PRPJ) is positive, all terms larger in magnitude than the value are printed. If the value of PRPA (or PRPJ) is negative, all terms smaller in magnitude than the value are printed. The default value requests no printout. PARAM,IRES,1 must be present for these parameters to be effective. The PA matrix contains the internal loads transmitted to the downstream superelement. The PJ matrix contains external loads applied on the

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superelement; that is, it has the same content as the data produced by the Case Control command OLOAD. All of this data may be obtained on restart using the SELR Case Control command option. A related parameter is IRES.

PRTCSTM — See **CHECKOUT**.

PRTEQXIN — See **CHECKOUT**.

PRTGPDT — See **CHECKOUT**.

PRTGPL — See **CHECKOUT**.

PRTGPTT — See **CHECKOUT**.

PRTMAXIM — Default = NO

PRTMAXIM controls the printout of the maximums of applied loads, single-point forces of constraint, multipoint forces of constraint, and displacements. The printouts are titled “MAXIMUM APPLIED LOADS”, “MAXIMUM SPCFORCES”, “MAXIMUM MPCFORCES”, and “MAXIMUM DISPLACEMENTS”.

PRTMGG — See **CHECKOUT**.

PRTPG — See **CHECKOUT**.

PRTRESLT — Default = YES

PRTRESLT controls the printout of the resultants of applied loads, single-point forces of constraint, and multipoint forces of constraint. The printouts are titled “OLOAD RESULTANTS”, “SPCFORCE RESULTANTS”, and “MPCFORCE RESULTANTS”.

Related Topics

Browse Simcenter Nastran documentation

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Q & R – Parameters

Q — Default = 0.0

Q specifies the dynamic pressure. Q must be specified in aeroelastic response analysis (SOL 146), and the default value will cause a User Fatal Message.

QSETREM — Default = YES

Specifies whether unused q-set DOF are removed when an external superelement is created with the EXTSEOUT case control command.

If QSETREM = YES, unused q-set DOF are removed from the definition of the external superelement.

If QSETREM = NO, all q-set DOF are retained in the definition of the external superelement. If the model is a mechanism, specifying QSETREM = NO is necessary to retain mechanism modes. However, specifying QSETREM = NO can lead to slow processing times for stresses, strains, and forces, especially if a large number of q-set DOF are defined, but are not associated with calculated modes.

RDCNT — Default = YES

Activates RDMODES substructuring for normal modes solutions that include contact.
If RDCNT = YES, use RDMODES substructuring.
If RDCNT = NO, do not use RDMODES substructuring.

For additional information, see the “Recursive Domain Normal Modes Analysis (RDMODES)” section of the *Parallel Processing Guide*.

RDFLUID — Default = NO

Determines if the fluid modes are calculated using the Lanczos method (default) or using the multi-level Recursive Domain Normal Modes method (RDMODES) when the RDMODES method is used with a fluid-structure model.

RDFLUID=YES

The RDMODES method is used to compute the fluid modes. The RDFLUID=YES option is generally beneficial for large models and requesting many modes (1M+ DOF and 1000+ modes).

RDFLUID=NO

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The Lanczos method is used to compute the fluid modes.

RDMTRAK Default = 0.9

Specifies the threshold value for the mode-tracking algorithm in SOL 414,110.

The mode tracking algorithm in SOL 414,110 uses the MAC criterion. If the MAC criterion is above the threshold value specified by the RDMTRAK parameter, the modes are paired. Otherwise, they are not.

RDMTRAK=0.0

Omits generation of the Campbell diagram.

Setting RDMTRAK to 0.0 is useful when you request the Coleman transformation on large models and the memory needed to compute the Campbell diagram is too high for the computer resources. The Coleman transformation supports recombination of 3D cyclic symmetry, so mode shape paring (calculation of MAC criteria) is done on all 3D recombined mode shapes unless you set RDMTRAK to 0.0.

0.0 < RDMTRAK < 1.0

Sets the threshold value.

RDQSOL Default = LANC

Determines if the reduced eigenvalue problem in the RDMODES is calculated using the Lanczos method (default) or using the in-core dense method.

RDQSOL=LANC

The Lanczos method is used to compute the reduced eigenvalue problem. It is suitable for most models.

RDQSOL=INCO

The in-core dense eigensolver is used to compute the reduced eigenvalue problem. This option is generally beneficial for dense models requesting a large number of modes. For example, models that include external superelements.

RDRESVEC Default = YES

Determines if accelerated residual vector calculation option is enabled or disabled when using the multi-level Recursive Domain Normal Modes method (RDMODES). See the *Simcenter Nastran Parallel Processing Guide* for details.

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RDRESVEC=YES
The accelerated residual vector calculation option is enabled, which takes advantage of the rdsparse option, and is more efficient than the original one in terms of computational time and I/O usage. The residual vectors with the accelerated calculation may differ slightly from the original, which cannot be used in conjunction with rdsparse.

RDRESVEC=NO
The accelerated residual vector calculation option is disabled. When disabled, the rdsparse option will be disabled automatically and full results are computed which typically increases runtime considerably.

RDSPARSE — Default = YES

Determines if the sparse eigenvector data recovery option is enabled or disabled when using the DMP, multi-level Recursive Domain Normal Modes method (RDMODES). See the *Simcenter Nastran Parallel Processing Guide* for details.

RDSPARSE=YES
The sparse eigenvector recovery option is enabled. The software automatically determines if the sparse eigenvector recovery option occurs based on the number of DOF in the output request.

RDSPARSE=NO
The sparse eigenvector recovery option is disabled (reverts to the data recovery used before NX Nastran 7). When disabled, full results are computed which typically increases runtime considerably.

RESLTOPT — Default = 8

RESLTOPT's default value provides component-level force summary output for model checkout (PARAM, CHECKOUT, YES), loads generation (OLOAD output), and SPC and MPC forces. Setting RESLTOPT to a value of 1 produces abbreviated output formats only.

RESVEC — Requests residual vectors for DOF associated with applied loads.

The default behavior for requesting residual vectors is described on the **RESVEC** case control command. This behaviour occurs when neither the RESVEC case control command, or any residual vector parameters are defined. To alter the default behaviour, it is recommended that you use the RESVEC case control command, and not the parameter inputs.

If both the RESVEC case control command, and any residual vector parameter inputs are defined, the RESVEC case control command takes precedence, and the parameter settings are ignored.

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If residual vector parameter inputs are defined, and the RESVEC case control command is undefined, the parameters will take precedence over the defaults described on the **RESVEC** case control command.

To turn the residual vector request off, you can either include the case control command RESVEC=NO, or you can include the parameter settings PARAM,RESVEC,NO and PARAM,RESVINER,NO.

The residual vector parameter inputs request the following:

PARAM setting	Description
PARAM,RESVEC,YES	Requests residual vectors for DOF associated with applied loads. Also requests residual vectors for all DOF included in USET,U6 and SEUSET,U6 bulk entries.
PARAM,RESVINER,YES	Requests residual vectors for DOF associated with the 3 translational and 3 rotational rigid body motion inertial loads.
PARAM,RESVSO,YES	Reorthogonalizes static shapes with mode shapes. If the stiffness is well-conditioned, this is not required.
PARAM,RESVSE.YES	Prints strain energy of the static shapes.
PARAM,RESVSLI,YES	Removes linearly dependent shapes.

RESVINER

Requests residual vectors for DOF associated with the 3 translational and 3 rotational rigid body motion inertial loads.

The default behavior for requesting residual vectors is described on the **RESVEC** case control command remarks. This behavior occurs when neither the RESVEC case control command, or any residual vector parameters are defined. To alter the default behavior, it is recommended that you use the RESVEC case control command, and not the parameter inputs.

See related parameter **RESVEC**.

RESVPGF

Default = 1.E-6

This parameter is used to filter out small terms in the residual vectors. The default is 1.E-6, meaning that terms in the residual vectors whose absolute values are smaller than 1.E-6 will get filtered out before they are used. To eliminate filtering of residual vectors, you can set RESVPGF to 0.0.

RESVRAT

Default = 1.E8

Value used to filter out residual vector trial vectors that have little independent content. To filter out more residual vector trial vectors, decrease the value.

RESVSE

Default = NO

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See related parameter **RESVEC**.

RESVSLI — Default = YES

RESVSO — Default = YES

See related parameter **RESVEC**.

RGBEAMA
RGBEAME
RGLCRIT
RGSPRGK

In SOL 401, to compute large displacement effects and thermal expansion, the software internally replaces the RBE2 and RBAR elements with either a stiff beam element or a stiff spring element. A coincident grid tolerance is used to determine if a beam or a spring is used. For the RBAR, if the distance between the connecting grids is less than the tolerance, the stiff spring formulation is used. For the RBE2, if the distance between the grid defined in the GN field on the RBE2 entry, and any of the grids defined in the GM fields on the RBE2 entry, is less than the tolerance, the stiff spring formulation is used. You can optionally define the coincident grid tolerance explicitly with the parameter RGLCRIT. By default, it is automatically computed by the software:

RGLCRIT=Coincident Grid Tolerance = MAX(1E-3*LMODEL, 1E-3) (units=length)

where LMODEL is the largest dimension of the model determined by the software.

You can optionally define the beam stiffness and area explicitly using the parameters RGBEAME and RGBEAMA, respectively. By default, they are automatically computed by:

RGBEAME=Beam Stiffness = 1e+2 * EMAX (units=force/length^2)

RGBEAMA=Beam Area = (LMODEL * 1e-2)^2 (units=length^2)

where EMAX is the largest Young's modulus in the model. If no material is specified in the model, EMAX is set to 1.0E12.

You can optionally define the spring stiffness explicitly using the parameter RGSPRGK. By default, it is automatically computed by:

RGSPRGK=Spring Stiffness = EMAX * LMODEL (units = force/length)

RLOOPNEW — Default = NO

For frequency response and complex eigenvalue rotor dynamic analysis, specify PARAM,RLOOPNEW,YES to include gyroscopic and circulation terms in the mass, damping, and stiffness matrices when the analysis is performed in the fixed reference system.

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RMSINT — Default = LINEAR for the trapezoidal approximation.

RMSINT specifies the interpolation method for numerical integration when computing both root mean square (RMS) and number of zero crossings about the mean frequency (N0) from power spectral density (PSD) function.

RMSINT = LINEAR requests the trapezoidal approximation, which is the existing Simcenter Nastran approach.

RMSINT = LOGLOG requests the log-log interpolation.

RMSSF — Default = 1.0

The value for RMSSF must be greater than 0.0.

RMSSF scales the RMS and CRMS random results. For example, if PARAM,RMSSF,2.0 is specified, the random results are scaled to 2σ. if PARAM,RMSSF,3.0 is specified, the random results are scaled to 3σ.

If the RMSSF describer is specified on a RANDOM case control command, the RMSSF describer takes precedence over the RMSSF parameter.

RMXPANEL — Default = 0

When the RMAXMIN case control command is defined in a SOL 112 run, this parameter can be used to process the data recovery in a series of smaller “panels” rather than the entire output set at once. This will reduce the overall amount of scratch disk space required for the run.

“PSIZE” is the number of time steps to process in a single panel. The high water disk space used will be a function of psize, and not of the total output request. A good initial value for psize is the greater of 100 or the square root of the number of time steps. Setting psize =0 (default) will disable the panel method and process the data recovery for all the time steps at once. The START, END, and RMXTRAN=YES keyword options on the RMAXMIN command are not supported when using RMXPANEL. The RMXPANEL method is not compatible with PARAM,DDRMM,-1.

RMXTRAN — Default = NO

Used in conjunction with the RMAXMIN case control to allow the user to specify whether transient results are output along with the computed RMAXMIN results. If RMXTRAN=NO (the default value), transient results are not output if RMAXMIN is activated. IF RMXTRAN=YES, transient results will be output according to the DISPLACEMENT, STRESS, and FORCE case control entries.

ROTCMRF — Default = 0.0

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Specifies the reference rotor speed that is used to compute the reduced modal basis for a SOL 107 rotor dynamic solve with complex modal reduction. Solutions at other reference rotor speeds are computed from the reduced modal basis. Expect the most accurate rotor dynamic analysis results at reference rotor speeds close to the reference rotor speed specified with ROTCMRF.

If rotor superelement reduction is requested with ROTSE bulk entries, unsymmetric a-set reduction is performed at the reference rotor speed specified with ROTCMRF.

The units for the reference rotor speed specified with ROTCMRF are the same as those specified in the RUNIT field of the ROTORD bulk entry.

ROTCOUP — No default

Specifies the coupling points for each rotor in a SOL 107 – 112 rotor dynamic analysis.

- If the model contains a single rotor, specify the coupling point with PARAM,ROTCOUP,*gridid*, where *gridid* is the grid ID of the coupling point.
- If the model contains multiple rotors, specify the coupling points with PARAM,ROTCOUP,*setid*, where *setid* is the identification number of a SET case control command that lists the grid ID of the coupling point for each rotor.

For a grid listed on the ROTCOUP parameter specification or in a referenced SET command to be a valid coupling point, it must also be listed on a ROTORB bulk entry.

ROTCSV — No default.

Defines the unit number of a CSV (.csv) comma-separated ASCII file to which Campbell diagram results from a complex eigenvalue rotor dynamics analysis are written. Requires:

```
ASSIGN OUTPUT4 = 'filename,csv', UNIT = unit_number, FORM
= FORMATTED
```

ROTCSV is not supported in SOL 414 rotor dynamic analysis.

ROTGPF — No default.

Defines the unit number of a GPF (.gpf) file to which Campbell diagram results from a complex eigenvalue rotor dynamics analysis are written. Requires:

```
ASSIGN OUTPUT4 = 'filename.gpf', UNIT = unit_number, FORM
= FORMATTED
```


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ROTGPF is not supported in SOL 414 rotor dynamic analysis.

ROTSYNC Default=YES

When ROTSYNC=NO, synchronous analysis is not performed in a rotor dynamic complex eigenvalue analysis.

ROTSYNC is not supported in SOL 414 rotor dynamic analysis.

RPOSTS1 Default = 0

Specifies the output format for random results. Set to 0 for SORT2 and 1 for SORT1.

If the RPOSTS1 describer is specified on a RANDOM case control command, the RPOSTS1 describer takes precedence over the RPOSTS1 parameter.

RSATT Default = NO

PARAM,RSATT,YES specifies that if static loads or U3 USET bulk entries are present, SOL 103 will compute attachment modes. RSATT is only used by Simcenter 3D Response Dynamics. The attachment modes are written to the OP2 file when PARAM,RSOPT,1 and PARAM,POST,-2 are defined.

RSCON Default = NO

PARAM,RSCON,YES specifies that if U2 USET bulk entries are present, SOL 103 will compute constraint modes. RSCON is only used by Simcenter 3D Response Dynamics. The constraint modes are written to the OP2 file when PARAM,RSOPT,1 and PARAM,POST,-2 are defined.

RSOPT Default = 0

PARAM,RSOPT,1 and PARAM,POST,-2 specifies that extra data blocks associated with Simcenter 3D Response Dynamics are written to the OP2 file.

RSPECTRA Default = -1

RSPECTRA = 0 requests that response spectra be calculated for transient analysis. See “User Interface for Response Spectra Generation” in the *Simcenter Nastran Advanced Dynamic Analysis User’s Guide*. Response spectra will be calculated for any superelements or the residual structure for which other output requests are present in the same run. Any punch data produced is sent to the standard Simcenter Nastran PUNCH file. Related parameters are TABID and RSPRINT.

RSPRINT Default = 0

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RSPRINT controls the printout of tabulated values of response spectra. RSPRINT = -1 suppresses the printout. The related parameter is RSPECTRA.

RVMASS Default = NO

In a residual vector calculation, PARAM,RVMASS,YES adds a small mass to DOF that has stiffness but no mass. The addition of the small mass does not affect the mass of the model.

RVMTV Default = NO

Requests the addition of modal truncation vectors to the set of residual vectors for a component mode reduction of an external superelement. To initiate component mode reduction, include q-set DOF to represent the reduced modes.

PARAM,RVMTV,YES adds modal truncation vectors to the set of residual vectors.

PARAM,RVMTV,NO (default) omits modal truncation vectors from the set of residual vectors.

The software produces six residual vectors due to the inertial loads. The six residual vectors describe an overall rigid body motion of the component. When you request component mode reduction, as a best practice, specify PARAM,RVMTV,YES. Doing so produces mass-related residual vectors for the boundary DOF, which ensures that acceleration is transmitted through the component correctly.

Related Topics

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S – Parameters

S1G, S1M Default = -1

PARAM,S1i,+1 requests the sorting and/or filtering of the element stresses selected on the DTI,INDTA entry. Stresses at grid points (S1G) and/or in material coordinate systems (S1M) based on the parameters BIGER, NUMOUT, SRTOPT, and SRTELTYP may be requested. The S1G and S1M options also require the presence of PARAM,CURV,1.

Parameter	Quantity	Coordinate System	Location	Elements
S1M≥0	Stresses	Material	Element centers	CQUAD4, CTRIA3
S1G≥0	Stresses	Material	Grid points to which elements connect	CQUAD4, CTRIA3

NUMOUT, in conjunction with BIGER, controls the amount of stress output.

1. NUMOUT = +N requests that N element stresses be printed (or punched) for each element type.
2. NUMOUT=0 outputs all elements in a group when one or more exceeds BIGER. Some of the elements will have stresses smaller than BIGER. This is conceptually the same as describing an element set in case control, and limiting output in this manner. Stress files obtained with element group filtering may be used for xy plotting and other postprocessor options with DMAP alters. By contrast, the stress file when NUMOUT = -2 is more discontinuous, and may not be used for xy plotting.
3. NUMOUT = 0 does not sort but filters according to BIGER by element group. In static analysis, an element group is defined as all case control selected elements for a given load case for SORT1 output. For SORT2 output, an element group is defined as the data for a given element type for all load cases. In transient analysis, an element group is defined as all case control selected elements at a given time for SORT1 output. For SORT2 output an element group is defined as the data for a given element at all time steps. The element group option applies only to output types described above for PARAM,S1. This option is not available with output types selected by PARAMs S1G and S1M.
4. NUMOUT = -1 requests that stresses be sorted and only those stresses with an absolute value that is greater than BIGER will be output.
5. NUMOUT = -2 (the default) does not sort but filters according to BIGER. Related parameters include BIGER, NOELOF, NOELOP, and NOGPF.

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BIGER controls the elements for which stresses will be printed. Elements with stresses that are smaller in absolute value than BIGER will not be output. The quantity tested is element type dependent. Related parameters include CURV, NUMOUT, S1G, and S1M. SRTOPT controls the scanning option to be performed.

SRTOPT Value	Description
0	Filter/sort on maximum magnitude.
1	Filter/sort on minimum magnitude.
2	Filter/sort on maximum algebraic value.
3	Filter/sort on minimum algebraic value.

SRTELTYP controls the element type to be processed, as described in the following table.

SRTELTYP Value	Description
0	All element types will be processed.
> 0	Only element type SRTELTYP will be processed.

NUMOUT1 and BIGER1 serve the same function as NUMOUT and BIGER except that they apply only to composite element stresses and do not require PARAM,S1i,=1.

NUMOUT2 and BIGER2 serve the same function as NUMOUT and BIGER except that they apply only to composite element failure indices and do not require PARAM,S1i,=1.

S1AG, S1AM

Default = -1

See **CURV**.

SCRSPEC

Default = -1 (SOLs 103, 115 and 187 only)

SCRSPEC=0 requests that structural response be calculated for response spectra input in normal modes analysis. See “Response Spectrum Application” in the *Simcenter Nastran Advanced Dynamic Analysis User’s Guide* for a discussion of this capability. The scaled response calculations are made for elements and grid points of the residual structure only. The responses are summed with the ABS, SRSS, NRL, or NRLO convention, depending on the value of PARAM,OPTION. If the SRSS, NRL, or NRLO options are used, close natural frequencies will be summed by the ABS convention, where close natural frequencies meet the inequality

$f_{i+1} < \text{CLOSE} \cdot f_i$

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Both PARAM,OPTION and PARAM,CLOSE may be set in any subcase, allowing summation by several conventions in a single run.

In pre-Simcenter Nastran 1.0, the NRL option was modified slightly to correspond to the NAVSEA-0908-LP-000-3010 specification. NRLO provides the V69 NRL.

SDAMPUP Default = NO

In a modal frequency response solution (SOL 111) which includes elements with frequency dependent stiffness (PELAST and PBUSHT entries), determines if natural frequencies are updated before modal damping is computed.

= NO Natural frequencies and eigenvectors are not recomputed using the frequency dependent stiffness. Instead, they are computed one time, using the nominal stiffness values defined on PELAS or PBUSH. Modal damping is then computed using these less accurate frequencies.

= YES The software updates the frequency-dependent modal stiffness and recomputes the natural frequency and eigenvector while remaining in modal coordinates (h-set). This more accurate frequency is used to compute the modal damping for each frequency response calculation. This option improves accuracy, but may also increase runtime substantially, particularly if the number of modes is large. See Damping in Modal Frequency Response in the *Basic Dynamic Analysis User's Guide*.

SECOMB Default = NO

By default, the results from superelements will not be combined for a superelement analysis. By specifying PARAM,SECOMB,YES, the results from superelements will be combined for a superelement analysis with the following conditions/restrictions:

- Instead of outputting results by subcase, the output results for all subcases will be unioned and output with the residual subcase. Results that are excluded are: (1) any XY results, (2) MPF (i.e. modal participation factors), (3) modal energies, and (4) SDISPLACEMENT/SVELOCITY/SACCELERATION.
- Output for each superelement is still defined based off of the output request specified in the superelement subcase. If the constraint mode method of enforced motion is being used and relative displacement/velocity/acceleration results are requested, the requests specified for the residual will take precedence when SECOMB is set to YES.
- This applies to both internal and external superelements defined in both the main bulk data and as a partitioned superelement.
- If partitioned superelements have grid or element labeling conflicts, PARAM,SECOMB,YES is not valid and results combination will not be allowed.

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- If complex displacement, velocity, or acceleration output is requested, the requests must be all real/imaginary or all magnitude/phase. If a mix of real/imaginary and magnitude/phase is requested, results combination will not be allowed.
- PARAM,OMID,YES is not valid if PARAM,SECOMB,YES is valid.
- If the OMODES case control command is used, OMODES must be the same for all superelements.

SEMAP Default = SEMAP

SEMAPOPT Default = 42

SEMAPPRT Default = 3

The superelement map (SEMAP table) contains several lists useful for determining how the program has partitioned superelement models. It is printed automatically each time this table is generated. It consists of three major parts:

GPM – The Grid Point Map contains a list of each grid point, its interior superelement placement, and the SEID of all grid points connected directly to it by elements. Three tables follow that summarize the connectivity between superelements sorted on grid point sequence, SEID, and the number of connections.

ISM – The Individual Superelement Map lists the interior grid points, exterior grid and scalar points, elements, and time and storage space estimates for each superelement.

SDT – The Superelement Definition Table contains the SEID of every superelement in the model, the processing order, and a pictorial representation of the superelement tree.

SEMAP, SEMAPOPT, and SEMAPPRT are used to control the amount of output that is printed and other special features. The possible values for SEMAP are shown in the following table.

SEMAP Value	Output and Application
SEMAP (Default)	ISM, SDT. The lengthy GPM is suppressed. This is the appropriate value for use after the model is stable and only minor changes are to be made.
SEMAPALL	GPM, ISM, SDT. All tables are printed. This value is useful on the initial debug run of a model and when making extensive modeling changes.

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SEMAP Value	Output and Application
SEMAPCON	Only the summary tables of the GPM and the estimation data is output. This is a useful value when iterating to an economic partitioning scheme for large, complex models.
SEMAPEST	Only the estimation data is printed. This is useful when evaluating several alternative partitioning schemes.
SEMAPPUN	No output is printed. The exterior grid points of the superelement with a SEID that is input on SEMAPOPT are placed on a CSUPER entry image on the PUNCH file, allowing the superelement to be used as an external superelement. If SEMAPOPT > 0, the superelement entry is given an SSID of SEMAPOPT. If SEMAPOPT < 0, the exterior points listed are those of the residual structure, but the CSUPER entry is given an SSID of |SEMAPOPT|.
Other special features are available with parameters SEMAPOPT and SEMAPPRT. They are fully described under parameters OPT1 and OPT2 in the description of the TABPRT module in the *Simcenter Nastran DMAP Programmer's Guide*. If the default value of SEMAP is used, the other two parameters may be used to further refine this output, as described in the *Simcenter Nastran DMAP Programmer's Guide* under the TABPRT module description. The printing of the SEMAP table can be avoided by the use of PARAM,SEMAPPRT,-1.	

SEMATPRT Default = 0

Controls which external superelement reduced matrices are written to the .f06 file in a SOL 414,103 superelement creation run and when you edit a superelement.

0: None

1: Reduced matrices

2: Reduced matrices and element load matrix

The reduced matrices depend on whether the reference frame for the rotor dynamic analysis is fixed or rotating.

- For a fixed reference frame, reduced stiffness, mass, gyroscopic, circulatory forces, and either viscous or structural (hysteretic) damping matrices are written to the .f06.
- For a rotating reference frame, reduced stiffness, mass, Coriolis, centrifugal, and either viscous or structural (hysteretic) damping matrices are written to the .f06.

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SENSUOO Default = NO

By default, in dynamic sensitivity analysis in SOL 200, displacements at the o-set due to pseudo-loads do not include any effect due to inertia and damping. If PARAM,SENSUOO,YES is specified then these effects will be computed in a quasi-static manner and included in the sensitivity analysis.

SEOP2CV Default = 0

Use to override the effect of PARAM,OP2FMT when the .op2 file is created by the EXTSEOUT, FRFOUT, ADAMSMNF (ADMOUT = YES), and MBDEXPORT (RECVROP2 = YES) case control commands.

0: No data conversion. All data in 64-bit format.

1: Convert data to 32-bit format.

2: Allow PARAM,OP2FMT setting to control .op2 file format.

SEP1XOVR Default = 0

The old and new location of moved shell grid points are printed if SEP1XOVR = 16 when the RSSCON shell-to-solid element connector is used. By default, the moved shell grid points are not printed, SEP1XOVR = 0. See the description of **TOLRSC** for more details.

SEQOUT Default = 0

See **OLDSEQ** .

SERST Default = AUTO (structured solutions only)

By default, in the structured solutions, all restarts are considered automatic (see “Restarts”in the *Simcenter Nastran User’s Guide*). If none of the following Case Control commands are entered, then SEALL=ALL is the default action: SEMG, SELG, SEKR, SELR, SELA, SEMA, SEMR, and SEALL.

The user may wish to partition the analysis into several runs, in which case the above commands may be used in the same manner they are used in the unstructured solutions. By default, the restart will proceed in automatic fashion for each command, regenerating only that data that is affected by modifications in the Bulk Data and Case Control or changes in upstream superelements. If the user wishes to overwrite the data, even if it is not affected by modifications to the data, then PARAM,SERST,MANUAL must be entered.

With PARAM,SERST,AUTO or MANUAL, all superelements will be processed through Phase 0 (see “Superelement Analysis” in the *Simcenter Nastran User’s Guide*). This

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phase includes execution of the sequencer module (SEQP), initial superelement processing (SEP1), and initial geometry processing (GP1 and GP2) modules, which can result in significant CPU overhead. If this overhead is not desired, then PARAM,SERST,SEMI will limit Phase 0 and Phase 1 to only those superelements specified on the SEMG, SELG, SEKR, SELR, SELA, SEMA, SEMR, and SEALL Case Control commands. If none of these commands is entered, then execution will skip Phase 0 and 1.

In the modal solution sequences (SOLs 110, 111, 112, 145, 146, and 200), the modes of the residual structure are automatically computed in Phase 2 if any SE-type command (e.g., SEMG=n) is requested for the residual structure. If PARAM,SERST,SEMI and no SE-type command is specified for the residual structure, then, by default, its modes will not be recomputed. This logic is intended for restarts from SOL 103 into one of the modal solutions. If, however, the modes have not already been computed or need to be recomputed, then PARAM,SERST,RSMDS must be specified to force the calculation of the residual structure modes.

If PARAM,SERST,SEDR is specified, then Phases 0, 1, and 2 will be skipped. This option is intended for data recovery (Phase 3) runs only.

The options of SEMI, RSMDS, and SEDR are intended for models that are defined on more than one database; i.e., superelements are defined on separate databases (multiple MASTER DBsets) and processed in separate runs. Also, with this technique, databases are attached with the DBLOCATE File Management statement rather than the RESTART File Management statement. In general, these options are not recommended because they disable the automatic restart capability, which compromises the database integrity to the same degree as in the unstructured solution sequences.

SESDAMP Default = NO

Modal damping is calculated for superelements if PARAM,SESDAMP,YES is specified. An SDAMPING case control command that selects a TABDMP1 bulk entry must also be specified in the subcase of the desired superelement. By default, modal damping is added to viscous damping (B). If you insert PARAM,KDAMP,-1 (or PARAM,KDAMPFL,-1 for fluid superelements), modal damping is added to structural damping (K4).

In SOL 414,103, for a superelement of a rotor calculated in the rotating frame, or a superelement of a non-rotating part calculated in the fixed reference frame, with boundary conditions, and modal damping defined by SDAMPING and TABDMP1, if you define SESDAMP=YES, the software edits and updates the damping matrix of the superelement.

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Note

The PARAM, KDAMP parameter does not apply to SOL 414 solutions. To add modal damping to a superelement damping matrix, use PARAM, SESDAMP,YES.

SESEF

Default = -1 (SOLs 103 and 115 only)

If SESEF = 0 in superelement normal modes analysis, the fraction of total strain energy for a superelement in each of the system's modes is output in the vector SESEFA for tip superelements and in SESEFG for nontip superelements. If SESEF = 1, strain energy fractions are output, and expansion of the eigenvectors from a-set size to g-set is branched over for tip superelements. If SESEF = -1 (the default value), no strain energy fractions are computed.

Output requests must be present in order for strain energy fractions to be calculated. If SESEF = 1, there will be no other output results for tip superelements.

SFEF70

Default = NO

Requests the import of a fluid-structure coupling matrix created by SFE AKUSMOD when YES. When NO, the coupling matrix is not imported.

SHIFT1

Default=-1.234

This sets the negative shift used when computing massless mechanism constraint modes. For a very stiff model (1000. hz for the first flexible mode), consider using a larger value.

SHLDAMP

Default=SAME

Determines if the structural damping coefficient (GE) defined on the PSHELL MIDi materials are used.

SAME: The structural damping coefficient (GE) defined on a PSHELL MID1 material will be used by all MIDi for that PSHELL.

DIFF: The structural damping coefficient (GE) defined on each PSHELL MIDi will be used. When each PSHELL MIDi is used as described, any structural damping coefficient (GE) values which are blank default to zero.

SIGMA

Default = 0.0

The radiant heat flux is proportional to SIGMA $\cdot (T_{grid} + T_{ABS})^4$, where SIGMA is the Stefan-Boltzmann constant, T_{grid} is the temperature at a grid point, and T_{ABS} is the scale factor for absolute temperature and may be specified by PARAM,TABS. These parameters must be given in units consistent with the rest of the data in the model.

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The value for SIGMA in SI units is 5.67×10^{-8} watts/m$^2K^4$. The default value causes radiant heat effects to be discarded.

SKPAMP Default = 0

For SOLs 145, 146, and 200, SKPAMP = -1 suppresses all unsteady aerodynamic calculations. The automatic restart of the structured solution sequences performs a similar function without this parameter. Specifying it ensures suppression of the calculations, regardless of the determination of the automatic restart.

SLOOPID Default = 0 (SOL 129 and 159 only)

In a nonlinear transient analysis (SOLs 129 and 159) restart, SLOOPID identifies the initial conditioning previous nonlinear analysis run (SOLs 106 and 153 respectively). Setting SLOOPID greater than 0 will cause SOLs 129 and 159 to start from the static deformed position.

SLOSHCMS Default = 0

In a SOL 103 or SOL 111 analysis that uses the defined fluid mass method to model an incompressible fluid volume with a sloshing (vibrating) surface, specifying PARAM,SLOSHCMS,n does the following:

- If n > 0, the software performs component mode synthesis (CMS) reduction on the sloshing surface and retains n modes.

  For models that contain multiple incompressible fluid volumes with sloshing surfaces, n modes are retained for each sloshing surface.
- If n = 0, the software does not perform CMS reduction on the sloshing surfaces.

When the SLOSHCMS field on the INCMPFL bulk entry for an incompressible fluid volume with a sloshing surface is specified along with a PARAM,SLOSHCMS,n specification, for the sloshing surface associated with the INCMPFL bulk entry, the SLOSHCMS field specification takes precedence.

For example, if your model contains multiple incompressible fluid volumes with sloshing surfaces and you want to perform CMS reduction on only one of them, you can enter a value in the SLOSHCMS field of the INCMPFL bulk entry associated with the sloshing surface where you want to perform the CMS reduction and specify PARAM,SLOSHCMS,0 or omit the SLOSHCMS parameter specification.

PARAM,SLOSHCMS,n is applicable when using subspace iteration, dry modes plus Ritz vectors, or ARPACK to solve a defined fluid mass problem.

SMALLQ Default = 1.0E-10

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By default, Simcenter Nastran does not remove unused or excess q-set degrees-of-freedom. If you are performing component mode synthesis, as a best practice, specify PARAM,SMALLQ,0.0 to remove all of the unused q-set degrees-of-freedom.

SNORM

Default = 20.0

When multiple shell elements are connected at a grid point and the shell element topology is curved, the local shell element normal for each element connected to that grid point will be different. The SNORM parameter is a threshold to determine if the software should use a single vector average of the local shell element normals at that grid point, or use the local normals such that each element has its own normal at the grid point.

Average grid point normals are generated for the CQUAD4, CTRIA3, CQUADR, and CTRIAR elements. The computation of a average grid point normal includes all shell elements including the CQUAD8 and CTRIA6 elements. The resulting grid point normal is used in all element calculations of the CQUAD4, CTRIA3, CQUADR, and CTRIAR elements.

When SNORM > 0.0, the software procedure is as follows (see **Figure 6-8**):

1. Find the shell elements connected to a grid point. As an example, assume there are four elements.

2. Compute the local normal of each element at the grid point corner such that you have norm_1, norm_2, norm_3, and norm_4.

3. Compute a single average vector of all the normals (average of norm_1, norm_2, norm_3, and norm_4).

4. Compute the angle between each norm_i vector and the single average vector.

5. If all of these angles is smaller than SNORM, the software uses the single average vector in all element calculations of the CQUAD4, CTRIA3, CQUADR, and CTRIAR elements.

If you have requested SNORMPRT>0, if any of these angles is larger than SNORM, the software prints "NOT CALCULATED".

For SOL 401, if a grid point is on a free shell element edge (that is, not enclosed by shell elements), the software does not apply the SNORM threshold check described above. If you have requested SNORMPRT>0, the software prints "NOT CALCULATED" for these locations.

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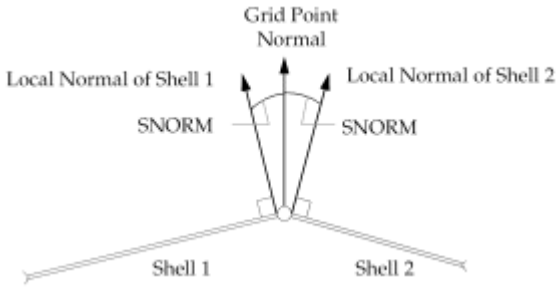
SNORM	Tolerance in Degrees
> 0.0	The shell normal check described above occurs for the CQUAD4, CTRIA3, CQUADR, and CTRIAR elements. If you have defined SNORM bulk entries, those normals overwrite a software generated normal.
= 0.0	The shell normal check is turned off. You can define normals with the SNORM bulk entry.
< 0.0	The shell normal check is turned off. SNORM bulk entries are ignored.

Figure 6-8: Average Grid Point Normal for Adjacent Shell Elements



In SOL 402, SNORM > 0.0 can also request the generation of unique grid point tangents for adjacent CBEAM, CBAR, and CROD elements.

If SNORM = 0.0, the SNORM bulk entry is ignored.

SNORM = 0.0 or SNORM < 0.0 have the same effect.

SNORMPRT Default = -1

PARAM,SNORMPRT,>0 writes the grid point normals of the model in the basic coordinate system to the.f06 and/or.pch files.

SNORMPRT	Switch to Print Out Normals
≤ 0	No output.
1	Print out to the punch file (.pch).
2	Print out to the print file (.f06).
3	Print out to the punch (.pch) and print file (.f06).

SOFTEXIT Default = NO

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In SOL 200, if soft convergence is achieved during optimization, before completing the maximum number of design iterations, the user may request an exit with PARAM,SOFTEXIT,YES.

SOLID — Default = 0

SOLID also allows several models to be stored in the same graphics database created by PARAM,POST,0.

SOLVSUB — Default: 1

Valid only when PARAM,CORROPT,YES exists.

0: Skip eigenvalue solution.

1: Perform eigenvalue solution.

SPCGEN — Default = 0

SPCGEN requests the creation of SPCi bulk entries from the identified singularities. See **AUTOSPC**.

SPCSTR — Default = NO

Used for fluid/structure analysis only.

When surface vibrations (displacement, velocity, or acceleration) on the structure are known and defined as an input excitation, you may only need the structural DOFs in order to enforce this loading to the acoustic mesh. You can set the SPCSTR parameter to YES to constrain the structural DOFs in the analysis set, but only after the structural excitation is transferred over to the fluid.

All structural DOFs are constrained by using SPCs, and SPCDs are applied as the enforced motion.

For SOL 108 Direct Frequency Response, when the model consists of FRF components only, and vibrations (displacement, velocity, or acceleration) are defined on all component DOFs (SPC or SPCD), the SPCSTR parameter must be set to YES. All structural DOFs are constrained and the SPC/SPCDs are applied as enforced motion. If the SPCSTR parameter is not set to YES, a fatal error occurs.

SRCOMPS — Default = NO

SRCOMPS controls the computation and printout of ply strength ratios. If SRCOMPS = YES, ply strength ratios are output for composite elements that have failure indices requested.

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For shell composites, when PARAM,SRCOMPS,YES is defined, the software computes both the failure index and the strength ratio.

For solid composites, when PARAM,SRCOMPS,YES is defined, the software only computes the strength ratio, and not both the failure index and the strength ratio.

SRTELTYP Default = 0

See **S1, S1G, S1M**.

SRTOPT Default = 0

See **S1, S1G, S1M**.

STABAO Default = NO

In a FEM Adaptive Order (FEMAO) acoustic analysis, the parameter STABAO requests a stabilization of the Gauss points for linear and pseudo-parabolic elements with order=1 (excluding the CPYRAM element). The stabilization can improve the solution accuracy by reducing the accumulation of dispersion error at higher frequencies (pollution effect).

=NO The stabilization is off.

=YES The stabilization is on.

STABFEM Default = NO

In a FEM acoustic or vibro-acoustic analysis, the parameter STABFEM requests a stabilization of the Gauss points for linear elements with order=1 (excluding the CPYRAM element). The stabilization can improve the solution accuracy by reducing the accumulation of dispersion error at higher frequencies (pollution effect).

=NO The stabilization is off.

=YES The stabilization is on.

START Default = 0

See **OLDSEQ** .

STIME Default = 0.0 (SOLs 129 and 159 only)

In SOLs 129 and 159 restarts from previous SOLs 129 and 159 runs, the user provides t_N , where t_N is the last time step of the subcase to be continued with a new or changed subcase in the new run. Thus, the loading and printout will start from t_N as though the original run contained the new subcase data.

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STRUCTMP	Default = 0 See **FLUIDMP**.
SUBID	Default = 1 (SOLs 106 and 153) In SOL 106, by default, the restart proceeds from the last LOOPID in the last subcase. SUBID may be used to specify an earlier subcase by specifying the sequential number (for SEID = 0) of the subcase. In SOLs 106 and 153, PARAM,LOOPID may also be specified for an earlier LOOPID.
SUPER	Default = 0 (nonsuperelement sequences) Default = -1 (superelement sequences) See **OLDSEQ**.
SWPANGLE	Default = 1.0 The angular increment in degrees at which failure indices and strength ratios are computed and output for laminates in SOL 108 and 111.

Related Topics

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T – Parameters

TABID

Default = 2

TABID controls the punch output for response spectra. A related parameter is **RSPECTRA**.

TABS

Default = 0.0

TABS is used to convert units of the temperature input (°F or °C) to the absolute temperature (°R or °K). Specify:

PARAM,TABS,273.16	When Celsius is used
PARAM,TABS,459.69	When Fahrenheit is used

For SOL 153 and 159 models which include radiation, the software uses PARAM,TABS to convert input temperatures to the absolute scale for the radiation computation. Note that the resulting output temperatures are consistent with the input temperature units, and do not include PARAM,TABS.

For SOL 401, if TABS is undefined, TABS=0.0 is used.

For SOL 402, if TABS is undefined, the default depends on the temperature units on PARAM,UNITSYS ("C"=Celsius; "F"=Fahrenheit; "K"=Kelvin; "R"=Rankine) or the length units if temperature units are undefined ("m", "cm" or "mm" = Celsius; "in" or "ft" = Fahrenheit). If PARAM,UNITSYS is undefined, TABS=0.0 is used.

Refer to the Bulk Data entry, **"CREEP"** for a creep analysis with SOLs 106 or 153. Refer to PARAM,SIGMA for heat transfer analysis.

TENSOQD

Default = -1

For SOL 401, controls if the tension-only shell element conversion behavior from a shell to a shear panel is on or off. The tension-only shell element initially behaves in shell element status. The software converts the element to shear panel status according to the conversion threshold you define in the SC and SDIR fields on the PSHLPNL bulk entry. By default, the shell to shear panel conversion behavior is on. You can define PARAM,TENSOQD,1 to turn the conversion behavior off, and the element remains as a shell element.

-1 (default): The conversion behavior is on.

1: The conversion behavior is off.

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TESTNEG

Default = –2 for Newton's method in SOLs 106 and 153

Default = 1 for Arc-length method in SOLs 106 and 153

In nonlinear static analysis, this parameter specifies the action to take when negative terms are encountered on the factor diagonal of matrix decomposition. Negative terms indicate that the differential stiffness has introduced a structural instability. The instability may be real (structural buckling) or mathematical (the current iteration appears unstable, but a stable solution exists).

TESTNEG	Results
-1	Stop if negative terms occur.
1 or 0	Continue if negative terms occur.
-2	If negative terms exist, do not use differential stiffness.
2	Do not use differential stiffness.

TINY

Default = 1.E-3

Any elements with strain energy that is less than a TINY percentage of the total strain energy for any superelement will not be printed or made available for postprocessing. TINY may also be used to suppress the printing of small numbers in the constraint check matrix $[E_{mh}]$ described in "Geometry Processing in SubDMAP PHASEO" in the *Simcenter Nastran User's Guide*.

TIREAPP

Default = NO

Applicable for a SOL 111 modal frequency response analysis of a rotating structure when the equations of motion of the structure are written in the rotating reference frame.

Not applicable for SOL 414,111.

YES: Include in the equations of motion Coriolis and spin softening terms due to rotation and, if applicable, differential stiffness from a preceding static subcase.

NO: Use SOL 111 as follows:

- If a ROTORD bulk entry is present, perform a SOL 111 rotor dynamic modal frequency response analysis as described in the *Simcenter Nastran Rotor Dynamics User's Guide (legacy)*.

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Note

In a SOL 111 rotor dynamic modal frequency response analysis, the equations of motion can include differential stiffness that arises due to centrifugal force only.

- If a ROTORD bulk entry is not present, perform a standard SOL 111 modal frequency response analysis as described in the Modal Frequency Response Analysis (SOL 111) section of the *Simcenter Nastran Basic Dynamic Analysis User's Guide*.

TOLRSC Default = 0.05

When the RSSCON shell-to-solid element connector is used, the connecting grid points of the shell element are moved on to the solid face if the grid points are close enough. The tolerable distance of the shell grid point to the solid edge or face is $\epsilon \cdot h$ where h is the height of the solid edge; see the sample figure below. The relative tolerance is user modifiable using the parameter.

```
PARAM,TOLRSC,  ε
```

The default for the relative tolerance is $\epsilon = 0.05$. Rigid body invariance is satisfied with double-precision accuracy if the shell grid points are adjusted.

TOQDGAUS Default = 0

For SOL 401 only, the tension-only shell element initially behaves in shell element status. The software converts the element to shear panel status according to the conversion threshold you define in the SC and SDIR fields on the PSHLPNL entry. For example, if you define SDIR=PSMIN, the software does the conversion when the minimum principal stress is greater than the threshold value you define in the SC field. The TOQDGAUS parameter selects from the following criteria computations:

0: The software converts an element to shear panel status when the average value from all of its Gauss points satisfies the criteria.

1: The software converts an element to shear panel status when the value at *any* of its Gauss points satisfies the criteria.

TORSIN Determines if the torsional mass moment of inertia is included on the CROD and CBAR mass matrices.

0: No torsional mass moment of inertia for CROD and CBAR (default).

1: Include torsional mass moment of inertia for CROD and CBAR.

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	2: Include torsional mass moment of inertia for CBAR. 3: Include torsional mass for CROD.
TSTATIC	Default = -1 (SOLs 129 and 159 only) If TSTATIC = 1, a static solution may be obtained while ignoring inertial and damping forces. This option is available only with the adaptive time-stepping method (see METHOD = "ADAPT" on the Bulk Data entry, **TSTEPNL**).

Related Topics

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U – Parameters

UNITSYS No default

UNITSYS is used alone or, if applicable, in combination with a UNITTIME specification to specify the units in which the physical quantities in the Simcenter Nastran input file are expressed.

Because Simcenter Nastran is a unitless program, when a pre-processor such as Simcenter 3D Pre/Post imports a Simcenter Nastran input file, it must determine the consistent set of units in which the physical quantities in the input file are expressed. To provide this information, include a UNITSYS specification or, if applicable, a UNITSYS specification in combination with a UNITTIME specification. The UNITSYS and UNITTIME specifications have no impact on the Simcenter Nastran solution sequence. Simcenter Nastran assumes that the set of units in which the physical quantities in the Simcenter Nastran input file are expressed are consistent and no conversions are performed.

The character string in a UNITSYS specification has the format of *force unit-length unit* or *force unit-length unit-temperature unit*. Examples are as follows:

```
PARAM,UNITSYS,N-M

PARAM,UNITSYS,N-M-R

PARAM,UNITSYS,N-M-C
```

Time units are in seconds by default, but for certain UNITSYS specifications, time units can be overridden by a UNITTIME specification.

The following table shows all of the supported UNITSYS and UNITTIME combinations.

UNITSYS specification	UNITTIME specification	Force unit	Length unit	Time unit	Temp unit	Derived mass unit
PARAM,UNITSYS,N-M PARAM,UNITSYS,N-M-C PARAM,UNITSYS,N-M-F PARAM,UNITSYS,N-M-K PARAM,UNITSYS,N-M-R	Ignored by Simcenter 3D Pre/Post if present	N	m	s	blank = C C F K R	kg
PARAM,UNITSYS,N-MM PARAM,UNITSYS,N-MM-C PARAM,UNITSYS,N-MM-F	PARAM,UNITTIME,S or not specified	N	mm	s	blank = C C	t

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UNITSYS specification	UNITTIME specification	Force unit	Length unit	Time unit	Temp unit	Derived mass unit
PARAM,UNITSYS,N-MM-K PARAM,UNITSYS,N-MM-R	See(1)				F K R	
PARAM,UNITSYS,N-MM PARAM,UNITSYS,N-MM-C PARAM,UNITSYS,N-MM-F PARAM,UNITSYS,N-MM-K PARAM,UNITSYS,N-MM-R	PARAM,UNITTIME,H See(1)	N	mm	hr	blank = C C F K R	N-hr^2/mm
PARAM,UNITSYS,MN-MM PARAM,UNITSYS,MN-MM-C PARAM,UNITSYS,MN-MM-F PARAM,UNITSYS,MN-MM-K PARAM,UNITSYS,MN-MM-R	Ignored by Simcenter 3D Pre/Post if present	mN	mm	s	blank = C C F K R	kg
PARAM,UNITSYS,CN-CM PARAM,UNITSYS,CN-CM-C PARAM,UNITSYS,CN-CM-F PARAM,UNITSYS,CN-CM-K PARAM,UNITSYS,CN-CM-R	Ignored by Simcenter 3D Pre/Post if present	cN	cm	s	blank = C C F K R	kg
PARAM,UNITSYS,UN-UM PARAM,UNITSYS,UN-UM-C PARAM,UNITSYS,UN-UM-F PARAM,UNITSYS,UN-UM-K PARAM,UNITSYS,UN-UM-R	PARAM,UNITTIME,MS See(2)	µN	µm	ms	blank = C C F K R	mg
PARAM,UNITSYS,KGF-M PARAM,UNITSYS,KGF-M-C PARAM,UNITSYS,KGF-M-F PARAM,UNITSYS,KGF-M-K PARAM,UNITSYS,KGF-M-R	Ignored by Simcenter 3D Pre/Post if present	kgf	m	s	blank = C C F K R	kgf-s^2/m
PARAM,UNITSYS,KGF-MM PARAM,UNITSYS,KGF-MM-C PARAM,UNITSYS,KGF-MM-F PARAM,UNITSYS,KGF-MM-K PARAM,UNITSYS,KGF-MM-R	Ignored by Simcenter 3D Pre/Post if present	kgf	mm	s	blank = C C F K R	kgf-s^2/mm

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UNITSYS specification	UNITTIME specification	Force unit	Length unit	Time unit	Temp unit	Derived mass unit
PARAM,UNITSYS,LBF-FT PARAM,UNITSYS,LBF-FT-C PARAM,UNITSYS,LBF-FT-F PARAM,UNITSYS,LBF-FT-K PARAM,UNITSYS,LBF-FT-R	Ignored by Simcenter 3D Pre/Post if present	lbf	ft	s	blank = F C F K R	slug
PARAM,UNITSYS,LBF-IN PARAM,UNITSYS,LBF-IN-C PARAM,UNITSYS,LBF-IN-F PARAM,UNITSYS,LBF-IN-K PARAM,UNITSYS,LBF-IN-R	Ignored by Simcenter 3D Pre/Post if present	lbf	in	s	blank = F C F K R	lbf-s^2/in
PARAM,UNITSYS,PDL-FT PARAM,UNITSYS,PDL-FT-C PARAM,UNITSYS,PDL-FT-F PARAM,UNITSYS,PDL-FT-K PARAM,UNITSYS,PDL-FT-R	Ignored by Simcenter 3D Pre/Post if present	pdl	ft	s	blank = F C F K R	lbm
(1)Upon import, an error occurs in Simcenter 3D Pre/Post if PARAM,UNITTIME,MS is specified. (2)Upon import, an error occurs in Simcenter 3D Pre/Post if PARAM,UNITTIME,MS is not present, or PARAM,UNITTIME,S or PARAM,UNITTIME,H is specified.						

Symbols for the units are as follows.

Force	Symbol
Newton	N
centi-Newton	cN
milli-Newton	mN
micro-Newton	µN
kilogram-force	kgf
pound-force	lbf
poundal	pdl

Length	Symbol
meter	m
centimeter	cm
millimeter	mm
micrometer	µm
foot	ft
inch	in

Time	Symbol
second	s
millisecond	ms
hour	hr

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Temperature	Symbol
Celsius	C
Fahrenheit	F
Kelvin	K
Rankine	R

Mass	Symbol
kilogram	kg
milligram	mg
tonne	t
slug	slug
pound-mass	lbm

UNITTIME Default = S

UNITTIME is used in combination with the UNITSYS parameter to specify the unit of time in which the physical quantities in the Simcenter Nastran input file are expressed. For more information, see **UNITSYS**.

UNITTIME specification	Time unit
`PARAM,UNITTIME,MS`	millisecond
`PARAM,UNITTIME,S`	second
`PARAM,UNITTIME,H`	hour

UNSYMF Default = No

In SOL 106, nonlinear statics, PARAM,UNSYMF,YES is required to include damping effects in the calculation of complex eigenvalues. See PARAM,**NMLOOP**.

UPDTBSH Default = NO

UPDTBSH controls the update of boundary shapes generated by auxiliary boundary model analysis in SOL 200. By default, the auxiliary boundary models and shapes are generated only once at the initial design cycle and will not be updated in subsequent cycles even if the shape of the primary model is changing. PARAM,UPDTBSH,YES requests that the auxiliary models and shapes are updated and reanalyzed at every cycle.

USETPRT Default = -1

USETSEL Default = 0

USETPRT controls the tabular printout of the degree-of-freedom sets. See **"Degree-of-Freedom Sets"**.

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Sequence	Print	USETPRT
None	None(default)	-1
Internal	Row sort only	0
	Column sort only	1
	Row and Column sort	2
External	Row sort only	10
	Column sort only	11
	Row and Column sort	12

The degrees-of-freedom can be listed in ascending order according to their internal or external sequence number, but not both. The external sequence number is the grid, scalar, or extra point identification number. The internal sequence number is the number assigned after resequencing (see PARAM,**OLDSEQ**).

For a given sequence there are two types of tables that may be printed: row sort and column sort. For row sort, a table is printed for each set selected by USETSEL. Here is an example of row sort (USETPRT=0 or 10):

```
                  U S E T   D E F I N I T I O N   T A B L E   ( I N T E R N A L   S E Q U E N C E ,   R O W   S O R T )
                                                    A   DISPLACEMENT SET
            -1-         -2-         -3-         -4-         -5-         -6-         -7-         -8-         -9-         -10-

1=          2-1         2-2
```

For column sort, a single table is printed for the following sets: SB, SG, L, A, F, N, G, R, O, S, M, E. Here is an example of column sort (USETPRT = 1 or 11):

```
U S E T   D E F I N I T I O N   T A B L E   ( I N T E R N A L   S E Q U E N C E ,   C O L U M N   S O R T )

  EXT GP. DOF  INT DOF  INT GP.    SB    SG    L     A     F     N     G     R     O     S     M     E
-------------------------------------------------------------------------------------------------------
      1 - 1      1-       1 G       1                               1     1               1
        - 2      2-                 2                               2     2               2
        - 3      3-                       1                         3     3               3
        - 4      4-                       2                         4     4               4
        - 5      5-                       3                         5     5               5
        - 6      6-                       4                         6     6               6
```

USETSEL specifies the sets which will be printed in the row sort (USETPRT = 0 or 10). In order to select specific sets to be printed, you must sum their corresponding decimal equivalent numbers. For example, sets A, L, and R are selected with USETSEL = 128 + 256 + 8 = 392. For column sort, a single table is printed for the following sets: SB, SG, L, A, F, N, G, R, O, S, M, E. Here is an example of column sort (USETPRT = 1 or 11):

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USETSEL	Sets Printed
-1	All sets as defined in **"Degree-of-Freedom Sets"**.
0	Mutually exclusive sets only; i.e., sets M, SB, SG, O, Q, R, C, B, E, and A.

USETSTRi Input-character-default=' '

USETSTR1 through USETSTR4 specifies the sets that will be printed by the specification of parameters USETPRT and USETSEL. Any set in Degree-of-Freedom Sets may be specified. A ":" is used as a separator. In the following example, the m-set (degrees-of-freedom eliminated by multipoint constraints) and s-set (degrees-of-freedom eliminated by single point constraints) are specified.

Example: PARAM,USETSTR1,M:S

Related Topics

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V – Parameters

VMFACT

Default = 4.0

The VMFACT parameter is applicable when VMOPT=2.

In the absence of an RSMETHOD specification, VMFACT specifies the number of dry modes that the software calculates as a multiple of the number of wet modes specified by the eigenmethod (EIGR/EIGRL selection) as follows:

$$ND_{dry} = \text{int}\left(VMFACT \times ND_{wet} + 0.5\right)$$

where:

ND_{dry}
Number of dry modes to calculate

ND_{wet}
Number of wet modes specified in the ND field of the eigenmethod (EIGR/EIGRL selection)

$VMFACT$
Value of the VMFACT parameter

If the eigenmethod (EIGR/EIGRL selection) specifies a frequency range over which to calculate the wet modes, the software uses the VMFACT specification to calculate the frequency range over which to calculate the dry modes as follows:

- The lower limit of the frequency range is retained. That is:

  $$F1_{dry} = F1_{wet}$$

  where:

  $F1_{dry}$
  Lower frequency range limit to use in the dry modes calculation

  $F1_{wet}$
  Lower frequency range limit for the wet modes calculation as specified in the F1 field of the eigenmethod (EIGR/EIGRL selection)

- The upper limit to the frequency range is calculated as follows:

  $$F2_{dry} = VMFACT \times F2_{wet}$$

  where:

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$F1_{dry}$
Upper frequency range limit to use in the dry modes calculation

$F1_{wet}$
Upper frequency range limit for the wet modes calculation as specified in the F2 field of the eigenmethod (EIGR/EIGRL selection)

$VMFACT$
Value of the VMFACT parameter

If an RSMETHOD case control command is present, it takes precedence over an explicit VMFACT specification or the VMFACT default behavior.

Note

In pre-Simcenter Nastran 2206 versions of the software, if you specified a single eigenmethod (EIGR/EIGRL selection), the software used it to calculate both the wet and dry modes. Thus, the number of wet and dry modes were identical.

Beginning with Simcenter Nastran 2206, the software automatically applies the VMFACT default value to calculate the dry modes when a single eigenmethod (EIGR/EIGRL selection) is present. Thus, for input files with a single eigenmethod (EIGR/EIGRL selection) present, the results obtained running Simcenter Nastran 2206 will typically differ from those obtained running a pre-Simcenter Nastran 2206 version of the software. To retain the pre-Simcenter Nastran 2206 behavior for such cases, add a VMFACT = 1.0 specification to the input file.

VMOPT Default = 0

The VMOPT parameter is applicable when the fluid is modeled by MFLUID bulk entries that are referenced by a MFLUID case control command.

VMOPT = 0 (default)
The modes of the structure only are first computed without virtual mass (VM) effects to obtain Ritz vectors to use in a generalized coordinate solution, similar in concept to the older Generalized Dynamic Reduction method. The modes are re-computed in the generalized basis after the VM effects are added. When an o-set exists (usually because of the presence of ASETi-type entries, or for super element reduction), the VM effects are added to the component modes (o-set level). When there is not an o-set the VM effects are added in the residual structure Phase II operations (a-set level), after the structure-mass-only eigensolution. The rigid body mass of the VM is output automatically, computed about the location of the grid point listed on PARAM, GRDPNT. Because this option results in a very densely populated fluid mass matrix, it is best suited for models:

- That are relatively small in size.

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- Where VMOPT = 2 yields inaccurate results.

VMOPT = 1

The virtual fluid mass will be included in the mass matrix at the same time as all other mass elements. In other words, the component modes will reflect the virtual mass. By default, VM is included after the component modes are computed. This is the most expensive option in terms of computer resource requirements and computation time. It is best suited for models:

- That are relatively small in size.
- Where the VM is in one super element only, with most elements in the super element wetted.
- Where VMOPT = 2 yields inaccurate results.

VMOPT = 2

This is the more efficient method and is summarized in the following steps:

1. Compute the normal modes of the structure without the fluid mass. These modes are referred to as the *dry* modes.
2. Generate reduced VM using the reduced, modal basis from the dry modes.
3. Compute the normal modes of the reduced VM problem from step 2. The resulting modes are referred to as the *wet* modes.

Distributed memory parallel (DMP) execution is available for both the dry modes calculation (step 1) and the VM generation (step 2) to reduce overall solution time.
The following DMP methods can be used to compute dry modes: RDMODES and FDMODES, GDMODES and HDMODES with GPART = 1. The DMP global iterative solver is also enabled for VM generation by specifying ITER = YES on the NASTRAN statement.

For certain models, such as those where the fluid mass is much greater than the structural mass, VMOPT = 2 may yield inaccurate results. Therefore, for such models, use VMOPT = 3.

Note

The VMOPT = 2 option should not be used with MAX normalization specified on the EIGRL bulk entry. VMOPT = 2 will only yield correct results with MASS normalization.

VMOPT = 3

This option is available for a more efficient normal modes computation when the virtual fluid mass is very large. The VMOPT=3 option is similar to the VMOPT=0 option, but instead of using the Lanczos method for the modes computation and applying the sum of virtual + structural mass in one step, the VMOPT=3 option:

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- Uses the virtual fluid mass eigenvalue method which uses the implicitly restarted Arnoldi method.
- Efficiently applies the large fluid mass in conjunction with the modes computation, which avoids the computation and application of the dense mass matrix in a single step.

This approach significantly accelerates the process compared to the traditional Lanczos method. However, it may occasionally yield results that are slightly off (by less than 0.5%) when compared to the Lanczos method.

Note

The virtual fluid mass eigenvalue method uses the implicitly restarted Arnoldi method. You can increase the value of the following parameters in the bulk data section of your input file if you encounter convergence issues:

- PARAM,BSIZE,*number of basis vectors*
- PARAM,VMITER,*maximum # of iterations*

VREF Default = 1.0

In modal flutter analysis, the velocities are divided by VREF to convert units or to compute flutter indices.

Related Topics

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W & X – Parameters

W3 (or W3FL), W4 (or W4FL)

Default = 0.0

Specifies the frequencies in radians per unit time at which to convert structural (hysteretic) damping to viscous damping in the transient response solutions and SOL 414 solutions except SOL 414,101 Maneuvers. The software adds the converted structural damping to the other viscous damping contributions.

Note

For solutions where the use of the DAMPING case control command and bulk entry are supported, if the input file contains both explicit parameter specifications such as PARAM,W3 and a referenced DAMPING bulk entry definition, the values in the W3, W4, and WMODAL fields (or the default values for these field if left blank) on the DAMPING bulk entry take precedence.

The structural damping to viscous damping conversion depends on the solution type:

- For SOLs 109, 112, 129, 159, or a modal-based analysis that writes an interface file for external multibody dynamics or control system software, the software converts the structural damping to viscous damping as follows:

$$\frac{G}{W3}\left[K^{1}\right]+\frac{GEFACT}{W4}\left(\left[K4^{2}\right]+\left[K^{4}\right]\right)$$

- For SOLs 401, 414,103, 414,110, 414,111, and 414,129, the software converts the structural damping to viscous damping as follows:

$$\frac{G}{W3}\left[K^{1}\right]+\frac{1}{W4}\left(\left[K4^{2}\right]+\left[K^{4}\right]\right)$$

Note

For SOL 414 solutions:

- To convert structural damping to viscous damping for a stator, use PARAM,W3 and PARAM,W4.
- To convert structural damping to viscous damping for a rotor, use the W3 and W4 fields on the ROTORD bulk entry.

For SOL 414,129, the software uses a value of 1 for W4 for a rotor and stator, except in the following cases:

- Rotor modeled as a 1D beam.

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- Rotor represented by a superelement.

In both of these cases, the software uses the value you set for W4.

- For SOL 402, the software does not use the W3 and W4 values and converts the structural damping to viscous damping as follows:

$$\left[K4^{2}\right] + \left[K^{4}\right]$$

where:

$W3$ is the frequency at which to convert $\left[K^{1}\right]$ (PARAM,W3 for structures and PARAM,W3FL for fluids, or the W3 field on a DAMPING bulk entry).

$W4$ is the frequency at which to convert $\left[K4^{2}\right]$ and $\left[K^{4}\right]$ (PARAM,W4 for structures and PARAM,W4FL for fluids, or the W4 field on a DAMPING bulk entry).

G is the overall structural damping coefficient (PARAM,G for structures and PARAM,GFL for fluids, or the G field on a DAMPING bulk entry).

$GEFACT$ is the scale factor that is applied to the K42GG direct input stiffness and assembled elemental damping matrices (GEFACT field on a DAMPING bulk entry).

$\left[K^{1}\right]$ is the structural stiffness matrix plus K2GG direct input stiffness matrix for the structure.

$\left[K4^{2}\right]$ is the K42GG direct input stiffness matrix for the structure.

$\left[K^{4}\right]$ is the assembled elemental damping matrix (GE field on material, property, or element bulk entries) such that:

$$\left[K^{4}\right] = \sum G_E \left[K_E\right]$$

Specifying W3 = 0.0 (the default) causes the $\left[K^{1}\right]$ term to be omitted from the conversion.

Specifying W4 = 0.0 (the default) causes the $\left[K4^{2}\right]$ and $\left[K^{4}\right]$ terms to be omitted from the conversion.

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For the fluid portion of a coupled fluid-structure analysis, W3 and W4 are applied to the structural portion of the model, and W3FL and W4FL are applied to the fluid portion of the model.

In SOLs 129 and 159, W4 can vary between subcases. However, the linear portion of the model uses only the W4 value from the first subcase, and the values in the subsequent subcases are applied to the nonlinear portion of the model.

The WMODAL parameter can also be used to select the structural to viscous damping conversion method in the modal transient response solutions and in some SOL 414 solutions. For more information, see WMODAL below.

WMODAL Default = NO

For the modal transient response solutions and SOL 414,103 Eigenvalue and Superelement Reduction, SOL 414,110 Complex Modal Analysis, and SOL 414,111 Harmonic Response solutions, specifies the structural (hysteretic) damping to viscous damping conversion method.

= NO The software uses the conversion methods described under the W3 and W4 parameter description above.

= YES The software uses the WMODAL conversion methods described below.

Note

For solutions where the use of the DAMPING case control command and bulk entry are supported, if the input file contains both explicit parameter specifications such as PARAM,WMODAL,YES and a referenced DAMPING bulk entry definition, the values in the W3, W4, and WMODAL fields (or the default values for these field if left blank) on the DAMPING bulk entry take precedence.

The structural damping to viscous damping conversion depends on the solution type:

- For all supported solutions except SOL 414 solutions, when WMODAL = YES, the software ignores the W3 and W4 settings and converts the structural damping to viscous damping as follows:

$$\left[B'_{qq}\right] = \begin{bmatrix} \frac{1}{\sqrt{\omega_1}} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sqrt{\omega_2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \vdots & \frac{1}{\sqrt{\omega_q}} \end{bmatrix} [\phi]^T \left(G\left[K^1\right] + GEFACT\left(\left[K4^2\right] + \left[K^4\right]\right)\right) [\phi] \begin{bmatrix} \frac{1}{\sqrt{\omega_1}} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sqrt{\omega_2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \vdots & \frac{1}{\sqrt{\omega_q}} \end{bmatrix}$$

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- For SOLs 414,103, 414,110, and 414,111, when WMODAL = YES, the software converts the structural damping to viscous damping as follows:

$$[B'_{qq}] = \begin{bmatrix} \frac{1}{\sqrt{\omega_1}} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sqrt{\omega_2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \vdots & \frac{1}{\sqrt{\omega_q}} \end{bmatrix} [\phi]^T \left(\frac{G}{W3}[K^1] + \frac{1}{W4}[K4^2] + \frac{1}{W4}[K^4] \right) [\phi] \begin{bmatrix} \frac{1}{\sqrt{\omega_1}} & 0 & \cdots & 0 \\ 0 & \frac{1}{\sqrt{\omega_2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \vdots & \frac{1}{\sqrt{\omega_q}} \end{bmatrix}$$

where:

W3 is the frequency at which to convert $[K^1]$ (PARAM,W3 for structures and PARAM,W3FL for fluids, or the W3 field on a DAMPING bulk entry).

W4 is the frequency at which to convert $[K4^2]$ and $[K^4]$ (PARAM,W3 for structures and PARAM,W3FL for fluids, or the W3 field on a DAMPING bulk entry).

$[B'_{qq}]$ is the matrix of converted structural damping in modal space.

$[\phi]$ is the matrix of eigenvectors (mode shapes).

G is the overall structural damping coefficient (PARAM,G for structures and PARAM,GFL for fluids, or the G field on a DAMPING bulk entry).

$GEFACT$ is the scale factor that is applied to the K42GG direct input stiffness and assembled elemental damping matrices (GEFACT field on a DAMPING bulk entry).

$[K^1]$ is the structural stiffness matrix plus K2GG direct input stiffness matrix for the structure.

$[K4^2]$ is the K42GG direct input stiffness matrix for the structure.

$[K^4]$ is the assembled elemental damping matrix (GE field on material, property, or element bulk entries) such that:

$$[K^4] = \sum G_E [K_E]$$

ω_i are the modal frequencies.

The software adds the converted structural damping to the other viscous damping contributions as follows:

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$$[B_{qq}] = [\phi]^T \left([B^1] + [B^2] + \text{ALPHA1} \cdot [M^1] + \text{ALPHA2} \cdot \left([K^1] + [K^d] \right) \right) [\phi] + [B'_{qq}]$$

where:

$[B_{qq}]$ is the viscous damping matrix in modal space.

$[B^1]$ is the viscous damping matrix from elements (CVISC, CDAMPi) and B2GG direct input damping.

$[B^2]$ is the viscous damping matrix from B2PP direct input and transfer functions.

ALPHA1 is the mass matrix scaling factor for Rayleigh (proportional) damping (PARAM,ALPHA1 for structures and PARAM,ALPHA1FL for fluids, or the ALPHA1 field on a DAMPING bulk entry).

ALPHA2 is the stiffness matrix scaling factor for Rayleigh (proportional) damping (PARAM,ALPHA2 for structures and PARAM,ALPHA2FL for fluids, or the ALPHA1 field on a DAMPING bulk entry).

$[M^1]$ is the structural mass matrix plus M2GG direct input mass for the structure.

$[K^1]$ is the structural stiffness matrix plus K2GG direct input stiffness matrix for the structure.

$[K^d]$ is the structural differential stiffness matrix.

When rigid body modes are present, the software partitions the corresponding diagonal entries out of the calculation. You can use the FZERO parameter to identify rigid body modes. For more information, see **FZERO**.

The software can use the $[B_{qq}]$ matrix directly in a modal transient response analysis (SOL 112), or in a multibody dynamics or control system simulation when a normal modes analysis (SOL 103), modal frequency response analysis (SOL 111), or modal transient response analysis (SOL 112) input file contains either the ADAMSMNF or MBDEXPORT case control commands. For a multibody dynamics simulation, the damping matrix written to the interface file contains only the values along the diagonal of the $[B_{qq}]$ matrix. For a control system simulation, the state-space representation written to the interface file uses all of $[B_{qq}]$.

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The WMODAL parameter converts structural damping to viscous damping in the residual structure and external superelements.

When the WMODAL parameter is used in SOL 145, the software uses the full $[B_{qq}]$ matrix.

WRTMAT Default: 1

PARAM, WRTMAT is valid only for when PARAM,CORROPT,YES is defined.

1: Write reduced A-SET mass (MXX) and stiffness (KXX) matrices to OP2 file.

2: In addition to MXX and KXX, write back expansion matrix (GOA) from A-SET to Display Set to OP2 file.

WTMASALL Default = YES

Applicable to incompressible fluid problems modeled using the defined fluid mass method.

Specify PARAM,WTMASALL,YES to multiply the fluid mass matrix by the value of the WTMASS parameter when the fluid mass matrix is generated.

Specify PARAM,WTMASALL,NO to multiply the fluid mass matrix by the value of WTMASSFL parameter when the fluid mass matrix is generated.

WTMASS Default = 1.0

The structural mass matrix is multiplied by the value of WTMASS when it is generated.

For incompressible fluid problems, the WTMASS parameter value is used as follows:

- When using the virtual fluid mass method, only the structural mass matrix is multiplied by the value of the WTMASS parameter when it is generated.
- When using the defined fluid mass method and PARAM,WTMASALL,YES (the default) is specified, both the structural and fluid mass matrices are multiplied by the value of the WTMASS parameter when they are generated.
- When using the defined fluid mass method and PARAM,WTMASALL,NO is specified, the structural mass matrix is multiplied by the value of the WTMASS parameter when it is generated and the fluid mass matrix is multiplied by the value of the WTMASSFL parameter when it is generated.

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For information about using WTMASS in SOL 414 solutions to convert the values of mass parameters that are expressed in weight units, see Converting weight units to mass units.

WTMASSFL Default = 1.0

Applicable to incompressible fluid problems modeled using the defined fluid mass method when PARAM,WTMASALL,NO is specified.

The fluid mass matrix is multiplied by the value of the WTMASSFL parameter when the fluid mass matrix is generated.

XFLAG Default = 2

By default (XFLAG = 2), when temperature loads and element deformations are present, the element strain energy for the linear elements is calculated using the following equation:

$$E = \frac{1}{2}u^T K_e u - \frac{1}{2}u^T P_{et}$$

where u is the deformation, K_e is the element stiffness and P_{et} is the element load vector for temperature differences and element deformations. This formula is the same strain energy calculation used for nonlinear elements.

If XFLAG is set to 0, the element strain energy for linear elements is calculated using the following equation:

$$E = \frac{1}{2}u^T K_e u - u^T P_{et}$$

Related Topics

Browse Simcenter Nastran documentation

Simcenter Nastran Quick Reference Guide
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Parameter Applicability Tables

Table 6-21 lists parameter applicability to the solution sequences (SOLs 101 through 115).

Table 6-22 lists parameter applicability to the solution sequences (SOLs 116 through 414).

The following nomenclature is used:

B	Must be specified in the Bulk Data Section only.
E	May be specified in either the Bulk Data and/or Case Control Section.
blank	Is not supported for that solution number.

Note: Footnotes are included for some parameters and solutions. For example, B(1) is shown for some parameters supported by SOL 401.

Table 6-21: PARAMeter Names in the SOLs

PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
ACOUT		E			E	E	E	E	E	E		
ACSYM						B			B			
ADPCON				E								
ADSTAT							B			B		
ADTOQDLD												
AESDISC												
AESMAXIT												
AESMETH												

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining SYSTEM(811)=1. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
AESRNDM												
AESTOL												
AFNORM		B							B	B		
AFZERO												
ALPHA1					B	B	B	B	B	B		
ALPHA2					B	B	B	B	B	B		
ALPHA1FL					B	B	B	B	B	B		
ALPHA2FL					B	B	B	B	B	B		
ALTRED	B		B								B	
AMPCZ	B	B	B	B	B	B	B	B	B	B	B	B
ASCOUP		E			E	E	E	E	E	E		
ASING	E	E	E	E	E	E	E	E	E	E	E	E
AUNITS												
AUTOADJ												
AUTOMPC	B	B	B	B	B	B	B	B	B	B	B	B
AUTOSEEL	E	E	E	E	E	E	E	E	E	E	E	E
AUTOSPC	E	E	E	E	E	E	E	E	E	E	E	E
AUTOSPCR				E								
AUTOSPRT		E			E	E	E	E	E	E		
BAILOUT	E	E	E	E	E	E	E	E	E	E	E	E
BDMNCON												
BETA												
BIGER	E	E	E								E	E

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
BIGER1	E	E	E								E	E
BIGER2	E	E	E								E	E
BLTGLSCL												
BLTTNSCL												
BOLTFACT	B	B	B		B	B	B	B	B	B		
BRSYMFAC					E	E	E	E	E			
BSHDAMP					B	B	B	B	B	B		
BSHFORCE						E			E			
BUCKLE				E								
CA1	E	E	E	E	E	E	E	E	E	E	E	E
CA2	E	E	E	E	E	E	E	E	E	E	E	E
CB1	E	E	E	E	E	E	E	E	E	E	E	E
CB2	E	E	E	E	E	E	E	E	E	E	E	E
CDIF												
CDITER	B											
CDPCH	B											
CDPRT	B											
CHECKOUT	E	E	E	E	E	E	E	E	E	E	E	E
CK1	E	E	E	E	E	E	E	E	E	E	E	E
CK2	E	E	E	E	E	E	E	E	E	E	E	E
CK3	E	E	E	E	E	E	E	E	E	E	E	E
CK41	E	E	E	E	E	E	E	E	E	E	E	E
CK42	E	E	E	E	E	E	E	E	E	E	E	E

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
CLOSE		B										B
CM1	E	E	E	E	E	E	E	E	E	E	E	E
CM2	E	E	E	E	E	E	E	E	E	E	E	E
CNTASET	E											
CNTSCL		B	B			B			B	B		
COLEMAN												
COLPHEXA												
COMPMATT				B								
CONFAC	B	B	B	B	B	B	B	B	B	B	B	B
CORROPT		E										
COUPMASS	E	E	E	E	E	E	E	E	E	E	E	E
CP1, CP2	E		E	E		E	E		E	E	E	
CURV	E	E	E								E	E
CURVPLOT	E	E	E	E							E	E
DBALL	E	E	E	E	E	E	E	E	E	E	E	E
DBCCONV	E	E	E	E	E	E	E	E	E	E	E	E
DBCDIAG	E	E	E	E	E	E	E	E	E	E	E	E
DBCOVWRT	E	E	E	E	E	E	E	E	E	E	E	E
DBDICT	B	B	B	B	B	B	B	B	B	B	B	B
DBDN	E	E	E	E	E	E	E	E	E	E	E	E
DBDRPRJ	B	B	B	B	B	B	B	B	B	B	B	B
DBDRVER	B	B	B	B	B	B	B	B	B	B	B	B
DBRCV	E	E	E	E	E	E	E	E	E	E	E	E

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
DBUP	E	E	E	E	E	E	E	E	E	E	E	E
DDRMM									B	B		
DESPCH												
DESPCH1												
DFREQ						B			B			
DIGITS	E	E			E	E	E	E	E	E	E	E
DOF123		E										
DOPT	E	E	E								E	E
DPEPS												
DPREONLY												
DSNOKD			B									
DSZERO	B	B	B									
DUCTFMAX						B						
DYNSPCF		E			E	E	E	E	E	E		E
ELITASPC	B											
EPPRT	E	E	E	E	E	E	E	E	E	E	E	E
EPZERO	E	E	E	E	E	E	E	E	E	E	E	E
ERROR	E	E	E	E	E	E	E	E	E	E	E	E
EST	E	E	E	E	E	E	E	E	E	E	E	E
EXTBEMI	E	E										
EXTBEMO	E	E										
EXTDR	B	B			B	B	B	B	B	B		
EXTDROUT	B	B			B	B	B	B	B	B		

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
EXTDRUNT	B	B			B	B	B	B	B	B		
EXTOUT	B	B	B	B	B	B	B	B	B	B		
EXTRCV	E	E	E	E	E	E	E	E	E	E	E	E
EXTUNIT	B	B	B	B	B	B	B	B	B	B	B	B
F56	B	B	B	B	B	B	B	B	B	B	B	B
FACTOR	B	B	B	B	B	B	B	B	B	B	B	B
FCTCi					B	B	B	B	B	B		
FCTMi					B	B	B	B	B	B		
FCTZi	B				B	B	B	B	B	B		
FDINTOPT						B			B			
FHPOST												
FHPNST												
FIXDIFFK		B										
FIXEDB	E	E										
FKSYMFAC				E								
FLEXINCR												
FLEXINV						B			B			
FLUIDMP									E			
FLUIDSE		B			B	B	B	B	B	B		
FOLLOWK		E	E	E	E	E	E	E	E	E		E
FRQDEPO						B			B			
FRRU									B			
FRUMIN									B			

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
FULLGPF						E	E		E	E		
FZERO		E				E	E	E	E	E		
G		B[2]			B	B	B	B	B	B		
GDAMPF		B			B	B	B	B	B	B		
GEOMU	E	E	E	E	E	E	E	E	E	E	E	E
GFL					B	B	B	B	B	B		
GPACAO						B			B			
GPECT	E	E	E	E	E	E	E	E	E	E	E	E
GRDPNT	E	E	E	E	E	E	E	E	E	E	E	E
GUSTAERO												
HARMLIN												
HEATSTAT	B											
HFREQ								B	B	B		
HFREQFL								B	B	B		
IFP	E	E	E	E	E	E	E	E	E	E	E	E
IFTM									B			
INEMETH					B	B	B					
INREL = -1	E		E									
INREL = -2	E											
INRLM		E			E	E	E	E	E	E		E
IRES	E		E	E		E	E		E	E	E	
ITAPE	B	B	B									
IUNIT	B	B	B									

Footnotes:

[1] For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no [1] footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

[2] Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

[3] For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

[4] Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
K6ROT	E	E	E	E	E	E	E	E	E	E	E	E
KDAMP		B						B	B	B		
KDAMPFL								B	B	B		
KDIAG				E								
KGGCPCH	B											
KGGLPCH	B	B	B	B	B	B	B	B	B	B	B	B
LAMLIN		B										
LANGLE				B								
LFREQ								B	B	B		
LFREQFL								B	B	B		
LGDISP				E								
LGSTRN												
LMDYN		E			E	E	E	E	E	E		
LMFACT	E	E	E	E	E	E	E	E	E	E		
LMODES		E						B	B	B		
LMODESFL								B	B	B		
LMSTAT	E		E	E								
LOADU	E	E	E	E	E	E	E	E	E	E	E	E
LOOPID				E								
MACH												
MATCTE												
MATNL												
MAXLP												

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
MAXRATIO	E	E	E	E	E	E	E	E	E	E	E	E
MDLDAMP		B										
MDOF						B	B					
MDSTYPE		B				B			B			
MECHFIL		E						E	E	E		
MECHFIX		E						E	E	E		
MECHPRT		E										
MEMSTRES												
MESH	B											
METHCMRS		E			E	E	E	E	E	E		
MGRID						B	B					
MMFIL						E	E					
MODACC									B	B		
MODEL	B	B	B	B	B	B	B	B	B	B	B	B
MODTRK					B	B	B	B	B	B		
MPCX	B	B	B	B	B	B	B	B	B	B	B	B
MPCZERO	E	E	E	E	E	E	E	E	E	E	E	E
MPIBUFF						B			B			
MTREPSI					B							
MTREPSR					B							
MRRFCTD					B							
MTRFCTV					B							
MTRFMAX					B							

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
MTRRMAX					B							
MTRSKIP					B							
NASPDV												
NASPRT												
NDAMP												
NFDINTPL						B			B			
NINTPTS	E	E	E								E	E
NLAYERS				E								
NLHTLS				E								
NLMAX						E	E					
NLMIN						E	E					
NLTOL				E								
NMLOOP				E	E	E	E	E	E	E		
NOCOMPS	E	E	E	E							E	E
NOELOF	E	E	E								E	E
NOELOP	E	E	E								E	E
NOFISR	E	E	E	E							E	E
NOGPF	E	E	E								E	E
NOMSGSTR	E	E	E								E	E
NONCUP		B							B	B		
NQSET		E			E	E	E	E	E	E		E
NUMOUT	E	E	E								E	E
NUMOUT1	E	E	E								E	E

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
NUMOUT2	E	E	E								E	E
OCMP	E	E	E	E							E	E
ODS		B							B			
OEE	E	E	E	E							E	E
OEF	E	E	E	E	E	E	E	E	E	E	E	E
OEFX	E	E	E	E	E	E	E	E	E	E	E	E
OEPT	E	E	E	E	E	E	E	E	E	E	E	E
OES	E	E	E	E	E	E	E	E	E	E	E	E
OESE	E	E	E	E	E	E	E	E	E	E	E	E
OESX	E	E	E	E	E	E	E	E	E	E	E	E
OG	E	E	E	E							E	E
OGEOM	B	B	B	B	B	B	B	B	B	B	B	B
OGPF	E	E	E	E							E	E
OGPS	E	E	E				E			E	E	E
OIBULK	B	B	B	B	B	B	B	B	B	B	B	B
OLDSEQ	B	B	B	B	B	B	B	B	B	B	B	B
OMAXR	E	E	E	E	E	E	E	E	E	E	E	E
OMID	B	B	B		B	B	B	B	B	B	B	B
OMPT	E	E	E	E	E	E	E	E	E	E	E	E
OP2FMT	B	B	B	B	B	B	B	B	B	B	B	B
OPG	E		E	E		E	E		E	E	E	
OPGEOM												
OPGTKG												

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
OPPHIB												
OPPHIPA												
OPTEXIT												
OPTION		E										E
OQG	E	E	E	E	E	E	E	E	E	E	E	E
OSWELM	B	B	B	B	B	B	B	B	B	B	B	B
OSWPPT	B	B	B	B	B	B	B	B	B	B	B	B
OUG	E	E	E	E	E	E	E	E	E	E	E	E
OUGCORD	E	E	E	E	E	E	E	E	E	E	E	E
OUMU		E										E
OUNIT1	E	E	E	E	E	E	E	E	E	E	E	E
OUNIT2	E	E	E	E	E	E	E	E	E	E	E	E
OUTOPT	E	E	E	E							E	E
PANELMP									E			
PATVER	E	E	E	E	E	E	E	E	E	E	E	E
PBRPROP	B	B	B	B	B	B	B	B	B	B	B	B
PDRMSG	E	E	E	E	E	E	E	E	E	E	E	E
PEWOPT						B			B			
PHIBGN					B	B	B	B	B	B		
PHIDEL					B	B	B	B	B	B		
PHINUM					B	B	B	B	B	B		
PKRSP										B		
PLTMSG	E	E	E	E	E	E	E	E	E	E	E	E

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
POST	E	E	E	E	E	E	E	E	E	E	E	E
POSTEXT	E	E	E	E	E	E	E	E	E	E	E	E
POSTHDF5												
POSTOPT												
POSTU	E	E	E	E	E	E	E	E	E	E	E	E
PREFDB							E	E	E	E		
PRGPST	E	E	E	E	E	E	E	E	E	E	E	E
PRINT												
PROUT	E	E	E	E	E	E	E	E	E	E	E	E
PRPA	E		E									
PRPJ	E		E			E	E		E	E	E	
PRTCSTM	E	E	E	E	E	E	E	E	E	E	E	E
PRTEQXIN	E	E	E	E	E	E	E	E	E	E	E	E
PRTGPDT	E	E	E	E	E	E	E	E	E	E	E	E
PRTGPL	E	E	E	E	E	E	E	E	E	E	E	E
PRTGPTT	E	E	E	E	E	E	E	E	E	E	E	E
PRTMAXIM	E	E	E		E	E	E	E	E	E	E	E
PRTMGG	E	E	E	E	E	E	E	E	E	E	E	E
PRTPG	E	E	E	E	E	E	E	E	E	E	E	E
PRTRESLT	E	E	E		E	E	E	E	E	E	E	E
Q												
QSETREM	B	B			B	B	B	B	B	B	B	B
RDCNT	E	E							E	E		

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
RDFLUID		B							B			
RDMTRAK												
RDQSOL		B							B			
RDRESVEC		B							B	B		
RDSPARSE		B							B			
RESLTOPT	E	E	E	E	E	E	E	E	E	E	E	E
RESVEC		E		E				E	E	E		
RESVINER		E		E				E	E	E		
RESVPGF		E		E				E	E	E		
RESVRAT		E		E				E	E	E		
RESVSE		E		E				E	E	E		
RESVSLI		E		E				E	E	E		
RESVSO		E		E				E	E	E		
RGBEAMA												
RGBEAME												
RGLCRIT												
RGSPRGK												
RLOOPNEW					B	B		B	B			
RMSINT						B			B			
RMSSF						B			B			
RMXPANEL										E		
RMXTRAN										B		
ROTCMRF					E							

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
ROTCOUP					B	B	B	B	B	B		
ROTCSV					B			B				
ROTGPF					B			B				
ROTSYNC					B			B				
RPOSTS1						B			B			
RSATT		B										
RSCON		B										
RSOPT		B										
RSPECTRA							E			E		
RSPRINT							E			E		
RVMASS		E		E				E	E	E		
RVMTV		E		E				E	E	E		
S1AG	E	E	E	E							E	E
S1AM	E	E	E	E							E	E
S1G	E	E	E	E							E	E
S1M	E	E	E	E							E	E
SCRSPEC		B										B
SDAMPUP									B			
SECOMB	B	B	B	B	B	B	B	B	B	B		
SEMAP	B	B	B	B	B	B	B	B	B	B	B	B
SEMAPOPT	B	B	B	B	B	B	B	B	B	B	B	B
SEMAPPRT	B	B	B	B	B	B	B	B	B	B	B	B
SEMATPRT												B

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
SENSUOO												
SEOP2CV	B	B			B	B	B	B	B	B		
SEP1XOVR	B	B	B	B	B	B	B	B	B	B	B	B
SEQOUT	B	B	B	B	B	B	B	B	B	B	B	B
SERST	B	B	B	B	B	B	B	B	B	B	B	B
SESDAMP		E			E	E	E	E	E	E		E
SESEF		E										E
SFEF70		E			E	E	E	E	E	E		
SHIFT1		E						E	E	E		
SHLDAMP					B	B	B	B	B	B		
SIGMA												
SKPAMP												
SLOOPID												
SLOSHCMS		E							E			
SMALLQ	B	B	B	B	B	B	B	B	B	B	B	B
SNORM	B	B	B	B	B	B	B	B	B	B	B	B
SNORMPRT	B	B	B	B	B	B	B	B	B	B	B	B
SOFTEXIT												
SOLID	B	B	B	B	B	B	B	B	B	B	B	B
SOLVSUB		E										
SPCGEN	E	E	E	E	E	E	E	E	E	E	E	E
SPCSTR						B						
SRCOMPS	B	B	B	B	B	B	B	B	B	B	B	B

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
SRTELTYP	E	E	E								E	E
SRTOPT	E	E	E								E	E
STABAO						B			B			
STABFEM		B				B			B			
START	B	B	B	B	B	B	B	B	B	B	B	B
STIME												
STRUCTMP									E			
SUBID				E								
SUPER	B	B	B	B	B	B	B	B	B	B	B	B
SWPANGLE						E			E			
TABID							E			E		
TABS												
TENSOQD												
TESTNEG				E								
TINY	E	E	E								E	E
TIREAPP									B			
TOLRSC	B	B	B	B	B	B	B	B	B	B	B	B
TOQDGAUS												
TORSIN	B	B	B	B	B	B	B	B	B	B	B	B
TSTATIC												
UNITSYS	B	B	B	B	B	B	B	B	B	B	B	B
UNITTIME	B	B	B	B	B	B	B	B	B	B	B	B
UNSYMF				B								

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (101 through 115)											
	101	103	105	106	107	108	109	110	111	112	114	115
UPDTBSH												
USETPRT	E	E	E	E	E	E	E	E	E	E	E	E
USETSEL	E	E	E	E	E	E	E	E	E	E	E	E
USETSTRi	E	E	E	E	E	E	E	E	E	E	E	E
VMFACT		E							E	E		
VMOPT		E	E	E	E	E	E	E	E	E	E	E
VREF												
W3		E(2)					E		E(2)	E		
W3FL							E			E		
W4		E(2)					E		E(2)	E		
W4FL							E			E		
WMODAL		E(2)							E(2)	E		
WRTMAT		E										
WTMASALL		B(4)							B(4)			
WTMASS	E	E	E	E	E	E	E	E	E	E	E	E
WTMASSFL		B(4)							B(4)			
XFLAG	E		E								E	

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

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Table 6-22: PARAMeter Names in the SOLs

PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
ACOUT										E			
ACSYM										E			
ADPCON			E				E	E					
ADSTAT													
ADTOQDLD											B		
AESDISC				B						B			
AESMAXIT				B						B			
AESMETH				B						B			
AESRNDM				B						B			
AESTOL				B						B			
AFNORM											B	B	
AFZERO											B		
ALPHA1		B	B		B	B		B		B	B	B	B
ALPHA2		B	B		B	B		B		B	B	B	B
ALPHA1FL		B	B		B	B		B		B			
ALPHA2FL		B	B		B	B		B		B			
ALTRED													
AMPCZ	B	B	B				B	B	B				
ASCOUP										E			
ASING	E	E	E	E	E	E	E	E	E	E			
AUNITS				B						B			
AUTOADJ										B			

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
AUTOMPC	B	B	B				B	B	B				
AUTOSEEL	E	E	E	E	E	E	E	E	E	E			
AUTOSPC	E	E	E	E	E	E	E	E	E	E			
AUTOSPCR							E						
AUTOSPRT					E	E				E			
BAILOUT	E	E	E	E	E	E	E	E	E	E	B		
BDMNCON										B			
BETA			E					E					
BIGER	E			E			E		E	E			
BIGER1	E			E			E		E	E			
BIGER2	E			E			E		E	E			
BLTGLSCL											B		
BLTTNSCL											B		
BOLTFACT													
BRSYMFAC													
BSHDAMP		B		B		B				B			
BUCKLE													
CA1	E	E	E	E	E	E	E	E	E	E			
CA2	E	E	E	E	E	E	E	E	E	E			
CB1	E	E	E	E	E	E	E	E	E	E			
CB2	E	E	E	E	E	E	E	E	E	E	B(1)		
CDIF										E			
CDITER													

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
CDPCH													
CDPRT													
CHECKOUT	E	E	E	E	E	E	E	E	E	E			
CK1	E	E	E	E	E	E	E	E	E	E			
CK2	E	E	E	E	E	E	E	E	E	E	B(1)		
CK3	E	E	E	E	E	E	E	E	E	E			
CK41	E	E	E	E	E	E	E	E	E	E			
CK42	E	E	E	E	E	E	E	E	E	E			
CLOSE									B				
CM1	E	E	E	E	E	E	E	E	E	E			
CM2	E	E	E	E	E	E	E	E	E	E	B(1)		
CNTASET													
CNTSCL													
COLEMAN													B
COLPHEXA											B		
COMPMATT													
CONFAC	B	B	B	B	B	B	B	B	B	B			
CORROPT													
COUPMASS	E	E	E	E	E	E	E	E	E	E	B(1)		
CP1, CP2	E	E	E	E		E	E	E	E	E			
CURV	E			E			E		E	E			
CURVPLOT	E			E					E	E			
DBALL		E	E	E	E	E	E	E	E	E			

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
DBCCONV		E	E	E	E	E	E	E	E	E			
DBCDIAG		E	E	E	E	E	E	E	E	E			
DBCOVWRT		E	E	E	E	E	E	E	E	E			
DBDICT		B	B	B	B	B	B	B	B	B			
DBDN		E	E	E	E	E	E	E	E	E			
DBDRPRJ		B	B	B	B	B	B	B	B	B			
DBDRVER		B	B	B	B	B	B	B	B	B			
DBRCV		E	E	E	E	E	E	E	E	E			
DBUP		E	E	E	E	E	E	E	E	E			
DDRMM						B				E			
DESPCH										E			
DESPCH1										E			
DFREQ		B				B				B			B
DIGITS		E	E	E	E	E		E	E				
DOF123													
DOPT				E			E		E	E			
DPEPS										B			
DPREONLY									B				
DSNOKD										B			
DSZERO									B				
DUCTFMAX													
DYNSPCF		E			E	E			E	E			
ELITASPC													

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
EPPRT		E	E	E	E	E	E	E	E	E			
EPZERO		E	E	E	E	E	E	E	E	E			
ERROR		E	E	E	E	E	E	E	E	E			
EST		E	E	E	E	E	E	E	E	E			
EXTBEMI													
EXTBEMO													
EXTDR									B				
EXTDROUT									B				
EXTDRUNT									B				
EXTOUT			B	B	B	B	B	B	B	B			
EXTRCV		E	E	E	E	E	E	E	E	E			
EXTUNIT		B	B	B	B	B	B	B	B	B			
F56	B	B	B	B	B	B	B	B	B	B	B	B	B
FACTOR		B	B	B	B	B	B	B	B	B			
FCTCi													
FCTMi													
FCTZi													
FDINTOPT													
FHPOST													B
FHPNST													B
FIXDIFFK													
FIXEDB									E	E			
FKSYMFAC													

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
FLEXINCR				B									
FLEXINV													
FLUIDMP										E			
FLUIDSE									B	B			
FOLLOWK	E								E	E			
FRQDEPO						B				B			
FRRU													
FRUMIN													
FULLGPF													
FZERO					E	E							
G		B	E		B(3)	B		B		B	B		B
GDAMPF		B	B		B	B		B		B			
GEOMU		E	E	E	E	E	E	E	E	E			
GFL						B				B			
GPACAO													
GPECT		E	E	E	E	E	E	E	E	E			
GRDPNT		E	E	E	E	E	E	E	E	E	B(1)		
GUSTAERO					B	B				B			
HARMLIM													B
HEATSTAT													
HFREQ					B	B				B			
HFREQFL						B				B			
IFP		E	E	E	E	E	E	E	E	E			

Footnotes:

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(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
IFTM						B							
INEMETH													
INREL = -1										B			
INREL = -2										B			
INRLM	E	E	E	E	E	E		E		E			
IRES		E	E	E	E	E	E	E		E			
ITAPE									B	B			
IUNIT									B	B			
K6ROT	E	E	E	E	E	E	E	E	E	E	B	B	B
KDAMP					B					B			
KDAMPFL						B				B			
KDIAG							E						
KGGCPCH													
KGGLPCH	B	B	B				B	B	B	B			
LAMLIN													
LANGLE			B				B	B					
LFREQ					B	B				B			
LFREQFL						B				B			
LGDISP			E				E	E			B	B	B
LGSTRN											B	B	
LMDYN													
LMFACT													
LMODES					B	B			E	B			

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
LMODESFL						B				B			
LMSTAT													
LOADU		E	E				E	E	E	E			
LOOPID			E				E	E					
MACH					B	B				B			
MATCTE											B	B	B
MATNL											B	B	B
MAXLP			E					E					
MAXRATIO		E	E	E	E	E	E	E	E	E	B	B	B
MDLDAMP													
MDOF													
MDSTYPE													
MECHFIL													
MECHFIX													
MECHPRT													
MEMSTRES									E				
MESH							B	B		B			
METHCMRS		E	E		E	E		E		E			
MGRID													
MMFIL													
MODACC						B				B			
MODEL		B	B	B	B	B	B	B	B	B			
MODTRK													

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
MPCX		B	B	B	B	B	B	B	B	B			
MPCZERO	E	E	E	E	E	E	E	E	E	E			
MPIBUFF													
MTREPSI													
MTREPSR													
MRRFCTD													
MTRFCTV													
MTRFMAX													
MTRRMAX													
MTRSKIP													
NASPDV										B			
NASPRT										B			
NDAMP			E					E					
NFDINTPL													
NINTPTS				E			E		E	E			
NLAYERS			E								B	B	B
NLHTLS							E						
NLMAX													
NLMIN													
NLTOL								E					
NMLOOP													
NOCOMPS				E			E		E	E			
NOELOF				E			E		E	E			

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
NOELOP				E			E		E	E			
NOFISR				E			E		E	E		B	
NOGPF				E			E		E	E			
NOMSGSTR				E			E		E	E			
NONCUP										B			
NQSET		E	E	E	E	E		E	E				
NUMOUT				E			E		E	E			
NUMOUT1				E			E		E	E			
NUMOUT2				E			E		E	E			
OCMP									E	E			
ODS													
OEE	E								E	E			
OEF	E	E	E	E	E	E	E	E	E	E			
OEFX	E	E	E	E	E	E	E	E	E	E			
OEPT	E	E	E	E	E	E	E	E	E	E			
OES	E	E	E	E	E	E	E	E	E	E			
OESE	E	E	E	E	E	E	E	E	E	E			
OESX	E	E	E	E	E	E	E	E	E	E			
OG	E			E			E		E	E			
OGEOM	B	B	B	B	B	B	B	B	B	B	B	B	
OGPF	E			E					E	E			
OGPS	E			E					E	E			
OIBULK	B	B	B	B	B	B	B	B	B	B	B	B	

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
OLDSEQ		B	B	B	B	B	B	B	B	B			
OMAXR		E	E	E	E	E	E	E	E	E	B(1)		
OMID	B	B		B	B	B			B	B			
OMPT	E	E	E	E	E	E	E	E	E	E	B(1)		
OP2FMT	B	B	B	B	B	B	B	B	B	B	B	B	
OPG	E	E	E		E	E		E		E	B(1)		
OPGEOM	E			B	B	B				B			
OPGTKG	E			B	B	B				B			
OPPHIB					B	B				B			
OPPHIPA					B	B				B			
OPTEXIT										B			
OPTION									E				
OQG	E	E	E	E	E	E	E	E	E	E			
OSWELM	B	B	B	B	B	B			B	B			
OSWPPT	B	B	B	B	B	B			B	B			
OUG	E	E	E	E	E	E	E	E	E	E			
OUGCORD	E	E	E	E	E	E	E	E	E	E	B(1)		
OUMU									E	E			
OUNIT1		E	E	E	E	E	E	E	E	E			
OUNIT2		E	E	E	E	E	E	E	E	E			
OUTOPT				E			E		E	E			
PANELMP										E			
PATVER		E	E	E	E	E	E	E	E	E			

Footnotes:

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(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
PBRPROP	B	B	B	B	B	B	B	B	B	B			
PDRMSG		E	E	E	E	E	E	E	E	E			
PEWOPT													
PHIBGN													
PHIDEL													
PHINUM													
PKRSP													
PLTMSG		E	E	E	E	E	E	E	E	E			
POST		E	E	E	E	E	E	E	E	E	B(1)	B	B
POSTEXT	E	E	E	E	E	E	E	E	E	E	B(1)		
POSTHDF5													B
POSTOPT											B(1)		
POSTU		E	E	E	E	E	E	E	E	E			
PREFDB										E			
PRGPST		E	E	E	E	E	E	E	E	E	B(1)		
PRINT					B					B			
PROUT		E	E	E	E	E	E	E	E	E	B(1)		
PRPA				E			E			E			
PRPJ		E		E		E	E	E		E			
PRTCSTM		E	E	E	E	E	E	E	E	E			
PRTEQXIN		E	E	E	E	E	E	E	E	E			
PRTGPDT		E	E	E	E	E	E	E	E	E			
PRTGPL		E	E	E	E	E	E	E	E	E			

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
PRTGPTT		E	E	E	E	E	E	E	E	E			
PRTMAXIM		E	E	E	E	E	E	E	E	E			
PRTMGG		E	E	E	E	E	E	E	E	E			
PRTPG		E	E	E	E	E	E	E	E	E			
PRTRESLT		E	E	E	E	E	E	E	E	E			
Q				B	B	B				B			
QSETREM		B	B	B	B	B		B	B				
RDCNT													
RDFLUID													
RDMTRAK													B
RDQSOL													
RDRESVEC										B			
RDSPARSE													
RESLTOPT	E	E	E	E	E	E	E	E	E	E			
RESVEC						E			E	E			
RESVINER						E			E	E			
RESVPGF						E			E	E			
RESVRAT						E			E	E			
RESVSE						E			E	E			
RESVSLI						E			E	E			
RESVSO						E			E	E			
RGBEAMA											B		
RGBEAME											B		

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
RGLCRIT											B		
RGSPRGK											B		
RLOOPNEW													
RMSINT													
RMSSF													
RMXPANEL													
RMXTRAN													
ROTCMRF													
ROTCOUP													
ROTCSV													
ROTGPF													
ROTSYNC													
RPOSTS1													
RSATT													
RSCON													
RSOPT													
RSPECTRA													
RSPRINT													
RVMASS						E			E	E			
RVMTV						E			E	E			
S1AG				E			E		E	E			
S1AM				E			E		E	E			
S1G				E			E		E	E			

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
S1M				E			E		E	E			
SCRSPEC									B				
SDAMPUP													
SECOMB			B	B		B	B	B		B			
SEMAP		B	B	B	B	B	B	B	B	B			
SEMAPOPT		B	B	B	B	B	B	B	B	B			
SEMAPPRT		B	B	B	B	B	B	B	B	B			
SEMATPRT													B
SENSUOO										B			
SEOP2CV			B	B	B	B		B	B				B
SEP1XOVR		B	B	B	B	B	B	B	B	B			
SEQOUT		B	B	B	B	B	B	B	B	B			
SERST		B	B	B	B	B	B	B	B	B			
SESDAMP	E	E	E		E				E				B
SESEF									E				
SFEF70										E			
SHIFT1													
SHLDAMP		B		B		B				B			
SIGMA								B					
SKPAMP					B	B			B	B			
SLOOPID			E										
SLOSHCMS													
SMALLQ		B	B	B	B	B	B	B	B	B			

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
SNORM		B	B	B	B	B	B	B	B	B		B	B
SNORMPRT		B	B	B	B	B	B	B	B	B			
SOFTEXIT										B			
SOLID		B	B	B	B	B	B	B	B	B			
SOLVSUB													
SPCGEN		E	E	E	E	E	E	E	E	E			
SPCSTR													
SRCOMPS	B	B	B	B	B	B	B	B	B	B		B	
SRTELTYP				E			E		E	E			
SRTOPT				E			E		E	E			
STABAO													
STABFEM													
START		B		B	B	B	B	B	B	B			
STIME			E					E					
STRUCTMP										E			
SUBID							E						
SUPER		B	B	B	B	B	B	B	B	B			
SWPANGLE													
TABID													
TABS							E	E		E	B	B	
TENSOQD											B		
TESTNEG							E						
TINY				E			E		E	E	B(1)		

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
TIREAPP													
TOLRSC		B	B	B	B	B	B	B	B	B			
TOQDGAUS											B		
TORSIN	B	B	B	B	B	B	B	B	B	B			
TSTATIC			E					E					
UNITSYS	B	B	B	B	B	B	B	B	B	B	B	B	B
UNITTIME	B	B	B	B	B	B	B	B	B	B	B	B	B
UNSYMF													
UPDTBSH										E			
USETPRT		E	E	E	E	E	E	E	E	E			
USETSEL		E	E	E	E	E	E	E	E	E			
USETSTRi		E	E	E	E	E	E	E	E	E			
VMFACT													
VMOPT		E	E	E	E	E	E	E	E	E			
VREF					B					B			
W3			B					E		B	B	B	B
W3FL										B			
W4			B					E		B	B		B
W4FL										B			
WMODAL					B								B
WRTMAT													
WTMASALL													
WTMASS		E	E	E	E	E	E	E	E	E	B(1)		B

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

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PARAMeter Name	Solution Numbers (116 through 414)												
	116	118	129	144	145	146	153	159	187	200	401	402	414
WTMASSFL													
XFLAG				E						E			

Footnotes:

(1) For SOL 401 in versions prior to Simcenter Nastran 2406, these parameters included "E" in the applicability table, indicating that they supported both a bulk data and/or a case control definition. Beginning with Simcenter Nastran 2406, by default, you must specify these parameters only in the bulk data or a fatal error will occur. You can revert to the previous behavior by defining **SYSTEM(811)=1**. This system cell setting allows you to define these parameters in the bulk data and/or the case control without causing a fatal error. Note that the SOL 401 parameters with "B" in the table (that is, no (1) footnote) must always be defined in the bulk data regardless of the system cell 811 setting.

(2) Only applicable when used with the ADAMSMNF or MBDEXPORT case control commands.

(3) For SOL 145 when using the PK-Method, PARAM,G specifications and GE field specifications on material, property, or element bulk entries are ignored. Using PARAM,WMODAL,YES and PARAM,FZERO converts the structural (hysteretic) damping to viscous damping.

(4) Applicable to incompressible fluid problems modeled using the defined fluid mass method when a sloshing surface is present.

Related Topics

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7. Item Codes

Item Codes Overview

Item codes are integer numbers assigned to specific output quantities such as the third translational component of displacement, the major principal stress at Z1 in a CQUAD4 element, or the torque in a CBAR element. Item codes have the following applications:

- DRESP1 entry for Design Sensitivity and Optimization (SOL 200).
- X-Y Plotting commands. See “Plotting” in the *Simcenter Nastran User’s Guide*.
- DTI,INDTA entry for stress sorting.

The items codes include only the elements tested and supported by the applications listed above. The elements that are not included are not supported for these applications.

For an application not listed above, for example, if you need to understand the output schema of any element in the .op2 file, you should use the OES and OEF data block descriptions in the DMAP programmers guide. The OES and OEF data block descriptions are comprehensive and include all elements.

The following tables provide item codes for:

- Element Stress or Strain.
- Element Force.
- Fluid Virtual Mass Pressure.
- Heat Transfer Flux.
- Slideline Contact Output.
- Element Strain Energy.

The following superscripts appear in the tables and indicate:

1. Data for components marked with the superscript (1) are included in the datablock MES output from module DRMS1. See the *Simcenter Nastran DMAP Programmer’s Guide*.
2. Composite Element Stresses and Failure Indices. See “Overview of Laminated Composite Materials” in the *Simcenter Nastran User’s Guide*.

Shell element item code remarks

1. The CENTER and CORNER options on the STRESS, STRAIN, and FORCE case control commands determine the CQUAD4, CQUADR, and CTRIAR item code tables. If CENTER is requested (default), the item code tables labeled as "Center Only" should be used. If CORNER is requested, the item code tables labeled as "Center and Corners" should be used.

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For example, when CENTER is requested for a CQUAD4 element, the CQUAD(33) item code format should be used. When CORNER is requested for a CQUAD4 element, the CQUAD(144) item code format should be used.

The CENTER and CORNER options on the STRESS, STRAIN, and FORCE case control commands are ignored by the CTRIA3, CTRIA6, and CQUAD8 elements. The CTRIA3 element item code table is "Center Only". The CTRIA6 and CQUAD8 item code tables are "Center and Corner".

When including shell element item codes on a DRESP1 bulk entry for Design Sensitivity and Optimization (SOL 200), do not mix elements which will produce a different item code table on the same DRESP1 stress, strain or force response.

2. Linear format shell element item codes are output in the element coordinate system. One exception; when PARAM,CURV,1 is defined, stress for the CQUAD4 and CTRIA3 elements is output in the material coordinate system.
3. Composite item codes are output in the material coordinate system defined on the element, and not the material coordinate system of the individual plies.
4. Nonlinear format shell element item codes are output in the element coordinate system.
5. Hyperelastic shell element item codes are output in the CID defined on the PLPLANE.

Solid element item code remarks

1. Linear format solid element item codes are output in the material system.
2. Nonlinear format solid element item codes are output in the element coordinate system.
3. Hyperelastic solid element item codes are output in the basic coordinate system.

Related Topics

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Element Stress (or Strain) Item Codes

All item codes refer to stresses (or strains) unless otherwise denoted.

If output is magnitude/phase, the magnitude replaces the real part, and the phase replaces the imaginary part. Strain item codes are equivalent to stress item codes. However, strain is computed for only some elements. See "Element Summary – Small Strain Elements" in the *Simcenter Nastran Element Library*.

Note

The items codes include only the elements tested and supported by the applications listed in **Item Codes Overview**. For an application not listed, for example, if you need to understand the output schema of any element in the .op2 file, you should use the OES and OEF data block descriptions in the DMAP programmers guide. The OES and OEF data block descriptions are comprehensive and include all elements.

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CBAR (34)	2[1]	End A-Point C	2[1]	End A-Point C	RM
	3[1]	End A-Point D	3[1]	End A-Point D	RM
	4[1]	End A-Point E	4[1]	End A-Point E	RM
	5[1]	End A-Point F	5[1]	End A-Point F	RM
	6[1]	Axial	6[1]	Axial	RM
	7	End A maximum	7[1]	End A-Point C	IP
	8	End A minimum	8[1]	End A-Point D	IP
	9	Safety margin in tension	9[1]	End A-Point E	IP
	10[1]	End B-Point C	10[1]	End A-Point F	IP
	11[1]	End B-Point D	11[1]	Axial	IP
	12[1]	End B-Point E	12[1]	End B-Point C	RM
	13[1]	End B-Point F	13[1]	End B-Point D	RM
	14	End B maximum	14[1]	End B-Point E	RM
	15	End B minimum	15[1]	End B-Point F	RM
	16	Safety margin in compression	16[1]	End B-Point C	IP
			17[1]	End B-Point D	IP
			18[1]	End B-Point E	IP
			19[1]	End B-Point F	IP

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CBAR (100) Intermediate Stations	2	Station Distance/Length	2	Station Distance/Length	RM
	3	Point C	3	Point C	RM
	4	Point D	4	Point D	RM
	5	Point E	5	Point E	RM
	6	Point F	6	Point F	RM
	7	Axial	7	Axial	RM
	8	Maximum	8	Maximum	RM
	9	Minimum	9	Minimum	RM
	10	Margin of Safety (Item codes above are given for End A. For codes 2 through 10 at intermediate stations add (K-1)*9 where K is the station number, and for codes at End B, K=number of stations plus 1.)	10	Point C	IP
			11	Point D	IP
			12	Point E	IP
			13	Point F	IP
			14	Axial	IP
			15	Maximum	IP
			16	Minimum (Item codes above are given for End A. For codes 2 through 16 at intermediate stations add (K-1)*15 where K is the station number, and for codes at End B, K=number of stations plus 1.)	IP

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CBEAM (2) Linear Format	2	External grid point ID	2	External grid point ID	
	3	Station distance/length	3	Station distance/length	
	4[1]	Long. Stress at Point C	4[1]	Long. Stress at Point C	RM
	5[1]	Long. Stress at Point D	5[1]	Long. Stress at Point D	RM
	6[1]	Long. Stress at Point E	6[1]	Long. Stress at Point E	RM
	7[1]	Long. Stress at Point F	7[1]	Long. Stress at Point F	RM
	8	Maximum stress	8[1]	Long. Stress at Point C	IP
	9	Minimum stress	9[1]	Long. Stress at Point D	IP
	10	Safety margin in tension	10[1]	Long. Stress at Point E	IP
	11	Safety margin in compression Item codes are given for End A. Addition of the quantity (K-1)*10 to the item code points to the same information for other stations, where K is the station number. K=11 for End B, and K=2 through 10 for intermediate stations.	11[1]	Long. Stress at Point F Item codes are given for End A. Addition of the quantity (K-1)*10 to the item code points to the same information for other stations, where K is the station number. K=11 for End B, and K=2 through 10 for intermediate stations.	IP

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CBEAM (94) Nonlinear Format	2	External grid point ID	Not applicable		
	3	C (Character)			
	4	Long. Stress at point C			
	5	Equivalent stress			
	6	Total strain			
	7	Effective plastic strain			
	8	Effective creep strain (Item codes 3 through 8 are repeated for points D, E, and F. Then the entire record (from 2 through N) is repeated for End B of the element.)			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CBEND (69)	2	External grid point ID	2	External grid point ID	
	3	Circumferential angle	3	Circumferential angle	
	4[1]	Long. Stress at Point C	4[1]	Long. Stress at Point C	RM
	5[1]	Long. Stress at Point D	5[1]	Long. Stress at Point D	RM
	6[1]	Long. Stress at Point E	6[1]	Long. Stress at Point E	RM
	7[1]	Long. Stress at Point F	7[1]	Long. Stress at Point F	RM
	8	Maximum stress	8[1]	Long. Stress at Point C	IP
	9	Minimum stress	9[1]	Long. Stress at Point D	IP
	10	Safety margin in tension	10[1]	Long. Stress at Point E	IP
	11	Safety margin in compression (Item codes are given for End A. Item codes 12 through 21 point to the same information for end B.)	11[1]	Long. Stress at Point F (Item codes are given for End A. Item codes 12 through 21 point to the same information for End B.)	IP

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CBUSH (102) Linear Format	2	Translation-x	2	Translation-x	RM
	3	Translation-y	3	Translation-y	RM
	4	Translation-z	4	Translation-z	RM
	5	Rotation-x	5	Rotation-x	RM
	6	Rotation-y	6	Rotation-y	RM
	7	Rotation-z	7	Rotation-z	RM
			8	Translation-x	IP
			9	Translation-y	IP
			10	Translation-z	IP
			11	Rotation-x	IP
			12	Rotation-y	IP
			13	Rotation-z	IP

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Element Name (Code)	Real Stresses or Strains	
	Item Code	Item
CBUSH (226) Nonlinear Format	5	Stress Translational X
	6	Stress Translational Y
	7	Stress Translational Z
	8	Strain Translational X
	9	Strain Translational Y
	10	Strain Translational Z
	14	Stress Rotational X
	15	Stress Rotational Y
	16	Stress Rotational Z
	17	Strain Rotational X
	18	Strain Rotational Y
	19	Strain Rotational Z

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CBUSH1D (40)	1	Element ID	Not applicable		
	2	Axial force			
	3	Axial displacement			
	4	Axial velocity			
	5	Axial stress			
	6	Axial strain			
	7				
	8				

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CDUM3	2[1]	S1	2[1]	S1	RM
thru	3[1]	S2	3[1]	S2	RM
	4[1]	S3	4[1]	S3	RM
CDUM9	5[1]	S4	5[1]	S4	RM
(55-61)	6[1]	S5	6[1]	S5	RM
	7[1]	S6	7[1]	S6	RM
	8[1]	S7	8[1]	S7	RM
	9[1]	S8	9[1]	S8	RM
	10[1]	S9	10[1]	S9	RM
			11[1]	S1	IP
			12[1]	S2	IP
			13[1]	S3	IP
			14[1]	S4	IP
			15[1]	S5	IP
			16[1]	S6	IP
			17[1]	S7	IP
			18[1]	S8	IP
			19[1]	S9	IP

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CELAS1 (11)	2[1]	Stress	2[1]	Stress	RM
			3[1]	Stress	IP
CELAS2 (12)	2[1]	Stress	2[1]	Stress	RM
			3[1]	Stress	IP
CELAS3 (13)	2[1]	Stress	2[1]	Stress	RM
			3[1]	Stress	IP

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CGAP (86)	2	Normal x	Not applicable		
	3	Shear y			
	4	Shear z			
	5	Axial u			
	6	Shear v			
	7	Shear w			
	8	Slip v			
	9	Slip w			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CHEXA (67) Linear Format	2	Stress coordinate system	2	Stress coordinate system	
	3	Coordinate type (Character)	3	Coordinate type (Character)	
	4	Number of active points	4	Number of active points	
	5	External grid ID (0=center)	5	External grid ID (0=center)	
	6[1]	Normal x	6[1]	Normal x	RM
	7[1]	Shear xy	7[1]	Normal y	RM
	8	First principal	8[1]	Normal z	RM
	9	First principal x cosine	9[1]	Shear xy	RM
	10	Second principal x cosine	10[1]	Shear yz	RM
	11	Third principal x cosine	11[1]	Shear zx	RM
	12	Mean pressure	12[1]	Normal x	IP
	13	von Mises or octahedral shear stress	13[1]	Normal y	IP
	14[1]	Normal y	14[1]	Normal z	IP
	15[1]	Shear yz	15[1]	Shear xy	IP
	16	Second principal	16[1]	Shear yz	IP
	17	First principal y cosine	17[1]	Shear zx	IP
	18	Second principal y cosine	18-121	Repeat items 5 through 17 for 8 corners	
	19	Third principal y cosine			
	20[1]	Normal z			
	21[1]	Shear zx			
	22	Third principal			
	23	First principal z cosine			
	24	Second principal z cosine			
	25	Third principal z cosine			
	26-193	repeat items 5 through 25 for 8 corners			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CHEXA (93) Nonlinear	2	Grid/Gauss	Not applicable		
	3	External grid ID (0=center)			
	4	Stress-X			
	5	Stress-Y			
	6	Stress-Z			
	7	Stress-XY			
	8	Stress-YZ			
	9	Stress-ZX			
	10	Equivalent stress			
	11	Effective plastic strain			
	12	Effective creep strain			
	13	Strain-X			
	14	Strain-Y			
	15	Strain-Z			
	16	Strain-XY			
	17	Strain-YZ			
	18	Strain-ZX			
	19-146	Repeat items 3 through 18 for 8 corners			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CHEXA (269) Composite Center Only	2	Lamina number	2	Lamina number	
	3	Fiber location	3	Fiber location	
	4	0	4	0	
	5	Normal-1	5	Normal-1	RM
	6	Normal-2	6	Normal-2	RM
	7	Normal-3	7	Normal-3	RM
	8	Shear-12	8	Shear-12	RM
	9	Shear-23	9	Shear-23	RM
	10	Shear-13	10	Shear-13	RM
	11	von Mises	11	Normal-1	IP
			12	Normal-2	IP
			13	Normal-3	IP
			14	Shear-12	IP
			15	Shear-23	IP
			16	Shear-13	IP
If CPLYBT is specified on the stress or strain request, repeat words 3-11 once for real stresses and strains, or repeat words 3-16 once for complex stresses and strains. If CPLYBMT is specified on the stress or strain request, repeat words 3-11 twice for real stresses and strains, or repeat words 3-16 twice for complex stresses and strains.					

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CHEXA (269) Composite Center and Corners	2	Lamina number	2	Lamina number	
	3	Fiber location	3	Fiber location	
	4	0	4	0	
	5	Normal-1	5	Normal-1	RM
	6	Normal-2	6	Normal-2	RM
	7	Normal-3	7	Normal-3	RM
	8	Shear-12	8	Shear-12	RM
	9	Shear-23	9	Shear-23	RM
	10	Shear-13	10	Shear-13	RM
	11	von Mises	11	Normal-1	IP
	12	Grid 1	12	Normal-2	IP
	13-19	Same as items 5-11	13	Normal-3	IP
	20	Grid 2	14	Shear-12	IP
	21-27	Same as items 5-11	15	Shear-23	IP
	28	Grid 3	16	Shear-13	IP
	29-35	Same as items 5-11	17	Grid 1	
	36	Grid 4	18-29	Same as items 5-16	
	37-43	Same as items 5-11	30	Grid 2	
			31-42	Same as items 5-16	
			43	Grid 3	
			44-55	Same as items 5-16	
			56	Grid 4	
			57-68	Same as items 5-16	

If CPLYBT is specified on the stress or strain request, repeat words 3-43 once for real stresses and strains, or repeat words 3-68 once for complex stresses and strains.

If CPLYBMT is specified on the stress or strain request, repeat words 3-43 twice for real stresses and strains, or repeat words 3-68 twice for complex stresses and strains.

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CHEXAFD (202) Hyperelastic with 8 grid points	2	Grid/Gauss	Not applicable		
	3	Grid/Gauss ID (0=center)			
	4	Cauchy stress-X			
	5	Cauchy stress-Y			
	6	Cauchy stress-Z			
	7	Cauchy stress-XY			
	8	Cauchy stress-YZ			
	9	Cauchy stress-ZX			
	10	Pressure $p=\frac{1}{3}(\sigma_x+\sigma_y+\sigma_z)$			
	11	Volumetric strain J-1			
	12	Logarithmic strain-X			
	13	Logarithmic strain-Y			
	14	Logarithmic strain-Z			
	15	Logarithmic strain-XY			
	16	Logarithmic strain-YZ			
	17	Logarithmic strain-ZX			
	18-122	Repeat items 3 through 17 for 7 Gauss/grid points			
CHEXAFD (207) Hyperelastic with 20 grid points	2-17	Same as CHEXAFD (202)	Not applicable		
	18-407	Repeat items 3 through 17 for 26 Gauss points			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CONROD (10) Linear Format	2[1]	Axial stress	2[1]	Axial stress	RM
	3	Axial safety margin	3[1]	Axial stress	IP
	4[1]	Torsional stress	4[1]	Torsional stress	RM
	5	Torsional safety margin	5[1]	Torsional stress	IP
CONROD (92) Nonlinear Format	2	Axial stress	Not applicable		
	3	Equivalent stress			
	4	Total strain			
	5	Effective plastic strain			
	6	Effective creep strain			
	7	Linear torsional stress			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPENTA (68) Linear Format	2	Stress coordinate system	2	Stress coordinate system	
	3	Coordinate type (Character)	3	Coordinate type (Character)	
	4	Number of active points	4	Number of active points	
	5	External grid ID	5	External grid ID	
		(0=center)		(0=center)	
	6[1]	Normal x	6[1]	Normal x	RM
	7[1]	Shear xy	7[1]	Normal y	RM
	8	First principal	8[1]	Normal z	RM
	9	First principal x cosine	9[1]	Shear xy	RM
	10	Second principal x cosine	10[1]	Shear yz	RM
	11	Third principal x cosine	11[1]	Shear zx	RM
	12	Mean pressure	12[1]	Normal x	IP
	13	von Mises or Octahedral shear stress	13[1]	Normal y	IP
	14[1]	Normal y	14[1]	Normal z	IP
	15[1]	Shear yz	15[1]	Shear xy	IP
	16	Second principal	16[1]	Shear yz	IP
	17	First principal y cosine	17[1]	Shear zx	IP
	18	Second principal y cosine	18-95	Repeat items 5 through 17 for 6 corners	
	19	Third principal y cosine			
	20[1]	Normal z			
	21[1]	Shear zx			
	22	Third principal			
	23	First principal z cosine			
	24	Second principal z cosine			
	25	Third principal z cosine			
	26-151	Repeat items 5 through 25 for 6 corners			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPENTA (91) Nonlinear Format	2	Grid/Gauss	Not applicable		
	3	External grid ID (0=center)			
	4	Normal x stress			
	5	Normal y stress			
	6	Normal z stress			
	7	Shear xy stress			
	8	Shear yz stress			
	9	Shear zx stress			
	10	Equivalent stress			
	11	Effective plastic strain			
	12	Effective creep strain			
	13	Normal x strain			
	14	Normal y strain			
	15	Normal z strain			
	16	Shear xy strain			
	17	Shear yz strain			
	18	Shear zx strain			
	19-114	Repeat items 3 through 18 for 6 corners			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPENTA (270) Composite Center Only	2	Lamina number	2	Lamina number	
	3	Fiber location	3	Fiber location	
	4	0	4	0	
	5	Normal-1	5	Normal-1	RM
	6	Normal-2	6	Normal-2	RM
	7	Normal-3	7	Normal-3	RM
	8	Shear-12	8	Shear-12	RM
	9	Shear-23	9	Shear-23	RM
	10	Shear-13	10	Shear-13	RM
	11	von Mises	11	Normal-1	IP
			12	Normal-2	IP
			13	Normal-3	IP
			14	Shear-12	IP
			15	Shear-23	IP
			16	Shear-13	IP
If CPLYBT is specified on the stress or strain request, repeat words 3-11 once for real stresses and strains, or repeat words 3-16 once for complex stresses and strains. If CPLYBMT is specified on the stress or strain request, repeat words 3-11 twice for real stresses and strains, or repeat words 3-16 twice for complex stresses and strains.					

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPENTA (270) Composite Center and Corners	2	Lamina number	2	Lamina number	
	3	Fiber location	3	Fiber location	
	4	0	4	0	
	5	Normal-1	5	Normal-1	RM
	6	Normal-2	6	Normal-2	RM
	7	Normal-3	7	Normal-3	RM
	8	Shear-12	8	Shear-12	RM
	9	Shear-23	9	Shear-23	RM
	10	Shear-13	10	Shear-13	RM
	11	von Mises	11	Normal-1	IP
	12	Grid 1	12	Normal-2	IP
	13-19	Same as items 5-11	13	Normal-3	IP
	20	Grid 2	14	Shear-12	IP
	21-27	Same as items 5-11	15	Shear-23	IP
	28	Grid 3	16	Shear-13	IP
	29-35	Same as items 5-11	17	Grid 1	
			18-29	Same as items 5-16	
			30	Grid 2	
			31-42	Same as items 5-16	
			43	Grid 3	
			44-55	Same as items 5-16	

If CPLYBT is specified on the stress or strain request, repeat words 3-35 once for real stresses and strains, or repeat words 3-55 once for complex stresses and strains.

If CPLYBMT is specified on the stress or strain request, repeat words 3-35 twice for real stresses and strains, or repeat words 3-55 twice for complex stresses and strains.

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPENTAFD (204) Hyperelastic with 6 grid points	2-17	Same as CHEXAFD (201)	Not applicable		
	18-92	Repeat items 3 through 17 for 5 Gauss points			
CPENTAFD (209) Hyperelastic with 15 grid points	2-17	Same as CHEXAFD (201)	Not applicable		
	18-317	Repeat items 3 through 17 for 20 Gauss points			

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN3 (271) Triangle plane strain Linear Format (Center Only)	2	Normal stress in x	2	Normal stress in x	RM
	3	Normal stress in y	3	Normal stress in x	IP
	4	Normal stress in z	4	Normal stress in y	RM
	5	In-plane shear stress	5	Normal stress in y	IP
	6	Von Mises stress	6	Normal stress in z	RM
			7	Normal stress in z	IP
			8	In-plane shear stress	RM
			9	In-plane shear stress	IP

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN4 (272) Quadrilateral plane strain Linear Format (Center and Corners)	3	Grid ID, 0 for centroid	3	Grid ID, 0 for centroid	
	4	Normal stress in x	4	Normal stress in x	RM
	5	Normal stress in y	5	Normal stress in x	IP
	6	Normal stress in z	6	Normal stress in y	RM
	7	In-plane shear stress	7	Normal stress in y	IP
	8	Von Mises stress	8	Normal stress in z	RM
	9	Grid ID, 1 for corner 1	9	Normal stress in z	IP
	10-14	Same as 4 through 8 for corner 1	10	In-plane shear stress	RM
	15	Grid ID, 2 for corner 2	11	In-plane shear stress	IP
	16-20	Same as 4 through 8 for corner 2	12	Grid ID, 1 for corner 1	
	21	Grid ID, 3 for corner 3	13-20	Same as 4 through 11 for corner 1	
	22-26	Same as 4 through 8 for corner 3	21	Grid ID, 2 for corner 2	
	27	Grid ID, 4 for corner 4	22-29	Same as 4 through 11 for corner 2	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
	28-32	Same as 4 through 8 for corner 4	30	Grid ID, 3 for corner 3	
			31-38	Same as 4 through 11 for corner 3	
			39	Grid ID, 4 for corner 4	
			40-47	Same as 4 through 11 for corner 4	

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN6 (273) Triangle plane strain Linear Format (Center and Corners)	3	Grid ID, 0 for centroid	3	Grid ID, 0 for centroid	
	4	Normal stress in x	4	Normal stress in x	RM
	5	Normal stress in y	5	Normal stress in x	IP
	6	Normal stress in z	6	Normal stress in y	RM
	7	In-plane shear stress	7	Normal stress in y	IP
	8	Von Mises stress	8	Normal stress in z	RM
	9	Grid ID, 1 for corner 1	9	Normal stress in z	IP
	10-14	Same as 4 through 8 for corner 1	10	In-plane shear stress	RM
	15	Grid ID, 2 for corner 2	11	In-plane shear stress	IP
	16-20	Same as 4 through 8 for corner 2	12	Grid ID, 1 for corner 1	
	21	Grid ID, 3 for corner 3	13-20	Same as 4 through 11 for corner 1	
	22-26	Same as 4 through 8 for corner 3	21	Grid ID, 2 for corner 2	
			22-29	Same as 4 through 11 for corner 2	
			30	Grid ID, 3 for corner 3	
			31-38	Same as 4 through 11 for corner 3	

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN8 (274) Quadrilateral plane strain Linear Format (Center and Corners)	3	Grid ID, 0 for centroid	3	Grid ID, 0 for centroid	
	4	Normal stress in x	4	Normal stress in x	RM
	5	Normal stress in y	5	Normal stress in x	IP
	6	Normal stress in z	6	Normal stress in y	RM
	7	In-plane shear stress	7	Normal stress in y	IP
	8	Von Mises stress	8	Normal stress in z	RM
	9	Grid ID, 1 for corner 1	9	Normal stress in z	IP
	10-14	Same as 4 through 8 for corner 1	10	In-plane shear stress	RM

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
	15	Grid ID, 2 for corner 2	11	In-plane shear stress	IP
	16-20	Same as 4 through 8 for corner 2	12	Grid ID, 1 for corner 1	
	21	Grid ID, 3 for corner 3	13-20	Same as 4 through 11 for corner 1	
	22-26	Same as 4 through 8 for corner 3	21	Grid ID, 2 for corner 2	
	27	Grid ID, 4 for corner 4	22-29	Same as 4 through 11 for corner 2	
	28-32	Same as 4 through 8 for corner 4	30	Grid ID, 3 for corner 3	
			31-38	Same as 4 through 11 for corner 3	
			39	Grid ID, 4 for corner 4	
			40-47	Same as 4 through 11 for corner 4	

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS3 (275) Triangle plane stress Linear Format (Center Only)	2	Normal stress in x	2	Normal stress in x	RM
	3	Normal stress in y	3	Normal stress in x	IP
	4	Normal stress in z	4	Normal stress in y	RM
	5	In-plane shear stress	5	Normal stress in y	IP
	6	Von Mises stress	6	Normal stress in z	RM
			7	Normal stress in z	IP
			8	In-plane shear stress	RM
			9	In-plane shear stress	IP

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS4 (276) Quadrilateral plane stress Linear Format (Center and Corners)	3	Grid ID, 0 for centroid	3	Grid ID, 0 for centroid	
	4	Normal stress in x	4	Normal stress in x	RM
	5	Normal stress in y	5	Normal stress in x	IP
	6	Normal stress in z	6	Normal stress in y	RM
	7	In-plane shear stress	7	Normal stress in y	IP
	8	Von Mises stress	8	Normal stress in z	RM
	9	Grid ID, 1 for corner 1	9	Normal stress in z	IP
	10-14	Same as 4 through 8 for corner 1	10	In-plane shear stress	RM
	15	Grid ID, 2 for corner 2	11	In-plane shear stress	IP
	16-20	Same as 4 through 8 for corner 2	12	Grid ID, 1 for corner 1	
	21	Grid ID, 3 for corner 3	13-20	Same as 4 through 11 for corner 1	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
	22-26	Same as 4 through 8 for corner 3	21	Grid ID, 2 for corner 2	
	27	Grid ID, 4 for corner 4	22-29	Same as 4 through 11 for corner 2	
	28-32	Same as 4 through 8 for corner 4	30	Grid ID, 3 for corner 3	
			31-38	Same as 4 through 11 for corner 3	
			39	Grid ID, 4 for corner 4	
			40-47	Same as 4 through 11 for corner 4	

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS6 (277) Triangle plane stress Linear Format (Center and Corners)	3	Grid ID, 0 for centroid	3	Grid ID, 0 for centroid	
	4	Normal stress in x	4	Normal stress in x	RM
	5	Normal stress in y	5	Normal stress in x	IP
	6	Normal stress in z	6	Normal stress in y	RM
	7	In-plane shear stress	7	Normal stress in y	IP
	8	Von Mises stress	8	Normal stress in z	RM
	9	Grid ID, 1 for corner 1	9	Normal stress in z	IP
	10-14	Same as 4 through 8 for corner 1	10	In-plane shear stress	RM
	15	Grid ID, 2 for corner 2	11	In-plane shear stress	IP
	16-20	Same as 4 through 8 for corner 2	12	Grid ID, 1 for corner 1	
	21	Grid ID, 3 for corner 3	13-20	Same as 4 through 11 for corner 1	
	22-26	Same as 4 through 8 for corner 3	21	Grid ID, 2 for corner 2	
			22-29	Same as 4 through 11 for corner 2	
			30	Grid ID, 3 for corner 3	
			31-38	Same as 4 through 11 for corner 3	

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS8 (278) Quadrilateral plane stress Linear Format (Center and Corners)	3	Grid ID, 0 for centroid	3	Grid ID, 0 for centroid	
	4	Normal stress in x	4	Normal stress in x	RM
	5	Normal stress in y	5	Normal stress in x	IP
	6	Normal stress in z	6	Normal stress in y	RM
	7	In-plane shear stress	7	Normal stress in y	IP
	8	Von Mises stress	8	Normal stress in z	RM

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
	9	Grid ID, 1 for corner 1	9	Normal stress in z	IP
	10-14	Same as 4 through 8 for corner 1	10	In-plane shear stress	RM
	15	Grid ID, 2 for corner 2	11	In-plane shear stress	IP
	16-20	Same as 4 through 8 for corner 2	12	Grid ID, 1 for corner 1	
	21	Grid ID, 3 for corner 3	13-20	Same as 4 through 11 for corner 1	
	22-26	Same as 4 through 8 for corner 3	21	Grid ID, 2 for corner 2	
	27	Grid ID, 4 for corner 4	22-29	Same as 4 through 11 for corner 2	
	28-32	Same as 4 through 8 for corner 4	30	Grid ID, 3 for corner 3	
			31-38	Same as 4 through 11 for corner 3	
			39	Grid ID, 4 for corner 4	
			40-47	Same as 4 through 11 for corner 4	

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN3 (281) Triangle plane strain Nonlinear format (Center Only)	2	Normal stress in x	Not applicable		
	3	Normal stress in y			
	4	Normal stress in z			
	5	In-plane shear stress			
	6	Equivalent stress			
	7	Effective plastic/inelastic strain			
	8	Effective creep strain			
	9	Strain in x			
	10	Strain in y			
	11	Strain in z			
	12	In-plane shear strain			

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN6 (283) Triangle plane strain Nonlinear format (Center and Corners)	3	Grid ID, 0 for centroid	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Equivalent stress			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
	9	Effective plastic/inelastic strain			
	10	Effective creep strain			
	11	Strain in x			
	12	Strain in y			
	13	Strain in z			
	14	In-plane shear strain			
	Words 3 through 14 repeat 3 times for corner locations.				

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN8 (284) Quadrilateral plane strain Nonlinear format (Center and Corners)	3	Grid ID, 0 for centroid	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Equivalent stress			
	9	Effective plastic/inelastic strain			
	10	Effective creep strain			
	11	Strain in x			
	12	Strain in y			
	13	Strain in z			
	14	In-plane shear strain			
	Words 3 through 14 repeat 4 times for corner locations.				

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS3 (285) Triangle plane stress Nonlinear format (Center Only)	2	Normal stress in x	Not applicable		
	3	Normal stress in y			
	4	Normal stress in z			
	5	In-plane shear stress			
	6	Equivalent stress			
	7	Effective plastic/inelastic strain			
	8	Effective creep strain			
	9	Strain in x			
	10	Strain in y			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
	11	Strain in z			
	12	In-plane shear strain			

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS6 (287) Triangle plane stress Nonlinear format (Center and Corners)	3	Grid ID, 0 for centroid	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Equivalent stress			
	9	Effective plastic/inelastic strain			
	10	Effective creep strain			
	11	Strain in x			
	12	Strain in y			
	13	Strain in z			
	14	In-plane shear strain			
	Words 3 through 14 repeat 3 times for corner locations.				

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS8 (288) Quadrilateral plane stress Nonlinear format (Center and Corners)	3	Grid ID, 0 for centroid	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Equivalent stress			
	9	Effective plastic/inelastic strain			
	10	Effective creep strain			
	11	Strain in x			
	12	Strain in y			
	13	Strain in z			
	14	In-plane shear strain			
	Words 3 through 14 repeat 4 times for corner locations.				

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN4 (289) Quadrilateral plane strain Nonlinear format (Center and Corners)	3	Grid ID, 0 for centroid	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Equivalent stress			
	9	Effective plastic/inelastic strain			
	10	Effective creep strain			
	11	Strain in x			
	12	Strain in y			
	13	Strain in z			
	14	In-plane shear strain			
	Words 3 through 14 repeat 4 times for corner locations.				

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS4 (290) Quadrilateral plane stress Nonlinear format (Center and Corners)	3	Grid ID, 0 for centroid	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Equivalent stress			
	9	Effective plastic/inelastic strain			
	10	Effective creep strain			
	11	Strain in x			
	12	Strain in y			
	13	Strain in z			
	14	In-plane shear strain			
	Words 3 through 14 repeat 4 times for corner locations.				

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN3 (291)	3	Grid ID	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
Triangle plane strain Hyperelastic (Grid)	6	Normal stress in z			
	7	In-plane shear stress			
	8	Pressure			
	9	Volume strain			
	10	Strain in x			
	11	Strain in y			
	12	Strain in z			
	13	In-plane shear strain			
	14-35	Words 3 through 13 repeat 2 times.			

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN4 (292) Quadrilateral plane strain Hyperelastic (Grid)	3	Grid ID	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Pressure			
	9	Volume strain			
	10	Strain in x			
	11	Strain in y			
	12	Strain in z			
	13	In-plane shear strain			
	14-46	Words 3 through 13 repeat 3 times.			

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN6 (293) Triangle plane strain Hyperelastic (Grid)	3	Grid ID	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Pressure			
	9	Volume strain			
	10	Strain in x			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
	11	Strain in y			
	12	Strain in z			
	13	In-plane shear strain			
	14-35	Words 3 through 13 repeat 2 times.			

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTN8 (294) Quadrilateral plane strain Hyperelastic (Grid)	3	Grid ID	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Pressure			
	9	Volume strain			
	10	Strain in x			
	11	Strain in y			
	12	Strain in z			
	13	In-plane shear strain			
	14-46	Words 3 through 13 repeat 3 times.			

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS3 (295) Triangle plane stress Hyperelastic (Grid)	3	Grid ID	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Pressure			
	9	Volume strain			
	10	Strain in x			
	11	Strain in y			
	12	Strain in z			
	13	In-plane shear strain			
	14-35	Words 3 through 13 repeat 2 times.			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS4 (296) Quadrilateral plane stress Hyperelastic (Grid)	3	Grid ID	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Pressure			
	9	Volume strain			
	10	Strain in x			
	11	Strain in y			
	12	Strain in z			
	13	In-plane shear strain			
	14-46	Words 3 through 13 repeat 3 times.			

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS6 (297) Triangle plane stress Hyperelastic (Grid)	3	Grid ID	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			
	8	Pressure			
	9	Volume strain			
	10	Strain in x			
	11	Strain in y			
	12	Strain in z			
	13	In-plane shear strain			
	14-35	Words 3 through 13 repeat 2 times.			

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPLSTS8 (298) Quadrilateral plane stress Hyperelastic	3	Grid ID	Not applicable		
	4	Normal stress in x			
	5	Normal stress in y			
	6	Normal stress in z			
	7	In-plane shear stress			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
(Grid)	8	Pressure			
	9	Volume strain			
	10	Strain in x			
	11	Strain in y			
	12	Strain in z			
	13	In-plane shear strain			
	14-46	Words 3 through 13 repeat 3 times.			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPYRAM (255) Linear Format	2	Stress/Strain coordinate system	2	Stress coordinate system	
	3	Coordinate type (Character)	3	Coordinate type (Character)	
	4	Number of active points	4	Number of active points	
	5	External grid ID	5	External grid ID	
		(0=center)		(0=center)	
	6	Normal x	6	Normal x	RM
	7	Shear xy	7	Normal y	RM
	8	First principal	8	Normal z	RM
	9	First principal x cosine	9	Shear xy	RM
	10	Second principal x cosine	10	Shear yz	RM
	11	Third principal x cosine	11	Shear zx	RM
	12	Mean pressure	12	Normal x	IP
	13	von Mises or octahedral shear stress	13	Normal y	IP
	14	Normal y	14	Normal z	IP
	15	Shear yz	15	Shear xy	IP
	16	Second principal	16	Shear yz	IP
	17	First principal y cosine	17	Shear zx	IP
	18	Second principal y cosine	18-82	Repeat items 5 through 17 for five corners	
	19	Third principal y cosine			
	20	Normal z			
	21	Shear zx			
	22	Third principal			
	23	First principal z cosine			
	24	Second principal z cosine			
	25	Third principal z cosine			
	26-130	Repeat items 5 through 25 for five corners			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPYRAM (256) Nonlinear Format	3	0 (Result at Center)	Not applicable		
	4	Stress in X			
	5	Stress in Y			
	6	Stress in Z			
	7	Stress in XY			
	8	Stress in YZ			
	9	Stress in ZX			
	10	Equivalent stress			
	11	Equivalent plastic strain			
	12	Effective creep strain			
	13	Strain in X			
	14	Strain in Y			
	15	Strain in Z			
	16	Strain in XY			
	17	Strain in YZ			
	18	Strain in ZX			
	19	Grid 1 ID			
	20–34	Same as items 4–18 for corner 1			
	35	Grid 2 ID			
	36–50	Same as items 4–18 for corner 2			
	51	Grid 3 ID			
	52–66	Same as items 4–18 for corner 3			
	67	Grid 4 ID			
	68–82	Same as items 4–18 for corner 4			
	83	Grid 5 ID			
	84–98	Same as items 4–18 for corner 5			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CPYRAMFD (257) Pyramid with 5-grids (258) Pyramid with 13-grids Hyperelastic	2	Grid	Not applicable		
	3	Grid ID			
	4	Stress in X			
	5	Stress in Y			
	6	Stress in Z			
	7	Stress in XY			
	8	Stress in YZ			
	9	Stress in ZX			
	10	Pressure $p=\frac{1}{3}(\sigma_x+\sigma_y+\sigma_z)$			
	11	Volumetric strain			
	12	Strain in X			
	13	Strain in Y			
	14	Strain in Z			
	15	Strain in XY			
	16	Strain in YZ			
	17	Strain in ZX			
	18-77	Repeat items 3 through 17 for 5 grid points			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUAD4 (33) Linear Format Center Only	2	Z1=Fiber Distance 1	2	Z1=Fiber Distance 1	
	3[1]	Normal x at Z1	3[1]	Normal x at Z1	RM
	4[1]	Normal y at Z1	4[1]	Normal x at Z1	IP
	5[1]	Shear xy at Z1	5[1]	Normal y at Z1	RM
	6	Shear angle at Z1	6[1]	Normal y at Z1	IP
	7	Major principal at Z1	7[1]	Shear xy at Z1	RM
	8	Minor principal at Z1	8[1]	Shear xy at Z1	IP
	9	von Mises or maximum shear at Z1	9	Z2=Fiber distance 2	
	10	Z2=Fiber distance 2	10[1]	Normal x at Z2	RM
	11[1]	Normal x at Z2	11[1]	Normal x at Z2	IP
	12[1]	Normal y at Z2	12[1]	Normal y at Z2	RM
	13[1]	Shear xy at Z2	13[1]	Normal y at Z2	IP
	14	Shear angle at Z2	14	Shear xy at Z2	RM
	15	Major principal at Z2	15	Shear xy at Z2	IP
	16	Minor principal at Z2			
	17	von Mises or maximum shear at Z2			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUAD4 (144) Linear Format Center and Corners	1	EID	1	EID	
	2	CEN/	2	CEN/	
	3	4	3	4	
	4	Z1-Fiber distance	4	Z1-Fiber distance	
	5	Normal x at Z1	5	Normal x at Z1	RM
	6	Normal y at Z1	6	Normal x at Z1	IP
	7	Shear xy at Z1	7	Normal y at Z1	RM
	8	Shear angle at Z1	8	Normal y at Z1	IP
	9	Major principal at Z1	9	Shear xy at Z1	RM
	10	Minor principal at Z1	10	Shear xy at Z1	IP
	11	von Mises or maximum shear at Z1	11	Z2-Fiber distance	
	12	Z2-Fiber distance	12	Normal x at Z2	RM
	13	Normal x at Z2	13	Normal x at Z2	IP
	14	Normal y at Z2	14	Normal y at Z2	RM
	15	Shear xy at Z2	15	Normal y at Z2	IP
	16	Shear angle at Z2	16	Shear xy at Z2	RM
	17	Major principal at Z2	17	Shear xy at Z2	IP
	18	Minor principal at Z2	18	Grid 1	
	19	von Mises or maximum shear at Z2	19-32	Same as 4 through 17 for corner 1	
	20	Grid 1	33	Grid 2	
	21-36	Same as items 4 through19 for corner 1	34-47	Same as items 4 through 17 for corner 2	
	37	Grid 2	48	Grid 3	
	38-53	Same as items 4 through19 for corner 2	49-62	Same as items 4 through 17 for corner 3	
	54	Grid 3	63	Grid 4	
	55-70	Same as items 4 through 19 for corner 3	64-77	Same as items 4 through 17 for corner 4	
	71	Grid 4			
	72-87	Same as items 4 through 19 for corner 4			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUAD4[2]	2	Lamina Number	2	Lamina Number	
(95)	3	Normal-1	3	Normal-1	RM
Composite	4	Normal-2	4	Normal-2	RM
Center Only	5	Shear-12	5	Shear-12	RM
	6	Shear-1Z	6	Shear-1Z	RM
	7	Shear-2Z	7	Shear-2Z	RM
	8	Shear angle	8	Normal-1	IP
	9	Major principal	9	Normal-2	IP
	10	Minor principal	10	Shear-12	IP
	11	Maximum shear	11	Shear-1Z	IP
			12	Shear-2Z	IP

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUAD4 (90) Nonlinear Format Center Only	2	Z1=Fiber Distance 1 (Plane stress only)	Not applicable		
	3	Stress-X (at Z1, if plane stress)			
	4	Stress-Y (at Z1, if plane stress)			
	5	Stress-Z (Plane strain only)			
	6	Stress-XY (at Z1, if plane stress)			
	7	Equivalent stress (at Z1, if plane stress)			
	8	Plastic strain (at Z1, if plane stress)			
	9	Creep strain (at Z1, if plane stress)			
	10	Strain-X (at Z1, if plane stress)			
	11	Strain-Y (at Z1, if plane stress)			
	12	Strain-Z (plane strain only)			
	13	Strain-XY (at Z1, if plane stress)			
	14-25	Items 2 through 13 repeated for fiber distance Z2 (plane stress only)			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUAD4 (254) Nonlinear Format Center and Corners	3	0 (Result at Center)	Not applicable		
	4	Z1=Fiber Distance			
	5	Stress in X at Z1			
	6	Stress in Y at Z1			
	7	Stress in Z at Z1			
	8	Stress in XY at Z1			
	9	Equivalent stress at Z1			
	10	Effective plastic/nlelastic strain at Z1			
	11	Effective creep strain at Z1			
	12	Strain in X at Z1			
	13	Strain in Y at Z1			
	14	Strain in Z at Z1			
	15	Strain in XY at Z1			
	16	Z2=Fiber Distance			
	17-27	Items 5 through 15 repeat for fiber distance Z2			
	28	Grid 1 ID			
	29–52	Same as items 4–27 for corner 1			
	53	Grid 2 ID			
	54–77	Same as items 4–27 for corner 2			
	78	Grid 3 ID			
	79–102	Same as items 4–27 for corner 3			
	103	Grid 4 ID			
	104–127	Same as items 4–27 for corner 4			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUAD8 (64) Linear Format Center and Corners		Same as CQUAD4(144)		Same as CQUAD4(144)	
CQUAD8[2] (96) Composite Center Only	2-11	Same as CQUAD4(95)	2-12	Same as CQUAD4(95)	
CQUAD8 (241) Nonlinear Format Center and Corners		Same as CQUAD4(254)	Not applicable		

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUADFD (201) Hyperelastic with 4 grid points Center and Corners	2	Grid/Gauss	Not applicable		
	3	Grid/Gauss ID (0=center)			
	4	Cauchy stress-X			
	5	Cauchy stress-Y			
	6	Cauchy stress-Z			
	7	Cauchy stress-XY			
	8	Pressure $p=\frac{1}{3}(\sigma_x+\sigma_y+\sigma_z)$			
	9	Volumetric strain J-1			
	10	Logarithmic strain-X			
	11	Logarithmic strain-Y			
	13	Logarithmic strain-XY			
	14-46	Items 3 through 13 repeated for 3 Gauss points			
CQUADFD (208) Hyperelastic with 5-9 grid points Center and Corners	2-13	Same as CQUADFD (201)	Not applicable		
	14-101	Repeat items 3 through 13 for 8 Gauss points			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUADR (82) Center and Corners		Same as CQUAD4(144)		Same as CQUAD4(144)	
CQUADR (228) Center only (not supported by old formulation)		Same as CQUAD4(33)		Same as CQUAD4(33)	
CQUADR^2 (232) Composite Center Only (not supported by old formulation)	2-11	Same as CQUAD4(95)	2-12	Same as CQUAD4(95)	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUADX4 (243) Linear Format Center and Corners Grid and Gauss	3	0 (Result at Center)	3	0 (Result at Center)	
	4	Radial	4	Radial	RM
	5	Azimuthal	5	Radial	IP
	6	Axial	6	Azimuthal	RM
	7	Shear stress	7	Azimuthal	IP
	8	Maximum principal	8	Axial	RM
	9	Maximum shear	9	Axial	IP
	10	von Mises or octahedral	10	Shear	RM
	11	Grid 1 ID	11	Shear	IP
	12–18	Same as items 4 through 10 for corner 1	12	Grid 1 ID	
	19	Grid 2 ID	13–20	Same as items 4 through 11 for corner 1	
	20–26	Same as items 4 through 10 for corner 2	21	Grid 2 ID	
	27	Grid 3 ID	22-29	Same as items 4 through 11 for corner 2	
	28–34	Same as items 4 through 10 for corner 3	30	Grid 3 ID	
	35	Grid 4 ID	31-38	Same as items 4 through 11 for corner 3	
	36–42	Same as items 4 through 10 for corner 4	39	Grid 4 ID	
			40-47	Same as items 4 through 11 for corner 4	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUADX4 (247) Nonlinear Format Center and Corners	3	0 (Result at Center)	Not applicable		
	4	Radial Stress			
	5	Azimuthal Stress			
	6	Axial Stress			
	7	Shear Stress			
	8	Equivalent Stress			
	9	Effective Plastic Strain			
	10	Effective Creep Strain			
	11	Radial Strain			
	12	Azimuthal Strain			
	13	Axial Strain			
	14	Shear Strain			
	15	Grid 1 ID			
	16–26	Same as items 4–14 for corner 1			
	27	Grid 2 ID			
	28–38	Same as items 4–14 for corner 2			
	39	Grid 3 ID			
	40–50	Same as items 4–14 for corner 3			
	51	Grid 4 ID			
	52–62	Same as items 4–14 for corner 4			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUADX4FD (262) Hyperelastic Grid or Gauss Center and Corners	2	Cord Type (GRID or GAUS)	Not applicable		
	3	Grid ID			
	4	Radial Stress			
	5	Azimuthal Stress			
	6	Axial Stress			
	7	Shear Stress			
	8	Pressure			
	9	Volumetric Strain			
	10	Radial Strain			
	11	Azimuthal Strain			
	12	Axial Strain			
	13	Shear Strain			
	14-46	Items 3 through 13 repeated for 4 Grid/Gauss Points			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUADX8 (245) Linear Format Grid and Gauss Center and Corners	3	0 (Result at Center)	3	0 (Result at Center)	
	4	Radial	4	Radial	RM
	5	Azimuthal	5	Radial	IP
	6	Axial	6	Azimuthal	RM
	7	Shear stress	7	Azimuthal	IP
	8	Maximum principal	8	Axial	RM
	9	Maximum shear	9	Axial	IP
	10	von Mises or octahedral	10	Shear	RM
	11	Grid 1 ID	11	Shear	IP
	12–18	Same as items 4 through 10 for corner 1	12	Grid 1 ID	
	19	Grid 2 ID	13–20	Same as items 4 through 11 for corner 1	
	20–26	Same as items 4 through 10 for corner 2	21	Grid 2 ID	
	27	Grid 3 ID	22-29	Same as items 4 through 11 for corner 2	
	28–34	Same as items 4 through 10 for corner 3	30	Grid 3 ID	
	35	Grid 4 ID	31-38	Same as items 4 through 11 for corner 3	
	36–42	Same as items 4 through 10 for corner 4	39	Grid 4 ID	
			40-47	Same as items 4 through 11 for corner 4	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUADX8 (249) Nonlinear Format	3	0 (Result at Center)	Not applicable		
	4	Radial Stress			
	5	Azimuthal Stress			
	6	Axial Stress			
	7	Shear Stress			
	8	Equivalent Stress			
	9	Effective Plastic Strain			
	10	Effective Creep Strain			
	11	Radial Strain			
	12	Azimuthal Strain			
	13	Axial Strain			
	14	Shear Strain			
	15	Grid 1 ID			
	16–26	Same as items 4–14 for corner 1			
	27	Grid 2 ID			
	28–38	Same as items 4–14 for corner 2			
	39	Grid 3 ID			
	40–50	Same as items 4–14 for corner 3			
	51	Grid 4 ID			
	52–62	Same as items 4–14 for corner 4			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUADX8FD Hyperelastic Center and Corners (264)	3	Grid ID	Not applicable		
	4	Radial Stress			
	5	Azimuthal Stress			
	6	Axial Stress			
	7	Shear Stress			
	8	Pressure			
	9	Volumetric Strain			
	10	Radial Strain			
	11	Azimuthal Strain			
	12	Axial Strain			
	13	Shear Strain			
	14-46	Items 3 through 13 repeated for 4 corner Grid Locations			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CQUADX8FD Hyperelastic Gauss Locations (266)	3	Grid ID	Not applicable		
	4	Radial Stress			
	5	Azimuthal Stress			
	6	Axial Stress			
	7	Shear Stress			
	8	Pressure			
	9	Volumetric Strain			
	10	Radial Strain			
	11	Azimuthal Strain			
	12	Axial Strain			
	13	Shear Strain			
	14-101	Items 3 through 13 repeated 8 times for a total of 9 Gauss locations			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CROD (1) Linear Format		Same as CONROD(10)		Same as CONROD(10)	
CROD (89) Nonlinear Format		Same as CONROD(92)		Not applicable	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CSHEAR (4)	2	Maximum shear	2	Maximum shear	RM
	3[1]	Average shear	3	Maximum shear	IP
	4	Safety margin	4[1]	Average shear	RM
			5[1]	Average shear	IP

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTETRA (39) Linear Format	2	Stress coordinate system	2	Stress coordinate system	
	3	Coordinate type (Character)	3	Coordinate type (Character)	
	4	Number of active points	4	Number of active points	
	5	External grid ID (0=center)	5	External grid ID (0=center)	
	6[1]	Normal x	6[1]	Normal x	RM
	7[1]	Shear xy	7[1]	Normal y	RM
	8	First principal	8[1]	Normal z	RM
	9	First principal x cosine	9[1]	Shear xy	RM
	10	Second principal x cosine	10[1]	Shear yz	RM
	11	Third principal x cosine	11[1]	Shear zx	RM
	12	Mean pressure	12[1]	Normal x	IP
	13	von Mises or octahedral shear stress	13[1]	Normal y	IP
	14[1]	Normal y	14[1]	Normal z	IP
	15[1]	Shear yz	15[1]	Shear xy	IP
	16	Second principal	16[1]	Shear yz	IP
	17	First principal y cosine	17[1]	Shear zx	IP
	18	Second principal y cosine	18-69	Repeat items 5 through 17 for four corners	
	19	Third principal y cosine			
	20[1]	Normal z			
	21[1]	Shear zx			
	22	Third principal			
	23	First principal z cosine			
	24	Second principal z cosine			
	25	Third principal z cosine			
	26-109	Repeat items 5 through 25 for four corners			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTETRA (85) Nonlinear Format	2	Grid/Gauss	Not applicable		
	3	External grid ID (0=center)			
	4	Stress-X			
	5	Stress-Y			
	6	Stress-Z			
	7	Stress-XY			
	8	Stress-YZ			
	9	Stress-ZX			
	10	Equivalent stress			
	11	Effective plastic strain			
	12	Effective creep strain			
	13	Strain-X			
	14	Strain-Y			
	15	Strain-Z			
	16	Strain-XY			
	17	Strain-YZ			
	18	Strain-ZX			
	19-82	Repeat items 3 through 18 for four corners			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTETRAFD (205) Hyperelastic with 4 grid points	2-17	Same as CHEXAFD (202)	Not applicable		
CTETRAFD (210) Hyperelastic with 10 grid points	2-17	Same as CHEXAFD (202)	Not applicable		
	18-77	Repeat items 3 through 17 for 4 Gauss points			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRAX3 (242) Linear Format Center and Corners Grid or Gauss	3	0 (Result at Center)	3	0 (Result at Center)	
	4	Radial	4	Radial	RM
	5	Azimuthal	5	Radial	IP
	6	Axial	6	Azimuthal	RM
	7	Shear stress	7	Azimuthal	IP
	8	Maximum principal	8	Axial	RM
	9	Maximum shear	9	Axial	IP
	10	von Mises or octahedral	10	Shear	RM
	11	Grid 1 ID	11	Shear	IP
	12–18	Same as items 4 through 10 for corner 1	12	Grid 1 ID	
	19	Grid 2 ID	13–20	Same as items 4 through 11 for corner 1	
	20–26	Same as items 4 through 10 for corner 2	21	Grid 2 ID	
	27	Grid 3 ID	22-29	Same as items 4 through 11 for corner 2	
	28–34	Same as items 4 through 10 for corner 3	30	Grid 3 ID	
			31-38	Same as items 4 through 11 for corner 3	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRAX3 (246) Nonlinear Format Center and Corners	3	0 (Result at Center)	Not applicable		
	4	Radial Stress			
	5	Azimuthal Stress			
	6	Axial Stress			
	7	Shear Stress			
	8	Equivalent Stress			
	9	Effective Plastic Strain			
	10	Effective Creep Strain			
	11	Radial Strain			
	12	Azimuthal Strain			
	13	Axial Strain			
	14	Shear Strain			
	15	Grid 1 ID			
	16–26	Same as items 4–14 for corner 1			
	27	Grid 2 ID			
	28–38	Same as items 4–14 for corner 2			
	39	Grid 3 ID			
	40–50	Same as items 4–14 for corner 3			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRAX3FD Hyperelastic (261) Grid and Gauss	3	Grid ID	Not applicable		
	4	Radial Stress			
	5	Azimuthal Stress			
	6	Axial Stress			
	7	Shear Stress			
	8	Pressure			
	9	Volumetric Strain			
	10	Radial Strain			
	11	Azimuthal Strain			
	12	Axial Strain			
	13	Shear Strain			
	14-35	Items 3 through 13 repeated for 3 corner Grid/Gauss Locations			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRAX6 (244) Linear Format Grid and Gauss Center and Corners	3	0 (Result at Center)	3	0 (Result at Center)	
	4	Radial	4	Radial	RM
	5	Azimuthal	5	Radial	IP
	6	Axial	6	Azimuthal	RM
	7	Shear stress	7	Azimuthal	IP
	8	Maximum principal	8	Axial	RM
	9	Maximum shear	9	Axial	IP
	10	von Mises or octahedral	10	Shear	RM
	11	Grid 1 ID	11	Shear	IP
	12–18	Same as items 4 through 10 for corner 1	12	Grid 1 ID	
	19	Grid 2 ID	13–20	Same as items 4 through 11 for corner 1	
	20–26	Same as items 4 through 10 for corner 2	21	Grid 2 ID	
	27	Grid 3 ID	22-29	Same as items 4 through 11 for corner 2	
	28–34	Same as items 4 through 10 for corner 3	30	Grid 3 ID	
			31-38	Same as items 4 through 11 for corner 3	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRAX6 (248) Nonlinear Format Center and Corners	3	0 (Result at Center)	Not applicable		
	4	Radial Stress			
	5	Azimuthal Stress			
	6	Axial Stress			
	7	Shear Stress			
	8	Equivalent Stress			
	9	Effective Plastic Strain			
	10	Effective Creep Strain			
	11	Radial Strain			
	12	Azimuthal Strain			
	13	Axial Strain			
	14	Shear Strain			
	15	Grid 1 ID			
	16–26	Same as items 4–14 for corner 1			
	27	Grid 2 ID			
	28–38	Same as items 4–14 for corner 2			
	39	Grid 3 ID			
	40–50	Same as items 4–14 for corner 3			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRAX6FD (263) Center and Corners Grid or Gauss Hyperelastic	2	Cord Type (GRID or GAUS)	Not applicable		
	3	Grid ID			
	4	Radial Stress			
	5	Azimuthal Stress			
	6	Axial Stress			
	7	Shear Stress			
	8	Pressure			
	9	Volumetric Strain			
	10	Radial Strain			
	11	Azimuthal Strain			
	12	Axial Strain			
	13	Shear Strain			
	14-35	Items 3 through 13 repeated for 3 Grid/Gauss Points			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRIA3 (74) Linear Format Center Only	2-17	Same as CQUAD4(33)	2-15	Same as CQUAD4(33)	
CTRIA3[2] (97) Composite Center Only	2-11	Same as CQUAD4(95)	2-12	Same as CQUAD4(95)	
CTRIA3 (88) Nonlinear Format Center Only	2-25	Same as CQUAD4(90)	Not applicable		

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRIA6 (75) Linear Format Center and Corners	2-70	Same as CQUAD4(144)	2-62	Same as CQUAD4(144)	
CTRIA6[2] (98) Composite Center Only	2-11	Same as CQUAD4(95)	2-12	Same as CQUAD4(95)	
CTRIA6 (240) Nonlinear Format Center and Corners	3-102	Same as CQUAD4(254)	Not applicable		

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRIAFD (206) Hyperelastic with 3 grid points	2-13	Same as CQUADFD(201)	Not applicable		
CTRIAFD (211) Hyperelastic with 6 grid points	2-13	Same as CQUADFD(201)	Not applicable		
	14-35	Items 3 through 12 repeated for 2 Gauss points			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTRIAR (70) Center and Corners	1-70	Same as CQUAD4(144)	2-62	Same as CQUAD4(144)	
CTRIAR (227) Center Only (not supported by old formulation)	2-17	Same as CQUAD4(33)	2-15	Same as CQUAD4(33)	
CTRIAR^2 (233) Composite Center Only (not supported by old formulation)	2-11	Same as CQUAD4(95)	2-12	Same as CQUAD4(95)	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag./ Phase
CTUBE (3) Linear Format		Same as CONROD (10)		Same as CONROD(10)	
CTUBE (87) Nonlinear Format		Same as CONROD(92)	Not applicable		

Related Topics

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Element Force Item Codes

All items are element forces (or moments) unless otherwise indicated.

Note

The items codes include only the elements tested and supported by the applications listed in **Item Codes Overview**. For an application not listed, for example, if you need to understand the output schema of any element in the .op2 file, you should use the OES and OEF data block descriptions in the DMAP programmers guide. The OES and OEF data block descriptions are comprehensive and include all elements.

Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CBAR (34)	2[1]	Bending End A plane 1	2[1]	Bending End A plane 1	RM
	3[1]	Bending End A plane 2	3[1]	Bending End A plane 2	RM
	4[1]	Bending End B plane 1	4[1]	Bending End B plane 1	RM
	5[1]	Bending End B plane 2	5[1]	Bending End B plane 2	RM
	6[1]	Shear plane 1	6[1]	Shear plane 1	RM
	7[1]	Shear plane 2	7[1]	Shear plane 2	RM
	8[1]	Axial force	8[1]	Axial force	RM
	9[1]	Torque	9[1]	Torque	RM
			10[1]	Bending End A plane 1	IP
			11[1]	Bending End A plane 2	IP
			12[1]	Bending End B plane 1	IP
			13[1]	Bending End B plane 2	IP
			14[1]	Shear plane 1	IP
			15[1]	Shear plane 2	IP
			16[1]	Axial force	IP
			17[1]	Torque	IP

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CBAR (100)	2	Station Distance/ Length	2	Station Distance/ Length	RM
	3	Bending Moment Plane 1	3	Bending Moment Plane 1	RM
	4	Bending Moment Plane 2	4	Bending Moment Plane 2	RM
	5	Shear Force Plane 1	5	Shear Force Plane 1	RM
	6	Shear Force Plane 2	6	Shear Force Plane 2	RM
	7	Axial	7	Axial	RM
	8	Torque	8	Torque	RM
	9	Item codes are given for End A. Addition of the quantity (K-1) * 7 to the item code points to the same information for other stations, where K is the station number. K=8 for End B and 2 through 7 for intermediate stations.	9	Bending Moment Plane 1	IP
			10	Bending Moment Plane 2	IP
			11	Shear Force Plane 1	IP
			12	Shear Force Plane 2	IP
			13	Axial	IP
			14	Torque (Item codes above are given for End A. For codes 2 through 14 at intermediate stations add (K-1) * 13 and K is the station number, and for codes at End B, K+number of stations plus 1.)	IP

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CBEAM (2)	2	External grid point ID	2	External grid point ID	
	3	Station distance/length	3	Station distance/length	
	4[1]	Bending moment plane 1	4[1]	Bending moment plane 1	RM
	5[1]	Bending moment plane 2	5[1]	Bending moment plane 2	RM
	6[1]	Web shear plane 1	6[1]	Web shear plane 1	RM
	7[1]	Web shear plane 2	7[1]	Web shear plane 2	RM
	8[1]	Axial force	8[1]	Axial force	RM
	9[1]	Total torque	9[1]	Total torque	RM
	10[1]	Warping torque	10[1]	Warping torque	RM
		(Item codes are given for End A. Addition of the quantity (K-1) 9 to the item code points to the same information for other stations, where K is the station number. K=11 for End B and 2 through 10 for intermediate stations.)	11[1]	Bending moment plane 1	IP
			12[1]	Bending moment plane 2	IP
			13[1]	Web shear plane 1	IP
			14[1]	Web shear plane 2	IP
			15[1]	Axial force	IP
			16[1]	Total torque	IP
			17[1]	Warping torque	IP
				(Item codes are given for End A. Addition of the quantity (K-1) 16 to the item code points to the same information for other stations, where K is the station number. K=11 for End B and 2 through 10 for intermediate stations.)	

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CBEAR (280)	2	Force-x	2	Force-x	RM
	3	Force-y	3	Force-y	RM
	4	Force-z	4	Force-z	RM
	5	Moment-x	5	Moment-x	RM
	6	Moment-y	6	Moment-y	RM
	7	Moment-z	7	Moment-z	RM
			8	Force-x	IP
			9	Force-y	IP
			10	Force-z	IP
			11	Moment-x	IP
			12	Moment-y	IP
			13	Moment-z	IP

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CBEND (69)	2	External grid point ID	2	External grid point ID	
	3[1]	Bending moment plane 1	3[1]	Bending moment plane 1	RM
	4[1]	Bending moment plane 2	4[1]	Bending moment plane 2	RM
	5[1]	Shear plane 1	5[1]	Shear plane 1	RM
	6[1]	Shear plane 2	6[1]	Shear plane 2	RM
	7[1]	Axial force	7[1]	Axial force	RM
	8[1]	Torque (Item codes are given for End A. Item codes 9 through 15 point to the same information for End B.)	8[1]	Torque	RM
			9[1]	Bending moment plane 1	IP
			10[1]	Bending moment plane 2	IP
			11[1]	Shear plane 1	IP
			12[1]	Shear plane 2	IP
			13[1]	Axial force	IP
			14[1]	Torque (Item codes are given for End A. Item codes 15 through 27 point to the same information for End B.)	IP

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CBUSH (102) Linear Format	2	Force-x	2	Force-x	RM
	3	Force-y	3	Force-y	RM
	4	Force-z	4	Force-z	RM
	5	Moment-x	5	Moment-x	RM
	6	Moment-y	6	Moment-y	RM
	7	Moment-z	7	Moment-z	RM
			8	Force-x	IP
			9	Force-y	IP
			10	Force-z	IP
			11	Moment-x	IP
			12	Moment-y	IP
			13	Moment-z	IP

Element Name (Code)	Real Element Forces	
	Item Code	Item
CBUSH (226) Nonlinear Format	2	Force-x
	3	Force-y
	4	Force-z
	11	Moment-x
	12	Moment-y
	13	Moment-z

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CDAMP1 (20)		Same as CELAS1		Same as CELAS1	
CDAMP2 (21)		Same as CELAS1		Same as CELAS1	
CDAMP3 (22)		Same as CELAS1		Same as CELAS1	
CDAMP4 (23)		Same as CELAS1		Same as CELAS1	

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CDUM3 thru CDUM9 (55-61)	2[1]	F1	2[1]	F1	RM
	3[1]	F2	3[1]	F2	RM
	4[1]	F3	4[1]	F3	RM
	5[1]	F4	5[1]	F4	RM
	6[1]	F5	6[1]	F5	RM
	7[1]	F6	7[1]	F6	RM
	8[1]	F7	8[1]	F7	RM
	9[1]	F8	9[1]	F8	RM
	10[1]	F9	10[1]	F9	RM
			11[1]	F1	IP
			12[1]	F2	IP
			13[1]	F3	IP
			14[1]	F4	IP
			15[1]	F5	IP
			16[1]	F6	IP
			17[1]	F7	IP
			18[1]	F8	IP
			19[1]	F9	IP

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CELAS1	2[1]	Force	2[1]	Force	RM
(11)			3[1]	Force	IP
CELAS2 (12)		Same as CELAS1		Same as CELAS1	
CELAS3 (13)		Same as CELAS1		Same as CELAS1	
CELAS4 (14)		Same as CELAS1		Same as CELAS1	

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CFAST (119)	2	mz bending End A plane 1	2	mz bending End A plane 1	RM
	3	my bending End A plane 2	3	my bending End A plane 2	RM
	4	mz bending End B plane 1	4	mz bending End B plane 1	RM
	5	my bending End B plane 2	5	my bending End B plane 2	RM
	6	fy shear force plane 1	6	fy shear force plane 1	RM
	7	fz shear force plane 2	7	fz shear force plane 2	RM
	8	fx axial force	8	fx axial force	RM
	9	mx torque	9	mx torque	RM
			10	mz bending End A plane 1	IP
			11	my bending End A plane 2	IP
			12	mz bending End B plane 1	IP
			13	my bending End B plane 2	IP
			14	fy shear force plane 1	IP
			15	fz shear force plane 2	IP
			16	fx axial force	IP
			17	mx torque	IP

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CONROD	2[1]	Axial force	2[1]	Axial force	RM
(10)	3[1]	Torque	3[1]	Axial force	IP
			4[1]	Torque	RM
			5[1]	Torque	IP

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CQUAD4 (33) Center Only	2[1]	Membrane force x	2[1]	Membrane force x	RM
	3[1]	Membrane force y	3[1]	Membrane force y	RM
	4[1]	Membrane force xy	4[1]	Membrane force xy	RM
	5[1]	Bending moment x	5[1]	Bending moment x	RM
	6[1]	Bending moment y	6[1]	Bending moment y	RM
	7[1]	Bending moment xy	7[1]	Bending moment xy	RM
	8[1]	Shear x	8[1]	Shear x	RM
	9[1]	Shear y	9[1]	Shear y	RM
			10[1]	Membrane force x	IP
			11[1]	Membrane force y	IP
			12[1]	Membrane force xy	IP
			13[1]	Bending moment x	IP
			14[1]	Bending moment y	IP
			15[1]	Bending moment xy	IP
			16[1]	Shear x	IP
			17[1]	Shear y	IP

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CQUAD4[2] (95) Composite Center Only	2-3	Theory or blank		Not applicable	
	4	Lamina number			
	5	FP (failure index for direct stresses)			
	6	Failure mode for			
		Maximum strain theory			
	7	FB or -1 (failure index for interlaminar shear-stress)			
	8	MAX of FP, FB or -1			
	9	Failure flag			

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CQUAD4 (144) Center and Corners	1	EID	1	EID	
	2	CEN/	2	CEN/	
	3	4	3	4	
	4	Membrane x	4	Membrane x	RM
	5	Membrane y	5	Membrane y	RM
	6	Membrane xy	6	Membrane xy	RM
	7	Bending x	7	Bending x	RM
	8	Bending y	8	Bending y	RM
	9	Bending xy	9	Bending xy	RM
	10	Shear x	10	Shear x	RM
	11	Shear y	11	Shear y	RM
	12	Grid 1	12	Membrane x	IP
	13-20	Same as 4 through 11 for corner 1	13	Membrane y	IP
			14	Membrane xy	IP
	21	Grid 2	15	Bending x	IP
	22-29	Same as 4 through 11 for corner 2	16	Bending y	IP
			17	Bending xy	IP
	30	Grid 3	18	Shear x	IP
	31-38	Same as 4 through 11 for corner 3	19	Shear y	IP
			20	Grid 1	
	39	Grid 4	21-36	Same as 4 through 19 for corner 1	
	40-47	Same as 4 through 11 for corner 4	37	Grid 2	
			38-53	Same as 4 through 19 for corner 2	
			54	Grid 3	
			55-70	Same as 4 through 19 for corner 3	
			71	Grid 4	
			72-87	Same as 4 through 19 for corner 4	

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CQUAD8 (64) Center and Corners		Same as CQUAD4(144)		Same as CQUAD4(144)	
CQUAD8[2] (96) Composite Center Only		Same as CQUAD4(95)		Same as CQUAD4(95)	

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CQUADR (82) Center and Corners		Same as CQUAD4(144)		Same as CQUAD4(144)	
CQUADR (228) Center Only (not supported by old formulation)		Same as CQUAD4(33)		Same as CQUAD4(33)	
CQUADR^2 (232) Composite Center Only (not supported by old formulation)		Same as CQUAD4(95)		Same as CQUAD4(95)	

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CROD (1)		Same as CONROD(10)		Same as CONROD(10)	

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CSHEAR (4)	2[1]	Force 4 to 1	2[1]	Force 4 to 1	RM
	3[1]	Force 2 to 1	3[1]	Force 2 to 1	RM
	4[1]	Force 1 to 2	4[1]	Force 1 to 2	RM
	5[1]	Force 3 to 2	5[1]	Force 3 to 2	RM
	6[1]	Force 2 to 3	6[1]	Force 2 to 3	RM
	7[1]	Force 4 to 3	7[1]	Force 4 to 3	RM
	8[1]	Force 3 to 4	8[1]	Force 3 to 4	RM
	9[1]	Force 1 to 4	9[1]	Force 1 to 4	RM
	10[1]	Kick force on 1	10[1]	Force 4 to 1	IP
	11[1]	Shear 12	11[1]	Force 2 to 1	IP
	12[1]	Kick force on 2	12[1]	Force 1 to 2	IP
	13[1]	Shear 23	13[1]	Force 3 to 2	IP
	14[1]	Kick force on 3	14[1]	Force 2 to 3	IP
	15[1]	Shear 34	15[1]	Force 4 to 3	IP
	16[1]	Kick force on 4	16[1]	Force 3 to 4	IP
	17[1]	Shear 41	17[1]	Force 1 to 4	IP
			18[1]	Kick force on 1	RM
			19[1]	Shear 12	RM
			20[1]	Kick force on 2	RM
			21[1]	Shear 23	RM
			22[1]	Kick force on 3	RM
			23[1]	Shear 34	RM
			24[1]	Kick force on 4	RM
			25[1]	Shear 41	RM
			26[1]	Kick force on 1	IP
			27[1]	Shear 12	IP
			28[1]	Kick force on 2	IP
			29[1]	Shear 23	IP
			30[1]	Kick force on 3	IP
			31[1]	Shear 34	IP
			32[1]	Kick force on 4	IP
			33[1]	Shear 41	IP

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CTRIA3 (74) Center Only		Same as CQUAD4(33)		Same as CQUAD4(33)	
CTRIA3[2] (97) Composite Center Only		Same as CQUAD4(95)		Same as CQUAD4(95)	
CTRIA6 (75) Center and Corners	1-38	Same as CQUAD4(144)	1-70	Same as CQUAD4(144)	
CTRIA6[2] (98) Composite Center Only		Same as CQUAD4(95)		Same as CQUAD4(95)	

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CTRIAR (70) Center and Corners	1-38	Same as CQUAD4(144)	1-70	Same as CQUAD4(144)	
CTRIAR (227) Center Only (not supported by old formulation)	2-9	Same as CQUAD4(33)	2-17	Same as CQUAD4(33)	
CTRIAR^2 (233) Composite Center Only (not supported by old formulation)	2-9	Same as CQUAD4(95)		Not applicable	

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Element Name Code	Real Element Forces		Complex Element Forces		
	Item Code	Item	Item Code	Item	Real/Mag. or Imag/ Phase
CTUBE (3)		Same as CONROD(10)		Same as CONROD(10)	
CVISC (24)		Not applicable		Same as CONROD(10)	
CWELDP (118) ELPAT or PARTPAT CWELDC (117) ELIMID or GRIDID with MSET=OFF CWELD (200) ELIMID or GRIDID with MSET=ON	2	mz bending End A plane 1	2	mz bending End A plane 1	RM
	3	my bending End A plane 2	3	my bending End A plane 2	RM
	4	mz bending End B plane 1	4	mz bending End B plane 1	RM
	5	my bending End B plane 2	5	my bending End B plane 2	RM
	6	fy shear force plane 1	6	fy shear force plane 1	RM
	7	fz shear force plane 2	7	fz shear force plane 2	RM
	8	fx axial force	8	fx axial force	RM
	9	mx torque	9	mx torque	RM
			10	mz bending End A plane 1	IP
			11	my bending End A plane 2	IP
			12	mz bending End B plane 1	IP
			13	my bending End B plane 2	IP
			14	fy shear force plane 1	IP
			15	fz shear force plane 2	IP
			16	fx axial force	IP
			17	mx torque	IP

Related Topics

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Fluid Virtual Mass Pressure Item Codes

Element Name	Real Fluid Pressure		Complex Fluid Pressure		
	Code	Item	Code	Item	Real/Mag. or Imag./Phase
Plate	2	Fluid pressure	2	Pressure	RM
Family			3	Pressure	IP

Element Name (Code)	Code	Item
Heat Transfer Elements	2	Element type
	3	
	4	x gradient
	5	y gradient
	6	z gradient
	7	x flux
	8	y flux
	9	z flux
CHBDYE (107)	4	Applied load
	5	Free convection
	6	Forced convection
	7	Radiation
	8	Total
CHBDYG (108)	Same as CHBDYE	Same as CHBDYE
CHBDYP (109)	Same as CHBDYE	Same as CHBDYE

Related Topics

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Slideline Contact Item Codes

Element Name (Code)	Real Element Data	
	Item Code	Item
CSLIFID (116)	1	Secondary grid point
	2	Contact region identification number
	3	Primary grid 1
	4	Primary grid 2
	5	Surface coordinate
	6	Normal force
	7	Shear force
	8	Normal stress
	9	Shear stress
	10	Normal gap
	11	Slip
	12	Slip ratio (Shear force/u*normal force)
	13-14	Slip code (Character)

Related Topics

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Element Strain Energy Item Codes

Element Name	Real Element Data	
	Item Code	Item
Element groups A and B	2	Element strain energy
Element groups A and B	3	Percent of total energy
Element group A	4	Element strain energy density

Remarks

Element group A includes elements CBAR, CBEAM, CBEND, CHEXA, CONROD, CPENTA, CQUAD4, CQUADR, CQUADX4, CQUADX8, CROD, CSHEAR, CTETRA, CTRAX3, CTRAX6, CTRIA3, CTRIA6, CTRIAR, and CTUBE. Element group B includes elements of CELAS1, CELAS2, CELAS3, and CGAP.

Related Topics

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8. Degree-of-Freedom Sets

Degree-of-Freedom Set Definitions

Each degree-of-freedom is a member of one mutually exclusive set. Set names have the following definitions.

Set Name	Definition
m	degrees-of-freedom eliminated by multipoint constraints*
sb**	degrees-of-freedom eliminated by single-point constraints that are included in boundary condition changes and by the AUTOSPC feature
sg**	degrees-of-freedom eliminated by single-point constraints that are specified on the PS field on GRID Bulk Data entries
o	degrees-of-freedom omitted by structural matrix partitioning
q	generalized degrees-of-freedom for dynamic reduction or component mode synthesis
r	reference degrees-of-freedom used to determine free body motion
c	degrees-of-freedom that are free during component mode synthesis or dynamic reduction
b	degrees-of-freedom fixed during component mode analysis or dynamic reduction
e	extra degrees-of-freedom introduced in dynamic analysis
sa	permanently constrained aerodynamic degrees-of-freedom
k	aerodynamic degrees-of-freedom
u1 – u9	user defined degrees-of-freedom

*These are the dependent degrees-of-freedom in a multipoint constraint.

** Strictly speaking, sb and sg are not exclusive with respect to one another. Degrees-of-freedom may exist in both sets simultaneously. Since these sets are not used explicitly in the solution sequences, this need not concern the user. However, those who use these sets in their own DMAPs should avoid redundant specifications when using these sets for partitioning or merging operations. That is, a degree-of-freedom should not be specified on both a PS field of a GRID entry (sg set) and on a selected SPC entry (sb set). Redundant specifications will cause UFM 2120 in the VEC module and behavior listed in the *Simcenter Nastran DMAP Programmer's Guide* for the UPARTN module. These sets are exclusive, however, from the other mutually exclusive sets.

Each degree-of-freedom is also a member of one or more combined sets called "supersets." Supersets have the following definitions:

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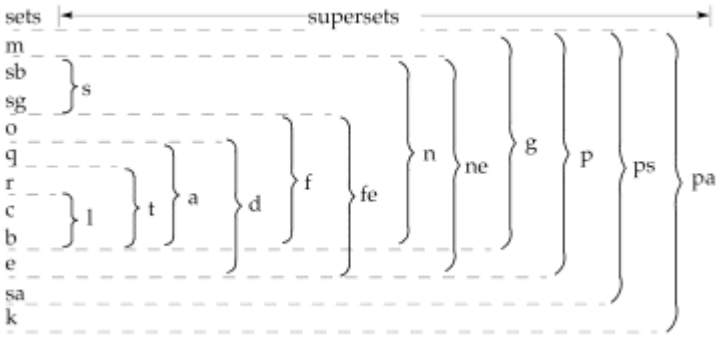

Set Name (+ indicates union of two sets)	Meaning
s = sb + sg	all degrees-of-freedom eliminated by single point constraints
l = b + c	the structural degrees-of-freedom remaining after the reference degrees-of-freedom are removed (degrees-of-freedom left over)
t = *l* + r	the total set of physical boundary degrees-of-freedom for superelements
a = t + q	the set assembled in superelement analysis
d = a + e	the set used in dynamic analysis by the direct method
f = a + o	unconstrained (free) structural degrees-of-freedom
fe = f + e	free structural degrees-of-freedom plus extra degrees-of-freedom
n = f + s	all structural degrees-of-freedom not constrained by multipoint constraints
ne = n + e	all structural degrees-of-freedom not constrained by multipoint constraints plus extra degrees-of-freedom
g = n + m	all structural (grid) degrees-of-freedom including scalar degrees-of-freedom
p = g + e	all physical degrees-of-freedom
ps = p + sa	physical and constrained (SPCi) aerodynamic degrees-of-freedom
pa = ps + k	physical set for aerodynamics
(fr = o) + *l*	statically independent set minus the statically determinate supports (fr = f – q – r)
v = o + c + r	the set free to vibrate in dynamic reduction and component mode synthesis

The a-set and o-set are created in the following ways:

1. If only OMITi entries are present, then the o-set consists of degrees-of-freedom listed explicitly on OMITi entries. The remaining f-set degrees-of-freedom are placed in the b-set, which is a subset of the a-set.

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2. If ASETi or QSETi entries are present, then the a-set consists of all degrees-of-freedom listed on ASETi entries and any entries listing its subsets, such as QSETi, SUPORTi, CSETi, and BSETi entries. Any OMITi entries are redundant. The remaining f-set degrees-of-freedom are placed in the o-set.
3. If there are no ASETi, QSETi, or OMITi entries present but there are SUPORTi, BSETi, or CSETi entries present, then the entire f-set is placed in the a-set and the o-set is not created.
4. There must be at least one explicit ASETi, QSETi, or OMITi entry for the o-set to exist, even if the ASETi, QSETi, or OMITi entry is redundant.

In dynamic analysis, additional vector sets are obtained by a modal transformation derived from real eigenvalue analysis of the a-set. These sets are as follows:

ξ_o = rigid body (zero frequency) modal degrees-of-freedom

ξ_f = finite frequency modal degrees-of-freedom

ξ_i = $\xi_o + \xi_f$, the set of all modal degrees-of-freedom

One vector set is defined that combines physical and modal degrees-of-freedom:

u_h = $\xi_i + u_e$, the set of all modal degrees-of-freedom

The membership of each degree-of-freedom can be printed by use of the Bulk Data entries PARAM,USETPRT and PARAM,USETSEL.

Related Topics

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Degree-of-Freedom Set Bulk Data Entries

Degrees-of-freedom are placed in sets as specified by the user on the following Bulk Data entries:

Name	Bulk Data Entry Name
m	MPC, MPCADD, RBAR, RBE1, RBE2, RBE3, RROD, RSPLINE, RTRPLT
sb	SPC, SPC1, SPCADD, BNDGRID, (PARAM,AUTOSPC,YES)
sg	GRID, GRDSET (PS field)
o	OMIT, OMIT1, GRID (SEID field), SESET
q	QSET, QSET1
r	SUPORT, SUPORT1
c	CSET, CSET1
b	BSET, BSET1
e	EPOINT
sa	CAEROi
k	CAEROi
a	ASET, ASET1, Superelement exterior degrees-of-freedom, CSUPEXT
u0 – u9	USET, USET1

In superelement analysis, the appropriate entry names are preceded by the letters SE, and have a field reserved for the superelement identification number. This identification is used because a boundary (exterior) grid point may be in one mutually exclusive set in one superelement and in a different set in the adjoining superelement. The SE-type entries are internally translated to the following types for the referenced superelement.

Entry Type	Equivalent Type
SEQSETi	QSETi
SESUP	SUPORT
SECSETi	CSETi
SEBSETi	BSETi

Related Topics

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User Defined Degree-of-Freedom Sets

The user defined degree-of-freedom sets U1 – U9 are created with the USET and USET1 bulk entries. They are typically used with DMAP alters.

Be aware that the user defined sets U2 - U8 are used by Simcenter Nastran in some special cases. You may use U2 – U8 as long as it doesn't conflict with these cases.

Set	Special Case Description	Solution Sequences
U1	No conflict exists. You may use in all situations.	
U2	Used with the RSCON parameter to select dof for SOL 103 constraint mode creation. See the RSCON parameter.	103
	Used by the Simcenter 3D FE Model Update product (SOL 200, ANALYSIS=MODES) when the parameter PARAM, OPTEXIT, 8 exists.	200
U3	Used with the RSATT parameter to select dof for SOL 103 attachment mode creation. See the RSATT parameter.	103
U4 and U5	Optionally used with the AUTOMPC parameter to add and remove dof to the m-set, respectively. See the AUTOMPC parameter.	101–129, 153–187
U6	Optionally used with the RESVEC parameter to apply a unit load for residual vector creation. See the RESVEC parameter.	103, 106–112, 145, 146, 187, 200
U7 and U8	Used with the MBDEXPORT case control command to define the input/output dof when exporting a control system interface file. See the MBDEXPORT case control command.	103, 115
U9	No conflict exists. You may use in all situations.	

Related Topics

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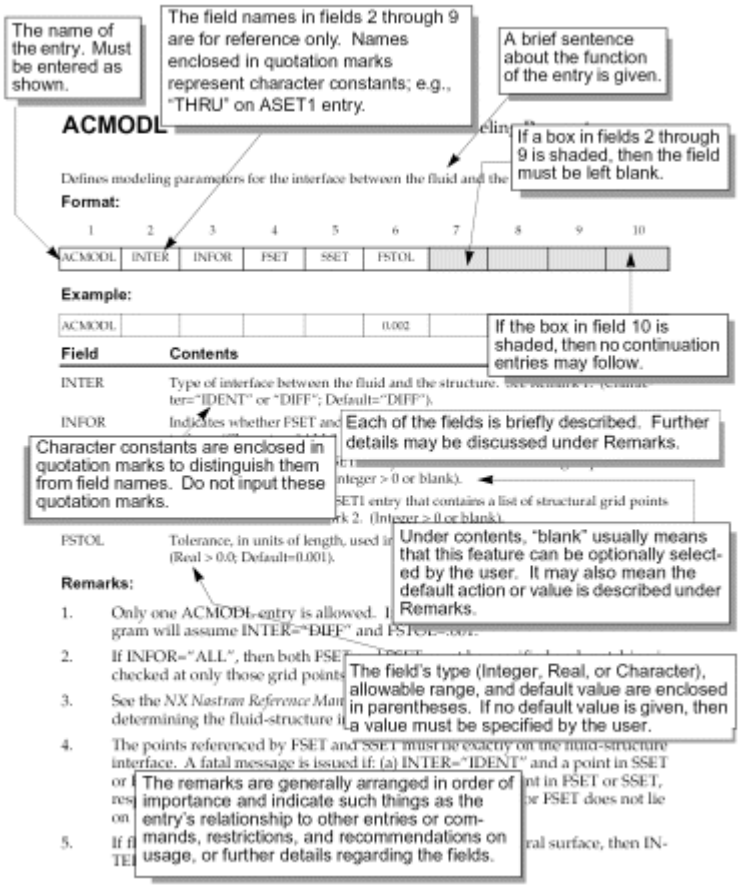
9. Bulk Entries

Key to Descriptions



Related Topics

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The Bulk Data Section

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The Bulk Data Section contains entries that specify model geometry, element connectivity, element and material properties, constraints (boundary conditions), and loads. Some entries, such as loads and constraints, are selected by an appropriate Case Control command.

Entries are prepared in either fixed or free field format. The descriptions in this section show only the fixed format. Entries that are used by the MSGMESH program are not included in this guide.

Related Topics

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Bulk Entry Descriptions

Each bulk entry is described as follows:

Description

A brief sentence about the function of the entry is given.

Format

The name of the entry is given in the first field. Subsequent fields are described under the Field and Contents section. Shaded fields must be left blank. If field 10 is shaded, then no continuation entries are permitted. Character strings enclosed in quotation marks must be specified without the quotation marks as shown in the example.

Example

A typical example is given.

Field and Contents

Each of the fields 2 through 9 that is named in the Format section is briefly described under Contents. The field's type (e.g., Integer, Real, or Character), allowable range, and default value are enclosed in parentheses. The field must be specified by the user if no default value is given.

Remarks

The remarks in the Remarks Section are generally arranged in order of importance and indicate such things as how the bulk entry is selected in the Case Control Section, its relationship to other entries, restrictions and recommendations on its use, and further descriptions of the fields.

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Related Topics

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Bulk Data Syntax Rules

Real, Integer, and Character Input Data

The three types of input data are described as follows:

Integer	Cannot contain a decimal point. By default, integers are limited to 8 digits. Integers can be defined up to 11 characters if you define nastran system(525)=1 and use large field formatted bulk entries (described below). The maximum ID of integers in this case is 99999999999. Setting system cell 525 to "1" is not supported by SOL 402, or when the MBDEXPORT case control is defined.
Real	Must contain a decimal point.
Character	Can be alphanumeric. The first character cannot be numeric. It must be 8 characters or less in length. Imbedded blanks are not allowed.

Real numbers may be entered in a variety of ways. For example, the following are all acceptable versions of the real number seven:

7.0	.7E1	0.7+1
.70+1	7.E+0	70.-1

Free and Fixed Field Format Summary

Bulk data inputs have 80 characters on each line and the lines are divided into fields. How the lines are divided depends upon the data format and if you are using **small fields** or **large fields**.

- Free format uses commas to separate the data fields. When you are using small fields with free format, a maximum of 8 characters is allowed per field. When you are using large fields with free format, some of the fields are extended to a maximum of 16 characters per field.
- Fixed format data is aligned in fixed columns of specific width. When you are using small fields, the fixed column width is 8 characters. When you are using large fields, some of the fields are extended to a fixed width of 16 characters.

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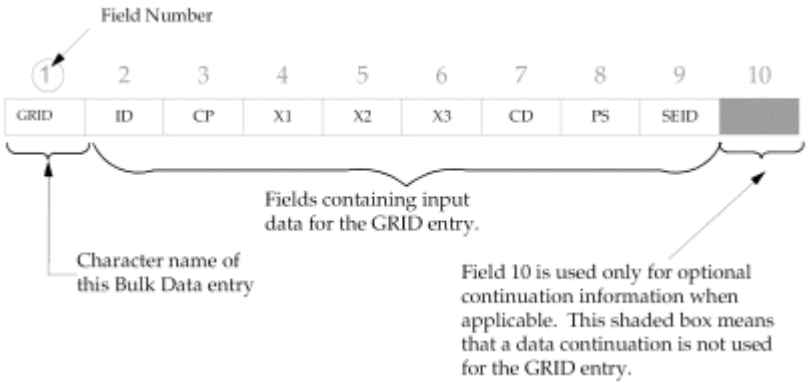
Bulk Entry Summary

In the Quick Reference Guide, each row of a bulk entry definition contains 10 fields. The first field in the first row contains the character name of the bulk entry (e.g., GRID, CBAR, MAT1, etc.). Then fields two through nine contain data. The tenth field in each row is reserved for a continuation character when more lines follow the first.

For example, consider the format of the GRID bulk entry.

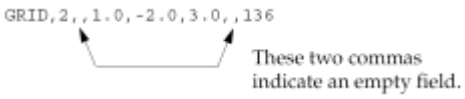


Free Field Format

In free field format, data fields are separated by commas. For example, the GRID entry

1	2	3	4	5	6	7	8	9	10
GRID	2		1.0	-2.0	3.0		136		

appears in free field format with small fields as:



The rules for free field format are as follows:

- To skip one field, use two commas in succession. To skip two fields, use three commas in succession (and so on).
- Integers and characters must be eight characters or less. Exceeding eight characters will cause a fatal error. This also applies to data defined in large fields.
- Free field format allows continuation data on a single input line. For example, the following SPC1 bulk entry is defined on a single line:

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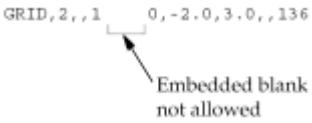
```
SPC1,1,123456,109,110,111,112,113,114,+LINE2,+LINE2,115,116,117
```

Removing the continuation fields will also work for this example:

```
SPC1,1,123456,109,110,111,112,113,114,115,116,117
```

See **Continuations**.

- Free field data cannot contain embedded blanks. The following is an example of an embedded blank:



- A dollar sign terminates the entry and comments may follow.

Fixed Field Format

In fixed format, the data fields are aligned in fixed columns of specific width.

When you are using small fields, the column width is a fixed 8 characters. For example, the GRID entry

1	2	3	4	5	6	7	8	9	10
GRID	2		1.0	-2.0	3.0		136		

appears in fixed field format with small fields as:

```
GRID           2              1.0    –2.0     3.0              136
```

The first two lines in this example are used to show the column alignment.

The rules for fixed field format are as follows:

- Data in the first and in the last field of each input line must be left justified.
- Data in the other columns can be defined anywhere within the bounds of a specific column.
- Integers and characters must be eight characters or less. Exceeding eight characters will cause a fatal error. This also applies to data defined in large fields.
- You cannot enter more characters than the field width.
- A dollar sign terminates the entry and comments may follow.

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Small Fields

Small fields have a maximum of 8 characters per field:

1	2	3	4	5	6	7	8	9	10
<-8 Char->	<-8 Char->	<-8 Char->	<-8 Char->	<-8 Char->	<-8 Char->	<-8 Char->	<-8 Char->	<-8 Char->	<-8 Char->

The following is an example of the GRID entry with small fields:

1	2	3	4	5	6	7	8	9	10
GRID	2		1.0	-2.0	3.0		136		

The fixed field input for this example is:

```
GRID            2              1.0    -2.0     3.0             136
```

The free field input for this example is:

```
GRID,2,,1.0,-2.0,3.0,,136
```

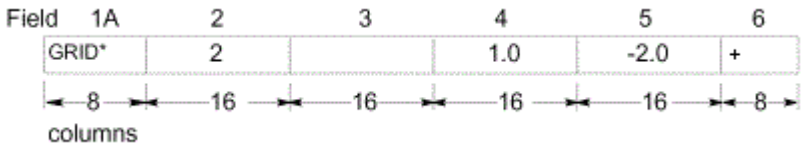
See **Continuations** for more examples.

Large Fields

Large field format is used when the small field format does not provide enough significant digits. The first and last field of each line contains eight characters, and the fields in between contain 16 characters. Large field entries are denoted by an asterisk (*) immediately following the character string in the first field of the first line and also in the first field of any continuation lines.

The following is an example of the GRID bulk entry example in large field format:

First Line:



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Second Line:

The fixed field input for this example is:

```
GRID*                   2                                1.0             –2.0+
*                     3.0                                136
```

The free field input for this example is:

```
GRID*,2,,1.0,-2.0,+
*,3.0,,136
```

See **Continuations** for more examples.

Related Topics

Browse Simcenter Nastran documentation

Continuations

Continuations are used when data continues beyond the first input line. There are several ways to define continuations. Each is described below using the following example bulk entry:

Example Bulk Entry:

1	2	3	4	5	6	7	8	9	10
PBAR	PID	MID	A	I1	I2	J	NSM		
	C1	C2	D1	D2	EI	E2	F1	F2	
	K1	K2	I12						

Small Field Continuation Format 1:

You can enter + or * in the last field of a line and + at the beginning of the next line. For example:

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PBAR	39	6	2.9	1.86	2.92	.48			+
+	0.	0.	0.	1.	1.	1.	1.	0.	+
+	.86	.86							

or

PBAR	39	6	2.9	1.86	2.92	.48			*
+	0.	0.	0.	1.	1.	1.	1.	0.	*
+	.86	.86							

Small Field Continuation Format 2:

You can simply leave the continuation fields blank. For example:

PBAR	39	6	2.9	1.86	2.92	.48			
	0.	0.	0.	1.	1.	1.	1.	0.	
	.86	.86							

Small Field Continuation Format 3:

You can enter a unique character string in the last field of a line, then a continuation line begins with the same character string. In the example below, +PB1 at the end of the first line is also included at the beginning of the second line, +PB2 at the end of the second line is also included at the beginning of the third line, and so on.

This format allows the continuation lines to be unsorted in the bulk data section. This was useful when punch cards where used and sorting them was difficult.

PBAR	39	6	2.9	1.86	2.92	.48			+PB1
+PB1	0.	0.	0.	1.	1.	1.	1.	0.	+PB2
+PB2	.86	.86							

Large Field Continuation Format 1:

You can enter + or * in the last field of a line and the first field on the continuation must include * to designate the large field format. For example:

PBAR*	39	6	2.9	1.86	+
*	2.92	.48			+
*	0.	0.	0.	1.	+

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*	1.	1.	1.	0.	+
*	.86	.86			

or

PBAR*	39	6	2.9	1.86	*
*	2.92	.48			*
*	0.	0.	0.	1.	*
*	1.	1.	1.	0.	*
*	.86	.86			

Large Field Continuation Format 2:

You can simply leave the last continuation fields blank, and the first field must include the * to designate the large field format. For example:

PBAR*	39	6	2.9	1.86	
*	2.92	.48			
*	0.	0.	0.	1.	
*	1.	1.	1.	0.	
*	.86	.86			

Large Field Continuation Format 3:

You can enter a unique character string in the last field of a line, then the next line begins with * plus the same character string to designate the large field format. In the large field example below, PB1 at the end of the first line is also included at the beginning of the second line, PB2 at the end of the second line is also included at the beginning of the third line, and so on.

This format allows the continuation lines to be unsorted in the bulk data section. This was useful when punch cards where used and sorting them was difficult.

PBAR*	39	6	2.9	1.86	+PB1
*PB1	2.92	.48			+PB2
*PB2	0.	0.	0.	1.	+PB3
*PB3	1.	1.	1.	0.	+PB4
*PB4	.86	.86			

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Replication

Replication is a limited data generation capability which may be used in a fixed or free-field format for small field input and in some limited cases for large field input. (The *Simcenter Nastran User's Guide* contains additional information on replication.)

1. Duplication of fields from the preceding entry is accomplished by coding the symbol =.
2. Duplication of all trailing fields from the preceding entry is accomplished by coding the symbol ==.
3. Incrementing a value from the previous entry is indicated by coding *x or *(x), where x is the value of the increment. "x" should be a real number for real fields or an integer for integer fields.
4. Repeated replication is indicated by coding =n or =(n), where n is the number of images to be generated using the values of the increments on the preceding entry.
5. Data items may be enclosed within parentheses or the parentheses may be deleted.
6. If NASTRAN MESH is specified in the File Management Section, MSGMESH includes the capabilities.
   - Continuation entry fields may be incremented or decremented.
   - Repeated replication is indicated by coding =(n) in field 1, where n is the number of entry images to be generated using the values of increments from the current or preceding replication entry.

     Entered entries:

```
GRID,101 ,17,1 .0,10.5,,17,3456
=(4), *(1),=,*(0.2),==$
```

     Generated entries:

GRID	101	17	1.0	10.5		17	3456		
GRID	102	17	1.2	10.5		17	3456		
GRID	103	17	1.4	10.5		17	3456		
GRID	104	17	1.6	10.5		17	3456		
GRID	105	17	1.8	10.5		17	3456		

   - A blank in field 1 indicates immediate continuation entry replication. The default continuation entry increment is 1. Example:

```
BSET1,123,1,2,3,4,5,6,7
,,*7,*7,*7,*7,*7,*7,*7
=(3)
```

     Generated entries:

BSET1	123	1	2	3	4	5	6	7	+00001
++00001		8	9	10	11	12	13	14	+00002

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++00002		15	16	17	18	19	20	21	+00003
++00003		22	23	24	25	26	27	28	+00004
++00004		29	30	31	32	33	34	35	+00005

- An “=(D)” in field 1 indicates delayed continuation entry replication. A maximum of 9 entries may be replicated as a group. The default continuation entry increment is 10. Example:

  Entered entries:

```
CTRIA3,10,1,1,10,11/+C1
=(D),*(1),=,=,*(1),*(1)/*(20)
+C1,,,2.0,1.0,1.0
=(2),==
```

  Generated entries:

CTRIA3	10	1	1	10	11				+C1
+C1			2.0	1.0	1.0				
CTRIA3	11	1	1	11	12				+C21
+C21			2.0	1.0	1.0				
CTRIA3	12	1	1	12	13				+C41
+C41			2.0	1.0	1.0				

- Parentheses are optional on replication entries and an equal sign may replace an asterisk.

  The following is an example of the use of replication, automatic continuation field generation, and the free field format.

```
GRID,101 ,17,1.0,10.5,,17,3456
=,*1,=,*0.2, *(0.1), == $ COMMENTS MAY APPEAR AFTER $
=3
EIGR,13,GIV,,30.
,MASS
CBAR,1 ,1 ,101 ,102,0.,0.,1.,,+0
=,*1,=,*1,*1====*1
+0,56
*1,=$
```

  The above free-field entries will generate the following Bulk Data in the 8-column format, as seen in the SORTED BULK DATA ECHO:

> **Note**
>
> A “,” should always be used after the “*1” for the continuation increment even if fixed field format is being used.

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CBAR	1	1	101	102	0.	0.	1.		+0
+0	56								
CBAR	2	1	102	103	0.	0.	1.		+1
+1	56								
EIGR	13			30.					+000001
++000001	MASS								
GRID	101	17	1.0	10.5		17	3456		
GRID	102	17	1.2	10.6		17	3456		
GRID	103	17	1.4	10.7		17	3456		
GRID	104	17	1.6	10.8		17	3456		
GRID	105	17	1.8	10.9		17	3456		

The automatically generated continuation entries start with the number 1, are incremented by 1, and are padded with zeros and plus signs as shown above. If this feature is used, it is the user's responsibility not to enter continuation entries that also use this convention. In particular, data generated on another run and then written to the PUNCH file with the ECHO=PUNCH, will cause problems when introduced into other data with blank continuation fields.

Related Topics

Browse Simcenter Nastran documentation

Bulk Entry Summary

This section contains a summary of all bulk entries under the following headings:

- Geometry
- Elements
- Materials
- Material Selection by Property
- Constraints and Partitioning
- Loads
- Solution Control
- Miscellaneous

Related Topics

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Geometry

Grid Points

GRDSET	Defines default options for fields 3, 7, 8, and 9 of all GRID entries.
GRID	Defines the location of a geometric grid point, the directions of its displacement, and its permanent single-point constraints.
IMPERF	Defines grid point position deltas for SOL 401.
IMPRADD	Defines a union of grid point position delta sets defined with multiple IMPERF bulk entries for SOL 401.
ROTORB	Defines bearing grids for the stationary portion of a model in a rotor dynamics solution.
ROTORG	Defines the rotating portion of a model in a rotor dynamics solution.
SEQEP	Redefines the sequence of extra points to optimize bandwidth.
SEQGP	Used to manually order the grid points and scalar points of the problem. This entry is used to redefine the sequence of grid and scalar points to optimize bandwidth.

Coordinate Systems

BAROR	Defines default values for field 3 and fields 6 through 8 of the CBAR entry.
BEAMOR	Defines default values for field 3 and fields 6 through 8 of the CBEAM entry.
CORDiC	Cylindrical coordinate system definition.
CORDiR	Rectangular coordinate system definition.
CORDiS	Spherical coordinate system definition.
CORD3G	Defines a general coordinate system using three rotational angles as functions of coordinate values in the reference coordinate system.
MATCID	Overrides the material coordinate system for CHEXA, CPENTA, CTETRA, and CPYRAM solid elements when the elements reference a PSOLID property. Also overrides the material coordinate system for CHEXA and CPENTA solid elements when the elements reference a PCOMPS property.

Scalar Points

EPOINT	Defines extra points for use in dynamic problems.
SEQEP	Redefines the sequence of extra points to optimize bandwidth.
SEQGP	Grid and scalar point number re-sequencing.
SPOINT	Defines scalar points.

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Monitor Points

MONPNT2	Defines an integrated load monitor point at a named element, and outputs component of stress, strain, or force at that element for SOL 101 and SOL 103.
MONPNT3	Defines an integrated load monitor point at a chosen point and sums select grid point forces for SOL 101 and SOL 103.

Fluid Points

ACMODL	Defines modeling parameters for the interface between the fluid and the structure.
GRID	Defines fluid points in coupled fluid-structural analysis.

Cyclic Symmetry

CYAX	Lists grid points that lie on the axis of symmetry in cyclic symmetry analysis.
CYCAXIS	Default cylindrical coordinate system for a cyclic model in SOLs 401, 402, and 414.
CYJOIN	Defines the boundary points of a segment in cyclic symmetry problems.

Superelement Analysis

CSUPER	Defines the grid or scalar point connections for identical or mirror image superelements or superelements from an external source. These are all known as secondary superelements.
CSUPEXT	Assigns exterior points to a superelement.
EXTRN	Defines a boundary connection for an external superelement.
GRID	Defines interior points for a superelement.
POINT	Define geometric points for use with the SELOC entry.
RELEASE	Defines degrees-of-freedom for superelement exterior grid points that are not connected to the superelement.
SEBNDRY	Defines a list of grid points in a partitioned superelement for the automatic boundary search between a specified superelement or between all other superelements in the model.
SEBULK	Defines superelement boundary search options and a repeated, mirrored, or collector superelement.
SECONCT	Explicitly defines grid and scalar point connection procedures for a partitioned superelement.

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SEDAMP	Defines viscous and structural (hysteretic) damping for superelements in SOL 414.
SEELT	Reassigns superelement boundary elements to an upstream superelement.
SEEXCLD	Defines grid points that will be excluded during the attachment of a partitioned superelement.
SELABEL	Defines a label or name to be printed in the superelement output headings.
SELOC	Defines a partitioned superelement relocation by listing three non-colinear points in the superelement and three corresponding points not belonging to the superelement.
SEMAT	Defines the location and orientation of a superelement in SOL 414.
SEMCF	Defines scaling factors for superelement matrices in SOL 414.
SEMPLN	Defines a mirror plane for mirroring a partitioned superelement.
SENAME	References a DESC bulk entry that contains a description of the superelement in SOL 414.
SEOUT	Requests results calculation for result type and grid point combinations specified on OUTMGT bulk entries in superelement creation runs during a SOL 414,103 with a transient response subcase, SOL 414,111, or SOL 414,129 analysis system run.
SEQSEP	Used with the CSUPER entry to define the correspondence of the exterior grid points between an identical or mirror-image superelement and its primary superelement.
SESET	Defines interior grid points for a superelement.
SETREE	Specifies superelement reduction order.
SEUNIT	Defines conversion factors to convert the superelement matrices into the units of the residual structure in SOL 414.

Aeroelastic Control Points

AECOMP	Defines a component for use in aeroelastic monitor point definition.
AECOMPL	Defines a component for use in aeroelastic monitor point definition as a union of other components.
MONPNT1	Defines an integrated load monitor point at a point (x,y,z) in a user defined coordinate system.
UXVEC	Specification of a vector of aerodynamic control point (extra point) values.

Related Topics

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Elements

Line Elements

BAROR	Default for orientation and property for CBAR.
BEAMOR	Default for orientation and property for CBEAM.
BOLT	Selects the CBEAM/CBAR elements to be included in the bolt pre-load calculation.
CBAR	Defines a simple beam element.
CBEAM	Defines a beam element.
CBEND	Connection definition for a curved beam.
CFAST	Defines a shell patch connection with direct stiffness input.
CONROD	Defines a rod element without reference to a property entry.
CROD	Defines a tension-compression-torsion element.
CTUBE	Defines a tension-compression-torsion tube element.
CWELD	Defines a weld or fastener connecting two surface patches or points.
PBAR	Defines the properties of a simple beam element (CBAR entry).
PBARL	Defines the properties of a simple beam element (CBAR entry) by cross-sectional dimensions.
PBCOMP	Alternate form of the PBEAM entry to define properties of a uniform cross-sectional beam referenced by a CBEAM entry.
PBEAM	Defines the properties of a beam element (CBEAM entry). This element may be used to model tapered beams.
PBEAML	Defines the properties of a beam element by cross-sectional dimensions.
PBEND	Defines the properties of a curved beam, curved pipe, or elbow element (CBEND entry).
PFAST	Defines properties for CFAST connector.
PROD	Defines the properties of a rod element (CROD entry).
PTUBE	Defines the properties of a thin-walled cylindrical tube element (CTUBE entry).
PWELD	Defines the properties of connector (CWELD) elements.
SWLDPRM	Define parameters for CWELD/CFAST connectors.

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Surface Elements

CPLSTNi	Defines a plane strain quadrilateral or triangular element for use in linear or nonlinear analysis.
CPLSTSi	Defines a plane stress quadrilateral or triangular element for use in linear or nonlinear analysis.
CQUAD	Defines a plane-strain quadrilateral element with up to nine grid points for use in fully nonlinear (i.e., large strain and large rotation) hyperelastic analysis.
CQUAD4	Defines an isoparametric membrane-bending or plane-strain quadrilateral plate element.
CQUAD8	Defines a curved quadrilateral shell or plane-strain element with eight grid points.
CQUADR	Defines an isoparametric membrane and bending quadrilateral plate element.
CSHEAR	Defines a shear panel element.
CTRIA3	Defines an isoparametric membrane-bending or plane-strain triangular plate element.
CTRIA6	Defines a curved triangular shell element or plane-strain elemetn with six grid points.
CTRIAR	Defines an isoparametric membrane-bending triangular plate element. However, this element does not include membrane-bending coupling. It is a companion to the CQUADR element.
PCOMP	Defines the properties of an n-ply composite material laminate.
PCOMPG	Defines the properties of an n-ply composite material laminate with global ply IDs.
PCOMPG1	Defines the properties of an n-ply composite material laminate which includes global ply IDs for SOLs 401 and 402.
PGPLSN	Defines the properties for generalized plane strain elements. Valid for SOL 401.
PLPLANE	Defines the properties of a fully nonlinear (i.e., large strain and large rotation) hyperelastic plane-strain or axisymmetric element.
PPLANE	Defines the properties of plane stress elements or plane strain elements.
PSHEAR	Defines the properties of a shear panel (CSHEAR entry).
PSHELL	Defines the membrane, bending, transverse shear, and coupling properties of thin shell elements.
PSHL3D	Defines the properties of 3D-shell elements.
PSHLPNL	Defines the properties for a tension-only shell elements.
SNORM	Defines a surface normal vector at a grid point for CQUAD4, CQUADR, CTRIA3, and CTRIAR shell elements.

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Solid Elements

CHEXA	Defines the connections of the six-sided solid element with eight to twenty grid points.
CHEXCZ	Defines the connections of the six-sided cohesive element with eight to twenty grid points.
CPENTA	Defines the connections of a five-sided solid element with six to fifteen grid points.
CPENTCZ	Defines the connections of a five-sided cohesive element with six to fifteen grid points.
CPYRAM	Defines the connection of the five-sided solid element with five to thirteen grid points.
CTETRA	Defines the connections of the four-sided solid element with four to ten grid points.
PCOMPS	Defines the properties of an n-ply composite material laminate for CHEXA and CPENTA solid elements.
PLSOLID	Defines a fully nonlinear (i.e., large strain and large rotation) hyperelastic solid element.
PSOLCZ	Defines the properties of cohesive elements (CHEXCZ and CPENTCZ entries).
PSOLID	Defines the properties of solid elements (CHEXA, CPENTA, and CTETRA entries).

Scalar and Bushing Elements

CBUSH	Defines a generalized spring-and-damper structural element that may be nonlinear or frequency dependent.
CBUSH1D	Defines the connectivity of a one-dimensional spring and viscous damper element.
CBUSH2	Defines a generalized spring-and-damper structural element.
CELASi	Connection definition for scalar spring, also property definition for i=2 or 4.
GENEL	Defines a general element.
PBUSH	Defines the nominal property values for a generalized spring-and-damper structural element.
PBUSH1D	Defines linear and nonlinear properties of a one-dimensional spring and damper element (CBUSH1D entry).
PBUSH2	Defines the properties for CBUSH2 elements.
PBUSHT	Defines the frequency-dependent properties or the stress-dependent properties for a generalized spring and damper structural element.

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PELAS	Specifies the stiffness, damping coefficient, and stress coefficient of a scalar elastic (spring) element (CELAS1 or CELAS3 entry).
PELAST	Defines the frequency dependent properties for a PELAS Bulk Data entry.

Axisymmetric Elements

CCHOCK3	Defines a triangular chocking element.
CCHOCK4	Defines a quadrilateral chocking element.
CCHOCK6	Defines a triangular chocking element with midside nodes.
CCHOCK8	Defines a quadrilateral chocking element with midside nodes.
CQUADX4	Defines an isoparametric and axisymmetric quadrilateral cross-section ring element for use in linear and fully nonlinear (i.e., large strain and large rotations) hyperelastic analysis.
CQUADX8	Defines an isoparametric and axisymmetric quadrilateral cross-section ring element with midside nodes for use in linear and fully nonlinear (i.e., large strain and large rotations) hyperelastic analysis.
CTRAX3	Defines an isoparametric and axisymmetric triangular cross-section ring element for use in linear and fully nonlinear (i.e., large strain and large rotations) hyperelastic analysis.
CTRAX6	Defines an isoparametric and axisymmetric triangular cross-section ring element with midside nodes for use in linear and fully nonlinear (i.e., large strain and large rotations) hyperelastic analysis.
FOU3	Coupling element between a three dimensional representation of a portion of a structure and an axisymmetric representation of the remainder of the structure in SOL 414.
PCHOCK	Defines the properties of a chocking element described on a CCHOCKi entry.
PSOLID	Defines the properties of axisymmetric elements (CQUADX4, CQUADX8, CTRAX3, and CTRAX6 entries).

Rigid Elements

RBAR	Defines a rigid bar with six degrees-of-freedom at each end.
RBE1	Defines a rigid body connected to an arbitrary number of grid points.
RBE2	Defines a rigid body with independent degrees-of-freedom that are specified at a single grid point and with dependent degrees-of-freedom that are specified at an arbitrary number of grid points.
RBE3	Defines the motion at a reference grid point as the weighted average of the motions at a set of other grid points.

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RROD	Defines a pin-ended element that is rigid in translation.
RSPLINE	Defines multipoint constraints for the interpolation of displacements at grid points.
RSSCON	Defines multipoint constraints to model clamped connections of shell-to-solid elements.
RTRPLT	Defines a rigid triangular plate.

Mass Elements

CMASSi	Connection definition for scalar mass, also property definition for i=2 or 4.
CONM1	Defines a 6 x 6 symmetric mass matrix at a geometric grid point.
CONM2	Defines concentrated mass at a grid point.
NSM	Defines a set of non-structural mass by ID.
NSM1	Alternate form for NSM entry. Defines non-structural mass entries by VALUE, ID list.
NSMADD	Non-structural mass set combination. Defines non-structural mass as the sum of the sets listed.
NSML	Defines a set of lumped non-structural mass by ID.
NSML1	Alternate form for NSML entry. Defines lumped non-structural mass entries by VALUE, ID list.
PMASS	Specifies the mass value of a scalar mass element (CMASS1 or CMASS3 entries).

Damping Elements

CBUSH1D	See line elements.
CDAMPi	Connection definition for a scalar damper, also property definition for i=2 or 4.
CVISC	Defines a viscous damper element.
PBUSH1D	See line elements.
PDAMP	Specifies the damping value of a scalar damper element using defined CDAMP1 or CDAMP3 entries.
PDAMP5	Defines the damping multiplier and references the material properties for damping. CDAMP5 is intended for heat transfer analysis only.
PDAMPT	Defines the frequency-dependent properties for a PDAMP Bulk Data entry.
PVISC	Defines properties of a one-dimensional viscous damping element (CVISC entry).

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Fluid and Acoustic Elements

ACADAPT	Invokes the FEMAO method for both uncoupled acoustic and coupled vibro-acoustic analysis.
ACORDER	Defines the polynomial order for the FEMAO method.
ACTRAD	Defines transfer admittance for the FEMAO method.
CAABSF	Defines a frequency-dependent acoustic absorber element in coupled fluid-structural analysis.
CHEXA	Connection definition for a hexahedron element in coupled fluid-structural analysis.
CPENTA	Connection definition for a pentahedron element in coupled fluid-structural analysis.
CPYRAM	Connection definition for a pyramid element in coupled fluid-structural analysis.
CTETRA	Connection definition for a tetrahedron element in coupled fluid-structural analysis.
ELIST	Defines a list of structural elements for virtual fluid mass.
PAABSF	Defines the properties of a frequency-dependent acoustic absorber element.
PAABSF1	Defines the properties of a frequency-dependent acoustic impedance or admittance element.
PACTRAD	Defines the property of transfer admittance for acoustics.
PANEL	Selects sets of structural grid points, elements, or physical properties that define one or more panels.
PSOLID	Defines the properties of solid elements (CHEXA, CPENTA, CPYRAM, CTETRA entries).
VISGRP	Defines poro-elastic visualization elements.

Heat Transfer Elements

BDYOR	Defines default values for the CHBDYP, CHBDYG, and CHBDYE entries.
CHBDYi	Connection definition for surface element (CHBDYE, CHBDYG, CHBDYP).
PHBDY	A property entry referenced by CHBDYP entries to give auxiliary geometric information for boundary condition surface elements.

The following elastic elements may also be used as heat conduction elements.

Linear:	CBAR, CROD, CONROD, CTUBE, CBEAM, CBEND.
Membrane:	CTRIA3, CTRIA6, CQUAD4, CQUAD8.
Solid:	CTETRA, CHEXA, CPENTA.

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Dummy Elements

ADUMi	Defines attributes of the dummy elements ($1 \le i \le 9$).
CDUMi	Defines a dummy element ($1 \le i \le 9$).
PDUMi	Defines the properties of a dummy element ($1 \le i \le 9$). Referenced by the CDUMi entry.
PLOTEL	Defines a one-dimensional dummy element for use in plotting.
PLOTEL3	Defines the connectivity of a triangular visualization element with three grid points for use in post-processing.
PLOTEL4	Defines the connectivity of a quadrilateral visualization element with four grid points for use in post-processing.
PLOTEL6	Defines the connectivity of a triangular visualization element with three to six grid points for use in post-processing.
PLOTEL8	Defines the connectivity of a quadrilateral visualization element with five to eight grid points for use in post-processing.
PLOTHEX	Defines the connectivity of a six-sided hexahedron visualization element with eight to twenty grid points for use in post-processing.
PLOTPEN	Defines the connectivity of a five-sided pentahedron visualization element with six to fifteen grid points for use in post-processing.
PLOTPYR	Defines the connectivity of a five-sided pyramid visualization element with five to thirteen grid points for use in post-processing.
PLOTTET	Defines the connectivity of a four-sided tetrahedron visualization element with four to ten grid points for use in post-processing.

Contact, Glue, and Gap Elements

BCONP	Defines the parameters for a contact region and its properties.
BCPROP	Defines a surface-to-surface contact or glue region by property IDs of shell elements.
BCPROPS	Defines a surface-to-surface contact or glue region by PSOLID Property ID.
BCRPARA	Defines parameters for a surface-to-surface contact region.
BCTADD	Defines a surface-to-surface contact set as a union of contact sets defined on BCTSET entries.
BCTPARM	Control parameters for the surface-to-surface contact algorithm.
BCTSET	Defines contact pairs of a 2D or 3D contact set.
BEDGE	Defines a glue or contact edge region.
BFRIC	Defines frictional properties between two bodies in contact.

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BGADD	Defines a surface-to-surface glue set as a union of glue sets defined on BGSET entries.
BGPARM	Control parameters for the glue algorithm.
BGSET	Defines glued contact pairs.
BLSEG	Defines a curve that consists of a number of line segments via grid numbers that may come in contact with another body.
BSURF	Defines a 3D contact or glue region by shell element IDs.
BSURFS	Defines a 3D contact or glue region by the faces of the CHEXA, CPENTA or CTETRA elements.
BVTADD	Defines a virtual thread contact set as a union of virtual thread contact sets defined on BVTSET entries.
BVTPAR	Defines virtual thread contact parameters for SOL 402.
BVTSET	Defines a virtual thread contact set for SOL 402.
BWIDTH	Defines widths or thicknesses for line segments in 3-D or 2-D slideline contact defined in the corresponding BLSEG Bulk Data entry.
CGAP	Defines a gap or friction element.
PGAP	Defines the properties of the gap element (CGAP entry).

Crack Tip Elements

CRAC2D	Defines a two-dimensional crack tip element.
CRAC3D	Defines a three-dimensional crack tip element.
CRAKTP	Specifies information related to a crack tip.
PRAC2D	Defines the properties and stress evaluation techniques to be used with the CRAC2D structural element.
PRAC3D	Defines the properties of the CRAC3D structural element.
VCEV	Defines virtual crack tip extension vectors.

Aerodynamic Elements

AEFACT	Defines real numbers for aeroelastic analysis.
AELINK	Defines relationships between or among AESTAT and AESURF entries.
AELIST	Defines a list of aerodynamic elements to undergo the motion prescribed with the AESURF Bulk Data entry for static aeroelasticity.
AESTAT	Specifies rigid body motions to be used as trim variables in static aeroelasticity.

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AESURF	Specifies an aerodynamic control surface as a member of the set of aerodynamic extra points.
AESURFS	Optional specification of the structural nodes associated with an aerodynamic control surface that has been defined on an AESURF entry.
CAERO1	Defines an aerodynamic macro element (panel) in terms of two leading edge locations and side chords.
CAERO2	Defines aerodynamic slender body and interference elements for Doublet-Lattice aerodynamics.
CAERO3	Defines the aerodynamic edges of a Mach Box lifting surface. If no cranks are present, this entry defines the aerodynamic Mach Box lifting surface.
CAERO4	Defines an aerodynamic macro element for Strip theory.
CAERO5	Defines an aerodynamic macro element for Piston theory.
CSSCHD	Defines a scheduled control surface deflection as a function of Mach number and angle of attack.
PAERO1	Defines associated bodies for the panels in the Doublet-Lattice method.
PAERO2	Defines the cross-sectional properties of aerodynamic bodies.
PAERO3	Defines the number of Mach boxes in the flow direction and the location of cranks and control surfaces of a Mach box lifting surface.
PAERO4	Defines properties of each strip element for Strip theory.
PAERO5	Defines properties of each strip element for Piston theory.

Aerodynamic to Structure Interconnection

SPLINE1	Defines a surface spline for interpolating motion and/or forces for aeroelastic problems on aerodynamic geometries defined by regular arrays of aerodynamic points.
SPLINE2	Defines a beam spline for interpolating motion and/or forces for aeroelastic problems on aerodynamic geometries defined by regular arrays of aerodynamic points.
SPLINE3	Defines a constraint equation for aeroelastic problems. Useful for control surface constraints.
SPLINE4	Defines a curved surface spline for interpolating motion and/or forces for aeroelastic problems on general aerodynamic geometries using either the Infinite Plate, Thin Plate or Finite Plate splining method.
SPLINE5	Defines a 1D beam spline for interpolating motion and/or forces for aeroelastic problems on aerodynamic geometries defined by irregular arrays of aerodynamic points.

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Rotor Dynamics Elements

CBEAR	Defines speed-dependent bearing connectivity.
CBEAR2	Defines bearing connection in SOL 414.
CGEAR	Defines a geared connection between rotors in SOL 414.
CLINK	Defines a connection between coincident grid points in the same rotor or different rotors that rotate at the same speed and in the same direction to which misalignment parameters can be applied in SOL 414,111 and SOL 414,129.
FOU3	Coupling element between a three dimensional representation of a portion of a structure and an axisymmetric representation of the remainder of the structure in SOL 414.
PBEAR	Defines speed-dependent stiffness, viscous damping, and mass matrices for bearing connection.
PBEAR2	Defines properties for bearing connection in SOL 414.
PGEAR	Defines the properties of a geared connection between rotors in SOL 414.

Kinematic joints (SOL 402 only)

CJOINT	Defines the joint topology.
DESC	Defines the joint description.
DRIVER	Defines a driver load or enforced displacement on kinematic joints.
JCON	Defines fixation times or liberation times for kinematic joints.
JCONADD	Defines a kinematic joint constraint set as a union of single kinematic joint constraint sets defined on JCON entries.
PJOINT	Defines the properties of the joint.
PJOINT2	Defines additional properties for the joint, like a spring or damper behavior.
FLXADD	Defines a flexible slider set as a union of single flexible sliders defined on FLXSLI entries.
FLXSLI	Defines a flexible slider.

Element Addition and Removal

ELAR	Defines the addition and removal of elements.
ELAR2	Defines the addition and removal of elements.
ELARADD	Defines an element add/remove set as a union of ELAR and ELAR2 bulk entries.

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Materials

Isotropic

MAT1	Defines the material properties for linear isotropic materials.
MAT1F	Specifies frequency-dependent isotropic material properties for SOLs 108 and 111.
MAT4	Defines the constant or temperature-dependent thermal material properties for conductivity, heat capacity, density, dynamic viscosity, heat generation, reference enthalpy, and latent heat associated with a single-phase change.
MATCZ	Defines damage model and material properties for cohesive elements (CHEXCZ and CPENTCZ).
MATHE	Specifies hyperelastic (rubber-like) material properties for nonlinear analysis.
MATHEM	Specifies Mullins effect (Ogden-Roxburgh model) on hyperelastic (MATHE) material.
MATHEV	Specifies viscoelastic effect (Holzapfel model) on hyperelastic (MATHE) material.
MATHP	Specifies material properties for use in fully nonlinear (i.e., large strain and large rotation) hyperelastic analysis of rubber-like materials (elastomers).
RADM	Defines the radiation properties of a boundary element for heat transfer analysis.

Anisotropic

MAT2	Defines the material properties for linear anisotropic materials for two-dimensional elements.
MAT2F	Specifies frequency-dependent material properties on MAT2 entry fields via TABLEDi entries for SOLs 108 and 111.
MAT3	Defines linear orthotropic material properties for axisymmetric, plane stress, and plane strain elements. Not valid for use with chocking elements.
MAT5	Defines the thermal material properties for anisotropic materials.
MAT8	Defines the material property for an orthotropic material for isoparametric shell elements.
MAT9	Defines the material properties for linear, temperature-independent, anisotropic materials for solid isoparametric elements (see the PSOLID entry description).
MAT9F	Specifies frequency-dependent material properties on MAT9 entry fields via TABLEDi entries for SOLs 108 and 111.
MAT11	Defines linear orthotropic material properties for axisymmetric, plane stress, plane strain, and solid elements.

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MATDMG	Defines damage-related material properties for progressive ply failure in composite solid elements. Used in combination with MAT1 or MAT11 entries. Valid for SOL 401 and SOL 402 only.
MATFT	Defines material properties for use with composite ply failure theories.

Temperature Dependent

MATTi	Table references for temperature-dependent MATi materials.
MATTC	Specifies temperature-dependent creep coefficients on CREEP entry fields via TABLEM1 entries. Valid for SOL 402.
RADMT	Specifies table references for temperature dependent RADM entry radiation boundary properties.
TABLEMi i=1,2,3,4	Tabular functions for generating temperature-dependent material properties.
TABLEST	Table references for temperature dependent MATS1 materials.
TEMP	Defines temperature at grid points for determination of thermal loading, temperature-dependent material properties, or stress recovery.
TEMPD	Defines a temperature value for all grid points of the structural model that have not been given a temperature on a TEMP entry.
TEMPPi	Defines a temperature field for surface elements.
TEMPRB	Defines a temperature field for the CBAR, CBEAM, CBEND, CROD, CTUBE, and CONROD elements for determination of thermal loading, temperature-dependent material properties, or stress recovery.

Stress Dependent

CREEP	Defines creep characteristics based on experimental data or known empirical creep law.
MATG	Defines the material properties for gasket materials.
MATS1	Specifies stress-dependent material properties for use in applications involving nonlinear materials.
MPLAS	Defines plastic material properties for SOLs 401 and 402.
MUCRP	Defines the material properties for the user-defined creep subroutines in SOL 401.
MVPLAS	Defines creep or visco-plastic material properties for SOLs 401 and 402.

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TABLEM5	Defines a nonlinear function relating the shear damage to the thermodynamic force for progressive ply failure with the UD damage model in SOL 401 and SOL 402.
TABLES1	Defines a tabular function for stress-dependent material properties such as the stress-strain curve (MATS1 entry), creep parameters (CREEP entry) and hyperelastic material parameters (MATHP entry).

Strain Dependent

MATSR	Specifies strain-rate dependent properties for use with a MATS1 entry. Valid for SOL 402.

Fluid

ACMODPE	Defines modeling parameters for a fluid to poro-elastic to structure interface
ACTEMP	Retrieves from an external file spatially-varying intensive property data for fluids in standard FEM and FEMAO acoustic analysis.
ACTRTM	Defines transfer admittance for vibro-acoustics.
MAT10	Defines material properties for fluid elements in coupled fluid-structural analysis.
MAT10C	Defines constant or nominal material properties for fluid or absorber elements in coupled fluid-structural analysis.
MATF10C	Defines tabular material properties for fluid or absorber elements in coupled fluid-structural analysis.
MATPOR	Defines acoustic absorber material properties.
MATVST	Defines visco-thermal material properties for fluid elements.
MFLUID	Defines the properties of an incompressible fluid volume for the purpose of generating a virtual mass matrix.
PACTRTM	Defines the property of transfer admittance for vibro-acoustics.

Damping

DAMPING	Specifies structural damping, Rayleigh (proportional) damping, and, for transient response analysis or a modal-based analysis that writes an interface file for external multibody dynamics or control system software, the structural (hysteretic) damping to viscous damping conversion method.
DAMPTAB	Damping definition for SOL 401 modal frequency response subcase.

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User defined material

MATMM	Defines a Multimech microstructure material.
MUMAT	Defines the material properties for the user defined material subroutine. Valid for SOL 401 and SOL 402.

Related Topics

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Constraints and Partitioning

Single Point Constraints

GRID	Includes single point constraint definition.
GRDSET	Includes deafult for single point constraints.
SPC	Defines a set of single-point constraints and enforced motion—enforced displacements in static analysis and enforced displacements, velocities or acceleration in dynamic analysis.
SPC1	Defines a set of single point constraints.
SPCADD	Defines a single-point constraint set as a union of single-point constraint sets defined on SPC or SPC1 entries.
SPCOFFi	Defines degrees-of-freedom to be excluded from the AUTOSPC operation.

Multipoint Constraints

CWELD	Defines a weld or fastener connecting two surface patches or points.
CYCADD	Combines cyclic symmetry sets in SOLs 401, 402, and 414.
CYCISC	Defines a connection between cyclic symmetry sectors for SOL 414.
CYCSET	Defines pairs for cyclic symmetry in SOLs 401, 402, and 414.
MPC	Defines a linear relationship for two or more degrees-of-freedom.
MPCADD	Combines multipoint constraint sets.
PWELD	Defines the properties of connector (CWELD) elements.
RBAR	Defines multipoint constraints for rigid bar.
RBEi	Defines multipoint constraints for RBE1, RBE2, and RBE3.
RROD	Defines multipoint constraints for rigid rod.
RSPLINE	Defines multipoint constraints for spline element.
RTRPLT	Defines multipoint constraints for rigid triangular plate.

Partitioning

ASET	Defines degrees-of-freedom in the analysis set (a-set).
ASET1	Defines degrees-of-freedom in the analysis set (a-set).
CSUPEXT	Assigns exterior points to a superelement.
GRID	Defines interior points for a superelement.

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OMIT	Defines degrees-of-freedom to be excluded (o-set) from the analysis set (a-set).
OMIT1	Defines degrees-of-freedom to be excluded (o-set) from the analysis set (a-set).
RELEASE	Defines degrees-of-freedom for superelement exterior grid points that are not connected to the superelement.
ROTSE	Defines the modal reduction type and additional a-set grids for a rotor superelement.
SEELT	Reassigns superelement boundary elements to an upstream superelement.
SESET	Defines interior grid points for a superelement.

Free Body Supports

CYSUP	Defines fictitious supports for cyclic symmetry analysis.
SUPORTi	Defines degrees-of-freedom for determinate reactions.

Component Mode Boundary Conditions

BNDFIX	Defines analysis set (a-set) degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.
BNDFIX1	Defines analysis set (a-set) degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.
BNDFREE	Defines analysis set (a-set) degrees-of-freedom to be free (c-set) during component modes calculations.
BNDFREE1	Defines analysis set (a-set) degrees-of-freedom to be free (c-set) during component modes calculations.
BSET	Defines the analysis set (a-set) degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.
BSET1	Defines the analysis set (a-set) degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.
CSET	Defines the analysis set (a-set) degrees-of-freedom to be free (c-set) during component modes calculations.
CSET1	Defines the analysis set (a-set) degrees-of-freedom to be free (c-set) during component modes calculations.
QSET	Defines the generalized degrees-of-freedom (q-set) to be used for component mode synthesis.
QSET1	Defines the generalized degrees-of-freedom (q-set) to be used for component mode synthesis.
SEBSET	Defines the boundary degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.

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SEBSET1	Defines the fixed boundary points for a superelement.
SECSET	Defines the boundary degrees-of-freedom to be free (c-set) during component mode synthesis calculations.
SECSET1	Defines the boundary degrees-of-freedom to be free (c-set) during component mode synthesis calculations.
SENQSET	Defines the number of internally generated scalar points for superelement dynamic reduction.
SEQSET	Defines the generalized degrees-of-freedom of the superelement to be used in component mode synthesis.
SEQSET1	Defines the generalized degrees-of-freedom of the superelement to be used in component mode synthesis.
SESUP	Defines the determinate reaction superelement degrees-of-freedom in a free-body analysis.

User Sets

DEFUSET	Defines the new names for degree-of-freedom sets.
SEUSET	Defines a degree-of-freedom set for a superelement.
SEUSET1	Defines a degree-of-freedom set for a superelement.
USET	Defines a degree-of-freedom set.
USET1	Defines a degrees-of-freedom set.

Miscellaneous

FMH3D	Defines the coupling of a Fourier multi-harmonic region with a 3D structure in SOL 402 and 414.

Related Topics

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Loads

Static Loads

ACCEL	Defines static acceleration loads, which may vary over a region of the structural model. The load variation is based on the tabular input defined on this Bulk Data entry.
ACCEL1	Defines static acceleration loads at individual grid points.
BOLTFRC	Defines preload force for a set of bolts.
BOLTLD	Combines sets of bolt loads and optionally scales the corresponding bolt preload forces.
CLOAD	Defines a static load as a linear combination of previously calculated superelement loads defined by the LSEQ entry in nonlinear static analysis (SOL 106 or 153).
DEFORM	Defines the enforced axial deformation for one-dimensional elements for use in statics problems.
FORCE	Defines a static concentrated force at a grid point by specifying a vector.
FORCEi	Defines a concentrated load at grid point.
GRAV	Defines acceleration vectors for gravity or other acceleration loading.
LOAD	Defines a static load as a linear combination of load sets..
LOADCYH	Defines the harmonic coefficients of a static or dynamic load for use in cyclic symmetry analysis.
LOADCYN	Defines a physical static or dynamic load for use in cyclic symmetry analysis.
LOADCYT	Specifies loads as a function of azimuth angle by references to tables that define scale factors of loads versus azimuth angles. This entry is used only when STYPE = “AXI” on the CYSYM entry.
MLINK	Defines static misalignment parameters for a CLINK bulk entry in SOL 414,101 and SOL 414,129 solutions, and in the static preload subcases for SOL 414,103, SOL 414,110, and SOL 414,111 solutions.
MLOAD	Defines a misalignment load in SOL 414.
MOMENT	Defines a static concentrated moment at a grid point by specifying a scale factor and a vector that determines the direction.
MOMENTi	Defines moment at grid point.
PLOAD	Defines a uniform static pressure load on a triangular or quadrilateral surface comprised of surface elements and/or the faces of solid elements.
PLOAD1	Defines concentrated, uniformly distributed, or linearly distributed applied loads to the CBAR or CBEAM elements at user-chosen points along the axis. For the CBEND element, only distributed loads over an entire length may be defined.

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PLOAD2	Defines a uniform static pressure load applied to CQUAD4, CSHEAR, or CTRIA3 two-dimensional elements.
PLOAD4	Defines a pressure load on a face of a CHEXA, CPENTA, CTETRA, CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, or CQUADR element.
PLOADE1	Defines a surface traction acting on an edge of a CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8, CPLSTS3, CPLSTS4, CPLSTS6, or CPLSTS8 element.
PLOADFP	Defines fluid penetration load for SOL 401 and SOL 402.
PLOADX1	Defines a surface traction to be used with the CQUADX4, CQUADX8, CTRAX3, and CTRAX6 axisymmetric elements.
RFORCE	Defines a static loading condition due to an angular velocity and/or acceleration for an entire model.
RFORCE1	Defines a static loading condition due to an angular velocity and/or acceleration for a portion of a model.
RFORCE2	Defines a static loading condition due to an angular velocity and/or acceleration for SOL 101 maneuver load rotor dynamic analysis.
RVDOF	Specifies the degrees-of-freedom where unit loads are applied to obtain static solutions for use in residual vector computations.
RVDOF1	Specifies the degrees-of-freedom where unit loads are applied to obtain static solutions for use in residual vector computations.
RVEL	Specifies CBUSH, CBUSH1D, CDAMPi, CELASi, and CVISC elements for which to obtain static solutions for use in residual vector computations.
SELOAD	Maps loads from an external superelement to a specified load set for the residual structure.
SLOAD	Defines concentrated static loads on scalar or grid points.
SPCD	Defines an enforced displacement value for static analysis and an enforced motion value (displacement, velocity or acceleration) in dynamic analysis.
TEMP	Defines temperature at grid points for determination of thermal loading, temperature-dependent material properties, or stress recovery.
TEMPD	Defines a temperature value for all grid points of the structural model that have not been given a temperature on a TEMP entry.
TEMPEX	Defines a time independent temperature set using a .bun file for SOL 401 only.
TEMPPi	Defines a temperature field for surface elements.
TEMPRB	Defines a temperature field for the CBAR, CBEAM, CBEND, CROD, CTUBE, and CONROD elements for determination of thermal loading, temperature-dependent material properties, or stress recovery.

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Dynamic Loads

ACSRCE	Defines the power versus frequency curve for a simple acoustic source.
DAREA	Defines scale (area) factors for static and dynamic loads. In dynamic analysis, DAREA is used in conjunction with RLOADi and TLOADi entries.
DELAY	Defines the time delay term τ in the equations of the dynamic loading function.
DLOAD	Defines a dynamic loading condition for frequency response or transient response problems as a linear combination of load sets defined via RLOAD1 or RLOAD2 entries for frequency response or TLOAD1 or TLOAD2 entries for transient response.
DPHASE	Defines the phase lead term θ in the equation of the dynamic loading function.
DRIVER	Defines a driver load or enforced displacement on kinematic joints.
DTEMP	Defines a time dependent temperature set for SOL 401 only.
DTEMPEX	Defines a time dependent temperature set using a .bun file for SOL 401 only.
LOADCYH	Defines the harmonic coefficients of a static or dynamic load for use in cyclic symmetry analysis.
LOADCYN	Defines a physical static or dynamic load for use in cyclic symmetry analysis.
LSEQ	Defines a sequence of static load sets.
MLINK	Defines misalignment parameters for a CLINK bulk entry in SOL 414,111 and SOL 414,129.
MLOAD	Defines a misalignment load in SOL 414
NLRGAP	Defines a nonlinear transient radial (circular) gap.
NOLINi	Nonlinear transient load definition.
RLOADi	Frequency dependent excitation definition.
RLOADEX	Defines frequency-dependent dynamic forces obtained from a file for use in frequency response problems.
SPCF	Defines enforced displacement on structural grid points for a frequency response analysis using frequency-dependent displacement results stored in an .op2 file.
SPDIR	Defines base excitation for DDAM.
TLOADi	Time dependent excitation definition.
TABLEDi	Tabular functions for generating dynamic loads.
UNBA	Defines an unbalance load in SOL 414.

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Acoustic Loads

ACDUCT	Defines an acoustic duct mode.
ACFAN	Defines fan noise for a frequency response analysis.
ACNDUCT	Defines an acoustic anechoic end duct.
ACPLNW	Defines an acoustic plane wave source.
ACPNVEL	Defines panel normal velocity on fluid free surfaces.
ACPOLE1	Defines an acoustic monopole source.
ACPOLE2	Defines an acoustic dipole source.
ACPRESS	Defines an enforced pressure value for a frequency response or real eigenvalue analysis.
ACRADS	Defines an acoustic radiating surface from an external *.scd5 or *.sc_h5 file for use in a frequency response analysis.
ACSPO2	Defines a surface dipole from an external sc_h5 file in HDF5 format for use in a frequency response analysis.
ACUPOLE	Defines a unit acoustic monopole source.
ACVPO4	Defines an acoustic volume quadrupole source from an external *.cgns file for use in a frequency response analysis.
ACVSTT	Defines visco-thermal losses for acoustics and vibro-acoustics.
ALOAD	Defines an acoustic loading condition for frequency response analysis as a linear combination of load sets defined with ACPOLE1, ACPOLE2, ACPLNW, ACPNVEL, and ACPRESS bulk entries.
FREQH	Defines harmonics and sub-harmonics for fan noise.
PACDUCT	Defines the cross sectional properties for an acoustics duct mode.
PACFAN	Defines the properties of fan noise frequency analysis.
PACSPO2	Defines the properties of a surface dipole for a frequency response analysis.
PACVPO4	Defines the properties of an acoustic volume quadrupole source for a frequency response analysis.

Heat Transfer Loads

CONV	Specifies a free convection boundary condition for heat transfer analysis through connection to a surface element (CHBDYi entry).
CONVM	Specifies a forced convection boundary condition for heat transfer analysis through connection to a surface element (CHBDYi entry).

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PCONV	Specifies the free convection boundary condition properties of a boundary condition surface element used for heat transfer analysis.
PCONVM	Specifies the forced convection boundary condition properties of a boundary condition surface element used for heat transfer analysis.
QBDY1	Defines a uniform heat flux into CHBDYj elements.
QBDY2	Defines grid point heat flux into CHBDYj elements.
QBDY3	Defines a uniform heat flux load for a boundary surface.
QHBDY	Defines a uniform heat flux into a set of grid points.
QVECT	Defines thermal vector flux from a distant source into a face of one or more CHBDYi boundary condition surface elements.
QVOL	Defines a rate of volumetric heat addition in a conduction element.
RADBC	Specifies a CHBDYi element face for the application of radiation boundary conditions.
RADBND	Specifies Planck's second radiation constant and the wavelength breakpoints used for radiation exchange problems.
RADCAV	Identifies the characteristics of each radiant enclosure.
RADLST	Identifies the individual CHBDYi surface elements that comprise the entire radiation enclosure.
RADMTX	Provides the $F_{ji} = A_j F_{ji}$ exchange factors for all the faces of a radiation enclosure specified in the corresponding RADLST entry.
RADSET	Specifies which radiation cavities are to be included for radiation enclosure analysis.
SLOAD	Defines concentrated static loads on scalar or grid points.
TEMP	Defines temperature at grid points for determination of thermal loading, temperature-dependent material properties, or stress recovery.
TEMPBC	Defines the temperature boundary conditions for heat transfer analysis. This applies to steady-state and transient conditions.
TEMPD	Specifies default initial temperature at grid points.
VIEW	Defines radiation cavity and shadowing for radiation view factor calculations.
VIEW3D	Defines parameters to control and/or request the Gaussian Integration method of view factor calculation for a specified cavity.

Related Topics

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Solution Control

Buckling Analysis

EIGB	Defines data needed to perform buckling analysis.
EIGRL	Defines data needed to perform real eigenvalue (vibration or buckling) analysis with the Lanczos method.

Eigenvalue Analysis

EIGC	Defines data needed to perform complex eigenvalue analysis.
EIGP	Defines poles that are used in complex eigenvalue extraction by the Determinant method.
EIGR	Defines data needed to perform real eigenvalue analysis.
EIGRL	Defines data needed to perform real eigenvalue (vibration or buckling) analysis with the Lanczos method.
MDSBULK	Selects mode set component representation for use in a residual model.
RDMODES	Solution options for Recursive Domain Normal Modes method (RDMODES) in SOL 103 and SOL 111.

Cyclic Symmetry

CYSYM	Defines parameters for cyclic symmetry analysis.

Frequency Response

FREQi	Defines a set of frequencies to be used in the solution of frequency response problems.
FREQV	Defines frequency range of interest for a vibro-acoustic transfer vector (VATV) response analysis.
FRFFLEX	Defines the dynamic flexibility matrix for a direct-input, frequency-dependent component.
FRFOMAP	Assigns an output type to the results calculated from FRFOTM bulk entries.
FRFOTM	Defines a direct-input, frequency-dependent output transformation matrix.
FRFSTIF	Defines the dynamic stiffness matrix for a direct-input, frequency-dependent component.
OMEGAi	Defines a set of rotational speeds to be used in a SOL 414,110 complex modal analysis.

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PEAKOUT	Specifies the criteria that are used to filter the peak responses from all the resonances in frequency response results.
RDMODES	Solution options for Recursive Domain Normal Modes method (RDMODES) in SOL 103 and SOL 111.
SLVCNTL	Solution control options for SOL 108 and SOL 111.
TABLED6	Defines complex tabular data.
TABDMP1	Defines modal damping as a tabular function of natural frequency.

Acoustic Transfer Vectors

ATVBULK	Specifies the Fortran unit number for the OUTPUT2 file that contains the acoustic transfer vector (ATV) results, and the element, grid point, and element property offsets to be used in an ATV system run.
ATVFS	Defines the interface that couples the acoustic transfer vector (ATV) to the structure in the ATV system run.
VATVBK	Defines the Fortran unit number for the ASSIGN INPUTT2 file that contains the vibro-acoustic transfer vector (VATV) results to be used in a VATV response analysis.
VATVFS	Defines the structural surfaces of a pressure boundary for the vibro-acoustic transfer vector (VATV) calculation.

Random Response

RANDPEX	Defines load set power spectral density factors for use in random analysis from an external HDF5 format file.
RANDPS	Defines load set power spectral density factors for use in random analysis.
RANDT1	Defines time lag constants for use in random analysis autocorrelation function calculation.
RCROSS	Defines a pair of response quantities for computing the cross-power spectral density and cross-correlation functions in random analysis.
RCROSSC	Defines a pair of response quantities for computing the cross-power spectral density and cross-correlation functions in random analysis for composites.
TABRND1	Defines power spectral density as a tabular function of frequency for use in random analysis. This is referenced by the RANDPS entry.

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Rotor Dynamics

NLCNTL2	Defines solution control parameters for SOL 402 and SOL 414.
ROTORD	Defines rotor dynamics solution options.
ROTPARM	Defines solution control parameters for complex eigenvalue rotor dynamic analysis. Valid for SOLs 107 and 110 only.
SENSCLC	Monitors the gap between a node on a rotor and a group of nodes on a stator. Valid for SOL 414 only.
SPEBE2	Defines rotational speed for CBEAR2 bearing connections.

Transient Response

TIC	Defines values for the initial conditions of variables used in structural transient analysis.
TSTEP	Defines time step intervals at which a solution will be generated and output in transient analysis.
TSTEP1	Defines the time step intervals at which a solution will be generated and output in SOL 401.
TSTEPNL	Defines parametric controls and data for nonlinear transient structural or heat transfer analysis. TSTEPNL is intended for SOLs 129, 159, and 99.

Multi-Step Nonlinear Analysis

NLCNTLG	Defines global solution control parameters for SOL 402 and SOL 414.
NLCNTL2	Defines subcase solution control parameters for SOL 402 and SOL 414.

Nonlinear Static Analysis

ITER	Defines options for the iterative solver in SOLs 101, 106, 108, 111 and 153.
MATOVR	Globally turns off creep and/or plasticity effects in groups of elements. Applicable to SOL 401 only.
NLARCL	Defines control parameters for SOL 401 nonlinear buckling.
NLCNTL	Defines solution control parameters for SOL 401.
NLPARM	Defines a set of parameters for nonlinear static analysis iteration strategy.
NLPCI	Defines a set of parameters for the arc-length incremental solution strategies in nonlinear static analysis (SOL 106).

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Optimization (SOL 200 Only)

BNDGRID	Specifies a list of grid point identification numbers on design boundaries or surfaces for shape optimization (SOL 200).
DCONADD	Defines the design constraints for a subcase as a union of DCONSTR entries.
DCONSTR	Defines design constraints.
DDVAL	Defines real, discrete design variable values for discrete variable optimization.
DEQATN	Defines a design variable for design optimization.
DESVAR	Defines a design variable for design optimization.
DLINK	Relates one design variable to one or more other design variables.
DMNCON	Defines a manufacturing constraint for SOL 200 topology optimization.
DMRLAW	Defines the relation between material properties and the normalized material density for SOL 200 topology optimization.
DOPTPRM	Overrides default values of parameters used in design optimization.
DRESP1	Defines a set of structural responses that is used in the design either as constraints or as an objective.
DRESP2	Defines equation responses that are used in the design, either as constraints or as an objective.
DRESP3	Defines SOL 200 design responses to be evaluated in an external user-supplied program, that are used in the design either as constraints or as an objective.
DSCREEN	Defines screening data for constraint deletion.
DTABLE	Defines a table of real constants that are used in equations (see the DEQATN entry).
DTI,DFRFNC	Frequency function input for use in DRESP2 & DRESP3.
DVBSHAP	Associates a design variable identification number to a linear combination of boundary shape vectors from a particular auxiliary model.
DVCREL1	Defines the relation between a connectivity property and design variables.
DVCREL2	Defines the relation between a connectivity property and design variables with a user-supplied equation.
DVGRID	Defines the relationship between design variables and grid point locations.
DVMREL1	Defines the relation between a material property and design variables.
DVMREL2	Defines the relation between a material property and design variables with a user-supplied equation.
DVPREL1	Defines the relation between an analysis model property and design variables.

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DVPREL2	Defines the relation between an analysis model property and design variables with a user-supplied equation.
DVSHAP	Defines a shape basis vector by relating a design variable identification number (DVID) to columns of a displacement matrix.
DVTREL1	Selects the elements for the software to automatically create a fill ratio design variable and other related data for topology optimization
MODTRAK	Specifies parameters for mode tracking in design optimization (SOL 200).

Aerodynamic Matrix Generation

MKAERO1	Provides a table of Mach numbers (m) and reduced frequencies (k) for aerodynamic matrix calculation.
MKAERO2	Provides a list of Mach numbers (m) and reduced frequencies (k) for aerodynamic matrix calculation.

Aeroelastic Stability Analysis

DIVERG	Defines Mach numbers (m) for a divergence analysis in SOLs 144 and 200.
FLFACT	Used to specify density ratios, Mach numbers, reduced frequencies, and velocities for flutter analysis.
FLUTTER	Defines data needed to perform flutter analysis.

Aeroelastic Response Analysis

AEDW	Defines a downwash vector associated with a particular control vector of the associated aerodynamic configuration (AECONFIG). From this downwash vector, a force vector on the aerodynamic grids will be defined for use in nonlinear static aeroelastic trim.
AEFORCE	Defines a vector of absolute forces (it will **not** be scaled by dynamic pressure) associated with a particular control vector.
AEPARM	Defines a general aerodynamic trim variable degree of freedom (aerodynamic extra point).
AEPRESS	Defines a vector of pressure/unit dynamic pressure associated with a particular control vector.
GUST	Defines a stationary vertical gust for use in aeroelastic response analysis.
TABRNDG	Defines the power spectral density (PSD) of a gust for aeroelastic response analysis.
TRIM	Specifies constraints for aeroelastic trim variables. The SPLINE1 and SPLINE4 entries need to be here for the finite plate spline.

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Aerodynamic Parameters

AERO	Gives basic aerodynamic parameters for unsteady aerodynamics.
AEROS	Defines basic parameters for static aeroelasticity.

Response Spectrum Analysis

RSAPPLY	Selects RSPOPT, RSPECTR, and RSPLOC bulk data entries that define data and parameters for a response spectrum application run.
RSPECTR	Specifies the fraction of critical damping values that correspond to the response spectra specified on TABLED1 bulk entries.
RSPLOC	When used in a response spectrum application run, specifies the use of the relative method rather than the large mass method, and specifies the grid points at which to apply the response spectrum and the direction in which to apply the response spectrum.
RSPOPT	Specifies parameters for a response spectrum application run.

DDAM

DDAMCTR	Defines parameters for DDAM that are otherwise contained in a DDAM control file.
SELMODE	Defines mode selection parameters for a DDAM solution.

Coupled Modes with Incompressible Fluids Analysis

INCMOPT	Specifies options for modeling incompressible fluids with fluid elements.
INCMPFL	Defines an incompressible fluid surface.

Related Topics

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Miscellaneous

Comments

$	Used to insert comments into the input file. Comment statements may appear anywhere within the input file.

Delete

/	Removes entries on restart.

Parameters

PARAM	Specifies values for parameters used in solution sequences or user-written DMAP programs.

Groups and Sets

GROUP	Defines a group of grid points, elements, or physical properties.
SET1	Defines a list of structural grid points, elements, or DRESPi (i=1,2,3) responses. SET1 entries are referenced by SPLINEi entries, PANEL entries, XYOUTPUT, and the P2RSET option on DOPTPRM entries.
SET2	Defines a list of structural grid points in terms of aerodynamic macro elements. SET2 entries are referenced by SPLINEi entries.
SET3	Defines a list of structural grid points, elements, or physical properties. SET3 entries are referenced by ACMODL entries and PANEL entries.

Direct Matrix Input

CONM1	Defines a 6x6 mass matrix at a geometric grid point.
DMI	Defines matrix data blocks.
DMIG	Defines direct input matrices related to grid, extra, and/or scalar points.
DMIG,UACCEL	Defines rigid body accelerations in the basic coordinate system.
TF	Defines a dynamic transfer function.

Direct Matrix Input for Aeroelasticity

DMIJ	Defines direct input matrices related to collation degrees-of-freedom (js-set) of aerodynamic mesh points for CAERO1, CAERO3, CAERO4 and CAERO5 and for the

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	slender body elements of CAERO2. These include W2GJ, FA2J and input pressures and downwashes associated with AEPRESS and AEDW entries.
DMIJI	Defines direct input matrices related to collation degrees-of-freedom (js-set) of aerodynamic mesh points for the interference elements of CAERO2.
DMIK	Defines direct input matrices related to physical (displacement) degrees-of-freedom (ks-set) of aerodynamic grid points.

Tabular Input

DTI	Defines table data blocks.
DTI,DFRFNC	Frequency function input for use in DRESP2 & DRESP3. SOL 200 only.
DTI,ESTDATA	Provides override data for time and space estimation for superelement processing operations.
DTI,INDTA	Specifies or overrides default item codes for the sorting and filtering of element stresses, strains, and forces.
DTI,SELOAD	Usage overrides automatic generation of DTI,SELOAD definitions created through the use of the EXTSEOUT case control command.
DTI,SETREE	Defines a superelement tree that determines the superelement processing order.
DTI,SPECSEL	Correlates spectra lines specified on TABLED1 entries with damping values.
DTI,SPSEL	Correlates output requests with frequency and damping ranges.
DTI,UNITS	Specifies the system of units.
TABDMP1	Defines modal damping as a tabular function of natural frequency.
TABLEDi	Tabular functions for generating dynamic loads.
TABLEMi	Tabular functions for generating temperature-dependent material properties.
TABLES1	Defines a tabular function for stress-dependent material properties such as the stress-strain curve (MATS1 entry), creep parameters (CREEP entry) and hyperelastic material parameters (MATHP entry).
TABRND1	Defines power spectral density as a tabular function of frequency for use in random analysis. Referenced by the RANDPS entry.

Output Control

BOUTPUT	Defines secondary nodes at which output is requested.
CBARAO	Defines a series of points along the axis of a bar element (CBAR entry) for stress and force recovery output.
ECHOOFF	Marks the point or points in the input file to deactivate printed echo of the Bulk Data.

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ECHOON	Marks the point or points in the input file to activate printed echo of the Bulk Data.
OUTMGT	Defines an XY plot request in SOL 402, SOL 414,103 with a transient response subcase, SOL 414,111, or SOL 414,129.
TSTEP	Specifies time step intervals for data recovery in transient response.
TSTEPNL	Specifies time step intervals for data recovery in nonlinear transient response.

Solution Control

ITER	Defines options for the iterative solver in SOLs 101, 106, 108, 111 and 153.

End of Input

ENDDATA	Designates the end of the Bulk Data Section.

Include File

INCLUDE	Inserts an external file into the input file. The INCLUDE statement may appear anywhere within the input data file.

Related Topics

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Element Utilization Summary

Use these tables to identify which combinations of element, property, and material bulk entries you can use to obtain a particular material behavior in a given solution.

1. The **Element Applicability Table** shows which commonly used element bulk entries are valid in solutions 101 through 401.

   For information on how to interpret this table, see **Key to Element Applicability Table**.

   > **Note**
   >
   > For a comprehensive listing of elements supported for use in solutions 401 and 402, in the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*, see Element Overview.
   >
   > For a comprehensive listing of elements supported for use in solution 414, in the *Simcenter Nastran Rotor Dynamics User's Guide*, see Supported bulk entries.

2. The **Element-Property-Material Cross-Reference Tables** show which combinations of material bulk entries and property bulk entries are valid for a given element bulk entry.

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For example, suppose you want to perform a SOL 106 plasticity analysis on a thin-walled structure. The *Element Applicability for Select Elements in Solutions 101–401 Table* indicates that the CQUAD4 and CTRIA3 shell elements can be used to model plasticity in SOL 106. Because you know that MATS1 bulk entries are required to model plasticity, the table *Shell Elements: Supplemental Materials by Primary Material, Property, and Element* indicates that the CQUAD4 and CTRIA3 elements must reference PSHELL bulk entries that in turn reference MAT1, MAT2, or MAT8 bulk entries.

Related Topics

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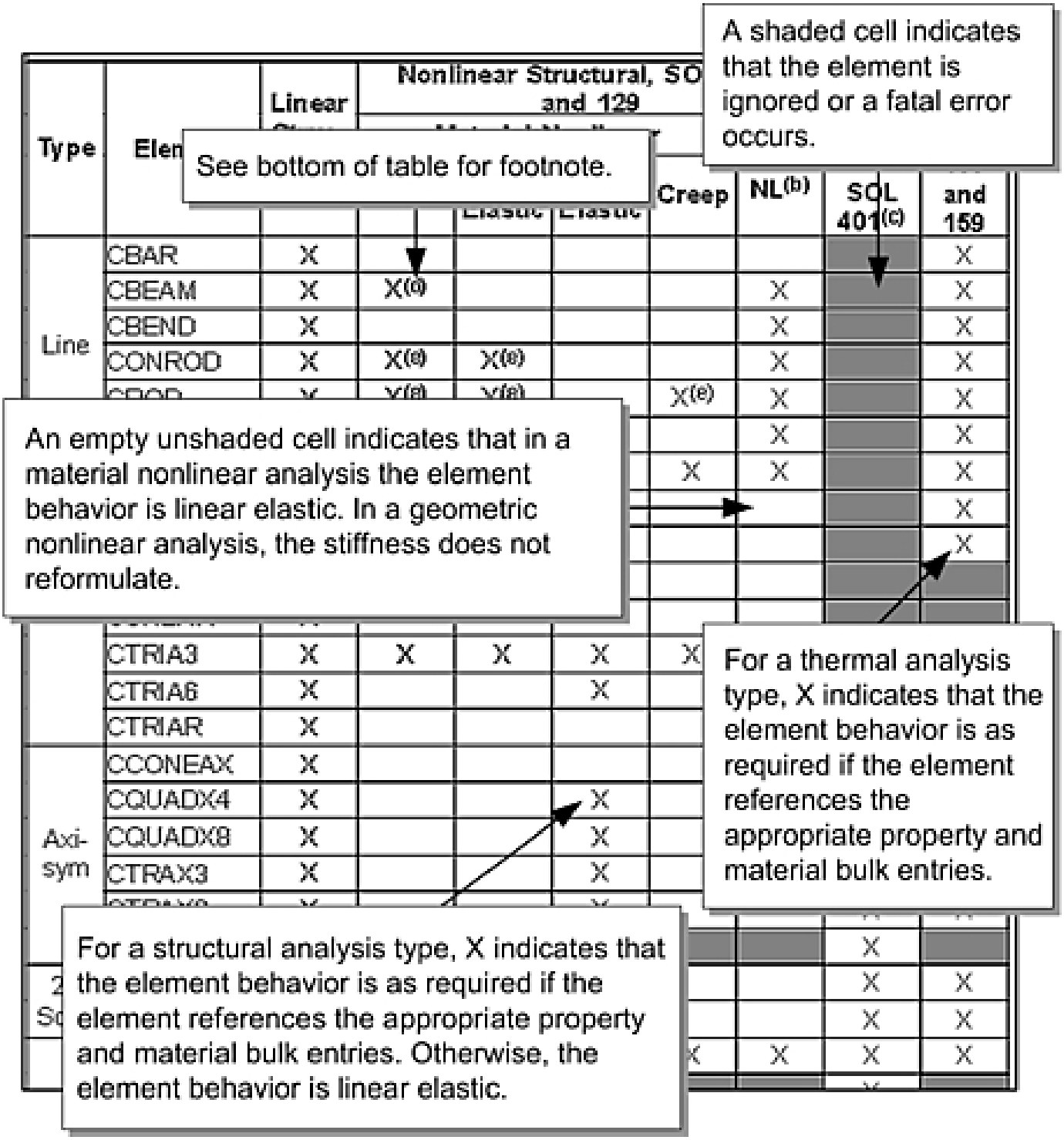
Key to Element Applicability Table



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Element Applicability Table

Table 9-23: Element Applicability for Select Elements in Solutions 101–401

Type	Element	Linear Structural SOLs[a]	Nonlinear Structural, SOLs 106 and 129					Multi-step Structural SOL 401[c]	Heat Trans SOLs 153 and 159
			Material Nonlinear				Geom NL[b]		
			Plastic	NL Elastic	Hyper-Elastic	Creep			
Line	CBAR	X						X	X
	CBEAM	X	X[d]				X	X	X
	CBEND	X					X		X
	CONROD	X	X[e]	X[e]			X		X
	CROD	X	X[e]	X[e]		X[e]	X	X	X
	CTUBE	X	X[e]	X[e]			X		X
Shell	CQUAD4	X	X	X	X	X	X	X[g]	X
	CQUAD8	X			X			X	X
	CQUADR	X						X	X
	CRAC2D	X							
	CSHEAR	X						X	
	CTRIA3	X	X	X	X	X	X	X[g]	X
	CTRIA6	X			X			X	X
	CTRIAR	X						X	X
Axi-sym	CQUADX4	X			X			X	X
	CQUADX8	X			X			X	X
	CTRAX3	X			X			X	X
	CTRAX6	X			X			X	X
	CCHOCKi							X	
2D Solid	CPLSTNi	X[f]						X	X
	CPLSTSi	X[f]						X	X
3D Solid	CHEXA	X	X	X	X	X	X	X	X
	CHEXCZ							X	

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Type	Element	Linear Structural SOLs[a]	Nonlinear Structural, SOLs 106 and 129					Multi-step Structural SOL 401[c]	Heat Trans SOLs 153 and 159
			Material Nonlinear				Geom NL[b]		
			Plastic	NL Elastic	Hyper-Elastic	Creep			
	CPENTA	X	X	X	X	X	X	X	X
	CPENTCZ							X	
	CPYRAM	X	X	X		X	X	X	X
	CRAC3D	X							
	CTETRA	X	X	X	X	X	X	X	X

[a] Includes SOLs 101, 103, 105, 107, 108, 109, 110, 111, 112, 114, 115, 116, 118, 144, 145, 146, 187, and 200.
[b] Small strain only. For large strain, refer to the hyperelasticity column.
[c] SOL 401 supports linear, plasticity, creep, and geometric nonlinear structural analysis.
[d] Elastic-perfectly plastic material behavior used to model the ends of beams as plastic hinges.
[e] Nonlinear capability for axial deflections only.
[f] Not supported for use in SOLs 114, 115, 116, 118, 144, 145, and 146.
[g] The software automatically converts CQUAD4 and CTRIA3 into CQUADR and CTRIAR, respectively.

For a comprehensive listing of elements supported for use in solutions 401 and 402, in the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*, see Element Overview.

For a comprehensive listing of elements supported for use in solution 414, in the *Simcenter Nastran Rotor Dynamics User's Guide*, see Supported bulk entries.

Related Topics

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Element-Property-Material Cross-Reference Tables

Table 9-24: Element-Property-Material Cross-Reference Tables

Element Type	Elements	Table
Line	CBAR, CBEAM, CBEND, CONROD, CROD, CTUBE	Table 9-25
Shell	CQUAD4, CQUAD8, CQUADR, CRAC2D, CSHEAR, CTRIA3, CTRIA6, CTRIAR	Table 9-26
Axisymmetric	CCHOCK3, CCHOCK4, CCHOCK6, CCHOCK8, CQUADX4, CQUADX8, CTRAX3, CTRAX6	Table 9-27
2D Solid	CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8, CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8	Table 9-28
3D Solid	CHEXA, CHEXCZ, CPENTA, CPENTCZ, CPYRAM, CRAC3D, CTETRA	Table 9-29
Special Purpose	CAABSF, CAERO1, CAERO2, CAERO3, CAERO4, CAERO5, CBEAR, CBEAR2, CBUSH, CBUSH1D, CDAMP1, CDAMP2, CDAMP3, CDAMP4, CDAMP5, CDUM1, CDUM2, CDUM3, CDUM4, CDUM5, CDUM6, CDUM7, CDUM8, CDUM9, CELAS1, CELAS2, CELAS3, CELAS4, CFAST, CGAP, CHBDYE, CHBDYG, CHBDYP, CMASS1, CMASS2, CMASS3, CMASS4, CONM1, CONM2, CONV, CONVM, CVISC, CWELD, RBAR, RBE1, RBE2, RBE3	Table 9-30

Table 9-25: Line Elements: Supplemental Materials by Primary Material, Property, and Element

Element	Property	Primary Material	Supplemental Material												
			MATi												CREEP
			CRP	DMG	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	
CBAR	PBAR	MAT1					X								
		MAT4								X					
		MAT5									X				
	PBARL	MAT1	X(1)			X(1)	X								
		MAT4								X					
		MAT5									X				
CBEAM	PBEAM	MAT1				X(2)	X								
		MAT4								X					
		MAT5									X				
	PBEAML	MAT1	X(1)			X(1)	X								
		MAT4								X					
		MAT5									X				
	PBCOMP	MAT1				X(2)	X								
		MAT4								X					
		MAT5									X				
CBEND	PBEND	MAT1					X								
		MAT4								X					

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Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DM G	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
		MAT5									X				
CONROD	N/A	MAT1				X(3)	X								
		MAT4								X					
		MAT5									X				
CROD	PROD	MAT1				X(3)	X								X(2)
		MAT4								X					
		MAT5									X				
CTUBE	PTUBE	MAT1				X(3)	X								
		MAT4								X					
		MAT5									X				

(0) Valid for SOL 401.

(1) For material nonlinear analysis, elastic-perfectly plastic material behavior only. Strain hardening is not supported. The ends of the element are modeled as plastic hinges.

(2) Nonlinear capability for axial deflections only.

For information on MATSR, MUCRP, and MUMAT material bulk entries, see the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*.

Table 9-26: Shell Elements: Supplemental Materials by Primary Material, Property, and Element

Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DM G	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
CQUAD4 CTRIA3	PSHELL	MAT1				X	X								X
		MAT2				X		X							X
		MAT4								X					
		MAT5									X				
		MAT8				X						X			X
	PCOMP	MAT1					X								
		MAT2						X							
		MAT8										X			
	PCOMPG	MAT1					X								
		MAT2						X							
		MAT8										X			
	PCOMPG1	MAT1					X								

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Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DMG	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
		MAT9											X		
		MAT11												X	
	PLPLANE	MATHP													
	N/A	MFLUID													
CQUAD8 CTRIA6	PSHELL	MAT1					X								
		MAT2						X							
		MAT4								X					
		MAT5									X				
		MAT8										X			
	PCOMP	MAT1					X								
		MAT2						X							
		MAT8										X			
	PCOMPG	MAT1					X								
		MAT2						X							
		MAT8										X			
	PCOMPG1	MAT1					X								
		MAT9											X		
		MAT11												X	
	PLPLANE	MATHP													
CQUADR CTRIAR	PSHELL	MAT1					X								
		MAT2						X							
		MAT4								X					
		MAT5									X				
		MAT8										X			
	PCOMP	MAT1					X								
		MAT2						X							
		MAT8										X			
	PCOMPG	MAT1					X								
		MAT2						X							
		MAT8										X			
	PCOMPG1	MAT1					X								
		MAT9											X		
		MAT11												X	

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Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DM G	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
CRAC2D	PRAC2D	MAT1					X								
		MAT2						X							
		MAT8										X			
CSHEAR	PSHEAR	MAT1													
For information on MATSR, MUCRP, and MUMAT material bulk entries, see the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*.															

Table 9-27: Axisymmetric Elements: Supplemental Materials by Primary Material, Property, and Element

Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DM G	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
CCHOCK3 CCHOCK4 CCHOCK6 CCHOCK8	PCHOCK	MAT1	X			X	X								X
		MAT11	X			X								X	X
CQUADX4 CQUADX8 CTRAX3 CTRAX6	PSOLID	MAT1	X				X								
		MAT3							X						
		MAT4								X					
		MAT5									X				
		MAT9	X										X		
		MAT11	X											X	
	PLSOLID	MATHP													
For information on MATSR, MUCRP, and MUMAT material bulk entries, see the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*.															

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Table 9-28: 2D Solid Elements: Supplemental Materials by Primary Material, Property, and Element

Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DMG	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
CPLSTN3 CPLSTN4 CPLSTN6 CPLSTN8	PPLANE	MAT1	X				X								
		MAT3							X						
		MAT4								X					
		MAT5									X				
		MAT11	X											X	
	PGPLSN	MAT1	X				X								
		MAT11	X											X	
CPLSTS3 CPLSTS4 CPLSTS6 CPLSTS8	PPLANE	MAT1	X				X								
		MAT3							X						
		MAT4								X					
		MAT5									X				
		MAT11	X											X	
For information on MATSR, MUCRP, and MUMAT material bulk entries, see the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*.															

Table 9-29: 3D Solid Elements: Supplemental Materials by Primary Material, Property, and Element

Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DMG	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
CHEXA CPENTA	PSOLID	MAT1	X			X	X								X
		MAT4								X					
		MAT5									X				
		MAT9	X			X							X		X
		MAT10													
		MAT11	X			X								X	X
		MATPOR													
	PLSOLID	MATHP													
	PCOMPS	MAT1	X	X	X		X								
		MAT9	X		X								X		
		MAT11	X	X	X									X	
CHEXCZ CPENTCZ	PSOLCZ	MAT1					X								
		MAT11												X	

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Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DMG	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
		MATCZ													
CPYRAM	PSOLID	MAT1	X			X	X								X
		MAT4								X					
		MAT5									X				
		MAT9	X			X							X		X
		MAT10													
		MAT11	X			X								X	X
		MATPOR													
CRAC3D	PRAC3D	MAT1					X								
		MAT9											X		
CTETRA	PSOLID	MAT1	X			X	X								X
		MAT4								X					
		MAT5									X				
		MAT9	X			X							X		X
		MAT10													
		MAT11	X			X								X	X
		MATPOR													
	PLSOLID	MATHP													

For information on MATSR, MUCRP, and MUMAT material bulk entries, see the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*.

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Table 9-30: Special Purpose Elements: Supplemental Materials by Primary Material, Property, and Element

Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DMG	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
CAABSF	PAABSF	N/A													
CAEROi [1]	PAEROi [1]	N/A													
CBEAR	PBEAR	N/A													
CBEAR2	PBEAR2	N/A													
CBUSH	PBUSH [2]	N/A													
CBUSH1D	PBUSH1D	N/A													
CBUSH2	PBUSH2	N/A													
CDAMP1	PDAMP	N/A													
CDAMP2	N/A	N/A													
CDAMP3	PDAMP	N/A													
CDAMP4	N/A	N/A													
CDAMP5	PDAMP5	MAT4								X					
		MAT5									X				
CDUMi [3]	PDUMi [3]	[4]													
CELAS1	PELAS	N/A													
CELAS2	N/A	N/A													
CELAS3	PELAS	N/A													
CELAS4	N/A	N/A													
CFAST	PFAST	N/A													
CGAP	PGAP	N/A													
CHBDYE	N/A	N/A													
CHBDYG	N/A	N/A													
CHBDYP	PHBDY	N/A													
CLINK	N/A	N/A													
CMASS1	PMASS	N/A													
CMASS2	N/A	N/A													
CMASS3	PMASS	N/A													
CMASS4	N/A	N/A													
CONMi [5]	N/A	N/A													
CONV	PCONV	MAT4								X					
CONVM	PCONVM	MAT4								X					
CVISC	PVISC	N/A													

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Element	Property	Primary Material	Supplemental Material												
			MATi												
			CRP	DMG	FT	S1	T1	T2	T3	T4	T5	T8	T9	T11	CREEP
CWELD	PWELD	MAT1													
FOU3	N/A	N/A													
RBAR	N/A	N/A													
RBEi [6]	N/A	N/A													

[1] i = 1 through 5

[2] Use a PBUSHT entry with the same PID as a PBUSH entry to define frequency dependent or stress dependent properties.

[3] i = 1 through 9

[4] Per user definition.

[5] i = 1 and 2

[6] i = 1, 2, and 3

For information on MATSR, MUCRP, and MUMAT material bulk entries, see the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*.

Related Topics

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$

Comment

Used to insert comments into the input file. Comment statements may appear anywhere within the input file.

Format:

$ followed by any characters out to column 80.

Example:

```
$ TEST FIXTURE-THIRD MODE
```

Remarks:

1. Comments are ignored by the program.
2. Comments will appear only in the unsorted echo of the Bulk Data.

Related Topics

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/

Delete

Removes entries on restart.

Format:

1	2	3	4	5	6	7	8	9	10
/	K1	K2							

Example:

/	4								

Field	Contents
K1	Sorted sequence number of first entry in sequence to be removed. (Integer > 0)
K2	Sorted sequence number of last entry in sequence to be removed. (Integer > 0; Default = K1)

Remarks:

1. This entry causes Bulk Data entries having sort sequence numbers K1 through K2 to be removed from the Bulk Data. The sort sequence numbers appear in the output of the previous run under the sorted Bulk Data echo.
2. If K2 is blank, only entry K1 is removed from the Bulk Data.
3. If the current execution is not a restart, the entries marked for deletion are ignored.
4. This entry can only be used in the rigid formats and the structured solution sequences. See “Executive Control Statements”, under the SOL statement.
5. K2 may be specified as larger than the actual sequence number of the last entry. This is convenient when deleting entries to the end of the Bulk Data Section.

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/

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10. Bulk Entries A—B

ACADAPT

Order Adaptation Rule for Acoustics FEM Adaptive Order Solution (FEMAO) Method

Invokes the FEMAO method for both uncoupled acoustic and coupled vibro-acoustic analyses.

Format:

1	2	3	4	5	6	7	8	9	10
ACADAPT	RULE								

Example:

ACADAPT	COARSE								

Fields:

Field	Contents
RULE	Type of adaptation rule. (Character; "COARSE", "STANDARD", or "FINE"; Default = "STANDARD")

Remarks:

1. ACADAPT is supported in SOL 108 and SOL 111 with FEMAO for both uncoupled acoustics and coupled vibro-acoustic analyses.

Related Topics

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ACCEL

Define Static Acceleration Loads

Defines static acceleration loads, which may vary over a region of the structural model. The load variation is based on the tabular input defined on this Bulk Data entry.

Format:

1	2	3	4	5	6	7	8	9	10
ACCEL	SID	CID	N1	N2	N3	DIR			
	LOC1	VAL1	LOC2	VAL2	Continues in groups of 2				

Example:

ACCEL	100	2	0.0	1.0	0.0	X			
	0.0	1.0	33.1	2.0					

Fields:

Field	Contents
SID	Load set identification number. (Integer>0)
CID	Coordinate system identification number. (Integer>0: Default=0)
Ni	Components of the acceleration vector measured in coordinate system CID. (Real; at least one Ni ≠ 0.0)
DIR	Component direction of acceleration variation. (Character; one of X, Y or Z)
LOCi	Location along direction DIR in coordinate system CID for specification of a load scale factor. (Real)
VALi	The load scale factor associated with location LOCi. (Real)

Remarks:

1. For all grids of the model, the acceleration vector is defined by **a** = *VAL* · **N** where **N** is the vector with components N1, N2, and N3. The magnitude of **a** is equal to *VAL* times the magnitude of **N**. The scale factor *VAL* for each grid is found by linearly interpolating the DIR coordinate of the grid between table values *LOCi*/*VALi*. If the GRID point coordinate in coordinate system CID is outside the range of the table, *VAL* is determined from the closer of *VAL1* or *VALn*. (See the following figure.)
2. This type of acceleration load may be combined with other loads such as FORCE, MOMENT, GRAV and ACCEL1 loads only by specification on a LOAD bulk data entry. That is, the SID on an ACCEL entry may not be the same as that on any other load entry.

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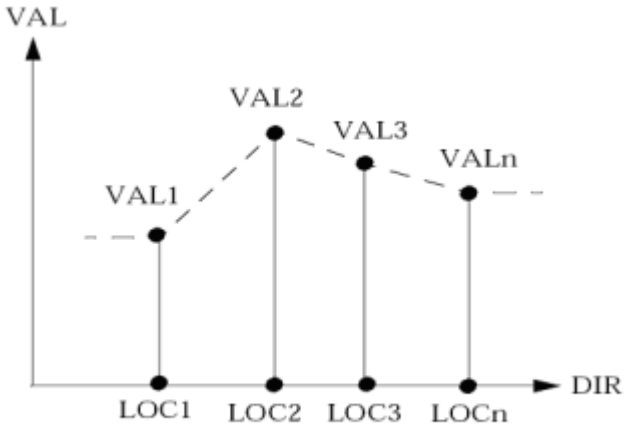
3. This acceleration load does not include effects due to mass on scalar points.
4. A blank CID entry or a CID of zero references the basic coordinate system.
5. The DIR field must contain one of the characters X, Y, or Z. The DIR direction defines the direction of acceleration load variation along direction 1, 2, or 3 respectively of coordinate system CID.
6. If a single pair of {LOCi, VALi} data is specified, the acceleration load is applied as a constant with no variation by location.
7. When applying a load with a GRAV, ACCEL, or ACCEL1 entry to axisymmetric elements, any component in the radial direction is treated as a radial acceleration load.
8. When a cylindrical or spherical system is selected in the CID field, the software applies the load at each grid point using a local triad computed relative to that location. Therefore, the load direction varies from one grid point to the next. **Figure 12-54** on the CORD1C entry and **Figure 12-56** on the CORD1S entry illustrates how the local triad is computed at a grid point with cylindrical and spherical systems, respectively. Note that before version 2506, the software treated a cylindrical or spherical system selected in the CID field as a Cartesian system, and the resulting load direction was constant at all grid points.
9. The ACCEL bulk entry is not supported for non-partitioned superelements.



Definition of Load Scale Factor vs Location

Related Topics

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ACCEL1

Define Static Acceleration Loads

Defines static acceleration loads at individual grid points.

Format:

1	2	3	4	5	6	7	8	9	10
ACCEL1	SID	CID	A	N1	N2	N3			
	GRIDID1	GRIDID2	etc						

Example:

ACCEL1	100	2	100.0	0.0	1.0	0.0			
	1	2	3	4	5	6	7	8	
	9	10	THRU	100	BY	5	200	250	

Fields:

Field	Contents
SID	Load set identification number. (Integer>0)
CID	Coordinate system identification number. (Integer>0: Default=0)
A	Acceleration vector scale factor. (Real)
Ni	Components of the acceleration vector measured in coordinate system CID. (Real; at least one Ni ≠ 0.0)
GRIDIDi	List of one or more GRID point identification numbers. Key words “THRU” and “BY” can be used to assist the listing. (Integer>0)

Remarks:

1. The acceleration vector is defined by $\mathbf{a} = A \cdot \mathbf{N}$ where **N** is the vector with components N1, N2, and N3. The magnitude of **a** is equal to *A* times the magnitude of **N**.
2. The SID on an ACCEL1 entry can be the same as the SID defined on other ACCEL1 entries. Although, the SID on an ACCEL1 entry cannot be the same as the SID defined on other types of load entries. For example, you cannot use the same SID on ACCEL1 and FORCE entries. The LOAD bulk entry is required to combine these loads.
3. This acceleration load does not include effects due to mass on scalar points.
4. A blank CID entry or a CID of zero references the basic coordinate system.

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5. The ACCEL1 card must contain at least one GRIDID.
6. The ACCEL1 bulk entry is not supported for non-partitioned superelements.
7. When applying a load with a GRAV, ACCEL, or ACCEL1 entry to axisymmetric elements, any component in the radial direction is treated as a radial acceleration load.
8. When a cylindrical or spherical system is selected in the CID field, the software applies the load at each grid point using a local triad computed relative to that location. Therefore, the load direction varies from one grid point to the next. **Figure 12-54** on the CORD1C entry and **Figure 12-56** on the CORD1S entry illustrates how the local triad is computed at a grid point with cylindrical and spherical systems, respectively. Note that before version 2506, the software treated a cylindrical or spherical system selected in the CID field as a Cartesian system, and the resulting load direction was constant at all grid points.

Related Topics

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ACCONV

Convected Acoustic Flow

Defines convected acoustic flow.

Format:

1	2	3	4	5	6	7	8	9	10
ACCONV	ID	UNIT	TID						

Example:

ACCONV	999	89	200						

Fields:

Field	Contents
ID	Identification number referenced by the ACCONV case control command. See Remark 2. (Integer>0; No default)
UNIT	Unit number of an SCH5 file that contains node-based flow velocity tables. See Remark 3. (Integer>0; No default)
TID	Identification number of a table in the SCH5 file referenced by the UNIT field that contains node-based flow velocity data (x,y,z). (Integer>0; No default)

Remarks:

1. The ACCONV entry is only supported in a SOL 108 FEMAO Acoustic analysis.
2. ACCONV should not be specified in an acoustic analysis with incident sources.
3. The unit number you enter in the UNIT field must be defined on an ASSIGN statement in the File Management System (FMS) of your input file.
4. Multiple ACCONV bulk entries with the same ID are supported. Each ACCONV entry can reference different flow velocity tables (TID) in either the same SCH5 file or different SCH5 files. A unique UNIT number must be used for each SCH5 file if multiple files are referenced.

Related Topics

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ACCONVD

Default Convected Acoustic Flow

Defines the default convected acoustic flow.

Format:

1	2	3	4	5	6	7	8	9	10
ACCONVD	ID	VX	VY	VZ					

Example:

ACCONVD	999	0.0	-50.0	0.0					

Fields:

Field	Contents
ID	Identification number referenced by the ACCONV case control command. See Remark 2. (Integer>0; No default)
VX, VY, VZ	Default flow velocities. (Real; Default=0.0)

Remarks:

1. The ACCONVD entry is only supported in a SOL 108 FEMAO Acoustic analysis.
2. ACCONVD should not be specified in an acoustic analysis with incident sources.
3. Only one unique ACCONVD bulk entry with a specific ID is allowed.

Related Topics

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ACDUCT

Duct Mode Definition

Defines duct modes for a frequency response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
ACDUCT	SID	PID	WTYPE	MTYPE	FORM				
	MODX1	MODY1	X1	Y1					

Examples:

ACDUCT	200	300			PHASE				
	1	2	0.1	60.0					

Fields:

Field	Contents
SID	Load set identification number reference by either the ALOAD case command or the ALOAD bulk entry. (Integer > 0; No default)
PID	Property ID reference by the PID field PACDUCT bulk entry. (Integer > 0; No default)
WTYPE	Type of duct modes. (Character; Default=SPECIFIC) If WTYPE=SPECIFIC, specific duct modes are defined. If WTYPE=DISTRIB, distributed duct modes are defined.
MTYPE	Defines the units of the duct mode amplitude. (Character; Default=PRESS) If MTYPE=PRESS, duct mode amplitude is a pressure unit. If MTYPE=INTENS, duct mode amplitude is an intensity unit. If MTYPE=POWER, duct mode amplitude is a power unit.
FORM	Defines the complex data format for the duct mode. (Character; Default=REAL) If FORM=REAL, the data is in rectangular format (real and imaginary). If FORM=PHASE, the data is in polar format (magnitude and phase).

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Field	Contents
MODX1	Mode number relative to the first section coordinate. (Integer) The section coordinates depend on how GTYPE is defined on the associated PACDUCT bulk entry. IF GTYPE=RECTANG, MODX1 is relative to the X direction. IF GTYPE=CIRCULAR, MODX1 is relative to the Azimuthal direction. IF GTYPE=ANNULAR, MODX1 is relative to the Azimuthal direction.
MODY1	Mode number relative to the second section coordinate. (Integer) The section coordinates depend on how GTYPE is defined on the associated PACDUCT bulk entry. IF GTYPE=RECTANG, MODY1 is relative to the Y direction. IF GTYPE=CIRCULAR, MODY1 is relative to the radial direction. IF GTYPE=ANNULAR, MODY1 is relative to the radial direction.
X1, Y1	Defines the complex data parts for a duct mode. (Real or Integer; Default=0.0) If FORM=REAL, X1 defines the real part and Y1 defines the imaginary part. If FORM=PHASE, X1 defines the magnitude and Y1 defines the phase. For X1 or Y1 and either FORM value, if an integer is defined, it is the identification number of a TABLEDi bulk entry which defines values as a function of frequency. If a real value is defined for X1 or Y1, the value remains constant as the solution frequency changes.

Remarks:

1. SID is referenced in a subcase by the ALOAD case control command, or by the ALOAD bulk entry when combining multiple loads.
2. MODX1, MODY1 only apply to WTYPE= SPECIFIC.
3. The ACDUCT entry is only supported in SOL 108 uncoupled acoustic and coupled vibro-acoustic solutions and requires the FEM Adaptive Order (FEMAO) method.

Related Topics

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ACFAN

Fan Noise Definition

Defines fan noise for a frequency response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
ACFAN	SID	PID	LUNF	LDID	LCID	WINDOW	CID	ROTATION	

Examples:

ACFAN	200	111	25	3	2	RECTANG	2		

Fields:

Field	Contents
SID	Load set identification number referenced by either the ALOAD case control command, or the ALOAD bulk entry. (Integer > 0; No default)
PID	Property identification number referenced by the PID field on the PACFAN bulk entry. (Integer > 0; No default)
LUNF	Logical unit number for an HDF5 file containing frequency-dependent force excitations. See Remark 1. (Integer>0; No default)
LDID	Identification number of a load definition in the HDF5 file. See Remark 1. (Integer>0; No default)
LCID	Load case identification number of a specific load that is included in the loading definition referenced by LDID. See Remark 1. (Integer>0; No default)
WINDOW	Window method for Fast Fourier transformation. (Character; Default = "RECTANG") ="HANNING" for general purposes. ="RECTANG" when the input signal is periodic within the chosen time segment, for example, when you are analyzing special sinusoids whose frequencies coincide with obtained frequencies of the analysis.
CID	Coordinate system identification number for the location and orientation of the fan. See Remark 2. (Integer≥-1; Default = -1)
ROTATION	Fan rotation direction. (Integer≥0 or "BLANK"; Default = "BLANK") BLANK = 0

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Field	Contents
	0 = Rotation from X to Y 1 = Rotation from Y to X

Remarks:

1. The logical unit number LUNF must reference an existing ASSIGN SC_H5 File Management System (FMS) statement. The transient force excitation definition referenced by LDID must be contained in this file including the degrees of freedom and time-steps at which the excitations are applied.
2. When CID= -1 (default), the coordinate system defined with the fan data in the HDF5 file is used to locate and orient the fan. When CID= 0, the basic coordinate system is used to locate and orient the fan. For other CID values, the referenced CORDi bulk entry is used to locate and orient the fan.

Related Topics

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ACMODL

Fluid-Structure Interface Modeling Parameters

Defines modeling parameters for the interface between the fluid and the structure.

Format:

1	2	3	4	5	6	7	8	9	10
ACMODL		INFOR	FSET	SSET	NORMAL		OVLPANG	SRCHUNIT	
	INTOL	AREAOP			CTYPE				

Example:

ACMODL					0.25		15.0		
	0.10				WEAK				

Fields:

Field	Contents
INFOR	Meaning of the SID entered on the FSET and SSET fields. (Character = ELEMENTS, PID, or SET3; For default behavior, see Remark 3) See Remark 4.
FSET	SID of a SET1 or SET3 bulk entry that defines the fluid elements to include in the coupling evaluation. A negative sign in front of the SID instructs the software to include all fluid elements minus those associated with the SET1 or SET3 bulk entry in the coupling evaluation. (Integer ≠ 0; Default = Include all fluid elements in the coupling evaluation) See Remarks 4 and 5.
SSET	SID of a SET1 or SET3 bulk entry that defines the structural elements for the coupling evaluation. A negative sign in front of the SID instructs the software to include all structural elements minus those associated with the SET1 or SET3 bulk entry in the coupling evaluation. (Integer ≠ 0; Default = Include all structural elements in the coupling evaluation) See Remarks 4 and and 5.
NORMAL	Value used to determine the outward normal search distance. If the fluid and structural element free faces are within the search distance of one another and they satisfy the angular tolerance, they are coupled. (Real > 0.0; Default = 0.5) See Remarks 6 and 7.
OVLPANG	Angular tolerance in degrees used to decide whether a fluid free face and a structural face can be considered as overlapping. If the angle between the normals of the fluid and structural faces exceeds this value, they are not coupled. (0.0 < Real ≤ 90.0; Default = 60.0)
SRCHUNIT	Specifies how the values in the NORMAL and INTOL fields are used. (Character = ABS or REL; Default = REL) See Remark 6.

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Field	Contents
INTOL	Value used to determine the inward normal search distance. If the fluid and structural element free faces are within the search distance of one another, they are coupled. (Real > 0.0; Default = 0.2) See Remarks 6 and 7.
AREAOP	Alternative fluid-structure coupling method selection. (Integer = 0 or 1; Default = 0) See Remark 8. 0 for the recommended method. 1 for the RBE3 method.
CTYPE	Fluid-structure coupling type (only supported by new acoustics method introduced in NX Nastran 11). (Character = STRONG or WEAK; Default = STRONG) See Remark 9. STRONG for two-way coupling. IWEAK for one-way coupling, where the effect of the fluid on the structure is assumed to be negligible.

Remarks:

1. Only one ACMODL entry is allowed. If this entry is not present, defaults will be used.
2. The ACMODL entry is supported in solutions 103, 107-112, and 200.
3. If the INFOR field is blank and the FSET and SSET fields are also blank, all elements are included in the coupling evaluation. However, if the FSET and SSET fields are specified, you must specify the INFOR field.
4. The INFOR field setting depends on whether SET1 or SET3 bulk entries are referenced in the FSET and SSET fields.

   For SET1 bulk entries:

   - Enter ELEMENTS if the SET1 bulk entries list element IDs.
   - Enter PID if the SET1 bulk entries list property IDs.

   For SET3 bulk entries, enter SET3. The TYPE field setting on the SET3 bulk entries determines whether the SET3 bulk entries list element IDs, property IDs, or grid IDs.
5. When SET1 or SET3 bulk entries list property IDs, the elements that are selected include all of the elements that reference one of the listed property IDs. When SET3 bulk entries list grid IDs, the elements that are selected include all of the elements whose connectivity includes at least one of the listed grid IDs.

   The PSOLID property ID is supported for selecting fluid elements. The PSOLID, PSHELL, PCOMP, and PCOMPG property IDs are supported for selecting structural elements. Solid composite elements using the PCOMPS property cannot be coupled to the fluid.

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6. How the software uses the values in the NORMAL and INTOL fields depend on the SRCHUNIT field setting as follows:

   - ABS means use the values in the NORMAL and INTOL fields as the physical distances (in the unit of distance for the model) to search as follows:

     $$L_{outward} = \alpha_{normal}$$

     $$L_{inward} = \alpha_{intol}$$

     where:

$L_{outward}$	Search distance in the outward direction
L_{inward}	Search distance in the inward direction
α_{normal}	Value in the NORMAL field
α_{intol}	Value in the INTOL field

   - REL means use the values in the NORMAL and INTOL fields as scaling factors that are applied to the maximum edge length of a fluid element free face such that the physical distance (in the unit of distance for the model) to search is as follows:

     $$L_{outward} = \alpha_{normal} L_{max}$$

     $$L_{inward} = \alpha_{intol} L_{max}$$

     where:

$L_{outward}$	Search distance in the outward direction
L_{inward}	Search distance in the inward direction
α_{normal}	Value in the NORMAL field
α_{intol}	Value in the INTOL field
L_{max}	Maximum edge length of the fluid element free face

   The following figure shows the outward and inward normal search distances for a quadrilateral free face of a fluid element.

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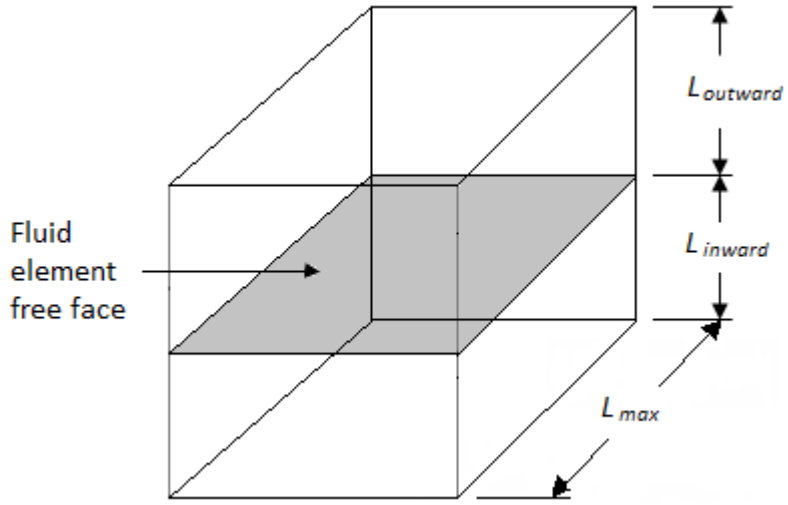

7. For each fluid element free face, the search for a structural free face with which to couple continues until either a structural free face is identified or the search distance is reached. For additional information, see Two-way coupling in the *Simcenter Nastran Acoustics User's Guide*.

8. AREAOP = 0 selects the recommended coupling method.

   AREAOP = 1 selects an alternate method that applies an area correction and removes parallel disconnected faces from the coupling.

9. For a two-way coupled solution (CTYPE = STRONG), the governing equation is as follows:

$$\begin{bmatrix} M_s & 0 \\ -A^T & M_f \end{bmatrix} \begin{Bmatrix} \ddot{u}_s \\ \ddot{p} \end{Bmatrix} + \begin{bmatrix} B_s & 0 \\ 0 & B_f \end{bmatrix} \begin{Bmatrix} \dot{u}_s \\ \dot{p} \end{Bmatrix} + \begin{bmatrix} K_s & A \\ 0 & K_f \end{bmatrix} \begin{Bmatrix} u_s \\ p \end{Bmatrix} = \begin{Bmatrix} P_s \\ P_f \end{Bmatrix}$$

   where the subscript *s* denotes structure and *f* denotes fluid.

   For a one-way solution (CTYPE = WEAK), the fluid-structure coupling matrix, *A*, is omitted from the stiffness formulation and the governing equation simplifies to:

$$\begin{bmatrix} M_s & 0 \\ -A^T & M_f \end{bmatrix} \begin{Bmatrix} \ddot{u}_s \\ \ddot{p} \end{Bmatrix} + \begin{bmatrix} B_s & 0 \\ 0 & B_f \end{bmatrix} \begin{Bmatrix} \dot{u}_s \\ \dot{p} \end{Bmatrix} + \begin{bmatrix} K_s & 0 \\ 0 & K_f \end{bmatrix} \begin{Bmatrix} u_s \\ p \end{Bmatrix} = \begin{Bmatrix} P_s \\ P_f \end{Bmatrix}$$

   Because the fluid-structure coupling matrix is omitted from the stiffness formulation, the structural displacements, u_s are solved for directly from the following equation:

$$[M_s]\{\ddot{u}_s\} + [B_s]\{\dot{u}_s\} + [K_s]\{u_s\} = \{P_s\}$$

   The fluid pressure, *p*, is then solved for directly from the following equation:

$$[M_f]\{\ddot{p}\} + [B_f]\{\dot{p}\} + [K_f]\{p\} = \{P_f\} + [A]^T\{\ddot{u}_s\}$$

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Related Topics

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ACMODPE

Poro-Elastic interface

Defines coupling between a poro-elastic region, a fluid region, and a structural region.

Format:

1	2	3	4	5	6	7	8	9	10
ACMODPE		INFOR	FSET	SSET	PESET	NORMAL		OVLPANG	
	SCHUNIT	INTOL	AREAOP			CTYPE			

Example:

ACMODPE						0.25		15.0	
						WEAK			

Fields:

Field	Contents
INFOR	Defines the meaning of the SID entered on the FSET and SSET fields. (Character; ELEMENTS, PID or SET3) See Remark 3.
FSET	Selects the ID of a SET3 entry listing the fluid elements at the interface. (Integer or blank) See Remark 3. If FSET=0 is defined, no fluid elements are selected for the interface. If a negative sign is included in front of the ID, the corresponding fluid elements are excluded. If blank, all fluid elements are considered.
SSET	Selects the ID of a SET3 entry listing the structural elements at the interface. (Integer or blank) See Remark 3. If SSET=0 is defined, no structural elements are selected for the interface. If a negative sign is included in front of the ID, the corresponding structural elements are excluded. If blank, all structural elements are considered.

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Field	Contents
PESET	Selects the ID of a SET3 entry listing the poro-elastic elements at the interface. (Integer or blank) See Remark 3. If a negative sign is included in front of the ID, the corresponding poro-elastic elements are excluded. If blank, all poro-elastic elements are considered.
NORMAL	Outward normal search distance to detect fluid-structure interface. (Real > 0.0; Default = 0.5) See Remark 4. If SRCHUNIT = REL, then NORMAL is a ratio of the height of the fluid box in the outward normal direction to the fluid surface to the maximum edge length of the fluid free face. If SRCHUNIT = ABS, then NORMAL is the outward search distance in the model/absolute units.
OVLPANG	Angular tolerance in degrees used to decide whether a fluid free face and a structural face can be considered as overlapping. If the angle between the normals of the fluid and structural faces exceeds this value, they cannot be coupled. (0.0 < Real ≤ 90.0; Default = 60.0)
SCHUNIT	Search units. (Character; Default = REL) See Remark 4. If SRCHUNIT = ABS, then the model units are absolute. If SRCHUNIT = REL, then the relative model units are based on element size.
INTOL	Inward normal search distance to detect fluid-structure interface. (Real > 0.0; Default = 0.20) See Remark 4. If SRCHUNIT = REL, then INTOL is a ratio of the height of the fluid box in the inward normal direction to the fluid surface to the maximum edge length of the fluid free face. If SRCHUNIT = ABS, then INTOL is the inward search distance in the model/absolute units.
AREAOP	Alternative fluid-structure coupling method selection. (Integer; Default=0) See Remark 5. 0 = The recommended method is used. 1 = The RBE3 method is used.
CTYPE	Fluid-structure coupling type. (Character; Currently only STRONG is used) CTYPE = STRONG, two-way coupling is used.

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Remarks:

1. Only one ACMODPE entry is allowed. If this entry is not present, defaults will be used.
2. The ACMODPE entry is supported in the direct frequency response (SOL 108) and the modal frequency response (SOL 111) solutions when the FEM Adaptive Order (FEMAO) is requested. This entry also supports SOL 108 solutions that contain an acoustic transfer vector (ATV) or vibro-acoustic transfer vector (VATV).
3. The SET3 entry selected in the FSET, SSET, or PESET fields can list element IDs, physical property IDs, or grid IDs. The TYPE field on the SET3 entries defines how the software should interpret the items selected by the SET3 entries. If property IDs or grid IDs are selected, the software will determine the associated elements. The PSOLID property ID is supported for selecting fluid and poro-elastic elements. The PSOLID, PSHELL, PCOMP, and PCOMPG property IDs are supported for selecting structural elements. Solid composite elements using the PCOMPS property cannot be coupled to the fluid.
4. The fields NORMAL and INTOL are interpreted as follows:

   If SRCHUNIT = REL, then NORMAL is a ratio of the outward height of the bounding box to the maximum edge length of the fluid free face. That is, if L is the largest edge of the fluid free face, the height H of the bounding box used to search for structural faces will be NORMAL * L. INTOL applies similarly, but the inward direction.

   If SRCHUNIT = ABS, then NORMAL defines the outward height of the fluid bounding box in the model/absolute units. INTOL defines the inward height of the fluid bounding box in the model/absolute units.

   If SRCHUNIT = ABS and NORMAL or INTOL are blank, then SRCHUNIT will be reset to REL and their corresponding default value is used.
5. AREAOP = 0 selects the default, recommended Simcenter Nastran coupling option.

   AREAOP = 1 selects an alternate option, which applies an area correction and removes parallel disconnected faces from the coupling.

Related Topics

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ACNDUCT

Anechoic End Duct Definition

Defines an anechoic end duct for a frequency response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
ACNDUCT	ID	PID							

Examples:

ACNDUCT	200	300							

Fields:

Field	Contents
SID	The identification number of anechoic end duct. (Integer > 0; No default)
PID	Property ID reference to the PID field PACDUCT bulk entry. (Integer > 0; No default)

Remarks:

1. The ACNDUCT entry is only supported in SOL 108 uncoupled acoustic and coupled vibro-acoustic solutions and requires the FEM Adaptive Order (FEMAO) method. It can used with and without a duct modes boundary condition defined with the ACDUCT bulk entry.

Related Topics

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ACORDER

Order for Acoustics FEM Adaptive Order (FEMAO) Method

Defines the polynomial order for the FEMAO method.

Format:

1	2	3	4	5	6	7	8	9	10
ACORDER	ORDER	NUMBER							

Examples:

ACORDER	MINIMUM	2							

ACORDER	MAXIMUM	8							

Fields:

Field	Contents
ORDER	Type of adaptive order. (Character = "MINIMUM" or "MAXIMUM"). See Remark 3.
NUMBER	Adaptive order number. (1 ≤ Integer ≤ 10) If ORDER = "MINIMUM", the default value for NUMBER = 2. If ORDER = "MAXIMUM", the default value for NUMBER = 10.

Remarks:

1. ACORDER is supported in SOL 108 and SOL 111 with FEMAO for both uncoupled acoustics and coupled vibro-acoustic analyses.
2. The specified order is applied to the entire model.
3. If ACORDER has no arguments or does not exist, the software uses all of the following arguments:
   a. ORDER = "MINIMUM" and NUMBER = 2.
   b. ORDER = "MAXIMUM" and NUMBER = 10.

Related Topics

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ACPLNW

Acoustic Plane Wave Source

Defines an acoustic plane wave source.

Format:

1	2	3	4	5	6	7	8	9	10
ACPLNW	SID	FORM	A	TR/TM	TI/TP				
	CID1	X1	X2	X3	CID2	N1	N2	N3	

Examples:

ACPLNW	200	PHASE		0.5	20.				
	2	100.0	0.0	0.0	3	0.0	1.		

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
FORM	Format to define the plane wave amplitude. (Character; "REAL" or "PHASE"; Default = REAL) If FORM = REAL, define the plane wave amplitude in rectangular format (real and imaginary). If FORM = PHASE, define the plane wave amplitude in polar format (magnitude and phase).
A	Scale factor for TR, TI, or TM. (Real; Default = 1.0)
TR	Real part of plane wave amplitude. (Real or Integer; Default = 0.0) For a constant real amplitude, enter a real value. For a frequency-dependent amplitude, enter the ID of the TABLEDi bulk entry that contains the real amplitude as a function of frequency.
TM	Magnitude of plane wave amplitude. (Real or Integer; Default = 0.0) For a constant amplitude, enter a real value.

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Field	Contents
	For a frequency-dependent amplitude, enter the ID of the TABLEDi bulk entry that contains the magnitude as a function of frequency.
TI	Imaginary part of plane wave amplitude. (Real or Integer; Default = 0.0) For a constant imaginary amplitude, enter a real value. For a frequency-dependent amplitude, enter the ID of the TABLEDi bulk entry that contains the imaginary amplitude as a function of frequency.
TP	Phase angle in degrees of plane wave amplitude. (Real or Integer; Default = 0.0) For a constant phase angle, enter a real value. For a frequency-dependent phase angle, enter the ID of the TABLEDi bulk entry that contains the phase angle as a function of frequency.
CID1	Identification number of coordinate system in which the location of plane wave is defined. (Integer ≥ 0 or blank; Default = blank). See Remark 3.
X1, X2, X3	Location of plane wave in coordinate system CID1. (Real; Default= 0.0)
CID2	Identification number of coordinate system in which the direction of plane wave is defined. (Integer ≥ 0 or blank; Default = blank). See Remark 3.
N1, N2, N3	Direction of plane wave in coordinate system CID2. (Real; Default= 0.0)

Remarks:

1. ACPLNW is supported by the FEM Adaptive Order (FEMAO) method.
2. The SID must be unique with respect to ALOAD.
3. A CID1 or CID2 of zero or blank (the default) references the basic coordinate system.
4. N1, N2, and N3 cannot be all equal to zero.

Related Topics

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ACPNVEL

Acoustic Panel Normal Velocity

Defines panel normal velocity on fluid free surfaces.

Format:

1	2	3	4	5	6	7	8	9	10
ACPNVEL	SID	FORM	A	TR/TM	TI/TP			BID	

Examples:

ACPNVEL	200	REAL		0.2	0.5			300	

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
FORM	Format to define the panel normal velocity. (Character; "REAL" or "PHASE"; Default = REAL) If FORM = REAL, define the panel normal velocity in rectangular format (real and imaginary). If FORM = PHASE, define the panel normal velocity in polar format (magnitude and phase).
A	Scale factor for TR, TI, or TM. (Real; Default = 1.0)
TR	Real part of velocity. (Real or Integer; Default = 0.0) For a constant velocity, enter a real value. For a frequency-dependent velocity, enter the ID of the TABLEDi bulk entry that contains the velocity as a function of frequency.
TM	Magnitude of velocity. (Real or Integer; Default = 0.0) For a constant velocity, enter a real value. For a frequency-dependent velocity, enter the ID of the TABLEDi bulk entry that contains the velocity as a function of frequency.
TI	Imaginary part of velocity. (Real or Integer; Default = 0.0)

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Field	Contents
	For a constant velocity, enter a real value. For a frequency-dependent velocity, enter the ID of the TABLEDi bulk entry that contains the velocity as a function of frequency.
TP	Phase angle in degrees of velocity. (Real or Integer; Default = 0.0) For a constant phase angle, enter a real value. For a frequency-dependent phase angle, enter the ID of the TABLEDi bulk entry that contains the phase angle as a function of frequency.
BID	Identification number of BSURFS bulk entry that normal velocity is applied to. (Integer > 0; No default)

Remarks:

1. ACPNVEL is supported by the FEM Adaptive Order (FEMAO) method.
2. The SID must be unique with respect to ALOAD.

Related Topics

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ACPOLE1

Acoustic Monopole Source

Defines an acoustic monopole source

Format:

1	2	3	4	5	6	7	8	9	10
ACPOLE1	SID	TYPE	FORM	A	TR/TM	TI/TP			
	CID	X1	X2	X3					

Examples:

ACPOLE1	200	POWER	PHASE	2.0	300	60.0			

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
TYPE	Monopole source type. (Character; "AMP" or "POWER"; Default = AMP) If TYPE = AMP, define the monopole source in terms of source amplitude. If TYPE = POWER, define the monopole source in terms of power. If TYPE = VDISP, define the monopole source in terms of volume displacement. If TYPE = VVEL, define the monopole source in terms of volume velocity. If TYPE = VACC, define the monopole source in terms of volume acceleration.
FORM	Format to define the monopole amplitude or power. (Character; "REAL" or "PHASE"; Default = REAL) If FORM = REAL, define the monopole amplitude, volume displacement, volume velocity, or volume acceleration in rectangular format (real and imaginary). If FORM = PHASE, define the monopole amplitude, power, volume displacement, volume velocity, or volume acceleration in polar format (magnitude and phase). See Remark 3.
A	Scale factor for TR, TI, or TM. (Real; Default = 1.0)

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Field	Contents
TR	Real part of the amplitude, volume displacement, volume velocity, or volume acceleration of monopole source. (Real or Integer; Default = 0.0) For a constant source type, enter a real value. For a frequency-dependent source type, enter the ID of the TABLEDi bulk entry that contains the source type as a function of frequency.
TM	Magnitude of the amplitude, power, volume displacement, volume velocity, or volume acceleration of monopole source. (Real or Integer; Default = 0.0) For a constant source type, enter a real value. For a frequency-dependent source type, enter the ID of the TABLEDi bulk entry that contains the source type as a function of frequency.
TI	Imaginary part of the amplitude, volume displacement, volume velocity, or volume acceleration of monopole source. (Real or Integer; Default = 0.0) For a constant source type, enter a real value. For a frequency-dependent source type, enter the ID of the TABLEDi bulk entry that contains the source type as a function of frequency.
TP	Phase angle in degrees of the amplitude, power, volume displacement, volume velocity, or volume acceleration of monopole source. (Real or Integer; Default = 0.0) For a constant source type, enter a real value. For a frequency-dependent source type, enter the ID of the TABLEDi bulk entry that contains the source type as a function of frequency.
CID	Identification number of coordinate system in which the location of monopole is defined. (Integer ≥ 0; See Remark 4 for default behavior)
X1, X2, X3	Location of monopole source in coordinate system CID. (Real; Default= 0.0)

Remarks:

1. ACPOLE1 is supported by the FEM Adaptive Order (FEMAO) method.
2. The SID must be unique with respect to ALOAD.
3. If TYPE = POWER, the FORM can only be defined as PHASE.
4. A CID of zero or blank (the default) references the basic coordinate system.
5. You can also define a monopole source with the ACSRCE bulk entry. However, ACSRCE is not supported by FEMAO.

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Related Topics

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ACPOLE2

Acoustic Dipole Source

Defines an acoustic dipole source

Format:

1	2	3	4	5	6	7	8	9	10
ACPOLE2	SID	FORM	CID1	X1	X2	X3			
	CID2	A1	TR1/TM1	TI1/TP1	A2	TR2/TM2	TI2/TP2		
		A3	TR3/TM3	TI3/TP3					

Examples:

ACPOLE2	300	REAL	2	1.0	1.0	1.0			
			0.1	0.1		0.2	0.2		

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
FORM	Format to define the dipole moment. (Character; "REAL" or "PHASE"; Default = REAL) If FORM = REAL, define the dipole moment in rectangular format (real and imaginary). If FORM = PHASE, define the dipole moment in polar format (magnitude and phase).
CID1	Identification number of coordinate system in which the location of dipole is defined. (Integer ≥ 0 or blank; Default = blank). See Remark 3.
X1, X2, X3	Location of dipole source in coordinate system CID1. (Real; Default = 0.0)
CID2	Identification number of coordinate system in which the dipole moment is defined. (Integer ≥ 0 or blank; Default = blank). See Remark 3.
Ai	The scale factor for TRi, TIi, or TMi. (Real; Default = 1.0)
TRi	Real part of dipole moment in the coordinate system CID2. (Real or Integer; Default = 0.0) For a constant moment, enter a real value.

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Field	Contents
	For a frequency-dependent moment, enter the ID of the TABLEDi bulk entry that contains the moment as a function of frequency.
TMi	Magnitude of dipole moment in the coordinate system CID2. (Real or Integer; Default = 0.0) For a constant moment, enter a real value. For a frequency-dependent moment, enter the ID of the TABLEDi bulk entry that contains the moment as a function of frequency.
TIi	Imaginary part of dipole moment in the coordinate system CID2. (Real or Integer; Default = 0.0) For a constant moment, enter a real value. For a frequency-dependent moment, enter the ID of the TABLEDi bulk entry that contains the moment as a function of frequency.
TPi	Phase angle in degrees of dipole moment. (Real or Integer; Default = 0.0) For a constant phase angle, enter a real value. For a frequency-dependent phase angle, enter the ID of the TABLEDi bulk entry that contains the phase angle as a function of frequency.

Remarks:

1. ACPOLE2 is supported by the FEM Adaptive Order (FEMAO) method.
2. The SID must be unique with respect to ALOAD.
3. A CID1 or CID2 of zero or blank (the default) references the basic coordinate system.

Related Topics

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ACPRESS

Enforced Pressure Value on Grids

Defines an enforced pressure value for a frequency response or real eigenvalue analysis.

Format:

1	2	3	4	5	6	7	8	9	10
ACPRESS	SID	FORM	A	TR/TM	TI/TP			G1	
	G2	G3	"THRU"	G4	"BY"	n	G5	G6	
	-etc.-								

Examples:

ACPRESS	200	REAL		1.0	-3.0			5001	

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
FORM	Format to define the pressure value. (Character; "REAL" or "PHASE"; Default = REAL) If FORM = REAL, define the pressure in rectangular format (real and imaginary). If FORM = PHASE, define the pressure in polar format (magnitude and phase).
A	Scale factor for TR, TI, or TM. (Real; Default = 1.0)
TR	Real part of pressure. (Real or Integer; Default = 0.0) For a constant pressure, enter a real value. For a frequency-dependent pressure, enter the ID of the TABLEDi bulk entry that contains the pressure as a function of frequency.
TM	Magnitude of the pressure. (Real or Integer; Default = 0.0) For a constant pressure, enter a real value. For a frequency-dependent pressure, enter the ID of the TABLEDi bulk entry that contains the pressure as a function of frequency.
TI	Imaginary part of the pressure. (Real or Integer; Default = 0.0)

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Field	Contents
	For a constant pressure, enter a real value. For a frequency-dependent pressure, enter the ID of the TABLEDi bulk entry that contains the pressure as a function of frequency.
TP	Phase angle in degrees of the pressure. (Real or Integer; Default = 0.0) For a constant phase angle, enter a real value. For a frequency-dependent phase angle, enter the ID of the TABLEDi bulk entry that contains the pressure as a function of frequency.
Gi	Grid identification number. (Integer > 0; No default)
G3 "THRU" G4	Grid identification number. (G4 > G3; Integer > 0; No default)
"BY"	Specifies an increment for a THRU specification. (Character; No default)
n	Increment for THRU range. (Integer > 0; No default)

Remarks:

1. ACPRESS is supported in the following solutions:
   - SOL 103, when the enforced pressure is zero.
   - SOL 108 acoustic and vibro-acoustic analyses.
   - SOL 111 acoustic and vibro-acoustic analyses.
2. ACPRESS is supported by the FEM Adaptive Order (FEMAO) method.
3. The SID must be unique with respect to ALOAD.
4. The continuation field is optional.

Related Topics

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ACRADS

Acoustic Radiating Surface Load

Defines an acoustic radiating surface from an external scd5 or sc_h5 file for use in a frequency response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
ACRADS	ID	TYPE	UNITNO	FTYPE	LCID	LDIDP	LDIDV	LDIDD	
	CID								

Example:

ACRADS	301	89	201	0	2	101	201	0	
	101								

Fields:

Field	Contents
ID	Load set identification number. (Integer > 0; No default). See Remark 1 and 2.
TYPE	Acoustic radiating surface type. (Character = "POTEN" or "KIRCH"; No default).
UNITNO	Unit number of the sc_h5 or scd5 file. (Integer > 0; No default). See Remark 3.
FTYPE	Type of file. (Integer ≥ 0; Default = 0). See Remark 4. 0 = SCD5 1 = SC_H5
LCID	Load case identification number of a specific loading in the loading description. (Integer > 0)
LDIDP	Identification number of the table in the SCH5 file specified by the UNIT field that contains the Pressure data. (Integer ≥ 0; Default = 0). See Remark 5.
LDIDV	Identification number of the table in the SCH5 file specified by the UNIT field that contains the Velocity data. (Integer ≥ 0; Default = 0). See Remark 5.
LDIDD	Identification number of the table in the SCH5 file specified by the UNIT field that contains the Density data. (Integer ≥ 0; Default = 0). See Remark 5.
CID	Coordinate system identification number for the location and orientation of the radiating surface. (Integer≥0; Default = 0).

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Remarks:

1. ACRADS is supported in the following solutions and requires the FEM Adaptive Order (FEMAO) method:
   - SOL 108 for uncoupled acoustic and coupled vibro-acoustic analyses .
   - SOL 111 for coupled vibro-acoustic analyses.
2. Multiple ACRADS entries with the same ID are allowed.
3. The logical unit number UNITNO must reference an existing ASSIGN SC_H5 or ASSIGN USERFILE statement in the File Management Section (FMS).
4. SCD5 file type is used only for TYPE= ”POTEN”(Potentials Acoustic Radiating Surface type). SC_H5 file type is used only for TYPE= ”KIRCH” (Kirchhoff Acoustic Radiating Surface type).
5. LDIDP, LDIDV and LDIDD refer to DTI, DISTL bulk entries that contain the loading description contained in the SC_H5 file referenced by an ASSIGN, File Management section entry. LDIDP refers to Pressure data set, LDIDV refers to Velocity data set, and LDIDD refers to Density data set.

Related Topics

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ACSPO2

Surface Dipole

Defines a surface dipole from an external sc_h5 file in HDF5 format for use in a frequency response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
ACSPO2	SID	PID	SET1ID	LUNF	LDID	LCID			

Example:

ACSPO2	200	111	2	25	100	3			

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0; No default)
PID	Parameter set identification number of PACSPO2. (Integer > 0; No default)
SET1ID	Identification number of a SET1 bulk entry that defines a list of element faces at which the loads are to be applied. See Remark 2.
LUNF	Logical unit number for HDF5 file containing frequency-dependent force excitations. (0 < Integer ≤ 9999; No default) See Remark 3.
LDID	Applied load descriptor identification number of a loading description contained in the HDF5 file referenced by LUNF. (0 < Integer ≤ 9999; No default) See Remark 3.
LCID	Load case identification number of a specific loading in the loading description. (Integer>0) See Remark 3.

Remarks:

1. ACSPO2 is supported in the following solutions and requires the FEM Adaptive Order (FEMAO) method:
   - SOL 108 for uncoupled acoustic and coupled vibro-acoustic analyses .
   - SOL 111 for coupled vibro-acoustic analyses.
2. The SET1ID references a SET1 bulk entry to define the fluid element faces. The element faces on the BSURFS bulk entry must all be on fluid elements. If the SET1ID = BLANK, the solver will use the face definition in the file specified by LUNF.

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3. The logical unit number LUNF must reference an existing ASSIGN SC_H5 File Management System (FMS) statement. The frequency-dependent force excitation description referenced by LDID must be contained in this file that also includes the DOF and frequencies at which the excitations are applied.

   If the frequencies of the force excitation matrix do not match the solution frequencies, interpolation will be used to obtain the applied forces at the solution frequency. No extrapolation will be performed if the solution frequency is less than or greater than the frequency range of the force excitation matrix, and the values of the force excitation matrix at the corresponding end point will be used.

   A DTI, DISTL bulk entry will be used to define the loading description referenced by LDID. The LCID identifier must correspond to a specific loading within the descriptor LDID.

Related Topics

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ACSRCE

Acoustic Source Specification

Defines source strength as a function of frequency for a simple acoustic source of the form:

$$\text{Source Strength} = Q(f) = A\left[\frac{1}{2\pi f}\sqrt{\frac{8\pi C\,P(f)}{\rho}}\right]e^{j(\theta + 2\pi f\tau)}$$

$$C^2 = B/\rho$$

Format:

1	2	3	4	5	6	7	8	9	10
ACSRCE	SID	EXCITEID	DELAY	DPHASE	TP	RHO	B		

Examples:

ACSRCE	103	11	0.001		12	1.0	15.0		

ACSRCE	103	11	104	45.0	12	1.0	15.0		

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
EXCITEID	Identification number of a DAREA or SLOAD entry that lists each degree of freedom to apply the excitation and the corresponding scale factor, *A*, for the excitation. (Integer > 0)
DELAY	Time delay, τ. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 6)
	If real entry, value of τ for all degrees of freedom in EXCITEID entry.
	If integer entry, identification number of a DELAY entry that contains values of τ for all degrees of freedom in EXCITEID entry. See Remark 7.
DPHASE	Phase angle, θ, in degrees. See Remark 8. (Real or Integer > 0 or blank; for default behavior, see Remark 6)
	If real entry, value of θ for all degrees of freedom in EXCITEID entry.
	If integer entry, identification number of a DPHASE entry that contains values of θ for all degrees of freedom in EXCITEID entry. See Remark 7.
TP	Power as a function of frequency, *P(f)*. (Real > 0.0 or Integer > 0)

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Field	Contents
	If real entry, value of *P(f)* used over all frequencies for all degrees of freedom in EXCITEID entry. If integer entry, identification number of a TABLEDi entry that defines *P(f)* for all degrees of freedom in EXCITEID entry.
RHO	Density of the fluid. (Real > 0.0)
B	Bulk modulus of the fluid. (Real > 0.0)

Remarks:

1. Acoustic sources must be selected with DLOAD = SID in the case control section.
2. SID must be unique for all ACSRCE, RLOADi, RLOADEX, TLOADi, and SELOAD entries.
3. If ACSRCE and RLOADi entries are combined with a DLOAD bulk entry, the identification numbers of the TABLEDi selected with the TP field on the ACSRCE entries must be different from the identification numbers of the TABLEDi selected with the TC and TD fields on RLOAD1 entries, and the TB and TP fields on RLOAD2 entries.
4. For additional remarks, see the RLOAD1 entry description.
5. The referenced EXCITEID, DELAY, and DPHASE entries must specify fluid points only.
6. If any of DELAY and DPHASE fields are blank or zero (either integer zero or real zero), the corresponding value for τ and θ used by the software is real zero.
7. For degrees of freedom in the EXCITEID entry that are not specified on the DELAY entry, the software uses real zero as the value for τ. For degrees of freedom in the EXCITEID entry that are not specified on the DPHASE entry, the software uses real zero as the value for θ.
8. The software converts the phase angle, θ, to radians.

Related Topics

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ACTEMP

Properties of Acoustic Fluids

Retrieves from a SCH5 file spatially varying fluid property data for acoustic analysis with standard FEM or FEMAO meshes.

Format:

1	2	3	4	5	6	7	8	9	10
ACTEMP	ID	R	GAMMA	PRESSURE	UNIT	DATACODE	TID1	TID2	

Examples:

ACTEMP	50	8.313E3	1.38		89	SS&P	100	101	

ACTEMP	75	8.313E3	1.4	1.2E-2	99	T&CP	200		

Fields:

Field	Contents
ID	Identification number. See Remark 1. (Integer > 0; No default)
R	Gas constant for the fluid. See Remark 2. (Real > 0.0; No default)
GAMMA	Isentropic expansion factor for the fluid. See Remark 2. (Real > 0.0; No default)
PRESSURE	Constant pressure. See Remark 3. (Real > 0.0; No default)
UNIT	Unit number of the SCH5 file. See Remark 4. (Integer > 0; No default)
DATACODE	Fluid property combination to retrieve. (Character = "T&MD", "T&P", "T&CP", "SS&MD", "SS&P", "SS&CP"; No default) = "T&MD" for temperature and mass density. = "T&P" for temperature and pressure. = "T&CP" for temperature and constant pressure. = "SS&MD" for speed of sound and mass density. = "SS&P" for speed of sound and pressure. = "SS&CP" for speed of sound and constant pressure.

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Field	Contents
TID1	Identification number of the table in the SCH5 file specified by the UNIT field that contains the data for the first property listed in the DATACODE specification. See Remark 3. (Integer > 0; No default)
TID2	Identification number of the table in the SCH5 file specified by the UNIT field that contains the data for the second property listed in the DATACODE specification. See Remark 3. (Integer > 0; No default)

Remarks:

1. If multiple ACTEMP bulk entries are present, each one must have a unique ID.
2. The R and GAMMA fields must be specified.
3. The DATACODE field specification determines which two of the TID1, TID2, and PRESSURE fields need to be specified.
   - If the DATACODE specification is "T&CP" or "SS&CP":
     - Use the PRESSURE field to specify the constant pressure.
     - Use the TID1 field to retrieve the temperature or speed of sound data, whichever is applicable.

     The software ignores the TID2 field.
   - If the DATACODE specification is "T&MD", "T&P", "SS&MD", or "SS&P":
     - Use the TID1 field to retrieve the temperature or speed of sound data, whichever is applicable.
     - Use the TID2 field to retrieve the mass density or pressure data, whichever is applicable.

     The software ignores the PRESSURE field.
4. In the File Management section, use an ASSIGN statement to assign the data file to the unit number specified in the UNIT field.

Related Topics

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ACTRAD

Acoustic Transfer Admittance

Defines transfer admittance for acoustics

Format:

1	2	3	4	5	6	7	8	9	10
ACTRAD	SID	PID	BIDS	BIDT	AREA	SDIST	ANGLE		

Examples:

ACTRAD	300	400	5	6		0.001	30.		

Fields:

Field	Contents
SID	Identification number of transfer admittance. (Integer > 0)
PID	Identification number of the PACTRAD bulk entry. (Integer > 0 or blank; Default = blank). See Remark 3. If PID = blank, the transfer admittance continuity is defined between the source and target region.
BIDS	Acoustic source region identification number for transfer admittance pairing. (Integer > 0; Default = blank)
BIDT	Acoustic target region identification number for transfer admittance pairing. (Integer > 0; Default = blank)
AREA	Area correction factor. (Real ≥ 0.0; Default = 0.0). See Remark 3.
SDIST	Search distance for transfer admittance region. (Real > 0.0; Default = 10.0)
Angle	Angular tolerance in degrees used to decide whether the acoustic source and target region can be considered as overlapping. If the angle between the normal of the acoustic source and target region exceeds this value, they cannot be coupled. (Real > 0.0; Default = 60.0)

Remarks:

1. ACTRAD is supported by the standard FEM and Finite Element Method Adaptive Order (FEMAO) method.

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2. The SID must be unique with respect to ACTRAD.
3. If PID = blank, the AREA input is ignored.

   If AREA = blank or 0.0, the solver automatically computes the area correction factor.

   If AREA > 0.0, the input value is used.

Related Topics

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ACTRTM

Vibro-Acoustic Transfer Admittance

Defines transfer admittance for vibro-acoustics.

Format:

1	2	3	4	5	6	7	8	9	10
ACTRTM	SID	PID	ACID	SSID					

Examples:

ACTRTM	300	400	5	6					

Fields:

Field	Contents
SID	Identification number of vibro-acoustic transfer admittance. See Remark 2. (Integer > 0).
PID	Identification number of the PACTRTM bulk entry. (Integer > 0; No default).
ACID	Acoustic region identification number for vibro-acoustic transfer admittance pairing. (Integer > 0; No default)
SSID	Structural region identification number for vibro-acoustic transfer admittance pairing. (Integer > 0; No default)

Remarks:

1. ACTRTM is supported by SOL 108 and SOL 111 strongly coupled vibro-acoustic solutions.
2. The SID must be unique with respect to ACTRTM.

Related Topics

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ACUPOLE

Unit Acoustic Monopole Source

Defines a unit acoustic monopole source

Format:

1	2	3	4	5	6	7	8	9	10
ACUPOLE	SID	TYPE	GID						

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
TYPE	Monopole source type. (Character; "AMP", "POWER", "VDISP", "VVEL", or "VACC"; Default = AMP) If TYPE = AMP, define the monopole source in terms of source amplitude. If TYPE = POWER, define the monopole source in terms of power. If TYPE = VDISP, define the monopole source in terms of volume displacement. If TYPE = VVEL, define the monopole source in terms of volume velocity. If TYPE = VACC, define the monopole source in terms of volume acceleration.
GID	Grid ID where the monopole source is located. (Integer > 0; No default)

Remarks:

1. ACUPOLE is supported by the FEM Adaptive Order (FEMAO) method.

Related Topics

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ACVPO4

Volume Quadrupole

Defines an acoustic volume quadrupole source from an external *.cgns file for use in a frequency response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
ACVPO4	SID	PID	SET1ID	SAMP	SMIN	SMAX	TRUNC		

Example:

ACVPO4	999	89	2	NO	1	1500	1.5E-3		

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0; No default) See Remark 3.
PID	Parameter set identification number of PACVPO4. (Integer > 0; No default)
SET1ID	Identification number of a SET1 bulk entry that defines a list of 3D elements at which the loads are to be applied.
SAMP	Time samples range selection type. (Character; "YES" or "NO"; Default = "NO") If SAMP = "NO", use all CGNS files in the specified location. If SAMP = "YES", use SMIN and SMAX.
SMIN	Time sample number start. (Integer ≥ 1; Default = 1) See Remark 4.
SMAX	Time sample number end. (Integer ≥ 1; Default = 1000) See Remark 4.
TRUNC	Truncation distance from envelope. (Real ≥ 0; Default = 0)

Remarks:

1. ACVPO4 is only supported in SOL 108 and SOL 111 acoustic or vibro-acoustic analyses and requires the FEM Adaptive Order (FEMAO) method.
2. A CFD General Notation System (CGNS) file with a *.cgns file extension is only supported.
3. The software allows you to define multiple ACVPO4 bulk entries. The SIDs must be unique with respect to ALOAD.

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4. SMIN and SMAX define the range of time samples. SMAX must be greater than SMIN.

Related Topics

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ACVSTT

Visco-Thermal Treatment

Defines visco-thermal losses for acoustics and vibro-acoustics.

Format:

1	2	3	4	5	6	7	8	9	10
ACVSTT	SID	MID	METHOD	SETID RID	SECTION	DIM1	DIM2	MAXFREQ	

Examples:

ACVSTT	1	1000	LRF	1	RECT	66.0	77.0	88.0	

Fields:

Field	Contents
SID	Identification number of visco-thermal treatment. (Integer > 0)
MID	Identification number of visco-thermal fluid material. (Integer > 0; No default)
METHOD	Modeling method. (Character; "LRF" or "BLI"; Default = "LRF") = LRF for low-reduced frequency. The LRF model assumes a constant pressure at the cross section of the domain. = BLI for boundary layer impedance. The BLI model applies equivalent complex impedance at the external boundary of the domain.
SETID	For METHOD = "LRF", use identification number of a SET1 or SET3 bulk entry that defines a list of solid fluid element faces at which the visco-thermal treatment is to be applied. (Integer > 0; No default)
RID	For METHOD = "BLI", use identification number of a BSURFS bulk entry that defines the free-face surface. (Integer > 0; No default)
SECTION	For METHOD = "LRF", specify section type. (Character; Default = "SLIT") = "SLIT" for layered slit. Specify the clearance l.

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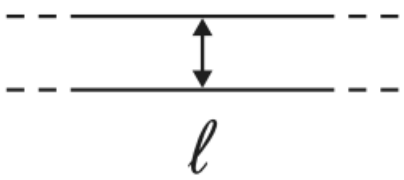
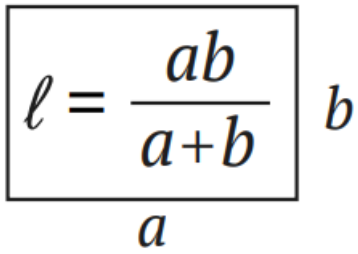
Field	Contents

Figure 10-9: Layered slit



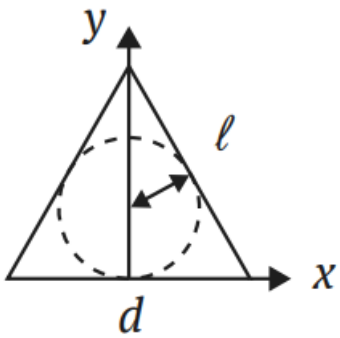
= "RECT" for rectangle. Specify width a and height b for the software to compute l.

Figure 10-10: Rectangular section



= "TRIA" for equilateral triangle. Specify the side length d of the triangle for the software to compute l.

Figure 10-11: Equilateral triangle



= "CIRC" for circle. Specify the radius l.

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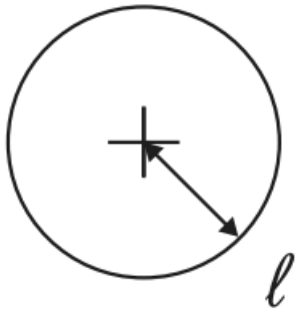
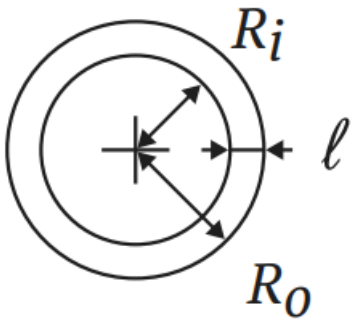
Field	Contents

Figure 10-12: Circular section



= "RING" for ring. Specify inner radius R_i and outer radius R_o for the software to compute l. R_o must be greater than R_i.

Figure 10-13: Ring section



Remarks:

1. SOL 108 and SOL 111 acoustic or coupled vibro-acoustic solutions support ACVSTT bulk entries. These entries require the FEM Adaptive Order (FEMAO) method.
2. The SID must be unique with respect to ACVSTT.

Related Topics

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ADUMi

Dummy Element Attributes

Defines attributes of the dummy elements ($1 \le i \le 9$).

Format:

1	2	3	4	5	6	7	8	9	10
ADUMi	NG	NC	NP	ND	ELNM				

Example:

ADUM2	8	2	1	3	CTRIM6				

Fields:

Field	Contents
NG	Number of grid points connected by a DUMi dummy element. (Integer > 0)
NC	Number of additional fields (Ai) on the CDUMi connection entry. (Integer ≥ 0)
NP	Number of additional fields (Ai) on the PDUMi property entry. (24 ≥ Integer ≥ 0)
ND	Number of displacement components at each grid point used in the generation of the differential stiffness matrix. Zero implies no differential stiffness. (Integer 3 or 6)
ELNM	The name of the element connection and property entry. In the example above, the connection entry is named “CTRIM6” and the property entry is named “PTRIM6”.

Related Topics

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AECOMP

Component for an Integrated Load Monitor Point

Defines a component for use in aeroelastic monitor point definition.

Format:

1	2	3	4	5	6	7	8	9	10
AECOMP	NAME	LISTTYPE	LISTID1	LISTID2	LISTID3	LISTID4	LISTID5	LISTID6	
	LISTID7	-etc.-							

Example:

AECOMP	WING	AELIST	1001	1002					

Fields:

Field	Contents
NAME	A character string of up to eight characters identifying the component. (Character)
LISTTYPE	One of CAERO or AELIST for aerodynamic components and SET1 for structural components. Aerodynamic components are defined on the aerodynamic ks-set mesh while the structural components are defined on the g-set mesh. See Remarks 2 and 3.
LISTIDi	The identification number of either SET1, AELIST or CAEROi entries that define the set of grid points that comprise the component. See Remarks 2 and 3.

Remarks:

1. The identification name must be unique among all AECOMP and AECOMPL entries.
2. If the component is defined on the structure, the LISTIDs must refer to SET1 entry(ies) that define the list of associated GRID points. For the AELIST or CAERO option, the LISTID's must refer to AELIST or CAERO i entries, respectively. Note that for DLM models (CAERO1/2), the set of points defined by the AELIST are the box identification numbers. For example, if the control surface's grids are desired, the same AELIST used for the AESURF can be referred to here. An AECOMP component must be defined as either an aerodynamic mesh component or a structural component. The two mesh classes cannot be combined into a single component.
3. The AECOMPL entry can be used to combine AECOMP entries into new components. When combining components, the structural and aerodynamic classes must be kept separate.

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Related Topics

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AECOMPL

Component for an Integrated Load Monitor Point

Defines a component for use in aeroelastic monitor point definition as a union of other components.

Format:

1	2	3	4	5	6	7	8	9	10
AECOMPL	NAME	LABEL1	LABEL2	LABEL3	LABEL4	LABEL5	LABEL6	LABEL7	
	LABEL8	-etc.-							

Example:

AECOMPL	HORIZ	STAB	ELEV	BALANCE					

Fields:

Field	Contents
NAME	A character string of up to eight characters identifying the component. (Character)
LABELi	A string of 8 characters referring to the names of other components defined by either AECOMP or other AECOMPL entries. (Character. See **Bulk Data Syntax Rules**.)

Remarks:

1. The identification name must be unique among all AECOMP and AECOMPL entries.
2. The AECOMPL entry can be used to combine AECOMP entries into new components. When combining components, the structural and aerodynamic classes must be kept separate.

Related Topics

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AEDW

Parametric Normal Wash Loading for Aerodynamics

Defines a downwash vector associated with a particular control vector of the associated aerodynamic configuration (AECONFIG). From this downwash vector, a force vector on the aerodynamic grids will be defined for use in nonlinear static aeroelastic trim.

Format:

1	2	3	4	5	6	7	8	9	10
AEDW	MACH	SYMXZ	SYMXY	UXID	DMIJ	DMIJI			

Example:

AEDW	0.90	SYMM	ASYMM	101	ALP1				

Fields:

Field	Contents
MACH	The Mach number for this force. See Remark 2 (Real ≥ 0.0, ≠ 1.0).
SYMXZ,SYMXY	The symmetry of this force vector. One of SYMM, ASYMM or ANTI (Character).
UXID	The identification number of a UXVEC entry that defines the control parameter vector associated with this downwash vector.
DMIJ	The name of a DMI or DMIJ entry that defines the downwash.
DMIJI	The name of a DMIJI entry that defines the CAERO2 interference element downwashes.

Remarks:

1. AEDW, AEFORCE and AEPRESS are associated with the current AECONFIG using either Case Control (if in the main Bulk Data Section) or using the BEGIN AECONFIG=<config> if in a partition of the Bulk Data.
2. The DMIJ field refers to either a DMI or a DMIJ entry. The DMIJI is only applicable to CAERO2 and is only required if nonzero downwash (j-set) input is needed on the interference body elements.
3. Mach numbers > 1.0 require that the supersonic aerodynamic option be available.

Related Topics

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AEFACT

Aerodynamic Lists

Defines real numbers for aeroelastic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
AEFACT	SID	D1	D2	D3	D4	D5	D6	D7	
	D8	D9	-etc.-						

Example:

AEFACT	97	.3	.7	1.0					

Fields:

Field	Contents
SID	Set identification number. (Unique Integer > 0)
Di	Number. (Real)

Remarks:

1. AEFACT entries must be selected by a CAEROi or PAEROi entry.
2. Embedded blank fields are not allowed.
3. To specify division points, there must be one more division point than the number of divisions.
4. When referenced by the CAERO3 entry, AEFACT defines the aerodynamic grid points. The ID number of the first point defined by each AEFACT entry is the value of the CAERO3 ID that selected the AEFACT entry. The ID of each following point defined on the AEFACT is incremented by 1.

Related Topics

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AEFORCE

Parametric Force for Aerodynamics

Defines a vector of absolute forces (it will **not** be scaled by dynamic pressure) associated with a particular control vector. This force vector may be defined on either the aerodynamic mesh (ks-set) or the structural mesh (g-set). The force vector will be used in nonlinear static aeroelastic trim.

Format:

1	2	3	4	5	6	7	8	9	10
AEFORCE	MACH	SYMXZ	SYMXY	UXID	MESH	FORCE	DMIK		

Example:

AEFORCE	0.90	SYMM	ASYMM	101	AERO		BETA		

Fields:

Field	Contents
MACH	The Mach number for this force. See Remark 2 (Real ≥ 0.0, ≠ 1.0).
SYMXZ, SYMXY	The symmetry of this force vector. One of SYMM, ASYMM or ANTI (Character).
UXID	The identification number of a UXVEC entry that defines the control parameter vector associated with this downwash vector.
MESH	One of AERO or STRUCT that declares whether the force vector is defined on the aerodynamic ks-set mesh or the structural g-set mesh. See Remark 3.
FORCE	The ID of a FORCE/MOMENT set that defines the vector. See Remark 3 (Integer > 0 if MESH=STRUCT).
DMIK	The name of a DMIK entry that defines the aerodynamic force vector. See Remark 3 (Character; required if MESH=AERO).

Remarks:

1. AEDW, AEFORCE and AEPRESS are associated with the current AECONFIG.
2. Mach numbers > 1.0 require that the supersonic aerodynamic option be available.
3. If the vector is defined on the structure, normal FORCEi and MOMENTi data are used. They will be subject to normal reduction to the solution set. For the AERO mesh option, the DMIK Bulk Data are used. Any forces associated with the permanently SPC'd degrees-of-freedom (which are dependent on the type of aerodynamic model being used) will be ignored.

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Related Topics

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AELINK

Links Aeroelastic Variables

Defines relationships between or among AESTAT and AESURF entries, such that:

$$u^D + \sum_{i=1}^{n} \mathrm{Ci}\ u_i^I = 0.0$$

u^D = dependent variable

and

u^I_i = independent variable

Format:

1	2	3	4	5	6	7	8	9	10
AELINK	ID	LABLD	LABL1	C1	LABL2	C2	LABL3	C3	
	LABL4	C4	-etc.-						

Example:

AELINK	10	INBDA	OTBDA	-2.0					

Fields:

Field	Contents
ID	Trim set identification number. (Integer > 0)
LABLD	Character string to identify the dependent aerodynamic variable. (Character)
LABLi	Character string to identify the i-th independent aerodynamic variable. (Character)
Ci	Linking coefficient for the i-th variable. (Real)

Remarks:

1. The AELINK entry (or entries) is selected by the TRIM=ID in Case Control.
2. This entry constrains the dependent variable to be a linear combination of the independent variables.
3. LABLD data must be unique for a given ID (i.e., the variable cannot be constrained twice).

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4. LABLD and LABLi data refer to AESURF or AESTAT Bulk Data entries.

Related Topics

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AELIST

Aerodynamic Elements List

Defines a list of aerodynamic elements to undergo the motion prescribed with the AESURF Bulk Data entry for static aeroelasticity.

Format:

1	2	3	4	5	6	7	8	9	10
AELIST	SID	E1	E2	E3	E4	E5	E6	E7	
	E8	-etc.-							

Example:

AELIST	75	1001	THRU	1075	1101	THRU	1109	1201	
	1202								

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
Ei	List of aerodynamic boxes generated by CAERO1 entries to define a surface. (Integer > 0 or “THRU”)

Remarks:

1. These entries are referenced by the AESURF entry.
2. When the “THRU” option is used, all intermediate grid points must exist. The word “THRU” may not appear in field 3 or 9 (2 or 9 for continuations).
3. Intervening blank fields are not allowed.

Related Topics

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AEPARM

General Controller for Use in Trim

Defines a general aerodynamic trim variable degree-of-freedom (aerodynamic extra point). The forces associated with this controller will be derived from AEDW, AEFORCE and AEPRESS input data.

Format:

1	2	3	4	5	6	7	8	9	10
AEPARM	ID	LABEL	UNITS						

Example:

AEPARM	5	THRUST	LBS						

Fields:

Field	Contents
ID	Controller identification number. (Integer > 0)
LABEL	Controller name. See Remark 1 (Character).
UNITS	Label used to describe the units of the controller values. (Character)

Remarks:

1. Controller LABELs that comprise the unique set relative to all the AESURF, AESTAT and AEPARM entries will define the set of trim variable degrees-of-freedom for the aeroelastic model.
2. Unit labels are optional and are only used to label outputs. No units will be associated with the controller if left blank.

Related Topics

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AEPRESS

Parametric Pressure Loading for Aerodynamics

Defines a vector of pressure/unit dynamic pressure associated with a particular control vector. From this pressure vector, a force vector on the aerodynamic grids will be defined for use in nonlinear static aeroelastic trim.

Format:

1	2	3	4	5	6	7	8	9	10
AEPRESS	MACH	SYMXZ	SYMXY	UXID	DMIJ	DMIJI			

Example:

AEPRESS	0.90	SYMM	ASYMM	101	ALP1				

Fields:

Field	Contents
MACH	The Mach number for this force. See Remark 2. (Real ≥ 0.0, ≠ 1.0)
SYMXZ, SYMXY	The symmetry of this force vector. One of SYMM, ASYMM or ANTI (Character).
UXID	The identification number of a UXVEC entry that defines the control parameter vector associated with this downwash vector.
DMIJ	The name of a DMI or DMIJ entry that defines the pressure per unit dynamic pressure.
DMIJI	The name of a DMIJI entry that defines the CAERO2 interference element downwashes.

Remarks:

1. AEDW, AEFORCE, and AEPRESS are associated with the current AECONFIG using Case Control.
2. Mach numbers > 1.0 require that the supersonic aerodynamic option be available.
3. The DMIJ field refers to either a DMI or a DMIJ entry. The DMIJI is only applicable to CAERO2 and is only required if nonzero downwash (j-set) input is needed on the interference body elements.

Related Topics

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AERO

Aerodynamic Physical Data

Gives basic aerodynamic parameters for unsteady aerodynamics.

Format:

1	2	3	4	5	6	7	8	9	10
AERO	ACSID	VELOCITY	REFC	RHOREF	SYMXZ	SYMXY			

Example:

AERO	3	1.3+4	100.	1.-5	1	-1			

Fields:

Field	Contents
ACSID	Aerodynamic coordinate system identification. See Remark 2 . (Integer ≥ 0; Default is the basic coordinate system.)
VELOCITY	Velocity for aerodynamic force data recovery and calculation of reduced frequency. See Remark 5 . (Real)
REFC	Reference length for reduced frequency. (Real)
RHOREF	Reference density. (Real)
SYMXZ	The symmetry key for the aero coordinate x-z plane. See Remark 6 (Integer=+1 for symmetry, 0 for no symmetry, and -1 for antisymmetry; Default=0).
SYMXY	The symmetry key for the aero coordinate x-y plane that can be used to simulate ground effect. (Integer=-1 for symmetry, 0 for no symmetry, and +1 for antisymmetry; Default=0)

Remarks:

1. This entry is required for aerodynamic problems. Only one AERO entry is allowed.
2. The ACSID must be a rectangular coordinate system. Flow is in the positive x-direction.
3. Set SYMXY=-1 to simulate ground effect.
4. PARAM,WTMASS does not affect aerodynamic matrices. RHOREF must be input in mass units.
5. VELOCITY is used only in aeroelastic response analysis, and it must be equal to V on the GUST Bulk Data entry.

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6. The symmetry fields on this entry are only used if neither of the Case Control commands (AESYMXY, AESYMXZ) are supplied. If either Case Control command is supplied, even the defaults from Case will override these Bulk Data entries. The Case Control symmetry is the preferred means of declaring the flow and geometric symmetry for aeroelastic analysis.

Related Topics

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AEROS

Static Aeroelasticity Physical Data

Defines basic parameters for static aeroelasticity.

Format:

1	2	3	4	5	6	7	8	9	10
AEROS	ACSID	RCSID	REFC	REFB	REFS	SYMXZ	SYMXY		

Example:

AEROS	10	20	10.	100.	1000.	1			

Fields:

Field	Contents
ACSID	Aerodynamic coordinate system identification. See Remark 2 . (Integer ≥ 0; Default is the basic coordinate system).
RCSID	Reference coordinate system identification for rigid body motions. (Integer ≥ 0; Default is the basic coordinate system.)
REFC	Reference chord length. (Real > 0.0)
REFB	Reference span. (Real > 0.0)
REFS	Reference wing area. (Real > 0.0)
SYMXZ	The symmetry key for the aero coordinate x-z plane. See Remark 6 . (Integer=+1 for symmetry, 0 for no symmetry, and -1 for antisymmetry; Default=0).
SYMXY	The symmetry key for the aero coordinate x-y plane that can be used to simulate ground effects. (Integer=+1 for antisymmetry, 0 for no symmetry, and -1 for symmetry; Default=0)

Remarks:

1. This entry is required for static aeroelasticity problems. Only one AEROS entry is allowed.
2. The ACSID must be a rectangular coordinate system. Flow is in the positive x-direction (T1).
3. The RCSID must be a rectangular coordinate system. All AESTAT degrees-of-freedom defining trim variables will be defined in this coordinate system.
4. REFB should be full span, even on half-span models.

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5. REFS should be half area on half-span models.
6. The symmetry fields on this entry are only used if neither of the Case Control commands (AESYMXY, AESYMXZ) are supplied. If either Case Control command is supplied, even the defaults from Case will override these Bulk Data entries. The Case Control symmetry is the preferred means of declaring the flow and geometric symmetry for aeroelastic analysis.

Related Topics

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AESTAT

Static Aeroelasticity Trim Variables

Specifies rigid body motions to be used as trim variables in static aeroelasticity.

Format:

1	2	3	4	5	6	7	8	9	10
AESTAT	ID	LABEL							

Example:

AESTAT	5001	ANGLEA							

Fields:

Field	Contents
ID	Identification number of an aerodynamic trim variable degree-of-freedom. See Remark 1 . (Integer > 0).
LABEL	An alphanumeric string of up to eight characters used to identify the degree-of-freedom. See Remark 1 (Character. See **Bulk Data Syntax Rules**.)

Remarks:

1. The degrees-of-freedom defined with this entry represent rigid body motion in the reference coordinate system defined on the AEROS entry. The standard labels that define the various rigid body motions are given below.

Table 10-31: Standard Labels Defining Rigid Body Motions

LABEL	Degree-of-Freedom Motion	Description
ANGLEA	u_r (R2)	Angle of Attack
SIDES	u_r (R3)	Angle of Sideslip
ROLL	$\mathring{u}_r$ (R1)	Roll Rate
PITCH	$\mathring{u}_r$ (R2)	Pitch Rate
YAW	$\mathring{u}_r$ (R3)	Yaw Rate
URDD1	$\ddot{u}_r$ (T1)	Longitudinal (See Remark 3 .)
URDD2	$\ddot{u}_r$ (T2)	Lateral

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LABEL	Degree-of-Freedom Motion	Description
URDD3	$\ddot{u}_r$ (T3)	Vertical
URDD4	$\ddot{u}_r$ (R1)	Roll
URDD5	$\ddot{u}_r$ (R2)	Pitch
URDD6	$\ddot{u}_r$ (R3)	Yaw

These reserved names may be defined on the AEPARM entry instead. See the AEPARM, AEPRESS, and AEFORCE entries.

2. The degrees-of-freedom defined with this entry are variables in the static aeroelastic trim solution, unless they are constrained by referencing them with a TRIM Bulk Data entry.
3. If a label other than those above is specified, then the user must either generate the corresponding forces with an AELINK or via a DMI Bulk Data entry along with a DMAP alter that includes the DMIIN module and additional statements to merge into the appropriate matrices. Or, using AEPARM and AEDW, AEPRESS, and/or AEFORCE, you can accomplish this purpose without the need for any alters.

Related Topics

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AESURF

Aerodynamic Control Surface

Specifies an aerodynamic control surface as a member of the set of aerodynamic extra points. The forces associated with this controller will be derived from rigid rotation of the aerodynamic model about the hinge line(s) and from AEDW, AEFORCE and AEPRESS input data. The mass properties of the control surface can be specified using an AESURFS entry.

Format:

1	2	3	4	5	6	7	8	9	10
AESURF	ID	LABEL	CID1	ALID1	CID2	ALID2	EFF	LDW	
	CREFC	CREFS	PLLIM	PULIM	HMLLIM	HMULIM	TQLLIM	TQULIM	

Example:

AESURF	6001	ELEV	100	100	200	200			
	10.0	180.0			-1.4E4	1.2E4	20	30	

Fields:

Field	Contents
ID	Controller identification number. (Integer > 0)
LABEL	Controller name, see Remark 1 (Character. See **Bulk Data Syntax Rules**.)
CIDi	Identification number of a rectangular coordinate system with a y-axis that defines the hinge line of the control surface component. (Integer > 0)
ALIDi	Identification of an AELIST Bulk Data entry that identifies all aerodynamic elements that make up the control surface component. (Integer > 0)
EFF	Control surface effectiveness. See Remark 4 (Real ≠ 0.0; Default=1.0).
LDW	Linear downwash flag. See Remark 2 (Character, one of LDW or NOLDW; Default=LDW).
CREFC	Reference chord length for the control surface. (Real > 0.0; Default=1.0)
CREFS	Reference surface area for the control surface. (Real > 0.0; Default=1.0)
PLLIM, PULIM	Lower and upper deflection limits for the control surface in radians. (Real; Default=± π/2)

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Field	Contents
HMLLIM, HMULIM	Lower and upper hinge moment limits for the control surface in force-length units. (Real, Default=no limit)
TQLLIM, TQULIM	Set identification numbers of TABLEDi entries that provide the lower and upper deflection limits for the control surface as a function of the dynamic pressure. (Integer > 0, Default=no limit)

Remarks:

1. The IDs on AESURF, AESTAT, and AEPARM entries are ignored. AESURFS can be used to define mass properties of the control surface.
2. The degrees-of-freedom defined on this entry represent a rigid body rotation of the control surface components about their hinge lines. In the default LDW (Linear DownWash) case, the downwash due to a unit perturbation of the control surface will be computed as part of the database. In the NOLDW case, the user must prescribe the controller's effects by direct definition of the induced forces using the AEPRESS, AEDW and/or AEFORCE entries.
3. Either one or two control surface components may be defined.
4. If EFF is specified, then the forces produced by this surface are modified by EFF (e.g., to achieve a 40% reduction, specify EFF=0.60).
5. The continuation is not required.
6. The CREFC and CREFS values are only used in computing the nondimensional hinge moment coefficients.
7. Position limits may be specified using either PiLIM or TQiLIM, but not both.
8. Position and hinge moment limits are not required.

Related Topics

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AESURFS

Structural Grids on an Aerodynamic Control Surface

Optional specification of the structural nodes associated with an aerodynamic control surface that has been defined on an AESURF entry. The mass associated with these structural nodes define the control surface moment(s) of inertia about the hinge line(s).

Format:

1	2	3	4	5	6	7	8	9	10
AESURFS	ID	LABEL		LIST1		LIST2			

Example:

AESURFS	6001	ELEV		6002		6003			

Fields:

Field	Contents
ID	Controller identification number. See Remark 1 (Integer > 0).
LABEL	Controller name. See Remark 1 (Character. See **Bulk Data Syntax Rules**.)
LISTi	Identification number of a SET1 entry that lists the structural grid points that are associated with this component of this control surface. (Integer > 0)

Remarks:

1. The LABEL on the AESURFS entry must match one on an AESURF entry. The ID is ignored.
2. The mass of the GRID points listed on the SETi entries is used to compute the mass moment of inertia of the control surface about its i'th hinge line. The presence of these data will allow the hinge moments to include the inertial forces in the computations. These data are optional, and, if omitted, result in hinge moments which include only the applied, aeroelastically corrected, forces.
3. These data will be associated with a structural superelement by grid list or partitioned SUPER=<seid> if the AESURFS is defined in the main bulk data section.

Related Topics

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ALOAD

Acoustic Load Combination with Unit Scale Factors

Defines an acoustic loading condition for frequency response as a linear combination of load sets defined with the ACDUCT, ACFAN, ACPOLE1, ACPOLE2, ACPLNW, ACPNVEL, ACPRESS, ACSPO2, or ACVPO4 bulk entries.

Format:

1	2	3	4	5	6	7	8	9	10
ALOAD	SID			L1	L2	L3	L4	L5	
	L6	L7	L8	L9	L10	L11	L12	L13	
	-etc.-								

Example:

ALOAD	200			8	9	200			

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
Li	Load set identification numbers defined on entry types listed above. (Integer > 0; No default)

Remarks:

1. Dynamic load sets must be selected in the Case Control section with ALOAD = SID.
2. Load set IDs (Li) must be unique.
3. An ALOAD entry may not reference a set identification number defined by another ALOAD entry.

Related Topics

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AMLREG

Automatically Matched Layer (AML) Region Definition

Defines automatically matched layer region for acoustic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
AMLREG	RID	SID	Name/Descriptor						
	NL	RADSURF	INFID1	INFID2	INFID3				

Example:

AMLREG	1	10	AML REGION 1						
	7	AML	12	13	14				

Fields:

Field	Contents
RID	AML region identification number. (Integer > 0) See Remark 2.
SID	Surface identification number. (Integer > 0) See Remark 3.
Name/ Descriptor	The Name/Descriptor is an optional character string. (Character; 48 characters maximum; See **Bulk Data Syntax Rules**) See Remark 4.
NL	Number of layers of extrusion to be formed by solver. (Integer > 0; Default = 5)
RADSURF	Radiation surface type. (Character; Default = AML) If RADSURF = AML, the pressure and velocities on the AML boundary are used to compute results in the far field. If RADSURF = PHYB, the pressure and velocities on the physical boundary (that is, all free fluid faces with the exception of faces on the AML and the infinite planes) are used to compute results in the far field. If RADSURF = NONE, the region does not radiate.
INFIDi	Identification number of an infinite plane. Up to three (3) infinite planes are considered for an AML region. The infinite planes are used when acoustic results are to be computed exterior to the AML region. (Integer ≥ 0; Default = 0) See Remark 5.

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Remarks:

1. AMLREG is supported by the FEM Adaptive Order (FEMAO) method.
2. RID must be unique with respect to all other AMLREG bulk entries. AMLREG definitions are only valid for analyses with AML method.
3. The SID references a BSURFS bulk entry to define the fluid element faces used to describe the AML region. The element faces on the BSURFS entry **must** all be on fluid elements.
4. Because of the name/descriptor field, free-field and large field data formats are not supported for this bulk entry.
5. If INFIDi = 0, there is no infinite plane for this AML region. If INFIDi > 0, INFIDi references an IPLANE bulk entry to define the infinite plane. If infinite planes exist, then every AML region free edge must belong to an infinite plane. Each and every AML region free edge must belong to one and only one infinite plane. No other element edges in the AML region except free edges can belong to an infinite plane.

Related Topics

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ASET

Degrees-of-freedom for the a-set

Defines degrees-of-freedom in the analysis set (a-set).

Format:

1	2	3	4	5	6	7	8	9	10
ASET	ID1	C1	ID2	C2	ID3	C3	ID4	C4	

Example:

ASET	16	2	23	3516	1	4			

Fields:

Field	Contents
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component numbers. (Integer zero or blank for scalar points, or any unique combinations of the Integers 1 through 6 for grid points with no embedded blanks.)

Remarks:

1. Degrees-of-freedom specified on this entry form members of the mutually exclusive a-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
2. When ASET, ASET1, QSET, and/or QSET1 entries are present, all degrees-of-freedom not otherwise constrained (e.g., SPCi or MPC entries) will be placed in the omitted set (o-set).
3. In nonlinear analysis, all degrees-of-freedom attached to nonlinear elements must be placed in the a-set. In other words, if the ASET or ASET1 entry, is specified then all nonlinear degrees-of-freedom must be specified on the ASET or ASET1 entry.

Related Topics

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ASET1

Degrees-of-freedom for the a-set, Alternate Form of ASET Entry

Defines degrees-of-freedom in the analysis set (a-set).

Format:

1	2	3	4	5	6	7	8	9	10
ASET1	C	ID1	ID2	ID3	ID4	ID5	ID6	ID7	
	ID8	ID9	ID10	-etc.-					

Example:

ASET1	345	2	1	3	10	9	6	5	
	7	8							

Alternate Format and Example:

ASET1	C	ID1	"THRU"	ID2					
ASET1	123456	7	THRU	109					

Fields:

Field	Contents
C	Component number. (Integer zero or blank for scalar points, or any unique combinations of the Integers 1 through 6 for grid points with no embedded blanks.)
IDi	Grid or scalar point identification numbers. (Integer > 0; for THRU option, ID1 < ID2)

Remarks:

1. Degrees-of-freedom specified on this entry form members of the a-set that are exclusive from other sets defined by Bulk Data entries. See **"Degree-of-Freedom Sets"** for a list of these entries.
2. When ASET, ASET1, QSET, and/or QSET1 entries are present, all degrees-of-freedom not otherwise constrained (e.g., SPCi or MPC entries) will be placed in the omitted set (o-set).
3. If the alternate format is used, all points in the sequence ID1 through ID2 are not required to exist, but there must be at least one degree-of-freedom in the a-set for the model, or a fatal error will result. Any points implied in the THRU that do not exist will collectively produce a warning message but will otherwise be ignored.

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4. In nonlinear analysis, all degrees-of-freedom attached to nonlinear elements must be placed in the a-set. In other words, if the ASET or ASET1 entry is specified, then all nonlinear degrees-of-freedom must be specified on the ASET or ASET1 entry.

Related Topics

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ATVBK

Panel/Grid Contribution with ATV

Defines the FORTAN unit number for the OUTPUT2 file created in ATV calculation and defines the element, grid, and properties ID offset.

The ATVBK bulk entry is available to compute the PANCON, GRDCON, and MODCON contribution for a weakly or strongly-coupled fluid/structural interaction. You specify the pre-computed ATV data in an OP2 file.

Format:

1	2	3	4	5	6	7	8	9	10
ATVBK		MEDIA	OPTION		UNITNO	EOFFSET	GOFFSET	POFFSET	

Examples:

ATVBK			2		33	1000	2000	20	

Fields:

Field	Contents
MEDIA	File type that contains the ATV matrix. (Character; “OP2”; Default= "OP2") IF MEDIA = "OP2", the ATV matrix and information are from an OP2 file.
OPTION	The options to use ATV matrix. (Character; “CON”; Default= "CON") If OPTION = "CON", the ATV matrix is used in panel/grid contribution computation only.
UNITNO	FORTRAN unit number for the OUTPUT2. See Remark 2. (Integer > 0; No default)
EOFFSET	Element ID offset. See Remark 1. (Integer; Default = 0)
GOFFSET	Grid point ID offset. See Remark 1. (Integer; Default = 0)
POFFSET	Element property ID offset. See Remark 1. (Integer; Default=0)

Remarks:

1. The user has to create the element ID offset, grid point ID offset and property ID offset to make sure there is no conflict with the IDs of elements, grids, and properties with any other part of the model. Otherwise a user fatal error is issued.

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For OPTION = 1, the grid numbering and microphone element IDs of the fluid model in the solution should be consistent with the grid numbering and microphone element IDs in the ATV .op2 file.

2. When specifying UNITNO, an appropriate ASSIGN OUTPUT2 statement must be present in the File Management Section (FMS) for the absolute value of UNITNO.
3. An .op2 file created from an ATV calculation run using the LP-64 nastran executable cannot be used in an ILP-64 nastran executable system run, and vice versa.

Related Topics

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ATVBULK

Selects ATV results for an ATV system run

Specifies the Fortran unit number for the OUTPUT2 file that contains the ATV results, and the element, grid point, and element property offsets to be used in an ATV system run.

Format:

1	2	3	4	5	6	7	8	9	10
ATVBULK		UNITNO	EOFFSET	GOFFSET	POFFSET				

Example:

ATVBULK		33	1000	2000	20				

Fields:

Field	Contents
UNITNO	Fortran unit number for the OUTPUT2 file that contains the ATV results. See Remarks 1 and 2. (Integer > 0; No default)
EOFFSET	Element ID offset. See Remark 3. (Integer > 0; Default = 0)
GOFFSET	Grid point ID offset. See Remark 3. (Integer > 0; Default = 0)
POFFSET	Element property ID offset. See Remark 3. (Integer > 0; Default = 0)

Remarks:

1. For specifying UNITNO, an appropriate ASSIGN OUTPUT2 statement must be present in the File Management Section of the input file.
2. An OUTPUT2 file that is created during a LP-64 Nastran executable ATV creation run cannot be used in an ILP-64 Nastran executable system run and vice versa. Note that beginning in NX Nastran 12, only the ILP-64 executable is available.
3. The element, grid, and element property IDs associated with the ATV results must be unique from element, grid, and element property IDs in the system run. To assure that they are unique, specify appropriate element, grid, and element property ID offsets.

Related Topics

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ATVFS

Coupling interface definition for an ATV

Defines the interface that couples the acoustic transfer vector (ATV) to the structure in the ATV system run.

Format:

1	2	3	4	5	6	7	8	9	10
ATVFS	SID		ID1	ID2	ID3	ID4	ID5	ID6	
	ID7	ID8	-etc-						

Example:

ATVFS	100		201	202					

Fields:

Field	Contents
SID	Set identification number of the fluid element faces that define the coupling interface. See Remark 1. (Integer > 0; No default)
IDi	Identification number of BSURFS bulk entries that list the fluid element faces that define the coupling interface. (Integer>0; No default)

Remarks:

1. SID is referenced by the ATVFS describer specification on the ATVOUT case control command.

Related Topics

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BAROR

CBAR Entry Default Values

Defines default values for field 3 and fields 6 through 8 of the CBAR entry.

Format:

1	2	3	4	5	6	7	8	9	10
BAROR		PID			X1	X2	X3	OFFT	

Example:

BAROR		39			0.6	2.9	-5.87		

Alternate Format and Example

BAROR		PID			G0			OFFT	

BAROR		39			18				

Fields:

Field	Contents
PID	Property identification number of the PBAR entry. (Integer > 0 or blank)
X1, X2, X3	Components of orientation vector $\vec{v}$, from GA, in the displacement coordinate system at GA. (Real)
G0	Alternate method to supply the orientation vector $\vec{v}$, using grid point G0. The direction of $\vec{v}$ is from GA to G0. $\vec{v}$ is then translated to End A. (Integer > 0; G0 ≠ GA or GB)
OFFT	Three character input which designates the coordinate systems used to define the cross section orientation vector and the two end offset vectors. Not supported by SOLs 402 and 414. See Remark 5. (Character; Default="GGG")

Remarks:

1. The contents of fields on this entry will be assumed for any CBAR entry whose corresponding fields are blank.
2. Only one BAROR entry is allowed.

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3. For an explanation of bar element geometry, see “CBAR Element”in the *Simcenter Nastran Element Library*.

4. If field 6 is an integer, then G0 is used to define the orientation vector and X2 and X3 must be blank. If field 6 is real or blank, then X1, X2, and X3 are used.

5. The OFFT field is a three character input that designates the coordinate systems (CSYS) used to define the cross section orientation vector and the two end offset vectors. With the default of "GGG", a displacement (global) CSYS is used for all three.

   - The first character defines the CSYS used for the cross section orientation vector. When "G" is defined as the first character, the displacement (global) CSYS of grid point A is used. When "B" is defined as the first character, the basic CSYS is used.
   - The second character defines the CSYS used for the offset vector at end A. When "G" is defined as the second character, the displacement (global) CSYS of grid point A is used. When "O" is defined as the second character, the element CSYS is used. See CBAR remarks for a definition of the element CSYS.
   - The third character defines the CSYS used for the offset vector at end B. When "G" is defined as the third character, the displacement (global) CSYS of grid point B is used. When "O" is defined as the third character, the element CSYS is used. See CBAR remarks for a definition of the element CSYS.

   The table below lists the supported character inputs. A fatal error will occur if you enter a character input that is not listed.

Input	Orientation CSYS	Offset End A CSYS	Offset End B CSYS
GGG	Global[A]	Global[A]	Global[B]
BGG	Basic	Global[A]	Global[B]
GGO	Global[A]	Global[A]	Offset
BGO	Basic	Global[A]	Offset
GOG	Global[A]	Offset	Global[B]
BOG	Basic	Offset	Global[B]
GOO	Global[A]	Offset	Offset
BOO	Basic	Offset	Offset

   A: The displacement (global) CSYS of grid point A is used.

   B: The displacement (global) CSYS of grid point B is used.

6. If G0 is used to define the orientation vector and the OFFT field is defined, the software will ignore the first character defined in the OFFT field, and will use the second and third characters as described in Remark 5. Three characters are still required in the OFFT field for this case.

Related Topics

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BCONP

Slideline Contact Parameters

Defines the parameters for a slideline contact region and its properties.

Format:

1	2	3	4	5	6	7	8	9	10
BCONP	ID	SOURCE	TARGET		SFAC	FRICID	PTYPE	CID	

Example:

BCONP	95	10	15		1.0	33	1		

Fields:

Field	Contents
ID	Contact region identification number. See Remark 1 (Integer > 0).
SOURCE	Source region identification number. See Remark 2 (Integer > 0).
TARGET	Target region identification number. See Remark 3 (Integer > 0).
SFAC	Stiffness scaling factor. SFAC is used to scale the penalty values automatically calculated by the program. See Remark 4 (Real > 0.0, Default=1.0).
FRICID	Contact friction identification number. See Remark 5 (Integer > 0 or blank).
PTYPE	Penetration type. See Remark 6 . (Integer 1 or 2, Default=1) 1: unsymmetrical (source penetration only—Default) 2: symmetrical
CID	Coordinate system identification number to define the slideline plane vector and the slideline plane of contact. See Remark 7 . (Integer > 0, Default=0, which means the basic coordinate system.)

Remarks:

1. The ID field must be unique with respect to all other BCONP identification numbers.

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2. The referenced SOURCE is the identification number in the BLSEG Bulk Data entry. This is the source line. The width of each source segment must also be defined to get proper contact stresses. See the Bulk Data entry, **"BWIDTH"** for details on specifying widths.

3. The referenced TARGET is the identification number in the BLSEG Bulk Data entry. This is the target line. For symmetrical penetration, the width of each target segment must also be defined. See the Bulk Data entry, **"BWIDTH"** for details on specifying widths.

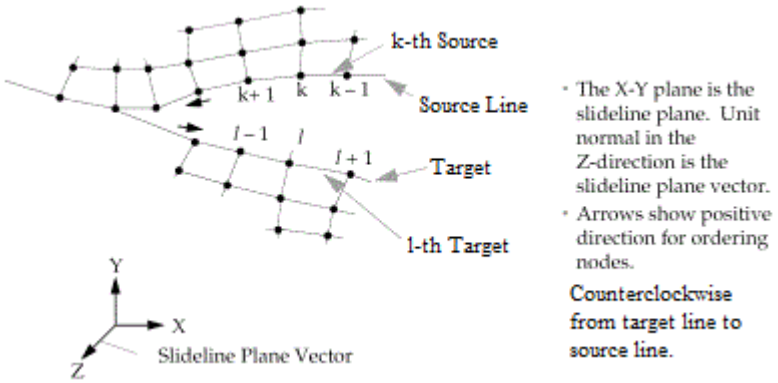
4. SFAC may be used to scale the penalty values automatically calculated by the program. The program calculates the penalty value as a function of the diagonal stiffness matrix coefficients that are in the contact region. In addition to SFAC, penalty values calculated by the program may be further scaled by the ADPCON parameter. (See the ADPCON parameter description for more details). The penalty value is then equal to k*SFAC*|ADPCON|, where k is a function of the local stiffness. It should be noted that the value in SFAC applies to only one contact region, whereas the ADPCON parameter applies to all the contact regions in the model.

5. The referenced FRLCLD is the identification number of the BFRLC Bulk Data entry. The BFRLC defines the frictional properties for the contact region.

6. In an unsymmetrical contact algorithm, only source nodes are checked for penetration into target segments. This may result in target nodes penetrating the source line. However, the error depends only on the mesh discretization. In symmetric penetration, both source and target nodes are checked for penetration. Thus, no distinction is made between source and target. Symmetric penetration may be up to thirty percent more expensive than the unsymmetric penetration.

7. In **Figure 10-14**, the unit vector in the Z-axis of the coordinate system defines the slideline plane vector. The slideline plane vector is normal to the slideline plane. Relative motions outside the slideline plane are ignored, and therefore must be small compared to a typical target segment. In a target segment, the direction from target node 1 to target node 2 gives the tangential direction (t). The normal direction for a target segment is obtained by the cross product of the slideline plane vector with the unit tangent vector (i.e., n=z x t). The definition of the coordinate system should be such that the normal direction must point toward the source region. For symmetric penetration, the normals of target segments and source segments must face each other. This is generally accomplished by traversing from target line to source line in a counterclockwise or clockwise fashion depending on whether the slideline plane vector forms a right-hand or left-hand coordinate system with the slideline plane.

Figure 10-14: A Typical Finite Element Slideline Contact Region



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Related Topics

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BCPROP

Surface-to-Surface Contact or Glue Region Definition by Property ID

Defines a surface-to-surface contact or glue region by property IDs of shell elements.

Format:

1	2	3	4	5	6	7	8	9	10
BCPROP	ID	IP1	IP2	IP3	IP4	IP5	IP6	IP7	
	IP8	IP9	IP10	-etc-					

Example:

BCPROP	3	1	5	8	3	22			

Fields:

Field	Contents
ID	Identification number of a contact or glue region. See Remark 2 and 3. (Integer > 0)
IPi	ID of a PSHELL, PCOMP, or PCOMPG entry (Integer > 0)

Remarks:

1. The continuation field is optional.
2. BCPROP is a collection of one or more shell property IDs. BCPROP defines a contact or glue region which may act as a source or target.
3. The ID must be unique with respect to all other BEDGE, BLSEG, BSURFS, BSURF, BCPROP, and BCPROPS entries.

Related Topics

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BCPROPS

Contact or glue region definition by PSOLID or PCOMPS property ID

Defines a surface-to-surface contact or glue region by PSOLID or PCOMPS property ID.

Format:

1	2	3	4	5	6	7	8	9	10
BCPROPS	ID	IP1	IP2	IP3	IP4	IP5	IP6	IP7	
	IP8	IP9	IP10	-etc-					

Example:

BCPROPS	3	1	5	8	3	22			

Fields:

Field	Contents
ID	Identification number of a contact or glue region. See Remark 2 and 3. (Integer > 0)
IPi	PSOLID or PCOMPS entry ID. (Integer > 0)

Remarks:

1. The continuation field is optional.
2. BCPROPS is a collection of one or more solid property IDs. BCPROPS defines a contact or glue region formed by the free faces of the solid elements (CHEXA, CPENTA, CPYRAM, or CTETRA) and may act as a source or target.
3. The ID must be unique with respect to all other BEDGE, BLSEG, BSURFS, BSURF, BCPROP, and BCPROPS entries.
4. Only the solid element free faces within the given set of PIDs are considered in the contact and/or glue algorithm. These free faces are automatically determined by the software.
5. The collapsed CHEXA element used in a SOL 401 crack simulation is not supported in a glue or contact region.

Related Topics

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BCRPARA

Contact Face and Edge Region Parameters

Defines parameters for a contact face or edge region.

Format:

1	2	3	4	5	6	7	8	9	10
BCRPARA	CRID	SURF	OFFSET	TYPE	MGP				

Example:

BCRPARA	1	TOP	0.02						

Fields:

Field	Contents
CRID	Contact region ID. (Integer > 0)
SURF	Indicates the contact side of shell elements. See Remark 1. (Character="TOP" or "BOT"; Default="TOP")
OFFSET	Offset distance for an edge or surface contact region. For SOLs 101, 103, 111, 112, and 401: Real, Default = 0.0. See SOL 402 Remark 2.
TYPE	Indicates whether a 3D contact region is a rigid (target) surface or is a shell coating on solid elements. (Character="FLEX" or "RIGID". Default="FLEX"). The TYPE field is only supported by SOL 402. TYPE=RIGID is only supported on shell element face regions defined with the BSURF and BCPROP entries.
MGP	Primary grid point for a target contact region with TYPE=RIGID or when the rigid-target algorithm is used. The primary grid point may be used to control the motion of a rigid surface. (Integer ≥ 0, Default=0) This is only supported by SOL 402.

Remarks:

1. The BCRPARA bulk entry is not required. When it is not present, the default values are used.
2. SURF is used to define the contact side of shell element regions. When SURF is "TOP", the contact side is consistent with the shell element normal and when "BOT" the opposite. In a SOL 101 (including consecutive solutions 103, 111 and 112), SURF must be defined so that source and target contact sides either face one another to represent a separation condition, or oppose one another to represent an interference condition.

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Remarks related to SOL 402:

1. The BCRPARA entry can be used to define target region parameters only. The target region consists of faces or edges. The BCRPARA entry cannot be used to define source region parameters. The source region consists of grid points.
2. The OFFSET parameter is ignored.
3. TYPE=RIGID is only supported on shell element face regions defined with the BSURF and BCPROP entries.

Related Topics

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BCTADD

Surface-to-Surface Contact Set Combination (SOLs 101, 103, 111, 112, 401, and 402)

Defines a surface-to-surface contact set as a union of contact sets defined on BCTSET entries.

Format:

1	2	3	4	5	6	7	8	9	10
BCTADD	CSID	SI	S2	S3	S4	S5	S6	S7	
	S8	S9	-etc-						

Example:

BCTADD	12	1	3	6	5	9			

Fields:

Field	Contents
CSID	Contact set identification number. (Integer > 0)
Si	Identification numbers of contact sets defined via BCTSET entries. (Integer > 0)

Remarks:

1. Multiple contact sets (BCTSETs) with unique CSIDs can be combined by including their CSIDs on a BCTADD entry. The BCTADD entry has its own unique CSID which is used on the BCSET case control command. Multiple contact sets with their own CSID are necessary when different parameters (BCTPARM) are needed for different contact sets.
2. To include several contact sets defined via BCTSET entries in a model, BCTADD must be used to combine the contact sets. CSID in BCTADD is then selected with the Case Control command BCSET.
3. Si must be unique and may not be the identification of this or any other BCTADD entry.

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Related Topics

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BCTPAR2

Contact Parameters for SOL 402

Defines contact parameters for SOL 402.

Format:

1	2	3	4	5	6	7	8	9	10
BCTPAR2	CSID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4	PARAM5	VALUE5	-etc-				

Example:

BCTPAR2	101	NSIDE	1	INITPENE	1	SEGNORM	0		
	ACTIVE	1	DISP	1	TIED	0	FRICMOD	5	
	FPARA1	.1	FPARA2	.2					

Fields:

Field	Contents
CSID	Contact set ID. Parameters defined in this command apply to contact set CSID defined by a BCTSET entry. (Integer > 0)
PARAMi	Name of the BCTPAR2 parameter. Allowable names are given in the parameter listing below. See Remark 8. (Character)
VALUEi	Value of the parameter. See Table 10-32 for the parameter listing. (Real or Integer)

Table 10-32: General Parameters

Name	Description
NSIDE	Flags single-sided or double-sided contact. NSIDE is not applicable for 2D contact. (Integer; Default=1) 1: Contact surfaces are single-sided. 2: Contact surfaces are double-sided.
INIPENE	Flags how initial penetrations or gaps are handled. See remarks 3, 4, and 5. (Integer 0, 1, 2, 3, 12, or 13; Default=1)

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Name	Description
	0 or 1: Contact is evaluated exactly as the geometry is modeled. No corrections will occur for gaps or penetrations. 2: Penetrations will be reset to a new initial condition in which there is no interference. 3: Gaps and penetrations are both reset to a new initial condition in which there is an interference provided by GAPVAL. 12: Penetrations will be reset to a new initial condition in which there is no interference for the most penetrated node. 13: Gaps and penetrations are both reset to a new initial condition in which there is an interference provided by GAPVAL for the closest node (or the one with the most penetration if any). **Note** When NSIDE = 2, you must use the default setting of INIPENE (0 or 1), otherwise a fatal error will occur.
ADJUST	Request for the software to remove initial overclosure and/or separation by moving grid points. (Real; No default) The software uses an adjustment tolerance to determine whether to move the source grid points on to the target faces. If ADJUST = 0.0, only the grid points that are initially overclosed are adjusted as long as the over-closure is below the characteristic distance*. If ADJUST > 0.0, in addition to the overclosed grid points described in ADJUST=0.0, the adjustment is also applied to grid points that are initially open with a gap smaller than the value of ADJUST * characteristic distance*. If ADJUST < 0.0, the software will adjust both the initially overclosed and the initially separated grid points when either is smaller than the absolute value of ADJUST. When you define the ADJUST parameter, the software uses the minimum and maximum search distance (MINDi and MAXDi) on the BCTSET bulk entry. MINDi and MAXDi define a search range in which the software uses to create or not create contact elements. The software searches starting at the source side element face in the normal direction towards the target grid points. The software creates a contact element when the target grid points are within the MINDi and MAXDi search range. By default, MINDi = -(characteristic distance*/5.0) and MAXDi= +(characteristic distance*/5.0).

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Name	Description
	You must define MAXDi greater than MINDi. You can define MINDi as a negative value when there is an interference condition modeled as overlapping regions. SOL 402 ignores MINDi and MAXDi when you have not defined the ADJUST parameter. The software computes the characteristic distance as 5.0*MAXLEN, and MAXLEN is the maximum edge length from all target element faces in a contact pair. The software reports the computed characteristic distance for each contact pair in the .f06 file.
SHLTHK	Shell thickness and Z-offset flag (Integer; Default=0). See Remark 10. 0: Shell thickness and Z-offset are both included in the contact problem. This option is only for shell elements referencing the PSHELL property, and not the composite properties. 1: Shell thickness and Z-offset are not included, and contact is evaluated at the midsurface of the geometry.
GAPVAL	Specifies a constant gap distance between the contact source region and the contact target region. Negative GAPVAL means that initial penetrations will be eliminated. See remarks 4 and 9. (Real, Default = 0.0; Unit = Length)
PDEPTH	Distance (penetration or gap) used to limit contact detection for target contact surface. (Real >= 0.0; Default=5 * the size of the greatest target face; Unit = Length) If PDEPTH > 0.0, then contact element is created only if the distance (penetration or gap) is less than or equal to PDEPTH.
SEGNORM	Indicates whether a continuous (interpolated) contact segment normal (and associated smoothed surface) is used for the contact surfaces. (Integer; Default=0) 0: SEGNORM=1 if NSIDE=1, SEGNORM=-1 if NSIDE=2. 1: Continuous segment normal is used. -1: Continuous segment normal is not used.
OFFSET	Default offset distance for contact regions. See Remarks 9 and 10. (Real; Default=0.0; Unit = Length)
ACTIVE	Requests that the software ramp the removal of initial penetrations. See Remark 11. (Integer; Default=1). 0 - The software eliminates initial penetrations by linearly ramping them over the subcase time steps or increments. 1 - The software eliminates initial penetrations immediately at the start of the subcase.

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Name	Description
	The descriptions above apply when contact is defined in the first subcase (or defined in the global subcase). For subcases that are not the first, the software always ramps the initial penetrations beginning with the status from the previous subcase and ending with the new status. Note that when contact is defined in a subcase that is not the first, the ACTIVE parameter setting does change the behavior in the first subcase. See Example 2 in Remark 11.

Table 10-33: Contact algorithm

Name	Description
DISP	Selects the displacement formulation used for the contact set. (Integer; The DISP parameter default and behavior depend on if the LGDISP and LGSTRN parameters are defined. See Remark 7.) 1: Use small displacement formulation (contact conditions are not updated). 2: Use large displacement formulation (contact conditions are updated).
TIED	Indicates whether contact regions in each contact pair are tied together in the normal direction. (Integer; Default=0) 0: Not tied. 1: Tied. 2: Remains tied after contact is active.
TIEDTOL	Tolerance used to determine whether contact nodes are tied to the target region when TIED=1 is specified. A contact node is tied to its target region if the distance between them is less than or equal to TIEDTOL. (Real; Default=0.0; Unit = Length)
EXTFAC	Factor for extending contact surfaces beyond their boundaries. The amount of extension is given by this factor multiplied by the length of the contact segments (Real ≥ 0; Unitless). The default value depends on how the DISP parameter is defined. If DISP = 1 (small sliding), the default is EXTFAC = 0.1. If DISP = 2 (large sliding), the default is EXTFAC = 0.025.
NPENAL2	Function/table that describes the relation between the normal distance and the contact pressure. See Remark6. (Table units: Ordinate=Length; Abscissa=Pressure)

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Name	Description
NCMOD	Normal contact modulus. See Remark 6. (Real -2.0, 0.0, or > 0.0; Default=0.0; Unit =Pressure/Length). 0.0: Fully rigid contact. -2.0: An automatic value is computed from the stiffnesses of both contact supports and the average element size of the contact target support. > 1.0E-16: define the stiffness to use.
NCMODF	Scales the computed normal contact stiffness when NCMOD = -2. (Real; Default=1.0) You can use the NCMODF parameter to scale the computed normal contact stiffness. For example, if you define NCMOD=-2.0, and the computed normal stiffness results in some final contact penetration, you could define NCMODF=100.0, and the software will multiply the computed normal stiffness by a factor of 100.0. The software ignores the NCMODF parameter if you explicitly define a stiffness value to the NCMOD parameter.
CFACTOR1	Compliance factor. See Remark 6. (Real >0.0; By default, the software does not scale the contact compliance; Unit = Length^3/Force)
ENFR	Turns contact friction on or off when contact is activated. (Integer; Default=0) 0: Friction is off during contact activation. 1: Friction is on during contact activation. When a new contact set is defined in a subcase that is not the first subcase, the contact condition is activated in that subcase. During contact activation, the software ramps the normal contact force by the number of increments defined for that subcase. By default, friction is not considered during contact activation, but you can optionally define ENFR=1 to request that the software include friction during contact activation. For example, if you are relying on contact friction to establish equilibrium due to a minimal constraint condition. We recommend that you define the new contact set in a subcase that has no additional new loads to allow the contact problem to become established. You can define additional loads in the subcases following the contact activation subcase. The ENFR parameter does not apply to contact defined in the first subcase, for example, when contact is defined globally. For this case, you can use the ACTIVE parameter on

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Name	Description
	the BCTPAR2 entry to request whether the contact normal force in the first subcase is ramped or stepped. The contact friction is not included in the first subcase at time=0.0.
SLFO	Requests a slide distance computation which accounts for shell offsets with large displacements. 0 : Standard slide distance computation. 1 : Slide distance computation for shell offsets and large displacements. When you have requested contact slide distance output with the BCRESULTS case control command, and your contact regions include shell elements with offsets, and your analysis includes large displacements, we recommend that you define SLFO=1. For all other cases, we recommend the default, standard slide distance computation.

Table 10-34: Friction Parameters

Name	Description
FRICMOD	Type of friction model (0 ≤ Integer ≤ 5; Default=0).

FRICMOD value	Description
0	Constant friction. Uses the friction coefficient FRICi specified for each contact pair in the BCTSET entry. All friction parameters of the BCTPAR2 entry (FPARA1, FPARA2, VCRIT, TCRIT, CFTE, CFVE, CFNF) are ignored.
1	Friction depends on time, velocity or temperature. Uses the friction coefficient FPARA1 on the BCTPAR2 entry. The friction coefficient is computed as being the product of the constant friction value defined by FPARA1 and existing functions depending on time (CFNF), temperature (CFTE) or velocity (CFVE). μ(t,T,v)= FPARA1 * CFVE(v) * CFTE(T) * CFNF(t) where t=time, v=velocity and T=temperature. If a table is not defined, its contribution to the formula is ignored. All other friction parameters of the BCTPAR2 entry (FPARA2, VCRIT, TCRIT) and the BCTSET entry (FRICi) are ignored.
2	Infinite friction.

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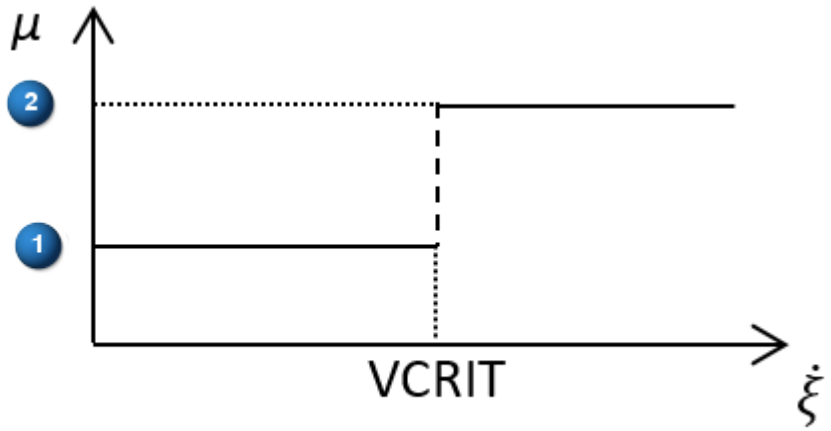
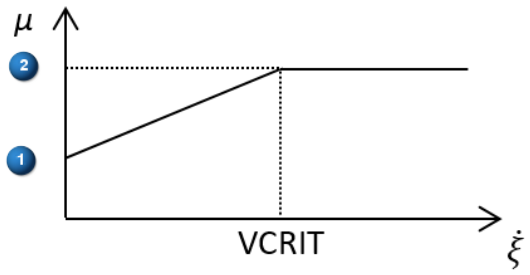
Name	Description

FRICMOD value	Description
	Uses an infinite friction coefficient. All friction parameters of the BCTPAR2 entry (FPARA1, FPARA2, VCRIT, TCRIT, CFTE, CFVE, CFNF) and the BCTSET entry (FRICi) are ignored.
3	Friction depends on sliding velocity.  Uses two different friction coefficients FPARA1 (1) and FPARA2 (2), depending on the sliding velocity: • A static friction coefficient FPARA1 at low sliding velocity $\dot{\xi} < \epsilon_v$ where ϵ_v = VCRIT. • A dynamic friction coefficient FPARA2 at larger sliding velocity $\dot{\xi} > \epsilon_v$. All other friction parameters of the BCTPAR2 entry (TCRIT, CFTE, CFVE, CFNF) and the BCTSET entry (FRICi) are ignored.
4	Friction depends on a linear law of the sliding velocity. 

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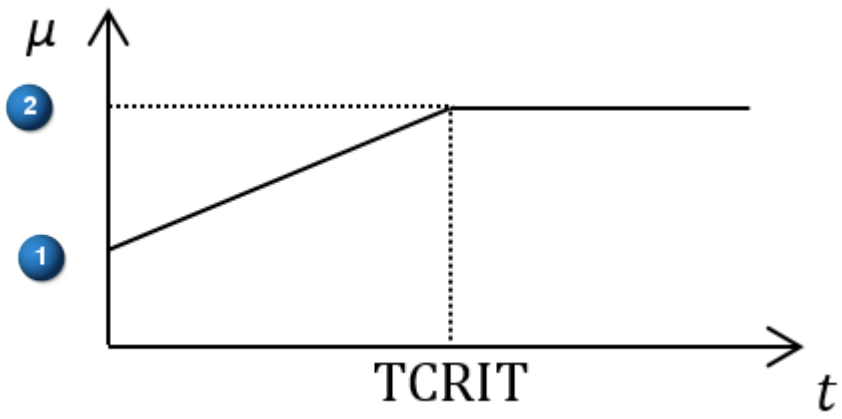
Name	Description

FRICMOD value	Description
	Uses a friction coefficient that varies linearly with the sliding velocity between two values FPARA1 (1) and FPARA2 (2), up to a critical sliding velocity VCRIT after which it remains constant. All other friction parameters of the BCTPAR2 entry (TCRIT, CFTE, CFVE, CFNF) and the BCTSET entry (FRICi) are ignored.
5	Friction depends on a linear law of the time.  Uses a friction coefficient that varies linearly with time between two values FPARA1 (1) and FPARA2 (2), up to a transition time TCRIT after which it remains constant. All other friction parameters of the BCTPAR2 entry (VCRIT, CFTE, CFVE, CFNF) and the BCTSET entry (FRICi) are ignored.

FPARA1	Friction parameter 1 (Real ≥ 0; Unitless).
FPARA2	Friction parameter 2 (Real ≥ 0; Unitless).
VCRIT	Critical sliding velocity (Real > 0; Units = Length/Time).
TCRIT	Transition Time (Real > 0; Units = Time)
CFVE	Friction coefficient vs sliding velocity table. (Integer; Table units: Ordinate=Unitless; Abscissa= Length/Time).
CFNF	Friction coefficient vs time table. (Integer; Table units: Ordinate=Unitless; Abscissa= Time).
CFTE	Friction coefficient vs temperature table. (Integer; Table units: Ordinate=Unitless; Abscissa= Temperature).

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Table 10-35: Regularization models

Name	Description
To avoid convergence problems when the contact starts sliding, we recommend that you introduce a regularization, either in terms of displacements or sliding velocity. This ensures a smooth transition of the friction stress between sliding and no sliding.	
STFR	Stiffness between the friction stress and the relative displacement. When the friction stress becomes equal to the pressure multiplied by the friction coefficient, this stiffness has no more influence. (Real -2.0, 0.0 or > 0; Default= -2.0; Unit =Pressure/Length). -2.0: an automatic value is computed using the stiffnesses of both contact supports and the average element size of the contact target support. 0.0: disabled (abrupt transition between sticking and sliding; not recommended). > 0.0: define the stiffness to use. This parameter is mutually exclusive with TOL. See Remark 12.
STFRF	Scales the computed tangential contact stiffness when STFR = -2. (Real; Default=1.0) You can use the STFRF parameter to scale the computed tangential contact stiffness. For example, you could define STFRF=100.0, and the software will multiply the computed tangential stiffness by a factor of 100.0. The software ignores the STFRF parameter if you explicitly define a stiffness value in the STFR parameter. See Remark 12.
TOL	Use a regularized friction coefficient μ_R as a function of the sliding velocity $\dot{\xi}$.

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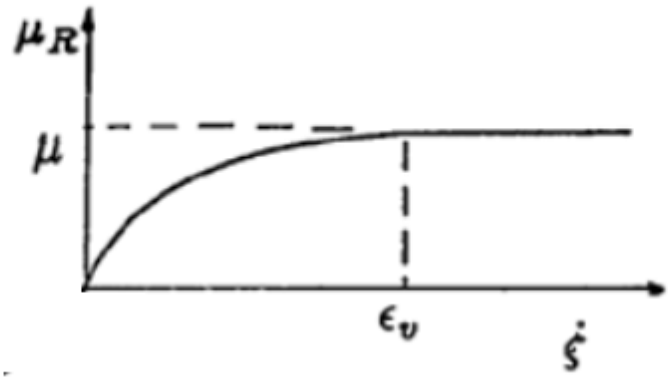
Name	Description

$$\mu_R(\dot{\xi}) = \begin{cases} \mu\left(2 - \frac{|\dot{\xi}|}{\varepsilon_v}\right)\frac{\dot{\xi}}{\varepsilon_v} & |\dot{\xi}| < \varepsilon_v \\ \mu\frac{\dot{\xi}}{|\dot{\xi}|} & |\dot{\xi}| \geq \varepsilon_v \end{cases}$$



where ϵ_v = TOL.

(Real >0; no default; Unit =Length/Time).

Once the sliding velocity becomes greater than or equal to the critical sliding velocity TOL, this regularization has no more influence. Depending on the friction model, this friction coefficient can have additional dependencies with the sliding velocity, time, and/or temperature.

This parameter is mutually exclusive with STFR.

This regularization model is available for ANALYSIS=DYNAMICS subcases only. It is not available for static analyses because the velocities must be computed by the solver, or when the friction coefficient is infinite (FRICMOD=2).

PRCS — Normal regularization type. (Real; Default = -3).

> 0.0: define the regularization value to use.

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Name	Description
	-1: use the characteristic stiffness of the whole structure. -2: use the characteristic stiffness of each contact support instead of the stiffness of the whole structure. -3: automatic choice: if the characteristic stiffnesses of both contact supports are very different, the characteristic stiffness of each support is used, otherwise the characteristic stiffness of the whole structure is used.

Note

Both regularization parameters (STFR and TOL) are mutually exclusive and a FATAL error is generated if both are specified. If none is specified, the solver uses displacement regularization with STFR=1.E8. To disable regularization, you must specify STFR=0.

Table 10-36: Stabilization Damping Parameters

Name	Description
CTDAMP	The CTDAMP parameter is available to request the stabilization damping option. This is useful, for example, when you are relying on the contact condition to prevent rigid body conditions, but the contact condition is not active. It can also be helpful in general to stabilize contact problems that are not converging. (Integer; Default=0) 0 - No stabilization damping. 1 - Stabilization damping applied only in the first subcase and it is ramped down to zero by the end of the subcase. The stabilization damping is only applied this way when the entire pair status is open. 2 - Stabilization damping applied always as long as the entire pair status is open. 3 - Stabilization damping applied always regardless of the time and the pair status. 4 - Stabilization damping applied only in the first subcase regardless of the entire pair status and it is ramped down to zero by the end of the subcase.
CTDAMPN	If you request stabilization damping with the CTDAMP parameter, the CTDAMPN parameter is available to either scale the normal damping value which the software automatically computes, or to define the normal damping value explicitly. (Real≠0.0; Default=1.0; Units depend on the sign of CTDAMPN) For CTDAMPN > 0.0, the normal damping value = CTDAMPN * automatically computed damping value. CTDAMPN is unitless in this case.

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Name	Description
	For CTDAMPN < 0.0, the normal damping value = ABS(CTDAMPN). CTDAMPN has units of pressure/velocity in this case.
CTDAMPT	If you request stabilization damping with the CTDAMP parameter, the CTDAMPT parameter is available to either scale the tangential damping value which the software automatically computes, or to define the normal damping value explicitly. (Real≠0.0; Default=0.001; Units depend on the sign of CTDAMPT) For CTDAMPT > 0.0, the tangential damping value = CTDAMPT * automatically computed damping value. CTDAMPT is unitless in this case. For CTDAMPT < 0.0, the tangential damping value = ABS(CTDAMPT). CTDAMPT has units of pressure/velocity in this case.
OPNTOL	Open gap tolerance scale factor. (Real≠0.0; Default=1.0; Units depend on the sign of OPNTOL) For OPNTOL > 0.0, the contact damping is applied to the contact elements that have a gap value less than or equal to OPNTOL * characteristic length, but greater or equal than 0.0. The contact damping is 0.0 if the gap is greater than OPNTOL * characteristic length. OPNTOL is unitless in this case. For OPNTOL < 0.0, the contact damping is applied to the contact elements that have a gap value less than or equal to ABS(OPNTOL), but greater or equal than 0.0. The contact damping is 0.0 if the gap is greater than ABS(OPNTOL). OPNTOL has units of length in this case.

Table 10-37: Damping models

Name	Description
In dynamic analysis (ANALYSIS=DYNAMICS), you can add a contact tangential viscous pressure. The viscous pressure is equal to the product of the contact velocity (in normal, tangential, or both directions) with a function of the normal distance.	
DPARA1	Normal velocity coefficient (INTEGER specifies a TABLEDi table whose units are a coefficient as a function of the normal penetration distance; Table units: Ordinate=Pressure/Speed [Force*Time/Length^3]; Abscissa= Length). For a given normal penetration distance, the value of the table is then multiplied by the normal velocity to produce a viscous pressure.
DPARA2	Tangential sliding velocity coefficient (INTEGER specifies a TABLEDi table whose units are a coefficient as a function of the normal penetration distance; Table units: Ordinate=Pressure/Speed [Force*Time/Length^3]; Abscissa= Length). For a given normal penetration distance, the value of the table is then multiplied by the tangential sliding velocity to produce a viscous pressure.

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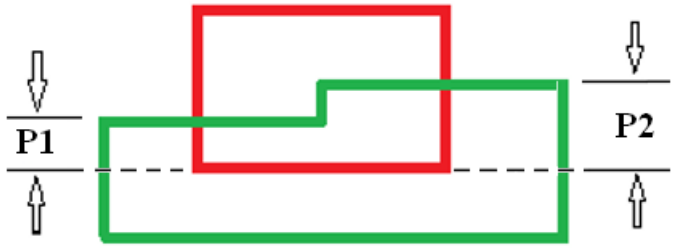
Remarks:

1. Field 9 on line 1 must be blank. Beginning on the continuation lines, fields 2 to 9 can be used for 4 pairs of PARAM/VALUE.
2. The BCTPAR2 bulk entry is not required. When it is not present, the default values are used. At least one parameter must be defined when a BCTPAR2 entry exists.
3. Contact offsets defined with the OFFSET parameter are applied before the software removes any gaps or penetrations as a result of the INIPENE setting. You can optionally use INIPENE=3 and GAPVAL to define a consistent gap or penetration to the contact set. Shell thicknesses are ignored.
4. GAPVAL and OFFSET are mutually exclusive. GAPVAL can only be used when INIPENE=3 or 13. And OFFSET is not allowable when INIPENE=3 or 13.
5. The following example demonstrates the software behavior with INIPENE, the combination of INIPENE+OFFSET, and the combination of INIPENE+GAPVAL. The red and green areas represent two 3D solid meshes that are modeled with interference and included in a contact definition.



SOL 402 applies the INIPENE settings first, then takes OFFSET or GAPVAL into account. GAPVAL can only be used when INIPENE=3 or 13, and OFFSET is not allowable when INIPENE=3 or 13.

For the following examples, consider the modeled penetrations are P1=-4.0 and P2=-5.0. After the software applies the INIPENE, OFFSET, or GAPVAL settings, the initial penetrations are reset to P1* and P2*, where a negative value is a penetration and a positive value is a gap.

	Settings	Description	P1*	P2*
1	INIPENE or OFFSET are not defined	The modeled penetrations P1 and P2 are not reset to a new initial condition.	-4.0	-5.0
2	INIPENE is set to 2 (or 3)	All penetrations (or gaps when 3 is defined) are reset to 0.0.	0.0	0.0
3	Same as (2) except OFFSET=0.1	The penetrations are reset to 0.0, then the OFFSET of 0.1 is applied.	-0.1	-0.1
4	INIPENE is set to 12 (or 13)	The penetrations (or gaps when 13 is defined) are reset relative to the closest nodes. The closest nodes have either the largest modeled penetration, or if no penetrations exist, the smallest modeled gap. The largest penetration for this example is P2=5.0. Therefore, the penetration of P2 is reset from 5.0 to	1.0	0.0

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	Settings	Description	P1 *	P2 *
		0.0, and the penetration of P1 is reset from 4.0 to a gap of 1.0 (P2 - P1 = 5.0 - 4.0 = 1.0).		
5	Same as (4) except INIPENE=12 and OFFSET=0.1	The penetration of P2 is reset to 0.0 (see description of 4 above), then the OFFSET of 0.1 is applied. The penetration of P1 is reset to a gap of 1.0, then the OFFSET is applied (1.0 - OFFSET = 1.0 - 0.1 = 0.9).	0.9	-0.1
6	INIPENE=13 and GAPVAL=1.0	The penetration of P2 is reset to 0.0 (see description of 4 above), then the GAPVAL of 1.0 is applied. The penetration of P1 is reset to a gap of 1.0, then the GAPVAL of 1.0 is applied (1.0 + GAPVAL = 1.0 + 1.0 = 2.0).	2.0	1.0

6. NPENAL2, NCMOD, and CFACTOR1 are mutually exclusive.

7. The DISP parameter setting and behavior depend on whether the LGDISP and LGSTRN parameters are defined:

DISP is undefined (default behavior):		
	LGDISP=-1 and LGSTRN=-1	DISP=1. Small sliding behavior is assumed and contact conditions do not update.
	LGDSP=1 or LGSTRN=1	DISP=2. If large sliding occurs, contact conditions will update.
DISP=1 is defined:		
	Any LGDISP and LGSTRN settings	DISP=1. Small sliding behavior is assumed and contact conditions do not update.
DISP=2 is defined:		
	LGDISP=1 or LGSTRN=1	DISP=2. If large sliding occurs, contact conditions will update.
	LGDISP=-1 and LGSTRN=-1	A FATAL error is generated. DISP=2 cannot be used without first taking large displacements or strains into account.

8. Contact parameters cannot be changed when you perform a restart run.

9. GAPVAL and OFFSET are mutually exclusive.

10. Before Simcenter Nastran 2020.1, neither the shell thickness or ZOFFS defined on the PSHELL entry were included in a contact problem. To include their effects, you were required to define OFFSET on the BCTPAR2 entry as .5 * shell thickness + ZOFFS, for example.

    Since shell thickness and ZOFFS are now included by default (SHLTHK=0), if your input files created before Simcenter Nastran 2020.1 still include an OFFSET on the BCTPAR2 representing the shell thickness, ZOFFS, or both, you will need to do one of the following:
    - remove the OFFSET definition from the BCTPAR2 entry and leave SHLTHK=0,
    - leave the OFFSET definition and define SHLTHK=1,

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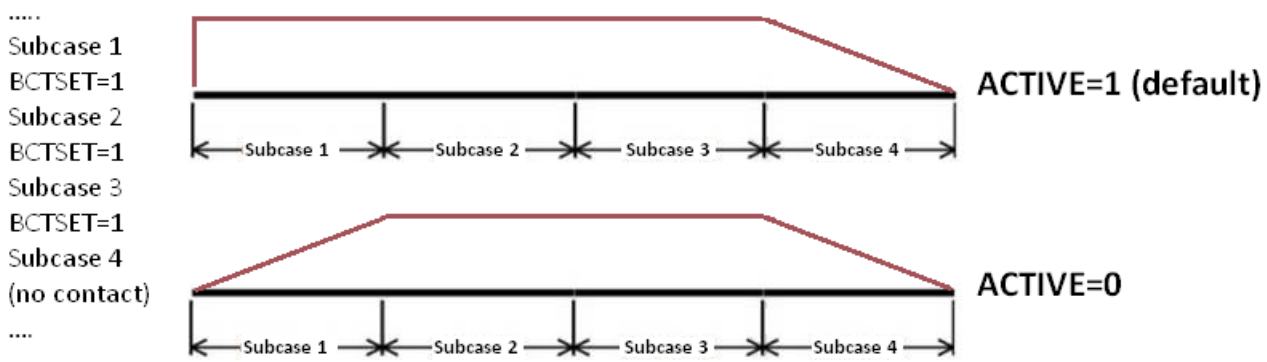
● or leave the OFFSET definition and define NLCNTLG,MODEVERS,20192, for example. The MODEVERS parameter will revert to the software behavior and defaults to a previous version number. Since the MODEVERS parameter will affect other defaults in the software, it should be used with care.

11. The following examples demonstrate how the ACTIVE parameter changes the removal of initial penetrations.

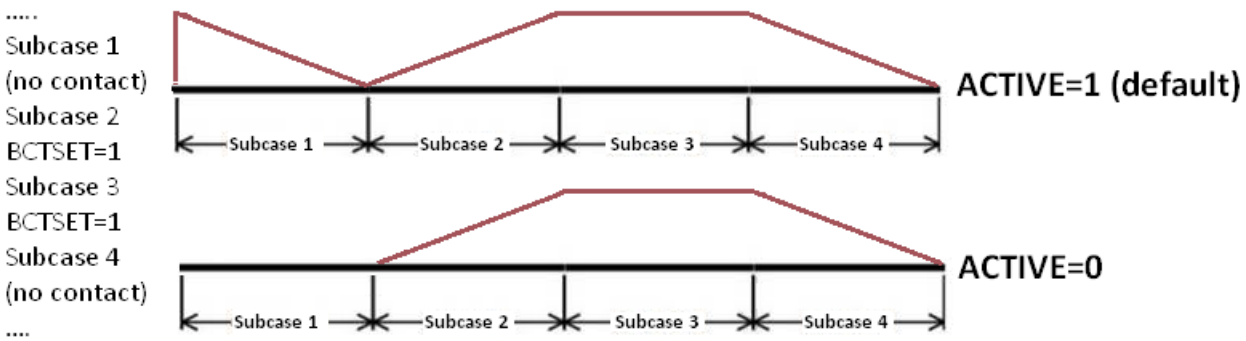
- **Example 1 (contact defined in Subcases 1,2,3 and no contact in Subcase 4):**

  Example 1 illustrates how the ACTIVE parameter changes the removal of initial penetrations when a contact set is defined in the first subcase.
  ● When ACTIVE=1, the software eliminates initial penetrations immediately at the start of the subcase.
  ● When ACTIVE=0, the software eliminates initial penetrations by linearly ramping them over the subcase time steps or increments.



- **Example 2 (contact defined in Subcases 2,3 and no contact in Subcases 1,4):**

  Example 2 illustrates how the ACTIVE parameter changes the behavior in the first subcase although the first subcase does not include a contact set.
  ● When ACTIVE=1, the software eliminates initial penetrations immediately at the start of the first subcase. Since contact is actually not defined for subcase 1, the penetrations are ramped back to their initial state before subcase 1 ends. Then in subcase 2, the software eliminates initial penetrations again by linearly ramping them over the time steps or increments defined in subcase 2.
  ● When ACTIVE=0, the software eliminates initial penetrations by linearly ramping them over the time steps or increments defined in subcase 2.



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12. To determine whether the tangential stiffness defined with STFR and STFRF is appropriate, you need to determine the elastic deformation result due to the tangential stiffness. This quantity is reported under the **SLIDING** heading in the contact messages that you request with the MSGLVLC parameter on the NLCNTL2 bulk entry. See SOL 402 - Contact status output in the *Multi-Step Nonlinear User's Guide*.

    Note that the slide distance result requested with the BCRESULTS case control command is the sum of slipping contact when the tangential forces become larger than the friction force, and tangential elastic deformation when the friction condition is sticking. This output does not separate the two computations.

Related Topics

Browse Simcenter Nastran documentation

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BCTPARM

Contact Parameters (SOLs 101, 103, 111, 112, and 401)

Control parameters for the contact algorithm.

Format:

1	2	3	4	5	6	7	8	9	10
BCTPARM	CSID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4	PARAM5	VALUE5	-etc-				

Example:

BCTPARM	1	PENN	10.0	PENT	0.5	CTOL	0.001		
	SHLTHK	1							

Fields:

Field	Contents
CSID	Contact set ID. Parameters defined in this command apply to contact set CSID defined by a BCTSET entry. (Integer > 0)
PARAMi	Name of the BCTPARM parameter. Allowable names are given in the parameter listing below. (Character)
VALUEi	Value of the BCTPARM parameter. Allowable values are given in the parameter listing below. (Real or Integer)

Parameters for SOLs 101, 103, 111, and 112

Table 10-38: SOLs 101, 103, 111, and 112 - General Parameters

Name	Description
REFINE	Requests that the software refine the mesh on the source region during the solution to be more consistent with the target side mesh. (Integer; Default=2) 0 - Refinement does not occur. 2 - Refinement occurs.
INTORD	Determines the number of contact evaluation points for a single element edge or face on the source region. (Integer; Default=2)

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Name	Description
	The number of contact evaluation points is dependent on the value of INTORD, and on the type of element face. See the table in Remark 4 for specific values. 1 - The reduced number of contact evaluation points is used. 2 - Use an increased number of contact evaluation points. 3 - Use a high number of contact evaluation points.
RESET	Flag to indicate if the contact status for a specific subcase is to start from the final status of the previous subcase. (Integer; Default=0) See Remark 9 . 0 - Starts from previous subcase. 1 - Starts from initial state.
PREVIEW	Requests the export of a bulk data representation of the element edges and faces where contact elements are created. (Integer; Default=0) See Remark 6. 0 - The bulk data export does not occur. 1 - The bulk data export occurs.

Table 10-39: SOLs 101, 103, 111, and 112 - Initial Penetration Parameters

Name	Description
INIPENE*	Use when the goal is for a pair of contact regions to be initially touching without interference, but due to the faceted nature of finite elements around curved geometry, some of the element edges or faces may have a slight gap or penetration. (Integer; Default=0) See Remarks 7 and 8. 0 or 1 - Contact is evaluated exactly as geometry is modeled. No corrections will occur for gaps or penetrations. 2 - Penetrations will be reset to a new initial condition in which there is no interference.

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Name	Description
	3 - Gaps and penetrations are both reset to a new initial condition in which there is no interference.
SHLTHK*	Shell thickness offset flag. (Integer; Default=0) 0 - Includes half shell thickness as surface offset. This option is only for shell elements referencing a PSHELL property. 1 - Shell thickness is excluded and contact is evaluated at mid-plane.
ZOFFSET	Determines if the shell element z-offset is included in the contact solution. (Integer; Default=0) 0 - Includes the shell z-offset when determining the contact surfaces. 1 - Does not include the shell z-offset when determining the contact surfaces.

Table 10-40: SOLs 101, 103, 111, and 112 - Stiffness Parameters

Name	Description
AUTOSCAL*	Scales the automatically calculated penalty factors PENN and PENT either up or down. (Real>0.0; Default=1.0) Will not scale explicitly defined value of PENN or PENT. AUTOSCAL can be used to scale the stiffness of specific contact pairs if convergence issues occur. See Remark 2.
TANSCL*	Scales the automatically calculated tangential penalty factor PENT either up or down. Will not scale explicitly defined value of PENT. TANSCL can be used to scale the stiffness of specific contact pairs if convergence issues occur. (Real>0.0; Default=1.0). See Remark 2.
PENN*	Penalty factor for normal direction. (Real>0.0; Default=Automatically Calculated) PENN and PENT are automatically calculated by default. See Remark 2. When PENT is defined but PENN is undefined, PENN = 10 * PENT.
PENT*	Penalty factor for transverse direction. (Real>0.0; Default=Automatically Calculated) PENN and PENT are automatically calculated by default. See Remark 2. When PENN is defined but PENT is undefined, PENT = PENN / 10.
PENTYP*	Changes unit definition for PENN and PENT. (Integer; Default=1). See Remark 2. 1 - PENN and PENT are entered as units of 1/Length.

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Name	Description
	2 - PENN and PENT are entered as units of Force/(Length x Area).
OPNSTF*	Open contact stiffness scale factor. (Real ≥ 0.0; Default = 1.0) The open contact stiffness is computed by OPNSTF * closed stiffness.

Table 10-41: SOLs 101, 103, 111, and 112 - Convergence Parameters

Name	Description
CTOL	Contact force convergence tolerance. (Real > 0.0; Default=0.01)
MAXS	Maximum number of iterations for a status loop. (Integer>0; Default=20)
MAXF	Maximum number of iterations for a force loop. (Integer>0; Default=10)
NCHG	Allowable number of contact changes for convergence. (Real≥0.0 or Integer≥1; Default=0.02). See Remark 3.
CSTRAT	Prevents all of the contact elements from becoming inactive. (Integer>0; Default=0) See Remark 5. 0 - All contact elements can become inactive. 1 - The software will reduce the likelihood of all of the contact elements becoming inactive.

* Can be defined on local BCTPARM entries. The BCTPARM bulk entries associated to individual BCTSET bulk entries, which are then combined with a BCTADD bulk entry, define local parameters. A local parameter definition overrides a global definition.

See “Contact Control Parameters – BCTPARM” in the *Simcenter Nastran User’s Guide* for more information on the BCTPARM options.

Parameters for SOL 401

Table 10-42: SOL 401 - General Parameters

Name	Description
METHOD	Selects the contact algorithm. (Integer; Default=2) 1 - Pure Penalty Method.

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Name	Description
	2 - Augmented Lagrangian Method.
INTRFC*	Selects the interface behavior between the source and target region. (Integer; Default=1) 0 - Inactive No interaction between source and target region. 1 - Standard Standard segment-segment contact algorithm with large sliding (pairing updates) and friction. 2 - No Sliding Does not allow sliding by assuming infinite friction. Source and target regions can separate. 3 - Bonded/Tied No relative displacement is permitted in both normal and tangential directions. 4 - No Separation No separation in the normal direction but large frictionless sliding is permitted in the tangential directions. 5 - Small sliding Simplified contact algorithm with no pairing updates and optional friction.
REFINE	Requests that the software refine the mesh on the source region during the solution to be more consistent with the target side mesh. (Integer; Default=2) 0 - Refinement does not occur. 2 - Refinement occurs.
INTORD	Determines the number of contact evaluation points for a single element edge or face on the source region. (Integer; Default=2) The number of contact evaluation points is dependent on the value of INTORD, and on the type of element face. See the table in Remark 4 for specific values. 1 - The reduced number of contact evaluation points is used. 2 - Use an increased number of contact evaluation points. 3 - Use a high number of contact evaluation points.

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Table 10-43: SOL 401 - Initial Penetration Parameters

Name	Description
INIPENE*	Use when the goal is for a pair of contact regions to be touching without interference, but due to the faceted nature of finite elements around curved geometry, some of the element edges or faces may have a slight gap or penetration. (Integer; Default=0) See Remarks 7 and 8. 0 - Contact is evaluated exactly as the geometry is modeled. No corrections will occur for gaps or penetrations. 1 – Gaps will be ignored. 2 - Penetrations will be reset to a new initial condition in which there is no interference. 3 - Gaps and penetrations are both reset to a new initial condition in which there is no interference. For contact pairs that are introduced after the first subcase, the initial penetration is evaluated based on the undeformed configuration. The initial penetration for these pairs is NOT evaluated based on the configuration present at the time of addition of the contact pairs.
IPENRAMP*	Requests that the software ramp the removal of initial penetrations in a constant time subcase. For example, a preload subcase is a constant time subcase. (Integer; Default=1) 0 - The removal of initial penetrations is not ramped, but instead, they are eliminated immediately at the start of the subcase. 1 - The initial penetrations are ramped by the load factor in the constant time subcase. Ramping the removal of initial penetrations can help convergence.
GAPVAL*	Optionally defines a constant gap distance between the source region and the target region when INIPENE=3. A negative GAPVAL value indicates that an initial penetration exists that the software will eliminated. (Real; Default = 0.0)
OFFSET*	Constant offset distance for a contact pair. (Real; Default = 0.0) This value overwrites an offset defined for a region on the BCRPARA entry. Note: If you define both the OFFSET and INIPENE parameters, the software adds the offset *after* it evaluates any gaps or penetrations as a result of the INIPENE setting.
SHLTHK*	Shell thickness offset flag. (Integer; Default=0)

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Name	Description
	0 – Includes half shell thickness as surface offset. This option is only for shell elements referencing a PSHELL property. 1 – Shell thickness is excluded and contact is evaluated at mid-plane.
ZOFFSET	Determines if the shell element z-offset is included in the contact solution. (Integer; Default=0) 0 - Includes the shell z-offset when determining the contact surfaces. 1 - Does not include the shell z-offset when determining the contact surfaces.
ADJUST*	Request for the software to remove initial separation by moving grid points. (Real; No default) The software uses an adjustment tolerance to move the source grid points on to the target faces when the absolute value of the initial separation distance is below the adjustment tolerance. If ADJUST = 0.0, only the grid points that are initially overclosed are adjusted as long as the over-closure is below the characteristic length. If ADJUST > 0.0, in addition to the overclosed grid points described in ADJUST=0.0, the adjustment is also applied to grid points that are initially open with a gap smaller than the value of ADJUST * characteristic length. If ADJUST < 0.0, the software will adjust both the initially overclosed and the initially separated grid points when either is smaller than the absolute value of ADJUST. Note: ADJUST works from the grid point locations and INIPENE works from the Gauss point locations. If both are defined, ADJUST is first applied, then INIPENE.
TZPENE*	Requests that the software ramp the removal of initial penetrations in a time-assigned subcase. TZPENE defines the end time when initial penetrations are eliminated. (Real≥0.0; Default = 0.0) For example, if a subcase start time is 5.0, end time is 10.0, and TZPENE=7.2, the initial penetrations are eliminated starting from the beginning of the subcase (5.0) until time=7.2. Ramping the removal of initial penetrations can help convergence. If TZPENE=0.0, the removal of initial penetrations is not ramped, but instead, they are eliminated immediately at the start of the subcase.

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Table 10-44: SOL 401 - Stiffness Parameters

Name	Description
AUTOSCAL*	Scales the automatically calculated normal penalty factor PENN either up or down. Will not scale explicitly defined value of PENN. AUTOSCAL can be used to scale the stiffness of specific contact pairs if convergence issues occur. (Real>0.0; Default=1.0). See Remark 2.
TANSCL*	Scales the automatically calculated tangential penalty factor PENT either up or down. Will not scale explicitly defined value of PENT. TANSCL can be used to scale the stiffness of specific contact pairs if convergence issues occur. (Real>0.0; Default=1.0). See Remark 2.
PENN*	Penalty factor for normal direction. PENN and PENT are automatically calculated by default. (Real>0.0; Default=Automatically Calculated) See Remark 2. When PENT is defined but PENN is undefined, PENN = 10 * PENT.
PENT*	Penalty factor for transverse direction. PENN and PENT are automatically calculated by default. (Real>0.0; Default=Automatically Calculated) See Remark 2. When PENN is defined but PENT is undefined, PENT = PENN / 10.
PENTYP*	Changes the unit definition for PENN and PENT. (Integer; Default=1) See Remark 2. 1- PENN and PENT are entered as units of 1/Length. 2 - PENN and PENT are entered as units of Force/(Length x Area).
OPNSTF*	Open contact stiffness scale factor. (Real ≥ 0.0; Default = 1.0E-6) The open contact stiffness is computed by OPNSTF * closed stiffness. See Remark 11.
PENADAPT*	Option to adaptively vary the contact stiffness between iterations. (Integer; Default=0) 0 – Penalty factor is not adaptively modified by the program. 1 – Penalty factor is adaptively modified. 2 - Penalty factor is adaptively modified in a range larger than PENADAPT=1. Generally, PENADAPT=1 is a good choice for most problems. However, for some cases you may need a wider range of penalty factors. For these cases, you should consider using PENADAPT=2.

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Table 10-45: SOL 401 - Friction Parameters

Name	Description
FRICMOD*	Friction behavior selection. (Integer; Default=1) 0 - The tangential stiffness is fixed throughout the subcase. You specify the friction coefficient FRICi on the BCTSET entry. 1 - The tangential stiffness is adaptively modified at every iteration as a function of contact pressure. The tangential stiffness is computed as: (FRICi * contact pressure) / SCRIT where FRICi is the coefficient of friction defined on the BCTSET entry, and SCRIT is the critical slip parameter defined on the BCTPARM entry. 2 - Friction is computed using the following product which optionally depends on time, velocity or temperature: μ(t,T,v)= FPARA1 * CFVE(v) * CFTE(T) * CFNF(t) FPARA1, CFNF, CFTE, and CFVE are all parameters defined on the BCTPARM entry. FRICi on the BCTSET entry is ignored. FPARA1 defines a constant friction value, and CFNF, CFTE, and CFVE are integer inputs which reference TABLEDi entries defining friction as a function of time, temperature, and velocity, respectively. If any of CFNF, CFTE, or CFVE are undefined, their contribution is ignored. 3 - Two different friction coefficients are used depending on the sliding velocity. This friction model uses the parameters VCRIT, FPARA1, and FPARA2 on the BCTPARM entry. FRICi on the BCTSET entry is ignored. VCRIT defines the critical sliding velocity, FPARA1 (1 in the image below) defines the friction coefficient when the sliding velocity < VCRIT, and FPARA2 (2 in the image below) defines the friction coefficient when the sliding velocity > VCRIT.

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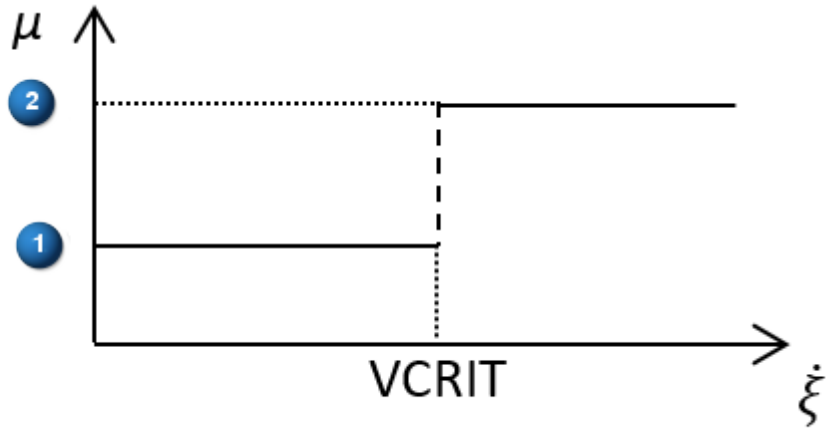
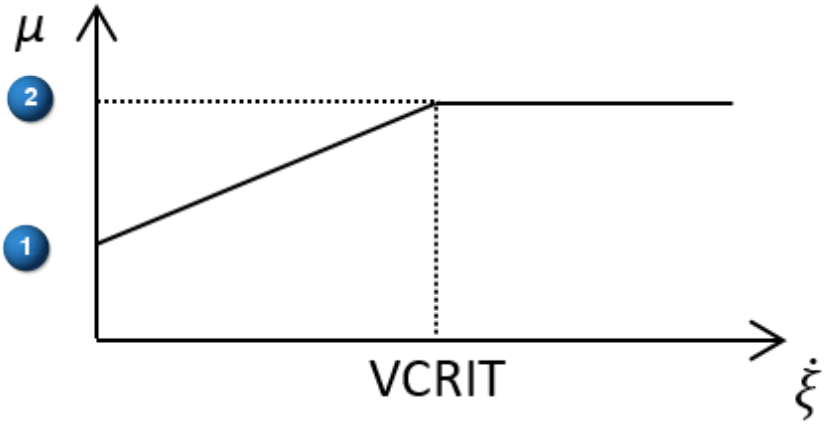
Name	Description



4 - Friction coefficient varies linearly with the sliding velocity.

This friction model uses the parameters FPARA1, VCRIT, and FPARA2 on the BCTPARM entry. FRICi on the BCTSET entry is ignored.

FPARA1 defines the friction coefficient at zero velocity, VCRIT defines the critical sliding velocity, and FPARA2 defines the friction coefficient at the critical sliding velocity. The friction varies linearly between FPARA1 (1 in the image below) and FPARA2 (2 in the image below), up to the critical sliding velocity VCRIT after which it remains constant.



SCRIT* Defines the critical slip when FRICMOD=1. (Real≠0.0; Default=0.005)

For SCRIT > 0.0: Critical slip = SCRIT * characteristic length for each pair.

For SCRIT < 0.0: Critical slip = ABS(SCRIT) for each pair.

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Name	Description
FPARA1*	Friction coefficient when FRICMOD = 3 or 4. (Real ≥ 0.0; No default)
FPARA2*	Friction coefficient when FRICMOD = 3 or 4. (Real ≥ 0.0; No default)
VCRIT*	Critical velocity when FRICMOD = 3 or 4. (Real > 0.0; No default)
CFVE*	ID of TABLEDi entry defining the friction coefficient as function of sliding velocity when FRICMOD = 2. (Integer)
CFNF*	ID of TABLEDi entry defining the friction coefficient as function of time when FRICMOD = 2. (Integer)
CFTE*	ID of TABLEDi entry defining the friction coefficient as function of temperature when FRICMOD = 2. (Integer)
FRICDLY *	Option to delay contact friction to help alleviate convergence problems. (Integer; Default=0) 0 - Friction is not delayed. 1 - Friction is delayed and applied in the subsequent step after the contact has been established.

Table 10-46: SOL 401 - Displacement Parameters

Name	Description
DISCAL	Displacement scaling option. (Integer; Default=1) 0 - No scaling will be done. 1 - Scaling will be done if required during every iteration. A check will be performed after every displacement increment to see if the incremental displacement would cause penetration between the source and target regions. If the penetrations exceed DISTOL, the entire incremental displacement will be scaled back to limit the penetrations in the model.
DISTOL	Tolerance for displacement scaling feature. (Real>0.0; Default = 1.0* characteristic length)
DISP*	Selects the small or large sliding contact formulation. (Integer; Default=0) 0 – Contact pairing is automatically updated by the software based on the amount of relative sliding in the pair. 1 – Small sliding formulation. Contact pairing is not updated with sliding displacements and is based on initial geometry.

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Name	Description
	2 – Large sliding formulation. Contact pairing is updated with sliding displacements. Note: When DISP=1, GUPDATE is internally set to 0 (no pairing update). When DISP=2, GUPDATE is set to 4 (update every iteration). When DISP=0 (AUTO), GUDPATE is internally set to 3 for small displacement solutions (LGDISP = -1), and GUDPATE is set to 2 for large displacement solutions (LGDISP = 1).

Table 10-47: SOL 401 - Sliding Parameters

Name	Description
GUPDATE*	Geometry update flag. (Integer; Default=2 when PARAM, LGDISP=1; Default=3 when PARAM, LGDISP=-1) 0 - Contact geometry updates will not be done during the analysis. 1 - Contact geometry updates occur when SLIP > (GUPTOL *Average element length). 2 - Contact geometry updates occur at the start of each step and when SLIP > (GUPTOL * Average element length). 3 - Contact geometry updates occur once a step. 4 - Contact geometry updates occur every iteration.
GUPTOL*	Geometry update tolerance. (Real>0.0; Default = 0.1 * characteristic length) If the relative sliding distance between the source and target regions exceeds this tolerance, a geometry update will be initiated with large displacement.

Table 10-48: SOL 401 - Convergence Parameters

Name	Description
CNTCONV	Contact convergence criteria. (Integer; Default=1) 1 - The contact convergence criteria is based on PTOL. 2 - The convergence criteria is based on CTOL.
PTOL*	Contact penetration tolerance used when CNTCONV=1. (Real≠0.0; Default = 0.01 *characteristic length)

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Name	Description
	If the contact penetrations exceed the penetration tolerance, an extra augmentation loop is performed. If the penetrations are below this tolerance, the augmentation loop is considered converged. In addition, if the global solution convergence criteria is satisfied, then the time step is considered converged. For PTOL > 0.0: Penetration tolerance = PTOL * characteristic length for each pair. For PTOL < 0.0: Penetration tolerance = ABS(PTOL) for each pair. See SOL 401 - Contact convergence in the *Simcenter Nastran Multi-Step Nonlinear User's Guide*.
CTOL	Contact augmentation traction convergence used when CNTCONV=2. (Real >0.0; Default=0.05) The augmentation loop convergence criteria can be based on traction convergence. See SOL 401 - Contact convergence in the *Simcenter Nastran Multi-Step Nonlinear User's Guide*.
DPTOL	Contact relative displacement tolerance. (Real >0.0; Default=0.05) The software only enforces this check for convergence if you did not request the displacement convergence criteria by including "U" with the CONV parameter on the NLCNTL bulk entry. See SOL 401 - Contact convergence in the *Simcenter Nastran Multi-Step Nonlinear User's Guide*.
RCTOL	Iterative contact force convergence. (Real>0.0; Default = 0.05) See SOL 401 - Contact convergence in the *Simcenter Nastran Multi-Step Nonlinear User's Guide*.
RCONSM	Eliminates contact pairs with a small force magnitude from the pair based iterative contact force convergence check. (Real>0.0; Default = 0.01) You request the pair based force convergence check by defining the system cell setting **SYSTEM(773)=1**. The software removes pairs from the pair based force convergence check when their force magnitude is smaller than the product of (RCONSM x maximum force magnitude among all pairs). See SOL 401 - Contact convergence in the *Simcenter Nastran Multi-Step Nonlinear User's Guide*.
DCTOL	Stabilization damping traction removal convergence used when the stabilization damping option is requested with the CTDAMP parameter. (Real > 0.0; Default=0.01)

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Name	Description
	When the stabilization damping option is requested, the software will ramp the damping tractions down as it iterates at a solution point until convergence is satisfied. See SOL 401 - Contact convergence in the *Simcenter Nastran Multi-Step Nonlinear User's Guide*.
MAXS	Maximum number of augmentation (outer) loops. If the augmentation loop has not converged in MAXS number of iterations, the solution will proceed to the next step if the global convergence criteria has been met. Setting MAXS=1 selects a pure penalty formulation. (Integer > 0; Default = 20)

Table 10-49: SOL 401 - Stabilization Parameters

Name	Description
CTDAMP*	The CTDAMP parameter is available to request the stabilization damping option when you are relying on the contact condition to prevent rigid body conditions, but the contact condition is not active. (Integer; Default=0) See Remark 11. 0 - No stabilization damping. 1 - Stabilization damping applied only in the first subcase and it is ramped down to zero by the end of the subcase. The stabilization damping is only applied this way when the entire pair status is open. 2 - Stabilization damping applied always as long as the pair status is open. 3 - Stabilization damping applied always regardless of the time and the pair status.
CTDAMPN*	If you request stabilization damping with the CTDAMP parameter, the CTDAMPN parameter is available to either scale the normal damping value which the software automatically computes, or to define the normal damping value explicitly. (Real≠0.0 or Integer; Default=1.0; units = pressure/velocity) For CTDAMPN > 0.0, the normal damping value = CTDAMPN * automatically computed damping value. For CTDAMPN < 0.0, the normal damping value = ABS(CTDAMPN). Alternately, CTDAMPN can be an integer value that selects a TABLEDi entry defining a normal damping coefficient as a function of the normal penetration distance. The table ordinate unit is Pressure/Velocity, and the abscissa unit is Length.
CTDAMPT*	If you request stabilization damping with the CTDAMP parameter, the CTDAMPT parameter is available to either scale the tangential damping value which the software

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Name	Description
	automatically computes, or to define the normal damping value explicitly. (Real≠0.0 or Integer; Default=1.0; units = pressure/velocity) For CTDAMPT > 0.0, the tangential damping value = CTDAMPT * automatically computed damping value. For CTDAMPT < 0.0, the tangential damping value = ABS(CTDAMPT). Alternately, CTDAMPT can be an integer value that selects a TABLEDi entry defining a tangential damping coefficient as a function of the normal penetration distance. The table ordinate unit is Pressure/Velocity, and the abscissa unit is Length.
OPNTOL*	Open gap tolerance scale factor. (Real≠0.0; Default=1.0) For OPNTOL > 0.0, the open contact stiffness (OPNSTF * closed stiffness) is applied to the contact elements that have a gap value less than or equal to OPNTOL * characteristic length, but greater or equal than 0.0. The contact element stiffness is 0.0 if the gap is greater than OPNTOL * characteristic length. For OPNTOL < 0.0, the open contact stiffness (OPNSTF * closed stiffness) is applied to the contact elements that have a gap value less than or equal to ABS(OPNTOL), but greater or equal than 0.0. The contact element stiffness is 0.0 if the gap is greater than ABS(OPNTOL).

* Can be defined on local BCTPARM entries. The BCTPARM bulk entries associated to individual BCTSET bulk entries, which are then combined with a BCTADD bulk entry, define local parameters. A local parameter definition overrides a global definition.

See “Contact Control Parameters – BCTPARM” in the *Simcenter Nastran User’s Guide* for more information on the BCTPARM options.

Remarks:

1. In SOLs 101, 103, 111, and 112, all of the parameters are supported for surface-to-surface contact definitions. For edge-to-edge contact definitions, the parameters CTOL, MAXF, MAXS, NCHG, INIPENE, PENN, PENT, PENTYP, RESET, CSTRAT, PREVIEW, TANSCL, and AUTOSCAL are supported.
2. The penalty factors PENN and PENT are automatically calculated by default. The automatic calculation is turned off if either PENN or PENT are defined.

   When PENTYP=1 (default) is defined, PENN and PENT have units of 1/(Length), and the contact element stiffness is calculated by K = e*E*dA where e represents PENN or PENT, E is the modulus value, and dA is area. A physical interpretation is that it is equivalent to the axial stiffness of a rod with area dA, modulus E, and length 1/e.

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When PENTYP=2 is defined, PENN and PENT become a spring rate per area Force/(Length x Area), and the contact element stiffness is calculated as K=e*dA. The spring rate input is a more explicit way of entering contact stiffness since it is not dependent on the modulus value.

The penalty factors influence the rate of convergence, and to a lesser extent, the accuracy of the contact solution. The automatic penalty factor calculation works well for most instances, but manual adjustments may be necessary, particularly if a contact problem fails to converge. See "Tips for Setting PENN and PENT" in the *Simcenter Nastran User's Guide* for tips on adjusting penalty factors.

3. In SOLs 101, 103, 111, and 112, if NCHG is a real number and is < 1.0, the software treats it as a fraction of the number of active contact elements in each outer loop of the contact algorithm. The number of active contact elements is evaluated at each outer loop iteration.

   If NCHG is an integer ≥ 1, the value defines the allowable number of contact changes.

   If NCHG = 0, no contact status changes can exist.

   Consider defining a lower NCHG value than the default when a large number of pairs are defined in a "stack up" type of configuration where the cumulative effect of a small contact element status change within some of the pairs will impact the contact element status of the other pairs.

4. A higher number of contact evaluation points can be used to increase the accuracy of a contact solution. Inaccuracies sometimes appear in the form of nonuniform contact pressure and stress results. There may be a penalty associated with using more evaluation points since the time for a contact problem to converge may be longer. The table below shows how the number of contact evaluation points is dependent on the element type, and how it can be adjusted using the INTORD option. The "Face Type" column applies to shell elements, and to the solid element with the associated face type.

	Number of Contact Evaluation Points		
Face Type	INTORD=1	INTORD=2	INTORD=3
Linear Triangle	1	3	7
Parabolic Triangle	3	7	12
Linear Quad	1	4	9
Parabolic Quad	4	9	16

5. In SOLs 101, 103, 111, and 112, under certain conditions, all of the contact elements could become inactive which may lead to singularities. Setting the parameter CSTRAT=1 will reduce the likelihood of all contact elements becoming inactive.

6. In SOLs 101, 103, 111, and 112, setting the PREVIEW parameter to "1" requests a bulk data representation of the element edges and faces where contact elements are created. The software will write a bulk data file containing dummy shell element entries for face locations, and dummy PLOTEL entries for edge locations. Dummy GRID, property and material entries are also written. You can import the file into a preprocessor to display both source and target contact locations. The preview file has the naming convention

```
<input_file_name>_cnt_preview_<subcaseid>_<contactsetid>.dat
```


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7. The following applies to the initial contact condition* when INIPENE= 2 or 3:

   * The initial contact condition is the contact status before the solution iterates on the contact condition. This is the contact status before the solution has applied loads.

   - If shell element contact regions exist and SHLTHK=0 (default) on the BCTPARM entry, the shell element thickness is applied *before* the software evaluates any gaps or penetrations as a result of the INIPENE setting. For example, when INIPENE=2, penetrations are reset to a new initial condition in which there is no interference. This includes the penetrations as a result of the shell thickness. When INIPENE=3, since penetrations and gaps are reset to a new initial condition, the shell thickness is not considered by the contact condition.
   - If shell element contact regions exist, and a shell element offset is defined with the ZOFF field on the element entry, and ZOFFSET=0 (default) on the BCTPARM entry, the shell element offset is added *after* the software removes any gaps or penetrations as a result of the INIPENE setting.

8. If an offset is defined with the OFFSET parameter on the BCTPARM or BCRPARA entry, the region offset is added *after* the software removes any gaps or penetrations as a result of the INIPENE setting.

9. In SOLs 101, 103, 111, and 112, when a bolt preload is defined on CBEAM or CBAR elements and contact is in the model, the software runs an initial preload solution to compute the strain in the bolt before applying any service loads. When RESET=1, for all consecutive solutions for the same subcase as the preload or for consecutive subcases, the software applies the computed bolt strain from the initial preload solution, and it uses the final contact status from the initial bolt preload solution as the starting contact status.

   In addition, if the set ID of any one of the BOLTLD, SPC, or MPC case control commands changes from one subcase to the next, the contact conditions will be recomputed in the new subcase. That is, the contact status in the new subcase will not be the final contact status from the previous subcase. This will occur regardless of how the RESET parameter is defined on the BCTPARM bulk entry.

10. In SOLs 101, 103, 111, and 112, if the set ID of any one of the BOLTLD, SPC, or MPC case control commands changes from one subcase to the next, the contact conditions will be recomputed in the new subcase. That is, the contact status in the new subcase will not be the final contact status from a previous subcase. This will occur regardless of how the RESET parameter is defined.

11. In SOL 401, you should specify either contact damping with the CTDAMP parameter, or the open stiffness value with the OPNSTF parameter, but not both. If both are defined, then the open stiffness is set to 0.0 and the damping is used.

Related Topics

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BCTSET

Contact Set Definition

Defines contact pairs of a 2D or 3D contact set (SOLs 101, 103, 105, 108, 111, 112, 401, and 402).

Format:

1	2	3	4	5	6	7	8	9	10
BCTSET	CSID	SID1	TID1	FRIC1	MIND1	MAXD1	DID		
		SID2	TID2	FRIC2	MIND2	MAXD2			
		-etc-							

SOL 101 Example:

BCTSET	4	3	2	0.0	0.0	0.01			
		1	8	0.0	-0.001	0.005			

SOL 402 Example:

BCTSET	4	3	2	0.0	0.0	0.01	111		
		1	8	0.0	-0.001	0.005			

Fields:

Field	Contents
CSID	Contact set identification number. (Integer > 0)
SIDi	Source region (contactor) identification number for contact pair i. (Integer > 0)
TIDi	Target region identification number for contact pair i. (Integer > 0)
FRICi	Static coefficient of friction for contact pair i. (Real; Default=0.0)
MINDi	Minimum search distance for contact. (Real; Default=0.0) See Remark 6. See SOL 402 Remark 1.
MAXDi	Maximum search distance for contact. (Real; Default=0.0) See Remark 6. See SOL 402 Remark 1.
DID	Identification number of a DESC bulk entry, which includes a contact set description. Only supported by SOL 402.

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Remarks:

1. CSID is selected by the case control command BCSET. If BCSET is not specified, the model will not include contact.
2. The following table summarizes all supported contact conditions.

Table 10-50: Contact Summary

Type	Description	Solution Support
Edge-to-Edge	Contact between the edges of axisymmetric elements (CTRAX3, CQUADX4, CTRAX6, CQUADX8), plane stress elements (CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8), and plane strain elements (CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8).	Solution 101 and consecutive solutions 103, 105, 108, 111, and 112 (elements in XZ or XY plane). Solutions 401 and 402.
Surface-to-Surface	Contact between shell element faces (CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, CQUADR) and solid element faces (CHEXA, CPYRAM, CPENTA, CTETRA).	Solution 101 and consecutive solutions 103, 105, 108, 111, and 112. Solution 401. Solution 402 (solid element faces only: CHEXA, CPYRAM, CPENTA, CTETRA).

3. The parameters for a contact set are defined by the BCTPARM entry.
4. SID and TID are contact regions defined via BCPROP, BCPROPS, BSURF, or BSURFS entries for 3D contact, and with BEDGE or BLSEG entry for 2D contact.
5. SIDi can be equal to TIDi to define a self-contacting pair.
6. The following rules apply to MINDi and MAXDi:
   - Both default to “0”, but at least one must be explicitly defined as non-zero.
   - MAXDi must be greater than MINDi (MINDi<MAXDi).
   - Negative values are allowed.
7. A converged SOL 101 contact condition can be included in a consecutive normal mode solution (SOL 103). The normal mode solution results, which included the SOL 101 contact conditions, can then be used in an optional dynamic response calculation (SOLs 111 and 112).

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Similarly, a converged SOL 101 contact condition can be included in a consecutive linear buckling solution (SOL 105), or in a direct frequency response solution (SOL 108).

See the chapter on contact in the *Simcenter Nastran User's Guide* for more information.

8. When defining contact regions and pairs on geometry which is not tangent continuous, creating single contact regions which cross corner transitions can result in nonuniform stress results around the corners. It is recommended to break these areas into multiple regions and pairs. See the chapter on contact in the *Simcenter Nastran User's Guide* for more information.
9. Contact definitions defined on composite solid faces which are perpendicular to the stack direction (edge faces) may produce poor stress continuity. If the contact definition is between edge faces belonging to different PCOMPS definitions, and if the number of plies on each PCOMPS definition is small and the same, and the ply thicknesses are similar, the stress continuity should be fairly smooth. This also applies to the results requested with the BCRESULTS case control command.
10. If the contact solution is having problems converging, the fields on the BCTPARM bulk data entry can be modified from their defaults. See the BCTPARM bulk data entry, and the chapter on surface contact in the Simcenter Nastran User's Guide for more information.
11. Contact conditions can be included in SMP and RDMODES solutions. See "Recursive Domain Normal Modes Analysis (RDMODES)" in the *Parallel Processing Guide*. Contact conditions cannot be included in a GDMODES, FDMODES, or HDMODES DMP solution.
12. Multiple contact sets (BCTSETs) with unique CSID's can be combined by including their CSID's on a BCTADD entry. The BCTADD entry has its own unique CSID which will be used in the BCSET case control command. The reason to have multiple contact sets with their own CSID is when different parameters (BCTPARM) are needed for different contact sets.
13. If multiple eigenvalue subcases (solutions 103 or 105) include STATSUB commands to select linear static subcases, and the linear static subcases include glue and/or linear contact definitions, the BCSET and/or BGSET case control commands must be specified in the global case. Failure to do this can result in erroneous results.
14. Contact conditions can be included in a model with main bulk data type superelements, but the contact regions must be in the residual. Contact conditions are not supported in a model with partitioned type superelements. Main bulk data and partitioned type superelements are both described in Chapter 2 of the *Simcenter Nastran Superelement User's Guide*.
15. You cannot restart from a solution which included contact conditions. You cannot add contact conditions when restarting.
16. The combination of a bolt preload in which the bolt is meshed with solid elements, the element iterative solver, and contact conditions is not supported.

Remarks related to SOL 402:

1. When you define the ADJUST parameter on the BCTPAR2 bulk entry, the software uses the minimum and maximum search distance (MINDi and MAXDi). The software searches starting at the source side element face in the normal direction towards the target grid points. The software creates a contact element when the target grid points are within the MINDi and MAXDi search range.

   By default for SOL 402, MINDi = -(characteristic distance/5.0) and MAXDi= +(characteristic distance/5.0), where the software computes the characteristic distance as 5.0*MAXLEN, and MAXLEN is

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maximum edge length from all target element faces in a contact pair. The software reports the computed characteristic distance for each contact pair in the .f06 file.

You must define MAXDi greater than MINDi. You can define MINDi as a negative value when there is an interference condition modeled as overlapping regions. SOL 402 ignores MINDi and MAXDi when you have not defined the ADJUST parameter.

Related Topics

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BDYOR

CHBDYi Entry Default Values

Defines default values for the CHBDYP, CHBDYG, and CHBDYE entries.

Format:

1	2	3	4	5	6	7	8	9	10
BDYOR	TYPE	IVIEWF	IVIEWB	RADMINF	RADMIDB		PID	GO	
	CE	E1	E2	E3					

Example:

BDYOR	AREA4	2	2	3	3		10		

Fields:

Field	Contents
TYPE	Default surface type. See Remark 2 (Character).
IVIEWF	Default identification number of the front VIEW entry. (Integer > 0 or blank)
IVIEWB	Default identification number of the back VIEW entry. (Integer > 0 or blank)
RADMIDF	Default identification number of a RADM entry for the front face. (Integer ≥ 0 or blank)
RADMIDB	Default identification number of a RADM entry for the back face. (Integer ≥ 0 or blank)
PID	Default PHBDY property entry identification number. (Integer > 0 or blank)
GO	Default orientation grid point. (Integer ≥ 0; Default=0)
CE	Default coordinate system for defining the orientation vector. (Integer ≥ 0 or blank)
E1, E2, E3	Default components of the orientation vector in coordinate system CE. The origin of this vector is grid point G1 on a CHBDYP entry. (Real or blank)

Remarks:

1. Only one BDYOR entry may be specified in the Bulk Data Section.
2. TYPE specifies the type of CHBDYi element surface; allowable values are: POINT, LINE, REV, AREA3, AREA4, ELCYL, FTUBE, AREA6, AREA8, and TUBE.
3. IVIEWF and IVIEWB are specified for view factor calculations only. (See the VIEW entry).

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4. GO is only used from BDYOR if neither GO nor the orientation vector is defined on the CHBDYP entry and GO is > 0.
5. E1, E2, E3 is not used if GO is defined on either the BDYOR entry or the CHBDYP entry.

Related Topics

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BEAMOR

CBEAM Entry Default Values

Defines default values for field 3 and fields 6 through 8 of the CBEAM entry.

Format:

1	2	3	4	5	6	7	8	9	10
BEAMOR		PID			X1	X2	X3	OFFT	

Example:

BEAMOR		39			0.6	2.9	-5.87		

Alternate Format and Example:

BEAMOR		PID			G0			OFFT	

BEAMOR		39			86				

Fields:

Field	Contents
PID	Property identification number of the PBEAM or PBCOMP entry. (Integer > 0 or blank)
X1, X2, X3	Components of orientation vector $\vec{v}$, from GA, in the displacement coordinate system at GA. (Real)
G0	Alternate method to supply the orientation vector $\vec{v}$ using grid point G0. Direction of $\vec{v}$ is from GA to G0. $\vec{v}$ is then transferred to end A. (Integer > 0; G0 ≠ GA or GB)
OFFT	Three character input which designates the coordinate systems used to define the cross section orientation vector and the two end offset vectors. Not supported by SOLs 402 and 414. See Remark 5. (Character; Default="GGG")

Remarks:

1. The contents of fields on this entry will be assumed for any CBEAM entry with corresponding fields that are blank.
2. Only one BEAMOR entry is allowed.

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3. For an explanation of beam element geometry, see the CBEAM entry description.
4. If X1 or G0 is an integer, G0 is used. If X1 or G0 is blank or real, then X1, X2, X3 is used.
5. The OFFT field is a three character input that designates the coordinate systems (CSYS) used to define the cross section orientation vector and the two end offset vectors. With the default of "GGG", a displacement (global) CSYS is used for all three.

   - The first character defines the CSYS used for the cross section orientation vector. When "G" is defined as the first character, the displacement (global) CSYS of grid point A is used. When "B" is defined as the first character, the basic CSYS is used.
   - The second character defines the CSYS used for the offset vector at end A. When "G" is defined as the second character, the displacement (global) CSYS of grid point A is used. When "O" is defined as the second character, the element CSYS is used. See CBEAM remarks for a definition of the element CSYS.
   - The third character defines the CSYS used for the offset vector at end B. When "G" is defined as the third character, the displacement (global) CSYS of grid point B is used. When "O" is defined as the third character, the element CSYS is used. See CBEAM remarks for a definition of the element CSYS.

   The table below lists the supported character inputs. A fatal error will occur if you enter a character input that is not listed.

Input	Orientation CSYS	Offset End A CSYS	Offset End B CSYS
GGG	Global[A]	Global[A]	Global[B]
BGG	Basic	Global[A]	Global[B]
GGO	Global[A]	Global[A]	Offset
BGO	Basic	Global[A]	Offset
GOG	Global[A]	Offset	Global[B]
BOG	Basic	Offset	Global[B]
GOO	Global[A]	Offset	Offset
BOO	Basic	Offset	Offset

   A: The displacement (global) CSYS of grid point A is used.

   B: The displacement (global) CSYS of grid point B is used.
6. If G0 is used to define the orientation vector and the OFFT field is defined, the software will ignore the first character defined in the OFFT field, and will use the second and third characters as described in Remark 5. Three characters are still required in the OFFT field for this case.

Related Topics

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BEDGE

Defines a Region of Element Edges

Defines a region of element edges for glue (BGSET entry), contact (BCTSET entry), fluid pressure load (PLOADFP entry), or wetted edges for a Co-simulation (CSMSET entry).

Format:

1	2	3	4	5	6	7	8	9	10
BEDGE	ID	EID1	GID1,1	GID1,2		EID2	GID2,1	GID2,2	
		EID3	GID3,1	GID3,2		-etc-			

Example:

BEDGE	14	101	15	190		5	42	65	
		23	57	82		16	132	4	
		94	255	26		18	39	97	

Fields:

Field	Contents
ID	Edge region identification number. (Integer > 0) See Remark 1.
EIDi	Element ID. (Integer > 0)
GIDi,1	First grid point identification number for EIDi. (Integer > 0) See Remark 2.
GIDi,2	Second grid point identification number for EIDi. (Integer > 0) See Remark 2.

Remarks:

1. The ID must be unique with respect to all other BEDGE, BLSEG, BSURFS, BSURF, BCPROP, and BCPROPS entries.
2. With edge-to-surface or edge-to-edge gluing:
   - With edge-to-surface gluing, a shell edge region defined with the BEDGE entry is glued to a face region defined with a BSURF, BSURFS, BCPROP, or BCPROPS entry.
   - With edge-to-edge gluing, an edge region defined with the BEDGE entry is glued to another edge region defined by a BEDGE entry or by a BLSEG entry.
   - All glue pairs are defined with the BGSET entry.

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- The grid point IDs on the BEDGE entry used to define a glue edge region must be included in the connectivity of the shell elements CQUAD4, CQUADR, CQUAD8, CTRIA3, CTRIAR, CTRIA6, the axisymmetric elements CTRAX3, CQUADX4, CTRAX6, CQUADX8, the plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8, or the plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8. Axisymmetric, plane stress and plane strain elements can all be paired. Edges on 3D solid elements are not supported.

3. With edge-to-edge contact:
   - Edge-to-edge contact can be defined between the edges of axisymmetric elements, plane stress elements, and plane strain elements. Shell elements are not supported. The BEDGE entry is paired to another edge region defined by a BEDGE entry or by a BLSEG entry. The edge-to-edge contact pairs are defined with the BCTSET entry.
   - The grid point IDs on the BEDGE entry used to define a contact edge region must be included in the connectivity of the axisymmetric elements CTRAX3, CQUADX4, CTRAX6, CQUADX8, of the plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8, or the plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8. Note that shell element edges and 3D solid element edges are not supported.
4. BEDGE can be used in SOLs 401 and 402 to define a region of element edges that can potentially receive a fluid penetrating pressure load defined with the PLOADFP bulk entry.
5. BEDGE can be used to define the wetted edges in the context of SOL 401 Multistep Nonlinear STAR CCM+ Co-simulation. Wetted edges represent structural edges that become a fluid boundary in the SOL 401 Multiphysics Co-Simulation with STAR CCM+.
6. For edge-to-edge glue with heat transfer, the plane stress, plane strain, axisymmetric solids, and shell elements are supported. These elements can be combined with any permutation. Note that 3D solid element edges are not supported.
7. For parabolic elements, GIDi,1 and GIDi,2 would represent the vertex/corner grids.
8. The generalized plane strain element used in SOL 401 is not supported in a glue or contact region.

Remark related to SOL 402

Edge-to-edge contact can be defined between the edges of shells.

Related Topics

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BFRIC

Slideline Contact Friction

Defines frictional properties between two bodies in contact.

Format:

1	2	3	4	5	6	7	8	9	10
BFRIC	FID		FSTIF	MU1					

Example:

BFRIC	33			0.3					

Fields:

Field	Contents
FID	Friction identification number. See Remark 1 (Integer > 0).
FSTIF	Frictional stiffness in stick. See Remarks 2 and 3 (Real > 0.0; Default=automatically selected by the program.)
MU1	Coefficient of static friction. (Real > 0.0)

Remarks:

1. This identification number must be unique with respect to all other friction identification numbers. This is used in the FRICID field of the BCONP Bulk Data entry.
2. The value of frictional stiffness requires care. A method of choosing its value is to divide the expected frictional strength (MU1 * expected normal force) by a reasonable value of the relative displacement that may be allowed before slip occurs. The relative value of displacement before slip occurs must be small compared to expected relative displacements during slip. A large stiffness value may cause poor convergence, while too small a value may cause poor accuracy.

   Frictional stiffness specified by the user is selected as the initial value. If convergence difficulties are encountered during the analysis, the frictional stiffness may be reduced automatically to improve convergence.
3. The stiffness matrix for frictional slip is unsymmetrical. However, the program does not use the true unsymmetrical matrix, but uses the symmetric terms instead. This is to avoid using the unsymmetrical solver to reduce CPU time.

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Related Topics

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BGADD

Surface-to-Surface Glue Set Combination

Defines a surface-to-surface glue set as a union of glue sets defined on BGSET entries.

Format:

1	2	3	4	5	6	7	8	9	10
BGADD	GSID	SI	S2	S3	S4	S5	S6	S7	
	S8	S9	-etc-						

Example:

BGADD	12	1	3	6	5	9			

Fields:

Field	Contents
GSID	Glue set identification number. (Integer > 0)
Si	Identification numbers of glue sets defined via BGSET entries. (Integer > 0)

Remarks:

1. Multiple glue sets (BGSETs) with unique GSIDs can be combined by including their GSIDs on a BGADD entry. The BGADD entry has its own unique GSID which is used on the BGSET case control command. Multiple glue sets with their own GSID are necessary when different parameters (BGPARM) are needed for different glue sets.
2. To include several glue sets defined via BGSET entries in a model, BGADD must be used to combine the glue sets. GSID on BGADD is then selected with the BGSET case control command.
3. Si must be unique and may not be the identification of this or any other BGADD entry.
4. Glue definitions are supported in all solution sequences except for SOLs 144–146. In a SOL 153/159 heat transfer analysis, glue definitions are treated as a constant near infinite conductivity connection.

Related Topics

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BGPARM

Glue Parameters

Control parameters for the glue algorithm.

Format:

1	2	3	4	5	6	7	8	9	10
BGPARM	GSID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4	PARAM5	VALUE5	-etc.-				

Example:

BGPARM	4	INTORD	2	PENN	10.0	PENT	10.0		
	REFINE	0							

Fields:

Field	Contents
GSID	Glue set ID. Parameters defined in this command apply to glue set GSID defined by a BGSET entry. (Integer > 0)
PARAMi	Name of the BGPARM parameter. Allowable names are given in the parameter listing below. (Character)
VALUEi	Value of the parameter. See below for the parameter listing. (Real or Integer)

Table 10-51: BGPARM Parameters:

Name	Description
GLUETYPE*	Selects the glue formulation for surface-to-surface glue. (Default=2) See Remark 2. Edge-to-surface and edge-to-edge glue pairs always use GLUETYPE=2. 1 - Normal and tangential springs will be used to define the connections. See Remark 1. 2 - A “weld like” connection will be used to define the connections.
PENTYP*	Changes how glue element stiffness and “conductance” are calculated. (Default=1) See Remark 2.
PENN*	Penalty factor for normal direction when GLUETYPE=1. (Default=100.0 when PENT is undefined; Default=10.0 * PENT when PENT is defined) See Remarks 1 and 2.

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Name	Description
PENT*	Penalty factor for transverse direction when GLUETYPE=1. (Default=0.1 * PENN) See Remarks 1 and 2.
PENGLUE*	Penalty factor when GLUETYPE=2. (Default=1) See Remarks 1 and 2.
INTORD	Determines the number of glue points for a single element face on the source region. Edge-to-surface glue pairs always use INTORD=3. 1 - Low order 2 - Medium order (default) 3 - High order
REFINE	Requests that the software refine the mesh on the source region during the solution to be more consistent with the target side mesh. 0 - Refinement does not occur. 2 - Refinement occurs. (default).
PREVIEW	Requests the export of a bulk data representation of the element edges and faces where glue elements are created. See Remark 6. 0 - The bulk data export does not occur (default). 1 - The bulk data export occurs.
ESOPT	Changes the edge-to-surface glue stiffness. Shell theory does not account for changes in shell thickness or normal strains perpendicular to the plane of the shell element. The ESOPT parameter gives you a choice of how to handle the linking of the zero normal strains which exist in the shell element associated with the edge, to the surface being glued. 0 - Strains in the plane of the surface being glued in the direction perpendicular to the edge are not constrained by the glue stiffness (default). 1- Strains in the plane of the surface being glued in the direction perpendicular to the edge are constrained by the glue stiffness.
SLIDE*	Requests the sliding glue in SOL 401. Sliding glue includes normal stiffness but no tangential stiffness. Gaps between the glued surfaces are preserved as small sliding occurs. 0 - Sliding glue is off (default). 1 - Sliding glue is on.

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Name	Description
	Note: If large sliding is desired, a contact condition with the INTRFC=4 defined on the BCTPARM bulk entry will bond the normal direction and permit large frictionless sliding. The sliding glue formulation is not supported with edge-to-edge or edge-to-surface glue conditions defined on 3D shell elements.
BGOPT**	Source node rotation constraint option in SOLs 402 and 414. 0 - The best option is automatically chosen according to the source and target regions defined for the glue pair (default). 1 - For the nodes in the source region, the rotations are not constrained. 2 - For the nodes in the source region, the rotations are constrained using target nodal rotations. This option can be used for instance when you glue two regions of shells. 3 - For the nodes in the source region, the rotations are constrained using target mean nodal displacements. This option can be used for instance when you glue shells to faces of volumes. See also **Allowed BGOPT options combinations**.

***** Can be defined on global and local BGPARM entries.

****** Can be defined *only* on local BGPARM entries.

Neither * or **: Can be defined *only* on a global BGPARM entry.

The BGPARM bulk entry, which uses the same GSID entered on the BGSET case control command, defines global parameters.

The BGPARM bulk entries associated to individual BGSET bulk entries, which are then combined with a BGADD bulk entry, define local parameters. If a parameter is defined as both global and local, the local parameter definition overrides the global definition.

Global and local input example:

```
CASE CONTROL
$GSID on the BGSET case control references GSID on BGADD
BGSET = 108
...
BULK DATA
$Global Glue Parameters
BGPARM 108 REFINE 0GLUETYPE 1
...
```


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```
$Local Glue Set definitions
BGSET 1 1 2
BGSET 2 3 4
5 6
...
$Local Glue Sets are combined with BGADD
BGADD 108 1 2
...
$Local Glue Parameters
BGPARM 1 PENN 80 PENT 80
BGPARM 2 PENN 110 PENT 110
...
```

Remarks:

1. The following table summarizes the parameters and solution support for the different glue types.

Glue Type	Parameter and Solution Support
Surface-to-Surface Glue	All parameters except ESOPT are supported. Surface-to-surface glue is supported in structural solutions, in heat transfer solutions, and for gluing fluid surfaces to other fluid surfaces in a coupled acoustics analysis. In a coupled acoustics analysis, GLUETYPE, PENTYP, and PENT are all ignored, and PENN is described in Remark 3.
Edge-to-Surface Glue	Only the parameters PENTYPE, PENGLUE, PREVIEW, and ESOPT are supported. GLUETYPE=2 formulation is always used. Edge-to-surface glue is supported in structural solutions except solution 402, and in heat transfer solutions.
Edge-to-Edge Glue	Only the parameters PENTYPE, PENGLUE, and PREVIEW are supported. GLUETYPE=2 formulation is always used. Edge-to-edge glue is supported in structural solutions, and in heat transfer solutions.

2. GLUETYPE has unique penalty factor inputs in structural solutions and in the heat transfer solutions. Heat transfer solutions ignore the GLUETYPE and PENT settings and internally use GLUETYPE=1. These inputs and units are described below.

GLUETYPE=2 (default for structural solutions)	The glue penalty stiffness is defined by PENGLUE. A physical interpretation of this glue connection is a beam like element. (Heat transfer analysis always uses GLUETYPE=1)	
Structural solutions:	PENTYP=1 (default)	PENGLUE is a unitless scale factor of the beam like element stiffness. The averaged modulus of the elements associated with the source side region are used when computing this stiffness.

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	PENTYP=2	PENGLUE is the beam like element modulus (E) with units of Force/Area.
Heat transfer solutions:	Always uses GLUETYPE=1. See Remark 4.	

GLUETYPE=1	The glue penalty stiffness is defined by PENN and PENT.	
	PENTYP=1	**Structural solutions**: PENN and PENT have units of 1/(length), and the glue element stiffness is calculated by K = e*E*dA where e represents PENN or PENT, E is an average modulus (averaged over elements associated with the source side region), and dA is area. A physical interpretation is that it is equivalent to the stiffness of a rod with area dA, modulus E, and length 1/e.
		Heat transfer solutions: PENT is ignored. PENN has the units of 1/(length), and "conductance" at the glue connection is calculated as $C = e*k_{avg}*dA*100$ where e represents PENN, k_{avg} is an average of the thermal conductivity (k) values defined on the MAT4 entries (averaged over elements associated with the source side region), and dA is area. A physical interpretation is that it is equivalent to the axial "conductance" of a rod with area dA, conductivity k_{avg}, and length 1/e. See Remark 4.
	PENTYP=2	**Structural solutions**: PENN and PENT become a spring rate per area Force/(Length x Area), and the glue element stiffness is calculated as K=e*dA. The spring rate input is a more explicit way of entering glue stiffness since it is not dependent on the average modulus.
		Heat transfer solutions: PENT is ignored. PENN has the units of (thermal conductivity*length)/area, and the "conductance" at the glue connection is calculated as C = e*dA, where e represents PENN, and dA is the area. See Remark 4.

3. When gluing fluid surfaces to other fluid surfaces for a coupled acoustics analysis, GLUETYPE, PENTYPE, PENT are all ignored, and PENN is used to calculate the acoustics penalty matrix K:

$$K = e * \frac{1}{\rho} * dA$$

where e is PENN, ρ is the average density of all fluid elements in the model, and dA is the surface

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area. The K matrix for an acoustic element is defined by

$$K = \frac{1}{\rho}\int_V \nabla N^T \nabla N \, dV$$

For a small fluid column (tube of length L and cross-sectional area dA), the K matrix can be written as

$$K = \frac{1}{L} * \frac{1}{\rho} * dA$$

Therefore, the penalty factor PENN (e) can be interpreted as 1/L.

Edge-to-surface glue pairs can not be used as acoustics glue connections.

4. Glue is supported in the heat transfer solutions 153 and 159, and in a SOL 101 heat transfer/structural combined solution.

   The SOL 101 combined solution requires both a heat transfer and a structure subcase. See the HEATSTAT parameter for information. Since these subcases represent different analysis types and glue has a different meaning in each, to define a unique set of glue parameters (penalty values) for each subcase, you must create unique glue sets for each subcase. Specifically, you must create a unique BGSET case control, BGSET bulk entry, and BGPARM bulk entry for each subcase. You select the same glue regions in each of the glue sets.

5. The BGPARM bulk entry is not required. When it is not present, the default values are used. At least one parameter should be defined when a BGPARM entry exists.

6. Setting the PREVIEW parameter to "1" requests a bulk data representation of the element edges and faces where glue elements are created. The software will write a bulk data file containing dummy shell element entries for face locations, and dummy PLOTEL entries for edge locations. Dummy GRID, property and material entries are also written. You can import the file into a preprocessor to display both source and target glue locations. The preview file has the naming convention

```
<input_file_name>_glue_preview_<subcaseid>_<gluesetid>.dat
```

Remarks for SOL 401:

1. SOL 401 only supports GLUETYPE=2. If you request GLUETYPE=1, the software will continue the solution using GLUETYPE=2.

Remarks for SOLs 402 and 414:

1. SOLs 402 and 414 only supports BGOPT. Other parameters are ignored.

   The glue is therefore always a perfect weld glue (GLUETYPE=2).

2. The following table describes the available BGOPT options according to the source and target regions defined in the glue set GSID (BGSET entry).

Glue Type	Source	Target	Allowed BGOPT values
Edge-to-Edge Glue	BEDGE	BEDGE	1

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Glue Type	Source	Target	Allowed BGOPT values
			2 0 (default); same as BGOPT=2
Edge-to-Surface Glue	BEDGE	BSURF	2 3 0 (default); same as BGOPT=3
Surface-to-Surface Glue	BSURF	BSURF	1 2 3 0 (default); same as BGOPT=3
	BSURF	BSURFS	1 3 0 (default); same BGOPT=3

For the following source and target combinations, the **BGOPT** option is ignored.

Glue Type	Source	Target
Edge-to-Surface Glue	BSURFS	BEDGE
	BEDGE	BSURFS
Surface-to-Surface Glue	BSURFS	BSURFS
	BSURFS	BSURF

Related Topics

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BGSET

Glue Contact Set Definition

Defines glued contact pairs.

Format:

1	2	3	4	5	6	7	8	9	10
BGSET	GSID	SID1	TID1	SDIST1		EXT1			
		SID2	TID2	SDIST2		EXT2			
		-etc-							

Example:

BGSET	4	1	2						
		4	3						

Fields:

Field	Contents
GSID	Glue set identification number. (Integer > 0)
SIDi	Source region identification number for glue pair i. (Integer > 0)
TIDi	Target region identification number for glue pair i. (Integer > 0)
SDISTi	Search distance for glue regions (Real > 0.0); (Default=10.0)
EXTi	Extension factor for target region (SOLs 402 and 414 only).

Remarks:

1. SDIST is used to glue the closest elements on the source and target regions. The software searches from the source region.

   From edge regions and solid face regions, the software searches using SDIST in the outward normal direction. In addition, the software searches a small distance in the inward normal direction in order to glue faces which may interfere due to meshing irregularities.

   From shell element face regions, the software searches using SDIST in both the top and bottom normal directions.

2. The following table summarizes all supported glue conditions.

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Table 10-52: Glue Summary

Type	Description	Solution Support
Edge-to-Edge	Glue between the edges of shell, axisymmetric, plane stress, and plane strain elements. Elements being glued must be in the same plane.	All solutions except SOLs 144–146. SOLs 402 and 414 do not support gluing of shell edges. In heat transfer solutions, they are treated as constant conductivity connections. See Remark 3. They cannot be used to represent fluid-to-fluid glue connections in a coupled acoustic analysis.
Edge-to-Surface	Glue between shell element edges and shell or solid element faces.	All solutions except SOLs 144–146, 402, and 414. In heat transfer solutions, they are treated as constant conductivity connections. See Remark 3. They cannot be used to represent fluid-to-fluid glue connections in a coupled acoustic analysis.
Surface-to-Surface	Glue between shell or solid element faces and shell or solid element faces.	All solutions except for SOLs 144–146. In heat transfer solutions, they

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		are treated as constant conductivity connections. See Remark 3. They can be used to represent fluid-to-fluid glue connections in a coupled acoustic analysis.

3. Glue is supported in the heat transfer solutions 153 and 159, and in a SOL 101 heat transfer/structural combined solution.

   The SOL 101 combined solution requires both a heat transfer and a structure subcase. See the HEATSTAT parameter for information. Since these subcases represent different analysis types and glue has a different meaning in each, to define a unique set of glue parameters (penalty values) for each subcase, you must create unique glue sets for each subcase. Specifically, you must create a unique BGSET case control, BGSET bulk entry, and BGPARM bulk entry for each subcase. You select the same glue regions in each of the glue sets.

4. Shell offsets are not accounted for in any glue definitions. The gluing occurs at the grid locations and not at the offset location.

5. When defining glue regions and pairs on geometry which is not tangent continuous, creating single glue regions which cross corner transitions can result in non-uniform stress results around the corners. It is recommended to break these regions into multiple regions and pairs. See the chapter on constraints in the *Simcenter Nastran User's Guide* for more information on glue definitions.

6. Glue definitions defined on composite solid faces which are perpendicular to the stack direction (edge faces) may produce poor stress continuity. If the glue definition is between edge faces belonging to different PCOMPS definitions, and if the number of plies on each PCOMPS definition is small and the same, and the ply thicknesses are similar, the stress continuity should be fairly smooth. This also applies to the results requested with the BGRESULTS case control command.

7. SID and TID are glue regions defined with BCPROP, BCPROPS, BSURF, or BSURFS entries for glue surfaces, and with BEDGE or BLSEG entries for glue edges. SOLs 401, 402, and 414 only support the BEDGE entry for edge glue regions.

8. With edge-to-surface gluing, an edge region defined with a BEDGE or BLSEG entry is glued to a face region defined with a BSURF, BSURFS, BCPROP, or BCPROPS entry. The edge or the face region can be defined as either the source (SID) or the target (TID). SOLs 401, 402, and 414 only support the BEDGE entry for edge glue regions.

9. If multiple eigenvalue subcases (solutions 103 or 105) include STATSUB commands to select linear static subcases, and the linear static subcases include glue and/or linear contact definitions, the BCSET and/or BGSET case control commands must be specified in the global case. Failure to do this can result in erroneous results.

10. Glue conditions can be included in a model with main bulk data type superelements. Each glue pair must be contained in a single superelement or in the residual. That is, the glue regions cannot

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cross superelement boundaries. Glue conditions are not supported in a model with partitioned type superelements. Main bulk data and partitioned type superelements are both described in Chapter 2 of the *Simcenter Nastran Superelement User's Guide*.

11. When gluing solid element faces, a glue condition is not created if the dot product of the element face normals is greater or equal to 0. Including SYSTEM(516)=1 will override this and allow the glue condition to be created if the search distance is satisfied.
12. You cannot add glue conditions when restarting. You can restart from a solution which included glue conditions, but only to request addition output.
13. For SOLs 106 and 129, the software does not update any type of glue condition when large displacements are requested (LGDISP=1). By default, the solution will end with a fatal error when glue conditions and large displacements are requested. If the glue condition is in a location of the model that will not experience large rotations, the glue condition should be reasonably accurate. For this case, you can define SYSTEM(784)=1 to allow the solution to continue.

Remarks related to SOL 401:

1. Edge-to-edge glue conditions on shell element edges and edge-to-surface glue conditions are not updated when large displacements are requested (PARM,LGDISP,1). By default, the solution will end with a fatal error when shell edge glue conditions and large displacements are requested. If the glue condition is in a location of the model that will not experience large rotations, the glue condition should be reasonably accurate. For this case, you can define SYSTEM(784)=1 to allow the solution to continue.

Remarks related to SOLs 402 and 414:

1. SOLs 402 and 414 use a node-to-face gluing. The source region is converted to a group of nodes, which is then glued to the target region faces.
2. SOLs 402 and 414 only support the BEDGE entry for edge glue regions. The BLSEG entry is not supported.
3. The default value for SDISTi is equal to the sum of the length of the bounding box containing all the nodes of the model divided by 1000.
4. All the source nodes that are closer than SDISTi are moved to their projection on the target face to avoid gaps and parasitic moments in case of large rotations of the face. This will slightly modify the mesh.
5. The EXTi extension value is equal to the element size in the parameter space -1 to 1, or 0 to 1, depending on the element type (0.0 < Real < 0.25). For example, a value of 0.01 corresponds to 1% of the element size in all directions. (Default = 0.01)

Related Topics

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BLSEG

Defines Glue or Contact Edge Region, or Curve for Slideline Contact

Defines a glue or contact edge region or a curve for slideline contact via grid numbers.

Format 1: (Formats 1 and 2 cannot be combined on the same line)

1	2	3	4	5	6	7	8	9	10
BLSEG	ID	G1	G2	G3	G4	G5	G6	G7	

Format 2:

BLSEG	ID	G1	“THRU”	G2	“BY”	INC			

Continuation Format 1: (Continuation formats 1 and 2 cannot be combined on a single continuation line)

	G8	G9	G10	G11	–etc.–				

Continuation Format 2:

	G8	“THRU”	G9	“BY”	INC				

Example:

BLSEG	14	101	THRU	190	BY	5			
	46	23	57	82	9	16			
	201	THRU	255						
	93	94	95	97					

Fields:

Field	Contents
ID	Edge region or curve identification number. See Remark 1 (Integer > 0).
Gi	Grid point identification numbers in a continuous topological order. See Remark 2 (Integer > 0).
INC	Grid point identification number increment. See Remark 3 for default (Integer or blank).

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Remarks:

1. The ID must be unique with respect to all other BEDGE, BLSEG, BSURFS, BSURF, BCPROP, and BCPROPS entries.
2. An edge region or curve consists of one or more line segments defined between consecutive grid points. The grid points defining the edge region or curve must be entered in a continuous topological order. If an edge region or curve forms a closed loop, for example, the grids around the perimeter of a cylinder edge, the last grid point identification number should be the same as the first grid point number.
3. When selecting grid points in a range using “THRU”, the default increment value is 1 if grid numbers are increasing or -1 if grid numbers are decreasing.
4. With edge-to-surface gluing:
   - With edge-to-surface gluing, an edge region defined with the BLSEG entry is glued to a face region defined with a BSURF, BSURFS, BCPROP, or BCPROPS entry. The edge-to-surface glue pairs are defined with the BGSET entry.
   - The grid point IDs on the BLSEG entry used to define a glue edge region can only be part of the CQUAD4, CQUADR, CQUAD8, CTRIA3, CTRIAR and CTRIA6 element connectivity.
   - The BWIDTH bulk entry is ignored.
5. With slideline contact:
   - A corresponding BWIDTH Bulk Data entry may be required to define the width/thickness of each line segment. If the corresponding BWIDTH is not present, the width/thickness for each line segment is assumed to be unity.
   - Each line segment has a width in a 3-D slideline and a thickness in a 2-D slideline contact to calculate contact stresses. The width/thickness of each line segment is defined via the BWIDTH Bulk Data entry. The ID in BLSEG must be same as the ID specified in the BWIDTH. That is, there must be a one to one correspondence between BLSEG and BWIDTH. BWIDTH Bulk Data entry may be omitted only if the width/thickness of each segment is unity.
   - The normal to the segment is determined by the cross product of the slideline plane vector (i.e., the Z direction of the coordinate system defined in the ‘CID’ field of the BCONP Bulk Data entry) and the tangential direction of the segment. The tangential direction is the direction from node 1 to node 2 of the line segment.

Related Topics

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BNDFIX

Fixed Boundary Degrees of Freedom

Defines analysis set (a-set) degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.

Format:

1	2	3	4	5	6	7	8	9	10
BNDFIX	ID1	C1	ID2	C2	ID3	C3	ID4	C4	

Example:

BNDFIX	2	135	14	6					

Fields:

Field	Contents
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component number. (Integer zero or blank for scalar points, or any unique combinations of the integers 1 through 6 for grid points. No embedded blanks.)

Remarks:

1. BSET and BNDFIX entries are equivalent.
2. If there are no BNDFREEi or BNDFIXi entries present, all a-set points are considered fixed during component mode analysis. If there are only BSETi entries present, any a-set degrees-of-freedom not listed are placed in the free boundary set (c-set). If there are both BNDFREEi or BNDFIXi entries present, the c-set degrees-of-freedom are defined by the BNDFREEi entries, and any remaining a-set points are placed in the b-set.
3. Degrees-of-freedom specified on this entry form members of the mutually exclusive b-set. They may not be specified on other entries that define mutually exclusive sets. See the **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES, then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

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Related Topics

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BNDFIX1

Fixed Boundary Degrees of Freedom, Alternate Form of BNDFIX Entry

Defines analysis set (a-set) degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.

Format:

1	2	3	4	5	6	7	8	9	10
BNDFIX1	C	ID1	ID2	ID3	ID4	ID5	ID6	ID7	
	ID8	ID9	ID10	-etc.-					

Example:

BNDFIX1	2	135	14	6	23	24	25	26	
	122	127							

Alternate Format and Example:

BNDFIX1	C	ID1	"THRU"	ID2					
BNDFIX1	3	6	THRU	32					

Fields:

Field	Contents
C	Component numbers. (Integer zero or blank for scalar points, or any unique combinations of the integers 1 through 6 for grid points with no embedded blanks.)
IDi	Grid or scalar point identification numbers. (Integer > 0; For "THRU" option, ID1< ID2)

Remarks:

1. BSET1 and BNDFIX1 entries are equivalent.
2. If there are no BNDFREEi or BNDFIXi entries present, all a-set points are considered fixed during component mode analysis. If there are only BNDFIXi entries present, any a-set degrees-of-freedom not listed are placed in the free boundary set (c-set). If there are both BNDFREEi or BNDFIXi entries present, the c-set degrees-of-freedom are defined by the BNDFREEi entries, and any remaining a-set points are placed in the b-set.

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3. Degrees-of-freedom specified on this entry form members of the mutually exclusive b-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES, then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

Related Topics

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BNDFREE

Free Boundary Degrees-of-Freedom

Defines analysis set (a-set) degrees-of-freedom to be free (c-set) during component modes calculations.

Format:

1	2	3	4	5	6	7	8	9	10
BNDFREE	ID1	C1	ID2	C2	ID3	C3	ID4	C4	

Example:

BNDFREE	124	1	5	23	6	16			

Fields:

Field	Contents
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component numbers. (Integer zero or blank for scalar points, or any unique combination of the integers 1 through 6 for grid points with no embedded blanks.)

Remarks:

1. CSET and BNDFREE entries are equivalent.
2. If there are no BNDFREEi or BNDFIXi entries present, all a-set degrees-of-freedom are considered fixed during component modes analysis. If there are only BNDFIXi entries present, any a-set degrees-of-freedom not listed are placed in the free boundary set (c-set). If there are both BNDFREEi or BNDFIXi entries present, the c-set degrees-of-freedom are defined by the BNDFREEi entries, and any remaining a-set points are placed in the b-set.
3. Degrees-of-freedom specified on this entry form members of the mutually exclusive c-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

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Related Topics

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BNDFREE1

Free Boundary Degrees-of-Freedom, Alternate Form of BNDFREE Entry

Defines analysis set (a-set) degrees-of-freedom to be free (c-set) during component modes calculations.

Format:

1	2	3	4	5	6	7	8	9	10
BNDFREE1	C	ID1	ID2	ID3	ID4	ID5	ID6	ID7	
	ID8	ID9	-etc.-						

Example:

BNDFREE1	124	1	5	7	6	9	12	122	
	127								

Alternate Formats and Example:

BNDFREE1	C	ID1	“THRU”	ID2					
BNDFREE1	3	6	THRU	32					

BNDFREE1		“ALL”							
BNDFREE1		ALL							

Fields:

Field	Contents
C	Component number. (Integer zero or blank for scalar points, or any unique combination of the Integers 1 through 6 for grid points with no embedded blanks.)
IDi	Grid or scalar point identification number. (Integer > 0; For THRU option, ID1 < ID2)

Remarks:

1. CSET1 and BNDFREE1 entries are equivalent.
2. If there are no BNDFREEi or BNDFIXi entries present, all a-set degrees-of-freedom are considered fixed during component modes analysis. If there are only BNDFIXi entries present, any a-set degrees-of-freedom not listed are placed in the free boundary set (c-set). If there are both BNDFREEi or BNDFIXi entries present, the c-set degrees-of-freedom are defined by the BNDFREEi entries, and any remaining a-set points are placed in the b-set.

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3. Degrees-of-freedom specified on this entry form members of the mutually exclusive c-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

Related Topics

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BNDGRID

Boundary Grid Points

Specifies a list of grid point identification numbers on design boundaries or surfaces for shape optimization (SOL 200).

Format:

1	2	3	4	5	6	7	8	9	10
BNDGRID	C	GP1	GP2	GP3	GP4	GP5	GP6	GP7	
	GP8	-etc.-							

Example:

BNDGRID	123	41	42	43	44	45	46	47	
	49								

Alternate Format and Example:

BNDGRID	C	GP1	"THRU"	GP2					
BNDGRID	123	41	THRU	49					

Fields:

Field	Contents
C	Component number (any unique combination of integers 1 through 6 with no embedded blanks). See Remark 1 .
GPi	Shape boundary grid point identification number. (0 < Integer < 1000000; For THRU option, GP1 < GP2)

Remarks:

1. C specifies the components for the listed grid points for which boundary motion is prescribed.
2. Multiple BNDGRID entries may be used to specify the shape boundary grid point identification numbers.
3. Both fixed and free shape boundary grid point identification numbers are listed on this entry.
4. The degrees-of-freedom specified on BNDGRID entries must be sufficient to statically constrain the model.

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5. Degrees-of-freedom specified on this entry form members of the mutually exclusive s-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.

Related Topics

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BOLT

Bolt definition

Selects the elements to be included in the bolt preload calculation.

Format for ETYPE = 1:

1	2	3	4	5	6	7	8	9	10
BOLT	BID	ETYPE	EID1	EID2	EID3	EID4	EID5	EID6	
	EID7	"THRU"	EID8	"BY"	INC				
	-etc-								

Format for ETYPE = 2:

1	2	3	4	5	6	7	8	9	10
BOLT	BID	ETYPE	CSID	IDIR	G1	G2	G3	G4	
	G5	"THRU"	G6	"BY"	INC				
	-etc-								

Format for ETYPE = 3:

1	2	3	4	5	6	7	8	9	10
BOLT	BID	ETYPE	CSID	IDIR	GP				
	EID1	EID2	EID3	EID4	EID5	EID6	EID7	EID8	
	EID9	"THRU"	EID10	"BY"	INC				
	-etc-								

Examples:

ETYPE = 1

BOLT	4	1	11						

ETYPE = 2

BOLT	8	2	4	3	12	23	55	128	
	133	THRU	165						

ETYPE = 3

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BOLT	9	3	1	3	148				
	56	24	43	21	73	52	62	41	
	106	THRU	202						

Fields:

Field	Contents
BID	Bolt identification number. (Integer > 0)
ETYPE	Element type. (Integer; No default) = 1 to model bolts with CBAR and CBEAM elements in SOLs 101, 103, 105, 107 through 112, 401, and 402. = 2 to model bolts with CHEXA, CPENTA, and CTETRA elements in SOLs 101, 103, 105, 107 through 112, or CHEXA, CPENTA, CTETRA, CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8, CTRAX3, CTRAX6, CQUADX4, and CQUADX8 elements in SOLs 401 and 402. = 3 legacy method to model bolts with CHEXA, CPENTA, CPYRAM, and CTETRA elements in SOLs 401 and 402. Requires the system cell 750.
“BY”	Specifies an increment when using THRU option. (Character)
INC	Increment used with THRU option. (Integer; Default = 1) INC > 0 can be defined, for example ...106,THRU,126,BY,INC,2 INC < 0 can be defined, for example ...126,THRU,106,BY,INC,-2

Fields for ETYPE = 1:

Field	Contents
EIDi	Selects element identification numbers to include in the bolt preload calculation. See Remark 2. (Integer > 0, or using “THRU”; EID7 < EID8 for THRU option; No default)

Fields for ETYPE = 2:

Field	Contents
CSID	Identification number of the coordinate system used to define the bolt axis. For the basic coordinate system, CSID = 0. (Integer ≥ 0; See Remark 7 for default behavior.)
IDIR	Direction of bolt axis relative to CSID. (Integer; See Remark 7 for default behavior.) = 1 for the X direction

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Field	Contents
	= 2 for the Y direction = 3 for the Z direction See SOL 401 and 402 ETYPE=2 remarks.
Gi	Identification numbers of grid points that form a cross section through the bolt. See Remarks 3 and 4. See SOL 401 and 402 ETYPE=2 remarks. (Integer ≥ 0; No default)

Fields for ETYPE = 3:

Field	Contents
EIDi	Selects element identification numbers to include in the bolt preload calculation. All elements representing the bolt must be included in EIDi, and have the same PID. (Integer > 0, or using "THRU"; EID7 < EID8 for THRU option; No default)
CSID	Identification number of the coordinate system used to define the bolt axis. For the basic coordinate system, CSID = 0. (Integer ≥ 0; Default=0)
IDIR	Direction of bolt axis relative to CSID. See SOL 401 Remark 4. See also SOL 402 Remarks 1, 2, 3, and 8. (Integer; Default = 0) = 0 for automatic determination of the bolt axis. = 1 for the X direction = 2 for the Y direction = 3 for the Z direction
GP	The identification number of the grid point where the bolt cross sectional area is calculated. See SOL 401 Remark 4. See also SOL 402 Remarks 4 and 5. (Integer > 0 or blank)

Remarks:

1. Each BOLT entry defines a single physical bolt which can be composed of multiple elements.
2. If multiple CBAR and CBEAM elements are used to model a bolt, only one of the elements needs to be listed on the BOLT entry. Enter the element ID in the EID1 field. If multiple CBAR and CBEAM elements are listed, the software will only use the element in the EID1 field for the bolt preload calculation.
3. Any grid point listed in the Gi or GP fields must be included in the connectivity of elements that are used to model the bolt.

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4. Gi must select enough GRID entries to define a cross section through the bolt. The selected Gi can only be included on a single BOLT entry. The grids can be listed in any order on the BOLT entry. For parabolic elements, mid-nodes must also be listed. Any Gi listed on a BOLT entry cannot be used in the connectivity of a solid composite element or be included in an SPC.

   The software splits the bolt mesh by duplicating each Gi on the BOLT entry. The identification numbers for the duplicated grid points start at the highest user-defined grid ID in the model plus one and continue sequentially higher.

   A pressure load cannot be applied to any face of an element if the connectivity of the element includes a Gi listed on a BOLT entry.

5. As a best practice, select GP such that it is near the middle of the cross section of the bolt.

6. In SOL 105, both the bolt preload and service load are scaled to determine the buckling load.

7. For ETYPE=2, if your bolt axis is parallel with one of the basic coordinate system axes, you can leave both the CSID and IDIR fields blank, and the software will automatically determine the bolt axis. If your bolt axis is *not* parallel with one of the basic axes, you must define the CSID and IDIR fields. If the CSID and IDIR fields are blank and the bolt axis is not aligned with one of the three basic coordinate axes, the software will end with a fatal error.

8. For ETYPE=2, you select a set of grid points (Gi) to define the cross section through the bolt axis where the software cuts the bolt. The software will end with a fatal error if the grid points listed in Gi are not coplanar, or if the normal vector to the plane defined by Gi is not parallel with the resulting bolt axis coordinate (IDIR).

9. For SOL 101, when you define a bolt with ETYPE=2, the software cuts the bolt in half, it creates new grid points resulting in grid pairs at the cut, it distributes the opposing axial bolt force across the grids on each side of the cut, and it first solves the bolt preload to determine the axial displacement of each bolt half.

   In the service load portion of the solution, the software then maintains the relative axial displacement at the bolt cut determined from the preload portion with either a glue or an MPC condition, and applies any service loads you have defined.

   - By default, the software uses a glue approach at the cut to hold the grid pairs in their relative deformed state. For the preload portion, the software zeroes the translational glue stiffness in the bolt axial direction but keeps the rotational stiffness. Because the glue condition couples the rotational and translational DOFs, a non-zero axial stiffness remains. As a result, the glue approach allows bending loads to transfer from one side of the bolt cut to the other during the preload portion.

     For the service load portion, the software holds the grid pairs at the cut in their relative deformed state with a large translational and rotational glue stiffness. As a result, bending effects are transfered at the bolt cut during the service load portion also.

     With the glue approach, you can recover bolt forces using the glue results requested with the BGRESULTS command.

   - You can select the pre-NX Nastran 12 MPC based approach by defining SYSTEM(610)=10. Because no axial MPC condition is present during the preload portion, there is no mechanism to transfer bending loads at the bolt cut.

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For the service load portion, the software holds the grid pairs in their relative deformed state with rotational and translational MPC conditions between the two sides of the bolt cut. As a result, bending effects are transfered at the bolt cut in the service load portion.

With the MPC approach, you recover bolt forces using the MPC force results requested with the MPCFORCES command.

Remarks related to SOL 401:

1. All of the BOLT bulk entries referenced in the same input file must be TYPE=1, or all TYPE=2. A fatal error will occur if BOLT entries with a combinations of TYPE=1 and TYPE=2 are referenced.
2. ETYPE=1 remark.

   For bolts with ETYPE=1, if multiple CBAR or CBEAM elements are used to model a bolt, only one of the elements needs to be listed on the BOLT entry. Although, if you are requesting results with the BOLTRESULTS case control command and you would like these results stored for all of these CBAR or CBEAM elements, then you should list all of the elements on the BOLT entry.
3. ETYPE=2 remarks.

   - You select a set of grid points (Gi) to define the cross section through the bolt axis where the software cuts the bolt. The software will end with a fatal error if the grid points listed in Gi are not coplanar, or if the normal vector to the plane defined by Gi is not parallel with the resulting bolt axis coordinate (IDIR).
   - If your bolt axis is parallel with one of the basic coordinate system axes, you can leave both the CSID and IDIR fields blank, and the software will automatically determine the bolt axis. If your bolt axis is *not* parallel with one of the basic axes, you must define the CSID and IDIR fields. If the CSID and IDIR fields are blank and the bolt axis is not aligned with one of the three basic coordinate axes, the software will end with a fatal error.
   - When defining a bolt preload on the axisymmetric elements CTRAX3, CTRAX6, CQUADX4 and CQUADX8, the axis of the axisymmetric bolt must be aligned with the axisymmetric axis of rotation defined by the basic coordinate system. For example, if the axisymmetric elements are defined on the X-Z plane, the axis of the bolt must be aligned with the Z-axis.
   - The axis of an axisymmetric bolt must be aligned with the axisymmetric axis of rotation defined by the basic coordinate system. For example, if the axisymmetric elements are defined on the X-Z plane, the axis of the bolt must be aligned with the Z-axis.

     When you define a bolt with ETYPE=2, the software cuts the bolt perpendicular to the bolt axis. As a result, when you define the axisymmetric bolt, you list the grid point IDs in the Gi fields on the BOLT entry where you want the software to cut the bolt. Because the bolt axis is parallel with the axisymmetric axis, you do not need to define the CSID or IDIR fields on the BOLT entry.
   - When you select the ETYPE=2 format, the software uses a nonlinear MPC approach by default. You can use **system cell 750** to change to the glue approach, which was the default approach used before Simcenter Nastran 2021.2.
4. ETYPE=3 remarks:

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• ETYPE=3 is a legacy method to model bolts with CHEXA, CPENTA, CPYRAM, and CTETRA elements. By default, the software will end with a fatal error when a ETYPE=3 bolt is defined. You must define the system cell 750 to allow the solution to proceed. See **system cell 750** for more information.

• When CSID and IDIR are blank, the software automatically determines the coordinate system and the bolt axis.

• When GP blank, the software automatically determines a middle location to compute the cross sectional area.

5. Composite solid elements (PCOMPS property card) are not supported for preloaded bolts.
6. New BOLT bulk entries cannot be added in a restart solution.

Remarks related to SOL 402:

1. When IDIR = 0 or blank (default), the direction of the bolt axis is automatically determined by the software to coincide with minimum principal moment of inertia of the bolt. CSID is ignored when IDIR = 0 or blank.
2. When IDIR is not specified, internal faces along the bolt cut are connected using contact elements. When it is specified, each face of the bolt cut are connected to an RBE3 element and both elements are linked to the same primary node.
3. If IDIR >0, the cut cannot be perpendicular to the bolt axis.
4. The software splits the bolt mesh at the grid point entered in the GP field.
5. If GP = 0 or blank, the location of the bolt plane is automatically determined by the program.
6. When ETYPE=1, if multiple CBAR and CBEAM elements are used to model a bolt, only one of the elements needs to be listed on the BOLT entry. Enter the element ID in the EID1 field. If multiple CBAR and CBEAM elements are listed, the software will only use the element in the EID1 field for the bolt preload calculation.

   If you are requesting results with the BOLTRESULTS case control command, regardless of whether you list one or multiple elements on the BOLT entry, the software only stores these results on the element in the EID1 field.
7. ETYPE=2 remarks.

   • ETYPE=2 is the recommended method to model bolts.

   • When you use the ETYPE=2 bolt, you select a set of grid points (Gi) to define the cross section through the bolt axis where the software cuts the bolt. The software will end with a fatal error if the grid points listed in Gi are not coplanar, or if the normal vector to the plane defined by Gi is not parallel with the resulting bolt axis coordinate (IDIR).

   • If your bolt axis is parallel with one of the basic coordinate system axes, you can leave both the CSID and IDIR fields blank, and the software will automatically determine the bolt axis. If your bolt axis is *not* parallel with one of the basic axes, you must define the CSID and IDIR fields. If the CSID and IDIR fields are blank and the bolt axis is not aligned with one of the three basic coordinate axes, the software will end with a fatal error.

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- When defining a bolt preload on the axisymmetric elements CTRAX3, CTRAX6, CQUADX4 and CQUADX8, the axis of the axisymmetric bolt must be aligned with the axisymmetric axis of rotation defined by the basic coordinate system. For example, if the axisymmetric elements are defined on the X-Z plane, the axis of the bolt must be aligned with the Z-axis. Because the bolt axis is parallel with a basic coordinate axis, you do not need to define the CSID or IDIR fields.

8. ETYPE=3 is a legacy method to model bolts with CHEXA, CPENTA, CPYRAM, and CTETRA elements. By default, the software will end with a fatal error when a ETYPE=3 bolt is defined. You must define the system cell 750 to allow the solution to proceed. See **system cell 750** for more information.
9. If a BOLT bulk entry was ignored in an initial run because it was not referenced by a preload bulk entry, no preload bulk entry can be added in the restart run.
10. New BOLT bulk entries cannot be added in a restart solution.

Related Topics

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BOLTFRC

Defines bolt preload force, displacement, or strain.

Defines bolt preload force, displacement, or strain.

Format:

1	2	3	4	5	6	7	8	9	10
BOLTFRC	SID	TYPE	D	LEN					
	B1	B2	B3	B4	B5	B6	B7	B8	
	B9	THRU	B10						
	B11	B12	-etc-						

Example:

BOLTFRC	4	DISP	.0025						
	1	4	6	9	10	26	32	34	
	37	43	51						

Fields:

Field	Contents
SID	Bolt preload set identification number (Integer; No default)
TYPE	TYPE=DISP or LOAD (Character; No default)
D	Preload value. (Real; No default) When TYPE=DISP, D is a preload displacement. When TYPE=LOAD, D is a preload force.
LEN	Optional bolt length when TYPE=DISP in SOL 401. (Real; Software automatically computes the length if LEN is undefined.)
Bi	Bolt identification numbers (BID) defined by BOLT bulk entries. (Integer>0 or use “THRU” option. For “THRU” option, B7<B8)

Remarks:

1. Bolt preload is supported in SOLs 101, 103, 105, 107 through 112, 401, and 402.

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SOLs 101, 103, 105, 107 through 112 support bolts modeled with the 1D elements CBAR and CBEAM, and the 3D solid elements CHEXA, CPENTA, and CTETRA.

SOLs 401 and 402 support bolts modeled with the 1D elements CBAR and CBEAM, the 3D solid elements CHEXA, CPENTA, and CTETRA, the plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, and CPLSTS8, and the axisymmetric elements CTRAX3, CTRAX6, CQUADX4, and CQUADX8.

See the **BOLT** bulk entry.

2. The software will fatal if multiple BOLTFRC entries with the same SID are referenced.
3. The following tables summarize the supported combinations of the ETYPE field on the BOLT bulk entry and the TYPE field on the BOLTFRC bulk entry.

SOLs 101, 103, 105, 107 through 112	TYPE=DISP	TYPE=LOAD
ETYPE=1	Supported	Supported
ETYPE=2	Supported	Supported

SOL 401	TYPE=DISP	TYPE=LOAD
ETYPE=1	Supported	Supported
ETYPE=2	Supported[a]	Supported

SOL 402	TYPE=DISP	TYPE=LOAD
ETYPE=1	Supported[a]	Supported
ETYPE=2	Supported[a]	Supported

[a] Displacement is applied as a shortening at the cut and LEN field is ignored.

[b] Displacement is used along with LEN to find the initial strain value (DISP/LEN).

4. A BID can only be referenced once in a subcase. This includes BID refererenced on a single BOLTFRC entry, and BID on multiple BOLTFRC entries.
5. For ETYPE=1 bolts, you can define a bolt preload as an axial load or displacement. For either of these definitions, the result is an axial preload strain in which the software will store, and reapply in consecutive preload steps in a sequence, and in consecutive sequentially dependent subcases. The software continues to reapply the axial preload strain until you either define a new bolt preload step or subcase which redefines the axial preload strain, or you define a nonsequentially dependent subcase.

   Note that in consecutive steps and subcases, the total axial strain on a preloaded bolt includes the axial preload strain, along with strain resulting from any other loads you apply, including preloads on neighboring bolts.
6. For ETYPE=1 bolts, when a bolt preload is defined as a load, the software computed axial preload strain includes the effects of any other loads you included in the preload step (thermal loads, contact

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forces, preloaded bolts from a previous preload step, and conditions from a previous subcase). For example, if you compare the results from a solution with only bolt preloads defined with the results from the same model with the addition of thermal loads and contact, the bolt force will be the same, but the final bolt strain needed to achieve the bolt force will be different.

Remarks related to SOLs 401 and 402:

1. For ETYPE=2 bolt, when you apply the bolt preload as a displacement, the software treats this as incremental shortening. That is, the specified shortening value is added to an existing bolt shortening (if a previous preload was applied) and the net shortening is enforced.
2. A bolt preload force equal to 0.0 is supported. The zero preload is useful, for example, to predict the residual axial bolt strain after applying a bolt preload, applying service loads, then finally unloading the bolt to the zero axial force condition.

Related Topics

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BOLTLD (bulk entry)

Bolt Preload Combining and Scaling

Combines and optionally scales bolt preload sets.

Format:

1	2	3	4	5	6	7	8	9	10
BOLTLD	SID	S	S1	L1	S2	L2	S3	L3	
	S4	L4	-etc-						

Example:

BOLTLD	101	0.5	1.0	3	6.2	4			

Fields:

Field	Contents
SID	Set identification number (SID) of BOLTLD bulk entry. (Integer>0)
S	Overall scale factor. (Real)
Si	Scale factor for individual Li. (Real)
Li	SID of BOLTFRC bulk entries. (Integer>0)

Remarks:

1. Bolt preload is supported in SOLs 101, 103, 105, 107 through 112, 401, and 402.

   SOLs 101, 103, 105, 107 through 112 support bolts modeled with the 1D elements CBAR and CBEAM, and the 3D solid elements CHEXA, CPENTA, and CTETRA.

   SOLs 401 and 402 support bolts modeled with the 1D elements CBAR and CBEAM, the 3D solid elements CHEXA, CPENTA, and CTETRA, the plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, and CPLSTS8, and the axisymmetric elements CTRAX3, CTRAX6, CQUADX4, and CQUADX8.

   See the **BOLT** bulk entry.

2. The applied preload value P_i for each BOLT bulk entry is given by:

   $P_i = S * Si * P_{Li}$

   where P_{Li} is the preload value defined on a BOLTFRC bulk entry Li, Si is the individual scale factor corresponding to Li, and S is the overall scale factor.

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3. A BID can only be referenced once in a subcase. This includes BID refererenced on a single BOLTFRC entry, and BID on multiple BOLTFRC entries.
4. The following rules apply when BOLTFRC, BOLTLD, or BOLTSEQ bulk entries are defined using the same SID.
   - If a BOLTSEQ entry is referenced by the BOLTLD case control command, BOLTFRC and BOLTLD entries with the same SID as the BOLTSEQ are only used if they are referenced in a sequence defined on the BOLTSEQ entry.
   - If a BOLTLD bulk entry is referenced by the BOLTLD case control command or by the BOLTSEQ bulk entry, BOLTFRC entries with the same SID as the BOLTLD entry are used if they are referenced on the BOLTLD entry.
   - The software will fatal if multiple BOLTFRC entries with the same SID are referenced.
   - Only one BOLTSEQ entry can be referenced. The software will fatal if multiple BOLTSEQ entries with the same SID are referenced.
   - Only one BOLTLD entry can be referenced. The software will fatal if multiple BOLTLD entries with the same SID are referenced.
5. Li must be unique. However, a single Li on a BOLTLD bulk entry can reference multiple BOLTFRC bulk entries with unique IDs. When this occurs, all the preload forces defined by these entries sharing the same SID are scaled by the corresponding individual scale factor Si and the overall scale factor S.
6. The bolt identification numbers (BID) in all BOLTFRC bulk entries selected by a single BOLTLD bulk entry must be unique.
7. The SID of each BOLTLD bulk entry must be unique.
8. Inertia relief (PARAM,INREL) is not supported with bolt preload.
9. The combination of a bolt preload in which the bolt is meshed with solid elements, the element iterative solver, and contact conditions is not supported.
10. Enforced displacements defined with non-zero SPC or SPCD bulk entries can be included with a bolt preload. When an SPCD is defined with a bolt preload, system cell 581 must be set to 1 to override an error trap that exists in the software. The software behaviour changes during the bolt preload solution steps for various input scenarios (with and without contact defined; enforced displacements defined with SPCD entries or with SPC entries). See “Enforced Displacements and Bolt Preloads” in the *User’s Guide* for a description of the software behavour.
11. When a bolt preload is defined on CBEAM or CBAR elements and contact is in the model, the software runs an initial preload solution to compute the strain in the bolt before applying any service loads. When RESET=1 is defined on the BCTPARM bulk entry for the contact condition, for all consecutive solutions for the same subcase as the preload or for consecutive subcases, the software applies the computed bolt strain from the initial preload solution, and it uses the final contact status from the initial bolt preload solution as the starting contact status.

Remarks related to SOL 402:

1. If the bolt load is defined in an ANALYSIS=PRELOAD subcase, the bolt load will be linearly ramped starting from 0.0 and ending with the bolt load defined for the subcase.

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2. If the bolt load is defined in a subcase which does not include ANALYSIS=PRELOAD (for example, in a subcase which includes ANALYSIS=STATIC), the bolt load will be applied as a function of time. The bolt load for this case begins with the value from the end of the previous subcase, and ends with the bolt load defined for the current subcase.

Related Topics

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BOLTSEQ

Specifies a bolt preload sequence in SOL 401.

Specifies a bolt preload sequence.

Format:

1	2	3	4	5	6	7	8	9	10
BOLTSEQ	SID								
	S_NOi	B_IDi	NINCi						

Examples:

BOLTSEQ	100								
	1	11	3						
	2	22	2						

Fields:

Field	Contents
SID	Bolt preload set identification number (Integer; No default)
S_NOi	Sequence order number for the BOLTLD and BOLTFRC IDs to be applied. (Integer; No default)
B_IDi	SID of BOLTLD or BOLTFRC bulk entries defining a bolt preload. (Integer; No default)
NINCi	Number of increments in which to ramp up the bolt preload defined in BOLTLD or BOLTFRC entries. (Integer; Default = 1)

Remarks:

1. BOLTSEQ is only supported in SOL 401.
2. A bolt preload force equal to 0.0 is supported. The zero preload is useful, for example, to predict the residual axial bolt strain after applying a bolt preload, applying service loads, then finally unloading the bolt to the zero axial force condition.
3. The S_NO values must be sequential.
4. If NINCi is defined on the BOLTSEQ entry, that value overrides the value defined on the TSTEP1 increment.
5. The parameter MSGLVLB on the NLCNTL bulk entry controls the printed bolt preload output to the .f06 file.

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6. The following rules apply when BOLTFRC, BOLTLD, or BOLTSEQ bulk entries are defined using the same SID.

   - If a BOLTSEQ entry is referenced by the BOLTLD case control command, BOLTFRC and BOLTLD entries with the same SID as the BOLTSEQ are only used if they are referenced in a sequence defined on the BOLTSEQ entry.
   - If a BOLTLD bulk entry is referenced by the BOLTLD case control command or by the BOLTSEQ bulk entry, BOLTFRC entries with the same SID as the BOLTLD entry are used if they are referenced on the BOLTLD entry.
   - The software will fatal if multiple BOLTFRC entries with the same SID are referenced.
   - Only one BOLTSEQ entry can be referenced. The software will fatal if multiple BOLTSEQ entries with the same SID are referenced.
   - Only one BOLTLD entry can be referenced. The software will fatal if multiple BOLTLD entries with the same SID are referenced.

7. You can define a bolt preload as an axial force, displacement, or strain on the BOLTFRC bulk entry. For any of these definitions, the result is an axial preload strain in which the software will store, and reapply in consecutive preload steps in a sequence, and in consecutive sequentially dependent subcases. The software continues to reapply the axial preload strain until you either define a new bolt preload step or subcase which redefines the axial preload strain, or you define a nonsequentially dependent subcase.

   Note that in consecutive steps and subcases, the total axial strain on a preloaded bolt includes the axial preload strain, along with strain resulting from any other loads you apply, including preloads on neighboring bolts.

8. When a bolt preload is defined as a force, the software computed axial preload strain includes the effects of any other loads you included in the preload step (thermal loads, contact forces, preloaded bolts from a previous preload step, and conditions from a previous subcase). For example, if you compare the results from a solution with only bolt preloads defined with the results from the same model with the addition of thermal loads and contact, the bolt force will be the same, but the final bolt strain needed to achieve the bolt force will be different.

Related Topics

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BOUTPUT

Output for Slideline Contact

Defines secondary nodes at which output is requested.

Format:

1	2	3	4	5	6	7	8	9	10
BOUTPUT	ID	G1	G2	G3	G4	G5	G6	G7	

Alternate Format:

BOUTPUT	ID	G1	"THRU"	G2	"BY"	INC			

The Continuation Entry formats may be used more than once and in any order. They may also be used with either format above.

Continuation Entry Format 1:

	G8	G9	G10	G11	-etc.-				

Continuation Entry Format 2:

	G8	"THRU"	G9	"BY"	INC				

Example:

BOUTPUT	15	5	THRU	21	BY	4			
	27	30	32	33					
	35	THRU	44						
	67	68	72	75	84	93			

Format and Example Using "ALL" (No continuation entry is allowed):

BOUTPUT	ID	ALL							
BOUTPUT	15	ALL							

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Fields:

Field	Contents
ID	Contact region identification number of a BCONP entry for which output is desired. (Integer > 0)
Gi	Secondary node numbers for which output is desired. (Integer > 0)
INC	Grid point identification number increment. See Remark (Integer or blank).

Remarks:

For automatic generation of grid numbers, the default increment value is 1 if grid numbers are increasing or -1 if grid numbers are decreasing (i.e., the user need not specify BY and the increment value).

Related Topics

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BSET

Fixed Boundary Degrees of Freedom

Defines analysis set (a-set) degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.

Format:

1	2	3	4	5	6	7	8	9	10
BSET	ID1	C1	ID2	C2	ID3	C3	ID4	C4	

Example:

BSET	2	135	14	6					

Fields:

Field	Contents
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component number. (Integer zero or blank for scalar points, or any unique combinations of the integers 1 through 6 for grid points. No embedded blanks.)

Remarks:

1. BSET and BNDFIX entries are equivalent.
2. If there are no CSETi or BSETi entries present, all a-set points are considered fixed during component mode analysis. If there are only BSETi entries present, any a-set degrees-of-freedom not listed are placed in the free boundary set (c-set). If there are both BSETi and CSETi entries present, the c-set degrees-of-freedom are defined by the CSETi entries, and any remaining a-set points are placed in the b-set.
3. Degrees-of-freedom specified on this entry form members of the mutually exclusive b-set. They may not be specified on other entries that define mutually exclusive sets. See the **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES, then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

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Related Topics

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BSET1

Fixed Boundary Degrees of Freedom, Alternate Form of BSET Entry

Defines analysis set (a-set) degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.

Format:

1	2	3	4	5	6	7	8	9	10
BSET1	C	ID1	ID2	ID3	ID4	ID5	ID6	ID7	
	ID8	ID9	ID10	-etc.-					

Example:

BSET1	2	135	14	6	23	24	25	26	
	122	127							

Alternate Format and Example:

BSET1	C	ID1	"THRU"	ID2					
BSET1	3	6	THRU	32					

Fields:

Field	Contents
C	Component numbers. (Integer zero or blank for scalar points, or any unique combinations of the integers 1 through 6 for grid points with no embedded blanks.)
IDi	Grid or scalar point identification numbers. (Integer > 0; For "THRU" option, ID1< ID2)

Remarks:

1. BSET1 and BNDFIX1 entries are equivalent.
2. If there are no CSETi or BSETi entries present, all a-set points are considered fixed during component mode analysis. If there are only BSETi entries present, any a-set degrees-of-freedom not listed are placed in the free boundary set (c-set). If there are both BSETi and CSETi entries present, the c-set degrees-of-freedom are defined by the CSETi entries, and any remaining a-set points are placed in the b-set.

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3. Degrees-of-freedom specified on this entry form members of the mutually exclusive b-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES, then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

Related Topics

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BSURF

Contact or Glue or Co-Simulation (Wetted) Region using Shell Elements

Defines a contact or glue or Co-Simulation (wetted) region using shell elements.

Format:

1	2	3	4	5	6	7	8	9	10
BSURF	ID	EID1	EID2	EID3	EID4	EID5	EID6	EID7	
	EID8	EID9	EID10	-etc-					

Alternate Format:

BSURF	ID	EID1	THRU	EID2	BY	INC			

The Continuation Entry formats may be used more than once and in any order. They may also be used with either format above.

Continuation Entry Format 1:

	EID8	EID9	EID10	EID11	-etc.-				

Continuation Entry Format 2:

	EID8	“THRU”	EID9	“BY”	INC				

Example:

BSURF	15	5	THRU	21	BY	4			
	27	30	32	33					
	35	THRU	44						
	67	68	70	85	92				

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Fields:

Field	Contents
ID	Identification number of a contact or glue or wetted region. See Remarks 2 and 3. (Integer > 0)
EIDi	Element identification numbers of shell elements (CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, and CQUADR entries). (Integer > 0)
INC	Identification number increment. See Remark 4 (Integer or blank).

Remarks:

1. The continuation field is optional.
2. BSURF is a collection of one or more shell elements (CTRIA3, CTRIA6, CTRIAR, CQUAD4,CQUAD8, CQUADR). BSURF defines a contact or glue region which may act as a source or target. It can also be used to define wetted faces in the context of SOL 401 Multistep Nonlinear STAR CCM+ Co-simulation. Wetted faces represent structural faces that become a fluid boundary in the SOL 401 Multiphysics Co-Simulation with STAR CCM+.
3. The ID must be unique with respect to all other BEDGE, BLSEG, BSURFS, BSURF, BCPROP, and BCPROPS entries.
4. For automatic generation of element ids, the default increment value is 1 if element numbers are increasing or -1 if element numbers are decreasing (i.e., the user need not specify BY and the increment value).
5. Understanding shell element normals, and making sure those in the same regions are consistent is very important to ensuring that contact elements will be created as expected. See the chapter on contact in the Simcenter Nastran User's Guide for more information.

Related Topics

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BSURFS

Contact or Glue or Co-Simulation (Wetted) Region using Solid Element Faces

Defines a glue or contact or Co-Simulation (wetted) region using solid element faces.

Format:

1	2	3	4	5	6	7	8	9	10
BSURFS	ID				EID1	G1	G2	G3	
	EID2	G1	G2	G3	EID3	G1	G2	G3	
	-etc-								

Example:

BSURFS	7				10	15	16	20	
	15	16	17	21	20	17	18	22	
	11	19	20	24	16	20	21	25	
	21	21	22	26					

Fields:

Field	Contents
ID	Identification number of a contact or glue or wetted region. See Remarks 2 and 3. (Integer > 0)
EIDi	Element identification numbers of solid elements (CHEXA, CPENTA, CPYRAM, or CTETRA entries). (Integer > 0)
G1 - G3	Identification numbers of 3 corner grid points on the face (triangular or quadrilateral) of the solid element. (Integer > 0)

Remarks:

1. The continuation field is optional.
2. BSURFS is a collection of one or more element faces on solid elements. BSURFS defines a contact or glue region which may act as a source or target. It can be used to define wetted faces in the context of SOL 401 Multistep Nonlinear STAR CCM+ Co-simulation. Wetted faces represent structural faces that become a fluid boundary in the SOL 401 Multiphysics Co-Simulation with STAR CCM+.
3. The ID must be unique with respect to all other BEDGE, BLSEG, BSURFS, BSURF, BCPROP, and BCPROPS entries.

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4. The collapsed CHEXA element used in a SOL 401 crack simulation is not supported in a glue or contact region.

Related Topics

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BVTADD

Virtual Thread Contact Set Combination

Defines a virtual thread contact set as a union of virtual thread contact sets defined on BVTSET entries.

Format:

1	2	3	4	5	6	7	8	9	10
BVTADD	SID	BID1	BID2	BID3	BID4	BID5	BID6	BID7	
	BID8	BID9	BID10	-etc-					

Example:

BVTADD	12	1	3	6	5	9			

Fields:

Field	Contents
SID	Virtual thread contact set identification number. (Integer > 0)
BIDi	Identification numbers of virtual thread contact sets defined via BVTSET entries. (Integer > 0)

Remarks:

1. Multiple virtual thread contact sets (BVTSETs) with unique SIDs can be combined by including their CSIDs on a BVTADD entry. The BVTADD entry has its own unique SID which is used on the BVSET case control command. Multiple virtual thread contact sets with their own SID are necessary when different parameters (BVTPAR) are needed for different virtual thread contact sets.
2. SID must be unique and may not be the identification of this or any other BVTADD entry.

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Related Topics

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BVTPAR

Virtual Thread Contact Parameters for SOL 402

Defines virtual thread contact parameters for SOL 402.

Format:

1	2	3	4	5	6	7	8	9	10
BVTPAR	CSID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4	PARAM5	VALUE5	-etc-				

Example:

BVTPAR	101	NSIDE	1	INITPENE	1	SEGNORM	0		
	ACTIVE	1	DISP	1	TIED	0	FRICMOD	5	
	FPARA1	.1	FPARA2	.2					

Fields:

Field	Contents
CSID	Virtual thread contact set ID. (Integer > 0)
PARAMi	Name of the BVTPAR2 parameter. Allowable names are given in the parameter listing below. (Character)
VALUEi	Value of the parameter. (Real or Integer)

Table 10-53: General Parameters

Name	Description
PDEPTH	Distance (penetration or gap) used to limit contact detection for target contact surface. (Real ≥ 0.0; Default=5 * the size of the greatest target face; Unit = Length) If PDEPTH > 0.0, then contact element is created only if the distance (penetration or gap) is less than or equal to PDEPTH.
SEGNORM	Indicates whether a continuous (interpolated) contact segment normal (and associated smoothed surface) is used for the contact surfaces. (Integer; Default=0) 0: SEGNORM=1 if NSIDE=1, SEGNORM=-1 if NSIDE=2. 1: Continuous segment normal is used.

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Name	Description
	-1: Continuous segment normal is not used.

Table 10-54: Contact algorithm

Name	Description
TIED	Indicates whether contact regions in each contact pair are tied together in the normal direction. (Integer; Default=0) 0: Not tied. 1: Tied. 2: Remains tied after contact is active.
TIEDTOL	Tolerance used to determine whether contact nodes are tied to the target region when TIED=1 is specified. A contact node is tied to its target region if the distance between them is less than or equal to TIEDTOL. (Real; Default=0.0; Unit = Length)
EXTFAC	Factor for extending contact surfaces beyond their boundaries. The amount of extension is given by this factor multiplied by the length of the contact segments (Real ≥ 0; Unitless). The default value depends on how the DISP parameter is defined. If DISP = 1 (small sliding), the default is EXTFAC = 0.1. If DISP = 2 (large sliding), the default is EXTFAC = 0.025.
NPENAL2	Function/table that describes the relation between the normal distance and the contact pressure. See Remark 3. (Table units: Ordinate=Length; Abscissa=Pressure)
NCMOD	Normal contact modulus. See Remark 3. (Real -2.0, 0.0, or > 0.0; No default; Unit =Pressure/Length). -2.0: an automatic value is computed from the stiffnesses of both contact supports and the average element size of the contact target support. 0.0: fully rigid contact. > 1.0E-16: define the stiffness to use.
CFACTOR1	Compliance factor. See Remark 3. (Real >0.0; Default=1.0E-06; Unit = Length^3/Force)

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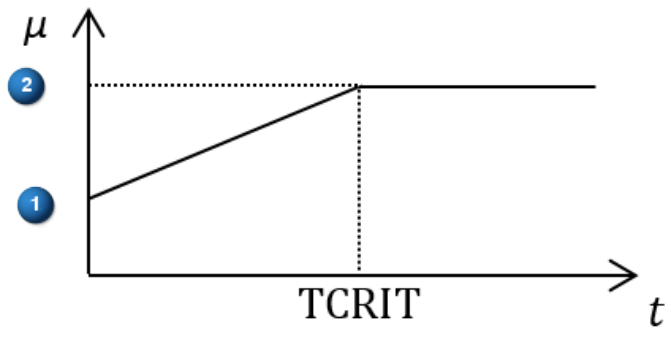
Table 10-55: Friction models

Name	Description
FRICMOD	Type of friction model (0 ≤ Integer ≤ 5; Default=0).

FRICMOD value	Description
0	Constant friction. Uses the friction coefficient FRICi specified for each contact pair in the BCTSET entry. All friction parameters of the BCTPAR2 entry (FPARA1, FPARA2, VCRIT, TCRIT, CFTE, CFVE, CFNF) are ignored.
1	Friction depends on time, velocity or temperature. Uses the friction coefficient FPARA1 on the BCTPAR2 entry. The friction coefficient is computed as being the product of the constant friction value defined by FPARA1 and existing functions depending on time (CFNF), temperature (CFTE) or velocity (CFVE). μ(t,T,v)= FPARA1 * CFVE(v) * CFTE(T) * CFNF(t) where t=time, v=velocity and T=temperature. If a table is not defined, its contribution to the formula is ignored. All other friction parameters of the BCTPAR2 entry (FPARA2, VCRIT, TCRIT) and the BCTSET entry (FRICi) are ignored.
2	Infinite friction. Uses an infinite friction coefficient. All friction parameters of the BCTPAR2 entry (FPARA1, FPARA2, VCRIT, TCRIT, CFTE, CFVE, CFNF) and the BCTSET entry (FRICi) are ignored.
5	Friction depends on a linear law of the time. 

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Name	Description

FRICMOD value	Description
	Uses a friction coefficient that varies linearly with time between two values FPARA1 (1) and FPARA2 (2), up to a transition time TCRIT after which it remains constant. All other friction parameters of the BCTPAR2 entry (VCRIT, CFTE, CFVE, CFNF) and the BCTSET entry (FRICi) are ignored.

FPARA1	Friction parameter 1 (Real ≥ 0; Unitless).
FPARA2	Friction parameter 2 (Real ≥ 0; Unitless).
TCRIT	Transition Time (Real > 0; Units = Time)
CFVE	Friction coefficient vs sliding velocity table. (Integer; Table units: Ordinate=Unitless; Abscissa= Length/Time).
CFNF	Friction coefficient vs time table. (Integer; Table units: Ordinate=Unitless; Abscissa= Time).
CFTE	Friction coefficient vs temperature table. (Integer; Table units: Ordinate=Unitless; Abscissa= Temperature).

Table 10-56: Regularization models

Name	Description
To avoid convergence problems when the contact starts sliding, we recommend that you introduce a regularization, either in terms of displacements or sliding velocity. This ensures a smooth transition of the friction stress between sliding and no sliding.	
STFR	Stiffness between the friction stress and the relative displacement. When the friction stress becomes equal to the pressure multiplied by the friction coefficient, this stiffness has no more influence. (Real -2.0, 0.0 or > 0; Default= -2.0; Unit =Pressure/Length). -2.0: an automatic value is computed using the stiffnesses of both contact supports and the average element size of the contact target support. 0.0: disabled (abrupt transition between sticking and sliding; not recommended). > 0.0: define the stiffness to use. This parameter is mutually exclusive with TOL.
PRCS	Normal regularization type. (Real; Default = -3). > 0.0: define the regularization value to use.

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Name	Description
	-1: use the characteristic stiffness of the whole structure. -2: use the characteristic stiffness of each contact support instead of the stiffness of the whole structure. -3: automatic choice: if the characteristic stiffnesses of both contact supports are very different, the characteristic stiffness of each support is used, otherwise the characteristic stiffness of the whole structure is used.

Note

Both regularization parameters (STFR and TOL) are mutually exclusive and a FATAL error is generated if both are specified. If none is specified, the solver uses displacement regularization with STFR=1.E8. To disable regularization, you must specify STFR=0.

Table 10-57: Damping models

Name	Description
In dynamic analysis (ANALYSIS=DYNAMICS), you can add a contact tangential viscous pressure. The viscous pressure is equal to the product of the contact velocity (in normal, tangential, or both directions) with a function of the normal distance.	
DPARA1	Normal velocity coefficient (INTEGER specifies a TABLEDi table whose units are a coefficient as a function of the normal penetration distance; Table units: Ordinate=Pressure/Speed [Force*Time/Length^3]; Abscissa= Length). For a given normal penetration distance, the value of the table is then multiplied by the normal velocity to produce a viscous pressure.
DPARA2	Tangential sliding velocity coefficient (INTEGER specifies a TABLEDi table whose units are a coefficient as a function of the normal penetration distance; Table units: Ordinate=Pressure/Speed [Force*Time/Length^3]; Abscissa= Length). For a given normal penetration distance, the value of the table is then multiplied by the tangential sliding velocity to produce a viscous pressure.

Remarks:

1. Field 9 on line 1 must be blank. Beginning on the continuation lines, fields 2 to 9 can be used for 4 pairs of PARAM/VALUE.
2. The BVTPAR bulk entry is not required. When it is not present, the default values are used. At least one parameter must be defined when a BVTPAR entry exists.
3. NPENAL2, NCMOD, and CFACTOR1 are mutually exclusive.
4. Parameters cannot be changed when you perform a restart run.

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Related Topics

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BVTSET

Virtual Thread Contact Set Definition

Defines a virtual thread contact set (SOL 402).

Format:

1	2	3	4	5	6	7	8	9	10
BVTSET	VTSID	OUTSID	INSID	FRIC	N1	N2		DID	
	PITCH	ANGLE	THDOPT						

Example:

BVTSET	4	3	2	0.0	102	104			
	.23	23.2							

Fields:

Field	Contents
VTSID	Virtual thread contact set identification number. (Integer>0)
OUTSID	Selects the hole region ID defined with the BSURFS bulk entry. See Remarks. (Integer > 0)
INSID	Selects the bolt region ID defined with the BSURFS bulk entry. See Remarks. (Integer > 0)
FRIC	Defines static coefficient of friction for the virtual thread contact. This value can be overwritten using the FRICMOD parameter on the BVTPAR bulk entry. The direction of the frictional force is parallel with the thread. (Real; Default=0.0)
N1 N2	Defines the bolt center axis and direction. You select either two grid points (GRID bulk entry) or two points (POINT bulk entry) located on the bolt center axis. The order you define N1 and N2 changes the contact normals for the THDOPT=-1 option. See Remarks. (Integer > 0)
DID	Identification number of a DESC bulk entry, which includes a contact set description.
PITCH	Defines the thread pitch (*P*) as shown in the following image. (Real > 0.0)

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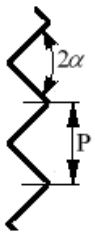
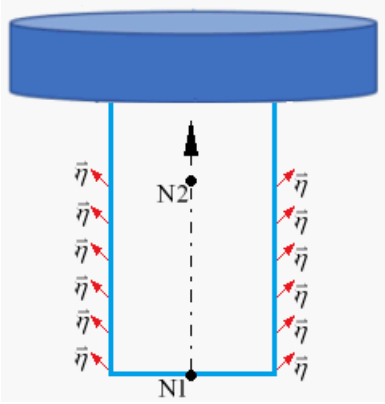
Field	Contents
	
ANGLE	Defines half the thread angle in degrees (a) as shown in the previous image. (0.0 < Real < 90.0)
THDOPT	Defines how the software creates the contact normals. (Integer; -1 or 0; Default=-1) For all of the THDOPT choices described below, the locations where the software creates the contact conditions is the same. The difference in the options is how the normals are computed at these locations. • THDOPT=-1 (Default) The software uses the thread angle (a) and the N1 to N2 vector to determine a consistent contact normal for all locations on the bolt surface. The thread pitch is ignored in this case. If N1 is defined on the bottom of the bolt and N2 higher on the bolt, the software uses the contact normal relative to the top of the thread (bolt in tension) in all locations. The following image illustrates this case.  If N1 is defined higher on the bolt and N2 is defined on the bottom of the bolt, the software uses the contact normal relative to the bottom of the thread (bolt in compression) in all locations. The following image illustrates this case.

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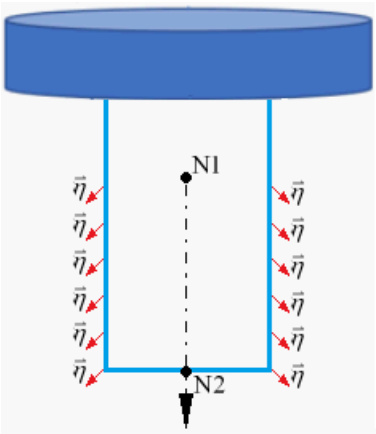
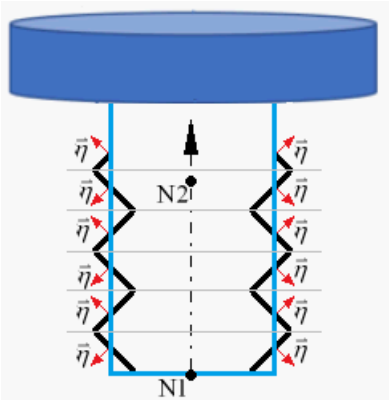
Field	Contents



- THDOPT=0

  The software uses the thread angle and pitch to determine the alternating contact normals for the top and bottom of the threads. For this case, you must define N1 on the bottom of the bolt and N2 higher on the bolt. The helix of the thread is not represented.



Remarks:

1. The outer and inner regions must be cylindrical and defined using BSURFS bulk entries. All 3D solid elements are supported: CHEXA, CPENTA, CTETRA, CPYRAM, including either linear or parabolic. Mixed-order elements are not supported.
2. The surfaces in the bolt and hole regions can be separated (gap) or interfering (penetration). The software will reset these conditions to a 0.0 gap/penetration condition at the beginning of the solution.
3. The software does not update the virtual thread contact condition as the solution proceeds. As a result, the bolt and hole surfaces should only experience small relative sliding. Although, the entire bolt and hole assembly can experience large displacement and rotation.
4. You can approximate the flexibility (normal stiffness) of the threads using the parameters NCMOD, NPENAL2, and CFACTOR1 on the BVTPAR bulk entry.

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Related Topics

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BWIDTH

Boundary Line Segment Width or Thickness

Defines widths or thicknesses for line segments in 3-D or 2-D slideline contact defined in the corresponding BLSEG Bulk Data entry.

Format:

1	2	3	4	5	6	7	8	9	10
BWIDTH	ID	W1	W2	W3	W4	W5	W6	W7	

Alternate Format:

BWIDTH	ID	W1	"THRU"	W2	"BY"	INC			

The Continuation Entry formats may be used more than once and in any order. They may also be used with either format above.

Continuation Entry Format 1:

	W8	W9	W10	W11	-etc.-				

Continuation Entry Format 2:

	W8	"THRU"	W9	"BY"	INC				

Example:

BWIDTH	15	2.0	THRU	5.0	BY	1.0			
	2.0	2.0	2.0	2.0					
	35.	THRU	44.						
	1.5	3.4	7.6	0.4	0.7				

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Fields:

Field	Contents
ID	BLSEG entry identification number. (Integer > 0)
Wi	Width values for the corresponding line segments defined in the BLSEG entry. See Remark 1 (Real > 0.0).
INC	Width value increment. See Remark 2 (Real or blank).

Remarks:

1. BWIDTH may be omitted if the width of each segment defined in the BLSEG entry is unity. The number of widths to be specified is equal to the number of segments defined in the corresponding BLSEG entry.
2. The default value for INC is 1.0 if the width is increasing or -1.0 if the width is decreasing. That is, the user need not specify BY and the increment value. If the number of widths specified is less than the number of segments defined in the corresponding BLSEG entry, the width for the remaining segments is assumed to be equal to the last width specified.
3. If there is only one grid point in the corresponding BLSEG entry, there is no contributory area associated with the grid point. To compute correct contact stresses, an area may be associated with a single grid point by specifying the area in field W1.

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11. Bulk Entries CA—CM

CAABSF

Frequency-Dependent Acoustic Absorber Element

Defines a frequency-dependent acoustic absorber element in coupled fluid-structural analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CAABSF	EID	PID	G1	G2	G3	G4			

Example:

CAABSF	44	38	1	10	20				

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number that matches a PAABSF or PAABSF1 entry. (Integer > 0; Default = EID). See Remark 2.
Gi	Grid point identification number of fluid connection points. (Integer ≥ 0 or blank)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. When CAABSF references:
   - The physical property PAABSF, the software supports the constant equivalent structural damping coefficient B and stiffness coefficient K.

     Point and line impedance are only supported through CAABSF that references PAABSF if the system cell 617 (ACFORM) is set to 0. ACFORM = 0 reverts to the NX Nastran 10 acoustics behavior.
   - The physical property PAABSF1, the software also supports the constant real and imaginary impedance and admittance on a 2D fluid free surface.
3. If only G1 is specified then a point impedance is assumed. If G1 and G2 are specified then a line impedance is assumed. If G1, G2, and G3 are specified, then an impedance is associated with the

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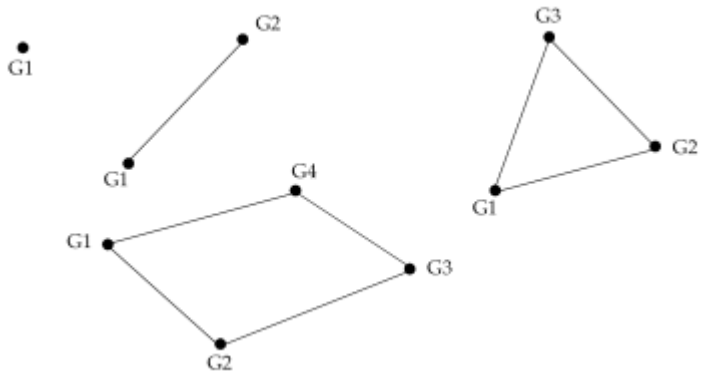
area of the triangular face. If G1 through G4 are specified, then an impedance is associated with the quadrilateral face. See Figure 11-15.

4. The CAABSF element must connect entirely to fluid points on the fluid-structure boundary.

Figure 11-15: Four Types of CAABSF Elements



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CAERO1

Aerodynamic Panel Element Connection

Defines an aerodynamic macro element (panel) in terms of two leading edge locations and side chords. This is used for the Doublet-Lattice theory for subsonic aerodynamics and, if licensed, the ZONA51 theory for supersonic aerodynamics.

Format:

1	2	3	4	5	6	7	8	9	10
CAERO1	EID	PID	CP	NSPAN	NCHORD	LSPAN	LCHORD	IGID	
	X1	Y1	Z1	X12	X4	Y4	Z4	X43	

Example:

CAERO1	1000	1		3			2	1	
	0.0	0.0	0.0	1.0	0.2	1.0	0.0	0.8	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PAERO1 entry; used to specify associated bodies. Required even if there are no associated bodies. (Integer > 0)
CP	Coordinate system for locating points 1 and 4. (Integer ≥ 0; Default = 0)
NSPAN	Number of spanwise boxes; if a positive value is given, NSPAN equal divisions are assumed. If zero or blank, a list of division points is given by LSPAN, field 7. (Integer ≥ 0)
NCHORD	Number of chordwise boxes; if a positive value is given NCHORD, equal divisions are assumed; if zero or blank, a list of division points is given at LCHORD, field 8. (Integer ≥ 0)
LSPAN	ID of an AEFACT data entry containing a list of division points for spanwise boxes. Used only if the NSPAN field 5 is zero or blank. (Integer > 0)
LCHORD	ID of an AEFACT data entry containing a list of division points for chordwise boxes. Used only if the NCHORD, field 6, is zero or blank. (Integer > 0)
IGID	Interference group identification; aerodynamic elements with different IGIDs are uncoupled. (Integer > 0) See Remark 8 for restriction.
X1, Y1, Z1	Location of points 1 and 4 in coordinate system CP. (Real)

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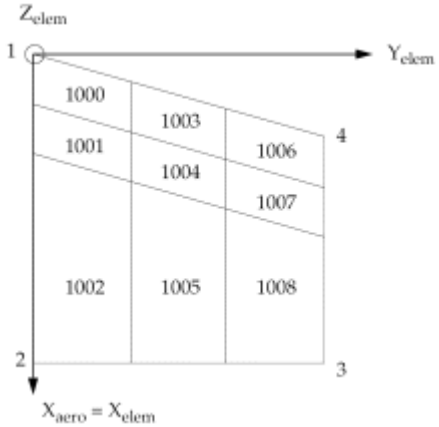
Field	Contents
X4, Y4, Z4	
X12, X43	Edge chord lengths in aerodynamic coordinate system. (Real ≥ 0.0, but not both zero.)

Remarks:

1. The boxes and corner point nodes are numbered sequentially, beginning with EID. The user should be careful to ensure that all box and corner point node numbers are unique. There can be overlapping ID's between the structural and aerodynamic model, but post processors may not be able to display any results. Also, non-unique corner ID's are allowed, but most post processor do not support this capability.
2. The number of division points is one greater than the number of boxes. Thus, if NSPAN=3, the division points are 0.0, 0.333, 0.667, 1.000. If the user supplies division points, the first and last points need not be 0.0 and 1.0 (in which case the corners of the panel would not be at the reference points).
3. A triangular element is formed if X12 or X43=0.0.
4. The element coordinate system is right-handed as shown in **Figure 11-16**.
5. The continuation is required.
6. It is recommended that NCHORD or LCHORD be chosen such that the typical box chord length Δx satisfies the condition $\Delta x < 0.08V/f$. Recent studies indicate that .02 V/f is needed to get converged stability derivatives. Where V is the minimum velocity and f, in hertz, is the maximum frequency to be analyzed (see the *Simcenter Nastran Aeroelastic Analysis User's Guide)*.
7. This entry can be used for two different aerodynamic theories. The Doublet-Lattice theory is used for subsonic and, if licensed, the ZONA51 theory for supersonic. The proper theory is selected based on the specification of Mach number on the MKAEROi or TRIM entry.

Figure 11-16: Element Coordinate System for Aerodynamic Panel



8. CAERO1 and CAERO2 panels should be defined in an IGID increasing order. For example, if you have a CAERO1 or CAER02 card with EID1 and ... IGID1, and another with EID2 and ... IGID2, IGID2 must

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be greater than IGID1, independent of whether EID1 < or > EID2. Failure to follow this restriction will create an erroneous pressure distribution resulting in incorrect flutter results.

Related Topics

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CAERO2

Aerodynamic Body Connection

Defines aerodynamic slender body and interference elements for Doublet-Lattice aerodynamics.

Format:

1	2	3	4	5	6	7	8	9	10
CAERO2	EID	PID	CP	NSB	NINT	LSB	LINT	IGID	
	X1	Y1	Z1	X12					

Example:

CAERO2	1500	2	100		4	99		1	
	-1.0	100.	-30.	175.					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PAERO2 entry. (Integer > 0)
CP	Coordinate system for locating point 1. (Integer ≥ 0; Default = 0)
NSB	Number of slender body elements. If NSB > 0, then NSB equal divisions are assumed; if zero or blank, specify a list of divisions in LSB. (Integer ≥ 0)
NINT	Number of interference elements. If NINT > 0, then NINT equal divisions are assumed; if zero or blank, specify a list of divisions in LINT. (Integer ≥ 0)
LSB	ID of an AEFACT Bulk Data entry for slender body division points; used only if NSB is zero or blank. (Integer ≥ 0)
LINT	ID of an AEFACT data entry containing a list of division points for interference elements; used only if NINT is zero or blank. (Integer ≥ 0)
IGID	Interference group identification. Aerodynamic elements with different IGIDs are uncoupled. (Integer ≥ 0) See remark 8 for restriction.
X1, Y1, Z1	Location of point 1 in coordinate system CP. (Real)
X12	Length of body in the x-direction of the aerodynamic coordinate system. (Real ≥ 0)

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Remarks:

1. Point 1 is the leading point of the body.
2. All CAERO1 (panels) and CAERO2 (bodies) in the same group (IGID) will have aerodynamic interaction.
3. At least one interference element is required for the aerodynamic body specified by this entry.
4. The beams and connection points are numbered sequentially beginning with EID. The user should be careful to ensure that all aero elements and connection point IDs are unique. Overlapping IDs between structure and aerodynamic models are allowed, but will prevent results visualization in most post processors.

   Old rules regarding numbering among Z, ZY, Y bodies and CAERO1 no longer apply: arbitrary ordering is allowed.
5. At least two slender body elements are required for each aerodynamic body.
6. Interference elements are only intended for use with panels.
7. Determining the size of the j-set (i.e., the number of aerodynamic elements) is essential to input D1JE and D2JE matrices. Use the following expressions for locating the proper row in the two matrices:

   J = Number of boxes = + Number of I-elements, z
   = + 2*(Number of I-elements, zy)
   = + Number of I-elements, y
   = + Number of S-elements, z
   = + 2*(Number of S-elements, zy)
   = + Number of S-elements, y

   where I-elements denote interference and S-elements denote slender body.
8. CAERO1 and CAERO2 panels should be defined in an IGID increasing order. For example, if you have a CAERO1 or CAER02 card with EID1 and ... IGID1, and another with EID2 and ... IGID2, IGID2 must be greater than IGID1, independent of whether EID1 < or > EID2. Failure to follow this restriction will create an erroneous pressure distribution resulting in incorrect flutter results.

Related Topics

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CAERO3

Aerodynamic Panel Element Configuration

Defines the aerodynamic edges of a Mach Box lifting surface. If no cranks are present, this entry defines the aerodynamic Mach Box lifting surface.

Format:

1	2	3	4	5	6	7	8	9	10
CAERO3	EID	PID	CP	LISTW	LISTC1	LISTC2			
	X1	Y1	Z1	X12	X4	Y4	Z4	X43	

Example:

CAERO3	2000	2001	0	22	33				
	1.0	0.0	0.0	100.	17.	130.	0.	100.	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PAERO3 entry. (Integer > 0)
CP	Coordinate system for locating points 1 and 4. (Integer ≥ 0; Default = 0)
LISTW	Identification number of an AEFACT entry that lists (x,y) pairs for structural interpolation of the wing. (Integer > 0)
LISTC1, LISTC2	Identification number of AEFACT entries that list (x,y) pairs for control surfaces, if they exist. (Integer ≥ 0)
X1, Y1, Z1, X4, Y4, Z4	Location of points 1 and 4 in coordinate system CP. (Real)
X12, X43	Edge chord lengths in the aerodynamic coordinate system. (Real ≥ 0, X12 ≠ 0)

Remarks:

1. EID must be unique with respect to all other element identification numbers.
2. The (x,y) pairs on LISTW, LISTC1 and LISTC2 AEFACT entries are in the aero element coordinate system (see Figure 11-17). The (x,y) pairs define a set of aerodynamic grid points that are independent of the Mach number and are selected by the user to be representative of the platform and motions of interest. The (x,y) pairs must be sufficient in number and distribution such that: the surface spline

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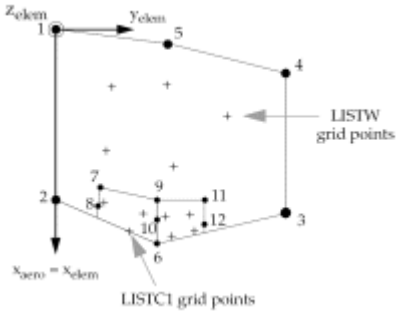
provides an accurate interpolation between them and the Mach Box centers that are variously located on the planform as a function of Mach number. A complete description of the Mach Box Method is given in the *Simcenter Nastran Aeroelastic Analysis User's Guide*.

3. The (x,y) pairs are numbered sequentially, beginning with EID for LISTW, then LISTC1, and finally for LISTC2. On SPLINEi entries, the box numbers (BOX1 and BOX2 on SPLINE1, ID1 and ID2 on SPLINE2, and UKID on SPLINE3) refer to the (x,y) pair sequence number appropriate for the surface (primary, or first or second control) being splined.
4. If cranks and/or control surfaces exist, their locations are given on the PAERO3 entry.
5. The numbering system and coordinate system are shown below:

Figure 11-17: CAERO3 Element Configuration



Planform Corners		Control	
1	Leading edge, inboard	7	Hinge line, inboard
2	Trailing edge, inboard	8	On inboard edge (usually at trailing edge)
3	Trailing edge, outboard	9	Hinge line, outboard
4	Leading edge, outboard	10	On outboard edge (usually at trailing edge)
Cranks		**Control (if two)**	
5	Leading edge	9	Hinge line, inboard
6	Trailing edge	10	On inboard edge (usually at trailing edge)
		11	Hinge line, outboard
		12	On outboard edge (usually at trailing edge)

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CAERO4

Aerodynamic Macro-Strip Element Connection

Defines an aerodynamic macro element for Strip theory.

Format:

1	2	3	4	5	6	7	8	9	10
CAERO4	EID	PID	CP	NSPAN	LSPAN				
	X1	Y1	Z1	X12	X4	Y4	Z4	X43	

Example:

CAERO4	6000	6001	100		315				
	0.0	0.0	0.0	1.0	0.2	1.0	0.0	0.8	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PAERO4 entry. (Integer > 0)
CP	Coordinate system for locating points 1 and 4. (Integer ≥ 0; Default = 0)
NSPAN	Number of strips; if a positive value is given, NSPAN equal strips are assumed. If zero or blank, LSPAN must be specified. (Integer ≥ 0)
LSPAN	ID of an AEFACT entry containing a list of division points for strips. Used only if NSPAN is zero or blank. (Integer > 0)
X1, Y1, Z1, X4, Y4, Z4	Location of points 1 and 4 in coordinate system CP. (Real)
X12, X43	Edge chord lengths in aerodynamic coordinate system. (Real ≥ 0.0, and not both zero.)

Remarks:

1. The strips are numbered sequentially, beginning with EID. The user must ensure that all strip numbers are unique and greater than structural grid, scalar, and extra point IDs.
2. The number of division points is one greater than the number of boxes. Thus, if NSPAN = 3, the division points are 0.0, 0.333, 0.667, and 1.000. If the user supplies division points, the first and last points need not be 0.0 and 1.0 (in which case the corners of the panel would not be at the reference points).

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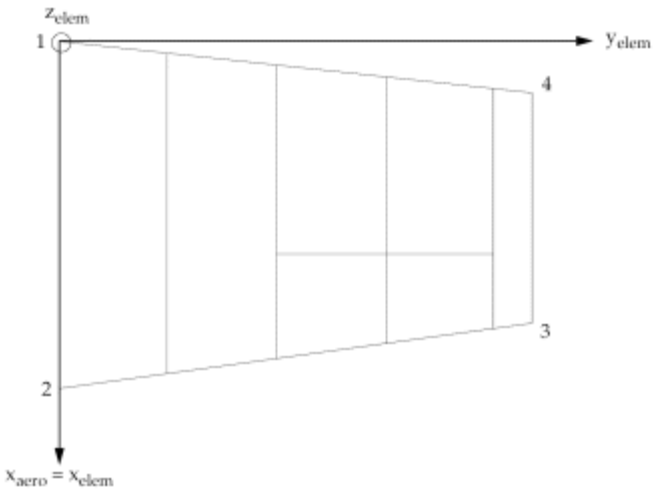
3. A triangular element is formed if X12 or X43 = 0.

Figure 11-18: CAERO4 Element Connection



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CAERO5

Aerodynamic Panel Element Configuration

Defines an aerodynamic macro element for Piston theory.

Format:

1	2	3	4	5	6	7	8	9	10
CAERO5	EID	PID	CP	NSPAN	LSPAN	NTHRY	NTHICK		
	X1	Y1	Z1	X12	X4	Y4	Z4	X43	

Example:

CAERO5	6000	6001	100		315	0	0		
	0.0	0.0	0.0	1.0	0.2	1.0	0.	0.8	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PAERO5 entry. (Integer > 0)
CP	Coordinate system for locating points 1 and 4. (Integer ≥ 0; Default = 0)
NSPAN	Number of strips. If a positive value is given, equal strips are assumed. If zero or blank, then LSPAN must be specified. (Integer)
LSPAN	ID of an AEFACT entry containing a list of division points for strips. Used only if NSPAN is zero or blank. (Integer > 0)
NTHRY	Parameter to select Piston or van Dyke's theory. (Integer = 0, 1, or 2; Default = 0) Blank - Piston theory is used to compute $\bar{C}_1$ and $\bar{C}_2$ or 1 1 = van Dyke's theory is used to compute $\bar{C}_1$ and $\bar{C}_2$ with no sweep correction (secΛ = 1.0). 2 = van Dyke's theory is used to compute $\bar{C}_1$ and $\bar{C}_2$ with a sweep correction based on the actual Λ.
NTHICK	Parameter to select thickness integrals input. (Integer ≥ 0; Default = 0)

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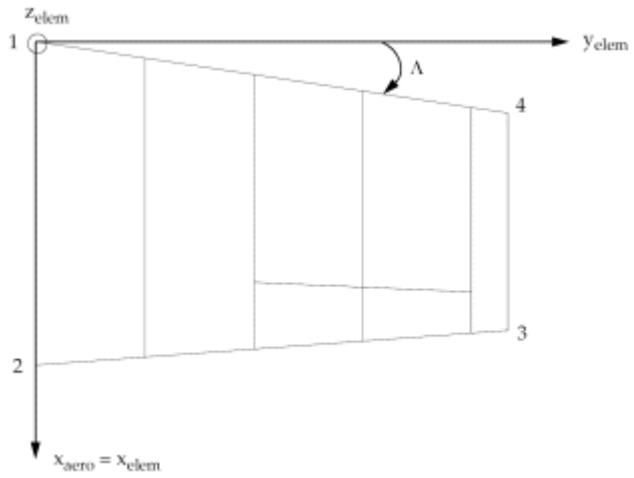
Field	Contents
	0 or Blank - Thickness integrals are computed internally. >0 - Thickness integrals are input directly and are the ID number of an AEFACT entry that lists the *I* and/or *J* integrals.
X1, Y1, Z1, X4, Y4, Z4	Location of points 1 and 4 in coordinate system CP. (Real)
X12, X43	Edge chord lengths in aerodynamic coordinate system. (Real ≥ 0; X12 and X43 cannot both be zero.)

Figure 11-19: CAERO5 Element Configuration



Remarks:

1. The strips are numbered sequentially, beginning with EID. The user must ensure that all strip numbers are unique and different from structural grid IDs.
2. The number of division points is one greater than the number of boxes. Thus, if NSPAN=3, the division points are 0.0, 0.333, 0.667, 1.000. If the user supplies division points, the first and last points need not be 0.0 and 1.0 (in which case the corners of the panel would not be at the reference points).
3. A triangular element is formed if X12 or X43 = 0.0.
4. $\bar{C}_1 = m / (m^2 - \sec^2 \Lambda)^{1/2}$

   $\bar{C}_2 = [m^4 (\gamma + 1) - 4 \sec^2 \Lambda (m^2 - \sec^2 \Lambda)] / [4(m^2 - \sec^2 \Lambda)^2]$

   where:

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m	=	Mach number
γ	=	Specific heat ratio
Λ	=	Leading edge sweep angle

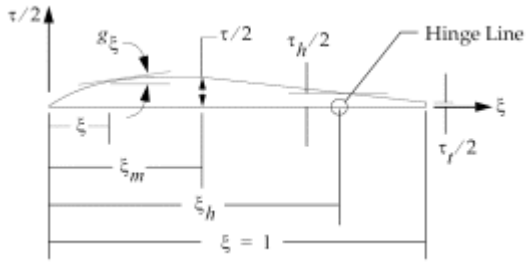
When sec Λ = 0.0, Piston theory coefficients are obtained (NTHRY = 1).

When sec Λ = 1.0, van Dyke coefficients are obtained (NTHRY = blank or 0).

When sec Λ ≠ 0.0 or ≠ 1.0, sweep corrections are included (NTHRY = 2).

5. *I* and *J* thickness integral definitions:

Figure 11-20: CAERO5 I and J Thickness Integral Definitions



$$g_\xi \equiv \frac{dg}{d\xi} = \text{slope of airfoil semithickness}$$

$$I_1 = \int_0^1 g_\xi d\xi \qquad J_1 = \int_{\xi_h}^1 g_\xi d\xi$$

$$I_2 = \int_0^1 \xi g_\xi d\xi \qquad J_2 = \int_{\xi_h}^1 \xi g_\xi d\xi$$

$$I_3 = \int_0^1 \xi^2 g_\xi d\xi \qquad J_3 = \int_{\xi_h}^1 \xi^2 g_\xi d\xi$$

$$I_4 = \int_0^1 g_\xi{}^2 d\xi \qquad J_4 = \int_{\xi_h}^1 g^2{}_\xi d\xi$$

$$I_5 = \int_0^1 \xi g_\xi{}^2 d\xi \qquad I_5 = \int_{\xi_h}^1 \xi g^2{}_\xi d\xi$$

$$I_6 = \int_0^1 \xi^2 g_\xi{}^2 d\xi \qquad J_6 = \int_{\xi_h}^1 \xi^2 g_\xi{}^2 d\xi$$

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CBAR

Simple Beam Element Connection

Defines a simple beam element.

Format:

1	2	3	4	5	6	7	8	9	10
CBAR	EID	PID	GA	GB	X1	X2	X3	OFFT	
	PA	PB	W1A	W2A	W3A	W1B	W2B	W3B	

Example:

CBAR	2	39	7	3	0.6	18.	26.		
		513							

Alternate Format and Example:

CBAR	EID	PID	GA	GB	G0			OFFT	
	PA	PB	W1A	W2A	W3A	W1B	W2B	W3B	

CBAR	2	39	7	6	105				
		513							

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Property identification number of a PBAR or PBARL entry. (Integer > 0 or blank; Default = EID unless BAROR has a nonzero entry in field 3.)
GA, GB	Grid point identification numbers of connection points. (Integer > 0; GA ≠ GB)
X1, X2, X3	Components of orientation vector , from GA, in the displacement coordinate system at GA. (Real)
G0	Alternate method to supply the orientation vector

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Field	Contents
	$\vec{v}$ using grid point G0. The direction of $\vec{v}$ is from GA to G0. $\vec{v}$ is then translated to End A. (Integer > 0; G0 ≠ GA or GB)
OFFT	Three character input that designates the coordinate systems used to define the cross section orientation vector and the two end offset vectors. See Remark 9. (Character; Default="GGG")
PA, PB	Pin flags for bar ends A and B, respectively. Used to remove connections between the grid point and selected degrees-of-freedom of the bar. The degrees-of-freedom are defined in the element's coordinate system (see Figure 11-21). The bar must have stiffness associated with the PA and PB degrees-of-freedom to be released by the pin flags. For example, if PA = 4 is specified, the PBAR entry must have a value for the torsional constant J. (Up to 5 of the unique integers 1 through 6 anywhere in the field with no embedded blanks; Integer > 0.)
W1A, W2A, W3A, W1B, W2B, W3B	Components of offset vectors $\vec{w}_a$ and $\vec{w}_b$, respectively (see Figure 11-21) in displacement coordinate systems at points GA and GB, respectively. See Remarks 7 and 8. (Real; Default = 0.0)

Note

See the BEAMOR entry for default options for field 3 and fields 6 through 8.

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. The CBAR is referred to as a simple beam since it does not have the full capabilities of the CBEAM, for example, varying cross-section and warping. See the 1D Elements Chapter in the *Simcenter Nastran Element Library Reference* for details on both CBAR and CBEAM elements.
3. Figure 11-21 and Figure 11-22 define bar element geometry with and without offsets.

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Figure 11-21: CBAR Element Geometry with Offsets

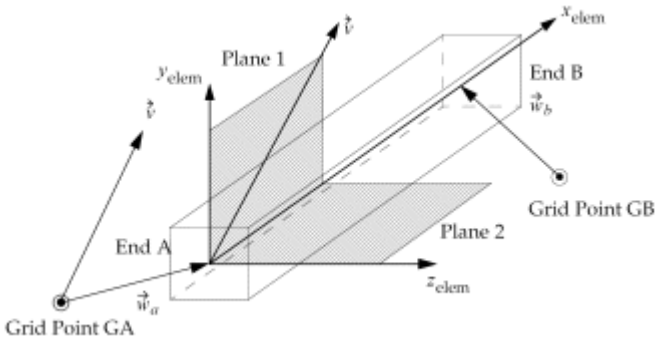
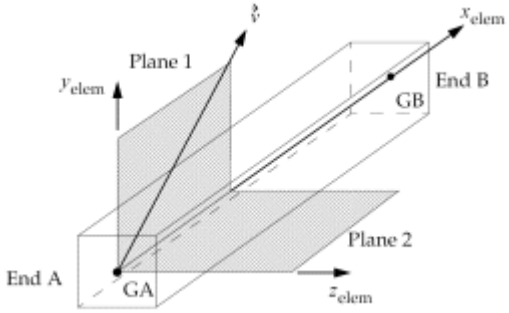


Figure 11-22: CBAR Element Geometry without Offsets



4. Figure 11-23 defines the elemental force and moment sign convention.

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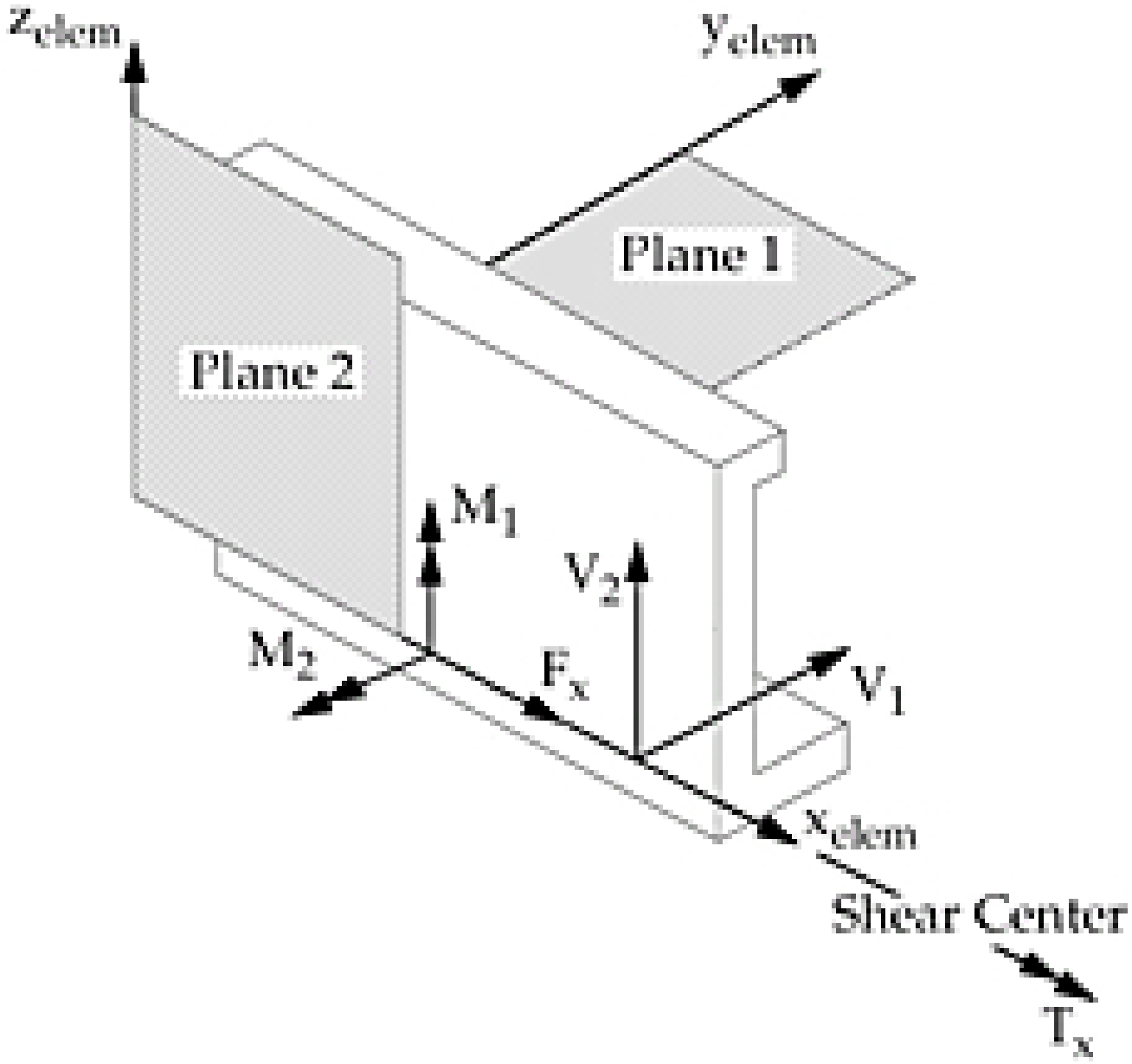
Figure 11-23: CBAR Element Internal Forces and Moments



5. The continuation may be omitted if there are no pin flags or offsets.
6. For the case where field 9 is blank and not provided by the BAROR entry, if an integer is specified in field 6, then G0 is used; if field 6 is blank or real, then X1, X2, X3 is used.
7. Element offsets are defined in a Cartesian system located at the connecting grid point. The components of the offsets are always defined in units of translation, even if the displacement coordinate system is cylindrical or spherical. For example, in Figure 11-24, the grid point displacement coordinate system is cylindrical, and the offset vector is defined using Cartesian coordinates u1, u2, and u3 in units of translation.

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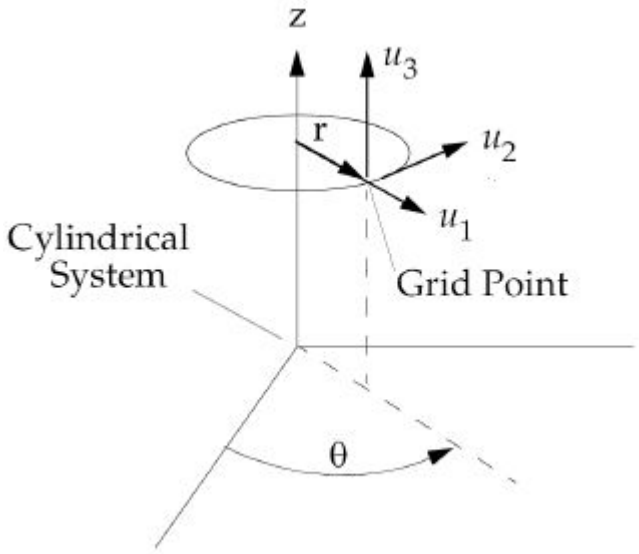
Figure 11-24: OFFSET Coordinates



8. Offset vectors are not affected by thermal loads.

9. The OFFT field is a three character input which designates the coordinate systems (CSYS) used to define the cross section orientation vector and the two end offset vectors. With the default of "GGG", a displacement (global) CSYS is used for all three.

   - The first character defines the CSYS used for the cross section orientation vector. When "G" is defined as the first character, the displacement (global) CSYS of grid point A is used. When "B" is defined as the first character, the basic CSYS is used.
   - The second character defines the CSYS used for the offset vector at end A. When "G" is defined as the second character, the displacement (global) CSYS of grid point A is used. When "O" is defined as the second character, the element CSYS is used. See Remark 3 for a definition of the element CSYS.
   - The third character defines the CSYS used for the offset vector at end B. When "G" is defined as the third character, the displacement (global) CSYS of grid point B is used. When "O" is defined as the third character, the element CSYS is used. See Remark 3 for a definition of the element CSYS.

   The table below lists the supported character inputs. A fatal error will occur if you enter a character input that is not listed.

Input	Orientation CSYS	Offset End A CSYS	Offset End B CSYS
GGG	Global[A]	Global[A]	Global[B]
BGG	Basic	Global[A]	Global[B]
GGO	Global[A]	Global[A]	Offset
BGO	Basic	Global[A]	Offset
GOG	Global[A]	Offset	Global[B]
BOG	Basic	Offset	Global[B]

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Input	Orientation CSYS	Offset End A CSYS	Offset End B CSYS
GOO	Global[A]	Offset	Offset
BOO	Basic	Offset	Offset

A: The displacement (global) CSYS of grid point A is used.

B: The displacement (global) CSYS of grid point B is used.

10. If G0 is used to define the orientation vector and the OFFT field is defined, the software will ignore the first character defined in the OFFT field, and will use the second and third characters as described in Remark 9. Three characters are still required in the OFFT field for this case.

Remarks related to SOLs 402 and 414:

1. For SOLs 402 and 414, CBAR is the same as the CBEAM element; therefore, CBAR takes the shear effect into account, just as the CBEAM element does.
2. For SOL 414 solutions, if your model requires a non-axisymmetric profile (T, C, U, I), do not use the CBAR element and the corresponding PBARL physical property. Instead, define a CBEAM element with PBEAML physical properties. The PBARL physical property assumes that the shear center corresponds to the neutral axis, as explained in Overview of 1D (Line) Elements. PBARL defined with a T, C, I, I profile causes an error in SOL 414 solutions.

Related Topics

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CBARAO

Auxiliary Output Points Along Bar Element Axis (CBAR Entry)

Defines a series of points along the axis of a bar element (CBAR entry) for stress and force recovery output.

Format:

1	2	3	4	5	6	7	8	9	10
CBARAO	EID	SCALE	X1	X2	X3	X4	X5	X6	

Example:

CBARAO	1065	FR	0.2	0.4	0.6	0.8			

Alternate Format and Example:

CBARAO	EID	SCALE	NPTS	X1	DELTAX				
CBARAO	1065	FR	4	0.2	0.2				

Fields:

Field	Contents
EID	Element identification of a CBAR entry. (Integer > 0)
SCALE	Defines scale of Xi values. (Character = “LE” or “FR”)
Xi	Series of locations along element axis for stress and force data recovery. (Real > 0)
DELTAX	Incremental distance along element axis. (Real)
NPTS	Number of stress recovery points, not including the end points. (Integer > 0)

Remarks:

1. This entry defines intermediate locations on the axis of selected CBAR elements for additional data recovery. The values of Xi are actual distances along the length if SCALE = “LE”. If SCALE = “FR”, the values of Xi are ratios of actual distances to the bar length. A PLOAD1 Bulk Data entry for the CBAR element in question must be present to obtain intermediate data recovery.
2. When the alternate format is used, a series of locations Xi = X[i-1]+DELTAX, i = 1, 2, ..., NPTS is generated.
3. If a CBARAO or PLOAD1 entry is specified and stress and/or force output is requested, then the stresses and/or forces will be calculated at each location Xi and output as a separate line. The force and stress

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values at the end points of the beam will always be output. This output format will be used for all beam and bar elements.

4. Intermediate loads on the element defined by the PLOAD1 entry will be accounted for in the calculation of element stresses and forces. If no PLOAD1 entry is defined for the element, the shear forces are constant, the moments are linear, and it is not necessary that the user define additional points.

5. For each bar element, either the basic format or the alternate format, but not both, may be used. A maximum of six internal points can be specified with the basic form. The end points must not be listed because data will be generated for them, as explained in Remark 3 If more than six unequally spaced internal points are desired, it is advisable to subdivide the bar into two or more elements.

Related Topics

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CBEAM

Beam Element Connection

Defines a beam element.

Format:

1	2	3	4	5	6	7	8	9	10
CBEAM	EID	PID	GA	GB	X1	X2	X3	OFFT	
	PA	PB	W1A	W2A	W3A	W1B	W2B	W3B	
	SA	SB							

Example:

CBEAM	2	39	7	13	8.2	6.1	-5.6		
		513		3.0					
	8	5							

Alternate Format and Example:

CBEAM	EID	PID	GA	GB	G0			OFFT	
	PA	PB	W1A	W2A	W3A	W1B	W2B	W3B	
	SA	SB							

CBEAM	2	39	7	13	105				
		513							

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Property identification number of PBEAM, PBCOMP or PBEAML entry. (Integer > 0; Default=EID)
GA, GB	Grid point identification numbers of connection points. (Integer > 0; GA ≠ GB)
X1, X2, X3	Components of orientation vector $\vec{v}$, from GA, in the displacement coordinate system at GA. (Real)

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Field	Contents
G0	Alternate method to supply the orientation vector $\vec{v}$ using grid point G0. Direction of $\vec{v}$ is from GA to G0. $\vec{v}$ is then transferred to end A. (Integer > 0; G0 ≠ GA or GB)
OFFT	Three character input which designates the coordinate systems (CSYS) used to define the cross section orientation vector and the two end offset vectors. See Remark 10. (Character; Default="GGG")
PA, PB	Pin flags for beam ends A and B, respectively; used to remove connections between the grid point and selected degrees-of-freedom of the beam. The degrees-of-freedom are defined in the element's coordinate system and the pin flags are applied at the offset ends of the beam (see **Figure 11-25**). The beam must have stiffness associated with the PA and PB degrees-of-freedom to be released by the pin flags. For example, if PA = 4, the PBEAM entry must have a nonzero value for the torsional constant J. (Up to five of the unique integers 1 through 6 with no embedded blanks.)
W1A, W2A, W3A, W1B, W2B, W3B	Components of offset vectors, measured in the displacement coordinate systems at grid points A and B, from the grid points to the end points of the axis of shear center. See Remarks **7** and **8**. (Real; Default = 0.0)
SA, SB	Scalar or grid point identification numbers for the ends A and B, respectively. The degrees-of-freedom at these points are the warping variables $d\theta/dx$. (Integers > 0 or blank)

Note

See the BEAMOR entry for default options for field 3 and fields 6 through 8.

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. The CBEAM element x-axis is defined by the line from End A to End B. When no offsets are defined, End A and End B are coincident with the grid point locations GA and GB. When offsets are defined, End A and End B shift by the offset.

   The orientation vector $\vec{v}$ and the element x-axis define the reference plane 1. The element y-axis lies on reference plane 1 and is perpendicular to the element x-axis. The element z-axis is perpendicular to the element x-axis and element y-axis using a right-handed coordinate direction. The reference plane 2 is defined by the x-axis and the z-axis. The PBEAM and PBEAML cross sectional properties are defined relative to reference plane 1 (element y-axis) and reference plane 2 (element z-axis). The cross-sectional shear center is always coincident with the element x-axis.

   Note that if G0 is used to define the orientation vector, the vector is always from the grid point location GA to G0, even when an offset is defined at End A.

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Figures Figure 11-25 and Figure 11-26 show CBEAM element geometry, forces and moments.

For an additional explanation of the beam element, see “CBEAM Element” in the *Simcenter Nastran Element Library*.

Figure 11-25: CBEAM Element Geometry System

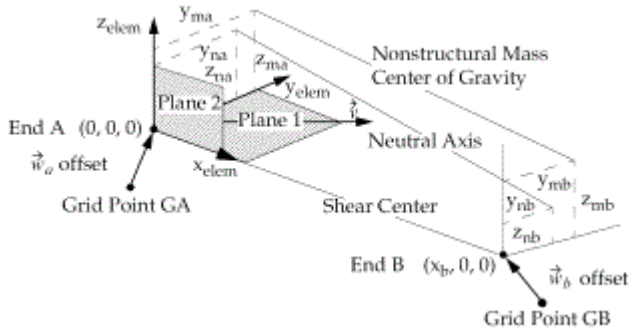
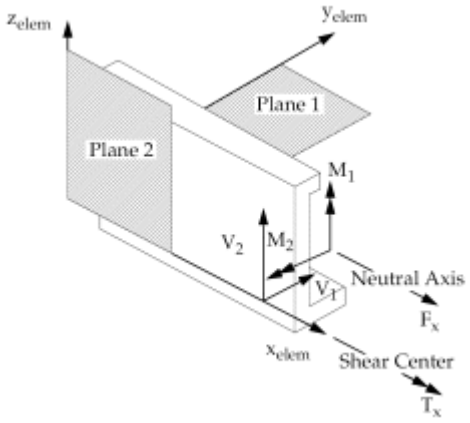


Figure 11-26: CBEAM Internal Element Forces and Moments



3. If field 6 is an integer, then G0 is used. If field 6 is blank or real, then X1, X2, X3 is used.
4. G0 cannot be located at GA or GB.
5. The rules for the continuation rows are:
   - Both continuations may be omitted if there are no pin flags, offsets, or warping variables.
   - If the second continuation is used, the first continuation must be included, even if all fields are blank.
   - If the second continuation is omitted, torsional stiffness due to warping of the cross section will not be considered.
6. If warping is allowed (SA and SB > 0), then SA and SB must be defined with SPOINT or GRID entries. If GRID entries are used, the warping degree-of-freedom is attached to the first (T1) component.
7. Offset vectors are subject to the following limitations.
   - Offset vectors are not affected by thermal loads.

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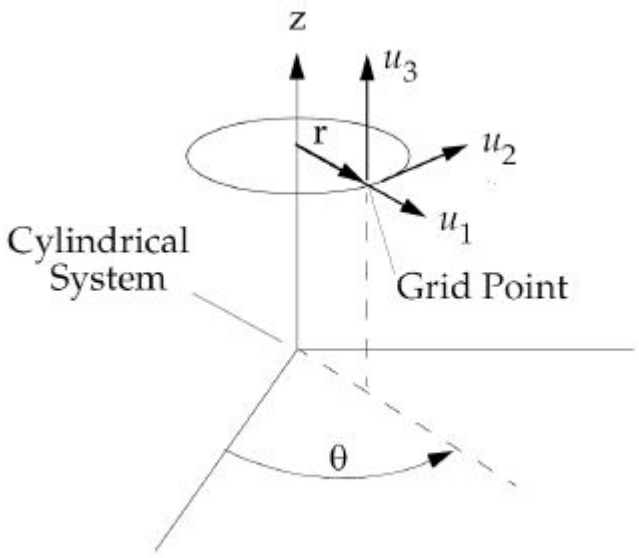
- The specification of offset vectors is not recommended with nonlinear solutions, SOLs 106, 129, 153, and 159. When geometry nonlinear conditions exist (PARAM,LGDISP,1), the offset vectors remain parallel to their original orientation when computing the differential stiffness. Also, when material nonlinear conditions are defined with the MATS1 entry, the offset vectors may produce incorrect results. As a result, the specification of offset vectors is not permitted in these solutions. However, setting SYSTEM(463) to 1 will disable the FATAL error message and the analysis will be allowed to continue (with offset vectors remaining parallel to their original orientation). The use of this system cell may generate incorrect results. Loading conditions that generate follower forces should not be used when SYSTEM(463) = 1. Use PARAM,FOLLOWK,NO if follower boundary conditions are specified.

8. Element offsets are defined in a Cartesian system located at the connecting grid point. The components of the offsets are always defined in units of translation, even if the displacement coordinate system is cylindrical or spherical. For example, in **Figure 11-27**, the grid point displacement coordinate system is cylindrical, and the offset vector is defined using Cartesian coordinates u1, u2, and u3 in units of translation.

Figure 11-27: OFFSET Coordinates



9. Pin flags PA and PB will produce a fatal error in SOLs 106 and 129 when nonlinear material conditions are defined with the MATS1, MATTi, or CREEP entries. They can be used if the only nonlinearity is geometric.

10. The OFFT field is a three character input that designates the coordinate systems (CSYS) used to define the cross section orientation vector and the two end offset vectors. With the default of "GGG", a displacement (global) CSYS is used for all three.

- The first character defines the CSYS used for the cross section orientation vector. When "G" is defined as the first character, the displacement (global) CSYS of grid point A is used. When "B" is defined as the first character, the basic CSYS is used.
- The second character defines the CSYS used for the offset vector at end A. When "G" is defined as the second character, the displacement (global) CSYS of grid point A is used. When "O" is defined as the second character, the element CSYS is used. See Remark 2 for a definition of the element CSYS.

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- The third character defines the CSYS used for the offset vector at end B. When "G" is defined as the third character, the displacement (global) CSYS of grid point B is used. When "O" is defined as the third character, the element CSYS is used. See Remark 2 for a definition of the element CSYS.

The table below lists the supported character inputs. A fatal error will occur if you enter a character input that is not listed.

Input	Orientation CSYS	Offset End A CSYS	Offset End B CSYS
GGG	Global[A]	Global[A]	Global[B]
BGG	Basic	Global[A]	Global[B]
GGO	Global[A]	Global[A]	Offset
BGO	Basic	Global[A]	Offset
GOG	Global[A]	Offset	Global[B]
BOG	Basic	Offset	Global[B]
GOO	Global[A]	Offset	Offset
BOO	Basic	Offset	Offset

A: The displacement (global) CSYS of grid point A is used.

B: The displacement (global) CSYS of grid point B is used.

11. If G0 is used to define the orientation vector and the OFFT field is defined, the software will ignore the first character defined in the OFFT field, and will use the second and third characters as described in Remark 10. Three characters are still required in the OFFT field for this case.

Remarks related to SOLs 402 and 414:

1. PA and PB are ignored.
2. SA and SB are grid points, not scalar points.

Related Topics

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CBEAR

Speed-Dependent Bearing Connection

Defines speed-dependent bearing connection. Valid for SOL 414 rotor dynamic solutions and legacy rotor dynamics solutions (SOL 101 and SOLs 107-112).

Format:

1	2	3	4	5	6	7	8	9	10
CBEAR	EID	PID	GA	GB					

Example:

CBEAR	1	2	100	101					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PBEAR entry. (Integer > 0)
GA, GB	Grid point identification number of bearing connection points. See Remark 3. (Integer > 0; GA ≠ GB)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. You must define a GROUP bulk entry for each rotor that has corresponding CBEAR entries. On each GROUP entry, list the CBEAR entries. Associate each GROUP entry with a rotor on the BRGSETi field on the ROTORD entry.
3. See the *Rotor Dynamics User's Guide (legacy)* for information on modeling bearings including bearings that carry thrust loads in only one axial direction.
4. For information about modeling bearings for SOL 414 rotor dynamics solutions, see *Simcenter Nastran Rotor Dynamics User's Guide (SOL 414)*.

Related Topics

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CBEAR2

Bearing Connection

Defines bearing connection in SOL 414.

Format:

1	2	3	4	5	6	7	8	9	10
CBEAR2	EID	PID	GA	GB	CID				

Example:

CBEAR2	1	2	100	101					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PBEAR2 entry. (Integer > 0; No default)
GA, GB	Grid point identification number of bearing connection points. (Integer > 0; GA ≠ GB; No default)
CID	Identification number of the bearing coordinate system. (Integer ≥ 0; Default = 0 or blank for the basic coordinate system)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.

Related Topics

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CBEND

Curved Beam or Pipe Element Connection

Defines a curved beam, curved pipe, or elbow element.

Format:

1	2	3	4	5	6	7	8	9	10
CBEND	EID	PID	GA	GB	X1	X2	X3	GEOM	

Example:

CBEND	32	39	17	19	6.2	5.1	-1.2		

Alternate Format and Example:

CBEND	EID	PID	GA	GB	G0			GEOM	

CBEND	32	39	17	19	106				

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Property identification number of a PBEND entry. (Integer > 0; Default = EID)
GA, GB	Grid point identification numbers of connection points. (Integer > 0; GA ≠ GB)
X1, X2, X3	Components of orientation vector $\vec{v}$, from GA, in the displacement coordinate system at GA. (Real)
G0	Alternate method to supply the orientation vector $\vec{v}$ using grid point G0. Direction of $\vec{v}$ is from GA to G0. $\vec{v}$ is then translated to end A. (Integer > 0; G0 ≠ GA or GB)
GEOM	Flag to select specification of the bend element. See Remark 3 (1 < Integer < 4)

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Remarks:

1. Element identification numbers must be unique with respect to all other element identification numbers.
2. For an additional explanation of the CBEND element, see the PBEND entry description. Figure 11-28 and Figure 11-29 define the element coordinate system and internal forces and moments.

Figure 11-28: CBEND Element Coordinate System

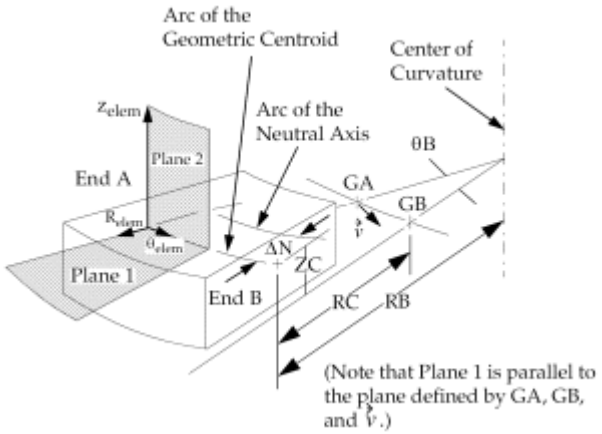
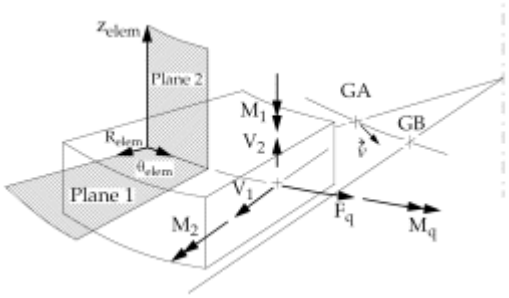
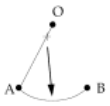
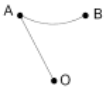


Figure 11-29: CBEND Element Internal Forces and Moments



3. The options for element connection to GA, GB using GEOM are shown below.

Table 11-58: GEOM Options

Configuration	GEOM	Description
	1	The center of curvature lies on the line AO (or its extension) or vector $\vec{v}$.
	2	The tangent of centroid arc at end A is parallel to line AO or vector $\vec{v}$. Point O (or vector $\vec{v}$) and the arc $\overline{AB}$ must be on the same side of the chord $\overline{AB}$.

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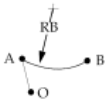
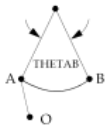
Configuration	GEOM	Description
	3	The bend radius (RB) is specified on the PBEND entry: Points A, B, and O (or vector $\vec{v}$) define a plane parallel or coincident with the plane of the element arc. Point O (or vector $\vec{v}$) lies on the opposite side of line AB from the center of the curvature.
	4	THETAB is specified on the PBEND entry. Points A, B, and O (or vector $\vec{v}$) define a plane parallel or coincident with the plane of the element arc. Point O (or vector $\vec{v}$) lies on the opposite side of line AB from the center of curvature.

Related Topics

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CBUSH

Generalized Spring-and-Damper Connection

Defines a generalized spring-and-damper structural element that may be nonlinear or frequency dependent.

Format:

1	2	3	4	5	6	7	8	9	10
CBUSH	EID	PID	GA	GB	GO/X1	X2	X3	CID	
	S	OCID	S1	S2	S3				

Example 1:

Non-coincident grid points

CBUSH	39	6	1	100	75				

Example 2:

GB not specified

CBUSH	39	6	1					0	

Example 3:

Coincident grid points

CBUSH	39	6	1	100				6	

Example 4:

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Non-coincident grid points with fields 6 through 9 blank and a spring-damper offset

CBUSH	39	6	1	600					
	0.25	10	0.	10.	10.				

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PBUSH or PBUSHT entry. (Integer > 0; Default = EID)
GA, GB	Grid point identification number of connection points. See Remark 6 and Remark 1 for SOL 414. (Integer GA > 0)
Xi	Components of orientation vector $\vec{v}$, from GA, in the displacement coordinate system at GA. (Real)
GO	Alternate method to supply vector $\vec{v}$ using grid point GO. Direction of $\vec{v}$ is from GA to GO. $\vec{v}$ is then transferred to end A. See Remark 3 . (Integer > 0)
CID	Element coordinate system identification. A 0 means the basic coordinate system. If CID is blank, then the element coordinate system is determined from GO or Xi. See Figure 11-30 and Remark 3 . (Integer ≥ 0 or blank)
S	Location of spring damper. See Figure 11-30. (0.0 ≤ Real ≤ 1.0; Default = 0.5)
OCID	Coordinate system identification of spring-damper offset. See Remark 10 . (Integer ≥ -1; Default = -1, which means the offset point lies on the line between GA and GB according to Figure 11-30.)
S1, S2, S3	Location of spring-damper in the OCID coordinate system if OCID ≥ 0. See Figure 11-32 and Remark 10 . (Real)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. Figure 11-30 shows the bush element geometry when CID is undefined.

   When CID ≥ 0, the element coordinate system is defined only by the coordinate system selected in the CID field, and is not dependent on the location of GA or GB. See Figure 11-31 for an example.

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3. CID ≥ 0 overrides GO and Xi. Then the element x-axis is along T1, the element y-axis is along T2, and the element z-axis is along T3 of the CID coordinate system. If the CID refers to a cylindrical coordinate system or a spherical coordinate system, then grid GA is used to locate the system. For cylindrical or spherical coordinate systems, if GA falls on the z-axis used to define them, it is recommended that another CID be selected to define the element x-axis.
4. For non-coincident grids, when GO or X1, X2, X3 is given and no CID is specified, the line AB is the element x-axis and the orientation vector $\vec{v}$ lies in the x-y plane (similar to the CBEAM element).
5. For non-coincident grids, if neither GO or X1, X2, X3 is specified and no CID is specified, then the line AB is the element x-axis, and the y-axis and z-axis are considered undefined. This option is valid only when K1 (or B1) or K4 (or B4) or both are specified on the PBUSH entry (but K2, K3, K5, K6 or B2, B3, B5, B6 are not specified). If K2, K3, K5, or K6 (or B2, B3, B5, or B6) are specified, a fatal message will be issued.
6. If GA and GB are coincident, or if GB is blank, then CID must be specified. When GB is blank, a grounded spring and damper is created at GA.
7. If you have modeled GA and GB as coincident and one or both of these grid points have a location coordinate system defined with the CP field on the GRID bulk entry, when the software transforms the grid point locations into the basic coordinates, as a result of floating point math, this transformation is not exact and GA and GB may no longer be exactly coincident. To prevent any unwanted coupling from occurring in this case, you can constrain the rotational DOF using an SPC entry.
8. Only CBUSH elements in the residual structure that do not attach to any omitted degree-of-freedom can reference a PID identifying both a PBUSH entry and a PBUSHT entry.
9. Element impedance output is computed in the CID coordinate system. The impedances in this system are uncoupled.
10. If OCID = -1 or blank (default) then S is used and S1, S2, S3 are ignored. If OCID ≥ 0, then S is ignored and S1, S2, S3 must be defined.

Figure 11-30: CBUSH Element with CID undefined

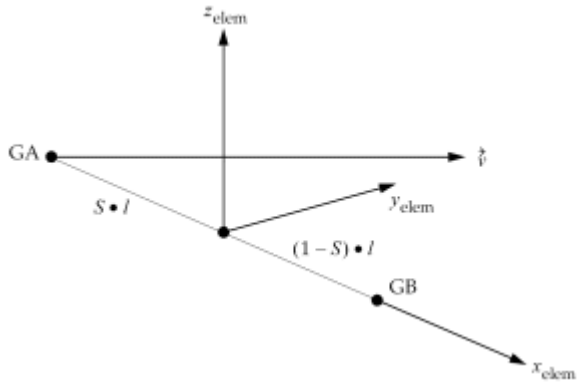

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Figure 11-31: CBUSH Element Coordinate System with CID defined

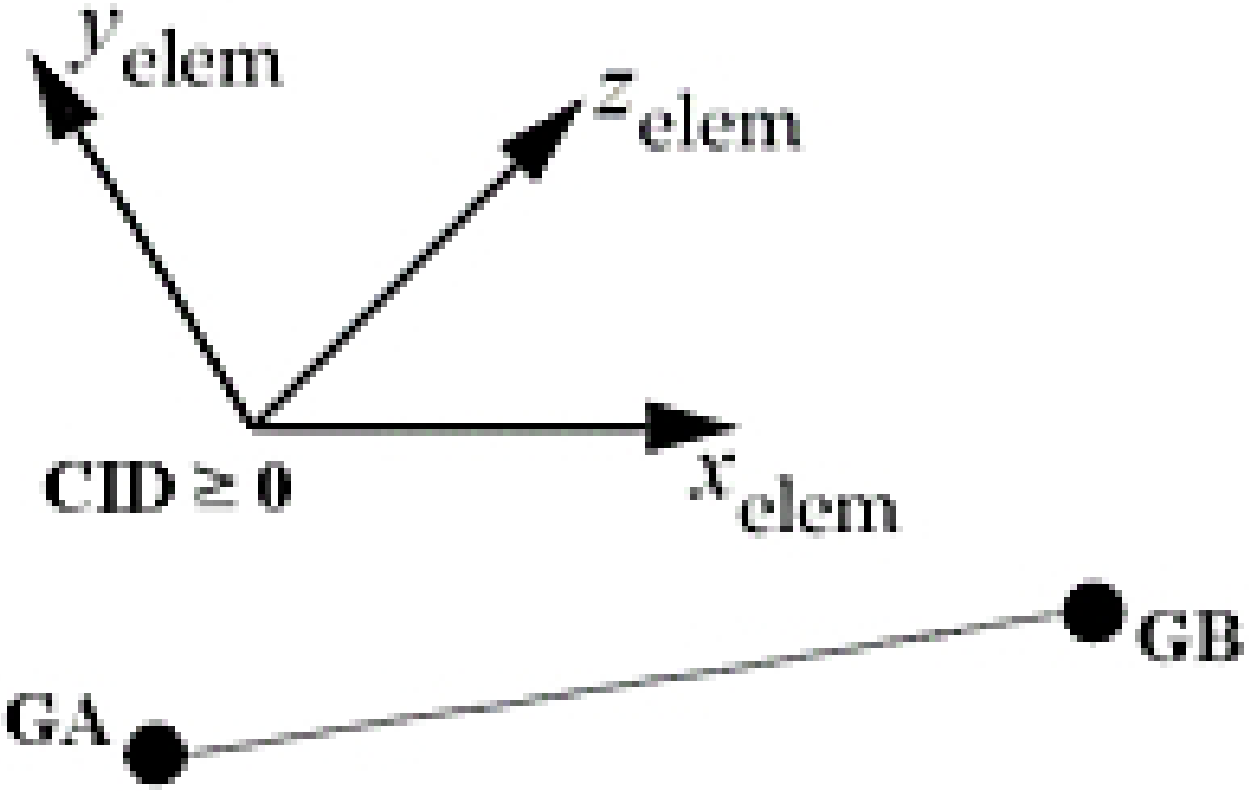
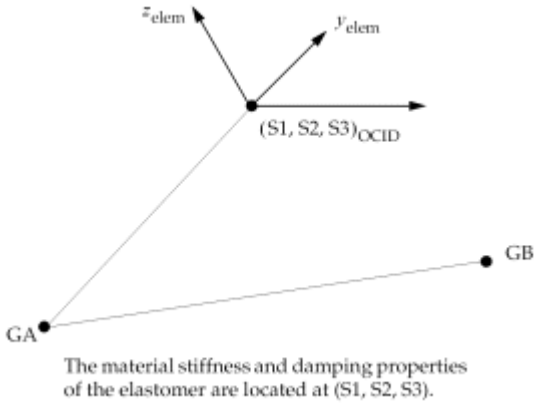


Figure 11-32: Definition of Offset S1, S2, S3



11. When CID ≥ 0, the element coordinate system is defined by the coordinate system selected in the CID field, and is not dependent on the location of GA or GB. See Figure 11-31. The force output is computed as Ke (UB – UA), where UB and UA are in element coordinates. If UA > UB, a compressive force will result. This is unlike the GO or Xi options, where relative positive elongation is tension and relative negative elongation is compression.

12. The effect of PARAM, G and the damping coefficient GE (defined on the PBUSH entry) are included in the bushing stiffness for force and stress calculation in frequency and transient analyses. The effects of PARAM,G can be excluded using PARAM,GDAMPF,NO. If PARAM,W3 is not specified, PARAM,G is

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ignored in a transient analysis. If PARAM,W4 is not specified, GE is ignored in a transient analysis. See **Parameters and Parameter Applicability Tables**. For more details on element force and stress calculation, see the *Simcenter Nastran Element Library Reference*.

GDAMPF is not supported for SOL 401, 402, or 414. For SOL 401 when ANALYSIS = DYNAMICS, the effect of PARAM,G is included in the force and stress computation. For SOL 414, the effect of PARAM,G is not included in the force and stress computation. For SOL 402, PARAM,G is not supported.

13. The STRESS case control command should be used to request the element force and stress output for the CBUSH and CBUSH1D elements.

Remarks for SOLs 401 and 402

1. For SOL 401, the CBUSH element does not support large displacement effects when PARAM,LGDISP,1 is defined. You can include the CBUSH element in a solution with PARAM,LGDISP,1 defined, but it should not be located in a region of the model where large rotations occur. SOL 402 does not have this limitation.

Remarks for SOL 414:

1. For a SOL 414,111 Harmonic Response solution, if you reference a PID that identifies a PBUSHT bulk entry, you must define CBUSH with two grid points.

Related Topics

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CBUSH1D

Rod Type Spring-and-Damper Connection

Defines the connectivity of a one-dimensional spring and viscous damper element.

Format:

1	2	3	4	5	6	7	8	9	10
CBUSH1D	EID	PID	GA	GB	CID				

Example:

CBUSH1D	35	102	108	112					

Fields:

Field	Contents	Default Values
EID	Element identification number.	Required
PID	Property identification number of a PBUSH1D entry. (Integer > 0).	EID
GA	Grid point id of first grid.	Required
GB	Grid point id of second grid	Blank
CID	Coordinate system id. (Integer ≥ 0)	Blank

Remarks:

1. For noncoincident grids GA ≠ GB and if CID is blank, the line GA to GB is the element axis. In geometric nonlinear analysis, the element axis (line GA to GB) follows the deformation of grids GA and GB. See **Figure 11-33**.
2. If CID ≥ 0 is specified, the x-axis of the CID coordinate system is the element axis. In geometric nonlinear analysis, the element axis (x-axis of CID) remains fixed for this case.
3. If GA and GB are coincident or if GB is blank, then CID ≥ 0 must be specified and the element axis is the x-axis of CID.

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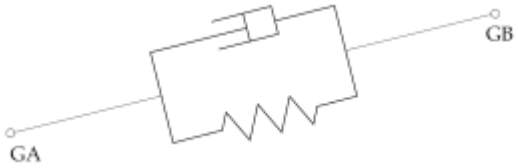
Figure 11-33: Spring and Damper Element



4. The STRESS case control command should be used to request the element force and stress output for the CBUSH and CBUSH1D elements.

Related Topics

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CBUSH2

Generalized Spring-and-Damper Connection

Defines a generalized spring-and-damper structural element for SOL 414.

Format:

1	2	3	4	5	6	7	8	9	10
CBUSH2	EID	PID	GA	GB	CID				

Example:

CBUSH2	3	6	1	100	5				

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PBUSH2 entry. (Integer > 0; Default = EID)
GA, GB	Grid point identification number of connection points. (Integer > 0; No default)
CID	Element coordinate system identification number. See Remark 2. (Integer ≥ 0; Default = 0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. A blank CID entry or a CID of zero references the basic coordinate system.
3. When a CBUSH2 bulk entry references a PBUSH2 bulk entry that contains parameters that reference TABLEST bulk entries, the CBUSH2 bulk entry must also reference a SPEBE2 bulk entry.

Related Topics

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CCHOCK3

2D Chocking Triangular Element Connection

Defines a 2D chocking triangular element connection. Valid for SOL 401 axisymmetric analysis only.

Format:

1	2	3	4	5	6	7	8	9	10
CCHOCK3	EID	PID	G1	G2	G3		THETA		
					GAP1	GAP2	GAP3		

Example:

CCHOCK3	19	9	1215	1216	1342				
						0.001	0.0		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PCHOCK bulk entry. (Integer > 0; No default)
Gi	Grid point identification number of connection points. (Integer > 0; No default)
THETA	Material property orientation angle in degrees. See Remark 4. (Real; Default = 0.0)
GAPi	Initial gap opening at grid point Gi. See Remark 6. (Real or blank; See Remark 5 for default behavior)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

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3. The basic coordinate system is the reference coordinate system for stress and strain output.
4. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures **Figure 11-34** and **Figure 11-35** show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.
5. If GAPi is blank, the value for GAPT on the corresponding PCHOCK entry is used for the initial gap opening at grid point Gi. If GAPi and GAPT on the corresponding PCHOCK entry are both specified, the GAPi specification takes precedence. The software computes the total initial gap opening at grid point Gi as the product of GAPi (or GAPT) and the value for NGAP on the corresponding PCHOCK entry.
6. If the value for GAPi or, if applicable, the value for GAPT on the corresponding PCHOCK entry is negative, the software uses the value to calculate an initial compressive hoop strain.
7. For any grid point Gi selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the displacement coordinate system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis (φ-axis for a spherical system) of the displacement coordinate system must be in the same direction as the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system must be in the same direction as the Y-axis of the basic coordinate system.
8. GPSTRESS and GPSTRAIN output are not supported.
9. The behavior of chocking elements depend on whether large displacements are enabled.
   - When large displacements are not enabled, the gap status at the beginning of the analysis is used throughout the analysis and the stiffness does not reformulate unless plasticity is enabled. When plasticity is enabled, the stiffness reformulates to account for plasticity effects only.
   - When large displacements are enabled, the solution is iterative. The initial gap status is used for the first iteration. The gap status at the end of each iteration is updated and the stiffness is reformulated. When plasticity is enabled, the stiffness reformulates to account for plasticity effects, as well as changes in gap status.

     To enable large displacements, specify PARAM,LGDISP,1 in the input file.
10. At any Gauss point location where there is no gap, the contribution to the elemental stiffness matrix for the chocking element from that Gauss point is based on the axisymmetric formulation of the chocking element. At any Gauss point location where there is a gap, the contribution to the elemental stiffness matrix for the chocking element from that Gauss point is based on the axisymmetric formulation of the chocking element with the E_θ, $v_{\theta r}$, and $v_{\theta z}$ elastic constants reduced by a factor of 1×10^6 so that the element stiffness becomes essentially identical to a plane stress formulation with σ_θ as the out-of-plane normal stress.

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Figure 11-34: CCHOCK3 Element Geometry and Coordinate Systems, XZ-Plane

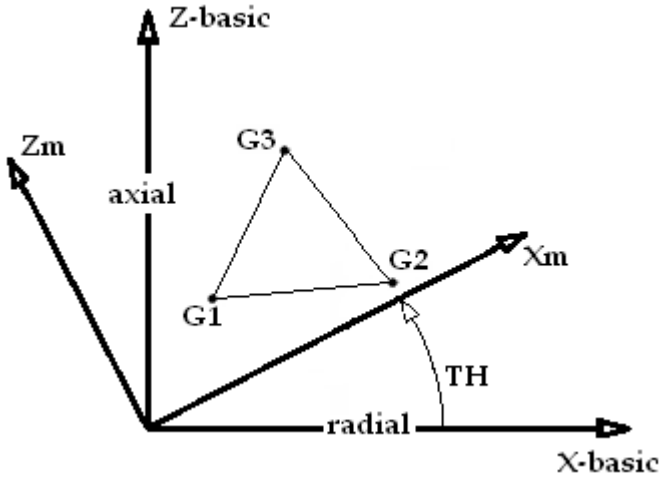
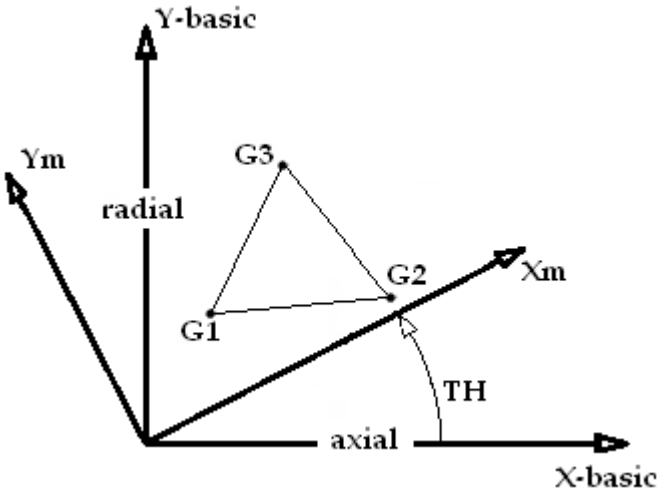


Figure 11-35: CCHOCK3 Element Geometry and Coordinate Systems, XY-Plane



Related Topics

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CCHOCK4

2D Chocking Quadrilateral Element Connection

Defines a 2D chocking quadrilateral element connection. Valid for SOL 401 axisymmetric analysis only.

Format:

1	2	3	4	5	6	7	8	9	10
CCHOCK4	EID	PID	G1	G2	G3	G4	THETA		
					GAP1	GAP2	GAP3	GAP4	

Example:

CCHOCK4	19	9	1215	1216	1342	1222			
					0.001		0.0	0.002	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PCHOCK bulk entry. (Integer > 0; No default)
Gi	Grid point identification number of connection points. (Integer > 0; No default)
THETA	Material property orientation angle in degrees. See Remark 4. (Real; Default = 0.0)
GAPi	Initial gap opening at grid point Gi. See Remark 6. (Real or blank; See Remark 5 for default behavior)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

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3. The basic coordinate system is the reference coordinate system for stress and strain output.
4. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures **Figure 11-36** and **Figure 11-37** show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.
5. If GAPi is blank, the value for GAPT on the corresponding PCHOCK entry is used for the initial gap opening at grid point Gi. If GAPi and GAPT on the corresponding PCHOCK entry are both specified, the GAPi specification takes precedence. The software computes the total initial gap opening at grid point Gi as the product of GAPi (or GAPT) and the value for NGAP on the corresponding PCHOCK entry.
6. If the value for GAPi or, if applicable, the value for GAPT on the corresponding PCHOCK entry is negative, the software uses the value to calculate an initial compressive hoop strain.
7. For any grid point Gi selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the displacement coordinate system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis (φ-axis for a spherical system) of the displacement coordinate system must be in the same direction as the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system must be in the same direction as the Y-axis of the basic coordinate system.
8. GPSTRESS and GPSTRAIN output are not supported.
9. The behavior of chocking elements depend on whether large displacements are enabled.
   - When large displacements are not enabled, the gap status at the beginning of the analysis is used throughout the analysis and the stiffness does not reformulate unless plasticity is enabled. When plasticity is enabled, the stiffness reformulates to account for plasticity effects only.
   - When large displacements are enabled, the solution is iterative. The initial gap status is used for the first iteration. The gap status at the end of each iteration is updated and the stiffness is reformulated. When plasticity is enabled, the stiffness reformulates to account for plasticity effects, as well as changes in gap status.

     To enable large displacements, specify PARAM,LGDISP,1 in the input file.
10. At any Gauss point location where there is no gap, the contribution to the elemental stiffness matrix for the chocking element from that Gauss point is based on the axisymmetric formulation of the chocking element. At any Gauss point location where there is a gap, the contribution to the elemental stiffness matrix for the chocking element from that Gauss point is based on the axisymmetric formulation of the chocking element with the E_θ, $v_{\theta r}$, and $v_{\theta z}$ elastic constants reduced by a factor of 1×10^6 so that the element stiffness becomes essentially identical to a plane stress formulation with σ_θ as the out-of-plane normal stress.

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Figure 11-36: CCHOCK4 Element Geometry and Coordinate Systems, XZ-Plane

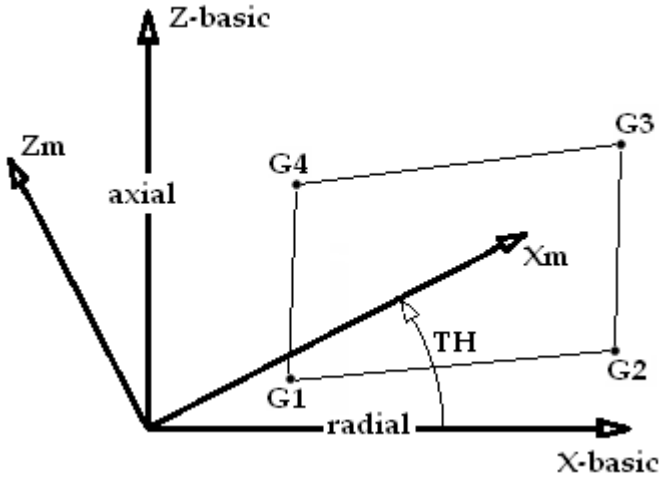
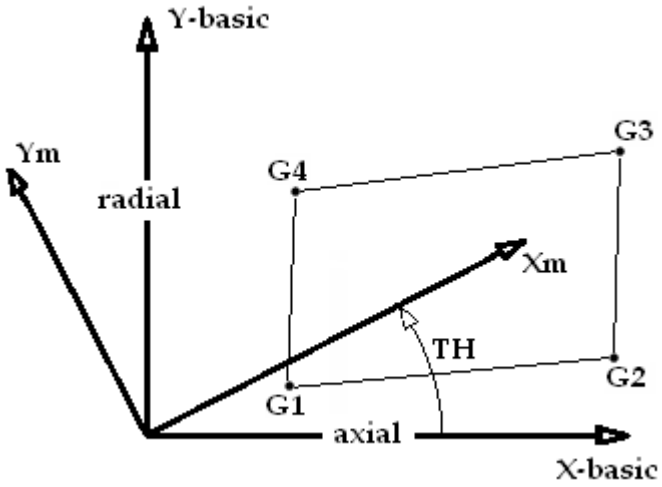


Figure 11-37: CCHOCK4 Element Geometry and Coordinate Systems, XY-Plane



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CCHOCK6

2D Chocking Triangular Element Connection

Defines a 2D chocking triangular element connection. Valid for SOL 401 axisymmetric analysis only.

Format:

1	2	3	4	5	6	7	8	9	10
CCHOCK6	EID	PID	G1	G2	G3	G4	G5	G6	
			THETA		GAP1	GAP2	GAP3		

Example:

CCHOCK6	237	1	1215	1217	1342	1216	1289	1290	
			30.0			0.001	0.0		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PCHOCK bulk entry. (Integer > 0; No default)
Gi	Grid point identification number of connection points. (Integer > 0; No default)
THETA	Material property orientation angle in degrees. See Remark 4. (Real; Default = 0.0)
GAPi	Initial gap opening at grid point Gi. See Remark 6. (Real or blank; See Remark 5 for default behavior)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

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3. The basic coordinate system is the reference coordinate system for stress and strain output.
4. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures **Figure 11-38** and **Figure 11-39** show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.
5. If GAPi is blank, the value for GAPT on the corresponding PCHOCK entry is used for the initial gap opening at grid point Gi. If GAPi and GAPT on the corresponding PCHOCK entry are both specified, the GAPi specification takes precedence. The software computes the total initial gap opening at grid point Gi as the product of GAPi (or GAPT) and the value for NGAP on the corresponding PCHOCK entry.
6. If the value for GAPi or, if applicable, the value for GAPT on the corresponding PCHOCK entry is negative, the software uses the value to calculate an initial compressive hoop strain.
7. For any grid point Gi selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the displacement coordinate system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis (φ-axis for a spherical system) of the displacement coordinate system must be in the same direction as the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system must be in the same direction as the Y-axis of the basic coordinate system.
8. GPSTRESS and GPSTRAIN output are not supported.
9. The behavior of chocking elements depend on whether large displacements are enabled.
   - When large displacements are not enabled, the gap status at the beginning of the analysis is used throughout the analysis and the stiffness does not reformulate unless plasticity is enabled. When plasticity is enabled, the stiffness reformulates to account for plasticity effects only.
   - When large displacements are enabled, the solution is iterative. The initial gap status is used for the first iteration. The gap status at the end of each iteration is updated and the stiffness is reformulated. When plasticity is enabled, the stiffness reformulates to account for plasticity effects, as well as changes in gap status.

     To enable large displacements, specify PARAM,LGDISP,1 in the input file.
10. At any Gauss point location where there is no gap, the contribution to the elemental stiffness matrix for the chocking element from that Gauss point is based on the axisymmetric formulation of the chocking element. At any Gauss point location where there is a gap, the contribution to the elemental stiffness matrix for the chocking element from that Gauss point is based on the axisymmetric formulation of the chocking element with the E_θ, $v_{\theta r}$, and $v_{\theta z}$ elastic constants reduced by a factor of 1×10^6 so that the element stiffness becomes essentially identical to a plane stress formulation with σ_θ as the out-of-plane normal stress.

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Figure 11-38: CCHOCK6 Element Geometry and Coordinate Systems, XZ-Plane

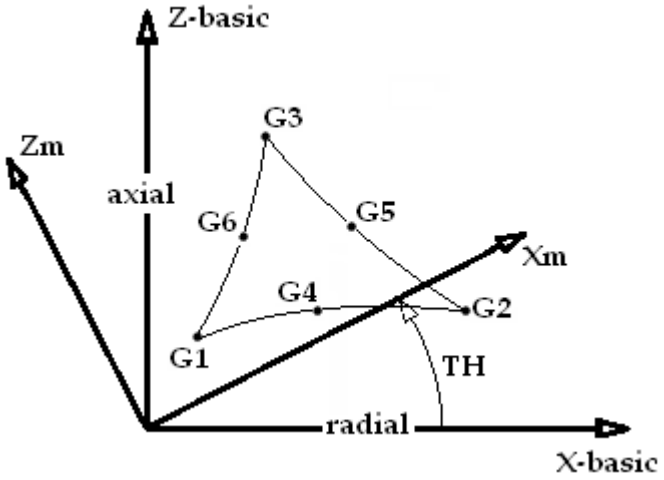
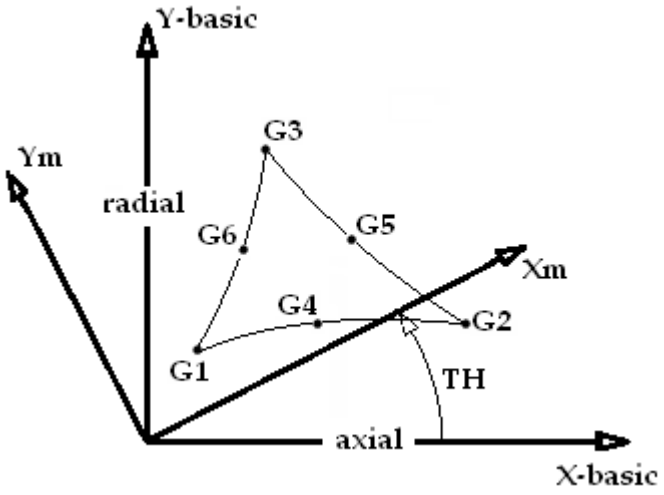


Figure 11-39: CCHOCK6 Element Geometry and Coordinate Systems, XY-Plane



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CCHOCK8

2D Chocking Quadrilateral Element Connection

Defines a 2D chocking quadrilateral element connection. Valid for SOL 401 axisymmetric analysis only.

Format:

1	2	3	4	5	6	7	8	9	10
CCHOCK8	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	THETA		GAP1	GAP2	GAP3	GAP4	

Example:

CCHOCK8	19	9	1215	1217	1442	1440	1216	1886	
	1441	1385	30.0		0.001		0.0	0.002	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PCHOCK bulk entry. (Integer > 0; No default)
Gi	Grid point identification number of connection points. (Integer > 0; No default)
THETA	Material property orientation angle in degrees. See Remark 4. (Real; Default = 0.0)
GAPi	Initial gap opening at grid point Gi. See Remark 6. (Real or blank; See Remark 5 for default behavior)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

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3. The basic coordinate system is the reference coordinate system for stress and strain output.
4. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures **Figure 11-40** and **Figure 11-41** show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.
5. If GAPi is blank, the value for GAPT on the corresponding PCHOCK entry is used for the initial gap opening at grid point Gi. If GAPi and GAPT on the corresponding PCHOCK entry are both specified, the GAPi specification takes precedence. The software computes the total initial gap opening at grid point Gi as the product of GAPi (or GAPT) and the value for NGAP on the corresponding PCHOCK entry.
6. If the value for GAPi or, if applicable, the value for GAPT on the corresponding PCHOCK entry is negative, the software uses the value to calculate an initial compressive hoop strain.
7. For any grid point Gi selected on axisymmetric, plane stress, plane stain, or chocking elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the displacement coordinate system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis (φ-axis for a spherical system) of the displacement coordinate system must be in the same direction as the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system must be in the same direction as the Y-axis of the basic coordinate system.
8. GPSTRESS and GPSTRAIN output are not supported.
9. The behavior of chocking elements depend on whether large displacements are enabled.
   - When large displacements are not enabled, the gap status at the beginning of the analysis is used throughout the analysis and the stiffness does not reformulate unless plasticity is enabled. When plasticity is enabled, the stiffness reformulates to account for plasticity effects only.
   - When large displacements are enabled, the solution is iterative. The initial gap status is used for the first iteration. The gap status at the end of each iteration is updated and the stiffness is reformulated. When plasticity is enabled, the stiffness reformulates to account for plasticity effects, as well as changes in gap status.

     To enable large displacements, specify PARAM,LGDISP,1 in the input file.
10. At any Gauss point location where there is no gap, the contribution to the elemental stiffness matrix for the chocking element from that Gauss point is based on the axisymmetric formulation of the chocking element. At any Gauss point location where there is a gap, the contribution to the elemental stiffness matrix for the chocking element from that Gauss point is based on the axisymmetric formulation of the chocking element with the E_θ, $v_{\theta r}$, and $v_{\theta z}$ elastic constants reduced by a factor of 1×10^6 so that the element stiffness becomes essentially identical to a plane stress formulation with σ_θ as the out-of-plane normal stress.

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Figure 11-40: CCHOCK8 Element Geometry and Coordinate Systems, XZ-Plane

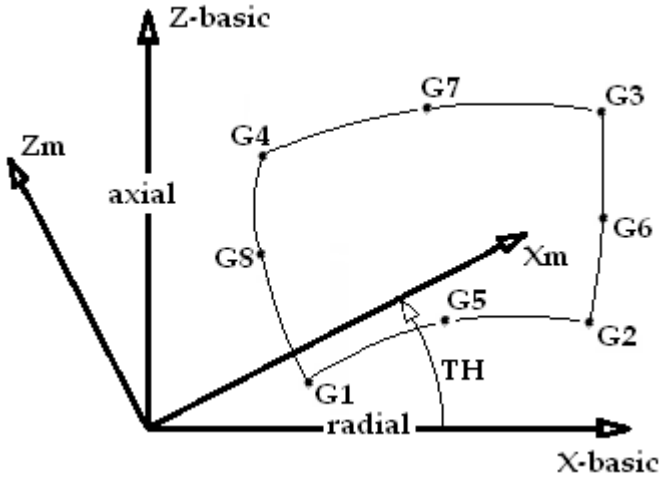
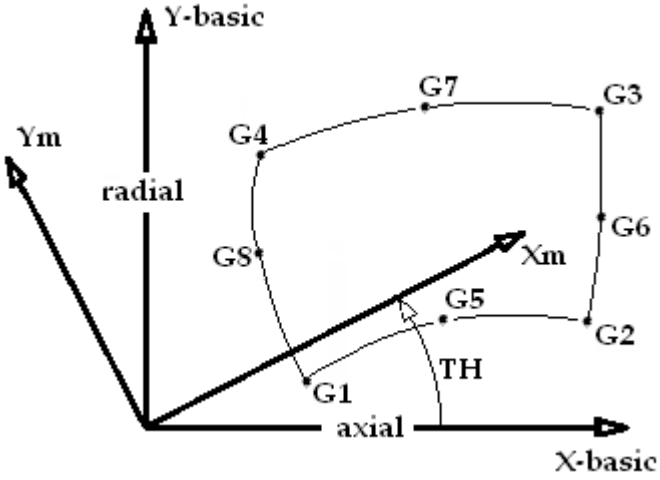


Figure 11-41: CCHOCK8 Element Geometry and Coordinate Systems, XY-Plane



Related Topics

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CDAMP1

Scalar Damper Connection

Defines a scalar damper element.

Format:

1	2	3	4	5	6	7	8	9	10
CDAMP1	EID	PID	G1	C1	G2	C2			

Example:

CDAMP1	19	6	0		23	2			

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Property identification number of a PDAMP property entry. (Integer > 0; Default = EID)
G1, G2	Geometric grid point identification number. (Integer ≥ 0)
C1, C2	Component number. (0 ≤ Integer ≤ 6; 0 or up to six unique integers, 1 through 6 may be specified in the field with no embedded blanks. 0 applies to scalar points and 1 through 6 apply to grid points.)

Remarks:

1. Scalar points may be used for G1 and/or G2, in which case the corresponding C1 and/or C2 must be zero or blank. Zero or blank may be used to indicate a grounded terminal G1 or G2 with a corresponding blank or zero C1 or C2. A grounded terminal is a point with a displacement that is constrained to zero.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. The two connection points (G1, C1) and (G2, C2), must be distinct.
4. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.
5. When CDAMP1 is used in heat transfer analysis, it generates a lumped heat capacity.
6. A scalar point specified on this entry need not be defined on an SPOINT entry.

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7. If Gi refers to a grid point then Ci refers to degrees-of-freedom(s) in the displacement coordinate system specified by CD on the GRID entry.

Remarks related to SOLs 402 and 414:

G1 and G2 must be grid points, i.e., they cannot be scalar points.

Related Topics

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CDAMP2

Scalar Damper Property and Connection

Defines a scalar damper element without reference to a material or property entry.

Format:

1	2	3	4	5	6	7	8	9	10
CDAMP2	EID	B	G1	C1	G2	C2			

Example:

CDAMP2	16	2.98	32	1					

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
B	Value of the scalar damper. (Real)
G1, G2	Geometric grid point identification number. (Integer ≥ 0)
C1, C2	Component number. (0 ≤ Integer ≤ 6; 0 or up to six unique integers, 1 through 6 may be specified in the field with no embedded blanks. 0 applies to scalar points and 1 through 6 apply to grid points.)

Remarks:

1. Scalar points may be used for G1 and/or G2, in which case the corresponding C1 and/or C2 must be zero or blank. Zero or blank may be used to indicate a grounded terminal G1 or G2 with a corresponding blank or zero C1 or C2. A grounded terminal is a point with a displacement that is constrained to zero.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. The two connection points (G1, C1) and (G2, C2), must be distinct.
4. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.
5. When CDAMP2 is used in heat transfer analysis, it generates a lumped heat capacity.
6. A scalar point specified on this entry need not be defined on an SPOINT entry.

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7. If Gi refers to a grid point then Ci refers to degrees-of-freedom(s) in the displacement coordinate system specified by CD on the GRID entry.

Remarks related to SOLs 402 and 414:

G1 and G2 must be grid points, i.e., they cannot be scalar points.

Related Topics

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CDAMP3

Scalar Damper Connection to Scalar Points Only

Defines a scalar damper element that is connected only to scalar points.

Format:

1	2	3	4	5	6	7	8	9	10
CDAMP3	EID	PID	S1	S2					

Example:

CDAMP3	16	978	24	36					

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Property identification number of a PDAMP entry. (Integer > 0; Default = EID)
S1, S2	Scalar point identification numbers. (Integer ≥ 0; S1 ≠ S2)

Remarks:

1. S1 or S2 may be blank or zero, indicating a constrained coordinate.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. Only one scalar damper element may be defined on a single entry.
4. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.
5. When CDAMP3 is used in heat transfer analysis, it generates a lumped heat capacity.
6. A scalar point specified on this entry need not be defined on an SPOINT entry.

Related Topics

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CDAMP4

Scalar Damper Property and Connection to Scalar Points Only

Defines a scalar damper element that is connected only to scalar points, without reference to a material or property entry.

Format:

1	2	3	4	5	6	7	8	9	10
CDAMP4	EID	B	S1	S2					

Example:

CDAMP4	16	-2.6	4	9					

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
B	Scalar damper value. (Real)
S1, S2	Scalar point identification numbers. (Integer ≥ 0; S1 ≠ S2)

Remarks:

1. S1 or S2 may be blank or zero, indicating a constrained coordinate.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. Only one scalar damper element may be defined on a single entry.
4. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.
5. If this entry is used in heat transfer analysis, it generates a lumped heat capacity.
6. A scalar point specified on this entry need not be defined on an SPOINT entry.

Related Topics

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CDAMP5

Scalar Damper with Material Property

Defines a damping element that refers to a material property entry and connection to grid or scalar points. This element is intended for heat transfer analysis only.

Format:

1	2	3	4	5	6	7	8	9	10
CDAMP5	EID	PID	G1	G2					

Example:

CDAMP5	1	4	10	20					

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Identification number of a PDAMP5 property entry. (Integer > 0; Default = EID)
G1, G2	Grid or scalar point identification numbers. (Integer ≥ 0 and G1 ≠ G2)

Remarks:

1. G1 or G2 may be blank or zero indicating a constraint.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. CDAMP5 generates a lumped heat capacity in heat transfer analysis.
4. A scalar point specified on CDAMP5 need not be defined on an SPOINT entry.

Related Topics

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CDUMi

Dummy Element Connection

Defines a dummy element ($1 \le i \le 9$)

Format:

1	2	3	4	5	6	7	8	9	10
CDUMi	EID	PID	G1	G2	G3	G4	-etc.-		
	A1	A2	-etc.-						

Example:

CDUM2	114	108	2	5	6	8	11		
	2.4		3.E4	2		50			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PDUMi entry. See Remark 2 . (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer > 0, G1 ≠ G2 ... ≠ GN)
Ai	Additional fields. (Real or Integer)

Remarks:

1. The user must write the associated element subroutines for matrix generation, stress recovery, etc., and perform a link edit to replace the dummy routines.
2. If no property entry is required, PID may contain the material identification number.
3. Additional entries are defined in the user-written element routines.
4. CDUM1 is replaced by the CTRIAX6 element. If CDUM1 is used, User Fatal Message 307 will be issued.

Related Topics

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CELAS1

Scalar Spring Connection

Defines a scalar spring element.

Format:

1	2	3	4	5	6	7	8	9	10
CELAS1	EID	PID	G1	C1	G2	C2			

Example:

CELAS1	2	6			8	1			

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Property identification number of a PELAS entry. (Integer > 0; Default = EID)
G1, G2	Geometric grid point identification number. (Integer ≥ 0)
C1, C2	Component number. (0 ≤ Integer ≤ 6; blank or zero if scalar point)

Remarks:

1. Scalar points may be used for G1 and/or G2, in which case the corresponding C1 and/or C2 must be zero or blank. Zero or blank may be used to indicate a grounded terminal G1 or G2 with a corresponding blank or zero C1 or C2. A grounded terminal is a point with a displacement that is constrained to zero. If only scalar points and/or ground are involved, it is more efficient to use the CELAS3 entry.
2. In solutions 153 and 159, the software calculates heat conduction for CELAS1 elements as

   $q = K \times \Delta T$

   where K is the value entered on the PELAS entry. Both G1 and G2 must be defined, and C1 and C2 must be “1” when grid point ID’s are entered for G1 and G2.
3. Element identification numbers should be unique with respect to all other element identification numbers.
4. The two connection points (G1, C1) and (G2, C2) must be distinct.
5. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.

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6. A scalar point specified on this entry need not be defined on an SPOINT entry.
7. If Gi refers to a grid point, then Ci refers to degree-of-freedom in the displacement coordinate system specified by CD on the GRID entry.
8. The effect of PARAM,G and the damping coefficient GE (defined on the PELAS entry) are included in the spring stiffness for force and stress calculation in frequency and transient analyses. The effects of PARAM,G can be excluded using PARAM,GDAMPF,NO. If PARAM,W3 is not specified, PARAM,G is ignored in a transient analysis. If PARAM,W4 is not specified, GE is ignored in a transient analysis. See **Parameters and Parameter Applicability Tables**. For more details on element force and stress calculation, see the *Simcenter Nastran Element Library Reference*.

   GDAMPF is not supported for SOL 401, 402, or 414. For SOL 401 when ANALYSIS = DYNAMICS, the effect of PARAM,G is included in the force and stress computation. For SOL 414, the effect of PARAM,G is not included in the force and stress computation. For SOL 402, PARAM,G is not supported.

Remarks related to SOLs 402 and 414:

1. G1 and G2 must be grid points, i.e., they cannot be scalar points.

Related Topics

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CELAS2

Scalar Spring Property and Connection

Defines a scalar spring element without reference to a property entry.

Format:

1	2	3	4	5	6	7	8	9	10
CELAS2	EID	K	G1	C1	G2	C2	GE	S	

Example:

CELAS2	28	6.2+3	32		19	4			

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
K	Stiffness of the scalar spring. See Remarks 2 and 11. (Real)
G1, G2	Geometric grid point or scalar identification number. (Integer ≥ 0)
C1, C2	Component number. (0 ≤ Integer ≤ 6; blank or zero if scalar point)
GE	Damping coefficient. See Remarks 9 and 10. (Real)
S	Stress coefficient. (Real)

Remarks:

1. Scalar points may be used for G1 and/or G2, in which case the corresponding C1 and/or C2 must be zero or blank. Zero or blank may be used to indicate a grounded terminal G1 or G2 with a corresponding blank or zero C1 or C2. A grounded terminal is a point with a displacement that is constrained to zero. If only scalar points and/or ground are involved, it is more efficient to use the CELAS4 entry.
2. In solutions 153 and 159, the software calculates heat conduction for CELAS2 elements as

   $q = K \times \Delta T$

   where K is the value entered on the CELAS2 entry. Both G1 and G2 must be defined, and C1 and C2 must be “1” when grid point ID’s are entered for G1 and G2.
3. Element identification numbers should be unique with respect to all other element identification numbers.

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4. The two connection points (G1, C1) and (G2, C2) must be distinct.
5. The element stress is computed by multiplying the stress coefficient S with the recovered element force.
6. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.
7. A scalar point specified on this entry need not be defined on an SPOINT entry.
8. If Gi refers to a grid point, then Ci refers to degree-of-freedom in the displacement coordinate system specified by CD on the GRID entry.
9. To obtain the damping coefficient GE, multiply the critical damping ratio ζ by 2.0. That is, $\mathrm{GE} = 2\zeta = 2(b/b_{cr})$. See Viscous and Structural Damping in the *Simcenter Nastran Basic Dynamic Analysis User's Guide*.
10. The effect of PARAM, G and the damping coefficient GE are included in the spring stiffness for force and stress calculation in a frequency and transient analyses. The effects of PARAM,G can be excluded using PARAM,GDAMPF,NO. If PARAM,W3 is not specified, PARAM,G is ignored in a transient analysis. If PARAM,W4 is not specified, GE is ignored in a transient analysis. See **Parameters and Parameter Applicability Tables**. For more details on element force calculation see the *Simcenter Nastran Element Library Reference*.

    GDAMPF is not supported for SOL 401, 402, or 414. For SOL 401 when ANALYSIS = DYNAMICS, the effect of PARAM,G is included in the force and stress computation. For SOL 414, the effect of PARAM,G is not included in the force and stress computation. For SOL 402, PARAM,G is not supported.
11. Rotational stiffness should be specified as moment per radian.

Remarks related to SOLs 402 and 414:

1. G1 and G2 must be grid points, i.e., they cannot be scalar points.
2. S is ignored.

Related Topics

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CELAS3

Scalar Spring Connection to Scalar Points Only

Defines a scalar spring element that connects only to scalar points.

Format:

1	2	3	4	5	6	7	8	9	10
CELAS3	EID	PID	S1	S2					

Example:

CELAS3	19	2	14	15					

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Property identification number of a PELAS entry. (Integer > 0; Default = EID)
S1, S2	Scalar point identification numbers. (Integer ≥ 0; S1 ≠ S2)

Remarks:

1. S1 or S2 may be blank or zero, indicating a constrained coordinate.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. Only one scalar spring element may be defined on a single entry.
4. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.
5. A scalar point specified on this entry need not be defined on an SPOINT entry.
6. In solutions 153 and 159, the software calculates heat conduction for CELAS3 elements as

   $q = K \times \Delta T$

   where K is the value entered on the PELAS entry. Both S1 and S2 must be defined.
7. The effect of PARAM,G and the damping coefficient GE (defined on the PELAS entry) are included in the spring stiffness for force and stress calculation in frequency and transient analyses. The effects of PARAM,G can be excluded using PARAM,GDAMPF,NO. If PARAM,W3 is not specified, PARAM,G is ignored in a transient analysis. If PARAM,W4 is not specified, GE is ignored in a transient analysis. See

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Parameters and Parameter Applicability Tables. For more details on element force calculation, see the *Simcenter Nastran Element Library Reference*.

GDAMPF is not supported for SOL 401. For SOL 401 when ANALYSIS = DYNAMICS, the effect of PARAM,G is included in the force and stress computation.

Related Topics

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CELAS4

Scalar Spring Property and Connection to Scalar Points Only

Defines a scalar spring element that is connected only to scalar points, without reference to a property entry.

Format:

1	2	3	4	5	6	7	8	9	10
CELAS4	EID	K	S1	S2					

Example:

CELAS4	42	6.2-3	2						

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
K	Stiffness of the scalar spring. See Remarks 8 and 9. (Real)
S1, S2	Scalar point identification numbers. (Integer ≥ 0; S1 ≠ S2)

Remarks:

1. S1 or S2, but not both, may be blank or zero indicating a constrained coordinate.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. The structural damping coefficient GE is not available with CELAS4. The effect of PARAM,G is included in the spring stiffness for force calculations in frequency and transient analyses. The effects of PARAM,G can be excluded using PARAM,GDAMPF,NO. If PARAM,W3 is not specified, PARAM,G is ignored in a transient analysis. See **Parameters and Parameter Applicability Tables**. For more details on element force calculation, see the *Simcenter Nastran Element Library Reference*.

   GDAMPF is not supported for SOL 401. For SOL 401 when ANALYSIS = DYNAMICS, the effect of PARAM,G is included in the force and stress computation.
4. No stress coefficient is available with CELAS4.
5. Only one scalar spring element may be defined on a single entry.
6. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.

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7. A scalar point specified on this entry need not be defined on an SPOINT entry.
8. Rotational stiffness should be specified as moment per radian.
9. In solutions 153 and 159, the software calculates heat conduction for CELAS4 elements as

   $q = K \times \Delta T$

   where K is the value entered on the CELAS4 entry. Both S1 and S2 must be defined.

Related Topics

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CFAST

Defines Shell Patch Connection

Defines a shell patch connection with direct stiffness input.

Format:

1	2	3	4	5	6	7	8	9	10
CFAST	EID	PID	TYPE	IDA	IDB	GS	GA	GB	
	XS	YS	ZS						

Example using PROP:

1	2	3	4	5	6	7	8	9	10
CFAST	10	25	PROP	2	4	105			

Example using ELEM:

1	2	3	4	5	6	7	8	9	10
CFAST	10	25	ELEM	36	46	105			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PFAST bulk data entry (Integer > 0; Default = EID)
TYPE	Determines if shell element patches A and B are defined by element or property identification numbers. (Character) See Remark 1. If TYPE = 'PROP', IDA,IDB define pshell property identification numbers. If TYPE = 'ELEM', IDA,IDB define shell element identification numbers.
IDA,IDB	Property id (for type PROP) or Element id (for type ELEM) defining patches A and B. IDA and IDB should be unique. (Integer > 0)
GS	Grid point used to determine the location of the connection on patches A and B, only when GA,GB are blank or 0. See Remark 2 . (Integer > 0 or blank)

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Field	Contents
GA,GB	Grid point used to determine the location of the connection on patches A and B. See Remark 2 . (Integer > 0 or blank)
XS,YS,ZS	Determines the coordinate location of the connection on patches A and B, only if GA,GB,GS are all 0 or blank. See Remark 2 . (Real or blank)

Remarks:

1. The CFAST element defines a weld like connection between two shell element patches. The CFAST element TYPE = “PROP” and “ELEM” are similar to the CWELD element TYPE = “PARTPAT” and “ELPAT”, respectively, except that the CFAST element stiffness is entered directly on the KTi and KRi fields of the PFAST entry. The CWELD element stiffness is derived from an axial rod of diameter D, where D is specified on the PWELD bulk entry. The diameter D specified on the PFAST entry is only used to determine the CFAST connectivity on the patches (grid points GHA1-GHA4 on patch A and GHB1-GHB4 on patch B). See the remarks on the **CWELD** bulk entry for the CFAST element connectivity information.
2. GA/GB, or GS, or XS/YS/ZS, in that order of precedence, determine the location(s) of the CFAST connection on surface patches A and/or B. The connection location on patch A is a specification of grid point GA, or a projection of GS or XS/YS/ZS normal to surface patch A. The location on patch B is a specification of grid point GB, or a projection of GS or XS/YS/ZS normal to surface patch B. When GS or XS/YS/ZS are used, a normal projection must exist in order to define a valid connection element. GS or XS/YS/ZS do not need to lie on either surface patch A or B. If GS or XS/YS/ZS are used to define the connection location, grid points are internally created at the connection location with an ID. The numbering of the internal IDs is describe in Remark 3.
3. When GA and GB are specified, the displacements for these grids will be printed/stored as requested on the DISPLACEMENT case control command. If these points are not specified, they will be generated by the software, and the displacements for these grids can only be written to the .f06 file with grid IDs calculated as follows:
   Grid ID = Value of SYSTEM(178) + (CWELD ID) + (end ID)
   where the end ID is 1 or 2 for the two ends. SYSTEM(178) defaults to 101,000,000.
   The new grid IDs are only used for labeling the displacement output and do not exist in the analysis or internal tables. This displacement result is not written to the .op2 or punch file.

Related Topics

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CGAP

Gap Element Connection

Defines a gap or friction element.

Format:

1	2	3	4	5	6	7	8	9	10
CGAP	EID	PID	GA	GB	X1	X2	X3	CID	

Example:

CGAP	17	2	110	112	5.2	0.3	-6.1		

Alternate Format and Example:

CGAP	EID	PID	GA	GB	G0			CID	

CGAP	17	2	110	112	13				

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PGAP entry. (Integer > 0; Default = EID)
GA, GB	Connected grid points at ends A and B. (Integers > 0; GA ≠ GB)
X1, X2, X3	Components of the orientation vector $\vec{v}$, from GA, in the displacement coordinate system at GA. See Remark 2. (Real)
G0	Alternate method to supply the orientation vector $\vec{v}$ using grid point G0. Direction of $\vec{v}$ is from GA to G0. See Remark 2. (Integer > 0)
CID	Element coordinate system identification number. CID must be specified if GA and GB are coincident (distance from GA to GB<10^{-4}), or if an interference condition is desired . See Remarks 2 and 6. (Integer ≥ 0 or blank)

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Remarks:

1. The CGAP element is treated as a nonlinear gap element when used in the nonlinear solutions 106, 129, 153, and 159, in which the gap conditions update as the nonlinear solution iterates.

   For linear solutions (solution 101 and consecutive solutions 103, 111 and 112) with contact defined (BCSET card exists), CGAP elements are treated as linear contact elements if the system cell (412) OLDGAPS is set to 0 (default). If the system cell OLDGAPS is set to 1, the CGAP will be treated the same as a linear spring element (as described below). See the chapter on surface contact in the *Simcenter Nastran User's Guide* for more information.

   For linear solutions without contact defined, and for all other solution sequences, the CGAP will be treated the same as a linear spring element, and remains linear with the initial stiffness. This stiffness depends on the value for the initial gap opening (U0 field in the PGAP entry).

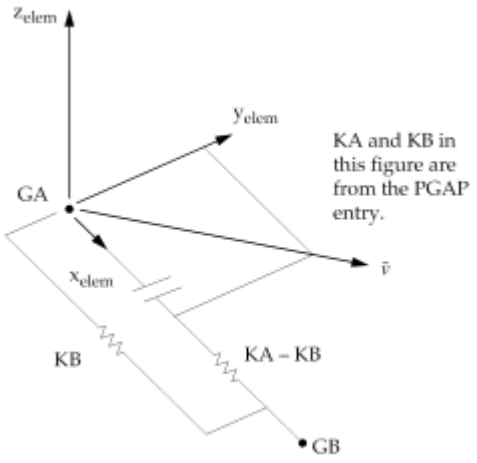
2. The gap element coordinate system is defined by one of two following methods:

   - If the coordinate system (CID field) is specified, the element coordinate system is established using that coordinate system, in which the element x-axis is in the T1 direction and the y-axis in the T2 direction. The orientation vector $\vec{v}$ will be ignored in this case. The basic coordinate system is used when CID=0.

     If an interference condition is desired, the CID positive x-axis (gap direction) should point from the "GB side" to the "GA-side" of the gap.

   - If the CID field is blank and the grid points GA and GB are not coincident (distance from A to B ≥ 10–4), then the line AB is the element x-axis and the orientation vector $\vec{v}$ lies in the x-y plane. See Figure 11-42.

Figure 11-42: CGAP Element Coordinate System



3. The element coordinate system does not rotate as a result of deflections.

4. Initial gap openings are specified on the PGAP entry and not derived from the separation distance between GA and GB.

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5. Forces, which are requested with the STRESS Case Control command, are output in the element coordinate system. F_x is positive for compression.
6. If CID is being used to define the element coordinate system and the CID refers to either a cylindrical or spherical coordinate system, then grid GA will be used to define the system. If grid GA lies on the z-axis of the cylindrical or spherical coordinate system it is recommended that a different coordinate system be used for this element.
7. See PARAM,CDITER for an alternative approach.

Remarks related to SOL 402:

1. X1, X2, and X3 are ignored.

Related Topics

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CGEAR

Gear Connection

Defines a geared connection between rotors in SOL 414.

Format:

1	2	3	4	5	6	7	8	9	10
CGEAR	EID	PID	GA	GB			CIDA	CIDB	

Example:

CGEAR	100	1	101	102			1	2	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PGEAR entry. (Integer > 0; No default)
GA	Identification number of the grid point that lies at the center of gear A. (Integer > 0; GA ≠ GB; No default)
GB	Identification number of the grid point that lies at the center of gear B. (Integer > 0; GA ≠ GB; No default)
CIDA	Identification number of the coordinate system for gear A. (Integer ≥ 0; Default = 0 or blank for the basic coordinate system)
CIDB	Identification number of the coordinate system for gear B. (Integer ≥ 0; Default = 0 or blank for the basic coordinate system)

Remarks:

1. GA and GB must belong to different rotors. Thus, they must be listed on different ROTORG entries.
2. The Z-axis of the CIDA and CIDB coordinate systems must coincide with the axis of rotation for Gear A and Gear B, respectively.

Related Topics

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CHBDYE

Geometric Surface Element Definition (Element Form)

Defines a boundary condition surface element with reference to a heat conduction element.

Format:

1	2	3	4	5	6	7	8	9	10
CHBDYE	EID	EID2	SIDE	IVIEWF	IVIEWB	RADMIDF	RADMIDB		

Example:

CHBDYE	2	10	1	3	3	2	2		

Fields:

Field	Contents
EID	Surface element identification number for a specific side of a particular element. See Remarks 1 and 10. (Unique Integer > 0 among all elements)
EID2	A heat conduction element identification number. (Integer > 0)
SIDE	A consistent element side identification number. See Remarks 6 , 7 , 8, and 9. (1 ≤ Integer ≤ 6)
IVIEWF	A VIEW entry identification number for the front face of the surface element. (Integer > 0, see Remark 2 for the default.)
IVIEWB	A VIEW entry identification number for the back face of the surface element. (Integer > 0, see Remark 2 for the default.)
RADMIDF	RADM identification number for the front face of surface element. (Integer ≥ 0, see Remark 2 for the default.)
RADMIDB	RADM identification number for the back face of the surface element. (Integer ≥ 0, see Remark 2 for the default.)

Remarks:

1. EID is a unique elemental ID associated with a particular surface element. EID2 identifies the general heat conduction element being considered for this surface element.
2. The defaults for IVIEWF, IVIEWB, RADMIDF, and RADMIDB may be specified on the BDYOR entry. If a particular field is blank both on the CHBDYE entry and the BDYOR entry, then the default is zero.

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3. For the front face of shell elements, the right-hand rule is used as one progresses around the element surface from G1 to G2 to ... Gn. For the edges of shell elements or the ends of line elements, an outward normal is used to define the front surface. For example, when a radiation boundary condition is defined on SIDE=1 of a shell element, a positive EID indicates that IVIEWF is relative to the top of the shell.
4. If the surface element is to be used in the calculation of view factors, it must have an associated VIEW entry.
5. All conduction elements to which any boundary condition is to be applied must be individually identified with the application of one of the surface element entries: CHBDYE, CHBDYG, or CHBDYP.
6. Side selection for solid elements:

   The sides of the solid elements are numbered consecutively according to the order of the grid point numbers on the solid element entry. The sides of solid elements are either quadrilaterals or triangles. For each element type, tabulate the grid points (gp) at the corners of each side.

Table 11-59: 8-node or 20-node CHEXA

side	gp	gp	gp	gp
1	4	3	2	1
2	1	2	6	5
3	2	3	7	6
4	3	4	8	7
5	4	1	5	8
6	5	6	7	8

Table 11-60: CPENTA

side	gp	gp	gp	gp
1	3	2	1	
2	1	2	5	4
3	2	3	6	5
4	3	1	4	6
5	4	5	6	

Table 11-61: CPYRAM

side	gp	gp	gp	gp
1	4	3	2	1
2	1	2	5	

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side	gp	gp	gp	gp
3	2	3	5	
4	3	4	5	
5	4	1	5	

Table 11-62: CTETRA

side	gp	gp	gp
1	3	2	1
2	1	2	4
3	2	3	4
4	3	1	4

7. Side selection for shell elements:

SIDE=1 selects the face area of the element defined by the corner grids in the connectivity. SIDE = 2 through 4 for a TRI or 2 through 5 for a QUAD select the individual edge areas around the perimeter such that SIDE=2 selects the edge between the first and second grid points, SIDE=3 selects the edge between the second and the third, and so on. See **"CHBDYG"** for surface type definition. The thickness is that of the shell element. Note that midside grids are ignored. See Remark 3.

8. Side selection for line elements:

SIDE=1 selects the outside surface area of the element between the end grid points excluding the end areas. SIDE=2 and SIDE=3 select the end areas at the first and second grid points in the element connectivity, respectively. The area calculation is consistent with the element geometry.

9. Side selection for axisymmetric elements:

The axisymmetric elements CTRAX3, CTRAX6, CQUADX4, and CQUADX8 have revolved sides. SIDE=1 is determined by the element connectivity in the table below. The other sides are determined by the same connectivity pattern around the perimeter.

Table 11-63: Axisymmetric rev sides

Element	Number of rev sides	SIDE=1
CTRAX3	3	G1 and G2
CTRAX6	3	G1, G4, and G2
CQUADX4	4	G1 and G2
CQUADX8	4	G1, G5, and G2

10. The EID field on all CHBDYi entries is entered as positive integer, but the EID referenced on certain load or boundary condition entries can be entered as a +EID or -EID to represent the front or back of the CHBDYi entry, respectively. For example, entering a -EID on a QVECT entry applies a heat flux

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opposite to the CHBDYi normal orientation vector. The table below summarizes the thermal load and boundary conditions, if they support CHBDYi entires, and if so, if they support a negative EID.

Loads	Description	CHBDYi Supported	Negative EID Supported
QVECT	Directional heat flux from a distant source.	Yes	Yes
QVOL	Volumetric internal heat generation.	No	
QHBDY	Heat flux applied to an area defined on grid points.	No	
QBDY1	Heat flux applied to surface elements.	Yes	Yes
QBDY2	Heat flux applied to grid points associated with a surface element.	Yes	No
QBDY3	Heat flux applied to surface elements with control node capability.	Yes	Yes
SLOAD	Power into a grid or scalar point.	No	
NOLIN1	Nonlinear transient load as a tabular function.	No	
NOLIN2	Nonlinear transient load as a product of two variables.	No	
NOLIN3	Nonlinear transient load as a positive variable raised to a power.	No	
NOLIN4	Nonlinear transient load as a negative variable raised to a power.	No	
Boundary Conditions			
CONV	CONV Free convection	Yes	No
CONVM	CONVM Forced convection (fluid “element”)	Yes, but CHBDYP only	No
RADBC	Radiation exchange with space	Yes	Yes*
RADSET	Radiation exchange within an enclosure	No	
RADLST	Listing of Enclosure Radiation Faces	Yes	Yes*

* RADBC and RADLST entries do support a negative EID, although the front and back IVIEWi and RADMIDi fields already allow for this control, and a negative EID is typically not needed. IVIEWF and RADMIDF are associated with the CHBDYi normal orientation vector and IVIEWB and RADMIDB with the opposite. For radiation problems, if the RADMIDF or RADMIDB is zero, default radiant properties assume perfect black body behavior.

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CHBDYG

Geometric Surface Element Definition (Grid Form)

Defines a boundary condition surface element without reference to a property entry.

Format:

1	2	3	4	5	6	7	8	9	10
CHBDYG	EID		TYPE	IVIEWF	IVIEWB	RADMIDF	RADMIDB		
	G1	G2	G3	G4	G5	G6	G7	G8	

Example:

CHBDYG	2		AREA4	3	3	2	2		
	100	103	102	101					

Fields:

Field	Contents
EID	Surface element identification number. (Unique Integer > 0 among all elemental entries)
TYPE	Surface type. See Remark 3 . (Character)
IVIEWF	A VIEW entry identification number for the front face. (Integer > 0; see Remark 2 for the default.)
IVIEWB	A VIEW entry identification number for the back face. (Integer > 0; see Remark 2 for the default.)
RADMIDF	RADM identification number for front face of the surface element. (Integer > 0; see Remark 2 for the default.)
RADMIDB	RADM identification number for the back face ofthe surface element. (Integer ≥ 0; see Remark 2 for the default.)
Gi	Grid point IDs of grids bounding the surface. (Integer > 0)

Remarks:

1. EID is a unique ID associated with a particular surface element as defined by the grid points.
2. The defaults for TYPE, IVIEWF, IVIEWB, RADMIDF, and RADMIDB may be specified on the BDYOR entry. If a particular field is blank on both the CHBDYG entry and the BDYOR entry, then the default is zero.

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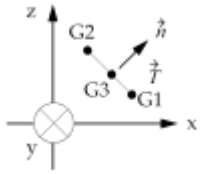
3. TYPE specifies the kind of element surface; allowed types are: REV, AREA3, AREA4, AREA6, and AREA8. See Figure 11-43, Figure 11-45, and Figure 11-46.

   - TYPE = REV

     The “REV” type, which is used with elements CTRAX3, CTRAX6, CQUADX4, and CQUADX8, has two primary grid points whose location depends on the element orientation.

     ◦ When the elements are oriented on the XZ plane, the two primary grid points must lie on the XZ plane of the basic coordinate system with X > 0. The line between the grid points G1 and G2 cannot be parallel to the X axis. A midside grid point G3 is optional and supports convection or heat flux from the edge of elements CTRAX6 and CQUADX8. The defined area is a conical section with Z as the axis of symmetry. A property entry is required for convection, radiation, or thermal vector flux. Automatic view factor calculations with VIEW data are not supported for the REV option.

Figure 11-43: Normal Vector for CHBDYG Element of Type “REV”, XZ plane orientation



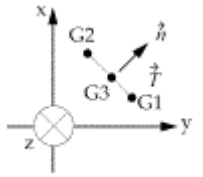
The unit normal lies in the XZ plane, and is given by

$$\vec{n} = (\vec{e}_y \times \vec{T}) / \left|\vec{e}_y \times \vec{T}\right|.$$

$\vec{e}_y$ is the unit vector in the Y direction.

◦ When the elements are oriented on the XY plane, the two primary grid points must lie on the XY plane of the basic coordinate system with Y > 0. The line between the grid points G1 and G2 cannot be parallel to the Y axis. A midside grid point G3 is optional and supports convection or heat flux from the edge of elements CTRAX6 and CQUADX8. The defined area is a conical section with X as the axis of symmetry. A property entry is required for convection, radiation, or thermal vector flux. Automatic view factor calculations with VIEW data are not supported for the REV option.

Figure 11-44: Normal Vector for CHBDYG Element of Type “REV”, XY plane orientation



The unit normal lies in the XY plane, and is given by

$$\vec{n} = (\vec{e}_z \times \vec{T}) / \left|\vec{e}_z \times \vec{T}\right|$$

$\vec{e}_z$ is the unit vector in the Z direction.

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- TYPE = AREA3, AREA4, AREA6, or AREA8

  These types have three and four primary grid points, respectively, that define a triangular or quadrilateral surface and must be ordered to go around the boundary. A property entry is required for convection, radiation, or thermal vector flux.

Figure 11-45: TYPE Examples

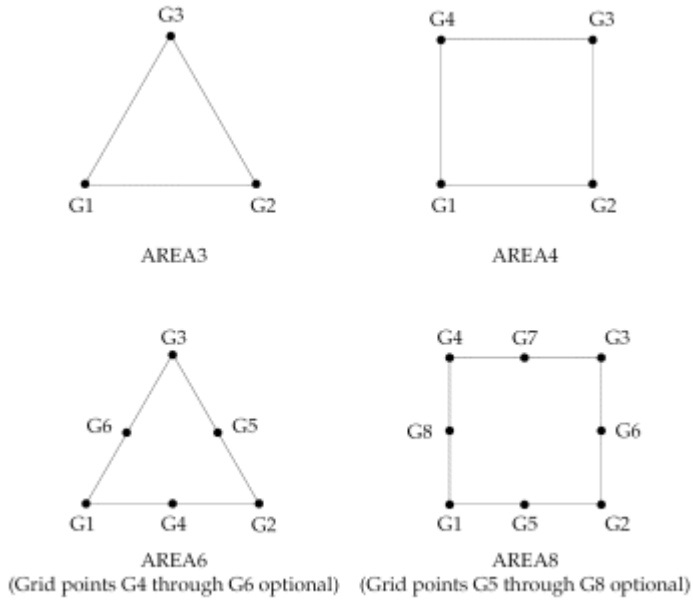
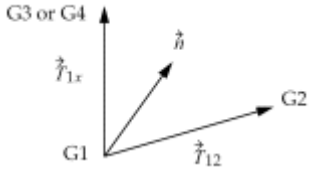


Figure 11-46: Normal Vector for CHBDYG Element of Types "AREAi"



The unit normal vector is given by

$$\vec{n} = \frac{(\vec{T}_{12} \times \vec{T}_{1x})}{|\vec{T}_{12} \times \vec{T}_{1x}|}$$

(G3 is used for triangles, and G4 is used for quadrilaterals.)

4. For defining the front face, the right-hand rule is used on the sequence G1 to G2 to ... Gn of grid points.
5. If the surface element is to be used in the calculation of view factors, it must have an associated VIEW entry.
6. All conduction elements to which any boundary condition is to be applied must be individually identified with one of the surface element entries: CHBDYE, CHBDYG, or CHBDYP.

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7. See the remarks on the **CHBDYE** bulk entry for application of boundary conditions using CHBDYG entries and a discussion of front and back faces.

Related Topics

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CHBDYP

Geometric Surface Element Definition (Property Form)

Defines a boundary condition surface element with reference to a PHBDY entry.

Format:

1	2	3	4	5	6	7	8	9	10
CHBDYP	EID	PID	TYPE	IVIEWF	IVIEWB	G1	G2	G0	
	RADMIDF	RADMIDB	GMID	CE	E1	E2	E3		

Example:

CHBDYP	2	5	POINT	2	2	101		500	
	3	3			0.0	0.0	1.0		

Fields:

Field	Contents
EID	Surface element identification number. (Unique Integer > 0 among all element identification numbers)
PID	PHBDY property entry identification numbers. (Integer > 0)
TYPE	Surface type. See Remark 3 . (Character).
IVIEWF	VIEW entry identification number for the front face. (Integer > 0 or blank)
IVIEWB	VIEW entry identification number for the back face. (Integer > 0 or blank)
G1, G2	Grid point identification numbers of grids bounding the surface. (Integer > 0)
GO	Orientation grid point. (Integer ≥ 0; Default = 0)
RADMIDF	RADM entry identification number for the front face. (Integer ≥ 0 or blank)
RADMIDB	RADM entry identification number for the back face. (Integer ≥ 0 or blank)
GMID	Grid point identification number of a midside node if it is used with the line type surface element.
CE	Coordinate system for defining orientation vector. (Integer ≥ 0; Default = 0)
Ei	Components of the orientation vector in coordinate system CE. The origin of the orientation vector is grid point G1. (Real or blank)

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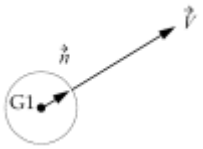
Remarks:

1. EID is a unique ID associated with a particular surface element as defined by the grid point(s).
2. The defaults for PID, TYPE, IVIEWF, IVIEWB, GO, RADMIDF, RADMIDB, CE, and Ei may be specified on the BDYOR entry. If a particular field is blank on both the CHBDYP entry and the BDYOR entry, then the default is zero.
3. TYPE specifies the kind of element surface; the allowed types are: "POINT," "LINE," "ELCYL," "FTUBE," and "TUBE." For TYPE = "FTUBE" and TYPE = "TUBE," the geometric orientation is completely determined by G1 and G2; the GO, CE, E1, E2, and E3 fields are ignored.

   - TYPE = "POINT"

     TYPE = "POINT" has one primary grid point, requires a property entry, and the normal vector Vi must be specified if thermal flux is to be used.

     Figure 11-47: Normal Vector for CHBDYP Element of Type "POINT" (See Remarks 4 and 5.)

     The unit normal vector is given by

     $\hat{n} = \vec{V} / |\vec{V}|$

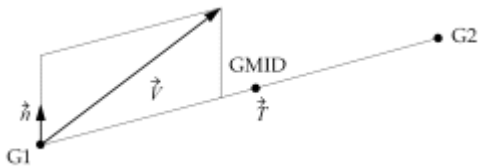
     where

     $\vec{V}$ is specified in the Ei field and given in the basic system at the referenced grid point. See Remarks 4 and 5 for the determination of $\vec{V}$.

   - TYPE = "LINE," "FTUBE," or "TUBE"

     The TYPE = "LINE" type has two primary grid points, requires a property entry, and the vector is required. TYPE = "FTUBE" and TYPE = "TUBE" are similar to TYPE = "LINE" except they can have linear taper with no automatic view factor calculations. GMID is an option for the TYPE = "LINE" surface element only and is ignored for TYPE = "FTUBE" and "TUBE".

     Figure 11-48: Normal Vector for CHBDYP Element with TYPE="LINE",TYPE="FTUBE", or TYPE="TUBE"(See Remarks 4 and 5.)

     The unit normal lies in the plane $\vec{V}$ and $\vec{T}$, is perpendicular to $\vec{T}$, and is given by:

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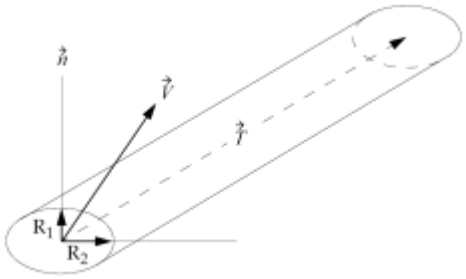
$$\vec{h} = \frac{\vec{T} \times (\vec{V} \times \vec{T})}{\left| \vec{T} \times (\vec{V} \times \vec{T}) \right|}$$

- TYPE = “ELCYL”

  TYPE = “ELCYL” (elliptic cylinder) has two connected primary grid points and requires a property entry. The vector must be nonzero. Automatic view factor calculations are not available.

Figure 11-49: Normal Vector for CHBDYP Element of TYPE=“ELCYL”(See Remarks 4 and 5.)



The same logic is used to determine $\vec{h}$ as for TYPE = LINE. The “radius” R_1 is in the $\vec{h}$direction, and R2 is the perpendicular to $\vec{h}$ and $\vec{T}$ (see fields 7 and 8 of the PHBDY entry).

4. For TYPE = “POINT,” TYPE = “LINE,” and TYPE = “ELCYL,” geometric orientation is required. The required information is sought in the following order:

- If GO > 0 is found on the CHBDYP entry, it is used.
- Otherwise, if a nonblank CE is found on the CHBDYP continuation entry, this CE and the corresponding vectors E1, E2, and E3 are used.
- If neither of the aboveis found, the same information is sought in the same way from the BDYOR entry.
- If none of the above apply, a warning message is issued.

5. The geometric orientation can be defined by either GO or the vector E1, E2, E3.

- If GO > zero:

  For a TYPE = “POINT” surface, the normal to the front face is the vector from G1 to GO. For the TYPE = “LINE” surface, the plane passes through G1, G2, GO and the right-hand rule is used on this sequence to get the normal to the front face. For a TYPE = “ELCYL” surface the first axis of the ellipse lies on the G1, G2, GO plane, and the second axis is normal to this plane. For TYPE = “FTUBE” or “TUBE” surface, no orientation is required, and GO is superfluous.

- If GO is zero:

  For a TYPE = “POINT” surface, the normal to the front face is the orientation vector. For the TYPE = “LINE” surface, the plane passes through G1, G2, and the orientation vector; the front face is based on the right-hand rule for the vectors G2-G1 and the orientation vector. For a TYPE = “ELCYL” surface, the first axis of the ellipse lies on the G1, G2, orientation vector plane, and the second axis is normal to this plane.

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6. The continuation entry is optional.
7. If the surface element is to be used in the calculation of view factors, it must have an associated VIEW entry.
8. All conduction elements to which any boundary condition is to be applied must be individually identified with the application of one of the following surface element entries: CHBDYE, CHBDYG, or CHBDYP.

Related Topics

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CHEXA

Six-Sided Solid Element Connection

Defines the connections of the six-sided solid element with eight to twenty grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CHEXA	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	G10	G11	G12	G13	G14	
	G15	G16	G17	G18	G19	G20			

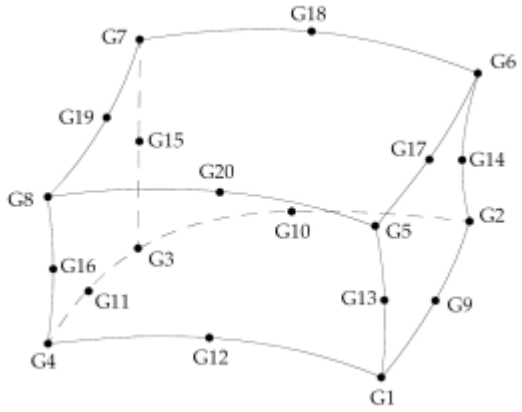
Example:

CHEXA	71	4	3	4	5	6	7	8	
	9	10	0	0	30	31	53	54	
	55	56	57	58	59	60			

Fields:

Field	Contents	Type	Default
EID	Element identification number.	Integer > 0	Required
PID	Property identification number of a PSOLID, PLSOLID, PCOMPS, or PMIC entry.	Integer > 0	Required
Gi	Grid point identification numbers of connection points.	Integer ≥ 0 or blank	Required

Figure 11-50: CHEXA Element Connection



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Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. Grid points G1 through G4 must be given in consecutive order about one quadrilateral face. G5 through G8 must be on the opposite face with G5 opposite G1, G6 opposite G2, etc.
3. The edge points G9 to G20 are optional. Any or all of them may be deleted. If the ID of any edge connection point is left blank or set to zero (as for G9 and G10 in the input example), the equations of the element are adjusted to give correct results for the reduced number of connections. Corner grid points cannot be deleted. The element is an isoparametric element (with shear correction) in all cases.
4. Components of stress are output in the material coordinate system. See the remarks on the **PSOLID** bulk entry for hyperelastic and nonlinear exceptions. See the remarks on the **PCOMPS** bulk entry for composite laminate exception.
5. The second continuation is optional.
6. Except when used as a hyperelastic element or as a composite laminate solid element, the element coordinate system for the CHEXA element is defined in terms of the three vectors R, S, and T, which join the centroids of opposite faces.

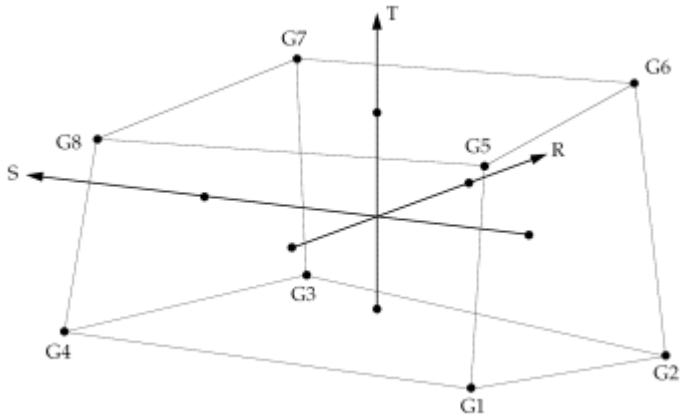
   R vector joins the centroids of faces G4-G1-G5-G8 and G3-G2-G6-G7.

   S vector joins the centroids of faces G1-G2-G6-G5 and G4-G3-G7-G8.

   T vector joins the centroids of faces G1-G2-G3-G4 and G5-G6-G7-G8.

   The origin of the coordinate system is located at the intersection of these vectors. The X, Y, and Z axes of the element coordinate system are chosen as close as possible to the R, S, and T vectors and point in the same general direction. (Mathematically speaking, the coordinate system is computed in such a way that if the R, S, and T vectors are described in the element coordinate system, a 3 x 3 positive-definite symmetric matrix would be produced.)

Figure 11-51: CHEXA Element R, S, and T Vectors



7. It is recommended that the edge points be located within the middle third of the edge.

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8. For hyperelastic elements, the plot codes are specified under the CHEXAFD element name in **"Item Codes"**.

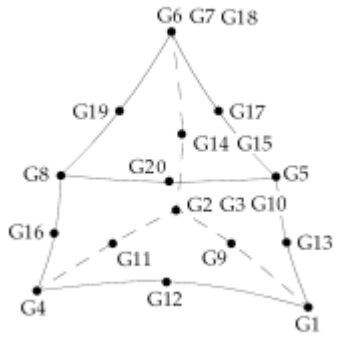
9. By default, all twelve edges of the element are considered straight unless any of G9 through G20 are specified.

10. Collapsed CHEXA elements are supported for use in crack simulations using SOL 401 and a third-party software. To allow for collapsed CHEXA elements, specify PARAM,COLPHEXA,YES. Note that the collapsed CHEXA element is not supported in a glue or contact region.

    Any face of a CHEXA element can be collapsed to an edge. The edge of the collapsed face represents the crack front.

    For example, **Figure 11-52** shows the CHEXA element of **Figure 11-50** with the G2–G14–G6–G18–G7–G15–G3–G10 face collapsed so that the G2–G14–G6 edge and the G3–G15–G7 edge become the crack front. Alternately, the G2–G14–G6–G18–G7–G15–G3–G10 face could be collapsed so that the G2–G10–G3 edge and the G6–G18–G7 edge would become the crack front.

Figure 11-52: Collapsed CHEXA Element



Two options are available for specifying a CHEXA element with a collapsed face:

- In Format 1, 15 unique grid IDs are specified in the 20 grid ID fields of the CHEXA bulk entry. Format 1 is typically used for elastic material models. With Format 1, mid-side grids can move to the quarter-span locations closest to the crack front.

  For the collapsed CHEXA element shown in **Figure 11-52**, the Format 1 specification is as follows:

1	2	3	4	5	6	7	8	9	10
CHEXA	EID	PID	G1	G2	G2	G4	G5	G6	
	G6	G8	G9	G2	G11	G12	G13	G14	
	G14	G16	G17	G6	G19	G20			

  where the same grid ID is entered in the G2, G3, and G10 fields, another grid ID is entered in both the G14 and G15 fields, and another grid ID is entered in the G6, G7, and G18 fields.

- In Format 2, 20 unique grid IDs are specified in the 20 grid ID fields of the CHEXA bulk entry. However, eight of the grid IDs do not have unique coordinates. Format 2 is typically used for elasto-plastic material models. With Format 2, mid-side grids remain at the mid-span locations.

  For the collapsed CHEXA element shown in **Figure 11-52**, the Format 2 specification is as follows:

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1	2	3	4	5	6	7	8	9	10
CHEXA	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	G10	G11	G12	G13	G14	
	G15	G16	G17	G18	G19	G20			

where the grids entered in the G2, G3, and G10 fields would share the same coordinates, the grids entered in the G14 and G15 fields would share the same coordinates, and the grids entered in the G6, G7, and G18 fields would share the same coordinates. Unlike Format 1 where grids in the CHEXA element connectivity are merged, Format 2 does not merge coincident grids in the CHEXA element connectivity. Thus, these grids can move independently of one another.

Remarks related to SOLs 402 and 414:

1. Output results will be done in the material coordinate system. If those axes are not defined, the material coordinate system is initially defined as the basic coordinate system. The material coordinate system always rotates with the element.

Related Topics

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Simcenter Nastran Quick Reference Guide
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CHEXCZ

Six-Sided Cohesive Element Connection

Defines the connections of the six-sided cohesive element with eight to twenty grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CHEXCZ	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	G10	G11	G12	G13	G14	
	G15	G16	G17	G18	G19	G20			

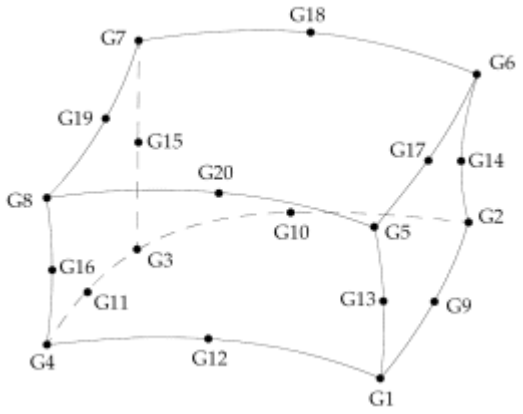
Example:

CHEXCZ	15	1	3	4	5	6	7	8	
	9	10		0	30	31	53	54	
	55	56	57	58	59	60			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PSOLCZ. (Integer > 0; No default)
Gi	Grid point identification numbers of connection points. (Integer > 0 or blank; See Remark 3 for default behavior)

Figure 11-53: CHEXCZ Element Connection



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Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. Grid points G1 through G4 must be given in consecutive order about one quadrilateral face. This is the bottom face of the element. G5 through G8 must be on the opposite face with G5 opposite G1, G6 opposite G2, G7 opposite G3, and G8 opposite G4. This is the top face of the element. The normal direction is defined relative to the bottom surface.
3. The edge points G9 to G20 are optional. Any or all of them may be deleted. If the ID of any edge connection point is left blank or set to zero (G9 and G10 in the input example), the equations of the element are adjusted to give correct results for the reduced number of connections. Corner grid points cannot be deleted.
4. If G15 through G20 are omitted, the second continuation line is not required.
5. For best results, locate edge grid points within the middle third of the edge.
6. For information regarding the material coordinate system, see **PSOLCZ**.

Remarks related to SOL 402:

1. Grid points G13 through G16 are dummy nodes: they will get the average displacement values of adjacent nodes.

Related Topics

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CJOINT

Kinematic Joint Connection (SOL 402 and SOL 414,129)

Defines a kinematic joint connection

Format:

1	2	3	4	5	6	7	8	9	10
CJOINT	EID	PID	G1	G2	G3		CID1	CID2	
	DID								

Example:

CJOINT	111	203	31	74	75		2		
	111								

Fields:

Field	Contents
EID	Element identification number. See Remark 1. (Integer > 0).
PID	Property identification number of a PJOINT and/or PJOINT2 entry. (Integer > 0; Default = EID).
G1, G2	Grid point identification numbers of connection points. (Integers > 0, all unique.)
G3	Grid point identification numbers of the third connection point. See Remark 2. (Integer > 0, unique.) (Optional).
CID1	Kinematic joint coordinate system identification number. See Remark 3. (Integer ≥ 0; If blank, the global Cartesian coordinate system is assumed).
CID2	Kinematic joint second coordinate system. See Remark 4. (Integer > 0) (Optional).
DID	Property identification number of a DESC entry. See Remark 5 (Integer > 0) (Optional).

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. G3 is not used for kinematic joints of type `SPHE`, `UNIVSL`, `FIXED`, or `CONVEL`.

   For other kinematic joints, if omitted, a grid point is created at the G2 position.

   For more information, see the PJOINT entry.

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3. CID1 coordinate system is:

   Required for kinematic joints of type REVO, UNIV, and CONV. If omitted, a Cartesian coordinate system is created at the G1 position.

   Required if G1 and G2 are coincident for kinematic joints of type SLID, CYLD, SLI, and SCRE. If omitted, a Cartesian coordinate system is created at the G1 position.

   For more information, see the **PJOINT** entry.

4. CID2 coordinate system is required for kinematic joints of type UNIVSL or CONVEL.

   If omitted, a Cartesian coordinate system is created at the G2 position.

   For more information, see the **PJOINT** entry.

5. The **DESC** entry is used to define the name of the kinematic joint and is useful when you post-process time history curves. If omitted, a name is defined with the EID property and the TYPE property of the PJOINT entry.

Related Topics

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CLINK

Connection used for misalignment definition, bearing definition for cyclic symmetry models, or linking rotating and non-rotating parts

For SOL 414 solutions, defines a connection that you can use to do the following:

- Apply misalignment parameters (all SOL 414 solutions).
- Link a grid point on the rotor axis to a grid point on the cyclic sector of the rotor in a fixed or rotating frame (all SOL 414 solutions).
- Link a grid point on the rotor axis to a grid point on the cyclic sector of the rotor in a fixed or rotating frame with or without the Coleman method (SOL 414,110 Complex Modal Analysis).
- Link fixed and rotating reference frames (SOL 414,111 Harmonic Response and SOL 414,129 Transient Response).

Format:

1	2	3	4	5	6	7	8	9	10
CLINK	EID	GA	GB	OPT	CSYS				

Example:

CLINK	100	54	220	2	1				

Fields:

Field	Contents
EID	Element identification number. See Remark 1. (Integer > 0)
GA, GB	Grid point identification number of connection points. See Remarks 2, 3, 7 and 9. (Integer > 0; GA ≠ GB; No default)
OPT	Misalignment, reference frame, or cyclic symmetry linking option. (Integer 1, 2, 3, 5, or 6; Default = 2) OPT = 1 for translational misalignment and axial offset. See Remark 4. OPT = 2 for translational and rotational misalignment. See Remark 4. OPT = 3 for reference frame definition. See Remarks 7 and 8. OPT = 5 for linking a cyclic symmetry sector without a Coleman transformation. See Remark 9.

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Field	Contents
	OPT = 6 for linking a cyclic symmetry sector with a Coleman transformation. See Remark 9.
CSYS	Coordinate system identification number. See Remark 10. (Integer; Default = 0 for the basic coordinate system)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. GA and GB must be coincident.
3. An axial offset (MSTZ field in the MLINK bulk entry) is a displacement in the Z-direction defined as:

   GB - GA

   where:

   - GB is the second grid point.
   - GA is the first grid point.

   Therefore, be sure to define GA and GB in the order that will provide the intended offset. For example, if GA is 1.2 on the Z-axis and GB is 1.8 on the Z-axis, the offset is 0.6 (1.8 - 1.2).
4. For misalignment, the connection can be between coincident grid points in the same rotor, coincident grid points in different rotors that rotate at the same speed and in the same direction, or coincident grid points between a bearing and rotor.
5. If an MLINK bulk entry does not reference the CLINK bulk entry, the CLINK connection behaves as a rigid connection.
6. Unlike FOU3 bulk entries, the software calculates output for CLINK bulk entries.
7. For SOL 414,111 and SOL 414,129, when OPT = 3, define GA and GB as follows:

   - GA must be a stator grid point.

     The GA grid point listed in the GA field is associated with the supporting structure and the motion of the supporting structure is defined relative to a fixed coordinate basis.
   - GB must be a rotor grid point.

     The GB grid point listed in the GB field is associated with the rotor and the motion of the rotor is defined relative to a rotating coordinate basis that is fixed to the rotor.

   This allows you to model systems with one non-symmetric rotor and a non-symmetric supporting structure.
8. In SOL 414,111 Harmonic Response solutions, when you add CLINK with option 3 to analyze a non-symmetric rotor (rotating reference frame with RMETHOD ROTORD defined) and non-symmetric stator (fixed reference frame), you must add the HARCOEFF case control command.
9. When OPT = 5 or 6, define GA and GB as follows:

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- GA must be a 3D grid point (such as a bearing or stator grid point) on the rotor axis.
- GB must be a grid point on the cyclic sector.

OPT = 5 is valid in all SOL 414 solutions.

OPT = 6 is valid when the Coleman transformation is activated (PARAM,COLEMAN,1).

10. Define the CSYS field in one of the following ways depending on how you define the OPT field.

- When OPT = 1 or 2, the CSYS field specification on the CLINK bulk entry and the RID field specification on the MLINK bulk entry define whether the misalignment load is static or dynamic.
  - Static misalignment loads are fixed relative to the supporting structure.

    For example, bearing misalignment can cause static misalignment loads.

    To define a static misalignment load, on the MLINK bulk entry, leave the RID field blank, and on the CLINK bulk entry, specify the coordinate system for the static misalignment in the CSYS field.
  - Dynamic misalignment loads are fixed relative to the rotor.

    For example, runout in a shaft can cause dynamic misalignment loads.

    To define a dynamic misalignment load, on the CLINK bulk entry, leave the CSYS field blank, and on the MLINK bulk entry, specify the rotor for which the dynamic misalignment load is defined in the RID field.
- When OPT = 5 or 6 and the rotor is not oriented along the global Z-axis, define the CSYS field with a coordinate system where the local Z-axis points into the direction of the rotor axis.

Related Topics

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CLOAD

Static Load Combination for Superelement Loads (Superposition)

Defines a static load as a linear combination of previously calculated superelement loads defined by the LSEQ entry in nonlinear static analysis (SOL 106 or 153).

Format:

1	2	3	4	5	6	7	8	9	10
CLOAD	CID	S	S1	IDV1	S2	IDV2	S3	IDV3	
	S4	IDV4	-etc.-						

Example:

CLOAD	25	1.0	25.0	10	-1.0	101	2.2-1	604	
	-62.0	62							

Fields:

Field	Contents
CID	Combination identification number. (Integer > 0)
S	Scale factor. (Real)
Si	Scale factors. (Real)
IDVi	Identification numbers of load vectors (EXCITEID of a selected LSEQ entry) calculated for a superelement loads entry. (Integer > 0)

Remarks:

1. The CLOAD entry must be selected in the residual solution subcase of the case control with CLOAD = CID and must be used if loads are applied to upstream superelements in SOL 106 or 153.
2. The load vector defined is given by

$$\{P\} = S\sum_{i} Si\{P_{\text{IDVi}}\}$$

3. The IDVi field refers to a previously calculated load vector for the superelement via the LSEQ approach. That is, a LOADSET keyword must have been selected in Case Control that in turn refers to one or more LSEQ entries in the Bulk Data Section. The IDVi refers to the EXCITEID of such LSEQ entries. For more details, see the **"LSEQ"** and the **"LOADSET"**.

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4. In the CID or IDV fields, a CLOAD entry may not reference an identification number defined by another CLOAD entry.

Related Topics

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CMASS1

Scalar Mass Connection

Defines a scalar mass element.

Format:

1	2	3	4	5	6	7	8	9	10
CMASS1	EID	PID	G1	C1	G2	C2			

Example:

CMASS1	32	6	2	1					

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Property identification number of a PMASS entry. (Integer > 0; Default = EID)
G1, G2	Geometric grid or scalar point identification number. (Integer ≥ 0)
C1, C2	Component number. (0 ≤ Integer ≤ 6; blank or zero if scalar point)

Remarks:

1. Scalar points may be used for G1 and/or G2, in which case the corresponding C1 and/or C2 must be zero or blank. Zero or blank may be used to indicate a grounded terminal G1 or G2 with a corresponding blank or zero C1 or C2. A grounded terminal is a point with a displacement that is constrained to zero. If only scalar points and/or ground are involved, it is more efficient to use the CMASS3 entry.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. The two connection points (G1, C1) and (G2, C2) must not be coincident.
4. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.
5. A scalar point specified on this entry need not be defined on an SPOINT entry.
6. If Gi refers to a grid point then Ci refers to degrees-of-freedom(s) in the displacement coordinate system specified by CD on the GRID entry.

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7. Scalar elements input coupled mass matrices when the second pair of fields is entered. When uncoupled point masses are desired input only the first pair of fields. When a coupled mass matrix is requested the submatrix added has M on the diagonal, and -M on the off-diagonal. The element is not checked for internal constraints, which is the user's responsibility if desired. There are instances where elements with internal constraints are desired, although not frequently. To identify the presence of internal constraints caused by coupled mass, inspect GPWG output, OLOAD output due to GRAV loads, and rigid body modes of free structures. Some forms of coupled mass will cause coupling of rigid body translational mass terms in GPWG output, and poor rigid body modes in modal analysis.

Remarks related to SOLs 402 and 414:

1. G1 and G2 must be grid points, i.e., they cannot be scalar points.

Related Topics

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CMASS2

Scalar Mass Property and Connection

Defines a scalar mass element without reference to a property entry.

Format:

1	2	3	4	5	6	7	8	9	10
CMASS2	EID	M	G1	C1	G2	C2			

Example:

CMASS2	32	9.25	6	1					

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
M	Value of the scalar mass. See Remark 2 for SOL 414. (Real)
G1, G2	Geometric grid or scalar point identification number. (Integer ≥ 0)
C1, C2	Component number. (0 ≤ Integer ≤ 6; blank or zero if scalar point.)

Remarks:

1. Scalar points may be used for G1 and/or G2, in which case the corresponding C1 and/or C2 must be zero or blank. Zero or blank may be used to indicate a grounded terminal G1 or G2 with a corresponding blank or zero C1 or C2. A grounded terminal is a point with a displacement that is constrained to zero. If only scalar points and/or ground are involved, it is more efficient to use the CMASS4 entry.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. The two connection points (G1, C1) and (G2, C2) must be distinct. Except in unusual circumstances, one of them will be a grounded terminal with blank entries for Gi and Ci.
4. For a discussion of the scalar elements, see Overview of 0D (Scalar) Elements in the *Simcenter Nastran Element Library*.
5. A scalar point specified on this entry need not be defined on an SPOINT entry.
6. If Gi refers to a grid point then Ci refers to degrees-of-freedom(s) in the displacement coordinate system specified by CD on the GRID entry.

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7. See Remark 7 for **"CMASS1"**.

Remarks related to SOLs 402 and 414:

1. G1 and G2 must be grid points, i.e., they cannot be scalar points.
2. For SOL 414 solutions, if you set the M (mass) parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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CMASS3

Scalar Mass Connection to Scalar Points Only

Defines a scalar mass element that is connected only to scalar points.

Format:

1	2	3	4	5	6	7	8	9	10
CMASS3	EID	PID	S1	S2					

Example:

CMASS3	13	42	62						

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
PID	Property identification number of a PMASS entry. (Integer > 0; Default = EID)
S1, S2	Scalar point identification numbers. (Integer ≥ 0; S1 ≠ S2)

Remarks:

1. S1 or S2 may be blank or zero, indicating a constrained coordinate.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. Only one scalar mass element may be defined on a single entry.
4. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.
5. A scalar point specified on this entry need not be defined on an SPOINT entry.

Related Topics

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CMASS4

Scalar Mass Property and Connection to Scalar Points Only

Defines a scalar mass element that is connected only to scalar points, without reference to a property entry.

Format:

1	2	3	4	5	6	7	8	9	10
CMASS4	EID	M	S1	S2					

Example:

CMASS4	23	14.92		23					

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
M	Scalar mass value. (Real)
S1, S2	Scalar point identification numbers. (Integer ≥ 0; S1 ≠ S2)

Remarks:

1. S1 or S2 may be blank or zero, indicating a constrained coordinate. This is the usual case.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. Only one scalar mass element may be defined on a single entry.
4. For a discussion of the scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.
5. A scalar point specified on this entry need not be defined on an SPOINT entry.

Related Topics

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12. Bulk Entries CO—CY

CONM1

Concentrated Mass Element Connection, General Form

Defines a 6 x 6 symmetric mass matrix at a geometric grid point.

Format:

1	2	3	4	5	6	7	8	9	10
CONM1	EID	G	CID	M11	M21	M22	M31	M32	
	M33	M41	M42	M43	M44	M51	M52	M53	
	M54	M55	M61	M62	M63	M64	M65	M66	

Example:

CONM1	2	22	2	2.9	6.3				
	4.8	28.6							
		28.6						28.6	

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
G	Grid point identification number. (Integer > 0)
CID	Coordinate system identification number for the mass matrix. (Integer ≥ 0)
Mij	Mass matrix values. See Remark 3 for SOL 414. (Real)

Remarks:

1. For a less general means of defining concentrated mass at grid points, see the CONM2 entry description.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. The mass matrix has the form:

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$$\begin{bmatrix} M_{11} & & & & & \\ M_{21} & M_{22} & & \text{symmetric} & & \\ M_{31} & M_{32} & M_{33} & & & \\ M_{41} & M_{42} & M_{43} & M_{44} & & \\ M_{51} & M_{52} & M_{53} & M_{54} & M_{55} & \\ M_{61} & M_{62} & M_{63} & M_{64} & M_{65} & M_{66} \end{bmatrix}$$

Remarks related to SOLs 402 and 414:

1. Only the diagonal terms of the mass matrix Mii ($1 \le i \le 6$) are used.
2. CID is ignored, i.e., mass components Mii are assumed to be in the displacement coordinate system of grid point G.
3. For SOL 414 solutions, if you set the mass parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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CONM2

Concentrated Mass Element Connection, Rigid Body Form

Defines a concentrated mass at a grid point.

Format:

1	2	3	4	5	6	7	8	9	10
CONM2	EID	G	CID	M	X1	X2	X3		
	I11	I21	I22	I31	I32	I33			

Example:

CONM2	2	15	6	49.7					
	16.2		16.2			7.8			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
G	Grid point identification number. (Integer > 0)
CID	Coordinate system identification number. For CID of -1 see X1, X2, X3 below. (Integer ≥ -1; Default = 0)
M	Mass value. See Remark 2 for SOL 414. (Real)
X1, X2, X3	Offset distances from the grid point to the center of gravity of the mass in the coordinate system defined in field 4, unless CID = -1, in which case X1, X2, X3 are the coordinates, not offsets, of the center of gravity of the mass in the basic coordinate system. (Real)
Iij	Mass moments of inertia measured at the mass center of gravity in the coordinate system defined by field 4. If CID = -1, the basic coordinate system is implied. See Remark 2 for SOL 414. (For I11, I22, and I33; Real ≥ 0.0; for I21, I31, and I32; Real)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. For a more general means of defining concentrated mass at grid points, see the CONM1 entry description.

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3. The continuation is optional.
4. If CID = -1, offsets are internally computed as the difference between the grid point location and X1, X2, X3. The grid point locations may be defined in a nonbasic coordinate system. In this case, the values of Iij must be in a coordinate system that parallels the basic coordinate system.
5. The form of the inertia matrix about its center of gravity is taken as:

$$\begin{bmatrix} M & & & & \\ & M & & \text{symmetric} & \\ & & M & & \\ & & & I11 & \\ & & & -I21 & I22 \\ & & & -I31 & -I32 & I33 \end{bmatrix}$$

where

$$\begin{aligned} M &= \int \rho dV \\ I11 &= \int \rho(x_2^2 + x_3^2)dV \\ I22 &= \int \rho(x_1^2 + x_3^2)dV \\ I33 &= \int \rho(x_1^2 + x_2^2)dV \\ I21 &= \int \rho(x_1 x_2)dV \\ I31 &= \int \rho(x_1 x_3)dV \\ I32 &= \int \rho(x_2 x_3)dV \end{aligned}$$

and

x_1, x_2, x_3 are components of distance from the center of gravity in the coordinate system defined in field 4.

The negative signs for the off-diagonal terms are supplied automatically. A warning message is issued if the inertia matrix is nonpositive definite, since this may cause fatal errors in dynamic analysis modules.

6. If CID ≥ 0, then X1, X2, and X3 are defined by a local Cartesian system, even if CID references a spherical or cylindrical coordinate system. This is similar to the manner in which displacement coordinate systems are defined.
7. See “Overview of Coordinate Systems in Simcenter Nastran” in the *Simcenter Nastran User’s Guide* for a definition of coordinate system terminology.

Remarks related to SOL 401:

1. X1, X2, and X3 are ignored, i.e., the center of gravity of the mass is assumed to be at the grid point.
2. CID is ignored, i.e., mass moments of inertia components Iii ($1 \le i \le 3$) are assumed to be in the displacement coordinate system of grid point G.

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3. I21, I31, and I32 are ignored. The displacement coordinate system for grid point G should coincide with the principal axes of the concentrated mass.

Remarks related to SOL 414:

1. The CID must be the same as the CID for the rotor. Otherwise, a fatal error occurs.
2. For SOL 414 solutions, if you set the mass parameters in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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CONROD

Rod Element Property and Connection

Defines a rod element without reference to a property entry.

Format:

1	2	3	4	5	6	7	8	9	10
CONROD	EID	G1	G2	MID	A	J	C	NSM	

Example:

CONROD	2	16	17	4	2.69				

Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
G1, G2	Grid point identification numbers of connection points. (Integer > 0; G1 ≠ G2)
MID	Material identification number. (Integer > 0)
A	Area of the rod. (Real)
J	Torsional constant. (Real)
C	Coefficient for torsional stress determination. (Real)
NSM	Nonstructural mass per unit length. See Remark 3 for SOL 414. (Real)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. For structural problems, MID must reference a MAT1 material entry.
3. For heat transfer problems, MID must reference a MAT4 or MAT5 material entry.

Remarks related to SOLs 402 and 414:

1. J and C are ignored.

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2. CONROD defines a truss element with no torsional stiffness, i.e., only axial force is transmitted by the element. Rotational degrees of freedom are introduced in order to connect with shells but they do not contribute to the element stiffness.
3. For SOL 414 solutions, if you set the mass parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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CONV

Heat Boundary Element Free Convection Entry

Specifies a free convection boundary condition for heat transfer analysis through connection to a surface element (CHBDYi entry).

Format:

1	2	3	4	5	6	7	8	9	10
CONV	EID	PCONID	FLMND	CNTRLND	TA1	TA2	TA3	TA4	
	TA5	TA6	TA7	TA8					

Example:

CONV	2	101	3	201	301				

Fields:

Field	Contents
EID	CHBDYG, CHBDYE, or CHBDYP surface element identification number. (Integer > 0)
PCONID	Convection property identification number of a PCONV entry. (Integer > 0)
FLMND	Point for film convection fluid property temperature. (Integer ≥ 0; Default = 0)
CNTRLND	Control point for free convection boundary condition. (Integer ≥ 0; Default = 0)
TAi	Ambient points used for convection. (Integer > 0 for TA1 and Integer ≥ 0 for TA2 through TA8; Default for TA2 through TA8 is TA1.)

Remarks:

1. The basic exchange relationship can be expressed in one of the following forms:
   - $q = H \cdot (T - \text{TAMB})^{\text{EXPF}}(T - \text{TAMB})$, CNTRLND = 0
   - $q = (H \cdot u_{\text{CNTRLND}})(T - \text{TAMB})^{\text{EXPF}}(T - \text{TAMB})$, CNTRLND ≠ 0
   - $q = H \cdot (T^{\text{EXPF}} - \text{TAMB}^{\text{EXPF}})$, CNTRLND = 0
   - $q = (H \cdot u_{\text{CNTRLND}})(T^{\text{EXPF}} - \text{TAMB}^{\text{EXPF}})$, CNTRLND ≠ 0

   EXPF is specified on the PCONV entry.

   (See **"PCONV"** for additional clarification of forms.)
2. The continuation entry is not required.

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3. CONV is used with an CHBDYi (CHBDYG, CHBDYE, or CHBDYP) entry having the same EID.
4. The temperature of the film convection point provides the look up temperature to determine the convection film coefficient. If FLMND=0, the reference temperature has several options. It can be the average of surface and ambient temperatures, the surface temperature, or the ambient temperature, as defined in the FORM field of the PCONV Bulk Data entry.
5. If only one ambient point is specified then all the ambient points are assumed to have the same temperature. If midside ambient points are missing, the temperature of these points is assumed to be the average of the connecting corner points.
6. See the Bulk Data entry, **"PCONV"** , for an explanation of the mathematical relationships involved in free convection and the reference temperature for convection film coefficient.

Related Topics

Browse Simcenter Nastran documentation

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CONVM

Heat Boundary Element Forced Convection Entry

Specifies a forced convection boundary condition for heat transfer analysis through connection to a surface element (CHBDYi entry).

Format:

1	2	3	4	5	6	7	8	9	10
CONVM	EID	PCONID	FLMND	CNTMDOT	TA1	TA2			

Example:

CONVM	101	1	201	301	20	21			

Fields:

Field	Contents
EID	CHBDYP element identification number. (Integer > 0)
PCONID	Convection property identification number of a PCONVM entry. (Integer > 0)
FLMND	Point used for fluid film temperature. (Integer ≥ 0; Default = 0)
CNTMDOT	Control point used for controlling mass flow. (Integer > 0)
TA1, TA2	Ambient points used for convection. (Integer > 0 for TA1 and Integer ≥ 0 for TA2; Default for TA2 is TA1.)

Remarks:

1. CONVM is used with an CHBDYP entry of type FTUBE having the same EID.
2. The temperature of the fluid film point may be specified to determine the material properties for the fluid. If FLMND = 0, the reference temperature has several options. It can be the average of surface and ambient temperatures, the surface temperatures, or the ambient temperature, as defined in the FORM field of the PCONVM Bulk Data entry.
3. CNTMDOT must be set to the desired mass flow rate (mdot) to effect the advection of energy downstream at an mdot $\cdot C_p \cdot T$ rate. In addition to the effect that mdot has on the transfer of thermal energy in the streamwise direction, this control point value is also used in computing the tube Reynolds number and subsequently the forced convection heat transfer coefficient if requested. This enables the fluid stream to exchange heat with its surroundings.
4. If only the first ambient point is specified, then the second ambient point is assumed to have the same temperature.

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5. See the Bulk Data entry, **"PCONVM"** , for an explanation of the mathematical relationships available for forced convection and the reference temperature for fluid material properties.

Related Topics

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CORD1C

Cylindrical Coordinate System Definition, Form 1

Defines a cylindrical coordinate system using three grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CORD1C	CIDA	G1A	G2A	G3A	CIDB	G1B	G2B	G3B	

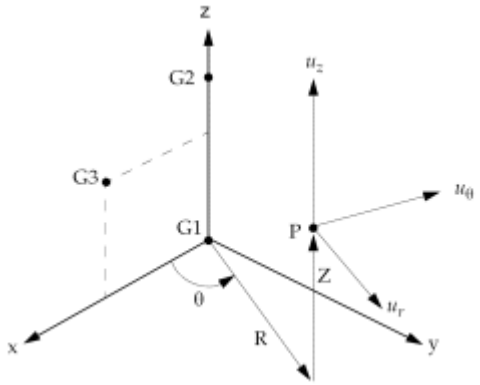
Example:

CORD1C	3	16	32	19					

Fields:

Field	Contents
CIDA, CIDB	Coordinate system identification number. (Integer > 0)
GiA, GiB	Grid point identification numbers. (Integer > 0; G1A ≠ G2A ≠ G3A; G1B ≠ G2B ≠ G3B;)

Figure 12-54: CORD1C Definition



Remarks:

1. Coordinate system identification numbers on all CORD1R, CORD1C, CORD1S, CORD2R, CORD2C, CORD2S and CORD3G entries must be unique.
2. One or two coordinate systems may be defined on a single entry.
3. GiA and GiB must be defined in coordinate systems with definitions that do not involve the coordinate system being defined. The first point is the origin, the second lies on the z-axis, and the third lies in

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the plane of the azimuthal origin. The three grid points GiA (or GiB) must be non-colinear and not coincident.

4. The location of a grid point (P in **Figure 12-54**) in this coordinate system is given by (R, θ, Z) where θ is measured in degrees.
5. The displacement coordinate directions at P are dependent on the location of P as shown above by (u_r, u_θ, u_z).
6. It is recommended that points on the z-axis not have their displacement directions defined in this coordinate system. See the discussion of cylindrical coordinate systems in "Overview of Coordinate Systems in Simcenter Nastran" in the *Simcenter Nastran User's Guide*.

Related Topics

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CORD1R

Rectangular Coordinate System Definition, Form 1

Defines a rectangular coordinate system using three grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CORD1R	CIDA	G1A	G2A	G3A	CIDB	G1B	G2B	G3B	

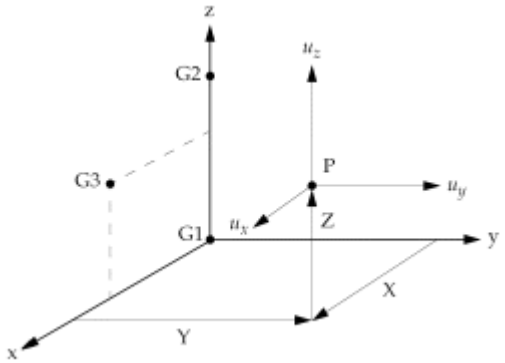
Example:

CORD1R	3	16	32	19					

Fields:

Field	Contents
CIDA, CIDB	Coordinate system identification number. (Integer > 0)
GiA, GiB	Grid point identification numbers. (Integer > 0; G1A ≠ G2A ≠ G3A and G1B ≠ G2B ≠ G3B)

Figure 12-55: CORD1R Definition



Remarks:

1. Coordinate system identification numbers on all CORD1R, CORD1C, CORD1S, CORD2R, CORD2C, CORD2S and CORD3G entries must be unique.
2. One or two coordinate systems may be defined on a single entry.

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3. GiA and GiB must be defined in coordinate systems with definitions that do not involve the coordinate system being defined. The first point is the origin, the second lies on the z-axis, and the third lies in the x-z plane. The three grid points GiA (or GiB) must be noncolinear and not coincident.
4. The location of a grid point (P in **Figure 12-55**) in this coordinate system is given by (X, Y, Z).
5. The displacement coordinate directions at P are shown above by (u_x, u_u, u_z).

Related Topics

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CORD1S

Spherical Coordinate System Definition, Form 1

Defines a spherical coordinate system by reference to three grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CORD1S	CIDA	G1A	G2A	G3A	CIDB	G1B	G2B	G3B	

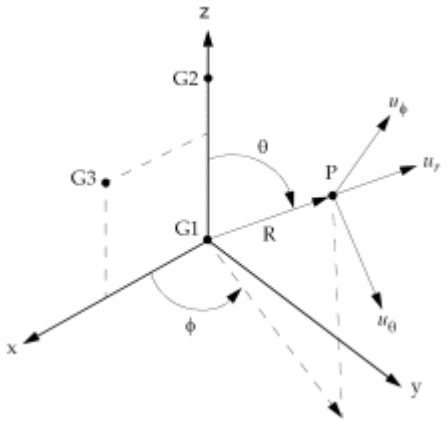
Example:

CORD1S	3	16	32	19					

Fields:

Field	Contents
CIDA, CIDB	Coordinate system identification numbers. (Integer > 0)
GiA, GiB	Grid point identification numbers. (Integer > 0; G1A ≠ G2A ≠ G3A and G1B ≠ G2B ≠ G3B)

Figure 12-56: CORD1S Definition



Remarks:

1. Coordinate system identification numbers on all CORD1R, CORD1C, CORD1S, CORD2R, CORD2C, CORD2S and CORD3G entries must be unique.
2. One or two coordinate systems may be defined on a single entry.

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3. GiA and GiB must be defined in coordinate systems with a definition that does not involve the coordinate system being defined. The first point is the origin, the second lies on the z-axis, and the third lies in the plane of the azimuthal origin. The three grid points GiA (or GiB) must be noncolinear and not coincident.
4. The location of a grid point (P in **Figure 12-56**) in this coordinate system is given by (R, θ, φ) where θ and φ are measured in degrees.
5. The displacement coordinate directions at P are dependent on the location of P as shown above by (u_r, u_θ, u_φ).
6. It is recommended that points on the z-axis not have their displacement directions defined in this coordinate system. See the discussion of spherical coordinate systems in "Overview of Coordinate Systems in Simcenter Nastran" in the *Simcenter Nastran User's Guide*.

Related Topics

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CORD2C

Cylindrical Coordinate System Definition, Form 2

Defines a cylindrical coordinate system using the coordinates of three points.

Format:

1	2	3	4	5	6	7	8	9	10
CORD2C	CID	RID	A1	A2	A3	B1	B2	B3	
	C1	C2	C3						

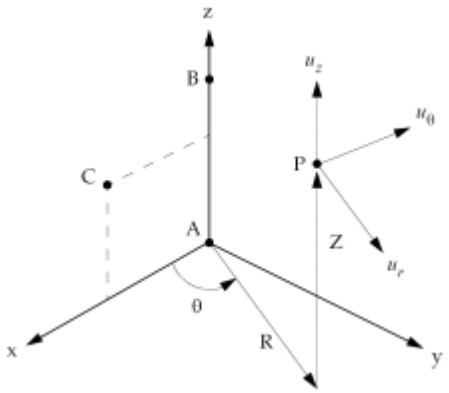
Example:

CORD2C	3	17	-2.9	1.0	0.0	3.6	0.0	1.0	
	5.2	1.0	-2.9						

Fields:

Field	Contents
CID	Coordinate system identification number. (Integer > 0)
RID	Identification number of a coordinate system that is defined independently from this coordinate system. (Integer ≥ 0; Default = 0 is the basic coordinate system.)
Ai, Bi, Ci	Coordinates of three points in coordinate system defined in field 3. (Real)

Figure 12-57: CORD2C Definition



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Remarks:

1. Coordinate system identification numbers on all CORD1R, CORD1C, CORD1S, CORD2R, CORD2C, CORD2S and CORD3G entries must be unique.
2. The three points [(A1, A2, A3), (B1, B2, B3), (C1, C2, C3)] must be unique and noncolinear. Non-colinearity is checked by the geometry processor. The first point defines the origin. The second point defines the direction of the z-axis. The third lies in the plane of the azimuthal origin. The reference coordinate system must be independently defined.
3. The continuation entry is required.
4. If RID is zero or blank, the basic coordinate system is used.
5. The location of a grid point (P in **Figure 12-57**) in this coordinate system is given by (R, θ, Z), where θ is measured in degrees.
6. The displacement coordinate directions at P are dependent on the location of P as shown above by u_r, u_θ, u_z).
7. It is recommended that points on the z-axis not have their displacement directions defined in this coordinate system. See the discussion of cylindrical coordinate systems in “Overview of Coordinate Systems in Simcenter Nastran” in the *Simcenter Nastran User’s Guide*.
8. If any CORD2C, CORD2R, or CORD2S entry is changed or added on restart, then a complete re-analysis is performed. Therefore, CORD2C, CORD2R, or CORD2S changes or additions are not recommended on restart.

Related Topics

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CORD2R

Rectangular Coordinate System Definition, Form 2

Defines a rectangular coordinate system using the coordinates of three points.

Format:

1	2	3	4	5	6	7	8	9	10
CORD2R	CID	RID	A1	A2	A3	B1	B2	B3	
	C1	C2	C3						

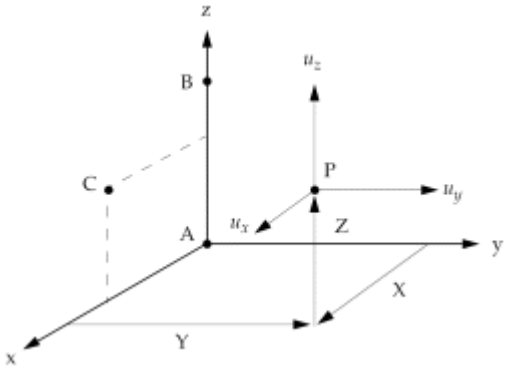
Example:

CORD2R	3	17	-2.9	1.0	0.0	3.6	0.0	1.0	
	5.2	1.0	-2.9						

Fields:

Field	Contents
CID	Coordinate system identification number. (Integer > 0)
RID	Identification number of a coordinate system that is defined independently from this coordinate system. (Integer ≥ 0; Default = 0, which is the basic coordinate system.)
Ai, Bi, Ci	Coordinates of three points in coordinate system defined in field 3. (Real)

Figure 12-58: CORD2R Definition



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Remarks:

1. Coordinate system identification numbers on all CORD1R, CORD1C, CORD1S, CORD2R, CORD2C, CORD2S, and CORD3G entries must be unique.
2. The three points [(A1, A2, A3), (B1, B2, B3), (C1, C2, C3)] must be unique and noncolinear. Noncolinearity is checked by the geometry processor. The first point defines the origin. The second defines the direction of the z-axis. The third point defines a vector which, with the z-axis, defines the x-z plane. The reference coordinate system must be independently defined.
3. The continuation entry is required.
4. If RID is zero or blank, the basic coordinate system is used.
5. The location of a grid point (P in the **Figure 12-58**) in this coordinate system is given by (X, Y, Z).
6. The displacement coordinate directions at P are shown by (u_x, u_y, u_z).
7. If any CORD2C, CORD2R, or CORD2S entry is changed or added on restart, then a complete re-analysis is performed. Therefore, CORD2C, CORD2R, or CORD2S changes or additions are not recommended on restart.

Related Topics

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CORD2S

Spherical Coordinate System Definition, Form 2

Defines a spherical coordinate system using the coordinates of three points.

Format:

1	2	3	4	5	6	7	8	9	10
CORD2S	CID	RID	A1	A2	A3	B1	B2	B3	
	C1	C2	C3						

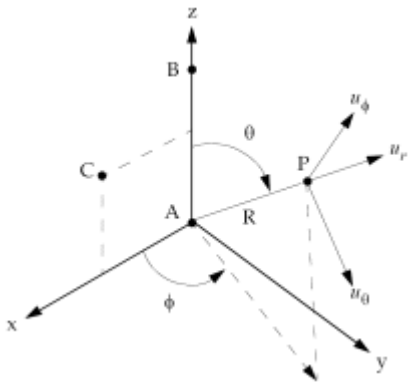
Example:

CORD2S	3	17	-2.9	1.0	0.0	3.6	0.0	1.0	
	5.2	1.0	-2.9						

Fields:

Field	Contents
CID	Coordinate system identification number. (Integer > 0)
RID	Identification number of a coordinate system that is defined independently from this coordinate system. (Integer ≥ 0; Default = 0 is the basic coordinate system.)
Ai, Bi, Ci	Coordinates of three points in coordinate system defined in field 3. (Real)

Figure 12-59: CORD2S Definition



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Remarks:

1. Coordinate system identification numbers on all CORD1R, CORD1C, CORD1S, CORD2R, CORD2C, CORD2S, and CORD3G entries must all be unique.
2. The three points [(A1, A2, A3), (B1, B2, B3), (C1, C2, C3)] must be unique and noncolinear. Noncolinearity is checked by the geometry processor. The first point defines the origin. The second point defines the direction of the z-axis. The third lies in the plane of the azimuthal origin. The reference coordinate system must be independently defined.
3. The continuation entry is required.
4. If RID is zero or blank, the basic coordinate system is used.
5. The location of a grid point (P in **Figure 12-59**) in this coordinate system is given by (R, θ, φ), where θ and φ are measured in degrees.
6. The displacement coordinate directions at P are shown above by (u_r, u_θ, u_φ).
7. It is recommended that points on the z-axis not have their displacement directions defined in this coordinate system. See the discussion of spherical coordinate systems in “Overview of Coordinate Systems in Simcenter Nastran” in the *Simcenter Nastran User’s Guide*.
8. If any CORD2C, CORD2R, or CORD2S entry is changed or added on restart, then a complete re-analysis is performed. Therefore, CORD2C, CORD2R, or CORD2S changes or additions are not recommended on restart.

Related Topics

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CORD3G

General Coordinate System

Defines a general coordinate system using three rotational angles as functions of coordinate values in the reference coordinate system. The CORD3G entry is used with the MAT9 entry to orient material principal axes for 3-D composite analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CORD3G	CID	METHOD		THETAID1	THETAID2	THETAID3	CIDREF		

Example:

CORD3G	100	E313		110	111	112	0		

Fields:

Field	Contents
CID	Coordinate system identification number. See Remark 1. (Integer > 0)
METHOD	E313 or S321 for Euler angle rotation in 313 sequence or space-fixed rotation in 321 sequence. See Remark 2. (Character; Default = "E313")
THETAID	Identification number for DEQATN Bulk Data entry which defines the three angles (in radians) measured from reference coordinates to the general material coordinate system. See Remark 3. (Integer > 0)
CIDREF	Identification number for the coordinate system from which the orientation of the general coordinate system is defined. (Integer > 0)

Remarks:

1. CID must be unique with respect to all other coordinate systems. CID cannot be referenced on GRID entries.
2. Three Euler angles specify the rotation of the CORD3G coordinate axes (xyz) with respect to the local Cartesian coordinate axes (XYZ) in CIDREF as follows: first rotate about Z-axis by q_1, next rotate about rotated x-axis by q_2, and then rotate about rotated z-axis by q_3. On the other hand, the space-fixed rotations in 321 sequence specify all the rotations about the fixed coordinate axes: first rotate about Z by q_1, next about Y by q_2, then about X by q_3.
3. The three rotations define a coordinate transformation which transforms position vectors in the reference coordinate system into the general coordinate system.

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Related Topics

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CPENTA

Five-Sided Solid Element Connection

Defines the connections of a five-sided solid element with six to fifteen grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CPENTA	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	G10	G11	G12	G13	G14	
	G15								

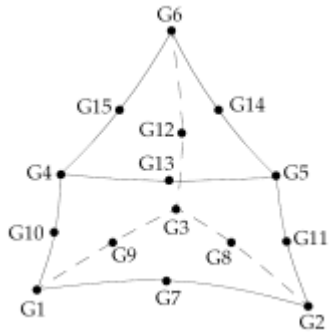
Example:

CPENTA	112	2	3	15	14	4	103	115	
	5	16	8				120	125	
	130								

Fields:

Field	Contents	Type	Default
EID	Element identification number.	Integer > 0	Required
PID	Property identification number of a PSOLID, PLSOLID, PCOMPS, or PMIC entry.	Integer > 0	Required
Gi	Identification numbers of connected grid points.	Integer ≥ 0 or blank	Required

Figure 12-60: CPENTA Element Connection



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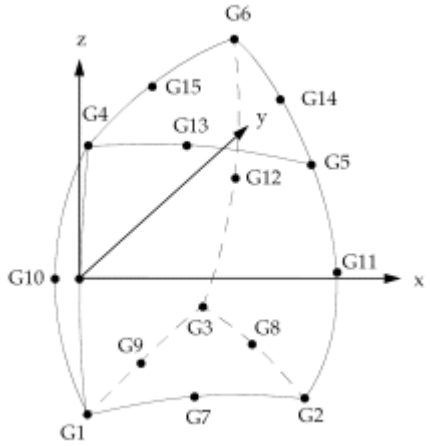
Remarks:

1. Element ID numbers must be unique with respect to all other element ID numbers.
2. The topology of the diagram must be preserved; i.e., G1, G2, and G3 define a triangular face, G1, G10, and G4 are on the same edge, etc.
3. The edge grid points, G7 to G15, are optional. Any or all of them may be deleted. In the example shown, G10, G11, and G12 have been deleted. The continuations are not required if all edge grid points are deleted.
4. Components of stress are output in the material coordinate system. See remarks on the **PSOLID** bulk entry for hyperelastic and nonlinear exceptions. See remarks on the **PCOMPS** bulk entry for composite laminate exception.
5. Except when used as a hyperelastic element or as a composite laminate solid element, the element coordinate system for the CPENTA element is defined as follows: The origin of the coordinate system is located at the midpoint of the straight line connecting the points G1 and G4. The Z axis points toward the triangle G4-G5-G6 and is oriented somewhere between the line joining the centroids of the triangular faces and a line perpendicular to the midplane. The midplane contains the midpoints of the straight lines between the triangular faces. The X and Y axes are perpendicular to the Z axis and point in a direction toward, but not necessarily intersecting, the edges G2 through G5 and G3 through G6, respectively.

Figure 12-61: CPENTA Element Coordinate System



6. It is recommended that the edge grid points be located within the middle third of the edge.
7. For hyperelastic elements, the plot codes are specified under the CPENTAFD element name in **"Item Codes"**.
8. By default, all of the nine edges of the element are considered straight unless any of G7 through G15 are specified.

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Remarks related to SOLs 402 and 414:

1. Output results will be done in the material coordinate system. If those axes are not defined, the material coordinate system is initially defined as the basic coordinate system. The material coordinate system always rotates with the element.

Related Topics

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CPENTCZ

Five-Sided Cohesive Element Connection

Defines the connections of a five-sided cohesive element with six to fifteen grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CPENTCZ	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	G10	G11	G12	G13	G14	
	G15								

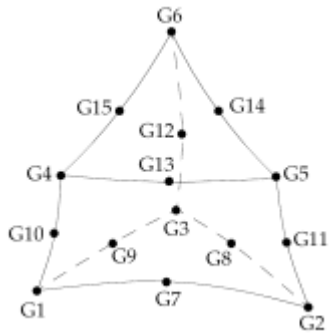
Example:

CPENTCZ	112	2	3	15	14	4	103	115	
	5	16	8	0			120	125	
	130								

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PSOLCZ entry. (Integer > 0; No default)
Gi	Identification numbers of connected grid points. (Integer ≥ 0 or blank; See Remark 3 for default behavior)

Figure 12-62: CPENTCZ Element Connection



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Remarks:

1. Element identification numbers must be unique with respect to all other element identification numbers.
2. Grid points G1 through G3 must define one triangular face. This is the bottom face. G4 through G6 must be on the opposite face with G4 opposite G1, G5 opposite G2, and G6 opposite G3. This is the top face. The normal direction is defined relative to the bottom surface.
3. The edge points G7 to G15 are optional. Any or all of them may be omitted. If the ID of any edge connection point is left blank or set to zero (G10, G11, and G12 in the input example), the equations of the element are adjusted to give correct results for the reduced number of connections. Corner grid points cannot be omitted.
4. If all edge grid points are omitted, the first and second continuation lines are not required. If G15 is the only edge grid point that is omitted, the second continuation line is not required.
5. For best results, locate edge grid points within the middle third of the edge.
6. For information regarding the material coordinate system, see **PSOLCZ**.

Remarks related to SOL 402:

1. Grid points G10 through G12 are dummy nodes: they will get the average displacement values of adjacent nodes.

Related Topics

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CPLSTN3

Plane Strain Triangular Element Connection

Defines a plane strain triangular element for use in linear or nonlinear analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CPLSTN3	EID	PID	G1	G2	G3	THETA			

Example:

CPLSTN3	111	203	31	74	75	30.0			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PPLANE, PLPLANE, or PGPLSN entry. (Integer > 0; Default = EID)
Gi	Grid point identification numbers of connection points. (Unique integers > 0)
THETA	Material property orientation angle in degrees. See Remark 8. THETA is ignored for hyperelastic elements. (Real; Default = 0.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - A plane parallel to the XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.
   - A plane parallel to the XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. The reference coordinate system for the output of linear stress/strain is the material coordinate system.

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4. The element behaves linearly if used in a SOL 106 or SOL 129 analysis.
5. The cyclic solution types (SOLs 114,115,116,118) and the aero solution types (SOLs 144,145,146) are not supported.
6. GPSTRESS or GPSTRAIN output is not supported.
7. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
8. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures Figure 12-63 and Figure 12-64 show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.

Figure 12-63: CPLSTN3 Element Geometry and Coordinate Systems, XZ-Plane

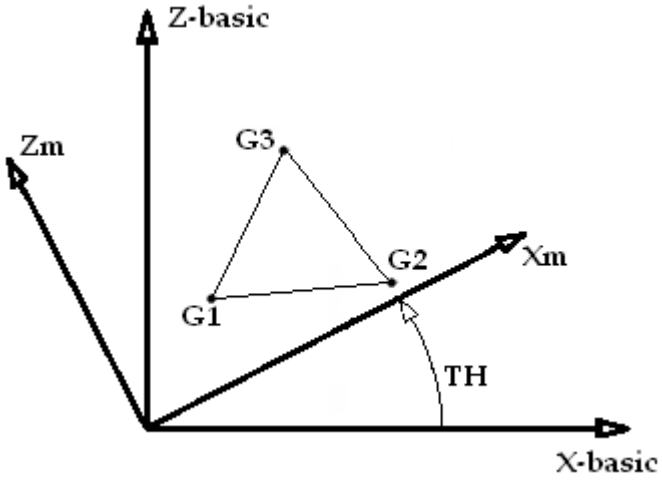

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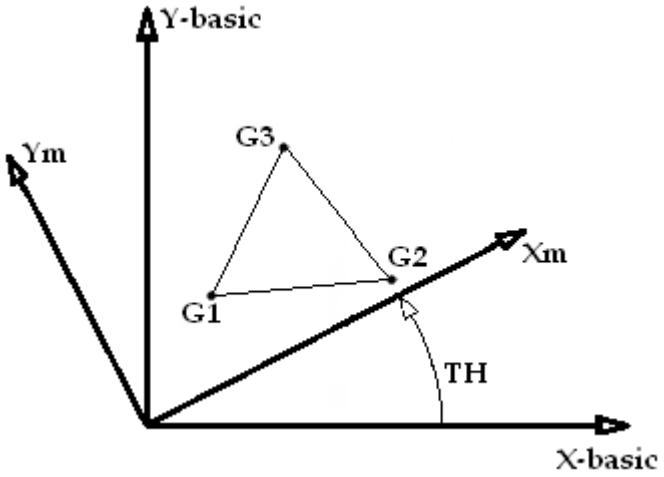
Figure 12-64: CPLSTN3 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The SPCFORCES output results are always given in force unit using a unitary thickness.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

Related Topics

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CPLSTN4

Plane Strain Quadrilateral Element Connection

Defines a plane strain quadrilateral element for use in linear or nonlinear analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CPLSTN4	EID	PID	G1	G2	G3	G4	THETA		

Example:

CPLSTN4	111	203	31	74	75	32	15.0		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PPLANE, PLPLANE, or PGPLSN entry. (Integer > 0; Default = EID)
Gi	Grid point identification numbers of connection points. (Unique integers > 0)
THETA	Material property orientation angle in degrees. See Remark 11. THETA is ignored for hyperelastic elements. (Real; Default = 0.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - A plane parallel to the XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.
   - A plane parallel to the XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. Grid points G1 through G4 must be ordered consecutively around the perimeter of the element.
4. All interior angles must be less than 180°.

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5. The element coordinate system is the basic coordinate system.
6. The reference coordinate system for the output of linear stress/strain is the material coordinate system.
7. The element behaves linearly if used in a SOL 106 or SOL 129 analysis.
8. The cyclic solution types (SOLs 114,115,116,118) and the aero solution types (SOLs 144,145,146) are not supported.
9. GPSTRESS or GPSTRAIN output is not supported.
10. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
    - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
    - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
11. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures **Figure 12-65** and **Figure 12-66** show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.

Figure 12-65: CPLSTN4 Element Geometry and Coordinate Systems, XZ-Plane

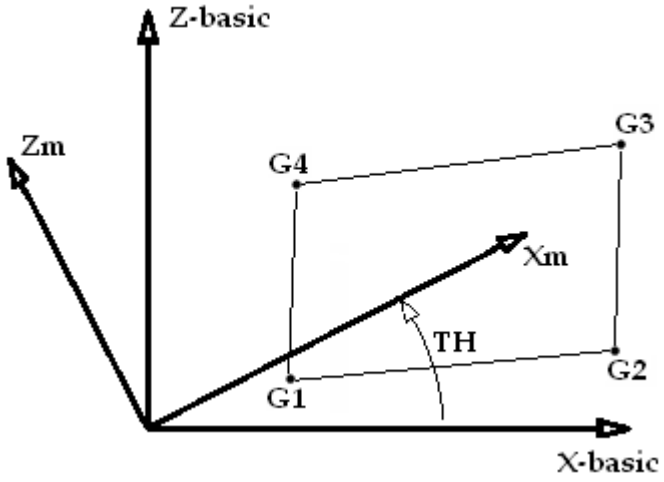

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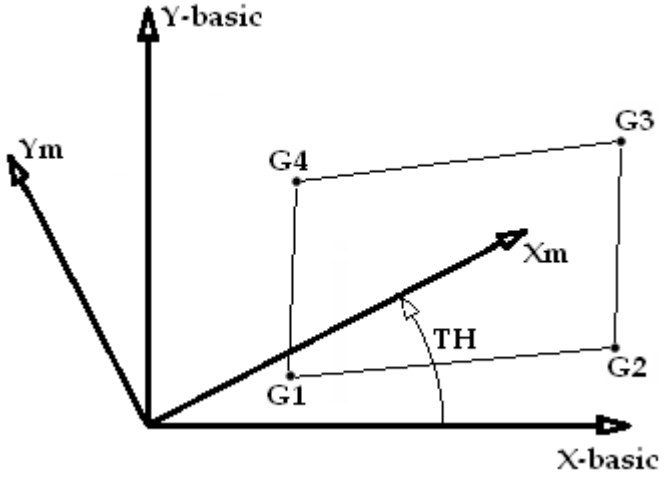
Figure 12-66: CPLSTN4 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The SPCFORCES output results are always given in force unit using a unitary thickness.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

Related Topics

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CPLSTN6

Plane Strain Triangular Element Connection

Defines a plane strain triangular element for use in linear or nonlinear analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CPLSTN6	EID	PID	G1	G2	G3	G4	G5	G6	
	THETA								

Example:

CPLSTN6	302	3	31	33	71	32	51	52	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PPLANE, PLPLANE, or PGPLSN entry. (Integer > 0; Default = EID)
Gi	Grid point identification number of connected points. G1, G2, G3 are the corner grid points and G2, G4, G6 are the midside grid points. (Unique integers > 0)
THETA	Material property orientation angle in degrees. See Remark 10. THETA is ignored for hyperelastic elements. (Real; Default = 0.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - A plane parallel to the XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.
   - A plane parallel to the XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

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3. Grid points G1 through G6 must be numbered as shown in CTRIA6.
4. The element coordinate system is the basic coordinate system.
5. The reference coordinate system for the output of linear stress/strain is the material coordinate system.
6. The element behaves linearly if used in a SOL 106 or SOL 129 analysis.
7. The cyclic solution types (SOLs 114,115,116,118) and the aero solution types (SOLs 144,145,146) are not supported.
8. GPSTRESS or GPSTRAIN output is not supported.
9. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
10. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures Figure 12-67 and Figure 12-68 show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.

Figure 12-67: CPLSTN6 Element Geometry and Coordinate Systems, XZ-Plane

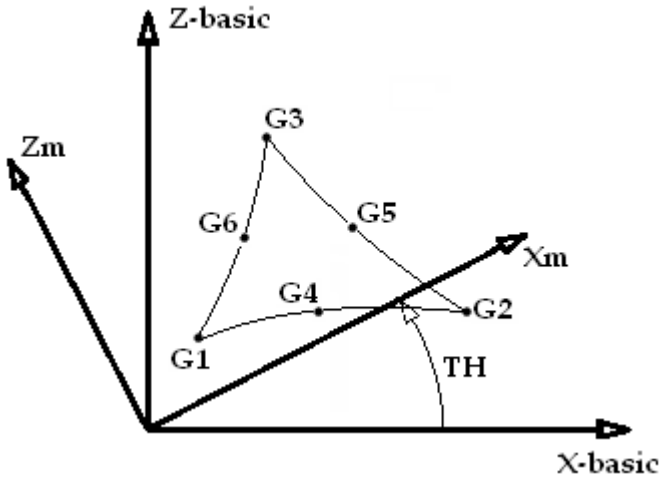

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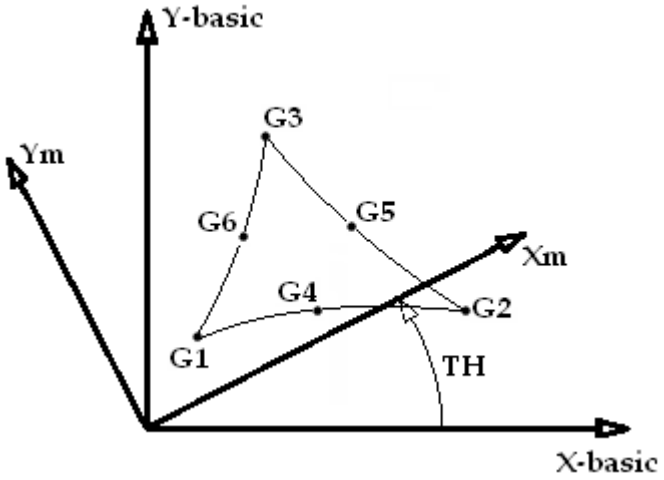
Figure 12-68: CPLSTN6 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The SPCFORCES output results are always given in force unit using a unitary thickness.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

Related Topics

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CPLSTN8

Plane Strain Quadrilateral Element Connection

Defines a plane strain quadrilateral element for use in linear or nonlinear analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CPLSTN8	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	THETA						

Example:

CPLSTN8	207	3	31	33	73	71	32	51	
	53	72	45.0						

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PPLANE, PLPLANE, or PGPLSN entry. (Integer > 0; Default = EID)
Gi	Grid point identification number of connected points. G1 to G4 are the corner grid points and G5 to G8 are the midside grid points. (Unique integers > 0)
THETA	Material property orientation angle in degrees. See Remark 10. THETA is ignored for hyperelastic elements. (Real; Default = 0.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - A plane parallel to the XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.
   - A plane parallel to the XY-plane of the basic coordinate system.

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The software automatically determines the orientation.

3. Grid points G1 through G8 must be numbered as shown in CQUAD8.
4. The element coordinate system is the basic coordinate system.
5. The reference coordinate system for the output of linear stress/strain is the material coordinate system.
6. The element behaves linearly if used in a SOL 106 or SOL 129 analysis.
7. The cyclic solution types (SOLs 114,115,116,118) and the aero solution types (SOLs 144,145,146) are not supported.
8. GPSTRESS or GPSTRAIN output is not supported.
9. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
10. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures Figure 12-69 and Figure 12-70 show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.

Figure 12-69: CPLSTN8 Element Geometry and Coordinate Systems, XZ-Plane

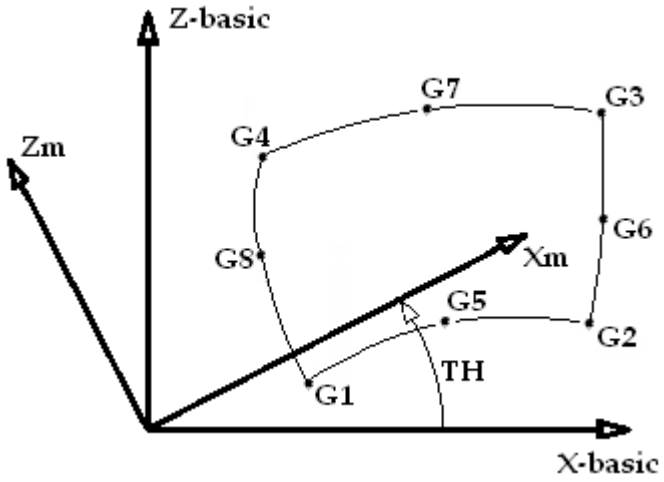

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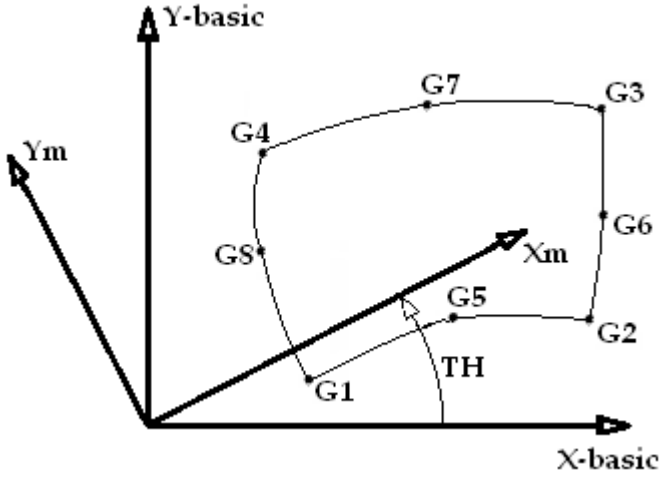
Figure 12-70: CPLSTN8 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The SPCFORCES output results are always given in force unit using a unitary thickness.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

Related Topics

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CPLSTS3

Plane Stress Triangular Element Connection

Defines a plane stress triangular element for use in linear or nonlinear analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CPLSTS3	EID	PID	G1	G2	G3		THETA		
				TFLAG	T1	T2	T3		

Example:

CPLSTS3	111	203	31	74	75		30.0		
				1	1.2	1.2	1.2		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PPLANE or PLPLANE entry. (Integer > 0; Default = EID)
Gi	Grid point identification numbers of connection points. (Unique integers > 0)
THETA	Material property orientation angle in degrees. See Remark 10. THETA is ignored for hyperelastic elements. (Real; Default = 0.0)
TFLAG	Flag that specifies how Ti is used to define element thickness. (Integer 0 or 1; Default = 0)
T1,T2,T3	Thickness or thickness multiplier at corner grid points. If TFLAG = 0, Ti is the thickness at the corresponding corner grid point. (Real ≥ 0.0; Default = T on the PPLANE or PLPLANE entry if T > 0.0, otherwise default = 0.0) If TFLAG = 1, Ti is a multiplier of T on the PPLANE or PLPLANE entry at the corresponding corner grid point. (Real ≥ 0.0; Default = 1.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.

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2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - A plane parallel to the XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.
   - A plane parallel to the XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. A fatal error occurs if T1 = T2 = T3 = 0.0.
4. The element coordinate system is the basic coordinate system.
5. The reference coordinate system for the output of linear stress/strain is the material coordinate system.
6. The element behaves linearly if used in a SOL 106 or SOL 129 analysis.
7. The cyclic solution types (SOLs 114,115,116,118) and the aero solution types (SOLs 144,145,146) are not supported.
8. GPSTRESS or GPSTRAIN output is not supported.
9. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
10. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures **Figure 12-71** and **Figure 12-72** show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.

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Figure 12-71: CPLSTS3 Element Geometry and Coordinate Systems, XZ-Plane

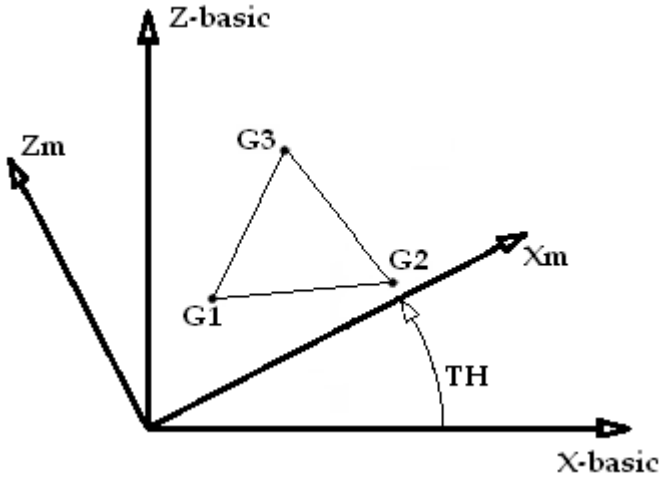
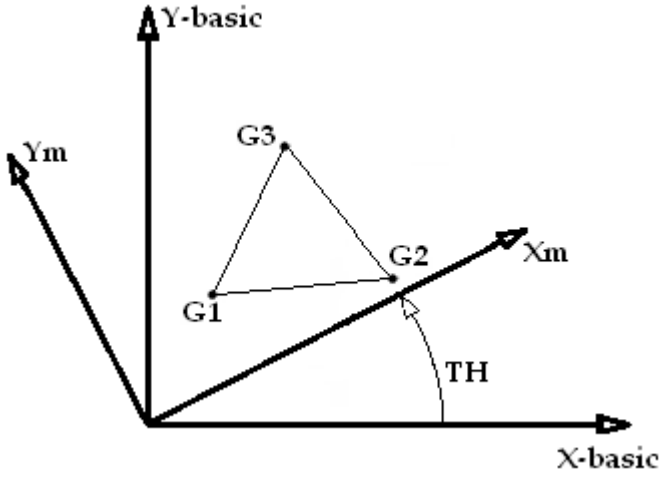


Figure 12-72: CPLSTS3 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The grid points must be defined in the XY-plane of the global coordinate system.
2. The SPCFORCES output results are always in force unit using the thickness(es) of the element or the thickness defined in the PPLANE bulk data card.
3. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:

   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

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Related Topics

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CPLSTS4

Plane Stress Quadrilateral Element Connection

Defines a plane stress quadrilateral element for use in linear or nonlinear analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CPLSTS4	EID	PID	G1	G2	G3	G4	THETA		
				TFLAG	T1	T2	T3	T4	

Example:

CPLSTS4	111	203	31	74	75	32	30.0		
					0.8	0.8	0.8	0.8	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PPLANE or PLPLANE entry. (Integer > 0; Default = EID)
Gi	Grid point identification numbers of connection points. (Unique integers > 0; No default)
THETA	Material property orientation angle in degrees. See Remark 10. THETA is ignored for hyperelastic elements. (Real; Default = 0.0)
TFLAG	Flag that specifies how Ti is used to define element thickness. (Integer 0 or 1; Default = 0)
T1,T2,T3,T4	Thickness or thickness multiplier at corner grid points. If TFLAG = 0, Ti is the thickness at the corresponding corner grid point. (Real ≥ 0.0; Default = T on the PPLANE or PLPLANE entry if T > 0.0, otherwise default = 0.0) If TFLAG = 1, Ti is a multiplier of T on the PPLANE or PLPLANE entry at the corresponding corner grid point. (Real ≥ 0.0; Default = 1.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.

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2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - A plane parallel to the XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.
   - A plane parallel to the XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. A fatal error occurs if T1 = T2 = T3 = T4 = 0.0.
4. The element coordinate system is the basic coordinate system.
5. The reference coordinate system for the output of linear stress/strain is the material coordinate system.
6. The element behaves linearly if used in a SOL 106 or SOL 129 analysis.
7. The cyclic solution types (SOLs 114,115,116,118) and the aero solution types (SOLs 144,145,146) are not supported.
8. GPSTRESS or GPSTRAIN output is not supported.
9. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
10. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures **Figure 12-73** and **Figure 12-74** show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.

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Figure 12-73: CPLSTS4 Element Geometry and Coordinate Systems, XZ-Plane

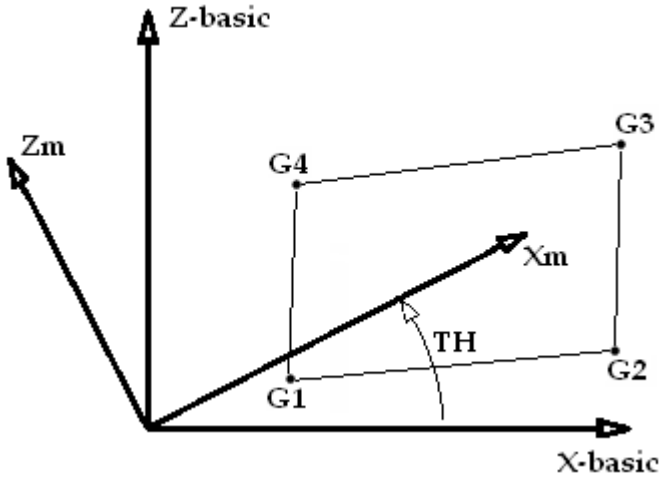
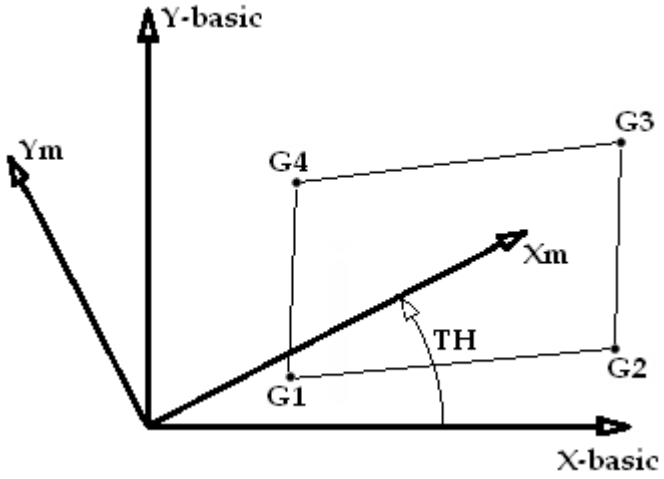


Figure 12-74: CPLSTS4 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The grid points must be defined in the XY-plane of the global coordinate system.
2. The SPCFORCES output results are always in force unit using the thickness(es) of the element or the thickness defined in the PPLANE bulk data card.
3. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:

   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

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CPLSTS6

Plane Stress Triangular Element Connection

Defines a plane stress triangular element for use in linear or nonlinear analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CPLSTS6	EID	PID	G1	G2	G3	G4	G5	G6	
			THETA	TFLAG	T1	T2	T3		
					TM4	TM5	TM6		

Example:

CPLSTS6	302	3	31	33	71	32	51	52	
			25.0	1	0.1	0.2	0.3		
					0.15	0.25	0.2		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PPLANE or PLPLANE entry. (Integer > 0; Default = EID)
Gi	Grid point identification number of connected points. (Unique integers > 0; No default)
THETA	Material property orientation angle in degrees. See Remark 13. THETA is ignored for hyperelastic elements. (Real; Default = 0.0)
TFLAG	Flag that specifies whether Ti and TMi are thicknesses that override the thickness specified on the PPLANE or PLPLANE entry (TFLAG = 0) or are factors that scale the thickness specified on the PPLANE or PLPLANE entry (TFLAG = 1). (Integer 0 or 1; Default = 0)
T1,T2,T3	Thickness override or thickness scaling factor at corner grid points. See Remark 3. If TFLAG = 0, Ti overrides the thickness specified on the PPLANE or PLPLANE entry at the corresponding corner grid point. (Real ≥ 0.0; Default = thickness specified on the PPLANE or PLPLANE entry; if thickness is not specified on the PPLANE or PLPLANE entry and the Ti field is blank, a fatal error occurs) If TFLAG = 1, Ti scales the thickness specified on the PPLANE or PLPLANE entry at the corresponding corner grid point. See Remark 4. (Real ≥ 0.0; Default = 1.0; if thickness

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Field	Contents
	is not specified on the PPLANE or PLPLANE entry and the Ti field is blank, a fatal error occurs)
TM4,TM5, TM6	Thickness override or thickness scaling factor at mid-side grid points. See Remarks 3 and 5.
	If TFLAG = 0, TMi overrides the thickness specified on the PPLANE or PLPLANE entry at the corresponding mid-side grid point. (Real ≥ 0.0; Default = thickness linearly interpolated from the thicknesses at the corner grid points of the same element edge)
	If TFLAG = 1, TMi scales the thickness specified on the PPLANE or PLPLANE entry at the corresponding mid-side grid point. See Remark 4. (Real ≥ 0.0; Default = thickness linearly interpolated from the thicknesses at the corner grid points of the same element edge)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - A plane parallel to the XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.
   - A plane parallel to the XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. At least one grid point must have a non-zero thickness. Otherwise, a fatal error occurs.
4. If TFLAG = 1, the thicknesses that the software uses at the grid points are calculated as follows:

$$t_1 = \mathrm{T1} \times \mathrm{T}\, t_2 = \mathrm{T2} \times \mathrm{T}\, t_3 = \mathrm{T3} \times \mathrm{T}\, t_4 = \mathrm{TM4} \times \mathrm{T}$$
$$t_5 = \mathrm{TM5} \times \mathrm{T}\, t_6 = \mathrm{TM6} \times \mathrm{T}$$

   where:

t_1, t_2, t_3, t_4, t_5, t_6	Thicknesses at G1, G2, G3, G4, G5, and G6, respectively.
T1, T2, T3, TM4, TM5, TM6	Values in the T1, T2, T3, TM4, TM5, and TM6 fields, respectively.
T	Thickness specified on the PPLANE or PLPLANE entry.

5. If a TMi field is specified, along the corresponding element edge, the software uses the TMi value and the thicknesses at the corner grid points on the same element edge to create a quadratic representation of the thickness along the element edge.
6. Grid points G1 through G6 must be ordered as shown in Figures Figure 12-75 and Figure 12-76.

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7. The element coordinate system is the basic coordinate system.
8. The reference coordinate system for the output of linear stress/strain is the material coordinate system.
9. The element behaves linearly if used in a SOL 106 or SOL 129 analysis.
10. The cyclic solution types (SOLs 114,115,116,118) and the aeroelastic solution types (SOLs 144,145,146) are not supported.
11. GPSTRESS and GPSTRAIN output are not supported.
12. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
    - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
    - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
13. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures **Figure 12-75** and **Figure 12-76** show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.

Figure 12-75: CPLSTS6 Element Geometry and Coordinate Systems, XZ-Plane

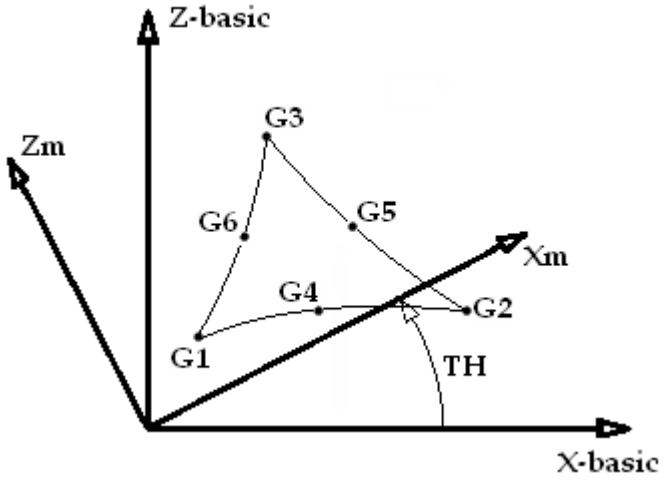

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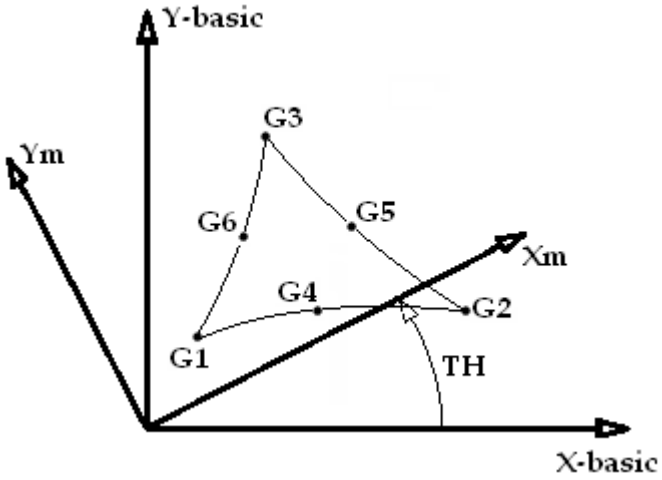
Figure 12-76: CPLSTS6 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The grid points must be defined in the XY-plane of the global coordinate system.
2. The SPCFORCES output results are always in force unit using the thickness(es) of the element or the thickness defined in the PPLANE bulk entry.
3. The TM4, TM5, and TM6 fields are ignored. Thus, the thickness along the element edges can be constant or can vary linearly only.
4. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

Related Topics

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CPLSTS8

Plane Stress Quadrilateral Element Connection

Defines a plane stress quadrilateral element for use in linear or nonlinear analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CPLSTS8	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	THETA	TFLAG	T1	T2	T3	T4	
					TM5	TM6	TM7	TM8	

Example:

CPLSTS8	207	3	31	33	73	71	32	51	
	53	72	70.0	1	0.5	0.5	0.5	0.5	
					0.5	0.75	0.75	0.6	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PPLANE or PLPLANE entry. (Integer > 0; Default = EID)
Gi	Grid point identification number of connected points. (Unique integers > 0; No default)
THETA	Material property orientation angle in degrees. See Remark 13. THETA is ignored for hyperelastic elements. (Real; Default = 0.0)
TFLAG	Flag that specifies whether Ti and TMi are thicknesses that override the thickness specified on the PPLANE or PLPLANE entry (TFLAG = 0) or are factors that scale the thickness specified on the PPLANE or PLPLANE entry (TFLAG = 1). (Integer 0 or 1; Default = 0)
T1,T2,T3,T4	Thickness override or thickness scaling factor at corner grid points. See Remark 3. If TFLAG = 0, Ti overrides the thickness specified on the PPLANE or PLPLANE entry at the corresponding corner grid point. (Real ≥ 0.0; Default = thickness specified on the PPLANE or PLPLANE entry; if thickness is not specified on the PPLANE or PLPLANE entry and the Ti field is blank, a fatal error occurs) If TFLAG = 1, Ti scales the thickness specified on the PPLANE or PLPLANE entry at the corresponding corner grid point. See Remark 4. (Real ≥ 0.0; Default = 1.0; if thickness

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Field	Contents
	is not specified on the PPLANE or PLPLANE entry and the Ti field is blank, a fatal error occurs)
TM5,TM6, TM7,TM8	Thickness override or thickness scaling factor at mid-side grid points. See Remarks 3 and 5.
	If TFLAG = 0, TMi overrides the thickness specified on the PPLANE or PLPLANE entry at the corresponding mid-side grid point. (Real ≥ 0.0; Default = thickness linearly interpolated from the thicknesses at the corner grid points of the same element edge)
	If TFLAG = 1, TMi scales the thickness specified on the PPLANE or PLPLANE entry at the corresponding mid-side grid point. See Remark 4. (Real ≥ 0.0; Default = thickness linearly interpolated from the thicknesses at the corner grid points of the same element edge)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - A plane parallel to the XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.
   - A plane parallel to the XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. At least one grid point must have a non-zero thickness. Otherwise, a fatal error occurs.
4. If TFLAG = 1, the thicknesses that the software uses at the grid points are calculated as follows:

$$t_1 = \mathrm{T1} \times \mathrm{T}\, t_2 = \mathrm{T2} \times \mathrm{T}\, t_3 = \mathrm{T3} \times \mathrm{T}\, t_4 = \mathrm{T4} \times \mathrm{T}$$
$$t_5 = \mathrm{TM5} \times \mathrm{T}\, t_6 = \mathrm{TM6} \times \mathrm{T}\, t_7 = \mathrm{TM7} \times \mathrm{T}\, t_8 = \mathrm{TM8} \times \mathrm{T}$$

   where:

t_1, t_2, t_3, t_4, t_5, t_6, t_7, t_8	Thicknesses at G1, G2, G3, G4, G5, G6, G7, and G8, respectively.
T1, T2, T3, T4, TM5, TM6, TM7, TM8	Values in the T1, T2, T3, T4, TM5, TM6, TM7, TM8 fields, respectively.
T	Thickness specified on the PPLANE or PLPLANE entry.

5. If a TMi field is specified, along the corresponding element edge, the software uses the TMi value and the thicknesses at the corner grid points on the same element edge to create a quadratic representation of the thickness along the element edge.

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6. Grid points G1 through G8 must be ordered as shown in Figures Figure 12-77 and Figure 12-78.
7. The element coordinate system is the basic coordinate system.
8. The reference coordinate system for the output of linear stress/strain is the material coordinate system.
9. The element behaves linearly if used in a SOL 106 or SOL 129 analysis.
10. The cyclic solution types (SOLs 114,115,116,118) and the aeroelastic solution types (SOLs 144,145,146) are not supported.
11. GPSTRESS and GPSTRAIN output are not supported.
12. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
    - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
    - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
13. For orthotropic materials, the THETA angle orients the principal material coordinates of the element relative to the basic coordinates. Figures Figure 12-77 and Figure 12-78 show the positive sense of the THETA angle when the element is defined on the XZ-plane and XY-plane, respectively.

Figure 12-77: CPLSTS8 Element Geometry and Coordinate Systems, XZ-Plane

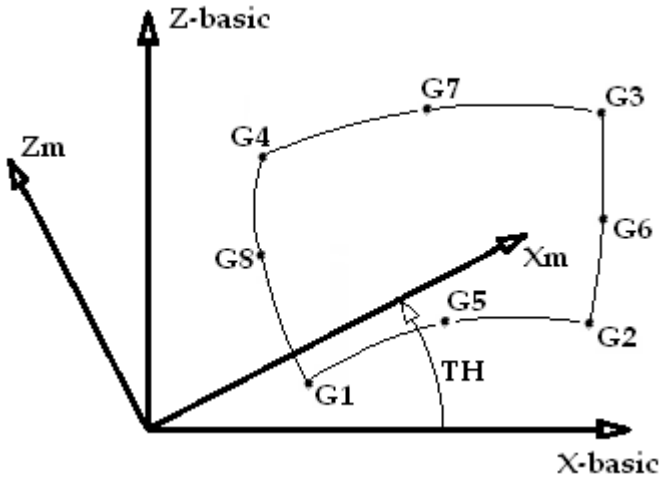

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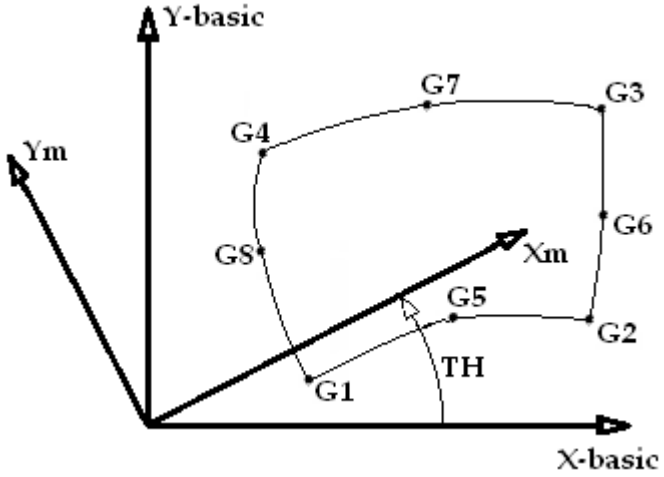
Figure 12-78: CPLSTS8 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The grid points must be defined in the XY-plane of the global coordinate system.
2. The SPCFORCES output results are always in force unit using the thickness(es) of the element or the thickness defined in the PPLANE bulk entry.
3. The TM5, TM6, TM7, and TM8 fields are ignored. Thus, the thickness along the element edges can be constant or can vary linearly only.
4. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.

Related Topics

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CPYRAM

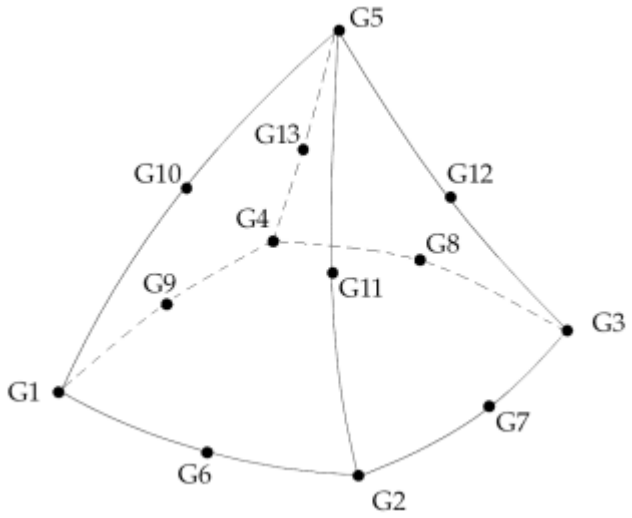
Five-Sided Solid Element Connection

Defines the connection of the five-sided solid element with five to thirteen grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CPYRAM	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	G10	G11	G12	G13		

Figure 12-79: CPYRAM Element Connection



Example:

CPYRAM	111	203	31	32	33	34	35	36	
	37	38	39	40	41	42	43		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PSOLID, PLSOLID, or PMIC entry. (Integer > 0; Default=EID)
Gi	Grid point identification numbers of connected points. (Unique integers > 0)

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Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. Grid points G1 through G4 must be given in consecutive order about one quadrilateral face. The other four faces are triangles. G5 is the vertex and must be opposite with the quadrilateral face.
3. The edge points G6 to G13 are optional. Any or all of them may be deleted. If the ID of any edge connection point is left blank or set to zero, the equations of the element are adjusted to give correct results for the reduced number of connections. Corner grid points cannot be deleted. The element is an isoparametric element in all cases.
4. Components of stress are output in the material coordinate system.
5. The CPYRAM element coordinate system is the same as the basic coordinate system.
6. It is recommended that the edge grids be located within the middle third of the edge.
7. The CPYRAM element does not support hyperelasticity in SOL 106 and 129.
8. By default, all eight edges of the element are considered straight unless any of G6 through G13 is specified.

Remarks related to SOLs 402 and 414:

1. Output results will be done in the material coordinate system. If those axes are not defined, the material coordinate system is initially defined as the basic coordinate system. The material coordinate system always rotates with the element.

Related Topics

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CQUAD

Fully Nonlinear Plane Strain Element

Defines a plane strain quadrilateral element with up to nine grid points for use in fully nonlinear (i.e., large strain and large rotation) hyperelastic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CQUAD	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9						

Example:

CQUAD	111	203	31	74	75	32			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PLPLANE entry. (Integer > 0)
G1, G2, G3, G4	Identification numbers of connected corner grid points. Required data for all four grid points. (Unique Integers > 0)
G5, G6, G7, G8	Identification numbers of connected edge grid points. Optional data for any or all four grid points. (Integer ≥ 0 or blank)
G9	Identification number of center grid point. Optional. (Integer ≥ 0 or blank)

Remarks:

1. Element identification numbers should be unique with respect to all other element IDs of any kind.
2. Grid points G1 to G9 must be numbered as shown and must lie on a plane.
3. It is recommended that the edge points be located within the middle third of the edge.
4. Plot codes are specified under the CQUADFD element name in **"Item Codes"** .
5. Stresses and strains are output in the coordinate system identified by the CID field of the PLPLANE entry.

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Remarks related to SOL 402:

1. The CPLSTN* elements are recommended instead of the CQUAD entry.

Related Topics

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CQUAD4

Quadrilateral Plate Element Connection

Defines an isoparametric membrane-bending or plane strain quadrilateral plate element.

Format:

1	2	3	4	5	6	7	8	9	10
CQUAD4	EID	PID	G1	G2	G3	G4	THETA or MCID	ZOFFS	
		TFLAG	T1	T2	T3	T4			

Example:

CQUAD4	111	203	31	74	75	32	2.6	0.3	
		1	1.77	2.04	2.09	1.80			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PCOMP, PCOMPG, PLPLANE, PMIC, PSHELL or PSHLPNL entry. (Integer > 0; Default = EID)
Gi	Grid point identification numbers of connection points. (Integers > 0, all unique.)
THETA	Material property orientation angle in degrees. THETA is ignored for hyperelastic elements. See Figure 12-81. (Real; Default = 0.0)
MCID	Material coordinate system identification number. The x-axis of the material coordinate system is determined by projecting the x-axis of the MCID coordinate system (defined by the CORDij entry or zero for the basic coordinate system) onto the surface of the element. MCID is ignored for hyperelastic elements. See Remark 12. (Integer ≥ 0; If blank, then THETA = 0.0 is assumed.)

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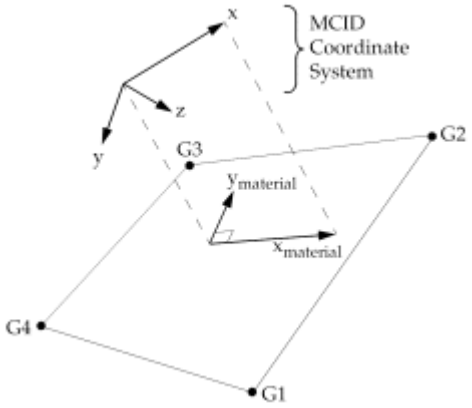
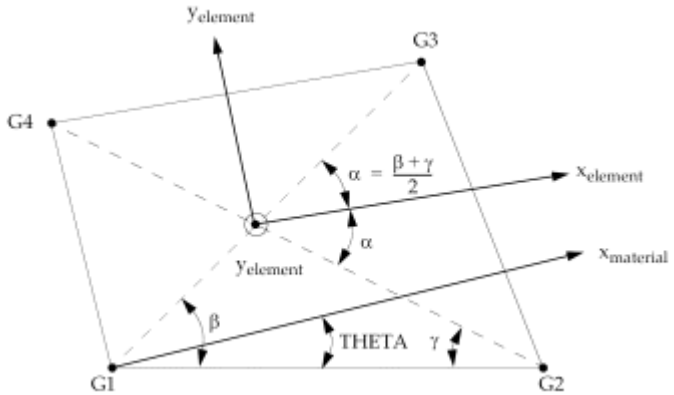
Field	Contents

Figure 12-80: MCID Coordinate System Definition



ZOFFS	Offset from the surface of grid points to the element reference plane. ZOFFS is ignored for hyperelastic elements. See Remarks **8** and **9**. (Real)
TFLAG	Integer Flag which specifies how Ti is used to define thickness of element. (0, 1, or blank)
Ti	Thickness of element at grid points G1 through G4 if TFLAG=0 or blank. If TFLAG=1, thickness becomes a product of Ti and the thickness on the PSHELL or PSHLPNL entry. Ti is ignored for hyperelastic elements. See Remark **6**. (Real > 0.0 or blank. See Remark **4** for the default.)

Figure 12-81: CQUAD4 Element Geometry and Coordinate Systems



Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. Grid points G1 through G4 must be ordered consecutively around the perimeter of the element.

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3. All interior angles must be less than 180°.
4. The continuation is optional. If it is not supplied, then T1 through T4 will be set equal to the value of T on the PSHELL entry.
5. When the PID refers to either a PCOMP or PCOMPG entry, the software converts the PCOMP/PCOMPG entry to equivalent PSHELL/MAT2 entries. If Ti is defined, they override or scale the values on the equivalent PSHELL entry. It should be noted that ply-stress and ply-strain recovery uses the ply thicknesses defined on the PCOMP/PCOMPG entries, and not Ti.
6. By default, a fatal error occurs if a Ti value is zero. To allow for Ti values of zero, use System Cell 495.
7. The reference coordinate system for the output of stress, strain and element force depends on the element type.
   - For CQUAD4 elements which are not hyperelastic, the reference coordinate system for output is the element coordinate system. For SOL 106 geometric nonlinear analysis, the nonlinear stresses are output in deformed element coordinate system.
   - For hyperelastic elements, the stress and strain are output according to CID on the PLPLANE entry.
8. Elements may be offset from the connection points by means of ZOFFS. Other data, such as material matrices and stress fiber locations, are given relative to the reference plane. A positive value of ZOFFS implies that the element reference plane is offset a distance of ZOFFS along the positive z-axis of the element coordinate system. If the ZOFFS field is used, then both the MID1 and the MID2 fields must be specified on the PSHELL entry referenced by PID on the CQUAD4 entry.

   ZOFFS should not be used when differential stiffness is calculated in solutions 105 and 200, and solutions 103, 107 - 112 using STATSUB, since the differential stiffness calculation does not include the offset vectors.

   ZOFFS is ignored in heat transfer solutions 153 and 159 including any solution subcase which includes the case control ANALYSIS=HEAT.
9. ZOFFS has the following caveats in nonlinear solutions 106 and 129:
   - For geometry nonlinear only, the differential stiffness calculation does not include the offset vectors. The exclusion of the offset in the differential stiffness will have an adverse effect on convergence, however, if the solution converges, the results will be correct. In a nonlinear solution, equilibrium (and convergence) is achieved when internal and external forces balance. The offset is correctly accounted for in the internal and external force calculations even though it is not accounted for in the differential stiffness.
   - ZOFFS should not be defined on elements which use MATS1 nonlinear material definitions.
   - ZOFFS is ignored for hyperelastic elements.
   - ZOFFS should not be used in nonlinear solutions 106 and 129 in combination with thermal loads.
10. For finite deformation hyperelastic analysis, the plot codes are given by the CQUADFD element name in **“Item Codes”**.
11. By default, all four edges of the element are considered straight.
12. If the x-direction of a material coordinate system is perpendicular or very close to being perpendicular to the element face, a projection is either impossible or unpredictable. A fatal error will occur when

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the angle between a shell element face normal and the x-direction of the material coordinate system (MCID) is less than the value specified by System Cell 489.

13. The MCID should be less than 99,999,999 when SYSTEM(525)=0 (default), and less than 99,999,999,999 when SYSTEM(525)=1.

14. For SOLs 401 and 402, see Shell elements in the Multi-Step Nonlinear User's Guide for details.

Remarks related to SOLs 402 and 414:

1. Supported only if system cell 370 is turned on. Otherwise, a fatal error is issued: use CQUADR instead or turn on the system cell.

Related Topics

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CQUAD8

Curved Quadrilateral Shell Element Connection

Defines a curved quadrilateral shell or plane strain element with eight grid points.

Note

Do not use the CQUAD8 element in SOL 414 models because SOL 414 analyses expect the deformation of the shell to be linear.

Format:

1	2	3	4	5	6	7	8	9	10
CQUAD8	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	T1	T2	T3	T4	THETA or MCID	ZOFFS	
	TFLAG								

Example:

CQUAD8	207	3	31	33	73	71	32	51	
	53	72	0.125	0.025	0.030	0.025	30.	0.03	

Fields:

Field	Contents
EID	Element identification number. See Remark 1. (Integer > 0)
PID	Property identification number of a PSHELL, PCOMP, PCOMPG, or PLPLANE entry. (Integer > 0; No default)
G1, G2, G3, G4	Identification numbers of connected corner grid points. Required data for all four grid points. See Remark 2. (Unique integers > 0; No default)
G5, G6, G7, G8	Identification numbers of connected edge grid points. Required data for all four grid points unless the element is hyperelastic. See Remarks 2, 3, and 4. (Unique integers ≥ 0; No default)
T1, T2, T3, T4	If TFLAG=0 or the TFLAG field is blank, thickness of the element at grid points G1 through G4. If TFLAG=1, scaling factor used to compute the thickness of the element at grid points G1 through G4 as the product of Ti and the thickness specified on the PSHELL bulk entry. Ti is ignored for hyperelastic elements. See Remarks 5 and 6. (Real > 0.0; If blank, use the thickness specified on the PSHELL bulk entry as the default)

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Field	Contents
THETA or MCID	If input is real or the field is blank, material orientation angle (THETA) in degrees. If input is integer, identification number of a coordinate system (MCID) from which THETA is calculated. Use a CORDij bulk entry to define the coordinate system. To use the basic coordinate system, enter integer zero. THETA or MCID are ignored for hyperelastic elements. See Remarks 7, 8, 9, and 10. (Real or integer ≥ 0; Default = 0.0)
ZOFFS	Offset from the surface of grid points to the element reference plane. See Remark 11. ZOFFS is ignored for hyperelastic elements. (Real; Default = 0.0)
TFLAG	Flag that specifies how the values in the Ti fields are used to define the thickness of element. (Integer = 0 or 1; Default = 0)

Remarks:

1. Element identification numbers should be unique with respect to the element identification numbers of all other elements regardless of element type.
2. Grid points G1 to G8 must be numbered as shown in Figure 12-83.
3. For best results, locate mid-side grid points within the middle third of the edge. If a mid-side grid point is located at the quarter point of the edge, a fatal error may occur because of a divide-by-zero error, or if a fatal error does not occur, the calculated stresses may be meaningless.
4. By default, a fatal error occurs if a G1 through G8 value is not specified. However, if the element is hyperelastic and there are no mid-side grid points, the element is processed identically to the hyperelastic CQUAD4 element. The variable grid point CQUAD8 is supported with the pre-Simcenter Nastran 2019.2 formulation by specifying SYSTEM(700)=1, except for SOLs 401 and 402 which always use the newer formulation and ignore the system cell setting.
5. When the PID refers to either a PCOMP or PCOMPG bulk entry, the software converts the PCOMP/PCOMPG bulk entry to equivalent PSHELL/MAT2 bulk entries. If Ti are defined, depending on the TFLAG setting, they either override or scale the values on the equivalent PSHELL bulk entry. The software uses the ply thicknesses defined on the PCOMP/PCOMPG bulk entries, to recover ply stresses and strains.
6. A fatal error occurs if a Ti value is zero. To allow for Ti values of zero, specify SYSTEM(495)=1.
7. If MCID is specified, the software uses the following procedure to calculate the value for THETA, which it then uses to determine the material coordinate system for the element as described in Remark 9.

   The following notation is used in this remark and Figure 12-82:

   $(x_{mcid}, y_{mcid}, z_{mcid})$
   Axes of the MCID coordinate system

   $(\hat{x}_{mcid}, \hat{y}_{mcid}, \hat{z}_{mcid})$
   Unit vectors aligned with the axes of the MCID coordinate system

   $(x_{mref}, y_{mref}, z_{mref})$
   Axes of the material coordinate system relative to the reference plane

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$\left(\hat{x}_{mref}, \hat{y}_{mref}, \hat{z}_{mref}\right)$

Unit vectors aligned with the axes of the material coordinate system relative to the reference plane

a. Because the element can be curved and warped, the software begins by creating a reference plane for the element. The unit normal to the reference plane is calculated as follows:

$$\hat{z}_{mref} = \frac{\vec{r}_{G1G3} \times \vec{r}_{G2G4}}{\left\|\vec{r}_{G1G3} \times \vec{r}_{G2G4}\right\|}$$

where $\vec{r}_{G1G3}$ is a vector that extends from G1 to G3, and $\vec{r}_{G2G4}$ is a vector that extends from G2 to G4.

b. The x-axis of the MCID coordinate system is projected along the $\hat{z}_{mref}$ direction onto the reference plane to obtain the x-axis of the material coordinate system on the reference plane.

c. The $\vec{r}_{G1G2}$ vector is projected along the $\hat{z}_{mref}$ direction onto the reference plane to obtain the $\vec{r}_{G1'G2'}$ vector.

d. The software calculates the angle θ in degrees from the scalar dot product as follows:

$$\theta = \cos^{-1}\left(\frac{\hat{x}_{mref} \cdot \vec{r}_{G1'G2'}}{\left\|\vec{r}_{G1'G2'}\right\|}\right)$$

e. To determine if angle θ lies in the 3rd or 4th quadrants, the software evaluates the algebraic sign of $\sin\theta$ from the vector cross product as follows:

$$(\sin\theta)\hat{z}_{mref} = \frac{\vec{r}_{G1'G2'}}{\left\|\vec{r}_{G1'G2'}\right\|} \times \hat{x}_{mref}$$

If $\sin\theta < 0$, the software adds 180° to the calculated value of θ.

f. The software uses θ as the value for THETA.

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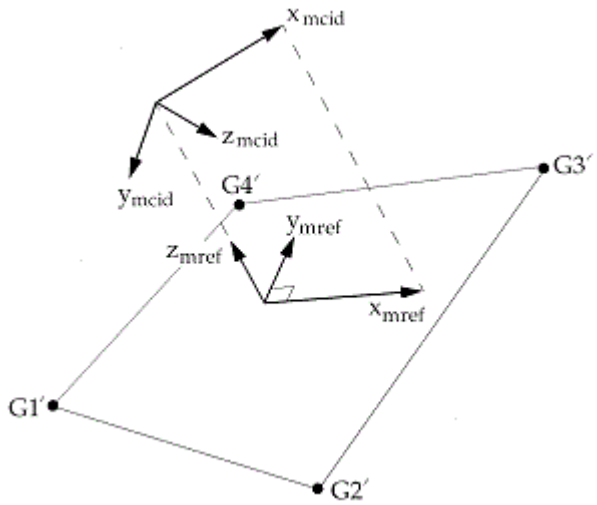
Figure 12-82: Projection of MCID Coordinate System X-Axis onto the Reference Plane



8. If the x-axis of the MCID coordinate system is perpendicular or very close to being perpendicular to the reference plane, projecting the x-axis of the MCID coordinate system onto the reference plane is either impossible or unpredictable. Thus, a fatal error occurs when the angle between the normal to the reference plane and the x-axis of the MCID coordinate system is less than the value specified by System Cell 489.

9. The software uses the following procedure to calculate the orientation of the material coordinate system.

   The following notation is used in this remark and Figures Figure 12-82 and Figure 12-83:

   $\left(x_{mref}, y_{mref}, z_{mref}\right)$
   Axes of the material coordinate system relative to the reference plane

   $\left(\hat{x}_{mref}, \hat{y}_{mref}, \hat{z}_{mref}\right)$
   Unit vectors aligned with the axes of the material coordinate system relative to the reference plane

   $\left(x_m, y_m, z_m\right)$
   Axes of the material coordinate system

   $\left(\hat{x}_m, \hat{y}_m, \hat{z}_m\right)$
   Unit vectors aligned with the axes of the material coordinate system

   a. Because the element can be curved and warped, the software begins by creating a reference plane for the element. The unit normal to the reference plane is calculated as follows:

$$\hat{z}_{mref} = \frac{\vec{r}_{G1G3} \times \vec{r}_{G2G4}}{\left\| \vec{r}_{G1G3} \times \vec{r}_{G2G4} \right\|}$$

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where $\vec{r}_{G1G3}$ is a vector that extends from G1 to G3, and $\vec{r}_{G2G4}$ is a vector that extends from G2 to G4.

b. The $\vec{r}_{G1G2}$ vector is projected along the $\hat{z}_{mref}$ direction onto the reference plane to obtain the $\vec{r}_{G1'G2'}$ vector.

c. The $\vec{r}_{G1'G2'}$ vector is rotated THETA degrees about $\hat{z}_{mref}$. The direction of the rotated $\vec{r}_{G1'G2'}$ vector is the direction of the x-axis of the material coordinate system on the reference plane.

d. To account for curvature and warping, at each corner grid point and the element centroid, the x-axis of the material coordinate system on the reference plane is rotated in the plane that it forms with the unit normal to the surface at that point, $\hat{n}_{Gi}$, so that it is tangent to the surface. The rotated axis is the x-axis of the material coordinate system at that point.

e. The z-axis of the material coordinate system at each corner grid point and the element centroid is the normal to the surface at that point. Thus, it follows that:

$$\hat{z}_m = \hat{n}_{Gi}$$

f. The unit vector in the direction of the y-axis of the material coordinate system at each corner grid point and the element centroid is obtained as follows:

$$\hat{y}_m = \hat{z}_m \times \hat{x}_m$$

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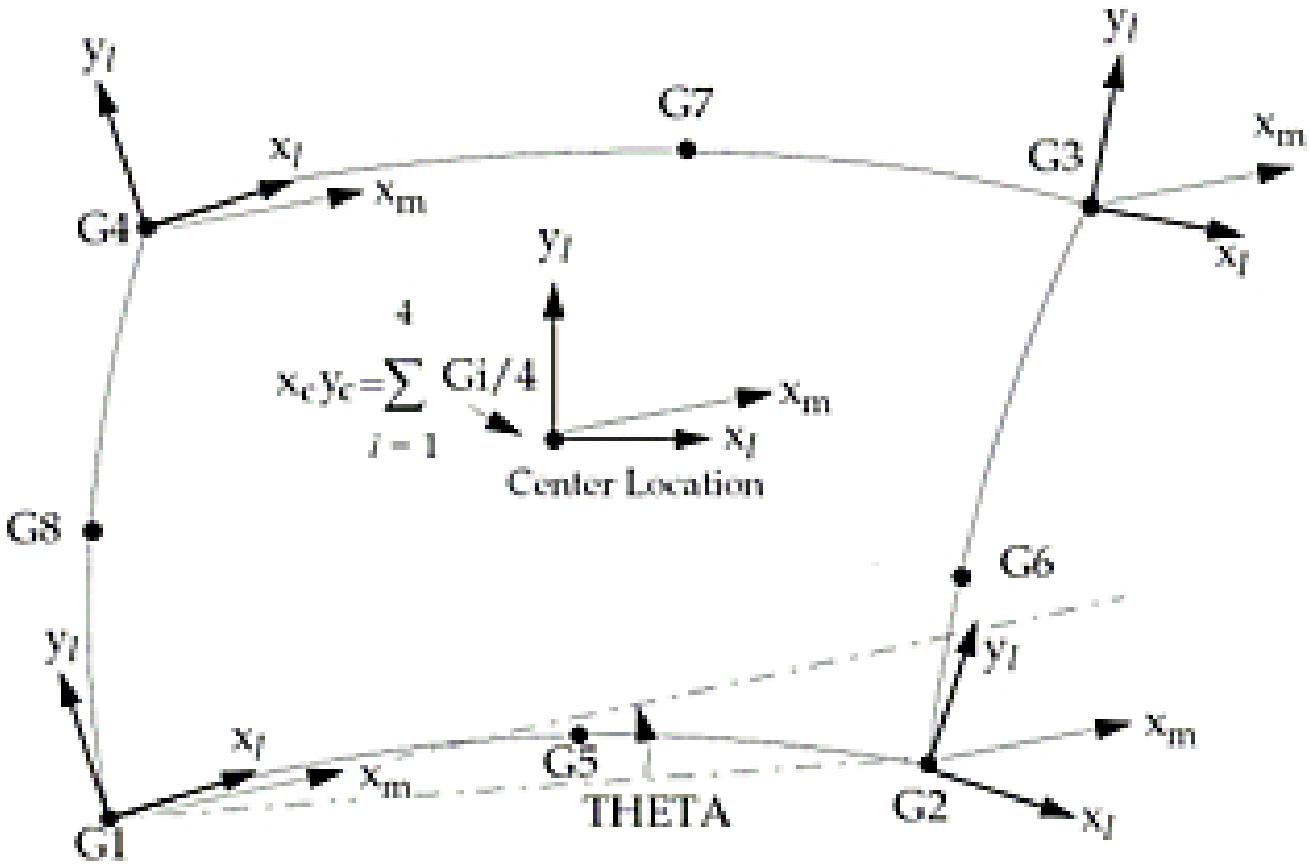
Figure 12-83: CQUAD8 Local and Material Coordinate Systems and Connectivity



10. MCID must be less than 99,999,999 when SYSTEM(525)=0 (default), and less than 99,999,999,999 when SYSTEM(525)=1.

11. Data such as material matrices and stress fiber locations, are given relative to the reference plane. A positive value of ZOFFS implies that the reference plane is offset a distance of ZOFFS along the positive z-axis of the local coordinate system. If the ZOFFS field is used, then both the MID1 and the MID2 fields must be specified on the PSHELL bulk entry referenced by the CQUAD8 element.

ZOFFS is ignored in heat transfer SOLs 153 and 159 including any solution subcase which includes the case control ANALYSIS=HEAT.

12. The software uses the following procedure to obtain the local coordinate system at each corner grid point and the element centroid.

The following notation is used in this remark and Figures **Figure 12-83** and **Figure 12-84**:

(x_l, y_l, z_l)

Axes of the local coordinate system

$(\hat{x}_l, \hat{y}_l, \hat{z}_l)$

Unit vectors aligned with the axes of the local coordinate system

(ξ, η, ζ)

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Parametric axes

$\left(\hat{\xi}, \hat{\eta}, \hat{\zeta}\right)$

Unit vectors aligned with the parametric axes

a. The software creates unit vectors $\hat{\xi}$ and $\hat{\eta}$ that are tangent to the ξ- and η-axes, respectively, at each corner grid point and the element centroid.

b. In the plane formed by each $\hat{\xi}$ and $\hat{\eta}$ pair, the software creates vector $\vec{A}$ which bisects $\hat{\xi}$ and $\hat{\eta}$.

c. Vector $\vec{A}$ is rotated 45° degree about $-\hat{n}_{Gi}$ to obtain the x-axis of the local coordinate system, where $\hat{n}_{Gi}$ is the unit vector normal to the surface of the element at Gi.

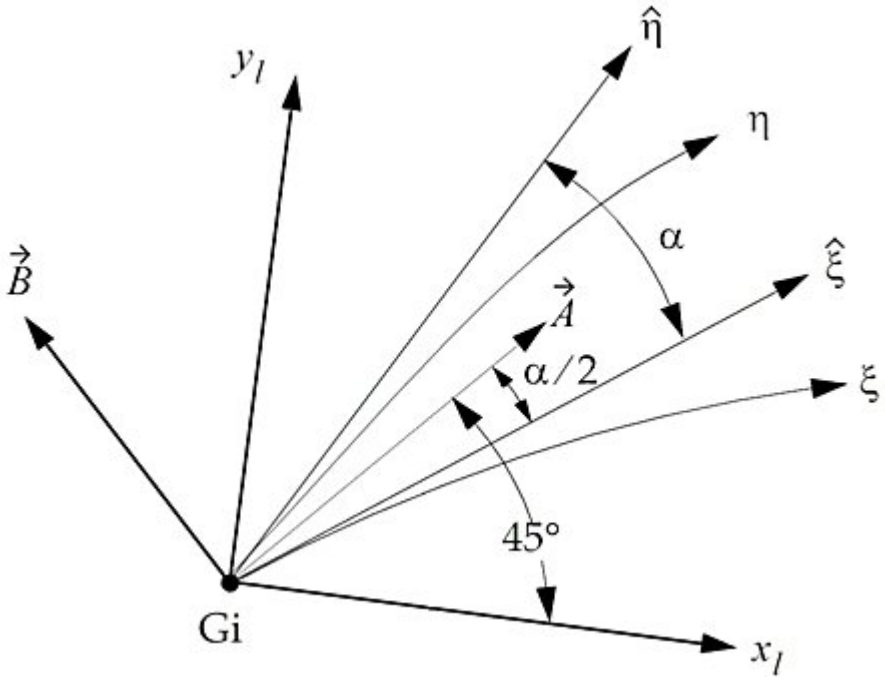
d. The z-axis of the local coordinate system is normal to the surface at Gi. Thus, it follows that:

$$\hat{z}_l = \hat{n}_{Gi}$$

e. The unit vector in the direction of the y-axis of the local coordinate system is obtained as follows:

$$\hat{y}_l = \hat{z}_l \times \hat{x}_l$$

Figure 12-84: Local Coordinate System



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13. For SOLs 401 and 402, stress and strain are output at the corner grid points and the element centroid in the material coordinate system. See Shell elements in the Multi-Step Nonlinear User's Guide for details.

    For other solutions, stress and strain are output at the corner grid points and the element centroid in the local coordinate system. **Figure 12-83** shows the local coordinate system at these points. For hyperelastic elements, the stress and strain are output in the coordinate system identified by the CID field on the PLPLANE bulk entry.

14. The CQUAD8 element formulation uses 8-noded serendipity shape functions for the translational DOF, and 9-noded Lagrange shape functions for the rotational DOF.

    By default, four Gauss points (2 x 2 integration) are used to integrate the membrane and transverse shear stiffness, and nine Gauss points (3 x 3 integration) are used to integrate the bending and coupled membrane-bending stiffness.

15. For hyperelastic elements the plot codes are specified under the CQUADFD element name in “Item Codes”.

16. To revert to the pre-Simcenter Nastran 2019.2 formulation, include SYSTEM(700)=1 in your input file. Because the new formulation is the only formulation in SOLs 401 and 402, the software ignores a SYSTEM(700)=1 specification in a SOL 401 or 402 solve.

17. For SOLs 401 and 402, see Shell elements in the Multistep User's Guide for details.

Related Topics

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CQUADR

Quadrilateral Plate Element Connection

Defines an isoparametric membrane and bending quadrilateral plate element. This element is less sensitive to initial distortion and extreme values of Poisson's ratio than the CQUAD4 element. It is a companion to the CTRIAR element.

Format:

1	2	3	4	5	6	7	8	9	10
CQUADR	EID	PID	G1	G2	G3	G4	THETA or MCID	ZOFFS	
		TFLAG	T1	T2	T3	T4			

Example:

CQUADR	82	203	31	74	75	32	2.6		
			1.77	2.04	2.09	1.80			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PSHELL, PCOMP, PCOMPG or PSHLPNL entry. (Integer > 0; Default = EID)
Gi	Grid point identification numbers of connection points. (Integers > 0, all unique)
THETA	Material property orientation angle in degrees. See Figure 12-86. (Real; Default = 0.0)
MCID	Material coordinate system identification number. The x-axis of the material coordinate system is determined by projecting the x-axis of the MCID coordinate system (defined by the CORDij entry or zero for the basic coordinate system) onto the surface of the element. See Remark 13. (Integer ≥ 0; If blank, then THETA = 0.0 is assumed.)

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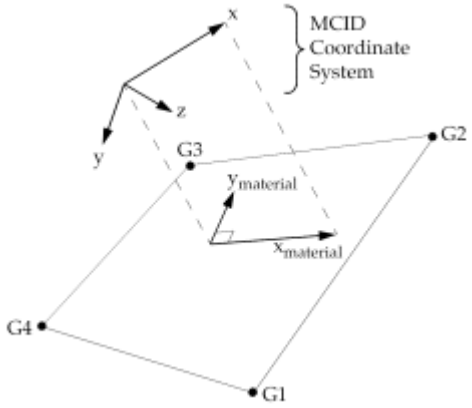
Field	Contents

Figure 12-85: MCID Coordinate System Definition



ZOFFS	Offset from the surface of grid points to the element reference plane. See Remark 8. (Real)
TFLAG	Integer Flag which specifies how Ti is used to define thickness of element. (0, 1, or blank)
Ti	Thickness of element at grid points G1 through G4 if TFLAG=0 or blank. If TFLAG=1, thickness becomes a product of Ti and the thickness on the PSHELL or PSHLPNL entry. See Remark 6. (Real > 0.0 or blank. See Remark 4 for the default.)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. Grid points G1 through G4 must be ordered consecutively around the perimeter of the element.
3. All the interior angles must be less than 180°.
4. The continuation is optional. If it is not supplied, then T1 through T4 will be set equal to the value of T on the PSHELL entry.
5. When the PID refers to either a PCOMP or PCOMPG entry, the software converts the PCOMP/PCOMPG entry to equivalent PSHELL/MAT2 entries. If Ti is defined, they override or scale the values on the equivalent PSHELL entry. It should be noted that ply-stress and ply-strain recovery uses the ply thicknesses defined on the PCOMP/PCOMPG entries, and not Ti.
6. By default, a fatal error occurs if a Ti value is zero. To allow for Ti values of zero, use System Cell 495.
7. Stresses and strains are output in the element coordinate system at the centroid unless the CORNER or CUBIC entry is specified on the STRESS case control card. Specifying CORNER or CUBIC will result in stresses at grid points G1 through G4.
8. Elements may be offset from the connection points by means of ZOFFS. Other data, such as material matrices and stress fiber locations, are given relative to the reference plane. A positive value of ZOFFS

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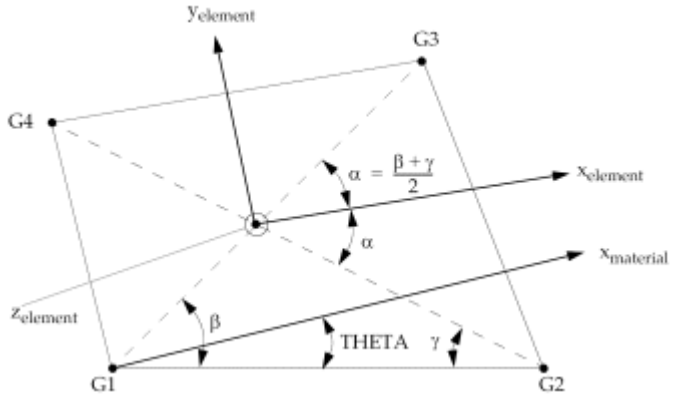
implies that the element reference plane is offset a distance of ZOFFS along the positive z-axis of the element coordinate system. If the ZOFFS field is used, then both the MID1 and the MID2 fields must be specified on the PSHELL entry referenced by PID on the CQUADR entry.

ZOFFS is ignored in heat transfer solutions 153 and 159 including any solution subcase which includes the case control ANALYSIS=HEAT.

9. The rotational degree-of-freedom normal (drilling) to the element are active in the CQUADR element formulation. Results will be inaccurate if the drilling degree-of-freedom of interior grids is constrained. At the boundary of a model, the drilling degree-of-freedom must be constrained to achieve edge boundary conditions such as a fixed or a symmetrical boundary.
10. As compared to the CQUAD4 element, the CQUADR element:
    - Accounts for normal rotational DOF to provide improved membrane accuracy.
    - Is less sensitive to high aspect ratios and values of Poisson's ratio near 0.5.
11. The CQUADR element is supported in all structural and heat transfer solutions. If used in SOL 106 or 129, the CQUADR element always behaves geometrically linearly.
12. The CTRIAR element is the triangular companion to the CQUADR element and should be used instead of CTRIA3 or CTRIA6.
13. If the x-direction of a material coordinate system is perpendicular or very close to being perpendicular to the element face, a projection is either impossible or unpredictable. A fatal error will occur when the angle between a shell element face normal and the x-direction of the material coordinate system (MCID) is less than the value specified by System Cell 489.
14. The MCID should be less than 99,999,999 when SYSTEM(525)=0 (default), and less than 99,999,999,999 when SYSTEM(525)=1.

Figure 12-86: CQUADR Element Geometry and Coordinate Systems



Remarks related to SOLs 401, 402, and 414:

1. Output results will be done in the material coordinate system. If those axes are not defined, the material coordinate system is parallel to the G1 - G2 edge. See Shell elements in the Multi-Step Nonlinear User's Guide for details.

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Related Topics

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CQUADX4

Axisymmetric Quadrilateral Element Connection

Defines an isoparametric and axisymmetric quadrilateral cross-section ring element for use in linear and fully nonlinear (i.e., large strain and large rotations) hyperelastic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CQUADX4	EID	PID	G1	G2	G3	G4	TH		

Example:

CQUADX4	203	100	10	20	30	40	30.0		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PMIC, PSOLID or PLSOLID entry. (Integer > 0; Default=EID)
Gi	Grid point identification numbers of connected points. (Unique integers > 0)
TH	Material property orientation angle in degrees. See Remark 4. TH is ignored for hyperelastic elements. (Real; Default = 0.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. When axisymmetric elements are defined on the XZ-plane, X is the radial direction, and Z is the axial direction. The grid points defining these elements must have X ≥ 0. See **Figure 12-88**.

   When axisymmetric elements are defined on the XY-plane, Y is the radial direction, and X is the axial direction. The grid points defining these elements must have Y ≥ 0. See **Figure 12-89**.

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4. For orthotropic materials, the TH angle orients the principal material coordinates of the element relative to the basic coordinates. **Figure 12-88** and **Figure 12-89** show the positive sense of the TH angle when the element is defined on the XZ-plane and XY-plane, respectively.
5. The initial element coordinate system is the basic coordinate system.
6. See the remarks on the **PSOLID** entry for information on stress and strain output coordinate systems.
7. The CQUADX4, CQUADX8, CTRAX3, and CTRAX6 elements behave linearly in SOL 106 and 129 when not hyperelastic.
8. For hyperelastic element, the plot codes are specified under the CQUADX4FD element name in **Item Codes Overview**.
9. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
10. A concentrated load (e.g. FORCE entry) at Gi is the total of the force around the circumference. Reaction force and applied load output are the same. For example, to apply a distributed load of 135.0 Newton/mm on a single grid where the radius is 0.5 mm:

    The value entered on a FORCE entry = (Distributed force * 2 * π * Radius)

    = 135.0 N/mm * 2 * π * 0.5 mm

    = 424.1 N

    See **system cell 587** for information on the pre-version 10 axisymmetric element behavior.

Figure 12-87: CQUADX4 Element Idealization

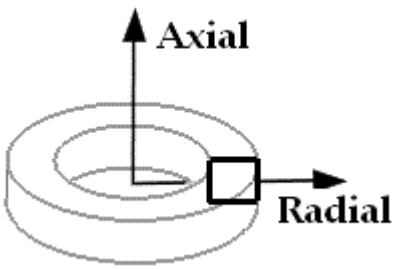

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Figure 12-88: CQUADX4 Element Geometry and Coordinate Systems, XZ-Plane

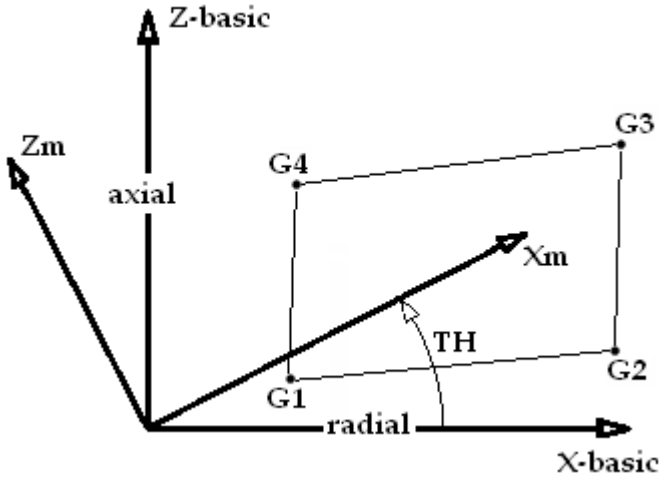
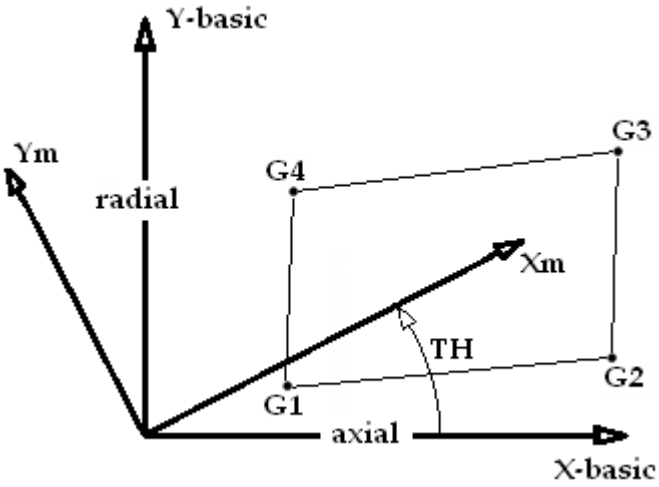


Figure 12-89: CQUADX4 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The SPCFORCES output results for the axisymmetric elements are always given in force unit per 2*pi radians and not per radian. System cell 587 is ignored.

Related Topics

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CQUADX8

Axisymmetric Quadrilateral Element Connection

Defines an isoparametric and axisymmetric quadrilateral cross-section ring element with midside nodes for use in linear and fully nonlinear (i.e., large strain and large rotations) hyperelastic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CQUADX8	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	TH						

Example:

CQUADX8	203	100	10	20	30	40	50	60	
	70	80	30.0						

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PSOLID or PLSOLID entry. (Integer > 0; Default=EID)
Gi	Grid point identification numbers of connected points (midside grids cannot be eliminated). (Unique integers > 0)
TH	Material property orientation angle in degrees. See Remark 4. TH is ignored for hyperelastic elements. (Real; Default = 0.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. When axisymmetric elements are defined on the XZ-plane, X is the radial direction, and Z is the axial direction. The grid points defining these elements must have X ≥ 0. See Figure 12-91.

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When axisymmetric elements are defined on the XY-plane, Y is the radial direction, and X is the axial direction. The grid points defining these elements must have Y ≥ 0. See **Figure 12-92**.

4. For orthotropic materials, the TH angle orients the principal material coordinates of the element relative to the basic coordinates. **Figure 12-91** and **Figure 12-92** show the positive sense of the TH angle when the element is defined on the XZ-plane and XY-plane, respectively.
5. The initial element coordinate system is the basic coordinate system.
6. See the remarks on the **PSOLID** entry for information on stress and strain output coordinate systems.
7. The CQUADX4, CQUADX8, CTRAX3, and CTRAX6 elements behave linearly in SOL 106 and 129 when not hyperelastic.
8. For hyperelastic element, the plot codes are specified under the CQUADX8FD element name in **Item Codes Overview**.
9. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
10. A concentrated load (e.g. FORCE entry) at Gi is the total of the force around the circumference. Reaction force and applied load output are the same. For example, to apply a distributed load of 135.0 Newton/mm on a single grid where the radius is 0.5 mm:

    The value entered on a FORCE entry = (Distributed force * 2 * π * Radius)

    = 135.0 N/mm * 2 * π * 0.5 mm

    = 424.1 N

    See **system cell 587** for information on the pre-version 10 axisymmetric element behavior.

Figure 12-90: CQUADX8 Element Idealization

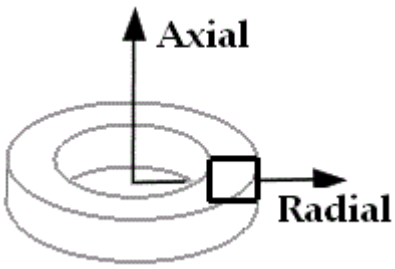

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Figure 12-91: CQUADX8 Element Geometry and Coordinate Systems, XZ-Plane

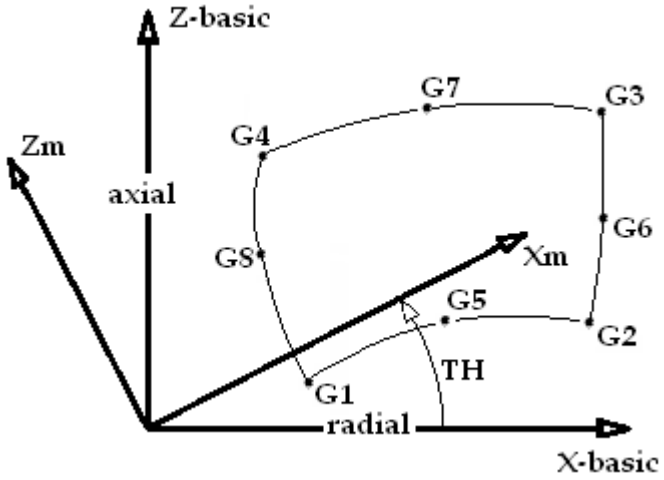
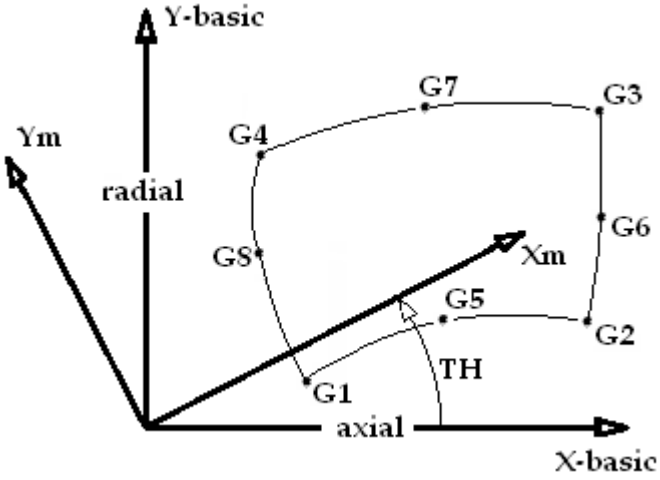


Figure 12-92: CQUADX8 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The SPCFORCES output results for the axisymmetric elements are always given in force unit per 2*pi radians and not per radian. System cell 587 is ignored.

Related Topics

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CRAC2D

Two-Dimensional Crack Tip Element

Defines a two-dimensional crack tip element.

Format:

1	2	3	4	5	6	7	8	9	10
CRAC2D	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	G10	G11	G12	G13	G14	
	G15	G16	G17	G18					

Example:

CRAC2D	114	108	2	5	6	8	7	11	
	12	14	16	17		20	22		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PRAC2D entry. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer ≥ 0, G11 through G18 may be blank.)

Remarks:

1. The following is a dummy element and requires the presence of one Bulk Data entry of the form:

ADUM8	18	0	5	0	CRAC2D				

2. The element should be planar. Significant deviations will produce fatal errors.
3. Grid points G1 through G10 are required while grid points G11 through G18 are optional for the quadrilateral form of the element.
4. The stresses and stress intensity factors are calculated assuming that G2 and G10 are coincident. Deviations from this will produce erroneous results.
5. For the symmetric half-crack option, grid points G1 through G7 are required while grid points G11 through G14 are optional. Grid points G8 through G10 and G15 through G18 must not be present for this option.

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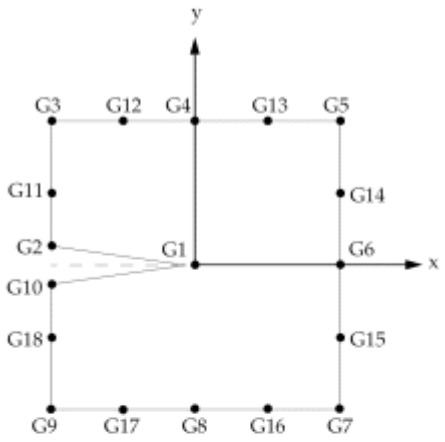
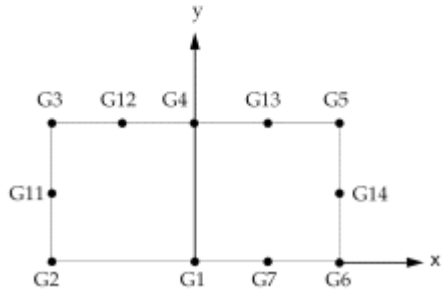
6. The ordering conventions for the full-crack and half-crack options are shown in Figure 12-93.
7. The stress output is interpreted as described in "Two-Dimensional Crack Tip Element (CRAC2D)" in the *Simcenter Nastran Element Library*.

Figure 12-93: CRAC2D Element Connection for Full and Symmetric Options



(a) Full Crack Option



(b) Symmetric Half-Crack Option

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CRAC3D

Three-Dimensional Crack Tip Element

Defines a three-dimensional crack tip element.

Format:

1	2	3	4	5	6	7	8	9	10
CRAC3D	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	G10	G11	G12	G13	G14	
	G15	G16	G17	G18	G19	G20	G21	G22	
	G23	G24	G25	G26	G27	G28	G29	G30	
	G31	G32	G33	G34	G35	G36	G37	G38	
	G39	G40	G41	G42	G43	G44	G45	G46	
	G47	G48	G49	G50	G51	G52	G53	G54	
	G55	G56	G57	G58	G59	G60	G61	G62	
	G63	G64							

Example:

CRAC3D	113	101	2	5	7	8	4	10	
	11	14	15	17		3	6	9	
	12		16		102	105	107	108	
	104	110	111	114	115	117		103	
	106	109	112		116		202	205	
	207	208	204	210	211	214	215	217	
	225	226							

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PRAC3D entry. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer ≥ 0)

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Remarks:

1. The following is a dummy element and requires the presence of one Bulk Data entry of the form:

ADUM9	64	0	6	0	CRAC3D				

2. Element identification numbers should be unique with respect to all other element identification numbers.
3. This element, including grid point numbering conventions, is shown in **Figure 12-94** and **Figure 12-95**. grid points G1 through G10, and G19 through G28 are required; midside and surface grid points G11 through G18, G29 through G36, and G37 through G64 are optional. Either all or none of grid points G37 through G46 should be present. A fatal error message will be issued for partial connectivity.
4. For the symmetric half-crack option grid points G1 through G7, and G19 through G25 are required, whereas grid points G11 through G14, G29 through G32, and G37 through G42 are optional. Grid points G8 through G10, G15 through G18, G26 through G28, G33 through G36, G43 through G46, G51 through G55, and G60 through G64 should not be specified to invoke this option.
5. It is recommended that both the faces (formed by grid points G1 through G18 and grid points G19 through G36) and the midplane (formed by grid points G37 through G46) be planar. It is also recommended that midside grid points G37 through G46 be located within the middle third of the edges.
6. The midside nodes on both the faces should be defined in pairs. For example, if grid point G11 is not defined, then grid point G29 should not be defined and vice versa.
7. The stresses and stress intensity factors are calculated with the assumptions that grid points G2 and G10, G20 and G28, and G38 and G46 are coincident. Deviation from this condition will produce erroneous results.
8. The stress output is interpreted as described in Three-Dimensional Crack Tip Element (CRAC3D)in the *Simcenter Nastran Element Library*.

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Figure 12-94: CRAC3D Solid Crack Tip Element with Required Connection Points Only

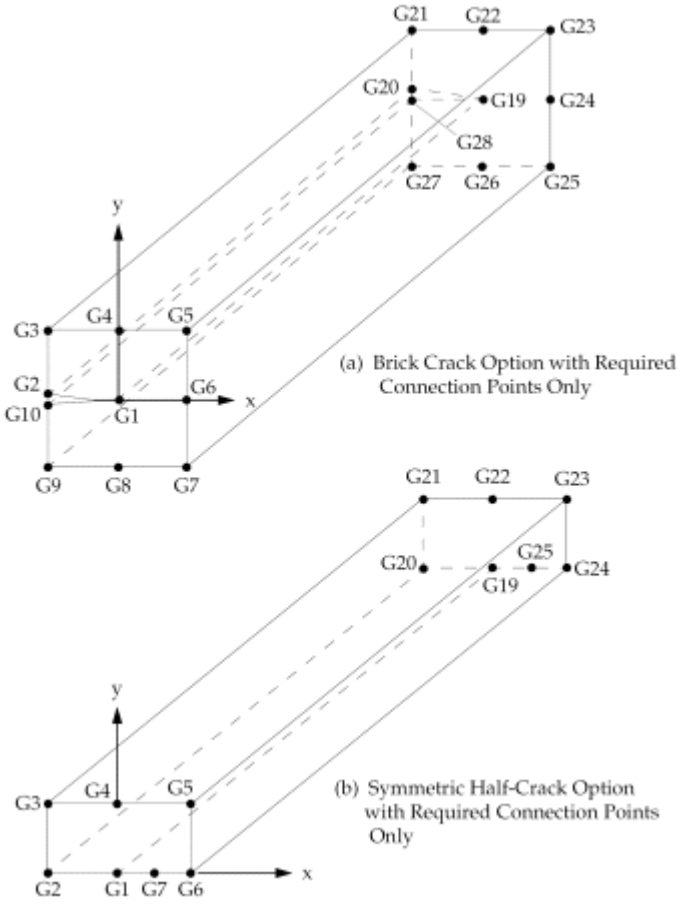

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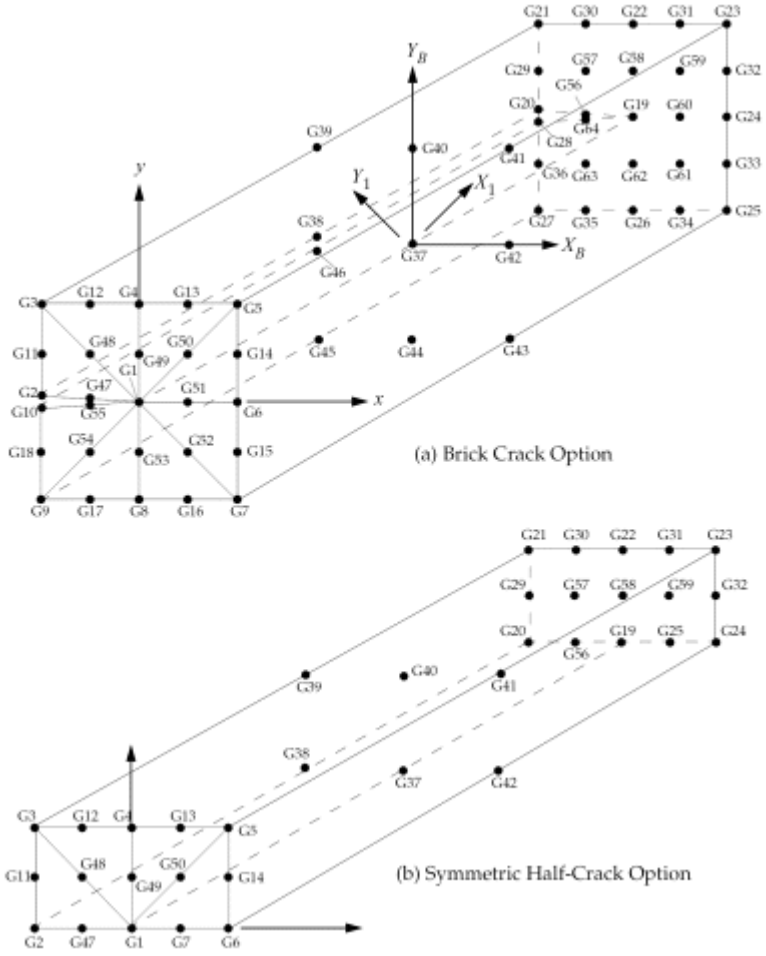
Figure 12-95: CRAC2D Solid Crack Tip Element with All Connection Points



Related Topics

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CRAKTP

Crack Tip Specification

Specifies information related to a crack tip.

Format:

1	2	3	4	5	6	7	8	9	10
CRAKTP	SID	NR							
	G1	VCEV1	G3	VCEV2	G3	VCEV3	G4	VCEV4	
	G5	VCEV5	etc.						

Example:

CRAKTP	1	3							
	11	1	12	2	13	3	16	4	
	17	9	52	8	57	9			

Fields:

Field	Contents
SID	Crack tip identification number. (Integer > 0)
NR	Number of rings to compute j-integral. (0 < Integer ≤ 10)
Gi	Grid point identification number for the crack tip. (Integer > 0)
VCEVi	Virtual crack extension vector identification number. (Integer > 0)

Related Topics

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CREEP

Creep Characteristics

Defines creep characteristics based on experimental data or known empirical creep law. This entry will be activated if a MAT1, MAT2, MAT8, MAT9, or MAT11 entry with the same MID is used and the NLPARM entry is prepared for creep analysis. The creep formulation is principally suited for isotropic materials and, in general, when used with anisotropic materials may produce incorrect results. However, slightly anisotropic materials may produce acceptable results.

Format:

1	2	3	4	5	6	7	8	9	10
CREEP	MID	T0	EXP	FORM	TIDKP	TIDCP	TIDCS	THRESH	
	TYPE	a	b	c	d	e	f	g	

Example:

CREEP	8	1100.		CRLAW					
	121	6.985-6	2.444	7.032-4	0.1072	6.73-9	0.1479	3.0	

Fields:

Field	Contents
MID	Material identification number of a MAT1, MAT2, MAT8, MAT9, or MAT11 entry. (Integer > 0)
T0	Reference temperature at which creep characteristics are defined. See Remark 2. (Real; Default = 0.0)
EXP	Temperature-dependent term, $e^{(-\Delta H/(R \cdot T0))}$, in the creep rate expression. See Remark 2. (0.0 < Real ≤ 1.0; Default = 1.0E-9)
FORM	Form of the input data defining creep characteristics. (Character: "CRLAW" for empirical creep law, or "TABLE" for tabular input data of creep model parameters.)
TIDKP, TIDCP, TIDCS	Identification number of a TABLES1 entry, which defines the creep model parameters $K_p(\sigma)$, $C_p(\sigma)$, and $C_s(\sigma)$, respectively. See Remarks 3 through 5. (Integer > 0)
THRESH	Threshold limit for creep process. Threshold stress under which creep does not occur is computed as THRESH multiplied by Young's modulus. (0.0 < Real < 1.0E-3; Default = 1.0E-5)

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Field	Contents
TYPE	Identification number of the empirical creep law type. See Remark 1. (Integer: 111, 112, 121, 122, 211, 212, 221, 222, or 300)
a through g	Coefficients of the empirical creep law specified in TYPE. Continuation should not be specified if FORM = "TABLE". See Remark 1. (Real)

Remarks:

1. Two classes of empirical creep law are available. **Creep Law Class 1** is expressed as:

Equation 12-4:

$$\varepsilon^c(\sigma, t) = A(\sigma)[1 - e^{-R(\sigma)t}] + K(\sigma)t$$

Parameters $A(\sigma)$, $R(\sigma)$, and $K(\sigma)$ are specified in the following form, as recommended by the Oak Ridge National Laboratory:

Parameter	Function 1	Digit	Function 2	Digit
A(σ)	$a\sigma^b$	i=1	$ae^{b\sigma}$	i=2
R(σ)	$ce^{d\sigma}$	j=1	$c\sigma^d$	j=2
K(σ)	$e \cdot [\sinh(f\sigma)]^g$	k=1	$ee^{f\sigma}$	k=2

TYPE=ijk where i, j, and k are digits equal to 1 or 2, according to the desired function in the table above. For example, TYPE=122 defines $A(\sigma) = a\sigma^b$, $R(\sigma) = c\sigma^d$, and $K(\sigma) = ee^{f\sigma}$.

Creep Law Class 2 (TYPE=300) is expressed as:

Equation 12-5:

$$\varepsilon^c(\sigma, t) = a\sigma^b t^d$$

where the values of b and d must be defined as follows:

$$1.0 < b < 8.0$$

and

$$0.2 < d < 2.0$$

The coefficient g should be blank if TYPE = 112, 122, 222, or 212 and c, e, f, and g should be blank if TYPE = 300. The coefficients a through g are dependent on the structural units; caution must be exercised to make these units consistent with the rest of the input data.

2. Creep law coefficients a through g are usually determined by least squares fit of experimental data, obtained under a constant temperature. This reference temperature at which creep behavior is characterized must be specified in the T0 field if the temperature of the structure is different from this

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reference temperature. The conversion of the temperature input (°F or °C) to °K (degrees Kelvin) must be specified in the PARAM,TABS entry as follows:

PARAM,TABS,273.16 (If Celsius is used.)

PARAM,TABS,459.69 (If Fahrenheit is used.)

When the correction for the temperature effect is required, the temperature distribution must be defined in the bulk entries TEMP, TEMPP1, and/or TEMPRB, which are selected by the Case Control command TEMP(LOAD) = SID within the subcase.

From the thermodynamic considerations, the creep rate is expressed as:

Equation 12-6:

$$\dot{\varepsilon}^{C} = \dot{\varepsilon}_{A}(e^{-\Delta H/RT})$$

where:

ΔH = energy of activation

R = gas constant (= 1.98 cal/mole °K)

T = absolute temperature (°K)

$\dot{\varepsilon}_{A}$ = strain/sec per activation

If the creep characteristics are defined at temperature T0, the creep rate at temperature T is corrected by the following factor:

Equation 12-7:

$$\frac{\dot{\varepsilon}^{c}}{\dot{\varepsilon}_{0}^{c}} = \mathrm{EXP}^{\left(\frac{T0}{T}-1\right)}$$

where:

$\dot{\varepsilon}^{c}$ = corrected creep rate

$\dot{\varepsilon}_{0}^{c}$ = creep rate at T0

$\mathrm{EXP}^{\left(\frac{T0}{T}-1\right)}$ = correction factor

3. If the creep model parameters K_p, C_p, and C_s are to be specified with FORM = “TABLE”, then TABLES1 entries (with IDs that appear in TIDxx fields) must be provided in the bulk section. In this case, the continuation should not be specified.
4. Creep model parameters K_p, C_p, and C_s represent parameters of the uniaxial rheological model as shown in **Figure 12-96**.

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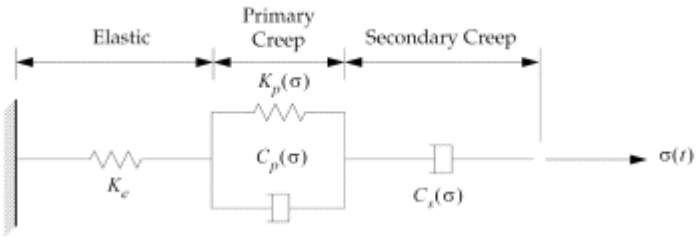
Tabular values (X_i, Y_i) in the TABLES1 entry correspond to (σ_i, K_{pi}), (σ_i, C_{pi}), and (σ_i, C_{si}) for the input of K_p, C_p, and C_s, respectively. For linear viscoelastic materials, parameters K_p, C_p, and C_s are constant and two values of σi must be specified for the same value of K_{pi}, C_{pi}, and C_{si}.

Figure 12-96: CREEP Parameter Idealization



Creep model parameters, as shown in Figure 12-97 through Figure 12-99 below, must have positive values. If the table look-up results in a negative value, the value will be reset to zero and a warning message (TABLE LOOK-UP RESULTS IN NEGATIVE VALUE OF CREEP MODEL PARAMETER IN ELEMENT ID=****) will be issued.

Figure 12-97: K_p Versus σ Example for CREEP

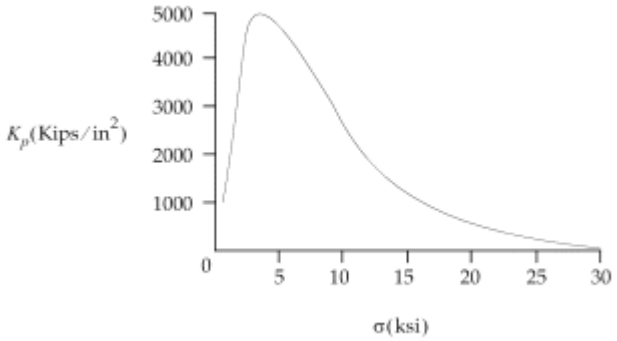


Figure 12-98: C_p Versus σ Example for CREEP

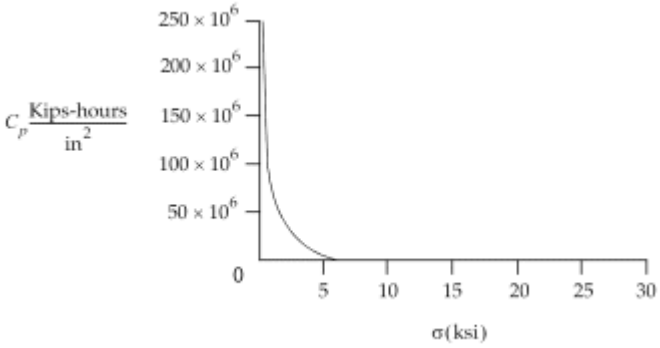


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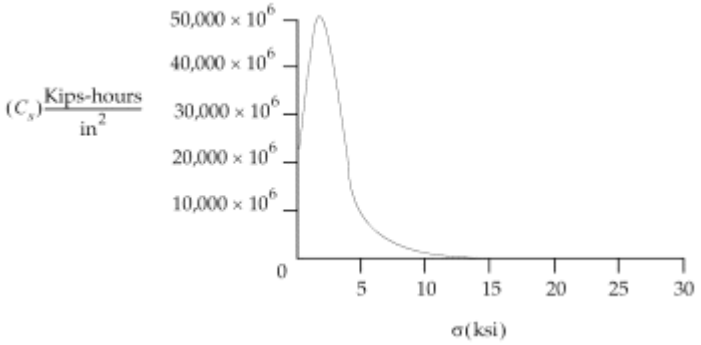
Figure 12-99: C_s Versus σ Example for CREEP



5. Creep analysis requires an initial static solution at *t = 0*, which can be obtained by specifying a subcase that requests an NLPARM entry with DT = 0.0.

Remarks related to SOL 402:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. MID is restricted to the material identification number of a MAT1 entry. CREEP and MAT1 entries model an elastic-creep material with only elastic strain, thermal strain and creep strain effects.
3. Only FORM=CRLAW is allowed.
4. T0, EXP, TIDKP, TIDCP, TIDCS, and THRESH are ignored.
5. If CREEP, MAT1 and MATS1 reference the same MID (plastic-creep material), elastic, thermal, plastic and creep strain effects will be taken into account. In that case, only isotropic hardening rule is allowed in the MATS1 entry.

Related Topics

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CROD

Rod Element Connection

Defines a tension-compression-torsion element.

Format:

1	2	3	4	5	6	7	8	9	10
CROD	EID	PID	G1	G2					

Example:

CROD	12	13	21	23					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PMIC or PROD entry. (Integer > 0; Default = EID)
G1, G2	Grid point identification numbers of connection points. (Integer > 0; G1 ≠ G2)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. See **"CONROD"** for alternative method of rod definition.
3. Only one element may be defined on a single entry.

Remark related to SOL 401:

1. The CROD defines a straight cylindrical element with constant axial and torsional stiffness along its length. You define the cross section area (A), the torsional constant (J), and reference a MAT1 material entry on the PROD entry.

   The CROD supports large displacements and rotations when large displacements are requested with PARAM,LGDISP,1. It does not support material nonlinear conditions.

   You can request axial force and torque output with the FORCE case control command. The positive directions for the forces are indicated in the figure.

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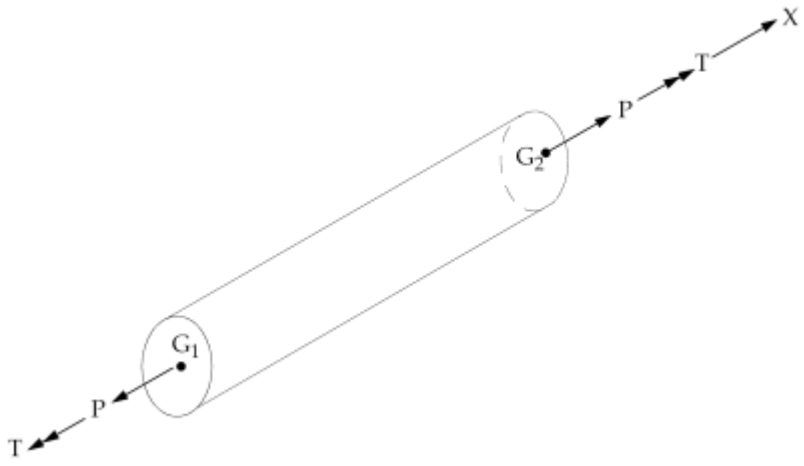
Figure 12-100: Rod Element Coordinate System and Element Forces



You can also request axial and torsional stress output with the STRESS case control command. The positive stress directions are the same as those indicated for the force vectors in the figure above.

Remarks related to SOLs 402 and 414:

1. CROD defines a truss element with no torsional stiffness, i.e., only axial force is transmitted by the element. Rotational degrees of freedom are introduced in order to connect with shells but they do not contribute to the element stiffness.

Related Topics

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CSET

Free Boundary Degrees-of-Freedom

Defines analysis set (a-set) degrees-of-freedom to be free (c-set) during component modes calculations.

Format:

1	2	3	4	5	6	7	8	9	10
CSET	ID1	C1	ID2	C2	ID3	C3	ID4	C4	

Example:

CSET	124	1	5	23	6	16			

Fields:

Field	Contents
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component numbers. (Integer zero or blank for scalar points, or any unique combination of the integers 1 through 6 for grid points with no embedded blanks.)

Remarks:

1. CSET and BNDFREE entries are equivalent.
2. If there are no CSETi or BSETi entries present, all a-set degrees-of-freedom are considered fixed during component modes analysis. If there are only BSETi entries present, any a-set degrees-of-freedom not listed are placed in the free boundary set (c-set). If there are both BSETi and CSETi entries present, the c-set degrees-of-freedom are defined by the CSETi entries, and any remaining a-set points are placed in the b-set.
3. Degrees-of-freedom specified on this entry form members of the mutually exclusive c-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

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Related Topics

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CSET1

Free Boundary Degrees-of-Freedom, Alternate Form of CSET Entry

Defines analysis set (a-set) degrees-of-freedom to be free (c-set) during component modes calculations.

Format:

1	2	3	4	5	6	7	8	9	10
CSET1	C	ID1	ID2	ID3	ID4	ID5	ID6	ID7	
	ID8	ID9	-etc.-						

Example:

CSET1	124	1	5	7	6	9	12	122	
	127								

Alternate Formats and Example:

CSET1	C	ID1	"THRU"	ID2					
CSET1	3	6	THRU	32					

CSET1		"ALL"							
CSET1		ALL							

Fields:

Field	Contents
C	Component number. (Integer zero or blank for scalar points, or any unique combination of the Integers 1 through 6 for grid points with no embedded blanks.)
IDi	Grid or scalar point identification number. (Integer > 0; For THRU option, ID1 < ID2)

Remarks:

1. CSET1 and BNDFREE1 entries are equivalent.
2. If there are no CSETi or BSETi entries present, all a-set degrees-of-freedom are considered fixed during component modes analysis. If there are only BSETi entries present, any a-set degrees-of-freedom not listed are placed in the free boundary set (c-set). If there are both BSETi and CSETi entries present, the c-set degrees-of-freedom are defined by the CSETi entries, and any remaining a-set points are placed in the b-set.

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3. Degrees-of-freedom specified on this entry form members of the mutually exclusive c-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

Related Topics

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CSHEAR

Shear Panel Element Connection

Defines a shear panel element.

Format:

1	2	3	4	5	6	7	8	9	10
CSHEAR	EID	PID	G1	G2	G3	G4			

Example:

CSHEAR	3	6	1	5	3	7			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PSHEAR entry. (Integer > 0; Default = EID)
Gi	Grid point identification numbers of connection points. (Integer > 0; G1 ≠ G2 ≠ G3 ≠ G4)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. Grid points G1 through G4 must be ordered consecutively around the perimeter of the element.
3. All interior angles must be less than 180°.

Figure 12-101: CSHEAR Element Connection and Coordinate System

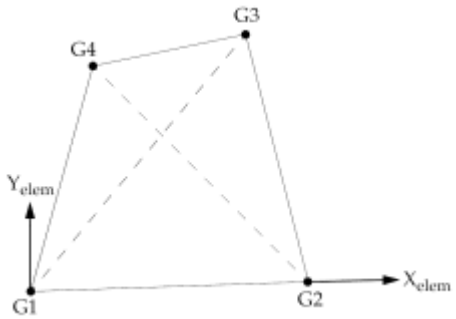

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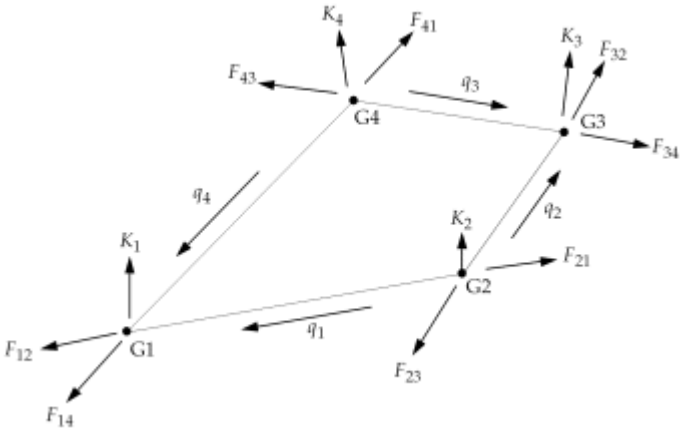
Figure 12-102: CSHEAR Element Corner Forces and Shear Flows



Related Topics

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CSMADD

Wetted Face and Edge Set Combination

Combines multiple CSMSET entries into a single set of wetted faces and edges for a SOL 401 FSI Co-Simulation with STAR CCM+.

Format:

1	2	3	4	5	6	7	8	9	10
CSMADD	CSID	CSID1	CSID2	CSID3	CSID4	CSID5	CSID6	CSID7	
	CSID8	CSID9	-etc-						

Example:

CSMADD	12	1	3	5	7	9			

Fields:

Field	Contents
CSID	ID referenced by the CMSET case control command. (Integer > 0)
CSIDi	IDs of CSMSET bulk entries. (Integer > 0)

Remarks:

1. CSMADD combines multiple CSMSET bulk entries into a single set of wetted faces and edges for a SOL 401 FSI Co-Simulation with STAR CCM+. The individual CSMSET bulk entries select the wetted regions defined with the BSURF entry for shell elements, the BSURFS entry for solid element faces, or the BEDGE entry for element edges.
2. The CSID defined on the CSMADD, and all CSIDi referencing CSMSET entries, must all be unique.

Related Topics

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CSMSET

Wetted Face and Edge Set

Defines a set of wetted faces and edges for a SOL 401 FSI Co-Simulation with STAR CCM+.

Format:

1	2	3	4	5	6	7	8	9	10
CSMSET	CSID	BID	DESCID						

Example:

CSMSET	1	100	200						

Fields:

Field	Contents
CSID	Co-simulation (Wetted) region ID. (Integer > 0). Must be unique with respect to other CSMSET bulk enties.
BID	ID of BSURF, BSURFS, or BEDGE bulk entry.
DESCID	ID of DESC bulk entry containing a description of the CSMSET. This ID is optional. If blank, no description is used for the CSMSET. (Integer≥0, Default=0)

Remarks:

1. CSMSET defines a set of wetted faces and edges for a SOL 401 FSI Co-Simulation with STAR CCM+.
2. The wetted region can be defined with the BSURF entry for shell elements, the BSURFS entry for solid element faces, or the BEDGE entry for element edges.

Related Topics

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CSSCHD

Aerodynamic Control Surface Schedule Input

Defines a scheduled control surface deflection as a function of Mach number and angle of attack.

Format:

1	2	3	4	5	6	7	8	9	10
CSSCHD	SID	AESID	LALPHA	LMACH	LSCHD				

Example:

CSSCHD	5	ELEV	12	15	25				

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
AESID	ID of an AESURF Bulk Data entry to which the schedule is being attached.
LALPHA	ID of an AEFACT Bulk Data entry containing a list of angles of attack (in radians) at which schedule information is provided. (Integer > 0, Default = no angle information provided.)
LMACH	ID of an AEFACT Bulk Data entry containing a list of Mach numbers at which schedule information is provided. (Integer > 0, default = no Mach information provided.)
LSCHD	ID of an AEFACT Bulk Data entry which contains the scheduling information. See Remarks 4 and 5 (Integer > 0, no Default).

Remarks:

1. Control system schedules must be selected with the Case Control command CSSCHD = SID.
2. The AESID cannot appear on an AELINK or TRIM Bulk Data entry for the same subcase.
3. The control surface deflection is computed using a linear interpolation for the Mach number provided on the associated TRIM entry and the angle of attack derived as part of the trim calculation.
4. The LSCHD data are provided as a list of deflections (in radians) as a function of Mach numbers and angles of attack. If there are NMACH Mach numbers and NALPHA angles of attack, the first NALPHA deflections are for the first Mach number, the next NALPHA are for the second Mach number, and so on, until the last NALPHA deflections are for the final Mach number.

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5. if LALPHA is blank, LSCHD contains NMACH deflections to define the Mach schedule. If LMACH is blank, LSCHD contains NALPHA deflections to define the angle of attack schedule.
6. LALPHA and LMACH cannot be simultaneously blank. If LALPHA or LMACH are not blank, at least two values of angle of attack or Mach number must be defined in order to perform interpolation.
7. If the Mach number or angle of attack is outside the range specified by the tabulated values, the value at the table end is used. That is, data are not extrapolated.

Related Topics

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CSUPER

Secondary Superelement Connection

Defines the grid or scalar point connections for identical or mirror image superelements or superelements from an external source. These are all known as secondary superelements.

Format:

1	2	3	4	5	6	7	8	9	10
CSUPER	SSID	PSID	GP1	GP2	GP3	GP4	GP5	GP6	
	GP7	GP8	-etc.-						

Example:

CSUPER	120003	21	3	6	4	10			

Fields:

Field	Contents
SSID	Coded identification number for secondary superelement. See Remark 1 . (Integer > 0)
PSID	Identification number for referenced primary superelement. See Remark 2 . (Integer > 0 or blank)
GPi	Grid or scalar point identification numbers of the exterior points of the secondary superelement. See Remark 3 . (Integer > 0)

Remarks:

1. The value of SSID is written in the form XXX0000 + n, where n is the referenced secondary superelement identification number and n must be less than 10000 and XXX is a displacement component sign reversal code as follows:

   The sign reversal code specifies the displacement component(s) normal to the plane of the mirror through which the reflection is to be made

Blank or 0	No reversal for identical superelement. If PSID is preceded by a minus sign and there is no xxx code on SSID, then a z-reversal mirror is generated.	
1 2 3	x-reversal y-reversal z-reversal}	Mirror Images
12	x and y-reversal	Identical Images

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23 31	y and z-reversal z and x-reversal}	
123	x, y, and z-reversal	Mirror Images

2. If PSID = 0 or blank, the superelement boundary matrices are obtained from an external source (such as a database or external file). See also PARAM, EXTOUT.

   If PSID ≠ 0, the secondary superelement is identical to, or is a mirror image of, a primary superelement.

3. For identical or mirror image superelements, the grid point IDs, GPi, may appear in any order. However, if they are not in the same order as the external GRIDs of the primary superelement, then the SEQSEP entry is also required. In case of external superelements, the GRID IDs must be in the order that the terms in the associated matrices occur in.

4. Image superelements and their primaries must be congruent. The identical or mirror image superelement must have the same number of exterior grid points as its primary superelement. The exterior grid points of the image superelement must have the same relative location to each other as do the corresponding points of the primary superelement. The global coordinate directions of each exterior grid point of the image superelement must have the same relative alignment as those of the corresponding grid points of the primary superelement. If congruency is not satisfied because of round-off, then the tolerance may be adjusted with PARAM,CONFAC or DIAG 37.

5. For superelements from an external source, please refer to PARAMS EXTDR,EXTDROUT, EXTDRUNT, EXTOUT,, and EXTUNIT.

Related Topics

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CSUPEXT

Superelement Exterior Point Definition

Assigns exterior points to a superelement.

Format:

1	2	3	4	5	6	7	8	9	10
CSUPEXT	SEID	GP1	GP2	GP3	GP4	GP5	GP6	GP7	

Example:

CSUPEXT	2	147	562	937					

Alternate Format and Example:

CSUPEXT	SEID	GP1	"THRU"	GP2					

CSUPEXT	5	12006	THRU	12050					

Fields:

Field	Contents
SEID	Identification number of a primary superelement. (Integer > 0)
GPi	Grid or scalar point identification number in the downstream superelement or residual structure. (Integer > 0 or "THRU"; for "THRU" option, GP1 < GP2)

Remarks:

1. Grid or scalar points are connected (that is, are exterior) to a superelement only if they are connected by structural, rigid, or plot elements. MPC entries are not regarded as elements. This entry is a means of providing connectivity for this purpose.
2. Open sets are allowed with the "THRU" option.
3. Scalar points may be interior to the residual structure (SEID = 0) only.
4. This entry may be applied only to the primary superelements. The CSUPER entry is used for secondary superelements (identical image, mirror image, and external superelements).

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Related Topics

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CTETRA

Four-Sided Solid Element Connection

Defines the connections of the four-sided solid element with four to ten grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CTETRA	EID	PID	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	G10					

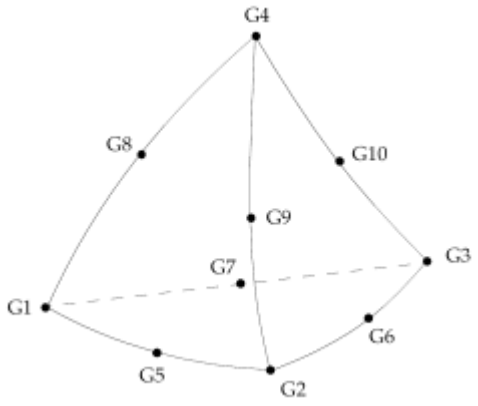
Example:

CTETRA	112	2	3	15	14	4	103	115	
	5	16	8	27					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0; Required)
PID	Property identification number of a PSOLID, PLSOLID, or PMIC entry. (Integer > 0)
Gi	Identification numbers of connected grid points. (Integer > 0 or blank)

Figure 12-103: CTETRA Element Connection



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Remarks:

1. Element ID numbers must be unique with respect to all other element ID numbers.
2. The topology of the diagram must be preserved, i.e., G1, G2, G3 define a triangular face; G1, G8, and G4 are on the same edge, etc.
3. The edge points, G5 to G10, are optional. Any or all of them may be deleted. If the ID of any edge connection point is left blank or set to zero, the equations of the element are adjusted to give correct results for the reduced number of connections. Corner grid points cannot be deleted. The element is an isoparametric element in all cases.
4. Components of stress are output in the material coordinate system. See remark 6 on the PSOLID bulk entry for hyperelastic and nonlinear exceptions.
5. For nonhyperelastic elements, the element coordinate system is derived from the three vectors R, S, and T, which join the midpoints of opposite edges.

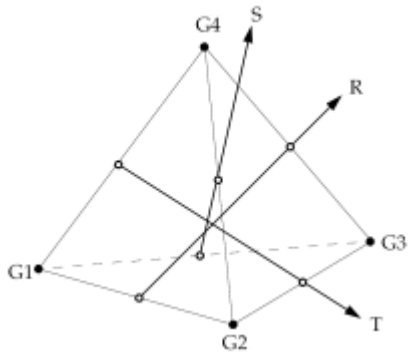
   R vector joins midpoints of edges G1-G2 and G3-G4.

   S vector joins midpoints of edges G1-G3 and G2-G4.

   T vector joins midpoints of edges G1-G4 and G2-G3.

   The origin of the coordinate system is located at G1. The element coordinate system is chosen as close as possible to the R, S, and T vectors and points in the same general direction. (Mathematically speaking, the coordinate system is computed in such a way that, if the R, S, and T vectors are described in the element coordinate system, a 3x3 positive definite symmetric matrix would be produced.)

Figure 12-104: CTETRA Element R, S, and T Vectors



6. It is recommended that the edge points be located within the middle third of the edge.
7. For hyperelastic elements, the plot codes are specified under the CTETRAFD element name in **"Item Codes"**.
8. By default, all of the six edges of the element are considered straight unless any of G5 through G10 are specified.

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Remarks related to SOLs 402 and 414:

1. Output results will be done in the material coordinate system. If those axes are not defined, the material coordinate system is initially defined as the basic coordinate system. The material coordinate system always rotates with the element.

Related Topics

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CTRAX3

Axisymmetric Triangular Element Connection

Defines an isoparametric and axisymmetric triangular cross-section ring element for use in linear and fully nonlinear (i.e., large strain and large rotations) hyperelastic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CTRAX3	EID	PID	G1	G2	G3	TH			

Example:

CTRAX3	203	100	10	20	30	30.0			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PSOLID or PLSOLID entry. (Integer > 0; Default=EID)
Gi	Grid point identification number of connected points. (Unique integers > 0)
TH	Material property orientation angle in degrees. See Remark 4. TH is ignored for hyperelastic elements. (Real; Default = 0.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. When axisymmetric elements are defined on the XZ-plane, X is the radial direction, and Z is the axial direction. The grid points defining these elements must have X ≥ 0. See Figure 12-106.

   When axisymmetric elements are defined on the XY-plane, Y is the radial direction, and X is the axial direction. The grid points defining these elements must have Y ≥ 0. See Figure 12-107.

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4. For orthotropic materials, the TH angle orients the principal material coordinates of the element relative to the basic coordinates. **Figure 12-106** and **Figure 12-107** show the positive sense of the TH angle when the element is defined on the XZ-plane and XY-plane, respectively.
5. The initial element coordinate system is the basic coordinate system.
6. See the remarks on the **PSOLID** entry for information on stress and strain output coordinate systems.
7. The CQUADX4, CQUADX8, CTRAX3, and CTRAX6 elements behave linearly in SOL 106 and 129 when not hyperelastic.
8. For hyperelastic element, the plot codes are specified under the CTRAX3FD element name in **Item Codes Overview**.
9. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
10. A concentrated load (e.g. FORCE entry) at Gi is the total of the force around the circumference. Reaction force and applied load output are the same. For example, to apply a distributed load of 135.0 Newton/mm on a single grid where the radius is 0.5 mm:

    The value entered on a FORCE entry = (Distributed force * 2 * π * Radius)

    = 135.0 N/mm * 2 * π * 0.5 mm

    = 424.1 N

    See **system cell 587** for information on the pre-version 10 axisymmetric element behavior.

Figure 12-105: CTRAX3 Element Idealization

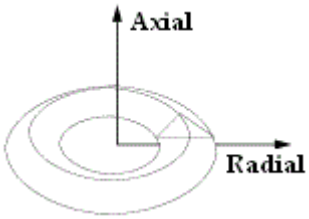

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Figure 12-106: CTRAX3 Element Geometry and Coordinate Systems, XZ-Plane

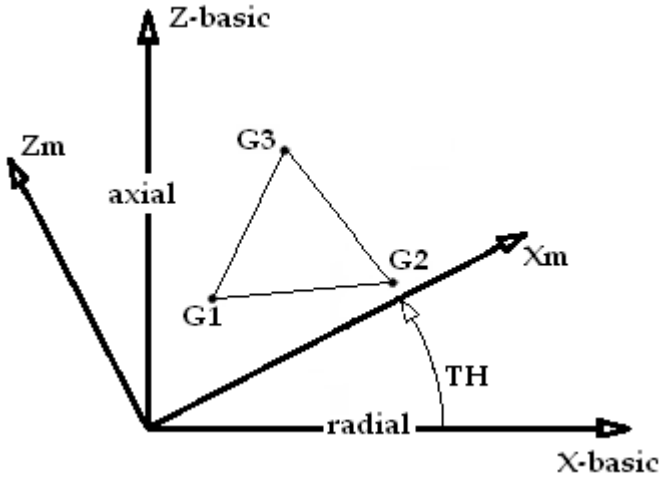
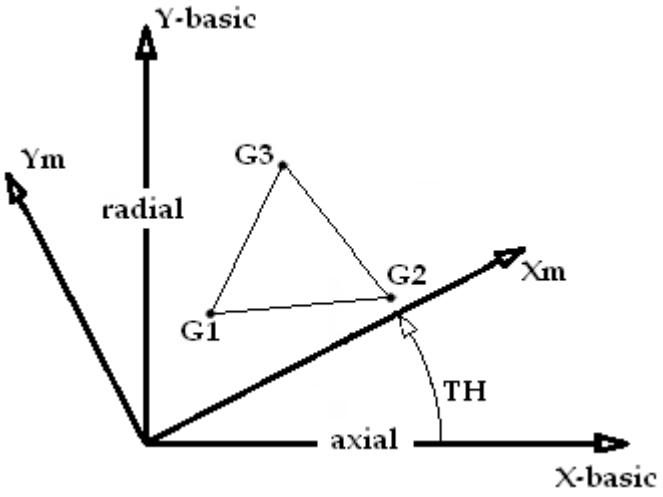


Figure 12-107: CTRAX3 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The SPCFORCES output results for the axisymmetric elements are always given in force unit per 2*pi radians and not per radian. System cell 587 is ignored.

Related Topics

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CTRAX6

Axisymmetric Triangular Element Connection

Defines an isoparametric and axisymmetric triangular cross-section ring element with midside nodes for use in linear and fully nonlinear (i.e., large strain and large rotations) hyperelastic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CTRAX6	EID	PID	G1	G2	G3	G4	G5	G6	
	TH								

Example:

CTRAX6	203	100	10	20	30	40	50	60	
	30.0								

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PSOLID or PLSOLID entry. (Integer > 0; Default=EID)
Gi	Grid point identification numbers of connected points (midside grids cannot be eliminated). (Unique integers > 0)
TH	Material property orientation angle in degrees. See Remark 4. TH is ignored for hyperelastic elements. (Real; Default = 0.0)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. All of the grid points in the element connectivity must lie in a plane. The plane can be one of the following:
   - XZ-plane of the basic coordinate system.
   - XY-plane of the basic coordinate system.

   The software automatically determines the orientation.
3. When axisymmetric elements are defined on the XZ-plane, X is the radial direction, and Z is the axial direction. The grid points defining these elements must have X ≥ 0. See Figure 12-109.

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When axisymmetric elements are defined on the XY-plane, Y is the radial direction, and X is the axial direction. The grid points defining these elements must have Y ≥ 0. See **Figure 12-110**.

4. For orthotropic materials, the TH angle orients the principal material coordinates of the element relative to the basic coordinates. **Figure 12-109** and **Figure 12-110** show the positive sense of the TH angle when the element is defined on the XZ-plane and XY-plane, respectively.
5. The initial element coordinate system is the basic coordinate system.
6. See the remarks on the **PSOLID** entry for information on stress and strain output coordinate systems.
7. The CQUADX4, CQUADX8, CTRAX3, and CTRAX6 elements behave linearly in SOL 106 and 129 when not hyperelastic.
8. For hyperelastic element, the plot codes are specified under the CTRAX6FD element name in **Item Codes Overview**.
9. For any grid point (Gi) selected on axisymmetric, plane stress, or plane stain elements, if you select a displacement coordinate system with the CD field on the GRID entry, you must orient the system according to the following rules:
   - If the elements are defined on the XY-plane of the basic coordinate system, the Z-axis is the out-of-plane direction. In this case, you must orient the Z-axis (φ-axis for a spherical system) of the displacement coordinate system to be parallel with the Z-axis of the basic coordinate system.
   - If the elements are defined on the XZ-plane of the basic coordinate system, the Y-axis is the out-of-plane direction. In this case, you must orient the Y-axis (θ-axis for the cylindrical and spherical systems) of the displacement coordinate system to be parallel with the Y-axis of the basic coordinate system.
10. A concentrated load (e.g. FORCE entry) at Gi is the total of the force around the circumference. Reaction force and applied load output are the same. For example, to apply a distributed load of 135.0 Newton/mm on a single grid where the radius is 0.5 mm:

    The value entered on a FORCE entry = (Distributed force * 2 * π * Radius)

    = 135.0 N/mm * 2 * π * 0.5 mm

    = 424.1 N

    See **system cell 587** for information on the pre-version 10 axisymmetric element behavior.

Figure 12-108: CTRAX6 Element Idealization

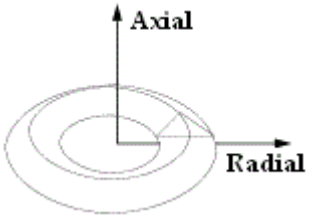

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Figure 12-109: CTRAX6 Element Geometry and Coordinate Systems, XZ-Plane

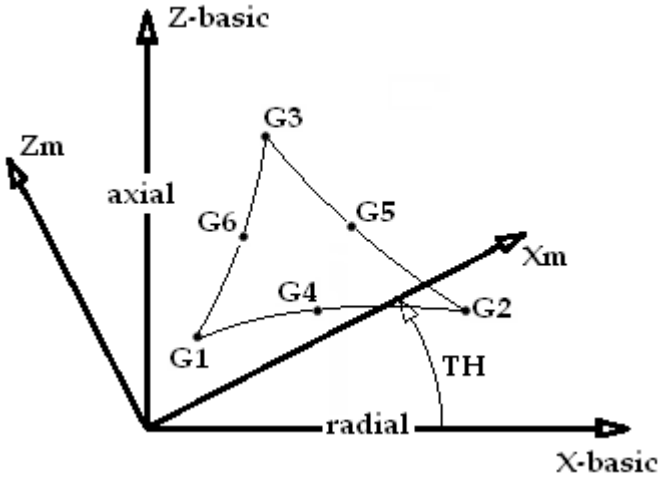
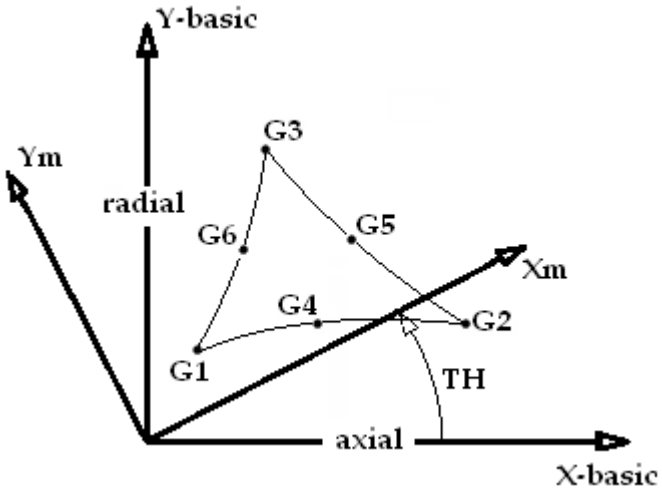


Figure 12-110: CTRAX6 Element Geometry and Coordinate Systems, XY-Plane



Remarks related to SOL 402:

1. The SPCFORCES output results for the axisymmetric elements are always given in force unit per 2*pi radians and not per radian. System cell 587 is ignored.

Related Topics

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CTRIA3

Triangular Plate Element Connection

Defines an isoparametric membrane-bending or plane strain triangular plate element.

Format:

1	2	3	4	5	6	7	8	9	10
CTRIA3	EID	PID	G1	G2	G3	THETA or MCID	ZOFFS		
		TFLAG	T1	T2	T3				

Example:

CTRIA3	111	203	31	74	75	3.0	0.98		
			1.77	2.04	2.09				

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PCOMP, PCOMPG, PLPLANE, PMIC, or PSHELL entry. (Integer > 0; Default = EID)
Gi	Grid point identification numbers of connection points. (Integers > 0, all unique)
THETA	Material property orientation angle in degrees. THETA is ignored for hyperelastic elements. (Real; Default = 0.0)
MCID	Material coordinate system identification number. The x-axis of the material coordinate system is determined by projecting the x-axis of the MCID coordinate system (defined by the CORDij entry or zero for the basic coordinate system) onto the surface of the element. MCID is ignored for hyperelastic elements. See Remark 11. (Integer ≥ 0; if blank, then THETA = 0.0 is assumed.)

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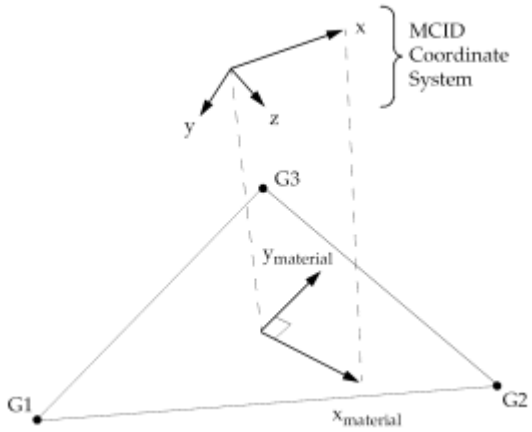
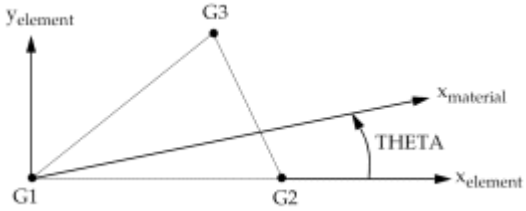
Field	Contents

Figure 12-111: MCID Coordinate System Definition



ZOFFS	Offset from the surface of grid points to the element reference plane. See Remarks 5 and 6. ZOFFS is ignored for hyperelastic elements. (Real)
TFLAG	Integer Flag which specifies how Ti is used to define thickness of element. (0, 1, or blank)
Ti	Thickness of element at grid points G1 through G3 if TFLAG=0 or blank. If TFLAG=1, thickness becomes a product of Ti and the thickness on the PSHELL card. Ti is ignored for hyperelastic elements. See Remark 4. (Real > 0.0 or blank. See Remark 2 for default.)

Figure 12-112: CTRIA3 Element Geometry and Coordinate Systems



Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. The continuation is optional. If it is not supplied, then T1 through T3 will be set equal to the value of T on the PSHELL entry.
3. When the PID refers to either a PCOMP or PCOMPG entry, the software converts the PCOMP/PCOMPG entry to equivalent PSHELL/MAT2 entries. If Ti is defined, they override or scale the values on the equivalent PSHELL entry. It should be noted that ply-stress and ply-strain recovery uses the ply thicknesses defined on the PCOMP/PCOMPG entries, and not Ti.

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4. By default, a fatal error occurs if a Ti value is zero. To allow for Ti values of zero, use System Cell 495.

5. Elements may be offset from the connection points by means of ZOFFS. Other data, such as material matrices and stress fiber locations, are given relative to the reference plane. A positive value of ZOFFS implies that the element reference plane is offset a distance of ZOFFS along the positive z-axis of the element coordinate system. If the ZOFFS field is used, then both the MID1 and the MID2 fields must be specified on the PSHELL entry referenced by PID on the CTRIA3 entry.

   ZOFFS should not be used when differential stiffness is calculated in solutions 105 and 200, and solutions 103, 107 - 112 using STATSUB, since the differential stiffness calculation does not include the offset vectors.

   ZOFFS is ignored in heat transfer solutions 153 and 159 including any solution subcase which includes the case control ANALYSIS=HEAT.

6. ZOFFS has the following caveats in nonlinear solutions 106 and 129:

   - For geometry nonlinear only, the differential stiffness calculation does not include the offset vectors. The exclusion of the offset in the differential stiffness will have an adverse effect on convergence, however, if the solution converges, the results will be correct. In a nonlinear solution, equilibrium (and convergence) is achieved when internal and external forces balance. The offset is correctly accounted for in the internal and external force calculations even though it is not accounted for in the differential stiffness.
   - ZOFFS should not be defined on elements which use MATS1 nonlinear material definitions.
   - ZOFFS is ignored for hyperelastic elements.
   - ZOFFS should not be used in nonlinear solutions 106 and 129 in combination with thermal loads.

7. The reference coordinate system for the output of stress, strain and element force depends on the element type.

   - For CTRIA3 elements which are not hyperelastic, the reference coordinate system for output is the element coordinate system. For SOL 106 geometry nonlinear analysis, the nonlinear stresses are output in deformed element coordinate system.
   - For hyperelastic elements, the stress and strain are output according to CID on the PLPLANE entry.

8. For hyperelastic elements, the plot codes are specified under the CTRIAFD element name in **"Item Codes"**.

9. System Cell 218 allows the user to control the minimum vertex angle for TRIA3 elements at which USER WARNING MESSAGE 5491 is issued. The default value is 10.0 degrees.

10. By default, all of the three edges of the element are considered straight.

11. If the x-direction of a material coordinate system is perpendicular or very close to being perpendicular to the element face, a projection is either impossible or unpredictable. A fatal error will occur when the angle between a shell element face normal and the x-direction of the material coordinate system (MCID) is less than the value specified by System Cell 489.

12. The MCID should be less than 99,999,999 when SYSTEM(525)=0 (default), and less than 99,999,999,999 when SYSTEM(525)=1.

13. For SOLs 401 and 402, see Shell elements in the Multi-Step Nonlinear User's Guide for details.

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Remarks related to SOLs 402 and 414:

1. Supported only if system cell 370 is turned on. Otherwise, a fatal error is issued: use CQUADR instead or turn on the system cell.

Related Topics

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CTRIA6

Curved Triangular Shell Element Connection

Defines a curved triangular shell or plane strain element with six grid points.

Format:

1	2	3	4	5	6	7	8	9	10
CTRIA6	EID	PID	G1	G2	G3	G4	G5	G6	
	THETA or MCID	ZOFFS	T1	T2	T3	TFLAG			

Example:

CTRIA6	302	3	31	33	71	32	51	52	
	45	0.03	0.02	0.025	0.025				

Fields:

Field	Contents
EID	Element identification number. See Remark 1. (Integer > 0)
PID	Property identification number of PSHELL, PCOMP, PCOMPG, or PLPLANE entry. (Integer > 0; No default)
G1, G2, G3	Identification numbers of connected corner grid points. Required data for all three grid points. See Remark 2. (Unique integers > 0; No default)
G4, G5, G6	Identification numbers of connected edge grid points. Required data for all three grid points unless the element is hyperelastic. See Remarks 2, 3, and 4. (Unique integers ≥ 0; No default)
THETA or MCID	If input is real or the field is blank, material orientation angle (THETA) in degrees. If input is integer, identification number of a coordinate system (MCID) from which THETA is calculated. Use a CORDij bulk entry to define the coordinate system. To use the basic coordinate system, enter integer zero. THETA or MCID are ignored for hyperelastic elements. See Remarks 7, 8, 9, and 10. (Real or integer ≥ 0; Default = 0.0)
ZOFFS	Offset from the surface of grid points to the element reference plane. See Remark 11. ZOFFS is ignored for hyperelastic elements. (Real)
TFLAG	Integer Flag which specifies how Ti is used to define thickness of element. (0, 1, or blank)
Ti	If TFLAG=0 or the TFLAG field is blank, thickness of the element at grid points G1 through G4. If TFLAG=1, scaling factor used to compute the thickness of the element

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Field	Contents
	at grid points G1 through G4 as the product of Ti and the thickness specified on the PSHELL bulk entry. Ti is ignored for hyperelastic elements. See Remarks 5 and 6. (Real > 0.0; If blank, use the thickness specified on the PSHELL bulk entry as the default)

Remarks:

1. Element identification numbers should be unique with respect to the element identification numbers of all other elements regardless of element type.
2. Grid points G1 through G6 must be numbered as shown in Figure 12-114.
3. For best results, locate mid-side grid points within the middle third of the edge.
4. By default, a fatal error occurs if a G1 through G6 value is not specified. However, if the element is hyperelastic and there are no mid-side grid points, the element is processed identically to the hyperelastic CTRIA3 element. The variable grid point CTRIA6 is supported with the pre-Simcenter Nastran 2019.2 formulation by specifying SYSTEM(700)=1, except for SOLs 401 and 402 which always use the newer formulation and ignore the system cell setting.
5. When the PID refers to either a PCOMP or PCOMPG bulk entry, the software converts the PCOMP/PCOMPG bulk entry to equivalent PSHELL/MAT2 bulk entries. If Ti are defined, depending on the TFLAG setting, they either override or scale the values on the equivalent PSHELL bulk entry. The software uses the ply thicknesses defined on the PCOMP/PCOMPG bulk entries, to recover ply stresses and strains.
6. A fatal error occurs if a Ti value is zero. To allow for Ti values of zero, specify SYSTEM(495)=1.
7. If MCID is specified, the software uses the following procedure to calculate the value for THETA, which it then uses to determine the material coordinate system for the element as described in Remark 9.

   The following notation is used in this remark and Figure 12-113:

   $(x_{mcid}, y_{mcid}, z_{mcid})$
   Axes of the MCID coordinate system

   $(\hat{x}_{mcid}, \hat{y}_{mcid}, \hat{z}_{mcid})$
   Unit vectors aligned with the axes of the MCID coordinate system

   $(x_{mref}, y_{mref}, z_{mref})$
   Axes of the material coordinate system relative to the reference plane

   $(\hat{x}_{mref}, \hat{y}_{mref}, \hat{z}_{mref})$
   Unit vectors aligned with the axes of the material coordinate system relative to the reference plane

   a. Because the element can be curved and warped, the software begins by creating a reference plane for the element. The unit normal to the reference plane is calculated as follows:

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$$\hat{z}_{mref} = \frac{\vec{r}_{G1G2} \times \vec{r}_{G1G3}}{\left\| \vec{r}_{G1G2} \times \vec{r}_{G1G3} \right\|}$$

where $\vec{r}_{G1G2}$ is a vector that extends from G1 to G2, and $\vec{r}_{G1G3}$ is a vector that extends from G1 to G3.

b. The x-axis of the MCID coordinate system is projected along the $\hat{z}_{mref}$ direction onto the reference plane to obtain the x-axis of the material coordinate system on the reference plane.

c. The $\vec{r}_{G1G2}$ vector is projected along the $\hat{z}_{mref}$ direction onto the reference plane to obtain the $\vec{r}_{G1'G2'}$ vector.

d. The software calculates the angle θ in degrees from the scalar dot product as follows:

$$\theta = \cos^{-1}\left(\frac{\hat{x}_{mref} \cdot \vec{r}_{G1'G2'}}{\left\| \vec{r}_{G1'G2'} \right\|} \right)$$

e. To determine if angle θ lies in the 3rd or 4th quadrants, the software evaluates the algebraic sign of $\sin\theta$ from the vector cross product as follows:

$$(\sin\theta)\hat{z}_{mref} = \frac{\vec{r}_{G1'G2'}}{\left\| \vec{r}_{G1'G2'} \right\|} \times \hat{x}_{mref}$$

If $\sin\theta < 0$, the software adds 180° to the calculated value of θ.

f. The software uses θ as the value for THETA.

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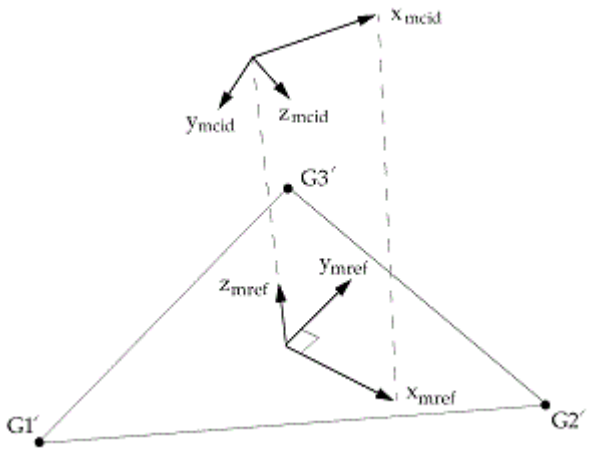
Figure 12-113: Projection of MCID Coordinate System X-Axis onto the Reference Plane



8. If the x-axis of the MCID coordinate system is perpendicular or very close to being perpendicular to the reference plane, projecting the x-axis of the MCID coordinate system onto the reference plane is either impossible or unpredictable. Thus, a fatal error occurs when the angle between the normal to the reference plane and the x-axis of the MCID coordinate system is less than the value specified by System Cell 489.

9. The software uses the following procedure to calculate the orientation of the material and local coordinate systems.

The following notation is used in this remark and Figures **Figure 12-113** and **Figure 12-114**:

$(x_{mref}, y_{mref}, z_{mref})$
Axes of the material coordinate system relative to the reference plane

$(\hat{x}_{mref}, \hat{y}_{mref}, \hat{z}_{mref})$
Unit vectors aligned with the axes of the material coordinate system relative to the reference plane

(x_m, y_m, z_m)
Axes of the material coordinate system

$(\hat{x}_m, \hat{y}_m, \hat{z}_m)$
Unit vectors aligned with the axes of the material coordinate system

(x_l, y_l, z_l)
Axes of the local coordinate system

$(\hat{x}_l, \hat{y}_l, \hat{z}_l)$
Unit vectors aligned with the axes of the local coordinate system

a. Because the element can be curved and warped, the software begins by creating a reference plane for the element. The unit normal to the reference plane is calculated as follows:

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$$\hat{z}_{mref} = \frac{\vec{r}_{G1G2} \times \vec{r}_{G1G3}}{\left\| \vec{r}_{G1G2} \times \vec{r}_{G1G3} \right\|}$$

where $\vec{r}_{G1G2}$ is a vector that extends from G1 to G2, and $\vec{r}_{G1G3}$ is a vector that extends from G1 to G3.

b. The $\vec{r}_{G1G2}$ vector is projected along the $\hat{z}_{mref}$ direction onto the reference plane to obtain the $\vec{r}_{G1'G2'}$ vector.

c. The $\vec{r}_{G1'G2'}$ vector is rotated THETA degrees about $\hat{z}_{mref}$. The direction of the rotated $\vec{r}_{G1'G2'}$ vector is the direction of the x-axis of the material coordinate system on the reference plane.

d. To account for curvature and warping, at each corner grid point and Gauss point, the x-axis of the material coordinate system on the reference plane is rotated in the plane that it forms with the unit normal to the surface at that point, $\hat{n}_{Gi}$, so that it is tangent to the surface. The rotated axis is the x-axis of the material coordinate system at that point.

e. The z-axis of the material coordinate system at each corner grid point and Gauss point is the normal to the surface at that point. Thus, it follows that:

$$\hat{z}_m = \hat{n}_{Gi}$$

f. The unit vector in the direction of the y-axis of the material coordinate system at each corner grid point and Gauss point is obtained as follows:

$$\hat{y}_m = \hat{z}_m \times \hat{x}_m$$

g. The x-axis of the local coordinate system at each corner grid point and Gauss point is tangent to the line of constant η that passes through the point.

h. The z-axis of the local coordinate system at each corner grid point and Gauss point is the normal to the surface at that point. Thus, it follows that:

$$\hat{z}_l = \hat{z}_m = \hat{n}_{Gi}$$

i. The y-axis of the local coordinate system at each corner grid point and Gauss point is obtained as follows:

$$\hat{y}_l = \hat{z}_l \times \hat{x}_l$$

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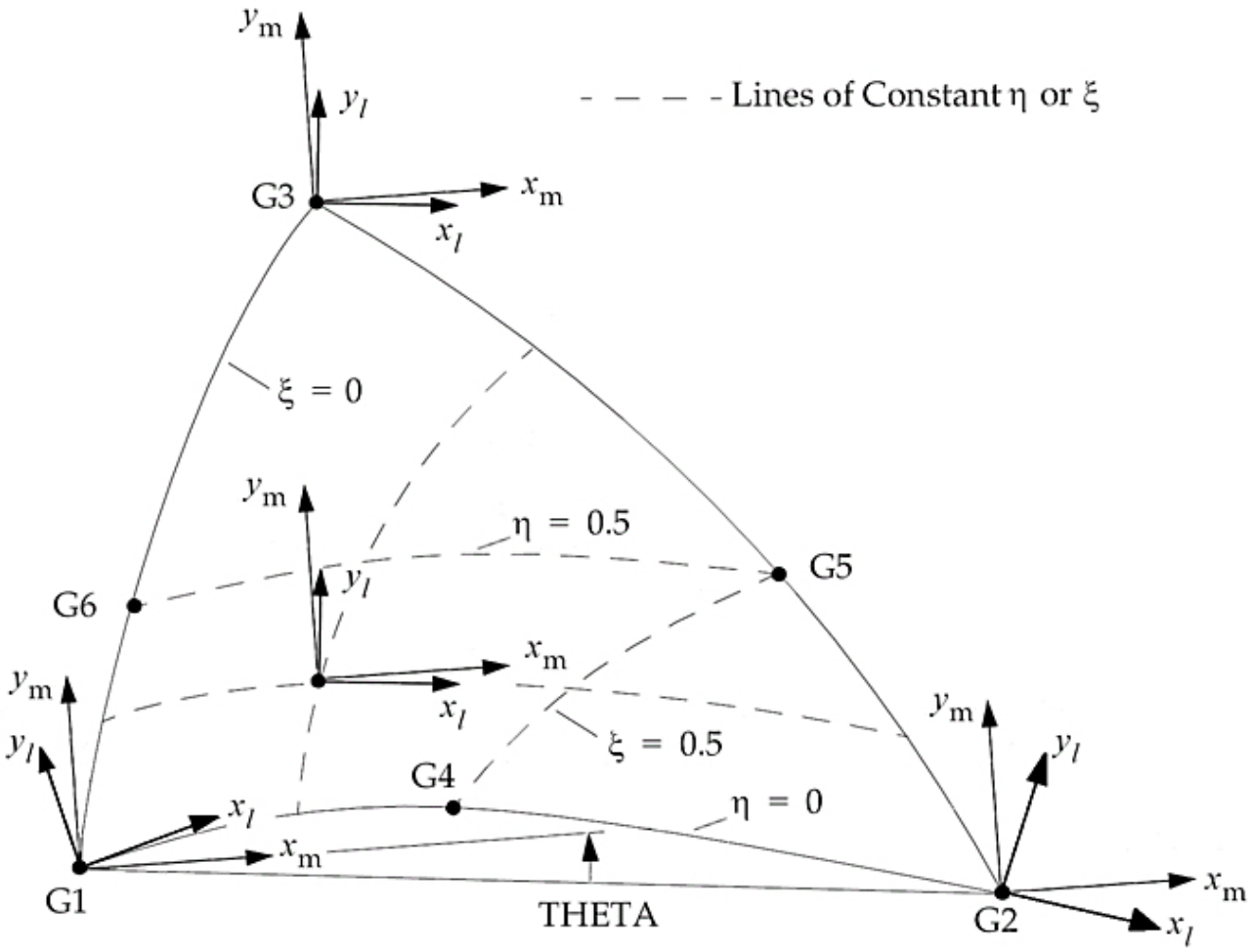
Figure 12-114: CTRIA6 Connectivity and Local and Material Coordinate Systems



10. MCID must be less than 99,999,999 when SYSTEM(525)=0 (default), and less than 99,999,999,999 when SYSTEM(525)=1.

11. Data such as material matrices and stress fiber locations, are given relative to the reference plane. A positive value of ZOFFS implies that the reference plane is offset a distance of ZOFFS along the positive z-axis of the local coordinate system. If the ZOFFS field is used, then both the MID1 and the MID2 fields must be specified on the PSHELL bulk entry referenced by the CTRIA6 element.

ZOFFS is ignored in heat transfer SOLs 153 and 159 including any solution subcase which includes the case control ANALYSIS=HEAT.

12. Stress and strain are output at the corner grid points and Gauss points in the local coordinate system. Figure 12-114 shows the local coordinate system at these points. For hyperelastic elements, the stress and strain are output in the coordinate system identified by the CID field on the PLPLANE bulk entry.

13. The CTRIA6 formulation uses 6-noded serendipity shape functions for the translational and rotational DOF.

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By default, three Gauss points are used to integrate the membrane, bending, transverse shear, and coupled membrane-bending stiffness.

14. For hyperelastic elements, the plot codes are specified under the CTRIAFD element name in “Item Codes”.
15. To revert to the pre-Simcenter Nastran 2019.2 formulation, include SYSTEM(700)=1 in your input file. Because the new formulation is the only formulation in SOLs 401 and 402, the software ignores a SYSTEM(700)=1 specification in a SOL 401 or 402 solve.
16. For SOLs 401 and 402, see Shell elements in the Multi-Step Nonlinear User's Guide for details.

Related Topics

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CTRIAR

Triangular Plate Element Connection

Defines an isoparametric membrane-bending triangular plate element. It is a companion to the CQUADR element.

Format:

1	2	3	4	5	6	7	8	9	10
CTRIAR	EID	PID	G1	G2	G3	THETA or MCID	ZOFFS		
		TFLAG	T1	T2	T3				

Example:

CTRIAR	111	203	31	74	75	3.0			
			1.77	2.04	2.09				

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PSHELL, PCOMP, or PCOMPG entry. (Integer > 0; Default = EID)
G1, G2, G3	Grid point identification numbers of connection points. (Integers > 0, all unique)
THETA	Material property orientation angle in degrees. (Real; Default = 0.0)
MCID	Material coordinate system identification number. The x-axis of the material coordinate system is determined by projecting the x-axis of the MCID coordinate system (defined by the CORDij entry or zero for the basic coordinate system) onto the surface of the element. See Remark 10. (Integer ≥ 0; if blank, then THETA = 0.0 is assumed.)

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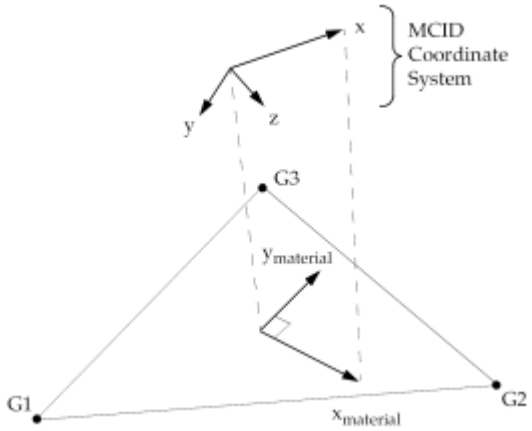
Field	Contents

Figure 12-115: MCID Coordinate System Definition



ZOFFS	Offset from the surface of grid points to the element reference plane. See Remark 9. (Real)
TFLAG	Integer Flag which specifies how Ti is used to define thickness of element. (0, 1, or blank)
Ti	Thickness of element at grid points G1 through G3 if TFLAG=0 or blank. If TFLAG=1, thickness becomes a product of Ti and the thickness on the PSHELL card. See Remark 4. (Real > 0.0 or blank. See Remark 2 for the default.)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. The continuation is optional. If it is not supplied, then T1 through T3 will be set equal to the value of T on the PSHELL entry.
3. When the PID refers to either a PCOMP or PCOMPG entry, the software converts the PCOMP/PCOMPG entry to equivalent PSHELL/MAT2 entries. If Ti is defined, they override or scale the values on the equivalent PSHELL entry. It should be noted that ply-stress and ply-strain recovery uses the ply thicknesses defined on the PCOMP/PCOMPG entries, and not Ti.
4. By default, a fatal error occurs if a Ti value is zero. To allow for Ti values of zero, use System Cell 495.
5. Stresses and strains are output in the element coordinate system at the centroid unless the CORNER or CUBIC entry is specified on the STRESS case control card. Specifying CORNER or CUBIC will result in stresses at the centroid and corners. See **"Shell element item code remarks"** at the beginning of the item codes chapter.
6. The rotational degree-of-freedom normal (drilling) to the element are active in the CTRIAR element formulation. Results will be inaccurate if the drilling degree-of-freedom of interior grids

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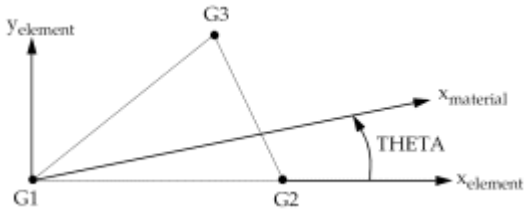
is constrained. At the boundary of a model, the drilling degree-of-freedom must be constrained to achieve edge boundary conditions such as a fixed or a symmetrical boundary.

7. As compared to the CTRIA3 element, the CTRIAR element:
   - Accounts for normal rotational DOF to provide improved membrane accuracy.
   - Is less sensitive to high aspect ratios and values of Poisson's ratio near 0.5.
8. The CTRIAR element is supported in all structural and heat transfer solutions. If used in SOL 106 or 129, the CTRIAR element always behaves geometrically linearly.
9. Elements may be offset from the connection points by means of ZOFFS. Other data, such as material matrices and stress fiber locations, are given relative to the reference plane. A positive value of ZOFFS implies that the element reference plane is offset a distance of ZOFFS along the positive z-axis of the element coordinate system. If the ZOFFS field is used, then both the MID1 and the MID2 fields must be specified on the PSHELL entry referenced by PID on the CTRIAR entry.

   ZOFFS is ignored in heat transfer solutions 153 and 159 including any solution subcase which includes the case control ANALYSIS=HEAT.
10. If the x-direction of a material coordinate system is perpendicular or very close to being perpendicular to the element face, a projection is either impossible or unpredictable. A fatal error will occur when the angle between a shell element face normal and the x-direction of the material coordinate system (MCID) is less than the value specified by System Cell 489.
11. The MCID should be less than 99,999,999 when SYSTEM(525)=0 (default), and less than 99,999,999,999 when SYSTEM(525)=1.

Figure 12-116: CTRIAR Element Geometry and Coordinate Systems



Remarks related to SOLs 401, 402, and 414:

1. Output results will be done in the material coordinate system. If those axes are not defined, the material coordinate system is parallel to the G1 - G2 edge. See Shell elements in the Multi-Step Nonlinear User's Guide for details.

Related Topics

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CTUBE

Tube Element Connection

Defines a tension-compression-torsion tube element.

Format:

1	2	3	4	5	6	7	8	9	10
CTUBE	EID	PID	G1	G2					

Example:

CTUBE	12	13	21	23					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PTUBE entry. (Integer > 0; Default = EID)
G1, G2	Grid point identification numbers of connection points. (Integer > 0; G1 ≠ G2)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. Only one tube element may be defined on a single entry.
3. The tube element has only axial and torsional stiffness, and only axial and torsional stresses. The only active degrees of freedom are displacements in the direction of the tube and rotation (torsion) about the axis of the tube. If you include the tube element in a buckling solution, you will not get buckling modes that involve bending since this element cannot bend.

Related Topics

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CVISC

Viscous Damper Connection

Defines a viscous damper element.

Format:

1	2	3	4	5	6	7	8	9	10
CVISC	EID	PID	G1	G2					

Example:

CVISC	21	6327	29	31					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
PID	Property identification number of a PVISC entry. (Integer > 0; Default = EID)
G1, G2	Grid point identification numbers of connection points. (Integer > 0; G1 ≠ G2)

Remarks:

1. Element identification numbers should be unique with respect to all other element identification numbers.
2. Only one viscous damper element may be defined on a single entry.

Related Topics

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CWELD

Weld or Fastener Element Connection

Defines a weld or fastener connecting two surface patches or points.

ELPAT Format: (See Remark 2)

1	2	3	4	5	6	7	8	9	10
CWELD	EWID	PWID	GS	"ELPAT"	GA	GB			
	SHIDA	SHIDB							
	XS	YS	ZS						

ELPAT Example:

CWELD	7	29	233	ELPAT					
	15	28							

PARTPAT Format: (See Remark 2)

1	2	3	4	5	6	7	8	9	10
CWELD	EWID	PWID	GS	"PARTPAT"	GA	GB			
	PIDA	PIDB							
	XS	YS	ZS						

PARTPAT Example:

CWELD	8	30	133	PARTPAT					
	101	201							

GRIDID Format: (See Remark 3)

1	2	3	4	5	6	7	8	9	10
CWELD	EWID	PWID	GS	"GRIDID"	GA	GB	SPTYP		
	GA1	GA2	GA3	GA4	GA5	GA6	GA7	GA8	
	GB1	GB2	GB3	GB4	GB5	GB6	GB7	GB8	
	XS	YS	ZS						

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GRIDID Example:

CWELD	7	29	233	GRIDID			QT		
	15	28	31	35	46	51	55	60	
	3	5	8						

ELEMID Format: (See Remark 4)

1	2	3	4	5	6	7	8	9	10
CWELD	EWID	PWID	GS	"ELEMID"	GA	GB			
	SHIDA	SHIDB							
	XS	YS	ZS						

ELEMID Example:

CWELD	3	28	354	ELEMID					
	15	16							

ALIGN Format: (See Remark 5)

1	2	3	4	5	6	7	8	9	10
CWELD	EWID	PWID		"ALIGN"	GA	GB			

ALIGN Example:

CWELD	7	29		ALIGN	103	259			

Fields:

Field	Contents
EWID	CWELD element identification number. See Remark 6. (Integer > 0)
PWID	Property identification number of a PWELD entry. (Integer > 0)
"ELPAT"	Designates the input format as “ELPAT”. See Remark 2. (Character; No default)
"PARTPAT"	Designates the input format as “PARTPAT”. See Remark 2. (Character; No default)
"GRIDID"	Designates the input format as “GRIDID”. See Remark 3. (Character; No default)
"ELEMID"	Designates the input format as “ELEMID”. See Remark 4. (Character; No default)
"ALIGN"	Designates the input format as “ALIGN”. See Remark 5. (Character; No default)

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Field	Contents
GA, GB	Grid identification numbers used to define the location of the connector on surface patch A and surface patch B, respectively when format is “ELPAT”, “PARTPAT”, “ELEMID”, or GRIDID. When format is “ALIGN”, GA and GB are required, and they must be vertex nodes of shell elements. See Remark 1. (Integer > 0 or blank)
GS	Grid point identification number used to define the end locations of the connector when GA/GB are undefined. The specific meaning of GS depends on whether it is used to define a location on a shell element patch when format is “ELPAT” , “PARTPAT”, “ELEMID”, or “GRIDID”, or is used to define a point location when the format is “ELEMID”, or “GRIDID”. See Remarks 1–6 for format specific definitions of GS. (Integer > 0)
XS,YS,ZS	Defines the location of the connector on a shell element patch when format is “ELPAT” , “PARTPAT”, “ELEMID”, or “GRIDID”, and when GA,/GB and GS are all undefined. See Remark 1. (Integer > 0)
PIDA	The physical property identification number of a PSHELL bulk data entry used to define surface patch A when format is “PARTPAT”. (Integer > 0)
PIDB	The physical property identification number of a PSHELL bulk data entry used to define surface patch B when format is “PARTPAT”. (Integer > 0)
SHIDA	The shell element identification number used to define surface patch A when format is “ELPAT” or “ELEMID”. (Integer > 0)
SHIDB	The shell element identification number used to define surface patch B when format is “ELPAT” or “ELEMID”.(Integer > 0)
SPTYP	String indicating the type of surface patches A and B when format is “GRIDID”. SPTYP=”QQ”, “TT”, “QT”, “TQ”, “Q” or “T”. Indicates quadrilateral (“Q”) or triangular (“T”) surface patch. See Remark 3. (Character)
GAi	Grid identification numbers defining surface patch A when format is “GRIDID”. GA1 to GA3 are required. See Remark 3. (Integer > 0)
GBi	Grid identification numbers defining surface patch B when format is “GRIDID”. See Remark 3. (Integer > 0)

Remarks:

1. GA/GB, or GS, or XS/YS/ZS, in that order of precedence, determine the location(s) of the CWELD connection on surface patches A and/or B, when format is “ELPAT”, “PARTPAT”, “ELEMID”, or “GRIDID”. The connection location on patch A is a specification of grid point GA, or a projection of GS or XS/YS/ZS normal to surface patch A. The location on patch B is a specification of grid point GB, or a projection of GS or XS/YS/ZS normal to surface patch B. When GS or XS/YS/ZS are used, a normal projection must exist in order to define a valid connection element. GS or XS/YS/ZS do not need to lie on either surface patch A or B. If GS or XS/YS/ZS are used to define the connection location, grid points are internally created at the connection location with an ID. The numbering of the internal IDs is described on the parameter **OSWPPT** for the formats “ELEMID” and “GRIDID” formats, and in Remark 8 for the formats “ELPAT” and “PARTPAT”.

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2. If the “ELPAT” format is used, the shell element selected on SHIDA along with those connected and the shell element specified on the SHIDB along with those connected will create shell element patches A and B, respectively. The “PARTPAT” format works similarly, except that the shell elements for patches A and B are selected with the PSHELL property ID specified on PIDA and PIDB, respectively. See **Figure 12-117**.

   For both “ELPAT” and “PARTPAT”, shell element patches A and B are connected together with a weld of diameter D specified on the PWELD bulk entry. The centers of the weld connection GA and GB on patches A and B are determined as described in remark 1. Virtual grid points GHA1-GHA4 on patch A and GHB1-GHB4 on patch B are created at the corners of a square having an area equivalent to a circle using the diameter D. See **Figure 12-118**. The points on each patch are connected together using constraint equations represented by the dashed green lines in **Figure 12-119**. The diameter of the CWELD should be entered such that all GHAi and GHBi locations are either within the element in which GA and GB are projected on to, or an element directly adjacent. For example, if any GAi location is within the gray area in **Figure 12-119**, CWELD creation may fail. Parameters on the **SWLDPRM** bulk entry can be used to adjust tolerances and defaults when CWELD creation fails.

Figure 12-117: Patch-to-Patch Connection Defined with Format ELPAT or PARTPAT

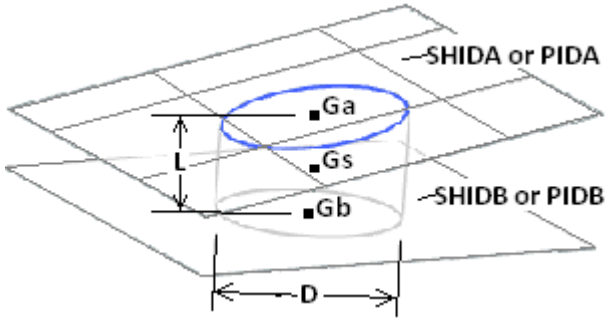
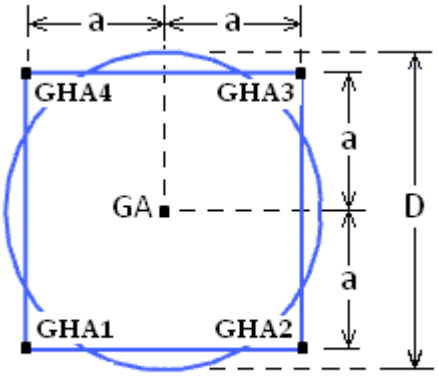


Figure 12-118: GHAi and GHBi Locations with Format ELPAT or PARTPAT



$$a = \sqrt{\pi}\frac{D}{4}$$

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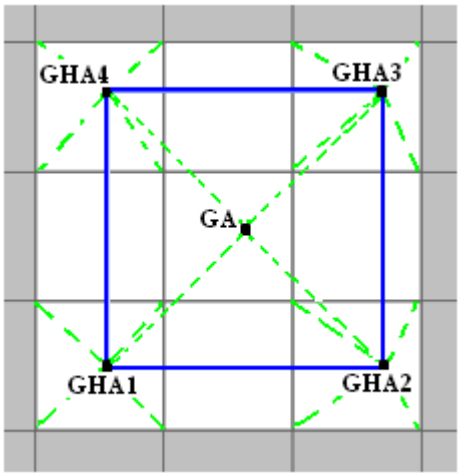
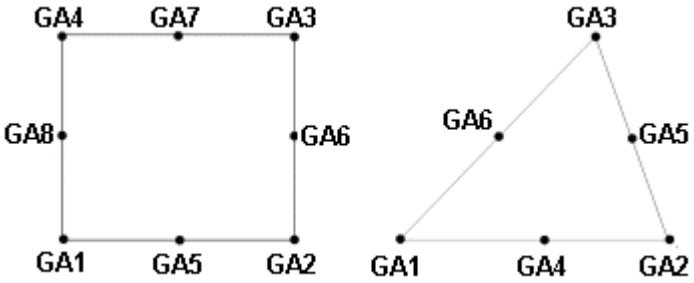
Figure 12-119: Weld Connections with Format ELPAT or PARTPAT



3. The format “GRIDID” defines either a point to element face or an element face to element face connection. For the point to element face connection, you must define GS and GAi. Then it is assumed that GS is a shell vertex grid and GAi are grids describing an element face. For the element face to element face connection, you must define GS, GAi and GBi. Then GAi describes the first element face and GBi the second element face.

   GAi are required for the “GRIDID” format. At least 3 and at most 8 grid IDs may be specified for GAi and GBi, respectively. Triangular and quadrilateral element definition sequences apply for the order of GAi and GBi, see **Figure 12-120**. Missing midside nodes are allowed. The envelop that is defined by the grids must represent a valid finite element shape, and not just a string of grids. The envelop may encapsulate more than one finite element.

Figure 12-120: Quadrilateral and Triangular Surface Patches Defined with Format GRIDID



SPTYP defines the type of surface patches to be connected. SPTYP is required for the “GRIDID” format to identify quadrilateral or triangular patches. Allowable combinations are:

SPTYP	Description
QQ	Connects a quadrilateral surface patch A (Q4 to Q8) with a quadrilateral surface patch B (Q4 to Q8).
QT	Connects a quadrilateral surface patch A (Q4 to Q8) with a triangular surface patch B (T3 to T6).

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SPTYP	Description
TT	Connects a triangular surface patch A (T3 to T6) with a triangular surface patch B (T3 to T6).
TQ	Connects a triangular surface patch A (T3 to T6) with a quadrilateral surface patch B (Q4 to Q8).
Q	Connects the shell vertex grid GS with a quadrilateral surface patch A (Q4 to Q8) if surface patch B is not specified
T	Connects the shell vertex grid GS with a triangular surface patch A (T3 to T6) if surface patch B is not specified

4. The “ELEMID” format defines a point to patch connection, GS to SHIDA or a patch to patch connection, SHIDA to SHIDB.
5. The “ALIGN” format defines a point to point connection. GA and GB are required, and they must be vertex nodes of shell elements. GA and GB are not required for the other formats.
6. CWELD defines a flexible connection between two surface patches, between a point and a surface patch, or between two shell vertex grid points. See Figure 12-121 through Figure 12-123.

Figure 12-121: Patch-to-Patch Connection Defined with Format GRIDID or ELEMID

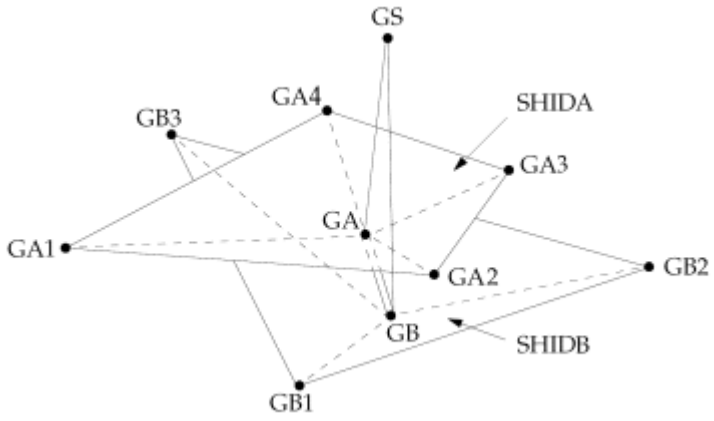

Figure 12-122: Point-to-Patch Connection Defined with Format GRIDID or ELEMID

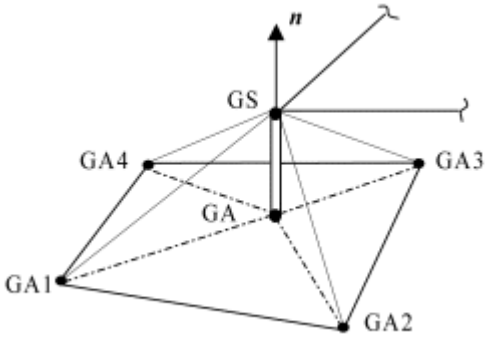

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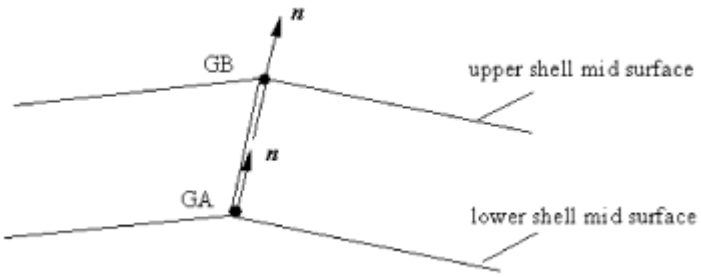
Figure 12-123: Point-to-Point Connection Defined with Format ALIGN



7. Forces and moments are output in the element coordinate system as shown in Figure 12-124. The output of the forces and moments including the sign convention is the same as on the CBAR element.

   The element x-axis is in the direction of GA to GB, and is computed using the x-components at GA and GB:

$$\overrightarrow{e_1} = \frac{x_B - x_A}{||x_B - x_A||}$$

   If GA and GB are coincident such that |*GB-GA*| < *1.0E-04 * characteristic length* of shell A, the element x-axis is defined by the normal vector from shell A, where:

   - For a quadrilateral shell element, the characteristic length is the average of the diagonal lengths.
   - For a triangular shell element, the characteristic length is the average of the three side lengths.
   - The normal vector from shell A for either a quadrilateral or triangular shell element is the cross product of the vector from G2 to G1 and the vector from G3 to G1.

     To compute the element y-axis, the software first determines which of the three basic coordinate directions $\hat{\mathbf{b}}_1,\ \hat{\mathbf{b}}_2,\ \hat{\mathbf{b}}_3$ has the *smallest* projection with the element x-axis. The basic coordinate direction determined $\hat{\mathbf{b}}_i$ and the element x-axis unit vector $\hat{\mathbf{e}}_1$ are used in the following equation to compute the y-axis orthogonal to the x-axis:

$$\overrightarrow{e_2} = \hat{\mathbf{b}}_i - \hat{\mathbf{e}}_1 \left(\frac{\hat{\mathbf{e}}_1 \cdot \hat{\mathbf{b}}_i}{\hat{\mathbf{e}}_1 \cdot \hat{\mathbf{e}}_1} \right)$$

     and the y-axis unit vector is computed:

$$\hat{\mathbf{e}}_2 = \frac{\overrightarrow{e}_2}{\left|\left|\overrightarrow{e}_2\right|\right|}$$

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For example, if an element x-axis unit vector in basic coordinates is $\widehat{\mathbf{e}}_1 = .6i + .6j + .3k$, the z-coordinate has the smallest component and has the smallest projection. For this case, $\widehat{\mathbf{b}}_i = \widehat{\mathbf{b}}_3 = 0.0i + 0.0\,j + 1.0k$ would be used to compute the y-axis.

The software computes the element z-axis unit vector by computing the cross product of the x and y-axis unit vectors, and dividing by the resultant of the same cross product:

$$\widehat{\mathbf{e}}_3 = \frac{\widehat{\mathbf{e}}_1 \times \widehat{\mathbf{e}}_2}{\left|\left|\widehat{\boldsymbol{e}}_1 \times \widehat{\mathbf{e}}_2\right|\right|}$$

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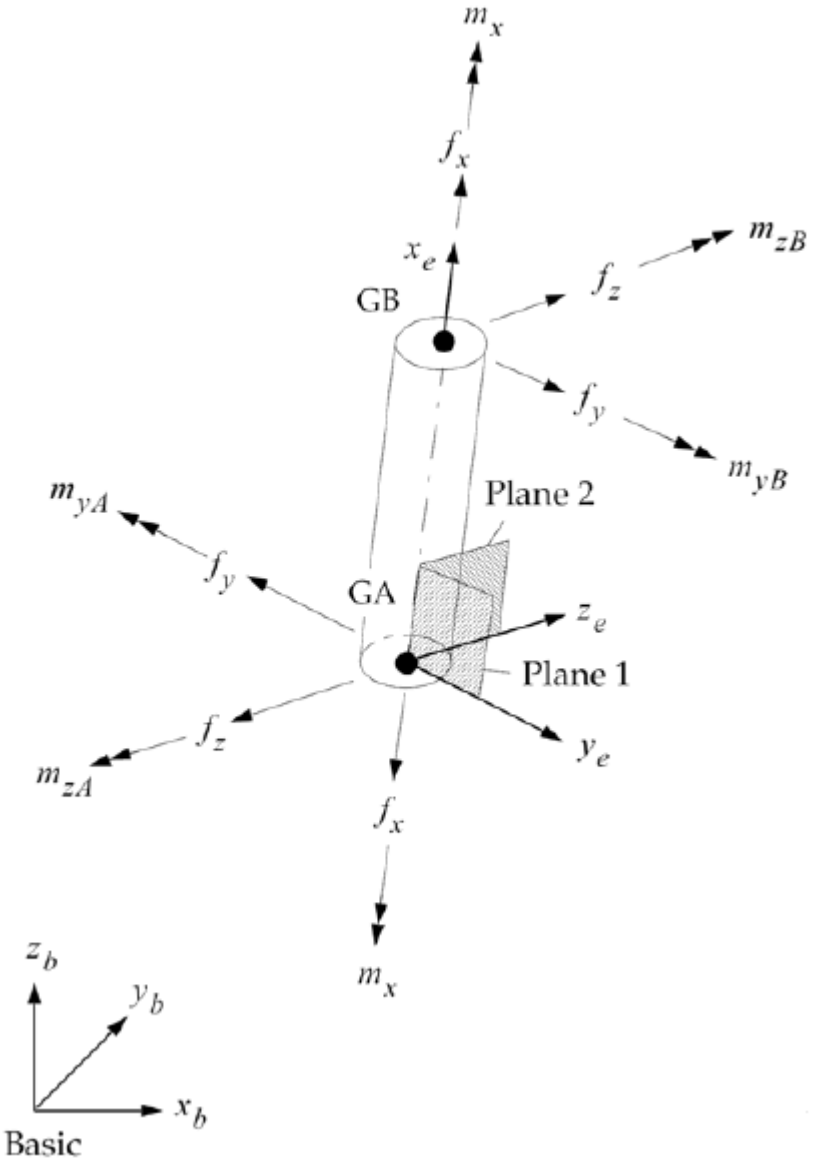
Figure 12-124: Element Coordinate System and Sign Convention of Element Forces



where,

f_x is the axial force.

f_y is the shear force for plane 1.

f_z is the shear force for plane 2.

m_x is the torque.

M_{yA} is the bending moment at end A for plane 2.

M_{yB} is the bending moment at end B for plane 2.

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M_{zA} is the bending moment at end A for plane 1.

M_{zB} is the bending moment at end B for plane 1.

8. When the format is ELPAT or PARTPAT and GA and GB are specified, the displacements for these grids will be printed/stored as requested on the DISPLACEMENT case control command. If these points are not specified, they will be generated by the software, and the displacements for these grids can only be written to the .f06 file with grid IDs calculated as follows:
Grid ID = Value of SYSTEM(178) + (CWELD ID) + (end ID)
where the end ID is 1 or 2 for the two ends. SYSTEM(178) defaults to 101,000,000.
The new grid IDs are only used for labeling the displacement output and do not exist in the analysis or internal tables. This displacement result is not written to the .op2 or punch file.

Related Topics

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CYAX

Grid Points on Axis of Symmetry

Lists grid points that lie on the axis of symmetry in cyclic symmetry analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CYAX	G1	G2	G3	G4	G5	G6	G7	G8	
	G9	G10	-etc.-						

Example:

CYAX	27	152	THRU	160	192	11			

Fields:

Field	Contents
Gi	A list of grid points on the axis of symmetry. (Integer > 0 or Character “THRU”)

Remarks:

1. The displacement coordinate system (see CD field on theGRID entry) for a grid point lying on the axis of symmetry must be a rectangular system with the z-component of motion aligned with the axis of symmetry. The positive axis of symmetry is defined so that the azimuthal angle from positive side 1 to side 2 of a segment is in the same direction as the angle from T1 to T2 for the axis point. This is consistent with the left- or right-hand rule.
2. If the dihedral symmetry option (STYPE = “DIH” on the CYSYM entry) is selected, the y-axis must be perpendicular to side 1.
3. Grid points lying on the axis of symmetry may be constrained by SPCs but not by MPCs. If the number of segments is three or more, SPCs must be applied to both components 1 and 2 or to neither, and SPCs must be applied to both components 4 and 5 or to neither in order to satisfy symmetry. In addition, the degrees-of-freedom (not constrained by SPCs, if any) at these grid points must be in the analysis set (a-set). If all degrees-of-freedom of grid points on the axis of symmetry are constrained by SPCs (including heat transfer, where there is only one degree-of-freedom), the grid point should not be listed on the CYAX entry.
4. Grid points lying on the axis of symmetry must not be defined on side 1 or on side 2 by means of a CYJOIN entry.
5. The word “THRU” must not appear in fields 2 or 9.

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Related Topics

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CYCADD

Combines Cyclic Symmetry Sets

Combines cyclic symmetry sets (CYCSET) in SOLs 401, 402, and 414, and combines interstage connections (CYCISC) in all SOL 414 solutions except for SOL 414,111 (harmonic response).

Format:

1	2	3	4	5	6	7	8	9	10
CYCADD	SID	S1	S2	S3	S4	S5	S6	S7	
	S8	S9	S7	-etc-					

Example:

CYCADD	12	1	3	6	5	9			

Fields:

Field	Contents
SID	Cyclic set (CYCSET) or interstage connection (CYCISC) identification number. (Integer>0)
Si	Identification numbers of cyclic sets defined with CYCSET or CYCISC bulk entries. (Integer>0)

Remarks:

1. Multiple cyclic sets (CYCSET bulk entries) and multiple interstage connections (CYCISC bulk entries) with unique SIDs can be combined by including their SIDs on a CYCADD bulk entry. The CYCADD bulk entry has its own unique SID which is referenced by a CYCSET case control command.
2. The CYCSET set identification numbers must be unique and may not be the identification of the CYCADD entry.
3. SOL 401 does not support the cyclic symmetry definition in the Fourier subcase.
4. For SOL 401, if you are modeling multiple cyclic sectors (stages), each with a different number of segments (NSEG value on the CYCSET bulk entry), you must use the STAGE field on the CYCSET bulk entry to define a unique identification number (ID) for each stage. All CYCSET entries having the same stage ID must also have the same NSEG value. CYCSET bulk entries that do not identify a stage ID must all have the same NSEG value.

   SOL 401 supports multiple stages and different sectors (NSEG) with the static, preload, dynamic and modal subcases. The cyclic modes subcase does not support multiple stages, and requires that all CYCSET entries have the same number of sectors (NSEG).

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5. For SOLs 402 and 414, the value defined in the NSEG field on the CYCSET entries referenced by a CYCADD entry can be different; CYCADD can collect CYCSET with different NSEG.

Related Topics

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CYCAXIS

Default Cylindrical CSYS for a Cyclic Model

Default cylindrical coordinate system for a cyclic model in SOLs 401, 402, and 414.

Format:

1	2	3	4	5	6	7	8	9	10
CYCAXIS	CSID								

Example:

CYCAXIS	101								

Fields:

Field	Contents
CSID	Identification number of the default cylindrical coordinate system for a SOL 401, 402, and 414 cyclic model. (Integer≥0).

Remarks:

1. The CYCAXIS bulk entry is required to define the default cylindrical coordinate system for the automatic coupling selected with the CYCSET case control. The origin of this cylindrical system must be at the center of the revolution, and the z-axis must be consistent with the axial direction.
2. The z-axis of every cylindrical coordinate system referenced by the CYCSET entry must have the same origin and direction as the z-axis of the default coordinate system selected on the CYCAXIS bulk entry.

Related Topics

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CYCISC

Creates an Interstage Connection

Connects two cyclic symmetry sectors on different cyclic symmetry components in an assembly by coupling the stages in each. You can use CYCISC in all SOL 414 solutions except for SOL 414,111 (harmonic response).

Format:

1	2	3	4	5	6	7	8	9	10
CYCISC	ID	SID	TID	SDIST					

Example:

CYCISC	3	5	5	0.05					

Fields:

Field	Contents
ID	CYCISC set identification number. (Integer > 0)
SID	Source region identification number. (Integer > 0; No default)
TID	Target region identification number. (Integer > 0; No default)
SDIST	Search distance for CYCISC pairs. See Remark 5. (Real; Default = 1.E-3 x DIAG in the tangential direction and 1.E-4 x DIAG in the radial direction)

Remarks:

1. You can use CYCISC in all SOL 414 solutions except for SOL 414,111 (harmonic response).
2. To do the automatic coupling, the software internally rotates the target region grids into the source region grids, refines the mesh both on the source and target, and then creates MPC equations using the target as the dependent DOF and the source as the independent DOF.

   The MPC equations are created between any source and target region grids within the specified search distance (SDIST) using a weighted area method. From each source grid, the search occurs in both the positive and negative theta DOF direction. The SDIST field is required for each pair of regions and calculated by default if not defined.
3. For SOL 414,101 (maneuvers), SOL 414,103 (eigenvalue and superelement reduction) for static and transient preload subcases, SOL 414,110 (complex modal) for static preload subcases, and SOL 414,129 (transient response), only harmonic=0 is considered for the cyclic symmetry.

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4. Only one harmonic (in the HARMONICS case control set) can be defined for the solution. If several harmonics are defined, only the lowest harmonic number is taken into account, and the solver writes a warning message to the .f06 file.
5. DIAG is the size of a software-computed box surrounding the model. The DIAG value is derived from the SDIST value.

Related Topics

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CYCSET

Defines Cyclic Symmetry Pairs

Defines pairs for cyclic symmetry in SOLs 401, 402, and 414.

Format:

1	2	3	4	5	6	7	8	9	10
CYCSET	CYSID	NSEG	CSID	STAGE	GROUPID	CBN	CYPARTOL		
	SID1	TID1	SDIST1						
	SID2	TID2	SDIST2						
	-etc-								

Example:

CYCSET	4	11	6	64	102	0			
	1	2							
	4	3							

Fields:

Field	Contents
CYSID	CYCSET set identification number. (Integer > 0)
NSEG	Number of segments. See Remarks 6 and 7. (Integer > 1; No default)
CSID	Identification number of a cylindrical coordinate system. (Integer>0; Default = CSYS selected with the CYCAXIS entry)
STAGE	Stage identification number. See Remark 6. (Integer > 0; No default)
GROUPID	Identification number of the GROUP entry for the stage. See Remarks 9, 10, and 12. (Integer > 0; No default)
CBN	SOL 414 only for cyclic sectors connected with the CYCISC connection (interstage connection for multi-stage cyclic symmetry). Controls whether the software creates boundary conditions between sectors. (Integer 0 or 1; Default = 1) 0 = Do not create boundary conditions between sectors. Use this option when the sectors are not contiguous.

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Field	Contents
	1 = Create boundary conditions between sectors. Use this option when the sectors are contiguous, such as sectors on a ring.
CYPARTOL	Parametric tolerance to determine if a rotated target grid point location lies within the matched source element. See Remark 13. (0.0 < Real < 0.1; Default =0.005)
SIDi	Identification number of the source region. See Remark 1. (Integer > 0; No default)
TIDi	Identification number of the target region. See Remark 1. (Integer > 0; No default)
SDISTi	Search distance tolerance for CYCSET pairs. (Real; Default =0.001) SOL 401 uses SDIST as the normal search distance. SOLs 402 and 414 compute the normal search distance using SDIST x DIAG, where DIAG is the size of a software-computed box surrounding the model.

Remarks:

1. The SIDi and TIDi fields select the BSURFS and BCPROPS bulk entries. These entries define regions on the faces of the CHEXA, CPENTA, CPYRAM, and CTETRA elements.
2. The source region selected in a pair must have a smaller positive theta location than the target region.
3. To do the automatic coupling, Simcenter Nastran internally rotates the target region grids into the source region grids, refines the mesh on both the source and target, and then creates MPC equations using the target as the dependent DOF and the source as the independent DOF.

   The MPC equations are created between any source and target region grids within the specified search distance (SDISTi) using a weighted area method. From each source grid, the search occurs in both the positive and negative theta DOF direction. The SDIST field is required for each pair of regions.
4. The following applies to a cylindrical coordinate system selected with the CSID field:
   - The Z-axis of the selected coordinate system must be parallel with the Z-axis of the CYCAXIS definition.
   - The origin of the selected coordinate system does *not* need to be coincident with the origin of the CYCAXIS definition, but it must lie on the Z-axis of the CYCAXIS definition.
   - The orientation of the R and θ axis of the selected coordinate system does *not* need to be parallel with the R and θ axis of the CYCAXIS definition.
   - When the CSID field is undefined, the CYCAXIS definition is used by default.
5. The software will use the number of segments (NSEG) field to compute the angle between the source and target faces. For example, if a 30-degree segment is modeled, NSEG would be 12 (=360/30).
6. For SOL 401, if you are modeling multiple cyclic sectors (stages), each with a different number of segments (NSEG), you must use the STAGE field to define a unique identification number (ID) for

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each stage. All cyclic sectors having the same stage ID must also have the same NSEG value. CYCSET entries that do not identify a stage ID must all have the same NSEG value. SOL 401 supports multiple stages and different sectors (NSEG) with the static, preload, dynamic and modal subcases. The cyclic modes and frequency response subcases do not support multiple stages, and require that all CYCSET entries have the same number of sectors (NSEG).

7. For SOLs 402 and 414, the value defined in the NSEG field on the CYCSET entries referenced by a CYCADD entry can be different; CYCADD can collect CYCSET with different NSEG.

8. SOL 401 does not support the cyclic symmetry definition in the Fourier subcase.

9. For SOL 401, if your model is a combination of 2D axisymmetric elements and a sector of 3D solid elements, and you listed the elements in your 3D sector (3D solid elements and CONMi mass elements) on a GROUP bulk entry referenced in the GROUP field, the software will automatically scale the material properties and applied loads on the 3D cyclic sector by NSEG. The scaling creates a (2π) consistent representation for both the 2D axisymmetric and 3D cyclic sector portions of the model. There is no connectivity check for elements in the GROUP referenced on the CYCSET entry. Properties for all elements in the group will be scaled.

The combined 2D/3D cyclic sector models with and without scaling are supported by the static, preload, dynamic and modal subcases. The cyclic modes subcase is not supported for the combined model.

You can disable the automatic scaling by defining the parameter setting CY3DSCL=0 on the NLCNTLG bulk entry. By default, CY3DSCL=1 and the automatic scaling occurs.

The software scales the following loads: FORCE, FORCE1, FORCE2, MOMENT, MOMENT1, MOMENT2, PLOAD, PLOAD4, FORCDST, PLOADFP, and BOLTFRC. Note the following rules for loads defined with the PLOADFP and BOLTFRC entries. The software will issue a fatal error if these are violated.

- For the fluid penetration pressure load defined with the PLOADFP entry, all of the elements in a region must either be in a cyclic sector or not in a cyclic sector. The software scales the pressure applied to these elements when they are all in a cyclic sector.
- Similarly, for bolt preloads of ETYPE= 2 and 3, all of the elements representing the bolts referenced by a BOLTFRC entry must be in a cyclic sector or not in a cyclic sector. The software scales the preload force when all of the elements are in a cyclic sector.

10. For SOLs 402 and 414, the GROUP entry referenced in the GROUPID field selects CHEXA, CPENTA, CPYRAM, and CTETRA elements that define the cyclic sector, 2D elements, RBE2 and RBE3 elements, bushing elements, FOU3 elements, and bearing elements. For bearing elements, also see Remark 11.

11. For SOL 414, use only CBEAR2 with cyclic symmetry sectors, but do not include CBEAR2 in the CYCSET group of elements. Do not use CBEAR with cyclic symmetry sectors.

For cyclic symmetry models, define the CBEAR2 elements on the rotor axis and connect the bearing grid points to the cyclic symmetry sector grid points using CLINK elements with one of the following options:

- CLINK element with OPT=5 if the Coleman transformation is not requested.
- CLINK element with OPT=6 if the Coleman transformation is requested (PARAM,COLEMAN,1).

12. When you request the Coleman transformation (PARAM,COLEMAN,1), you must reference a GROUPID field and a single stage number for all of the cyclic symmetry sectors.

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If the GROUPID field consists of groups of grid points and groups of element faces, the number of segments (NSEG) in each cyclic symmetry sector must be the same.

13. The CYPARTOL field is available to adjust the parametric coordinate tolerance.

    The software creates a parametric coordinate frame on each source element face where the element corners are bounded by the coordinates -1.0 to +1.0. It projects each target grid point to a source element to compute the parametric coordinates for the projected location. The software includes a 0.005 tolerance, by default, in case the projected location is slightly outside of the source element face. As a result, each projected target grid point must have a parametric coordinate within -1.005 or +1.005, or the grid point is not coupled. The grid points that fail to project to an element face will always be on the mesh boundary.

    The software prints the following in the .f06 file showing the total number of target grid points (nodes) and the number of target grid points that were paired.

```
*** USER INFORMATION MESSAGE 23230 (WRTGEOM4CYC)
NUMBER OF NODES ON CYLIC TARGET FACE - 10130
NUMBER OF NODES ON CYCLIC TARGET FACE PAIRED WITH NODES ON CYCLIC
SOURCE FACE - 10128
NUMBER OF CYCLIC MPC CONDITIONS FORMED FOR STATICS/HARMONIC=0 SOLUTION
- 30384
```

    If you see a difference in the total number and the number paired, you can consider increasing the parametric tolerance in the CYPARTOL field.

14. For SOL 414 solutions, as a best practice, do not use the BGSET case control command and bulk entry in the same cyclic symmetry sector. Instead, define different cyclic symmetry sectors (CYCSET) with the same number of segments (NSEG), define a different stage ID, and connect the two sectors using the interstage junction (CYCISC).

Related Topics

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CYJOIN

Cyclic Symmetry Boundary Points

Defines the boundary points of a segment in cyclic symmetry problems.

Format:

1	2	3	4	5	6	7	8	9	10
CYJOIN	SIDE	C	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	-etc.-					

Example:

CYJOIN	1	T2	7	9	16	THRU	33	64	
	72	THRU	89						

Fields:

Field	Contents
SIDE	Side identification. (Integer 1 or 2)
C	Type of coordinate system used on boundaries of dihedral or axisymmetry problems. See Remark 3 . (Character: “T1”, “T2”, “T3”, “R”, “C”, “S”)
Gi	Grid or scalar point identification numbers. (Integer > 0 or Character “THRU”)

Remarks:

1. CYJOIN entries are used only for cyclic symmetry problems. The CYSYM entry must be used to specify rotational, dihedral, or axisymmetry.
2. For rotational or axisymmetry problems, there must be one logical entry for SIDE = 1 and one for SIDE = 2. The two lists specify grid points to be connected; therefore, both lists must have the same length.
3. For dihedral problems, side 1 refers to the boundary between segments and side 2 refers to the middle of a segment. For dihedral and/or AXI type of symmetry, the grid point degree-of-freedom that is normal to the boundary must be specified in field 3 as “T1”, “T2”, or “T3”. (“R”, rectangular, and “C”, cylindrical, are the same as “T2”while “S”, spherical, is the same as “T3”). For scalar and extra points with one degree-of-freedom, these should be specified as blank, “T2”, or “T3” if they are to have the same sign, and “T1”, if the two connected points are to be opposite in sign.
4. All components of displacement at boundary points are connected to adjacent segments except those constrained by SPCi, MPC, or OMITi entries.

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5. The points on the axis of symmetry of the model, defined in the CYAX entry must not be defined as a side 1 or side 2 point by means of this entry.
6. The word “THRU” may not appear in fields 4 or 9 of the parent entry and fields 2 or 9 of the continuation entries.
7. All grid points that are implicitly or explicitly referenced must be defined.
8. For rotational and axisymmetry problems, the displacement coordinate systems must be consistent between sides 1 and 2. This is best satisfied by the use of a spherical or cylindrical coordinate system.

Related Topics

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CYSUP

Fictitious Supports for Cyclic Symmetry

Defines fictitious supports for cyclic symmetry analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CYSUP	GID	C							

Example:

CYSUP	16	1245							

Fields:

Field	Contents
GID	Grid point identification number. (Integer > 0)
C	Component numbers. (Any unique combination of the Integers 1 through 6 with no embedded blanks.)

Remarks:

1. Components of motion defined on this entry may not appear on SPC, SPC1, OMIT, OMIT1 entries, or in rigid elements or multipoint constraints as dependent degrees-of-freedom.
2. Supports are applied at the grid point identified in field 2 to prevent rigid body motions in static analysis, or to define rigid body modes in eigenvalue analysis. All degrees-of-freedom should be at a single grid point. In other words, there can only be one such supported grid point in the model. The supports are applied only to the cyclic components of order k=0 or k=1. In order to satisfy conditions of symmetry, certain restrictions are placed on the location of the grid point and the orientation of its displacement coordinate system, as shown in the following table:

Symmetry Option (STYPE on the CYSYM entry)	ROT	ROT	DIH	DIH	DIH
Number of Segments, N	2	≥3	1	2	≥3

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Location of Grid Point	See Note c.	See Note d.	Side 1	Side 1	Side 1
Special Restrictions on Displacement Coordinate System	See Notes a and e.	See Note b.	None	See Note a.	See Note b.

Note

a. T3 axis must be parallel to axis of symmetry.

b. The displacement coordinate system at the referenced grid point must be cylindrical with the z-axis along the axis of symmetry.

c. Any location except on side 2.

d. Any location except on the axis of symmetry or on side 2.

e. If the grid point is on the axis of symmetry, the displacement coordinate system must be rectangular.

3. If the number of segments, N, is 1 (in the case of DIH symmetry) or 2 (in the case of ROT or AXI symmetry), it is important that the rotational components referenced in field 3 be elastically connected to the structure. If N ≥ 2 (in the case of DIH symmetry) or N ≥ 3 (in the case of ROT or AXI symmetry), it is not important, because in this case the supports for rigid body rotation are actually applied to translational motions.
4. if N ≥ 3, supports will be applied to both the 1 and 2 (inplane-translational) components, if either is referenced, and to both the 4 and 5 (out-of-plane rotational) components, if either is referenced. If component 6 is supported, component 2 should not appear on OMIT or OMIT1 entries.
5. The restrictions noted in Remarks 2 and 4 are related to symmetry requirements. For N ≥ 3, symmetry requires that the supports be symmetrical (or antisymmetrical), with respect to any plane passing through the axis of symmetry. For the DIH options, N = 1 and N = 2, symmetry requires that the supports be symmetrical (or antisymmetrical) with respect to the plane(s) of symmetry. For the ROT option, N = 2, symmetry requires that a support be either parallel or perpendicular to the axis of symmetry.
6. GID must be a grid point and not a scalar point.

Related Topics

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CYSYM

Cyclic Symmetry Parameters

Defines parameters for cyclic symmetry analysis.

Format:

1	2	3	4	5	6	7	8	9	10
CYSYM	NSEG	STYPE							

Example:

CYSYM	6	ROT							

Fields:

Field	Contents
NSEG	Number of segments. (Integer > 0)
STYPE	Symmetry type. (Character: “ROT”, “DIH”, or “AXI”)

Remarks:

1. STYPE = “AXI” is a special case of STYPE = “ROT” used to model axisymmetric structures.
2. If STYPE = “AXI”, then all grid points must lie on side 1, side 2, or the axis. Also, plate elements with midside grid points may not be defined. See “Cyclic Symmetry” in the *Simcenter Nastran User’s Guide*.

Related Topics

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13. Bulk Entries D—E

DAMPING

Damping Definition

Specifies the following:

- Structural (hysteretic) damping
- Rayleigh (proportional) damping
- Structural (hysteretic) damping to viscous damping conversion method for transient response analysis and modal-based analysis that writes an interface file for external multibody dynamics or control system software.

Format:

1	2	3	4	5	6	7	8	9	10
DAMPING	SID	G	ALPHA1	ALPHA2		GEFACT	MATGE		
	W3	W4	WMODAL						

Example:

DAMPING	100	0.1				0.05			
			YES						

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
G	Overall structural damping coefficient. See Remark 6. (Real ≥ 0.0; Default = 0.0)
ALPHA1	Scale factor that is applied to the mass matrix for Rayleigh damping. See Remarks 7 and 8. (Real ≥ 0.0; Default = 0.0)
ALPHA2	Scale factor that is applied to the stiffness matrix for Rayleigh damping. See Remarks 7 and 8. (Real ≥ 0.0; Default = 0.0)
GEFACT	Scale factor that is applied to the K42GG direct input stiffness and assembled elemental damping matrices. See Remarks 9 and 10. (Real ≥ 0.0; Default = 1.0)
MATGE	Application method for the structural damping coefficient defined in the G field. (Character = "USEG" or blank)

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Field	Contents
	"USEG" to apply the structural damping coefficient defined in the G field to only those 2D and 3D elements that reference a material bulk entry where the GE field is not specified. Leave blank to apply the structural damping coefficient defined in the G field to all 2D and 3D elements. See Remark 11.
W3, W4	Structural damping to viscous damping conversion factors in radians per unit time. See Remark 12. (Real ≥ 0.0; Default = 0.0)
WMODAL	Structural damping to viscous damping conversion method that uses the solved modal frequencies as the conversion factors. See Remark 13. (Character; Default = "NO") "YES" to use the WMODAL conversion method and ignore the W3 and W4 fields "NO" to use the W3 and W4 fields and conversion method

Remarks:

1. The set identification number must be unique with respect to all other DAMPING bulk entry set identification numbers.
2. The set identification number is referenced by a DAMPING case control command.
3. When the input file contains both a DAMPING bulk entry and a functionally equivalent parameter specification, the DAMPING bulk entry takes precedence.

   For example, if the input file contains PARAM,W3,50.0 and the W3 field on the DAMPING bulk entry contains 30.0, the software uses 30.0 as the W3 conversion frequency. For the same example, if the W3 field on the DAMPING bulk entry is left blank, the software uses 0.0 as the W3 conversion value, which causes the $[K^1]$ term to be omitted from the conversion.
4. When the W3 and W4 fields are specified and the WMODAL field is set to YES, the software ignores the W3 and W4 settings and uses the WMODAL method to convert the structural damping to viscous damping. For more information, see **WMODAL**.
5. The interpretation of the G, ALPHA1, ALPHA2, W3, and W4 fields depend on whether the STRUCTURE or FLUID describer is specified on the DAMPING case control command that references the DAMPING bulk entry as follows:

Field	Interpretation when referenced by DAMPING(STRUCTURE)	Interpretation when referenced by DAMPING(FLUID)
G	G	GFL
ALPHA1	ALPHA1	ALPHA1FL
ALPHA2	ALPHA2	ALPHA2FL
W3	W3	W3FL
W4	W4	W4FL

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6. For detailed information about how the software uses the value in the G field to define structural (hysteretic) damping, see **W3 (or W3FL), W4 (or W4FL)** and **WMODAL**.
7. The ALPHA1 (or ALPHA1FL) and ALPHA2 (or ALPHA2FL) parameters require complex input, where the imaginary part must be set to 0.0. However, when you specify the ALPHA1 and ALPHA2 fields on the DAMPING bulk entry, the values you enter are equivalent to the real part of an ALPHA1 (or ALPHA1FL) and ALPHA2 (or ALPHA2FL) parameter specification, respectively, and the software automatically takes the imaginary part to be 0.0.
8. For detailed information about how the software uses the values in the ALPHA1 and ALPHA2 fields to define Rayleigh damping, see **ALPHA1, ALPHA2** and **ALPHA1FL, ALPHA2FL**.
9. For detailed information about how the software uses the value in the GEFACT field to scale the K42GG direct input stiffness and assembled elemental damping matrices, see **W3 (or W3FL), W4 (or W4FL)** and **WMODAL**.
10. The GE fields on material, property, and element bulk entries allow you to define structural (hysteretic) damping for elements individually, whereas the overall structural damping coefficient (PARAM,G or the G field on the DAMPING bulk entry) is applied uniformly to all elements. The GEFACT field allows you to scale all structural damping specified by the GE fields uniformly with a single GEFACT specification, therefore avoiding editing numerous material, property, and element bulk entries.

    For example, you can turn off the structural damping specified by the GE fields by setting the GEFACT field to 0.0.
11. When the MATGE field is blank and a 2D or 3D element references a material bulk entry where the GE field is specified, the total structural damping is the sum of the contributions from the GE field of the material bulk entry and the G field of the DAMPING bulk entry.
12. For detailed information about how the software uses the values in the W3 and W4 fields to convert structural (hysteretic) damping to viscous damping, see **W3 (or W3FL), W4 (or W4FL)**.
13. For detailed information about how the software converts structural (hysteretic) damping to viscous damping when the WMODAL field is set to "YES", see **WMODAL**.

Remarks related to SOL 401:

1. For SOL 401, you can include the DAMPING=n case control command in the modal frequency response subcase to select the DAMPTAB bulk entry and the DAMPING bulk entry. You can define both the DAMPTAB entry and the DAMPING entry, and they can share the same SID. See Damping definition in the Modal frequency response subcase (SOL 401) topic in the Multi-Step Nonlinear User's Guide for details on all supported damping in this subcase.

Related Topics

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DAMPTAB

For SOL 401 only - Damping Definition for Modal Frequency Response Subcase

Damping definition for SOL 401 modal frequency response subcase.

Format:

1	2	3	4	5	6	7	8	9	10
DAMPTAB	SID	T(G)	T(ALPHA1)	T(ALPHA2)	T(MDAMP)				

Example:

DAMPTAB	100	20	32	34	54				

Fields:

Field	Contents
SID	Set identification number. (Integer>0)
T(G)	Identification number of a TABLEDi entry defining solution frequencies versus structural damping coefficients. (Integer>0)
T(ALPHA1)	Identification number of a TABLEDi entry defining solution frequencies versus the proportional damping mass matrix scale factor. (Integer>0)
T(ALPHA2)	Identification number of a TABLEDi entry defining solution frequencies versus the proportional damping stiffness matrix scale factor. (Integer>0)
T(MDAMP)	Identification number of a TABDMP1 entry defining natural frequencies versus modal damping values (critical damping ratios only). (Integer>0)

Remarks:

1. The set identification number must be unique with respect to all other DAMPING bulk entry set identification numbers.
2. The set identification number is referenced by a DAMPING case control command.
3. The modal frequency response subcase supports proportional viscous damping, modal viscous damping, and structural damping. Viscous damping defined with the CDAMPi, CVISC, CBUSH, or CBUSH1D bulk entries is not supported.
   - To define damping as frequency dependent, you must define the DAMPTAB bulk entry.

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- To define your structural damping and proportional viscous damping as a constant, you can use the DAMPING bulk entry, or you can define the parameter values G, ALPHA1, and ALPHA2. The parameter definition does not require the DAMPING=n case control command.
- You can define damping as a combination of the three options (DAMPTAB bulk entry, DAMPING bulk entry, and parameters) and the software uses the following precedence to select one of the proportional viscous damping definitions, one of the modal viscous damping definitions, and one of the structural damping definitions. For example, the software can use the parameter definition for the structural damping, and use the DAMPTAB definition for the modal damping.

  1. DAMPTAB entry.
  2. DAMPING entry.
  3. Parameters.

- The software computes the damping as follows.
  - Structural damping:

    You can define the global structural damping coefficient G_f, which the software uses to compute the structural damping k_i^{damp} for each mode:

    $$k_i^{damp} = \mathrm{G}_f k_i$$

    where k_i is the modal stiffness for the ith mode.

    The subscript f on the damping coefficient indicates that it can be defined dependent on the solution frequency.

    When you define frequency dependent structural damping, you define data tables representing the solution frequencies versus the structural damping. The software uses the actual solution frequencies you define on the FREQi entries to interpolate the data in the table. As a result, a unique structural damping value is used at each solution frequency.

  - Viscous proportional damping:

    You can define $\mathrm{ALPHA1}_f$, which the software uses to compute a viscous damping proportional to the mass of each mode m_i, and $\mathrm{ALPHA2}_f$, which the software uses to compute a viscous damping proportional to the stiffness of each mode k_i. The subscript f indicates that ALPHA1 and ALPHA2 can be defined dependent on the solution frequency.

    The proportional damping is computed as:

    $$b_i^p = \mathrm{ALPHA1}_f \cdot m_i + \mathrm{ALPHA2}_f \cdot k_i$$

    When you define frequency dependent proportional damping, you define data tables representing the solution frequencies versus ALPHA1 or ALPHA2. The software uses the actual

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solution frequencies you define on the FREQi entries to interpolate the data in the table. As a result, a unique ALPHA1 and ALPHA2 value is used at each solution frequency.

- Viscous modal damping:

  You define viscous modal damping with a data table containing angular natural frequencies versus critical damping ratios. The software uses the actual natural frequencies computed in the normal modes subcase to interpolate the data in the table. As a result, each mode has a unique, interpolated critical damping ratio which is applied consistently for all solution frequencies.

  The software uses the interpolated critical damping ratio g_i to compute the modal damping for each mode:

  $$b_i^{\mathrm{m}} = g_i \cdot b_i^{cr}$$

  where the critical damping b_i^{cr} for each mode is computed as

  $$b_i^{cr} = 2m_i\omega_i^{natfreq}$$

  and $\omega_i^{natfreq}$ is the angular natural frequency for the mode.

  The total viscous damping for each mode is the sum of the proportional and modal damping:

  $$b_i^{total} = b_i^{\mathrm{m}} + b_i^{p}$$

Remarks related to SOL 401:

1. For SOL 401, you can include the DAMPING=n case control command in the modal frequency response subcase to select the DAMPTAB bulk entry and the DAMPING bulk entry. You can define both the DAMPTAB entry and the DAMPING entry, and they can share the same SID. See Damping definition in the Modal frequency response subcase (SOL 401) topic in the Multi-Step Nonlinear User's Guide for details on all supported damping in this subcase.

Related Topics

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DAREA

Load Scale Factor

Defines scale (area) factors for static and dynamic loads. In dynamic analysis, DAREA is used in conjunction with RLOADi and TLOADi entries.

Format:

1	2	3	4	5	6	7	8	9	10
DAREA	SID	P1	C1	A1	P2	C2	A2		

Example:

DAREA	3	6	2	8.2	15	1	10.1		

Fields:

Field	Contents
SID	Identification number. (Integer>0)
Pi	Grid, extra, or scalar point identification number. (Integer>0)
Ci	Component number. (Integer 1 through 6 for grid point; blank or 0 for extra or scalar point) See Remark 3.
Ai	Scale factor. (Real)

Remarks:

1. One or two scale factors may be defined on a single entry.
2. Refer to RLOAD1, RLOAD2, TLOAD1, TLOAD2, or ACSRCE entries for the formulas that define the scale factor Ai in dynamic analysis.
3. Component numbers refer to the displacement coordinate system defined with the CD field on the GRID entry.
4. In dynamic analysis, DAREA entries may be used with LSEQ Bulk Data entries if LOADSET is specified in Case Control. The LSEQ and static load entries will be used to internally generate DAREA entries.
5. If DAREA is referenced by a GUST entry, Pi must be defined. However, it is only used if selected through a DLOAD Case Control command. WG from the GUST entry is used instead of Ai when requested via a GUST entry.
6. All DAREA entries corresponding to all grid and scalar points are automatically converted internally by the program to equivalent FORCE/MOMENT/SLOAD entries (as appropriate) if there are no LSEQ Bulk

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Data entries. The DAREA entries can only be used in static analysis if there are no LSEQ Bulk Data entries.

7. In superelement analysis, DAREA may be used to specify loads not only on the interior points of the residual, but also on the interior points of upstream superelements if there are no LSEQ Bulk Data entries.

Related Topics

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DCONADD

Design Constraint Set Combination

Defines the design constraints for a subcase as a union of DCONSTR entries.

Format:

1	2	3	4	5	6	7	8	9	10
DCONADD	DCID	DC1	DC2	DC3	DC4	DC5	DC6	DC7	
	DC8	-etc.-							

Example:

DCONADD	10	4	12						

Fields:

Field	Contents
DCID	Design constraint set identification number. (Integer>0)
DCi	DCONSTR entry identification number. (Integer>0)

Remarks:

1. The DCONADD entry is selected by a DESSUB or DESGLB Case Control command.
2. All DCi must be unique from other DCi.

Related Topics

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DCONSTR

Design Constraints

Defines design constraints.

Format:

1	2	3	4	5	6	7	8	9	10
DCONSTR	DCID	RID	LALLOW or LTID	UALLOW or UTID	LOWFQ	HIGHFQ			

Examples:

DCONSTR	10	4	1.25						

DCONSTR	300	101	–0.5	5.53					
DCONSTR	300	201	11	12					
DCONSTR	300	207	11	5.53					

Fields:

Field	Contents
DCID	Design constraint set identification number. (Integer>0)
RID	DRESPi entry identification number. (Integer>0)
LALLOW or LTID	Lower bound on the response quantity (Real) or TABLEDi ID (Integer) (Default = -1.0E20). See Remarks 4 and 10. See the remarks on the DRESP1 Bulk Data entry.
UALLOW or UTID	Upper bound on the response quantity (Real) or TABLEDi ID (Integer) (Default = 1.0E20). See Remark 4. See the remarks on the DRESP1 Bulk Data entry.
LOWFQ	Low end of frequency range in Hz. See Remark 9. (Real ≥ 0.0, Default = 0.0)
HIGHFQ	High end of frequency range in Hz (Real ≥ LOWFQ, Default = 1.0E+20). See Remark 9.

Remarks:

1. The DCONSTR entry may be selected in the Case Control section by the DESSUB or DESGLB command.

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2. DCID may be referenced by the DCONADD Bulk Data entry.
3. For a given DCID, the associated RID can be referenced only once.
4. If LALLOW and/or UALLOW are defined as real values, they are interpreted as constant.

   If defined as integers, they are interpreted as the ID of a TABLED1, TABLED2, TABLED3, or TABLED4 bulk entry which defines the bound as a function of frequency. When spawning multiple constraints over a frequency set from a single DCONSTR input, the program loops over the frequencies in the frequency set to determine the corresponding bound for each frequency.
5. The units of LALLOW and UALLOW must be consistent with the referenced response defined on the DRESPi entry. If RID refers to an "EIGN" response, then the imposed bounds must be expressed in the units of the eigenvalues, $(\text{radian/time})^2$. If RID refers to a "FREQ" response, then the imposed bounds must be expressed in cycles/time.
6. LALLOW and UALLOW are unrelated to the stress limits specified on the MATi entry.
7. For selection of DOT as the optimizer (the default), the constraints are computed as follows:

   g = (LALLOW – r) / GNORM for lower bound constraints

   g = (r – UALLOW) / GNORM for upper bound constraints

   where

$$\text{GNORM} = \begin{cases} |\text{LALLOW}| \text{ for lower bounds if } |\text{LALLOW}| > \text{GSCAL} \\ |\text{UALLOW}| \text{ for upper bounds if } |\text{UALLOW}| > \text{GSCAL} \\ \text{GSCAL otherwise} \end{cases}$$

   GSCAL is specified on the DOPTPRM entry. (Default = 0.001)

   If the alternate optimizer Siemens Design Optimization (SDO) is selected by setting system(425)=1, then a true normalization is performed, and the normalized constraint values always vary between -1.0 and +1.0 in a symmetric manner. Due to the method of normalization employed with the use of this optimizer, there is no need to account for a small denominator. Note also that the "Maximum Value of Constraint" column in the "Summary of Design Cycle History" table in the f06 file will also contain different values based on which optimizer is selected. The normalization for the Analytic Design System follows the below expressions:

   If a bound is sandwiched between zero and r:

   g=(LALLOW–r) / (|r|+GMAX) or g=(r–UALLOW) / (|r|+GMAX)

   Otherwise:

   g=(LALLOW–r) / (|2*LALLOW–r|+GMAX)

   or

   g=(r–UALLOW) / (|2*UALLOW–r|+GMAX)

   where GMAX is described with the DOPTPRM bulk entry.

   The only special case gives a constraint value of 0.0, thus no difficulties arise with a zero denominator.

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8. As Remark 7 indicates, when DOT is selected as the optimizer, small values of UALLOW and LALLOW require special processing and should be avoided. Bounds of exactly zero are particularly troublesome. This can be avoided by using a DRESP2 entry that offsets the constrained response from zero.

9. LOWFQ and HIGHFQ fields are functional only for DRESP1 response types FRDISP, FRVELO, FRACCL, FRSPCF, FRFORC, FRSTRE, PRES, PSDDISP, PSDVELO, and PSDACCL. The bounds provided in LALLOW and UALLOW are applicable to a response only when the value of the response of the forcing frequency falls between the LOWFQ and HIGHFQ. If the ATTB field of the DRESP1 entry is not blank, LOWFQ and HIGHFQ are ignored.

10. For responses where 0.0 is a natural lower bound that can never be violated, such as in the case of the von Mises equivalent stress, it is best to leave the lower bound to the default. The reason for this is that a lower bound of 0.0, when satisfied, does not add to solution value and can mislead the algorithm. For example, consider a feasible design where the maximum von Mises equivalent stress value is at 80% of the upper bound, but some elements are stressed very little. If a lower bound of 0.0 is prescribed, the stress values will then be very close to the constraint surfaces for these latter elements, and the constraint values will come out as far more critical (closer to zero) than for the actually stressed elements for which the constraint values are negative because the stresses are in the feasible region.

Related Topics

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DDAMCTR

DDAM Control Parameters

Defines parameters for DDAM that are otherwise contained in a DDAM control file.

Format:

1	2	3	4	5	6	7	8	9	10
DDAMCTR		VESSEL	LOCATION	TYPE	GMIN	FUNITS	AUNITS		
	AXISFA	AXISV	UNITNO						

Example:

DDAMCTR		SURFACE	HULL	ELASTIC	10.0	4.448	0.0254		
	X	Z	21						

Fields:

Name	Description
VESSEL	Type of ship. (Character = "SURFACE" or "SUBMERG"; Default = "SURFACE") • VESSEL = "SURFACE" for a surface ship. • VESSEL = "SUBMERG" for a submarine.
LOCATION	Equipment location. (Character = "DECK", "HULL", or "SHELL"; Default = "HULL") • LOCATION = "DECK" for deck-mounted. • LOCATION = "HULL" for hull-mounted. • LOCATION = "SHELL" for shell-mounted.
TYPE	Material behavior. (Character = "ELASTIC" or "ELPL"; Default = "ELASTIC") • TYPE = "ELASTIC" for elastic. • TYPE = "ELPL" for elastic/plastic.
GMIN	Minimum acceleration value in G's that is used in the shock design acceleration calculation for each mode. See Remark 1. (Real ≥ 0.0; Default = 6.0)
FUNITS	Force conversion factor. See Remark 2. (Real > 0.0; Default = 1.0)
AUNITS	Acceleration conversion factor. See Remark 3. (Real > 0.0; Default = 1.0)

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Name	Description
AXISFA	Fore/aft direction in the basic coordinate system. See Remark 4. (Character = "X", "Y", or "Z"; No default)
AXISV	Vertical direction in the basic coordinate system. See Remark 4. (Character = "X", "Y", or "Z"; No default)
UNITNO	Unit number for the external coefficient file. See Remarks 5 and 6. (Integer ≥ 0; Default = 0)

Remarks:

1. For information about how the software uses the value in the GMIN field to calculate the shock design acceleration for each mode, see "NAVSHOCK Overview" in the "DDAM" section of the *Simcenter Nastran Advanced Dynamic Analysis User's Guide*.
2. The NAVSHOCK program requires that forces be expressed in units of lbf. If the units of force in your Simcenter Nastran model are not lbf, you must specify a force conversion factor in the FUNITS field. The value that you specify in the FUNITS field should be the numerical value for force in the units of the Simcenter Nastran model that is physically equivalent to 1 lbf.

   For example, if the units for force in your Simcenter Nastran model are N, then you specify 4.448 in the FUNITS field because 4.448 N = 1 lbf.
3. The NAVSHOCK program requires that accelerations be expressed in units of in/sec^2. If the units of acceleration in your Simcenter Nastran model are not in/sec^2, you must specify an acceleration conversion factor in the AUNITS field. The value that you specify in the AUNITS field should be the numerical value for acceleration in the units of the Simcenter Nastran model that is physically equivalent to 1 in/sec^2.

   For example, if the units for acceleration in your Simcenter Nastran model are mm/sec^2, then you specify 25.4 in the AUNITS field because 24.4 mm/sec^2 = 1 in/sec^2.
4. The fore/aft direction and the vertical direction must be orthogonal.

   For example, you cannot select the X-direction for both the fore/aft direction and the vertical direction.

   Based on your specification for the fore/aft and vertical directions, the software automatically uses the direction you did not specify as the athwartship direction.

   For example, if you specify the X-direction for the fore/aft direction and the Z-direction for the vertical direction, the software uses the Y-direction as the athwartship direction.
5. When you specify the UNITNO field, you must use an unreserved unit number.

   For a listing of the reserved unit numbers, see "ASSIGN" in the "File Management Statements" section of the *Simcenter Nastran Quick Reference Guide*.
6. When the UNITNO field is zero or blank, the software uses the built-in shock design coefficients. For a listing of the built-in shock design coefficients, see "Coefficients" in the "DDAM" section of the *Simcenter Nastran Advanced Dynamic Analysis User's Guide*.

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Related Topics

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DDVAL

Discrete Design Variable Values

Define real, discrete design variable values for discrete variable optimization.

Format:

1	2	3	4	5	6	7	8	9	10
DDVAL	ID	DVAL1	DVAL2	DVAL3	DVAL4	DVAL5	DVAL6	DVAL7	

Alternate Format:

DDVAL	ID	DVAL1	“THRU”	DVAL	“BY”	INC			

The Continuation Entry formats may be used more than once and in any order. They may also be used with either format above.

Continuation Entry Format 1:

	DVAL8	DVAL9	DVAL10	DVAL11	-etc.-				

Continuation Entry Format 2::

	DVAL8	“THRU”	DVAL9	“BY”	INC				

Example:

1	2	3	4	5	6	7	8	9	10
DDVAL	110	0.1	0.2	0.3	0.5	0.6	0.4		
	07	thru	1.0	by	0.05				
	1.5	2.0							

Fields:

Field	Contents
ID	Unique discrete value set identification number. (Integer>0)
DVALi	Discrete values. (Real, or “THRU” or “BY”)
INC	Discrete value increment. (Real)

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Remarks:

1. DDVAL entries must be referenced by a DESVAR entry in the DDVAL field (field 8).
2. Trailing fields on a DDVAL record can be left blank if the next record is of type DVALi “THRU” DVALj “BY” INC. Also fields 7 - 9 must be blank when the type DVALi “THRU” DVALj “BY” INC is used in fields 2 - 6 and fields 8 - 9 must be blank when the type DVALi “THRU” DVALj “BY” INC is used in fields 3 - 7 for the first record. Embedded blanks are not permitted in other cases.
3. The DVALi sequence can be random.
4. The format DVALi “THRU” DVALj “BY” INC defines a list of discrete values, e.g., DVALi, DVALi+INC, DVALIi+2.0*INC, ..., DVALj. The last discrete DVALj is always included, even if the range is not evenly divisible by INC.

Related Topics

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DEFORM

Static Element Deformation

Defines enforced axial deformation for one-dimensional elements for use in statics problems.

Format:

1	2	3	4	5	6	7	8	9	10
DEFORM	SID	EID1	D1	EID2	D2	EID3	D3		

Example:

DEFORM	1	535	.05	536	-.10				

Fields:

Field	Contents
SID	Deformation set identification number. (Integer>0)
EIDi	Element number. (Integer>0)
Di	Deformation. (Real; positive value represents elongation.)

Remarks:

1. The referenced element must be one-dimensional (CROD, CONROD, CTUBE, CBAR, CBEAM).
2. Deformation sets must be selected in the Case Control Section with DEFORM = SID.
3. One to three enforced element deformations may be defined on a single entry.
4. The DEFORM entry, when called by the DEFORM Case Control command, is applicable to linear static, inertia relief, differential stiffness, and buckling (Solutions 101, 105, 114, and 200) and will produce fatal messages in other solution sequences. Use SPCD to apply enforced displacements in solution sequences for which DEFORM does not apply.

Related Topics

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DEFUSET

Degree-of-Freedom Set Name Definition

Defines new names for degree-of-freedom sets.

Format:

1	2	3	4	5	6	7	8	9	10
DEFUSET	OLD1	NEW1	OLD2	NEW2	OLD3	NEW3	OLD4	NEW4	

Example:

DEFUSET	U2	X	U4	Y	U3	Z			

Fields:

Field	Contents
OLDi	Default set name. (One to four characters)
NEWi	New set name. (One to four characters)

Remarks:

1. From one to four set names may be specified on a single entry.
2. OLDi must refer to any of the set names given in **"Degree-of-Freedom Sets"**. It is recommended that OLDi refer only to the set names U1 through U6. If sets PA or PS are referenced, a user fatal message is issued.
3. All NEWi names must be unique with respect to all other set names.
4. The DEFUSET entry is optional since default set names exist for all displacement sets.
5. The DEFUSET entry must be present in the Bulk Data Section in all restarts.
6. The user defined degree-of-freedom sets U1 – U9 are created with the USET and USET1 bulk entries. Be aware that the user defined sets U2 - U8 are used by Simcenter Nastran in some special cases. You may use U2 – U8 as long as it doesn't conflict with these cases. See **"User Defined Degree-of-Freedom Sets"** for the list of these special cases.

Related Topics

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DELAY

Dynamic Load Time Delay

Defines the time delay term τ in the equations of the dynamic loading function.

Format:

1	2	3	4	5	6	7	8	9	10
DELAY	SID	P1	C1	T1	P2	C2	T2		

Example:

DELAY	5	21	6	4.25	7	6	8.1		

Fields:

Field	Contents
SID	Identification number of the DELAY entry. (Integer>0)
Pi	Grid, extra, or scalar point identification number. (Integer>0)
Ci	Component number. (Integer 1 through 6 for grid point, blank or 0 for extra point or scalar point.)
Ti	Time delay τ for designated point Pi and component Ci. (Real)

Remarks:

1. One or two dynamic load time delays may be defined on a single entry.
2. SID must also be specified on a RLOAD1, RLOAD2, TLOAD1, TLOAD2, or ACSRCE entry. See those entry descriptions for the formulas that define the manner in which the time delay τ is used.
3. A DAREA and/or LSEQ entry should be used to define a load at Pi and Ci.
4. In superelement analysis, DELAY entries may only be applied to loads on points in the residual structure.

Related Topics

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DEQATN

Design Equation Definition

Defines one or more equations for use in design sensitivity analysis.

Format:

1	2	3	4	5	6	7	8	9	10
DEQATN	EQID	EQUATION							
	EQUATION (Cont.)								

Example:

DEQATN	14	F1 (A, B, C, D, R) = A + B · C – (D ** 3 + 10.0) + sin (PI (1 · R))	
	+ A**2/ (B – C) ; F = A + B – F1 · D		

Fields:

Field	Contents
EQID	Unique equation identification number. (Integer>0)
EQUATION	Equation(s). See Remarks. (Character)

Remarks:

1. EQUATION is a single equation or a set of nested equations and is specified in fields 3 through 9 on the first entry and may be continued on fields 2 through 9 on the continuation entries. On the continuation entries, no commas can appear in columns 1 through 8. All data in fields 2 through 9 must be specified in columns 9 through 72. The large-field format is not allowed.

   A single equation has the following format:

   variable-1 (x1, x2, ..., xn)=expression-1

   A set of nested equations is separated by semicolons and has the format:

   variable-1 (x1, x2, ..., xn)=expression-1;
   variable-2=expression-2;
   variable-3=expression-3;
   etc.
   variable-m=expression-m

   Expression-i is a collection of constants, real variables, and real functions, separated by operators, and must produce a single real value. (x1, x2, ..., xn) is the list of all the variable names (except variable-i) that appear in all expressions. Variable-i may be used in subsequent expressions. The last equation,

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variable-m=expression-m, provides the value that is returned to the Bulk Data entry that references EQID; e.g., DRESP2. The example above represents the following mathematical equations:

$$F1 = A + B \cdot C - (D^3 + 10) + \sin(\mathrm{PI}(1 \cdot R)) + \frac{A^2}{B - C}$$

$$F = A + B + F1 \cdot D$$

where SIN and PI are intrinsic functions. See Remark 4 .

2. EQUATION may contain embedded blanks. EQUATION must contain less than 32,000 nonblank characters. If more characters are required for use with a DRESP2 entry, the DRESP2 can be divided into two or more DRESP2 entries with a primary DRESP2 referencing subsequent DRESP2s.

3. The syntax of the expressions follows FORTRAN language standards. The allowable arithmetic operations are shown in Table 13-64 in the order of execution precedence. Parenthesis are used to change the order of precedence. Operations within parentheses are performed first with the usual order of precedence being maintained within the parentheses.

Table 13-64: DEQATN Entry Operators

Operator	Operation	Sample Expressions	Interpreted As
-, +	Negative or Positive immediately preceded by exponentiation	X * * -Y	X * * (-Y)
* *	Exponentiation	-X * * Y	(-X * * Y)
-, +	Negative or Positive	-X-Y	(-X)-Y
*, /	Multiplication or Division	X * Y-Z	(X * Y)-Z
+, -	Addition or Subtraction	X+Y	X+Y

4. The expressions may contain intrinsic functions. Table 13-65 contains the format and descriptions of functions that may appear in the expressions. The use of functions that may be discontinuous must be used with caution because they can cause discontinuous derivatives. These are ABS, DIM, MAX, MIN, and MOD. For examples and further details see the *Simcenter Nastran DMAP Programmer's Guide*.

Table 13-65: DEQATN Entry Functions

Format	Description	Mathematical Expressions
ABS(x)	absolute value	|x|
ACOS(x)	arccosine	$\cos^{-1} x$
ACOSH(x)	hyperbolic arccosine	$\cosh^{-1} x$
ASIN(x)	arcsine	$\sin^{-1} x$
ASINH(x)	hyperbolic arcsine	$\sinh^{-1} x$

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Format	Description	Mathematical Expressions
ATAN(x)	arctangent	$\tan^{-1} x$
ATAN2(x,y)	arctangent of quotient	$\tan^{-1} (x/y)$
ATANH(x)	hyperbolic arctangent	$\tanh^{-1} x$
ATANH2(x,y)	hyperbolic arctangent of quotient	$\tanh^{-1} (x/y)$
AVG(X_1, X_2, .., Xn)	average	$\frac{1}{n}\sum_{i=1}^{n} X_i$
COS(x)	cosine	cos x
COSH(x)	hyperbolic cosine	cosh x
DB(P, PREF)	sound pressure in decibel	$20.0 \cdot \log\left(\frac{P}{PREF}\right)$
DBA(P, PREF, F)	sound pressure in decibel (perceived)	$20.0 \cdot \log\left(\frac{P}{PREF}\right) + 10.0 \cdot \log(Ta1) + 10.0 \cdot \log(Ta2)$
DIM(x,y)	positive difference	x-MIN(x,y)
EXP(x)	exponential	e^x
INVDB(DB, PREF)	inverse Db	$10^{\left(\frac{DB}{20.0} + \log PREF\right)}$
INVDBA(DBA, PREF, F)	inverse Dba	$10^{\left(\frac{DBA - 10.0 \cdot \log(Ta1) - 10.0 \cdot \log(Ta2)}{20.0}\right)}$
LOG(x)	natural logarithm	$\log_e x$
LOG10(x)	common logarithm	$\log_{10} x$
LOGX(x,y)	base x logarithm	$\log_x y$
MAX(x_1, x_2, ..)	maximum	maximum of x_1, etc.
MIN(x_1, x_2, ...)	minimum	minimum of x_1, etc.
MOD(x,y)	remainder (modulo)	$x - y \cdot (\text{INT } (x / y))$
PI(x)	multiples of pi (π)	$x \cdot \pi$
RSS(X_1, ..., Xn)	square root of sum of squares	$\sqrt{\sum_{i=1}^{n} X_i^2}$

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Format	Description	Mathematical Expressions
SIN(x)	sine	sin x
SINH(x)	hyperbolic sine	sinh x
SQRT(x)	square root	$\sqrt{x}$
SSQ(X_1, ..., Xn)	sum of squares	$\sum_{i=1}^{n} X_i^2$
SUM(X_1, ..., Xn)	summation	$\sum_{i=1}^{n} X_i$
TAN(x)	tangent	tan x
TANH(x)	hyperbolic tangent	tanh x

$X_1, X_2, .., Xn, P$ = structure responses or acoustic pressure

PREF = reference pressure

F = forcing frequency

DB = acoustic pressure in Decibel

DBA = perceived acoustic pressure in Decibel

T_a1 = $\frac{K3 \cdot F^4}{(F^2 + P2^2)(F^2 + P3^2)}$

T_a2 = $\frac{K1 \cdot F^4}{(F^2 + P1^2)^2 (F^2 + P4^2)^2}$

$K1$ = 2.242882e+16

$K3$ = 1.562339

$P1$ = 20.598997

$P2$ = 107.65265

$P3$ = 737.86223

$P4$ = 12194.22

5. If the DEQATN entry is referenced by the:

- DVPREL2 entry, then X_i represents the DVIDj and LABLk fields.
- DRESP2 entry, then X_i represents the DVIDj, LABLk, NRm, Gp, DPIPq, DCICr, DMIMs, DPI2Pt, DCI2Cu, DMI2Mv, and NRRw fields in that order.

6. The DMAP logical operators NOT, AND, OR, XOR, and XQV cannot be used as X_i names.

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7. Input errors on the DEQATN entry often result in poor messages. Substituting a “[” for a parenthesis or violating the restriction against large field format are examples. Known messages are UFM 215, SFM 233 and UFM 5199. If any of these messages are encountered then, review the DEQATN entry input.
8. Intrinsic functions MAX, MIN, RSS, SSQ, SUM, and AVG are limited to <100 arguments. If more arguments are desired, the functions may be concatenated.
9. Arithmetic is carried out using the type of the input data. For example, in the expression:

   X = *A***(1/2)

   both values in the exponent are integers so that the value returned for the exponent is calculated using integer arithmetic or 1/2 = 0. In this case 1/2 should be replaced by (.5).

Related Topics

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DESC

Description

Defines a description for a set of kinematic joints or contact set in SOL 402, acoustic definitions in SOL 108 and SOL 111, or superelements in SOL 414.

Format:

1	2	3	4	5	6	7	8	9	10
DESC	DID	DESCRIPTION							

Example:

DESC	111	Joint of the left arm							

Fields:

Field	Contents
DID	Identification number. (Integer > 0)
DESCRIPTION	A string of no more than 56 characters (fields 3 through 9). (Character)

REMARKS:

1. In Simcenter 3D Pre/Post, the DESCRIPTION populates the contact set **Name** option.

Related Topics

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DESVAR

Design Variable

Defines a design variable for design optimization.

Format:

1	2	3	4	5	6	7	8	9	10
DESVAR	ID	LABEL	XINIT	XLB	XUB	DELXV	DDVAL		

Example:

DESVAR	2	BARA1	35.0	10.	100.	0.2			

Fields:

Field	Contents
ID	Unique design variable identification number. (Integer>0)
LABEL	User-supplied name for printing purposes. (Character. See **Bulk Data Syntax Rules**.)
XINIT	Initial value. (Real, XLB ≤ XINIT ≤ XUB)
XLB	Lower bound. (Real, Default = -1.0E+20)
XUB	Upper bound. (Real, Default = +1.0E+20)
DELXV	Fractional change allowed for the design variable during approximate optimization. (Real>0.0; for Default see Remark 2 .)
DDVAL	ID of a DDVAL entry that provides a set of allowable discrete values. (Blank or Integer>0; Default=blank for continuous design variables. See Remark 3 .)

Remarks:

1. DELXV can be used to control the change in the design variable during one optimization cycle.
2. If DELXV is blank, the default is taken from the specification of the DELX parameter on the DOPTPRM entry. If DELX is not specified, then the default is 1.0.
3. If the design variable is to be discrete (Integer>0 in DDVAL field), and if either of the XLB and/or XUB bounds are wider than those given by the discrete list of values on the corresponding DDVAL entry, XLB and/or XUB will be replaced by the minimum and maximum discrete values.
4. See also bulk data input for DVCREL1, DVCREL2, DVGEOM, DVGRID, DVMREL1, DVMREL2, DVPREL1, DVPREL2 in regard to associating the design variables with structural quantities.

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5. When associating design variables with structural quantities by way of the bulk data listed in Remark 4, it is useful to scale the design variables so as to make all design variables the same order of magnitude as far as possible, for purposes of healthier optimization. For example, Young's modulus and material density values are usually orders of magnitude apart, and if design variables were associated with each of these, the derivative for the first may end up being numerically negligible as compared to the second. By scaling the design variables properly, you can ensure that the effect of a given design variable does not get disregarded during optimization.

Related Topics

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DIVERG

Divergence Analysis Data

Defines Mach numbers (m) for a divergence analysis in SOLs 144 and 200.

Format:

1	2	3	4	5	6	7	8	9	10
DIVERG	SID	NROOT	M1	M2	M3	M4	M5	M6	
	M7	M8	-etc.-						

Example:

DIVERG	70	2	.5	.8	.9				

Fields:

Field	Contents
SID	Unique set identifier. (Integer>0)
NROOT	Number of divergence roots that are to be output and their eigenvectors printed. (Integer; Default = 1)
Mi	Mach number. (Real ≥ 0.0)

Remarks:

1. The DIVERG entry is referenced in Case Control by “DIVERG = SID”.
2. The NROOT lowest divergence dynamic pressures are printed. If there are fewer than NROOT pressures, all available dynamic pressures are printed.
3. Mi values must be distinct.
4. A blank Mach number field terminates the input.

Related Topics

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DLINK

Multiple Design Variable Linking

Relates one design variable to one or more other design variables.

Format:

1	2	3	4	5	6	7	8	9	10
DLINK	ID	DDVID	C0	CMULT	IDV1	C1	IDV2	C2	
	IDV3	C3	-etc.-						

Example:

DLINK	10	2	0.1	0.33	2	2.0	6	-1.0	
	8	7.0							

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
DDVID	Dependent design variable identification number. (Integer>0)
C0	Constant term. (Real; Default = 0.0)
CMULT	Constant multiplier. (Real; Default = 1.0)
IDVi	Independent design variable identification number. (Integer>0)
Ci	Coefficient i corresponding to IDVi. (Real)

Remarks:

1. DLINK defines the relationship

$$\text{DDVID} = C0 + \text{CMULT}\sum_{i} \text{Ci} \cdot \text{IDVi}$$

2. This capability provides a means of linking physical design variables such as element thicknesses to nonphysical design variables such as the coefficients of interpolating functions.
3. CMULT provides a simple means of scaling the Ci. For example if Ci = 1/7, 2/7, 4/7, etc. is desired, then CMULT = 1/7 and Ci = 1, 2, 4, etc., may be input.
4. An independent IDVi must not occur on the same DLINK entry more than once.

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5. ID is for user reference only.
6. If a design variable is specified as dependent on one DLINK entry, then it cannot be specified as independent on another DLINK entry.

Related Topics

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DLOAD

Dynamic Load Combination or Superposition

Defines a dynamic loading condition for frequency response or transient response analyses by using a linear combination of load sets. You combine the sets with:

- RLOAD1, RLOAD2, or RLOADEX bulk entries for a frequency response analysis.
- TLOAD1 or TLOAD2 bulk entries for a transient response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
DLOAD	SID	S	S1	L1	S2	L2	S3	L3	
	S4	L4	-etc.-						

Example:

DLOAD	17	1.0	2.0	6	-2.0	7	2.0	8	
	-2.0	9							

Fields:

Field	Contents
SID	Load set identification number. (Integer>0)
S	Overall scale factor. (Real)
Si	Scale factors. (Real)
Li	Load set identification numbers of RLOAD1, RLOAD2, TLOAD1, TLOAD2, and ACSRCE entries. (Integer>0)

Remarks:

1. Dynamic load sets must be selected in the Case Control section with DLOAD = SID.
2. The load vector $\{P\}$ that a DLOAD entry produces is the linear combination of the load sets $\{P_{Li}\}$ for the individual TLOADi or RLOADi entries. Thus, $\{P\}$ can be written mathematically as follows:

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$$\{P\} = S\left(\sum_i Si\,\{P_{Li}\}\right)$$

where S is the overall scale factor and Si are the scale factors for the individual load sets.

3. Each Li must be unique from any other Li on the same entry.
4. SID must be unique from all TLOADi, RLOADi, ACSRCE, and RLOADEX entries.
5. Nonlinear transient load sets (NOLINi entries) may not be specified on DLOAD entries. NOLINi entries are selected separately in the Case Control section by the NONLINEAR command.
6. A DLOAD entry may not reference a set identification number defined by another DLOAD entry.
7. TLOAD1 and TLOAD2 loads may be combined only through the use of the DLOAD entry.
8. RLOAD1, RLOAD2, and RLOADEX loads may be combined only through the use of the DLOAD entry.

Remarks related to SOLs 402 and 414 :

1. For SOL 402 (beginning with Simcenter Nastran 2021.2) and SOL 414,129 Transient Response (beginning with Simcenter Nastran 2206), the DLOAD=*n* case control command that selects either the TLOAD1 or DLOAD bulk entries must be defined globally (above all subcases). A fatal error occurs if you define the DLOAD=*n* command in a subcase. You can use the system cell setting NASTRAN SYSTEM(767)= 0 to turn off the fatal error and revert to the previous behavior.
2. DLOAD may be used in a nonlinear static or transient analysis to combine time-dependent loads.
3. Li can only be the load set identification numbers of TLOAD1 entries.
4. Both LOAD and DLOAD can be used in the same analysis to define some loads with constant magnitude and some time-dependent loads.
5. For SOLs 402 and 414, before Simcenter Nastran 2212, when you combined rotational inertia loads defined with the RFORCE or RFORCE1 on the LOAD or DLOAD bulk entries, the software scaled the angular velocity ($\vec{\omega}$) and acceleration ($\vec{\alpha}$) by both the time varying scale factor *f(t)*, and by the scale factors S and Si on the LOAD or DLOAD entry.

   The software now uses S and S_i on the LOAD or DLOAD entry to scale the rotational force, and not the angular velocity and acceleration. The angular velocity and acceleration are still scaled by the time varying scale factor *f(t)*:

   $$\vec{\omega}_s = f(t)\,\vec{\omega}\,\vec{\alpha}_s = f(t)\;\vec{\alpha}$$

   Because the software applies S and S_i the same way for all loads, the total load from *i* load sets is now computed using:

   $$\{P\} = S\sum_i S_i\,\{P_{Li}\}$$

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where $\{P_{Li}\}$ represents all load sets referenced on the LOAD or DLOAD entry, including rotational inertia loads.

You can revert to the previous scaling by defining SYSTEM(767)=0. When SYSTEM(767)=0, the angular velocity and acceleration are scaled using:

$$\vec{\omega}_s = S\ S_i\ f(t)\ \vec{\omega}\,\vec{\alpha}_s = S\ S_i\ f(t)\ \vec{\alpha}$$

Related Topics

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DMI

Direct Matrix Input

Defines matrix data blocks. Generates a matrix of the following form:

$$[\text{NAME}] = \begin{bmatrix} X_{11} & X_{12} & \cdots & X_{1n} \\ X_{21} & X_{22} & \cdots & X_{2n} \\ \cdot & \cdot & \cdot & \cdot \\ X_{m1} & \cdots & \cdots & X_{mn} \end{bmatrix}$$

where the elements X_{ij} may be real ($X_{ij} = A_{ij}$) or complex ($X_{ij} = A_{ij} + iB_{ij}$). The matrix is defined by a single header entry and one or more column entries. Only one header entry is required. A column entry is required for each column with nonzero elements. Any unspecified elements of the matrix are taken to be zero.

Header Entry Format:

1	2	3	4	5	6	7	8	9	10
DMI	NAME	"0"	FORM	TIN	TOUT		m	n	

General Column Entry Format for Real Matrices:

In this data entry format, you selectively specify elements in a column of a real matrix. Because elements having a value of zero can be excluded, it is useful for defining sparsely populated real matrices.

DMI	NAME	J	I1	A(I1,J)	A(I1+1,J)	A(I1+2,J)	-etc.-	I2	
	A(I2,J)	A(I2+1,J)	A(I2+2,J)	-etc.-					

Unabridged Column Entry Format for Real Matrices:

In this data entry format, you specify all the elements in a column of a real matrix. It is useful for defining highly populated real matrices.

DMI	NAME	J	"1"	A(1,J)	A(2,J)	A(3,J)	A(4,J)	A(5,J)	
	A(6,J)	A(7,J)	-etc.-						

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General Column Entry Format for Complex Matrices:

In this data entry format, you selectively specify elements in a column of a complex matrix. Because elements having a value of zero can be excluded, it is useful for defining sparsely populated complex matrices.

DMI	NAME	J	I1	A(I1,J)	B(I1,J)	A(I1+1,J)	B(I1+1,J)	A(I1+2,J)	
	B(I1+2,J)	-etc.-	I2	A(I2,J)	B(I2,J)	A(I2+1,J)	B(I2+1,J)	A(I2+2,J)	
	B(I2+2,J)	-etc.-							

Unabridged Column Entry Format for Complex Matrices:

In this data entry format, you specify all the elements in a column of a complex matrix. It is useful for defining highly populated complex matrices.

DMI	NAME	J	"1"	A(1,J)	B(1,J)	A(2,J)	B(2,J)	A(3,J)	
	B(3,J)	A(4,J)	B(4,J)	-etc.-					

General Column Entry Format Examples:

$$[BBB] = \begin{bmatrix} 1.0 & 0.0 \\ 3.0 & 6.0 \\ 5.0 & 0.0 \\ 0.0 & 8.0 \end{bmatrix}$$

DMI	BBB	0	2	1	1		4	2	
DMI	BBB	1	1	1.0	3.0	5.0			
DMI	BBB	2	2	6.0	4	8.0			

$$[QQQ] = \begin{bmatrix} 1.0 + 2.0i & 0.0 + 0.0i \\ 3.0 + 0.0i & 6.0 + 7.0i \\ 5.0 + 6.0i & 0.0 + 0.0i \\ 0.0 + 0.0i & 8.0 + 9.0i \end{bmatrix}$$

DMI	QQQ	0	2	3	3		4	2	
DMI	QQQ	1	1	1.0	2.0	3.0	0.0	3	
	5.0	6.0							
DMI	QQQ	2	2	6.0	7.0	4	8.0	9.0	

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Unabridged Column Entry Format Examples:

$$[BBB] = \begin{bmatrix} 1.0 & 0.0 \\ 3.0 & 6.0 \\ 5.0 & 0.0 \\ 0.0 & 8.0 \end{bmatrix}$$

DMI	BBB	0	2	1	1		4	2	
DMI	BBB	1	1	1.0	3.0	5.0	0.0		
DMI	BBB	2	1	0.0	6.0	0.0	8.0		

$$[QQQ] = \begin{bmatrix} 1.0 + 2.0i & 0.0 + 0.0i \\ 3.0 + 0.0i & 6.0 + 7.0i \\ 5.0 + 6.0i & 0.0 + 0.0i \\ 0.0 + 0.0i & 8.0 + 9.0i \end{bmatrix}$$

DMI	QQQ	0	2	3	3		4	2	
DMI	QQQ	1	1	1.0	2.0	3.0	0.0	5.0	
	6.0	0.0	0.0						
DMI	QQQ	2	1	0.0	0.0	6.0	7.0	0.0	
	0.0	8.0	9.0						

Fields:

Field	Contents
NAME	Name of the matrix. See Remark 1. NAME is used to reference the data block in the DMAP sequence. (One to eight alphanumeric characters, the first of which must be alphabetic.)
FORM	Form of matrix. (Integer) 1 = Square matrix (not symmetric) 2 = General rectangular matrix 3 = Diagonal matrix (elements on the diagonal are stored in a column vector having m rows) 4 = Lower triangular factor 5 = Upper triangular factor 6 = Symmetric matrix

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Field	Contents
	8 = Identity matrix (m = number of rows, n = m)
TIN	Type of matrix being input. (Integer) 1 = Real, single precision (one field used per element) 2 = Real, double precision (one field used per element) 3 = Complex, single precision (two fields used per element) 4 = Complex, double precision (two fields used per element)
TOUT	Type of matrix being output. (Integer) 0 = Set by precision cell 1 = Real, single precision 2 = Real, double precision 3 = Complex, single precision 4 = Complex, double precision
m	Total number of rows in the matrix. (Integer>0)
n	Total number of columns in the matrix. Not required if FORM 3 or 8. (Integer>0)
“0”	Indicates a header entry. (Integer)
J	Specified column number of the matrix. (Integer>0)
“1”	Designates element entry begins with the first row of column J. (Integer)
I1, I2, etc.	Row number in column J where a single element is to be defined or the beginning of a string of elements is to be defined. See Remark 12. (Integer>0)
A(Ix,J)	Real part of the element (see TIN). (Real)
B(Ix,J)	Imaginary part of the element (see TIN). (Real)

Remarks:

1. In order to use the DMI feature, the user must write a DMAP, or make alterations to a solution sequence that includes the DMIIN module. See the *Simcenter Nastran DMAP Programmer’s Guide*. All rules governing the use of data blocks in DMAP sequences apply.
2. The total number of DMIs and DTIs may not exceed 1000.
3. Field 3 of the header entry must contain an integer of zero (0).

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4. For symmetric matrices, the entire matrix must be input.
5. Leading and trailing zeros in a column do not have to be entered. However, a blank field between nonzero fields on this entry is not equivalent to a zero. If a zero input is required, the appropriate type zero must be entered (i.e. 0.0 or 0.0D0).
6. Complex input must have both the real and imaginary parts entered if either part is nonzero (i.e. the zero component must be input explicitly).
7. If A(Ix,J) is followed by "THRU" in the next field and an integer row number "IX" after THRU, then A(Ix,J) will be repeated in each row through IX. The "THRU" must follow an element value. For example, the entries for a real matrix RRR would appear as follows:

1	2	3	4	5	6	7	8	9	10
DMI	NAME	J	I1	A(I1,J)			I1	A(I2,J)	
DMI	RRR	1	2	1.0	THRU	10	12	2.0	

These entries will cause the first column of the matrix RRR to have a zero in row 1, the values 1.0 in rows 2 through 10, a zero in row 11, and 2.0 in row 12.

8. Each column must be a single logical entry. The terms in each column must be specified in increasing row number order.
9. The "FORM" options 4, 5, and 8 are nonstandard forms and may be used only in conjunction with the modules indicated in Table 13-66.

Table 13-66: DMI FORM Options

FORM	Matrix Description	Modules			
		ADD	FBS	MATPRN	MPYAD
4	Lower Triangular Factor		X	X	
5	Upper Triangular Factor		X	X	
8	Identity	X	X	X	X

10. Form 3 matrices are converted to Form 6 matrices, which may be used by any module.
11. Form 7 matrices may not be defined on this entry.
12. I1 must be specified. I2, etc. are not required if their matrix elements follow the preceding element in the next row of the matrix. For example, in the column entry for column 1 of QQQ, neither I2 nor I3 is specified.
13. The DMIG entry is more convenient for matrices with rows and columns that are referenced by grid or scalar point degrees-of-freedom.

Related Topics

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DMIG

Direct Matrix Input at Points

Defines direct input matrices related to grid, extra, and/or scalar points. The matrix is defined by a single header entry and one or more column entries. A column entry is required for each column with nonzero elements.

Header Entry Format:

1	2	3	4	5	6	7	8	9	10
DMIG	NAME	"0"	IFO	TIN	TOUT	POLAR		NCOL	

Column Entry Format:

DMIG	NAME	GJ	CJ		G1	C1	A1	B1	
	G2	C2	A2	B2	-etc.-				

Example:

DMIG	STIF	0	1	3	4				
DMIG	STIF	27	1		2	3	3.+5	3.+3	
	2	4	2.5+10	0.	50		1.0	0.	

Fields:

Field	Contents
NAME	Name of the matrix. See Remark 1 . (One to eight alphanumeric characters, the first of which is alphabetic.)
IFO	Form of matrix input. (Integer) 1 = Square 2 = Rectangular 6 = Symmetric 9 = Rectangular You must specify IFO equals:

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Field	Contents
	1 for matrices selected by the A2GG Case Control commands. 1 or 6 for matrices selected by the B2PP, K2PP, and M2PP Case Control commands. 6 for matrices selected by the B2GG, K2GG, M2GG, and K42GG Case Control commands. 9 for matrices selected by the P2G Case Control commands. For SOL 414, see Remarks **3** and **4**.
TIN	Type of matrix being input. (Integer) 1 = Real, single precision (One field is used per element.) 2 = Real, double precision (One field is used per element.) 3 = Complex, single precision (Two fields are used per element.) 4 = Complex, double precision (Two fields are used per element.)
TOUT	Type of matrix that will be created: (Integer) 0 = Set by precision system cell (Default) 1 = Real, single precision 2 = Real, double precision 3 = Complex, single precision 4 = Complex, double precision
POLAR	Input format of Ai, Bi. (Integer=blank or 0 indicates real, imaginary format; Integer>0 indicates amplitude, phase format.)
NCOL	Number of columns in a rectangular matrix. Used only for IFO = 9. See Remarks **5**. and **6**. (Integer>0)
GJ	Grid, scalar or extra point identification number for a column index. (Integer>0)
CJ	Component number for a grid point GJ. (0<Integer ≤ 6; blank or zero if GJ is a scalar or extra point.)
Gi	Grid, scalar, or extra point identification number for a row index. (Integer>0)
Ci	Component number for Gi for a grid point. (0<CJ ≤ 6 ; blank or zero if Gi is a scalar or extra point.)
Ai, Bi	Real and imaginary (or amplitude and phase) parts of a matrix element. If the matrix is real (TIN = 1 or 2), then Bi must be blank. (Real)

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Field	Contents
	For SOL 414, see Remark 5.

Remarks:

1. Matrices defined on this entry may be used in dynamics by selection in the Case Control with K2PP = NAME, or B2PP = NAME, or M2PP = NAME for $[K_{pp}]$, $[B_{pp}]$, or $[M_{pp}]$, respectively. Matrices may also be selected for all solution sequences with K2GG = NAME, or B2GG = NAME, or M2GG = NAME, or K42GG = NAME. The g-set matrices are added to the structural matrices before constraints are applied, while p-set matrices are added in dynamics after constraints are applied. Load matrices may be selected by P2G = NAME for dynamic and superelement analyses.
2. The header entry containing IFO, TIN and TOUT is required. Each nonnull column is started with a GJ, CJ pair. The entries for each row of that column follows. Only nonzero terms need be entered. The terms may be input in arbitrary order. A GJ, CJ pair may be entered more than once, but input of an element of the matrix more than once will produce a fatal message.
3. Field 3 of the header entry must contain an integer 0.
4. For symmetric matrices (IFO = 6), a given off-diagonal element may be input either below or above the diagonal. While upper and lower triangle terms may be mixed, a fatal message will be issued if an element is input both below and above the diagonal.
5. The recommended format for rectangular matrices requires the use of NCOL and IFO = 9. The number of columns in the matrix is NCOL. (The number of rows in all DMIG matrices is always either p-set or g-set size, depending on the context.) The GJ term is used for the column index. The CJ term is ignored.
6. If NCOL is not used for rectangular matrices, two different conventions are available:
   - If IFO = 9, GJ and CJ will determine the sorted sequence, but will otherwise be ignored; a rectangular matrix will be generated with the columns submitted being in the 1 to N positions, where N is the number of logical entries submitted (not counting the header entry).
   - If IFO = 2, the number of columns of the rectangular matrix will be equal to the index of the highest numbered non-null column (in internal sort). Trailing null columns of the g- or p-size matrix will be truncated.
7. The matrix names must be unique among all DMIGs.
8. TIN should be set consistent with the number of decimal digits required to read the input data adequately. For a single-precision specification on a short-word machine, the input will be truncated after about eight decimal digits, even when more digits are present in a double-field format. If more digits are needed, a double precision specification should be used instead. However, note that a double precision specification requires a “D” type exponent even for terms that do not need an exponent. For example, unity may be input as 1.0 in single precision, but the longer form 1.0D0 is required for double precision.
9. On long-word machines, almost all matrix calculations are performed in single precision and on short-word machines, in double precision. It is recommended that DMIG matrices also follow these conventions for a balance of efficiency and reliability. The recommended value for TOUT is 0, which instructs the program to inspect the system cell that measures the machine precision at run time and

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sets the precision of the matrix to the same value. TOUT = 0 allows the same DMIG input to be used on any machine. If TOUT is contrary to the machine type specified (for example, a TOUT of 1 on a short-word machine), unreliable results may occur.

10. If any DMIG entry is changed or added on restart then a complete re-analysis is performed. Therefore, DMIG entry changes or additions are not recommended on restart.

Remark related to SOLs 401 and 402:

1. SOL 401 supports the direct matrix input of stiffness, damping, and mass on grid points. SOL 402 supports the direct matrix input of mass on grid points. See the remarks on the **K2GG**, **B2GG**, and **M2GG** case control commands.

Remarks related to SOL 414 solutions:

1. In SOL 414 models, you can replace non-rotating components (that is, components with no gyroscopic effect) with DMIG components. If you apply the DMIG component on a rotor node, gyroscopic effect is not taken into account.
2. SOL 414 supports the direct matrix input of stiffness (**K2GG** case control command), viscous damping (**B2GG** case control command), and mass (**M2GG** case control command).
3. The matrices can be non-symmetric (square, IFO=1) or direct symmetric (IFO=6).
   - For the non-symmetric (square, IFO=1) matrices:

     The stiffness (K2GG), viscous damping (B2GG), and mass (M2GG) matrices are supported on grid points only.

     The square (IFO=1) mass matrix must be fully symmetric.
   - For the direct symmetric (IFO=6) matrices:

     The stiffness (K2GG), viscous damping (B2GG), and mass (M2GG) matrices are supported on grid points or scalar points (SPOINT).
4. If the SOL 414 model uses some scalar points to define the matrices:
   - The model must include scalar points and at least one stiffness matrix.
   - To process the matrix data, SOL 414 attempts to group the matrices by DOFs, using the stiffness matrix K as the primary matrix.
     - For each stiffness matrix K, SOL 414 searches for a mass matrix M that shares the same DOFs (or a subset of the DOFs) as the given stiffness matrix DOFs to create a K-M group to process.

       If one of the DOFs in matrix M does not have a match in matrix K, the M matrix is not grouped with the K matrix. Fewer DOFs in the M matrix than in the K matrix do not make a difference.
     - If the model contains damping matrices, for each stiffness matrix K, SOL 414 searches for the damping matrix B that shares the same DOFs (or a subset of the DOFs) as the given stiffness matrix DOFs to complete the K-M group and make final K-M-B groups of matrices to process (one K-M-B group for each stiffness matrix).

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If one of the DOFs in matrix B does not have a match in matrix K, matrix B is not grouped with the K-M group. Fewer DOFs in matrix B than in matrix K do not make a difference. Scalar points in matrix B are excluded from the search process.

Thus, only the K-M or K-M-B groups are analyzed and output.

Other ungrouped matrices are discarded with a warning.

Linear combinations of DMIG tables in K2GG, B2GG, or M2GG case control commands that contain SPOINT are blocked with an error message.

5. SOL 414 does not take Bi values (imaginary part of matrix element) into account.

Related Topics

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DMIG,UACCEL

Direct Matrix Input of Enforced Static Acceleration

Defines rigid body accelerations in the basic coordinate system.

Format:

1	2	3	4	5	6	7	8	9	10
DMIG	UACCEL	"0"	"9"	TIN				NCOL	
DMIG	UACCEL	L			G1	C1	X1		
	G2	C2	X2		G3	C3	X3		

Example:

DMIG	UACCEL	0	9	1				4	
DMIG	UACCEL	2			2	3	386.4		
DMIG	UACCEL	3			2	4	3.0		
DMIG	UACCEL	4			2	6	1.0		

Fields:

Field	Contents
TIN	Type of matrix being input. (Integer 1 or 2) 1 = Real, single precision. (one field is used per element)2 = Real, double precision. (one field is used per element)
NCOL	Number of columns. See Remark 2 . (Integer>0; Default is the number of columns specified)
L	Load sequence number. (Integer>0)
Gi	Grid point identification number of a single reference point. (Integer>0)
Ci	Component number for Gi in the basic coordinate system. See Remark 4 . (0<Integer ≤ 6)
Xi	Value of enforced acceleration term in the basic coordinate system. (Real)

Remarks:

1. DMIG,UACCEL is an optional entry when PARAM,INREL,-1 is specified in SOLs 101 or 200. If DMIG,UACCEL is present, the loads applied to the structure are the sum of the conventional applied

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loads plus the inertia loads resulting from the rigid body accelerations defined on this entry. If it is not present, conventional inertia relief calculations are performed.

2. The load sequence number interpretation depends on the value of the NCOL field. The recommended method is to set it equal to the number of loading conditions. The load sequence number L is then the sequence number of the subcase to which the applied acceleration will be applied.
3. The grid point identification number listed on Gi defines a single grid point on the model where loads will be applied to cause the enforced acceleration state. Gi must also appear on a SUPORT Bulk Data entry and a PARAM,GRDPNT entry. In superelement analysis, it must be a residual structure point exterior to all superelements.
4. The Xi value is the enforced acceleration at grid point Gi. The translation and rotation components are in consistent units and will be applied in the basic coordinate system regardless of the displacement coordinate system specified for Gi (CD field on GRID entry).
5. Only nonzero terms need be entered.
6. See the *Simcenter Nastran User's Guide* for the theoretical basis of inertia relief with superelements.
7. If any DMIG entry is changed or added on restart then a complete re-analysis is performed. Therefore, DMIG entry changes or additions are not recommended on restart.

Related Topics

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DMIJ

Direct Matrix Input at js-Set of the Aerodynamic Mesh

Defines direct input matrices related to collation degrees of freedom (js-set) of aerodynamic mesh points for CAERO1, CAERO3, CAERO4 and CAERO5 and for the slender body elements of CAERO2. These include W2GJ, FA2J and input pressures and downwashes associated with AEPRESS and AEDW entries. The matrix is described by a single header entry and one or more column entries. A column entry is required for each column with nonzero elements. For entering data for the interference elements of a CAERO2, use DMIJI or DMI.

Header Entry Format:

1	2	3	4	5	6	7	8	9	10
DMIJ	NAME	"0"	IFO	TIN	TOUT	POLAR		NCOL	

Column Entry Format:

DMIJ	NAME	GJ	CJ		G1	C1	A1	B1	
	G2	C2	A2	B2	-etc.-				

Example:

DMIJ	ALPH1	0	9	2	0			1	
DMIJ	ALPH1	1	1		1	1	.1		
	2	1	.1						

Fields:

Field	Contents
NAME	Name of the matrix. See Remark 1 . (One to eight alphanumeric characters, the first of which is alphabetic.)
IFO	Form of matrix being input. (Integer) 1 = Square 9 or 2 = Rectangular 6 = Symmetric
TIN	Type of matrix being input. (Integer)

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Field	Contents
	1 = Real, single precision (one field s used per element)
	2 = Real, double precision (one field used per element)
	3 = Complex, single precision (two fields used per element)
	4 = Complex, double precision (two fields used per element)
TOUT	Type of matrix being created: (Integer)
	0 = Set by precision system cell (Default)
	1 = Real, single precision
	2 = Real, double precision
	3 = Complex, single precision
	4 = Complex, double precision
POLAR	Input format of Ai, Bi. (Integer = blank or 0 indicates real, imaginary format. Integer>0 indicates magnitude, phase format.)
NCOL	Number of columns in a rectangular matrix. Used only for IFO = 9. (Integer>0)
GJ	Grid, scalar or extra point identification number for a column index. (Integer>0)
CJ	Component number for grid point GJ. (0<Integer ≤ 6; blank or zero if GJ is a scalar or extra point.)
Gi	Grid, scalar, or extra point identification number for a row index. (Integer>0)
Ci	Component number for Gi for a grid point. (0<CJ ≤6; blank or zero if Gi is a scalar or extra point.)
Ai, Bi	Real and imaginary (or amplitude and phase) parts of a matrix element. If the matrix is real (TIN = 1 or 2), then Bi must be blank. (Real)

Remarks:

1. Matrices defined on this entry are referenced in static aeroelastic analysis by reference on AEDW and/or AEPRESS entries. In that paradigm, a single column is required. Also, DMIJ may also be used for the W2GJ and FA2J entries. Again, a single column is required. If both DMI and DMIJ are specified for W2GJ or FA2J, the DMI entry will be used. DMI may NOT be used for AEDW and AEPRESS.
2. The js-set DOFs for each aerodynamic theory are limited to the six-DOF paradigm (3 translations and 3 rotations). However, particular DOFs are permanently SPC'd based on the theory's ability to support those degrees of freedom. Unlike the DMIG entry, DMIJ data will be partitioned to the j-set, not reduced. No warnings are issued about truncated data.The j-set DOFs for each aerodynamic element/

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grid are highly method dependent. The following data define the known set, but the j-set definition is somewhat arbitrary in the general (external aerodynamics) paradigm.

Entry Type	COMP				
	1	2	3	5	6
CAERO1			X		
CAERO2-Y		X			
CAERO2-Z			X		
CAERO2-ZY		X	X		

3. The header entry containing IFO, TIN and TOUT is required. Each nonnull column is started with a GJ, CJ pair. The entries for each row of that column then follow. Only nonzero terms need be entered. The terms may be input in arbitrary order. A GJ, CJ pair may be entered more than once, but the input of an element of the matrix more than once results in a fatal message.
4. Field 3 of the header entry must contain an integer 0.
5. For symmetric matrices (very rare in the j-set) (IFO=6), a given off-diagonal element may be input either below or above the diagonal. While upper and lower triangle terms may be mixed, a fatal message will be issued if an element is input both above and below the diagonal.
6. The recommended format for rectangular matrices requires the use of NCOL and IFO = 9. The number of columns in the matrix is NCOL. (The number of rows in all DMIJ matrices is always the js-set size—the union of the j-set and the permanently SPC'd partition.) The GJ term is used for the column index. the CJ term is ignored.
7. If NCOL is not used for rectangular matrices, two different conventions are available:
   - If IFO = 9, GJ and CJ will determine the sorted sequence, but will otherwise be ignored; a rectangular matrix will be generated with the columns submitted being in the 1 to N positions, where N is the number of logical entries submitted (not counting the header entry).
   - If IFO = 2, the number of columns of the rectangular matrix will be equal to the index of the highest numbered nonnull column (in internal sort). Training null columns of the js-size matrix will be truncated.
8. The matrix names must be unique among all DMIJ.
9. TIN should be consistent with the number of decimal digits required to read the input data adequately. For a single precision specification on a short word machine, the input will be truncated after about eight decimal digits, even when more digits are present in a double field format. If more digits are needed, a double precision specification should be used instead. However, not that a double precision specification requires a "D" type exponent even for terms that do not need an exponent. For example, unity may be input as 1.0 in single precision, but the longer form 1.0D0 is required for double precision.
10. On long word machines, almost all matrix calculations are performed in single precision and, on short word machines, in double precision. It is recommended that DMIJ matrices also follow these conventions for a balance of efficiency and reliability. The recommended value for TOUT is 0, which instructs the program to inspect the system cell that measures the machine precision at run time and

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sets the precision of the matrix to the same value. TOUT = 0 allows the same DMIJ input to be used on any machine. If TOUT is contrary to the machine type specified, unreliable results may occur.

11. If any DMIJ entry is changed or added on restart then a complete reanalysis is performed. Therefore, DMIJ entry changes or additions are not recommended on restart.

Related Topics

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DMIJI

Direct Matrix Input at js-Set of the Interference Body

Defines direct input matrices related to collation degrees of freedom (js-set) of aerodynamic mesh points for the interference elements of CAERO2. These include W2GJ, FA2J and input pressures and downwashes associated with AEPRESS and AEDW entries. The matrix is described by a single header entry and one or more column entries. A column entry is required for each column with nonzero elements. For entering data for the slender elements of a CAERO2 or a CAERO1, 3, 4 or 5 use DMIJ or DMI.

Header Entry Format:

1	2	3	4	5	6	7	8	9	10
DMIJI	NAME	“0”	IFO	TIN	TOUT	POLAR		NCOL	

Column Entry Format:

DMIJ	NAME	GJ	CJ		G1	C1	A1	B1	
	G2	C2	A2	B2	-etc.-				

Example:

DMIJI	ALPH1	0	9	2	0			1	
DMIJI	ALPH1	1	1		1	1	.1		
	2	1	.1						

Fields:

Field	Contents
NAME	Name of the matrix. See Remark 1 . (One to eight alphanumeric characters, the first of which is alphabetic.)
IFO	Form of matrix being input. (Integer) 1 = Square 9 or 2 = Rectangular 6 = Symmetric
TIN	Type of matrix being input. (Integer) 1 = Real, single precision (one field is used per element)

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Field	Contents
	2 = Real, double precision (one field is used per element) 3 = Complex, single precision (two fields used per element) 4 = Complex, double precision (two fields used per element)
TOUT	Type of matrix being created. (Integer) 0 = Set by precision system cell (Default) 1 = Real, single precision 2 = Real, double precision 3 = Complex, single precision 4 = Complex, double precision
POLAR	Input format of Ai, Bi. (Integer = blank or 0 indicates real, imaginary format. Integer>0 indicated magnitude, phase format.)
NCOL	Number of columns in a rectangular matrix. Used only for IFO = 9. (Integer>0)
GJ	Grid, scalar or extra point identification number for a column index. (Integer>0)
CJ	Component number for grid point GJ. (0<Integer ≤ 6; blank or zero if GJ is a scalar or extra point.)
Gi	Grid, scalar, or extra point identification number for a row index. (Integer>0)
Ci	Component number for Gi for a grid point. (0<CJ ≤ 6; blank or zero if Gi is a scalar or extra point.)
Ai, Bi	Real and imaginary (or amplitude and phase) parts of a matrix element. If the matrix is real (TIN = 1 or 2), then Bi must be blank. (Real)

Remarks:

1. Matrices defined on this entry are referenced in static aeroelastic analysis by reference on AEDW and/or AEPRESS entries. In that paradigm, a single column is required. DMI may NOT be used for AEDW and AEPRESS.
2. The js-set DOFs for each aerodynamic theory are limited to the six-DOF paradigm (3 translations and 3 rotations). However, particular DOFs are permanently SPC'd based on the theory's ability to support those degrees of freedom. Unlike the DMIG entry, DMIJI data will be partitioned to the j-set, not reduced. No warnings are issued about truncated data. The j-set DOFs for each aerodynamic element/grid are highly method dependent. The following data define the known set, but the j-set definition is somewhat arbitrary in the general (external aerodynamics) paradigm.

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Entry Type	COMP				
	1	2	3	5	6
CAERO2-Y		X			
CAERO2-Z			X		
CAERO2-ZY		X	X		

3. The header entry containing IFO, TIN and TOUT is required. Each nonnull column is started with a GJ, CJ pair. The entries for each row of that column then follow. Only nonzero terms need be entered. The terms may be input in arbitrary order. A GJ, CJ pair may be entered more than once, but the input of an element of the matrix more than once results in a fatal message.
4. Field 3 of the header entry must contain an integer 0.
5. For symmetric matrices (very rare in the j-set!) (IFO=6), a given off-diagonal element may be input either below or above the diagonal. While upper and lower triangle terms may be mixed, a fatal message will be issued if an element is input both above and below the diagonal.
6. The recommended format for rectangular matrices requires the use of NCOL and IFO = 9. The number of columns in the matrix is NCOL. (The number of rows in all **DMIJI** matrices is always the js-set size—the union of the j-set and the permanently SPC'd partition.) The GJ term is used for the column index. The CJ term is ignored.
7. If NCOL is not used for rectangular matrices, two different conventions are available:
   - If IFO = 9, GJ and CJ will determine the sorted sequence, but will otherwise be ignored; a rectangular matrix will be generated with the columns submitted being in the 1 to N positions, where N is the number of logical entries submitted (not counting the header entry).
   - If IFO = 2, the number of columns of the rectangular matrix will be equal to the index of the highest numbered nonnull column (in internal sort). Trailing null columns of the js-size matrix will be truncated.
8. The matrix names must be unique among all DMIJI.
9. TIN should be consistent with the number of decimal digits required to read the input data adequately. For a single precision specification on a short word machine, the input will be truncated after about eight decimal digits, even when more digits are present in a double field format. If more digits are needed, a double precision specification should be used instead. However, note that a double precision specification requires a "D" type exponent even for terms that do not need an exponent. For example, unity may be input as 1.0 in single precision, but the longer form 1.0D0 is required for double precision.
10. On long word machines, almost all matrix calculations are performed in single precision and, on short word machines, in double precision. It is recommended that DMIJ matrices also follow these conventions for a balance of efficiency and reliability. The recommended value for TOUT is 0, which instructs the program to inspect the system cell that measures the machine precision at run time and sets the precision of the matrix to the same value. TOUT = 0 allows the same DMIJI input to be used on any machine. If TOUT is contrary to the machine type specified, unreliable results may occur.
11. If any DMIJ entry is changed or added on restart then a complete reanalysis is performed. Therefore, DMIJ entry changes or additions are not recommended on restart.

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Related Topics

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DMIK

Direct Matrix Input at ks-Set of the Aerodynamic Mesh

Defines direct input matrices related to physical (displacement) degrees of freedom (ks-set) of aerodynamic grid points. These include WKK, WTFACT and input forces associated with AEFORCE entries. The matrix is described by a single header entry and one or more column entries. A column entry is required for each column with nonzero elements.

Header Entry Format:

1	2	3	4	5	6	7	8	9	10
DMIK	NAME	"0"	IFO	TIN	TOUT	POLAR		NCOL	

Column Entry Format:

DMIK	NAME	GJ	CJ		G1	C1	A1	B1	
	G2	C2	A2	B2	-etc.-				

Example:

DMIK	ALPH1	0	9	2	0			1	
DMIK	ALPH1	1	1		1	1	1.0		
	2	1	1.0						

Fields:

Field	Contents
NAME	Name of the matrix. See Remark 1 . (One to eight alphanumeric characters, the first of which is alphabetic.)
IFO	Form of matrix being input. (Integer) 1 = Square 9 or 2 = Rectangular 6 = Symmetric
TIN	Type of matrix being input. (Integer) 1 = Real, single precision (One field is used per element)

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Field	Contents
	2 = Real, double precision (One field is used per element) 3 = Complex, single precision (Two fields are used per element) 4 = Complex, double precision (Two fields are used per element)
TOUT	Type of matrix being created: (Integer) 0 = Set by precision system cell (Default) 1 = Real, single precision 2 = Real, double precision 3 = Complex, single precision 4 = Complex, double precision
POLAR	Input format of Ai, Bi. (Integer = blank or 0 indicates real, imaginary format. Integer>0 indicated magnitude, phase format.)
NCOL	Number of columns in a rectangular matrix. Used only for IFO = 9. (Integer>0)
GJ	Grid, scalar or extra point identification number for column index. (Integer>0)
CJ	Component number for grid point GJ. (0<Integer ≤ 6; blank or zero if GJ is a scalar or extra point.)
Gi	Grid, scalar, or extra point identification number for row index. (Integer>0)
Ci	Component number for Gi for a grid point. (0<CJ ≤ 6; blank or zero if Gi is a scalar or extra point.)
Ai, Bi	Real and imaginary (or amplitude and phase) parts of a matrix element. If the matrix is real (TIN = 1 or 2), then Bi must be blank. (Real)

Remarks:

1. Matrices defined on this entry are referenced in static aeroelastic analysis by reference on AEFORCE entries. In that paradigm, a single column is required. Also, DMIK may also be used for the WKK and WTFACT entries. If both DMI and DMIK are specified for WKK or WTFACT, the DMI entry will be used. DMI may NOT be used for AEFORCE.
2. The ks-set DOFs for each aerodynamic theory are limited to the six-DOF paradigm (3 translations and 3 rotations). However, particular DOFs are permanently SPC'd based on the theory's ability to support those degrees of freedom. Unlike the DMIG entry, DMIK data will be partitioned to the k-set, not reduced. No warnings are issued about truncated data. The k-set DOFs for each aerodynamic element/grid are highly method dependent. The following data define the known set, but the j-set definition is somewhat arbitrary in the general (external aerodynamics) paradigm.

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Entry Type	COMP				
	1	2	3	5	6
CAERO1			X	X	
CAERO2-Y		X		X	
CAERO2-Z			X		X
CAERO2-ZY		X	X	X	X

3. The header entry containing IFO, TIN and TOUT is required. Each nonnull column is started with a GJ, CJ pair. The entries for each row of that column then follow. Only nonzero terms need be entered. The terms may be input in arbitrary order. A GJ, CJ pair may be entered more than once, but the input of an element of the matrix more than once results in a fatal message.
4. Field 3 of the header entry must contain an integer 0.
5. For symmetric matrices (IFO=6), a given off-diagonal element may be input either below or above the diagonal. While upper and lower triangle terms may be mixed, a fatal message will be issued if an element is input both above and below the diagonal.
6. The recommended format for rectangular matrices requires the use of NCOL and IFO = 9. The number of columns in the matrix is NCOL. (The number of rows in all DMIK matrices is always the ks-set size—the union of the k-set and the permanently SPC'd partition). The GJ term is used for the column index. The CJ term is ignored.
7. If NCOL is not used for rectangular matrices, two different conventions are available:
   - If IFO = 9, GJ and CJ will determine the sorted sequence, but will otherwise be ignored; a rectangular matrix will be generated with the columns submitted being in the 1 to N positions, where N is the number of logical entries submitted (not counting the header entry).
   - If IFO = 2, the number of columns of the rectangular matrix will be equal to the index of the highest numbered nonnull column (in internal sort). Training null columns of the js-size matrix will be truncated.
8. The matrix names must be unique among all DMIK
9. TIN should be consistent with the number of decimal digits required to read the input data adequately. For a single precision specification on a short word machine, the input will be truncated after about eight decimal digits, even when more digits are present in a double field format. If more digits are needed, a double precision specification should be used instead. However, not that a double precision specification requires a "D" type exponent even for terms that do not need an exponent. For example, unity may be input as 1.0 in single precision, but the longer form 1.0D0 is required for double precision.
10. On long word machines, almost all matrix calculations are performed in single precision and, on short word machines, in double precision. It is recommended that DMIK matrices also follow these conventions for a balance of efficiency and reliability. The recommended value for TOUT is 0, which instructs the program to inspect the system cell that measures the machine precision at run time and sets the precision of the matrix to the same value. TOUT = 0 allows the same DMIK input to be used on any machine. If TOUT is contrary to the machine type specified, unreliable results may occur.

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11. If any DMIK entry is changed or added on restart then a complete reanalysis is performed. Therefore, DMIK entry changes or additions are not recommended on restart.

Related Topics

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DMNCON

Manufacturing Constraint for Topology Optimization

Defines a manufacturing constraint for topology optimization.

Additive Manufacturing (ADDM) Format:

1	2	3	4	5	6	7	8	9	10
DMNCON	ID	ADDM							
	ANGLE	MIND	X	Y	Z	N1	N2	N3	

Casting Die Direction(s) (CDID) Format:

1	2	3	4	5	6	7	8	9	10
DMNCON	ID	CDID							
	X	Y	Z	N1	N2	N3			
	D1	D2	D3	D21	D22	D23			

Checker-boarding control (CHBC) Format:

1	2	3	4	5	6	7	8	9	10
DMNCON	ID	CHBC							
	OFF-FLAG								

Cyclic symmetry (SYMC) Format:

1	2	3	4	5	6	7	8	9	10
DMNCON	ID	SYMC							
	X	Y	Z	N1	N2	N3			
	M1	M2	M3	NSECT					

Extrusion along a straight line (EXTC) Format:

1	2	3	4	5	6	7	8	9	10
DMNCON	ID	EXTC							
	N1	N2	N3						

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Minimum size (MINS) Format:

1	2	3	4	5	6	7	8	9	10
DMNCON	ID	MINS							
	Size								

Maximum size (MAXS) Format:

1	2	3	4	5	6	7	8	9	10
DMNCON	ID	MAXS							
	Size								

Planar symmetry (SYMP) Format:

1	2	3	4	5	6	7	8	9	10
DMNCON	ID	SYMP							
	X	Y	Z	N1	N2	N3			

Examples:

DMNCON	500	SYMP							
	0.0	1.0	0.0	0.7071	0.7071	1.0			

DMNCON	200	EXTC							
	1.0	0.0	0.0						

Fields:

Field	Contents
General Fields:	
ID	Identification number. (Integer>0)
TYPE	Type of manufacturing constraint. (TYPE examples: ADDM, CHBC, EXTC) See Remark 4. (Character; No default)
Fields for Additive Manufacturing (ADDM):	
Note that either ANGLE or MIND can be blank, indicating there is no such constraint.	
ANGLE	Maximum angle from the normal that points away from the base plate, for overhangs, in degrees. (Real)

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Field	Contents
MIND	Minimum allowed size. (Real).
X,Y,Z	Coordinates of a point on a plane for the base plate. (Real)
Ni	Components of a vector normal to the base plate in the direction of material addition. (Real)

Fields for Casting die direction(s) (CDID):

X,Y,Z	Coordinates of a point on the casting plane. (Real)
Ni	Components of a vector normal to the casting plane. (Real)
Di	Components of a vector that defines the first direction of mold removal. (Real)
D2i	Components of an optional vector that defines a second direction of mold removal. If these components D2i are all 0.0 or blank, then only the first direction is used. (Real: default=0.0,0.0,0.0)

Fields for Checker-boarding control (CHBC):

OFF-FLAG	This field is used only to turn off the default checker-boarding control, achieved by entering a negative real number in this field. (Real) The checker-boarding control manufacturing constraint is on by default, even when you have not defined the DMNCON bulk entry. To disable checker-boarding control, you must define the DMNCON bulk entry with TYPE= CHBC and a negative real number in the OFF-FLAG field. Any other choices for the OFF-FLAG field will result in the checker-boarding control remaining on by default.

Fields for Cyclic symmetry (SYMC):

X,Y,Z	Coordinates of a point on the rotation axis. (Real)
Ni	Components of a vector defining the axis of rotation. (Real)
Mi	Components of a vector perpendicular to the axis of rotation, defining the first boundary of the first symmetry sector. The software will sweep this vector counter clockwise by the angle = 360/NSECT to determine the boundaries of the consecutive symmetry sectors. (Real)
NSECT	The number of sectors that would fill the 360 degrees. (Positive or negative integer) When NSECT is a positive integer, the symmetry is that of repeated sectors. When NSECT is a negative integer, the symmetry is that of reflected sectors.

Fields for Extrusion along a straight line (EXTC):

Ni	Components of a vector defining the extrusion direction. (Real)

Fields for Min and Max size (MINS or MAXS):

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Field	Contents
Size	Minimum or maximum member size. (Real)
Fields for Planar symmetry (SYMP):	
X,Y,Z	Coordinates of a point on the symmetry plane. (Real)
Ni	Components of a vector normal to the symmetry plane. (Real)

Remarks:

1. Multiple DMNCON bulk entries are supported if you would like to define a variety of manufacturing conditions. For example, you can have multiple DMNCON entries with TYPE=SYMP, and could also add another DMNCON entry with TYPE=MAXS.
2. The manufacturing conditions currently apply to all of the active elements selected for topology optimization.
3. Except for TYPE=SYMP, the parameter BDMNCON defines the design cycle at which the software will start application of the manufacturing constraints during the SOL 200 topology optimization solution. The resulting delay has been found to improve the quality of results in topology optimization with manufacturing constraints. The BDMNCON parameter default is the smaller of 10 cycles, or the value of the DESMAX parameter minus 3, but not smaller than 2. The DESMAX parameter defines the maximum number of design cycles allowed, and is defined on the DOPTPRM bulk entry. SYMP is applied from the beginning.
4.

TYPE	Description
Additive (ADDM)	Additive manufacturing builds material layer by layer. The previous layers in a build-up must be sufficient to support consecutive layers. When building up overhanging geometry, if the overhang is built in an aggressive angle, there is a possibility that the structure can collapse as a result of cantilevered material. You define a maximum overhang angle and/or a minimum size applicable to any slender members, as well as the manufacturing direction. Overhang angles are measured relative to the positive manufacturing direction vector.
Casting die direction (CDID)	Parts that are cast should not have any pockets that cannot be formed or protrusions that would interfere with the mold pieces from parting, or with the part coming out of the mold. You define the coordinates of a point on the casting plane, a vector normal to the casting plane, a vector along which one part of the mold travels when separating, and another, optional, vector along which the other part of the mold, if it exists, travels when separating.
Checker-boarding control (CHBC)	Checker-boarding is a condition where the software removes material in an alternating pattern similar to a checker board, when simpler finite elements are used. It is undesirable because it does not represent an optimal distribution of material and the results are difficult to

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TYPE	Description
	manufacture. The checker-boarding control constraint can be used to help prevent this condition from occurring. OFF-FLAG < 0.0, the checker-boarding control is turned off. OFF-FLAG ≥ 0.0, the checker-boarding control is left on. * Note: The software applies the checker-boarding control by default even when you have not defined a DMNCON bulk entry. If you want to disable the checker-boarding control, you must define the DMNCON bulk entry with TYPE=CHBC and a negative real number in the OFF-FLAG field.
Cyclic symmetry (SYMC)	You provide a full circular model that has repeated cyclic symmetry sectors. The mesh on each sector does not need to match. The software will work to symmetrize the normalized mass density (NMD) values on each sector. You define the axis of rotation, the number of sectors (NSECT) within 360 degrees, and relative to a 0-degree symmetry plane, you define a point on that symmetry plane and a radial vector perpendicular to the axis of rotation along one of the edges of the symmetry plane. The software sweeps the 0-degree symmetry plane counter clockwise by the angle = 360 / | NSECT | to determine the consecutive symmetry planes. When NSECT is a positive integer, the software treats your model as having repeated sectors. When NSECT is a negative integer, the software treats your model as having reflected sectors. With repeated symmetry, each sector is similar to every other sector. With reflected symmetry, every sector is the mirror image of the previous sector. You can use reflected symmetry only with an even number of sectors.
Extrusion along a straight line (EXTC)	Extruded parts must have material continuity in the extrusion direction. You define the straight line extrusion direction.
Min size (MINS)	This allows you to control the minimum member size for created “slender” members. You define the minimum cross-sectional "thickness" size.
Max size (MAXS)	This allows you to control the maximum member size. For example, if you define the maximum member size, truss members created by the optimization process will not be any "thicker" than the specified size.
Planar symmetry (SYMP)	With this condition, your model must be meshed on both sides of the symmetry plane. The mesh on each side of the symmetry plane does not need to match. The software works to symmetrize the NMD values on both sides of a given symmetry plane. You define a point on the symmetry plane and a vector normal to the symmetry plane.

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DMRLAW

Material relation law definition for SOL 200 topology optimization.

Defines the relation between material properties and the normalized mass density (NMD) for SOL 200 topology optimization.

Format:

1	2	3	4	5	6	7	8	9	10
DMRLAW	ID	TYPE	TYPEPRM						

Example:

DMRLAW	200	SIMP	3.5						

DMRLAW	200	SIMP							

DMRLAW	302	CUBIC							

Fields:

Field	Contents
ID	Material relation law ID. (Integer > 0)
TYPE	Type of law. See Remarks 2 and 3 for the supported law types. (Character; Default=RAMP, with TYPEPRM=5.0)
TYPEPRM	Defines *p* for the SIMP penalty law, and *q* for the RAMP penalty law. See Remark 3. (Real; Default is p=3.0 for the SIMP law and q=5.0 for the RAMP law)

Remarks:

1. Only a single DMRLAW bulk entry is allowed.
2. A lattice in this context is a thin truss-like repeating cell structure that is manufactured, for example, using additive manufacturing. Each lattice type has a unique mass density to stiffness relationship. With the “LINEAR“ type (see below), the software auto-generates the design variables for topology optimization such that a linear relationship exists between the NMD and the Young’s Modulus. In contrast, lattices, as well as SIMP and RAMP, impose non-linear relations.

   Currently you can only request a single DMRLAW bulk entry for all elements you have designated as active with one or more DVTREL1 entries. The software will adjust design variable to property relations accordingly.

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The following types of lattice cells and penalty functions are supported:

TYPE Field Value:	Lattice Type
OCTET	Octet
CUBIC	Cubic
BCC	BCC
FCC	FCC
OCTAHDRL	Octahedral
BCCCUBIC	BCC+Cubic
FCCCUBIC	FCC+Cubic
BCCFCC	BCC+FCC
BFCUBIC	BCC+FCC+Cubic
LINEAR	See Remark 2.
SIMP	See Remark 3.
RAMP	See Remark 3.

3. In addition to the lattice types, you also have the option to define a mass density to stiffness relationship using one of two following penalty laws. These penalty laws define the effective Young's modulus as a function of the normalized mass density μ.

   - For TYPE=SIMP, the SIMP law has the form:

$$E(\mu) = \mu^{p} E_{0}$$

     where p is a penalty parameter defined with the TYPEPRM field. If you leave the TYPEPRM field blank, the software uses the default value of p=3.0.

   - For TYPE=RAMP, the RAMP law has the form:

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$$E(\mu) = \frac{\mu}{1 + q(1 - \mu)} E_0$$

where q is the penalty parameter defined with the TYPEPRM field. If you leave the TYPEPRM field blank, the software uses the default value of q=5.0.

Related Topics

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DOPTPRM

Design Optimization Parameters

Overrides default values of parameters used in design optimization.

Format:

1	2	3	4	5	6	7	8	9	10
DOPTPRM	PARAM1	VAL1	PARAM2	VAL2	PARAM3	VAL3	PARAM4	VAL4	
	PARAM5	VAL5	-etc.-						

Example:

DOPTPRM	IPRINT	5	DESMAX	10					

Fields:

Field	Contents
PARAMi	Name of the design optimization parameter. Allowable names are given in Table 13-67. (Character)
VALi	Value of the parameter. (Real or Integer) See Table 13-67.

Remarks:

1. Only one DOPTPRM entry is allowed in the Bulk Data Section.

Table 13-67: PARAMi Names and Descriptions

Name	Description, Type, and Default Value
APRCOD	Approximation method to be used. 1 = Direct Linearization; 2=Mixed Method based on response type; 3 = Convex Linearization. APRCOD = 1 is recommended for shape optimization problems. (Integer 1, 2, or 3; Default = 2)
CONV1	Relative criterion to detect convergence. If the relative change in objective between two optimization cycles is less than CONV1, optimization is terminated. (Real>0.0; Default = 0.0001)
CONV2	Absolute criterion to detect convergence. If the absolute change in objective between two optimization cycles is less than CONV2, optimization is terminated. (Real>0.0; Default = 1.0E-20)
CONVDV	Relative convergence criterion on design variables. (Real>0.0; Default = 0.001)

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Name	Description, Type, and Default Value
CONVPR	Relative convergence criterion on properties. (Real>0.0; Default = 0.001)
CT	Constraint tolerance. Constraint is considered active if current value is greater than CT. (Real<0.0; Default = -0.03)
CTMIN	Constraint is considered violated if current value is greater than CTMIN.(Real>0.0; Default = 0.003)
DABOBJ*	Maximum absolute change in objective between ITRMOP consecutive iterations (see ITRMOP) to indicate convergence at optimizer level. F0 is the initial objective function value. See Remark 2. (Real>0.0; Default = MAX[0.001*ABS(F0), 0.0001])
DELB	Relative finite difference move parameter. (Real>0.0; Default = 0.0001)
DELOBJ*	Maximum relative change in objective between ITRMOP consecutive iterations to indicate convergence at optimizer level. See Remark 2. (Real>0.0; Default = 0.001)
DELP	Fractional change allowed in each property during any optimization design cycle. This provides constraints on property moves. (Real>0.0; Default = 0.2)
DELX	Fractional change allowed in each design variable during any optimization cycle. (Real>0.0; Default = 1.0) See DXMIN for absolute minimum change allowed per cycle for design variables.
DESMAX	Maximum number of design cycles (not including FSD cycle) to be performed. (Integer ≥ 0; Default = 5) While the default for DESMAX is 5 cycles, large problems often require a significantly higher number of cycles. For example, this could be somewhere between 20 and 50, depending on many factors. The most significant factors are the time required for a full analysis pass, the number of design variables, and the nonlinearity of the constraints.
DISBEG	Design cycle ID for discrete variable processing initiation. Discrete variable processing analysis is carried out for every design cycle after DISBEG. (Integer>=0, default = 0=the last design cycle) If fully stressed design (FSD), discrete variable processing can start only after the last FSD cycle.
DISCOD	Discrete Processing Method: (Integer 1, 2, 3 or 4; Default = 1) 1: Design of Experiments 2: Conservative Discrete Design 3: Rounding up to the nearest design variable 4: Rounded off to the nearest design variable
DOBJ1*	Relative change in objective attempted on the first optimization iteration. Used to estimate initial move in the one-dimensional search. Updated as the optimization progresses. See Remark 2. (Real>0.0; Default = 0.1)

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Name	Description, Type, and Default Value
DOBJ2*	Absolute change in objective attempted on the first optimization iteration. See Remark 2. (Real>0.0; Default = 0.2*(F0))
DPMIN	Minimum move limit imposed. (Real>0.0; Default = 0.01)
DX1*	Maximum relative change in a design variable attempted on the first optimization iteration. Used to estimate the initial move in the one-dimensional search. Updated as the optimization progresses. See Remark 2. (Real>0.0; Default = 0.01)
DX2*	Absolute change in a design variable attempted on the first optimization iteration. See Remark 2. (Real>0.0; Default = 0.2*MAX[X(I)])
DXMIN	Minimum absolute limit on design variable move (Real>0.0; Default = 0.05). See DELX for allowable relative change per cycle in design variables.
EDVOUT	Defines the fraction of DVEREL1-generated design variables with increased and decreased values to be printed into the Design Variable History at the end of the f06 file, as well as at requested cycles. This fraction is applied separately to the increased and to the decreased DVEREL1-generated design variables. For example, when the default of 0.001 is used, both 0.1% of the most increased and 0.1% of the most decreased DVEREL1-generated design variables will be printed. A value of 1.0 requests to print all DVEREL1-generated design variables that have either increased or decreased. (0.0<Real≤1.0; Default = 0.001) Note: The increase or decrease in the value of a DVEREL1 generated design variable is based on the relative change between the “initial” and the “best design” cycles. Note: The design variables that are not generated by the DVEREL1 entry are all printed in full.
FSDALP	Relaxation parameter applied in Fully Stressed Design (Real, 0.0<FSDALP ≤ 1.0, Default = 0.9)
FSDMAX	Specifies the number of Fully Stressed Design Cycles that are to be performed (Integer, Default = 0)
GMAX	Maximum constraint violation allowed at the converged optimum. (Real>0.0; Default = 0.005)
GSCAL	Constraint normalization factor. See Remarks under the DSCREEN and DCONSTR entries. (Real>0.0; Default = 0.001)
IGMAX	If IGMAX = 0, only gradients of active and violated constraints are calculated. If IGMAX>0, up to NCOLA gradients are calculated including active, violated, and near active constraints. (Integer≥0; Default = 0)
IPRINT*	Print control during approximate optimization phase. Increasing values represent increasing levels of optimizer information. See Remark 2. (0 ≤ Integer ≤ 7; Default = 0) 0 no output (Default)

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Name	Description, Type, and Default Value
	1 internal optimization parameters, initial information, and results 2 same as 1, plus objective function and design variables at each iteration 3 same as 2, plus constraint values and identification of critical constraints 4 same as 3, plus gradients 5 same as 4, plus search direction 6 same as 5, plus scaling factors and miscellaneous search information 7 same as 6, plus one-dimensional search information
IPRNT1*	If IPRNT1 = 1, print scaling factors for design variable vector. See Remark 2. (Integer 0 or 1; Default = 0)
IPRNT2*	If IPRNT2 = 1, print miscellaneous search information. If IPRNT2 = 2, turn on print during one-dimensional search process. (Warning: This may lead to excessive output.) See Remark 2. (Integer 0, 1, or 2; Default = 0)
ISCAL	Design variables are rescaled at every ISCAL iteration. Set ISCAL = -1 to turn off scaling. (Integer; Default=NDV (number of design variables))
ITMAX*	Maximum number of iterations allowed at optimizer level during each design cycle. See Remark 2. (Integer; Default = 40)
ITRMOP*	Number of consecutive iterations for which convergence criteria must be satisfied to indicate convergence at the optimizer level. See Remark 2. (Integer; Default = 2)
ITRMST*	Number of consecutive iterations for which convergence criteria must be met at the optimizer level to indicate convergence in the Sequential Linear Programming Method. See Remark 2. (Integer>0; Default = 2)
IWRITE*	FORTRAN unit for print during approximate optimization phase. Default value for IWRITE is set to the FORTRAN unit for standard output. See Remark 2. (Integer>0, Default=6 or value of SYSTEM(2).)
JPRINT*	Sequential Linear Programming subproblem print. If JPRINT>0, IPRINT is turned on during the approximate linear subproblem. See Remark 2. (Default = 0)
JTMAX*	Maximum number of iterations allowed at the optimizer level for the Sequential Linear Programming Method. This is the number of linearized subproblems solved. See Remark 2. (Integer≥0; Default = 20)
JWRITE*	If JWRITE>0, file number on which iteration history will be written. See Remark 2. (Integer>0; Default = 0)
METHOD	DOT Optimization Method: (Integer 1, 2, or 3; Default = 1) 1: Modified Method of Feasible Directions. (Default) 2: Sequential Linear Programming

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Name	Description, Type, and Default Value
	3: Sequential Quadratic Programming For Siemens Analytic Design System (SADS), all selections will automatically revert to a proprietary, linear programming-based algorithm. To request the SADS optimizer, system cell 425 in Table 1-1 of the NASTRAN Statement section should be set to 1.
MXCRTRSP	Control the data written to the csv file, or to request for the software not to output the file. (Integer; Default=5) -1: The csv file is not created. 0: For a SOL 200 Design Optimization solution, the file includes the design objective values and the design variable values for each design cycle. For a SOL 200 Topology Optimization solution, the file only includes the design objective values for each design cycle. 1-20: In addition to design objective values (and the design variable values for a Design Optimization solution), the file includes design constraint values. The number you enter (1-20) is the number of critical constraints that the software will output. The critical constraints are the most violated constraints, or the feasible constraints closest to their bound, for each design cycle.
OBJCNV	Convergence method: (Integer 0 or 1; Default=0) 0: The software considers the solution converged when the objective, the design property, and the design variable checks have each converged. 1: The software considers the solution converged when only the objective check has converged. See Convergence Tests in the *Simcenter Nastran Design and Optimization User's Guide* for more information.
P1	Determines the design cycles in which output is printed. (Integer ≥ 0; Default = 0) Initial results are always printed prior to the first approximate optimization. If an optimization task is performed, final results are always printed for the final analysis unless PARAM,SOFTEXIT,YES is specified. These two sets of print are not controllable. n: Print at every n-th design cycle.
P2	Items to be printed at the design cycles defined by P1 (Integer; Default = 1) 0: No print. 1: Print objective and design variables. (Default) 2: Print properties. 4: Print constraints. 8: Print responses. 16: Print weight as a function of a material ID (note that this is not a design quantity so that only inputs to the approximate design are available).

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Name	Description, Type, and Default Value
	n: Sum of desired items. For example, P2 = 10 means print properties and responses.
P2RSET	ID of a SET1 bulk entry listing constrained responses to be printed if retained (the SET1 lists DRESPi ID numbers). The default is to print all retained constrained responses. Constrained responses are those responses referenced directly or indirectly by a DCONSTR bulk entry. Retained constrained responses are the constrained responses which the software determines to be relevant in a design cycle in terms of being critical in driving the optimization process.
PLVIOL	Flag for handling of property limit violation. By default, the job will terminate with a user fatal message if the property derived from design model (DVPRELi, DVMRELi, DVCRELi) exceeds the property limits. Setting PLVIOL to a non-zero number will cause the program to issue a user warning message by ignoring the property limits violation and proceed with the analysis. (Integer; Default=0)
PTOL	Maximum tolerance on differences allowed between the property values on property entries and the property values calculated from the design variable values on the DESVAR entry (through DVPRELi relations). PTOL is provided to trap ill-posed design models. (The minimum tolerance may be specified on user parameter DPEPS. See **"Parameters and Parameter Applicability Tables"**) (Real>0.0; Default = 1.0E+35)
STPSCL	Scaling factor for shape finite difference step sizes, to be applied to all shape design variables. (Real>0.0; Default = 1.0)

2. The parameter names with "*" are supported only by the DOT Optimization Method.

Related Topics

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DPHASE

Dynamic Load Phase Lead

Defines the phase lead term θ in the equation of the dynamic loading function.

Format:

1	2	3	4	5	6	7	8	9	10
DPHASE	SID	P1	C1	TH1	P2	C2	TH2		

Example:

DPHASE	4	21	6	2.1	8	6	7.2		

Fields:

Field	Contents
SID	Identification number of DPHASE entry. (Integer>0)
Pi	Grid, extra, or scalar point identification number. (Integer>0)
Ci	Component number. (Integers 1 through 6 for grid points; zero or blank for extra or scalar points)
THi	Phase lead θ in degrees. (Real)

Remarks:

1. One or two dynamic load phase lead terms may be defined on a single entry.
2. SID must be referenced on a RLOADi or ACSRCE entry. Refer to the RLOAD1, RLOAD2, or ACSRCE entry for the formulas that define how the phase lead θ is used.
3. A DAREA and/or LSEQ entry should be used to define a load at Pi and Ci.
4. In superelement analysis, DPHASE entries may only be applied to loads on points in the residual structure.

Related Topics

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DRESP1

Design Response Quantities

Defines a set of structural responses that are used for the objective and/or design constraints, or for sensitivity analysis purposes.

Format:

1	2	3	4	5	6	7	8	9	10
DRESP1	ID	LABEL	RTYPE	PTYPE	REGION	ATTA	ATTB	ATT1	
	ATT2	ATT3	ATT4	-etc.-					

DRESP1	1	DX1	STRESS	PROD	2	3		102	
	103								

Fields:

Field	Contents
ID	Unique entry identifier. (Integer>0)
LABEL	User-defined label. (Character. See **Bulk Data Syntax Rules**.)
RTYPE	Response type. See **Table 13-68**. (Character)
PTYPE	Type of entities listed under the ATTi fields. Depending on the response type, the following values are allowed (See Remarks **6**, **37**, and **39**): • Element flag (PTYPE = ELEM). • Property entry name (for example, PTYPE = PSHELL). • Panel flag for ERP responses (PTYPE = PANEL). • RANDPS ID for random analysis. • PTYPE=GROUP for a volume response. See Remark **27**. • Blank for grid point responses. "ELEM" or property name options are used only with element type responses (stress, strain, force, etc.) to identify the relevant element IDs, or the property type and

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Field	Contents
	relevant property IDs. (Character: “ELEM”, “PBAR”, “PSHELL”, “PCOMP”, “PANEL”, etc.). PTYPE = RANDPS ID is used when RTYPE = “PSDDISP” or “PSDVELO” or “PSDACCL”.
REGION	Region identifier for constraint screening. See Remark 10 for defaults. (Integer>0)
ATTA	Response attribute. See Table 13-68. See Remark 7. (Integer>0, or Blank if specified)
ATTB	Response attribute. See Table 13-68. (Integer>0 or Real>0.0 or Character or Blank)
ATTi (i=1,2,3...)	List of response attributes. See Table 13-68. (Integer>0, or Blank if specified)

Table 13-68: Design Response Attributes

Response Type (RTYPE)	Response Attributes		
	ATTA (Integer>0, or Blank if specified)	**ATTB (Integer>0 or Real>0.0 or Character or Blank)**	**ATTi (i=1,2,3...) (Integer>0, or Blank if specified)**
CEIG	Complex Eigenvalue Mode Number	ALPHA or OMEGA (Character) (Default=ALPHA) See Remark 24.	Blank
CFAILURE See Remark 2.	Failure Criterion Item Code	LAMINA Number (Integer>0; Default = 1). See Remark 3.	PCOMP or PCOMPG Property entry PID(s) See Remark 6.
CMPLNCE See Remark 40.	Blank	Blank	Blank
COG	Center of gravity location component (Single integer 1, 2, or 3) See Remark 31.	Blank	Blank
CSTRAIN See Remark 2.	Strain Item Code See Remark 1.	LAMINA Number (Integer>0; Default = 1). See Remark 3.	PCOMP or PCOMPG Property entry PID(s) See Remark 6.
CSTRESS See Remark 2.	Stress Item Code See Remark 1.	LAMINA Number (Integer>0; Default = 1). See Remark 3.	PCOMP or PCOMPG Property entry PID(s) See Remark 6.
DISP	Displacement Component	Blank	Grid ID(s) See Remark 33.
DWEIGHT See Remarks 28 and 30.	Blank	Blank	Blank
EIGN See Remark 19.	Normal Modes Mode Number	Approximation Code (Integer 1 or 2) See Remark 20.	Blank
ERP	For ERP magnitude: Blank or 0	Frequency Value (Blank or Real>0.0 or Character)	Panel related SET3 IDs

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Response Type (RTYPE)	Response Attributes		
	ATTA (Integer>0, or Blank if specified)	ATTB (Integer>0 or Real>0.0 or Character or Blank)	ATTi (i=1,2,3...) (Integer>0, or Blank if specified)
	For ERP density: 1 See Remark 38.	See Remarks 16 and 22.	
ESE	Strain Energy Item Code See Remark 23.	Blank	Property entry PID(s) See Remark 6.
FLUTTER	Blank	Blank	See Remark 15.
FORCE	Force Item Code See Remark 1.	Blank	Property entry PID(s) See Remark 6.
FRACCL	Acceleration Component See Remarks 8 and 9.	Frequency Value (Blank, Real>0.0 or Character) See Remarks 16 and 22.	Grid ID(s) See Remark 33.
FRDISP	Displacement Component See Remarks 8 and 9.	Frequency Value (Blank, Real>0.0 or Character) See Remarks 16 and 22.	Grid ID(s) See Remark 33.
FREQ See Remark 19.	Normal Modes Mode Number	Approximation Code (Integer 1 or 2) See Remark 20.	Blank
FRFORC	Force Item Code See Remark 1.	Frequency Value (Blank or Real>0.0 or Character) See Remarks 16 and 22 .	Property entry PID(s) See Remark 6.
FRSPCF	SPC Force Component See Remarks 8 and 9.	Frequency Value (Blank, Real>0.0 or Character) See Remarks 16 and 22.	Grid ID(s) See Remark 33.
FRSTRE	Stress Item Code See Remark 1.	Frequency Value. (Blank or Real>0.0 or Character) See Remarks 16 and 22 .	Property entry PID(s) See Remark 6.
FRVELO	Velocity Component . See Remarks 8 and 9.	Frequency Value. (Blank, Real>0.0 or Character) See Remarks 16 and 22.	Grid ID(s) See Remark 33.
LAMA See Remark 21.	Buckling Mode Number	Approximation Code (Integer 1 or 2). See Remark 20.	Blank
PRES	Acoustic Pressure Component. See Remark 26.	Frequency Value (Real>0.0) See Remark 16.	Grid ID(s) See Remark 33.
PSDACCL See Remarks 32, 34, 35, 36.	Acceleration Component See Remark 8.	Frequency Value. (Blank, Real>0.0 or Character) See Remarks 16 and 22.	Grid ID(s) See Remark 33.
PSDDISP See Remarks 32, 34, 35, 36.	Displacement Component See Remark 8.	Frequency Value. (Blank, Real>0.0 or Character) See Remarks 16 and 22.	Grid ID(s) See Remark 33.
PSDVELO See Remarks 32, 34, 35, 36.	Velocity Component See Remark 8.	Frequency Value. (Blank, Real>0.0 or Character) See Remarks 16 and 22.	Grid ID(s) See Remark 33.
RMSACCL	Acceleration Component See Remark 8.	RANDPS ID (Integer>0)	Grid ID(s) See Remark 33.

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Response Type (RTYPE)	Response Attributes		
	ATTA (Integer>0, or Blank if specified)	ATTB (Integer>0 or Real>0.0 or Character or Blank)	ATTi (i=1,2,3...) (Integer>0, or Blank if specified)
See Remarks 32, 36.			
RMSDISP See Remarks 32, 36.	Displacement Component See Remark 8.	RANDPS ID (Integer>0)	Grid ID(s) See Remark 33.
RMSVELO See Remarks 32, 36.	Velocity Component See Remark 8.	RANDPS ID (Integer>0)	Grid ID(s) See Remark 33.
SPCFORCE	SPC Force Component	Blank	Grid ID(s) See Remark 33.
STABDER See Remark 14.	AESTAT or AESURF Entry ID	Restraint Flag. (Integer 0 or 1) See Remark 13.	Component
STRAIN	Strain Item Code See Remark 1.	Blank	Property entry PID(s) See Remark 6.
STRESS	Stress Item Code See Remark 1.	Blank	Property entry PID(s) See Remark 6.
TACCL	Acceleration Component	Time Value (Blank, Real>0.0, or Character) See Remarks 17 and 22.	Grid ID(s) See Remark 33.
TDISP	Displacement Component	Time Value (Blank, Real>0.0, or Character) See Remarks 17 and 22.	Grid ID(s) See Remark 33.
TFORC	Force Item Code See Remark 1.	Time Value. (Blank or Real>0.0 or Character) See Remarks 17 and 22 .	Property entry PID(s) See Remark 6.
TRIM	AESTAT or AESURF Entry ID	Blank	Blank
TSPCF	SPC Force Component	Time Value. (Blank, Real>0.0, or Character) See Remarks 17 and 22.	Grid ID(s) See Remark 33.
TSTRE	Stress Item Code See Remark 1.	Time Value. (Blank or Real>0.0 or Character) See Remarks 17 and 22 .	Property entry PID(s) See Remark 6.
TVELO	Velocity Component	Time Value (Blank, Real>0.0, or Character) See Remarks 17 and 22.	Grid ID(s) See Remark 33.
VOLUME	Blank	Blank	SEIDi, ALL, or the ID of a GROUP entry See Remark 27.
WEIGHT See Remarks 28 and 29.	Row Number (1 ≤ ROW ≤ 6)	Column Number (Integer) (1 ≤ COL ≤ 6)	SEIDi or All or blank

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Remarks:

1. Stress, strain, and force item codes can be found in **"Item Codes"**. For stress or strain item codes that have dual meanings, such as von Mises or maximum shear, the option specified in the Case Control Section will be used; i.e., STRESS(VONM) or STRESS(MAXS). Note that item codes may vary depending on whether CENTER or CORNER is chosen with the relevant output request in case control.
2. RTYPE = "CSTRESS", "CSTRAIN", and "CFAILURE" are used only with the PCOMP and PCOMPG entries. "CSTRESS" and "CSTRAIN" item codes are described under Table 1 (Element Stress/Strain Item Codes) in **"Item Codes"**. "CFAILURE" item codes are described under Table 2 (Element Force Item Codes) in **"Item Codes"**. Only force item codes that refer to failure indices of direct stress and interlaminar shear stress are valid.

   The CFAILURE response type requires the following specifications on the applicable entries:

   - Failure theory in the FT field on PCOMP entry.
   - Allowable bonding shear stress in the SB field on PCOMP entry.
   - Stress limits in the ST, SC, and SS fields on all MATi entries.
3. ATTB is used only for responses of composite laminae, dynamics, complex eigenvalue, and stability derivatives. For other responses, this field must be blank. When including the ID of a PCOMPG entry in a DRESP1 response, the lamina number entered in the ATTB field is relative to the order of the ply definitions (1,2,3, ..) and is not the global ply id.
4. All grids associated with a DRESP1 entry are considered to be in the same region for screening purposes. Only up to NSTR displacement constraints (see DSCREEN entry) per group per load case will be retained in the design optimization phase.
5. DRESP1 identification numbers must be unique with respect to DRESP2 and DRESP3 identification numbers.
6. If PTYPE = "ELEM", the ATTi correspond to element identification numbers. If RTYPE="ERP" and PTYPE = blank or "PANEL", the ATTi correspond to one or more panel related SET3 IDs (as in the PANEL bulk data).
7. If RTYPE = "DISP", "TDISP", "TVELO", "TACCL", "TSPCF", "TFORC" or "TSTRE", multiple component numbers (any unique combination of the digits 1 through 6 with no embedded blanks) may be specified on a single entry. Multiple response components may not be used on any other response types.
8. For grid frequency response type DRESP1, i.e. those starting with "FR", "RMS", or "PSD", only one component number may be specified in the ATTA field. Numbers 1 through 6 correspond to real (or magnitude) components and 7 through 12 imaginary (or phase) components. If more than one component for the same grid is desired, then a separate entry is required.

   Note that, in case the response is used for a constraint, the choice of response computation format (real/imaginary vs. magnitude/phase) will affect constraint values. See also Remark 9.
9. Real/imaginary representation is the default for complex response types. Magnitude/phase representation must be requested by the corresponding Case Control command; e.g., DISP(PHASE) = ALL.

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10. REGION is used for constraint screening. The NSTR field on DSCREEN entries gives the maximum number of constraints retained for each region per load case.

    IF RTYPE = “WEIGHT”, “VOLUME”, “LAMA”, “EIGN” or “FREQ”, no REGION identification number should be specified. For all other responses, if the REGION field is left blank, the default specified in **Table 13-69** is used. Usually, the default value is appropriate.

    If the REGION field is not blank, all the responses on this entry as well as all responses on other DRESP1 entries that have the same RTYPE and REGION identification number will be grouped into the same region.

Table 13-69: Default Regions for Design Sensitivity Response Types

Response Type	Default Region
EIGN	No region
FREQ	No region
LAMA	No region
VOLUME	No region
WEIGHT	No region
DISP	One region per DRESP1 entry
ERP	One region per DRESP1 entry
FLUTTER	One region per DRESP1 entry
FRACCL	One region per DRESP1 entry
FRDISP	One region per DRESP1 entry
FRSPCF	One region per DRESP1 entry
FRVELO	One region per DRESP1 entry
PRES	One region per DRESP1 entry
SPCFORCE	One region per DRESP1 entry
TACCL	One region per DRESP1 entry
TDISP	One region per DRESP1 entry
TSPCF	One region per DRESP1 entry
TVELO	One region per DRESP1 entry
OTHER	One region per PROPERTY entry. If PTYPE = “ELEM”, then one region per DRESP1 entry

11. REGION is valid only among the same type of responses. Responses of different types will never be grouped into the same region, even if they are assigned the same REGION identification number by the user.

12. If RTYPE = “WEIGHT” or “VOLUME”, field ATTi = “ALL” implies total weight/volume of all superelements except external superelements.

13. RTYPE = “STABDER” identifies a stability derivative response. ATTB is the restraint flag for the stability derivative. ATTB = 0 means unrestrained, and ATTB = 1 means restrained. For example, ATTA = 4000, ATTB = 0, and ATT1 = 3 reference the unrestrained C_z derivative for the AESTAT (or AESURF) entry ID = 4000.

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14. Aeroelastic static subcases that reference only constrained STABDER responses must come after any aeroelastic static subcases that reference other types of constrained responses or reference a combination of other constrained responses including STABDER responses.

15. RTYPE = “FLUTTER” identifies a set of damping responses. The set is specified by ATTi:

    ATT1 = Identification number of a SET1 entry that specifies a set of modes.

    ATT2 = Identification number of an FLFACT entry that specifies a list of densities.

    ATT3 = Identification number of an FLFACT entry that specifies a list of Mach numbers.

    ATT4 = Identification number of an FLFACT entry that specifies a list of velocities.

    If the flutter analysis is type PKNL, it is necessary to put PKNL in the PTYPE field of this entry.

16. For RTYPE = “ERP”, FRDISP”, “FRVELO”, “FRACCL”, “FRSPCF”, “FRFORC”, “FRSTRE”, “PRES”, “PSDDISP”, “PSDVELO”, and “PSDACCL”, a real value for ATTB specifies a frequency value in cycles per unit time. If ATTB is specified, then the responses are evaluated at the closest frequency selected by the OFREQ command. The default for ATTB is all frequencies selected by the OFREQ command. See Remark 22 for additional ATTB options.

17. For RTYPE = “TDISP”, “TVELO”, “TACCL”, “TSPCF”, “TFORC”, and “TSTRE”, ATTB specifies a time value. If ATTB is specified, then the responses are evaluated at the closest time selected by the OTIME command. The default for ATTB is all time steps selected by the OTIME command.

18. Intermediate station responses on CBAR elements due to PLOAD1 and/or CBARAO entries may not be defined on the DRESP1 entry.

19. RTYPE = “EIGN” refers to normal modes response in terms of eigenvalue (radian/time)**2 while RTYPE = “FREQ” refers to normal modes response in terms of natural frequency or units of cycles per unit time.

20. For RTYPE = LAMA, EIGN or FREQ, the response approximation used for optimization can be individually selected using the ATTB field. (Approximation Code = 1 = direct linearization, = 2 = Inverse Linearization).

21. For RTYPE = LAMA, the response can be negative. For example, the buckling load factor for a bar under a tensile load would negative.

22. Character input for ATTB is available for RTYPE of ERP, FRDISP, FRVELO, FRACCL, FRSPCF, PSDDISP, PSDVELO, PSDACCL, TDISP, TVELO, TACCL, TSPCF, FRFORC, FRSTRE, TFORC, and TSTRE. The character input represents a mathematical function and the options for character input are SUM, AVG, SSQ, RSS, MAX and MIN. The expression of mathematical function is shown as follows:

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$$\text{SUM}(X_1, X_2, \ldots, Xn) = \sum_{i=1}^{n} X_i$$

$$\text{AVG}(X_1, X_2, \ldots, Xn) = \sum_{i=1}^{n} X_i / n$$

$$\text{SSQ}(X_1, X_2, \ldots, Xn) = \sum_{i=1}^{n} X_i^2$$

$$\text{RSS}(X_1, X_2, \ldots, Xn) = \sqrt{\sum_{i=1}^{n} X_i^2}$$

MAX (X_1, X_2, ..., X_n) = Largest value among X_i (i=1 to n)

MIN (X_1, X_2, ..., X_n) = Smallest value among X_i (i=1 to n)

where X_i is the response for a forcing frequency. For example

DRESP1,10,DX1,FRSTRE,ELEM,,3,AVG,10

yields a response which is equal to the average stress for element 10. The average is done by first adding up the component 3 stress of element 10 for all forcing frequencies (all time steps for transient response), and then dividing by the number of forcing frequencies. Note that the response computed is considered as type 2 response. **Therefore, if referenced on a DRESP2, the ID of such DRESP1 (ATTB with character input) must be listed following the DRESP2 keyword**.

Such DRESP1 may only be referenced from a DRESP2 (not from a DRESP3), and may only be nested one level. That is, the referencing DRESP2 may not be referenced further from another DRESP2 or a DRESP3.

(Important note: If any such integrated response DRESP1 is either erroneously or intentionally referred to as DRESP1 from within a DRESP2 or DRESP3, the program will spawn multiple responses over the frequency set, or, for the objective function, take only the value corresponding to the first frequency in the set, not the integrated response, whether or not the DRESP1 is DRSPAN related.)

23. Element strain energy item codes can be found in **Element Strain Energy Item Codes**. Only element strain energy and element strain energy density can be referenced on a DRESP1 entry.

24. For RTYPE=CEIG, the allowable character inputs are ALPHA (for the real component) and OMEGA (for the imaginary component), with ALPHA being the default.

25. For RTYPE=RMSDISP, RMSVELO, or RMSACCL the ATTB specifies the appropriate RANDPS ID.

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26. For RTYPE=PRES, if you enter an integer other than 1 or 7 in the ATTA field (acoustic pressure component), the software will reset the value to 1 (if less than 7) or 7 (if greater than 7 and less than 13).

In addition, the pressure response is evaluated based on the format (real/imaginary or magnitude/phase) specified on the PRESSURE case control command, except in the following cases:
● If DISPLACEMENT output is request in the case control section.
● If both the RTYPE=PRES and the RTYPE=FRDISP responses are defined on DRESP1 entries (FRDISP requests the frequency displacement response).

For those two case, the format defined for the PRESSURE case control command is disregarded and the format (real/imaginary or magnitude/phase) specified on the DISPLACEMENT case control command (or its default format if not present) is used and a warning issued.

27. You can create a design objective or design constraint for volume for any SOL 200 problems. The volume objective and volume constraint must be defined in the global level (above the subcases). When "ALL" is defined in the ATT1 field, all of the elements are considered when computing the volume. You can optionally limit the volume response to a subset of the elements by entering “GROUP” in the PTYPE field, and an integer into the ATT1 field, where the integer value is the ID defined on a GROUP bulk entry that lists the elements to include in the volume computation. The GROUP option is not supported with superelements.

For topology optimization, the volume response represents the sum of the volume of the elements included in the optimization (active) and not included (frozen). The volume of the active elements changes during the solution as the normalized mass density (NMD) design variable changes, and is computed by:

$$Volume_{active} = \sum (NMD_i * Volume_{geometric})$$

where $Volume_{geometric}$ is the element geometric volume, and NMD_i is the normalized mass density.

28. The software computes the WEIGHT and DWEIGHT response types with the Grid Point Weight Generator. The resulting response is either a mass or a weight depending on how you define density on the MATi entries.

- When you define the density on the MATi entries as a mass density (mass/volume), the WEIGHT and DWEIGHT response types are a mass response.
- When you define the density on the MATi entries as a weight density (weight/volume), the WEIGHT and DWEIGHT response types are a weight response.

If you define density on the MATi entries as a weight density and you have requested a dynamics analysis in any subcase, or you have body loads defined with the GRAV, ACCEL, ACCEL1, RFORCE, RFORCE1 entries which depend on the mass, you must use the WTMASS parameter to convert weight to mass for these applications. The software applies the WTMASS parameter to convert the mass for all mass applications except for the actual WEIGHT or DWEIGHT responses. These response values are always the unconverted value from the Grid Point Weight Generator, which is only based on how you defined the density.

See Weight Center of Gravity and Moment of Inertia Check in the *Simcenter Nastran User's Guide*.

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29. The WEIGHT response type is obtained from the Grid Point Weight Generator for a reference point (see the parameter GRDPNT). For SOL 200, if PARAM,GRDPNT is either not defined, equal to zero, or not a defined grid point, the reference point is taken as the origin of the basic coordinate system. Fields ATTA and ATTB refer to the row and column numbers of the rigid body mass (or weight) matrix, which is partitioned as follows:

$$[W] = \begin{bmatrix} W_x & W_{12} & W_{13} & W_{14} & W_{15} & W_{16} \\ W_{21} & W_y & W_{23} & W_{24} & W_{25} & W_{26} \\ W_{31} & W_{32} & W_z & W_{34} & W_{35} & W_{36} \\ W_{41} & W_{42} & W_{43} & I_x & W_{45} & W_{46} \\ W_{51} & W_{52} & W_{53} & W_{54} & I_y & W_{56} \\ W_{61} & W_{62} & W_{63} & W_{64} & W_{65} & I_z \end{bmatrix}_{6 \times 6}$$

The default value of ATTA and ATTB is 3, which specifies the Wz value. Typically the value of Wx, Wy, and Wz each equal the mass (or weight) of the structure. Although, if directional mass is defined, which is possible using the CONM1, CMASSi, or DMI bulk entries, the values can be different. If you are uncertain about the mass (or weight) matrix value to use in your response, include the parameter GRDPNT to print the mass (or weight) matrix in the .f06 file.

The first field of ATTi: ATT1 = “ALL” implies total mass (or weight) of all superelements except external superelements. When the ATTi fields contain SEIDi, each SEIDi refers to a superelement identification number. SEIDi= “0” refers to the residual superelement. The default for this first field of ATTi, ATT1, is blank, which is equivalent to “ALL”.

30. The DWEIGHT response type is only supported in topology optimization. It requests a mass (or weight) response computed for only the elements designated as active using the DVTREL1 bulk entry. See the remarks on **DVTREL1**. The computed mass (or weight) for the DWEIGHT response does not include any directional mass defined with the CMASSi, CONMi, and DMI bulk entries. As a result, the DWEIGHT response M (or W) is such that M=Mx=My=Mz (or W=Wx=Wy=Wz).

The WEIGHT response type is also supported in topology optimization to request a mass or weight response. The WEIGHT response type is computed using all elements in the model, independent if they are active or frozen, and independent if they can participate in topology optimization or not. In addition, mass defined with the CMASSi, CONMi, and DMI bulk entries is also included with the WEIGHT response type.

WEIGHT and DWEIGHT responses cannot exist simultaneously in the same problem.

31. The COG response type computes the location of the center of gravity and reports its location in one of the three basic coordinate axis. You define one of the three basic coordinate axis in the ATTA field as either 1 for the X axis, 2 for the Y axis, or 3 for the Z axis. By default, the reference location of the center of gravity is the basic coordinate system origin. You can specify a different reference location by selecting a grid point ID with the GRDPNT parameter (PARAM,GRDPNT,ID).

For example, if you entry 2 in the ATTA field and the GRDPNT parameter is undefined, the COG response will be the Y-coordinate location of the center of gravity relative to the basic coordinate system origin. You can define three independent DRESP1 entries to define a COG response for each center of gravity coordinate (X, Y, and Z).

The COG response type should not be used when direction dependent mass is defined with the CONM1, CMASSi, or DMIG bulk entries.

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32. PSD and RMS type DRESP1 responses are meaningful only for those random loading frequency response analysis subcases listed in the RANDPS bulk data referred to by the relevant RANDOM ID.

33. Multiple grid point IDs in the ATTi fields are supported.

34. Since PSD type DRESP1 responses are a summation over the sets of relevant subcases correlated in the RANDPS bulk data, a constraint on a PSD type DRESP1 response should be specified only in the lead subcase of such a batch of subcases. Any PSD response specified otherwise will be ignored. For example, if the subcases referred to by RANDPS ID=75 are (3,3); (4,4); (5,5); and (3,5), where the last two numbers indicate cross-correlation among two subcases, any PSD response for this particular RANDPS ID should be placed in subcase 3, as that is the lead subcase.

35. If a PSD type response is to appear in a DRESP2 or DRESP3 response together with responses from other subcases, it should then be assigned to the lead subcase defined in item 28 above by an appropriate DRSPAN case control command and a related set.

36. Models with superelements will either ignore any RMS and PSD type DRESP1 design responses, or will fail in the presence of such DRESP1.

37. For PBUSHT, PDAMPT, and PELAST property types, currently use PBUSH, PDAMP, or PELAS, as appropriate, in the PTYPE field, since the latter have the same IDs as the corresponding frequency dependent properties.

38. ATTA can be 0 for ERP magnitude responses. This is equivalent to leaving it blank.

39. Currently, for ERP design responses, any entry or blank in the PTYPE field defaults to PANEL.

40. The CMPLNCE response type requests the compliance of the finite element model and is a measure of the flexibility of the model in loaded regions. It is only computed in a ANALYSIS=STATICS subcase. It is defined as the dot product of strain and stress or as twice the strain energy. In general, compliance would correspond to the integration of the dot product of infinitesimal increments of strain with similar increments of stress.

    The following items are not supported when a CMPLNCE response type is requested:
    - The design variable to grid point relation (DVGRID bulk entry) defined on grid points that are included on a rigid element (RBE*i*).
    - Inertia relief (PARAM,INREL,-1 or PARAM,INREL,-2).
    - The direct method for sensitivity analysis (PARAM,AUTOADJ,NO). You must use the adjoint method (PARAM,AUTOADJ,YES), which is the default.

Related Topics

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DRESP2

Design Equation Response Quantities

Defines equation responses that are used for the objective and/or design constraints, or for sensitivity analysis purposes.

Format:

1	2	3	4	5	6	7	8	9	10
DRESP2	ID	LABEL	EQID or FUNC	REGION					
	"DESVAR"	DVID1	DVID2	DVID3	DVID4	DVID5	DVID6	DVID7	
		DVID8	-etc.-						
	"DTABLE"	LABL1	LABL2	LABL3	LABL4	LABL5	LABL6	LABL7	
		LABL8	-etc.-						
	"DFRFNC"	DFRFID1	DFRFID2	DFRFID3	DFRFID4	DFRFID5	DFRFID6	DFRFID7	
		DFRFID8	-etc.-						
	"DRESP1"	NR1	NR2	NR3	NR4	NR5	NR6	NR7	
		NR8	-etc.-						
	"DNODE"	G1	C1	G2	C2	G3	C3		
		G4	C4	etc.					
	"DVPREL1"	DPIP1	DPIP2	DPIP3	DPIP4	DPIP5	DPIP6	DPIP7	
		DPIP8	DPIP9	-etc.-					
	"DVCREL1"	DCIC1	DCIC2	DCIC3	DCIC4	DCIC5	DCIC6	DCIC7	
		DCIC8	DCIC9	-etc.-					
	"DVMREL1"	DMIM1	DMIM2	DMIM3	DMIM4	DMIM5	DMIM6	DMIM7	
		DMIM8	DMIM9	-etc.-					
	"DVPREL2"	DPI2P1	DPI2P2	DPI2P3	DPI2P4	DPI2P5	DPI2P6	DPI2P7	
		DPI2P8	DPI2P9	-etc.-					
	"DVCREL2"	DCI2C1	DCI2C2	DCI2C3	DCI2C4	DCI2C5	DCI2C6	DCI2C7	
		DCI2C8	DCI2C9	-etc.-					
	"DVMREL2"	DMI2M1	DMI2M2	DMI2M3	DMI2M4	DMI2M5	DMI2M6	DMI2M7	
		DMI2M8	DMI2M9	-etc.-					
	"DRESP2"	NRR1	NRR2	NRR3	NRR4	NRR5	NRR6	NRR7	
		NRR8	-etc.-						

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Example:

DRESP2	1	LBUCK	5	3					
	DESVAR	101	3	4	5	1	205	209	
		201							
	DTABLE	PI	YM	L					
	DRESP1	14	1	4	22	6	33	2	
	DNODE	14	1	4	1	22	3		
		2	1	43	1				
	DVPREL1	101	102						
	DVCREL1	201	202						
	DVMREL1	301							
	DVPREL2	401	402						
	DVCREL2	501							
	DVMREL2	601	602	603					
	DRESP2	50	51						

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
LABEL	User-defined label. (Character. See **Bulk Data Syntax Rules**.)
EQID	DEQATN entry identification number. (Integer>0)
FUNC	Function to be applied to the arguments. See Remark **8**. (Character)
REGION	Region identifier for constraint screening. See Remark **5**. (Integer>0)
“DESVAR”	Flag indicating DESVAR entry identification numbers. (Character)
DVIDi	DESVAR entry identification number. (Integer>0)
“DTABLE”	Flag indicating that the labels for the constants in a DTABLE entry follow. (Character)
LABLj	Label for a constant in the DTABLE entry. (Character. See **Bulk Data Syntax Rules**.)
“DFRFNC”	Flag indicating DFRFNC ID numbers to be found in DTI, DFRFNC table records.
DFRFIDi	ID number for a DFRFNC record in the DTI, DFRFNC table. Usually there will be a single DFRFNC ID entry here. However, as long as there is a match in the number of frequencies for which the DRESP2 is to be evaluated, there may be multiple DFRFNC ID numbers.
“DRESP1”	Flag indicating DRESP1 entry identification numbers. (Character)

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Field	Contents
NRk	DRESP1 entry identification number. (Integer>0)
“DNODE”	Flag indicating grid point and component identification numbers. (Character)
Gm	Identification number for any grid point in the model. (Integer>0)
Cm	Component number of grid point Gm. (1 ≤ Integer ≤ 3)
DVPREL1	Flag indicating DVPREL1 entry identification number. (Character)
DPIPi	DVPREL1 entry identification number. (Integer>0)
DVCREL1	Flag indicating DVCREL1 entry identification number. (Character)
DCICi	DVCREL1 entry identification number. (Integer>0)
DVMREL1	Flag indicating DVPREL2 entry identification number. (Character)
DMIMi	DVMREL1 entry identification number. (Integer>0)
DVPREL2	Flag indicating DVPREL2 entry identification number. (Character)
DPI2Pi	DVPREL2 entry identification number. (Integer>0)
DVCREL2	Flag indicating DVCREL2 entry identification number. (Character)
DCI2Ci	DVCREL2 entry identification number. (Integer>0)
DVMREL2	Flag indicating DVMREL2 entry identification number. (Character)
DMI2Mi	DVMREL2 entry identification number. (Integer>0)
DRESP2	Flag indicating other DRESP2 entry identification number. (Character)
NRRk	DRESP2 entry identification number. (Integer>0)

Remarks:

1. DRESP2 entries may only reference DESVAR, DTABLE, DRESP1, DNODE, DVPREL1, DVCREL1, DVMREL1, DVPREL2, DVCREL2, and DVMREL2 entries. They may also reference other DRESP2 entries. However, a DRESP2 entry cannot reference itself directly or recursively.
2. Referenced DRESP1 entries cannot span analysis types or superelements, except when such DRESP1 are specified in DRSPAN referenced sets, in which case they may span subcases and analysis types (see also Remark **11** below, and the remarks on the **DRSPAN** case control command).
3. DRESP2 entries must have unique identification numbers with respect to DRESP1 and DRESP3 entries.
4. The “DESVAR”, “DTABLE”, “DFRFNC”, “DRESP1”, “DNODE”, “DVPREL1”, “DVCREL1” and “DVMREL1”, “DVPREL2”, DVCREL2”, “DVMREL2”, and “DRESP2” flags in field 2 must appear in the order given above. Any of these words, along with the identification numbers associated with them, may be omitted if they are not involved in this DRESP2 relationship. However, at least one of these twelve types of arguments must exist.
5. The REGION field follows the same rules as for the DRESP1 entries. DRESP1, DRESP2, and DRESP3 responses will never be contained in the same region, even if they are assigned the same REGION

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identification number. The default is to put all responses referenced by one DRESP2 entry in the same region.

6. The variables identified by DVIDi, LABLj, NRk, the Gm, Cm pairs, DPIPi, DCICm, DMIMn, DPI2Po, DCI2Cp, DMI2Mq, and NRRu are assigned (in that order) to the variable names (x1, x2, x3, etc.) specified in the left-hand side of the first equation on the DEQATN entry referenced by EQID. In the example below,

   DESVARs 101 and 3 are assigned to arguments A and B.DTABLEs PI and YM are assigned to arguments C and D.Grid 14, Component 1 is assigned to argument R.

DRESP2	1	LBUCK	5	3					
	DESVAR	101	3						
	DTABLE	PI	YM						
	DNODE	14	1						
DEQATN	5	F1(A, B, C, D, R)=A+B*C-(D**3+10.0)+sin(C*R)							

7. (Gm, Cm) can refer to any grid component and is no longer limited to a designed grid component.

8. The FUNC attributes can be used in place of the EQID and supports the functions shown in the following table:

Function	Description
SUM	Sum of the arguments
AVG	Average of the arguments
SSQ	Sum of the squares of the arguments
RSS	Square root of the sum of the squares of the arguments
MAX	The largest value of the argument list
MIN	The smallest value of the argument list

   When EQID has character input, the DEQATN entry is no longer needed. The functions are applied to all arguments on the DRESP2 regardless of the type.

9. The number of arguments of a DEQATN can be more than the number of values defined on the DRESP2 if the DRESP1s referenced have RTYPE with ‘FR’ prefix. Arguments are still positional. The extra arguments in the DEQATN must appear at the end of the argument list. The discrepancy is resolved internally with the forcing frequency(ies) associated with DRESP1s. An example is shown as follows:

DRESP1	10	FDISP1	FRDISP			1	10.	1001	
DRESP1	20	FDISP2	FRDISP			1	20.	1001	
DRESP2	30	AVGFD	100						
	DRESP1	10	20						
DEQATN	100	AVG(D1,D2,F1,F2) = (D1/F1+D2/F2)*0.5							

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In the above example, the DEQATN has two more additional terms than have been defined on the DRESP2. The first additional term is the forcing frequency (in hertz) of the first DRESP1 ID on the DRESP2. The second additional term is the forcing frequency of second DRESP1 ID in the list. When all DRESP1s involved have the same frequency, the user is not required to name all the additional terms in the argument list of DEQATN.

10. The DRESP1 IDs listed under a DRESP2 response may only point to DRESP1s with the same number of responses, in case the DRESP1s are spawned for multiple frequencies or time steps.
11. When referenced DRESP2 entries are in fact DRESP1 that are mathematical functions (integrated responses) described in Remark 20 for DRESP1, they may again span subcases and analysis types, if specified in DRSPAN referenced sets.

Related Topics

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DRESP3

Externally Computed Design Responses

Defines SOL 200 design responses to be evaluated in an external user-supplied program. The differences from the DRESP2 data are indicated in bold. Such responses are then used for the objective and/or design constraints, or for sensitivity analysis purposes.

Format:

1	2	3	4	5	6	7	8	9	10
DRESP3	ID	LABEL	**GROUP**	**TYPE**	REGION				
	"DESVAR"	DVID1	DVID2	DVID3	DVID4	DVID5	DVID6	DVID7	
		DVID8	-etc.-						
	"DTABLE"	LABL1	LABL2	LABL3	LABL4	LABL5	LABL6	LABL7	
		LABL8	-etc.-						
	"DFRFNC"	DFRFID1	DFRFID2	DFRFID3	DFRFID4	DFRFID5	DFRFID6	DFRFID7	
		DFRFID8	-etc.-						
	"DRESP1"	NR1	NR2	NR3	NR4	NR5	NR6	NR7	
		NR8	-etc.-						
	"DNODE"	G1	C1	G2	C2	G3	C3		
		G4	C4	etc.					
	"DVPREL1"	DPIP1	DPIP2	DPIP3	DPIP4	DPIP5	DPIP6	DPIP7	
		DPIP8	DPIP9	-etc.-					
	"DVCREL1"	DCIC1	DCIC2	DCIC3	DCIC4	DCIC5	DCIC6	DCIC7	
		DCIC8	DCIC9	-etc.-					
	"DVMREL1"	DMIM1	DMIM2	DMIM3	DMIM4	DMIM5	DMIM6	DMIM7	
		DMIM8	DMIM9	-etc.-					
	"DVPREL2"	DPI2P1	DPI2P2	DPI2P3	DPI2P4	DPI2P5	DPI2P6	DPI2P7	
		DPI2P8	DPI2P9	-etc.-					
	"DVCREL2"	DCI2C1	DCI2C2	DCI2C3	DCI2C4	DCI2C5	DCI2C6	DCI2C7	
		DCI2C8	DCI2C9	-etc.-					
	"DVMREL2"	DMI2M1	DMI2M2	DMI2M3	DMI2M4	DMI2M5	DMI2M6	DMI2M7	
		DMI2M8	DMI2M9	-etc.-					
	"DRESP2"	NRR1	NRR2	NRR3	NRR4	NRR5	NRR6	NRR7	
		NRR8	-etc.-						
	"USRDATA"	**String**	**(character strings in user-defined format for any use in the evaluator)**						
		-etc.-							

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Example:

DRESP3	32	OBJ1	OBJ1GRP	REG1					
	DESVAR	1	3	4	5	1	205	209	
		201							
	DTABLE	PI	YM	L					
	DRESP1	14	1	4	22	6	33	2	
	DNODE	14	1	4	1	22	3		
		2	1	43	1				
	DVPREL1	101	102						
	DVCREL1	201	202						
	DVMREL1	301							
	DVPREL2	401	402						
	DVCREL2	501							
	DVMREL2	601	602	603					
	DRESP2	50	51						
	USRDATA	102,ABCD	EF,(X-Y)						

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
LABEL	User-defined label which is referenced by a constraint bulk data entry or by the objective function. (Character. See **Bulk Data Syntax Rules**.)
GROUP	Selects a specific external response routine. (Character) See Remark **2**.
TYPE	Refers to a specific user-created response calculation type in the external function evaluator. See Remark **3**. (Character)
REGION	Region identifier for constraint screening. See Remark **5**. (Integer>0)
“DESVAR”	Flag indicating DESVAR entry identification numbers. (Character)
DVIDi	DESVAR entry identification number. (Integer>0)
“DTABLE”	Flag indicating that the labels for the constants in a DTABLE entry follow. (Character)
LABLj	Label for a constant in the DTABLE entry. (Character. See **Bulk Data Syntax Rules**.)
“DFRFNC”	Flag indicating DFRFNC ID numbers to be found in DTI, DFRFNC table records.
DFRFIDi	ID number for a DFRFNC record in the DTI, DFRFNC table. Usually there will be a single DFRFNC ID entry here. However, as long there is a match in the number of frequencies for which the DRESP3 is to be evaluated, there may be multiple DFRFNC ID numbers.

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Field	Contents
“DRESP1”	Flag indicating DRESP1 entry identification numbers. (Character)
NRk	DRESP1 entry identification number. (Integer>0)
“DNODE”	Flag indicating grid point and component identification numbers. (Character)
Gm	Identification number for any grid point in the model. (Integer>0)
Cm	Component number of grid point Gm. (1 ≤ Integer ≤ 3)
DVPREL1	Flag indicating DVPREL1 entry identification number. (Character)
DPIPi	DVPREL1 entry identification number. (Integer>0)
DVCREL1	Flag indicating DVCREL1 entry identification number. (Character)
DCICi	DVCREL1 entry identification number. (Integer>0)
DVMREL1	Flag indicating DVPREL2 entry identification number. (Character)
DMIMi	DVMREL1 entry identification number. (Integer>0)
DVPREL2	Flag indicating DVPREL2 entry identification number. (Character)
DPI2Pi	DVPREL2 entry identification number. (Integer>0)
DVCREL2	Flag indicating DVCREL2 entry identification number. (Character)
DCI2Ci	DVCREL2 entry identification number. (Integer>0)
DVMREL2	Flag indicating DVMREL2 entry identification number. (Character)
DMI2Mi	DVMREL2 entry identification number. (Integer>0)
DRESP2	Flag indicating DRESP2 entry identification number. (Character)
NRRk	DRESP2 entry identification number. (Integer>0)
“USRDATA”	Flag indicating user input data (Character).

Remarks:

1. DRESP3 entries must have unique identification numbers with respect to DRESP1 and DRESP2 entries.
2. The “GROUP” field must reference the group defined on a FMS CONNECT statement.
3. The “TYPE” field is used to select one type of response evaluation, since more than one option can exist in the external user-created routine.
4. The “DESVAR”, “DTABLE”, “DFRFNC”, “DRESP1”, “DNODE”, “DVPREL1”, “DVCREL1” and “DVMREL1”, “DVPREL2”, DVCREL2”, “DVMREL2”, “DRESP2”, and USRDATA flags in field 2 must appear in the order given above. Any of these words, along with the identification numbers associated with them, may be omitted if they are not involved in this DRESP3 relationship. However, at least one of these types of arguments must exist.

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5. The REGION field is used with the DSCREEN bulk entry in the constraint screening process as discussed in the section titled "Constraint Regionalization and Deletion" in chapter one of the book *Simcenter Nastran Design and Optimization User's Guide*. If REGION is defined, all responses which share the REGION id and have the same response type will be evaluated together according to the DSCREEN bulk entry which also has the same response type.

   The REGION field follows the same rules as for the DRESP1 and DRESP2 entries. DRESP1, DRESP2 or DRESP3 responses will never be contained in the same region, even if they are assigned the same REGION identification number. The default is to put all responses referenced by one DRESP3 entry in the same region.

6. The string input for USRDATA is limited to 32000 characters.

7. The DRESP1 IDs listed under a DRESP3 response may only point to DRESP1s with the same number of responses, in case the DRESP1s are spawned for multiple frequencies or time steps.

8. Template Fortran subroutines are provided for DRESP3, together with Simcenter Nastran, wherein the user can modify these subroutines and add, if necessary, new called subroutines to complete a customized external program. More information is supplied with the template subroutines as to the data they receive by way of the DRESP3 bulk data. The external program can then be built using the files provided for this purpose. While the default name for this external program will be dr3serv, the user can modify the build file to change this to a desired name.

9. The external program described in Remark 8 is referenced by way of three entities: a) the CONNECT statement in the data file, which provides an alias for the path name, b) a separate file which resolves this alias by providing the corresponding path name, and c) the "nastran" statement which refers to this separate file by the gmconn option (i.e. gmconn = filename).

10. Referenced DRESP1 entries cannot span analysis types or superelements, except when such DRESP1 are specified in DRSPAN referenced sets, in which case they may span subcases and analysis types (see the remarks on the **DRSPAN** case control command).

11. DRESP1 that are mathematical functions described in Remark 20 for DRESP1 may not be referenced by DRESP3.

12. See Defining Design Responses with External Programs in the Design Sensitivity and Optimization User's Guide, and the **CONNECT** File Management statement.

Related Topics

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DRIVER

Driver load for Kinematic Joint (SOL 402 and SOL 414,129)

Defines a driver load or enforced displacement on kinematic joints.

Format 1: (Formats 1 and 2 cannot be combined on the same line)

1	2	3	4	5	6	7	8	9	10
DRIVER	DID	TYPE	P					EID1	
	EID2	EID3	EID4	EID5	–etc.–				

Format 2

1	2	3	4	5	6	7	8	9	10
DRIVER	DID	TYPE	P					EID1	
	EID2	“THRU”	EID3	“BY”	INC				

Continuation Format 1: (Continuation formats 1 and 2 cannot be combined on the same line)

	EID6	EID7	EID8	EID9	–etc.–				

Continuation Format 2:

	EID4	“THRU”	EID5	“BY”	INC				

Fields:

Field	Contents
DID	Driver load identification number. See Remark 4. (Integer > 0).
TYPE	Driver load or enforced displacement type. See Remark 1, Remark 2 and Remark 3. (Character = "FORC", "TORQ", "DISP", or "ROT").
P	Magnitude of the driver load. See Remark 5. (Real > 0.0; Unit = Force, Length, or Radian). (Optional).
EIDi	Kinematic joint identification number defined on a CJOINT bulk entry. (Integers > 0, all unique.)
THRU	Specifies a range of CJOINT entry IDs. (Optional)

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Field	Contents
BY	Specifies an increment when using THRU option. (Optional)
INC	Increment used with THRU option. (Integer > 0; Default=1)

Remarks:

1. FORC or DISP load types can only be applied on INLINE, SLIDER, SLIUNV, CYLDR, or SCREW kinematic joints.

   TORQ or ROT enforced displacement types can only be applied on REVOLUTE, CYLDR, or SCREW kinematic joints.

   For more information, see the **PJOINT** entry.

2. The SCREW and CYLDR kinematic joints support a combination of a driver load type FORC or DISP and a driver load type of TORQ or ROT. When a combination of load types is applied to the same SCREW or CYLDR kinematic joint, you must define two DRIVER bulk entries with unique DIDs (one for each load type). For example, a DRIVER entry with DID=5 can have a load type DISP, and a second DRIVER entry with DID=7 can have a load type ROT, and both DRIVER entries would include the same kinematic joint ID in the EID field. See remarks 4 and 5.

3. SPHERE, FIXED, UNIVSEL, or CONVEL kinematic joints do not support drivers.

4. A driver will be active only if its DID number is selected:

   - by the LOAD or DLOAD case control command.
   - by a combination of loads listed on the LOAD or DLOAD bulk entry.

5. A driver time-assigned load definition is defined with the TLOAD1 bulk entry. You must define the following in the TYPE field on the TLOAD1 bulk entry for each driver type:

   - For the drivers DISP or ROT, you must define TYPE=1 on the TLOAD1 bulk entry.
   - For the drivers FORC or ROT, you must define TYPE=0 on the TLOAD1 bulk entry.

   Note

   For SOL 414,129 Transient Response solutions, DRIVER can be set to ROT, and TLOAD1 can be set to Type 2 (enforced velocity) to apply an angular velocity.

Related Topics

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DSCREEN

Design Constraint Screening Data

Defines screening data for constraint deletion.

Format:

1	2	3	4	5	6	7	8	9	10
DSCREEN	RTYPE	TRS	NSTR						

Example:

DSCREEN	STRESS	-0.7	2						

Fields:

Field	Contents
RTYPE	Response type for which the screening criteria apply. See Table 13-70 and Remark 3 . (Character)
TRS	Truncation threshold. (Real; Default = -0.5)
NSTR	Maximum number of constraints to be retained per region per load case. See Remark 3 . (Integer>0; Default = 20)

Remarks:

1. Displacement and displacement derivative constraints associated with one particular load case are grouped by the specification of DRESP1 entries. From each group, a maximum of NSTR constraints are retained per load case.

Table 13-70: Default Region Specifications for RTYPE

RTYPE	Default Region Specification		RTYPE	Default Region Specification
CFAILURE	Property ID		FRSPC	DRESP1
CSTRAIN	Property ID		FRSTRE	Property ID
CSTRESS	Property ID		FRVELO	DRESP1
DISP	DRESP1		SPCFORCE	DRESP1
DRESP3	DRESP3		STRAIN	Property ID
EQUA	DRESP2		STRESS	Property ID
ESE	Property ID		TACCL	DRESP1
FLUTTER	DRESP1		TDISP	DRESP1

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RTYPE	Default Region Specification		RTYPE	Default Region Specification
FORCE	Property ID		TFORC	Property ID
FRACCL	DRESP1		TSPC	DRESP1
FRDISP	DRESP1		TSTRE	Property ID
FRFORC	Property ID		TVELO	DRESP1

2. Stress-strain constraints are grouped by the property; i.e., all elements belonging to the set of PIDs specified under ATTi on a DRESPi entry are regarded as belonging to the same region. In superelement sensitivity analysis, if the property (PID) is defined in more than one superelement, then separate regions are defined. A particular stress constraint specification may be applied to many elements in a region generating many stress constraints, but only up to NSTR constraints per load case will be retained.
3. If RTYPE = “WEIGHT”, “VOLUME”, “EIGN”, “FREQ”, “LAMA”, “TRIM”, or “STABDER” then NSTR is unlimited and must be left blank. TRS is available for all response types.
4. If a certain type of constraint exists but no corresponding DSCREEN entry is specified, all the screening criteria used for this type of constraint will be furnished by the default values.
5. Constraints can be retained only if they are greater than TRS. See the Remarks under the **“DCONSTR”** entry for a definition of constraint value.
6. Constraint screening is applied to each superelement.

Related Topics

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DTABLE

Table Constants

Defines a table of real constants that are used in equations (see DEQATN entry).

Format:

1	2	3	4	5	6	7	8	9	10
DTABLE	LABL1	VALU1	LABL2	VALU2	LABL3	VALU3	LABL4	VALU4	
	LABL5	VALU5	LABL6	VALU6	-etc.-				

Example:

DTABLE	PI	3.142	H	10.1	E	1.0E6	T	0.1	
	G	5.5E5	B	100.					

Fields:

Field	Contents
LABLi	Label for the constant. (Character. See **Bulk Data Syntax Rules**.)
VALUi	Value of the constant. (Real)

Remarks:

1. Only one DTABLE entry may be specified in the Bulk Data Section.
2. LABLi are referenced by the LABi on the DVPREL2, DRESP2, or DRESP3 entries.

Related Topics

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DTEMP

Time dependent temperature set definition for SOLs 401, 402, and 414,129.

Defines a time dependent temperature set.

Format:

1	2	3	4	5	6	7	8	9	10
DTEMP	SID								
	T1	TID1	T2	TID2	T3	TID3	T4	TID4	
	T5	TID5	-etc.-						

Example:

DTEMP	200								
	0.	1	45.	2	50.	3	60.	4	
	105.	5	160.	6	170.	7	180..	8	

Fields:

Field	Contents
SID	Unique identification number specified in the DTEMP case control command. (Integer>0)
Ti	Times in increasing order of magnitude. See Remark 5. (Real ≥ 0.0) T1 < T2 < T3 <...< Tn
TIDi	SID of TEMP or TEMPD bulk entries (Integer>0).

Remarks:

1. SID must be unique to all other DTEMP entries.
2. The SID of the DTEMP bulk entry can only be selected with the DTEMP case control command. The TEMPERATURE case control command cannot be used to select the DTEMP bulk entry.
3. The same TIDi can be used multiple times on the same DTEMP entry in order to define the same temperature conditions for multiple times.
4. All TID on a DTEMP entry must have temperatures assigned to the same set of grid points.
5. For solution times less than T1, the software uses the temperatures at T1. For solution times greater than Tn, the software uses the temperatures at Tn.

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Related Topics

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DTEMPEX

Time-dependent temperature set for SOLs 401 and 402

Defines a time-dependent temperature set for SOLs 401 and 402 that is obtained from an external file.

Format:

1	2	3	4	5	6	7	8	9	10
DTEMPEX	SID	UNID	FILETYPE	TID					

Example:

DTEMPEX	300	22	FILETYPE	120					

Fields:

Field	Contents
SID	Unique identification number specified in the DTEMP case control command. (Integer>0)
UNID	FORTRAN unit number listed on the ASSIGN statement for a .unv file, a .bun file, or a .scd5 file. See Remark 3. (Integer>0)
FILETYPE	Specifies the type of external temperature data file. (Character; Default=BUN) FILETYPE="BUN" The external file is either a .bun or a .unv file. This option is supported by both SOLs 401 and 402. FILETYPE="SCD5" The external file is an .scd5 file. This option is only supported by SOL 401. It is not supported by SOL 402.
TID	Identification number of a TABLED1 bulk entry that contains a mapping of the times in a universal file to the solution times stored in the database. See Remark 4. (Integer > 0; See Remark 5 for default behavior)

Remarks:

1. SID must be unique to all other DTEMPEX bulk entries.
2. The SID of the DTEMPEX bulk entry can only be selected with the DTEMP case control command. The TEMPERATURE case control command cannot be used to select the DTEMPEX bulk entry.

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3. For example, set the UNID field to 22 for the following ASSIGN statements:

```
ASSIGN BUN='temperature0.bun' UNIT=22 FORM=UNFORM STATUS=OLD
```

```
ASSIGN UNV='temperature0.unv' UNIT=22 FORM=FORM STATUS=OLD
```

```
ASSIGN SC_H5='temperature0.scd5' UNIT=22 FORM=UNFORM STATUS=OLD
```

4. You can optionally map the time values that the software reads from a universal file and then writes to the database for the SOL 401 or 402 structural solution. To use this capability, you use the TID field to select a TABLED1 bulk entry which maps the time values in the external universal file. For values of time in the universal file where a tabular data point does not exist in the TABLED1 bulk entry, the software interpolates the tabular data to determine the value of time to store in the database for the structural solution. For information on the available interpolation options, see **TABLED1**.

   Depending on the application, you may be able to use this capability to:

   - Conserve computational resources.
   - Circumvent the six significant digit limit for the numerical data in universal files that results from the FORTRAN E13.5 formatting of universal files.

   For more information, see Mapping times in universal files in the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)*.

5. When the TID field is blank, SOLs 401 and 402 use the times contained in the external file.

Remarks for SOL 402:

1. If a solution time defined with the TSTEP1 bulk entry is less or greater than the times defined in your external .bun or .unv file (Ext_Time1 - to - Ext_Time2):

   - When a solution time < Ext_Time1, the software uses the temperatures referenced by the TEMPERATURE(INIT) case control command.
   - When a solution time > Ext_Time2, the software uses the temperatures at Ext_Time2 in your external file.

2. SOL 402 only supports FILETYPE="BUN".

Related Topics

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DTI

Direct Table Input

Defines table data blocks.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	NAME	“0"	T1	T2	T3	T4	T5	T6	
	V01	V02	-etc.-						
DTI	NAME	IREC	V1	V2	V3	V4	V5	V6	
	V7	V8	V9	V10	-etc.-	“ENDREC”			

Example: (The first logical entry is the header entry.)

DTI	XXX	0	3	4	4096		1	0	
	1.2	2.3							
DTI	XXX	1	2.0	-6	ABC	6.000	-1	2	
	4	-6.2	2.9	1	DEF	-1	ENDREC		

Fields:

Field	Contents
NAME	Any character string that will be used in the DMAP sequence to reference the data block. See Remark 1 . (Character; the first character must be alphabetic.)
Ti	Trailer values. (Integer ≥ 0; Default = 32767)
IREC	Record number. (Integer>0)
V0i, Vi	Value. (Integer, Real, Character or blank)
“ENDREC”	Flags the end of the string of values (V0i or Vi) that constitute record IREC. (Character)

Remarks:

1. The user defines the data block and therefore must write a DMAP (or ALTER a solution sequence), which includes the DTIIN modules, in order to use the DTI feature. See the *Simcenter Nastran DMAP Programmer’s Guide*. All of the rules governing the use of data blocks in DMAP sequences apply.
2. All fields following ENDREC must be blank.

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3. The entry using IREC = 0 is called the header entry and is an optional entry. The values T1 through T6 go to a special record called the trailer. Other values on the optional continuation go to the header record. If the header entry or the trailer is not specified, T1 through T6 = 32767. On this entry, “ENDREC” may be used only if there is at least one continuation.
4. In addition to the optional header entry, there must be one logical entry for each record in the table. Null records require no entries.
5. “ENDREC” is used to input blank values at the end of a record. If “ENDREC” is not specified, the string for a record ends with the last nonblank field.
6. The maximum number of DMI and DTI data blocks is 1000.
7. If Ti is not an integer, a machine-dependent error will be issued that may be difficult to interpret.

Related Topics

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DTI,DFRFNC

Frequency function input for use in DRESP2 & DRESP3

Provides a way to input functions of frequency so that they can be automatically looped over when spawning multiple responses over the frequency set when using DRESP2 or DRESP3. Such functions can be, for example, frequency dependent DRESP1 response bounds. More than one such function can be entered in sequence.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	DFRFNC	"0"							
DTI	DFRFNC	IREC	FID	FTYPE	V1	V2	V3	V4	
	V5	V6	V7	V8	-etc.-				

Example:

DTI	DFRFNC	0								
DTI	DFRFNC	1	325	LISTING	0.036	0.038	0.0395	0.043		
		0.049	0.054							
DTI	DFRFNC	2	513	EQUATION	750					
DTI	DFRFNC	3	517	EQUATION	755					
DTI	DFRFNC	4	647	CONSTANT	12.34					
DTI	DFRFNC	5	116	INCRMENT	3.0	0.2				

Fields:

Field	Contents
IREC	Logical DFRFNC record number. The first DTI,DFRFNC record should be 0, and unreferenced. Each later DTI,DFRFNC record should point to a separate DFRFNC ID (FID).
FID	The ID of the DFRFNC record as referenced from a DRESP2 or DRESP3 type response.
FTYPE	DFRFNC function input format type. There are currently four different ways of inputting DFRFNC function values: CONStant; EQUAtion; INCRment; LISTing. (It is sufficient to enter the first four letters as shown by capitals here). The format of the rest of the data for each DFRFNC record depends on the FTYPE for that record.
V1	For FTYPE = CONS, V1 is the constant real value of the function. For FTYPE = EQUA, V1 is the ID (EQID) of the equation by which the function values are to be computed, with the frequency values in the relevant frequency set being the

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Field	Contents
	only variable in the equation. Thus, the equation loops over the frequencies in the relevant frequency set, and computes the corresponding DFRFNC function value for each frequency in the set. For FTYPE = INCR, V1 is the starting value of the incrementation, i.e. the value of the frequency function corresponding to the first frequency in the relevant frequency set. For FTYPE = LIST, V1 is the first value of the frequency function, i.e. the value of the frequency function corresponding to the first frequency in the relevant frequency set.
V2	For FTYPE = INCR, V2 is the value of the increment by which the function of the frequency will change. The program automatically increments the function value for the actual number of frequencies in the relevant frequency set. For FTYPE = LIST, V2 is the value of the frequency function corresponding to the second frequency in the relevant frequency set.
Vi	These are currently valid only for FTYPE = LIST. Vi corresponds to the value of the frequency function corresponding to the ith frequency in the frequency set.

Remarks:

1. As DTI type table data makes no check on the validity of the data, the user should exercise care in input. It is anticipated that this input will be replaced or paralleled by bulk data input DFRFNC in the future.
2. When a DFRFNC ID is referenced by a DRESP2 or DRESP3 type bulk data, this is normally to enable the spawning of multiple DRESP2 or DRESP3 data over the relevant frequency set. Such spawning of multiple responses is triggered by nested DRESP1 data within the DRESP2 or DRESP3, where such DRESP1 responses are spawned over the frequency set by leaving the ATTB field blank in the DRESP1 input.
3. DFRFNC normally should not be used for cases where multiple DRESP2 or DRESP3 responses are not going to be spawned over the relevant frequency set. Simple table input of a constant value is sufficient in these cases.
4. Normally, the relevant frequency set is determined by the subcase which indirectly references the DRESP2 or DRESP3 response in question. However, for DRESP2 or DRESP3 responses which contain DRSPAN related DRESP1 responses spawned over one or more frequency sets, it is required that the referenced frequency sets for the subcases to which these DRESP1 belong have the same number of frequencies. If the FTYPE = EQUA option is used, the frequency values will be picked up from the frequency set belonging to the last subcase to be analyzed in the relevant DRSPAN domain.

Related Topics

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DTI,DISTL

Load description

Provides load description in an SC_H5 file.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	DISTL	LIDID	LDID	DESCRIPT	DESCRIPT (cont.)	DESCRIPT (cont.)	DESCRIPT (cont.)	DESCRIPT (cont.)	
	-etc.-	"ENDREC"							

Example:

DTI	DISTL	3	3	LOADING	AT ENGIN	E NACELL	E ON POR	T WING	
	ENDREC								

Fields:

Field	Contents
LDID	Load description identification number. This appears in both fields 3 and 4 and must be the same value. (Integer > 0; No default) See Remark 2.
DESCRIPT	Alphanumeric description of the loading. The description continues until ENDREC is encountered in a field. (Character; 1024 characters maximum; No default) See Remark 3.

Remarks:

1. DTI,DISTL is only supported in SOL 108 and SOL 111.
2. DTI,DISTL is referenced by an RLOADEX, RANDPEX, SPCF, or MDSBULK bulk entry when you access loading data from an SC_H5 file. LDID must be unique with respect to all other DTI,DISTL bulk entries.
3. DESCRIPT is the loading description contained in the SC_H5 file referenced by an RLOADEX, RANDPEX, SPCF, or MDSBULK bulk entry. The loading description (that is, the dataset name) defined in the SC_H5 file must be in all uppercase letters.

Related Topics

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DTI,ESTDATA

Superelement Estimation Data Overrides

Provides override data for time and space estimation for superelement processing operations.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	ESTDATA	"0"							
	kd1	vd1	kd2	vd2	-etc.-				

The next entries are repeated for any superelement for which estimate data overrides are desired. IREC must be incremented by 1.

DTI	ESTDATA	IREC	SEFLAG	SEID	k1	v1	k2	v2	
	k3	v3	-etc.-						

Example:

DTI	ESTDATA	0							
	NOMASS	-1							
DTI	ESTDATA	1	SE	10	C1	5.5	C3	4.5	
	C7	7.3							

Fields:

Field	Contents
kdi	Keyword for estimation parameter. (Character from Table 13-71.)
vdi	Value assigned to the estimation parameter kdi. (The type given in Table 13-71.)
IREC	Record number beginning with 1. (Integer>0)
SEFLAG	SEFLAG = "SE" or "SEID" indicates the next field containing a superelement identification number. (Character)
SEID	Superelement identification number. (Integer>0)
ki	Keyword for override of estimation parameter for indicated superelement. (Character from Table 13-71.)
vi	Value for keyword ki. (Type depends on ki as shown in the Table 13-71 .)

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Table 13-71: DTI,ESTDATA Input Parameter Descriptions

Input Parameters				Meaning and Comments
Keyword	Type	Default Value	Math Symbol	
CRMS*	Real	-1.0	C	Number of active columns in $[K_{oo}]$.
FCRMS*	Real	0.10		If FCRMS ≤ 0.0, FCRMS is used (c/o).
C1	Real	6.0	c_1	Average number of degrees-of-freedom per grid point in o-set.
C3	Real	8.0	c_3	Average number of connections per grid point.
C4	Real	0.15	c_4	I/O time (seconds) per block transferred.
C5	Real	6.0	c_5	Average number of effective degrees-of-freedom per grid point in a-set.
C6	Real	1.2	c_6	Total CPU factor.
C7	Real	6.0	c_7	Number of equivalent KGG data blocks for space estimation.
WF	Real	-1.0	W	If WF ≤ 0.0 then use available working storage in units of single-precision words.
NOMASS	Integer	1		If NOMASS ≠ 1 then exclude mass terms from estimates.
TSEX	Real	0.5 (min)		Threshold limit for CPU.
SSEX	Real	50.0(blocks)		Threshold limit for space.
TWALLX	Real	5.0 (min)		Threshold limit for wall time.
BUFSIZ	Integer	Machine Buffsize	B	Buffsize.
ML	Real	MachineLoop Time	M	Arithmetic time for the multiply/add loop. See the *Simcenter Nastran Installation and Operations Guide*.
CONIO	Integer	Machinel/O ratio		I/O count/CPU equivalence
PREC	Integer	1 or 2		Machine Word Length (1 = long, 2 = short).
NLOADS	Integer	1	N_L	Number of loading conditions
SETYPE	Character	"T"		Superelement type (T = Tip)
CMAX	Real	-1.0	C_{max}	Maximum bandwidth

Parameters Obtained from SEMAP

NGI	Number of interior grid points.
NPE	Number of exterior grid points.
NS	Number of scalar points
NE	Number of elements.

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Derived Parameters

$O=C1 \cdot NG1$	Size of o-set.
$A=C5(NPE - NS) + NS$	Size of a-set.
T = BUFFSIZE/PREC	Number of matrix terms in a buffer.

Estimation Equations

For each superelement, estimates of CPU time and disk space are made using the following equations.

Table 13-72: Equations Used for CPU Time and Disk Space Estimate

Printout Symbol	Math Symbol	Equations
TD	T_1	$T_1 = 1/2 \cdot M \cdot O \cdot C^2$
TFBS	T_2	$T_2 = 2 \cdot M \cdot C \cdot O \cdot a$
TMAA	T_3	$T_3 = M \cdot O \cdot a^2$ (set to 0.0 if NOMASS ≠ +1)
TSE	T_{SE}	$T_{SE} = C_6(T_1 + T_2 + T_3)$
SLOO	S_1	$S_1 = O \cdot C \cdot \frac{\text{PREC}}{b}$
SGO	S_2	$S_2 = O \cdot a \cdot \frac{\text{PREC}}{B}$
SKGG	S_3	$S_3 = 36(NG_i + NG_e - NS)(c_3 + 1.0)\left(\frac{\text{PREC}}{B}\right)$
SSE	S_{SE}	$S_{SE} = S_1 + S_2 + c_7 \cdot S_3$
PASSES	p	$\text{FBS passes} = p = a \cdot O \cdot \frac{\text{PREC}}{WF}$
BKSTRN	BT	Blocks Transferred = $BT = 2 \cdot p \cdot S_1 + S_2 + p \cdot S_2$ (Last term omitted if NOMASS ≠ +1)
TWALL	T_W	Wall Time = $T_W = T_{SE} + c_4 \cdot BT$

Remarks:

1. In the superelement solution sequences, this data is stored automatically.
2. The header record continuation entries are optional if no global override data is to be specified. In this case, the complete header entry is optional.

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- Active column data can come from one of several places. The value for CRMS is determined as follows:
- RMS from the entry when IREC>0 and field 4 is “SE”.
- RMS from entries with IREC = 0.
- Computed bandwidth when PARAM,OLDSEQ is specified.
- If FCRMS is specified when IREC>0 and field 4 is “SE”, then CRMS = FCRMS · O.
- If FCRMS is specified when IREC = 0, then CRMS = FCRMS · O.
- CRMS = 0.1 · O.

3. If CMAX is not specified, then it is defaulted to CRMS.
4. In the example above, mass terms are excluded for all superelements and new values are given for parameters C1, C3, and C7 for Superelement 10 only.
5. The estimates for TSEX, SSEX, and TWALLX are not printed unless at least one estimate exceeds the threshold.

Related Topics

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DTI,INDTA

Stress, Strain and/or Force Sort/Filter Item Code Override

Specifies or overrides default item codes for the sorting and filtering of element stresses, strains, and forces.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	INDTA	“0"							

To specify/override items for a sort of stress quantities:

DTI	INDTA	“1"	B1	C1	B2	C2	“ENDREC”		

To specify/override items for a sort of force quantities:

DTI	INDTA	“2"	B1	C1	B2	C2	“ENDREC”		

Example:

DTI	INDTA	0							

To specify/override items for a sort of stress quantities:

DTI	INDTA	1	64	18	75	18	ENDREC		

To specify/override items for a sort of force quantities:

DTI	INDTA	2	34	2	2	4	ENDREC		

Fields:

Field	Contents
Bi	Element type identification number. See the table in **“Item Codes”** for allowable values. (Integer>0)
Ci	Item code identification number for the stress, strain, or force quantity on which the sort or filter is to be performed. See the table in the **“Item Codes”** for allowable values. (Integer)

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Remarks:

1. This table is recognized only in SOLs 101, 103, 105, 106, 108, 109, 111, 112, 114, 115, 144, 153, and for stress quantities only. One or more of the user parameters S1, S1G, or S1M must be specified with a value greater then or equal to zero in order to request sorting and/or filtering. See user parameter S1 in **"Parameters and Parameter Applicability Tables"** . In order to sort force or strain quantities, a DMAP Alter is required.
2. If the Ci value is -1, the element type will be suppressed on the output file. An example of this feature could be as follows: If an element type is to be sorted on two different values and output twice, this can be accomplished by two calls to the STRSORT module with two unique DTI tables. However, other element types will be printed twice. This additional print can be suppressed by setting their sort codes to -1.
3. **Table 13-73** lists the elements currently that are sortable. In addition, the element type identification number, the default stress output quantity, and the associated stress code identification numbers are provided. If this entry is not specified, then the stresses are sorted based on the default quantity given in **Table 13-73**.

   The following should be noted:

   - The element type identification number is used internally by the program to differentiate element types.
   - The stress code identification number is merely the word number in the standard printed output for the stress quantity of interest. For example, the thirteenth word of stress output for the CHEXA element is the octahedral shear stress. For this element type, the element identification number and the grid point ID each count as a separate word. Stress codes for the elements are tabulated in **"Item Codes"** .
   - By default, stress sorting for the membrane and plate elements will be performed on the Hencky-von Mises stress. For maximum shear stress, the STRESS (MAXS) Case Control command should be specified.

Table 13-73: Sortable Elements

Element	Element Type ID Number	Default Stress Output Quantity and Identification Number	
		Quantity	Stress Code ID Number
CBAR	34	Maximum stress at end B	14
CBEAM	2	Maximum stress at end B	108
CBEND	69	Maximum stress at end B	20
CONROD	10	Axial stress	2
CELAS1	11	Stress	2
CELAS2	12	Stress	2
CELAS3	13	Stress	2
CHEXA	67	Hencky-von Mises or Octahedral stress	13
CQUAD4	33	Maximum shear or Hencky-von Mises stress at Z_2	17
CQUAD4*	144	Maximum shear or Hencky-von Mises stress at Z_2	19

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Element	Element Type ID Number	Default Stress Output Quantity and Identification Number	
		Quantity	Stress Code ID Number
CQUAD8	64	Maximum shear or Hencky-von Mises stress at Z_2	19
CQUADR	82	Maximum shear or Hencky-von Mises stress at Z_2	19
CPENTA	68	Octahedral stress	13
CROD	1	Axial stress	2
CSHEAR	4	No default	---
CTETRA	39	No default	---
CTRIA3	74	Maximum shear or Hencky-von Mises stress at Z_2	17
CTRIA6	75	Maximum shear or Hencky-von Mises stress at Z_2	19
CTRIAR	70	Maximum shear or Hencky-von Mises stress at Z_2	19
CTUBE	3	Axial stress	2

* CORNER output

Related Topics

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DTI,SELOAD

External Superelement Load Set IDs

Usage overrides automatic generation of DTI,SELOAD definitions created through the use of the EXTSEOUT case control command.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	SELOAD	LIDSE	EXCSE	"0"	DESCRIPT	DESCRIPT (cont.)	DESCRIPT (cont.)	DESCRIPT (cont.)	
	-etc.-	"ENDREC"							

Example:

DTI	SELOAD	1	100	0	FORCE_AT	_GRID_2	ENDREC	ELEM_100	

Fields:

Field	Contents
LIDSE	Load set identification number in the external superelement. See Remark 1. (Integer > 0)
EXCSE	Excitation identification number in the external superelement. See Remark 1. (Integer > 0)
DESCRIPT	Alphanumeric description of the loading. The description continues until ENDREC is encountered in a field. The description must be defined such that the first non-blank character in each field is not numeric. If the first non-blank character in a field is numeric, the field may be interpreted as numeric. If the field then contains any alphabetic character, an error will be reported. If the data in the field is all numeric, it will be shifted to the right. Imbedded blanks within a field may produce an error. To avoid this problem, use underscores to represent blank characters. (Character)

Remarks:

1. The values of LIDSE and EXCSE depend on the describers specified on the EXTSEOUT case control command in the external superelement creation run.
   - For a static solution, two options are available. If the loading is referenced in the case control using the LOAD case control command, LIDSE and EXCSE are both equal to the set identification number of the LOAD case control command. If the loading is referenced in the case control using the LOADSET case control command, LIDSE and EXCSE are the values corresponding to the values

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of LID and EXCITEID, respectively, on each LSEQ bulk entry referenced by the LOADSET case control command. If both LOAD and LOADSET case control commands exist, LOADSET takes precedence.

- For a dynamic solution, LIDSE and EXCSE are both equal to the EXCITEID value for each load (not enforced motion) defined on RLOADi and TLOADi bulk entries, regardless of whether or not they are referenced in the case control. The use of LOADSET and LSEQ is identical to that for a static solution.
- Thermal loads and enforced motion loads defined using the SPCD bulk data method are not supported.

2. The use of DTI,SELOAD is optional. If a DTI,SELOAD bulk entry does not exist for an external superelement load set (LIDSE), one will be automatically generated. The description (DESCRIPT) of the automatically generated entry will have the form “SE_LOAD_xx”, where “xx” is the character representation of the external superelement load set.
3. DTI,SELOAD can be considered for use when the EXTSEOUT case control command specifies one of the following data storage options: MATDB (or MATRIXDB), DMIGDB, DMIGOP2 = unit, or MATOP4 (or MATRIXOP4) = unit. The data storage option DMIGPCH is only applicable to static analysis and requires the use of the P2G case control command. See the **EXTSEOUT** case control command for additional information.

Related Topics

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DTI,SETREE

Superelement Tree Definition

Defines a superelement tree that determines the superelement processing order.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	SETREE	"1"	SEUP1	SEDOWN1	SEUP2	SEDOWN2	SEUP3	SEDOWN3	
	SEUP4	SEDOWN4	SEUP5	SEDOWN5	-etc.-				

Example:

DTI	SETREE	1	1	14	2	14	3	14	
	4	14	14	0					

Fields:

Field	Contents
SEUPi	Identification number of the superelement upstream from SEDOWNi. (Integer>0)
SEDOWNi	Identification number of the superelement into which SEUPi is assembled. (Integer ≥ 0)

Remarks:

1. SETREE entries or the DTI,SETREE entry are required for multi-level superelement configurations.
2. If an DTI,SETREE entry is provided, then SETREE entries are not required.
3. If both SETREE entries and a DTI,SETREE entry exist, then the DTI,SETREE entry will be ignored.
4. Superelements not referenced on the SETREE or DTI,SETREE entry will have the residual structure as their downstream superelement.
5. If this entry is not present, the superelement tree and the processing order are determined automatically.
6. A superelement identification may appear only once in a SEUPi field.
7. On restart, if a superelement identification does not appear in a SEUPi field, its matrices will not be assembled, even though they may be present in the database.
8. See the *Simcenter Nastran Superelement User's Guide* for a description of user-designated trees.
9. This entry is stored in the database automatically. Once stored, the Bulk Data entry may be removed from the input file.

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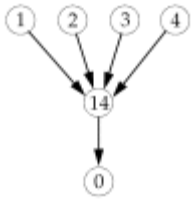
10. In the example above, the following superelement tree is defined:

Figure 13-125: Sample Superelement Tree



Related Topics

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DTI,SPECSEL

Response Spectra Input Correlation Table

Correlates spectra lines specified on TABLED1 entries with damping values.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	SPECSEL	RECNO		TYPE	TID1	DAMP1	TID2	DAMP2	
	TID3	DAMP3	TID4	DAMP4	TID5	DAMP5	-etc.-		

Example:

DTI	SPECSEL	1		A	1	.02	2	.04	
	3	.06							
DTI	SPECSEL	3		V	4	.01			

Fields:

Field	Contents
RECNO	Spectrum number. (Integer>0)
TYPE	Type of spectrum. (Character: “A” for acceleration, “V” for velocity, or “D” for displacement.)
TIDi	TABLED1 entry identification number. (Integer>0)
DAMPi	Damping value assigned to TIDi. (Real)

Remarks:

1. The RECNO is the number of the spectrum defined by this entry. It is referenced on DLOAD Bulk Data entries.
2. The TIDi, DAMPi pairs list the TABLEDl entry, which defines a line of the spectrum and the damping value assigned to it. The damping value is in the units of fraction of critical damping.
3. This entry is placed in the database automatically. Once stored, the Bulk Data entry may be removed from the input file.

Related Topics

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DTI,SPSEL

Response Spectra Generation Correlation Table

Correlates output requests with frequency and damping ranges.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	SPSEL	RECNO	DAMPL	FREQL	G1	G2	G3	G4	
	G5	G6	G7	-etc.-					

Example:

DTI	SPSEL	1	2	1	11	12			
DTI	SPSEL	2	4	3	1	7	11	12	
	13	14							

Fields:

Field	Contents
DAMPL	Identification number of the FREQ, FREQ1, or FREQ2 Bulk Data entry that specifies the list of damping values. (Integer>0)
FREQL	Identification number of the FREQi Bulk Data entry that specifies the list of frequencies. (Integer>0)
Gi	Grid point number where response spectra will be calculated. (Integer>0)
RECNO	Record number of spectra to be generated. (Sequential integer beginning with 1.)

Remarks:

1. This table is used in SOLs 109, and 112.
2. Damping values are in the units of fraction of critical damping.
3. Output of response spectra requires the use of the XYPLOT...SPECTRA(RECNO)/Gi... command, where Gi is restricted to the grid points listed on the (RECNO) record of this entry.
4. See the *Simcenter Nastran User's Guide* for example problems using this feature.
5. The SPSEL table is stored in the database automatically in SOLs 109 and 112. Once stored, the Bulk Data entry may be removed from the input file.

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DTI,UNITS

Specifies the system of units.

Specifies the system of units.

Format:

1	2	3	4	5	6	7	8	9	10
DTI	UNITS	"1"	MASS	FORCE	LENGTH	TIME	TEMPERATURE		

Example:

DTI	UNITS	1	KG	N	M	S	F		

Fields:

Field	Contents
MASS	Defines the mass units Acceptable mass inputs are: KG - kilogram LBM – pound-mass (0.45359237 kg) SLUG – slug (14.5939029372 kg) GRAM – gram (1E-3 kg) OZM – ounce-mass (0.02834952 kg) KLBM – kilo pound-mass (1000 lbm) (453.59237 kg) MGG – megagram (1E3 kg) MG – milligram (1E-6 kg) MCG – microgram (1E-9 kg) NG – nanogram (1E-12 kg) UTON – U.S. ton (907.18474 kg)

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Field	Contents
	SLI – slinch (175.1268352 kg)
FORCE	Defines the force units Acceptable force inputs are: N – Newton LBF – pound-force (4.44822161526 N) KGF – kilograms-force (9.80665 N) OZF – ounce-force (0.2780139 N) DYNE – dyne (1E-5 N) KN – kilonewton (1E3 N) KLBF – kilo pound-force (1000 lbf) (4448.22161526 N) MN – millinewton (1E-3 N) MCN – micronewton (1E-6 N) NN – nanonewton (1E-9 N) CN – centinewton[1] (1E–2 N) P – poundal[1] (0.138254954 N)
LENGTH	Defines the length units Acceptable length inputs are: M – meter KM – kilometer (1E3 m) CM – centimeter (1E-2 m) MM – millimeter (1E-3 m) MI – mile (1609.344 m) FT – foot (0.3048 m)

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Field	Contents
	IN – inch (25.4E-3 m) MCM – micrometer (1E-6 m) NM – nanometer (1E-9 m) A – Angstrom (1E-10 m) YD – yard (0.9144 m) ML – mil (25.4E-6 m) MCI – microinch (25.4E-9 m)
TIME	Defines the time units Acceptable time inputs are: S – second H – hour (3600.0 sec) MIN-minute (60.0 sec) MS – millisecond (1E-3 sec) MCS – microsecond (1E-6 sec) NS – nanosecond (1E-9 sec) D – day (86.4E3 sec)
TEMPERATURE	Defines the temperature units[2] Acceptable temperature inputs are: F – Fahrenheit R – Rankin C – Celsius K – Kelvin

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[1]: not supported by MNF

[2]: not supported by MNF or RFI

Remarks:

1. Since DTI,UNITS determines all units for the MNF and RFI, the units defined in WTMASS, which are important for units consistency in Simcenter Nastran, are ignored in the output to these files. For example, if the model mass is kilograms, force in Newtons, length in meters, and time in seconds, then WTMASS would equal 1 ensuring that Simcenter Nastran works with the consistent set of kg, N, and m. The units written to the MNF or RFI would be: “DTI,UNITS,1,KG,N,M,S”.

Related Topics

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DVBSHAP

Design Variable to Boundary Shapes

Associates a design variable identification number to a linear combination of boundary shape vectors from a particular auxiliary model.

Format:

1	2	3	4	5	6	7	8	9	10
DVBSHAP	DVID	AUXMID	COL1	SF1	COL2	SF2	COL3	SF3	

Example:

DVBSHAP	4	1	1	1.6					

Fields:

Field	Contents
DVID	Design variable identification number of a DESVAR entry. (Integer>0)
AUXMID	Auxiliary model identification number. (Integer>0)
COLi	Load sequence identification number from AUXMODEL Case Control command. (Integer>0)
SFi	Scaling factor for load sequence identification number. (Real; Default = 1.0)

Remarks:

1. Design variable DVID must be defined on a DESVAR entry.
2. Multiple references to the same DVID and/or COLi will result in the vector addition of the referenced boundary shape vectors.
3. Multiple DVBSHAP entries may be specified.

Related Topics

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DVCREL1

Design Variable to Connectivity Property Relation

Defines the relation between a connectivity property and design variables.

Format:

1	2	3	4	5	6	7	8	9	10
DVCREL1	ID	TYPE	EID	CPNAME	CPMIN	CPMAX	C0		
	DVID1	COEF1	DVID2	COEF2	DVID3	COEF3	-etc.-		

Example:

DVCREL1	5	CQUAD4	1	ZOFFS		1.0			
	1	1.0							

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
TYPE	Name of an element connectivity entry, such as “CBAR”, CQUAD4”, etc. (Character; case sensitive)
EID	Element Identification number. (Integer>0)
CPNAME	Name of connectivity property, such as “X1”, X2”, “X3”, “ZOFFS”, etc. (Character)
CPMIN	Minimum value allowed for this property. If CPNAME references a connectivity property that can only be positive, then the default value of CPMIN is 1.0E-15. Otherwise, it is -1.0E35. See Remark 4. (Real)
CPMAX	Maximum value allowed for this property. See Remark 4. (Real; Default =1.0E+20)
C0	Constant term of relation. (Real; Default = 0.0)
DVIDi	DESVAR entry identification number. (Integer>0)
COEFi	Coefficient of linear relation. (Real)

Remarks:

1. The relationship between the connectivity property and design variables is given by:

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$$CP_j = C_0 + \sum_i COEF_i \times DVID_i$$

2. The continuation entry is required.
3. The fifth field of the entry, CPNAME, only accepts string characters. These string values must be the same as those given in the connectivity entry descriptions in this Guide. For example, if the plate offset is to be designed (CQUAD4, CTRIA3, etc), ZOFFS (case insensitive) must be specified on the CPNAME field.
4. The CPMIN and CPMAX values may be violated at a given design cycle if there is other, more critical violation by response constraints. Since design variables may never violate their lower and upper bounds, it may be best, if possible and reasonable, to configure the property bounds accordingly, so that they are not violated even for infeasible designs.

Related Topics

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DVCREL2

Design Variable to Connectivity Property Relation

Defines the relation between a connectivity property and design variables with a user-supplied equation.

Format:

1	2	3	4	5	6	7	8	9	10
DVCREL2	ID	TYPE	EID	CPNAME	CPMIN	CPMAX	EQID		
	“DESVAR”	DVID1	DVID2	DVID3	-etc.-				
	“DTABLE”	LABL1	LABL2	LABL3	-etc.-				

Example:

DVCREL2	1	CBAR	100	X1	0.05	1.0	100		
	DESVAR	1001							
	DTABLE	X10							

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
TYPE	Name of an element connectivity entry, such as “CBAR”, CQUAD4”, etc. (Character)
EID	Element Identification number. (Integer>0)
CPNAME	Name of connectivity property, such “X1”, X2”, “X3”, “ZOFFS”, etc. (Character)
CPMIN	Minimum value allowed for this property. If CPNAME references a connectivity property that can only be positive, then the default value of CPMIN is 1.0E-20. Otherwise, it is -1.0E35. See Remark 4. (Real)
CPMAX	Maximum value allowed for this property. See Remark 4. (Real; Default =1.0E+20)
EQID	DEQATN entry identification number. (Integer>0)
“DESVAR”	DESVAR flag. Indicates that the IDs of DESVAR entries follow. (Character)
DVIDi	DESVAR entry identification number. (Integer>0)
“DTABLE”	DTABLE flag. Indicates that the LABLs for the constants in a DTABLE entry follow. This field may be omitted if there are no constants involved in this relation. (Character)
LABLi	Label for a constant on the DTABLE entry. (Character. See **Bulk Data Syntax Rules**.)

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Remarks:

1. The variable identified by DVIDi and LABLi correspond to variable names (x1, x2, etc.) listed in the left-hand side of the first equation on the DEQATN entry identified by EQID. The variable names x1 through xN (where $N = m + n$) are assigned in the order DVID1, DVID2, ..., DVIDm, LABL1, LABL2, ..., LABLn.
2. If both “DESVAR” and “DTABLE” are specified in field 2, “DESVAR” must appear first.
3. The fifth field of the entry, CPNAME, only accepts string characters. These string values must be the same as those given in the connectivity entry descriptions in this Guide. For example, if the plate offset is to be designed (CQUAD4, CTRIA3, etc.), ZOFFS (case insensitive) must be specified on the CPNAME field.
4. The CPMIN and CPMAX values may be violated at a given design cycle if there is other, more critical violation by response constraints. Since design variables may never violate their lower and upper bounds, it may be best, if possible and reasonable, to configure the property bounds accordingly, so that they are not violated even for infeasible designs.

Related Topics

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DVEREL1

Automatic shell element thickness design variable

Automatic shell element thickness design variable.

Format:

1	2	3	4	5	6	7	8	9	10
DVEREL1	ID	TYPE	GRPID	PNAME	PMIN	PMAX			
	DVID1	COEF1							

Example:

DVEREL1	701	PSHELL	1001	T	0.1	10.0			
	4053	0.1							

The referenced GROUP bulk entry may look like the following.

```
...
GROUP, 1001, shell_thickness,+
    +, ELEM, 1001, THRU, 1871
    +, ELEM, 2001, THRU, 2589
    ...
```

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
TYPE	The property type used by the elements selected on the associated GROUP entry. The software will auto-generate design variables for these elements. Currently only PSHELL is supported. This field is case sensitive. (Character; Default=PSHELL)
GRPID	ID of a GROUP bulk entry which selects the shell elements CQUADR, CQUAD4, CQUAD8, CTRIAR, CTRIA3, CTRIA6 for which the software will auto-generate a design variable associated with the thickness. See the remarks and the example. (Integer>0)
PNAME	Property name or field position of the property entry. Currently, only "T" is supported. (Character or Integer; Default=T)
PMIN	Minimum value allowed for the associated properties. (Real; Default=1.0E-15)
PMAX	Maximum value allowed for the associated properties. (Real; Default =1.0E+20)

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Field	Contents
DVID	Identification number for the first auto-generated PSHELL, DESVAR, and DVPREL1 entry. When the default is used, the design variable IDs will be the same as the IDs of the associated shell elements. (Integer>0; Default=0)
COEF	The coefficient in the expression P = COEF*DV. (Real; Default =1)

Remarks:

1. The GROUP entry referenced by the DVEREL1 entry must currently use TYPEi=ELEM. The software ignores any unsupported elements in the ELEM lists, and it ignores elements selected on a GROUP entry with TYPEi=PROP.
2. For each element selected by the DVEREL1 and associated GROUP entries, the software automatically associates a new design variable with the element thickness. This results in a separate independent thickness design variable for each element selected. The starting thickness value is determined as follows:

   - If the shell element definition in the input file includes corner thicknesses, the software will compute an average thickness for the element.

   - If corner thicknesses are not defined, the thickness on the original PSHELL in the input file is used.
3. The relationship between the connectivity property and design variables is given by:

$$P = COEF \times DVID$$

4. A GROUP ID may only be referenced by a single DVEREL1.
5. The ID of each auto-generated DESVAR entry must be unique with respect to any existing DESVAR or other DVEREL1 generated DESVAR. These IDs cannot duplicate any existing DVPREL1 or PSHELL IDs either. Therefore, care should be taken when defining the DVID1 field on the DVEREL1 entry.
6. If you request an .op2 file with the PARAM,POST,-1 or PARAM,POST-2 definitions, the shell element thicknesses at the element grid points computed for the best design cycle will be written to the OSHT1 datablock in the .op2 file at the end of the analysis. You can thus display the thickness values in a post processor which supports the OSHT1 datablock.
7. The EDVOUT parameter is available on the DOPTPRM bulk entry to control what fraction of DVEREL1 generated design variables are printed. The printing of variables occurs in the Design Variable History at the end of the f06 file, as well as at requested cycles. The fraction you assign the EDVOUT parameter is applied separately to the increased and to the decreased DVEREL1 generated design variables. For example, when the default of 0.001 is used, both 0.1% of the most increased and 0.1% of the most decreased DVEREL1 generated design variables will be printed. A value of 1.0 requests to print all DVEREL1 generated design variables which have either increased or decreased.

   To determine the most increased and the most decreased thickness variables, the software evaluates the variables using their percentage change between the best and initial design cycles. For example, if the thickness of variable one decreases from 1.0 to 0.8 or 20%, and the thickness of variable two decreases from 3.0 to 2.55 or 15%, variable one will have priority for printing.

   Note that the design variables which are not generated by the DVEREL1 entry are all printed in full.

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Related Topics

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DVGRID

Design Variable to Grid Point Relation

Defines the relationship between design variables and grid point locations.

Format:

1	2	3	4	5	6	7	8	9	10
DVGRID	DVID	GID	CID	COEFF	N1	N2	N3		

Example:

DVGRID	3	108	5	0.2	0.5	0.3	1.0		

Fields:

Field	Contents
DVID	DESVAR entry identification number. (Integer>0)
GID	Grid point (GRID) or geometric point (POINT) identification number. (Integer>0)
CID	Coordinate system identification number. (Integer ≥ 0; Default = 0)
COEFF	Multiplier of the vector defined by Ni. (Real; Default = 0.0)
Ni	Components of the vector measured in the coordinate system defined by CID. (Real; at least one Ni ≠ 0.0)

Remarks:

1. A CID of zero or blank (the default) references the basic coordinate system.
2. Multiple references to the same grid ID and design variable result in vectorial addition of the participation vectors defined by CID, COEFF, and Ni. There is no restriction on the number of DVGRID entries that may reference a given grid (GID) or design variable (DVID).
3. The coordinate update equation is given as

$$\{g\}_i - \{g\}_i^0 = \sum_j \mathrm{COEFF}_j \left(X_{\mathrm{DVID}j} - X^0_{\mathrm{DVID}j} \right) \{N\}_j$$

where $\{g\}_i$ is the location of the *i-th* grid, $[g_x g_y g_z]^T$.

The vector $\{N\} = [N_x N_y N_z]^T$ is determined from CID and Ni. Note that it is a change in a design variable from its initial value X^0, and not the absolute value of the design variable itself, that represents a change in a grid point location, $\{g\}_i - \{g\}^0_i$.

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4. The DVGRID entry defines the participation coefficients (basis vectors) of each design variable for each of the coordinates affected by the design process in the relationship

$$\{\Delta g\}_i = \sum_j \{T\}_{ij} \cdot \Delta X_j$$

5. DVGRID entries that reference grid points on MPCs or RSSCON entries produce incorrect sensitivities. Often the sensitivities are 0.0 which may result in a warning message indicating zero gradients which may be followed by UFM 6499. Other rigid elements produce correct results.

Related Topics

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DVMREL1

Design Variable to Material Relation

Defines the relation between a material property and design variables.

Format:

1	2	3	4	5	6	7	8	9	10
DVMREL1	ID	TYPE	MID	MPNAME	MPMIN	MPMAX	C0		
	DVID1	COEF1	DVID2	COEF2	DVID3	COEF3	-etc.-		

Example:

DVMREL1	5	MAT1	1	RHO	0.05	1.0			
	1	1.0							

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
TYPE	Name of a material property type. The supported types are "MAT1", "MAT2", "MAT3", "MAT8", "MAT9", "MAT10", and "MAT11". (Character)
MID	Material Identification number of a MAT1, MAT2, MAT3, MAT8, MAT9, MAT10, or MAT11 bulk entry. (Integer>0)
MPNAME	Name of material property, such “E”, “RHO”, etc. (Character)
MPMIN	Minimum value allowed for this property. If MPNAME references a material property that can only be positive, then the default value for MPMIN is 1.0E-15. Otherwise, it is -1.0E35. See Remark 4. (Real)
MPMAX	Maximum value allowed for this property. See Remark 4. (Real; Default = 1.0E+20)
C0	Constant term of relation. (Real, Default = 0.0)
DVIDi	DESVAR entry identification number. (Integer>0)
COEFi	Coefficient of linear relation. (Real)

Remarks:

1. The relationship between the material property and design variables is given by:

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$$MP_i = C_0 + \sum_i COEF_i \cdot DVID_i$$

2. The continuation entry is required.
3. The fifth field of the entry, MPNAME, only accepts string characters. It must refer to the same name as appears in the **"Bulk Data Entries"** for various material properties. For example, if the isotropic material density is to be design, RHO (case insensitive) must be specified on the MPNAME field.
4. The MPMIN and MPMAX values may be violated at a given design cycle if there is other, more critical violation by response constraints. Since design variables may never violate their lower and upper bounds, it may be best, if possible and reasonable, to configure the property bounds accordingly, so that they are not violated even for infeasible designs.

Related Topics

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DVMREL2

Design Variable to Material Relation

Defines the relation between a material property and design variables with a user-supplied equation.

Format:

1	2	3	4	5	6	7	8	9	10
DVMREL2	ID	TYPE	MID	MPNAME	MPMIN	MPMAX	EQID		
	“DESVAR”	DVID1	DVID2	DVID3	-etc.-				
	“DTABLE”	LABL1	LABL2	LABL3	-etc.-				

Example:

DVMREL2	5	MAT1	1	E	0.05	1.0	100		
	DESVAR	1	2						
	DTABLE	E0							

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
TYPE	Name of a material property type. The supported types are "MAT1", "MAT2", "MAT3", "MAT8", "MAT9", "MAT10", and "MAT11". (Character)
MID	Material Identification number of a MAT1, MAT2, MAT3, MAT8, MAT9, MAT10, or MAT11 bulk entry. (Integer>0)
MPNAME	Name of material property, such “E”, “RHO”, etc. (Character)
MPMIN	Minimum value allowed for this property. If MPNAME references a material property that can only be positive, then the default value for MPMIN is 1.0E-20. Otherwise, it is -1.0E35. See Remark 4. (Real)
MPMAX	Maximum value allowed for this property. See Remark 4. (Real; Default = 1.0E+20)
EQID	DEQATN entry identification number. (Integer>0)
“DESVAR”	DESVAR flag. Indicates that the IDs of DESVAR entries follow. (Character)
DVIDi	DESVAR entry identification number. (Integer>0)

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Field	Contents
“DTABLE”	DTABLE flag. Indicates that the LABLs for the constants in a DTABLE entry follow. This field may be omitted if there are no constants involved in this relation. (Character)
LABLi	Label for a constant on the DTABLE entry. (Character. See **Bulk Data Syntax Rules**.)

Remarks:

1. The variables identified by DVIDi and LABLi correspond to variable names (x1, x2, etc.) listed in the left-hand side of the first equation on the DEQATN entry identified by EQID. The variable names x1 through xN (where $N = m + n$) are assigned in the order DVID1, DVID2, ..., DVIDm, LABL1, LABL2, ..., LABLn.
2. If both “DESVAR” and “DTABLE” are specified in field 2, “DESVAR” must appear first.
3. The fifth field of the entry, MPNAME, only accepts string characters. It must refer to the same name as appears in the **“Bulk Data Entries”** for various material properties. For example, if Young’s modulus is to be defined as a function of design variables, E (case insensitive) must be specified in the MPNAME field.
4. The MPMIN and MPMAX values may be violated at a given design cycle if there is other, more critical violation by response constraints. Since design variables may never violate their lower and upper bounds, it may be best, if possible and reasonable, to configure the property bounds accordingly, so that they are not violated even for infeasible designs.

Related Topics

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DVPREL1

Design Variable to Property Relation

Defines the relation between an analysis model property and design variables.

Format:

1	2	3	4	5	6	7	8	9	10
DVPREL1	ID	TYPE	PID	PNAME/FID	PMIN	PMAX	C0	FREQ	
	DVID1	COEF1	DVID2	COEF2	DVID3	-etc.-			

Examples:

DVPREL1	12	PBAR	612	6	0.2	3.0			
	4	0.25	20	20.0	5	0.3			

DVPREL1	73	PELAST	319	TKID			0.725	3.5	
	18	0.375	28	1.52					

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
TYPE	Name of a property entry, such as “PBAR”, “PBEAM”, etc. (Character)
PID	Property entry identification number. (Integer>0)
PNAME/FID	Property name, such as “T”, “A”, or field position of the property entry, or word position in the element property table of the analysis model. (Character or Integer ≠ 0)
PMIN	Minimum value allowed for this property. If FID references a stress recovery location, then the default value for PMIN is -1.0 + 35. PMIN must be explicitly set to a negative number for properties that may be less than zero (for example, field ZO on the PCOMP entry). See Remark 9. (Real; Default = 1.0E-15)
PMAX	Maximum value allowed for this property. See Remark 9. (Real; Default = 1.0E+20)
C0	Constant term of relation. (Real; Default = 0.0)
FREQ	Frequency at which a frequency dependent property given by PNAME/FID is defined. See Remark 8. (Real)

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Field	Contents
DVIDi	DESVAR entry identification number. (Integer>0)
COEFi	Coefficient of linear relation. (Real)

Remarks:

1. The relationship between the analysis model property and design variables is given by:

$$P_i = C0 + \sum_i \text{COEFi} \cdot \text{DVIDi}$$

2. The continuation entry is required.
3. PTYPE = “PBEND” is not supported. PTYPE= “PBEAML” is not supported with the FID option.
4. FID may be either a positive or a negative number. However, for PTYPE=“PBEAM” or “PBUSH”, FID must be negative. If FID>0, it identifies the field position on a property entry. If FID<0, it identifies the word position of an entry in the element property table. For example, to specify the area of a PBAR, either FID = +4 or FID = -3 can be used. In general, use of PNAME is recommended.
5. Designing PBEAML or PBEAM requires specification of both property name and station. Table 13-74 shows several examples.

   Table 13-74:

PTYPE	Property Name	END A	END B	i-th Station
PBEAML	DMI1	DIM1 or DIM1(A)	DIM1(B)	DIM1(i)
PBEAM	A	A or A(A)	A(B)	A(i)

   Only stations that are input on a PBEAM or PBEAML entry can be referenced by a DVPREL1. For example, using an END B property name on a DVPREL1 entry when the referenced PBEAM is a constant section is not allowed.
6. The PWELD and PFAST property types are not supported.
7. The PPLANE property type is not supported.
8. FREQ (Field 9) is applicable only for TYPE=PBUSHT/PDAMPT/PELAST type properties. A FREQ value must currently match an analysis frequency, though there is no need to match a TABLED1 entry. Properties should be designed at a minimum of two frequencies for a TABLED1 referenced by a designed frequency dependent property PNAME/FID, because these tables are currently re-generated, not added to.
9. The PMIN and PMAX values may be violated at a given design cycle if there is other, more critical violation by response constraints. Since design variables may never violate their lower and upper bounds, it may be best, if possible and reasonable, to configure the property bounds accordingly, so that they are not violated even for infeasible designs.

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Related Topics

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DVPREL2

Design Variable to Property Relation

Defines the relation between an analysis model property and design variables with a user-supplied equation.

Format:

1	2	3	4	5	6	7	8	9	10
DVPREL2	ID	TYPE	PID	PNAME/FID	PMIN	PMAX	EQID	FREQ	
	“DESVAR”	DVID1	DVID2	DVID3	-etc.-				
	“DTABLE”	LABL1	LABL2	LABL3	-etc.-				

Examples:

DVPREL2	13	PBAR	712	5	0.2	4			
	DESVAR	4	11	13	5				
	DTABLE	PI	YM						

DVPREL2	17	PBUSHT	122	TBID2	0.01	0.03	200	6.0	
	DESVAR	111	112	113	114				
	DTABLE	XYZ11	XYZ13	XYZ15					

Fields:

Field	Contents
ID	Unique identification number. (Integer>0)
TYPE	Name of a property entry, such as PBAR, PBEAM, etc. (Character)
PID	Property entry identification number. (Integer>0)
PNAME/FID	Property name, such as “T”, “A”, or field position of the property entry, or word position in the element property table of the analysis model. (Character or Integer ≠ 0)
PMIN	Minimum value allowed for this property. If FID references a stress recovery location field, then the default value for PMIN is -1.0+35. PMIN must be explicitly set to a negative number for properties that may be less than zero (for example, field ZO on the PCOMP entry). See Remark 10. (Real; Default = 1.E-20)
PMAX	Maximum value allowed for this property. See Remark 10. (Real; Default = 1.0E20)

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Field	Contents
EQID	DEQATN entry identification number. (Integer>0)
FREQ	Frequency at which a frequency dependent property given by PNAME/FID is defined. See Remark 9. (Real)
“DESVAR”	DESVAR flag. Indicates that the IDs of DESVAR entries follow. (Character)
DVIDi	DESVAR entry identification number. (Integer>0)
“DTABLE”	DTABLE flag. Indicates that the LABLs for the constants in a DTABLE entry follow. This field may be omitted if there are no constants involved in this relation. (Character)
LABLi	Label for a constant on the DTABLE entry. (Character. See “Bulk Data Syntax Rules”.)

Remarks:

1. The variables identified by DVIDi and LABLi correspond to variable names (x1, x2, etc.) listed in the left-hand side of the first equation on the DEQATN entry identified by EQID. The variable names x1 through xN (where N = m+n) are assigned in the order DVID1, DVID2, ..., DVIDn, LABL1, LABL2, ..., LABLm.
2. If both “DESVAR” and “DTABLE” are specified in field 2, “DESVAR” must appear first.
3. FID may be either a positive or a negative number. However, for PTYPE=“PBEAM” or “PBUSH”, FID must be negative. If FID>0, it identifies the field position on a property entry. If FID<0, it identifies the word position of an entry in EPT. For example, to specify the area of a PBAR, either FID = +4 or FID = -3 may be used. In general, use of PNAME is recommended.
4. PTYPE = “PBEND” is not supported.
5. Designing PBEAM requires specification of both property name and station. Table 13-75 shows one example.

   Table 13-75:

PTYPE	Property Name	END A	END B	i-th Station
PBEAM	A	A or A(A)	A(B)	A(i)

   Only stations that are input on a PBEAM entry can be referenced by a DVPREL2. For example, using an END B property name on a DVPREL2 entry when the referenced PBEAM is a constant section is not allowed.
6. PWELD and PFAST property types are not supported.
7. PBARL and PBEAML property types are not supported.
8. The PPLANE property type is not supported.
9. FREQ (Field 9) is applicable only for TYPE=PBUSHT/PDAMPT/PELAST type properties. A FREQ value must currently match an analysis frequency, though there is no need to match a TABLED1 entry. Properties should be designed at a minimum of two frequencies for a TABLED1 referenced by a

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designed frequency dependent property PNAME/FID, because these tables are currently re-generated, not added to.

10. The PMIN and PMAX values may be violated at a given design cycle if there is other, more critical violation by response constraints. Since design variables may never violate their lower and upper bounds, it may be best, if possible and reasonable, to configure the property bounds accordingly, so that they are not violated even for infeasible designs.

Related Topics

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DVSHAP

Design Variable to Basis Vector(s)

Defines a shape basis vector by relating a design variable identification number (DVID) to columns of a displacement matrix.

Format:

1	2	3	4	5	6	7	8	9	10
DVSHAP	DVID	COL1	SF1	COL2	SF2	COL3	SF3		

Example:

DVSHAP	2	1	2.0	4	1.0				

Fields:

Field	Contents
DVID	Design variable identification number on the DESVAR entry. (Integer>0)
COLi	Column number of the displacement matrix. See Remark 2 . (1 ≤ Integer ≤ maximum column number in the displacement matrix.)
SFi	Scaling factor applied to the COLi-th column of the displacement matrix. (Real; Default = 1.0)

Remarks:

1. DVID must be defined on a DESVAR entry.
2. COLi must be a valid column number in the displacement matrix.
3. Multiple references to the same DVID and/or COLi will result in a linear combination of displacement vectors. In the example above, the shape basis vector is a linear combination of the fourth column and twice the first column.
4. The displacement matrix must have been created by Simcenter Nastran and be available on a database, which is attached via the DBLOCATE FMS statement shown below:

```
ASSIGN    DISPMAT=’ physical filename of MASTER DBset ’
DBLOCATE    DATABLK=(UG/UGD,GEOM1/GEOM1D,GEOM2/GEOM2D) ,
            LOGICAL=DISPMAT
```


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Related Topics

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DVTREL1

Element Selection for Topology Optimization

Selects the elements for the software to automatically create design variables and other related data for topology optimization.

Format:

1	2	3	4	5	6	7	8	9	10
DVTREL1	ID	LABEL	GRPID	STATE	DSVFLG				
	DVID1								

Example:

DVTREL1	200		24						
	30000								

Fields:

Field	Contents
ID	Identification number of DVTREL1 entry. (Integer>0)
LABEL	Optional user-defined label. (Character)
GRPID	ID of GROUP bulk entry that specifies a list of elements. The software associates the following types of elements with topology optimization: The 3D solid elements CHEXA, CPENTA, CPYRAM, CTETRA. and the shell elements CTRIA3, CTRIA6. CTRIAR, CQUAD4, CQUAD8, CQUADR. Any other element types or IDs for non-existent elements listed on a GROUP entry are ignored. (Integer>0)
STATE	Specifies that the elements referenced in the GRPID field are either ACTIVE or FROZEN. The software modifies the active elements, but not the frozen elements. (Character; Default =ACTIVE). See Remark 1 for examples demonstrating how the STATE field is used.
DSVFLG	Flag set by the software when you are recycling design variables from a previous solution for restart purposes. (Integer=0 or 1; Default=0) DSVFLG=0 or blank (Default): Value used in a first pass optimization solution. This value triggers the software to create new design variables and other relevant data for the elements designated as ACTIVE.

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Field	Contents
	DSVFLG=1: Value used in a restart solution. This value triggers the software to use already existing design variables and their values created in a previous solution. Normally, the software sets this value of 1 for DSVFLG automatically when writing the updated bulk data into the punch file. See Remark 2.
DVID1	Option to control the identification numbers for the auto-generated design variables. (Integer≥0; Default=0) If DVID1=0 or blank (Default), the design variable IDs will be equal to the IDs of the associated elements plus an automatically calculated offset to ensure no conflicts occur with any other design variables. If DVID1>0, see Remark 3 for the description.

Remarks:

1. The STATE field allows you to optionally specify that the elements referenced in the GROUP field are either ACTIVE or FROZEN. The software creates a design variable for each active element, but not for frozen elements. The STATE field default is ACTIVE. The following examples demonstrate how the STATE field is used.

   - If you define one or multiple DVTREL1 entries and FROZEN is defined in the STATE field for all DVTREL1 entries, the topology optimizable elements that are not referenced by a DVTREL1 entry are active. This scenario is useful when you are designating most of your model as active except for a few elements. You only need to reference the few frozen elements on a DVTREL1 entry.
   - If you define one or multiple DVTREL1 entries and all use the default for the STATE field (ACTIVE), only those elements designated as ACTIVE are active. Elements that are not referenced by a DVTREL1 entry are frozen.
   - If you define multiple DVTREL1 entries and some use the default for the STATE field (ACTIVE) and some have FROZEN defined, the elements designated as ACTIVE are active. An exception to this is when an element is referenced by multiple DVTREL1 entries and is designated as both FROZEN and as ACTIVE. Such elements are frozen. All other elements designated as FROZEN along with any elements that are not selected in any DVTREL1 referenced GROUP data are also frozen.

   Note that other elements that are not 3D solids or shell elements can be included in the model and are included in the analysis. The software simply does not create design data for these elements, so they remain constant during the topology optimization process.

2. You can optionally recycle design variable values from a previous solution to use them in a restart. The procedure is as follows.

   - In the original solution, your DVTREL1 entries will use the default value DSVFLG=0 and the software auto-generates new design variables with initial values of 1.0. Also, in your input file

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for the original optimization solution, you will request a punch file output containing the bulk data for the best design cycle, using the following command in case control:

```
ECHO=SORT, PUNCH(BSTBULK)
```

At the end of the original solution, the punch file that the software creates will include an entire BEGIN BULK section which includes everything from your original BEGIN BULK section but with some changes and additions. Specifically, your DVTREL1 entries in the punch file will now have DSVFLG=1 defined, and the punch file will include the explicit design variable values corresponding to the best design cycle found in the original solution.

- For the restart solution, you will create a new input file which uses the BEGIN BULK contents from the punch file created in original solution.
- If you wish to restart from the last design cycle instead of the best one, you will need to use the following ECHO case control command in your original input file:

```
ECHO=SORT,PUNCH(NEWBULK)
```

When the last design cycle is the best one, this selection does not matter. However, it is not possible to know this until the original solution is complete.

3. The DVID1 field allows you to optionally control the identification numbers for the auto-generated design variables. The software auto-generates a DESVAR and other various data for each element that is active. The software uses the DVID1 field differently for the following STATE field scenarios.
   - When FROZEN is defined in the STATE field for all DVTREL1 entries, the design variable ID numbering for the active elements begins with the largest DVID1 found among all DVTREL1 entries. If none of the DVTREL1 entries have the DVID1 field defined, the element IDs of the active elements become the design variable IDs with an automatic offset applied if a conflict occurs.
   - When ACTIVE is defined in the STATE field for a DVTREL1 entry, the design variable ID numbering for the elements referenced by that DVTREL1 entry begins with the value defined in the DVID1 field. The numbering then continues incrementally for the remainder of the active elements referenced by that DVTREL1 entry. If DVID1 is undefined, the element IDs of the active elements become the design variable IDs with an automatic offset applied if a conflict occurs.

Related Topics

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ECHOOFF

Deactivate Printed Echo

Marks the point or points in the input file to deactivate printed echo of the Bulk Data.

Format:

1	2	3	4	5	6	7	8	9	10
ECHOOFF									

Example:

ECHOOFF									

Remarks:

1. This entry may also be used in the Executive Control and Case Control Sections. It is described in the **"Case Control Commands"**.
2. The companion to this entry is the ECHOON entry.

Related Topics

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ECHOON

Activate Printed Echo

Marks the point or points in the input file to activate printed echo of the Bulk Data.

Format:

1	2	3	4	5	6	7	8	9	10
ECHOON									

Example:

ECHOON									

Remarks:

1. This entry may also be used in the Executive Control and Case Control Sections. It is described in "Case Control Commands".
2. The companion to this entry is the ECHOOFF entry.

Related Topics

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EIGB

Buckling Analysis Set

Defines data needed to perform buckling analysis.

Format:

1	2	3	4	5	6	7	8	9	10
EIGB	SID	METHOD	L1	L2	ND	NDP	NDN		
	NORM	G	C						

Example:

EIGB	13	IRAM			10				
	MAX								

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
METHOD	Eigenvalue extraction method. See Remark 7. (Character: "IRAM", "SUBS", “SINV”; No default)
	IRAM Implicitly restarted Arnoldi method
	SUBS Subspace iteration method
	SINV Enhanced inverse power method
L1	Lower bound of eigenvalue extraction range. Valid when METHOD = "SINV". (Real with L1 < L2; No default)
L2	Upper bound of eigenvalue extraction range. Valid when METHOD = "SINV". (Real ≥ 0.0 with L1 < L2; No default and must be defined for METHOD = "SINV")
ND	Number of desired eigenvalues. Valid when METHOD = “IRAM” or “SUBS”. See Remark 8. (Integer > 0; No default)
NDP	Number of desired positive eigenvalues. Valid when METHOD = "SINV". (Integer > 0; No default)
NDN	Number of desired negative eigenvalues. Valid when METHOD = "SINV". (Integer > 0; No default)
NORM	Eigenvector normalization method. See Remark related to SOL 402 1. (Character: “MAX” or “POINT”; Default = “MAX”)

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Field	Contents	
	MAX	Normalize eigenvectors by scaling the largest component in the analysis set to unity.
	POINT	Normalize eigenvectors by scaling the component defined in the G and C fields to unity. If the value for the component defined in the G and C fields is zero, it cannot be scaled and NORM defaults to MAX.
G	Grid or scalar point identification number. Used when NORM = "POINT". (Integer > 0; No default)	
C	Component number. Used when NORM = "POINT" and G is a geometric grid point. (1 ≤ Integer ≤ 6; No default)	

Remarks:

1. The EIGB entry must be selected by a METHOD case control command that references the SID of the EIGB entry.
2. Each eigenvalue is the factor by which the pre-buckled state of stress is multiplied to produce the buckled shape described by the corresponding eigenvector.
3. The continuation entry is optional. If the continuation is not specified, then NORM = "MAX" normalization is performed.
4. When NORM = "MAX", components that are not in the analysis set may have values larger than unity.
5. The SINV method uses Sturm sequence techniques to ensure that all eigenvalues in the range have been found.
6. The SINV method uses a convergence tolerance is 10^{-6}.

   See the *Simcenter Nastran Theoretical Manual* for a discussion of convergence criteria.
7. In addition to using an EIGB bulk entry to specify the eigenmethod, you can also use the EIGRL bulk entry to specify the Lanczos method as the eigenmethod. When you want to calculate many eigenvalues, which is not typical in a linear buckling analysis because only the lowest buckling load is of any practical interest, the Lanczos method offers better computational performance than the other eigenmethods. However, the Lanczos and enhanced inverse power methods may not calculate negative eigenvalues. Thus, they are not as reliable as the implicitly restarted Arnoldi and subspace iteration methods. Therefore, as a best practice, use the implicitly restarted Arnoldi or subspace iteration method.

   For additional information about the eigenmethods applicable to a SOL 105 linear buckling solution, see Eigenvalue Extraction Methods in the *Simcenter Nastran User's Guide*.
8. When METHOD = "IRAM" or "SUBS", the software calculates the ND lowest magnitude positive and negative eigenvalues.

Remarks related to SOL 402:

1. Only NORM = "MAX" is supported.

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Related Topics

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EIGC

Complex Eigenvalue Extraction Data

Defines data needed to perform complex eigenvalue analysis.

Format:

1	2	3	4	5	6	7	8	9	10
EIGC	SID	METHOD				EPS	ND0		

Continuation lines are used to define search regions or shift points for the Inverse Power (INV) method, Complex Lanczos (CLAN) methods, Iterative Schur-Rayleigh-Ritz (ISRR) method, and ARPACK (ARPK) method. A maximum of 10 continuation lines can be specified except for the ARPACK (ARPK) method, which allows for a single continuation line only. Continuation lines are not recommended or required for the Hessenberg (HESS) methods.

Continuation Line Format for Inverse Power Method:

1	2	3	4	5	6	7	8	9	10
	ALPHAAj	OMEGAAj	ALPHABj	OMEGABj	Lj	NEj	NDj		

Continuation Line Format for Complex Lanczos Methods:

1	2	3	4	5	6	7	8	9	10
	SHIFTRj	SHIFTIj	MBLKSZj	IBLKSZj	KSTEPSj		NDj		

Continuation Line Format for Iterative Schur-Rayleigh-Ritz Method:

1	2	3	4	5	6	7	8	9	10
	SHIFTR1	SHIFTI1				ISRRFLG	ND1		

Continuation Line Format for ARPACK Method:

1	2	3	4	5	6	7	8	9	10
	SHIFTR	SHIFTI		NCV	MXITER		ND1		

Examples:

EIGC	14	HESS					6		

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EIGC	15	INV							
			3.0	4.0	2.5	10			

EIGC	16	CLAN					20		

EIGC	17	CLAN							
		+5.6					4		
		-5.5					3		

EIGC	18	ISRR							
	-1.0	0.0				1	10		

EIGC	19	ARPK							
	-1.0	1.0					20		

Fields:

Field	Contents	
SID	Set identification number. (Unique Integer>0)	
METHOD	Method of complex eigenvalue extraction. (Character: “HESS”, “INV”, “CLAN”, “ISRR”, or "ARPK"; No default)	
	HESS	Hessenberg methods. See Remarks 1 and 5.
	INV	Inverse Power method. See Remark 1 and Remark 4.
	CLAN	Complex Lanczos methods. See Remarks 1, 1, and 5.
	ISRR	Iterative Schur-Rayleigh-Ritz method. See Remark 1.
	ARPK	ARPACK method. See Remark 1.
EPS	Convergence tolerance. (Real ≥ 0.0)	
	Default values are:	
	= 10^{-15} if METHOD = HESS	
	= 10^{-4} if METHOD = INV	
	= 10^{-8} if METHOD = CLAN	
	= 10^{-8} if METHOD = ISRR	
	Value set by software if METHOD = ARPK	

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Field	Contents
	For SOLs 402 and 414, see Remark 7.
ND0	Number of eigenvalues and/or eigenvectors desired. See Remarks 3 and 6. (Integer > 0 or blank; No default) See also the EIGRL bulk entry for more information on the allowed combinations of the V1,V2 frequency range and this ND field.

Continuation Line Fields for Inverse Power Method:

Each continuation line defines a rectangular search region in the complex plane. Each rectangular search region is defined from end points A and B, and width L as indicated in Figure 13-126. Search regions can overlap. Eigenvalues in overlapping search regions will not be extracted more than once.

Field	Contents
ALPHAAj	Real part of end point A of the j^{th} search region in rad/time. (Real; Default = 0.0)
OMEGAAj	Imaginary part of end point A of the j^{th} search region in rad/time. (Real; Default = 0.0)
ALPHABj	Real part of end point B of the j^{th} search region in rad/time. (Real; Default = 0.0)
OMEGABj	Imaginary part of end point B of the j^{th} search region in rad/time. (Real; Default = 0.0)
Lj	Width of the j^{th} search region in rad/time. (Real; Default = 1.0)
NEj	Estimated number of eigenvalues in the j^{th} search region. (Integer > -1; Default = 0)
NDj	Desired number of eigenvalues and eigenvectors to extract from the j^{th} search region. See Remark 3. (Integer > 0; Default = 3 x NEj)

Continuation Line Fields for Complex Lanczos Method:

Each continuation line specifies a shift point and the number of eigenvalues and eigenvectors to compute from the shift. For Single Vector Complex Lanczos method the MBLKSZj and IBLKSZj fields are ignored.

Field	Contents
SHIFTRj	Real part of the j^{th} shift point in rad/time. (Real; Default = 0.0)
SHIFTIj	Imaginary part of the j^{th} shift point in rad/time. (Real; Default = 0.0)
MBLKSZj	Maximum block size for the j^{th} shift point in rad/time. (Integer > 0; Default = 7)
IBLKSZj	Initial block size for the j^{th} shift point in rad/time. (Integer > 0) Default values are: = 1 if N < 1000 = 2 if 1000 ≤ N < 50,000 = 4 if 50,000 ≤ N < 100,000

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Field	Contents
	= 5 if N ≥ 100,000 where N is the linearized problem size. See Remark 10.
KSTEPSj	Frequency of solve for the j^{th} shift point in rad/time. (Integer > 0; Default = 5)
NDj	Desired number of eigenvalues and eigenvectors to extract from the j^{th} shift point. See Remark 3. (Integer > 0; No default)

Continuation Line Fields for Iterative Schur-Rayleigh-Ritz Method:

A single continuation line is used to specify a shift point that the software uses to avoid singularity during decomposition and the number of eigenvectors to compute from the shift point.

Field	Contents	
SHIFTR1	Real part of the shift point in rad/time if ISRRFLG = 1. Blank otherwise. (Real or blank)	
SHIFTI1	Imaginary part of the shift point in rad/time if ISRRFLG = 1. Blank otherwise. (Real or blank)	
ISRRFLG	ISRR method instruction flag. ISRRFLG values may be added to obtain a combination of settings. For example ISRRFLG = 323 would indicate options “1”, “2” and a maximum subspace of 10 vectors (1 + 2 + 10*32 = 323). (Integer)	
	0 (Default)	All of the options below are disabled.
	1	Reserves fields 2 (SHIFTR1) and 3 (SHIFTI1) for shift point specification. The shift point does not redefine the search region, but is only used during decomposition to avoid a singularity. Specifying a shift point is recommended to obtain better performance.
	2	Forces the out-of-core path in the code.
	4	Overrides system cell 405.
	8	Forces balanced iteration for real unsymmetric problems only.
	16	Forces generation of starting vectors for quadratic problems from the values found for the linear case when damping is ignored. In all other cases, the starting vectors are randomly generated.
	M * 32	Forces the maximum size of the subspace to M vectors.
ND1	Desired number of eigenvectors to extract. (Integer > 0; No default)	

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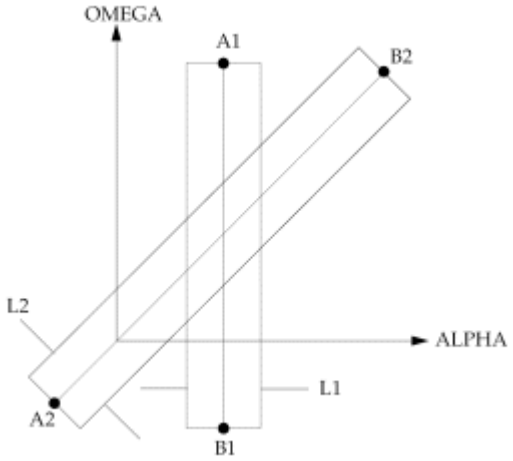
Figure 13-126: Sample Search Regions for Inverse Power Method



Continuation Line Fields for ARPACK Method:

The continuation line specifies a shift point and the number of eigenvalues and eigenvectors to compute from the shift. One continuation line is allowed.

Field	Contents
SHIFTR	Real part of the shift point in rad/time. (Real; Default = 1.0)
SHIFTI	Imaginary part of the shift point in rad/time. (Real; Default = 0.0)
NCV	Maximum number of basis vectors. (Integer; See Remark 11 for the default behavior)
MXITER	Maximum number of Arnoldi update iterations. (Integer; Default = 30)
ND1	Desired number of eigenvalues and eigenvectors to extract from the shift point. (Integer > 0; No default)

Remarks:

1. A summary of the supported complex eigenvalue methods is provide in the following table.

Method	Description and Applicability
Hessenberg	The Hessenberg methods are generally reliable and economical for small to moderate-size problems. The Hessenberg methods compute all eigenvalues and ND0 eigenvectors. The QZ Hessenberg method is used by default. The applicability of the QR Hessenberg and LR Hessenberg methods has been superseded by the QZ Hessenberg method. However, the QR Hessenberg and LR Hessenberg methods can still be selected with system cell 108. See Remark 2.

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Method	Description and Applicability
	This method is not supported in SOL 402.
Inverse Power	The applicability of the Inverse Power method has been superseded by the Block Complex Lanczos method. This method is not supported in SOLs 402 and 414.
Complex Lanczos	The Block Complex Lanczos method is more reliable and will not accept inaccurate eigenvalues, which the Single Vector Complex Lanczos method has a tendency to do. Given the same input, the Block Complex Lanczos method may often accept fewer eigenvalues. The Block Complex Lanczos method is used by default. The applicability of the Single Vector Complex Lanczos method has been superseded by the Block Complex Lanczos method. However, the Single Vector Complex Lanczos method can still be selected with system cell 108. See Remark 2 and Remark 2.
ISRR	The Iterative Schur-Rayleigh-Ritz method works well on sparse matrices, confines the search region to a circle centered on the origin of the complex plane, and provides some reliability that all modes within the circle have been found. This method is not supported in SOLs 402 and 414.
ARPACK	The ARPACK method is an implementation of the Arnoldi method that is effective at calculating clustered eigenvalues and eigenvectors for large sparse or structured matrices. The use of system cell 108 is not supported when METHOD = ARPK. This method is not supported in SOLs 402 and 414.

2. The EIGC entry must be selected in the case control section with the command CMETHOD = SID. The complex eigenvalue method that is used depends on the METHOD field of the EIGC entry, but, optionally, can be overridden by specifying system cell 108.

SYSTEM(108)	Specification(1)
0 (Default)	Use QZ Hessenberg method if METHOD = HESS, Inverse Power method if METHOD = INV, Block Complex Lanczos method if METHOD = CLAN, or Iterative Schur-Rayleigh-Ritz method if METHOD = ISRR
1	Force QR Hessenberg method (with spill, mass matrix must be non-singular) (2)
2	Force Single Vector Complex Lanczos method(3)
4	Force Block Complex Lanczos method
8	Debugging output for Lanczos methods
16	Turn off block reduction in Block Complex Lanczos method
32	Turn off block augmentation in Block Complex Lanczos method

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SYSTEM(108)	Specification[1]
64	Turn off full orthogonality in Block Complex Lanczos method
128	Turn off preprocessing of initial vectors in Block Complex Lanczos method
256	Force LR Hessenberg method (without spill, mass matrix must be non-singular)[2]
512	Force QZ Hessenberg method
65536	Use semi-algebraic sort on imaginary part of roots

[1] In this table, "force" implies that the method selected by the system cell will be used even if another method has been selected on an EIGC entry. Sums of these values will produce two or more actions at once, when feasible. As the numbers get larger, the function is more developer-oriented than user-oriented.
[2] The applicability of this method has been superseded by the QZ Hessenberg method.
[3] The applicability of this method has been superseded by the Block Complex Lanczos method.

System cell 108 is ignored in SOL 402

3. For all methods, if no continuation lines are present, then ND0 must be specified and the software will calculate ND0 eigenvalues and eigenvectors near the origin of the complex plane. If continuation lines are present, the ND0 field must be left blank.

   Related method-specific considerations for METHOD = HESS and METHOD = ISRR are as follows:

   - If METHOD = HESS and continuation lines are present, the value specified in the ND1 field is the only value on the continuation lines that is used. All other values specified on the continuation lines are ignored.
   - If METHOD = ISRR, only the value in the ND1 field will be used.

4. See the *Simcenter Nastran Theoretical Manual* for a discussion of convergence criteria and the search procedure for the Inverse Power method.

5. DIAG 12 prints diagnostics for the Inverse Power, Block Complex Lanczos, Single Vector Complex Lanczos, QZ Hessenberg, QR Hessenberg, LR Hessenberg, and ISRR methods.

6. The normalized eigenvectors may be output with the SDISPLACEMENT and DISPLACEMENT case control commands.

7. The eigenvectors are normalized by making the component with the largest magnitude a unit value for the real part and a zero value for the imaginary part. All eigenvectors are then scaled accordingly.

8. The SVD method is provided for DMAP applications. If used in SOL 107 or 110, and mass or damping terms are present, a user fatal error occurs. The SVD operation decomposes the input stiffness matrix K into the factors U, S, and V. U and V are collections of vectors of orthogonal functions. S is a rectangular matrix with terms on the diagonal of its left partition. The factors satisfy the equation K = USV', where V' is the complex conjugate transpose of V. The ND1 value has a meaning for the SVD functions which differs from the eigensolution.

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ND1	OUTPUT
> 0	All vectors of U and V are output.
= 0	U and V are returned in a purged state.
< 0	S is returned as a square matrix whose number of columns is equal to the minimum number of rows or columns of the input matrix. U and V are truncated to be conformable with S. This reduces the computational effort required to solve rectangular input matrices by providing a partial solution for the most interesting vectors.

9. For DMAP applications there are input parameters, not present in the solution sequences, that may be used to replace the function of the EIGC and CMETHOD entries.
10. The linearized problem size is either twice the dimension of K, M, and B if all three are present, or equal to the dimension of K, M, or B if only two of the three are present.
11. For the ARPACK method, the default maximum number of basis vectors is the desired number of modes plus 20.
12. A fatal error occurs if both the ND0 field and ND1 field are specified.

Remarks related to SOLs 402 and 414:

1. (SOL 402) If EIGC is used alone (without an EIGR or EIGRL bulk entry), a direct complex value extraction method is used. If EIGC is used in conjunction with EIGR/EIGRL, a modal complex value extraction method is used. With this modal method, a symmetrical real undamped eigenvalue problem is first solved in order to generate a real modal basis. The matrices are then projected in this basis and the reduced real or complex eigenvalue problem is solved by the QR algorithm. The number of real eigenvalues specified on the EIGR/EIGRL bulk entry should be larger than (or equal to) the number of complex eigenvalues specified on the EIGC bulk entry.
2. (SOL 402) This method uses a Single Vector Complex Lanczos, and system cell 108 is ignored.
3. (SOL 402) NDJ of the continuation card will overwrite ND0.
4. (SOLs 402 and 414) Only one continuation line is allowed. If more lines are detected, a fatal message will occur.
5. (SOL 414) For METHOD, you can select HESS or CLAN and the results will be the same. HESS is recommended for smaller models, and CLAN is recommended for larger models.
6. (SOL 414) If you set IA12 on the NLCNTL2 bulk entry and you omit a value for ND0 (number of eigenvalues and/or eigenvectors) on the EIGC bulk entry, the software uses 1. A warning message is written to the .f06 file if system cell 675 is set to a value greater than 1.
7. (SOLs 402 and 414) Use only the default for the EPS convergence tolerance. SOLs 402 and 414 cannot use any other value.

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Alternate Format for Complex Lanczos Methods:

Each continuation line specifies a shift point and the number of eigenvalues and eigenvectors to compute from the shift. For Single Vector Complex Lanczos method the MBLKSZj and IBLKSZj fields are ignored.

1	2	3	4	5	6	7	8	9	10
EIGC	SID	METHOD				EPS	ND0		
	KEYWORD1=<value>, KEYWORD2=<value>, KEYWORD3=<value>, etc.								

Allowable keywords include:

Keyword	Contents
METHOD	Method of complex eigenvalue extraction. See Alternate Format Remark 2. (Character: "CLAN"; No default)
EPS	Convergence tolerance. See Alternate Format Remark 2. (Real ≥ 0.0; Default = 10^{-8})
SHIFTRj	Real part of the j^{th} shift point in rad/time.(Real; Default = 0.0)
SHIFTIj	Imaginary part of the j^{th} shift point in rad/time. (Real; Default = 0.0)
MBLKSZj	Maximum block size for the j^{th} shift point in rad/time. (Integer > 0; Default = 7)
IBLKSZj	Initial block size for the j^{th} shift point in rad/time. (Integer > 0) Default values are: = 1 if N < 1000 = 2 if 1000 ≤ N < 50,000 = 4 if 50,000 ≤ N < 100,000 = 5 if N ≥ 100,000 where N is the linearized problem size.
KSTEPSj	Frequency of solve for the j^{th} shift point in rad/time. (Integer > 0; Default = 5)
NDj	Desired number of eigenvalues and eigenvectors to extract from the j^{th} shift point. See Alternate Format Remark 2. (Integer > 0; No default)

Alternate Format Examples:

EIGC	1	CLAN							
	SHIFTR1=0.0, SHIFTI1=20.0, MBLKSZ1=5, IBLKSZ1=2, ND1=12, EPS=1.E-12								

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EIGC	3	CLAN							
	SHIFTR1=5.0, SHIFTI1=10.0, ND1=5								
	SHIFTR2=0.0, SHIFTI2=20.0, ND2=8, MBLKSZ2=5, IBLKSZ2=4								

Alternate Format Remarks:

1. The first field of the keyword-driven continuation entry must be blank.
2. If METHOD is specified on the continuation entry, the METHOD field on the original entry must be left blank. If EPS is specified on the continuation entry, the EPS field on the original entry must be left blank. If ND1 is specified on the continuation entry, the ND0 field on the original entry must be left blank.

Related Topics

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EIGP

Poles in Complex Plane

Defines poles that are used in complex eigenvalue extraction by the Determinant method.

Format:

1	2	3	4	5	6	7	8	9	10
EIGP	SID	ALPHA1	OMEGA1	M1	ALPHA2	OMEGA2	M2		

Example:

EIGP	15	-5.2	0.0	2	6.3	5.5	3		

Fields:

Field	Contents
SID	Set identification number. (Integer>0)
ALPHAi, OMEGAi	Coordinates of point in complex plane. (Real)
Mi	Multiplicity of complex root at pole defined by point at ALPHAi and OMEGAi. (Integer>0)

Remarks:

1. The EIGP entry defines poles in the complex plane that are used with an associated EIGC entry having the same set number.
2. The units of ALPHAi and OMEGAi are radians per unit time.
3. Poles are used only in the determinant method. (METHOD = “DET” on the EIGC entry).
4. One or two poles may be defined on a single entry.
5. See *The NASTRAN Theoretical Manual*.

Related Topics

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EIGR

Real Eigenvalue Extraction Data

Defines data needed to perform real eigenvalue analysis.

Format:

1	2	3	4	5	6	7	8	9	10
EIGR	SID	METHOD	F1	F2	NE	ND			
	NORM	G	C						

Example:

EIGR	13	LAN				12			

Fields:

Field	Contents	
SID	Set identification number. (Unique Integer > 0)	
METHOD	Method of eigenvalue extraction.(Character: "LAN", "AHOU", "SUBS", "INV", SINV", "GIV", "MGIV", "AGIV", "HOU", or "MHOU"; No default)	
	Modern methods:	
	LAN	Lanczos method
	AHOU	Automatic selection of HOU or MHOU method. See Remark 14.
	SUBS	Subspace iteration method
	Obsolete Methods:	
	INV	Inverse Power method.
	SINV	Inverse Power method with enhancements.
	GIV	Givens method of tridiagonalization.
	MGIV	Modified Givens method.
	AGIV	Automatic selection of GIV or MGIV method. See Remark 14.
	HOU	Householder method of tridiagonalization.
	MHOU	Modified Householder method.
F1, F2, NE, ND	See F1, F2, NE, and ND Options .	

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Field	Contents	
NORM	Method for normalizing eigenvectors. (Character: “MASS,” “MAX,” “AF”, or “POINT”; Default = “MASS”)	
	MASS	Normalize to unit value of the generalized mass. (Default)
	MAX	Normalize to unit value of the largest component in the analysis set.
	AF	Amplitude-Frequency normalization method. See Remark 16. Supported when METHOD = LAN or AHOU.
	POINT	Normalize to a positive or negative unit value of the degree-of-freedom defined by the G and C fields. POINT is not supported for METH = LAN.
G	Grid or scalar point identification number. Required entry when NORM = POINT only. (Integer > 0; See Remark 17 for default behavior)	
C	Component number. Required entry when NORM = POINT and G is a geometric grid point. (1 ≤ Integer ≤ 6; See Remark 17 for default behavior)	

F1, F2, NE, and ND Options:

Field	LAN	SUBS	INV or SINV	GIV, MGIV, HOU, or MHOU
F1, F2	See F1, F2, and ND Options for METHOD = LAN or SUBS	Same as LAN method. See F1, F2, and ND Options for METHOD = LAN or SUBS	Frequency range of interest. F1 must be input. For METHOD = SINV, if ND is blank, then F2 must be input. (Real ≥ 0.0)	Frequency range of interest. Eigenvectors are found with natural frequencies that lie in the range between F1 and F2. If ND is not blank, at most ND eigenvectors are found. (Real ≥ 0.0; F1 < F2)
NE	Not used	Block size. (Integer > 0; See Remark 22 for default behavior)	Estimate of number of roots in range. Required for METHOD = INV. Not used by METHOD = SINV. (Integer > 0)	Not used
ND	See F1, F2, and ND Options for	See F1, F2, and ND Options for METHOD = LAN or SUBS	Desired number of roots. If this field is blank and METHOD = SINV,	Desired number of eigenvectors. If ND is zero, the number of eigenvectors is

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Field	LAN	SUBS	INV or SINV	GIV, MGIV, HOU, or MHOU
	METHOD = LAN or SUBS		then all roots between F1 and F2 are searched with a limit of 600 roots. (Integer > 0, Default is 3 ≠ NE for METHOD = INV only)	determined from F1 and F2. If all three are blank, then ND is automatically set to one more than the number of degrees-of-freedom listed on SUPORTi entries. (Integer ≥ 0; Default = 0)

F1, F2, and ND Options for METHOD = LAN or SUBS:

F1	F2	ND	Number and Type of Roots Found
F1	F2	ND	Lowest ND or all in range, whichever is smaller
F1	F2	Blank	All in range
F1	Blank	ND	Lowest ND in range [F1,+∞]
F1	Blank	Blank	Lowest in range [F1,+∞]
Blank	Blank	ND	Lowest ND in [-∞,+∞]
Blank	Blank	Blank	Lowest root
Blank	F2	ND	Lowest ND below F2
Blank	F2	Blank	All below F2

Remarks:

1. The EIGR entry must be selected with the Case Control command METHOD = SID.
2. See Eigenvalue Extraction Methods in the *Simcenter Nastran Basic Dynamic Analysis User's Guide* for a discussion of method selection.
3. The units of F1 and F2 are cycles per unit time.
4. The continuation entry is optional. If the continuation entry is not specified, then mass normalization is performed.

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5. The contemporary methods are LAN, AHOU, and SUBS. The other methods are in a maintenance-only status, with no enhancements planned for them. They may be eliminated in a future release.

6. The LAN method is the most general-purpose method, and may be used on both small- and large-size problems. It takes advantage of sparsity of input matrices, leading to greater efficiency on large-size problems. Because Lanczos performance is tuned for medium to large problems, this has caused difficulties with very small problems. Thus, by default, on problems with fewer than 20 degrees-of-freedom when the LAN method is selected, the method is switched to AHOU. The criteria for automatic switching is controlled by SYSTEM(359) on the NASTRAN entry. The NE, G, and C fields are ignored for the LAN method. The EIGRL entry is an alternative means to select the LAN method. It has several other input options for special cases. When both and EIGRL and EIGR have the same SID, and that SID is selected by a METHOD command, the EIGRL entry takes precedence.

7. The SUBS method is a good alternative to the LAN method when a relatively small number of modes (< 50) are to be extracted. For such cases, SUBS requires less computational effort than LAN.

8. The AHOU method is competitive with the LAN method when there are small, dense matrices and many eigenvectors are required. This most commonly occurs when static or dynamic reduction is performed. The AHOU method does not take advantage of matrix sparsity, so that computation cost rises with the cube of the number of degrees-of-freedom. The AHOU method responds to all permitted values for all the other fields except NE, which is ignored.

9. All methods require a positive semi-definite (PSD) mass matrix for stable solutions. The mass matrix may be tested for this condition for all methods of solution by setting SYSTEM(303). A value of "-4" should be sufficient to identify problem matrices. A fatal error exit is taken when it is not met. All Simcenter Nastran metric elements are designed to produce PSD mass matrices. CMASSi elements, DMIG matrices selected by the M2GG command, and matrices input via INPUTT4 are special methods that allow addition of non-PSD terms by use of non-metric element input. If none of this type of special input is present and the fatal error exit is taken you may have encountered an error in a metric element. Contact a Customer Support representative for corrective action in this case.

10. The LAN and AHOU methods allow singular, but positive semi-definite (PSD) mass matrices.

11. The tridiagonal methods include the xGIV and xHOU methods, where "x" is described in the following comments. All tridiagonal methods compute all eigenvalues, and the number of eigenvectors specified by the Fi and ND fields, as described in the above table.

12. If "x" is blank (for example, the HOU method is selected) the mass matrix must be non-singular.

13. If "x" is M (for example, the MHOU method is selected) the mass matrix may be singular. A modified, shifted problem is solved in an inverse basis with this method. Some precision in the solution and longer computation time is exchanged for a more stable solution.

14. If "x" is A (for example, the AHOU method is selected) an attempt is made to solve the problem without shifting, in the interest of cost reduction and maximum precision. If the mass matrix is determined to be poorly conditioned for inversion the basis is automatically shifted with the modified method.

15. If NORM = MAX, components that are not in the analysis set may have values larger than unity.

16. In AF (amplitude-frequency) normalization, the normalized eigenvectors are given by:

    $(C / \omega \Delta_{max}) \{x\}$

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where C is an optional user-defined scale factor, ω is the natural frequency for the mode in Hz, Δ_{max} is the magnitude of the maximum grid point translation for the mode, and $\{x\}$ is the un-normalized eigenvector for the mode. The optional user-defined scale factor, C, is specified with the AFNORM parameter.

17. If NORM = POINT, the selected degree-of-freedom should be in the analysis set (a-set). If the selected degree-of-freedom is not in the analysis set, the software uses NORM = MAX. The displacement value at the selected degree-of-freedom is positive or negative unity. If either or both of the G and C fields are blank when NORM = POINT, a fatal error occurs.
18. The SINV method is an enhanced version of the INV method. It uses Sturm sequence number techniques to make it more likely that all roots in the range have been found. It is generally more reliable and more efficient than the INV method.
19. For the INV and SINV methods, convergence is achieved at 10^{-6}. Convergence is tested by other criteria for the other methods.
20. For the SINV method only, if F2 is blank, the first shift will be made at F1, and only one eigensolution above F1 is calculated. If there are no modes below F1, it is likely that the first mode will be calculated. If there are modes below F1 (including rigid body modes defined by SUPORT entries), a mode higher than the first mode above F1 may be calculated.
21. When F1, F2, and ND are all zero or blank, ND is reset to 1. A User Warning Message is produced for this condition, which is interpreted as likely to be due to an inadvertent omission by the user.
22. The default block size is the maximum of the following values:
    - Twice the value specified in the ND field.
    - The value in the ND field plus 30.

    If the maximum of these values exceeds the number of degrees-of-freedom in the analysis set (a-set), the software uses the number of degrees-of-freedom in the a-set as the default block size.
23. When you use the defined mass method to perform a dynamic analysis of a structure that contains an incompressible fluid, use the INCMOPT bulk entry to select the subspace iteration eigenvalue extraction method, rather than using a METHOD case control command that references an EIGR bulk entry where METHOD = SUBS. For more information, see Defined Fluid Mass in the *Simcenter Nastran Advanced Dynamic Analysis User's Guide*.

Remarks related to SOLs 402 and 414:

1. Only METHOD=LAN is supported.
2. NE is ignored.

Related Topics

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EIGRL

Real Eigenvalue Extraction Data, Lanczos Method

Defines data needed to perform real eigenvalue (vibration or buckling) analysis with the Lanczos method.

Format:

1	2	3	4	5	6	7	8	9	10
EIGRL	SID	V1	V2	ND	MSGLVL	MAXSET	SHFSCL	NORM	
	option_1 = value_1 option_2 = value_2, etc.								

Example:

EIGRL	1	0.1	3.2	10					
	NORM=MAX NUMS=2								

Fields:

Field	Contents	
SID	Set identification number. (Unique Integer>0)	
V1, V2	For vibration analysis: frequency range of interest. For buckling analysis: eigenvalue range of interest. See Remarks 4 and 21, and SOL 414 Remark 3. (Real or blank, $-5. \times 10^{16} \le V1 < V2 \le 5. \times 10^{16}$)	
ND	Number of roots desired. See Remark 4. (Integer>0 or blank)	
MSGLVL	Diagnostic level. ($0 \le$ Integer ≤ 4; Default = 0)	
MAXSET	Number of vectors in block or set. Default is machine dependent. See Remark 14.	
SHFSCL	Estimate of the first flexible mode natural frequency. See Remark 10. (Real or blank)	
NORM	Method for normalizing eigenvectors (Character: “MASS”, “MAX”, or “AF”)	
	MASS	Normalize to unit value of the generalized mass. Not available for buckling analysis. (Default for normal modes analysis.)
	MAX	Normalize to unit value of the largest displacement in the analysis set. Displacements not in the analysis set may be larger than unity. (Default for buckling analysis.)
	AF	Amplitude-Frequency normalization method. See Remark 20.

Note: The following options (ALPH, Fi, and NUMS) may be specified using the option=value format on the continuation row.

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Field	Contents
ALPH	Specifies a constant for the calculation of the frequencies (Fi) at the upper boundary segments for the fdmodes and hdmodes parallel methods or the multisegment serial method based on the following formula: (0.0 < Real ≤ 1.0; Default = 1.0). ALPH < 1.0, $Fi = V1 + (V2 - V1)\frac{1-\text{ALPH}^{i}}{1. - \text{ALPH}^{N}}$ ALPH = 1.0, $Fi = V1 + (V2 - V1)\left(\frac{i}{N}\right)$ The number of frequency segments N is defined as follows: FDMODES: N is defined either by the NUMSEG keyword or the NASTRAN NUMSEG statement and is limited to 16. See Remark 13. HDMODES: N is defined by the NCLUST keyword. See Remark 13. Multisegment serial method: N is defined either by the NUMS field on the EIGRL entry or the NUMSEG keyword. See Remark 19.
Fi	Frequencies at the upper boundary of each i-th segment for the fdmodes parallel method or the multisegment serial method. Fi is optionally used instead of ALPH to explicitly define the boundaries. Fi cannot be used with hdmodes. The number of Fi boundaries = N – 1 where the number of segments N is defined above in the description of ALPH. (Real or blank; V1 < F1 < F2 <...< F15 < V2)
NUMS	Number of frequency segments for the multisegment serial method. (0 < Integer ≤ 16; Default = 1) See Remark 19.
option=value	Assigns a value to the options above except for SID. ALPH, NUMS, and Fi must be specified in this format. V1, V2, ND, MSGLVL, MAXSET, SHFSCL, and NORM may be specified in this format as long as their corresponding field is blank in the parent entry.

Remarks:

1. Real eigenvalue extraction data sets must be selected with the Case Control command METHOD = SID.
2. The units of V1 and V2 are cycles per unit time in vibration analysis, and are eigenvalues in buckling analysis. Each eigenvalue is the factor by which the prebuckling state of stress is multiplied to produce buckling in the shape defined by the corresponding eigenvector.
3. NORM = “MASS” is ignored in buckling analysis and NORM = “MAX” will be applied.

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4. The roots are found in order of increasing magnitude; that is, those closest to zero are found first. V2 should be set to a physically meaningful upper bound on the desired eigenvalues, or omitted. Extremely large values of V2 may result in a fatal message, depending on the model.

   If ND is specified, it is possible that the requested eigenvalues include some but not all from a set of multiple eigenvalues (for example, if ND=3 and there are six rigid modes). In this case, the modal space spanned by the eigenvectors is not uniquely defined, which is undesirable in frequency or transient response. In this situation, the eigensolution is computed as requested, accompanied by a warning message. Alternatively, setting system(509)=1 allows the READ module to increase the number of computed eigenpairs automatically to include the entire set.

   The number and type of roots to be found can be determined from Table 13-76.

   Table 13-76: Number and Type of Roots Found with EIGRL Entry

V1	V2	ND	Number and Type of Roots Found
V1	V2	ND	Lowest ND or all in range, whichever is smaller.
V1	V2	blank	All in range
V1	blank	ND	Lowest ND in range [V1,+∞]
V1	blank	blank	Lowest root in range [V1,+∞]
blank	blank	ND	Lowest ND roots in [-∞,+∞]
blank	blank	blank	Lowest root. See Remark 11.
blank	V2	ND	Lowest ND roots below V2
blank	V2	blank	All below V2

5. In vibration analysis, if V1<0.0, the negative eigenvalue range will be searched. (Eigenvalues are proportional to Vi squared; therefore, the negative sign would be lost.) This is a means for diagnosing improbable models. In buckling analysis, negative V1 and/or V2 require no special logic.

6. Eigenvalues are sorted on order of magnitude for output. An eigenvector is found for each eigenvalue.

7. MSGLVL controls the amount of diagnostic output during the eigenvalue extraction process. The default value of zero suppresses all diagnostic output. A value of one prints eigenvalues accepted at each shift. Higher values result in increasing levels of diagnostic output.

8. MAXSET is used to limit the maximum block size. It is otherwise set by the region size or by ND with a maximum size of 15. It may also be reset if there is insufficient memory available. The default value is recommended.

9. In vibration analysis, if V1 is blank, all roots less than zero are calculated. Small negative roots are usually computational zeroes which indicate rigid body modes. Finite negative roots are an indication of modeling problems. If V1 is set to zero, negative eigenvalues are not calculated.

10. A specification for SHFSCL may improve performance, especially when large mass techniques are used in enforced motion analysis. Large mass techniques can cause a large gap between the rigid body and flexible frequencies. If this field is blank, a value for SHFSCL is estimated automatically.

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11. On occasion, it may be necessary to compute more roots than requested to ensure that all roots in the range have been found. However, this method will not output the additional roots.

12. NASTRAN SYSTEM(146) provides options for I/O in sparse method only:

Table 13-77: SYSTEM(146) Options

SYSTEM(146)	Description
2	Increase memory reserved for sparse method by approximately 100%.
3	Increase memory reserved for sparse method by approximately 300%.
4	Increase memory reserved for sparse method by approximately 400%.

13. For the fdmodes distributed parallel method, the frequency range between V1 and V2 is subdivided into segments and analyzed in parallel. V1 and V2 must be defined, and the number of frequency segments must be specified with the NUMSEG keyword or the NASTRAN NUMSEG statement. NUMSEG should equal the number of processors which is defined with the DMP keyword. If NUMSEG and DMP are not the same, DMP will overwrite the NUMSEG value. The upper frequencies of each segment may be generated automatically by ALPH or specified directly in Fi. If both are specified, then Fi takes precedence over ALPH as long as they are consistent. ALPH, if multiplied by 100, may also be specified on FRQSEQ keyword of the NASTRAN statement. For best performance, ND should not be used with fdmodes. The segment boundaries are distributed automatically in units of frequency, not cycles.

    For the gdmodes distributed parallel method, the geometry is partitioned into segments and analyzed in parallel. ND, V1 and V2 can all be used with gdmodes. On the rare occasion that the geometry partitioning fails, gdmodes will revert to the fdmodes method if the keyword gpart=1 (default).

    For the hdmodes distributed parallel method, the frequency range between V1 and V2 is subdivided into segments, as in fdmodes. Also as in fdmodes, V1 and V2 must be defined. However, the number of segments equals the value of the NCLUST keyword. If desired, the upper frequencies of each segment may be adjusted using ALPH; they cannot be specified directly in Fi. For best performance, ND should not be used with hdmodes.

    The rdmodes shared or distributed memory parallel method uses substructuring technology for very large scale normal modes problems. The RDMODES approach computes close approximations to the normal modes and its subsequent modal frequency response analysis. The rdmodes method requires V2. This method is not supported in buckling solutions.

Table 13-78: EIGRL Input Summary for DMP Methods

	Modal Solution (SOL 103)	Buckling Solution (SOL 105)
fdmodes	V1 and V2 are required. ND is not supported.	V1 and V2 are required. ND is not supported.
gdmodes	ND, V1 and V2 can all be used. See Table 13-76.	ND, V1 and V2 can all be used. If ND is not specified, both V1 and V2 must be specified.

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hdmodes	V1 and V2 are required. ND is not supported.	V1 and V2 are required. ND is not supported.
rdmodes	V2 is required.	Not supported.

14. Increasing MAXSET may improve performance for large problems where a large number of eigenvalues are being found. The default is 7 on all machines. SYSTEM(263) may be set in an rcfile to effectively modify the default; however the setting on the EIGRL entry always takes precedence.

15. SYSTEM(196), keyword SCRSAVE, controls reuse of scratch files when segment logic is invoked. SYSTEM(196) is useful only when multiple frequency segments are requested on a Lanczos run. (Multiple frequency segments can be requested via the NUMS field in the EIGRL entry and by SYSTEM(197).) Each frequency segment requires a minimum of three scratch files. When multiple frequency segments are used on a single processor computer then each frequency segment is solved serially. In this case, it makes sense to let segment #2 use the scratch files which were used by segment #1 since work for segment #1 has been completed (otherwise it wouldn't be working on #2). Similarly, when work for segment #2 is finished, segment #3 should be able to use #2's scratch files. SYSTEM(196)=1 allows such file reuse and is considered a safe default on Version 70 and later systems.

16. The new buckling shift logic in Version 70.5 tends to shift to 1.0 first. The logic may have difficulty finding the lowest ND roots if a problem requests a small number of roots (ND) when there are thousands of roots below 1. In this case either the loading should be scaled, SHFSCL specified, or a smaller frequency range requested.

17. Because Lanczos performance is tuned for medium to large problems, this has caused difficulties with very small problems. Thus, by default, on problems with fewer than 20 degrees-of-freedom when the LAN method is selected, the method is switched to AHOU. The criteria for automatic switching is controlled by SYSTEM(359) on the NASTRAN entry.

18. The parameter input VMOPT=2 should not be used with MAX normalization. VMOPT=2 will only yield correct results with MASS normalization.

19. The multisegment serial method subdivides the frequency range between V1 and V2 into segments, as in fdmodes. Unlike fdmodes, the multisegment method analyzes each segment in turn on one processor only. V1 and V2 must be defined, and ND must not be defined. The number of segments may be specified in NUMS, with the NUMSEG keyword, or with the NASTRAN NUMSEG statement. If both NUMS and NUMSEG are specified, NUMS takes precedence. Unlike the parallel methods, the multisegment serial method is not meant to decrease solution times. It is sometimes used to solve an eigenvalue solution which otherwise has numerical problems.

20. In AF (amplitude-frequency) normalization, the normalized eigenvectors are given by:

$(C / \omega \Delta_{max}) \{x\}$

where C is an optional user-defined scale factor, ω is the natural frequency for the mode in Hz, Δ_{max} is the magnitude of the maximum grid point translation for the mode, and $\{x\}$ is the un-normalized eigenvector for the mode. The optional user-defined scale factor, C, is specified with the AFNORM parameter.

21. Negative eigenvalues can occur in a SOL 103 analysis if the model contains sufficient preloading to cause buckling. If the V1 field is left blank, the software assumes that all of the eigenvalues are

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non-negative. Because of this, depending on the magnitude of a negative eigenvalue relative to shift scale, a negative eigenvalue may be filtered out or not computed.

22. In SOL 401, for the cyclic and Fourier subcase types, a unique modal solution occurs for the requested harmonic numbers. As a result of the inherent symmetry with these solutions, mode pairs exist for specific harmonic numbers. For a cyclic subcase, mode pairs exist for harmonic numbers 1 through N/2 -1. For the Fourier subcase, mode pairs exist for all harmonic numbers > 0. The software automatically outputs the mode pairs for these subcase types for the modes requested with the EIGRL entry.

    For example, if you request 10 modes on the EIGRL entry:

    For a cyclic subcase:

    For harmonic index 0 and N/2, 10 modes are computed.

    For harmonic numbers 1 through N/2 -1, 20 modes are computed (10 distinct modes).

    For a Fourier subcase:

    For harmonic index 0, 10 modes are computed.

    For harmonic index > 0, 20 modes are computed (10 distinct modes).

Remarks related to SOLs 402 and 414:

1. ALPH, Fi and NUMS are ignored.
2. For SOL 414, NORM=AF is not supported. If you define NORM=AF, the software will use NORM=MAX.
3. For SOL 414,110, defining V1=0 can help to improve performance.
4. For SOL 414,110, when you solve cyclic symmetry (CYCSET) without the Coleman transformation (PARAM,COLEMAN,0 or PARAM,COLEMAN not defined), you must define the METHOD case control command and EIGRL bulk entry in addition to the CMETHOD case control command and EIGC bulk entry. Otherwise, the software cannot writes an error message.

Related Topics

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ELAR

Element Add/Remove Set Definition (SOL 401)

Defines element addition and removal for elements selected in a group.

General Format:

1	2	3	4	5	6	7	8	9	10
ELAR	ESID	TYPE	AC	AD	RC	RD	STNFREE		
	GRPID								

TYPE=TIME Format:

1	2	3	4	5	6	7	8	9	10
ELAR	ESID	TIME	AC	AD	RC	RD	STNFREE		
	GRPID								

TYPE=RINELE Format:

1	2	3	4	5	6	7	8	9	10
ELAR	ESID	RINELE			RC	RD			
	GRPID								

TYPE=RMECHE Format:

1	2	3	4	5	6	7	8	9	10
ELAR	ESID	RMECHE			RC	RD			
	GRPID								

Examples:

ELAR	100	TIME	0.0	0.1	0.6	0.1	NO		
	101								

ELAR	100	RINELE			0.03	-1			
	101								

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ELAR	100	RMECHE			0.03	0.0			
	101								

Fields:

Field	Contents
ESID	Element add/remove set identification number. (Integer > 0)
TYPE	Addition/Removal criteria type. (Character) TYPE = TIME: Element addition or removal based on time. TYPE = RINELE: Element removal based on inelastic strain. TYPE = RMECHE: Element removal based on mechanical strain.
GRPID	Identification of a GROUP bulk entry, which selects the elements for adding or removing. The referenced GROUP bulk entry must have TYPEi = ELEM. (Integer > 0)
Fields for TYPE=TIME:	
AC	Addition time. (Real ≥ 0.0 or blank)
AD	Addition delta time. (Integer = -1 or Real ≥ 0.0; Default = 0.0)
RC	Removal time. (Real > AC+AD or blank)
RD	Removal delta time. (Integer = -1 or Real ≥ 0.0; Default = 0.0)
STNFREE	Addition with or without strain. (Character, Default = NO) STNFREE=YES Strain free addition. Only supported with TYPE=TIME. The grid point locations and the resulting element geometry at the add time (AC field) becomes the initial undeformed reference state for the added strain-free elements. Deformation after the add time will then begin to strain these elements. The elements are added at time=AC immediately with zero strain. The delta time=AD is always 0.0 for the strain free addition. If you define AD≠0.0, the software will force it to 0.0 and issue a warning. STNFREE=NO Addition with strain. The initial geometry at time=0.0 is the undeformed reference state for the added elements. The element is added from time=AC (start) to time=AC+AD (fully added) with non-zero strain (assuming the connecting grid points have deformed from their initial state).
Fields for TYPE=RINELE or RMECHE:	
AC	Ignored.
AD	Ignored.

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Field	Contents
RC	Removal strain (Effective strain). (Real > 0.0 or blank)
RD	Removal delta time. (Integer = -1 or Real ≥ 0.0; Default = 0.0)

Remarks:

1. ESID is selected either by the ELAR case control command or the ELARADD bulk entry.
2. See Element addition and removal in the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)* for the list of elements supported for addition and removal.
3. For the static and dynamic subcases:
   - Element addition and removal by time, and element removal by strain are supported.
   - Element addition or removal is not supported in a statics subcase in which nonlinear buckling is requested with the NLARCL=ID case control command.
   - The ELAR case control command can be defined above the subcase level (global) or in a subcase.

     An ELAR command in a statics or dynamic subcase and a definition at the global level are equivalent. That is, if you define the ELAR command in a subcase, the software internally converts it to a global definition. As a result, the element addition or removal will occur in the static or dynamic subcase where the requested time or strain occurs, and not necessarily the subcase in which the ELAR command was defined.

     If you define multiple ELAR definitions globally and/or in subcases, the software combines and applies all ELAR sets (global + subcase definitions) as a global set.
   - For TYPE=TIME, AC and RC define the addition time and removal time, respectively.

     If the time defined for AC or RC is in the time range for a subcase, but not precisely a solution time defined by the TSTEP1 bulk entry, a new solution time will be created for the element addition or removal.

     If AC or RC are not in the time range of one of the subcases, the software will not add or remove the associated elements.
   - For TYPE = RINELE, you enter the effective inelastic strain in the RC field in which the element will be removed. The strain value you define includes strain contributions from both plasticity and creep materials if both are defined on an element. Plastic and creep strain output can be requested with the PLSTRN and CRSTRN case control commands, respectively.
   - For TYPE = RMECHE, you enter the effective mechanical strain in the RC field in which the element will be removed. The strain value you define includes the elastic and inelastic strains. Elastic strain output can be requested with the ELSTRN case control command. For example, if an element has both plastic and creep strain, the effective mechanical strain is the sum of the effective elastic, effective plastic, and effective creep strains. If the material is a user-defined material (UMAT), the effective mechanical strain is the equivalent mechanical strain computed from mechanical strain components.

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- The addition delta time (AD field) and removal delta time (RD field) are time intervals used to add or remove the element mass, damping, and stiffness. The delta time options slow the transition of adding or removing elements, which helps to prevent convergence problems.

  The delta time begins at the add/remove time and continues after. If the delta time is greater than the time remaining in the subcase, it will be adjusted to equal the remaining time in the subcase.

  If you define -1 in the AD or RD fields, the software will ramp the element addition or removal over the remaining subcase time.

  When the AD and RD fields are undefined, an element addition or removal occurs instantly.

4. For preload subcases:

   - The ELAR case control command must be defined in the preload subcase.
   - Only element addition using TYPE=TIME is supported. The fields AC, AD, and STNFREE are ignored. The software internally assigns AC to be the subcase time (time is constant for a preload subcase), AD=0.0, and STNFREE=YES.

     Because element removal is not allowed in the preload subcase, an error is reported if RC or RD are defined. An element added in a preload subcase can be removed in a later static or dynamic subcase.
   - A preload subcase can include an element addition set (ELAR command) or a bolt preload set (BOLDLD command), but it cannot include both.
   - Elements included in a bolt definition cannot be added or removed in any subcase (static, dynamic, preload).
   - A preload subcase (with and without the ELAR command) can include a new thermal load set (TEMPERATURE command) and/or a new contact set (BCSET command). No other new load sets are allowed in a preload subcase. Load sets defined globally or in previous subcases are maintained in the preload subcase.

5. Element addition and removal is not supported when large strain is requested.

Related Topics

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ELAR2

Alternative Element Add/Remove Set Definition (SOL 401)

Defines element addition and removal for the elements listed.

General Format:

1	2	3	4	5	6	7	8	9	10
ELAR2	ESID	TYPE							
	EID1	AC1	AD1	RC1	RD1	STNFREE1			
	EID2	AC2	AD2	RC2	RD2	STNFREE2			
	...	...	...	...	...	...			
	EIDi	ACi	ADi	RCi	RDi	STNFREEi			

TYPE=TIME Format:

1	2	3	4	5	6	7	8	9	10
ELAR2	ESID	TIME							
	EID1	AC1	AD1	RC1	RD1	STNFREE1			
	EID2	AC2	AD2	RC2	RD2	STNFREE2			
	...	...	...	...	...	...			
	EIDi	ACi	ADi	RCi	RDi	STNFREEi			

TYPE=RINELE Format:

1	2	3	4	5	6	7	8	9	10
ELAR2	ESID	RINELE							
	EID1			RC1	RD1				
	EID2			RC2	RD2				
	...			...	...				
	EIDi			RCi	RDi				

TYPE=RMECHE Format:

1	2	3	4	5	6	7	8	9	10
ELAR2	ESID	RMECHE							
	EID1			RC1	RD1				

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1	2	3	4	5	6	7	8	9	10
	EID2			RC2	RD2				
	...			...	...				
	EIDi			RCi	RDi				

Examples:

ELAR2	100	TIME							
	101	0.0	0.1	0.6	0.1	NO			
	102	0.0	0.1	0.5	0.1	NO			
	103	0.0	0.1	0.4	0.1	YES			

ELAR2	100	RINELE							
	101			0.03	-1				
	102			0.03	0.0				
	103			0.03	-1				

ELAR2	100	RMECHE							
	101			0.03	-1				
	102			0.03	0.0				
	103			0.03	-1				

Fields:

Field	Contents
ESID	Element add/remove set identification number. (Integer > 0)
TYPE	Addition/Removal criteria type. (Character)
	TYPE = TIME: Element addition or removal based on time.
	TYPE = RINELE: Element removal based on inelastic strain.
	TYPE = RMECHE: Element removal based on mechanical strain.
EIDi	Element ID. (Integer>0)
Fields for TYPE=TIME:	
AC	Addition time. (Real ≥ 0.0 or blank)
AD	Addition delta time. (Integer = -1 or Real ≥ 0.0; Default=0.0)

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Field	Contents
RC	Removal time. (Real > AC+AD or blank)
RD	Removal delta time. (Integer = -1 or Real≥ 0.0; Default=0.0)
STNFREE	Addition with or without strain. (Character, Default = NO) STNFREE=YES Strain free addition. Only supported with TYPE=TIME. The grid point locations and the resulting element geometry at the add time (AC field) becomes the initial undeformed reference state for the added strain-free elements. Deformation after the add time will then begin to strain these elements. The elements are added at time=AC immediately with zero strain. The delta time=AD is always 0.0 for the strain free addition. If you define AD≠0.0, the software will force it to 0.0 and issue a warning. STNFREE=NO Addition with strain. The initial geometry at time=0.0 is the undeformed reference state for the added elements. The element is added from time=AC (start) to time=AC+AD (fully added) with non-zero strain (assuming the connecting grid points have deformed from their initial state).
Fields for TYPE=RINELE or RMECHE:	
AC	Ignored.
AD	Ignored.
RC	Removal strain (Effective strain). (Real > 0.0 or blank)
RD	Removal delta time. (Integer = -1 or Real ≥0.0; Default=0.0)

Remarks:

1. ESID is selected either by the ELAR case control command or the ELARADD bulk entry.
2. See Element addition and removal in the *Simcenter Nastran Multi-Step Nonlinear User's Guide (SOL 401 and SOL 402)* for the list of elements supported for addition and removal.
3. For the static and dynamic subcases:
   - Element addition and removal by time, and element removal by strain are supported.
   - Element addition or removal is not supported in a statics subcase in which nonlinear buckling is requested with the NLARCL=ID case control command.
   - The ELAR case control command can be defined above the subcase level (global) or in a subcase.

     An ELAR command in a statics or dynamic subcase and a definition at the global level are equivalent. That is, if you define the ELAR command in a subcase, the software internally converts it to a global definition. As a result, the element addition or removal will occur in the static or dynamic subcase where the requested time or strain occurs, and not necessarily the subcase in which the ELAR command was defined.

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If you define multiple ELAR definitions globally and/or in subcases, the software combines and applies all ELAR sets (global + subcase definitions) as a global set.

- For TYPE=TIME, AC and RC define the addition time and removal time, respectively.

  If the time defined for AC or RC is in the time range for a subcase, but not precisely a solution time defined by the TSTEP1 bulk entry, a new solution time will be created for the element addition or removal.

  If AC or RC are not in the time range of one of the subcases, the software will not add or remove the associated elements.

- For TYPE = RINELE, you enter the effective inelastic strain in the RC field in which the element will be removed. The strain value you define includes strain contributions from both plasticity and creep materials if both are defined on an element. Plastic and creep strain output can be requested with the PLSTRN and CRSTRN case control commands, respectively.

- For TYPE = RMECHE, you enter the effective mechanical strain in the RC field in which the element will be removed. The strain value you define includes the elastic and inelastic strains. Elastic strain output can be requested with the ELSTRN case control command. For example, if an element has both plastic and creep strain, the effective mechanical strain is the sum of the effective elastic, effective plastic, and effective creep strains. If the material is a user-defined material (UMAT), the effective mechanical strain is the equivalent mechanical strain computed from mechanical strain components.

- The addition delta time (AD field) and removal delta time (RD field) are time intervals used to add or remove the element mass, damping, and stiffness. The delta time options slow the transition of adding or removing elements, which helps to prevent convergence problems.

  The delta time begins at the add/remove time and continues after. If the delta time is greater than the time remaining in the subcase, it will be adjusted to equal the remaining time in the subcase.

  If you define -1 in the AD or RD fields, the software will ramp the element addition or removal over the remaining subcase time.

  When the AD and RD fields are undefined, an element addition or removal occurs instantly.

4. For preload subcases:

- The ELAR case control command must be defined in the preload subcase.

- Only element addition using TYPE=TIME is supported. The fields AC, AD, and STNFREE are ignored. The software internally assigns AC to be the subcase time (time is constant for a preload subcase), AD=0.0, and STNFREE=YES.

  Because element removal is not allowed in the preload subcase, an error is reported if RC or RD are defined. An element added in a preload subcase can be removed in a later static or dynamic subcase.

- A preload subcase can include an element addition set (ELAR command) or a bolt preload set (BOLDLD command), but it cannot include both.

- Elements included in a bolt definition cannot be added or removed in any subcase (static, dynamic, preload).

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- A preload subcase (with and without the ELAR command) can include a new thermal load set (TEMPERATURE command) and/or a new contact set (BCSET command). No other new load sets are allowed in a preload subcase. Load sets defined globally or in previous subcases are maintained in the preload subcase.

5. Element addition and removal is not supported when large strain is requested.

Related Topics

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ELARADD

Defines a union of element add/remove sets (SOL 401)

Defines an element add/remove set as a union of ELAR and ELAR2 bulk entries.

Format:

1	2	3	4	5	6	7	8	9	10
ELARADD	ESID	S1	S2	S3	...	Si			

Example:

1	2	3	4	5	6	7	8	9	10
ELARADD	101	1	2	3	4				

Fields:

Field	Contents
ESID	Element add/remove set identification number. (Integer > 0)
Si	Identification numbers of element add/remove sets defined with ELAR / ELAR2 entries. (Integer > 0)

Remarks:

1. ESID can be selected by the ELAR case control command.

Related Topics

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ELIST

Element List

Defines a list of structural elements for virtual fluid mass.

Format:

1	2	3	4	5	6	7	8	9	10
ELIST	LID	E1	E2	E3	E4	E5	E6	E7	
	E8	E9	E10	-etc.-					

Example:

ELIST	3	51	-62	68	THRU	102	122		

Fields:

Field	Contents
LID	Identification number of list. (Integer>0)
Ei	Identification number of a structural element. See Remark 1 . for the meaning of the negative sign. The string “THRU” may be used to indicate that all existing elements between those referenced in the preceding and succeeding fields are in the list. (Integer ≠ 0 or “THRU”)

Remarks:

1. If the ELIST entry is referenced by field 6 of an MFLUID entry, the wetted side of the element is determined by the presence or absence of a minus sign preceding the element’s ID on the ELIST entry. A minus sign indicates that the fluid is on the side opposite to the element’s positive normal as determined by applying the right-hand rule to the sequence of its corner points. If the “THRU” option is used, then immediately preceding and succeeding elements must have the same sign.
2. Large open sections with “THRU” should be avoided.
3. The word “THRU” may not appear in fields 3 or 9 on the parent row, or in fields 2 or 9 in any continuation, i.e. no leading or trailing THRU is allowed on the parent entry or on any continuations.

Related Topics

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ENDDATA

Bulk Data Delimiter

Designates the end of the Bulk Data Section.

Format:

ENDDATA

Remarks:

1. ENDDATA is optional.

Related Topics

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EPOINT

Extra Point List

Defines extra points for use in dynamics problems.

Format:

1	2	3	4	5	6	7	8	9	10
EPOINT	ID1	ID2	ID3	ID4	ID5	ID6	ID7	ID8	

Example:

EPOINT	3	18	1	4	16	2			

Alternate Format and Example:

EPOINT	ID1	"THRU"	ID2						
EPOINT	17	THRU	43						

Fields:

Field	Contents
IDi	Extra point identification number. (1000000>Integer>0; for "THRU" option, ID1<ID2).

Remarks:

1. The extra point identification number must be unique with respect to all other identification numbers defined with GRID, SPOINT, EPOINT, and POINT bulk entries for the direct methods of solution. For modal methods, they must be larger than the number of eigenvectors retained for analysis.
2. EPOINT is used to define coordinates used in transfer function definitions (see the TF and DMIG entries).
3. If the alternate format is used, extra points ID1 through ID2 are also defined to be extra points.
4. See the *Simcenter Nastran Advanced Dynamic Analysis User's Guide* for a discussion of extra points.

Related Topics

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EXTRN

Partitioned External Superelement Connection

Defines a boundary connection for an external superelement.

Format:

1	2	3	4	5	6	7	8	9	10
EXTRN	GID1	C1	GID2	C2	GID3	C3	GID4	C4	
	-etc.-		GID6	"THRU"	GID7	C6	-etc.-		

Example:

EXTRN	1001	123	1120	123456	1201	123			

Fields:

Field	Contents
GIDi	Grid identification number to which the exterior superelement matrices will be connected.
Ci	Component numbers. (Integer 0, blank, or 1 for scalar points; Integers 1 through 6 with no embedded blanks for grids.)

Remarks:

1. EXTRN can only be specified in partitioned Bulk Data Sections and is ignored in the main Bulk Data Section.
2. Connection grids must be specified in the partitioned Bulk Data Section following BEGIN SUPER = SEID.
3. "THRU" may be specified only in fields 3, 5, or 7.
4. Pairs of blank fields may be entered to allow easier modification of the EXTRN entry.

Related Topics

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14. Bulk Entries F—L

FLFACT

Aerodynamic Physical Data

Used to specify density ratios, Mach numbers, reduced frequencies, and velocities for flutter analysis.

Format:

1	2	3	4	5	6	7	8	9	10
FLFACT	SID	F1	F2	F3	F4	F5	F6	F7	
	F8	F9	-etc.-						

Example:

FLFACT	97	.3	.7	3.5					

Alternate Format and Example:

FLFACT	SID	F1	"THRU"	FNF	NF	FMID			
FLFACT	201	.200	THRU	.100	11	.133333			

Fields:

Field	Contents
SID	Set identification number. (Unique Integer > 0)
Fi	Aerodynamic factor. (Real)
FNF	Final aerodynamic factor. (Real)
NF	Number of aerodynamic factors. (Integer > 0)
FMID	Intermediate aerodynamic factors. See Remark 4 . (Real)

Remarks:

1. Only the factors selected by a FLUTTER entry will be used.
2. Embedded blank fields are not allowed in the first format above.

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3. The factors must be specified in the order in which they are to be used within the looping of flutter analysis.
4. FMID must lie between F1 and FNF; otherwise, FMID will be set to (*F*1+*FNF*)/2. then

$$Fi = \frac{F1(FNF - FMID) + FNF(FNF - F1)(i-1)}{(FNF - FMID) + (FMID - F1)(i-1)}$$

where i = 1, 2, ..., NF

The use of FMID (middle factor selection) allows unequal spacing of the factors.

$$FMID = \frac{2 \cdot F1 \cdot FNF}{F1 + FNF}$$

gives equal values to increments of the reciprocal of Fi.

5. If method = PK and this entry specifies velocities, then the velocities must be non-zero. Input of negative values produces eigenvector results at a velocity equal to the positive value of the input. Input of positive values provide eigenvalues results without eigenvectors.

Related Topics

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FLUTTER

Aerodynamic Flutter Data

Defines data needed to perform flutter analysis.

Format:

1	2	3	4	5	6	7	8	9	10
FLUTTER	SID	METHOD	DENS	MACH	RFREQ	IMETH	NVALUE	EPS	

Example:

FLUTTER	19	K	119	219	319	S	5	1.-4	

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
METHOD	Flutter analysis method. (Character: “K” for K-method, “PK” for PK method, “PKNL” for PK method with no looping, “KE” for the K-method restricted for efficiency.)
DENS	Identification number of an FLFACT entry specifying density ratios to be used in flutter analysis. (Integer > 0)
MACH	Identification number of an FLFACT entry specifying Mach numbers (m) to be used in flutter analysis. (Integer > 0)
RFREQ (or VEL)	Identification number of an FLFACT entry specifying reduced frequencies (k) to be used in flutter analysis; for the PK and PKNL method, the velocities FLFACT entry is specified in this field. (Integer > 0)
IMETH	Choice of interpolation method for aerodynamic matrix interpolation. Used in the “K” and “KE” methods only. See Remark 6 . (Character: “L” = linear, “S” = surface; Default = “L”.)
NVALUE	Number of eigenvalues beginning with the first eigenvalue for output and plots. [Integer > 0; Default is the number of modal degrees-of-freedom (u_h)]
EPS	Convergence parameter for k. Used in the PK and PKNL methods only. See Remark 4 . (Real > 0.0; Default = 10^{-3})

Remarks:

1. The FLUTTER entry must be selected with the Case Control command FMETHOD = SID.

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2. The density is given by DENS · RHOREF, where RHOREF is the reference value specified on the AERO entry and DENS is the density ratio specified on the FLFACT entry.

3. The reduced frequency is given by $k = (REFC \cdot \omega/2 \cdot V)$, where REFC is given on the AERO entry, ω is the circular frequency, and V is the velocity. If k= 0.0, as specified on the FLFACT entry, then only the K-method may be specified and the Inverse Power method of eigenvalue extraction (INV on the EIGC entry) must be used. Aeroelastic divergence analysis is more appropriately performed using the PK-method or PKNL method.

4. An eigenvalue is accepted in the PK and PKNL methods when:

$$|k - k_{estimate}| < EPS \qquad \text{for } k_{estimate} < 1.0$$

$$|k - k_{estimate}| < EPS \cdot k_{estimate} \qquad \text{for } k_{estimate} \geq 1.0$$

5. When the PK or PKNL method is selected, physical displacements will only be generated for the velocities on the FLFACT that are specified as negative values of the requested velocity. Also, structural damping as specified on the GE field of MATi entries is ignored.

6. If IMETH = "L", a linear interpolation is performed on reduced frequencies at the Mach numbers specified on the FLFACT entry using the MKAEROi entry Mach number that is closest to the FLFACT entry Mach number. For IMETH = "S", a surface interpolation is performed across Mach numbers and reduced frequencies. For METHOD = "PK" or "PKNL", linear interpolation is always performed.

7. For the "K", "KE" and "PK" methods, all combinations of the FLFACT entry are analyzed. For the "PKNL" method, only ordered pairs are analyzed; i.e., (M_1, V_1), (M_2, V_2)...(M_n, V_n). For the PKNL method, equal number of densities, Mach numbers and velocities must be specified.

8. Only the "PK" and "PKNL" methods are supported for design sensitivity and optimization.

Related Topics

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FLXADD

Kinematic Joint Constraint Combination (SOL 402)

Defines a flexible sliders set as a union of single flexible sliders defined on FLXSLI entries.

Format:

1	2	3	4	5	6	7	8	9	10
FLXADD	FIID	F1	F2	F3	F4	F5	F6	F7	
	F8	F9	-etc-						

Example:

FLXADD	101	2	3	9					

Fields:

Field	Contents
FID	Combination set identification number. (Integer > 0)
Fi	Identification numbers of flexible sliders defined via FLXSLI entries. (Integer > 0; FID ≠ Fi)

Remarks:

1. Combination sets must be selected with the Case Control command FLXSLI=FID.
2. No Fi may be the identification number of a constraint set defined by another FLXADD entry.
3. The Fi values must be unique.
4. FLXADD entries take precedence over FLXSLI entries. If both have the same set ID, only the FLXADD entry will be used.

Related Topics

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FLXSLI

Flexible Slider Definition (SOL 402)

Defines a flexible slider.

Format:

1	2	3	4	5	6	7	8	9	10
FLXSLI	FID	TYPE	NGRPID	BGRPID	SGID	PROJ	NLIM		
	DTYPE	DGID	DVAL	DTID	FOP	CF	TOL	KCF	

Example:

FLXSLI	111	FLEX	11	12					
	FORC	77	1.0	22					

Fields:

Field	Contents
FID	Flexible slider identification number. See Remark 5. (Integer > 0).
TYPE	Slider type. You can define a `FLEX`, `CYLF`, `PRIF`, or `TWIF` type.
NGRPID	ID of GROUP bulk entry that specifies a list of grid points that slide along the slider. (Integer >0).
BGRPID	ID of GROUP bulk entry that specifies a list of beams that define the slider trajectory. (Integer >0).
SGID	Sensor grid point. See Remark 1, Remark 2, and Remark 3. (Integer >0). (Optional).
PROJ	Projection option. Valid options: 0-NO, 1-YES. (Character or Integer, Default= 0-NO). (Optional).
NLIM	Number of beams of the BGRPID group to take into account. (Integer > 0; Default = all the beams of the BGRPID group). (Optional).
DTYPE	Driver type to apply on the slider. Valid types: 1-FORC, 2-DISP, 3-SENSOR. (Character or Integer). (Optional)
DGID	Grid point identification number on which the driver applies. See Remark 4. (Integer >0). (Optional). Applies when a DTYPE is specified.

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Field	Contents
DVAL	Driver load or enforced displacement value. (Real > 0.0; Unit = Force or Length). (Optional). Applies when DTYPE is set to 1-FORC or 2-DISP.
DTID	Table ID for driver load. (Integer > 0). (Optional). Applies when DTYPE is set to 1-FORC or 2-DISP.
FOP	Friction option. Valid options: 0-NO, 2-INF, 3-VELO, 4-DISP. (Character or Integer, Default= 0-NO).
CF	Friction coefficient. (Real > 0.0; Unitless). (Optional). Applies when FOP = 3-VELO or 4-DISP.
TOL	Regularization velocity. (Real > 0.0; Unit= Length / Second). (Optional). Applies when FOP = 3-VELO.
KCF	Regularization stiffness. (Real > 0.0; Unit= Force / Length). (Optional). Applies when FOP = 4-DISP.

Remarks:

1. If SGID is omitted, a grid point will be created if DTYP = 1-FORC or 2-DISP is specified.
2. If SGID is specified, but DTYP is omitted, SGID will be ignored and no output request will be computed.
3. SGID grid point will be used,
   - either by the driver for DTYP = 1-FORC or 2-DISP.
   - either by an output request if the FXLRESULTS Case Control command is set to ALL or n where n is a set of grid points that contains the SGID number.
4. If the DGID grid point does not belong to the NGRPID group, the solver will add it to this group.
5. FID entry must be selected in a FLXSLI Control Case command.

Related Topics

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FMH3D

Defines 2D-3D Coupling in SOLs 402 and 414

Defines the coupling of a Fourier multi-harmonic region with a 3D structure in SOLs 402 and 414. Not supported in a superelement creation run.

Format:

1	2	3	4	5	6	7	8	9	10
FMH3D	FMHID	SID	TID						
	ANGLEI	ANGLED	N	DIST	SDISN	SDIST			

Example:

FMH3D	5	200	300						

Fields:

Field	Contents
FMHID	Set identification number. (Integer > 0)
SID	Identification number of BEDGE bulk entry that defines an element edge region on the 2D Fourier mesh (CQUADX4, CQUADX8, CTRAX3, and CTRAX6 elements). (Integer >0)
TID	Identification number of BSURFS bulk entry that defines an element face region on the 3D mesh. Region parameters can also be referenced by defining the BCRPARA bulk entry with the same ID. (Integer >0)
ANGLEI	Angle in the circumferential direction beginning at the 2D edge region, and ending at a location near the beginning edge of the 3D solid face region. You must define both ANGLEI and ANGLED, or neither. (Real>0.0; Default=Auto computed by software)
ANGLED	Angle in the circumferential direction beginning at the location determined by ANGLEI, then measured to a location near the ending edge of the 3D solid face region. You must define both ANGLEI and ANGLED, or neither. (Real>0.0; Default=Auto computed by software)
N	For the grid points referenced by the 2D region, N is the number of evenly distributed copies of virtual grid points in the arc defined by ANGLED. N and DIST cannot both be defined. A fatal error will occur if both are defined. (Integer>0; Default=blank, which selects the fully automatic coupling unless DIST is defined). See the remarks for a description of the coupling methods.
DIST	For the grid points referenced by the 2D region, DIST is the distance between the copies of virtual grid points in the arc defined by ANGLED. The software uses DIST to compute a number of grid points N such that ALPHAD*RADIUS / (N+1) < DIST. N and DIST cannot

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Field	Contents
	both be defined. A fatal error will occur if both are defined. (Real>0.0; Default=blank which selects the fully automatic coupling unless N is defined). See the remarks for a description of the coupling methods.
SDISN	Normal search distance. The normal distance from a virtual node to the solid element face must be less than SDISN for the software to create a glue condition at that location. (Real>0.0; Default = 0.001*DIAG, where DIAG is the software computed diagonal of the mesh region selected in the TID field)
SDIST	Tangential search distance. For virtual nodes that are beyond the projected area of the 3D solid face region, the tangential distance from a virtual node to the solid element face must be less than SDIST for the software to create a glue condition at that location. (Real>0.0; Default = 0.001*DIAG, where DIAG is the software computed diagonal of the mesh region selected in the TID field)

Remarks:

1. The ID in the FMHID field should be referenced by the FMH3D=n case control command.
2. Multiple FMH3D bulk entries with the same set identification number can be defined.
3. FMH3D requests the coupling to connect a Fourier axisymmetric mesh edge to a 3D solid element mesh face. The 3D solid element mesh can represent a cyclic symmetry portion of the structure, or a full 360-degree representation.

   You create an element edge region on the Fourier mesh and a face region on the 3D mesh, and then reference the regions in the SID and TID fields, respectively. The software creates virtual nodes during the solution along a face that is parallel to the 3D face region. The virtual nodes are copies of the nodes in the 2D edge region. The software glues the virtual nodes to the nodes on the 3D face and connects the virtual nodes to the nodes on the 2D edge using constraint conditions similar to how the new FOU3 elements couple.
4. The first row defining the FMHID, SID, and TID fields is required for both the fully automatic and the manual methods. All other fields depend on the method and are described below.

   - **Fully automatic method**

     You can select the fully automatic method by leaving the N, DIST, ANGLEI, and ANGLED fields blank. The software will automatically determine the number and location of the virtual grid points, and ignores the SDISN and SDIST fields.

     The software does the following when you use the fully automatic method:

     - Positions the virtual grid points in optimum locations
     - Automatically accounts for skew angles (when the 2D axisymmetric edge is not parallel with the 3D mesh edges).
     - Avoids coupling redundancies on the same 3D face.
     - Avoids coupling redundancies with the dependent degrees-of-freedom on a cyclic symmetry boundary condition.

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- Accounts for different mesh sizes on the 3D face and the 2D axisymmetric mesh.
- Accounts for elements that include midside grid points (higher order elements).

- **Manual method**

  You can select the manual method by defining either the N or the DIST field. You can optionally define the ANGLEI, ANGLED, SDISN and SDIST fields when using the manual method, or leave them blank and the software will use their default values.

  The software does the following when you use the manual method:

  - To prevent over-constraining the elements due to the creation of too many virtual grid points, the software associates each virtual grid point with the closest possible 3D grid point.
  - The software avoids coupling redundancies, particularly in the dependent degrees-of-freedom on a cyclic symmetry boundary condition.

5. For SOL 414 solutions that use FMH3D, follow these best practices when you define the harmonics:

- For SOL 414,101 Maneuvers and SOL 414,129 Transient Response Analysis solutions, the cyclic wave (harmonic index) is always 0.

  In a Static Preload subcase, or a SOL 414,101 or SOL 414,129 solution where cyclic wave (harmonic index) is always 0, the Fourier harmonic must be 0, and, optionally multipliers of *N*, which represents the number of sectors of cyclic symmetry, *N*, 2*N*, 3*N*.

- For SOL 414,103 Eigenvalue Analysis and SOL 414,110 Complex Modal Analysis solutions, to ensure that the junction of the 2D-3D coupling is represented correctly, you must ensure that the cyclic wave (harmonic index defined in the HARMONICS case control command) and Fourier harmonics (FHAR) coincide in each subcase.
- In a Modal subcase (SOL 414,103) or Complex Modal subcase (SOL 414,110), the Fourier harmonic must coincide with the cyclic wave (harmonic index). Optionally, the other harmonics can be *N-XX*, *N+XX*, 2*N-XX*, 2*N+XX*, 3*N-XX*, 3*N+XX*, where:
  - *N* is the number of sectors.
  - *XX* is the single cyclic wave (harmonic index).

SOL 414,103 Superelement Reduction and Eigenvalue Analysis solutions do not support 2D-3D coupled models for the creation of superelements (EXTSEOUT).

For more information, see Examples for SOL 414,110 Complex Modal Analysis: 2D Fourier shaft coupled to a cyclic symmetry sector using FMH3D.

Related Topics

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FORCDST

Forces and moments to be distributed

Defines forces and moments at a point that the software will automatically distribute to element faces or edges in the form of a specific traction. Only valid for SOL 401.

Format:

1	2	3	4	5	6	7	8	9	10
FORCDST	SID	RID	CID	PLOC	G	O1	O2	O3	
	F	FN1	FN2	FN3	M	MN1	MN2	MN3	

Example:

FORCDST	3	6	100	ORIG					
		322.4	219.3	530.5		14.5	25.4	12.4	

Fields:

Field	Contents
SID	Set identification number. See **Remark 2**. (Integer>0)
RID	Region identification number. See **Remark 3**. (Integer>0)
CID	Coordinate system identification number used to define the components FNi and MNi, the location of FNi and MNi when PLOC="ORIG", and the components of Oi. See **Remark 4**. (Integer ≥ 0; Default=0 selects the basic CS)
PLOC	Reference location of the point load. See **Remark 5**. (Character: ORIG, CNTRD, or GRID)
G	Grid identification number when PLOC="GRID". See **Remark 5**. (Integer>0)
Oi	Components of an offset vector relative to the coordinate system selected by CID. It moves the load point from the location defined by PLOC to a new offset location. (Real; Default=0.0.)
F	Scale factor of force components. See **Remark 6**. (Real; No default)
FNi	Components of a force vector relative to the coordinate system selected by CID. (Real; at least one FNi ≠ 0.0.)
M	Scale factor of moment components. See **Remark 6**. (Real; No default)
MNi	Components of a moment vector relative to the coordinate system selected by CID. (Real; at least one MNi ≠ 0.0.)

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Remarks:

1. The software uses the forces and moments that you define at the single point on the FORCDST bulk entry to automatically compute and apply a traction to a surface or edge patch. The traction variation relative to the software computed patch coordinate system is:

$$\overrightarrow{T} = \overrightarrow{T}_1\hat{\varepsilon}_1 + \overrightarrow{T}_2\hat{\varepsilon}_2 + \overrightarrow{T}_3\hat{\varepsilon}_3$$

$$\overrightarrow{T}_3 = C_1 + C_2x_1^L + C_3x_2^L$$

$$\overrightarrow{T}_1 = C_4 + C_6x_2^L$$

$$\overrightarrow{T}_2 = C_5 - C_6x_1^L$$

where $\left(\hat{\varepsilon}_1, \hat{\varepsilon}_2, \hat{\varepsilon}_3\right)$ are the software computed unit vectors which define the patch coordinate system, and x_1^L, x_2^L are the coordinates of the grid points on the patch with respect to the patch coordinate system.

2. For a time-unassigned load, you reference the SID of one or multiple FORCDST bulk entry and any other mechanical loads with the LOAD=n case control command either in a subcase or above the subcases (globally). Tme-unassigned loads with different SID can also be combined and optionally scaled with the LOAD bulk entry.

   For a time-assigned load, you reference the SID of one or multiple FORCDST bulk entry with the EXCITEID on the TLOAD3 bulk entry. The TLOAD3 entry includes six TID fields that you can use to independently scale the three force and the three moment components with time by referencing a unique TABLEDi entry for each. You can also reference the same TABLEDi entry for any or all TID fields to define consistent load variations. The FORCDST entry is the only load you can select with the TLOAD3 entry. All other time-assigned mechanical loads are selected by the EXCITEID of the TLOAD1 bulk entry, which scales all referenced loads consistently with time. You can combine TLOAD1 and TLOAD3 entries into a single load set with the DLOAD bulk entry.

   Note that the SID of a FORCDST bulk entry and the SID of any other mechanical load such as the FORCE entry can only be the same when the SID is selected as a time-unassigned load. For a time-assigned load, the SID of FORCDST bulk entries must be unique to the SID of all other mechanical loads.

3. The RID field selects a region of element edges or faces where the software computes and applies the traction. The regions are defined with the BSURFS, BSURF, and BEDGE bulk entries. The BSURFS entry creates a face region on the solid elements CHEXA, CPENTA, CPYRAM, and CTETRA. The BSURF entry creates a face region on the shell elements CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, and CQUADR. The BEDGE entry creates a edge region on the axisymmetric elements CTRAX3, CQUADX4, CTRAX6, CQUADX8, the plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8, and the chocking elements CCHOCK3, CCHOCK4, CCHOCK6, CCHOCK8.

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4. CID=0 or blank (the default) references the basic coordinate system.
5. The location options for PLOC are “ORIG” for the origin of the coordinate system selected by CID (field 4), “CNTRD” for the centroid of the surface or edge region selected by RID (field 3), or “GRID” for the location of the grid point selected in G (field 6). The software automatically computes the centroid location. The PLOC field does not have a default, so it must be defined.
6. The software multiplies the scale factor F by the components FN1, FN2, FN3. The scale factor F must be defined, and at least one of FNi must be defined.

   The software multiplies the scale factor M by the components MN1, MN2, MN3. The scale factor M must be defined, and at least one of MNi must be defined.
7. For axisymmetric element edges, the forces and moments defined on the FORCDST entry are the total of the forces and moments applied to the full theoretical 360 degree area. For plane stress element edges, the forces and moments defined on the FORCDST entry are the total of the forces and moments applied to the full theoretical area of thickness t. For example, the total forces and moments defined on the FORCDST entry would be the same whether you want the traction computed and applied to solid or shell element faces, or to element edges.
8. For the SOL 401 modal frequency response subcase, the FORCDST entry cannot be referenced by the EXCITEID field on a RLOADi entry.

Related Topics

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FORCE

Static Force

Defines a static concentrated force at a grid point by specifying a vector.

Format:

1	2	3	4	5	6	7	8	9	10
FORCE	SID	G	CID	F	N1	N2	N3		

Example:

FORCE	2	5	6	2.9	0.0	1.0	0.0		

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Grid point identification number. (Integer > 0)
CID	Coordinate system identification number. (Integer ≥ 0; Default = 0)
F	Scale factor. (Real)
Ni	Components of a vector measured in coordinate system defined by CID. (Real; at least one Ni ≠ 0.0.)

Remarks:

1. The static force applied to grid point G is defined by $\mathbf{f} = F \cdot \mathbf{N}$ where **N** is the vector with components N1, N2, and N3 defined in fields 6, 7 and 8. The magnitude of **f** is equal to *F* times the magnitude of **N**.
2. In the static solution sequences, the load set ID must be selected by the Case Control command LOAD = SID. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
3. A CID of zero or blank (the default) references the basic coordinate system.
4. For scalar points see SLOAD.
5. For interpretation of a FORCE entry in an axisymmetric analysis, see the listing for the axisymmetric element type.

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Remarks related to SOLs 402 and 414:

1. To apply a force with constant magnitude, SID is selected by Case Control command LOAD = SID for both static and transient analyses.
2. To apply a time-dependent force, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by Case Control command DLOAD.

Related Topics

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FORCE1

Static Force, Alternate Form 1

Defines a static concentrated force at a grid point by specification of a magnitude and two grid points that determine the direction.

Format:

1	2	3	4	5	6	7	8	9	10
FORCE1	SID	G	F	G1	G2				

Example:

FORCE1	6	13	-2.93	16	13				

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Grid point identification number. (Integer > 0)
F	Magnitude of the force. (Real)
G1, G2	Grid point identification numbers. (Integer > 0; G1 and G2 may not be coincident.)

Remarks:

1. The static force applied to grid point G is defined by $\mathbf{f} = F \cdot \mathbf{n}$ where **n** is a unit vector parallel to a vector from G1 to G2.
2. In the static solution sequences, the load set ID must be selected by the Case Control command LOAD=SID. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
3. The follower force effects due to loads from this entry are included in the stiffness in all linear solution sequences that calculate a differential stiffness. The linear solution sequences are SOLs 103, 105, 107 to 112, 115 and 116. (See also the parameter FOLLOWK.) Follower force effects are included in the force balance in the nonlinear static and nonlinear transient dynamic solution sequences, SOLs 106, 129, 153, and 159, if geometric nonlinear effects are turned on with PARAM,LGDISP,1. The follower force stiffness is included in the nonlinear static solution sequences (SOLs 106 and 153) but not in the nonlinear transient dynamic solution sequences (SOLs 129 and 159).

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For SOL 401, follower force effects are included in the force balance in the nonlinear static solution if geometric nonlinear effects are turned on with PARAM,LGDISP,1. For additional information, see the NLCNTL bulk entry.

Remarks related to SOLs 402 and 414:

1. The force is a follower force in large displacement analysis (i.e., PARAM,LGDISP,1). For small displacments, the direction of the force is defined by the original positions of G1 and G2 and does not change.
2. The force effects are included in the stiffness matrix only if a non-symmetric solver is used and the geometric nonlinear effects are turned on with PARAM,LGDISP,1.
3. To apply a force with constant magnitude, SID is selected by Case Control command LOAD = SID for both static and transient analysis.
4. To apply a time-dependent force, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by Case Control command DLOAD.

Related Topics

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FORCE2

Static Force, Alternate Form 2

Defines a static concentrated force at a grid point by specification of a magnitude and four grid points that determine the direction.

Format:

1	2	3	4	5	6	7	8	9	10
FORCE2	SID	G	F	G1	G2	G3	G4		

Example:

FORCE2	6	13	-2.93	16	13	17	13		

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Grid point identification number. (Integer > 0)
F	Magnitude of the force. (Real)
Gi	Grid point identification numbers. (Integer > 0; G1 and G2 may not be coincident; G3 and G4 cannot be coincident.)

Remarks:

1. The direction of the force is parallel to the cross product of vectors from G1 to G2 and G3 to G4.
2. In the static solution sequences, the load set ID must be selected by the Case Control command LOAD=SID. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
3. The follower force effects due to loads from this entry are included in the stiffness in all linear solution sequences that calculate a differential stiffness. The linear solution sequences are SOLs 103, 105, 107 to 112, 115 and 116. (See also the parameter FOLLOWK.) Follower force effects are included in the force balance in the nonlinear static and nonlinear transient dynamic solution sequences, SOLs 106, 129, 153, and 159, if geometric nonlinear effects are turned on with PARAM,LGDISP,1. The follower force stiffness is included in the nonlinear static solution sequences (SOLs 106 and 153) but not in the nonlinear transient dynamic solution sequences (SOLs 129 and 159).

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For SOL 401, follower force effects are included in the force balance in the nonlinear static solution if geometric nonlinear effects are turned on with PARAM,LGDISP,1. For additional information, see the NLCNTL bulk entry.

Remarks related to SOLs 402 and 414:

1. The force is a follower force in large displacement analysis (i.e., PARAM,LGDISP,1). For small displacments, the direction of the force is defined by the original positions of G1 and G2 and does not change.
2. The force effects are included in the stiffness matrix only if a non-symmetric solver is used and the geometric nonlinear effects are turned on with PARAM,LGDISP,1.
3. To apply a force with constant magnitude, SID is selected by Case Control command LOAD = SID for both static and transient analysis.
4. To apply a time-dependent force, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by Case Control command DLOAD.

Related Topics

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FOU3

Coupling Element Connection

Connects a grid point with six DOF to a grid point used in the connectivity of a Fourier axisymmetric element for SOLs 402 and 414.

Format:

1	2	3	4	5	6	7	8	9	10
FOU3	EID	GA	GB	OPT					

Example:

FOU3	2	84	293	1					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
GA	Grid point identification number of the 3D grid point. (Integer > 0.0)
GB	Grid point identification number of the 2D Fourier multi-harmonic grid point. (Integer > 0.0)
OPT	Coupling option. (Integer="1", "2", or "5"; No default) OPT=1: With this option, GA and GB must be coincident, and the translational DOF are coupled (R,θ,Z). OPT=2: With this option, GA must be on the symmetry axis, and GA and GB must have the same axial coordinate. A rigid connection is created between the translational DOF on the 3D node and the Fourier node. This option links the 3D node at r=0.0, to a rigid disk, where the radius of the disk is determined by the coordinate of the Fourier node. OPT=5: With this option, GA must be on the symmetry axis, and GA and GB must have the same axial coordinate. All six DOF of the 3D grid point are coupled (translational and rotational DOF). This option links the 3D node at r=0.0, to the mean displacement of a disk, where the radius of the disk is determined by the coordinate of the Fourier node. The Fourier disk is not rigid and is free to deform.

Related Topics

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FREQ

Frequency List

Defines a set of frequencies to be used in the solution of frequency response problems.

Format:

1	2	3	4	5	6	7	8	9	10
FREQ	SID	F1	F2	F3	F4	F5	F6	F7	
	F8	F9	F10	-etc.-					

Example:

FREQ	3	2.98	3.05	17.9	21.3	25.6	28.8	31.2	
	29.2	22.4	19.3						

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
Fi	Frequency value in units of cycles per unit time. (Real ≥ 0.0)

Remarks:

1. Frequency sets must be selected with the Case Control command FREQUENCY = SID.
2. All FREQi entries with the same frequency set identification numbers will be used. The software automatically searches for duplicate frequencies. When a duplicate is found, the software keeps the lower frequency in the pair and removes the higher frequency.

   f_N and f_{N-1} are considered duplicates if:

   $$|f_N - f_{N-1}| < \text{DFREQ} \cdot |f_{MAX} - f_{MIN}|,$$

   where DFREQ is a user parameter, with a default of 10^{-5}. f_{MAX} and f_{MIN} are the maximum and minimum excitation frequencies of the combined FREQi entries.
3. In modal analysis, solutions for modal DOFs from rigid body modes at zero excitation frequencies may be discarded. Solutions for nonzero modes are retained.

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FREQ1

Frequency List, Alternate Form 1

Defines a set of frequencies to be used in the solution of frequency response problems by specification of a starting frequency, frequency increment, and the number of increments desired.

Format:

1	2	3	4	5	6	7	8	9	10
FREQ1	SID	F1	DF	NDF					

Example:

FREQ1	6	2.9	0.5	13					

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
F1	First frequency in set. (Real ≥ 0.0)
DF	Frequency increment. (Real > 0.0)
NDF	Number of frequency increments. (Integer > 0; Default = 1)

Remarks:

1. FREQ1 entries must be selected with the Case Control command FREQUENCY = SID.
2. The units for F1 and DF are cycles per unit time.
3. The frequencies defined by this entry are given by

   $$f_i = F1 + DF \cdot (i-1)$$

   where i = 1 to (NDF + 1).
4. All FREQi entries with the same frequency set identification numbers will be used. The software automatically searches for duplicate frequencies. When a duplicate is found, the software keeps the lower frequency in the pair and removes the higher frequency.

   f_N and f_{N-1} are considered duplicates if:

   $$|f_N - f_{N-1}| < \text{DFREQ} \cdot |f_{MAX} - f_{MIN}|,$$

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where DFREQ is a user parameter, with a default of 10^{-5}. f_{MAX} and f_{MIN} are the maximum and minimum excitation frequencies of the combined FREQi entries.

5. In modal analysis, solutions for modal DOFs from rigid body modes at zero excitation frequencies may be discarded. Solutions for nonzero modes are retained.

Related Topics

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FREQ2

Frequency List, Alternate Form 2

Defines a set of frequencies to be used in the solution of frequency response problems by specification of a starting frequency, final frequency, and the number of logarithmic increments desired.

Format:

1	2	3	4	5	6	7	8	9	10
FREQ2	SID	F1	F2	NF					

Example:

FREQ2	6	1.0	8.0	6					

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
F1	First frequency. (Real > 0.0)
F2	Last frequency. (Real > 0.0, F2 > F1)
NF	Number of logarithmic intervals. (Integer > 0; Default = 1)

Remarks:

1. FREQ2 entries must be selected with the Case Control command FREQUENCY = SID.
2. The units for F1 and F2 are cycles per unit time.
3. The frequencies defined by this entry are given by:

$$f_i = F1 \cdot e^{(i-1)d}$$

where:

$$d = \frac{1}{NF} \ln \frac{F2}{F1}$$

and $i = 1, 2, \ldots, (NF+1)$.

In the example above, the list of frequencies will be 1.0, 1.4142, 2.0, 2.8284, 4.0, 5.6569 and 8.0 cycles per unit time.

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4. All FREQi entries with the same frequency set identification numbers will be used. The software automatically searches for duplicate frequencies. When a duplicate is found, the software keeps the lower frequency in the pair and removes the higher frequency.

   f_N and f_{N-1} are considered duplicates if:

   $$|f_N - f_{N-1}| < \text{DFREQ} \cdot |f_{MAX} - f_{MIN}|,$$

   where DFREQ is a user parameter, with a default of 10^{-5}. f_{MAX} and f_{MIN} are the maximum and minimum excitation frequencies of the combined FREQi entries.

5. In modal analysis, solutions for modal DOFs from rigid body modes at zero excitation frequencies may be discarded. Solutions for nonzero modes are retained.

Related Topics

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FREQ3

Frequency List, Alternate 3

Defines a set of excitation frequencies for modal frequency-response solutions by specifying number of excitation frequencies between two modal frequencies.

Format:

1	2	3	4	5	6	7	8	9	10
FREQ3	SID	F1	F2	TYPE	NEF	CLUSTER			

Example:

FREQ3	6	20.0	200.0	LINEAR	10	2.0			

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
F1	Lower bound of modal frequency range in cycles per unit time. F1 is required. (Real ≥ 0.0 for TYPE = LINEAR and Real > 0.0 for TYPE = LOG)
F2	Upper bound of modal frequency range in cycles per unit time. If F2=F1, F1 is the only excitation frequency resulting from that FREQ3 entry. (Real > 0.0, F2 ≥ F1, Default = F1)
TYPE	LINEAR or LOG. Specifies linear or logarithmic interpolation between frequencies. (Character; Default = "LINEAR")
NEF	Number of excitation frequencies within each subrange including the end points. The first subrange is between F1 and the first modal frequency within the bounds. The second subrange is between first and second modal frequencies between the bounds. The last subrange is between the last modal frequency within the bounds and F2. (Integer > 1, Default = 10)
CLUSTER	Specifies clustering of the excitation frequency near the end points of the range. See Remark 6 . (Real > 0.0; Default = 1.0)

Remarks:

1. FREQ3 applies only to modal frequency-response solutions (SOLs 111, 146, and 200) and is ignored in direct frequency response solutions.
2. FREQ3 entries must be selected with the Case Control command FREQUENCY = SID.

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3. In the example above, there will be 10 frequencies in the interval between each set of modes within the bounds 20 and 2000, plus 10 frequencies between 20 and the lowest mode in the range, plus 10 frequencies between the highest mode in the range and 2000.

4. Since the forcing frequencies are near structural resonances, it is important that some amount of damping be specified.

5. All FREQi entries with the same frequency set identification numbers will be used. The software automatically searches for duplicate frequencies. When a duplicate is found, the software keeps the lower frequency in the pair and removes the higher frequency.

   f_N and f_{N-1} are considered duplicates if:

   $$|f_N - f_{N-1}| < \text{DFREQ} \cdot |f_{MAX} - f_{MIN}|,$$

   where DFREQ is a user parameter, with a default of 10^{-5}. f_{MAX} and f_{MIN} are the maximum and minimum excitation frequencies of the combined FREQi entries.

6. CLUSTER is used to obtain better resolution near the modal frequencies where the response varies the most. CLUSTER > 1.0 provides closer spacing of excitation frequency towards the ends of the frequency range, while values of less than 1.0 provide closer spacing towards the center of the frequency range. For example, if the frequency range is between 10 and 20, NEF = 11, TYPE = "LINEAR"; then, the excitation frequencies for various values of CLUSTER would be as shown in **Table 14-79**.

   $$\hat{f}_k = \frac{1}{2}(\hat{f}_1 + \hat{f}_2) + \frac{1}{2}(\hat{f}_2 - \hat{f}_1)|\xi|^{1/\text{CLUSTER}} \cdot \text{SIGN}(\xi)$$

   where

ξ	= $-1 + 2(k-1)/(\text{NEF}-1)$ is a parametric coordinate between -1 and 1
k	= varies from 1 to NEF (k = 1, 2, ..., NEF)
$\hat{f}_1$	= is the lower limit of the frequency subrange
$\hat{f}_2$	= is the upper limit o the subrange
$\hat{f}_k$	= is the k-th excitation frequency
$\hat{f}$	= is the frequency, or the logarithm of the frequency, depending on the value specified for TYPE

Table 14-79: CLUSTER Usage Example

Excitation Frequency Number	ξ	CLUSTER				
		c=0.25	c-0.50	c-1.0	c-2.0	c-4.0
		Excitation Frequencies in Hertz				
1	-1.0	10.00	10.0	10.0	10.00	10.00
2	-0.8	12.95	11.8	11.0	10.53	10.27

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Excitation Frequency Number	ξ	CLUSTER				
		c=0.25	c-0.50	c-1.0	c-2.0	c-4.0
		Excitation Frequencies in Hertz				
3	-0.6	14.35	13.2	12.0	11.13	10.60
4	-0.4	14.87	14.2	13.0	11.84	11.02
5	-0.2	14.99	14.8	14.0	12.76	11.66
6	0.0	15.00	15.0	15.0	15.00	15.00
7	0.2	15.01	15.2	16.0	17.24	18.34
8	0.4	15.13	15.8	17.0	18.16	18.98
9	0.6	15.65	16.8	18.0	18.87	19.40
10	0.8	17.05	18.2	19.0	19.47	19.73
11	1.0	20.00	20.0	20.0	20.00	20.00

7. In design optimization (SOL 200), the excitation frequencies are derived from the natural frequencies computed at each design cycle.
8. In modal analysis, solutions for modal DOFs from rigid body modes at zero excitation frequencies may be discarded. Solutions for nonzero modes are retained.

Related Topics

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FREQ4

Frequency List, Alternate Form 4

Defines a set of frequencies used in the solution of modal frequency-response problems by specifying the amount of "spread" around each natural frequency and the number of equally spaced excitation frequencies within the spread.

Format:

1	2	3	4	5	6	7	8	9	10
FREQ4	SID	F1	F2	FSPD	NFM				

Example:

FREQ4	6	20.0	200.0	0.30	21				

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
F1	Lower bound of frequency range in cycles per unit time. (Real ≥ 0.0, Default = 0.0)
F2	Upper bound of frequency range in cycles per unit time. (Real > 0.0, F2 > F1, Default = 1.0E20)
FSPD	Frequency spread, +/- the fractional amount specified for each mode which occurs in the frequency range F1 to F2. (1.0 > Real > 0.0, Default = 0.10)
NFM	Number of evenly spaced frequencies per "spread" mode. (Integer > 0; Default = 3; If NFM is even, NFM + 1 will be used.)

Remarks:

1. FREQ4 applies only to modal frequency-response solutions (SOLs 111, 146, and 200) and is ignored in direct frequency-response solutions.
2. FREQ4 entries must be selected with the Case Control command FREQUENCY=SID.
3. There will be NFM excitation frequencies between (1–*FSPD*) f_N and (1+*FSPD*) f_N, for each natural frequency in the range F1 to F2.
4. In the example above there will be 21 equally spaced frequencies across a frequency band of $0.7 \cdot f_N$ to $1.3 \cdot f_N$ for each natural frequency that occurs between 20 and 2000. See **Figure 14-127** for the definition of frequency spread.

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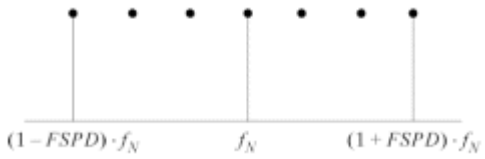
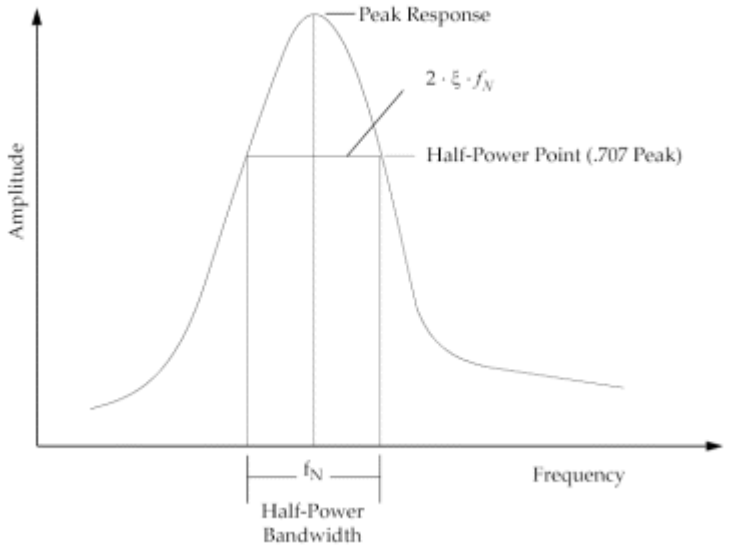
Figure 14-127: Frequency Spread Definition



Excitation frequencies may be based on natural frequencies that are not within the range (F1 and F2) as long as the calculated excitation frequencies are within the range. Similarly, an excitation frequency calculated based on natural frequencies within the range (F1 through F2) may be excluded if it falls outside the range.

5. The frequency spread can be used also to define the half-power bandwidth. The half-power bandwidth is given by $2 \cdot \xi \cdot f_N$, where ξ is the damping ratio. Therefore, if FSPD is specified equal to the damping ratio for the mode, NFM specifies the number of excitation frequency within the half-power bandwidth. See Figure 14-128 for the definition of half-power bandwidth.

Figure 14-128: Half-Power Bandwidth Definition



6. Since the forcing frequencies are near structural resonances, it is important that some amount of damping be specified.

7. All FREQi entries with the same frequency set identification numbers will be used. The software automatically searches for duplicate frequencies. When a duplicate is found, the software keeps the lower frequency in the pair and removes the higher frequency.

   f_N and f_{N-1} are considered duplicates if:

$$\left| f_N - f_{N-1} \right| < \text{DFREQ} \cdot \left| f_{MAX} - f_{MIN} \right|,$$

   where DFREQ is a user parameter, with a default of 10^{-5}. f_{MAX} and f_{MIN} are the maximum and minimum excitation frequencies of the combined FREQi entries.

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8. In design optimization (SOL 200), the excitation frequencies are derived from the natural frequencies computed at each design cycle.
9. In modal analysis, solutions for modal DOFs from rigid body modes at zero excitation frequencies may be discarded. Solutions for nonzero modes are retained.

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FREQ5

Frequency List, Alternate Form 5

Defines a set of frequencies used in the solution of modal frequency-response problems by specification of a frequency range and fractions of the natural frequencies within that range.

Format:

1	2	3	4	5	6	7	8	9	10
FREQ5	SID	F1	F2	FR1	FR2	FR3	FR4	FR5	
	FR6	FR7	-etc.-						

Example:

FREQ5	6	20.0	200.0	1.0	0.6	0.8	0.9	0.95	
	1.05	1.1	1.2						

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
F1	Lower bound of frequency range in cycles per unit time. (Real ≥ 0.0; Default = 0.0)
F2	Upper bound of frequency range in cycles per unit time. (Real > 0.0, F2 > F1, Default = 1.0E20)
FRi	Fractions of the natural frequencies in the range F1 to F2. (Real > 0.0)

Remarks:

1. FREQ5 applies only to modal frequency-response solutions (SOLs 111, 146, and 200) and is ignored in direct frequency response solutions.
2. FREQ5 entries must be selected with the Case Control command FREQUENCY=SID.
3. The frequencies defined by this entry are given by

$$f_i = FRi \cdot f_{N_i}$$

where f_{Ni} are the natural frequencies in the range F1 through F2.

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4. In the example above, the list of frequencies will be 0.6, 0.8, 0.9, 0.95, 1.0, 1.05, 1.1, and 1.2 times each natural frequency between 20 and 2000. If this computation results in excitation frequencies less then F1 and greater than F2, those computed excitation frequencies are ignored.

   Excitation frequencies may be based on natural frequencies that are not within the range (F1 and F2) as long as the calculated excitation frequencies are within the range. Similarly, an excitation frequency calculated based on natural frequencies within the range (F1 through F2) may be excluded if it falls outside the range.

5. Since the forcing frequencies are near structural resonances, it is important that some amount of damping be specified.

6. All FREQi entries with the same frequency set identification numbers will be used. The software automatically searches for duplicate frequencies. When a duplicate is found, the software keeps the lower frequency in the pair and removes the higher frequency.

   f_N and f_{N-1} are considered duplicates if:

   $$|f_N - f_{N-1}| < \text{DFREQ} \cdot |f_{MAX} - f_{MIN}|,$$

   where DFREQ is a user parameter, with a default of 10^{-5}. f_{MAX} and f_{MIN} are the maximum and minimum excitation frequencies of the combined FREQi entries.

7. In design optimization (SOL 200), the excitation frequencies are derived from the natural frequencies computed at each design cycle.

8. In modal analysis, solutions for modal DOFs from rigid body modes at zero excitation frequencies may be discarded. Solutions for nonzero modes are retained.

Related Topics

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FREQH

Harmonics for fan noise

Defines harmonics and sub-harmonics for fan noise.

Format:

1	2	3	4	5	6	7	8	9	10
FREQH	SID	TYPE	HTYPE	PID	START	END	STEP	NSUB	

Example:

FREQH	200	SINGLE	SHAFT	200	2				

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
TYPE	Harmonic or sub-harmonic type (Character; Default=SINGLE) If TYPE=SINGLE, single step. If TYPE=LINEAR, linear step. IF TYPE = LSUB, linear step with sub-harmonics.
HTYPE	The type of harmonic passing definition: SHAFT or BLADE (Character; Default=SHAFT)
PID	Identification number of PACFAN bulk entry. (Integer>0; No default)
START	The start harmonic number. (Integer > 0; Default = 1)
END	The end harmonic number. Only applies when TYPE = LINEAR or LSUB. (Integer>1; No default)
STEP	The harmonic increment. Only applies when TYPE = LINEAR or LSUB. (Integer>0; Default = 1)
nSub	The number of sub-harmonics between the harmonics. Only applies when TYPE = LSUB. (Integer>0; Default= 1)

Remarks:

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Related Topics

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FREQV

Frequency range for VATV response analysis

Defines frequency range of interest for a Vibro-Acoustic Transfer Vector (VATV) response analysis. The exact frequencies are derived from the VATV matrix computed frequencies in the specified frequency range.

Format:

1	2	3	4	5	6	7	8	9	10
FREQV	SID	FMIN	FMAX						

Example:

FREQV	6	10.	600.						

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
FMIN	Lower bound of frequency range in cycles per unit time. (Real ≥ 0.0; Default = 0.0)
FMAX	Upper bound of frequency range in cycles per unit time. (Real > 0.0; FMAX > FMIN; Default = 1.0E20)

Remarks:

1. VATV response analysis will only use frequencies specified by the FREQV card. All other FREQi bulk entries will be ignored.

Related Topics

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FRFFLEX

Direct-Input, Frequency-Dependent Component, Flexibility Entry

Defines the dynamic flexibility matrix at grid and scalar points for a frequency-dependent component in a frequency response analysis.

The matrix is defined by a single header entry and a column entry for each column in the matrix. The matrix must be square. The numerical data that defines the dynamic flexibility vs. frequency is contained in TABLED6 bulk entries. The dynamic flexibility data on the TABLED6 bulk entries can be defined in terms of displacement, velocity, or acceleration per unit force, or angular displacement, velocity, or acceleration per unit moment.

Header Entry Format:

1	2	3	4	5	6	7	8	9	10
FRFFLEX	ID	"0"	TYPE	SYMFLAG	LSCALE	FSCALE	PSCALE	QSCALE	
	EPS								

Column 1 Format:

FRFFLEX	ID	GRID1	COMP1						
	GRID1	COMP1	TID11	GRID2	COMP2	TID21			
	GRID3	COMP3	TID31	-etc.-					

Column 2 Format:

FRFFLEX	ID	GRID2	COMP2						
	GRID1	COMP1	TID12	GRID2	COMP2	TID22			
	GRID3	COMP3	TID32	-etc.-					

Column "n" Format:

FRFFLEX	ID	GRIDn	COMPn						
	GRID1	COMP1	TID1n	GRID2	COMP2	TID2n			
	GRID3	COMP3	TID3n	-etc.-					

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Example:

The following example represents a 4 x 4 unsymmetric dynamic flexibility matrix for a component that is connected to the residual structure at grid 80 (components 1 and 3) and grid 200 (components 2 and 3). Assume the unit basis of the FE model and the FRFFLEX are identical.

FRFFLEX	101	0	VELO	1					
FRFFLEX	101	80	1						
	80	1	11	80	3	21			
	200	2	31	200	3	41			
FRFFLEX	101	80	3						
	80	1	12	80	3	22			
	200	2	32	200	3	42			
FRFFLEX	101	200	2						
	80	1	13	80	3	23			
	200	2	33	200	3	43			
FRFFLEX	101	200	3						
	80	1	14	80	3	24			
	200	2	34	200	3	44			

Fields:

Field	Contents
ID	Unique identification number. (Integer > 0)
TYPE	Type of data in TABLED6 bulk entries. See Remarks 3 and 4. (Integer = 1, 2, or 3, or character: DISP, VELO, or ACCE; Default = 1) TYPE = 1 or DISP for displacement per unit force vs. frequency or angular displacement per unit moment vs. frequency TYPE = 2 or VELO for velocity per unit force vs. frequency or angular velocity per unit moment vs. frequency TYPE = 3 or ACCE for acceleration per unit force vs. frequency or angular acceleration per unit moment vs. frequency
SYMFLAG	Symmetry flag. See Remark 5. (Integer = 0 or 1; Default = 0) SYMFLAG = 0 for a symmetric matrix SYMFLAG = 1 for an unsymmetric matrix

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Field	Contents
LSCALE	Length scaling factor. See Remark 6. (Real > 0.0; Default = 1.0)
FSCALE	Force scaling factor. See Remark 6. (Real > 0.0; Default = 1.0)
PSCALE	Pressure scaling factor. See Remark 6. (Real > 0.0; Default = 1.0)
QSCALE	Acoustic source strength scaling factor. See Remark 6. (Real > 0.0; Default = 1.0)
EPS	Filter for normalized singular values. See Remark 7. (Real > 0.0; Default= 1.0E-11)
GRIDi	Identification number of the i^{th} connection grid or scalar point and component combination. See Remark 8. (Integer > 0; No default)
COMPi	Component of the connection grid or scalar point listed in the GRIDi field. (Any one of the integers 1 through 6 for structural grid points; 1 for fluid grid points; 0, 1, or blank for scalar points; see Remark 9 for default behavior)
TIDij	Identification number of TABLED6 bulk entry that gives the flexibility, c_{ij}. See Remark 10. (Integer > 0; No default)

Remarks:

1. The FRFIN case control command references FRFFLEX bulk entries.
2. FRFFLEX and FRFSTIF bulk entries cannot have the same identification number.
3. The software uses data expressed in terms of displacement per unit force or angular displacement per unit moment for dynamic flexibility. However, the software permits the data in TABLED6 bulk entries to be expressed in terms of displacement, velocity, or acceleration per unit force, or angular displacement, velocity or acceleration per unit moment. Thus, if the TABLED6 data is expressed in terms of velocity per unit force, acceleration per unit force, angular velocity per unit moment, or angular acceleration per unit moment, the software performs the following conversions to obtain data expressed in terms of displacement per unit force or angular displacement per unit moment:
   - If TYPE = 2 or VELO, the software divides the tabular data by $i\omega$.
   - If TYPE = 3 or ACCE, the software divides the tabular data by $-\omega^2$.
4. Angular measures must be in radians and the unit of time used to define velocity, acceleration, angular velocity, and angular acceleration must be the same as the unit of time used to define frequency.

   For example, for angular velocity per unit moment vs. frequency data, if the unit of time is seconds, angular velocity must be in rad/sec and the frequency must be in Hz.
5. If SYMFLAG = 0, specify entries for the lower tridiagonal of the matrix only.
6. Use the LSCALE, FSCALE, PSCALE, and QSCALE fields to convert the units of the FRFFLEX to the units of the FE model. If TYPE = 2 or 3, the software converts the TABLED6 data to displacement per unit force or angular displacement per unit moment prior to unit conversion.

   The relationship between the TABLED6 units, frequency response analysis units, and the scaling factors are as follows:

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$L_{FE\ model}$ = LSCALE · $L_{FRFFLEX}$

$F_{FE\ model}$ = FSCALE · $F_{FRFFLEX}$

$P_{FE\ model}$ = PSCALE · $P_{FRFFLEX}$

$Q_{FE\ model}$ = QSCALE · $Q_{FRFFLEX}$

For example, suppose that the FE model requires the units for dynamic flexibility to be mm/N. If the FRFFLEX data is in in/lbf, use LSCALE = 25.4 and FSCALE = 4.448.

As a second example, suppose that the FE model requires the units for dynamic flexibility to be rad/N-mm. If the FRFFLEX data is in rad/lbf-in, use LSCALE = 25.4 and FSCALE = 4.448.

After the software performs the units conversion, it inverts the direct input flexibility matrix to find the corresponding values for stiffness. It then adds the stiffness values to the dynamic stiffness of the model.

7. The software uses the EPS value when singular value decomposition is selected to invert the dynamic flexibility matrix into a dynamic stiffness matrix. Because the dynamic flexibility matrix may be rank deficient, the software removes near-zero singular values prior to inversion. To do so, after the software calculates the eigenvalues, it normalizes them with respect to the eigenvalue that has the largest value. It then removes any eigenvalue whose normalized value is less than the EPS value. To select singular value decomposition as the inversion method, specify PARAM,FLEXINV,SVD.
8. Only DOF that connect the direct-input, frequency-dependent component to the residual structure are valid GRIDi and COMPi combinations.
9. If the COMPi field corresponds to a GRIDi field that references a structural grid point, you must specify a component number in the COMPi field.

   If the COMPi field corresponds to a GRIDi field that references a fluid grid point, you must enter "1" in the COMPi field.

   If the COMPi field corresponds to a GRIDi field that references a scalar point, you can enter "0" or "1" in the COMPi field or leave the COMPi field blank.
10. Flexibility, c_{ij}, represents the contribution to the displacement, velocity, or acceleration at the i^{th} grid or scalar point and component combination that results from a unit force at the j^{th} grid or scalar point and component combination.

Related Topics

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FRFOMAP

Assigns Output Type to FRFOTM Results

Assigns an output type to the results calculated from FRFOTM bulk entries.

Format:

1	2	3	4	5	6	7	8	9	10
FRFOMAP	ID								
	YTYPE1	YGRID1	YCOMP1	YGRID2	YCOMP2	YGRID3	YCOMP3		
		YGRID4	YCOMP4	-etc.-					
	YTYPE2	YGRID1	YCOMP1	YGRID2	YCOMP2	YGRID3	YCOMP3		
		YGRID4	YCOMP4	-etc.-					
	-etc.-								

Alternate Format:

Use to assign the same type to all output grid / component combinations.

FRFOMAP	ID								
	YTYPE1	"ALL"							

Example:

This is a continuation of the example provided with the FRFOTM bulk entry. The following FRFOMAP defines the output at grid 100 (component 2) as displacement and the output at grid 150 (component 3) as velocity.

FRFOMAP	101								
	DISP	100	2						
	VELO	200	3						

Fields:

Field	Contents
ID	Identification number of the corresponding FRFOTM bulk entry. See Remark 1. (Integer > 0)
YTYPEj	Type of output. (Integer = 0, 1, 2, 3, or 4, or character; Default = 0)

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Field	Contents
	= 0 or NONE for user-defined = 1 or DISP for displacement or pressure = 2 or VELO for velocity = 3 or ACCE for acceleration = 4 or LOAD for load
YGRIDi	Identification number of the i^{th} output grid or scalar point and component combination. See Remark 2. (Integer > 0; No default)
YCOMPi	Component of the grid or scalar point listed in the YGRIDi field. (Any one of the integers 1 through 6 for structural grid points; 1 for fluid grid points; 0, 1, or blank for scalar points; see Remark 3 for default behavior)
ALL	Assigns the specified YTYPEj to all YGRIDi and YCOMPi combinations in the FRFOTM output.

Remarks:

1. Although the use of FRFOMAP bulk entries is optional, an FRFOMAP bulk entry can be specified for each FRFOTM bulk entry.
2. Only YGRIDi and YCOMPi combinations that are listed on the corresponding FRFOTM bulk entry are valid entries. The software assigns YTYPE = NONE to any YGRIDi and YCOMPi combinations that are listed on the corresponding FRFOTM bulk entry, but are not listed on the FRFOMAP bulk entry.
3. If the YCOMPi field corresponds to a YGRIDi field that references a structural grid point, you must specify a component number in the YCOMPi field.

   If the YCOMPi field corresponds to a YGRIDi field that references a fluid grid point, you must enter "1" in the YCOMPi field.

   If the YCOMPi field corresponds to a YGRIDi field that references a scalar point, you can enter "0" or "1" in the YCOMPi field or leave the YCOMPi field blank.

Related Topics

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FRFOTM

Output Transformation Matrix Definition

Defines a frequency-dependent output transformation matrix (OTM) that the software uses during results recovery to calculate results for grid and scalar points that are not part of the FE model as follows:

$$\{y(\omega)\} = [OTM(\omega)]\{x(\omega)\}$$

where $\{x(\omega)\}$ are values derived from the results of the FE solve.

The OTM is defined by a single header entry and a column entry for each column.

Header Entry Format:

1	2	3	4	5	6	7	8	9	10
FRFOTM	ID	"0"	LSCALE	FSCALE	PSCALE	QSCALE			

Column 1 Format:

FRFOTM	ID	XGRID1	XCOMP1	XTYPE1					
	YGRID1	YCOMP1	TID11	YGRID2	YCOMP2	TID21			
	YGRID3	YCOMP3	TID31	-etc.-					

Column 2 Format:

FRFOTM	ID	XGRID2	XCOMP2	XTYPE2					
	YGRID1	YCOMP1	TID12	YGRID2	YCOMP2	TID22			
	YGRID3	YCOMP3	TID32	-etc.-					

Column "n" Format:

FRFOTM	ID	XGRIDn	XCOMPn	XTYPEn					
	YGRID1	YCOMP1	TID1n	YGRID2	YCOMP2	TID2n			
	YGRID3	YCOMP3	TID3	-etc.-					

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Note

The column format for each non-null column begins with an XGRIDj, XCOMPj, and XTYPEj combination. The continuation of the column format lists all of the YGRIDi and YCOMPi combinations that the XGRIDj, XCOMPj, and XTYPEj combination contributes to. YGRIDi and YCOMPi combinations that do not contribute can be omitted from the list. The order that the YGRIDi and YCOMPi combinations are listed is arbitrary.

Example:

The following example represents a 2 row by 4 column OTM where the output at grid 100 (component 2) and grid 150 (component 3) are calculated as a linear combination of the displacements at grid 80 (components 1 and 3) and the velocities at grid 200 (components 2 and 3). Assume that the unit basis of the FE model and the OTM are identical.

FRFOTM	101	0							
FRFOTM	101	80	1	DISP					
	100	2	11	150	3	21			
FRFOTM	101	80	3	DISP					
	100	2	12	150	3	22			
FRFOTM	101	200	2	VELO					
	100	2	13	150	3	23			
FRFOTM	101	200	3	VELO					
	100	2	14	150	3	24			

Fields:

Field	Contents
ID	Unique identification number. (Integer > 0)
LSCALE	Length scaling factor. See Remark 4. (Real > 0.0; Default = 1.0)
FSCALE	Force scaling factor. See Remark 4. (Real > 0.0; Default = 1.0)
PSCALE	Pressure scaling factor. See Remark 4. (Real > 0.0; Default = 1.0)
QSCALE	Acoustic source strength scaling factor. See Remark 4. (Real > 0.0; Default = 1.0)
XGRIDj	Identification number of the j^{th} input grid or scalar point. See Remark 5. (Integer > 0; No default)
XCOMPj	Component of the grid or scalar point listed in the XGRIDj field. See Remark 5. (Any one of the integers 1 through 6 for structural grid points; 1 for fluid grid points; 0, 1, or blank for scalar points; see Remark 6 for default behavior)
XTYPEj	Result type of the XGRIDj and XCOMPj combination. See Remarks 8 and 9. (Integer = 1, 2, or 3, or character; Default = 1)

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Field	Contents
	= 0 or LOAD for force or acoustic source strength = 1 or DISP for displacement or pressure = 2 or VELO for velocity = 3 or ACCE for acceleration
YGRIDi	Identification number of the i^{th} output grid or scalar point. See Remark 5. (Integer > 0; No default)
YCOMPi	Component of the grid or scalar point listed in the YGRIDi field. See Remark 5. (Any one of the integers 1 through 6 for structural grid points; 1 for fluid grid points; 0, 1, or blank for scalar points; see Remark 10 for default behavior)
TIDij	Identification number of TABLED6 bulk entry that corresponds to the i^{th} output grid or scalar point and component combination, and the j^{th} input grid or scalar point, component, and result type combination. See Remark 7. (Integer > 0; No default)

Remarks:

1. The FRFIN case control command references FRFOTM bulk entries.
2. Unlike the FRFFLEX and FRFSTIF bulk entries, the FRFOTM bulk entry does not add stiffness to the model. Its only use is to recover results for the DOF specified by YGRIDi and YCOMPi combinations. The intended use for the FRFOTM bulk entry is to recover results for grid or scalar point, component, and result type combinations on a substructure that is defined by FRFFLEX or FRFSTIF bulk entries. However, other applications are possible.
3. The OTM relates the input at XGRIDj, XCOMPj, and XTYPEj combinations to the output at YGRIDi and YCOMPi combinations as follows:

$$\{y(\omega)\} = [OTM(\omega)]\{x(\omega)\}$$

   The OTM is an M x N matrix, where M is the total number of YGRIDi and YCOMPi combinations and N is the total number of XGRIDj, XCOMPj, and XTYPEj combinations.
4. Use the LSCALE, FSCALE, PSCALE, and QSCALE fields to convert the units of the FE model to the units of the OTM as follows:

   $L_{FE\ model} = \text{LSCALE} \cdot L_{OTM}$

   $F_{FE\ model} = \text{FSCALE} \cdot F_{OTM}$

   $P_{FE\ model} = \text{PSCALE} \cdot P_{OTM}$

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$Q_{FE\ model}$ = QSCALE · Q_{OTM}

For example, enter 1000.0 in the LSCALE field if the unit of length in the FE model is millimeters and the unit of length in the OTM is meters.

5. Any XGRIDj and XCOMPj combination that is used as an input DOF on an FRFOTM bulk entry cannot be used as a YGRIDi and YCOMPi combination on any FRFOTM bulk entry in the input file. In addition, the output from one OTM cannot be the input to another OTM.

6. If the XCOMPj field corresponds to a XGRIDj field that references a structural grid point, you must specify a component number in the XCOMPj field.

   If the XCOMPj field corresponds to a XGRIDj field that references a fluid grid point, you must enter "1" in the XCOMPj field.

   If the XCOMPj field corresponds to a XGRIDj field that references a scalar point, you can enter "0" or "1" in the XCOMPj field or leave the XCOMPj field blank.

7. Angular measures must be in radians and the unit of time used to define velocity, acceleration, angular velocity, and angular acceleration must be the same as the unit of time used to define frequency.

   For example, for moment per unit angular velocity vs. frequency data, if the unit of time is seconds, angular velocity must be in rad/sec and the frequency must be in Hz.

8. XTYPEj = 0 or LOAD is a valid selection only if a FRFFLEX or FRFSTIF bulk entry with the same ID as the FRFOTM bulk entry is present, and the corresponding XGRIDj and XCOMPj combination are DOF that connect a direct-entry, frequency-dependent component to the FE model.

9. If XTYPEj = 0 or LOAD, the software uses the FRFFLEX or FRFSTIF bulk entry with the same ID as the FRFOTM bulk entry and the displacement results from the FE solve to calculate the load to use in the OTM calculation.

   If XTYPEj = 1 or DISP, the software uses:

   - In a structural frequency response analysis, the displacement result from the FE solve.
   - In an acoustic frequency response analysis, the pressure result from the FE solve.

   If XTYPEj = 2 or VELO, the software uses the displacement result from the FE solve multiplied by $i\omega$.

   If XTYPEj = 3 or ACCE, the software uses the displacement result from the FE solve multiplied by $-\omega^2$.

10. If the YCOMPi field corresponds to a YGRIDi field that references a structural grid point, you must specify a component number in the YCOMPi field.

    If the YCOMPi field corresponds to a YGRIDi field that references a fluid grid point, you must enter "1" in the YCOMPi field.

    If the YCOMPi field corresponds to a YGRIDi field that references a scalar point, you can enter "0" or "1" in the YCOMPi field or leave the YCOMPi field blank.

Related Topics

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FRFSTIF

Direct-Input, Frequency-Dependent Component, Stiffness Entry

Defines the dynamic stiffness matrix at grid and scalar points for a frequency-dependent component in a frequency response analysis.

The matrix is defined by a single header entry and a column entry for each column in the matrix. The matrix must be square. The numerical data that defines the dynamic stiffness vs. frequency is contained in TABLED6 bulk entries. The dynamic stiffness data on the TABLED6 bulk entries can be defined in terms of force per unit displacement, velocity, or acceleration, or moment per unit angular displacement, velocity, or acceleration.

Header Entry Format:

1	2	3	4	5	6	7	8	9	10
FRFSTIF	ID	"0"	TYPE	SYMFLAG	LSCALE	FSCALE	PSCALE	QSCALE	

Column 1 Format:

FRFSTIF	ID	GRID1	COMP1						
	GRID1	COMP1	TID11	GRID2	COMP2	TID21			
	GRID3	COMP3	TID31	-etc.-					

Column 2 Format:

FRFSTIF	ID	GRID2	COMP2						
	GRID1	COMP1	TID12	GRID2	COMP2	TID22			
	GRID3	COMP3	TID32	-etc.-					

Column "n" Format:

FRFSTIF	ID	GRIDn	COMPn						
	GRID1	COMP1	TID1n	GRID2	COMP2	TID2n			
	GRID3	COMP3	TID3n	-etc.-					

Example:

The following example represents a 4 x 4 unsymmetric dynamic stiffness matrix for a component that is connected to the residual structure at grid 80 (components 1 and 3) and grid 200 (components 2 and 3). Assume the unit basis of the FE model and the FRFSTIF are identical.

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FRFSTIF	101	0	VELO	1					
FRFSTIF	101	80	1						
	80	1	11	80	3	21			
	200	2	31	200	3	41			
FRFSTIF	101	80	3						
	80	1	12	80	3	22			
	200	2	32	200	3	42			
FRFSTIF	101	200	2						
	80	1	13	80	3	23			
	200	2	33	200	3	43			
FRFSTIF	101	200	3						
	80	1	14	80	3	24			
	200	2	34	200	3	44			

Fields:

Field	Contents
ID	Unique identification number. (Integer > 0)
TYPE	Type of data in TABLED6 bulk entries. See Remarks 3 and 4. (Integer = 1, 2, or 3, or character: DISP, VELO, or ACCE; Default = 1) TYPE = 1 or DISP for force per unit displacement vs. frequency or moment per unit angular displacement vs. frequency TYPE = 2 or VELO for force per unit velocity vs. frequency or moment per unit angular velocity vs. frequency TYPE = 3 or ACCE for force per unit acceleration vs. frequency or moment per unit angular acceleration vs. frequency
SYMFLAG	Symmetry flag. See Remark 5. (Integer = 0 or 1; Default = 0) SYMFLAG = 0 for a symmetric matrix SYMFLAG = 1 for an unsymmetric matrix
LSCALE	Length scaling factor. See Remark 6. (Real > 0.0; Default = 1.0)
FSCALE	Force scaling factor. See Remark 6. (Real > 0.0; Default = 1.0)
PSCALE	Pressure scaling factor. See Remark 6. (Real > 0.0; Default = 1.0)
QSCALE	Acoustic source strength scaling factor. See Remark 6. (Real > 0.0; Default = 1.0)

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Field	Contents
GRIDi	Identification number of the i^{th} connection grid or scalar point and component combination. See Remark 7. (Integer > 0; No default)
COMPi	Component of the connection grid or scalar point listed in the GRIDi field. (Any one of the integers 1 through 6 for structural grid points; 1 for fluid grid points; 0, 1, or blank for scalar points; see Remark 8 for default behavior)
TIDij	Identification number of TABLED6 bulk entry that gives the stiffness, k_{ij}. See Remark 9. (Integer > 0; No default)

Remarks:

1. The FRFIN case control command references FRFSTIF bulk entries.
2. FRFSTIF and FRFFLEX bulk entries cannot have the same identification number.
3. The software uses data expressed in terms of force per unit displacement or moment per unit angular displacement for dynamic stiffness. However, the software permits the data in TABLED6 bulk entries to be expressed in terms of force per unit displacement, velocity, or acceleration, or moment per unit angular displacement, velocity or acceleration. Thus, if the TABLED6 data is expressed in terms of force per unit velocity, force per unit acceleration, moment per unit angular velocity, or moment per unit angular acceleration, the software performs the following conversions to obtain data expressed in terms of force per unit displacement or moment per unit angular displacement:
   - If TYPE = 2 or VELO, the software multiplies the tabular data by $i\omega$.
   - If TYPE = 3 or ACCE, the software multiplies the tabular data by $-\omega^2$.
4. Angular measures must be in radians and the unit of time used to define velocity, acceleration, angular velocity, and angular acceleration must be the same as the unit of time used to define frequency.

   For example, for moment per unit angular velocity vs. frequency data, if the unit of time is seconds, angular velocity must be in rad/sec and the frequency must be in Hz.
5. If SYMFLAG = 0, specify entries for the lower tridiagonal of the matrix only.
6. Use the LSCALE, FSCALE, PSCALE, and QSCALE fields to convert the units of the FRFSTIF to the units of the FE model. If TYPE = 2 or 3, the software converts the TABLED6 data to force per unit displacement or moment per unit angular displacement prior to unit conversion.

   The relationship between the FRFSTIF units, FE model units, and the scaling factors are as follows:

   $L_{FE\ model} = \text{LSCALE} \cdot L_{FRFSTIF}$

   $F_{FE\ model} = \text{FSCALE} \cdot F_{FRFSTIF}$

   $P_{FE\ model} = \text{PSCALE} \cdot P_{FRFSTIF}$

   $Q_{FE\ model} = \text{QSCALE} \cdot Q_{FRFSTIF}$

   For example, suppose that the FE model requires the units for dynamic stiffness to be N/mm. If the FRFSTIF data is in lbf/in, use LSCALE = 25.4 and FSCALE = 4.448.

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As a second example, suppose that the FE model requires the units for dynamic stiffness to be N-mm/rad. If the FRFSTIF data is in lbf-in/rad, use LSCALE = 25.4 and FSCALE = 4.448.

After the software performs the units conversion, it adds the values to the dynamic stiffness of the model.

7. Only DOF that connect the direct-input, frequency-dependent component to the residual structure are valid GRIDi and COMPi combinations.

8. If the COMPi field corresponds to a GRIDi field that references a structural grid point, you must specify a component number in the COMPi field.

   If the COMPi field corresponds to a GRIDi field that references a fluid grid point, you must enter "1" in the COMPi field.

   If the COMPi field corresponds to a GRIDi field that references a scalar point, you can enter "0" or "1" in the COMPi field or leave the COMPi field blank.

9. Stiffness, k_{ij}, represents the contribution to the force at the i^{th} grid or scalar point and component combination that results from a unit displacement, velocity, or acceleration at the j^{th} grid or scalar point and component combination.

Related Topics

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GENEL

General Element

Defines a general element.

Format:

1	2	3	4	5	6	7	8	9	10
GENEL	EID		UI1	CI1	UI2	CI2	UI3	CI3	
	UI4	CI4	UI5	CI5	-etc.-				

UIm – The last item in the UI list will appear in one of fields 2, 4, 6, or 8.

	"UD"		UD1	CD1	UD2	CD2	-etc.-		

UDn –The last item in the UD list will appear in one of fields 2, 4, 6, or 8.

	"K" or "Z"	KZ11	KZ21	KZ31	-etc.-	KZ22	KZ32		
	-etc.-		KZ33	KZ43	-etc.-				

KZmm – The last item in the K or Z matrix will appear in one of fields 2 through 9.

	"S"	S11	S12	-etc.-		S21	-etc.-		

Smn – The last item in the S matrix will appear in one of fields 2 through 9.

Example:

GENEL	629		1	1	13	4	42	0	
	24	2							
	UD		6	2	33	0			
	Z	1.0	2.0	3.0	4.0	5.0	6.0	7.0	
	8.0	9.0	10.0						
	S	1.5	2.5	3.5	4.5	5.5	6.5	7.5	
	8.5								

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Fields:

Field	Contents
EID	Unique element identification number. (Integer > 0)
UIi, CIi, UDj, CDj	Identification numbers of coordinates in the UI or UD list, in sequence corresponding to the [K], [Z], and [S] matrices. UIi and UDi are grid point numbers, and CIi and CDj are the component numbers. If a scalar point is given, the component number is zero. (Integer ≥ 0)
KZij	Values of the [K] or [Z] matrix ordered by columns from the diagonal, according to the UI list. (Real)
Sij	Values of the [S] matrix ordered by rows according to the UD list. (Real)
"UD", "K", "Z", and "S"	Character strings that indicate the start of data belonging to the UD list or the [K], [Z], or [S] matrices.

Remarks:

1. The stiffness approach:

$$\begin{Bmatrix} f_1 \\ f_d \end{Bmatrix} = \begin{bmatrix} K & -KS \\ -S^T K & S^T KS \end{bmatrix} \begin{Bmatrix} u_i \\ u_d \end{Bmatrix}$$

The flexibility approach:

$$\begin{bmatrix} u_i \\ f_d \end{bmatrix} = \begin{bmatrix} Z & S \\ -S^T & O \end{bmatrix} \begin{bmatrix} f_i \\ u_d \end{bmatrix} \begin{bmatrix} u_i \\ f_d \end{bmatrix} = \begin{bmatrix} Z & S \\ -S^T & O \end{bmatrix} \begin{bmatrix} f_i \\ u_d \end{bmatrix} \begin{bmatrix} u_i \\ f_d \end{bmatrix} = \begin{bmatrix} Z & S \\ -S^T & O \end{bmatrix} \begin{bmatrix} f_i \\ u_d \end{bmatrix}$$

where

$\{u_i\} = [u_{i1}, u_{i2}, \ldots, u_{im}]^T$

$\{u_d\} = [u_{d1}, u_{d2}, \ldots, u_{dn}]^T$

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$$[KZ] = [K] \text{ or } [Z] = \begin{bmatrix} KZ11 & \cdots & \cdots & \cdots \\ KZ21 & KZ22 & \cdots & \cdots \\ KZ31 & KZ32 & \cdots & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ KZ_{m1} & \cdots & \cdots & KZ_{mm} \end{bmatrix} \text{ and } [KZ]^T = [KZ]$$

$$[S] = \begin{bmatrix} S11 & \cdots & S_{1n} \\ S21 & \cdots & \cdots \\ S31 & \cdots & \cdots \\ \vdots & \vdots & \vdots \\ S_{m1} & \cdots & S_{mn} \end{bmatrix}$$

The required input is the $\{u_i\}$ list and the lower triangular portion of $[K]$ or $[Z]$. Additional input may include the $\{u_d\}$ list and $[S]$. If $[S]$ is input, $\{u^d\}$ must also be input. If $\{u_d\}$ is input but $[S]$ is omitted, $[S]$ is internally calculated. In this case, $\{u_d\}$ must contain six and only six degrees-of-freedom.

The forms shown above for both the stiffness and flexibility approaches assume that the element is a free body with rigid body motions that are defined by $\{u_i\} = [S]\{u_d\}$. See "General Element Capability (GENEL)" in the *Simcenter Nastran Element Library*.

2. When the stiffness matrix K is input, the number of significant digits should be the same for all terms.
3. Double-field format may be used for input of K or Z.
4. The DMIG entry or the INPUTT4 module offer alternative methods for inputting large matrices.
5. The general element entry in the example above defines the following:

   $[u_i] = [1\text{–}1, 13\text{–}4, 42, 24\text{–}2]^T$

   $\{u_d\} = [6\text{–}2, 33]^T$

   where i-j means the j-th component of grid point i. Points 42 and 33 are scalar points.

$$[Z] = \begin{bmatrix} 1.0 & 2.0 & 3.0 & 4.0 \\ 2.0 & 5.0 & 6.0 & 7.0 \\ 3.0 & 6.0 & 8.0 & 9.0 \\ 4.0 & 7.0 & 9.0 & 10.0 \end{bmatrix} \quad [S] = \begin{bmatrix} 1.5 & 2.5 \\ 3.5 & 4.5 \\ 5.5 & 6.5 \\ 7.5 & 8.5 \end{bmatrix}$$

Remarks for SOL 414:

1. The following configurations are allowed:
   - Stiffness matrix [K] and the U_i list
   - Flexibility matrix [Z] and the U_i list

   The U_i list, U_d list, and stiffness matrix [K] must be used with the [S] matrix.

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Related Topics

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GRAV

Acceleration or Gravity Load

Defines acceleration vectors for gravity or other acceleration loading.

Format:

1	2	3	4	5	6	7	8	9	10
GRAV	SID	CID	A	N1	N2	N3	MB		

Example:

GRAV	1	3	32.2	0.0	0.0	-1.0			

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
CID	Coordinate system identification number. (Integer ≥ 0; Default = 0)
A	Acceleration vector scale factor. (Real)
Ni	Acceleration vector components measured in coordinate system CID. (Real; at least one Ni ≠ 0.0)
MB	Indicates whether the CID coordinate system is defined in the main Bulk Data Section (MB = -1) or the partitioned superelement Bulk Data Section (MB = 0). Coordinate systems referenced in the main Bulk Data Section are considered stationary with respect to the assembly basic coordinate system. See Remark 10 . (Integer; Default = 0)

Remarks:

1. The acceleration vector is defined by **a** = $A \cdot$ **N** where **N** is the vector with components N1, N2, and N3. The magnitude of **a** is equal to *A* times the magnitude of **N**. The static loads generated by this entry are in the direction of **a**.
2. A CID of zero references the basic coordinate system.
3. Acceleration or gravity loads may be combined with “simple loads” (e.g. FORCE, MOMENT, SELOAD) only by specification on a LOAD entry. That is, the SID on a GRAV entry may not be the same as that on a simple load entry.
4. In the static solution sequences, the load set ID must be selected by the Case Control command LOAD = SID. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.

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5. At most 100 GRAV entries can be selected in a given run either by Case Control or the LOAD Bulk Data entry. Multiples or reflections of a given acceleration or gravity load can be economically accomplished by use of the LOAD Bulk Data entry.
6. In cyclic symmetry solution sequences, the T3 axis of the coordinate system referenced in field 3 must be parallel to the axis of symmetry. In dihedral cyclic symmetry (where STYPE = "DIH" on the CYSYM entry), the T1 axis must, in addition, be parallel to Side 1 of segment 1R of the model.
7. For image superelements, the coordinate system must be rotated if the image is rotated relative to its primary superelement.
8. Acceleration or gravity loads do not include effects due to mass on scalar points.
9. The RFORCE entry may be used to specify rotational accelerations.
10. The coordinate systems in the main Bulk Data Section are defined relative to the assembly basic coordinate system which is fixed. This feature is useful when a superelement defined by a partitioned Bulk Data Section is rotated or mirrored and the gravity load is more conveniently defined in terms of coordinates which are fixed. If a superelement defined by a partitioned Bulk Data Section is rotated using an SELOC bulk entry, the GRAV loads do not necessarily update on restart because nothing in the partitioned superelement has changed. To force the GRAV loads to update correctly they can be assigned new SID's in the restart run.
11. When applying a load with a GRAV, ACCEL, or ACCEL1 entry to axisymmetric elements, any component in the radial direction is treated as a radial acceleration load.
12. When a cylindrical or spherical system is selected in the CID field, the software applies the load at each grid point using a local triad computed relative to that location. Therefore, the load direction varies from one grid point to the next. **Figure 12-54** on the CORD1C entry and **Figure 12-56** on the CORD1S entry illustrates how the local triad is computed at a grid point with cylindrical and spherical systems, respectively. Note that before version 2506, the software treated a cylindrical or spherical system selected in the CID field as a Cartesian system, and the resulting load direction was constant at all grid points.

Remarks related to SOLs 402 and 414:

1. MB is ignored.
2. To apply an acceleration or gravity load with constant magnitude, SID is selected by Case Control command LOAD = SID for both static and transient analyses.
3. To apply a time-dependent acceleration or gravity load, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by Case Control command DLOAD.

Related Topics

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GRDSET

GRID Entry Defaults

Defines default options for fields 3, 7, 8, and 9 of all GRID entries.

Format:

1	2	3	4	5	6	7	8	9	10
GRDSET		CP				CD	PS	SEID	

Example:

GRDSET		16				32	3456		

Fields:

Field	Contents
CP	Identification number of coordinate system in which the location of the grid points are defined. (Integer ≥ 0 or blank)
CD	Identification number of coordinate system in which the displacements, degrees-of-freedom, constraints, and solution vectors of the grid point are defined. (Integer ≥ -1 or blank)
PS	Permanent single-point constraints on the grid point. (Any combination of Integers 1 through 6 with no embedded blanks, or blank.)
SEID	Superelement identification number. (Integer ≥ 0 or blank)

Remarks:

1. The contents of fields 3, 7, 8, or 9 of this entry are assumed for the corresponding fields of any GRID entry whose field 3, 7, 8, and 9 are blank. If any of these fields on the GRID entry are blank, the default option defined by this entry occurs for that field. If no permanent single-point constraints are desired, one of the coordinate systems is basic, or the grid is assigned to the residual structure then the default may be overridden on the GRID entry by making one of fields 3, 7, 8, or 9 zero (rather than blank). Only one GRDSET entry may appear in the Bulk Data Section.
2. The primary purpose of this entry is to minimize the burden of preparing data for problems with a large amount of repetition (e.g., two-dimensional pinned-joint problems).
3. At least one of the fields CP, CD, PS, or SEID must be specified.

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Related Topics

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GRID

Grid Point

Defines the location of a geometric grid point, the directions of its displacement, and its permanent single-point constraints.

Format:

1	2	3	4	5	6	7	8	9	10
GRID	ID	CP	X1	X2	X3	CD	PS	SEID	

Example:

GRID	2	3	1.0	-2.0	3.0		316		

Fields:

Field	Contents
ID	Grid point identification number. (Integer > 0)
CP	Identification number of coordinate system in which the location of the grid point is defined. (Integer ≥ 0 or blank*)
X1, X2, X3	Location of the grid point in coordinate system CP. (Real; Default = 0.0)
CD	Identification number of coordinate system in which the displacements, degrees-of-freedom, constraints, and solution vectors are defined at the grid point. In Remarks related to SOLs 402 and 414, see Remark 2. (Integer ≥ -1 or blank *)
PS	Permanent single-point constraints associated with the grid point. (Any of the Integers 1 through 6 with no embedded blanks, or blank*.)
SEID	Superelement identification number. (Integer ≥ 0; Default = 0)

* See the GRDSET entry for default options for the CP, CD, PS, and SEID fields.

Remarks:

1. The grid point identification number must be unique with respect to all other identification numbers defined with GRID, SPOINT, EPOINT, and POINT bulk entries.
2. The meaning of X1, X2, and X3 depends on the type of coordinate system CP as follows (see the CORDij entry descriptions):

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Type	X1	X2	X3
Rectangular	X	Y	Z
Cylindrical	R	θ(degrees)	Z
Spherical	R	θ(degrees)	φ(degrees)

See “Overview of Coordinate Systems in Simcenter Nastran” in the *Simcenter Nastran User’s Guide* for a definition of coordinate system terminology.

3. The collection of all CD coordinate systems defined on all GRID entries is called the global coordinate system. All degrees-of-freedom, constraints, and solution vectors are expressed in the global coordinate system. See the chapter “Understanding Coordinate Systems” in the *Simcenter Nastran User's Guide* for more details.
4. The SEID field can be overridden by use of the SESET entry.
5. CD = -1 defines a fluid grid point (see Defining the Fluid in the *Simcenter Nastran Acoustics Simcenter Nastran User's Guide*). This type of point may only connect the CAABSF, CHEXA, CPENTA, and CTETRA elements to define fluid elements.
6. A zero (or blank if the GRDSET entry is not specified) in the CP and CD fields refers to the basic coordinate system.

Remarks related to SOLs 402 and 414:

1. SEID is ignored.
2. For SOL 414, the solver uses the coordinate system identified in CD for enforced displacements and for boundary conditions. The solver does not use the coordinate system identified in CD to output results in that coordinate system.
   - For contour plot (spatial results), nodal output is always expressed in structural axes.
   - For XY plots (OUTMGT), nodal output is expressed in structural axes by default, but you can define a local coordinate system to express the XY plots in that coordinate system.

   SOL 402 takes the CD into account in the output .op2 file.

Related Topics

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GROUP

Group definition

Defines a group of grid points, elements, or physical properties.

Format:

1	2	3	4	5	6	7	8	9	10
GROUP	ID	Descriptor							
	META	Meta Information							
	TYPEi	List of IDs							

Example:

1	2	3	4	5	6	7	8	9	10
GROUP	10	Assembly AA4							
	META	100 RPM							
	META	Optionally continue the meta data							
	GRID	1	2	3	4	5	6	7	
		8							
	GRID	10	THRU	20					
	GRID	100	THRU	200					
	GRID	341	THRU	360	BY	2			
	ELEM	30	THRU	40					
	PROP	ALL							

Fields:

Field	Contents
ID	Group identification number. (0 < Integer)
META	Optional character data to store with the group definition. META can be continued on multiple continuation lines. The META keyword in the 2nd field must appear on every line where metadata is entered. (Character)
TYPEi	Designates the type of IDs defined in a specific row, or in a set of rows. Multiple TYPE can exist on the same GROUP entry. (Character)

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Field	Contents
	"GRID" Designates the row includes grid point IDs. "ELEM" Designates the row includes element IDs. "PROP" Designates the row includes property IDs.

Remarks:

1. Because of the descriptor field, free field data format is not supported for this bulk entry. For large field fixed format, the maximum number of characters is reduced to 48.
2. Not all of the identification numbers in a THRU range have to correspond to elements, grid points, or physical property IDs. For example, elements 2, 4, 7, and 9 may be selected with 2 THRU 9, even if elements 3, 5, 6, and 8 do not exist. This is called an open set. It should be noted that large open sets can cause higher computational costs.
3. The following restrictions exist when using a THRU range:
   - You cannot include an ID after a THRU range in the same row. For example, the following input will produce an error:

1	2	3	4	5	6	7	8	9	10
GROUP	10	Example of ID after a THRU range							
	ELEM	59	THRU	61	64				

   - You cannot include multiple THRU ranges in the same row. For example, the following input will produce an error:

1	2	3	4	5	6	7	8	9	10
GROUP	10	Example of multiple THRU ranges in the same row							
	ELEM	59	THRU	61	84	THRU	102		

   - You cannot include THRU in column 8. For example, the following input will produce an error:

1	2	3	4	5	6	7	8	9	10
GROUP	10	Example of THRU in column 8							
	ELEM	12	14	18	21	88	THRU	105	

Remarks related to SOLs 402 and 414:

1. Descriptor, META and Meta information are ignored.

Related Topics

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GUST

Aerodynamic Gust Load Description

Defines a stationary vertical gust for use in aeroelastic response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
GUST	SID	DLOAD	WG	X0	V				

Example:

GUST	133	61	1.0	0.	1.+4				

Fields:

Field	Contents
SID	Gust set identification number. (Integer > 0)
DLOAD	Set identification number of a TLOADi or RLOADi entry that defines the time or frequency dependence. (Integer > 0)
WG	Scale factor (gust velocity/forward velocity) for gust velocity. (Real ≠ 0.0)
X0	Streamwise location in the aerodynamic coordinate system of the gust reference point. (Real)
V	Velocity of vehicle. See Remark 5 . (Real > 0.0)

Remarks:

1. The GUST entry must be selected with the Case Control command GUST = SID.
2. The gust angle is in the +z direction of the aerodynamic coordinate system. The value is

   $$WG \cdot T\left(t - \frac{X - X0}{V}\right)$$

   where *T* is the tabular function.
3. In random analysis, a unit gust velocity (WG = 1/velocity) is suggested. The actual rms value is entered on the TABRNDG entry.
4. X0 and V may not change between subcases under one execution.
5. V must be equal to VELOCITY on the AERO Bulk Data entry.

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Related Topics

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INCMOPT

Incompressible Fluid Options

Specifies options for modeling incompressible fluids with fluid elements.

Format:

1	2	3	4	5	6	7	8	9	10
INCMOPT	METHOD	SF	SFTOL	BSIZE	NITER	RBFLMASS	VMESH		
	GOFFSET	EOFFSET							

Example:

INCMOPT	DRYMODES	2.5	5.0			YES	YES		
	1000								

Fields:

Field	Contents
METHOD	Selects the method for the modal calculation with the incompressible fluid. See Remark 5. (Character = "SUBS", "ARPK", or "DRYMODES"; Default = "SUBS") **SUBS** Use the subspace iteration method **ARPK** Use the implicitly restarted Arnoldi method **DRYMODES** Use the dry modes method
SF	Scale factor applied to the frequency range (or number of modes) for the modal calculation with the incompressible fluid to determine the frequency range (or number of modes) for the modal calculation without the incompressible fluid. See Remarks 3 and 6. (Real ≥ 1.5; Default = 2.0)
SFTOL	Tolerance (in degrees) for determining whether a free surface is sufficiently flat to perform the modal analysis with the incompressible fluid. See Remarks 7 and 8. (Real > 0.0; Default = 1.0)
BSIZE	Number of eigenvectors calculated with subspace iteration or implicitly restarted Arnoldi method. See Remarks 9 and 10. (Integer > 0; Default = 30)
NITER	Maximum allowable number of iterations for convergence of subspace iteration or implicitly restarted Arnoldi method. See Remark 9. (Integer > 0; Default = 20)

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Field	Contents
RBFLMASS	Compute rigid body incompressible fluid mass and output it to the .f06 file. (Character: "YES" or "NO"; Default = "NO")
VMESH	Output visualization mesh for a sloshing surface. See Remark 11. (Character = "YES" or "NO", default = "NO")
GOFFSET	Grid ID offset for internally generated PLOTEL visualization element mesh. See Remark 12. (Integer ≥ 0; Default = ID of fluid grid being duplicated plus the maximum ID of the structural grids, fluid grids, and scalar points in the model)
EOFFSET	Element ID offset for internally generated PLOTEL visualization element mesh. See Remark 13. (Integer ≥ 0; Default = maximum ID of the structural and fluid element IDs in the model plus one)

Remarks:

1. If an INCMOPT bulk entry is present in an input file where an incompressible fluid is defined, the software uses the INCMOPT bulk entry. The software ignores the presence of an INCMOPT bulk entry if an incompressible fluid is not defined.
2. The presence of multiple INCMOPT bulk entries in an input file causes a fatal error.
3. When METHOD = SUBS or ARPK, the software ignores the SF field.
4. When METHOD = SUBS or ARPK, on the corresponding EIGRL bulk entry:
   - The ND field must be specified. Otherwise, the software terminates the solve.
   - The V1 and V2 fields are ignored.
5. When METHOD = SUBS and the eigensolution fails, the software automatically retries the eigensolution using the implicitly restarted Arnoldi method. Although the implicitly restarted Arnoldi method is less efficient than subspace iteration, it is stable and an exact numerical solution.
6. The number of modes that the software calculates for the model without the incompressible fluid, as a multiple of the number of modes that the software calculates for the model with the incompressible fluid (as specified by the eigenmethod (EIGR/EIGRL selection)), is as follows:

$$ND_{without} = \text{ceiling}\left(SF \times ND_{with}\right)$$

where the ceiling function rounds the argument to the next highest integer and:

$ND_{without}$	Number of modes to calculate without the incompressible fluid
ND_{with}	Number of modes specified in the ND field of the eigenmethod (EIGR/EIGRL selection)
SF	Value in the SF field

If the eigenmethod (EIGR/EIGRL selection) specifies a frequency range over which to calculate the modes with the incompressible fluid, the software uses the value in the SF field to calculate the frequency range over which to calculate the modes without the incompressible fluid as follows:

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- The lower limit of the frequency range is retained. That is:

  $$F1_{without} = F1_{with}$$

  where:

  $F1_{without}$ Lower frequency range limit to use in the modal calculation without the incompressible fluid

  $F1_{with}$ Lower frequency range limit for the modal calculation with the incompressible fluid as specified in the eigenmethod (EIGR/EIGRL selection)

- The upper limit to the frequency range is calculated as follows:

  $$F2_{without} = SF \times F2_{with}$$

  where:

  $F2_{without}$ Upper frequency range limit to use in the modal calculation without the incompressible fluid

  $F2_{with}$ Upper frequency range limit for the modal calculation with the incompressible fluid as specified in the eigenmethod (EIGR/EIGRL selection)

  SF Value in the SF field

7. For a free surface to be sufficiently flat to perform the modal analysis with the incompressible fluid, the maximum angle difference in the calculated normals between any two grid points on the free surface must be less than the value in the SFTOL field.

   A free surface is defined with a BSURFS bulk entry. For more information, see **INCMPFL**.

8. If SFTOL ≥ 90°, the software does not check the flatness of the free surface.

9. When METHOD = DRYMODES, the software ignores the BSIZE and NITER fields.

10. The number of eigenvectors that the software calculates is the value in the ND field of the eigenmethod (EIGR/EIGRL selection) plus the value specified in the BSIZE field. Unless the default block size creates problems, as a best practice, leave the BSIZE field blank and allow the software to use the default setting.

11. When the model contains a sloshing surface (SURF = SLOSH on the INCMPFL bulk entry), the software outputs the PLOTEL visualization element mesh that it creates for the sloshing surface to the .op2 file. When VMESH = YES, the software also outputs the PLOTEL visualization element mesh to an ASCII file named *<input_file>_plotel.dat* and saves the file in the folder with the other files for the solution.

12. When the software generates a PLOTEL visualization element mesh for a sloshing surface, it duplicates the fluid grid points on the sloshing surface and uses the duplicate grids to define the connectivity of the PLOTEL visualization elements. The grid IDs that the software assigns to the duplicate grid points depend on GOFFSET as follows:

    - If GOFFSET is greater than the maximum ID of the structural grids, fluid grids, and scalar points in the model, the ID for a duplicate grid point is the ID of the fluid grid being duplicated plus GOFFSET.

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- If GOFFSET is 0 or blank, the ID for a duplicate grid point is the ID of the fluid grid being duplicated plus the maximum ID of the structural grids, fluid grids, and scalar points in the model.
- If GOFFSET is nonzero and less than or equal to the maximum ID of the structural grids, fluid grids, and scalar points in the model, the software issues a fatal error message.

13. The software uses EOFFSET to determine the element IDs for the internally generated PLOTEL visualization element mesh as follows:

- If EOFFSET is greater than the maximum ID of the structural and fluid elements in the model, the software begins the numbering of the PLOTEL visualization element IDs at EOFFSET plus one.
- If EOFFSET is 0 or blank, the software begins the numbering of the PLOTEL visualization element IDs at the maximum ID of the structural and fluid element IDs in the model plus one.
- If EOFFSET is nonzero and less than or equal to the maximum ID of the structural and fluid elements in the model, the software issues a fatal error message.

Related Topics

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INCMPFL

Incompressible Fluid Volume

Defines the type of volume for an incompressible fluid.

Format:

1	2	3	4	5	6	7	8	9	10
INCMPFL	SURF	SID	GRAV	SLOSHCMS					

Example:

INCMPFL	SLOSH	10	9810.0	15					

Fields:

Field	Contents
SURF	Specifies whether the incompressible fluid volume has a free surface that is sloshing or non-sloshing, or is completely enclosed by the structure. (Character = "SLOSH", "FREE", or "ENCL"; No default) **SLOSH** Allow sloshing by applying a gravitational potential normal to the free surface **FREE** Disallow sloshing by applying a zero (gauge) pressure constraint to the free surface **ENCL** Treat the fluid as completely enclosed by the structure and without a free surface
SID	Identification number of a BSURFS bulk entry that defines the free surface. See Remark 4. (Integer > 0; No default)
GRAV	Acceleration due to gravity. See Remark 5. (Real > 0.0; No default)
SLOSHCMS	Number of modes retained for the sloshing surface after component mode synthesis (CMS) reduction of the sloshing surface. See Remarks 5, 6, and 7. (Integer > 0; See Remark 8 for default behavior)

Remarks:

1. If an INCMPFL bulk entry is present in an input file where an incompressible fluid volume is defined, the software uses the INCMPFL bulk entry. The software ignores the presence of an INCMPFL bulk entry if an incompressible fluid volume is not defined.

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2. An incompressible fluid volume is a finite element mesh of solid elements such as CHEXA elements whose connectivity is defined by fluid grid points. Fluid grid points are GRID bulk entries with the CD field set to -1.
3. The input file can contain multiple INCMPFL bulk entries. Use an INCMPFL bulk entry for each incompressible fluid volume to ensure proper modes calculation.

   For example, if a structure stores two fluids separately such as with a fuel and an oxidizer, use one INCMPFL bulk entry for the fuel and another INCMPFL bulk entry for the oxidizer.
4. When SURF = ENCL, the software ignores the SID field.
5. When SURF = FREE or ENCL, the software ignores the GRAV and SLOSHCMS fields.
6. After the software performs CMS reduction on the sloshing surface, it then uses the retained modes for the sloshing surface in the wetted modes calculation for the structure.
7. If both the SLOSHCMS field and PARAM,SLOSHCMS,n are specified, the SLOSHCMS field specification takes precedence for the sloshing surface associated with the INCMPFL bulk entry.
8. When the SLOSHCMS field is blank, the default behavior is as follows:
   - If the SLOSHCMS parameter is not specified, the software does not perform CMS reduction on the sloshing surface associated with the INCMPFL bulk entry and retains all of the modes for the sloshing surface.
   - If PARAM,SLOSHCMS,n with n > 0 is specified, the software performs CMS reduction on the sloshing surface associated with the INCMPFL bulk entry and retains n modes for the sloshing surface.
   - If PARAM,SLOSHCMS,0 is specified, the software does not perform CMS reduction on the sloshing surface associated with the INCMPFL bulk entry and retains all of the modes for the sloshing surface.

Related Topics

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IMPERF

SOL 401 Grid Point Position Deltas

Defines grid point position deltas for SOL 401.

Format:

1	2	3	4	5	6	7	8	9	10
IMPERF	SID	CS							
		GRID 1	ΔX1	ΔY1	ΔZ1				
		GRID 2	ΔX2	ΔY2	ΔZ2				
		-etc-							

Example:

IMPERF	2	0							
		1	-0.008	0.0367	0.234				
		2	0.001	-0.0005	0.067				

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
CS	Coordinate system in which deltas are specified. (Integer > 0; Default=0)
GRIDi	Grid point identification number. (Integer > 0)
ΔXi, ΔYi, ΔZi	Grid point position deltas (Real; No default)

Remarks:

1. Only supported in SOL 401.
2. You can use the IMPERF bulk entry to adjust the X,Y,Z location of a grid point with a positive or negative delta.
3. The IMPERF case control command which references a IMPERF bulk entry must be defined globally (above the subcases). As a result, the updated location of the referenced grid points applies to all subcases.
4. The grid point deltas are also referred to as imperfections since they can be used in a nonlinear buckling solution to induce a buckling condition.

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Related Topics

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IMPRADD

SOL 401 Union of Grid Point Position Delta Sets

Defines a union of grid point position delta sets defined with multiple IMPERF bulk entries for SOL 401.

Format:

1	2	3	4	5	6	7	8	9	10
IMPRADD	SID	S	S1	I1	S2	I2	S3	I3	
	S4	I4	-etc-						

Example:

IMPRADD	1	1.000	1.000	1	1.2	2	0.25	3	
	-1.5	7							

Fields:

Field	Contents
SID	Overall set identification number referenced on the IMPERF case control command. (Integer > 0)
S	Overall scale factor (Real, Default=1.0)
Si	Scale factor for individual IMPERF bulk entries. (Real, Default=1.0)
Ii	Set identification numbers of individual IMPERF bulk entries. (Integer > 0)

Remarks:

1. Only supported in SOL 401.
2. SID must be unique to other IMPRADD entries.
3. The IMPRADD entry allows you to combine multiple IMPERF bulk entries into a single set and optionally scale the referenced individual sets either independently or collectively.

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Related Topics

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INCLUDE

Insert External File

Inserts an external file into the input file. The INCLUDE statement may appear anywhere within the input data file.

Format:

INCLUDE ‘filename’

Describer:

filename	External file to be inserted. The ‘*directory_path*/*filename*’ must begin and end with the single quote character.

Example:

The following INCLUDE statement is used to obtain the Bulk Data from another file called MYBULK.DATA:

```
SOL 101
CEND
TITLE = STATIC ANALYSIS
LOAD = 100
BEGIN BULK
INCLUDE ‘MYBULK.DATA’
ENDDATA
```

Remarks:

1. INCLUDE statements may be nested; that is, INCLUDE statements may appear inside the external file. The nested depth level must not be greater than 10.
2. Each line has a 72 character limit. Multiple lines can be used when file names are long. The entire string must begin and end with the single quote character.

   For example,

   D:\folder1\folder2\folder3\folder4\folder5\folder6\folder7\folder8\folder9\mydata.dat

   can be defined on multiple lines:

   ```
   INCLUDE ‘D:\folder1\folder2\folder3\folder4\folder5\folder6\folder7\fold
   er8\folder9\mydata.dat'
   ```


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The following input format is also supported:

```
INCLUDE ‘D:\folder1\folder2\folder3\
         folder4\folder5\
     folder6\folder7\
        folder8\folder9\mydata.dat'
```

Related Topics

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INITADD

Combines multiple initial stress or initial strain sets in SOL 401

Combines multiple initial stress or initial strain sets in SOL 401.

Format:

1	2	3	4	5	6	7	8	9	10
INITADD	SID								
	SID1	SID2	SID3	SID4	SID5	SID6	SID7	SID8	
	SID9	SID10	-etc.-						

Example:

INITADD	80								
	100	101	102	103	104	105	106	107	
	108								

Fields:

Field	Contents
SID	Set identification number for INITADD bulk entry. (Integer > 0)
SIDi	Set identification number of INITS bulk entries. (Integer > 0)

Remarks:

1. The ID of all INITS and INITADD entries must be unique.
2. The INITS case control must be defined above the subcases (globally). If you want to include the effects of initial stress or strain in a modal subcase, you should include a static subcase before the sequentially dependent modal subcase.
3. The INITS entries selected by the INITADD entry must be all TYPE=STRESS or all TYPE=STRAIN. You cannot mix initial stress and initial strain definitions in the same input file.
4. If data is defined on the same grid or element corner location, a fatal error will occur.

Related Topics

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INITS

Defines initial stress or strain state in SOL 401.

Defines initial stress or strain state in SOL 401.

Format of initial stress defined on grid points (TYPE=STRESS and LOC=GRID):

1	2	3	4	5	6	7	8	9	10
INITS	SID	STRESS	GRID	CSYS					
	GRID ID	Sxx	Syy	Szz	Sxy	Syz	Szx		
	GRID ID	Sxx	Syy	Szz	Sxy	Syz	Szx		
	...	...	...	...	...	...	...		

Format of initial stress defined at element grid locations (TYPE=STRESS and LOC = NOE):

1	2	3	4	5	6	7	8	9	10
INITS	SID	STRESS	NOE	CSYS					
	Elem ID	GRID ID	Sxx	Syy	Szz	Sxy	Syz	Szx	
	Elem ID	GRID ID	Sxx	Syy	Szz	Sxy	Syz	Szx	
	...	...	...	...	...	...	...	...	

Format of initial strain defined on grid points (TYPE=STRAIN and LOC=GRID):

1	2	3	4	5	6	7	8	9	10
INITS	SID	STRAIN	GRID	CSYS					
	GRID ID	Exx	Eyy	Ezz	Exy	Eyz	Ezx		
	GRID ID	Exx	Eyy	Ezz	Exy	Eyz	Ezx		
	...	...	...	...	...	...	...		

Format of initial strain at element grid locations (TYPE=STRAIN and LOC = NOE):

1	2	3	4	5	6	7	8	9	10
INITS	SID	STRAIN	NOE	CSYS					
	Elem ID	GRID ID	Exx	Eyy	Ezz	Exy	Eyz	Ezx	
	Elem ID	GRID ID	Exx	Eyy	Ezz	Exy	Eyz	Ezx	
	...	...	...	...	...	...	...	...	

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Example:

1	2	3	4	5	6	7	8	9	10
INITS	SID	STRAIN	NOE	0					
	12	3	.0045	.0342	.00015				
	12	28	.00012	.0035	.00015				
	12	65	.00012	.0035	.00015				
	12	72	.00012	.0035	.00015				

Fields:

Field	Contents
SID	Initial stress or strain set identification number. (Integer>0; also see Remark 3)
TYPE	Defines if the defined data is initial stress or initial strain. There is no default value for this field. This field must be defined. (Character) = STRAIN for initial strain. = STRESS for initial stress.
LOC	The location where the data is defined. There is no default value for this field. This field must be defined. (Character) = GRID for data on grid points. = NOE for data on an element at grid locations. This can include corner and/or midside grid locations.
CSYS	Selects a coordinate system to define the initial stress or strain orientation. (Integer) = -1 Material coordinate system is used. = 0 Basic coordinate system is used. (Default) > 0 Selects a specific CSYS.
Exx,Eyy,Ezz, Exy,Eyz,Ezx	Engineering normal strains and engineering shear strains. (Real)
Sxx,Syy,Szz, Sxy,Syz,Szx,	Engineering normal stresses and engineering shear stresses. (Real)
When LOC=GRID:	

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Field	Contents
GRID ID	Grid point ID where data is applied. (Integer>0)
When LOC=NOE:	
Elem ID	Element where the data is applied. (Integer>0)
GRID ID	Grid point ID defining a location on the element. (Integer>0)

Remarks:

1. The ID of all INITS and INITADD entries must be unique.
2. The option to define an initial stress or strain is available on all elements in SOL 401 except for the plane strain elements including the generalized plane strain elements, solid composite elements defined with the PCOMPS entry, and rigid elements.
3. The INITS case control must be defined above the subcases (globally), and is reapplied in each static subcase. If you want to include the effects of initial stress or strain including any other load in a normal modes subcase, you should include a static subcase before the sequentially dependent normal modes subcase.
4. For the plane stress and axisymmetric elements, if you select a coordinate system other than the basic system in the CSYS field on the INITS entry, the software first transforms the data into the basic system, then uses the components consistent with the formats described below.

   For the plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8, both in-plane and out-of-plane initial *strain* values are used by the software. Although, only in-plane initial *stress* values are used. For example, when the plane stress elements are defined on the XY plane and the basic coordinate system is used to define the initial stress or strain, Sxx, Syy, Sxy or Exx, Eyy, Ezz, Exy should be defined. For elements defined on the XZ plane, Sxx, Szz, Szx or Exx, Eyy, Ezz, Ezx should be defined.

   For the axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6, in-plane (radial and axial) and out-of-plane (theta) initial stress or strain values are used by the software. For example, when the axisymmetric elements are defined on the XY plane and the basic coordinate system is used to define the initial stress or strain, Sxx, Syy, Szz, Sxy or Exx, Eyy, Ezz, Exy should be defined. For elements defined on the XZ plane, Sxx, Syy, Szz, Szx or Exx, Eyy, Ezz, Ezx should be defined.
5. Initial stress and strain can be defined on a subset of the model. The software assumes a value of 0.0 at the locations where data is undefined. An exceptiion is when data is undefined at a mid-side grid point, and data is defined at both or either of the related corners. In this case, the software will interpolate a value for that mid-side grid point.
6. The software converts initial stress to strain using the elastic modulus. Stress/strain data and the yield point on the MATS1 bulk entry, if defined, are not used to convert stress to strain.
7. Multiple INITS bulk entries can be defined, each with a unique ID, then combined with the INITADD entry. The INITADD entry is selected with the ID on the INITS case control. The INITS entries selected by the INITADD entry must be all TYPE=STRESS or all TYPE=STRAIN. As a result, you cannot mix initial stress and initial strain definitions in the same input file.
8. If data is defined on the same grid or element location, a fatal error will occur.

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Related Topics

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INITSO

Defines initial strain offsets in SOL 401.

Defines initial strain offsets in SOL 401.

Format:

Format of initial offset strain defined on grid points (TYPE=STRAIN and LOC=GRID):

1	2	3	4	5	6	7	8	9	10
INITS	SID	STRAIN	GRID	CSYS					
	GRID ID	Exx	Eyy	Ezz	Exy	Eyz	Ezx		
	GRID ID	Exx	Eyy	Ezz	Exy	Eyz	Ezx		
	...	...	...	...	...	...	...		

Format of initial offset strain at element grid locations (TYPE=STRAIN and LOC = NOE):

1	2	3	4	5	6	7	8	9	10
INITS	SID	STRAIN	NOE	CSYS					
	Elem ID	GRID ID	Exx	Eyy	Ezz	Exy	Eyz	Ezx	
	Elem ID	GRID ID	Exx	Eyy	Ezz	Exy	Eyz	Ezx	
	...	...	...	...	...	...	...	...	

Example:

1	2	3	4	5	6	7	8	9	10
INITS	3	STRAIN	NOE	0					
	12	3	.0045	.0342	.00015				
	12	28	.00012	.0035	.00015				
	12	65	.00012	.0035	.00015				
	12	72	.00012	.0035	.00015				

Fields:

Field	Contents
SID	Initial stress or strain set identification number. (Integer>0; also see Remark 3)
TYPE	Offset strains can only be defined as TYPE=STRAIN. This field must be defined. (Character)

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Field	Contents
LOC	The location where the data is defined. There is no default value for this field. This field must be defined. (Character) = GRID for data on grid points. = NOE for data on an element at grid locations. This can include corner and/or midside grid locations.
CSYS	Selects a coordinate system to define the initial stress or strain orientation. (Integer) = -1 Material coordinate system is used. = 0 Basic coordinate system is used. (Default) > 0 Selects a specific CSYS.
Exx,Eyy,Ezz, Exy,Eyz,Ezx	Engineering offset strains representing the symmetric portion of the strain tensor. (Real)
When LOC=GRID:	
GRID ID	Grid point ID where data is applied. (Integer>0)
When LOC=NOE:	
Elem ID	Element where the data is applied. (Integer>0)
GRID ID	Grid point ID defining a location on the element. (Integer>0)

Remarks:

1. The ID of all INITSO and INITADD entries must be unique.
2. Multiple INITSO bulk entries can be defined, each with a unique ID, then combined with the INITADD entry. The INITADD entry is selected with the ID on the INITSO case control. The INITADD entry must include either all INITSO, or all INITS.
3. The option to define an initial stress or strain is available on all elements in SOL 401 except for the plane strain elements including the generalized plane strain elements, solid composite elements defined with the PCOMPS entry, rigid, beam, and shell elements.
4. The INITSO case control must be defined above the subcases (globally), and is reapplied in each static subcase. If you want to include the effects of initial stress or strain including any other load in a normal modes subcase, you should include a static subcase before the sequentially dependent normal modes subcase.
5. For the plane stress and axisymmetric elements, if you select a coordinate system other than the basic system in the CSYS field on the INITS entry, the software first transforms the data into the basic system, then uses the components consistent with the formats described below.

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For the plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8, both in-plane and out-of-plane initial *strain* values are used by the software.

For the axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6, in-plane (radial and axial) and out-of-plane (theta) initial offset strain values are used by the software.

6. Initial offset strains can be defined on a subset of the model. The software assumes a value of 0.0 at the locations where data is undefined. An exceptiion is when data is undefined at a mid-side grid point, and data is defined at both or either of the related corners. In this case, the software will interpolate a value for that mid-side grid point.

Related Topics

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IPLANE

Infinite Plane Definition

Defines infinite plane for acoustic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
IPLANE	IPID	TYPE	SID	CID	CSP				

Example:

IPLANE	100		20						

IPLANE	200		0	10	1				

Fields:

Field	Contents
IPID	Identification number of infinite plane. (Integer > 0; No default)
TYPE	Type of reflective plane. See Remark 1. (Integer 0 or 1; Default = 0)
	0 = symmetric
	1 = anti-symmetric
SID	Surface region identification number. (Integer ≥ 0; See Remark 2 for default behavior)
CID	Rectangular coordinate system identification number. (Integer ≥ 0; Default = 0 for basic coordinate system)
CSP	Plane of coordinate system. (Integer 1, 2, or 3; No default)
	1 = YZ
	2 = ZX
	3 = XY

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Remarks:

1. The TYPE field designates the infinite plane as a zero velocity or a zero pressure acoustic reflection boundary. You can select up to three infinite planes with the AMLREG entry. If you select multiple infinite planes, they must be perpendicular to one another.
   - When TYPE = 0, the infinite plane is a rigid, reflective boundary in which the velocity is zero. This is also known as a symmetric acoustic boundary. For example, air-to-ground is a zero velocity reflective boundary.
   - When TYPE = 1, the infinite plane is a pressure release reflective boundary in which the pressure is zero. This is also known as an anti-symmetric acoustic boundary. For example, air-to-water is a zero pressure reflective boundary.
2. The infinite plane is defined as follows:
   - SID > 0

     When SID > 0, the SID references a BSURFS bulk entry containing element faces that are used to define the flat infinite plane. The referenced BSURFS entry **must** contain element faces in a flat plane.
   - SID = 0 or blank

     When SID = 0 or blank, a rectangular coordinate system defined by the CID and CSP fields is used to define the flat infinite plane.

   If SID > 0, and the CID and CSP fields are specified, the SID specification takes precedence.
3. The AML region defined by the AMLREG bulk entry (that references this IPLANE entry) **must** not protrude through the infinite plane.

Related Topics

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ITER

Iterative Solver Options

Defines options for the global iterative solver in SOLs 101, 106, 108, 111, 153 and the CASI element iterative solver in SOLs 101 and 401.

Format:

1	2	3	4	5	6	7	8	9	10
ITER	SID								
	OPTION1VALUE1 OPTION2VALUE2 -etc.-								

Example:

ITER	100		
	ITSEPS=1.0E0-7, MSGFLG=YES, PRECOND=BICWELL, IPAD=3		

Fields:

Field	Contents	
Fields for both the global and element iterative solution:		
SID	Set identification number. (Integer > 0).	
PRECOND	Preconditioner option for global and CASI element iterative solution. (Global default = “BIC” for real analysis and “BICCMPLX” for complex analysis. Element default is to leave option blank.) See Remark 5.	
	Global iterative solution preconditioner options:	
	J	Jacobi
	JS	Jacobi with diagonal scaling.
	C	Incomplete Cholesky.
	CS	Incomplete Cholesky with diagonal scaling.
	RIC	Reduced incomplete Cholesky.
	RICS	Reduced incomplete Cholesky with diagonal scaling.
	BIC	Block incomplete Cholesky for real problems.
	BICCMPLX	Block incomplete Cholesky for complex problems.
	USER	User given preconditioning. For direct frequency response (SOL 108): a decomposition will be done for 1st frequency and the factor will

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Field	Contents	
		be used for all subsequent frequencies as a preconditioner with the iterative solver. Other solutions require a DMAP alter. Please refer to the *Simcenter Nastran Numerical Methods User's Guide* description of the SOLVIT module.
	CASI element iterative solution preconditioner options:	
	STD	Standard preconditioning option for the element iterative solver.
	PRIMAL	Optional preconditioning option for the element iterative solver. This option may produce faster convergence for models which contain a large number of elements with bad aspect ratios.
	blank	The default preconditioning option is used for the element iterative solver: – The STD option is used by default when contact conditions are *not* defined. – The PRIMAL option is used by default when contact conditions are defined.
CONV	Convergence criterion. (Character; Default = "AREX")	
	Global iterative solution convergence options:	
	AR	$\|\|r\|\|/\|\|b\|\|$ where r is the residual vector of current iteration and b is the initial load vector; internal criterion.
	GE	Alternative convergence criterion using geometric progression and the differences between two consecutive solution updates; internal criterion.
	AREX	Same criterion as AR but with the additional consideration of the external convergence criterion. See Remark 4 . (Default).
	GEEX	Same criterion as GE but with the additional consideration of the external convergence criterion. See Remark 4 .
	CASI element iterative solution convergence options:	
	AR	For relative reduction in the residual 2-norm.
	GE	For relative reduction in the residual preconditioned norm.
	AREX	For one of the other or energy norm of error. Also each must fall below a looser stopping threshold (Default).
	GEEX	For relative reduction in the residual infinity norm.
MSGFLG	Message flag. (Character; Default = "NO")	
	YES	Messages will be printed for each iteration.
	NO	Only minimal messages will be printed from the iterative solver (Default).

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Field	Contents
ITSEPS	User-given convergence parameter epsilon for the global and element iterative solvers. (Real>0.0; Global Iterative Solver Default=1.0E-6; CASI Element Iterative Solver Default=1.0E-8) See remark 7 for information on setting ITSEPS with the element iterative solver.
ITSMAX	Maximum number of iterations for the global and element iterative solvers. (Integer > 0; Global Iterative Solver Default = N/4 where N is the number of rows in the matrix; Element Iterative Solver Default = 1000.) See Remark 1.
Fields specific to the global iterative solution:	
IPAD	Global iterative solution padding value for RIC, RICS, BIC, and BICCMPLX preconditioning. The default values are recommended. (Integer)
	Default = 0 PRECOND = "RIC" or "RICS"
	Default = 2 PRECOND = "BIC" and purely three-dimensional models.
	Default = 3 PRECOND = "BIC" and mixed element models.
	Default = 5 PRECOND = "BICCMPLX".
IEXT	Global iterative solution extraction level in reduced or block incomplete Cholesky factorization. The default values are recommended. (Integer = 0–7)
	Default = 0 Purely three-dimensional model. Requires USET and SIL.
	Default = 1 Purely two-dimensional model. Heuristic block structure.
	Default = 2 Mixed element types including rigid elements.
PREFONLY	Specifies early termination of the global iterative solver. (Integer = 0 or -1; Default = 0)
	0 Runs to completion (Default).
	-1 Terminates after preface giving resource estimates.
Field specific to the CASI element iterative solver:	
ZPIVOT	Used to adjust the singularity tolerance during the preconditioner phase of an element iterative solution. (Real>0; Default=1.0e-9, Default=1.0e-10 when contact (BCSET) or glue (BGSET) conditions exist)

Remarks:

1. The global iterative solver is supported in solutions 101, 106, 108, 111, and 153. The CASI element iterative solver is supported in solutions 101 and 401. The ITER bulk entry is optional and only required to override the defaults. The ITER bulk entry is selected by the SMETHOD = SID case control command. Note that, if the ITER bulk entry is defined, the default CONV for the serial global iterative solver in SOLs 108 and 111 is GEEX instead of AREX.
2. The global iterative solver is requested by specifying ITER=YES on the NASTRAN statement.
3. The input requirements to request the CASI element iterative solver are summarized as follows.

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- To request the CASI element iterative solver for SOL 101, you define both ITER=YES and ELEMITER=YES on the NASTRAN statement.
- To request the CASI element iterative solver for SOL 401, you define SOLVER=ELEMITER on the NLCNTL entry.

See **system cell 216**(ITER) and **system cell 399**(ELEMITER) on the NASTRAN statement.

4. The global and CASI element iterative solvers compute the external epsilon as follows:

$$\varepsilon = \frac{(r, x)}{(b, x)}$$

where r is the final residual vector, x is the final solution vector and b is the initial load vector (r, x) indicates the inner product of r and x and (b, x) indicates the inner product of b and x.

5. When using the global iterative solver with the SMP or DMP parallel solution options, USER preconditioning is recommended with SMP, and Jacobi preconditioning with DMP.

6.

CASI element iterative solver restrictions in SOL 101
Superelements are not supported.
The CQUAD8, CTRIA6, and RBE2 elements are only supported if the parameter setting ELITASPC=YES is defined.
The CQUAD4 and CTRIA3 elements are supported only if the parameter K6ROT is defined (results obtained will be the same as if the parameter snorm = 0.0 was entered).
Inertia relief is not supported.
DMIG is not supported.
Differential stiffness is not supported.
The combination of a bolt preload in which the bolt is meshed with solid elements, contact conditions, and the element iterative solver is not supported.
If you define the BOLT bulk entry using the ETYPE=2 bolt, instead of the glue based approach, the software will revert to an MPC approach.
GROUNDCHECK and WEIGHTCHECK case control requests are not supported.
The DMP parallel solution options are not supported. The SMP parallel solution option is supported.

CASI element iterative solver restrictions in SOL 401
The combination of a bolt preload in which the bolt is meshed with solid elements and contact conditions is not supported.
GROUNDCHECK and WEIGHTCHECK case control requests are not supported.
The DMP parallel solution options are not supported. The SMP parallel solution option is supported.

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7. The CASI element iterative solver has a default convergence tolerance of 1.0E-8. The equation to be solved is $Kx=b$, the residual is $r=b-K(x_0)$, where x_0 is the current solution. The solution stops iterating when the minimum of the following four error measures is below the specified value:

   - The relative reduction in the 2-Norm of the residual (the 2-Norm of r divided by the 2–Norm of b).
   - The relative reduction in the preconditioned norm of the residual.
   - The relative reduction in the infinity norm of the residual.
   - The estimated energy norm of the error.

   For each error measure, the software also applies a looser tolerance than the tolerance you define here.

   This table shows how the convergence tolerance affects the accuracy of results:

Convergence Tolerance	Typical Results
1e-5	Most displacement within 1% accuracy. Unreliable stresses.
1e-6	All displacements within 1% accuracy. Some stresses off 10–20%.
1e-7	Some stresses off as much as 5%.
1e-8	All stresses (even the small values) within 1%.

8. For the continuation rows defined for either the global or the element iterative solver, no commas can appear in columns 1 through 8, and the data in fields 2 through 9 must be specified in columns 9 through 72. The large-field format is not allowed.

Related Topics

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JCON

Kinematic Joint Constraint (SOL 402 and SOL 414,129)

Defines fixation times or liberation times for kinematic joints

Format 1: (Formats 1 and 2 cannot be combined on the same line)

1	2	3	4	5	6	7	8	9	10
JCON	JCID	GRPID	TYPE1	T					

Format 2

1	2	3	4	5	6	7	8	9	10
JCON	JCID	GRPID	TYPE21	T1	T2	T3			
			TYPE22	R1	R2	R3			

Example1:

JCON	111	203	JNTFIX	0.1					

Example2:

JCON	111	203	REMFXDT	0.1	0.2	0.3			
			REMFXDR	0.4	0.5	0.6			

Fields:

Field	Contents
JCID	Identification number. See Remark 1. (Integer > 0).
GRPID	Group identification number. See Remark 2 and Remark 3.(Integer > 0; Default = EID).
TYPE1	String to define a JNTFIX joint fixation constraint or a JNTLIB joint liberation constraint. (Character).
T	Fixation time for JNTFIX or release time for JNTLIB constraint type. (Real >0.0) Only applies for a JNTFIX or JNTLIB constraint type.
TYPE21	String to define a REMFXDT fixation constraint in translation or a REMFXDR fixation constraint in rotation. (Character).

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Field	Contents
	Applies only for FIXED joint types. For more information, see the **PJOINT** entry.
TYPE22	String to define an additional REMFXDR fixation constraint in rotation or an additional REMFXDT fixation constraint in translation when TYPE11 already defines a REMFXDT or a REMFXDR respectively. (Character). (Optional). Applies only for FIXED joint types. For more information, see the **PJOINT** entry.
T1, T2, T3	Fixation times for the three REMFXDT translation DOFs or for the three REMFXDR rotation DOFs. (Real > 0.0) T2 and T3 are optional.
R1, R2, R3	Additional fixation times for REMFXDT or REMFXDR constraint types. (Real > 0.0) R2 and R3 are optional.

Remarks:

1. JCON entry must be selected in a JCONSET control case command.
2. The GROUP selected by the GRPID field must be one of the following types:
   - TYPE "ELEM" and contain CJOINT kinematic joints.
   - TYPE "PROP" and contain PJOINT kinematic joint properties.
3. For REMFXDT and REMFXDR constraint types, the group must only contain FIXED joints.

   For more information, see the **PJOINT** entry.

Related Topics

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JCONADD

Kinematic Joint Constraint Combination (SOL 402 and SOL 414,129)

Defines a kinematic joint constraint set as a union of single kinematic joint constraint sets defined in JCON bulk entries.

Format:

1	2	3	4	5	6	7	8	9	10
JCONADD	JCID	J1	J2	J3	J4	J5	J6	J7	
	J8	J9	-etc-						

Example:

JCONADD	101	2	3	9					

Fields:

Field	Contents
JCID	Combination set identification number. (Integer > 0)
Ji	Identification numbers of constraint sets defined via JCON bulk entries. (Integer > 0; JCID ≠ Ji)

Remarks:

1. Combination sets must be selected with the case control command JCONSET=JCID.
2. No Ji may be the identification number of a constraint set defined by another JCONADD bulk entry.
3. The Ji values must be unique.
4. JCONADD bulk entries take precedence over JCON bulk entries. If both have the same set ID, only the JCONADD bulk entry is used.

Related Topics

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LOAD

Static Load Combination (Superposition)

Defines a static load as a linear combination of load sets defined with ACCEL, ACCEL1, DAREA, DRIVER, FORCE, FORCE1, FORCE2, GRAV, MOMENT, MOMENT1, MOMENT2, PLOAD, PLOAD1, PLOAD2, PLOAD4, PLOADE1, PLOADX1, QBDY1, QBDY2, QBDY3, QHBDY, QVECT, QVOL, RFORCE, RFORCE1, SELOAD, and SLOAD entries.

Format:

1	2	3	4	5	6	7	8	9	10
LOAD	SID	S	S1	L1	S2	L2	S3	L3	
	S4	L4	-etc.-						

Example:

LOAD	101	-0.5	1.0	3	6.2	4			

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
S	Overall scale factor. (Real)
Si	Scale factor on Li. (Real)
Li	Load set identification numbers of the individual ACCEL, FORCE, PLOAD and so on entries. (Integer > 0)

Remarks:

1. The load vector $\{P\}$ that a LOAD entry produces is the linear combination of the load sets $\{P_{Li}\}$ for the individual ACCEL, FORCE, PLOAD and so on entries. Thus, $\{P\}$ can be written mathematically as follows:

$$\{P\} = S\left(\sum_i Si\,\{P_{Li}\}\right)$$

where S is the overall scale factor and Si are the scale factors for the individual load sets.

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2. The load set identification number for each individual load set (Li) must be unique.
3. ACCEL, ACCEL1, and GRAV entries cannot have the same set identification number as any other loading entries. If you want to combine these with other static loads, you must use the LOAD bulk entry.
4. In the static solution sequences, the load set ID must be selected by the Case Control command LOAD=SID. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
5. A LOAD entry may not reference a set identification number defined by another LOAD entry.
6. Because centrifugal forces are not linear in angular velocity and therefore cannot be superposed, include only a single RFORCE or RFORCE1 entry that defines angular velocity on a LOAD entry.

   For example, both of the following RFORCE specifications appear to define the same angular velocity.

```
RFORCE, 11, 999, .159, 1., 0., 0.
RFORCE, 12, 999, .159, 0., 1., 0.
RFORCE, 13, 999, .159, 0., 0., 1.
```

```
RFORCE, 11, 999, .159, 1., 1., 1.
```

   However, if you include the three RFORCE entries on a LOAD entry, the centrifugal force that the software calculates is different from the correct centrifugal force that the software calculates if you include the single RFORCE entry on the LOAD entry.

Remarks related to SOLs 402 and 414:

1. LOAD may be used in a static or transient analysis to combine loads with constant magnitude.
2. Both LOAD and DLOAD can be used in the same analysis to define some loads with constant magnitude and some time-dependent loads.

Related Topics

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LOADCYH

Harmonic Load Input for Cyclic Symmetry

Defines the harmonic coefficients of a static or dynamic load for use in cyclic symmetry analysis.

Format:

1	2	3	4	5	6	7	8	9	10
LOADCYH	SID	S	HID	HTYPE	S1	L1	S2	L2	

Example:

LOADCYH	10	1.0	7	C	0.5	15			

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
S	Scale Factor. (Real)
HID	Harmonic. See Remark 6. (Integer or blank)
HTYPE	Harmonic type. (Character: “C”, “S”, “CSTAR” “SSTAR”, “GRAV”, “RFORCE”, or blank)
Si	Scale factor on Li. (Real)
Li	Load set identification number. See Remark 10. (Integer > 0)

Remarks:

1. The LOADCYH entry is selected with the Case Control command LOAD = SID.
2. If HTYPE is blank, the load will be applied to all applicable types in the problem.
3. If HTYPE is “GRAV” or “RFORCE”, GRAV or RFORCE entry loading will be used. Harmonic loads for appropriate available harmonics will be generated automatically in these cases.
4. L1 and L2 may reference LOAD entries. However, the LOAD entry in such a case must not reference load sets defined via RFORCE and/or GRAV entries.
5. If L1 refers to a set ID defined by an SPCD entry, the same ID must not reference any nonzero loading specified by the other Bulk Data loading entries given via FORCE, MOMENT, FORCE1, MOMENT1, FORCE2, MOMENT2, PLOAD, PLOAD1, PLOAD2, PLOAD4, and SLOAD entries. In other words, an enforced deformation loading via the SPCD entry is specified by a load set ID L1 or L2 and a zero magnitude loading via a load (such as FORCE) with the same ID.

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6. If HTYPE is “GRAV” or “RFORCE”, the entry in HID will be ignored and therefore may be blank. S2 and L2 must be blank for this case.
7. Load set IDs L1 or L2 may not be referenced by load set ID L1 or L2 of LOADCYN entries.
8. If HTYPE = “C”, “S”, “CSTAR”, or “SSTAR”, the load on component (HTYPE) of harmonic (HID) is L = S(S1 · L1 + S2 · L2).
9. S must be nonzero. In addition, either S1 or S2 must be nonzero.
10. L1 and L2 may reference any of the static or dynamic loading entries including GRAV and RFORCE.

Related Topics

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LOADCYN

Physical Load Input for Cyclic Symmetry

Defines a physical static or dynamic load for use in cyclic symmetry analysis.

Format:

1	2	3	4	5	6	7	8	9	10
LOADCYN	SID	S	SEGID	SEGTYPE	S1	L1	S2	L2	

Example:

LOADCYN	10	1.0	1	R	0.5	17			

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
S	Scale Factor. (Real)
SEGID	Segment identification number. (Integer)
SEGTYPE	Segment type. (Character: “R”, “L”, or blank)
Si	Scale Factors. (Real)
Li	Load set ID numbers. See Remark 8. (Integer > 0)

Remarks:

1. The LOADCYN entry is selected by the LOAD Case Control command.
2. If SEGTYPE is blank, both R and L segments will be used in DIH-type symmetry.
3. L1 and L2 may reference LOAD entries. However, the LOAD entry in such a case must not reference load sets defined via RFORCE and/or GRAV entries.
4. If L1 refers to a set ID defined by SPCD loading entry, the same ID must not reference any nonzero loading specified by the other Bulk Data loading entries given via FORCE, MOMENT, FORCE1, MOMENT1, FORCE2, MOMENT2, PLOAD, PLOAD1, PLOAD2, PLOAD4, and SLOAD entries. In other words, an enforced deformation loading via the SPCD entry is specified by a load set ID L1 or L2 and a zero magnitude loading via a load (such as FORCE) with the same ID.
5. Load set IDs L1 or L2 may not be referenced by load set ID L1 or L2 of LOADCYH entries.
6. The load on the segment (or half-segment) is $L = S(S1 \cdot L1 + S2 \cdot L2)$.

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7. S must be nonzero. In addition, either S1 or S2 must be nonzero.
8. L1 and L2 may reference any of the static or dynamic loading entries except GRAV and RFORCE.

Related Topics

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LOADCYT

Table Load Input for Cyclic Symmetry

Specifies loads as a function of azimuth angle by references to tables that define scale factors of loads versus azimuth angles. This entry is used only when STYPE = "AXI" on the CYSYM entry.

Format:

1	2	3	4	5	6	7	8	9	10
LOADCYT	SID	TID1	LSET1	METHOD1	TID2	LSET2	METHOD2		

Example:

LOADCYT	10	19	27		21	26	1		

Fields:

Field	Contents	
SID	Load set identification number. (Integer > 0)	
TIDi	Table ID for table load input for load set Li. See Remark 3. (Integer > 0)	
LSETi	Load set Li. (Integer > 0)	
METHODi	Method of interpolation. See Remark 5. (Integer: 0 or 1, Default = 0)	
	0	interpolate the load with the Fourier coefficients specified in the table up to the specified number of harmonics. (Default)
	1	interpolate the magnitude of the load at corresponding grid points in all segments.

Remarks:

1. The LOADCYT entry is selected by the LOAD Case Control command.
2. The load set ID given in fields 4 or 7 of this entry may refer to FORCE, MOMENT, PLOAD, PLOAD2, PLOAD4, SPCD, TEMP, or TEMPP1 Bulk Data entries.
3. Either TABLED1 or TABLED2 type tabular data of azimuth angle (Xi) versus scale factors (Yi) may be used. The azimuth angle values must be in degrees.
4. The scale factors given in the tables referenced by TIDi entries will be applied only to the magnitudes of the loads defined by LOADSET IDs given in fields 4 or 7.
5. For grid point loading entries, (like FORCE, MOMENT, SPCD, and TEMP Bulk Data entries) METHODi = 1 option should be used. For element loading entries (like PLOAD, PLOAD2, PLOAD4, and TEMPP1

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Bulk Data entries) either METHODi = 0 or METHODi = 1 option can be used. In particular, if harmonic output of element stresses under temperature loading via TEMPP1 Bulk Data entry, METHODi = 0 option should be used to specify TEMPP1 load set.

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LSEQ

Static Load Set Definition

Defines a sequence of static load sets.

Format:

1	2	3	4	5	6	7	8	9	10
LSEQ	SID	EXCITEID	LID	TID					

Example:

LSEQ	100	200	1000	1001					

Fields:

Field	Contents
SID	Set identification of the set of LSEQ entries. See Remark 5 . (Integer > 0)
EXCITEID	The EXCITEID set identification assigned to this static load vector. See Remark 5 . (Integer > 0)
LID	Load set identification number of a set of static load entries such as those referenced by the LOAD Case Control command. (Integer > 0 or blank)
TID	Temperature set identification of a set of thermal load entries such as those referenced by the TEMP(LOAD) Case Control command. (Integer > 0 or blank)

Remarks:

1. LSEQ will not be used unless selected in the Case Control Section with the LOADSET command.
2. The number of static load vectors created for each superelement depends upon the type of analysis. In static analysis, the number of vectors created is equal to the number of unique EXCITEID IDs on all LSEQ entries in the Bulk Data; in dynamic analysis, the number of vectors created is equal to the number of unique EXCITEID IDs on all RLOAD1, RLOAD2, TLOAD1, TLOAD2, ACSRCE, and SELOAD entries in the Bulk Data.
3. EXCITEID may be referenced by CLOAD, RLOAD1, RLOAD2, TLOAD1, TLOAD2, and SELOAD entries in order to apply the static load in nonlinear, static and dynamic analysis.
4. Element data recovery for thermal loads is not currently implemented in dynamics.
5. The SID-EXCITEID number pair must be unique with respect to similar pairs on all other LSEQ entries in the Bulk Data.

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6. In a nonsuperelement analysis, LID and TID cannot both be blank. In superelement analysis, they may both be blank as long as static loads are prescribed in the upstream superelements.
7. It is no longer necessary to employ LOADSET/LSEQ data to specify static loading data for use in dynamic analysis. In the absence of LSEQ Bulk Data entries, all static loads whose load set IDs match the EXCITEID IDs on all RLOAD1, RLOAD2, TLOAD1, TLOAD2, ACSRCE, and SELOAD entries in the Bulk Data are automatically processed.

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15. Bulk Entries M—N

MAT1

Isotropic Material Property Definition

Defines the material properties for linear isotropic materials.

Format:

1	2	3	4	5	6	7	8	9	10
MAT1	MID	E	G	NU	RHO	A	TREF	GE	
	ST	SC	SS	MCSID					

Example:

MAT1	17	3.+7		0.33	4.28	6.5-6	5.37+2	0.23	
	20.+4	15.+4	12.+4	1003					

Fields:

Field	Contents
MID	Material identification number. See Remark 10. (Integer > 0)
E	Young's modulus. (Real ≥ 0.0 or blank)
G	Shear modulus. (Real ≥ 0.0 or blank)
NU	Poisson's ratio. (-1.0 < Real ≤ 0.5 or blank)
RHO	Mass density. See Remark 6 and SOL 414 Remark 6. (Real; Default = 0.0)
A	Thermal expansion coefficient. (Real)
TREF	Reference temperature for the calculation of thermal loads, or a temperature-dependent thermal expansion coefficient. See Remarks 9 and 10. (Real; Default = 0.0 if A is specified.)
GE	Structural element damping coefficient (dimensionless). See Remark 9. (Real)

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Field	Contents
ST, SC, SS	Stress limits for tension, compression, and shear are optionally supplied, used only to compute margins of safety in certain elements; and have no effect on the computational procedures. (Real ≥ 0.0 or blank)
MCSID	Material coordinate system identification number. Used only for PARAM,CURV processing. See **Parameters and Parameter Applicability Tables** for more information. (Integer ≥ 0 or blank)

Remarks:

1. The material identification number must be unique for all MAT1, MAT2, MAT3, and MAT9 entries.
2. The following rules apply when E, G, or NU are blank in solutions other than SOL 401.
   - E and G may not both be blank.
   - If NU and E, or NU and G, are both blank, then both are set to 0.0.
   - If only one E, G, or NU is blank, then it will be computed from the equation: $E = 2 \cdot (1 + NU) \cdot G$. If this is not desired, then the MAT2 entry is recommended. If E, G, or NU are made temperature dependent by the MATT1 entry, then the equation is applied to the initial values only.

   Note that these rules are validated at the beginning of the solution, even when a MATS1 entry is used.
3. For solutions other than SOL 401, if values are specified for all of the properties E, G, and NU, then it is recommended that the following relationship be satisfied:

$$\left| 1 - \frac{E}{2 \cdot (1 + NU) \cdot G} \right| < 0.01$$

   If this relationship is not desired, then the MAT2 entry is recommended.

   It should also be noted that some of the properties are not applied in the stiffness formulation of certain elements as indicated in **Table 15-80**. Therefore, it is recommended that only the applicable properties be specified for a given element.

Table 15-80: Material Property Usage Versus Element Types

Element Entry	E	NU	G
CROD CBEAM CBAR	Extension and Bending	Not Used	Torsion Transverse Shear
CQUADi CTRIAi	Membrane, including In-plane Shear, and Bending		Transverse Shear
CSHEAR	Not Used		Shear
CRAC2D	All Terms		Not Used
CHEXA	All Terms		Not Used

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Element Entry	E	NU	G
CPENTA CTETRA CRAC3D			
CQUADX4 CQUADX8 CTRAX3 CTRAX6	Radial, Axial, Circumferential	All Coupled Ratios	Shear

4. SOL 401 uses the relationship G = E / [2 (1 + NU)] to compute E, NU, or G.
   - If only E is specified, the software sets NU = 0.0, and computes G.
   - If only G is specified, the software sets E, NU, and G to 0.0, and ignores the specified G.
   - If only E and NU are specified, the software computes G.
   - If only E and G are specified, the software computes NU.
   - If only NU and G are specified, the software computes E.
   - If E, G, and NU are specified, the software uses E and NU to compute G, and ignores the specified G.
5. MAT1 materials may be made temperature-dependent by use of the MATT1 entry. In SOL 106, linear and nonlinear elastic material properties in the residual structure will be updated as prescribed under the TEMPERATURE case control command.
6. The mass density RHO will be used to compute mass for all structural elements automatically.
7. Weight density may be used in field 6 if the value 1/g is entered on the PARAM,WTMASS entry, where g is the acceleration of gravity (see **Parameters and Parameter Applicability Tables** for more information).
8. MCSID must be nonzero if PARAM,CURV is specified to calculate stresses or strains at grid points on plate and shell elements only.
9. TREF and GE are ignored if the MAT1 entry is referenced by a PCOMP or PCOMPG entry.
10. When PCOMP or PCOMPG bulk entries are used, the MATi IDs must be less than 100,000,000. In addition, when only PSHELL bulk entries exist and MID4 is greater than 400,000,000, the field A has a special format. It must be defined as (C11+C12+C13)(*alpha*), where *alpha* is the MID4 thermal expansion coefficient. If MIDs larger than 99,999,999 are used, PARAM,NOCOMPS,-1 must be specified to obtain stress output.
11. TREF is used for the calculation of thermal strain. See the remarks on the **TEMPERATURE** case control command for solution specific information.
12. If PARAM,W4 is not specified, GE is ignored in transient analysis. For more information, see **Parameters and Parameter Applicability Tables**.

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Remarks related to SOLs 402 and 414:

1. ST, SC, SS, and MCSID are ignored.
2. E and NU may not be both blank.
3. If E, G and NU are all specified, G is ignored.
4. If GE ≠ 0 is defined, its value is assumed to be the viscosity in the viscous-elastic Kelvin-Voigt model (a model that has a purely viscous damper in parallel with a purely elastic spring). The material is no longer elastic but viscous-elastic. Its behavior thus depends on elastic strain rates.
5. If TREF is blank, the reference temperature for the calculation of thermal loads is obtained from the TEMPD entry referenced by the TEMPERATURE(INITIAL) set selection. If TEMPD is also not specified, TREF=0.0 is used.
6. For SOL 414 solutions, if you set the mass parameters in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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MAT1F

Isotropic Material Frequency Dependence

Specifies frequency-dependent material properties on MAT1 entry fields via TABLEDi entries for SOLs 108 and 111.

Format:

1	2	3	4	5	6	7	8	9	10
MAT1F	MID	T(E)	T(G)	T(NU)	T(RHO)			T(GE)	

Example:

MAT1F	33	15							

Fields:

Field	Contents
MID	Material property identification number that matches the identification number on a MAT1 entry. (Integer > 0).
T(E)	Identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 entry for the Young's modulus. (Integer ≥ 0 or blank)
T(G)	Identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 entry for the shear modulus. (Integer ≥ 0 or blank)
T(NU)	Identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 entry for the Poisson's ratio. (Integer ≥ 0 or blank)
T(RHO)	Identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 entry for the mass density. (Integer ≥ 0 or blank)
T(GE)	Identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 entry for the structural damping coefficient. (Integer ≥ 0 or blank)

Remarks:

1. MAT1F entry is supported in direct and modal frequency response solutions. It is supported in vibro-acoustics analysis including FEMAO.
2. Frequency dependence is supported by the following elements:
   - Solid elements CTETRA, CHEXA, CPENTA, and CPYRAM using the PSOLID property.
   - Shell elements CTRIA3, CQUAD4, CTRIA6, CQUAD8, CTRIAR, and CQUADR using the PSHELL property.

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- Bar element CBAR with the PBAR property.
- Beam element CBEAM with the PBEAM property.

3. If you define both a MAT1F and a MATT1 with the same ID in a modal or direct frequency response solution, MAT1F will take precedence, and MATT1 is ignored. For SOLs other than modal or direct frequency response, MAT1F is not supported, so the software uses the MATT1.
4. When the MAT1F entry is defined, at least one of the fields T(E), T(G), T(NU), T(RHO), or T(GE) must contain a value > 0.
5. Fields 3, 4, 5, 6, and 9 of this entry correspond to the same field numbers on the referenced MAT1 entry. The value in a field of the MAT1 entry is replaced or modified by the table referenced in the corresponding field of this entry. In the example shown, E is modified by TABLEDi 15 for MAT1 33. Blank or zero entries indicate that there is no frequency dependence defined.
6. When you specify only two of the elastic constants E, G, or NU, the software calculates the third using G = E/2(1 + NU). If you specify E and G as frequency-dependent, the calculated value of NU must satisfy -1.0 < NU ≤ 0.5. Otherwise, the value of NU associated with the corresponding MAT1 bulk entry is used. The value of NU associated with the MAT1 bulk entry is either explicitly specified on the MAT1 bulk entry or is calculated from the E and G values on the MAT1 bulk entry.
7. If you define all three of the elastic constants E, G, or NU on the MAT1 bulk entry, the software checks that these values satisfy the isotropic relationship G = E/2(1 + NU). It prints a warning message if the relationship is violated and continues with the solution using the values that you defined. The software does not check that the values defined on the MAT1F bulk entry at each solution frequency satisfy the isotropic relationship.

Related Topics

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MAT2

Shell Element Anisotropic Material Property Definition

Defines the material properties for linear anisotropic materials for two-dimensional elements.

Format:

1	2	3	4	5	6	7	8	9	10
MAT2	MID	C11	C12	C13	C22	C23	C33	RHO	
	A1	A2	A3	TREF	GE	ST	SC	SS	
	MCSID								

Example:

MAT2	13	6.2+3			6.2+3		5.1+3	0.056	
	6.5-6	6.5-6		-500.0	0.002	20.+5			
	1003								

Fields:

Field	Contents
MID	Material identification number. See Remark 12. (Integer > 0)
Cij	The material property matrix. (Real)
RHO	Mass density. See Remark 3 for SOL 414. (Real; Default = 0.0)
Ai	Thermal expansion coefficient vector. (Real)
TREF	Reference temperature for the calculation of thermal loads, or a temperature-dependent thermal expansion coefficient. See Remarks 9 and 10. (Real or blank)
GE	Structural element damping coefficient (dimensionless). See Remarks 9 and 11. (Real)
ST, SC, SS	Stress limits for tension, compression, and shear are optionally supplied (these are used only to compute margins of safety in certain elements) and have no effect on the computational procedures. (Real or blank)
MCSID	Material coordinate system identification number. Used only for PARAM,CURV processing. See Parameters and Parameter Applicability Tables for more information. (Integer ≥ 0 or blank)

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Remarks:

1. The material identification numbers must be unique for all MAT1, MAT2, MAT3, and MAT9 entries.
2. MAT2 materials may be made temperature dependent by use of the MATT2 entry. In SOL 106, linear and nonlinear elastic material properties in the residual structure will be updated as prescribed under the TEMPERATURE Case Control command.
3. The mass density, RHO, will be used to automatically compute mass for all structural elements.
4. The convention for the Cij in fields 3 through 8 are represented by the matrix relationship

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} = \begin{bmatrix} C11 & C12 & C13 \\ C12 & C22 & C23 \\ C13 & C23 & C33 \end{bmatrix} \left(\begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{Bmatrix} - (T - T_0) \begin{Bmatrix} A1 \\ A2 \\ A3 \end{Bmatrix} \right)$$

5. If this entry is referenced by the MID3 field (transverse shear) on the PSHELL, then C13, C23, and C33 must be blank. See *The NASTRAN Theoretical Manual*.
6. MCSID must be nonzero if PARAM,CURV is specified to extrapolate element centroid stresses or strains to grid points on plate and shell elements only. CQUAD4 element corner stresses are not supported by PARAM,CURV.
7. If the MAT2 entry is referenced by a PCOMP or PCOMPG entry, the transverse shear flexibility for the referenced lamina is zero.
8. Unlike the MAT1 entry, data from the MAT2 entry is used directly without adjustment of equivalent E, G, or NU values.
9. TREF and GE are ignored if this entry is referenced by a PCOMP or PCOMPG entry.
10. TREF is used for the calculation of thermal strain. See the remarks on the **TEMPERATURE** case control command for solution specific information.
11. If PARAM,W4 is not specified, GE is ignored in transient analysis.
12. PCOMP or PCOMPG entries generate MATi entries greater than 100,000,000. Explicitly specified MAT2 IDs must not conflict with internally generated MAT2 IDs. When PCOMP or PCOMPG bulk data entries are used, the MATi IDs must be less than 100,000,000. Furthermore, when only PSHELL bulk data entries exist and the ID of a MAT2 referenced on the PSHELL MID4 field is greater than 400,000,000, A1, A2, and A3 are a special format. They are [*Cij*] · [*alpha*], and not [*alpha*]. If MIDs larger than 99,999,999 are used, PARAM,NOCOMPS,-1 must be specified to obtain stress output.

Remarks related to SOLs 402 and 414:

1. GE is ignored.
2. If TREF is blank, the reference temperature for the calculation of thermal loads is obtained from the TEMPD entry referenced by the TEMPERATURE(INITIAL) set selection. If TEMPD is also not specified, TREF=0.0 is used.

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3. For SOL 414 solutions, if you set the mass parameters in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

Browse Simcenter Nastran documentation

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MAT2F

Shell Element Anisotropic Frequency Dependent Material

Specifies frequency-dependent material properties on MAT2 entry fields via TABLEDi entries for SOLs 108 and 111.

Format:

1	2	3	4	5	6	7	8	9	10
MAT2F	MID	T(C11)	T(C12)	T(C13)	T(C22)	T(C23)	T(C33)		
					T(GE)				

Example:

MAT2F	33	15	10	8					

Fields:

Field	Contents
MID	Material property identification number that matches the identification number on a MAT2 entry. (Integer > 0).
T(Cij)	Identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 entry for the frequency-dependent material property matrix. (Integer ≥ 0 or blank)
T(GE)	Identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 entry for the structural damping coefficient. (Integer ≥ 0 or blank)

Remarks:

1. MAT2F entry is supported in direct and modal frequency response solutions. It is supported in vibro-acoustics analysis including a FEMAO solution.
2. MAT2F is supported by the following shell elements CTRIA3, CQUAD4, CTRIA6, CQUAD8, CTRIAR, and CQUADR using the PSHELL property.
3. If you define both a MAT2F and a MATT2 with the same ID in a modal or direct frequency response solution, MAT2F will take precedence, and MATT2 is ignored. For SOLs other than modal or direct frequency response, MAT2F is not supported, so the software uses the MATT2.
4. The fields on this entry correspond to the same fields on the referenced MAT2 entry. The value in a field of the MAT2 entry is replaced or modified by the table referenced in the corresponding field of this entry.

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Related Topics

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MAT3

Orthotropic Material Property Definition for Axisymmetric, Plane Stress and Plane Strain Elements

Defines linear orthotropic materials for the axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6; the plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8; and the plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.

Note

The use of MAT11 is recommended for defining orthotropic materials for axisymmetric elements and 2D solid elements.

Format:

1	2	3	4	5	6	7	8	9	10
MAT3	MID	EX	EY	EZ	NUXY	NUYZ	NUZX	RHO	
	GXY		GZX	AX	AY	AZ	TREF	GE	

Example:

MAT3	23	1.0+7	1.1+7	1.2+7	.3	.25	.27	1.0-5	
			2.5+6	1.0-4	1.0-4	1.1-4	68.5	.23	

Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
EX, EY, EZ	Young's moduli in the X-, Y-, and Z-principal material directions. (Real > 0.0)
NUXY, NUYZ, NUZX	Poisson's ratios: (Real) NUXY = Poisson's ratio for strain in the Y-principal material direction when stressed in the X-principal material direction. NUYZ = Poisson's ratio for strain in the Z-principal material direction when stressed in the Y-principal material direction. NUZX = Poisson's ratio for strain in the X-principal material direction when stressed in the Z-principal material direction.
RHO	Mass density. See Remark 1 for SOL 414. (Real; Default = 0.0)
GZX	Shear modulus when elements are defined on the XZ-plane of the basic coordinate system. (Real > 0.0)

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Field	Contents
GXY	Shear modulus when elements are defined on the XY-plane of the basic coordinate system. (Real > 0.0)
AX, AY, AZ	Thermal expansion coefficients in the X-, Y-, and Z-principal material directions. (Real)
TREF	Reference temperature for the calculation of thermal loads or a temperature-dependent thermal expansion coefficient. See Remark 6. (Real or blank)
GE	Structural element damping coefficient (dimensionless). See Remark 7. (Real)

Remarks:

1. The material identification number must be unique with respect to the collection of all MAT1, MAT2, MAT3, MAT8, MAT9, and MAT11 entries.
2. MAT3 materials may be made temperature dependent by use of the MATT3 entry. In SOL 106, linear and nonlinear elastic material properties in the residual structure will be updated as prescribed under the TEMPERATURE case control command.
3. Note the following:

$$\frac{\nu_{xy}}{E_x}=\frac{\nu_{yx}}{E_y} \qquad \frac{\nu_{xz}}{E_x}=\frac{\nu_{zx}}{E_z} \qquad \frac{\nu_{yz}}{E_y}=\frac{\nu_{zy}}{E_z}$$

   See Understanding the MAT3 Bulk Entry in the *Simcenter Nastran User's Guide* for the MAT3 strain-stress relations for the axisymmetric, plane stress, and plane strain elements.
4. A warning message will be issued if any value of NUXY or NUYZ has an absolute value greater than 1.0.
5. For CPLSTNi, CPLSTSi, CQUAX4, CQUADX8, CTRAX3, and CTRAX6 elements, the X,Y,Z–principal material directions on the MAT3 bulk entry correspond to the X_m,Y_m,Z_m–directions on the elements, respectively.
6. TREF is used for the calculation of thermal strain. See the remarks on the **TEMPERATURE** case control command for solution specific information.
7. If PARAM,W4 is not specified, GE is ignored in a transient analysis.

Remarks related to SOL 402:

1. GE is ignored.
2. If TREF is blank, the reference temperature for the calculation of thermal loads is obtained from the TEMPD entry referenced by the TEMPERATURE(INITIAL) set selection. If TEMPD is also not specified, TREF=0.0 is used.

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Remarks related to SOL 414:

1. For SOL 414 solutions, if you set the mass parameters in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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Simcenter Nastran Quick Reference Guide
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MAT4

Heat Transfer Material Properties, Isotropic

Defines the constant or temperature-dependent thermal material properties for conductivity, heat capacity, density, dynamic viscosity, heat generation, reference enthalpy, and latent heat associated with a single-phase change.

Format:

1	2	3	4	5	6	7	8	9	10
MAT4	MID	K	CP	RHO	H	μ	HGEN	REFENTH	
	TCH	TDELTA	QLAT						

Example:

MAT4	1	204.	.900	2700.					

Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
K	Thermal conductivity. (Blank or Real > 0.0)
CP	Heat capacity per unit mass at constant pressure (specific heat). (Blank or Real ≥ 0.0)
RHO	Density. (Real > 0.0 or blank; Default = 1.0)
H	Free convection heat transfer coefficient. (Real or blank)
μ	Dynamic viscosity. See Remark 2. (Real > 0.0 or blank)
HGEN	Heat generation capability used with QVOL entries. (Real ≥ 0.0; Default = 1.0)
REFENTH	Reference enthalpy. (Real or blank)
TCH	Lower temperature limit at which phase change region is to occur. (Real or blank)
TDELTA	Total temperature change range within which a phase change is to occur. (Real ≥ 0.0 or blank)
QLAT	Latent heat of fusion per unit mass associated with the phase change. (Real > 0.0 or blank)

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Remarks:

1. MAT4 may specify material properties for any conduction elements as well as properties for a forced convection fluid (see CONVM). MAT4 also provides the heat transfer coefficient for free convection (see CONV).
2. For a forced convection fluid, μ must be specified.
3. The material identification number may be the same as a structural material (for example, MAT1, MAT2, MAT3, MAT8, MAT9, MAT11 entries) but must be unique with respect to other MAT4 or MAT5 entries.
4. REFENTH is the enthalpy corresponding to zero temperature if the heat capacity CP is a constant. If CP is obtained through a TABLEM lookup, REFENTH is the enthalpy at the first temperature in the table.
5. Properties specified on the MAT4 entry may be defined as temperature dependent by use of the MATT4 entry.

Related Topics

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MAT5

Thermal Material Property Definition

Defines the thermal material properties for anisotropic materials.

Format:

1	2	3	4	5	6	7	8	9	10
MAT5	MID	KXX	KXY	KXZ	KYY	KYZ	KZZ	CP	
	RHO	HGEN							

Example:

MAT5	24	.092			.083		0.20	0.2	
	2.00								

Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
Kij	Thermal conductivity. (Real)
CP	Heat capacity per unit mass. (Real ≥ 0.0 or blank)
RHO	Density. (Real > 0.0; Default = 1.0)
HGEN	Heat generation capability used with QVOL entries. (Real ≥ 0.0; Default = 1.0)

Remarks:

1. X, Y, and Z on the MAT5 fields designate the material coordinates defined on the element or property entries. The general thermal conductivity matrix has the form:

$$K = \begin{bmatrix} KXX & KXY & KXZ \\ KXY & KYY & KYZ \\ KXZ & KYZ & KZZ \end{bmatrix}$$

The thermal conductivity matrix for axisymmetric (see Remark 2), plane stress, and plane strain elements defined on the XZ-plane has the form:

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$$K = \begin{bmatrix} KXX & KXZ \\ KXZ & KZZ \end{bmatrix}$$

The thermal conductivity matrix for axisymmetric (see Remark 2), plane stress, and plane strain elements defined on the XY-plane has the form:

$$K = \begin{bmatrix} KXX & KXY \\ KXY & KYY \end{bmatrix}$$

The axisymmetric, plane stress, and plane strain elements defined on the XZ plane of the basic coordinate system have different material coordinates than elements defined on the XY plane. As a result, MAT5/MATT5 entries defined for elements in one plane should not be used for elements in the other plane. See the remarks on the element entries, for example CQUADX4, for the definition of the material coordinates.

2. The material identification number may be the same as a structural material (for example, MAT1, MAT2, MAT3, MAT8, MAT9, MAT11 entries) but must be unique with respect to other MAT4 or MAT5 entries.
3. MAT5 materials may be made temperature-dependent by use of the MATT5 entry.

Related Topics

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MAT8

Shell Element Orthotropic Material Property Definition

Defines the material property for an orthotropic material for isoparametric shell elements.

Format:

1	2	3	4	5	6	7	8	9	10
MAT8	MID	E1	E2	NU12	G12	G1Z	G2Z	RHO	
	A1	A2	TREF	Xt	Xc	Yt	Yc	S	
	GE	F12	STRN						

Example:

MAT8	171	30.+6	1.+6	0.3	2.+6	3.+6	1.5+6	0.056	
	28.-6	1.5-6	155.0	1.+4	1.5+4	2.+2	8.+2	1.+3	
	1.-4		1.0						

Fields:

Field	Contents
MID	Material identification number. See Remark 8. (0 < Integer)
E1	Modulus of elasticity in longitudinal direction, also defined as the fiber direction or 1-direction. (Real ≠ 0.0)
E2	Modulus of elasticity in lateral direction, also defined as the matrix direction or 2-direction. (Real ≠ 0.0)
NU12	Poisson's ratio (ϵ_2/ϵ_1 for uniaxial loading in 1-direction). Note that $\nu_{21}=\epsilon_1/\epsilon_2$ for uniaxial loading in 2-direction is related to ν_{12}, E_1, and E_2 by the relation $\nu_{12}E_2=\nu_{21}E_1$. (Real)
G12	In-plane shear modulus. (Real ≥ 0.0; Default = 0.0)
G1Z	Transverse shear modulus for shear in 1-Z plane. (Real > 0.0; Default for SOL 402 = G12 x 1000.0; Default for other solutions implies an infinite shear modulus.)
G2Z	Transverse shear modulus for shear in 2-Z plane. (Real > 0.0; Default for SOL 402 = G12 x 1000.0; Default for other solutions implies an infinite shear modulus.)
RHO	Mass density. See Remark 3 for SOL 414. (Real; Default = 0.0)
Ai	Thermal expansion coefficient in i-direction. (Real)

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Field	Contents
TREF	Reference temperature for the calculation of thermal loads, or a temperature-dependent thermal expansion coefficient. See Remarks 4 and 6. (Real or blank)
Xt, Xc	Allowable stresses or strains in tension and compression, respectively, in the longitudinal direction. Required if failure index is desired. See the FT field on the PCOMP or PCOMPG entry. (Real > 0.0; Default value for Xc is Xt.)
Yt, Yc	Allowable stresses or strains in tension and compression, respectively, in the lateral direction. Required if failure index is desired. (Real > 0.0; Default value for Yc is Yt.)
S	Allowable stress or strain for in-plane shear. See the FT field on the PCOMP or PCOMPG entry. (Real > 0.0)
GE	Structural damping coefficient (dimensionless). See Remarks 6 and 8. (Real)
F12	Interaction term in the tensor polynomial theory of Tsai-Wu. Default = 0.0. See the FT field on the PCOMP or PCOMPG entry. See Remark 5. (Real)
STRN	For the maximum strain theory only (see STRN in PCOMP or PCOMPG entry). Indicates whether Xt, Xc, Yt, Yc, and S are stress or strain allowables. (Real = 1.0 for strain allowables; blank (Default) for stress allowables.)

Remarks:

1. If G1Z and G2Z values are specified as zero or blank, then transverse shear flexibility calculations will not be performed, which is equivalent to zero shear flexibility (i.e., infinite shear stiffness).
2. MAT8 materials may be made temperature dependent by use of the MATT8 entry.
3. An approximate value for G1Z and G2Z is the in-plane shear modulus G12. If test data are not available to accurately determine G1Z and G2Z for the material and transverse shear calculations are deemed essential; the value of G12 may be supplied for G1Z and G2Z. In SOL 106, linear and nonlinear elastic material properties in the residual structure will be updated as prescribed in the TEMPERATURE Case Control command.
4. Xt, Yt, and S are required for composite element failure calculations when requested in the FT field of the PCOMP or PCOMPG entry. Xc and Yc are also used but not required for failure calculations. Regardless if the FT field is defined or not, you must specify values for Xt, Yt and S if any of Xt, Xc, Yt, Yc, and S are specified.
5. If you enter a nonzero value for F_{12} and the stability criterion below is not satisfied, the software uses $F_{12} = 0.0$.

$$\left(\frac{1}{X_t X_c}\right)\left(\frac{1}{Y_t Y_c}\right) - F_{12}^2 > 0$$

6. TREF and GE are ignored if this entry is referenced by a PCOMP or PCOMPG entry.
7. TREF is used for the calculation of thermal strain. See the remarks on the TEMPERATURE case control command for solution specific information.
8. If PARAM,W4 is not specified, GE is ignored in transient analysis.

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9. When PCOMP or PCOMPG bulk entries are used, the MATi IDs must be less than 100,000,000. Furthermore, when only PSHELL bulk entries exist and MID4 is greater than 400,000,000, A1 and A2 have a special format. They must be defined as A1=C11(*alpha1*)+C12(*alpha2*) and A2=C21(*alpha1*)+C22(*alpha2*) where *alpha1* and *alpha2* are the MID4 thermal expansion coefficients. If MIDs larger than 99,999,999 are used, PARAM,NOCOMPS,-1 must be specified to obtain stress output.

Remarks related to SOLs 402 and 414:

1. GE is ignored.
2. If TREF is blank, the reference temperature for the calculation of thermal loads is obtained from the TEMPD entry referenced by the TEMPERATURE(INITIAL) set selection. If TEMPD is also not specified, TREF=0.0 is used.
3. For SOL 414 solutions, if you set the mass parameters in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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MAT9

Anisotropic Material Property Definition

Defines the material properties for linear, temperature-independent, anisotropic materials.

Format:

1	2	3	4	5	6	7	8	9	10
MAT9	MID	C11	C12	C13	C14	C15	C16	C22	
	C23	C24	C25	C26	C33	C34	C35	C36	
	C44	C45	C46	C55	C56	C66	RHO	A1	
	A2	A3	A4	A5	A6	TREF	GE		

Example:

MAT9	17	6.2+3						6.2+3	
					6.2+3				
	5.1+3			5.1+3		5.1+3	3.2	6.5-6	
	6.5-6					125.	.003		

Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
Cij	Elements of the 6 × 6 symmetric material property matrix in the material coordinate system. (Real)
RHO	Mass density. See Remark 3 for SOL 414. (Real; Default = 0.0)
Ai	Thermal expansion coefficient. (Real)
TREF	Reference temperature for the calculation thermal loads, or a temperature-dependent thermal expansion coefficient. See Remark 6. (Real or blank)
GE	Structural element damping coefficient (dimensionless). See Remark 7. (Real)

Remarks:

1. The material identification numbers must be unique for all MAT1, MAT2, MAT3, and MAT9 entries.

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2. MAT9 materials may be made temperature-dependent by use of the MATT9 entry. In nonlinear static analysis (e.g., SOL 106), linear and nonlinear elastic material properties in the residual structure will be updated as prescribed under the TEMPERATURE Case Control command.
3. The mass density RHO will be used to compute mass in a structural dynamics problem automatically.
4. The third continuation entry is optional.
5. The subscripts 1 through 6 refer to x, y, z, xy, yz, and zx of the material coordinate system (see the CORDM field on the PSOLID entry description). The stress-strain relationship is

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{Bmatrix} = \begin{bmatrix} C11 & C12 & C13 & C14 & C15 & C16 \\ & C22 & C23 & C24 & C25 & C26 \\ & & C33 & C34 & C35 & C36 \\ & & & C44 & C45 & C46 \\ & \text{symmetric} & & & C55 & C56 \\ & & & & & C66 \end{bmatrix} \left(\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} - \begin{bmatrix} A1 \\ A2 \\ A3 \\ A4 \\ A5 \\ A6 \end{bmatrix} (T - \text{TREF}) \right)$$

6. TREF is used for the calculation of thermal strain. See the remarks on the **TEMPERATURE** case control command for solution specific information.
7. If PARAM,W4 is not specified, GE is ignored in transient analysis.

Remarks related to SOLs 402 and 414:

1. GE is ignored.
2. If TREF is blank, the reference temperature for the calculation of thermal loads is obtained from the TEMPD entry referenced by the TEMPERATURE(INITIAL) set selection. If TEMPD is also not specified, TREF=0.0 is used.
3. For SOL 414 solutions, if you set the mass parameters in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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MAT9F

Solid Element Anisotropic Frequency Dependent Material

Specifies frequency-dependent material properties on MAT9 entry fields via TABLEDi entries for SOLs 108 and 111.

Format:

1	2	3	4	5	6	7	8	9	10
MAT9	MID	T(C11)	T(C12)	T(C13)	T(C14)	T(C15)	T(C16)	T(C22)	
	T(C23)	T(C24)	T(C25)	T(C26)	T(C33)	T(C34)	T(C35)	T(C36)	
	T(C44)	T(C45)	T(C46)	T(C55)	T(C56)	T(C66)			
							T(GE)		

Example:

MAT9F	17	3						28	
					12				
	32			100		62			
							26		

Fields:

Field	Contents
MID	Material property identification number that matches the identification number on a MAT9 entry. (Integer > 0).
T(Cij)	Identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 entry for the frequency-dependent material property matrix. (Integer ≥ 0 or blank)
T(GE)	Identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 entry for the structural damping coefficient. (Integer ≥ 0 or blank)

Remarks:

1. MAT9F entry is supported in direct and modal frequency response solutions. It is supported in vibro-acoustics analysis including a FEMAO solution.
2. MAT9F is supported by the solid elements CTETRA, CHEXA, CPENTA, and CPYRAM using the PSOLID property.

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3. If you define both a MAT9F and a MATT9 with the same ID in a modal or direct frequency response solution, MAT9F will take precedence, and MATT9 is ignored. For SOLs other than modal or direct frequency response, MAT9F is not supported, so the software uses the MATT9.
4. The fields on this entry correspond to the same field numbers on the referenced MAT9 entry. The value in a field of the MAT9 entry is replaced or modified by the table referenced in the corresponding field of this entry.

Related Topics

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MAT10

Fluid or Absorber Material Property Definition

Defines material properties for fluid or absorber elements in coupled fluid-structural analysis.

Format:

1	2	3	4	5	6	7	8	9	10
MAT10	MID	BULK	RHO	C	GE	GAMMA			
		TIDBULK	TIDRHO		TIDGE	TIDGAMMA			

Example:

MAT10	103	0.656	0.011						

Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
BULK	Bulk modulus. (Real > 0.0)
RHO	Mass density. (Real; Default = 0.0)
C	Speed of sound. (Real > 0.0)
GE	Fluid element damping coefficient. (Real or blank; Default = 0.0)
GAMMA	Ratio of imaginary bulk modulus to real bulk modulus. (Real or blank; Default = 0.0)
TIDBULK	Identification number of TABLEDi bulk entry containing bulk modulus versus frequency data. (Integer > 0 or blank; No default)
TIDRHO	Identification number of TABLEDi bulk entry containing mass density versus frequency data. (Integer > 0 or blank; No default)
TIDGE	Identification number of TABLEDi bulk entry containing damping coefficient versus frequency data. (Integer > 0 or blank; No default)
TIDGAMMA	Identification number of TABLEDi bulk entry containing ratio of imaginary bulk modulus to real bulk modulus versus frequency data. (Integer > 0 or blank; No default)

Remarks:

1. In solutions 103, 107-112, and 200, the MAT10 entry can be referenced by PSOLID entries.

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2. The material identification numbers must be unique for all MAT1, MAT2, MAT3, MAT9, and MAT10 entries.

3. BULK, RHO, and C are always used in modal calculations. Any two of BULK, RHO, and C must be specified. The third is then calculated from:

$$\text{BULK} = C^2 \cdot \text{RHO}$$

   If all three are specified, BULK and RHO are used.

4. A 3-D frequency dependent absorber can be defined by specifying a nonzero value in the GAMMA field. In this case, the solution becomes complex and frequency dependent with the imaginary part of the governing matrix equation representing acoustic damping. The TIDBULK, TIDRHO, TIDGE, and TIDGAMMA fields allow you to model frequency dependence of the bulk modulus, density, damping coefficient, and ratio of imaginary to real bulk modulus, respectively. See Frequency-dependent acoustic materials or absorbers in the *Acoustics User's Guide* for details.

5. Specifying any combination of GE, GAMMA, TIDBULK, TIDRHO, TIDGE, and TIDGAMMA is optional.

6. If all of TIDBULK, TIDRHO, TIDGE, and TIDGAMMA are left blank, frequency dependence is ignored and BULK, RHO, GE, and GAMMA are used at all frequencies. If any of TIDBULK, TIDRHO, TIDGE, and TIDGAMMA are specified, frequency dependence is assumed. When frequency dependence is assumed, BULK is used at all frequencies if TIDBULK is left blank, RHO is used at all frequencies if TIDRHO is left blank, GE is used at all frequencies if TIDGE is left blank, and GAMMA is used at all frequencies if TIDGAMMA is left blank.

7. If PARAM,W4FL is not specified, GE is ignored in transient analysis.

8. PARAM,OMACHPR,NO (default) selects the pre-NX Nastran 8.5 format for the MAT10 entry and any continuation fields that are specified are removed with the MAKEOLD DMAP module. PARAM,OMACHPR,YES selects the current Simcenter Nastran format for the MAT10 entry and enables the frequency-dependent material capability.

Related Topics

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MAT10C

Fluid or Absorber Material Property Definition in Complex Format

Defines constant or nominal material properties for fluid or absorber elements in coupled fluid-structural analysis.

Format:

1	2	3	4	5	6	7	8	9	10
MAT10C	ID	FORM	RHOR	RHOI	CR	CI			

Example:

MAT10C	100		3	-0.01	20.0	5			

Fields:

Field	Contents
ID	Material identification number. (Integer > 0)
FORM	Format to define the fluid density and speed of sound. (Character; "REAL" or "PHASE"; Default = REAL) If FORM = REAL, define the fluid density and speed of sound in rectangular format (real and imaginary). If FORM = PHASE, define the fluid density and speed of sound in polar format (magnitude and phase).
RHOR	Real part of density. (Real; Default = 0.0)
RHOI	Imaginary part of density. (Real; Default = 0.0)
CR	Real part of speed of sound. (Real; Default = 0.0)
CI	Imaginary part of speed of sound. (Real; Default = 0.0)

Remarks:

1. RHOR and RHOI cannot both be equal to zero.
2. CR and CI cannot both be equal to zero.

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Related Topics

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MAT11

Orthotropic Material Property Definition

Defines the orthotropic material properties for shell elements, shell composite elements, axisymmetric elements, 2D solid elements, 3D solid elements, and 3D solid composite elements.

Format:

1	2	3	4	5	6	7	8	9	10
MAT11	MID	E1	E2	E3	NU12	NU13	NU23	G12	
	G13	G23	RHO	A1	A2	A3	TREF	GE	

Example:

MAT11	101	2.1E7	2.2E7	2.3E7	0.31	0.29	0.33	2.1E6	
	2.2E6	2.3E6	0.34	0.35	0.36	0.37	0.38	0.39	

Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
E1	Modulus of elasticity in the 1–principal material direction. See Remark 1. (Real > 0.0; No default, must be defined)
E2	Modulus of elasticity in the 2–principal material direction. See Remark 1. (Real > 0.0; No default, must be defined)
E3	Modulus of elasticity in the 3–principal material direction. See Remark 1. (Real > 0.0; Default = E2)
NU12	Poisson's ratio (ϵ_2 / ϵ_1 for uniaxial loading in 1-principal material direction). (Real; No default, must be defined)
NU13	Poisson's ratio (ϵ_3 / ϵ_1 for uniaxial loading in 1-principal material direction). (Real; Default = 0.0)
NU23	Poisson's ratio (ϵ_3 / ϵ_2 for uniaxial loading in 2-principal material direction). (Real; Default = NU13)
G12	Shear modulus relating τ_{12} and γ_{12}. (Real > 0.0; No default, must be defined)
G13	Shear modulus relating τ_{13} and γ_{13}. (Real > 0.0; No default, must be defined)
G23	Shear modulus relating τ_{23} and γ_{23}. (Real > 0.0; No default, must be defined)
RHO	Mass density. (Real; Default = 0.0)

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Field	Contents
A1	Thermal expansion coefficient in 1–principal material direction. (Real; Default = 0.0)
A2	Thermal expansion coefficient in 2–principal material direction. (Real; Default = 0.0)
A3	Thermal expansion coefficient in 3–principal material direction. (Real; Default = 0.0)
TREF	Reference temperature for the calculation of thermal loads, or a temperature-dependent thermal expansion coefficient. (Real; Default = 0.0)
GE	Structural damping coefficient (dimensionless). (Real; Default = 0.0)

Remarks:

1. For CCHOCKi, CHEXA, CPENTA, CPLSTNi, CPLSTSi, CPYRAM, CQUAX4, CQUADX8, CTETRA, CTRAX3, and CTRAX6 elements, the 1,2,3–principal material directions on the MAT11 bulk entry correspond to the X_m,Y_m,Z_m–directions on the elements, respectively.
2. In general, NUij is not equal to NUji, but they are related as follows:

   NUij / Ei = NUji / Ej
3. Material stability requires that the following relations are satisfied:

   $Ei > NUij^2 Ej$

   and

   1-NU12 NU21 – NU23 NU32 – NU31 NU13 – 2 NU21 NU32 NU13 > 0.0
4. MAT11 materials may be made temperature dependent by use of the MATT11 entry.
5. TREF is used for the calculation of thermal strain. See the remarks on the **TEMPERATURE** case control command for solution specific information.

Remarks related to SOLs 402 and 414:

1. For SOL 402, GE is ignored.
2. For SOLs 402 and 414, if TREF is blank, the reference temperature for the calculation of thermal loads is obtained from the TEMPD entry referenced by the TEMPERATURE(INITIAL) set selection. If TEMPD is also not specified, TREF=0.0 is used.

Related Topics

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MATCID

Material Coordinate System for Solid Elements

Overrides the material coordinate system for CHEXA, CPENTA, CTETRA, and CPYRAM solid elements when the elements reference a PSOLID property.

Overrides the CORDM field on the PCOMPS entry for CHEXA and CPENTA solid elements when the elements reference a PCOMPS property.

Overrides the material coordinate system for CHEXCZ and CPENTCZ solid elements.

Format:

1	2	3	4	5	6	7	8	9	10
MATCID	CID	EID1	EID2	EID3	EID4	EID5	EID6	EID7	
	EID8	EID9	EID10	-etc.-					

Example:

MATCID	20	101	102	103	104	105			

Alternate Formats and Examples:

1	2	3	4	5	6	7	8	9	10
MATCID	CID	EID1	“THRU”	EID2					
MATCID	20	101	THRU	105					

1	2	3	4	5	6	7	8	9	10
MATCID	CID	EID1	“THRU”	EID2	“BY”	N			
MATCID	20	101	THRU	105	BY	2			

1	2	3	4	5	6	7	8	9	10
MATCID	CID	“ALL”							
MATCID	20	ALL							

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Fields:

Field	Contents
CID	Material coordinate system identification number. (Integer > -2) See Remarks 1, 2, and 3.
EIDi	Element identification number. (Integer > 0 or “ALL” or “THRU” or “THRU” with “BY”; For “THRU” options, EID1 < EID2)
N	Element selection increment for use with ”THRU” with “BY” option. (Integer ≥ 1)

Remarks:

1. For CHEXA, CPENTA, CTETRA, and CPYRAM solid elements that reference a PSOLID property, CID = -1 refers to the element coordinate system and CID = 0 refers to the basic coordinate system.
2. For CHEXA and CPENTA solid elements that reference a PCOMPS property, CID = -1 and CID = 0 are treated the same with both referring to the basic coordinate system. Ply stresses and strains are output in the ply coordinate system rather than the CID coordinate system.
3. For CHEXCZ and CPENTCZ solid elements that reference a PSOLCZ property, CID = -1 and CID = 0 are treated the same with both referring to the basic coordinate system.

Related Topics

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MATCRP

Creep Material Definition

Defines coefficients for creep constitutive model in SOLs 401 and 402.

Format:

1	2	3	4	5	6	7	8	9	10
MATCRP	MID	TYPE	THRESH	HARD					
	Value1	Value2	Value3	Value4					

Format for Bailey-Norton law (SOLs 401 and 402)

1	2	3	4	5	6	7	8	9	10
MATCRP	MID	“301”	THRESH	HARD					
	A	B		D					

Format for strain hardening law (SOLs 401 and 402)

1	2	3	4	5	6	7	8	9	10
MATCRP	MID	"1" or "STRNHARD"	THRESH						
	C1	C2	C3	C4					

Format for Norton law (SOLs 401 and 402)

1	2	3	4	5	6	7	8	9	10
MATCRP	MID	“2” or "NORTON"	THRESH						
	C1	C2		C4					

Format for generalized Garofalo law (SOLs 401 and 402)

1	2	3	4	5	6	7	8	9	10
MATCRP	MID	"3" or "GENGAROF"	THRESH						
	C1	C2	C3	C4					

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Example:

MATCRP	20	301		STRAIN	104	105			
	0.5	3.2		0.8					

Fields:

Field	Contents
MID	Identification number of a MAT1, MAT3, MAT9, or MAT11 bulk entry. See Remark 2. (Integer > 0)
THRESH	Factor that when multiplied by the elastic modulus yields the stress threshold below which creep does not occur. (0.0 < Real < 1.0E-3; Default = 1.0E-5)
HARD	Hardening rule. (Character) “STRAIN” for strain hardening (Default) “TIME” for time hardening
A,B,C, D	Coefficients for the "301" creep constitutive model. See Remark 3. (Real or Integer > 0) If real entry, value of coefficient used at all temperatures. If integer entry, identification number of TABLEM1 bulk entry that defines the coefficient as a function of temperature.
C1, C2, C3, C4	Coefficients for the SOLs 401 and 402 specific creep constitutive models. See SOLs 401 and 402 Remarks. (Real or Integer > 0) If real entry, value of coefficient used at all temperatures. If integer entry, identification number of TABLEM1 bulk entry that defines the coefficient as a function of temperature.

Remarks:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. A MAT1, MAT3, MAT9, or MAT11 bulk entry that is referenced by a MATCRP bulk entry can in turn reference a corresponding MATT1, MATT3, MATT9, or MATT11 bulk entry.
3. The A, B, and D coefficients are used to define the Bailey-Norton creep model as follows:

$$\varepsilon_e^c = A\sigma_e^B t^D$$

where ϵ^c is the effective creep strain, σ is effective stress, and t is time.

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A fatal error occurs if any of the A, B, or D fields are left blank.

Remarks related to SOLs 401 and 402:

1. The C1, C2, C3, and C4 coefficients are used to define the strain hardening creep model as follows:

$$\dot{\varepsilon}_{cr} = C_1 \sigma^{C_2} \varepsilon_{cr}^{C_3} e^{-C_4/T}$$

where ϵ_{cr} is the effective creep strain, σ is the effective stress, and T is the temperature.

A fatal error occurs if any of the C1, C2, C3 fields are left blank. If C4 is left blank, the temperature term is ignored.

2. The C1, C2, and C4 coefficients are used to define the Norton creep model as follows:

$$\dot{\varepsilon}_{cr} = C_1 \sigma^{C_2} e^{-C_4/T}$$

A fatal error occurs if either of C1 or C2 fields are left blank. If C4 is left blank, the temperature term is ignored.

3. The C1, C2, C3, and C4 coefficients are used to define the generalized Garofalo creep model as follows:

$$\dot{\varepsilon}_{cr} = C_1 \left[\sinh\left(C_2 \sigma\right)\right]^{C_3} e^{-C_4/T}$$

A fatal error occurs if any of the C1, C2, or C3 fields are left blank.

If C4 is left blank, the temperature term is ignored.

Related Topics

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MATCZ

Cohesive Element Material

Defines damage model and material properties for cohesive elements (CHEXCZ and CPENTCZ).

Format for Polynomial law (DLAW = "POLY"):

1	2	3	4	5	6	7	8	9	10
MATCZ	MID	DLAW	TANG	TAU	ADEL				
	K03S	K02	K01	GIC	GIIC	GIIIC	DCOU	Y0S	
	EXPN								

Format for Bi-Triangular law (DLAW = "BITR"):

1	2	3	4	5	6	7	8	9	10
MATCZ	MID	DLAW	TANG	TAU	ADEL				
	K03S	K02	K01	GIC	GIIC	GIIIC	DCOU	Y0S	

Format for Exponential law (DLAW = "EXPO"):

1	2	3	4	5	6	7	8	9	10
MATCZ	MID	DLAW	TANG	TAU	ADEL				
	K03S	K02	K01	GIC	GIIC	GIIIC	DCOU		

Examples:

MATCZ	1	POLY	1	1.5	3.0				
	4.0+8	1.0+8	1.0+8	0.32	0.48	0.48	1.0	0.1	
	0.5								

MATCZ	1	BITR	1						
	4.0+8	1.0+8	1.0+8	0.32	0.48	0.48	1.0	0.1	

MATCZ	1	EXPO	1	2.0					
	4.0+8	1.0+8	1.0+8	0.32	0.48	0.48	1.0		

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Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
DLAW	Damage model. (Character; Default = "POLY")
	"POLY" for the Polynomial model.
	"BITR" for the Bi-Triangular model.
	"EXPO" for the Exponential model.
TANG	Tangent of constitutive model. (Integer; Default = 1)
	1: Use the tangent stiffness when it is positive and use zero when it is negative.
	2: Use the secant stiffness.
	3: Use the tangent stiffness.
TAU	Time for delay. See Remark 3. (Real; Default = 0.0)
ADEL	Parameter for delay. See Remark 4. (Real > 0.0; Default = 1.0)
K03S	Out-of-plane normal stiffness in force per unit length^3. (Real > 0; No default)
K02	Transverse shear stiffness linked to the shear stress σ_{32}, in force per unit length^3. (Real > 0; No default)
K01	Transverse shear stiffness linked to the shear stress σ_{31}, in force per unit length^3. (Real > 0; No default)
GIC	Critical energy release rate for Mode I. (Real > 0; No default)
GIIC	Critical energy release rate for Mode II. (Real > 0; No default)
GIIIC	Critical energy release rate for Mode III. (Real > 0; No default)
DCOU	Coupling coefficient. (Real ≠ 0.0)
Y0S	Thermodynamic force. (GIC ≥ Real ≥ 0.0)
EXPN	Exponent. (Real > 0.0)

Remarks:

1. For SOL 401, you must define PARAM,MATNL,1 to activate damage estimation. SOL 402 does not have this requirement.
2. The material identification number must be unique with respect to the collection of all MAT1, MAT11, and MATCZ entries.

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3. Use a positive value to define time for delay. The software interprets negative values as zero time for delay.
4. If TAU ≤ 0.0, there is no time for delay and the ADEL field is ignored.

Remarks related to SOL 402:

1. For bi-triangular model (DLAW = BITR), if Y0S = 0.0, then the solver changes automatically the behavior into exponential.

Related Topics

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MATDMG

Material Properties for Progressive Ply Failure

Defines material properties and parameters for progressive ply failure in composite solid elements defined with the PCOMPS bulk entry. Used in combination with MAT11 entries that have the same MID. Valid for SOLs 401 and 402.

Format:

1	2	3	4	5	6	7	8	9	10
MATDMG	MID	PPFMOD							
	COEF1	COEF1	COEF1	-etc-					

Format for PPFMOD="UD":

1	2	3	4	5	6	7	8	9	10
MATDMG	MID	UD							
	Y012	YC12	YS12	YS22	Y11LIMT	Y11LIMC	KSIT	KSIC	
	B2	B3	A	LITK	BIGK	EXPN	TAU	ADEL	
	PLYUNI	TID	HBAR	DMAX	PE				

Format for PPFMOD="EUD":

1	2	3	4	5	6	7	8	9	10
MATDMG	MID	EUD							
	Y012	YC12	K	ALPHA	Y11LIMT	Y11LIMC	KSIT	KSIC	
	B2	B3	A	LITK	BIGK	EXPN	TAU	ADEL	
	USER	R01	HBAR	DMAX	DS	GIC	GIIC	GIIIC	

Example:

MATDMG	3	UD							
	0.1	12.0	10.0	0.08	5.7	1.0E16	8.0		
	0.3	0.5	0.9	20.0	750.0	0.4	0.0001		
			0.43						

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Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
PPFMOD	Progressive ply failure model. Allowable entries are: "UD" for the unidirectional ply model. "EUD" for the enhanced unidirectional ply model. (Character; No default)
If PPFMOD="UD"	
Y012	Energy threshold for shear damage d_{12}. (Real > 0; For default behavior, see Remark 2)
YC12	Critical value of energy for shear damage d_{12}. (Real > 0; For default behavior, see Remark 2)
YS12	Limit value of energy for shear damage d_{12}. (Real > 0.0; No default)
YS22	Limit energy in the matrix direction. (Real > 0.0; No default)
Y11LIMT	Energy threshold in tension for the fiber direction. (Real > 0.0; No default)
Y11LIMC	Energy threshold in compression for the fiber direction. (Real > 0.0; No default)
KSIT	Nonlinearity coefficient in tension for the fiber direction. See Remark 3. (Real; Default = 0.0)
KSIC	Nonlinearity coefficient in compression for the fiber direction. See Remark 3. (Real; Default = 0.0)
B2	Coupling coefficient. (Real ≥ 0.0; Default = 0.0)
B3	Coupling coefficient between the damage variables. (Real > 0.0; Default = 1.0)
A	Coupling coefficient. (Real ≥ 0.0; Default = 0.0)
LITK	Initial plastic threshold. (Real > 0.0; Default = 1.0E15)
BIGK	Parameter for Plastic Law. (Real ≥ 0.0; Default = 0.0)
EXPN	Exponent for Plastic Law. (Real ≥ 0.0; Default = 0.0)
TAU	Time constant. See Remark 4. (Real; Default = 0.0)
ADEL	Parameter for delay. See Remark 5. (Real > 0.0; Default = 1.0)
PLYUNI	Control parameter for the application of nonlinearity coefficients. (Integer = 0 or 1; Default = 0) 0: Nonlinearity coefficients are applied to stress.

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Field	Contents
	1: Nonlinearity coefficients are applied to strain.
TID	Table identification number of a TABLEM5 entry. Used to define a nonlinear damage evolution law for in-plane shear. Integer ≥ 0 or blank; for default behavior, see Remark 1)
HBAR	Transition thickness of ply. (Real ≥ 0.0; Default = 0.0)
DMAX	Maximum damage value at a Gauss point. (0.0 < Real ≤ 0.999; Default = 0.999)
PE	3D effect parameter. (Integer = 0 or 1; Default = 0) 0: Plane stress effect option. The effect of damage in the normal and shear stress associated with the out-of-plane direction are ignored. 1: 3D stress effect option. The effect of damage in the normal and shear stress associated with the out-of-plane direction are included.
If PPFMOD="EUD"	
Y012	Fiber/matrix debonding threshold. (Real > 0.0; No default).
YC12	Critical thermodynamic force. (Real > 0.0; No default).
K	Coupling coefficient to account for the contribution of thermodynamic force Y_d to fiber damage in compression. Y_d is thermodynamic force linked to shear stress σ_{12} and σ_{13}. (Real >=0.0; Default=0.0).
ALPHA	Parameter given by the user to define the relationship for three crack modes in rupture envelope. (Real > 0.0; Default=1.0).
Y11LIMT	Energy threshold in tension for the fiber direction. (Real > 0.0; No default).
Y11LIMC	Energy threshold in compression for the fiber direction. (Real > 0.0; No default).
KSIT	Nonlinearity coefficient in tension for the fiber direction. See Remark 3. (Real; Default = 0.0).
KSIC	Nonlinearity coefficient in compression for the fiber direction. See Remark 3. (Real; Default = 0.0).
B2	Coupling coefficient to account for the contribution of thermodynamic force $Y_{d'}$ to shear damage variable d. where d is the damage variable linked to shear stress σ_{12} and σ_{13}. $Y_{d'}$ is the thermodynamic force linked to normal stress σ_{22} and σ_{33}. (Real ≥ 0.0; Default = 0.0).
B3	Coupling coefficient between damage variables d and d'; (Real ≥ 0.0; Default = 0.0).
A	Coupling coefficient for plasticity; (Real>=0; Default=0.0).
LITK	Initial plastic threshold. (Real > 0.0; Default = 1.0E15).

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Field	Contents
BITK	Parameter for Plastic Law. (Real ≥ 0.0; Default = 0.0).
EXPN	Exponent for Plastic Law. (Real ≥ 0.0; Default = 0.0).
TAU	Time constant. See Remark 4. (Real; Default = 0.0).
ADEL	Parameter for delay. See Remark 5. (Real > 0.0; Default = 1.0).
USER	If USER is equal to 1, static damage w is limited to the maximum damage before taking into account time delay effect. If USER is equal to 0, the "delayed" damage d after taking into account time delay effect is limited to the maximum damage. (Integer 0, 1; Default = 0).
R01	Maximum transverse micro-cracking density. (Real; Default = 0.7).
HBAR	A critical thickness defining the threshold between thin and thick ply behavior. (Real ≥ 0.0; Default = 0.0).
DMAX	Maximum fiber damage value at a Gauss point. (0.0 < Real ≤ 0.999; Default = 0.999).
DS	Maximum diffuse damage value at a Gauss point. (0.0 < Real ≤ 0.999; Default = 0.55).
GIC	Critical mode I crack energy release rates for the ply. (Real > 0.0; No default).
GIIC	Critical mode II crack energy release rates for the ply. (Real > 0.0; No default).
GIIC	Critical mode III crack energy release rates for the ply. (Real > 0.0; No default).

Remarks:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. An error message is issued if the Y012 and YC12 fields are both blank, or if the Y012 field or the YC12 field is defined and the other is not.
3. The software uses the nonlinearity coefficients KSIT and KSIC to calculate the elastic modulus in the fiber direction as a nonlinear function of stress if PLYUNI = 0, or strain if PLYUNI = 1.
4. Use a positive value to define time delay. The software interprets negative values as zero time delay.
5. If TAU ≤ 0.0, there is no time delay and the ADEL field is ignored.
6. To define a nonlinear damage model for shear, you should leave the Y012 and YC12 fields blank and use the TID field to reference a TABLEM5 entry. On the TABLEM5 entry, you will include tabular data that defines how shear damage d12 varies as a function of thermodynamic force Y. An error message is issued if the Y012, YC12, and TID fields are all defined or are all blank, or if the Y012 field or the YC12 field is defined and the other is not.
7. For the UD model, the software uses the nonlinearity coefficients KSIT and KSIC to calculate the elastic modulus in the fiber direction as a nonlinear function of stress if PLYUNI= 0, or strain if PLYUNI = 1. For the EUD model, the software uses the nonlinearity coefficients KSIT and KSIC to calculate the elastic modulus in the fiber direction as a nonlinear function of strain.

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8. You should enter the time delay as a positive value. The software interprets a negative time delay value as zero.

9. If TAU ≤ 0.0, there is no time delay and the ADEL field is ignored.

10. The shear damage (d12) used for the UD ply failure model can be defined as a linear or nonlinear function of the thermodynamic force Y. You can specify the nonlinear function with the TABLEM5 bulk entry, which is referenced by the TID field on the MATDMG bulk entry. The TABLEM5 specifies the function *d12=f(sqrt(Y))*, where d12 is the y data, and sqrt(Y) is the x data.

11. The coupling coefficient A determines if the normal stress in the transverse direction σ_{22} and the normal stress in the out-of-plane direction σ_{33} are taken into consideration when plasticity occurs. For example, if A=0.0, then ϵ^p_{22} and ϵ^p_{33}, and the contribution of σ_{22} and σ_{33} to the yielding surface are ignored. The radius of the yielding surface follows the rule specified by $R_0+K\epsilon^{\gamma}_p$, where ϵ_p is the equivalent plastic strain, R_o is specified by LITK, K is specified by BIGK, and γ is specified by EXPN.

Related Topics

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MATF10C

Fluid or Absorber Material Property Definition in Complex Format

Defines tabular material properties for fluid or absorber elements in coupled fluid-structural analysis.

Format:

1	2	3	4	5	6	7	8	9	10
MATF10C	ID		TIDRHOR	TIDRHOI	TIDCR	TIDCI			

Example:

MATF10C	100		501	502	601	602			

Fields:

Field	Contents
ID	Material identification number. (Integer > 0)
TIDRHOR	Identification number of TABLEDi bulk entry which defines the real part of density. (Integer > 0; Default = 0)
TIDRHOI	Identification number of TABLEDi bulk entry which defines the imaginary part of density. (Integer > 0; Default = 0)
TIDCR	Identification number of TABLEDi bulk entry which defines the real part of speed of sound. (Integer > 0; Default=0)
TIDCI	Identification number of TABLEDi bulk entry which defines imaginary part of speed of sound. (Integer > 0; Default=0)

Remarks:

1. If TIDRHOR, TIDRHOI, TIDCR, or TIDCI field = blank, the corresponding value specified in MAT10C will be used in the computation.
2. TIDRHOR, TIDRHOI, TIDCR, or TIDCI cannot be all equal to zero.
3. MATF10C will be used only when MAT10C with the same ID is specified.
4. The FORM specified in MAT10C bulk entry will be applied to MATF10C bulk entry.

Related Topics

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MATFT

Material Properties for Composite Ply Failure Theories

Defines material properties for use with composite ply failure theories for shell and solid composites.

Format for Hill Failure Theory (FT = “HILL”)

1	2	3	4	5	6	7	8	9	10
MATFT	MID								
	FT	Xt	Xc	Yt	Yc	Zt	Zc	S12	
		S13	S23						

Format for Hoffman Failure Theory (FT = “HOFF”)

1	2	3	4	5	6	7	8	9	10
MATFT	MID								
	FT	Xt	Xc	Yt	Yc	Zt	Zc	S12	
		S13	S23						

Format for Tsai-Wu Failure Theory (FT = “TSAI”)

1	2	3	4	5	6	7	8	9	10
MATFT	MID								
	FT	Xt	Xc	Yt	Yc	Zt	Zc	S12	
		S13	S23	F12	F13	F23			

Format for Maximum Strain Failure Theory (FT = “STRN”)

1	2	3	4	5	6	7	8	9	10
MATFT	MID								
	FT	Xet	Xec	Yet	Yec	Zet	Zec	Se12	
		Se13	Se23						

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Format for Maximum Stress Failure Theory (FT = “STRS”)

1	2	3	4	5	6	7	8	9	10
MATFT	MID								
	FT	Xt	Xc	Yt	Yc	Zt	Zc	S12	
		S13	S23						

Format for Maximum Transverse Shear Stress Failure Theory (FT = “TS”)

1	2	3	4	5	6	7	8	9	10
MATFT	MID								
	FT	S13	S23						

Example:

1	2	3	4	5	6	7	8	9	10
MATFT	20								
	HILL	1.0E5	5.4E6	5.0E4	2.0E4	2.0E4	2.0E4		
	HOFF	7.2E6	5.6E6	5.2E6	4.4E6	4.0E5	3.5E5	2.1E6	
		2.1E6	2.1E6						

Fields:

Field	Contents
MID	Material identification number. See Remark 1. (Integer > 0)
FT	Ply failure theory. (Character; Default = No failure calculation is performed when blank.) Allowable entries are: “HILL” for the Hill failure theory. “HOFF” for the Hoffman failure theory. “TSAI” for the Tsai-Wu failure theory. “STRN” for the Maximum Strain failure theory. “STRS” for the Maximum Stress failure theory.

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Field	Contents
	"TS" for the Maximum Transverse Shear Stress failure theory. For a detailed explanation of each failure theory, see "Laminates" in the *Simcenter Nastran User's Guide*.
Xc, Yc, Zc	Allowable compressive stress in the longitudinal, lateral, and transverse principal material directions, respectively. (Real > 0.0; Default = Xc for Yc and Zc)
Xt, Yt, Zt	Allowable tensile stress in the longitudinal, lateral, and transverse principal material directions, respectively. (Real > 0.0; Default = Xt for Yt and Zt)
S12, S13, S23	Allowable in-plane shear stress (S12) and transverse shear stresses (S13 and S23). (Real > 0.0)
F12, F13, F23	Interaction terms in the tensor polynomial theory of Tsai-Wu. Required if the failure index is desired and either F12, F13, or F23 are non-zero. See Remark 4. (Real; Default = 0.0)
Xec, Yec, Zec	Allowable compressive strain in the longitudinal, lateral, and transverse principal material directions, respectively. (Real > 0.0; Default = Xec for Yec and Zec)
Xet, Yet, Zet	Allowable tensile strain in the longitudinal, lateral, and transverse principal material directions, respectively. (Real > 0.0; Default = Xet for Yet and Zet)
Se12, Se13, Se23	Allowable in-plane shear strain (Se12) and transverse shear strains (Se13 and Se23). (Real > 0.0 or blank)

Remarks:

1. The MID on a MATFT bulk entry must be the same as the MID on a MAT1, MAT9, or MAT11 bulk entry. Only one MATFT bulk entry is allowed per MID.

   The MATFT entry is required to define the allowable stress and strain values with the MAT9 and MAT11 material entries. For the MAT1 entry, if you use FT=HILL, HOFF, or TSAI, you can optionally specify the allowable stress on the MAT1 entry directly instead of the MATFT. If the allowables are defined on both the MAT1 and MATFT entries, the software applies the values on the MATFT.

   The MATFT entry is not supported with the MAT2 or MAT8 entry. You must define the allowable stress on the MAT2 entry, and the allowable stress or strain on the MAT8 entry.

2. Multiple failure theories can be defined on a single MATFT bulk entry, and each failure theory can only be specified once. The software applies the specific failure theory you select for the ply in the FTi field on the PCOMPS or PCOMPG1 entries.

3. If any failure modulus related to the transverse direction is not specified, failure indices and strength ratios are only calculated for the two-dimensional plane of the ply. If any failure modulus related to the longitudinal or lateral direction is not specified, failure indices and strength ratios are not calculated.

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4. If you enter a nonzero value for F_{12} and the first stability criterion below is not satisfied, the software uses $F_{12} = 0.0$. This behavior is the same for F_{23} and F_{13} using the second and third stability criterion below, respectively. Each is evaluated independently.

$$\left(\frac{1}{Xt\ Xc}\right)*\left(\frac{1}{Yt\ Yc}\right) - F_{12}^2 > 0$$
$$\left(\frac{1}{Zt\ Zc}\right)*\left(\frac{1}{Yt\ Yc}\right) - F_{23}^2 > 0$$
$$\left(\frac{1}{Xt\ Xc}\right)*\left(\frac{1}{Zt\ Zc}\right) - F_{13}^2 > 0$$

5. Failure index or strength ratio results for the maximum stress and maximum strain failure theories are ordered differently than the corresponding allowable values are listed on the MATFT bulk entry.

Order on MATFT bulk entry	Order in results file
Value corresponding to 11 component	Value corresponding to 11 component
Value corresponding to 22 component	Value corresponding to 22 component
Value corresponding to 33 component	Value corresponding to 33 component
Value corresponding to 12 component	Value corresponding to 12 component
Value corresponding to 13 component	Value corresponding to 23 component
Value corresponding to 23 component	Value corresponding to 13 component

6. For shell composites, only FT = HILL, HOFF, TSAI, and STRN are supported (FT = STRS and TS are not supported). In addition, the transverse material properties are ignored by shell elements.

   FT = HILL or HOFF: Zt, Zc, S13, and S23 are ignored.

   FT = TSAI: Zt, Zc, S13, S23, F13, and F23 are ignored.

   FT = STRN: Zet, Zec, Se13, and Se23 are ignored.

Related Topics

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MATG

Gasket Material Properties (SOL 402 only)

Defines the material properties for gasket materials.

Format:

1	2	3	4	5	6	7	8	9	10
MATG	MID	IDMEM	BEHAV	TABLD	TABLU1	TABLU2	TABLU3	TABLU4	
	TABLU5	TABLU6	TABLU7	TABLU8	TABLU9	TABLU10	YPRESS	EPL	
	GPL	GAP							

Example:

MATG	3	1	0	101	102	103	104		
							350	2000	
	1270	0							

Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
IDMEM	ID of MAT1 providing material behavior for the membrane (in-plane) direction. (Integer > 0)
BEHAV	Behavior type. Only type 0 is supported. (Integer)
TABLD	Identification number of a TABLES1 entry providing the main loading curve of the gasket (pressure vs. closure). (Integer > 0)
TABLUi	Identification number of a TABLES1 entry providing the unloading curve(s) of the gasket (pressure vs. closure). At least one unloading curve must be specified. Leave fields blank for tables that are not required. See Remark 3. (Integer > 0)
YPRESS	Initial yield pressure. See Remark 5. (Real >0.0)
EPL	Tensile modulus (pressure per unit length). (Real > 0.0)
GPL	Transverse shear modulus (force per unit area). (Real > 0.0)
GAP	Initial closure strain (unitless). A non-zero value of GAP shifts the pressure-strain curve in Figure 15-129 to the right by the value of GAP, as shown in Figure 15-130. (0.0≤Real<1.0; Default=0.0)

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Remarks:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. The material X-axis is the thickness direction unless it is explicitly defined by the CORDM field of the PSOLID bulk entry. However, no automatic detection of thicknesses is done.
3. MATG can only be used with CHEXA elements with 8 grid points or CPENTA elements with 6 grid points. There should be only one layer of elements in the direction of the gasket thickness.
4. MATG defines the compressive behavior in the thickness direction. The in-plane behavior is defined by the MAT1 entry referenced by IDMEM.
5. All unloading curves must have the same number of points as the elastic portion of the loading curve (i.e., up to the initial yield pressure). The first point must be defined at pressure=0.0 and the last point must coincide with a yield point on the loading curve. Not all yield points require unloading curves. The unloading behavior at intermediate points is interpolated from adjacent curves. See Figure 15-129.
6. MID, IDMEM, TABLD, TABLU1, YPRS, EPL, and GPL must be specified.
7. The initial yield stress must match a point on the loading curve of TABLD.
8. Closure (strain) is defined as the change in gasket thickness divided by the original gasket thickness.

Figure 15-129: Pressure-Closure Curves without Initial Gap (GAP)

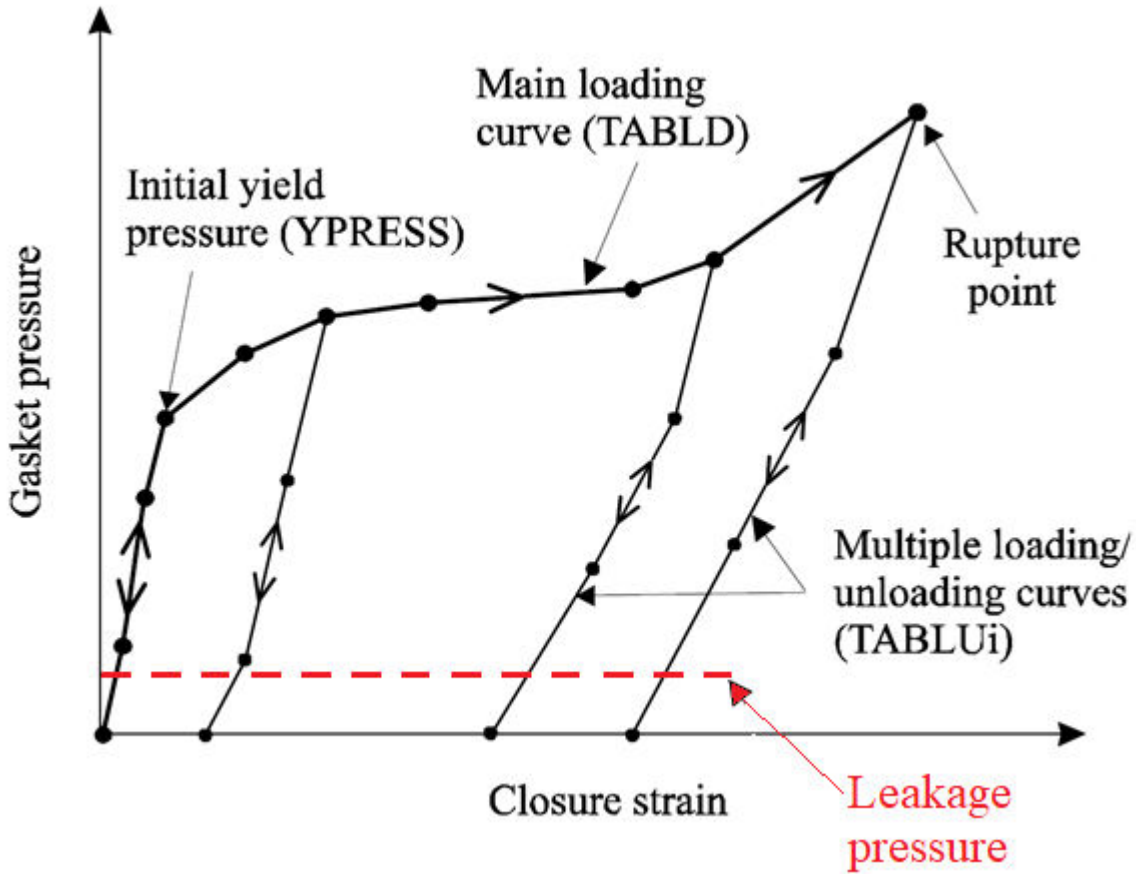

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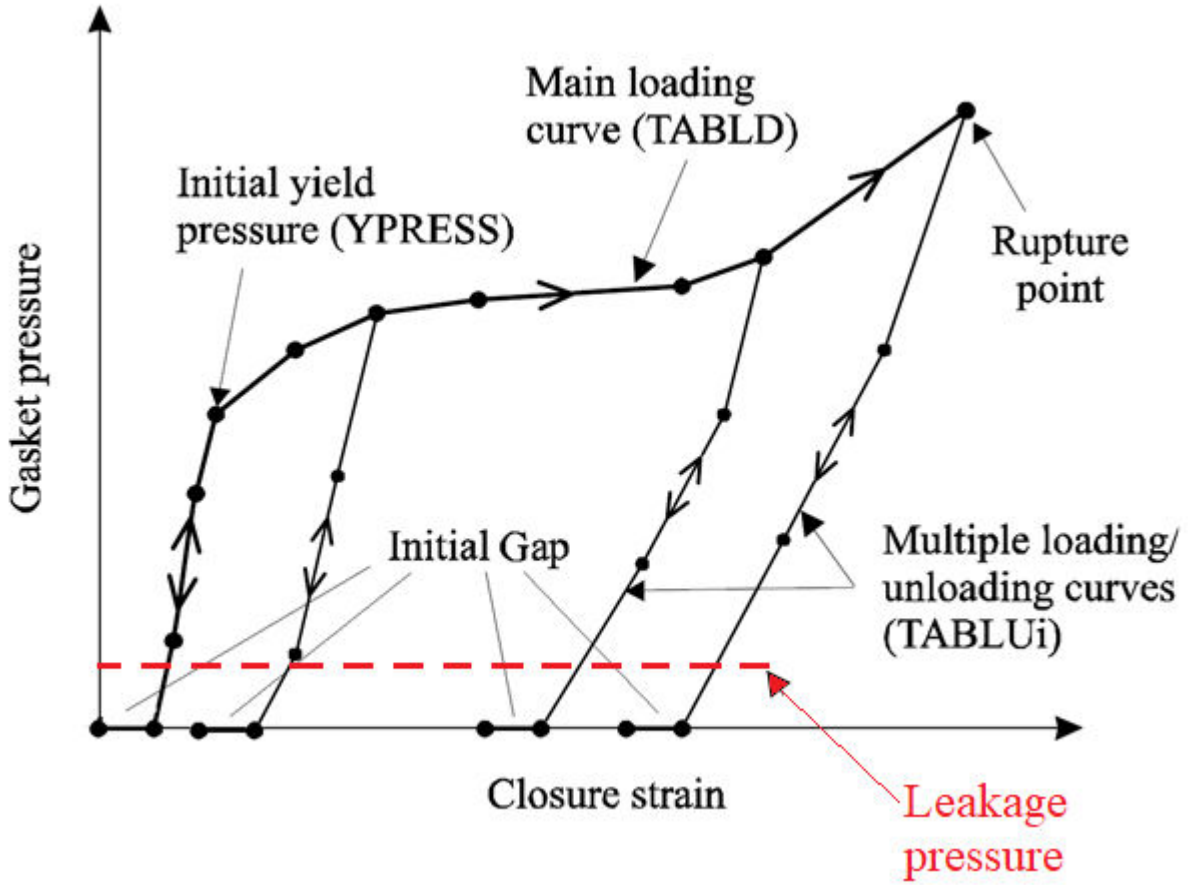
Figure 15-130: Pressure-Closure Curves with Initial Gap (GAP)



9. The software chooses the material X-axis as the thickness direction unless it is explicitly defined by the CORDM field of the PSOLID bulk entry. However, no automatic detection of thicknesses is done.

Related Topics

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MATHE

Hyperelastic Material Properties (SOL 402 only)

Specifies hyperelastic (rubber-like) material properties for SOL 402.

Format for Generalized Mooney-Rivlin model (default) (Model = Mooney):

1	2	3	4	5	6	7	8	9	10
MATHE	MID	Model		K	RHO	TEXP	TREF		
	C10	C01							
	C20	C11	C02						
	C30	C21	C12	C03					

Format for Ogden model or Hyperfoam model (Model = Ogden or Foam):

1	2	3	4	5	6	7	8	9	10
MATHE	MID	Model		K	RHO	TEXP	TREF		
	MU1	ALPHA1	BETA1						
	MU2	ALPHA2	BETA2	MU3	ALPHA3	BETA3			
	MU4	ALPHA4	BETA4	MU5	ALPHA5	BETA5			
	MU6	ALPHA6	BETA6	MU7	ALPHA7	BETA7			
	MU8	ALPHA8	BETA8	MU9	ALPHA9	BETA9			

Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
MODEL	Hyperelastic material model type (Character; Default = MOONEY) MOONEY: Generalized Mooney-Rivlin model OGDEN: Ogden model FOAM: Hyperfoam model
K	Specifies a bulk modulus. Not used in hyperfoam material. (Real or Integer > 0; Default = automatically set for nearly incompressible condition). If real entry, value is used at all temperatures.

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Field	Contents
	If integer entry, identification number of TABLEM1 bulk entry that defines the bulk modulus as a function of temperature.
RHO	Mass density in reference configuration. (Real; Default = 0.0)
TEXP	Coefficient of thermal expansion. (Real or Integer > 0; Default = 0.0) If real entry, value is used at all temperatures. If integer entry, identification number of TABLEM1 bulk entry that defines the thermal expansion as a function of temperature.
TREF	Reference temperature for the calculation of thermal strain. See the remarks on the **TEMPERATURE** case control command. (Real; Default = 0.0)
Cij	Material coefficients related to distortional deformation for generalized Mooney-Rivlin model. See Remarks **3** and **4**. (Real or Integer > 0; Default = 1.0 for C10 and C01, = 0.0 for other constants). If real entry, value is used at all temperatures. If integer entry, identification number of TABLEM1 bulk entry that defines the coefficients as a function of temperature.
MUk	Coefficients μ_k of the strain energy function for Ogden or hyperfoam material. See Remarks **5** and **6**. (Real or Integer > 0; Default = 0.0) If real entry, value is used at all temperatures. If integer entry, identification number of TABLEM1 bulk entry that defines the coefficients as a function of temperature.
ALPHAk	Coefficients α_k of the strain energy function for Ogden or hyperfoam material. See Remarks **5** and **6**. (Real or Integer > 0; Default = 0.0) If real entry, value is used at all temperatures. If integer entry, identification number of TABLEM1 bulk entry that defines the coefficients as a function of temperature.
BETAk	Coefficients β_k of the strain energy function for hyperfoam material. These fields are unused for the Ogden model. See Remark **6**. (Real or Integer > 0; Default = 0.0) If real entry, value is used at all temperatures. If integer entry, identification number of TABLEM1 bulk entry that defines the coefficients as a function of temperature.

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Remarks:

1. Hyperelastic materials *do not* require the parameter setting PARAM,MATNL,1 to switch material nonlinear on.
2. Minimum input lines: 1 for Mooney, 2 for Ogden or foam.
3. For the generalized Mooney-Rivlin material model, the strain energy function is

$$W\left(J,\bar{I}_1,\bar{I}_2\right)=\sum_{i+j=1}^{3}C_{ij}\left(\bar{I}_1-3\right)^i\left(\bar{I}_2-3\right)^j+0.5K(J-1)^2$$

with

$$\bar{I}_1=\bar{\lambda}_1^2+\bar{\lambda}_2^2+\bar{\lambda}_3^2$$

$$\bar{I}_2=\bar{\lambda}_1^2\bar{\lambda}_2^2+\bar{\lambda}_2^2\bar{\lambda}_3^2+\bar{\lambda}_3^2\bar{\lambda}_1^2$$

where:

- $\bar{\lambda}_i$ is the deviatoric stretch ratio defined as:

$$\bar{\lambda}_i=J^{-1/3}\lambda_i$$

- *K* and *J* are the bulk modulus and volume ratio, respectively.
- Note that the volume part of the strain energy function for the MATHE and MATHP bulk entries is different. Specifically, you define the bulk modulus *K* directly on the MATHE entry, but on the MATHP, you define *D1*, which is computed as *D1=2/K*.

4. For the Mooney-Rivlin model, the solver requires the linearized Poisson coefficient to be positive. The linearized Poisson coefficient *Nu* is computed based on the bulk modulus *K* and the two first material constants C_{01} and C_{10} using:

$$G=2\left(C_{01}+C_{10}\right)$$

$$Nu=\frac{3K-2G}{6K+2G}$$

where *G* is the shear modulus.

5. For the Ogden material model, the strain energy function is

$$W=\sum_{k=1}^{9}\frac{\mu_k}{\alpha_k}\left(\bar{\lambda}_1^{\alpha_k}+\bar{\lambda}_2^{\alpha_k}+\bar{\lambda}_3^{\alpha_k}-3\right)+0.5K(J-1)^2$$

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μ_k and α_k are material constants.

$\bar{\lambda}_i$ is the deviatoric stretch ratio defined as:

$$\bar{\lambda}_i = J^{-1/3}\lambda_{i,}$$

K and *J* are the bulk modulus and volume ratio, respectively.

Note that the Ogden strain energy function used in Simcenter Nastran is the same used in Ansys, but is different from the function used in Abaqus and Samcef, which is:

$$U = \sum_{i=1}^{N} \frac{2\mu_i}{\alpha_i^2}\left(\bar{\lambda}_1^{\alpha_i} + \bar{\lambda}_2^{\alpha_i} + \bar{\lambda}_3^{\alpha_i} - 3\right) + \sum_{i=1}^{N} \frac{1}{D_i}\left(J^{el} - 1\right)^{2i}$$

The difference is in the deviatoric part with the definition of μ_k used in the Simcenter Nastran function and μ_i used in the Abaqus and Samcef function. You can convert μ_i to μ_k with the following:

$$\mu_k = (2/\alpha_i) * \mu_i$$

A two-term Ogden model can be made equivalent to a simple Mooney-Rivlin model by defining μ_1=2C_{10}, α_1=2, μ_2=2C_{01}, and α_2=-2.

For small strains, the shear modulus *G* is related to the Ogden constants by

$$G = \frac{1}{2}\sum_{k=1}^{9} \mu_k\alpha_k$$

6. For the hyperfoam material model, the strain energy function is

$$W = \sum_{k=1}^{9} \frac{\mu_k}{\alpha_k}\left(\lambda_1^{\alpha_k} + \lambda_2^{\alpha_k} + \lambda_3^{\alpha_k} - 3 + \frac{1}{\beta_k}\left(J^{-\alpha_k\beta_k} - 1\right)\right)$$

μ_k, α_k, β_k are material constants,

and *J* is the volume ratio.

Note that the Ogden strain energy function used in Simcenter Nastran is the same used in Ansys, but is different from the function used in Abaqus and Samcef, which is:

$$U = \sum_{i=1}^{N} \frac{2\mu_i}{\alpha_i^2}\left(\bar{\lambda}_1^{\alpha_i} + \bar{\lambda}_2^{\alpha_i} + \bar{\lambda}_3^{\alpha_i} - 3\right) + \frac{1}{\beta_i}\left(\left(J^{el}\right)^{-\alpha_i\beta_i} - 1\right)$$

The difference is in the deviatoric part with the definition of μ_k used in the Simcenter Nastran function and μ_i used in the Abaqus and Samcef function. You can convert μ_i to μ_k with the following:

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$$\mu_k = (2/\alpha_i) * \mu_i$$

For small strains, the shear modulus *G* and bulk modulus *K* are related to the hyperfoam constants by

$$G = \frac{1}{2}\sum_{k=1}^{9}\mu_k\alpha_k$$

$$K = \sum_{k=1}^{9}\left(\beta_k + \frac{1}{3}\right)\mu_k\alpha_k$$

These moduli must be greater than zero, as a result, β_k should be greater than -1/3.

When all of the β_k are equal to each other, then β is related to the Poisson's ratio by

$$\beta = \frac{\nu}{1-2\nu}$$

7. For any of the hyperelastic material models, for almost incompressible materials, if you define your material properties such that the computed linearized Poisson coefficient *Nu* exceeds .495, the solution is likely to experience convergence problems. The software will report a warning message if *Nu* exceeds .495. You can reduce *Nu* by reducing the bulk modulus.

Nu is computed with the bulk modulus (*K*) and the shear modulus (*G*) by

$$Nu = \frac{3K-2G}{6K+2G} = \frac{(3K/G)-2}{(6K/G)+2}$$

For the Mooney-Rivlin model, *K* is specified directly. See Remark **4** for the computation of *G*.

For the Ogden model, *K* is specified directly. See Remark **5** for the computation of *G*.

For the Hyperfoam model, see Remark **6** for the computation of *K* and *G*.

Related Topics

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MATHEM

Mullins Effect on Hyperelastic Material (SOL 402)

Specifies Mullins effect (Ogden-Roxburgh model) on hyperelastic (MATHE) material.

Format

1	2	3	4	5	6	7	8	9	10
MATHEM	MID	R	M						

Example

MATHEM	11	2.0	22.0						

Fields:

Field	Contents
MID	Material identification number of a MATHE entry. (Integer > 0).
R, M	Material constants of the Ogden-Roxburgh model for the Mullins effect. (Real > 0.0)

Related Topics

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MATHEV

Viscoelastic Effect on Hyperelastic Material (SOL 402)

Specifies viscoelastic effect (Holzapfel model) on hyperelastic (MATHE) material.

Format:

1	2	3	4	5	6	7	8	9	10
MATHEV	MID	SHIFT	C1	C2	TREF				
	beta1	tau1		usage1					
	beta2	tau2		usage2					
	-etc.-								

Example:

MATHEV	11								
	0.5	0.7		0					
	0.3	0.6		1					

Fields:

Field	Contents
MID	Material identification number of the MATHE bulk entry. (Integer > 0)
SHIFT	Time-temperature superposition shift law. See Remarks 1, 2, and 3. (Integer, Default = 0) 0: No time-temperature superposition 1: Use WLF (Williams-Landel-Ferry) shift function 2: Use Arrhenius shift function
C1	Material-dependent constant for the WLF or Arrhenius shift functions. See Remarks 2 and 3. (Real > 0.0; No default)
C2	Material-dependent constant for the WLF or Arrhenius shift functions. See Remarks 2 and 3. (Real ≥ 0.0; Default = 0.0)

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Field	Contents
TREF	Reference temperature used by the WLF or Arrhenius shift function. (Real; For default behavior, see Remark 4)
beta(i)	Factor beta for chain (i) of the viscoelastic model. (Real > 0.0; No default)
tau(i)	Relaxation time for chain (i) of the viscoelastic model. (Real > 0.0; No default)
usage(i)	Usage of chain (i). (Integer, Default = 0)
	0: Chain is based on the deviatoric strain energy
	1: Chain is based on the volumetric strain energy
	2: Chain is based on the total strain energy

Remarks:

1. When the SHIFT field is blank or set to zero, the software ignores the C1, C2, and TREF fields.
2. The WLF shift function calculates the shift factor as follows:

$$\log_{10}\alpha_T = -\left[\frac{C_1\left(T - T_{ref}\right)}{C_2 + \left(T - T_{ref}\right)}\right]$$

where:

α_T Shift factor

C_1 Dimensionless material-dependent constant specified in the C1 field

C_2 Material-dependent constant specified in the C2 field that has the units of temperature

T Temperature of the material

T_{ref} Reference temperature specified in the TREF field

The temperature units for the values in the C2 and TREF fields must be the same as the temperature units for the input temperature of the material. If they are in °C (°F), include a PARAM,TABS,273.16 (PARAM,TABS,459.69) specification in the input file to convert the values in the C2 and TREF fields, and the temperature of the material to K (R).

3. The Arrhenius shift function calculates the shift factor as follows:

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$$\log_{10}\alpha_T = \begin{cases} C_1\left(\frac{1}{T} - \frac{1}{T_{ref}}\right) \text{ if } T \geq T_{ref} \\ C_2\left(\frac{1}{T} - \frac{1}{T_{ref}}\right) \text{ if } T < T_{ref} \end{cases}$$

where:

α_T	Shift factor
C_1	Material-dependent constant specified in the C1 field that has the units of temperature
C_2	Material-dependent constant specified in the C2 field that has the units of temperature
T	Temperature of the material
T_{ref}	Reference temperature specified in the TREF field

The temperature units for the values in the C1, C2, and TREF fields must be the same as the temperature units for the input temperature of the material. If they are in °C (°F), include a PARAM,TABS,273.16 (PARAM,TABS,459.69) specification in the input file to convert the values in the C1, C2, and TREF fields, and the temperature of the material to K (R).

4. If the TREF field is blank, the reference temperature for the calculation of thermal loads is obtained from the TEMPD entry referenced by the TEMPERATURE(INITIAL) set selection. If TEMPD is not specified, TREF defaults to 0.0.
5. If two Prony series are used (at least one USAGEi=0 or 1), Ogden and Hyperfoam Hyperelastic models are not available.
6. The number of terms in a Prony series is limited to 20.

Related Topics

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MATHP

Hyperelastic Material Properties

Specifies material properties for use in fully nonlinear (i.e., large strain and large rotation) hyperelastic analysis of rubber-like materials (elastomers).

Format:

1	2	3	4	5	6	7	8	9	10
MATHP	MID	A10	A01	D1	RHO	AV	TREF	GE	
		NA	ND						
	A20	A11	A02	D2					
	A30	A21	A12	A03	D3				
	A40	A31	A22	A13	A04	D4			
	A50	A41	A32	A23	A14	A05	D5		
	TAB1	TAB2	TAB3	TAB4				TABD	

Fields:

Field	Contents
MID	Identification number of a MATHP entry. (Integer > 0; No Default)
Aij	Material constants related to distortional deformation. (Real; Default = 0.0)
Di	Material constants related to volumetric deformation. (Real ≥ 0; Default for D1 is 10^3·(A10+A01); Default for D2 through D5 is 0.0)
RHO	Mass density in original configuration. (Real; Default = 0.0)
AV	Coefficient of volumetric thermal expansion. (Real; Default = 0.0)
TREF	Reference temperature for the calculation of thermal strain. See the remarks on the TEMPERATURE case control command. (Real; Default = 0.0)
GE	Structural damping element coefficient. (Real; Default = 0.0)
NA	Order of the distortional strain energy polynomial function. (0 < Integer ≤ 5; Default = 1)
ND	Order of the volumetric strain energy polynomial function. (0 < Integer ≤ 5; Default = 1)
TAB1	Table identification number of a TABLES1 entry that contains simple tension/compression data to be used in the estimation of the material constants Aij. xi values in the TABLES1 entry must be stretch ratios l/l_0 and yi values must be values of the engineering stress F/A_0. Stresses are negative for compression and positive for tension.

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Field	Contents
	If this convention is not followed the job may fail to converge or issue a traceback. (Integer > 0 or blank)
TAB2	Table identification number of a TABLES1 entry that contains equibiaxial tension data to be used in the estimation of the material constants Aij. xi values in the TABLES1 entry must be stretch ratios l/l_0. yi values must be values of the engineering stress F/A_0. l is the current length, F is the current force, A_0 is the cross-sectional area. In the case of pressure of a spherical membrane, the engineering stress is given by $Pr_0\lambda^2/2t_0$ where P=current value of the pressure and r_0, t_0=initial radius and thickness. (Integer > 0 or blank)
TAB3	Table identification number of a TABLES1 entry that contains simple shear data to be used in the estimation of the material constants Aij. xi values in the TABLES1 entry must be values of the shear tangent γ and yi values must be values of the engineering shear stress F/A_0 ·. (Integer > 0 or blank)
TAB4	Table identification number of a TABLES1 entry that contains pure shear data to be used in the estimation of the material constants Aij. xi and yi values in the TABLES1 entry must be stretch ratios $\lambda_1=l/l_0$ and values of the nominal stress F/A_0.l is the current length, F is the current force, l_0 and A_0 are the initial length and cross-sectional area, respectively in the 1-direction. (Integer > 0 or blank)
TABD	Table identification number of a TABLES1 entry that contains pure volumetric compression data to be used in the estimation of the material constants Di. xi values in the TABLES1 entry must be values of the volume ratio $J=\lambda^3$ where $\lambda=l/l_0$ is the stretch ratio in all three directions; yi values must be values of the pressure, assumed positive in compression. (Integer > 0 or blank)

Remarks:

1. The generalized Mooney-Rivlin strain energy function may be expressed as follows:

$$U\left(J,\bar{I}_1,\bar{I}_2\right)=\sum_{i+j=1}^{NA}A_{ij}\left(\bar{I}_1-3\right)^i\left(\bar{I}_2-3\right)^j+\sum_{i=1}^{ND}D_i(J-1-AV\left(T-T_0\right))^{2i}$$

   where:

   - $\bar{I}_1$ and $\bar{I}_2$ are the first and second distortional strain invariants, respectively.
   - J=det F is the determinant of the deformation gradient.
   - $2D_1=K$ and $2(A_{10}+A_{01})=G$ at small strains, in which K is the bulk modulus and G is the shear modulus.

     The model reduces to a Mooney-Rivlin material if NA=1, and to a Neo-Hookean material if NA = 1 and A_{01} = 0.0. See Remark 2. For Neo-Hookean or Mooney-Rivlin materials no continuation entry is required.
   - T is the current temperature and T_0 is the initial temperature.

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2. Conventional Mooney-Rivlin and Neo-Hookean materials are fully incompressible. Full incompressibility is not presently available but may be simulated with a large enough value of D_1. A value of D_1 higher than $10^3 \cdot (A_{10} + A_{01})$ is not recommended.

3. A_{ij} and D_i are obtained from least squares fitting of experimental data. A_{ij}, as well as D_i, are in the units of force per area. One or more of four experiments (TAB1 to TAB4) may be used to obtain A_{ij}. D_i may be obtained from pure volumetric compression data (TABD). If all TAB1 through TAB4 are blank, A_{ij} must be specified by the user. Parameter estimation, specified through any of the TABLES1 entries, supersedes the manual input of the parameters.

4. If *ND*=1 and a nonzero value of D_1 is provided or is obtained from experimental data in TABD, then the parameter estimation of the material constants A_{ij} takes compressibility into account in the cases of simple tension/compression, equibiaxial tension, and general biaxial deformation. Otherwise, full incompressibility is assumed in estimating the material constants.

5. For SOL 106, see "Hyperelastic Material" in the *Simcenter Nastran Basic Nonlinear Analysis User's Guide* for further details.

Remarks related to SOL 402:

1. Hyperelastic materials *do not* require the parameter setting PARAM,MATNL,1 to switch material nonlinear on.

2. The degree of the Mooney-Rivlin strain potential is limited to 4 for the distortional part such that coefficients A_{50}, A_{41}, A_{32}, A_{23}, A_{14} and A_{05} are ignored (*NA* should be at most equal to 4).

3. Regarding the volumetric part, the degree must be equal to 1 which means that *ND* has to be equal to 1 and that coefficients D_2, D_3, D_4, and D_5 are ignored.

4. The solver requires the linearized Poisson coefficient to be positive. The linearized Poisson coefficient *Nu* is computed based on the volumetric coefficient D_1 and the two first distortional constants A_{10} and A_{01}.

$$G = 2\left(A_{10} + A_{01}\right)$$

$$Nu = \frac{3D_1 - G}{6D_1 + G}$$

where *G* is the shear modulus.

Substituting $D_1 = K / 2$, *Nu* can be expressed in terms of *K* and *G*.

$$Nu = \frac{3K - 2G}{6K + 2G} = \frac{(3K/G) - 2}{(6K/G) + 2}$$

For almost incompressible materials, if you define your material properties such that the computed linearized Poisson coefficient *Nu* exceeds .495, the solution is likely to experience convergence problems. The software will report a warning message if *Nu* exceeds .495. You can reduce *Nu* by reducing *K*.

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5. Note that the volume part of the strain energy function for the MATHE and MATHP bulk entries is different. Specifically, you define the bulk modulus *K* directly on the MATHE entry, but on the MATHP, you define *D1*, which is computed as *D1=2/K*.
6. TABi and TABD may be used to obtain the material constants.
7. If TREF is blank, the reference temperature for the calculation of thermal loads is obtained from the TEMPD entry referenced by the TEMPERATURE(INITIAL) set selection. If TEMPD is also not specified, TREF=0.0 is used.

Related Topics

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MATOVR

Disable Nonlinear Effects in Elements

Used to globally turn off creep and/or plasticity effects in groups of elements. Applicable to SOL 401 only.

Format:

1	2	3	4	5	6	7	8	9	10
MATOVR	GRPID	PL	CR						

Example:

MATOVR	110		NO						

Fields:

Field	Contents
GRPID	Group identification number. See Remark 1. (Integer > 0)
PL	If nonlinear analysis is enabled, include plasticity effects. See Remark 2. (Character = “YES” or “NO”; Default = “YES”)
CR	If nonlinear analysis is enabled, include creep effects. See Remark 2. (Character = “YES” or “NO”; Default = “YES”)

Remarks:

1. The GROUP bulk entry referenced in the GRPID field must have TYPE = “ELEM” or “PROP”. If TYPE = “ELEM”, all the elements listed in the group are selected. If TYPE = “PROP”, all the elements that reference the physical properties listed in the group are selected.
2. If the MATNL parameter specification does not enable nonlinear analysis, specifying PL or CR = “YES” does not switch on plasticity or creep effects in the group of elements.
3. Multiple MATOVR bulk entries are permissible so long as the GRPID fields are unique.

Related Topics

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MATPOR

Porous Material Property Definition

Defines material properties for porous materials used as acoustic absorbers.

Format for Craggs model (MODEL = “CRAGGS”):

1	2	3	4	5	6	7	8	9	10
MATPOR	MID	MODEL	RHO	C					
	RES	POR	TORT						

Format for Delaney-Miki model (MODEL = “DELMIKI”):

1	2	3	4	5	6	7	8	9	10
MATPOR	MID	MODEL	RHO	C	FRAME				
	RES	POR		DENS					

Format for Johnson-Champoux-Allard model (MODEL = “JCA”):

1	2	3	4	5	6	7	8	9	10
MATPOR	MID	MODEL	RHO	C	FRAME	GAMMA	PR	MU	
	RES	POR	TORT	DENS	L1	L2	E/TIDE	NU/TIDNU	
	GE/TIDGE	FREQREF							

Examples:

MATPOR	20	CRAGGS	1.225E-9	340000.0					
	87.0E-6	0.97	2.52						

MATPOR	20	DELMIKI	1.225E-9	340000.0	LIMP				
	87.0E-6	0.97		31.0E-9					

MATPOR	20	JCA	1.225E-9	340000.0	LIMP	1.4	0.71	1.84E-8	
	87.0E-6	0.97	2.52	31.0E-9	0.037	0.119			

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Fields:

Field	Contents
MID	Material identification number. (Integer > 0)
MODEL	Porous material model. (Character; No default) See Remark 4.
	"CRAGGS" for the Craggs model.
	"DELMIKI" for the Delany-Miki model.
	"JCA" for the Johnson-Champoux-Allard model.
RHO	Density of the fluid. (Real > 0.0; No default)
C	Speed of sound in the fluid. (Real > 0.0; No default)
FRAME	Compliance of frame structure. (Character; No default) See Remark 3.
	"RIGID" for a rigid frame structure.
	"ELASTIC" for an elastic frame structure (poro-elastic material).
	"LIMP" for an extremely soft frame structure.
GAMMA	Ratio of constant pressure specific heat to constant volume specific heat. (Real > 0.0; No default)
PR	Prandtl number. (Real > 0.0; No default)
MU	Dynamic viscosity. (Real > 0.0; No default)
RES	Flow resistivity. (Real ≥ 0.0; No default)
POR	Porosity. (0.0 < Real ≤ 1.0; No default)
TORT	Tortuosity. (Real ≥ 1.0; Default = 1.0)
DENS	Density of the frame. (Real; Default = 0.0)
L1	Characteristic viscous length. (Real > 0.0; No default)
L2	Characteristic thermal length. (Real > 0.0; No default)
E/TIDE	Young's modulus of the frame structure when FRAME=ELASTIC. (Real > 0.0 or Integer > 0; No default)
	E: For a constant modulus at all frequencies, enter a real value.
	TIDE: For a frequency-dependent modulus, enter the ID (integer) of a TABLEDi bulk entry that defines the modulus as a function of frequency.

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Field	Contents
NU/TIDNU	Poisson's ratio of the frame structure when FRAME=ELASTIC. (Real > 0.0 or Integer > 0; No default) NU: For a constant Poisson's ratio at all frequencies, enter a real value. TIDNU: For a frequency-dependent Poisson's ratio, enter the ID (integer) of a TABLEDi bulk entry that defines Poisson's ratio as a function of frequency.
GE/TIDGE	Damping coefficient of the frame structure when FRAME=ELASTIC. (Real > 0.0 or Integer > 0; No default) GE: For a constant damping coefficient at all frequencies, enter a real value. TIDGE: For a frequency-dependent damping coefficient, enter the ID (integer) of a TABLEDi bulk entry that defines the damping coefficient as a function of frequency.
FREQREF	Reference frequency used when FRAME=ELASTIC, and E, NU, or GE are defined as frequency dependent. The reference frequency is used to determine a value of E, NU, or GE when computing the initial frame properties in a modal solution. (Real ≥ 1.0; No default)

Remarks:

1. MATPOR bulk entries can only be referenced by PSOLID bulk data entries and associated with CHEXA, CPENTA, CPYRAM, and CTETRA bulk data entries.
2. The software converts a MATPOR bulk entry to an equivalent fluid with equivalent mass density and speed of sound dependent bulk modulus. The equivalent fluid is calculated as follows: sqrt (bulk modulus / mass density).
3. The frame type describes how the software takes into account the structure used for the solid skeleton around air-filled pores.
   - FRAME = RIGID: The frame's properties are not taken into account in the derivation of equivalent mass density and bulk modulus.
   - FRAME = ELASTIC: The frame's elastic properties are taken into account. For this case, the software uses the full Biot model which includes both the acoustic fluid phase DOF and the structural solid phase DOFs.
   - FRAME = LIMP: The mass of the frame is taken into account in the computation of the equivalent mass density of the fluid; that is, the inertial effect is taken into account. However, the porous material is still represented as an equivalent fluid with only one pressure DOF per node and no extra structural displacement DOFs. This leads to more accurate results than FRAME = RIGID.
4. For a detailed explanation of each material model, see the *Acoustics User's Guide*.

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Related Topics

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MATS1

Material Stress Dependence

Specifies stress-dependent material properties.

Format:

1	2	3	4	5	6	7	8	9	10
MATS1	MID	TID	TYPE	H	YF	HR	LIMIT1	LIMIT2	
	STRMEAS								

Example:

MATS1	17	28	PLASTIC	0.0	1	1	2.+4		

Fields:

Field	Contents
MID	SOLs 106 and 129: Identification number of a MAT1, MAT2, MAT8, MAT9, or MAT11 entry. SOLs 401 and 402: Identification number of a MAT1.
TID	Identification number of a TABLES1 or TABLEST entry. If H is given, then this field must be blank. See Remark 3. (Integer > 0 or blank)
TYPE	Type of material nonlinearity. (Character: “NLELAST” or “PLASTIC” or “PLSTRN”) “NLELAST” for nonlinear elastic. Valid for SOLs 106, 129, and 401. See Remarks 1 and 3. “PLASTIC” for elasto-plastic. Valid for SOLs 106, 129, 401, and 402. See Remarks 2 and 3. “PLSTRN” for plastic strain. Valid for SOLs 401 and 402 only. See Remarks 2 and 3.
H	Work hardening slope (slope of stress versus plastic strain) in units of stress. For elastic-perfectly plastic cases, H = 0.0. For more than a single slope in the plastic range, the stress-strain data must be supplied on a TABLES1 entry referenced by TID, and this field must be blank. See Remark 2. (Real or blank; Default = 0.0 if TID field is blank)
YF	Yield function. See Remark 3. (Integer) 1 = von Mises (Default) 2 = Tresca

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Field	Contents
	3 = Mohr-Coulomb 4 = Drucker-Prager
HR	Hardening rule. See Remark 2. (Integer) 1 = Isotropic (Default) 2 = Kinematic 3 = Combined isotropic and kinematic hardening
LIMIT1	Initial yield point. See Table 15-81. (Real > 0.0 or blank)
LIMIT2	Internal friction angle, measured in degrees, for the Mohr-Coulomb and Drucker-Prager yield criteria. See Table 15-81. (0.0 ≤ Real < 45.0 or blank)
STRMEAS	Stress/strain measure of the TABLES1 or TABLEST data referenced by the TID field. Valid for SOLs 401 and 402 only. See Remark 4. (Character: "UNDEF", "ENG", “TRUE”, "CAUCHY" or blank)

Table 15-81: Yield Functions Versus LIMIT1 and LIMIT2

Yield Function (YF)	LIMIT1	LIMIT2
von Mises or Tresca	Initial Yield Stress in Tension, σ_y	Not used
Mohr-Coulomb or Drucker-Prager	2 x Cohesion, 2c (in units of stress)	Angle of Internal Friction φ (in degrees)

Remarks:

1. If TYPE = “NLELAST”, then MID may reference a MAT1 entry only. The stress-strain data in the TABLES1 entry is used to determine the stress for a given value of strain. The values H, YF, HR, LIMIT1, and LIMIT2 will not be used in this case.

   Thermo-elastic analysis with temperature-dependent material properties is available for isotropic linear and nonlinear elastic materials (TYPE = “NLELAST”) and anisotropic linear elastic materials. Four options for the constitutive relation exist. These options are listed in Table 15-82 along with the required bulk entries.

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Table 15-82: Constitutive Relations and Required Material Property Entries

Constitutive Relation	Required Bulk Data Entries
$\{\sigma\} = [G_e(T)]\{\varepsilon\}$	MATi and MATTi where i = 1, 2, 8, 9, or 11 (MATS1 is not required)
$\left\{\sigma\right\} = \frac{\overline{E}\left(\sigma,\varepsilon\right)}{E}[G_e(T)]\left\{\varepsilon\right\}$	MAT1, MATT1, MATS1, and TABLES1
$\left\{\sigma\right\} = \frac{\overline{E}\left(T,\sigma,\varepsilon\right)}{E}[G_e]\left\{\varepsilon\right\}$	MAT1, MATS1, TABLEST, and TABLES1
$\left\{\sigma\right\} = \frac{\overline{E}\left(T,\sigma,\varepsilon\right)}{E}[G_e(T)]\left\{\varepsilon\right\}$	MAT1, MATT1, MATS1, TABLEST, and TABLES1

In Table 15-82, $\{\sigma\}$ is the stress vector, $\{\epsilon\}$ is the strain vector, $[G_e]$ is the elasticity matrix, T is temperature, $\bar{E}$ is the effective elastic modulus, and E is the reference elastic modulus.

2. If TYPE = "PLASTIC" or "PLSTRN", either the table identification TID or the work hardening slope H may be specified, but not both. If the TID is omitted, the work hardening slope H must be specified unless the material is perfectly plastic. The work hardening slope H is related to the tangential modulus E_T by:

$$H = \frac{E_T}{1 - \frac{E_T}{E}}$$

where E is the elastic modulus and $E_T = d\sigma/d\epsilon$ is the slope of the stress-strain curve in the plastic region. See Figure 15-131.

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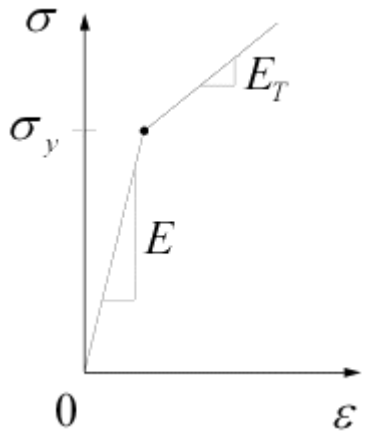
Figure 15-131: Stress vs. total strain curve when H is specified



3. If TID is specified, the tabular (Xi, Yi) data listed on the TABLES1 bulk entry represents a stress-strain curve. The strains are the Xi values and the stresses are the Yi values. The tabular data must conform to the following rules:

   - If TYPE = "NLELAST", the stress-strain curve can be defined in the first and third quadrants to accommodate differences in tension and compression. If the stress-strain curve is defined only in the first quadrant, then the first data point must be the origin, (X1, Y1) = (0.0, 0.0), and the behavior in compression is assumed to be the mirror image of the behavior in tension.
   - If TYPE = "PLASTIC", the TID cannot reference a TABLEST entry for SOL 106 or 129. The stress-strain curve must be defined in the first quadrant as indicated in **Figure 15-132**. The first data point must be the origin, (X1, Y1) = (0.0, 0.0). The second data point depends on the yield function as indicated in **Table 15-83**.

   Table 15-83: Second Data Point Versus Yield Function

Yield Function	Applicable SOLs	Second Data Point
von Mises	SOLs 106, 129, 401 and 402	(X2, Y2) = (ϵ_1, σ_y)
Tresca	SOLs 106 and 129	(X2, Y2) = (ϵ_1, σ_y)
Mohr-Coulomb	SOLs 106 and 129	(X2, Y2) = (ϵ_1, 2c)
Drucker-Prager	SOLs 106 and 129	(X2, Y2) = (ϵ_1, 2c)

   The slope of the line joining the first and second data points must equal to the value of *E* on the MAT1 entry. The work hardening slope, H_k, is calculated for each successive pair of data points in the plastic region from the following formula:

$$H_k = \frac{\sigma_{k+1} - \sigma_k}{\varepsilon^p_{k+1} - \varepsilon^p_k}$$

   where $\epsilon_k{}^p$ is the plastic strain at point *k*. The plastic strain at point *k* is related to the total strain at point *k* as follows:

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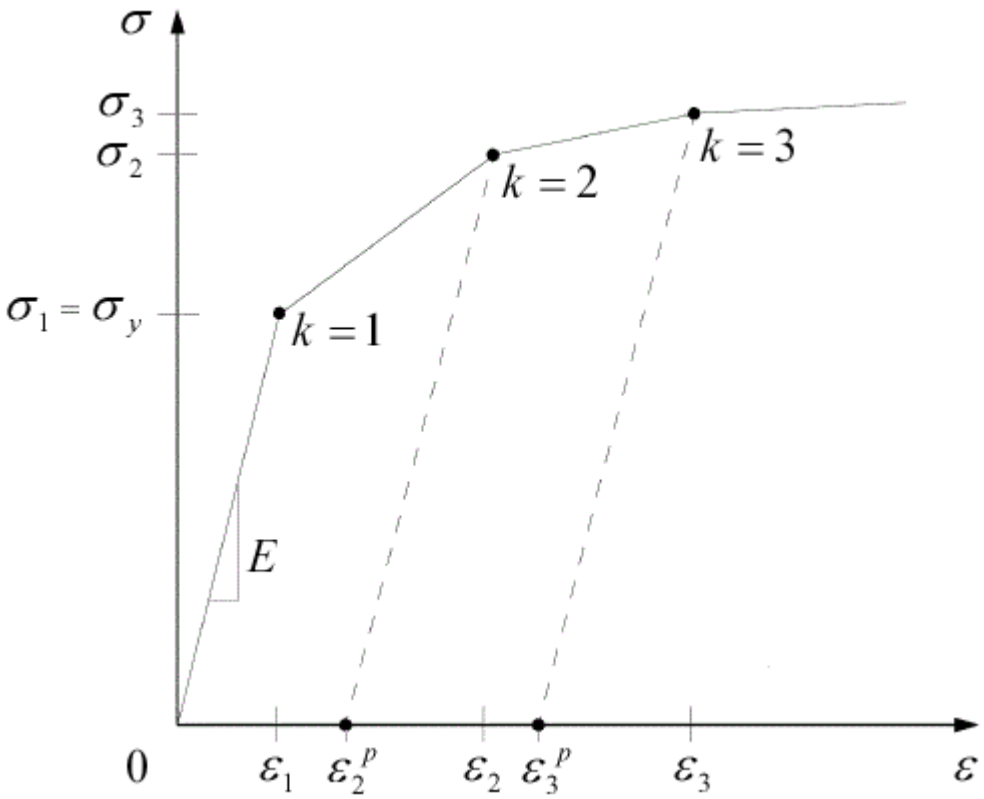
$$\varepsilon_k^p = \varepsilon_k - \frac{\sigma_k}{E}$$

For SOL 106, if the LIMIT1 value is different from the value for the yield strength in the TABLES1 data, the LIMIT1 value takes precedence for the purpose of calculating the strain at yield. However, the work hardening slopes, H_k, are still calculated from the data points on the TABLES1 entry.

Figure 15-132: Stress vs. total strain curve when TID is specified and TYPE = "PLASTIC"



- If TYPE = "PLSTRN", the TID can reference a TABLEST entry and the stress-plastic strain curve must be defined in the first quadrant as indicated in Figure Figure 15-133. The work hardening slope, H_k, is calculated for each successive pair of data points from the following formula:

$$H_k = \frac{\sigma_{k+1} - \sigma_k}{\varepsilon_{k+1}^p - \varepsilon_k^p}$$

where $\epsilon_k{}^p$ is the plastic strain at point k. The total strain at point k is related to the plastic strain at point k as follows:

$$\varepsilon_k = \varepsilon_k^p + \frac{\sigma_k}{E}$$

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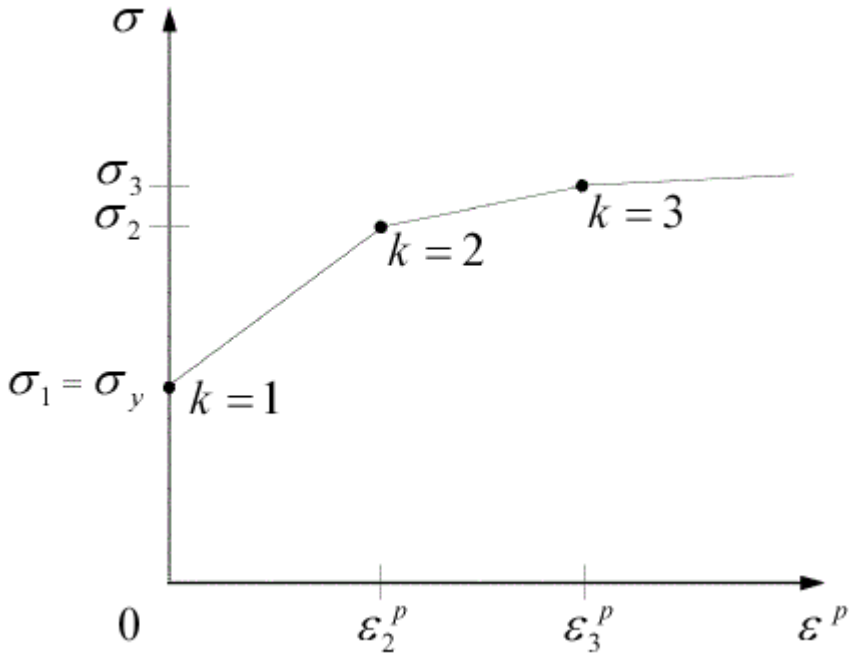
Figure 15-133: Stress vs. plastic strain curve when TID is specified and TYPE = “PLSTRN”



4. The software interprets the data on the TABLES1 entries as engineering stress and strain.
5. If MATS1 is specified, isotropic plasticity theory is used to calculate plastic strains regardless of elastic property type. For anisotropic elastic properties this may be unrealistic and should be used with caution. For more information, see the *Simcenter Nastran Handbook of Nonlinear Analysis*.

Remarks related to SOLs 401 and 402:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. LIMIT2 is ignored. LIMIT1 is only used if TID is blank and H is specified (i.e. for a bilinear plastic material).
3. Only YF = 1 (von Mises yield criterion) is supported. Selecting any other yield function results in a fatal error.
4. "ENG": the tables curves are defined in engineering measures. The curves will be converted to Cauchy measures if large strains (PARAM,LGSTRN,1) is defined. If small strains (PARAM,LGSTRN,0) is defined, curves are not converted.

   "Cauchy" or "TRUE": no conversion of the tables curves is performed, whatever the PARAM,LGSTRN is.

   "UNDEF" or blank: undefined measure. The "ENG" conversion above applies.

   For more information, see SOLs 401 and 402 - Stress-strain measures in the *Multi-Step Nonlinear User's Guide*.
5. When TID is defined for SOLs 401 and 402, stress-strain conversion from total to plastic strain and/or from Cauchy to Engineering might be required. This depends on how the parameter LGSTRN is defined, the type of table (TABLES1 or TABLEST), and if the elastic properties on the MATTi entry are a function of temperature. In some of these cases, a reference temperature is required, which

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is defined with either the TREF field on the MATi entry for SOLs 401 and 402, or with a TEMPD entry referenced by the TEMPERATURE(INITIAL) command for only SOL 402. This is the case when the elastic coefficients are a function of temperature and the hardening curve is not.

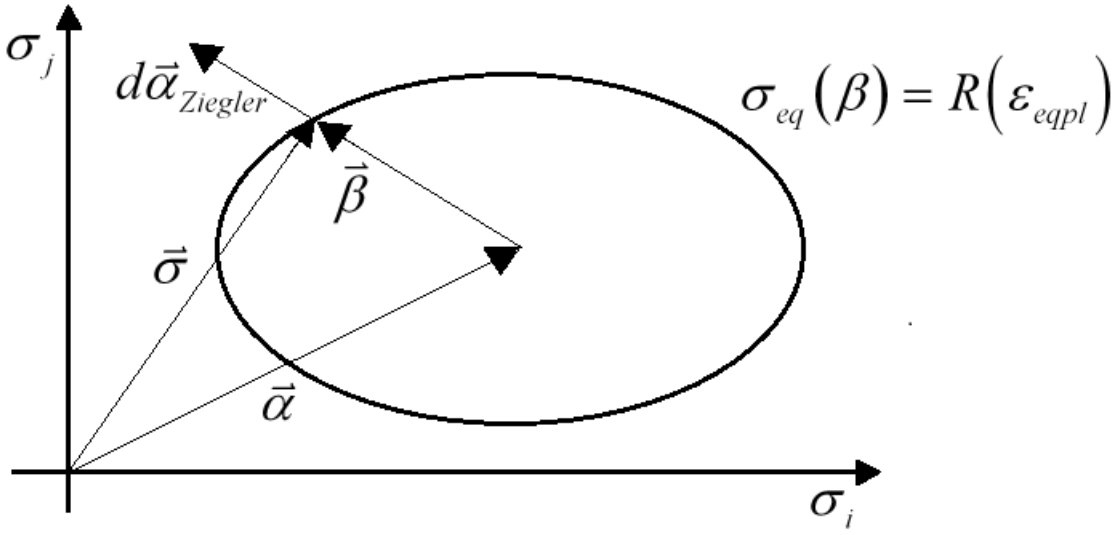
Remarks related to SOL 401:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. When a bilinear stress-strain representation is defined, all three hardening rules are supported.

   When a multilinear stress-strain representation is defined, only isotropic hardening is supported.
3. If the slope of the line joining the origin and second point in TABLES1 (for TYPE= "PLASTIC" and a given TID) is not equal to the value of E in the MAT1 entry, the software still uses E to calculate the linear elastic tangential matrix, but uses the slope as the elastic modulus in the following two conversions:
   • Converting from total to plastic strain.
   • Converting from Engineering to Cauchy stress/strain measure when large strains (PARAM,LGSTRN,1) is defined.
4. TYPE = "NLELAST" is supported by SOL 401.

Remarks related to SOL 402:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. If HR = 2 or 3, the Ziegler hardening is used.



where the stress values must be inside a yield surface such that,

$$\sigma_{eq}(\beta) \leq R(\varepsilon_{eqpl})$$

where β is a stress measured from the yield surface center α,

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σ_{eq} is a scalar equivalent stress measure,

ε_{eqpl} is a scalar equivalent plastic strain and R is a function describing the radius of the criterion.

If HR = 3, a mixed hardening factor of 0.5 is used.

3. The TYPE of material nonlinearity will be checked against the actual values contained in the TID table. If the values do not match the TYPE declaration, a fatal error will be issued and the computation will be stopped with an appropriate error message.
4. If a MATSR bulk data entry has the same MID as a MATS1 one, the TYPE defined in MATS1 is ignored and the type of the table referenced by TID in MATSR will be deducted from the table itself.

Related Topics

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MATSR

Material Strain Rate Dependence (SOLs 401 and 402)

Specifies strain-rate dependent properties for use with MATS1 entry with the same MID.

Format:

1	2	3	4	5	6	7	8	9	10
MATSR	MID	BVALUE	TSRATE	TID					

Examples:

MATSR	15	0.361	0.05						

MATSR	15			2					

Fields:

Field	Contents
MID	Identification number of a MATS1. (Integer > 0)
BVALUE	Strain-rate hardening parameter. Must be specified if TID is blank or zero. (Real or blank)
TSRATE	Transition strain rate. Must be specified if TID is blank or zero. (Real > 0.0 or blank)
TID	Identification number of a TABLEST entry. (Integer ≥ 0 or blank)

Remarks:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. You have two options to define the strain rate dependent plasticity.

   Option 1: **Static stress scaling**

   This option scales the static stress found from a elastic-plastic material defined with the MATS1 entry. The elastic-plastic material definition can also reference temperature dependent stress-strain data from the MATS1 entry. The software determines the current static yield stress σ_{stat} using the elastic-plastic material definition. The static yield stress value is then scaled using the following strain-rate scaling function.

$$\sigma_{dyn} = \sigma_{stat}\left(1 + \left(\frac{\dot{\varepsilon}_{pl,eq}}{D}\right)^{1/P}\right)$$

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where $\dot{\varepsilon}_{pl,eq}$ is the current equivalent plastic strain rate, D is a constant defined with the TSRATE field, and P is a constant defined with the BVALUE field.

Option 2: **Strain-rate dependent tables**

From the TID field on the MATSR bulk entry, you reference a TABLEST entry which then references multiple TABLES1 entries defining stress-strain tables for multiple plastic strain rates. In this case, the software does not use the stress-strain data referenced by the TID or H fields on the MATS1 entry. You cannot define a temperature dependency with this option. The MATS1 entry is still needed to define the type of nonlinearity (TYPE field), the hardening rule (HR field), and the initial yield point (LIMIT1 field).

Because plastic strain rate data is often provided at logarithmic intervals, the software internally computes the natural logarithm of both the current plastic strain rate value and the plastic strain rates you enter on the TABLEST entry. Because log(0.0) cannot be computed, if you are including stress-strain data for a plastic strain rate value of 0.0, you should enter a very small non-zero value instead of 0.0. For example, .0001.

Related Topics

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MATT1

Isotropic Material Temperature Dependence

Specifies temperature-dependent material properties on MAT1 entry fields via TABLEMi entries.

Format:

1	2	3	4	5	6	7	8	9	10
MATT1	MID	T(E)	T(G)	T(NU)	T(RHO)	T(A)		T(GE)	
	T(ST)	T(SC)	T(SS)						

Example:

MATT1	17	32				15			
	52								

Fields:

Field	Contents
MID	Material property identification number that matches the identification number on MAT1 entry. (Integer > 0)
T(E)	Identification number of a TABLEMi entry for the Young's modulus. (Integer ≥ 0 or blank)
T(G)	Identification number of a TABLEMi entry for the shear modulus. (Integer ≥ 0 or blank)
T(NU)	Identification number of a TABLEMi entry for the Poisson's ratio. (Integer ≥ 0 or blank)
T(RHO)	Identification number of a TABLEMi entry for the mass density. (Integer ≥ 0 or blank)
T(A)	Identification number of a TABLEMi entry for the thermal expansion coefficient. (Integer ≥ 0 or blank)
T(GE)	Identification number of a TABLEMi entry for the damping coefficient. (Integer ≥ 0 or blank)
T(ST)	Identification number of a TABLEMi entry for the tension stress limit. (Integer ≥ 0 or blank)
T(SC)	Identification number of a TABLEMi entry for the tension compression limit. (Integer ≥ 0 or blank)
T(SS)	Identification number of a TABLEMi entry for the tension shear limit. (Integer ≥ 0 or blank)

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Remarks:

1. To include temperature dependent material properties:
   - The identification number of the MATT1 must match the identification number of the associated MAT1 entry.
   - The identification number of the MAT1 and MATT1 entries must be referenced by an element property, for example, a PSOLID entry.
   - The TEMPERATURE case control command must select the material evaluation temperatures defined in the bulk data. See the remarks on the TEMPERATURE case control command.
2. For information on how the software calculates the temperature-dependent value of a material property, see the remarks for the TABLEMi bulk entry that the material property references.
3. A blank or zero in a material property field means that the material property is not temperature dependent. When this occurs, the software uses the corresponding value on the MAT1 bulk entry as the value for the material property.
4. Any quantity modified by this entry must have a value on the MAT1 entry. Initial values of E, G, or NU may be supplied according to Remark 3 on the MAT1 entry.
5. Table references must be present for each item that is temperature dependent. For example, it is not sufficient to give table references only for fields 3 and 4 (Young's modulus and shear modulus) if Poisson's ratio is temperature dependent.
6. SOL 401 computes the non-temperature dependent values E, NU, and G using the MAT1 definition, before it considers the temperature-dependent properties defined on the MATT1 entry. (See the SOL 401 remarks on the **MAT1** entry for the details of this computation.)

   Because of this, the software uses the MAT1 and MATT1 values to compute

   G(T) = E' / [2 (1 + NU')]

   where:

   - E' = E(T) if E(T) is specified on the MATT1 entry, otherwise E' = E.
   - NU' = NU(T) if NU(T) is specified on the MATT1 entry, otherwise NU' = NU.

   as follows:

   - If only E(T) is specified, E(T) and NU are used to compute G(T).
   - If only NU(T) is specified, NU(T) and E are used to compute G(T).
   - If only E(T) and NU(T) are specified, E(T) and NU(T) are used to compute G(T).
   - If only E(T) and G(T) are specified, E(T) and NU are used to compute G(T), and the specified G(T) is ignored.
   - If only NU(T) and G(T) are specified, E and NU(T) are used to compute G(T), and the specified G(T) is ignored.
   - If E(T), NU(T), and G(T) are specified, E(T) and NU(T) are used to compute G(T), and the specified G(T) is ignored.

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- If only G(T) is specified, E and NU are used to compute G, and the specified G(T) is ignored. As a result of this, G is not temperature dependent.

Remarks related to SOLs 402 and 414:

1. T(G) and T(GE) are ignored.
2. If T(RHO) is defined, a constant mass density is computed as follows:
   ● If TREF is defined on the MATi entry, the mass density T(RHO) is evaluated at temperature TREF.
   ● If TREF is undefined on the MATi entry, a default temperature defined with the TEMPD bulk entry must be referenced by the TEMPERATURE(INITIAL) set selection. Then, the mass density T(RHO) is evaluated at temperature TREF=TEMPD. If TEMPD is not specified either, a FATAL error will be generated.

   Note that T(RHO) is an exception, and other temperature dependent properties are evaluated at the temperatures referenced by the TEMPERATURE(LOAD) set selection.

Related Topics

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MATT2

Anisotropic Material Temperature Dependence

Specifies temperature-dependent material properties on MAT2 entry fields via TABLEMi entries.

Format:

1	2	3	4	5	6	7	8	9	10
MATT2	MID	T(C11)	T(C12)	T(C13)	T(C22)	T(C23)	T(C33)	T(RHO)	
	T(A1)	T(A2)	T(A3)		T(GE)	T(ST)	T(SC)	T(SS)	

Example:

MATT2	17	32					15		
	62								

Fields:

Field	Contents
MID	Material property identification number that matches the identification number on a MAT2 entry. (Integer > 0)
T(Cjk)	Identification number of a TABLEMi entry for the terms in the material property matrix. (Integer ≥ 0 or blank)
T(RHO)	Identification number of a TABLEMi entry for the mass density. (Integer ≥ 0 or blank)
T(Aj)	Identification number of a TABLEMi entry for the thermal expansion coefficient. (Integer ≥ 0 or blank)
T(GE)	Identification number of a TABLEMi entry for the damping coefficient. (Integer ≥ 0 or blank)
T(ST)	Identification number of a TABLEMi entry for the tension stress limit. (Integer ≥ 0 or blank)
T(SC)	Identification number of a TABLEMi entry for the tension compression limit. (Integer ≥ 0 or blank)
T(SS)	Identification number of a TABLEMi entry for the tension shear limit. (Integer ≥ 0 or blank)

Remarks:

1. To include temperature dependent material properties:

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- The identification number of the MATT2 must match the identification number of the associated MAT2 entry.
- The identification number of the MAT2 and MATT2 entries must be referenced by an element property, for example, a PSOLID entry.
- The TEMPERATURE case control command must select the material evaluation temperatures defined in the bulk data. See the remarks on the TEMPERATURE case control command.

2. For information on how the software calculates the temperature-dependent value of a material property, see the remarks for the TABLEMi bulk entry that the material property references.
3. A blank or zero in a material property field means that the material property is not temperature dependent. When this occurs, the software uses the corresponding value on the MAT2 bulk entry as the value for the material property.
4. Any quantity modified by this entry must have a value on the MAT2 entry.

Remarks related to SOLs 402 and 414:

1. T(GE) is ignored.
2. If T(RHO) is defined, a constant mass density is computed as follows:
   ● If TREF is defined on the MATi entry, the mass density T(RHO) is evaluated at temperature TREF.
   ● If TREF is undefined on the MATi entry, a default temperature defined with the TEMPD bulk entry must be referenced by the TEMPERATURE(INITIAL) set selection. Then, the mass density T(RHO) is evaluated at temperature TREF=TEMPD. If TEMPD is not specified either, a FATAL error will be generated.

   Note that T(RHO) is an exception, and other temperature dependent properties are evaluated at the temperatures referenced by the TEMPERATURE(LOAD) set selection.

Related Topics

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MATT3

Orthotropic Material Temperature Dependence for Axisymmetric, Plane Stress and Plane Strain Elements

Specifies temperature-dependent material properties on MAT3 entry fields via TABLEMi entries that are temperature dependent.

Format:

1	2	3	4	5	6	7	8	9	10
MATT3	MID	T(EX)	T(EY)	T(EZ)	T(NUXY)	T(NUYZ)	T(NUZX)	T(RHO)	
	T(GXY)		T(GZX)	T(AX)	T(AY)	T(AZ)		T(GE)	

Example:

MATT3	23	32		15					
			62						

Fields:

Field	Contents
MID	Material property identification number that matches the identification number on MAT3 entry. (Integer > 0)
T(EX), T(EY), T(EZ)	Identification number of a TABLEMi entry for the Young's moduli in the X,Y,Z-principal material directions. (Integer ≥ 0 or blank)
T(NUXY), T(NUYZ), T(NUZX)	Identification number of a TABLEMi entry for the Poisson's ratio. (Integer ≥ 0 or blank)
T(RHO)	Identification number of a TABLEMi entry for the mass density. (Integer ≥ 0 or blank)
T(GXY), T(GZX)	Identification number of a TABLEMi entry for the shear moduli. (Integer ≥ 0 or blank)
T(AX), T(AY), T(AZ)	Identification number of a TABLEMi entry for the thermal expansion coefficients in the X,Y,Z-principal material directions. (Integer ≥ 0 or blank)
T(GE)	Identification number of a TABLEMi entry for the damping coefficient. (Integer ≥ 0 or blank)

Remarks:

1. To include temperature dependent material properties:

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- The identification number of the MATT3 must match the identification number of the associated MAT3 entry.
- The identification number of the MAT3 and MATT3 entries must be referenced by an element property, for example, a PSOLID entry.
- The TEMPERATURE case control command must select the material evaluation temperatures defined in the bulk data. See the remarks on the TEMPERATURE case control command.

2. For information on how the software calculates the temperature-dependent value of a material property, see the remarks for the TABLEMi bulk entry that the material property references.
3. A blank or zero in a material property field means that the material property is not temperature dependent. When this occurs, the software uses the corresponding value on the MAT3 bulk entry as the value for the material property.
4. Any quantity modified by this entry must have a value on the MAT3 entry.

Remarks related to SOL 402:

1. T(GE) is ignored.
2. If T(RHO) is defined, a constant mass density is computed as follows:
   ● If TREF is defined on the MATi entry, the mass density T(RHO) is evaluated at temperature TREF.
   ● If TREF is undefined on the MATi entry, a default temperature defined with the TEMPD bulk entry must be referenced by the TEMPERATURE(INITIAL) set selection. Then, the mass density T(RHO) is evaluated at temperature TREF=TEMPD. If TEMPD is not specified either, a FATAL error will be generated.

   Note that T(RHO) is an exception, and other temperature dependent properties are evaluated at the temperatures referenced by the TEMPERATURE(LOAD) set selection.
3. Only TABLEM1, TABLEM2 and TABLEM4 entries may be used to define temperature-dependent properties.
4. TABLEM4 is ignored

Related Topics

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MATT4

Thermal Material Temperature Dependence

Specifies table references for temperature-dependent MAT4 material properties.

Format:

1	2	3	4	5	6	7	8	9	10
MATT4	MID	T(K)	T(CP)		T(H)	T(μ)	T(HGEN)		

Example:

MATT4	2	10	11						

Fields:

Field	Contents
MID	Identification number of a MAT4 entry that is temperature dependent. (Integer > 0)
T(K)	Identification number of a TABLEMi entry that gives the temperature dependence of the thermal conductivity. (Integer ≥ 0 or blank)
T(CP)	Identification number of a TABLEMi entry that gives the temperature dependence of the thermal heat capacity. (Integer ≥ 0 or blank)
T(H)	Identification number of a TABLEMi entry that gives the temperature dependence of the free convection heat transfer coefficient. (Integer ≥ 0 or blank)
T(μ)	Identification number of a TABLEMi entry that gives the temperature dependence of the dynamic viscosity. (Integer ≥ 0 or blank)
T(HGEN)	Identification number of a TABLEMi entry that gives the temperature dependence of the internal heat generation property for QVOL. (Integer ≥ 0 or blank)

Remarks:

1. To include temperature dependent material properties:
   - The identification number of the MATT4 must match the identification number of the associated MAT4 entry.
   - The identification number of the MAT4 and MATT4 entries must be referenced by an element property, for example, a PSOLID entry.
   - The TEMPERATURE case control command must select the material evaluation temperatures defined in the bulk data. See the remarks on the TEMPERATURE case control command.

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2. For information on how the software calculates the temperature-dependent value of a material property, see the remarks for the TABLEMi bulk entry that the material property references.
3. A blank or zero in a material property field means that the material property is not temperature dependent. When this occurs, the software uses the corresponding value on the MAT4 bulk entry as the value for the material property.

Related Topics

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MATT5

Thermal Anisotropic Material Temperature Dependence

Specifies temperature-dependent material properties on MAT5 entry fields via TABLEMi entries.

Format:

1	2	3	4	5	6	7	8	9	10
MATT5	MID	T(KXX)	T(KXY)	T(KXZ)	T(KYY)	T(KYZ)	T(KZZ)	T(CP)	
		T(HGEN)							

Example:

MATT5	24	73							

Fields:

Field	Contents
MID	Identification number of a MAT5 entry that is to be temperature dependent. (Integer > 0)
T(Kjk)	Identification number of a TABLEMi entry. The TABLEMi entry specifies temperature dependence of the matrix term. (Integer ≥ 0 or blank)
T(CP)	Identification number of a TABLEMi entry that specifies the temperature dependence of the thermal heat capacity. (Integer ≥ 0 or blank)
T(HGEN)	Identification number of a TABLEMi entry that gives the temperature dependence of the internal heat generation property for the QVOL entry. (Integer ≥ 0 or blank)

Remarks:

1. To include temperature dependent material properties:
   - The identification number of the MATT5 must match the identification number of the associated MAT5 entry.
   - The identification number of the MAT5 and MATT5 entries must be referenced by an element property, for example, a PSOLID entry.
   - The TEMPERATURE case control command must select the material evaluation temperatures defined in the bulk data. See the remarks on the TEMPERATURE case control command.
2. For information on how the software calculates the temperature-dependent value of a material property, see the remarks for the TABLEMi bulk entry that the material property references.

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3. A blank or zero in a material property field means that the material property is not temperature dependent. When this occurs, the software uses the corresponding value on the MAT5 bulk entry as the value for the material property.

Related Topics

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MATT8

Shell Orthotropic Material Temperature Dependence

Specifies temperature-dependent material properties on MAT8 entry fields via TABLEMi entries.

Format:

1	2	3	4	5	6	7	8	9	10
MATT8	MID	T(E1)	T(E2)	T(NU12)	T(G12)	T(G1Z)	T(G2Z)	T(RHO)	
	T(A1)	T(A2)		T(Xt)	T(Xc)	T(Yt)	T(Yc)	T(S)	
	T(GE)	T(F12)							

Example:

MATT8	20	10							
	11								

Fields:

Field	Contents
MID	Material property identification number that matches the identification number on MAT8 entry. (Integer > 0)
T(E1)	Identification number of a TABLEMi entry for Young's modulus 1. (Integer ≥ 0 or blank)
T(E2)	Identification number of a TABLEMi entry for Young's modulus 2. (Integer ≥ 0 or blank)
T(NU12)	Identification number of a TABLEMi entry for Poisson's ratio 12. (Integer ≥ 0 or blank)
T(G12)	Identification number of a TABLEMi entry for shear modulus 12. (Integer ≥ 0 or blank)
T(G1Z)	Identification number of a TABLEMi entry for transverse shear modulus 1Z. (Integer ≥ 0 or blank)
T(G2Z)	Identification number of a TABLEMi entry for transverse shear modulus 2Z. (Integer ≥ 0 or blank)
T(RHO)	Identification number of a TABLEMi entry for mass density. (Integer ≥ 0 or blank)
T(A1)	Identification number of a TABLEMi entry for thermal expansion coefficient 1. See Remark 3. (Integer ≥ 0 or blank)
T(A2)	Identification number of a TABLEMi entry for thermal expansion coefficient 2. See Remark 3. (Integer ≥ 0 or blank)

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Field	Contents
T(Xt)	Identification number of a TABLEMi entry for tension stress/strain limit 1. (Integer ≥ 0 or blank)
T(Xc)	Identification number of a TABLEMi entry for compression stress/strain limit1. (Integer ≥ 0 or blank)
T(Yt)	Identification number of a TABLEMi entry for tension stress/strain limit 2. (Integer ≥ 0 or blank)
T(Yc)	Identification number of a TABLEMi entry for compression stress/strain limit 2. (Integer ≥ 0 or blank)
T(S)	Identification number of a TABLEMi entry for shear stress/strain limit. (Integer ≥ 0 or blank)
T(GE)	Identification number of a TABLEMi entry for structural damping coefficient. (Integer ≥ 0 or blank)
T(F12)	Identification number of a TABLEMi entry for Tsai-Wu interaction term. (Integer ≥ 0 or blank)

Remarks:

1. To include temperature dependent material properties:
   - The identification number of the MATT8 must match the identification number of the associated MAT8 entry.
   - The identification number of the MAT8 and MATT8 entries must be referenced by an element property, for example, a PSOLID entry.
   - The TEMPERATURE case control command must select the material evaluation temperatures defined in the bulk data. See the remarks on the TEMPERATURE case control command.
2. For information on how the software calculates the temperature-dependent value of a material property, see the remarks for the TABLEMi bulk entry that the material property references.
3. A blank or zero in a material property field means that the material property is not temperature dependent. When this occurs, the software uses the corresponding value on the MAT8 bulk entry as the value for the material property.
4. Any quantity modified by this entry must have a value on the MAT8 entry.

Remarks related to SOLs 402 and 414:

1. T(GE) is ignored.
2. If T(RHO) is defined, a constant mass density is computed as follows:
   ● If TREF is defined on the MATi entry, the mass density T(RHO) is evaluated at temperature TREF.
   ● If TREF is undefined on the MATi entry, a default temperature defined with the TEMPD bulk entry must be referenced by the TEMPERATURE(INITIAL) set selection. Then, the mass density T(RHO)

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is evaluated at temperature TREF=TEMPD. If TEMPD is not specified either, a FATAL error will be generated.

Note that T(RHO) is an exception, and other temperature dependent properties are evaluated at the temperatures referenced by the TEMPERATURE(LOAD) set selection.

Related Topics

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MATT9

Solid Element Anisotropic Material Temperature Dependence

Specifies temperature-dependent material properties on MAT9 entry fields via TABLEMi entries.

Format:

1	2	3	4	5	6	7	8	9	10
MATT9	MID	T(C11)	T(C12)	T(C13)	T(C14)	T(C15)	T(C16)	T(C22)	
	T(C23)	T(C24)	T(C25)	T(C26)	T(C33)	T(C34)	T(C35)	T(C36)	
	T(C44)	T(C45)	T(C46)	T(C55)	T(C56)	T(C66)	T(RHO)	T(A1)	
	T(A2)	T(A3)	T(A4)	T(A5)	T(A6)		T(GE)		

Example:

MATT9	17	32			18			17	
				12					
				5			10		

Fields:

Field	Contents
MID	Material property identification number that matches the identification number on MAT9 entry. (Integer > 0)
T(Cjk)	Identification number of a TABLEMi entry for the terms in the material property matrix. (Integer ≥ 0 or blank)
T(RHO)	Identification number of a TABLEMi entry for the mass density. (Integer ≥ 0 or blank)
T(Aj)	Identification number of a TABLEMi entry for the thermal expansion coefficients. (Integer ≥ 0 or blank)
T(GE)	Identification number of a TABLEMi entry for the damping coefficient. (Integer ≥ 0 or blank)

Remarks:

1. To include temperature dependent material properties:
   - The identification number of the MATT9 must match the identification number of the associated MAT9 entry.

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- The identification number of the MAT9 and MATT9 entries must be referenced by an element property, for example, a PSOLID entry.
- The TEMPERATURE case control command must select the material evaluation temperatures defined in the bulk data. See the remarks on the TEMPERATURE case control command.

2. For information on how the software calculates the temperature-dependent value of a material property, see the remarks for the TABLEMi bulk entry that the material property references.
3. A blank or zero in a material property field means that the material property is not temperature dependent. When this occurs, the software uses the corresponding value on the MAT9 bulk entry as the value for the material property.
4. Any quantity modified by this entry must have a value on the MAT9 entry.
5. The continuation entries are optional.

Remarks related to SOLs 402 and 414:

1. T(GE) is ignored.
2. If T(RHO) is defined, a constant mass density is computed as follows:
   ● If TREF is defined on the MATi entry, the mass density T(RHO) is evaluated at temperature TREF.
   ● If TREF is undefined on the MATi entry, a default temperature defined with the TEMPD bulk entry must be referenced by the TEMPERATURE(INITIAL) set selection. Then, the mass density T(RHO) is evaluated at temperature TREF=TEMPD. If TEMPD is not specified either, a FATAL error will be generated.

   Note that T(RHO) is an exception, and other temperature dependent properties are evaluated at the temperatures referenced by the TEMPERATURE(LOAD) set selection.

Related Topics

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MATT11

Solid Orthotropic Material Temperature Dependence

Defines the temperature dependent orthotropic material properties for axisymmetric elements, 2D solid elements, 3D solid elements, and 3D solid composite elements.

Format:

1	2	3	4	5	6	7	8	9	10
MATT11	MID	T(E1)	T(E2)	T(E3)	T(NU12)	T(NU13)	T(NU23)	T(G12)	
	T(G13)	T(G23)	T(RHO)	T(A1)	T(A2)	T(A3)		T(GE)	

Example:

MATT11	20	10							
			11						

Fields:

Field	Contents
MID	Material property identification number that matches the identification number on MAT11 entry. (Integer > 0)
T(E1)	Identification number of a TABLEMi entry for modulus of elasticity in the 1-principal material direction. (Integer ≥ 0 or blank)
T(E2)	Identification number of a TABLEMi entry for modulus of elasticity in the 2-principal material direction. (Integer ≥ 0 or blank)
T(E3)	Identification number of a TABLEMi entry for modulus of elasticity in the 3-principal material direction. (Integer ≥ 0 or blank)
T(NU12)	Identification number of a TABLEMi entry for Poisson's ratio 12. (Integer ≥ 0 or blank)
T(NU13)	Identification number of a TABLEMi entry for Poisson's ratio 13. (Integer ≥ 0 or blank)
T(NU23)	Identification number of a TABLEMi entry for Poisson's ratio 23. (Integer ≥ 0 or blank)
T(G12)	Identification number of a TABLEMi entry for shear modulus 12. (Integer ≥ 0 or blank)
T(G13)	Identification number of a TABLEMi entry for shear modulus 13. (Integer ≥ 0 or blank)
T(G23)	Identification number of a TABLEMi entry for shear modulus 23. (Integer ≥ 0 or blank)
T(RHO)	Identification number of a TABLEMi entry for mass density. (Integer ≥ 0 or blank)

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Field	Contents
T(A1)	Identification number of a TABLEMi entry for thermal expansion coefficient in the 1-principal material direction. (Integer ≥ 0 or blank)
T(A2)	Identification number of a TABLEMi entry for thermal expansion coefficient in the 2-principal material direction. (Integer ≥ 0 or blank)
T(A3)	Identification number of a TABLEMi entry for thermal expansion coefficient in the 3-principal material direction. (Integer ≥ 0 or blank)
T(GE)	Identification number of a TABLEMi entry for structural damping coefficient. (Integer ≥ 0 or blank)

Remarks:

1. To include temperature dependent material properties:
   - The identification number of the MATT11 must match the identification number of the associated MAT11 entry.
   - The identification number of the MAT11 and MATT11 entries must be referenced by an element property, for example, a PSOLID entry.
   - The TEMPERATURE case control command must select the material evaluation temperatures defined in the bulk data. See the remarks on the TEMPERATURE case control command.
2. For information on how the software calculates the temperature-dependent value of a material property, see the remarks for the TABLEMi bulk entry that the material property references.
3. A blank or zero in a material property field means that the material property is not temperature dependent. When this occurs, the software uses the corresponding value on the MAT11 bulk entry as the value for the material property.
4. Any quantity modified by this entry must have a value on the MAT11 entry.

Remarks related to SOL 402:

1. T(GE) is ignored.
2. If T(RHO) is defined, a constant mass density is computed as follows:
   ● If TREF is defined on the MATi entry, the mass density T(RHO) is evaluated at temperature TREF.
   ● If TREF is undefined on the MATi entry, a default temperature defined with the TEMPD bulk entry must be referenced by the TEMPERATURE(INITIAL) set selection. Then, the mass density T(RHO) is evaluated at temperature TREF=TEMPD. If TEMPD is not specified either, a FATAL error will be generated.

   Note that T(RHO) is an exception, and other temperature dependent properties are evaluated at the temperatures referenced by the TEMPERATURE(LOAD) set selection.

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MATTC

Creep Material Temperature Dependence for SOL 402

Specifies temperature-dependent creep coefficients on CREEP entry fields via TABLEM1 entries for SOL 402.

Format:

1	2	3	4	5	6	7	8	9	10
MATTC	MID	T(a)	T(b)		T(d)				

Example:

MATTC	111	3	4						

Fields:

Field	Contents
MID	Material identification number of a CREEP entry. (Integer >0). See Remark 1.
T(a), T(b), T(d)	Identification number of a TABLEM1 entry that gives the temperature dependence of the creep coefficient. (Integer > 0). See Remarks 2 and 3.

Remarks:

1. Only TYPE=300 in CREEP entry can be used with MATTC entry.
2. The TABLEM1 entries must all have the same number of temperature points with the same temperature values.
3. In TABLEM1 entry for T(a), the YAXIS field may be LINEAR or LOG.
4. For a specific solution temperature, the software interpolates the coefficients such that the creep strain varies linearly with respect to the temperature points.

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MATVST

Visco-Thermal Fluid Material Property Definition

Defines visco-thermal material properties for fluid elements.

Format:

1	2	3	4	5	6	7	8	9	10
MATVST	MID	RHO	C	GAMMA	PR	MU			

Example:

MATVST	1000	1.225-9	3.400+5	1.4	0.71	1.84-8			

Fields:

Field	Contents
MID	Material identification number. See Remark 2. (Integer > 0)
RHO	Mass density. (Real ≥ 0.0; Default = 1.225)
C	Speed of sound in the fluid. (Real ≥ 0.0; Default = 3.40e2)
GAMMA	Ratio of constant pressure specific heat to constant volume specific heat. (Real ≥ 0.0; Default = 1.4)
PR	Prandtl number. (Real ≥ 0.0; Default = 0.71)
MU	Dynamic viscosity. (Real ≥ 0.0; Default = 1.84e-5)

Remarks:

1. SOL 108 and SOL 111 acoustic or coupled vibro-acoustic solutions support MATVST bulk entries. These entries require the FEM Adaptive Order (FEMAO) method.
2. The MID must be unique with respect to MATVST.
3. ACVSTT entries can only reference MATVST entries.
4. The defaults for RHO, C, and MU are for standard conditions and valid for metric models only. GAMMA and PR are dimensionless.

Related Topics

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MDSBULK

Selects mode set component representation for use in a residual model

Specifies the logical unit number for the SC_H5 file that contains a mode set representation along with element and grid point ID offsets to be used in the residual portion of the system model.

Format:

1	2	3	4	5	6	7	8	9	10
MDSBULK		UNITNO	EOFFSET	GOFFSET	LDIDT	LDIDR			

Example:

MDSBULK		33	1000	2000	20	5			

Fields:

Field	Contents
UNITNO	Logical unit number for the SC_H5 file that contains the mode set representation. See Remarks 1. (Integer > 0; No default)
EOFFSET	Element ID offset. See Remark 2. (Integer ≥ 0; Default = 0)
GOFFSET	Grid point ID offset. See Remark 2. (Integer ≥ 0; Default = 0)
LDIDT	Descriptor identification number of a description contained in the SC_H5 file referenced by UNITNO for translational mode shapes. See Remark 3. (0 ≤ Integer ≤ 9999; No default)
LDIDR	Descriptor identification number of a description contained in the SC_H5 file referenced by UNITNO for rotational mode shapes. See Remark 3. (0 ≤ Integer ≤ 9999; No default)

Remarks:

1. For specifying UNITNO, an appropriate ASSIGN SC_H5 statement must be present in the File Management Section of the input file. The DEFER keyword must exist.
2. The element and grid IDs associated with the mode set representation must be unique from element and grid IDs in the residual model. To ensure that they are unique, specify appropriate element and grid ID offsets.
3. The SC_H5 file must contain the descriptions referenced by LDIDT and LDIDR. Use a DTI,DISTL bulk entry to define the description referenced by LDIDT or LDIDR. The description in the SC_H5 file must be in uppercase letters. Either LDIDT or LDIDR may be 0 (that is, no data requested), but not both.

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Related Topics

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MFLUID

Fluid Volume Properties

Defines the properties of an incompressible fluid volume for the purpose of generating a virtual mass matrix.

Format:

1	2	3	4	5	6	7	8	9	10
MFLUID	SID	CID	ZFS	RHO	ELIST1	ELIST2	PLANE1	PLANE2	
	RMAX	FMEXACT							

Example:

MFLUID	3	2	15.73	1006.	3	4	S	N	
		100.							

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
CID	Identification number of rectangular coordinate system used to specify the orientation of the free surface (normal to X_3) and of planes of symmetry, if any. (Integer ≥ 0 or blank)
ZFS	Intercept of the free surface on the X_3 axis of the coordinate system referenced by CID. If X3 > ZFS then there is no fluid. See Remark 3. (Real; Default means that the free surface is located at an infinitely large positive value of X_3)
RHO	Density of the fluid. (Real)
ELIST1	Identification number of an ELIST bulk entry that lists the IDs of two-dimensional elements that can be wetted on one side by the fluid. Only those elements connected to at least one grid point below ZFS are wetted by the fluid. See Remarks 3 and 5. (Integer ≥ 0)
ELIST2	Identification number of an ELIST bulk entry that lists the IDs of two-dimensional elements that can be wetted on both sides by the fluid. Only those elements connected to at least one grid point below ZFS are wetted by the fluid. (Integer ≥ 0; ELIST1 + ELIST2 > 0)
PLANE1, PLANE2	Planes of symmetry, antisymmetry, or no symmetry. “S” means that plane 1, which is the plane containing the X_1 and X_3 axes of CID, is a plane of symmetry. “A” means that

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Field	Contents
	plane 1 is a plane of antisymmetry. “N” means that it is neither. See Remark **5.** Plane 2 uses “S”, “A”, or “N” for the X_2 and X_3 plane. (Character: “S”, “A”, or “N”)
RMAX	Characteristic length. Interactions between elements with separation that is greater than RMAX will be neglected. (Real > 0.0; Default = 1.0E 10)
FMEXACT	Exact integration is used if the distance between two elements is less than FMEXACT times the square root of the area of the larger element. Otherwise, center point integration is used by default. (Real; Default = 1.0E 15)

Remarks:

1. The MFLUID entry must be selected with the Case Control command MFLUID = SID.
2. Several MFLUID entries corresponding to different fluid volumes can be used simultaneously.
3. The wetted side of an element in ELIST1 is determined by the presence or absence of a minus sign preceding the element’s ID on the ELIST entry. A minus sign indicates that the fluid is on the side opposite to the element’s positive normal, as determined by applying the right-hand rule to the sequence of its corner points. The same element can appear on two ELIST entries, indicating that it forms a barrier between the unconnected fluids.
4. The fluid volume may be finite (interior) or infinite (exterior). The volume may be bounded by a free surface and one or two planes of structural symmetry. If structural symmetry is used, the structure must have the symmetric or antisymmetric boundary corresponding to the selection in fields 8 and 9. Interior fluids must have ELIST1 data and a free surface or plane of antisymmetry.
5. The planes of symmetry and/or antisymmetry defined in fields 8 and 9 must be planes of symmetry for the entire analysis. The user may apply appropriate structural boundary conditions at all grid points lying in these planes.
6. The current list of elements that may be placed in ELIST1 and ELIST2 include CTRIA3 and CQUAD4.
7. The continuation entry is optional.
8. If there is ELIST1 data and no free surface nor plane of antisymmetry, the program assumes a special form of external fluid. These special external fluids must have a CID (field 3) such that the origin of the fluid coordinate system is near the center of the enclosed volume, since the singularity for volume change will be placed at the origin. Special external fluids are supported only in SOLs 103, and 107 through 112. If used in conventional solution sequences, System Fatal Message 3001 results for file 205.
9. See the parameter VMOPT. VMOPT controls when the virtual mass is included in the mass matrix.
10. The SPCD method of enforced motion is not recommended for large models. Use the large mass method instead.

Related Topics

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MICPNT

Microphone Point Definition

Defines microphone point for acoustic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
MICPNT	EID	GID	Name/Descriptor						

Example:

MICPNT	100	246	Microphone point 100 on fluid grid 246.						

Fields:

Field	Contents
EID	Element ID number for microphone point. (Integer > 0) See Remark 1.
GID	Fluid grid point identification number. (Integer > 0) See Remark 2.
Name/ Descriptor	The Name/Descriptor is an optional character string. (Character; 48 characters maximum; See Bulk Data Syntax Rules.) See Remark 3.

Remarks:

1. A MICPNT is a 0D microphone element.
2. The GRID bulk entries referenced must be fluid point grids, which are designated on the GRID bulk entry by CD = -1.
3. Because of the name/descriptor field, free-field and large field data formats are not supported for this bulk entry.

Related Topics

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MKAERO1

Mach Number - Frequency Table

Provides a table of Mach numbers (m) and reduced frequencies (k) for aerodynamic matrix calculation.

Format:

1	2	3	4	5	6	7	8	9	10
MKAERO1	m1	m2	m3	m4	m5	m6	m7	m8	
	k1	k2	k3	k4	k5	k6	k7	k8	

Example:

MKAERO1	.1	.7							
	.3	.6	1.0						

Fields:

Field	Contents
mi	List of from 1 to 8 Mach numbers. (Real ≥ 0.0)
kj	List of from 1 to 8 reduced frequencies. (Real > 0.0)

Remarks:

1. Blank fields end the list, and thus cannot be used for 0.0.
2. All combinations of (mi, kj) will be used.
3. The continuation entry is required.
4. Multiple MKAERO1 entries are permitted.
5. For the lifting surface theories (Doublet-Lattice and Mach Box), the maximum value of kj should be less than one quarter of the number of boxes on a representative chord (i.e.,

   $$\mathrm{MAX(kj)} < \bar{C}/4\Delta x \mathrm{MAX(kj)} < \bar{C}/4\Delta x$$

   where $\bar{C}$ is the reference chord and Δx is a typical box chord length).
6. In SOL 146, the program selects only one value of mi to use in computing the dynamic response solution and, by default, the lowest value is selected. The PARAM,MACH,m entry may be specified to select a different value. If PARAM,MACH,m is specified, then the value of mi closest to m will be selected.

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7. The very low nonzero value of kj required for aeroelastic divergence analysis of restrained vehicles with the K- and KE-methods of flutter analysis must be included on this entry.

Related Topics

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MKAERO2

Mach Number - Frequency Table

Provides a list of Mach numbers (m) and reduced frequencies (k) for aerodynamic matrix calculation.

Format:

1	2	3	4	5	6	7	8	9	10
MKAERO2	m1	k1	m2	k2	m3	k3	m4	k4	

Example:

MKAERO2	.10	.30	.10	.60	.70	.30	.70	1.0	

Fields:

Field	Contents
mi	Mach numbers. (Real ≥ 0.0)
ki	Reduced frequencies. (Real > 0.0)

Remarks:

1. MKAERO2 will cause the aerodynamic matrices to be computed for the given sets of parameter pairs. Embedded blank pairs are skipped.
2. No continuation entries are allowed, but multiple MKAERO2 entries are permitted.
3. In SOL 146, the program selects only one value of mi to use in computing the dynamic response solution and, by default, the lowest value is selected. The PARAM,MACH,m entry may be specified to select a different value. If PARAM,MACH,m is specified, then the value of mi closest to m will be selected.
4. The very low nonzero value of ki required for aeroelastic divergence analysis of restrained vehicles with the K- and KE-methods of flutter analysis must be included on this entry.

Related Topics

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MLINK

Misalignment Definition

Defines misalignment parameters for a CLINK bulk entry in SOL 414 rotor dynamics.

Format:

1	2	3	4	5	6	7	8	9	10
MLINK	SID	EID	RID						
	ECCT	PHASET	ECCR	PHASER	MSTZ				

Example:

MLINK	20	100	1						
	0.1	30	0.4	15	0.75				

Fields:

Field	Contents
SID	MLINK identification number. (Integer > 0)
EID	CLINK identification number. (Integer > 0; No default)
RID	Rotor identification number. (Integer > 0; No default)
ECCT	Translational (radial) misalignment. See Remark 3. (Real ≥ 0.0; Default = 0.0)
PHASET	Phase of translational (radial) misalignment in degrees. (Real; Default = 0.0)
ECCR	Rotational (angular) misalignment in degrees. See Remark 3. (Real ≥ 0.0; Default = 0.0)
PHASER	Phase of rotational (angular) misalignment in degrees. (Real; Default = 0.0)
MSTZ	Axial offset that adds a new displacement that produces force. See Remarks 3 and 5. (Real; Default = 0.0)

Remarks:

1. Use an MLOAD case control command to select the MLINK bulk entry.
2. To define a static misalignment load, leave the RID field blank.
3. The geometric interpretation of the ECCR, ECCA, and MSTZ fields is shown in the following table.

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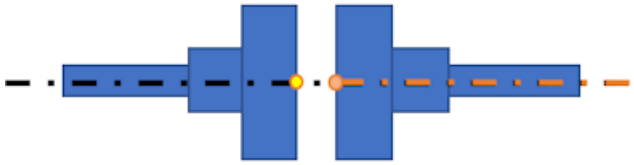
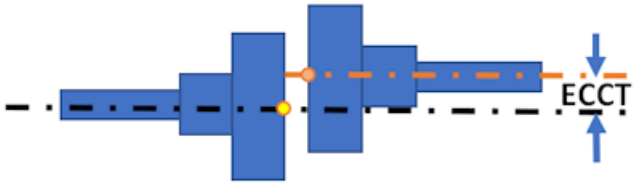
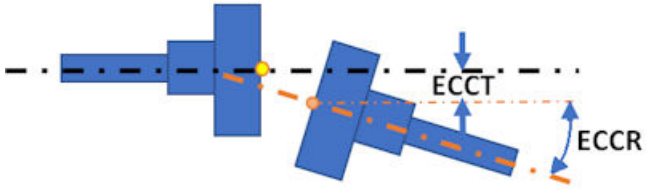
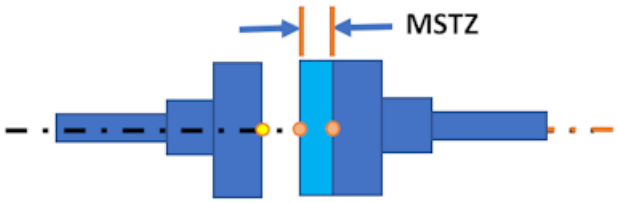
Case	Geometric interpretation
No misalignment or offset (1)	
Translational misalignment	
Translational and rotational misalignment	
Axial offset	
(1) The rotor halves are shown separated for clarity.	

4. Use the OPT field on the CLINK bulk entry that the MLINK bulk entry references and the ECCT, ECCR, and MSTZ fields on the MLINK bulk entry to specify any combination of translational misalignment, rotational misalignment, and axial offset as follows:

- For translational misalignment only, specify OPT = 1, enter a value in the ECCT field, and leave the ECCR and MSTZ fields blank.
- For translational misalignment and axial offset only, specify OPT = 1, enter values in the ECCT and MSTZ fields, and leave the ECCR field blank.
- For translational misalignment and rotational misalignment only, specify OPT = 2, enter values in the ECCT and ECCR fields, and leave the MSTZ field blank.
- For translational misalignment, rotational misalignment, and axial offset, specify OPT = 2 and enter values in the ECCT, ECCR, and MSTZ fields.
- For rotational misalignment only, specify OPT = 2, enter a value in the ECCR field, and leave the ECCT and MSTZ fields blank.
- For rotational misalignment and axial offset only, specify OPT = 2, enter values in the ECCR and MSTZ fields, and leave the ECCT field blank.

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- For axial offset only, specify either OPT = 1 or 2, enter a value in the MSTZ field, and leave the ECCT and ECCR fields blank.

5. Axial offset is a displacement in the Z-direction defined as:

   GB - GA

   where:

   - GB is the second grid point defined in CLINK.
   - GA is the first grid point defined in CLINK.

   The Z-direction is given by the basic or local coordinate system referenced in the CSYS field in the CLINK element.

6. In SOL 414,103 Eigenvalues and Superelement Reduction solutions, you cannot define the MLINK/MLOAD misalignment load in Nonlinear Preload subcases.

   In SOL 414,111 Harmonic Response solutions, you cannot define the MLINK/MLOAD misalignment load in Nonlinear Preload and Frequency Response subcases.

Related Topics

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MODTRAK

Mode Tracking Parameters

Specifies parameters for mode tracking in design optimization (SOL 200).

Format:

1	2	3	4	5	6	7	8	9	10
MODTRAK	SID	LOWRNG	HIGHRNG	MTFILTER					

Example:

MODTRAK	100	1	26	0.80					

Fields:

Field	Contents
SID	Set identification number that is selected in the Case Control Section with the MODTRAK command. See Remark 1. (Integer; No Default)
LOWRNG	Lowest mode number in range to search. See Remark 2. (Integer ≥ 0, Default = 0. If nonzero, LOWRNG < HIGHRNG.)
HIGHRNG	Highest mode number in range to search. See Remark 2. (Integer > 0, Default = number of eigenvalues extracted. If nonzero, LOWRNG < HIGHRNG.)
MTFILTER	Filtering parameter used in mode cross-orthogonality check. See Remark 3. (Real, Default = 0.9)

Remarks:

1. Only the designed modes for the subcase will be tracked. A designed mode is one that is used in the design model (in connection with either objective or constraints) and, therefore, identified on a DRESP1 entry.
2. The range of modes LOWRNG through HIGHRNG, inclusive, will be used to track the designed modes. If LOWRNG and HIGHRNG are both blank, then all computed modes will be used to search for the designed modes. Since large numbers of computed modes will result in higher computational costs, limiting the search range with LOWRNG and HIGHRNG is recommended.
3. Modes are considered to correlate if their mass normalized cross orthogonalities are greater than MTFILTER.

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Related Topics

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MOMENT

Static Moment

Defines a static concentrated moment at a grid point by specifying a scale factor and a vector that determines the direction.

Format:

1	2	3	4	5	6	7	8	9	10
MOMENT	SID	G	CID	M	N1	N2	N3		

Example:

MOMENT	2	5	6	2.9	0.0	1.0	0.0		

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Grid point identification number at which the moment is applied. (Integer > 0)
CID	Coordinate system identification number. (Integer ≥ 0 or blank)
M	Scale factor. (Real)
Ni	Components of the vector measured in the coordinate system defined by CID. (Real; at least one Ni ≠ 0.0)

Remarks:

1. The static moment applied to grid point G is defined by $\mathbf{m} = M \cdot \mathbf{N}$ where **N** is the vector with components N1, N2, and N3. The magnitude of **m** is equal to *M* times the magnitude of **N**.
2. In the static solution sequences, the load set ID (SID) is selected by the Case Control command, LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
3. A CID of zero or blank references the basic coordinate system.
4. For scalar points see SLOAD.

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Remarks related to SOLs 402 and 414:

1. To apply a moment with constant magnitude, SID is selected by Case Control command LOAD = SID for both static and transient analyses.
2. To apply a time-dependent moment, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by Case Control command DLOAD.

Related Topics

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MOMENT1

Static Moment, Alternate Form 1

Defines a static concentrated moment at a grid point by specifying a magnitude and two grid points that determine the direction.

Format:

1	2	3	4	5	6	7	8	9	10
MOMENT1	SID	G	M	G1	G2				

Example:

MOMENT1	6	13	-2.93	16	13				

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Grid point identification number at which the moment is applied. (Integer > 0)
M	Magnitude of moment. (Real)
G1, G2	Grid point identification numbers used to define the unit vector $\vec{n}$. (Integer > 0; G1 and G2 cannot be coincident.)

Remarks:

1. The static moment applied to grid point G is defined by $\mathbf{m} = M \cdot \mathbf{n}$ where **n** is a unit vector parallel to a vector from G1 to G2.
2. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
3. The follower force effects due to loads from this entry are included in the stiffness in all linear solution sequences that calculate a differential stiffness. The solution sequences are SOLs 103, 105, 107 to 112, 115 and 116 (see also the parameter **"FOLLOWK"**). In addition, follower force effects are included in the force balance in the nonlinear static and nonlinear transient dynamic solution sequences, SOLs 106, 129, 153, and 159, if geometric nonlinear effects are turned on with PARAM,LGDISP,1. The follower force stiffness is included in the nonlinear static solution sequences (SOLs 106 and 153) but not in the nonlinear transient dynamic solution sequences (SOLs 129 and 159).

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Remarks related to SOLs 402 and 414:

1. The moment is a follower moment in large displacement analysis (i.e., PARAM,LGDISP,1). For small displacement analysis, the direction of the moment is defined by the original positions of G1 and G2 and does not change.
2. The moment effects are included in the stiffness matrix only if a non-symmetric solver is used (NLCNTLG INLY=1) and the geometric nonlinear effects are turned on (PARAM,LGDISP,1).
3. To apply a moment with constant magnitude, SID is selected by Case Control command LOAD = SID for both static and transient analysis.
4. To apply a time-dependent moment, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by Case Control command DLOAD.

Related Topics

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MOMENT2

Static Moment, Alternate Form 2

Defines a static concentrated moment at a grid point by specification of a magnitude and four grid points that determine the direction.

Format:

1	2	3	4	5	6	7	8	9	10
MOMENT2	SID	G	M	G1	G2	G3	G4		

Example:

MOMENT2	6	13	-2.93	16	13	17	13		

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Grid point identification number at which the moment is applied. (Integer > 0)
M	Magnitude of moment. (Real)
Gi	Grid point identification numbers used to determine the unit vector $\vec{n}$. (Integer > 0; G1 and G2 cannot be coincident; G3 and G4 cannot be coincident.)

Remarks:

1. The static moment applied to grid point G is defined by $\mathbf{m} = M \cdot \mathbf{n}$ where **n** is a unit vector parallel to the cross product of the vectors from G1 to G2, and G3 to G4.
2. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
3. The follower force effects due to loads from this entry are included in the stiffness in all linear solution sequences that calculate a differential stiffness. The solution sequences are SOLs 103, 105, 107 to 112, 115 and 116 (see also the parameter **"FOLLOWK"**). In addition, follower force effects are included in the force balance in the nonlinear static and nonlinear transient dynamic solution sequences, SOLs 106, 129, 153, and 159, if geometric nonlinear effects are turned on with PARAM,LGDISP,1. The follower force stiffness is included in the nonlinear static solution sequences (SOLs 106 and 153) but not in the nonlinear transient dynamic solution sequences (SOLs 129 and 159).

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Remarks related to SOLs 402 and 414:

1. The moment is a follower moment in large displacement analysis (i.e., PARAM,LGDISP,1). For small displacement analysis, the direction of the moment is defined by the original positions of G1 and G2 and does not change.
2. The moment effects are included in the stiffness matrix only if a non-symmetric solver is used (NLCNTLG INLY=1) and the geometric nonlinear effects are turned on (PARAM,LGDISP,1).
3. To apply a moment with constant magnitude, SID is selected by Case Control command LOAD = SID for both static and transient analysis.
4. To apply a time-dependent moment, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by Case Control command DLOAD.

Related Topics

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MONPNT1

Integrated Load Monitor Point

Defines an integrated load monitor point at a point (x,y,z) in a user defined coordinate system. The integrated loads about this point over the associated nodes will be computed and printed for static aeroelastic analysis (SOL 144) and form integrated loads on the nonlinear static aeroelastic database.

Format:

1	2	3	4	5	6	7	8	9	10
MONPNT1	NAME	LABEL							
	AXES	COMP	CID	X	Y	Z			

Example:

MONPNT1	WING155	Wing Integrated Load to Butline 155							
	34	WING		0.0	155.0	15.0			

Fields:

Field	Contents
NAME	A character string of up to eight characters Identifying the monitor point. (Character)
LABEL	A string comprising no more than 56 characters (fields 3 through 9) that identifies and labels the monitor point.
AXES	Component axes about which to sum. (Any unique combination of the integers 1 thru 6 with no embedded blanks.)
COMP	The name of an AECOMP or AECOMPL entry that defines the set of grid points over which the monitor point is defined.
CID	The identification number of a coordinate system in which the (x,y,z) coordinates are defined.
X,Y,Z	The coordinates in the CID coordinate system about which the forces are to be summed. (Default = 0.0)

Remarks:

1. The Identification NAME must be unique among all MONPNT1 entries.
2. The LABEL is a 56 character string that should be unique among all MONPNT1 entries (the string is case insensitive). It is used as additional label information in the printed output.

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Related Topics

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MONPNT2

Integrated Load Monitor Point — Element Monitor Output Results Item

Defines an integrated load monitor point at a named element, and outputs component of stress, strain, or force at that element for SOL 101 and SOL 103 (SORT1 format only).

Format:

1	2	3	4	5	6	7	8	9	10
MONPNT2	NAME	LABEL							
	TABLE	TYPE	COMP	EID					

Example:

MONPNT2	MPT22	This is a MONPNT2 monitor point							
	STRAIN	CBAR	EX1A	1100082					
	STRESS	CBAR	SX1A	1100234					

Fields:

Field	Contents
NAME	A unique character string that identifies the monitor point. (Character; 8 characters maximum; No default)
LABEL	A string that identifies and labels the monitor point. (Character; 56 characters maximum; Optional)
TABLE	Type of output to be monitored. (Character; "STRESS", "FORCE", or "STRAIN"; No default)
TYPE	Element type. (Character; No default)
COMP	Component of the element type to be monitored. This is the COMP label for the particular table and element type. For a list of supported elements and their components, see **Stress monitor table** and **Force monitor table**. (Character; No default)
EID	Element ID. (Integer > 0; No default)

Remarks:

1. For a list of supported elements and their components, see **Stress monitor table** and **Force monitor table**.
2. Because of the name/descriptor field, free-field, and large field data formats are not supported for this bulk entry.

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3. For shell laminates, only the homogeneous stress/strain is output. The NOCOMPS parameter allows you to control the shell laminate output. If this parameter:

   = 0 or -1, The homogeneous stress/strain is output.

   = 1, No homogeneous stress/strain is output.

   Note

   Output for solid laminates and layer by layer output is not supported.

4. Output for superelements that are processed at the monitor point:

   - For internal and partitioned superelements, MONPNT2 results can be output for the residual or downstream superelements.
   - For external superelements, MONPNT2 results are only output for the residual superelement.

   Note

   The various elements defined for a MONPNT2 may not exist for all of the superelements in a model. When this occurs, the MONPNT2 is not processed for the superelement.

5. Invalid Table, Type, COMP, or EID generates a warning message and is ignored.

6. The software prints the results in a table titled `STRUCTURAL INTERNAL MONITOR POINT LOADS (MONPNT2)`

Stress monitor table

Element Name (Code)	Real Stresses/Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CBAR 34	SX1A/ EX1A	End A-Point C	SX1AR/ EX1AR	End A-Point C	RM
	SX2A/ EX2A	End A-Point D	SX2AR/ EX2AR	End A-Point D	RM
	SX3A/ EX3A	End A-Point E	SX3AR/ EX3AR	End A-Point E	RM
	SX4A/ EX4A	End A-Point F	SX4AR/ EX4AR	End A-Point F	RM
	AS/AE	Axial	ASR/AER	Axial	RM
	BMAXA/ EBMAXA	End A maximum	SX1AI/ EX1AI	End A-Point C	IP

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	BMINA/ EBMINA	End A minimum	SX2AI/ EX2AI	End A-Point D	IP
	MST/MST	Safety margin in tension	SX3AI/ EX3AI	End A-Point E	IP
	SX1B/EX1B	End B-Point C	SX4AI/ EX4AI	End A-Point F	IP
	SX2B/EX2B	End B-Point D	ASI/AEI	Axial	IP
	SX3B/EX3B	End B-Point E	SX1BR/ EX1BR	End B-Point C	RM
	SX4B/EX4B	End B-Point F	SX2BR/ EX2BR	End B-Point D	RM
	BMAXB/ EBMAXB	End B maximum	SX3BR/ EX3BR	End B-Point E	RM
	BMINB/ EBMINB	End B minimum	SX4BR/ EX4BR	End B-Point F	RM
	MSC	Safety margin in compression	SX1BI/ EX1BI	End B-Point C	IP
			SX2BI/ EX2BI	End B-Point D	IP
			SX3BI/ EX3BI	End B-Point E	IP
			SX4BI/ EX4BI	End B-Point F	IP

Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CBAR 100 Intermediate Stations	SD	Station Distance/Length	SD	Station Distance/ Length	RM
	SXC/EXC	Point C	SXCr/EXCr	Point C	RM
	SXD/EXD	Point D	SXDr/ EXDr	Point D	RM
	SXE/EXE	Point E	SXEr/EXEr	Point E	RM
	SXF/EXF	Point F	SXFr/EXFr	Point F	RM
	AS/AE	Axial	ASr/AEr	Axial	RM

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	SMAX/ EMAX	Maximum	SMAXr/ EMAXr	Maximum	RM
	SMIN/ EMIN	Minimum	SMINr/ EMINr	Minimum	RM
	MS	Margin of Safety (NDDL entitys above are given for End A. For codes 2 through 10 at intermediate stations add (K-1)*9 where K is the station number, and for codes at End B, K=number of stations plus 1.)	SXCi/EXCi	Point C	IP
			SXDi/EXDi	Point D	IP
			SXEi/EXEi	Point E	IP
			SXFi/EXFi	Point F	IP
			ASi/AEi	Axial	IP
			SMAXi/ EMAXi	Maximum	IP
			SMINi/ EMINi	Minimum (NDDL entitys above are given for End A. For codes 2 through 16 at intermediate stations add (K-1)*15 where K is the station number, and for codes at End B, K=number of stations plus 1.)	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CBEAM 2 Linear Format	GRID	External grid point ID	GRID	External grid point ID	
	SD	Station distance/length	SD	Station distance/ length	
	SXC/EXC	Long. Stress/Strain at Point C	SRCR/ ERCR	Long. Stress/Strain at Point C	RM
	SXD/EXD	Long. Stress/Strain at Point D	SXDR/ EXDR	Long. Stress/Strain at Point D	RM
	SXE/EXE	Long. Stress/Strain at Point E	SXER/ EXER	Long. Stress/Strain at Point E	RM

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	SXF/EXF	Long. Stress/Strain at Point F	SXFR/ EXFR	Long. Stress/Strain at Point F	RM
	SMAX/ EMAX	Maximum Stress/Strain	SXCI/EXCI	Long. Stress/Strain at Point C	IP
	SMIN/ EMIN	Minimum Stress/Strain	SXDI/EXDI	Long. Stress/Strain at Point D	IP
	MST	Safety margin in tension	SXEI/EXEI	Long. Stress/Strain at Point E	IP
	MSC	Safety margin in compression NDDL entitys are given for End A. Addition of the quantity (K-1)*10 to the NDDL entity points to the same information for other stations, where K is the station number. K=11 for End B, and K=2 through 10 for intermediate stations.	SXFI/EXFI	Long. Stress/Strain at Point F NDDL entitys are given for End A. Addition of the quantity (K-1)*10 to the NDDL entity points to the same information for other stations, where K is the station number. K=11 for End B, and K=2 through 10 for intermediate stations.	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CBEND 69	GRID	External grid point ID	GRID	External grid point ID	
	CA	Circumferential angle	CA	Circumferential angle	
	SC/EC	Long. Stress/Strain at Point C	SCR/ECR	Long. Stress/Strain at Point C	RM
	SD/ED	Long. Stress/Strain at Point D	SDR/EDR	Long. Stress/Strain at Point D	RM
	SE/EE	Long. Stress/Strain at Point E	SER/EER	Long. Stress/Strain at Point E	RM
	SF/EF	Long. Stress/Strain at Point F	SFR/EFR	Long. Stress/Strain at Point F	RM

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	SMAX/ EMAX	Maximum Stress/Strain	SCI/ECI	Long. Stress/Strain at Point C	IP
	SMIN/ EMIN	Minimum Stress/Strain	SDI/EDI	Long. Stress/Strain at Point D	IP
	MST	Safety margin in tension	SEI/EEI	Long. Stress/Strain at Point E	IP
	MSC	Safety margin in compression (NDDL entitys are given for End A. NDDL entitys 12 through 21 point to the same information for end B.)	SFI/EFI	Long. Stress/Strain at Point F (NDDL entitys are given for End A. NDDL entitys 12 through 21 point to the same information for End B.)	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CBUSH 102 Linear Format	TX	Translation-x	TXR	Translation-x	R
	TY	Translation-y	TYR	Translation-y	R
	TZ	Translation-z	TZR	Translation-z	R
	RX	Rotation-x	RXR	Rotation-x	R
	RY	Rotation-y	RYR	Rotation-y	R
	RZ	Rotation-z	RZR	Rotation-z	R
			TXI	Translation-x	I
			TYI	Translation-y	I
			TZI	Translation-z	I
			RXI	Rotation-x	I
			RYI	Rotation-y	I
			RZI	Rotation-z	I
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or

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					Imag./ Phase
CBUSH1D 40	FE	Axial force			
	UE	Axial displacement			
	VE	Axial velocity			
	AS	Axial stress			
	AE	Axial strain			
	EP				
	FAIL				
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CELAS1 11	S/E	Stress/Strain	SR/ER	Stress/Strain	RM
			SI/EI	Stress/Strain	IP
CELAS2 12	S/E	Stress/Strain	SR/ER	Stress/Strain	RM
			SI/EI	Stress/Strain	IP
CELAS3 13	S/E	Stress/Strain	SR/ER	Stress/Strain	RM
			SI/EI	Stress/Strain	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CGAP 86	CPX	Normal x	Not applicable		
	SHY	Shear y			
	SHZ	Shear z			
	AU	Axial u			
	SHV	Shear v			
	SHW	Shear w			
	SLV	Slip v			

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	SLP	Slip w			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CHEXA 67 Linear Format	CID	Stress coordinate system	CID	Stress coordinate system	
	CTYPE	Coordinate type (Character)	CTYPE	Coordinate type (Character)	
	NODEF	Number of active points	NODEF	Number of active points	
	GRID	External grid ID (0=center)	GRID	External grid ID (0=center)	
	SX/EX	Normal x	SXR/EXR	Normal x	RM
	TXY/ETXY	Shear xy	SYR/EYR	Normal y	RM
	P1/EP1	First principal	SZR/EZR	Normal z	RM
	P1X/P1X	First principal x cosine	TXYR/ ETXYR	Shear xy	RM
	P2X/P2X	Second principal x cosine	TYZR/ ETYZR	Shear yz	RM
	P3X/P3X	Third principal x cosine	TZXR/ ETZXR	Shear zx	RM
	PR/EPR	Mean pressure	SXI/EXI	Normal x	IP
	OCT/EOCT	von Mises or octahedral shear stress	SYI/EYI	Normal y	IP
	SY/EY	Normal y	SZI/EZI	Normal z	IP
	TYZ/ETYZ	Shear yz	TXYI/ ETXYI	Shear xy	IP
	P2/EP2	Second principal	TYZI/ ETYZI	Shear yz	IP
	P1Y/P1Y	First principal y cosine	TZXI/ ETZXI	Shear zx	IP
	P2Y/P2Y	Second principal y cosine			
	P3Y/P3Y	Third principal y cosine			

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	SZ/EZ	Normal z			
	TZX/ETZX	Shear zx			
	P3/EP3	Third principal			
	P1Z/P1Z	First principal z cosine			
	P2Z/P2Z	Second principal z cosine			
	P3Z/P3Z	Third principal z cosine			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CONROD 10 Linear Format	AS/AE	Axial Stress/Strain	ASR/AER	Axial Stress/Strain	RM
	MSA	Axial safety margin	ASI/AEI	Axial Stress/Strain	IP
	TS/TE	Torsional Stress/Strain	TSR/TER	Torsional Stress/ Strain	RM
	MST	Torsional safety margin	TSI/TEI	Torsional Stress/ Strain	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPENTA 68 Linear Format	CID	Stress coordinate system	CID	Stress coordinate system	
	CTYPE	Coordinate type (Character)	CTYPE	Coordinate type (Character)	
	NODEF	Number of active points	NODEF	Number of active points	
	GRID	External grid ID	GRID	External grid ID	
	SX/EX	Normal x	SXR/EXR	Normal x	RM
	TXY/ETXY	Shear xy	SYR/EYR	Normal y	RM
	P1/EP1	First principal	SZR/EZR	Normal z	RM
	P1X/P1X	First principal x cosine	TXYR/ ETXYR	Shear xy	RM

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	P2X/P2X	Second principal x cosine	TYZR/ ETYZR	Shear yz	RM
	P3X/P3X	Third principal x cosine	TZXR/ ETZXR	Shear zx	RM
	PR/EPR	Mean pressure	SXI/EXI	Normal x	IP
	OCT/EOCT	von Mises or Octahedral shear stress	SYI/EYI	Normal y	IP
	SY/EY	Normal y	SZI/EZI	Normal z	IP
	TYZ/ETYZ	Shear yz	TXYI/ ETXYI	Shear xy	IP
	P2/EP2	Second principal	TYZI/ ETYZI	Shear yz	IP
	P1Y/P1Y	First principal y cosine	TZXI/ ETZXI	Shear zx	IP
	P2Y/P2Y	Second principal y cosine			
	P3Y/P3Y	Third principal y cosine			
	SZ/EZ	Normal z			
	TZX/ETZX	Shear zx			
	P3/EP3	Third principal			
	P1Z/P1Z	First principal z cosine			
	P2Z/P2Z	Second principal z cosine			
	P3Z/P3Z	Third principal z cosine			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPLSTN3 271 Triangle plane strain Linear Format	SX	Normal stress in x	SXR	Normal stress in x	RM
	SY	Normal stress in y	SXI	Normal stress in x	IP
	SZ	Normal stress in z	SYR	Normal stress in y	RM
	SP	In-plane shear stress	SYI	Normal stress in y	IP
	SMAX	Von Mises stress	SZR	Normal stress in z	RM

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(Center Only)			SZI	Normal stress in z	IP
			SPR	In-plane shear stress	RM
			SPI	In-plane shear stress	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPLSTN4 272 Quadrilateral plane strain Linear Format (Center and Corners)	GRID	Grid ID, 0 for centroid	GRID	Grid ID, 0 for centroid	
	SX	Normal stress in x	SXR	Normal stress in x	RM
	SY	Normal stress in y	SXI	Normal stress in x	IP
	SZ	Normal stress in z	SYR	Normal stress in y	RM
	SP	In-plane shear stress	SYI	Normal stress in y	IP
	SMAX	Von Mises stress	SZR	Normal stress in z	RM
			SZI	Normal stress in z	IP
			SPR	In-plane shear stress	RM
			SPI	In-plane shear stress	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPLSTN6 273 Triangle plane strain Linear Format (Center and Corners)	GRID	Grid ID, 0 for centroid	GRID	Grid ID, 0 for centroid	
	SX	Normal stress in x	SXR	Normal stress in x	RM
	SY	Normal stress in y	SXI	Normal stress in x	IP
	SZ	Normal stress in z	SYR	Normal stress in y	RM
	SP	In-plane shear stress	SYI	Normal stress in y	IP
	SMAX	Von Mises stress	SZR	Normal stress in z	RM
			SZI	Normal stress in z	IP
			SPR	In-plane shear stress	RM
			SPI	In-plane shear stress	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		

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	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPLSTN8 274 Quadrilateral plane strain Linear Format (Center and Corners)	GRID	Grid ID, 0 for centroid	GRID	Grid ID, 0 for centroid	
	SX	Normal stress in x	SXR	Normal stress in x	RM
	SY	Normal stress in y	SXI	Normal stress in x	IP
	SZ	Normal stress in z	SYR	Normal stress in y	RM
	SP	In-plane shear stress	SYI	Normal stress in y	IP
	SMAX	Von Mises stress	SZR	Normal stress in z	RM
			SZI	Normal stress in z	IP
			SPR	In-plane shear stress	RM
			SPI	In-plane shear stress	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPLSTS3 275 Triangle plane stress Linear Format (Center Only)	SX	Normal stress in x	SXR	Normal stress in x	RM
	SY	Normal stress in y	SXI	Normal stress in x	IP
	SZ	Normal stress in z	SYR	Normal stress in y	RM
	SP	In-plane shear stress	SYI	Normal stress in y	IP
	SMAX	Von Mises stress	SZR	Normal stress in z	RM
			SZI	Normal stress in z	IP
			SPR	In-plane shear stress	RM
			SPI	In-plane shear stress	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPLSTS4 276	GRID	Grid ID, 0 for centroid	GRID	Grid ID, 0 for centroid	
	SX	Normal stress in x	SXR	Normal stress in x	RM

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Quadrilateral plane stress Linear Format (Center and Corners)	SY	Normal stress in y	SXI	Normal stress in x	IP
	SZ	Normal stress in z	SYR	Normal stress in y	RM
	SP	In-plane shear stress	SYI	Normal stress in y	IP
	SMAX	Von Mises stress	SZR	Normal stress in z	RM
			SZI	Normal stress in z	IP
			SPR	In-plane shear stress	RM
			SPI	In-plane shear stress	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPLSTS6 277 Triangle plane stress Linear Format (Center and Corners)	GRID	Grid ID, 0 for centroid	GRID	Grid ID, 0 for centroid	
	SX	Normal stress in x	SXR	Normal stress in x	RM
	SY	Normal stress in y	SXI	Normal stress in x	IP
	SZ	Normal stress in z	SYR	Normal stress in y	RM
	SP	In-plane shear stress	SYI	Normal stress in y	IP
	SMAX	Von Mises stress	SZR	Normal stress in z	RM
			SZI	Normal stress in z	IP
			SPR	In-plane shear stress	RM
			SPI	In-plane shear stress	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPLSTS8 278 Quadrilateral plane stress Linear Format	GRID	Grid ID, 0 for centroid	GRID	Grid ID, 0 for centroid	
	SX	Normal stress in x	SXR	Normal stress in x	RM
	SY	Normal stress in y	SXI	Normal stress in x	IP
	SZ	Normal stress in z	SYR	Normal stress in y	RM
	SP	In-plane shear stress	SYI	Normal stress in y	IP
	SMAX	Von Mises stress	SZR	Normal stress in z	RM

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(Center and Corners)			SZI	Normal stress in z	IP
			SPR	In-plane shear stress	RM
			SPI	In-plane shear stress	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CPYRAM 255 Linear Format	CID	Stress/Strain coordinate system	CID	Stress coordinate system	
	CTYPE	Coordinate type (Character)	CTYPE	Coordinate type (Character)	
	NODEF	Number of active points	NODEF	Number of active points	
	GRID	External grid ID	GRID	External grid ID	
		(0=center)		(0=center)	
	SX/EX	Normal x	SXR/EXR	Normal x	RM
	TXY/ETXY	Shear xy	SYR/EYR	Normal y	RM
	P1/EP1	First principal	SZR/EZR	Normal z	RM
	P1X/P1X	First principal x cosine	TXYR/ ETXYR	Shear xy	RM
	P2X/P2X	Second principal x cosine	TYZR/ ETYZR	Shear yz	RM
	P3X/P3X	Third principal x cosine	TZXR/ ETZXR	Shear zx	RM
	PR/EPR	Mean pressure	SXI/EXI	Normal x	IP
	OCT/EOCT	von Mises or octahedral shear stress	SYI/EYI	Normal y	IP
	SY/EY	Normal y	SZI/EZI	Normal z	IP
	TYZ/ETYZ	Shear yz	TXYI/ ETXYI	Shear xy	IP
	P2/EP2	Second principal	TYZI/ ETYZI	Shear yz	IP
	P1Y	First principal y cosine	TZXI/ ETZXI	Shear zx	IP

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	P2Y	Second principal y cosine			
	P3Y	Third principal y cosine			
	SZ/EZ	Normal z			
	TZX/ETZX	Shear zx			
	P3/EP3	Third principal			
	P1Z	First principal z cosine			
	P2Z	Second principal z cosine			
	P3Z	Third principal z cosine			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUAD4 33 Linear Format Center Only	FD1	Z1=Fiber Distance 1	FD1	Z1=Fiber Distance 1	
	SX1/EX1	Normal x at Z1	SX1R/ EX1R	Normal x at Z1	RM
	SY1/EY1	Normal y at Z1	SX1I/EX1I	Normal x at Z1	IP
	TXY1/EXY1	Shear xy at Z1	SY1R/ EY1R	Normal y at Z1	RM
	SA1/EA1	Shear angle at Z1	SY1I/EY1I	Normal y at Z1	IP
	SMJRP1/ EMJRP1	Major principal at Z1	TXY1R/ EXY1R	Shear xy at Z1	RM
	SMNRP1/ EMNRP1	Minor principal at Z1	TXY1I/ EXY1I	Shear xy at Z1	IP
	SMAX1/ EMAX1	von Mises or maximum shear at Z1	FD2/FD2	Z2=Fiber distance 2	
	FD2	Z2=Fiber distance 2	SX2R/ EX2R	Normal x at Z2	RM
	SX2/EX2	Normal x at Z2	SX2I/EX2I	Normal x at Z2	IP
	SY2/EY2	Normal y at Z2	SY2R/ EY2R	Normal y at Z2	RM
	TXY2/EXY2	Shear xy at Z2	SY2I/EY2I	Normal y at Z2	IP

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	SA2/EA2	Shear angle at Z2	TXY2R/ EXY2R	Shear xy at Z2	RM
	SMJRP2/ EMJRP2	Major principal at Z2	TXY2I/ EXY2I	Shear xy at Z2	IP
	SMNRP2/ EMNRP2	Minor principal at Z2			
	SMAX2/ EMAX2	von Mises or maximum shear at Z2			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUAD144 144 Linear Format Center and Corners	TERM	CEN/	TERM	CEN/	
	GRID	4	GRID	4	
	FD1	Z1-Fiber distance	FD1	Z1-Fiber distance	
	SX1/EX1	Normal x at Z1	SX1R/ EX1R	Normal x at Z1	RM
	SY1/EY1	Normal y at Z1	SX1I/EX1I	Normal x at Z1	IP
	TXY1/ ETXY1	Shear xy at Z1	SY1R/ EY1R	Normal y at Z1	RM
	A1	Shear angle at Z1	SY1I/EY1I	Normal y at Z1	IP
	MJRP1/ EMJRP1	Major principal at Z1	TXY1R/ ETXY1R	Shear xy at Z1	RM
	MNRP1/ EMNRP1	Minor principal at Z1	TXY1I/ ETXY1I	Shear xy at Z1	IP
	TMAX1/ ETMAX1	von Mises or maximum shear at Z1	FD2	Z2-Fiber distance	
	FD2	Z2-Fiber distance	SX2R/ EX2R	Normal x at Z2	RM
	SX2/EX2	Normal x at Z2	SX2I/EX2I	Normal x at Z2	IP
	SY2/EY2	Normal y at Z2	SY2R/ EY2R	Normal y at Z2	RM
	TXY2/ ETXY2	Shear xy at Z2	SY2I/EY2I	Normal y at Z2	IP

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	A2	Shear angle at Z2	TXY2R/ ETXY2R	Shear xy at Z2	RM
	MJRP2/ EMJRP2	Major principal at Z2	TXY2I/ ETXY2I	Shear xy at Z2	IP
	MNRP2/ EMNRP2	Minor principal at Z2			
	TMAX2/ ETMRP2	von Mises or maximum shear at Z2			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUAD4 95 Composite Center Only	PLY	Lamina Number	PLY	Lamina Number	
	SX1/EX1	Normal-1	SX1/EX1	Normal-1	RM
	SY1/EY1	Normal-2	SY1/EY1	Normal-2	RM
	T1/ET1	Shear-12	T1/ET1	Shear-12	RM
	SL1/EL1	Shear-1Z	SL1/EL1	Shear-1Z	RM
	SL2/EL2	Shear-2Z	SL2/EL2	Shear-2Z	RM
	A1	Shear angle	SX1I/EX1I	Normal-1	IP
	MJRP1/ EMJRP1	Major principal	SY1I/EY1I	Normal-2	IP
	MNRP1/ EMNRP1	Minor principal	T1I/ET1I	Shear-12	IP
	TMAX1/ ETMAX1	Maximum shear	SL1I/EL1I	Shear-1Z	IP
			SL2I/EL2I	Shear-2Z	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUAD8 64	TERM	CEN/	TERM	CEN/	
	GRID	4	GRID	4	
	FD1	Z1-Fiber distance	FD1	Z1-Fiber distance	

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Linear Format Center and Corners	SX1/EX1	Normal x at Z1	SX1R/ EX1R	Normal x at Z1	RM
	SY1/EY1	Normal y at Z1	SX1I/EX1I	Normal x at Z1	IP
	TXY1/ ETXY1	Shear xy at Z1	SY1R/ EY1R	Normal y at Z1	RM
	A1	Shear angle at Z1	SY1I/EY1I	Normal y at Z1	IP
	SMJRP1/ EMJRP1	Major principal at Z1	TXY1R/ ETXY1R	Shear xy at Z1	RM
	SMNRP1/ EMNRP1	Minor principal at Z1	TXY1I/ ETXY1I	Shear xy at Z1	IP
	TMAX1/ ETMAX1	von Mises or maximum shear at Z1	FD2	Z2-Fiber distance	
	FD2	Z2-Fiber distance	SX2R/ EX2R	Normal x at Z2	RM
	SX2/EX2	Normal x at Z2	SX2I/EX2I	Normal x at Z2	IP
	SY2/EY2	Normal y at Z2	SY2R/ EY2R	Normal y at Z2	RM
	TXY2/ ETXY2	Shear xy at Z2	SY2I/EY2I	Normal y at Z2	IP
	A2	Shear angle at Z2	TXY2R/ ETXY2R	Shear xy at Z2	RM
	SMJRP2/ EMJRP2	Major principal at Z2	TXY2I/ ETXY2I	Shear xy at Z2	IP
	SMNRP2/ EMNRP2	Minor principal at Z2			
	TMAX2/ ETMAX2	von Mises or maximum shear at Z2			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUAD8 96 Composite Center Only	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUADR 82 Center and Corners	TERM	CEN/	TERM	CEN/	
	GRID	4	GRID	4	
	FD1	Z1-Fiber distance	FD1	Z1-Fiber distance	
	SX1/EX1	Normal x at Z1	SX1R/ EX1R	Normal x at Z1	RM
	SY1/EY1	Normal y at Z1	SX1I/EX1I	Normal x at Z1	IP
	TXY1/ ETXY1	Shear xy at Z1	SY1R/ EY1R	Normal y at Z1	RM
	A1	Shear angle at Z1	SY1I/EY1I	Normal y at Z1	IP
	MJRP1/ EMJRP1	Major principal at Z1	TXY1R/ ETXY1R	Shear xy at Z1	RM
	MNRP1/ EMNRP1	Minor principal at Z1	TXY1I/ ETXY1I	Shear xy at Z1	IP
	TMAX1/ ETMAX1	von Mises or maximum shear at Z1	FD2	Z2-Fiber distance	
	FD2	Z2-Fiber distance	SX2R/ EX2R	Normal x at Z2	RM
	SX2/EX2	Normal x at Z2	SX2I/EX2I	Normal x at Z2	IP
	SY2/EY2	Normal y at Z2	SY2R/ EY2R	Normal y at Z2	RM
	TXY2/ ETXY2	Shear xy at Z2	SY2I/EY2I	Normal y at Z2	IP
	A2	Shear angle at Z2	TXY2R/ ETXY2R	Shear xy at Z2	RM
	MJRP2/ EMJRP2	Major principal at Z2	TXY2I/ ETXY2I	Shear xy at Z2	IP
	MNRP2/ EMNRP2	Minor principal at Z2			
	TMAX2/ ETMRP2	von Mises or maximum shear at Z2			

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUADR 228 Center only (not supported by old formulation)	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUADR 232 Composite Center Only (not supported by old formulation)	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUADX4 243 Linear Format Center and Corners Grid and Gauss	CTYPE	0 (Result at Center)	CTYPE	0 (Result at Center)	
	LOC	Location	LOC	Location	
	SX	Radial	SXR	Radial	RM
	SY	Azimuthal	SXI	Radial	IP
	SZ	Axial	SYR	Azimuthal	RM
	SS	Shear stress	SYI	Azimuthal	IP
	MAXP	Maximum principal	SZR	Axial	RM
	TMAX	Maximum shear	SZI	Axial	IP
	OCTS	von Mises or octahedral	SSR	Shear stress	RM
			SSI	Shear stress	IP

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Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CQUADX8 245 Linear Format Grid and Gauss Center and Corners	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CROD 1 Linear Format	Same as CONROD(10)	Same as CONROD(10)	Same as CONROD(10)	Same as CONROD(10)	
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CSHEAR 4	TMAX/ ETMAX	Maximum shear	TMAXR/ ETMAXR	Maximum shear	RM
	TAVG/ ETAVG	Average shear	TMAXI/ ETMAXI	Maximum shear	IP
	MS	Safety margin	TAVGR/ ETAVGR	Average shear	RM
			TAVGI/ ETAVGI	Average shear	IP
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CTETRA 39	CID	Stress coordinate system	CID	Stress coordinate system	

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Linear Format	CTYPE	Coordinate type (Character)	CTYPE	Coordinate type (Character)	
	NODEF	Number of active points	NODEF	Number of active points	
	GRID	External grid ID	GRID	External grid ID	
		(0=center)		(0=center)	
	SX/EX	Normal x	SXR/EXR	Normal x	RM
	TXY/ETXY	Shear xy	SYR/EYR	Normal y	RM
	P1/EP1	First principal	SZR/EZR	Normal z	RM
	P1X	First principal x cosine	TXYR/ ETXYR	Shear xy	RM
	P2X	Second principal x cosine	TYZR/ ETYZR	Shear yz	RM
	P3X	Third principal x cosine	TZXR/ ETZXR	Shear zx	RM
	PR/EPR	Mean pressure	SXI/EXI	Normal x	IP
	OCT/EOCT	von Mises or octahedral shear stress	SYI/EYI	Normal y	IP
	SY/EY	Normal y	SZI/EZI	Normal z	IP
	TYZ/ETYZ	Shear yz	TXYI/ ETXYI	Shear xy	IP
	P2/EP2	Second principal	TYZI/ ETYZI	Shear yz	IP
	P1Y	First principal y cosine	TZXI/ ETZXI	Shear zx	IP
	P2Y	Second principal y cosine			
	P3Y	Third principal y cosine			
	SZ/EZ	Normal z			
	TZX/ETZX	Shear zx			
	P3/EP3	Third principal			
	P1Z	First principal z cosine			
	P2Z	Second principal z cosine			

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	P3Z	Third principal z cosine			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CTRAX3 242 Linear Format Center and Corners Grid or Gauss	Same As CQUADX4 (243)	Same As CQUADX4 (243)	Same As CQUADX4 (243)	Same As CQUADX4 (243)	
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CTRAX6 244 Linear Format Grid and Gauss Center and Corners	Same As CQUADX4 (243)	Same As CQUADX4 (243)	Same As CQUADX4 (243)	Same As CQUADX4 (243)	
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CTRIA3 74 Linear Format Center Only	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	
CTRIA3 97 Composite Center Only	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase

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CTRIA6 75 Linear Format Center and Corners	Same as CQUAD8(64)	Same as CQUAD8(64)	Same as CQUAD8(64)	Same as CQUAD8(64)	
CTRIA6 98 Composite Center Only	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CTRIAR 70 Center and Corners	TERM	CEN/	TERM	CEN/	
	GRID	4	GRID	4	
	FD1	Z1-Fiber distance	FD1	Z1-Fiber distance	
	SX1/EX1	Normal x at Z1	SX1R/ EX1R	Normal x at Z1	RM
	SY1/EY1	Normal y at Z1	SX1I/EX1I	Normal x at Z1	IP
	TXY1/ ETXY1	Shear xy at Z1	SY1R/ EY1R	Normal y at Z1	RM
	A1	Shear angle at Z1	SY1I/EY1I	Normal y at Z1	IP
	MJRP1/ EMJRP1	Major principal at Z1	TXY1R/ ETXY1R	Shear xy at Z1	RM
	MNRP1/ EMNRP1	Minor principal at Z1	TXY1I/ ETXY1I	Shear xy at Z1	IP
	TMAX1/ ETMAX1	von Mises or maximum shear at Z1	FD2	Z2-Fiber distance	
	FD2	Z2-Fiber distance	SX2R/ EX2R	Normal x at Z2	RM
	SX2/EX2	Normal x at Z2	SX2I/EX2I	Normal x at Z2	IP
	SY2/EY2	Normal y at Z2	SY2R/ EY2R	Normal y at Z2	RM
	TXY2/ ETXY2	Shear xy at Z2	SY2I/EY2I	Normal y at Z2	IP
	A2	Shear angle at Z2	TXY2R/ ETXY2R	Shear xy at Z2	RM

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	MJRP2/ EMJRP2	Major principal at Z2	TXY2I/ ETXY2I	Shear xy at Z2	IP
	MNRP2/ EMNRP2	Minor principal at Z2			
	TMAX2/ ETMAX2	von Mises or maximum shear at Z2			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CTRIAR 227 Center Only (not supported by old formulation)	FD1	Z1=Fiber Distance 1	FD1	Z1=Fiber Distance 1	
	SX1/EX1	Normal x at Z1	SX1R/ EX1R	Normal x at Z1	RM
	SY1/EY1	Normal y at Z1	SX1I/EX1I	Normal x at Z1	IP
	TXY1/EXY1	Shear xy at Z1	SY1R/ EY1R	Normal y at Z1	RM
	SA1/EA1	Shear angle at Z1	SY1I/EY1I	Normal y at Z1	IP
	SMJRP1/ EMJRP1	Major principal at Z1	TXY1R/ EXY1R	Shear xy at Z1	RM
	SMNRP1/ EMNRP1	Minor principal at Z1	TXY1I/ EXY1I	Shear xy at Z1	IP
	SMAX1/ EMAX1	von Mises or maximum shear at Z1	FD2/FD2	Z2=Fiber distance 2	
	FD2	Z2=Fiber distance 2	SX2R/ EX2R	Normal x at Z2	RM
	SX2/EX2	Normal x at Z2	SX2I/EX2I	Normal x at Z2	IP
	SY2/EY2	Normal y at Z2	SY2R/ EY2R	Normal y at Z2	RM
	TXY2/EXY2	Shear xy at Z2	SY2I/EY2I	Normal y at Z2	IP
	SA2/EA2	Shear angle at Z2	TXY2R/ EXY2R	Shear xy at Z2	RM
	SMJRP2/ EMJRP2	Major principal at Z2	TXY2I/ EXY2I	Shear xy at Z2	IP
	SMNRP2/ EMNRP2	Minor principal at Z2			

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	TMAX2/ EMAX2	von Mises or maximum shear at Z2			
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CTRIAR 233 Composite Center Only (not supported by old formulation)	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	Same as CQUAD4(95)	
Element Name (Code)	Real Stresses or Strains		Complex Stresses or Strains		
	NDDL entity	Item	NDDL entity	Item	Real/ Mag. or Imag./ Phase
CTUBE 3 Linear Format	AS	Axial Stress			
	SE	Equivalent Stress			
	TE	Total Strain			
	EPS	Eff Plastic Strain			
	ECS	Eff Creep Strain			
	LTS	Linear Torsional Stress			

Force monitor table

Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CBAR 34	BM1A	Bending End A plane 1	BM1AR	Bending End A plane 1	RM
	BM2A	Bending End A plane 2	BM2AR	Bending End A plane 2	RM
	BM1B	Bending End B plane 1	BM1BR	Bending End B plane 1	RM

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	BM2B	Bending End B plane 2	BM2BR	Bending End B plane 2	RM
	TS1	Shear plane 1	TS1R	Shear plane 1	RM
	TS2	Shear plane 2	TS2R	Shear plane 2	RM
	AF	Axial force	AFR	Axial force	RM
	TRQ	Torque	TRQR	Torque	RM
			BM1AI	Bending End A plane 1	IP
			BM2AI	Bending End A plane 2	IP
			BM1BI	Bending End B plane 1	IP
			BM2BI	Bending End B plane 2	IP
			TS1I	Shear plane 1	IP
			TS2I	Shear plane 2	IP
			AFI	Axial force	lP
			TRQI	Torque	IP
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CBAR 100	SD	Station Distance/ Length	SD	Station Distance/ Length	RM
	BM1	Bending Moment Plane 1	BM1r	Bending Moment Plane 1	RM
	BM2	Bending Moment Plane 2	BM2r	Bending Moment Plane 2	RM
	TS1	Shear Force Plane 1	TP1r	Shear Force Plane 1	RM
	TS2	Shear Force Plane 2	TP2r	Shear Force Plane 2	RM
	AF	Axial	AFr	Axial	RM
	TRQ	Torque	TrQR	Torque	RM
		NDDL entitys are given for End A. Addition of the quantity (K-1) * 7 to the NDDL entity points to the same information for	BM1i	Bending Moment Plane 1	IP

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		other stations, where K is the station number. K=8 for End B and 2 through 7 for intermediate stations.			
			BM2i	Bending Moment Plane 2	IP
			TS1i	Shear Force Plane 1	IP
			TS2i	Shear Force Plane 2	IP
			AFi	Axial	IP
			TRQi	Torque	IP
				(NDDL entitys above are given for End A. For codes 2 through 14 at intermediate stations add (K-1) * 13 and K is the station number, and for codes at End B, K+number of stations plus 1.)	
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CBEAM 2	GRID	External grid point ID	GRID	External grid point ID	
	SD	Station distance/ length	SD	Station distance/length	
	BM1	Bending moment plane 1	BM1R	Bending moment plane 1	RM
	BM2	Bending moment plane 2	BM2R	Bending moment plane 2	RM
	TS1	Web shear plane 1	TS1R	Web shear plane 1	RM
	TS2	Web shear plane 2	TS2R	Web shear plane 2	RM
	AF	Axial force	AFR	Axial force	RM
	TTRQ	Total torque	TTRQR	Total torque	RM
	WTRQ	Warping torque	WTRQR	Warping torque	RM

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		(NDDL entitys are given for End A. Addition of the quantity (K-1) 9 to the NDDL entity points to the same information for other stations, where K is the station number. K=11 for End B and 2 through 10 for intermediate stations.)	BM1I	Bending moment plane 1	IP
			BM2I	Bending moment plane 2	IP
			TS1I	Web shear plane 1	lP
			TS2I	Web shear plane 2	IP
			AFI	Axial force	IP
			TTRQI	Total torque	IP
			WTRQI	Warping torque	lP
				(NDDL entitys are given for End A. Addition of the quantity (K-1) 16 to the NDDL entity points to the same information for other stations, where K is the station number. K=11 for End B and 2 through 10 for intermediate stations.)	
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CBEAR 280	TX	Force-x	TXR	Force-x	RM
	TY	Force-y	TYR	Force-y	RM
	TZ	Force-z	TZR	Force-z	RM
	RX	Moment-x	RXR	Moment-x	RM
	RY	Moment-y	RYR	Moment-y	RM
	RZ	Moment-z	RZR	Moment-z	RM
			TXI	Force-x	IP
			TYI	Force-y	IP
			TZI	Force-z	IP
			RXI	Moment-x	IP
			RYI	Moment-y	IP

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			RZI	Moment-z	IP
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CBEND 69	GRID	External grid point ID	GRID	External grid point ID	
	BM1	Bending moment plane 1	BM1R	Bending moment plane 1	RM
	BM2	Bending moment plane 2	BM2R	Bending moment plane 2	RM
	TS1	Shear plane 1	TS1R	Shear plane 1	RM
	TS2	Shear plane 2	TS2R	Shear plane 2	RM
	AF	Axial force	AFR	Axial force	RM
	TRQ	Torque	TRQR	Torque	RM
		(NDDL entitys are given for End A. NDDL entitys 9 through 15 point to the same information for End B.)	BM1I	Bending moment plane 1	IP
			BM2I	Bending moment plane 2	IP
			TS1I	Shear plane 1	IP
			TS2I	Shear plane 2	IP
			AFI	Axial force	IP
			TRQi	Torque	IP
				(NDDL entitys are given for End A. NDDL entitys 15 through 27 point to the same information for End B.)	
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CBUSH	FX	Force-x	FXR	Force-x	RM

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102 Linear Format	FY	Force-y	FYR	Force-y	RM
	FZ	Force-z	FZR	Force-z	RM
	MX	Moment-x	MXR	Moment-x	RM
	MY	Moment-y	MYR	Moment-y	RM
	MZ	Moment-z	MZR	Moment-z	RM
			FXI	Force-x	IP
			FYI	Force-y	IP
			FZI	Force-z	IP
			MXI	Moment-x	IP
			MYI	Moment-y	IP
			MZI	Moment-z	IP

Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CDAMP1 20 CDAMP2 21 CDAMP3 22 CDAMP4 23	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	
	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	
	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	
	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	

Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CDUM3 thru CDUM9 (55-61)	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	Same as CELAS1(11)	

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Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CELAS1 11	F	Force	FR	Force	RM
			FI	Force	IP
CELAS2 12		Same as CELAS1		Same as CELAS1	
CELAS3 13		Same as CELAS1		Same as CELAS1	
CELAS4 14		Same as CELAS1		Same as CELAS1	
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CFAST 119	BM1A	mz bending End A plane 1	BM1AR	mz bending End A plane 1	RM
	BM2A	my bending End A plane 2	BM2AR	my bending End A plane 2	RM
	BM1B	mz bending End B plane 1	BM1BR	mz bending End B plane 1	RM
	BM2B	my bending End B plane 2	BM2BR	my bending End B plane 2	RM
	TS1	fy shear force plane 1	TS1R	fy shear force plane 1	RM
	TS2	fz shear force plane 2	TS2R	fz shear force plane 2	RM
	AF	fx axial force	AFR	fx axial force	RM
	TRQ	mx torque	TRQR	mx torque	RM
			BM1AI	mz bending End A plane 1	IP
			BM2AI	my bending End A plane 2	IP
			BM1BI	mz bending End B plane 1	IP

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			BM2BI	my bending End B plane 2	IP
			TS1I	fy shear force plane 1	IP
			TS2I	fz shear force plane 2	IP
			AFI	fx axial force	IP
			TRQI	mx torque	IP
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CONROD 10	AF	Axial force	AFR	Axial force	RM
	TRQ	Torque	AFI	Axial force	IP
			TRQR	Torque	RM
			TRQI	Torque	lP
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CQUAD4 33 Center Only	MX	Membrane force x	MXR	Membrane force x	RM
	MY	Membrane force y	MYR	Membrane force y	RM
	MXY	Membrane force xy	MXYR	Membrane force xy	RM
	BMX	Bending moment x	BMXR	Bending moment x	RM
	BMY	Bending moment y	BMYR	Bending moment y	RM
	BMXY	Bending moment xy	BMXYR	Bending moment xy	RM
	TX	Shear x	TXR	Shear x	RM
	TY	Shear y	TYR	Shear y	RM
			MXI	Membrane force x	IP
			MYI	Membrane force y	IP
			MXYI	Membrane force xy	IP
			BMXI	Bending moment x	IP
			BMYI	Bending moment y	IP
			BMXYI	Bending moment xy	IP

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			TXI	Shear x	IP
			TYI	Shear y	IP
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CQUAD4 144 Center and Corners	TERM	CEN/	TERM	CEN/	
	GRID	4	GRID	4	
	MX	Membrane x	MXR	Membrane x	RM
	MY	Membrane y	MYR	Membrane y	RM
	MXY	Membrane xy	MXYR	Membrane xy	RM
	BMX	Bending x	BMXR	Bending x	RM
	BMY	Bending y	BMYR	Bending y	RM
	BMXY	Bending xy	BMXYR	Bending xy	RM
	TX	Shear x	TXR	Shear x	RM
	TY	Shear y	TYR	Shear y	RM
			MXI	Membrane x	IP
			MYI	Membrane y	IP
			MXYI	Membrane xy	IP
			BMXI	Bending x	IP
			BMYI	Bending y	IP
			BMXYI	Bending xy	IP
			TXI	Shear x	IP
			TYI	Shear y	IP
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CQUAD8 64 Center and Corners	Same as CQUAD144(144)	Same as CQUAD144(144)	Same as CQUAD144(144)	Same as CQUAD144(144)	
Element Name Code	Real Element Forces		Complex Element Forces		

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	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CQUADR 82 Center and Corners	Same as CQUAD144(144)	Same as CQUAD144(144)	Same as CQUAD144(144)	Same as CQUAD144(144)	
CQUADR 228 Center Only (not supported by old formulation)	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CROD 1	Same as CONROD(10)	Same as CONROD(10)	Same as CONROD(10)	Same as CONROD(10)	
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CSHEAR 4	F41	Force 4 to 1	F41R	Force 4 to 1	RM
	F21	Force 2 to 1	F21R	Force 2 to 1	RM
	F12	Force 1 to 2	F12R	Force 1 to 2	RM
	F32	Force 3 to 2	F32R	Force 3 to 2	RM
	F23	Force 2 to 3	F23R	Force 2 to 3	RM
	F43	Force 4 to 3	F43R	Force 4 to 3	RM
	F34	Force 3 to 4	F34R	Force 3 to 4	RM
	F14	Force 1 to 4	F14R	Force 1 to 4	RM
	KF1	Kick force on 1	F41I	Force 4 to 1	IP
	S12	Shear 12	F21I	Force 2 to 1	IP
	KF2	Kick force on 2	F12I	Force 1 to 2	IP
	S23	Shear 23	F32I	Force 3 to 2	IP
	KF3	Kick force on 3	F23I	Force 2 to 3	IP

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	S34	Shear 34	F43I	Force 4 to 3	IP
	KF4	Kick force on 4	F34I	Force 3 to 4	IP
	S41	Shear 41	F14I	Force 1 to 4	IP
			KF1R	Kick force on 1	RM
			S12R	Shear 12	RM
			KF2R	Kick force on 2	RM
			S23R	Shear 23	RM
			KF3R	Kick force on 3	RM
			S34R	Shear 34	RM
			KF4R	Kick force on 4	RM
			S41R	Shear 41	RM
			KF1I	Kick force on 1	IP
			S12I	Shear 12	IP
			KF2I	Kick force on 2	IP
			S23I	Shear 23	IP
			KF3I	Kick force on 3	IP
			S34I	Shear 34	IP
			KF4I	Kick force on 4	IP
			S41I	Shear 41	IP
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CTRIA3 74 Center Only	Same as CQUAD 4(33)	Same as CQUAD4(33)	Same as CQUAD4(3 3)	Same as CQUAD4(33)	
CTRIA6 75 Center and Corners	Same as CQUAD1 44(144)	Same as CQUAD144(144)	Same as CQUAD14 4(144)	Same as CQUAD144(144)	
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase

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CTRIAR 70 Center and Corners	Same as CQUAD144(144)	Same as CQUAD144(144)	Same as CQUAD144(144)	Same as CQUAD144(144)	
CTRIAR 227 Center Only (not supported by old formulation)	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	Same as CQUAD4(33)	
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CTUBE 3	AF	Axial force	AFR	Axial force	RM
	TRQ	Torque	AFI	Axial force	IP
			TRQR	Torque	RM
			TEQI	Torque	lP
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CVlSC 24	Same as CONROD(10)	Same as CONROD(10)	Same as CONROD(10)	Same as CONROD(10)	
Element Name Code	Real Element Forces		Complex Element Forces		
	NDDL entity	Item	NDDL entity	Item	Real/Mag. or Imag/ Phase
CWELDP (118) ELPAT or PARTPAT CWELDC (117) ELIMID or GRIDID with MSET=OFF CWELD (200) ELIMID or GRIDID with MSET=ON	BM1A	mz bending End A plane 1	BM1AR	mz bending End A plane 1	RM
	BM2A	my bending End A plane 2	BM2AR	my bending End A plane 2	RM
	BM1B	mz bending End B plane 1	BM1BR	mz bending End B plane 1	RM
	BM2B	my bending End B plane 2	BM2BR	my bending End B plane 2	RM
	TS1	fy shear force plane 1	TS1R	fy shear force plane 1	RM
	TS2	fz shear force plane 2	TS2R	fz shear force plane 2	RM

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	AF	fx axial force	AFR	fx axial force	RM
	TRQ	mx torque	TRQR	mx torque	RM
			BM1AI	mz bending End A plane 1	IP
			BM2AI	my bending End A plane 2	IP
			BM1BI	mz bending End B plane 1	IP
			BM2BI	my bending End B plane 2	IP
			TS1I	fy shear force plane 1	IP
			TS2I	fz shear force plane 2	IP
			AFI	fx axial force	IP
			TRQI	mx torque	IP

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MONPNT3

Integrated Load Monitor Point — Sums Grid Point Forces

Defines an integrated load monitor point at a chosen point and sums select grid point forces for SOL 101 and SOL 103.

Format:

1	2	3	4	5	6	7	8	9	10
MONPNT3	NAME	LABEL							
	AXES	GRIDGRP	ELEMGRP	CP	X1	X2	X3	XFLAG	
	CD								

Example:

MONPNT3	MPT33	This is a MONPNT3 monitor point.							
	246	1001		2002	1.0	2.0	3.0	MD	
	1003								

Fields:

Field	Contents
NAME	A unique character string that identifies the monitor point. (Character; 8 characters maximum; No default)
LABEL	A string that identifies and labels the monitor point. (Character; 56 characters maximum; Optional)
AXES	Component axes about which to sum. (Integer; Any unique combination of the integers 1 through 6 with no embedded blanks; No default)
GRIDGRP	GROUP entry that has a list of grids to be included in the monitor point. (Integer; No default). See Remark 1.
ELEMGRP	GROUP entry that has a list of elements to process at the monitor point. (Integer; Optional). See Remark 2.
CP	The identification number of a coordinate system in which the X1, X2, and X3 coordinates are defined. (Integer; Default = 0; Optional). See Remark 5.
X1, X2, X3	The coordinates in the CP coordinate system about which the forces are to be summed. (Real; Default = 0.0).
XFLAG	Exclusion flag excludes the indicated Grid Point Force types from summation at the monitor point. (Default = blank (no type excluded)).

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Field	Contents
	If XFLAG = "S", SPC forces are excluded. If XFLAG = "M", MPC forces are excluded. If XFLAG = "A", "L", or "P", applied loads, which includes thermal loads, are excluded. If XFLAG = "D", DMIGs at the monitored point are excluded.
CD	The identification number of a coordinate system in which the results are output. (Integer ≥ 0; Default = Coordinate system specified by the CP field)

Remarks:

1. Requirements for grids that are included in the monitor point:
   - You select a set of grids to sum forces and moments about the specified point.
   - You must use only GRIDs in the GROUP definition. MONPNT3 does not process SPOINTs, EPOINTs, and fluid grid points when they are in the GROUP definition.
2. Requirements for elements that are processed at the monitor point:
   - When ELEMGRP is defined for any MONPNT3 bulk entry, you must define the GPFORCE=ALL case control command for all subcases, and the request must be for the SORT1 output format. You must either define GPFORCE=ALL in each subcase, or above the subcases (globally). When ELEMGRP is blank, no contributions are made from the set of elements attached to the grid. The elements attached to the grid are excluded.
3. Output for superelements that are processed at the monitor point:
   - For internal and partitioned superelements, MONPNT3 results can be output for the residual or downstream superelements.
   - For external superelements, MONPNT3 results are only output for the residual superelement.

   Note

   The various GROUPs and CPs defined for a MONPNT3 may not exist for all of the superelements in a model. When this occurs, the MONPNT3 is not processed for the superelement. A warning message indicates that the MONPNT3 is not processed.
4. Invalid grids or elements do not generate error or warning messages.
5. A CP of zero (the default) references the basic coordinate system.
6. In a structure, you can apply MONPNT3 to determine the internal shear, moment, torque, and thermal loads.

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Example

A long and slender component is divided and all grids that reside on the division are placed in the GRIDGRP group. Based on the values of the ELEMGRP and XFLAG fields, the software outputs various internal load resultants.

- ELEMGRP contains all elements connected to the GRIDGRP on the outer part of the split:
  - XFLAG is blank:

    The software calculates the internal load with the elements connected to the GRIDGRP that are not included in ELEMGRP. This produces resultants on the inside of the split and contains any loads applied to the GRIDGRP from any other source.
  - XFLAG = SMAD:

    The software calculates the internal load with the elements listed in ELEMGRP. This produces resultants on the outer part of the split and does not contain loads applied to the GRIDGRP from any other source.

7. The software prints the results in a table titled `STRUCTURAL INTEGRATED FREE BODY MONITOR POINT LOADS (MONPNT3)`. The total loads in this table include the thermal loads unless you set XFLAG = A.

Related Topics

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MPC

Multipoint Constraint

Defines a multipoint constraint equation of the form

$$\sum_i Aj\, u_j = 0$$

where u_j represents degree-of-freedom Cj at grid or scalar point Gj.

Format:

1	2	3	4	5	6	7	8	9	10
MPC	SID	G1	C1	A1	G2	C2	A2		
		G3	C3	A3	-etc.-				

Example:

MPC	3	28	3	6.2	2		4.29		
		1	4	-2.91					

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
Gj	Identification number of grid or scalar point. (Integer > 0)
Cj	Component number. (Any one of the Integers 1 through 6 for grid points; blank or zero for scalar points.)
Aj	Coefficient. (Real; Default = 0.0 except A1 must be nonzero.)

Remarks:

1. Multipoint constraint sets must be selected with the Case Control command MPC = SID.
2. The first degree-of-freedom (G1, C1) in the sequence is defined to be the dependent degree-of-freedom. A dependent degree-of-freedom assigned by one MPC entry cannot be assigned dependent by another MPC entry or by a rigid element.
3. Forces of multipoint constraint may be recovered in all solution sequences, except SOL 129, with the MPCFORCES Case Control command.

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4. The m-set degrees-of-freedom specified on this entry may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.

5. By default, the grid point connectivity created by the MPC and MPCADD entries is not considered during resequencing, (see the description of parameter **"OLDSEQ"**). In order to consider the connectivity during resequencing, SID must be specified on the PARAM,MPCX entry. Using the example above, specify PARAM,MPCX,3.

Related Topics

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MPCADD

Multipoint Constraint Set Combination

Defines a multipoint constraint set as a union of multipoint constraint sets defined via MPC entries.

Format:

1	2	3	4	5	6	7	8	9	10
MPCADD	SID	S1	S2	S3	S4	S5	S6	S7	
	S8	S9	-etc.-						

Example:

MPCADD	101	2	3	1	6	4			

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
Sj	Set identification numbers of multipoint constraint sets defined via MPC entries. (Integer > 0)

Remarks:

1. Multipoint constraint sets must be selected with the Case Control command MPC = SID.
2. The Sj must be unique and may not be the identification number of a multipoint constraint set defined by another MPCADD entry.
3. MPCADD entries take precedence over MPC entries. If both have the same SID, only the MPCADD entry will be used.
4. By default, the grid point connectivity created by the MPC and MPCADD entries is not considered during resequencing, (see **"OLDSEQ"** for more information). In order to consider the connectivity during resequencing, SID must be specified on the PARAM,MPCX entry. Using the example above, specify PARAM,MPCX,101.

Related Topics

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MPLAS

Plastic material for SOLs 401, 402, and 414

Defines plastic material properties for SOLs 401, 402, 414,103, and 414,129.

General Format:

1	2	3	4	5	6	7	8	9	10
MPLAS	MID	Name							
	"YIELD"	Y-TYPE	Y-PAR1	Y-PAR2	...				
	"ISOH"	I-TYPE	I-PAR1	I-PAR2	...				
	"KINH"	K-TYPE	K-PAR1	K-PAR2	...				

Yield Function "YIELD" options:

MISES: Von Mises

	"YIELD"	"MISES"							

HILL: Hill, anisotropic

	"YIELD"	"HILL"	R11	R22	R33	R12	R23	R31	

GCIRON: Gray Cast Iron

	"YIELD"	"GCIRON"							

PLASGEN: Plasticity

	"YIELD"	"PLASGEN"	FLIX	FLIY	FLIZ	MLIX	MLIY	MLIZ	
			NF	ADEL	TAU	DDEN			
			FRUX	FRUY	FRUZ	MRUX	MRUY	MRUZ	

Isotropic Hardening Rule "ISOH" options:

TABULAR: Using a multilinear stress-strain curve

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	"ISOH"	"TABULAR"	I-TID	MODE	STRTYPE	STRMEAS		NUPL	
	"ISOH"	"TABULAR"	I-TID	"TC"	"PLS" or "TOT"	"ENG" or "TRUE"			
	"ISOH"	"TABULAR"	I-TID	"T"	"PLS" or "TOT"	"ENG" or "TRUE"		NUPL	
	"ISOH"	"TABULAR"	I-TID	"C"	"PLS" or "TOT"	"ENG" or "TRUE"			

EXPSAT: Using an exponential saturation stress-strain function

	"ISOH"	"EXPSAT"	Y0	NUM_IH	Q1	B1	Q2	B2	
			Q3	B3	Q4	B4	Q5	B5	
			Q6	B6	...	...	...	...	

Kinematic Hardening Rule "KINH" options:

SUBLAY: Sublayer-Besseling kinematic hardening using a multilinear stress-strain curve

	"KINH"	"SUBLAY"	K-TID	MODE	STRTYPE	STRMEAS			
	"KINH"	"SUBLAY"	K-TID	"TC"	"PLS" or "TOT"	"ENG" or "TRUE"			

CHAB: Chaboche nonlinear kinematic hardening using an exponential saturation stress-strain function

	"KINH"	"CHAB"		NUM_KH	C1	G1	C2	G2	
			C3	G3	C4	G4	C5	G5	
			C6	G6	...	...	...	...	

Gray Cast Iron Format:

1	2	3	4	5	6	7	8	9	10
MPLAS	MID	Name							
	"YIELD"	"GCIRON"							
	"ISOH"	"TABULAR"	TID tension	"T"	"PLS" or "TOT"	"ENG" or "TRUE"		NUPL	
			TID compression	"C"	"PLS" or "TOT"	"ENG" or "TRUE"			

Example:

1	2	3	4	5	6	7	8	9	10
MPLAS	1	Properties for cast iron material							+
+	YIELD	GCIRON							+

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1	2	3	4	5	6	7	8	9	10
+	ISOH	TABULAR	50	T	PLS	TRUE		60	+
+			55	C	PLS	TRUE			

Fields:

Field	Contents
MID	Identification number of a MAT1, MAT9, or MAT11 entry. (Integer > 0)
Name	Optional character string. (Character; 56 characters maximum)

"YIELD": Yield Function fields

"YIELD"	The "YIELD" field designates the beginning of a yield data row. (Character)
Y-TYPE	Defines the yield type. (Character)

"MISES": Von Mises yield potential function, for isotropic plasticity:

$$f\left(\sigma_{ij}, Y\right) = 1/2[\left(\sigma_{11} - \sigma_{22}\right)^2 + \left(\sigma_{22} - \sigma_{33}\right)^2 + \left(\sigma_{11} - \sigma_{33}\right)^2 + 6\left({\sigma_{12}}^2 + {\sigma_{23}}^2 + {\sigma_{31}}^2\right)] - Y^2 = 0$$

The Y-PARi fields are ignored.

"**HILL**": Hill yield potential function, for anisotropic plasticity:

$$f\left(\sigma_{ij}, Y\right) = [F\left(\sigma_{22} - \sigma_{33}\right)^2 + G\left(\sigma_{33} - \sigma_{11}\right)^2 + H\left(\sigma_{11} - \sigma_{22}\right)^2 + 2L{\sigma_{23}}^2 + 2M{\sigma_{31}}^2 + 2N{\sigma_{12}}^2] - Y^2 = 0$$

where F, G, H, L, M, N are functions described in the Y-PARi field descriptions below.

"GCIRON": Cast iron yield type.

Cast iron has a different behavior in traction and compression. In tension, it is associated with the Rankine yield criterion (a cube in the 3D space of principal stresses). In compression, it is associated with the Von Mises yield criterion (a cylinder in the 3D space of principal stress). The Y-PARi fields are ignored.

"PLASGEN": Plasticity yield function type for the rupture limits of forces and moments. See Remark 8.

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Field	Contents
Y-PARi	Parameters associated with specific yield types (Real for a constant value that does not depend on temperature; Integer>0 that references the ID of a TABLEM1 defining temperature dependence). **If Y-TYPE= "HILL"**, the parameters define the six stress ratios Y-PAR1=R_{11}, Y-PAR2=R_{22}, Y-PAR3=R_{33}, Y-PAR4=R_{12}, Y-PAR5=R_{23}, and Y-PAR6=R_{13}, where $$R_{11} = \frac{Y_{0,11}}{Y_0} \quad R_{22} = \frac{Y_{0,22}}{Y_0} \quad R_{33} = \frac{Y_{0,33}}{Y_0}$$ $$R_{12} = \sqrt{3}\frac{Y_{0,12}}{Y_0} \quad R_{23} = \sqrt{3}\frac{Y_{0,23}}{Y_0} \quad R_{13} = \sqrt{3}\frac{Y_{0,13}}{Y_0}$$ Y_0 is the yield stress defined in the referenced stress-strain curve, and $Y_{0,ij}$ are the direction specific yield stress values. The stress ratios are used in the following coefficients, and included in the Hill yield potential function described in the Y-TYPE field above: $$F = \frac{1}{2}\left(\frac{1}{{R_{22}}^2} + \frac{1}{{R_{33}}^2} - \frac{1}{{R_{11}}^2}\right) \quad G = \frac{1}{2}\left(\frac{1}{{R_{33}}^2} + \frac{1}{{R_{11}}^2} - \frac{1}{{R_{22}}^2}\right)$$ $$H = \frac{1}{2}\left(\frac{1}{{R_{22}}^2} + \frac{1}{{R_{11}}^2} - \frac{1}{{R_{33}}^2}\right)$$ $$L = \frac{3}{2}\left(\frac{1}{{R_{23}}^2}\right) \quad M = \frac{3}{2}\left(\frac{1}{{R_{13}}^2}\right) \quad N = \frac{3}{2}\left(\frac{1}{{R_{12}}^2}\right)$$

"ISOH": Isotropic Hardening fields

"ISOH"	The "ISOH" field designates the beginning of an isotropic hardening data row. (Character) The isotropic hardening denotes that the yield surface in the 3D principal stress space expands in an isotropic way. It is commonly represented as a 2D yield surface or the uniaxial stress-strain diagram as shown below.

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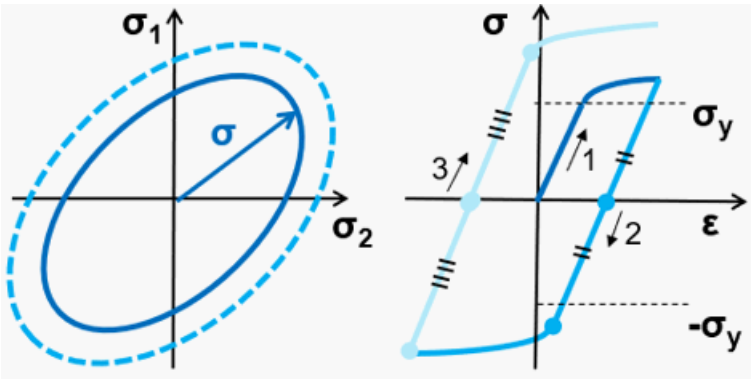
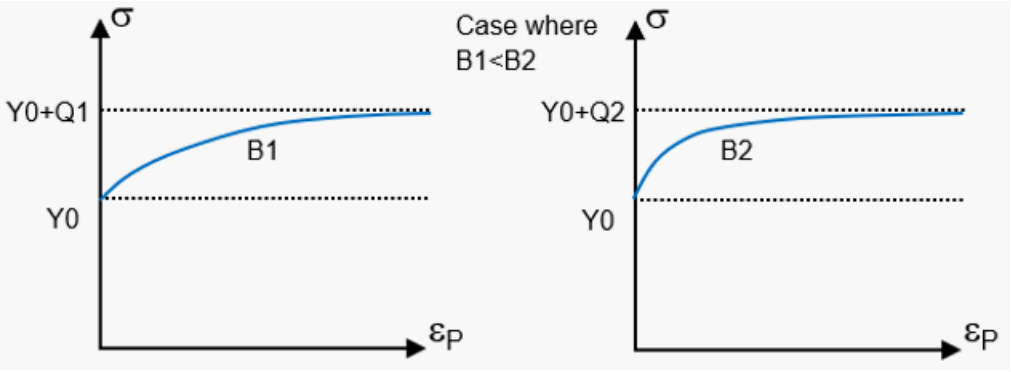
Field	Contents
	Pure Isotropic Hardening
	Yield surface in σ_1, σ_2 / Uniaxial Stress-Strain
	
I-TYPE	Defines the data format of the stress-strain curve. (Character)
	"TABULAR": The stress-strain data is defined on the TABLES1 entry. See rows below for the related parameters.
	It can be combined with Chaboche kinematic hardening, but not with Sublayer kinematic hardening. See Remark 7.
	"EXPSAT": The stress-plastic strain data is defined with the following analytical exponential saturation of the following yield stress function, where the data is determined from an isothermal uniaxial tensile test:

$$\sigma_{uniaxial} = Y_0 + \sum_{j=1}^{NUM_IH} \{Q_j \left[1 - EXP\left(-B_j \varepsilon_{eq,pl}\right)\right]\}$$

with j=1,.., NUM_IH and Max(NUM_IH)=20.



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Field	Contents
	It can be combined only with Chaboche kinematic hardening, but not with Sublayer kinematic hardening. See Remark 7. Note that: - The strain in the yield stress function is always expressed as plastic strain, and the STRTYPE field is ignored. - The software does not make any conversion of the plastic strain from engineering to true measure, so the software assumes true measure and ignores the STRMEAS field. - Q_j can be negative to allow the definition of cyclic softening. See Remark 4. - "EXPSAT" can be combined with MVPLAS (creep) only when Chaboche kinematic hardening is also selected. See Remark 7. See rows below for related parameters.
If I-TYPE="TABULAR"	
I-TID	Identification number of a TABLES1 or TABLEST entry that defines the stress-strain data. The TABLES1 entry defines a single set of data. The TABLEST entry references multiple sets of data, each defined with the TABLES1 entry, for temperature dependence. See Remark 3. (Integer>0)
MODE	Defines the hardening data direction. (Character; Default="TC") If Y-TYPE = "MISES" or "HILL": "TC": Hardening data is defined tensile only, but is also applied to compression. If Y-TYPE = "GCIRON" (both "T" and "C" rows are required): "T": Hardening data applied in the tensile direction. "C": Hardening data applied in the compressive direction.
STRTYPE	Defines the strain data type. See Remark 3. (Character; Default="PLS") "PLS": Plastic strain data. "TOT": Combined elastic + plastic strain data. The solver uses STRTYPE=PLS, so if you define STRTYPE=TOT, the software will automatically convert the data to PLS.

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Field	Contents
STRMEAS	Stress/strain measure of the TABLES1 or TABLEST data referenced by the TID field. (Character; Default="TRUE") "TRUE": True stress-strain data. "ENG": Engineering stress-strain data. When a large strain analysis is requested (PARAM, LGSTRN, 1), the software uses STRMEAS=TRUE. As a result, if you define STRMEAS=ENG, the software will automatically convert the data to true measure. In case of a small strain analysis (PARAM, LGSTRN, 0), the software ignores the STRMEAS field. In this case, if you define either engineering or true data, the software will not do a conversion. For more information, see SOLs 401 and 402 - Stress-strain measures in the *Multi-Step Nonlinear User's Guide*.
NUPL	If Y-TYPE= "GCIRON" and DATADIR= "T": Defines the plastic Poisson's ratio in tension for the grey cast iron yield type. When defined as an integer, it is the identification number of a TABLEM1 entry that defines NUPL as temperature dependent. (-1.0 < Real < 0.5, Integer>0; Default=0.04)
If I-TYPE="EXPSAT"	
Y0	Initial yield stress in units of pressure. (Real >0 or Integer > 0) If real entry, value of coefficient used at all temperatures. If integer entry, identification number of TABLEM1 bulk entry that defines the coefficient as a function of temperature.
NUM_IH	Number of terms of the series from no terms (NUM_IH=0) up to the maximum (NUM_IH=20). If NUM_IH=0, only yield stress is defined here and there is no isotropic hardening. (0≤Integer≤20)
Qi, Bi	Parameters for isotropic hardening with exponential saturation law. Q_i is in units of pressure. B_i is dimensionless. (Real; $B_i \geq 0.0$; or Integer > 0) If real entry, value of coefficient used at all temperatures. If integer entry, identification number of TABLEM1 bulk entry that defines the coefficient as a function of temperature.
"KINH": Kinematic Hardening fields	
"KINH"	The "KINH" field designates the beginning of a kinematic hardening data row. (Character)

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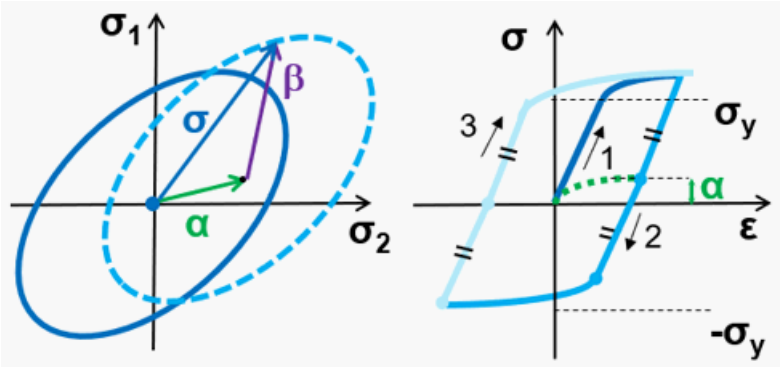
Field	Contents
	The kinematic hardening denotes that the yield surface in the 3D principal stress space moves by the back-stress tensor **α**. It is commonly represented as a 2D yield surface or the uniaxial stress-strain diagram as shown below.

Pure Kinematic Hardening	
Yield surface in σ_1, σ_2	Uniaxial Stress-Strain



K-TYPE	"SUBLAY": Sublayer or Besseling kinematic hardening. The K-TID field references stress-strain data defining a multilinear stress-strain curve. A full cyclic load defined with either enforced displacements or forces provides a closed stress-strain cycle. This hardening rule cannot be coupled to isotropic hardening. For this reason, this hardening rule can be used to study the Bauschinger effect for stabilized cycles that give closed cycles. See the associated parameters below for selecting the stress-strain data, as well the type of measure and type of strain. See Remark 5.

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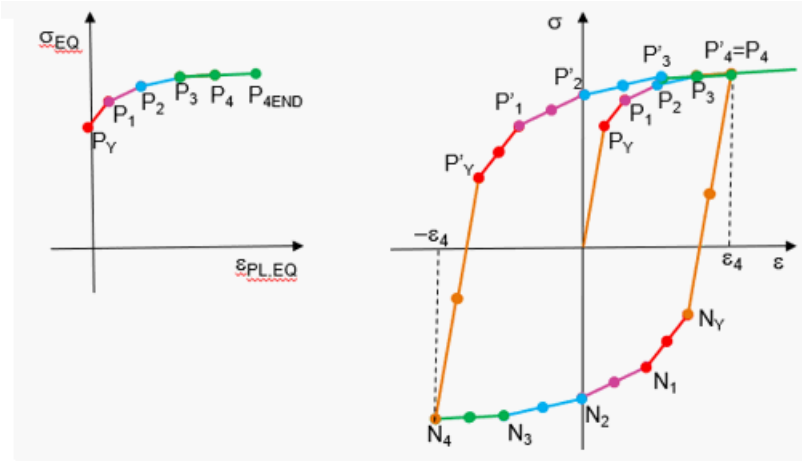
Field	Contents

Sublayer Kinematic Hardening	
Multilinear Stress vs. Plastic Strain	Stress-Strain result under cyclic loading



"CHAB": Chaboche nonlinear kinematic hardening. It is a nonlinear kinematic hardening rule in the sense that the back-stress is associated to a nonlinear expression of the plastic strain:

$$[\dot{\alpha}] = \sum_{k}^{NUM_KH} [\dot{\alpha}_k] = \sum_{k}^{NUM_KH} \left(\frac{2}{3} C_k [\dot{\varepsilon}_{pl}] + G_k [\alpha_k] \dot{\varepsilon}_{pl,eq} + \frac{1}{C_k} \frac{dC_k}{dT} \dot{T} [\alpha_k] \right)$$

- The C_k coefficients are associated to the linear terms with the Prager method.
- The G_k coefficients are associated to the nonlinear terms.
- K=1, ..., NUM_KH and Max(NUM_KH)=20.
- The anisothermal term with the $\dot{T}$ factor is called the Rice term.

Note that Chaboche kinematic hardening is a superposition of Armstrong-Frederick kinematic hardening models.

Combined isotropic and kinematic hardening:

The Chaboche kinematic hardening can be combined with tabular isotropic hardening (I-TYPE = "TABULAR") or exponential saturated isotropic hardening (I-TYPE = "EXPSAT"). The combination is a simple superposition. See Remark 7.

The following applies to both EXPSAT and Chaboche:
- The software always expects plastic strain, it ignores the STRTYPE field, and it does no conversion of the strain.
- The software always expects true stress-strain data, it ignores the STRMEAS field, and it does no conversion of the data.

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Field	Contents
	With an isothermal uniaxial tensile test, the following equation is solved with a series of exponential saturation functions:

$$\alpha_{uniaxial} = \sum_k \left\{ \frac{C_k}{G_k} \left[1 - EXP\left(-G_k \varepsilon_{eq,pl} \right) \right] \right\}$$

Then in combination with the isotropic hardening exponential saturated form gives:

$$\sigma_{uniaxial} = Y_0 + \sum_j \left\{ Q_j \left[1 - EXP\left(-B_j \varepsilon_{eq,pl} \right) \right] \right\} + \sum_k \left\{ \frac{C_k}{G_k} \left[1 - EXP\left(-G_k \varepsilon_{eq,pl} \right) \right] \right\}$$

Q_j can be positive or negative. If Q_j is positive, the isotropic hardening part has a hardening behavior, and if negative, it has a softening behavior.

Chaboche kinematic hardening has an advantage over sublayer kinematic hardening since it can be combined with isotropic hardening. The combination allows you to study the Bauschinger effect for non-stabilized cycles, where the stiffening or softening effect is present, and to study the asymptotic closing of the cycle.

See Remarks 4, 5, and 7.

If K-TYPE= "SUBLAY"

K-TID	Identification number of a TABLES1 or TABLEST entry that defines the stress-plastic strain data. The hardening slope (H_i) of each successive stress-plastic strain segment must be decreasing (H1>H2>H3>....etc.), and the hardening slope of any segment cannot be zero or negative (H_i>0.0). The TABLES1 entry defines a single set of data. The TABLEST entry references multiple sets of data, each defined with the TABLES1 entry, for temperature dependence. Each set of data at each temperature must have the same number of points. For SOL 401, the maximum number of data points in the stress-plastic strain curve on the TABLES1 entry is 27. See Remark 3. (Integer>0)
MODE	"TC": Hardening data is defined tensile only, but is also applied to compression.
STRTYPE	Defines the strain data type. (Character; Default="PLS") "PLS": Plastic strain data. "TOT": Combined elastic + plastic strain data. The solver uses STRTYPE=PLS, so if you define STRTYPE=TOT, the software will automatically convert the data to PLS.

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Field	Contents
STRMEAS	Stress/strain measure of the TABLES1 or TABLEST data referenced by the TID field. (Character; Default = "TRUE”) "TRUE": True stress-strain data. "ENG": Engineering stress-strain data. When a large strain analysis is requested (PARAM, LGSTRN, 1), the software uses STRMEAS=TRUE. As a result, if you define STRMEAS=ENG, the software will automatically convert the data to true measure. In case of a small strain analysis (PARAM, LGSTRN, 0), the software ignores the STRMEAS field. In this case, you can define either engineering or true data, and the software will not do a conversion. For more information, see SOLs 401 and 402 - Stress-strain measures in the *Multi-Step Nonlinear User's Guide*.
If K-TYPE="CHAB"	
NUM_KH	Number of terms of the series (1≤Integer≤20).
C_k, G_k	Parameters for kinematic hardening. C_k is in units of pressure, and G_k is dimensionless. (Real, $C_k > 0.0$, $G_k \geq 0.0$; or Integer > 0) If real entry, value of coefficient used at all temperatures. If all G_k=0.0, it is equivalent to the linear kinematic hardening rule. If integer entry, identification number of a TABLEM1 bulk entry that defines the coefficient as a function of temperature.
"PLASGEN": Plasticity fields	
FLIX, FLIY, FLIZ, MLIX, MLIY, MLIZ	Limit forces and moments of elasticity in the X, Y, and Z directions. For more information, see *Generalized Plasticity Behavior in bushing* in the Simcenter Samcef help.
NF	Yield function referring to the TABLES1 bulk entry, which describes an amplitude function of displacement. For more information, see *Generalized Plasticity Behavior in bushing* in the Simcenter Samcef help.
ADEL	Time constant, τ_c. See Remark **10**. (Real ≥ 0.0; No default)
TAU	Scaling factor, a_c. See Remark **10**. (Real; Default = 1.0)
DDEN	Critical value, *CRIT*, in the ellipsoid criterion at which rupture is predicted to initiate. See Remarks **9** and **10**. (Real; Default = 1.0)
FRUX, FRUY, FRUZ, MRUX, MRUY, MRUZ	Strength values that are used in the ellipsoid criterion for axial bushings. See Remark **9**. (Real ≠ 0.0; No default)

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Remarks:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. The gray cast iron yield type (YTYPE=GCIRON) has the following requirements:
   - Both a tension (MODE="T") and a compression (MODE="C") row must be defined, and both sets of data must both be defined in the first quadrant (positive values).
   - NUPL must be defined for the tension data row (MODE="T").
   - All character inputs must be defined: YTYPE="GCIRON", "ISOH", I-TYPE="TABULAR", STRTYPE="PLS", and STRMEAS="TRUE".
   - The second hardening row must have fields 2 and 3 blank.
3. For STRTYPE=PLS, the tabular (Xi, Yi) data listed on the TABLES1 bulk entry represents a plastic strain versus stress curve. The plastic strains are the Xi values and the stresses are the Yi values.

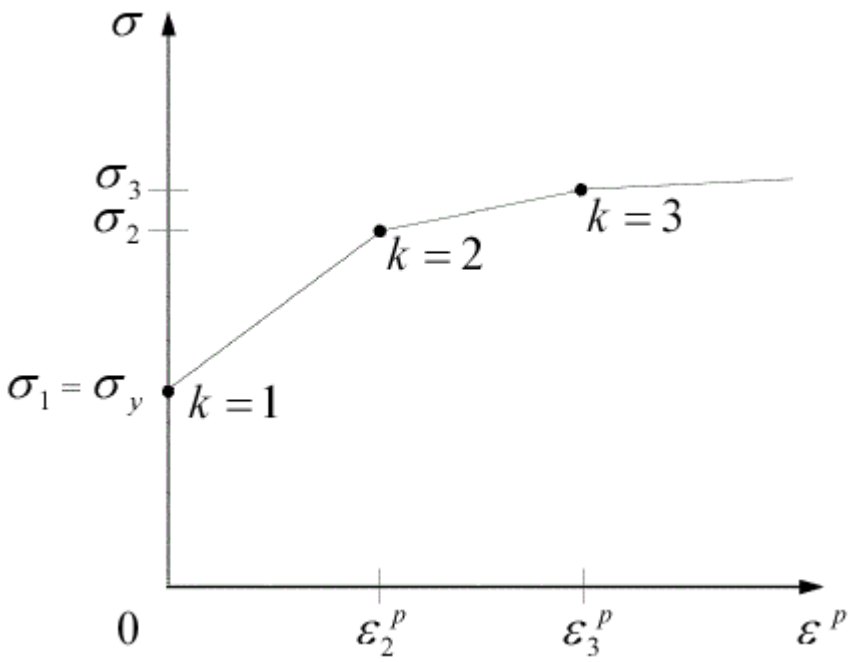
   The stress-plastic strain curve must be defined in the first quadrant as indicated in Figure 15-134, where ε_k^p is the plastic strain at each data point *k*.

   The first point for STRTYPE=PLS is the yield point. The total strain at each point *k* is related to the plastic strain *k* as follows:

$$\varepsilon_{total} = \varepsilon_k^p + \frac{\sigma_k}{E}$$

   where E is the modulus defined on the MATi bulk entry.

Figure 15-134: Stress vs. plastic strain curve



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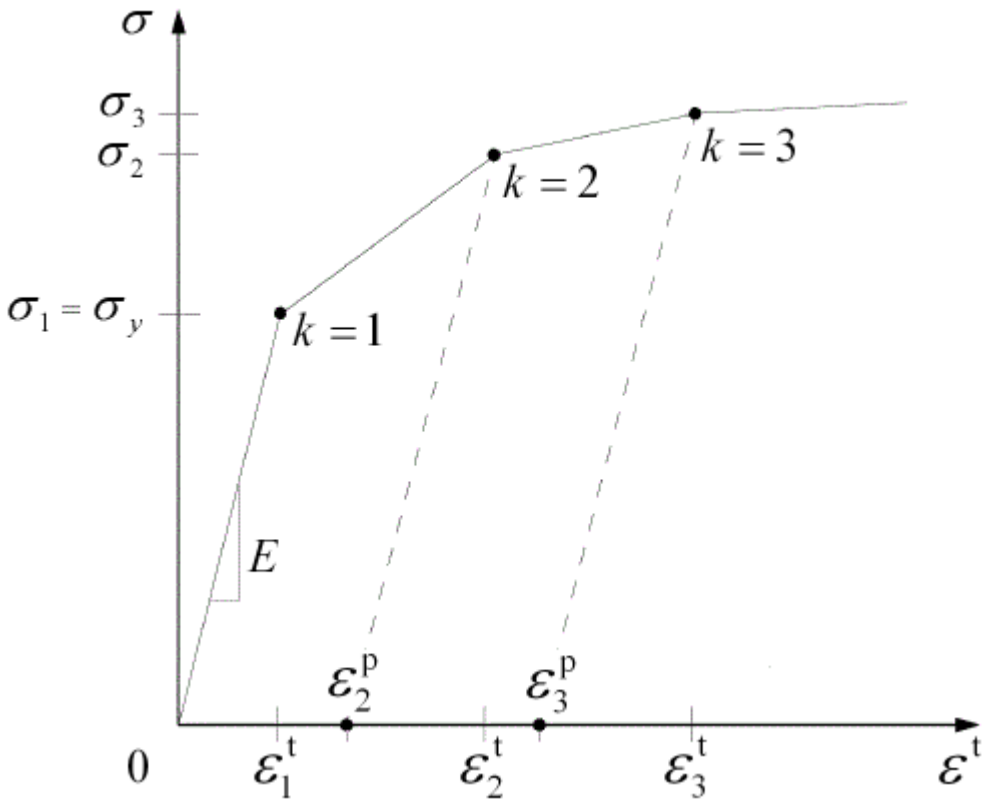
For STRTYPE=TOT, the tabular (Xi, Yi) data listed on the TABLES1 bulk entry represents the total strain versus stress curve. The total strains are the Xi values and the stresses are the Yi values.

The stress-total strain curve must be defined in the first quadrant as indicated in **Figure 15-135**, where ε_k^t is the total strain at each data point *k*.

The first point for STRTYPE=TOT is (0.0, 0.0) and the second point is the yield point.

Figure 15-135: Stress vs. total strain curve



4. The following two cases demonstrate Chaboche hardening:

Case 1 demonstrates Chaboche hardening with a single term and perfect plastic isotropic hardening.

Case 2 demonstrates the combination of Chaboche hardening with multiple terms, plus the exponential saturated isotropic hardening (I-TYPE = "EXPSAT") by superposition.

Case 1: Chaboche hardening with a single term, and perfect plastic isotropic hardening.	
Initial yield stress	Y_0 =300 MPa
Chaboche coefficients for *a*	C_1=30000 MPa
	G_1=60

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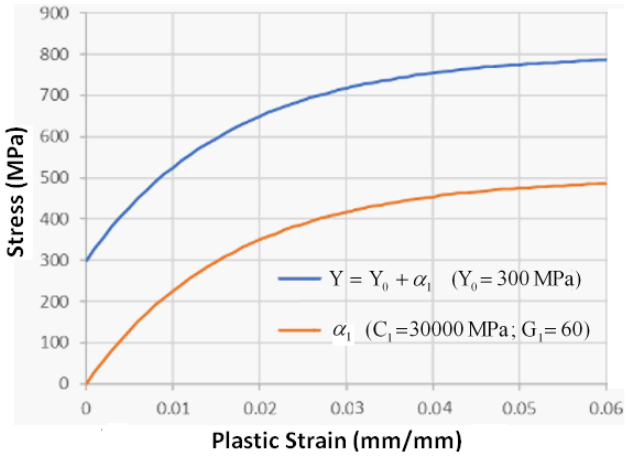
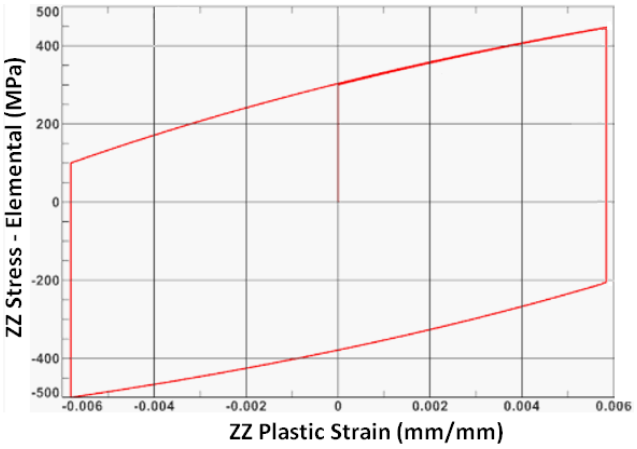
Isothermal uniaxial tensile test	Isothermal Cyclic Test under force load
	

Case 2: Combination of Chaboche hardening with multiple terms and the exponential saturated isotropic hardening (I-TYPE = "EXPSAT") by superposition. Results for both exponential saturated isotropic hardening with Q_j>0 and softening with Q_j< 0 are included.	
Initial yield stress	Y_0 =1.30E+02 MPa
Chaboche coefficients for a_1	C_1=4.14E+05 MPa
	G_1=20000
Chaboche coefficients for a_2	C_2=8.86E+04 MPa
	G_2=800
Chaboche coefficients for a_3	C_3=3.14E+03 MPa
	G_3=9
First exponential saturation function coefficients.	Q_1=-5.00E01 MPa
	B_1=10
Second exponential saturation function coefficients.	Q_2=-2.00E+01 MPa
	B_2=5

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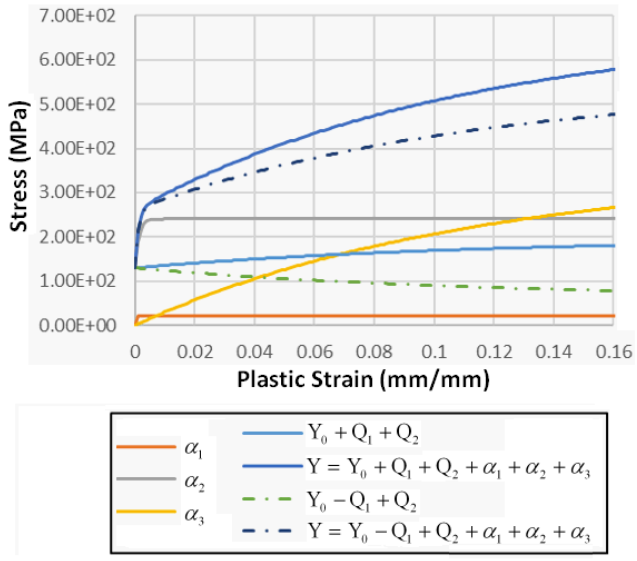
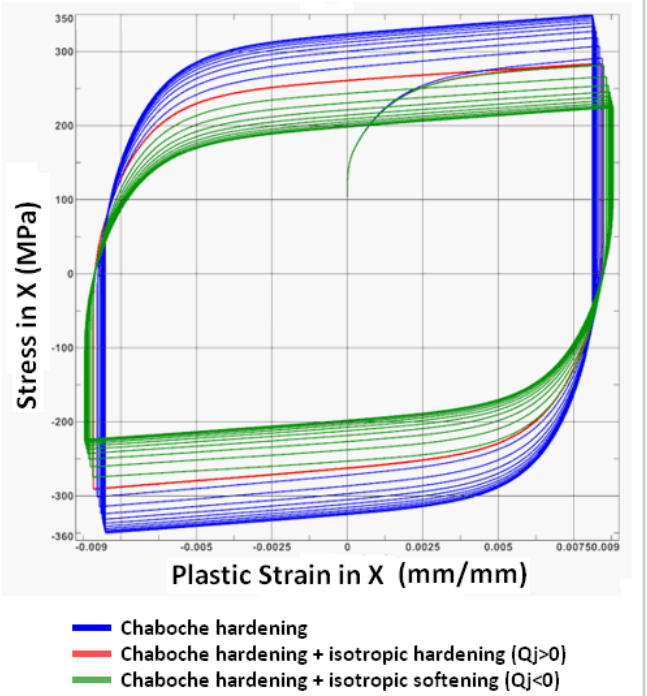
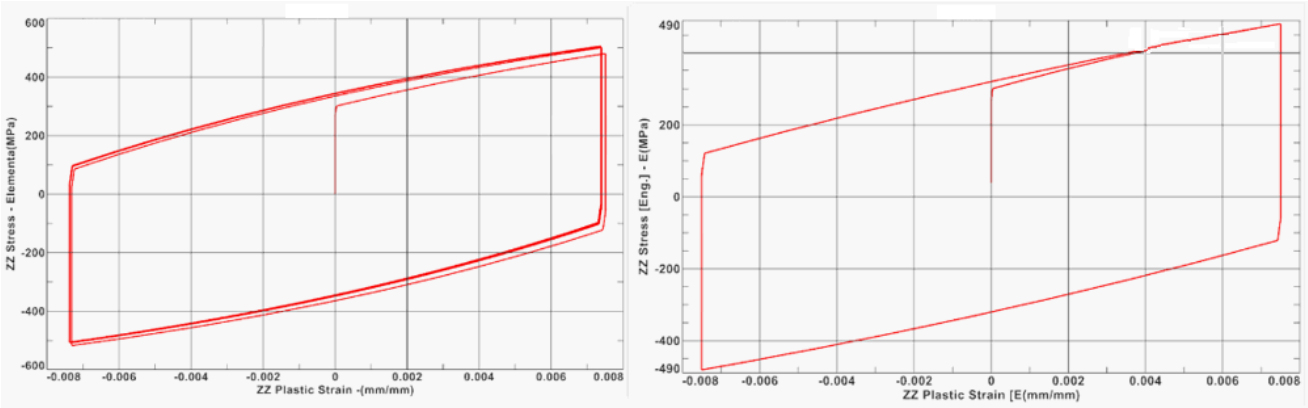
Isothermal uniaxial tensile test	Isothermal Cyclic Test under force load
	

5. The following isothermal, uniaxial, and cyclic loading cases compare Chaboche and Sublayer hardening. The plots on the left use Chaboche hardening with a single term (perfect plastic isotropic hardening). The plots on the right use Sublayer hardening. All plots use the same stress-strain data.

Load case 1: Symmetric enforced displacement cycles.



Load case 2: Symmetric force cycles.

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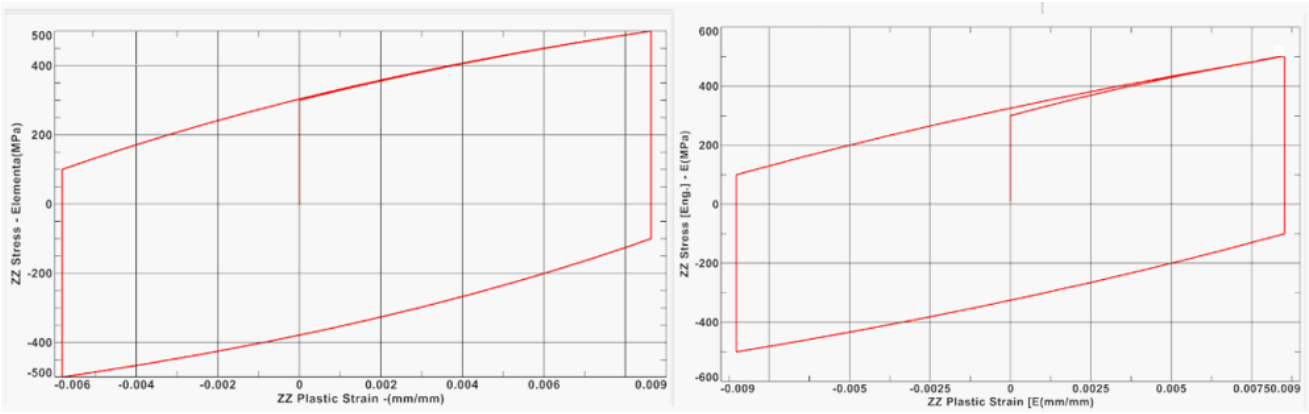
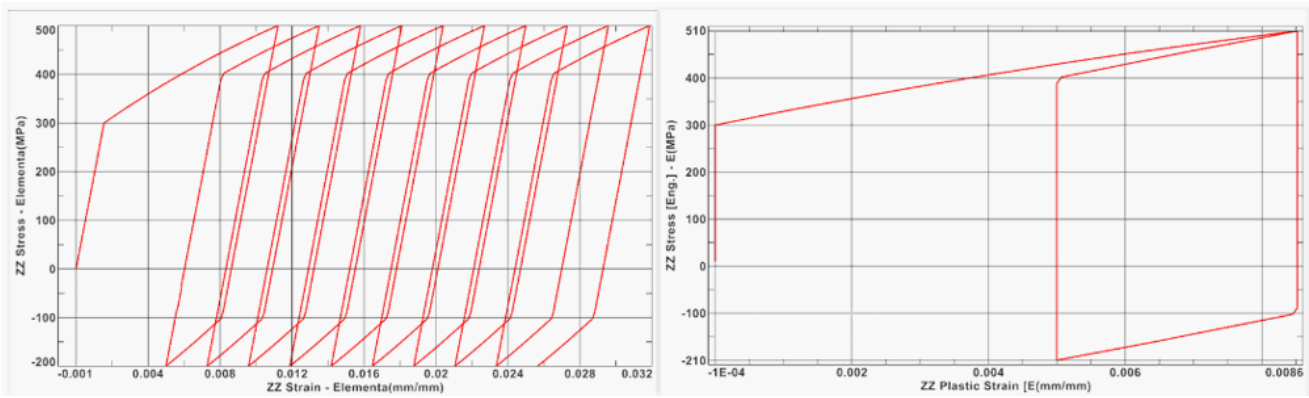


Load case 3: Unsymmetric force cycles.



Note that for the third load case, the Chaboche hardening plot shows strain increments at each cycle (mechanical ratcheting), and the Sublayer hardening plot shows that the strain remains in closed cycles without increment.

6. When TID is defined for SOLs 401 and 402, stress-strain conversion from total to plastic strain and/or from Cauchy to Engineering might be required. This depends on how the parameter LGSTRN is defined, the type of table (TABLES1 or TABLEST), and if the elastic properties on the MATTi entry are a function of temperature. In some of these cases, a reference temperature is required, which is defined with either the TREF field on the MATi entry for SOLs 401 and 402, or with a TEMPD entry referenced by the TEMPERATURE(INITIAL) command for only SOL 402. This is the case when the elastic coefficients are a function of temperature and the hardening curve is not.

7. The following table summarizes all of the possible plastic material definitions with the MPLAS bulk entry. Those with "YES" in the last column can be combined with a creep material defined with the MVPLAS bulk entry.

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Table 15-84: Plasticity model options (MPLAS bulk entry)

Case	YIELD[a]	Isotropic hardening (ISOH)[b]				Kinematic hardening (KINH)[c]		Combine with Creep[d]
			TABULAR					
		EXPSAT	TC	T	C	SUBLAY	CHAB	
1	MISES		X					YES[d]
2		X						NO
3		X					X	YES[d]
4			X				X	YES[d]
5						X		NO
6	HILL		X					YES[d]
7		X						NO
8		X					X	YES[d]
9			X				X	YES[d]
10						X		NO
11	GCIRON			X	X			NO
12	PLASGEN							NO

[a] You select a yield function for plasticity using the "YIELD" row on the MPLAS entry.

- MISES selects the von Mises yield function for isotropic plasticity.
- HILL selects the Hill yield function for anisotropic plasticity.
- GCIRON selects the cast iron yield function.
- PLASGEN selects force and moment limits for plasticity in the PBUSH2 properties (axial bushings) for CBUSH2 elements.

[b] You select a isotropic hardening rule for plasticity in the "ISOH" row on the MPLAS entry:

- EXPSAT selects the exponential saturation isotropic hardening rule. It can be defined alone, or combined with the Chaboche kinematic hardening rule.
- TABULAR selects a isotropic hardening rule defined with stress-strain data. It can be defined alone, or combined with the Chaboche kinematic hardening rule.

[c] You select a kinematic hardening rule for plasticity in the "KINH" row on the MPLAS entry:

- SUBLAY selects the Sublayer-Besseling kinematic hardening rule. It cannot be combined with an isotropic hardening rule.

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- CHAB selects the Chaboche kinematic hardening rule. It *must* be combined with an isotropic hardening rule.

[d] The following table summarizes the input options for the combined plastic and creep material. When you combine a plastic and creep material, the plastic material is limited to the options in the table above, which are summarized in the "MPLAS definition" column below. The creep law choices listed in the "MVPLAS definition" column below have no restrictions. That is, any creep law can be combined with any of the plastic materials listed in the "MPLAS definition" column.

Table 15-85: Combined plasticity and creep model options (MPLAS and MVPLAS entries)

MPLAS definition	MVPLAS definition
(ISOH=TABULAR) + (YIELD=MISES or HILL)	(LAW=1 or 2 or 3 or 301 or USER)[e] + (ANISO=HILL or MPLAS or Blank)[f]
(ISOH=TABULAR) + (KINH=CHAB) + (YIELD=MISES or HILL)	
(ISOH=EXPSAT) + (KINH=CHAB) + (YIELD=MISES or HILL)	

[e] You select the creep law using the "LAW" row on the MVPLAS entry.

[f] You select the effective stress for creep using the "ANISO" row on the MVPLAS entry.

- HILL selects the Hill effective stress for anisotropic creep.

  Note that when both creep and plasticity select HILL independently (YIELD=HILL on MPLAS and ANISO=HILL on MVPLAS), you will define a unique set of Hill ratios for plasticity, and a unique set of Hill ratios for creep. For plasticity, the Hill ratios refer to the Yield point (Y_0). For creep, the Hill ratios are associated to the stress (q^*) of a reference level of strain and strain rate. See the Y-PARi field description on MPLAS and the Rii field description on MVPLAS for details.

- MPLAS selects the same anisotropic effective stress defined on the MPLAS entry for the plastic material. You can only use this option when HILL is defined in the "YIELD" row on the MPLAS entry. This definition ensures consistency for the combined plastic and creep materials.

- Blank indicates that no ANISO row is defined on the MVPLAS entry. For this case, the software uses the von Mises effective stress for creep. When you define the ANISO row, HILL or MPLAS must be defined in that row or a fatal error occurs.

8. Only SOL414,103 Eigenvalues (with a Transient Preload subcase) and SOL 414,129 Transient Response solutions can use the PLASGEN yield criterion. If you define PLASGEN in any other solution, the software writes an error message.

When you define the PLASGEN yield criterion for SOL 414,103 and SOL 414,129 solutions, you cannot define any other yield criterion. If you define any other yield criterion, the software writes an error message.

9. For axial bushings, the ellipsoid criterion for the rupture criterion is given as follows:

$$CRIT = \left(\frac{F_x}{\text{FRUX}}\right)^2 + \left(\frac{F_y}{\text{FRUY}}\right)^2 + \left(\frac{F_z}{\text{FRUZ}}\right)^2 + \left(\frac{M_x}{\text{MRUX}}\right)^2 + \left(\frac{M_y}{\text{MRUY}}\right)^2$$

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where *CRIT* is the threshold at which yield or fracture of the bushing is predicted to initiate, and FRUX, FRUY, FRUZ, MRUX, MRUY, and MRUZ are the strengths of the bushing relative to a single load.

For example, FRUX is the load at which yield or fracture occurs when the bushing is loaded by a force in the X-direction.

If any of the FRUX, FRUY, FRUZ, MRUX, MRUY, or MRUZ strength values are not defined, the software sets the value of the quotient for the missing strength value to zero when it evaluates the ellipsoid criterion.

For a theoretical background of the plasticity criterion, see *Generalized Plasticity Behavior* in the Simcenter Samcef help.

10. After the critical value for the ellipsoid criterion is reached, the static damage, d^s, is set to unity. The true damage is obtained from the following equation:

$$\dot{d} = \frac{1}{\tau_c}\left(1 - e^{-a_c(d^s - d)_+}\right)$$

Related Topics

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MUCRP

Defines the material properties for the user-defined creep subroutines in SOLs 401 and 402

Defines the material properties for the user-defined creep subroutines in SOLs 401 and 402

Format:

1	2	3	4	5	6	7	8	9	10
MUCRP	MID	MODNAME1	MODNAME2	NUMSTAT		SETID	THRESH		
	"INTEGER"	INTEGER1	INTEGER2	INTEGER3	...	...	...	...	
	"REAL"	REAL1	REAL2	REAL3	...	...	...	...	
	"TABLES1"	TIDS1	TIDS2	TIDS3	...	...	...	...	
	"TABLEST"	TIDST1	TIDST2	TIDST3	...	...	...	...	
	"TABLEM1"	TIDM1	TIDM2	TIDM3	...	...	...	...	

Example:

1	2	3	4	5	6	7	8	9	10
MUCRP	1	CREEPMOD		5					+
+	INTEGER	0	1	1	0				+
+	REAL	5.0E-4	0.0	0.0	1.06				+
+	TABLEM1	101	102						+

Fields:

Field	Contents
MID	Identification number of a MAT1, MAT9, or MAT11 entry. (Integer > 0)
MODNAME1	Material model name. (Character, length≤8)
MODNAME2	Material model name. (Character, length≤8)
NUMSTAT	Number of state variables. (Integer, default=0)
SETID	Identification number of a SET1 entry which lists the state variables for Simcenter Nastran to output when the STATVAR case control is defined. SETID=0 requests that all state variables be output. (Integer, default=0)
THRESH	Factor that when multiplied by the elastic modulus yields the stress threshold below which creep does not occur. (0.0 < Real < 1.0E-3; Default = 1.0E-12)
INTEGERi	Material constants that are type INTEGER. (Integer)

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Field	Contents
REALi	Material constants that are type REAL. (Real) Double precision is allowed. for example, 1.2345D6, but Simcenter Nastran will convert them to REAL before storing.
TIDSi	Identification number of a TABLES1 entry. (Integer≥0)
TIDSTi	Identification number of a TABLEST entry. (Integer≥0)
TIDMi	Identification number of a TABLEM1 entry. (Integer≥0)

Remarks:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. Before using user defined creep models, you must link to the shared library, which is built with user defined creep subroutine NUCREEP.
3. MUCRP can be used independently, or it can be referenced by the MVPLAS entry which has "USER" defined in the LAW field. Only the later can be used to define an anisotropic creep.
4. For SOL 402, user creep materials are supported for all element types.

   In SOL 401, the following elements are supported. If you reference an element not listed below for the user creep materials, a fatal message will occur.

   · 3D solids elements CTETRA, CHEXA, CPENTA, CPYRAM.

   · Axisymmetric elements CQUADX4, CQUADX8, CTRAX3, CTRAX6.

   · Plane strain elements CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8.

   · Plane strain elements that reference the generalized plane strain property (PGPLSN).

   · Plane stress elements CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8.

   · Chocking elements CCHOCK3, CCHOCK4, CCHOCK6, CCHOCK8.

   · Shell elements CTRIA6, CQUAD8, CTRIAR, CQUADR.
5. SOL 402 can output a maximum of 64 state variables. If the number of state variables for output is larger than 64:

   • When SETID=0 (default) is used, the software will output the first 64 variables defined on the MUCRP entry, and report a warning.

   • When SETID references a SET1 entry listing more than 64 variables for output, the software will output the first 64 listed on the SET1 entry, and report a warning.
6. If a REAL, INTEGER, TABLES1, TABLEST, or TABLEM1 row is repeated, the data type in field 2 (INTEGER, REAL, TABLES1, TABLEST, TABLEM1) is optional in the consecutive row, provided the previous row is fully defined. In this case, the previous row will have data in fields 3 through 9.
7. For a REAL, INTEGER, TABLES1, TABLEST, or TABLEM1 row, a blank field means the end of the line. All of the data after the blank field is ignored.

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8. For TABLES1, TABLEST, or TABLEM1, 0 means no table is specified; for the table ID larger than 0, a valid table ID must be input, and the table ID must belong to an existing table bulk entry.

9. You can optionally use the MODNAME1 and MODNAME2 character fields in your external subroutine however you like. For example, you may have multiple material models in the same routine, and these fields could be used to branch off to a specific model. Each field must have no more than 8 characters.

   For TABLES1, TABLEST, or TABLEM1, 0 means no table is specified; for the table ID larger than 0, a valid table ID must be input, the table ID must belong to an existing table bulk data entry.

Related Topics

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MUMAT

Defines the material properties for the user defined material subroutine in SOLs 401 and 402.

Defines the material properties for a user defined material subroutine in SOLs 401 and 402.

Format:

1	2	3	4	5	6	7	8	9	10
MUMAT	MID	MODNAME1	MODNAME2	NUMSTAT	MATNAME	SETID			
	REAL	REAL1	REAL2	REAL3	...				
	INTEGER	INTEGER1	INTEGER2	INTEGER3	...				
	TABLES1	TIDS1	TIDS2	TIDS3	...				
	TABLEST	TIDST1	TIDST2	TIDST3	...				
	TABLEM1	TIDM1	TIDM2	TIDM3	...				

Example:

1	2	3	4	5	6	7	8	9	10
MUMAT	21	MODELA	MODELB	5	TESTMAT1	6			+
+	INTEGER	98	187	574	...				+
+	REAL	1.111	2.222	R3	...				+
+	TABLES1	101	102	103	104	105	106	107	+
+	TABLES1	111	112	113	114	115	116	117	+
+		121	122	123	124	125	126	127	+
+	TABLES1	131	132	133	134	135	136	137	+
+	TABLEST	303	202	203	204	205	206	207	+
+		211	212	213	214	215			+
+	TABLEM1	301	TIDST2	TIDST3	...				+
+	TABLES1	148	149						+

Fields:

Field	Contents
MID	Identification number of a MAT1, MAT2, MAT8, MAT9, MAT11 or MATCZ entry. (Integer > 0)
MODNAME1	Material model name. (Character)
MODNAME2	Material model name. (Character)

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Field	Contents
NUMSTAT	Number of state variables. (Integer, default=0)
MATNAME	Material name. (Character)
SETID	Identification number of a SET1 entry which lists the state variables for Simcenter Nastran to output. SETID=0 requests that all state variables be output. (Integer, default=0)
REALi	Material constants that are type REAL. (Real) Double precision is allowed. for example, 1.2345D6, but Simcenter Nastran will convert them to REAL before storing.
INTEGERi	Material constants that are type INTEGER. (Integer)
TIDSi	Identification number of a TABLES1 entry. (Integer>0)
TIDSTi	Identification number of a TABLEST entry. SOL401 only. (Integer>0)
TIDMi	Identification number of a TABLEM1 entry. (Integer>0)

Remarks:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on. An exception for SOL 401 is when MUMAT is used to define Hooke's matrix for the initial elastic stiffness (IOPER=3 in the NXUMAT arguments). For that linear material definition, SOL 401 uses the MUMAT definition regardless of the PARAM, MATNL setting.
2. If a REAL, INTEGER, TABLES1, TABLEST, or TABLEM1 row is repeated, the data type in field 2 (REAL, INTEGER, TABLES1, TABLEST, TABLEM1) is optional in the consecutive row as long as the previous row is fully defined. That is, the previous row has data in fields 3 through 9.
3. You can optionally use the MODNAME1, MODNAME2, and MATNAME character fields however you would like to apply them in your external subroutine. For example, you may have multiple material models in the same routine, and these fields could be used to branch to a specific model.
4. SOL 402 can output a maximum of 64 state variables. If the number of state variables for output is larger than 64:

   • When SETID=0 (default) is used, the software will output the first 64 variables defined on the MUMAT entry, and report a warning.

   • When SETID references a SET1 entry listing more than 64 variables for output, the software will output the first 64 listed on the SET1 entry, and report a warning.

Related Topics

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MVPLAS

Creep (or Visco-Plastic) material properties for SOLs 401 and 402

Defines creep (or visco-plastic) material properties for SOLs 401 and 402.

Format:

1	2	3	4	5	6	7	8	9	10
MVPLAS	MID	Name							
	"LAW"	L-TYPE	L-PAR1	L-PAR2	...				
	"ANISO"	A-TYPE	A-PAR1	A-PAR2	...				

CREEP LAW:

GENERALIZED STRAIN HARDENING LAW

	"LAW"	"1" or "STRNHARD"	THRESH		C1	C2	C3	C4	

GENERALIZED NORTON CREEP LAW

	"LAW"	"2" or "NORTON"	THRESH		C1	C2		C4	

GENERALIZED GAROFALO LAW

	"LAW"	"3" or "GENGAROF"	THRESH		C1	C2	C3	C4	

TIME HARDENING NORTON-BAILEY LAW

	"LAW"	"301" or "TIMEHARD"	THRESH		A	B		D	

USER CREEP LAW (uses the data defined on the MUCRP bulk entry)

	"LAW"	"USER"							

ANISOTROPIC OPTIONS:

Hill using R_{ij} data defined on MVPLAS entry

	"ANISO"	"HILL"	R11	R22	R33	R12	R23	R31	

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Hill using R_{ij} data on MPLAS entry (creep and plasticity combined)

	"ANISO"	"MPLAS"							

Fields:

Field	Contents
MID	Identification number of a MAT1, MAT9, or MAT11 entry. (Integer > 0)
Name	Optional character string. (Character; 56 characters maximum)
"LAW"	The "LAW" field designates the beginning of a creep law data row. (Character)
L-TYPE	Defines the Creep Law type. (Character). "STRNHARD" or "1": Strain hardening law. See Remark 1. $$\dot{\varepsilon}_{cr} = C_1 \sigma^{C_2} \varepsilon_{cr}^{C_3} e^{-C_4/T}$$ "NORTON" or "2": Norton law. See Remark 2. $$\dot{\varepsilon}_{cr} = C_1 \sigma^{C_2} e^{-C_4/T}$$ "GENGAROF" or "3": Generalized Garofalo law. See Remark 3. $$\dot{\varepsilon}_{cr} = C_1 [\sinh(C_2 \sigma)]^{C_3} e^{-C_4/T}$$ "TIMEHARD" or "301": Time hardening Norton-Bailey law. See Remark 4. $$\dot{\varepsilon}_{cr} = AD\sigma^B t^{D-1}$$ "USER": User Defined Law. The data defined on the MUCRP bulk entry is used.
L-PAR1	First creep law parameter For all creep laws except "USER", L-PAR1 defines the parameter THRESH. THRESH: Factor that when multiplied by the elastic modulus, yields the stress threshold below that creep does not occur. (0.0 < Real < 1.0E-3; Default = 1.0E-12) If LAW=USER, you define THRESH on the MUCRP bulk entry with the same ID as the MVPLAS entry. THRESH has the same default (1.0E-12) on both MUCRP and MVPLAS.

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Field	Contents
L-PAR2	Not used.
L-PAR3 to L-PAR6	If the law is a generalized form, L-PAR3, L-PAR4, L-PAR5, and L-PAR6 are C1, C2, C3, and C4, respectively in the specific creep law. If the law is "TIMEHARD", L-PAR3, L-PAR4, and L-PAR6 are A, B, and D, respectively in the time hardening Norton-Bailey law. If real entry, the value is used at all temperatures. If integer entry, it is the identification number of a TABLEM1 bulk entry that defines the coefficient as a function of temperature. The units and input requirements are as follows: ● "STRNHARD" or "1": Units: C1 [1/time]; C2 [unitless]; C3 [unitless]; C4 [absolute temperature]. A fatal error occurs if either C1 or C2 fields are left blank. If C4 is left blank, the temperature term is ignored. ● "NORTON" or "2": Units: C1 [1/time]; C2 [unitless]; C4 [absolute temperature]. A fatal error occurs if either C1 or C2 fields are left blank. If C4 is left blank, the temperature term is ignored. ● "GENGAROF" or "3": Units: C1 [1/time]; C2 [1/stress]; C3 [unitless]; C4 [absolute temperature]. A fatal error occurs if any of the C1, C2, or C3 fields are left blank. If C4 is left blank, the temperature term is ignored. ● "TIMEHARD" or "301": Units: A [1/time]; B [unitless]; D [unitless]. A fatal error occurs if any of the A, B, or D fields are left blank.
"ANISO"	The "ANISO" field designates the beginning of an anisotropic data row. (Character) When no anisotropic data row is defined, the software applies von Mises stress to the creep laws. You can optionally define the anisotropic data row to request for the software to apply the Hill anisotropic stress computation. ● If you are defining anisotropic creep without a combined plasticity definition (MVPLAS and no MPLAS), you should use the "HILL" option described below. ● If you are defining both anisotropic creep and plasticity (MVPLAS and MPLAS), and you want to apply *different* Hill coefficients for creep and for plasticity, you should use the "HILL" option.

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Field	Contents
	● If you are defining both anisotropic creep and plasticity (MVPLAS and MPLAS), and you want to apply the *same* Hill coefficients for both creep and plasticity, you should use the "MPLAS" option.
MPLAS	Requests anisotropic creep when both creep and plasticity use the same Hill stress ratios. You define the Hill stress ratios on the MPLAS bulk entry (not on MVPLAS). The ANISO="MPLAS" definition ensures consistency for the combined plasticity and creep with no differences in the yield and creep potentials.
HILL	Requests anisotropic creep. You define the Hill stress ratios on MVPLAS. This option refers to the same Hill potential defined with plasticity, but instead of referring to the Yield stress (Y), when using the Hill option with creep alone, it refers to the stress (q): $$f(\sigma_{ij}, q) = [F(\sigma_{22} - \sigma_{33})^2 + G(\sigma_{33} - \sigma_{11})^2 + H(\sigma_{11} - \sigma_{22})^2 + 2L{\sigma_{23}}^2 + 2M{\sigma_{31}}^2 + 2N{\sigma_{12}}^2] - q^2 = 0$$ Where F, G, H, L, M and N are functions of the Hill Stress Ratios. $$F = \frac{1}{2}\left(\frac{1}{{R_{22}}^2} + \frac{1}{{R_{33}}^2} - \frac{1}{{R_{11}}^2}\right) \quad G = \frac{1}{2}\left(\frac{1}{{R_{33}}^2} + \frac{1}{{R_{11}}^2} - \frac{1}{{R_{22}}^2}\right)$$ $$H = \frac{1}{2}\left(\frac{1}{{R_{22}}^2} + \frac{1}{{R_{11}}^2} - \frac{1}{{R_{33}}^2}\right)$$ $$L = \frac{3}{2}\left(\frac{1}{{R_{23}}^2}\right) \quad M = \frac{3}{2}\left(\frac{1}{{R_{13}}^2}\right) \quad N = \frac{3}{2}\left(\frac{1}{{R_{12}}^2}\right)$$

If ANISO=HILL, you will define the Hill stress ratios R11, R22, R33, R12, R23, R31 on MVPLAS.

R11, R22, R33, R12, R23, R31	If the "HILL" option is used, defines the six stress ratios. (Real if constant, or an Integer that references the ID of a TABLEM1 defining temperature dependence). Experimental input data: In pure anisotropic creep, the Hill stress ratios are not defined exactly as those for plasticity: ● For plasticity, the Hill ratios refer to the Yield point (Y_0).

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Field	Contents

$$\{\sigma = Y_0;\ \ \varepsilon_{pl} = 0.0\}$$

● For creep, the reference point is associated to the stress (q^*) of a reference level of strain and strain rate.

$$\left\{\sigma = q^*;\ \ \varepsilon_{cr} = \varepsilon_{cr}{}^*;\ \ \dot{\varepsilon}_{cr} = \dot{\varepsilon}_{cr}{}^*\right\}$$

In terms of the creep potential function, it turns out that the stress ratios refer to the equivalent stress q_{Hill}.

$$q_{Hill} = \left[F(\sigma_{22} - \sigma_{33})^2 + G(\sigma_{33} - \sigma_{11})^2 + H(\sigma_{11} - \sigma_{22})^2 + 2L\sigma_{23}{}^2 + 2M\sigma_{13}{}^2 + 2N\sigma_{12}{}^2\right]^{1/2}$$

where F, G, H, L, M, N are the previously described coefficients, and the stress ratios are:

When i=j

$$R_{ii} = \frac{\sigma_{ii}}{q_{Hill}}$$

and when i≠j,

$$R_{ij} = \sqrt{3}\frac{\sigma_{ij}}{q_{Hill}}$$

Remarks:

1. You must define the parameter setting PARAM,MATNL,1 to globally switch material nonlinear on.
2. The strain hardening creep law uses the coefficients C1, C2, C3, and C4 as follows:

   $$\dot{\varepsilon}_{cr} = C_1\sigma^{C_2}\varepsilon_{cr}^{C_3}e^{-C_4/T}$$

   where ϵ_{cr} is the effective creep strain, σ is the effective stress, and T is the temperature.

   A fatal error occurs if any of the C1, C2, C3 fields are left blank. If C4 is left blank, the temperature term is ignored.
3. The Norton creep law uses the coefficients C1, C2, and C4 as follows:

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$$\dot{\varepsilon}_{cr} = C_1 \sigma^{C_2} e^{-C_4/T}$$

A fatal error occurs if either of C1 or C2 fields are left blank. If C4 is left blank, the temperature term is ignored.

4. The Garofalo creep law uses the coefficients C1, C2, C3, and C4 as follows:

$$\dot{\varepsilon}_{cr} = C_1 \left[\sinh\left(C_2 \sigma\right)\right]^{C_3} e^{-C_4/T}$$

A fatal error occurs if any of the C1, C2, or C3 fields are left blank.

If C4 is left blank, the temperature term is ignored.

5. The time-hardening Norton-Bailey creep law uses the coefficients A, B, and D as follows:

$$\dot{\varepsilon}_{cr} = AD\sigma^B t^{D-1}$$

A fatal error occurs if any of the A, B, or D fields are left blank.

Note that because time is explicitly defined, this law is limited and only accurate if the stress and the temperature are constant.

The Norton-Bailey law can be also defined in the strain hardening law. See Remark 5.

6. The Norton-Bailey strain-hardening law, where time is an effective time, is defined as:

$$\dot{\varepsilon}_{cr} = D\left[A\sigma^B \varepsilon_{cr}^{D-1}\right]^{1/D}$$

Practically, this is a particular case of the "STRNHARD" law, which including the additional temperature factor, gives the generalized form:

$$\dot{\varepsilon}_{cr} = C_1 \sigma^{C_2} \varepsilon_{cr}^{C_3} e^{-C_4/T}$$

The equivalence between strain hardening Norton-Bailey and generalized strain hardening is given by the following relations:

$$C_1 = DA^{1/D}; \quad C_2 = B/D; \quad C_3 = (D-1)/D; \quad C_4 = 0$$

7. In the creep equations, T is always expressed in Kelvin temperature unit. As a result, PARAM,TABS is used to make the conversion from whatever unit to Kelvin if necessary. TABS is added to the temperature value, and to the coefficient C4. When TABS > 0.0, the exponent in the creep equations has the form:

$$e^{-(C4+TABS)/(T+TABS)}$$

8. The following table summarizes all of the possible plastic material definitions with the MPLAS entry. Those with "YES" in the last column can be combined with a creep material defined with the MVPLAS bulk entry.

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Table 15-86: Plasticity model options (MPLAS entry)

Case	YIELD[a]	Isotropic hardening (ISOH)[b]				Kinematic hardening (KINH)[c]		Combine with Creep[d]
			TABULAR					
		EXPSAT	TC	T	C	SUBLAY	CHAB	
1	MISES		X					YES[d]
2		X						NO
3		X					X	YES[d]
4			X				X	YES[d]
5						X		NO
6	HILL		X					YES[d]
7		X						NO
8		X					X	YES[d]
9			X				X	YES[d]
10						X		NO
11	GCIRON			X	X			NO

[a] You select a yield function for plasticity using the "YIELD" row on the MPLAS entry.

- MISES selects the von Mises yield function for isotropic plasticity.
- HILL selects the Hill yield function for anisotropic plasticity.
- GCIRON selects the cast iron yield function.

[b] You select a isotropic hardening rule for plasticity in the "ISOH" row on the MPLAS entry:

- EXPSAT selects the exponential saturation isotropic hardening rule. It can be defined alone, or combined with the Chaboche kinematic hardening rule.
- TABULAR selects a isotropic hardening rule defined with stress-strain data. It can be defined alone, or combined with the Chaboche kinematic hardening rule.

[c] You select a kinematic hardening rule for plasticity in the "KINH" row on the MPLAS entry:

- SUBLAY selects the Sublayer-Besseling kinematic hardening rule. It cannot be combined with an isotropic hardening rule.
- CHAB selects the Chaboche kinematic hardening rule. It *must* be combined with an isotropic hardening rule.

[d] The following table summarizes the input options for the combined plastic and creep material. When you combine a plastic and creep material, the plastic material is limited to the options in the table above, which are summarized in the "MPLAS definition" column below. The creep law choices

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listed in the "MVPLAS definition" column below have no restrictions. That is, any creep law can be combined with any of the plastic materials listed in the "MPLAS definition" column.

Table 15-87: Combined plasticity and creep model options (MPLAS and MVPLAS entries)

MPLAS definition	MVPLAS definition
(ISOH=TABULAR) + (YIELD=MISES or HILL)	(LAW=1 or 2 or 3 or 301 or USER)[e] + (ANISO=HILL or MPLAS or Blank)[f]
(ISOH=TABULAR) + (KINH=CHAB) + (YIELD=MISES or HILL)	
(ISOH=EXPSAT) + (KINH=CHAB) + (YIELD=MISES or HILL)	

[e] You select the creep law using the "LAW" row on the MVPLAS entry.

[f] You select the effective stress for creep using the "ANISO" row on the MVPLAS entry.

- HILL selects the Hill effective stress for anisotropic creep.

  Note that when both creep and plasticity select HILL independently (YIELD=HILL on MPLAS and ANISO=HILL on MVPLAS), you will define a unique set of Hill ratios for plasticity, and a unique set of Hill ratios for creep. For plasticity, the Hill ratios refer to the Yield point (Y_0). For creep, the Hill ratios are associated to the stress (q^*) of a reference level of strain and strain rate. See the Y-PARi field description on MPLAS and the Rii field description on MVPLAS for details.

- MPLAS selects the same anisotropic effective stress defined on the MPLAS entry for the plastic material. You can only use this option when HILL is defined in the "YIELD" row on the MPLAS entry. This definition ensures consistency for the combined plastic and creep materials.

- Blank indicates that no ANISO row is defined on the MVPLAS entry. For this case, the software uses the von Mises effective stress for creep. When you define the ANISO row, HILL or MPLAS must be defined in that row or a fatal error occurs.

Related Topics

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NLARCL

SOL 401 Nonlinear Buckling Control

Defines control parameters for SOL 401 nonlinear buckling.

Format:

1	2	3	4	5	6	7	8	9	10
NLARCL	ID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4	PARAM5	VALUE5	-etc-				

Example:

NLARCL	1	TYPE	CRIS	MINALR	0.25	MAXALR	4.0		
	SCALE	0.0							

Fields:

Field	Contents
SID	Identification number. (Integer > 0)
PARAMi	Name of the NLARCL parameter. Allowable names are given in the parameter listing below. (Character)
VALUEi	Value of the parameter. (Real, Integer, or Character)

NLARCL Parameters:

Table 15-88:

Name	Description
TYPE	= RIKS selects the Riks arc-length method = MRIKS selects the modified Riks arc-length method (Default) = CRIS selects the Crisfield arc-length method
MINALR	Minimum allowable arc-length adjustment ratio between increments for the adaptive arc-length method. (0.0<Real≤1.0; Default=0.25)
MAXALR	Maximum allowable arc-length adjustment ratio between increments for the adaptive arc-length method. (Real≥1.0, Default=4.0)

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Name	Description
MAXR	Defines the overall upper and lower bounds on the load increment /arc-length in the subcase. (Real>0.0; Default = 20.0)
SCALE	Scale factor for controlling loading contribution in the arc-length constraint. (Real>0.0; Default = 0.0)
DESITER	Desired number of iterations for convergence to be used for the adaptive arc-length adjustment. (Integer>0, Default=12)
MXINC	Maximum number of controlled load increments done in the arc-length subcase (Integer>0; Default=20)
LDFACIN	Initial load factor. This load factor will be used to compute initial arc-length (Real>0, Default=1.0).
NOUTAL	Skip factor for output of the incremental results. Output always occurs at the final increment. For example, if you define NOUTAL=2, output occurs at every other converged solution increment and for the final increment. If you define NOUTAL=0, output only occurs at the final increment. (Integer≥0; Default=1)
MXLDFAC	Maximum value of load-factor at which solution will be terminated. (Real, Default = 1.0)

Remarks:

1. To request the nonlinear buckling analysis in SOL 401, your statics subcase should include the standard ANALYSIS=STATICS command and the NLARCL command. The ID on the NLARCL command selects the NLARCL bulk entry which defines the nonlinear buckling parameters.
2. The NLARCL case control command in a statics subcase is the trigger which the software uses to start the nonlinear buckling analysis. The referenced NLARCL bulk entry is also required, even when the default values are used.
3. The nonlinear buckling statics subcase must be either the first subcase, or the last in a sequence of static subcases. A nonlinear buckling statics subcase can only be followed by a modal subcase.
   a. If the nonlinear buckling statics subcase is the first subcase, all of the loads defined in the current subcase are incrementally applied by the software during the arc-length solution.
   b. If the nonlinear buckling statics subcase is the last subcase and it is sequentially dependent, the loads applied in the previous subcase are held constant in the current subcase. The difference between the load defined in the nonlinear buckling statics subcase and the load from the previous subcase is computed. This load difference is incrementally applied by the software during the arc-length solution.

Related Topics

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NLCNTL

Strategy Parameters for SOL 401

Defines solution control parameters for SOL 401.

Format:

1	2	3	4	5	6	7	8	9	10
NLCNTL	ID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4	PARAM5	VALUE5	-etc-				

Example:

NLCNTL	1	MAXITER	30	CONV	UPW	MAXDIV	2		
	SOLVER	ELEMITER							

Fields:

Field	Contents
SID	Identification number. (Integer > 0)
PARAMi	Name of the NLCNTL parameter. Allowable names are given in the parameter listing below. (Character)
VALUEi	Value of the parameter. (Real, Integer, or Character)

NLCNTL Parameters:

Table 15-89: General Convergence and Iteration Control Parameters

Name	Description
See Error functions and weighted normalization in the *Multi-Step Nonlinear User's Guide* for details regarding the convergence error functions.	
CONV	Specifies the convergence criteria. See Remarks 8 and 9. (Character = "U", "P", "W", or any combination; See Remark 8 for default behavior).
EPSP	Error tolerance on force. (Real > 0.0; Default = 1.0E-2)
EPSU	Error tolerance on displacement. (Real > 0.0; Default = 1.0E-2)
EPSW	Error tolerance on work. (Real > 0.0; Default = 1.0E-6)
FPPUPDT	Controls when the software updates the fluid pressure load defined with the PLOADFP bulk entry. (Integer; Default = 1)

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Name	Description
	0: The fluid pressure application is only updated at the beginning of every time step using the final contact status from the previous time step. When FPPUPDT=0, for the first time step in a solution when there is no previous contact status, the fluid pressure is not applied to any elements including elements that are not in a contact region. With this option, you will need to include enough time steps for the coupled contact and fluid pressure to become established and achieve a steady state. 1: The fluid pressure is updated for every solution iteration that occurs in a single time step. This choice will keep the fluid pressure current with the changing contact status. Defining FPPUPDT=1 is more computational, and it can cause contact convergence issues if the fluid pressure and the contact conditions are strongly coupled.
MAXBIS	Maximum number of bisections allowed. (0 < Integer ≤ 10; Default = 5)
MAXDIV	Number of divergences before solution is assumed to diverge. (Integer > 0; Default = 3)
MAXITER	Maximum number of iterations per time step. (Integer > 0; Default = 25)
MAXQN	Maximum number of quasi-Newton correction vectors to be saved. (Integer ≥ 0; Default = 10)
NORMP	Defines the norm criteria for the force convergence error function. (Character = L1, L2, or LINF to select a norm criteria of 1, 2, or infinity, respectively; Default = "L2")
NORMU	Defines the norm criteria for the displacement convergence error function. (Character = L1, L2,or LINF to select a norm criteria of 1, 2, or infinity, respectively; Default = "L2")
REFP	Optionally replaces the denominator in the force convergence error function. (Real > 0.0)
REFU	Optionally replaces the denominator in the displacement convergence error function. (Real > 0.0)
RTOLB	When your model includes beams, bars, or shell elements that have rotational degrees-of-freedom, RTOLB limits the incremental rotation ($\Delta\theta x$, $\Delta\theta y$, $\Delta\theta z$) in a single iteration. (Real > 0.0; Default=20.0 degrees) The software bisects a time increment if either of the following criteria are satisfied: **1**. If $\|\Delta\theta\|$ at any rotational DOF exceeds RTOLB. **2**. If the average value of all $\|\Delta\theta\|$ in the model exceeds RTOLB/2. When the software limits the rotations in an increment, it helps convergence by limiting the model orientation from becoming inconsistent with the current stiffness matrix. You can limit the maximum number of bisections in an iteration with the MAXBIS parameter. By default, MAXBIS=5.

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Table 15-90: Bolt Preload Parameters

Name	Description
EPSBOLT	Convergence tolerance for a nonzero bolt preload. Supported for ETYPE=1 bolts. See Remark 10. (Real > 0.0; Default = 1.0E-3)
ITRBOLT	Maximum number of bolt iterations before the bolt preload calculation is considered non-converged. Supported for ETYPE=1 bolts. See Remark 10. (Integer > 0; Default = 20)
MISFBLT	Limits the bolt strain from one preload increment to the next. Supported for ETYPE=1 bolts. (Real > 1.0E-6; No default. The software does not apply MISFBLT when it is undefined.) MISFBLT is useful, for example, if you model a bolted joint with a gap condition. The bolt preload algorithm increments the bolt strain as the joint compresses. When a gap is present, before the joint is compressed, the axial bolt force will be relatively low. As a result, the preload algorithm will increase the strain increments. Once the gap is closed and the joint begins to compress, the bolt and contact forces will change quickly, possibly causing the contact conditions to destabilize. By limiting the bolt preload strain increments, you can reduce the chance of destabilization, and ultimately help achieve convergence. At the start of a bolt preload solution, the software uses your requested preload force and the bolt geometry to estimate a bolt strain. The estimated strain (ϵ_{es}) is computed with the assumption that everything is rigid except for the bolt. The software then computes the maximum allowable strain increment using the MISFBLT parameter. $\Delta\epsilon_{max} = \epsilon_{es}$ * MISFBLT At each preload increment, the software compares $\Delta\epsilon_{max}$ with the next computed preload increment $\Delta\epsilon^{i+1}$. The software uses the smaller of the two at each preload increment.
MSGLVLB	Diagnostic output for bolt preload. (Integer = 0, 1, or 2; Default = 1) 0: Bolt summary messages upon convergence. 1: Bolt summary messages at every bolt preload iteration. 2: Bolt summary messages and load/tolerance messages per bolt at every bolt preload iteration. MSGLVLB=1 and 2 requires MSGLVL=1. If MSGLVL= 0, MSGLVLB is also forced to 0, even if MSGLVLB=1 or 2 are defined. MSGLVLB 1 and 2 also output a table of bolt data showing the forces, moments and strains per bolt. This is only after the bolts converge.
ZERBOLT	Convergence tolerance for a zero bolt preload. Supported for ETYPE=1 bolts. See Remark 10. (Real>0.0; Default=1.0E-7)

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Table 15-91: General Solution Parameters

Name	Description
INACCN	When an ANALYSIS=DYNAMICS subcase is the first subcase of the solution, or it follows a statics or preload subcase, INACCN controls whether an initial acceleration is computed. (Character = "YES" or "NO"; Default = "NO") YES: Initial acceleration is computed. NO: Initial acceleration is not computed. The acceleration is computed by: $$a_0 = \frac{\Delta F - Ku_0 - Cv_0}{M}$$ where, ΔF is the load increment at the start of the subcase. u_0 is the displacement defined as an initial condition. v_0 is the velocity defined as an initial condition. If the initial acceleration is not computed, it is assumed to be zero.
INERTIA	Optionally turns off the inertial effects for a dynamic subcase. (Character ="ON" or "OFF"; Default ="ON") ON: The inertia force term is turned on. The general equation of motion for this analysis is: $$M\ddot{u} + B\dot{u} + Ku = P(t)$$ OFF: Turns off the inertia force term $M\,\ddot{u}(t)$. The general equation of motion for this analysis is: $$B\dot{u} + Ku = P(t)$$ When INERTIA=OFF, body loads defined with the bulk entries such as RFORCE or GRAV will continue to use the mass to compute the loads.

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Name	Description
LOADOFF	Turns off categories of mechanical loads. (Character = “NONE”, “BODY”, “BOUNDRY”, or “BOTH”; Default = “NONE”) NONE: No loads are turned off. BODY: Loads defined with the GRAV, RFORCE, RFORCE1, ACCEL, and ACCEL1 are turned off. BOUNDRY: Loads defined with the PLOAD, PLOAD1, PLOAD2, PLOAD4, PLOADE1, PLOADX1, FORCE, FORCE1, FORCE2, MOMENT, MOMENT1, MOMENT2, and SLOAD are turned off. BOTH: Both the BODY and BOUNDRY loads listed above are turned off.
LSTCONV	Controls the output of results at the last converged step when you did not request output for that time step. (Character = “ON” or “OFF”; Default = “ON”) ON: The software always outputs results at the converged last step, even when you did not request output for that step. OFF: The software only outputs results at the converged last step when you requested output for that step on the TSTEP1 bulk entry. For either LSTCONV=ON or LSTCONV=OFF, the software always outputs results at the time steps that converged and for which you requested output on the TSTEP1 entry.
LVAR	Specifies whether time unassigned loads are ramped or stepped. (Character = "RAMP" or "STEP"; Default = "RAMP")
MSGLVL	Diagnostic level. (Integer = 0, 1, or 2; Default = 1) 0: No additional diagnostic output 1: Convergence information is output for each iteration 2: Convergence information is output for each iteration plus the location and value of the maximum translational displacement. In addition, the solution accuracy indicators table is output. See Remark 16.
NRRFOUT	Controls the number of residual force vectors written to universal files. (Integer ≥ -2; Default = 0) You can open universal files in Simcenter 3D Pre/Post using the companion results import and display contour plots of the results. Contour displays of the residual vector components or magnitude can help you determine locations where the residual force is high. For example, you may find the largest residual occurs in a contact region. If the solution continues to iterate and not converge, you might consider ending the solution,

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Name	Description
	temporarily softening the contact stiffness in that location to stabilize the problem, and then restarting the solution. 0: No output occurs. N (where N>0): The software writes a *.unv file after each iteration of the current time step. It keeps the *.unv files corresponding to the last N iterations of the current time step by removing the oldest file when a new iteration and new *.unv file begins. This continues until the current time step converges, in which case the software removes all *.unv files associated with that time step. This process repeats for each time step. -1: The software writes a*.unv file after each iteration of the current time step. It keeps the *.unv files corresponding to all iterations of the current time step until the time step converges, in which case it removes the files associated with that time step. This process repeats for each time step. -2: The software writes a *.unv file after each iteration of each time step. It does not remove the *.unv files after a time step converges. At the end of the solution, the *.unv files from all iterations and time steps remain.
RFVAR	Specifies how the software interpolates the time variation for the rotational loads RFORCE or RFORCE1. (Character = "OMEGA" or "FORCE"; Default ="OMEGA") See SOL 401 - Interpolating rotational inertia loads in the *Multi-Step Nonlinear User's Guide*. OMEGA: The software interpolates the time variation using $f(t)$. As a result, if a linear time variation is defined, the angular velocity will vary linearly. Since $\vec{\omega}$ is squared in the force computation, the force will not vary linearly. FORCE: The software interpolates the time variation using $f(t)^2$. The software then takes the square root of the interpolated value before applying it to the angular velocity. As a result, if a linear time variation is defined, the force will vary linearly, but $\vec{\omega}$ will not.
SOLVER	Specifies the solver for the statics, preload, and transient dynamics subcases when matrices are symmetric. See Remarks 11 and 12. (Character = "MUMPS", "SPARSE", "PARDISO", or "ELEMITER"; Default = "MUMPS")
THRMST	Includes thermal strain in a static analysis (Character = "YES" or "NO"; Default = "YES")
TVAR	Specifies whether time unassigned temperature loads are ramped or stepped. (Character = "RAMP" or "STEP"; Default = "RAMP")

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Table 15-92: Stiffness Control Parameters

Name	Description
FOLLOWK	Includes follower stiffness. Follower stiffness is computed from the follower loads defined with the FORCE1, FORCE2, PLOAD, or PLOAD4 entries. (Character = “YES” or “NO”; Statics default = “NO”; Modal default = “YES”)
KUPDATE	Stiffness update strategy. (Integer; Default = 0) -1: Initial stiffness approach. Never update the stiffness. 0: Auto stiffness update. See Remarks **18** and **19**. 1: Full Newton-Raphson. Update the stiffness at each iteration. >1: Quasi Newton-Raphson, and KUPDATE is the number of iterations before a stiffness update in the same time step. -3: Update the stiffness only at iteration 1 of each new time step. See Remark **20**.
MSFAC	Matrix stabilization factor. (Real; Default = 1.0E-10; Unitless)
MSTAB	Indicates whether the stiffness matrix stabilization feature is used. See Remark **17**. (Integer; Default = 0) 0 – Matrix stabilization is not used 1 – Matrix stabilization is used
SPINK	Include spin softening. See Remark **13**. (Character = “YES” or “NO”; Statics default = “NO”; Modal default = “YES”)
STFOPTN	Material stiffness matrix option. (Integer ≥ 0; Default = 3 with small strains (PARAM,LGSTRN,0); Default = 2 with large strain (PARAM,LGSTRN,1)) 1: The elastic stiffness matrix is used. 2: The tangent stiffness matrix is used. 3: The elastic stiffness matrix is used to start each subcase, the elastic stiffness matrix is used if a stiffness update is requested prior to the beginning of a new time step, and the tangent stiffness matrix is used at any intermediate stiffness update.
STKFRST	Controls whether the software includes stress stiffening at the first iteration of a time step, or waits until the second iteration. (Character = "YES" or "NO"; default = "YES") YES: Stress stiffening is included at the first iteration of a time step.

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Name	Description
	NO: Stress stiffening is not included at the first iteration of a time step. Instead, the software waits until the second iteration to update the stiffness as a result of stress stiffening. Note: Stress stiffening is only applied when either large displacements (PARAM,LGDISP,1) or large strain (PARAM,LGSTRN,1) is requested, and stress stiffening is requested (STRESSK=YES).
STRESSK	Include stress stiffening. (Character = "YES" or "NO"; Statics default = "YES"; Modal default = "YES")
TSTEPK	Stiffness is updated at the beginning of the time step. Applicable only if KUPDATE>1. (Character = "YES" or "NO"; Default = "NO")

Table 15-93: Contact Parameters

Name	Description
CNTMDIV	Number of permissible contact divergences before bisection is initiated. (Integer > 0; Default = 9) Friction defined with the FRICi field on the BCTSET bulk entry along with sliding contact conditions can result in unsymmetric stiffness terms in the global stiffness matrix. Including the unsymmetric stiffness terms will slow the solution performance, although the unsymmetric terms are sometimes necessary to achieve solution convergence. You can use the FSYMTOL, KSYM and KSYMTOL parameters to control if the unsymmetric terms are included.
FSYMTOL	Contact friction coefficient threshold to control if the sliding contact stiffness includes the unsymmetric stiffness term. (Real ≥ 0.0; Default = 0.2) When the maximum FRICi for all contact pairs is less than or equal to FSYMTOL, the unsymmetric stiffness term is not included, and a symmetric stiffness is generated. When the maximum FRICi for all contact pairs is greater than FSYMTOL, the unsymmetric stiffness term is included, and an unsymmetric stiffness is generated.
KMODSCL	When sequencing from a nonlinear subcase (PRELOAD, STATIC, DYNAMICS) that includes contact to a modal subcase (MODES, CYCMODES, FOURIER), the parameter KMODSCL scales the software computed normal contact stiffness for the modal subcase. You can use the KMODSCL parameter with any of the KMODTN parameter options described below. (Real > 0.0; Default = 1.0)
KMODTN	When sequencing from a nonlinear subcase (PRELOAD, STATIC, DYNAMICS) that includes contact to a modal subcase (MODES, CYCMODES, FOURIER), the parameter KMODTN allows you to select a different tangential contact stiffness for the modal subcase than was used in the previous nonlinear subcase. It can be used, for example, when the nonlinear subcase required frictionless contact, yet the modal subcase needs a tangential

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Name	Description

contact stiffness to prevent spurious rigid body modes. KMODTN is only used in the modal subcases. (Integer; Default = 1)

The KMODTN settings allow you to independently control the tangential contact stiffness of the contact pairs with friction (**A**), and without friction (**B**).

A: Contact pairs with friction are defined by INTRFC=1, 2*, or 5 and FRICMOD=0 or 1 on the BCTPARM bulk entry with FRICi>0.0 on the BCTSET bulk entry.

*When INTRFC=2, KMODTN=1, 2, 3, or 4, and FRICi>0.0 on the BCTSET bulk entry, the software applies the stiffness described in 1A. When INTRFC=2, KMODTN=5, and FRICi>0.0 on the BCTSET bulk entry, the software applies the stiffness described in 5.

B: Contact pairs without friction are defined either by INTRFC=4 on BCTPARM, or by INTRFC=1, 2, or 5 and FRICMOD=0 or 1 on the BCTPARM bulk entry with FRICi=0.0 on the BCTSET bulk entry.

KMODTN Setting	Friction status	Tangential contact stiffness in modal subcase
1	**A**	The software applies a uniform tangential contact stiffness of (0.1 * normal stiffness) in a modal subcase.
	B	The software does not apply tangential contact stiffness in a modal subcase.
2	**A**	The software applies the actual final tangential contact stiffness from the end of a nonlinear subcase in a modal subcase. The applied tangential contact stiffness can be a mixture of sticking and sliding stiffness values.
	B	The software does not apply tangential contact stiffness in a modal subcase.
3	**A**	The software applies the actual final tangential contact stiffness from the end of a nonlinear subcase in a modal subcase. The applied tangential contact stiffness can be a mixture of sticking and sliding stiffness values.
	B	The software applies a uniform tangential contact stiffness of (0.1 * normal stiffness) in a modal subcase.
4	**A**	The software applies a uniform tangential contact stiffness of (0.1 * normal stiffness) in a modal subcase.
	B	
5	**A**	The software replaces the final stiffness of any active contact element at the end of the nonlinear subcase with a no-separation, tied-contact stiffness in both the normal and
	B	

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Name	Description

KMODTN Setting	Friction status	Tangential contact stiffness in modal subcase
		tangential directions, regardless of the friction value. The tied-contact stiffness applied in the modal subcase is similar in magnitude to the normal and tangential stiffness resulting from a glue condition.

KSYM — When an unsymmetric material stiffness is generated as described in the FSYMTOL description, KSYM controls if the unsymmetric matrix is symmetrized. (Character = “AUTO”, “SYM”; or “UNSYM”; Default = “AUTO”)

SYM: The unsymmetric matrix is always symmetrized.

UNSYM: The unsymmetric matrix is never symmetrized.

AUTO: The unsymmetric matrix is symmetrized if the KSYMTOL tolerance is satisfied as described below.

KSYMTOL — Tolerance for symmetrizing the unsymmetric global stiffness when KSYM=AUTO. (Real ≥ 0.0; Default = 0.001)

If K_{full} is the unsymmetric global stiffness, the symmetric part K_{sp} is:

$$K_{sp} = \frac{1}{2}\left(K_{\text{full}} + K_{\text{full}}^{\text{T}}\right)$$

The unsymmetric part K_{up} is:

$$K_{up} = K_{\text{full}} - K_{sp}$$

The ratio of the unsymmetric part to the full matrix is:

$$r = \frac{||K_{up}||}{||K_{\text{full}}||}$$

The ratio r is a measure of the matrix asymmetry. It is compared with the KSYMTOL.

If r < KSYMTOL, the unsymmetric matrix is symmetrized.

If r > KSYMTOL, the unsymmetric matrix is used.

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Name	Description
MSGLVLC	Diagnostic output for contact. (Integer = 0, 1 or 2; Default = 1) 0: No additional diagnostic output. 1: Convergence information is printed for each iteration. Changes in contact status are also printed. Before the first iteration, a summary of the number of sliding/sticking/active/inactive elements is printed. The number of augmentations and number of pair updates is also printed. 2: Everything described in option 1 and after each iteration, detailed information for every contact pair is printed.
USOLVER	Selects the unsymmetric solver. (Character = “MUMPS”, “SPARSE”, or “PARDISO”; Default = “MUMPS”) Specifies the solver for the statics, preload, and transient dynamics subcases when matrices are unsymmetric. See Remark 11.

Table 15-94: Plasticity and Creep Control Parameters

Name	Description
CRCERAT	For the ratio of maximum creep increment to elastic strain method, the ratio of maximum creep increment to elastic strain that is used to calculate the next time step. Valid for creep analysis only. (Real ≥ 0.0; Default = 0.4)
CRCINC	For the maximum creep increment method, the maximum creep increment that is used to calculate the next time step. Valid for creep analysis only. (Real ≥ 0.0; Default = 1.0E-4)
CREEP	Include creep effects. (Character = “YES” or “NO”; Default = “YES”)
CRICOFF	When used with TSCCR, creep strain increment below which the next time step is the product of the current time step and CRMFMX. When used with CRLIMR, the difference in creep strain increment below which the next time step is the product of the current time step and EQMFMX. Valid for creep analysis only. (0.0 < Real < 1.0; Default = 1.0E-6)
CRINFAC	Integration factor used to calculate incremental creep strain. Valid for creep analysis only. See Remarks 5 and 6. (0.0 ≤ Real ≤ 1.0; Default = 0.5) **Note** 1.0: backward Euler method (implicit) 0.5: trapezoidal rule

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Name	Description
	0.0: forward Euler method (explicit)
CRLIMR	Requests the CRLIMR time stepping method for creep analysis. The software uses the CRLIMR value to adjust the integration error used for adjusting the creep strain increment. (Real ≥ 0.0; Default = 1.0) By default if you do not define either the CRLIMR or the TSCCR parameter, the software uses the CRLIMR method with CRLIMR=1.0. See Remark 21 for all possible CRLIMR and TSCCR input combinations and the resulting SOL 401 behavior. See Creep analysis in the *Simcenter Nastran Multi-Step Nonlinear User's Guide* for details.
CRMFMN	Minimum time step multiplying factor. If the next time step calculated by the adaptive time stepping algorithm is smaller than the product of the current time step and CRMFMN, the software restarts the current time step with half of the current time increment. Only used with TSCCR. Valid for creep analysis only. (0 ≤ Real ≤ 1.0; Default = 0.1)
CRMFMX	Maximum time step multiplying factor. See the CRICOFF parameter for additional information. Only used with TSCCR. Valid for creep analysis only. (Real ≥ 1.0; Default = 5.0)
CRTEABS	Maximum absolute truncation error. Valid for creep analysis only. (0.0 ≤ Real < 1.0; Default = 1.0E-4)
CRTECO	For the error truncation method, use CRTEABS to calculate the next time step if the creep strain is less than CRTECO, and use CRTEREL to calculate the next time step if the creep strain is greater than CRTECO. Valid for creep analysis only. (0.0 ≤ Real < 1.0; Default = 1.0E-4)
CRTEREL	Maximum relative truncation error. Valid for creep analysis only. (0.0 ≤ Real < 1.0; Default = 0.01)
PLASTIC	Include plasticity effects. See Remark 14. (Character = “YES” or “NO”; Default = “YES”)
PLLIM	Maximum equivalent plastic strain increment allowed within each time step. When PLLIM>0.0, the solution can adjust the time step increment as described in Remark 15. (Real ≥ 0.0; Default=0.0, which disables the capability)
PLLIMF	Multiplying factor of PLLIM when adjusting the time step increment. See Remark 15. (Real > 0.0; Default=1.0)
PLSHUT	Enables the hydrostatic pressure plasticity switch in a preload, statics, or dynamics subcase. (Character; Default=NO) NO = The hydrostatic pressure plasticity switch is disabled for the subcase YES = The hydrostatic pressure plasticity switch is enabled for the subcase.

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Name	Description
	In addition to defining PLSHUT=YES for a subcase, you must also define the PLSHSOL parameter to "YES" on the NLCNTLG bulk entry to turn the switch on globally (above the subcase level).

Table 15-95: Time Integration Parameters

Name	Description
ALFA	HHT method parameter. (0.0 ≤ Real ≤ 1/3; Default = 0.05; Unitless).
BETA	Newmark method parameter one. (Real > 0.0; Default = 0.25; Unitless).
GAMA	Newmark method parameter two. (Real > 0.0; Default = 0.5 ; Unitless).
TETA	Generalized alpha method parameter. (Real; Default = -0.97 for the generalized-theta method and 0.33 for the modified generalized-theta method; Unitless).
TINTMTH	Integration method. (Character = “NEWMARK", "HHT", "GENALP", and "MGENALP"; Default = "NEWMARK"). “NEWMARK" = Newmark method "HHT" = Hilber-Hughes-Taylor (HHT) method "GENALP" = Generalized alpha method "MGENALP" = Modified generalized alpha method
RHOINF	Modified generalized alpha method parameter. (0.0 ≤ Real ≤ 1.0; Default = 0.8) Defines the spectral radius at infinity ρ_∞.

Table 15-96: Time Step Control Parameters

Name	Description
AUTOTIM	Automatic timing scheme ON/OFF parameter. When AUTOTIM=ON (default), automatic time step strategy will be used. Setting AUTOTIM=OFF turns off the automatic time stepping scheme. (Character = "ON" or "OFF"; Default="ON")
DTINIT	Initial time step or constant time step. (Real > 0.0) If you explicitly define DTINIT, the software uses your value for the initial time step. Otherwise, the following defaults are used: ● Creep material is defined: Default = The MINIMUM of either 0.01 or the subcase duration*0.01.

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Name	Description
	● No creep material is defined: Default = The time steps defined with TSTEP1. ● UMAT material is defined: Default = The MINIMUM of either 0.01 or the subcase duration*0.01.
DTMAX	Maximum time step. If DTMAX is set to 0.0 (default), the software accepts the next time step. If DTMAX is nonzero and the next time step is larger than DTMAX, the software uses DTMAX as the next time step. Otherwise, the next time step is compared to DTMIN. (Real ≥ 0.0; Default = 0.0)
DTMIN	Minimum time step. If the next time step is larger than DTMIN, the software accepts the next time step. (0.0 ≤ Real ≤ DTMAX; Default = 0.001*DTINIT) For creep analysis, when used with TSCCR, if the next time increment is smaller than DTMIN, the software restarts the current time step with half of the current time increment. When used with CRLIMR, if the next time increment is smaller than DTMIN, the analysis stops. For plastic analysis, when used with PLLIM, if the next time increment is smaller than DTMIN, the analysis stops.
DTSBCDT	Option to use DTINIT or not in a new subcase. A subcase with a zero start time always uses DTINIT. (Integer; Default = 0) 0: Use the time step prediction at the end of the previous subcase. If the predicted time step is zero, use DTINIT. 1: Use DTINIT
EQMFMIN	Minimum time step factor for equilibrium iteration criterion. (0.0 < Real < 1.0; Default=0.476)
EQMFMX	Maximum time step factor for equilibrium iteration criterion. (Real > 1.0; Default=1.9)
TSCCR	Specifies the time stepping method for creep analysis. (Integer or blank; Default = 12) By default if you do not define either the CRLIMR or the TSCCR parameter, the software uses the CRLIMR method with CRLIMR=1.0. See Remark 21 for all possible CRLIMR and TSCCR input combinations and the resulting SOL 401 behavior. See Creep analysis in the *Simcenter Nastran Multi-Step Nonlinear User's Guide* for details. -1: Turns off the adaptive time stepping method specified by the TSCCR parameter for creep analysis. The time steps are determined from the TSTEP1 bulk entry and other time step controls. 0: Use the constant time step defined with DTINIT.

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Name	Description
	1: Use adaptive time stepping based on the error truncation method only. 2: Use adaptive time stepping based on the ratio of maximum creep increment to elastic strain method only. 3: Use adaptive time stepping based on the maximum creep increment method only. 12: Use adaptive time stepping based on both the error truncation method and the ratio of maximum creep increment to elastic strain method. 13: Use adaptive time stepping based on both the error truncation method and the ratio of maximum creep increment method. 23: Use adaptive time stepping based on both the ratio of maximum creep increment to elastic strain method and the maximum creep increment method. 123: Use adaptive time stepping based on the error truncation method, the ratio of maximum creep increment to elastic strain method, and the maximum creep increment method. Note: When you select one of the TSCCR adaptive time stepping options that is based on a combination of methods, the software calculates a value for the next time step from each selected method. The software compares the values and uses the smallest as the next time step.
TSCEQ	Flag to turn ON/OFF stepping based on equilibrium iterations. (Character = "ON" or "OFF"; Default="OFF")
TSCUMAT	Flag to activate stepping based on UMAT supplied time increments. (Character = "ON" or "OFF"; Default="ON")
UMFMIN	Minimum time step factor for UMAT stepping. (Real > 0.0; Default=0.2)
UMFMX	Maximum time step factor for UMAT stepping. (Real > 0.0; Default=5.0)

Remarks:

1. The NLCNTL bulk entry must be selected with the NLCNTL = SID case control command.
2. NLCNTL case control commands can be placed inside subcases. Because each NLCNTL case control command can point to a different NLCNTL bulk entry, the NLCNTL parameter settings can vary from subcase to subcase.
3. A fatal error occurs when a PARAMi field is defined and the corresponding VALUEi field is left blank.
4. If an NLCNTL parameter is applicable to a certain type of analysis, but it is not defined on an NLCNTL bulk entry, the default value for the parameter is used.
5. Incremental creep strain, $\Delta\epsilon^c$, is calculated from the generalized trapezoidal rule as follows:

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$$\Delta\varepsilon^c = \Delta t\left[(1-\beta)\dot{\varepsilon}^c_t + \beta\,\dot{\varepsilon}^c_{t+\Delta t}\right]$$

where Δt is the current time step, $\dot{\epsilon}^c{}_t$ is the creep strain rate at t, $\dot{\epsilon}^c{}_{t+\Delta t}$ is the creep strain rate at $t+\Delta t$, and β is the integration factor specified with the CRINFAC parameter.

6. If your external creep material subroutine does not define derivatives (that is, the NUCREEP arguments D_EFF_CR_SN(2) and D_EFF_CR_SN(5) are zero), you must use explicit integration by defining CRINFAC=0.0. Otherwise, if you request implicit integration by defining CRINFAC>0.0, a fatal error will occur unless you turn off creep material computations in a subcase by setting the CREEP parameter to NO (CREEP=NO).

If you do not want to calculate creep under a specific temperature and you want to define CRINFAC>0.0, you must reinitialize all partial derivatives to a small value (for example, 1.0e-20).

7. Solution times are the times specified by the TENDi and NINCi fields on TSTEP1 bulk entries. At all times during the creep analysis, if the next time step would result in skipping over a solution time, the software truncates the next time step so that a solve occurs at the solution time. If a time step is truncated to avoid skipping over a solution time, the truncated time step is not subject to the any minimum time step requirement.

8. You can define any combination of U, P, and W with the CONV parameter to request which of displacement (U), force (P), and work (W) the software will consider for the convergence criteria. The tolerances (EPSU, EPSP, and EPSW) define the convergence criteria. All the requested criteria (combination of U, P, and/or W) are satisfied upon convergence.

If you *do not* define force (P) on the CONV parameter, a solution is converged when both of the following are satisfied (1 and 2):

1. All of the checks requested with CONV (either displacement, work, or both) if you define CONV, or the work check if you do not define CONV (default behavior).

2. Either of the following is satisfied (a or b):

a. The L2 norm of the residual for the current iteration is less than the L2 norm of the residual from the first iteration for the time step. This is represented with **RMAG** in the .f06 output.

b. The error in force is less than EPSP.

If you define force (P) on the CONV parameter (either CONV=P, UP, PW or UPW), a solution is converged when all the checks requested with CONV (displacement, load, work) are satisfied, and RMAG is no longer checked.

See SOL 401 - Error functions and weighted normalization in the Multi-Step Nonlinear User's Guide.

9. If you have large rotations that are driven by an enforced motion load, it is recommended that you include the displacement convergence error check by including "U" on the CONV parameter, and tighten the displacement error tolerance with the EPSU parameter. For example, EPSU=1.E-4.

10. With bolts defined with ETYPE=1, when the software determines that a bolt preload is nonzero, if the difference between the software computed preload and the user-defined preload is less than EPSBOLT, the bolt preload calculation is considered converged. If the difference is greater than EPSBOLT, the preload strain is modified accordingly for the next bolt preload iteration. The iterations continue until either convergence is satisfied, or the number of iterations reaches ITRBOLT.

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When the software determines that a requested preload is zero, it uses ZERBOLT to check for convergence by checking if P/AE < ZERBOLT, where P is the axial bolt force from an applied axial strain value, A is the area of the bolt cross section, and E is the modulus of the material selected for the elements on the bolt. EPSBOLT is ignored for a zero bolt preload.

Note that the software considers a bolt preload force to be zero when F/AE < ZERBOLT, where F is the preload force you requested on the BOLTFRC entry. A preload of 0.0 always satisfies this, but it is also possible for the preload to be nonzero, yet A and E are large enough for the software to determine that the zero preload convergence checking should be used.

11. SOL 401 uses the MUltifrontal Massively Parallel sparse direct Solver (MUMPS) by default for all subcases and for symmetric and unsymmetric matrices.

You use the SOLVER parameter to select the solver for symmetric matrices for the nonlinear subcases (statics, preload, transient dynamics). The other symmetric solver choices are the sparse direct solver, the element iterative solver, and the PARDISO solver. When the element iterative solver is selected, you can optionally define the SMETHOD case control command and the ITER bulk entry to alter the default options available on the ITER entry.

You use the USOLVER parameter to select the solver for unsymmetric matrices for the nonlinear subcases. The other unsymmetric solver choices are the sparse direct solver and the PARDISO solver.

You can request a different solver from one subcase to the next, but with the following exceptions.

- The transient dynamics subcase doesn’t support the element iterative solver. If the element iterative solver is requested for a transient dynamics subcase, it will revert to the sparse solver.
- The normal modes, cyclic modes, and Fourier modes subcases always use the solver selected with the system cell 810 which defaults to MUMPS.

By default, the software automatically selects either the symmetric or unsymmetric solver at each iteration depending if unsymmetric terms actually exist and how significant those terms are. Contact friction and damping defined for dynamics can create unsymmetric terms. Solving with unsymmetric matrices is more costly. You can eliminate the unsymmetric terms and solve with a symmetric solver when the unsymmetric terms are not significant to the result and/or convergence. You use the FSYMTOL, KSYM and KSYMTOL parameters on the NLCNTL entry to control if the unsymmetric terms are included.

12. MUMPS, by default, does not save the stiffness factors to the database from one subcase to the next. It must recompute the stiffness factors (matrix decomposition) at the start of each subcase. For models that are mostly linear and do not require many stiffness updates, that include many subcases, and use KUPDATE = 0 or -1, the stiffness factor updates at the start of each subcase can reduce performance.

A workaround is to use the COREMEM keyword to reserve a percentage of total memory as dynamic memory used for retaining matrix factors. For example, COREMEM=20.0X. See MUMPS memory option (SOL 401) in the Multi-Step Nonlinear User's Guide.

Problems that are nonlinear include, for example, contact conditions, large displacements, large strain, and nonlinear materials. For these problems, a stiffness update is often required for most iterations, and as a result, the stiffness factor update at the end of each subcase is often needed for convergence. For nonlinear problems, the COREMEM keyword definition will typically not help the solution performance.

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Note

When using SOL 401 in the Simcenter 3D Multiphysics environment, MUMPS, by default, does not save the stiffness factors to the database from one thermo-mechanical coupling step to the next. Similar to the subcase, MUMPS must recompute the stiffness factors at the start of each coupling step. If your problem is mostly linear, you are using KUPDATE = 0 or -1, and many coupling steps are required, you should consider defining the COREMEM keyword. For example, COREMEM=20.0X.

13. If spin softening is requested with the SPINK parameter, and a rotational force is defined with the RFORCE or RFORCE1 entry, if a grid point is selected in the G field on the RFORCE or RFORCE1 entry to define the rotation center, and this grid point is not used in the connectivity of any element, all translational DOF for this grid point must be constrained.

14. If PARAM,MATNL is set to 1 (all the material nonlinearities turned on), you can use PLASTIC,NO to exclude plastic effects for a specific subcase.

    If PARAM,MATNL is set to -1 (all the material nonlinearities turned off), the PLASTIC parameter status has no effect on the solve. So, you cannot use PLASTIC,YES to enable plasticity effects in a subcase while material nonlinearities are turned off by default in the solution.

PARAM, MATNL	NLCNTL PLASTIC	Is plasticity considered in the solve?
-1	Yes	No (default)
-1	No	No
1	Yes	Yes
1	No	No

15. The parameters PLLIM and PLLIMF are used with the software computed equivalent plastic strain increment ($\Delta\epsilon^p$) to determine whether a time step increment is decreased or increased. The following time step control is only active when PLLIM > 0.0:

    When $\Delta\epsilon^p$ > PLLIM, the solver restarts the current time step with the following decreased time increment:

    $\Delta t_{next}=\Delta t*\min[\ PLLIMF*PLLIM/\Delta\epsilon^p\ ;\ 0.9\ ]$

    When $\Delta\epsilon^p \leq$ PLLIM after a converged time step, the solver starts the next time step with the following time increment:

    $\Delta t_{next}=\Delta t*\max[\ PLLIMF*PLLIM/\Delta\epsilon^p\ ;\ 1.0\ ]$

16. When MSGLVL=2, the following solution accuracy indicators table is printed to the .f06 file.

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```
================================================================================
                S O L U T I O N   A C C U R A C Y   I N D I C A T O R S
                    FOR CONTACT DAMPING AND STIFFNESS STABILIZATION
================================================================================
                          INCREMENTAL    %OF INTERNAL        INTERNAL     %OF INTERNAL
                         INTERNAL WORK       WORK          FORCE SUM         FORCE
                         -------------  -------------     -------------  -------------
TOTAL                     2.10226E-01                      6.36106E+01
CONTACT DAMPING           3.35523E-04    1.59601E-03       1.66000E-02    2.60963E-04
STIFFNESS STABILIZATION   3.43939E-02    1.63604E-01       2.16145E-01    3.39794E-03
================================================================================
```

The software prints the table after each time step converges. The table shows the total force and work for the model, the force and work attributed to each stabilization mechanism, and the percentage of force and work each mechanism contributes to the total. You want the force and work from each of the mechanisms to be insignificant relative to the total. We recommend a value on the order of 1% or smaller. The contact damping referenced in the table is requested with the CTDAMP parameter on the BCTPARM bulk entry, and the stiffness stabilization referenced in the table is requested with the MSTAB parameter on the NLCNTL bulk entry. If you have requested the output but did not request contact damping and/or matrix stabilization, the table is still printed but with zero values.

17. The software ignores the stiffness matrix stabilization request (MSTAB=1) when the element iterative solve is used (SOLVER=ELEMITER). See Matrix stabilization in the *Multi-Step Nonlinear User's Guide*.

18. When contact is defined, the software internally sets KUPDATE=1, except KUPDATE=0 is used for any of the following:

a. When every contact pair has INTRFC=5 defined on the BCTPARM bulk entry.

b. When every contact pair has INTRFC=3 or 4 and GUPDATE=0 defined on the BCTPARM bulk entry.

c. When all pairs are defined as a combination of a and b.

19. When an arc-length method is requested for a nonlinear buckling analysis (NLARCL command), and the new Newton-Raphson iteration strategy is used (system(779)=1, default strategy), the software always sets KUPDATE=1 internally.

20. KUPDATE=-3 is recommended when a creep material is the only nonlinear condition, you have requested the explicit creep integration method, and you have temperature-dependent materials (TEMP(LOAD) case control and MATTi bulk entries).

21. You use the CRLIMR or the TSCCR parameter to select the creep adaptive time stepping method. Their settings determine if the CRLIMR method is used, if the TSCCR method is used, or if no creep time stepping method is used.

The following includes the possible input combinations and the resulting software behavior.

- By default if you do not define either the CRLIMR or the TSCCR parameter, the software uses the CRLIMR method with CRLIMR=1.0.
- When you explicitly define both the CRLIMR and the TSCCR parameters:

  If you define CRLIMR > 0.0 and any TSCCR setting, the software uses the CRLIMR method and ignores the TSCCR setting.

  If you define CRLIMR = 0.0 and TSCCR ≠ -1, the software uses the TSCCR method.

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If you define CRLIMR = 0.0 and TSCCR = -1, the software does not use a creep time stepping method.

- When you explicitly define only the CRLIMR parameter:

  If you define CRLIMR > 0.0, the software uses the CRLIMR method.

  If you define CRLIMR = 0.0, the software uses the TSCCR method with the TSCCR = 12 default.

- When you explicitly define only the TSCCR parameter:

  If you define TSCCR ≠ -1, the software uses the TSCCR method.

  If you define TSCCR = -1, the software uses the CRLIMR method using the CRLIMR = 1.0 default.

Related Topics

Browse Simcenter Nastran documentation

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NLCNTL2

Strategy Parameters for SOLs 402 and 414

Defines solution control parameters for SOLs 402 and 414. The parameters can vary from one subcase to another.

Format:

1	2	3	4	5	6	7	8	9	10
NLCNTL2	ID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4	PARAM5	VALUE5	-etc-				

Fields:

Field	Contents
SID	Identification number. (Integer > 0)
PARAMi	Name of the NLCNTL2 parameter. Allowable names are given in the parameter listing below. (Character)
VALUEi	Value of the parameter. (Real, Integer, or Character)

NLCNTL2 Parameters:

Table 15-97: Analysis Control

Name	Description
DISLIM	Maximum displacement. (Real > 0.0; Default = ∞ ; Unit = Length).
ROTLIM	Maximum rotation. (Real > 0.0; Default = 0.1 if PARAM,LGDISP,0, else ∞ ; Unit = radians).
DEFLIM	Maximum deformation. (Real > 0.0; Default = 0.1 if PARAM,LGDISP,0, else ∞. ; Unitless)
MADI	Maximal allowed displacements variation. (Real ; Default = -1; Unit = Length)
DIPR	Specifies whether or not to predict the displacement at the first iteration of a time step using the displacement rate (Δdisplacement / Δtime) from the previous time step. DIPR applies to all time steps except for the first time step of a subcase. (Integer 0 or 1; Default depends on load type) DIPR = 0: The unknown displacements at the first iteration of a new time step are assumed to be zero. Enforced displacement increments are applied if they are defined. DIPR = 1: The displacement rate from the previous time step is used to predict the unknown displacements at the first iteration of a new time step.

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Name	Description
	The default depends if time-assigned loads are defined for the subcase: Default = 0 is used if the subcase includes time-assigned loads defined with the DLOAD bulk entry. Because time-assigned loads can have a varying displacement rate, it is more conservative for the software to not predict the unknown displacements. Default = 1 is used if the subcase does not include time-assigned loads. Because time-unassigned loads defined with the LOAD bulk entry are linearly ramped by the increments in a subcase, the displacement rate remains constant and predictable from one time step to the next. DIPR = 1 can still be helpful when time-assigned loads are defined as long as the displacement rate from one time step to the next is not changing drastically. Using smaller time steps can help. With intricate nonlinear situations, for example, convergence issues with both creep and plasticity, it might be best to use DIPR=0. Because DIPR=0 could affect accuracy when these intricate nonlinear situations exists, you can consider a tighter energy convergence tolerance such as PRCE=1E-4 to increase accuracy.
NORM	Computes the norm of the internal or external forces before or after assembly. This is useful when only thermal loads are introduced because all the forces would be in a balanced state after assembly. (Integer 0 or 1; Default = 0). 0 = after assembly. 1 = before assembly.
MATSYM	Requests symmetric or unsymmetric stiffness, damping and mass matrices for a complex modes subcase. (Character = “1” or “0”). This parameter is currently ignored by the software. Unsymmetric matrices are always used in a complex modes subcase, and as a result, the unsymmetric solver is always used.
STAB	Stabilization factor. (Real ≥ 0.0; Default = 0.0; Unitless).
INERTIA	Take inertia into account in nonlinear dynamic subcases. (Character = “ON”, “OFF”, or "STAT"; Default = “ON”). “ON" = Account for inertia in nonlinear dynamic subcases. “OFF" = Ignore inertia in nonlinear dynamic subcases. “STAT" = Ignore inertia and damping in nonlinear dynamic subcases.
IAR0	Specify if data must be saved at the beginning of the computation of a sequentially dependent subcase (SEQDEP=YES). (Integer -1 or 0; Default = 0). -1 = no save.

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Name	Description
	0 = save.
ITHE	Requests how the software manages thermal loads to improve convergence. (Integer -1 or 0; Default = 0) The appropriate ITHE parameter option can significantly improve the solution convergence. Either option will produce the same results if the solution converges. -1 = Free thermal expansion You should use this option if your thermal loads produce unconstrained free thermal deformation resulting with little or no thermal stress. 0 = Constrained thermal expansion You should use this option if your model is constrained such that your thermal loads produce little free thermal deformation but significant thermal stress.
RFVAR	Specifies how the software interpolates the ramping load factor $f(t)$ for a time-unassigned rotational load defined with the RFORCE or RFORCE1 bulk entry. (Character = "OMEGA" or "FORCE"; Default ="OMEGA") See SOL 402 - Interpolating rotational inertia loads in the *Multi-Step Nonlinear User's Guide*. OMEGA: The software interpolates the time variation using $f(t)$. As a result, if a linear time variation is defined, the angular velocity will vary linearly. Since $\vec{\omega}$ is squared in the force computation, the force will not vary linearly. FORCE: The software interpolates the time variation using $f(t)^2$. The software then takes the square root of the interpolated value before applying it to the angular velocity. As a result, if a linear time variation is defined, the force will vary linearly, but $\vec{\omega}$ will not.
ISTO	SOL 414,111 only, and only when an RMETHOD case control command is not present. Stop run option. (Integer = 2 or 3; Default = 2 or 3 depending on the presence of a METHOD case control command) 2 = Stop the run after estimating how much space to reserve in the central memory and on the disk. (2 is the default when a METHOD case control command is not present.) 3 = Stop the run after a complete or partial calculation of the dynamic response. (3 is the default when a METHOD case control command is present.)
IRDY	SOL 414,111 only, and only when an RMETHOD case control command is not present.

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Name	Description
	Solution method. (Integer = 0 or 1; Default = 0 or 1 depending on the presence of a METHOD case control command) 0 = Use modal superposition to calculate the forced harmonic response and residuals or reactions. (0 is the default when a METHOD case control command is present.) 1 = Use direct calculation of the forced harmonic response. (1 is the default when a METHOD case control command is not present.)
BALF	SOL 414,111 only, and only when an RMETHOD case control command is not present. Solution frequency option. (Integer = 0 or 1; Default = 0) 0 = Calculate the harmonic response at the solution frequencies. 1 = Calculate the harmonic response at the solution frequencies and modal frequencies that lie within the range of solution frequencies. **Note** This parameter can be used only when damping is present in the structure and IRDY = 0.
LL2	Dimension in double-precision words of the space required to store the laws of excitation. If the dimension is too small, the software issues a warning and the run is interrupted. (Integer > 0; Default = 20000)
LVAR	Specifies whether time-unassigned mechanical loads are ramped or stepped. A fatal error occurs by default if you define LVAR=STEP. See Remark 12. (Character "RAMP" or "STEP"; Default = "RAMP").
TVAR	Specifies whether time-unassigned temperature loads are ramped or stepped. A fatal error occurs by default if you define TVAR=STEP. See Remark 12. (Character = “RAMP” or “STEP”; Default = “RAMP”)

Table 15-98: Plasticity and Creep Control Options

Name	Description
CREEP	Include creep effects. See Remark 5. (Character = “YES” or “NO”; Default = “YES”)
CRICOFF	Creep strain increment threshold used when the CRLIMR time stepping method is requested (CRLIMR>0.0). Valid for creep analysis only. See Creep analysis in the *Simcenter Nastran Multi-Step Nonlinear User's Guide* for details. (0.0 < Real < 1.0; Default = 1.0E-6)
CRINFAC	Integration factor used to calculate incremental creep strain. Valid for creep analysis only. See Remarks 7 and 8. (0.0 ≤ Real ≤ 1.0; Default = 1.0; Unitless).

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Name	Description
	Note 1.0: backward Euler method (implicit) 0.5: trapezoidal rule 0.0: forward Euler method (explicit)
CRLIMR	Requests the CRLIMR time stepping method for creep analysis. The software uses the CRLIMR value to adjust the integration error used for adjusting the creep strain increment. See Creep analysis in the *Simcenter Nastran Multi-Step Nonlinear User's Guide* for details. (Real ≥ 0.0; Default = 0.0)
PLASTIC	Include plasticity effects. See Remark **6**. (Character = “YES” or “NO”; Default = “YES”)
PLLIM	SOL 402 only. Maximum equivalent plastic strain increment allowed within each time step. The solution will bisect the time step if the plastic strain increment is greater than PLLIM. (Real ≥ 0.0; Default=0.0 which disables the capability) See Remark **10**.
PLSHUT	SOL 402 only. Enables the hydrostatic pressure plasticity switch in a preload, statics, or dynamics subcase. (Character; Default=NO) NO = The hydrostatic pressure plasticity switch is disabled for the subcase. YES = The hydrostatic pressure plasticity switch is enabled for the subcase. In addition to defining PLSHUT=YES for a subcase, you must also define the PLSHSOL parameter to "YES" on the NLCNTLG bulk entry to turn the switch on globally (above the subcase level).

Table 15-99: Time Integration

Name	Description
TINTMTH	Integration method. (Character = “NEWMARK", "HHT", "GENMID", and "MGENALP"; Default = "MGENALP"). “NEWMARK" = Newmark implicit predictor-corrector. "HHT" = Hilber-Hughes-Taylor (HHT) implicit predictor-corrector.

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Name	Description
	"GENMID" = Generalized midpoint. "MGENALP" = Modified generalized alpha.
BETA	Newmark method parameter 1. (Real > 0.0; Default = 0.25; Unitless).
GAMA	Newmark method parameter 2 . (Real > 0.0; Default = 0.5 ; Unitless).
ALFA	HHT method parameter. (0.0 ≤ Real ≤ 1/3; Default = 0.05; Unitless).
TETA	Generalized midpoint method parameter. (0.5 ≤ Real ≤ 1.0 ; Default = 0.55; Unitless).
RHOINF	Modified generalized alpha method parameter. (0.0 ≤ Real ≤ 1.0; Default = 0.8) Defines the spectral radius at infinity ρ_∞.

Table 15-100: Equilibrium Iteration and Convergence

Name	Description
ILNS	Line search. (Integer 0, positive value, negative value; Default = 0). 0 = No line search. N > 0 = line search method based on minimizing the energy (activated at Newton iteration N). N < 0 = line search method based on minimizing the residue (activated at Newton iteration N).
PRLN	Line search convergence tolerance. (Real > 0.0; Default = 100*PRCR if ILNS > 0, 0.1 if ILNS < 0 ; Unitless).
AMIN	Minimum line search factor. (Real > 0.0; Default = 0.5 if ILNS > 0, 0.1 if ILNS < 0 ; Unitless).
AMAX	Maximum line search factor. (Real > 0.0; Default = 1 / AMIN ; Unitless).
MAXBIS	SOL 402 only. Maximum number of time step reductions allowed when automatic time stepping is activated, which is the default for AUTOTIM. (Integer ≥ 0; Default = 0 which is no limit)
ITMA	Maximum number of iterations for an implicit time step. (Integer > 0; Default = 20 for SOL 402; Default = 10 for SOL 414).
PRCR	Relative force tolerance. (Real > 0.0; Default = 0.001; Unitless).
REFP	Reference force. (Real > 0.0; Default = 1 ; Unit = Force).
PRCQ	Precision threshold for the displacement/temperature variation (to be compared to TESQ value in the .f06 listing file) (Real > 0.0; Default = 0.1 ; Unitless).

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Name	Description
REFU	Reference displacement. (Real > 0.0; Default = 1 ; Unit = Length).
PRCE	Relative energy tolerance. (Real > 0.0; Default = .001; Unitless).
PRDR	Relaxed relative force tolerance. (Real > 0.0; Default = PRCR; Unitless). The software will first try to converge using the default tolerance value defined with the parameter PRCR. If convergence is difficult and the software iterates more than the maximum number of iterations, it then considers the relaxed criteria (PRDR). To enable the relaxed force tolerance, you must define PRDR>PRCR.
PRDE	Relaxed relative energy tolerance. (Real > 0.0; Default = PRCE; Unitless) The software will first try to converge using the default tolerance value defined with the parameter PRCE. If convergence is difficult and the software iterates more than the maximum number of iterations, it then considers the relaxed criteria (PRDE). To enable the relaxed energy tolerance, you must define PRDE>PRCE.
REFE	Reference energy. (Real > 0.0; Default = 1 ; Unit = Energy).
IT1K[a]	Iteration for the first stiffness update. (Integer > 0; Default[b] = 1)
IT2K[a]	Iteration for the second stiffness update. (Integer > 1; Default[b] = IT1K + 1)
IT3K[a]	Iteration frequency for stiffness updates. (Integer > 0; Default[b] = 1)

[a] For each time step, IT1K is the iteration for the first stiffness update, IT2K is the iteration for the second stiffness update, and IT3K is the iteration frequency for stiffness updates until the time step converges. As a result, a stiffness update occurs at the following iterations:

IT1K, IT2K, IT2K + IT3K, IT2K + 2× IT3K, etc.

[b] Note that at least one of IT1K, IT2K, or IT3K must be explicitly defined for their defaults to apply. For example, if only IT1K is defined, the software will use the defaults of IT2K and IT3K. See KUPDATE description below.

KUPDATE	Stiffness update strategy. (Integer; Default = 0) If any of the parameters IT1K, IT2K, or IT3K are explicitly defined, KUPDATE is ignored and the ITiK parameters are used. When neither KUPDATE or ITiK parameters are explicitly defined, the KUPDATE=0 default is used. >1: Quasi Newton-Raphson. KUPDATE is the number of iterations before another stiffness update in the same time step. 1: Full Newton-Raphson. Update the stiffness at each iteration. 0: Same as KUPDATE=1.

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Name	Description
	-1: Initial stiffness approach. Never update the stiffness. -3: Update the stiffness only at iteration 1 of each new time step. See Remark 11. -4: The software automatically selects the stiffness update strategy according to the following: For a pure linear analysis* or if explicit creep is the only nonlinearity, and • you have temperature-dependent materials (MATTi bulk entries), the software uses KUPDATE=-3. • you *do not* have temperature-dependent materials (MATTi bulk entries), the software uses KUPDATE=-1. For all other cases (nonlinear conditions exists), the software uses KUPDATE=1. *A pure linear analysis means no large displacement or strain (PARAM,LGDISP,-1 and PARAM,LGSTRN,0), no other nonlinear materials except creep, and no large sliding contact (DISP=1 setting BCTPAR2 bulk entry).
PLAS	Choice for the integration of the constitutive law at the first iteration of a time step for a plastic material. (Integer -1, -2, -3, or 1; Default = 1). -1 = elastic tangent matrix, plastic computation of the variation of stresses. -2 = elastic tangent matrix, elastic computation of the variation of stresses. -3 = plastic tangent matrix, elastic computation of the variation of stresses. 1 = plastic tangent matrix, plastic computation of the variation of stresses.
OTRE	Allow an easier convergence when no external loads are applied (rigid body motion). (Integer 0, 1, or 2; Default = 1). 0= the option is disabled. 1= • if both energy (PRCE) and displacement (PRCQ) convergence criteria are satisfied for iterations 2 to 4, • and if the force (PRCR) criteria is smaller than 10 times its defined tolerance, the step converges.

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Name	Description
	2= in addition to OTRE=1, the REFP criteria must be satisfied at the 1st iteration.

Table 15-101: Automatic Time Stepping

Name	Description
AUTOTIM	Automatic time stepping activation. See Remark 9. (Character = “ON” or “OFF”; Default = “ON”). "OFF" = fixed time steps. "ON" = automatic time stepping.
DTINIT	Size of the first time step. See Remark 4. (Real > 0.0; Default = automatically computed ; Unit = Time).
DTMIN	Minimum time step if ANALYSIS = STATICS or DYNAMICS. (Real > 0.0; Default = DTINIT / 1E05 ; Unit = Time). Minimum load factor if ANALYSIS = PRELOAD. (Real > 0.0; Default = 1E-03; Unit = Unitless).
DTMAX	Maximum time step if ANALYSIS = STATICS or DYNAMICS. (Real > 0.0; Default = ∞ ; Unit = Time). Maximum load factor if ANALYSIS = PRELOAD. (Real > 0.0; Default = ∞; Unit = Unitless).
EQMFMX	Maximum increase ratio for next time step. (Real ≥ 1.0; Default = 1.9; Unitless).
EQMFMIN	Minimum decrease ratio for next time step. (0.0 < Real < 1.0; Default = 0.476; Unitless).
ERCD	Activation of time step criterion based on the absolute error of the displacement/ temperature change. (Integer 0 or 1; Default = 0). 0 = no activation. 1 = activation.
PRED	If ERCD=1, value of absolute allowable displacement/temperature change per time step. (Real > 0.0; No default; Unit = Length).
ITEREF	Number of iterations. (Integer >0; Default = 0.6 * ITMA).
TSDYN	Integration error control. (Character = “ON” or “OFF”; Default = “OFF”).
HPRCO	Time step limit for the integration error for integration schemes with automatic time stepping (Newmark, HHT, Generalized midpoint and Generalized-α). (Real ≥ 0.0; Default = 0 ; Unit = Time).

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Name	Description
PRCO	Integration error threshold for integration schemes with automatic time stepping (Newmark, HHT, Generalized midpoint and Generalized-α). (Real > 0.0; Default = 0.001; Unitless).
RJPZ	Zero pivot time step rejection. (Character = “ON” or “OFF”; Default = “OFF”).
RJPN	Negative pivot time step rejection. (Integer = 0, 1,2, 3; Default = 0). 0 = never. 1 = if last iteration only. 2 = never but print state at last iteration. 3 = always.
TSCVSC	Viscous material integration error control. (Character = “ON” or “OFF”; Default = “OFF”).
VSCOTE	Viscous material integration error threshold. (Real ≥ 0.0; Default = 0.1; Unitless)
VSCOSN	Viscous material minimum stress normalization factor. (Integer>0; Default = 1).

Table 15-102: Internal Restart Options

Name	Description
RSUB	SOL 402 only. Restart computation flag for subcases with the SEQDEP=NO case control command. (Integer -1, 0 or positive value). -1 = the initial state is the one at the end of the previous subcase. 0 = the initial state is the one at the initial time. (Default). N > 0 = Restart computation after subcase N.

Table 15-103: Contact Options

Name	Description
RELC	Relaxation factor for contact. See Remark 1. (Real > 0.0; Default = 1.0 ; Unitless).
DCON	Characteristic length for contact. See Remark 2. (Real > 0.0; Default = automatically computed; Unit = Length).
PRCS	Regularization factor p for the augmented Lagrange multipliers. See Remark 3. (Real > 0.0; Default = automatically computed; Unitless).

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Name	Description
PRCF	SOL 402 only. Precision threshold for the contact forces variation (to be compared to TESC value in the .f06 listing file) (Real > 0.0; Default = 0.1 ; Unitless).
KMODTN	When sequencing from a nonlinear subcase (PRELOAD, STATIC, DYNAMICS) that includes contact to a modal subcase (MODES, CYCMODES, FOURIER), the parameter KMODTN allows you to select a different tangential contact stiffness for the modal subcase than was used in the previous nonlinear subcase. It can be used, for example, when the nonlinear subcase required frictionless contact, yet the modal subcase needs a tangential contact stiffness to prevent spurious rigid body modes. KMODTN is only used in the modal subcases. (Integer; Default = 1) The KMODTN settings allow you to independently control the tangential contact stiffness of the contact pairs with friction (**A**), and without friction (**B**). **A**: Contact pairs *with* friction are defined by FRICi>0.0 on the BCTSET bulk entry or FRICMOD>0 on the BCTPAR2 bulk entry. **B**: Contact pairs *without* friction are defined by FRICi=0.0 on the BCTSET bulk entry and FRICMOD=0 on the BCTPAR2 bulk entry.

KMODTN Setting	Friction status	Tangential contact stiffness in modal subcase
1	**A**	The software holds the contact pairs tangentially in a modal subcase.
	B	The software does not apply tangential contact stiffness in a modal subcase.
2	**A**	The software applies the actual final tangential contact stiffness from the end of a nonlinear subcase in a modal subcase. The applied tangential contact stiffness can be a mixture of sticking and sliding stiffness values.
	B	The software does not apply tangential contact stiffness in a modal subcase.
3	**A**	The software applies the actual final tangential contact stiffness from the end of a nonlinear subcase in a modal subcase. The applied tangential contact stiffness can be a mixture of sticking and sliding stiffness values.
	B	The software holds the contact pairs tangentially in a modal subcase.

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<table>
<tr><th>Name</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><th>KMODTN Setting</th><th>Friction status</th><th>Tangential contact stiffness in modal subcase</th></tr>
<tr><td rowspan="2">4</td><td>A</td><td>The software holds the contact pairs tangentially in a modal subcase.</td></tr>
<tr><td>B</td><td>The software holds the contact pairs tangentially in a modal subcase.</td></tr>
</table>
</td></tr>
<tr><td>MSGLVLC</td><td>Diagnostic output for contact. (Integer = 0, 1, 2, 11 or 12; Default = 11)

See SOL 402 - Contact status output in the Multi-Step Nonlinear User's Guide.

0 = No additional diagnostic output.

1 = Information related to each contact pair is printed for each iteration. The information includes the contact force, normal distance, sliding distance, and the number of inactive/open/sticking/slipping contact elements. This is a more concise listing as compared to the MSGLVLC=2 option.

2 = Everything described in option 1 but expanded. This option also lists the residual forces and the variations in the Lagrange multipliers.

11 = Same as option 1, but only printed in the case of non-convergence. (Default)

12 = Same as option 2, but only printed in the case of non-convergence.</td></tr>
</table>

Table 15-104: Diagnostic Listing File Printout

Name	Description
IMPG	Printing frequency of general results (number of plastic elements, ...). (Integer > 0; Default = 10).
IMPR	Number of the greatest residuals to be printed in listing file. (Integer ≥ 0; Default = 0).
IMPV	Maximum number of pivots (negative or zero) that may be printed by the solver in the listing file. (Integer ≥ 0; Default = 5).

Table 15-105: Result Storage Control

Name	Description
IA4	SOL 414,111 and SOL 414,129 only. Storage cycle for element stresses. Element stresses are stored after frequency steps that are multiples of IA4 in a SOL 414,111 harmonic response analysis or after time steps that

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Name	Description
	are multiples of IA4 in a SOL 414,129 transient response analysis. (Integer > 0; Default = 100) **Note** Because the software calculates stress results from displacements results, IA4 must be a multiple of IA16. For example, if IA16 is 8, IA4 can be 8, 16, 24, 32, and so on.
IA16	SOL 402, SOL 414,111, and SOL 414,129 only. Storage cycle for grid point results. Grid point results are stored after frequency steps that are multiples of IA16 in a SOL 414,111 harmonic response analysis or after time steps that are multiples of IA16 in a SOL 402 or a SOL 414,129 transient response analysis. (Integer > 0; Default = 1 if NOUT = ALL on the TSTEP1 bulk entry; Default = 9999999 if NOUT ≠ ALL on the TSTEP1 bulk entry) The relationship between the IA16 parameter and the NOUT field on the TSTEP1 bulk entry is as follows: NOUT saves results at a frequency relative to the time steps defined on TSTEP1. IA16=n saves results at a frequency of n between two NOUT time steps. For example, if a computation goes from 0 to 1.0 in 10 increments, the solution points will be (.1, .2, .3, .4, .5, .6, .7, .8, .9, 1.0). If NOUT=5 and IA16=2, the NOUT=5 request will output results every 5 increments, or at .5 and 1.0. The IA16=2 request will output results ever two increments between the NOUT increments, or every two from (.1 to .5) which is .2 and .4, and every two increments from (.5 to 1.0) which is .7 and .9. So output will occur at 0.2 (due to IA16), 0.4 (due to IA16), 0.5 (due to NOUT), 0.7 (due to IA16), 0.9 (due to IA16), and 1.0 (due to NOUT). IA16 and IA19 must be multiples of each other. See note under IA19 description.
IA19	SOL 402, SOL 414,111, and SOL 414,129 only. Storage cycle for XY plotting results requested by OUTMGT bulk entries. Results are stored after frequency steps that are multiples of IA19 in a SOL 414,111 harmonic response analysis or after time steps that are multiples of IA16 in a SOL 402 or a SOL 414,129 transient response analysis. (Integer > 0; Default = 1) **Note** Because the software uses grid point results for XY plotting, IA16 and IA19 must be a multiples of each other (depending on which is larger). Since IA16 is typically used to output results at all grid points for fewer solution points, and IA19 is used to output results at specific DOF for many solution points, IA19 is typically defined smaller than IA16.

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Name	Description
	For example, if IA16=8, IA19 can be 1, 2, 4, 8 (or 16, 24, 32, etc., if you want IA19 larger than IA16).
IA12	SOL 414,111 frequency response subcase only. Controls the complex eigenfrequencies and eigenmodes calculations. (Integer > 0; Default = 0 (no complex modes are calculated or stored)) Complex eigenmodes and eigenvalues are computed every IA12 frequency steps. When you set IA12, you must also set the CMETHOD (complex eigenvalue extraction) case control to request the EIGC (complex eigenvalue extraction) bulk entry. Otherwise, a fatal error occurs. When your model uses linear bearings, you can set the mpi402 keyword for distributed-memory parallel (DMP) processing to reduce computation time. However, when your model uses nonlinear bearings, parallel computations are prohibited. **Note** When you set IA12, the software uses a Direct or Modal solution scheme depending on the presence of the EIGRL (real eigenvalues) and the METHOD case control. You can use the results in a post-processing application to generate Campbell diagrams.

Remarks:

1. RELC can trigger more iterations in contact solving. Values smaller than 1.0 will be under-relaxation. Values larger than 1.0 (over-relaxation) are allowed but for advanced users.
2. Default for DCON is automatically computed from the size of the elements.
3. Default for PRCS is automatically computed from the average stiffness of the whole model.
4. Default for DTINIT is automatically computed by the software.
5. If PARAM,MATNL is set to 1 (all material nonlinearities enabled), you can use CREEP,NO to exclude creep effects for a specific subcase.

   If PARAM,MATNL is set to -1 (all material nonlinearities disabled), the CREEP parameter status has no effect on the solve. Thus, you cannot use CREEP,YES to enable creep effects in a subcase when material nonlinearities are disabled.

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PARAM,MATNL	NLCNTL2 CREEP	Is creep considered in the solve?
-1	Yes	No (default)
-1	No	No
1	Yes	Yes
1	No	No

6. If PARAM,MATNL is set to 1 (all material nonlinearities enabled), you can use PLASTIC,NO to exclude plastic effects for a specific subcase.

   If PARAM,MATNL is set to -1 (all material nonlinearities disabled), the PLASTIC parameter status has no effect on the solve. Thus, you cannot use PLASTIC,YES to enable plasticity effects in a subcase when material nonlinearities are disabled.

PARAM,MATNL	NLCNTL2 PLASTIC	Is plasticity considered in the solve?
-1	Yes	No (default)
-1	No	No
1	Yes	Yes
1	No	No

7. Incremental creep strain, $\Delta\epsilon^c$, is calculated from the generalized trapezoidal rule as follows:

$$\Delta\varepsilon^c = \Delta t \left[(1-\beta)\dot{\varepsilon}^c_t + \beta\,\dot{\varepsilon}^c_{t+\Delta t}\right]$$

   where Δt is the current time step, $\dot{\epsilon}^c_t$ is the creep strain rate at t, $\dot{\epsilon}^c_{t+\Delta t}$ is the creep strain rate at $t+\Delta t$, and β is the integration factor specified with the CRINFAC parameter.

8. If your external creep material subroutine does not define derivatives (that is, the NUCREEP arguments D_EFF_CR_SN(2) and D_EFF_CR_SN(5) are zero), you must use explicit integration by defining CRINFAC=0.0. Otherwise, if you request implicit integration by defining CRINFAC>0.0, a fatal error will occur unless you turn off creep material computations in a subcase by setting the CREEP parameter to NO (CREEP=NO).

   If you do not want to calculate creep under a specific temperature and you want to define CRINFAC>0.0, you must reinitialize all partial derivatives to a small value (for example, 1.0e-20).

9. If AUTOTIM="OFF", ITEREF, TSDYN, PRCO, HPRCO, TSCVSC, VSCOTE, VSCOSN, RJPZ, and RJPN will be ignored.

10. When plastic deformation becomes significant, you may need to decrease the load increments to capture the changing material behavior and to help convergence. The PLLIM parameter is available for you to define the maximum equivalent plastic strain increment allowed within a time step. When the plastic strain increment is greater than PLLIM, the software will adjust the current time step increment. The plastic strain increment limit is disabled by default, so you must define the PLLIM parameter greater than 0.0 to enable the limit. The software uses an associated flow rule where the

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plastic potential function, *g*, is the same as the yield function, *f*. The components of the plastic strain increment are given by:

$$d\varepsilon_{ij}^{P} = d\lambda \frac{\partial f}{\partial \sigma_{ij}}$$

Then the equivalent plastic strain increment in a time step is:

$$\Delta \bar{\varepsilon}_{P} = \sqrt{\frac{2}{3} \Delta \epsilon^{P} \cdot \Delta \epsilon^{P}}$$

11. KUPDATE=-3 is recommended when a creep material is the only nonlinear condition, you have requested the explicit creep integration method, and you have temperature-dependent materials (TEMP(LOAD) case control and MATTi bulk entries).

12. For SOL 402, beginning with Simcenter Nastran 2021.2, a fatal error occurs by default if you define the parameter settings LVAR=STEP or TVAR=STEP. You can use the system cell setting NASTRAN SYSTEM(767)= 0 to turn off the fatal error, and the solution will proceed using these settings. Please be aware that the STEP approach can adversely effect convergence.

Related Topics

Browse Simcenter Nastran documentation

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NLCNTLG

Strategy Parameters for SOLs 401, 402, and 414

Defines solution global control parameters for SOLs 401, 402, and 414.

Format:

1	2	3	4	5	6	7	8	9	10
NLCNTLG	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3	PARAM4	VALUE4	
	PARAM5	VALUE5	-etc-						

Example:

NLCNTLG	RESO	5							

Fields:

Field	Contents
PARAMi	Name of the NLCNTLG parameter. Allowable names are given in the parameter listing below. (Character)
VALUEi	Value of the parameter. (Real, Integer, or Character)

Parameters

Name	Description
Global parameter for SOL 401:	
CHOCK	Defines the chocking element gap status update frequency when large displacements are off with LGDISP=-1. (Character; Default = YES) YES: The software computes an updated gap status and associated stiffness at the end of every time step. NO: The software computes the gap status in the first iteration, and uses the resulting element stiffness for the entire solution.
CY3DSCL	Controls the scaling of material properties and loads for the elements included in a group selected by the CYCSET bulk entry. The scaling is useful when a model includes both 2D axisymmetric elements and a sector of 3D solid elements. (Integer; Default=1) 1: Material properties and load scaling is turned on.

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Name	Description
	0: Material properties and load scaling is turned off.
RESXTRP	When you use the improved architecture in a small strain analysis with nonlinear materials, controls how the software calculates the grid point stresses for the elements that support the improved architecture. See the note below regarding shell elements. (Character; Default = ON) RESXTRP = ON: The software does the following: **1.** Calculates the strain at the Gauss points. **2.** Using the material law (constitutive relation), calculates the stress at the Gauss points from the strain at the Gauss points. **3.** Using the shape functions, extrapolates the stress and strain at the Gauss points to the corresponding grid points. RESXTRP = OFF: The software does the following: **1.** Calculates the strain at the Gauss points. **2.** Using the shape functions, extrapolates the strain at the Gauss points to the corresponding grid points. **3.** Using the material law (constitutive relation), calculates the stress at the grid points from the strain at the grid points. When your model is dominated by the elements that support the improved architecture, specifying RESXTRP = ON (the default) reduces the total data I/O. Both RESXTRP options produce identical grid point total strains, but the grid point stresses can differ. For an explanation about why this occurs, for the list of elements that support the improved architecture, and other information about the improved architecture, see Improved architecture for small strain problems (SOL 401) in the *Multi-Step Nonlinear User's Guide*. **Note** For the shell elements CTRIA3, CTRIAR, CQUAD4, CQUADR, CTRIA6, and CQUAD8 referencing a PSHELL property, the software always calculates the grid point stresses using the RESXTRP,ON behavior. The RESXTRP setting is ignored by the shell elements in the improved architecture.
Global parameter for SOL 402:	
BOLTMETH	Defines how the software maintains a bolt preload condition with a bolt defined as ETYPE=2 or 3 on the BOLT bulk entry. (Character; Default described below)

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Name	Description
	The software maintains the grid pairs at the bolt cut location in their relative deformed (preloaded) state using one of the following options. • SE3D: The preload condition is maintained on the two sides of the bolt cut using RBE3 elements and a sensor element. This is the default if you defined the bolt axis with the IDIR field on the BOLT entry. • CONT: The preload condition is maintained on the two sides of the bolt cut using a contact condition. This is the default if you leave the IDIR field undefined on the BOLT entry.
FHMETH	For the axisymmetric multiharmonic analysis, selects if the harmonics are coupled or uncoupled, and if the negative harmonics are computed. (Integer; Default=1) 1 = The zero, positive, and negative harmonics are computed and are fully coupled. 2 = The zero, positive, and negative harmonics are computed and are uncoupled. 3 = The zero, positive, and negative harmonics are computed, and positive and negative pairs are coupled. 11 = The zero and positive harmonics are computed and are coupled. 12 = The zero and positive harmonics are computed and are uncoupled. 13 = Same as FHMETH=12. When nonlinear conditions are present (material nonlinear, large displacements, large strain, contact), coupled harmonics becomes more important to achieve convergence. Some nonlinear problems can still converge with the uncoupled option, but an increase in iterations may be required. The results of a converged coupled analysis or a converged uncoupled analysis will be the same. Options 1 and 2 are the more common choices. The fully coupled option is much more computational than the uncoupled option. For linear problems, the harmonics are uncoupled, and you can reliably use the uncoupled option to improve the performance. If you understand your loading and the resulting deformation, you can consider removing the negative harmonics to reduce the model size with the options 11, 12, and 13. These options can also be used in a normal modes subcase to avoid duplicate modes. Note that you only request the positive harmonics with the FHAR command, and the software automatically computes the negative harmonics based on the FHMETH option. For example, if harmonics FHAR=0, 1, 2 are requested, for FHMETH=1, 2, or 3, the harmonics -2, -1, 0, 1, 2 are computed, and for FHMETH=11, 12, or 13, the harmonics 0, 1, 2 are computed.

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Name	Description
Global parameters for SOLs 402 and 414:	
NSTK	Defines how the software handles redundant constraints. (Integer; Default=0) When grid points are included in a combination of MPC, contact, glue, cyclic symmetry conditions, their degrees-of-freedom become over-constrained because they are dependent on multiple conditions. This can occur, for example, on the edges between contact, glue, and cyclic symmetry conditions. 0 = By default, the software leaves one of the dependencies but sets the others to zero indicated by the redundant Lagrange multiplier listing and by the zero pivot messages in the .f06 file. In general, if a solution runs to completion and converges, you can ignore the redundant Lagrange multiplier messages in the .f06 file. If a solution has convergence issues and possibly errors, you can consider NSTK=1. 1 = The software replaces the redundant conditions with diagonal stiffness terms in the tangent matrix. The result is similar to replacing the hard constraints with spring elements. The software no longer sees the multiple conditions as a conflict and the Lagrange multipliers can remain without the need for setting them to 0.0. For SOL 414 solutions, you can use NLCNTLG,NSTK,1 to avoid redundant constraints that can occur in gluing and cyclic symmetry conditions. Avoiding redundant constraints improves the performance of the simulation, especially in SOL 414,103 and SOL 414,110 solutions. To determine whether redundant constraints exist in your simulation, look for null pivots in the .f06 file. See SOL 402 - Redundant constraints in the *Multi-Step Nonlinear User's Guide*.
RESO	Solver type. See Remark 1. (Integer 1, 5, or 6; Default = 6). 1 = Skyline 5 = Sparse (supported with sequential or SMP) 6 = MUMPS (supported with sequential, SMP, and DMP)
INLY	Requests the symmetric or unsymmetric solver for statics, dynamics, and preload subcases (Integer 0 or 1; Default = 0). 0 = Symmetric solver is selected. 1 = Unsymmetric solver is selected. The complex modes subcase always uses the unsymmetric solver. The buckling, normal modes, cyclic normal modes, and Fourier normal modes subcases always use the symmetric solver.

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Name	Description
	The unsymmetric solver is useful when matrices are actually unsymmetric. For example, contact friction can produce an unsymmetric stiffness. When INLY=0, matrices are forced to be symmetric. When INLY=1, the full matrices are used regardless if unsymmetry exists or not.
KINI	Requests checks of the initial stiffness matrix. (Integer 1 or 2; Default = 2). 1 = The checks are performed. 2 = The checks are not performed. The software does the following checks when KINI=1: • Produces the largest, smallest, and mean diagonal values of the stiffness matrix. If the ratio between the smallest and largest diagonals is greater than 1.E8, numerical problems can occur during the solve. A warning message is printed to the .f06 file. • Finds the degrees-of-freedom in the stiffness matrix that are equal to zero. A zero can indicate, for example, a degree-of-freedom that is not connected properly to the structure, an unconnected degree-of-freedom (that is, a mass defined on a grid point), a beam stiffness of 0, or a Lagrange multiplier that is overly constrained. • Removes the degrees-of-freedom that are not connected and inverts the stiffness matrix. This check identifies degrees-of-freedom with zero pivots. A zero pivot related to a Lagrange multiplier indicates a redundant constraint (two rigid bodies in parallel). A zero pivot related to a grid point corresponds to a kinematic mode, which is subsequently computed. This helps you to verify the boundary conditions.
IREF	Choice of the initial static and/or steady-state computation. See Remark 2. (Integer 0, 1, 10, or 11; Default = 0). 0 = No initial static or steady-state computation. 1 = Initial static computation only. 10 = Steady-state computation only (dynamic subcase). 11 = Initial Static and steady-state computation (dynamic subcase).
Global parameters for SOLs 401 and 402:	
STRCONV	Choice of the stress-strain curves conversion from Engineering to Cauchy. See Remarks 3 and 4. (Integer 0 or 1; Default = 1). 0 = Exact method. 1 = Standard method.

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Name	Description
PLSHSOL	Enables the hydrostatic pressure plasticity switch globally. (Character; Default=NO) NO = The hydrostatic pressure plasticity switch is disabled for all subcases. YES = The hydrostatic pressure plasticity switch is enabled for any preload, statics, or dynamics subcases that also include the parameter setting PLSHUT=YES on the NLCNTL for SOL 401 or NLCNTL2 for SOL 402.
Global parameters for SOL 414:	
BOWCASE	Activates the calculation of unbalances due to a static deformation (*bow case scenario* or *rotor bow effect*). In Remarks related to SOL 414, see Remarks 1, 2, and 3. (Integer 0, 1, or -1; Default = 0). 0 = Bow case is not activated. 1 = Bow case is activated by static and thermal loads. -1 = Bow case is activated by thermal loads only.
MPKTRA	Adapts the stiffness of mapped surfaces, which are glued surfaces and boundary conditions (faces) in cyclic symmetry. (Real >0.0; No default) If NSTK is set to 0 and you define MPKTRA, the software updates NSTK to 1. The unit of MPKTRA is stiffness (Force/Length). You must set MPKTRA high enough to adapt the stiffness, but setting it too high can result in redundant constraints.
Advanced parameters for SOLs 402 and 414:	
PRECPIVO	Threshold to detect zero pivots. (Real >0.0; Default = 1.0E-08).
NKINE	Number of kinematic modes to save as output. (Integer >0; Default=10).
MODEVERS	Lets you go back to a previous behavior of a parameter when the default of this parameter has changed. See Remark 5. (Integer >0; No default).
External restart parameters for SOLs 401 and 402:	
See "SOL 401 Restart" and "SOL 402 Restart" in the *Multi-Step Nonlinear User's Guide* for a detailed description of restarts.	
RSTGEN	For the initial run, requests that restart data be saved for static, dynamic, and preload subcases. SOL 401 saves the restart data in the output OP2 file. For SOL 402, the data is saved in the output OP2 file, the Samcef .sdb and .adb database files, and the Samcef output .u18 file. (Character; Default = NO). YES = Restart data is saved. NO = Restart data is not saved.

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Name	Description
RSTUNIT=*n*	For the restart run, defines the unit number of the external restart file. The unit number must also be specified on an ASSIGN statement that references the physical file name of the .op2. (Integer≥161; Default=161)
RSTFROM =*n*	For the restart run, defines the subcase ID to restart from. This subcase ID was defined in the initial run input file. RSTFROM has no default and must be defined. (Integer>0; No default).
EXEFROM	For the restart run, defines the subcase ID to execute from. This subcase ID is defined in the restart run input file. EXEFROM has no default and must be defined. For SOL 401, it is recommended to define EXEFROM≥RSTFROM. (Integer>0; No default) EXEFROM=RSTFROM is a special case in SOL 401. See "Special case: EXEFROM = RSTFROM" in the *Multi-Step Nonlinear User's Guide*.
MDLVAL	For the restart run, allows you to turn off the model validation check (Character; Default = "YES"). YES = Request model validation. NO = Do not request model validation.

External file generation parameter for SOL 414:

Unlike SOLs 401 and 402, SOL 414 solutions do not use the RSTGEN parameter to generate restart files. SOL 414 solutions use RSTGEN to save intermediate files that are necessary for some post-processing functions.

RSTGEN — In SOL 414 solutions, you can use the RSTGEN parameter to save intermediate files in the following two situations, not related to restarting a solution:

- When the model contains one or more cyclic symmetry sectors, use RSTGEN=YES to keep the intermediate files necessary to recombine the results in 3D.
- When the assembly contains superelements, use RSTGEN=YES to keep the intermediate files necessary to recover results on the original structure.

Remarks:

1. Default for RESO:

Before Simcenter Nastran 2022.1, the sparse solver was used by default for sequential and SMP runs (that is, runs that are not using DMP), and MUMPS was used by default for DMP runs (that is, parallel runs requested with the keyword MPI402). Now all runs use MUMPS by default. You can revert to the previous default behavior by defining the system cell setting SYSTEM(777)=0. You can also explicitly select the solver by defining RESO=2 for sparse or RESO=6 for MUMPS.

By default, the software allocates half of the total memory you specify with the memory keyword to the main executable and the other half to the MUMPS solver. You can adjust the percentage of

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memory allocated to MUMPS with the mumpsmem=*n* keyword, where *n* is an integer between 1 and 99 (default = 50). Particularly if MUMPS does not have enough memory to work in-core.

2. No initial static or steady-state computation are performed by default.

   Initial static computation are recommended if the model contains SPC with non-zero initial imposed displacements or nodal temperature values TEMP(LOAD) that are not equal to the initial temperature values TEMP(INIT) at the beginning of the simulation or of the non-sequentially-dependent subcases.

   Steady-state computation is useful if a simulation contains a dynamic subcase and transient initial velocities conditions TIC V0. The solver will then compute consistent nodal velocities on the whole model if the you only impose them on a few degrees-of-freedom.

3. You use the STRCONV parameter to select the following stress-strain conversion options:

   Standard method:

   In the standard method, we assume small strains and a Poisson's ratio of 0.5 for the plastic part.

Type	Conversion
Engineering Stress → True Stress	$\sigma_i = \sigma_{Ni}(1+\varepsilon_{Ni})$
Engineering Strain → Log (True) Strain	$\varepsilon_i = \ln(1+\varepsilon_{Ni})$

   Exact method:

Type	Conversion
Engineering Stress → True Stress	$\sigma_i = \frac{\sigma_{Ni}}{\left(1+\varepsilon_{Ni}^{L}\right)^2}$
Engineering Strain → Log (True) Strain	$\varepsilon_i = \ln(1+\varepsilon_{Ni})$

   where,

   σ_i and ε_i are the true stress and strain,

   σ_{Ni} and ε_{Ni} are the nominal (engineering) stress and strain defined on the TABLES1 bulk entry,

   and ε_{Ni}^{L} is the lateral nominal strain computed by;

$$\varepsilon_{Ni}^{L} = \varepsilon_{Ni}^{Le} + \varepsilon_{Ni}^{Lp} = -\nu\varepsilon_{Ni}^{e} - \frac{1}{2}\varepsilon_{Ni}^{p}$$

   where,

   ε_{Ni}^{Le} is the lateral elastic nominal strain,

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ε_{Ni}^{Lp} is the lateral plastic nominal strain,

ε_{Ni}^{e} is the elastic nominal strain,

and ε_{Ni}^{p} is the plastic nominal strain.

If plastic strain is defined in the TABLES1 entry, the nominal modulus E_N on the MAT1 entry is used to compute the elastic nominal strain (ε_{Ni}^{e}). If total strain is defined on the TABLES1 entry, the slope of the line joining the first and the second point is used to compute the nominal modulus E_N which is then used to compute the elastic nominal strain.

4. For cast iron plasticity, STRCONV is ignored and different conversion formulas are used for compression and tension curve.

   The conversion method for tensile hardening data in cast iron plasticity:

$$\varepsilon^l = \ln\left(1+\varepsilon^T\right) \sigma = S(1-\nu\varepsilon^e - \nu^p\varepsilon^p)^{-2}$$

   where v^p is the plastic Poisson's ratio in tension defined with the NUPL field on the MPLAS bulk entry.

   The conversion method for compressive hardening data in cast iron plasticity:

$$\varepsilon^l = \left|\ln\left(1-\varepsilon^T\right)\right| \sigma = S\left(1+\nu\varepsilon^e + \frac{1}{2}\varepsilon^p\right)^{-2}$$

   For the compression curve, the engineering total strain ϵ^T should not be larger than 1.0.

5. You can revert back to the defaults of the previous release with the MODEVERS=20191 value.

   Note

   With MODEVERS, you can only revert back to the defaults of the parameters that *have* a different default between two releases of the software.

   Doing so, you will change *all* these defaults: you cannot change the default for one specific parameter.

Remarks related to SOL 414

1. The BOWCASE parameter applies to SOL 414,111 Harmonic Response with a Static Loads, Constraints preload subcase.
2. When BOWCASE is activated (value of 1 or -1), the unbalance load caused by a static deformation is calculated and applied to the rotor, synchronously with the different frequencies.
3. When BOWCASE is set to -1 (thermal loads only), the geometric stiffening still considers all loads defined in the Static Loads, Constraints preload subcase, but the software uses only the bow case caused by the temperature loads to compute the unbalance load.

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4. Define RSTGEN=YES to save intermediate files that are necessary for the following post-processing functions:
   - For cyclic symmetry sectors, allows you to recombine the results in 3D.
   - For SOL 414 superelements, allows you to recover results on the original structure.

Related Topics

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NLPARM

Parameters for Nonlinear Static Analysis Control

Defines a set of parameters for nonlinear static analysis iteration strategy.

Format:

1	2	3	4	5	6	7	8	9	10
NLPARM	ID	NINC	DT	KMETHOD	KSTEP	MAXITER	CONV	INTOUT	
	EPSU	EPSP	EPSW	MAXDIV	MAXQN	MAXLS	FSTRESS	LSTOL	
	MAXBIS				MAXR		RTOLB		

Example:

NLPARM	15	5		ITER					

Fields:

Field	Contents
ID	Identification number. (Integer > 0)
NINC	Number of increments. (0 < Integer < 1000; Default=10 with SOL 106; Default =1 with SOLs 153 and 159 when ANALYSIS case control command = “HEAT”)
DT	Incremental time interval for creep analysis. See Remark 3. (Real ≥ 0.0; Default = 0.0 for no creep)
KMETHOD	Method for controlling stiffness updates. See Remark 4. (Character = “AUTO”, “SEMI”; or “ITER”; Default = “AUTO”)
KSTEP	Number of iterations before the stiffness update for ITER method. See Remark 5. (Integer ≥ 1; Default = 5)
MAXITER	Limit on number of iterations for each load increment. See Remark 6. (Integer > 0; Default = 25)
CONV	Flags to select convergence criteria. See Remark 7. (Character = “U”, “P”, “W”, or any combination; Default = “PW”)
INTOUT	Intermediate output flag. See Remark 8. (Character = “YES”, “NO”, or “ALL”; Default = NO)
EPSU	Error tolerance for displacement (U) criterion. See Remark 17. (Real > 0.0; Default = 1.0E-3)
EPSP	Error tolerance for load (P) criterion. See Remark 17. (Real > 0.0; Usual default = 1.0E-3)

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Field	Contents
EPSW	Error tolerance for work (W) criterion. See Remark 17. (Real > 0.0; Usual default = 1.0E-7)
MAXDIV	Limit on probable divergence conditions per iteration before the solution is assumed to diverge. See Remark 10. (Integer ≠ 0; Default = 3)
MAXQN	Maximum number of quasi-Newton correction vectors to be saved on the database. See Remark 11. (Integer ≥ 0; Default = MAXITER)
MAXLS	Maximum number of line searches allowed for each iteration. See Remark 12. (Integer ≥ 0; Default = 4)
FSTRESS	Fraction of effective stress $\mathrm{FSTRESS} \cdot \bar{\sigma}$ used to limit the subincrement size in the material routines. See Remark 13. (0.0 < Real < 1.0; Default = 0.2)
LSTOL	Line search tolerance. See Remark 12. (0.01 < Real < 0.9; Default = 0.5)
MAXBIS	Maximum number of bisections allowed for each load increment. See Remark 14. (-10 ≤ MAXBIS ≤ 10; Default = 5)
MAXR	Maximum ratio for the adjusted arc-length increment relative to the initial value. See Remark 15. (1.0 ≤ MAXR ≤ 40.0; Default = 20.0)
RTOLB	Maximum value of incremental rotation (in degrees) allowed per iteration to activate bisection. See Remark 16. (Real > 2.0; Default = 20.0)

Remarks:

1. The NLPARM entry is selected by the Case Control command NLPARM = ID. Each solution subcase requires an NLPARM command.
2. In cases of static analysis (DT = 0.0) using Newton methods, NINC is the number of equal subdivisions of the load change defined for the subcase. Applied loads, gravity loads, temperature sets, enforced displacements, etc., define the new loading conditions. The differences from the previous case are divided by NINC to define the incremental values. In cases of static analysis (DT = 0.0) using arc-length methods, NINC is used to determine the initial arc-length for the subcase, and the number of load subdivisions will not be equal to NINC. In cases of creep analysis (DT > 0.0), NINC is the number of time step increments.
3. The unit of DT must be consistent with the unit used on the CREEP entry that defines the creep characteristics. Total creep time for the subcase is DT multiplied by the value in the field NINC; i.e., DT*NINC.
4. The stiffness update strategy is selected in the KMETHOD field.
   - If the AUTO option is selected, the software automatically selects the most efficient strategy based on convergence rates. At each step the number of iterations required to converge is estimated. The stiffness is updated if:
     - The estimated number of iterations to converge exceeds MAXITER.

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- The estimated time required for convergence with the current stiffness exceeds the estimated time required for convergence with updated stiffness.
- The solution diverges.

For additional information, see Remarks 10 and 14.

- If the SEMI option is selected, for each load increment the software:
  - Performs a single iteration based upon the new load.
  - Updates the stiffness matrix.
  - Resumes the normal AUTO option.
- If the ITER option is selected and:
  - KSTEP < MAXITER, the software updates the stiffness at every KSTEP iteration and upon convergence.
  - KSTEP = MAXITER (the Modified Newton-Raphson iteration strategy), the software updates the stiffness only upon convergence.
  - KSTEP > MAXITER, the software does not update the stiffness.
  - KSTEP = 1 (the Newton-Raphson iteration strategy), the software updates the stiffness after each iteration.

5. For AUTO and SEMI options, the stiffness matrix is updated on convergence if KSTEP is less than the number of iterations that were required for convergence with the current stiffness. See Remark 7.

6. The number of iterations for a load increment is limited to MAXITER. If the solution does not converge in MAXITER iterations, the load increment is bisected and the analysis is repeated. If the load increment cannot be bisected (i.e., MAXBIS is attained or MAXBIS = 0) and MAXDIV is positive, the best attainable solution is computed and the analysis is continued to the next load increment. If MAXDIV is negative, the analysis is terminated.

7. The test flags (U = displacement error, P = load equilibrium error, and W = work error) and the tolerances (EPSU, EPSP, and EPSW) define the convergence criteria. All the requested criteria (combination of U, P, and/or W) are satisfied upon convergence.

When KMETHOD=AUTO or SEMI, EPSP is used to determine if the stiffness is updated, even when P is not included in the CONV field. This can have an effect on convergence or results. See “Update Principles” in Section 3.3.1 of the *Simcenter Nastran Handbook of Nonlinear Analysis*.

8. INTOUT controls the output requests for displacements, element forces and stresses, etc. YES or ALL must be specified in order to be able to perform a subsequent restart from the middle of a subcase. The options are summarized in the following two tables.

Table 15-106: For the Newton family of iteration methods (NLPCI command is not specified)

INTOUT	Output Processed
YES or ALL	For every computed load increment. This includes user defined increments and software increments as a result of bisection.
NO	For the last load of the subcase.

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Table 15-107: For the arc-length methods (NLPCI command is specified)

INTOUT	Output Processed
YES	For user-specified load increments.
ALL	For every computed and user-specified load increment.
NO	For the last load of the subcase.

- For arc-length methods (i.e., when the NLPCI command is specified) the computed load increment in general is not going to be equal to the user-specified load increment, and is not known in advance. The option ALL allows the user to obtain solutions at the desired intermediate load increments.

If the YES or ALL options are specified and the output is directed to an .xdb file, an error message is issued and the run terminates. This occurs because the .xdb file is incapable of storing multiple outputs of the same data block. If intermediate results are required, use the YES or ALL option and direct the output to an .op2 file. If intermediate results are not required, use the NO option and direct the output to either an .op2 file or an .xdb file.

9. Displacement convergence is not supported when KMETHOD=ITER and KSTEP=1 are defined (which selects the Full Newton-Raphson method). If displacement convergence is requested ("U" on the CONV field) when KMETHOD=ITER and KSTEP=1 are defined, the program will use work convergence instead (EPSW).

10. The ratio of energy errors before and after the iteration is defined as divergence rate (E^i), i.e.,

$$E^i = \frac{\{\Delta u^i\}^T \{R^i\}}{\{\Delta u^i\}^T \{R^{i-1}\}}$$

Depending on the divergence rate, the number of diverging iteration (NDIV) is incremented as follows:

If $E^i \geq 1$ or $E^i < -10^{12}$, then NDIV=NDIV+2

If $-10^{12} < E^i < -1$, then NDIV=NDIV+1

The solution is assumed to diverge when NDIV ≥ |MAXDIV|. If the solution diverges and the load increment cannot be further bisected (i.e., MAXBIS is attained or MAXBIS is zero), the stiffness is updated based on the previous iteration and the analysis is continued. If the solution diverges again in the same load increment while MAXDIV is positive, the best solution is computed and the analysis is continued to the next load increment. If MAXDIV is negative, the analysis is terminated on the second divergence.

11. The BFGS update is performed if MAXQN > 0. As many as MAXQN quasi-Newton vectors can be accumulated. The BFGS update with these QN vectors provides a secant modulus in the search direction. If MAXQN is reached, no additional ON vectors will be accumulated. Accumulated QN vectors are purged when the stiffness is updated and the accumulation is resumed.

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12. The line search is performed as required, if MAXLS > 0. In the line search, the displacement increment is scaled to minimize the energy error. The line search is not performed if the absolute value of the relative energy error is less than the value specified in LSTOL.

    EPSW is used to determine if line search is performed, even when W is not included in the CONV field. This can have an effect on convergence or results. See "Other Provisions for Line Search" in section 3.4.5 of the *Simcenter Nastran Handbook of Nonlinear Analysis*.

13. The number of subincrements in the material routines (elastoplastic and creep) is determined so that the subincrement size is approximately

    $\mathrm{FSTRESS} \cdot \overline{\sigma}$ (equivalent stress).

    FSTRESS is also used to establish a tolerance for error correction in the elastoplastic material; i.e.,

    $$error\ in\ yield\ function < FSTRESS \cdot \overline{\sigma}$$

    If the limit is exceeded at the converging state, the program will exit with a fatal message. Otherwise, the stress state is adjusted to the current yield surface.

14. The number of bisections for a load increment/arc-length is limited to the absolute value of MAXBIS. Different actions are taken when the solution diverges depending on the sign of MAXBIS. If MAXBIS is positive, the stiffness is updated on the first divergence, and the load is bisected on the second divergence. If MAXBIS is negative, the load is bisected every time the solution diverges until the limit on bisection is reached. If the solution does not converge after |MAXBIS| bisections, the analysis is continued or terminated depending on the sign of MAXDIV. See Remark **10**.

15. MAXR is used in the adaptive load increment/arc-length method to define the overall upper and lower bounds on the load increment/arc-length in the subcase; i.e.,

    $$\frac{1}{\mathrm{MAXR}} \leq \frac{\Delta I_n}{\Delta I_o} \leq \mathrm{MAXR}$$

    where Δl_n is the arc-length at step n and Δl_o is the original arc-length. The arc-length method for load increments is selected by an NLPCI Bulk Data entry. This entry must have the same ID as the NLPARM Bulk Data entry.

16. The bisection is activated if the incremental rotation for any degree-of-freedom ($\Delta\theta_x$, $\Delta\theta_y$, or $\Delta\theta_z$) exceeds the value specified by RTOLB. This bisection strategy is based on the incremental rotation and controlled by MAXBIS.

17. Default tolerance sets are determined based on model type and desired accuracy. Accuracy is under user control and can be specified on the PARAM, NLTOL entry. NLTOL's value is used only if the CONV, EPSU, EPSP and EPSW fields are blank, and if NINC is set to a value of 10 or larger. Otherwise, the NLTOL selection will be overridden. The tables below list tolerances according to NLTOL selections:

Table 15-108: Default Tolerances for Static Nonlinear SOL 106 Models Without Gaps, Contact or Heat Transfer

NLTOL	Designation	CONV	EPSU	EPSP	EPSW
0	Very high	PW	_____	1.0E-3	1.0E-7
1	High	PW	_____	1.0E-2	1.0E-3
2	Engineering	PW	_____	1.0E-2	1.0E-2

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NLTOL	Designation	CONV	EPSU	EPSP	EPSW
3	Prelim Design	PW	_____	1.0E-1	1.0E-1
None	Engineering	PW	_____	1.0E-2	1.0E-2

Table 15-109: Default Tolerances for Static Nonlinear SOL 106 Models With Gaps or Contact (Enter NLTOL Values of 0 or 2 Only or Omit the Parameter)

NLTOL	Designation	CONV	EPSU	EPSP	EPSW
0	Very high	PW	_____	1.0E-3	1.0E-7
2	Engineering	PW	_____	1.0E-3	1.0E-5
None	Engineering	PW	_____	1.0E-3	1.0E-5

Table 15-110: Default Tolerances for Static Nonlinear SOL 106 or 153 Models With Heat Transfer (Enter NLTOL Value of 0 Only or Omit the Parameter)

NLTOL	Designation	CONV	EPSU	EPSP	EPSW
0	Very high	PW	_____	1.0E-3	1.0E-7
None	Very high	PW	_____	1.0E-3	1.0E-7

Related Topics

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NLPCI

Parameters for Arc-Length Methods in Nonlinear Static Analysis

Defines a set of parameters for the arc-length incremental solution strategies in nonlinear static analysis (SOL 106). This entry will be used if a subcase contains an NLPARM command (NLPARM = ID).

Format:

1	2	3	4	5	6	7	8	9	10
NLPCI	ID	TYPE	MINALR	MAXALR	SCALE		DESITER	MXINC	

Example:

NLPCI	10	CRIS	1.0	1.0			12	10	

Fields:

Field	Contents
ID	Identification number of an associated NLPARM entry. (Integer > 0)
TYPE	Constraint type. See Remark 2. (Character: “CRIS”, “RIKS”, or “MRIKS”; Default = “CRIS”)
MINALR	Minimum allowable arc-length adjustment ratio between increments for the adaptive arc-length method. See Remarks 3 and 4. (0.0 < Real ≤ 1.0; Default = 0.25)
MAXALR	Maximum allowable arc-length adjustment ratio between increments for the adaptive arc-length method. See Remarks 3 and 4. (Real ≥ 1.0; Default = 4.0)
SCALE	Scale factor (w) for controlling the loading contribution in the arc-length constraint. (Real ≥ 0.0; Default = 0.0)
DESITER	Desired number of iterations for convergence to be used for the adaptive arc-length adjustment. See Remarks 3 and 4. (Integer > 0; Default = 12)
MXINC	Maximum number of controlled increment steps allowed within a subcase. See Remark 5. (Integer > 0; Default = 20)

Remarks:

1. The NLPCI entry is selected by the Case Control command NLPARM = ID. There must also be an NLPARM entry with the same ID. However, for creep analysis (DT ≠ 0.0 in NLPARM entry), the arc-length methods cannot be activated, and the NLPCI entry is ignored if specified. The NLPCI entry is not recommended for heat transfer analysis in SOL 153.
2. The available constraint types are as follows:

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TYPE = “CRIS”:

$$\left\{u_n^i - u_n^0\right\}\left\{u_n^i - u_n^0\right\} + w^2(\mu^i - \mu^0)^2 = \Delta l_n^2$$

TYPE = “RIKS”:

$$\left\{u_n^i - u_n^{i-1}\right\}^T\left\{u_n^1 - u_n^0\right\} + w^2\Delta\mu^i = 0$$

TYPE = “MRIKS”:

$$\left\{u_n^i - u_n^{i-1}\right\}^T\left\{u_n^{i-1} - u_n^0\right\} + w^2\Delta\mu^i(\mu^{i-1} - \mu^0) = 0$$

where:

w	=	the user-specified scaling factor (SCALE)
μ	=	the load factor
Δl	=	the arc-length

The constraint equation has a disparity in the dimension by mixing the displacements with the load factor. The scaling factor (w) is introduced as user input so that the user can make constraint equation unit-dependent by a proper scaling of the load factor μ. As the value of w is increased, the constraint equation is gradually dominated by the load term. In the limiting case of infinite w, the arc-length method is degenerated to the conventional Newton’s method.

3. The MINALR and MAXALR fields are used to limit the adjustment of the arc-length from one load increment to the next by:

$$\text{MINALR} \le \frac{\Delta l_{\text{new}}}{\Delta l_{\text{old}}} \le \text{MAXALR}$$

The arc-length adjustment is based on the convergence rate (i.e., number of iterations required for convergence) and the change in stiffness. For constant arc-length during analysis, use MINALR = MAXALR = 1.

4. The arc-length Δl for the variable arc-length strategy is adjusted based on the number of iterations that were required for convergence in the previous load increment (I_{max}) and the number of iterations desired for convergence in the current load increment (DESITER) as follows:

$$\Delta l_{new} = \sqrt{\frac{\text{DESITER} \cdot \Delta l_{old}}{I_{max}}}$$

5. The MXINC field is used to limit the number of controlled increment steps in case the solution never reaches the specified load. This field is useful in limiting the number of increments computed for a collapse analysis.

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NLRGAP

Nonlinear Transient Load Proportional to Gap

Defines a nonlinear transient radial (circular) gap.

Format:

1	2	3	4	5	6	7	8	9	10
NLRGAP	SID	GA	GB	PLANE	TABK	TABG	TABU	RADIUS	

Example:

NLRGAP	21	3	4	XY	3	10	6	1.6	

Fields:

Field	Contents
SID	Nonlinear load set identification number. (Integer > 0)
GA	Inner (e.g., shaft) grid for radial gap. (Integer > 0)
GB	Outer (e.g., housing) grid for radial gap. (Integer > 0)
PLANE	Radial gap orientation plane: XY, YZ, or ZX. (Character, Default = XY.)
TABK	Table ID of gap stiffness vs. time. (Integer > 0)Table ID of gap stiffness vs. penetration. (Integer < 0)
TABG	Table ID for radial gap clearance as function of time. (Integer > 0)
TABU	Table ID for radial coefficient of friction as function of time. (Integer > 0)
RADIUS	Shaft radius. (Real ≥ 0.0, Default = 0.0)

Remarks:

1. NLRGAP must be selected with the Case Control command NONLINEAR = SID.
2. Multiple NLRGAP entries with the same SID are allowed.
3. The NLRGAP is not an element, but a nonlinear load similar to the NOLINi Bulk Data entries. It computes the relative displacements of GA and GB in the selected plane and applies appropriate nonlinear loads to simulate the radial contact.
4. The degrees-of-freedom in the XY, YZ, or ZX planes (depending on the PLANE) of GA and GB must be members of the solution set. This means the e-set for modal formulation and the d-set for direct

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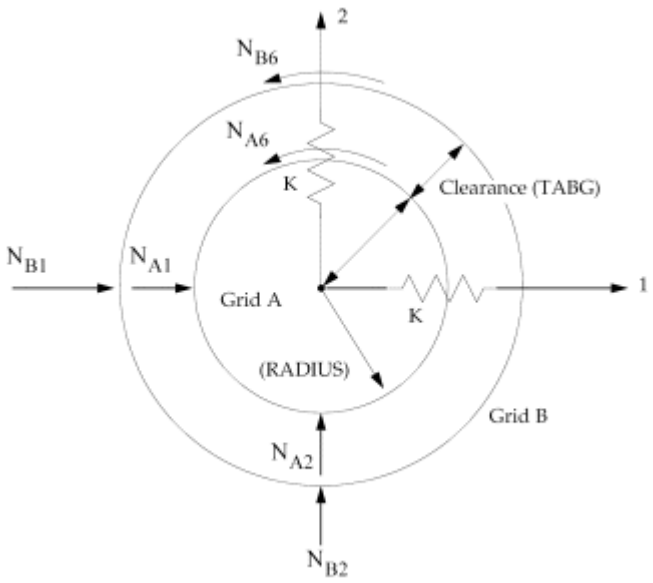
formulation. If RADIUS is > 0.0, then the in-plane rotation degree-of-freedom must also be in the solution set.

5. As with the NOLINi entries, the NLRGAP is limited to use in direct transient response solution sequences.
6. The XY, YZ and ZX planes are relative to the displacement coordinates systems of GA and GB. GA and GB should be coincident grids with parallel displacement coordinate systems. Simcenter Nastran does not check or enforce this. Wrong answers can occur if this rule is not followed.
7. If TABK is negative, the corresponding TID on the referenced TABLEDi entry is input positive and not negative.
8. The shaft radius is used only for the computation of friction induced torque.
9. In the underlying equations, a positive coefficient of friction is consistent with counter-clockwise shaft rotation from axis 1 towards axis 2 (anti-clockwise). A negative coefficient of friction is consistent with clockwise shaft rotation from axis 2 towards axis 1 (clockwise). See **Figure 15-136**.
10. Nonlinear forces for the grids referenced on the NLRGAP can be output with the NLLOAD Case Control command. See **Figure 15-136** for the sign conventions.

Figure 15-136: Radial Gap Orientation and Nonlinear Load Sign Conventions



Related Topics

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NOLIN1

Nonlinear Transient Load as a Tabular Function

Defines nonlinear transient forcing functions of the form

Function of displacement: $P_i(t) = S \cdot T(u_j(t))$

Function of velocity: $P_i(t) = S \cdot T(\dot{u}_j(t))$

where $u_j(t)$ and $\dot{u}_j(t)$ are the displacement and velocity at point GJ in the direction of CJ.

$u_j(t)$ is temperature in SOL 159.

Format:

1	2	3	4	5	6	7	8	9	10
NOLIN1	SID	GI	CI	S	GJ	CJ	TID		

Example:

NOLIN1	21	3	4	2.1	3	10	6		

Fields:

Field	Contents
SID	Nonlinear load set identification number. (Integer > 0)
GI	Grid, scalar, or extra point identification number at which nonlinear load is to be applied. (Integer > 0)
CI	Component number for GI. (0 < Integer ≤ 6; blank or zero if GI is a scalar or extra point.)
S	Scale factor. (Real)
GJ	Grid, scalar, or extra point identification number. (Integer > 0)
CJ	Component number for GJ according to the following table:

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Type of Point	Displacement	Velocity
Grid	1 ≤ Integer ≤ 6	11 ≤ Integer ≤ 16
Scalar	Blank or zero	Integer = 10
Extra	Blank or zero	Integer = 10

TID Identification number of a TABLEDi entry. (Integer > 0)

Remarks:

1. Nonlinear loads must be selected with the Case Control command NONLINEAR = SID.
2. Nonlinear loads may not be referenced on DLOAD entry.
3. All degrees-of-freedom referenced on NOLIN1 entries must be members of the solution set. This means the e-set (EPOINT entry) for modal formulation and the d-set for direct formulation.
4. Nonlinear loads as a function of velocity (Equation 2) are denoted by components ten greater than the actual component number; i.e., a component of 11 is component 1 (velocity). The velocity is determined by

$$\dot{u}_{j,t} = \frac{u_{j,t} - u_{j,t-1}}{\Delta t}$$

where Δt is the time step interval and $u_{j,t-1}$ is the displacement of GJ-CJ for the previous time step.

5. See the *Advanced Dynamic Analysis User's Guide* for information on using NOLINi bulk entries in solutions 109 and 112. See the *Handbook of Nonlinear Analysis* for information on using NOLINi bulk entries in solution 129. See the *Thermal Analysis User's Guide* for information on using NOLINi bulk entries in solution 159.

Related Topics

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NOLIN2

Nonlinear Transient Load as the Product of Two Variables

Defines nonlinear transient forcing functions of the form

$$P_i(t) = S \cdot X_j(t) \cdot X_k(t)$$

where $X_j(t)$ and $X_k(t)$ can be either displacement or velocity at points GJ and GK in the directions of CJ and CK.

$X_j(t)$ and $X_k(t)$ are temperatures in SOL 159.

Format:

1	2	3	4	5	6	7	8	9	10
NOLIN2	SID	GI	CI	S	GJ	CJ	GK	CK	

Example:

NOLIN2	14	2	1	2.9	2	1	2		

Fields:

Field	Contents
SID	Nonlinear load set identification number. (Integer > 0)
GI	Grid, scalar, or extra point identification number at which nonlinear load is to be applied. (Integer > 0)
CI	Component number for GI. (0 ≤ Integer ≤ 6; blank or zero if GI is a scalar or extra point.)
S	Scale factor. (Real)
GJ, GK	Grid, scalar, or extra point identification number. (Integer > 0)
CJ, CK	Component number for GJ, GK according to the following table:

Type of Point	Displacement	Velocity
Grid	1 ≤ Integer ≤ 6	11 ≤ Integer ≤ 16
Scalar	Blank or zero	Integer = 10
Extra	Blank or zero	Integer = 10

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Remarks:

1. Nonlinear loads must be selected with the Case Control command NONLINEAR=SID.
2. Nonlinear loads may not be referenced on a DLOAD entry.
3. All degrees-of-freedom referenced on NOLIN2 entries must be members of the solution set. This means the e-set for modal formulation and the d-set for direct formulation.
4. GI-CI, GJ-CJ, and GK-CK may be the same point.
5. Nonlinear loads may be a function of displacement ($X = u$) or velocity ($X = \dot{u}$). Velocities are denoted by a component number ten greater than the actual component number; i.e., a component of 10 is component 0 (velocity). The velocity is determined by

$$\dot{u}_t = \frac{u_t - u_{t-1}}{\Delta t}$$

   where $\Delta t$is the time step interval and u_{t-1} is the displacement of GJ-CJ or GK-CK for the previous time step.
6. See the *Advanced Dynamic Analysis User's Guide* for information on using NOLINi bulk entries in solutions 109 and 112. See the *Handbook of Nonlinear Analysis* for information on using NOLINi bulk entries in solution 129. See the *Thermal Analysis User's Guide* for information on using NOLINi bulk entries in solution 159.

Related Topics

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NOLIN3

Nonlinear Transient Load as a Positive Variable Raised to a Power

Defines nonlinear transient forcing functions of the form

$$P_i(t) = \begin{cases} S \cdot [X_j(t)]^A, X_j(t) > 0 \\ 0, X_j(t) \le 0 \end{cases}$$

where X_j (t) may be a displacement or a velocity at point GJ in the direction of CJ.

$X_j(t)$ is temperature in SOL 159.

Format:

1	2	3	4	5	6	7	8	9	10
NOLIN3	SID	GI	CI	S	GJ	CJ	A		

Example:

NOLIN3	4	102		-6.1	2	15	-3.5		

Fields:

Field	Contents
SID	Nonlinear load set identification number. (Integer > 0)
GI	Grid, scalar, or extra point identification number at which the nonlinear load is to be applied. (Integer > 0)
CI	Component number for GI. (0 ≤ Integer ≤ 6; blank or zero if GI is a scalar or extra point.)
S	Scale factor. (Real)
GJ	Grid, scalar, extra point identification number. (Integer > 0)
CJ	Component number for GJ according to the following table:

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Type of Point	Displacement	Velocity
Grid	1 ≤ Integer ≤ 6	11 ≤ Integer ≤ 16
Scalar	Blank or zero	Integer = 10
Extra	Blank or zero	Integer = 10

Field	Contents
A	Exponent of the forcing function. (Real)

Remarks:

1. Nonlinear loads must be selected with the Case Control command NONLINEAR = SID.
2. Nonlinear loads may not be referenced on a DLOAD entry.
3. All degrees-of-freedom referenced on NOLIN3 entries must be members of the solution set. This means the e-set for modal formulation and the d-set for direct formulation.
4. Nonlinear loads may be a function of displacement (X_j=u_j) or velocity (X_j=u_j). Velocities are denoted by a component number ten greater than the actual component number; e.g., a component of 16 is component 6 (velocity). The velocity is determined by

$$\dot{u}_{j,t} = \frac{u_{j,t} - u_{j,t-1}}{\Delta t}$$

   where Δt is the time step interval and $u_{j,t-1}$ is the displacement of GJ-CJ for the previous time step.
5. Use a NOLIN4 entry for the negative range of X_j (t).
6. See the *Advanced Dynamic Analysis User's Guide* for information on using NOLINi bulk entries in solutions 109 and 112. See the *Handbook of Nonlinear Analysis* for information on using NOLINi bulk entries in solution 129. See the *Thermal Analysis User's Guide* for information on using NOLINi bulk entries in solution 159.

Related Topics

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NOLIN4

Nonlinear Transient Load as a Negative Variable Raised to a Power

Defines nonlinear transient forcing functions of the form

$$P_i(t) = \begin{cases} -S \cdot [-X_j(t)]^A, X_j(t) < 0 \\ 0, X_j(t) \geq 0 \end{cases}$$

where $X_j(t)$ may be a displacement or a velocity at point GJ in the direction of CJ.

$X_j(t)$ is temperature in SOL 159.

Format:

1	2	3	4	5	6	7	8	9	10
NOLIN4	SID	GI	CI	S	GJ	CJ	A		

Example:

NOLIN4	2	4	6	2.0	101		16.3		

Fields:

Field	Contents
SID	Nonlinear load set identification number. (Integer > 0)
GI	Grid, scalar, or extra point identification number at which nonlinear load is to be applied. (Integer > 0)
CI	Component number for GI. (0 < Integer ≤ 6; blank or zero if GI is a scalar or extra point.)
S	Scale factor. (Real)
GJ	Grid, scalar, or extra point identification number. (Integer > 0)
CJ	Component number for GJ according to the following table:

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Type of Point	Displacement	Velocity
Grid	1 ≤ Integer ≤ 6	11 ≤ Integer ≤ 16
Scalar	Blank or zero	Integer = 10
Extra	Blank or zero	Integer = 10

Field	Contents
A	Exponent of forcing function. (Real)

Remarks:

1. Nonlinear loads must be selected with the Case Control command NONLINEAR = SID.
2. Nonlinear loads may not be referenced on a DLOAD entry.
3. All degrees-of-freedom referenced on NOLIN4 entries must be members of the solution set. This means the e-set for modal formulation and the d-set for direct formulation.
4. Nonlinear loads may be a function of displacement (X_j=u_j) or velocity X_j=u_j). Velocities are denoted by a component number ten greater than the actual component number; i.e., a component of 10 is component 0 (velocity). The velocity is determined by

$$\dot{u}_{j,\,t} = \frac{u_{j,\,t} - u_{j,\,t-1}}{\Delta t}$$

   where Δt is the time step interval and $u_{j,t-1}$ is the displacement of GJ-CJ for the previous time step.
5. Use a NOLIN3 entry for the positive range of $X_j(t)$.
6. See the *Handbook of Nonlinear Analysis* for more information on NOLINi bulk entries.
7. See the *Advanced Dynamic Analysis User's Guide* for information on using NOLINi bulk entries in solutions 109 and 112. See the *Handbook of Nonlinear Analysis* for information on using NOLINi bulk entries in solution 129. See the *Thermal Analysis User's Guide* for information on using NOLINi bulk entries in solution 159.

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NSM

Define Non-Structural Mass

Defines a set of non-structural mass by ID.

Format:

1	2	3	4	5	6	7	8	9	10
NSM	SID	TYPE	ID	VALUE	ID	VALUE	ID	VALUE	
	ID	VALUE	ID	VALUE	etc.				

Example:

NSM	11	PSHELL	16	0.25					

Fields:

Field	Contents
SID	Non-structural mass set identification number. (Integer > 0)
TYPE	Set points to either property entries or element entries. Properties are: PBAR, PBARL, PBEAM, PBEAML, PBCOMP, PCOMP, PCOMPG, PCOMPG1, PPLANE, PROD, CONROD, PBEND, PSHEAR, PSHELL, PTUBE, and PRAC2D. ELEMENT list of individual element IDs of element that can have NSM. (Character)
ID	Property or Element ID. (Integer > 0)
VALUE	Non-structural mass per unit area for 2D elements, or non-structural mass per unit length for 1D elements. (Real)

Remarks:

1. Non-structural mass sets must be selected with Case Control command NSM = SID.
2. The ELSUM Case Control command will give a summary of both structural and non-structural mass by element or property type.
3. If you specify non-structural mass on a property entry and also specify it on a NSM bulk entry (or a NSMADD bulk entry that lists the NSM bulk entry) that references the property entry, the software sums the non-structural mass value on the property entry and the non-structural mass value on the NSM bulk entry.

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NSM1

Define Non-Structural Mass

Alternate form for NSM entry. Defines non-structural mass entries by VALUE, ID list.

Format:

1	2	3	4	5	6	7	8	9	10
NSM1	SID	TYPE	VALUE	ID	ID	ID	ID	ID	
	ID	ID	ID	etc.					

Example:

NSM1	11	ELEMENT	0.46	1243	1532				

Alternate forms:

NSM1	SID	TYPE	VALUE	ID	THRU	ID			

NSM1	SID	TYPE	VALUE	ALL					

NSM1	SID	TYPE	VALUE	ID	THRU	ID	BY	N	

Fields:

Field	Contents
SID	Non-structural mass set identification number. (Integer > 0)
TYPE	Set points to either property entries or element entries. Properties are: PBAR, PBARL, PBEAM, PBEAML, PBCOMP, PCOMP, PCOMPG, PCOMPG1, PPLANE, PROD, CONROD, PBEND, PSHEAR, PSHELL, PTUBE, and PRAC2D. ELEMENT list of individual element IDs of element that can have NSM. (Character)
ID	Property or Element ID. (Integer > 0 or “ALL” or “THRU” or “BY” or N (the BY increment))
VALUE	Non-structural mass per unit area for 2D elements, or nonstructural mass per unit length for 1D elements. (Real)

Remarks:

1. Non-structural mass sets must be selected with Case Control command NSM = SID.

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2. Defining TYPE = PSHELL, PCOMP, PCOMPG, PCOMPG1, or PPLANE applies mass to all of these types.
   Defining TYPE = PBEAM, PBEAML, or PBCOMP applies mass to all of these types.
   Defining TYPE = PBAR or PBARL applies mass to both of these types.

   For example, the input "NSML1,12,PSHELL,12.5, ALL" will apply non-structural mass to the elements associated to all PSHELL, all PCOMP, all PCOMPG, all PCOMPG1, and all PPLANE properties.
3. The ELSUM Case Control command will give a summary of both structural and non-structural mass by element or property type.
4. If you specify non-structural mass on a property entry and also specify it on a NSM1 bulk entry (or a NSMADD bulk entry that lists the NSM1 bulk entry) that references the property entry, the software sums the non-structural mass value on the property entry and the non-structural mass value on the NSM1 bulk entry.

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NSMADD

Define Non-Structural Mass Set Combination

Non-structural mass set combination. Defines non-structural mass as the sum of the sets listed.

Format:

1	2	3	4	5	6	7	8	9	10
NSMADD	SID	S1	S2	S3	S4	S5	S6	S7	
	S8	S9	S10	etc.					

Example:

NSMADD	7	30	31	32	33				

Fields:

Field	Contents
SID	Non-structural mass set identification number. (Integer > 0)
Si	Identification numbers of non-structural mass sets defined via NSM, NSM1, NSML, or NSML1 entries. (Integer > 0)

Remarks:

1. The non-structural mass sets must be selected with the Case Control command NSM = SID.
2. No Si may be the identification number of a non-structural mass set defined by another NSMADD entry.
3. The NSMADD bulk entry take precedence over NSM, NSM1, NSML, or NSML1 entries. If both have the same set ID, only the NSMADD entry will be used.
4. The ELSUM Case Control command will give a summary of both structural and non-structural mass by element or property type.

Related Topics

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NSML

Define Lumped Non-Structural Mass

Defines a set of lumped non-structural mass by ID.

Format:

1	2	3	4	5	6	7	8	9	10
NSML	SID	TYPE	ID	VALUE	ID	VALUE	ID	VALUE	
	ID	VALUE	ID	VALUE	etc.				

Example:

NSML	23	PSHELL	6	0.22					

Fields:

Field	Contents
SID	Non-structural mass set identification number. (Integer > 0)
TYPE	Set points to either property entries or element entries. Properties are: PBAR, PBARL, PBEAM, PBEAML, PBCOMP, PCOMP, PCOMPG, PCOMPG1, PPLANE, PROD, CONROD, PBEND, PSHEAR, PSHELL, PTUBE, and PRAC2D. ELEMENT list of individual element IDs of element that can have NSM. (Character)
ID	Property or Element ID. (Integer > 0)
VALUE	A lumped mass value to be distributed. See Remark 2. (Real)

Remarks:

1. If TYPE = ELEMENT is used, line element (CBAR, CBEAM, CBEND, CROD, CTUBE, and CONROD) IDs cannot be mixed with area element (CQUAD4, CQUAD8, CQUADR, CTRIA3, CTRIA6, CTRIAR, CSHEAR, and CRAC2D) IDs.
2. VALUE is the total non-structural mass to be distributed across all IDs.
   - Area element calculation (for example, CQUAD4):
     Mass for a particular element = (Element area) x (VALUE / ∑Element area for all IDs)
   - Line element calculation (for example, CBEAM):
     Mass for a particular element = (Element length) x (VALUE / ∑Element length for all IDs)
3. Non-structural mass sets must be selected with the case control command NSM = SID.

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4. The ELSUM case control command will give a summary of both structural and non-structural mass by element or property type.
5. If you specify non-structural mass on a property entry and also specify it on an NSML bulk entry (or an NSMADD bulk entry that lists the NSML bulk entry) that references the property entry, the software sums the non-structural mass value on the property entry and the non-structural mass value on the NSML bulk entry.

Related Topics

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NSML1

Define Lumped Non-Structural Mass

Alternate form for NSML entry. Defines lumped non-structural mass entries by VALUE, ID list.

Format:

1	2	3	4	5	6	7	8	9	10
NSML1	SID	TYPE	VALUE	ID	ID	ID	ID	ID	
	ID	ID	ID	etc.					

Example:

NSML1	13	ELEMENT	0.72	156	286				

Alternate forms:

1	2	3	4	5	6	7	8	9	10
NSML1	SID	TYPE	VALUE	ID1	THRU	ID2			

1	2	3	4	5	6	7	8	9	10
NSML1	SID	TYPE	VALUE	ALL					

1	2	3	4	5	6	7	8	9	10
NSML1	SID	TYPE	VALUE	ID1	THRU	ID2	BY	N	

Fields:

Field	Contents
SID	Non-structural mass set identification number. (Integer > 0)
TYPE	Set points to either property entries or element entries. Properties are: PBAR, PBARL, PBEAM, PBEAML, PBCOMP, PCOMP, PCOMPG, PCOMPG1, PPLANE, PROD, CONROD, PBEND, PSHEAR, PSHELL, PTUBE, and PRAC2D. ELEMENT list of individual element IDs of element that can have NSM. (Character)
ID	Property or Element ID. (Integer > 0 or “ALL” or “THRU” (ID1 < ID2 for “THRU” option) or “BY” or N (N > 0))
VALUE	A lumped mass value to be distributed. See Remark 2. (Real)

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Remarks:

1. If TYPE = ELEMENT is used, line element (CBAR, CBEAM, CBEND, CROD, CTUBE, and CONROD) IDs cannot be mixed with area element (CQUAD4, CQUAD8, CQUADR, CTRIA3, CTRIA6, CTRIAR, CSHEAR, and CRAC2D) IDs.
2. VALUE is the total non-structural mass to be distributed across all IDs.
   - Area element calculation (for example, CQUAD4):
     Mass for a particular element = (Element area) x (VALUE / ∑Element area for all IDs)
   - Line element calculation (for example, CBEAM):
     Mass for a particular element = (Element length) x (VALUE / ∑Element length for all IDs)
3. Alternate input forms using THRU, BY, and N can be included on different continuation lines except for ALL. ALL can only be used on a single line input.
4. Non-structural mass sets must be selected with the case control command NSM = SID.
5. Defining TYPE = PSHELL, PCOMP, PCOMPG, PCOMPG1, or PPLANE applies mass to all of these types.
   Defining TYPE = PBEAM, PBEAML, or PBCOMP applies mass to all of these types.
   Defining TYPE = PBAR or PBARL applies mass to both of these types.

   For example, the input "NSML1,12,PSHELL,12.5, ALL" will apply non-structural mass to the elements associated to all PSHELL, all PCOMP, all PCOMPG, all PCOMPG1, and all PPLANE properties.
6. The ELSUM case control command will give a summary of both structural and non-structural mass by element or property type.
7. If you specify non-structural mass on a property entry and also specify it on a NSML1 bulk entry (or a NSMADD bulk entry that lists the NSML1 bulk entry) that references the property entry, the software sums the non-structural mass value on the property entry and the non-structural mass value on the NSML1 bulk entry.

Related Topics

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16. Bulk Entries O—P

OMEGA

Rotational Speed Definition

Defines a set of rotational speeds at which to perform a SOL 414,110 complex modal analysis.

Format:

1	2	3	4	5	6	7	8	9	10
OMEGA	SID	W1	W2	W3	W4	W5	W6	W7	
	W8	W9	-etc.-						

Example:

OMEGA	5	0.0	500.0	1000.0	1500.0	2000.0			

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
Wi	Rotational speed in rpm. (Real ≥ 0.0; No default)

Remarks:

1. The OMEGA bulk entry is used when the RSTEP field on the ROTORD bulk entry is set to 0.0.
2. Include an OMEGA = SID specification in the case control to select an OMEGA bulk entry with the same SID. If multiple OMEGA, OMEGA1, and OMEGA2 bulk entries are present with the same SID, the rotational speeds that the software uses are the union of the sets defined by each of the OMEGA, OMEGA1, and OMEGA2 bulk entries.

   From the set of rotational speeds, the software automatically searches for duplicate values. The software considers rotational speeds to be duplicates if the rotational speeds satisfy the following inequality:

   $$|W_n - W_{n-1}| < DFREQ \cdot |W_{\max} - W_{\min}|$$

   where:

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W_n, W_{n-1} Adjacent rotational speeds when the rotational speeds are ordered according to magnitude

$DFREQ$ Value for the DFREQ parameter (default = 10^{-5})

$W_{\min}, W_{\max}$ The minimum and maximum rotational speeds in the set of rotational speeds

When a duplicate is found, the software keeps the lower rotational speed in the pair and removes the higher rotational speed.

Related Topics

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OMEGA1

Rotational Speed Definition, Alternate Form 1

Defines a set of rotational speeds at which to perform a SOL 414,110 complex modal analysis from starting speed, speed increment, and number of speed increments specifications.

Format:

1	2	3	4	5	6	7	8	9	10
OMEGA1	SID	W1	DW	NDW					

Example:

OMEGA1	5	0.0	500.0	5					

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
W1	First rotational speed in rpm. (Real ≥ 0.0; No default)
DW	Rotational speed increment. (Real ≥ 0.0; No default)
NDW	Number of rotational speed increments. (Integer > 0; Default = 1)

Remarks:

1. The OMEGA1 bulk entry is used when the RSTEP field on the ROTORD bulk entry is set to 0.0.
2. The rotational speeds defined by this entry are given by:

   $$\omega_i = W1 + DW\,(i - 1)$$

   where $i = 1, 2, \ldots\ldots, (1 + NDW)$.
3. Include an OMEGA = SID specification in the case control to select an OMEGA1 bulk entry with the same SID. If multiple OMEGA, OMEGA1, and OMEGA2 bulk entries are present with the same SID, the rotational speeds that the software uses are the union of the sets defined by each of the OMEGA, OMEGA1, and OMEGA2 bulk entries.

   From the set of rotational speeds, the software automatically searches for duplicate values. The software considers rotational speeds to be duplicates if the rotational speeds satisfy the following inequality:

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$$|W_n - W_{n-1}| < DFREQ \cdot |W_{\max} - W_{\min}|$$

where:

W_n, W_{n-1}	Adjacent rotational speeds when the rotational speeds are ordered according to magnitude
$DFREQ$	Value for the DFREQ parameter (default = 10^{-5})
$W_{\min}, W_{\max}$	The minimum and maximum rotational speeds in the set of rotational speeds

When a duplicate is found, the software keeps the lower rotational speed in the pair and removes the higher rotational speed.

Related Topics

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OMEGA2

Rotational Speed Definition, Alternate Form 2

Defines a set of rotational speeds at which to perform a SOL 414,110 complex modal analysis from starting speed, final speed, and number of logarithmic speed increments specifications.

Format:

1	2	3	4	5	6	7	8	9	10
OMEGA1	SID	W1	W2	NW					

Example:

OMEGA1	5	1.0	8.0	6					

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
W1	First rotational speed in rpm. (Real > 0.0; No default)
W2	Last rotational speed in rpm. (Real > 0.0; W2 > W1; No default)
NW	Number of logarithmic rotational speed increments. (Integer > 0; Default = 1)

Remarks:

1. The OMEGA2 bulk entry is used when the RSTEP field on the ROTORD bulk entry is set to 0.0.
2. The rotational speeds defined by this entry are given by:

$$\omega_i = W1 \cdot e^{(i-1)d}$$

where:

$$i = 1, 2, \ldots\ldots, (1 + NW)$$

$$d = \left(\frac{1}{NW}\right) \ln\left(\frac{W2}{W1}\right)$$

For the example, the software calculates the rotational speeds to be 1.0, 1.4142, 2.0, 2.8284, 4.0, 5.6569 and 8.0 rpm.

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3. Include an OMEGA = SID specification in the case control to select an OMEGA2 bulk entry with the same SID. If multiple OMEGA, OMEGA1, and OMEGA2 bulk entries are present with the same SID, the rotational speeds that the software uses are the union of the sets defined by each of the OMEGA, OMEGA1, and OMEGA2 bulk entries.

   From the set of rotational speeds, the software automatically searches for duplicate values. The software considers rotational speeds to be duplicates if the rotational speeds satisfy the following inequality:

$$|W_n - W_{n-1}| < DFREQ \cdot |W_{\max} - W_{\min}|$$

   where:

W_n, W_{n-1}	Adjacent rotational speeds when the rotational speeds are ordered according to magnitude
$DFREQ$	Value for the DFREQ parameter (default = 10^{-5})
$W_{\min}, W_{\max}$	The minimum and maximum rotational speeds in the set of rotational speeds

   When a duplicate is found, the software keeps the lower rotational speed in the pair and removes the higher rotational speed.

Related Topics

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OMIT

Omitted Degrees-of-Freedom

Defines degrees-of-freedom to be excluded (o-set) from the analysis set (a-set).

Format:

1	2	3	4	5	6	7	8	9	10
OMIT	ID1	C1	ID2	C2	ID3	C3	ID4	C4	

Example:

OMIT	16	2	23	3516			1	4	

Fields:

Field	Contents
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component numbers. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points; zero or blank for scalar points.)

Remarks:

1. The degrees-of-freedom specified on this entry form members of the mutually exclusive o-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
2. Up to 24 degrees-of-freedom may be specified on a single entry.
3. In many cases it may be more convenient to use OMIT1, ASET, or ASET1 entries.
4. In nonlinear analysis, degrees-of-freedom attached to nonlinear elements must be placed in the a-set. In other words, nonlinear degrees-of-freedom must not be specified on OMIT or OMIT1 entries.

Related Topics

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OMIT1

Omitted Degrees-of-Freedom, Alternate Form 1

Defines degrees-of-freedom to be excluded (o-set) from the analysis set (a-set).

Format:

1	2	3	4	5	6	7	8	9	10
OMIT1	C	G1	G2	G3	G4	G5	G6	G7	
	G8	G9	G10	-etc.-					

Example:

OMIT1	3	2	1	3	10	9	6	5	
	7	8							

Alternate Format and Example:

OMIT1	C	G1	"THRU"	G2					
OMIT1	0	17	THRU	109					

Fields:

Field	Contents
C	Component numbers. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points; zero or blank for scalar points.)
Gi	Grid or scalar point identification number. (Integer > 0; for "THRU" option, G1 < G2.)

Remarks:

1. The degrees-of-freedom specified on this entry form members of the mutually exclusive o-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
2. If the alternate format is used, not all points in the range G1 through G2 have to be defined. Undefined points will collectively produce a warning message but will otherwise be ignored.
3. In nonlinear analysis, degrees-of-freedom attached to nonlinear elements must be placed in the a-set. In other words, nonlinear degrees-of-freedom must not be specified on OMIT or OMIT1 entries.

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Related Topics

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OUTMGT

XY Plot Definition

Defines an XY plot request for nodes and elements in SOL 402, SOL 414,103 Eigenvalues and Superelement Reduction with a Nonlinear Preload subcase, SOL 414,111 Harmonic Response, or a SOL 414,129 Transient Response solution. XY plot requests for bearings and bushings apply only to CBEAR2 and CBUSH2 elements.

Format:

1	2	3	4	5	6	7	8	9	10
OUTMGT	ID	TYPE	SID	RES	COMP	LOC		CID	
	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3	PARAM4	VALUE4	
	PARAM5	VALUE5	-etc-						

Examples:

OUTMGT	100	NODE	654	ACCEL	Y			103	

OUTMGT	100	ELEGRP	458	STRESS	VM	TBM			

The following OUTMGT example for a CBEAR2 element requests the following output:

- Stiffness coefficients K11 and K22
- Damping coefficients C11 and C22
- Relative displacements (all components)
- Relative velocities in translation
- Force in the X direction and moment in the X direction
- Relative displacement elliptic orbit parameters (in a SOL 414,111 Harmonic Response analysis)
- Radial displacement and force (in a SOL 414,129 Transient Response analysis)

OUTMGT	100	ELEM	25	CBEAR2					
	K11	1	K22	1	C11	1	C22	1	
	DISP	123456	VELOC	123	FORCE	14	ORBITD	1	
	RDISP	1	RFORCE	1					

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Fields:

Field	Contents
ID	Identification number of the OUTMGT bulk entry. This number identifies a set of OUTMGT bulk entries that are defined with the same ID. Add this ID to the OUTMGT case control command. This allows you to create multiple OUTMGT bulk entries per one OUTMGT case control command. (Integer > 0; No default)
TYPE	Type of identification number in the SID field. (Character: "NODE", "NODGRP"; "ELEM", or "ELEGRP"; No default) "NODE" for a grid point identification number. "NODGRP" for a group of grid point identification numbers. "ELEM" for an element identification number. See Remark 10. "ELEGRP" for a group of element identification numbers.
SID	Identification number of grid point, grid point group, element, or element group for which to create the plot or plots. (Integer > 0) See Remarks 9 and 10.
RES	Result type to plot as a function of frequency or time. (Character: "DISP", "VELOC", "ACCEL", "REAC", "ORBITP", "STRESS", "FORCE", "CBEAR2", "CBUSH2", "CJOINT"; No default). See Remark 2. "DISP" for displacement. Valid when TYPE field is set to "NODE" or "NODGRP". "VELOC" for velocity. Valid when TYPE is set to "NODE" or "NODGRP". "ACCEL" for acceleration. Valid when TYPE is set to "NODE" or "NODGRP". "REAC" for force and moment reactions. Valid when TYPE is set to "NODE" or "NODGRP". "ORBITP" for semi-major axis, semi-minor axis, and attitude angle of an elliptic orbit. Valid when TYPE is set to "NODE" or "NODGRP". See Remark 3. "STRESS" for stresses on elements. Valid when TYPE is set to "ELEM" or "ELEGRP". See Remarks 4 and 10. "FORCE" for forces on elements. Valid when TYPE is set to "ELEM" or "ELEGRP". See Remarks 4 and 10. "CBEAR2" for results on bearing elements. Valid when TYPE is set to "ELEM" or "ELEGRP". See Remark 4.

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Field	Contents
	"CBUSH2" for results on bearing elements. Valid when TYPE is set to "ELEM" or "ELEGRP". See Remark 4.
	"CJOINT" for results on kinematic joints. Valid when TYPE is set to "ELEM" or "ELEGRP". See Remark 11.
COMP	Component to plot. See Remark 5. (Character: "X", "Y", "Z", "RX", "RY", "RZ", "ALL", "ALLTRANS", or "ALLROT"; No default)
	"X", "Y", "Z", "ALL", or "ALLTRANS" for displacement, velocity, acceleration, reaction, or force components.
	"RX", "RY", "RZ", "ALL", or "ALLROT" for angular displacement, angular velocity, angular acceleration, or moment components.
	"XX", "XY", "XZ", "YZ", "ZZ", "VM", "ALL" for stress results on components
	Leave blank when RES is set to "CBEAR2", "CBUSH2", or "CJOINT".>
LOC	Location of the stress results on 2D shell elements. (Character: "T", "B", "M", "TB", "TM", "MB", "TBM"; No default)
	"T" for top locations.
	"B" for bottom locations.
	"M" for middle locations.
	"TB" for top and bottom locations.
	"TM" for top and middle locations.
	"MB" for middle and bottom locations.
	"TBM" for top, middle, and bottom locations.
CID	Coordinate system identification number for grid point results. (Integer ≥ 0; Default = 0 or blank for the basic coordinate system)
	Leave blank when TYPE is set to "ELEM" or "ELEMGRP".
PARAMi	Name of the parameter. Allowable names are given in the parameter listing below. (Character; No default)
VALUEi	Value of the parameter. (Real, Integer, or Character)

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Parameters:

CBEAR2, CBUSH2, and CJOINT.

Note
OUTMGT does not support CBEAR or CBUSH.

Name	Description	Allowable Values
Kij	Stiffness coefficient of CBEAR2 element, where i = 1, 2, 3, 4, 5, 6 and j = 1, 2, 3, 4, 5, 6.	"1" plots the stiffness coefficient as a function of frequency or time. "0" omits the plot of the stiffness coefficient.
Cij	Viscous damping coefficient of CBEAR2 element, where i = 1, 2, 3, 4, 5, 6 and j = 1, 2, 3, 4, 5, 6.	"1" plots the viscous damping coefficient as a function of frequency or time. "0" omits the plot of the viscous damping coefficient.
Mij	Mass coefficient of CBEAR2 element, where i = 1, 2, 3, 4, 5, 6 and j = 1, 2, 3, 4, 5, 6.	"1" plots the mass coefficient as a function of frequency or time. "0" omits the plot of the mass coefficient.
DISP	Relative displacements and rotations	Component number 1, 2, 3, 4, 5, or 6 as a function of frequency or time. See Remark 6.
VELOC	Relative velocities and angular velocities.	Component number 1, 2, 3, 4, 5, or 6 as a function of frequency or time. See Remark 6.
FORCE	Relative forces and moments	Component number 1, 2, 3, 4, 5, or 6 as a function of frequency or time. See Remark 6.
RDISP	Radial displacement Applies only to SOL 414,129 for CBEAR2 and CBUSH2.	"1" plots the radial displacement as a function of time. "0" omits the radial displacement plot.
RFORCE	Radial force Applies only to SOL 414,129 for CBEAR2 and CBUSH2.	"1" plots the radial force as a function of time. "0" omits the radial force plot.
ORBITD	Relative displacement elliptic orbit parameters for semi-major axis, semi-minor axis, and attitude angle for CBEAR2 and CBUSH2.	"1" plots relative displacement elliptic orbit parameters as a function of frequency for semi-major axis, semi-minor axis, and attitude angle. "0" omits the plots.

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Name	Description	Allowable Values
	Applies only to SOL 414,111.	
ORBITR	Relative rotation elliptic orbit parameters for semi-major axis, semi-minor axis, and attitude angle for CBEAR2 and CBUSH2. Applies only to SOL 414,111.	"1" plots relative rotation elliptic orbit parameters as a function of frequency or time for semi-major axis, semi-minor axis, and attitude angle. "0" omits the plots.
ORBITF	Relative force elliptic orbit parameters for semi-major axis, semi-minor axis, and attitude angle for CBEAR2 and CBUSH2. Applies only to SOL 414,111.	"1" plots relative force elliptic orbit parameters as a function of frequency for semi-major axis, semi-minor axis, and attitude angle. "0" omits the plots.
ORBITM	Relative moment elliptic orbit parameters for semi-major axis, semi-minor axis, and attitude angle for CBEAR2 and CBUSH2. Applies only to SOL 414,111.	"1" plots relative moment elliptic orbit parameters as a function of frequency for semi-major axis, semi-minor axis, and attitude angle. "0" omits the plots.
DAMAGE	Bearing rupture for CBUSH2 element. Applies only to SOL 414,129 for CBUSH2 only.	"1" plots bearing rupture as a function of time. "0" omits the plots.

Remarks:

1. Use an OUTMGT case control command to select OUTMGT bulk entries. All OUTMGT bulk entries that have the same set identification number as the set identification number listed on the OUTMGT bulk entry are selected.
2. OUTMGT produces output for CBEAR2 and CBUSH2, but not for CBEAR and CBUSH.
3. Specify "ORBITP" to request plots for the semi-major axis, semi-minor axis, and attitude angle of an elliptic orbit as a function of frequency. The elliptic orbit is the locus of points that a grid point on the axis of a rotor passes through as the rotor simultaneously rotates and deforms.

   The "ORBITP" result type is valid for SOL 414,111 only.
4. The "ELEM" and "ELEMGRP" TYPE is valid for SOL 414,111 Harmonic Response, SOL 414,103 Eigenvalue and Superelement Reduction (with a transient preload subcase), and SOL 414,129 Transient Response solutions only.
5. Specify "ALL" to request plots for all translational and rotational components.

   Specify "ALLTRANS" to request plots for all translational components.

   Specify "ALLROT" to request plots for all rotational components.

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6. For CBEAR2, CBUSH2, and CJOINT, translation and rotation components are represented by integers 1 through 6, and may be placed in the field with no embedded blanks.

7. In a SOL 414,103 with a transient response subcase, SOL 414,111, or SOL 414,129 analysis with superelements system run, you can use the SEOUT capability to obtain plots of displacement, velocity, and acceleration as a function of frequency or time for grid points in superelements that are not boundary grid points, without having to define retained grid points. To do so, include an OUTMGT case control command and one or more OUTMGT bulk entries in the input files for the SOL 414,103 superelement creation runs. On the OUTMGT bulk entries, specify the result type and grid point combinations for which you want to create the plots. For the system run, include an SEOUT bulk entry in the input file. During the system run, the software calculates the results for the result type and grid point combinations and writes the results to the result file where they can be accessed by a post-processor.

8. You can use the IA19 parameter on the NLCNTL2 bulk entry to control the output frequency for the OUTMGT request. For example, if you define IA19=1, the results requested with the OUTMGT command are output at all solution steps.

   You can use the IA16 parameter on the NLCNTL2 bulk entry to refine the output frequency for the DISPLACEMENT, VELOCITY, ACCELERATION, and SPCFORCE request (and any other results requests such as STRESS, STRAIN, etc.). The IA16 definition applies to all computed solution steps, including additional steps created by the software as a result of the automatic stepping methods.

   See the IA16 and IA19 parameters on the NLCNTL2 bulk entry for more information.

9. For groups that include both nodes and elements, the solver uses only the required entities and ignores the others. For example, if a group contains elements and nodes and you use the group for a result type that requires only elements, the nodes in the group are ignored.

10. OUTMGT supports the following types of elements.

Type of element	Supported elements	Supported components
1D elements	CBEAM CBAR CROD CONROD	Force: X, Y, Z RX, RY, RZ, ALLTRANS, ALLROT, ALL Stress: VM
2D elements	CQUAD8 CQUAD4 CTRIA6 CTRIA3 CTRIAR CQUADR	Stress: XX, XY, YY, VM, ALL You can also select the top, middle, or bottom location on the shell.
3D elements	CTETRA(4) CTETRA(10) CTETRA(mixed order)	Stress: XX, XY, XZ, YY, YZ, ZZ, VM, ALL

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Type of element	Supported elements	Supported components
	CHEXA(8) CHEXA(20) CPENTA(6) CPENTA(15) CPYRAM(5) CPYRAM(13) CPYRAM(mixed order)	
Kinematic joints	CJOINT	DISP, FORCE, VELOC: For DISP, FORCE, and VELOC, you can select components 1, 2, 3, 4, 5, and 6.

The axisymmetric structural elements (CTRAX3, CTRAX6, CQUADX4, and CQUADX8) are not valid.

If an existing group with invalid elements is used, those elements are ignored.

11. For SOL 414,129 Transient Response analyses, RES can be CJOINT for a kinematic joint element.

Related Topics

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PAABSF

Frequency-Dependent Absorbers Element Property

Defines the properties of a frequency-dependent acoustic absorber element.

Format:

1	2	3	4	5	6	7	8	9	10
PAABSF	PID	TZREID	TZIMID	S	A	B	K	RHOC	

Example:

PAABSF	44	38	47						

Fields:

Field	Contents
PID	Property identification number that matches the identification number of the corresponding CAABSF entry. (Integer > 0)
TZREID	Identification number of a TABLEDi entry that defines the resistance as a function of frequency. The real part of the impedance. See Remark 1 . (Integer > 0)
TZIMID	Identification number of a TABLEDi entry that defines the reactance as a function of frequency. The imaginary part of the impedance. See Remark 1 . (Integer > 0)
S	Impedance scale factor. (Real; Default = 1.0)
A	Area factor when 1 or 2 grid points are specified on the CAABSF entry. (Real > 0.0; Default = 1.0)
B	Equivalent structural damping coefficient. (Real ≥ 0.0; Default = 0.0)
K	Equivalent structural stiffness coefficient. (Real ≥ 0.0; Default = 0.0)
RHOC	Constant used in data recovery for calculating an absorption coefficient. RHO is the media density, and C is the speed of sound in the media. (Real; Default = 1.0)

Remarks:

1. At least one of the four fields TZREID, TZIMID, B, or K must be specified.
2. If only one grid point is specified on the CAABSF entry, then the impedance $Z(f)=Z_R + iZ_I$ is the total impedance at the point. If two grids are specified, then the impedance is the impedance per unit length. If three or four points are specified, then the impedance is the impedance per unit area. $Z_R(f)$=TZREID(f)+B and $Z_I(f)$=TZIMID(f)–$KI(2\pi f)$.

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3. The resistance represents a damper quantity B. The reactance represents a quantity of the type ($\omega M - K/\omega$). The impedance is defined as $Z = p/\dot{u}$ where p is the pressure and $\dot{u}$ is the velocity.

4. The impedance scale factor S is used in computing element stiffness and damping terms as:

$$k = \frac{A}{S} \cdot \frac{2\pi f Z_I(f)}{Z_R^2 + Z_I^2} \int \text{(of shape functions)}$$

$$b = \frac{A}{S} \cdot \frac{Z_R(f)}{Z_R^2 + Z_I^2} \int \text{(of shape functions)}$$

5. The output for the element is specified by the STRESS Case Control command and consists of the resistance, reactance, and absorption coefficient. The absorption coefficient is defined as:

$$a = \frac{4(Z_R/\rho c)}{(Z_R/\rho c + 1)^2 + (Z_I/\rho c)^2}$$

Related Topics

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PAABSF1

Acoustic Impedance or Admittance Element Property

Defines the properties of a frequency-dependent acoustic impedance or admittance element.

Format:

1	2	3	4	5	6	7	8	9	10
PAABSF1	PID	PTYPE	FORM	ZR/ZM	ZI/ZP				

Example:

PAABSF1	100	IMP		-0.01	20.0				

Fields:

Field	Contents
PID	Property identification number that matches the identification number of the corresponding CAABSF entry. (Integer > 0)
PTYPE	Type of the acoustic element property. (Character; "IMP" or "ADM"; Default = IMP) If PTYPE = IMP, define the impedance property. If PTYPE = ADM, define the admittance property.
FORM	Format to define the impedance or admittance. (Character; Default = REAL) If FORM = REAL, define the impedance or admittance in rectangular format (real and imaginary). If FORM = PHASE, define the impedance or admittance in polar format (magnitude and phase).
ZR	Real part of impedance or admittance. (Real or Integer; Default = 0.0) For a constant impedance or admittance, enter a real value. For a frequency-dependent impedance or admittance, enter the ID of the TABLEDi bulk entry that contains the impedance or admittance as a function of frequency. See Remark 2.
ZM	Magnitude of impedance or admittance. (Real or Integer; Default = 0.0)

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Field	Contents
	For a constant impedance or admittance, enter a real value. For a frequency-dependent impedance or admittance, enter the ID of the TABLEDi bulk entry that contains the magnitude as a function of frequency.
ZI	Imaginary part of impedance or admittance. (Real or Integer; Default = 0.0) For a constant impedance or admittance, enter a real value. For a frequency-dependent impedance or admittance, enter the ID of the TABLEDi bulk entry that contains the impedance or admittance as a function of frequency. See Remark 2.
ZP	Phase angle in degrees of impedance or admittance. (Real or Integer; Default = 0.0) For a constant phase angle, enter a real value. For a frequency-dependent phase angle, enter the ID of the TABLEDi bulk entry that contains the phase angle as a function of frequency.

Remarks:

1. CAABSF in combination with PAABSF1 is supported by the FEM Adaptive Order (FEMAO) method.
2. ZR/ZM and ZI/ZP cannot both be equal to zero.

Related Topics

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PACDUCT

Properties of a Duct Section

Defines the geometric properties of a duct section.

Format:

1	2	3	4	5	6	7	8	9	10
PACDUCT	PID	BID	OFFSET	GTYPE	DIM1	DIM2	DID	CID	
	XLOC	YLOC	ZLOC	XVEC	YVEC	ZVEC			

Examples:

PACDUCT	200	300		RECTANG	10.0	10.0			
	10.0	0.0	0.0	1.0	0.0	0.0			

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
BID	ID of a BSURFS bulk entry which selects the fluid element faces where the duct section boundary is applied. (Integer > 0)
OFFSET	Offset of the duct section location. (Real; Default = 0.0)
GTYPE	Boundary shape of the duct section. The options are CIRCULAR, ANNULAR, or RECTANG which define a circular, annular, or rectangular boundary, respectively. (Character; Default=CIRCULAR)
DIM1, DIM2	Defines the duct section dimensions. (Real; No Default) If GTYPE= RECTANG, DIM1 is the width and DIM2 is the height. If GTYPE = CIRCULAR, DIM1 is the radius and DIM2 is ignored. If GTYPE = ANNULAR, DIM1 is the external radius and DIM2 is the internal radius.
DID	Identification number of a DESC bulk entry for your description of the duct boundary. (Integer >0)
CID	Local coordinate ID to define the origin and transverse axis of the duct section. (Integer; Default=0)

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Field	Contents
XLOC, YLOC, ZLOC	Origin of the duct section relative to the coordinate system selected in the CID field.
XVEC, YVEC, XVEC	Coordinates of a vector defining the transverse direction (that is, the duct axis) relative to the coordinate system selected in the CID field. (Real, Default=0.0)

Remarks:

1. PID must be unique with respect to all other PACDUCT property identification numbers.
2. The PACDUCT entry is only supported in SOL 108 uncoupled acoustic and coupled vibro-acoustic solutions and requires the FEM Adaptive Order (FEMAO) method.

Related Topics

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PACFAN

Properties of a fan noise

Defines the properties of fan noise frequency analysis.

Format:

1	2	3	4	5	6	7	8	9	10
PACFAN	PID	TYPE	RPM	nBlades					

Examples:

PACFAN	200	TONAL	200	2					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
TYPE	FAN noise type (Character; Default=TONAL) If TYPE=TONAL, all fan blades have the same force. Define tonal components only. If TYPE=TBROAD, define tonal and broadband components. Fan blades may have different forces.
RPMID	Fan rotational speed in revolutions per minute. (Real >0.0; Default = 0.0).
nBlades	Apply the same forces for each fan blade. (Integer>0; Default=1)

Related Topics

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PACSPO2

Surface Dipole Property

Defines the properties of a surface dipole for a frequency response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
PACSPO2	PID		TYPE	EDGE					

Example:

PACSPO2	200		INCOMP						

Fields:

Field	Contents
PID	A property identification number that matches the identification number of the corresponding ACSPO2 bulk entry. (Integer > 0; No default)
TYPE	Scattering flow type. (Character; Default = "INCOMP") If TYPE = "COMP", define dipole sources and scattering in compressible flow. If TYPE = "INCOMP", define dipole sources and scattering in incompressible flow.
EDGE	Edge correction. (Character; Default = "NO") If EDGE = "YES", use edge correction. If EDGE = "NO", do not use edge correction.

Remarks:

1. ACSPO2 in combination with PACSPO2 requires the FEM Adaptive Order (FEMAO) method.

Related Topics

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PACTRAD

Property of Acoustic Transfer Admittance

Defines the property of transfer admittance for acoustics.

Format:

1	2	3	4	5	6	7	8	9	10
PACTRAD	PID		A1R	A1I	A2R	A2I	A3R	A3I	
	A4R	A4I	A5R	A5I	A6R	A6I			

Examples:

PACTRAD	300		0.3	0.3	0.4	0.4	0.5	0.5	
	0.6	0.6	0.1	0.1	0.2	0.2			

Fields:

Field	Contents
PID	Identification number of the parameters of acoustic transfer admittance. (Integer > 0)
AiR	Real parameters for transfer admittance. (Real or Integer; Default = 0.0)
(i = 1,2,3...)	For a constant transfer admittance, enter a real value.
	For a frequency-dependent transfer admittance, enter the ID of the TABLEDi bulk entry that contains the transfer admittance as a function of frequency. See Remark 3.
AiI	Imaginary parameters for transfer admittance. (Real or Integer; Default = 0.0)
(i = 1,2,3...)	For a constant transfer admittance, enter a real value.
	For a frequency-dependent transfer admittance, enter the ID of the TABLEDi bulk entry that contains the transfer admittance as a function of frequency. See Remark 3.

Remarks:

1. The PID must be unique with respect to each PACTRAD.
2. AiR and AiI cannot both be equal to zero, except for α_3 and α_6.
3. Ai are the transfer admittance coefficients that define the relationship of normal velocity and pressure between the two surfaces.

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In its most general form, this relationship is expressed as

$$\begin{bmatrix} v_{n1} \\ v_{n2} \end{bmatrix} = \begin{bmatrix} \alpha_1 & \alpha_2 \\ \alpha_4 & \alpha_5 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} + \begin{bmatrix} \alpha_3 \\ \alpha_6 \end{bmatrix}$$

where:

- v_{n1} is the normal velocity on the nodes of the first face selection or normal velocity on the positive sides of the element selection.
- v_{n2} is the normal velocity on the nodes of the second face selection or normal velocity on the negative sides of the element selection.
- p_1 is the pressure on the nodes of the first face selection or pressure on the positive sides of the element selection.
- p_2 is the pressure on the nodes of the second face selection or pressure on the negative sides of the element selection.
- α_1, α_2, α_4, and α_5 are complex admittance coefficients.
- α_3 and α_6 are complex source coefficients.

The six coefficients α_1 through α_6 are determined by the nature of the relation (see *Acoustics User's Guide)*. Because these coefficients have the dimension [velocity/pressure], they are called *transfer admittance coefficients*.

Related Topics

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PACTRTM

Vibro-Acoustic Transfer Admittance

Defines the property of transfer admittance for vibro-acoustics.

Format:

1	2	3	4	5	6	7	8	9	10
PACTRTM	PID	FORM	A1R	A1I	A2R	A2I	A4R	A4I	
	A5R	A5I							

Examples:

PACTRTM	300		0.3	0.3	0.4	0.4	0.5	0.5	
	0.6	0.6							

Fields:

Field	Contents
PID	Identification number of the vibro-acoustic transfer admittance coefficients. (Integer > 0)
FORM	Format of the AiR / AiI parameters. (Real / Imaginary = 0; Amplitude / Phase = 1; Default = 0) If FORM = 0, define the vibro-acoustic transfer admittance in rectangular format (real and imaginary). If FORM = 1, define the vibro-acoustic transfer admittance in polar format (amplitude and phase).
AiR (i = 1,2,4, or 5)	Depending on the FORM setting, real part or magnitude for vibro-acoustic transfer admittance. (Real or Integer; Default = 0.0) To define as a constant, enter a real value. To define as frequency-dependent, enter the ID of a TABLEDi bulk entry that contains the vibro-acoustic transfer admittance as a function of frequency. See Remark 2.
AiI (i = 1,2,4, or 5)	Depending on the FORM setting, imaginary part or phase for vibro-acoustic transfer admittance. (Real or Integer; Default = 0.0) To define as a constant, enter a real value.

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Field	Contents
	To define as frequency-dependent, enter the ID of a TABLEDi bulk entry that contains the vibro-acoustic transfer admittance as a function of frequency. See Remark 2.

Remarks:

1. The PID must be unique with respect to each PACTRTM.
2. AiR (or AiM) and AiI (or AiP):
   - Cannot both be equal to zero.
   - Can be TIDAiR (or TIDAiM) and TIDAiI (or TIDAiP) for table IDs.
3. The format can be as follows:
   - PID/FORM/A1R, TIDA1R/A1I, TIDA1I/A2R, TIDA2R/A2I, TIDA2I/A4R, TIDA4I/A4I, TIDA4I/A5R, TIDA5R/A5I, or TIDA5I.
   - PID/FORM/A1M, TIDA1M/A1P, TIDA1P/A2M, TIDA2M/A2P, TIDA2P/A4M, TIDA4M/A4P, TIDA4P/A5M, TIDA5M/A5P, or TIDA5P.
4. Ai are the transfer admittance coefficients that define the relationship of normal velocity and pressure between two surfaces.

   In its most general form, this relationship is expressed as:

$$\begin{Bmatrix} v_{n1} \\ v_{n2} \end{Bmatrix} = \begin{bmatrix} \alpha_1 & \alpha_2 \\ \alpha_4 & \alpha_5 \end{bmatrix} \begin{Bmatrix} p_1 \\ p_2 \end{Bmatrix}$$

   where:

   - v_{n1} is the inlet velocity.
   - v_{n2} is the outlet velocity.
   - α_1, α_2, α_4, and α_5 are complex admittance coefficients.
   - p_1 is the inlet pressure.
   - p_2 is the outlet pressure.

   The four α coefficients are determined by the nature of the relation. Because these coefficients have the dimension [velocity/pressure], they are called *transfer admittance coefficients*.

Related Topics

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PACVPO4

Volume Quadrupole Properties

Defines the properties of an acoustic volume quadrupole source for a frequency response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
PACVPO4	PID	LUNF	SEP	LENGTH	TIME	WINDOW	DID		

Example:

PACVPO4	999	89	2	M	S	HANNING	999		

Fields:

Field	Contents
PID	A property identification number that matches the identification number of the corresponding ACVPO4 bulk entry. (Integer > 0; No default)
LUNF	Logical unit number for CGNS file that contains time-dependent velocities. (0 < Integer ≤ 9999; No default). See Remark 1.
SEP	CGNS file name separator. (Integer; No default) If SEP = 1, use files that match the underscore (_) separator. If SEP = 2, use files that match the ampersand (@) separator.
LENGTH	CGNS file length units. (Character; Default = "M"). See Remark 2.
TIME	CGNS file time units. (Character; Default = "S"). See Remark 2.
WINDOW	Windowing method for Fast Fourier transformation. (Character; Default = "HANNING"). = "HANNING" for general purposes. = "RECTANG" when the input signal is periodic within the chosen time segment, for example, when you are analyzing special sinusoids whose frequencies coincide with obtained frequencies of the analysis. = "HAMMING" for general purposes when the dynamic range is about 50 dB. = "KAISERBE" or "BLACKMAN" when you need to distinguish between distinct harmonic signals.

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Field	Contents
	= "FLATTOP" for calibration procedures and for those situations where the correct amplitude measurements are important.
DID	Property identification number of a DESC bulk entry (Integer > 0; No default). See Remark 3.

Remarks:

1. You must specify a CGNS file in the File Management Section (FMS) using an appropriate ASSIGN USERFILE statement and reference it in the PACVPO4 bulk entry.
2. When CGNS file does not contain units, specify the length units in "M", "MM", "CM", "IN", or "FT" and time units in "S", "MIN", or "H".
3. The DESC bulk entry refers to the loading description that is stored in the CGNS file. For example, when the description field of the DESC bulk entry shows the value “Velocity”, the software queries the CGNS file for the “VelocityX", "VelocityY" and "VelocityZ” loads.

Related Topics

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PAERO1

Aerodynamic Panel Property

Defines associated bodies for the panels in the Doublet-Lattice method.

Format:

1	2	3	4	5	6	7	8	9	10
PAERO1	PID	B1	B2	B3	B4	B5	B6		

Example:

PAERO1	1	3							

Fields:

Field	Contents
PID	Property identification number referenced by a CAERO1 entry. (Integer > 0)
Bi	Identification number of CAERO2 entries for associated bodies. (Integer ≥ 0 or blank)

Remarks:

1. The associated bodies must be in the same aerodynamic group, as specified in the IGID field on CAERO2 entry.
2. If there are no bodies, the entry is still required (with Bi fields blank).
3. The Bi numbers above must appear on a CAERO2 entry to define these bodies completely.

Related Topics

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PAERO2

Aerodynamic Body Properties

Defines the cross-sectional properties of aerodynamic bodies.

Format:

1	2	3	4	5	6	7	8	9	10
PAERO2	PID	ORIENT	WIDTH	AR	LRSB	LRIB	LTH1	LTH2	
	THI1	THN1	THI2	THN2	THI3	THN3			

Example:

PAERO2	2	Z	6.0	1.0	22	91	100		
	1	3							

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
ORIENT	Orientation flag. Type of motion allowed for bodies. Refers to the aerodynamic coordinate system of ACSID. See AERO entry. (Character = “Z”, “Y”, or “ZY”)
WIDTH	Reference half-width of body and the width of the constant width interference tube. (Real > 0.0)
AR	Aspect ratio of the interference tube (height/width). (Real > 0.0)
LRSB	Identification number of an AEFACT entry containing a list of slender body half-widths at the end points of the slender body elements. If blank, the value of WIDTH will be used. (Integer > 0 or blank)
LRIB	Identification number of an AEFACT entry containing a list of slender body half-widths at the end points of the interference elements. If blank, the value of WIDTH will be used. (Integer > 0 or blank)
LTH1, LTH2	Identification number of AEFACT entries for defining θ arrays for interference calculations. (Integer ≥ 0)
THIi, THNi	The first and last interference element of a body to use the θ_1 array; the others use the θ_2 array. (Integer ≥ 0)

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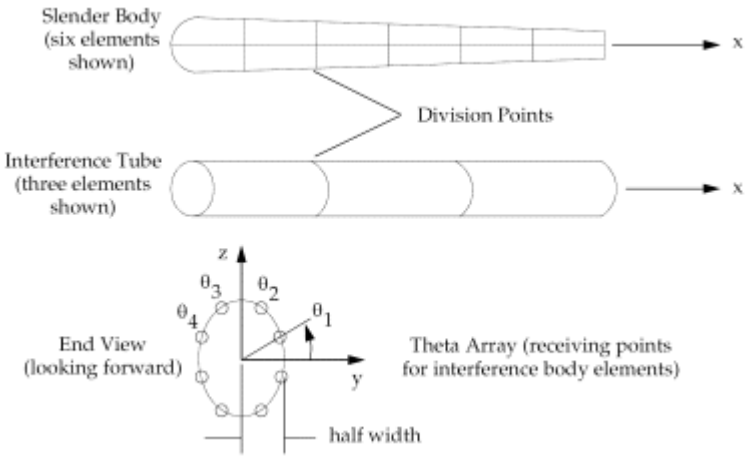
Remarks:

1. The half-widths (given on AEFACT entries referenced in fields 6 and 7) are specified at division points. The number of entries on an AEFACT entry used to specify half-widths must be one greater than the number of elements.
2. The half-width at the first point (i.e., the nose) on a slender body is usually 0.0; thus, it is recommended (but not required) that the LRSB data is supplied with a zero first value.
3. THIi and THNi are interference element numbers on a body. The first element is one for each body.
4. A body is represented by a slender body surrounded by an interference tube. The slender body creates the downwash due to the motion of the body, while the interference tube represents the effects upon panels and other bodies.

Figure 16-137: Idealization of Aerodynamic Body



5. The angles θ_1 and θ_2 are input in degrees using the aerodynamic element coordinate system as the reference coordinate system for defining the theta points.
6. Distribution of the theta points need not be uniform. A theta point must be placed a finite distance from any aerodynamic box edge; preferably the box edge would be equidistant from any two theta points. This aerodynamic coordinate system is defined on the AERO Bulk Data entry.
7. For half models, the theta arrays LTH1 and LTH2 should encompass a full 360 degree range.

Related Topics

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PAERO3

Aerodynamic Panel Property

Defines the number of Mach boxes in the flow direction and the location of cranks and control surfaces of a Mach box lifting surface.

Format:

1	2	3	4	5	6	7	8	9	10
PAERO3	PID	NBOX	NCTRL		X5	Y5	X6	Y6	
	X7	Y7	X8	Y8	X9	Y9	X10	Y10	
	X11	Y11	X12	Y12					

Example:

PAERO3	2001	15	1		0.	65.			
	78.	65.	108.	65.	82.	97.5	112.	97.5	
	86.	130.	116.	130.					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
NBOX	Number of Mach boxes in the flow direction. (0 < Integer < 50)
NCTRL	Number of control surfaces. (Integer 0, 1, or 2)
X5 through Y12	Locations of points 5 through 12, which are in the aerodynamic coordinate system, to define the cranks and control surface geometry. (Real)

Remarks:

1. For an illustration of the geometry, see the CAERO3 entry description.
2. If Y5 ≤ 0.0, there is no leading edge crank. Also, if Y6 ≤ 0.0, there is no trailing edge crank.
3. If NCTRL = 0, no continuations are required. If NCTRL = 1 or 2, then NCTRL continuations are required.
4. Y7 ≥ Y8, Y9 ≥ Y10, and Y11 ≥ Y12.
5. The number of Mach boxes in the spanwise direction (NSB) may be found from the following formula:

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$$NSB = \text{INT}\left[\frac{\beta \cdot y_{max}}{\left(\frac{x_{max}}{\text{NBOX} + 0.5}\right)} + 0.5\right]$$

where

$\beta = \sqrt{M^2 - 1}$

x_{max} = maximum chordwise direction

y_{max} = maximum spanwise direction

NBOX = initial number of boxes specified in field 3

The number of Mach boxes in the streamwise direction may then be computed from:

$$\text{NBOX} = \text{INT}\left[\frac{x_{max}}{\left(\frac{\beta \cdot y_{max}}{NSB - 0.5}\right)} + 0.999\right]$$

The number of chordwise boxes specified by the user (NBOX ≥ 50) will be replaced by a floating point number (usually slightly higher than NBOX). The method contracts the mesh equally in both dimensions until a box edge lies on the surface tip. This mesh size is then used to compute the number of chordwise boxes.

Note

A minimum of seven Mach boxes in the flow direction (NBOX) is recommended.

Related Topics

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PAERO4

Aerodynamic Strip Properties

Defines properties of each strip element for Strip theory.

Format:

1	2	3	4	5	6	7	8	9	10
PAERO4	PID	CLA	LCLA	CIRC	LCIRC	DOC1	CAOC1	GAPOC1	
	DOC2	CAOC2	GAPOC2	DOC3	CAOC3	GAPOC3	-etc.-		

Example:

PAERO4	6001	1	501	0	0	0.0	0.0	0.0	
	0.50	0.25	0.02	0.53	0.24	0.0			

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
CLA	Select Prandtl-Glauert correction. (Integer = -1, 0, 1; Default = 0)
	-1 Compressibility correction made to lift curve slope data for a reference Mach number.
	0 No correction and no list needed. (Default)
	+1 No correction and lift curve slope provided by a list as a function of strip location and Mach number.
LCLA	Identification number of the AEFACT entry that lists the lift curve slope on all strips for each Mach number on the MKAEROi entry. See Remark PAERO4 below. (Integer = 0 if CLA = 0, > 0 if CLA ≠ 0)
CIRC	Select Theodorsen's function $C(k)$ or the number of exponential coefficients used to approximate $C(k)$. (Integer = 0, 1, 2, 3; Default = 0. Must be zero if CLA ≠ 0.)
	0 Theodorsen function.
	1, 2, 3 Approximate function with b_0, b_1, β_1, ..., b_n, β_n n = 1, 2, 3.
LCIRC	Identification number of the AEFACT entry that lists the b, β values for each Mach number. See Remark PAERO4, PAERO4, and PAERO4 below; variable b's and β's for each mi on the MKAEROi entry. (Integer = 0 if CIRC = 0, > 0 if CIRC ≠ 0)

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Field	Contents
DOCi	d/c = distance of the control surface hinge aft of the quarter-chord divided by the strip chord (Real ≥ 0.0)
CAOCi	c_a/c = control surface chord divided by the strip chord. (Real ≥ 0.0)
GAPOCi	g/c = control surface gap divided by the strip chord. (Real ≥ 0.0)

Remarks:

1. PAERO4 is required for Strip theory with three fields (DOCi, CAOCi, GAPOCi) specified per strip.
2. If CLA = -1, lift curve slope data at one Mach number are needed on the AEFACT entry.
3. If CAOCi = 0.0, there is no control surface.
4. If GAPOCi = 0.0, there is no slot flow.
5. If GAPOCi < 0.01, then 0.01 is used.
6. Embedded blank fields are not permitted.
7. The following table lists the lift curve slope or lag function selection and the AEFACT entry formats used for Strip theory:

Table 16-111: Strip Theory Function Selections and AERACT Entry Formats

Theodorsen Function	Data Type Input	Parameter Combinations				Number of Words	Entry Format Index
		CLA	LCLA	CIRC	LCIRC		
Exact	Lift Curve Slope $c_{l_{\alpha i}} = 2\pi$	0	0	0	0	No AEFACT entry required	
	$c_{l_{\alpha i}}$ Input, Uses Prandtl-Glauert Correction	-1	ID	0	0	(NSTRIP+1)	a.
	$c_{l_{\alpha i}}$ Input, for All m's on MKAERO Entry	1	ID	0	0	(NSTRIP+1)* NMACH	b.
Approximate Coefficients	b_{0i}, b_{1i}, β_{1i}, etc.	0	0	1	ID	4*NMACH	c.
		0	0	2	ID	6*NMACH	d.
		0	0	3	ID	8*NMACH	e.

Entry Format:

•

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$$\text{AEFACT, ID, } m_1, c_{l\alpha_1}, c_{l\alpha_2}, \ldots, c_{l\alpha_{\text{NSTRIP}}}$$

- $$\text{AEFACT, ID, } m_1, c_{l\alpha_{11}}, c_{l\alpha_{21}}, \ldots, c_{l\alpha_{\text{NSTRIP1}}},$$
  $$m_2, c_{l\alpha_{11}}, c_{l\alpha_{12}}, c_{l\alpha_{21}} c_{l\alpha_{22}}, \ldots, c_{l\alpha_{\text{NSTRIP1}}}, c_{l\alpha_{\text{NSTRIP2}}},$$ for all m on MKAEROi data entry
- AEFACT, ID, m_1, b_{01}, b_{11}, β_{11}, m_2, b_{02}, b_{12}, P_{12}, m_3, etc.
- AEFACT, ID, m_1, b_{01}, b_{11}, β_{11}, b_{21}, β_{21}, m_2, etc.
- AEFACT, ID, m_1, b_{01}, b_{11}, β_{11}, b_{21}, β_{21}, b_{31}, β_{31}, m_2, etc.

8. A control surface rotation is positive when the trailing edge moves in the negative z-direction of the aerodynamic element coordinate system; see the *Simcenter Nastran Aeroelastic Analysis User's Guide*.

Related Topics

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PAERO5

Aerodynamic Panel Property

Defines properties of each strip element for Piston theory.

Format:

1	2	3	4	5	6	7	8	9	10
PAERO5	PID	NALPHA	LALPHA	NXIS	LXIS	NTAUS	LTAUS		
	CAOC1	CAOC2	CAOC3	CAOC4	CAOC5				

Example:

PAERO5	7001	1	702	1	701	1	700		
	0.0	0.0	5.25	3.99375	0.0				

Fields:

Field	Contents
PID	Property identification number. (Unique Integer > 0)
NALPHA	Number of angle of attack (α) values to be input for each Mach number (mi) on the MKAERO1 or MKAERO2 entry. (Integer > 0)

NALPHA	Meaning
1	α is the same value for all strips; enter one value, in units of degrees, on the AEFACT entry for each Mach number.
Number of Strips	α is different for each strip; enter α's, in units of degrees, in the following order: m_1, α_1, α_2, ..., m_2, α_1, α_2, ..., etc.

LALPHA	ID number of the AEFACT entry that lists the α's for the strips at each Mach number in the MKAERO1 or MKAERO2 entry. (Integer > 0)
NXIS	Number of dimensionless chord coordinates ξ to be input. (Integer ≥ 0, Default = 0)

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NXIS	Meaning
0	No ξ's are required. (Default)
1	ξ's are the same for all strips; enter values for one strip on the AEFACT entry (ξ_h if NTHICK > 0, or ξ_m and ξ_h if NTHICK = 0)
Number of Strips	ξ's have to be input for each strip (ξ_{h1}, ξ_{h2}, ..., ξ_{hNSPAN}, if NTHICK> 0, or ξ_{m1}, ξ_{h1}, ξ_{m2}, ξ_{h2}, ..., ξ_{mNSPAN}, ξ_{hNSPAN}, β_{hNSPAN} if NTHICK = 0)

LXIS	Identification number of AEFACT entry that lists the ξ values for the strip in order indicated by values of NXIS and NTHICK. (Integer = 0 if c_a = 0 and NTHICK > 0 or LXIS > 0 if c_a = 0 and/or NTHICK = 0)
NTAUS	Parameter used to select the number of thickness ratio (τ) values to be input. (Integer ≥ 0, Default = 0)

NTAUS	Meaning
0	No τ's are required. (Default)
1	τ's are the same for all strips; enter (τ_1, τ_{h1}, τ_{t1}) values for one strip on AEFACT entry.
Number of Strips	τ's must to be input for each strip on an AEFACT entry in the following order: (τ_1, τ_{h1}, τ_{t1}, τ_2, τ_{h2}, τ_{t2}, ..., τ_{NSPAN}, τ_{hNSPAN}, τ_{tNSPAN})

LTAUS	Identification number of AEFACT entry that lists the τ values for the strips. (Integer = 0 or blank if NTAUS = 0, LTAUS > 0 if NTAUS > 0)
CAOCi	c_a/c = control surface chord divided by the strip chord. (Real ≥ 0.0)

Remarks:

1. The continuation entry is required for Piston theory with one entry (CAOCi) per strip.
2. Embedded blank fields are not allowed on the continuation entry.
3. If CAOCi = 0.0, there is no control surface.
4. Table 16-112 lists the thickness data input and AEFACT entry format used for Piston theory.

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Table 16-112: Thickness Data Input and AEFACT Entry Format for Piston Theory

Type of Input	Parameter Combinations						Number of Words	Entry Format Index
	CAOC	NGHICK	NXIS	LXIS	NTAUS	LTAUS		
No control surfaces, Integrals input are same for all strips	0.0	ID[(a)]	0	0	0	0	6	a.
With control surfaces, Integrals input, same hinge on all strips	≠ 0.0	ID[(b)]	1	ID[(c)]	0	0	121	b.c.
With control surfaces, Integrals input, variable hinge	≠ 0.0	ID[(b)]	NSTRIP	ID[(d)]	0	0	12NSTRIP	b.d.
No control surfaces, thickness inputs are same for all strips	0.0	0	1	ID[(f)]	1	ID[(e)]	32	e.f.
With control surfaces, thickness inputs are same for all strips	≠ 0.0	0	1	ID[(f)]	1	ID[(e)]	32	e.f.
With control surfaces, thickness inputs vary for strips	≠ 0.0	0	NSTRIP	ID[(h)]	NSTRIP	ID[(g)]	3*NSTRIP 2*NSTRIP	g.h.

Entry Format

a. AEFACT, ID, I_1, I_2, I_3, I_4, I_5, I_6

b. AEFACT, ID, I_1, ..., I_6, J_1, ..., $J_6 I_1$, I_2, I_3, I_4, I_5, I_6

c. AEFACT, ID, ξ_h

d. AEFACT, ID, ξ_{h1}, ξ_{h2}, ξ_{h3}, ..., $\xi_{h\text{NSTRIP}}$

e. AEFACT, ID, τ_m, τ_h, τ_t

f. AEFACT, ID, ξ_m, ξ_h

g. AEFACT, ID, τ_{m1}, τ_{h1}, τ_{t1}, τ_{m2}, τ_{h2}, τ_{t2}, ..., $\tau_{m\text{NSTRIP}}$, $\tau_{h\text{NSTRIP}}$, $\tau_{t\text{NSTRIP}}$

τ_{m1}, τ_{h1}, τ_{t1}, τ_{m2}, τ_{h2}, τ_{t2}, ..., $\tau_{m\text{NSTRIP}}$, $\tau_{h\text{NSTRIP}}$, $\tau_{t\text{NSTRIP}}$

h. AEFACT, ID, ξ_{m1}, ξ_{h1}, ξ_{m2}, ξ_{h2}, ..., $\xi_{m\text{NSTRIP}}$, $\xi_{h\text{NSTRIP}}$

5. The following table lists the angle-of-attack distribution and AEFACT entry formats used for Piston theory.

Type of Distribution	Parameter Combinations		Number of Words	Entry Format Index
	NALPHA	LALPHA		
Equal angle of attack on all strips	1	ID	2*NMACH	a.
Unequal angle of attack	NSTRIP	ID	(1 + NSTRIP) * NMACH	b.

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Entry Format

a. AEFACT, ID, m_1, α_1, m_2, α_2, ...,

b. AEFACT, ID, m_1, α_{11}, α_{21}, α_{31}, ..., $\alpha_{NSTRIP1}$, m_2, α_{12}, α_{22}, ..., $\alpha_{NSTRIP2}$, m_2, etc., for all m on MKAEROi entry.

c. A control surface rotation is positive when the trailing edge moves in the negative z-direction of the aerodynamic element coordinate system; see the *Simcenter Nastran Aeroelastic Analysis User's Guide*.

Related Topics

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PANEL

Panel Definition for Coupled Fluid-Structural Analysis

Selects sets of structural grid points, elements, or physical properties that define one or more panels.

Format:

1	2	3	4	5	6	7	8	9	10
PANEL	NAME1	SID1	NAME2	SID2	NAME3	SID3	NAME4	SID4	
	NAME5	SID5	-etc.-						

Example:

PANEL	BKDOOR	103							

Fields:

Field	Contents
NAMEi	Panel label. (Character. See **Bulk Data Syntax Rules**.)
SIDi	Identification number of a SET1 or SET3 bulk entry that lists the structural grid points, elements, or physical properties of the panel. (Integer > 0)

Remarks:

1. The ERP case control command requests results on elements. As a result, the panels used for this request are a selection of shell elements or solid element faces. See the remarks on the ERP command for the input requirements for this request.
2. The MODCON or PANCON case control commands request results on grid points. As a result, the panels used for these requests are a selection of structural grid points.
   - If a set of grid points is referenced, the set must include only structural grid points. The panel will consist of all the grid points in the referenced set. SET1 and SET3 bulk entries are used to define sets of grid points.
   - If a set of elements is referenced, the set must include only structural elements. The panel will consist of all the grid points that are connection points for these elements. SET3 bulk entries are used to define sets of elements.
   - If a set of physical property identifiers is referenced, the physical properties must be referenced by structural elements. The panel will consist of all grid points that are connection points for the structural elements referencing the physical properties included in the set. SET3 bulk entries are used to define sets of physical property identifiers.

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3. If the referenced SET1 or SET3 bulk entries include structural grid points, the sets must include at least four grid points for quadrilateral faces and three grid points for triangular faces.
4. It is recommended that all of the connection points for a given element belong to the same panel.
5. NAMEi is used only for labeling the output of the panel modal participation factors (refer to the MODCON and PANCON case control commands). See Defining Panels in the *Simcenter Nastran Acoustics Simcenter Nastran User's Guide*.
6. When used with the ERP case control command, you must select the geometry defining the PANEL with the SET3 bulk entry. See the ERP command remarks for the input requirements.

Related Topics

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PARAM

Parameter

Specifies values for parameters used in solution sequences or user-written DMAP programs.

Format:

1	2	3	4	5	6	7	8	9	10
PARAM	N	V1	V2						

Example:

PARAM	IRES	1							

Fields:

Field	Contents
N	Parameter name (one to eight alphanumeric characters, the first of which is alphabetic).
V1, V2	Parameter value based on parameter type, as follows:

Type	V1	V2
Integer	Integer	Blank
Real, single-precision	Real	Blank
Character	Character	Blank
Real, double-precision	Double-precision real	Blank
Complex, single-precision	Real or blank	Real or blank
Complex, double-precision	Double-precision real	Double-precision real

Remarks:

1. See **"Parameters and Parameter Applicability Tables"** for a list of parameters used in solution sequences that may be set by the user on PARAM entries.
2. If the large field entry format is used, the second physical entry must be present, even though fields 6 through 9 are blank.

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Related Topics

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PBAR

Simple Beam Property

Defines the properties of a simple beam element (CBAR entry).

Format:

1	2	3	4	5	6	7	8	9	10
PBAR	PID	MID	A	I1	I2	J	NSM		
	C1	C2	D1	D2	E1	E2	F1	F2	
	K1	K2	I12						

Example:

PBAR	39	6	2.9		5.97				
			2.0	4.0					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. See Remarks 2 and 3. (Integer > 0)
A	Cross sectional area. See Remark 5. (Real; Default = 0.0)
I1, I2	Second moment of the cross sectional area. See Remarks 4 and 5. (Real ≥ 0.0, I1 x I2 ≥ $I12^2$; Default = 0.0)
I12	Mixed second moment of the cross sectional area. See Remarks 4 and 5. (Real ≥ 0.0; Default = 0.0)
J	Torsional constant of the cross sectional area. See Remark 5. (Real ≥ 0.0; Default = 0.0)
NSM	Nonstructural mass per unit length. In Remarks related to SOLs 402 and 414, see Remark 2. (Real)
Ci, Di, Ei, Fi	Coordinates of stress recovery points. See Remark 8. (Real; Default = 0.0)
K1	Transverse shear factor for bending in the element XY-plane. See Remark 7. (Real; Default = 0.0)
K2	Transverse shear factor for bending in the element XZ-plane. See Remark 7. (Real; Default = 0.0)

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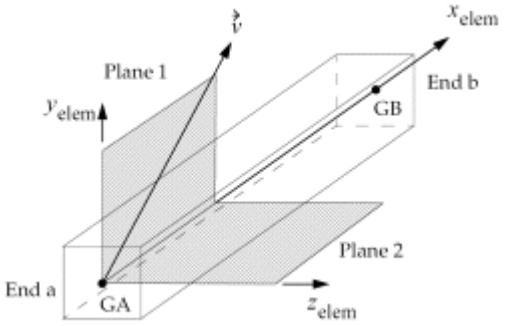
Remarks:

1. Both continuation entries may be omitted.
2. For structural problems, MID must reference a MAT1 material entry.
3. For heat transfer problems, MID must reference a MAT4 or MAT5 material entry.
4. I1 and I2 are also referred to as moments of inertia. I12 is also referred to as a product of inertia.
5. The following figure shows the element coordinate system for the PBAR entry, where the y_{elem} and z_{elem} axes are neutral axes.

Figure 16-138: PBAR Element Coordinate System



The following table shows the formulas for cross sectional properties.

Field	Formula
A	$\int_{Area} dA$
I1	$\int_{Area} y_{elem}^2 dA$
I2	$\int_{Area} z_{elem}^2 dA$

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Field	Formula
I12	$\int_{Area} y_{elem} z_{elem} dA$
J	$\int_{Area} r^2 dA = \int_{Area} (y_{elem}^2 + z_{elem}^2) dA$

For additional information about the CBAR element geometry, see the CBAR entry description.

6. By definition, the shear center and neutral axes coincide.
7. The specification of the K1 and K2 fields determine the type of beam model that the software uses:
   - For a Timoshenko beam, specify real nonzero values for the K1 and K2 fields, where the values in the K1 and K2 fields are the shear stiffness factors in the element Y- and Z-directions, respectively.
   - For an Euler-Bernoulli beam, specify the K1 and K2 fields as real zero or leave the fields blank.

   Timoshenko beams account for transverse shear in the beam deflection calculation, whereas Euler-Bernoulli beams do not.

   If you specify one of the K1 or K2 fields as real nonzero and the other as real zero or leave it blank, the software uses the Timoshenko beam model for the deflection calculation in the plane of the real nonzero shear stiffness factor and uses the Euler-Bernoulli beam model for the deflection calculation in the plane of the real zero or blank shear stiffness factor. For example, if you specify K1 = 0.5 and K2 = 0.0, the software uses the Timoshenko beam model for the deflection in the element XY-plane, and uses the Euler-Bernoulli beam model for the deflection calculation in the element XZ-plane

   For unsymmetric bending, that is I12 ≠ 0.0, the software uses the Euler-Bernoulli beam model regardless of the K1 and K2 field specifications.

   The K1 and K2 fields must be blank if the A field is set to real zero (the default).
8. C1 and C2 are the y_{elem} and z_{elem} coordinates, respectively, of the C stress recovery point, D1 and D2 are the y_{elem} and z_{elem} coordinates, respectively, of the D stress recovery point, and so on. The stresses are computed at both ends of the bar.
9. For response spectra analysis, use the CBEAM element rather than the CBAR element as the CBAR element can produce inaccurate results.

Remarks related to SOLs 402 and 414:

1. The PBAR entry cannot be used for elasto-plastic beam elements. In that case, you can use the PBARL entry. See the remarks on the PBARL entry for the supported cross sections.
2. The NSM field is supported by SOLs 402 and 414 to apply nonstructural mass to the element. For SOL 414 only, you can use the WTMASS parameter to scale all mass in your model including any nonstructural mass defined. For example, you can use the WTMASS parameter to convert applied

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mass from weight units (Newtons, pound-force) to mass units (kilograms, pound-mass). All mass in your model must be defined using consistent units.

Related Topics

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PBARL

Simple Beam Cross-Section Property

Defines the properties of a simple beam element (CBAR entry) by cross-sectional dimensions.

Format:

1	2	3	4	5	6	7	8	9	10
PBARL	PID	MID	GROUP	TYPE					
	DIM1	DIM2	DIM3	DIM4	DIM5	DIM6	DIM7	DIM8	
	DIM9	-etc.-	NSM						

Example:

PBARL	39	6		I					
	14.0	6.0	0.5	0.5	0.5				

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. (Integer > 0)
GROUP	Cross-section group. See Remarks 6 and 10. (Character; Default = “MSCBML0”)
TYPE	Cross-section type. See Remarks 6 and 10 and Figure 16-139. (Character: “ROD”, “TUBE”, “I”, “CHAN”, “T”, “BOX”, “BAR”, “CROSS”, “H”, “T1”, “I1”, “CHAN1”, “Z”, “CHAN2”, “T2”, “BOX1”, “HEXA”, “HAT”, “HAT1” for GROUP = “MSCBML0”)
DIMi	Cross-sectional dimensions. (Real > 0.0 for GROUP = “MSCBMLO”)
NSM	Nonstructural mass per unit length. NSM is specified after the last DIMi. (Real ≥ 0; Default = 0.0)

Remarks:

1. For structural problems, PBARL entries must reference a MAT1 material entry.
2. PID must be unique with respect to all other PBAR and PBARL property identification numbers.
3. See CBAR entry for a discussion of bar element geometry.
4. For heat-transfer problems, the MID must reference a MAT4 or MAT5 material entry.

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5. For response spectra analysis on stress recovery coefficients, the CBEAM element should be used because results for the CBAR element will not be accurate.
6. The GROUP is associated with an FMS CONNECT statement that specifies the evaluator. A reserved GROUP name is “MSCBML0”. Users may create their own cross-section types. Each of the types will require one or more subroutines to convert DIMi information to geometric property information contained on a PBAR entry and optimization information.
7. An equivalent PBAR entry is created from the PBARL entry. Any sorted echo request will also cause printout and/or punch of the derived PBAR.
8. The cross sectional properties on the equivalent PBAR entry are calculated using hard-coded formulas for each section type. To use the mesh-based Pilkey method to calculate the cross sectional properties, specify PARAM,PBRPROP,YES.
9. For the formulas that the software uses to calculate the cross-section properties and, if applicable, the shear center location, see Using Supplied Beam and Bar Libraries in the *Simcenter Nastran Element Library Reference*.
10. For GROUP = “MSCBML0”, the cross-sectional properties, shear flexibility factors, and stress recovery points (C, D, E, and F) are computed using the TYPE and DIMi as shown in **Figure 16-139**. The origin of element coordinate system is centered at the shear center of the cross-section oriented as shown. Except for SOL 402, The PBARL does not account for offsets between the neutral axis and the shear center. Therefore, the CHAN, CHAN1 and CHAN2 cross-sections may produce incorrect results. The PBEAML is recommended in those cases.
11. C, D, E, and F in **Figure 16-139** are the locations of stress recovery on a cross-section.

Remarks related to SOLs 402 and 414:

1. Only the axisymmetric cross section types “ROD” and “TUBE” are supported on the PBARL entry. You can use the CBEAM and PBEAML entries if you need to define the non-axisymmetric cross section types "BOX", "BAR", “I”, “I1”, “CHAN”, “CHAN1”, “T”, “Z”, or "L".
2. The cross section types “T1”, “T2”, “CHAN2”, “CROSS”, “H”, “BOX1”, “HEXA”, “HAT”, and “HAT1” are not fully supported. If you define these types, the software automatically creates an equivalent PBEAM entry but does not perform stress-recovery. For TYPE = “T1”, “T2”, “CHAN2”, and “H”, equivalent supported profiles exist on the PBEAML entry in other axes (“T”, “CHAN”, “CHAN1”, “I” or “I1”).
3. TYPE="BAR", "BOX", "CHAN", "CHAN1", "I", "I1", "T", "TUBE", and "Z" can be used with a nonlinear material (for example, plastic or temperature-dependent). If a plastic material is used (MATS1 entry with TYPE="PLASTIC"), the behavior is not limited to elastic-perfectly plastic.

   The material behavior is calculated by numerical integration in the central section of the element. It should be constant for the whole element. To have a behavior close to a plastic hinge, you must use a finer mesh.
4. TYPE=“ROD” is not supported with nonlinear materials. As an alternative, you can use a TUBE section with a small DIM2 dimension.
5. For TYPE = “I”, “I1”, “CHAN”, “CHAN1”, “Z”, “T”, “BOX”, and “TUBE”, the stress recovery locations are located at the element's mid-thickness instead of the outer skin. Therefore, small deviations can be observed compared to other Nastran solutions like SOL 401.

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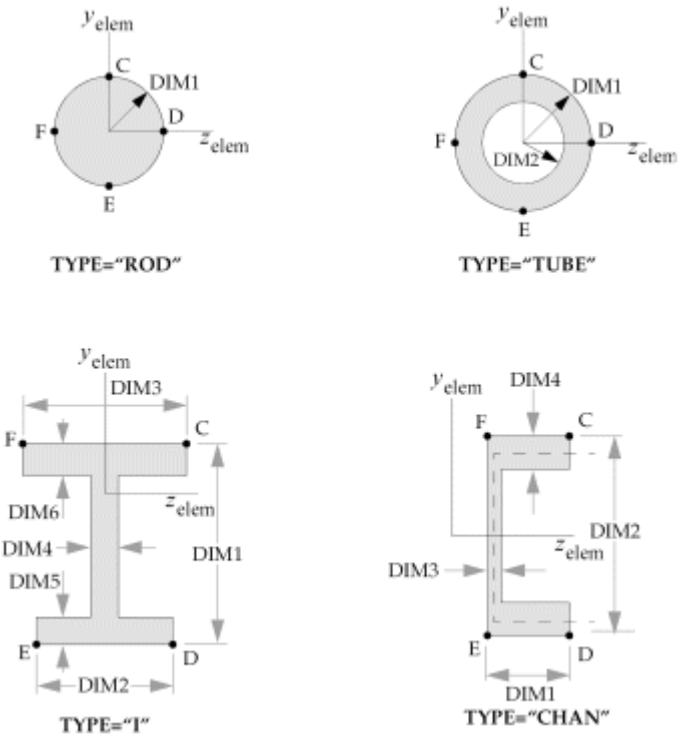
6. Nonstructural mass (NSM) defined on the PBARL entry is not supported.

Figure 16-139: Definition of Cross-Section Geometry and Stress Recovery Points for GROUP = "MSCBML0"



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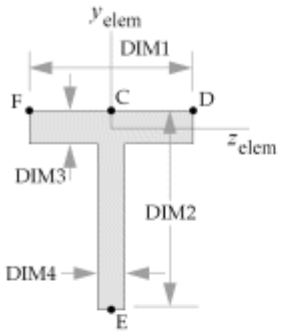
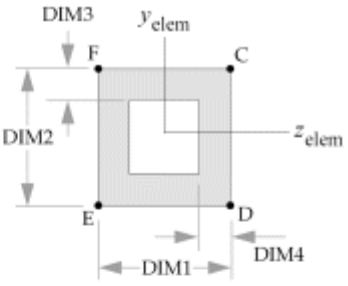
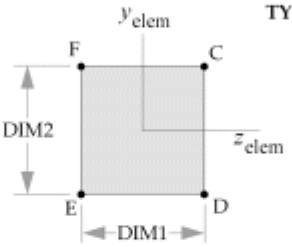


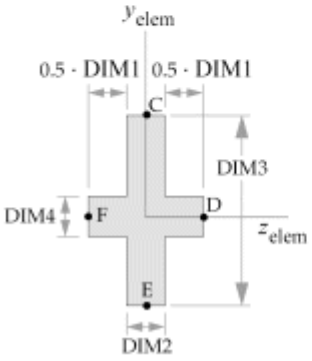
TYPE="T"



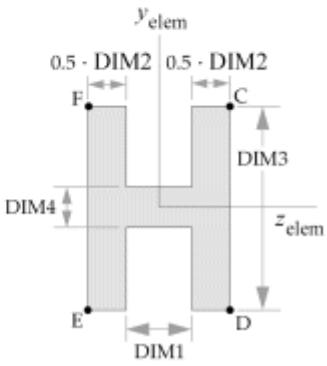
TYPE="BOX"



TYPE="BAR"



TYPE="CROSS"



TYPE="H"

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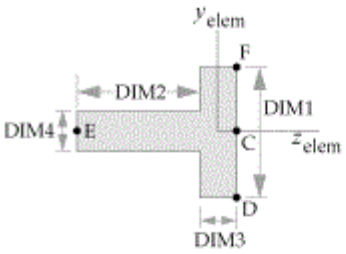
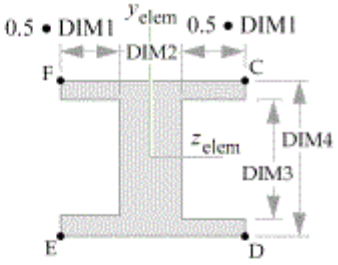
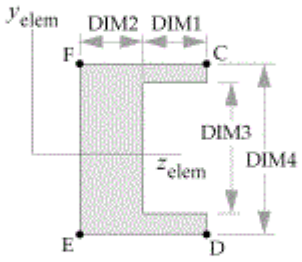


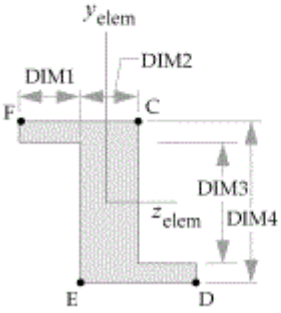
TYPE="T1"



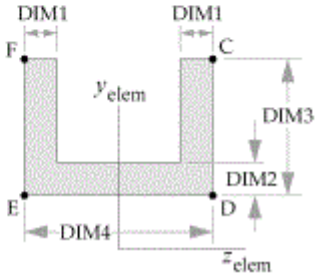
TYPE="I1"



TYPE="CHAN1"



TYPE="Z"



TYPE="CHAN2"

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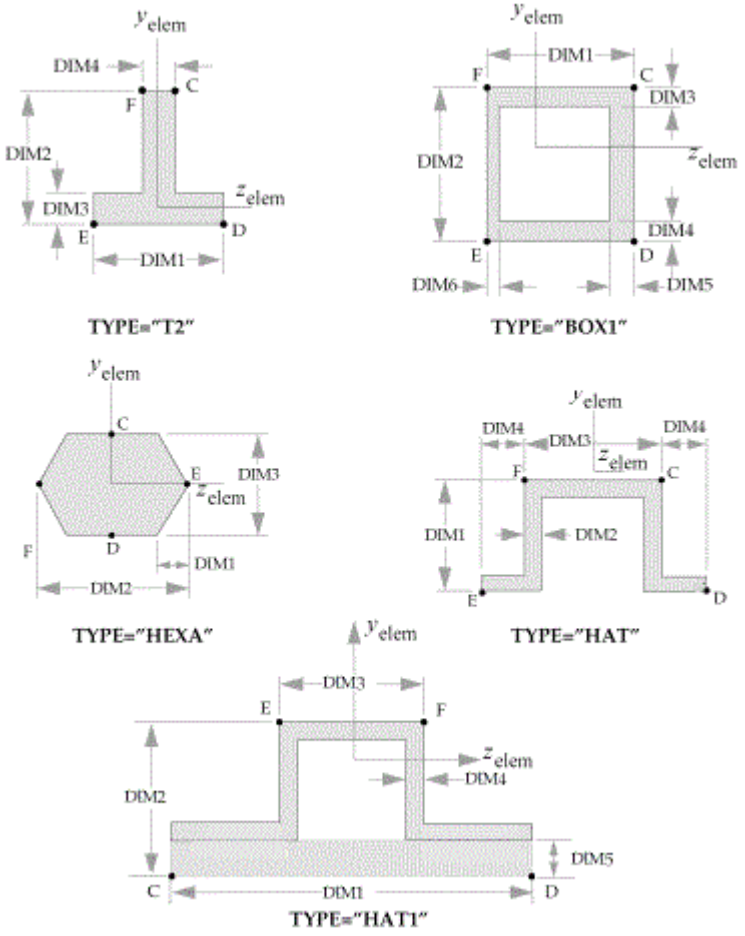

Related Topics

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PBCOMP

Beam Property (Alternate Form of PBEAM)

Alternate form of the PBEAM entry to define properties of a uniform cross-sectional beam referenced by a CBEAM entry. This entry is also used to specify lumped areas of the beam cross section for nonlinear analysis and/or composite analysis.

Format:

1	2	3	4	5	6	7	8	9	10
PBCOMP	PID	MID	A	I1	I2	I12	J	NSM	
	K1	K2	M1	M2	N1	N2	SYMOPT		
	Y1	Z1	C1	MID1					
	Y2	Z2	C2	MID2					
	-etc.-								

Example:

PBCOMP	39	6	2.9						
							1		
	-0.5	1.2	0.1	18					
	0.2	0.9	0.15						

Fields:

Field	Contents
PID	Property identification number. See Remark 1 . (Integer > 0)
MID	Material identification number. See Remarks 2 and 5. (Integer > 0)
A	Area of beam cross section. (Real > 0.0)
I1	Area moment of inertia in plane 1 about the neutral axis. See Remark 6 . (Real > 0.0)
I2	Area moment of inertia in plane 2 about the neutral axis. See Remark 6 . (Real > 0.0)
I12	Area product of inertia. See Remark 6 . (Real; Default = 0.0, but $I1 \cdot I2 - (I12)^2 > 0.0$)
J	Torsional constant. See Remark 6 . (Real > 0.0; Default = 0.0)
NSM	Nonstructural mass per unit length. (Real > 0.0; Default = 0.0)
K1, K2	Shear stiffness factor K in K*A*G for plane 1 and plane 2. See Remark 4 . (Real > 0.0; Default = 1.0)

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Field	Contents
M1, M2	The (y,z) coordinates of center of gravity of nonstructural mass. See the figure in the CBEAM entry description. (Real; Default = 0.0)
N1, N2	The (y,z) coordinates of neutral axis. See the figure in the CBEAM entry description. (Real; Default = 0.0)
SYMOPT	Symmetry option to input lumped areas for the beam cross section. See **Figure 16-141** and Remark **7** . (0 ≤ Integer ≤ 5; Default = 0)
Yi, Zi	The (y,z) coordinates of the lumped areas in the element coordinate system. See Remark **1** . (Real)
Ci	Fraction of the total area for the i-th lumped area. (Real > 0.0; Default = 0.0)
MIDi	Material identification number for the i-th integration point. See Remark **5** . (Integer > 0)

Remarks:

1. The PID number must be unique with respect to other PBCOMP entries as well as PBEAM entries. The second continuation entry may be repeated 18 more times. If SECTION = 5 a maximum of 21 continuation entries is allowed; i.e., a maximum of 20 lumped areas may be input. If SECTION = 1 through 4, the total number of areas input plus the total number generated by symmetry must not exceed 20. If these are not specified, the program defaults, as usual, to the elliptically distributed eight nonlinear rods. See **Figure 16-140**.

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Figure 16-140: PBCOMP Entry SYMOPT Type Examples with 8 Lumped Areas

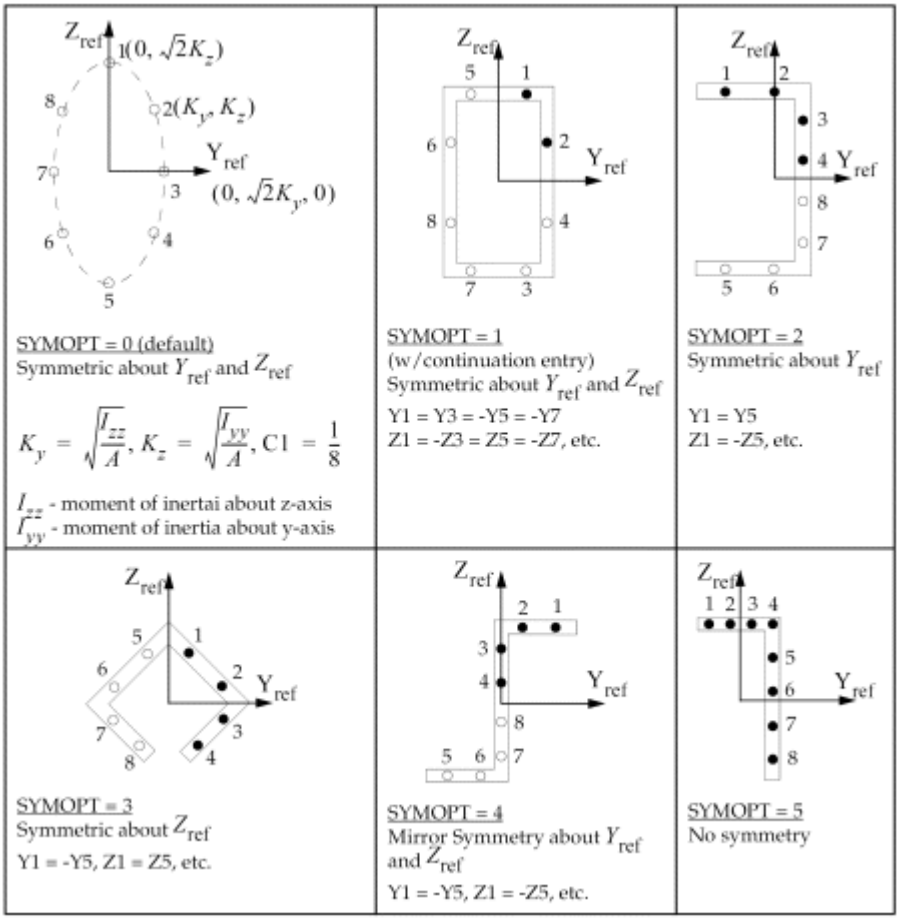

Figure Example Notes:

Integration points (lumped area) are numbered 1 through 8.

User-specified points are denoted by ● and the program default point is denoted by ○.

2. For structural problems, MID and MIDi must reference a MAT1 material entry. For material nonlinear analysis, the material should be perfectly plastic since the plastic hinge formulation is not valid for strain hardening. For heat transfer problems, MID and MIDi must reference a MAT4 or MAT5 material entry.

3. For the case where the user specifies I1, I2 and I12 on the parent entry, the stress-output location may also be specified on continuation entries. The (y,z) coordinates specified on these entries will serve as stress output locations with the corresponding Ci's set to 0. Stress output is provided at the first four lumped area locations only. If one of the symmetry options is used and fewer than four lumped areas are input explicitly, the sequence of output locations in the imaged quadrants is shown in **Figure 16-140**. For one specific example in the model shown in Remark **7 Figure 16-141**), output can be obtained at points 1 and 2 and in the image points 3 and 4.

4. Blank fields for K1 and K2 are defaulted to 1.0. If a value of 0.0 is used for K1 and K2, the transverse shear stiffness becomes rigid and the transverse shear flexibilities are set to 0.0.

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5. The values E_0 and G_0 are computed based on the value of MID on the parent entry. MID is will follow the same symmetry rules as Ci depending on the value of SECTION. If the MIDi field on a continuation entry is blank, the value will be that of MID on the parent entry. MIDi values may be input on continuations without the corresponding Yi, Zi, and Ci values to allow different stress-strain laws.

6. If the lumped cross-sectional areas are specified, fields I1, I2, and I12 will be ignored. These and other modified values will be calculated based on the input data (Yi, Zi, Ci, MIDi) as follows:

$$y_{NA} = \frac{\sum_{i=1}^{n} \text{Yi Ci Ei}}{\sum_{i=1}^{n} \text{Ci Ei}}$$

$$z_{NA} = \frac{\sum_{i=1}^{n} \text{Zi Ci Ei}}{\sum_{i=1}^{n} \text{Ci Ei}}$$

$$\bar{A} = A \sum_{i=1}^{n} \frac{\text{Ci Ei}}{\text{E}_0}$$

$$\bar{I}_1 = A \sum_{i=1}^{n} \frac{\text{Ci Ei (Yi} - y_{\text{NA}})^2}{\text{E}_o}$$

$$\bar{I}_2 = A \sum_{i=1}^{n} \frac{\text{Ci Ei (Yi} - z_{\text{NA}})^2}{\text{E}_o}$$

$$\bar{I}_{12} = A \sum_{i=1}^{n} \frac{\text{Ci Ei(Yi} - y_{NA})(\text{Zi} - z_{NA})}{\text{E}_o}$$

$$\bar{J} = \text{J} \sum_{i=1}^{n} \frac{\text{Ci Gi}}{\text{G}_o}$$

where n is the number of lumped cross-sectional areas specified.

7. As can be seen from **Figure 16-140**, if the user chooses to leave the SECTION field blank, the program defaults to the elliptically distributed eight nonlinear rods, similar to the PBEAM entry. For this particular case it is illegal to supply Ci and MIDi values. For a doubly symmetric section (SECTION = 1), if the lumped areas are specified on either axis, the symmetry option will double the areas. For example, for the section shown in **Figure 16-141**, points 2 and 4 are coincident and so are points 6 and 8. In such cases, it is recommended that users input the value of area as half of the actual value at point 2 to obtain the desired effect.

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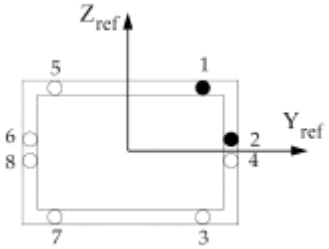
Figure 16-141: Doubly Symmetric PBCOMP Section



8. For SECTION = 5, at least one Yi and one Zi must be nonzero.

Related Topics

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PBEAM

Beam Property

Defines the properties of a beam element (CBEAM entry). This element may be used to model tapered beams.

Format:

1	2	3	4	5	6	7	8	9	10
PBEAM	PID	MID	A(A)	I1(A)	I2(A)	I12(A)	J(A)	NSM(A)	
	C1 (A)	C2 (A)	D1 (A)	D2 (A)	E1 (A)	E2 (A)	F1 (A)	F2 (A)	

The next two continuations are repeated for each intermediate station and for end B (X/Xb = 1.0) as described in Remark 4. SO and X/XB must be specified.

	SO	X/XB	A	I1	I2	I12	J	NSM	
	C1	C2	D1	D2	E1	E2	F1	F2	

The last two continuations are:

	K1	K2	S1	S2	NSI(A)	NSI(B)	CW(A)	CW(B)	
	M1(A)	M2(A)	M1(B)	M2(B)	N1(A)	N2(A)	N1(B)	N2(B)	

Example:

In the following example, the beam is tapered with a cross sectional area of 2.9 at end A and a cross sectional area of 5.3 at end B.

PBEAM	39	6	2.9	3.5	5.97				
			2.0	-4.0					
	YES	1.0	5.3	56.2	78.6				
			2.5	-5.0					
			1.1		2.1		0.21		
					0.5		0.0		

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Fields:

Field	Contents	Default Values
PID	Property identification number. (Integer > 0)	Required
MID	Material identification number. See Remarks 1 and 2. (Integer > 0)	Required
A(A)	Cross sectional area of the beam at end A. (Real > 0.0)	Required
I1(A)	Second moment of the cross sectional area at end A for bending in plane 1 about the neutral axis. See Remarks 9 and 10. (Real > 0.0)	Required
I2(A)	Second moment of the cross sectional area at end A for bending in plane 2 about the neutral axis. See Remarks 9 and 10. (Real > 0.0)	Required
I12(A)	Mixed second moment of the cross sectional area at end A. See Remarks 9 and 10. (Real ≥ 0.0)	0.0
J(A)	Torsional constant of the cross sectional area at end A. See Remark 9. (Real ≥ 0.0, but > 0.0 if warping is present)	0.0
NSM(A)	Nonstructural mass per unit length at end A. See SOLs 402 and 414 Remark 3.	0.0
Ci(A), Di(A) Ei(A), Fi(A)	Coordinates of stress recovery points for end A with i = 1 corresponding to the y_{elem} coordinate and i = 2 corresponding to the z_{elem} coordinate. See Figure 16-142. (Real)	$y_{elem} = z_{elem} = 0.0$
SO	Stress output request option at intermediate locations and end B. See Remarks 4 and 8. (Character) "YES" for stresses recovered at points Ci, Di, Ei, and Fi on the next continuation. "YESA" for stresses recovered at points with the same y_{elem} and z_{elem} location as end A. "NO" for no stresses or forces are recovered at the specific X/XB locations.	Required
X/XB	Distance from end A in the element coordinate system divided by the length of the element. See Figure 16-142 and Remark 4. (Real > 0.0)	Required
A, I1, I2, I12, J, NSM	Cross sectional area, second moments, polar moment, and nonstructural mass at an intermediate station or at end B. See Remark 4. See SOLs 402 and 414 Remark 3. (Real; J > 0.0 if warping is present)	

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Field	Contents	Default Values
Ci, Di, Ei, Fi	Coordinates of stress recovery points at X/XB with i = 1 corresponding to the y_{elem} coordinate and i = 2 corresponding to the z_{elem} coordinate. See Figure 16-142. The values are fiber locations for stress data recovery. (Real)	
K1	Transverse shear factor for bending in the element plane 1. See Remark 12. (Real)	1.0
K2	Transverse shear factor for bending in the element plane 2. See Remark 12. (Real)	1.0
S1, S2	Shear relief coefficient due to taper for plane1 and plane 2. (Real)	0.0, 0.0
NSI(A), NSI(B)	Nonstructural mass moment of inertia per unit length about nonstructural mass center of gravity at ends A and B. See Figure 16-142. (Real)	0.0, same as end A
CW(A), CW(B)	Warping coefficient for end A and end B. See Remark 11. (Real)	0.0, same as end A
M1(A), M2(A), M1(B), M2(B)	y_{elem} and z_{elem} coordinates of the center of gravity of nonstructural mass for end A and end B. See Figure 16-142. (Real)	0.0 (no offset from shear center), same values as end A
N1(A), N2(A),N1(B), N2(B)	y_{elem} and z_{elem} coordinates of neutral axis for end A and end B. See Figure 16-142. (Real)	0.0 (no offset from shear center), same values as end A

Remarks:

1. For structural analysis, MID must reference a MAT1 material entry. For material nonlinear analysis, MID may also reference a MATS1 entry, but the material properties must be defined as elastic-perfectly plastic using only H = 0.0. If a MATS1 is referenced which is not elastic-perfectly plastic with H = 0.0, a message is produced stating the material will be treated as elastic. Also, only one-eighth of the length at each end of the element abides by material nonlinear law. That is, the element is modeled as a plastic hinge. Any other type of material property specification may yield inaccurate results.
2. For heat transfer analysis, MID must reference a MAT4 or MAT5 material entry.
3. If SO is “YESA” or “NO”, the third continuation entry, which contains the fields C1 through F2, must be omitted. If SO is “YES”, the continuation for Ci, Di, Ei, and Fi must be the next entry.
4. The rules for the continuations entries are:
   - The second and third continuation entries, which contain fields SO through F2, may be repeated nine more times for intermediate X/XB values and for end B (X/Xb=1.0). The order of these continuation pairs is independent of the X/XB value; however, one value of X/XB must be 1.0, corresponding to end B.

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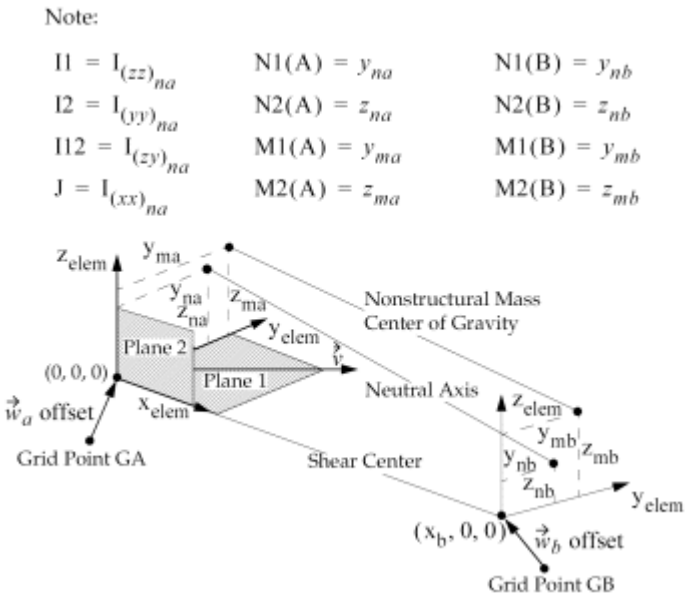
- For the nonlinear solutions 106, 129, and 401, intermediate values are permitted if, for example, the beam cross section properties change in the middle. Although, stress output requests are only supported at the ends A and B. The stress output requests at the intermediate locations (0 < X/XB < 1.0) are ignored for these nonlinear solutions.
- The fourth and fifth continuation entries, which contain fields K1 through N2(B), are optional and may be omitted if the default values are appropriate.

5. If any fields 4 through 9 are blank on the continuation with the value of X/XB = 1.0, then the values for A, I1, I2, I12, J and NSM are set to the values given for end A. For the continuations that have intermediate values of X/XB between 0.0 and 1.0 and use the default option (any of the fields 4 through 9 are blank), a linear interpolation between the values at ends A and B is performed to obtain the missing section properties.
6. If SO is "YES", blank fields are defaulted to 0.0.
7. Blank fields for K1, K2 are defaulted to 1.0. If a value of 0.0 is used for K1 and K2, the transverse shear flexibilities are set to 0.0 and field G on the MAT1 entry selected by MID must be nonzero.
8. If force is desired at the intermediate stations and/or end B, put "YES" or "YESA" in the SO field for the intermediate stations and/or end B.
9. Figure 16-142 describes the PBEAM element coordinate system.

Figure 16-142: PBEAM Element Coordinate System



The following table shows the formulas for cross sectional properties.

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Field	Formula
A	$\int_{Area} dA$
I1	$\int_{Area} y_{elem}^2 dA$
I2	$\int_{Area} z_{elem}^2 dA$
I12	$\int_{Area} y_{elem} z_{elem} dA$
J	$\int_{Area} r^2 dA = \int_{Area} \left(y_{elem}^2 + z_{elem}^2\right) dA$

10. The product of I1 and I2 must be greater than I12 squared. That is, I1 x I2 > $I12^2$. Otherwise, a fatal message is issued.

11. The warping coefficient CW is represented in the following differential equation for torsion of a beam about the axis of the shear centers:

$$G\frac{d}{dx}\left(J\frac{d\theta}{dx}\right) - E\frac{d^2}{dx^2}\left(CW\frac{d^2\theta}{dx^2}\right) = m$$

where:

G = Shear modulus

J = Polar moment of the cross sectional area

E = Young's modulus

θ = Angle of rotation at any cross section

m = Applied torsional moment per unit length

CW has units of $(length)^6$.

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12. The specification of the K1 and K2 fields determine the type of beam model that the software uses:

    - For a Timoshenko beam, specify real nonzero values for the K1 and K2 fields.
    - For an Euler-Bernoulli beam, specify the K1 and K2 fields as real zero.

    Timoshenko beams account for transverse shear in the beam deflection calculation, whereas Euler-Bernoulli beams do not.

    If you specify one of the K1 or K2 fields as real nonzero or leave it blank and the other as real zero, the software uses the Timoshenko beam model for the deflection calculation in the plane of the real nonzero or blank shear stiffness factor and uses the Euler-Bernoulli beam model for the deflection calculation in the plane of the real zero shear stiffness factor. For example, if you specify K1 = 0.5 and K2 = 0.0, the software uses the Timoshenko beam model for the deflection in the element plane 1, and uses the Euler-Bernoulli beam model for the deflection calculation in the element plane 2.

13. I1 and I2 are also referred to as moments of inertia. I12 is also referred to as a product of inertia.

Remarks related to SOLs 402 and 414:

1. PBEAM cannot be used for elasto-plastic beam elements. In that case, use the PBEAML entry.
2. Cross sections can only be defined at ends A and B. A linear variation is assumed between the ends. Intermediate stations are not supported.
3. The NSM(A) field is supported by SOLs 402 and 414 to apply nonstructural mass to the element. The software applies the value for NSM(A) as a constant along the beam. For SOL 414 only, you can use the WTMASS parameter to scale all mass in your model, including any nonstructural mass defined. For example, you can use the WTMASS parameter to convert applied mass from weight units (Newtons, pound-force) to mass units (kilograms, pound-mass). All mass in your model must be defined using consistent units.

Related Topics

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PBEAML

Beam Cross-Section Property

Defines the properties of a beam element by cross-sectional dimensions.

Format:

Note: n = number of dimensions and m = number of intermediate stations.

1	2	3	4	5	6	7	8	9	10
PBEAML	PID	MID	GROUP	TYPE					
	DIM1(A)	DIM2(A)	-etc.-	DIMn(A)	NSM(A)	SO(1)	X(1)/XB	DIM1(1)	
	DIM2(1)	-etc.-	DIMn(1)	NSM(1)	SO(2)	X(2)/XB	DIM1(2)	DIM2(2)	
	-etc.-	DIMn(2)	-etc.-	NSM(m)	SO(m)	X(m)/XB	DIM1(m)	-etc.-	
	DIMn(m)	NSM(m)	SO(B)	1.0	DIM1(B)	DIM2(B)	-etc.-	DIMn(B)	
		NSM(B)							

Example:

PBEAML	99	21		T					
	12.0	14.8	2.5	2.6		NO	0.4	6.0	
	7.0	1.2	2.6		YES	0.6	6.0	7.8	
	5.6	2.3		YES					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. (Integer > 0)
GROUP	Cross-section group. (Character; Default = “MSCBML0”)
TYPE	Cross-section shape. See Remark 4. (Character: “ROD”, “TUBE”, “L”, “I”, “CHAN”, “T”, “BOX”, “BAR”, “CROSS”, “H”, “T1”, “I1”, “CHAN1”, “Z”, “CHAN2”, “T2”, “BOX1”, “HEXA”, “HAT”, “HAT1” for GROUP = “MSCBML0”)
DIMi(A), DIMi(B)	Cross-section dimensions at end A and B. (Real > 0.0 for GROUP = “MSCBML0”)
NSM(A), NSM(B)	Nonstructural mass per unit length. In Remarks related to SOLs 402 and 414, see Remark 8. (Real ≥ 0.0; Default = 0.0)

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Field	Contents
SO(j),SO(B)	Stress output request option for intermediate station j and end B. (Character; Default = "YES") YES: Stresses recovered at all points on next continuation and shown in **Figure 16-143** as C, D, E, and F. NO: No stresses or forces are recovered.
X(j)/XB	Distance from end A to intermediate station j in the element coordinate system divided by the length of the element. (Real > 0.0; Default = 1.0)
NSM(j)	Nonstructural mass per unit length at intermediate station j. (Real ≥ 0.0; Default = 0.0)
DIMi(j)	Cross-section dimensions at intermediate station j. (Real > 0.0 for GROUP = "MSCBML0")

Remarks:

1. For structural problems, PBEAML entries must reference a MAT1 material entry. For material nonlinear analysis, MID may also reference a MATS1 entry, but the material properties must be defined as elastic-perfectly plastic using only H = 0.0. If a MATS1 is referenced which is not elastic-perfectly plastic with H=0.0, a message is produced stating the material will be treated as elastic. Also, only one-eighth of the length at each end of the element abides by material nonlinear law; i.e., the element is modeled as a plastic hinge. Any other type of material property specification may yield inaccurate results.
2. PID must be unique with respect to all other PBEAM and PBEAML property identification numbers.
3. For heat-transfer problems, the MID must reference a MAT4 or MAT5 material entry.
4. See the PBEAM entry description for a discussion of beam-element geometry.
5. If any of the fields NSM(B), DIMi(B) are blank on the continuation entry for End B, the values are set to the values given for end A. For the continuation entries that have values of X(j)/XB between 0.0 and 1.0 and use the default option (blank field), a linear interpolation between the values at ends A and B is performed to obtain the missing field.
6. The GROUP is associated with a FMS CONNECT statement, which specifies the evaluator. A reserved GROUP name is "MSCBML0". Users may create their own cross-section types. Each of the types will require a one or more subroutines to convert DIMi information to geometric property information contained on a PBEAM entry.
7. For GROUP = "MSCBML0", the cross-sectional properties, shear flexibility factors and stress recovery points are computed using the TYPE and DIMi as shown in **Figure 16-143**. The element coordinate system is located at the shear center.
8. For the formulas that the software uses to calculate the cross-section properties and, if applicable, the shear center location, see Using Supplied Beam and Bar Libraries in the *Simcenter Nastran Element Library Reference*.

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9. An equivalent PBEAM entry is created from the PBEAML entry. Any sorted echo request will also cause printout and/or punch of the derived PBEAM. The equivalent PBEAM does not include warping coefficients.

10. The cross sectional properties on the equivalent PBEAM entry are calculated using hard-coded formulas for each section type. To use the mesh-based Pilkey method to calculate the cross sectional properties, specify PARAM,PBRPROP,YES.

11. Beams can have no more than 14 dimensions per station. The total number of dimensions at all stations must be less than 200. The transfer of data with the beam server is limited to 4000 words.

    None of these limits are exceeded with the Simcenter Nastran beam library, but a user defined beam library could.

    There is a further limit that the NSM values input on the PBARL and PBEAML must be ≥ 0. No other property types have this limit.

12. C, D, E, and F in **Figure 16-143** are the locations of stress recovery on a cross-section.

Remarks related to SOLs 402 and 414:

1. The “ROD”, "TUBE", "BOX", "BAR", “L”, “I”, “I1”, “CHAN”, “CHAN1”, “T”, and “Z” cross section types are supported.

2. Beginning with version 2506, the origin of the supported cross sections is at the shear center, which is consistent with other solutions such as SOLs 103 and 401. See the remarks on the **CBEAM** for the definition of the element coordinate system.

3. The cross section types “T1”, “T2”, “CHAN2”, “CROSS”, “H”, “BOX1”, “HEXA”, “HAT”, and “HAT1” are not fully supported. If you define these types, the software automatically creates an equivalent PBEAM entry but does not perform stress-recovery. For TYPE = “T1”, “T2”, “CHAN2”, and “H”, equivalent supported profiles exist in other axes (“T”, “CHAN”, “CHAN1”, “I” or “I1”).

4. Cross sections can only be defined at ends A and B. A linear variation is assumed between the ends. Intermediate stations are not supported.

5. TYPE="BAR", "BOX", "CHAN", "CHAN1", "I", "I1", "L", "T", "TUBE", and "Z" can be used with a nonlinear material (for example, plastic or temperature-dependent). If a plastic material is used (MATS1 entry with TYPE="PLASTIC"), the behavior is not limited to elastic-perfectly plastic.

   The material behavior is calculated by numerical integration in the central section of the element. It should be constant for the whole element. To have a behavior close to a plastic hinge, you must use a finer mesh.

6. TYPE=”ROD” is not supported with nonlinear materials. As an alternative, you can use a TUBE section with a small DIM2 dimension.

7. For TYPE = “I”, “I1”, “CHAN”, “CHAN1”, “L”, “Z”, “T”, “BOX” and “TUBE”, the stress recovery locations are located at the element's mid-thickness instead of the outer skin. Therefore, small deviations can be observed compared to other solutions like SOL 401.

8. Nonstructural mass (NSM) defined on the PBEAML is not supported.

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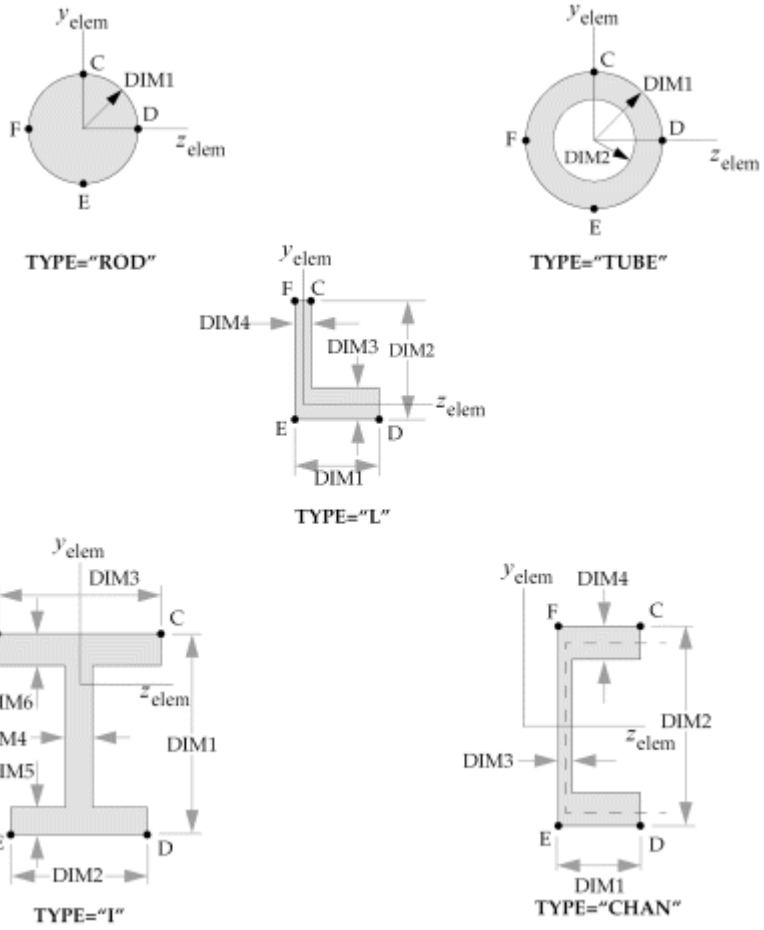
Figure 16-143: Definition of Cross-Section Geometry and Stress Recovery Points for GROUP = "MSCBML0"



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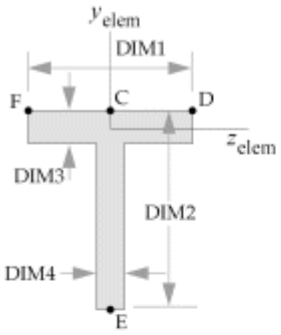
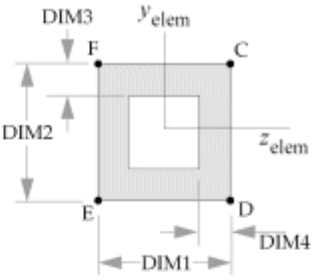
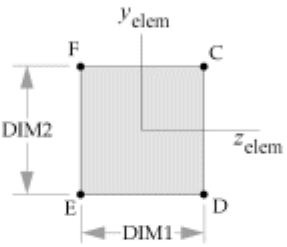


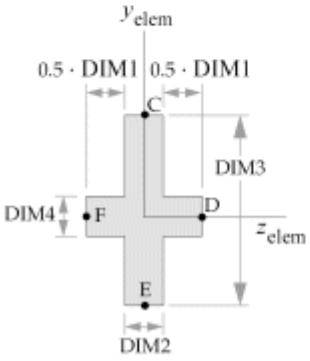
TYPE="T"



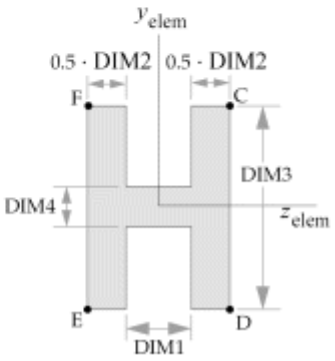
TYPE="BOX"



TYPE="BAR"



TYPE="CROSS"



TYPE="H"

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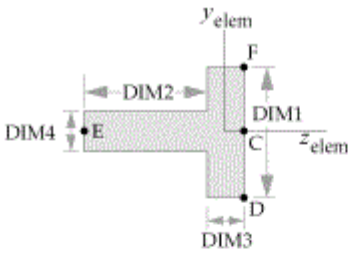

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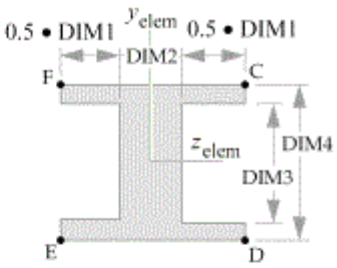
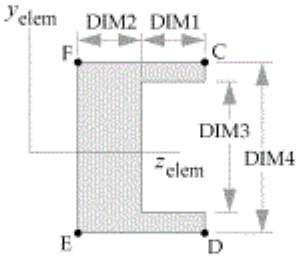
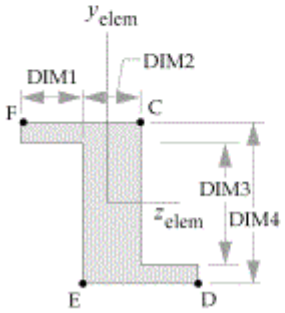


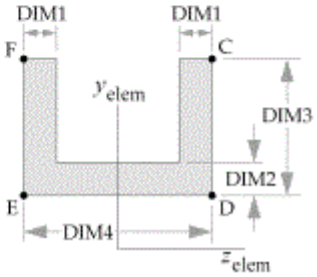
TYPE="I1"



TYPE="CHAN1"



TYPE="Z"



TYPE="CHAN2"

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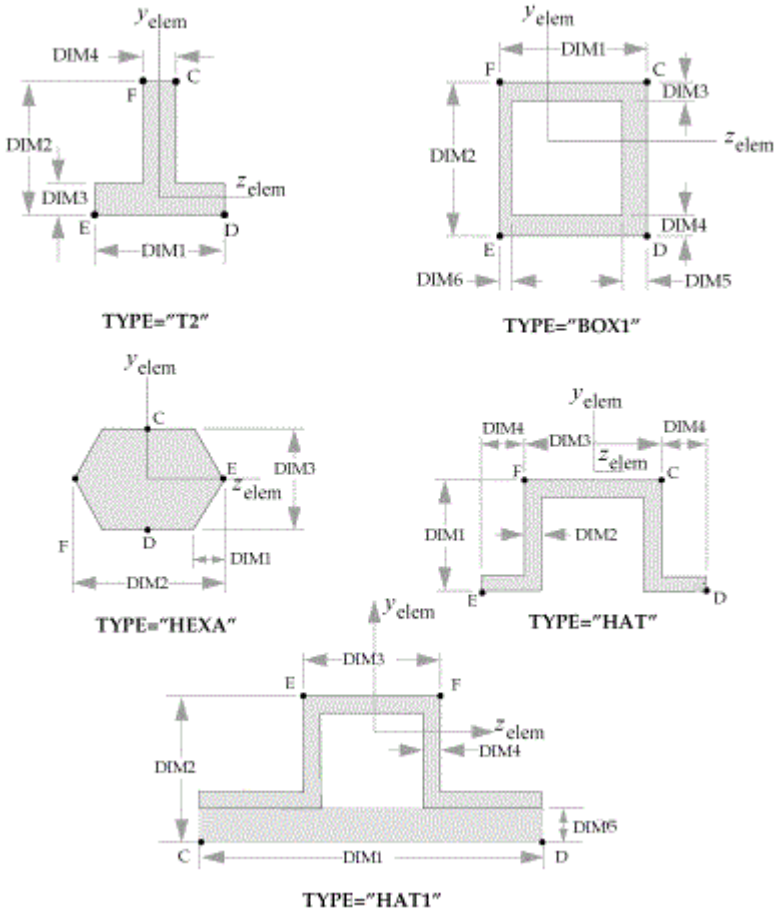

Related Topics

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PBEAR

Bearing Property Definition

Defines stiffness, viscous damping, and mass matrices for bearing connection. Applicable to all rotor dynamics solution types (SOLs 101, 107, 108, 109, 110, 111, 112, and 414).

Format:

1	2	3	4	5	6	7	8	9	10
PBEAR	PID	TYPE	TXX	TXY	TYX	TYY		NOMVAL1	

Continuation Line Formats:

1	2	3	4	5	6	7	8	9	10
		"COM"	C1	C1Z	D1O				
		TYPEZ	TXZ	TYZ	TZX	TZY	TZZ	NOMVAL2	
		"COMZ"	C2	C2Z	D2O				
		TYPER	TRXRX	TRXRY	TRYRX	TRYRY		NOMVAL3	
		"COMR"	C3	C3Z	D3O				

The ordering of continuation lines is arbitrary.

Examples:

PBEAR	5	K	1001	1002	25.0	27.5			
		B	2001	2002	2003	2004			

PBEAR	5	KD	1001	1002	25.0	27.5		1.0E-1	
		BD	4001	4002	4002	4001			
		COM	1.0	0.0	1.0E-3				
		KDZ	2001	2002	2001	2002	2003	2.0E-1	
		BDZ	5001	5002	5001	5002	5003		
		COMZ	0.0	1.0	0.0				
		KDR	3001	3002	3002	1.0E5		1.0E-2	
		BDR	6001	6002	6002	1.0E2			
		COMR	1.0	0.0	0.0				

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Fields:

Field	Contents
PID	Property identification number which is referenced by a CBEAR entry. (Integer > 0)
TYPE	Type of data in the TXX, TXY, TYX, and TYY fields on the same line. (Character: “K”, “B”, “M”, “KD”, “KF”, “BD”, “BF”, “MD”, “MF”) • If TYPE = “K”, specifies constant stiffness or stiffness that is a function of rotor speed. • If TYPE = “B”, specifies constant viscous damping or viscous damping that is a function of rotor speed. • If TYPE = “M”, specifies constant mass or mass that is a function of rotor speed. See Remark 3 for SOL 414. • If TYPE = “KD”, specifies constant stiffness or stiffness that is a function of rotor speed and relative displacement. • If TYPE = “KF”, specifies constant stiffness or stiffness that is a function of rotor speed and relative force. • If TYPE = “BD”, specifies constant viscous damping or viscous damping that is a function of rotor speed and relative displacement. • If TYPE = “BF”, specifies constant viscous damping or viscous damping that is a function of rotor speed and relative force. • If TYPE = “MD”, specifies constant mass or mass that is a function of rotor speed and relative displacement. See Remark 3 for SOL 414. • If TYPE = “MF”, specifies constant mass or mass that is a function of rotor speed and relative force. See Remark 3 for SOL 414. If TYPE = “KD”, “KF”, “BD”, “BF”, “MD”, “MF”, see Remark 1 and Remark 2 . See also Remark 12.
TXX, TXY, TYX, TYY	Stiffness, viscous damping, or mass matrix entry. See Remark 4. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 5) If TYPE = “K”, “B”, “M” • If real entry, value of stiffness, viscous damping, or mass matrix entry used for all rotor speeds. • If integer entry, identification number of a TABLEDi bulk entry that defines the stiffness, viscous damping, or mass matrix entry as a function of rotor speed. See Remark 6. If TYPE = “KD”, “KF”, “BD”, “BF”, “MD”, “MF”:

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Field	Contents
	• If real entry, value of stiffness, viscous damping, or mass matrix entry used for all rotor speeds and relative displacements or rotor speeds and relative forces. • If integer entry, identification number of a TABLEST bulk entry that defines the stiffness, viscous damping, or mass matrix entry as a function of rotor speed and relative displacement or rotor speed and relative force. The TABLEST bulk entry references a series of TABLEDi bulk entries. The TABLEDi bulk entries contain tabular data of stiffness, viscous damping, or mass vs. rotor speed at constant values of relative displacement or relative force. See Remark 6 and Remark 9.
NOMVAL1	Valid if TYPE = “KD”, “KF”, “BD”, “BF”, “MD”, “MF”. Field is ignored if TYPE = “K”, “B”, “M”. See Remark 7. (Real ≥ 0.0; Default = 0.0) • For SOLs 101, 108, 109, 111, and 112, defines the relative displacement or relative force that is used to either directly compute the bearing stiffness, viscous damping, or mass, or initiate iteration for the bearing stiffness, viscous damping, or mass. See Remark 8. • For SOLs 107, 110, defines the relative displacement or relative force that is used to directly compute the bearing stiffness, viscous damping, or mass.
“COM”	COM flag. Indicates that coefficients for composite relative displacement radial equation or composite relative force radial equation follow. (Character)
C1, C1Z, D1O	Coefficients for composite relative displacement radial equation or composite relative force radial equation. See Remark 10 and Remark 11. (Real; Defaults are C1 = 1.0, C1Z = 0.0, D1O = 0.0)
TYPEZ	Type of data in the TXZ, TYZ, TZX, TZY, and TZZ fields on the same line. (Character: “KZ”, “BZ”, “MZ”, “KDZ”, “KFZ”, “BDZ”, “BFZ”, “MDZ”, “MFZ”) • If TYPE = “KZ”, specifies constant stiffness or stiffness that is a function of rotor speed. • If TYPE = “BZ”, specifies constant viscous damping or viscous damping that is a function of rotor speed. • If TYPE = “MZ”, specifies constant mass or mass that is a function of rotor speed. See Remark 3 for SOL 414. • If TYPE = “KDZ”, specifies constant stiffness or stiffness that is a function of rotor speed and relative displacement. • If TYPE = “KFZ”, specifies constant stiffness or stiffness that is a function of rotor speed and relative force. • If TYPE = “BDZ”, specifies constant viscous damping or viscous damping that is a function of rotor speed and relative displacement. • If TYPE = “BFZ”, specifies constant viscous damping or viscous damping that is a function of rotor speed and relative force. • If TYPE = “MDZ”, specifies constant mass or mass that is a function of rotor speed and relative displacement. See Remark 3 for SOL 414.

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Field	Contents
	• If TYPE = “MFZ”, specifies constant mass or mass that is a function of rotor speed and relative force. See Remark 3 for SOL 414. If TYPE = “KDZ”, “KFZ”, “BDZ”, “BFZ”, “MDZ”, “MFZ”, see Remark 1 and Remark 2.
TXZ, TYZ, TZX, TZY, TZZ	Stiffness, viscous damping, or mass matrix entry. See Remark 4. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 5) If TYPE = “KZ”, “BZ”, “MZ”: • If real entry, value of stiffness, viscous damping, or mass matrix entry used for all rotor speeds. • If integer entry, identification number of a TABLEDi bulk entry that defines the stiffness, viscous damping, or mass matrix entry as a function of rotor speed. See Remark 6. If TYPE = “KDZ”, “KFZ”, “BDZ”, “BFZ”, “MDZ”, “MFZ”: • If real entry, value of stiffness, viscous damping, or mass matrix entry used for all rotor speeds and relative displacements or rotor speeds and relative forces. • If integer entry, identification number of a TABLEST bulk entry that defines the stiffness, viscous damping, or mass matrix entry as a function of rotor speed and relative displacement or rotor speed and relative force. The TABLEST bulk entry references a series of TABLEDi bulk entries. The TABLEDi bulk entries contain tabular data of stiffness, viscous damping, or mass vs. rotor speed at constant values of relative displacement or relative force. See Remark 6 and Remark 9.
NOMVAL2	Valid if TYPE = “KDZ”, “KFZ”, “BDZ”, “BFZ”, “MDZ”, “MFZ”. Field is ignored if TYPE = “KZ”, “BZ”, “MZ”. See Remark 7. (Real ≥ 0.0; Default = 0.0) • For SOLs 101, 108, 109, 111, and 112, defines the relative displacement or relative force that is used to either directly compute the bearing stiffness, viscous damping, or mass, or initiate iteration for the bearing stiffness, viscous damping, or mass. See Remark 8. • For SOLs 107, 110, defines the relative displacement or relative force that is used to directly compute the bearing stiffness, viscous damping, or mass.
“COMZ”	COMZ flag. Indicates that coefficients for composite relative displacement axial equation or composite relative force axial equation follow. (Character)
C2, C2Z, D2O	Coefficients for composite relative displacement axial equation or composite relative force axial equation. See Remark 10 and Remark 11. (Real; Defaults are C2 = 0.0, C2Z = 1.0, D2O = 0.0)
TYPER	Type of data in the TRXRX, TRXRY, TRYRX, and TRYRY fields on the same line. (Character: “KR”, “BR”, "MR", “KDR”, “KFR”, “BDR”, “BFR”, “MDR”, “MFR”)

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Field	Contents
	• If TYPE = “KR”, specifies constant stiffness or stiffness that is a function of rotor speed. • If TYPE = “BR”, specifies constant viscous damping or viscous damping that is a function of rotor speed. • If TYPE = “MR”, specifies constant mass or mass that is a function of rotor speed. See Remark 3 for SOL 414. • If TYPE = “KDR”, specifies constant stiffness or stiffness that is a function of rotor speed and relative displacement. • If TYPE = “KFR”, specifies constant stiffness or stiffness that is a function of rotor speed and relative force. • If TYPE = “BDR”, specifies constant viscous damping or viscous damping that is a function of rotor speed and relative displacement. • If TYPE = “BFR”, specifies constant viscous damping or viscous damping that is a function of rotor speed and relative force. • If TYPE = “MDR”, specifies constant mass or mass that is a function of rotor speed and relative displacement. See Remark 3 for SOL 414. • If TYPE = “MFR”, specifies constant mass or mass that is a function of rotor speed and relative force. See Remark 3 for SOL 414. If TYPE = “KDR”, “KFR”, “BDR”, “BFR”, “MDR”, “MFR”, see Remark 1 and Remark 2.
TRXRX, TRXRY, TRYRX, TRYRY	Stiffness or viscous damping matrix entry. See Remark 4. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 5) If TYPE = “KR”, “BR”, “MR”: • If real entry, value of stiffness, viscous damping, or mass matrix entry used for all rotor speeds. • If integer entry, identification number of a TABLEDi bulk entry that defines the stiffness, viscous damping, or mass matrix entry as a function of rotor speed. See Remark 6. If TYPE = “KDR”, “KFR”, “BDR”, “BFR”, “MDR”, “MFR”: • If real entry, value of stiffness, viscous damping, or mass matrix entry used for all rotor speeds and relative displacements or rotor speeds and relative forces. • If integer entry, identification number of a TABLEST bulk entry that defines the stiffness, viscous damping, or mass matrix entry as a function of rotor speed and relative displacement or rotor speed and relative force. The TABLEST bulk entry references a series of TABLEDi bulk entries. The TABLEDi bulk entries contain tabular data of stiffness, viscous damping, or mass vs. rotor speed at constant values of relative displacement or relative force. See Remark 9.

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Field	Contents
NOMVAL3	Valid if TYPE = "KDR", "KFR", "BDR", "BFR", "MDR", "MFR". Field is ignored if TYPE = "KR", "BR", "MR". See Remark 7. (Real ≥ 0.0; Default = 0.0) • For SOLs 101, 108, 109, 111, and 112, defines the relative displacement or relative force that is used to either directly compute the bearing stiffness, viscous damping, or mass, or initiate iteration for the bearing stiffness, viscous damping, or mass. See Remark 8. • For SOLs 107, 110, defines the relative displacement or relative force that is used to directly compute the bearing stiffness, viscous damping, or mass.
"COMR"	COMR flag. Indicates that coefficients for composite relative displacement rotational equation or composite relative force rotational equation follow. (Character)
C3, C3Z, D3O	Coefficients for composite relative displacement rotational equation or composite relative force rotational equation. See Remark 10 and Remark 11. (Real; Defaults are C3 = 1.0, C3Z = 0.0, D3O = 0.0)

Remarks:

1. On a PBEAR bulk entry, all of the property types must be of the same type. That is, all of the TYPE fields on a PBEAR bulk entry must be specified as either:
   - Constant or speed dependent with the TYPE fields set to K, KZ, KR, B, BZ, BR, M, MZ, and MR.
   - Constant or speed and displacement dependent with the TYPE fields set to KD, KDZ, KDR, BD, BDZ, BDR, MD, MDZ, and MDR.
   - Constant or speed and force dependent with the TYPE fields set to KF, KFZ, KFR, BF, BFZ, BFR, MF, MFZ, and MFR.

   An error occurs if a PBEAR bulk entry has multiple property types.
2. *Relative displacement* is the displacement between the two coincident grids used to define the CBEAR connection. *Relative force* is the force carried through the CBEAR connection. The relative force is calculated from the relative displacement.
3. Define speed-dependent bearing properties in the same units for speed that are specified in the RUNIT field of the ROTORD bulk entry.
4. "X", "Y", and "Z", in TXX, TXY, TYX, TYY, TXZ, TYZ, TZX, TZY, and TZZ, refer to the X, Y, and Z-axes of the coordinate system referenced in the RCORDi field of the ROTORD bulk entry. "RX" and "RY" in TRXRX, TRXRY, TRYRX, and TRYRY refer to rotation about the X-axis and rotation about the Y-axis of the coordinate system referenced in the RCORDi field of the ROTORD bulk entry. In a rotor dynamic analysis, a rotor's axis of rotation must be aligned with the Z-axis of the coordinate system referenced in the RCORDi field of the ROTORD bulk entry.
5. If any of TXX, TXY, TYX, TYY, TXZ, TYZ, TZX, TZY, TZZ, TRXRX, TRXRY, TRYRX, or TRYRY fields are blank or zero (either integer zero or real zero), the software uses real zero as the value for the corresponding field.

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6. In TABLEDi bulk entries, enter the rotor speed data in the units that are specified in the RUNIT field of the ROTORD bulk entry.

7. If a NOMVALi is defined in both a stiffness, damping, and mass row, the value in the stiffness row is used.

8. When TYPE = “KD”, “KF”, “BD”, “BF”, “MD”, “MF”, “KDZ”, “KFZ”, “BDZ”, “BFZ”, “MDZ”, “MFZ”, “KDR”, “KFR”, “BDR”, “BFR”, “MDR”, or “MFR” for SOL 101, 108, and 111, the method used to determine the stiffness, viscous damping, or mass for CBEAR elements depends on the value of MAXITER.

   - If MAXITER = 0, the value specified in the NOMVALi field is used to directly compute the bearing stiffness, viscous damping, and mass.
   - If MAXITER ≠ 0, the value specified in the NOMVALi field is used to initiate iteration over relative displacement or relative force. Consecutive iterations use the relative displacement and relative force from the previous iteration to update the bearing stiffness, viscous damping, and mass. The relative displacement or relative force magnitude on the CBEAR element at each iteration is compared with the value from the previous iteration to evaluate convergence. When iterating over relative displacement, convergence is reached when:

     |(Disp (old) - Disp (new)) / Disp (old)| < THRSHOLD

     When iterating over relative force, convergence is reached when:

     |(Force (old) - Force (new)) / Force (old)| < THRSHOLD

     Iterating stops when either convergence is met, or the number of iterations exceeds MAXITER.

     MAXITER and THRSHOLD are specified on the ROTORD bulk entry.

9. On a TABLEST bulk entry referenced by a PBEAR bulk entry, the values for relative displacement or relative force that correspond to the TABLEDi bulk entries must be consistently defined for all TABLEDi bulk entries. For example, a valid entry is as follows:

```
PBEAR   789     KD      6891    6892    6893    6894
$
$ TABLEST tables for PBEAR
$
TABLEST 6891
        0.0     7891    0.1     8891    0.3     9891    ENDT
TABLEST 6892
        0.0     7892    0.1     8892    0.3     9892    ENDT
TABLEST 6893
        0.0     7893    0.1     8893    0.3     9893    ENDT
TABLEST 6894
        0.0     7894    0.1     8894    0.3     9894    ENDT
```

The following entry is invalid:

```
PBEAR   789     KD      6891    6892    6893    6894
$
$ TABLEST tables for PBEAR
$
TABLEST 6891
```


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```
        0.0     7891    0.1     8891    0.3     9891    ENDT
TABLEST 6892
        0.0     7892    0.1     8892    0.4     9892    ENDT
TABLEST 6893
        0.0     7893    0.1     8893    0.3     9893    ENDT
TABLEST 6894
        0.0     7894    0.1     8894    0.3     9894    ENDT
```

The following entry is also invalid:

```
PBEAR   789     KD      6891    6892    6893    6894
$
$ TABLEST tables for PBEAR
$
TABLEST 6891
        0.0     7891    0.1     8891    0.3     9891    ENDT
TABLEST 6892
        0.0     7892    0.1     8892    0.3     9892    ENDT
TABLEST 6893
        0.0     7893    0.1     8893    ENDT
TABLEST 6894
        0.0     7894    0.1     8894    0.3     9894    ENDT
```

However, the tabular data entered on the TABLEDi bulk entries that are referenced by a TABLEST bulk entry do not need to have the same range.

10. *Composite relative displacements* are linear combinations of radial and axial relative displacements. The software calculates the composite relative displacements from the following equations:

$\Delta_1 = C1\ \Delta_r + C1Z\ \Delta_z + D1O$
$\Delta_2 = C2\ \Delta_r + C2Z\ \Delta_z + D2O$
$\Delta_3 = C3\ \Delta_r + C3Z\ \Delta_z + D3O$

where Δ_1 is the composite radial relative displacement, Δ_2 is the composite axial relative displacement, Δ_3 is the composite rotational relative displacement, Δ_r is the radial relative displacement, and Δ_z is the axial relative displacement. The D1O, D2O, and D3O coefficients represent preload displacements.

The software calculates Δ_r as follows:

$\Delta_r = (\Delta_x^2 + \Delta_y^2)^{1/2}$

where Δ_x is the relative displacement in the x-direction and Δ_y is the relative displacement in the y-direction.

The software calculates Δ_z as follows:

$\Delta_z = \Delta_{GB} - \Delta_{GA}$

where Δ_{GA} and Δ_{GB} are the axial displacements of the grids listed on the CBEAR bulk entry.

For bearing properties that are speed and displacement-dependent, the software uses the composite relative displacements to look up values for bearing stiffness, viscous damping, and mass. The matrix for bearing stiffness, viscous damping, or mass is as follows:

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$$\begin{Bmatrix} f_x \\ f_y \\ f_z \\ f_{rx} \\ f_{ry} \end{Bmatrix} = \begin{bmatrix} T_{xx}(\Delta_1,\Omega) & T_{xy}(\Delta_1,\Omega) & T_{xz}(\Delta_2,\Omega) & 0 & 0 \\ T_{yx}(\Delta_1,\Omega) & T_{yy}(\Delta_1,\Omega) & T_{yz}(\Delta_2,\Omega) & 0 & 0 \\ T_{zx}(\Delta_2,\Omega) & T_{zy}(\Delta_2,\Omega) & T_{zz}(\Delta_2,\Omega) & 0 & 0 \\ 0 & 0 & 0 & T_{rxrx}(\Delta_3,\Omega) & T_{rxry}(\Delta_3,\Omega) \\ 0 & 0 & 0 & T_{ryrx}(\Delta_3,\Omega) & T_{ryry}(\Delta_3,\Omega) \end{bmatrix} \begin{Bmatrix} \Delta_x \\ \Delta_y \\ \Delta_z \\ \Delta_{rx} \\ \Delta_{ry} \end{Bmatrix}$$

where $T_{ij}(\Delta_i, \Omega)$ are stiffness, viscous damping, or mass matrix entries as a function of composite relative displacement and angular speed of the rotor, Ω. If the matrix entries represent viscous damping, Δ_x, Δ_y, and Δ_z represent translational velocities, and Δ_{rx} and Δ_{ry} represent angular velocities. If the matrix entries represent mass, Δ_x, Δ_y, and Δ_z represent translational accelerations, and Δ_{rx} and Δ_{ry} represent angular accelerations.

If the calculated relative displacement is less than zero, the software automatically resets it to zero when it looks up values for bearing stiffness, viscous damping, and mass.

11. *Composite relative forces* are linear combinations of radial and axial relative forces. The software calculates the composite relative forces from the following equations:

F_1 = C1 F_r + C1Z F_z + D1O
F_2 = C2 F_r + C2Z F_z + D2O
F_3 = C3 F_r + C3Z F_z + D3O

where F_1 is the composite radial relative force, F_2 is the composite axial relative force, F_3 is the composite rotational relative force, F_r is the radial relative force, and F_z is the axial relative force. The D1O, D2O, and D3O coefficients represent preload forces.

The software calculates F_r as follows:

$F_r = (F_x^2 + F_y^2)^{1/2}$

where F_x is the relative force in the x-direction and F_y is the relative force in the y-direction.

For bearing properties that are speed and force-dependent, the software uses the composite relative forces to look up values for bearing stiffness, viscous damping, and mass. The matrix for bearing stiffness, viscous damping, or mass is as follows:

$$\begin{Bmatrix} f_x \\ f_y \\ f_z \\ f_{rx} \\ f_{ry} \end{Bmatrix} = \begin{bmatrix} T_{xx}(F_1,\Omega) & T_{xy}(F_1,\Omega) & T_{xz}(F_2,\Omega) & 0 & 0 \\ T_{yx}(F_1,\Omega) & T_{yy}(F_1,\Omega) & T_{yz}(F_2,\Omega) & 0 & 0 \\ T_{zx}(F_2,\Omega) & T_{zy}(F_2,\Omega) & T_{zz}(F_2,\Omega) & 0 & 0 \\ 0 & 0 & 0 & T_{rxrx}(F_3,\Omega) & T_{rxry}(F_3,\Omega) \\ 0 & 0 & 0 & T_{ryrx}(F_3,\Omega) & T_{ryry}(F_3,\Omega) \end{bmatrix} \begin{Bmatrix} \Delta_x \\ \Delta_y \\ \Delta_z \\ \Delta_{rx} \\ \Delta_{ry} \end{Bmatrix}$$

where $T_{ij}(F_i, \Omega)$ are stiffness, viscous damping, or mass matrix entries as a function of composite relative force and angular speed of the rotor, Ω. If the matrix entries represent viscous damping, Δ_x, Δ_y, and Δ_z represent translational velocities, and Δ_{rx} and Δ_{ry} represent angular velocities. If the matrix

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entries represent mass, Δ_x, Δ_y, and Δ_z represent translational accelerations, and Δ_{rx} and Δ_{ry} represent angular accelerations.

If the calculated relative force is less than zero, the software automatically resets it to zero when it looks up values for bearing stiffness, viscous damping, and mass.

12. Only the following combinations of TYPE are allowed:
    - Constant of speed dependent: K, KZ, KR, B, BZ, BR, M, MZ, MR.
    - Constant or speed and displacement dependent: KD, KDZ, KDR, BD, BDZ, BDR, MD, MDZ, MDR.
    - Constant or speed and force dependent: KF, KFZ, KFR, BF, BFZ, BFR, MF, MFZ. MFR.

    Mixing the different options on the same PBEAR entry is not allowed.

Remarks related to SOL 414:

1. PBEAR entries are valid in SOL 414 subject to the following restrictions:
    - If TYPE = “KD”, “KF”, “BD”, “BF”, “MD”, or “MF”, the NOMVAL1 field must be specified.
    - If TYPE = “KDZ”, “KFZ”, “BDZ”, “BFZ”, “MDZ”, or “MFZ”, the NOMVAL2 field must be specified.
    - If TYPE = “KDR”, “KFR”, “BDR”, “BFR”, “MDR”, or “MFR”, the NOMVAL3 field must be specified.
2. Coefficients for composite relative displacement are not supported in SOL 414.
3. For SOL 414 solutions, if you set the mass parameters (M, MD, MF, MZ, MDZ, MFZ, MR, MDR, MFR) in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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PBEAR2

Bearing Property Definition

Defines properties for bearing connections in SOL 414.

Format:

1	2	3	4	5	6	7	8	9	10
PBEAR2	PID	TYPE							
	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3	PARAM4	VALUE4	
	PARAM5	VALUE5	PARAM6	VALUE6	-etc-				

Examples:

PBEAR2	5	BEAR							
	K11	45.0	K12	10	K16	12.5	K21	20	
	K22	53.2	K33	20.0	K61	8.3	ROTF	1	

PBEAR2	5	ROLL							
	TYPE	11	NROL	8	RADB	1.25	RAD1	25.0	
	RAD2	35.0	MAT1	10	MAT2	10	MAT3	20	
	CR1	1.35	CR2	1.35					

PBEAR2	5	USER							
	CAG21	0.25	CAG11	110	CAG34	1.7	CAG33	16.0	
	CAG32	24.0	CAG43	56.0	CAG41	18.0	CAG54	0.83	
	CAG22	1.00							

Fields:

Field	Contents
PID	Property identification number that is referenced by a CBEAR2 entry. (Integer > 0)
TYPE	Type of bearing. (Character; No default) • TYPE = "BEAR" for a bearing whose mechanical properties are defined through direct-entry. • TYPE = "ROLL" for a rolling-element bearing.

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Field	Contents
	• TYPE = "SQUE" for a squeeze-film damper bearing. • TYPE = "HYDRO" for a hydrodynamic bearing. • TYPE = "USER" for a bearing whose mechanical properties are obtained from an external executable file.
PARAMi	Name of a parameter. (Character) For a listing of allowable names, see: • **Parameters for Direct-Entry Bearings** for TYPE = "BEAR". • **Parameters for Rolling-Element Bearings** for TYPE = "ROLL". • **Parameters for Squeeze-Film Damper Bearings** for TYPE = "SQUE". • **Parameters Common to all Hydrodynamic Bearings**, **Parameters for Cylindrical Hydrodynamic Bearings**, **Parameters for Hydrodynamic Pivoted-Pad Bearings**, and **Parameters for Hydrodynamic Multi-Lobe Bearings** for TYPE = "HYDRO". • **Parameters for User-Defined Bearings** for TYPE = "USER".
VALUEi	Value of the parameter. (See the parameter for real, integer, or character entry requirement and default behavior.)

General Remarks:

1. The parameters can be listed in any order.

Parameters for Direct-Entry Bearings (TYPE = "BEAR"):

Name	Description
Kij	Constant value for stiffness matrix entry where i = 1,2,...6 and j = 1,2,...6. See Remark 1. (Real; Default = 0.0)
Cij	Constant value for viscous damping matrix entry where i = 1,2,...6 and j = 1,2,...6. See Remark 1. (Real; Default = 0.0)
Mij	Constant value for mass matrix entry where i = 1,2,...6 and j = 1,2,...6. See Remarks 1 and 12. (Real; Default = 0.0)
Sij	Constant value for flexibility matrix entry where i = 1,2,...6 and j = 1,2,...6. See Remark 1. (Real; Default = 0.0)
FKij	Identification number of TABLED1 bulk entry that defines the value for the stiffness matrix entry as a function of rotor speed, frequency, or time, where i = 1,2,...6 and j = 1,2,...6. See Remarks 1, 2, and 3. (Integer > 0 or blank; No default)

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Name	Description
CKij	Identification number of TABLED1 bulk entry that defines the value for the viscous damping matrix entry as a function of rotor speed, frequency, or time, where i = 1,2,...6 and j = 1,2,...6. See Remarks 1, 2, and 3. (Integer > 0 or blank; No default)
FMij	Identification number of TABLED1 bulk entry that defines the value for the mass matrix entry as a function of rotor speed, frequency, or time, where i = 1,2,...6 and j = 1,2,...6. See Remarks 1, 2, 3, 6, and 12. (Integer > 0 or blank; No default)
FSij	Identification number of TABLED1 bulk entry that defines the value for the flexibility matrix entry as a function of rotor speed, frequency, or time, where i = 1,2,...6 and j = 1,2,...6. See Remarks 1, 2, 3, 4, 5, and 6. (Integer > 0 or blank; No default)
HYST	Hysteretic damping factor. See Remarks 7, 8, and 10. (Real; Default = 0.0)
AVIS	Proportional damping factor. See Remarks 9 and 10. (Real; Default = 0.0)
ROTF	Flag that designates the independent variable of the TABLEDi bulk entries that are referenced by FKij, CKij, and FMij parameters to be rotor speed. (Integer = 1; No default)
FREQ	Flag that designates the independent variable of the TABLEDi bulk entries that are referenced by FKij, CKij, and FMij parameters to be frequency. (Integer = 1; No default)
TIME	Flag that designates the independent variable of the TABLEDi bulk entries that are referenced by FKij, CKij, and FMij parameters to be time. (Integer = 1; No default)

Remarks for Direct-Entry Bearings (TYPE = "BEAR"):

1. When both a constant value and a TABLED1 bulk entry are specified for a mass, viscous damping, or stiffness matrix entry, the software sums the value obtained from the TABLED1 lookup and the constant value.
2. When FKij, CKij, FMij, or FSij parameters are specified, an ROTF, FREQ, or TIME parameter must also be specified.
3. When FKij, CKij, FMij, or FSij parameters are specified as speed-dependent, use a SPEBE2 bulk entry to define the rotational speed of the bearing.
4. You cannot define both stiffness and flexibility coefficients for the same bearing.
5. The flexibility matrix is the inverse of the stiffness matrix.
6. For the Kij, Cij, Mij, Sij, FKij, CKij, FMij, and FSij, parameters, express angles in radians and angular velocities such as rotor speeds in radians/sec.
7. The HYST parameter (hysteretic damping) applies only to ROTF (rotation speed dependent) and FREQ (frequency dependent) parameter flags.
8. Hysteretic damping is applied when HYST is defined and the value is greater than or less than 0.

   To define hysteretic damping in the bearing using the HYST parameter, use the following table as a guideline for which parameters to specify.

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Kij/Sij	FKij/FSij	Cij	CKij	Hysteretic damping formulation with G = 1 / HYST (*)
x				Kij + j * G * Kij
x		x		Kij + j * Cij (**)
x			x	Kij + j * CKij (**)
x	x			Kij + FKij + j * G * Kij + j * G * FKij
	x			FKij + j * G * FKij
	x	x		FKij + j * Cij (**)
	x		x	FKij + j * CKij (**)

(*) G is the inverse of the value of the HYST parameter.

(**) When you define the HYST parameter on PBEAR2:

- If viscous damping is defined (constant Cij or function CKij), hysteretic damping is triggered, but the value of HYST is always 1.
- If viscous damping is not defined, the HYST parameter is used.

9. To define the proportional damping factor in the bearing using the AVIS parameter, use the following table as a guideline for which parameters to specify.

Kij/Sij	FKij/FSij	Cij	CKij	Contribution to viscous damping
x				AVIS * Kij
	x			AVIS * FKij
		x		AVIS parameter is ignored. The contribution to the viscous damping is already provided by Cij.
			x	AVIS parameter is ignored. The contribution to the viscous damping is already provided by CKij.

10. When both the HYST and AVIS parameters are specified, the AVIS parameter takes precedence and the HYST parameter is ignored.

11. The mass matrix (as a constant value (Mij) or as a function of rotor speed or frequency (FMij)) does not contribute to the global mass of the model.

12. If you set the mass parameters (Mij, FMij, MPIi, IPIi) in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Parameters for Rolling-Element Bearings (TYPE = "ROLL"):

Name	Description
TYPE	Type of rolling-element bearing. (Integer = 11, 12, 13, or 14; No default)

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Name	Description
	TYPE = 11 for single-row, deep-groove ball bearing TYPE = 12 for single-row, cylindrical roller bearing TYPE = 13 for single row, angular-contact ball bearing TYPE = 14 for single row, tapered roller bearing
NROL	Number of rolling elements. (Integer > 0; No default)
RADB	Radius of the rolling element. See Remark 2. (Real > 0.0; No default)
L	Working length of cylindrical or tapered roller. See Remark 3. (Real > 0.0; No default)
CL	Radial clearance between the rolling element and the races. See Remark 2. (Real ≥ 0.0; Default = 0.0)
RAD1	Inside radius of the bearing. See Remark 2. (Real > 0.0; No default)
RAD2	Outside radius of the bearing. See Remark 2. (Real > 0.0; No default)
CR1	Radius of curvature for the inner race. (Real ≥ RADB; No default)
CR2	Radius of curvature for the outer race. (Real ≥ RADB; No default)
MAT1	Identification number of MAT1 bulk entry that defines the material property for the inner race. (Integer > 0; No default)
MAT2	Identification number of MAT1 bulk entry that defines the material property for the outer race. (Integer > 0; No default)
MAT3	Identification number of MAT1 bulk entry that defines the material property for the rolling element. (Integer > 0; No default)
CANG	Initial contact angle in radians for single row, angular-contact ball bearings and single row, tapered roller bearings. See Remark 4. (Real; Default = 0.0)
PRED	Initial axial displacement due to preload for single row, angular-contact ball bearings and single row, tapered roller bearings. See Remark 4. (Real; Default = 0.0)
HERT	Palmgren's constant. See Remark 5. (Real; No default)

Remarks for Rolling-Element Bearings (TYPE = "ROLL"):

1. RADB, RAD1, RAD2, and CL are defined in the following figure:

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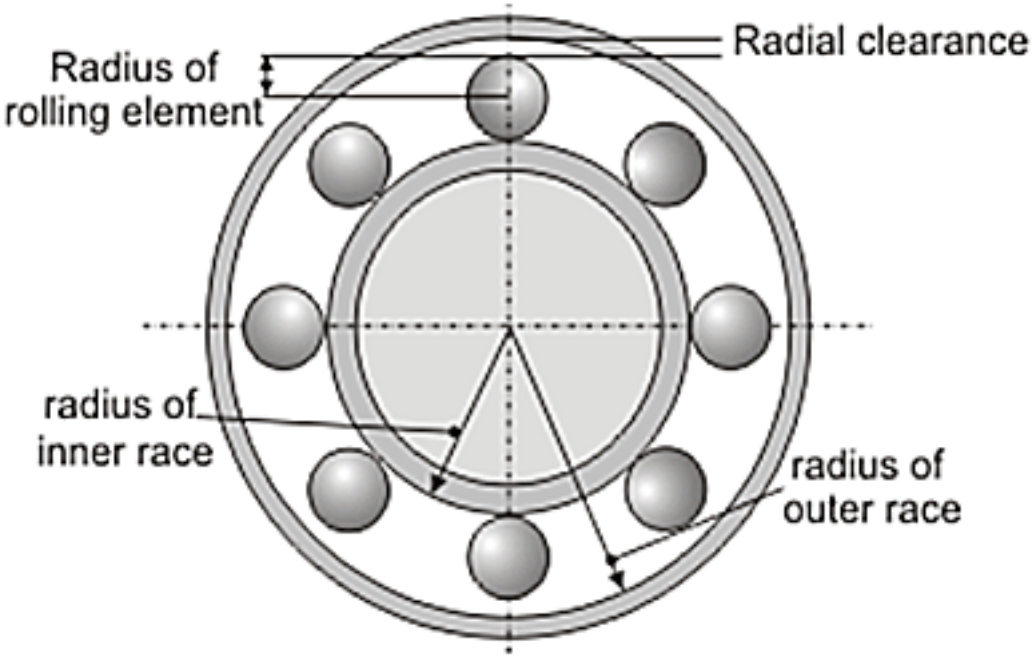

2. Of the RADB, RAD1, RAD2, and CL parameters, specify three. The software calculates the fourth parameter automatically.
3. For ball bearings (TYPE = 11 or 13), the software ignores the L parameter if it is specified.
4. For deep-groove ball bearings and cylindrical roller bearings (TYPE = 11 or 12), the software ignores the CANG and PRED parameters if they are specified.
5. The software calculates the Hertz stiffness constant as follows:
   - For ball bearing types (TYPE = 11 or 13), if the HERT parameter is not specified, the software uses Houpert's formula to calculate the Hertz stiffness. Houpert's formula uses the values for the CR1, CR2, MAT1, MAT2, and MAT3 parameters.
   - For roller bearing types (TYPE = 12 or 14), the software calculates the Hertz stiffness constant from the values for the HERT and L parameters, where the HERT parameter is Palmgren's constant.

Parameters for Squeeze-Film Damper Bearings (TYPE = "SQUE"):

Name	Description
TYPE	Type of squeeze-film damper. (Integer = 11, 12, 13, 14, 15, 17, or 18; No default) TYPE = 11 for squeeze-film with feed and drain TYPE = 12 for squeeze-film with feed hole only TYPE = 13 for no end seals present with specified inlet and outlet pressures TYPE = 14 for no end seals present with specified inlet pressure

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Name	Description
	TYPE = 15 for natural squeeze-film damper TYPE = 17 for end seals present with fluid leakage, the groove at mid-span TYPE = 18 for end seals present with fluid leakage, the groove at an end
DIA	Shaft diameter. See Remark 1. (Real > 0.0; No default)
CL	Nominal squeeze-film gap. See Remark 1. (Real > 0.0; No default)
L	Axial length of squeeze-film gap. See Remark 1. (Real > 0.0; No default)
MU	Viscosity of fluid. (Real > 0.0; No default)
PSAT	Saturation pressure of fluid. (Real; No default)
PIN	Inlet pressure of fluid. (Real; No default)
POUT	Outlet pressure of fluid. See Remark 2. (Real; No default)
THIN	Angular location of the feed hole in degrees relative to the X-axis of the bearing coordinate system. See Remarks 3 and 4. (Real; Default = 0.0)
TOUT	Angular location of the drain hole in degrees relative to the X-axis of the bearing coordinate system. See Remarks 3 and 4. (Real; Default = 180.0)
LEAK	Seal leakage factor. (0.0 ≤ Real ≤ 1.0; Default = 0.0) LEAK = 0.0 for full leakage and short bearing theory LEAK = 1.0 for no leakage and long bearing theory
KR	Contact stiffness. See Remark 5. (Real; No default)

Remarks for Squeeze-Film Damper Bearings (TYPE = "SQUE"):

1. DIA, CL, and L are defined in the following figure:

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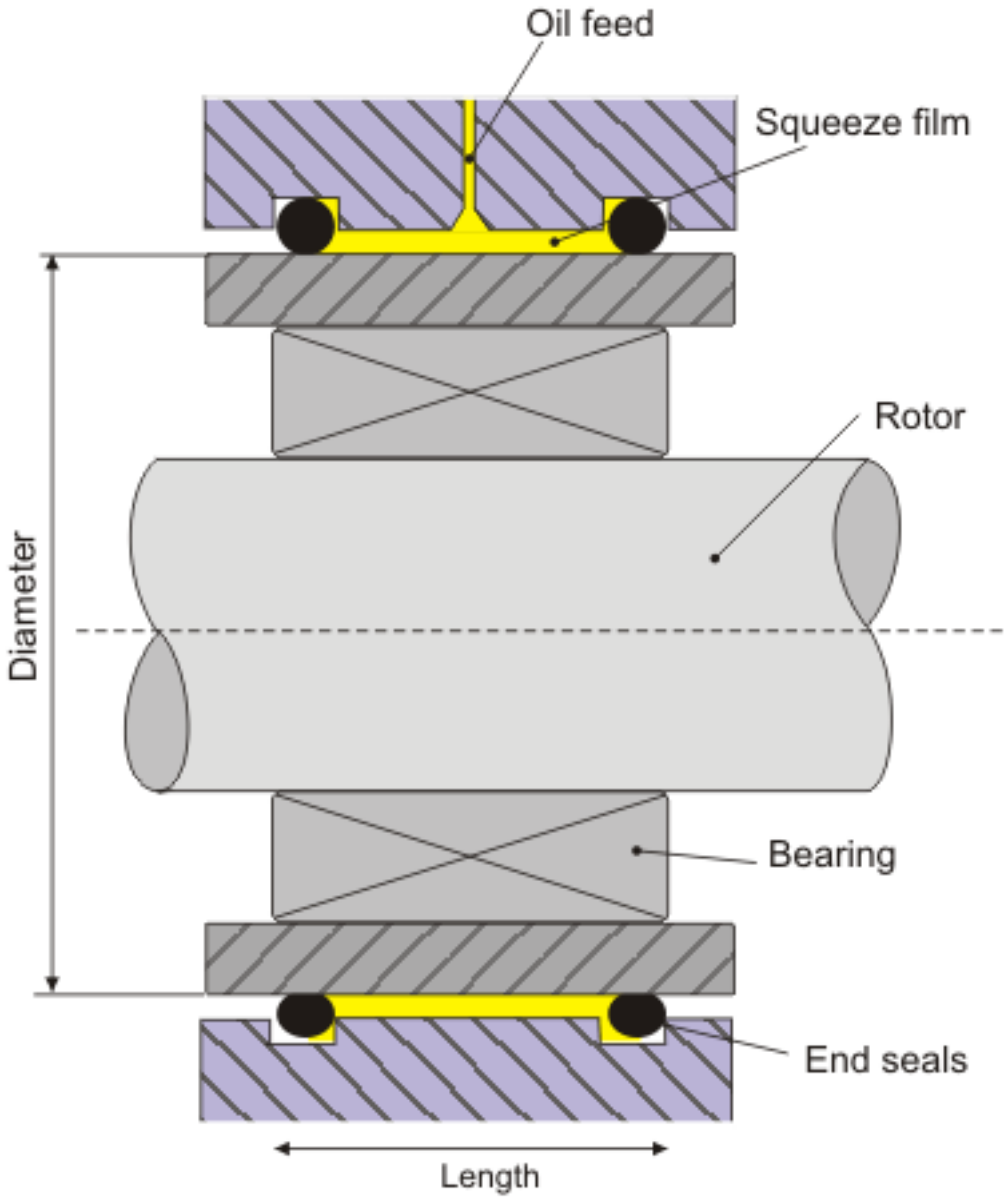

2. For squeeze-film damper bearings where TYPE = 17 or 18, the software ignores the POUT parameter if it is specified.
3. THIN and TOUT are measured counterclockwise from the X-axis of the rotor as shown in the following figure:

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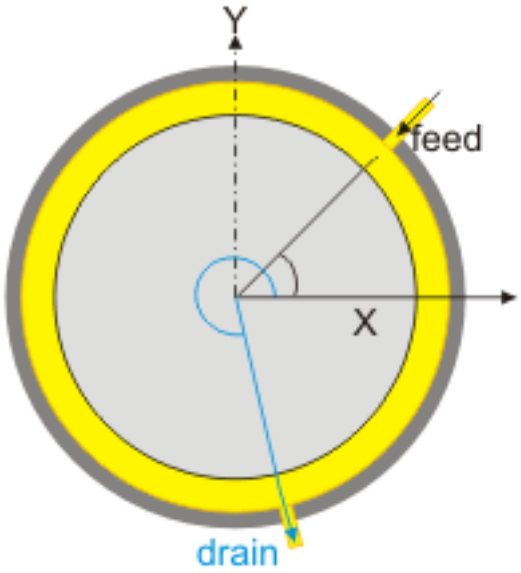

4. For natural squeeze-film damper bearings (TYPE = 15), the software ignores the THIN and TOUT parameters if they are specified.
5. The software ignores the KR parameter for all squeeze-film damper bearing types except the natural squeeze-film damper bearing type (TYPE = 15).

Parameters Common to all Hydrodynamic Bearings (TYPE = "HYDRO"):

Name	Description
DIA	Rotor diameter. See Remark 1. (Real > 0.0; No default)
CL	Radial gap between the rotor and the bearing inside diameter. See Remark 1. (Real > 0.0; No default)
L	Axial length of hydrodynamic bearing. See Remark 1. (Real > 0.0; No default)
MU	Viscosity of fluid. (Real > 0.0; No default)
PSAT	Saturation pressure of fluid. (Real; No default)
NPAT	Mesh density in the circumferential direction. (Integer > 0; Default = 720)
OMEG	Rotational speed at time zero in rad/s. (Real; Default = 0.0)

Remarks for all Hydrodynamic Bearings (TYPE = "HYDRO"):

1. DIA, CL, and L are defined in the following figure:

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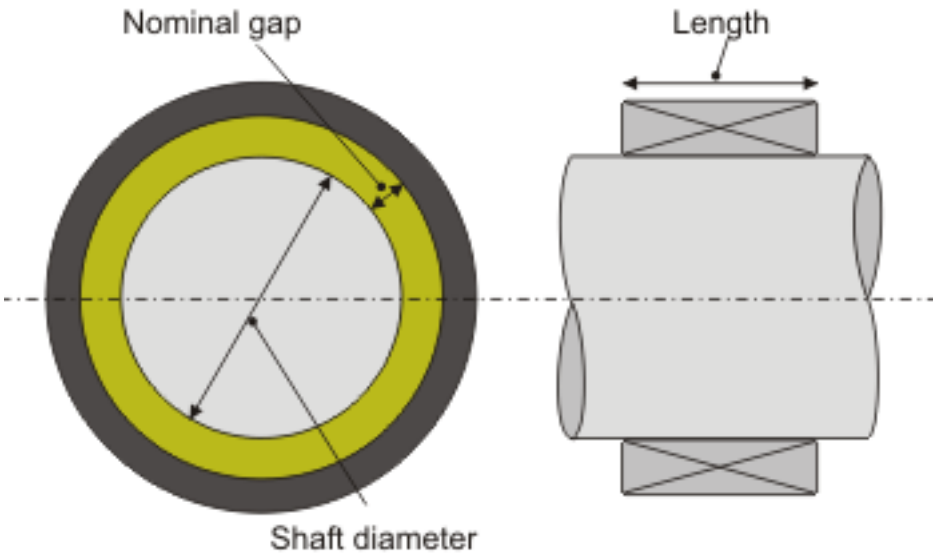

Parameters for Cylindrical Hydrodynamic Bearings (TYPE = "HYDRO"):

Name	Description
MESA	Misalignment flag. (Integer = 0 or 1; Default = 0) MESA = 0 to ignore misalignment (short bearing theory) MESA = 1 to account for misalignment by solving the 2D Reynolds equation
NPAZ	Mesh density in the axial direction. (Integer > 0; No default)

Parameters for Hydrodynamic Pivoted-Pad Bearings (TYPE = "HYDRO"):

Name	Description
CPRE	Pivot gap. See Remark 2. (Real > 0.0; No default)
CAVI	Cavitation flag. (Integer = 0 or 1; Default = 0) CAVI = 0 to account for cavitation CAVI = 1 to assume no cavitation occurs
KR	Rotational contact stiffness when the rotor contacts one of the bearing pads. (Real; No default)
PADS	Number of pads. See Remarks 3 and 4. (0 < Integer ≤ 8; No default)
ALPi	Relative position in degrees of the pivot relative to the beginning of the i^{th} bearing pad where i is the pad number. See Remark 1. (Real; No default)

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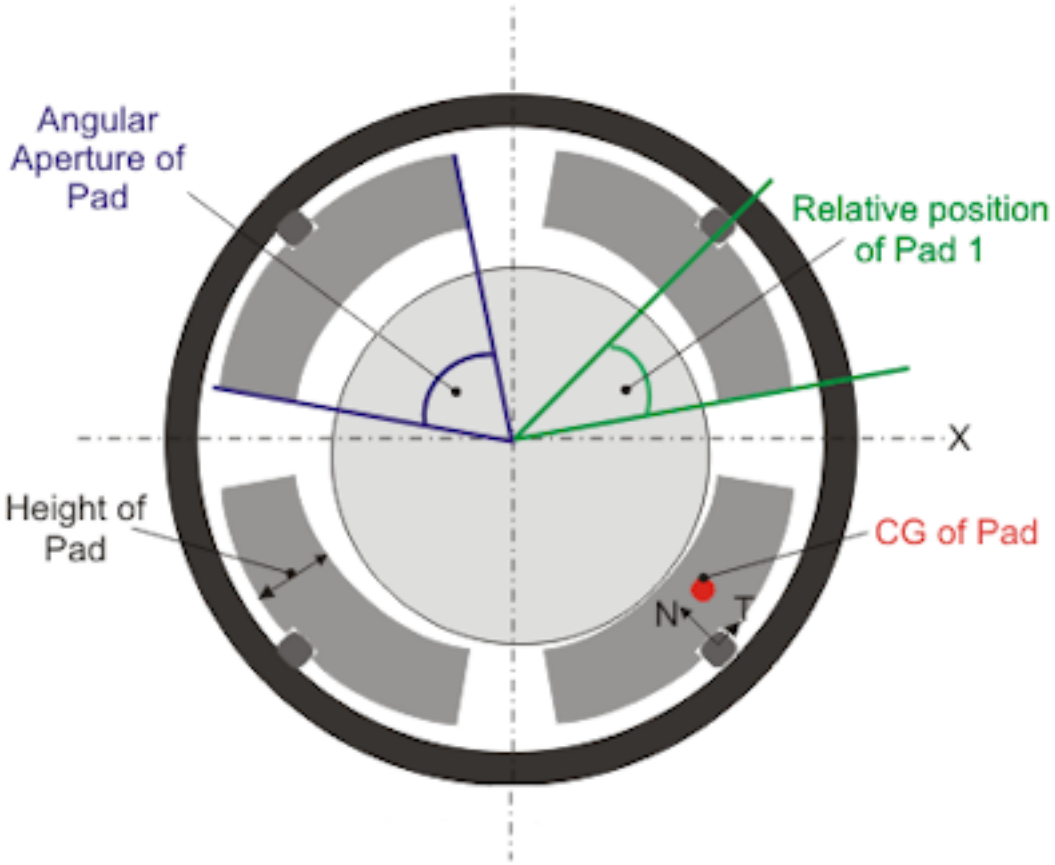
Name	Description
BETi	Angular aperture of the i^{th} bearing pad in degrees where i is the pad number. See Remark 1. (Real; No default)
THEi	Position of the i^{th} pivot with respect to the element X-axis in degrees where i is pad number. See Remark 1. (Real; No default)
MPIi	Mass of the i^{th} bearing pad where i is the pad number. See Remark 5. (Real; No default)
IPIi	Moment of inertia of the i^{th} bearing pad about the pivot where i is the pad number. See Remark 5. (Real; No default)
HPIi	Height of the i^{th} bearing pad where i is the pad number. See Remark 1. (Real > 0.0; No default)
PTNi	Tangential distance from the pivot to the center of mass of the i^{th} bearing pad where i is the pad number. See Remark 1. (Real; Default = 0.0)
PNMi	Radial distance from the pivot to the center of mass of the i^{th} bearing pad where i is the pad number. See Remark 1. (Real > 0.0; Default = HPIi/2)

Remarks for Hydrodynamic Pivoted-Pad Bearings (TYPE = "HYDRO"):

1. ALPi, BETi, THEi, HPIi, PTNi, and PNMi are defined in the following figure:



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2. The geometric preload is the difference between the value for the CL parameter and the value for the CPRE parameter.
3. The physical dimensions and inertial properties of each pad must be specified unless all of the pads have identical physical dimensions and inertial properties.

   For example, if the hydrodynamic pivoted-pad bearing contains two differing pads, the following parameters must be defined: ALP1, BET1, THE1, MPI1, IPI1, HPI1, PTN1, PNM1, ALP2, BET2, THE2, MPI2, IPI2, HPI2, PTN2, and PNM2.
4. If all of the pads have the same physical dimensions and inertial properties, use the PADS parameter to specify the number of pads. Use the ALP1, BET1, THE1, MPI1, IPI1, HPI1, PTN1, and PNM1 parameters to specify the physical dimensions and inertial properties.
5. If you set the mass parameters (MPIi, IPIi) in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Parameters for Hydrodynamic Multi-Lobe Bearings (TYPE = "HYDRO"):

Name	Description
PIN	Inlet pressure of fluid. (Real; No default)
CAVI	Cavitation flag. (Integer = 0 or 1; Default = 0) CAVI = 0 to account for cavitation CAVI = 1 to assume no cavitation occurs
NPAZ	Mesh density in the axial direction. (Integer > 0; No default)
CPRE	Radius of lobe centers circle. See Remark 1. (Real; No default)
LOBE	Number of lobes. See Remarks 2, 3, and 4. (0 < Integer ≤ 8; No default)
ALOB	Angular aperture of lobe in degrees. See Remark 1. (Real; No default)
BLOB	Angle of groove in degrees. See Remark 1. (Real; No default)
ALPi	Position of the centerline with respect to the beginning of the i^{th} lobe in degrees where i is the lobe number. See Remark 1. (Real; No default)
BETi	Angular aperture of the i^{th} lobe in degrees where i is the lobe number. See Remark 1. (Real; No default)
THEi	Position of the i^{th} lobe with respect to the element X-axis in degrees where i is the lobe number. See Remark 1. (Real; No default)
MESA	Misalignment flag. (Integer = 0 or 1; Default = 0) MESA = 0 to ignore misalignment (short bearing theory) MESA = 1 to account for misalignment by solving the 2D Reynolds equation

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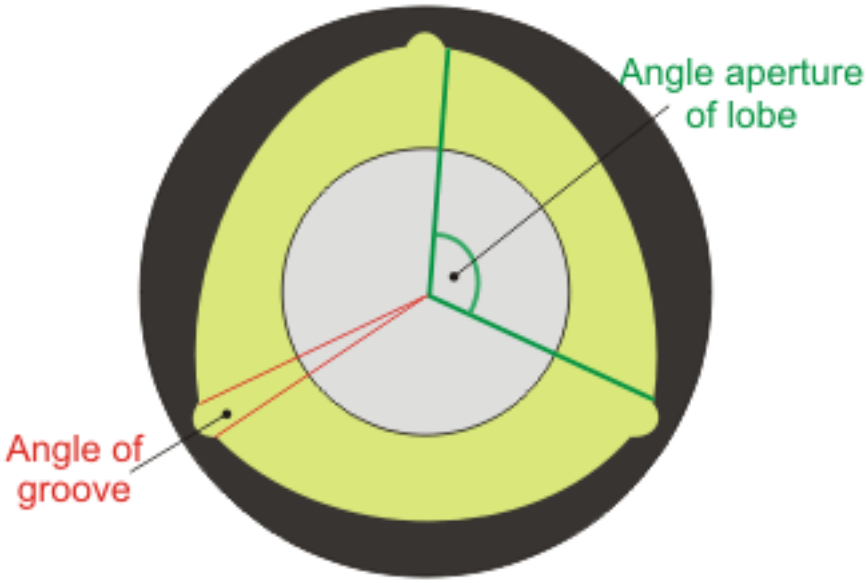
Remarks for Hydrodynamic Multi-Lobe Bearings (TYPE = "HYDRO"):

1. CPRE, ALOB, BLOB, ALPi, BETi, and THEi are defined in the following figure:



2. The physical dimensions of each lobe must be specified unless all of the lobes have identical physical dimensions and inertial properties.

   For example, if the hydrodynamic multi-lobe bearing contains two differing lobes, the following parameters must be defined: ALP1, BET1, THE1, ALP2, BET2, and THE2.

3. For a hydrodynamic multi-lobe bearing with a single lobe, the software does not use the ALP1, BET1, and THE1 parameters. Thus, they do not need to be specified.

4. If all of the lobes have the same physical dimensions, use the LOBE parameter to specify the number of lobes and the ALOB and BLOB parameters to specify the physical dimensions. If the physical dimensions of the lobes differ, use the ALPi, BETi, and THEi parameters to specify the physical dimensions of the lobes.

Parameters for User-Defined Bearings (TYPE = "USER"):

Name	Description
CAG11	Required parameter that defines the identification number of the user-defined bearing. (Integer; No default)
CAGij	Numerical arguments passed to the external executable file. (Real; No default)
i = 2,3,4,5,6	
j = 1,2,3,4	

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Remarks for User-Defined Bearings (TYPE = "USER"):

1. If the user-defined bearing requires the bearing speed to evaluate the bearing properties, you must include a SPEBE2 bulk entry for the rotor (or rotors) that the bearing supports.
2. You can list the CAG11 and CAGij parameters in any order.
3. You must specify the CAG11 parameter, but you can specify any combination of the CAGij parameters. For example, it is valid to specify the CAG11, CAG24, CAG33, and CAG61 parameters only.
4. SOL 414 does not support the WTMASS parameter with user-defined bearings, so you must enter all mass parameters in mass units, not weight units.

Related Topics

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PBEND

Curved Beam or Pipe Element Property

Defines the properties of a curved beam, curved pipe, or elbow element (CBEND entry).

Format:

1	2	3	4	5	6	7	8	9	10
PBEND	PID	MID	A	I1	I2	J	RB	THETAB	
	C1	C2	DI	D2	E1	E2	F1	F2	
	K1	K2	NSM	RC	ZC	DELTAN			

Example:

PBEND	39	1	0.8	0.07	0.04	0.04	10.		
	0.5	0.4	-0.5	0.4					
	0.6	0.6				0.1			

Alternate Format and Example for Elbows and Curved Pipes:

PBEND	PID	MID	FSI	RM	T	P	RB	THETAB	
	SACL	ALPHA	NSM	RC	ZC	FLANGE			
	KX	KY	KZ		SY	SZ			

PBEND	39	1	1	0.5	0.02	10.	10.		
					0.1				

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. See Remarks 1 and 2. (Integer > 0)
A	Area of the beam cross section. (Real > 0.0)
I1, I2	Area moments of inertia in planes 1 and 2. (Real > 0.0)
J	Torsional constant. (Real > 0.0)
FSI	Flag selecting the flexibility and stress intensification factors. See“Flexibility and stress intensification factors” in the *Simcenter Nastran Element Library*. (Integer = 1-6)

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Field	Contents
RM	Mean cross-sectional radius of the curved pipe. (Real > 0.0)
T	Wall thickness of the curved pipe. (Real ≥ 0.0; RM + T/2 < RB)
P	Internal pressure. (Real)
RB	Bend radius of the line of centroids. (Real. Optional, see CBEND entry.)
THETAB	Arc angle of element. (Real, in degrees. Optional, see CBEND entry.)
Ci, Di, Ei, Fi	The r,z locations from the geometric centroid for stress data recovery. See Remark 7. (Real)
K1, K2	Shear stiffness factor K in K*A*G for plane 1 and plane 2. (Real)
NSM	Nonstructural mass per unit length. (Real)
RC	Radial offset of the geometric centroid from points GA and GB. See Figure 16-144. (Real)
ZC	Offset of the geometric centroid in a direction perpendicular to the plane of points GA and GB and vector v. See Figure 16-144. See Remark 8. (Real)
DELTAN	Radial offset of the neutral axis from the geometric centroid, positive is toward the center of curvature. See Figure 16-144. See Remark 8 . (Real; Default is described in Remark 4.)
SACL	Miter spacing at center line. See Figure 16-147 and Remark 9 . (Real > 0.0)
ALPHA	One-half angle between the adjacent miter axes (Degrees). (Real) Required for FSI=5 with miter bend. See Figure 16-147 and Remark 9.
FLANGE	For FSI=5, defines the number of flanges attached. See Remark 10. (Integer; Default=0)
KX	For FSI=6, the user defined flexibility factor for the torsional moment. (Real ≥ 1.0) See Remark 11.
KY	For FSI=6, the user defined flexibility factor for the out-of-plane bending moment. (Real ≥ 1.0) See Remark 11.
KZ	For FSI=6, the user defined flexbility factor for the in-plane bending moment. (Real ≥ 1.0) See Remark 11.
SY	For FSI=6, the user defined stress intensificatation factor for the out-of-plane bending. (Real ≥ 1.0)
SZ	For FSI=6, the user defined stress intensification factor for the in-plane bending. (Real ≥ 1.0)

Remarks:

1. For structural problems, MID must reference a MAT1 material entry.
2. For heat transfer problems, MID must reference a MAT4 or MAT5 material entry.
3. The transverse shear stiffness in planes 1 and 2 are K1*A*G and K2*A*G, respectively. The default values for K1 and K2 on the first format are zero, which means the transverse shear flexibilities

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(1/Ki*A*G) are set equal to zero. Transverse shear stiffness for the alternate format are automatically calculated for the curved pipe.

4. The neutral axis radial offset from the geometric centroid is default to the

$$\Delta N = \frac{I1}{A \cdot RB}$$

It is recommended that the default be used whenever

$$\frac{(RB)^2 A}{I1} < 15$$

in which case the default value of ΔN is within 5% of the exact expression for circular or rectangular cross sections. For the alternate format, the neutral axis offset is calculated from an analytical expression for a hollow or solid circular cross section.

The user may compute an exact value for *N* as follows:

$$\Delta N = \frac{RB}{1 + \frac{(RB)^2 A}{Z}}$$

where

$$Z = \int \frac{r^2 dA}{1 + \frac{r}{RB}}$$

The integration is carried out over the cross section of the element.

5. If T is zero, a solid circular cross section of radius RM is assumed and FSI must be 1.
6. If the first format is used, third-order moments are neglected for the consistent mass matrix. These moments are zero whenever the cross section of the beam is symmetric about both the r and z axes.
7. If the circular cross-sectional property entry format is used, the stress points are automatically located at the points indicated in **Figure 16-144**.
8. Offset vectors are treated like rigid elements and are therefore subject to the same limitations.
   - Offset vectors are not affected by thermal loads.
   - The specification of offset vectors is not recommended in solution sequences that compute differential stiffness because the offset vector remains parallel to its original orientation. (Differential stiffness is computed in buckling analysis provided in SOLs 105 and 200 and also nonlinear analysis provided in SOLs 106, 129, 153, and 159 with PARAM,LGDISP,1.)
9. For FSI=5, an elbow bend will be assumed if ALPHA is not defined (i.e. blank). If ALPHA is defined and SACL is not defined, then the widely spaced miter bend factors will be used. When both SACL and ALPHA are defined, the closely spaced miter bend factors will be used if the spacing at center line is less than RM*(1+tan(ALPHA) where RM is the mean cross-sectional radius of the cuved pipe.
10. The FLANGE option is only supported when FSI=5 for elbow and closely spaced miter bend.

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11. If a value less than 1 is entered for KX, KY, or KZ, its value will be reset to 1.0, a warning will be issued and the solve will continue.

Figure 16-144: PBEND Circular Cross Section

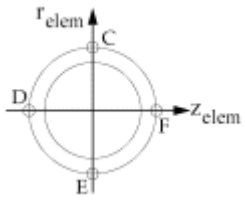

Figure 16-145: PBEND Element Coordinate System

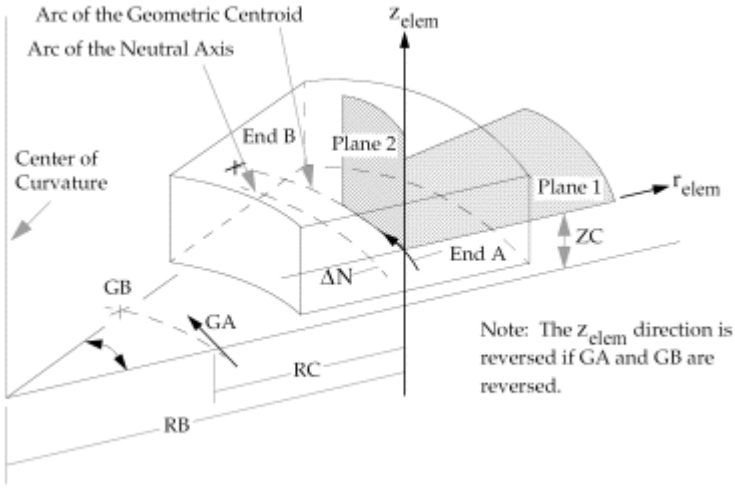

Figure 16-146: PBEND Circular Cross Section Element Coordinate System

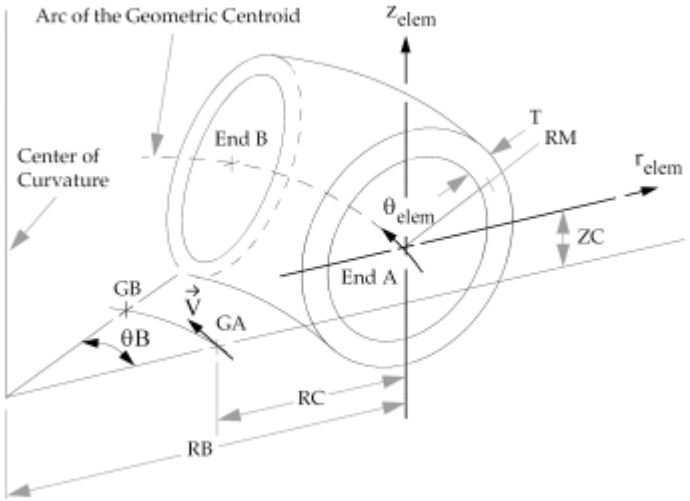

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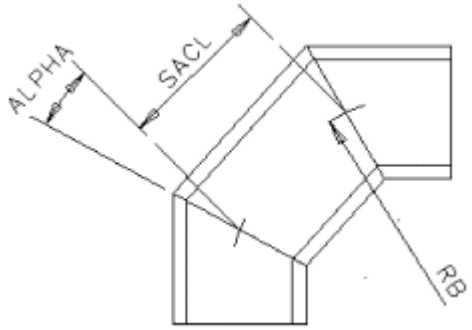
Figure 16-147: Definition of SACL and ALPHA



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Where:

for a closely spaced miter:

$$R_B = \frac{(SACL)\cot(ALPHA)}{2}$$

for a widely spaced miter:

$$R_B = \frac{R_M(1 + \cot(ALPHA))}{2}$$

Related Topics

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PBUSH

Generalized Spring-and-Damper Property

Defines the nominal property values for a generalized spring-and-damper structural element.

Format:

1	2	3	4	5	6	7	8	9	10
PBUSH	PID	"K"	K1	K2	K3	K4	K5	K6	
		"B"	B1	B2	B3	B4	B5	B6	
		"GE"	GE1	GE2	GE3	GE4	GE5	GE6	
		"RCV"	SA	ST	EA	ET			

Example 1: Stiffness and structural damping are specified.

PBUSH	35	K	4.35	2.4				3.1	
		GE	.06						
		RCV	7.3	3.3					

Example 2: Damping force per unit velocity is specified.

PBUSH	35	B	2.3						

Example 3: All damping types are specified.

PBUSH	2	K	1000.0		1000.0				
		B	0.02		0.02				
		GE	0.01		0.02				
		RCV	1.0	1.0					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
"K"	Flag indicating that the next 1 to 6 fields are stiffness values in the element coordinate system. (Character)
Ki	Nominal stiffness values in directions 1 through 6. See Remarks 2, 3 and 9. (Real; Default = 0.0)

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Field	Contents
“B”	Flag indicating that the next 1 to 6 fields are force-per-velocity damping. (Character)
Bi	Nominal damping coefficient in units of force per unit velocity. (Real; Default = 0.0)
“GE”	Flag indicating that the next fields, 1–6, are structural damping constants. (Character)
GEi	Nominal structural damping coefficient (dimensionless) in directions 1–6. See Remarks 2 and 3. (Real; Default = 0.0)
“RCV”	Flag indicating that the next 1 to 4 fields are stress or strain coefficients. (Character)
SA	Stress recovery coefficient in the translational component numbers 1 through 3. (Real; Default = 1.0)
ST	Stress recovery coefficient in the rotational component numbers 4 through 6. (Real; Default = 1.0)
EA	Strain recovery coefficient in the translational component numbers 1 through 3. (Real; Default = 1.0)
ET	Strain recovery coefficient in the rotational component numbers 4 through 6. (Real; Default = 1.0)

Remarks:

1. Ki, Bi, or GEi can optionally be made frequency dependent for a direct or a modal frequency response solution with the PBUSHT entry.

   Ki can also optionally be defined as a nonlinear force-deflection table for a nonlinear solution with the PBUSHT entry.

2. The nominal values are used for all analysis types except when both the PBUSH and PBUSHT entries are used for a frequency response or a nonlinear solution. When a PBUSHT is defined for a modal frequency response, the normal modes are computed using the nominal Ki values defined on the PBUSH entry. The frequency-dependent values defined on the PBUSHT entry are then used at every excitation frequency.
3. If PARAM,W4 is not specified, GEi is ignored in transient analysis.
4. The element stresses are computed by multiplying the stress coefficients with the recovered element forces.
5. The element strains are computed by multiplying the strain coefficients with the recovered element displacements.
6. The “K”, “B”, “GE”, or “RCV” entries may be specified in any order.
7. Applicable fields refer to directions in the element’s coordinate system.
8. (Not applicable to SOLs 402 and 414) For upward computability, if ONLY GE1 is specified on ALL PBUSH entries and GEi, i=2 – 6 are blank on ALL PBUSH entries, then a single structural damping for each PBUSH applied to all defined Ki for each PBUSH is assumed. If ANY PBUSH entry has a GEi, i=2 – 6 specified, then the GEi fields are considered variable on ALL PBUSH entries.

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9. Rotational stiffness should be specified as moment per radian.

Remark related to SOLs 402 and 414:

1. “RCV” entries are not supported.
2. In SOL 414 linear analyses, if GEi values are defined, hysteretical damping is generated as a GEi x Ki value.

Related Topics

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PBUSH1D

Rod Type Spring-and-Damper Property

Defines linear and nonlinear properties of a one-dimensional spring and damper element (CBUSH1D entry).

Format:

1	2	3	4	5	6	7	8	9	10
PBUSH1D	PID	K	C	M		SA	SE		
	"SHOCKA"	TYPE	CVT	CVC	EXPVT	EXPVC	IDTS		
			IDETS	IDECS	IDETSD	IDECSD			
	"SPRING"	TYPE	IDT	IDC	IDTDU	IDCDU			
	"DAMPER"	TYPE	IDT	IDC	IDTDV	IDCDV			
	"GENER"		IDT	IDC	IDTDU	IDCDU	IDTDV	IDCDV	

Example:

PBUSH1D	35	3000.	200.	300.					
	SHOCKA	TABLE	2.2	1.2	1.		200		

The continuation entries are optional. The four options, SHOCKA, SPRING, DAMPER, and GENER can appear in any order

Fields:

Field	Contents	Default
PID	Property identification number. (Integer > 0).	Required
K	Stiffness. (Real ≥ 0).	See Remark 1.
C	Viscous damping. (Real ≥ 0).	See Remarks 1 and 2.
M	Total mass of the element. See Remark 3 for SOL 414. (Real ≥ 0).	Blank
SA	Stress recovery coefficient [1/area]. (Real ≥ 0).	Blank
SE	Strain recovery coefficient [1/length]. (Real ≥ 0).	Blank
SHOCKA	Character string specifying that the next 10 fields are coefficients of the following force versus velocity/displacement relationship. (Character).	

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Field	Contents	Default
	$F(u, v) = C_v \cdot S(u) \cdot \text{sign}(v) \cdot \lvert v \rvert^{\text{EXPV}}$ The force F, the displacement u, and the velocity v, are in the axial direction of the damper. The axis of the damper is defined by the two connecting grid points GA and GB on the CBUSH1D Bulk Data entry. The displacement u and the velocity v, are the relative displacement and the relative velocity with respect to the grid point GA. The scale factor S(u) must be defined with a table or with an equation.	
TYPE	Character string indicating the type of definition. (Character). For TYPE = EQUAT, the fields IDETS, IDECS, IDETSD, and IDECSD are identification numbers of DEQATN entries. For TYPE = TABLE the field IDTS is an identification number of a TABLEDi entry. If no character string is provided (blanks), TYPE = TABLE is set.	TABLE
CVT	Viscous damping coefficient C_V for tension v > 0, force per unit velocity. (Real).	Required for SHOCKA
CVC	Viscous damping coefficient C_V for compression v > 0, force per unit velocity. (Real).	CVT
EXPVT	Exponent of velocity EXPV for tension v > 0.(Real).	1.
EXPVC	Exponent of velocity EXPV for compression v < 0. (Real).	EXPVT
IDTS	Identification number of a TABLEDi entry for tension and compression if TYPE = TABLE. The TABLEDi entry defines the scale factor S, versus displacement u.	Required for SHOCKA and TYPE=TABLE
IDETS	Identification number of a DEQATN entry for tension if TYPE = EQUAT. The DEQATN entry defines the scale factor S, versus displacement u, for tension u > 0.	Required for SHOCKA and TYPE=EQUAT
IDECS	Identification number of a DEQATN entry for compression if TYPE = EQUAT. The DEQATN entry defines the scale factor S, versus displacement u, for compression u < 0.	IDETS
IDETSD	Identification number of a DEQATN entry for tension if TYPE = EQUAT. The DEQATN entry defines the derivative of the scale factor S, with respect to the displacement u, for tension u>0.	Required for SHOCKA and TYPE=EQUAT
IDECSD	Identification number of a DEQATN entry for compression if TYPE = EQUAT. The DEQATN entry defines the derivative of the scale factor S, with respect to the displacement u, for compression u < 0.	IDETSD
SPRING	Character string specifying that the next 5 fields define a nonlinear elastic spring element in terms of a force versus displacement relationship. (Character).	

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Field	Contents	Default
	$F(u) = F_T(u)$	
	Tension is u > 0 and compression is u < 0.	
DAMPER	Character string specifying that the next 5 fields define a nonlinear viscous element in terms of a force versus velocity relationship. (Character).	
	$F(v) = F_T(u)$	
	Tension is v > 0 and compression is v < 0.	
GENER	Character string specifying that the next 7 fields define a general nonlinear elastic spring and viscous damper element in terms of a force versus displacement and velocity relationship. (Character). For this element, the relationship can only be defined with TYPE = EQUAT.	
	$F(u, v) = F_T(u, v)$	
	Tension is u > 0 and compression is u < 0.For SPRING, DAMPER, and GENER, the remaining fields are	
TYPE	Character string indicating the type of definition. (Character). For TYPE = EQUAT the following fields are identification numbers of DEQATN entries. For TYPE = TABLE the following field is an identification number of a TABLEDi entry. TYPE is ignored for GENER.	Required for SPRING or DAMPER
IDT	Identification number of a DEQATN entry for tension if TYPE = EQUAT. Identification number of a TABLEDi entry for tension and compression if TYPE = TABLE.	Required for SPRING, DAMPER, and GENER
IDC	Identification number of a DEQATN entry for compression if TYPE = EQUAT. Is ignored for TYPE = TABLE.	IDT
IDTDU	Identification number of a DEQATN entry for tension if TYPE = EQUAT. The DEQATN entry defines the derivative of the force F with respect to the displacement u, for tension u > 0. For SPRING and GENER only.	Required if TYPE=EQUAT
IDCDU	Identification number of a DEQATN entry for compression if TYPE = EQUAT. The DEQATN entry defines the derivative of the force F with respect to the displacement u, for compression u < 0. For SPRING and GENER only.	IDTDU
IDTDV	Identification number of a DEQATN entry for tension if TYPE = EQUAT. The DEQATN entry defines the derivative of the force F	Required if TYPE=EQUAT

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Field	Contents	Default
	with respect to the velocity v, for tension v > 0. For DAMPER and GENER only.	
IDCDV	Identification number of a DEQATN entry for compression if TYPE = EQUAT. The DEQATN entry defines the derivative of the force F with respect to the velocity v, for compression v < 0. For DAMPER and GENER only.	IDCDT

Remarks:

1. Either the stiffness K or the damping C must be specified.
2. The damping C and mass M are ignored in static solution sequences.
3. The parameters defined on the continuation entries are used in nonlinear solution sequences only.
4. The linear parameters K and C are used in all solution sequences unless parameters on continuation entries are defined and a nonlinear solution sequence is used. Then, the parameters K and C are used for initial values in the first iteration of the first load step and the parameters from continuation entries overwrite the linear parameters thereafter. When SHOCKA, SPRING or GENER are specified, K is overwritten. When SHOCKA, DAMPER or GENER is specified, C is overwritten.
5. PBUSH1D may only be referenced by CBUSH1D elements in the residual structure which do not attach to omitted degrees-of-freedom.
6. The continuation entries SHOCKA, SPRING, DAMPER and GENER may be specified in any order. If more than one continuation entry is defined, then the forces of SHOCKA, SPRING, etc. are added. Multiple continuation entries of the same kind are not allowed, for example, multiple SPRING continuation entries.
7. For TYPE = TABLE, values on the TABLEDi entry are for tension and compression. If table values $f(u)$ are provided only for positive values u > 0, then it is assumed that $f(-u) = -f(u)$.
8. For TYPE = EQUAT, the equations for tension and compression can be different. If the identification numbers for compression are left blank, it is assumed that the equation for tension is also valid for compression.

Remarks related to SOLs 402 and 414:

1. Only “SPRING” and "DAMPER" are supported with the following fields: PID, K, C, M, TYPE and IT. SA and SE are ignored.
2. Only TYPE=TABLE (or blank) is supported since the DEQATN bulk entry is not supported. Therefore, IDC, IDTDU and IDCDU are ignored.
3. For SOL 414 solutions, if you set the mass (M) parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

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Related Topics

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PBUSH2

Nonlinear Spring and Damper Property

Defines the properties for CBUSH2 elements. CBUSH2 elements are valid in SOL 414.

Format:

1	2	3	4	5	6	7	8	9	10
PBUSH2	PID	TYPE							
	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3	PARAM4	VALUE4	
	PARAM5	VALUE5	-etc.-						

Example:

PBUSH2	2	AXIAL							
	KTX	50.0	KTY	25.0	KTZ	40.0	PLAST	1	

Fields:

Field	Contents
PID	Property identification number that matches the identification number on a CBUSH2 entry. (Integer > 0)
TYPE	Bushing type. Valid entries are "AXIAL" and "RADIAL". (Character; Default = "AXIAL")
PARAMi	Name of the parameter. Allowable names are given in the parameter listing below. (Character; No default)
VALUEi	Value of the parameter. (Real, Integer, or Character)

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Parameters:

Table 16-113: Eccentricity

Name	Description
LOCA	Location of local stiffness. 0.0 corresponds to the first node, 1.0 corresponds to the second node, 0.5 corresponds to midway between the first and second nodes, and so on. (0.0 ≤ Real ≤ 1.0; Default = 0.5)
ECC1, ECC2, ECC3	Eccentricities relative to the first node. (Real; No default)
FRM2	Identification number of the coordinate system relative to which the eccentricities defined. (Integer≥0; Default is the structural axes)

Table 16-114: Damping

Name	Description
DAMP	Damping forces are always computed. (Integer = 0 or 1; Default = 0) DAMP = 0 means compute damping forces. DAMP = 1 means force damping forces to zero when elastic forces are zero. For axial bushings, this test is applied to each component. For radial bushings, the test is applied to the radial and axial components.

Table 16-115: Axial bushing (Cartesian coordinates)

Name	Description
KTX, KTY, KTZ	Stiffness in translation. (Real; No default)
KRX, KRY, KRZ	Stiffness in rotation per unit radian. (Real; No default)
CTX, CTY, CTZ	Viscous damping in translation. See Remarks 2 and 3. (Real; No default)
CRX, CRY, CRZ	Viscous damping in rotation per unit radian. See Remarks 2 and 3. (Real; No default)
HCTX, HCTY, HCTZ	Hysteretic damping component in translation. See Remarks 2 and 3. (Real; No default)
HCRX, HCRY, HCRZ	Hysteretic damping in rotation per unit radian. See Remarks 2 and 3. (Real; No default)
FETX, FETY, FETZ	Identification number of a TABLED1 entry that represents elastic force in translation. See Remark 10. (Integer > 0; No default)

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Name	Description
FERX, FERY, FERZ	Identification number of a TABLED1 entry that represents elastic moment in rotation. See Remark 10. (Integer > 0; No default)
FATX, FATY, FATZ	Identification number of a TABLED1 entry that represents viscous damping force in translation. See Remarks 3 and 10. (Integer > 0; No default)
FARX, FARY, FARZ	Identification number of a TABLED1 entry that represents viscous damping moment in rotation. See Remarks 3 and 10. (Integer > 0; No default)

Table 16-116: Radial bushing (cylindrical coordinates)

Name	Description
KTR, KTZ	Radial (KTR) or axial (KTZ) stiffness in translation. (Real; No default)
KRR, KRZ	Radial (KRR) or axial (KRZ) stiffness in rotation per unit radian. (Real; No default)
CTR, CTZ	Radial (CTR) or axial (CTZ) viscous damping in translation. (Real; No default)
CRR, CRZ	Radial (CRR) or axial (CRZ) viscous damping in rotation per unit radian. (Real; No default)
FETR	Identification number of a TABLED1 or TABLEST entry that represents the radial elastic force in translation as a function of radial displacement, or radial displacement and rotational speed. See Remarks 4, 5, 6, and 10. (Integer > 0; No default)
FERR	Identification number of a TABLED1 or TABLEST entry that represents the radial elastic moments in rotation as a function of radial displacement, or radial displacement and rotational speed. See Remarks 4, 5, 6, and 10. (Integer > 0; No default)
FATR	Identification number of a TABLED1 or TABLEST entry that represents the radial viscous damping in translation as a function of relative speed, or relative speed and rotational speed. See Remarks 4, 5, 6, and 10. (Integer > 0; No default)
FARR	Identification number of a TABLED1 or TABLEST entry that represents the radial viscous damping in rotation as a function of relative speed, or relative speed and rotational speed. See Remarks 4, 5, 6, and 10. (Integer > 0; No default)
FETZ	Identification number of a TABLED1 entry that represents the axial elastic force in translation as a function of radial displacement. See Remarks 4 and 10. (Integer > 0; No default)
FERZ	Identification number of a TABLED1 entry that represents the axial elastic moments in rotation as a function of radial displacement. See Remark 4 and 10. (Integer > 0; No default)
FATZ	Identification number of a TABLED1 entry that represents the axial viscous damping in translation as a function of relative speed. See Remark 4 and 10. (Integer > 0; No default)
FARZ	Identification number of a TABLED1 entry that represents the axial viscous damping in rotation as a function of relative speed. See Remark 4 and 10. (Integer > 0; No default)

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Table 16-117: Ellipsoid criterion

Name	Description
FRUX, FRUY, FRUZ, MRUX, MRUY, MRUZ	Strength values that are used in the ellipsoid criterion for axial bushings. See Remarks 7 and 9. (Real ≠ 0.0; No default)
FRUR, FRUZ, MRUR, MRUZ	Strength values that are used in the ellipsoid criterion for radial bushings. See Remark 7. (Real ≠ 0.0; No default)

Table 16-118: Rupture with damage

Name	Description
TAU	Time constant, τ_c. See Remarks 8 and 9. (Real ≥ 0.0; No default)
ADEL	Scaling factor, a_c. See Remarks 8 and 9. (Real; Default = 1.0)
DDEN	Critical value, *CRIT*, in the ellipsoid criterion at which rupture is predicted to initiate. See Remarks 7, 8, and 9. (Real; Default = 1.0)

Table 16-119: MASS

Name	Description
MASS	Value of the mass to apply to the bushing element. One half of the mass is applied to each node. See Remark 12. (Real > 0.0; No default)

Table 16-120: Plasticity

Name	Description
PLAST	Identification number of the MAT1 and MPLAS bulk entries. The MAT1 bulk entry defines the isotropic material properties, and the MPLAS bulk entry defines the plastic properties. Both MAT1 and MPLAS must use the same MID. See Remark 11. (Integer >1; No default)

Remarks:

1. Specify rotations in rad and rotational speeds in rad/time.
2. You can define viscous damping or hysteretic damping, but not both.

   However, if you want to simulate viscous damping and hysteretic damping, you can create two bushing elements in parallel and define viscous damping for one and hysteretic damping for the other.

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Note

If you define both viscous damping and hysteretic damping for the same bushing element, the software uses only the defined hysteretic parameters.

3. If you define the PLAST field to account for plasticity in the PBUSH2 properties, do not define these viscous and hysteretic damping parameters. Plasticity in bushing elements does not use damping.

4. To represent FETR, FETZ, FERR, or FERZ as a function of radial displacement, reference a TABLED1 bulk entry.

   To represent FATR, FATZ, FARR, or FARZ as a function of relative speed, reference a TABLED1 bulk entry.

5. To represent FETR or FERR as a function of both radial displacement and rotational speed, reference a TABLEST bulk entry.

   To represent FATR or FARR as a function of both relative speed and rotational speed, reference a TABLEST bulk entry.

6. If the PBUSH2 bulk entry contains a parameter that references a TABLEST bulk entry, the CBUSH2 bulk entries that reference the PBUSH2 bulk entry must also reference SPEBE2 bulk entries.

7. For axial bushings, the ellipsoid criterion is given as follows:

$$CRIT = \left(\frac{F_x}{\mathrm{FRUX}}\right)^2 + \left(\frac{F_y}{\mathrm{FRUY}}\right)^2 + \left(\frac{F_z}{\mathrm{FRUZ}}\right)^2 + \left(\frac{M_x}{\mathrm{MRUX}}\right)^2 + \left(\frac{M_y}{\mathrm{MRUY}}\right)^2$$

   For radial bushings, the ellipsoid criterion is given as follows:

$$CRIT = \left(\frac{F_x^2 + F_y^2}{\mathrm{FRUR}^2}\right) + \left(\frac{F_z}{\mathrm{FRUZ}}\right)^2 + \left(\frac{M_x^2 + M_y^2}{\mathrm{MRUR}^2}\right) + \left(\frac{M_z}{\mathrm{MRUZ}}\right)^2$$

   where *CRIT* is the threshold at which yield or fracture of the bushing is predicted to initiate, and FRUX, FRUY, FRUZ, FRUR, MRUX, MRUY, MRUZ, and MRUR are the strengths of the bushing relative to a single load.

   For example, FRUX is the load at which yield or fracture occurs when the bushing is loaded by a force in the X-direction.

   If any of the FRUX, FRUY, FRUZ, FRUR, MRUX, MRUY, MRUZ, or MRUR strength values are not defined, the software sets the value of the quotient for the missing strength value to zero when it evaluates the ellipsoid criterion.

8. After the critical value for the ellipsoid criterion is reached, the static damage, d^s, is set to unity. The true damage is obtained from the following equation:

$$\dot{d} = \frac{1}{\tau_c}\left(1 - e^{-a_c(d^s - d)_+}\right)$$

9. The ellipsoid criterion parameters for axial bushings (FRUX, FRUY, FRUZ, MRUX, MRUY, and MRUZ), and the rupture damage parameters (ADEL, TAU, and DDEN) are the same as the ellipsoid criterion

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and rupture damage parameters that you can define in the PLASGEN yield function. When you define them in the PLASGEN yield function, do not also define them in the PBUSH2 bulk entry.

10. The following parameters are supported only in SOLs 414,111 and 414,129:

    FETX, FETY, FETZ
    FERX, FERY, FERZ
    FATX, FATY, FATZ
    FARX, FARY, FARZ
    FETR, FETZ, FERR, FERZ
    FATR, FATZ, FARR, FARZ

11. You can use PLAST only in SOL 414,103 with a Transient Preload subcase and SOL 414, 129 Transient Response.

12. If you set the MASS parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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PBUSHT

Frequency Dependent or Nonlinear Force Deflection Spring and Damper Property

Defines the frequency dependent properties or the stress dependent properties for a generalized spring and damper structural element.

Format:

1	2	3	4	5	6	7	8	9	10
PBUSHT	PID	"K"	TKID1	TKID2	TKID3	TKID4	TKID5	TKID6	
		"B"	TBID1	TBID2	TBID3	TBID4	TBID5	TBID6	
		"GE"	TGEID1	TGEID2	TGEID3	TGEID4	TGEID5	TGEID6	
		"KN"	TKNID1	TKIND2	TKNID3	TKIND4	TKIND5	TKIND6	
		"BW"	TBWID1	TBWID2	TBWID3	TBWID4	TBWID5	TBWID6	

Example:

PBUSHT	2	K	100	101					
		B	102	103					
		GE	104	105					

Fields:

Field	Contents
PID	Property identification number that matches the identification number on a PBUSH entry. (Integer > 0)
"K"	Flag indicating that the next 1 to 6 fields are stiffness vs. frequency table identification numbers. (Character)
TKIDi	Identification number of a TABLEDi entry that defines the stiffness vs. frequency relationship. See Remark 11. (Integer ≥ 0; Default = 0)
"B"	Flag indicating that the next 1 to 6 fields are force per unit velocity vs. frequency table identification numbers. (Character)
TBIDi	Identification number of a TABLEDi entry that defines the force per unit velocity vs. frequency relationship. (Integer ≥ 0; Default = 0)
"GE"	Flag indicating that the next fields 1–6 are structural damping vs. frequency table identification numbers. (Character)
TGEIDi	Identification number of a TABLEDi entry that defines the non-dimensional structural damping vs. frequency relationship. (Integer ≥ 0; Default = 0)

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Field	Contents
“KN”	Flag indicating that the next 1 to 6 entries are nonlinear force vs. deflection table identification numbers. (Character)
TKNIDi	Identification number of a TABLEDi entry that defines the nonlinear force vs. deflection relationship. See Remark 13. (Integer ≥ 0; Default = 0)
“BW”	Flag indicating that the next 1 to 6 entries are nonlinear damping force vs. velocity table identification numbers. (Character)
TBWIDi	Identification number of a TABLEDi entry that defines the nonlinear damping force vs. velocity relationship. See Remark 14. (Integer ≥ 0; Default = 0)

Remarks:

1. The “K”, “B”, and “GE” entries are associated with the same entries on the PBUSH entry.
2. Only CBUSH elements in the residual structure that do not attach to any omitted degrees-of-freedom can reference a PID identifying both a PBUSH entry and a PBUSHT entry.
3. The PBUSHT is used by frequency response and nonlinear solutions. For all other solutions, the nominal values defined on the PBUSH are used.
4. When frequency dependent stiffness is included in a modal frequency response, the modes are computed using the nominal Ki values. The frequency-dependent values are used at every excitation frequency. See the parameter SDAMPUP for more options.
5. The “K”, “B”, “GE”, “KN”, and "BW" entries may be specified in any order.
6. The PBUSHT entry is ignored in all solution sequences except frequency response analysis, nonlinear analysis, and SOL 414 rotor dynamic frequency and transient response analysis.
7. For nonlinear analysis, only the “KN” field is used.
8. For frequency response analysis, only the “K”, “B” and “GE” fields are used.
9. For SOL 414 rotor dynamics analysis, “K”, “B” and “GE” fields are used in frequency response analysis, and “KN” and “BW” fields are used in transient response analysis.
10. (Not applicable to SOLs 402 and 414) For upward computability, if ONLY TGEID1 is specified on ALL PBUSHT entries and TGEIDi, i=2 – 6 are blank on ALL PBUSHT entries, then a single structural damping table for each PBUSHT applied to all defined Ki for each PBUSH is assumed. If ANY PBUSH entry has a TGEIDi, i=2 – 6 specified, then the GEi fields on the PBUSH and the TGEIDi fields on the PBUSHT are considered variable on ALL PBUSH and PBUSHT entries.
11. Rotational stiffness should be specified as moment per radian.
12. With designed frequency dependent properties in SOL 200, currently only TABLED1 can be used, and the use of TABLED1 for this case is limited to the LINEAR, LINEAR default options for XAXIS and YAXIS. For frequency dependent properties not associated with design variables, other options and other TABLEDi can be used.

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13. If the force vs. deflection relationship is defined for positive deflections only and the deflection is negative, the software looks up the value for the force that it uses to compute the stiffness at the absolute value of the negative deflection.

    If the force vs. deflection relationship is defined for negative deflections only and the deflection is positive, the software looks up the value for the force that it uses to compute the stiffness at the opposite value of the deflection.

14. The units for the tables that define the nonlinear damping force vs. velocity are as follows:
    - For translational DOF, force vs. linear velocity.
    - For rotational DOF, moment vs. angular velocity in rad/s.

Remarks related to SOL 402:

1. Only the “KN” and the "BW" entries are supported.

Remarks related to SOL 414:

1. The “GE” entry is not supported.
2. For SOL 414,101 and SOL 414,110, the software uses the values that are defined on the PBUSH bulk entry and ignores the PBUSHT bulk entry.
3. For SOL 414,111, the software uses only the values associated with the "B" and "K" fields on the PBUSHT bulk entry and ignores the PBUSH bulk entry.
4. For SOL 414,129, the software uses only the values associated with the "BW" and "KN" fields on the PBUSHT bulk entry and ignores the PBUSH bulk entry.
5. For SOL 414,103, the software can use the tables associated with the "BW" and "KN" fields on the PBUSHT bulk entry, but it considers them only as a constant value (by evaluating the force versus deflection table at the initial zero displacement state).

Related Topics

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PCHOCK

Defines Properties for 2D Chocking Elements

Defines properties for CCHOCKi elements.

Format:

1	2	3	4	5	6	7	8	9	10
PCHOCK	PID	MID	NGAP	GAPT					

Example:

PCHOCK	100	1	6	1.0					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number for a MAT1 or MAT11 bulk entry. (Integer > 0; No default)
NGAP	Number of gaps. (Integer > 0; No default)
GAPT	Initial gap opening. See Remarks 1, 2, and 3. (Real; Default = 0.0)

Remarks:

1. The value for GAPT is used for all grid points Gi in the connectivity of chocking elements that reference the PCHOCK entry, except for those grid points Gi where GAPi on the chocking element bulk entry is specified. For those grid points, the GAPi value takes precedence.
2. If the value for GAPT or, if applicable, the value for GAPi on the corresponding CCHOCKi entry is negative, the software uses the value to calculate an initial compressive hoop strain.
3. Figure 16-148 shows a possible application for chocking elements and how the initial gap opening GAPT is defined.

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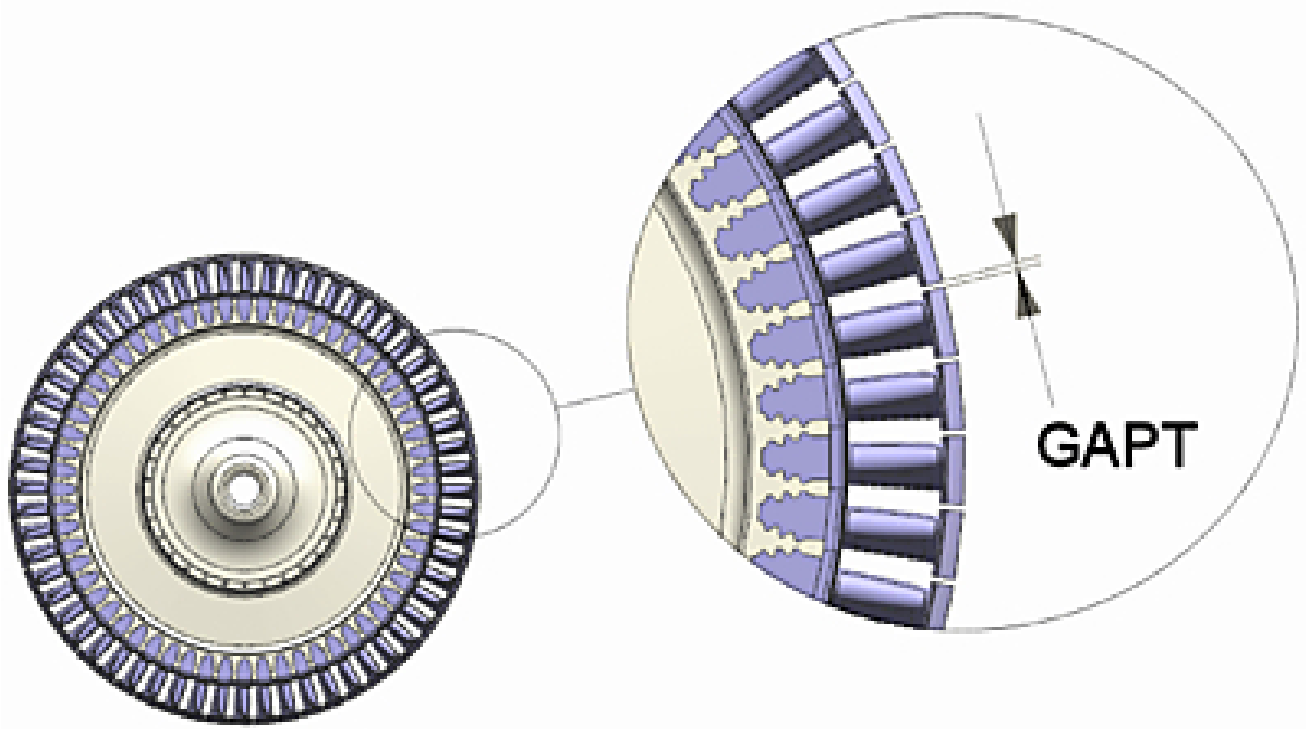
Figure 16-148: Initial Gap Opening



Related Topics

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PCOMP

Layered Composite Element Property

Defines the properties of an n-ply composite material laminate.

Format:

1	2	3	4	5	6	7	8	9	10
PCOMP	PID	Z0	NSM	SB	FT	TREF	GE	LAM	
	MID1	T1	THETA1	SOUT1	MID2	T2	THETA2	SOUT2	
	MID3	T3	THETA3	SOUT3	-etc.-				

Example:

PCOMP	181	-0.224	7.45	10000.0	HOFF				
	171	0.056	0.0	YES			45.0		
			-45.0				90.0		

Fields:

Field	Contents
PID	Property identification number. (0 < Integer < 10000000)
Z0	Distance from the reference plane to the bottom surface. See Remark 16. (Real; Default = -0.5 times the element thickness.)
NSM	Nonstructural mass per unit area. See Remark 5 for SOL 414. (Real)
SB	Allowable shear stress of the bonding material (allowable interlaminar shear stress). Required if FT is specified. (Real > 0.0) See Remarks 13 and 23.
FT	Failure theory. The following theories are allowed (Character; Default = No failure calculation is performed when blank.): “HILL” for the Hill theory. “HOFF” for the Hoffman theory. “TSAI” for the Tsai-Wu theory. “STRN” for the Maximum Strain theory.

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Field	Contents	
	See the Laminates chapter in the *Simcenter Nastran User's Guide* for a detailed explanation of each theory.	
TREF	Reference temperature. See Remark 5. (Real; Default = 0.0)	
GE	Structural damping coefficient (dimensionless). See Remark 6. (Real; Default = 0.0)	
LAM	Laminate Options. (Character or blank, Default = blank). See Remarks 4 and 18.	
	"Blank"	All plies must be specified and all stiffness terms are developed.
	"SYM"	Only plies on one side of the laminate centerline are specified. The plies are numbered starting with 1 for the bottom ply. If the laminate contains an odd number of plies, then model the center ply as half the thickness of the actual center ply.
	"MEM"	All plies must be specified, but only membrane terms (MID1 on the derived PSHELL entry) are computed.
	"BEND"	All plies must be specified, but only bending terms (MID2 on the derived PSHELL entry) are computed.
	"SMEAR"	All plies must be specified, stacking sequence is ignored, on the derived PSHELL entry, MID1=MID2, MID3=MID4=0 where 0 is the same as blank, and TS/T = 12I/T**3 = 1.0.
	"SMCORE"	Face plies on one side of the laminate and the core are specified to define a laminate that is symmetric about the midplane of the core. The core is specified last. When calculating face sheet stiffness, stacking sequence of the face sheets is ignored.
MIDi	Material ID of the various plies. The plies are identified by serially numbering them from 1 at the bottom layer. The MIDs can refer to MAT1, MAT2, MAT8 bulk entries. See Remark 2. (0 < Integer < 99999999 or blank, except MID1 must be specified.)	
Ti	Thicknesses of the various plies. See Remark 2. (Real or blank, except T1 must be specified.)	
THETAi	Orientation angle of the longitudinal direction of each ply with the material axis of the element. (If the material angle on the element connection entry is 0.0, the material axis and side 1-2 of the element coincide.) The plies are to be numbered serially starting with 1 at the bottom layer. The bottom layer is defined as the surface with the largest -Z value in the element coordinate system. (Real; Default = 0.0)	
SOUTi	Controls individual ply stress and strain print or punch output. See Remarks 8 and 9. (Character: "YES" or "NO"; Default = "NO")	

Remarks:

1. PID must be unique with respect to all PCOMP, PCOMPG, and PSHELL entries.
2. The default for MIDi+1, ..., MIDn is the last defined MIDi. In the example above, MID(PLY1) is the default for MID(PLY2), MID(PLY3), and MID(PLY4). The same logic applies to Ti.

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3. Composite shell elements do not support nonlinear elastic materials defined with the MATS1 bulk entry.

4. At least one of the four values (MIDi, Ti, THETAi, SOUTi) must be present for a ply to exist. The minimum number of plies is one.

5. A temperature-dependent material defined with the combined MATi and MATTi entries can be referenced for a ply material (MIDi field on the PCOMP entry). For linear solutions, when computing the equivalent PSHELL and MAT2 entries from the PCOMP definition, the software uses TREF defined on the PCOMP entry to evaluate any temperature-dependent material properties for the plies. TREF defaults to 0.0 if undefined. The TEMPERATURE(INIT) case control command is not used in this phase of the solution, although it must be defined. Otherwise, the software will ignore the temperature-dependent material properties and use the properties on the referenced MATi entry. After the software creates the equivalent PSHELL and MAT2 entries, if a thermal load was defined with the TEMPERATURE(LOAD) case control command, the software will use the TEMPERATURE(INIT) command to compute thermal strains as described in the remarks on the TEMPERATURE case control command.

   By default, SOL 106 behaves as described above. Although, if PARAM, COMPMATT, YES is defined, SOL 106 will use the temperatures selected with the TEMPERATURE(LOAD) command to evaluate temperature-dependent material properties for the plies when computing the equivalent PSHELL and MAT2 entries. A unique TEMPERATURE(LOAD) command in each subcase will result in the recomputing of the equivalent PSHELL and MAT2 entries. As described above for the linear solutions, the TEMPERATURE(INIT) case control command is also required in SOL 106 in order for the software to use the temperature-dependent material properties when computing the equivalent PSHELL and MAT2 entries.

6. GE given on the PCOMP entry will be used for the element and the values supplied on material entries for individual plies are ignored. You are responsible for supplying the equivalent damping value on the PCOMP entry. GE is ignored in a transient analysis if PARAM,W4 is not specified. See the parameter **W4**.

7. Ply stress and strain on the individual lamina is output in the ply coordinate system, where the ply coordinate system is the element material coordinate system rotated by THETAi on each ply.

   For SOLs 101, 103, 105, 106, 108, 109, 111, 112, 114, 129, 144, and 200, interlaminar shear stress is output in the element material coordinate system (not the ply coordinate system). The X-axis of the element material coordinate system is defined with the THETA/MCID field on the element entry. For example, the CQUAD4 entry.

   For SOLs 401, 402, and 414, interlaminar shear stress is output in the same ply coordinate system as stress and strain output on the individual lamina.

8. The parameter NOCOMPS determines whether stress and/or strain recovery is at the composite ply layers (default), on the equivalent PSHELL, or both. See the parameter **NOCOMPS**. The STRESS and/or STRAIN case control commands are required for any of these recovery options. When ply results are requested, stress and/or strain are computed at the middle of each ply. To print the ply stress and/or strain results, the case control command request must include the “PRINT” option (default). To punch these results, the case control command request must include the “PUNCH” option. SOUTi=YES should then be defined on any ply definitions in which you would like print or punch output. The SOUTi entry is not used in the computing or printing of failure indices. See Remark **10**.

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9. Stress and strain output for individual plies are available in all superelement static and normal modes analysis and requested by the STRESS and STRAIN case control commands.

10. To compute STRESS failure index, the following must be present:

    a. STRESS case control command.

    b. The parameter NOCOMPS set to 1 (default) or 0.

    c. SB and FT (= to HILL, HOFF or TSAI) on the PCOMP bulk entry.

    d. Stress allowables Xt, Xc, Yt, Yc, and S on all referenced MAT8 bulk entries.

    e. Stress allowables ST, SC, and SS on all referenced MAT1 bulk entries.

    To compute STRAIN failure index, the following must be present:

    a. STRESS case control command.

    b. The parameter NOCOMPS set to 1 (default) or 0.

    c. SB and FT (= STRN) on the PCOMP bulk entry.

    d. Strain allowables Xt, Xc, Yt, Yc, S, and STRN=1.0 on all referenced MAT8 bulk entries.

    By default, failure index output prints in the f06 file even when using the PLOT or PUNCH describers on the STRESS and STRAIN case control commands. The parameter entry PARAM,NOFISR,1 can be used to turn off the printing of the failure index output. See the parameter **NOFISR**.

11. To output strength ratio, the failure index output conditions listed in Remark **10** must exist, and the parameter SRCOMPS must equal "YES". See the parameter **SRCOMPS**.

12. Stress resultant output can be requested with the FORCE case control command.

13. The failure index of the bonding material is calculated by:
    $FI_{bonding} = (\max(\tau_{1z}, \tau_{2z})/$ allowable bonding stress).
    The allowable bonding stress is defined on the SB field. The strength ratio for the bonding material is:
    $SR_{bonding} = (1 / FI_{bonding})$.

14. The software automatically creates equivalent PSHELL and MATi entries from a PCOMP definition. You can optionally include a sorted echo request to print the derived PSHELL and MATi entries in User Information Message 4379, or to the punch file.

15. The parameter NOCOMPS controls if stress and strain are computed for the composite elements, the equivalent homogeneous element, or both. See the parameter **NOCOMPS**. The software designates the equivalent homogeneous elements with a MID1 or MID2 ID greater than or equal to 10^8 on the PSHELL entry. Homogeneous stresses are based upon a smeared representation of the laminate's properties and in general will be significantly different than the more accurate lamina stresses available from PCOMP-based elements.

16. You can use the Z0 field on the PCOMP/PCOMPG or the ZOFFS field on the element entries to offset the midplane of the laminate away from the plane in which the connecting grid points are defined. Your choice effects the reference plane used for computing element forces, homogeneous strains, homogeneous stresses, and contact conditions.

    - By default (when you have not defined Z0 or ZOFFS), the midplane of the laminate and the reference plane are both located on the surface defined by the connecting grid points. See example A in **Figure 16-149**.

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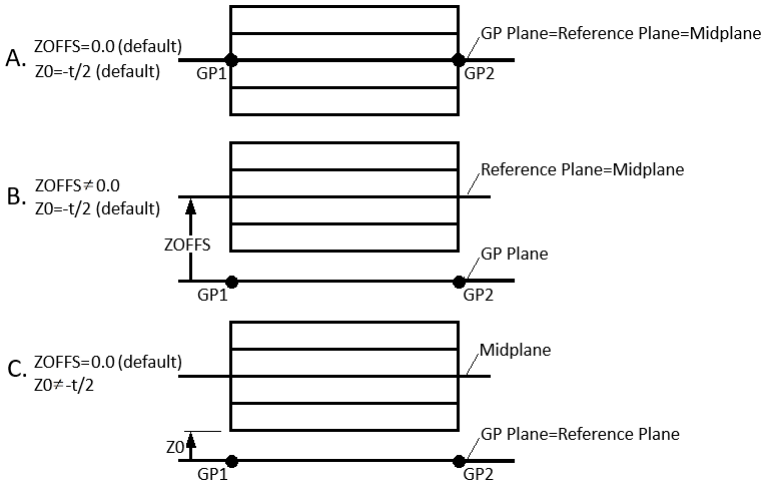
- When you define the ZOFFS field on the element entries, the reference plane and laminate midplane are offset together from the surface defined by the connecting grid points in the element Z-direction. See example B in **Figure 16-149**.
- When you define the Z0 field on the PCOMP/PCOMPG entries, only the laminate midplane is offset from the surface defined by the connecting grid points in the element Z-direction. The reference plane remains on the surface defined by the connecting grid points. See example C in **Figure 16-149**.

Figure 16-149: Laminate offsets with Z0 or ZOFFS



The global behavior of the model and stiffness are the same for both B and C when they are defined physically equivalent (that is, you define Z0=ZOFFS–t/2). Both cases will result in the same displacements, homogenous strains, ply strains, and ply stresses.

- When you use the default strain computation location on the STRAIN case control command (STRCUR describer=reference plane), the homogenous strains for both B and C are computed at different reference plane locations and are different, although the curvature is the same.

  You can include the FIBER describer on the STRAIN command to request the strain computation location at the top and bottom of the laminate (Z1 and Z2), and the homogenous strains will be the same for B and C. Note that with the FIBER option, curvature is not output.
- Because shell resultants (element forces) for both B and C are computed relative to different reference plane locations, the shell resultant output, particularly the moments, are different.
- Because the homogenous stresses requested by specifying PARAM, NOCOMPS, -1 or 0 are computed from the shell resultants, they are also different for the two options. Homogenous stresses are computed with the assumption that the reference plane is at the midplane (neutral axis) of the laminate. As a result, homogenous stresses are incorrect when you use the Z0 field to define an offset.

In general, we recommend that you use the ZOFFS field on the element entries to define the offset since the shell resultants (element forces) are computed relative to the reference plane location at the midplane of the laminate, and the homogenous stresses are correct.

In addition, Z0 has the limitation that it is ignored for determining the offset position of the reference plane for contact calculation. To include an offset of the reference plane in the contact calculation,

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use ZOFFS on the corresponding element entry instead of Z0. The contact faces can also be offset relative to the reference plane based on the element thickness using the SHLTHK parameter on BCTPARM. By default, the contact faces are computed at a distance of +/- 0.5 times the element thickness from the reference plane, but element thickness can be ignored in the contact calculation by setting SHLTHK = 1.

17. An unsymmetrical layup or the use of Z0 to specify an unsymmetrical layup, is not recommended in buckling analysis or the calculation of differential stiffness. Also, Z0 should not be used to specify an unsymmetrical layup.
18. The SYM option for the LAM option computes the complete stiffness properties while specifying half the plies. The MEM, BEND, SMEAR and SMCORE options provide the following special purpose stiffness calculations: MEM option only considers membrane effects, BEND option only considers bending effects, SMEAR ignores stacking sequence and is intended for cases where the sequence is not yet known, SMCORE allows simplified modeling of a sandwich panel with equal face sheets and a central core.
19. Element output for the SMEAR and SMCORE options are produced using the PARAM NOCOMPS -1 methodology that suppresses ply stress/strain results and prints results for the equivalent homogeneous element.
20. When the PCOMP or PCOMPG bulk entries are included in a distributed parallel method, the gpart keyword used for selecting the partitioning method must be gpart=1.
21. PCOMP is supported in all solutions except SOL 153 or 159 heat transfer analysis.
22. For elements referencing a PCOMP/PCOMPG, stress and strain output for the individual lamina is supported in solutions 101, 103, 105, 106, 108, 109, 111, 112, 114, 129, 144, 200, 401 and 402. For SOLs 106 and 129, individual lamina stress and strain are only output when a linear analysis is run (LGDISP = -1). In other solutions (including 106 and 129), stress and strain can only be recovered for the equivalent PSHELL that the software creates for the laminate.
23. Interlaminar failure is computed using the out-of-plane normal and the shear stresses. For the solid elements, all of the interlaminar stress components exist while only the shear stresses exist for the shell elements (due to the plane stress hypothesis in shell theories). For this reason, only the SB field is available on the PCOMP, PCOMPG, and PCOMPG1 bulk entries, but both the SB and NB fields are available on the PCOMPS entry.

Remarks related to SOL 401:

1. LAM is ignored.
2. PID must be unique with respect to all PCOMP, PCOMPG, PCOMPG1, and PSHELL bulk entries.
3. The software automatically creates an equivalent PCOMPG1 bulk entry for each PCOMP bulk entry. Thus, all the behaviors are aligned with the PCOMPG1 bulk entry. For example, only ply stress and strain results are computed for stress and strain output requests.

Remarks related to SOLs 402 and 414:

1. GE and SOUTi are ignored.

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2. TREF is ignored. Instead, the TREF of the material bulk entry of each ply is taken into account.
3. Z0 is added to the offset introduced on the element bulk entry (ZOFFS). If element forces are required, bending moments are given with respect to the updated neutral plane.
4. In SOL 402, the LAM field must either be left blank, or set to "SYM". The other LAM field options are not supported and will result in an error if defined.
5. For SOL 414 solutions, if you set the nonstructural mass (NSM) parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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PCOMPG

Layered Composite Element Property with global ply IDs

Defines the properties of an n-ply composite material laminate that includes global ply IDs.

Format:

1	2	3	4	5	6	7	8	9	10
PCOMPG	PID	Z0	NSM	SB	FT	TREF	GE	LAM	
	GPLYIDi	MIDi	Ti	THETAi	SOUTi				

Example:

PCOMPG	73	–2.E-4	0.0	8.E+7	TSAI				
	101	1	1.E-4	0.	YES				
	102		1.E-4	0.	YES				
	103		1.E-4	0.	YES				
	104		1.E-4	0.	YES				

Fields:

Field	Contents
PID	Property identification number. (0 < Integer < 10000000)
Z0	Distance from the reference plane to the bottom surface. See Remark 16. (Real; Default = -0.5 times the element thickness.)
NSM	Nonstructural mass per unit area. See Remark 4 for SOL 414. (Real)
SB	Allowable shear stress of the bonding material (allowable interlaminar shear stress). Required if FT is specified. (Real > 0.0) See Remarks 13 and 23.
FT	Failure theory. The following theories are allowed (Character; Default = No failure calculation is performed when blank.): “HILL” for the Hill theory “HOFF” for the Hoffman theory “TSAI” for the Tsai-Wu theory “STRN” for the Maximum Strain theory

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Field	Contents	
	See the Laminates chapter in the *Simcenter Nastran User's Guide* for a detailed explanation of each theory.	
TREF	Reference temperature. See Remark 5. (Real; Default = 0.0)	
GE	Structural damping coefficient (dimensionless). See Remark 8. (Real; Default = 0.0)	
LAM	Laminate Options. (Character or blank, Default = blank). See Remark 18.	
	"Blank"	All plies must be specified and all stiffness terms are developed.
	"MEM"	All plies must be specified, but only membrane terms (MID1 on the derived PSHELL entry) are computed.
	"BEND"	All plies must be specified, but only bending terms (MID2 on the derived PSHELL entry) are computed.
	"SMEAR"	All plies must be specified, stacking sequence is ignored, on the derived PSHELL entry, MID1=MID2, MID3=MID4=0 where 0 is the same as blank, and TS/T = 12I/T**3 = 1.0.
	"SMCORE"	Face plies on one side of the laminate and the core are specified to define a laminate that is symmetric about the midplane of the core. The core is specified last. When calculating face sheet stiffness, stacking sequence of the face sheets is ignored.
GPLYIDi	Global ply IDs. See Remark 2. (Integer > 0)	
MIDi	Material ID of the various plies. The plies are identified by serially numbering them from 1 at the bottom layer. The MIDs must refer to MAT1, MAT2, MAT8, and MAT11 (Sol 402 only) bulk entries. See Remark 4. (0 < Integer < 99999999 or blank, except MID1 must be specified.)	
Ti	Thicknesses of the various plies. See Remark 4. (Real or blank, except T1 must be specified.)	
THETAi	Orientation angle of the longitudinal direction of each ply with the material axis of the element. (If the material angle on the element connection entry is 0.0, the material axis and side 1-2 of the element coincide.) The plies are to be numbered serially starting with the first listed at the bottom layer. The bottom layer is defined as the surface with the largest -Z value in the element coordinate system. (Real; Default = 0.0)	
SOUTi	Controls individual ply stress and strain print or punch output. See Remarks 9 and 10. (Character: "YES" or "NO"; Default = "NO")	

Remarks:

1. PID must be unique with respect to all PCOMP, PCOMPG, and PSHELL entries.
2. Each global ply identification number GPLYIDi in a single PCOMPG entry should be unique.

   The global ply identification numbers (GPLYIDi) are reused across different PCOMPG bulk entries in order to post-process relative ply layers with common GPLYIDi.

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3. Composite shell elements do not support nonlinear elastic materials defined with the MATS1 bulk entry.

4. The default for MIDi+1, ..., MIDn is the last defined MIDi. In the example above, MID(PLY1) is the default for MID(PLY2), MID(PLY3), and MID(PLY4). The same logic applies to Ti.

5. A temperature dependent material defined with the combined MATi and MATTi entries can be referenced for a ply material (MIDi field on the PCOMPG entry). For linear solutions, when computing the equivalent PSHELL and MAT2 entries from the PCOMPG definition, the software uses TREF defined on the PCOMPG entry to evaluate any temperature dependent material properties for the plies. TREF defaults to 0.0 if undefined. The TEMPERATURE(INIT) case control command is not used in this phase of the solution, although it must be defined, otherwise the software will ignore the temperature dependent material properties and use the properties on the referenced MATi entry. After the software creates the equivalent PSHELL and MAT2 entries, if a thermal load was defined with the TEMPERATURE(LOAD) case control command, the software will use the TEMPERATURE(INIT) command to compute thermal strains as described in the remarks on the TEMPERATURE case control command.

   By default, SOL 106 behaves as described above. Although, if PARAM, COMPMATT, YES is defined, SOL 106 will use the temperatures selected with the TEMPERATURE(LOAD) command to evaluate temperature dependent material properties for the plies when computing the equivalent PSHELL and MAT2 entries. A unique TEMPERATURE(LOAD) command in each subcase will result in the recomputing of the equivalent PSHELL and MAT2 entries. As described above for the linear solutions, the TEMPERATURE(INIT) case control command is also required in SOL 106 in order for the software to use the temperature dependent material properties when computing the equivalent PSHELL and MAT2 entries.

6. At least one of the four values (MIDi, Ti, THETAi, SOUTi) must be present for a ply to exist. The minimum number of plies is one.

7. Ply stress and strain on the individual lamina is output in the ply coordinate system, where the ply coordinate system is the element material coordinate system rotated by THETAi on each ply.

   For SOLs 101, 103, 105, 106, 108, 109, 111, 112, 114, 129, 144, and 200, interlaminar shear stress is output in the element material coordinate system (not the ply coordinate system). The X-axis of the element material coordinate system is defined with the THETA/MCID field on the element entry. For example, the CQUAD4 entry.

   For SOLs 401, 402, and 414, interlaminar shear stress is output in the same ply coordinate system as stress and strain output on the individual lamina.

8. GE given on the PCOMPG entry will be used for the element and the values supplied on material entries for individual plies are ignored. You are responsible for supplying the equivalent damping value on the PCOMPG entry. GE is ignored in a transient analysis if PARAM,W4 is not specified. See the parameter **W4**.

9. The parameter NOCOMPS determines if stress and/or strain recovery is at the composite ply layers (default), on the equivalent PSHELL, or both. See the parameter **NOCOMPS**. The STRESS and/or STRAIN case control commands are required for any of these recovery options. When ply results are requested, stress and/or strain are computed at the middle of each ply. To print the ply stress and/or strain results, the case control command request must include the “PRINT” option (default). To punch these results, the case control command request must include the “PUNCH” option. SOUTi=YES

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should then be defined on any ply definitions in which you would like print or punch output. The SOUTi entry is not used in the computing or printing of failure indices. See Remark 11.

10. Stress and strain output for individual plies are available in all superelement static and normal modes analysis and requested by the STRESS and STRAIN case control commands.

11. To output STRESS failure index, the following must be present:

    a. STRESS case control command.

    b. The parameter NOCOMPS set to 1 (default) or 0.

    c. SB and FT (= to HILL, HOFF or TSAI) on the PCOMPG Bulk Data entry.

    d. Stress allowables Xt, Xc, Yt, Yc, and S on all referenced MAT8 Bulk Data entries.

    e. Stress allowables ST, SC, and SS on all referenced MAT1 Bulk Data entries.

    To output STRAIN failure index, the following must be present:

    a. STRESS case control command.

    b. The parameter NOCOMPS set to 1 (default) or 0.

    c. SB and FT (= STRN) on the PCOMPG Bulk Data entry.

    d. Strain allowables Xt, Xc, Yt, Yc, S, and STRN=1.0 on all referenced MAT8 Bulk Data entries.

    By default, failure index output prints in the f06 file even when using the PLOT or PUNCH describers on the STRESS and STRAIN case control commands. The parameter entry PARAM,NOFISR,1 can be used to turn off the printing of the failure index output. See the parameter NOFISR.

12. To output strength ratio, the failure index output conditions listed in Remark 11 must exist, and the parameter SRCOMPS must equal “YES”. See the parameter SRCOMPS.

13. The failure index of the bonding material is calculated by:
    $FI_{bonding} = (\max(\tau_{1z}, \tau_{2z})/$ allowable bonding stress).
    The allowable bonding stress is defined on the SB field. The strength ratio for the bonding material is:
    $SR_{bonding} = (1 / FI_{bonding})$.

14. The software automatically creates equivalent PSHELL and MATi entries from a PCOMPG definition. You can optionally include a sorted echo request to print the derived PSHELL and MATi entries in User Information Message 4379, or to the punch file.

15. The parameter NOCOMPS controls if stress and strain are computed for the composite elements, the equivalent homogeneous element, or both. See the parameter NOCOMPS. The software designates the equivalent homogeneous elements with a MID1 or MID2 ID greater than or equal to 10^8 on the PSHELL entry. Homogeneous stresses are based upon a smeared representation of the laminate’s properties and in general will be significantly different than the more accurate lamina stresses available from PCOMP-based elements.

16. You can use the Z0 field on the PCOMP/PCOMPG or the ZOFFS field on the element entries to offset the midplane of the laminate away from the plane in which the connecting grid points are defined. Your choice effects the reference plane used for computing element forces, homogeneous strains, homogeneous stresses, and contact conditions.

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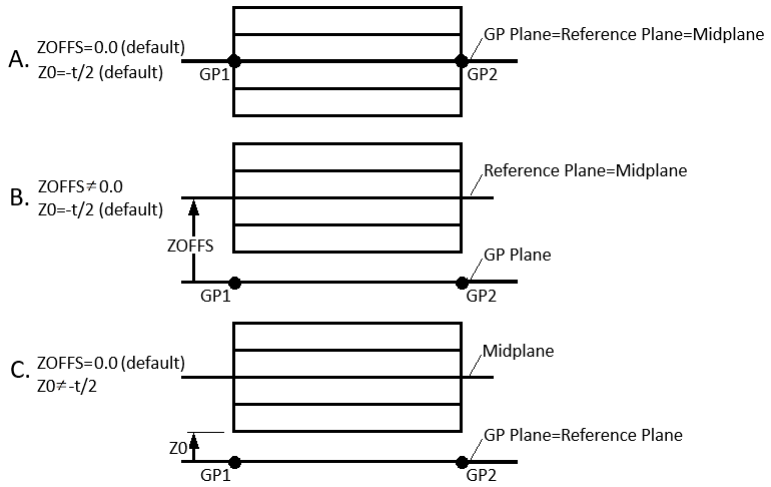
- By default (when you have not defined Z0 or ZOFFS), the midplane of the laminate and the reference plane are both located on the surface defined by the connecting grid points. See example A in **Figure 16-149**.
- When you define the ZOFFS field on the element entries, the reference plane and laminate midplane are offset together from the surface defined by the connecting grid points in the element Z-direction. See example B in **Figure 16-149**.
- When you define the Z0 field on the PCOMP/PCOMPG entries, only the laminate midplane is offset from the surface defined by the connecting grid points in the element Z-direction. The reference plane remains on the surface defined by the connecting grid points. See example C in **Figure 16-149**.

Figure 16-150: Laminate offsets with Z0 or ZOFFS



The global behavior of the model and stiffness are the same for both B and C when they are defined physically equivalent (that is, you define Z0=ZOFFS–t/2). Both cases will result in the same displacements, homogenous strains, ply strains, and ply stresses.

- When you use the default strain computation location on the STRAIN case control command (STRCUR describer=reference plane), the homogenous strains for both B and C are computed at different reference plane locations and are different, although the curvature is the same.

  You can include the FIBER describer on the STRAIN command to request the strain computation location at the top and bottom of the laminate (Z1 and Z2), and the homogenous strains will be the same for B and C. Note that with the FIBER option, curvature is not output.
- Because shell resultants (element forces) for both B and C are computed relative to different reference plane locations, the shell resultant output, particularly the moments, are different.
- Because the homogenous stresses requested by specifying PARAM, NOCOMPS, -1 or 0 are computed from the shell resultants, they are also different for the two options. Homogenous stresses are computed with the assumption that the reference plane is at the midplane (neutral axis) of the laminate. As a result, homogenous stresses are incorrect when you use the Z0 field to define an offset.

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In general, we recommend that you use the ZOFFS field on the element entries to define the offset since the shell resultants (element forces) are computed relative to the reference plane location at the midplane of the laminate, and the homogenous stresses are correct.

In addition, Z0 has the limitation that it is ignored for determining the offset position of the reference plane for contact calculation. To include an offset of the reference plane in the contact calculation, use ZOFFS on the corresponding element entry instead of Z0. The contact faces can also be offset relative to the reference plane based on the element thickness using the SHLTHK parameter on BCTPARM. By default, the contact faces are computed at a distance of +/- 0.5 times the element thickness from the reference plane, but element thickness can be ignored in the contact calculation by setting SHLTHK = 1.

17. An unsymmetrical layup or the use of Z0 to specify an unsymmetrical layup, is not recommended in buckling analysis or the calculation of differential stiffness. Also, Z0 should not be used to specify an unsymmetrical layup.
18. The MEM, BEND, SMEAR and SMCORE options provide the following special purpose stiffness calculations: MEM option only considers membrane effects, BEND option only considers bending effects, SMEAR ignores stacking sequence and is intended for cases where the sequence is not yet known, SMCORE allows simplified modeling of a sandwich panel with equal face sheets and a central core.
19. Element output for the SMEAR and SMCORE options are produced using the PARAM NOCOMPS -1 methodology that suppresses ply stress/strain results and prints results for the equivalent homogeneous element.
20. When the PCOMP or PCOMPG bulk entries are included in a distributed parallel method (DMP), the gpart keyword used for selecting the partitioning method must be gpart=1.
21. PCOMPG is supported in all solutions except SOL 153 or 159 heat transfer analysis.
22. For elements referencing a PCOMP/PCOMPG, stress and strain output for the individual lamina is supported in solutions 101, 103, 105, 106, 108, 109, 111, 112, 114, 129, 144, 200, 401 and 402. For SOLs 106 and 129, individual lamina stress and strain are only output when a linear analysis is run (LGDISP = -1). In other solutions (including 106 and 129), stress and strain can only be recovered for the equivalent laminate. That is, output on the equivalent PSHELL created by the software.
23. Interlaminar failure is computed using the out-of-plane normal and the shear stresses. For the solid elements, all of the interlaminar stress components exist while only the shear stresses exist for the shell elements (due to the plane stress hypothesis in shell theories). For this reason, only the SB field is available on the PCOMP, PCOMPG, and PCOMPG1 bulk entries, but both the SB and NB fields are available on the PCOMPS entry.

Remarks related to SOL 401:

1. LAM is ignored.
2. PID must be unique with respect to all PCOMP, PCOMPG, PCOMPG1, and PSHELL bulk entries.
3. The software automatically creates an equivalent PCOMPG1 bulk entry for each PCOMPG bulk entry. Thus, all the behaviors are aligned with the PCOMPG1 bulk entry. For example, only ply stress and strain results are computed for stress and strain output requests.

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Remarks related to SOLs 402 and 414:

1. GE, LAM and SOUTi are ignored.
2. TREF is ignored. Instead, the TREF of the material bulk entry of each ply is taken into account.
3. Z0 is added to the offset introduced on the element bulk entry (ZOFFS). If element forces are required, bending moments are given with respect to the updated neutral plane.
4. For SOL 414 solutions, if you set the nonstructural mass (NSM) parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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PCOMPG1

Layered Composite Element Property with global ply IDs, Alternate form for SOLs 401, 402, and 414

Defines the properties of an n-ply composite material laminate which includes global ply IDs for SOLs 401, 402, and 414.

Format:

1	2	3	4	5	6	7	8	9	10
PCOMPG1	PID	Z0	NSM	SB		TREF	GE		
	GPLYIDi	MIDi	TRi	THETAi	FTi		SOUTi		

Example:

PCOMPG1	20		0.0	10000					
	1	1	0.02	45.	TSAI		YES		
	2	1	0.03	-45.	HILL		YES		
	3	2	0.05	0.	TSAI		YES		
	4	1	0.03	-45.	HILL		YES		
	5	1	0.02	45.	TSAI		YES		

Fields:

Field	Contents
PID	Property identification number. (0 < Integer < 10000000)
Z0	Distance from the reference plane to the bottom surface. See SOL 401 Remark 1, and SOLs 402 and 414 Remark 2. (Real; Default = -0.5 times the element thickness.)
NSM	Nonstructural mass per unit area. See Remark 3 for SOL 414. (Real)
SB	Allowable inter-laminar shear stress of the bonding material. Required if FTi is specified. See Remarks 10 and 14. (Real > 0.0)
TREF	Reference temperature. (Real; Default = 0.0)
GE	Damping coefficient. See Remark 6. (Real; Default = 0.0)
GPLYIDi	Global ply IDs. For information on stacking order. See Remark 2. (Integer > 0; No default)
MIDi	Material ID of the various plies. The MIDs must refer to MAT1, MAT2, MAT8, MAT9, or MAT11 bulk entries. For progressive ply failure, the MIDs must refer to MAT11 bulk entries. (Integer > 0; For default behavior, see Remark 3)
TRi	Ply thickness. See Remark 4. (Real > 0.0; No default)

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Field	Contents
THETAi	THETAi angle is the longitudinal direction of each ply with the material axis of the element. (Real; Default = 0.0, plan to follow the existing definition) .
FTi	Ply failure theory. The following theories are supported. (Character; Default = No failure calculation is performed when blank.) “HILL” for the Hill failure theory. “HOFF” for the Hoffman failure theory. “TSAI” for the Tsai-Wu failure theory. “STRN” for the Maximum Strain failure theory.
SOUTi	Controls individual ply stress, strain, and damage output. See Remark 7 and Remark 8. Allowable entries are: “NO” for do not compute. “YES” for compute. (Character; Default = “NO”)

Remarks related to SOLs 401, 402, and 414:

1. The software uses the THETAi specifications combined with the angle in the element specification (for example CQUAD4) to define the orientation of the material coordinates.
2. The ply stacking order is the same as the order of the continuation lines. The ply defined in the first continuation line is ply 1. The ply defined in the second continuation line is ply 2, and so on. Thus, the stacking order is independent of the GPLYIDi values.
3. MID1 must be defined. If all other MIDi are blank, they default to the value in the MID1 field. If multiple, but not all, MIDi fields are defined, the blank fields default to the last defined MIDi. For example, if a laminate consists of six plies and MID1 and MID4 are defined, MID1 will be used for the first three plies, and MID4 will be used for the last three plies.
4. FTi is failure theory for the ith ply.
5. Ply stress and strain results are always computed in the ply coordinate system.
6. GE given on the PCOMPG1 entry will be used for the element and the values supplied on material entries for individual plies are ignored. You are responsible for supplying the equivalent damping value on the PCOMPG1 entry. GE is ignored in a transient analysis if PARAM,W4 is not specified. See the parameter W4.
7. To request that ply stress and/or strain be computed, the STRESS and/or STRAIN case control command must be defined with the appropriate PRINT, PUNCH, or PLOT output option, and the SOUTi field must equal “YES”. See the remarks on the STRESS and STRAIN case control commands.

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8. To compute STRESS failure index, the following must be present:
   - STRESS case control command.
   - SB and FTi (= to HILL, HOFF or TSAI).
   - Stress allowables on the referenced MAT1, MAT9, MAT11, or MATFT bulk entry.
   - SOUTi=YES

   To compute STRAIN failure index, the following must be present:
   - STRESS case control command.
   - SB and FTi (= STRN).
   - Strain allowables on the referenced MATFT bulk entry.
   - SOUTi=YES

   By default, failure index output prints in the f06 file even when using the PLOT or PUNCH describers on the STRESS and STRAIN case control commands. The parameter entry PARAM,NOFISR,1 can be used to turn off the printing of the failure index output. See the parameter NOFISR.
9. To output strength ratio, the failure index output conditions listed in Remark 7 must exist, and the parameter SRCOMPS must equal “YES”. See the parameter SRCOMPS.
10. The failure index of the bonding material is calculated by:
    $FI_{bonding}$ = (max(τ_{1z}, τ_{2z}) / allowable bonding stress).
    The allowable bonding stress is defined on the SB field. The strength ratio for the bonding material is:
    $SR_{bonding}$ = (1 / $FI_{bonding}$).
11. The software uses TREF defined on the MATi entry referenced by a ply for the ply reference temperature. The software ignores TREF on the PCOMPG1 entry.
12. The equivalent homogeneous stresses are not supported for PCOMPG1.
13. PCOMPG1 is only supported in SOLs 401 and 402
14. Interlaminar failure is computed using the out-of-plane normal and the shear stresses. For the solid elements, all of the interlaminar stress components exist while only the shear stresses exist for the shell elements (due to the plane stress hypothesis in shell theories). For this reason, only the SB field is available on the PCOMP, PCOMPG, and PCOMPG1 bulk entries, but both the SB and NB fields are available on the PCOMPS entry.

Remark related to SOL 401:

1. Z0 is added to the offset introduced on the element bulk entry (ZOFFS) for a total offset of the reference plane. If element forces are requested, bending moments are given with respect to the updated reference plane. The contact faces can also be offset relative to the reference plane based on the element thickness using the SHLTHK parameter on BCTPAR2. By default, the contact faces are computed at a distance of +/- 0.5 times the element thickness from the reference plane, but element thickness can be ignored in the contact calculation by setting SHLTHK = 1.

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Remarks related to SOLs 402 and 414:

1. GE and SOUTi are ignored.
2. Z0 is added to the offset introduced on the element bulk entry (ZOFFS) for a total offset of the reference plane. If element forces are requested, bending moments are given with respect to the updated reference plane. The contact faces can also be offset relative to the reference plane based on the element thickness using the SHLTHK parameter on BCTPAR2. By default, the contact faces are computed at a distance of +/- 0.5 times the element thickness from the reference plane, but element thickness can be ignored in the contact calculation by setting SHLTHK = 1.
3. For SOL 414 solutions, if you set the nonstructural mass (NSM) parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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PCOMPS

Layered Composite Element Property for Solid Elements

Defines the properties of an n-ply composite material laminate for CHEXA and CPENTA solid elements.

Format:

1	2	3	4	5	6	7	8	9	10
PCOMPS	PID	CORDM	PSDIR	SB	NB	TREF	GE		
	GPLYIDi	MIDi	TRi	THETAi	FTi	ILFTi	SOUTi		

Example:

PCOMPS	20	2	13	10000.					
	1	1	0.02	45.	TSAI	NB	YES		
	2	1	0.03	–45.	HILL	SB	YES		
	3	2	0.05	0.	TSAI	SB	YES		
	4	1	0.03	–45.	HILL	SB	YES		
	5	1	0.02	45.	TSAI	NB	YES		

Fields:

Field	Contents
PID	Property identification number. (0 < Integer < 10000000)
CORDM	Identification number of the coordinate system that the software uses to determine the orientation of the material coordinate system. See Remark 1. Enter “0” or leave blank to use the basic coordinate system. (Integer; Default = 0) The CORDM field is ignored if a MATCID bulk entry references the associated element ID. See Remark 2.
PSDIR	Two integers in which the first is the material X-axis and the second is the stack direction (Z-axis). Enter the X-, Y-, and Z-axes of CORDM as 1, 2, and 3, respectively. See Remark 1. (Integers; 12,13,21,23,31,32; Default = 13)
SB	Allowable inter-laminar shear stress of the bonding material. See Remark 17. (Real > 0.0; Default = Failure index not calculated)
NB	Allowable inter-laminar normal stress of the bonding material. See Remark 17. (Real > 0.0; Default = Failure index not calculated)
TREF	Reference temperature. (Real; Default = 0.0)

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Field	Contents
GE	Damping coefficient. See Remark 7. (Real; Default = 0.0)
GPLYIDi	Global ply IDs. For information on stacking order, see Remark 3. (Integer > 0; No default)
MIDi	Material ID of the various plies. The MIDs must refer to MAT1, MAT9, or MAT11 bulk entries. For progressive ply failure, the MIDs must refer to MAT11 bulk entries. (Integer > 0; For default behavior, see Remark 4)
TRi	Ply thickness ratio. See Remark 5. (Real > 0.0; No default)
THETAi	Ply orientation angle. (Real; Default = 0.0)
FTi	Ply failure theory. (Character; Default = No failure calculation is performed when blank.) See Remark 8. Allowable entries are: “HILL” for the Hill failure theory. “HOFF” for the Hoffman failure theory. “TSAI” for the Tsai-Wu failure theory. “STRN” for the Maximum Strain failure theory. “STRS” for the Maximum Stress failure theory. “TS” for the Maximum Transverse Shear Stress failure theory. “PFA” for progressive ply failure. See Remark 9. (Character; Default = No failure theory). Not supported in SOL 402.
ILFTi	Inter-laminar failure theory. Allowable entries are: “SB” for transverse shear stress failure index. “NB” for normal stress failure index. (Character; Default = No failure index)
SOUTi	Controls individual ply stress, strain, and damage output. See Remark 12. Allowable entries are: “NO” for do not compute. “YES” for compute. (Character; Default = “NO”)

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Remarks:

1. The software uses the CORDM and PSDIR specifications to define the orientation of the material coordinates. Additionally, it uses THETAi to define the individual ply coordinate systems.

   The software procedure to compute the material system is as follows:

   a. The following three skew vectors are computed at the elemental center using parametric coordinates.

   $$\vec{\xi} = \left(\frac{\partial x}{\partial \xi} + \frac{\partial y}{\partial \xi} + \frac{\partial z}{\partial \xi} \right) \vec{\eta} = \left(\frac{\partial x}{\partial \eta} + \frac{\partial y}{\partial \eta} + \frac{\partial z}{\partial \eta} \right)$$
   $$\vec{\zeta} = \left(\frac{\partial x}{\partial \zeta} + \frac{\partial y}{\partial \zeta} + \frac{\partial z}{\partial \zeta} \right)$$

   b. The software determines which skew vector most closely aligns with the CORDM axis vector specified by the second number in the PSDIR field (stacking direction).

   For the CPENTA, the skew vector $\vec{\zeta}$ corresponds to the direction normal to the two triangular faces. If the coordinate selected by the second number in the PSDIR field is not mostly aligned with $\vec{\zeta}$ (normal to the two triangular faces), a fatal error is reported.

   For the CHEXA, the coordinate selected by the second number in the PSDIR field can be aligned with any of the skew vectors (that is, normal to any of the element faces).

   c. Finally, the material coordinate system is computed:

   $$\vec{X}_m = \bar{\xi}\,\vec{Z}_m = \vec{X}_m \times \vec{\eta}\,\vec{Y}_m = \vec{Z}_m \times \vec{X}_m$$

   where $\vec{Z}_m$ is the stacking direction,

   and $\vec{X}_m$, $\vec{Y}_m$ are in-plane to the ply coordinates.

   The ply coordinate systems are determined as follows:

   a. $\vec{Z}_p$ and $\vec{Z}_m$ are equivalent.

   b. The software rotates $\vec{X}_m$ about $\vec{Z}_p$ through angle THETAi to obtain $\vec{X}_p$ for the ith ply. The positive sense of rotation for THETAi is about the positive $\vec{Z}_p$.

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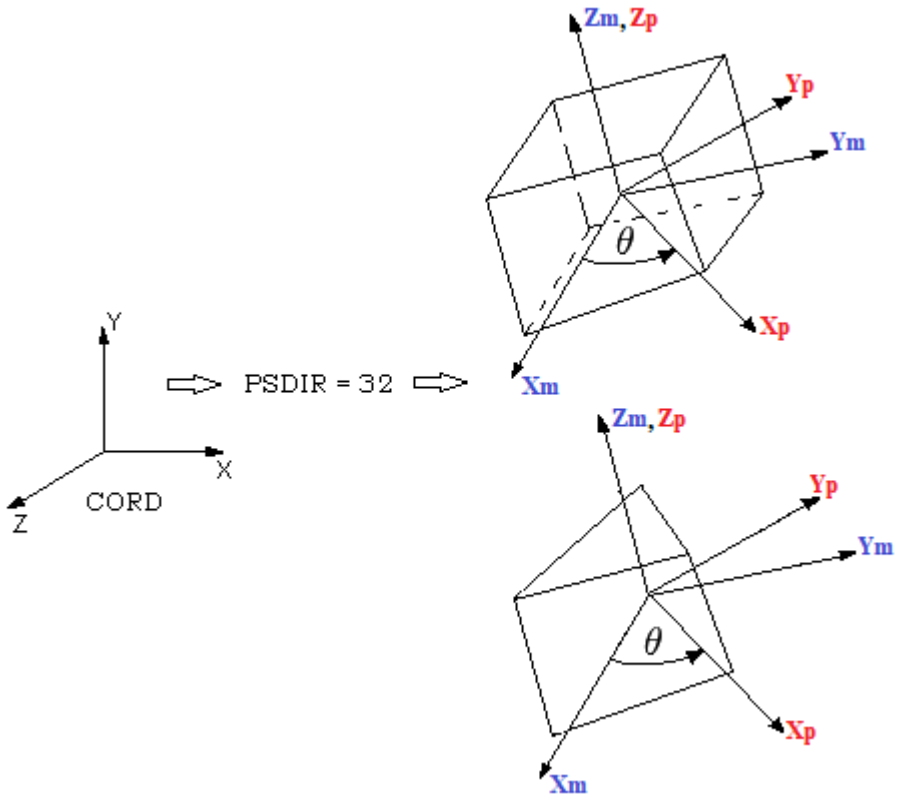
c. For each ply, the software computes $\vec{Y}_p$ so that it is orthogonal to $\vec{X}_p$ and $\vec{Z}_p$, and yields a right-hand coordinate system.

The figure below shows the material and ply coordinate systems for the i[th] ply in a CHEXA and a CPENTA element when PSDIR=32.



2. If a MATCID bulk entry references the associated element ID, the coordinate system selected with the MATCID bulk entry becomes the CORDM selection, and the CORDM field is ignored. The software uses the coordinate system selected with the MATCID bulk entry to determine the orientation of the material coordinate system as described in Remark 1.

3. The ply stacking order is the same as the order of the continuation lines. The ply defined in the first continuation line is ply 1. The ply defined in the second continuation line is ply 2, and so on. Thus, the stacking order is independent of the GPLYIDi values.

Because the orientation of an element coordinate system depends on element connectivity, the positive sense for a pair of parallel Z_m–axes can differ. To allow for a single PCOMPS definition to be applied across many elements, the software uses the positive sense of the CORDM axis that is used to determine the Z_m–axis for each element as the direction of increasing ply number. That is, ply 2 is further out the positive axis than ply 1.

4. MID1 must be defined. If all other MIDi are blank, they default to the value in the MID1 field. If multiple, but not all, MIDi fields are defined, the blank fields default to the last defined MIDi. For example, if a laminate consists of six plies and MID1 and MID4 are defined, MID1 will be used for the first three plies, and MID4 will be used for the last three plies.

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5. The software adjusts the thickness of the plies proportionate to their thickness ratios (TRi) such that the total thickness of the laminate at the element corners is equal to the distance between the grid points. The software adjusts the thickness by:

$$Ply_Thk(i) = \left(\frac{Elem_Thk}{Sum_of_TR(i)}\right) TR(i)$$

where *Ply_Thk(i)* is the adjusted ply thickness for the i-th ply, *Elem_Thk* is the average distance between the grid points, *Sum_of_TR(i)* is the sum of all thickness ratios, and *TR(i)* is the thickness ratio for the i-th ply.

For example, if you define four plies with thickness ratios 1.0, 2.0, 1.0, and 2.0, and the distance between the grid points is 3.0, the actual ply thicknesses would be .5, 1.0, .5, and 1.0, respectively.

6. The software uses the thickness ratios you defined in the TRi fields to compute the actual thickness of each ply in the laminate is adjusted proportionally.

7. GE given on the PCOMPS entry will be used for the element and the values supplied on material entries for individual plies are ignored. You are responsible for supplying the equivalent damping value on the PCOMPS entry. GE is ignored in a transient analysis if PARAM,W4 is not specified. See the parameter **W4**.

8. FTi is failure theory for the i^{th} ply. The material properties for all failure theories except progressive ply failure are specified by MATFT bulk entries. The material properties for progressive ply failure are specified by MATDMG bulk entries. For the i^{th} ply, the software references the MATFT or MATDMG bulk entry that has the same MID as that used for the MIDi field.

9. To considers in-plane progressive damage to a ply, use progressive ply failure. To consider inter-laminar failure, use the SB or NB options in the ILFTi field. To consider inter-laminar progressive failure and delamination, leave the ILFTi field blank and use cohesive elements to model the inter-laminar behavior.

10. To compute a ply and/or bonding failure index, the STRESS case control command must be present, SOUTi on the PCOMPS bulk entry must be set to “YES”, and the following must be defined.

    For a stress or strain ply failure index:

    a. FTi on the PCOMPS bulk entry.

    b. The stress or strain allowables on the referenced MATFT bulk entry.

    For a stress bonding failure index:

    a. ILFTi on the PCOMPS bulk entry.

    b. The stress allowables SB or NB on the PCOMPS bulk entry.

    By default, failure index output prints in the f06 file even when using the PLOT or PUNCH describers on the STRESS and STRAIN case control commands. The parameter entry PARAM,NOFISR,1 can be used to turn off the printing of the failure index output. See the parameter **NOFISR**.

11. Ply stress and strain results are always computed in the ply coordinate system.

12. To request that ply stress and/or strain be computed, the STRESS and/or STRAIN case control command must be defined with the appropriate PRINT, PUNCH, or PLOT output option, and the SOUTi field must equal “YES”. The STRESS and STRAIN commands also include the CPLYMID, CPLYBT, and

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CPLYBMT describers to specify stress or strain recovery at the bottom, middle, or top of the plies. See the remarks on the STRESS and STRAIN case control commands.

13. GPSTRESS or GPSTRAIN output is not supported.
14. Glue or contact definitions defined on composite solid faces which are perpendicular to the stack direction (edge faces) may produce poor stress continuity. If the glue/contact definition is between edge faces belonging to different PCOMPS definitions, and if the number of plies on each PCOMPS definition is small and the same, and the ply thicknesses are similar, the stress continuity should be fairly smooth. This also applies to the results requested with the BCRESULTS and BGRESULTS case control commands.
15. PCOMPS is supported in solutions 101, 103, 105, 108, 109, 111, 112, 401 and 402.
16. For SOLs 101, 103, 105, 108, 109, 111, and 112, the software uses TREF on the PCOMPS entry as the reference temperature for all plies. The software ignores TREF on the MATi entries referenced by the plies. See the remarks for linear solutions on the TEMPERATURE case control command for information on how the reference temperature is used to compute thermal strain in these solutions.

    For SOLs 401 and 402, the software uses TREF defined on the MATi entry referenced by a ply for the ply reference temperature. The software ignores TREF on the PCOMPS entry. See "Thermal Loads" in the *Multi-Step Nonlinear User's Guide* for information on how the reference temperature is used to compute thermal strain in SOL 401.
17. Interlaminar failure is created by out-plane normal and shear stresses. In the solid elements, all the interlaminar stress components are present while only the shear stresses exist in the shell elements (due to the plane stress hypothesis in shell theories).

    That's why SB and NB are both present in PCOMPS card while only SB is accepted in PCOMP, PCOMPG, and PCOMPG1 cards.

Remark related to SOLs 402 and 414:

1. GE and SOUTi are ignored.
2. PFA progressive ply failure is not supported.
3. TREF is ignored. Instead, the TREF of the material bulk entry of each ply is taken into account.

Related Topics

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PCONV

Convection Property Definition

Specifies the free convection boundary condition properties of a boundary condition surface element used for heat transfer analysis.

Format:

1	2	3	4	5	6	7	8	9	10
PCONV	PCONID	MID	FORM	EXPF					

Example:

PCONV	3	2	0	.25					

Fields:

Field	Contents
PCONID	Convection property identification number. (Integer > 0)
MID	Material property identification number. (Integer > 0)
FORM	Type of formula used for free convection. (Integer 0, 1, 10, 11, 20, or 21; Default = 0)
EXPF	Free convection exponent as implemented within the context of the particular form that is chosen. See Remark 3. (Real ≥ 0.0; Default = 0.0)

Remarks:

1. Every surface to which free convection is to be applied must reference a PCONV entry. PCONV is referenced on the CONV Bulk Data entry.
2. MID must reference a MAT4 entry that specifies the convection heat transfer coefficient (H).
3. EXPF is the free convection temperature exponent.
   - If FORM = 0, 10, or 20, EXPF is an exponent of (T – TAMB), where the convective heat transfer is represented as

$$q = H \cdot u_{\text{CNTRLND}} \cdot (T - \text{TAMB})^{\text{EXPF}} \cdot (T - \text{TAMB}) \ .$$

   - If FORM = 1, 11, or 21,

$$q = H \cdot u_{\text{CNTRLND}} \cdot (T^{\text{EXPF}} - \text{TAMB}^{\text{EXPF}})$$

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where T represents the elemental grid point temperatures and TAMB is the associated ambient temperature.

4. FORM specifies the formula type and the reference temperature location used in calculating the convection film coefficient if FLMND = 0.

   - If FORM = 0 or 1, the reference temperature is the average of element grid point temperatures (average) and the ambient point temperatures (average).
   - If FORM = 10 or 11, the reference temperature is the surface temperature (average of element grid point temperatures).
   - If FORM = 20 or 21, the reference temperature is the ambient temperature (average of ambient point temperatures).

Related Topics

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PCONVM

Forced Convection Property Definition

Specifies the forced convection boundary condition properties of a boundary condition surface element used for heat transfer analysis.

Format:

1	2	3	4	5	6	7	8	9	10
PCONVM	PCONID	MID	FORM	FLAG	COEF	EXPR	EXPPI	EXPPO	

Example:

PCONVM	3	2	1	1	.023	0.80	0.40	0.30	

Fields:

Field	Contents
PCONID	Convection property identification number. (Integer > 0)
MID	Material property identification number. (Integer > 0)
FORM	Type of formula used for convection. (Integer = 0, 1, 10, 11, 20, or 21; Default = 0)
FLAG	Flag for mass flow convection. (Integer = 0 or 1; Default = 0)
COEF	Constant coefficient used for forced convection. (Real > 0.0)
EXPR	Reynolds number convection exponent. (Real ≥ 0.0; Default = 0.0)
EXPPI	Prandtl number convection exponent for heat transfer into the working fluid. (Real ≥ 0.0; Default = 0.0)
EXPPO	Prandtl number convection exponent for heat transfer out of the working fluid. (Real ≥ 0.0; Default = 0.0)

Remarks:

1. Every surface to which forced convection is applied must reference a PCONVM entry. PCONVM is referenced on the CONVM entry.
2. MID references a MAT4 entry that specifies material properties of the working fluid at the temperature of the point FLMND. FLMND is specified on the CONVM entry.
3. The material properties are used in conjunction with the average diameter and mass flow rate (mdot). MID references the material properties and supplies the fluid conductivity (k), heat capacity (cp), and viscosity (μ) needed to compute the Reynolds (Re) and Prandtl (Pr) numbers as follows:

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$$\mathrm{Re} = (4 \cdot |\mathrm{mdot}|) / (\pi \cdot \mathrm{diameter} \cdot \mu)$$
$$\mathrm{Pr} = \mathrm{cp} \cdot \mu / k$$

4. FORM controls the type of formula used in determination of the forced convection film coefficient h. There are two cases:
   - If FORM = 0, 10, or 20 than $h = \mathrm{coef} \cdot \mathrm{Re}^{EXPR} \cdot \mathrm{Pr}^{EXPP}$.
   - If FORM = 1, 11, or 21 then the above h is multiplied by k and divided by the average hydraulic diameter.
   - FORM also specifies the reference temperature used in calculating material properties for the fluid if FLMND = 0.
   - If FORM = 0 or 1, the reference temperature is the average of element grid point temperatures (average) and the ambient point temperature (average).
   - If FORM = 10 or 11, the reference temperature is the surface temperature (average of element grid point temperatures).
   - If FORM = 20 or 21, the reference temperature is the ambient temperature (average of ambient point temperature).
5. In the above expression, EXPP is EXPPI or EXPPO, respectively, for heat flowing into or out of the working fluid. This determination is performed internally.
6. FLAG controls the convective heat transfer into the downstream point (the second point as identified on the CHBDYi statement is downstream if mdot is positive).
   - FLAG = 0, no convective flow (stationary fluid).
   - FLAG = 1, convective energy flow that is consistent with the Streamwise Upwind Petrov Galerkin (SUPG) element formulation.
7. No phase change or internal heat generation capabilities exist for this element.

Related Topics

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PDAMP

Scalar Damper Property

Specifies the damping value of a scalar damper element using defined CDAMP1 or CDAMP3 entries.

Format:

1	2	3	4	5	6	7	8	9	10
PDAMP	PID1	B1	PID2	B2	PID3	B3	PID4	B4	

Example:

PDAMP	14	2.3	2	6.1					

Fields:

Field	Contents
PIDi	Property identification number. (Integer > 0)
Bi	Force per unit velocity. (Real)

Remarks:

1. Damping values are defined directly on the CDAMP2 and CDAMP4 entries, and therefore do not require a PDAMP entry.
2. A structural viscous damper, CVISC, may also be used for geometric grid points.
3. Up to four damping properties may be defined on a single entry.
4. For a discussion of scalar elements, see “Overview of 0D (Scalar) Elements” in the *Simcenter Nastran Element Library*.

Related Topics

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PDAMP5

Scalar Damper Property for CDAMP5

Defines the damping multiplier and references the material properties for damping. CDAMP5 is intended for heat transfer analysis only.

Format:

1	2	3	4	5	6	7	8	9	10
PDAMP5	PID	MID	B						

Example:

PDAMP5	2	3	4.0						

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number of a MAT4 or MAT5 entry. (Integer > 0)
B	Damping multiplier. (Real > 0.0)

Remarks:

1. B is the mass that multiplies the heat capacity CP on the MAT4 or MAT5 entry.

Related Topics

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PDAMPT

Frequency-Dependent Damper Property

Defines the frequency-dependent properties for a PDAMP Bulk Data entry.

Format:

1	2	3	4	5	6	7	8	9	10
PDAMPT	PID1	TBID1							

Example:

PDAMPT	12	34							

Fields:

Field	Contents
PID	Property identification number that matches the identification number on a PDAMP entry. (Integer > 0)
TBID1	Identification number of a TABLEDi entry that defines the damping force per-unit velocity versus frequency relationship. (Integer ≥ 0; Default = 0)

Remarks:

1. PDAMPT may only be referenced by CDAMP1 or CDAMP3 elements in the residual structure, which do not attach to any omitted degrees-of-freedom.
2. The PDAMPT entry is ignored in all solution sequences except frequency response analysis (including within SOL 200).
3. With designed frequency dependent properties in SOL 200, currently only TABLED1 can be used, and the use of TABLED1 for this case is limited to the LINEAR, LINEAR default options for XAXIS and YAXIS. For frequency dependent properties not associated with design variables, other options and other TABLEDi can be used.

Related Topics

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PDUMi

Dummy Element Property

Defines the properties of a dummy element (1 ≤ i ≤ 9). Referenced by the CDUMi entry.

Format:

1	2	3	4	5	6	7	8	9	10
PDUMi	PID	MID	A1	A2	A3	A4	A5	A6	
	A7	-etc.-							

Example:

PDUM3	108	2	2.4	9.6	1.E4	15.		3.5	
	5		2						

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. (Integer > 0)
Aj	Additional fields. (Real or Integer)

Remarks:

1. The additional fields are defined in the user-written element subroutines.

Related Topics

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PEAKOUT

Peak Response Filtering Criteria

Specifies the filtering criteria and the DOF whose frequency responses are filtered to obtain the set of frequencies at which the peak responses occur.

Format:

1	2	3	4	5	6	7	8	9	10
PEAKOUT	ID	NPEAK	NEAR	FAR	LFREQ	HFREQ	DBREF		
	TYPE	GID1	CID1	CUTOFF1	GID2	CID2	CUTOFF2		
		GID3	CID3	CUTOFF3	.....	.....	.....		

Examples:

PEAKOUT	101	3	3.0	25.0	10.0	200.0			
	DISP	1023	2	2.3E-2	2033	1	1011		

PEAKOUT	1003	5	3.0	25.0	10.0	200.0			
	DISP	1023	2	2.3E-2	2033	1	1011		
	PRESDBA	8001	1	2001	8002	1	2002		

Fields:

Field	Contents
ID	Identification number. See Remark 1. (Integer > 0; No default)
NPEAK	Number of peak responses to select from the responses that remain after the LFREQ, HFREQ, CUTOFFi, and NEAR criteria are applied to the frequency response results. (Integer > 0; Default = 5)
NEAR	Minimum allowable frequency difference between adjacent peak responses in cycles per unit time. See Remark 2. (Real ≥ 0.0; Default = 0.0)
FAR	Frequency difference used to retain responses that are eliminated by the NPEAK criterion, but are at much different frequencies than the peak responses that are retained after applying the LFREQ, HFREQ, CUTOFFi, NEAR, and NPEAK criteria in cycles per unit time. See Remark 3. (Real > NEAR; Default = 1.0E6)
LFREQ	Frequency below which the frequency response results are excluded from the set of frequencies at which peak responses occur in cycles per unit time. See Remark 4. (Real ≥ 0.0; Default = 0.0)

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Field	Contents
HFREQ	Frequency above which the frequency response results are excluded from the set of frequencies at which peak responses occur in cycles per unit time. See Remark 4. (Real > LFREQ; Default = 1.0E6)
DBREF	Reference value for dB calculations. See Remark 5. (Real > 0.0; Default = 1.0E-20)
TYPE	Response type. See Remark 1. (Character = "DISP", "VELO", "ACCE", "PRES", "PRESDB", "PRESDBA"; No default) = "DISP" for displacement response. = "VELO" for velocity response. = "ACCE" for acceleration response. = "PRES" for pressure response. = "PRESDB" for pressure response in dB. See Remark 5. = "PRESDBA" for pressure response in dBA. See Remark 5.
GIDi	Grid point identification number. (Integer > 0; No default)
CIDi	Component identification number. (Integer: 1 thru 6 for structural grid points, 1 for fluid grid points and scalar points; No default)
CUTOFFi	Response magnitude below which frequency response results are excluded from the set of frequencies at which peak responses occur. (Real > 0.0 or integer > 0; No default) Real input defines a constant value for the cutoff. Integer input specifies the identification number of a TABLED1, TABLED2, TABLED3, or TABLED4 bulk entry that defines a frequency-dependent cutoff.

Remarks:

1. A PEAKOUT bulk entry can contain multiple types. When this occurs, the set of frequencies at which peak responses occur for each DOF is the union of the frequencies identified for each type for the DOF.

   For example, suppose that a PEAKOUT case control command has two types. If 100 Hz, 145 Hz, 176 Hz, and 232 Hz satisfy the criteria for one type for a given DOF, and 85 Hz, 133 Hz, 176 Hz, and 210 Hz satisfy the criteria for the other type for the same DOF, the set of frequencies at which peak responses occur for the DOF is 85 Hz, 100 Hz, 133 Hz, 145 Hz, 176 Hz, 210 Hz, and 232 Hz.

   Multiple PEAKOUT bulk entries with the same ID are also allowed. When this occurs, the set of frequencies at which peak responses occur for a given DOF is the union of the peak responses identified by each PEAKOUT bulk entry for the DOF.

   For example, suppose that a second PEAKOUT bulk entry with the same ID is present. If 125 Hz, 182 Hz, and 258 Hz satisfy the criteria for this PEAKOUT bulk entry for the same DOF, the set of

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frequencies at which peak responses occur for the DOF is 85 Hz, 100 Hz, 125 Hz, 133 Hz, 145 Hz, 176 Hz, 182 Hz, 210 Hz, 232 Hz, and 258 Hz.

2. Use the NEAR criterion to retain only the peak response when multiple responses are clustered over a small frequency range. The software applies the NEAR criterion after it applies the LFREQ, HFREQ, and CUTOFFi criteria to the frequency response as follows:

   a. The software identifies the lowest frequency response from those retained after the LFREQ, HFREQ, and CUTOFFi criteria are applied. Let f_1 be the frequency at which this response occurs.

   b. The software identifies the next lowest frequency response. Let f_2 be the frequency at which this response occurs.

   c. If $f_1 + \Delta f_{NEAR} < f_2$, where Δf_{NEAR} is the frequency difference defined by the value in the NEAR field, both responses are retained. Then the software repeats the process by comparing the response at f_2 to the next lowest frequency response which occurs at f_3.

   d. If $f_2 \le f_1 + \Delta f_{NEAR}$, the software excludes whichever of the two responses has the smaller magnitude response. Then the software repeats the process by comparing the response that it retains to the next lowest frequency response which occurs at f_3.

   e. The software continues the process until all the responses that are retained after applying the LFREQ, HFREQ, and CUTOFFi criteria are evaluated.

3. Use the FAR criterion to retain as peak responses those responses that are at much different frequencies than the peak responses that are retained after applying the NPEAK criterion. The software applies the FAR criterion as follows:

   a. The software creates frequency bands about each peak response that are retained after applying the NPEAK criterion. The range of the band about the i^{th} peak response is $f_i - \Delta f_{FAR} < f < f_i + \Delta f_{FAR}$, where Δf_{FAR} is the frequency difference defined by the value in the FAR field.

   b. The software retains as peak responses any responses that are eliminated because of the NPEAK criterion that lie outside the frequency bands, but are within LFREQ $\le f \le$ HFREQ. Thus, it is possible to have more frequencies in the set of frequencies at which peak responses occur for a given DOF than the number of peak responses specified in the NPEAK field.

4. If the slope of the frequency response curve is descending at LFREQ, the software treats the response at LFREQ as a possible peak response, even though it typically does not coincide with a peak in the frequency response. If the slope of the frequency response curve is ascending at HFREQ, the software treats the response at HFREQ as a possible peak response, even though it typically does not coincide with a peak in the frequency response. Thus, they are retained as peak responses if they satisfy the CUTOFFi, NEAR, and NPEAK criteria, or the FAR criterion.

5. The dB and dBA values are calculated as follows:

$$dB = 20\log_{10}\left(\frac{P}{P_{ref}}\right)$$

$$dBA = 20\log_{10}\left(\frac{P}{P_{ref}}\right) + 20\log_{10}\left(R_A\left(f\right)\right) + 2$$

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where *P* is the sound pressure level and P_{ref} is the sound pressure reference level given by the value in the DBREF field. The A-weighting function is given as follows:

$$R_A(f) = \frac{12194^2 \cdot f^4}{\left(f^2 + 20.6^2\right)\sqrt{\left(f^2 + 107.7^2\right)\left(f^2 + 737.9^2\right)}\left(f^2 + 12194^2\right)}$$

where *f* is the frequency in Hz. Thus, the unit of time must be seconds in order for the A-weighted results to be correct.

Related Topics

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PELAS

Scalar Elastic Property

Specifies the stiffness, damping coefficient, and stress coefficient of a scalar elastic (spring) element (CELAS1 or CELAS3 entry).

Format:

1	2	3	4	5	6	7	8	9	10
PELAS	PID1	K1	GE1	S1	PID2	K2	GE2	S2	

Example:

PELAS	7	4.29	0.06	7.92	27	2.17	0.0032		

Fields:

Field	Contents
PIDi	Property identification number. (Integer > 0)
Ki	Elastic property value. See Remarks 8, 9, and 10. (Real)
GEi	Structural damping coefficient (dimensionless). See Remarks 6 and 8, and for SOLs 402 and 414, see Remark 1. (Real)
Si	Stress coefficient. For SOLs 402 and 414, see Remark 2. (Real)

Remarks:

1. Be careful using negative spring values.
2. Spring values are defined directly on the CELAS2 and CELAS4 entries, and therefore do not require a PELAS entry.
3. One or two elastic spring properties may be defined on a single entry.
4. The element stress is computed by multiplying the stress coefficient S with the recovered element force.
5. For a discussion of scalar elements, see *Overview of 0D (Scalar) Elements* in the *Simcenter Nastran Element Library*.
6. If PARAM,W4 is not specified, GEi is ignored in transient analysis. See **Parameters and Parameter Applicability Tables**.

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7. To obtain the damping coefficient GE, multiply the critical damping ratio ζ by 2.0. That is, $\mathrm{GE} = 2\zeta = 2(b/b_{cr})$. See Viscous and Structural Damping in the *Simcenter Nastran Basic Dynamic Analysis User's Guide*.
8. If PELAST is used in conjunction with PELAS, the nominal values here should be Ki > 0 and/or GEi > 0, as applicable, as otherwise the corresponding PELAST entries will be ignored.
9. Rotational stiffness should be specified as moment per radian.
10. In solutions 153 and 159, the software calculates heat conduction for CELAS1 and CELAS3 elements as

    q = K x ΔT

    where K is the value entered on the PELAS entry.

Remarks related to SOLs 402 and 414:

1. If PARAM,W4 is not specified, GEi is ignored.

   If PARAM,W4 is specified, GEi is considered as equivalent viscous damping with the formulation GE/W4.
2. Si are ignored.

Related Topics

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PELAST

Frequency or Displacement Dependent Elastic Property

Defines frequency or displacement dependent properties for a PELAS Bulk Data entry.

Format:

1	2	3	4	5	6	7	8	9	10
PELAST	PID	TKID	TGEID	TKNID					

Example:

PELAST	44	38							

Fields:

Field	Contents
PID	Property identification number that matches the identification number on a PELAS entry. (Integer > 0)
TKID	Identification number of a TABLEDi entry that defines the force per unit displacement vs. frequency relationship. (Integer > 0; Default = 0)
TGEID	Identification number of a TABLEDi entry that defines the nondimensional structural damping coefficient vs. frequency relationship. (Integer > 0; Default = 0)
TKNID	Identification number of a TABLEDi entry that defines the nonlinear force vs. displacement relationship. (Integer > 0; Default = 0)

Remarks:

1. The PELAST entry may only be referenced by CELAS1 or CELAS3 elements in the residual structure which do not attach to any omitted degrees-of-freedom.
2. The nominal values defined on the PELAS entry are used for all analysis types except frequency response and nonlinear analysis.
3. When frequency dependent stiffness is included in a modal frequency response, the modes are computed using the nominal Ki values. The frequency-dependent values are used at every excitation frequency. See the parameter SDAMPUP for more options.
4. The following table summarizes the usage PELAST entry in various solution sequences.

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Field	Frequency Response	Nonlinear	Linear (Non-Frequency Response)
TKID	Used	Ignored	Ignored
TGEID	Used	Ignored	Ignored
TKNID	Ignored	Used	Ignored

5. With designed frequency dependent properties in SOL 200, currently only TABLED1 can be used, and the use of TABLED1 for this case is limited to the LINEAR, LINEAR default options for XAXIS and YAXIS. For frequency dependent properties not associated with design variables, other options and other TABLEDi can be used.

Remarks related to SOLs 401 and 402:

1. TKID and TGEID are ignored.

Related Topics

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PFAST

Define Properties for CFAST Connector

Defines properties for CFAST connector.

Format:

1	2	3	4	5	6	7	8	9	10
PFAST	PID	D	MCID	MFLAG	KT1	KT2	KT3	KR1	
	KR2	KR3	MASS	GE					

Example:

1	2	3	4	5	6	7	8	9	10
PFAST	44	.85	11		130000	54000	11500		

Fields:

Field	Contents
PID	Property identification number of a PFAST bulk data entry (Integer > 0).
D	Diameter of the virtual connector. Used to locate the virtual grids on the shell element patch. (Real > 0)
MCID	Specifies a material coordinate system in which KTi and KRi are applied. (Integer >= -1 or blank. Default = -1) See Remark 1.
MFLAG	Determines MCID is treated as absolute or relative. (Integer 0 or 1, Default = 0) When MFLAG = 0, MCID is a relative coordinate system. When MFLAG = 1, MCID is an absolute coordinate system. See Remark 1.
KTi	X,Y,Z translational stiffness values. (Real, Default = 0.0)
KRi	X,Y,Z rotational stiffness values. (Real, Default = 0.0)
MASS	Mass of connector. (Real, Default = 0.0)
GE	Structural damping coefficient (dimensionless). (Real, Default = 0.0)

Remarks:

1. The CFAST element coordinates and stiffness directions depend on how MCID and MFLAG are defined:

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A. When MCID = –1 (default), MFLAG is ignored. The element x-axis is defined from GA to GB. If zero length (GA and GB are coincident), the element x-axis is defined by the normal from shell A. Similar to the CWELD element, the element y-axis is perpendicular to the element x-axis, and is oriented relative to the closest basic coordinate axis. If two basic coordinate axis are equally close, the first of x, y, then z is taken. The element z-axis is the cross product of the element x- and y-axis. KT1, KT2, and KT3 are applied in the element x-, y-, and z- axis, respectively.

B. When MCID > 0 and MFLAG=0, the element x-axis is defined as described in “A”. The cross product of the element x-axis into the MCID y-axis determines the element z-axis. The element y-axis is the cross product of the element z- and x-axis. KT1, KT2, and KT3 are applied in the element x-, y-, and z-axis, respectively.

C. When MCID ≥ 0 and MFLAG = 1 (absolute), the element stiffness values KT1, KT2, and KT3 are applied in the material coordinates, although, the element forces are computed in the coordinates as described in “A”.

Related Topics

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PGAP

Gap Element Property

Defines the properties of the gap element (CGAP entry).

Format:

1	2	3	4	5	6	7	8	9	10
PGAP	PID	U0	F0	KA	KB	KT	MU1	MU2	
	TMAX	MAR	TRMIN						

Example:

PGAP	2	.025	2.5	1.E6		1.E6	0.25	0.25	

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
U0	Initial gap opening. See Figure 16-152. (Real; Default = 0.0)
F0	Preload. See Figure 16-152. (Real ≥ 0.0; Default = 0.0)
KA	Axial stiffness for the closed gap; i.e., $U_a - U_b > U0$. See Figure 16-152. (Real ≥ 0.0)
KB	Axial stiffness for the open gap; i.e., $U_a - U_b < U0$. See Figure 16-152. See Remark 2 . (Real ≥ 0.0; Default = 1.0E-14 · KA)
KT	Transverse stiffness when the gap is closed. See Figure 16-153. It is recommended that KT ≥ (0.1 · KA). (Real ≥ 0.0; Default = MUI · KA)
MU1	Coefficient of static friction (μ_s) for the adaptive gap element or coefficient of friction in the y transverse direction (μ_y) for the nonadaptive gap element. See Remark 3 and Figure 16-153. (Real ≥ 0.0; Default = 0.0)
MU2	For SOL 106 only, coefficient of kinetic friction (μ_k) for the adaptive gap element or coefficient of friction in the z transverse direction (μ_z) for the nonadaptive gap element. See Remark 3 and Figure 16-153. (Real ≥ 0.0 for the adaptive gap element, MU2 ≤ MU1; Default = MU1)
TMAX	For SOL 106 only, maximum allowable penetration used in the adjustment of penalty values. The positive value activates the penalty value adjustment. See Remark 4 (Real; Default = 0.0).

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Field	Contents
MAR	For SOL 106 only, maximum allowable adjustment ratio for adaptive penalty values KA and KT. See Remark 5 ($1.0 < \text{Real} < 10^6$; Default = 100.0).
TRMIN	For SOL 106 only, fraction of TMAX defining the lower bound for the allowable penetration. See Remark 6 . ($0.0 \le \text{Real} \le 1.0$; Default = 0.001)

Remarks related to SOL 106:

1. Figure 16-151, Figure 16-152, and Figure 16-153 show the gap element and the force-displacement curves used in the stiffness and force computations for the element.
2. For most contact problems, KA (penalty value) should be chosen to be three orders of magnitude higher than the stiffness of the neighboring grid points. A much larger KA value may slow convergence or cause divergence, while a much smaller KA value may result in inaccurate results. The value is adjusted as necessary if TMAX > 0.0.
3. When the gap is open, there is no transverse stiffness. When the gap is closed and there is friction, the gap has the elastic stiffness (KT) in the transverse direction until the friction force is exceeded and slippage starts to occur.
4. There are two kinds of gap elements in SOL 106: adaptive gap and nonadaptive gap. If TMAX≥0.0, the adaptive gap element is selected by the program. When TMAX=0.0, penalty values will not be adjusted, but other adaptive features will be active (i.e., the gap-induced stiffness update, gap-induced bisection, and subincremental process). The value of TMAX = -1.0 selects the nonadaptive (old) gap element. The recommended allowable penetration TMAX is about 10% of the element thickness for plates or the equivalent thickness for other elements that are connected to the gap.
5. The maximum adjustment ratio MAR is used only for the adaptive gap element. Upper and lower bounds of the adjusted penalty are defined by

$$\frac{K^{init}}{\text{MAR}} \le K \le K^{init} \cdot \text{MAR}$$

   where K^{init} is either KA or KT.
6. TRMIN is used only for the penalty value adjustment in the adaptive gap element. The lower bound for the allowable penetration is computed by TRMIN * TMAX. The penalty values are decreased if the penetration is below the lower bound.

Figure 16-151: The CGAP Element Coordinate System

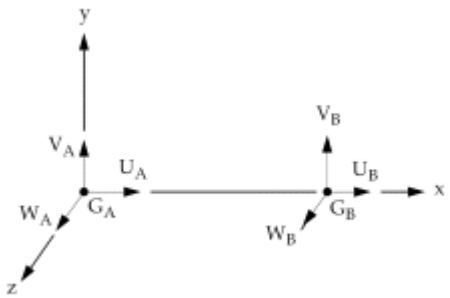

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Figure 16-152: CGAP Element Force-Deflection Curve for Nonlinear Analysis

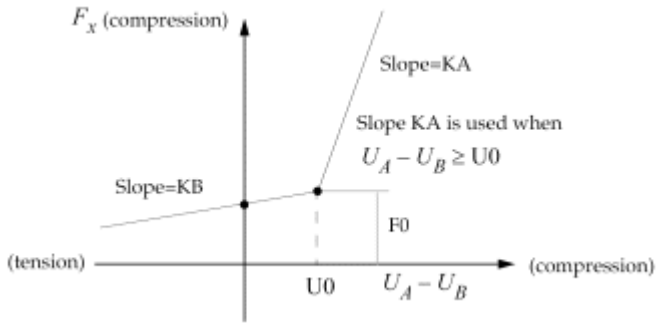
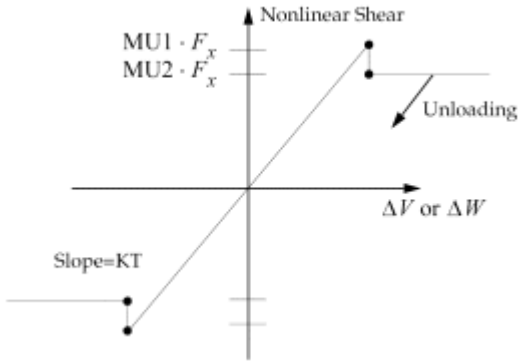


Figure 16-153: Shear Force for CGAP Element



7. If U0 is specified negative and GA and GB are not coincident, then the direction for closing must be controlled by the use of the CID field on the CGAP entry.

Remarks related to SOL 101 and consecutive solutions 103, 111 and 112:

1. The behavior of CGAP elements in SOL 101 depends on if a contact set is defined (BCSET=n case control command) and the system cell (412) setting.

- When a contact set is defined (a BCSET=n case control command exists) and system cell (412) is set to 0 (default), CGAP elements are treated as linear contact elements. You can use CGAP elements this way with active contact conditions (BCTSET bulk entry also exists), or without (no BCTSET bulk entry exists). In the case where no active contact conditions are defined, "n" on the BCSET=n case control command can point to a BCTPARM bulk entry which optionally defines PENN, PENT or PENTYP for the CGAP/linear contact elements, or to nothing if a BCTPARM bulk entry does not exist (an integer value for "n" is still required in this case).

  When the CGAP elements are treated as linear contact elements, the CGAP element will use the U0, MU1, KA, and KT fields on the PGAP bulk entry. The fields F0, MU2, TMAX, MAR and TRMIN are ignored.

  The following rules apply to the CGAP contact definition.

  - If KA is defined, it is used for the contact penalty factor in the normal direction.

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- If KA is undefined and KB is defined, KB is used for the contact penalty factor in the normal direction.
- If KA and KB are both undefined, the PENN field on the associated BCTPARM is used for the contact penalty factor in the normal direction. The automatic penalty factor calculation used for surface-to-surface contact does not apply to CGAP elements.
- If KA, KB, and PENN are all undefined, the default value of PENN is used for the contact penalty factor in the normal direction. PENTYP=2 should not be used in this case.
- If KT is undefined, the contact penalty factor in the transverse direction is calculated from the PENT and PENTYP fields on the associated BCTPARM.
- When NASTRAN SYSTEM(464) = 1, KA and KB are ignored, and PENN and PENT defined on the BCTPARM bulk entry are used for the penalty factor in the normal and transverse directions.
- MU1 is used as the coefficient of friction in both transverse directions.

- When a contact set is not defined (no BCSET=n command exists), or if a contact set is defined (BCSET=n case control command exists) but system cell (412) is set to 1, CGAP elements behave as a linear spring element. The software assigns either the value of KA or KB to the spring stiffness depending on the value entered in the U0 field (initial gap) on the PGAP bulk entry.
  - If the U0 field is defined as 0.0 (default) or a negative value, the software assigns KA to the spring stiffness.
  - If the U0 field is defined as a positive value, the software assigns KB to the spring stiffness.

Remarks related to SOL 402:

1. KT, MU1, MU2, TMAX, MAR and TRMIN are not supported. To model friction, use contact elements instead of CGAP elements.

Related Topics

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PGEAR

Gear Property Definition

Defines the properties of a geared connection between rotors in SOL 414.

Format:

1	2	3	4	5	6	7	8	9	10
PGEAR	PID	NMOD	ZA	ZB	ANGPRE	KR			
		CONA	CONB	HELA	HELB				

Example:

PGEAR	1	10.0	75	25	14.5	0.0			
		0.0	0.0	20.0	20.0				

Fields:

Field	Contents
PID	Property identification number that is referenced by a CGEAR entry. (Integer > 0)
NMOD	Module of the gears. See Remark 1. (Real > 0.0; No default)
ZA	Number of teeth on gear A. (Integer > 0; No default)
ZB	Number of teeth on gear B. (Integer > 0; No default)
ANGPRE	Pressure angle in degrees. (Real > 0.0)
KR	Mesh stiffness coefficient in force per unit length. See Remark 3. (Real > 0.0; No default)
CONA	Pitch cone angle for gear A in degrees. See Remarks 4 and 5. (Real; Default = 0.0)
CONB	Pitch cone angle for gear B in degrees. See Remarks 4 and 5. (Real; Default = 0.0)
HELA	Helix angle for gear A in degrees. See Remarks 5 and 6. (Real; Default = 0.0)
HELB	Helix angle for gear B in degrees. See Remarks 5 and 6. (Real; Default = 0.0)

Remarks:

1. Module is the pitch diameter of the gear divided by the number of teeth.
2. With the CGEAR and PGEAR bulk entries, you can model spur gears, helical gears, and internal gears. For all gear types, you are limited to fixed axes of rotation for the gears. Thus, if you model an

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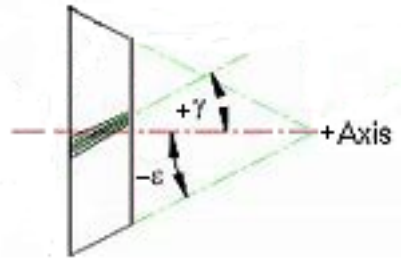
epicyclic gear train, the carrier of the planet gears must be fixed. See Remark 4 for information on how to model internal gears.

3. The mesh stiffness field is optional. The software considers the gears as rigid if the mesh stiffness field is left blank.

4. To model internal gears, specify 0.0 degrees as the pitch cone angle of the pinion and 180.0 degrees as the pitch cone angle of the internal gear.

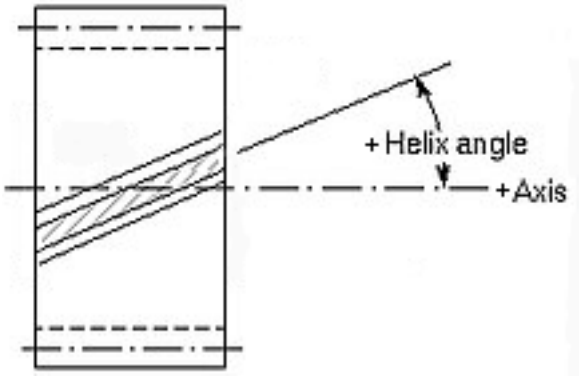
5. Figure 16-154 depicts the negative sense of the pitch cone angle, ϵ, and the positive sense of the helix angle for a bevel gear, γ, relative to the positive sense of the axis of rotation.

Figure 16-154: Negative Sense of Pitch Cone Angle and Positive Sense of Helix Angle for Bevel Gears



6. Figure 16-155 depicts the positive sense of the helix angle for a helical gear relative to the positive sense of the axis of rotation.

Figure 16-155: Positive Sense of Helix Angle for Helical Gears



7. The angular speed of rotor B is related to the angular speed of rotor A as follows:

$$\omega_B = \left(\frac{Z_A}{Z_B}\right)\omega_A$$

The pitch radii for the gears are as follows:

$$r_A = \frac{m \cdot Z_A}{2\left|\cos\beta_A\right|} \; r_B = \frac{m \cdot Z_B}{2\left|\cos\beta_B\right|}$$

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where:

Z_A	Number of teeth on gear A
Z_B	Number of teeth on gear B
m	Module of the gears
β_A	Pitch cone angle for gear A
β_B	Pitch cone angle for gear B

Related Topics

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PGPLSN

Generalized Plane Strain Element Property for SOL 401

Defines the properties of generalized plane strain elements for SOL 401.

Format:

1	2	3	4	5	6	7	8	9	10
PGPLSN	PID	MID	CGID	T					
	KN	KR1	KR2						

Example:

PGPLSN	100	1	85	10.0					

Fields:

Field	Contents
PID	Property identification number. See Remarks 1 and 2. (Integer > 0)
MID	Identification number of a MAT1 or MAT11 entry. (Integer > 0; No default)
CGID	Identification number of control grid point. (Integer > 0; No default)
T	Undeformed element thickness. (Real > 0.0; No default)
KN	Optional user-specified additive normal stiffness relative to the planar area defined by the mesh of generalized plane strain elements. (Real ≥ 0.0 or Integer > 0; Default = 0.0) If real entry, value of stiffness at all times. If integer entry, identification number of a TABLEDi entry that contains value of stiffness as a function of time.
KRi	Optional user-specified additive rotational stiffness in the units moment/radian about the i^{th}-axis of the displacement coordinate system for the control grid point. See Remark 3. (Real ≥ 0.0 or Integer > 0; Default = 0.0) If real entry, value of stiffness at all times. If integer entry, identification number of a TABLEDi entry that contains value of stiffness as a function of time.

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Remarks:

1. All PGPLSN property bulk entries must have unique identification numbers with respect to all other property bulk entries.
2. PGPLSN bulk entries can only be referenced by CPLSTN3, CPLSTN4, CPLSTN6, and CPLSTN8 element bulk entries for SOL 401. When these elements reference a PGPLSN bulk entry, their generalized plane strain formulation is used.
3. Any planar area contains a pair of principal axes. The values for the optional user-specified additive rotational stiffness should be defined about axes that pass through the control point and are parallel to the principal axes of the planar area defined by the mesh of generalized plane strain elements. To do this, specify a displacement coordinate system for the control grid point and define two of the axes for the displacement coordinate system as parallel to the principal axes of the planar area as follows:
   - If the axisymmetric plane is the XY-plane of the basic coordinate system, define the X- and Y-axes of the displacement coordinate system as parallel to the principal axes of the planar area.
   - If the axisymmetric plane is the XZ-plane of the basic coordinate system, define the X- and Z-axes of the displacement coordinate system as parallel to the principal axes of the planar area.
4. The generalized plane strain element is not supported in a glue or contact region.

Related Topics

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PHBDY

CHBDYP Geometric Element Definition

A property entry referenced by CHBDYP entries to give auxiliary geometric information for boundary condition surface elements.

Format:

1	2	3	4	5	6	7	8	9	10
PHBDY	PID	AF	D1	D2					

Example:

PHBDY	2	.02	1.0	1.0					

Fields:

Field	Contents
PID	Property identification number. (Unique Integer among all PHBDY entries). (Integer > 0)
AF	Area factor of the surface used only for CHBDYP element TYPE = “POINT”, TYPE = “LINE”, TYPE = “TUBE”, or TYPE = “ELCYL”. For TYPE = “TUBE”, AF is the constant thickness of the hollow tube. (Real > 0.0 or blank)
D1, D2	Diameters associated with the surface. Used with CHBDYP element TYPE = “ELCYL”, “TUBE”, and “FTUBE”. (Real > 0.0 or blank; Default for D2 = D1)

Remarks:

1. The PHBDY entry is used with CHBDYP entries.
2. AF
   - For TYPE = “POINT” surfaces, AF is the area.
   - For TYPE = “LINE” or TYPE = “ELCYL” surfaces, AF is the effective width: area = AF · (length).
   - For TYPE = “FTUBE” and outer TYPE = “TUBE” surfaces

$$\text{area} = \pi \cdot \left(\frac{D1 + D2}{2}\right) \cdot \sqrt{(LGTH)^2 + \left(\frac{D1 - D2}{2}\right)^2}$$

3. D1 and D2 are used only with TYPE = “ELCYL”, TYPE=“TUBE”, and TYPE=“FTUBE” surfaces.
   - For TYPE=“ELCYL” surfaces, D1 and D2 are the two diameters associated with the ellipse.

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- For TYPE="FTUBE" and outer TYPE="TUBE" surfaces, D1 and D2 are the diameters associated with the first and second grid points, respectively.
- For inner TYPE="TUBE" surfaces, the diameters are reduced by twice the thickness ($2 \cdot AF$).

Related Topics

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PJOINT

Kinematic Joint Properties (SOL 402 and SOL 414,129)

Defines properties for kinematic joints.

Format:

1	2	3	4	5	6	7	8	9	10
PJOINT	PID	TYPE	KT	KR	CF	FR0	TOL	KCF	
	LR	LX	LY	LZ	LBL	PITCH	OPT		

Example:

PJOINT	111	REVO	1.1	1.2					
	3.3								

Fields:

Field	Contents
PID	Property identification number. (Integer > 0).
TYPE	Kinematic joint type. You can define a REVOLUTE, an INLINE, a SLIDER, a SPHERE, a CYLDR, a SLIUNV, a FIXED, a SCREW, an UNIVSEL, or a CONVEL joint type.
KT	Translational stiffness. (Real > 0.0; Unit = Force / Length). (Optional).
KR	Rotational stiffness. (Real > 0.0; Unit = Force * Length / Radian). (Optional).
CF	Friction coefficient. (Real > 0.0; Unitless). (Optional).
FR0	Tightening force for friction calculation. See Remark 1. (Real > 0.0; Unit = Force). (Optional).
TOL	Regularization velocity. See Remark 1 and Remark 2. (Real > 0.0; Unit = Length / Second). (Optional).
KCF	Regularization stiffness. See Remark 1 and Remark 2. (Real > 0.0; Unit = Force / Length). (Optional).
LR	Length radius. See Remark 1 and Remark 3. (Real > 0.0; Unit = Length). (Optional).
LX, LY, LZ	Length radii. See Remark 1 and Remark 3. (Real > 0.0; Unit = Length). (Optional).
LIBL	Liberation length. (Real > 0.0; Unit = Length). (Optional).

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Field	Contents
PITCH	Pitch of the SCREW joint. (Real > 0.0; Unit = Length).
OPT	Type of identification between the DOFs of the nodes. See Remark 4. (Integer =1 or 2; Default=2).

Joint type	Joint properties
INLINE	CF, FR0, TOL, KCF, LIBL
SLIDER	KT, KR, CF, FR0, TOL, KCF, LX, LY, LZ, LIBL, OPT
CYLDR	KT, KR, CF, FR0, TOL, KCF, LX, LIBL
SLIUNV	KT, KR, CF, FR0, TOL, KCF, LT, LZ, LIBL
SCREW	KT, KR, PITCH
REVOLUTE	KT, KR, CF, FR0, TOL, KCF, LR, OPT
SPHERE	KT, CF, FR0, TOL, KCF, LR
FIXED	KT, KR
UNIVSEL	KT, KR, OPT
CONVEL	OPT

Remarks:

1. FR0, TOL, KCF, LR, LX, LY, LZ are not used if CF is omitted.
2. TOL and KCF are mutually exclusive.
3. For the SLIDER joint, you must define all three LX, LY, and LZ values; otherwise, LX, LY, and LZ will be ignored.

   For the SLIUNV joint, you must define both LY and LZ values; otherwise, LY and LZ will be ignored.
4. With OPT=1, there is a Boolean identification between the DOFs of the definition nodes.

   With OPT=2, Lagrangian multipliers are created.

   OPT=2 is the default.
5. For the REVOLUTE joint, the hinge always rotates about the X-axis on the basic or local coordinate system.

Related Topics

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PJOINT2

Kinematic Joint Additional Properties (SOL 402 and SOL 414,129)

Defines spring and/or damper properties for the kinematic joint.

Format:

1	2	3	4	5	6	7	8	9	10
PJOINT2	PID	SPR1TRA	K1T	K1TTID	SK1TTID				
		SPR1ROT	K1R	K1RTID	SK1RTID				
		DAM1TRA	B1T	B1TTID	SB1TTID				
		DAM1ROT	B1R	B1RTID	SB1RTID				

Example:

PJOINT2	111	SPR1TRA	1.1						

Fields:

Field	Contents
PID	Property identification number. See Remark 1. (Integer > 0).
SPR1TRA, SPR1ROT, DAM1TRA, DAM1ROT	Names of spring or damper properties that you can define in translation or rotation. See Remark 2. (Character).

Field	Contents
KT	Translational stiffness in unconstrained direction. (Real > 0.0; Unit = Force / Length). (Optional).
K1TTID	Table ID for force vs. displacement law. (Integer > 0). (Optional).
SK1TTID	Table ID for time dependent stiffness force scale factor. (Integer > 0). (Optional).
K1R	Rotational stiffness in unconstrained direction. (Real > 0.0; Unit = Force * Length / Radian). (Optional).
K1RTID	Table ID for torque-angular vs. displacement law. (Integer > 0). (Optional).
SK1RTID	Table ID for time dependent stiffness torque scale factor. (Integer > 0). (Optional).
B1T	Translational damping in unconstrained direction. (Real > 0.0; Unit = Force * second / Length). (Optional).

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Field	Contents
B1TTID	Table ID for force vs. velocity law. (Integer > 0). (Optional).
SB1TTID	Table ID for time dependent damping force scale factor. (Integer > 0). (Optional).
B1R	Rotational damping in unconstrained direction. (Real > 0.0; Unit = Force * second * Length / Radian). (Optional).
B1RTID	Table ID for torque-angular vs. velocity law. (Integer > 0). (Optional).
SB1RTID	Table ID for time dependent damping torque scale factor. (Integer > 0). (Optional).

Remarks:

1. The PID must be the same as the PID value of the **PJOINT** entry.
2. In a PJOINT2 entry, you can define up to four spring and damper companions.

Related Topics

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PLOAD

Static Pressure Load

Defines a uniform static pressure load on a triangular or quadrilateral surface comprised of surface elements and/or the faces of solid elements.

Format:

1	2	3	4	5	6	7	8	9	10
PLOAD	SID	P	G1	G2	G3	G4			

Example:

PLOAD	1	-4.0	16	32	11				

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
P	Pressure. (Real)
Gi	Grid point identification numbers. (Integer > 0; G4 may be zero or blank.)

Remarks:

1. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
2. The grid points define either a triangular or a quadrilateral surface to which a pressure is applied. If G4 is zero or blank, the surface is triangular.
3. In the case of a triangular surface, the assumed direction of the pressure is computed according to the right-hand rule using the sequence of grid points G1, G2, G3 illustrated in **Figure 16-156**.

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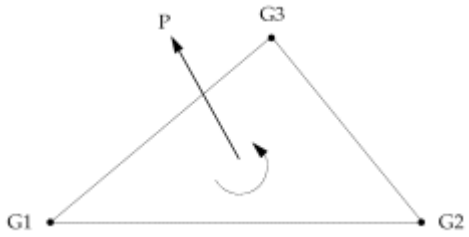
Figure 16-156: Pressure Convention for Triangular Surface of Surface Elements and/or the Faces of Solid Elements

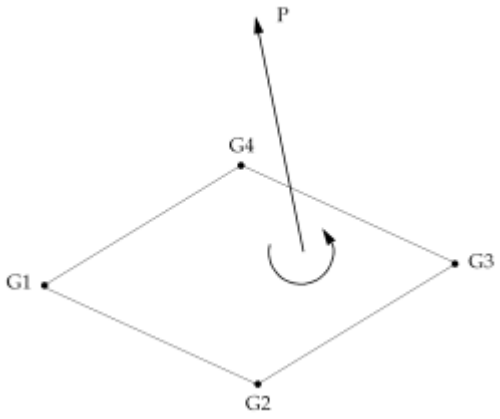


The total load on the surface (see Figure 16-157), AP, is divided into three equal parts and applied to the grid points as concentrated loads. A minus sign in field 3 reverses the direction of the load.

4. In the case of a quadrilateral surface, the grid points G1, G2, G3, and G4 should form a consecutive sequence around the perimeter. The right-hand rule is applied to find the assumed direction of the pressure. Four concentrated loads are applied to the grid points in approximately the same manner as for a triangular surface. The following specific procedures are adopted to accommodate irregular and/or warped surfaces:

- The surface is divided into two sets of overlapping triangular surfaces. Each triangular surface is bounded by two of the sides and one of the diagonals of the quadrilateral.
- One-half of the pressure is applied to each triangle, which is then treated in the manner described in Remark 2 .

Figure 16-157: Pressure Convention for Quadrilateral Surface of Surface Elements and/or the Faces of Solid Elements



5. The follower force effects due to loads from this entry are included in the stiffness in all linear solution sequences that calculate a differential stiffness. The linear solution sequences are SOLs 103, 105, 107 to 112, 115 and 116 (see also the parameter **"FOLLOWK"**). Follower force effects are included in the force balance in the nonlinear static and nonlinear transient dynamic solution sequences, SOLs 106, 129, 153, and 159, if geometric nonlinear effects are turned on with PARAM,LGDISP,1. The follower force stiffness is included in the nonlinear static solution sequences (SOLs 106 and 153) but not in the nonlinear transient dynamic solution sequences (SOLs 129 and 159).

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For SOL 401, follower force effects are included in the force balance in the nonlinear static solution if geometric nonlinear effects are turned on with PARAM,LGDISP,1. For additional information, see the NLCNTL bulk entry.

Related Topics

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PLOAD1

Applied Load on CBAR, CBEAM or CBEND Elements

Defines concentrated, uniformly distributed, or linearly distributed applied loads to the CBAR or CBEAM elements at user-chosen points along the axis. For the CBEND element, only distributed loads over an entire length may be defined.

Format:

1	2	3	4	5	6	7	8	9	10
PLOAD1	SID	EID	TYPE	SCALE	X1	P1	X2	P2	

Example:

PLOAD1	25	1065	MY	FRPR	0.2	2.5E3	0.8	3.5E3	

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
EID	CBAR, CBEAM, or CBEND element identification number. (Integer > 0)
TYPE	Load type. (Character: “FX”, “FY”, “FZ”, “FXE”, “FYE”, “FZE”, “MX”, “MY”, “MZ”, “MXE”, “MYE”, “MZE”)
SCALE	Determines scale factor for X1, X2. (Character: “LE”, “FR”, “LEPR”, “FRPR”)
X1, X2	Distances along the CBAR, CBEAM, or CBEND element axis from end A. (Real; X2 may be blank; $0 \le X1 \le X2$). See Remark 1 for SOLs 402 and 414.
P1, P2	Load factors at positions X1, X2. (Real or blank)

Remarks:

1. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
2. If $X2 \neq X1$, a linearly varying distributed load will be applied to the element between positions X1 and X2, having an intensity per unit length of bar equal to P1 at X1 and equal to P2 at X2, except as noted in Remarks 8 and 9.
3. If X2 is blank or equal to X1, a concentrated load of value P1 will be applied at position X1.

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4. If P1 = P2 and X2 ≠ X1, a uniform distributed load of intensity per unit length equal to P1 will be applied between positions X1 and X2 except as noted in Remarks 8 and 9.

5. Load TYPE is used as follows to define loads:

   "FX", "FY" or "FZ": Force in the x, y, or z direction of the basic coordinate system.

   "MX", "MY" or "MZ": Moment in the x, y, or z direction of the basic coordinate system.

   "FXE", "FYE" or "FZE": Force in the x, y, or z direction of the element's coordinate system.

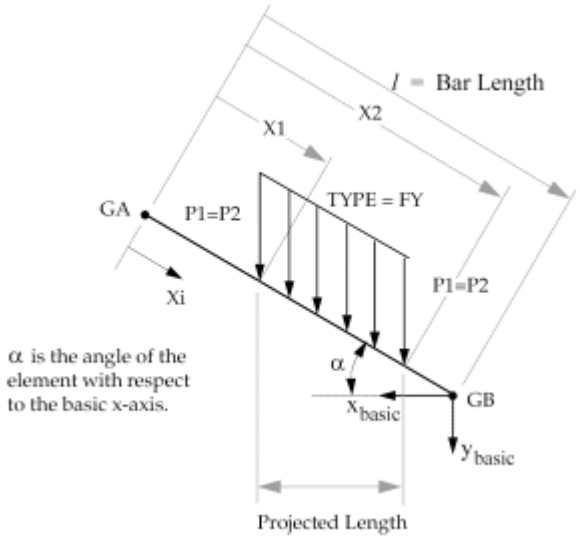
   "MXE", "MYE" or "MZE": Moment in the x, y, or z direction of the element's coordinate system.

6. If SCALE = "LE" (length), the xi values are actual distances along the element axis, and, if $X2 \neq X1$, then Pi are load intensities per unit length of the element.

7. If SCALE = "FR" (fractional), the xi values are ratios of the distance along the axis to the total length, and (if $X2 \neq X1$) Pi are load intensities per unit length of the element.

8. If SCALE = "LEPR" (length projected), the xi values are actual distances along the element axis, and (if $X2 \neq X1$) the distributed load is input in terms of the projected length of the element.

Figure 16-158: PLOAD1 Convention on Beam or Bar Elements



If SCALE = "LE", the total load applied to the bar is P1 ($X2 - X1$) in the y-basic direction.

If SCALE = "LEPR", the total load applied to the bar is P1 ($X2 - X1$)cosα in the y-basic direction.

9. If SCALE = "FRPR" (fractional projected), the Xi values are ratios of the actual distance to the length of the bar (CBAR entry), and if $X2 \neq X1$, then the distributed load is specified in terms of the projected length of the bar.

10. Element identification numbers for CBAR, CBEAM, and CBEND entries must be unique.

11. For the CBEND element, the following coordinate equivalences must be made for the element coordinates

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$$R_{elem} = X_{elem}$$
$$\theta_{elem} = Y_{elem}$$

12. Only distributed loads applied over the entire length of the CBEND element may be applied.
13. Projected loads are not applicable to the CBEND element.
14. Loads on CBEAM elements defined with PLOAD1 entries are applied along the line of the shear centers.
15. If a CBARAO or PLOAD1 entry is specified and stress and/or force output is requested, then the stresses and/or forces will be calculated at each location Xi and output as a separate line. The force and stress locations Xi=0 and Xi = *l* will always be output. This output format will be used for all beam and bar elements.
16. If on the TYPE field of the PLOAD1 entry, the element coordinate system direction (e.g. TYPE = FYE) option is selected, then the projection (i.e. SCALE = FRPR or LEPR) option is ignored and the result is the same as the SCALE = FR (or LE) option.

Remarks related to SOLs 402 and 414:

1. SOLs 402 and 414 support only distributed loads, not concentrated loads or moments. Therefore, X2 cannot be blank, and X2 cannot be equal to X1.
2. If X2 is not equal to X1, a distributed load will be applied over the entire length of the element. That is, only X1 = 0.0 and X2 = 1.0 or X2 is equal to the length of the element is allowed. This option is limited to TYPE=FX, FY, or FZ.
3. SCALE = “LE” (length) and SCALE = “FR” (fractional) are the only supported scale factors options.
4. To apply a load with a constant magnitude (with respect to time), SID is selected by the case control command LOAD = SID for both static and transient analysis.
5. To apply a time-dependent load, SID is referenced by the EXCITEID field on the TLOAD1 entry. Time-dependent loads are selected in the case control by the DLOAD case control command.

Related Topics

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PLOAD2

Uniform Normal Pressure Load on a Surface Element

Defines a uniform static pressure load applied to CQUAD4, CQUADR, CSHEAR, CTRIA3, or CTRIAR two-dimensional elements.

Format:

1	2	3	4	5	6	7	8	9	10
PLOAD2	SID	P	EID1	EID2	EID3	EID4	EID5	EID6	

Example:

PLOAD2	21	-3.6		4	16		2		

Alternate Format and Example:

PLOAD2	SID	P	EID1	“THRU”	EID2				
PLOAD2	1	30.4	16	THRU	48				

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
P	Pressure value. (Real)
EIDi	Element identification number. (Integer ≥ 0 or blank; for the “THRU” option, EID1 < EID2.)

Remarks:

1. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
2. At least one positive EID must be present on each PLOAD2 entry.
3. If the alternate form is used, all elements EID1 through EID2 must be two-dimensional.
4. The direction of the pressure is computed according to the right-hand rule using the grid point sequence specified on the element entry. Refer to the PLOAD entry.

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5. All referenced elements must exist (closed list) for residual only runs and are not required to exist (open list) for superelement runs; and they cannot be hyperelastic for either.

6. The follower force effects due to loads from this entry are included in the stiffness in all linear solution sequences that calculate a differential stiffness. The solution sequences are SOLs 103, 105, 107 to 112, 115 and 116 (see also the parameter **"FOLLOWK"**). In addition, follower force effects are included in the force balance in the nonlinear static and nonlinear transient dynamic solution sequences, SOLs 106, 129, 153, and 159, if geometric nonlinear effects are turned on with PARAM,LGDISP,1. The follower force stiffness is included in the nonlinear static solution sequences (SOLs 106 and 153) but not in the nonlinear transient dynamic solution sequences (SOLs 129 and 159).

Remarks related to SOLs 402 and 414:

1. To apply a pressure load with constant magnitude, SID is selected by Case Control command LOAD = SID for both static and transient analyses.

2. To apply a time-dependent pressure load, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by Case Control command DLOAD.

Related Topics

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PLOAD4

Pressure Load on Shell and Solid Element Faces

Defines a pressure load on a face of a CHEXA, CPENTA, CTETRA, CPYRAM, CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, or CQUADR element.

Format:

1	2	3	4	5	6	7	8	9	10
PLOAD4	SID	EID	P1	P2	P3	P4	G1	G3 or G4	
	CID	N1	N2	N3					

Example:

PLOAD4	2	1106	10.0	8.0	5.0		48		
	6	0.0	1.0	0.0					

Alternate Format and Example

(See Remark 8):

PLOAD4	SID	EID1	P1	P2	P3	P4	“THRU”	EID2	
	CID	N1	N2	N3					

PLOAD4	2	1106	10.0	8.0	5.0		THRU	1143	
	6	0.0	1.0	0.0					

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
EID EID1 EID2	Element identification number. (Integer > 0; for the “THRU” option, EID1 < EID2)
P1, P2, P3, P4	Load per unit surface area (pressure) at the corners of the face of the element. (Real or blank; Default for P2, P3, and P4 is P1.)
G1	Identification number of a grid point connected to a corner of the face. Required data for solid elements only. (Integer > 0 or blank)

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Field	Contents
G3	For CHEXA, CPYRAM, or CPENTA quadrilateral faces, G3 is the identification number of a grid point connected to a corner diagonally opposite to G1. Required for quadrilateral faces of CHEXA, CPYRAM and CPENTA elements only. For CPYRAM element triangle faces, G1 and G3 are adjacent corner nodes on the quadrilateral face, and the load is applied on the triangular face which includes those grids. For CPENTA element triangle faces, G3 must be omitted.
G4	Identification number of the CTETRA grid point located at the corner; this grid point may not reside on the face being loaded. This is required data and is used for CTETRA elements only. (Integer > 0)
CID	Coordinate system identification number. See Remark 2. (Integer ≥ 0; Default = 0)
N1, N2, N3	Components of vector measured in coordinate system defined by CID. Used to define the direction (but not the magnitude) of the load intensity. See Remark 2. (Real)

Remarks:

1. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
2. The continuation entry is optional. If fields 2, 3, 4, and 5 of the continuation entry are blank, the load is assumed to be a pressure acting normal to the face. If these fields are not blank, the load acts in the direction defined in these fields. Note that if CID is a curvilinear coordinate system, the direction of loading may vary over the surface of the element. The load intensity is the load per unit of surface area, not the load per unit of area normal to the direction of loading.
3. For the faces of solid elements, the direction of positive pressure (defaulted continuation) is inward. For triangular and quadrilateral faces, the load intensity P1 acts at grid point G1 and load intensities P2, P3, (and P4) act at the other corners in a sequence determined by applying the right-hand rule to the outward normal.
4. For plate elements, the direction of positive pressure (defaulted continuation) is in the direction of positive normal, determined by applying the right-hand rule to the sequence of connected grid points. The load intensities P1, P2, P3, (and P4) act respectively at corner points G1, G2, G3, (and G4) for triangular and quadrilateral elements. (See plate connection entries.)
5. If P2, P3, and P4 are blank fields, the load intensity is uniform and equal to P1. P4 has no meaning for a triangular face and may be left blank in this case.
6. Equivalent grid point loads are computed by linear or bilinear interpolation of load intensity followed by numerical integration using isoparametric shape functions. Note that a uniform load intensity will not necessarily result in equal equivalent grid point loads.
7. G1 and G3 are ignored for CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, and CQUADR elements.

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8. The alternate format is available only for CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, and CQUADR elements. The continuation entry may be used in the alternate format.

9. For triangular faces of CPENTA elements, G1 is an identification number of a corner grid point that is on the face being loaded and the G3 or G4 field is left blank. For faces of CTETRA elements, G1 is an identification number of a corner grid point that is on the face being loaded and G4 is an identification number of the corner grid point that is not on the face being loaded. Since a CTETRA has only four corner points, this point G4 will be unique and different for each of the four faces of a CTETRA element.

10. All referenced elements must exist (closed list) for residual only runs and are not required to exist (open list) for superelement runs; and they cannot be hyperelastic for either.

11. If fields 3 through 5 of the continuation entry are not blank, the load is assumed to have a fixed direction. If fields 2 through 5 of the continuation entry are left blank, the load is assumed to be a pressure load. In this case, follower force effects are included in the stiffness in all linear solution sequences that calculate a differential stiffness. The linear solution sequences are SOLs 103, 105, 107 to 112, 115 and 116 (see also the parameter **"FOLLOWK"**). Follower force effects are included in the force balance in the nonlinear static and nonlinear transient dynamic solution sequences, SOLs 106, 129, 153, and 159, if geometric nonlinear effects are turned on with PARAM,LGDISP,1. The follower force stiffness is included in the nonlinear static solution sequences (SOLs 106 and 153) but not in the nonlinear transient dynamic solution sequences (SOLs 129 and 159).

    For SOL 401, follower force effects are included in the force balance in the nonlinear static solution if geometric nonlinear effects are turned on with PARAM,LGDISP,1. For additional information, see the NLCNTL bulk entry.

Remarks related to SOLs 402 and 414:

1. To apply a pressure load with constant magnitude (with respect to time), SID is selected by Case Control command LOAD = SID for both static and transient analyses.

2. To apply a time-dependent pressure load, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by Case Control command DLOAD.

Related Topics

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PLOADE1

Edge Load on Plane Strain and Plane Stress Elements

Defines a surface traction that acts along the edge of CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8, CPLSTS3, CPLSTS4, CPLSTS6, or CPLSTS8 elements. The magnitude of the surface traction can be constant or vary linearly along the edge of CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8, CPLSTS3, and CPLSTS4 elements. The magnitude of the surface traction can be constant or vary linearly or quadratically along the edge of CPLSTS6 and CPLSTS8 elements.

Format:

1	2	3	4	5	6	7	8	9	10
PLOADE1	SID	EID	PA	PB	GA	GB	THETA	PM	

Example:

PLOADE1	200	35	3.5	10.5	10	30	20.		

Fields:

Field	Contents
SID	Load set identification number. See Remark 1. (Integer > 0)
EID	Element identification number. (Integer > 0)
PA	Magnitude of the surface traction at grid point GA in force per unit area. (Real; No default)
PB	Magnitude of the surface traction at grid point GB in force per unit area. (Real; Default = PA)
GA, GB	Corner grid points. GA and GB are any two adjacent corner grid points of the element. (Integer > 0; No default)
THETA	Angle between the direction of the surface traction and inward normal to the element edge in degrees. See Remarks 2, 3, and 4. (Real; Default = 0.0)
PM	Magnitude of the surface traction at the mid-side grid point between GA and GB in force per unit area. (Real; For default behavior, see Remark 5)

Remarks:

1. In static solution sequences, the load set ID (SID) is selected by the LOAD case control command. In dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the LOADSET case control command.

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2. THETA is measured relative to the inward normal to the element edge that connects GA and GB as shown in Figures Figure 16-159, Figure 16-160, and Figure 16-161. The positive sense of rotation for THETA is about the negative Y-axis when the element lies in the XZ-plane, and is about the negative Z-axis when the element lies in the XY-plane.

Figure 16-159: Surface Traction on Linear Plane Stress (CPLSTS3, CPLSTS4) and Plane Strain (CPLSTN3, CPLSTS4) Element Edges

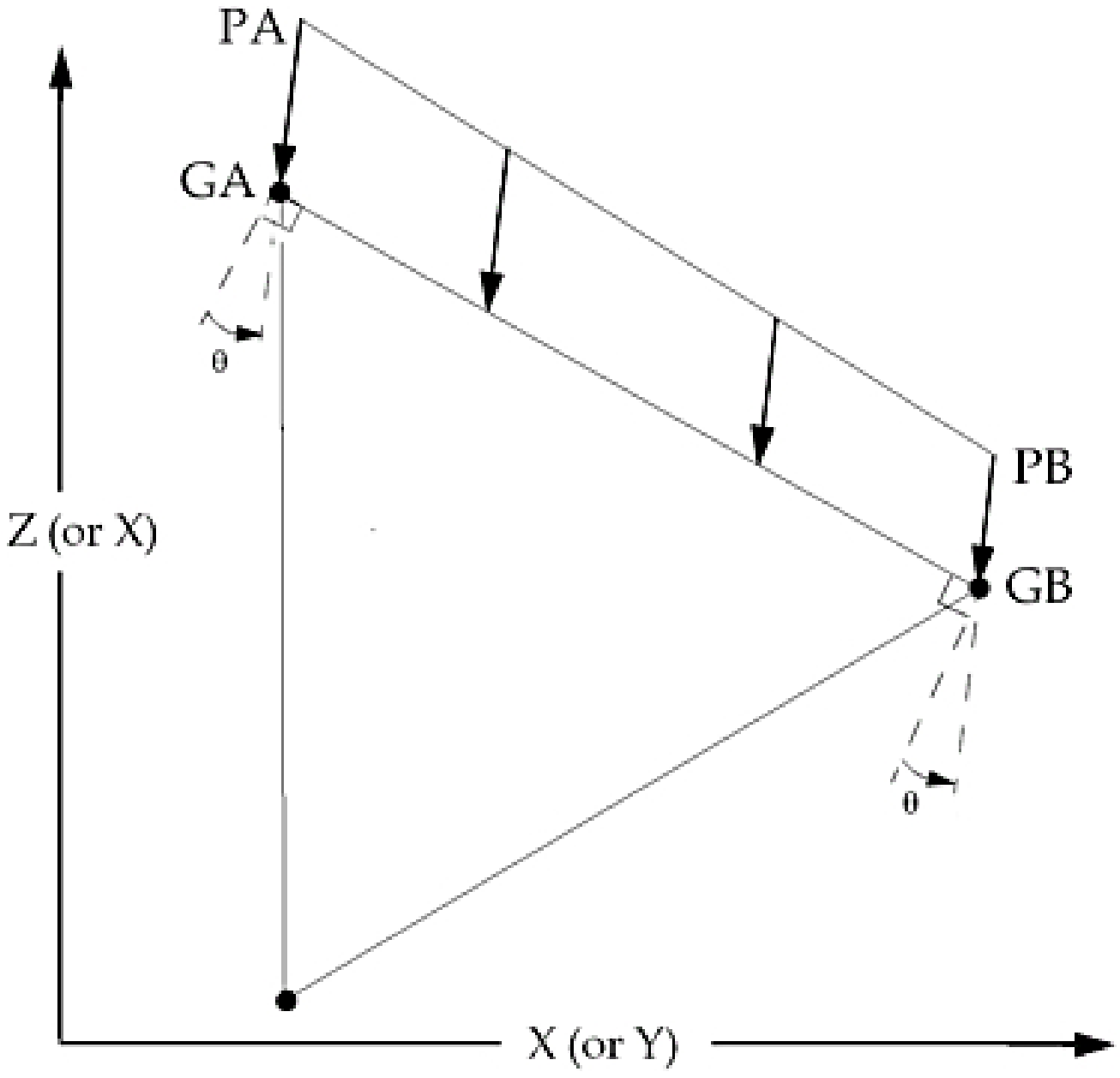

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Figure 16-160: Surface Traction on Quadratic Plane Stress (CPLSTS6, CPLSTS8) Element Edges

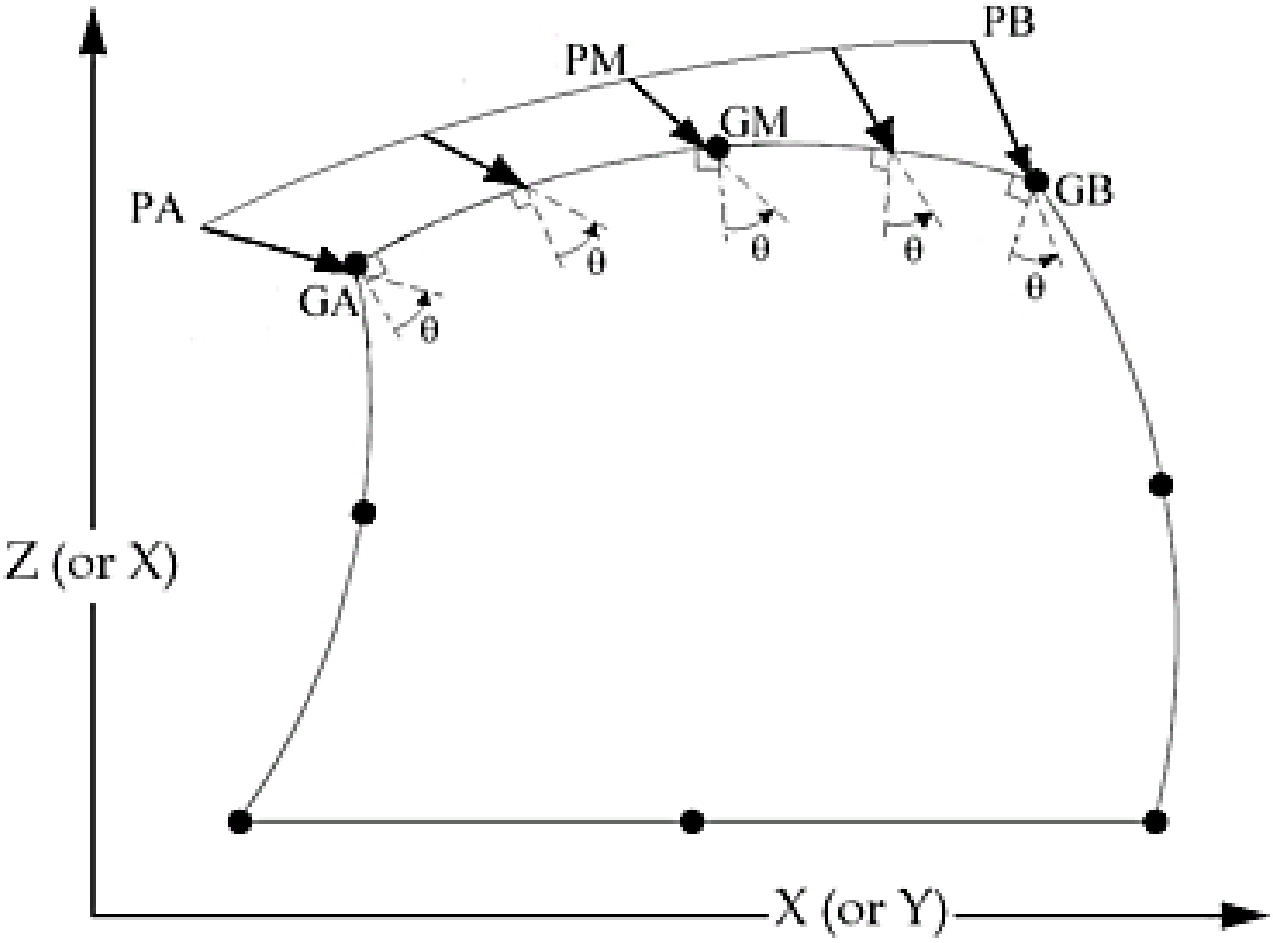

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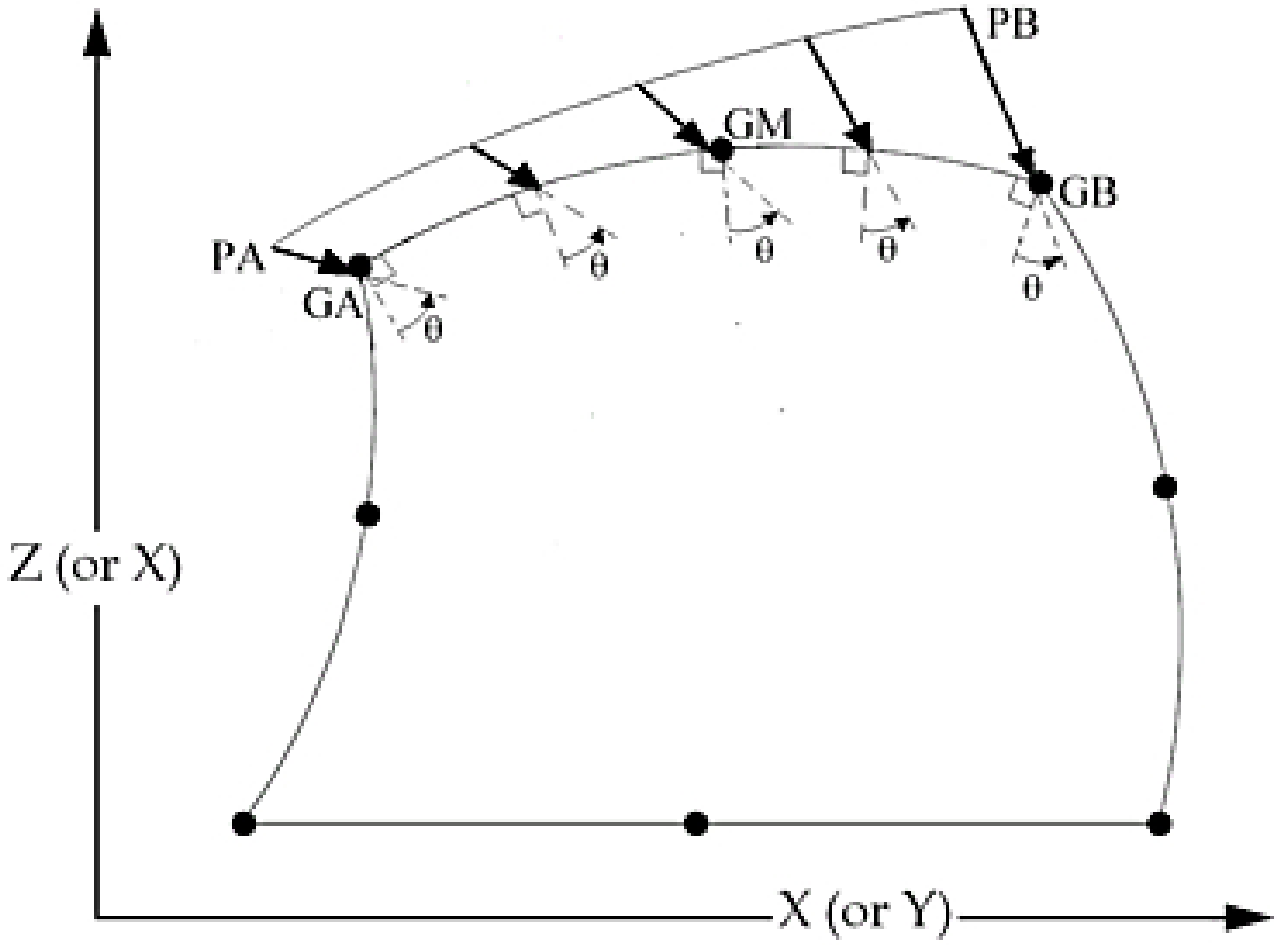
Figure 16-161: Surface Traction on Quadratic Plane Strain (CPLSTN6, CPLSTN8) Element Edges



3. Because THETA is a constant, the normal and the tangential components of the surface traction are a fixed ratio.
4. The direction in which the surface traction acts depends on THETA and the sign of the surface traction magnitude. For example, if -90.0 < THETA < 90.0, the normal component of the surface traction is directed inward when the magnitude of the surface traction is positive, and is directed outward when the magnitude of the surface traction is negative. If 90.0 < THETA < 270.0, the normal component of the surface traction is directed outward when the magnitude of the surface traction is positive, and is directed inward when the magnitude of the surface traction is negative.
5. The software obtains the equation that represents how the magnitude of the surface traction is distributed as follows:
   - For CPLSTN3, CPLSTN4, CPLSTN6, CPLSTN8, CPLSTS3, and CPLSTS4 elements, the software ignores the PM field and fits a linear equation through the surface traction magnitudes at GA and GB.
   - CPLSTS6 and CPLSTS8 elements, the software fits a quadratic equation through the surface traction magnitudes at GA, GB, and the mid-side node between GA and GB. If the PM field is undefined, the software fits a linear equation through the surface traction magnitudes at GA and GB.
6. Table 16-121 shows the default follower force behavior for each analysis type.

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Table 16-121: Follower Effects Default Behavior

Analysis Type	Follower Force	Follower Stiffness
SOLs 103, 105, 107, 108, 109, 110, 111, and 112 without STATSUB	Excluded	Excluded
SOLs 103, 105, 107, 108, 109, 110, 111, and 112 with STATSUB	Excluded	Included(1)
SOL 401 with ANALYSIS = MODES, CYCMODES, or FOURIER	Excluded	Included(2)
SOL 401 small displacement (PARAM,LGDISP,-1)	Excluded	Excluded
SOL 401 large displacement (PARAM,LGDISP,1)	Included	Excluded(2)

(1) To exclude follower stiffness, specify PARAM,FOLLOWK,NO.
(2) To override the default behavior, use the FOLLOWK parameter on the NLCNTL bulk entry.

Remarks related to SOL 402:

1. To define a surface traction as time-independent, use LOAD = SID in the case control.
2. To define a surface traction as time-dependent, reference the SID in the EXCITEID field of a TLOAD1 entry and include a DLOAD case control command that references the TLOAD1 entry.
3. THETA is ignored.
4. PM is ignored.

Related Topics

Browse Simcenter Nastran documentation

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PLOADFP

Fluid penetrating pressure load for SOLs 401 and 402

Defines a fluid penetrating pressure load for SOLs 401 and 402.

Format:

1	2	3	4	5	6	7	8	9	10
PLOADFP	SID	PRESSURE	WETID	GRSTART	CPCRIT	OPT			

Example:

PLOADFP	10	1.0E3	101	102	0.1				

Fields:

Field	Contents
SID	Unique load set identification number. (Integer > 0)
PRESSURE	Normal pressure value. (Real)
WETID	Region ID of a BSURF, BSURFS, BEDGE, BCPROP, BCPROPS containing element edges and faces that can potentially receive the fluid pressure. (Integer>0)
GRSTART	ID of a GROUP bulk entry listing starting grid point IDs. See remarks for default behavior. (Integer>0)
CPCRIT	Critical contact pressure. See Remark 11. (Real≥0.0; Default=1.0E-6)
OPT	Wet region definition. SOL 402 only. See Remark 3. (Integer) 1: Contact based method. The source and target side contact pressures are used to determine the application of the fluid pressure. 2: Topology based method. Only the source side contact pressures are used to determine the application of the fluid pressure. The locations receiving the fluid pressure on the source are then projected and applied to the target side DOF as an equivalent load.

Remarks:

1. See "Fluid penetrating pressure load" under the Loads chapter in the *Multi-Step Nonlinear User's Guide* for a more complete description of the Fluid penetrating pressure load.
2. A fluid pressure load is applied when both of the following are satisfied.

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- The element edges or faces you select to potentially receive the fluid pressure must have a contact pressure lower than a critical contact pressure you define, or they should not be part of any contact set.
- The fluid pressure application must have an open traceable path back to at least one starting grid point. That is, at least one of the grid points connected to the element edge or face must either be a starting grid point, or it must be connected to an element in which the fluid pressure is active.

  The software uses only corner grid points for the starting locations, and ignores any midside grid points you select in the GRSTART field. As a result, your group must include at least one corner grid point.

3. For element edges and for the faces of solid elements, the direction of positive pressure is inward.

   For shell elements, the direction of the positive pressure is in the direction of the positive normal, which is determined by applying the right-hand rule to the sequence of the connected grid points.

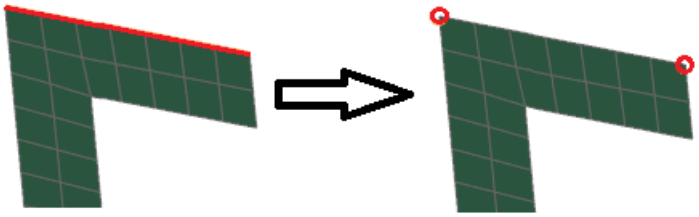
4. You can define the PLOADFP entry as a time-unassigned load, which you select using the LOAD=n case control command, or as a time-assigned load, which you select using the DLOAD=n case control command.

   The LVAR parameter on the NLCNTL bulk entry is available to request whether the software should ramp time-unassigned loads (default) by the number of increments (NINC) defined on the TSTEP1 bulk entry for the subcase. The ramping helps convergence by reducing the load increments. You can optionally turn the ramping off by setting LVAR=STEP on the NLCNTL bulk entry.

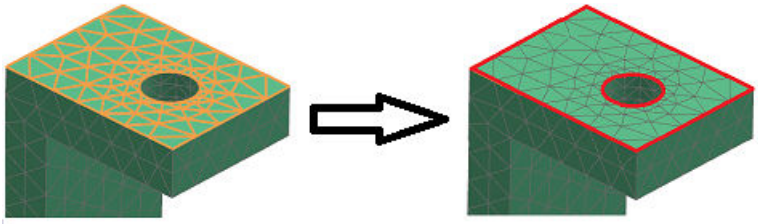
5. You can define multiple PLOADFP entries with the same ID. This allows you to define a PLOADFP for multiple regions, and then reference all with a single load set ID. See "Loads" in the *Multi-Step Nonlinear User's Guide* for details.

6. If you have not selected starting grid points in the GRSTART field, the software determines the starting grids automatically using the grid points on the free boundaries of your region. The following two figures demonstrate the default start boundary for 2D and 3D regions.

2D region default starting boundary



3D region default start boundary



7. For SOL 401, when the software determines whether an element edge or face receives the fluid pressure, it does not consider the region itself. That is, it does not matter if the actual region is

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selected by a contact set. The software is only considering the element edge or face itself, and whether that edge or face has a contact pressure that is lower than the critical contact pressure. This is helpful because, when you define your fluid pressure, you can select element edges or faces in a potentially wetted region regardless of how you define your contact regions, and vice versa.

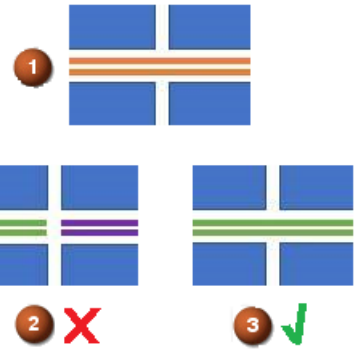
See the SOL 402 remarks below.

8. For SOL 401, the parameter FPPUPDT is available on the NLCNTL bulk entry to control when the software updates the fluid pressure load as a result of the changing contact status. By default, the software never applies the fluid pressure during the first time step, including on element edges and faces that are not in a contact region. For more information, see the NLCNTL bulk entry and "Fluid penetrating pressure load" in the Multi-Step Nonlinear User's Guide.
9. The OPRESS case control command includes the FPP describer to request the fluid pressure output for the 2D and 3D regions referenced by the PLOADFP entry.
10. Fluid pressure penetration is not supported in a Simcenter 3D Multiphysics solution.
11. The value of CPCRIT corresponds to the maximum contact pressure below which the contact is considered open. It is an absolute value in pressure units and not a scaling factor. CPCRIT is ignored by element edges and faces that are not included in a contact region; however, they still must be in the traceable path to receive the fluid pressure.

Remarks related to SOL 402:

1. If the element regions you reference with the PLOADFP entry are also included in contact regions, SOL 402 requires that the contact definition should be defined with only a single contact pair on one BCTSET bulk entry.

   For example, in the image below, 1 represents the fluid pressure regions referenced by the PLOADFP, 2 represents an incorrect contact definition in which two contact pairs are defined (four contact regions), and 3 represents the correct contact definition in which one contact pair is defined (two contact regions).



2. When you define multiple PLOADFP bulk entries that reference different regions in the WETID field, you cannot include the same element edge or face in multiple regions.

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- To vary the PLOADFP settings by region in the same subcase, you must define multiple PLOADFP entries that reference unique WETID regions that do not share any element edge or face in common.
- To vary the PLOADFP settings for the same regions from one subcase to the next, you can reference the same region IDs in new PLOADFP entries defined in each subcase.

3. With the default wet region definition (OPT=1), when the meshes on the contact source and target are not matching, the contact status and the resulting fluid pressure area may not be exactly the same on the source and target. This difference in the fluid pressure area is typically small, but when the pressure is large, this difference becomes more significant.

   You can use the topology-based option (OPT=2) to request for the software to apply the same fluid pressure on the source and target. The topology option uses only the source side contact pressures to determine the application of the fluid pressure. The locations receiving the fluid pressure on the source are then projected and applied to the target side DOF as an equivalent load.

   With the topology-based option, the software does a mapping of source faces to target face candidates at the start of the solution using the undeformed geometry. Then it uses this initial set of face candidates at each iteration for the fluid pressure projection. Because the set of face candidates is not updated again, the topology option should not be used if significant sliding occurs between the source and target regions.

Related Topics

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PLOADX1

Pressure Load on Axisymmetric and Chocking Elements

Defines a surface traction that acts along the edge of CTRAX3, CTRAX6, CQUADX4, or CQUADX8 axisymmetric elements, and CCHOCKi chocking elements. The magnitude of the surface traction can be constant or vary linearly along the edge.

Format:

1	2	3	4	5	6	7	8	9	10
PLOADX1	SID	EID	PA	PB	GA	GB	THETA		

Example:

PLOADX1	200	35	3.5	10.5	10	30	20.		

Fields:

Field	Contents
SID	Load set identification number. See Remark 1. (Integer > 0)
EID	Element identification number. (Integer > 0)
PA	Magnitude of the surface traction at grid point GA in force per unit area. (Real; No default)
PB	Magnitude of the surface traction at grid point GB in force per unit area. (Real; Default = PA)
GA, GB	Corner grid points. GA and GB are any two adjacent corner grid points of the element. (Integer > 0; No default)
THETA	Angle between the direction of the surface traction and inward normal to the element edge in degrees. See Remarks 4. (Real; Default = 0.0)

Remarks:

1. In the static solution sequences, the load set ID (SID) is selected by the LOAD case control command. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the LOADSET case control command.
2. The software obtains the equation that represents how the magnitude of the surface traction is distributed by fitting a linear equation through the surface traction magnitudes at GA and GB.

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3. For chocking elements, the software uses the circumference minus the product of the gap size and number of gaps when it calculates the forces that act at the grid points that lie on the edge defined by GA and GB.

4. THETA is measured relative to the inward normal to the element edge that connects GA and GB as shown in Figures Figure 16-162 and Figure 16-163. The positive sense of rotation for THETA is about the negative Y-axis when the element lies in the XZ-plane, and is about the negative Z-axis when the element lies in the XY-plane.

Figure 16-162: Surface Traction on Linear Axisymmetric (CTRAX3, CTRAX4) and Chocking (CCHOCK3, CCHOCK4) Element Edges

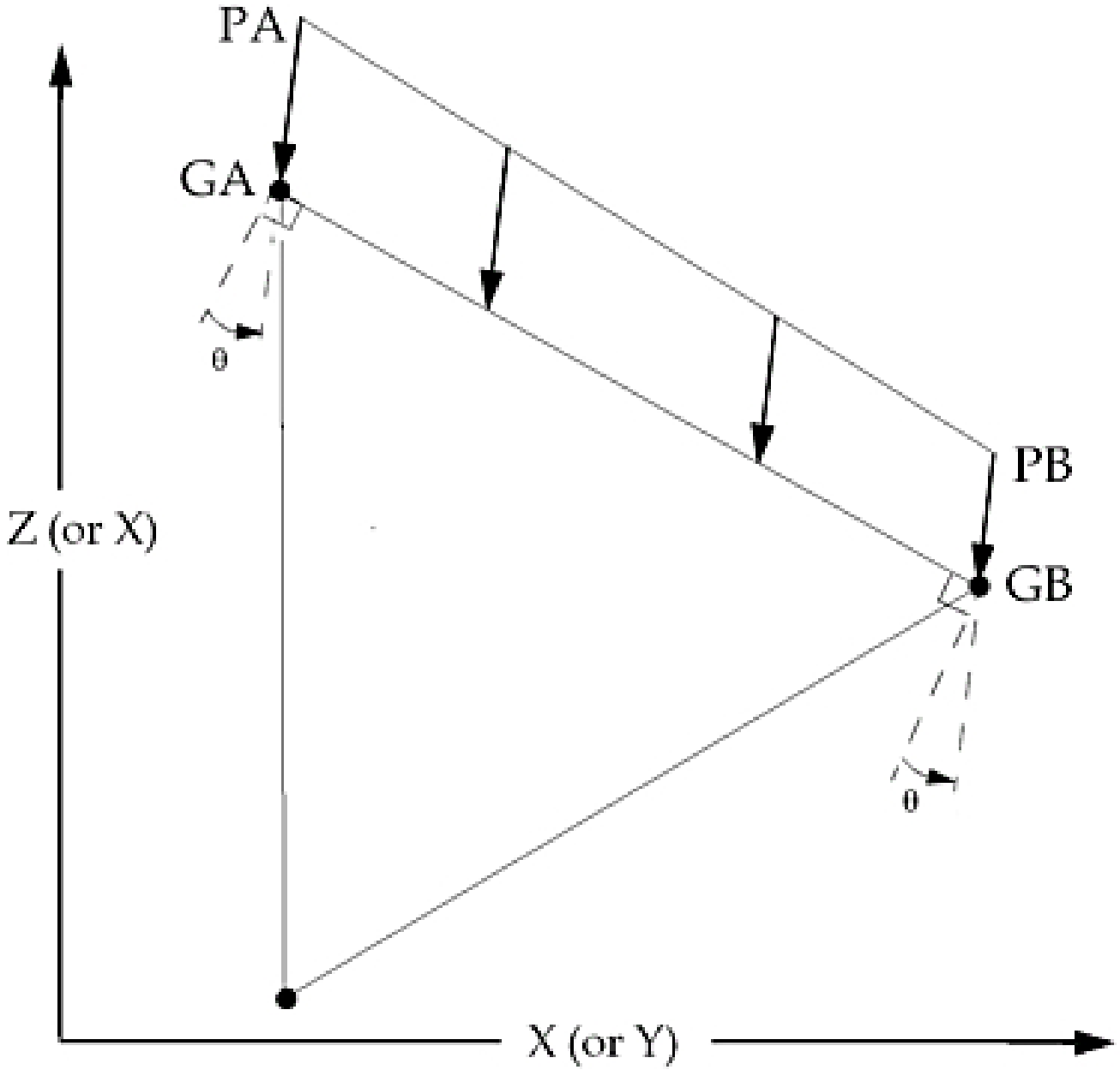

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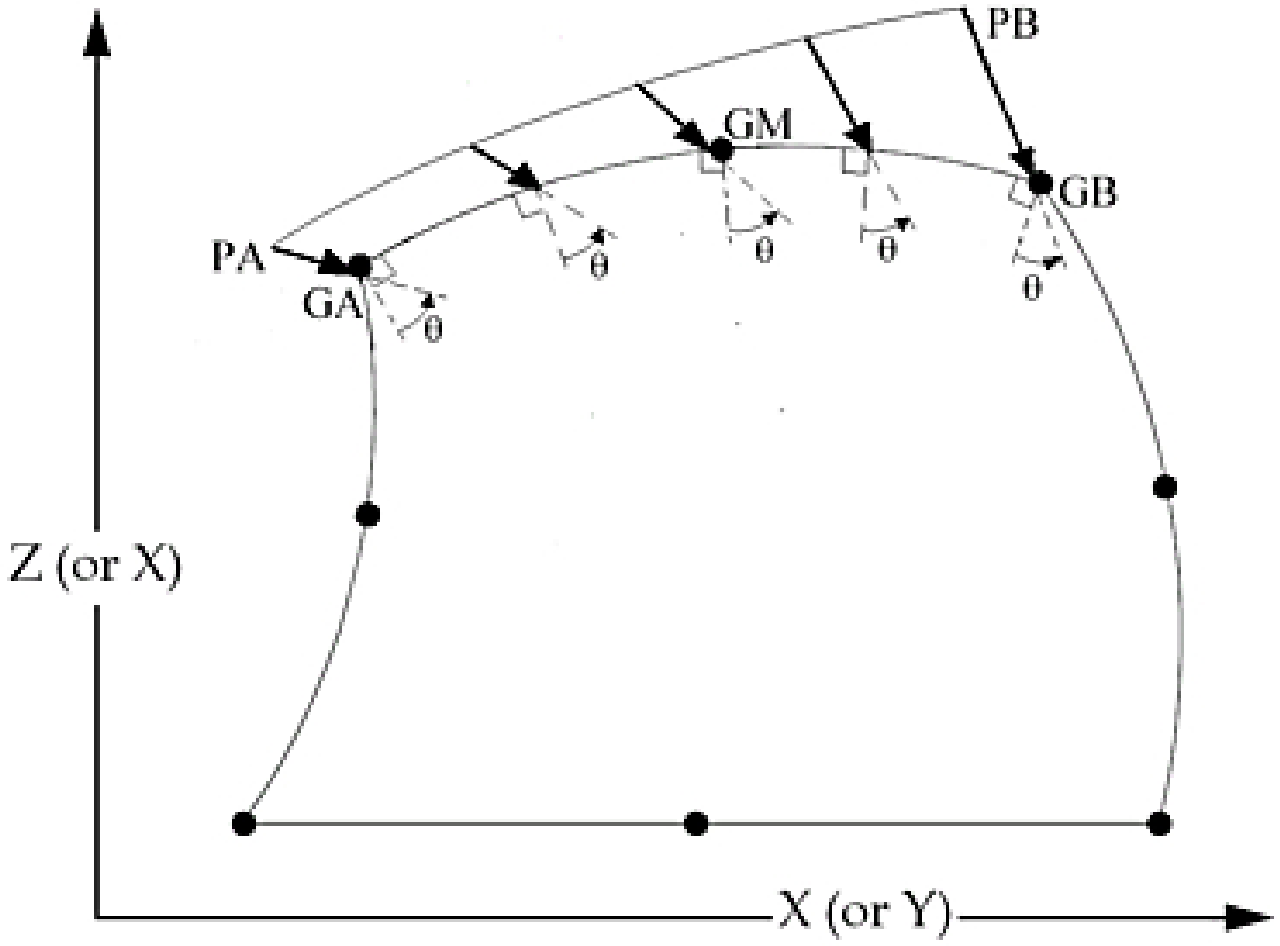
Figure 16-163: Surface Traction on Quadratic Axisymmetric (CTRAX6, CTRAX8) and Chocking (CCHOCK6, CCHOCK8) Element Edges



5. Table 16-122 shows the default follower force behavior for each analysis type.

Table 16-122: Follower Effects Default Behavior

Analysis Type	Follower Force	Follower Stiffness
SOLs 103, 105, 107, 108, 109, 110, 111, and 112 without STATSUB	Excluded	Excluded
SOLs 103, 105, 107, 108, 109, 110, 111, and 112 with STATSUB	Excluded	Included(1)
SOL 401 with ANALYSIS = MODES, CYCMODES, or FOURIER	Excluded	Included(2)
SOL 401 small displacement (PARAM,LGDISP,-1)	Excluded	Excluded

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Analysis Type	Follower Force	Follower Stiffness
SOL 401 large displacement (PARAM,LGDISP,1)	Included	Excluded(2)

(1) To exclude follower stiffness, specify PARAM,FOLLOWK,NO.
(2) To override the default behavior, use the FOLLOWK parameter on the NLCNTL bulk entry.

Remarks related to SOLs 402 and 414:

1. To apply a pressure load with constant magnitude (with respect to time), SID is selected by case control command LOAD = SID for both static and transient analysis.
2. To apply a time-dependent pressure load, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by case control command DLOAD.

Related Topics

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PLOTEL

Dummy Plot or Visualization Element

Defines a one-dimensional dummy element for use in plotting or visualization element for use in post-processing.

Format:

1	2	3	4	5	6	7	8	9	10
PLOTEL	EID	G1	G2						

Example:

PLOTEL	29	35	16						

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
G1, G2	Grid point identification numbers of connection points. (Integer > 0; G1 ≠ G2)

Remarks:

1. This element is not used in the model during any of the solution phases of a problem. It is used to simplify plotting of structures with large numbers of colinear grid points, where the plotting of each grid point along with the elements connecting them would result in a confusing plot.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. Only one PLOTEL element may be defined on a single entry.
4. In superelement analysis, PLOTELs, as well as other elements such as CBAR, CQUAD4, etc., will affect the formation of the superelement tree. The PLOTEL EIDs will also appear in the superelement map output; see the description of PARAM,SEMAPPRT in **"Parameters and Parameter Applicability Tables"**.
5. Only grid points connected by structural elements appear on structure plots. This does not include points connected only by rigid or general elements or MPCs. A plot element in parallel with elements that do not plot will cause these points to be present.

Related Topics

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PLOTEL3

Triangular Visualization Element

Defines the connectivity of a triangular visualization element with three grid points for use in post-processing.

Format:

1	2	3	4	5	6	7	8	9	10
PLOTEL3	EID	G1	G2	G3					

Example:

PLOTEL3	29	35	16	42					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer > 0; All Gi must be unique)

Remarks:

1. The element connectivity is the same as a CTRIA3 element.
2. Element identification numbers should be unique with respect to all other PLOTi bulk entries.
3. Grid point IDs G1 through G3 are mandatory.

Related Topics

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PLOTEL4

Quadrilateral Visualization Element

Defines the connectivity of a quadrilateral visualization element with four grid points for use in post-processing.

Format:

1	2	3	4	5	6	7	8	9	10
PLOTEL4	EID	G1	G2	G3	G4				

Example:

PLOTEL4	29	35	16	42	23				

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer > 0; All Gi must be unique)

Remarks:

1. The element connectivity is the same as a CQUAD4 element.
2. Element identification numbers should be unique with respect to all other PLOTi bulk entries.
3. Grid point IDs G1 through G4 are mandatory.

Related Topics

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PLOTEL6

Triangular Visualization Element

Defines the connectivity of a triangular visualization element with three to six grid points for use in post-processing.

Format:

1	2	3	4	5	6	7	8	9	10
PLOTEL6	EID	G1	G2	G3	G4	G5	G6		

Example:

PLOTEL6	29	35	16	42	23	57	39		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer > 0; All Gi must be unique)

Remarks:

1. The element connectivity is the same as a CTRIA6 element.
2. Element identification numbers should be unique with respect to all other PLOTi bulk entries.
3. Grid point IDs G1 through G3 are mandatory. Grid point IDs G4 through G6 are optional.

Related Topics

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PLOTEL8

Quadrilateral Visualization Element

Defines the connectivity of a quadrilateral visualization element with five to eight grid points for use in post-processing.

Format:

1	2	3	4	5	6	7	8	9	10
PLOTEL8	EID	G1	G2	G3	G4	G5	G6	G7	
	G8								

Example:

PLOTEL8	29	35	16	42	23	57	39	45	
	48								

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer > 0; All Gi must be unique)

Remarks:

1. The element connectivity is the same as a CQUAD8 element.
2. Element identification numbers should be unique with respect to all other PLOTi bulk entries.
3. Grid point IDs G1 through G4 are mandatory. Grid point IDs G5 through G8 are optional.

Related Topics

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PLOTHEX

Six-Sided Hexahedron Visualization Element

Defines the connectivity of a six-sided hexahedron visualization element with eight to twenty grid points for use in post-processing.

Format:

1	2	3	4	5	6	7	8	9	10
PLOTHEX	EID	G1	G2	G3	G4	G5	G6	G7	
	G8	G9	G10	G11	G12	G13	G14	G15	
	G16	G17	G18	G19	G20				

Example:

PLOTHEX	29	35	16	42	23	57	39	44	
	73	65	61	47	33	64	32	49	
	36	51	19	12	45				

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer > 0; All Gi must be unique)

Remarks:

1. The element connectivity is the same as a CHEXA element.
2. Element identification numbers should be unique with respect to all other PLOTi bulk entries.
3. Grid point IDs G1 through G8 are mandatory. Grid point IDs G9 through G20 are optional.

Related Topics

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PLOTPEN

Five-Sided Pentahedron Visualization Element

Defines the connectivity of a five-sided pentahedron visualization element with six to fifteen grid points for use in post-processing.

Format:

1	2	3	4	5	6	7	8	9	10
PLOTPEN	EID	G1	G2	G3	G4	G5	G6	G7	
	G8	G9	G10	G11	G12	G13	G14	G15	

Example:

PLOTPEN	29	35	16	42	23	57	39	44	
	73	65	61	47	33	64	32	49	

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer > 0; All Gi must be unique)

Remarks:

1. The element connectivity is the same as a CPENTA element.
2. Element identification numbers should be unique with respect to all other PLOTi bulk entries.
3. Grid point IDs G1 through G6 are mandatory. Grid point IDs G7 through G15 are optional.

Related Topics

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PLOTPYR

Five-Sided Pyramid Visualization Element

Defines the connectivity of a five-sided pyramid visualization element with five to thirteen grid points for use in post-processing.

Format:

1	2	3	4	5	6	7	8	9	10
PLOTPYR	EID	G1	G2	G3	G4	G5	G6	G7	
	G8	G9	G10	G11	G12	G13			

Example:

PLOTPYR	29	35	16	42	23	57	39	44	
	73	65	61	47	33	64			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer > 0; All Gi must be unique)

Remarks:

1. The element connectivity is the same as a CPYRAM element.
2. Element identification numbers should be unique with respect to all other PLOTi bulk entries.
3. Grid point IDs G1 through G5 are mandatory. Grid point IDs G6 through G13 are optional.

Related Topics

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PLOTTET

Four-Sided Tetrahedron Visualization Element

Defines the connectivity of a four-sided tetrahedron visualization element with four to ten grid points for use in post-processing.

Format:

1	2	3	4	5	6	7	8	9	10
PLOTTET	EID	G1	G2	G3	G4	G5	G6	G7	
	G8	G9	G10						

Example:

PLOTTET	29	35	16	42	23	57	39	44	
	73	65	61						

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
Gi	Grid point identification numbers of connection points. (Integer > 0; All Gi must be unique)

Remarks:

1. The element connectivity is the same as a CTETRA element.
2. Element identification numbers should be unique with respect to all other PLOTi bulk entries.
3. Grid point IDs G1 through G4 are mandatory. Grid point IDs G5 through G10 are optional.

Related Topics

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PLPLANE

Fully Nonlinear Plane Element Properties

Defines the properties of a fully nonlinear (i.e., large strain and large rotation) hyperelastic plane strain.

Format:

1	2	3	4	5	6	7	8	9	10
PLPLANE	PID	MID	CID	STR					

Example:

PLPLANE	203	204	201						

Fields:

Field	Contents
PID	Element property identification number. (Integer > 0)
MID	Identification number of a MATHP or MATHE entry. See Remark 4. (Integer > 0)
CID	Identification number of a coordinate system defining the plane of deformation. See Remark 3. (Integer ≥ 0; Default = 0)
STR	Location of stress and strain output. (Character: “GAUS” or “GRID”, Default = “GRID”)

Remarks:

1. All PLPLANE property entries should have unique identification numbers with respect to all other property entries.
2. PLPLANE can be referenced by CQUAD, CQUAD4, CQUAD8, CTRIA3, and CTRIA6 entries in solutions 106 and 129. The CQUAD4, CQUAD8, CTRIA3, and CTRIA6 entries are treated as hyperelastic plane strain elements.
3. The plane strain hyperelastic elements CQUAD, CQUAD4, CQUAD8, CTRIA3, CTRIA6 must lie on the x-y plane of the basic coordinate system. Stress and strain is output in the CID coordinate system for these elements.
4. For SOL 106, MID must refer to a MATHP entry.

Remarks related to SOL 402:

1. PLPLANE can be referenced by CPLSTN3, CPLSTN4, CPLSTN6, and CPLSTN8 entries.

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2. STR is ignored. Stresses and strains are output at grid points.
3. The plane strain elements must lie on the XZ-plane or on the XY-plane of the basic coordinate system.

Related Topics

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PLSOLID

Fully Nonlinear Solid Element Properties

Defines a fully nonlinear (i.e., large strain and large rotation) hyperelastic solid or axisymmetric element.

Format:

1	2	3	4	5	6	7	8	9	10
PLSOLID	PID	MID	STR						

Example:

PLSOLID	20	21							

Fields:

Field	Contents
PID	Element property identification number. (Integer > 0)
MID	Identification number of a MATHP or MATHE entry. See Remark 3. (Integer > 0)
STR	Location of stress and strain output. (Character: “GAUS” or “GRID”, Default = “GRID”)

Remarks:

1. PLSOLID can be referenced by a CHEXA, CPENTA, CTETRA, CTRAX3, CTRAX6, CQUADX4, or CQUADX8 entry.
2. Stress and strain are output in the basic coordinate system.
3. For SOL 106, MID must refer to a MATHP entry.

Remarks related to SOL 402:

1. STR is ignored. Stresses and strains are output at grid points.

Related Topics

Browse Simcenter Nastran documentation

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PMASS

Scalar Mass Property

Specifies the mass value of a scalar mass element (CMASS1 or CMASS3 entries).

Format:

1	2	3	4	5	6	7	8	9	10
PMASS	PID1	M1	PID2	M2	PID3	M3	PID4	M4	

Example:

PMASS	7	4.29	6	13.2					

Fields:

Field	Contents
PIDi	Property identification number. (Integer > 0)
Mi	Value of scalar mass. See Remark 1 for SOL 414. (Real)

Remarks:

1. Mass values are defined directly on the CMASS2 and CMASS4 entries, and therefore do not require a PMASS entry.
2. Up to four mass values may be defined by this entry.
3. For a discussion of scalar elements, see Overview of 0D (Scalar) Elements in the *Simcenter Nastran Element Library*.

Remarks related to SOL 414:

1. For SOL 414 solutions, if you set the scalar mass (Mi) parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

Browse Simcenter Nastran documentation

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PMIC

Dummy Property for Microphone Elements

Defines dummy property for microphone elements for acoustic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
PMIC	ID								

Example:

PMIC	1								

Fields:

Field	Contents
ID	Microphone element property identification number. (Integer > 0) See Remark 1.

Remarks:

1. ID is referenced by the PID of elements to designate them as microphone elements. The allowable element types are CROD, CTRIA3, CQUAD4, CHEXA, CPENTA, CTETRA, and CPYRAM. PMIC entries must have unique identification numbers with respect to all other property entries.

Related Topics

Browse Simcenter Nastran documentation

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POINT

Defines a point location

Defines a point location.

Format:

1	2	3	4	5	6	7	8	9	10
POINT	ID	CP	X1	X2	X3				

Example:

POINT	12	1	1.	2.	5.				

Fields:

Field	Contents	Type	Default
ID	Point identification number.	Integer ≥ 0	Required
CP	Identification number of coordinate system in which the location of point is defined.	Integer ≥ 0	0
X1, X2, X3	Location of the point in coordinate system CP.	Real	0.0

Remarks:

1. The point identification number must be unique with respect to all other identification numbers defined with GRID, SPOINT, EPOINT, and POINT bulk entries.
2. There are no degrees-of-freedom assigned to a point.
3. ID of POINTs must be unique with respect to ID of GRID entries.

Related Topics

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PPLANE

Properties of Plane Stress or Plane Strain Elements

Defines the properties of plane stress elements or plane strain elements.

Format:

1	2	3	4	5	6	7	8	9	10
PPLANE	PID	MID	T	NSM	FOROPT				

Examples:

PPLANE	203	204	0.11						

PPLANE	1	3	0.5	0.0	1				

Fields:

Field	Contents
PID	Element property identification number. (Integer > 0)
MID	Material identification number. (Integer >0)
T	Thickness for plane stress elements. If T is blank or zero, the thickness must be specified for Ti on the CPLSTS3, CPLSTS4, CPLSTS6, and CPLSTS8 entries. T is ignored for plane strain elements. (Real ≥ 0.0 or blank)
NSM	Nonstructural mass per unit area. (Real)
FOROPT	Formulation option number. See Remark 7. (Integer; Default = 0)

Remarks:

1. All PPLANE property entries should have unique identification numbers with respect to all other property entries.
2. The entry is referenced by the CPLSTS3, CPLSTS4, CPLSTS6, CPLSTS8, CPLSTN3, CPLSTN4, CPLSTN6, and CPLSTN8 entries via PID.
3. For structural problems, MID must reference a MAT1, MAT3, or MAT11 material property entry.
4. For heat transfer problems, MID must reference a MAT4 or MAT5 material property entry.
5. In a model where there are many plane stress elements with different thicknesses, it is recommended that the thickness values be specified in the element entries (Ti value in CPLSTS3, CPLSTS4, CPLSTS6,

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CPLSTS8) instead of assigning one PPLANE entry to one thickness value and resulting in many PPLANE entries.

6. The fields on the PPLANE cannot be associated to design variables in SOL 200.
7. The following lists the integration options (FOROPT field) for all supported solutions.

FOROPT	Linear Solutions	SOL 401 Small Strain	SOL 401 Large Strain	SOL 402[b]
Blank or 0 (default)	Bubble Function for CPLSTN4/ CPLSTS4. Standard for all other plane stress/ strain elements.	Bubble Function for CPLSTN4/ CPLSTS4. Standard for all other plane stress/ strain elements.	Bubble Function for CPLSTN4/ CPLSTS4. Standard for all other plane stress/ strain elements.	Bubble Function for CPLSTN4/ CPLSTS4. Standard for all other plane stress/ strain elements.
1	Mean Dilatational for CPLSTN4/ CPLSTN6/ CPLSTN8. Standard for all other plane stress/ strain elements.	NA[a]	NA[a]	NA[a]
2	NA[a]	NA[a]	Standard for all elements.	Standard for all elements.
3	NA[a]	NA[a]	NA[a]	U/P Formulation for all plane strain elements. Standard for all plane stress elements.

a. NA indicates not applicable. If a value is defined that is not applicable, the software uses the default value.

b. For SOL 402, the FOROPT values of 0, 2, 3 are relative to the default integration scheme (NG=0) described in SOL 402 remark 2. To select the “more integration” option (NG=3), the FOROPT choices would be 30, 32, and 33. See SOL 402 remark 2 below.

Remarks related to SOL 402:

1. NSM is ignored.
2. FOROPT is coded as: 10*NG + BUBBLE where:

   BUBBLE Indicates bubble modes.

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0= (Default), 4 bubble modes are used in CPLSTN4 and CPLSTS4.

2 = Standard formulation (no bubble mode) is used.

3 = U/P formulation is used: 1 Lagrange multiplier for element CPLSTN4 and CPLSTN3, 3 Lagrange multipliers for element CPLSTN8 and CPLSTN6.

NG Indicates the integration scheme.

0 or 2 = (Default) Less integration: 2x2 integration points for CPLSTN4, CPLSTN8, CPLSTS4, and CPLSTS8; 1 integration points for CPLSTN3 and CPLSTS3, and 3 integration points for CPLSTN6 and CPLSTS6.

3 = More integration: 3x3 integration points for CPLSTN4, CPLSTN8, CPLSTS4 and CPLSTS8; 3 integration points for CPLSTN3 and CPLSTS3; and 6 integration points for CPLSTN6 and CPLSTS6.

Related Topics

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PRAC2D

CRAC2D Element Property

Defines the properties and stress evaluation techniques to be used with the CRAC2D structural element.

Format:

1	2	3	4	5	6	7	8	9	10
PRAC2D	PID	MID	T	IPLANE	NSM	GAMMA	PHI		

Example:

PRAC2D	108	2	0.10	0	.17	.50	180.		

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. (Integer > 0)
T	Element thickness. (Real > 0.0)
IPLANE	Plane strain or plane stress option. Use 0 for plane strain; 1 for plane stress. (Integer = 0 or 1)
NSM	Nonstructural mass per unit area. (Real ≥ 0.0; Default = 0)
GAMMA	Exponent used in the displacement field. See Remark 4 . (Real; Default = 0.5)
PHI	Angle (in degrees) relative to the element x-axis along which stress intensity factors are to be calculated. See Remark 4 . (Real; Default = 180.0)

Remarks:

1. All PRAC2D property entries should have unique identification numbers with respect to all other property entries.
2. PRAC2D entry may refer to MAT1, MAT2, or MAT8 material property entries.
3. For plane strain analysis, only MAT1 type data should be used.
4. Nondefault values for GAMMA and PHI have not been tested. Therefore, the default value should be used.

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Related Topics

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PRAC3D

CRAC3D Element Property

Defines the properties of the CRAC3D structural element.

Format:

1	2	3	4	5	6	7	8	9	10
PRAC3D	PID	MID	GAMMA	PHI					

Example:

PRAC3D	108	2	.50	180.					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. (Integer > 0)
GAMMA	Exponent used in the displacement field. See Remark 3 . (Real; Default = 0.5)
PHI	Angle (in degrees) relative to the element x axis along which stress intensity factors are to be calculated. See Remark 3 . (Real; Default = 180.0)

Remarks:

1. All PRAC3D property entries should have unique identification numbers with respect to all other property entries.
2. Either isotropic (MAT1) or anisotropic (MAT9) material entries may be referenced.
3. Nondefault values for GAMMA and PHI have not been tested. Therefore, the default value should be used.

Related Topics

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PROD

Rod Property

Defines the properties of a rod element (CROD entry).

Format:

1	2	3	4	5	6	7	8	9	10
PROD	PID	MID	A	J	C	NSM			

Example:

PROD	17	23	42.6	17.92	4.2356	0.5			

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. See Remarks 2 and 3. (Integer > 0)
A	Area of the rod. (Real)
J	Torsional constant. (Real)
C	Coefficient to determine torsional stress. (Real; Default = 0.0)
NSM	Nonstructural mass per unit length. See Remark 3 for SOL 414. (Real)

Remarks:

1. PROD entries must all have unique property identification numbers.
2. For structural problems, MID must reference a MAT1 material entry.
3. For heat transfer problems, MID must reference a reference MAT4 or MAT5 entry.
4. The formula used to calculate torsional stress is

$$\tau = \frac{CM_\theta}{J}$$

where M_θ is the torsional moment.

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Remarks related to SOLs 402 and 414:

1. J and C are ignored.
2. MID can reference a nonlinear material (MATS1 bulk entry).
3. For SOL 414 solutions, if you set the nonstructural mass (NSM) parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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PSHEAR

Shear Panel Property

Defines the properties of a shear panel (CSHEAR entry).

Format:

1	2	3	4	5	6	7	8	9	10
PSHEAR	PID	MID	T	NSM	F1	F2	F3	F4	

Example:

PSHEAR	17	23	42.6	17.92	4.236	0.5			

SOL 401 example:

PSHEAR	17	23	42.6	17.92	4.236	0.5	1.1	1.2	

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number of a MAT1 entry. (Integer > 0)
T	Thickness of shear panel. (Real ≠ 0.0)
NSM	Nonstructural mass per unit area. (Real)
F1	For solutions other than SOL 401, effectiveness factor for extensional stiffness along edges 1-2 and 3-4. For SOL 401, effective extensional stiffness along edge 1-2, or edges 1-2 and 3-4 if F3 is undefined. See Remark 2 and SOL 401 Remark 1. (Real ≥ 0.0; Default = 0.0)
F2	For solutions other than SOL 401, effectiveness factor for extensional stiffness along edges 2-3 and 1-4. For SOL 401, effective extensional stiffness along edge 2-3, or edges 2-3 and 1-4 if F4 is undefined. See Remark 2 and SOL 401 Remark 1. (Real ≥ 0.0; Default = 0.0)

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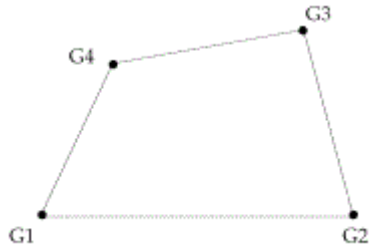
Field	Contents
F3	For SOL 401 only, effective extensional stiffness along edge 3-4. (Real ≥ 0.0; Default = F1) See SOL 401 Remark 1.
F4	For SOL 401 only, effective extensional stiffness along edge 4-1. (Real ≥ 0.0; Default = F2) See SOL 401 Remark 1.

Remarks:

1. All PSHEAR entries should have unique identification numbers with respect to all other property entries.
2. The effective extensional area is defined by means of equivalent rods on the perimeter of the element. If F1 ≤ 1.01, the areas of the rods on edges 1-2 and 3-4 are set equal to (F1 · T · PA)/(L12 + L34) where PA is the panel surface area-half the vector cross product area of the diagonals-and L12, L34 are the lengths of sides 1-2 and 3-4. Thus, if F1 = 1.0, the panel is fully effective for extension in the 1-2 direction. If F1 > 1.01, the areas of the rods on edges 1-2 and 3-4 are each set equal to 0.5 · F1 · T^2.

Figure 16-164: Extensional Area for Shear Panel



Thus, if F1 = 30, the effective width of skin contributed by the panel to the flanges on edges 1-2 and 3-4 is equal to 15T. The significance of F2 for edges 2-3 and 1-4 is similar.

3. Poisson's ratio coupling for extensional effects is ignored.

Remark for SOL 401:

1. For SOL 401, the shear panel element is equivalent to the element used in the linear solutions such as SOL 101, but with the additional option to define F3 and F4. The element resists the action of tangential forces applied to its edges and resists the action of normal forces if the effective extensional stiffness is defined. The element formulation is defined with the effective extensional area, which is equivalent to rods on the perimeter of the element. You define the effective extensional stiffness with the F1, F2, F3, and F4 fields, which the software uses to compute the effective extensional areas A_i:

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$$A_1 = \begin{cases} \frac{F_1 T A_P}{L_{12}+L_{34}} & \text{if } F_1 \le 1.01 \\ \frac{1}{2} F_1 T^2 & \text{if } F_1 > 1.01 \end{cases}$$

$$A_2 = \begin{cases} \frac{F_2 T A_P}{L_{23}+L_{41}} & \text{if } F_2 \le 1.01 \\ \frac{1}{2} F_2 T^2 & \text{if } F_2 > 1.01 \end{cases}$$

$$A_3 = \begin{cases} \frac{F_3 T A_P}{L_{12}+L_{34}} & \text{if } F_3 \le 1.01 \\ \frac{1}{2} F_3 T^2 & \text{if } F_3 > 1.01 \end{cases}$$

$$A_4 = \begin{cases} \frac{F_4 T A_P}{L_{23}+L_{41}} & \text{if } F_4 \le 1.01 \\ \frac{1}{2} F_4 T^2 & \text{if } F_4 > 1.01 \end{cases}$$

where T is the element thickness, A_p is the panel surface area defined by half the vector cross product area of the diagonals, and L_{12}, L_{34}, L_{23}, and L_{41} are the lengths of sides 1-2, 3-4, 2-3, and 4-1, respectively.

Related Topics

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PSHELL

Shell Element Property

Defines the membrane, bending, transverse shear, and coupling properties of thin shell elements.

Format:

1	2	3	4	5	6	7	8	9	10
PSHELL	PID	MID1	T	MID2	12I/T**3	MID3	TS/T	NSM	
	Z1	Z2	MID4						

Example:

PSHELL	203	204	1.90	205	1.2	206	0.8	6.32	
	+.95	-.95							

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID1	Material identification number for the membrane. (Integer ≥ 0 or blank)
T	Default membrane thickness for Ti on the connection entry. If T is blank then the thickness must be specified for Ti on the CQUAD4, CTRIA3, CQUAD8, and CTRIA6 entries. (Real or blank)
MID2	Material identification number for bending. (Integer ≥ -1 or blank)
12I/T**3	Bending moment of inertia ratio, $12I/T^3$. Ratio of the actual bending moment inertia of the shell, I, to the bending moment of inertia of a homogeneous shell, $T^3/12$. The default value is for a homogeneous shell. (Real > 0.0; Default = 1.0)
MID3	Material identification number for transverse shear. (Integer > 0 or blank; unless MID2 > 0, must be blank.)
TS/T	Transverse shear thickness ratio, T_s/T. Ratio of the shear thickness, (T_s), to the membrane thickness of the shell, T. The default value is for a homogeneous shell. (Real > 0.0; Default = .833333)
NSM	Nonstructural mass per unit area. See Remark 10 for SOL 414. (Real)

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Field	Contents
Z1, Z2	Fiber distances for stress calculations. The positive direction is determined by the right-hand rule, and the order in which the grid points are listed on the connection entry. See Remark 11 for defaults. (Real or blank)
MID4	Material identification number for membrane-bending coupling. See Remark 6. (Integer > 0 or blank, must be blank unless MID1 > 0 and MID2 > 0, may not equal MID1 or MID2.)

Remarks:

1. All PSHELL property entries should have unique identification numbers with respect to all other property entries.
2. The structural mass is calculated from the density using the membrane thickness and membrane material properties.
3. The results of leaving an MID field blank (or MID2 = -1) are:

MID1	No membrane or coupling stiffness.
MID2	No bending, coupling, or transverse shear stiffness.
MID3	No transverse shear flexibility.
MID4	No membrane-bending coupling unless ZOFFS is specified on the connection entry. See Remark 6.
MID2=-1	See Remark 12.
	Note: MID1 and MID2 must be specified if the ZOFFS field is also specified on the connection entry.

4. The continuation entry is not required.
5. The structural damping (GE on the MATi entry) is obtained from MID1 material.
6. The following should be considered when using MID4.
   - The MID4 field should be left blank if the material properties are symmetric with respect to the middle surface of the shell. If the element centerline is offset from the plane of the grid points but the material properties are symmetric, the preferred method for modeling the offset is by use of the ZOFFS field on the connection entry. Although the MID4 field may be used for this purpose, it may produce ill-conditioned stiffness matrices (negative terms on factor diagonal) if done incorrectly.
   - Only one of the options MID4 or ZOFFS should be used; if both methods are specified the effects are cumulative. Since this is probably not what the user intended, unexpected answers will result. Note that the mass properties are not modified to reflect the existence of the offset when the ZOFFS and MID4 methods are used. If the weight or mass properties of an offset plate are to be used in an analysis, the RBAR method must be used to represent the offset. See “Plate and Shell Elements” in the *Simcenter Nastran Element Library*.

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- The effects of MID4 are not considered in the calculation of differential stiffness. Therefore, it is recommended that MID4 be left blank in buckling analysis.

7. This entry is referenced by the CTRIA3, CTRIA6, CTRIAR, CQUAD4, CQUAD8, and CQUADR entries via PID.
8. For structural problems, MIDi must reference a MAT1, MAT2, or MAT8 material property entry.
9. If the transverse shear material MID3 references a MAT2 entry, then G33 must be zero. If MID3 references a MAT8 entry, then G1Z and G2Z must not be zero.
10. For heat transfer problems, MIDi must reference a MAT4 or MAT5 material property entry.
11. The default for Z1 is -T/2, and for Z2 is +T/2. T is the local plate thickness defined either by T on this entry or by membrane thicknesses at connected grid points, if they are input on connection entries.
12. For plane strain analysis, set MID2=-1 and set MID1 to reference a MAT1 entry. In-plane loads applied to plane strain elements are interpreted as line-loads with a value equal to the load divided by the thickness. Thus, if a thickness of "1.0" is used, the value of the line-load equals the load value. Pressure can be approximated with multiple line loads where the pressure value equals the line-load divided by the length between the loads.
13. A PSHELL entry with a MID1 or MID2 greater than or equal to 10^8 requires the parameter NOCOMPS to be 0 or -1 for any stress or strain recovery. See the parameter **NOCOMPS** for more information.
14. For a material nonlinear property, MID1 must reference a MATS1 entry and be the same as MID2, unless a plane strain (MID2 = -1) formulation is desired. Also, MID3 cannot reference a MATS1 entry.
15. If transverse shear flexibility is specified for a model with curved shells where the loading is dominated by twist, results will not converge and may be inaccurate. PARAM,SNORM should be set for this unique model condition.

Remarks related to SOL 401:

1. See Shell elements in the Multi-Step Nonlinear User's Guide for details.
2. Z1 and Z2 are ignored.
3. Both MID1 and MID2 must be greater than 0 but they can be different. If MID3 is blank, then MID3=MID2. MID4 is used to represent the coupling effect of material asymmetry. For the coupling effect due to geometry asymmetry, you can define the ZOFFS field on the CTRIAR, CQUADR, CTRIA6, CQUAD8, CQUAD4, or CTRIA3 entries.
4. MIDi must reference a MAT1, MAT2, MAT8, MAT9, or MAT11 entry.
5. For material nonlinear, MID1 and MID2 must be the same, MID3 is ignored, and MID4 must be blank. The global parameter NLAYERS can be used in material nonlinear analysis to define the number of integration layers through the thickness. NLAYERS can only be 3, 5, 7, or 9.
6. 12I/T**3 and TS/T are only taken into account for elastic materials.

Remarks related to SOLs 402 and 414:

1. See Shell elements in the *Multi-Step Nonlinear User's Guide* for details.

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2. 12I/T**3 and TS/T are only taken into account for elastic materials.
3. Z1 and Z2 are ignored.
4. MID1 must be specified and must be greater than 0.
5. If MID1 is a non-linear material, you must use MID1=MID2=MID3 and leave MID4 blank. Otherwise, MIDi can be any elastic material MATi and can potentially all be different from each other.
6. MID2=-1 to change a shell element into a plane strain element is not supported. The CPLSTNx or CPLSTRx elements should be used in that case instead of the shell element bulk entry.
7. Incompatible modes are not used for 4-node shell elements.
8. If MID3=0, MID2=MID3.
9. If MID2=MID3=MID4=0, the element has no rotational DOF and thus no bending stiffness. It simply has membrane stiffness. Otherwise, the element has membrane and bending stiffness.
10. For SOL 414 solutions, if you set the nonstructural mass (NSM) parameter in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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PSHL3D

3D-Shell Element Property (SOLs 402 and 414 only)

Defines the properties of 3D-shell elements.

Format:

1	2	3	4	5	6	7	8	9	10
PSHL3D	PID	MID	T						

Example:

PSHL3D	2	5	0.05						

Fields:

Field	Contents
PID	Property identification number referenced by CQUADR, CTRIAR, CQUAD8, and CTRIA6 bulk entries. (Integer > 0)
MID	Identification number of a MAT1, MATHP or MATHE entry. (Integer > 0)
T	Default shell thickness. If T is blank, then the thickness must be specified for Ti on the CQUADR, CTRIAR, CQUAD8, and CTRIA6 entries. (Real > 0.0 or blank)

Remarks:

1. PID is the property identification number referenced by CQUADR, CTRIAR, CQUAD8, and CTRIA6 bulk entries.

Related Topics

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PSHLPNL

Tension-Only Shell Element Property

Defines the properties of tension-only shell elements for SOL 401.

Format:

1	2	3	4	5	6	7	8	9	10
PSHLPNL	PID	MID1	T	MID2	12I/T**3	MID3	TS/T	NSM	
			MID4						
	SDIR	SC	F1	F2	F3	F4			

Example:

PSHLPNL	203	204	1.90	205	1.2	206	0.8	6.32	
	PSMIN	100.0	4.236	0.5					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID1	Material identification number of a MAT1 bulk entry for the membrane when the conversion behavior is on (PARAM,TENSOQD,-1 default). See Remark 16. (Integer > 0; No default)
T	Default membrane thickness for Ti on the connection entry. If T is blank, the thickness must be specified for Ti on the CQUAD4 or CQUADR entries. (Real or blank)
MID2	Material identification number for bending. (Integer > 0; No default)
12I/T**3	Bending moment of inertia ratio, $12I/T^3$. Ratio of the actual bending moment inertia of the shell, I, to the bending moment of inertia of a homogeneous shell, $T^3/12$. The default value is for a homogeneous shell. (Real > 0.0; Default = 1.0)
MID3	Material identification number for transverse shear. (Integer > 0 or blank; Default=MID2)
TS/T	Transverse shear thickness ratio, T_s/T. Ratio of the shear thickness, (T_s), to the membrane thickness of the shell, T. The default value is for a homogeneous shell. (Real > 0.0; Default = .833333)
NSM	Nonstructural mass per unit area. (Real; No default)

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Field	Contents
MID4	Material identification number for membrane-bending coupling. See Remark 10. (Integer > 0 or blank, may not equal MID1 or MID2.)
SDIR	Criteria to determine if the element behaves as a shell element or as a shear panel. (Character; Default=PSMIN)

Character input	Criteria
"PSMIN"	Minimum principal stress < SC.
"ESYSX"	σ_{xx} in the element coordinate system < SC.
"ESYSY"	σ_{yy} in the element coordinate system < SC.
"MSYSX"	σ_{xx} in the material coordinate system < SC.
"MSYSY"	σ_{yy} in the material coordinate system < SC.

SC	Stress value used in the above criteria. (Real; Default=0.0)
F1	Effective extensional stiffness along edge 1-2. (Real ≥ 0.0; Default = 1.01)
F2	Effective extensional stiffness along edge 2-3. (Real ≥ 0.0; Default = 1.01)
F3	Effective extensional stiffness along edge 3-4. (Real ≥ 0.0; Default = F1)
F4	Effective extensional stiffness along edge 4-1. (Real ≥ 0.0; Default = F2)

Remarks:

1. For SOL 401, the tension-only shell element initially behaves in shell element status. The software converts the element to shear panel status according to the conversion threshold defined in the SC and SDIR fields. When it is in shell element status, it is consistent with the SOL 401 shell element defined with the PSHELL property. When it is in shear panel status, it is equivalent to the CSHEAR element used in the linear solutions such as SOL 101 but with the additional option to define F3 and F4.

When the conversion from shell to shear panel occurs, the software does a stiffness update regardless of the KUPDATE setting you defined on the NLCNTL bulk entry. Once the software converts the element from shell to shear panel it will not revert it back to shell status, unless you created a new sequentially independent subcase. Then the element begins again in shell element status.

The conversion threshold can be satisfied at an iteration before a time step has converged, and the conversion from shell to shear panel will occur during the iterations at the current time step.

By default, the shell to shear panel conversion behavior is on. You can define the parameter setting PARAM,TENSOQD,1 to turn the conversion behavior off, and the element remains as a shell element. In this situation, the element can behave nonlinearly (material and geometric).

When the conversion behavior is on, the tension-only shell element does not consider any material or geometric nonlinear conditions. Although, the software will compute the nonlinearities for any other element types that you include with the tension-only shell element.

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Note that when the element is converted to shear panel status, it no longer includes any rotational stiffness. Any loads or constraints defined to the rotational DOF on the tension-only shell are no longer applied when the element is in shear panel status.

You can define the ESHLPNL case control command to request status and conversion time output for the tension-only shell element.

2. The PSHLPNL entry is not supported by SOLs 402 and 414. For solutions other than SOLs 401, 402, and 414 (for example, SOLs 101, 103, and 106), the software considers the PSHLPNL entry the same as a PSHELL entry (that is, it is a shell element with no conversion to shear panel status).

3. When the element is in shear panel status, it resists the action of tangential forces applied to its edges, and the action of normal forces if the effective extensional stiffness is defined. The element formulation is defined with the effective extensional area which is equivalent to rods on the perimeter of the element. You define the effective extensional stiffness with the F1, F2, F3, and F4 fields on the PSHLPNL entry, which the software uses to compute the effective extensional areas A_i:

$$A_1 = \begin{cases} \frac{F_1 T A_P}{L_{12}+L_{34}} & \text{if } F_1 \le 1.01 \\ \frac{1}{2} F_1 T^2 & \text{if } F_1 > 1.01 \end{cases}$$

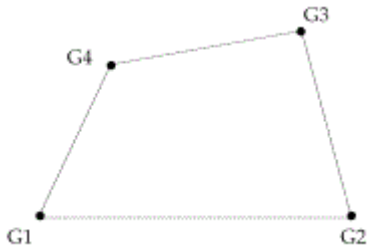
$$A_2 = \begin{cases} \frac{F_2 T A_P}{L_{23}+L_{41}} & \text{if } F_2 \le 1.01 \\ \frac{1}{2} F_2 T^2 & \text{if } F_2 > 1.01 \end{cases}$$

$$A_3 = \begin{cases} \frac{F_3 T A_P}{L_{12}+L_{34}} & \text{if } F_3 \le 1.01 \\ \frac{1}{2} F_3 T^2 & \text{if } F_3 > 1.01 \end{cases}$$

$$A_4 = \begin{cases} \frac{F_4 T A_P}{L_{23}+L_{41}} & \text{if } F_4 \le 1.01 \\ \frac{1}{2} F_4 T^2 & \text{if } F_4 > 1.01 \end{cases}$$

where T is the element thickness, A_p is the panel surface area defined by half the vector cross product area of the diagonals, and L_{12}, L_{34}, L_{23}, and L_{41} are the lengths of sides 1-2, 3-4, 2-3, and 4-1, respectively.

Figure 16-165: Extensional Area for Shear Panel



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4. When thermal loads are applied and the element is in shear panel status, the effective extensional rods will expand or contract due to the thermal load, but there is no change in shear deformation due to the thermal load.
5. When the element is in the shear panel status, the Poisson's ratio coupling for extensional effects is ignored.
6. All PSHLPNL property entries must have unique identification numbers with respect to all other property entries.
7. The structural mass is calculated as a shell element. The density for the material referenced in MID1 is used to compute the mass.
8. Both MID1 and MID2 must be greater than 0 but they can be different. If MID3 is blank, then MID3=MID2. MID4 is used to represent the coupling effect of material asymmetry. For the coupling effect due to geometry asymmetry, you can define the ZOFFS field on the CQUADR or CQUAD4 entries.
9. The software uses the structural damping you defined (GE on the MATi entry) on the material referenced on MID1.
10. Consider the following when using MID4.
    - Leave the MID4 field blank if the material properties are symmetric with respect to the middle surface of the shell. If the element centerline is offset from the plane of the grid points but the material properties are symmetric, the preferred method for modeling the offset to use the ZOFFS field on the connection entry. Although you can use the MID4 field for this purpose, it may produce ill-conditioned stiffness matrices (negative terms on factor diagonal) if done incorrectly.
    - Only one of the options MID4 or ZOFFS should be used; if both methods are specified, the effects are cumulative. Because this is probably not what you intended, unexpected answers will result. Note that the mass properties are not modified to reflect the existence of the offset when the ZOFFS and MID4 methods are used. If the weight or mass properties of an offset plate are to be used in an analysis, the RBAR method must be used to represent the offset.
    - The effects of MID4 are not considered in the calculation of differential stiffness. Therefore, it is recommended that MID4 be left blank in buckling analysis.
11. If the transverse shear material MID3 references a MAT2 entry, G33 must be zero. If MID3 references a MAT8 entry, G1Z and G2Z must not be zero.
12. If transverse shear flexibility is specified for a model with curved shells where the loading is dominated by twist, results will not converge and may be inaccurate. PARAM,SNORM should be set for this unique model condition.
13. MIDi must reference a MAT1, MAT2, MAT8, MAT9, or MAT11 entry.
14. If you turn the conversion behavior off (PARAM,TENSOQD,1) and have included material nonlinear, MID1 and MID2 must be the same, MID3 is ignored, and MID4 must be blank. The global parameter NLAYERS can be used in material nonlinear analysis to define the number of integration layers through the thickness. NLAYERS can only be 3, 5, 7, or 9.
15. 12I/T**3 and TS/T are only taken into account for elastic materials.

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16. If you set PARAM,TENSOQD,-1 (conversion behavior is on, which is the default), the element can behave as a shell element or shear panel element, and MID1 must reference an isotropic material (MAT1 bulk entry).

    If you set PARAM,TENSOQD,1 (conversion behavior is off), the element always behaves as a shell element, and MID1 can reference an isotropic material (MAT1 bulk entry) or an anisotropic material (MAT2 or MAT8, MAT9, or MAT11 bulk entry).

Related Topics

Browse Simcenter Nastran documentation

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PSOLCZ

Properties of Cohesive Elements

Defines the properties of cohesive elements (CHEXCZ and CPENTCZ entries).

Format:

1	2	3	4	5	6	7	8	9	10
PSOLCZ	PID	MID	CORDM	THICK					

Example:

PSOLCZ	1	4	3	0.75					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Identification number of a MAT1, MAT11, or MATCZ entry. See Remark 2. (Integer > 0; No default)
CORDM	Identification number of the material coordinate system. Enter “0” or leave blank to use the basic coordinate system. See Remarks 3 and 4. (Integer; Default = 0)
THICK	Thickness of cohesive element. See Remark 2. (Real > 0.0; Default = 1.0)

Remarks:

1. PSOLCZ entries should have unique identification numbers with respect to all other property entries.
2. Three stiffness values are associated with cohesive elements: K01 and K02 are transverse shear stiffness values; K03S is the out-of-plane normal stiffness value.

   The physical thickness of a cohesive element as defined by the geometric coordinates of the grids is not used to determine the stiffness of the cohesive element.

   If the PSOLCZ entry references a MAT1 entry, the software calculates the stiffness of the cohesive elements to be K01 = G / THICK, K02 = G / THICK, and K03S = E / THICK.

   If the PSOLCZ entry references a MAT11 entry, the software calculates the stiffness of the cohesive elements to be K01 = G13 / THICK, K02 = G23 / THICK, and K03S = E3 / THICK.

   If the PSOLCZ entry references a MATCZ entry, you specify the stiffness of the cohesive element directly in the K01, K02, and K03S fields of the MATCZ entry and the THICK field is ignored.

   The unit of stiffness for cohesive elements is force per unit length cubed.

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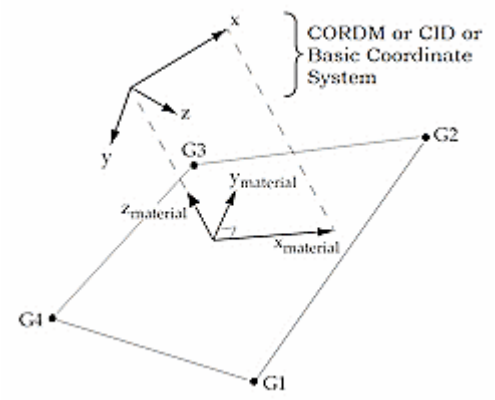
3. The Z-axis of the material coordinate system for a cohesive element is normal to the plane defined by grids G1-G2-G3-G4 for a CHEXCZ element or the plane defined by grids G1-G2-G3 for a CPENTCZ element. The positive sense of the Z-axis is the same direction as the vector obtained by crossing a vector directed from G1 to G2 with a vector directed from G1 to G4 for the CHEXCZ element or a vector directed from G1 to G3 for the CPENTCZ element.

   The X-axis of the material coordinate system for a cohesive element depends on whether a MATCID entry references the element.

   - If a MATCID entry references the element, the X-axis of the material coordinate system is the projection of the X-axis of the coordinate system listed in the CID field of the MATCID entry onto the plane defined by grids G1-G2-G3-G4 for a CHEXCZ element or the plane defined by grids G1-G2-G3 for a CPENTCZ element.
   - If a MATCID entry does not reference the element, the X-axis of the material coordinate system depends on how the CORDM field is specified.
     - If the CORDM field is specified, but not set to "0", the X-axis of the material coordinate system is the projection of the X-axis of the CORDM coordinate system onto the plane defined by grids G1-G2-G3-G4 for a CHEXCZ element or the plane defined by grids G1-G2-G3 for a CPENTCZ element.
     - If the CORDM field is blank or is set to "0", the X-axis of the material coordinate system is the projection of the X-axis of the basic coordinate system onto the plane defined by grids G1-G2-G3-G4 for a CHEXCZ element or the plane defined by grids G1-G2-G3 for a CPENTCZ element.

   The Y-axis of the material coordinate system lies in the plane defined by grids G1-G2-G3-G4 for a CHEXCZ element or the plane defined by grids G1-G2-G3 for a CPENTCZ element, and together with the X-axis and Z-axis forms a right-handed coordinate system.



   The software cannot project the X-axis of the CORDM, CID, or basic coordinate system onto the plane of a cohesive element if it is parallel to the Z-axis of the material coordinate system for the cohesive element. If they are parallel, a fatal error results.

4. If MID references a MAT11 or MATCZ entry, then CORDM defines the principal material coordinates.

5. Use the CZRESULTS case control command to request results for CHEXCZ and CPENTCZ elements. Results are output for corner grid points only.

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6. A cohesive element can use temperature-dependent materials MAT1 and MATT1, or MAT11 and MATT11. However, logarithmic interpolation is not supported if a TABLEM1 is referenced. The software does not issue a warning message and reverts to linear interpolation if a logarithmic AXIS is specified.

Related Topics

Browse Simcenter Nastran documentation

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PSOLID

Properties of Solid Elements

Defines the properties of solid elements (CHEXA, CPENTA, CPYRAM, CQUADX4, CQUADX8, CTETRA, CTRAX3, and CTRAX6 entries).

Format:

1	2	3	4	5	6	7	8	9	10
PSOLID	PID	MID	CORDM	IN	STRESS	ISOP	FCTN		

Example:

PSOLID	2	100	6	TWO	GRID	REDUCED			

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Identification number of a MAT1, MAT3, MAT4, MAT5, MAT9, MAT10, MAT10C, MATF10C, MAT11, or MATPOR entry. See Remarks 2 and 4. (Integer > 0 or blank if FCTN = "PFLUIDEX")
CORDM	Identification number of the material coordinate system. See Remarks 5, 8, and 7. (Integer; Default = 0, which is the basic coordinate system)
IN	Controls the element integration order and options for bubble function degrees-of-freedom. The control varies by element and solution type. See Remarks 9, 10, 11, and 12. (Integer, Character, or blank)
STRESS	Location selection for stress and strain output. See Remarks 7, 9, and 13. (Integer, Character, or blank)
ISOP	Controls the option for reduced shear integration. The control varies by element and solution type. See Remarks 9, 10, 11, and 12. (Integer, Character, or blank)
FCTN	Fluid element flag. See Remark 15. (Character: “PFLUID” indicates a fluid element, "PFLUIDEX" indicates a fluid element that references an external file for the fluid properties, “SMECH” indicates a structural element; Default = “SMECH”)

Remarks:

1. PSOLID entries must have unique identification numbers with respect to all other property entries.

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2. Isotropic (MAT1 or MAT4), axisymmetric solid orthotropic (MAT3), anisotropic (MAT5, MAT9), fluid (MAT10), and 3-D solid orthotropic (MAT11) material properties may be referenced. If FCTN = “PFLUID”, MID must reference a MAT10 entry. If FCTN = “PFLUIDEX”, MID should be left blank. If FCTN = “PFLUIDEX” and MID is specified, the MID specification is ignored.
3. In solutions 103, 107–112, and 200, if an axisymmetric element references a PSOLID entry, the PSOLID entry cannot reference a MAT10 entry.
4. For material nonlinear analysis using SOL 106 or SOL 129, the MATS1 entry can be used with the solid elements CHEXA, CPENTA, CPYRAM, and CTETRA only.
5. The material coordinate system (CORDM) may be the basic system (0 or blank), any defined system (Integer > 0), or the element coordinate system (-1). See the CHEXA, CPENTA, CPYRAM or CTETRA entry for the definition of the element coordinate system. The default value for CORDM is zero unless it is overridden by the NASTRAN statement with the CORDM keyword. See **The nastran Command**.

   If a MATCID bulk entry references the associated element ID, the coordinate system selected with the MATCID bulk entry becomes the material coordinate system, and the CORDM field is ignored.
6. The element coordinate system for the CQUADX4, CQUADX8, CTRAX3, and CTRAX6 elements is the basic coordinate system. The material coordinate system on these elements is defined by “TH” on the element definition, and not by CORDM.
7. Components of stress are output in the material coordinate system except:

   - Stress output labeled "NONLINEAR STRESSES", which is output in the element coordinate system. For SOL 106 geometric nonlinear analysis, the nonlinear stresses are output in deformed element coordinate system.
   - Hyperelastic element stress, which is always output in the basic coordinate system, including when in the stress output labeled "NONLINEAR STRESSES".
   - For SOL 401, 402, and 414, stress and strain are output in the material coordinate system, which defaults to the basic coordinate system if CORDM is not specified.
   - For SOL 401, an output coordinate system exception exists for the cyclic modes subcase. For the solid elements that represent a cyclic symmetry fundamental segment, stress and strain are output in the cylindrical coordinate system selected on the CYCAXIS bulk entry. This exception only applies to the stress and strain output from the cyclic modes subcase. Stress and strain output for a cyclic symmetry fundamental segment from all other subcase types is in the material coordinate system.
8. If MID references a MAT9 or MAT11 entry, CORDM defines the material property coordinate system for Cij.
9. The tables following the PSOLID remarks indicate the allowed options and combination of options. If a combination not found in the table is used, a warning message is issued and default values are assigned for all options.
10. For CHEXA and CPENTA elements with no midside grids, reduced shear integration with bubble functions (ISOP = blank or “REDUCED” and IN = blank or “BUBBLE”) is the default. This is recommended setting because it minimizes shear locking and Poisson’s ratio locking and does not cause modes of deformation that lead to no strain energy. The effects of using non-default values are as follows:

    a. IN = “THREE” or 3 produces an overly stiff element.

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b. If IN = "TWO" or 2 and the element has midside grids, modes of deformation may occur that lead to no strain energy.

c. Standard isoparametric integration (ISOP = "FULL" or 1 and IN = "TWO" or 2; or "THREE" or 3) produces an element overly stiff in shear. This type of integration is more suited to nonstructural problems.

11. IN = "BUBBLE" is not allowed for CTETRA elements or for CHEXA and CPENTA elements with midside grids.

12. For CTETRA and fluid elements (FCTN = "PFLUID"), standard isoparametric integration (ISOP = "FULL" or 1 and IN = "TWO" or 2; or "THREE" or 3) is the default and the only option available.

13. For linear solutions: Stress and strain output may be requested at the Gauss points (STRESS = "GAUSS" or 1) on the CHEXA and CPENTA elements with no midside grids, on the CTETRA and CPYRAM elements with or without midside grids, and on the CTRAX3, CTRAX6, CQUADX4, CQUADX8 elements.

For nonlinear solutions SOL 106 and SOL 129: Stress and strain output may be requested at the Gauss points on the CHEXA, CPENTA, CPYRAM and CTETRA elements with or without midside grids.

For SOL 401, stress and strain can be requested at the Gauss points on CHEXA, CPENTA, CPYRAM and CTETRA elements with or without midside grids using the GSTRESS, GSTRAIN, GTHSTRN, and GELSTRN case control commands.

14. The Gauss point locations for the solid elements are documented in Elements for Nonlinear Analysis in the *Simcenter Nastran Basic Nonlinear Analysis User's Guide*.

15. If FCTN = "PFLUIDEX", the fluid properties are obtained from an external file. The external file is referenced through the file unit number specified on the ACTEMP bulk entry. The ACTEMP bulk entry is referenced by an ACTEMP case control command.

Remarks related to SOLs 402 and 414:

1. FCTN is not supported.

2. Two integration schemes are available, less integration and more integration, typically 2x2x2 and 3x3x3 for CHEXA elements. They can be used in elements with or without midside-grids.

3. Hexa and elements formulation is enriched with EAS modes. ISOP parameter allows to neglect these modes or use traditional bubble modes. For all elements, the U/P formulation is the default if hyperelastic material is used.

4. STR is ignored. In output results, stresses and strains are extrapolated from Gauss points to grid points. This extrapolation can lead to values that do not exactly satisfy the material law.

5. CORDM=-1 is not supported. If CORDM=-1 is defined, the basic coordinate system is used for the material coordinates.

CHEXA Entry Options

CHEXA Entry Options – Linear Solutions[1]				
CHEXA	**Integration**	**IN**	**STRESS (Default: GRID)**	**ISOP**

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8 Node	2x2x2 Reduced Shear with Bubble Function (Default)	BUBBLE or Blank or 0 (Default)	Blank or GRID or GAUSS or 1	Blank or 0 or REDUCED (Default)(2)
	2x2x2 Reduced Shear Only	TWO or 2	Blank or GRID	
	2x2x2 Standard Isoparametric			FULL or 1
	3x3x3 Reduced Shear Only	THREE or 3		Blank or 0 or REDUCED
	3x3x3 Standard Isoparametric			FULL or 1
9 -20 Node	2x2x2 Reduced Shear Only	TWO or 2	Blank or GRID	Blank or 0 or REDUCED
	2x2x2 Standard Isoparametric			FULL or 1
	3x3x3 Reduced Shear Only	Blank or 0 or THREE or 3 (Default)		Blank or 0 or REDUCED (Default)(2)
	3x3x3 Standard Isoparametric	THREE or 3		FULL or 1

1. All solutions except SOLs 106, 129, 153/159 with ANALYSIS=STRUCTURE, 401, 402, and 414.

2. REDUCED is the default only for structural elements (FCTN = "SMECH").

CHEXA Entry Options – SOL 401

CHEXA	Small Strain		Large Strain		STRESS / ISOP
	Integration	IN(1)	Integration	IN	Ignored for small and large strain.
8 Node	2x2x2 with 9 Bubble modes	BUBBLE or Blank or 0 (Default)	2x2x2 with 9 Bubble modes	BUBBLE or Blank or 0 (Default)	
	2x2x2 Standard Isoparametric	TWO or 2	2×2×2 with B-Bar method	1	
			2x2x2 Standard Isoparametric	2	
9 -20 Node	3x3x3 Standard Isoparametric	Not Used(2)	2x2x2 Standard Isoparametric	Blank or 0 (Default)	
			3x3x3 Standard Isoparametric	3	

1. If IN = 3 is specified with small strain, a value of IN = 0 is used instead. A warning message is issued in the f06 file.

2. A valid entry must be specified because the PSOLID entry may be used by other element types.

CHEXA Entry Options – SOLs 402 and 414

CHEXA	Integration	IN	STRESS	Enrichment	ISOP
8 Node	2x2x2	Blank or 0 or 2 or TWO (Default)	Ignored	24 EAS modes	0
				9 BUBBLE modes	Blank or 1 (Default[1])
	3x3x3	THREE or 3		No enrichment	2
				U/P formulation with 1 Lagrange multiplier	3
9 -20 Node	2x2x2	Blank or 0 or 2 or TWO (Default[1])		Ignored	Blank or 0 (Default)
				Ignored	1
	3x3x3	THREE or 3		Ignored	2

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				U/P formulation with 4 Lagrange multipliers	3

1. The parameter MODEVERS is available on the NLCNTLG bulk entry to revert SOL 402 and 414 behavior to a previous version. For example, specifying MODEVERS, 20201 changes the behavior back to the default values before version 2020.2.

CHEXA Entry Options – Basic Nonlinear Solutions(1)

CHEXA	Integration	IN	STRESS (Default: GRID)	ISOP
8 Node(3)	2x2x2 Reduced Shear with Bubble Function (Default)	BUBBLE or Blank or 0 (Default)	Blank or GRID or GAUSS or 1	Blank or 0 or REDUCED (Default)(2)
	2x2x2 Reduced Shear Only	TWO or 2		
	2x2x2 Standard Isoparametric			FULL or 1
9 -20 Node	3x3x3 Standard Isoparametric	Not Used(4)		Not Used(4)

1. SOLs 106, 129, and 153/159 with ANALYSIS = STRUCTURE. When running a linear solution (geometric nonlinearity off and no nonlinear material), the **"CHEXA Entry Options – Linear Solutions"** table should be used.
2. REDUCED is the default only for structural elements (FCTN = "SMECH").
3. If IN = 3 is specified, a value of IN = 0 is used instead because IN = 3 is not supported. A warning message is also issued in the f06 file.
4. A valid entry must be specified because the PSOLID entry may be used by other element types.

CPENTA Entry Options

CPENTA Entry Options – Linear Solutions(1)

CPENTA	Integration	IN	STRESS (Default: GRID)	ISOP
6 Node	2x3 Reduced Shear with Bubble Function (Default)	Blank or 0 or BUBBLE (Default)	Blank or GRID or GAUSS or 1	Blank or 0 or REDUCED (Default) (2)
	2x3 Reduced Shear Only	TWO or 2	Blank or GRID	
	2x3 Standard Isoparametric			FULL or 1
	3x7 Reduced Shear Only	THREE or 3		Blank or 0 or REDUCED
	3x7 Standard Isoparametric			FULL or 1
7-15 Node	2x3 Reduced Shear Only	TWO or 2	Blank or GRID	Blank or 0 or REDUCED
	2x3 Standard Isoparametric			FULL or 1
	3x7 Reduced Shear Only (Default)	Blank or 0 or THREE or 3 (Default)		Blank or 0 or REDUCED (Default) (2)
	3x7 Standard Isoparametric	THREE or 3		FULL or 1

1. All solutions except SOLs 106, 129, 153/159 with ANALYSIS=STRUCTURE, 401, 402, and 414.
2. REDUCED is the default only for structural elements (FCTN = "SMECH").

CPENTA Entry Options – SOL 401

CPENTA	Small Strain		Large Strain		STRESS / ISOP
	Integration	IN(1)	Integration	IN	Ignored for small and large strain.
6 Node	2x3 with Bubble Function (Default)	BUBBLE or Blank or 0 (Default)	1x2 Standard Isoparametric	Ignored	

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	2x3 Standard Isoparametric	TWO or 2			
7 -15 Node	3x7 Standard Isoparametric	Not Used(2)	3x3 Standard Isoparametric		

1. If IN = 3 is specified, a value of IN = 0 is used instead. A warning message is issued in the f06 file.
2. A valid entry must be specified because the PSOLID entry may be used by other element types.

CPENTA Entry Options – SOLs 402 and 414

CPENTA	Integration(1)	IN	STRESS	Enrichment	ISOP
6 Node	1x2	Blank or 0 or 2 or TWO (Default)	Ignored	Ignored	Blank or 0 (Default)
				Ignored	1
	3x3	THREE or 3		Ignored	2
				U/P formulation with 1 Lagrange multiplier	3
7 - 15 Node	3x2	Blank or 0 or 2 or TWO (Default[2])		Ignored	Blank or 0 (Default)
				Ignored	1
	7x3	THREE or 3		Ignored	2
				U/P formulation with 6 Lagrange multipliers	3

1. First position is the number of integration points in the triangle, last is in the perpendicular direction.
2. The parameter MODEVERS is available on the NLCNTLG bulk entry to revert SOLs 402 and 414 behavior to a previous version. For example, specifying MODEVERS, 20201 changes the behavior back to the default values before version 2020.2.

CPENTA Entry Options – Basic Nonlinear Solutions[1]

CPENTA	Integration	IN	STRESS (Default: GRID)	ISOP
6 Node(3)	2x3 Reduced Shear with Bubble Function (Default)	Blank or 0 or BUBBLE (Default)	Blank or GRID or GAUSS or 1	Blank or 0 or REDUCED (Default) (2)
	2x3 Reduced Shear Only	TWO or 2		
	2x3 Standard Isoparametric			FULL or 1
7–15 Node	3x7 Standard Isoparametric	Not used(4)		Not used(4)

1. SOLs 106, 129, and 153/159 with ANALYSIS = STRUCTURE. When running a linear solution (geometric nonlinearity off and no nonlinear material), the **"CPENTA Entry Options – Linear Solutions"** table should be used.
2. REDUCED is the default only for structural elements (FCTN = "SMECH").
3. If IN = 3 is specified, a value of IN = 0 will be used instead because IN = 3 is not supported. A warning message will also be issued in the F06 file.
4. A valid entry must be specified because the PSOLID card may be used by other element types.

CPYRAM Entry Options

CPYRAM Entry Options – Linear Solutions(1)

CPYRAM	Integration	IN	STRESS (Default: GRID)	ISOP

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	Structural	Heat Transfer			
5 Node	2x2x2 Bedrosian[4] Isoparametric	2x2x2 Bedrosian[4] Isoparametric	Blank or 0 (Default) or 2	Blank or GRID or GAUSS or 1	Blank or 2 or 0 (Default)[2]
	3x3x3 Bedrosian[4] Isoparametric		1[3] or 3		
6–13 Node	2x2x2 Bedrosian[4] Isoparametric	3x3x3 Bedrosian[4] Isoparametric	Blank or 0 (Default) or 2		
	3x3x3 Bedrosian[4] Isoparametric		1[3] or 3		

1. All solutions except SOLs 106, 129, 153/159 with ANALYSIS=STRUCTURE, 401, 402, and 414.

2. The default value of Blank or 0 uses a standard formulation. A value of 2 specifies the B-BAR option for nearly incompressible materials.

3. For CPYRAM, IN = 1 produces an overly stiff element.

4. For additional information, see G. Bedrosian, "Shape Functions and Integration Formulas for Three-Dimensional Finite Element Analysis", *International Journal for Numerical Methods in Engineering*, vol. 35, 95–108 (1992).

CPYRAM Entry Options – SOL 401

CPYRAM	Integration	IN	STRESS	ISOP
5 Node	2x2x2 Bedrosian[2] Isoparametric (Default)	Blank or 0 (Default) or 2	Ignored for both small and large strain.	Ignored for small and large strain.
	3x3x3 Bedrosian[2] Isoparametric	1[1] or 3		
6-13 Node	2x2x2 Bedrosian[2] Isoparametric (Default)	Blank or 0 (Default) or 2		
	3x3x3 Bedrosian[2] Isoparametric	1[1] or 3		

1. For CPYRAM, IN = 1 produces an overly stiff element.

2. For additional information, see G. Bedrosian, "Shape Functions and Integration Formulas for Three-Dimensional Finite Element Analysis", *International Journal for Numerical Methods in Engineering*, vol. 35, 95–108 (1992).

CPYRAM Entry Options – SOLs 402 and 414

CPYRAM	Integration[1]	IN	STRESS	Enrichment	ISOP
5 Node	5	Blank or 0 or 2 or TWO (Default)	Ignored	Ignored	Blank or 0 (Default)
				Ignored	1
	8	THREE or 3		Ignored	2
				U/P formulation with 1 Lagrange multiplier	3
6 - 13 Node	8	Blank or 0 or 2 or TWO (Default[2])		Ignored	Blank or 0 (Default)
				Ignored	1
	8	THREE or 3		Ignored	2
				U/P formulation with 1 Lagrange multiplier	3

1. Total number of integration points.

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2. The parameter MODEVERS is available on the NLCNTLG bulk entry to revert SOL 402 and 414 behavior to a previous version. For example, specifying MODEVERS, 20201 changes the behavior back to the default values before version 2020.2.

CPYRAM Entry Options – Basic Nonlinear Solutions(1)

CPYRAM	Integration	IN	STRESS (Default: GRID)	ISOP
5 Node	2x2x2 Bedrosian(4) Isoparametric	Blank or 0 (Default) or 2	Blank or GRID or GAUSS or 1	Blank or 2 or 0 (Default)(2)
	3x3x3 Bedrosian(4) Isoparametric	1(3) or 3		
6–13 Node	2x2x2 Bedrosian(4) Isoparametric	Blank or 0 (Default) or 2		
	3x3x3 Bedrosian(4) Isoparametric	1(3) or 3		

1. SOLs 106, 129, and 153/159 with ANALYSIS = STRUCTURE. Only for geometric nonlinearity. When geometric nonlinearity is off, the **"CPYRAM Entry Options – Linear Solutions"** table should be used.
2. The default value of Blank or 0 uses a standard formulation. A value of 2 specifies the B-BAR option for nearly incompressible materials.
3. For CPYRAM, IN = 1 produces an overly stiff element.
4. For additional information, see G. Bedrosian, "Shape Functions and Integration Formulas for Three-Dimensional Finite Element Analysis", *International Journal for Numerical Methods in Engineering*, vol. 35, 95–108 (1992).

CTETRA Entry Options

CTETRA Entry Options – Linear Solutions(1)

CTETRA	Integration	IN	STRESS (Default: GRID)	ISOP
4 Node	1-Point Standard Isoparametric (Default)	Blank or 0 or TWO or 2 (Default)	Blank or GRID or GAUSS or 1	Blank or FULL
	5-Point Standard Isoparametric	THREE or 3		
5–10 Node	5-Point Standard Isoparametric	Not used(2)		

1. All solutions except SOLs 106, 129, 153/159 with ANALYSIS=STRUCTURE, 401, 402, and 414.
2. A valid entry must be specified because the PSOLID card may be used by other element types.

CTETRA Entry Options – SOL 401

CTETRA	Small Strain		Large Strain		STRESS / ISOP
	Integration	IN	Integration	IN	Ignored for small and large strain.
4 Node	1-Point Standard Isoparametric (Default)	Blank or 0 or TWO or 2 (Default)	1-Point Standard Isoparametric	Ignored	
	5-Point Standard Isoparametric	THREE or 3			
5–10 Node	4-Point Standard Isoparametric (Default)	Blank or 0 or TWO or 2 (Default)	4-Point Standard Isoparametric		
	5-Point Standard Isoparametric	THREE or 3			

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CTETRA Entry Options – SOLs 402 and 414					
CTETRA	**Integration[1]**	**IN**	**STRESS**	**Enrichment**	**ISOP**
4 Node	1	Blank or 0 or 2 or TWO (Default)	Ignored	Ignored	Blank or 0 (Default)
				Ignored	1
	4	THREE or 3		Ignored	2
				U/P formulation with 1 Lagrange multiplier	3
5 - 10 Node	4	Blank or 0 or 2 or TWO (Default[2])		Ignored	Blank or 0 (Default)
				Ignored	1
	8	THREE or 3		Ignored	2

1. Total number of integration points.
2. The parameter MODEVERS is available on the NLCNTLG bulk entry to revert SOL 402 and 414 behavior to a previous version. For example, specifying MODEVERS, 20201 changes the behavior back to the default values before version 2020.2.

CTETRA Entry Options – Basic Nonlinear Solutions[1]				
CTETRA	**Integration**	**IN**	**STRESS (Default: GRID)**	**ISOP**
4 Node	1-Point Standard Isoparametric (Default)	Blank or 0 or TWO or 2 (Default)	Blank or GRID or GAUSS or 1	Blank or FULL
	5-Point Standard Isoparametric	THREE or 3		
5–10 Node	4-Point Standard Isoparametric (Default)	Blank or 0 (Default)		
	5-Point Standard Isoparametric	THREE or 3		

1. SOLs 106, 129, and 153/159 with ANALYSIS = STRUCTURE. When running a linear solution (geometric nonlinearity off and no nonlinear material), the **"CTETRA Entry Options – Linear Solutions"** table should be used.

Axisymmetric Element Options – Linear Solutions

Axisymmetric Element Options – Linear Solutions[1]				
	Integration	**IN**	**STRESS**	**ISOP**
CQUADX4	2x2 with 5 bubble modes	Blank or 0 or BUBBLE (Default)	Blank or GRID or GAUSS or 1	Ignored
	2x2 with B-Bar method	BBAR or 1		
	2x2 Standard Isoparametric	TWO or 2		
CTRAX3	3-Point Standard Isoparametric	Blank or 0 (Default)		
	3-Point with B-Bar method	BBAR or 1		
CTRAX6	7-Point Standard Isoparametric	Blank or 0 (Default)		
	7-Point with B-Bar method	BBAR or 1		
CQUADX8	3x3 Standard Isoparametric	Blank or 0 (Default)		

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	3x3 with B-Bar method	BBAR or 1		

1. All solutions except SOLs 401, 402, and 414.

Axisymmetric Element Options – SOL 401

Axisymmetric Element Options – SOL 401 small strain and large strain					
	Small Strain		**Large Strain**		
	Integration	**IN**	**Integration**	**IN**	**STRESS / ISOP**
CQUADX4	2x2 with 5 bubble modes	Blank or 0 or BUBBLE (Default)	2x2 with 5 bubble modes	Blank or 0 or BUBBLE (Default)	Ignored for small and large strain.
	2x2 with B-Bar method	BBAR or 1	2x2 with B-Bar method	BBAR or 1	
	2x2 Standard Isoparametric	TWO or 2	2x2 Standard Isoparametric	TWO or 2	
CTRAX3	3-Point Standard Isoparametric	Blank or 0 (Default)	1-Point Standard Isoparametric	Ignored	
	3-Point with B-Bar method	BBAR or 1			
CTRAX6	7-Point Standard Isoparametric	Blank or 0 (Default)	3-Point Standard Isoparametric	Ignored	
	7-Point with B-Bar method	BBAR or 1			
CQUADX8	3x3 Standard Isoparametric	Blank or 0 (Default)	2x2 Standard Isoparametric	Blank or 0 (Default)	
	3x3 with B-Bar method	BBAR or 1	3x3 Standard Isoparametric	THREE or 3	

Axisymmetric Elements Options – SOLs 402 and 414(2)

	Integration	**IN**	**STRESS**	**Enrichment**	**ISOP**
CQUADX4	2x2	Blank or 0 or 2 or TWO (Default)	Ignored	4 BUBBLE modes	Blank or 0 (Default)
	3x3	THREE or 3		No enrichment	2
				U/P formulation with 1 Lagrange multiplier	3
CQUADX8	2x2	Blank or 0 or 2 or TWO (Default)		Ignored	Blank, 0, 1 or 2 (Default)
	3x3	THREE or 3		U/P formulation with 3 Lagrange multipliers	3
CTRAX3	1(1)	Blank or 0 or 2 or TWO (Default)		Ignored	Blank, 0, 1 or 2 (Default)
	3(1)	THREE or 3		U/P formulation with 3 Lagrange multipliers	3
CTRAX6	3(1)	Blank or 0 or 2 or TWO (Default)		Ignored	Blank, 0, 1 or 2 (Default)

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	6[1]	THREE or 3		U/P formulation with 3 Lagrange multipliers	

1. Total number of integration points.
2. In SOL 414, axisymmetric elements always behave as Fourier axisymmetric elements.

Related Topics

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PTUBE

Tube Property

Defines the properties of a thin-walled cylindrical tube element (CTUBE entry).

Format:

1	2	3	4	5	6	7	8	9	10
PTUBE	PID	MID	OD	T	NSM	OD2			

Example:

PTUBE	2	6	6.29	0.25					

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. See Remarks 3 and 4. (Integer > 0)
OD	Outside diameter of tube. (Real > 0.0)
T	Thickness of tube. (Real; T ≤ OD/2.0)
NSM	Nonstructural mass per unit length. (Real)
OD2	Diameter of tube at second grid point (G2) on CTUBE entry. (Real; Default = OD)

Remarks:

1. If T is zero, a solid circular rod is assumed.
2. PTUBE entries must all have unique property identification numbers.
3. For structural problems, MID must reference a MAT1 material entry.
4. For heat transfer problems, MID must reference a MAT4 or MAT5 material entry.
5. Tapered OD tubes with constant wall thickness are available for heat transfer only. The effective diameter is given by:

$$D_{effective} = T + \frac{D_2 - D_1}{log_e\left(\frac{D_2 - T}{D_1 - T}\right)}$$

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where:

D_1 = OD

D_2 = $\begin{cases} \text{OD2 if OD2} \neq 0 \\ \text{OD if OD2} = 0 \text{ or blank} \end{cases}$

Related Topics

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PVISC

Viscous Damping Element Property

Defines properties of a one-dimensional viscous damping element (CVISC entry).

Format:

1	2	3	4	5	6	7	8	9	10
PVISC	PID1	CE1	CR1		PID2	CE2	CR2		

Example:

PVISC	3	6.2	3.94						

Fields:

Field	Contents
PIDi	Property identification number. (Integer > 0)
CE1, CE2	Viscous damping values for extension in units of force per unit velocity. (Real)
CR1, CR2	Viscous damping values for rotation in units of moment per unit velocity. (Real)

Remarks:

1. Viscous properties are material independent; in particular, they are temperature independent.
2. One or two viscous element properties may be defined on a single entry.

Related Topics

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PWELD

WELD Element Property

Defines the properties of connector (CWELD) elements.

Format:

1	2	3	4	5	6	7	8	9	10
PWELD	PID	MID	D			MSET		TYPE	
	LDMIN	LDMAX							

Example:

PWELD	100	3	1.0						

Fields:

Field	Contents
PID	Property identification number. (Integer > 0)
MID	Material identification number. See Remark 1 . (Integer > 0)
D	Diameter of the connector. See Remark 1 . (Real > 0)
MSET	Flag to eliminate m-set degrees of freedom.See Remark 2 . (Character: “ON” or “OFF”; Default= “OFF”).
TYPE	Character string indicating the type of connection. See Remark 3 . (Character: ”SPOT” or blank; Default=blank, “SPOT”= spot weld connector, blank= general connector)
LDMIN	Smallest ratio of weld length to diameter. See Remark 4 . (Real > 0; Default=0.2)
LDMAX	Largest ratio of weld length to diameter. See Remark 4 . (Real > 0; Default=5.0)

Remarks:

1. Material MID, diameter D, and the weld length are used to calculate the stiffness of the connector in 6 directions. MID can only refer to the MAT1 Bulk Data entry.

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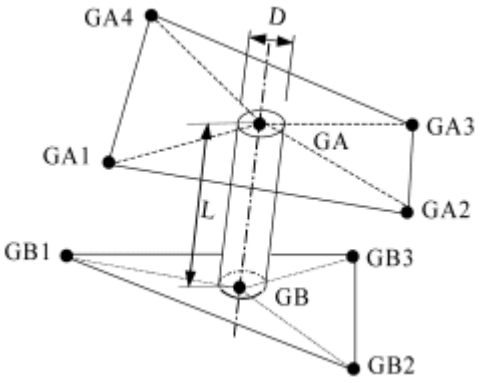
Figure 16-166: Length and Diameter of the Weld Connector



2. MSET is only used for formats “GRIDID” and “ELEMID”. MSET=”ON” generates explicit m-set constraints. MSET=”OFF” (default) incorporates constraints at the element stiffness matrix level avoiding explicit m-set constraint equations. For example, if a patch-to-patch connection is specified with the formats “GRIDID” or “ELEMID” on the CWELD entry, and MSET=ON is specified, 2x6 explicit constraints are generated which connect the 6 degrees of freedom of GA to the translational degrees of freedom of GAI and the 6 degrees of freedom of GB to GBI. The 2x6 degrees of freedom of GA and GB are put into the m-set. The constraints are labeled “RWELD”. The identification numbers of the generated RWELD constraint elements start with an offset specified on PARAM, OSWELM. If MSET=OFF is specified, the 2x6 constraint equations are built into the stiffness matrix of the CWELD element thereby condensing the 2x6 degrees of freedom of GA and GB. MSET=OFF is only available for the patch-to-patch connection (GRIDID or ELEMID).
3. When TYPE= “SPOT”, and the format on the CWELD entry is either “ELEMID”, “ELPAT”, or “PARTPAT”, the effective length for the stiffness of the CWELD element is set to L_e=1/2(t_A+t_B) regardless of the distance GA to GB. t_A and t_B are the shell thicknesses of shell *A* and *B* respectively. For all other cases, the effective length of the CWELD element is equal to the true length, the distance of GA to GB, as long as the ratio of length to diameter is in the range $0.2 \le L/D \le 5.0$. If *L* is below the range, the effective length is set to $L_e = 0.2D$ and if *L* is above the range, the effective length is set to $L_e = 5.0D$.
4. When TYPE is blank, the weld length is the distance from GA to GB, as long as the L/D ratio is within LDMIN≤L/D≤LDMAX. If the ratio is below LDMIN, the weld length is set to LDMIN*D, and if the ratio is above LDMAX, the weld length is set to LDMAX*D.

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17. Bulk Entries Q—S

QBDY1

Boundary Heat Flux Load for CHBDYj Elements, Form 1

Defines a uniform heat flux into CHBDYj elements.

Format:

1	2	3	4	5	6	7	8	9	10
QBDY1	SID	Q0	EID1	EID2	EID3	EID4	EID5	EID6	

Example:

QBDY1	109	1.-5	721						

Alternate Format and Example:

QBDY1	SID	Q0	EID1	"THRU"	EID2				
QBDY1	109	1.-5	725	THRU	735				

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
Q0	Heat flux into element. (Real)
EIDi	CHBDYj element identification numbers. (Integer ≥ 0 or "THRU". For "THRU" option EID2 >EID1.)

Remarks:

1. QBDY1 entries must be selected with the Case Control command LOAD=SID in order to be used in static analysis. The total power into an element is given by the equation:

$$P_{in} = (\text{Effective area}) \cdot Q0$$

2. QBDY1 entries must be referenced on a TLOAD entry for use in transient analysis. The total power into an element is given by the equation:

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$$P_{in}((t) = (\text{Effective area}) \cdot Q0 \cdot F(t - \tau))$$

where the function of time $F(t–\tau)$ is specified on a TLOAD1 or TLOAD2 entry.

3. The sign convention for Q0 is positive for heat input.

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QBDY2

Boundary Heat Flux Load for CHBDYj Elements, Form 2

Defines grid point heat flux into CHBDYj elements.

Format:

1	2	3	4	5	6	7	8	9	10
QBDY2	SID	EID	Q01	Q02	Q03	Q04	Q05	Q06	
	Q07	Q08							

Example:

QBDY2	109	721	1.-5	1.-5	2.-5	2.-5			

Fields:

Field	Contents
SID	Load set identification number. (Integer >0)
EID	Identification number of an CHBDYj element. (Integer > 0)
Q0i	Heat flux at the i-th grid point on the referenced CHBDYj element. (Real or blank)

Remarks:

1. QBDY2 entries must be selected with the Case Control command LOAD=SID in order to be used in static analysis. The total power into each point i on an element is given by

$$P_i = AREA_i \cdot Q0i$$

2. QBDY2 entries must be referenced on a TLOAD entry for use in transient analysis. All connected grid points will have the same time function but may have individual delays. The total power into each point i on an element is given by

$$P_i(t) = AREA_i \cdot Q0i \cdot F(t - \tau_i)$$

   where $F(t–\tau_i)$ is a function of time specified on a TLOAD1 or TLOAD2 entry.

3. The sign convention for Q0i is positive for heat flux input to the element.

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QBDY3

Boundary Heat Flux Load for a Surface

Defines a uniform heat flux load for a boundary surface.

Format:

1	2	3	4	5	6	7	8	9	10
QBDY3	SID	Q0	CNTRLND	EID1	EID2	EID3	EID4	EID5	
	EID6	etc.							

Example:

QBDY3	2	20.0	10	1	THRU	50	BY	2	

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
Q0	Thermal heat flux load, or load multiplier. Q0 is positive for heat flow into a surface. (Real)
CNTRLND	Control point for thermal flux load. (Integer ≥ 0; Default=0)
EIDi	CHBDYj element identification numbers. (Integer > 0 or “THRU” or “BY”)

Remarks:

1. QBDY3 entries must be selected in Case Control (LOAD=SID) to be used in steady state. The total power into a surface is given by the equation:
   - if CNTRLND ≤ 0 then P_{in}=(Effective area) $\cdot Q0$
   - if CNTRLND > 0 thenP_{in}=(Effective area) $\cdot Q0 \cdot u_{\text{CNTRLND}}$

     where u_{CNTRLND} is the temperature of the control point and is used as a load multiplier.
2. In transient analysis SID is referenced by a TLOADi Bulk Data entry through the DAREA entry. A function of time $F(t-\tau)$ defined on the TLOADi multiplies the general load, with τ specifying time delay. The load set identifier on the TLOADi entry must be selected in Case Control (DLOAD=SID) for use in transient analysis. If multiple types of transient loads exist, they must be combined by the DLOAD Bulk Data entry.
3. The CNTRLND multiplier cannot be used with any higher-order elements.

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4. When using “THRU” or “BY”, all intermediate CHBDYE, CHBDYG, or CHBDYP elements must exist.

Related Topics

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QHBDY

Boundary Heat Flux Load

Defines a uniform heat flux into a set of grid points.

Format:

1	2	3	4	5	6	7	8	9	10
QHBDY	SID	FLAG	Q0	AF	G1	G2	G3	G4	
	G5	G6	G7	G8					

Example:

QHBDY	2	AREA4	20.0		101	102	104	103	

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
FLAG	Type of face involved (must be one of the following: “POINT”, “LINE”, “REV”, “AREA3”, “AREA4”, “AREA6”, “AREA8”)
Q0	Magnitude of thermal flux into face. Q0 is positive for heat into the surface. (Real)
AF	Area factor depends on type. (Real > 0.0 or blank)
Gi	Grid point identification of connected grid points. (Integer > 0 or blank)

Remarks:

1. The continuation entry is optional.
2. For use in steady state analysis, the load set is selected in the Case Control Section (LOAD=SID).
3. In transient analysis SID is referenced by a TLOADi Bulk Data entry through the DAREA entry. A function of time defined on the TLOADi entry multiplies the general load. specifies time delay. The load set identifier on the TLOADi entry must be selected in Case Control (DLOAD=SID) for use in transient analysis. If multiple types of transient loads exist, they must be combined by the DLOAD Bulk Data entry.
4. The heat flux applied to the area is transformed to loads on the points. These points need not correspond to an HBDY surface element.
5. The flux is applied to each point i by the equation

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$$P_i = AREA_i \cdot Q0$$

6. The number of connected points for the types are 1 (POINT), 2 (LINE, REV), 3 (AREA3), 4 (AREA4), 4-6 (AREA6), 5-8 (AREA8).
7. The area factor AF is used to determine the effective area for the POINT and LINE types. It equals the area and effective width, respectively. It is not used for the other types, which have their area defined implicitly and must be left blank.
8. The type of face (FLAG) defines a surface in the same manner as the CHBDYi data entry. For physical descriptions of the geometry involved, see the CHBDYG discussion.

Related Topics

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QSET

Generalized Degrees-of-Freedom

Defines generalized degrees-of-freedom (q-set) to be used for component mode synthesis.

Format:

1	2	3	4	5	6	7	8	9	10
QSET	ID1	C1	ID2	C2	ID3	C3	ID4	C4	

Example:

QSET	15	123456	1	7	9	2	105	6	

Fields:

Field	Contents
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component number. (Integer zero or blank for scalar points or any unique combination of the Integers 1 through 6 for grid points with no embedded blanks.)

Remarks:

1. Degrees-of-freedom specified on this entry form members of the mutually exclusive q-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
2. Degrees-of-freedom specified on QSET and QSET1 entries are automatically placed in the a-set.
3. When ASET, ASET1, QSET, and/or QSET1 entries are present, all degrees-of-freedom not otherwise constrained (e.g., SPCi or MPC entries) will be placed in the omitted set (o-set).

Related Topics

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QSET1

Generalized Degrees-of-Freedom (Alternate Form of QSET Entry)

Defines generalized degrees-of-freedom (q-set) to be used for component mode synthesis.

Format:

1	2	3	4	5	6	7	8	9	10
QSET1	C	ID1	ID2	ID3	ID4	ID5	ID6	ID7	
	ID8	ID9	-etc.-						

Example:

QSET1	123456	1	7	9	22	105	6	22	
	52	53							

Alternate Format and Example:

QSET1	C	ID1	"THRU"	ID2					
QSET1	0	101	THRU	110					

Fields:

Field	Contents
C	Component number. (Integer zero or blank for scalar points or any unique combination of the Integers 1 through 6 for grid points with no embedded blanks.)
IDi	Grid or scalar point identification number. (Integer > 0; For THRU option, ID1 < ID2.)

Remarks:

1. Degrees-of-freedom specified on this entry form members of the mutually exclusive q-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
2. Degrees-of-freedom specified on QSET and QSET1 entries are automatically placed in the a-set.
3. When ASET, ASET1, QSET, and/or QSET1 entries are present, all degrees-of-freedom not otherwise constrained (e.g., SPCi or MPC entries) will be placed in the omitted set (o-set).

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QVECT

Thermal Vector Flux Load

Defines thermal vector flux from a distant source into a face of one or more CHBDYi boundary condition surface elements.

Format:

1	2	3	4	5	6	7	8	9	10
QVECT	SID	Q0	TSOUR	CE	E1 or TID1	E2 or TID2	E3 or TID3	CNTRLND	
	EID1	EID2	-etc.-						

Example:

QVECT	10	20.0	1000.0		1.0	1.0	1.0	101	
	20	21	22	23					

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
Q0	Magnitude of thermal flux vector into face. (Real or blank)
TSOUR	Temperature of the radiant source. (Real or blank)
CE	Coordinate system identification number for thermal vector flux. (Integer ≥ 0 or blank)
Ei	Vector components (direction cosines in coordinate system CE) of the thermal vector flux. (Real; Default=0.0)
TIDi	TABLEDi entry identification numbers defining the components as a function of time. (Integer > 0)
CNTRLND	Control point. (Integer ≥ 0; Default=0)
EIDi	Element identification number of a CHBDYE, CHBDYG, or CHBDYP entry. (Integer ≠ 0 or “THRU”)

Remarks:

1. The continuation entry is required.
2. If the coordinate system CE is not rectangular, then the thermal vector flux is in different directions for different CHBDYi elements. The direction of the thermal vector flux over an element is aligned to be in the direction of the flux vector at the geometric center of the element. The geometric

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center is measured using the grid points and includes any DISLIN specification on the VIEW entry for TYPE=LINE CHBDYi elements. The flux is presumed to be uniform over the face of each element; i.e., the source is relatively distant.

3. For use in steady-state analysis, the load set is selected in the Case Control Section (LOAD=SID). The total power into an element is given by:

   - If CNTRLND=0 then, $P_{in}=-\alpha A\ (\vec{e}\cdot\vec{n})\cdot Q0$.
   - If CNTRLND > 0 then, $P_{in}=-\alpha A\ (\vec{e}\cdot\vec{n})\cdot Q0\cdot u_{\text{CNTRLND}}$.

   where

α	=	face absorptivity (supplied from a RADM statement).
A	=	face area as determined from a CHBDYi surface element.
$\vec{e}$	=	vector of direction cosines E1, E2, E3.
$\vec{n}$	=	face normal vector. See CHBDYi entries.
$\vec{e}\cdot\vec{n}$	=	0 if the vector product is positive, (i.e., the flux is coming from behind the face).
u_{cntrlnd}	=	temperature value of the control point used as a load multiplier.

4. If the absorptivity is constant, its value is supplied by the ABSORP field on the RADM entry. If the absorptivity is not a constant, the thermal flux is assumed to have a wavelength distribution of a black body at the temperature TSOUR.

   - For a temperature-dependent absorptivity, the element temperature is used to determine α.
   - For a wavelength-dependent absorptivity, the integration of the flux times is computed for each wavelength band. The sum of the integrated thermal fluxes over all the wavelength bands is Q0. The wave bands are specified with the RADBND entry.
   - The user has the responsibility of enforcing Kirchhoff's laws.

5. In transient analysis, SID is referenced by a TLOADi Bulk Data entry through the DAREA specification. A function of time $F(t-\tau)$ defined on the TLOADi entry multiplies the general load. τ provides any required time delay. $F(t-\tau)$ is a function of time specified on the TLOADi entry. The value of is calculated for each loaded grid point. The load set identifier on the TLOADi entry must be selected in Case Control (DLOAD=SID) for use in transient analysis. If multiple types of transient loads exist, they must be combined by the DLOAD Bulk Data entry.

   The total power into an element is given by:

   - If CNTRLND=0 then, $P_{in}= -\alpha A(\vec{e}(t)\cdot\vec{n})\cdot Q0\cdot F(t-\tau)$.
   - If CNTRLND > 0 then,

     $P_{in}= -\alpha A(\vec{e}(t)\cdot\vec{n})\cdot F(t-\tau)Q0\cdot u_{\text{CNTRLND}}$.

6. If the referenced face is of TYPE=ELCYL, the power input is an exact integration over the area exposed to the thermal flux vector.

7. If the referenced face is of TYPE=REV, the thermal flux vector must be parallel to the axis of symmetry if an axisymmetric boundary condition is to be maintained.

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8. When applied to a surface element associated with a radiation enclosure cavity, any incident energy that is not absorbed ($\alpha < 1.0$) is lost from the system and is not accounted for in a reflective sense ($\alpha + \rho = 1.0$).

Related Topics

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QVOL

Volume Heat Addition

Defines a rate of volumetric heat addition in a conduction element.

Format:

1	2	3	4	5	6	7	8	9	10
QVOL	SID	QVOL	CNTRLND	EID1	EID2	EID3	EID4	EID5	
	EID6	etc.							

Example:

QVOL	5	10.0	101	10	12	11	9		

Fields:

Field	Contents
SID	Load set identification. (Integer > 0)
QVOL	Power input per unit volume produced by a heat conduction element. (Real)
CNTRLND	Control point used for controlling heat generation. (Integer ≥ 0; Default=0)
EIDi	A list of heat conduction elements. (Integer > 0 or “THRU” or “BY”)

Remarks:

1. EIDi has material properties (MAT4) that include HGEN, the element material property for heat generation, which may be temperature dependent. This association is made through the element EID. If HGEN is temperature dependent, it is based on the average element temperature.
2. QVOL provides either the constant volumetric heat generation rate or the load multiplier. QVOL is positive for heat generation. For steady-state analysis, the total power into an element is
   - If CNTRLND=0, then P_{in}=volume · HGEN · QVOL.
   - If CNTRLND > 0, then P_{in}=volume · HGEN · QVOL · u_{CNTRLND}.

     where u_{CNTRLND}. is the temperature multiplier.
3. For use in steady-state analysis, the load set is selected in the Case Control Section (LOAD=SID).
4. In transient analysis SID is referenced by a TLOADi Bulk Data entry. A function of time $F\,[t - \tau]$ defined on the TLOADi entry multiplies the general load where τ specifies time delay. The load set identifier

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on the TLOADi entry must be selected in Case Control (DLOAD=SID) for use in transient analysis. If multiple types of transient loads exist, they must be combined by the DLOAD Bulk Data entry.

5. For “THRU” or “BY”, all intermediate referenced heat conduction elements must exist.
6. The CNTRLND multiplier cannot be used with any higher-order elements.

Related Topics

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RADBC

Space Radiation Specification

Specifies an CHBDYi element face for application of radiation boundary conditions.

Format:

1	2	3	4	5	6	7	8	9	10
RADBC	NODAMB	FAMB	CNTRLND	EID1	EID2	EID3	-etc.-		

Example:

RADBC	5	1.0	101	10					

Fields:

Field	Contents
NODAMB	Ambient point for radiation exchange. (Integer > 0)
FAMB	Radiation view factor between the face and the ambient point. (Real ≥ 0.0)
CNTRLND	Control point for radiation boundary condition. (Integer ≥ 0; Default=0)
EIDi	CHBDYi element identification number. (or “THRU” or “BY”)

Remarks:

1. The basic exchange relationship is:
   - if CNTRLND=0, then $q=\sigma \cdot FAMB \cdot (\epsilon_e T^4_e - \alpha_e T^4_{amb})$
   - if CNTRLND > 0, then

     $q=\sigma \cdot FAMB \cdot u_{CNTRLND} \cdot (\epsilon_e T^4_e - \alpha_e T^4_{amb})$
2. NODAMB is treated as a black body with its own ambient temperature for radiation exchange between the surface element and space. No surface element that is a member of a radiation enclosure cavity may also have a radiation boundary condition applied to it.
3. Two PARAM entries are required when stipulating radiation heat transfer:
   - TABS defines the absolute temperature scale; this value is added internally to any specified temperature given in the problem. Upon solution completion, this value is subtracted internally from the solution vector.
   - SIGMA (σ) is the Stefan-Boltzmann constant.

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4. RADBC allows for surface radiation to space in the absence of any cavity behavior. The emissivity and absorptivity are supplied from a RADM entry.
5. When using “THRU” or “BY”, all intermediate referenced CHBDYi surface elements must exist.

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RADBND

Radiation Wavelength Band Definition

Specifies Planck's second radiation constant and the wavelength breakpoints used for radiation exchange problems.

Format:

1	2	3	4	5	6	7	8	9	10
RADBND	NUMBER	PLANCK2	λ1	λ2	λ3	λ4	λ5	λ6	
	λ7	etc.							

Example:

RADBND	6	14388.0	1.0	2.0	4.0	8.0	12.0		

Fields:

Field	Contents
NUMBER	Number of radiation wave bands. See Remarks. (Integer > 1)
PLANCK2	Planck's second radiation constant. See Remarks. (Real > 0.0)
λi (LAMBDAi)	Highest wavelength of the i-th wave band. See Remarks. (Real ≥ 0.0)

Remarks:

1. Only one RADBND entry may be specified in the Bulk Data Section and must always be used in conjunction with the RADM entry.
2. PLANCK2 has the units of wavelength times temperature. The same units of length must be used for LAMBDAi as for PLANCK2. The units of temperature must be the same as those used for the radiating surfaces. For example: 25898. μm °R or 14388. μm °K.
3. The first wavelength band extends from 0 to LAMBDA1 and the last band extends from LAMBDAn to infinity, where n=NUMBER – 1.
4. Discontinuous segments in the emissivity versus wavelength piecewise linear curve must be treated as a wavelength band of zero width.
5. LAMBDAi must be specified in ascending order, and all LAMBDAi fields where i is greater than or equal to NUMBER must be blank.

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RADCAV

Radiation Cavity Identification

Identifies the characteristics of each radiant enclosure.

Format:

1	2	3	4	5	6	7	8	9	10
RADCAV	ICAVITY	ELEAMB	SHADOW	SCALE	PRTPCH	NFECI	RMAX		
	SET11	SET12	SET21	SET22	SET31	SET32	-etc.-		

Example:

RADCAV	1	1					.99		
	3	5	4	5	7	5			

Fields:

Field	Contents
ICAVITY	Unique cavity identification number associated with enclosure radiation. (Integer > 0)
ELEAMB	CHBDYi surface element identification number for radiation if the view factors add up to less than 1.0. (Unique Integer > 0 among all CHBDYi elements or blank.)
SHADOW	Flag to control third body shading calculation during view factor calculation for each identified cavity. (Character="YES" or "NO"; Default="YES")
SCALE	View factor that the enclosure sum will be set to if a view factor is greater than 1.0. (0.0 ≤ Real ≤ 1.0; Default=0.0)
PRTPCH	Facilitates the blocking of view factor printing and punching onto RADLST and RADMTX entries. (Integer=0, 1, 2, or 3; Default=0):
NFECI	Controls whether finite difference or contour integration methods are to be used in the calculation of view factors in the absence of a VIEW3D Bulk Data entry. (Character="FD" or "CONT"; See Remark 4 . for default.)
RMAX	Subelement area factor. See Remark 5 . (Real ≥ 0.0; Default=0.1)
SETij	Set identification pairs for the calculation of global view factors. Up to 30 pairs may be specified (i=1 to 2 and j=1 to 30). (Integer > 0)

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PRTPCH	Print/Punch
0 (default)	print and punch
1	no print
2	no punch
3	no print or punch

Remarks:

1. For the surfaces of an incomplete enclosure (view factors add up to less than 1.0), a complete enclosure may be achieved (SUM=1.0) by specifying an ambient element, ELEAMB. When multiple cavities are defined, each cavity must have a unique ambient element if ambient elements are desired. No elements can be shared between cavities.
2. Third-body shadowing is ignored in the cavity if SHADOW="NO". In particular, if it is known a priori that there is no third-body shadowing, SHADOW=NO overrides KSHD and KBSHD fields on the VIEW Bulk Data entry as well as reduces the calculation time immensely.
3. The view factors for a complete enclosure may add up to slightly more than 1.0 due to calculation inaccuracies. SCALE can be used to adjust all the view factors proportionately to acquire a summation equal to the value specified for SCALE. If SCALE is left blank or set to 0.0, no scaling is performed.
4. If the VIEW3D Bulk Data entry is not specified, the view factors are calculated using finite difference and contour integration methods. If NFECI="FD", then all view factors are calculated using the finite difference technique. NFECI="CONT" invokes contour integration for all view factor calculations. If NFECI is blank, the program selects a method to use between any two particular elements based on RMAX.
5. The comparison value for RMAX is equal to A_s / d^2_{rs} where A_s is the area of a subelement and d_{rs} is the distance between two subelements r and s for which view factors are being computed. When NFECI is blank, the program selects the contour integral method only if $A_s / d^2_{rs} >$ RMAX.
6. When a number of elements are grouped together and considered as a conglomerate surface, view factors can be calculated between these groups. These are referred to as global view factors. The SET1 Bulk Data entry is used to define the conglomerate. When using this feature, negative EIDs are not allowed.
7. If a RADLST and RADMTX entry exists for this cavity ID, new view factors are not computed and the existing RADLST and RADMTX are used in the thermal analysis.

Related Topics

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RADLST

Listing of Enclosure Radiation Faces

Identifies the individual CHBDYi surface elements that comprise the entire radiation enclosure.

Format:

1	2	3	4	5	6	7	8	9	10
RADLST	ICAVITY	MTXTYP	EID1	EID2	EID3	EID4	EID5	EID6	
	EID7	-etc.-							

Example:

RADLST	2	1	2	3	4	5	6	7	

Fields:

Field	Contents
ICAVITY	Unique cavity identification number that links a radiation exchange matrix with its listing of enclosure radiation faces. (Integer > 0)
MTXTYP	Type of radiation exchange matrix used for this cavity. (Integer ≤ 4 and ≠ 0; Default=1 for an enclosure without an ambient element. Default=4 for an enclosure with an ambient element as specified on the RADCAV entry.) 1: Symmetric view factor matrix [F] and nonconservative radiation matrix [R]. 2: Symmetric exchange factor matrix [ℑ] and conservative radiation matrix [R]. 3: Unsymmetrical exchange factor matrix [ℑ] and conservative radiation matrix [R]. 4: Symmetric view factor matrix [F] and conservative radiation matrix [R]. -n: The first n CHBDYi elements may lose energy to space but the remainder may not. Symmetric exchange factor matrix [F] and nonconservative radiation matrix [R].
EIDi	Identification numbers for the CHBDYi elements in this radiation cavity. (Integer ≠ 0 or "THRU")

Remarks:

1. A radiation EIDi list isolates those CHBDYi surface element faces that are to communicate in a radiation enclosure. View-factor calculation and RADMTX formation for an enclosure is performed only for (or among) those faces identified within the same RADCAV.

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2. A radiation exchange matrix (RADMTX) can only reference one radiative face list (RADLST). The companion RADCAV, RADLST, and RADMTX must share a unique ICAVITY.
3. For each EIDi, the appropriate CHBDYi element is located, and the proper RADM entry ID field found.
4. If the radiation exchange matrix or any radiation boundary conditions are available from an external source, the RADMTX must be user generated.
5. Multiple RADLST entries may be specified.

Related Topics

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RADM

Radiation Boundary Material Property

Defines the radiation properties of a boundary element for heat transfer analysis.

Format:

1	2	3	4	5	6	7	8	9	10
RADM	RADMID	ABSORP	EMIS1	EMIS2	EMIS3	EMIS4	EMIS5	EMIS6	
	EMIS7	-etc.-							

Example:

RADM	11		.45	.33	.29	.20	.17	.13	

Fields:

Field	Contents
RADMID	Material identification number. (Integer > 0)
ABSORP	Surface absorptivity or the temperature function curve multiplier if ABSORP is variable. See Remark 2 . (0.0 ≤ Real ≤ 1.0)
EMISi	Surface emissivity at wavelength LAMBDAi or the temperature function curve multiplier if EMISi is variable (See the RADBND entry.) (0.0 ≤ Real ≤ 1.0)

Remarks:

1. The RADM entry is directly referenced only by one of the CHBDYE, CHBDYG, or CHBDYP type surface element entries.
2. For radiation enclosure problems, ABSORP is set equal to emissivity. For QVECT loads, absorptivity is specified by ABSORP.
3. If there is more than one EMISi, then:
   - There must be a RADBND entry.
   - The number of EMISi may not exceed the number of LAMBDAi on the RADBND entry.
   - The emissivity values are given for a wavelength specified by the corresponding LAMBDAi on the RADBND entry. Within each discrete wavelength band, the emissivity is assumed to be constant.
   - At any specific wavelength and surface temperature, the absorptivity is exactly equal to the emissivity.
4. To perform any radiation heat transfer exchange, the user must furnish PARAM entries for:

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- TABS to define the absolute temperature scale.
- SIGMA σ to define the Stefan-Boltzmann constant in appropriate units.

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RADMT

Radiation Boundary Material Property Temperature Dependence

Specifies table references for temperature dependent RADM entry radiation boundary properties.

Format:

1	2	3	4	5	6	7	8	9	10
RADMT	RADMID	T(A)	$T(\epsilon_1)$	$T(\epsilon_2)$	$T(\epsilon_3)$	$T(\epsilon_4)$	$T(\epsilon_5)$	$T(\epsilon_6)$	
	$T(\epsilon_7)$	-etc.-							

Example:

RADMT	11		1	2	3	4	5	6	

Fields:

Field	Contents
RADMID	Material identification number. (Integer > 0)
T(A)	TABLEMj identifier for surface absorptivity. (Integer ≥ 0 or blank)
$T(\epsilon_i)$	TABLEMj identifiers for surface emissivity. (Integer ≥ 0 or blank)

Remarks:

1. The basic quantities on the RADM entry of the same RADMID are always multiplied by the corresponding tabular function.
2. Tables T(A) and $T(\epsilon_i)$ have an upper bound that is less than or equal to one and a lower bound that is greater than or equal to zero.
3. The TABLEMj enforces the element temperature as the independent variable. Blank or zero fields means there is no temperature dependence of the referenced property on the RADM entry.

Related Topics

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RADMTX

Radiation Exchange Matrix

Provides the $F_{ji}=A_j f_{ji}$ exchange factors for all the faces of a radiation enclosure specified in the corresponding RADLST entry.

Format:

1	2	3	4	5	6	7	8	9	10
RADMTX	ICAVITY	INDEX	Fi,j	Fi+1,j	Fi+2,j	Fi+3,j	Fi+4,j	Fi+5,j	
	Fi+6,j	-etc.-							

Example:

RADMTX	2	1	0.0	0.1	0.2	0.2	0.3	0.2	

Fields:

Field	Contents
ICAVITY	Unique cavity identification number that links a radiation exchange matrix with its listing of enclosure radiation surface elements. (Integer > 0)
INDEX	Column number in the matrix. (Integer > 0)
$F_{k,j}$	If symmetric, the matrix values start on the diagonal (i=j) and continue down the column (k=i + 1, i + 2, etc.). If unsymmetrical, the values start in row (i=1). i refers to EIDi on the RADLST entry. (Real ≥ 0)

Remarks:

1. If the matrix is symmetric, only the lower triangle is input, and i=j=INDEX. If the matrix is unsymmetrical, i=1, and j=INDEX.
2. Only one ICAVITY may be referenced for those faces that are to be included in a unique radiation matrix.
3. Coefficients are listed by column with the number of columns equal to the number of entries in the RADLST.
4. All faces involved in any radiation enclosure must be defined with an CHBDYi element.

Related Topics

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RADSET

Identifies a Set of Radiation Cavities

Specifies which radiation cavities are to be included for radiation enclosure analysis.

Format:

1	2	3	4	5	6	7	8	9	10
RADSET	ICAV1	ICAV2	ICAV3	ICAV4	ICAV5	ICAV6	ICAV7	ICAV8	
	ICAV9	-etc.-							

Example:

RADSET	1	2	3	4					

Fields:

Field	Contents
ICAVi (ICAVITYi)	Unique identification number for a cavity to be considered for enclosure radiation analysis. (Integer > 0)

Remarks:

1. For multiple radiation cavities, RADSET specifies which cavities are to be included in the analysis.

Related Topics

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RANDPEX

Power Spectral Density (PSD) Specification from an External File

References a PSD specification from an external sc_h5 file in HDF5 format for use in a random analysis.

Format:

1	2	3	4	5	6	7	8	9	10
RANDPEX	SID	LUNF	LDID	FAXIS	PAXIS				

Example:

RANDPEX	5	101	7						

Fields:

Field	Contents
SID	Random analysis set identification number. (Integer > 0) See Remark 2.
LUNF	Logical unit number for HDF5 file containing PSD specification. (Integer > 0; No default)
LDID	Identification number of a loading description contained in the HDF5 file referenced by LUNF. (Integer > 0; No default) See Remark 5.
FAXIS	Specifies a linear or logarithmic interpolation for the frequency axis. (Character; “LINEAR” or “LOG”; Default = ”LINEAR”)
PAXIS	Specifies a linear or logarithmic interpolation for the PSD data axis. (Character; “LINEAR” or “LOG”; Default = ”LINEAR”)

Remarks:

1. RANDPEX is only supported in SOL 108 and SOL 111.
2. The set identification numbers must be selected with the case control command RANDOM = SID. SID must be unique with respect to all other RANDPS and RANDPEX bulk entries. In other words, a RANDOM case control command cannot reference both RANDPEX and RANDPS bulk entries; it can only reference one or the other.
3. An entire diagonal PSD matrix will be referenced. This matrix will contain one PSD function for each subcase. Only auto spectral density calculations will be performed and the subcases in the referenced PSD matrix must conform to the subcases of the force response calculations performed prior to the random analysis.
4. For a discussion of random analysis, see the *Advanced Dynamic Analysis User’s Guide*.

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5. The LDID entry references a DTI, DISTL bulk entry that contains the description string of the required PSD matrix specification in the HDF5 file referenced by LUNF.

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RANDPS

Power Spectral Density Specification

Defines load set power spectral density factors for use in random analysis having the frequency-dependent form:

$$S_{jk}(F) = (X + iY)G(F)$$

Format:

1	2	3	4	5	6	7	8	9	10
RANDPS	SID	J	K	X	Y	TID			

Examples:

RANDPS	5	3	7	2.0	2.5	4			

RANDPS	5	3	7	2.0	2.5	7.5			

Fields:

Field	Contents
SID	Random analysis set identification number. (Integer > 0)
J	Subcase identification number of the excited load set. (Integer > 0)
K	Subcase identification number of the applied load set. (Integer with K ≥ J)
X, Y	Components of the complex scaling factor. (Real)
TID	*G(F)* specification. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 3) If real entry, value of *G(F)* used over all frequencies. If integer entry, identification number of a TABRND1 or TABRNDG entry that defines *G(F)*.

Remarks:

1. Set identification numbers must be selected with the Case Control command (RANDOM = SID).
2. For auto spectral density, J = K, X must be greater than zero and Y must be equal to zero.
3. *G(F)* = 1.0 if the TID entry field is left blank, if TID = 0.0 (real), or if TID = 0 (integer).

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4. RANDPS may only reference subcases included within a single loop (a change in direct matrix input is not allowed).
5. See the *Simcenter Nastran Advanced Dynamic Analysis User's Guide* for a discussion of random analysis.
6. In the case of cyclic symmetry, Solution Sequence 118, J and K must refer to the coded subcase IDs. See "Using Cyclic Symmetry" in the *Simcenter Nastran User's Guide* for the coding procedure.
7. In superelement analysis, J and K must refer to the superelement subcases. For example, if superelement 10 has SUBCASEs 1 and 2 and superelement 20 has SUBCASEs 3 and 4, then a separate RANDPS entry is required for each superelement, even though X, Y, and TID may be identical.
8. For uncoupled PSDF (no K > J entries), only one K = J entry is allowed for a unique value of J. For coupled PSDF (some K > J entries), any number of entries are allowed.

Related Topics

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RANDT1

Autocorrelation Function Time Lag

Defines time lag constants for use in random analysis autocorrelation function calculation.

Format:

1	2	3	4	5	6	7	8	9	10
RANDT1	SID	N	T0	TMAX					

Example:

RANDT1	5	10	3.2	9.6					

Fields:

Field	Contents
SID	Random analysis set identification number. (Integer > 0)
N	Number of time lag intervals. (Integer > 0)
T0	Starting time lag. (Real ≥ 0.0)
TMAX	Maximum time lag. (Real > T0)

Remarks:

1. Time lag sets must be selected with the Case Control command RANDOM=SID.
2. At least one RANDPS or RANDPEX entry must be present with the same set identification number.
3. The time lags defined on this entry are given by

$$T_i = T0 + \frac{\text{TMAX} - T0}{N}(i - 1),\ i = 1, \text{N} + 2$$

4. See the *Simcenter Nastran Advanced Dynamic Analysis User's Guide* for a discussion of random analysis.

Related Topics

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RBAR

Rigid Bar

Defines a rigid bar with six degrees-of-freedom at each end.

Format:

1	2	3	4	5	6	7	8	9	10
RBAR	EID	GA	GB	CNA	CNB	CMA	CMB	ALPHA	

Example:

RBAR	5	1	2	234	123				

Fields:

Field	Contents
EID	Element identification number.
GA, GB	Grid point identification number of connection points. (Integer > 0)
CNA, CNB	Component numbers of independent degrees-of-freedom in the global coordinate system for the element at grid points GA and GB. See Remark 1 . (Integers 1 through 6 with no embedded blanks, or zero or blank.)
CMA, CMB	Component numbers of dependent degrees-of-freedom in the global coordinate system assigned by the element at grid points GA and GB. See Remarks 2 and 3. (Integers 1 through 6 with no embedded blanks, or zero or blank.)
ALPHA	Thermal expansion coefficient.

Remarks:

1. The total number of components in CNA and CNB must equal six; for example, CNA=1236, CNB=34. Furthermore, they must jointly be capable of representing any general rigid body motion of the element.
2. If both CMA and CMB are zero or blank, all of the degrees-of-freedom not in CNA and CNB will be made dependent; i.e., they will be made members of the m-set.
3. The m-set coordinates specified on this entry may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. Element identification numbers should be unique with respect to all other element identification numbers.

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5. Rigid elements, unlike MPCs, are not selected through the Case Control Section.
6. Forces of multipoint constraint may be recovered in all solution sequences, except SOL 129, with the MPCFORCES Case Control command.
7. Rigid elements are ignored in heat transfer problems.
8. See R-Type Elements" in the *Simcenter Nastran Element Library* for a discussion of rigid elements.
9. The RIGID case control command must be assigned to "LAGRAN" to use ALPHA for the calculation of thermal expansion. See the RIGID case control command for details.
10. The TEMP(LOAD) and TEMP(INIT) value used on RBAR, RROD, and RTRPLT elements is an average calculated from the grid point values. On RBE1, RBE2 and RBE3 elements, an average TEMP(LOAD) and TEMP(INIT) is calculated for each leg of the element using the values on the independent/dependent grid pairs such that each leg can have a different thermal strain if the temperatures vary at the grids.

    The rigid element thermal strains are calculated by
    $\epsilon_{thermal} = \alpha(AVGTEMP(LOAD) - AVGTEMP(INIT))$.

    If TEMP(LOAD) or TEMP(INIT) are not defined, they are assumed to be zero.

Remarks related to SOLs 402 and 414:

1. The RBAR only supports CNA or CNB equal to '123456' and CMA or CMB equal to '0' or blank.
2. Large rotations are taken into account when PARAM,LGDISP,1 or PARAM,LGSTRN,1 is specified.

Related Topics

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RBE1

Rigid Body Element, Form 1

Defines a rigid body connected to an arbitrary number of grid points.

Format:

1	2	3	4	5	6	7	8	9	10
RBE1	EID	GN1	CN1	GN2	CN2	GN3	CN3		
		GN4	CN4	GN5	CN5	GN6	CN6		
	"UM"	GM1	CM1	GM2	CM2	GM3	CM3		
		GM4	CM4	-etc.-	ALPHA				

Example:

RBE1	59	59	123	60	456				
	UM	61	246						

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
GNi	Grid points at which independent degrees-of-freedom for the element are assigned. (Integer > 0)
CNi	Independent degrees-of-freedom in the global coordinate system for the rigid element at grid point(s) GNi. See Remark 1 . (Integers 1 through 6 with no embedded blanks.)
"UM"	Indicates the start of the degrees-of-freedom belonging to m-set. (Character)
GMj	Grid points at which dependent degrees-of-freedom are assigned. (Integer > 0)
CMj	Dependent degrees-of-freedom in the global coordinate system at grid point(s) GMj. (Integers 1 through 6 with no embedded blanks.)
ALPHA	Thermal expansion coefficient. (Real; Default = 0.0)

Remarks:

1. The total number of components in CN1 to CN6 must equal six; for example, CN1=123, CN2=3, CN3=2, CN4=3. Furthermore, they must jointly be capable of representing any general rigid body motion of the element.

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2. The first continuation entry is not required if there are fewer than four GN points.
3. Dependent degrees-of-freedom assigned by one rigid element may not also be assigned dependent by another rigid element or by a multipoint constraint.
4. A degree-of-freedom cannot be both independent and dependent for the same element. However, both independent and dependent components can exist at the same grid point.
5. Element identification numbers should be unique with respect to all other element identification numbers.
6. Rigid elements, unlike MPCs, are not selected through the Case Control Section.
7. Forces of multipoint constraint may be recovered in all solution sequences, except SOL 129, with the MPCFORCES Case Control command.
8. Rigid elements are ignored in heat transfer problems.
9. See R-Type Elements in the *Simcenter Nastran Element Library* for a discussion of rigid elements.
10. The m-set coordinates specified on this entry may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
11. The RIGID case control command must be assigned to "LAGRAN" to use ALPHA for the calculation of thermal expansion. See the RIGID case control command for details.
12. The TEMP(LOAD) and TEMP(INIT) value used on RBAR, RROD, and RTRPLT elements is an average calculated from the grid point values. On RBE1, RBE2 and RBE3 elements, an average TEMP(LOAD) and TEMP(INIT) is calculated for each leg of the element using the values on the independent/dependent grid pairs such that each leg can have a different thermal strain if the temperatures vary at the grids.

    The rigid element thermal strains are calculated by
    $\epsilon_{thermal} = \alpha(\text{AVGTEMP(LOAD)} - \text{AVGTEMP(INIT)})$.

    If TEMP(LOAD) or TEMP(INIT) are not defined, they are assumed to be zero.

Related Topics

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RBE2

Rigid Body Element, Form 2

Defines a rigid body with independent degrees-of-freedom that are specified at a single grid point and with dependent degrees-of-freedom that are specified at an arbitrary number of grid points.

Format:

1	2	3	4	5	6	7	8	9	10
RBE2	EID	GN	CM	GM1	GM2	GM3	GM4	GM5	
	GM6	GM7	GM8	-etc.-	ALPHA				

Example:

RBE2	9	8	12	10	12	14	15	16	
	20								

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
GN	Identification number of grid point to which all six independent degrees-of-freedom for the element are assigned. (Integer > 0)
CM	Component numbers of the dependent degrees-of-freedom in the global coordinate system at grid points GMi. (Integers 1 through 6 with no embedded blanks.)
GMi	Grid point identification numbers at which dependent degrees-of-freedom are assigned. (Integer > 0)
ALPHA	Thermal expansion coefficient. (Real; Default = 0.0)

Remarks:

1. The components indicated by CM are made dependent (members of the m-set) at all grid points GMi.
2. Dependent degrees-of-freedom assigned by one rigid element may not also be assigned dependent by another rigid element or by a multipoint constraint.
3. Element identification numbers should be unique with respect to all other element identification numbers.
4. Rigid elements, unlike MPCs, are not selected through the Case Control Section.

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5. Forces of multipoint constraint may be recovered in all solution sequences, except SOL 129, with the MPCFORCES Case Control command.
6. Rigid elements are ignored in heat transfer problems.
7. See R-Type Elements in the *Simcenter Nastran Element Library* for a discussion of rigid elements.
8. The m-set coordinates specified on this entry may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
9. The RIGID case control command must be assigned to "LAGRAN" to use ALPHA for the calculation of thermal expansion. See the RIGID case control command for details.
10. The TEMP(LOAD) and TEMP(INIT) value used on RBAR, RROD, and RTRPLT elements is an average calculated from the grid point values. On RBE1, RBE2 and RBE3 elements, an average TEMP(LOAD) and TEMP(INIT) is calculated for each leg of the element using the values on the independent/dependent grid pairs such that each leg can have a different thermal strain if the temperatures vary at the grids.

    The rigid element thermal strains are calculated by
    $\epsilon_{thermal} = \alpha(AVGTEMP(LOAD) - AVGTEMP(INIT))$.

    If TEMP(LOAD) or TEMP(INIT) are not defined, they are assumed to be zero.

Remarks related to SOLs 402 and 414:

1. Large rotations are taken into account when PARAM,LGDISP,1 or PARAM,LGSTRN,1 is specified.

Related Topics

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RBE3

Interpolation Constraint Element

Defines the motion at a reference grid point as the weighted average of the motions at a set of other grid points.

Format:

1	2	3	4	5	6	7	8	9	10
RBE3	EID		REFGRID	REFC	WT1	C1	G1,1	G1,2	
	-etc.-	WT2	C2	G2,1	G2,2	-etc.-	WT3	C3	
	G3,1	G3,2	-etc.-	WT4	C4	G4,1	G4,2	-etc.-	
	"UM"	GM1	CM1	GM2	CM2	GM3	CM3		
		GM4	CM4	GM5	CM5	-etc.-			
	"ALPHA"	ALPHA							

Example:

RBE3	14		100	1234	1.0	123	1	3	
	5	4.7	1	2	4	6	5.2	2	
	7	8	9	5.1	1	15	16		
	UM	100	14	5	3	7	2		
	ALPHA	17.3E-6							

Fields:

Field	Contents
EID	Element identification number. Unique with respect to other rigid elements. (Integer > 0)
REFGRID	Reference grid point identification number. (Integer > 0)
REFC	Component numbers at the reference grid point. (Any of the Integers 1 through 6 with no embedded blanks.)
WTi	Weighting factor for components of motion on the following entry at grid points Gi,j. (Real)
Ci	Component numbers with weighting factor WTi at grid points Gi,j. (Any of the Integers 1 through 6 with no embedded blanks.)

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Field	Contents
Gi,j	Grid points with components Ci that have weighting factor WTi in the averaging equations. (Integer > 0)
"UM"	Indicates the start of the degrees-of-freedom belonging to the m-set. The default action is to assign only the components in REFC to the m-set. (Character)
GMi	Identification numbers of grid points with degrees-of-freedom in the m-set. (Integer > 0)
CMi	Component numbers of GMi to be assigned to the m-set. (Any of the Integers 1 through 6 with no embedded blanks.)
"ALPHA"	Indicates that the next field is the coefficient of thermal expansion. (Character)
ALPHA	Coefficient of thermal expansion. See Remark 9. (Real; Default = 0.0)

Remarks:

1. It is recommended that for most applications only the translation components 123 be used for Ci. An exception is the case where the Gi,j are colinear. A rotation component may then be added to one grid point to stabilize its associated rigid body mode for the element.
2. Blank spaces may be left at the end of a Gi,j sequence.
3. The default for "UM" should be used except in cases where the user wishes to include some or all REFC components in displacement sets exclusive from the m-set. If the default is not used for "UM":
   - The total number of components in the m-set (i.e., the total number of dependent degrees-of-freedom defined by the element) must be equal to the number of components in REFC (four components in the example).
   - The components specified after "UM" must be a subset of the components specified under REFC and (Gi,j, Ci).
   - The coefficient matrix $[R_m]$ described in "Multipoint Constraint Operations in SubDMAPs SEKR and SEMR2 and SEMRB" in the *Simcenter Nastran User's Guide* must be nonsingular. PARAM,CHECKOUT in SOLs 101 through 200 may be used to check for this condition.
4. Dependent degrees-of-freedom assigned by one rigid element may not also be assigned dependent by another rigid element or by a multipoint constraint.
5. Rigid elements, unlike MPCs, are not selected through the Case Control Section.
6. Forces of multipoint constraint may be recovered in all solution sequences, except SOL 129, with the MPCFORCES Case Control command.
7. Rigid elements are ignored in heat transfer problems.
8. The m-set coordinates specified on this entry may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
9. The RIGID case control command must be assigned to "LAGRAN" to use ALPHA for the calculation of thermal expansion. See the RIGID case control command for details.

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10. The TEMP(LOAD) and TEMP(INIT) value used on RBAR, RROD, and RTRPLT elements is an average calculated from the grid point values. On RBE1, RBE2 and RBE3 elements, an average TEMP(LOAD) and TEMP(INIT) is calculated for each leg of the element using the values on the independent/dependent grid pairs such that each leg can have a different thermal strain if the temperatures vary at the grids.

    The rigid element thermal strains are calculated by
    $\epsilon_{thermal} = \alpha(AVGTEMP(LOAD) - AVGTEMP(INIT))$.

    If TEMP(LOAD) or TEMP(INIT) are not defined, they are assumed to be zero.

Remarks related to SOLs 402 and 414:

1. UM, all GMi and CMi are ignored.
2. Large rotations are taken into account when PARAM,LGDISP,1 or PARAM,LGSTRN,1 is specified.
3. Only cases with REFC equal to 123 or 123456 or 456 are taken into account.
4. ALPHA is ignored.

Related Topics

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RCROSS

Cross-Power Spectral Density and Cross-Correlation Function Output

Defines a pair of response quantities for computing the cross-power spectral density and cross-correlation functions in random analysis.

Format:

1	2	3	4	5	6	7	8	9	10
RCROSS	SID	RTYPE1	ID1	COMP1	RTYPE2	ID2	COMP2	CURID	

Examples:

RCROSS	20	DISP	50	2	STRESS	150	8	4	

RCROSS	200	PRESS	101	1	DISP	50	2		

Fields:

Field	Contents
SID	Case control RCROSS identification number for cross-power spectral density and cross-correlation functions. (Integer>0)
RTYPEi	Type of response quantity. At least one field must be selected. See Remark 2. (Character or blank)
IDi	Element, grid, or scalar point identification number. (Integer > 0)
COMPi	Component code (item) identification number. See Remark 3. (Integer > 0)
CURID	Curve identification number. See Remark 5. (Integer > 0 or blank)

Remarks:

1. This entry is required for computing the cross-power spectral density and cross-correlation functions. SID must be selected with the case control command (RCROSS=SID). Fields RTYPE1, ID1, and COMP1 represent the first response quantity; fields RTYPE2, ID2, and COMP2 represent the second response quantity.
2. The keywords for field RTYPEi are listed as follows:

Keyword	Meaning
DISP	Displacement Vector

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VELO	Velocity Vector
ACCEL	Acceleration Vector
OLOAD	Applied Load Vector
SPCF	Single-point Constraint Force Vector
MPCF	Multi-point Constraint Force Vector
STRESS	Element Stress
STRAIN	Element Strain
FORCE	Element Force
PRESS	Acoustic Pressure

If either RTYPE1 or RTYPE2 is blank, then the blank field takes the default from the defined field.

3. For elements, the item code COMPi represents a component of the element stress, strain or force and is described in Tables “Element Stress-Strain Item Codes” and “Element Force Item Codes”. For an item having both a real and imaginary part, the code of the real part must be selected. This is required for computing both the cross-power spectral density function and cross-correlation function.

   For grid points, the item code is one of 1,2,3,4,5, and 6, which represent the mnemonics T1, T2, T3, R1, R2, and R3, respectively. For scalar points or PRESS, always use 1.

4. Elements defined as a laminate cannot be selected by the RCROSS entry. The RCROSSC entry should be used to request cross-power spectral density and cross-correlation function output for shell composites defined with the PCOMP and PCOMPG property entries, and solid element composites defined with the PCOMPS property entry.

5. Field CURID is optional. It is for your convenience to identify the output by using a single index.

Related Topics

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RCROSSC

Cross-Power Spectral Density and Cross-Correlation Function Output for Shell and Solid Element Composites

Defines a pair of response quantities for computing the cross-power spectral density and cross-correlation functions in random analysis for shell and solid element composites.

Format:

1	2	3	4	5	6	7	8	9	10
RCROSSC	SID	RTYPE1	ID1	COMP1	PLY1				
		RTYPE2	ID2	COMP2	PLY2			CURID	

Examples:

RCROSSC	20	DISP	50	2					
		STRESS	150	8	2			4	

RCROSSC	200	PRESS	101	1					
		STRESS	150	8	2			4	

Fields:

Field	Contents
SID	RCROSS case control command identification number. (Integer>0)
RTYPEi	Response quantity. (Character; For default behavior, see Remark 2)
IDi	Element, grid, or scalar point identification number. (Integer > 0; No default)
COMPi	Component (item) code identification number. See Remark 3. (Integer > 0; No default)
PLYi	Ply number. (Integer ≥ 0; For default behavior, see Remark 4)
CURID	Optional curve identification number. See Remark 5. (Integer > 0; No default)

Remarks:

1. The RCROSSC entry is used to request cross-power spectral density and cross-correlation function output for shell composites defined with the PCOMP and PCOMPG property entries, and solid element composites defined with the PCOMPS property entry.
2. The keywords for field RTYPEi are listed as follows:

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Keyword	Meaning
DISP	Displacement Vector
VELO	Velocity Vector
ACCEL	Acceleration Vector
OLOAD	Applied Load Vector
SPCF	Single-point Constraint Force Vector
MPCF	Multi-point Constraint Force Vector
STRESS	Element Stress
STRAIN	Element Strain
FORCE	Element Force
PRESS	Acoustic Pressure

If either the RTYPE1 or RTYPE2 field is blank, then the blank field defaults to the response quantity listed in the defined field.

3. For elements, the component (item) code COMPi represents a component of the element stress, strain or force as described in the “Element Stress (or Strain) Item Codes” and “Element Force Item Codes” tables. For an item having both a real and imaginary part, the code of the real part must be selected. This is required for computing both the cross-power spectral density function and cross-correlation function.

   For grid points, the item code is one of 1,2,3,4,5, and 6, which represent T1, T2, T3, R1, R2, and R3, respectively.

   For scalar points or PRESS, always use 1.

4. PLY1 and PLY2 cannot both be zero or blank. If it is desired to have both zero, use the RCROSS bulk entry.

   For a non-composite element, grid point, or scalar point, specify PLYi = 0 or leave the PLYi field blank. For a non-composite element, grid point, or scalar point, if PLYi > 0, the ply number specification is ignored.

   For a composite element, if PLYi = 0 or the PLYi field is blank or PLYi is greater than the actual number of plies, the ply number specification is ignored.

5. To identify the output with a single index, specify the index in the CURID field.

Related Topics

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RDMODES

Solution options for RDMODES

Solution options for RDMODES.

Format:

1	2	3	4	5	6	7	8	9	10
RDMODES	SID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4							

Example:

RDMODES	110								

RDMODES	110	RDSCALE	8.0						

Fields:

Field	Contents
SID	RDMODES set selection. See Remark 1. (Integer>0)
PARAM*i*	Name of RDMODES parameter. Allowable names are given in the parameter listing below. (Character)
VALUE*i*	Value of the parameter. (Real, Integer, or Character)

RDMODES Parameters:

Table 17-123:

Name	Description
RDSCALE	A factor to adjust the frequency range of the RDMODES substructures. (Real>0.0, Default=5.0) Increasing RDSCALE increases the solution accuracy, but at the cost of increased computation time. In most circumstances, values in the range of 2.0 to 10.0 are acceptable.
OLDSEQ	Used to specify the method of creating the substructures. (Integer>0; Default=210)

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Name	Description

OLDSEQ	Method description
10	Metis with super-nodal and grid compressions.
11	MLV with super-nodal and grid compressions.
110	Metis with super-nodal compression.
111	MLV with super-nodal compression.
210	Metis with grid compression.
211	MLV with grid compression.

RDQSOL — Defines the method for calculating the reduced eigenvalues. Applicable to old RDMODES only. (Character; Default=LANC)

LANC: The Lanczos method is used. It is suitable for most models.

INCO: The in-core dense eigensolver is used. It is beneficial for dense models requiring a large number of modes.

VERSION — Selects either the new RDMODES (beginning with Simcenter Nastran 2020.1) or old RDMODES (prior to Simcenter Nastran 2020.1). See Remark 2. (Character; Default=NEW)

NEW: The new RDMODES method is used.

OLD: The old RDMODES method is used.

Remarks:

1. The SID defined on the RDMODES bulk entry must be referenced by the METHOD=SID case control command. An EIGRL bulk entry with the same SID as the RDMODES entry is also required.
2. Specifying VERSION=NEW (the default) is recommended. However, VERSION=NEW currently does not support the parameter ODS, SOL 401, or contact conditions. For these cases specify VERSION=OLD.
3. For the new RDMODES (VERSION=NEW), SMP is beneficial, and a combination of SMP and DMP is beneficial. For many modes requests, SMP is preferred over DMP. When DMP has been specified, a combination of SMP and DMP is suggested, particularly on a workstation or a cluster of multi-processor nodes.

Related Topics

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RELEASE

Superelement Boundary Grid Point Release

Defines degrees-of-freedom for superelement exterior grid points that are not connected to the superelement.

Format:

1	2	3	4	5	6	7	8	9	10
RELEASE	SEID	C	G1	G2	G3	G4	G5	G6	
	G7	G8	-etc.-						

Example:

RELEASE	15	456	3	7	11	2	156	9	
	152	162							

Alternate Formats and Example:

RELEASE	SEID	C	G1	"THRU"	G2				
RELEASE	6	2	15	THRU	127				

RELEASE	SEID	C	"ALL"						
RELEASE	127	156	ALL						

Fields:

Field	Contents
SEID	Superelement identification number. (Integer > 0)
C	Component number. (Any unique combination of the Integers 1 through 6 with no embedded blanks.)
Gi	Grid point identification numbers. (Integer > 0, "THRU", or "ALL"; For THRU option, G1 < G2.)

Remarks:

1. A grid point referenced on this entry must be an exterior grid point of the superelement referenced on the entry.

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2. In the first alternate format, all grid points in the sequence G1 through G2 are not required to be exterior grid points. Such grid points will collectively produce a warning message but will otherwise be ignored.
3. If the “ALL” option is used, all exterior grid points are released for the referenced degrees-of-freedom.
4. The RELEASE entry is applicable to only the superelement solution sequences (SOLs 101 through 200). It may not reference the residual structure (SEID=0).
5. This entry is not supported for partitioned superelements.

Related Topics

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RLCYC1

Frequency Response Dynamic Excitation for Cyclic Symmetry, Form 1

Defines a frequency-dependent dynamic load of the form:

$$P(f) = A\left[\mathrm{S_{Rk}}\cos\left(k\left(n-1\right)2\pi/N\right) + \mathrm{S_{Ik}}\sin\left(k\left(n-1\right)2\pi/N\right)\right]\left(C(f)+iD(f)\right)e^{i\theta}$$

for use in a SOL 401 frequency response subcase with cyclic symmetry or a SOL 414,111 harmonic response solution with cyclic symmetry, where

k is the harmonic number,

n is the sector number,

and *N* is the total number of sectors. The other terms are defined in the field descriptions below.

Format:

1	2	3	4	5	6	7	8	9	10
RLCYC1	SID	EXCITEID		DPHASE	TC	TD	TYPE	TS	

Examples:

RLCYC1	2	3		90.0	1	2		12	

Fields:

Field	Contents
SID	Load set identification number of the load set referenced by the DLOAD case control command. (Integer > 0) See Remarks 1 and 2.
EXCITEID	Identification number of a static load set that selects the degree-of-freedom to apply the excitation and the corresponding scale factor, *A*. (Integer > 0) See SOL 414 Remarks 2 and 3.
DPHASE	Phase angle, <*θ*>, in degrees. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 3). If real entry, the software applies the value of <*θ*> to all degree-of-freedom in EXCITEID entry. See SOL 414 Remark 4.

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Field	Contents
	If integer entry, identification number of DPHASE entries that define values of <θ> for individual degree-of-freedom in EXCITEID entry.
TC	<*C(f)*> (Real or Integer ≥ 0 or blank; for default behavior, see Remark 3). If real entry, value of <*C(f)*> used over all frequencies for all degree-of-freedom in EXCITEID entry. If integer entry, identification number of a TABLEDi entry that defines <*C(f)*> for all degree-of-freedom in EXCITEID entry.
TD	Scale factor for the imaginary part <*D(f)*> (Real or Integer ≥ 0 or blank; for default behavior, see Remark 3) If real entry, value of <*D(f)*> used over all frequencies for all degree-of-freedom in EXCITEID entry. If integer entry, identification number of a TABLEDi entry that defines <*D(f)*> for all degree-of-freedom in EXCITEID entry.
TYPE	Type of dynamic excitation. (Integer or character; Default = 0) See Remarks 4 and 5. See SOL 414 Remarks 2 and 3.
TS	Table ID of a TABLEH2 bulk entry defining the scale factors S_{Rk} and S_{Ik} at each harmonic *k*. See Remark 6 and SOL 414 Remark 6. (Integer > 0; no default)

Remarks:

1. SID must be unique for all RLCYC1 and RLCYC2 bulk entries. Multiple RLCYC1 and RLCYC2 excitations may be combined using a DLOAD bulk entry.
2. Dynamic excitation sets must be selected with DLOAD = SID in the case control. The DLOAD = SID command that selects a frequency-dependent load set must be defined in the frequency response subcase.
3. If any of the DPHASE, TC, and TD fields are blank or defined with a zero (either the integer value 0 or the real value 0.0), the corresponding value for <θ>, <*C(f)*>, and <*D(f)*> used by the software is 0.0 (real value).
4. The type of dynamic excitation is specified by the TYPE field according to the following table:

TYPE	Type of dynamic excitation
0, L, LO, LOA, or LOAD	Applied load (force or moment) (Default) See SOL 414 Remarks 2 and 3.
1, D, DI, DIS, or DISP	Enforced displacement defined with the SPCD bulk entry.

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TYPE	Type of dynamic excitation
2, V, VE, VEL, or VELO	Enforced velocity with the SPCD bulk entry.
3, A, AC, ACC, or ACCE	Enforced acceleration with the SPCD bulk entry.

5. For SOL 401 and TYPE = 0, EXCITEID may reference DAREA, FORCE, FORCE1, FORCE2, GRAV, MOMENT, MOMENT1, MOMENT2, PLOAD, PLOAD1, PLOAD2, PLOAD4, PLOADE1, PLOADX1, RFORCE, and RFORCE1 bulk entries.

   See SOL 414 Remark 2.

6. The software ignores a harmonic requested for the CYCLIC modes subcase greater than N/2. Similarly, the software ignores any scale factors S_{Rk} and S_{Ik} defined on the TABLEH2 bulk entry where the harmonic (k) is greater than N/2. This is because the harmonics from k=0 to N/2 for an even number of sectors N, and from k=0 to (N-1)/2 for an odd number of sectors represent the solution for all harmonics from 0 to N. See Frequency-dependent cyclic loads in the Multistep User's Guide for more information.

Remarks for SOL 414

1. RLCYC1 and RLCYC2 are supported in SOL 414,111 Harmonic Response solutions, which must include a nonlinear preload subcase.
2. The EXCITEID may reference the FORCE and PLOAD bulk entries. It does not support enforced motion loads defined with the SPCD bulk entry.
3. TYPE must be 0 or empty.
4. DPHASE must be defined as a real value.
5. SOL 414 Harmonic Response solutions use a single harmonics index <*k*>.
6. You request the solution harmonics using the HARMONICS and the SET case control commands. The software ends with a fatal error if you request a harmonic index larger than N/2 for an even number of segments N or larger than (N-1)/2 for an odd number of segments. In addition, the software ignores any scale factors S_{Rk} and S_{Ik} defined on the TABLEH2 bulk entry where the harmonic (k) is larger than N/2 for an even number of segments or larger than (N-1)/2 for an odd number of segments.

Related Topics

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RLCYC2

Frequency Response Dynamic Excitation for Cyclic Symmetry, Form 2

Defines a frequency-dependent dynamic load of the form:

$$P(f) = A\left[\mathrm{S_{Rk}}\cos\left(k\left(n-1\right)2\pi/N\right) + \mathrm{S_{Ik}}\sin\left(k\left(n-1\right)2\pi/N\right)\right]B(f)e^{i(\phi(f)+\theta)}$$

for use in a SOL 401 frequency response subcase with cyclic symmetry or a SOL 414,111 harmonic response solution with cyclic symmetry, where

k is the harmonic number,

n is the sector number,

and *N* is the total number of sectors. The other terms are defined in the field descriptions below.

Format:

1	2	3	4	5	6	7	8	9	10
RLCYC2	SID	EXCITEID		DPHASE	TB	TP	TYPE	TS	

Examples:

RLCYC2	2	3		30.0	1	2		12	

Fields:

Field	Contents
SID	Load set identification number of the load set referenced by the DLOAD case control command. (Integer > 0) See Remarks 1 and 2. xid2210474
EXCITEID	Identification number of a static load set that selects the degree-of-freedom to apply the excitation and the corresponding scale factor, *A*. (Integer > 0) See SOL 414 Remarks 2 and 3.
DPHASE	Phase angle, <*θ*>, in degrees. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 3).

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Field	Contents
	If real entry, the software applies the value of <θ> to all degree-of-freedom in EXCITEID entry. See SOL 414 Remark 4. If integer entry, identification number of DPHASE entries that define values of <θ> for individual degree-of-freedom in EXCITEID entry.
TB	Scale factor for the magnitude part <*B(f)*> that references the TABLEDi bulk entry of frequency-dependent scale factors. (Real or Integer ≥ 0; No default) If real entry, value of <*B(f)*> used over all frequencies for all degrees of freedom in EXCITEID entry. If integer entry, identification number of a TABLEDi entry that defines <*B(f)*> for all degrees of freedom in the EXCITEID entry.
TP	<ϕ*(f)*> in degrees. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 3). If real entry, value of <ϕ*(f)*> used over all frequencies for all degrees of freedom in EXCITEID entry. If integer entry, identification number of a TABLEDi entry that defines <ϕ*(f)*> for all degrees of freedom in EXCITEID entry.
TYPE	Type of dynamic excitation. (Integer or character; Default = 0) See Remarks 4 and 5. See SOL 414 Remarks 2 and 3.
TS	Table ID of a TABLEH2 bulk entry defining the scale factors S_{Rk} and S_{Ik} at each harmonic *k*. See Remark 6 and SOL 414 Remark 6. (Integer > 0; no default)

Remarks:

1. SID must be unique for all RLCYC1 and RLCYC2 bulk entries. Multiple RLCYC1 and RLCYC2 excitations may be combined using a DLOAD bulk entry.
2. Dynamic excitation sets must be selected with DLOAD = SID in the case control. The DLOAD = SID command that selects a frequency-dependent load set must be defined in the frequency response subcase.
3. If either of the DPHASE or TP fields are blank or defined with a zero (either the integer value 0 or the real value 0.0), the corresponding value for <θ> or <ϕ*(f)*> used by the software is 0.0 (real value).
4. The type of dynamic excitation is specified by the TYPE field according to the following table:

TYPE	Type of dynamic excitation
0, L, LO, LOA, or LOAD	Applied load (force or moment) (Default) See SOL 414 Remarks 2 and 3.
1, D, DI, DIS, or DISP	Enforced displacement defined with the SPCD bulk entry.

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TYPE	Type of dynamic excitation
2, V, VE, VEL, or VELO	Enforced velocity with the SPCD bulk entry.
3, A, AC, ACC, or ACCE	Enforced acceleration with the SPCD bulk entry.

5. For SOL 401 and TYPE = 0, EXCITEID may reference DAREA, FORCE, FORCE1, FORCE2, GRAV, MOMENT, MOMENT1, MOMENT2, PLOAD, PLOAD1, PLOAD2, PLOAD4, PLOADE1, PLOADX1, RFORCE, and RFORCE1 bulk entries.

   See SOL 414 Remark 2.

6. The software ignores a harmonic requested for the CYCLIC modes subcase greater than N/2. Similarly, the software ignores any scale factors S_{Rk} and S_{Ik} defined on the TABLEH2 bulk entry where the harmonic (k) is greater than N/2. This is because the harmonics from k=0 to N/2 for an even number of sectors N, and from k=0 to (N-1)/2 for an odd number of sectors represent the solution for all harmonics from 0 to N. See Frequency-dependent cyclic loads in the Multistep User's Guide for more information.

Remarks for SOL 414

1. RLCYC1 and RLCYC2 are supported in SOL 414,111 Harmonic Response solutions, which must include a nonlinear preload subcase.
2. The EXCITEID may reference the FORCE and PLOAD bulk entries. It does not support enforced motion loads defined with the SPCD bulk entry.
3. TYPE must be 0 or empty.
4. DPHASE must be defined as a real value.
5. SOL 414 Harmonic Response solutions use a single harmonics index <*k*>.
6. You request the solution harmonics using the HARMONICS and the SET case control commands. The software ends with a fatal error if you request a harmonic index larger than N/2 for an even number of segments N or larger than (N-1)/2 for an odd number of segments. In addition, the software ignores any scale factors S_{Rk} and S_{Ik} defined on the TABLEH2 bulk entry where the harmonic (k) is larger than N/2 for an even number of segments or larger than (N-1)/2 for an odd number of segments.

Related Topics

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RFORCE

Rotational Force

Defines a static loading condition due to an angular velocity and/or acceleration.

Format:

1	2	3	4	5	6	7	8	9	10
RFORCE	SID	G	CID	A	R1	R2	R3	METHOD	
	RACC	MB							

Example:

RFORCE	2	5		-6.4	0.0	0.0	1.0	2	
	1.0								

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Point location through which the rotation vector acts. The location can be defined with the identification number of a GRID or POINT bulk entry. (Integer ≥ 0; Default = 0 which is the origin of the coordinate system defined in CID)
CID	Coordinate system defining the components of the rotation vector. See Remark 17. (Integer ≥ 0, Default = 0)
A	Scale factor of the angular velocity in revolutions per unit time. (Real; No default)
R1, R2, R3	Rectangular components of the rotation vector $\vec{R}$. The vector defined will pass through point G. (Real; $R1^2 + R2^2 + R3^2 > 0.0$; Defaults R1 = 1.0, R2 = 0.0, R3 = 0.0)
METHOD	Method used to compute centrifugal forces due to angular velocity. For angular acceleration, see Remarks 7 and 20. (Integer = 1 or 2; Default = 1)
RACC	Scale factor of the angular acceleration in revolutions per unit time squared. (Real; Default = 0.0)
MB	Indicates whether the CID coordinate system is defined in the main Bulk Data Section (MB = −1) or the partitioned superelement Bulk Data Section (MB = 0). Coordinate systems referenced in the main Bulk Data Section are considered stationary with respect to the assembly basic coordinate system. See Remark 16. (Integer; Default = 0)

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Remarks:

1. In **Figure 17-167**, the force vector at point Gi is given by

$$\left\{\vec{F}\right\}_i = -[m]_i \left[\vec{\omega} \times \left(\vec{\omega} \times \left(\vec{r}_i - \vec{r}_a\right)\right) + \vec{\alpha} \times \left(\vec{r}_i - \vec{r}_a\right)\right]$$

where $[m]_i$ is a 3 × 3 translational mass matrix at point Gi.

2. $\left(A \cdot \vec{R}\right)$ and $\left(RACC \cdot \vec{R}\right)$ have the units of revolutions per unit time and revolutions per unit time squared, respectively. The software multiplies each by 2π to compute ω and α:

Angular velocity	$\vec{\omega} = 2\pi\left(A \cdot \vec{R}\right) = \frac{(radians)}{(unit\,time)}$
Angular acceleration	$\vec{\alpha} = 2\pi\left(RACC \cdot \vec{R}\right) = \frac{(radians)}{(unit\,time)^2}$

Note: The equation for $\left\{\vec{F}\right\}_i$ will have additional terms if the mass is offset and METHOD = 1 is selected.

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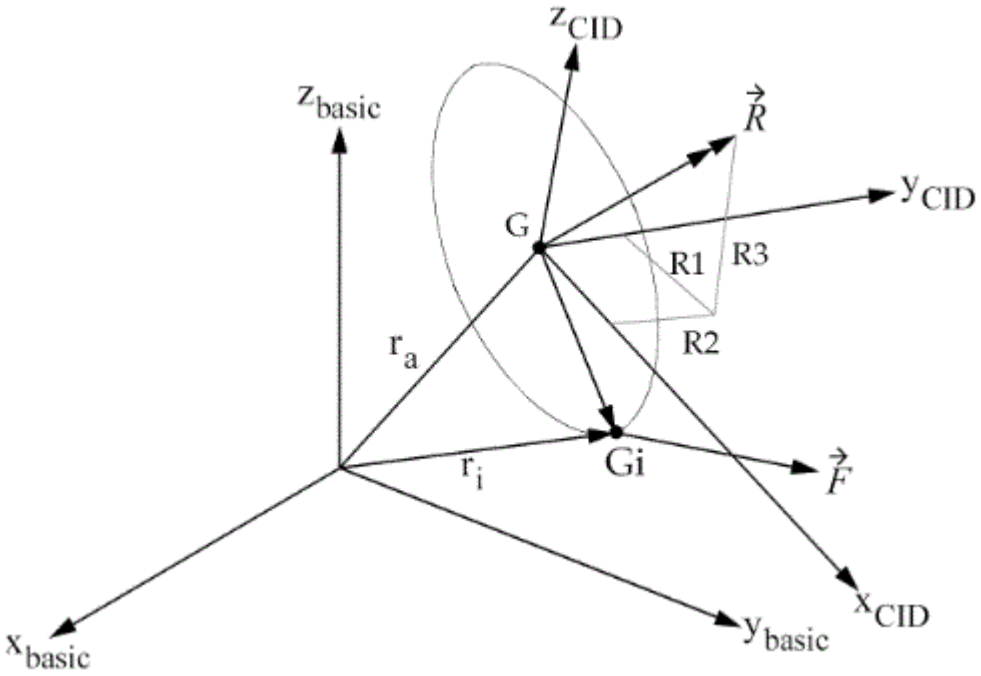
Figure 17-167: RFORCE Vector at Point Gi



3. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
4. G = 0 signifies that the rotation vector acts through the origin of the basic coordinate system.
5. CID = 0 (Default) signifies that the rotation vector is defined in the basic coordinate system.
6. The load vector generated by this entry can be printed with an OLOAD command in the Case Control Section.
7. METHOD = 1 yields correct results only when there is no coupling in the mass matrix. This occurs when the lumped mass option is used with or without the ZOFFS option (see the CQUAD4 entry for a description of ZOFFS).

   METHOD = 2 yields correct results for lumped or coupled mass matrix only if the ZOFFS option is not used. The acceleration terms due to the mass offset (X1, X2, X3) on the CONM2 entry are not computed with METHOD = 2. All the possible combinations of mass matrices and offset and the correct method to be used are shown below.

	No Offset	Offset
Lumped	METHOD = 1 or METHOD = 2	METHOD = 1
Coupled	METHOD = 2	Neither

8. In cyclic symmetry analyses, the T3 axis of the basic coordinate system must be coincident with the axis of symmetry. In the DIH type of cyclic symmetry, the T1 axis also must be parallel to side 1 of segment 1R of the model.

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9. For superelement analysis, G should reference a residual structure point that is exterior to all superelements. If it is not exterior to a superelement, then centrifugal loads will not be generated for that superelement. However, in cyclic analysis, User Fatal Message 4347 will be issued.

10. In a geometric nonlinear static analysis (SOL 106 and 401 when PARAM LDGISP is set to +1), this type of loading is a follower force type of loading. However, the orientation of coordinate system CID is not updated.

11. In nonlinear static solutions when there is more than one increment (INC) specified on the NLPARM entry for a given subcase, the load vector resulting from the RFORCE input (and not the angular velocity vector) is scaled linearly. This means that loading by increments in the angular velocity can only be achieved by having subcases where the RFORCE loading is applied in a single increment.

12. The continuation entry is optional.

13. Forces due to angular acceleration (RACC) are computed with METHOD = 2 even if METHOD = 1 is specified.

14. Loads derived from this entry do not include effects due to mass specified for scalar points.

15. The follower force effects due to loads from this entry are included in the stiffness in all linear solution sequences that calculate a differential stiffness. The solution sequences are SOLs 103, 105, 107 to 112, 115 and 116 (see also the parameter **FOLLOWK**). Follower force effects are included in the force balance in the nonlinear static and nonlinear transient dynamic solution sequences, SOLs 106, 129, 153, and 159, if geometric nonlinear effects are turned on with PARAM,LGDISP,1. The follower force stiffness is included in the nonlinear static solution sequences (SOLs 106 and 153) but not in the nonlinear transient dynamic solution sequences (SOLs 129 and 159).

    For SOL 401, follower force effects are included in the force balance in the nonlinear static solution if geometric nonlinear effects are turned on with PARAM,LGDISP,1. For additional information, see the NLCNTL bulk entry.

16. The coordinate systems in the main Bulk Data Section are defined relative to the assembly basic coordinate system which is fixed. This feature is useful when a superelement defined by a partitioned Bulk Data Section is rotated or mirrored and the gravity load is more conveniently defined in terms of coordinates which are fixed.

17. If CID is not a rectangular coordinate system, RFORCE will treat it as if it were and unexpected answers may result.

18. Follower force stiffness (param,followk,yes) is supported for method 2 only.

19. When you reference RFORCE or RFORCE1 entries on a LOAD entry, because centrifugal forces are not linear in angular velocity and therefore cannot be superposed, reference only a single RFORCE or RFORCE1 entry to define the angular velocity.

    For example, both of the following RFORCE specifications appear to define the same angular velocity.

```
RFORCE, 11, 999, .159, 1., 0., 0.
RFORCE, 12, 999, .159, 0., 1., 0.
RFORCE, 13, 999, .159, 0., 0., 1.
```

```
RFORCE, 11, 999, .159, 1., 1., 1.
```


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However, if you reference the three RFORCE entries on a LOAD entry, the centrifugal force that the software calculates is different from the correct centrifugal force that the software calculates if you reference the single RFORCE entry on the LOAD entry.

20. SOL 401 always uses METHOD = 2. As a result, the METHOD field is ignored by SOL 401.

21. In SOL 401, if a rotational force is defined with the RFORCE or RFORCE1 entry, and spin softening is requested with the SPINK parameter on the NLCNTL entry, if the rotation center is defined with a grid point which is not used in the connectivity of any element, all translational DOF for this grid point must be constrained.

22. In SOL 401, loads selected with the LOAD=n case control command are time-unassigned. Time-unassigned loads are ramped in a subcase when the LVAR parameter on the NLCNTL entry is set to RAMP (default). The following rules apply to the time-unassigned and ramped RFORCE and RFORCE1 entries.

- The RFVAR parameter is available on the NLCNTL bulk entry to change how the software interpolates the ramping load factor. See SOL 401 - RFORCE and RFORCE1 scaling in the *Multi-Step Nonlinear User's Guide* for details.
- RFORCE and RFORCE1 entries cannot be defined in the same subcase.
- If multiple RFORCE1 entries are defined in a subcase, the grids associated with each RFORCE1 entry must be mutually exclusive.
- The rotational load vector direction defined with the R1, R2, R3 fields and the coordinate system selected with the CID field cannot change from subcase to subcase.
- The scale factors A and RACC can change, and RFORCE and RFORCE1 entries can be added or removed, as long as the rotational load vectors and coordinate systems do not change.

23. In SOL 402, loads selected with the LOAD=n case control command are time-unassigned and are ramped in a subcase. When you define a time-unassigned rotational inertia load with the RFORCE or RFORCE1 entries, the RFVAR parameter is available on the NLCNTL2 bulk entry to change how the software interpolates the ramping load factor. See SOL 402 - RFORCE and RFORCE1 scaling in the *Multi-Step Nonlinear User's Guide* for details.

24. When RFORCE or RFORCE1 entries are referenced by the EXCITEID field on a TLOAD1 entry, the software uses the data on the associated TABLEDi to scale the computed force (F) in all solutions except for SOLs 401, 402, and 414.

- For SOL 401, when RFORCE or RFORCE1 entries are referenced by the EXCITEID field on a TLOAD1 entry, the software checks the RFVAR parameter on the NLCNTL bulk entry. See SOL 401 - RFORCE and RFORCE1 scaling in the *Multi-Step Nonlinear User's Guide* for details.
- For SOLs 402 and 414, when RFORCE or RFORCE1 entries are referenced by the EXCITEID field on a TLOAD1 entry, the data on the associated TABLEDi scale the angular velocity (ω) and acceleration (α), which are used to compute an inertia force in the equation $F = -[m] [\omega \times (\omega \times r)) + \alpha \times r]$. Since ω is squared in the force computation, the resulting scaling is not linearly related to the computed force (F).

25. For SOL 401, beginning with Simcenter Nastran 2021.2, a new body-load computation method is available for RFORCE and RFORCE1 loads. With this method, the load vector is computed on an element basis rather than a grid point basis. The resulting applied load is the same for the element basis and the previous grid point basis. With the new method, for the force vector described in

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Remark 1, $[m]_i$ becomes the element mass matrix and $\{F\}_i$ becomes the element contribution to the load vector.

The motivation for this change is related to a new grid point force output format requested with the GPFORCE case control command. The system cell 771 is available to change both the body-load computation method and the GPFORCE output format to the pre-Simcenter Nastran 2021.2 behavior. See **system cell 771**.

26. For SOL 401, if you want to add or change a rotational load in a restart run, you cannot add a new grid point to define the location of a new rotation vector. The software requires that the geometry information (grids and elements) in the restart run must be consistent with the initial run. As a result, you can use the POINT entry with the RFORCE and RFORCE1 bulk entries in a SOL 401 restart solution. A POINT entry can be added in the restart run even though it did not exist in the initial run.

Remarks related to SOLs 402 and 414:

1. METHOD and MB are ignored.

2. To apply rotational force with constant magnitude, SID is selected by case control command LOAD=SID for both static and transient analyses.

3. To apply a time-dependent rotational load, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by case control command DLOAD.

4. Only one RFORCE can be applied in an analysis.

5. CID must be a rectangular coordinate system.

6. For SOL 402 and 414, scaling is done on the angular velocity (A) and the angular acceleration (RACC). The equivalent force is not computed ahead of time but in the solver.

7. For SOL 402, if you want to add or change a rotational load in a restart run, you cannot add a new grid point to define the location of a new rotation vector. The software requires that the geometry information (grids and elements) in the restart run must be consistent with the initial run. As a result, you can use the POINT entry with the RFORCE and RFORCE1 bulk entries in a SOL 401 restart solution. A POINT entry can be added in the restart run even though it did not exist in the initial run.

Related Topics

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RFORCE1

Rotational Force with repeated SID

Defines a static loading condition due to an angular velocity and/or acceleration.

Format:

1	2	3	4	5	6	7	8	9	10
RFORCE1	SID	G	CID	A	R1	R2	R3	METHOD	
	RACC	MB	GROUPID						

Example:

RFORCE1	2	5		-6.4	0.0	0.0	1.0	2	
	1.0								

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Point location through which the rotation vector acts. The location can be defined with the identification number of a GRID or POINT bulk entry. (Integer ≥ 0; Default = 0 which is the origin of the coordinate system defined in CID)
CID	Coordinate system defining the components of the rotation vector. See Remark 17. (Integer ≥ 0, Default = 0)
A	Scale factor of the angular velocity in revolutions per unit time. (Real; No default)
R1, R2, R3	Rectangular components of rotation vector $\vec{R}$. The vector defined will pass through point G. (Real; $R1^2 + R2^2 + R3^2 > 0.0$; Defaults R1 = 1.0, R2 = 0.0, R3 = 0.0)
METHOD	Method used to compute centrifugal forces due to angular velocity. For angular acceleration, see Remarks 7 and 20. (Integer = 1 or 2; Default = 2)
RACC	Scale factor of the angular acceleration in revolutions per unit time squared. (Real; Default = 0.0)
MB	Indicates whether the CID coordinate system is defined in the main Bulk Data Section (MB = –1) or the partitioned superelement Bulk Data Section (MB = 0). Coordinate

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Field	Contents
	systems referenced in the main Bulk Data Section are considered stationary with respect to the assembly basic coordinate system. See Remark 16. (Integer; Default = 0)
GROUPID	Group identification number. The GROUP entry referenced in the GROUPID field selects the grid points to which the load is applied. See Remark 27. (Integer > 0)

Remarks:

1. The RFORCE and RFORCE1 entries have the following difference.
   - The RFORCE entry applies loads that result from rotational velocities and accelerations to all grid points.
   - The RFORCE1 entry applies loads that result from rotational velocities and accelerations to select grid points. The grid points are selected using a GROUP bulk entry.
2. In Figure 17-168, the force vector at point Gi is given by

$$\left\{\vec{F}\right\}_i = -[m]_i \left[\vec{\omega} \times \left(\vec{\omega} \times \left(\vec{r}_i - \vec{r}_a\right)\right) + \vec{\alpha} \times \left(\vec{r}_i - \vec{r}_a\right)\right]$$

where $[m]_i$ is a 3 × 3 translational mass matrix at point Gi.

$\left(A \cdot \vec{R}\right)$ and $\left(RACC \cdot \vec{R}\right)$ have the units of revolutions per unit time and revolutions per unit time squared, respectively. The software multiplies each by 2π to compute ω and α:

Angular velocity	$\vec{\omega} = 2\pi\left(A \cdot \vec{R}\right) = \frac{(radians)}{(unit\,time)}$
Angular acceleration	$\vec{\alpha} = 2\pi\left(RACC \cdot \vec{R}\right) = \frac{(radians)}{(unit\,time)^2}$

Note: The equation for $\left\{\vec{F}\right\}_i$ will have additional terms if the mass is offset and METHOD = 1 is selected.

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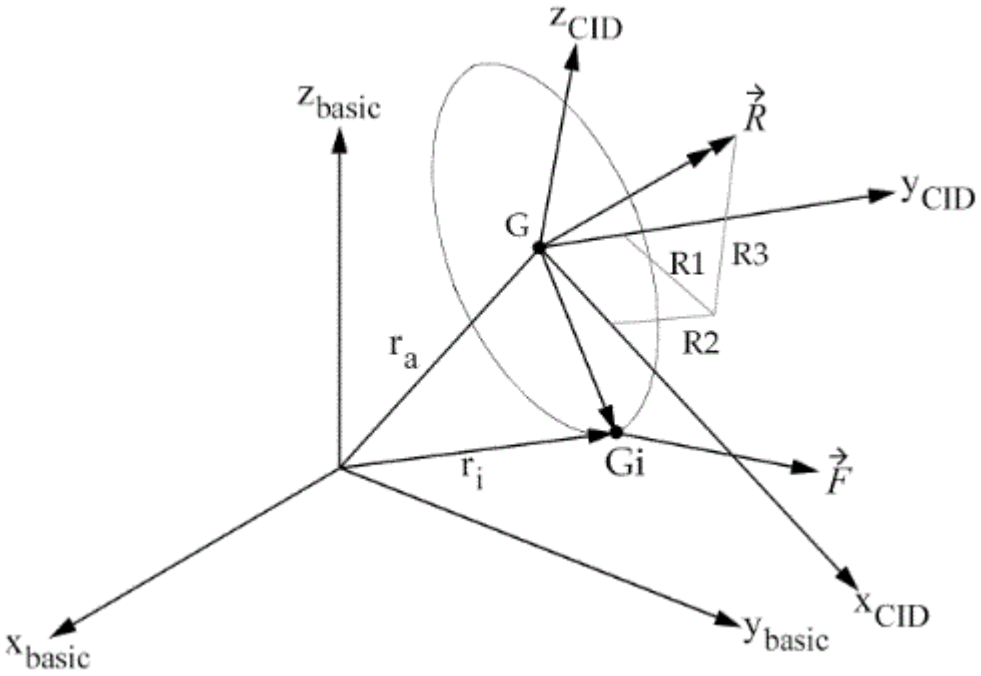
Figure 17-168: RFORCE1 Vector at a Point



3. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
4. G = 0 signifies that the rotation vector acts through the origin of the basic coordinate system.
5. CID = 0 (Default) signifies that the rotation vector is defined in the basic coordinate system.
6. The load vector generated by this entry can be printed with an OLOAD command in the Case Control Section.
7. METHOD = 1 yields correct results only when there is no coupling in the mass matrix. This occurs when the lumped mass option is used with or without the ZOFFS option (see the CQUAD4 entry for a description of ZOFFS).

   METHOD = 2 yields correct results for lumped or coupled mass matrix only if the ZOFFS option is not used. The acceleration terms due to the mass offset (X1, X2, X3) on the CONM2 entry are not computed with METHOD = 2. All the possible combinations of mass matrices and offset and the correct method to be used are shown below.

	No Offset	Offset
Lumped	METHOD = 1 or METHOD = 2	METHOD = 1
Coupled	METHOD = 2	Neither

8. In cyclic symmetry analyses, the T3 axis of the basic coordinate system must be coincident with the axis of symmetry. In the DIH type of cyclic symmetry, the T1 axis also must be parallel to side 1 of segment 1R of the model.

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9. For superelement analysis, G should reference a residual structure point that is exterior to all superelements. If it is not exterior to a superelement, then centrifugal loads will not be generated for that superelement. However, in cyclic analysis, User Fatal Message 4347 will be issued.

10. In a geometric nonlinear static analysis (SOL 106 and 401 when PARAM LDGISP is set to +1), this type of loading is a follower force type of loading. However, the orientation of coordinate system CID is not updated.

11. In nonlinear static solutions when there is more than one increment (INC) specified on the NLPARM entry for a given subcase, the load vector resulting from the RFORCE1 input (and not the angular velocity vector) is scaled linearly. This means that loading by increments in the angular velocity can only be achieved by having subcases where the RFORCE1 loading is applied in a single increment.

12. The continuation entry is optional.

13. Forces due to angular acceleration (RACC) are computed with METHOD = 2 even if METHOD = 1 is specified.

14. Loads derived from this entry do not include effects due to mass specified for scalar points.

15. The follower force effects due to loads from this entry are included in the stiffness in all linear solution sequences that calculate a differential stiffness. The solution sequences are SOLs 103, 105, 107 to 112, 115 and 116 (see also the parameter **FOLLOWK**). Follower force effects are included in the force balance in the nonlinear static and nonlinear transient dynamic solution sequences, SOLs 106, 129, 153, and 159, if geometric nonlinear effects are turned on with PARAM,LGDISP,1. The follower force stiffness is included in the nonlinear static solution sequences (SOLs 106 and 153) but not in the nonlinear transient dynamic solution sequences (SOLs 129 and 159).

    For SOL 401, follower force effects are included in the force balance in the nonlinear static solution if geometric nonlinear effects are turned on with PARAM,LGDISP,1. For additional information, see the NLCNTL bulk entry.

16. The coordinate systems in the main Bulk Data Section are defined relative to the assembly basic coordinate system which is fixed. This feature is useful when a superelement defined by a partitioned Bulk Data Section is rotated or mirrored and the gravity load is more conveniently defined in terms of coordinates which are fixed.

17. If CID is not a rectangular coordinate system, RFORCE1 will treat it as if it were and unexpected answers may result.

18. Follower force stiffness (param,followk,yes) is supported for method 2 only.

19. When you reference RFORCE or RFORCE1 entries on a LOAD entry, because centrifugal forces are not linear in angular velocity and therefore cannot be superposed, reference only a single RFORCE or RFORCE1 entry to define the angular velocity.

    For example, both of the following RFORCE specifications appear to define the same angular velocity.

```
RFORCE, 11, 999, .159, 1., 0., 0.
RFORCE, 12, 999, .159, 0., 1., 0.
RFORCE, 13, 999, .159, 0., 0., 1.
```

```
RFORCE, 11, 999, .159, 1., 1., 1.
```


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However, if you reference the three RFORCE entries on a LOAD entry, the centrifugal force that the software calculates is different from the correct centrifugal force that the software calculates if you reference the single RFORCE entry on the LOAD entry.

20. SOL 401 always uses METHOD = 2. As a result, the METHOD field is ignored by SOL 401.

21. In SOL 401, if a rotational force is defined with the RFORCE or RFORCE1 entry, and spin softening is requested with the SPINK parameter on the NLCNTL entry, if the rotation center is defined with a grid point which is not used in the connectivity of any element, all translational DOF for this grid point must be constrained.

22. In SOL 401, loads selected with the LOAD=n case control command are time-unassigned. Time-unassigned loads are ramped in a subcase when the LVAR parameter on the NLCNTL entry is set to RAMP (default). The following rules apply to the time-unassigned and ramped RFORCE and RFORCE1 entries.

- The RFVAR parameter is available on the NLCNTL bulk entry to change how the software interpolates the ramping load factor. See SOL 401 - RFORCE and RFORCE1 scaling in the *Multi-Step Nonlinear User's Guide* for details.
- RFORCE and RFORCE1 entries cannot be defined in the same subcase.
- If multiple RFORCE1 entries are defined in a subcase, the grids associated with each RFORCE1 entry must be mutually exclusive.
- The rotational load vector direction defined with the R1, R2, R3 fields and the coordinate system selected with the CID field cannot change from subcase to subcase.
- The scale factors A and RACC can change, and RFORCE and RFORCE1 entries can be added or removed, as long as the rotational load vectors and coordinate systems do not change.

23. In SOL 402, loads selected with the LOAD=n case control command are time-unassigned and are ramped in a subcase. When you define a time-unassigned rotational inertia load with the RFORCE or RFORCE1 entries, the RFVAR parameter is available on the NLCNTL2 bulk entry to change how the software interpolates the ramping load factor. See SOL 402 - RFORCE and RFORCE1 scaling in the *Multi-Step Nonlinear User's Guide* for details.

24. When RFORCE or RFORCE1 entries are referenced by the EXCITEID field on a TLOAD1 entry, the software uses the data on the associated TABLEDi to scale the computed force (F) in all solutions except for SOLs 401, 402, and 414.

- For SOL 401, when RFORCE or RFORCE1 entries are referenced by the EXCITEID field on a TLOAD1 entry, the software checks the RFVAR parameter on the NLCNTL bulk entry. See SOL 401 - RFORCE and RFORCE1 scaling in the *Multi-Step Nonlinear User's Guide* for details.
- For SOLs 402 and 414, when RFORCE or RFORCE1 entries are referenced by the EXCITEID field on a TLOAD1 entry, the data on the associated TABLEDi scale the angular velocity (ω) and acceleration (α), which are used to compute an inertia force in the equation $F = -[m] [\omega \times (\omega \times r)) + \alpha \times r]$. Since ω is squared in the force computation, the resulting scaling is not linearly related to the computed force (F).

25. For SOL 401, beginning with Simcenter Nastran 2021.2, a new body-load computation method is available for RFORCE and RFORCE1 loads. With this method, the load vector is computed on an element basis rather than a grid point basis. The resulting applied load is the same for the element basis and the previous grid point basis. With the new method, for the force vector described in

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Remark 1, $[m]_i$ becomes the element mass matrix and $\{F\}_i$ becomes the element contribution to the load vector.

The motivation for this change is related to a new grid point force output format requested with the GPFORCE case control command. The system cell 771 is available to change both the body-load computation method and the GPFORCE output format to the pre-Simcenter Nastran 2021.2 behavior. See **system cell 771**.

26. For SOL 401, if you want to add or change a rotational load in a restart run, you can not add a new grid point to define the location of a new rotation vector. The software requires that the geometry information (grids and elements) in the restart run must be consistent with the initial run. As a result, you can use the POINT entry with the RFORCE and RFORCE1 bulk entries in a SOL 401 restart solution. A POINT entry can be added in the restart run even though it did not exist in the initial run.
27. Grid points are either listed on the GROUP entry or are related to elements or properties listed on the GROUP entry as follows:
    - If TYPE = “GRID” on the GROUP entry, the GROUP entry lists the grid points to which the load is applied.
    - If TYPE = “ELEM” on the GROUP entry, the GROUP entry contains a list of elements. The load is applied to the grid points used in the connectivity of the listed elements.
    - If TYPE = “PROP” on the GROUP entry, the GROUP entry contains a list of properties. The load is applied to the grid points used in the connectivity of elements that reference the listed properties.

Remarks related to SOLs 402 and 414:

1. Only one RFORCE1 bulk entry can be applied in an analysis.
2. METHOD and MB are ignored.
3. To apply rotational force with constant magnitude, SID is selected by case control command LOAD = SID for both static and transient analyses.
4. To apply a time-dependent rotational load, SID is referenced by the field EXCITEID = SID in the TLOAD1 entry. Time-dependent loads are selected by case control command DLOAD.
5. CID must be a rectangular coordinate system.
6. Elements which are not selected with your GROUP bulk entry will not contribute to the resulting grid point mass used to compute the load. For example, if a grid point is used by elements A and B, but only element A is included in the GROUP, the mass of element A contributes to the grid point mass used to compute the RFORCE1 load. The mass of element B does not contribute. This behavior is unique to solutions 402 and 414. Using the same example, the other Simcenter Nastran solutions use the mass contribution of both elements A and B when computing the RFORCE1 load.
7. For SOL 402 and 414, scaling is done on the angular velocity (A) and the angular acceleration (RACC). The equivalent force is not computed ahead of time but in the solver.
8. For SOL 402 and 414, GROUPID must point to a group of elements or element properties and not to grid points.
9. For SOL 402, if you want to add or change a rotational load in a restart run, you cannot add a new grid point to define the location of a new rotation vector. The software requires that the geometry

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information (grids and elements) in the restart run must be consistent with the initial run. As a result, you can use the POINT entry with the RFORCE and RFORCE1 bulk entries in a SOL 401 restart solution. A POINT entry can be added in the restart run even though it did not exist in the initial run.

Related Topics

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RFORCE2

Rotational Force for rotor dynamics analyses using SOL 414,101 Maneuvers or SOL 101 Rotor Dynamic Analysis (legacy)

Defines a static loading condition due to an angular velocity and/or angular acceleration for maneuver load rotor dynamic analysis. Valid for SOL 414,101 Maneuvers and SOL 101 (legacy) only.

Format:

1	2	3	4	5	6	7	8	9	10
RFORCE2	SID	G	CID	A	R1	R2	R3	METHOD	
	RACC	MB	GYROP						

Example:

RFORCE2	2	5		−6.4	0.0	0.0	1.0	2	
	1.0		1						

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Point location through which the rotation vector acts. The location can be defined with the identification number of a GRID or POINT bulk entry. (Integer ≥ 0; Default = 0 which is the origin of the coordinate system defined in CID)
CID	Coordinate system defining the components of the rotation vector. See Remarks 6 and 7. (Integer ≥ 0, Default = 0)
A	Scale factor of the angular velocity in revolutions per unit time. (Real; Default = 1.0)
R1, R2, R3	Dimensionless rectangular components of the rotation vector $\vec{R}$ that passes through the point G. (Real; $R1^2 + R2^2 + R3^2 > 0.0$; Defaults R1 = 1.0, R2 = 0.0, R3 = 0.0)
METHOD	Method used to compute centrifugal forces. See Remarks 8 and 9. (Integer = 1 or 2; Default = 2)
RACC	Scale factor of the angular acceleration in revolutions per unit time squared. (Real; Default = 0.0)
MB	Indicates whether the CID coordinate system is defined in the main Bulk Data Section or the partitioned superelement Bulk Data Section. Coordinate systems referenced in the main Bulk Data Section are considered stationary with respect to the assembly basic coordinate system. See Remark 10. (Integer = −1 or 0; Default = 0)

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Field	Contents
	= –1 if the CID coordinate system is defined in the main Bulk Data Section = 0 if the CID coordinate system is defined in the partitioned superelement Bulk Data Section
GYROP	Force calculation option. (Integer = 0 or 1 or 2; Default = 0) = 0 to include centrifugal and gyroscopic force terms = 1 to include centrifugal force terms only = 2 to include gyroscopic force terms only

Remarks:

1. If RFORCE2 entries are used in any analysis other than SOL 414,101 Maneuvers or SOL 101 rotor dynamic analysis (legacy), an error message is issued and the run is terminated.

2. In Figure 17-169, the force vector at a point Gi is given by:

$$\left\{\vec{F}\right\}_i = -[m]_i \left[\vec{\omega} \times \left(\vec{\omega} \times \left(\vec{r}_i - \vec{r}_a\right)\right) + \vec{\alpha} \times \left(\vec{r}_i - \vec{r}_a\right)\right]$$

where $[m]_i$ is a 3 × 3 translational mass matrix at point Gi.

The angular velocity and angular acceleration vectors are given by:

Angular velocity	$\vec{\omega} = 2\pi\left(A \cdot \vec{R}\right) = \frac{(radians)}{(unit\,time)}$
Angular acceleration	$\vec{\alpha} = 2\pi\left(RACC \cdot \vec{R}\right) = \frac{(radians)}{(unit\,time)^2}$

where *A* is the angular velocity scale factor, *RACC* is the angular acceleration scale factor, and $\vec{R}$ is the dimensionless rotation vector.

The equation for $\left\{\vec{F}\right\}_i$ will have additional terms if the mass is offset and METHOD = 1 is selected.

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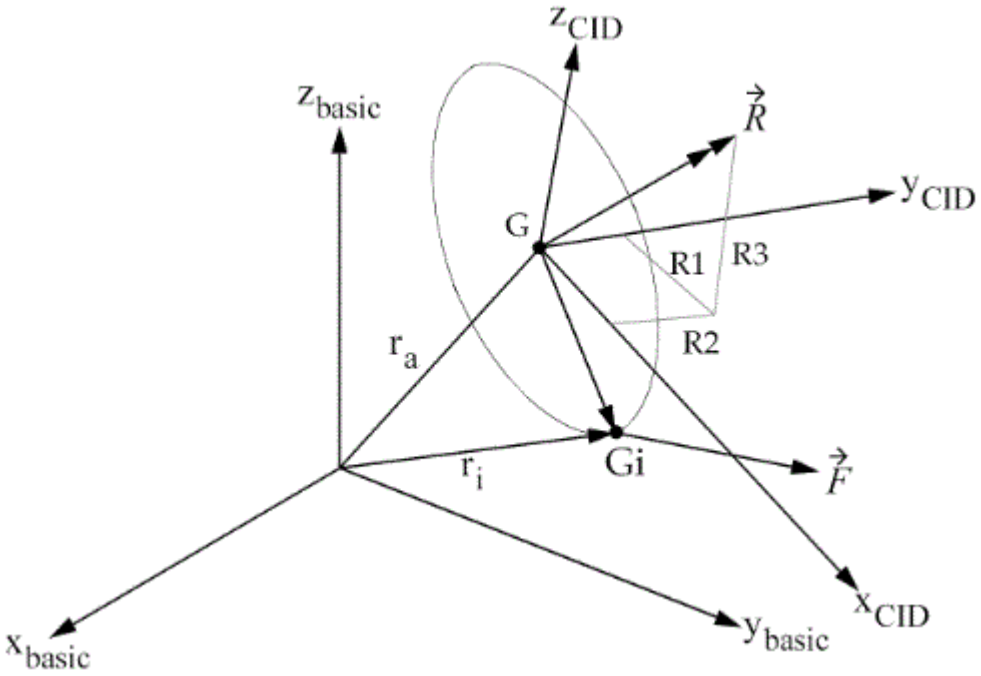
Figure 17-169: RFORCE2 Vector at a Point



3. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
4. The load vector generated by this entry can be printed with an OLOAD command in the Case Control Section.
5. G = 0 (Default) signifies that the rotation vector acts through the origin of the basic coordinate system.
6. CID = 0 (Default) signifies that the rotation vector is defined in the basic coordinate system.
7. If CID is not a rectangular coordinate system, RFORCE2 will treat it as if it were and unexpected answers may result.
8. METHOD = 1 yields correct results only when there is no coupling in the mass matrix. This occurs when the lumped mass option is used with or without the ZOFFS option (see the CQUAD4 entry for a description of ZOFFS).

   METHOD = 2 yields correct results for lumped or coupled mass matrix only if the ZOFFS option is not used. The acceleration terms due to the mass offset (X1, X2, X3) on the CONM2 entry are not computed with METHOD = 2. All the possible combinations of mass matrices and offset and the correct method to be used are shown below.

	No Offset	Offset
Lumped	METHOD = 1 or METHOD = 2	METHOD = 1
Coupled	METHOD = 2	Neither

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9. Forces due to angular acceleration (RACC) are computed with METHOD = 2 even if METHOD = 1 is specified.
10. The coordinate systems in the main Bulk Data Section are defined relative to the assembly basic coordinate system which is fixed. This feature is useful when a superelement defined by a partitioned Bulk Data Section is rotated or mirrored and the gravity load is more conveniently defined in terms of coordinates which are fixed.
11. For superelement analysis, G should reference a residual structure point that is exterior to all superelements. If it is not exterior to a superelement, then centrifugal loads will not be generated for that superelement. However, in cyclic analysis, User Fatal Message 4347 will be issued.
12. The continuation entry is optional.
13. Loads derived from this entry do not include effects due to mass specified for scalar points.
14. To model angular motion where the angular velocity and angular acceleration vectors have different directions, use two RFORCE2 entries that have the same SID. Use one entry to define the angular velocity vector, and use the other entry to define the angular acceleration vector.

Remarks related to SOL 414:

1. The GYROP field must be set to 0 or be blank.

Related Topics

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RLHARM

Information for frequency-dependent loads RLOAD1 and RLOAD2

Completes the definition of frequency-dependent loads RLOAD1 and RLOAD2 for use SOL 414,111 Harmonic Response solutions.

Format:

1	2	3	4	5	6	7	8	9	10
RLHARM	SID	IHARM							

Examples:

RLHARM	100	3							

Fields:

Field	Contents
SID	Set identification number of the frequency response dynamic excitation load set defined in an RLOAD1 or RLOAD2 bulk entry. (Integer > 0) See Remark 5.
IHARM<*i*>	Identification of the <*i*>th coefficient in the list of coefficients in the set identified by HARCOEFF. (Integer > 0) See Remark 6.

Remarks:

1. RLHARM is a companion bulk entry for the RLOAD1 (Real,Imaginary) and RLOAD2 (Magnitude,Phase) frequency-dependent loads. RLHARM identifies the RLOAD1 or RLOAD2 load set, and it selects one of the harmonics defined in the set identified by the HARCOEFF case control command.
2. RLHARM with RLOAD1/RLOAD2 and HARCOEFF applies in SOL 414,111 Harmonic Response solutions defined with a rotor (RMETHOD ROTORD). It does not apply in SOL 414,111 Harmonic Response solutions where no rotor is defined, so you cannot use RLHARM in a SOL 414,111 Harmonic Response solution with a nonlinear preload.
3. RLHARM is required in these circumstances:
   - Forces are defined using RLOAD1 or RLOAD2.
   - Harmonic coefficients are defined in the set identified by HARCOEFF.
4. When RLOAD1 and/or RLOAD2 is defined for a simulation in a rotating reference frame, both HARCOEFF and RLHARM are required.
5. SID must be unique for all RLHARM bulk entries, and it must match the SID of an RLOAD1 or RLOAD2.

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6. Identification of the <*i*>th coefficient in the list of coefficients in the set identified by HARCOEFF. In the following example:

```
SET 11 = 0.0, 1.0, 2.0
HARCOEFF = 77
RLOAD1,100,4011, 4012, 4013
RLHARM,100,3
```

IHARM=3 means the third coefficient in the list of coefficients in the set identified in HARCOEFF, which is 2.0. Therefore, the frequency of those loads will be 2ω.

There is no default value.

Related Topics

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RLOAD1

Frequency Response Dynamic Excitation, Form 1

Defines a frequency-dependent dynamic load of the form:

$$\{P(f)\} = \left\{A\left[C(f) + iD(f)\right]e^{i\{\theta - 2\pi f\tau\}}\right\}$$

for use in frequency response problems.

Format:

1	2	3	4	5	6	7	8	9	10
RLOAD1	SID	EXCITEID	DELAY	DPHASE	TC	TD	TYPE		

Examples:

RLOAD1	5	3	100	30.0	1				

RLOAD1	5	3	0.025	4	1	10			

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
EXCITEID	Identification number of the DAREA, SPCD, SPCF, SELOAD bulk entries, static load set, or thermal load set (for heat transfer analysis) that lists each degree-of-freedom to apply the excitation and the corresponding scale factor, *A*, for the excitation. See Remark 7, Remark 8, and SOL 414 Remark 1. (Integer > 0)
DELAY	Time delay, τ. See SOL 414 Remark 2. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 2). If real entry, value of τ for all degrees of freedom in EXCITEID entry. If integer entry, identification number of DELAY entries that contain values of τ for individual degrees of freedom in EXCITEID entry. See Remark 3.
DPHASE	Phase angle, θ, in degrees. See Remark 10 and SOL 414 Remark 3. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 2). If real entry, value of θ for all degrees of freedom in EXCITEID entry.

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Field	Contents
	If integer entry, identification number of DPHASE entries that contain values of θ for individual degrees of freedom in EXCITEID entry. See Remark 3.
TC	*C(f)* (Real or Integer ≥ 0 or blank; for default behavior, see Remark 2). If real entry, value of *C(f)* used over all frequencies for all degrees of freedom in EXCITEID entry. If integer entry, identification number of a TABLEDi entry that defines *C(f)* for all degrees of freedom in EXCITEID entry.
TD	*D(f)* (Real or Integer ≥ 0 or blank; for default behavior, see Remark 2). If real entry, value of *D(f)* used over all frequencies for all degrees of freedom in EXCITEID entry. If integer entry, identification number of a TABLEDi entry that defines *D(f)* for all degrees of freedom in EXCITEID entry.
TYPE	Defines the type of the dynamic excitation. See Remarks 7 and 8. (Integer, character or blank; Default = 0)

Remarks:

1. Dynamic excitation sets must be selected with DLOAD = SID in the case control section.
2. If any of DELAY, DPHASE, TC, and TD fields are blank or zero (either integer zero or real zero), the corresponding value for τ, θ, *C(f)*, and *D(f)* used by the software is real zero.
3. For degrees of freedom in the EXCITEID entry that are not specified on DELAY entries, the software uses real zero as the value for τ. For degrees of freedom in the EXCITEID entry that are not specified on DPHASE entries, the software uses real zero as the value for θ.
4. RLOAD1 excitations may be combined with RLOAD2 or RLOADEX excitations using a DLOAD bulk entry.
5. SID must be unique for all RLOAD1, RLOAD2, RLOADEX, TLOAD1, TLOAD2, ACSRCE, and SELOAD entries.
6. If ACSRCE and RLOADi entries are combined with a DLOAD bulk entry, the identification numbers of the TABLEDi selected with the TP field on the ACSRCE entries must be different from the identification numbers of the TABLEDi selected with the TC and TD fields on RLOAD1 entries, and the TB and TP fields on RLOAD2 entries.
7. The type of the dynamic excitation is specified by TYPE (field 8) according to the following table:

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TYPE	Type of dynamic excitation
0, L, LO, LOA, or LOAD	Applied load (force or moment) (Default)
1, D, DI, DIS, or DISP	Enforced displacement using SPC, SPCD, SPCF data
2, V, VE, VEL, or VELO	Enforced velocity using SPC, SPCD, or SPCF data
3, A, AC, ACC, or ACCE	Enforced acceleration using SPC, SPCD, or SPCF data

8. TYPE determines the manner in which EXCITEID is used by the software.

   - TYPE = 0: Applied load excitation.

     EXCITEID may reference DAREA, FORCEi, MOMENTi, RFORCEi, PLOAD, PLOAD1, QHBDY, QBDYi, QVECT, QVOL, and SELOAD entries.

   - TYPE = 1, 2, or 3: Enforced vibration excitation.

     - If EXCITEID references SPC, SPCD, or SPCF bulk entries, the software will use the SPCD method of enforced vibration. You directly specify:

       ◊ Displacements, velocities, or accelerations with SPC, SPCD, or SPCF bulk entries.

       ◊ For SPCF, the entire dynamic load equation is defined in the file referenced by the SPCF and any TC, TD, DELAY, and DPHASE values are ignored.

     - If EXCITEID references a load entry (DAREA, FORCEi, MOMENTi, PLOAD, PLOAD1, RFORCEi, and SELOAD entries), the software will use the large mass method of enforced vibration. You create a large mass and load at the grid and degree-of-freedom where the enforced vibration is desired. The software then computes the enforced vibration the same for TYPE = 1, 2, and 3.

       Type = 1, Enforced displacement = Dynamic load input / Large mass
       Type = 2, Enforced velocity = Dynamic load input / Large mass
       Type = 3, Enforced acceleration = Dynamic load input / Large mass

     See the “Enforced Motion” chapter of the *Simcenter Nastran Basic Dynamic Analysis User’s Guide* for details on both the SPCD and large mass methods of enforced vibration.

9. The legacy method of selecting a DAREA, FORCEi, MOMENTi, RFORCEi, PLOAD, and PLOAD1 entry for a dynamic loading required the LOADSET case control command which selected the LSEQ bulk entry, which in turn selected the load entry. Because the DAREA, FORCEi, MOMENTi, PLOAD, PLOAD1, and SPCD entries are now selected directly with the EXCITEID on the dynamic load entries DLOADi, TLOADi, and RLOADi, the LOADSET case control and LSEQ bulk entry are no longer required. The legacy method is still supported, and is described on the LSEQ entry remarks.

10. The software converts the phase angle, θ, to radians.

Remarks for SOL 414:

1. To set an enforced motion excitation load, EXCITEID can refer to a SPCD bulk entry only.
2. SOL 414 solutions ignore the DELAY field.
3. DPHASE can be integer or real if loads are applied on grid points. If loads are applied on elements, then DPHASE must be real.

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4. When you define RLOAD1 for SOL 414,111 Harmonic Response solutions for a rotor in the rotating reference frame (with RMETHOD ROTORD defined), you must add the companion RLHARM bulk entry and the HARCOEFF case control command to identify the harmonic coefficient that you want to associate with this frequency load. If the analysis is in the rotating reference frame and the RLHARM bulk entry is not defined, the software generates a FATAL error.
5. When you define RLOAD1 for SOL 414,111 Harmonic Response solutions for a rotor in the fixed reference frame (with RMETHOD ROTORD defined) at a frequency that is a multiplier of the reference frequency, add a companion RLHARM bulk entry and the HARCOEFF case control command to identify the harmonic coefficient that you want to associate with this frequency load.
6. When DPHASE is defined by components, the components of DPHASE are always associated to the global coordinate system. When you define a FORCE load to associate with RLOAD1, however, you can define the FORCE in the local coordinate system. Therefore, the DPHASE and FORCE components might not match.

Related Topics

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RLOAD2

Frequency Response Dynamic Excitation, Form 2

Defines a frequency-dependent dynamic excitation of the form:

$$\{P(f)\} = \left\{A \cdot B(f) e^{i[\phi(f)+\theta-2\pi f\tau]}\right\}$$

for use in frequency response problems.

Format:

1	2	3	4	5	6	7	8	9	10
RLOAD2	SID	EXCITEID	DELAY	DPHASE	TB	TP	TYPE		

Examples:

RLOAD2	5	3	0.2	15.0	7				

RLOAD2	5	3		6	7	8			

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
EXCITEID	Identification number of a DAREA entry, SPCD entry, SELOAD entry, static load set, or thermal load set (for heat transfer analysis) that lists each degree of freedom to apply the excitation and the corresponding scale factor, *A*, for the excitation. See Remark 7, Remark 8, and SOL 414 Remark 1. (Integer > 0)
DELAY	Time delay, τ. See SOL 414 Remark 2. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 2). If real entry, value of τ for all degrees of freedom in EXCITEID entry. If integer entry, identification number of DELAY entries that contain values of τ for individual degrees of freedom in EXCITEID entry. See Remark 3.
DPHASE	Phase angle, θ, in degrees. See Remark 10 and SOL 414 Remark 3. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 2). If real entry, value of θ for all degrees of freedom in EXCITEID entry.

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Field	Contents
	If integer entry, identification number of DPHASE entries that contain values of θ for individual degrees of freedom in EXCITEID entry. See Remark 3.
TB	*B(f)* (Real or Integer ≥ 0; No default) If real entry, value of *B(f)* used over all frequencies for all degrees of freedom in EXCITEID entry. If integer entry, identification number of a TABLEDi entry that defines *B(f)* for all degrees of freedom in EXCITEID entry.
TP	$\phi(f)$ in degrees. See Remark 10. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 2). If real entry, value of $\phi(f)$ used over all frequencies for all degrees of freedom in EXCITEID entry. If integer entry, identification number of a TABLEDi entry that defines $\phi(f)$ for all degrees of freedom in EXCITEID entry.
TYPE	Defines the type of the dynamic excitation. See Remark 7 and Remark 8. (Integer, character or blank; Default = 0)

Remarks:

1. Dynamic excitation sets must be selected with DLOAD = SID in the case control section.
2. If any of DELAY, DPHASE, and TP fields are blank or zero (either integer zero or real zero), the corresponding value for τ, θ, and $\phi(f)$ used by the software is real zero.
3. For degrees of freedom in the EXCITEID entry that are not specified on DELAY entries, the software uses real zero as the value for τ. For degrees of freedom in the EXCITEID entry that are not specified on DPHASE entries, the software uses real zero as the value for θ.
4. RLOAD2 excitations may be combined with RLOAD1 or RLOADEX excitations using a DLOAD bulk entry.
5. SID must be unique for all RLOAD1, RLOAD2, RLOADEX, TLOAD1, TLOAD2, ACSRCE, and SELOAD entries.
6. If ACSRCE and RLOADi entries are combined with a DLOAD bulk entry, the identification numbers of the TABLEDi selected with the TP field on the ACSRCE entries must be different from the identification numbers of the TABLEDi selected with the TC and TD fields on RLOAD1 entries, and the TB and TP fields on RLOAD2 entries.
7. The type of the dynamic excitation is specified by TYPE (field 8) according to the following table:

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TYPE	Type of dynamic excitation
0, L, LO, LOA or LOAD	Applied load (force or moment) (Default)
1, D, DI, DIS or DISP	Enforced displacement using SPC/SPCD data
2, V, VE, VEL or VELO	Enforced velocity using SPC/SPCD data
3, A, AC, ACC or ACCE	Enforced acceleration SPC/SPCD data

8. TYPE determines the manner in which EXCITEID is used by the software.

   - TYPE = 0 : Applied load excitation.

     EXCITEID may reference DAREA, FORCEi, MOMENTi, RFORCEi, PLOAD, PLOAD1, QHBDY, QBDYi, QVECT, QVOL, and SELOAD entries.

   - TYPE = 1, 2, or 3 : Enforced motion excitation.

     - If EXCITEID references SPC/SPCD entries, the software will use the SPCD method of enforced motion. You directly specify displacements, velocities, or accelerations with SPC/SPCD entries.
     - If EXCITEID references a load entry (DAREA, FORCEi, MOMENTi, PLOAD, PLOAD1, RFORCEi, and SELOAD entries), the software will use the large mass method of enforced motion. You create a large mass and load at the grid and degree-of-freedom where the enforced motion is desired. The software then computes the enforced motion the same for TYPE = 1, 2, and 3.

       Type = 1, Enforced displacement = Dynamic load input / Large mass
       Type = 2, Enforced velocity = Dynamic load input / Large mass
       Type = 3, Enforced acceleration = Dynamic load input / Large mass

     See the Enforced Motion chapter in the *Simcenter Nastran Basic Dynamic Analysis User's Guide* for details on both the SPCD and large mass methods of enforced motion.

9. The legacy method of selecting a DAREA, FORCEi, MOMENTi, RFORCEi, PLOAD, and PLOAD1 entry for a dynamic loading required the LOADSET case control command which selected the LSEQ bulk entry, which in turn selected the load entry. Since the FORCEi, MOMENTi, PLOAD, PLOAD1, and SPCD entries are now selected directly with the EXCITEID on the dynamic load entries DLOADi, TLOADi, and RLOADi, the LOADSET case control and LSEQ bulk entry are no longer required. The legacy method is still supported, and is described on the LSEQ entry remarks.

10. The software converts the phase angle, θ, and frequency-dependent function, $\phi(f)$, to radians.

Remarks for SOL 414:

1. To set an enforced motion excitation load, EXCITEID can refer to a SPCD bulk entry only.
2. SOL 414 solutions ignore the DELAY field.
3. DPHASE can be integer or real if the loads are applied on grid points. If loads are applied on elements, then DPHASE must be real.
4. When you define RLOAD2 for SOL 414,111 Harmonic Response solutions for a rotor in the rotating reference frame (with RMETHOD ROTORD defined), you must add the companion RLHARM bulk entry and the HARCOEFF case control command to identify the harmonic coefficient that you want to

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associate with this frequency load. If the analysis is in the rotating reference frame and the RLHARM bulk entry is not defined, the software generates a FATAL error.

5. When you define RLOAD2 for SOL 414,111 Harmonic Response solutions for a rotor in the fixed reference frame (with RMETHOD ROTORD defined) at a frequency that is a multiplier of the reference frequency, add a companion RLHARM bulk entry and the HARCOEFF case control command to identify the harmonic coefficient that you want to associate with this frequency load.
6. When DPHASE is defined by components, the components of DPHASE are always associated to the global coordinate system. When you define a FORCE load to associate with RLOAD2, however, you can define the FORCE in the local coordinate system. Therefore, the DPHASE and FORCE components might not match.

Related Topics

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RLOADEX

Frequency Response Dynamic Excitation, File Input

Defines frequency-dependent dynamic loads (forces or moments) obtained from a file of the form:

$$\{P(f)\} = \{A[F(f)]e^{i\{\theta-2\pi f\tau}\}$$

for use in frequency response problems.

Format:

1	2	3	4	5	6	7	8	9	10
RLOADEX	SID	DELAY	DPHASE	A	LUNF	LDIDT	LCIDT	SET1ID	
	LDIDR	LCIDR							

Examples:

RLOADEX	5		90.0	2.0	102	3	12	2	
	13	3							

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0; No default) See Remark 6.
DELAY	Time delay, τ. (Real or blank; for default behavior, see Remark 3) Value of τ for all degrees of freedom (DOFs) defined on the LUNF file.
DPHASE	Phase angle, θ, in degrees. (Real or blank; for default behavior, see Remark 3) Value of θ for all DOFs defined on the LUNF file.
A	Scale factor. (Real or blank; Default = 1.0)
LUNF	Logical unit number of an HDF5 file that contains frequency-dependent load excitations. (0 < Integer ≤ 9999; No default) See Remark 6.
LDIDT	Applied load descriptor identification number of a translational loading description contained in the HDF5 file referenced by LUNF. (0 ≤ Integer ≤ 9999; No default) See Remark 6.
LCIDT	Load case identification number of a specific translational loading in the loading description (Integer ≥ 0; No default). This descriptor can be equal to zero only when LDIDT is zero. See Remark 6.

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Field	Contents
SET1ID	Identification number of a SET1 bulk entry that defines a list of poro-elastic or structural grids at which the loads are to be extracted and applied to for all DOFs of the grid. A value of 0 implies that all structural grids contained in the HDF5 file will have loads applied. (Integer ≥ 0; Default = 0) See Remark 7.
LDIDR	Applied load descriptor identification number of a rotational loading description contained in the HDF5 file referenced by LUNF. (0 ≤ Integer ≤ 9999; No default) See Remark 6.
LCIDR	Load case identification number of a specific rotational loading in the loading description (Integer ≥ 0; No default). This descriptor can be equal to zero only when LDIDR is zero. See Remark 6.

Remarks:

1. RLOADEX with:
   - Structural grids is only supported in SOL 108 and SOL 111 with the standard (fixed low-order) FEM or FEM Adaptive Order method.
   - Poro-elastic grids is only supported in SOL 108 and SOL 111, and requires the FEM Adaptive Order method.
2. RLOADEX bulk entries must be selected with DLOAD = SID in the case control section.
3. If DELAY or DPHASE fields are blank, the corresponding value for τ or θ used by the software is real zero. The software converts the phase angle to radians.
4. RLOADEX excitations may be combined with RLOAD1 and RLOAD2 excitations using a DLOAD bulk entry.
5. The SID must be unique for all RLOADEX, RLOAD1, RLOAD2, TLOAD1, TLOAD2, ACSRCE, and SELOAD entries.
6. The logical unit number LUNF must reference an existing ASSIGN SC_H5 File Management System (FMS) statement. The frequency-dependent load excitation description referenced by LDIDT or LDIDR must be contained in this file that also includes the DOFs and frequencies at which the excitations are applied.

   If the frequencies of the load excitation matrix do not match the solution frequencies, linear interpolation will be used to obtain the applied loads at the solution frequency. No extrapolation will be performed if the solution frequency is less than or greater than the frequency range of the load excitation matrix, and the values of the load excitation matrix at the corresponding end point will be used.

   A separate DTI, DISTL bulk entry will be used to define the loading description referenced by LDIDT and LDIDR. The LCIDT and LCIDR identifiers must correspond to a specific loading within the descriptor LDIDT and LDIDR respectively. LDIDT and LDIDR cannot both be equal to zero (that is, no data requested). When LDIDT or LDIDR is equal to zero, the software ignores the corresponding LCIDT or LCIDR descriptor respectively.

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7. Poro-elastic grids are fluid grids that are referenced by acoustic elements and have poro-elastic materials assigned.

Related Topics

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ROTORB

Bearing Grids Selection

Selects the stationary grid from each pair of grid points that define the connectivity of bearing supports for a specific rotor.

Format 1: (Formats 1 and 2 cannot be combined on the same line)

1	2	3	4	5	6	7	8	9	10
ROTORB	RSETID	G1	G2	G3	G4	G5	G6	G7	

Format 2:

ROTORB	RSETID	G1	"THRU"	G2	"BY"	INC			

Continuation Format 1: (Continuation formats 1 and 2 cannot be combined on a single continuation line)

	G8	G9	G10	G11	–etc.–				

Continuation Format 2:

	G3	"THRU"	G4	"BY"	INC				

Example:

ROTORB	14	101	THRU	190	BY	5			
	46	23	57	82	9	16			
	201	THRU	255						
	93	94	95	97					

Fields:

Field	Contents
RSETID	References an RSETi on the ROTORD bulk entry. (Integer>0)
Gi	Grid points defining bearing locations on a rotor. (Integer>0)
THRU	Specifies a range of grid ID's. (Optional)

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Field	Contents
BY	Specifies an increment when using THRU option. (Optional)
INC	Increment used with BY option. (Integer>0; Default=1)

Remarks:

1. By default, there are no bearing locations on a rotor. The RSETID field is referred to by the RSETi field on the ROTORD bulk data entry. If a model contains multiple rotors, use a separate ROTORB entry for each rotor.

Remarks related to SOL 414:

1. ROTORB entries are not required and are ignored if present. This behavior occurs even when the model includes CBEAR elements.

Related Topics

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ROTORD

Define Rotor Dynamics Solution Options

Defines rotor dynamics solution options.

Format for SOL 414 Rotor Dynamics:

1	2	3	4	5	6	7	8	9	10
ROTORD	SID	RSTART	RSTEP	NUMSTEP	REFSYS		RUNIT		
		ORBEPS	ROTPRT	SYNC					
	RID1	RSET1	RSPEED1	RCORD1	W3_1	W4_1		BRGSET1	
	RID2	RSET2	RSPEED2	RCORD2	W3_2	W4_2		BRGSET2	
....									
	RIDi	RSETi	RSPEEDi	RCORDi	W3_i	W4_i		BRGSETi	
....									
	RID10	RSET10	RSPEED10	RCORD10	W3_10	W4_10		BRGSET10	

Example:

ROTORD	998	0.0	250.0	58	FIX	-1.0	CPS		
				1					
	1	11	1.0	1	0.0	0.0			
	2	12	2.0	1	0.0	0.0			

Format for SOL 101 and SOLs 107 through 112 Rotor Dynamics (legacy):

1	2	3	4	5	6	7	8	9	10
ROTORD	SID	RSTART	RSTEP	NUMSTEP	REFSYS	CMOUT	RUNIT	FUNIT	
	ZSTEIN	ORBEPS	ROTPRT	SYNC	ETYPE	EORDER	THRSHOLD	MAXITER	
	RID1	RSET1	RSPEED1	RCORD1	W3_1	W4_1	RFORCE1	BRGSET1	
	RID2	RSET2	RSPEED2	RCORD2	W3_2	W4_2	RFORCE2	BRGSET2	
....									
	RIDi	RSETi	RSPEEDi	RCORDi	W3_i	W4_i	RFORCEi	BRGSETi	
....									
	RID10	RSET10	RSPEED10	RCORD10	W3_10	W4_10	RFORCE10	BRGSET10	

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Example:

ROTORD	998	0.0	250.0	58	FIX	-1.0	CPS		
	NO								
	1	11		1	0.0	0.0	1	101	
	2	12		1	0.0	0.0		102	

Fields:

Field	Contents
SID	Set identifier for all rotors. Must be selected in the case control deck by RMETHOD = SID. (Integer > 0)
RSTART	Starting value of reference rotor speed. See Remark 3. (Real)
RSTEP	Step-size of reference rotor speed. See Remark 4. (Real ≠ 0.0 for SOLs 101, 107-112, 414,101; Real for all SOL 414 types)
NUMSTEP	Number of steps for reference rotor speed including RSTART. (Integer ≥ 0 for SOLs 107, 110, and 414,110. Integer > 0 for all other supported SOLs)
REFSYS	Reference system. (Character; Default = 'ROT') = 'FIX' analysis is performed in the fixed reference system. = 'ROT' analysis is performed in the rotational reference system, or in a mixed frame approach.
CMOUT	Determines the rotor speeds at which eigenvectors are calculated and output. (Real; Default = 0.0) = 0.0 no eigenvectors are calculated; no eigenvectors are output; no whirl directions are output. > 0.0 eigenvectors are calculated at the specified speed; eigenvectors are output at the specified speed; whirl direction is output at the specified speed. = -1.0 eigenvectors are calculated at all speeds; eigenvectors are output at all speeds; whirl directions are output at all speeds.
RUNIT	Units used for rotor speed inputs (CMOUT, RSTART and RSTEP), output (units for output list and Campbell diagram output), and speed-dependent bearing property tables. (Character; Default = 'RPM') = 'RPM' revolutions per minute. = 'CPS' cycles per second.

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Field	Contents
	= ‘HZ’ cycles per second.
	= ‘RAD’ radians per second.
FUNIT	Units used for frequency output (Campbell diagram output). (Character; Default = ‘RPM’)
	= ‘RPM’ revolutions per minute.
	= ‘CPS’ cycles per second.
	= ‘HZ’ cycles per second.
	= ‘RAD’ radians per second.
ZSTEIN	Option to incorporate Steiner’s inertia terms. (Character; Default = 'NO')
	= 'YES' Steiner’s inertia terms are included.
	= 'NO' Steiner’s inertia terms are not included.
ORBEPS	Threshold value for detection of whirl direction. (Real > 0.0; Default = 1.E-6)
ROTPRT	Controls .f06 output options. (Integer; Default = 0))
	= 0 no print.
	= 1 print generalized matrices; print final nonlinear bearing values at each frequency or time.
	= 2 print eigenvalue summary and eigenvectors at each RPM; print intermediate nonlinear bearing values for each iteration.
	= 3 combination of 1 and 2.
SYNC	Option to select synchronous or asynchronous analysis for frequency response analysis. (Integer; Default = 1)
	= 1 synchronous
	= 0 asynchronous
ETYPE	Excitation type. (Integer; Default = 1)
	= 1 Mass unbalanced. Specify mass unbalance = m x r on DLOAD bulk entry and the program will multiply by Ω^2.

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Field	Contents
	= 0 Force excitation. Specify force = m x r x Ω^2 on DLOAD bulk entry.
EORDER	Excitation order. (Real; Default 1.0) = 1.0 (Default) (modes crossing with 1P line in the fixed system) = 0.0 Forward whirl (modes crossing with 0P line in the rotating system) = 2.0 Backward whirl (modes crossing with 2P line in the rotating system)
THRSHOLD	Convergence threshold when iterating to determine bearing stiffness or viscous damping for CBEAR elements that have speed and displacement or speed and force dependent stiffness or viscous damping. (Real > 0.0; Default = 0.02)
MAXITER	Maximum number of iterations to determine bearing stiffness or viscous damping for CBEAR elements that have speed and displacement or speed and force dependent stiffness or viscous damping. A value of 0 implies that no iterations are performed. (Integer ≥ 0; Default = 10)
RIDi	Identification number of rotor i. (Integer > 0 with RID(i+1) > RIDi for all solutions except for SOL 414; RID(i+1) = RID(i) + 1 for SOL 414; Default = i) See SOL 414 Remark **10**.
RSETi	Refers to the RSETID value on the ROTORG, ROTORB, and ROTSE bulk entries for rotor RIDi. (Integer > 0 or blank if only one rotor in SOL 101 and SOLs 107-112)
RSPEEDi	Multiplier of reference rotor speed for rotor i. (Real ≠ 0.0 or Integer > 0 or blank; Default = 1.0) If real entry, value of multiplier at all reference rotor speeds. If integer entry, identification number of a TABLEDi entry that contains value of multiplier as a function of reference rotor speed, or rotor speed as a function of the sweeping parameter. See SOL 414 remarks for more information.
RCORDi	Identification number of the Cartesian coordinate system whose Z-axis is the axis of rotation axis for rotor i. (Integer; Default = 0 for basic coordinate system)
W3_i	Reference frequency for structural damping defined by PARAM,G for rotor i. (Real; Default = 0.0)
W4_i	Reference frequency for structural damping defined by GE parameter in MAT1 for rotor i. (Real; Default = 0.0)
RFORCEi	Points to RFORCE, RFORCE1, or RFORCE2 bulk entry for rotor I. (Integer; Default = 0 for no rotational force applied; a rotational force is required for differential stiffness to be calculated.)
BRGSETi	Identification number of a GROUP bulk entry that lists the CBEAR elements for the corresponding RIDi. Only the GROUP type ELEM is supported. (Integer ≥ 0 or blank; no default)

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Remarks related to SOL 101, 107 through 112, and 414:

1. There is a maximum limit of 10 rotors. Thus, there can be up to 11 continuation lines.
2. The RSTART, RSTEP, and NUMSTEP fields define the reference rotor speed. The value of an actual rotor is the product of the reference rotor speed and the value for the RSPEEDi field.
3. The direction of rotation is defined in accordance with the right-hand rule. *Positive* indicates rotation about the +Z-axis of the RCORDi coordinate system. *Negative* indicates rotation about the –Z-axis of the RCORDi coordinate system.
4. A negative value for RSTEP causes the reference rotor speed to algebraically decrease. For example, assume RSTEP = –200.0 rpm. If the current reference rotor speed is +1200 rpm, the reference rotor speed will be +1000 rpm after one step. After five additional steps the reference rotor speed would be 0 rpm. Additional steps would then cause the reference rotor speed to increase in the opposite rotational direction.
5. The W3 parameter defines the reference frequency for structural damping defined by PARAM,G.
6. The W4 parameter defines the reference frequency for structural damping defined by GE.
7. The rotor speed defined by RSTART, RSTEP, and NUMSTEP is called the reference rotor speed. Rotors with relative speed defined by RSPEED will spin at the defined factor multiplied by the reference rotor speed.

Remarks related to SOLs 101 and 107 through 112:

1. When the ROTCOUP parameter is specified and REFSYS = ROT, the equation of motion includes time-dependent terms and is solved at discrete azimuth angles. The software can either solve the equation of motion over a range of azimuth angles at a single rotor speed, or solve the equation of motion at a single azimuth angle over a range of rotor speeds.
   - To solve over a range of azimuth angles at a single rotor speed, use the PHIBGN, PHIDEL, and PHINUM parameters to specify the azimuth angle range, and use the RSTART field to specify the rotor speed.
   - To solve over a range of rotor speeds at a single azimuth angle, use the PHIBGN parameter to specify the azimuth angle, omit the PHIDEL and PHINUM parameters, and use the RSTART, RSTEP, and NUMSTEP fields to specify the rotor speed range. If you also omit the PHIBGN parameter, the solve is at an azimuth angle of zero because the default value for the PHIBGN parameter is zero.
2. The Steiner's term option (ZSTEIN field) should only be used when analyzing solid models in the fixed system.
3. The W3 and W4 parameters are required for all direct solutions. In the modal solutions, the eigenvalues are used as default. If the W3 and W4 parameters are defined for modal analysis, they will be used, but these parameters are not recommended for modal solutions.
4. The static centrifugal force is calculated for unit speed measured in rad/sec. On the RFORCE, RFORCE1, or RFORCE2 bulk entry, the unit of Hz is used, thus the conversion $1/(2\pi) = 0.159155$ must be used by the user.
5. For calculating frequency response using synchronous analysis, the rotation speeds are defined by the RSTART, RSTEP, and NUMSTEP fields on the ROTORD bulk entry. The frequencies corresponding

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to these rotation speeds must match the frequencies specified with the FREQUENCY case control command. If the frequencies differ, a fatal error is issued.

6. For calculating the frequency or transient response using asynchronous analysis, the unique rotation speed is defined by the RSTART field on the ROTORD bulk entry. The RSTEP and NUMSTEP fields in this case will be ignored. The frequency and dynamic load definitions are defined with the standard FREQ, DLOAD, RLOAD, etc. bulk entries for frequency response, and with the TSTEP, DLOAD, TLOAD, etc. bulk entries for transient response.
7. The SYNC, ETYPE, and EORDER fields are ignored during a SOL 107 or SOL 110 complex eigenvalue analysis. During a SOL 107 or SOL 110 complex eigenvalue analysis, if NUMSTEP = 0 and PARAM,ROTSYNC,YES (default) are specified, only synchronous analysis is performed. If NUMSTEP = 0 and PARAM,ROTSYNC,NO are specified, no analysis is performed.
8. For additional information on how the values in the THRSHOLD and MAXITER fields are used, see the PBEAR bulk entry.
9. ETYPE is applicable for synchronous analysis (SYNC = 1) only.

Remarks related to SOL 414:

1. RSETi always refers to a RSETID value of an existing ROTORG bulk entry.
2. If RSPEEDi is an integer entry, it references the identification number of a TABLEDi bulk entry that contains:
   - The values of the multiplier as a function of reference rotor speed if RSTEP ≠ 0.0.
   - The values of the rotor speed as a function of the sweeping parameter if RSTEP = 0.0.

     The sweeping parameter is specified with OMEGAi in SOL414,110, FREQi in SOL 414,111, and TSTEP in SOL 414,129.

   If RSPEEDi is a real entry, it specifies a multiplier factor of the rotation speed.
3. For direct calculation of critical speeds and the corresponding mode shapes in a SOL 414,110 analysis, specify NUMSTEP = 0.
4. For calculation of complex modes at prescribed rotor speeds in a SOL 414,110 analysis, you have two options:
   - Set RSTEP ≠ 0.0 to derive the rotor speeds from the RSTART, RSTEP, and NUMSTEP fields.
   - Set RSTEP = 0.0 to derive the rotor speeds from the OMEGA, OMEGA1, and OMEGA2 bulk entries.
5. For calculation of frequency response at prescribed rotor speeds in a SOL 414,111 analysis, you have two options:
   - Set RSTEP ≠ 0.0 to derive the rotor speeds from the RSTART, RSTEP, and NUMSTEP fields.
   - Set RSTEP = 0.0 to derive the rotor speeds from the FREQi bulk entries.
6. For calculation of transient response at prescribed rotor speeds in a SOL 414,129 analysis, you have two options:
   - Set RSTEP ≠ 0.0 to derive the rotor speeds from the RSTART, RSTEP, and NUMSTEP fields.

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- Set RSTEP = 0.0 to derive the rotor speeds from the TSTEPi bulk entries.

7. For SOL 414,101, RSTEP = 0.0 is not allowed.
8. For taking centrifugal loads into account for computations in the rotating reference frame, use RFORCE bulk entries with a value of = 0.159155 Hz (which is equivalent to 1 rad/s) defined on the rotor axis, in the direction of the rotor axis, in a preload subcase for SOL 414,110 or SOL 414,111.
9. RFORCE1 bulk entry can be used in SOL 414,101 and SOL 414,110.

   RFORCE2 bulk entry can be used in SOL 414,101.
10. When your model consists of an assembly of rotors, you must number the rotors in ascending order and consecutively starting with 1. For example, if your assembly consists of three rotors, they must be numbered as RID1=1, RID2=2, and RID3=3, and so on. Otherwise, the results of the analysis will be incorrect.

Related Topics

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ROTORG

Rotor Grids Selection

Selects the grids that define a rotor.

Format 1:

Formats 1 and 2 cannot be combined on the same line.

1	2	3	4	5	6	7	8	9	10
ROTORG	RSETID	G1	G2	G3	G4	G5	G6	G7	

Format 2:

ROTORG	RSETID	G1	"THRU"	G2	"BY"	INC			

Continuation Format 1:

Continuation formats 1 and 2 cannot be combined on a single continuation line.

	G8	G9	G10	G11	–etc.–				

Continuation Format 2:

	G3	"THRU"	G4	"BY"	INC				

Example:

ROTORG	14	101	THRU	190	BY	5			
	46	23	57	82	9	16			
	201	THRU	255						
	93	94	95	97					

Fields:

Field	Contents
RSETID	References an RSETi on the ROTORD bulk entry. (Integer>0)
Gi	Grids defining a rotor. (Integer>0)

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Field	Contents
THRU	Specifies a range of grid ID's. (Optional)
BY	Specifies an increment when using THRU option. (Optional)
INC	Increment used with BY option. (Integer>0; Default=1)

Remarks:

1. For a rotor dynamic analysis input file with ROTORG entries present, the software assumes grids listed on the ROTORG entries are rotating and grids not listed on the ROTORG entries are stationary. For a rotor dynamic analysis input file without ROTORG entries present, the software assumes all grids are rotating.
2. If a model contains both rotating and stationary portions, list the grids associated with each rotor on a unique ROTORG entry. On the ROTORD entry, have each RSETi field reference the RSETID field of the corresponding ROTORG entry.
3. If an entire model is rotating, you can do either of the following:
   - Include a ROTORG entry that contains all the grids and have the RSET1 field on the ROTORD entry reference the RSETID field of the ROTORG entry.
   - Do not include a ROTORG entry and leave the RSET1 field blank on the ROTORD entry. This option is not valid for SOL 414.

Related Topics

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ROTPARM

Parameter Specification for Rotor Dynamic Analysis

Defines solution control parameters for frequency response and complex eigenvalue in legacy rotor dynamic analyses. You cannot use the ROTPARM parameter specification for SOL 414 rotor dynamic solutions. For information about parameters for SOL 414,110 Complex Modal Analysis, SOL 414,111 Harmonic Response, and other SOL 414 solutions, see *Simcenter Nastran Rotor Dynamics User's Guide (SOL 414)*.

Format:

1	2	3	4	5	6	7	8	9	10
ROTPARM	SID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4	PARAM5	VALUE5	-etc-				

Example:

ROTPARM	1	SEFILT	YES	SETHRSH	20.0	KETHRSH	15.0		
	REFROT	1							

Fields:

Field	Contents
SID	Set identification number. See Remark 1. (Integer > 0)
PARAMi	Parameter name. Allowable names are given in the parameter listing below. (Character)
VALUEi	Value of the parameter. (Real, Integer, or Character)

ROTPARM Parameters:

Name	Description
ROTSEKE	Print energy calculations. (Character “YES” or “NO”; Default = “NO”)
REFROT	Identification number (RIDi on the ROTORD bulk entry) of the reference rotor. The reference rotor is used as the rotor whose strain or kinetic energy is used in the calculation of ratio of energy of the rotor to total energy of the system, and is used as the reference rotor for rotor speed calculations. See Remark 2. (Integer > 0; For default behavior, see Remark 3)
SEFILT	Identify modes that have a ratio of strain energy of the reference rotor to total strain energy of the system that exceeds the strain energy threshold value. See Remarks 4, 5, and 6. (Character “YES” or “NO”; Default = “NO”)

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Name	Description
KEFILT	Identify modes that have a ratio of kinetic energy of the reference rotor to total kinetic energy of the system that exceeds the kinetic energy threshold value. See Remarks 4, 5, and 6. (Character “YES” or “NO”; Default = “NO”)
SETHRSH	Strain energy threshold value in percent. (Real > 0.0; Default = 30.0)
KETHRSH	Kinetic energy threshold value in percent. (Real > 0.0; Default = 30.0)

Remarks:

1. The SID of the ROTPARM bulk entry must be the same as the SID of the ROTORD bulk entry.
2. If the REFROT field is not specified, to obtain meaningful results, on the ROTORD bulk entry, the RSPEEDi field for the reference rotor must be set to 1.0. The software does not trap other values.
3. For a system with a single rotor, the single rotor is the default reference rotor for the energy calculations in complex eigenvalues. For a system with multiple rotors, there is no default reference rotor.
4. If only SEFILT is specified, modes that pass the SEFILT criterion are identified by “***” in the summary line of the results. If only KEFILT is specified, modes that pass the KEFILT criterion are identified by “***” in the summary line of the results. If both SEFILT and KEFILT are specified, modes that pass either the SEFILT criterion or KEFILT criterion are identified by “***” in the summary line of the results.
5. When either or both SEFILT and KEFILT are specified, and MODTRK = 4 is specified, only the modes identified by “***” in the summary line of the results are tracked for use in a Campbell diagram.
6. Energy-based filtering of modes is valid for SOL 107 and SOL 110 complex eigenvalue rotor dynamic analysis only.

Related Topics

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ROTSE

Supplemental Rotor Superelement Definition

Defines the modal reduction type and additional a-set grids for a rotor superelement. Valid for SOLs 107, 108, and 109.

Format 1: (Formats 1 and 2 cannot be combined on the same line)

1	2	3	4	5	6	7	8	9	10
ROTSE	RSETID	TYPE	EVID						
	G1	G2	G3	G4	G5	G6	G7	G8	

Format 2:

ROTSE	RSETID	TYPE	EVID						
	G1	"THRU"	G2	"BY"	INC				

Continuation Format 1: (Continuation formats 1 and 2 cannot be combined on the same line)

	G9	G10	G11	G12	–etc.–				

Continuation Format 2:

	G3	"THRU"	G4	"BY"	INC				

Example:

ROTSE	5	CX	1001						
	101	THRU	190	BY	5				
	46	23	57	82	9	16			
	201	THRU	255						
	93	94	95	97					

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Fields:

Field	Contents
RSETID	References an RSETi on the ROTORD bulk entry. See Remark 1. (Integer>0)
TYPE	Modal reduction type. (Character: “RL”, “CX”) = “RL”, use real modal reduction. = “CX”, use complex modal reduction.
EVID	Eigenvalue extraction data set identification number. (Integer>0) For TYPE = “RL”, set identification number of an EIGRL bulk entry. For TYPE = “CX”, set identification number of an EIGC bulk entry.
Gi	Grids to remove from the o-set and place in the a-set. See Remark 2. (Integer > 0)
THRU	Specifies a range of grid ID's. (Optional)
BY	Specifies an increment when using THRU option. (Optional)
INC	Increment used with THRU option. (Integer > 0; Default=1)

Remarks:

1. The RSETID field is referred to by the RSETi field on the ROTORD bulk entry. If a model contains multiple rotors, use separate ROTSE bulk entries for each rotor.
2. In a rotor dynamic analysis, the a-set consists of any grids that are not listed on any ROTORG bulk entry and any grids that are listed on any ROTSE bulk entry.
3. If the a-set does not contain at least one unconstrained DOF, the run terminates.

Related Topics

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RROD

Rigid Pin-Ended Element Connection

Defines a pin-ended element that is rigid in translation.

Format:

1	2	3	4	5	6	7	8	9	10
RROD	EID	GA	GB	CMA	CMB	ALPHA			

Example:

RROD	14	1	2	2					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
GA, GB	Grid point identification numbers of connection points. (Integer > 0)
CMA, CMB	Component number of one and only one dependent translational degree-of-freedom in the global coordinate system assigned by the user to either GA or GB. (Integer 1, 2, or 3. Either CMA or CMB must contain the integer, and the other must be blank.)
ALPHA	Thermal expansion coefficient.

Remarks:

1. The m-set coordinates specified on this entry may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
2. Element identification numbers should be unique with respect to all other element identification numbers.
3. Rigid elements, unlike MPCs, are not selected through the Case Control Section.
4. Forces of multipoint constraint may be recovered in all solution sequences, except SOL 129, with the MPCFORCES Case Control command.
5. Rigid elements are ignored in heat transfer problems.
6. The degree-of-freedom selected to be dependent must have a nonzero component along the axis of the element. This implies that the element must have finite length.
7. See R-Type Elements in the*Simcenter Nastran Element Library* for a discussion of rigid elements.

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8. The RIGID case control command must be assigned to “LAGRAN” to use ALPHA for the calculation of thermal expansion. See the RIGID case control command for details.

9. The TEMP(LOAD) and TEMP(INIT) value used on RBAR, RROD, and RTRPLT elements is an average calculated from the grid point values. On RBE1, RBE2 and RBE3 elements, an average TEMP(LOAD) and TEMP(INIT) is calculated for each leg of the element using the values on the independent/dependent grid pairs such that each leg can have a different thermal strain if the temperatures vary at the grids.

   The rigid element thermal strains are calculated by
   $\epsilon_{thermal} = \alpha(\text{AVGTEMP(LOAD)} - \text{AVGTEMP(INIT)})$.

   If TEMP(LOAD) or TEMP(INIT) are not defined, they are assumed to be zero.

Related Topics

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RSAPPLY

Response Spectrum Data

Selects the RSPOPT, RSPECTR, and RSPLOC bulk data entries that define data and parameters for a response spectrum application run.

Format:

1	2	3	4	5	6	7	8	9	10
RSAPPLY	SID	RSPOPTID	RSPECID1	SCALE1	RSPLOID1	RSPECID2	SCALE2	RSPLOID2	
	RSPECID3	SCALE3	RSPLOID3	RSPECID4	SCALE4	RSPLOID4			
	RSPECID5	SCALE5	RSPLOID5	-etc.-					

Example:

RSAPPLY	5	100	200	1.0	220	201	1.2	221	
	202	1.0	221						

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
RSPOPTID	Identification number of an RSOPT bulk entry. (Integer > 0; See Remark 3 for default behavior)
RSPECIDi	Identification number of an RSPECTR bulk entry. (Integer > 0; No default)
SCALEi	Scale factor for the spectrum defined by the RSPECTR bulk entry listed in the RSPECIDi field. (Real; Default = 1.0)
RSPLOIDi	Identification number of an RSPLOC bulk entry. (Integer > 0; See Remark 4 for default behavior)

Remarks:

1. The set identification number must be unique with respect to all other RSAPPLY bulk entry set identification numbers.
2. The set identification number is referenced by an RSAPPLY case control command.
3. If the RSPOPTID field is left blank, the software uses the absolute value method as the summation method. Thus, leaving the RSPOPTID field blank is equivalent to referencing a RSPOPT bulk entry whose METHOD field is set to "ABS".

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4. If a RSPLOIDi field is left blank, the software uses the large mass method rather than the relative method.

Related Topics

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RSPECTR

Response Spectra Input Correlation Table

Specifies the fraction of critical damping values that correspond to the response spectra specified on TABLED1 bulk entries.

Format:

1	2	3	4	5	6	7	8	9	10
RSPECTR	SID	TYPE	DAMP1	TID1	DAMP2	TID2	DAMP3	TID3	
	DAMP4	TID4	-etc.-						

Example:

RSPECTR	5	A	0.1	100	0.2	101	0.3	102	
	0.4	103	0.5	104	0.6	105			

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
TYPE	Type of spectrum. (Character = "D", "V", or "A"; Default = "A") "D" for relative displacement as a function of frequency in Hz "V" for relative velocity as a function of frequency in Hz "A" for absolute acceleration as a function of frequency in Hz
DAMPi	Fraction of critical damping expressed as a decimal value that corresponds to the response spectrum defined in the TIDi field. (Real ≥ 0.0; No default)
TIDi	Identification number of the TABLEDi bulk entry that defines the response spectrum for the fraction of critical damping value specified in the DAMPi field. (Integer > 0; No default)

Remarks:

1. The set identification number must be unique with respect to all other RSPECTR bulk entry set identification numbers.
2. The set identification number is referenced in a RSPECIDi field of the RSAPPLY bulk entry.

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Related Topics

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RSPLINE

Interpolation Constraint Element

Defines multipoint constraints for the interpolation of displacements at grid points.

Format:

1	2	3	4	5	6	7	8	9	10
RSPLINE	EID	D/L	G1	G2	C2	G3	C3	G4	
	C4	G5	C5	G6	-etc.-				

Example:

RSPLINE	73	.05	27	28	123456	29		30	
	123	75	123	71					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
D/L	Ratio of the diameter of the elastic tube to the sum of the lengths of all segments. (Real > 0.0; Default=0.1)
Gi	Grid point identification number. (Integer > 0)
Ci	Components to be constrained. See Remark 2 . (Blank or any combination of the Integers 1 through 6.)

Remarks:

1. Displacements are interpolated from the equations of an elastic beam passing through the grid points.
2. A blank field for Ci indicates that all six degrees-of-freedom at Gi are independent. Since G1 must be independent, no field is provided for C1. Since the last grid point must also be independent, the last field must be a Gi, not a Ci. For the example shown G1, G3, and G6 are independent. G2 has six constrained degrees-of-freedom while G4 and G5 each have three.
3. Dependent (i.e., constrained) degrees-of-freedom assigned by one rigid element may not also be assigned dependent by another rigid element or by a multipoint constraint.
4. Degrees-of-freedom declared to be independent by one rigid body element can be made dependent by another rigid body element or by a multipoint constraint.

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5. EIDs must be unique.
6. Rigid elements (including RSPLINE), unlike MPCs, are not selected through the Case Control Section.
7. Forces of multipoint constraint may be recovered in all solution sequences, except SOL 129, with the MPCFORCES Case Control command.
8. Rigid elements are ignored in heat transfer problems.
9. See R-Type Elements in the *Simcenter Nastran Element Library* for a discussion of rigid elements.
10. The m-set coordinates specified on this entry may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
11. The constraint coefficient matrix is affected by the order of the Gi Ci pairs on the RSPLINE entry. The order of the pairs should be specified in the same order that they appear along the line that joins the two regions. If this order is not followed then the RSPLINE will have folds in it that may yield some unexpected interpolation results.
12. The independent degrees-of-freedom that are the rotation components most nearly parallel to the line joining the regions should not normally be constrained.
13. The RSPLINE has a limit of 100 grid points.

Related Topics

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RSPLOC

Response Spectrum Method

In a response spectrum application run, specifies:

- The use of the relative method rather than the large mass method.
- The grid points at which to apply the response spectrum and the direction in which to apply the response spectrum.

Format:

1	2	3	4	5	6	7	8	9	10
RSPLOC	SID	CID	DIR	GID1	GID2	GID3	GID4	GID5	
	GID6	GID7	-etc.-						

Example:

RSPLOC	150		3	210	211	315	68	83	
	202	435	434	436					

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
CID	Coordinate system identification number. (Integer ≥ 0; Default = 0 for the basic coordinate system)
DIR	Direction in the coordinate system listed in the CID field that the response spectrum is applied. (Integer = 1, 2, 3, 4, 5, or 6; Default = 1)
GIDi	Identification number of grid points at which the response spectrum is applied. See Remark 3. (Integer > 0; No default)

Remarks:

1. The set identification number must be unique with respect to all other RSPLOC bulk entry set identification numbers.
2. The set identification number is referenced in a RSPLOIDi field on the RSAPPLY bulk entry.
3. Each GIDi must be constrained in the direction specified by the DIR field by an SPC or SPC1 bulk entry.

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4. When you specify an RSPLOC bulk entry, the software automatically includes contributions to the response from residual vectors, if residual vectors were calculated during the modal analysis of the structure.

Related Topics

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RSPOPT

Response Spectrum Options

Specifies options for a response spectrum application run.

Format:

1	2	3	4	5	6	7	8	9	10
RSPOPT	SID	METHOD	FACTOR	OPTION	RIGRESP	VALUE1	VALUE2	MMZPA	

Example:

RSPOPT	100	SRSS	1.05						

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
METHOD	Modal combination summation method. See Remark 3. (Character = "ABS", "SRSS", "NRL", "CQC", "DSUM", "ROSE", "GRP", or "TEN"; Default = "ABS") "ABS" for absolute value method "SRSS" for square-root-of-sum-of-squares method "NRL" for Naval Research Laboratories method "CQC" for complete quadratic combination method "DSUM" for Nuclear Regulatory Commission double sum method "ROSE" for Rosenblueth method "GRP" for grouping method "TEN" for ten percent method
FACTOR	Method-dependent factor. See Remark 4. When the METHOD field is set to SRSS or NRL, the proximity of modes to sum using the ABS method. See Remark 5. (Real ≥ 0.0; Default = 1.0) When the METHOD field is set to DSUM or ROSE, the duration of the dynamic event. (Real ≥ 0.0; Default = 0.1)
OPTION	Multipoint spectra combination method. See Remarks 6 and 7. (Character = "ABS", "SRSS", "THIRTY", or "FORTY"; Default = "ABS") "ABS" for absolute value method "SRSS" for square-root-of-sum-of-squares method

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Field	Contents
	"THIRTY" for 100-30-30 method "FORTY" for 100-40-40 method
RIGRESP	Rigid response method. (Character = "GUPTA", "LINDLEY", or "NONE"; Default = "NONE") "GUPTA" for Gupta method "LINDLEY" for Lindley-Yow method "NONE" for no rigid response
VALUE1	Rigid response parameter. See Remark 8. (Real ≥ 0.0; No default) When the RIGRESP field is set to GUPTA, the first cutoff frequency in Hz. When the RIGRESP field is set to LINDLEY, the zero-period acceleration. See Remark 9.
VALUE2	Rigid response parameter. See Remark 8. (Real ≥ 0.0; No default) When the RIGRESP field is set to GUPTA, the second cutoff frequency in Hz. When the RIGRESP field is set to LINDLEY, the field is ignored.
MMZPA	Missing mass zero-period acceleration. See Remark 9. (Real ≥ 0.0; Default = 0.0)

Remarks:

1. The set identification number must be unique with respect to all other RSPOPT bulk entry set identification numbers.
2. The set identification number is referenced in the RSOPTID field of the RSAPPLY bulk entry.
3. For detailed information about the modal combination summing procedures, see "Theoretical Background of Shock and Response Spectrum Analysis" in the *Simcenter Nastran Advanced Dynamic Analysis User's Guide*.
4. The FACTOR field setting is ignored when using the ABS, CQC, GRP, and TEN methods.
5. Modes whose natural frequencies are close to one another may have their peak responses occur at nearly the same time and phase. Because of this, the SRSS and NRL methods contain a provision that sums close modal responses using the ABS method. The software uses the value in the FACTOR field to determine when to use the ABS method.

   Close natural frequencies are defined as natural frequencies that satisfy the following inequality:

   $$\omega_{i+1} < \mathrm{FACTOR} \cdot \omega_i$$

   where ω_i and ω_{i+1} are adjacent natural frequencies such that $\omega_i < \omega_{i+1}$.
6. The OPTION field is ignored when using a single-point spectra.

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7. For detailed information about the multipoint spectra combination options, see "Theoretical Background of Shock and Response Spectrum Analysis" in the *Simcenter Nastran Advanced Dynamic Analysis User's Guide*.
8. The VALUE1 and VALUE2 fields are ignored when the RIGRESP field is set to NONE.
9. Zero-period acceleration is a linear acceleration. Therefore, the units of the numerical value that you enter in the VALUE1 field should be consistent with the units for the model.

Related Topics

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RSSCON

Shell-to-Solid Element Connector

Defines multipoint constraints to model clamped connections of shell-to-solid elements.

Format:

1	2	3	4	5	6	7	8	9	10
RSSCON	RBID	TYPE	ES1	EA1	EB1	ES2	EA2	EB2	

Example:

RSSCON	110	GRID	11	12	13	14	15	16	

RSSCON	111	GRID	31	74	75				

RSSCON	115	ELEM	311	741					

Fields:

Field	Contents
RBID	Element identification number. (Integer > 0)
TYPE	Type of connectivity TYPE="ELEM" connection is described with element identification numbers. TYPE="GRID" connection is described with grid point identification numbers. (Character: "GRID" or "ELEM"; Default="ELEM")
ES1	Shell element identification number if TYPE="ELEM". Shell grid point identification number if TYPE="GRID". See **Figure 17-170**. (Integer > 0)
EA1	Solid element identification number if TYPE="ELEM". Solid grid point identification number if TYPE="GRID". (Integer > 0)
EB1	Solid grid-point identification number for TYPE="GRID" only. (Integer > 0 or blank)
ES2	Shell grid-point identification number for TYPE="GRID" only. (Integer > 0 or blank)
EA2	Solid grid-point identification number for TYPE="GRID" only. (Integer > 0 or blank)
EB2	Solid grid-point identification number for TYPE="GRID" only. (Integer > 0 or blank)

Remarks:

1. RSSCON generates a multipoint constraint that models a clamped connection between a shell and a solid element. The shell degrees-of-freedom are put in the dependent set (m-set). The translational

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degrees-of-freedom of the shell edge are connected to the translational degrees-of-freedom of the upper and lower solid edge. The two rotational degrees-of-freedom of the shell are connected to the translational degrees-of-freedom of the lower and upper edges of the solid element face. Poisson's ratio effects are considered in the translational degrees-of-freedom.

2. The shell grid point must lie on the line connecting the two solid grid points. It can have an offset from this line, which can not be more than 5% of the distance between the two solid grid points. The shell grid points that are out of the tolerance will not be constrained, and a fatal message will be issued. This tolerance is adjustable. Please see PARAM,TOLRSC and PARAM,SEPIXOVR.
3. When using the TYPE="ELEM" option:
   - The solid elements are CHEXA, CPENTA, and CTETRA with and without midside nodes. The shell elements are CQUAD4, CTRIA3, CQUADR, CTRIAR, CQUAD8, or CTRIA6.
   - Both the shell and solid elements have to belong to the same superelement. This restriction can be bypassed using SEELT entry to reassign the downstream boundary element to an upstream superelement.
   - It is not recommended to connect more than one shell element to the same solid using the ELEM option. If attempted, conflicts in the multipoint constraint relations may lead to UFM 6692.
4. When using TYPE="GRID" option:
   - The GRID option does not verify that the grids used are valid shell and/or solid grids.
   - The grids in the GRID option can be in different superelements. The shell grid must be in the upstream superelement.
5. It is recommended that the height of the solid element's face is approximately equal to the shell element's thickness of the shell. The shell edge should then be placed in the middle of the solid face.
6. The shell edge may coincide with the upper or lower edge of the solid face.
7. The RSSCON entry, unlike MPCs, cannot be selected through the Case Control Section.
8. Forces of multipoint constraints may be recovered in the linear structured solution sequences (SOLs 101 through 200) with the MPCFORCES Case Control command.
9. The RSSCON is ignored in heat-transfer problems.
10. The m-set coordinates (shell degrees-of-freedom) may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.

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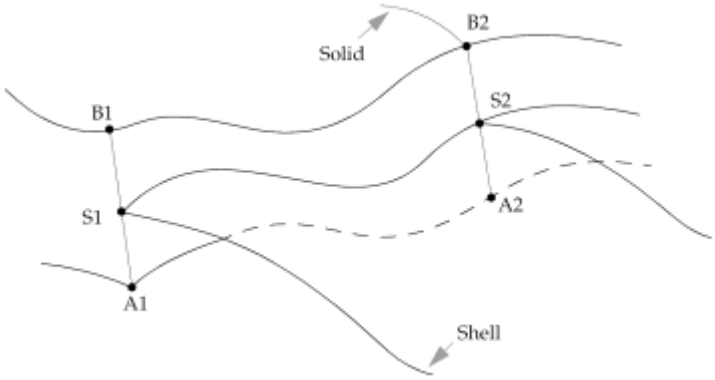
Figure 17-170: Shell Elements Connected to the Faces of Solid Elements



Related Topics

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RTRPLT

Rigid Triangular Plate

Defines a rigid triangular plate.

Format:

1	2	3	4	5	6	7	8	9	10
RTRPLT	EID	GA	GB	GC	CNA	CNB	CNC		
	CMA	CMB	CMC	ALPHA					

Example:

RTRPLT	7	1	2	3	1236	3	3		

Fields:

Field	Contents
EID	Element identification number.
GA, GB, GC	Grid point identification number of connection points.
CNA, CNB, CNC	Independent degrees-of-freedom in the global coordinate system for the element at grid points GA, GB, and GC, indicated by any of the Integers 1 through 6 with no embedded blanks. See Remark 1 . (Integer ≥ 0 or blank)
CMA, CMB, CMC	Component numbers of dependent degrees-of-freedom in the global coordinate system. (Any of the Integers 1 through 6 with no embedded blanks, or 0 or blank.)
ALPHA	Thermal expansion coefficient.

Remarks:

1. The total number of components in CNA, CNB, and CNC must equal six; for example, CNA=1236, CNB=3, CNC=3. Furthermore, they must jointly be capable of representing any general rigid body motion of the element.
2. If CMA, CMB, and CMC are all zero or blank or if the continuation entry is omitted, all of the degrees-of-freedom not in CNA, CNB, and CNC will be made dependent (i.e., they will be made members of the m-set).
3. Dependent degrees-of-freedom assigned by one rigid element may not also be assigned dependent by another rigid element or by a multipoint constraint.

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4. Element identification numbers should be unique with respect to all other element identification numbers.
5. Rigid elements, unlike MPCs, are not selected through the Case Control Section.
6. Forces of multipoint constraint may be recovered in all solution sequences, except SOL 129, with the MPCFORCES Case Control command.
7. Rigid elements are ignored in heat transfer problems.
8. See R-Type Elements in the *Simcenter Nastran Element Library* for a discussion of rigid elements.
9. The m-set coordinates specified on this entry may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
10. The RIGID case control command must be assigned to "LAGRAN" to use ALPHA for the calculation of thermal expansion. See the RIGID case control command for details.
11. The TEMP(LOAD) and TEMP(INIT) value used on RBAR, RROD, and RTRPLT elements is an average calculated from the grid point values. On RBE1, RBE2 and RBE3 elements, an average TEMP(LOAD) and TEMP(INIT) is calculated for each leg of the element using the values on the independent/dependent grid pairs such that each leg can have a different thermal strain if the temperatures vary at the grids.

    The rigid element thermal strains are calculated by
    $\epsilon_{thermal} = \alpha(\text{AVGTEMP(LOAD)} - \text{AVGTEMP(INIT)})$.

    If TEMP(LOAD) or TEMP(INIT) are not defined, they are assumed to be zero.

Related Topics

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RVDOF

Degree-of-Freedom Specification for Residual Vector Computations

Specifies the degrees-of-freedom where unit loads are applied to obtain static solutions for use in residual vector computations.

Format:

1	2	3	4	5	6	7	8	9	10
RVDOF	ID1	C1	ID2	C2	ID3	C3	ID4	C4	

Example:

RVDOF	900	1	200	3					

Fields:

Field	Contents
IDi	Grid point or scalar point identification number. (Integer > 0)
Ci	Component number. (0 ≤ Integer ≤ 6; up to six Unique Integers, 1 through 6, may be placed in the field with no embedded blanks. 0 applies to scalar points and 1 through 6 applies to grid points.)

Remarks:

1. The operation of the residual vector calculation with RVDOF is functionally equivalent to PARAM,RESVEC,YES when USET, U6 DOF are present. The RVDOF/RVDOF1 bulk entries can select both a-set and o-set DOF. This is different than the USET, U6 capability which requires USET, U6 to select a-set DOF, but requires SEUSET, U6 to select o-set DOF.
2. The unit loads applied to the interior points of a superelement due to RVDOFi bulk entries are passed downstream to the residual for the purpose of residual vector processing by all superelements in its downstream path. This produces more accurate results as compared to the results produced when USETi,U6 or SEUSETi,U6 bulk entries are used for residual vector processing. When USETi,U6 or SEUSETi,U6 bulk entries are used, unit loads on a superelement are not passed downstream for residual vector processing by the downstream superelements.

Related Topics

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RVDOF1

Degree-of-Freedom Specification for Residual Vector Computations

Specifies the degrees-of-freedom where unit loads are applied to obtain static solutions for use in residual vector computations.

Format:

1	2	3	4	5	6	7	8	9	10
RVDOF1	C	ID1	ID2	ID3	ID4	ID5	ID6	ID7	

Alternate Format:

1	2	3	4	5	6	7	8	9	10
RVDOF1	C	ID1	"THRU"	ID2					

Example:

RVDOF1	2	101	201	356					

Alternate Example:

RVDOF1	123456	7	THRU	109					

Fields:

Field	Contents
C	Component number. (0 ≤ Integer ≤ 6; up to six Unique Integers, 1 through 6, may be placed in the field with no embedded blanks. 0 applies to scalar points and 1 through 6 applies to grid points.)
IDi	Grid point or scalar point identification number. (Integer > 0)

Remarks:

1. The operation of the residual vector calculation with RVDOF is functionally equivalent to PARAM,RESVEC,YES when USET, U6 DOF are present. The RVDOF/RVDOF1 bulk entries can select both a-set and o-set DOF. This is different than the USET, U6 capability which requires USET, U6 to select a-set DOF, but requires SEUSET, U6 to select o-set DOF.

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2. The unit loads applied to the interior points of a superelement due to RVDOFi bulk entries are passed downstream to the residual for the purpose of residual vector processing by all superelements in its downstream path. This produces more accurate results as compared to the results produced when USETi,U6 or SEUSETi,U6 bulk entries are used for residual vector processing. When USETi,U6 or SEUSETi,U6 bulk entries are used, unit loads on a superelement are not passed downstream for residual vector processing by the downstream superelements.

Related Topics

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RVEL

Specification of Elements for Residual Vector Computations

Specifies CBUSH, CBUSH1D, CDAMPi, CELASi, and CVISC elements for which to obtain static solutions for use in residual vector computations.

Format:

1	2	3	4	5	6	7	8	9	10
RVEL	TYPE	ID1	ID2	ID3	ID4	ID5	ID6	ID7	

Examples:

RVEL	ELEM	3	6	8	9	15	7		

RVEL	ELEM	3	6	THRU	9	15			

Fields:

Field	Contents
TYPE	Specifies whether the identification numbers listed correspond to elements or properties. (Character: “ELEM” or “PROP”; No default)
IDi	Element identification number if TYPE = ELEM or property identification number if TYPE = PROP. See Remark 1. (Integer > 0 or “THRU”; No default)

Remarks:

1. If TYPE = PROP, all elements that reference the listed properties are available for residual vector computations.
2. Elements referenced by an RVEL bulk entry will have residual vectors calculated for opposing unit loads that are applied to the ends of the elements. Elements with stiffness or damping in multiple directions will have separate opposing loads applied for each direction.
3. RVEL bulk entries are only processed if the RESVEC case control command is present with the RVEL describer specified.
4. Continuations are not supported. Although, multiple RVEL bulk entries are permitted. For example, the following input is supported:

```
RVEL    ELEM    5       8       12      34      66      83      23
RVEL    ELEM    103     THRU    123
```


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5. The load vectors for elements contained in a superelement and specified by an RVEL bulk entry are carried to the next downstream superelement and used for residual vector computation in that downstream superelement.

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SEBNDRY

Superelement Boundary-Point Definition

Defines a list of grid points in a partitioned superelement for the automatic boundary search between a specified superelement or between all other superelements in the model.

Format:

1	2	3	4	5	6	7	8	9	10
SEBNDRY	SEIDA	SEIDB	GIDA1	GIDA2	GIDA3	GIDA4	GIDA5	GIDA6	
	GIDA7	GIDA8	-etc.-						

Example 1:

SEBNDRY	400	4	10	20	30	40			

Example 2:

SEBNDRY	400	ALL	10	20	30	THRU	40		

Fields:

Field	Contents
SEIDA	Partitioned superelement identification number. (Integer > 0)
SEIDB	Superelement identification. See Remark 2 . (Integer > 0 or Character “ALL” ; Default=“ALL”)
GIDAi	Identification number of a boundary grid point in superelement SEIDA. (Integer > 0 or “THRU”; For “THRU” option, G1 < G2.)

Remarks:

1. SEBNDRY may only be specified in the main Bulk Data Section and is not recognized after the BEGIN SUPER=n.
2. SEIDB may reference partitioned superelements or superelements in the main Bulk Data Section.

Related Topics

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SEBSET

Fixed Boundary Degrees-of-Freedom

Defines boundary degrees-of-freedom to be fixed (b-set) during component mode synthesis calculations.

Format:

1	2	3	4	5	6	7	8	9	10
SEBSET	SEID	ID1	C1	ID2	C2	ID3	C3		

Example:

SEBSET	5	2	135	14	6				

Fields:

Field	Contents
SEID	Superelement identification number. (Integer > 0)
Ci	Component number. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points; Integer zero or blank for scalar points)
IDi	Grid or scalar point identification numbers. (Integer > 0)

Remarks:

1. If there are no SECSETi or SEBSETi entries present, all boundary points are, by default, fixed during component mode analysis. If only SEBSETi are entries present, any boundary degrees-of-freedom not listed are placed in the free boundary set (c-set). If both SEBSETi and SECSETi entries are present, the c-set degrees-of-freedom are defined by the SECSETi entries and any remaining boundary points are placed in the b-set.
2. Degrees-of-freedom listed on SEBSETi entries must be exterior degrees-of-freedom of the superelement and may not be specified on SECSETi entries.
3. Degrees-of-freedom specified on this entry form members of the mutually exclusive b-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES, then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.

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- If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

Related Topics

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SEBSET1

Fixed Boundary Degrees-of-Freedom, Alternate Form of SEBSET

Defines boundary degrees-of-freedom to be fixed (b-set) during component mode calculations.

Format:

1	2	3	4	5	6	7	8	9	10
SEBSET1	SEID	C	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	-etc.-					

Example:

SEBSET1	5	2	135	14	6	23	24	25	
	122	127							

Alternate Format and Example:

SEBSET1	SEID	C	G1	“THRU”	G2				
SEBSET1	5	3	6	THRU	32				

Fields:

Field	Contents
SEID	Superelement identification number. (Integer > 0)
C	Component number. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points, 0 or blank for scalar points.)
Gi	Grid or scalar point identification numbers. (Integer > 0 or “THRU”; for THRU option G1 < G2.)

Remarks:

1. If there are no SECSETi or SEBSETi entries present, all boundary points are, by default, fixed during component mode analysis. If there are only SEBSETi entries present, any boundary degrees-of-freedom not listed are placed in the free boundary set (c-set). If there are both SEBSETi and SECSETi entries present, the c-set degrees-of-freedom are defined by the SECSETi entries, and any remaining boundary points are placed in the b-set.
2. Degrees-of-freedom listed on SEBSETi entries must be exterior degrees-of-freedom of the superelement and may not be specified on SECSETi entries.

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3. Degrees-of-freedom specified on this entry form members of the mutually exclusive b-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. If PARAM,AUTOSPC is YES, then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

Related Topics

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SEBULK

Superelement Connection

Defines superelement boundary search options and a repeated, mirrored, or collector superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SEBULK	SEID	TYPE	RSEID	METHOD	TOL	LOC	UNITNO		

Examples:

SEBULK	14	REPEAT	4	AUTO	1.0E-3				

SEBULK	101	FRFOP2		AUTO	1.0E-3		99		

Fields:

Field	Contents	
SEID	Partitioned superelement identification number. (Integer > 0)	
TYPE	Superelement type. (Character; No Default)	
	PRIMARY	Primary
	REPEAT	Identical. See Remark 1.
	MIRROR	Mirror. See Remarks 1 and 6.
	COLLCTR	Collector. See Remark 7.
	EXTERNAL	External. See Remark 8.
	EXTOP2	External using an OP2 file created in an earlier run. See Remark 8.
	EXTOP4	External using an OP4 file created in an earlier run.See Remark 8.
	FRFOP2	External using an OP2 file created in an earlier run. See Remarks 10 and 11.
	EXTSAM	External using SDB and U18 files created in an earlier run. See Remark 12.
RSEID	Identification number of the reference superelement, used if TYPE="REPEAT" and "MIRROR". (Integer≥0; Default=0)	
METHOD	Method to be used when searching for boundary grid points. (Character: "AUTO" or "MANUAL"; Default="AUTO")	

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Field	Contents
TOL	Location tolerance to be used when searching for boundary grid points. (Real; Default=10E-5)
LOC	Coincident location check option for manual connection option. (Character: “YES” or “NO”; Default=“YES”)
UNITNO	Fortran unit number for the OUTPUT2 or OP4 file (applicable and meaningful only when TYPE = ”EXTOP2”, “EXTOP4”, or "FRFOP2").

Remarks:

1. The TYPE=“REPEAT” or “MIRROR” does not include superelements upstream of the reference superelement. A repeated or mirrored superelement can have boundaries, loads, constraints, and reduction procedures that are different than the reference superelement.
2. METHOD=“MANUAL” requires SECONCT entries. SEBNDRY and SEEXCLD, which reference SEID, will produce a fatal message.
3. SECONCT, SEBNDRY, and SEEXCLD entries can be used to augment the search procedure and/or override the global tolerance.
4. For combined automatic and manual boundary search, the METHOD=“AUTO” should be specified and connections should be specified on a SECONCT entry.
5. TOL and LOC are the default values that can be modified between two superelements by providing the required tolerance on the SECONCT entry.
6. TYPE=“MIRROR” also requires specification of a SEMPLN entry.
7. TYPE=“COLLCTR” indicates a collector superelement, which does not contain any grids or scalar points.
8. For TYPE = “EXTERNAL”, “EXTOP2”, or “EXTOP4”, see discussion under the description of the EXTSEOUT case control command for employing external superelements using the two-step procedure.
   For employing external superelements using the three-step procedure, see discussion under the description of the EXTOUT parameter.
9. This entry will only work if PART superelements (BEGIN SUPER) or external superelements created by employing the EXTSEOUT case control entry exist.
10. Specify TYPE = "FRFOP2" for frequency-dependent boundary matrices that are valid for SOL 108 and 111 only.
11. For TYPE = "FRFOP2", the boundary DOF must connect to the residual structure (SEID=0).
12. Specify TYPE = "EXTSAM" for boundary matrices that are valid for SOL 414 only.

Related Topics

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SECONCT

Partitioned Superelement Boundary-Point Connection

Explicitly defines grid and scalar point connection procedures for a partitioned superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SECONCT	SEIDA	SEIDB	TOL	LOC					
	GIDA1	GIDB1	GIDA2	GIDB2	GIDA3	GIDB3	-etc.-		

Example:

SECONCT	10	20	1.0E-4	YES					
	1001	4001			2222	4444			

Alternate Format and Example:

SECONCT	SEIDA	SEIDB	TOL	LOC					
	GIDA1	'THRU'	GIDA2	GIDB1	'THRU'	GIDB2			

SECONCT	10	20							
	101	'THRU'	110	201	'THRU'	210			

Fields:

Field	Contents
SEIDA	Partitioned superelement identification number. (Integer > 0)
SEIDB	Identification number of superelement for connection to SEIDA. (Integer ≥ 0)
TOL	Location tolerance to be used when searching for or checking boundary grid points. (Real; Default=10E-5)
LOC	Coincident location check option for manual connection. (Character; "YES" or "NO"; Default="YES")
GIDAi	Identification number of a grid or scalar point in superelement SEIDA, which will be connected to GIDBi.
GIDBi	Identification number of a grid or scalar point in superelement SEIDB, which will be connected to GIDAi.

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Remarks:

1. SECONCT can only be specified in the main Bulk Data Section and is ignored after the BEGIN SUPER=n command.
2. TOL and LOC can be used to override the default values specified on the SEBULK entries.
3. The continuation entry is optional.
4. The (GIAi, GIBi) pair must both be grids or scalar points.
5. All six degrees-of-freedom of grid points will be defined as boundary degrees-of-freedom.
6. This entry will only work if PART superelements (BEGIN SUPER) exist.
7. Blank fields are allowed after the first GIDA1-GIDB1 pair. Blank fields must also occur in pairs. This remark does not apply to the alternate format.
8. For Alternate Format 1, the thru ranges must be closed sets. That is, all IDs listed between 101 and 110 in the example must exist in the model.

Related Topics

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SECSET

Free Boundary Degrees-of-Freedom

Defines boundary degrees-of-freedom to be free (c-set) during component mode synthesis calculations.

Format:

1	2	3	4	5	6	7	8	9	10
SECSET	SEID	ID1	C1	ID2	C2	ID3	C3		

Example:

SECSET	3	124	1	5	23	6	15		

Fields:

Field	Contents
SEID	Superelement identification number. (Integer > 0)
Ci	Component numbers. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points; Integer 0 or blank for scalar points.)
IDi	Grid or scalar point identification number. (Integer > 0)

Remarks:

1. Exterior grid and scalar points are, by default, fixed during component mode analysis and placed in the b-set unless listed on SECSETi or SESUP entries. Coordinates listed on this entry are considered free (c-set) during component mode calculations. Exterior grid and scalar points are determined by the program and listed in the SEMAP table output.
2. Degrees-of-freedom specified on this entry are assigned to the mutually exclusive c-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
3. There must be a sufficient number of degrees-of-freedom specified on SESUP entries to discard any free body modes of the superelement.
4. If PARAM,AUTOSPC is YES, then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

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Related Topics

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SECSET1

Free Boundary Degrees-of-Freedom, Alternate Form of SECSET

Defines boundary degrees-of-freedom to be free (c-set) during component mode synthesis calculations.

Format:

1	2	3	4	5	6	7	8	9	10
SECSET1	SEID	C	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	-etc.-					

Example:

SECSET1	5	2	135	14	6	23	24	25	
	122	127							

Alternate Formats and Example:

SECSET1	SEID	C	G1	“THRU”	G2				
SECSET1	5	3	6	THRU	32				

SECSET1	SEID		“ALL”						
SECSET1	SEID		ALL						

Fields:

Field	Contents
SEID	Superelement identification number. (Integer > 0)
C	Component numbers of degree-of-freedoms. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points; Integer 0 or blank for scalar points.)
Gi	Grid or scalar point identification number. (Integer > 0)

Remarks:

1. Exterior grid and scalar points are, by default, fixed during component mode analysis and placed in the b-set unless listed on SECSETi or SESUP entries. Degrees-of-freedom listed on this entry are considered free (c-set) during component mode calculations. Exterior grid and scalar points are determined automatically and listed in the SEMAP table output.

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2. If the alternate formats are used, the grid points Gi are not required to exist or to be exterior degrees-of-freedom and may be listed on SECSET1 entries. Points of this type will cause one warning message but will otherwise be ignored.
3. Degrees-of-freedom specified on this entry are assigned to the mutually exclusive c-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
4. There must be a sufficient number of degrees-of-freedom specified on SESUP entries to discard any free body modes of the superelement.
5. If PARAM,AUTOSPC is YES, then singular b-set and c-set degrees-of-freedom will be reassigned as follows:
   - If there are no o-set (omitted) degrees-of-freedom, then singular b-set and c-set degrees-of-freedom are reassigned to the s-set.
   - If there are o-set (omitted) degrees-of-freedom, then singular c-set degrees-of-freedom are reassigned to the b-set. Singular b-set degrees-of-freedom are not reassigned.

Related Topics

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SEDAMP

Superelement Damping

Defines viscous and structural (hysteretic) damping for superelements created using SOL 414,103. You can define SEDAMP only for superelements created without any damping in the external creation run.

Format:

1	2	3	4	5	6	7	8	9	10
SEDAMP	SEID	ALPHA2	ALPHA1	GE					

Example:

SEDAMP	10	0.03	0.05	0.02					

Fields:

Name	Description
SEID	Superelement identification number. (Integer > 0)
ALPHA2	Viscous damping coefficient for stiffness. See Remarks 1 and 2. (Real > 0.0 or Integer > 0; Default = 0.0)
ALPHA1	Viscous damping coefficient for mass. See Remark 1. (Real > 0.0; Default = 0.0)
GE	Structural damping coefficient. See Remark 3. (Real > 0.0; Default = 0.0)

Remarks:

1. The viscous damping matrix for the superelement is formed from a linear combination of the mass and stiffness matrices for the superelement as follows:

$$[B] = \alpha_1 [M] + \alpha_2 [K]$$

where:

$[B]$	Viscous damping matrix
$[M]$	Mass matrix
$[K]$	Stiffness matrix
α_1	Value in the ALPHA1 field

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α_2 Value in the ALPHA2 field

2. Integer entry references a TABLEDi bulk entry that defines the viscous damping coefficient for stiffness as a function of:
   - Frequency in a SOL 414,110 complex modal or SOL 414,111 harmonic response analysis.
   - Time in a SOL 414,103 eigenvalues and superelement reduction or SOL 414,129 transient response analysis.
3. The complex stiffness for the superelement is formed from the stiffness matrix for the superelement as follows:

$$\left[K'\right] = (1 + iGE)\left[K\right]$$

where:

$\left[K'\right]$ Complex stiffness matrix

$\left[K\right]$ Stiffness matrix

GE Value in the GE field

4. The software considers PARAM,ALPHA1 and PARAM,ALPHA2 as follows:
   - When you add SEDAMP and define PARAM,ALPHA1 and PARAM,ALPHA2, the software ignores PARAM,ALPHA1 and PARAM,ALPHA2 for the superelement.
   - When you omit SEDAMP and define PARAM,ALPHA1 and PARAM,ALPHA2, the software applies PARAM,ALPHA1, and PARAM,ALPHA2 to the superelement.

Related Topics

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SEELT

Superelement Boundary Element Reassignment

Reassigns superelement boundary elements to an upstream superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SEELT	SEID	EID1	EID2	EID3	EID4	EID5	EID6	EID7	

Example:

SEELT	2	147	562	937					

Alternate Format and Example:

SEELT	SEID	EID1	“THRU”	EID2					
SEELT	5	12006	THRU	12050					

Fields:

Field	Contents
SEID	Superelement identification number. (Integer > 0)
EIDi	Element identification numbers. (Integer > 0 or “THRU”; for “THRU” option EID1 < EID2.)

Remarks:

1. Elements connected entirely to the exterior points of an upstream superelement are called boundary elements and are assigned to the downstream superelement. The SEELT entry provides the means of reassigning the element to the upstream superelement. This entry may be applied to boundary elements only.
2. Open sets are allowed with the “THRU” option.
3. Elements processed with primary superelements will also be contained in any referencing secondary superelement.
4. EIDi may refer to plot elements, general elements, and structural elements.
5. This entry does not change the exterior grid point set of the superelement.
6. SEELT can only be specified in the main Bulk Data Section and is ignored after the BEGIN SUPER=n command.

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7. If the model contains a BEGIN SUPER, the SEELT entry will assign the specified elements to the SEID (not just the boundary elements). This is an alternative to using SESET. In this case, SEID=0 is a valid entry.

Related Topics

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SEEXCLD

Partitioned Superelement Exclusion

Defines grids that will be excluded during the attachment of a partitioned superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SEEXCLD	SEIDA	SEIDB	GIDA1	GIDA2	GIDA3	GIDA4	GIDA5	GIDA6	
	GIDA7	GIDA8	-etc.-						

Example1:

SEEXCLD	110	10	45	678	396				

Example 2:

SEEXCLD	400	ALL	10	20	30	THRU	40		

Fields:

Field	Contents
SEIDA	Partitioned superelement identification number. (Integer ≥ 0)
SEIDB	Superelement identification. (Integer ≥ 0 or Character="ALL", Default="ALL")
GIDAi	Identification number of a grid in superelement SEIDA to be excluded from connection to superelement SEIDB. (Integer > 0 or "THRU"; for "THRU" option G1 < G2.)

Remarks:

1. SEEXCLD can only be specified in the main Bulk Data Section and is ignored after the BEGIN SUPER=n command.
2. SEIDB may reference partitioned superelements or superelements defined in the main Bulk Data Section.
3. This entry will only work if PART superelements (BEGIN SUPER) exist.

Related Topics

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SELABEL

Superelement Output Label

Defines a label or name to be printed in the superelement output headings.

Format:

1	2	3	4	5	6	7	8	9	10
SELABEL	SEID	LABEL							

Example:

SELABEL	10	LEFT REAR FENDER, MODEL XYZ2000	

Fields:

Field	Contents
SEID	Partitioned superelement identification number. (Integer > 0)
LABEL	Label associated with superelement SEID for output headings. (Character. See **Bulk Data Syntax Rules**.)

Remarks:

1. SELABEL can only be specified in the main Bulk Data Section and is ignored after the BEGIN SUPER=n command.
2. Only one SELABEL per superelement may be specified.
3. The label will appear in all superelement output headings. However, in some headings the label may be truncated.
4. This entry will only work if PART superelements (BEGIN SUPER) exist.

Related Topics

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SELMODE

Mode selection for DDAM solution

Defines mode selection parameters for a SOL 187 DDAM solution

Format:

1	2	3	4	5	6	7	8	9	10
SELMODE		PARAM	VALUE						

Examples:

SELMODE		CUTOFF	80.0						

SELMODE		PERCENT	0.1						

SELMODE		TOP	10						

Fields:

Name	Description
PARAM	Name of the parameter. Allowable names are in the parameter listing below. (Character)
VALUE	Value of the parameter. (Real or Integer)

Parameters:

Name	Description
CUTOFF	CUTOFF: The software sorts the modes in decreasing order of modal weight (W_i) such that W_1 has the largest modal weight. Then the cumulative modal weight sum (CW_i) is computed for each mode such that $CW_1=W_1$, $CW_2=W_1+W_2$, $CW_3=W_1+W_2+W_3$, etc. The modes with (CW_i X 100% / Total Modal Weight > CUTOFF) are ignored for response computations (Real > 0.0; Default=0.0).
PERCENT	PERCENT: Modes with (Modal Weight / Total Modal Weight < PERCENT) are ignored for response computations (Real > 0.0; Default=0.0).
TOP	TOP: The top n modes with the largest modal weight are included. All other modes are ignored for response computations (Integer > 0; Default=All modes are included)

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Related Topics

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SELOAD

External Superelement Load Mapping to Residual

Maps loads from an external superelement to a specified load set for the residual structure.

Format:

1	2	3	4	5	6	7	8	9	10
SELOAD	LIDS0	SEID	LIDSE						

Example:

SELOAD	10010	100	10						

Fields:

Field	Contents
LIDS0	Identification number of the mapped load set to be used in the residual structure. (Integer > 0)
SEID	Partitioned identification number of the partitioned external superelement representing the external superelement. (Integer > 0)
LIDSE	Load set identification number used in the external superelement. (Integer > 0)

Remarks:

1. SELOAD can only be specified in the main bulk section and is ignored after the BEGIN SUPER = *n* command. Because this load is applied at the boundary grids of the external superelement, the external superelement boundary grids must be connected (via the SECONCT bulk entry for example) directly to a residual grid. If this is not done, the external superelement loading cannot be properly mapped to the residual and an error will be issued (e.g. user fatal message 2008; load set 2 references undefined grid point 201).

   As a best practice, create the external superelement using the component mode reduction method. The component mode reduction method requires the use of q-set scalar DOF to represent the component modes. The q-set DOF must also be connected to the corresponding scalar DOF in the residual via the SECONCT bulk entry. The scalar DOF in the external superelement and the residual are defined using the SPOINT bulk entry.
2. SELOAD only works if PART superelements (BEGIN SUPER) exist that represent external superelements created using the EXTSEOUT case control command.
3. LIDSE is the identification number of a load set defined in the external superelement creation run using the EXTSEOUT case control command.

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4. Multiple SELOAD commands can be used for the same SEID, but a specific LIDSE can only be mapped once.
5. LIDS0 must be unique for the residual structure.
6. The mapped load sets LIDS0s can be referenced by other load bulk entries in the system solution. Examples of load bulk entries include LOAD, LSEQ, TLOADi, and RLOADi.

Related Topics

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SELOC

Partitioned Superelement Location

Defines a partitioned superelement relocation by listing three non-colinear points in the superelement and three corresponding points not belonging to the superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SELOC	SEID	PA1	PA2	PA3	PB1	PB2	PB3		

Example:

SELOC	110	10	100	111	1010	112	30		

Fields:

Field	Contents
SEID	Partitioned identification number of the partitioned superelement. (Integer > 0)
PAi	Identification numbers of three non-colinear grids which are in the partitioned superelement. (Integer > 0)
PBi	Identification numbers of three grids (GRID entry) or points (POINT entry) defined in the main Bulk Data Section to which PAi will be aligned. (Integer > 0)

Remarks:

1. SELOC can only be specified in the main Bulk Data Section and is ignored after the BEGIN SUPER=n command.
2. The superelement will be rotated and translated for alignment of the GAi and GBi locations.
3. The PAi can be grid points only; PBi can be either GRIDs or POINTs.
4. PA1, PA2, and PA3 must be contained in superelement SEID.
5. PB1, PB2, and PB3 must be specified in the main Bulk Data Section. If they belong to a superelement that is also relocated, then the original (unmoved) positions of PB1, PB2, and PB3 are used.
6. PB1, PB2, and PB3 must have the same relative locations as PA1, PA2, and PA3.
7. Three grids or points are required even if the superelement connects to only one or two exterior grids.
8. Coordinate systems, global displacement directions, and element coordinate systems for the superelement will rotated and translated.

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9. The global coordinate directions of the boundary grid points of the upstream superelement will be transformed internally to the global coordinate directions of the attachment grid points in the downstream superelement. For displacement data recovery, the output will be in the original global coordinate system.
10. The translation and rotation of the superelement to the new position is accomplished by defining local rectangular coordinate systems based on the specified grid locations:
    - The local systems have their origin at PX1 and the x-axis points from PX1 to PX2.
    - The y-axis lies in the plane containing PX1, PX2, and PX3, is perpendicular to the x-axis, and points toward PX3.
    - The z-axis is defined by the cross product of the x-axis into the y-axis.
    - The rotation and translation transformation aligns the local system defined by the superelement grids with the local system defined by the main Bulk Data Section grids.
11. This entry will only work if PART superelements (BEGIN SUPER) exist.

Related Topics

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SEMAT

Superelement Positioning

Defines the location and orientation relative to the basic coordinate system for a superelement created using SOL 414,103.

Format:

1	2	3	4	5	6	7	8	9	10
SEMAT	SEID	R11	R12	R13	T1				
		R21	R22	R23	T2				
		R31	R32	R33	T3				

Example:

SEMAT	10	0.0	-1.0	0.0	2.8				
		1.0	0.0	0.0	0.0				
		0.0	0.0	1.0	-0.7				

Fields:

Name	Description
SEID	Superelement identification number. (Integer > 0)
Rij	Direction cosine between the i^{th} unit vector in the superelement coordinate system and the j^{th} unit vector in the basic coordinate system. (-1.0 ≤ Real ≤ 1.0; Default = 1.0 for i = j and default = 0.0 for i ≠ j)
Ti	Component of the position vector that locates the origin of the superelement coordinate system relative to the basic coordinate system. (Real; Default = 0.0)

Related Topics

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SEMCF

Superelement Scaling Factors

Defines scaling factors for superelement matrices created using SOL 414,103.

Format:

1	2	3	4	5	6	7	8	9	10
SEMCF	SEID	CK	CM	CG	CB	CC			

Example:

SEMCF	10	1.1		0.9	1.1				

Fields:

Name	Description
SEID	Superelement identification number. (Integer > 0)
CK	Scaling factor for stiffness matrix. See Remark 1. (Real > 0.0; Default = 1.0)
CM	Scaling factor for mass matrix. See Remark 1. (Real > 0.0; Default = 1.0)
CG	Scaling factor for gyroscopic matrix. See Remark 1. (Real; Default = 1.0)
CB	Scaling factor for damping matrix. See Remark 1. (Real > 0.0; Default = 1.0)
CC	Scaling factor for circulatory force matrix. See Remark 1. (Real; Default = 1.0)

Remarks:

1. The scaling factors are applied to the superelement matrices as follows:

$$[A'] = C\,[A]$$

where:

$[A']$	Scaled matrix
$[A]$	Unscaled matrix
C	Value in the scaling factor field for the matrix

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2. The scaling factors can be used to overweight or underweight matrices related to the superelement listed in the SIED field. For the gyroscopic and circulatory force matrices, the scale factor can also be used to eliminate those matrices from the solution. To do so, set the corresponding scale factor to 0.0.
3. The superelement matrices contain terms related to both translational and rotational DOF. Thus, the dimensionality of the elements in a matrix may differ. For example, a mass matrix typically contains elements that have the dimensionality of mass and others that have the dimensionality mass moment of inertia. Because of this, do not use a SEMCF bulk entry to convert the units of the superelement to those of the residual structure, rather use a SEUNIT bulk entry to convert the units of the superelement to those of the residual structure.

Related Topics

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SEMPLN

Superelement Mirror Plane

Defines a mirror plane for mirroring a partitioned superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SEMPLN	SEID	“PLANE”	P1	P2	P3				

Example:

SEMPLN	110	PLANE	12	45	1125				

Fields:

Field	Contents
SEID	Partitioned superelement identification number. (Integer > 0)
“PLANE”	Flag indicating that the plane is defined by three non-colinear points.
Pi	GRID or POINT entry identification numbers of three non-colinear points. (Integer > 0)

Remarks:

1. SEMPLN can only be specified in the main Bulk Data Section and is ignored after the BEGIN SUPER=n command.
2. Grids or points referenced on this entry must be defined in the main Bulk Data Section.

Related Topics

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SENAME

Superelement Description

References a DESC bulk entry that contains a description of the superelement created using SOL 414,103.

Format:

1	2	3	4	5	6	7	8	9	10
SENAME	SEID	DID							

Example:

SENAME	10	15							

Fields:

Name	Description
SEID	Superelement identification number. (Integer > 0)
DID	Identification number of a DESC bulk entry. (Integer > 0)

Related Topics

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SENQSET

Superelement Internal Generalized Degree-of-Freedom

Defines number of internally generated scalar points for superelement dynamic reduction.

Format:

1	2	3	4	5	6	7	8	9	10
SENQSET	SEID	N							

Example:

SENQSET	110	45							

Fields:

Field	Contents
SEID	Partitioned superelement identification number. See Remark 3 . (Integer > 0 or Character="ALL")
N	Number of internally generated scalar points for dynamic reduction generalized coordinates. (Integer > 0; Default=0)

Remarks:

1. SENQSET can only be specified in the main Bulk Data Section and is ignored after the BEGIN SUPER=n command.
2. SENQSET is only required if the user wants to internally generated scalar points used for dynamic reduction.
3. SEID="ALL" will automatically generate N q-set degrees-of-freedom for all superelements, except the residual structure (SEID=0). Specifying additional SENQSET entries for specific superelements will override the value of N specified on this entry.
4. If the user manually specifies q-set degrees-of-freedom using a SEQSETi or QSETi entries, then the internally generated scalar points will not be generated.
5. See PARAM,NQSET for an alternate method of specifying QSET degree-of-freedoms.
6. This entry will only work if PART superelements (BEGIN SUPER) exist.

Related Topics

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SENSCLC

Contact clearance sensor

Monitors the space between a grid point on a rotor in a fixed reference frame and a group of grid points on a stator. See Remark 1.

The SENSCLC bulk entry acts as a sensor for how much of the allowed clearance (estimated gap between the rotor and stator) is consumed when the rotor is in rotation.

You can define a SENSCLC bulk entry in SOL 414,101, SOL 414,103 with a transient preload subcase, SOL 414,103 and SOL 414,110 with static preloads, SOL 414,111, and SOL 414,129. See Remark 2.

Format:

1	2	3	4	5	6	7	8	9	10
SENSCLC	SID	GA	STATORGR	CL	CTYPE	OUTPUT	CID	RID	

Example:

SENSCLC	1	99	5	1.1	TIME	123	1	2	

Fields:

Field	Contents
SID	Identification number of the SENSCLC bulk entry. (Integer > 0; No default)
GA	ID of a grid point on the rotor (1D, 2D axisymmetric, 3D, or superelement). See Remark 3. (Integer > 0; No default)
STATORGR	ID of the GROUP bulk entry that identities the grid points on the stator. (Integer > 0; No default)
CL	Clearance definition that represents the available gap between GA (grid point on rotor) and STATORGR (group of grid points on stator). (Integer > 0 for constant radial clearance, or Real for a function; No default) CL can be: Integer value > 0 = ID of a TABLED1 representing a function of frequency, rotation speed, or time. See Remark 4. Real value > 0 = Constant radial clearance.
CTYPE	If CL is an ID of a TABLED1 representing a function, the type of dependent variable. See Remark 2. (No default)

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Field	Contents
	FREQ = frequency dependent ROTF = rotor speed dependent TIME = time dependent
OUTPUT	The output to generate for XY plots. (Possible values are 1, 2, 3, 12, 13, 23, 123. Default = "123") The XY plots are represented by a combination of the integers 1, 2, and 3 where: "1" = maximum gap or clearance "2" = residual gap or clearance "3" = radial clearance, which is the same as CL when CL is defined with a real value that represents radial clearance
CID	Coordinate system identification number. The coordinate system can be rectangular or cylindrical) as long as the Z-axis is the axis of rotation. (Integer; Default = 0 for the basic coordinate system) You can use this option when the clearance does not depend on the rotation speed.
RID	Rotor ID. See Remarks 1 and 5. (No default) The CID of the rotor and the rotation speed are retrieved automatically.

Remarks:

1. The SENSCLC bulk entry is valid for rotors in a fixed reference frame. You cannot define it for rotors in a rotating reference frame.
2. You can define the SENSCLC bulk entry in the following SOL 414 solutions.

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Clearance definition (CL)	Dependent variable (CTYPE)	SOL 414,101	SOL 414,103 (transient preload)	SOL 414,103 (static preload) SOL 414,110 (static preload)	SOL 414,111 (1)	SOL 414,129
Constant	N/A	X	X	X	X	X
Variable	ROTF	N/A	N/A	N/A	X	X
	FREQ	N/A	N/A	N/A	X	N/A
	TIME	N/A	X	N/A	N/A	X

(1) When you define a static preload in SOL 414,111 Harmonic Response solutions, you can define only a constant clearance.

Note

In SOL 414,103 and SOL 414,110 solutions with no preload, the SENSCLC bulk entry is ignored.

3. You can select the rotor grid point as follows:
   - 1D rotor model—Select a grid point on the rotor axis.
   - 2D axisymmetric rotor model—Select any grid point on the 2D Fourier rotor model. Creating an additional grid point on the rotor axis to link to an FOU3 element is not necessary.
   - 3D rotor model (not cyclic symmetry sector)—Select a grid point on the rotor axis. The grid point must be linked to the rotor grid point with an RBE3 element.
   - Superelement of a rotor—Select a retained grid point on the rotor axis.
4. When you define the clearance function in TABLED1, the software uses the following units:
   - Hz for frequency
   - rad/s for rotation speed
   - seconds for time
5. You can define either the CID or RID:
   - If the clearance is expressed as a function of the rotation speed, you must define the RID.
   - If RID is defined with CID, the software uses the RID.
   - If RID is not defined, and CID is not defined, then the CID absolute is the default.

Related Topics

Browse Simcenter Nastran documentation

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SEOUT

Superelement XY Plot Data

Requests results calculation for result type and grid point combinations specified on OUTMGT bulk entries in superelement creation runs during a SOL 414,103 with a transient response subcase, SOL 414,111, or SOL 414,129 analysis system run.

Format:

1	2	3	4	5	6	7	8	9	10
SEOUT	SEID	DISP	VELOC	ACCEL					

Example:

SEOUT	101	YES	NO	YES					

Fields:

Field	Contents
SEID	Superelement identification number. (Integer > 0)
DISP	Generate displacement data as a function of frequency or time data for grid points. (Character: "YES" or "NO"; No default) "YES" to generate displacement data. "NO" to not generate displacement data.
VELOC	Generate velocity data as a function of frequency or time data for grid points. (Character: "YES" or "NO"; No default) "YES" to generate velocity data. "NO" to not generate velocity data.
ACCEL	Generate acceleration data as a function of frequency or time data for grid points. (Character: "YES" or "NO"; No default) "YES" to generate acceleration data. "NO" to not generate acceleration data.

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Remarks:

1. You can use the SEOUT capability to obtain plots of displacement, velocity, and acceleration as a function of frequency or time for grid points in superelements that are not boundary grid points, without having to define retained grid points. This capability is supported in the following solution types:

   - SOL 414,103 with a transient response subcase
   - SOL 414,111
   - SOL 414,129 analysis with superelements system run

   To use this capability, include an OUTMGT case control command and one or more OUTMGT bulk entries in the input files for the SOL 414,103 superelement creation runs. On the OUTMGT bulk entries, specify the result type and grid point combinations for which you want to create the plots. For the system run, include an SEOUT bulk entry in the input file. During the system run, the software calculates the results for the result type and grid point combinations and writes the results to the result file where they can be accessed by a post-processor.

Related Topics

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SEQEP

Grid and Scalar Point Resequencing

Redefines the sequence of extra points to optimize bandwidth.

Format:

1	2	3	4	5	6	7	8	9	10
SEQEP	ID1	SEQID1	ID2	SEQID2	ID3	SEQID3	ID4	SEQID4	

Example:

SEQEP	5392	15.6	596	0.2	2	1.9	3	2	

Fields:

Field	Contents
IDi	Extra point identification number. (Integer > 0)
SEQIDi	Sequenced identification number. (Real or Integer > 0)

Remarks:

1. The real format is used to insert a point ID between two consecutively numbered and existing point IDs. In the example above, point ID 5392 is inserted between IDs 15 and 16 by specifying 15.6 for SEQID.
2. The SEQIDi numbers must be unique and may not be the same as a point IDi that is not being changed. No extra point IDi may be referenced more than once.
3. From one to four extra points may be specified on a single entry.
4. If an extra point IDi is referenced more than once, the last reference will determine its sequence.

Related Topics

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SEQGP

Grid and Scalar Point Resequencing

Used to manually order the grid points and scalar points of the problem. This entry is used to redefine the sequence of grid and scalar points to optimize bandwidth.

Format:

1	2	3	4	5	6	7	8	9	10
SEQGP	ID1	SEQID1	ID2	SEQID2	ID3	SEQID3	ID4	SEQID4	

Example:

SEQGP	5392	15.6	596	0.2	2	1.9	3	2	

Fields:

Field	Contents
IDi	Grid or scalar point identification number. (Integer > 0)
SEQIDi	Sequenced identification number. (Real > 0.0 or Integer > 0)

Remarks:

1. The real format is used to insert a point ID between two consecutively numbered and existing point IDs. In the example above, point ID 5392 is inserted between IDs 15 and 16 by specifying 15.6 for SEQID.
2. The SEQIDi numbers must be unique and may not be the same as a point IDi which is not being changed. No grid point IDi may be referenced more than once.
3. From one to four grid or scalar points may be resequenced on a single entry.
4. If a point IDi is referenced more than once, the last reference will determine its sequence.
5. Automatic resequencing is also available. See **"OLDSEQ"** .

Related Topics

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SEQSEP

Superelement Sequences

Used with the CSUPER entry to define the correspondence of the exterior grid points between an identical or mirror-image superelement and its primary superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SEQSEP	SSID	PSID	GP1	GP2	GP3	GP4	GP5	GP6	
	GP7	GP8	-etc.-						

Example:

SEQSEP	121	21	109	114	124	131			

Fields:

Field	Contents
SSID	Identification number for secondary superelement. (Integer > 0)
PSID	Identification number for the primary superelement. (Integer ≥ 0)
GPi	Exterior grid point identification numbers for the primary superelement. (Integer > 0)

Remarks:

1. This entry is not needed if the grid points listed on the CSUPER entry with the same SSID are in the order of the corresponding exterior grid points of the primary superelement.
2. In Figure 17-171, the exterior grid points of 10, 20, and 30 of SEID=1 correspond to the points 13, 12, and 11, respectively, of image SEID=2. The CSUPER entry may be defined alone or with a SEQSEP entry as shown in Figure 17-171.

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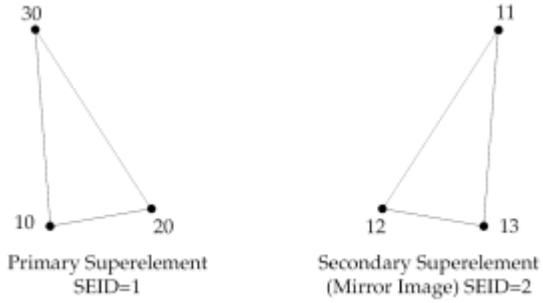
Figure 17-171: Grid Point Correspondence Between Primary and Secondary Superelements



CSUPER Entry Only:

1	2	3	4	5	6	7	8	9	10
CSUPER	2	1	13	12	11				

CSUPER and SEQSEP Entries:

CSUPER	2	I	11	12	13				
SEQSEP	2	1	30	20	10				

Related Topics

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SEQSET

Superelement Generalized Degrees-of-Freedom

Defines the generalized degrees-of-freedom of the superelement to be used in component mode synthesis.

Format:

1	2	3	4	5	6	7	8	9	10
SEQSET	SEID	ID1	C1	ID2	C2	ID3	C3		

Example:

SEQSET	15	1	123456	7	5	22	3		

Fields:

Field	Contents
SEID	Superelement identification number. Must be a primary superelement. (Integer > 0)
Ci	Component number. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points; Integer zero or blank for scalar points.)
IDi	Grid or scalar point identification numbers. Must be an exterior point. (Integer > 0)

Remarks:

1. Degrees-of-freedom specified on this entry may not be specified for another superelement.
2. Generalized degrees-of-freedom are interior to the residual structure.
3. Connectivity to the superelement is provided by this entry. There is no need to use a CSUPEXT entry for this purpose.
4. Degrees-of-freedom specified on this entry form members of the mutually exclusive q-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
5. This entry describes the set used for generalized degrees-of-freedom only for the SEID listed. Degrees-of-freedom listed on this entry must also be members of a downstream superelement. The set used for these variables in downstream superelements must be prescribed by user action using other entries. If they are scalar points, they are automatically in the residual structure, which is the recommended procedure. If they are grid points, it is the user's responsibility to place them in a downstream superelement.

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Generalized degrees-of-freedom of superelements that are also members of the residual structure are included as dynamic variables by placing them in the a-set. It is also necessary to place some or all residual structure physical degrees-of-freedom in the a-set to allow the boundary points to participate in the system mode shapes.

Grid points of downstream superelements used as generalized degrees-of-freedom may be used for advanced applications, such as omitting upstream generalized degrees-of-freedom from assembly into downstream superelements. Again, it is the user's responsibility to place these variables in the proper set in all downstream superelements of which they are members.

6. This entry may be applied only to primary superelements. The CSUPER entry automatically defines these degrees-of-freedom for secondary superelements.

Related Topics

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SEQSET1

Superelement Generalized Degrees-of-Freedom, Alternate Form

Defines the generalized degrees-of-freedom of the superelement to be used in component mode synthesis.

Format:

1	2	3	4	5	6	7	8	9	10
SEQSET1	SEID	C	G1	G2	G3	G4	G5	G6	
	G7	G8	-etc.-						

Example:

SEQSET1	15	123456	1	7	9	22	105	6	
	52	53							

Alternate Format and Example:

SEQSET1	SEID	C	G1	“THRU”	G2				
SEQSET1	16	0	101	THRU	110				

Fields:

Field	Contents
SEID	Superelement identification number. Must be a primary superelement. (Integer > 0)
C	Component numbers. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points; Integer 0 or blank for scalar points.)
Gi	Grid or scalar point identification numbers. Must be exterior points. (Integer > 0 or “THRU”; for THRU option G1 < G2.)

Remarks:

1. Degrees-of-freedom specified on this entry may not be specified for another superelement.
2. Generalized degrees-of-freedom are interior to the residual structure.
3. Connectivity to the superelement is provided by this entry. There is no need to use a CSUPEXT entry for this purpose.

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4. Degrees-of-freedom specified on this entry form members of a mutually exclusive set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.

5. This entry describes the set used for generalized degrees-of-freedom only for the SEID listed. Degrees-of-freedom listed on this entry must also be members of a downstream superelement. The set used for these variables in downstream superelements must be prescribed by user action using other entries. If they are scalar points, they are automatically in the residual structure, which is the recommended procedure. If they are grid points, it is the user's responsibility to place them in a downstream superelement.

   Generalized degrees-of-freedom of superelements that are also members of the residual structure are included as dynamic variables by placing them in the a-set. It is also necessary to place some or all residual structure physical degrees-of-freedom in the a-set, to allow the boundary points to participate in the system mode shapes.

   Grid points of downstream superelements used as generalized degrees-of-freedom may be used for advanced applications, such as omitting upstream generalized degrees-of-freedom from assembly into downstream superelements. Again, it is the user's responsibility to place these variables in the proper set in all downstream superelements of which they are members.

6. This entry may be applied only to primary superelements. The CSUPER entry automatically defines these entries for secondary superelements.

Related Topics

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SESET

Superelement Interior Point Definition

Defines interior grid points for a superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SESET	SEID	G1	G2	G3	G4	G5	G6	G7	

Example:

SESET	5	2	17	24	25	165			

Alternate Format and Example:

SESET	SEID	G1	"THRU"	G2					
SESET	2	17	THRU	165					

Fields:

Field	Contents
SEID	Superelement identification number. Must be a primary superelement. (Integer ≥ 0)
Gi	Grid or scalar point identification number. (0 < Integer; G1 < G2)

Remarks:

1. Interior grid points may also be defined via field 9 of the GRID and GRIDG Bulk Data entries. The SESET entry takes precedence over the SEID field on the GRID on GRIDG entries. SESET defines grid and scalar points to be included as interior to a superelement. SESET may be used as the primary means of defining superelements or it may be used in combination with SEELT entries which define elements interior to a superelement.
2. Gi may appear on an SESET entry only once.
3. Scalar points are ignored.
4. Open sets are allowed with the "THRU" option. Missing grid points (whether in "THRU" range or mentioned explicitly) are not identified.
5. All degrees-of-freedom for Gi are placed in the o-set of the superelement. See **"Degree-of-Freedom Sets"** .

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6. SESET can only be specified in the main Bulk Data Section and is ignored after the BEGIN SUPER=n command.

7. Continuations are not allowed. Multiple SESET entries with the same ID are permitted when multiple input lines are required. For example,

SESET	5	17	THRU	25	45	76	49	84	
SESET	5	102	107	165					

Related Topics

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SESUP

Fictitious Support

Defines determinate reaction superelement degrees-of-freedom in a free-body analysis.

Format:

1	2	3	4	5	6	7	8	9	10
SESUP	SEID	ID1	C1	ID2	C2	ID3	C3		

Example:

SESUP	5	16	215						

Fields:

Field	Contents
SEID	Superelement identification number. Must a primary superelement. (Integer > 0)
IDi	Grid or scalar point identification number. Must be exterior points. (Integer > 0)
Ci	Component numbers. (Integer zero or blank for scalar points; Any unique combination of the Integers 1 through 6 for grid points.)

Remarks:

1. The degrees-of-freedom specified on this entry form members of the mutually exclusive r-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
2. The Ci degrees-of-freedom must be exterior degrees-of-freedom of the SEID superelement.
3. See "Rigid Body Supports" in the *Simcenter Nastran User's Guide* for a discussion of supported degrees-of-freedom (members of the r-set).
4. There must be a sufficient number of degrees-of-freedom on SESUP entries to discard any free body modes of the superelement.
5. SESUP Bulk Data entries are not allowed for part (partitioned bulk data) superelements. Use the SUPORT Bulk Data records to identify component rigid body modes.

Related Topics

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SET1

Set Definition

Defines a list of structural grid points, elements, or DRESPi (i=1,2,3) responses.

SET1 entries are referenced by SPLINEi entries, PANEL entries, XYOUTPUT, and the P2RSET option on DOPTPRM entries.

Grid or Element ID Format:

1	2	3	4	5	6	7	8	9	10
SET1	SID	ID1	ID2	ID3	ID4	ID5	ID6	ID7	
	ID8	-etc.-							

Response ID Format:

1	2	3	4	5	6	7	8	9	10
SET1	SID	R1	R2	R3	R4	R5	R6	R7	
	R8	-etc.-							

Example:

SET1	3	31	62	93	124	16	17	18	
	19								

Alternate Formats and Example:

SET1	SID	ID1	"THRU"	ID2	ID3	"THRU"	ID4		
– or –									
SET1	SID	R1	"THRU"	R2	R3	"THRU"	R4		
SET1	6	32	THRU	50	60	70	73	78	
	80	THRU	86						

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Fields:

Field	Contents
SID	Unique identification number. (Integer > 0)
IDi	List of structural grid point or element identification numbers. (Integer > 0 or “THRU”; for the “THRU” option, G1 < G2.)
Ri	List of DRESPi (i=1,2,3) response IDs for the P2RSET option on the DOPTPRM bulk entry. (Integer > 0 or “THRU”; for the “THRU” option, R1 < R2.)

Remarks:

1. SET1 entries may be referenced by the SPLINEi entries, PANEL entries, XYOUTPUT, and the P2RSET option on the DOPTPRM bulk entry.
2. When using the “THRU” option for SPLINEi or PANEL data entries, all intermediate grid points must exist.
3. When using the “THRU” option for XYOUTPUT requests, missing grid points are ignored.
4. When using the “THRU” option for DRESPi requests, missing response IDs are ignored.
5. The SID must be unique from other SET1 and SET3 SIDs.
6. “THRU” cannot appear in Field 3 or 9. For continuations, “THRU” cannot appear in Fields 2 or 9.

Related Topics

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SET2

Grid Point List

Defines a list of structural grid points in terms of aerodynamic macro elements.

SET2 entries are referenced by SPLINEi entries.

Format:

1	2	3	4	5	6	7	8	9	10
SET2	SID	MACRO	SP1	SP2	CH1	CH2	ZMAX	ZMIN	

Example:

SET2	3	111	0.0	0.75	0.0	0.667	3.51		

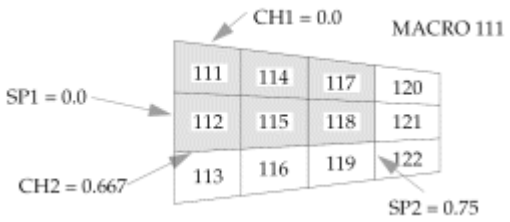
Fields:

Field	Contents
SID	Unique identification number. (Integer > 0)
MACRO	Element identification number of an aerodynamic macro element. (Integer > 0)
SP1, SP2	Lower and higher span division points defining the prism containing the set. (Real)
CH1, CH2	Lower and higher chord division points defining the prism containing the set. (Real)
ZMAX, ZMIN	Z-coordinates of top and bottom (using right-hand rule with the order of the corners as listed on a CAEROi entry) of the prism containing set. (Real)

Remarks:

1. The SET2 entry is referenced by the SPLINEi entry.
2. Every grid point within the defined prism and within the height range will be in the list. For example:

Figure 17-172: SET2 Entry Example



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The shaded area in Figure 17-172 defines the cross section of the prism for the sample data given above. Points exactly on the boundary may be missed; therefore, to get all the grid points within the area of the macro element, SP1=-.01, SP2=1.01, etc. should be used.

3. A zero value for ZMAX or ZMIN implies a value of infinity. Usually, ZMAX ≥ 0.0 and ZMIN ≤ 0.0.
4. To print the (internal) grid IDs found, use DIAG 18.

Related Topics

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SET3

Set Definition

Defines a list of structural grid points, elements, or physical properties.

SET3 entries are referenced by ACMODL entries and PANEL entries.

Grid ID Format:

1	2	3	4	5	6	7	8	9	10
SET3	SID	TYPE	ID1	ID2	ID3	ID4	ID5	ID6	
	ID7	-etc.-							

Example:

SET3	1	GRID	11	13	14	15	20	22	
	34	41							

Alternate Formats and Example:

SET3	SID	TYPE	ID1	“THRU”	ID2	ID3	“THRU”	ID4	
SET3	2	ELEM	20	THRU	33	36	THRU	44	
	49	THRU	62	91	THRU	110			

Fields:

Field	Contents
SID	Unique identification number. (Integer > 0)
TYPE	Set type. (Character: “GRID”, “ELEM”, “PROP”)
IDi	Identifying numbers of structural grids, elements, or physical properties. (Integer > 0)

Remarks:

1. The SID must be unique from other SET1 and SET3 SIDs.
2. By specifying the GRID field, the bulk entry referencing the SET3 bulk entry interprets the IDi as structural grid point IDs.
3. By specifying the ELEM field, the bulk entry referencing the SET3 bulk entry interprets the IDi as structural element IDs.

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4. By specifying the PROP field, the bulk entry referencing the SET3 bulk entry interprets the IDi as physical property IDs for structural elements.
5. “THRU” cannot appear in Field 4 or 9. For continuations, “THRU” cannot appear in Fields 2 or 9.

Related Topics

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SETREE

Superelement Tree Definition (Alternate Form of DTI,SETREE)

Specifies superelement reduction order.

Format:

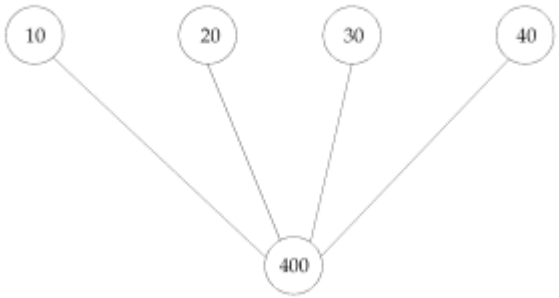
1	2	3	4	5	6	7	8	9	10
SETREE	SEID	SEUP1	SEUP2	SEUP3	SEUP4	SEUP5	SEUP6	SEUP7	
	SEUP8	SEUP9	-etc.-						

Example:

SETREE	400	10	20	30	40				

Fields:

Field	Contents
SEID	Partitioned superelement identification number of a downstream superelement. (Integer ≥ 0)
SEUPi	Identification number of superelements that are upstream of SEID. (Integer > 0)



Remarks:

1. SETREE entries or DTI,SETREE entry are required for multilevel superelement configurations.
2. At least one SETREE entry is required for each nontip superelement, including the residual structure (SEID=0). Multiple SETREE entries with the same SEID are allowed.
3. A superelement may appear only once in an SEUPi field on all SETREE entries.
4. If an DTI,SETREE entry is provided, then SETREE entries are not required.

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5. If both SETREE entries and a DTI,SETREE entry exist, then the DTI,SETREE entry will be ignored.
6. Superelements not referenced on the SETREE or DTI,SETREE entry will not be included in the analysis unless they are attached to grid points which belong to the residual structure. These superelements will be placed in the tree immediately above the residual structure.
7. The SETREE entry will only work if PART (BEGIN SUPER) superelements exist in the model. If there are no PARTs in the model, the SETREE entries will be ignored.

Related Topics

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SEUNIT

Superelement Unit Conversion

Defines conversion factors for length, mass, and time that the software uses to convert the superelement matrices into the units of the residual structure for superelements created using SOL 414,103.

Format:

1	2	3	4	5	6	7	8	9	10
SEUNIT	SEID	UNIL	UNIM	UNIT					

Example:

SEUNIT	10	1000.0	1.0	1.0					

Fields:

Name	Description
SEID	Superelement identification number. (Integer > 0)
UNIL	Conversion factor for length. See Remark 1. (Real > 0.0; Default = 1.0)
UNIM	Conversion factor for mass. See Remark 1. (Real > 0.0; Default = 1.0)
UNIT	Conversion factor for time. See Remark 1. (Real > 0.0; Default = 1.0)

Remarks:

1. The conversion factors are the values required to convert the superelement units into the units of the residual structure.

   For example, if the unit of length for the superelement is meters and the unit of length for the residual structure is millimeters, enter 1000.0 in the UNIL field. Similarly, if the unit of length for the superelement is inches and the unit of length for the residual structure is millimeters, enter 25.4 in the UNIL field.

Related Topics

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SEUSET

Superelement Degree-of-Freedom Set Definition

Defines a degree-of-freedom set for a superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SEUSET	SEID	SNAME	ID1	C1	ID2	C2	ID3	C3	

Example:

SEUSET	15	U4	1	123456	7	5	22	3	

Fields:

Field	Contents
SEID	Superelement identification number. (Integer > 0)
SNAME	Set name. (One to four characters or string “ZERO”, followed by the set name.)
IDi	Grid or scalar point identification numbers. (Integer > 0)
Ci	Component number. (Any unique combination of the Integers 1 through 6 with no embedded blank for grid points; Integer 0 or blank for scalar points.)

Remarks:

1. SNAME may refer to any of the set names given in **“Degree-of-Freedom Sets”** or their new names on the DEFUSET entry. However, in the Solution Sequences 100 through 200, it is recommended that SNAME refer only to the set names “U1” through “U9” or their new names on the DEFUSET entry.
2. If SNAME=“ZEROi”, where i is a set name, then the degrees-of-freedom are omitted from set i.
3. A maximum of 18 degrees-of-freedom may be designated on a single entry.
4. If degrees-of-freedom defined by SEUSET entries are found to be singular and AUTOSPC is requested for a degree-of-freedom that is also in a set that AUTOSPC may change, then the set defined by the SEUSET entry will be removed by the AUTOSPC operation. An avoidance is to use PARAM,AUTOSPC,NO.
5. The SEUSET entry is processed by the GP4 module with its effect appearing in the USET table. User-written DMAPs must therefore include the GP1 and GP4 modules if SEUSET entries are used.

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6. If a SEUSETi bulk entry lists a standard Simcenter Nastran set, such as S or M, the program may fail in the PARTN module with the message “SYSTEM FATAL MESSAGE 3007, ILLEGAL INPUT TO SUBROUTINE”. This entry should only reference new sets defined on DEFUSET bulk entries.
7. The user defined degree-of-freedom sets U1 – U9 are created with the SEUSET/SEUSET1 bulk entries. Be aware that the user defined sets U2 - U8 are used by Simcenter Nastran in some special cases. You may use U2 – U8 as long as it doesn't conflict with these cases. See **“User Defined Degree-of-Freedom Sets”** for the list of these special cases.

Related Topics

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SEUSET1

Superelement Degree-of-Freedom Set Definition, Alternate Form

Defines a degree-of-freedom set for a superelement.

Format:

1	2	3	4	5	6	7	8	9	10
SEUSET1	SEID	SNAME	C	G1	G2	G3	G4	G5	
	G6	G7	-etc.-						

Example:

SEUSET1	15	U4	1	12	15	17	22	25	
	52	53							

Alternate Format and Example:

SEUSET1	SEID	SNAME	C	G1	“THRU”	G2			
SEUSET1	15	U4	1	12	THRU	27			

Fields:

Field	Contents
SEID	Superelement identification number. (Integer > 0)
SNAME	Set name. (One to four characters or string “ZERO”, followed by the set name.)
C	Component numbers. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points; Integer 0 or blank for scalar points.)
Gi	Grid or scalar point identification number. (Integer > 0)

Remarks:

1. SNAME may refer to any of the set names given in **“Degree-of-Freedom Sets”** or their new names on the DEFUSET entry. However, in the Solution Sequences 100 through 200, it is recommended that SNAME refer only to the set names “U1” through “U9” or their new names on the DEFUSET entry.
2. If SNAME= “ZEROi”, where i is a set name, then the degrees-of-freedom are omitted from set i.
3. If the alternate format is used, all of the points G1 through G2 are assigned to the set.

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4. If degrees-of-freedom defined by SEUSET1 entries are found to be singular and AUTOSPC is requested for a degree-of-freedom that is also in a set that AUTOSPC may change, then the set defined by the SEUSET1 entry will be removed by the AUTOSPC operation. An avoidance is to use PARAM,AUTOSPC,NO.
5. The SEUSET1 entry is processed by the GP4 module with its effect appearing in the USET table. User-written DMAPs must therefore include the GP1 and GP4 modules if SEUSET1 entries are used.
6. If a SEUSETi bulk entry lists a standard Simcenter Nastran set, such as S or M, the program may fail in the PARTN module with the message “SYSTEM FATAL MESSAGE 3007, ILLEGAL INPUT TO SUBROUTINE”. This entry should only reference new sets defined on DEFUSET bulk entries.
7. The user defined degree-of-freedom sets U1 – U9 are created with the SEUSET/SEUSET1 bulk entries. Be aware that the user defined sets U2 - U8 are used by Simcenter Nastran in some special cases. You may use U2 – U8 as long as it doesn't conflict with these cases. See **“User Defined Degree-of-Freedom Sets”** for the list of these special cases.

Related Topics

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SLOAD

Static Scalar Load

Defines concentrated static loads on scalar or grid points.

Format:

1	2	3	4	5	6	7	8	9	10
SLOAD	SID	S1	F1	S2	F2	S3	F3		

Example:

SLOAD	16	2	5.9	17	-6.3	14	-2.93		

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
Si	Scalar or grid point identification number. (Integer > 0)
Fi	Load magnitude. (Real)

Remarks:

1. In the static solution sequences, the load set ID (SID) is selected by the Case Control command LOAD. In the dynamic solution sequences, SID must be referenced in the LID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
2. Up to three loads may be defined on a single entry.
3. If Si refers to a grid point, the load is applied to component T1 of the displacement coordinate system (see the CD field on the GRID entry).

Related Topics

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SLVCNTL

Solution options for frequency response and aeroelastic flutter analysis

Solution options for frequency response SOLs 108 and 111, and aeroelastic flutter SOL 145.

Format:

1	2	3	4	5	6	7	8	9	10
SLVCNTL	ID	PARAM1	VALUE1	PARAM2	VALUE2	PARAM3	VALUE3		
	PARAM4	VALUE4	-etc.-						

Examples:

SLVCNTL	110	USOLVER	PARDISO	MSGLVL	1				

Fields:

Field	Contents
ID	Unique identification number. (Integer > 0)
PARAM*i*	Name of the SLVCNTL parameter. Allowable names are given in the parameter listing below. (Character)
VALUE*i*	Value of the parameter. (Real, Integer, or Character)

SLVCNTL Parameters:

Table 17-124:

Name	Description
USOLVER	Unsymmetric solver choice. (Character; Default = MUMPS for SOLs 108, 111, and 145) SPARSE: Use the Sparse solver. PARDISO: Use the Pardiso solver. (Not valid for SOL 145) MUMPS: Use the MUltifrontal Massively Parallel sparse direct Solver (MUMPS).
SOLVER	Symmetric solver choice. Valid for SOL 108 and 111 only. (Character; Default = MUMPS for SOL 108 and SOL 111) SPARSE: Use the Sparse solver.

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Name	Description
	PARDISO: Use the Pardiso solver. MUMPS: Use MUMPS.
PTHRESH	When the Pardiso solver is selected, defines the exponent value for the pivoting threshold (also known as the perturbation threshold). For example, PTHRESH=12 defines a pivoting threshold of 1.0E-12. (Integer > 0; Default=13)
MSGLVL	When the Pardiso solver is selected, controls if solver messages are output to the .log file. (Integer ≥ 0; Default=0) 0: Messages are not output to the .log file. 1: Messages are output to the .log file. When MUMPS is selected, controls if solver messages are output to the .f04 and .f06 file. (Integer ≥ 0; Default=0) 0: Error messages are output to the .f06 file. 1: Error messages are output to the .f06 files. Warnings and main statistics are printed to the .f04 files. 2: Error messages are output to the .f06 files. Warnings and terse diagnostics (only first ten entries of arrays) are printed to the .f04 files. 3: Error messages are output to the .f06 files. Warnings and information on input, output parameters are printed to the .f04 files.

Remarks:

1. The SLVCNTL entry ID is not referenced by a case control command. If you define multiple SLVCNTL entries, the software uses the entry with the largest ID.

Related Topics

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SNORM

Surface Normal Vector at Grid Point

Defines a surface normal vector at a grid point for CQUAD4, CQUADR, CTRIA3, and CTRIAR shell elements.

In SOL 402, also defines a tangent vector at grid point for CBAR, CBEAM, and CROD elements.

Format:

1	2	3	4	5	6	7	8	9	10
SNORM	GID	CID	N1	N2	N3				

Example:

SNORM	3	2	0.	-1.	0.				

Fields:

Field	Contents
GID	Unique grid point identification number. (Integer > 0)
CID	Identification number of coordinate system in which the components of the normal (or tangent) vector are defined. See Remark 3. (Integer ≥ 0; Default=0 for the basic coordinate system)
Ni	Components of vector. The three components need not define a unit vector. (Real; Default=0.0)

Remarks:

1. The SNORM Bulk Data entry overrides any unique, internally-generated grid point normals that may have been requested with the user parameter SNORM , described in this guide.
2. The normal is used in CQUAD4, CQUADR, CTRIA3, and CTRIAR shell elements.

   The tangent is used in CBEAM, CBAR, and CROD elements only in SOL 402.

   For all other elements, the SNORM Bulk Data entry is ignored.
3. If CID is a cylindrical or spherical coordinate system, the components Ni are in the local tangent system at grid GID. For example, if CID=10 is a spherical coordinate system and normals must be defined pointing outwards in the radial direction of the sphere, see **Figure 17-173**, then the SNORM entries for all grids GID on the sphere are simply

   SNORM, GID, 10, 1., 0., 0.

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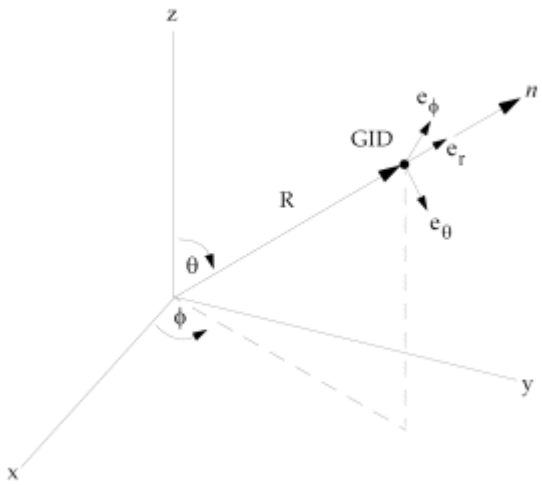
Figure 17-173:



Related Topics

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SPC

Single-Point Constraint

Defines a set of single-point constraints and enforced motion (enforced displacements in static analysis and enforced displacements, velocities or acceleration in dynamic analysis).

Format:

1	2	3	4	5	6	7	8	9	10
SPC	SID	G1	C1	D1	G2	C2	D2		

Example:

SPC	2	32	3	-2.6	5				

Fields:

Field	Contents
SID	Identification number of the single-point constraint set. (Integer > 0)
Gi	Grid or scalar point identification number. (Integer > 0)
Ci	Component number. (0 ≤ Integer ≤ 6; up to six Unique Integers, 1 through 6, may be placed in the field with no embedded blanks. 0 applies to scalar points and 1 through 6 applies to grid points.)
Di	Value of enforced motion for all degrees of freedom designated by Gi and Ci. (Real)

Remarks:

1. Single-point constraint sets must be selected with the Case Control command SPC=SID.
2. Degrees of freedom specified on this entry form members of the mutually exclusive s-set. They may not be specified on other entries that define mutually exclusive sets. See **Degree-of-Freedom Sets** for a list of these entries.
3. Single-point forces of constraint are recovered during stress data recovery.
4. From 1 to 12 degrees of freedom may be specified on a single entry.
5. Degrees of freedom on this entry may be redundantly specified as permanent constraints using the PS field on the GRID entry.
6. For reasons of efficiency, the SPCD entry is the preferred method for applying enforced motion rather than the Di field described here.
7. Rotational degrees of freedom are in radians.

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Remarks related to SOLs 108 and 111:

1. Constraining poro-elastic grids is only supported in SOL 108 and SOL 111, and requires the FEM Adaptive Order method. Poro-elastic grids are fluid grids that are referenced by acoustic elements and have poro-elastic materials assigned.
2. For additional limitations, see the *Poro-elastic material with MATPOR* section of the Defining porous or equivalent porous materials topic in the *Simcenter Nastran Acoustics User's Guide*.

Remarks related to SOLs 402 and 414:

1. To apply enforced displacements with constant magnitude, SPC or SPCD may be used. To apply time-dependent enforced displacements, velocity, or acceleration, SPCD must be used. See the SPCD entry.
2. If you define an SPC with a non-zero Di value, you must request an initial static computation with the IREF parameter of the NLCNTLG entry.
3. You should consider the following when applying a large enforced rotation with the SPC or SPCD entries:

   - The rotational components 4, 5,and 6 represent a pseudo rotation vector. The amplitude of the pseudo rotation vector defined with the SPC or SPCD entries can only be defined between [-π, +π] radians.
   - Applying an enforced rotation with the SPC or SPCD entries on one component and leaving the other two components free will not result in a unique definition of the rotation operator and can lead to unexpected results.

   To avoid these issues with the SPC or SPCD entries, we recommend that you instead apply large rotations using a kinematic joint with TYPE= REVOLUTE defined on the PJOINT bulk entry.

Related Topics

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SPC1

Single-Point Constraint, Alternate Form

Defines a set of single-point constraints.

Format:

1	2	3	4	5	6	7	8	9	10
SPC1	SID	C	G1	G2	G3	G4	G5	G6	
	G7	G8	G9	-etc.-					

Example:

SPC1	3	2	1	3	10	9	6	5	
	2	8							

Alternate Format and Example:

SPC1	SID	C	G1	“THRU”	G2				
SPC1	313	12456	6	THRU	32				

Fields:

Field	Contents
SID	Identification number of single-point constraint set. (Integer > 0)
C	Component numbers. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points. This number must be Integer 0 or blank for scalar points.)
Gi	Grid or scalar point identification numbers. (Integer > 0 or “THRU”; For “THRU” option, G1 < G2.)

Remarks:

1. Single-point constraint sets must be selected with the Case Control command SPC=SID.
2. Enforced displacements are available via this entry when used with the recommended SPCD entry.
3. Degrees-of-freedom specified on this entry form members of the mutually exclusive s-set. They may not be specified on other entries that define mutually exclusive sets. See **Degree-of-Freedom Sets** for a list of these entries.

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4. Degrees of freedom on this entry may be redundantly specified as permanent constraints using the PS field on the GRID entry.
5. If the alternate format is used, points in the sequence G1 through G2 are not required to exist. Points that do not exist will collectively produce a warning message but will otherwise be ignored.
6. Rotational degrees-of-freedom are in radians.

Remarks related to SOLs 108 and 111:

1. Constraining poro-elastic grids is only supported in SOL 108 and SOL 111, and requires the FEM Adaptive Order method. Poro-elastic grids are fluid grids that are referenced by acoustic elements and have poro-elastic materials assigned.
2. For additional limitations, see the *Poro-elastic material with MATPOR* section of the Defining porous or equivalent porous materials topic in the *Simcenter Nastran Acoustics User's Guide*.

Related Topics

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SPCADD

Single-Point Constraint Set Combination

Defines a single-point constraint set as a union of single-point constraint sets defined on SPC or SPC1 entries.

Format:

1	2	3	4	5	6	7	8	9	10
SPCADD	SID	S1	S2	S3	S4	S5	S6	S7	
	S8	S9	-etc.-						

Example:

SPCADD	101	3	2	9	1				

Fields:

Field	Contents
SID	Single-point constraint set identification number. (Integer > 0)
Si	Identification numbers of single-point constraint sets defined via SPC or by SPC1 entries. (Integer > 0; SID ≠ Si)

Remarks:

1. Single-point constraint sets must be selected with the Case Control command SPC=SID.
2. No Si may be the identification number of a single-point constraint set defined by another SPCADD entry.
3. The Si values must be unique.
4. SPCADD entries take precedence over SPC or SPC1 entries. If both have the same set ID, only the SPCADD entry will be used.

Related Topics

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SPCD

Enforced Motion Value

Defines an enforced displacement value for static analysis and an enforced motion value (displacement, velocity, or acceleration) in dynamic analysis.

Format:

1	2	3	4	5	6	7	8	9	10
SPCD	SID	G1	C1	D1	G2	C2	D2		

Example:

SPCD	100	32	436	-2.6	5		2.9		

Fields:

Field	Contents
SID	Set identification number of the SPCD entry. (Integer > 0)
Gi	Grid or scalar point identification number. (integer > 0)
Ci	Component numbers. (0 ≤ Integer ≤ 6; up to six unique Integers may be placed in the field with no embedded blanks.)
Di	Value of enforced motion for at Gi and Ci. (Real)

Remarks:

1. In the static solution sequences, the set ID of the SPCD entry (SID) is selected by the LOAD Case Control command.
2. In dynamic analysis, the selection of SID is determined by the presence of the LOADSET request in Case Control as follows:
   - There is no LOADSET request in Case Control

     SID is selected by the EXCITEID ID of an RLOAD1, RLOAD2, TLOAD1 or TLOAD2 Bulk Data entry that has enforced motion specified in its TYPE field
   - There is a LOADSET request in Case Control

     SID is selected by LID in the selected LSEQ entries that correspond to the EXCITEID entry of an RLOAD1, RLOAD2, TLOAD1 or TLOAD2 Bulk Data entry that has enforced motion specified in its TYPE field.

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3. A global coordinate (Gi and CI) referenced on this entry must also be referenced on a SPC or SPC1 Bulk Data entry and selected by the SPC Case Control command.
4. Values of Di will override the values specified on an SPC Bulk Data entry, if the SID is selected as indicated above.
5. The LOAD Bulk Data entry will not combine an SPCD load entry.
6. In static analysis, this method of applying enforced displacements is more efficient than the SPC entry when more than one enforced displacement condition is applied. It provides equivalent answers.
7. In dynamic analysis, this direct method of specifying enforced motion is more accurate, efficient and elegant than the large mass and Lagrange multiplier techniques.
8. Rotational degrees of freedom are in radians.

Remarks related to SOLs 108 and 111:

1. Constraining poro-elastic grids is only supported in SOL 108 and SOL 111, and requires the FEM Adaptive Order method. Poro-elastic grids are fluid grids that are referenced by acoustic elements and have poro-elastic materials assigned.
2. For additional limitations, see the *Poro-elastic material with MATPOR* section of the Defining porous or equivalent porous materials topic in the *Simcenter Nastran Acoustics User's Guide*.

Remarks related to SOLs 402 and 414:

1. A degree of freedom cannot have an imposed displacement in some subcases while being free in others. It should be imposed to a value (possibly 0.), in all subcases.
2. Constant displacement (LOAD):
   a. To enforce displacement with constant magnitude, SID is selected by Case Control command LOAD=SID for both static and transient analysis.
   b. Constant imposed displacements can be ramped (default) or stepped in each subcase, depending on the parameter LVAR on the NLCNTL2 entry for the current subcase.
3. Time dependent displacements (DLOAD):
   a. To enforce a time dependent displacement, SID is referenced by the field EXCITEID=SID in the TLOAD1 entry with TYPE=1. Time dependent loads are selected by Case Control command DLOAD.
   b. The Case Control command DLOAD must be global or must be defined in all subcases and point towards the same SPCD and TABLED1 entries. The time evolution cannot change from subcase to subcase.
4. Constant displacement and time-dependent displacements can be combined. If they involve the same degree of freedom, their respective displacements will be summed.
5. A global coordinate (Gi and Ci) referenced on this entry does not need to be referenced on a SPC or SPC1 entry.

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6. You should consider the following when applying a large enforced rotation with the SPC or SPCD entries:

   - The rotational components 4, 5,and 6 represent a pseudo rotation vector. The amplitude of the pseudo rotation vector defined with the SPC or SPCD entries can only be defined between [-π, +π] radians.
   - Applying an enforced rotation with the SPC or SPCD entries on one component and leaving the other two components free will not result in a unique definition of the rotation operator and can lead to unexpected results.

   To avoid these issues with the SPC or SPCD entries, we recommend that you instead apply large rotations using a kinematic joint with TYPE= REVOLUTE defined on the PJOINT bulk entry.

Related Topics

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SPCF

Enforced Motion Load Values, File Input

Defines enforced motion load values for a frequency response analysis using frequency-dependent motion (displacement, velocity, or acceleration) results that are stored in a file.

Format:

1	2	3	4	5	6	7	8	9	10
SPCF	SID	SET1ID	SUBC	UNITNO	FTYPE	LDIDT	LDIDR		

Examples:

SPCF	100	10	2	12	1	3			

Fields:

Field	Contents
SID	Set identification number of the SPCF bulk entry. (Integer > 0; No default) See Remark 2.
SET1ID	Identification number of a SET1 bulk entry that defines a list of poro-elastic or structural grids at which enforced motion loads are to be extracted and applied to for all degrees of freedom (DOFs) of the grid. A value of 0 implies that all structural grids contained in the file will have enforced motion loads applied. You must define corresponding SPCs for these grids in the Bulk Data section. (Integer ≥ 0; Default = 0) See Remark 6.
SUBC	For FTYPE = 0, subcase number of motion loads in one or more OUGV1 data blocks in OUTPUT2 file to be used for enforced motion loads on structural DOF. A value of 0 requests the first subcase encountered to be used. (Integer ≥ 0; Default = 0) See Remark 4. For FTYPE = 1, subcase number of a specific set of enforced motion loads in the LDIDT or LDIDR loading description. (Integer > 0) See Remark 12.
UNITNO	Logical unit number of the file that contains the frequency-dependent enforced motion loads to be applied to structural DOFs. (Integer > 0; No default) See Remarks 3, 4, and 5.
FTYPE	Type of file that contains the frequency-dependent motion loads. (Integer ≥ 0; Default = 0) 0 = OUTPUT2 1 = SC_H5
LDIDT	Applicable when FTYPE = 1.

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Field	Contents
	Load description identification number of a loading description contained in the SC_H5 file that is referenced by UNITNO for translational enforced motions. (0 ≤ Integer ≤ 9999; No default) See Remark 12.
LDIDR	Applicable when FTYPE = 1. Load description identification number of a loading description contained in the SC_H5 file that is referenced by UNITNO for rotational enforced motions. (0 ≤ Integer ≤ 9999; No default) See Remark 12.

Remarks:

1. SPCF with:
   - Structural grids is only supported in SOL 108 and SOL 111 with the standard (fixed low-order) FEM or FEM Adaptive Order method.
   - Poro-elastic grids is only supported in SOL 108 and SOL 111, and requires the FEM Adaptive Order method. Poro-elastic grids are fluid grids that are referenced by acoustic elements and have poro-elastic materials assigned.
2. The SID must be unique with respect to all other SPCF, SPC, or SPCD bulk entries. The selection of SID is determined by the presence of the LOADSET request in the Case Control section as follows:
   - If LOADSET is not present, SID is selected by the EXCITEIDID of an RLOAD1 bulk entry that has enforced motion loads specified in its TYPE field.
   - If LOADSET is present, SID is selected by LID in the selected LSEQ entries that correspond to the EXCITEID entry of an RLOAD1 bulk entry that has enforced motion loads specified in its TYPE field.
3. An appropriate ASSIGN OUTPUT2 or ASSIGN SC_H5 statement must be present in the File Management Section (FMS) for the value of UNITNO.
4. More than one OUGV1 data block on the OUTPUT2 file can exist and each OUGV1 data block can contain multiple subcases. SUBC refers to the subcase ID contained in one of these OUGV1 data blocks. If the subcase defined by SUBC cannot be found, an error will occur. If SUBC = 0, the first subcase encountered in the first OUGV1 data block found will be used.
5. You must ensure that the enforced motion loads defined in the OUGV1 data block on the OUTPUT2 file are in the same nodal displacement coordinate systems as the grids defined in your model.
6. The use of the SPCF bulk entry requires the use of the SPC bulk entry in the same way as required with the use of the SPCD bulk entry.
7. Enforced motion load values contained in the selected OUGV1 data block will override the values specified on an SPCD bulk entry if the SID is selected as indicated above.
8. The direct method of specifying enforced motion loads is more accurate, efficient, and refined than the large mass or Lagrange multiplier method.
9. Rotational DOFs are in radians.

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10. If the solution frequencies are within the frequency range of the motion loads in the file but do not match the frequencies, the solver performs linear interpolation. If the solution frequencies are below or above the frequency range of the enforced motion loads in the file, the enforced motion load values outside the range are assumed to be zero, and the solver does not extrapolate.
11. The type of enforced motion load, such as displacement, velocity, or acceleration, is determined by the TYPE field on the RLOAD1 bulk entry that references the SPCF bulk entry.
12. Requirements for SC_H5 file, loading description, and subcases:
    - The file must contain:
        - The frequency-dependent motion descriptions referenced by LDIDT and LDIDR.
        - The DOFs and frequencies at which the enforced motion loads are applied.
    - A separate DTI,DISTL bulk entry must define the loading description referenced by LDIDT or LDIDR.
    - The SUBC identifier must correspond to a specific loading within the LDIDT or LDIDR descriptor.

    Note

    LDIDT and LDIDR cannot both be equal to zero.

Related Topics

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SPCOFF

Excludes Degrees-of-Freedom from the AUTOSPC Operation

Defines a set of degrees-of-freedom to be excluded from the AUTOSPC operation. See “Constraint and Mechanism Problem Identification” in SubDMAP SEKR in the *Simcenter Nastran User’s Guide* for a description of the AUTOSPC operation.

Format:

1	2	3	4	5	6	7	8	9	10
SPCOFF	G1	C1	G2	C2	G3	C3	G4	C4	

Example:

SPCOFF	32	436	5	1					

Fields:

Field	Contents
Gi	Grid or scalar point identification number. (Integer > 0)
Ci	Component numbers. (Integer zero or blank for scalar points; Integers 1 through 6 with no embedded blanks for grid points.)

Remarks:

1. Degrees-of-freedom specified on this entry are to be excluded from the AUTOSPC operation. If any degree-of-freedom in this set is found to be singular, a warning message is issued and no constraint is applied.
2. Degrees-of-freedom that are specified as both SPC and SPCOFF will be considered as SPC.

Related Topics

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SPCOFF1

Excludes DOFs from AUTOSPC Processing, Alternate Form

Defines a set of degrees-of-freedom to be excluded from the AUTOSPC operation. See “Constraint and Mechanism Problem Identification” in “SubDMAP SEKR” in the *Simcenter Nastran User’s Guide* for a description of the AUTOSPC operation.

Format:

1	2	3	4	5	6	7	8	9	10
SPCOFF1	C	G1	G2	G3	G4	G5	G6	G7	
	G8	G9	-etc.-						

Example:

SPCOFF1	2	1	3	10	9	6	5	4	
	8								

Alternate Format and Example:

SPCOFF1	C	G1	“THRU”	G2					
SPCOFF1	12456	6	THRU	32					

Fields:

Field	Contents
C	Component numbers. (Any unique combination of the Integers 1 through 6 with no embedded blanks for grid points; Integer 0 or blank for scalar points.)
Gi	Grid or scalar point identification numbers. (Integer > 0 or “THRU”; for “THRU” option, G1 < G2.)

Remarks:

1. Degrees-of-freedom specified on this entry are to be excluded from the AUTOSPC operation. If any degree-of-freedom in this set is found to be singular, a warning message is issued and no constraint is applied.
2. Degrees-of-freedom that are both specified as SPC and SPCOFF will be considered as SPC.
3. If the alternate format is used, points in the sequence G1 through G2 are not required to exist. Points which do not exist will collectively produce a warning message but will otherwise be ignored.

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Related Topics

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SPDIR

Base Excitation Definition

Defines base excitation for DDAM.

Format:

1	2	3	4	5	6	7	8	9	10
SPDIR		CSID	"123"						

Example:

SPDIR		100	COMP						

Fields:

Name	Description
CSID	Output coordinate system. (Integer ≥ 0; Default = 0 for the basic coordinate system)
COMP	Component numbers for the base excitation. See Remark 1. (1 ≤ Integer ≤ 3; No default)

Remarks:

1. For DDAM, all three translational directions must be specified. That is, the COMP field must be specified as "123". If rotational components 4, 5, or 6 are included, they are ignored.

Related Topics

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SPEBE2

Bearing Speed Definition

Defines the bearing or connection speed for:

- CBEAR2 bearing connections when the PBEAR2 TYPE field is set to "BEAR" or "USER" and the bearing properties are speed-dependent.
- CBUSH2 bearing connections when the PBUSH2 bulk entry contains a parameter that references a TABLEST bulk entry.
- CLINK connections with option 3 connecting an asymmetric rotor and stator in SOL 414,129 Transient Response solutions.

Format:

1	2	3	4	5	6	7	8	9	10
SPEBE2	SID	EID	RID1	RID2	DIREC	SPEED			

Example of a single rotor:

SPEBE2	5	232	3						

Example of coaxial rotors:

SPEBE2	5	232	3	4	COUN				

Fields:

Field	Contents
SID	Speed definition identification number. (Integer > 0)
EID	Identification number of the bearing connection. (Integer > 0; No default)
RID1	Identification number of the rotor. (Integer > 0 or blank; No default)
RID2	Identification number of the second rotor when the bearing connection specified in the EID field connects coaxial rotors. (Integer > 0 or blank; No default)
DIREC	Bearing speed formula for coaxial rotors. See Remark 1. (Character="CORO" or "COUN"; Default="CORO") "CORO" for the same direction. See Remark 2.

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Field	Contents
	"COUN" for opposite directions. See Remark 2.
SPEED	Identification number of a TABLEDi entry that defines the bearing speed as a function of time or frequency. See Remarks 3, 4, and 5. (Integer > 0; No default)

Remarks:

1. The following table shows how the software calculates the bearing speed.

RIDi Specified	DIREC Specification	Bearing Speed(1)(2)
RID1 only	DIREC field specification is ignored	ω_{RID1}
RID1 and RID2	"CORO"	$\omega_{RID1} - \omega_{RID2}$ (3)
RID1 and RID2	"COUN"	$\omega_{RID1} + \omega_{RID2}$
Neither RID1 or RID2	DIREC field specification is ignored	Value for the SPEED field

(1) ω_{RID1} is the angular velocity of the rotor specified in the RID1 field, and ω_{RID2} is the angular velocity of the rotor specified in the RID2 field.

(2) Bearings are speed dependent when the PBEAR2 physical properties define the FKij, CKij, FMij, or FSij parameters with the ROTF (rotor speed) or FREQ (frequency) parameter.

(3) When the bearing is speed dependent, the speeds for the rotors selected by RID1 and RID2 define the matrix terms as follows:

$FKij(\omega_{RID1} - \omega_{RID2})$
$CKij(\omega_{RID1} - \omega_{RID2})$
$FSij(\omega_{RID1} - \omega_{RID2})$
$FMij(\omega_{RID1} - \omega_{RID2})$

Therefore, you must consider the order in which you define the rotors in RID1 and RID2 so that the matrix terms do not result in a negative value. For example, if the speed of one rotor is 80 rev/s and the speed of the other rotor is 100 rev/s, set RID1 to the rotor with the higher speed (100 rev/s in this example) and set RID2 to the rotor with the lower speed (80 rev/s in this example).

Using these example values for the two rotors, the frequency at which the matrix term Kij is defined becomes one of the following depending on which rotors you assign to RID1 and RID2.

- If RID1 = the rotor at 100 rev/s and RID2 = the rotor at 80 rev/s, the frequency at which the matrix term Kij is defined becomes

  (100 rev/s) - (80 rev/s) = 20 rev/s

- If you define the rotors for RID1 and RID2 in reverse so that RID1 = the rotor at 80 rev/s and RID2 = the rotor at 100 rev/s, the frequency at which the matrix term Kij is defined becomes

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RIDi Specified	DIREC Specification	Bearing Speed(1)(2)
(80 rev/s) - (100 rev/s) = -20 rev/s The negative value causes the Kij function to be out of bounds, and extrapolation of the function becomes K(0).		

2. For coaxial rotors (both RID1 and RID2 are specified), when the positive sense of the axis of rotations are in opposite directions, specify the DIREC field as indicated in the following table.

Physical Sense of Rotation	Positive Sense of Axis of Rotations	DIREC Specification
Same	Same	"CORO"
Same	Opposite	"COUN"
Opposite	Same	"COUN"
Opposite	Opposite	"CORO"

3. The software uses the SPEED field specification when the RID1 and RID2 fields are blank.
4. For rotor dynamic transient response analysis, use the SPEED field to reference a TABLEDi entry that contains bearing speed vs. time tabular data. For rotor dynamic frequency response or complex modal analysis, use the SPEED field to reference a TABLEDi entry that contains bearing speed vs. frequency tabular data.
5. The units for the value specified in the SPEED field or the table referenced in the SPEED field must be the same as the units for the RUNIT field in the ROTORD bulk entry.

Related Topics

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SPLINE1

Surface Spline Methods

Defines a surface spline for interpolating motion and/or forces for aeroelastic problems on aerodynamic geometries defined by regular arrays of aerodynamic points.

Format:

1	2	3	4	5	6	7	8	9	10
SPLINE1	EID	CAERO	BOX1	BOX2	SETG	DZ	METH	USAGE	
	NELEM	MELEM							

Example:

SPLINE1	3	111	115	122	14	0.			

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
CAERO	Aero-element (CAEROi entry ID) that defines the plane of the spline. (Integer > 0)
BOX1, BOX2	First and last box with motions that are interpolated using this spline. See Remarks 3 and 4. (Integer > 0, BOX2 ≥ BOX1)
SETG	Refers to the SETi entry that lists the structural grid points to which the spline is attached. (Integer > 0)
DZ	Linear attachment flexibility. See Remark 2. (Real ≥ 0.0; Default=0.0)
METH	Method for the spline fit. IPS,TPS or FPS. See Remark 1. (Character, Default=IPS)
USAGE	Spline usage flag to determine whether this spline applies to the force transformation, displacement transformation or both. FORCE, DISP or BOTH. See Remark 5. (Character, Default=BOTH)
NELEM	The number of FE elements along the local spline x-axis if using the FPS option. (Integer > 0; Default=10)
MELEM	The number of FE elements along the local spline y-axis if using the FPS option. (Integer > 0; Default=10)

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Remarks:

1. The default METHOD will result in the use of the Harder-Desmarais Infinite Plate Spline (IPS). The other options are the Thin Plate Spline (TPS) and the Finite Plate Spline (FPS). The continuation applies only to the FPS option and is required only if the defaults are not adequate.
2. The attachment flexibility (units of area) is used for smoothing the interpolation. If DZ=0.0, the spline will pass through all deflected grid points. If DZ is much greater than the spline area, a least squares plane fit will be applied. Intermediate values will provide smoothing.
3. The interpolated points (k-set) will be defined by aero boxes. The following figure shows the cells for which u_k is interpolated if BOX1=115 and BOX2=122.

Figure 17-174: SPLINE1 Entry Example

4. When using the Mach Box method, BOX1 and BOX2 refer to the ID number of the first and last aerodynamic grids (x,y pairs on the AEFACT entry) which will be used for interpolation to structural grids. BOX1 and BOX2 do not refer to Mach Boxes.
5. The USAGE field allows you to specify that the particular spline interpolant defined in this entry is to be used for just the force transformation (FORCE) or just the displacement transformation (DISP) or for both (BOTH).

   $Fg=[GPkg]^T \{Pk\}$ (FORCE/BOTH splines are in the transform)

   $Uk=[GDkg] \{Ug\}$ (DISP/BOTH splines are in the transform)

   In general, the two transforms are done with distinct matrices. Only when ALL splines are of type BOTH is the familiar transpose relationship $[GPgk]^T]=[GDkg]$ satisfied. The default behavior (BOTH for all splines) is compatible with pre-Simcenter Nastran 1.0 versions.

   In general, the USAGE field can be used to apply aerodynamic forces to the structure from aerodynamic panels that are intended NOT to move (USAGE=FORCE) or to apply structural displacements to aerodynamic grids whose forces are not to be applied to the structure (USAGE=DISP). The DISP option is somewhat esoteric in that you are then suggesting that the aeroelastic effect of the surface is important while its forces are not. (In other words, only the forces arising from tis effects on other surfaces is important.) While there may be circumstances where this is true, it is unlikely. Take care that you included all the FORCEs from aerodynamic panels that are important by including them in either FORCE or BOTH spline(s). Simcenter Nastran will NOT issue a warning unless ALL forces are omitted. All displacements may be omitted without warning (and is a means to perform “rigid aerodynamic” analyses).

Related Topics

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SPLINE2

Linear Spline

Defines a beam spline for interpolating motion and/or forces for aeroelastic problems on aerodynamic geometries defined by regular arrays of aerodynamic points.

Format:

1	2	3	4	5	6	7	8	9	10
SPLINE2	EID	CAERO	ID1	ID2	SETG	DZ	DTOR	CID	
	DTHX	DTHY		USAGE					

Example:

SPLINE2	5	8	12	24	60	0.	1.0	3	
	1.								

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
CAERO	Aero panel or body (CAEROi entry ID) that is to be interpolated. (Integer > 0)
ID1, ID2	First and last box or body element whose motions are interpolated using this spline. See Remark 6 when using the Mach Box method. (Integer > 0, ID2 ≥ ID1)
SETG	Refers to an SETi entry that lists the structural grid points to which the spline is attached. (Integer > 0)
DZ	Linear attachment flexibility. (Real ≥ 0.0; Default=0.0)
DTOR	Torsional flexibility ratio (*EI*/*GJ*). (Real > 0.0; Default=1.0; use 1.0 for bodies.)
CID	Rectangular coordinate system for which the y-axis defines the axis of the spline. Not used for bodies, CAERO2. (Integer ≥ 0)
DTHX, DTHY	Rotational attachment flexibility. DTHX is for rotation about the spline's x-axis (in-plane bending rotations); however, it is not used for bodies. DTHY is for rotation about the spline's y-axis (torsion); however, it is used for slope of bodies. (Real; Default=0.0)
USAGE	Spline usage flag to determine whether this spline applies to the force transformation, displacement transformation or both. FORCE, DISP or BOTH. See Remark 9 . (Character, Default=BOTH)

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Remarks:

1. The interpolated points (k-set) will be defined by aero boxes.
2. The spline axis for panels is the projection of the y-axis of coordinate system CID, projected onto the plane of the panel. For bodies, the spline axis is parallel to the x-axis of the aerodynamic coordinate system.
3. The flexibilities DZ, DTHX, and DTHY are used for smoothing. (Zero attachment flexibility values will imply rigid attachment (i.e., no smoothing, whereas negative values of DTHX or DTHY will imply infinity, therefore, no attachment). See the *Simcenter Nastran Aeroelastic Analysis User's Guide* for a discussion of special cases.
4. The continuation entry is required.
5. The SPLINE2 EID must be unique with respect to all SPLINEi entries.
6. When using the Mach Box method, ID1 and ID2 refer to the ID number of the first and last aerodynamic grids (x,y pairs on the AEFACT entry) which will be used for interpolation to the structural grids. ID1 and ID2 do not refer to Mach Boxes.
7. DTOR is the ratio of rotational to linear deflection and, in lieu of a more accurate estimate, a value of 1.0 is recommended. A different value may be used; e.g., if DTOR is much greater than 1.0, primarily rotational deflection will occur; if DTOR is much less than 1.0, primarily linear deflection will occur.
8. If a SPLINE2 element only references one grid point, the job will fail without a message in the GI Module.
9. The USAGE field allows you to specify that the particular spline interpolant defined in this entry is to be used for just the force transformation (FORCE) or just the displacement transformation (DISP) or for both (BOTH).

   $Fg=[GPkg]^T \{Pk\}$ (FORCE/BOTH splines are in the transform)

   $Uk=[GDkg] \{Ug\}$ (DISP/BOTH splines are in the transform)

   In general, the two transforms are done with distinct matrices. Only when ALL splines are of type BOTH is the familiar transpose relationship $[GPgk]^T]=[GDkg]$ satisfied. The default behavior (BOTH for all splines) is compatible with pre-Simcenter Nastran 1.0 versions.

   In general, the USAGE field can be used to apply aerodynamic forces to the structure from aerodynamic panels that are intended NOT to move (USAGE=FORCE) or to apply structural displacements to aerodynamic grids whose forces are not to be applied to the structure (USAGE=DISP). The DISP option is somewhat esoteric in that you are then suggesting that the aeroelastic effect of the surface is important while its forces are not. (In other words, only the forces arising from tis effects on other surfaces is important.) While there may be circumstances where this is true, it is unlikely. Take care that you included all the FORCEs from aerodynamic panels that are important by including them in either FORCE or BOTH spline(s). Simcenter Nastran will NOT issue a warning unless ALL forces are omitted. All displacements may be omitted without warning (and is a means to perform "rigid aerodynamic" analyses).

Related Topics

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SPLINE3

Aeroelastic Constraint Equation

Defines a constraint equation for aeroelastic problems. Useful for control surface constraints.

Format:

1	2	3	4	5	6	7	8	9	10
SPLINE3	EID	CAERO	BOXID	COMP	G1	C1	A1	USAGE	
	G2	C2	A2		-etc.-				

Example:

SPLINE3	7000	107	109	6	5	3	1.0		
	43	5	-1.0						

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
CAERO	Identification number of the macro-element on which the element to be interpolated lies. (Integer > 0)
BOXID	Identification number of the aerodynamic element; i.e., the box number. (Integer > 0)
COMP	The component of motion to be interpolated. See Remark 4 . (One of the Integers 1, 2, 3, 4, 5, or 6.)
Gi	Grid point identification number of the independent grid point. (Integer > 0)
Ci	Component numbers in the displacement coordinate system. (One of the Integers 1 through 6 for grid points, or 0 for scalar points.)
Ai	Coefficient of the constraint relationship. (Real)
USAGE	Spline usage flag to determine whether this spline applies to the force transformation, displacement transformation or both. FORCE, DISP or BOTH. See Remark 6 . (Character, Default=BOTH).

Remarks:

1. The independent grid points and components must refer to degrees-of-freedom in the g-set.
2. The constraint is given by

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$$u_d = \sum A_i u_i$$

where

u_d = value of the dependent component of the aerodynamic element

u_i = displacement at grid Gi, component Ci.

3. The SPLINE3 EID must be unique with respect to all SPLINEi entries.

4. The allowable components by CAEROi entry type are indicated by an "X" in the table below:

Entry Type	COMP				
	1	2	3	5	6
CAERO1			X	X	
CAERO2		X	X	X	X
CAERO3			X		
CAERO4			X	X	X
CAERO5			X	X	X
3D Geometry	X	X	X	X	X

COMP=2: lateral displacement

COMP=3 transverse displacement

COMP=5: pitch angle

COMP=6: relative control angle for CAERO4 and CAERO5 yaw angle for CAERO2.

For general 3D aerodynamic geometries the components numbers refer to axes of the Aerodynamic Coordinate System (u_x, u_y, u_z, θ_x, θ_y, θ_z).

5. For Strip theory and Piston theory, the COMP=6 control surface relative angle is positive when the trailing edge has a negative z-deflection in the element coordinate system (see the *Simcenter Nastran Aeroelastic Analysis User's Guide*.)

6. The USAGE field allows you to specify that the particular spline interpolant defined in this entry is to be used for just the force transformation (FORCE) or just the displacement transformation (DISP) or for both (BOTH).

$Fg=[GPkg]^T \{Pk\}$ (FORCE/BOTH splines are in the transform)

$Uk=[GDkg] \{Ug\}$ (DISP/BOTH splines are in the transform)

In general, the two transforms are done with distinct matrices. Only when ALL splines are of type BOTH is the familiar transpose relationship $GPgk]^T]=[GDkg]$ satisfied. The default behavior (BOTH for all splines) is compatible with pre-Simcenter Nastran 1.0 versions.

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In general, the USAGE field can be used to apply aerodynamic forces to the structure from aerodynamic panels that are intended NOT to move (USAGE=FORCE) or to apply structural displacements to aerodynamic grids whose forces are not to be applied to the structure (USAGE=DISP). The DISP option is somewhat esoteric in that you are then suggesting that the aeroelastic effect of the surface is important while its forces are not. (In other words, only the forces arising from tis effects on other surfaces is important.) While there may be circumstances where this is true, it is unlikely. Take care that you included all the FORCEs from aerodynamic panels that are important by including them in either FORCE or BOTH spline(s). Simcenter Nastran will NOT issue a warning unless ALL forces are omitted. All displacements may be omitted without warning (and is a means to perform “rigid aerodynamic” analyses).

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SPLINE4

Surface Spline Methods

Defines a curved surface spline for interpolating motion and/or forces for aeroelastic problems on general aerodynamic geometries using either the Infinite Plate, Thin Plate or Finite Plate splining method.

Format:

1	2	3	4	5	6	7	8	9	10
SPLINE4	EID	CAERO	AELIST		SETG	DZ	METH	USAGE	
	NELEM	MELEM							

Example:

SPLINE4	3	111	115		14	0.	IPS		

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
CAERO	Aero panel ID that defines the interpolation surface. (Integer > 0)
AELIST	Identification of an AELIST entry listing boxes with motions that are interpolated using this spline; see Remark 3 when using Mach Box method. (Integer > 0)
SETG	Refers to the SETi entry that lists the structural grid points to which the spline is attached. (Integer > 0)
DZ	Linear attachment flexibility. (Real ≥ 0.0)
METH	Method for the spline fit. One of IPS, TPS or FPS. See Remark 1 . (Character, Default=IPS)
USAGE	Spline usage flag to determine whether this spline applies to the force transformation, displacement transformation or both. FORCE, DISP or BOTH. See Remark 4 . (Character, Default=BOTH)
NELEM	The number of FE elements along the local spline x-axis if using the FPS option. (Integer > 0; Default=10)
MELEM	The number of FE elements along the local spline y-axis if using the FPS option. (integer > 0; Default=10)

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Remarks:

1. The default METHOD will result in the use of the Harder-Desmarais Infinite Plate Spline (IPS). The other options are the Thin Plate Spline (TPS) and the Finite Plate Spline (FPS). The continuation applies only to the FPS option and is required only if the defaults are not adequate.
2. The attachment flexibility (units of area) is used for smoothing the interpolation. If DZ=0.0, the spline will pass through all deflected grid points. If DZ is much greater than the spline area, a least squares plane fit will be applied. Intermediate values will provide smoothing.
3. When using the Mach Box method, the AELIST boxes refer to the ID number of the first and last aerodynamic grids (x,y pairs on the AEFACT entry) which will be used for interpolation to structural grids. They do not refer to Mach Boxes.
4. The USAGE field allows you to specify that the particular spline interpolant defined in this entry is to be used for just the force transformation (FORCE) or just the displacement transformation (DISP) or for both (BOTH).

   $Fg=[GPkg]^T \{Pk\}$ (FORCE/BOTH splines are in the transform)

   $Uk=[GDkg] \{Ug\}$ (DISP/BOTH splines are in the transform)

   In general, the two transforms are done with distinct matrices. Only when ALL splines are of type BOTH is the familiar transpose relationship $[GPgk]^T=[GDkg]$ satisfied. The default behavior (BOTH for all splines) is compatible with pre-Simcenter Nastran 1.0 versions.

   In general, the USAGE field can be used to apply aerodynamic forces to the structure from aerodynamic panels that are intended NOT to move (USAGE=FORCE) or to apply structural displacements to aerodynamic grids whose forces are not to be applied to the structure (USAGE=DISP). The DISP option is somewhat esoteric in that you are then suggesting that the aeroelastic effect of the surface is important while its forces are not.

   (In other words, only the forces arising from tis effects on other surfaces is important.) While there may be circumstances where this is true, it is unlikely. Take care that you included all the FORCEs from aerodynamic panels that are important by including them in either FORCE or BOTH spline(s). Simcenter Nastran will NOT issue a warning unless ALL forces are omitted. All displacements may be omitted without warning (and is a means to perform “rigid aerodynamic” analyses).

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SPLINE5

Linear Spline

Defines a 1D beam spline for interpolating motion and/or forces for aeroelastic problems on aerodynamic geometries defined by irregular arrays of aerodynamic points. The interpolating beam supports axial rotation and bending in the yz-plane.

Format:

1	2	3	4	5	6	7	8	9	10
SPLINE5	EID	CAERO	AELIST		SETG	DZ	DTOR	CID	
	DTHX	DTHY		USAGE					

Example:

SPLINE5	5	8	12		60			3	
	1.			BOTH					

Fields:

Field	Contents
EID	Element identification number. (Integer > 0)
CAERO	Aero panel or body (CAEROi entry ID) that is to be interpolated. (Integer > 0)
AELIST	Identification number of an AELIST entry that identifies the aerodynamic boxes whose motions are interpolated using this spline. See Remark 6 when using the Mach Box method. (Integer > 0)
SETG	Refers to an SETi entry that lists the structural grid points to which the spline is attached. (Integer > 0)
USAGE	Spline usage flag to determine whether this spline applies to the force transformation, displacement transformation or both. FORCE, DISP or BOTH. See Remark 8 . (Character, Default=BOTH)
CID	Rectangular coordinate system that defines the y-axis of the spline and the xy- and yz-planes for bending. Not used for bodies, CAERO2. (Integer ≥ 0)
DTHX, DTHY	Rotational attachment flexibility. DTHX is for rotation about the spline's x-axis (the bending rotations). DTHY is for rotation about the spline's y-axis (torsion); however, it is used for bending of bodies. (Real)

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Field	Contents
DZ	Linear attachment flexibility. (Real ≥ 0.0, Default=0.0)
DTOR	Torsional flexibility ratio (EI/GJ) for the bending in the zy-plane. This value is ignored for slender bodies since they have no torsion; see Remark 7 . (Real > 0.0; Default=1.0; ignored for CAERO2 bodies.)

Remarks:

1. The interpolated points (k-set) will be defined by aero boxes.
2. The spline axis for panels is the projection of the y-axis of coordinate system CID, projected onto the plane of the panel. For bodies, the interpolating beam (y-axis) is parallel to the x-axis of the aerodynamic coordinate system; the z-axis is taken from the referenced CID and x is made orthogonal.
3. The flexibilities DZ, DTHX and DTHY are used for smoothing. (Zero attachment flexibility values will imply rigid attachment (i.e., no smoothing, whereas negative values of DTHX or DTHY will imply infinity, therefore, no attachment.) See the *Simcenter Nastran Aeroelastic Analysis User's Guide* for a discussion of special cases.
4. The continuation entry is required.
5. The SPLINE5 EID must be unique with respect to all SPLINEi entries.
6. When using the Mach Box method, the AELIST entries refer to the ID numbers of aerodynamic grids (x,y pairs on the AEFACT entry) which will be used for interpolation to the structural grids. They do not refer to Mach Boxes.
7. DTOR is the ratio of axial rotational to bending deflection and, in lieu of a more accurate estimate, a value of 1.0 is recommended. A different value may be used; e.g., if DTOR is much greater than 1.0, primarily rotational deflection will occur; if DTOR is much less than 1.0, primarily linear deflection will occur. The values will affect the results only if the structural grids over constrain the motion of the interpolating beam. Slender bodies have no torsional motion, so these values will not be used for CAERO2 entries.
8. The USAGE field allows you to specify that the particular spline interpolant defined in this entry is to be used for just the force transformation (FORCE) or just the displacement transformation (DISP) or for both (BOTH).

   $Fg=[GPkg]^T \{Pk\}$ (FORCE/BOTH splines are in the transform)

   $Uk=[GDkg] \{Ug\}$ (DISP/BOTH splines are in the transform)

   In general, the two transforms are done with distinct matrices. Only when ALL splines are of type BOTH is the familiar transpose relationship $[GPgk]^T=[GDkg]$ satisfied. The default behavior (BOTH for all splines) is compatible with pre-Simcenter Nastran 1.0 versions.

   In general, the USAGE field can be used to apply aerodynamic forces to the structure from aerodynamic panels that are intended NOT to move (USAGE=FORCE) or to apply structural displacements to aerodynamic grids whose forces are not to be applied to the structure (USAGE=DISP). The DISP option is somewhat esoteric in that you are then suggesting that the aeroelastic effect of the surface is important while its forces are not. (In other words, only the forces

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arising from tis effects on other surfaces is important.) While there may be circumstances where this is true, it is unlikely. Take care that you included all the FORCEs from aerodynamic panels that are important by including them in either FORCE or BOTH spline(s). Simcenter Nastran will NOT issue a warning unless ALL forces are omitted. All displacements may be omitted without warning (and is a means to perform “rigid aerodynamic” analyses).

Related Topics

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SPOINT

Scalar Point Definition

Defines scalar points.

Format:

1	2	3	4	5	6	7	8	9	10
SPOINT	ID1	ID2	ID3	ID4	ID5	ID6	ID7	ID8	

Example:

SPOINT	3	18	1	4	16	2			

Alternate Format and Example:

SPOINT	ID1	“THRU”	ID2						
SPOINT	5	THRU	649						

Fields:

Field	Contents
IDi	Scalar point identification number. (0 < Integer; for “THRU” option, ID1 < ID2)

Remarks:

1. The scalar point identification number must be unique with respect to all other identification numbers defined with GRID, EPOINT, and POINT bulk entries. However, duplicate scalar point identification numbers are allowed in the input.
2. A scalar point defined by its appearance on the connection entry for a scalar element (see the CELASi, CMASSi, and CDAMPi entries) need not appear on an SPOINT entry.
3. This entry is used primarily to define scalar points appearing in single-point or multipoint constraint equations to which no scalar elements are connected.
4. If the alternate format is used, all scalar points ID1 through ID2 are defined.
5. For a discussion of scalar points, see “Understanding Scalar Points” in the *Simcenter Nastran User’s Guide*.

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SUPORT

Fictitious Support

Defines determinate reaction degrees-of-freedom in a free body.

Format:

1	2	3	4	5	6	7	8	9	10
SUPORT	ID1	C1	ID2	C2	ID3	C3	ID4	C4	

Example:

SUPORT	16	215							

Fields:

Field	Contents
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component numbers. (Integer 0 or blank for scalar points. Any unique combination of the Integers 1 through 6 for grid points with no embedded blanks.)

Remarks:

1. The SUPORT entry specifies reference degrees-of-freedom for rigid body motion. It is not intended to be used in place of a constraint (i.e., SPCi entry or PS on the GRID entry).
2. SUPORT and/or SUPORT1 entries are required to perform inertia relief in static analysis (SOL 101).

   In SOL 101, PARAM,INREL,-1 must also be specified or the SUPORTi entries will be treated as constraints.
3. Be careful not to spell SUPORT with two Ps.
4. Degrees-of-freedom specified on this entry form members of the mutually exclusive r-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
5. From 1 to 24 support degrees-of-freedom may be defined on a single entry.
6. See "Rigid Body Supports" in the *Simcenter Nastran User's Guide* for a discussion of supported degrees-of-freedom (members of the r-set).
7. An alternative to SUPORT is the SUPORT1 entry, which is requested by the SUPORT1 Case Control command.
8. SUPORT entries are not supported for SOL 106, 114–118, and 129.

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Related Topics

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SUPORT1

Fictitious Support, Alternate Form

Defines determinate reaction degrees-of-freedom (r-set) in a free body-analysis. SUPORT1 must be requested by the SUPORT1 Case Control command.

Format:

1	2	3	4	5	6	7	8	9	10
SUPORT1	SID	ID1	C1	ID2	C2	ID3	C3		

Example:

SUPORT1	5	16	215						

Fields:

Field	Contents
SID	Identification number of the support set. See Remark 1. (Integer > 0)
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component numbers. (Integer 0 or blank for scalar points. Any unique combination of the Integers 1 through 6 for grid points with no embedded blanks.)

Remarks:

1. The SUPORT and SUPORT1 entries specify reference degrees-of-freedom for rigid body motion. It is not intended to be used in place of a constraint; (i.e., SPCi entry or PS on the GRID entry).
2. SUPORT and/or SUPORT1 entries are required to perform inertia relief in static analysis; i.e., SOL 101.

   In SOL 101, PARAM,INREL,-1 must also be specified or the SUPORTi entries will be treated as constraints.
3. SUPORT1 must be requested by the SUPORT1 Case Control command. The degrees-of-freedom specified on SUPORT1 will be combined with those on the SUPORT entry.
4. Be careful not to spell SUPORT with two Ps.
5. Degrees-of-freedom specified on this entry form members of the mutually exclusive r-set. They may not be specified on other entries that define mutually exclusive sets. See **"Degree-of-Freedom Sets"** for a list of these entries.
6. From 1 to 18 support degrees-of-freedom may be defined on a single entry.

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7. See “Rigid Body Supports” in the *Simcenter Nastran User’s Guide* for a discussion of supported degrees-of-freedom (members of the r-set).
8. In superelement analysis, SUPORT1 may be specified for points belonging to the residual structure only.
9. SUPORT1 entries are not supported for SOL 106, 114–118, and 129.

Related Topics

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SWLDPRM

Parameters for CWELD/CFAST Connectors

Define parameters for CWELD/CFAST connectors.

Format:

1	2	3	4	5	6	7	8	9	10
SWLDPRM	PARAM1	VAL1	PARAM2	VAL2	PARAM3	VAL3	PARAM4	VAL4	
	PARAM5	VAL5	-etc.-						

Example:

1	2	3	4	5	6	7	8	9	10
SWLDPRM	CHKRUN	1	GSPROJ	17.0					

Fields:

Field	Contents
PARAMi	Parameter name. Parameters are listed below.
VALi	Value assigned to parameter.
	Parameter Descriptions
CHKRUN	Integer ≥ 0; Default=0 Determines if solution continues or ends after connector element generation. = 0 Solutions runs to completion. = 1 Solution stops after connector elements are generated.
GSMOVE	Integer ≥ 0; Default=0 Number of times GS (or XS,YS,ZS) is relocated if the projection to patches A and/or B is not found. GSMOVE only applies to the ELPAT and PARTPAT methods.
GSPROJ	Real; Default=20.0 Allowable angle between the normal projection from GS (or XS,YS,ZS) to patch A, and the normal projection from GS (or XS,YS,ZS) to patch B. Connectors are only created when this angle is less than GSPROJ. The angle check is skipped when GSPROJ=-1.0.
GSTOL	Real ≥ 0.0; Default=Ignored

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Field	Contents
	Allowable distance from GS (or XS,YS,ZS) to the intersection of patches A and B. Connectors are only created when the distance is less than GSTOL. Only applies to ELPAT and PARTPAT methods.
NREDIA	0 ≤ Integer ≤ 4; Default=0 Number of times the diameter is reduced in half if the diameter fails to project entirely onto patches A and/or B. The reduced diameter only applies to connection point creation; the CWELD stiffness is calculated with the original value. NREDIA only applies to the ELPAT and PARTPAT methods.
PROJTOL	0.0 ≤ Real ≤ 0.2; Default=0.0 Used to calculate the Tolerance = (PROJTOL) * (Size of shell element), which determines if a projected point located outside of the shell elements in patch A or B is used.
PRTSW	Integer ≥ 0; Default=0 Determines if CWELD/CFAST diagnostic information is written to the .f06 file. = 0 No diagnostic output. = 1 Diagnostic output is written to the .f06 file.
ANGPAT	Real ≥ 0.0; Default=18 degrees. The maximum angle in degrees between the normal of the primary weld element and the normal of any adjacent elements to be considered part of the A and B side patchs. Increasing the angle may generate patches which are very irregular and may not represent a good weld connection.
GHOFF	Integer ≥ 0; Default=0 Determines if a CWELD/CFAST can be created when GHA or GHB points do not project on to an element. = 0 CWELD/CFAST creation will fail if any of the GHA or GHB points do not project on to any element in the patch. = 1 If at least 1 of the GHA or GHB points project onto an element in the patch, the GHA or GHB points that do not project onto the patch will still be associated with the primary element.
DISPRT	Integer; Default=2 Determines if displacements are calculated at GA/GB.

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Field	Contents
	= 0 Displacements calculated for GA/GB in all cases. If GA/GB are not explicitly defined, they are only written to the .f06 file. = 1 Displacements will not be calculated for GA/GB for the case when GA/GB are not grid points (coordinates generated internally). = 2 Displacements will not be calculated for GA/GB in all cases.

Related Topics

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18. Bulk Entries T—V

TABDMP1

Modal Damping Table

Defines modal damping as a tabular function of natural frequency for modal solutions.

Format:

1	2	3	4	5	6	7	8	9	10
TABDMP1	TID	TYPE							
	f1	g1	f2	g2	f3	g3	-etc.-		

Example:

TABDMP1	2								
	2.5	0.01057	2.6	0.01362	ENDT				

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
TYPE	Specifies interpretation of damping values. (Character: “G”, “CRIT”, or “Q”; Default is “G”)
	“G” gi interpreted as structural damping values.
	“CRIT” gi interpreted as fraction of critical damping values.
	“Q” gi interpreted as quality or amplification factor values.
fi	Frequency value in cycles per unit time. (Real ≥ 0.0)
gi	Damping value corresponding to frequency fi. (Real)

Remarks:

1. TABDMP1 bulk entries are used to define modal damping for modal complex eigenvalue analysis (SOL 110), modal frequency response analysis (SOL 111), modal transient response analysis (SOL 112), the modal frequency response subcase (SOL 401, and, in rotor dynamics analyses, superelement reduction or edition (SOL 414,103), complex modal (SOL 414,110) analysis, and harmonic response (SOL 414,111).
2. TABDMP1 bulk entries are selected with the SDAMPING case control command.

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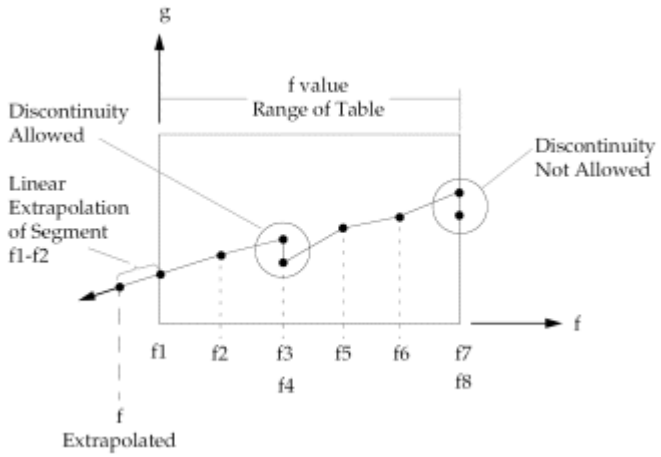
3. Damping values defined on a TABDMP1 bulk entry are interpolated or extrapolated to obtain the damping at the undamped natural frequency of each mode. Depending on the parameter KDAMP, the damping at each undamped natural frequency is used to calculate either the modal viscous damping or the modal structural damping for each mode. The modal viscous damping or modal structural damping is used in the equations of motion for response calculations. For modal transient response analysis (SOL 112), only modal viscous damping is valid. See Remark **10** for additional information on the KDAMP parameter.
4. The frequency values, fi, must be specified in either ascending or descending order, but not both.
5. Any fi or gi entry may be ignored by placing "SKIP" in either of the two fields used for that entry.
6. At least one continuation entry must be specified.
7. The damping at a specific frequency, g(*f*), is either linearly interpolated or linearly extrapolated from the damping values listed in the TABDMP1 bulk entry. To extrapolate, the last two end points are used. No warning messages are issued if table data is input incorrectly.
8. Except at the lowest and highest frequencies in the range of the table, damping values can be discontinuous. At a discontinuity, the damping value used is the average of the discontinuous damping values. For example, in **Figure 18-175**, the damping value used at f = f3 = f4 is g = (g3 + g4) / 2.

Figure 18-175: Extrapolation of and Discontinuity in Damping Values



9. The end of the table is indicated by the existence of "ENDT" in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag "ENDT".
10. PARAM,KDAMP is used to specify whether the damping values listed in the TABDMP1 bulk entry are used to calculate modal viscous damping or modal structural damping. See **Table 18-125** for more information on the use of PARAM,KDAMP.

> **Note**
>
> PARAM,KDAMP does not apply to SOL 414 rotor dynamics solutions.

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Table 18-125: Use of PARAM,KDAMP

	PARAM,KDAMP,1 (Default)	PARAM,KDAMP,-1
Damping values used to calculate →	Modal viscous damping, b_j	Modal structural damping, G_j
If TYPE = G	$b_j = g(f_j)\ b_{cr} / 2$	$G_j = g(f_j)$
If TYPE = CRIT	$b_j = g(f_j)\ b_{cr}$	$G_j = 2\ g(f_j)$
If TYPE = Q	$b_j = b_{cr} / (2\ g(f_j))$	$G_j = 1 / g(f_j)$
Modal damping force added to equation of motion in a transient response solution.	$b_j\ d\zeta_j/dt$	$iG_jk_j\zeta_j$
Modal damping force added to equation of motion in a frequency response solution.	$ib_j\ \Omega\zeta_j$ where Ω is an excitation frequency defined on the FREQ*i* bulk entries.	
where: b_j = Modal viscous damping for the j^{th} mode G_j = Modal structural damping for the j^{th} mode f_j = Undamped natural frequency for the j^{th} mode $g(f_j)$ = Damping value at f_j interpolated or extrapolated from gi damping values ζ_j = Modal coordinate for the j^{th} mode $b_{cr} = 2\ m_j\ \omega_j$ = Critical damping for the j^{th} mode with $\omega_j = 2\pi\ f_j$ m_j = Modal mass for the j^{th} mode		

For additional information, see “Formulation of Dynamic Equations in SubDMAP GMA” in the *Simcenter Nastran User’s Guide*.

11. The SOL 401 modal frequency response subcase only supports TYPE=CRIT and PARAM,KDAMP,1 .

Related Topics

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TABLED1

Dynamic Load Tabular Function, Form 1

Defines a tabular function for use in generating frequency-dependent and time-dependent dynamic loads.

Format:

1	2	3	4	5	6	7	8	9	10
TABLED1	TID	XAXIS	YAXIS	EXTRAP	LOEXT	HIEXT			
	x1	y1	x2	y2	x3	y3	-etc.-	"ENDT"	

Example:

TABLED1	32			1					
	-3.0	6.9	2.0	5.6	3.0	5.6	ENDT		

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
XAXIS, YAXIS	Specifies whether the interpolation is linear in Cartesian, semi-log, or log-log space. See Remark 9. (Character: "LINEAR" or "LOG"; Default = "LINEAR") For interpolation in Cartesian space, specify XAXIS = LINEAR and YAXIS = LINEAR. For interpolation in semi-log space, specify XAXIS = LINEAR and YAXIS = LOG, or XAXIS = LOG and YAXIS = LINEAR. See Remark 7. For interpolation in log-log space, specify XAXIS = LOG and YAXIS = LOG. See Remark 8.
EXTRAP	Extrapolation option. (Integer: -1, 0, 1, or 2; Default = 0) If EXTRAP = -1 (SOL 402 only), no extrapolation occurs. When xi goes outside of the table, yi becomes undefined. When TABLEDi is referenced by a force, a force=0.0 is applied. If it is referenced by an enforced displacement (SPCD), the displacement is released (free). If EXTRAP = 0, obtain table lookup values for $x < x_{min}$ and $x > x_{max}$ by applying the interpolation formula to the two starting and two ending data points. See Figure 18-176 and Remark 9. If EXTRAP = 1, use the value of the table field at x_{min} for the table lookup when $x < x_{min}$ and use the value of the table field at x_{max} for the table lookup when $x > x_{max}$.

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Field	Contents
	If EXTRAP = 2, use LOEXT for the table lookup when $x < x_{min}$ and use HIEXT for the table lookup when $x > x_{max}$.
LOEXT	For EXTRAP = 2, table lookup value when $x < x_{min}$. (Real; Default = 0.0)
HIEXT	For EXTRAP = 2, table lookup value when $x > x_{max}$. (Real; Default = 0.0)
xi	Tabular values of the independent domain. See Remarks 2 and 5. (Real; No default)
yi	Tabular values of the dependent domain. See Remark 5. (Real; Default = 0.0)
"ENDT"	Flag indicating the end of the table.

Remarks:

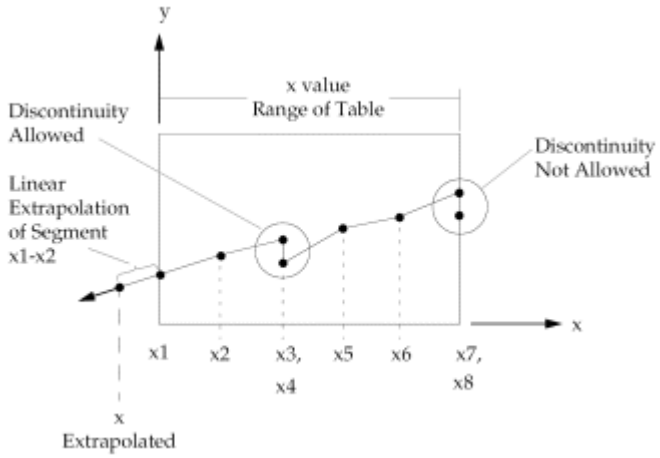
1. No warning messages are issued if tabular data is input incorrectly.
2. The values for xi must be listed in either ascending or descending order according to algebraic size. Thus, the values at the starting and ending data points are either x_{min} and x_{max}, respectively, or x_{max} and x_{min}, respectively.
3. Discontinuities may be specified at any point except the end points. For example, in Figure 18-176, a discontinuity is not allowed between points x7 and x8. In Figure 18-176, if YAXIS = LINEAR, the table lookup value at x3 is:

$$y = \frac{y3 + y4}{2}$$

If YAXIS = LOG, the table lookup value at x3 is:

$$y = \sqrt{y3 \cdot y4}$$

Figure 18-176: Example of Table Extrapolation and Discontinuity



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If in Figure 18-176 the discontinuity is at x2, x3, and EXTRAP = 0, the software uses data points (x1,y1) and (x2,y2) to calculate the linear extrapolation formula.

4. At least one continuation must be specified.
5. Any data point, (xi,yi), may be ignored by placing the character string “SKIP” in either the xi or yi field.
6. The end of the table is indicated by the existence of the character string “ENDT” in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag “ENDT”.
7. Specify XAXIS = LINEAR and YAXIS = LOG if the tabular data approximates a function of the form $y = be^{mx}$. Because such a function plots as a straight line in semi-log space with XAXIS = LINEAR and YAXIS = LOG, fewer data points are needed to obtain accurate interpolations.
8. Specify XAXIS = LOG and YAXIS = LOG if the tabular data approximates a function of the form $y = bx^{m}$. Because such a function plots as a straight line in log-log space, fewer data points are needed to obtain accurate interpolations.
9. The table lookup at x returns $y_T(x)$. The formula that the software uses to interpolate, and if EXTRAP = 0, extrapolate the tabular data for the table lookup at x depends on how the XAXIS and YAXIS fields are specified, as indicated in Table 18-126.

Table 18-126: Interpolation formula

XAXIS	YAXIS	$y_T(x)$
LINEAR	LINEAR	$\frac{xj - x}{xj - xi}yi + \frac{x - xi}{xj - xi}yj$
LOG	LINEAR	$\frac{\ln(xj/x)}{\ln(xj/xi)}yi + \frac{\ln(x/xi)}{\ln(xj/xi)}yj$
LINEAR	LOG	$\exp\left[\frac{xj - x}{xj - xi}\ln yi + \frac{x - xi}{xj - xi}\ln yj\right]$
LOG	LOG	$\exp\left[\frac{\ln(xj/x)}{\ln(xj/xi)}\ln yi + \frac{\ln(x/xi)}{\ln(xj/xi)}\ln yj\right]$
(xi,yi) and (xj,yj) are adjacent data points and, $xi < x < xj$ or $xi > x > xj$.		

10. Extrapolation is not used for Fourier transform methods. The function is zero outside the range of the table.
11. For frequency-dependent loads, xi is measured in cycles per unit time.
12. Values for xi must be positive if XAXIS = LOG, and values for yi must be positive if YAXIS = LOG. Otherwise, the software issues a fatal message.
13. TABLED1 can be used for designed frequency-dependent properties in SOL 200 if XAXIS = LINEAR and YAXIS = LINEAR. For frequency-dependent properties that are not associated with design variables, TABLED1 can be used with the other axis options, or the other TABLEDi can be used.

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Remarks related to SOLs 402 and 414:

1. XAXIS=LOG is not supported and will generate a FATAL error.
2. Discontinuities may be specified between any two points except the two starting points or two ending points. If a discontinuity is defined at x = X3 (with X3 = X4), the value of y for x = X3 is Y3. The value Y4 is used to compute y over the interval X4 < x ≤ X5.

Related Topics

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TABLED2

Dynamic Load Tabular Function, Form 2

Defines a tabular function for use in generating frequency-dependent and time-dependent dynamic loads. Also contains parametric data for use with the table.

Format:

1	2	3	4	5	6	7	8	9	10
TABLED2	TID	X1	EXTRAP	LOEXT	HIEXT				
	x1	y1	x2	y2	x3	y3	-etc.-	"ENDT"	

Example:

TABLED2	15	-10.5	2	-5.0	7.0				
	1.0	-4.5	2.0	-4.2	2.0	2.8	7.0	6.5	
	SKIP	SKIP	9.0	6.5	ENDT				

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
X1	Table parameter. See Remark 7. (Real)
EXTRAP	Extrapolation option. (Integer: -1, 0, 1, or 2; Default = 0) If EXTRAP = -1 (SOL 402 only), no extrapolation occurs. When xi goes outside of the table, yi becomes undefined. When TABLEDi is referenced by a force, a force=0.0 is applied. If it is referenced by an enforced displacement (SPCD), the displacement is released (free). If EXTRAP = 0, obtain table lookup values for $(x - X1) < x_{min}$ and $(x - X1) > x_{max}$ by linearly extrapolating the two starting and two ending data points. See Figure 18-177. If EXTRAP = 1, use the value of the table field at x_{min} for the table lookup when $(x - X1) < x_{min}$ and use the value of the table field at x_{max} for the table lookup when $(x - X1) > x_{max}$. If EXTRAP = 2, use LOEXT for the table lookup when $(x - X1) < x_{min}$ and use HIEXT for the table lookup when $(x - X1) > x_{max}$.
LOEXT	For EXTRAP = 2, table lookup value when $(x - X1) < x_{min}$. (Real; Default = 0.0)

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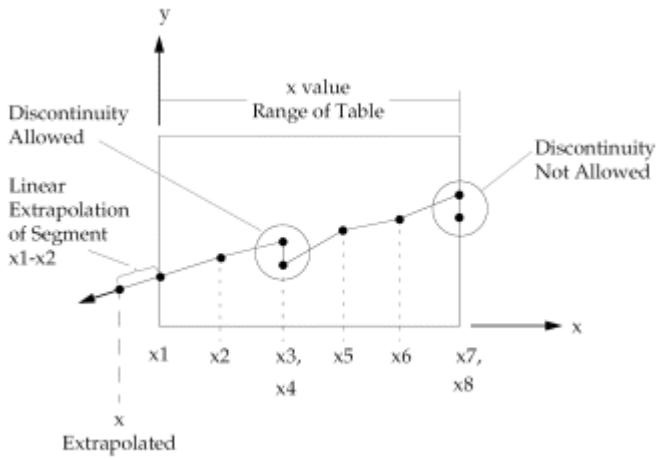
Field	Contents
HIEXT	For EXTRAP = 2, table lookup value when (x - X1) > x_{max}. (Real; Default = 0.0)
xi	Tabular values of the independent domain. See Remarks 2 and 5. (Real; No default)
yi	Tabular values of the dependent domain. See Remark 5. (Real; Default = 0.0)
"ENDT"	Flag indicating the end of the table.

Remarks:

1. No warning messages are issued if tabular data is input incorrectly.
2. The values for xi must be listed in either ascending or descending order according to algebraic size. Thus, the values at the starting and ending data points are either x_{min} and x_{max}, respectively, or x_{max} and x_{min}, respectively.
3. Discontinuities may be specified at any point except the end points. For example, in Figure 18-177, a discontinuity is not allowed between points x7 and x8. If y is evaluated at a discontinuity, then the average value of y is used. In Figure 18-177, the value of y at x = x3 is:

$$y = \frac{y3 + y4}{2}$$

Figure 18-177: Example of Table Extrapolation and Discontinuity



If in Figure 18-177 the discontinuity is at x2, x3, and EXTRAP = 0, the software uses data points (x1,y1) and (x2,y2) to calculate the linear extrapolation formula.

4. At least one continuation entry must be specified.
5. Any data point, (xi,yi), may be ignored by placing the character string "SKIP" in either the xi or yi field.
6. The end of the table is indicated by the existence of the character string "ENDT" in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag "ENDT".

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7. The table lookup at *x* returns:

   $$y_T\left(x - X1\right)$$

   The formula that the software uses to interpolate the tabular data for the table lookup at *x* is:

   $$\left(\frac{y_j - y_i}{x_j - x_i}\right)\left[\left(x - X1\right) - x_i\right] + y_i$$

   where (xi,yi) and (xj,yj) are adjacent data points and, xi < (*x* - X1) < xj or xi > (*x* - X1) > xj.

   If EXTRAP = 0, the formula that the software uses to extrapolate the tabular data for the table lookup is the same, except that the starting two or ending two data points are used for (xi,yi) and (xj,yj).

8. Extrapolation is not used for Fourier transform methods. The function is zero outside the range of the table.

9. For frequency-dependent loads, xi and X1 are measured in cycles per unit time.

Remarks related to SOLs 402 and 414:

1. Discontinuities may be specified between any two points except the two starting points or two ending points. If a discontinuity is defined for x = X3 (with X3 = X4), the value of y for x = X3 is Y3. The value Y4 is used to compute y over the interval X4 < x ≤ X5.

Related Topics

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TABLED3

Dynamic Load Tabular Function, Form 3

Defines a tabular function for use in generating frequency-dependent and time-dependent dynamic loads. Also contains parametric data for use with the table.

Format:

1	2	3	4	5	6	7	8	9	10
TABLED3	TID	X1	X2	EXTRAP	LOEXT	HIEXT			
	x1	y1	x2	y2	x3	y3	-etc.-	“ENDT”	

Example:

TABLED3	62	126.9	30.0	1					
	2.9	2.9	3.6	4.7	5.2	5.7	ENDT		

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
X1, X2	Table parameters. See Remark 7. (Real; X2 ≠ 0.0)
EXTRAP	Extrapolation option. (Integer: -1, 0, 1, or 2; Default = 0) If EXTRAP = -1 (SOL 402 only), no extrapolation occurs. When xi goes outside of the table, yi becomes undefined. When TABLEDi is referenced by a force, a force=0.0 is applied. If it is referenced by an enforced displacement (SPCD), the displacement is released (free). If EXTRAP = 0, obtain table lookup values for (x - X1) / X2 < x_{min} and (x - X1) / X2 > x_{max} by linearly extrapolating the two starting and two ending data points. See Figure 18-178. If EXTRAP = 1, use the value of the table field at x_{min} for the table lookup when (x - X1) / X2 < x_{min} and use the value of the table field at x_{max} for the table lookup when (x - X1) / X2 > x_{max}. If EXTRAP = 2, use LOEXT for the table lookup when (x - X1) / X2 < x_{min} and use HIEXT for the table lookup when (x - X1) / X2 > x_{max}.
LOEXT	For EXTRAP = 2, table lookup value when (x - X1) / X2 < x_{min}. (Real; Default = 0.0)
HIEXT	For EXTRAP = 2, table lookup value when (x - X1) / X2 > x_{max}. (Real; Default = 0.0)

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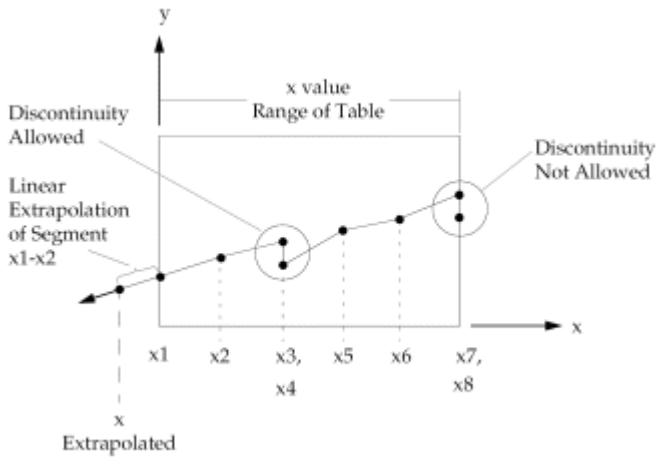
Field	Contents
xi	Tabular values of the independent domain. See Remarks 2 and 5. (Real; No default)
yi	Tabular values of the dependent domain. See Remark 5. (Real; Default = 0.0)
“ENDT”	Flag indicating the end of the table.

Remarks:

1. No warning messages are issued if tabular data is input incorrectly.
2. The values for xi must be listed in either ascending or descending order according to algebraic size. Thus, the values at the starting and ending data points are either x_{min} and x_{max}, respectively, or x_{max} and x_{min}, respectively.
3. Discontinuities may be specified at any point except the end points. For example, in Figure 18-178, a discontinuity is not allowed between points x7 and x8. If y is evaluated at a discontinuity, then the average value of y is used. In Figure 18-178, the value of y at *x* = x3 is:

$$y = \frac{y3 + y4}{2}$$

Figure 18-178: Example of Table Extrapolation and Discontinuity



If in Figure 18-178 the discontinuity is at x2, x3, and EXTRAP = 0, the software uses data points (x1,y1) and (x2,y2) to calculate the linear extrapolation formula.

4. At least one continuation entry must be present.
5. Any data point, (xi,yi), may be ignored by placing the character string “SKIP” in either the xi or yi field.
6. The end of the table is indicated by the existence of the character string “ENDT” in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag “ENDT”.
7. The table lookup at *x* returns:

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$$y_T\left(\frac{x-X1}{X2}\right)$$

The formula that the software uses to interpolate the tabular data for the table lookup at *x* is:

$$\left(\frac{y_j-y_i}{x_j-x_i}\right)\left[\left(\frac{x-X1}{X2}\right)-x_i\right]+y_i$$

where (xi,yi) and (xj,yj) are adjacent data points and, xi < (*x* - X1) / X2 < xj or xi > (*x* - X1) / X2 > xj.

If EXTRAP = 0, the formula that the software uses to extrapolate the tabular data for the table lookup is the same, except that the starting two or ending two data points are used for (xi,yi) and (xj,yj).

8. Extrapolation is not used for Fourier transform methods. The function is zero outside the range of the table.

9. For frequency-dependent loads, xi and X1 are measured in cycles per unit time. X2 is dimensionless.

Remarks related to SOLs 402 and 414:

1. Discontinuities may be specified between any two points except the two starting points or two end points. If a discontinuity is defined for x = X3 (with X3 = X4), the value of y for x = X3 is Y3. The value Y4 is used to compute y over the interval X4 < x ≤ X5.

Related Topics

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TABLED4

Dynamic Load Tabular Function, Form 4

Defines the coefficients of a power series for use in generating frequency-dependent and time-dependent dynamic loads. Also contains parametric data for use with the table.

Format:

1	2	3	4	5	6	7	8	9	10
TABLED4	TID	X1	X2	X3	X4				
	A0	A1	A2	A3	A4	A5	-etc.-	“ENDT”	

Example:

TABLED4	28	0.0	1.0	0.0	100.				
	2.91	-0.0329	6.51-5	0.0	-3.4-7	ENDT			

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
Xi	Table parameters. (Real; X2 ≠ 0.0; X3<X4)
Ai	Coefficients. (Real)
“ENDT”	Flag indicating the end of the table.

Remarks:

1. At least one continuation entry must be specified.
2. The end of the table is indicated by the existence of “ENDT” in the field following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag “ENDT”.
3. TABLED4 uses the algorithm

$$y = \sum_{i=0}^{N} \mathrm{Ai}\left(\frac{x - \mathrm{X1}}{\mathrm{X2}}\right)^{i}$$

where *x* is input and *y* is returned. N is the number of pairs.

Whenever *x*< X3, use X3 for *x*; whenever *x*> X4, use X4 for *x*. There are N+1 entries in the table. There are no error returns from this procedure.

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4. For frequency-dependent loads, xi and X1 are measured in cycles per unit time. Specify the units for Ai and X2 so that the units of all terms in the summation are consistent.

Related Topics

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TABLED6

Dynamic Load Tabular Function, Form 6

Defines a tabular function of the form $w(x) = u(x) + iv(x)$ or $w(x) = |u(x)|e^{iv(x)}$.

Format:

1	2	3	4	5	6	7	8	9	10
TABLED6	TID	TYPE	EXTRAP	ULOEXT	UHIEXT	VLOEXT	VHIEXT		
	x1	u1	v1	x2	u2	v2	-etc.-	"ENDT"	

Example:

TABLED6	101	MP	1						
	10.0	2.9	16.5	20.0	5.2	93.0			
	30.0	4.9	132.0	40.0	4.1	108.0	ENDT		

Fields:

Field	Contents
TID	Unique table identification number. See Remark 2. (Integer > 0)
TYPE	Complex format of response data. (Character: RI or MP; Default = RI) TYPE = RI for real / imaginary TYPE = MP for magnitude / phase
EXTRAP	Extrapolation option. See Remark 5. (Integer = 0, 1, or 2; Default = 0) If EXTRAP = 0, obtain table lookup values for $x < x_{min}$ and $x > x_{max}$ by linearly extrapolating the two starting and two ending data points. See Figure 18-179. If EXTRAP = 1, use the value of the table field at x_{min} for the table lookup when $x < x_{min}$ and use the value of the table field at x_{max} for the table lookup when $x > x_{max}$. If EXTRAP = 2, use ULOEXT and VLOEXT for the table lookup when $x < x_{min}$ and use UHIEXT and VHIEXT for the table lookup when $x > x_{max}$.
ULOEXT	Real part of table lookup value when $x < x_{min}$ if EXTRAP = 2 and TYPE = RI. (Real; Default = 0.0) Magnitude of table lookup value when $x < x_{min}$ if EXTRAP = 2 and TYPE = MP. (Real; Default = 0.0)

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Field	Contents
UHIEXT	Real part of table lookup value when $x > x_{max}$ if EXTRAP = 2 and TYPE = RI. (Real; Default = 0.0)
	Magnitude of table lookup value when $x > x_{max}$ if EXTRAP = 2 and TYPE = MP. (Real; Default = 0.0)
VLOEXT	Imaginary part of table lookup value when $x < x_{min}$ if EXTRAP = 2 and TYPE = RI. (Real; Default = 0.0)
	Phase of table lookup value when $x < x_{min}$ if EXTRAP = 2 and TYPE = MP. (Real; Default = 0.0)
VHIEXT	Imaginary part of table lookup value when $x > x_{max}$ if EXTRAP = 2 and TYPE = RI. (Real; Default = 0.0)
	Phase of table lookup value when $x > x_{max}$ if EXTRAP = 2 and TYPE = MP. (Real; Default = 0.0)
xi	Tabular values of the independent domain. See Remarks 3 and 7. (Real; No default)
ui	Real part of the dependent variable if TYPE = RI. (Real; Default = 0.0)
	Magnitude of the dependent variable if TYPE = MP. (Real ≥ 0.0; Default = 0.0)
vi	Imaginary part of the dependent variable if TYPE = RI. (Real; Default = 0.0)
	Phase of the dependent variable in degrees if TYPE = MP. (Real; Default = 0.0)
"ENDT"	Flag indicating the end of the table.

Remarks:

1. No warning messages are issued if tabular data is input incorrectly.
2. The TID must be unique for all TABLEDi bulk entries.
3. The values for xi must be listed in either ascending or descending order according to algebraic size. Thus, the values at the starting and ending data points are either x_{min} and x_{max}, respectively, or x_{max} and x_{min}, respectively.
4. Discontinuities are not allowed. See **Figure 18-179**.

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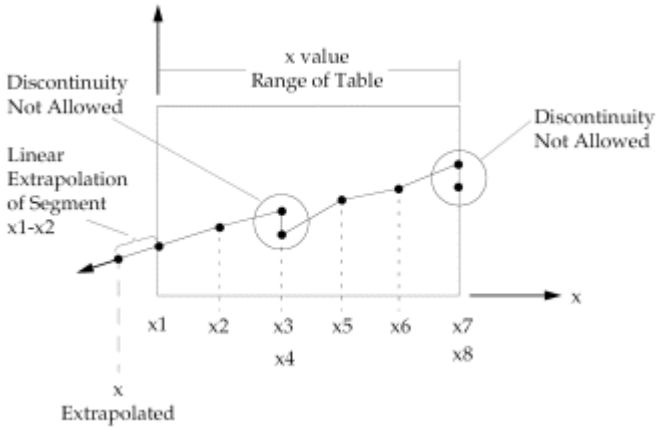
Figure 18-179: Example of Table Extrapolation and Discontinuity



5. When EXTRAP = 0, the two starting and two ending data points are linearly extrapolated to obtain table lookup values for $x < x_{min}$ and $x > x_{max}$. See **Figure 18-179**.
6. At least one continuation entry must be present.
7. Any (xi,ui,vi) combination may be ignored by placing the character string “SKIP” in one of the three fields.
8. The end of the table is indicated by the existence of the character string “ENDT” in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag “ENDT”.
9. TABLED6 interpolates the tabular data as follows:
   a. If TYPE = RI, the tabular data is converted to magnitude and phase.
   b. The magnitudes are linearly interpolated to obtain the lookup value.
   c. The phase angles are linearly interpolated to obtain the lookup value. For this calculation the software uses the smallest subtended angle between the phase angles that are used in the interpolation calculation.

For example, suppose a TABLED6 bulk entry is defined as follows:

```
TABLED6 100     MP      1
+
        10.0    2.9     16.5    20.0    5.2     93.0
+
        30.0    4.9     132.0   40.0    4.1     326.0   ENDT
```

The lookup value for the magnitude at x = 27.5 is:

$$u = 5.2 + \left(\frac{4.9 - 5.2}{30.0 - 20.0}\right)\left(27.5 - 20.0\right) = 4.975$$

The lookup value for the phase angle at x = 27.5 is:

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$$v = 93.0 + \left(\frac{132.0 - 93.0}{30.0 - 20.0}\right)\left(27.5 - 20.0\right) = 122.25$$

The lookup value for the magnitude at x = 35.0 is:

$$u = 4.9 + \left(\frac{4.1 - 4.9}{40.0 - 30.0}\right)\left(35.0 - 30.0\right) = 4.5$$

At x = 35.0, the smallest subtended angle between 132.0° and 326° is 166.0°. Thus, the lookup value for the phase at x = 35.0 is:

$$v = 132.0 + \left(\frac{-166.0}{40.0 - 30.0}\right)\left(35.0 - 30.0\right) = 49.0$$

10. For frequency-dependent loads, xi is measured in cycles per unit time.

Related Topics

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TABLEH2

Direct Harmonic Load Tabular Function, Form 2

Defines a tabular function for harmonic loads.

TABLEH2 represents the Fast Fourier Transform (FFT) of the angular distribution of the force along the circumference using the real part (SRk) and the imaginary part (SIk) for each harmonic.

Format:

1	2	3	4	5	6	7	8	9	10
TABLEH2	TID								
	H1	SR1	SI1	H2	SR2	SI2	-etc.-	“ENDT”	

Examples:

TABLEH2	11								
	0	0.	1.	1	0.	0.5			
	2	0.	0.5	ENDT					

TABLEH2	12								
	0	1.	0.	1	0.5	0.0			
	2	0.5	0.	ENDT					

Fields:

Field	Contents
TID	Unique table identification number for table referenced by the RLCYCi bulk data field TS. See Remark 3. (Integer > 0)
Hk	Harmonic index. See Remarks 4, 5, and 6. (Integer ≥ 0; no default)
SRk	Real scale factor for kth harmonic. (Real; Default = 1.0)
SIk	Imaginary scale factor for kth harmonic. (Real; Default = 1.0)
“ENDT”	Flag indicating the end of the table.

Remarks:

1. For every harmonic you list for the solution with the HARMONIC=n command in the case control, you must define the scale factors on the TABLEH2 entry.

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2. No warning messages are issued if tabular data is input incorrectly.
3. The TID must be unique for all TABLEHi bulk entries.
4. You must define a minimum of two harmonic indexes in the table.
5. The values for the harmonic indexes must be listed in ascending order according to algebraic size.
6. The list of harmonic indexes must be inclusive. For example, to enter a scale factor for harmonic index 3, you must also define harmonic index 0, 1, and 2.
7. At least one continuation entry must be present.
8. The end of the table is indicated by the existence of the character string "ENDT" in either of the two fields following the last entry. An error is issued if any continuations follow the entry containing the end-of-table flag "ENDT".

Related Topics

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TABLEM1

Material Property Table, Form 1

Uses tabular data to define temperature-dependent material properties. The value for the material property at a specific temperature is interpolated or extrapolated from the tabular data.

Format:

1	2	3	4	5	6	7	8	9	10
TABLEM1	TID	XAXIS	YAXIS	EXTRAP					
	x1	y1	x2	y2	x3	y3	-etc.-	“ENDT”	

Example:

TABLEM1	32								
	-3.0	6.9	2.0	5.6	3.0	5.6	ENDT		

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
XAXIS	Specifies a linear or logarithmic interpolation for the x-axis. See Remark 6. (Character: “LINEAR” or “LOG”; Default=“LINEAR”)
YAXIS	Specifies a linear or logarithmic interpolation for the y-axis. See Remark 6. (Character: “LINEAR” or “LOG”; Default=“LINEAR”)
EXTRAP	Extrapolation option. (Integer: 0 or 1; Default = 0) • If EXTRAP = 0, extrapolate the two starting data points to obtain the table look up when x < xi, and extrapolate the two ending data points to obtain the table look up when x > xi. • If EXTRAP = 1, use the value of the table field at the starting data point for the table look up when x < xi, and use the value of the table field at the ending data point for the table look up when x > xi.
xi, yi	Tabular values. xi are values for temperature. yi are values for the material property. (Real)
“ENDT”	Flag indicating the end of the table.

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Remarks:

1. xi must be in either ascending or descending order, but not both. No warning message is issued if tabular data is input incorrectly.
2. Discontinuities may be specified between any two points except the two starting points or two end points. For example, in **Figure 18-180**, discontinuities are allowed only between points x2 through x7.

   If the y-axis is a LINEAR axis, the value of y at x3 in **Figure 18-180** is:

   $y = (y_3 + y_4) / 2$

   If the y-axis is a LOG axis, the value of y at x3 in **Figure 18-180** is:

   $y = \sqrt{y_3 y_4}$
3. At least one continuation entry must be specified.
4. Any xi-yi pair may be ignored by placing "SKIP" in either of the two fields.
5. The end of the table is indicated by the existence of "ENDT" in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag "ENDT".
6. The tabular data is interpolated for values of *x* that lie within the range of the tabular data and, when EXTRAP = 0, the tabular data is extrapolated from the two starting or end points for values of *x* that lie outside the range of the tabular data.

   The value *y* that the software returns for the material property at temperature *x* depends on which MATTi bulk entry references the TABLEM1 bulk entry.

   - If a MATT1, MATT2, MATT3, MATT8, MATT9, or MATT11 structural material property references the TABLEM1 bulk entry:

     $$y = y_T(x)$$

   - If a MATT4 or MATT5 thermal material property references the TABLEM1 bulk entry:

     $$y = z y_T(x)$$

   where $y_T(x)$ is the table look up at *x*, and *z* is the value of the material property on the corresponding MATi entry.

   The equations that the software uses to interpolate or extrapolate the tabular data for each combination of XAXIS and YAXIS options are:

XAXIS option	YAXIS option	Table look up value, $y_T(x)$
LINEAR	LINEAR	$\frac{\mathrm{xj} - x}{\mathrm{xj} - \mathrm{xi}}\mathrm{yi} + \frac{x - \mathrm{xi}}{\mathrm{xj} - \mathrm{xi}}\mathrm{yj}$
LOG	LINEAR	$\frac{\ln(\mathrm{xj} / x)}{\ln(\mathrm{xj} / \mathrm{xi})}\mathrm{yi} + \frac{\ln(x / \mathrm{xi})}{\ln(\mathrm{xj} / \mathrm{xi})}\mathrm{yj}$

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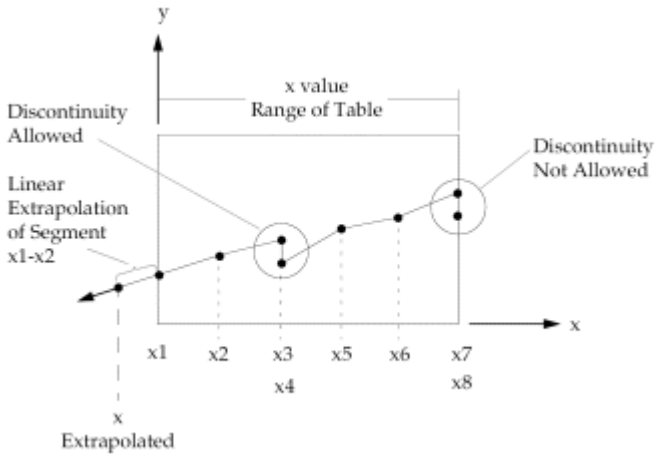
XAXIS option	YAXIS option	Table look up value, $y_T(x)$
LINEAR	LOG	$\exp\left[\frac{xj - x}{xj - xi}\ln yi + \frac{x - xi}{xj - xi}\ln yj\right]$
LOG	LOG	$\exp\left[\frac{\ln(xj/x)}{\ln(xj/xi)}\ln yi + \frac{\ln(x/xi)}{\ln(xj/xi)}\ln yj\right]$

where xj and yj follow xi and yi.

Figure 18-180: Example of Table Extrapolation and Discontinuity



7. If an axis is logarithmic, the tabular values on the axis must be positive. A fatal message will be issued if an axis has a tabular value ≤ 0.

Remarks related to SOLs 402 and 414:

1. XAXIS=LOG is not supported and will generate a fatal error.
2. Discontinuities may be specified between any two points except the two starting points or two ending points. If a discontinuity is defined for x = X3 (with X3 = X4), the value of y for x = X3 is Y3. The value Y4 is used to compute y over the interval X4 < x ≤ X5.

Related Topics

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TABLEM2

Material Property Table, Form 2

Uses shifted tabular data to define temperature-dependent material properties. The value for the material property at a specific temperature is the product of the interpolated or extrapolated table look up value and the value for the material property as specified on the corresponding MATi entry.

Format:

1	2	3	4	5	6	7	8	9	10
TABLEM2	TID	X1	EXTRAP						
	x1	y1	x2	y2	x3	y3	-etc.-		

Example:

TABLEM2	15	-10.5							
	1.0	-4.5	2.0	-4.5	2.0	2.8	7.0	6.5	
	SKIP	SKIP	9.0	6.5	ENDT				

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
X1	Shift parameter. (Real)
EXTRAP	Extrapolation option. (Integer: 0 or 1; Default = 0) • If EXTRAP = 0, extrapolate the two starting data points to obtain the table look up when x < xi, and extrapolate the two ending data points to obtain the table look up when x > xi. • If EXTRAP = 1, use the value of the table field at the starting data point for the table look up when x < xi, and use the value of the table field at the ending data point for the table look up when x > xi.
xi, yi	Tabular values. xi are values for temperature. yi are values for the material property. (Real)
“ENDT”	Flag indicating the end of the table.

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Remarks:

1. xi must be in either ascending or descending order, but not both. No warning message is issued if the tabular data is input incorrectly.
2. Discontinuities may be specified between any two points except the two starting points or two end points. For example, in **Figure 18-181** discontinuities are allowed only between points x2 through x7.

   If y is evaluated at a discontinuity, then the average value of y is used. For example, the value of y at x3 = (x – X1) in **Figure 18-181** is:

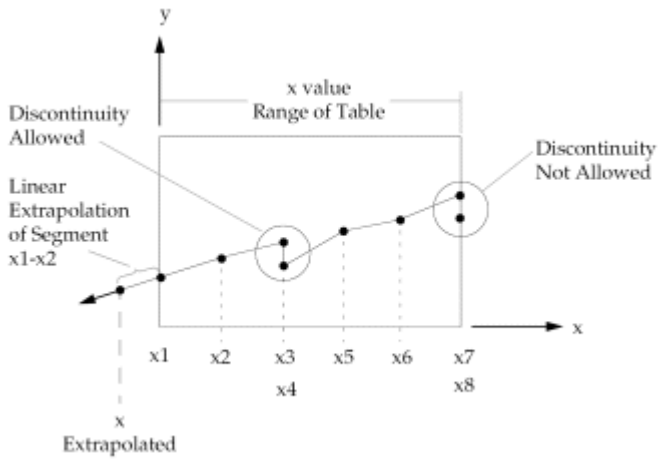
   $y = (y_3 + y_4) / 2$
3. At least one continuation entry must be specified.
4. Any xi-yi pair may be ignored by placing "SKIP" in either of the two fields.
5. The end of the table is indicated by the existence of "ENDT" in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag "ENDT".
6. The tabular data is linearly interpolated for values of (x – X1) that lie within the range of the tabular data and, when EXTRAP = 0, the tabular data is linearly extrapolated from the two starting or end points for values of (x – X1) that lie outside the range of the tabular data.

   The value y that the software returns for the structural or thermal material property at temperature x is:

   $$y = zy_T(x - \mathrm{X1})$$

   where $y_T(x - \mathrm{X1})$ is the table look up at (x – X1), and z is the value of the material property on the corresponding MATi entry.

Figure 18-181: Example of Table Extrapolation and Discontinuity



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Remarks related to SOLs 402 and 414:

1. Discontinuities may be specified between any two points except the two starting points or two ending points. If a discontinuity is defined for x = X3 (with X3 = X4), the value of y for x = X3 is Y3. The value Y4 is used to compute y over the interval X4 < x ≤ X5.

Related Topics

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TABLEM3

Material Property Table, Form 3

Uses shifted and scaled tabular data to define temperature-dependent material properties. The value for the material property at a specific temperature is the product of the interpolated or extrapolated table look up value and the value for the material property as specified on the corresponding MATi entry.

Format:

1	2	3	4	5	6	7	8	9	10
TABLEM3	TID	X1	X2	EXTRAP					
	x1	y1	x2	y2	x3	y3	-etc.-		

Example:

TABLEM3	62	126.9	30.0	1					
	2.9	2.9	3.6	4.7	5.2	5.7	ENDT		

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
X1	Shift parameter. See Remark 6. (Real; X2 ≠ 0.0)
X2	Scaling parameter. See Remark 6. (Real; X2 ≠ 0.0)
EXTRAP	Extrapolation option. (Integer: 0 or 1; Default = 0) • If EXTRAP = 0, extrapolate the two starting data points to obtain the table look up when x < xi, and extrapolate the two ending data points to obtain the table look up when x > xi. • If EXTRAP = 1, use the value of the table field at the starting data point for the table look up when x < xi, and use the value of the table field at the ending data point for the table look up when x > xi.
xi, yi	Tabular values. xi are values for temperature. yi are values for the material property. (Real)
“ENDT”	Flag indicating the end of the table.

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Remarks:

1. xi must be in either ascending or descending order, but not both. No warning message is issued if the tabular data is input incorrectly.
2. Discontinuities may be specified between any two points except the two starting points or two end points. For example, in Figure 18-182 discontinuities are allowed only between points x2 through x7.

   If y is evaluated at a discontinuity, then the average value of y is used. For example, the value of y at x3 = (x – X1) / X2 in Figure 18-182 is:

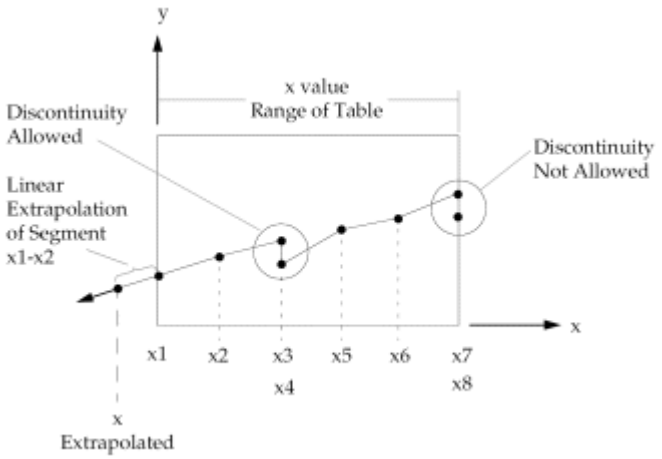
   $y = (y_3 + y_4) / 2$
3. At least one continuation entry must be specified.
4. Any xi-yi pair may be ignored by placing "SKIP" in either of the two fields.
5. The end of the table is indicated by the existence of "ENDT" in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag "ENDT".
6. The tabular data is linearly interpolated for values of (x – X1) / X2 that lie within the range of the tabular data and, when EXTRAP = 0, the tabular data is linearly extrapolated from the two starting or end points for values of (x – X1) / X2 that lie outside the range of the tabular data.

   The value y that the software returns for the structural or thermal material property at temperature x is:

$$y = z y_T\left(\frac{x - \mathrm{X1}}{\mathrm{X2}}\right)$$

   where y_T((x – X1) / X2) is the table look up at (x – X1) / X2, and z is the value of the material property on the corresponding MATi entry.

Figure 18-182: Example of Table Extrapolation and Discontinuity



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Remarks related to SOLs 402 and 414:

1. Discontinuities may be specified between any two points except the two starting points or two ending points. If a discontinuity is defined for x = X3 (with X3 = X4), the value of y for x = X3 is Y3. The value Y4 is used to compute y over the interval X4 < x ≤ X5.

Related Topics

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TABLEM4

Material Property Table, Form 4

Uses a power series to define temperature-dependent material properties. The value for the material property at a specific temperature is the product of the power series sum evaluated at that temperature, and the value for the material property as specified on the corresponding MATi entry.

Format:

1	2	3	4	5	6	7	8	9	10
TABLEM4	TID	X1	X2	X3	X4				
	A0	A1	A2	A3	A4	A5	-etc.-		

Example:

TABLEM4	28	0.0	1.0	0.0	100.				
	2.91	-0.0329	6.51-5	0.0	-3.4-7	ENDT			

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
X1, X2, X3, X4	Table parameters. (Real; X2 ≠ 0.0; X3 < X4)
Ai	Coefficients. (Real)
“ENDT”	Flag indicating the end of the table.

Remarks:

1. At least one continuation entry must be specified.
2. The end of the table is indicated by the existence of “ENDT” in the field following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag “ENDT”.
3. The value *y* that the software returns for the structural or thermal material property at temperature *x* is:

$$y = z \sum_{i=0}^{N} \mathrm{Ai}\left(\frac{x - \mathrm{X1}}{\mathrm{X2}}\right)^{i}$$

where *N* is the number of coefficients, and *z* is the value of the material property on the corresponding MATi entry.

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Whenever $x < X3$, use X3 for x; whenever $x > X4$, use X4 for x.

There are $(N + 1)$ entries in the table.

There are no error returns from this procedure.

Related Topics

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TABLEM5

Tabular Function for Progressive Ply Failure

For progressive ply failure with the UD damage model in SOLs 401 and 402, TABLEM5 is optionally used to define a nonlinear function relating the shear damage (d12) to the thermodynamic force (Y). TABLEM5 can only be referenced by the TID field on a MATDMG entry with PPFMOD=UD defined.

Format:

1	2	3	4	5	6	7	8	9	10
TABLEM5	TID			EXTRAP					
	X1	Y1	X2	Y2	X3	Y3	-etc.-	"ENDT"	

Example:

TABLEM5	1								
	0.0	1.0	1.0	2.0	2.0	3.0	ENDT		

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
EXTRAP	Extrapolation option. (Integer= 0 or 1; Default=0) • If EXTRAP = 0, extrapolate the two starting data points to obtain the table look up when x < xi, and extrapolate the two ending data points to obtain the table look up when x > xi. • If EXTRAP = 1, use the value of the table field at the starting data point for the table look up when x < xi, and use the value of the table field at the ending data point for the table look up when x > xi.
Xi, Yi	Tabular values. Xi are the thermodynamic force values. Yi are the the shear damage values. (Real)
“ENDT”	Flag indicating the end of the table.

Remarks:

1. The TABLEM5 entry can only be referenced by the TID field on the MATDMG entry. The TABLEM5 entry defines the shear damage nonlinear function *d12=f(sqrt(Y))*, where d12 is the yi data, and sqrt(Y) is the xi data. Y in the function is the thermodynamic force.

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2. The linear shear damage d12 is computed using Y012 and YC12, which you define on the MATDMG entry. Linear and nonlinear parameters can not be specified at the same time: An error message will be issued if the Y012, YC12, and TID fields on the MATDMG entry are either all defined or are all blank, or if the Y012 field or the YC12 field is defined and the other is not.
3. Xi must be in either ascending or descending order, but not both. No warning message is issued if tabular data is input incorrectly.
4. Discontinuities may be specified between any two points except the two starting points or the two end points. For example, in **Figure 18-183**, discontinuities are allowed only between points X2 through X7.

   The value of Y at X3 in **Figure 18-183** is:

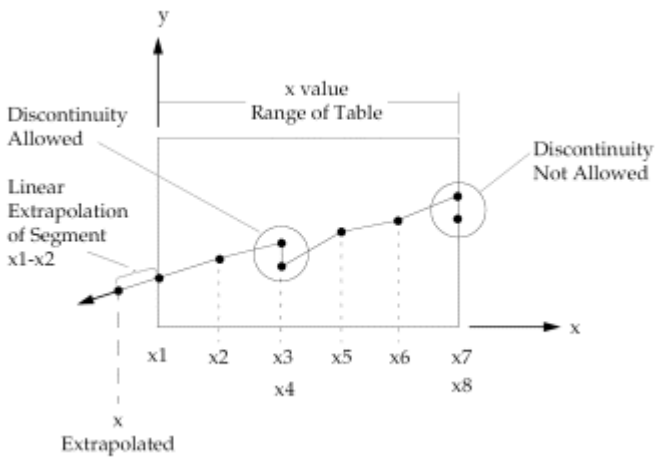
   $y = (y_3 + y_4) / 2$
5. At least one continuation entry must be specified.
6. Any xi, yi pair may be ignored by placing "SKIP" in either of the two fields.
7. The end of the table is indicated by the existence of "ENDT" in either of the two fields following the last entry. An error will occur if any continuations follow the entry containing the end-of-table flag "ENDT".
8. The tabular data is interpolated for values of *x* that lie within the range of the tabular data and, when EXTRAP=0, the tabular data is extrapolated from the two starting or end points for values of *x* that lie outside the range of the tabular data.

   The value *y* that the software returns for the shear damage at thermodynamic force x is:

   $$y = y_T(x)$$

   where $y_T(x)$ is the table look up at *x*. The table look-up is performed using linear interpolation within the table and linear extrapolation outside the table using the two starting or end points if EXTRAP = 0.

Figure 18-183: Example of Table Extrapolation and Discontinuity



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Remarks related to SOLs 402 and 414:

1. Discontinuities may be specified between any two points except the two starting points or two ending points. If a discontinuity is defined for x = X3 (with X3 = X4), the value of y for x = X3 is Y3. The value Y4 is used to compute y over the interval X4 < x ≤ X5.

Related Topics

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TABLES1

Material Property Table, Form 1

Defines a tabular function for stress-dependent material properties such as the stress-strain curve (MATS1 entry), creep parameters (CREEP entry) and hyperelastic material parameters (MATHP entry).

Format:

1	2	3	4	5	6	7	8	9	10
TABLES1	TID								
	x1	y1	x2	y2	x3	y3	-etc.-	“ENDT”	

Example:

TABLES1	32								
	0.0	0.0	.01	10000.	.02	15000.	ENDT		

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
xi, yi	Tabular values. (Real)
“ENDT”	Flag indicating the end of the table.

Remarks:

1. xi must be in either ascending or descending order, but not both.
2. At least one continuation entry must be present.
3. Any xi-yi pair may be ignored by placing “SKIP” in either of the two fields used for that entry.
4. The end of the table is indicated by the existence of “ENDT” in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag ENDT.
5. TABLES1 is used to input a curve in the form of

$$y = y_T(x)$$

where *x* is input to the table and *y* is returned. The table look-up is performed using linear interpolation within the table and linear extrapolation outside the table using the two starting or end points. See Figure 18-184. No warning messages are issued if table data is input incorrectly.

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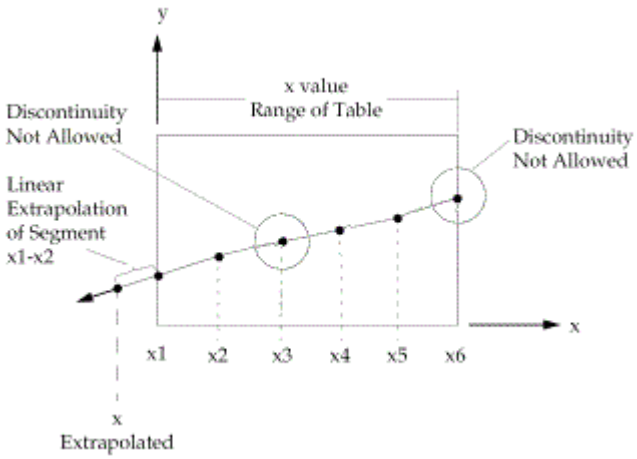
Figure 18-184: Example of Table Extrapolation



6. Discontinuities are not allowed.

Related Topics

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TABLEST

Table of tables

Uses a series of tables to represent a functional relationship that has two independent variables.

Format:

1	2	3	4	5	6	7	8	9	10
TABLEST	TID	EXTRAP							
	T1	TID1	T2	TID2	T3	-etc.-			

Example:

TABLEST	101	1							
	150.0	10	175.0	20	ENDT				

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
EXTRAP	Extrapolation option. See Remarks 7 and 3. (Integer; Default for SOL 106 = 0) SOL 106 supports option 0 or 1. See remarks related to SOLs 401, 402, and 414 for options and defaults. • If EXTRAP = 0, extrapolate for value when temperature or displacement or force is outside the range of Ti. • If EXTRAP = 1, use the value of the nearest table field when temperature or displacement or force is outside the range of Ti. • If EXTRAP = -1, for SOL 401 and 402 plastic materials defined with MATS1 or MPLAS only, no extrapolation. A fatal error occurs when the load temperature is outside the range of Ti.
Ti	Temperature or displacement or force that corresponds to the table referenced in field TIDi. (Real) See Remark 4.
TIDi	Table identification numbers of TABLES1 or TABLEDi bulk entries. (Integer > 0)

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Remarks:

1. TIDi must be unique with respect to all TABLES1 or TABLEDi and TABLEST table identification numbers.
2. Temperature or displacement or force values must be listed in ascending order.
3. The end of the table is indicated by the existence of ENDT in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag ENDT.
4. The following solutions can reference the TABLEST bulk entry:
   - For SOLs 106 and 401, MATS1 bulk entry to define temperature-dependent nonlinear elastic material.
   - For SOLs 401 and 402, MATS1 and MPLAS bulk entries to define temperature-dependent plasticity material.
   - For SOLs 401 and 402, MATSR bulk entry to define temperature-dependent strain rate material.
   - For SOL 414, PBEAR bulk entries to define bearing stiffness or viscous damping (TYPE = ”KD”,”KF”,BD”,”BF”) as a function of rotor speed and displacement or rotor speed and force. For this usage, the number of TIDi entries is limited to 100.
   - For SOL 414, PBUSH2 bulk entries to define:
     - Radial elastic forces in translation as a function of radial displacement and rotational speed (TYPE = "RADIAL" and PARAM = "FETR") or radial elastic moments in rotation as a function radial displacement and rotational speed (TYPE = "RADIAL" and PARAM = "FERR").
     - Radial viscous damping in translation or rotation as a function of relative speed and rotational speed (TYPE = "RADIAL" and PARAM = "FATR" or "FARR").
5. A function of two variables such as $z = f(x,y)$ can be approximated by a series of N single-variable functions:

   $z_1 = f_1(x) = f(x,Y_1)$
   $z_2 = f_2(x) = f(x,Y_2)$
   .....
   $z_i = f_i(x) = f(x,Y_i)$
   .....
   $z_N = f_N(x) = f(x,Y_N)$

   where Y_i are constants.

   The single-variable functions, $f_i(x)$, are approximated by the tabular data listed on the TABLES1 and TABLEDi bulk entries that are referenced in the TIDi fields of the TABLEST bulk entry. The constants, Y_i, are given by the values listed in the Ti fields of the TABLEST bulk entry.
6. The table lookup value for $f(X_0,Y_0)$ when Y_0 is within the range of data is obtained as follows:
   a. The bounding table fields are identified. The bounding table fields are the table fields, $f_j(x)$ and $f_k(x)$, whose fixed variables, Y_j and Y_k, are in closest proximity to Y_0 such that $Y_j < Y_0 < Y_k$.
   b. The tabular data for each bounding table field is interpolated to obtain $f_j(X_0)$ and $f_k(X_0)$.
   c. Points $(f_j(X_0),Y_j)$ and $(f_k(X_0),Y_k)$ are interpolated to find the value corresponding to Y_0.

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7. The table lookup value for $f(X_0,Y_0)$ when Y_0 is outside the range of data is obtained as follows:

   - If EXTRAP = 0

     a. The two table fields in closest proximity are identified. These table fields are the table fields, $f_j(x)$ and $f_k(x)$, whose fixed variables, Y_j and Y_k, are in closest proximity to Y_0 such that $Y_0 > Y_j > Y_k$ or $Y_0 < Y_j < Y_k$.

     b. The tabular data for these table fields is interpolated to obtain $f_j(X_0)$ and $f_k(X_0)$.

     c. Points $(f_j(X_0),Y_j)$ and $(f_k(X_0),Y_k)$ are extrapolated to find the value corresponding to Y_0.

   - If EXTRAP = 1

     a. The table field in closest proximity is identified. This table field is the table field, $f_j(x)$, whose fixed variable, Y_j is in closest proximity to Y_0.

     b. The tabular data for the table field is interpolated to obtain $f_j(X_0)$.

     c. $f_j(X_0)$ is used as the lookup value.

Remarks related to SOLs 401, 402, and 414:

1. For SOLs 401 and 402, TIDi must only reference TABLES1 tables.

   For SOL 414, TIDi can reference TABLES1 or TABLED1 tables.

2. EXTRAP field defaults and options:

   For SOL 401 temperature-dependent nonlinear elastic material (MATS1), default is EXTRAP=1, and EXTRAP=0 is an available option.

   For SOLs 401 and 402 temperature-dependent plasticity material (MATS1 and MPLAS), default is EXTRAP=1, and EXTRAP=-1 is an available option.

   For SOLs 401 and 402 temperature-dependent strain rate material (MATSR), EXTRAP=1 is always used.

   For SOL 414, EXTRAP=1 is always used.

3. For SOL 414, when the TABLEST bulk entry is referenced by a PBEAR bulk entry, the software behaves as if EXTRAP = 1, regardless of whether the EXTRAP field is left blank or EXTRAP = 0 is specified.

4. For SOL 414, when TABLEST is used with PBUSH2, Ti is the rotation speed.

Related Topics

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TABRND1

Power Spectral Density Table

Defines power spectral density as a tabular function of frequency for use in random analysis. Referenced by the RANDPS entry.

Format:

1	2	3	4	5	6	7	8	9	10
TABRND1	TID	XAXIS	YAXIS						
	f1	g1	f2	g2	f3	g3	-etc.-		

Example:

TABRND1	3								
	2.5	.01057	2.6	.01362	ENDT				

Fields:

Field	Contents
TID	Table identification number. (Integer > 0)
XAXIS	Specifies a linear or logarithmic interpolation for the x-axis. (Character: “LINEAR” or “LOG”; Default=“LINEAR”)
YAXIS	Specifies a linear or logarithmic interpolation for the y-axis. (Character: “LINEAR” or “LOG”; Default=“LINEAR”)
fi	Frequency value in cycles per unit time. (Real ≥ 0.0)
gi	Power spectral density. (Real)

Remarks:

1. The fi must be in either ascending or descending order, but not both.
2. Discontinuities may be specified between any two points except the two starting points or two end points. For example, in Figure 18-185 discontinuities are allowed only between points f2 through f7. Also, if g is evaluated at a discontinuity, then the average value of g is used. In Figure 18-185, the value of g at f=f3 is $g=(g3+g4)/2$. If the y-axis is a LOG axis then the jump at the discontinuity is evaluated as $y = \sqrt{y3y4}$.
3. At least two entries must be present.
4. Any fi-gi pair may be ignored by placing “SKIP” in either of the two fields used for that entry.

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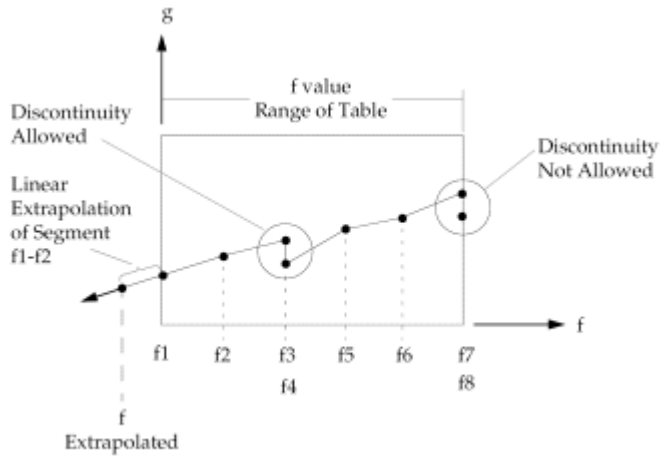
5. The end of the table is indicated by the existence of “ENDT” in either of the two fields following the last entry. An error is detected if any continuations follow the entry containing the end-of-table flag “ENDT”.

6. TABRND1 uses the algorithm

$$g = g_T(f)$$

where *f* is input to the table and *g* is returned. The table look-up is performed using linear interpolation within the table and linear extrapolation outside the table using the two starting or end points. See Figure 18-185. No warning messages are issued if table data is input incorrectly.

Figure 18-185: Example of Table Extrapolation and Discontinuity



7. For auto spectral density, the value of *g* returned must be greater than or equal to zero, as shown in Remark 6 .

8. Tabular values on an axis if XAXIS or YAXIS=LOG must be positive. A fatal message will be issued if an axis has a tabular value ≤ 0.

9. The algorithms used are:

XAXIS	YAXIS	f(x)
LINEAR	LINEAR	$\frac{f_{i+1}-f}{f_{i+1}-f_i}g_i+\frac{f-f_i}{f_{i+1}-f_i}g_{i+1}$
LOG	LINEAR	$\frac{\ln(f_{i+1}/f)}{\ln(f_{i+1}/f_i)}g_i+\frac{\ln(f/f_i)}{\ln(f_{i+1}/f_i)}g_{i+1}$
LINEAR	LOG	$\exp\left[\frac{f_{i+1}-f}{f_{i+1}-f_i}\ln g_i+\frac{f-f_i}{f_{i+1}-f_i}\ln g_{i+1}\right]$
LOG	LOG	$\exp\left[\frac{\ln(f_{i+1}/f)}{\ln(f_{i+1}/f_i)}\ln g_i+\frac{\ln(f/f_i)}{\ln(f_{i+1}/f_i)}\ln g_{i+1}\right]$

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where $f_i < f < f_{i+1}$

Related Topics

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TABRNDG

Gust Power Spectral Density

Defines the power spectral density (PSD) of a gust for aeroelastic response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
TABRNDG	TID	TYPE	L/U	WG					

Example:

TABRNDG	1020	1	1.3	.1					

Fields:

Field	Contents
TID	Table identification number. (Integer >0)
TYPE	PSD type: von Karman (TYPE=1) or Dryden model (TYPE=2). (Integer=1 or 2)
L/U	Scale of turbulence divided by velocity (units of time). See L/U in Remark 2 . (Real)
WG	Root-mean-square gust velocity. (Real)

Remarks:

1. This entry must be referenced by a RANDPS entry.
2. The power spectral density is given by

$$S_q(\omega) = 2(\mathrm{WG})^2(\mathrm{L}/\mathrm{U})\frac{1 + 2(p+1)k^2(\mathrm{L}/\mathrm{U})^2\omega^2}{[1 + k^2(\mathrm{L}/\mathrm{U})^2\omega^2]^{p+3/2}}$$

   where

Type	p	k
1=von Karman	1/3	1.339
2=Dryden	1/2	1.0

   and $\omega=2\pi f$. The units of S_q (ω) are velocity squared per frequency (f).
3. Other power spectral density functions may be defined using the TABRND1 entry.

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Related Topics

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TEMP

Grid Point Temperature Field

Defines temperature at grid points for determination of thermal loading, temperature-dependent material properties, or stress recovery.

Format:

1	2	3	4	5	6	7	8	9	10
TEMP	SID	G1	T1	G2	T2	G3	T3		

Example:

TEMP	3	94	316.2	49	219.8				

Fields:

Field	Contents
SID	Temperature set identification number. (Integer > 0)
Gi	Grid point identification number. (Integer > 0)
Ti	Temperature. (Real)

Remarks:

1. In the static solution sequences, the temperature set ID(SID) is selected by the Case Control command TEMP. In the dynamic solution sequences, SID must be referenced in the TID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
2. Set ID must be unique with respect to all other LOAD type entries if TEMP(LOAD) is specified in the Case Control Section.
3. From one to three grid point temperatures may be defined on a single entry.
4. If thermal effects are requested, all elements must have a temperature field defined either directly on a TEMPP1, TEMPP3, or TEMPRB entry or indirectly as the average of the connected grid point temperatures defined on the TEMP or TEMPD entries. Directly defined element temperatures always take precedence over the average of grid point temperatures.
5. If the element material is temperature dependent, its properties are evaluated at the average temperature.
6. Average element temperatures are obtained as a simple average of the connecting grid point temperatures when no element temperature data are defined. Gauss point temperatures are averaged for solid elements instead of grid point temperature.

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7. For steady state heat transfer analysis, this entry together with the TEMPD entry supplies the initialization temperatures for nonlinear analysis. The Case Control command TEMP(INIT)=SID requests selection of this entry. The temperature values specified here must be coincident with any temperature boundary conditions that are specified.
8. For transient heat transfer analysis, this entry together with the TEMPD entry supplies the initial condition temperatures. The Case Control command IC=SID requests selections of this entry. The temperature values specified here must be coincident with any temperature boundary condition specified.
9. In linear and nonlinear buckling analysis, the follower force effects due to loads from this entry are not included in the differential stiffness. See "Buckling Analysis in SubDMAP MODERS" in the *Simcenter Nastran User's Guide* and the *Simcenter Nastran Basic Nonlinear Analysis User's Guide*.

Remarks related to SOLs 402 and 414:

1. To apply constant temperature loads, SID is selected by Case Control command TEMP=SID for both static and transient analyses. TEMP(INIT)=SID is used to specify initial temperatures.
2. To apply time-dependent temperature loads, SID is referenced by the field EXCITEID=SID in the DTEMP entry. Time-dependent loads are selected by Case Control command DTEMP.
3. When you add TEMP(LOAD), you must also add TEMP(INIT).

Remarks related to SOL 414:

1. You can add TEMP(LOAD) only when the rotor is in the fixed reference frame. TEMP(LOAD) is not supported in rotors that are in the rotating reference frame.
2. For SOL 414,101 Transient Response or SOL 414,103 Eigenvalues and Superelement Reduction solutions, TEMP(LOAD) must be global, or it must be the same in all subcases.
3. For SOL 414,110 Complex Modal Analysis or SOL 414,111 Harmonic Response solutions with a static preload subcase:
   - When rotors are defined in fixed reference frames, TEMP(LOAD) must be global or be the same in all subcases
   - When rotors are defined in rotating reference frames, TEMP(LOAD) cannot be added.
4. For SOL 414,129 Transient Response solutions, TEMP(LOAD) has no restrictions.

Related Topics

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TEMPBC

Grid Point Temperatures

Defines the temperature boundary conditions for heat transfer analysis. Applies to steady-state and transient conditions.

Format:

1	2	3	4	5	6	7	8	9	10
TEMPBC	SID	TYPE	TEMP1	GID1	TEMP2	GID2	TEMP3	GID3	

Example:

TEMPBC	10	STAT	100.0	1	100.0	2	100.0	3	

Alternate Format and Example:

TEMPBC	SID	TYPE	TEMP1	GID1	"THRU"	GID2	"BY"	INC	
TEMPBC	20	STAT	100.0	4	THRU	50	BY	2	

Fields:

Field	Contents
SID	Temperature set identification number. (Integer > 0)
TYPE	Type of temperature boundary condition. See Remarks. (Character; Default="STAT"): STAT – Constant temperature boundary condition TRAN – Time-varying temperature boundary condition
TEMPi	Temperature (Real)
GIDi	Grid point identification number. (Integer > 0 or "THRU" or "BY")
INC	Grid point number increment. (Integer)

Remarks:

1. For a constant Boundary Condition (TYPE="STAT"), the temperature boundary load set (SID) is selected in the Case Control Section (SPC=SID). TYPE="STAT" may be used in both steady-state (SOL 153) and transient analysis (SOL 159).

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2. For transient analysis (SOL 159), a constant boundary condition should be specified using the SPC Bulk Data entry.
3. For a time-varying boundary condition (TYPE="TRAN"), SID is referenced by a TLOADi Bulk Data entry through the DAREA specification. TYPE="TRAN" is permitted only in transient analysis (SOL 159). A function of time $F(t - \tau)$ defined on the TLOADi entry multiplies the general load. τ provides any required time delay. The load set identifier on the TLOADi entry must be selected in Case Control (DLOAD=SID) for use in transient analysis.
4. In the alternate format, TEMP1 is the nodal temperature for the grid points GID1,GID1+INC,...,GID2. If "BY" and INC are not specified, then the grid point number increment is unity.
5. If TYPE="STAT", then no SPCi Bulk Data entries may be specified.
6. If TYPE="TRAN", then no CELAS2 or DAREA Bulk Data entries may be specified. Also, "U" must be specified in the CONV field on the TSTEPNL entry to obtain accurate results.
7. All TEMPBC entries in the Bulk Data Section must indicate either TYPE="STAT" or TYPE="TRAN" but not both.

Related Topics

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TEMPD

Grid Point Temperature Field Default

Defines a temperature value for all grid points of the structural model that have not been given a temperature on a TEMP entry.

Format:

1	2	3	4	5	6	7	8	9	10
TEMPD	SID1	T1	SID2	T2	SID3	T3	SID4	T4	

Example:

TEMPD	1	216.3							

Fields:

Field	Contents
SIDi	Temperature set identification number. (Integer > 0)
Ti	Default temperature value. (Real)

Remarks:

1. For structural analysis in the static solution sequences, the temperature set ID (SID) is selected by the Case Control command TEMP. In the dynamic solution sequences, SID must be referenced in the TID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
2. SIDi must be unique with respect to all other LOAD type entries if TEMP(LOAD) is specified in the Case Control Section.
3. From one to four default temperatures may be defined on a single entry.
4. If thermal effects are requested, all elements must have a temperature field defined either directly on a TEMPP1, TEMPP3, or TEMPRB entry or indirectly as the average of the connected grid point temperatures defined on the TEMP or TEMPD entries. Directly defined element temperatures always take precedence over the average of grid point temperatures.
5. If the element material is temperature dependent, its properties are evaluated at the average temperature.
6. Average element temperatures are obtained as a simple average of the connecting grid point temperatures when no element temperature data is defined.
7. For steady-state heat transfer analysis, this entry together with the TEMP entry supplies the initialization temperatures for nonlinear analysis. The Case Control command TEMP(INIT)=SID

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requests selection of this entry. The temperature values specified here must be coincident with any temperatures boundary conditions that are specified.

8. For transient heat transfer analysis, this entry together with the TEMP entry supplies the initial condition temperatures. The Case Control command IC=SID request selection of this entry. The temperature values specified here must be coincident with any temperature boundary conditions that are specified.
9. In linear and nonlinear buckling analysis, the follower force effects due to loads from this entry are not included in the differential stiffness. See “Buckling Analysis in SubDMAP MODERS” in the *Simcenter Nastran User’s Guide* and the *Simcenter Nastran Basic Nonlinear Analysis User’s Guide*.
10. For partitioned Bulk Data superelements and auxiliary models, TEMPD must be specified in all partitioned Bulk Data Sections.

Remarks related to SOLs 402 and 414:

1. To apply constant temperature loads, SID is selected by Case Control command TEMP=SID for both static and transient analyses. TEMP(INIT)=SID is used to specify initial temperatures.
2. To apply time-dependent temperature loads, SID is referenced by the field EXCITEID=SID in the DTEMP entry. Time-dependent loads are selected by Case Control command DTEMP.

Related Topics

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TEMPEX

Time-independent temperature set for SOLs 401 and 402

Defines a time-independent temperature set for SOLs 401 and 402 that is obtained from an external binary (.bun) or ASCII (.unv) universal file.

Format:

1	2	3	4	5	6	7	8	9	10
TEMPEX	SID	UNID	FILETYPE						

Example:

TEMPEX	200	21							

Fields:

Field	Contents
SID	Unique identification number specified in a TEMP(LOAD) or TEMP(INIT) case control command. (Integer>0)
UNID	FORTRAN unit number listed on the ASSIGN statement for a .unv file, a .bun file, or a .scd5 file. See Remark 3. (Integer>0)
FILETYPE	Specifies the type of external temperature data file. (Character; Default=BUN) FILETYPE="BUN" The external file is either a .bun or a .unv file. This option is supported by both SOLs 401 and 402. FILETYPE="SCD5" The external file is an .scd5 file. This option is only supported by SOL 401. It is not supported by SOL 402.

Remarks:

1. SID must be unique to all other TEMPEX entries.
2. The SID of the TEMPEX bulk entry can only be selected with a TEMP(LOAD) or TEMP(INIT) case control command. The DTEMP case control command cannot be used to select the TEMPEX bulk entry.
3. For example, set the UNID field to 22 for the following ASSIGN statements:

```
ASSIGN BUN='temperature0.bun' UNIT=22 FORM=UNFORM STATUS=OLD
ASSIGN UNV='temperature0.unv' UNIT=22 FORM=FORM STATUS=OLD
ASSIGN SC_H5='temperature0.scd5' UNIT=22 FORM=UNFORM STATUS=OLD
```


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Remark for SOL 402:

1. SOL 402 only supports FILETYPE="BUN".

Related Topics

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TEMPP1

Plate Element Temperature Field, Form 1

Defines a temperature field for plate, membrane, and combination elements (by an average temperature and a thermal gradient through the thickness) for determination of thermal loading, temperature-dependent material properties, or stress recovery.

Format:

1	2	3	4	5	6	7	8	9	10
TEMPP1	SID	EID1	TBAR	TPRIME	T1	T2			
	EID2	EID3	EID4	EID5	EID6	EID7	-etc.-		

Example:

TEMPP1	2	24	62.0	10.0	57.0	67.0			
	26	21	19	30					

Alternate Format and Example of Continuation Entry:

	EID2	"THRU"	EIDi	EIDj	"THRU"	EIDk			
	1	THRU	10	30	THRU	61			

Fields:

Field	Contents
SID	Temperature set identification number. (Integer > 0)
EIDi, EIDj, EIDk	Unique element identification number(s). (Integer > 0 or the continuation entries may have "THRU" in fields 3 and/or 6, in which case EID2 < EIDi and EIDj < EIDk.)
TBAR	Temperature at the element's reference plane as defined by ZOFFS on the connection entry. (Real)
TPRIME	Effective linear thermal gradient. Not used for membranes. (Real)
T1, T2	Temperatures for an additional membrane stress calculation at points defined on the element property entry (*Z*1 and *Z*2 field on PSHELL entry). See Remark 9. T1 may be specified on the lower surface and T2 on the upper surface for the CQUAD4, CQUAD8, CTRIA3, CTRIA6, CQUADR, and CTRIAR elements. These data are not used for membrane elements. See Remark 11. If T1 and T2 are blank, they are ignored, and the additional stress calculation does not occur. (Real)

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Remarks:

1. In the static solution sequences, the temperature set ID (SID) is selected by the Case Control command TEMP. In the dynamic solution sequences, SID must be referenced in the TID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
2. Set ID must be unique with respect to all other LOAD type entries if TEMP(LOAD) is specified in the Case Control Section.
3. If continuation entries are present, EID1 and elements specified on the continuation entry are used. Elements must not be specified more than once.
4. If thermal effects are requested, all elements must have a temperature field defined either directly on a TEMPP1 or TEMPRB entry or indirectly as the average of the connected grid point temperatures defined on the TEMP or TEMPD entries. Directly defined element temperatures always take precedence over the average of grid point temperatures.
5. For temperature field other than a constant gradient, the "effective gradient" for a homogeneous plate is

$$\text{TPRIME} = \frac{1}{I}\int_z T(z) z\, dz$$

   where *I* is the bending inertia and *z* is the distance from the neutral surface in the positive normal direction.
6. The "average" temperature for a homogeneous plate is

$$\text{TBAR} = \frac{1}{Volume}\int_{Volume} T dVolume$$

7. If the element material is temperature dependent, its properties are evaluated at the average temperature TBAR.
8. Large "THRU" ranges will lead to System Fatal Message 3008 ("Insufficient Core") and should be avoided, particularly for open sets.
9. If the element material is nonlinear then T1 and T2 should be left blank (see the MATS1 entry).
10. In linear and nonlinear buckling analysis, the follower force effects due to loads from this entry are not included in the differential stiffness. See "Buckling Analysis in SubDMAP MODERS" in the *Simcenter Nastran User's Guide* and the *Simcenter Nastran Basic Nonlinear Analysis User's Guide*.
11. If T1 or T2 are defined, an additional membrane stress is calculated by

    $\{\Delta\sigma\} = -[G_e]\{\alpha_e\}(T - T_{fiber})$

    where T is the explicitly entered temperatures T1,T2 and T_{fiber} is the outer fiber temperatures Tfiber1,Tfiber2 computed by Tfiber=TBAR + z·TPRIME (z is the distance from the center fiber). $[G_e]$ and $\{\alpha_e\}$ are evaluated at TBAR. The thermal expansion due to T1,T2 is assumed to be completely restrained by elastic stiffness.

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Related Topics

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TEMPP3

Plate Element Temperature Field, Form 3

TEMPP3 is no longer available. Use TEMPP1.

Related Topics

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TEMPRB

One-Dimensional Element Temperature Field

Defines a temperature field for the CBAR, CBEAM, CBEND, CROD, CTUBE, and CONROD elements for determination of thermal loading, temperature-dependent material properties, or stress recovery.

Format:

1	2	3	4	5	6	7	8	9	10
TEMPRB	SID	EID1	TA	TB	TP1A	TP1B	TP2A	TP2B	
	TCA	TDA	TEA	TFA	TCB	TDB	TEB	TFB	
	EID2	EID3	EID4	EID5	EID6	EID7	-etc.-		

Example:

TEMPRB	200	1	68.0	23.0	0.0	28.0		2.5	
	68.0	91.0	45.0		48.0	80.0	20.0		
	9	10							

Alternate Format and Example of Continuation Entry:

	EID2	“THRU”	EIDi	EIDj	“THRU”	EIDk			
	2	THRU	4	10	THRU	14			

Fields:

Field	Contents
SID	Temperature set identification number. (Integer > 0)
EIDi, EIDj, EIDk	Unique element identification number(s). (Integer > 0 or the second continuation entry may have “THRU” in fields 3 and/or 6 in which case EID2 < EIDi and EIDj < EIDk.)
TA, TB	Temperature at end A and end B on the neutral axis. (Real)
TPij	Effective linear gradient in direction i on end j; used with CBAR, CBEAM, and CBEND only. In Remarks related to SOLs 402 and 414, see Remark 2. (Real)
Tij	Temperature at point i as defined on the PBAR, PBEAM, and PBEND entries at end j. This data is used for stress recovery only with CBAR, CBEAM, and CBEND exclusively. (Real)

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Remarks:

1. In the static solution sequences, the temperature set ID (SID) is selected by the Case Control command TEMP. In the dynamic solution sequences, SID must be referenced in the TID field of an LSEQ entry, which in turn must be selected by the Case Control command LOADSET.
2. SID must be unique with respect to all other LOAD type entries if TEMP(LOAD) is specified in the Case Control Section.
3. If at least one nonzero or nonblank Tij is present, the point temperatures given are used for stress recovery. If no Tij values are given, linear temperature gradients are assumed for stress recovery. The Tij values are not used in the calculation of differential stiffness.
4. If the second (and succeeding) continuation is present, EID1 and elements specified on the second (and succeeding) continuations are used. Elements must not be specified more than once.
5. If thermal effects are requested, all elements must have a temperature field defined either directly on a TEMPP1 or TEMPRB entry or indirectly as the average of the connected grid point temperatures defined on the TEMP or TEMPD entries. Directly defined element temperatures always take precedence over the average of grid point temperatures.
6. The effective thermal gradients in the element coordinate system for the CBAR element are defined by the following integrals over the cross section. For end “A” (end “B” is similar).

$$\mathrm{TA} = \frac{1}{A}\int_A TA(y,z)\,dA$$

$$\mathrm{TP1A} = \frac{I_2}{\Delta}\int_A (y-y_n)TA(y,z)\,dA - \frac{I_{12}}{\Delta}\int_A (z-z_n)TA(y,z)\,dA$$

$$\mathrm{TP2A} = \frac{I_1}{\Delta}\int_A (z-z_n)TA(y,z)\,dA - \frac{I_{12}}{\Delta}\int_A (y-y_n)TA(y,z)\,dA$$

$$\Delta = I_1 I_2 - I_{12}^2$$

if I_{12}=0

$$\mathrm{TP1A} = \frac{1}{I_1}\int_A (y-y_n)TA(y,z)\,dA$$

$$\mathrm{TP2A} = \frac{1}{I_2}\int_A (z-z_n)TA(y,z)\,dA$$

where *TA*(*y*, *z* is the temperature at point y,z (in the element coordinate system) at end “A” of the bar. See the CBAR entry description for the element coordinate system: I_1, I_2, and $I_{12}$are the moments of inertia about the z and y axes, respectively. The temperatures are assumed to vary linearly along the length (x-axis). Note that if the temperature varies linearly over the cross section, then TP1A, TP1B, TP2A and TP2B are the actual gradients.

7. If the element material is temperature-dependent, the material properties are evaluated at the average temperature (*TA*+*TB*) / 2.

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8. In linear and nonlinear buckling analysis, the follower force effects due to loads from this entry are not included in the differential stiffness. See “Buckling Analysis in SubDMAP MODERS” in the *Simcenter Nastran User’s Guide* and the *Simcenter Nastran Basic Nonlinear Analysis User’s Guide*.
9. If any T_y is specified the stresses computed by the effective gradient are corrected by Δσ such that:

$$\sigma = \sigma\Big|_{T_A + y^{TPIA} + z^{TP2A}} + \Delta\sigma$$

where Δσ is in the form

$$\Delta\sigma = -\alpha E[T_{CA} - T_o - C_1 \cdot TPIA - C_2 \cdot TP2A] \text{ etc}$$

for CBAR and CBEAM

$$\Delta\sigma = -\alpha E[T_{CA} - T_o - (C_1 + \Delta N) \cdot TPIA - C_2 \cdot TP2A] \text{ etc}$$

for CBEND.

Remarks related to SOLs 402 and 414:

1. SOLs 402 and 414 do not support CBEND and CTUBE elements.
2. Only TA and TB are taken into account. The average is computed as (TA+TB)/2 and used for the simulation. Gradients are not taken into account.

Related Topics

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TF

Dynamic Transfer Function

Defines a dynamic transfer function of the form:

Equation 18-8:

$$(B0 + B1 \cdot p + B2 \cdot p^2)u_d + \sum_i (A0(i) + A1(i)p + A2(i)p^2)u_i = 0$$

Can also be used as a means of direct matrix input. See Remark **4** .

Format:

1	2	3	4	5	6	7	8	9	10
TF	SID	GD	CD	B0	B1	B2			
	G(1)	C(1)	A0(1)	A1(1)	A2(1)	-etc.-			

Example:

TF	1	2	3	4.0	5.0	6.0			
	3	4	5.0	6.0	7.0				

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
GD, G(i)	Grid, scalar, or extra point identification numbers. (Integer > 0)
CD, C(i)	Component numbers. (Integer zero or blank for scalar or extra points, any one of the Integers 1 through 6 for a grid point.)
B0, B1, B2A0(i), A1(i), A2(i)	Transfer function coefficients. (Real)

Remarks:

1. Transfer function sets must be selected with the Case Control command TFL=SID.
2. Continuation entries are optional.
3. The matrix elements defined by this entry are added to the dynamic matrices for the problem.

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4. The constraint relation given in **Equation 18-8** will hold only if no structural elements or other matrix elements are connected to the dependent coordinate u_d. In fact, the terms on the left side of **Equation 18-8** are simply added to the terms from all other sources in the row for u_d.
5. See the *Simcenter Nastran Advanced Dynamic Analysis User's Guide* for a discussion of transfer functions.
6. For each SID, only one logical entry is allowed for each GD, CD combination.
7. For heat transfer analysis, the initial conditions must satisfy **Equation 18-8**.

Related Topics

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TIC

Transient Initial Condition

Defines values for the initial conditions of variables used in structural transient analysis. Both displacement and velocity values may be specified at independent degrees-of-freedom. This entry may not be used for heat transfer analysis.

Format:

1	2	3	4	5	6	7	8	9	10
TIC	SID	G	C	U0	V0				

Example:

TIC	1	3	2	5.0	-6.0				

Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
G	Grid, scalar, or modal coordinate identification number (Integer>0). See Remark 4.
C	Component numbers. (Any one of the integers 1 through 6 for grid points, integer 0 or blank for scalar, and integer -1 for modal coordinates.)
U0	Initial displacement. (Real)
V0	Initial velocity. (Real) See Remark 4.

Remarks:

1. Transient initial condition sets must be selected with the case control command. Note the use of IC in the case control command versus TIC on the bulk entry. For heat transfer, the IC case control command selects TEMP or TEMPD bulk entries for initial conditions and not the TIC entry.
2. If no TIC set is selected in the case control section, all initial conditions are assumed to be zero.
3. Initial conditions for coordinates not specified on TIC bulk entries will be assumed zero.
4. In direct transient analysis (SOL 109 and 129) where the TIC bulk entry is selected by an IC or IC(PHYSICAL) case control command, G may reference only grid or scalar points. In modal transient analysis (SOL 112) where the TIC bulk entry is selected by an IC or IC(PHYSICAL) case control command, G may reference only grid or scalar points. In modal transient analysis (SOL 112) where the TIC bulk entry is selected by an IC(MODAL) case control command, G may reference only modal coordinates.

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5. The initial conditions for the independent degrees-of-freedom specified by this bulk entry are distinct from, and may be used in conjunction with, the initial conditions for the enforced degrees-of-freedom specified by TLOAD1 and/or TLOAD2 bulk entries.
6. Input at extra points will be ignored.

Related Topics

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TLOAD1

Transient Response Dynamic Excitation, Form 1

Defines a time-dependent dynamic load or enforced motion of the form:

$$\{P(t)\} = \{A \cdot F(t-\tau)\}$$

for use in transient response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
TLOAD1	SID	EXCITEID	DELAY	TYPE	TID	US0VS0	VS0AS0		

Examples:

TLOAD1	5	7	101	LOAD	13				

TLOAD1	5	7	0.005		13				

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
EXCITEID	Identification number of a load set (for example, the DAREA entry), SPCD entry, SELOAD entry, or thermal load set (for heat transfer analysis) that lists each degree-of-freedom to apply the excitation and the corresponding scale factor, *A*, for the excitation. See Remark 2 and Remark 3. (Integer > 0)
DELAY	Time delay, τ. (Real or Integer ≥ 0 or blank; for default behavior, see Remark 10) If real entry, value of τ for all degrees-of-freedom in EXCITEID entry. If integer entry, identification number of DELAY entries that contain values of τ for individual degrees-of-freedom in EXCITEID entry. See Remark 11.
TYPE	Defines the type of the dynamic excitation. See Remark 2 and Remark 3. (Integer, character or blank; Default = 0)
TID	Identification number of TABLEDi entry that defines *F(t)* for all degrees-of-freedom in EXCITEID entry. (Integer > 0)
US0VS0, VS0AS0	Specifies initial conditions when the TLOAD1 bulk entry is used to specify enforced motion. See Remark 12. (Real; for default behavior see Remark 13)

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Field	Contents
	• When the enforced motion is acceleration, use the US0VS0 field to specify the initial displacement and use the VS0AS0 field to specify the initial velocity. • When the enforced motion is velocity, use the US0VS0 field to specify the initial displacement and use the VS0AS0 field to specify the initial acceleration. • When the enforced motion is displacement, use the US0VS0 field to specify the initial velocity and use the VS0AS0 field to specify the initial acceleration.

Remarks:

1. Dynamic excitation sets must be selected with DLOAD = SID in the case control section.
2. The type of the dynamic excitation is specified by TYPE (field 5) according to the following table:

TYPE	Type of dynamic excitation
0, L, LO, LOA or LOAD	Applied load (force or moment) (Default)
1, D, DI, DIS, or DISP	Enforced displacement using large mass or SPC/SPCD/DRIVER data
2, V, VE, VEL or VELO	Enforced velocity using large mass or SPC/SPCD data
3, A, AC, ACC or ACCE	Enforced acceleration using large mass or SPC/SPCD data

3. TYPE determines the manner in which EXCITEID is used by the program.
   - TYPE = 0 : Applied load excitation.
   - TYPE = 1, 2, or 3 : Enforced motion excitation.
     - If EXCITEID references SPC/SPCD entries, the software will use the SPCD method of enforced motion. You directly specify displacements, velocities, or accelerations with SPC/SPCD entries.
     - If EXCITEID references a load entry (DAREA, FORCEi, MOMENTi, PLOAD, PLOAD1, RFORCEi, and SELOAD entries), the software will use the large mass method of enforced motion. You create a large mass and load at the grid and degree-of-freedom where the enforced motion is desired. The software then computes the enforced motion the same for TYPE = 1, 2, and 3.

       Type = 1, Enforced displacement = Dynamic load input / Large mass
       Type = 2, Enforced velocity = Dynamic load input / Large mass
       Type = 3, Enforced acceleration = Dynamic load input / Large mass

   See the Enforced Motion chapter in the *Simcenter Nastran Basic Dynamic Analysis User's Guide* for details on both the SPCD and large mass methods of enforced motion.
4. The legacy method of selecting a DAREA, FORCEi, MOMENTi, RFORCEi, PLOAD, and PLOAD1 entry for a dynamic loading required the LOADSET case control command which selected the LSEQ bulk entry, which in turn selected the load entry. Since the FORCEi, MOMENTi, PLOAD, PLOAD1, and SPCD entries are now selected directly with the EXCITEID on the dynamic load entries DLOADi, TLOADi, and RLOADi, the LOADSET case control and LSEQ bulk entry are no longer required. The legacy method is still supported, and is described on the LSEQ entry remarks.

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5. EXCITEID may reference sets containing QHBDY, QBDYi, QVECT, and QVOL entries when using the heat transfer option.
6. TLOAD1 loads may be combined with TLOAD2 loads using a DLOAD bulk entry.
7. SID must be unique for all TLOAD1, TLOAD2, RLOAD1, RLOAD2, RLOADEX, ACSRCE, and SELOAD entries.
8. If the heat transfer option is used, the referenced QVECT entry may also contain references to functions of time, and therefore *A* may be a function of time.
9. If TLOADi entries are selected in SOL 111 or SOL 146, a Fourier analysis is used to transform the time-dependent loads on the TLOADi entries to the frequency domain and then combine them with loads from RLOADi entries. Then the analysis is performed as a frequency response analysis but the solution and the output are converted to and printed in the time domain. See the “Fourier Transform” section of the *Simcenter Nastran Advanced Dynamic Analysis User’s Guide*.
10. If the DELAY field is blank or zero (either integer zero or real zero), the corresponding value for τ used by the software is real zero.
11. For degrees-of-freedom in the EXCITEID entry that are not specified on DELAY entries, the software uses real zero as the value for τ.
12. Specifying initial conditions with the US0VS0 and VS0AS0 fields is valid for SOL 109, SOL 112, and SOL 129 only.
13. If both the US0VS0 and VS0AS0 fields are left blank, the software initializes the numerical integration algorithm using the procedures described in the "Initial Conditions in Direct Transient Response Analysis" and "Initial Conditions in Modal Transient Response Analysis" sections of the *Simcenter Nastran Basic Dynamic Analysis User's Guide*.

    If only one of the US0VS0 and VS0AS0 fields is left blank, the software issues an error.
14. In SOLs 401 and 402, when RFORCE or RFORCE1 entries are referenced by the EXCITEID field on a TLOAD1 entry, the data on the associated TABLEDi, along with the scale factors S and Si on a DLOAD entry (if defined), scale the angular velocity (ω) and acceleration (α), which are used to compute an inertia force in the equation $F = [m] [\omega \times (\omega \times r)) + \alpha \times r]$. Since ω is squared in the force computation, the resulting scaling is not linearly related to the computed force (F). All other solutions scale the computed force (F).
15. In SOL 401, thermal loads, defined, for example, with the TEMPD, TEMP, or TEMPP1 bulk entries, cannot be referenced by the EXCITEID field in order to define a time-stepped thermal load. If you attempt to reference a thermal load with the EXCITEID field, the software will ignore the definition without issuing a warning or an error. Thermal loads can be selected in a subcase with the TEMP(LOAD) case control command.

Remarks related to SOLs 402 and 414:

1. For SOL 402, beginning with Simcenter Nastran 2021.2, the DLOAD=*n* case control command that selects either the TLOAD1 or DLOAD bulk entries must be defined globally (above all subcases). A fatal error occurs if you define the DLOAD=*n* command in a subcase. You can use the system cell setting NASTRAN SYSTEM(767)= 0 to turn off the fatal error and revert to the previous behavior.

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2. TLOAD1 may be used in a static or transient analysis to specify time-dependent loads. In a nonlinear static analysis, TLOAD1 is frequently used with TSTEP to apply incremental loads to achieve better convergence in the solution.

   In SOL 402, TLOAD1 may be used to define driver loads or enforced displacements.

3. SID may be selected directly with the Case Control command DLOAD = SID, or be combined with other TLOAD1 entries in a DLOAD entry. For the latter case, the SID in the DLOAD entry is then selected by the Case Control command DLOAD.

4. For TYPE = 0, EXCITEID may reference the SID in a FORCE, FORCE1, FORCE2, MOMENT, MOMENT1, MOMENT2, PLOAD, PLOAD1, PLOAD2, PLOAD4, PLOADX1, GRAV, RFORCE, TEMP, or TEMPD entry. SELOAD entries are not supported.

5. For TYPE = 1, EXCITED must reference the SID in an SPCD entry.

6. Delay is ignored.

7. For SOL 402, for all TYPE, EXCITED must reference the SID in an SPCD entry.

Related Topics

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TLOAD2

Transient Response Dynamic Excitation, Form 2

Defines a time-dependent dynamic excitation or enforced motion of the form:

$$\{P(t)\} = \begin{cases} 0 & , \ t < (T1 + \tau) \text{ or } t > (T2 + \tau) \\ A\bar{t}^{B} e^{C\bar{t}} \cos\left(2\pi F\bar{t} + \frac{P\pi}{180}\right) & , \ (T1 + \tau) \le t \le (T2 + \tau) \end{cases}$$

for use in a transient response problem, where:

$$\bar{t} = t - T1 - \tau$$

Format:

1	2	3	4	5	6	7	8	9	10
TLOAD2	SID	EXCITEID	DELAY	TYPE	T1	T2	F	P	
	C	B							

Examples:

TLOAD2	4	10	106		2.1	4.7	12.0		
	2.0								

TLOAD2	4	10	0.01		2.1	4.7	12.0		
	2.0	0.4							

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
EXCITEID	Identification number of a load set (for example, the DAREA entry), SPCD entry, SELOAD entry, or a thermal load set (for heat transfer analysis) that lists each degree-of-freedom to apply the excitation and the corresponding scale factor, *A*, for the excitation. See Remark 2 and Remark 3. (Integer > 0)
DELAY	Time delay, τ. (Real or Integer ≥ 0 blank; for default behavior, see Remark 6) If real entry, value of τ for all degrees-of-freedom in EXCITEID entry.

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Field	Contents
	If integer entry, identification number of DELAY entries that contain values of τ for individual degrees-of-freedom in EXCITEID entry. See Remark 7.
TYPE	Defines the type of the dynamic excitation. See Remark 2 and Remark 3. (Integer, character or blank; Default = 0)
T1	Time constant. (Real ≥ 0.0)
T2	Time constant. (Real; T2 > T1)
F	Frequency in cycles per unit time. (Real ≥ 0.0; Default = 0.0)
P	Phase angle in degrees. (Real; Default = 0.0)
C	Exponential coefficient. (Real; Default = 0.0)
B	Growth coefficient. (Real; Default = 0.0)

Remarks:

1. Dynamic excitation sets must be selected with DLOAD = SID in the case control section.
2. The type of the dynamic excitation is specified by TYPE (field 5) according to the following table:

TYPE	Type of dynamic excitation
0, L, LO, LOA or LOAD	Applied load (force or moment) (Default)
1, D, DI, DIS, or DISP	Enforced displacement using large mass or SPC/SPCD data
2, V, VE, VEL or VELO	Enforced velocity using large mass or SPC/SPCD data
3, A, AC, ACC or ACCE	Enforced acceleration using large mass or SPC/SPCD data

3. TYPE determines the manner in which EXCITEID is used by the program.
   - TYPE = 0: Applied load excitation.
   - TYPE = 1, 2, or 3: Enforced motion excitation.
     - If EXCITEID references SPC/SPCD entries, the software will use the SPCD method of enforced motion. You directly specify displacements, velocities, or accelerations with SPC/SPCD entries.
     - If EXCITEID references a load entry (DAREA, FORCEi, MOMENTi, PLOAD, PLOAD1, RFORCEi, and SELOAD entries), the software will use the large mass method of enforced motion. You create a large mass and load at the grid and degree-of-freedom where the enforced motion is desired. The software then computes the enforced motion the same for TYPE = 1, 2, and 3.

       Type = 1, Enforced displacement = Dynamic load input / Large mass
       Type = 2, Enforced velocity = Dynamic load input / Large mass
       Type = 3, Enforced acceleration = Dynamic load input / Large mass

   See the “Enforced Motion” chapter in the *Simcenter Nastran Basic Dynamic Analysis User’s Guide* for details on both the SPCD and large mass methods of enforced motion.

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4. The legacy method of selecting a FORCEi, MOMENTi, RFORCEi, PLOAD, and PLOAD1 entry for a dynamic loading required the LOADSET case control command which selected the LSEQ bulk entry, which in turn selected the load entry. Since the FORCEi, MOMENTi, PLOAD, PLOAD1, and SPCD entries are now selected directly with the EXCITEID on the dynamic load entries DLOADi, TLOADi, and RLOADi, the LOADSET case control and LSEQ bulk entry are no longer required. The legacy method is still supported, and is described on the LSEQ entry remarks.
5. EXCITEID (field 3) may reference sets containing QHBDY, QBDYi, QVECT, and QVOL entries when using the heat transfer option.
6. If the DELAY field is blank or zero (either integer zero or real zero), the corresponding value for τ used by the software is real zero.
7. For degrees-of-freedom in the EXCITEID entry that are not specified on DELAY entries, the software uses real zero as the value for τ.
8. TLOAD1 loads may be combined with TLOAD2 loads using a DLOAD bulk entry.
9. SID must be unique for all TLOAD1, TLOAD2, RLOAD1, RLOAD2, RLOADEX, ACSRCE, and SELOAD entries.
10. If the heat transfer option is used, the referenced QVECT entry may also contain references to functions of time, and therefore *A* may be a function of time.
11. If TLOADi entries are selected in SOL 111 or 146, a Fourier analysis is used to transform the time-dependent loads on the TLOADi entries to the frequency domain and them combine them with loads from RLOADi entries. Then the analysis is performed as a frequency response analysis but the solution and the output are converted to and printed in the time domain. In this case, *B* will be rounded to the nearest integer. See the “Fourier Transform” section of the *Simcenter Nastran Advanced Dynamic Analysis User’s Guide*.
12. The continuation entry is optional.
13. TLOAD2 is not supported in SOL 402.

Related Topics

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TLOAD3

Time-dependent load defined with the FORCDST entry in SOL 401

References a time-dependent load defined with the FORCDST entry in SOL 401. The load has the form:

$$\left\{\overrightarrow{P(t)}\right\} = \left\{\begin{matrix}\overrightarrow{F}_D\left(t\right)\\ \overrightarrow{M}_D\left(t\right)\end{matrix}\right\} = \left\{\begin{matrix}F^*\left\{\begin{matrix}FN1{*}TFN1(t-\tau)\\ FN2{*}TFN2(t-\tau)\\ FN3{*}TFN3(t-\tau)\end{matrix}\right\}\\ M^*\left\{\begin{matrix}MN1{*}TMN1(t-\tau)\\ MN2{*}TMN2(t-\tau)\\ MN3{*}TMN3(t-\tau)\end{matrix}\right\}\end{matrix}\right\}$$

where F, FNi, M, and MNi are defined on the FORCDST entry, and TFNi, TMNi, and τ are defined on the TLOAD3 entry. The time steps (t) are defined with the TSTEP1 entry.

Format:

1	2	3	4	5	6	7	8	9	10
TLOAD3	SID	EXCITEID	DELAY						
		TFN1	TFN2	TFN3		TMN1	TMN1	TMN1	

Example:

TLOAD3	5	7	0.01		13				
		100	101	102		103	104	105	

Fields:

Field	Contents
SID	Load set identification number. (Integer>0)
EXCITEID	Identification number of a FORCDST bulk entry. (Integer>0)
DELAY	Time delay τ for all degree-of-freedom referenced by the FORCDST entry. (Real≥0.0 or blank; Default=0.0).

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Field	Contents
TFNi	Identification number of TABLEDi entry that defines the time dependent scale factors for FNi force components defined on the FORCDST entry. (Blank or Integer>0; Default=1.0). See Remark 2.
TMNi	Identification number of TABLEDi entry that defines the time dependent scale factors for MNi moment components defined on the FORCDST entry. (Blank or Integer>0; Default=1.0). See Remark 2.

Remarks:

1. A single TLOAD3 entry can be selected directly by the DLOAD=SID case control command, or by the DLOAD bulk entry to combine and optionally scaled with other TLOAD3 or TLOAD1 bulk entries.
2. When a blank value is used for TFNi or TMNi, the software applies a unit scale factor for all times.
3. SID must be unique on all TLOAD1 and TLOAD3 entries.

Related Topics

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TRIM

Trim Variable Constraint

Specifies constraints for aeroelastic trim variables. The SPLINE1 and SPLINE4 entries need to be here for the finite plate spline.

Format:

1	2	3	4	5	6	7	8	9	10
TRIM	ID	MACH	Q	LABEL1	UX1	LABEL2	UX2	AEQR	
	LABEL3	UX3	-etc.-						

Example:

TRIM	1	0.9	100.	URDD3	1.0	ANGLEA	7.0	0.0	
	ELEV	0.2							

Fields:

Field	Contents
SID	Trim set identification number. (Integer > 0)
MACH	Mach number. (Real ≥ 0.0 and ≠ 1.0)
Q	Dynamic pressure. (Real > 0.0)
LABELi	The label identifying aerodynamic trim variables defined on an AESTAT or AESURF entry. (Character. See **Bulk Data Syntax Rules**.)
UXi	The magnitude of the aerodynamic extra point degree-of-freedom. (Real)
AEQR	Flag to request a rigid trim analysis (Real ≥ 0.0 and ≤ 1.0, Default1.0. A value of 0.0 provides a rigid trim analysis, see Remark **4**.

Remarks:

1. The TRIM entry must be selected with the Case Control command TRIM=SID.
2. The selected TRIM entry specifies the constrained values of the listed extra point degrees-of-freedom (“trim variables”) for a particular loading condition. These variables are listed on AESTAT and/or AESURF entries.
3. If MACH is less than 1.0, the Doublet-Lattice theory is used. If MACH is greater than 1.0, the ZONA51 theory is used if licensed to do so.

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4. AEQR=0.0 can be used to perform a rigid trim analysis (ignoring the effects of structural deformation on the loading). AEQR=1.0 provides standard aeroelastic trim analysis. Intermediate values are permissible, but have no physical interpretation (they may be useful for model checkout).

Related Topics

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TSTEP

Transient Time Step

Defines time step intervals at which a solution will be generated and output in transient analysis.

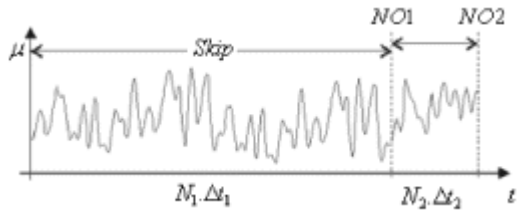
Format:

1	2	3	4	5	6	7	8	9	10
TSTEP	SID	N1	DT1	NO1					
		N2	DT2	NO2					
		-etc.-							

Example:

TSTEP	101	9000	.001	9000					
		1000	.001	1					

In the above example, 10000 time steps with increment .001 seconds are computed, the first 9000 time steps are skipped, and time steps 9001 thru 10000 are output.



Fields:

Field	Contents
SID	Set identification number. (Integer > 0)
Ni	Number of time steps of value DTi. (Integer ≥ 1)
DTi	Time increment. (Real > 0.0)
NOi	Skip factor for output. Every NOi-th step will be saved for output. (Integer > 0; Default=1)

Remarks:

1. TSTEP entries must be selected with the Case Control command TSTEP=SID.

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2. Note that TSTEP permits changes in the size of the time step during the course of the solution. In the above example, 10000 time steps with increment .001 seconds are computed, the first 9000 time steps are skipped, and time steps 9001 thru 10000 are output.
3. See "Guidelines for Effective Dynamic Analysis " in the *Simcenter Nastran Basic Dynamic Analysis User's Guide* for a discussion of considerations leading to the selection of time steps.
4. In modal frequency response analysis (SOLs 111 and 146), this entry is required only when TLOADi is requested; i.e., when Fourier methods are selected.
5. The maximum and minimum displacement at each time step and the SIL numbers of these variables can be printed by altering DIAGON(30) before the transient module TRD1 and by altering DIAGOFF(30) after the module. This is useful for runs that terminate due to overflow or excessive run times.
6. For heat transfer analysis in SOL 159, use the TSTEPNL entry.

Remarks related to SOLs 402 and 414:

1. Note that TSTEPNL is not supported in SOL 402.

Related Topics

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TSTEP1

Time step intervals for SOLs 401, 402, and 414.

Defines the time/solution points at which a solution will be computed and optionally output for SOLs 401, 402, and 414.

Format:

1	2	3	4	5	6	7	8	9	10
TSTEP1	SID	Tend1	Ninc1	Nout1					
		Tend2	Ninc2	Nout2					
		Tend3	Ninc3	Nout3					
		-etc.-							

Example:

TSTEP1	1	10.0	5	2					
		50.0	4	3					
		100	2	ALL					

In this example, assuming a start time*=0.0 for the subcase, the resulting time points are as follows. The time points, in which output occurs, are highlighted.

The 1st row has an end point of 10.0, 5 time points, and output at every 2nd time point.

Time Point 1	Time Point 2	Time Point 3	Time Point 4	Time Point 5
2.0	4.0	6.0	8.0	10.0

The 2nd row has an end time point of 50.0, 4 time points, and output at every 3rd time point.

Time Point 1	Time Point 2	Time Point 3	Time Point 4
20.0	30.0	40.0	50.0

The 3rd row has an end time of 100.0, 2 increments, and output at all time points.

Time Point 1	Time Point 2
75.0	100.0

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In the same example, assuming a start time*=5.0 for the subcase, the resulting time points for the first row are as follows.

*See the remarks for the definition of start time.

Time Point 1	Time Point 2	Time Point 3	Time Point 4	Time Point 5
6.0	7.0	8.0	9.0	10.0

The 2nd and 3rd row are the same:

Time Point 1	Time Point 2	Time Point 3	Time Point 4
20.0	30.0	40.0	50.0

Time Point 1	Time Point 2
75.0	100.0

Fields:

Field	Contents
SID	Identification number. (Integer > 0)
Tendi	End time (REAL≥ 0.0; $Tend_i < Tend_{i+1}$; Default = 1.0 for Tend1). See Remark **4**.
Ninci	Number of increments (Integer > 0; Default = 1).
Nouti	Controls the frequency of output. See **Example**. (Character = "YES", "END", "ALL", "CPLD", or Integer ≥ 0; Default = "YES"). See Remark **5**.

Remarks:

1. The TSTEP1 entry applies to:
   - Static and dynamic time-based subcases in SOLs 401 and 402.
   - Static and preload constant time subcases in SOLs 401 and 402. For example, a preload subcase always has the same end time and start time, and as a result, is constant time subcase.
2. The TSTEP1 entry must be selected with the TSTEP=SID case control command.
3. For SOL 401, the start time (Tstart) for a subcase is determined as follows:
   - If a subcase definition in the case control includes SEQDEP=NO, that subcase is not sequentially dependent (NSD). The start time for an NSD subcase is 0.0.

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- For a sequentially dependent (SD) static subcase (default), the final Tendi from a previous SD or NSD static subcase is the start time (Tstart) for the current SD subcase. If an SD subcase has no previous SD or NSD static subcases, the start time is 0.0 for that SD subcase.

4. A constant-time subcase has a TSTEP1 bulk entry defined with a Tend which is the same as the start time for that subcase. All bolt preload subcases are constant time. See Preload subcase.

Static subcases can be defined with a constant time if they are before all other subcases, and as a result, have Tend=0.0. See Static subcase.

The following conditions apply to a constant-time subcase:

- Only a single Tendi is permitted.
- Ninc can be defined to divide loads, temperatures, and contact offsets into incremental solution points. This is useful to help convergence. The initial increment of load, temperature, and contact offset will be 1/NINC.
- For a preload subcase, which includes a bolt preload sequence, the number of incremental solution points (Ninc) for each sequence can optionally be defined on the BOLTSEQ bulk entry. If Ninc is defined for a bolt preload sequence on the BOLTSEQ entry, it overrides the value defined on the TSTEP1 entry.
- A constant time subcase does not consider the time stepping criteria of plastic materials, creep materials, user defined materials (UMAT), time stepping based on equilibrium iterations (TSCEQ), or element addition and removal (ELAR).
- If the ELAR command is defined in a constant time subcase, element addition happens immediately at the start of the subcase (at load factor = 0), and element removal is not allowed.
- A constant time subcase does consider non-convergence bisection that activated by material or element, such as material integration, element distortion and negative Jacobian in large strain analysis.

5. Nouti controls the frequency of results output. The table below summarizes the input options. See SOL 402 remark 4.

Nout	Output frequency
YES (Default)	SOL 401: Output occurs at all time points defined by the TSTEP1 entry, and all solution points as a result of using Ninc in a constant time subcase. Output does not occur at any time/solution points created when the software subincrements. SOL 402: Except for the PRELOAD subcase, output occurs at all time points defined by TSTEP1. For the PRELOAD subcase, output occurs only at the final solution point. Output does not occur at any time/solution points created when the software subincrements.
END	SOL 401 and SOL 402: Output occurs at the end time point.
ALL	SOL 401 and SOL 402: Output occurs at all time points defined by the TSTEP1 entry, at all solution points as a result of using Ninc in a constant time subcase, and any time/solution points created when the software subincrements. See SOL 402 Remark 5.

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	Note: When Nouti=ALL in the context of the Simcenter 3D Multiphysics solution, the output is a combination from both the structural and coupled analyses.
Integer = 0	SOL 401 and SOL 402: No output occurs.
Integer > 0	SOL 401 and SOL 402: Output occurs at every Nout time point defined by the TSTEP1 entry, and at every Nout solution point as a result of using Ninc in a constant time subcase. See **Example**.
CPLD	SOL 401 only: Output occurs only at time/solution points of coupled analysis. This option can only be defined by the Simcenter 3D Multiphysics environment.

Remarks related to SOLs 402 and 414:

1. The starting time point (Tstart) for a time-based subcase is always the ending time point from the previous time-based subcase, regardless of the sequential dependency.
2. Both TSTEP and TSTEP1 entries cannot be used in the same model.
3. Even if NOUT=0, output will always occur at the very end of the subcase.
4. For the PRELOAD subcases only, output only occurs at the final solution point by default. However, output can occur at all solution points, if Nouti=ALL and Ninc>1.
5. If Nout=ALL is defined in any line of a TSTEP1 entry, the software will apply ALL to all lines in that entry. As a result, output will occur at all time/solution points defined by that TSTEP1 entry, and at all time/solution points created when the software subincrements.

Related Topics

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TSTEPNL

Parameters for Nonlinear Transient Analysis

Defines parametric controls and data for nonlinear transient structural or heat transfer analysis. TSTEPNL is intended for SOLs 129 and 159.

Format:

1	2	3	4	5	6	7	8	9	10
TSTEPNL	ID	NDT	DT	NO		KSTEP	MAXITER	CONV	
	EPSU	EPSP	EPSW	MAXDIV	MAXQN	MAXLS	FSTRESS		
	MAXBIS	ADJUST	MSTEP	RB	MAXR	UTOL	RTOLB	KDAMP	
	KUPDATE								

Example:

TSTEPNL	250	100	.01	1		2	10	PW	
	1.E-2	1.E-3	1.E-6	2	10	2	.02		
	5	5	0	0.75	16.0	0.1	20.		

Fields:

Field	Contents
ID	Identification number. (Integer > 0)
NDT	Number of time steps of value DT. See Remark 2 . (Integer ≥ 4)
DT	Time increment. See Remark 2 . (Real > 0.0)
NO	Time step interval for output. Every NO-th step will be saved for output. See Remark 3 . (Integer > 0; Default=1)
KSTEP	KSTEP is the number of converged bisection solutions between stiffness updates. (Integer > 0; Default=2)
MAXITER	Limit on number of iterations for each time step. See Remark 4 . (Integer ≠ 0; Default=10)
CONV	Flags to select convergence criteria. See Remark 5 . (Character: “U”, “P”, “W”, or any combination; Default=“PW.”)
EPSU	Error tolerance for displacement (U) criterion. (Real > 0.0; Default=1.0E-2)
EPSP	Error tolerance for load (P) criterion. (Real > 0.0; Default=1.0E-3)
EPSW	Error tolerance for work (W) criterion. (Real > 0.0; Default=1.0E-6)

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Field	Contents
MAXDIV	Limit on the number of diverging conditions for a time step before the solution is assumed to diverge. See Remark 6 . (Integer > 0; Default=2)
MAXQN	Maximum number of quasi-Newton correction vectors to be saved on the database. See Remark 7 . (Integer ≥ 0; Default=10)
MAXLS	Maximum number of line searches allowed per iteration. See Remark 7 . (Integer ≥ 0; Default=2)
FSTRESS	Fraction of effective stress $(\bar{\sigma})$ used to limit the subincrement size in the material routines. See Remark 8 . (0.0 < Real < 1.0; Default=0.2)
MAXBIS	Maximum number of bisections allowed for each time step. See Remark 9 . (-9 ≤ Integer ≤ 9; Default=5)
ADJUST	Time step skip factor for automatic time step adjustment. See Remark 10 . (Integer ≥ 0; Default=5)
MSTEP	Number of steps to obtain the dominant period response. See Remark 11 . (10 ≤ Integer ≤ 200; Default=variable between 20 and 40.)
RB	Define bounds for maintaining the same time step for the stepping function during the adaptive process. See Remark 11 . (0.1 ≤ Real ≤ 1.0; Default=0.75)
MAXR	Maximum ratio for the adjusted incremental time relative to DT allowed for time step adjustment. See Remark 12 . (1.0 ≤ Real ≤ 32.0; Default=16.0)
UTOL	Tolerance on displacement or temperature increment below which a special provision is made for numerical stability. See Remark 13 . (0.001 < Real ≤ 1.0; Default=0.1)
RTOLB	Maximum value of incremental rotation (in degrees) allowed per iteration to activate bisection. See Remark 14. (Real > 2.0; Default=20.0)
KDAMP	Option to include differential stiffness in calculating structural damping. (Character= "YES", "NO"; Default = "NO")
KUPDATE	Method for controlling stiffness updates. See Remark 15. (Character = "AUTO", "FULL"; Default = "AUTO")

Remarks:

1. The TSTEPNL Bulk Data entry is selected by the Case Control command TSTEP=ID. Each residual structure subcase requires a TSTEP entry and either applied loads via TLOADi data or initial values from a previous subcase. Multiple subcases are assumed to occur sequentially in time with the initial values of time and displacement conditions of each subcase defined by the end conditions of the previous subcase.
2. NDT is used to define the total duration for analysis, which is NDT * DT. Since DT is adjusted during the analysis , the actual number of time steps, in general, will not be equal to NDT). Also, DT is used only as an initial value for the time increment.

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3. For printing and plotting the solution, data recovery is performed at time steps 0, NO, 2 * NO, ..., and the last converged step. The Case Control command OTIME may also be used to control the output times.

4. The number of iterations for a time step is limited to MAXITER. If MAXITER is negative, the analysis is terminated when the divergence condition is encountered twice during the same time step or the solution diverges for five consecutive time steps. If MAXITER is positive, the program computes the best solution and continues the analysis until divergence occurs again. If the solution does not converge in MAXITER iterations, the process is treated as a divergent process. See Remark 6 .

5. The convergence test flags (U=displacement error test, P=load equilibrium error test, W=work error test) and the error tolerances (EPSU, EPSP, and EPSW) define the convergence criteria. All requested criteria (combination of U, P, and/or W) are satisfied upon convergence. Note that at least two iterations are necessary to check the displacement convergence criterion.

   When KUPDATE=AUTO, EPSP is used to detemine if the stiffness is updated, even when P is not included in the CONV field. This can have an effect on convergence or results. See “Update Principles” in Section 3.3.1 of the *Simcenter Nastran Handbook of Nonlinear Analysis*.

6. MAXDIV provides control over diverging solutions. Depending on the rate of divergence, the number of diverging solutions (NDIV) is incremented by 1 or 2. The solution is assumed to diverge when NDIV reaches MAXDIV during the iteration. If the bisection option is used (allowed MAXBIS times) the time step is bisected upon divergence. Otherwise, the solution for the time step is repeated with a new stiffness based on the converged state at the beginning of the time step. If NDIV reaches MAXDIV again within the same time step, the analysis is terminated.

7. Nonzero values of MAXQN and MAXLS will activate the quasi-Newton update and the line search process, respectively.

   EPSW is used to detemine if line search is performed, even when W is not included in the CONV field. This can have an effect on convergence or results. See “Other Provisions for Line Search” in section 3.4.5 of the *Simcenter Nastran Handbook of Nonlinear Analysis*.

8. The number of subincrements in the material routines is determined such that the subincrement size is approximately FSTRESS. FSTRESS is also used to establish a tolerance for error correction in elastoplastic material, i.e.,

   $$\text{error in yield function} < \text{FSTRESS} \cdot \text{yield stress}$$

   If the limit is exceeded at the converging state, the program will terminate with a fatal error message. Otherwise, the stress state is adjusted to the current yield surface, resulting in $\delta=0$.

9. The bisection process is activated when divergence occurs and MAXBIS $\neq 0$. The number of bisections for a time increment is limited to |MAXBIS|. If MAXBIS is positive and the solution does not converge after MAXBIS bisections, the best solution is computed and the analysis is continued to the next time step. If MAXBIS is negative and the solution does not converge in |MAXBIS| bisection, the analysis is terminated.

10. ADJUST controls the automatic time stepping. Since the automatic time step adjustment is based on the mode of response and not on the loading pattern, it may be necessary to limit the adjustable step size when the period of the forcing function is much shorter than the period of dominant response frequency of the structure. It is the user’s responsibility to ensure that the loading history is properly traced with the ADJUST option. The ADJUST option should be suppressed for the duration of short

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pulse loading. If unsure, start with a value for DT that is much smaller than the pulse duration in order to properly represent the loading pattern.

- If ADJUST=0, then the automatic adjustment is deactivated. This is recommended when the loading consists of short duration pulses.
- If ADJUST > 0, the time increment is continually adjusted for the first few steps until a good value of Δt is obtained. After this initial adjustment, the time increment is adjusted every ADJUST-th time step only.
- If ADJUST is one order greater than NDT, then automatic adjustment is deactivated after the initial adjustment.

11. MSTEP and RB are used to adjust the time increment during analysis. The recommended value of MSTEP for nearly linear problems is 20. A larger value (e.g., 40) is required for highly nonlinear problems. By default, the program automatically computes the value of MSTEP based on the changes in the stiffness.

The time increment adjustment is based on the number of time steps desired to capture the dominant frequency response accurately. The time increment is adjusted as follows:

$$\Delta t_{n+1} = f(r)\Delta t_n$$

where

$$r = \frac{1}{\text{MSTEP}}\left(\frac{2\pi}{\omega_n}\right)\left(\frac{1}{\Delta t_n}\right)$$

with

f	=	0.25	for	$r < 0.5 \cdot \text{RB}$
f	=	0.5	for	$0.5 \cdot \text{RB} \le r < \text{RB}$
f	=	1.0	for	$\text{RB} \le r < 2.0$
f	=	2.0	for	$2.0 \le r < 3.0/\text{RB}$
f	=	4.0	for	$r \ge 3.0/\text{RB}$

12. MAXR is used to define the upper and lower bounds for adjusted time step size, i.e.,

$$\text{MIN}\left(\frac{\text{DT}}{2^{\text{MAXBIS}}}, \frac{\text{DT}}{\text{MAXR}}\right) \le \Delta t \le \text{MAXR} \cdot \text{DT}$$

13. UTOL is a tolerance used to filter undesirable time step adjustments; i.e.,

$$\frac{\|\dot{U}_n\|}{\|\dot{U}\|_{max}} < \text{UTOL}$$

Under this condition no time step adjustment is performed in a structural analysis (SOL 129). In a heat transfer analysis (SOL 159) the time step is doubled.

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14. The bisection is activated if the incremental rotation for any degree-of-freedom ($\Delta\theta_x$, $\Delta\theta_y$, $\Delta\theta_z$) exceeds the value specified by RTOLB. This bisection strategy is based on the incremental rotation and controlled by MAXBIS.

15. The following stiffness update methods can be selected in the KUPDATE field.

    - If AUTO is selected, the program automatically selects the most efficient strategy based on convergence rates. At each step, the number of iterations required to converge is estimated. Stiffness is updated, if
      (i) the estimated number of iterations to converge exceeds MAXITER,
      (ii) the estimated time required for convergence with current stiffness exceeds the estimated time required for convergence with updated stiffness, and
      (iii) the solution diverges.

      See Remark 5.

    - If the FULL option is selected, the program updates the stiffness matrix at every Newton-Raphson iteration. For large deflection problems (PARAM,LGDISP > 0), the FULL option is recommended.

Related Topics

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UNBA

Unbalance Load

Defines an unbalance load in SOL 414.

Format:

1	2	3	4	5	6	7	8	9	10
UNBA	SID	G	MASS	ECC	VBAL	PHASEF	SPEED	RID	
		CID	MOMENT	THETA	MBAL	PHASEM			

Example:

UNBA	1	56	0.6	0.35		43.8	10	2	
			8.2	15.8		-14.0			

Fields:

Field	Contents
SID	Load set identification number. (Integer > 0)
G	Grid point identification number where the unbalance load is applied. (Integer > 0; No default)
MASS	Unbalance mass. See Remarks 1 and 6. (Real or blank; No default)
ECC	Eccentricity of unbalance mass. See Remark 1. (REAL > 0.0 or blank; No default)
VBAL	Identification number of TABLEDi entry that contains the unbalance as a function of time. See Remarks 1, 2, and 6. (Integer > 0 or blank; No default)
PHASEF	Initial angle of the eccentric mass that is defined by the MASS and ECC fields, or the VBAL field relative to the rotor coordinate system. (Real; Default = 0.0)
SPEED	Identification number of TABLEDi entry that contains the rotor speed in rad/s as a function of time. (Integer > 0 or blank; See Remark 3 for default)
RID	Rotor identification number. (Integer > 0; No default)
CID	Identification number of the rotor coordinate system. (Integer > 0; Default = 0 for the basic coordinate system)
MOMENT	Difference between the diametral moment and the polar moment of the rotor. See Remarks 1 and 6. (Real or blank; No default)
THETA	Rotor misalignment angle in rad. See Remark 1. (Real or blank; No default)

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Field	Contents
MBAL	Identification number of TABLEDi entry that contains the unbalance moment as a function of time. See Remarks 1, 4, 5, and 6. (Integer > 0 or blank; No default)
PHASEM	Initial angle of the eccentric mass that is defined by the MOMENT and THETA fields, or the MBAL field relative to the rotor coordinate system. (Real; Default = 0.0)

Remarks:

1. If the MASS, ECC, and VBAL fields are all specified, the VBAL field takes precedence and the software ignores the MASS and ECC field specifications.

   If the MOMENT, THETA, and VBAL fields are all specified, the MBAL field takes precedence and the software ignores the MOMENT and THETA field specifications.

2. Unbalance is the product of the unbalance mass and eccentricity.

3. The angular speed for the unbalance calculation depends on the SPEED field as follows:

   - If the SPEED field is blank, the software uses the speed of the rotor that contains the grid specified in the G field.
   - If the SPEED field is specified, the software uses the lookup value for the TABLEDi entry whose ID is specified in the SPEED field.

4. The software calculates the unbalance moment from the MOMENT and THETA field values as follows:

   $$M_{unbalance} = M\Theta\omega^2$$

   where M is the value in the MOMENT field, θ is the value in the THETA field, and ω is the rotor speed.

   The software calculates the unbalance moment from the MBAL field value as follows:

   $$M_{unbalance} = M_{bal}\omega^2$$

   where M_{bal} is the value for the MBAL field and ω is the rotor speed.

5. To account for an unbalance force only, do not specify the moment unbalance-related fields (MOMENT, THETA, MBAL, PHASEM).

   To account for an unbalance moment only, do not specify force unbalance-related fields (MASS, ECC, VBAL, PHASEF).

   To account for an unbalance force and moment, specify the force unbalance and moment unbalance-related fields (MASS, ECC, VBAL, PHASEF, MOMENT, THETA, MBAL, PHASEM).

6. If you set the mass parameters in weight units, add the **WTMASS** parameter to the input file to convert the weight units to mass units.

Related Topics

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USET

Degree-of-Freedom Set Definition

Defines a degree-of-freedom set.

Format:

1	2	3	4	5	6	7	8	9	10
USET	SNAME	ID1	C1	ID2	C2	ID3	C3		

Example:

USET	U4	333	26	17	0				

Fields:

Field	Contents
SNAME	Set name. (One to four characters, or the string “ZERO” followed by the set name.)
IDi	Grid or scalar point identification number. (Integer > 0)
Ci	Component number. (Integer zero or blank for scalar points, or any unique combinations of the Integers 1 through 6 for grid points with no embedded blanks.)

Remarks:

1. SNAME may refer to any of the set names given in **“Degree-of-Freedom Sets”** or their new names on the DEFUSET entry. However, it is recommended that SNAME refer only to the set names U1 through U9 or their new names on the DEFUSET entry. If set names a through v are used then the degrees-of-freedom may also have to be defined in the applicable super sets on other USETi entries.
2. If SNAME=“ZEROi”, where i is a set name, then the degrees-of-freedom are omitted from set i.
3. A maximum of 18 degrees-of-freedom may be designated on a single entry.
4. If degrees-of-freedom defined by USET entries are found to be singular and AUTOSPC is requested for a degree-of-freedom that is also in a set that AUTOSPC may change, then the set defined by the USET entry will be removed by the AUTOSPC operation. An avoidance is to use PARAM,AUTOSPC,NO.
5. The USET entry is processed by the GP4 module with its effect appearing in the USET table. User-written DMAPs must therefore include the GP1 and GP4 modules if USET entries are used.
6. If a USETi Bulk Data entry lists a standard Simcenter Nastran set, such as S or M, the program may fail in the PARTN module with the message “SYSTEM FATAL MESSAGE 3007, ILLEGAL INPUT TO SUBROUTINE”. This entry should only reference new sets defined on DEFUSET Bulk Data entries.

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7. The user defined degree-of-freedom sets U1 – U9 are created with the USET and USET1 bulk entries. Be aware that the user defined sets U2 - U8 are used by Simcenter Nastran in some special cases. You may use U2 – U8 as long as it doesn't conflict with these cases. See **"User Defined Degree-of-Freedom Sets"** for the list of these special cases.
8. Use the USETPRT and USETSEL parameters to request a tabular print out of degrees-of-freedom sets.

Related Topics

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USET1

Degree-of-Freedom Set Definition, Alternate Form

Defines a degrees-of-freedom set.

Format:

1	2	3	4	5	6	7	8	9	10
USET1	SNAME	C	ID1	ID2	ID3	ID4	ID5	ID6	
	ID7	ID8	-etc.-						

Example:

USET1	SB	345	2	1	36	5	9	7	
	40								

Alternate Format and Example:

USET1	SNAME	C	ID1	“THRU”	ID2				
USET1	SB	123	170	THRU	180				

Fields:

Field	Contents
SNAME	Set name. (One to four characters or the word “ZERO” followed by the set name.)
C	Component numbers. (Integer zero or blank for scalar points or any unique combinations of the Integers 1 through 6 for grid points with no embedded blanks.)
IDi	Grid or scalar point identification number. (Integer > 0; for “THRU” option, ID1 < ID2.)

Remarks:

1. SNAME may refer to any of the set names given in **“Degree-of-Freedom Sets”** or their new names on the DEFUSET entry. However, it is recommended that SNAME refer only to the set names U1 through U9 or their new names on the DEFUSET entry. If set names a through v are used then the degrees-of-freedom may also have to be defined in the applicable super sets on other USETi entries.
2. If SNAME= “ZEROi”, where i is a set name, then the degrees-of-freedom are omitted from set i.
3. If the alternate format is used, all of the points ID1 through ID2 are assigned to the set.

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4. If degrees-of-freedom defined by USET entries are found to be singular and AUTOSPC is requested for a degree-of-freedom that is also in a set that AUTOSPC may change, then the set defined by the USET entry will be removed by the AUTOSPC operation. An avoidance is to use PARAM,AUTOSPC,NO.
5. The USET1 entry is processed by the GP4 module with its effect appearing in the USET table. User-written DMAPs must therefore include the GP1 and GP4 modules if USET entries are used.
6. If a USETi Bulk Data entry lists a standard Simcenter Nastran set, such as S or M, the program may fail in the PARTN module with the message “SYSTEM FATAL MESSAGE 3007, ILLEGAL INPUT TO SUBROUTINE”. This entry should only reference new sets defined on DEFUSET Bulk Data entries.
7. The user defined degree-of-freedom sets U1 – U9 are created with the USET and USET1 bulk entries. Be aware that the user defined sets U2 - U8 are used by Simcenter Nastran in some special cases. You may use U2 – U8 as long as it doesn't conflict with these cases. See **“User Defined Degree-of-Freedom Sets”** for the list of these special cases.
8. Use the USETPRT and USETSEL parameters to request a tabular print out of degrees-of-freedom sets.

Related Topics

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UXVEC

Control Parameter State

Specification of a vector of aerodynamic control point (extra point) values. These data define the control positions corresponding to user defined nonlinear control forces that have been defined by AEDW, AEPRESS and AEFORCE entries. Only nonzero values need to be defined.

Format:

1	2	3	4	5	6	7	8	9	10
UXVEC	ID								
	LABEL1	UX1	LABEL2	UX2	-etc.-				

Example:

UXVEC	1001								
	THRUST	1.E4	ANGLEA	.015					

Fields:

Field	Contents
ID	Control vector identification number, see Remark 1. (Integer > 0)
LABELi	Controller name. This must correspond to an existing AESURF, AESTAT or AEPARM label. (Character. See **Bulk Data Syntax Rules**.)
UXi	The magnitude of the aerodynamic extra point degree of freedom (Real)

For ETOL, when the error estimate exceeds the value input for the ETOL entry

1. The contour method is employed to develop an improved view
2. The units of the user defined AEPARM controllers are implied by their use on this entry and the corresponding values on the force vector definition. The user must be self-consistent in all uses of the same controller. AESURF controllers are expressed in radians as are the rigid body angles ANGLEA and BETA. The rigid body rates, ROLL, PITCH and YAW are nondimensional rates pb/2V, qc/2V, rb/2V; respectively. V is the velocity and b and c are the reference span and chord lengths, respectively.
3. LABELs that are part of the UX vector that are omitted in the UXVEC specification are assigned a value of 0.0.

Related Topics

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VATVBK

VATV response analysis specification

Defines the Fortran unit number for the ASSIGN INPUTT2 file that contains the Vibro-Acoustic Transfer Vector (VATV) results to be used in a VATV response analysis.

Format:

1	2	3	4	5	6	7	8	9	10
VATVBK		UNITNO							

Example:

VATVBK		33							

Fields:

Field	Contents
UNITNO	Fortran unit number for the INPUTT2 file that contains the VATV results. See Remark 1. (Integer > 0; No default)

Remarks:

1. For specifying UNITNO, an appropriate ASSIGN INPUTT2 statement must be present in the File Management Section (FMS) for the absolute value of UNITNO.

Related Topics

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VATVFS

Structural surface definition for VATV calculation

Defines the structural surfaces of a pressure boundary for the Vibro-Acoustic Transfer Vector (VATV) calculation.

Format:

1	2	3	4	5	6	7	8	9	10
VATVFS	ID		SID1	SID2					

Example:

VATVFS	100		201	202					

Fields:

Field	Contents
ID	Identification number of a set of free surfaces (Integer > 0). See Remark 1.
SID1	Free surface identification number for 3D structural elements. (Integer > 0). The SID1 references a BSURFS bulk entry to define the structural element faces used to describe the free surface region. The element faces on the BSURFS bulk entry must all be on structural elements.
SID2	Free surface identification number for 2D structural elements. (Integer > 0). The SID2 references a BSURF bulk entry to define the structural element faces used to describe the free surface region. The element faces on the BSURF bulk entry must all be on structural elements.

Remarks:

1. ID can be referenced by the VATVFS describer specification on the VATVOUT case control command.

Related Topics

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VCEV

Vitual Crack Extension Vector

Defines virtual crack tip extension vectors.

Format:

1	2	3	4	5	6	7	8	9	10
VCEV	SID	CID							
	N11	N12	N13						
	N21	N22	N23						
	N31	N32	N33						
	N41	N42	N43						
	N51	N52	N53						
	N61	N62	N63						
	N71	N72	N73						

Example:

VCEV	1	5							
	1.0	0.0	0.0						
	0.707	0.707	0.0						
	0.5	0.866	0.0						
	0.0	1.0	0.0						
	–0.5	0.866	0.0						
	–0.707	0.707	0.0						
	–1.0	0.0	0.0						

Fields:

Field	Contents
SID	Virtual crack extension vector identification number. (Integer > 0)
CID	Coordinate system identification number. (Integer ≥ 0; Default = 0)
Ni1, Ni2, Ni3	Components of the i-th virtual crack extension vector in the CID coordinate system. (Real)

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Remarks:

1. Up to seven virtual crack extension vectors can be defined.
2. Vectors do not need to be normalized when they are entered on a VCEV bulk entry. The software will normalize them when computing the j-integral.

Related Topics

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VIEW

View Factor Definition

Defines radiation cavity and shadowing for radiation view factor calculations.

Format:

1	2	3	4	5	6	7	8	9	10
VIEW	IVIEW	ICAVITY	SHADE	NB	NG	DISLIN			

Example:

VIEW	1	1	BOTH	2	3	0.25			

Fields:

Field	Contents
IVIEW	Identification number. (Integer > 0)
ICAVITY	Cavity identification number for grouping the radiant exchange faces of CHBDYi elements. (Integer > 0)
SHADE	Shadowing flag for the face of CHBDYi element. (Character, Default="BOTH") NONE means the face can neither shade nor be shaded by other faces. KSHD means the face can shade other faces. KBSHD means the face can be shaded by other faces. BOTH means the face can both shade and be shaded by other faces. (Default)
NB	Subelement mesh size in the beta direction. (Integer > 0; Default=1)
NG	Subelement mesh size in the gamma direction. (Integer > 0; Default=1)
DISLIN	The displacement of a surface perpendicular to the surface. See **Figure 18-186**. (Real; Default=0.0)

Remarks:

1. VIEW must be referenced by CHBDYE, CHBDYG, or CHBDYP elements to be used.
2. ICAVITY references the cavity to which the face of the CHBDYi element belongs; a zero or blank value indicates this face does not participate in a cavity.

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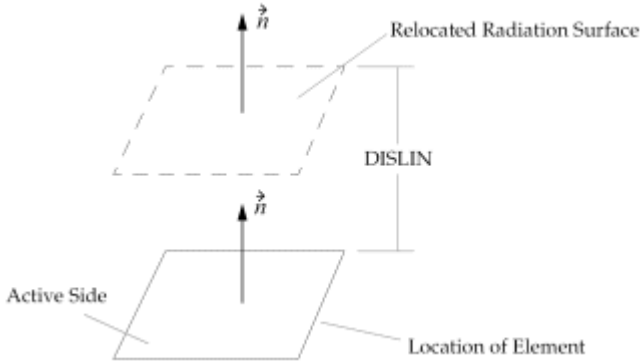
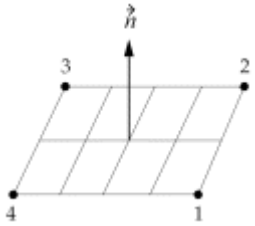
3. NB, NG, and DISLIN are used in the calculation of view factors by finite difference or contour integration techniques. They are not used with the VIEW3D entry.
4. A summary of the shadowing conditions can be requested by the PARAM,MESH,YES Bulk Data entry.
5. SHADE references shadowing for CHBDYi elements participating in a radiation cavity, the VIEW calculation can involve shadowing.
6. DISLIN should only be used with LINE type CHBDYE and CHBDYP surface elements. DISLIN > 0.0 means into the cavity. See Figure 18-186.

Figure 18-186: DISLIN Convention



7. NB and NG define the subelement mesh refinement when using the VIEW module (as opposed to the VIEW3D module) for the calculation of view factors.

Figure 18-187: Typical AREA4 surface element where NB=2 and NG=4



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VIEW3D

View Factor Definition - Gaussian Integration Method

Defines parameters to control and/or request the Gaussian Integration method of view factor calculation for a specified cavity.

Format:

1	2	3	4	5	6	7	8	9	10
VIEW3D	ICAVITY	GITB	GIPS	CIER	ETOL	ZTOL	WTOL	RADCHK	

Example:

VIEW3D	1	2	2	4		1.0E-6			

Fields:

Field	Contents
ICAVITY	Radiant cavity identification number on RADCAV entry. (Integer > 0)
GITB	Gaussian integration order to be implemented in calculating net effective view factors in the presence of third-body shadowing. (Integer 2, 3, 4, 5, 6 or 10; Default=4)
GIPS	Gaussian integration order to be implemented in calculating net effective view factors in the presence of self-shadowing. (Integer 2, 3, 4, 5, 6 or 10; Default=4)
CIER	Discretization level used in the semi-analytic contour integration method. (1 ≤ Integer ≤ 20; Default=4)
ETOL	Error estimate above which a corrected view factor is calculated using the semi-analytic contour integration method. (Real ≥ 0.0; Default=0.1)
ZTOL	Assumed level of calculation below which the numbers are considered to be zero. (Real ≥ 0.0; Default=1.E-10)
WTOL	Assumed degree of warpage above which the actual value of F_{ii} will be calculated. (0.0 ≤ Real ≤ 1.0; Default=0.01)
RADCHK	Type of diagnostic output desired for the radiation exchange surfaces. (Integer; Default=3) RADCHK=-1, No diagnostic output requested RADCHK=1, Grid table and element connectivity RADCHK=2, Surface Diagnostics - Surface type, area, skewness, taper, warpage, grid point sequencing, aspect ratio, and shading flags.

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Field	Contents
	RADCHK=3, Area, view factor, area-view factor product with error estimate, existence flags for partial self-shadowing, third-body shadowing with error estimate, and enclosure summations for view factor. (Default) RADCHK=0, Same as RADCHK=1, 2, and 3 RADCHK=12, Same as RADCHK=1 and 2 RADCHK=13, Same as RADCHK=1 and 3 RADCHK=23, Same as RADCHK=2 and 3

Remarks:

1. For ETOL, when the error estimate exceeds the value input for the ETOL entry, the contour method is employed to develop an improved view factor.
2. For ZTOL, the use of a geometry scale that results in small numerical values of *AiFij* should be avoided.
3. When WTOL is exceeded, the actual value of *Fii* will be calculated when using the adaptive view module. Warpage will not be considered in the calculation of factor *Fij*.

Related Topics

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VISGRP

Poro-Elastic Visualization Elements

Defines poro-elastic visualization elements

Format:

1	2	3	4	5	6	7	8	9	10
VISGRP	GID								

Example:

VISGRP	1								

Fields:

Field	Contents
GID	ID of a GROUP bulk entry that lists the IDs of PLOTEL elements visualization elements created inside the poro-elastic mesh. (Integer > 0)

Remarks:

1. The GROUP entry referenced by GID can select the PLOTEL elements created with the following bulk entries: PLOTEL, PLOTEL3, PLOTEL4, PLOTEL6, PLOTEL8, PLOTTET, PLOTHEX, PLOTPEN, and PLOTPYR.
2. To obtain structural displacement, velocity, and acceleration results at locations inside the poro-elastic domain, you can define PLOTEL elements inside the poro-elastic mesh, select them with a GROUP bulk entry, select the GROUP with the VISGRP entry, and define the DISPLACEMENT, VELOCITY, or ACCELERATION case control commands. The software computes the poro-elastic displacement, velocity, or acceleration on the PLOTEL element grid points. If you do not select PLOTEL elements with the GROUP/ VISGRP entries, the results on those PLOTEL element grid points will be zero, even when they are defined inside the poro-elastic mesh.

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